
M O D E L A N S W E R B A N K

RGUHS MD Community Medicine — Paper I



B A S I C S C I E N C E S , E P I D E M I O L O G Y A N D R E S E A R C H M E T H O D S

2 0 1 6 – 2 0 2 6

173 topics in this volume · from a 688-topic bank · Textbook-grounded model answers · cited Recent Advances

G R O U N D E D I N

Park's Textbook of Preventive and Social Medicine, 28e — the spine for every topic
Gordis Epidemiology 7e — epidemiological method and study design
Handbook of Epidemiology (Ahrens & Pigeot) — advanced epidemiological method
Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e — a second PSM voice on Indian practice
Sunder Lal — Textbook of Community Medicine — Indian community medicine practice
Plotkin's Vaccines, 7ed — vaccines and immunisation
Nelson & Williams — Infectious Disease Epidemiology — infectious disease epidemiology
Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e — public health and preventive medicine breadth
ICMR National Ethical Guidelines 2017 — research ethics in the Indian setting
Browner -- Designing Clinical Research (5e, 2022) — clinical research design
Routine Immunization I Manual For Medical Officers 2024 — current immunisation service delivery
Oxford Textbook of Public Health 7e — health systems and international health



Dr. Harshad Ramineni
@aurapharmlab

Contents

Concepts in public health and the social sciences.....7

Anti-smoking health education for illiterate adolescents.....	7
Approach to health education.....	14
Behaviour change communication.....	21
Contributions of Louis Pasteur to public health.....	29
Dark ages of medicine.....	38
De-professionalization of medicine.....	44
Social stratification.....	51
Natural history of disease — prevention.....	61
Communication process.....	69
Role of family in health and disease.....	75
Role of health education in disease prevention and control.....	83
Service and regulatory approaches to health education.....	91
John Snow's contributions to public health.....	98
Group discussion and group dynamics in health education.....	105
Relevance of anthropology in public health.....	113
Louis Pasteur's contribution to public health.....	122
Triple burden of diseases (onion peel phenomenon).....	130
Principles of health education.....	137
Health education — non-communicable disease control.....	143
Role of medico-social worker in community health.....	155
Women empowerment measures in India.....	163
Methods of communication — individual vs group approach.....	171
Social problems and their prevention.....	179
Behaviour change communication in disease control.....	189
Family in health and disease.....	201
Functions of health communication.....	210
Gender issues and women's health.....	216
Family life education.....	225
Health education approach for a primitive society.....	234
Health education to reduce tobacco hazards.....	243
Health as a relative concept.....	256
Methods of health education.....	263
Poverty and its impact on health.....	274
Problem-based learning in medical education.....	283
Role of leaders in information, education and communication.....	291
Socio-cultural factors in health and disease — social control.....	298

Ronald Ross.....	309
Sanitary awakening.....	316
Social engineering phase.....	325
Social factors influencing the health of people.....	332
Socio-economic status scale.....	342
Theory of contagion.....	354
Treatment adherence and non-adherence in long-term illnesses.....	361
Changing concepts in public health.....	374
Determinants and indicators of health status.....	381
Counselling in HIV/AIDS.....	392
Acculturation.....	402
Social marketing.....	408
Women abuse — a medico-social problem.....	414
Epidemiology.....	427
Epidemiological triad for non-communicable disease.....	428
Study design for breast self-examination training.....	435
Study design for prevalence of obesity.....	447
Nipah virus outbreak management.....	454
Cervical cancer screening programme.....	467
Cancer screening.....	479
Sero-epidemiology.....	490
Cancer registry.....	497
Monitoring versus surveillance.....	508
Bias in research studies.....	518
Mixed study designs.....	527
Adverse events following immunization surveillance.....	535
Criteria for judging causality.....	547
Randomised controlled trials.....	555
Disease surveillance.....	563
Broad Street cholera outbreak.....	576
Uses of epidemiology.....	583
Endemic index.....	590
Case-control vs cohort studies — risk estimation.....	597
Observational study designs.....	604
Validity and reliability in epidemiological research.....	612
Sensitivity and specificity of screening tests.....	620
Randomisation procedures in research.....	628
Experimental epidemiology — steps in RCTs.....	636

Framingham Heart Study.....	643
Burden of coronary heart disease.....	650
Investigation of a waterborne outbreak.....	663
Investigation of a food poisoning outbreak.....	674
Disease surveillance — role of the microbiology laboratory.....	681
Screening for diabetes mellitus.....	692
Mathematical models of infectious diseases.....	702
Cohort study — design, merits and demerits.....	715
Nested case-control study.....	722
Quasi-experimental studies.....	729
Role of surveillance in public health.....	737
Screening for cardiovascular disease in the community.....	750
Sensitivity and specificity issues in HIV screening.....	763
Sentinel surveillance.....	771
Environmental surveillance in reducing morbidity and mortality.....	778
Time trends in epidemiology.....	788
Vaccine efficacy and its impact on disease control.....	795
Web of causation.....	807
Community diagnosis.....	813
Likelihood ratio.....	821
WHO STEPwise approach to NCD risk factor surveillance (STEPS).....	828
Biostatistics.....	838
Cluster sampling technique.....	838
Cluster sampling.....	845
Coefficient of variation.....	853
Correlation and regression coefficients.....	861
Null hypothesis and tests of significance.....	873
Tests of significance in health research.....	880
Sampling methods.....	889
Standard normal curve in health research.....	897
Sampling error and standard error.....	904
Lot quality assurance sampling (LQAS).....	911
Non-parametric tests.....	918
Nonparametric tests of significance in statistics.....	925
Poisson distribution.....	931
Sampling techniques suitable for rapid survey.....	939
Tests of significance.....	946
Box plots in data presentation.....	953

Histogram in data presentation.....	962
Scatter diagram in data presentation.....	976
Life table.....	987
Regression analysis.....	994
Research methodology and ethics.....	1002
Focus group discussion.....	1003
Evidence-based health care.....	1011
Indian Council of Medical Research.....	1023
Meta-analysis.....	1032
Qualitative research methods.....	1045
Social mapping.....	1052
Good clinical practice — guidelines and informed consent.....	1064
Medical entomology and zoonoses.....	1075
Aedes aegypti index – interpretation and control.....	1076
Arthropod-borne diseases and transmission.....	1083
Zoonoses — classification and control.....	1092
One Health approach.....	1104
Itch mite life cycle and scabies.....	1118
Post-exposure prophylaxis for animal bites.....	1132
Challenges in control of arthropod-borne diseases.....	1142
Kyasanur Forest disease (KFD).....	1153
Chemical control of mosquito larvae.....	1164
Rodent-borne diseases and control measures.....	1175
Epidemiology and control of leptospirosis.....	1185
Epidemiology and control of avian influenza.....	1194
Genetically modified vectors in malaria control.....	1207
Vector bionomics in vector control.....	1216
Principles of arthropod control.....	1223
Emerging and re-emerging zoonotic diseases.....	1235
Mosquito control under the Urban Malaria Scheme.....	1247
Toxicity of insecticides.....	1257
Trombiculid mites.....	1265
Vectorial capacity.....	1274
Integrated vector control.....	1281
Environmental health and ecology.....	1291
Safe water for all.....	1291
Urban solid waste management.....	1304
Biological effects of radiation.....	1316

Biomedical Waste (Management and Handling) Rules 1998.....	1323
Climate change and health.....	1336
Hardness of water.....	1349
Carbon footprint and carbon credit.....	1357
Hospital waste management.....	1370
Surveillance of drinking water quality.....	1386
Global warming and its health effects.....	1397
Water pollution.....	1408
Air pollution — effects and control.....	1419
Five R's of waste management.....	1432
Slum dwellers — environmental health factors.....	1443
Town planning and community health.....	1457
Urbanisation and its impact on health.....	1468
Household water purification methods.....	1476
Temperature inversion.....	1483
Health implications of traditional urban planning.....	1489
Indices of thermal comfort in comfort zone.....	1502
Indices of thermal comfort.....	1511
Motor vehicle air pollution control.....	1521
Standards of ventilation.....	1533
Types of disinfection.....	1539
Water conservation.....	1551
Water quality standards.....	1559
Water-related diseases.....	1572
Planning healthy cities.....	1580
Air pollution control in city planning.....	1591
Ideal disinfectant and types of disinfection.....	1603
Solid waste (refuse) — sources and disposal.....	1613

Concepts in public health and the social sciences



Anti-smoking health education for illiterate adolescents

Asked in: Paper II 15-Nov-2021 — RGUHS MD Community Medicine

Definition — Health education is "the process by which individuals and groups of people learn to behave in a manner conducive to the promotion, maintenance or restoration of health" (*John M. Last; Park 28e, Ch 22, p 1014*). Propagating "Smoking Kills" to illiterate adolescents is therefore not a printing exercise but a planned **behaviour change communication** task: the same message has to be re-coded into non-print, spoken and visual channels, delivered through people the adolescent trusts, and carried from awareness through motivation to action (*Park 28e, Ch 22, pp 1016-1017*).

Audience analysis — what "illiterate adolescent" changes in the plan

Feature of the audience	Communication implication
Cannot read leaflets, pack warnings, hoardings or newspaper copy	Printed word is dropped as the primary carrier; the message must be spoken, shown, enacted or sung
Lack of visual literacy as well as health literacy — pictures are also mis-read	Aids must be pre-tested locally; use photographs and three-dimensional models rather than abstract graphics (<i>Sunder Lal, Ch 4, p 155; Park 28e, Ch 22, p 1023</i>)
Below-average educational level	Personal (interpersonal and group) communication is more influential than mass media at this educational level, whereas mass media is more effective among the above-average educated (<i>Park 28e, Ch 22, Table 2, pp 1020-1021</i>)
Adolescent: peer-driven, present-oriented, experimenting	Distant mortality is a weak motive; peer educators and immediate, visible harms carry more weight (<i>Sunder Lal, Ch 10, pp 620-622</i>)
Language, dialect and medical jargon are active barriers	Message in the local dialect, no technical vocabulary (<i>Sunder Lal, Ch 4, p 155</i>)
Often out of school, working or unengaged	Reach them in their own spaces — community, workplace, <i>Anganwadi</i> , youth club — not only in classrooms (<i>Park 28e, Ch 7, p 540</i>)

Principles of health education applied to this message

Principle (<i>Park 28e, Ch 22, pp 1018-1019</i>)	Application to "Smoking Kills" in illiterate adolescents
Credibility	Message must be scientifically true and compatible with local culture; delivered by a respected doctor, teacher or health worker, never by someone who smokes
Interest	Anchor to what adolescents care about — stamina in sport, breath, teeth, skin, smell, money spent, attractiveness — not to statistics
Participation	Adolescents help make the posters, act the street play, run the "no-tobacco" pledge; participation creates involvement and acceptance

Principle (Park 28e, Ch 22, pp 1018-1019)	Application to "Smoking Kills" in illiterate adolescents
Motivation	Awaken a <i>felt need</i> ; use positive incentives (a "carrot") rather than fear alone, as the carrot works better than the stick
Comprehension	Pitch to the mental capacity and literacy of the group; never use words strange or new to them
Reinforcement	Repeat the message at intervals in <i>different ways</i> — few learn it all in a single exposure
Learning by doing	"If I hear, I forget; if I see, I remember; if I do, I know" — role play and demonstration beat a lecture
Known to unknown	Start from the <i>bidi</i> the boy already sees at home and move to lung disease — concrete to abstract, simple to complicated
Setting an example	The educator must not smoke while teaching hazards of smoking
Leaders	Convince the village headman, teacher, <i>Panchayat</i> and peer leader first — they are the agents of change
Feedback	Modify message and channel from the audience's response

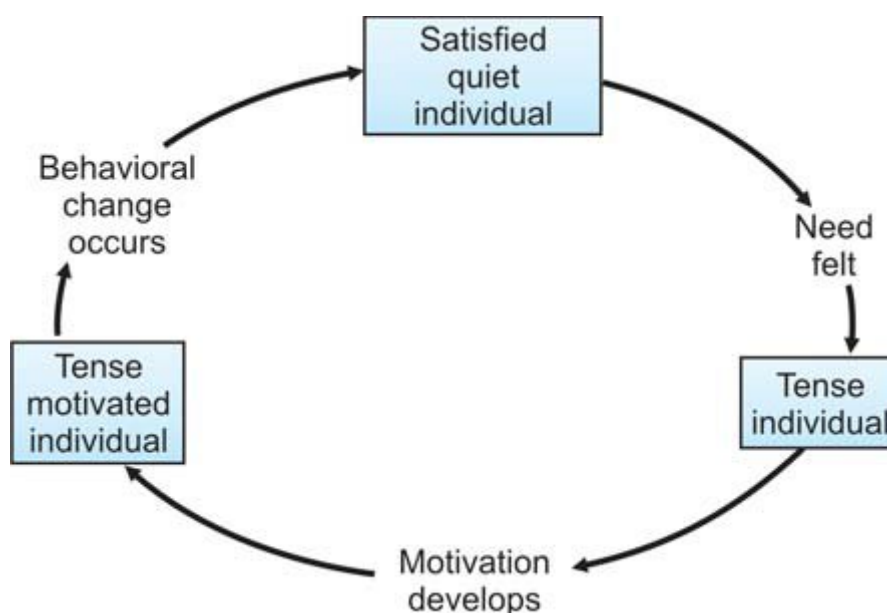
The message itself — translating "Smoking Kills" for this audience

- **Persuade, do not preach.** Persuasive communication is more effective than coercive or authoritative communication and can change lifestyle; the moment it is used to manipulate feelings it degenerates into propaganda (Park 28e, Ch 22, p 1014).
- **Make death concrete and near, not statistical.** About **one million people die every year in India of tobacco-related disease** (Jugal Kishore, Ch 1, p 605), and **half of those killed by tobacco die in middle age, 35-69 years, losing almost 25 years of non-smoker life expectancy** (Park 28e, Ch 20, p 991) — put in the adolescent's words, this is a father dying before his children are grown.
- **Add the harm the adolescent can see today** — breathlessness on running, cough, stained teeth, bad breath, money spent daily on *bidis*.
- **Include the family**, because it converts the adolescent into an advocate at home: second-hand smoke contains **at least 7,000 chemicals** and is a known human carcinogen classified by the International Agency for Research on Cancer, and **52 per cent of Indian adults were exposed to second-hand smoke at home** (Global Adult Tobacco Survey India 2010; Jugal Kishore, Ch 1, p 606).
- **Name the local product, not "tobacco" in the abstract** — *bidi, khaini, gutka, hookah* — and use the local dialect throughout.
- **Attach one clear, doable action** to every session: refuse the first cigarette; if already using, come to the Adolescent Friendly Health Clinic or tobacco cessation service.

Methods and audio-visual aids for a non-reading audience

Approach	Method	Aid used and why it works here
Individual	Personal contact, home visit, counselling at the sub-centre / Ayushman Arogya Mandir	Counselling implies choice, not force; best for the adolescent who has already started and needs to quit (Park 28e, Ch 22, p 1014)
Group	Group discussion — 6-12 members seated in a circle with a leader and recorder	Explicitly the method of choice where long-term compliance is involved, e.g. cessation of smoking (Park 28e, Ch 22, p 1021)
Group	Demonstration	High motivational value — e.g. the smoking-machine/cotton-wool tar demonstration, peak-flow before and after (Park 28e, Ch 22, p 1021)
Group	Role play (socio-drama) and <i>nukkad natak</i> street theatre	Values that cannot be put in words are dramatised; the audience is active, not passive (Park 28e, Ch 22, p 1022)
Group	Chalk-and-talk with flip charts (~25 × 30 cm, flashed one by one) and a flannelgraph (khadi/flannel over a board with sand-paper-backed cut-outs)	Group ≤ 30, talk 15-20 minutes; pictures carry the meaning where words fail, and the pre-arranged picture sequence keeps continuity (Park 28e, Ch 22, pp 1020-1021)
Mass	Radio and television , cinema slides, sound films	Purely auditory and combined audio-visual aids need no reading; radio reaches the widest population
Mass	Folk media — <i>nautanki</i> (Uttar Pradesh), <i>burrakatha</i> (Andhra Pradesh), <i>tamasha</i> (Western India), <i>kirtan</i> , <i>ghazal</i> , <i>qawwali</i> , puppet shows	Indigenous, culturally close, entirely oral-visual; the principal traditional channel for the rural population (Park 28e, Ch 21, pp 1010-1011; Ch 22, p 1024)
Mass	Health exhibition / museum with photographic panels and lit three-dimensional models, explained by a worker	Photographs humanise the message and outperform graphic panels; models outperform photographs (Park 28e, Ch 22, p 1023)

The **pictorial health warning** is the one printed carrier that still works, because it is a picture: rules effective 1 April 2016 mandate warnings on **85 per cent of the principal display area on both sides of the pack (60 per cent picture + 25 per cent text)** (Park 28e, Ch 7, p 567).

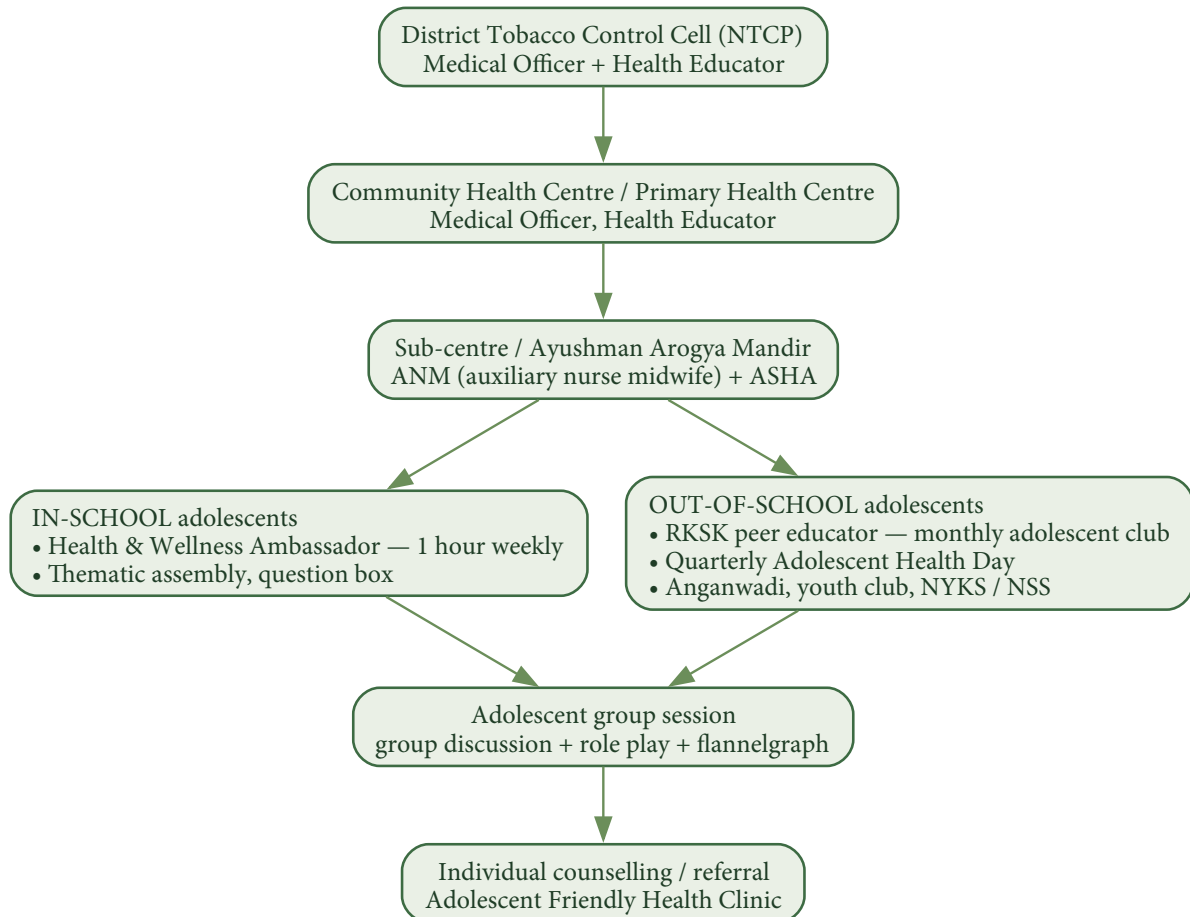


Dynamics of behavioural change (Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e, p.575)

Delivery platform — fitting the campaign into existing Indian programmes

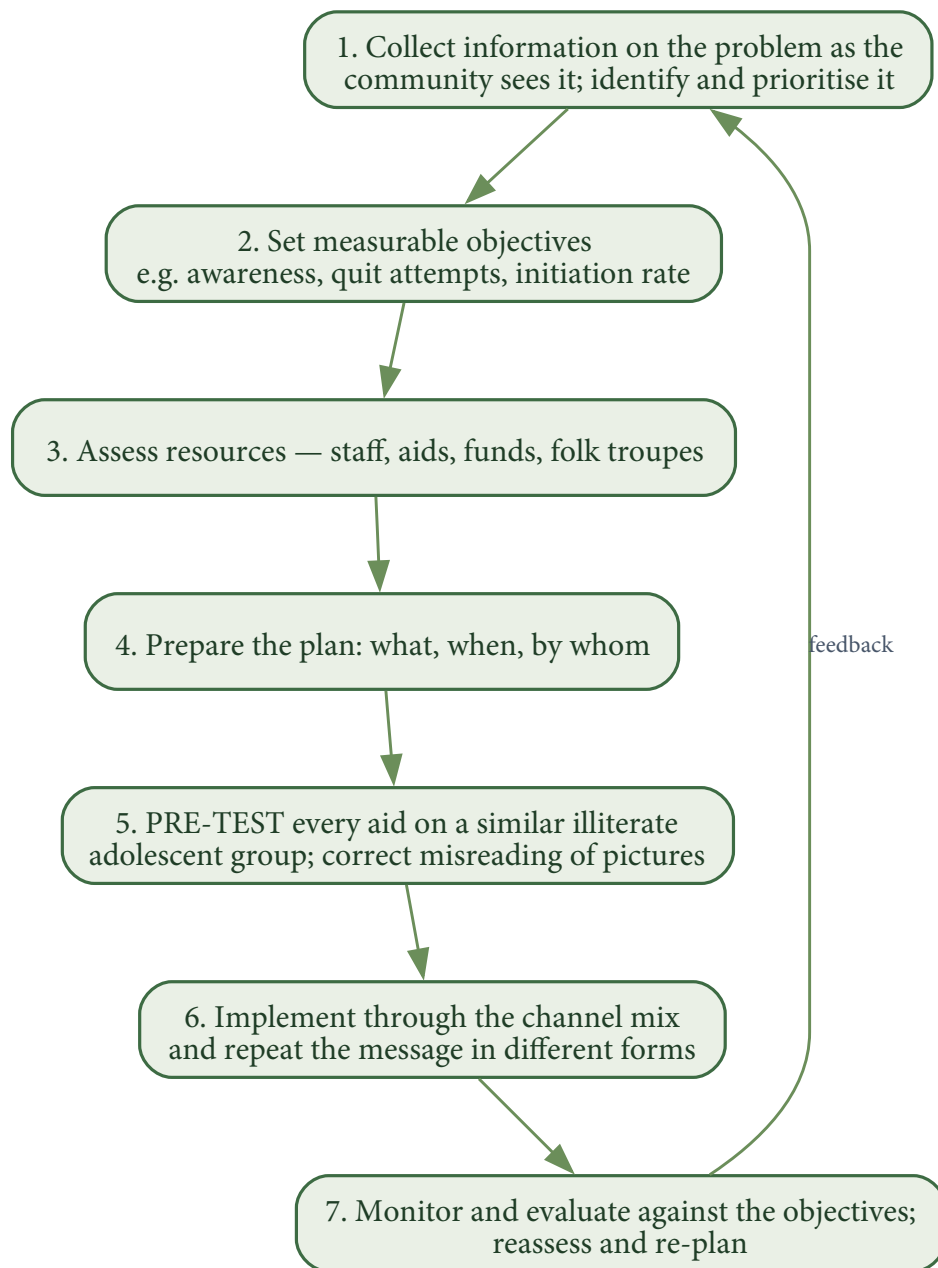
Platform	What it contributes to this campaign
National Tobacco Control Programme (NTCP) — pilot 2007-08 in 16 districts of 9 states, now 108 districts in 31 states	Public-awareness and mass-media campaigns for behaviour change; a school programme ; training of health and social workers, NGOs and school teachers; district tobacco control cells (Park 28e, Ch 7, p 567)
COTPA 2003 (Cigarettes and Other Tobacco Products Act)	The environment that backs the message: no sale to persons below 18 years , no sale within 100 yards of any school, college or educational institution , ban on direct and indirect advertisement, no smoking in public places (in force 2 October 2008) (Jugal Kishore, Ch 2, pp 868-869; Park 28e, Ch 7, p 567)
Rashtriya Kishor Swasthya Karyakram (RKSK), launched January 2014 for 253 million adolescents	Peer-led interventions in the adolescents' own spaces, covering substance misuse : peer educators run an adolescent health club at sub-centre level meeting monthly; quarterly Adolescent Health Days ; Adolescent Friendly Health Clinics ; social and behaviour change communication with focus on interpersonal communication ; school-based interventions (Park 28e, Ch 7, p 540; Sunder Lal, Ch 10, pp 621-622)
School Health Programme under Ayushman Bharat (2018)	Two trained teachers per school as Health and Wellness Ambassadors transacting health promotion as activities for one hour every week ; thematic school assemblies and a question box; students become "Health and Wellness Messengers" (Park 28e, Ch 9, p 685)
Ministry of Youth Affairs and Sports — <i>Nehru Yuva Kendra Sangathan</i> , National Service Scheme	Reaches out-of-school adolescents through youth clubs and life-skills education (Sunder Lal, Ch 10, p 621)

Platform	What it contributes to this campaign
Central Health Education Bureau (1956) and State Health Education Bureaux; DAVP, Press Information Bureau, All India Radio, Doordarshan	Production, pre-testing and supply of audio-visual material and mass-media time (Park 28e, Ch 22, p 1024)



Delivery route for the "Smoking Kills" message to illiterate adolescents

Planning, pre-testing and evaluation



Operational sequence of the campaign, following the steps in health education planning

Evaluation is continuous and measures the **change in knowledge, attitude and practice** produced by the education, plus programme performance and consumer satisfaction (Park 28e, Ch 22, p 1024; Sunder Lal, Ch 4, p 153). Practical indicators: proportion of adolescents correctly naming two harms of smoking (knowledge), proportion who say they will not accept a cigarette from a friend (attitude), current tobacco use prevalence and mean age at initiation in the target group (practice), sessions held per peer educator per month and pre-test-corrected aids in use (process).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Methods in health communication** (Park 28e, Ch 22, Fig 3) — three columns: *Individual approach* (personal contact, home visits, personal letters); *Group approach* (lectures, demonstrations, discussion methods — group discussion, panel discussion, symposium, workshop, conferences, seminars, role play); *Mass approach* (television, radio, newspaper, printed material, direct mailing, posters, health museums and exhibitions, folk methods, internet). Mark with a tick the ones usable without literacy.
- ▶ **Flannelgraph** — label: wooden board, khadi/rough flannel sheet, cut-out picture with sand-paper backing, sequence of cut-outs (healthy lung → smoker's lung → oral cancer → empty purse).

High-yield exam pointers

- ▶ Key discriminator sentence: **mass media reaches the widest population and works best among the above-average educated; personal (interpersonal and group) communication is more influential at average and below-average educational level** — hence the illiterate adolescent is reached mainly by group and folk methods (Park 28e, Ch 22, Table 2).
- ▶ Group discussion size **6-12**; chalk-and-talk group ≤ 30 , talk **15-20 minutes**; flip chart $\sim 25 \times 30$ cm.
- ▶ Learning-by-doing proverb: "*If I hear, I forget; if I see, I remember; if I do, I know.*"
- ▶ Audio-visual aid classification: **auditory** (radio, tape recorder, microphone) / **visual** — **not requiring projection** (poster, chart, flannelgraph, blackboard, leaflet, model, exhibit) / **visual** — **requiring projection** (slides, film strips) / **combined** (television, cinema, slide-tape).
- ▶ COTPA 2003 numbers: sale banned below **18 years** and within **100 yards** of an educational institution; pictorial warning **85 %** of principal display area (**60 % picture + 25 % text**), both sides, from **1 April 2016**.
- ▶ Dated Indian figure: **GATS-2 (2016-17)** — tobacco use fell 6 percentage points from **34.6 %** (GATS-1) to **28.6 %** of adults; **42.4 %** of men and **14.2 %** of women use tobacco in any form; *khaini* and *bidi* are the commonest products (Park 28e, Ch 20, p 991).
- ▶ **WHO MPOWER**: Monitor, Protect, Offer help to quit, Warn about the dangers, Enforce advertising bans, Raise taxes (Jugal Kishore, Ch 1, p 609).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The retrieved corpus gives the National Tobacco Control Programme coverage as 108 districts in 31 states and quotes Global Adult Tobacco Survey round 2 (2016-17); both are the latest figures the indexed textbooks carry, so quote each with its year rather than as a current value.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 22 Communication for Health Education; Ch 21 Health Information; Ch 7 Health Programmes in India; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics; Ch 20 Mental Health); Sunder Lal — Textbook of Community Medicine (Ch 4 Principles of Health Promotion and Education; Ch 10 RMNCH+A); Jugal Kishore — National Health Programmes of India (tobacco control, COTPA 2003, MPOWER); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e (Fig 29.1, dynamics of behavioural change). Named documents: Cigarettes and Other Tobacco Products Act (COTPA) 2003; Global Adult Tobacco Survey India, GATS-2, 2016-17; Operational Guidelines on School Health Programme under Ayushman Bharat, 2018; Rashtriya Kishor Swasthya Karyakram, 2014.

**Approach to health education**

Asked in: Paper I 14-Jul-2020 — RGUHS MD Community Medicine

Definition — There is no single accepted definition of health education; a variety exist, all treating it as a process for inducing behavioural change. *John M. Last* defines it as "the process by which individuals and groups of people learn to behave in a manner conducive to the promotion, maintenance or restoration of health", while the **National Conference on Preventive Medicine, USA** defines it as "a process that informs, motivates and helps people to adopt and maintain healthy practices and lifestyles, advocates environmental changes as needed to facilitate this goal and conducts professional training and research to the same end". Following the Alma-Ata Declaration (1978) the **dynamic definition** is "a process aimed at encouraging people to want to be healthy, to know how to stay healthy, to do what they can individually and collectively to maintain health and to seek help when needed" (Park 28e, Ch 22, pp 1014–1015).

Aims, objectives and the changing concept

- ▶ To encourage people to **adopt and sustain health-promoting lifestyles and practices**.
- ▶ To promote the **proper use of the health services** available to them.
- ▶ To arouse interest, provide new knowledge, improve skills and change attitudes so that people make rational decisions to solve their own problems.
- ▶ To stimulate **individual and community self-reliance and participation**, involving people at every step from identifying the problem to solving it (Park 28e, Ch 22, p 1015).

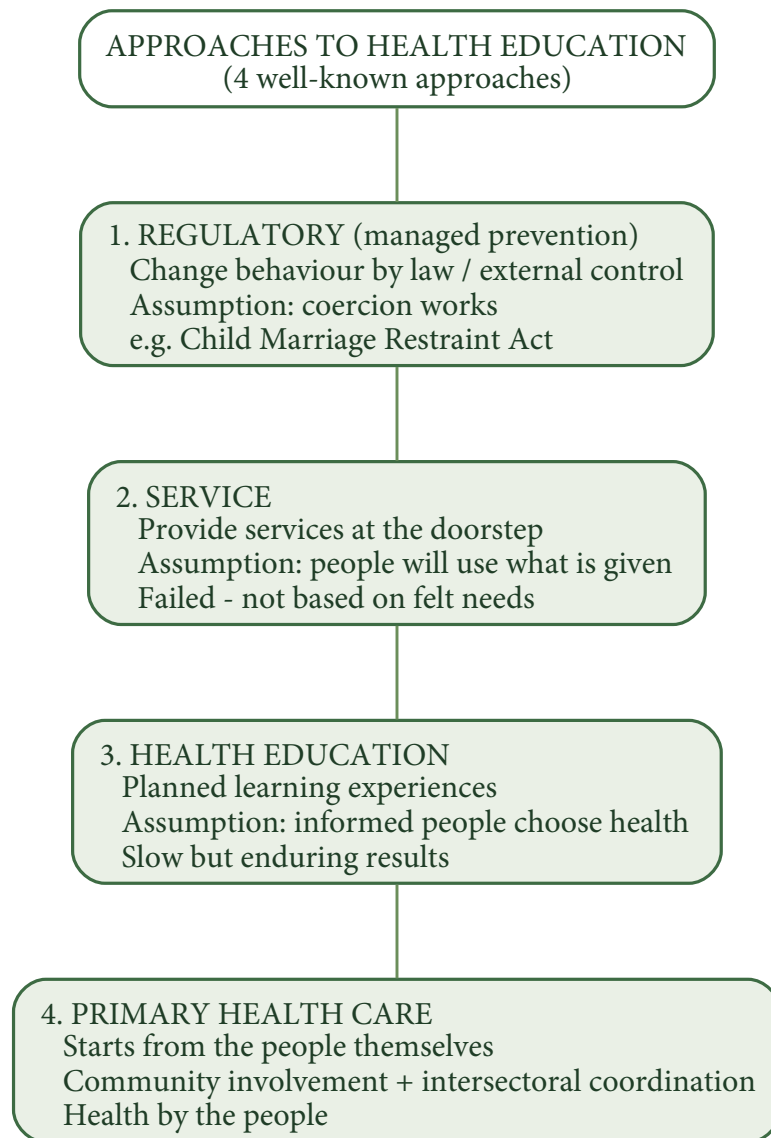
Since the Alma-Ata Declaration (1978) the emphasis has shifted from prevention of disease to promotion of healthy lifestyles; from modification of individual behaviour to modification of the **social environment** in which the individual lives; from community *participation* to community *involvement*; and towards promotion of individual and community **self-reliance**. Education is necessary but not sufficient — the target population must also have **access to proven preventive measures** (Park 28e, Ch 22, p 1015).

The four approaches to health education

#	Approach	What it does	Indian / classical example	Comment
1	Regulatory approach (managed prevention)	Any governmental intervention, direct or indirect, designed to alter human behaviour; laws and external controls placed on people, ranging from prohibition to imprisonment	<i>Child Marriage Restraint Act</i> , India; compulsory seat-belt use in western countries	Seems the simplest and quickest way, but has important failures — e.g. prohibition of alcohol
2	Service approach	Provides all the health services people need at their doorstep, on the assumption that people will use them to improve their own health	Water-seal latrines supplied free of cost by government in rural India	Proved a failure because it was not based on the felt needs of the people — many did not use them
3	Health education approach	Aims at informed, autonomous decision-making; people are educated through planned learning experiences and encouraged to make their own choice for a healthy life	Smoking, use of safe water supply, fertility control — problems solvable only by education	Consistent with democratic philosophy — it does not order the individual; results are slow but enduring
4	Primary health care approach	Radically new approach starting from the people , with their full participation and active involvement in planning and delivery of health services	"Health by the people"; "placing people's health in people's hands"	Rests on community involvement and intersectoral coordination ; a fundamental shift from the earlier approaches

(Park 28e, Ch 22, pp 1015–1016; principles of primary health care — Park 28e, Ch 2, p 49.)

The health education approach must be distinguished from propaganda: education is **behaviour-centred** — knowledge and skills are actively acquired, people are made to think for themselves and to appeal to reason — whereas propaganda is **information-centred**, knowledge is spoon-fed and passively received, it appeals to emotion and no change of attitude or behaviour is designed (differences drawn up by the Central Health Education Bureau, Government of India — Park 28e, Ch 22, Table 1, p 1016).



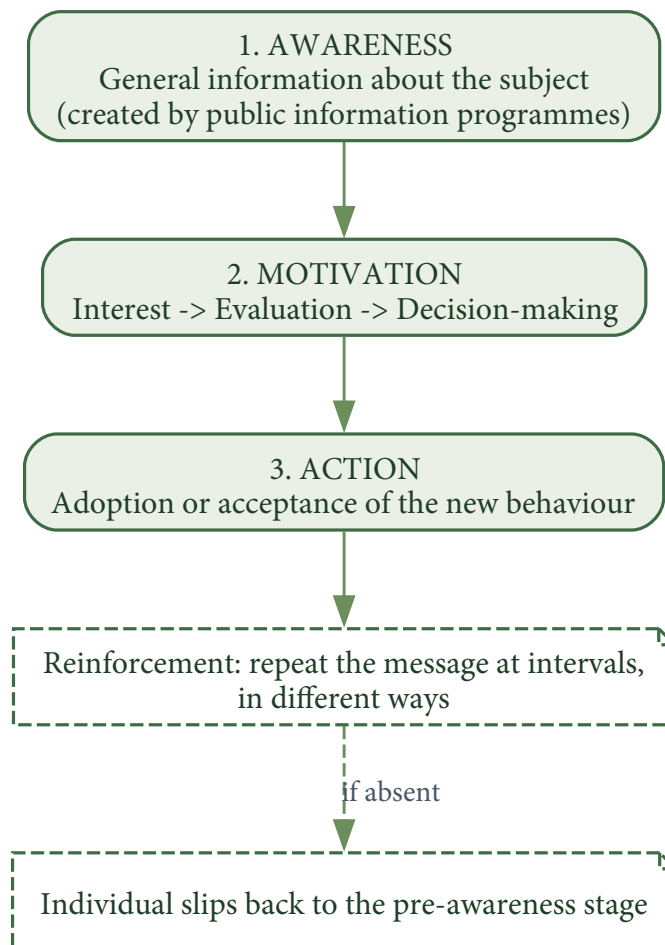
The four approaches to health education and the assumption underlying each

Models within the health education approach

Model	Underlying assumption	Method	Limitation / current status
Medical model	Knowledge transfer alone changes behaviour; social, cultural and psychological factors were thought unimportant	Dissemination of scientific facts by health professionals; primarily interested in the recognition and treatment of disease	Did not bridge the gap between knowledge and behaviour
Motivation model	People act when motivated, not merely when informed	Motivation as the force translating information into health action; because attitudes and behaviour patterns form early in life, health education is begun in the young population , in whom behaviour is more easily developed than in adults	Insufficient alone for today's complex problems — reducing smoking, small-family norm, raising age at marriage, eliminating dowry

Model	Underlying assumption	Method	Limitation / current status
Social intervention model	It is often not the individual but the social environment shaping his behaviour that must change; a new idea is not accepted until "legitimated" by the group (e.g. vasectomy or loop insertion is adopted far more readily where group support exists)	Uses group support and legitimation; based on precise knowledge of human ecology and of the interaction between cultural, biological, physical and social environmental factors	The current model , born of the failure of the motivation model

(Park 28e, Ch 22, pp 1016–1017.)



Adoption model - the stages an individual passes through before a health behaviour is adopted (Park 28e, Ch 22, Fig 2)

Principles that govern the approach

#	Principle	Operational meaning
1	Credibility	The message must be perceived as trustworthy — consistent with scientific fact and with local culture, the educational system and social goals
2	Interest	People listen only to what interests them; find the people's felt needs — a programme built on felt needs earns willing participation

#	Principle	Operational meaning
3	Participation	The key word; based on the psychological principle of active learning. Alma-Ata: people "have a right and duty to participate individually and collectively in the planning and implementation of their health care"
4	Motivation	Awakening the fundamental desire to learn; primary motives are inborn, secondary motives (praise, reward, recognition, rivalry) are created by outside incentives. Motivation is contagious
5	Comprehension	Teach within the mental capacity, literacy and educational level of the audience; never use words strange to the people
6	Reinforcement	Few learn all in a single period — repeat the message at intervals in different ways
7	Learning by doing	"If I hear, I forget; if I see, I remember; if I do, I know"
8	Known to unknown	Proceed from the concrete to the abstract, simple to complicated, easy to difficult, particular to general; use existing knowledge as the peg for new knowledge
9	Setting an example	The educator must practise what he teaches (an educator who smokes cannot teach the hazards of smoking)
10	Good human relations	Information, ideas and feelings pass most easily between people who have a good relationship
11	Feedback	A key concept of the systems approach — the educator modifies message and channel in the light of audience response
12	Leaders	We learn best from those we respect; penetrate the community through local leaders — village headman, school teacher, political worker — who act as agents of change

(Park 28e, Ch 22, pp 1018–1019.)

Delivering the approach — individual, group and mass methods

Approach	Methods	Merits	Limitations
Individual (interpersonal)	Personal contact and interview in clinic/health centre, home visits, personal letters	Most precise and specific; allows discussion, argument and persuasion; questions can be asked on specific interests; friendly and participatory	Numbers reached are small; reaches only those who come into contact with us
Group	Lectures, demonstrations, group discussion, panel discussion, symposium, workshop, conference, seminar, role play	Effective way of educating the community; group support legitimises the new behaviour; more influential with average and below-average educational level	Choice of subject must relate directly to the group's interest (talk to a woman in labour about child-birth and baby care, not tuberculosis control)

Approach	Methods	Merits	Limitations
Mass	Television, radio, newspaper, printed material, direct mailing, posters, health museums and exhibitions, folk methods, internet	Reaches the widest population; gets public attention; supports concentrated campaigns such as a health week or month	Chiefly creates awareness only; one-way, no feedback; more effective among those of above-average education

(Park 28e, Ch 22, pp 1020–1023, Fig 3 and Table 2.)

Combining methods is the rule — effective communication is seldom achieved by one method alone; the mix is chosen according to the objective, the behaviour to be influenced and the funds available.

Planning, evaluation, barriers and administration

Steps in planning a health education programme (Park 28e, Ch 22, p 1024):

- ▶ Collect information on the specific problem as **seen by the community**.
- ▶ Identify the problem.
- ▶ Decide priorities.
- ▶ Set goals and **measurable** objectives.
- ▶ Assess resources.
- ▶ Consider possible solutions.
- ▶ Prepare a plan of action — what will be done, when, and by whom.
- ▶ Implement the plan.
- ▶ Monitor and evaluate the degree to which objectives have been achieved.
- ▶ Reassess the planning process.

Evaluation means the **change in knowledge, attitude and practice (KAP)** produced by the health education — knowledge by standard questions, attitude by a *Likert* scale, practice by direct observation — assessed by the workers themselves during home visits and contacts and by exit interviews (Sunder Lal, Ch 4, pp 153–154).

Barrier to communication	Examples
Physiological	Difficulty in hearing, difficulty in expression
Psychological	Emotional disturbance, neurosis, level of intelligence, language or comprehension difficulty
Environmental	Noise, invisibility, congestion
Cultural	Illiteracy, level of knowledge, customs, beliefs, religion, attitudes, economic and social class differences, language variation, urban–rural difference

(Park 28e, Ch 21, p 1012.)

Who administers the approach in India (Park 28e, Ch 22, p 1024):

- ▶ **National** — the **Central Health Education Bureau**, established by the Government of India in 1956 in the Ministry of Health, New Delhi, to promote and coordinate health education work in the country.
- ▶ **State** — **State Health Education Bureaux** in the State Health Directorates.
- ▶ **Other official agencies** — Directorate of Advertising and Visual Publicity (DAVP), Press Information Bureau, Doordarshan, All India Radio.
- ▶ **Voluntary sector** — Indian Red Cross and other voluntary agencies.
- ▶ **International** — International Union for Health Education, headquarters Paris; its **South East Asia Regional Bureau** established in 1983 with headquarters at Bangalore.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **The communication process** (Park 28e, Ch 22, Fig 1, p 1013) — a horizontal chain of four boxes labelled exactly *Sender (source)* → *Message (content)* → *Channel (medium)* → *Receiver (audience)*, with a return arrow drawn beneath, from the receiver back to the sender, labelled *Feedback (effect)*.
- ▶ **Methods in health communication** (Park 28e, Ch 22, Fig 3, p 1020) — one top box *Health communication*, branching into three columns labelled *Individual approach* (personal contact, home visits, personal letters), *Group approach* (lectures, demonstrations, group discussion, panel discussion, symposium, workshop, conferences, seminars, role play) and *Mass approach* (television, radio, newspaper, printed material, direct mailing, posters, health museums and exhibitions, folk methods, internet).

High-yield exam pointers

- ▶ **Four approaches = R-S-H-P**: Regulatory (managed prevention) → Service → Health education → Primary health care. Name the failure of each: regulatory = alcohol prohibition; service = unused free water-seal latrines (not based on felt needs).
- ▶ **Three models** of health education: medical → motivation → social intervention. The medical model failed because it **did not bridge the knowledge-behaviour gap**.
- ▶ **Adoption model**: Awareness → Motivation (interest, evaluation, decision-making) → Action; without reinforcement the individual slips back to the pre-awareness stage.
- ▶ **12 principles**: credibility, interest, participation, motivation, comprehension, reinforcement, learning by doing, known to unknown, setting an example, good human relations, feedback, leaders.
- ▶ Alma-Ata 1978 shifts: disease prevention → lifestyle promotion; individual behaviour → social environment; participation → involvement; and towards **self-reliance**.
- ▶ Health education is **behaviour-centred**; propaganda is **information-centred** (Central Health Education Bureau table).
- ▶ Four barriers to communication: physiological, psychological, environmental, cultural.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only — the World Health Organization has established a new Division of Health Education and Health Promotion, and its former Division of Public Information and Education for Health is now called the Division of Public Information and Public Relations (Park 28e, Ch 22, p 1024). In India the current peripheral vehicle for delivering this approach is the Ayushman Arogya Mandir (erstwhile Ayushman Bharat Health and Wellness Centre, launched 2018), and information, education and communication (IEC) and behaviour change communication (BCC) services are listed among the essential health services provided at the sub-health-centre level (Park 28e, Ch 7, p 541; Sunder Lal, Ch 17, p 745).

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 22 (Communication for Health Education), pp 1013–1024; Ch 21 (Health Information and Basic Medical Statistics), p 1012; Ch 7 (Health Programmes in India), p 541; Ch 2 (Concept of Health and Disease), p 49. Sunder Lal — Textbook of Community Medicine, Ch 4 (Principles of Health Promotion and Education), pp 143–155; Ch 17 (Healthcare of the Community), p 745. WHO definition of health education (1969) and the Alma-Ata Declaration (1978) as cited in Park 28e.



Behaviour change communication

Asked in: Paper I 21-Jun-2024 — RGUHS MD Community Medicine

Definition — Behaviour change communication (BCC) is a systematic process that motivates individuals, families and communities to change their inappropriate or unhealthy behaviour, or to continue a healthy behaviour (Park 28e, Ch 7, p 499). It is the behaviour-directed end of the information, education and communication (IEC) continuum: it begins from the audience's existing knowledge, attitudes and practices, uses a planned mix of channels, and is judged by measured change in practice — not by the number of messages delivered.

BCC in relation to IEC and health education

Feature	Information, education and communication (IEC)	Behaviour change communication (BCC)
Primary aim	Create awareness and disseminate information on services and entitlements	Motivate a named change in practice, or sustain an already healthy practice
Starting point	The message the programme wishes to send	The audience's knowledge, attitudes, beliefs, practices, myths and misconceptions
Direction	Largely one-way (didactic); mass media dominant	Two-way (Socratic); interpersonal communication dominant, mass media supportive

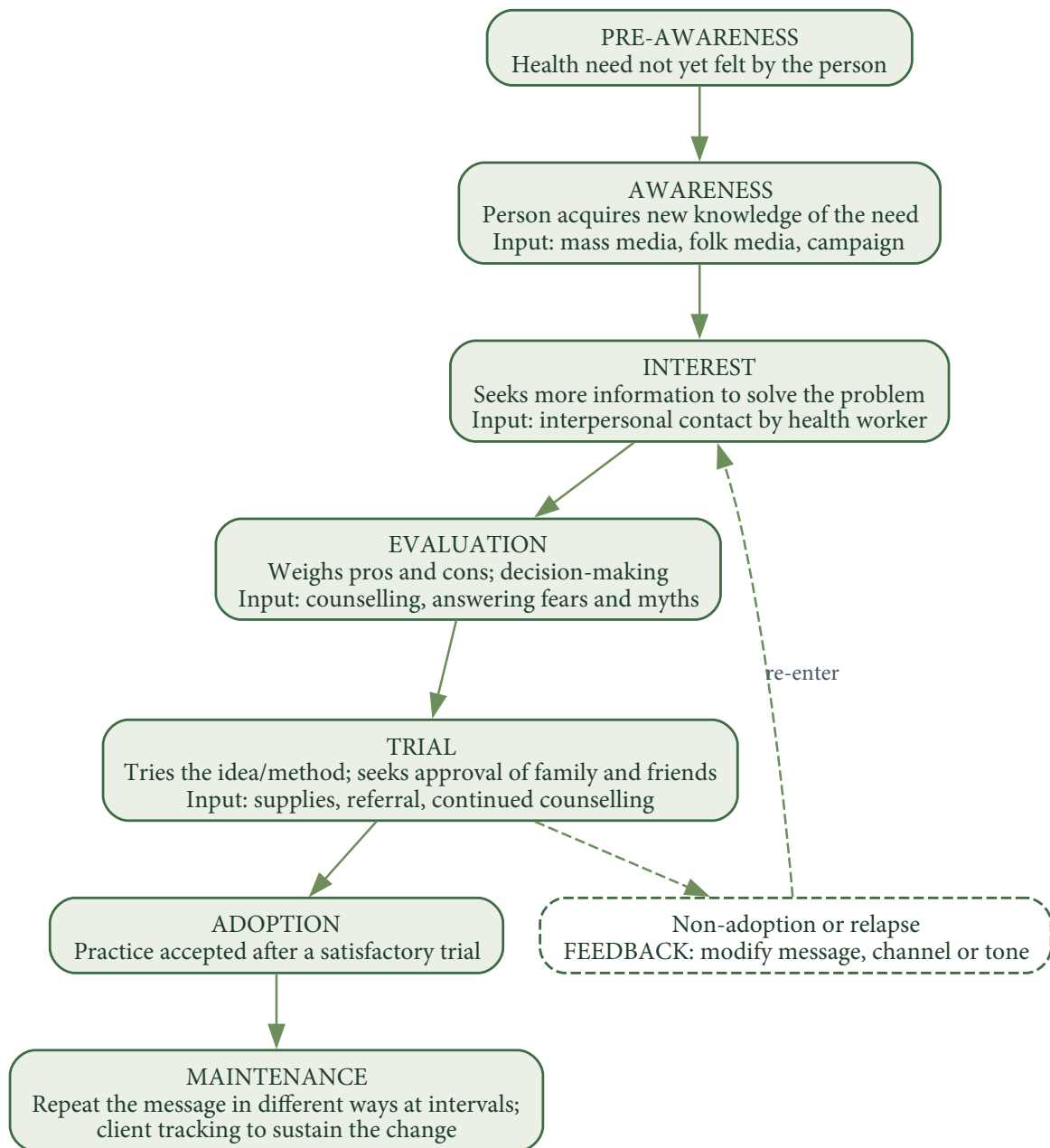
Feature	Information, education and communication (IEC)	Behaviour change communication (BCC)
Channel plan	"360-degree" spread — mass media, mid-media, social media, interpersonal communication (Sunder Lal, Ch 2, p 85)	Segmented channel mix chosen per audience segment, with client tracking and repetition
Unit of action	General public	Defined audience segment (e.g. currently married men with one child)
Outcome measured	Reach, exposure, recall	Change in the specified practice against a baseline and a dated target
Programme label	Under the National Tuberculosis Elimination Programme, IEC is used to mean advocacy, communication and social mobilisation (ACSM) (Jugal Kishore, Ch 1, p 304)	A named supportive intervention, e.g. under the National Vector Borne Disease Control Programme (Park 28e, Ch 7, p 493)

Objectives and the behaviour change pathway

Three programme objectives of BCC (Jugal Kishore, Ch 1, p 416):

- ▶ **Enhance awareness** of the health problem and of available services.
- ▶ **Promote attitudinal and value change** among target audiences, leading to informed decisions, modified behaviour and desirable practices.
- ▶ **Build intersectoral support** across society and among public and private health-care providers, and elicit commitment for action.

Communication produces three effects in sequence — a change in **knowledge**, then in **attitude**, then in **behaviour** (practice) — though the sequence is not invariable, and attitude change often *follows* an action rather than preceding it (Sunder Lal, Ch 4, p 149). This is why information alone fails: Park traces the shift from the **medical model** (knowledge transfer alone, which never bridged the knowledge-behaviour gap), through the **motivation model**, to the **social intervention model** (it is often the social environment, not the individual, that must change) and the **primary health care approach** (people's full participation); BCC is the operational form of the last two (Park 28e, Communication for Health Education, p 1016). The behavioural-science mainstays underpinning BCC interventions are the **Health Belief Model** (Becker, 1974), **Social Cognitive Theory** (Bandura, 2004) and the **Transtheoretical Model of Stages of Change** (Prochaska, 2002) (Oxford Textbook of Public Health 7e, Ch 15, p 605), with the **Diffusion of Innovations** model (Rogers) describing how a new behaviour spreads through a population (Oxford Textbook of Public Health 7e, Ch 33, pp 1216–1217).



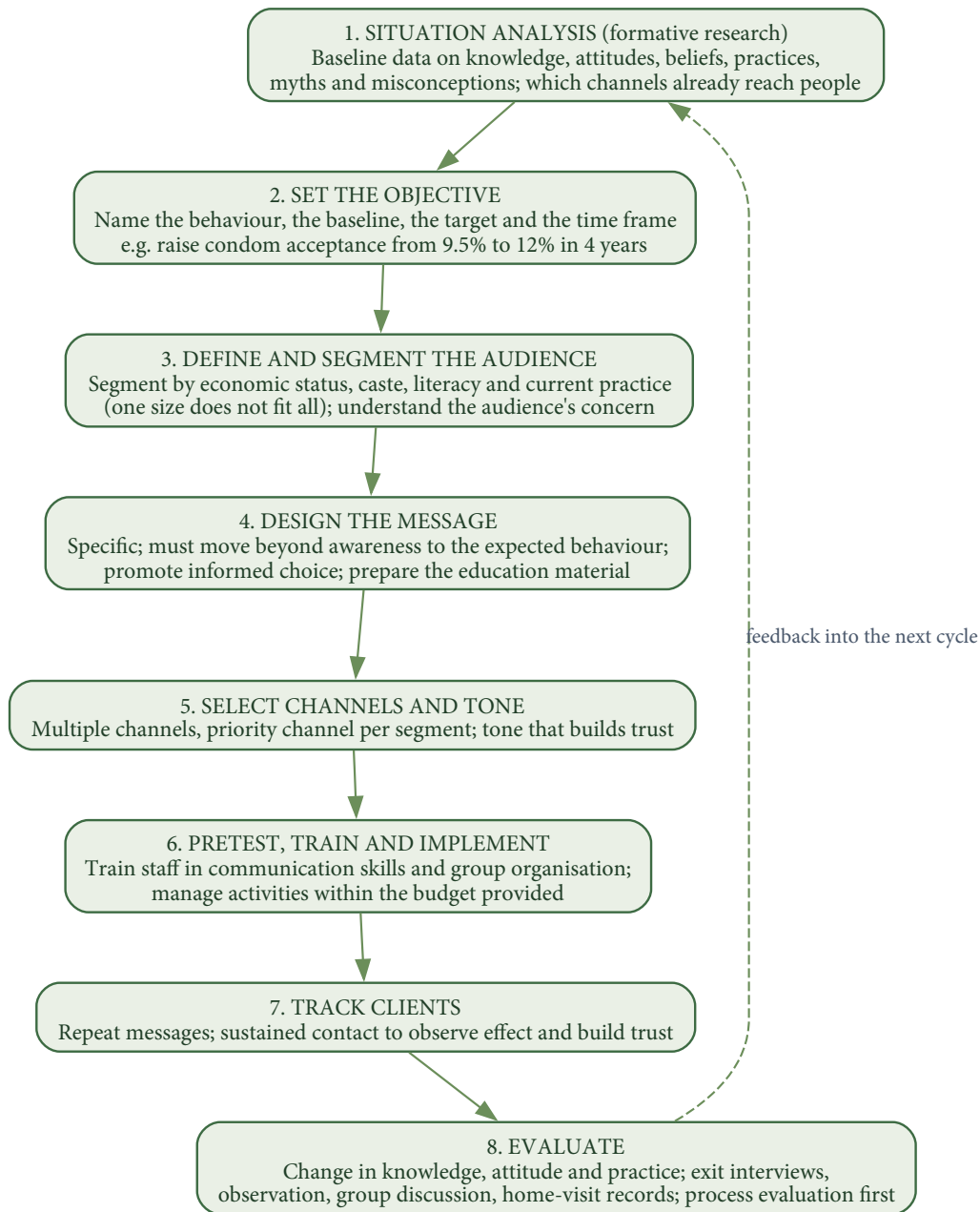
Pathway of behaviour change and the health-system input needed at each stage

Channels and methods of BCC

Channel	What it is / examples	Merits	Limitations
Interpersonal (individual) communication	One-to-one contact on home visits by the auxiliary nurse midwife (ANM), accredited social health activist (ASHA) or male health worker; personal letters	Most precise and specific; most interactive and participatory; message passes directly source → receiver; permits discussion, argument and persuasion; more influential at average and below-average educational levels	Numbers reached are small; reaches only those who come in contact; depends heavily on worker's communication skills

Channel	What it is / examples	Merits	Limitations
Group approach	Group discussion, demonstration, lecture, panel discussion, symposium, workshop, role play; <i>Mahila Swasthya Sangh</i> , <i>Mahila Mandal</i> , self-help groups, adolescent groups, monthly Village Health, Sanitation and Nutrition Committee meeting at the anganwadi centre	Two-way and consensus-building; valuable where long-term compliance is needed (smoking cessation, obesity reduction); demonstration has high motivational value	Effective size is limited — not fewer than 6 and not more than 12 members for a group discussion (Park 28e, Communication for Health Education, p 1021); needs a trained facilitator
Mass media	Television, radio, newspapers and printed matter, direct mailing, posters, health museums and exhibitions, internet	Reaches the widest population in the shortest time; gets public attention; supports concentrated week- or month-long campaigns; more effective among the above-average educated	One-way; carries messages only from the centre outwards; feedback mechanisms poorly organised; usually not effective at changing established behaviour by itself
Traditional / folk media	Folk dances and song, <i>Nautanki</i> and <i>Harikatha</i> (Uttar Pradesh), <i>Burrakatha</i> (Andhra Pradesh), puppet shows, street plays, health <i>melas</i>	Culturally close to the rural population; high acceptability and retention	Local reach only; message fidelity depends on the performer
Mid-media, social media and telecommunication	Hoardings, bus panels, wall paintings, rallies; social media; pre-caller tunes (used pan-India for COVID-19 appropriate behaviour)	Repeated low-cost exposure; point-to-point telecommunication behaves like interpersonal contact	Exposure without dialogue; unequal access

Planning, delivering and evaluating a BCC strategy



The BCC cycle as planned and evaluated at primary health centre and district level

Responsibility for planning and evaluating this cycle lies with the **medical officer in charge of the primary health centre and community health centre** (Sunder Lal, Ch 4, p 155). At national level, evaluation draws on the National Family Health Survey (NFHS) rounds 1–5 and on the behavioural surveillance surveys of the National AIDS Control Programme (Sunder Lal, Ch 4, p 154). A worked BCC evaluation matrix — the format expected in the answer — is shown below (Sunder Lal, Ch 4, Table 4.1, pp 154–155).

Area of concern	Current behaviour (2021)	Expected behaviour (2025)	Target audience	Message	Media and channel	Outcome indicator
Use of spacing methods by women	7.8% practise	10%	Currently married women with one child	Space births at least 3 years apart to improve maternal health and child survival	Interpersonal communication by female health worker; television; <i>Mahila Swasthya Sangh</i> groups	% of women with one or two children using spacing methods (annual worker survey)
Condom use by men	9.5% use	12%	Currently married men with one child	As above	Interpersonal communication by male health worker; opinion leaders; television	% of men using condoms (annual worker survey)

Principles of effective BCC, and the barriers it must overcome

Principle (Park 28e, Communication for Health Education, pp 1018–1019)	Application in BCC
Credibility	Message must be trustworthy — scientifically correct and compatible with local culture and social goals
Interest	Address felt needs ; if the message is not of interest the audience returns to the pre-awareness stage
Participation	Active involvement of the family and community; Alma-Ata affirms the people's right and duty to participate
Motivation	Use secondary motives — praise, rivalry, reward, recognition; motivation is contagious (a vasectomy acceptor is the best advertisement for male sterilisation)
Comprehension	Pitch to the audience's level of education and literacy; never use words strange to the people
Reinforcement	Repeat the message at intervals in different ways — few people learn all of it at once
Learning by doing	"If I hear I forget; if I see I remember; if I do I know" — demonstration and practice
Known to unknown	Proceed from the concrete to the abstract, simple to complicated, existing knowledge to new
Setting an example	The educator must practise what is taught
Good human relations	Sharing happens most easily between people with a good relationship
Feedback	Modify message and channel in the light of audience response — the key systems-approach concept

Principle (Park 28e, Communication for Health Education, pp 1018–1019)	Application in BCC
Leaders	Work through local leaders — village headman, school teacher, political worker — as agents of change

Barrier	Examples
Physiological	Difficulty in hearing; difficulty in expression
Psychological	Emotional disturbance, neurosis, level of intelligence, language or comprehension difficulty
Environmental	Noise, invisibility, congestion
Cultural	Illiteracy; level of knowledge and understanding; customs, beliefs, religion, attitudes; economic and social class differences; language variation; urban-educated worker facing a rural population
Operational blocks in the encounter (Sunder Lal, Ch 4, p 155)	Talking <i>at</i> the client and not listening; lack of empathy and trust; poor eye contact; low credibility of the source; medical jargon and dialect; frequent interruptions (telephone, files); low visual and health literacy; the worker looking bored or at the watch

Even where health services are readily available, social and cultural barriers can seriously obstruct behaviour change, and these must be identified and removed before messaging can work (Park 28e, Communication for Health Education, p 1012).

BCC in Indian national programmes

Programme	BCC content
National Vector Borne Disease Control Programme (NVBDCP)	BCC is a listed <i>supportive intervention</i> alongside case management and integrated vector management. Directed at: early recognition of signs and symptoms of malaria; early treatment-seeking from an appropriate provider; adherence to treatment regimens; protection of children and pregnant women; use of insecticide-treated and long-lasting insecticidal nets; acceptance of indoor residual spraying. Anti-Malaria Month is observed every June , before the monsoon transmission season (Park 28e, Ch 7, pp 493, 499)
National AIDS Control Programme (NACP-IV)	Communication directed at increasing knowledge of safe sexual behaviour among the general population, especially youth and women; sustaining behaviour change in high-risk and bridge populations; generating demand for care, support and treatment; and changing societal norms to reduce stigma and discrimination. Adolescence Education Programme runs 16-hour sessions in classes IX and XI (Park 28e, Ch 7, p 526)
National Tuberculosis Elimination Programme (formerly RNTCP)	IEC delivered as advocacy, communication and social mobilisation (ACSM) ; planning, channel choice and monitoring decentralised to state and district; a central multidisciplinary advocacy and IEC unit (Jugal Kishore, Ch 1, p 304)

Programme	BCC content
National Leprosy Eradication Programme	Decentralised state IEC plans using mass media, outdoor media (hoardings, wall paintings, rallies), rural media (interpersonal communication meetings, school talks and quizzes, folk media, exhibitions, health <i>melas</i>) and advocacy with <i>Zila Parishad</i> , <i>Mahila Mandals</i> and non-governmental organisations (Jugal Kishore, Ch 1, pp 466–467)
Reproductive and Child Health / RMNCH+A	National communication strategy resting on interpersonal communication for behaviour change at field level, decentralised production of material at state and district level, mass media for advocacy and awareness, greater NGO involvement, and capacity building at all levels (Sunder Lal, Ch 4, p 155)
Adolescent health	National communication strategy for adolescents developed with UNICEF — interpersonal communication, mass-media advocacy, decentralisation of IEC to states and districts, NGO and private-sector involvement, capacity building; the GATHER approach is used for counselling (Sunder Lal, Ch 10, p 623)
District Mental Health Programme	IEC/BCC through local and folk media and interpersonal communication, focused on early identification of mental illness, availability of treatment and reduction of stigma; 10 October observed as World Mental Health Day (Sunder Lal, Ch 15, pp 683–684)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- The communication process (Park, Fig. 1) — a horizontal chain with a return loop. Labels, in order: **Sender (source)** → **Message (content)** → **Channel (medium — interpersonal / mass media / folk media)** → **Receiver (audience)**, with a return arrow labelled **Feedback (effect)** running from Receiver back to Sender. Mark four blocks sitting on the channel and label them **Physiological, Psychological, Environmental, Cultural** — the barriers of communication.

High-yield exam pointers

- **Definition to reproduce verbatim:** "BCC is a systematic process that motivates individuals, families and communities to change their inappropriate or unhealthy behaviour, or to continue a healthy behaviour" (Park 28e, Ch 7, p 499).
- **Five elements of communication:** sender, receiver, message, channel, feedback. **Three media systems:** interpersonal, mass media, traditional/folk.
- **Adoption ladder mnemonic — A-I-E-T-A:** Awareness → Interest → Evaluation → Trial → Adoption. Park's compressed version: **Awareness → Motivation → Action.**
- **Four classes of barriers:** physiological, psychological, environmental, cultural.
- **Numbers to quote** as the communication gaps that justify BCC: group discussion 6–12 members; National Family Health Survey (NFHS) rounds 4 and 5, 2015–21 — 76.4% of children aged 12–23 months fully immunised, 23.3% of women aged 20–24 married before 18 years, oral rehydration salt use 60.6%, 38% of men and 8.9% of women using tobacco (as cited in Sunder Lal, Ch 4, p 154).

- **Programme link most often asked:** BCC is a *supportive intervention* under NVBDCP; under NTEP the same function is called ACSM.
- **Always write a BCC objective in SMART form:** named behaviour + baseline + target + deadline (e.g. condom acceptance 9.5% → 12% in 4 years).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the corpus treats IEC, health education and BCC as points on one continuum rather than as sharply separated activities, so the comparison table above is a teaching device drawn from the retrieved descriptions of each; the National AIDS Control Programme material retrieved is from NACP-IV, and the tuberculosis programme was renamed the National Tuberculosis Elimination Programme in 2020 while the retrieved text still uses "RNTCP".

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 7, Health Programmes in India; Ch 22, Communication for Health Education); Sunder Lal — Textbook of Community Medicine (Ch 2, Ch 4 Principles of Health Promotion and Education, Ch 10 RMNCH+A, Ch 15 Mental Health); Jugal Kishore — National Health Programmes of India; Oxford Textbook of Public Health 7e (behavioural determinants of health and disease; community intervention trials). National Family Health Survey-5 (2019–21) figures as cited in Sunder Lal. Coverage note: the indexed figure library returned no captioned plate matching the communication process or the behaviour change pathway, so both are given as a hand-drawn label box and as flowcharts instead.



Contributions of Louis Pasteur to public health

Asked in: Paper I 12-Nov-2021 — RGUHS MD Community Medicine

Definition — Louis Pasteur (1822–1895) was a French chemist and bacteriologist whose laboratory work displaced the concepts of disease causation in vogue up to his time — the supernatural theory, the theory of humors, contagion, the miasmatic theory and spontaneous generation — and put a demonstrable microbial aetiology in their place (Park 28e, Ch 2, p 51). In 1860 he demonstrated bacteria in air and **disproved spontaneous generation**; in 1873 he advanced the **germ theory of disease** (Park 28e, Ch 1, p 18).

His public health legacy rests on four pillars: the **germ theory** itself, the **principle of attenuation** that made vaccines manufacturable in the laboratory, **anti-rabies treatment** with the Pasteur Institutes that followed it, and **pasteurization**.

Chronology of Pasteur's landmark contributions

Year	Contribution	Public health significance
Early career, Lille	Research on fermentation ; proved it occurs only when microbes are present	Founded the new science of microbiology ; the wrong microbe gives sour milk or bitter wine (Park Plus, Ch 1, p 8)
1860	Swan-neck flask experiment ; bacteria demonstrated in air; spontaneous generation disproved	Removed the last rival aetiological theory and made a microbial cause of disease conceivable (Park 28e, Ch 1, p 18; Park Plus, Ch 1, p 8)
Applied from the wine work	Heating wine enough to kill most microbes without altering flavour — pasteurization	Transferred to milk; became a permanent public health measure for milk hygiene (Park Plus, Ch 1, p 8)
1873	Advanced the germ theory of disease	Shifted causal thinking from "bad air" to microbes; confirmed when Koch showed anthrax was bacterial in 1877 — the <i>golden age of bacteriology</i> (Park 28e, Ch 1, p 18)
Late 1870s	Attenuation of the chicken cholera bacterium (<i>Pasteurella multocida</i>)	First major advance after Jenner; proved standardized, reproducible vaccines could be "created" at will (Plotkin 7e, Ch 1, pp 21–23)
1881	Attenuated <i>Bacillus anthracis</i> and field-tested the vaccine in livestock	First effective live bacterial vaccine ; Greenfield performed similar work at the same time (Plotkin 7e, Ch 11, p 203)
6 July 1885	First successful human rabies vaccination — <i>Joseph Meister</i>	First effective post-exposure prophylaxis against a human viral disease (Plotkin 7e, Ch 50, p 1336)
1888	Pasteur Institute, Paris founded	Pasteur Institutes worldwide soon provided rabies vaccination — a precedent for organized vaccination well into the 20th century (Plotkin 7e, Ch 50, pp 1336–1337)

1. Fermentation, the swan-neck flask and the germ theory of disease

a. Fermentation — the birth of microbiology

- Pasteur proved that **fermentation takes place only when living microbes are present**: with the right microbe the desired product results, with the wrong microbe the wrong substance — **sour milk or bitter wine**. This founded the new branch of science, **microbiology** (Park Plus, Ch 1, p 8).

b. The swan-neck flask experiment — how spontaneous generation was disproved

- He poured **meat infusions** into flasks and drew each neck into a **long curved ("swan") neck** that would admit air but not dust; once heated, these infusions **remained sterile** so long as they were not exposed to dust.
- He then **opened the flasks on a dusty road** and resealed them: **all the opened flasks grew microorganisms**, while the **unopened flasks remained sterile** — air by itself does not generate life, the dust it carries does (Park Plus, Ch 1, p 8).

- This is the basis of the 1860 result — bacteria demonstrated in air, spontaneous generation disproved; germ theory followed in 1873, and Koch's 1877 demonstration that anthrax was bacterial confirmed it (Park 28e, Ch 1, p 18).

c. What the germ theory did for public health

- It opened the **golden age of bacteriology** — microbe after microbe was identified in quick succession: gonococcus 1847; typhoid bacillus and pneumococcus 1880; tubercle bacillus 1882; cholera vibrio 1883; diphtheria bacillus 1884 (Park 28e, Ch 1, p 18).
- The theory embodies a **one-to-one relationship** between causal agent and disease, giving the model *disease agent* → *man* → *disease* (Park 28e, Ch 2, p 51).
- **Preventive medicine got a firm foundation only after** the causative agents of disease were discovered and the germ theory established. Specific measures became possible — **quarantine, blocking transmission, disinfection, antiseptics, immunization and pasteurization** — yielding Pasteur's anti-rabies treatment (1883), cholera vaccine (1892), diphtheria antitoxin (1894) and anti-typhoid vaccine (1898), and then the elucidation of transmission routes (Bruce and the tsetse fly 1896; Ross and *Anopheles* 1898; Reed and *Aedes* in yellow fever 1900) (Park 28e, Ch 1, p 18).
- **Limitation** — the one-to-one model led many epidemiologists to a one-sided view of causation; not everyone exposed to tuberculosis develops tuberculosis. Modern medicine has therefore moved away from strict adherence to the germ theory towards the **epidemiological triad** and **multifactorial causation**, first mooted by *Pettenkofer* of Munich (1819–1901) (Park 28e, Ch 2, pp 51–52).

2. Where Pasteur sits in the history of public health

Park demarcates **four distinct phases** in the history of public health; Pasteur's germ theory is not itself a phase but the discovery that made disease-specific action possible.

Phase	Period	What the phase consisted of	Relation to Pasteur
a. Disease control phase	1880–1920	Public health of the 19th century was largely sanitary legislation and sanitary reform, not aimed at the control of any specific disease ; efforts directed entirely at general cleanliness, garbage and refuse disposal; quarantine conventions held to contain disease	Pasteur's germ theory arose within this phase and gave preventive medicine its firm foundation — it did not replace the sanitary era
b. Health promotion phase	1920–1960	"Health promotion" begins to take a share; the State is accepted as having a direct responsibility for the health of the individual; health departments expand into promotional activity	The specific measures Pasteur made possible — vaccines, antitoxins, antiseptics, disinfection, pasteurization — mature and are delivered at scale
c. Social engineering phase	from the 1960s	Chronic and behavioural disease, which could not be explained by the germ theory; social and behavioural priorities; the concept of risk factors	The germ theory is shown to be insufficient for the diseases of affluent societies

Phase	Period	What the phase consisted of	Relation to Pasteur
d. "Health for All" phase	1981–2000 AD	Response to the glaring contrast in health between developed and developing countries	—

(All rows: Park 28e, Ch 1, pp 17, 20–21.)

3. The principle of attenuation — the birth of laboratory-made vaccines

Vaccine	Organism / agent	Method of attenuation	Significance
Chicken cholera (late 1870s)	<i>Pasteurella multocida</i>	Modification of virulence through culture and passage ; ageing of the bouillon culture	First deliberate laboratory attenuation; showed vaccination need not depend on person-to-person or animal-to-animal transfer (Plotkin 7e, Ch 1, p 21)
Anthrax (1881)	<i>Bacillus anthracis</i>	Attenuation of the organism, then a controlled field test in livestock	First effective live bacterial vaccine; a veterinary and zoonosis-control intervention with direct human benefit (Plotkin 7e, Ch 11, p 203)
Rabies (1885)	Fixed rabies virus	Serial drying of spinal cords from rabid rabbits — long-dried cord is avirulent, progressively less-dried cord more virulent	First human post-exposure vaccine; at least 50 dogs had been protected by the protocol first (Plotkin 7e, Ch 50, p 1336)

The concepts Pasteur assembled — **attenuation, modification through passage, renewed virulence**, and the replacement of person-to-person vaccination by something safer and more consistent — are the foundations on which all subsequent live attenuated vaccines rest (Plotkin 7e, Ch 1, pp 21, 23). His rabies vaccine of 1885 was followed through the 1930s by antitoxins and vaccines against cholera, typhoid, plague, diphtheria, tetanus, anthrax and tuberculosis (Oxford Textbook of Public Health 7e, Ch 96, pp 3582–3583).

4. Anti-rabies treatment and the Pasteur Institutes

- ▶ Because rabies germs take **several weeks** to reach the brain after a bite, the vaccine could be given **after exposure** — unlike all previous vaccination, which had to precede exposure; only those actually bitten needed treatment (Park Plus, Ch 1, pp 8–9).
- ▶ **Joseph Meister**, a 9-year-old boy from Alsace bitten **14 times** by a rabid dog, reached Paris about **60 hours** after the attack and was inoculated on **6 July 1885** with rabid-rabbit spinal cord dried for **15 days**, then **12 further cords of increasing virulence** — **13 doses over about 10 days** (Plotkin 7e, Ch 50, pp 1336–1337; Ch 1, p 23).
- ▶ **Jean Baptiste Jupille**, aged 14, presented **6 days** after multiple bites and received the same course; **both survived** (Plotkin 7e, Ch 1, p 23).
- ▶ The **Pasteur method** was the *first generation* of rabies vaccination and was rapidly accepted despite major disagreement; the **Pasteur Institute of Paris was founded in 1888**, and Pasteur

Institutes worldwide soon delivered rabies vaccination as an organized service (Plotkin 7e, Ch 50, pp 1336–1337).

- For more than 70 years after Pasteur's original work, only nerve-tissue vaccines were available — modified by Fermi and Semple, who used phenol to partially or completely inactivate the virus; the phenol-inactivated Semple vaccine, prepared in sheep, goat or rabbit brain, remained in use in Asia and Africa. Adverse neurological reactions to brain-tissue vaccine, recognized since Pasteur's time, are why nerve-tissue vaccines were ultimately replaced by cell-culture vaccines (Plotkin 7e, Ch 50, pp 1344–1345).

5. Pasteurization — the enduring milk-hygiene contribution

Pasteurization grew directly out of the wine-spoilage problem: Pasteur heated wine enough to kill most of the microbes present without altering the flavour, chilling then preventing any survivors from multiplying, and the new process took his name (Park Plus, Ch 1, p 8). Applied to milk, pasteurization is the heating of milk to such temperatures and for such periods of time as are required to destroy any pathogens that may be present, while causing minimal changes in the composition, flavour and nutritive value (WHO, 1970) (Park 28e, Ch 11, p 797).

Method	Time-temperature	Cooling	Use
Holder (Vat) method	63–66 °C for at least 30 minutes	Quickly to 5 °C	Recommended for small and rural communities; going out of use in larger cities
HTST (High Temperature Short Time)	~72 °C for not less than 15 seconds	Rapidly to 4 °C	Most widely used; very large quantities per hour
UHT (Ultra-High Temperature)	125 °C for a few seconds, usually in 2 stages (second under pressure)	Rapidly cooled and bottled as quickly as possible	Long shelf-life packaged milk

(All rows: Park 28e, Ch 11, p 797.)

What pasteurization does and does not do

- Kills nearly 90 per cent of bacteria in milk, including the more heat-resistant tubercle bacillus and the Q fever organism.
- Does not kill thermophilic bacteria or bacterial spores — hence pasteurized milk is rapidly cooled to 4 °C and kept cold until it reaches the consumer.
- Hygienically produced pasteurized milk has a keeping quality of not more than 8–12 hours at 18 °C (Park 28e, Ch 11, p 797).
- It is not a sterilization process — its purpose is to destroy pathogenic microorganisms; the time-temperature relation for hot-water pasteurization of articles is generally 70 °C for 30 minutes (Park 28e, Ch 3, p 156).

Milk-borne diseases controlled — Joint FAO/WHO Expert Committee on Milk Hygiene, 1970 (Park 28e, Ch 11, p 797)

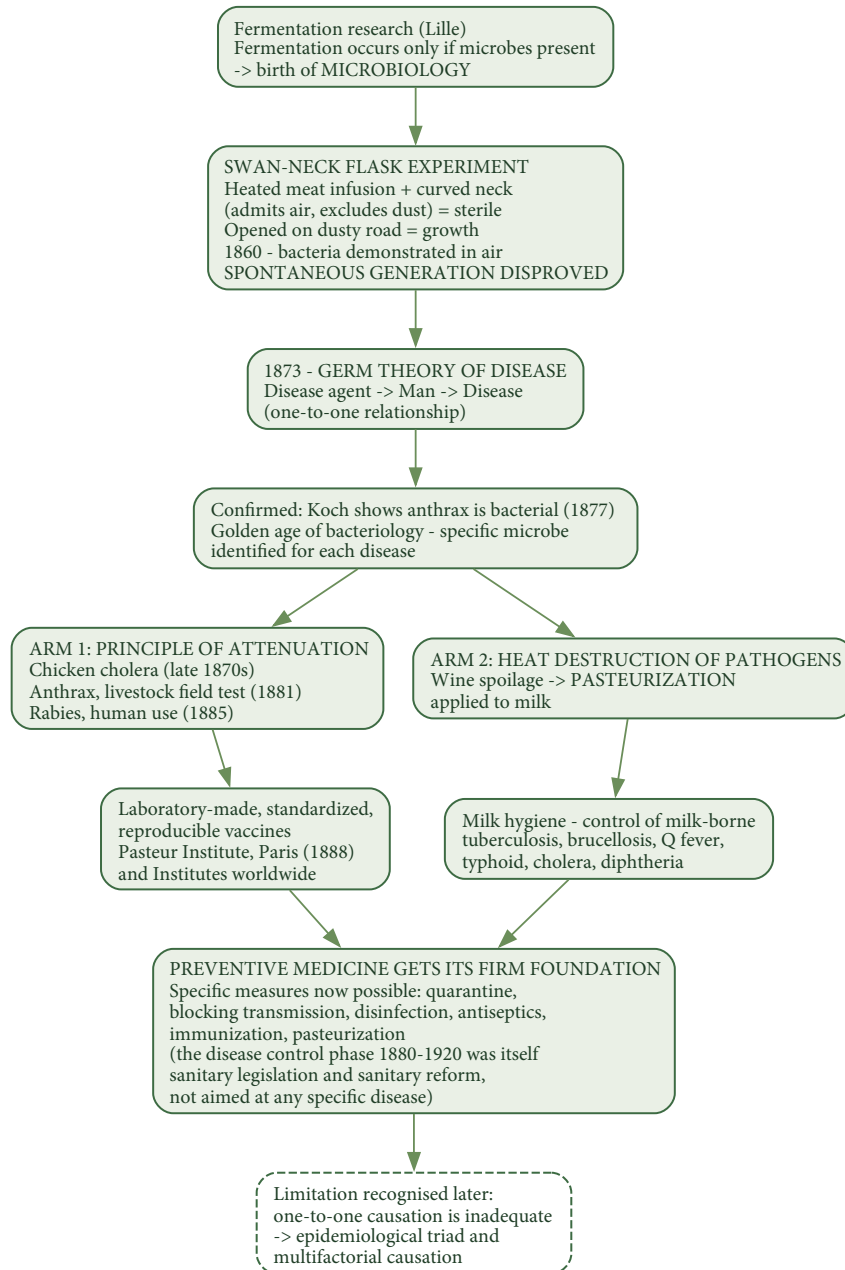
Group	Diseases
(1) Infections of animals transmissible to man — <i>primary importance</i>	Tuberculosis, brucellosis, streptococcal infections, staphylococcal enterotoxin poisoning, salmonellosis, Q fever
(1) ... — <i>lesser importance</i>	Cow-pox, foot and mouth disease, anthrax, leptospirosis, tick-borne encephalitis
(2) Infections primary to man transmitted through milk — <i>diarrhoeal</i>	Typhoid and paratyphoid fevers, shigellosis, cholera, enteropathogenic <i>Escherichia coli</i>
(2) ... — <i>non-diarrhoeal</i>	Streptococcal infections, staphylococcal food poisoning, diphtheria, tuberculosis, enteroviruses, viral hepatitis

Quality-control test	Basis	Standard
Methylene blue reduction test (before pasteurization)	Bacteria growing in milk decolourize methylene blue; sample held at 37 °C until blue colour disappears	Milk remaining blue longest is of best quality; grading scale by reduction time
Phosphatase test (after pasteurization)	Raw-milk phosphatase is destroyed at 60 °C for 30 minutes — a temperature closely matching pasteurization	Widely used to detect inadequate pasteurization or addition of raw milk
Standard plate count	Bacteriological quality of pasteurized milk	Limit of 30,000 bacteria per ml in most Western countries
Coliform count	Coliforms are completely destroyed by pasteurization	Coliforms absent in 1 ml; presence means improper pasteurization or post-pasteurization contamination

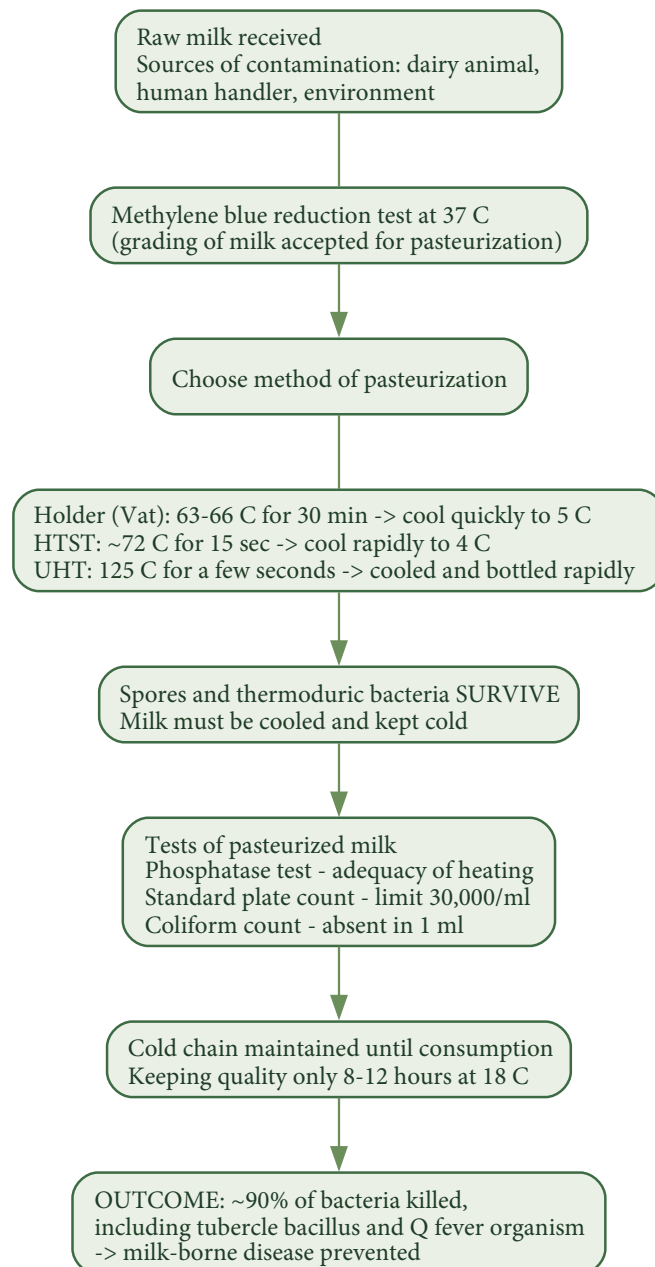
(All rows: Park 28e, Ch 11, pp 797–798.)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- Draw Pasteur's swan-neck flask experiment (Park Plus, Ch 1, p 8). Draw two flasks side by side, each with a long curved neck. Label the left flask "Heated meat infusion, swan neck intact — air enters, dust trapped in the bend — REMAINS STERILE"; label the right flask "Same infusion, opened on a dusty road and resealed — MICROBIAL GROWTH". Write the conclusion underneath: "Air alone does not generate life — the dust it carries does. Spontaneous generation disproved, 1860."
- Draw Pasteur's rabies inoculation principle (Plotkin 7e, Ch 50, p 1336). Draw a horizontal row of flasks labelled "Spinal cord of rabid rabbit, dried in potassium hydroxide"; label the left-most flask "Dried 15 days — avirulent — Dose 1", the intermediate flasks "Progressively less dried — increasing virulence", and the right-most "Dose 13 — most virulent"; write below the row "13 subcutaneous inoculations over ~10 days" and to the side "Joseph Meister, 9 years, bitten 14 times, treated 60 hours after exposure, 6 July 1885 — survived".



From Pasteur's laboratory to public health practice — the germ theory and its two operational arms



Pasteurization of milk - the operational pathway from raw milk to the consumer, with its quality-control checks

High-yield exam pointers

- **Five dates to write:** 1860 spontaneous generation disproved (swan-neck flask); 1873 germ theory advanced; 1881 anthrax vaccine field-tested in livestock; 6 July 1885 first human rabies vaccination (Joseph Meister); 1888 Pasteur Institute, Paris.
- **Swan-neck flask in one line:** heated infusion + curved neck admitting air but not dust = sterile; opened on a dusty road = growth. Also give **fermentation** → **microbiology** and **wine spoilage** → **pasteurization**.
- **Germ theory model** = *disease agent* → *man* → *disease*, a **one-to-one** relationship; modern medicine has moved away from it to the **epidemiological triad** and multifactorial causation (Pettenkofer).

- ▶ **Pasteurization temperatures** — memorise as 63/72/125: Holder 63–66 °C × 30 min, cooled to 5 °C; HTST 72 °C × 15 sec, cooled to 4 °C; UHT 125 °C × few seconds, rapidly cooled and bottled.
- ▶ **Phosphatase test** is the test of *adequacy* of pasteurization (phosphatase destroyed at 60 °C × 30 min); **methylene blue reduction test** is done *before* pasteurization; **standard plate count** 30,000/ml, coliforms absent in 1 ml.
- ▶ Pasteurization kills ~90 % of milk bacteria including **tubercle bacillus** and **Q fever organism**, but **not spores or thermophilic bacteria**; keeping quality 8–12 hours at 18 °C. Tuberculosis appears in BOTH FAO/WHO milk-borne groups.
- ▶ **Situate the contribution correctly**: Park's first phase, the **disease control phase (1880–1920)**, was itself sanitary legislation and sanitary reform, **not aimed at any specific disease**; Pasteur's germ theory gave **preventive medicine its firm foundation** and made specific measures possible. Next phase = **health promotional phase (1920–1960)**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park 28e lists "Pasteur's anti-rabies treatment (1883)" among the late-19th-century discoveries in preventive medicine, whereas Plotkin's Vaccines 7e and the Oxford Textbook of Public Health 7e both date the first human rabies vaccination to 1885 (Joseph Meister, 6 July 1885), so 1885 is used above for the first human use and 1883 should be understood as Park's date for the development of the treatment. For completeness at postgraduate level: Plotkin records that Pasteur's chicken cholera and anthrax results were less clear-cut than presented, that he withheld conflicting data from the Académie de Médecine, and that two earlier unpublicised human attempts preceded Meister — this is a historiographical caveat, not exam-answer content.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 1 — Man and Medicine: Towards Health for All, pp 17–18, 20–21; Ch 2 — Concept of Health and Disease, pp 51–52; Ch 3 — Principles of Epidemiology and Epidemiologic Methods, p 156; Ch 11 — Nutrition and Health, pp 797–798); Park Plus (Amrutha & Vijayalakshmi), Ch 1 — Man and Medicine: Towards Health for All, pp 8–9; Plotkin's Vaccines, 7ed (Ch 1 — A Short History of Vaccination, pp 21–23; Ch 11 — Anthrax Vaccines, p 203; Ch 50 — Rabies Vaccines, pp 1336–1337, 1344–1345); Oxford Textbook of Public Health, 7e (Ch 58 — Urbanization and Health, pp 2153, 2155–2156; Ch 96 — Immunization and Vaccination, pp 3582–3583); Joint FAO/WHO Expert Committee on Milk Hygiene, 1970, and WHO definition of pasteurization, 1970, as quoted in Park 28e.



Dark ages of medicine

Asked in: Paper I May 2018 — RGUHS MD Community Medicine

Definition — "Dark Ages of Medicine" is the name given to the medieval period, the roughly one thousand years between 500 and 1500 A.D., during which the science of medicine in Europe stopped advancing and reverted to primitive, supernatural practice after the fall of the Roman empire. Park describes the medieval period as "a time of great strife, of socio-political change, of regression and progression" — the phrase matters, because while European medicine regressed, Arab medicine and the monastic hospital preserved and even extended the inherited knowledge (Park 28e, Ch 1, p 16).

Place of the Dark Ages in the evolution of medicine

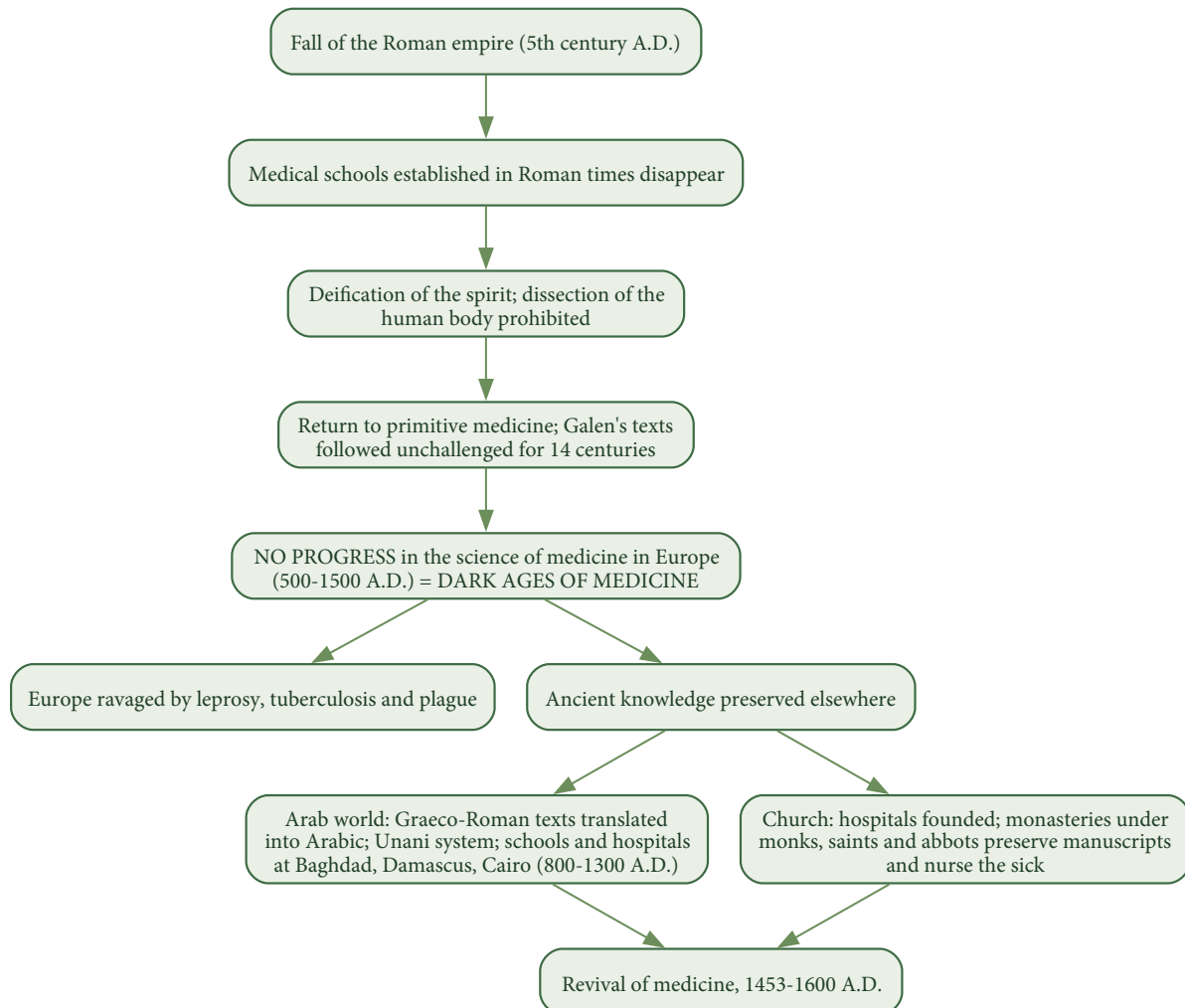
Period	Phase	Prevailing idea of disease causation	Marker of the phase
Prehistoric	Primitive medicine	Supernatural theory — disease as the work of spirits and displeased gods	Magico-religious healing
Antiquity to 500 A.D.	Ancient civilisations — Egyptian, Mesopotamian, Indian, Chinese, Greek, Roman	Theory of humors (Greek), <i>tridosha</i> (Ayurveda); Galen's predisposing, exciting and environmental factors	Hippocrates; Alexandria's museum, the first university, with a library of over 70,000 books (Park 28e, Ch 1, pp 15–16)
500–1500 A.D.	Middle Ages = Dark Ages of Medicine	Deification of the spirit; superstition and dogma	Medical schools of Roman times disappear; dissection prohibited (Park 28e, Ch 1, p 16)
1453–1600 A.D.	Revival of medicine	Theory of contagion (Fracastorius)	Age of individual scientific endeavour (Park 28e, Ch 1, p 17)
After 1500 A.D.	Dawn of scientific medicine	Anatomical and physiological	Vesalius 1543, Harvey 1628, Leeuwenhoek's microscope 1670, Jenner 1796
Mid-19th century onward	Great sanitary awakening, then germ theory and modern medicine	Filth/miasma, then specific micro-organisms	Chadwick's sanitary report; Public Health Act 1848, England (Park 28e, Ch 1, p 17)

In the sociological scheme of the eras of medicine the whole medieval period lies inside the **empirical health era (before 1850)**, in which the focus of medicine was the *symptom* and the underlying sciences were unknown (Sunder Lal, Ch 2, pp 66–67).

Why the medieval period is called the "Dark Ages of Medicine"

Determinant	Consequence for medicine
Fall of the Roman empire (5th century A.D.)	The medical schools established in Roman times disappeared; organised medical teaching in Europe ceased (Park 28e, Ch 1, p 16)

Determinant	Consequence for medicine
Deification of the spirit; rejection of the body as the accepted pattern of behaviour	Healing reverted to prayer, relics, exorcism and charms — a return to primitive medicine
Dissection of the human body was prohibited	Anatomy and physiology could not be studied; observation was replaced by authority
Uncritical authority of Galen	Galen's writings were accepted as standard medical textbooks for 14 centuries , unchallenged until the anatomist Vesalius in 1543 and the physiologist William Harvey in 1628 , almost 1500 years after his death (Park 28e, Ch 1, p 16)
War, famine, mass movement of people, insanitary crowded settlements	Europe was ravaged by leprosy, tuberculosis and plague (Park 28e, Ch 1, p 16)
Net effect	No progress in the science of medicine in Europe for about a thousand years



Why medicine regressed in medieval Europe, and where the ancient knowledge survived

Disease burden and the first public health responses of the period

- **Endemic and epidemic disease.** Leprosy, tuberculosis and plague dominated; the "Middle Ages" correspond to the *age of pestilence and famine*, with a low, unstable life expectancy at birth of 20–40 years (Oxford Textbook of Public Health 7e, Ch 21, pp 803–804).
- **The Black Death.** The second European plague pandemic of the Middle Ages is estimated to have caused **30–40 million deaths** (Oxford Textbook of Public Health 7e, Ch 76, pp 2883–2884).
- **Birth of quarantine.** From the **late fourteenth century**, first at **Ragusa (Dubrovnik)** and then across Italy, persons believed to be carrying a life-threatening communicable disease were physically segregated. **Forty days** free of disease was the period found necessary to clear an infected ship — the origin of the word *quarantine* (Oxford Textbook of Public Health 7e, Ch 2, p 182).
- **The Venetian model.** In the city-state of **Venice** arriving ships were not permitted to dock for 40 days and travellers at land borders were held in isolation for 40 days, to keep plague out — the first legislated recognition that disease crosses national frontiers (Oxford Textbook of Public Health 7e, Ch 75, p 2856).
- **Isolation of the sick**, the practice that later produced separate fever wards and fever hospitals, descends directly from this medieval segregation.
- **Counting of burials** as an epidemic early-warning system began in the Italian city states; its most developed descendant was **London's weekly Bills of Mortality** — sporadic in the sixteenth century, continuous from the early seventeenth century and then carrying attested causes of death, which gave **John Graunt** the raw material for *Natural and Political Observations* (1662), the beginning of epidemiology (Oxford Textbook of Public Health 7e, Ch 2, p 182).

Examination point: quarantine, isolation, notification and mortality registration — four permanent tools of communicable disease control — are all medieval inventions. This is precisely why the period is called one of *regression and progression* rather than of regression alone.

The counter-current — what progressed during the Dark Ages

A. Arab (Islamic) medicine — golden age 800–1300 A.D. While Europe was passing through the Dark Ages, the Arabs translated the Graeco-Roman medical literature into Arabic and so preserved the ancient knowledge. Borrowing largely from the Greeks and Romans they developed their own system, the **Unani system of medicine**, and founded schools of medicine and hospitals at Baghdad, Damascus, Cairo and other Muslim capitals. Osler's summary is the quotable line: "*The Arabs lit a brilliant torch from Grecian lamps*" (Park 28e, Ch 1, p 16).

Arab contribution	Detail
Rhazes (Abu Bakr, 865–925), Persian	Director of a large hospital in Baghdad and court physician; first to observe the pupillary reaction to light ; first to use mercurial purgatives ; published the first known book on children's diseases ; best regarded today for his book on smallpox and measles , which he distinguished clinically
Avicenna (Ibn Sina, 980–1037), Persian	Compiled the 21-volume encyclopaedia , the " Canon of Medicine "; responsible for elevating Islamic medicine to its zenith in the middle ages

Arab contribution	Detail
Pharmacology — the greatest Arab contribution	Seeking the "elixir of life" they created pharmaceutical chemistry, introduced a large number of herbal and chemical drugs, and invented the art of writing prescriptions ; introduced syrups, oils, poultices, plasters, pills, powders, alcoholates and aromatic waters. The words <i>drug</i> , <i>alcohol</i> , <i>syrup</i> and <i>sugar</i> are all Arabian
Transmission to India	The Unani-Tibb system, traced to ancient Greek medicine, was introduced into India by the Muslim rulers about the 10th century A.D. , and by the 13th century was firmly entrenched at Delhi, Aligarh, Lucknow and Hyderabad (Park 28e, Ch 1, p 14)

B. Christianity and the birth of the hospital. During the turbulent middle ages Christianity exerted a wholesome influence, and its spread led to the establishment of hospitals. Early medieval hospitals rarely specialised in treating the sick — patients were received to supply their bodily wants and cater to their spiritual needs. The **first hospital on record in England was built at York in 937 A.D.** A chain of hospitals then sprang up from Persia to Spain, with **more than 60 in Baghdad and 33 in Cairo**; Cairo's **Al Mansur** hospital had separate departments for different diseases, wards for both sexes, fountains to cool fever patients, libraries, and musicians and story-tellers for the sleepless (Park 28e, Ch 1, p 16).

C. The monasteries. Religious institutions headed by monks, saints and abbots admitted men and women of all ranks, including kings and queens. They both preserved the ancient knowledge and rendered active medical and nursing care to the sick (Park 28e, Ch 1, pp 16–17).

End of the Dark Ages — the revival of medicine, 1453–1600 A.D.

For many historians the revival of medicine spans 1453 to 1600 A.D.; it was an age of individual scientific endeavour (Park 28e, Ch 1, p 17).

Personality	Contribution that ended the Dark Ages
<i>Paracelsus</i> (1493–1541), Swiss	Revived medicine; born at a time " <i>when Europe stretched her limbs after a sleep of a thousand years in a bed of darkness</i> "; publicly burnt the works of Galen and Avicenna, attacked superstition and dogma, and turned medicine towards rational research
<i>Fracastorius</i> (1483–1553), Italian	Enunciated the theory of contagion — infection transferred by minute invisible particles; recognised sexual transmission of syphilis; founder of epidemiology
<i>Vesalius</i> (1514–1564), Brussels	Performed human dissections, demonstrated Galen's errors, raised anatomy to a science; called "the first man of modern science"; his <i>Fabrica</i> became a classic text
<i>Ambroise Pare</i> (1510–1590), French army surgeon	Advanced the art of surgery; "father of surgery"

Public health significance for the postgraduate

- It demonstrates that medical progress is not automatic — it stops when **observation is replaced by authority** and when dissection, autopsy and free enquiry are forbidden.

- ▶ It shows that **knowledge is preserved by institutions**, not individuals: translation (Arab), the hospital (Church) and the manuscript library (monastery) carried medicine across a thousand years.
- ▶ It is the historical origin of **quarantine, isolation and mortality registration**, still the backbone of the International Health Regulations and of outbreak response.
- ▶ It supplies the contrast that defines modern community medicine — from *symptom* and superstition in the empirical era to *community* and universal health coverage today (Sunder Lal, Ch 2, pp 66–67).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Horizontal timeline bar of the evolution of medicine** — mark and label, left to right: *Primitive medicine (supernatural theory) → Ancient civilisations: Hippocrates, Galen → MIDDLE AGES 500–1500 A.D. = DARK AGES OF MEDICINE* (shade this block and annotate below it: "medical schools disappear; dissection prohibited; leprosy, tuberculosis, plague") with a parallel band above it labelled "Golden age of Arabic medicine 800–1300 A.D." → *Revival of medicine 1453–1600 A.D. → Dawn of scientific medicine after 1500 A.D. → Great sanitary awakening, mid-19th century → Germ theory and modern medicine.*
- ▶ **Outline map — seats of medical learning during the Dark Ages** — label: *Baghdad* (more than 60 hospitals; Rhazes), *Damascus, Cairo* (33 hospitals; Al Mansur hospital), *Persia, Spain* (the chain of hospitals ran from Persia to Spain), *York, England* (first hospital on record, 937 A.D.), *Ragusa/Dubrovnik and Venice* (origin of 40-day quarantine, late 14th century).

High-yield exam pointers

- ▶ **Dates to write:** Middle Ages / Dark Ages of Medicine **500–1500 A.D.**; golden age of Arabic medicine **800–1300 A.D.**; revival of medicine **1453–1600 A.D.**; Galen unchallenged for **14 centuries** until Vesalius **1543** and Harvey **1628**.
- ▶ **The defining Park phrase, verbatim:** "a time of great strife, of socio-political change, of regression and progression" — always give both halves.
- ▶ **Three causes of regression:** fall of the Roman empire (medical schools vanish) → deification of the spirit → prohibition of dissection.
- ▶ **Three streams of progression:** Arab medicine (Rhazes, Avicenna, *Canon of Medicine*, Unani, pharmacology and prescriptions) — Christianity (hospitals; York 937 A.D.) — monasteries (manuscripts and nursing care).
- ▶ **Osler's quotable line:** "The Arabs lit a brilliant torch from Grecian lamps."
- ▶ **Public health yield:** quarantine born at Ragusa/Venice in the late 14th century, **40 days**; counting of burials → London's Bills of Mortality → John Graunt, 1662.
- ▶ **Who ended it:** Paracelsus burnt Galen and Avicenna; Fracastorius gave the theory of contagion; Vesalius made anatomy a science.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: "Dark Ages of Medicine" is a textbook label for medieval **European** medicine and is deliberately not applied to the same centuries in the Arab world, India or China, where medical learning continued — Park's own phrase "regression and progression" makes this qualification, and an answer that calls the whole world's medicine dark is factually wrong.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 1 (Man and Medicine: Towards Health for All), pp 13–19, and Ch 2 (Concept of Health and Disease); Oxford Textbook of Public Health 7e — Ch 2 (The history and development of public health in developed countries), Ch 21, Ch 75 and Ch 76; Sunder Lal — Textbook of Community Medicine, Ch 2 (Relationship of Social and Behavioural Sciences to Health and Disease), Table 2.1.

Coverage note: the indexed corpus carries the Dark Ages as a single short passage in Park 28e (Ch 1, p 16); the medieval epidemic and quarantine material was supplemented from the Oxford Textbook of Public Health 7e, and no captioned library figure for the history of medicine exists in the figure catalogue, so hand-drawn diagram labels are given instead.



De-professionalization of medicine

Asked in: Paper I 24-Sep-2025 — RGUHS MD Community Medicine

Definition — De-professionalization (deprofessionalization) of medicine is the planned transfer of a part of health care out of the exclusive control of the fully-qualified physician to auxiliaries, community-selected volunteers and the people themselves, so that "laymen have come to play a prominent role in the delivery of health care"; Park states that the practice of primary health care involves a good deal of de-professionalization of medicine (Park 28e, Ch 1, p 23). Sociologically it is the erosion of the *domination–deprivation* relationship, in which the medical profession occupies a dominant position by virtue of its expertise while the patient remains dependent for lack of specialised knowledge (Oxford Textbook of Public Health 7e, Ch 44). In public-health language it is the shift from **health care FOR the people to health care BY the people**, i.e. the current trend to "demedicalize" health and involve communities meaningfully (Park 28e, Ch 2, p 34).

Strategies of de-professionalization, with examples

#	Strategy	How it de-professionalizes care	Suitable example
1	Health team concept — use of auxiliaries	The physician of today is overworked; many of his functions can be performed by auxiliaries given suitable training. An auxiliary worker is "one who has less than full professional qualifications in a particular field and is supervised by a professional worker" (Park 28e, Ch 2, p 48)	Health worker (female) / auxiliary nurse midwife (ANM) at a sub-centre covering 5,000 population (3,000 in hilly, tribal, desert and difficult areas) giving essential obstetric, child and family-planning services (Sunder Lal, Ch 16, p 718)
2	Community-selected, locally-trained volunteers	They are selected by the local community and trained locally, serve free of charge, and overcome cultural and communication barriers (Park 28e, Ch 10, p 720)	Village Health Guides scheme, launched 1977, one guide per 1,000 population; and trained <i>dais</i> (Park 28e, Ch 2, p 34; Sunder Lal, Ch 17, p 744)
3	Accrediting a lay woman as first contact	A non-professional villager becomes the community's health activist, case-finder and treatment provider	ASHA (accredited social health activist) under the National Health Mission — the worked example below (Park 28e, Ch 2, p 34)
4	Task-shifting to a mid-level health provider	Clinical leadership of the peripheral team passes from the doctor to a non-MBBS provider	Community health officer (CHO) at the sub-health-centre Ayushman Arogya Mandir (erstwhile Health and Wellness Centre) — a BSc in community health, or a GNM nurse with BSc, or an Ayurveda practitioner, certified by a six-month community-health programme through IGNOU / state public health universities, leading the ANM–ASHA team (Sunder Lal, Ch 17, p 742)

#	Strategy	How it de-professionalizes care	Suitable example
5	Absorbing indigenous-system practitioners	Practitioners of indigenous medicine are counted within the new cadre of health workers forming the health team (Park 28e, Ch 1, p 23)	ASHA's drug kit deliberately contains both AYUSH and allopathic formulations (Park 28e, Ch 10, p 727)
6	Appropriate technology	Technology that is "scientifically sound, adaptable to local needs, acceptable to those who apply it and those for whom it is used, and that can be maintained by the people themselves in keeping with the principle of self-reliance" (Park 28e, Ch 10, p 720)	The ASHA drug kit and depot (oral rehydration therapy, iron-folic acid, disposable delivery kits, oral pills, condoms); the growth chart maintained by the anganwadi worker (Park 28e, Ch 10, p 727)
7	Self-care	"Health activities, including promotion, maintenance, treatment care and health-related decision making, carried out by individuals and families"; its generic attribute is its non-professional, non-bureaucratic, non-industrial character (Park 28e, Ch 2, p 34)	Patients taught to record their own blood pressure and blood sugar, so the load on the official health services is considerably reduced (Park 28e, Ch 2, p 34)
8	Community participation in planning and management	A community can participate in three ways — providing facilities, manpower, logistics and funds; being actively involved in planning, management and evaluation; and joining in and using the services (Park 28e, Ch 2, p 34)	Village Health Sanitation and Nutrition Committee (VHSNC) with an untied fund of Rs 10,000 per annum drawing the village health action plan (over 5.55 lakh established); Mahila Arogya Samiti for 50–100 urban households (Sunder Lal, Ch 17, p 744; Park 28e, Ch 7, p 535)
9	Delegating a supervised clinical task to a lay person	Diagnosis-linked treatment, previously the doctor's monopoly, is handed to a volunteer under protocol and supervision	ASHA treats minor ailments (diarrhoea, fever), gives first aid for minor injuries, and is the directly observed treatment, short-course (DOTS) provider under the National Tuberculosis Elimination Programme (formerly RNTCP) (Park 28e, Ch 10, p 727)
10	Demystifying medical knowledge through health education	Health education aims at people working actively with health workers in identifying their own health problems and in developing solutions and plans; jargon the patient cannot understand is itself a barrier (Park 28e, Ch 22, p 1019)	Village Health, Nutrition and Sanitation Day at the anganwadi each month, with at least four hours of contact between ANM, anganwadi worker, ASHA and beneficiaries (Sunder Lal, Ch 16, p 718)

#	Strategy	How it de-professionalizes care	Suitable example
11	Consumer empowerment and legal accountability	Lay individuals now share with, or often replace, the physician in deciding what is ethically acceptable; medical choices can no longer be made purely on the basis of medical science (Park 28e, Ch 12, p 837)	Consumer Protection Act 1986 brought medical services within its ambit, giving a forum for speedy redressal and six consumer rights — safety, to be informed, to choose, to be heard, to seek redressal, consumer education (Park 28e, Ch 12, p 837)

Worked example — the ASHA under the National Health Mission

Item	Detail
Selection	Preferably a resident of the village; a woman (married / widow / divorced) aged 25–45 years; formal education up to class 8; communication skills and leadership qualities; adequate representation of disadvantaged population groups (Park 28e, Ch 10, p 726; Ch 7, p 539)
Norm	One ASHA per 1,000 population; relaxed to one per habitation in tribal, hilly and desert areas; attached to the Ayushman Arogya Mandir team (Park 28e, Ch 7, p 539; Ch 10, p 726)
Numbers	Over 10.72 lakh ASHAs working, forming the base of the health human-resource pyramid (Sunder Lal, Ch 16, p 708)
Health-education role	Creates awareness on determinants of health — nutrition, basic sanitation and hygiene, healthy living, and available services; counsels on birth preparedness, safe delivery, breastfeeding and complementary feeding, immunisation, contraception, and reproductive tract / sexually transmitted infection (Park 28e, Ch 10, p 727)
Service role	Mobilises the community to immunisation, antenatal and postnatal check-up and supplementary nutrition; escorts pregnant women and children to the PHC / CHC / first referral unit; treats minor ailments and gives first aid; DOTS provider; depot holder for the drug kit (Park 28e, Ch 10, p 727)
Planning role	Works with the VHSNC of the gram panchayat to develop a comprehensive village health plan (Park 28e, Ch 10, p 727)
Supervision	Anganwadi worker and ANM act as resource persons for her training; ANM holds weekly/fortnightly review meetings with her (Park 28e, Ch 10, p 727)
Why it illustrates the concept	A woman with class-8 schooling and no professional qualification now performs case-finding, health education, counselling, drug distribution and directly observed treatment — tasks once the physician's monopoly — while the doctor supervises and receives referrals

The physician's changed role

- ▶ He can no longer restrict himself to the traditional role of **diagnoser of ailments, prescriber of pills and potions, and exciser of lumps** (Park 28e, Ch 1, p 23).
- ▶ He acquires five new roles — **educator, case-finder, preventer, counsellor, and agent of social change** (Park 28e, Ch 1, p 23).
- ▶ He becomes leader of a health team, defined as "a group of persons who share a common goal and common objectives, determined by community needs, towards the achievement of which

each member of the team contributes in accordance with her/his competence and skills, and respecting the functions of the other"; the auxiliary is an essential member of that team (Park 28e, Ch 2, p 48).

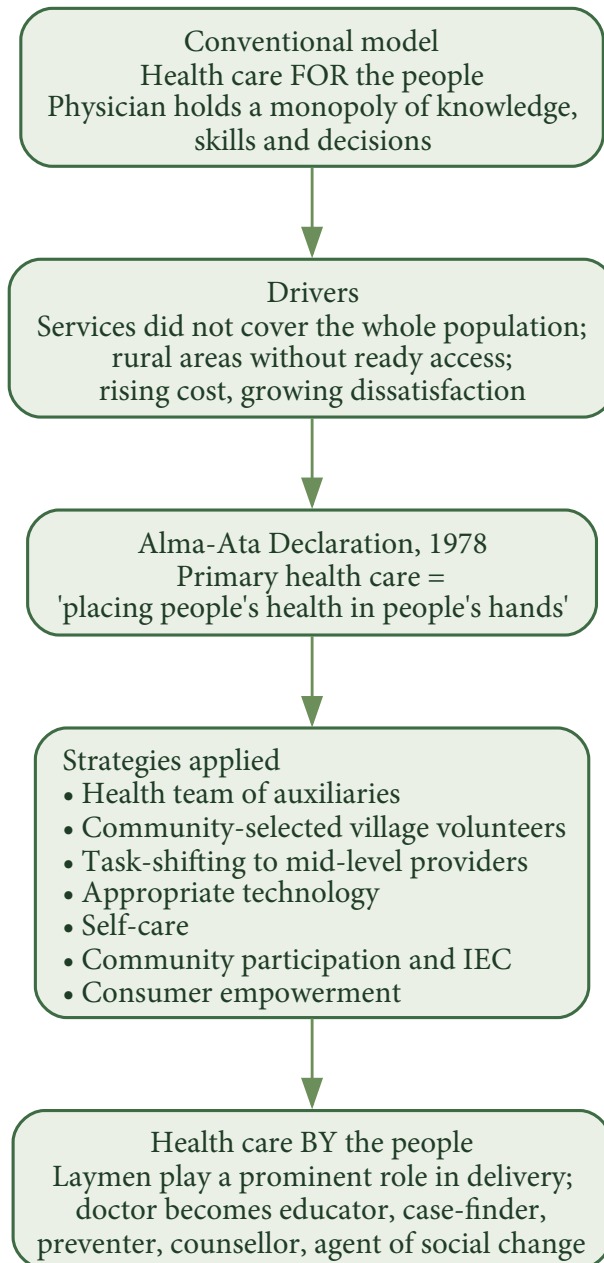
- ▶ The aim is the right "mix" of health personnel for full health coverage of the entire population; the mere presence of a variety of health professionals is not teamwork — the benefit comes from the proper division and combination of their operations (Park 28e, Ch 2, p 48).

Limitations and safeguards

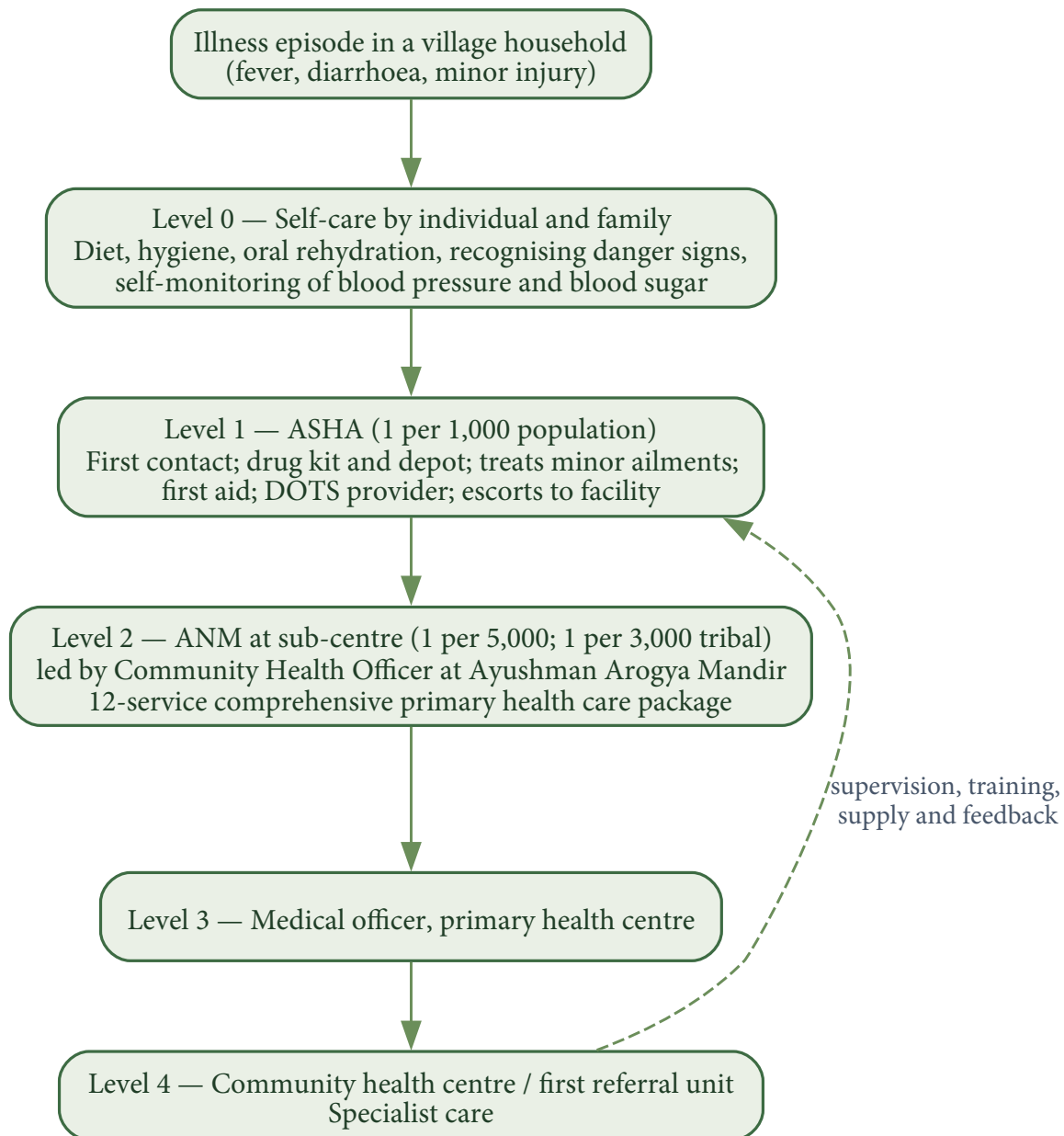
- ▶ **Resistance from within the profession** — in the health sector the greatest resistance to the health guide's involvement in primary health care came from the medical profession rather than from the lay public (Park 28e, Ch 2, p 34).
- ▶ **Social stratification** — traditional Indian society cut across rigid religion and caste lines has been a serious obstacle to complete community participation; no standard pattern can be prescribed and flexibility of approach is essential (Park 28e, Ch 2, p 34).
- ▶ **Quality depends wholly on training** — the effectiveness of services in India largely depends on the auxiliaries' skills and competence, which their basic training and continuing education determine (Sunder Lal, Ch 16, p 708); e.g. 3–5 day in-service modular training of sub-centre workers and ASHAs at the CHC on non-communicable diseases, with on-the-job support by PHC medical officers (Sunder Lal, Ch 8, pp 529–530).
- ▶ **It is delegation, not abandonment** — by definition the auxiliary is supervised by a professional worker, so an intact referral chain (sub-centre → PHC → CHC / first referral unit) and defined supervision are what make de-professionalization safe (Park 28e, Ch 2, p 48).
- ▶ **A curative culture works against it** — the obstacles listed for primary health care in India are shortage of health manpower, entrenchment of a curative culture within the existing health system, and a high concentration of health services and personnel in urban areas; each of these blunts the shift of care towards non-physician cadres (Park 28e, Ch 2, p 48).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Pyramid of health human resources in India (label exactly):** base — *People / families: self-care*; next — *ASHA (1:1,000) · Anganwadi worker (1:400–800) · Trained dai*; next — *ANM / multipurpose health worker at sub-centre (1:5,000; 1:3,000 hilly-tribal-desert)*; next — *Community health officer (mid-level health provider) at sub-centre Ayushman Arogya Mandir*; next — *Medical officer, primary health centre*; apex — *Specialists, community health centre / first referral unit*. Left-side arrow pointing up labelled "increasing professional qualification"; right-side arrow pointing down labelled "increasing de-professionalization and closeness to the community".



De-professionalization of medicine — the shift from health care FOR the people to health care BY the people



Delegated-care pathway for a common illness in a de-professionalized primary care system, with the supervisory return loop

High-yield exam pointers

- ▶ **Park's definitional sentence to reproduce:** "The practice of primary health care involves a good deal of deprofessionalization of medicine. Laymen have come to play a prominent role in the delivery of health care."
- ▶ **Physician's five new roles:** educator, case-finder, preventer, counsellor, agent of social change.
- ▶ **Numbers to write:** ASHA 1:1,000 (1 per habitation in tribal/hilly/desert), 25–45 years, class-8 schooling, >10.72 lakh working; anganwadi worker 1:400–800, 4 months' training, Rs 1,500/month honorarium; ANM/sub-centre 1:5,000 (1:3,000 hilly-tribal-desert); Village Health Guides scheme 1977, 1:1,000; VHSNC untied fund Rs 10,000/year; CHO certificate **6 months**.
- ▶ **Auxiliary worker (WHO):** "one who has less than full professional qualifications in a particular field and is supervised by a professional worker" — and WHO no longer uses the term "paramedical".

- ▶ **Appropriate technology (WHO):** scientifically sound, adaptable to local needs, acceptable to those who apply it and those for whom it is used, maintainable by the people themselves in keeping with self-reliance.
- ▶ **International example to quote:** China's **barefoot doctors** — community participation on an unprecedented scale, which greatly influenced India's village health guides (Park 28e, Ch 10, p 720).
- ▶ **Closing quotation (Henry Sigerist):** "the war against disease and for health cannot be fought by physicians alone. It is a people's war in which the entire population must be mobilized permanently."

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park 28e spells the term "deprofessionalization" and cites the **Consumer Protection Act 1986**; the retrieved corpus does not carry any later re-enactment of that Act, so quote it as the textbook does. Health and Wellness Centres have been officially renamed **Ayushman Arogya Mandir**, and RNTCP is now the National Tuberculosis Elimination Programme — write the current name first with the former name in brackets once.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 1 (Man and Medicine: Towards Health for All), Ch 2 (Concept of Health and Disease), Ch 7 (Health Programmes in India), Ch 10 (Health Care of the Community), Ch 12 (Medicine and Social Sciences), Ch 22 (Communication for Health Education); Sunder Lal — Textbook of Community Medicine, Ch 8, Ch 16 (Health Planning and Management), Ch 17 (Healthcare of the Community); Oxford Textbook of Public Health 7e, Ch 44 (Sociology and Psychology in Public Health); Declaration of Alma-Ata, 1978; Ayushman Bharat — Health and Wellness Centre (Ayushman Arogya Mandir) norms as carried in the above texts.

Coverage note: the corpus defines de-professionalization in a single short paragraph (Park 28e, Ch 1, p 23); the strategies above are assembled from the corpus's own primary health care, health team, community participation, appropriate technology and self-care passages, which is where the textbook itself locates them.



Social stratification

Asked in: Paper I Nov 2017 — RGUHS MD Community Medicine

Definition — **Social stratification** is a relatively fixed, hierarchical arrangement in society by which groups have different access to resources, power and perceived social worth; it is, in short, a system of **structured social inequality** (Sociology: The Essentials 6e, Ch 8, p 205). It arises when the **social differentiation** normally present in every group — the process by which different statuses develop — becomes organised into a ranked hierarchy. In community medicine the operational expression of a person's position in that hierarchy is **socio-economic status (SES)**, defined by Park as *the position that an individual or family occupies with reference to the prevailing average standards of cultural and material possessions, income, and participation in group activity of the community* (Park 28e, Ch 12, p 832).

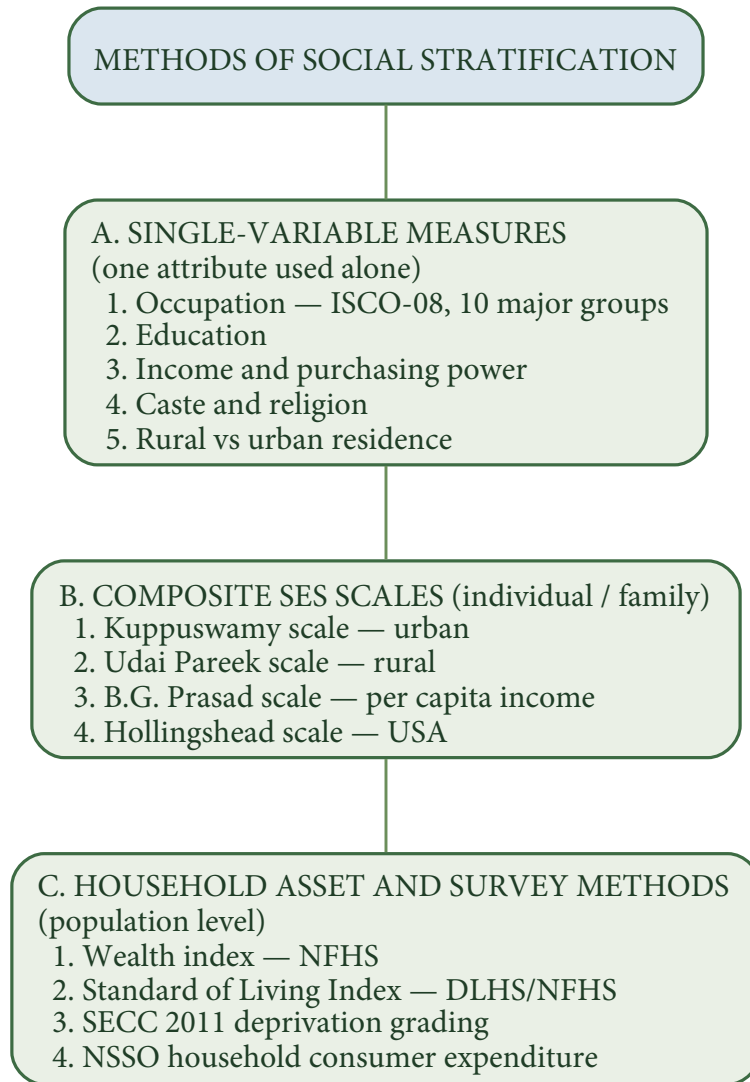
Allied terms that must be distinguished

Term	Meaning
Social differentiation	Process by which different statuses develop in any group or society; a prerequisite for, but not the same as, stratification (Sociology 6e, Ch 8, p 205)
Social class	The structural position a group holds relative to the economic, social, political and cultural resources of society; bound up with economic status, education, way of life, attitudes and exposure to stress (Park 28e, Ch 12, p 831)
Life chances	<i>Max Weber's</i> term for the opportunities people share by virtue of belonging to a given class (Sociology 6e, Ch 8, p 207)
Social mobility	Movement of an individual or family between class positions — upward with attainment of literacy, better occupation and higher income, or downward (Sunder Lal, Ch 2, p 75)
Open vs closed class system	In an open system movement up the social ladder is unrestricted and based on achievement; in a closed system, e.g. the Indian caste system, people do not change caste or religion (Park 28e, Ch 12, p 831)

Bases (systems) on which societies are stratified

System	Placement determined by	Indian/other example
Estate system	Power and property held by a single elite class	Feudal landholding societies (Sociology 6e, Ch 8, p 233)
Caste system	Birth — a closed system with endogamy, food taboos, ritual purity, caste-linked occupation, supported by religion; less rigid in urban and industrial areas	Indian caste and sub-caste hierarchy (Park 28e, Ch 12, pp 825, 831); <i>apartheid</i> South Africa (Sociology 6e, Ch 8, p 233)
Class system	Achievement — achieved status earned through acquisition of resources and power; boundaries blurred and mobility possible	Modern urban industrial societies (Sociology 6e, Ch 8, p 233)
Weber's three dimensions	<i>Class</i> (economic), <i>status</i> (prestige — cultural/social), <i>party</i> (power — political); a multidimensional view, against Marx's purely economic one	Basis of all modern composite SES scales (Sociology 6e, Ch 8, pp 223–224)

Methods used to stratify — overview



Methods available to stratify individuals, families and populations by social position

A. Single-variable measures and their limitations

Measure	How it stratifies	Limitations
Occupation — the most widely used single measure; graded by the <i>International Standard Classification of Occupations 2008 (ISCO-08)</i> : 436 unit groups → 130 minor → 43 sub-major → 10 major groups (managers; professionals; technicians and associate professionals; clerical support; service and sales; skilled agricultural, forestry and fishery; craft and related trades; plant and machine operators; elementary occupations; armed forces)	Occupation determines economic rewards, authority, obligations, degree of status, values and life-style (Park 28e, Ch 12, pp 831–832)	(1) Heterogeneous grouping — each class lumps unrelated occupations differing in education and family background; classification is one of convenience only. (2) Occupational mobility — people change occupations, and occupations themselves are promoted or demoted in the hierarchy, invalidating comparison over time. (3) Women — difficulty in classifying women alongside men; the same job title (e.g. "secretary") means very different things for a man and a woman. (4) India has no satisfactory occupational class system (Park 28e, Ch 12, pp 831–832)
Education	Measures the inculcation of values, knowledge and achievement of the individual	Reflects achievement rather than current material circumstances; correlates only imperfectly with income and occupation, so it identifies a different gradient (Park 28e, Ch 12, p 832; Oxford Textbook of Public Health 7e, Ch 9, pp 389–390)
Income / wealth	Groups people as upper, middle and lower class	People have great reluctance to state income truthfully ; income is often concealed and hard to verify; needs continual price-index updating to stay comparable (Park 28e, Ch 12, pp 831–832; Sunder Lal, Ch 2, p 74)
Purchasing power	May classify people better than occupation does	No standard graded instrument in Indian practice; still an indirect proxy for income (Park 28e, Ch 12, p 832)
Caste	Important basis of differentiation especially in rural India; caste groups are hierarchical and carry different degrees of social prestige	A closed system — no mobility, so it cannot register improvement in circumstances; less rigid in urban areas , where inter-caste relationships are not strictly hierarchical, so it discriminates poorly in cities (Park 28e, Ch 12, pp 825, 831)

Measure	How it stratifies	Limitations
Religion	Bears on socio-cultural patterns of living — habits, customs, cleanliness, eating, clothing, child care; infant mortality in India differed between Christians (lowest), Hindus and Muslims (highest)	Not hierarchical, therefore not a true stratifier; confounded by regional, educational and economic differences (Park 28e, Ch 12, p 832)
Rural vs urban residence	Rural and urban people differ in life-style, opinions and values, which bear on health and sickness status	A crude binary that conceals wide within-group inequality (Park 28e, Ch 12, p 832)

B. Composite socio-economic status scales for the individual or family

Scale	Components and grading	Limitations
<i>Kuppuswamy's</i> scale — urban areas (the most widely used in India)	Three components scored for the head of the family : education (professional/honours 7 → illiterate 1), occupation (legislators, senior officials and managers 10 → unemployed 1), monthly family income (7 bands scored 12, 10, 6, 4, 3, 2, 1; top band > ₹1,26,360/month in the January 2018 update). Total (max 29) → 26–29 = class I upper ; 16–25 = II upper middle ; 11–15 = III lower middle ; 5–10 = IV upper lower ; < 5 = V lower (Sunder Lal, Ch 2, p 74)	Validated for urban use only; grades the head of family alone , so a working spouse and the rest of the household are invisible; the income column becomes obsolete within a year and must be re-derived from the All-India Average Consumer Price Index for Industrial Workers; carries all the income-concealment and occupational-heterogeneity problems of its components (Park 28e, Ch 12, pp 832–833; Sunder Lal, Ch 2, p 74)
<i>Udai Pareek's</i> scale — rural areas	Nine components, each weighted: caste , occupation , education , land holding , social participation , family size , house type , farm power (draught animals/tractor) and material possession . Total → > 43 = I ; 33–42 = II ; 24–32 = III ; 13–23 = IV ; < 13 = V (Sunder Lal, Ch 2, p 75)	Applicable only to an agrarian rural household — land, draught animals and farm implements carry heavy weight, so landless, non-farming and urban families are misgraded; caste is scored as a variable , which is socially and ethically contentious; the material-possession list (bullock cart, radio, cycle) no longer discriminates between households (Sunder Lal, Ch 2, p 75)
<i>B.G. Prasad's</i> scale (1961), updated	Uses a single variable — per capita monthly income . Original 1961: class I ≥ ₹100, II ₹50–99, III ₹30–49, IV ₹15–29, V < ₹15. Updated to May 2021 : I ≥ ₹7,863; II ₹3,931–7,862; III ₹2,359–3,930; IV ₹1,179–2,358; V < ₹1,179 (Sunder Lal, Ch 2, p 75)	Unidimensional — ignores education and occupation, so two families with identical income but different education are graded alike; depends entirely on a self-reported, concealable income ; requires recalculation with every consumer price index base-year revision, and different published updates give different cut-offs for the same year (Sunder Lal, Ch 2, p 75)

Scale	Components and grading	Limitations
<i>Hollingshead's scale (USA)</i>	Education, occupation and residential address (Park 28e, Ch 12, p 832)	Developed for and validated in a Western urban setting; residential address is a poor discriminator in Indian cities where slum and affluent housing coexist within one locality (Park 28e, Ch 12, p 832)

C. Household asset and survey-based methods

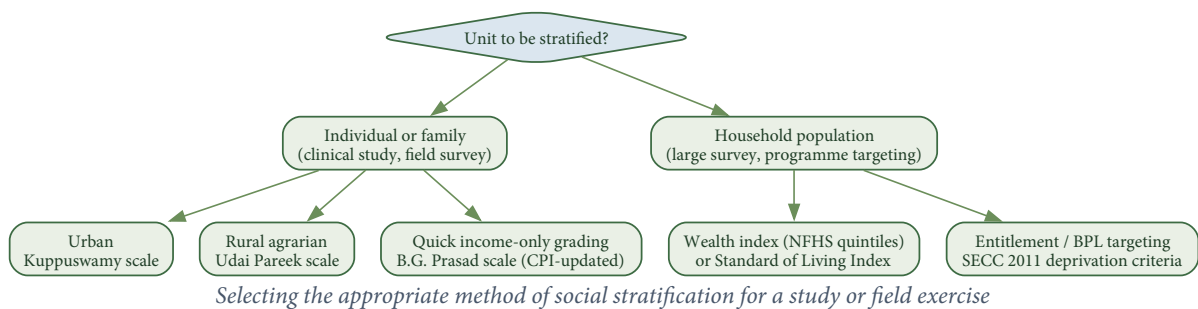
Method	Basis and grading	Limitations
Wealth index (introduced as a background characteristic in the <i>National Family Health Survey (NFHS)-3</i>)	Composite index of household asset and housing data: electrification, type of windows, drinking-water source, toilet facility, flooring, exterior walls, roofing, cooking fuel, house ownership, persons per sleeping room, bank/post-office account, and ownership of mattress, pressure cooker, chair, cot, table, fan, radio, television, sewing machine, telephone, computer, refrigerator, watch, bicycle, motorcycle, animal cart, car, water pump, thresher and tractor. Households ranked into quintiles (Park 28e, Ch 12, p 833)	Relative, not absolute — by construction 20% of the household population falls in each quintile nationally, so a "richest quintile" household in a poor state is not comparable to one in a rich state, and it is not necessarily 20% per quintile at state level; asset lists date rapidly and are not comparable between survey rounds (Park 28e, Ch 12, pp 833–834)
Standard of Living Index (SLI) — used in District Level Household Surveys and NFHS	Scores drinking-water source, house type, lighting, cooking fuel, toilet facility and ownership of listed items; total 0–40 → low ≤ 9, medium 10–19, high > 19 (Sunder Lal, Ch 2, pp 74–75)	Only three broad grades, so it is insensitive to gradients within a category; asset-based and therefore lags real change in income; item list is dated (Sunder Lal, Ch 2, pp 74–75)
<i>Socio-Economic and Caste Census (SECC) 2011</i>	Door-to-door enumeration across 640 districts; automatic exclusion on 14 parameters, automatic inclusion on 5 parameters (households without shelter, destitute living on alms, manual scavengers, primitive tribal groups, legally released bonded labourers) and grading on 7 deprivation criteria for rural households (Sunder Lal, Ch 2, p 75)	Designed to rank households for entitlement (below poverty line listing), not to describe a graded social hierarchy; a single 2011 enumeration, so it does not track mobility; urban grading uses a separate occupational-vulnerability list
Household consumer expenditure (<i>National Sample Survey Organisation</i>)	Sum of monetary values of all goods and services consumed by the household in the reference period; thin annual sample with a quinquennial (5-yearly) large-sample survey (Sunder Lal, Ch 2, p 75)	Yields a population distribution, not a ready individual class; long inter-survey interval; recall-dependent

Limitations common to all methods

- **No single best measure.** Education, occupation and income usually point in the same direction, but the strength of association varies between them because the underlying causal mechanisms differ; there is no "best single measure" (Oxford Textbook of Public Health 7e, Ch 9, pp 389–390).

- **Non-comparability between studies.** Social classification varies from country to country and means different things to different people, so associations of disease with social class vary with the investigator's concept of class and results of different studies cannot be pooled (Park 28e, Ch 3, p 89).
- **Categories are conveniences.** Every class is a heterogeneous grouping; membership is assigned for convenience of analysis, not because the members are truly alike (Park 28e, Ch 12, p 832).
- **Drift over time.** Social mobility of individuals and re-ranking of occupations both erode the validity of a classification used over years (Park 28e, Ch 12, p 832; Sunder Lal, Ch 2, p 75).
- **Every measure has a "snag" and must be examined critically and used with discrimination** depending on the subject matter of the enquiry (Park 28e, Ch 12, p 832).

Choosing the appropriate method



DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Social stratification pyramid.** Draw a five-tier pyramid, apex to base labelled Class I — Upper, Class II — Upper middle, Class III — Lower middle, Class IV — Upper lower, Class V — Lower. Feed three arrows into the left face labelled **Education, Occupation, Income** (the three Kuppuswamy inputs); mark the score bands 26–29, 16–25, 11–15, 5–10 and < 5 against the corresponding tiers; label the base "widest stratum" and the apex "narrowest stratum"; add a vertical double-headed arrow alongside labelled **Social mobility** (open class system) and a crossed-out arrow labelled **Caste — closed class system**.

High-yield exam pointers

- **Definition to write verbatim:** social stratification = a relatively fixed hierarchical arrangement by which groups have different access to resources, power and perceived social worth = structured social inequality.
- **Kuppuswamy cut-offs (memorise):** education max 7 + occupation max 10 + income max 12 = 29; classes 26–29 / 16–25 / 11–15 / 5–10 / < 5.
- **Pareek cut-offs:** > 43 / 33–42 / 24–32 / 13–23 / < 13; nine components — caste, occupation, education, land, social participation, family size, house, farm power, material possession.
- **Prasad (May 2021):** ≥ ₹7,863 / ₹3,931–7,862 / ₹2,359–3,930 / ₹1,179–2,358 / < ₹1,179 per capita per month; original 1961 figures ₹100 / 50 / 30 / 15.
- **SLI:** total score 0–40 → low ≤ 9, medium 10–19, high > 19. **Wealth index:** quintiles, 20% of household population per quintile nationally.

- ▶ ISCO-08: 436 unit → 130 minor → 43 sub-major → **10 major groups**.
- ▶ **Three limitations of occupation-based classification** (guaranteed sub-question): heterogeneous grouping, occupational mobility, difficulty in classifying women.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park 28e prints the Kuppaswamy income slabs recalculated to the 2007 value while Sunder Lal prints the January 2018 update and the May 2021 Prasad update, because the income column of both scales is re-derived from the All-India Average Consumer Price Index for Industrial Workers whenever the base year changes — always quote a slab together with the year of its update, since the class cut-off scores stay fixed but the rupee bands do not.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, NFHS-6 (2023–24), MoSPI Household Consumption Expenditure Survey (2022–23 and 2023–24), Labour Bureau CPI-IW (base 2016 = 100), Global Adult Tobacco Survey, WHO Health Equity Assessment Toolkit

Programme and policy updates (India)

- ▶ The textbook's "quinquennial NSSO consumer-expenditure survey" now runs as the **Household Consumption Expenditure Survey (HCES)**, conducted by the National Sample Survey Office under MoSPI; the 2023–24 round covered **261,953 households** against **101,651** in the 2011–12 (68th) round, so monthly per capita expenditure (MPCE) ranking now rests on a far larger sample than the textbook describes. Between those two rounds the food share of household spending fell from 53.4% to 47.8% in rural and 43.9% to 40.8% in urban areas, and the cereal share of expenditure roughly halved (rural 11.24% → 5.13%, urban 7.37% → 4.04%), the drop being sharpest in the poorest quintiles. [PMID 41946892]
- ▶ **HCES doubles as the between-census source for amenity-based deprivation grading**: the 2022–23 round (261,746 households across all 36 states and Union Territories) found **12.5% of Indian households — 13.4% of the population, about 162.7 million people — with no toilet at all**, 17.4% rural against 1.6% urban, roughly 93% of such households rural. The gradient runs along every standard stratifier: 27.9% in the lowest monthly-consumption quintile, 23.3% where the head of the household was not literate, 29.1% in Scheduled Tribe and 19.1% in Scheduled Caste households; about 71% of all those without a toilet live in just six states. [PMID 40536324]
- ▶ **The Alkire–Foster Multidimensional Poverty Index (MPI) is now being computed directly from NFHS data as a deprivation-based stratifier to set beside SECC 2011**: one such analysis across 29 states and Union Territories derived an all-India **rural MPI of 0.110** from NFHS 2019/21 — headcount 0.260, i.e. **26% of rural people multidimensionally poor**, each deprived on average in **42.4%** of the ten indicators (three dimensions — health, education, living standard) — lowest in Kerala and highest in Bihar. On regression, a 1% higher Scheduled Caste density and a 1% higher Scheduled Tribe density were each associated with a 0.14% higher rural MPI (significant at 5% and 10% respectively), and a 1% better local-governance score with a 0.32% lower MPI. Quote this as the authors' own calculation from NFHS, not as an official national estimate. [PMID 40007909]

Current national data

- ▶ **NFHS-6 (2023–24) fact sheets have been released by the MoHFW and are now the current round** — cite NFHS-6 rather than NFHS-5 (2019–21) when quoting wealth-index quintiles and other background characteristics; they record persistent undernutrition, rising urban overweight, declining exclusive breastfeeding and widening rural–urban and tribal inequities against gains in immunisation, antenatal care and reduction of diarrhoea and pneumonia, but carry **no mortality indicators**. [PMID 42554934]
- ▶ **Caste still behaves as a live health stratifier despite being a closed system:** across NFHS-3 to NFHS-5 the composite index of anthropometric failure gap between Hindu upper-caste and Scheduled Caste children narrowed only from 19.28 to 14.67 percentage points, and upper-caste versus Scheduled Tribe from 22.46 to 18.42 points, with a falling share of the gap explained by the measured child, maternal, household and regional variables — that is, a rising unmeasured residual [PMID 41821265]; in GATS 2016–17 (74,037 adults) tobacco use graded cleanly by social group — **ST 40.4% > SC 33.5% > OBC 26.5% > General 24.8%** — and decomposition of the ST-versus-General gap attributed 18% to region of residence, about 15% jointly to wealth and occupation, 4.6% to higher education and 2.7% to sex, leaving nearly one-third of the disparity unexplained. [PMID 41671315]

Guideline and legislative updates

- ▶ **The method for updating the B.G. Prasad scale has been standardised into a published seven-step template, and it explains why published cut-offs disagree for the same month:** the CPI-IW series now runs on base 2016 = 100, so three linking factors must be chained back to 1961 — $2.88 (2001 \leftrightarrow 2016) \times 4.63 (1982 \leftrightarrow 2001) \times 4.93 (1960 \leftrightarrow 1982)$ — giving a multiplication factor of $(\text{current CPI-IW} \times 2.88 \times 4.63 \times 4.93) \div 100$, applied to the 1961 class limits, rounded to the nearest ₹10, with the gaps then closed. All-India and centre-wise CPI-IW carry *different* linking factors, so for October 2023 the all-India index of 138.4 yields a factor of 90.98 and class I $\geq ₹9,100$, whereas the Coimbatore centre (index 138.0; factors $2.6 \times 4.49 \times 4.93$) yields $\geq ₹7,940$ — a defensible answer must state the index used, the centre and the month. [PMID 41451071]
- ▶ **Prasad scale, January 2024 (CPI-IW 138.9):** class I $\geq ₹9,131$; II ₹4,566–9,130; III ₹2,739–4,565; IV ₹1,370–2,738; V $< ₹1,370$ per capita per month, projected to $\geq ₹9,669$ by December 2024; CPI-IW is released about two months in arrears, which is why regression-projected slabs circulate for the intervening months. [PMID 40949512]
- ▶ **A new multidimensional Indian scale has been validated — Solomon's Socioeconomic Status Index (SESI, 2026)** — nine weighted domains: housing quality 20%, education 15%, occupation 15%, land 10%, vehicle 10%, business ownership 10%, durable assets 10%, locality type 5% and **digital access** 5%; tested in 1,247 households (Cronbach's α 0.84, test–retest r 0.89, CFI 0.94, RMSEA 0.051) and correlating with the modified Kuppaswamy scale ($r = 0.72$) and B.G. Prasad scale ($r = 0.68$). Its stated rationale — obsolete income bands, urban-centric assumptions and unidimensional frameworks — is precisely the limitation list above. [PMID 42362781]

Global and WHO developments

- ▶ **WHO's Health Equity Assessment Toolkit (HEAT, 2024 update, version 6.1) is the current standard for disaggregating a health indicator by an equity stratifier**, reporting each gap on both an absolute and a relative scale — difference, ratio, population attributable risk (PAR),

population attributable fraction (PAF), and the absolute and relative concentration indices.

Applied to India 2000–2022 (UN IGME data) on two of its dimensions, sex and wealth quintile: the sex gap in under-5 mortality closed from 9.6 to 0.2 per 1,000 live births, but the poorest-versus-richest quintile gap only narrowed from 81.9 to 28.9, the ratio held at about 2.9 throughout and the PAF barely moved (–49.9 to –48.8) — economic stratification, not sex, is now the binding inequality. [PMID 41624926]

- **Socio-economic inequality in early neonatal death has widened in India between 1993 and 2021** across five NFHS rounds: absolute wealth-quintile inequality in the early neonatal mortality rate rose from 18.3 to 18.4 per 1,000 and relative inequality from 1.9 to 2.9, even while both fell for late-neonatal, post-neonatal and child mortality. Most child deaths now fall in the two lowest wealth quintiles and among mothers with no schooling, and those children are not on track for the SDG early-neonatal and post-neonatal targets (neonatal mortality ≤ 12 and under-5 mortality ≤ 25 per 1,000 live births by 2030) — so a wealth-quintile or maternal-education stratifier remains obligatory in any child-survival analysis. [PMID 40316411]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 12 — Medicine and Social Sciences, pp 825, 831–834; Ch 3, p 89); Sunder Lal — Textbook of Community Medicine (Ch 2 — Relationship of Social and Behavioural Sciences to Health and Disease, pp 73–75); Sociology: The Essentials 6e (Ch 8 — Social Class and Social Stratification, pp 205–233); Oxford Textbook of Public Health 7e (Ch 9 — Socioeconomic inequalities in health, pp 389–390); International Standard Classification of Occupations 2008 (ISCO-08); National Family Health Survey wealth index (NFHS-3 onward, quintiles reported up to NFHS-5, 2019–21); Socio-Economic and Caste Census 2011; updated B.G. Prasad classification, May 2021.

Recent-advances sources: 41946892 — Food consumption patterns and their micronutrient content in India: evidence from the household consumption expenditure surveys, 2011-12 and 2023-24, *European Journal of Clinical Nutrition* 2026, DOI [10.1038/s41430-026-01732-3](https://doi.org/10.1038/s41430-026-01732-3); 40536324 — Prevalence and burden of no-toilet households in India: an analysis of 261,746 households in 36 states/Union Territories in 2022-2023, *Global Health Action* 2025, DOI [10.1080/16549716.2025.2511351](https://doi.org/10.1080/16549716.2025.2511351); 40007909 — Caste, local governance effectiveness, and multidimensional poverty in rural India, *Frontiers in Sociology* 2025, DOI [10.3389/fsoc.2025.1482825](https://doi.org/10.3389/fsoc.2025.1482825); 42554934 — Trends in child health indicators in India: insights from National Family Health Survey (NFHS)-6, *Indian Pediatrics* 2026, DOI [10.1007/s13312-026-00412-6](https://doi.org/10.1007/s13312-026-00412-6) (abstract-only); 41821265 — Why are Hindu Scheduled Caste and Scheduled Tribe children still at higher risk of anthropometric failure than upper-caste children in India? A chronological analysis, 2005-06 to 2019-21, *American Journal of Human Biology* 2026, DOI [10.1002/ajhb.70221](https://doi.org/10.1002/ajhb.70221) (abstract-only); 41671315 — Caste and tobacco use: decomposing inequalities using Global Adult Tobacco Survey, India, *PLoS One* 2026, DOI [10.1371/journal.pone.0341459](https://doi.org/10.1371/journal.pone.0341459); 41451071 — A seven-step process to update Dr. B.G. Prasad's socioeconomic status scale, *Indian Journal of Community Medicine* 2025, DOI [10.4103/ijcm.ijcm_825_23](https://doi.org/10.4103/ijcm.ijcm_825_23); 40949512 — B.G. Prasad socioeconomic scale 2024: an update using conventional approach and linear regression modelling, *Indian Journal of Community Medicine* 2025, DOI [10.4103/ijcm.ijcm_153_24](https://doi.org/10.4103/ijcm.ijcm_153_24); 42362781 — Development and validation of Solomon's socioeconomic status index (SESI): a multidimensional household classification tool for South Asia, *European Journal of Health Economics* 2026, DOI [10.1007/s10198-026-01959-8](https://doi.org/10.1007/s10198-026-01959-8) (abstract-only); 41624926 — Gender and economic inequality in under-5 mortality in India from 2000 to 2022, *International Journal of Preventive Medicine* 2025, DOI [10.4103/ijpvm.ijpvm_371_24](https://doi.org/10.4103/ijpvm.ijpvm_371_24); 40316411 — Trends in socioeconomic inequality in mortality during childhood between 1993 and 2021 in India, *BMJ Global Health* 2025, DOI [10.1136/bmjgh-2024-016386](https://doi.org/10.1136/bmjgh-2024-016386).

♦ ♦ ♦

Natural history of disease — prevention

Asked in: Paper I 06-Apr-2026 — RGUHS MD Community Medicine

Definition — The natural history of a disease is the sequence of phases through which a disease passes in an individual **in the absence of any intervention or treatment**, from the earliest interaction of agent, host and environment, through pathological change and clinical illness, to recovery, disability or death (Park 28e, Ch 2, pp 53-54; Sunder Lal, Ch 1, p 43). It is customarily described in two phases — **prepathogenesis**, the process in the environment, and **pathogenesis**, the process in man — and infectious disease is used as the principal model (Park 28e, Ch 2, pp 53-54). Knowing where a disease stands in this sequence identifies the exact point at which an intervention can be applied, and is therefore the scientific basis of the levels of prevention.

Phases of the natural history of disease

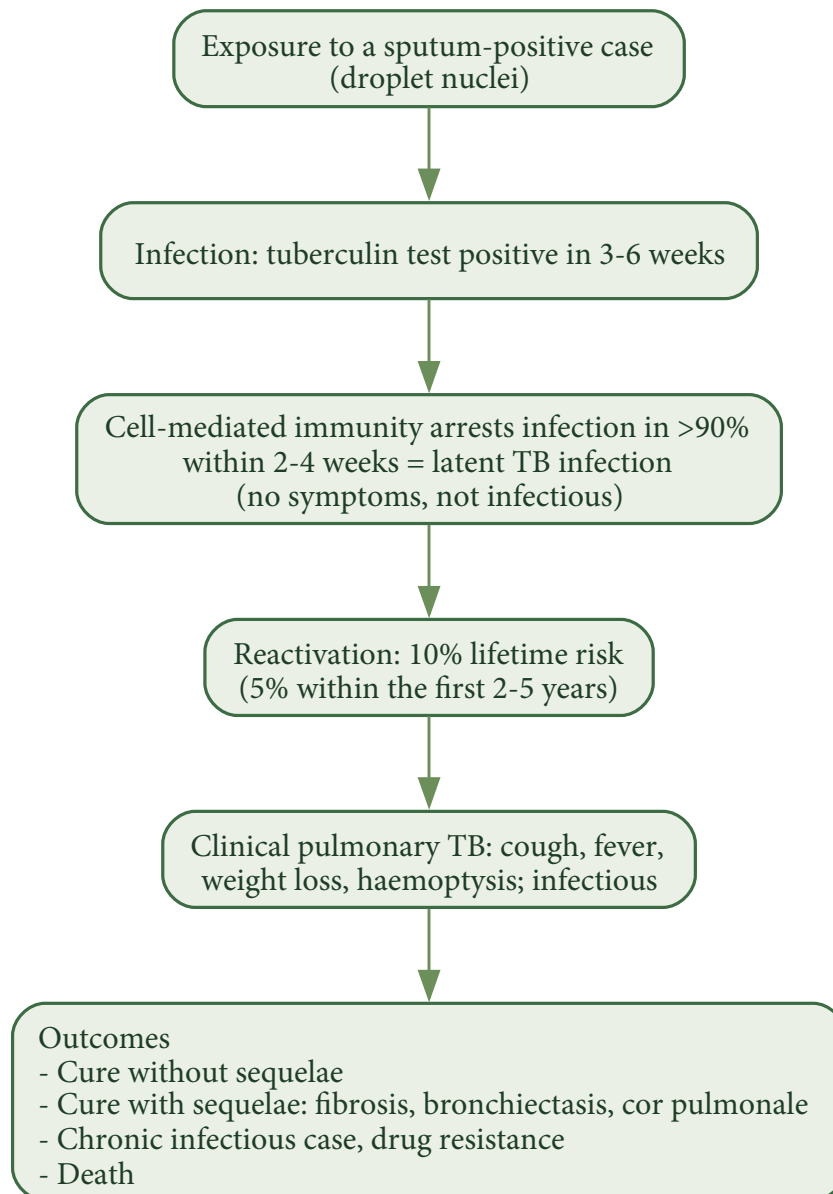
Phase	Events in the host	Defining concept	Detectable by
1. Prepathogenesis	Agent has not yet entered man; agent, host and favourable environmental factors coexist in balance. "Man in the midst of disease" / "man exposed to the risk of disease" (Park 28e, Ch 2, p 54)	Epidemiological triad — agent, host, environment. Their mere presence is insufficient; interaction of the three initiates disease and determines its distribution in the community (Park 28e, Ch 2, p 54)	Risk-factor surveys, environmental and entomological surveys
2a. Early pathogenesis (subclinical / presymptomatic)	Agent enters man and multiplies, or risk factors accumulate; tissue and biochemical changes begin. Host response is not predictable — infection may be clinical or subclinical, typical or atypical, or the host may become a carrier with or without clinical disease, as in diphtheria and hepatitis B (Park 28e, Ch 2, p 54)	Changes lie below the "clinical horizon" . In chronic disease (coronary heart disease, hypertension, cancer) this is the presymptomatic phase (Park 28e, Ch 2, pp 54-55)	Screening tests, serology, tuberculin test — never by symptoms
2b. Clinical stage	Begins when recognisable signs or symptoms appear; by then the disease is already well advanced into late pathogenesis (Park 28e, Ch 2, p 55)	Spectrum of disease — severity graded from mild to severe; in infections also called the gradient of infection (Park 28e, Ch 2, pp 57-58)	Clinical diagnosis, confirmatory investigation
2c. Outcome	Complete recovery without sequelae; recovery with sequelae (impairment, disability); chronic or carrier state; or death (Sunder Lal, Ch 1, p 44)	<i>Disease → impairment → disability → handicap</i> (e.g. accident → loss of foot → cannot walk → unemployed) (Park 28e, Ch 2, p 62)	Case-fatality, disability and rehabilitation records

Spectrum and iceberg. For almost every disease a spectrum of severity exists, *rabies* being an exception; **leprosy** is the classical example of the spectral concept (Park 28e, Ch 2, pp 57-58). The related **iceberg phenomenon** holds that the floating tip is what the physician sees — clinical cases — while the vast submerged portion is the hidden mass of latent, inapparent, presymptomatic and undiagnosed cases

and carriers; the "waterline" separates apparent from inapparent disease (Park 28e, Ch 2, p 58). In hypertension, diabetes, anaemia, malnutrition and mental illness the unknown morbidity far exceeds the known morbidity, and this hidden reservoir is the target of screening (Park 28e, Ch 2, p 58; Ch 4, p 163).

Suitable example — natural history of pulmonary tuberculosis

Phase	Events in tuberculosis
Prepathogenesis	Agent: <i>Mycobacterium tuberculosis</i> , expelled as droplet nuclei by a sputum-positive case; not transmitted by fomites (Park 28e, Ch 5, p 225). Host: infection rate in India rises from about 2% at 0-14 years to about 20% at 15-24 years; commoner in males; malnutrition predisposes; tuberculosis is not hereditary though inherited susceptibility matters (Park 28e, Ch 5, p 225). Environment: overcrowding, poor ventilation, poverty
Infection (early pathogenesis)	Time from receipt of infection to a positive tuberculin test is 3-6 weeks (Park 28e, Ch 5, p 225). Within 2-4 weeks the host mounts cell-mediated immunity that arrests infection in more than 90% of those infected (Sunder Lal, Ch 7, pp 280-282)
Latent tuberculosis infection	Bacilli viable but not multiplying; person looks and feels healthy and is not infectious ; detected only by tuberculin test. About 40% of the Indian population is infected (Sunder Lal, Ch 7, p 282; Ch 8, p 365) — a textbook iceberg
Progression / reactivation	Lifetime risk of reactivation in a healthy HIV-uninfected person is 10% , of which 5% develop disease within the first 2-5 years after infection. Risk rises more than 25 times in contacts of bacteriologically confirmed patients, 16-21 times with HIV co-infection and 3-4 times in diabetics and silicotics (Sunder Lal, Ch 8, p 365)
Clinical disease	Pulmonary or extrapulmonary tuberculosis with prolonged morbidity; patients remain infective as long as they are untreated, and effective antimicrobial treatment reduces infectivity by 90% within 48 hours (Park 28e, Ch 5, p 225)
Outcome	Cure without sequelae; cure with sequelae (fibrosis, bronchiectasis, cor pulmonale); chronic infectious case with drug resistance; or death



Natural history of pulmonary tuberculosis in an untreated host

Role of prevention in the natural history of a disease

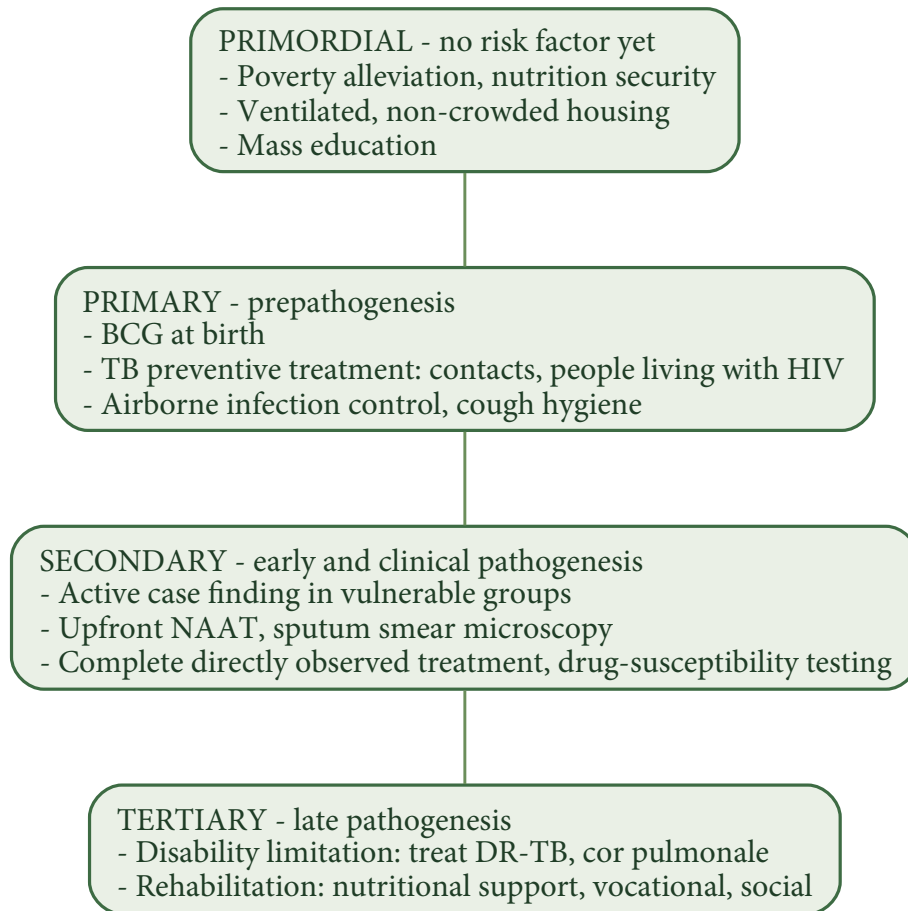
Prevention works by intercepting the natural history at a defined point; the levels of prevention form a continuum corresponding to the natural history of any disease, and are related to agent, host and environment (Park 28e, Ch 2, p 61). Authorities differ on the precise boundaries between levels, but the differences are semantic rather than substantive (Park 28e, Ch 2, pp 59-60).

Level	Point of interception in the natural history	Mode of intervention (Leavell and Clark)	Aim	Tuberculosis example
Primordial	Before the risk factor itself appears	Individual and mass education; national policy	Prevent emergence or development of risk factors in populations in which they have not yet appeared (Park 28e, Ch 2, p 60)	Poverty reduction, nutrition security, non-crowded ventilated housing, tobacco-free norms in children
Primary	Prepathogenesis phase — before disease onset	1. Health promotion (health education, environmental modification, nutritional intervention, lifestyle and behavioural change); 2. Specific protection (immunisation, specific nutrients, chemoprophylaxis, protection against occupational hazards, accidents, carcinogens, allergens, environmental hazards, consumer product safety) (Park 28e, Ch 2, pp 61-62)	"Action taken prior to the onset of disease which removes the possibility that a disease will ever occur" — reduces incidence (Park 28e, Ch 2, p 60)	BCG vaccination — the preventive component of the national tuberculosis programme (Park 28e, Ch 5, p 225); tuberculosis preventive treatment for household contacts and people living with HIV (Sunder Lal, Ch 8, p 365); airborne infection control
Secondary	Early pathogenesis is — subclinical or incipient disease, below the clinical horizon	3. Early diagnosis and treatment (screening tests, case-finding programmes, adequate treatment) (Park 28e, Ch 2, p 60)	"Action which halts the progress of a disease at its incipient stage and prevents complications"; arrests the process before irreversible change and reverses communicability — reduces prevalence by shortening duration (Park 28e, Ch 2, pp 60-61)	Case finding plus treatment — the curative component; the combination of case-finding and treatment is the most powerful weapon (Park 28e, Ch 5, p 225). Case-finding evolved from passive (1962-) to intensified (2001-), active (2017-) and population-based screening (2020-) (Park 28e, Ch 5, p 225); upfront nucleic acid amplification test (NAAT) for diagnosis (Park 28e, Ch 7, p 510)

Level	Point of interception in the natural history	Mode of intervention (Leavell and Clark)	Aim	Tuberculosis example
Tertiary	Late pathogenesis — advanced disease, sequelae	4. Disability limitation; 5. Rehabilitation (medical, vocational, social and psychological) (Park 28e, Ch 2, pp 60-61)	Limit impairment and disability, minimise suffering, promote the patient's adjustment to irremediable conditions — reduces disability and handicap (Park 28e, Ch 2, pp 60-61)	Treatment of drug-resistant disease and cor pulmonale; nutritional support of ₹500 per month by direct benefit transfer under Nikshay Poshan Yojana; vocational rehabilitation (Sunder Lal, Ch 8, pp 364-365; Park 28e, Ch 7, p 514)

Why the natural history determines the choice of intervention

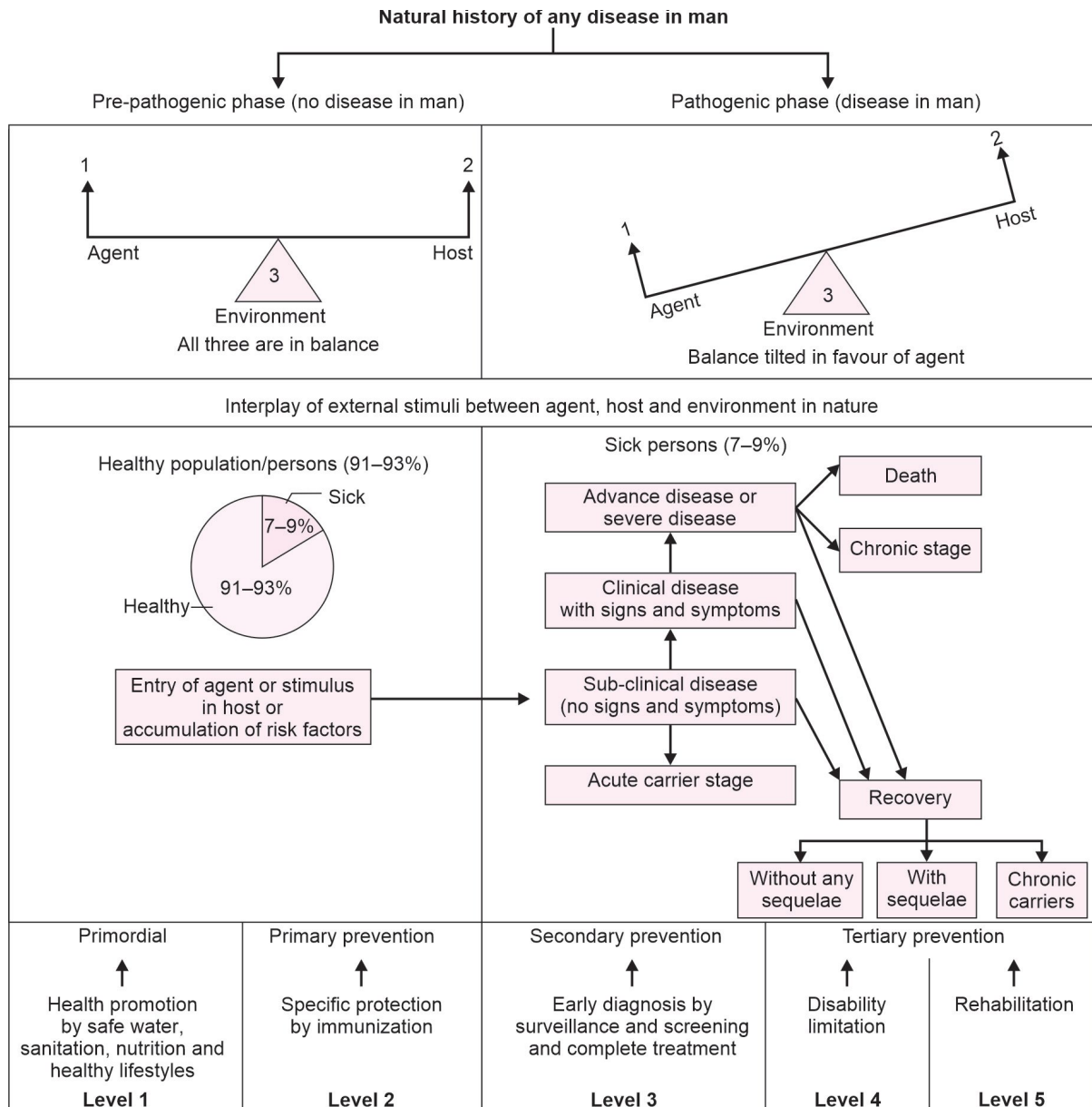
- ▶ The **point of interception decides the level** — an intervention placed in prepathogenesis is primary, the same measure applied after entry of the agent is secondary. Supply of food supplements to a family may be primary prevention for some members and secondary for others (Park 28e, Ch 2, pp 59-60).
- ▶ A **long subclinical phase makes screening worthwhile**; where the subclinical state cannot be detected, as in many chronic diseases of unknown aetiology, secondary prevention fails — this is the major deterrent to the study of chronic disease (Park 28e, Ch 2, p 58).
- ▶ **Diagnostic criteria must be pushed backwards** to early biochemical, morphological and functional changes that precede manifest signs, while changes are still reversible (Park 28e, Ch 2, p 62).
- ▶ Strictly, early diagnosis and treatment is not prevention because disease has already begun, but it is included in the schema because it intercepts the disease process — "stamping out the spark rather than calling the fire brigade" (Park 28e, Ch 2, p 62).
- ▶ **Disability prevention** operates across levels: reducing occurrence of impairment (primary), disability limitation by appropriate treatment (secondary), and preventing transition of disability to handicap (tertiary) (Park 28e, Ch 2, pp 62-63).
- ▶ Primordial and primary prevention contribute most to the health of the whole population and are the safest and cheapest, but all levels are complementary (Sunder Lal, Ch 1, p 45; Park 28e, Ch 2, p 60).



Levels of prevention applied along the natural history of tuberculosis

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Iceberg of disease** — draw an iceberg with a waterline. Label above the waterline: *clinical cases — what the physician sees*. Label the waterline: *"clinical horizon" / demarcation between apparent and inapparent disease*. Label below the waterline, from top down: *presymptomatic cases, latent and inapparent infection, undiagnosed cases, carriers, the hidden mass of disease*.
- **Epidemiological triad** — an equilateral triangle with *AGENT*, *HOST* and *ENVIRONMENT* at the three corners, sitting on a beam-balance: label one drawing *"all three in balance — prepathogenesis, no disease"* and a second *"balance tilted in favour of agent — pathogenesis, disease in man"*.



Natural history of disease and levels of prevention (Sunder Lal — Textbook of Community Medicine, Ch 1, p 44)

High-yield exam pointers

- ▶ Two phases only: **prepathogenesis** (process in the environment) and **pathogenesis** (process in man); the pathogenesis phase is divided by the **clinical horizon** into presymptomatic and clinical stages.
- ▶ Five **modes of intervention** (Leavell and Clark) in order: health promotion, specific protection, early diagnosis and treatment, disability limitation, rehabilitation — the first two are primary, the third secondary, the last two tertiary.
- ▶ One-line definitions: primary = "action taken prior to the onset of disease which removes the possibility that a disease will ever occur"; secondary = "action which halts the progress of a disease at its incipient stage and prevents complications"; primordial = prevention of the emergence of risk factors where they have not yet appeared.
- ▶ Effect on frequency: primordial and primary reduce **incidence**; secondary reduces **prevalence** (by shortening duration); tertiary reduces **disability**.

- ▶ Disability sequence: **disease** → **impairment** → **disability** → **handicap** (accident → loss of foot → cannot walk → unemployed).
- ▶ Tuberculosis numbers: tuberculin conversion in **3-6 weeks**; cell-mediated immunity contains **>90%** of infections; about **40%** of Indians infected; **10%** lifetime risk of disease (5% in the first 2-5 years); treatment cuts infectivity by **90% in 48 hours**.
- ▶ Spectrum exceptions and exemplars: **leprosy** is the classical spectral disease; **rabies** has essentially no spectrum.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the Revised National Tuberculosis Control Programme has been renamed the **National Tuberculosis Elimination Programme (NTEP)** (Sunder Lal, Ch 8, p 358), so use the current name in the answer; Sunder Lal quotes tuberculosis infection prevalence in India as "about 40%" in one chapter and "31.4%" in another, so quote 40% with the source rather than as an exact national figure.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 2 (Concept of Health and Disease, pp 53-63), Ch 4 (Screening for Disease, p 163), Ch 5 (Epidemiology of Communicable Diseases, pp 225, 246-247), Ch 7 (Health Programmes in India, pp 507-514); Sunder Lal — Textbook of Community Medicine, Ch 1 (pp 43-45), Ch 7 (pp 280-282, 331), Ch 8 (pp 358, 364-365); Leavell and Clark, *Preventive Medicine for the Doctor in His Community* (natural-history figure reproduced in Park 28e, Ch 2, p 54); National Tuberculosis Elimination Programme, Guidelines for Tuberculosis Preventive Treatment, July 2021 (as cited in Sunder Lal, Ch 8, p 365).



Communication process

Asked in: Paper I 12-Nov-2021 — RGUHS MD Community Medicine

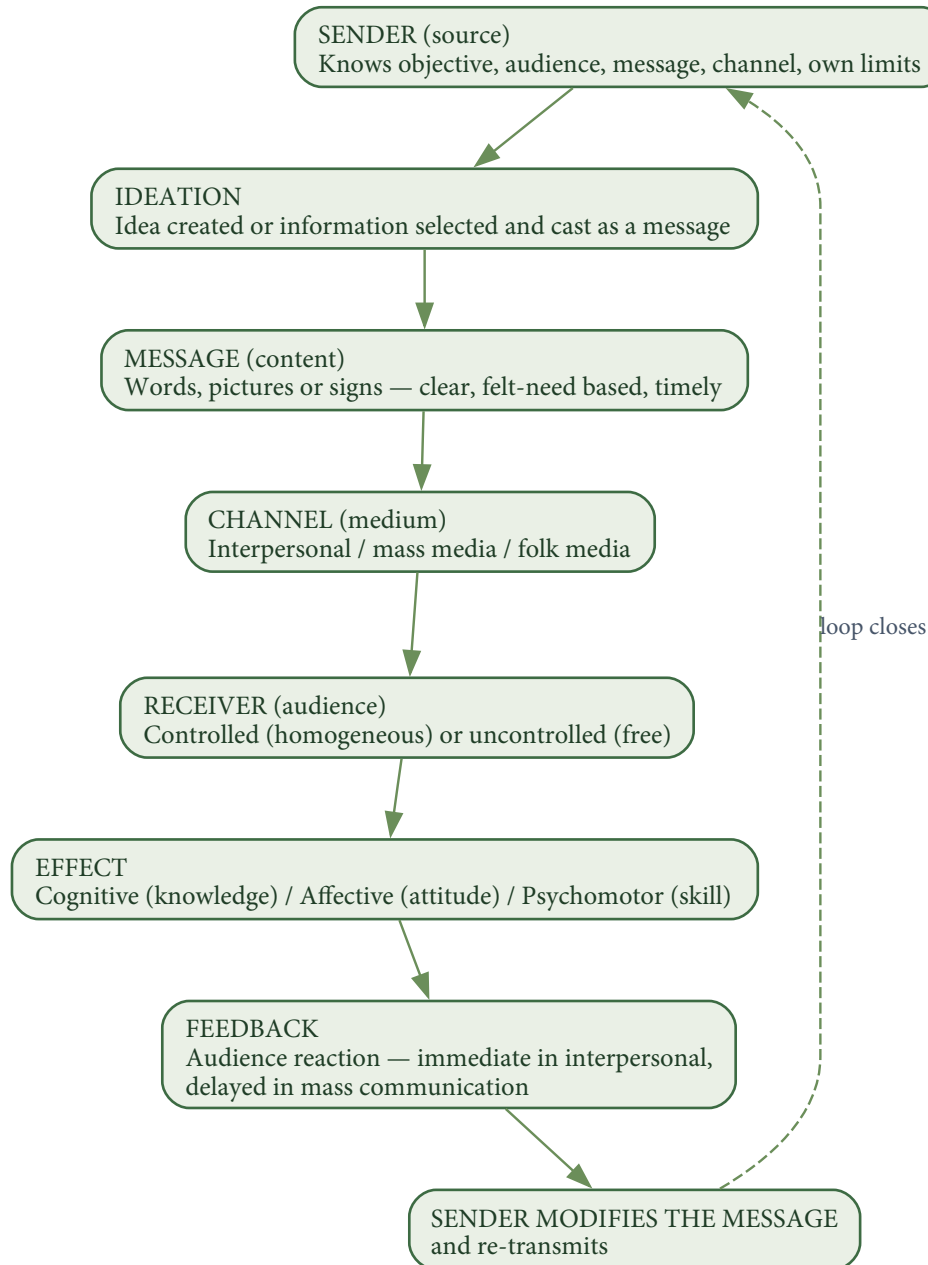
Definition — Communication is a **two-way process of exchanging or shaping ideas, feelings and information**; broadly, "the countless ways that humans have of keeping in touch with one another" (Park 28e, Ch 22, p 1010). It is more than a mere exchange of information — it is the process that paves the way for desired changes in human behaviour and for informed individual and community participation in achieving predetermined goals. Being the basis of all human interaction, it is a complex process built on five main components.

1. Elements (components) of the communication process

Element	What it is	What makes it effective
1. Sender (source, communicator)	The originator of the message; may be a person, a group, an institution or an organisation (e.g. the Government of India was the source of the <i>Hum do, hamare do</i> message)	Must know his objectives (clearly defined), his audience and its interests and needs, his message , his channels , and his own professional abilities and limitations . Impact depends on his social status (authority), knowledge and prestige in the community; the source must be credible, accessible, acceptable and legitimate (Park 28e, Ch 22, p 1013; Sunder Lal, Ch 4, p 151)
2. Message (content)	The information or "technical know-how" the communicator transmits for the audience to receive, understand, accept and act upon; may be in words, pictures or signs	Nine qualities (below). Even a well-chosen channel fails if the message is inadequate
3. Channel (medium)	The "physical bridge" between sender and receiver — the media of communication	Every channel has its own advantages and limitations; no single method suffices, so a combination of methods should be used (e.g. surgery is best taught by verbal explanation <i>plus</i> demonstration) (Park 28e, Ch 22, p 1011)
4. Receiver (audience)	A single person or a group; without an audience communication is "nothing more than mere noise". Their frame of mind — opinions, attitudes, prejudices — gives meaning to the message	Controlled audience = held together by a common interest, homogeneous, easier to address. Uncontrolled ("free") audience = heterogeneous, some present out of curiosity — a far greater challenge to the educator
5. Feedback (effect)	The flow of information from audience back to sender — the audience's reaction to the message	Immediate in interpersonal communication; delayed in mass communication. Obtained through opinion polls, attitude surveys and interviews. It rectifies transmission errors and lets the sender modify an unacceptable message and re-transmit it (Park 28e, Ch 22, p 1011)

Qualities of a good message — 1. in line with the objective(s); 2. in a language the people understand and find meaningful; 3. based on **felt needs**; 4. clear and understandable; 5. specific and accurate; 6. timely and adequate; 7. fitting the audience and the actions expected of them; 8. interesting; 9. culturally and socially appropriate. Transmitting **the right message to the right people at the right time** is the crucial factor in successful communication (Park 28e, Ch 22, p 1011).

Goal of the process — the ultimate aim of all communication is change in the desired direction in the receiver, expressed as three levels of learning objective: **cognitive** (increase in knowledge), **affective** (change in existing attitudes and behaviour patterns) and **psychomotor** (acquisition of new skills) (Park 28e, Ch 22, p 1010).



The communication process — the five elements and the feedback loop

2. Types of communication

Type	Description	Key point for the exam
One-way (didactic method)	Flow is one-way from communicator to audience; classic example is the classroom lecture	Drawbacks — knowledge is imposed , learning is authoritative , little audience participation, no feedback , and it does not influence human behaviour

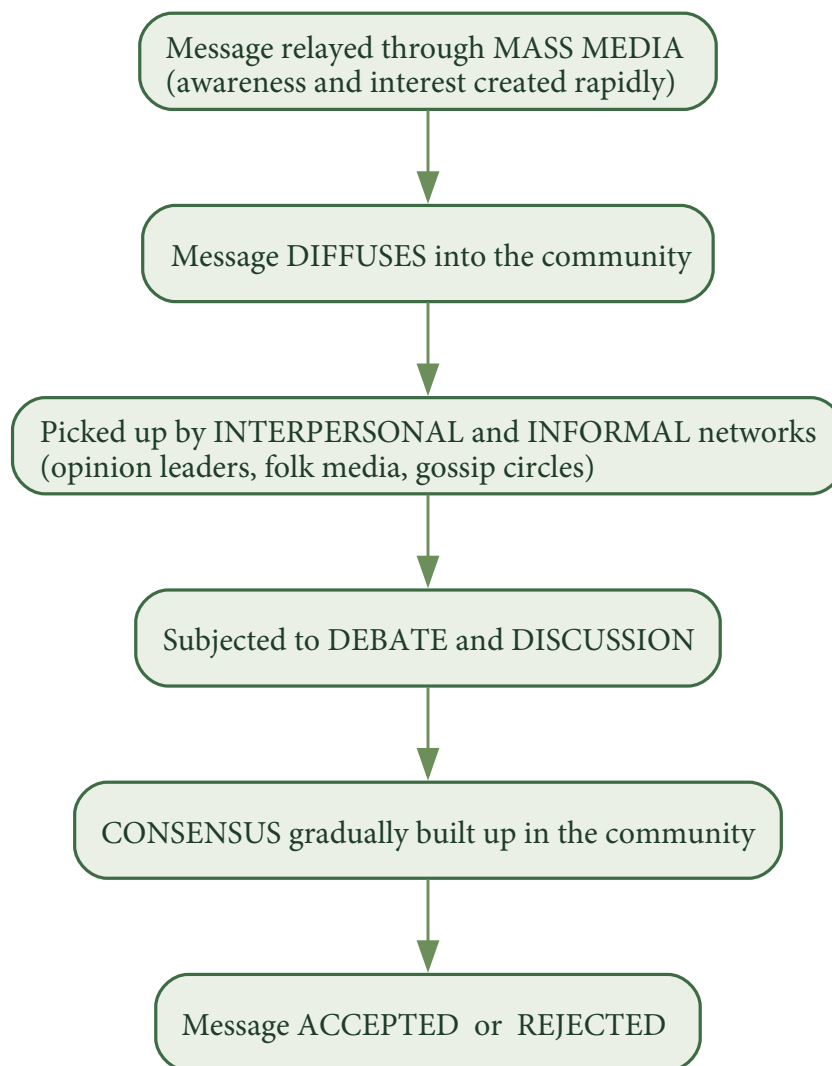
Type	Description	Key point for the exam
Two-way (Socratic method)	Both communicator and audience take part; the audience may raise questions and add its own information, ideas and opinions	Learning is active and democratic , and it is far more likely to influence behaviour — the method of choice in health education
Verbal (oral)	The oldest form of communication — by word of mouth; direct verbal communication is the most persuasive	Written communication is of comparatively recent origin and may not be as persuasive as the spoken word ; word of mouth may be loaded with hidden meanings, and medical jargon defeats it
Non-verbal	Occurs without words — bodily movements, posture, gestures, facial expression (smile, raised eyebrow, frown), staring, gazing; silence is also non-verbal communication and can speak louder than words	Combining verbal with non-verbal skill enriches the process; the <i>anganwadi</i> worker uses it in Integrated Child Development Services (ICDS) play-way activities to develop the cognitive, affective and psychomotor domains in young children (Sunder Lal, Ch 4, p 151)
Formal and informal	Formal follows the lines of authority; informal is the "grape vine" (e.g. gossip circles) present in every organisation	Informal channels become more active when formal channels fail to meet information needs
Visual	Charts and graphs, pictograms, tables, maps, posters	Overcomes illiteracy where the spoken word cannot
Telecommunication and internet	Communicating over distance by electromagnetic means. Radio, television and internet are mass media ; the telephone is a point-to-point system	Point-to-point systems are closer to interpersonal communication; satellites have caused an explosion of electronic communication

3. Media systems and the media mix

The total communication effort rests on **three media systems** (Park 28e, Ch 22, pp 1010–1011):

Media system	Examples	Advantages	Limitations
a. Interpersonal (face-to-face)	Home visit, clinic consultation, group discussion, counselling by the auxiliary nurse midwife (ANM) or accredited social health activist (ASHA)	Most common channel; being personal and direct it is the most persuasive and effective ; feedback is immediate; documented superiority over mass media in creating motivation and in influencing the undecided	Slow, small coverage, manpower-intensive
b. Mass media	Television, radio, printed media, internet	Reach a relatively larger population in a much shorter time	One-way channels — carry messages only from the centre to the periphery; feedback mechanisms are poorly organised; usually not effective in changing established modes of behaviour

Media system	Examples	Advantages	Limitations
c. Traditional or folk media	Folk dance, song, drama, puppet shows and <i>keerthan</i> ; <i>Nautanki</i> (Uttar Pradesh), <i>Burrakatha</i> (Andhra Pradesh) and their counterparts in Western India; <i>ghazal</i> and <i>qawwali</i> ; informal group channels such as religious meetings	Every community has its own network of folk media; they are culturally closest to the rural population , have preserved the cultural heritage, and carry high credibility	Local in reach; the message may be distorted by the performer



The two-step media mix — how a mass-media message is finally accepted or rejected by a community

4. Functions of health communication

Health communication must serve eight functions (Park 28e, Ch 22, pp 1012–1015):

- **Information** — its primary function; providing scientific knowledge about health problems and how to maintain and promote health. People rarely seek it although they have a right to know; it eliminates ignorance, prejudice and misinformation.

- ▶ **Education** — the basis of all education is communication; it increases knowledge, on the assumption that knowledge determines attitudes and attitudes determine behaviour.
- ▶ **Motivation** — the power that drives a person from within to translate health information into behaviour, assisting the individual from awareness and interest through to decision-making and adoption. Motivation may not be long-lasting and needs reinforcement.
- ▶ **Persuasion** — the art of winning friends and influencing people; more effective than coercion or authoritative communication. When deliberately used to manipulate feelings, attitudes and beliefs it degenerates into **propaganda or brain-washing**.
- ▶ **Counselling** — helps people understand and deal with their problems, reinforces motivation to change behaviour and gives support in crisis; it develops positive attitudes and is integral to all health care programmes.
- ▶ **Raising morale** — morale is the capacity of a team to pull together consistently; vertical, horizontal, internal and external communication is the first step in raising it.
- ▶ **Health development** — diffuses knowledge of development goals and prepares people for the roles expected of them; its role here is essentially **supportive**.
- ▶ **Health organisation** — communication is the lifeblood of an organisation. It flows **vertically** (downward from the top administrator through the hierarchy, and upward) and **horizontally** / **cross-wise** between equals at any level.

5. Barriers to communication

Health education often fails because of barriers between the educator and the community (Park 28e, Ch 22, p 1012):

Barrier	Examples	Corrective step
Physiological	Difficulty in hearing; difficulty in expression	Speak clearly and slowly, use hearing aids, use demonstration and visual aids
Psychological	Emotional disturbance, neurosis, level of intelligence, language or comprehension difficulty	Assess audience level first; use simple sequenced messages, repeat and reinforce
Environmental	Noise, invisibility, congestion	Choose a quiet, well-lit, uncrowded venue; correct seating and audio-visual placement
Cultural	Illiteracy; level of knowledge and understanding; customs, beliefs, religion, attitudes; economic and social class difference; language variation; urban-educator versus rural-audience gap	Use folk media and the local dialect; work through opinion leaders; build the message on the community's own felt needs

Practical blocks observed in the field consultation (Sunder Lal, Ch 4, p 155): talking *at* people and giving advice instead of listening; lack of empathy, trust and understanding; poor or absent eye contact; low credibility of the source; **medical jargon**; frequent interruptions (telephone calls, files); low visual and health literacy; and body language that signals boredom (looking at the watch, getting up mid-consultation).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Draw Fig. 1 — The communication process (Park 28e, Ch 22, p 1013), a horizontal chain with a return loop. Exact labels:

- ▶ **SENDER** (source / communicator)
- ▶ **MESSAGE** (content)
- ▶ **CHANNEL** (medium)
- ▶ **RECEIVER** (audience)
- ▶ **EFFECT** — cognitive / affective / psychomotor
- ▶ **FEEDBACK** — a dashed arrow running back from Receiver to Sender (label it "reaction of the audience")
- ▶ Mark **BARRIERS** — physiological, psychological, environmental, cultural — as arrows striking the Channel from outside

High-yield exam pointers

- ▶ **Five elements, mnemonic S-M-C-R-F** — Sender, Message, Channel, Receiver, Feedback (Park's Fig. 1).
- ▶ **Two-way (Socratic) beats one-way (didactic)**: the five drawbacks of the didactic method are knowledge imposed, authoritative learning, little participation, no feedback, no behaviour change.
- ▶ **Three media systems** — interpersonal (most persuasive, immediate feedback), mass media (widest and fastest but one-way and poor at changing behaviour), folk/traditional media (culturally closest).
- ▶ **Eight functions of health communication** — Information, Education, Motivation, Persuasion, Counselling, Raising morale, Health development, Health organisation.
- ▶ **Four barriers** — physiological, psychological, environmental, cultural.
- ▶ **Park's adoption model (Fig. 2) — 3 stages of behaviour change**: Awareness → **Motivation** (interest → evaluation → decision-making) → **Action** (adoption or acceptance). Interpersonal communication with friends, kinship groups and technical persons is what supports the decision at the evaluation stage.
- ▶ **Three learning objectives** — cognitive (knowledge), affective (attitude), psychomotor (skill).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park's text places this content in the chapter *Communication for Health Education* (28e, pp 1010–1014); some scanned copies carry the same pages under the preceding chapter heading, so page numbers rather than chapter number are the safer reference. The retrieved corpus does not name an author-attributed model (Shannon-Weaver, Berlo's SMCR, Lasswell), so none is asserted here; the five-element description above is Park's own.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e — Communication for Health Education (pp 1010–1015, 1019, 1024); Sunder Lal — Textbook of Community Medicine, Principles of Health Promotion and Education (pp 150–155). Coverage note: no captioned figure of the communication process exists in the indexed figure catalogue, so a "Diagram to draw" label box and two Graphviz flowcharts are given in place of a reproduced plate.



Role of family in health and disease

Asked in: Paper I 14-Jul-2020 — RGUHS MD Community Medicine

Definition — The family is the primary unit in all societies: a group of biologically related individuals living together and eating from a common kitchen (Park 28e, Ch 12, p 825). It differs from the **household**, all of whose members need not be blood relations (e.g. servants); in Indian enumeration the number of households in a house is decided by the number of kitchens (*chulahs*) (Sunder Lal, Ch 2, p 69). Social scientists separate the **family of origin** (into which one is born) from the **family of procreation** (which one sets up after marriage) (Park 28e, Ch 12, p 825).

The family as a unit of health and disease

The family is a...	Because	Consequence for health and disease
Biological unit	Members share a pool of genes	Familial and genetic disorders aggregate; consanguinity raises congenital malformation risk (Park 28e, Ch 12, pp 825, 828)
Social unit	Members share a common physical and social environment	Overcrowding, water, fuel, sanitation and food distribution act on all members together (Park 28e, Ch 12, p 825)
Cultural unit	It reflects the culture of the wider society and sets the behaviour and attitudes of its members	Child-rearing, dietary, hygiene and care-seeking beliefs are transmitted intact to the next generation (Park 28e, Ch 12, p 825)
Epidemiological unit	Shared environment + shared susceptibility	The family is the "playground" for tuberculosis, measles, mumps, rubella, chickenpox, diphtheria, scabies, dysentery, diarrhoea and enteric fever (Park 28e, Ch 12, p 828)
Unit of service and comprehensive medical care	Care, cost and compliance decisions are taken collectively	The family is the <i>fulcrum of health services</i> and the unit for family folders, screening and follow-up (Park 28e, Ch 12, pp 825, 829)

Types of family and their health significance

Basis	Type	Defining features	Health significance
Structure	Nuclear (elementary)	Married couple + dependent children in one dwelling; universal	Strain of child rearing when both parents work; children left with creches, <i>balwadis</i> , <i>anganwadis</i> or servants; more intimate husband–wife relationship; loss of grandparental support (Park 28e, Ch 12, pp 826-827; Sunder Lal, Ch 2, p 70)
	Joint / extended	Several married couples + children, common residence, common purse, common kitchen, common land; authority in the senior male; familial relations take primacy over marital ones	Merit — "union is strength": economic and social security for the old, sick, widowed, disabled and unemployed. Demerits — hinders personality development, minimal privacy to the couple, uncontrolled births (Park 28e, Ch 12, p 827; Sunder Lal, Ch 2, p 70)
	Three-generation	Grandparents + married sons + their offspring; common kitchen; arises when young couples cannot obtain separate housing	Grandparental child care, but the dependency ratio is high and care of the aged falls wholly on the family (Park 28e, Ch 12, p 827; Sunder Lal, Ch 2, p 70)
Authority	Patriarchal / matriarchal	Decision-making and descent through the male / female line	Determines who decides on care-seeking, food allocation and contraception; gender inequity is generated here (Sunder Lal, Ch 2, p 69)
Mating / marriage	Monogamous	One regular partner	Lowest risk of sexually transmitted infection and HIV (Sunder Lal, Ch 2, p 69)
	Polygamous / polyandrous	Multiple partners	Greater risk of sexually transmitted infection and HIV transmission (Sunder Lal, Ch 2, p 69)
	Endogamous (consanguineous) vs exogamous	Marriage within vs outside the in-group	Cousin marriage carries higher risk of transmission of genetic disorders; exogamy lower (Sunder Lal, Ch 2, p 69)

Family life cycle and family crises

Phase	Begins with	Ends with
I. Formation	Marriage	Birth of 1st child
II. Extension	Birth of 1st child	Birth of last child
III. Complete extension	Birth of last child	1st child leaves home
IV. Contraction	1st child leaves home	Last child leaves home of parents
V. Completed contraction	Last child has left home	1st spouse dies
VI. Dissolution	1st spouse dies	Death of survivor (extinction)

Source: basic model of nuclear family life cycle (Park 28e, Ch 12, Table 1, p 826).

- ▶ **Transitional crises** arise within the cycle itself — birth of the first child, loss of a spouse (Park 28e, Ch 12, p 826).
- ▶ **Non-transitional crises** are imposed from outside — war, uprooting and migration, disaster, sudden shift in economic status, mental disorder or incapacity of a member (Park 28e, Ch 12, p 826).
- ▶ Stress and health hazards rise when environmental change coincides with a **critical developmental period** — adolescence, first pregnancy, menopause, retirement (Park 28e, Ch 12, p 826).
- ▶ The model needs amendment in high-mortality settings, and "leaving home" has limited relevance in much of Asia and Africa; early death of a child or spouse, divorce and childlessness are the common variants (Park 28e, Ch 12, p 826).

Role of the family in HEALTH — the positive functions

Function	What the family does	Health importance
1. Child rearing	Physical care of the dependent young so they survive to adulthood; feeding, hygiene, sleep, clothing, discipline, habit training passed from one generation to the next	The commonest point of contact for the community health worker; traditional practices upheld by religious sanction are the chief obstacle to change (Park 28e, Ch 12, pp 827-828)
2. Socialization	A <i>latent</i> function — teaching the young the values, culture, beliefs and codes of conduct of society by example and precept	Determines lifestyle, tobacco and alcohol use, hand-washing, toilet use and oral hygiene; "seeds of non-communicable disease are sown in childhood" (Park 28e, Ch 12, p 828; Sunder Lal, Ch 2, p 70)
3. Personality formation	Early experience with father, mother and siblings shapes capacity to withstand stress and to interact with others; the family acts as a " placenta " — excluding some influences, modifying others, adding its own	Lays the foundation of physical, mental and social health of the child (Park 28e, Ch 12, p 828)
4. Care of dependent adults	(a) The sick and injured — the family gives front-line care, mainly the mother, and does more nursing than the hospital even in developed countries; (b) women in pregnancy and childbirth — maternity leave, supplementary diet, reduced work; (c) the aged and the handicapped	Attitudes to feared illness (e.g. leprosy) decide whether care is given or the patient is excluded; societal attitude to childbearing bears on stillbirth, prematurity, infant and maternal mortality; without family support no amount of medical care of the aged can succeed (Park 28e, Ch 12, p 828)
5. Stabilization of adult personality	A " shock absorber " for injury, illness, births, deaths, worry, anxiety and economic insecurity; provides release of tension and mental equilibrium	Buffers mental illness, alcoholism and narcotic addiction, and the "stress diseases" — peptic ulcer, colitis, high blood pressure, rheumatism, skin disease (Park 28e, Ch 12, p 828)

Function	What the family does	Health importance
6. Residence, economic and social care	Provides a clean and decent home (patrilocal or matrilocal), division of labour, an economic partnership and inheritance, and social care — status, protection from insult and defamation, regulation of marital, political and religious activity, and of sex relations through incest taboos	The family converts income and knowledge into health; it is also the first source of shelter, food and old-age security (Park 28e, Ch 12, p 827; Sunder Lal, Ch 2, p 70)
7. Biological / reproductive function	Bearing and rearing children; interruption of this function by contraception	Total fertility rate fell from 6.0 (1951) to 2.0 (National Family Health Survey-5, 2019-21), below the replacement level of 2.1 — but it remains higher in Bihar, Uttar Pradesh, Rajasthan, Jharkhand and Meghalaya, the high-focus states for the family planning programme (Sunder Lal, Ch 2, pp 69-70)

Role of the family in DISEASE — the pathogenic side

Family factor	Mechanism	Diseases / outcomes
Familial susceptibility	Shared gene pool + shared environment	Haemophilia, colour blindness, diabetes, mental illness; schizophrenia, psychoneurosis and some forms of mental deficiency show familial incidence (Park 28e, Ch 12, p 828)
Consanguinity	Endogamous, cousin marriage	Higher incidence of congenital malformation than in non-consanguineous marriage (Park 28e, Ch 12, p 828)
Common environment (family as epidemiological unit)	Crowding, common kitchen, shared fomites	Rapid intra-familial spread of tuberculosis, common cold, scabies, diphtheria, measles, mumps, rubella, chickenpox, dysentery, diarrhoea, enteric fever (Park 28e, Ch 12, p 828)
Large family size	Dilution of food, care and income	Episodes of infectious diarrhoea rise with family size; prevalence of malnutrition is increased in families with more than 4 children (Park 28e, Ch 9, p 652)
Broken family (parents separated, divorced, or death of one or both)	<i>John Bowlby's maternal deprivation</i> — one of the most dangerous pathogenic factors in child development; paternal separation and dual-parental separation act similarly	Psychopathic behaviour, immature personality, retardation of speech and intellect in later years; drift to prostitution, crime and vagrancy; orphanage placement (Park 28e, Ch 12, pp 828-829; Sunder Lal, Ch 2, p 71)

Family factor	Mechanism	Diseases / outcomes
Problem / high-risk family	Standards of life far below the accepted minimum; parents unable to meet children's physical and emotional needs. Underlying factors — personality and relationship problems, backwardness, poverty, illness, mental and emotional instability, character defects, marital disharmony; recognised as social pathology	Found in all social classes but commoner in the lower classes; children become victims of prostitution, crime and vagrancy. In Indian practice, large families, below-poverty-line families, broken families, addiction in the husband and prolonged illness of a member define the priority family for outreach (Park 28e, Ch 12, p 829; Sunder Lal, Ch 2, p 71)
Juvenile delinquency	Broken homes, death or separation of parents, step-mothers, poverty, alcoholism, parental neglect, ignorance of child care, too many children	Prevented chiefly by <i>improvement of family life</i> — preparing parents for parenthood and meeting children's needs — with schooling next in importance (Park 28e, Ch 9, p 690)
Gender inequity within the family	Family decides food allocation, schooling, and care-seeking	Adverse sex ratio, denial of cereals to young children on wrong beliefs, and delayed care for girls are ultimately family choices (Sunder Lal, Ch 2, p 72)
Social class of the family	Lower-class families are larger, women marry early and bear more children	Less use of hospital facilities, fewer doctor consultations, lower uptake of antenatal, postnatal, check-up and immunization services (Park 28e, Ch 12, p 834)

The family as the operating unit of health care in India

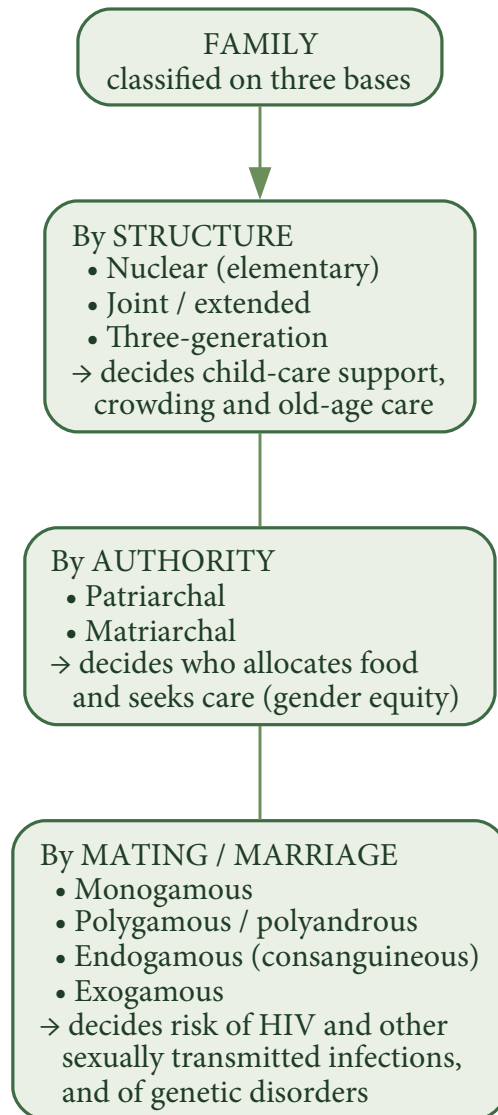
- **Family medicine** is health care centred on the family as the unit — from first contact to ongoing care of chronic problems, from prevention to rehabilitation — and is the right place for integrating preventive, promotive and curative services (Park 28e, Ch 1, p 22).
- **The first response to any illness comes from the family** — initiation and exclusivity of breastfeeding, home-available fluids and continued feeding in diarrhoea, smokeless kitchen, birth spacing, choice of institutional delivery, antenatal registration, immunization and completion of tuberculosis treatment are all family decisions (Sunder Lal, Ch 2, pp 72, 82).
- **Family-based records** connect every household to the system: the **household survey (family) register**, updated annually, forms the basis of the village and sub-centre action plan and of community health needs assessment (Sunder Lal, Ch 2, p 72). The accredited social health activist (ASHA) additionally maintains the eligible couple, antenatal, birth and death, immunization and home-based newborn care records, the community-based assessment checklist (CBAC) for early detection of non-communicable disease, and the **family folder** with individual health records (Sunder Lal, Ch 8, p 533).
- **Family-retained records** — the **Mother and Child Protection (MCP) card** of the Ministry of Women and Child Development with the Ministry of Health and Family Welfare — put the schedule of health actions up to 3 years of age into the family's own hands (Sunder Lal, Ch 2, pp 76-78).
- **Prioritised home visiting**: the female health worker, ASHA and *anganwadi* worker visit about 10 families a day and cover the whole area in 3 months, prioritising eligible couples, large families,

unregistered pregnancies, registered pregnancies with complications, unseen postnatal cases and low-birth-weight neonates (Sunder Lal, Ch 2, p 73).

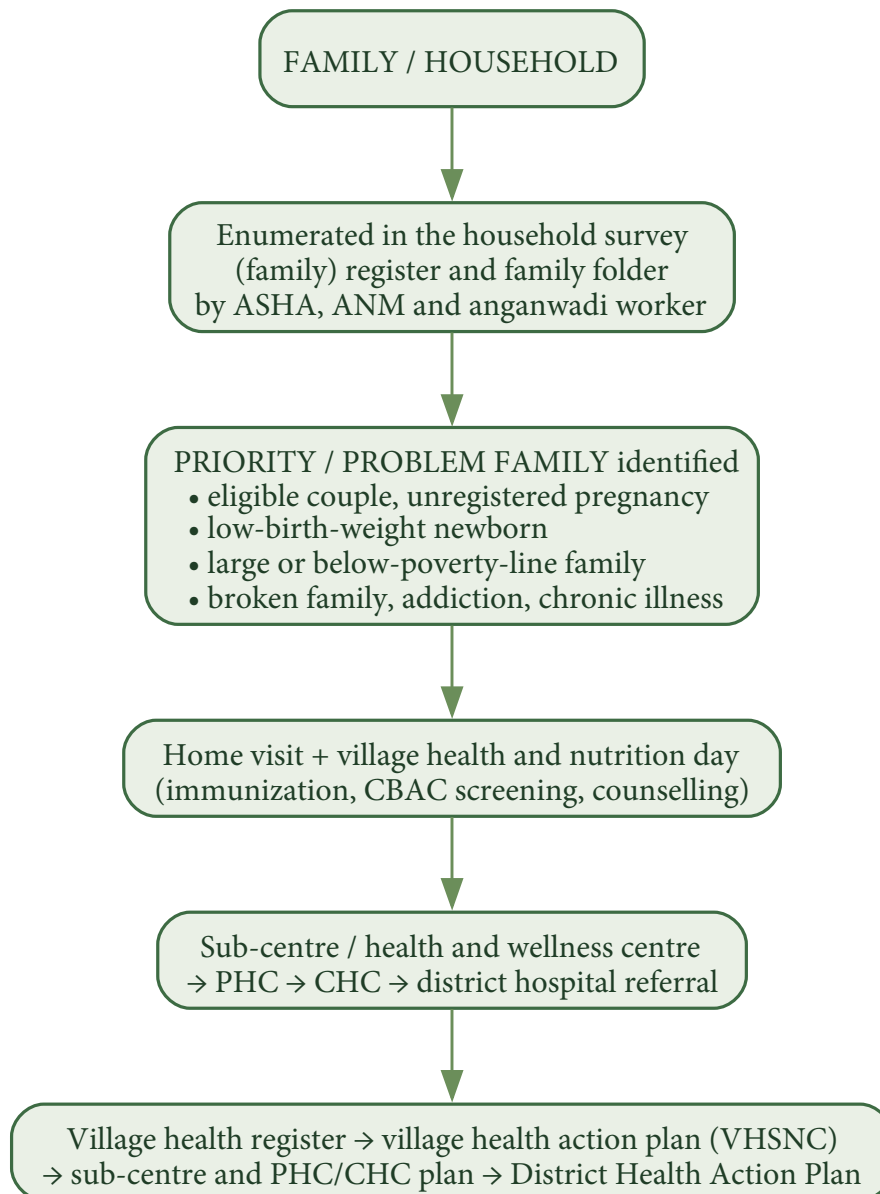
- ▶ **Barriers the family must cross** — poverty (economic barrier), illiteracy and poor health literacy, cultural beliefs reducing acceptability, and poor physical access to services (Sunder Lal, Ch 2, p 84).
- ▶ Hence, as Florence Nightingale said, "**the secret of national health lies in the homes of the people**" — the family is the fulcrum of health services (Park 28e, Ch 12, p 829).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **The family as a "placenta"** — draw the child at the centre inside an oval labelled FAMILY; arrows entering from outside labelled *society, culture, economy, infection, stress*; the oval labelled with three actions — *excludes some influences, modifies others, adds its own*; output arrow to the child labelled *foundation of physical, mental and social health*.
- ▶ **Family life cycle wheel** — six sectors labelled in order: *Formation → Extension → Complete extension → Contraction → Completed contraction → Dissolution*, with the transition event written on each spoke (marriage, birth of 1st child, birth of last child, 1st child leaves home, last child leaves home, 1st spouse dies, death of survivor).



Classification of the family and the health risk attached to each basis



The family as the operating unit of primary health care in India

High-yield exam pointers

- ▶ Park's eight roles of the family in health and disease (mnemonic C-S-P-C-S-F-B-P): Child rearing, Socialization, Personality formation, Care of dependent adults, Stabilization of adult personality, Familial susceptibility, Broken family, Problem families.
- ▶ Two metaphors that must appear: the family as a "**placenta**" (personality formation) and as a "**shock absorber**" (stabilization of adult personality).
- ▶ Six phases of the family life cycle: Formation → Extension → Complete extension → Contraction → Completed contraction → Dissolution.
- ▶ Broken family → *Bowlby's maternal deprivation*; problem family → social pathology.
- ▶ Dated Indian figures: total fertility rate 6.0 (1951) → 2.0 (NFHS-5, 2019-21), replacement level 2.1; malnutrition rises in families with **more than 4 children** (Park 28e, Ch 9).
- ▶ Family differs from household; number of households in a house = number of **kitchens** (*chulahs*).

- Closing quotation: Florence Nightingale — "the secret of national health lies in the homes of the people".

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park's family life-cycle model is drawn from low-mortality Western settings and its "children leaving home" phases apply imperfectly to India, and the family-record system named above has been progressively digitised under the Ayushman Bharat Digital Mission with the sub-centre registers merged into the Integrated Reproductive and Child Health Register version 2.0 (Sunder Lal, Ch 8, p 533).

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 12 (Medicine and Social Sciences, pp 825-829, 834), Ch 9 (Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 652, 690), Ch 1 (Man and Medicine: Towards Health for All, p 22); Sunder Lal — Textbook of Community Medicine, Ch 2 (Relationship of Social and Behavioural Sciences to Health and Disease, pp 69-84), Ch 8 (p 533), Ch 16 (pp 701-702); National Family Health Survey-5 India Report (2019-21).



Role of health education in disease prevention and control

Asked in: Paper I 07-Jul-2023 — RGUHS MD Community Medicine

Definition — Health education is "a process that informs, motivates and helps people to adopt and maintain healthy practices and lifestyles, advocates environmental changes as needed to facilitate this goal, and conducts professional training and research to the same end" (National Conference on Preventive Medicine, USA; Park 28e, Ch 22, p 1014). *Last* defines it as the process by which individuals and groups learn to behave in a manner conducive to the promotion, maintenance or restoration of health. It is not propaganda: propaganda instils ready-made slogans and spoon-fed knowledge, whereas health education changes knowledge, attitude and behaviour through the learner's own active participation (Park 28e, Ch 22, p 1016).

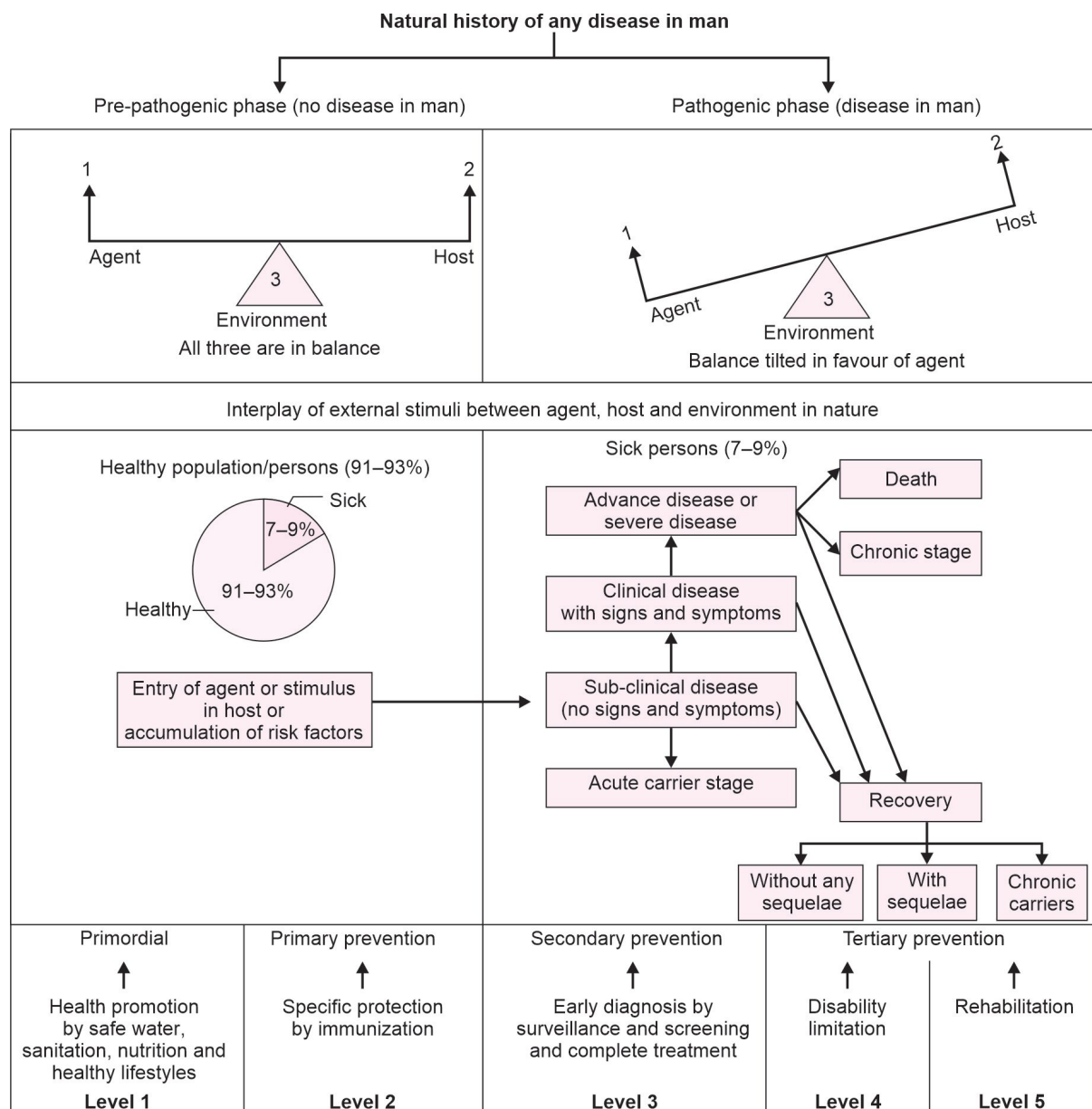
Why health education is central to disease control

- Most of the world's major health problems and premature deaths are **preventable through changes in human behaviour at low cost** (Park 28e, Ch 22, p 1012).
- **Drugs alone will not solve health problems** — without health education a person falls sick again and again from the same disease; the eradication of cholera, typhoid, malaria and tuberculosis in western countries turned on education (Park 28e, Ch 22, p 1018).
- **Education of the people about the prevention and control of locally endemic diseases is the first of the eight essential activities of primary health care** (Alma-Ata, 1978; Park 28e, Ch 22, p 1018).

- Technology without education fails: the malaria-eradication experience showed that **anti-malarial insecticide spray cannot solve the problem without health education** (Park 28e, Ch 22, p 1018).
- **Education alone is also insufficient** — the target population must have access to proven preventive measures or procedures (Park 28e, Ch 22, p 1015).

Aims relevant to disease control (WHO 1969 and Alma-Ata 1978; Park 28e, Ch 22, p 1015): encourage people to adopt and sustain health-promoting lifestyles and practices; promote proper use of the health services available to them; arouse interest, give new knowledge, improve skills and change attitudes so that people make rational decisions to solve their own problems; and stimulate individual and community self-reliance and participation at every step from identifying problems to solving them.

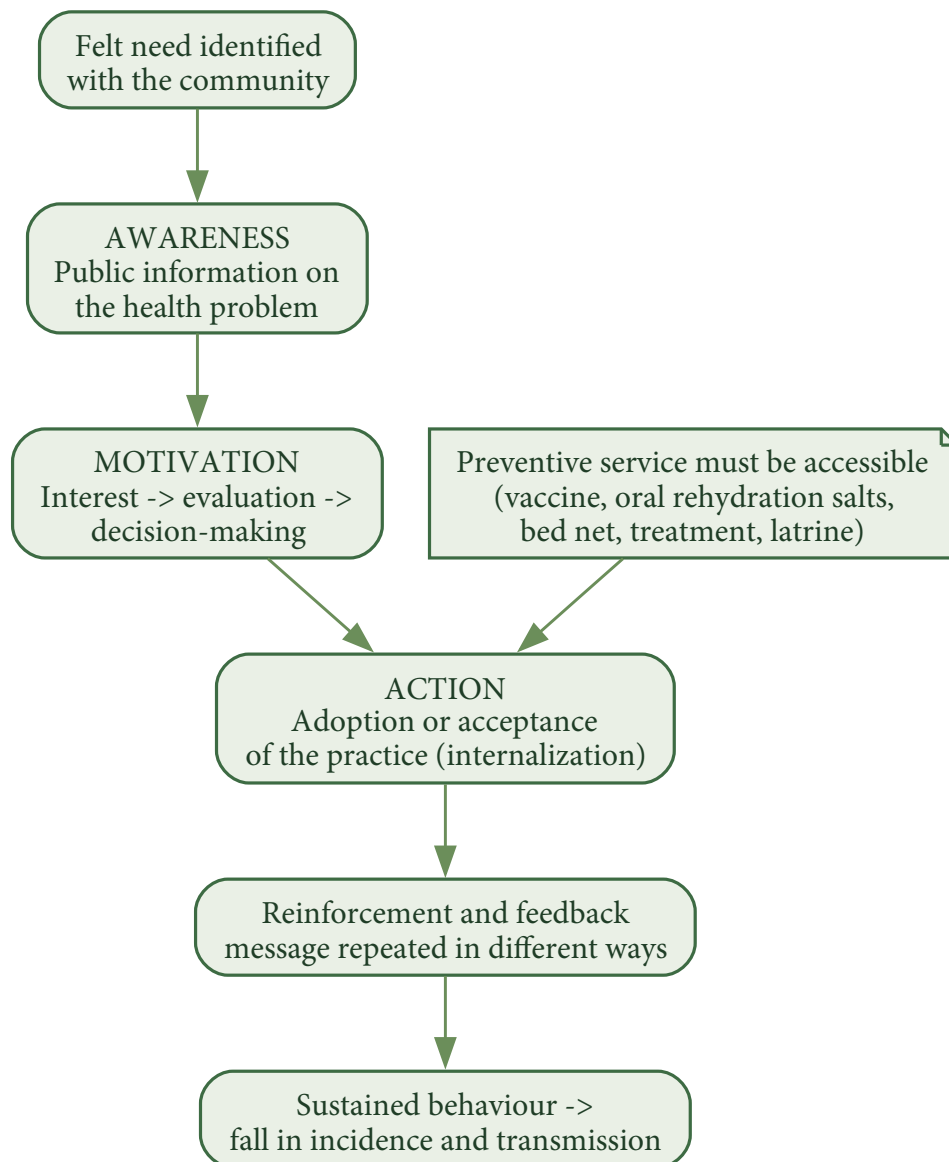
Role at each level of prevention



Natural history of disease and levels of prevention (Sunderlal, p.44)

Level of prevention	What health education contributes	Worked example
Primordial	Information, education and communication (IEC) and behaviour change communication (BCC) so that risk behaviours never emerge; behaviour is more easily developed in the young than in adults (Park 28e, Ch 22, p 1016)	Community-level education through mass media and interpersonal communication on the predominant risk factors — undernutrition, unsafe sex, unsafe water and bad sanitation, high blood pressure, high cholesterol, overweight and tobacco use (Sunder Lal, Ch 1, p 46)
Primary — health promotion	Establishing the time-honoured "best practices": personal hygiene, handwashing, tooth-brushing, avoiding tobacco and alcohol, regular exercise, healthy eating — learnt first in the family, then at school and workplace	Swachh Bharat Abhiyan generating demand for toilets in households, schools and <i>anganwadi</i> centres through awareness and behaviour change (Sunder Lal, Ch 1, pp 40–41, 46)
Primary — specific protection	Creating demand for, and correct use of, a proven preventive measure	The male health worker chlorinates public water sources and educates the community on the advantage and use of sanitary latrines and smokeless <i>chulhas</i> (Park 28e, Ch 10, p 742); resistance to latrine use is overcome only through health education (Ch 15, p 920)
Secondary — early diagnosis and treatment	Teaching danger signs and where to report; promoting uptake of screening; motivating regular and complete treatment	The auxiliary nurse midwife (ANM) checks whether cases under treatment for malaria, tuberculosis and leprosy are taking regular treatment and motivates defaulters; gives education, counselling, referral and follow-up for sexually transmitted and reproductive tract infection and HIV/AIDS; and delivers IEC for prevention and detection of non-communicable disease (Park 28e, Ch 10, p 740)
Tertiary — disability limitation and rehabilitation	Sustained adherence, self-care, family support and reduction of stigma	Promoting the family's self-reliance in child-rearing, self-care and care of the sick — the family is the first defence and the chief reliance for the well-being of its members (Park 28e, Ch 22, p 1018)

How education becomes disease-preventing behaviour



Adoption model — pathway from a health message to a fall in disease incidence

Mere awareness is of little value unless it leads to motivation, and motivation itself contains the stages of interest, evaluation and decision-making before adoption occurs; reliance on any one method alone is therefore likely to lead to failure (Park 28e, Ch 22, p 1017).

Role in named disease-control efforts

Disease or problem	Educational content	Contribution to control
Malaria and other vector-borne disease	Personal protection, source reduction, acceptance of indoor residual spray	Spray alone cannot control transmission without community acceptance (Park 28e, Ch 22, p 1018)

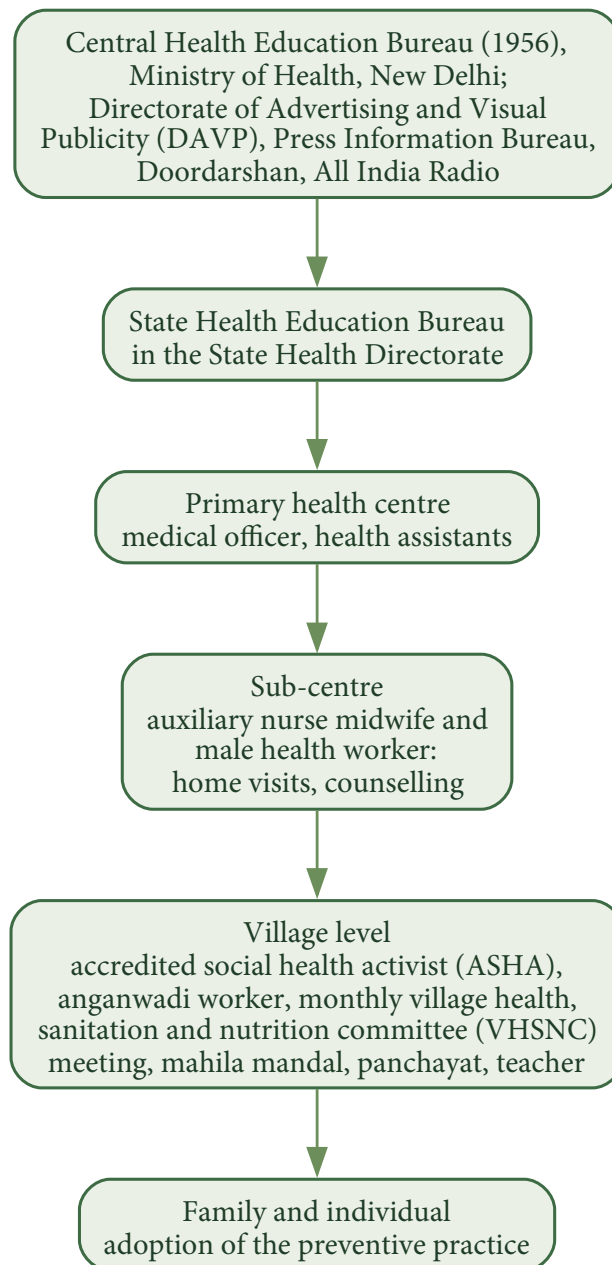
Disease or problem	Educational content	Contribution to control
Tuberculosis and leprosy	Symptom recognition, referral, importance of regular and complete treatment	The male health worker educates the community about control and preventive measures against communicable disease and about taking regular and complete treatment, and motivates defaulters (Park 28e, Ch 10, pp 741–742)
Diarrhoeal disease	Demonstration of oral rehydration technique, handwashing, safe water, disposal of liquid and solid wastes, home sanitation, use of sanitary latrines	Demonstration carries high motivational value in environmental sanitation and mother-and-child health (Park 28e, Ch 22, p 1021; Ch 10, p 742)
HIV/AIDS	Adolescent Education Programme — sixteen scheduled hours in classes IX and XI, life-skills training, Red Ribbon Clubs, targeted interventions, integrated counselling and testing centres	Comprehensive knowledge of HIV/AIDS is only 21.6% in adult women against 30.7% in adult men (NFHS-5), the gap these interventions target (Sunder Lal, Ch 8, p 347; Ch 10, p 624)
Non-communicable disease	Healthy diet, physical activity, avoiding tobacco and harmful alcohol use; group discussion where long-term compliance is needed (smoking cessation, obesity reduction)	Primordial and primary prevention of risk factors (Sunder Lal, Ch 8, pp 339–340; Park 28e, Ch 22, p 1021)
Scabies and skin infection	Demonstration of treatment and personal hygiene	Named by Park as a demonstration-responsive disease-control activity (Park 28e, Ch 22, p 1021)

Approaches, methods and principles

Approach (Park 28e, Ch 22, pp 1015–1016)	Basis	Limitation
Regulatory (managed prevention)	Governmental intervention altering behaviour by law — Child Marriage Restraint Act, compulsory seat belts	Simplest and quickest route, but has important failures, e.g. prohibition of alcohol
Service approach	Provide all needed health services at the doorstep, assuming people will then use them	Failed where not based on felt needs — free water-seal latrines went unused in rural areas
Health education approach	Planned learning experiences; people are informed, educated and encouraged to make their own choice for a healthy life	Results are slow, but enduring
Primary health care approach	People's full participation and active involvement in planning and delivery of health services	Requires genuine community involvement, not token participation

Method (Park 28e, Ch 22, pp 1020–1022)	Examples	Value in disease control
Individual (interpersonal)	Personal contact, home visits, personal letters, counselling	Permits discussion, argument and persuasion, and questions on specific interests; limitation is that numbers reached are small
Group	Lecture, demonstration, group discussion (not less than 6 and not more than 12 members seated in a circle with a group leader and a recorder), panel discussion, symposium, workshop, conference, seminar, role play	More influential at average and below-average educational levels; valued where long-term compliance is needed
Mass	Television, radio, newspaper, printed material, direct mailing, posters, health museums and exhibitions, folk methods, internet	Reaches the widest population for concentrated campaigns; more effective above average educational level

Principle (Park 28e, Ch 22, pp 1018–1019)	What it demands of the educator
Credibility	Message trustworthy, scientifically sound, compatible with local culture, educational system and social goals
Interest	Teach to the felt needs of the people; slogans they are not interested in are useless
Participation	Active involvement creates acceptance and feedback; without it programmes are unlikely to succeed
Motivation	Awaken the fundamental desire to learn; positive incentives in preference to threats
Comprehension	Pitch to the mental capacity of the audience; never use words strange and new to the people
Reinforcement	Repeat the message at intervals and in different ways
Learning by doing	"If I hear I forget, if I see I remember, if I do I know"
Known to unknown	Concrete to abstract, simple to complicated, easy to difficult
Setting an example	The educator must practise what he teaches
Good human relations	Friendliness, and sympathetic listening to the anxious or hostile
Feedback	Modify the message and the channel in the light of the audience's response
Leaders	Work through the village headman, school teacher or political worker — people learn best from those they respect



Delivery of health education through the Indian health system

Barriers and evaluation

Barrier	Illustration
Physiological	Difficulty in hearing, difficulty in expression
Psychological	Emotional disturbance, neurosis, level of intelligence, difficulty of language or comprehension
Environmental	Noise, invisibility, congestion, frequent interruptions
Cultural and social	Illiteracy, customs, beliefs, religion, attitudes, economic and social class differences, language and dialect, medical jargon, poor health and visual literacy
Relational	Only talking and not listening, lack of empathy and trust, poor eye contact, low credibility of the source

(Park 28e, Ch 22, p 1012; Sunder Lal, Ch 4, p 155)

Evaluation — the real test is not counting activities (clients contacted, camps held, film shows, handouts distributed) but measuring the **change in knowledge, attitude and practice (KAP)** produced (Sunder Lal, Ch 4, pp 153–154):

- ▶ **Knowledge** — standard questions on the specific problem.
- ▶ **Attitude** — a *Likert* scale measuring degree of agreement on a 3-, 5- or 7-point range.
- ▶ **Practice** — direct observation of behaviour, such as handwashing, nail cutting, cord cutting, cooking and feeding practice.
- ▶ **Design** — baseline data on current practice compared with repeat or annual survey data after a defined interval, plus exit interviews and home visits.
- ▶ **National sources** — National Family Health Survey rounds 1–5 and behavioural surveillance surveys under the National AIDS Control Programme; process evaluation is the most important component.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **The health communication process** (Park 28e, Ch 22, p 1013) — draw five linked boxes with a return arrow, labelled exactly: **Sender (source) → Message (content) → Channel (medium) → Receiver (audience) → Feedback (effect)**, the feedback arrow running back from receiver to sender. Annotate the message box with the attributes of a good message: in line with the objectives, meaningful, based on felt needs, clear and understandable, specific and accurate, timely and adequate, fitting the audience, interesting, culturally and socially appropriate.

High-yield exam pointers

- ▶ Education about the prevention and control of locally endemic diseases is the **FIRST** of the eight essential activities of primary health care (Alma-Ata, 1978).
- ▶ The two limiting statements the examiner looks for: "drugs alone will not solve health problems", and "education alone is insufficient — the target population must have access to proven preventive measures or procedures".
- ▶ Group discussion size: **not less than 6 and not more than 12 members**.
- ▶ Dated Indian figure: comprehensive knowledge of HIV/AIDS **21.6% in adult women vs 30.7% in adult men** (NFHS-5, 2019–21).
- ▶ **Central Health Education Bureau established 1956**, Ministry of Health, New Delhi.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park's chapter is titled "Communication for Health Education"; the Alma-Ata wording of the first essential activity is "education concerning prevailing health problems and the methods of preventing and controlling them", which Park renders as education about the prevention and control of locally endemic diseases — either wording is accepted in the exam.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 22 Communication for Health Education, pp 1012–1024; Ch 10 Health Care of the Community, pp 740–742; Ch 15 Environment and Health, p 920); Sunder Lal — Textbook of Community Medicine (Ch 1 Concept of Health and Disease, pp 40–46; Ch 4 Principles of Health Promotion and Education, pp 149–155; Ch 8, pp 339–340, 347; Ch 10, p 624); Alma-Ata Declaration 1978; WHO Technical Report Series 409 (1969); National Family Health Survey-5 (2019–21).



Service and regulatory approaches to health education

Asked in: Paper I Nov 2018 — RGUHS MD Community Medicine

Definition — Health education is a process for inducing behavioural change, its focus being on people and on action, and its aims being to make people adopt health-promoting lifestyles, use available health services properly, and become self-reliant (Park 28e, Ch 22, pp 1014-1015). Park describes **four well-known approaches to health education** — regulatory, service, health education and primary health care (Park 28e, Ch 22, pp 1015-1016). The **regulatory approach** seeks behaviour change through external legal control imposed by the State; the **service approach** seeks it by delivering services free at the people's doorstep, expecting availability alone to generate use.

The four approaches — where the two in question sit

#	Approach	Core instrument	Underlying assumption
1	Regulatory approach (also called <i>managed prevention</i> , coercive or legislative approach)	Governmental intervention, direct or indirect, designed to alter human behaviour — forms ranging from prohibition to imprisonment	People will change if the State compels them
2	Service approach	Provision of all needed health services at the people's doorstep, free of cost (tried by the Basic Health Services in the 1960s)	People will use services to improve their own health if the services are made available
3	Health education approach	Planned learning experiences so that an informed person makes his own choice	Autonomy over one's own life, including health affairs, produces enduring change
4	Primary health care approach	Full community participation and involvement in planning and delivery of services, with intersectoral coordination	People become self-reliant in matters of health once guided to identify their own problems and workable solutions

(Park 28e, Ch 22, pp 1015-1016)

Regulatory approach (managed prevention)

- ▶ **Definition** — regulation, in the context of health education, is *any governmental intervention, direct or indirect, designed to alter human behaviour*. Regulations are promulgated by the State through a variety of administrative agencies and may take many forms, from prohibition to imprisonment (Park 28e, Ch 22, p 1015).
- ▶ **Mechanism** — change is sought through a variety of **external controls or laws placed on people**; the individual is the object of the law, not a partner in it. Compliance is secured by penalty and enforcement rather than by conviction.
- ▶ **Attraction** — the legislative route appears the **simplest and quickest** way to bring about a desired change in society (Park 28e, Ch 22, p 1015), and where the behaviour is a discrete, enforceable act the effect can be large and immediate.
- ▶ **Reasons for failure of the coercive approach** (Park 28e, Ch 22, p 1015):
 - The **cause of disease, medical or social, cannot be eradicated by legislation**; at most the government can make laws to prevent a person spreading disease in his community, as with vaccination in an emergency.
 - In areas involving **personal choice** (diet, exercise, smoking) no government can legislate people into eating a balanced diet or not smoking — it amounts to taking away some of the rights of the individual.
 - A law is **difficult to enforce unless the majority are in favour of it** and unless it does not interfere with individual rights.
 - Enforcement needs a **vast administrative infrastructure and considerable expenditure**, even where the State has a duty to legislate (e.g. for acquired immunodeficiency syndrome, AIDS).
- ▶ **Where it is legitimately useful** — in emergencies and limited situations such as control of an epidemic disease or management of fairs and festivals, and to **reinforce the pressure to change collective behaviour** alongside education. Its fundamental objection remains that it runs counter to the **basic tenet of health education, that we do not force people to change** (Park 28e, Ch 22, p 1015).

Indian and international examples cited by Park

Example	Instrument	Outcome
Child Marriage Restraint Act of 1978 — raised the legal age of marriage from 15 to 18 years for girls and from 18 to 21 years for boys	Prohibitory law	Child marriages still in vogue in parts of Rajasthan, Madhya Pradesh and Uttar Pradesh (Park 28e, Ch 9, pp 697-698)
Compulsory seat belts (western countries)	Mandatory-use law	Clear, immediate fall in deaths and injuries (Park 28e, Ch 3, p 107)
Prohibition of alcohol	Total ban	Cited by Park as an important failure of law (Park 28e, Ch 22, p 1015)

Example	Instrument	Outcome
Compulsory sterilization campaign, 1976, India	Coercion in family planning	"Disastrous"; contributed to the Congress defeat in the 1977 elections (Park 28e, Ch 22, p 1015)
Cigarettes and Other Tobacco Products Act (COTPA), 2003	Comprehensive tobacco control legislation, paired with mass-media awareness under the National Tobacco Control Programme	Regulation deliberately combined with education, not used alone (Park 28e, Ch 7, p 567)

Service approach

- **Definition and origin** — this approach was tried by the **Basic Health Services in the 1960s**. It aimed at **providing all the health services needed by the people at their doorsteps**, on the assumption that people would use them to improve their own health (Park 28e, Ch 22, p 1016).
- **The vehicle** — "basic health services" was the term used by the United Nations Children's Fund (UNICEF) and the World Health Organization (WHO) in their joint health policy of **1965**, defined as *a network of coordinated, peripheral and intermediate health units capable of performing effectively a selected group of functions essential to the health of an area and assuring the availability of competent professional and auxiliary personnel to perform these functions* (Park 28e, Ch 10, pp 718-719).
- **Why it failed** — the approach was **not based on the felt needs of the people**. Park's classical illustration: when **water-seal latrines were provided by the government free of cost**, many **people in rural areas did not make use of them**, because it was **not their habit to use latrines** (Park 28e, Ch 22, p 1016).
- **The lesson** — *people will not accept a programme or service, even if it is offered free of cost, unless it is based on their felt needs* (Park 28e, Ch 22, p 1016).
- **Systemic drawbacks that compounded the failure** — the basic health services shared the handicaps of comprehensive health care: **lack of community participation, lack of intersectoral coordination, and dissociation from the socio-economic aspects of health**; primary health centres were understaffed and poorly supplied with medicines and equipment, and so did not enjoy the confidence of the people (Park 28e, Ch 10, pp 718-719).
- **Wider evidence of the same failure** — immunization has not gained universal acceptance despite effective vaccines, and programmes for maternal and child health, water supply, sanitary latrines and infant-rearing practice have proved abortive or only partially successful (Park 28e, Ch 12, p 811).

Point-by-point differentiation

Point of difference	Service approach	Regulatory approach
Alternative name	No separate name in Park; identified with the Basic Health Services of the 1960s	Managed prevention; coercive or legislative approach

Point of difference	Service approach	Regulatory approach
Core premise	Make the service available and free, and people will use it	Make the behaviour compulsory by law, and people will comply
Instrument used	Health services, supplies and facilities delivered at the doorstep	Acts, rules, statutory bans, licensing, fines and imprisonment
Agency	Health department and its peripheral field staff	The State, through a variety of administrative and enforcement agencies
Historical setting	Basic Health Services, 1960s; UNICEF/WHO joint health policy 1965	Long-standing; still the standard route for public health legislation
Direction of change	"Supplied" from provider to people (top-down provision)	"Imposed" from State on people (top-down compulsion)
Role of the individual	Passive recipient of a service	Passive subject of a law; no scope for personal choice
Motivating force	Free availability and physical access	Fear of penalty and enforcement
Basis in felt needs	Not based on felt needs — its central defect	Not based on felt needs either, and may actively conflict with them
Speed of effect	Slow and uncertain; depends on voluntary uptake	Fast where the act is discrete and enforceable
Durability of change	Poor — service falls into disuse once external supply stops	Poor once enforcement relaxes; no internalised conviction
Cost implication	High recurrent cost of infrastructure, supplies and staff	High cost of a vast administrative and enforcement infrastructure
Ethical concern	Paternalistic, but not coercive	Takes away some of the rights of the individual
Typical example	Free water-seal latrines that villagers did not use	Child Marriage Restraint Act; compulsory seat belts; 1976 sterilization campaign
Reason for failure	Ignored felt needs, community participation and intersectoral coordination	A law cannot eradicate the medical or social cause of disease, and cannot govern areas of personal choice
Legitimate place today	As the delivery arm of a programme once demand has been created by education	In emergencies, epidemic control and mass gatherings, and to reinforce collective behaviour change
Relation to health education	Provides the service that education creates demand for	Runs counter to the basic tenet that we do not force people to change

Complementary use — the practical conclusion

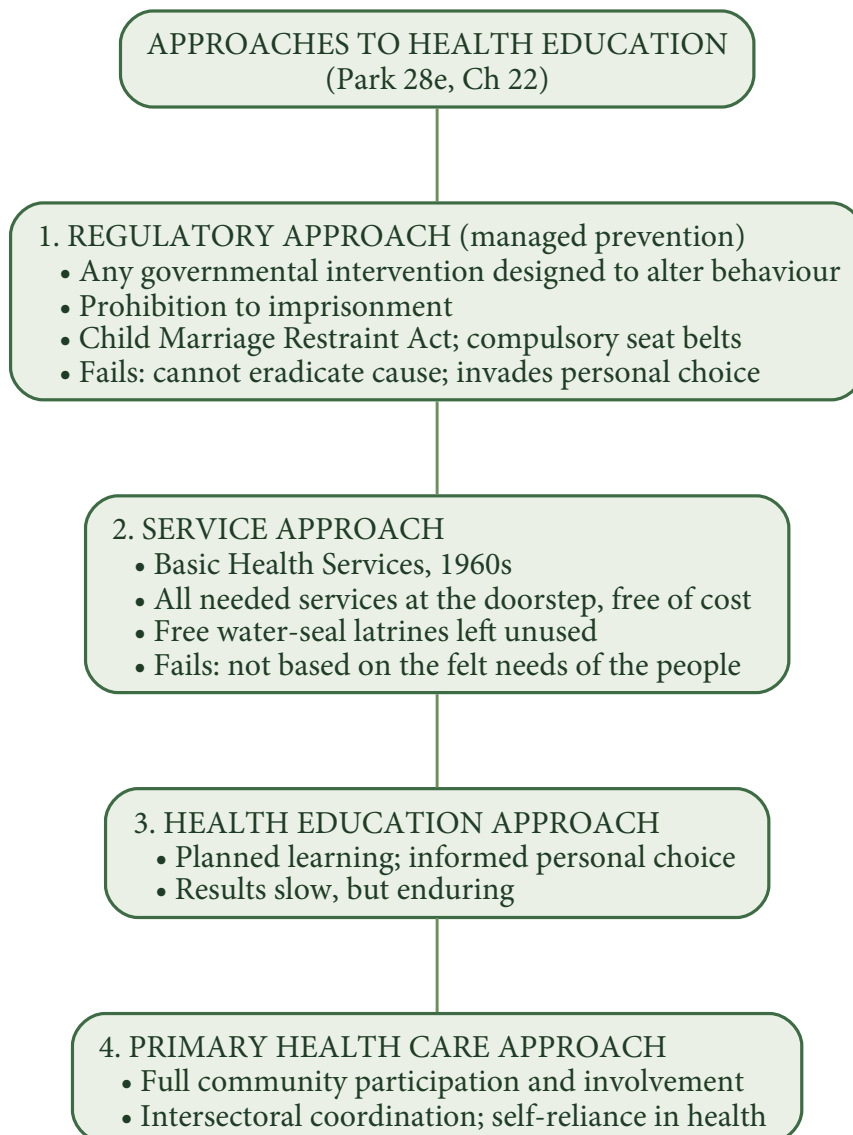
- ▶ Neither works alone: the service approach fails from **absent demand**, the regulatory approach from **absent consent**.
- ▶ Park's resolution is the third and fourth approaches — **planned learning experiences** so that the informed person chooses (slow but enduring), and the **primary health care approach**, starting

from the people with full participation and intersectoral coordination so that they become self-reliant in matters of health (Park 28e, Ch 22, p 1016).

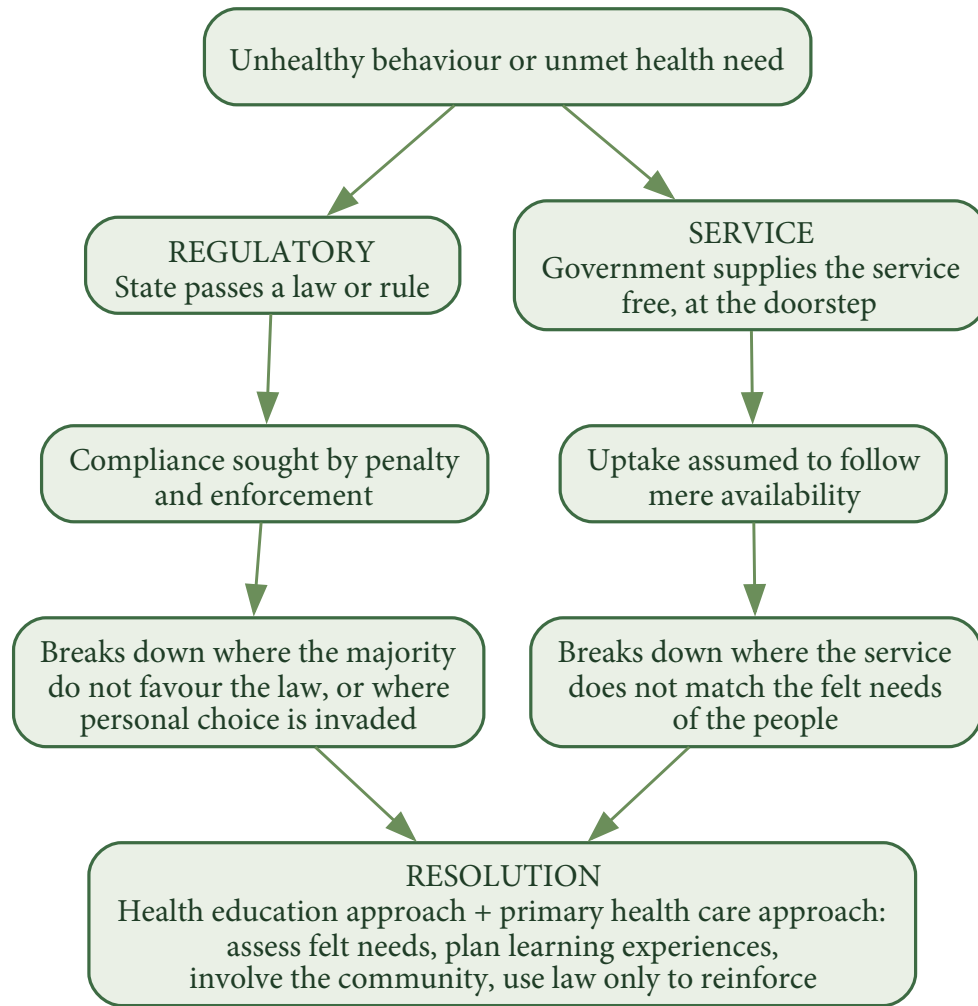
- ▶ A national programme uses all three levers together: the National Tobacco Control Programme pairs COTPA 2003 enforcement with mass-media campaigns for behaviour change and with service delivery through the health system (Park 28e, Ch 7, p 567).
- ▶ Legislation is best introduced **after** education has brought the majority round to it, since a law is difficult to enforce unless the majority favour it (Park 28e, Ch 22, p 1015).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Spectrum of coercion versus participation** — draw one horizontal arrow, label the left end "Coercion / change imposed" and the right end "Participation / change chosen", and mark four points along it in order: 1. Regulatory approach (law, penalty), 2. Service approach (free doorstep provision), 3. Health education approach (informed personal choice), 4. Primary health care approach (community involvement, self-reliance).

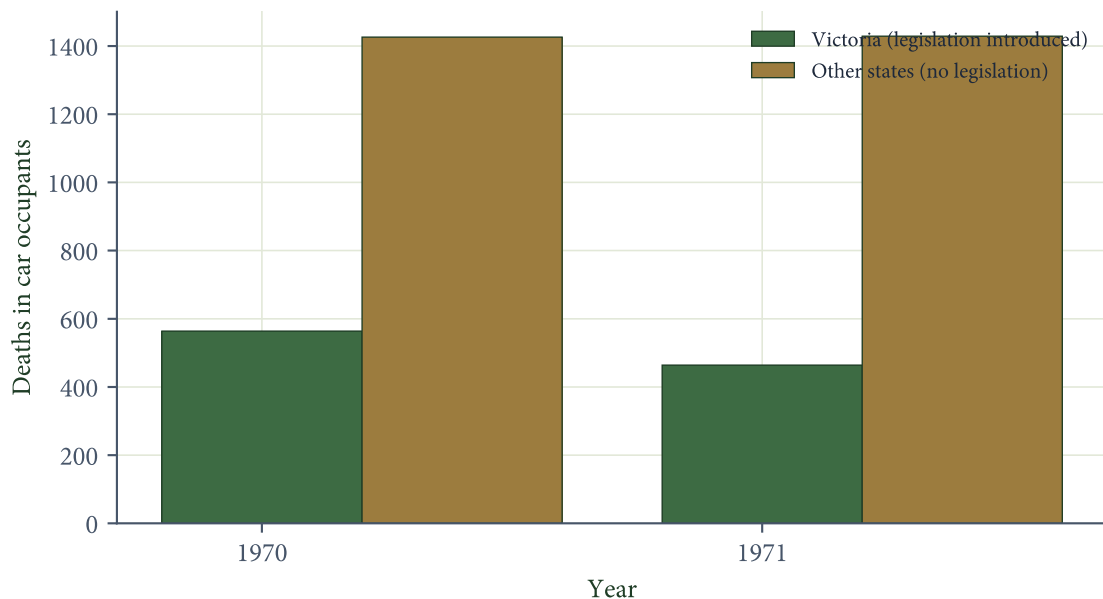


The four approaches to health education, with the two asked about at the top



How each approach tries to produce behaviour change, and why each broke down

lation can work where the act is discrete: compulsory seat-belt legislation, Victoria (Au



Deaths fell 17.7% in Victoria after compulsory seat-belt legislation, against a 0.2% rise in states without the law; injuries fell 14.8% versus a 1.0% rise. Source: Park's Textbook of Preventive and Social Medicine 28e, Ch 3, p 107 (Table 25).

High-yield exam pointers

- ▶ **Four approaches to health education (Park):** Regulatory → Service → Health education → Primary health care. Mnemonic "R-S-H-P" = *Rules, Services, Health education, People*.
- ▶ **Regulatory approach one-liner:** *any governmental intervention, direct or indirect, designed to alter human behaviour*; forms range from prohibition to imprisonment; also called **managed prevention**.
- ▶ **Service approach one-liner:** tried by the **Basic Health Services in the 1960s**; all services at the doorstep, free of cost; failed because it was **not based on felt needs**.
- ▶ **The two killer examples:** free **water-seal latrines** left unused (service approach failure) and the **1976 sterilization campaign** (regulatory approach failure).
- ▶ **"Basic health services", UNICEF/WHO 1965:** a network of coordinated, peripheral and intermediate health units performing a selected group of functions essential to the health of an area.
- ▶ **Closing line the examiner looks for:** the coercive approach runs counter to the basic tenet of health education — *in health education we do not force people to change*; law is used only to reinforce collective behaviour change.
- ▶ **Quotable figure:** Victoria seat-belt law 1971 — deaths 564 → 464 (−17.7%) against +0.2% in states without the law (Park 28e, Ch 3, p 107).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park still names the **Child Marriage Restraint Act**, superseded in current Indian law by the **Prohibition of Child Marriage Act, 2006** (Oxford Textbook of Public Health 7e, Ch 88), though the legal ages quoted by Park (18 years for girls, 21 for boys) are unchanged.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e — Ch 22 (Communication for Health Education, pp 1014-1016), Ch 10 (pp 718-719, basic health services), Ch 12 (p 811), Ch 9 (pp 697-698, Child Marriage Restraint Act), Ch 7 (p 567, National Tobacco Control Programme and COTPA 2003), Ch 3 (p 107, seat-belt legislation data); Oxford Textbook of Public Health 7e (Ch 88); UNICEF/WHO joint health policy 1965, as cited in Park 28e.



John Snow's contributions to public health

Asked in: Paper I 19-Nov-2020 — RGUHS MD Community Medicine

Definition — John Snow was an English physician and anaesthetist who studied the epidemiology of cholera in London from 1848 to 1854 and established the role of polluted drinking water in its spread (Park 28e, Ch 1, p 17). He is regarded as the father of modern epidemiology because he generated a causal hypothesis from the distribution of cases, tested it quantitatively against a comparison population, and acted on it to control an epidemic — all of this long before the birth of bacteriology, when *Vibrio cholerae* was still unknown (Park 28e, Ch 3, p 87; Gordis 7e, Ch 1, p 31).

The setting: London public health before Snow

Element of the setting	What it held	Why it mattered for Snow
Miasma theory (the ruling doctrine)	Disease arises from dirty living environments, unclean water and poor ventilation due to overcrowding — from noxious air and vapours	It drove sanitary reform but explained nothing; no control effort based on miasma had ever stopped a cholera outbreak (Oxford PH 7e, Ch 53, p 1951; Nelson & Williams, Ch 2, p 39)
Contagion theory	Poison passes directly from person to person; answered by isolation and quarantine	Reduced spread by ships but did not stop urban epidemics (Oxford PH 7e, Ch 53, p 1951)
Public Health Act 1848	First example of governmental involvement in public health affairs; fulfilled Johann Peter Frank's principle that the State is responsible for the health of its people	Despite it and improved sewage disposal, large cholera outbreaks continued in London in the 1850s (Park 28e, Ch 1, p 17; Oxford PH 7e, Ch 53, p 1951)
William Farr and the Registrar General's office	The world's first modern vital statistics system; Farr's careful documentation of deaths	Supplied the death data Snow used; earlier Registrar General reports had already hinted that water supply and cholera rates were linked (Lilienfeld, Ch 2, p 42; Handbook of Epidemiology, Ch 5, p 262)
London Epidemiological Society, 1850	Formed after a further cholera outbreak; Babington its first president; Snow a founding member alongside John Simon, Thomas Addison and Richard Bright	Institutionalised epidemiology as a discipline; Snow was then known chiefly as the anaesthetist who gave chloroform to Queen Victoria in childbirth (Lilienfeld, Ch 2, pp 42–43; Gordis 7e, Ch 1, p 29)

The Broad Street pump investigation, Golden Square 1854

- **The outbreak.** In the first week of September 1854, approximately 600 people living within a few blocks of the Broad Street pump in the Golden Square district of Soho died of cholera (Gordis 7e, Ch 1, p 29).
- **Descriptive step — the spot map.** Snow plotted every cholera death on a map of London by place of residence. The **spot map** showed at a glance the boundaries and the pattern of clustering of cases (Park 28e, Ch 3, pp 86–87).
- **Inference from clustering.** Because a cluster of cases on a spot map may suggest a common source of infection or a common risk factor shared by all the cases, the map focused attention on

the **common water pump in Broad Street** (now Broadwick Street) as the source (Park 28e, Ch 3, pp 86–87; Lilienfeld, Ch 2, p 43).

- ▶ **Hypothesis.** Snow hypothesised that **cholera was a water-borne disease**, transmitted by swallowed matter and not by foul air — a hypothesis framed long before bacteriology and without any knowledge of the causative organism (Park 28e, Ch 3, p 87).
- ▶ **Public health action.** Snow lobbied the local authority to shut down the pump; the **handle of the incriminated water pump was removed** and the epidemic abated (Lilienfeld, Ch 2, p 43; Park 28e, Ch 3, p 143).
- ▶ **The honest caveat.** Whether removal of the handle actually caused the epidemic to decline is uncertain, since the outbreak was already waning; the act's enduring value is as a demonstration of decisive control action (Lilienfeld, Ch 2, p 43).

The south London "natural experiment" — the quantitative proof

The Broad Street map generated a hypothesis; the comparison of two water companies **tested** it. Two companies supplied most homes south of the Thames and their mains were **interspersed**, so that adjoining houses in the same street drew water from different sources. During the 1849 epidemic both had drawn from the Thames, but the **Lambeth Company had since moved its intake upstream of the sewage discharge point**, while the **Southwark and Vauxhall Company continued to draw water downstream of it** (Lilienfeld, Ch 2, pp 43–44; Handbook of Epidemiology, Ch 5, pp 262–263).

Snow then did what is now called "**shoe-leather epidemiology**" — going from house to house, counting all cholera deaths in each house and determining which company supplied it — and calculated cholera death rates per 10,000 houses for the first seven weeks of the epidemic (Gordis 7e, Ch 1, p 29; Lilienfeld, Ch 2, p 45).

Water supply	No. of houses	Deaths from cholera	Deaths per 10,000 houses
Southwark and Vauxhall Co. (intake downstream of sewage)	40,046	1,263	315
Lambeth Co. (intake relocated upstream)	26,107	98	38
Other districts in London	256,423	1,422	56

(Gordis 7e, Ch 1, Table 1.5, p 31, from Snow on Cholera ; the same counts appear in Lilienfeld, Ch 2, Table 2.3, p 45)

- ▶ Cholera mortality in homes supplied by Southwark and Vauxhall was **eight to nine times** that in homes supplied by Lambeth — the exposed and unexposed groups differing only in the source of their water (Lilienfeld, Ch 2, p 45).
- ▶ Because exposure was allocated by the accident of which company had laid the main, and not by the investigator, this is the classical **natural experiment**; it is cited today as an early example of a **cohort study** applied to an infectious disease (Lilienfeld, Ch 2, p 45; Handbook of Epidemiology, Ch 5, p 262).

- Snow's account was published as *On the Mode of Communication of Cholera*, 2nd edition, London, 1855, while germ theory was still in its infancy (Handbook of Epidemiology, Ch 21, p 889; Oxford PH 7e, Ch 53, p 1951).

Snow's contributions to public health and epidemiology

Contribution	What Snow actually did	What it founded in modern practice
Established water-borne transmission of cholera	Showed polluted drinking water spread cholera in London, 1848–1854, and that contaminated water was the reservoir of the agent	Environmental sanitation and safe water supply as the primary control measure for cholera (Park 28e, Ch 1, p 17; Nelson & Williams, Ch 2, p 39)
The spot map	Plotted fatal cases by place of residence around Golden Square	Descriptive epidemiology by place ; today's disease mapping and geographic information systems (Park 28e, Ch 3, pp 86–87)
Hypothesis generation from distribution	Read a causal hypothesis out of the clustering of cases	The descriptive-to-analytical sequence that structures every outbreak investigation (Park 28e, Ch 3, p 87)
The natural experiment and rate comparison	Classified households by water company and compared cholera death rates per 10,000 houses	Analytical epidemiology; the comparison group, the denominator, and the forerunner of the cohort study (Lilienfeld, Ch 2, p 45; Handbook of Epidemiology, Ch 5, p 262)
Field enumeration ("shoe-leather epidemiology")	House-to-house ascertainment of deaths and of water source	Active case search and house-to-house survey in outbreak investigation (Gordis 7e, Ch 1, p 29)
Control without knowing the agent	Removed the pump handle although <i>V. cholerae</i> was unknown	The principle that removing a single essential or weakest link in the chain may control a disease even when the aetiology is incompletely known (Park 28e, Ch 3, p 143)
Evidence translated into policy	His data persuaded Farr and the authorities	Epidemiology as the basis of public health policy and legislation (Gordis 7e, Ch 1, p 31; Lilienfeld, Ch 2, p 45)
Anaesthesia	Administered chloroform to Queen Victoria during childbirth	Recognised separately as a founder of clinical anaesthesia (Gordis 7e, Ch 1, p 29)

Impact on policy and the enduring legacy

- **Change in vital registration.** Snow's data so convinced William Farr, the Registrar General, that he required the registrar of each district in south London to record **which water company supplied each house in which a person died of cholera** — surveillance data collected specifically to answer an epidemiological question (Gordis 7e, Ch 1, p 31).
- **Legislation.** Snow's report led to legislation requiring **all London water companies to filter their water by 1857**; he was elected President of the Medical Society of London (Lilienfeld, Ch 2, p 45).

- ▶ **Environmental health.** Relocation of the sources of drinking water successfully controlled the outbreak, making this a pioneering study in the identification of **environmental factors** in a disease outbreak (Oxford PH 7e, Ch 53, pp 1951–1952).
- ▶ **Vindication of the method.** Farr initially rejected Snow's hypothesis in favour of miasma, but was eventually convinced; his own analysis of the 1866 epidemic demonstrated that contaminated water was a risk for cholera (Nelson & Williams, Ch 1, p 26).
- ▶ **A balanced view.** Snow was later discredited for advocating against clean air legislation, a reminder that a pioneer's authority does not extend to every question (Lilienfeld, Ch 2, p 45).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Diagram 1 — Snow's spot map of Asiatic cholera, Golden Square, London 1854.** Labels to reproduce: street grid of the Golden Square / Soho district; **one dot = one cholera death**, plotted by place of residence; dense clustering of dots in the streets immediately around the centre; **Broad Street pump (marked X)** at the centre of the cluster; other public pumps in the vicinity marked and shown with few or no deaths around them; title "Spot map of Asiatic cholera in London".
- ▶ **Diagram 2 — Map of the two water supplies south of the Thames.** Labels: River Thames with the **sewage discharge point**; **Lambeth Company intake — upstream** of the discharge; **Southwark and Vauxhall Company intake — downstream** of the discharge; a block of streets south of the river with **interspersed mains**, adjacent houses labelled "Lambeth" and "S & V" alternately; box beneath giving the two rates, 38 and 315 deaths per 10,000 houses.

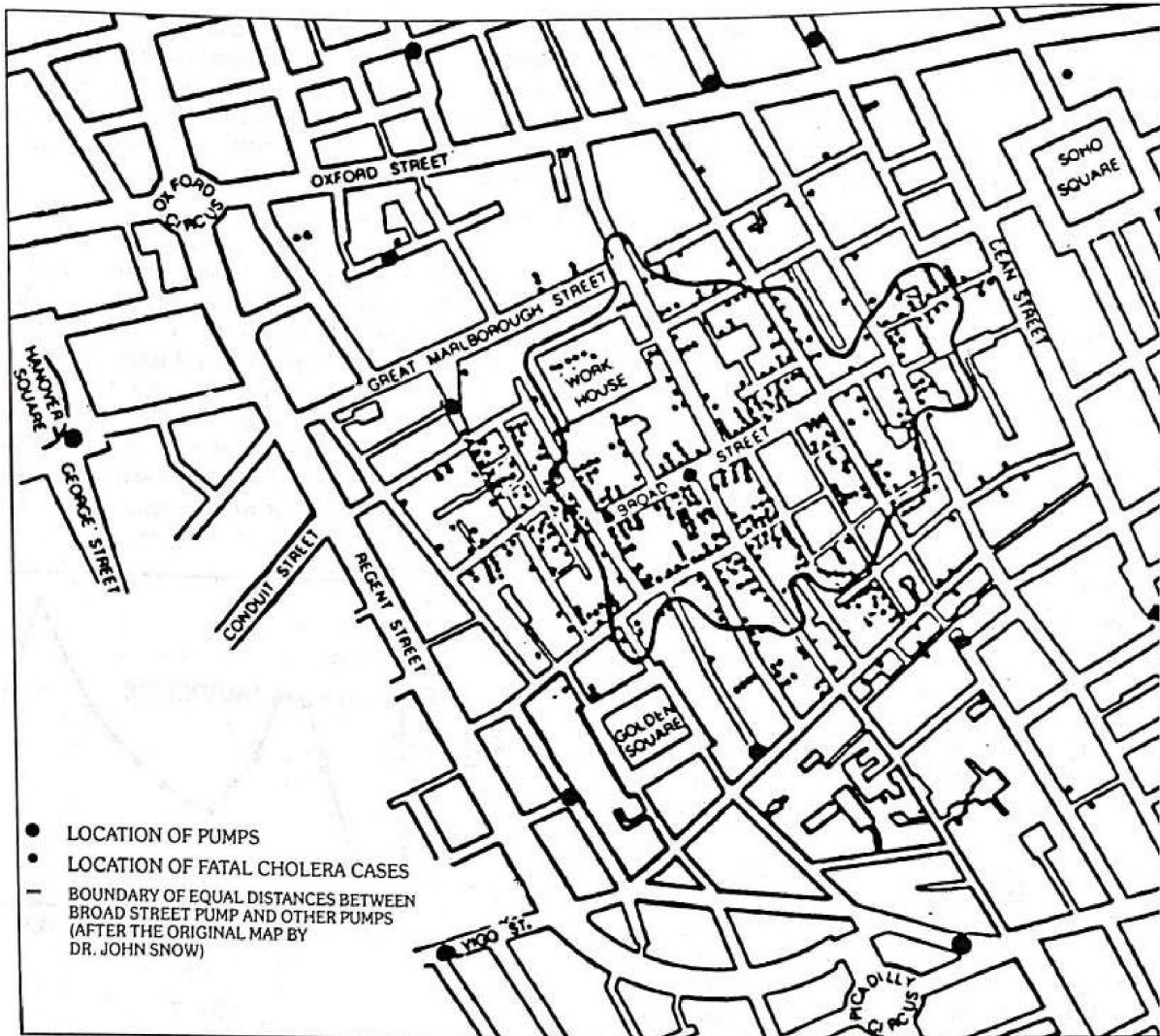
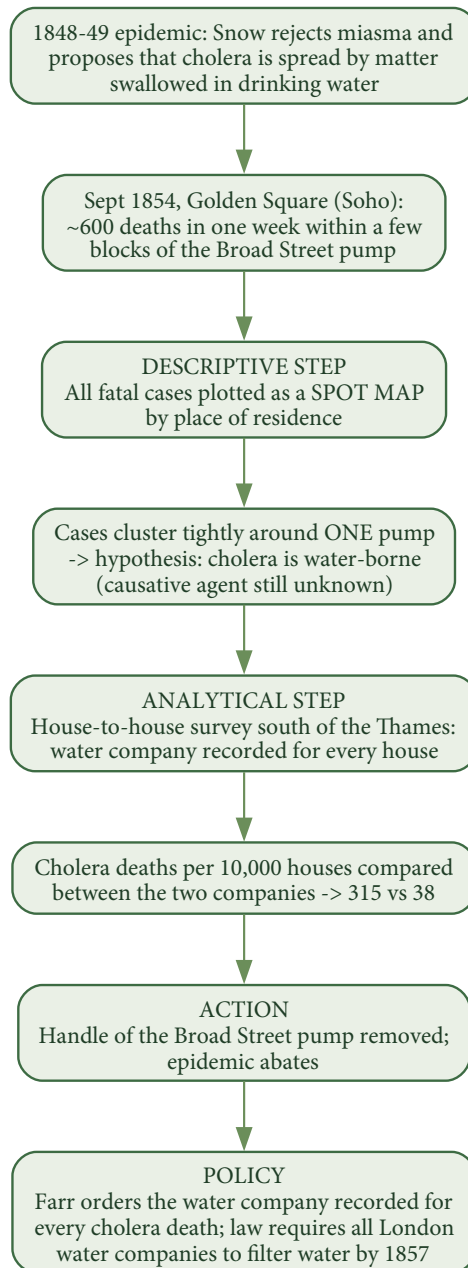
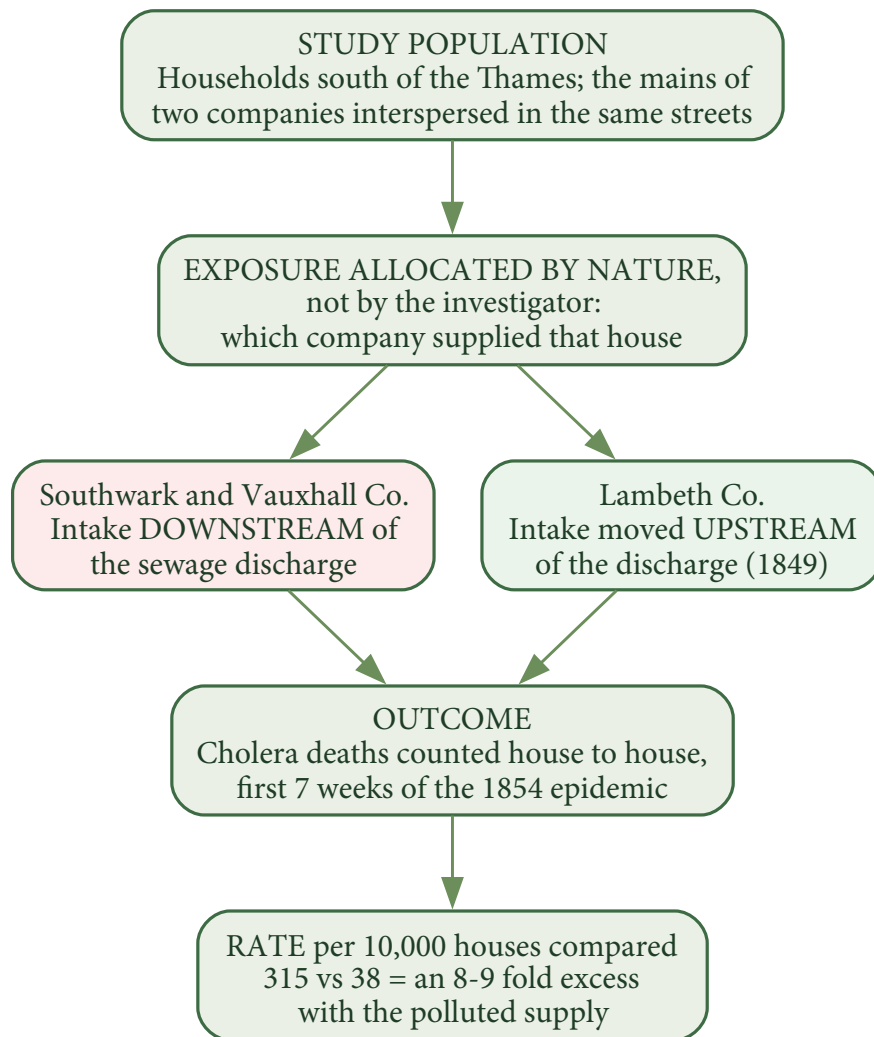


Fig 6 — Spot map of Asiatic cholera in London (Park's Textbook of Preventive and Social Medicine, 28e, p.87)



Sequence of John Snow's cholera investigations in London, 1848-1857



Structure of Snow's south London natural experiment - the forerunner of the cohort study

High-yield exam pointers

- ▶ Snow's two studies, one line each: the **Broad Street spot map (1854, Golden Square)** generated the hypothesis; the **south London water-company comparison** tested it quantitatively.
- ▶ **The table to write:** Southwark and Vauxhall 40,046 houses, 1,263 deaths, **315 per 10,000**; Lambeth 26,107 houses, 98 deaths, **38 per 10,000**; other districts 256,423 houses, 1,422 deaths, 56 per 10,000 — an **8–9 fold excess** (Gordis 7e, Table 1.5).
- ▶ **Quotable Park line:** removal of a **single essential or weakest link** in the chain may control a disease even when the aetiology is unknown — the classic example being Snow controlling cholera by removing the pump handle (Park 28e, Ch 3, p 143).
- ▶ **Do not confuse:** Park calls **cholera** "the father of public health"; **Snow** is the father of **modern epidemiology**.
- ▶ **Chronology anchors:** Public Health Act 1848 → Snow's cholera studies 1848–1854 → London Epidemiological Society 1850 → *On the Mode of Communication of Cholera*, 2nd ed. 1855 → compulsory filtration of London water 1857 → Pasteur's germ theory 1873 and Koch's cholera vibrio 1883, i.e. Snow's work preceded bacteriology by three decades.

- **Terms Snow gave epidemiology:** spot map, natural experiment, comparison group with a denominator, shoe-leather epidemiology, evidence-based control action.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: Lilienfeld's *Foundations of Epidemiology* reprints the same Snow counts with rates of 37 and 59 per 10,000 (against Gordis's 38 and 56), a difference of rounding in the same original data, so either set is acceptable provided the numerators and denominators quoted are the ones above; Snow's dates (1813–1858) are standard biographical fact but were not carried by the passages retrieved for this answer; and the Handbook of Epidemiology's cohort-study chapter transposes the two company labels against its own rates, so the Gordis and Lilienfeld tables are the safer source for the figures.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 1, Man and Medicine: Towards Health for All; Ch 3, Principles of Epidemiology and Epidemiologic Methods); Gordis Epidemiology 7e (Ch 1, Introduction); Lilienfeld's Foundations of Epidemiology (Ch 2, Threads of Epidemiologic History); Handbook of Epidemiology, Ahrens & Pigeot (Ch 1, Basic Concepts; Ch 5, Cohort Studies; Ch 21, Geographical Epidemiology); Oxford Textbook of Public Health 7e (Ch 53, Environmental Health Methods); Nelson & Williams, Infectious Disease Epidemiology (Ch 1, Early History of Infectious Disease; Ch 2, Epidemiology of Infectious Disease: General Principles). Primary source cited within these: Snow J. *On the Mode of Communication of Cholera*, 2nd edn, Churchill, London, 1855 (reprinted as *Snow on Cholera*, The Commonwealth Fund, New York, 1936).



Group discussion and group dynamics in health education

Asked in: Paper I 24-Sep-2025 — RGUHS MD Community Medicine

Definition — A **group** is "an aggregation of people interacting in a face-to-face situation", which contrasts sharply with a classroom audience (Park 28e, Ch 22, p 1021). **Group discussion** is a participatory, two-way method of health communication in which such a group deliberates on a felt problem, jointly sets its goals and arrives at a consensus solution under a facilitator (Sunder Lal, Ch 4, pp 146-147). **Group dynamics** is the interaction between group members, and of members with the facilitator or resource person, during that discussion; mapped as arrows showing who interacted with whom and how often, it becomes a **sociogram** (Sunder Lal, Ch 4, pp 147-148).

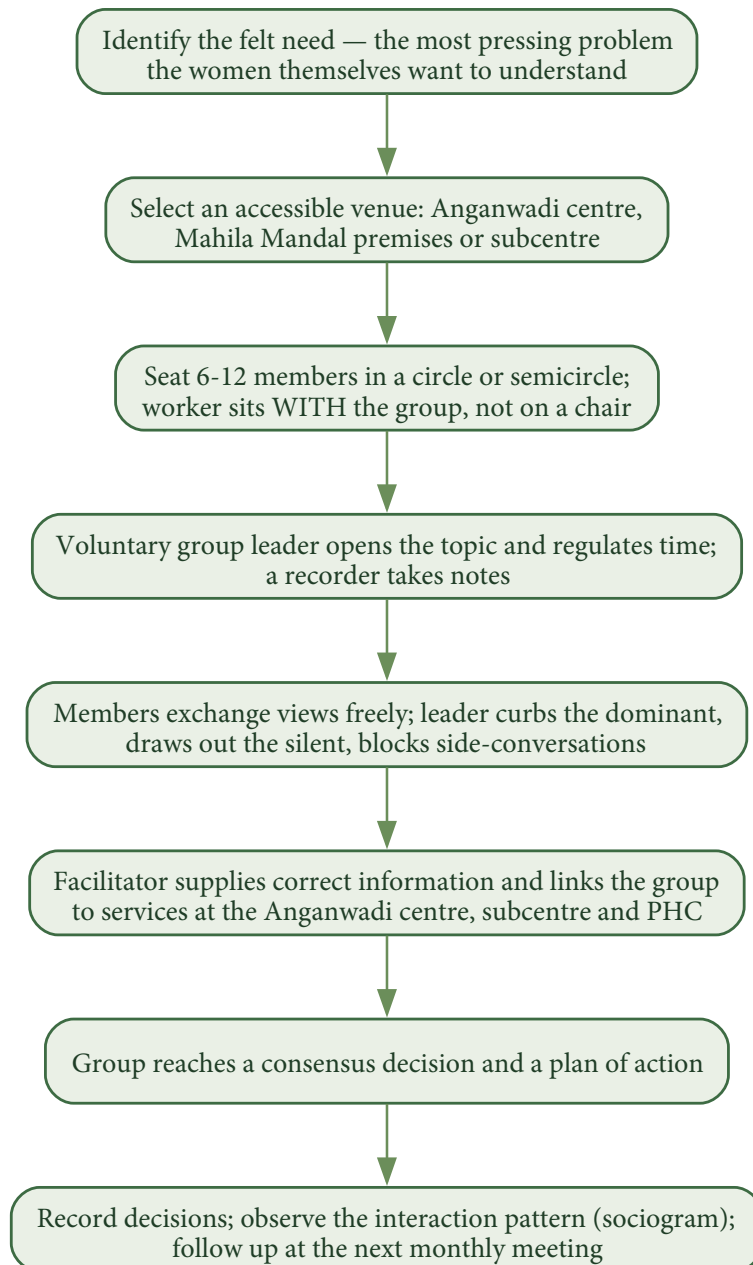
Place of group discussion, and why it works

Group teaching is the middle of the three approaches in health communication, and the choice of subject must relate directly to the interest of the group — one talks to a woman coming for delivery about child-birth and baby care, not about tuberculosis control (Park 28e, Ch 22, p 1020).

Property of group discussion	Consequence for health education
Free exchange of knowledge, ideas and opinions	Wider interaction among members than any other method allows (Park 28e, Ch 22, p 1021)
Group may reach decisions no individual member could have made alone	Decision rests on the ideas of ALL the people, not one person (Park 28e, Ch 22, p 1022)
A group decision is adopted more readily than a solitary decision	It has a binding effect on the individual member, who translates acceptance into action (Park 28e, Ch 22, p 1022)
Two-way (Socratic) traffic, based on felt needs	"Information trickles downwards and experience moves upwards" — the worker gets direct feedback (Mahajan & Gupta 4e, Ch 29, pp 584-585)
Especially valuable where long-term compliance is needed	Smoking cessation, obesity reduction (Park 28e, Ch 22, p 1021)

Structure and conduct of a group discussion

Element	Requirement
Size	Not less than 6 and not more than 12 members (Park 28e, Ch 22, p 1021); 5-15 in the conventional group method (Mahajan & Gupta 4e, Ch 29, p 580); <i>Mahila Swasthya Sangh</i> group 10-20 women (Sunder Lal, Ch 4, p 146)
Seating	All seated in a circle (or semicircle), each fully visible to all others; gives visual contact and a sense of equality (Park 28e, Ch 22, p 1021; Sunder Lal, Ch 4, p 146)
Position of the worker	Sits with the group on the mat, never on a chair acting as a teacher; facilitates, does not "talk at" people (Sunder Lal, Ch 4, p 147)
Group leader	Initiates the subject, guides the discussion, prevents side-conversations, encourages everyone to participate, sums up at the end; chosen voluntarily and regulates time (Park 28e, Ch 22, p 1021; Sunder Lal, Ch 4, p 147)
Recorder / rapporteur	Records the issues discussed and the agreements reached, and prepares the group report (Park 28e, Ch 22, pp 1021-1022; Mahajan & Gupta 4e, Ch 29, p 580)
Rules for members	Express ideas clearly and concisely; listen to what others say; do not interrupt when others are speaking; make only relevant remarks; accept criticism gracefully; help to reach conclusions (Park 28e, Ch 22, p 1022)
Prerequisites for success	Members can discuss freely, know one another beforehand, and have adequate resource material — printed matter, posters, demonstration material (Park 28e, Ch 22, p 1022; Sunder Lal, Ch 4, p 147)
Venue and hospitality	<i>Anganwadi</i> centre, <i>Mahila Mandal</i> premises or subcentre, easily accessible; refreshment optional and offered by the community, not from government funds (Sunder Lal, Ch 4, p 147)



Steps in conducting a community group discussion (mothers' meeting / Mahila Swasthya Sangh)

Group dynamics: member roles and their measurement

Every group has leaders, who are responsible for the solidarity of group behaviour and the morale of its people; groups have definite programmes and objectives, and their members think, feel and act together, so many community problems can be solved by group effort approached through group discussion (Park 28e, Ch 12, p 823).

Role played in the group	Behaviour	Action by the facilitator
Dominator	Monopolises the floor, gives no chance to others	Discourage and control through the group leader (Sunder Lal, Ch 4, p 147)

Role played in the group	Behaviour	Action by the facilitator
Silent / shy member	Does not participate at all	Encourage to speak; reassure that nothing is "wrong or right" (Sunder Lal, Ch 4, p 147)
Obstructor (negative role)	Disturbs and blocks the proceedings	Discourage the practice (Sunder Lal, Ch 4, p 147)
Positive / harmoniser	Gives information, seeks more information, promotes harmony	Reinforce; use as the natural group leader (Sunder Lal, Ch 4, p 147)
Hostile member	Antagonistic to the group or the topic	Handle without confrontation; keep discussion on the felt need
Side-talker	Engaged in mutual private discussion	Leader prevents side-conversations (Park 28e, Ch 22, p 1021)

Sociogram — arrows are drawn for every interaction (who spoke to whom, how many times); when the arrows are connected the resulting picture is a sociogram, and it distinguishes a well-conducted discussion (arrows radiating between all members) from a dominated one (Sunder Lal, Ch 4, pp 147-148; Mahajan & Gupta 4e, Ch 29, Fig 29.2, p 581).

Limitations of group discussion

- **Shy members do not take part**, so participation is unequal unless properly guided (Park 28e, Ch 22, p 1022).
- **Domination** by one or two members distorts the group decision (Park 28e, Ch 22, Fig 5, p 1022).
- Members may **deviate from the subject**, making the discussion irrelevant or unprofitable (Park 28e, Ch 22, p 1022).
- **"Group think"** discourages individuals from expressing opinions at odds with the majority; a designated devil's advocate helps overcome it (Oxford Textbook of Public Health 7e, Ch 31, pp 1151-1152).
- Social pressure of small-group interaction shapes what participants say, so the method is exploratory and not a substitute for a survey (Oxford Textbook of Public Health 7e, Ch 44, p 1636).
- Recording is difficult and analysis cumbersome — every statement, and even non-verbal communication, has to be captured (Mahajan & Gupta 4e, Ch 24, p 470).

Application in health education

Setting in India	Group and periodicity	Typical content
Village-level women's group	<i>Mahila Swasthya Sangh</i> (10-20 women) / <i>Mahila Mandal</i> ; mothers' meeting weekly , MSS meeting monthly, run by the <i>Anganwadi</i> worker and health worker (Sunder Lal, Ch 4, pp 146-147)	Breastfeeding, anaemia in pregnancy, sex-selective abortion and the PNDT Act, child sex ratio

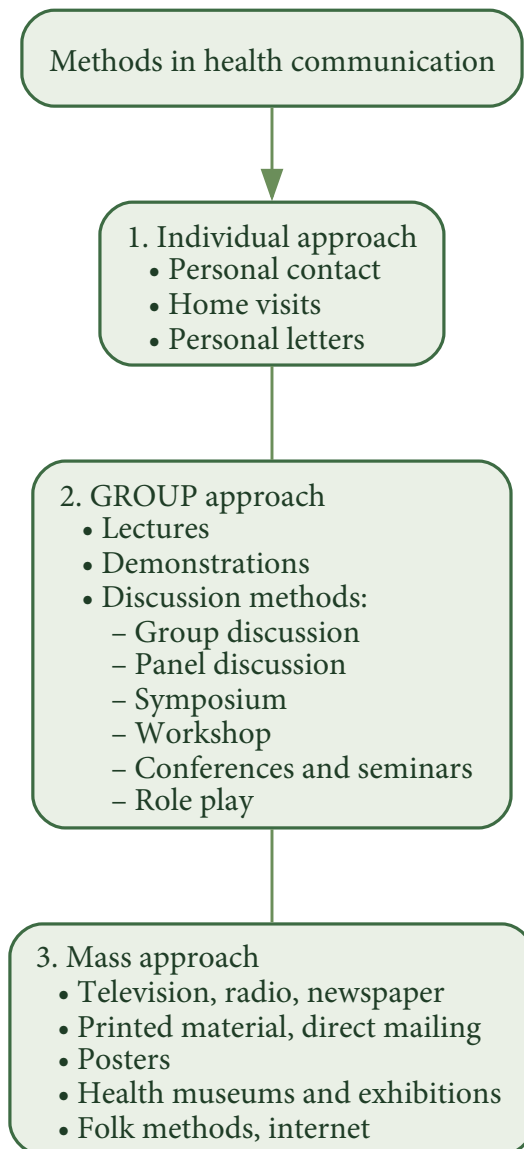
Setting in India	Group and periodicity	Typical content
Village committee	Village Health, Sanitation and Nutrition Committee (VHSNC) — meets monthly at the <i>Anganwadi</i> centre (Sunder Lal, Ch 4, p 147)	Village health plan, monitoring of ASHA, AWW and subcentre services
Urban slum	Mahila Arogya Samiti (MAS) — 10-12 members per 50-100 households (250-500 population), fortnightly/monthly meetings, annual grant Rs 5000 (Sunder Lal, Ch 17, p 758)	Peer education, community-based monitoring, referral linkage
Groups by life stage or exposure	School children, adolescent girls' and boys' groups, self-help groups, <i>Panchayats</i> , patients and industrial workers (Park 28e, Ch 22, p 1021; Sunder Lal, Ch 4, p 146)	Oral hygiene for school children, tuberculosis for patients, accidents for workers
Long-term behaviour change	Any small group with a shared risk	Smoking cessation and obesity reduction, where sustained compliance is needed (Park 28e, Ch 22, p 1021)
Formative research before an IEC campaign	Focus group discussion (FGD) — 6-8 homogeneous beneficiaries, a moderator plus a note-taker, session under 1-1½ hours (Sunder Lal, Ch 6, p 270; Mahajan & Gupta 4e, Ch 24, p 470)	Assess behaviour change communication (BCC) needs; social marketing of oral rehydration salts, contraceptives, immunisation

Why the FGD matters to the health educator — it captures "top of the mind" responses in the client's own words and tells us the "why" behind a problem, whereas a close-ended survey tells us only "what" the problem is (Sunder Lal, Ch 6, p 270). Its uses are to secure background information, generate project ideas, obtain programme feedback, develop the vocabulary for the education programme, frame questions for a formal schedule, and interpret available data (Mahajan & Gupta 4e, Ch 29, p 580). Members must be homogeneous for gender, age and social status — if men are present, women may not speak on sensitive topics (Mahajan & Gupta 4e, Ch 24, p 470).

Group discussion compared with the other group methods

Method	Participants	Distinguishing feature
Group discussion	6-12	Members themselves discuss; leader + recorder; consensus decision (Park 28e, Ch 22, p 1021)
Panel discussion	4-8 qualified panelists + chairman, before an audience	No agenda, no order of speaking, no set speeches; audience joins later; success depends on the chairman (Park 28e, Ch 22, p 1022)
Symposium	Several experts + chairman	A series of speeches; no discussion among the members ; audience questions and a chairman's summary at the end (Park 28e, Ch 22, p 1022)
Workshop	Small groups, each with a chairman and recorder	Four or more meetings with consultants; participants leave with a plan of action (Park 28e, Ch 22, p 1022)

Method	Participants	Distinguishing feature
Role play (socio-drama)	Group of about 25	Situation is dramatised, then discussed; especially useful for school children (Park 28e, Ch 22, pp 1022-1023)
Focus group discussion	6-8 homogeneous clients + moderator	Thematic, for formative research and BCC needs, not for teaching (Sunder Lal, Ch 6, p 270)



Place of group discussion among the methods of health communication (after Park 28e, Ch 22, Fig 3, p 1020)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Sociogram of a well-conducted group discussion** — draw a circle of 6-12 numbered members with the leader marked; arrows radiating from every member to every other member and to the facilitator. Labels: *group leader, recorder, members 1-8, interaction arrows, "all members participate"*.
- **Sociogram of a dominated group** — the same circle, but almost all arrows converging on and radiating from members No. 1 and No. 7, with the rest silent. Labels: *dominators (No. 1, No. 7), silent members, unequal participation*.

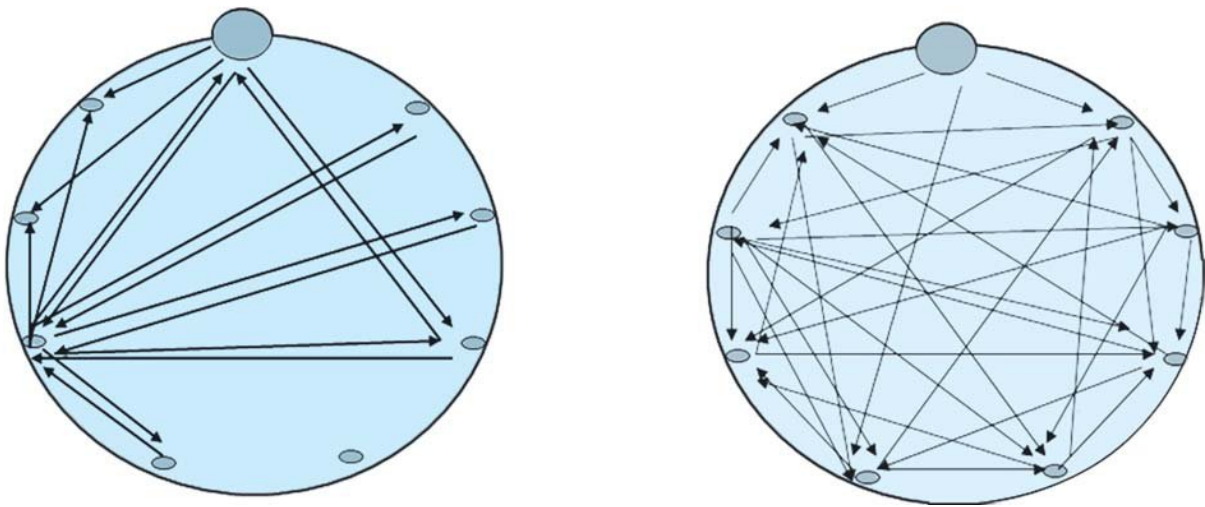


Fig 29.2 — Sociogram showing examples of Not a good discussion and Good discussion (Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e, p.581)

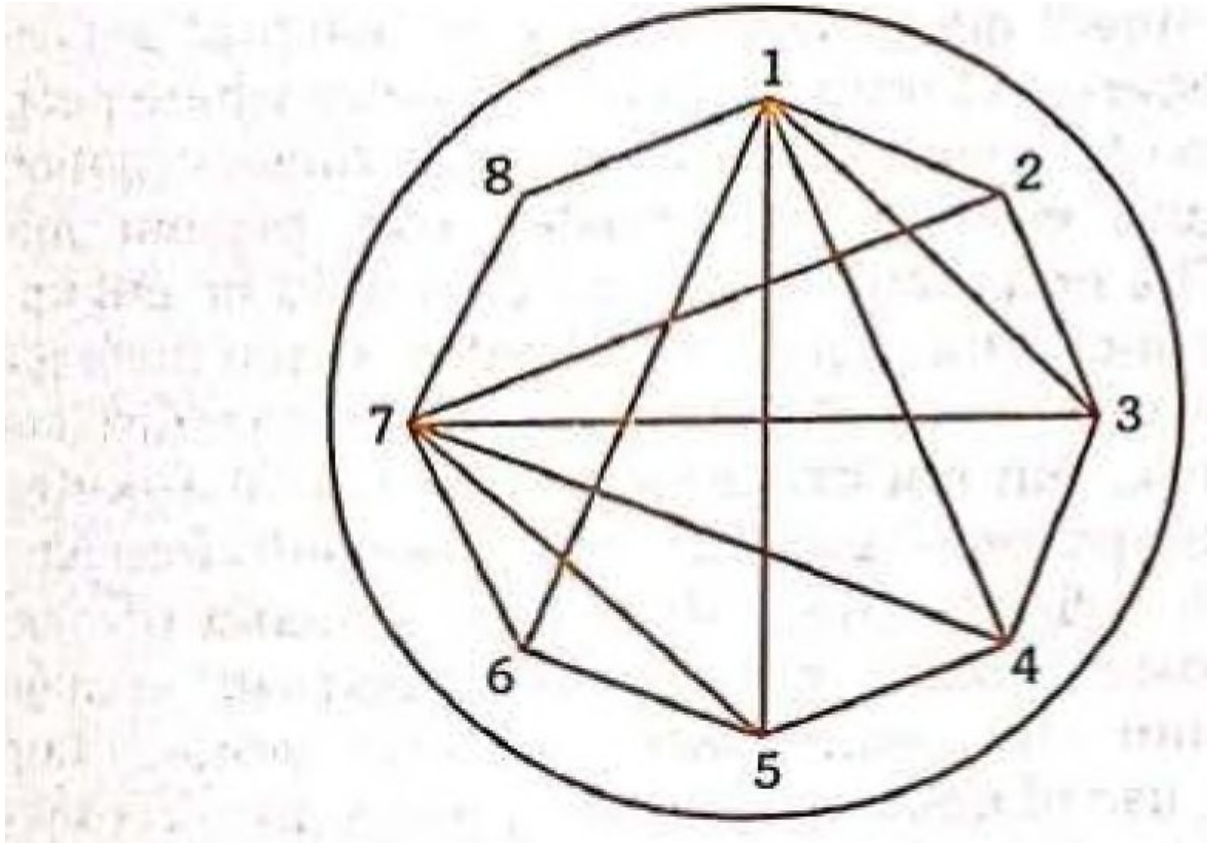


Fig 5 — , A dominated group. No.1 and No.7 dominate the discussion (Park's Textbook of Preventive and Social Medicine, 28e, p.1022)

High-yield exam pointers

- ▶ **Group** = "aggregation of people interacting in a face-to-face situation" (Park's own wording) — quote it as the opening line.
- ▶ **Numbers to write:** group discussion 6-12; panel discussion 4-8; role play about 25; FGD 6-8 participants for 1-1½ hours; MSS 10-20 women; MAS 10-12 members per 50-100 households.
- ▶ **Group dynamics** = interaction among members and with the facilitator, drawn as arrows → **sociogram**. One line, one figure, full marks.
- ▶ **Six member roles:** dominator, silent, obstructer, positive/harmoniser, hostile, side-talker — with the facilitator's counter-action for each.
- ▶ **Three functionaries:** group leader (initiates, guides, prevents side-conversations, sums up), recorder (issues and agreements), facilitator (listens, supplies information, links to services).
- ▶ **The single strongest sentence:** a group decision is adopted more readily than a solitary one and has a **binding effect** on the individual member — this is why group discussion changes attitudes and behaviour.
- ▶ FGD tells "why", a close-ended survey tells "what".

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only — Park describes the classical group-discussion technique while Sunder Lal describes how it is actually delivered in India through *Mahila Swasthya Sangh*, the Village Health, Sanitation and Nutrition Committee and *Mahila Arogya Samiti* under the National Health Mission; quote the Indian delivery platform after the classical technique, because the examiner expects both.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 22, Communication for Health Education, pp 1014-1023; Ch 12, Medicine and Social Sciences, pp 823-825; Ch 7, p 535); Sunder Lal — Textbook of Community Medicine (Ch 4, Principles of Health Promotion and Education, pp 146-148; Ch 6, p 270; Ch 17, p 758); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e (Ch 29, Information, Education, Communication and Training in Health, pp 580-585; Ch 24, p 470); Oxford Textbook of Public Health 7e (Ch 31, Qualitative research, pp 1151-1152; Ch 44, p 1636).



Relevance of anthropology in public health

Asked in: Paper I Nov 2018 — RGUHS MD Community Medicine

Definition — Anthropology (Greek *anthropos*, man + *logos*, science) is the study of the **physical, social and cultural history of man**, and of all the sciences that study man it comes nearest to being a **total study of man**; its branch of direct public-health interest, **medical anthropology**, "deals with the **cultural component in the ecology of health and disease**" (Park 28e, Ch 12, p 813). With sociology and social psychology it is classed as a **behavioural science** because it deals directly with human behaviour, and these should be regarded as basic sciences of medicine in the same sense as anatomy and physiology (Park 28e, Ch 12, pp 811-812).

Branches of anthropology and what each contributes to public health

Branch	What it studies	Public-health yield
Physical (biological) anthropology	Human evolution, racial differences, inheritance of bodily traits, growth and decay of the human organism; physical measurement of skull and skeleton and inherent racial characteristics (Park 28e, Ch 12, p 813; Sunder Lal, Ch 2, p 83)	The anthropometric method on which nutritional assessment and growth monitoring rest; distribution of inherited traits in endogamous groups
Social anthropology	Development and various types of social life; institutions, organisations, social groups and human relationships, past and present (Park 28e, Ch 12, p 813; Sunder Lal, Ch 2, p 83)	Family, caste, kinship and the village as units of health-service delivery; who in the household actually decides to seek care

Branch	What it studies	Public-health yield
Cultural anthropology	The total way of life of contemporary man — his ways of thinking, feeling and action — and cultural variation (Park 28e, Ch 12, p 813; Sunder Lal, Ch 2, p 83)	Beliefs about causation and cure, food habits, child rearing, hygiene — i.e. the raw material of every behaviour-change programme
Linguistic anthropology	Languages and their influence on social life (Sunder Lal, Ch 2, p 83)	Health messages in the local dialect and idiom; language variation is a listed cultural barrier to communication (Park 28e, p 1012)
Archaeological anthropology	Earlier human civilisations and societies from their physical remains (Sunder Lal, Ch 2, p 83)	Historical depth to disease patterns and to sanitary practice
Applied anthropology	Management and business, medical , forensic and education anthropology (Sunder Lal, Ch 2, p 83)	The operational arm — programme design, evaluation and community mobilisation

Domains in which anthropology is relevant to public health

- It answers the central question of community health — "Why do people behave as they do?" Social scientists in India are repeatedly asked to explain the failure of health measures, because the physician needs **two kinds of knowledge, medical and social** (Park 28e, Ch 12, pp 811-812).
- It explains **non-utilisation of services that are available and free**. Immunisation has not gained universal acceptance despite a wide range of vaccines; family planning is under-used; and programmes for mother and child health, water supply, sanitary latrines, dietary patterns and infant rearing "have all proved abortive or only partially successful" (Park 28e, Ch 12, p 811).
- **Community diagnosis and needs assessment**. Understanding a people's culture is essential to get a programme accepted — family planning acceptance requires thorough understanding of reproductive behaviour and of the community's knowledge, beliefs, attitudes and practices about reproduction (Sunder Lal, Ch 2, p 84).
- **Illness behaviour and the sick role**. It studies why people react differently to the **same** disease process or treatment regimen, and defines the **sick role** as a social position with its own rights and obligations, including the doctor-patient interaction (Park 28e, Ch 12, p 812; Oxford Textbook of Public Health 7e, Ch 44, p 1625).
- **Design of information, education and communication (IEC) and behaviour change communication (BCC)**. The listed **cultural barriers** are illiteracy, level of understanding, customs, beliefs, religion, attitudes, class differences, language variation and the gap between urban educators and a rural population; even where services are readily available these block behaviour change and must be identified and removed (Park 28e, p 1012).
- **Nutrition**. People classify foods as **hot, cold, heavy, light and windy**, and these classifications impose the restrictions that generate undernutrition in young children and in pregnant and lactating women; knowing them lets nutrition education work through acceptable foods and **effect the minimum change needed** (Sunder Lal, Ch 5, p 174).
- **Epidemiology**. Epidemiologists have forged a close alliance with social scientists, studying disease in relation to **social status, income, occupation, housing, overcrowding, social customs, habits and behaviour** — culture is a genuine exposure variable (Park 28e, Ch 12, p 812).

- **Reaching the unreached.** Tribal, migrant and slum populations are defined by a distinct way of life as much as by geography (Park 28e, Ch 13, pp 848-851).
- **It supplies the qualitative research toolkit** (methods table below).
- **It disciplines the worker against ethnocentrism.** **Ethnocentrism** is seeing things only from the viewpoint of one's own group; **cultural relativism** holds that a practice can be understood only in its cultural context (Sociology: The Essentials 6e, Ch 2, pp 65-66). Park's operational form of the same rule: **not all customs and beliefs are bad** — many rest on centuries of trial and error and have positive value (Park 28e, Ch 12, p 829).

Suitable examples — cultural factors in health and disease as observed in India

Domain	Belief or practice	Health consequence	Public-health response
Supernatural aetiology	Disease as the wrath of gods and goddesses — chickenpox as <i>Chhoti Mata</i> , where giving drugs is considered harmful, cases are not notified and <i>pujas</i> are performed instead (Park 28e, Ch 12, p 829)	Delayed treatment and under-notification , which corrupts surveillance data	Engage the community's own explanatory model before offering the biomedical one
Supernatural aetiology (contd.)	Breach of taboo (venereal disease), past sins (leprosy, tuberculosis), evil eye (children, warded off with charms, amulets and incantations by an exorcist), spirit or ghost intrusion (hysteria, epilepsy) (Park 28e, Ch 12, p 829)	Stigma, exorcist consulted first, long patient delay in leprosy and tuberculosis	Work with traditional and faith healers rather than against them
Environmental sanitation	Open-field defecation regarded as time-honoured and harmless — "latrines are meant for city dwellers"; waste water let into the street; refuse and cow dung heaped in front of houses (Park 28e, Ch 12, p 829)	Faeco-oral disease, fly and mosquito breeding	Sanitation drives must attack the belief , not only supply the pan and trap
Water supply	The well as the village meeting place, used for bathing, washing clothes and watering animals; tanks and ponds used for washing and drinking; "holy" rivers where people bathe and drink the raw water , and bottled "holy water" carried long distances to relatives (Park 28e, Ch 12, p 829)	Epidemics of cholera and gastroenteritis traced to these practices; step-wells and guinea-worm disease	Source protection plus culturally framed messaging on ritual and drinking water
Housing	<i>Kutcha</i> , damp, ill-lit, ill-ventilated houses with no windows for reasons of security ; no separate kitchen, latrine, bathroom or drainage; humans and animals under one roof (Park 28e, Ch 12, p 829)	Indoor air pollution, tuberculosis transmission, zoonoses	Housing standards that respect the security concern that removed the window

Domain	Belief or practice	Health consequence	Public-health response
Food habits	Vegetarianism honoured; pork abhorred by Muslims and beef by Hindus; the hot/cold classification (meat, fish, eggs, jaggery = heat-generating; curd, milk, vegetables, lemon = cooling); fasts; adulteration of milk partly sustained by the belief that boiling pure milk dries up the donor animal's secretion; men eat first and women last (Park 28e, Ch 12, p 830)	Restriction of protein-rich foods in illness and pregnancy; intra-family maldistribution driving anaemia and undernutrition in women and girls	Nutrition education phrased inside the hot/cold idiom
Mother and child health	Customs classified good / bad / unimportant / uncertain : <i>good</i> — prolonged breast-feeding, oil bath, massage, exposure to sun; <i>bad</i> — foods forbidden in pregnancy (eggs, meat, fish, milk, leafy vegetables), delivery by the untrained <i>dai</i> , branding of the skin, opium and drastic purgatives; <i>uncertain</i> — <i>kajal</i> on the eyelids, blamed for transmitting trachoma and other eye infections (Park 28e, Ch 12, p 830)	Maternal and neonatal morbidity; ocular infection	Promote the good, replace the bad, study the uncertain — this is social paediatrics
Personal hygiene	<i>Pan</i> with lime, with or without tobacco; smoking with the burning end of the cigar inside the mouth , a custom among villagers in Andhra Pradesh; hubble-bubble (shared <i>hookah</i>) smoking; the barber (<i>nai</i>) who does not sterilise his instruments; <i>purdah</i> ; sleeping on the ground; going barefoot (Park 28e, Ch 12, p 830)	Oral cancer; spread of tuberculosis by shared <i>hookah</i> ; blood-borne infection; tuberculosis reported to be higher among women observing <i>purdah</i> , who are also deprived of sunlight; insect bites; hookworm	Occupation- and custom-specific risk messages rather than generic advice
Sex and marriage	Menstruation regarded as a time of uncleanness when women are forbidden to pray or have intercourse; universality and earliness of marriage; the family considered incomplete without a male child (Park 28e, Ch 12, pp 830-831)	Bearing on contraceptive acceptance, on the population problem and on the sex ratio	Family planning counselling built on the community's reproductive norms
Caste and social class	Indian society as a rigid " closed-class " system with little social mobility; infant mortality differs by religion and caste through age-old habits affecting cleanliness, eating, clothing and child care (Park 28e, Ch 12, pp 831-832; Ch 9, p 674)	Reform meets resistance; health inequity tracks caste and class	Identify and work through existing social hierarchies and opinion leaders

Anthropological methods used in public health practice

Method	What is done	Public-health use
Ethnography	Immersion in the study context to observe everyday activities in their natural setting, producing a " thick description " and a holistic understanding of a group's way of life (Oxford Textbook of Public Health 7e, Ch 31, p 1148)	Explains a community's health culture in depth before a programme is designed
Participant observation	The researcher is personally involved in the action being observed (Abramson, Ch 15, p 145)	Observing how mothers actually prepare infant food or oral rehydration solution
Key-informant interviews and conversations	Members of the community express attitudes, perceptions, motivations, feelings and behaviour (Abramson, Ch 15, p 145)	Rapid appraisal of why a service is not used
Focus group discussion	A small group talks freely and spontaneously on a theme set by the investigator (Abramson, Ch 15, p 145); thematic discussions among beneficiaries about, for example, diarrhoea (Sunder Lal, Ch 6, p 269)	Perceptions and practices around a single illness or service
Transect / group walk	Walking through the village or slum to observe access, schools, <i>anganwadis</i> , the health facility, garbage and refuse disposal and waste-water management (Sunder Lal, Ch 6, p 269)	Spatial data for community diagnosis
Rapid ethnographic assessment / rapid assessment procedures	Informal interviews, key informants, focused small-group discussion, the nominal group technique and unstructured observation — "one does not need an advanced degree in anthropology" (Abramson, Ch 31, pp 310-311)	Programme-cycle speed; usable in outbreak and disaster settings
Triangulation	Use of more than one qualitative method, and independent analysis by more than one investigator, to confirm validity (Abramson, Ch 15, p 145; Ch 31, p 311)	Rigour, so that qualitative findings withstand scrutiny
Sequenced with quantitative work	In a study of incomplete immunisation in Haiti , qualitative methods identified the barriers to use of preventive services and these were then measured in a quantitative survey (Abramson, Ch 15, p 145)	The standard mixed-methods design for coverage evaluation

Worked example — tribal health in India

Over 104 million tribal people live in India across 705 tribes, 8.6 per cent of the population, marginal geographically, socio-economically and politically (Park 28e, Ch 13, p 848).

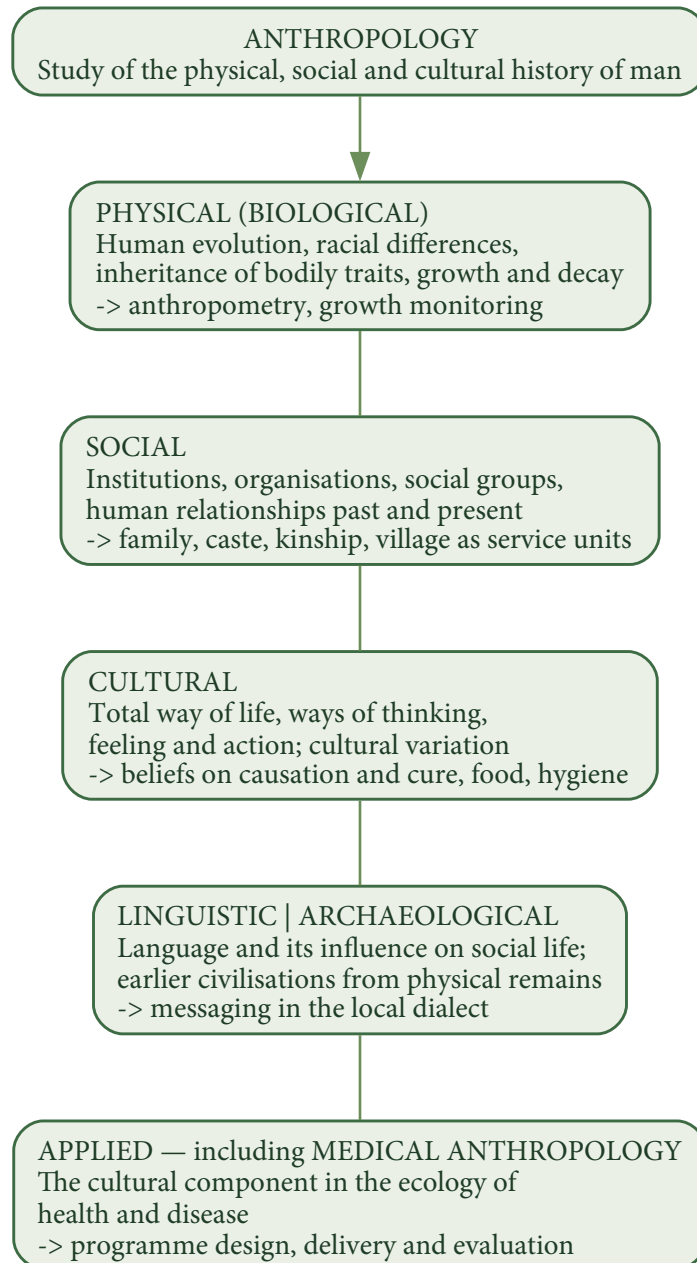
Indicator	Scheduled Tribe	All India	Source
Institutional delivery	68.0%	88.6%	NFHS-5, 2019-21 (Park 28e, Ch 13, p 848)
Under-five children stunted	40.9%	35.5%	NFHS-5, 2019-21 (Park 28e, Ch 13, p 848)

Indicator	Scheduled Tribe	All India	Source
Estimated prevalence of pulmonary tuberculosis (per 100,000)	703	256	Park 28e, Ch 13, p 849
Literacy (Census 2011)	59.0%	73.0%	Park 28e, Ch 13, p 848

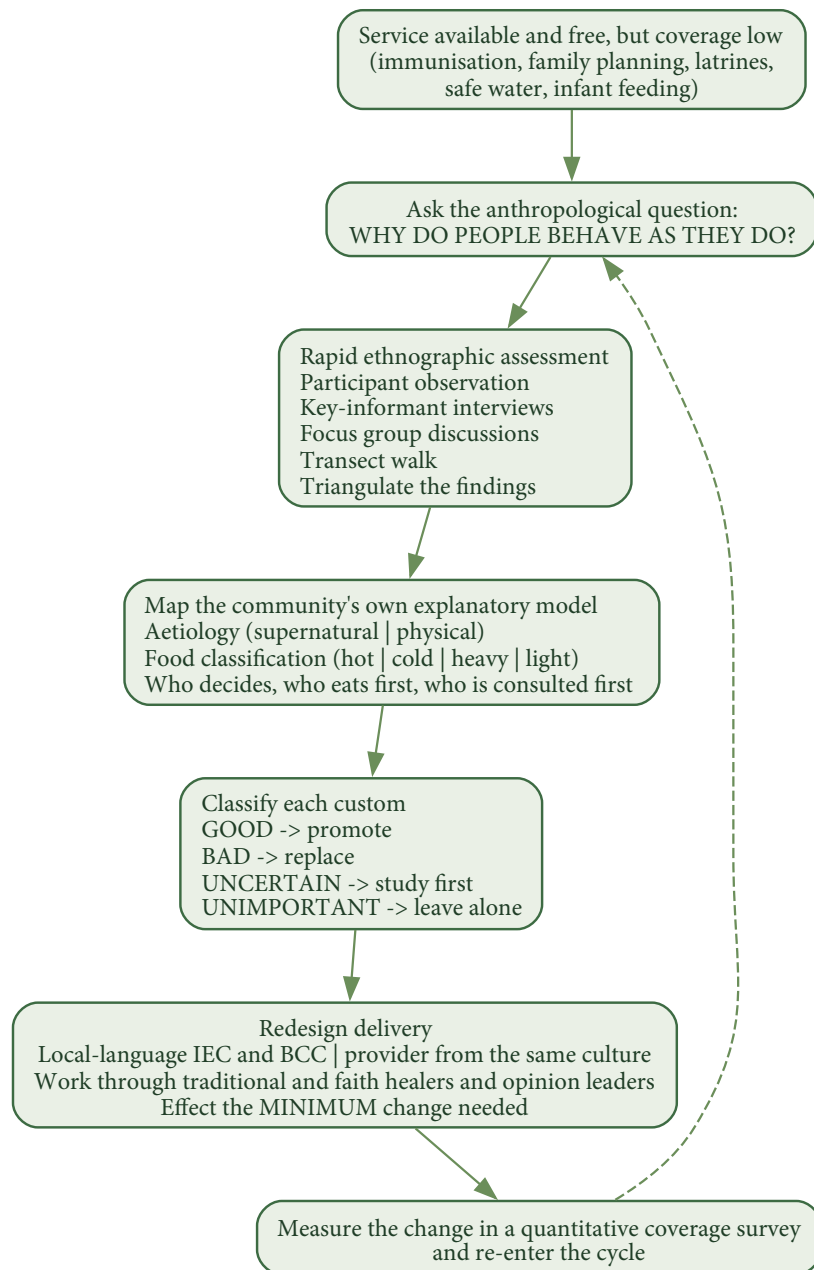
- **Relaxed service norms** — in tribal and hilly areas, one health sub-centre per 3,000 population, one primary health centre per 20,000 and one community health centre per 80,000 (Park 28e, Ch 13, p 850).
- **The provider should be of the culture** — the Expert Committee on Tribal Health (Ministry of Health and Family Welfare with the Ministry of Tribal Affairs) recommends that the provider be, as far as possible, a **local tribal**, trained and deployed locally, with the centre of gravity of the workforce **closer to the communities** rather than at the top (Park 28e, Ch 13, p 851).
- **Programmes co-opt the existing cultural structure** — the malaria strategy provides **hamlet-wise** rather than village-wise accredited social health activists (ASHAs), engages *anganwadi* workers, **faith healers** and local practitioners where an ASHA is not possible, and mobilises the community by **utilising traditional tools and practices** (Park 28e, Ch 7, pp 498-499).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Diagram 1 — social factors influencing the health of people** (after Park 28e, Ch 12, Fig 1, p 812). Draw a central circle labelled **HEALTH**. Feed four boxes into it with arrows: *Individual predispositions* (inheritance, perinatal events, previous illness, upbringing, childhood nutrition); *Ecological predispositions*; *Current circumstances* (income, family life, housing, employment, stresses and crises, social support); *Opportunities and situations* (access to health services, exposure to pathogens, availability of education, food, sanitation, alcohol, drugs, marriage partners). Above all four place one wide box labelled **SOCIAL / STRUCTURAL FACTORS** listing *demographic, economic, political, legal, religious, educational, technological*, with arrows down into the four boxes; add a **feedback arrow** from HEALTH back up to the social/structural box.



Branches of anthropology and the public-health question each answers



Feeding anthropological understanding into a health programme that the community is not using

High-yield exam pointers

- ▶ **Definition to write first:** medical anthropology deals with the **cultural component in the ecology of health and disease** (Park 28e, Ch 12, p 813); anthropology, sociology and social psychology are the three **behavioural sciences**.
- ▶ **Six divisions:** biological/physical, social, cultural, archaeological, linguistic, applied — the last contains medical anthropology (Sunder Lal, Ch 2, p 83).
- ▶ **The one-line justification:** the central question of community health is "Why do people behave as they do?" (Park 28e, Ch 12, p 811).
- ▶ **Customs are classified good / bad / unimportant / uncertain** — *kajal* on the eyelids (blamed for trachoma) is the classic **uncertain** custom; prolonged breast-feeding, oil bath, massage and sun exposure are the **good** ones (Park 28e, Ch 12, p 830).

- ▶ **Two examples that always earn marks:** chickenpox as *Chhoti Mata*, where drugs are believed harmful and **cases are not notified**; and shared **hubble-bubble** smoking spreading tuberculosis (Park 28e, Ch 12, pp 829-830).
- ▶ **Dated tribal figures:** 104 million tribal people, 705 tribes, 8.6% of the population; pulmonary tuberculosis **703 vs 256 per 100,000**; institutional delivery **68.0% vs 88.6%** (NFHS-5, 2019-21) (Park 28e, Ch 13, pp 848-849).
- ▶ **Closing principles:** avoid **ethnocentrism**, apply **cultural relativism**, effect the **minimum change** needed (Sociology: The Essentials 6e, Ch 2, pp 65-66; Sunder Lal, Ch 5, p 174).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park has no chapter headed "anthropology in public health"; this answer is assembled from its sections on the behavioural sciences and on cultural factors in health and disease (Ch 12), its tribal health chapter (Ch 13) and its cultural barriers to communication, with Sunder Lal for the six divisions and Abramson and the Oxford textbook for the qualitative methods. Park's illustrative customs are long-standing descriptions carried across editions and are undated, so quote them as illustrations, not as current survey findings.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 12 Medicine and Social Sciences, pp 811-832; Ch 13 Tribal Health in India, pp 848-851; Ch 9, p 674; Ch 7, pp 498-499; barriers of communication, p 1012); Sunder Lal — Textbook of Community Medicine (Ch 2, pp 83-84; Ch 5, p 174; Ch 6, p 269); Oxford Textbook of Public Health 7e (Ch 31 Qualitative research imagination, pp 1147-1148; Ch 44 Sociology and psychology in public health, pp 1625-1627); Abramson — Research Methods in Community Medicine (Ch 15, p 145; Ch 31, pp 310-311); Sociology: The Essentials 6e (Ch 2 Culture and the Media, pp 65-66). Indian figures from the National Family Health Survey-5 (2019-21) and Census 2011 as tabulated in Park 28e.



Louis Pasteur's contribution to public health

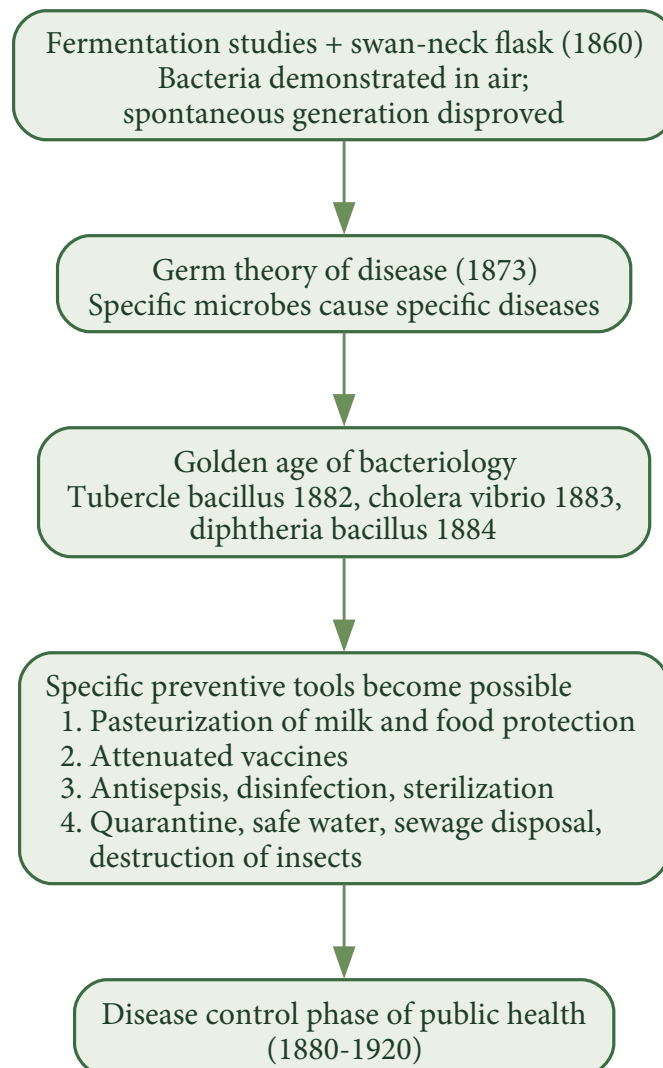
Asked in: Paper I 26-May-2025 — RGUHS MD Community Medicine

Definition — Louis Pasteur (1822–1895) was a French chemist-turned-bacteriologist whose laboratory work displaced the miasmatic and spontaneous-generation theories and established that specific living microbes cause specific diseases (Park 28e, Ch 1, p 18). His contribution to public health is not a single discovery but a chain: a new theory of causation, a new class of preventive tools (pasteurization, attenuated vaccines, antiseptics), and a new institution (the Pasteur Institute). Together these opened the **disease control phase of public health (1880–1920)** (Park 28e, Ch 1, pp 17, 20–21).

1. Foundation — disproving spontaneous generation and establishing the germ theory

Year	Pasteur's work	Public health consequence
Before 1860	Studies on fermentation — showed fermentation occurs only when microbes are present; the right microbe gives wine, the wrong one gives sour milk or bitter wine (Park Plus, Ch 1, p 8)	Founded the science of microbiology ; disease could now be thought of as a biological, not a chemical or supernatural, event
1860	Swan-neck flask experiment — heated infusions in flasks with long curved necks stayed sterile until exposed to dust; demonstrated bacteria are present in air and disproved spontaneous generation (Park 28e, Ch 1, p 18; Park Plus, Ch 1, p 8)	Microbes come from pre-existing microbes → contamination can be prevented , i.e. prevention became scientifically possible
1873	Advanced the “ germ theory of disease ” (Park 28e, Ch 1, p 18)	Replaced the supernatural theory, theory of humors, contagion theory, miasmatic theory and the theory of spontaneous generation
1877 (Koch, on Pasteur's foundation)	Robert Koch (1843–1910) showed anthrax was caused by a bacterium; the discoveries of Pasteur and Koch together confirmed the germ theory (Park 28e, Ch 1, p 18)	Opened the “ golden age of bacteriology ” — gonococcus 1847; typhoid bacillus and pneumococcus 1880; tubercle bacillus 1882; cholera vibrio 1883; diphtheria bacillus 1884

Two consequences follow for the discipline. First, **aetiology became specific**: attention shifted from “bad air” to a named agent, and medicine “shed the rags of dogma and superstition” (Park 28e, Ch 1, p 18). Second, **control became specific**: once the agent and its route were known, targeted measures — quarantine, safe water, pasteurization of milk, protection of foods, proper disposal of sewage, destruction of insects and disinfection — could be deployed instead of undifferentiated cleansing (Park 28e, Ch 1, p 18).



How Pasteur's laboratory findings became routine public health practice

2. Pasteurization — the single most widely applied Pasteurian public health measure

Pasteurization is the heating of milk to such temperatures and for such periods of time as are required to destroy any pathogens that may be present, while causing minimal changes in composition, flavour and nutritive value (WHO, 1970) (Park 28e, Ch 11, p 797). It began as Pasteur's solution to spoilage of wine — heating enough to kill most microbes without altering flavour — and was later applied to milk (Park Plus, Ch 1, p 8).

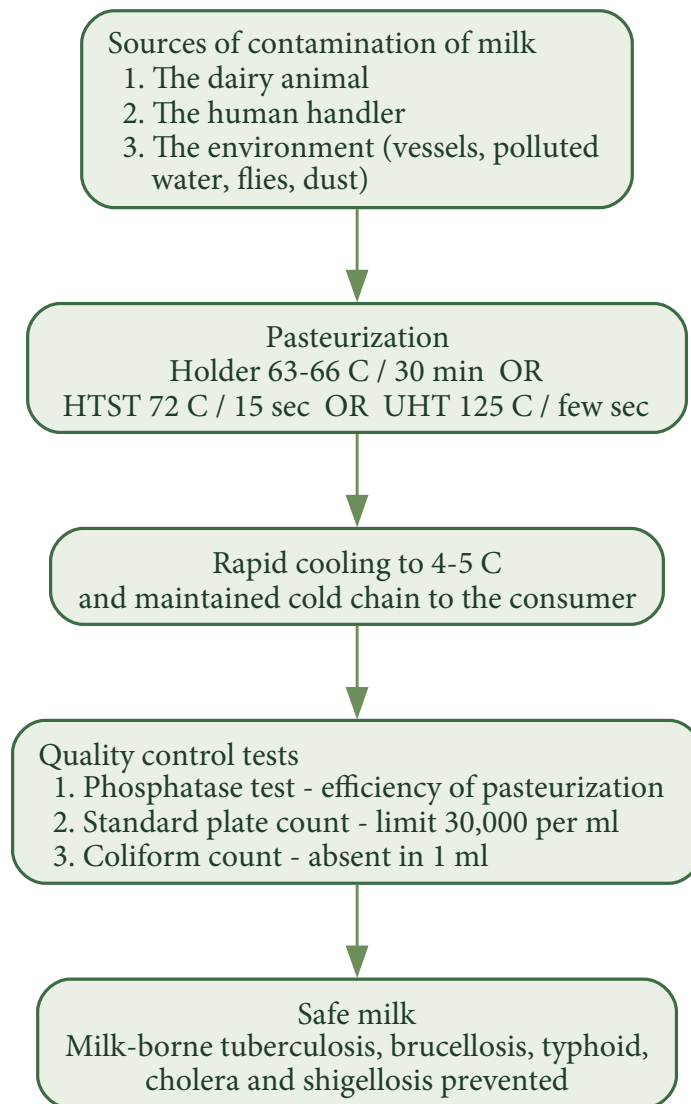
Method	Time-temperature	Cooling	Where used
Holder (Vat) method	63–66°C for at least 30 minutes	Quickly to 5°C	Recommended for small and rural communities; going out of use in larger cities
HTST (High Temperature Short Time)	Nearly 72°C for not less than 15 seconds	Rapidly to 4°C	Most widely used; very large volumes per hour

Method	Time–temperature	Cooling	Where used
UHT (Ultra-High Temperature)	125°C for a few seconds, usually in 2 stages	Rapid cooling, bottled at once	Long-life milk
Hot-water pasteurization (disinfection of articles)	70°C for 30 minutes	—	Not a sterilization process; destroys pathogens, not spores (Park 28e, Ch 3, p 156)

Public health point	Fact
Efficacy	Kills nearly 90% of bacteria in milk, including the more heat-resistant tubercle bacillus and Q fever organisms
Limitation	Does not kill thermoduric bacteria or bacterial spores; hence pasteurized milk must be held cold (4°C) until it reaches the consumer
Keeping quality	Hygienically produced pasteurized milk keeps not more than 8–12 hours at 18°C
Diseases prevented (FAO/WHO 1970 classification)	Animal infections transmissible to man — tuberculosis, brucellosis, streptococcal infection (primary importance); infections primary to man spread through milk — typhoid and paratyphoid, shigellosis, cholera, enteropathogenic <i>E. coli</i>, diphtheria, enteroviruses, viral hepatitis
Phosphatase test	Checks efficiency of pasteurization — raw milk phosphatase is completely destroyed at 60°C for 30 minutes; a positive test means inadequate pasteurization or addition of raw milk
Standard plate count	Bacteriological quality; most Western countries enforce a limit of 30,000 organisms per ml
Coliform count	Coliforms are completely destroyed by pasteurization; standard is absent in 1 ml — presence means improper pasteurization or post-pasteurization contamination

(Park 28e, Ch 11, pp 797–798)

Pasteurization is the classic **automatic-protective strategy** in public health — population-wide protection that does not depend on an individual's daily decision, alongside water fluoridation and infant immunization (Oxford Textbook of Public Health 7e, Ch 15, p 609).



The safe-milk pathway that pasteurization created

3. Vaccination — the principle of attenuation

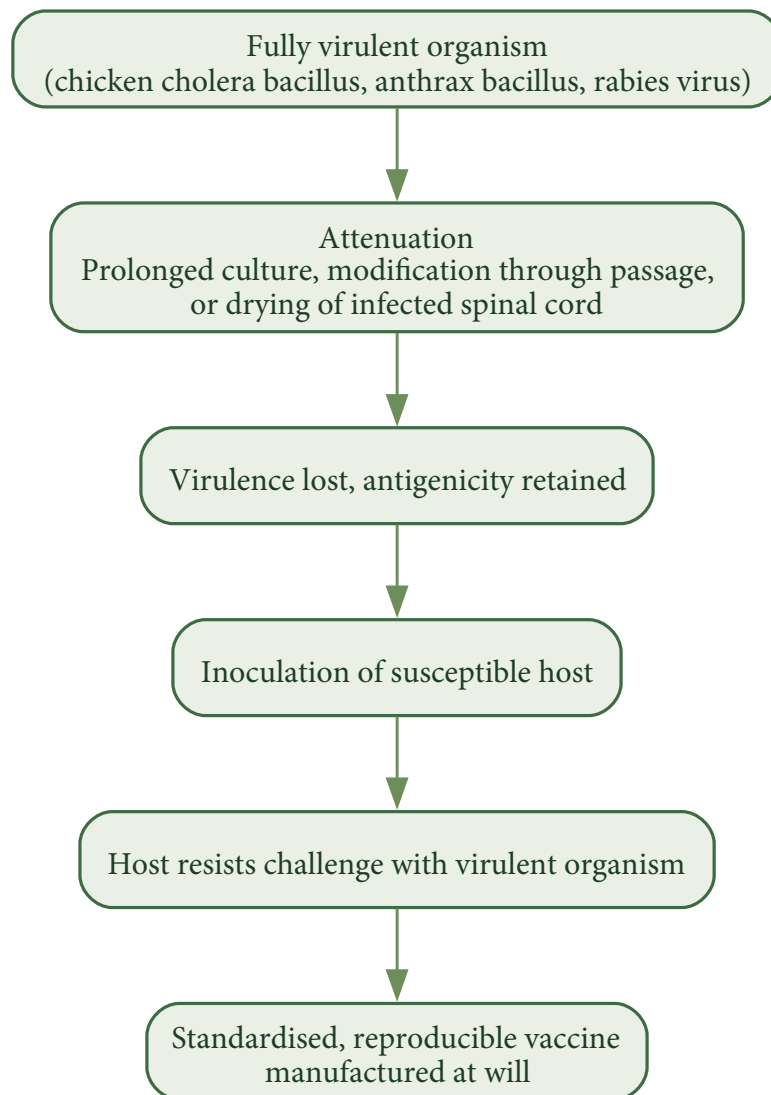
Pasteur's work on attenuation of the chicken cholera bacterium in the late 1870s was **the first major advance after Jenner's vaccination** (Plotkin's Vaccines 7e, Ch 1, p 21). Its importance is methodological: it proved that vaccines could be **created deliberately, standardised and reproduced at will**, replacing person-to-person or animal-to-animal inoculation with something safer and more consistent (Plotkin 7e, Ch 1, pp 21–23).

Vaccine	Basis	Public health significance
Chicken cholera (<i>Pasteurella multocida</i>), from 1878–79	Attenuation by modification through culture and passage	First demonstration that virulence could be reduced at will while immunogenicity was retained
Anthrax, early 1880s	Attenuated live bacterial vaccine	Pasteur's first effective live bacterial vaccine (Plotkin 7e, Ch 11, p 210); a generalisable, controlled experiment, unlike Jennerian vaccination

Vaccine	Basis	Public health significance
Rabies, first human use 1885	Serial suspensions of spinal cord from rabid rabbits, dried for varying periods; dogs inoculated from avirulent (long-dried) towards virulent cords resisted intracerebral challenge; at least 50 dogs protected	Listed as the 19th-century live attenuated human vaccine (rabies, 1885) in the outline of human vaccine development (Plotkin 7e, Ch 1, pp 25–26)

The rabies work is Pasteur’s specific gift to preventive practice — and its public health logic is worth stating exactly:

- ▶ Earlier vaccination required the vaccine to be given **before** exposure.
- ▶ Rabies has a long delay of several weeks between the bite and the germ reaching the brain (Park Plus, Ch 1, pp 8–9).
- ▶ That delay makes vaccination **after** the bite effective — the first true **post-exposure prophylaxis**.
- ▶ Consequently only those actually bitten by a rabid animal need treatment, which is the operating principle of every animal-bite management clinic today.
- ▶ **Joseph Meister**, a 9-year-old boy bitten 14 times and presenting some 60 hours later, received subcutaneous inoculation on **6 July 1885**, with 12 further inoculations of cords of increasing virulence — 13 doses over about 10 days — and survived (Plotkin 7e, Ch 1, p 23; Ch 50, p 1336).
- ▶ For **more than 70 years** after Pasteur’s original work, only **nerve tissue vaccines** were available; the Semple and Fermi phenol-inactivated modifications derive directly from his method (Plotkin 7e, Ch 50, pp 1344–1345).



Pasteur's principle of attenuation - the basis of every live attenuated vaccine

4. Antisepsis, disinfection and the laboratory era

- ▶ **Antiseptic surgery** — Lister (1827–1912), applying Pasteur's germ theory, reduced wound infections by prophylactic application of an antiseptic (**carbolic acid**) to wounds; Semmelweis had earlier shown the value of handwashing with antiseptic solutions against puerperal fever (Park 28e, Ch 3, pp 152–153). The Pasteurian era gave these empirical successes their explanation and generalised them.
- ▶ **Disinfection and sterilization entered routine practice** — the era from 1827 to 1912 is dated in Park's list of preventive medicine advances as that of "antiseptics and disinfectants" (Park 28e, Ch 1, p 18). Modern definitions of *disinfection*, *sterilization*, *antiseptic*, *asepsis* and *sanitizer* all rest on the germ theory (Park 28e, Ch 3, pp 152–153).
- ▶ **Development of laboratory methods** for the detection of disease was a further advance directly enabled by his work (Park 28e, Ch 1, p 18); public health acquired a laboratory arm, and interventions shifted toward eliminating bacteria — vaccination, pasteurization of milk and chlorination of municipal water (Oxford Textbook of Public Health 7e, Ch 58, pp 2155–2156).

- **Sequential specific vaccines and sera** followed his lead in the same generation — anti-rabies treatment, **cholera vaccine (1892)**, **diphtheria antitoxin (1894)** and **anti-typhoid vaccine (1898)** (Park 28e, Ch 1, p 18).

5. Institution building and international health

- The **Pasteur Institute** was founded in Paris in **1888**, funded largely by public donation raised for the anti-rabies vaccine, and flourished in both research and teaching (Park Plus, Ch 1, p 9; Plotkin 7e, Ch 50, p 1337; Textbook of International Health, Ch 2, p 84).
- Within a decade there were **Pasteur Institutes throughout the world providing rabies vaccination**, setting the precedent for organised vaccination services well into the 20th century (Plotkin 7e, Ch 50, p 1337).
- Beginning at **Saigon in 1891**, Pasteur Institutes were established across France's colonial empire in Africa, Asia and the Caribbean, and in Europe and the Middle East — the first true network of international public health laboratories. In these outlying laboratories **Yersin** worked on plague, **Laveran** on malaria and **Calmette** on BCG (Textbook of International Health, Ch 2, p 84).
- Scientific leadership in applying germ theory to identify the pathogens of communicable diseases occurred **first in France (the Institut Pasteur) and Germany (Koch)** — establishing the laboratory model of public health that other countries adopted (Oxford Textbook of Public Health 7e, Ch 2, p 194).

6. Critical appraisal — the germ theory's limitations

An examiner expects the contribution to be weighed, not only listed.

Strength	Corresponding limitation
Gave a specific, testable model of causation, ending centuries of empirical theories	Embodied a rigid one-to-one relationship between agent and disease — <i>Disease agent</i> → <i>Man</i> → <i>Disease</i> (Park 28e, Ch 2, p 51)
Made specific prevention (vaccination, pasteurization, disinfection) possible	Led many epidemiologists to a one-sided view of causation ; they “could not think beyond the germ theory”
Explained acute infectious disease well	Could not explain why not everyone exposed to tuberculosis or to beta-haemolytic streptococci develops disease — host and environmental factors matter equally, demanding the epidemiological triad of agent, host and environment
Dominated the bacteriological era (1850–1900), whose focus was bacteria or disease (Sunder Lal, Ch 2, pp 66–67)	Overshadowed Pettenkofer of Munich (1819–1901) , an early proponent of multifactorial causation, delaying that concept by decades (Park 28e, Ch 2, p 52)
Underpinned the disease control phase of public health	Chronic “diseases of civilization” — cancer, coronary heart disease, chronic bronchitis, mental illness — could not be explained by it nor prevented by isolation, immunization and sanitation, forcing the web of causation and risk factor concepts (Park 28e, Ch 1, pp 20–21; Ch 2, pp 52–53)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Pasteur's swan-neck flask experiment** — label: (1) flask containing boiled meat infusion; (2) long curved (swan) neck admitting air but trapping dust; (3) unopened flask — remains sterile; (4) flask opened on a dusty road and resealed — microbial growth; (5) conclusion — micro-organisms come from air-borne dust, not from spontaneous generation.
- ▶ **Epidemiological triad** — label the three vertices **Agent, Host, Environment**, with the caption “the broader model that had to replace the one-to-one germ theory” (Park 28e, Ch 2, pp 51–52).

High-yield exam pointers

- ▶ **Dates to write:** bacteria in air and spontaneous generation disproved **1860**; germ theory **1873**; Koch–anthrax **1877**; anthrax vaccine **1881**; first human rabies vaccination (**Joseph Meister**) **6 July 1885**; Pasteur Institute, Paris, **1888**.
- ▶ **Pasteurization temperatures (memorise verbatim):** Holder/Vat **63–66°C × 30 min**; HTST **72°C × 15 sec**; UHT **125°C × few sec**; hot-water pasteurization of articles **70°C × 30 min**.
- ▶ **Pasteurized-milk standards:** phosphatase destroyed at **60°C × 30 min**; standard plate count limit **30,000/ml**; coliforms **absent in 1 ml**; kills **~90%** of bacteria including tubercle bacillus and Q fever, not spores.
- ▶ **Rabies is the eponymous public health lesson:** long incubation → **post-exposure prophylaxis** is possible → treat only those bitten.
- ▶ **Germ theory model:** *Disease agent* → *Man* → *Disease*, a one-to-one relationship; its failure gave the **epidemiological triad**, then the **web of causation** and **risk factors**.
- ▶ **Placement in history:** Pasteur is the pivot of the **bacteriological/basic science era (1850–1900)** and the trigger of the **disease control phase (1880–1920)** of public health.
- ▶ **Pasteur established the method, Koch established the proof** — Pasteur was convinced microbes cause disease but never directly linked one microbe to one disease; Koch did (Park Plus, Ch 1, p 10).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park 28e lists “Pasteur's anti-rabies treatment (1883)” among late-19th-century preventive medicine advances, whereas Plotkin's Vaccines 7e dates the first human administration (Joseph Meister) to 6 July 1885 — quote 1885 for the first human use and 1882–83 for when the rabies work itself began, and either is acceptable if the context is stated.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 1 — Man and Medicine: Towards Health for All; Ch 2 — Concept of Health and Disease; Ch 3 — disinfection and sterilization; Ch 11 — milk hygiene and pasteurization); Plotkin's Vaccines, 7ed (Ch 1 — A Short History of Vaccination; Ch 11 — Anthrax Vaccines; Ch 50 — Rabies Vaccines); Oxford Textbook of Public Health 7e (Ch 2, Ch 15, Ch 58); Sunder Lal — Textbook of Community Medicine (Ch 2, Table 2.1 — historical eras of medicine as a social science); Textbook of International Health (Ch 2 — The Historical Origins of Modern International Health); Park Plus (Amrutha & Vijayalakshmi), Ch 1; FAO/WHO Joint Expert Committee on Milk Hygiene, 1970 (definition of pasteurization and classification of milk-borne diseases).



Triple burden of diseases (onion peel phenomenon)

Asked in: Paper II May 2019 — RGUHS MD Community Medicine

Definition — The **triple burden of disease** describes the **concurrent, often interacting** health burdens of (1) **communicable, maternal, neonatal and nutritional disease (CMNND)**, (2) a **rising burden of non-communicable disease (NCD)**, and (3) **substantial morbidity and mortality from injuries** — the characteristic profile of middle-income countries such as India (Oxford Textbook of Public Health 7e, Ch 1.4, p 238). The **onion peel phenomenon** is the metaphor for how these burdens succeed one another: **just as the layers of an onion are peeled off, old diseases wane and give place to new ones** — infectious diseases are replaced by non-infectious (non-communicable) diseases, and these in turn by personal and behavioural problems (Park Plus, Ch 24, p 397). Both terms are descriptions of the same underlying process, the **epidemiological (health) transition**; the triple burden is what results when the peeling is **incomplete** and the layers persist together.

The three components of the triple burden

Component	What it contains	Magnitude in India (with year)
1. The "unfinished agenda" — CMNND	Tuberculosis, malaria, diarrhoea, lower respiratory infection, HIV, neonatal disorders, maternal deaths, undernutrition and micronutrient deficiency	22.1% of all deaths in India, 2021 (Park 28e, Ch 6, p 422). Malaria: 0.17 million cases, 57.26% <i>P. falciparum</i> , 83 deaths in 2022 (Park 28e, Ch 10, p 723). 36% of under-fives stunted, 19% wasted, 32% underweight (NFHS-5, 2019–21)
2. The rising NCD burden	Cardiovascular disease, cancer, chronic respiratory disease, diabetes, mental illness	68% of all deaths in India, 2021 — cardiovascular disease 30%, chronic respiratory disease 14%, other NCDs 12%, cancers 9%, diabetes 3% (total deaths 1,30,46,300) (Park 28e, Ch 6, p 422). Probability of dying between ages 30 and 70 years from the four major NCDs = 24% (27% men, 20% women) (Park 28e, Ch 6, p 422)

Component	What it contains	Magnitude in India (with year)
3. Injuries	Road traffic injury, falls, drowning, burns, poisoning, self-harm, interpersonal violence	10% of all deaths in India, 2021 (Park 28e, Ch 6, p 422). Globally >4.6 million injury deaths in 2016 — road traffic injury >25%, self-harm 17.7%, falls 14.7% (Oxford 7e, Ch 9.4, pp 3056–3057)

The three layers **interact** rather than simply coexist: undernutrition in early life programmes adult diabetes and hypertension; diabetes worsens tuberculosis; antiretroviral therapy has made people with HIV live long enough to acquire NCDs; and injury and NCD share the same urbanisation, alcohol and tobacco pathways. This produces **multimorbidity** — the coexistence of two or more chronic conditions, whether physical NCD, mental disorder, or long-duration infection such as HIV or hepatitis C — for which vertically designed, single-disease programmes are poorly configured (Oxford 7e, Ch 1.4, Box 1.4.2, pp 238–239).

The epidemiological transition that produces the layers

The general shift from **acute infectious and deficiency diseases**, characteristic of underdevelopment, to **chronic non-communicable diseases**, characteristic of modernisation, is called the epidemiologic transition; it has two components — change in **population growth trajectory and age composition** (younger to older), and change in **patterns of mortality**, with rising life expectancy and a reordering of the leading causes of death (Park Plus, Ch 24, pp 395–396). *Omran* (1971) described three stages, and later (1998) three models of how fast a country moves through them.

Stage (Omran)	Mortality and life expectancy	Dominant causes
I. Age of pestilence and famine	High and fluctuating mortality; life expectancy at birth 20–40 years ; population growth not sustained	Epidemic infections (plague, cholera, smallpox), undernutrition, pregnancy-related illness, tuberculosis
II. Age of receding pandemics	Declining mortality; life expectancy rises from about 30 to 50 years and beyond; sustained population growth	Infections recede; shift begins from infectious to chronic non-infectious disease
III. Age of degenerative and man-made diseases	Low mortality; life expectancy >50 years ; morbidity overshadows mortality as the index of health	Cardiovascular disease, cancer, diabetes, injuries, behavioural and lifestyle disorders

Omran's three models are the **classical or Western model** (Western Europe, North America), the **accelerated model** (Japan, Eastern Europe) and the **delayed or contemporary model** (most developing countries, including India) (Oxford 7e, Ch 2.5, pp 803–804; Ch 8.8, p 3861).

Onion peel phenomenon — the two senses in which it is used

- As the sequence of transition (the usual exam sense). Like the layers of an onion, **old diseases wane off and give place to new ones**: infectious → non-infectious (non-communicable) → **personal and behavioural problems** such as substance abuse, violence, suicide, stress and lifestyle disorders (Park Plus, Ch 24, p 397). The metaphor stresses that the layers are **stacked, not separate** — removing one exposes the next.

- ▶ As an explanation for apparent temporal variation (descriptive epidemiology sense). When comparing disease occurrence across decades, an apparent rise may be artefactual; among the recognised reasons is the "onion peel phenomenon" — elimination or reduction of other conditions, so that competing causes are stripped away and the remaining condition surfaces and appears commoner (Sathe & Sathe, Ch 1, p 13). Other reasons in the same list are change in nomenclature, wider recognition and a higher index of suspicion, better availability of medical services and of diagnostic facilities, change in age or sex composition, and change in lifestyle.

Why India shows a triple and not a single burden: the outer layer is being peeled only partly and unevenly. The absolute burden from diarrhoea, lower respiratory infection, tuberculosis and neonatal disorders is falling but remains high, while the contribution of heart disease, stroke and diabetes is rising, and injuries are rising with motorisation — so India's health system faces a dual (indeed triple) challenge simultaneously (Park 28e, Ch 6, p 422). The COVID-19 pandemic was a reminder that infectious disease can emerge or re-emerge to threaten everyone, adding a fourth strand of emerging and re-emerging infection to the same stack (Park 28e, Ch 5, p 205).

Mechanisms driving the transition

- ▶ Demographic changes — falling mortality and fertility, ageing of the population.
- ▶ Biological factors — alteration in antigenic identity, emergence of drug-resistant strains, dual infection.
- ▶ Environmental factors — improved sanitation, safe water, sanitary waste disposal and better housing reducing communicable disease.
- ▶ Ionizing radiation — cancer.
- ▶ Insecticides and use of vehicles — air and water pollution.
- ▶ Overcrowding — rural-to-urban migration favouring droplet infection.
- ▶ Social, cultural and behavioural factors — changed community relationships and lifestyle.
- ▶ Practice of modern medicine — adverse effects of drugs and iatrogenic disease (Park Plus, Ch 24, p 396).

To these, unplanned urbanisation adds both faces of the burden at once: slum-dwellers carry tuberculosis, acute respiratory infection, diarrhoea and vector-borne disease and child undernutrition, while the same urbanisation drives dietary change, tobacco and alcohol use, physical inactivity, obesity, hypertension, air pollution, mental ill-health, violence and injury (Oxford 7e, Ch 1.4, pp 236–238).

Measuring the transition in India — the epidemiological transition ratio

Item	Content
Definition	Epidemiological transition ratio (ETR) = disability-adjusted life years (DALYs) caused by CMNND ÷ DALYs caused by NCDs + injuries
Interpretation	Ratio >1 = higher burden of CMNND; <1 = NCDs and injuries predominate. The lower the ratio, the further advanced the state
India, 1990 vs 2016	Most states had a ratio >1 in 1990; all states had a ratio <1 in 2016 — a major shift in the drivers of health loss

Item	Content
Range across states, 2016	Kerala 0.16 (furthest along) to Bihar 0.74 (double burden most acute)
Epidemiological transition level (ETL), 2016	0.56–0.75 = lowest ETL; 0.41–0.55 = lower-middle ; 0.31–0.40 = higher-middle ; ≤ 0.30 = highest ETL

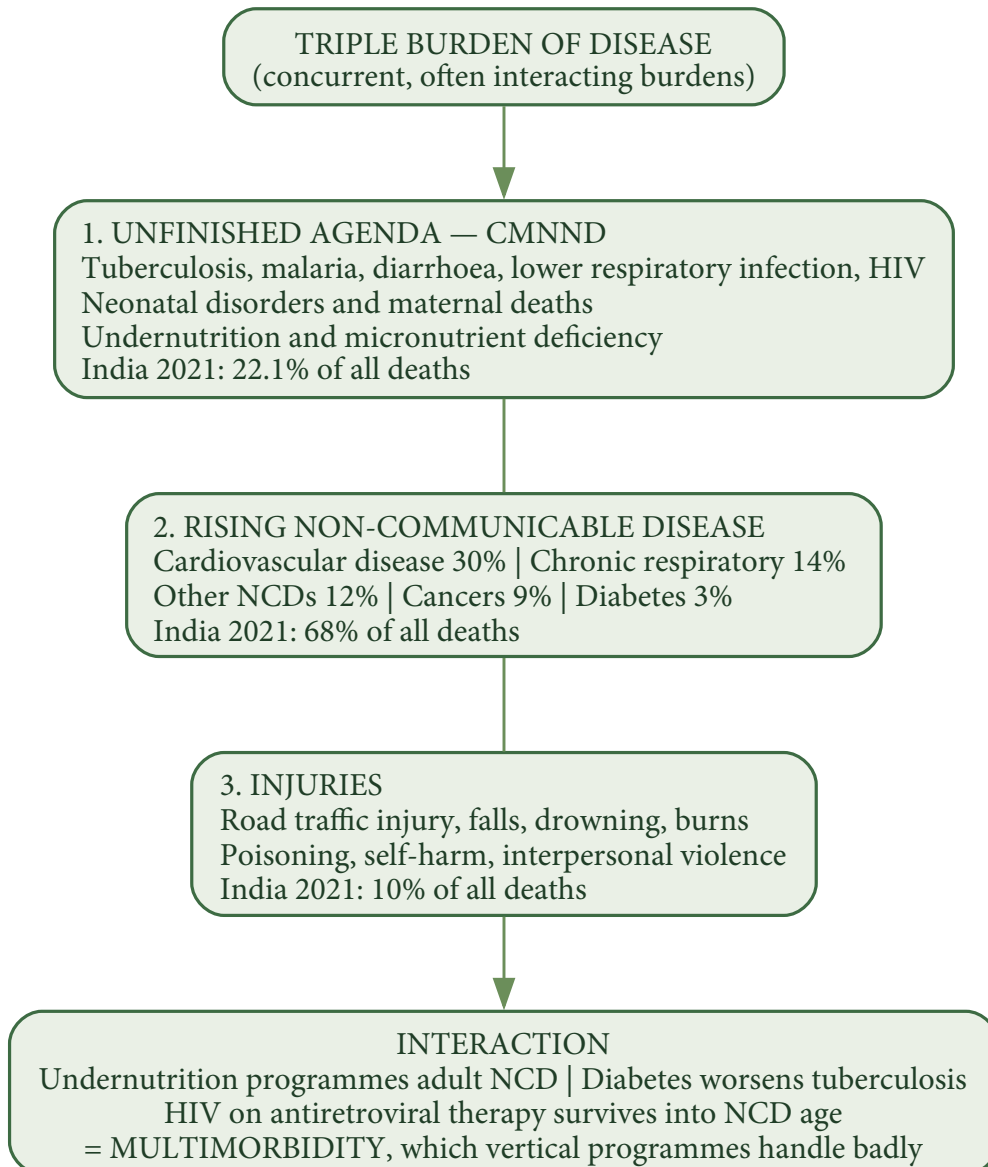
(Park 28e, Ch 6, p 422)

Public health implications

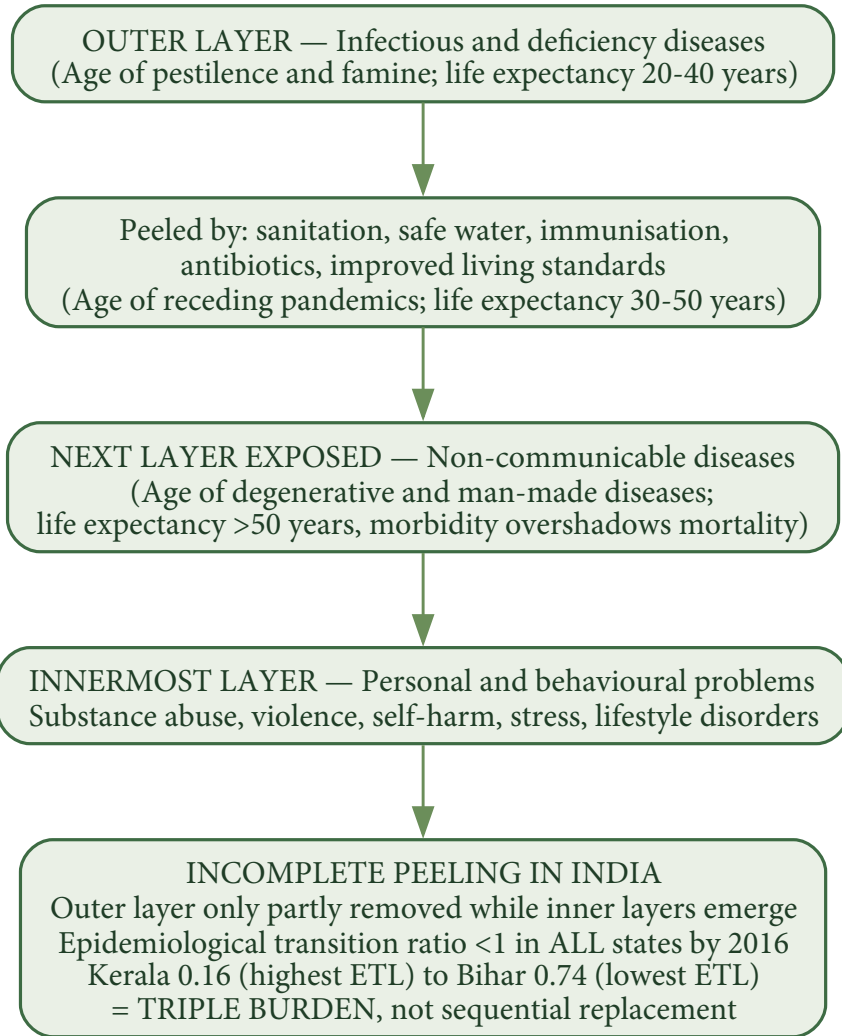
- ▶ **No dismantling of communicable disease programmes** — the unfinished agenda must be completed while NCD and injury capacity is built alongside, not instead.
- ▶ **Integration at the primary level** — screening, prevention and management of NCDs, tuberculosis and leprosy have been brought into **comprehensive primary health care** delivered through **Ayushman Arogya Mandir** (erstwhile Ayushman Bharat Health and Wellness Centre), launched 2018; **1,72,148** were operational as on 31 March 2024 (Park 28e, Ch 7, pp 535, 570).
- ▶ **Financial protection for chronic care** — Ayushman Bharat–Pradhan Mantri Jan Arogya Yojana gives ₹5 lakh per family per year to about **10.74 crore** poor and vulnerable families (Park 28e, Ch 7, p 570).
- ▶ **State-specific planning by ETL** — a lowest-ETL state such as Bihar needs a CMNND-weighted plan; a highest-ETL state such as Kerala needs an NCD-, injury- and ageing-weighted plan.
- ▶ **Intersectoral action** — water, sanitation, housing, road safety, tobacco and alcohol control, food policy and air quality lie outside the health department and must be engaged (Oxford 7e, Ch 1.4, p 248).
- ▶ **Strengthened information systems** — civil registration, cause-of-death certification and disease-burden estimation are needed to track which layer a district is in.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

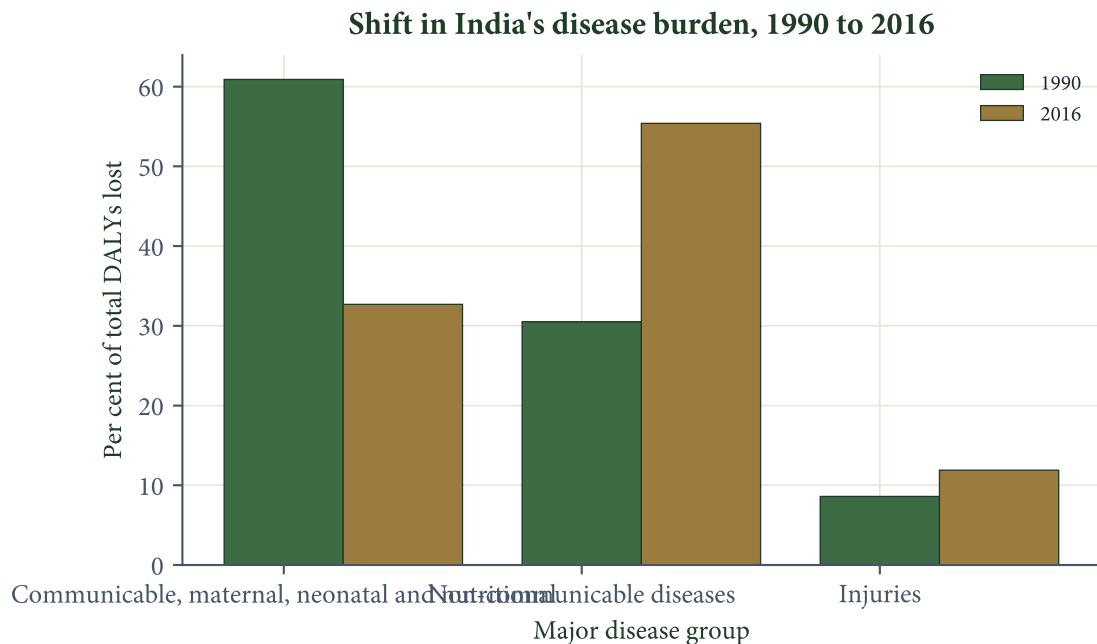
- ▶ **Diagram 1 — the onion peel of disease.** Draw four concentric rings with a knife/arrow peeling from the outside inwards. Label from outermost inwards: **Layer 1 — Communicable, maternal, neonatal and nutritional diseases**; **Layer 2 — Non-communicable diseases**; **Layer 3 — Injuries**; **Layer 4 (core) — Personal and behavioural problems**. Mark a curved arrow along the outside labelled "**old diseases wane and give place to new**", and a bracket across Layers 1–3 labelled "**TRIPLE BURDEN — layers coexist in India**".



The triple burden of disease — three concurrent, interacting layers and their Indian magnitude (2021)



Onion peel phenomenon — successive layers of the epidemiological transition and why they overlap in India



*Disease burden in India by major disease group, per cent of total disability-adjusted life years (DALYs) lost, 1990 versus 2016.
Source: ICMR, PHFI, IHME (2017), India State-Level Disease Burden Initiative, quoted in Sunder Lal, Ch 8, p 498.*

High-yield exam pointers

- ▶ **One-line definitions to write first:** triple burden = concurrent burdens of CMNND + NCD + injuries (Oxford 7e); onion peel = **old diseases wane and give place to new**, infectious → non-communicable → personal and behavioural (Park Plus).
- ▶ **India, 2021 proportional mortality (write all three):** NCDs 68%, CMNND and perinatal/nutritional 22.1%, injuries 10%; total deaths 1,30,46,300 (Park 28e, Ch 6, p 422).
- ▶ **DALY shift 1990 → 2016:** CMNND 60.9 → 32.7%, NCD 30.5 → 55.4%, injuries 8.6 → 11.9% (India State-Level Disease Burden Initiative, 2017).
- ▶ **ETR formula and its cut-offs:** DALYs from CMNND ÷ DALYs from (NCDs + injuries); **all states <1 by 2016; Kerala 0.16 → Bihar 0.74;** ETL bands 0.56–0.75 lowest, 0.41–0.55 lower-middle, 0.31–0.40 higher-middle, ≤0.30 highest.
- ▶ **Omran's three stages** — pestilence and famine → receding pandemics → degenerative and man-made diseases; **three models** — classical/Western, accelerated, delayed/contemporary (India).
- ▶ **30–70-year unconditional probability of dying from the four major NCDs in India = 24%** (27% men, 20% women).
- ▶ **Second meaning of onion peel** worth one line for extra marks: elimination or reduction of competing conditions makes another condition *appear* commoner over time (Sathe & Sathe, Ch 1, p 13).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The corpus contains no single textbook section that treats "triple burden" and "onion peel phenomenon" together; the triple-burden definition and its injury data come from the Oxford Textbook of Public Health 7e, the onion-peel definition from Park Plus and Sathe & Sathe, and all Indian magnitudes from Park's Textbook 28e, Sunder Lal and NFHS-5, each cited at the point of use. Ayushman Arogya Mandir is the current name of what was launched in 2018 as the Ayushman Bharat Health and Wellness Centre.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 6 Epidemiology of Chronic Non-communicable Diseases and Conditions, pp 422–423; Ch 5 Epidemiology of Communicable Diseases, pp 205, 426; Ch 7 Health Programmes in India, pp 535, 541, 570; Ch 10 Health Care of the Community, p 723); Oxford Textbook of Public Health 7e (Ch 1.4 Public health priorities in countries undergoing economic transition: the middle-income countries, pp 236–248; Ch 2.5 Health policy in developing countries, pp 803–804; Ch 8.8 Strategies for health services, p 3861; Ch 9.4 Injury prevention and control, pp 3056–3057; Measuring the health of populations: the Global Burden of Disease study methods, p 1461); Park Plus — Amrutha & Vijayalakshmi (Ch 24 Miscellaneous, "Epidemiologic transition and Onion-peel Phenomenon", pp 395–397); Sathe & Sathe — Epidemiology and Management for Health Care (Ch 1 General concepts, p 13); Sunder Lal — Textbook of Community Medicine (Ch 8, p 498). Data sources named with their year: ICMR, PHFI and IHME (2017) *India: Health of the Nation's States — The India State-Level Disease Burden Initiative*; National Family Health Survey-5 (2019–21); WHO proportional mortality estimates for India, 2021.

**Principles of health education**

Asked in: Paper I 06-Apr-2026 — RGUHS MD Community Medicine

Definition — Health education is "the process by which individuals and groups of people learn to behave in a manner conducive to the promotion, maintenance or restoration of health" (*John M. Last*, quoted in Park 28e, Ch 22, p 1014). It brings together the art and science of medicine with the principles and practice of general education, the link being the social and behavioural sciences — sociology, psychology and social anthropology (Park 28e, Ch 22, p 1018). Health education cannot be "given" by one person to another: teaching and learning is a **two-way transaction in human relations**, and the teacher cannot teach unless the pupil wants to learn (Park 28e, Ch 22, p 1018).

Why the principles matter

- ▶ Health education involves the **teaching, learning and inculcation of habits** directed at healthful living; the principles below are simply the principles of learning abstracted for use in health work (Park 28e, Ch 22, p 1018).
- ▶ Education alone is **necessary but not sufficient** — the target population must also have access to proven preventive measures or procedures (Park 28e, Ch 22, p 1015).
- ▶ Technology without education fails: anti-malarial insecticide spraying could not solve the malaria problem without health education (Park 28e, Ch 22, p 1018).

The twelve principles of health education, with suitable examples

#	Principle	What it means	Suitable example
1	Credibility	The degree to which the message is perceived as trustworthy by the receiver ; it must be consistent with scientific fact <i>and</i> with local culture, the educational system and social goals. Without trust and confidence in the communicator, no action follows.	A nutrition message that hands mothers "book recipes" conflicting with the family's cultural and economic conditions is seldom acted upon; messages built on locally available cereals, pulses and cooked foods are believed and followed (Sunder Lal, Ch 5, p 186).
2	Interest	People listen only to what is of interest to them; teaching must relate to their felt needs — the needs people themselves feel — which the educator must first find out. Where a group has needs it is unaware of, the educator must first create recognition of the need.	Vague slogans such as " <i>Take care of your health</i> " or " <i>eat nutritious food</i> " are as useless as simply asking people to be healthy; a message on child-birth and baby care to a mother attending for delivery is heard (Park 28e, Ch 22, pp 1019, 1020).
3	Participation	Based on the psychological principle of active learning ; people work with health workers in identifying their problems and in developing solutions. High participation creates involvement, personal acceptance and feedback.	<i>Alma-Ata Declaration</i> (1978): people "have a right and duty to participate individually and collectively in the planning and implementation of their health care"; without community participation as an integral part, health programmes are unlikely to succeed (Park 28e, Ch 22, p 1019).
4	Motivation	Awakening the fundamental desire to learn. Primary motives are inborn (hunger, sex, survival); secondary motives are created by outside forces or incentives — praise, love, rivalry, reward and punishment, recognition. Incentives may be positive (the carrot) or negative (the stick), and motivation is contagious.	Men who have already undergone vasectomy are among the best advertisements for male sterilization . An overweight woman told she may develop cardiovascular disease may not act; told she will look more charming, she accepts the advice (Park 28e, Ch 22, p 1019).
5	Comprehension	Teach within the level of understanding, education and literacy of the audience; never use words that are strange or new to them. Using words that cannot be understood is a major barrier to communication.	A doctor told a diabetic to "cut down starchy foods" — the patient had no idea what starchy foods were. Another prescribed "one teaspoonful three times a day" to a village woman who had never seen a teaspoon (Park 28e, Ch 22, p 1019).
6	Reinforcement	Few people learn all of a new message in a single exposure; repetition at intervals , and repetition in different ways, is necessary. Without reinforcement the individual slips back to the pre-awareness stage.	The same message on early initiation of breastfeeding repeated at the antenatal clinic, at the Village Health and Nutrition Day and on the postnatal home visit — spoken, demonstrated and shown on a flip-book (Park 28e, Ch 22, p 1019; Sunder Lal, Ch 4, p 153).

#	Principle	What it means	Suitable example
7	Learning by doing	Learning is an action process , not memorizing in the narrow sense.	The Chinese proverb " <i>If I hear, I forget; if I see, I remember; if I do, I know</i> " — hence a mother is made to prepare oral rehydration solution and to demonstrate handwashing herself rather than being lectured (Park 28e, Ch 22, p 1019).
8	Known to unknown	Proceed "from the concrete to the abstract", "from the particular to the general", "from the simple to the more complicated", "from the easy to the more difficult" — i.e. from the known to the unknown . Start where the people are, and use their existing knowledge as pegs on which to hang new knowledge .	Existing beliefs about religion and traditional medicine are used as the starting point to explain modern medicine, because the growth of knowledge from the unknown to the known is a long process full of obstacles and resistance, and quick results cannot be expected (Park 28e, Ch 22, p 1019).
9	Setting an example	The health educator must practise what he teaches , or the message is discounted.	An educator who lectures on the "hazards of smoking" but himself smokes, or who preaches the small-family norm while his own family is large, will not succeed (Park 28e, Ch 22, p 1019). School teachers who brush, wash hands and use toilets serve as role models in the <i>Child-to-Child</i> programme (Sunder Lal, Ch 10, p 615).
10	Good human relations	Sharing of information, ideas and feelings happens most easily between people who have a good relationship ; relationship-building goes hand in hand with communication skill.	In individual health teaching the educator first creates an atmosphere of friendship and lets the individual talk as much as possible, so that questions on specific personal interests are asked (Park 28e, Ch 22, pp 1019, 1020).
11	Feedback	A key concept of the systems approach — the educator modifies the elements of the system (message, channel) in the light of the audience's response. Feedback is of paramount importance for effective communication.	A knowledge–attitude–practice (KAP) re-survey and exit interviews after an information, education and communication (IEC) campaign show that the message was not understood, so the wording and the channel are changed (Park 28e, Ch 22, p 1019; Sunder Lal, Ch 4, p 153).
12	Leaders	We learn best from people we respect and regard ; the community is penetrated through local leaders, who are agents of change. Convince the leaders first and implementation becomes easy. Attributes: understands community needs; gives guidance and takes initiative; receptive to people's views; identifies with the community; selfless, honest, impartial, considerate, sincere; easily accessible; able to control and compromise community factions; skilled at eliciting cooperation and coordinating official and non-official organizations.	The village headman, the school teacher or the political worker is briefed and convinced first before a village-wide immunization or sanitation drive (Park 28e, Ch 22, p 1019).

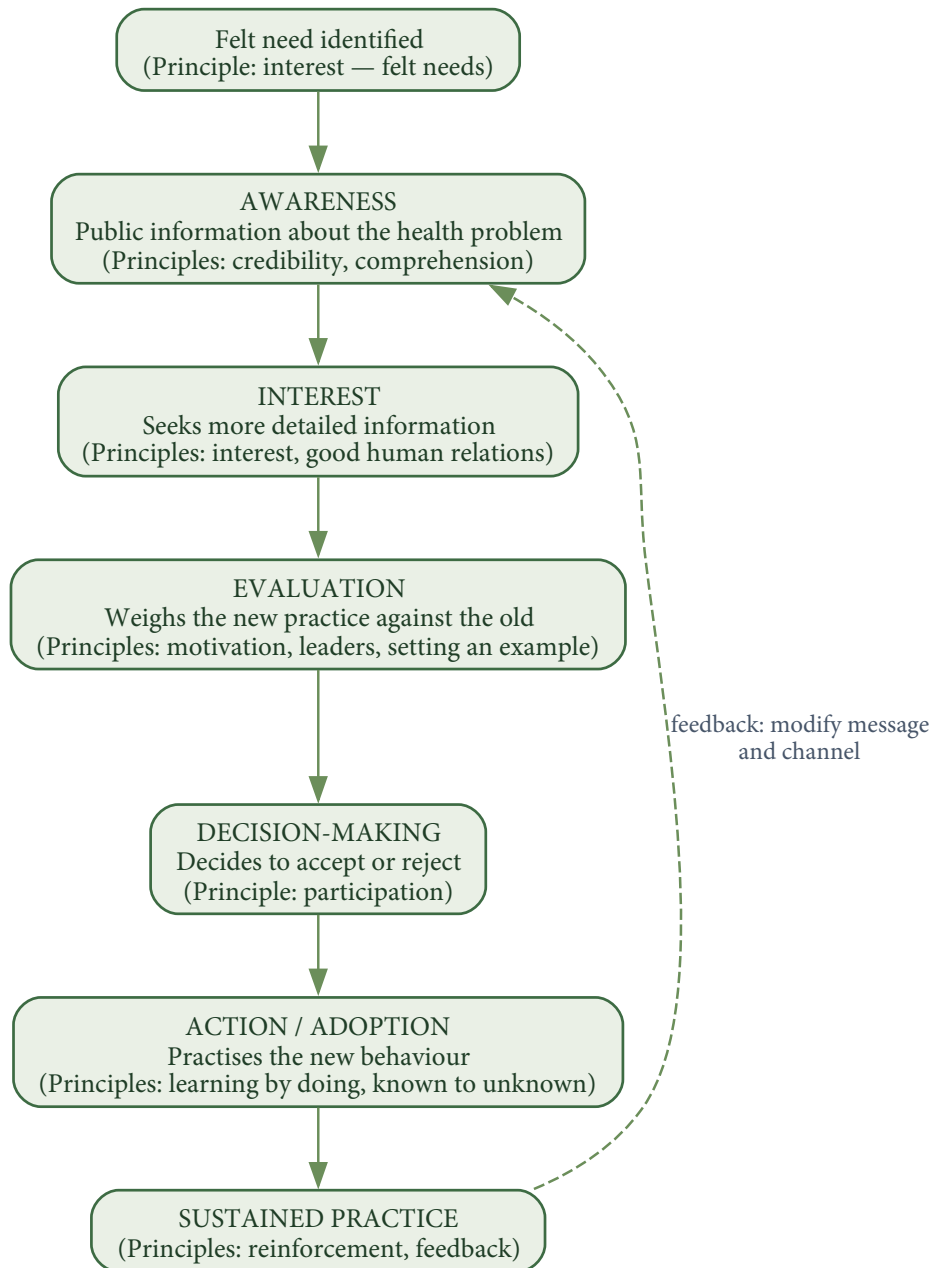
The same principles stated at programme level (Sunder Lal)

Principle	Content
I — Define objective, target audience and their concern	Give scientific health information to people so that they are empowered for self-reliance and action ; adequate health information is the right of everyone, and it has not reached the poor, illiterate and disadvantaged who need it most (Sunder Lal, Ch 4, p 145).
II — Frame the message	The message must be specific , and must not merely create awareness but lead on to behaviour change; e.g. in family planning, information on all methods is given to promote informed choice (Sunder Lal, Ch 4, p 153).
III — Select the right channel	Match channel to audience — television, radio, print, women's groups (<i>Mahila Swasthya Sangh</i>), male and female health workers, or inter-personal channels; always use multiple channels for maximum coverage (Sunder Lal, Ch 4, pp 145–146, 153). India's inter-personal channel is large: over 10.72 lakh accredited social health activists (ASHAs) and 13.4 lakh <i>Anganwadi</i> workers.
IV — Track and evaluate	Repeat messages, use a tone that creates trust (empathy) , keep contact with clients to sustain behaviour change, and finally evaluate the change in knowledge, attitude and practice (KAP) through home visits and exit interviews (Sunder Lal, Ch 4, p 153).

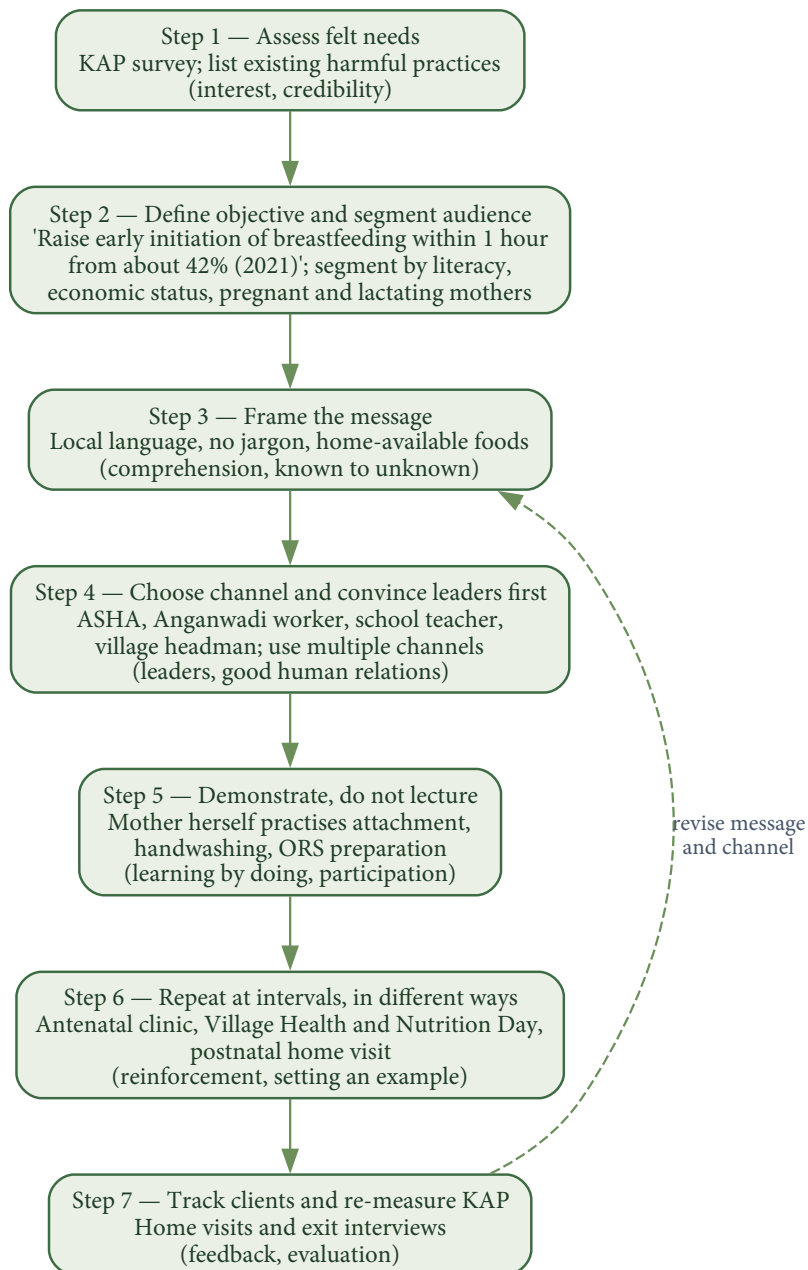
Sunder Lal's five key principles of communication, in order: (i) define your target audience; (ii) understand the concern of the audience; (iii) frame what you expect from the audience — the expected behaviour; (iv) select the right media; (v) find the right tone (Sunder Lal, Ch 4, p 153).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **DRAWN figure — the two-way communication loop (label exactly):** Sender / health educator → Message (specific, in local language) → Channel (inter-personal, print, mass media) → Receiver / audience → **Feedback arrow returning to the sender**; label "Barriers: jargon, illiteracy, cultural beliefs" alongside the channel arrow, and mark the feedback arrow "systems approach — modify message and channel".



The adoption sequence and the principle operating at each stage (Park 28e, Ch 22, p 1017)



Applying the principles — a health education session for mothers at an Anganwadi centre

High-yield exam pointers

- ▶ **Mnemonic for Park's 12 principles** — "CIP-M-CR-LK-S-GFL": Credibility, Interest, Participation, Motivation, Comprehension, Reinforcement, Learning by doing, Known to unknown, Setting an example, Good human relations, Feedback, Leaders.
- ▶ **Adoption sequence (Park Fig 2)**: Awareness → Interest → Evaluation → Decision-making → Action/Adoption; *motivation* covers the interest, evaluation and decision-making stages.
- ▶ **Known to unknown = four movements**: concrete → abstract; particular → general; simple → complicated; easy → difficult. Existing knowledge is the "peg" for new knowledge.
- ▶ **Quotable lines**: Last's definition of health education; "If I hear, I forget; if I see, I remember; if I do, I know"; Alma-Ata (1978) "right and duty to participate individually and collectively".

- ▶ **Motivation classification:** primary (inborn — hunger, sex, survival) vs secondary (praise, love, rivalry, reward and punishment, recognition); incentives positive (carrot) or negative (stick).
- ▶ **Two-line closing:** education is necessary but not sufficient — services must also be accessible; and health education is a two-way process, not a sermon.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: the workforce counts quoted above (10.72 lakh ASHAs, 13.4 lakh *Anganwadi* workers) and the breastfeeding figure of about 42% are given in Sunder Lal's textbook without a stated survey year for the workforce data, so quote them as textbook figures rather than as current official statistics; different textbooks also number and name Park's principles in slightly different order, and Sunder Lal restates the same content as four programme-level principles rather than twelve learning principles.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 22 (Communication for Health Education), pp 1014–1020, and Ch 21, p 1012; Sunder Lal — Textbook of Community Medicine, Ch 4 (Principles of Health Promotion and Education), pp 143–153, Ch 5 (Nutrition), p 186, and Ch 10, p 615; Declaration of Alma-Ata, 1978; WHO definition of health education, 1969.



Health education — non-communicable disease control

Asked in: Paper I 24-Sep-2025 — RGUHS MD Community Medicine

Definition — Health education is the process of providing scientific health information and learning experiences that enable people to adopt and sustain health-promoting behaviour of their own free choice. Its aims, adopted from the World Health Organization (WHO) definition of 1969 and the Alma-Ata Declaration of 1978, are to encourage health-promoting lifestyles, promote proper use of available health services, arouse interest and change attitudes so that people make rational decisions about their own problems, and stimulate individual and community self-reliance and participation (Park 28e, Ch 22, p 1015). Non-communicable diseases (NCDs) — cardiovascular disease, cancer, diabetes and chronic respiratory disease — are the group in which this tool matters most, because their principal causes are behaviours that only the person concerned can change.

Why health education is the pivotal tool in NCD control

- ▶ NCDs accounted for **60 per cent of all deaths in India in 2016**, and the country is in rapid health transition with a rising chronic-disease burden (Park 28e, Ch 7, p 564).
- ▶ The four major NCDs share **four modifiable behavioural risk factors** — tobacco use, unhealthy diet, physical inactivity and harmful use of alcohol — so a single set of educational messages simultaneously attacks several diseases (Park 28e, Ch 6, p 437).

- ▶ Because the causes are common, WHO recommends an **integrated programme for prevention and control of NCDs as part of primary health care**, attacking several risk factors at once rather than one disease at a time (Park 28e, Ch 6, p 436).
- ▶ Most of the world's major health problems and premature deaths are preventable through **changes in human behaviour at low cost**; health education is the only intervention that acts directly on that behaviour (Park 28e, Ch 21, p 1012).
- ▶ NCD risk factors act with a "**lag-time**" effect — today's mortality reflects past exposure — so education must begin long before disease appears, in young populations (Park 28e, Ch 6, p 438).

Behavioural risk factor	Core health-education message	Physiological risk factor it corrects	NCD outcome averted
Tobacco use — smoking and smokeless	Quit and avoid initiation; avoid household environmental tobacco smoke	Raised blood pressure, dyslipidaemia	Cardiovascular disease, cancers, chronic obstructive pulmonary disease
Unhealthy diet	Reduce salt, saturated and trans-fat and refined carbohydrate; increase fruit and vegetables	Raised blood pressure, raised cholesterol, raised blood sugar	Cardiovascular disease, stroke, stomach and colorectal cancer
Physical inactivity	Regular moderate or vigorous activity as part of normal daily life, from childhood onward	Overweight/obesity, raised blood sugar	Diabetes, cardiovascular disease, cancer
Harmful use of alcohol	Reduce volume, pattern and quality of drinking	Overweight, raised blood pressure	Cirrhosis, cancers, injury, cardiovascular disease
Air pollution, indoor and outdoor	Clean household fuel, avoidance of exposure	—	Chronic respiratory disease, asthma, cardiovascular disease

(Risk-factor to outcome chain: Park 28e, Ch 6, pp 434-435; risk-factor list Park 28e, Ch 2, p 57.)

Aims, approaches and the place of health education within health promotion

Following Alma-Ata the emphasis of health education shifted from prevention of disease to **promotion of healthy lifestyles**, from modification of individual behaviour to modification of the **social environment**, from community participation to **community involvement**, and towards individual and community **self-reliance** — precisely the shift NCD control requires (Park 28e, Ch 22, p 1015).

Approach/model	What it assumes	Value in NCD work
Medical model	Supplying scientific facts will make people act	Necessary but insufficient; does not bridge the knowledge-behaviour gap
Motivation model	Adoption of new behaviour is a process, not a single act, needing motivation	Explains why one-off NCD camps fail without reinforcement
Primary health care approach	People participate fully in planning and delivering their own services	The basis of NCD risk-factor control at the health and wellness centre

The five *Ottawa Charter* (1986) action areas place health education in its correct frame — it is the "develop personal skills" arm and must be supported by the other four: build healthy public policy, create supportive environments, strengthen community action and re-orient health services; the three strategies are to **advocate, enable and mediate** (Park 28e, Ch 2, p 50; Oxford Textbook of Public Health 7e, Ch 47, p 1735). Health education alone cannot achieve optimum health — the population must also have **access to proven preventive procedures** (Park 28e, Ch 21, p 1012).

Role at each level of prevention

Level	Educational role in NCDs	Concrete example
Primordial prevention	Prevent the emergence and spread of risk factors and lifestyles that have not yet become endemic; preserve traditional eating patterns and lifestyles	School and adolescent life-skills education on diet, activity, tobacco and alcohol before habits form (Park 28e, Ch 6, p 442)
Primary prevention	Change the community as a whole, not only individual subjects — shift the whole risk-factor distribution towards "biological normality"	Mass-media anti-tobacco campaigns, salt- and fat-reduction messages, cyclathons and yoga sessions at health and wellness centres (Park 28e, Ch 6, p 442; Ch 7, p 570)
Secondary prevention	Create awareness of early warning signals , promote self-examination, and secure uptake of population-based screening from age 30 years	Breast self-examination, oral self-examination, warning signals of cancer; opportunistic screening for hypertension and diabetes (Sunder Lal, Ch 8, pp 506, 524)
Tertiary prevention	Sustain treatment compliance, self-care and lifestyle change in the diagnosed; support rehabilitation and palliative care	Accredited social health activist (ASHA) follow-up visits for positive behaviour modification, treatment compliance and regular blood-pressure/blood-sugar checks (Park 28e, Ch 7, p 565)

Methods, settings and principles

Approach	Methods (Park 28e, Ch 22, Fig 3, p 1020)	Assets (Park 28e, Ch 22, Table 2, p 1021)
Individual	Personal contact, home visits, personal letters	Capitalises on warmth, understanding and knowledge of the communicator; allows questions and expression of fears; changes personal habits more readily; more influential at average and below-average educational level
Group	Lectures, demonstrations, group discussion, panel discussion, symposium, workshop, conference, seminar, role play	Permits involvement and two-way discussion; suited to defined groups such as school children, workers, diabetics
Mass	Television, radio, newspaper, printed material, direct mailing, posters, health museums and exhibitions, folk methods, internet	Reaches the widest population, gets public attention, supports concentrated campaigns of a week or month; more effective at above-average educational level

Mass media alone are **generally inadequate to change human behaviour** and must be combined with interpersonal and group methods (Park 28e, Ch 22, p 1023). Principles that decide success:

- ▶ **Credibility and interest** — the message must come from a trusted source and relate to the people's own interests, not to slogans they cannot act on.
- ▶ **Participation** — people work actively with health workers in identifying their own problems and planning solutions; without it health programmes are unlikely to succeed.
- ▶ **Motivation** — awaken the desire to learn; a motivated person spreads motivation through the group.
- ▶ **Comprehension** — pitch to the level of understanding, education and literacy of the audience; avoid strange words and medical jargon.
- ▶ **Reinforcement** — repeat at intervals in different ways; a single exposure leaves the individual back at the pre-awareness stage.
- ▶ **Learning by doing** — "if I hear I forget; if I see I remember; if I do I know."
- ▶ **Known to unknown; simple to complicated; concrete to abstract** — begin from existing knowledge as pegs on which to hang the new.
- ▶ **Setting an example** — the educator who smokes cannot teach tobacco cessation.
- ▶ **Feedback** — modify message and channel in the light of audience response.
- ▶ **Use of leaders** — the village headman, school teacher or political worker as agents of change.

(Principles: Park 28e, Ch 22, pp 1018-1019.)

Delivery through Indian national programmes

Under the **National Programme for Prevention and Control of Non-Communicable Diseases (NP-NCD)** — formerly the National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Diseases and Stroke (NPCDCS) — **primary prevention of major NCDs through health promotion and community empowerment for prevention of NCDs** are stated immediate objectives (Jugal Kishore, p 595).

Level	Health-education component
Sub-centre / health and wellness centre	Health promotion for behaviour change and counselling by camps, interpersonal communication, posters and banners in community, household, school, <i>Anganwadi</i> and workplace settings; key messages — healthy food, more physical activity, avoid tobacco and alcohol, warning signs of common cancers; population-based/opportunistic screening of all above 30 years and referral (Sunder Lal, Ch 8, pp 506, 530; Park 28e, Ch 7, p 564)
Primary health centre	Counselling with clinical diagnosis and treatment of hypertension and diabetes; screening of oral, breast and cervical cancer (visual inspection with acetic acid); identification of early warning signals
Community health centre (NCD clinic)	Diagnosis and management, counselling of diagnosed cases and families, supervision of village-level monthly clinics
District hospital and above	Referral care, follow-up chemotherapy, rehabilitation and physiotherapy; medical colleges and tertiary cancer centres add training of personnel and operational research (Sunder Lal, Ch 8, p 506)

Programme-specific educational components:

- ▶ **National Tobacco Control Programme** — public awareness and mass-media campaigns for behavioural change; training of health and social workers, non-governmental organisations and school teachers; a school programme; and tobacco cessation facilities, alongside enforcement of the Cigarettes and Other Tobacco Products Act (COTPA), 2003 (Park 28e, Ch 7, p 567).
- ▶ **National Cancer Control Programme** — a scheme funding non-governmental organisations for information, education and communication and early detection: primary prevention of cancer by health education, secondary prevention by screening and self-examination (Park 28e, Ch 7, p 566).
- ▶ **Rashtriya Kishor Swasthya Karyakram** — life-skills-based adolescence education using the school as the platform to counsel adolescents on NCD risk-factor modification, with peer educators and adolescent health days (Jugal Kishore, pp 182-183).
- ▶ **Village Health, Sanitation and Nutrition Day and ASHA home visits** — the monthly *Anganwadi*-based platform for group and interpersonal NCD messages and for community follow-up of diagnosed patients (Sunder Lal, Ch 17, p 744; Park 28e, Ch 7, p 565).

Barriers to effectiveness and evaluation

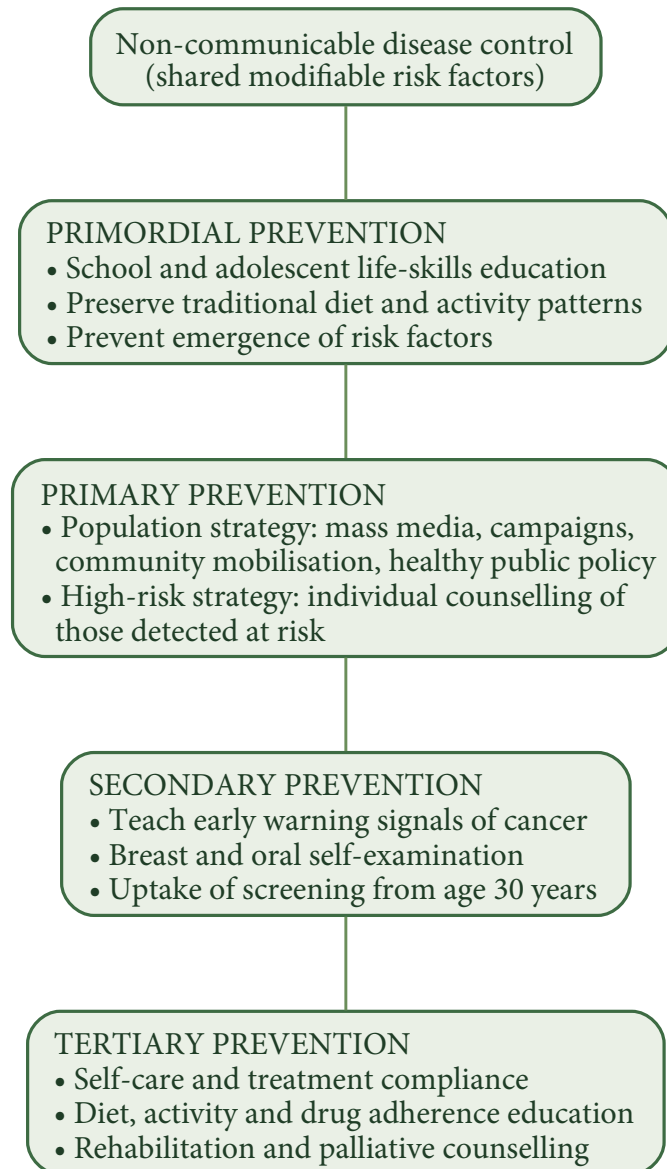
Barrier	Examples
Physiological	Difficulty in hearing or expression, especially in the elderly with NCDs
Psychological	Emotional disturbance, neurosis, level of intelligence, comprehension difficulty
Environmental	Noise, invisibility, congestion at the camp or clinic
Cultural	Illiteracy, customs, beliefs, religion, attitudes, economic and social class differences, language variation, urban–rural gap
Communicator-related	Talking at rather than listening, medical jargon, poor eye contact, low credibility, frequent interruptions, lack of health literacy

(Park 28e, Ch 21, p 1012; Sunder Lal, Ch 4, p 155.)

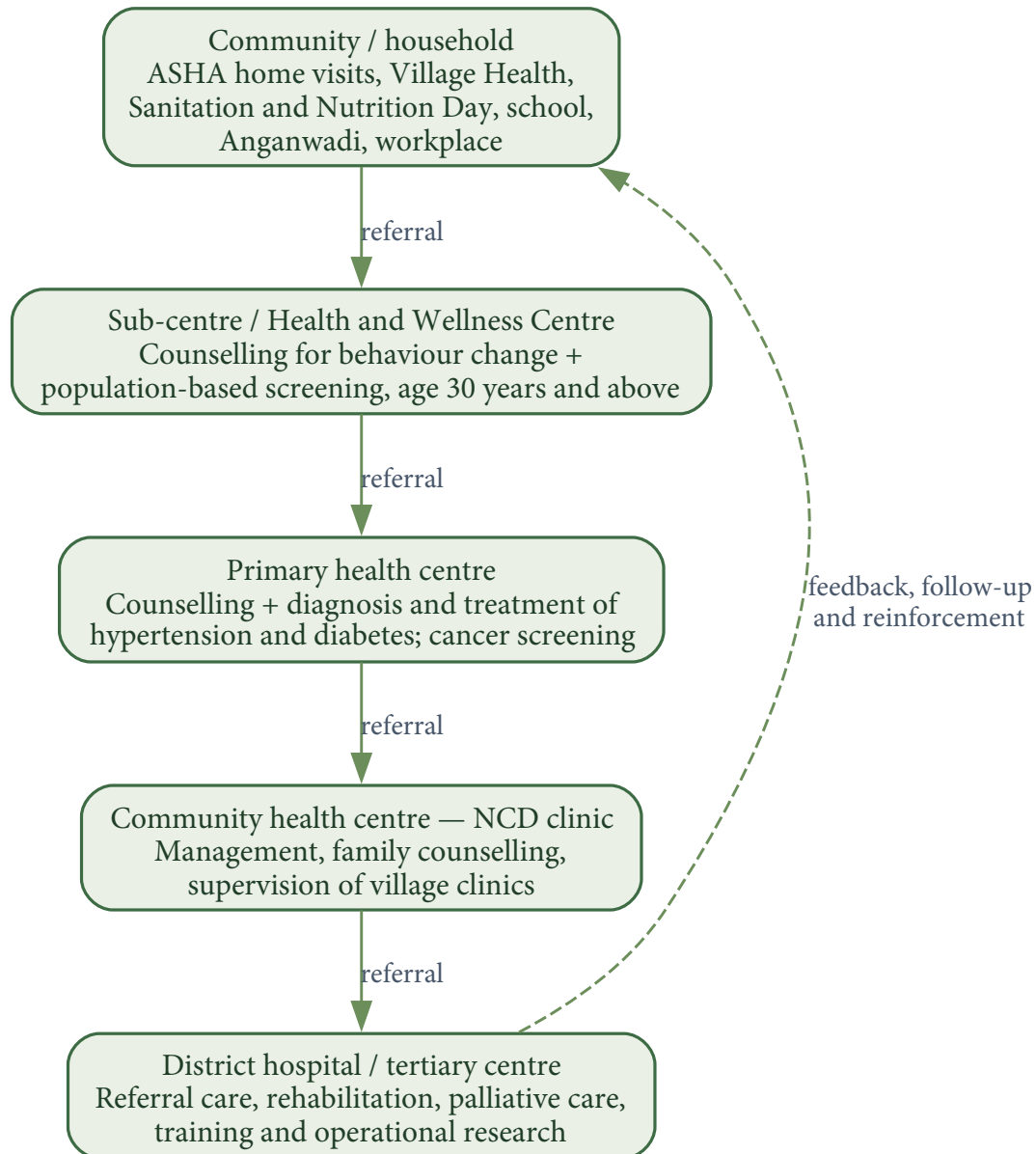
Evaluation of NCD health education means measuring the **change in knowledge, attitude and practice (KAP)** produced, assessed by workers during home visits and contacts and by exit interviews (Sunder Lal, Ch 4, p 153). Process indicators are sessions held, population covered and material distributed; outcome indicators are risk-factor prevalence measured by the WHO STEPS surveillance framework, screening uptake above 30 years, tobacco quit rates and treatment compliance, against the global targets of a 25 per cent relative reduction in premature NCD mortality by 2025 and a one-third relative reduction under the Global Action Plan extended to 2030 (Park 28e, Ch 6, p 437).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Adoption (behaviour change) model in health education** — label: *Awareness* (public information about the health need) → *Motivation* (interest → evaluation → decision-making) → *Action* (adoption or acceptance of the new behaviour), with a feedback arrow from Action back to Motivation labelled "reinforcement / repetition" (Park 28e, Ch 22, Fig 2, p 1017).
- ▶ **Risk-factor to outcome chain for NCDs** — three stacked columns: *Behavioural risk factors* (tobacco use — smoking and smokeless; alcohol use; unhealthy diet; physical inactivity; air pollution) → *Physiological risk factors* (overweight/obesity, raised blood pressure, raised blood sugar, raised total cholesterol/lipids) → *Disease outcomes* (cardiovascular disease, cancers, chronic respiratory disease, diabetes, chronic obstructive pulmonary disease and asthma, stroke, non-alcoholic fatty liver disease), with an arrow labelled "health education acts here" pointing at the first column (Park 28e, Ch 6, p 434).



Health education across the levels of prevention of non-communicable diseases



Delivery of NCD health education through the Indian health system under NP-NCD

High-yield exam pointers

- ▶ Four shared behavioural risk factors of the four major NCDs: **tobacco, unhealthy diet, physical inactivity, harmful alcohol use** (Park 28e, Ch 6, p 437).
- ▶ Four aims of health education (WHO 1969 / Alma-Ata 1978): healthy lifestyle, proper use of health services, rational decision-making, individual and community self-reliance.
- ▶ Adoption model: **Awareness → Motivation (interest, evaluation, decision-making) → Action (adoption)**.
- ▶ Twelve principles of health education — remember credibility, interest, participation, motivation, comprehension, reinforcement, learning by doing, known to unknown, setting an example, good human relations, feedback, leaders.
- ▶ Screening entry point in India: **all persons aged 30 years and above** for hypertension, diabetes and three common cancers (oral, breast, cervix).

- ▶ Breast self-examination message: "3 minutes, three fingers, once in 30 days" (Sunder Lal, Ch 8, p 524).
- ▶ Targets: 25 per cent relative reduction in premature NCD mortality by 2025; one-third relative reduction under the Global Action Plan extended to 2030. India, 2016: NCDs = 60 per cent of all deaths.
- ▶ Barriers to communication: physiological, psychological, environmental, cultural.
- ▶ Evaluation of health education = change in knowledge, attitude and practice (KAP).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the retrieved textbooks still use the older programme name NPCDCS (National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Diseases and Stroke), which the Ministry of Health and Family Welfare has since renamed the National Programme for Prevention and Control of Non-Communicable Diseases (NP-NCD); the service package and the health-promotion component described above are unchanged, and the India mortality share of 60 per cent is a 2016 figure quoted in Park 28e.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, MoHFW NP-NCD, NFHS-5, GATS-2, National NCD Monitoring Survey 2017-18, WHO HEARTS technical package, UN Political Declaration on NCDs and Mental Health 2025

Programme and policy updates (India)

- ▶ Health and Wellness Centres were renamed Ayushman Arogya Mandir (AAM) in November 2023, with the tagline *Arogyam Paramam Dhanam* ("health is the supreme wealth"), and the remit was widened from illness care to holistic wellness — lifestyle counselling, yoga and physical exercise, mental-health support, nutrition and elderly care under AYUSH principles; renaming in regional scripts was funded at ₹3,000 per facility (₹48 crore outlay). Field observations caution that where staffing and service packages do not change alongside, the renaming risks being cosmetic — the examinable point being that the health-promotion package, not the signboard, delivers NCD prevention [PMID 42257165].
- ▶ National health communication has shifted from Information, Education and Communication (IEC) to Social and Behaviour Change Communication (SBCC) with a digital arm. An analysis of the official AAM social-media handles (2022–24) found preventive and promotive content out-performed observance-day posts on X — a hypertension "regular check-ups" post drew 132 engagements from 868 reach, the highest of any post — while Facebook follower growth collapsed to a compound annual rate of 0.01 by 2024 and audiences skewed male (70.6% on X). This is direct evidence for the textbook rule that mass media alone are inadequate: digital reach is not behaviour change, and platform-specific audience skew is now a fresh barrier to add to the physiological–psychological–environmental–cultural list [PMID 42453216].

Current national data

- ▶ NCDs now account for about 63 per cent of all deaths in India (cardiovascular disease alone 28 per cent), superseding the 60 per cent figure of 2016 quoted in the textbooks; hypertension

affects 28.5 per cent and diabetes 9.3 per cent of Indian adults. A facility survey of 561 sub-centre AAMs across nine districts in the second half of 2024 found 94.1 per cent staffed by a community health officer or mid-level health provider, blood-pressure apparatus in 85 per cent, glucometers in 83.8 per cent and at least two essential antihypertensive and antidiabetic drugs in about 78 per cent each, with roughly 70 per cent of patients starting treatment at that level — the platform on which the counselling described above actually runs [PMID 40991221].

- **The national hypertension care cascade, dated: NFHS-5 (2019–21) gives 28 per cent age-standardised prevalence, 37 per cent diagnosed, 18 per cent on treatment and only 9 per cent controlled, against the National NCD Monitoring Survey 2017-18 figures of 29 per cent prevalence, 28 per cent aware, 15 per cent treated and 13 per cent controlled among the treated. The health-education finding within this is the sharpest available quantification of the knowledge-behaviour gap: among people aware of their hypertension but not taking medicines in 2023-24, half (50%; 505/1008) gave "did not feel sick" as the reason, and 39 per cent said no drug had been prescribed [PMID 41146104].**

Elimination, surveillance and disease status

- **The India Hypertension Control Initiative (IHCI), launched November 2017 on the WHO HEARTS technical package, has scaled to 89 districts across 15 states, registering 1,235,453 people with hypertension, of whom 1,046,512 remained under care with 44 per cent blood-pressure control. Control was highest at health and wellness centre/sub-centre level (57 per cent) and lowest at community health centres (26 per cent), community health centres carrying 1.86 times the risk of uncontrolled blood pressure (95% CI 1.84–1.88) — evidence that proximity, personal contact with the nurse or community health worker, and easy recall of defaulters (that is, the interpersonal method) outperform higher-technology facilities. Control fell from 68 per cent on a single reading to 52 per cent on two consecutive readings [PMID 41314840].**
- **Surveillance has moved onto the National NCD Portal, the digital backbone of population-based screening of the five common NCDs (diabetes, hypertension, and oral, breast and cervical cancer) from age 30, integrated with the Ayushman Bharat Digital Mission/ABHA identity and the AB-HWC portal; it is in use in 31 states and union territories, with 53.94 crore beneficiaries enrolled and 3.95 crore on treatment for hypertension or diabetes. Frontline users report that dual paper-plus-digital entry, server and connectivity failure, and uneven digital skills among ASHAs and ANMs are the operational limits — a modern addition to the classical communicator-related barriers [PMID 41555328].**

Guideline and legislative updates

- **Pictorial health warnings under COTPA continue to earn their place as a mass-media health-education tool. In GATS-2 (2016-17), among 9,398 adult smokers, 70.1 per cent had noticed warnings on tobacco packs; intention to quit was 53.1 per cent among those who noticed them versus 41.6 per cent among those who did not (adjusted OR 1.191, 95% CI 1.072–1.325), independent of prior quit attempts, knowledge of harms and nicotine dependence. Odds were lower in women and Scheduled Tribe respondents, which is the argument for supplementing pack warnings with interpersonal work in low-literacy and marginalised groups [PMID 42536835].**

- **Training the educator changes practice measurably.** A one-day, Government-of-India-module-based capacity-building programme for 202 ASHAs in rural Varanasi (2023) raised awareness of the COTPA Act from 38.1 to 65.4 per cent and produced significant gains across knowledge, attitude and practice domains ($p < 0.01$); at baseline only 19.8 per cent routinely performed symptom-based oral-cancer screening, and afterwards 75.7 per cent were willing to give tobacco-cessation counselling and 64.8 per cent to screen. Legal literacy of the frontline worker, not just clinical knowledge, is now an explicit training outcome [PMID 42374470].

Global and WHO developments

- **The Fourth UN High-Level Meeting on the Prevention and Control of NCDs and the Promotion of Mental Health and Well-being was held in September 2025**, and its Political Declaration — adopted by 175 Member States including India — treats community-driven strategies, early detection and meaningful engagement of people living with NCDs as essential rather than optional, aligned to Sustainable Development Goal target 3.4; the Fifth meeting is scheduled for 2031. India is among the few countries to have formally incorporated meaningful involvement of people living with NCDs into its National Programme for Prevention and Control of NCDs (2023), with state guidelines in Punjab and Maharashtra (2022) and Community Advisory Boards in Rajasthan and Andhra Pradesh — the post-Alma-Ata shift from community *participation* to community *involvement* now written into programme documents [PMID 42144248].
- **Knowledge of tobacco harms is high for cancer but persistently poor for cardiovascular disease, and is improving far too slowly to serve as an evaluation success.** Across 26 International Tobacco Control Project countries (2002–22) and 32 GATS countries (2008–20), knowledge that smoking causes lung cancer was 92.2 and 91.9 per cent respectively, but knowledge that it causes stroke only 75.0 and 66.1 per cent; average change across five disease indicators was +0.14 per cent per year. For KAP-based evaluation of NCD health education, this identifies cardiovascular risk messaging as the specific residual gap [PMID 42552006].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 2 Concept of Health and Disease; Ch 6 Epidemiology of Chronic Non-communicable Diseases and Conditions; Ch 7 Health Programmes in India; Ch 21 Health Information and Basic Medical Statistics; Ch 22 Communication for Health Education); Sunder Lal — Textbook of Community Medicine (Ch 4 Principles of Health Promotion and Education; Ch 8 Epidemiology of Communicable and Non-Communicable Diseases; Ch 17 Healthcare of the Community); Jugal Kishore — National Health Programmes of India (NPCDCS objectives and strategies; National Tobacco Control Programme; Rashtriya Kishor Swasthya Karyakram); Oxford Textbook of Public Health 7e (Ch 47 Health promotion, health education and the public's health); WHO Ottawa Charter for Health Promotion, 1986; WHO Global Action Plan for the Prevention and Control of NCDs, 2013-2020 extended to 2030; Cigarettes and Other Tobacco Products Act (COTPA), 2003.

Recent-advances sources: 42257165 — Renaming of Ayushman Bharat Health and Wellness centres as Ayushman Arogya Mandirs with a Tagline of 'Arogyam Paramam Dhanam' — A step towards holistic care, *Journal of Family Medicine and Primary Care* 2026, doi:[10.4103/jfmmpc.jfmmpc.1966.24](https://doi.org/10.4103/jfmmpc.jfmmpc.1966.24); 42453216 — Social media as a catalyst for health behaviour change: From traditional IEC to digital SBCC in the era of Ayushman Bharat, *Journal of Family Medicine and Primary Care* 2026, doi:[10.4103/jfmmpc.jfmmpc.1202.25](https://doi.org/10.4103/jfmmpc.jfmmpc.1202.25); 40991221 — Functionality of Ayushman Arogya Mandirs for hypertension & diabetes care in nine districts of India, *Indian Journal of Medical Research* 2025, doi:[10.25259/IJMR.1693.2025](https://doi.org/10.25259/IJMR.1693.2025); 41146104 — Changing patterns in health-seeking behaviour and prescription practices for treatment of hypertension in nine districts: insights from India Hypertension Control Initiative, 2018-19 and 2023-24, *BMC Public Health* 2025, doi:[10.1186/s12889-025-24898-0](https://doi.org/10.1186/s12889-025-24898-0); 41314840 — Scaling India Hypertension Control Initiative strategies to 15 states: treatment outcomes and risk factors for uncontrolled blood pressure, India: a cohort study, 2018-2021, *BMJ Open* 2025, doi:[10.1136/bmjopen-2025-106372](https://doi.org/10.1136/bmjopen-2025-106372); 41555328 — A qualitative study on users' perspective on the functioning of the National Non-Communicable Disease portal in Dakshina Kannada district, Karnataka, *BMC Health Services Research* 2026, doi:[10.1186/s12913-026-14008-0](https://doi.org/10.1186/s12913-026-14008-0); 42536835 — Exposure to Tobacco Health Warnings and Quitting Intentions Among Indian Smokers: Evidence from GATS-2, *Substance Use & Misuse* 2026, doi:[10.1080/10826084.2026.2709650](https://doi.org/10.1080/10826084.2026.2709650) (abstract-only); 42374470 — Effectiveness of a capacity-building training for knowledge and practices of tobacco control and oral cancer prevention: a pre-post study from rural northern India, *BMC Oral Health* 2026, doi:[10.1186/s12903-026-08866-7](https://doi.org/10.1186/s12903-026-08866-7) (abstract-only); 42144248 — Integrating meaningful involvement of people living with non-communicable diseases in India's National NCD Programme: lessons learnt and opportunities for global, regional and national contextualisation and adoption, *BMJ Global Health* 2026, doi:[10.1136/bmjgh-2025-020189](https://doi.org/10.1136/bmjgh-2025-020189); 42552006 — Changes in knowledge of the major harms of tobacco smoking and secondhand smoke exposure among adults who smoke: findings from 26 countries of the International Tobacco Control (ITC) Project (2002-2022) and 32 countries of the Global Adult Tobacco Survey (GATS) (2008-2020), *BMJ Open* 2026, doi:[10.1136/bmjopen-2025-107667](https://doi.org/10.1136/bmjopen-2025-107667) (abstract-only). Also used: Ministry of Health and Family Welfare, Ayushman Arogya Mandir renaming, November 2023; UN Political Declaration

on NCDs and Mental Health, September 2025.

♦ ♦ ♦

Role of medico-social worker in community health

Asked in: Paper I May 2018 — RGUHS MD Community Medicine

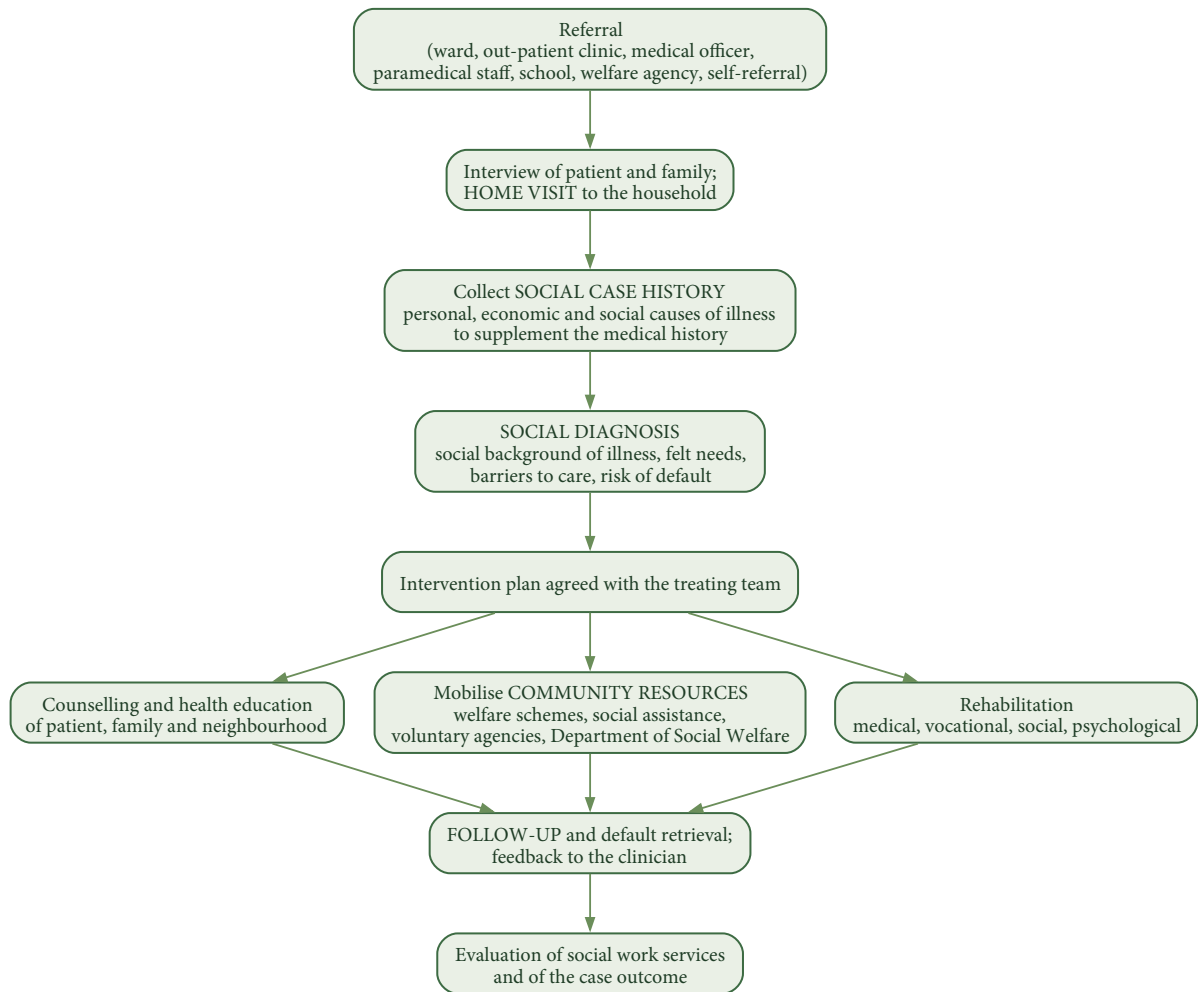
Definition — Medical social work is the branch of social work that uses "**case work**" as its main technique to find out the **social background of illness**; this information helps the doctor arrive at a **social diagnosis**, treat the illness and estimate the prognosis (Park 28e, Ch 12, p 836). The **medico-social worker (MSW)** is a worker trained in social case work and in the art of interviewing people, whose purpose is to help sick people *individual by individual*, through the best use of the patient's own capabilities and of **community resources**, in matters of personal and social adjustment in the community, including rehabilitation (Park 28e, Ch 12, pp 836–837). Medical social work began in England in **1895**, when a social worker called the *Almoner* was appointed at the **Royal Free Hospital, London** (Park 28e, Ch 12, p 836).

Placement, and why the role exists

Item	Detail
Training	Trained in social case work and in the art of interviewing people (Park 28e, Ch 12, pp 836–837)
Settings of employment	Hospitals, tuberculosis clinics, family planning clinics, cancer control centres, mental health services, maternal and child welfare services, and school and university health services (Park 28e, Ch 12, pp 836–837)
Core position	Forms the link between the institution where she is employed and the community (Park 28e, Ch 12, pp 836–837)
Place in the health team	Modern medicine is a combined effort of physicians, nurses, social workers , ASHA and others; each member must have a specific, recognised role and freedom to exercise her particular skill (Park 28e, Ch 2, p 48)
Rationale	Many community health problems are in essence social problems , and vice versa; the theme common to community health and the social sciences is human behaviour (Park 28e, Ch 12, pp 811–812)
Recognition	Now recognised as an essential professional colleague of the doctor in the analysis and correction of social and emotional factors, and relied upon for information of fundamental importance in formulating the complete diagnosis and directing treatment (Park 28e, Ch 12, p 837)

Core functions and the social case-work method

Three statutory-style functions define the post: (i) to **assess the need** for social work services of the individual, family and community; (ii) to **arrange or provide** social work services; and (iii) to **participate in overall planning and evaluation** of social work services (Sunder Lal, Ch 15, p 695). The services delivered are counselling, psychotherapy, rehabilitation (physical, social and occupational), and **prevention of disease and disability** (Sunder Lal, Ch 15, p 695).



The social case-work sequence followed by the medico-social worker

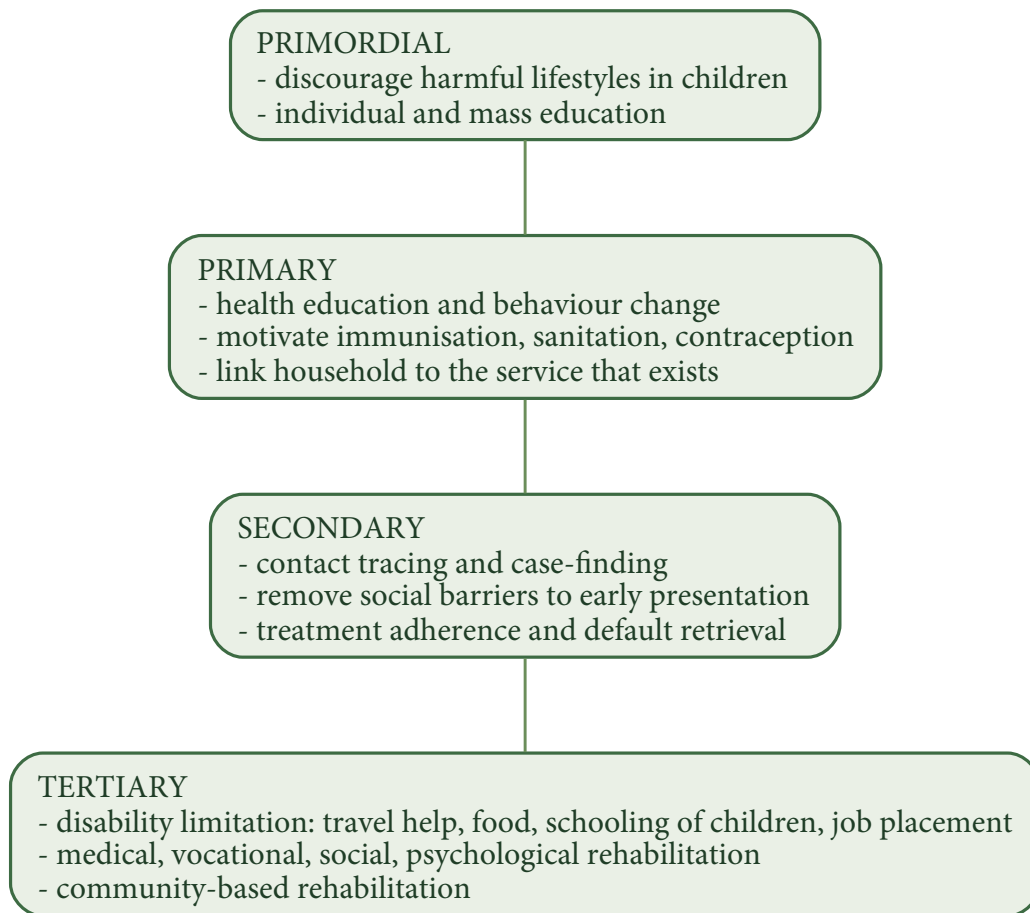
The principal work is to **visit the family and probe into the personal, economic and social causes of illness**, and to collect the social history of the patient with a view to supplementing the medical history; the worker then tries to secure help for the patient through community resources (Park 28e, Ch 12, pp 836–837). Social case work is most significant in **chronic diseases, mental disorders, HIV/AIDS, leprosy, tuberculosis, cancer and behavioural problems of children** (Sunder Lal, Ch 15, p 695).

Role in prevention of disease, by level of prevention

Level	What the medico-social worker contributes
Primordial	Discourages children and adolescents from adopting harmful lifestyles — smoking, faulty eating patterns, physical inactivity — the main intervention here being individual and mass education , which is the worker's own métier (Park 28e, Ch 2, p 60)
Primary — health promotion	Delivers the modes of intervention that need no drug: health education , environmental modification, nutritional intervention, and lifestyle and behavioural change (Park 28e, Ch 2, p 61); motivates uptake of immunisation, sanitation and contraception
Primary — specific protection	Motivates and follows up defaulters for immunisation and prophylaxis; ensures the household actually reaches the service it has been told about (Park 28e, Ch 22, p 1015)

Level	What the medico-social worker contributes
Secondary — early diagnosis	Traces contacts, brings in symptomatic family members, removes the social barriers (cost, distance, stigma, loss of wages) that delay presentation; supports early detection services such as counselling and testing for HIV (Park 28e, Ch 7, p 520)
Secondary — adequate treatment	Treatment adherence and default retrieval — adherence is assessed by the counsellor at every visit and support provided as required, with linkage to care and support centres for psycho-social support (Park 28e, Ch 7, p 523)
Tertiary — disability limitation	Identifies and meets the social and economic needs of patient and family: assistance to travel to and from the clinic, food grains, clothes, care of children and their education, and job placement , provided through voluntary agencies and Departments of Social Welfare (Park 28e, Ch 5, p 396)
Tertiary — rehabilitation	Helps in the rehabilitation of the patient in chronic disabling illness such as tuberculosis, leprosy and poliomyelitis (Park 28e, Ch 12, p 837); social work is one of the recognised non-medical disciplines of rehabilitation alongside physiotherapy, occupational therapy, speech therapy, psychology, education and vocational placement (Park 28e, Ch 2, p 63)

The four domains of rehabilitation the worker services are **medical** (restoration of function), **vocational** (restoration of the capacity to earn a livelihood), **social** (restoration of family and social relationships) and **psychological** (restoration of personal dignity and confidence) (Park 28e, Ch 2, p 63). In leprosy the target is **preventive rehabilitation** — preventing deformity and social and vocational disruption by early diagnosis and adequate treatment, so that debilitation never occurs and the uphill task of rehabilitation is never needed (Park 28e, Ch 5, p 396).



Where the medico-social worker acts along the levels of prevention

Role in promotion of health of the community

Ottawa Charter action area (Park 28e, Ch 2, p 50)	The worker's community-level contribution
Develop personal skills	Carries the health education objectives — encourage people to adopt and sustain health-promoting lifestyles, promote proper use of available health services, arouse interest and change attitudes for rational decisions, and stimulate self-reliance (Park 28e, Ch 22, p 1015)
Strengthen community action	Works through local leaders — village headman, school teacher, political worker — who are the agents of change; once leaders are convinced, implementation becomes easy (Park 28e, Ch 22, p 1019)
	Secures community participation , which creates a sense of involvement, personal acceptance and decision-making; the Alma-Ata Declaration states people have a right and duty to participate individually and collectively in planning and implementing their health care (Park 28e, Ch 22, p 1019)
Create supportive environments	Uncovers social pathology — the relation between disease and social conditions — through social surveys , the tool by which social pathology is disclosed (Park 28e, Ch 12, p 816)

Ottawa Charter action area (Park 28e, Ch 2, p 50)		The worker's community-level contribution
Build healthy public policy	Feeds social-survey evidence to programme managers; social problems (alcoholism, drug dependence, sexually transmitted disease, vagrancy, juvenile delinquency) are solved by social welfare programmes, social assistance and social legislation (Park 28e, Ch 12, p 816)	
Re-orient health services	Links the beneficiary to social security — protection against sickness, invalidity, maternity, old age and death, comprising social insurance and social assistance (Park 28e, Ch 12, pp 845–846)	

The behaviour-change sequence the worker drives is **awareness** → **motivation** (**interest, evaluation, decision-making**) → **action/adoption**; mere awareness is of little value unless it leads to motivation, and motivation is contagious — one motivated person may spread motivation through a whole group (Park 28e, Ch 22, p 1017).

Role in national health programmes in India

Programme / service	Defined role of the social worker or counsellor
District Mental Health Programme	A psychiatric social worker / trained social worker is one of the five members of the district technical team, alongside the psychiatrist or trained medical officer, clinical psychologist, psychiatric community nurse and record keeper (Sunder Lal, Ch 15, pp 682–683)
	Programme aims served directly: reducing stigma through changes in attitude and public education; treatment and rehabilitation within the community of patients discharged from the mental hospital; continuing mental health literacy (Sunder Lal, Ch 15, pp 682–683)
Child guidance clinics	Parental counselling, family therapy, vocational guidance, therapeutic group work with children, adolescents and parents; referrals come from hospitals, practitioners, schools, welfare organisations, parents and guardians (Sunder Lal, Ch 15, p 695)
National AIDS Control Programme	Integrated Counselling and Testing Centre (ICTC) : counselling and testing, basic information on transmission and prevention, promoting behavioural change, reducing vulnerability, and linking people living with HIV to prevention, care and treatment services (Park 28e, Ch 7, p 520)
	ART centre : adherence assessed by the counsellor at every visit; counselling on treatment adherence, nutrition, positive prevention and positive living; referral to care and support centres for psycho-social support and for availing social beneficiary schemes (Park 28e, Ch 7, p 523)
National Tuberculosis Elimination Programme	Health education for community awareness about tuberculosis and about completing treatment; interventions on nutrition, poverty and economic condition; incentives — the Nikshay Poshan Yojana payment is a monitored programme indicator (Jugal Kishore, Ch 1, p 280; Park 28e, Ch 7, p 514)

Programme / service	Defined role of the social worker or counsellor
National Leprosy Eradication Programme	Social assistance and social support to patient and family, delivered through voluntary agencies and Departments of Social Welfare; community-based rehabilitation as a strategy within general community development for rehabilitation, equalisation of opportunities and social inclusion of all people with disabilities (Park 28e, Ch 5, pp 396–397)
Rashtriya Kishor Swasthya Karyakram	Dedicated adolescent health counsellors provide facility-based counselling on nutrition, puberty, contraception, delaying marriage and childbearing, substance misuse, sexual abuse and mental health, plus outreach in schools, colleges, vocational training centres and at the Village Health and Nutrition Day (Park 28e, Ch 7, p 556)
Services for the disabled	Ministry of Social Justice and Empowerment schemes give up to 90% assistance in urban and 95% in rural areas to voluntary organisations for education, training and rehabilitation of the disabled and of persons recovering from mental illness, with emphasis on vocational guidance and training (Park 28e, Ch 9, p 689)

Constraints on the role

- ▶ **No sanctioned cadre at the periphery** — the worker is described for hospitals and specialised clinics; the sub-centre and primary health centre team as laid down consists of the medical officer, health assistants and health workers, with no medico-social worker post (Park 28e, Ch 10, pp 742–744).
- ▶ **Numbers and skill mix** — counsellor posts are programme-tied and unevenly filled; under the adolescent programme, ICTC counsellors had to be enrolled in 23 States/Union Territories to cover the gap (Park 28e, Ch 7, p 556).
- ▶ **Role dilution** — where the managerial and liaison functions of the primary health centre are neglected and doctors work almost exclusively in a curative role, community liaison — the worker's core output — is the first casualty (Sunder Lal, Ch 17, p 755).
- ▶ **Dependence on the voluntary sector** — much social assistance in leprosy and disability is routed through non-governmental organisations and Departments of Social Welfare rather than the health department, so continuity depends on grants (Park 28e, Ch 5, p 396; Ch 9, p 689).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Draw the **medico-social worker** as the two-way link between the institution and the community — two labelled boxes joined by a broad two-headed arrow with the worker astride it:

- ▶ Left box — **INSTITUTION**: hospital ward and out-patient department, tuberculosis clinic, family planning clinic, cancer control centre, mental health service, maternal and child welfare clinic, school and university health service
- ▶ Centre, on the arrow — **MEDICO-SOCIAL WORKER** (*trained in social case work and in the art of interviewing*)
- ▶ Right box — **COMMUNITY**: family and household, local leaders, panchayat, school, voluntary agencies, Department of Social Welfare, social security and welfare schemes
- ▶ Arrow label pointing **community** → **institution**: *social case history; personal, economic and social causes of illness; social diagnosis*
- ▶ Arrow label pointing **institution** → **community**: *health education, counselling, follow-up, default retrieval, rehabilitation, mobilised community resources*
- ▶ Caption beneath: *The medico-social worker forms the link between the institution where she is employed and the community*

High-yield exam pointers

- ▶ **Open with the origin and the technique** — medical social work began in **England, 1895**, with the **Almoner** at the **Royal Free Hospital, London**; its main technique is **social case work**, and its output is the **social diagnosis**.
- ▶ **One-line definition of the role** — "the medico-social worker forms the **link between the institution where she is employed and the community**"; write this sentence verbatim.
- ▶ **Principal work** — visit the family, probe the **personal, economic and social causes of illness**, collect the social history to **supplement the medical history**, and secure help through **community resources**.
- ▶ **Three functions (Sunder Lal)** — assess the need for social work services of individual, family and community; arrange or provide them; participate in overall **planning and evaluation**.
- ▶ **Four rehabilitation domains** — **medical** (function), **vocational** (livelihood), **social** (family and social relationships), **psychological** (dignity and confidence); in leprosy the goal is **preventive rehabilitation**.
- ▶ **Behaviour-change chain** — awareness → motivation (**interest, evaluation, decision-making**) → action/adoption; and the five Ottawa Charter action areas for the health-promotion half of the answer.
- ▶ **Named programme posts to quote** — **psychiatric social worker** in the five-member District Mental Health Programme team; **ICTC and ART counsellors** under the National AIDS Control Programme; **RKSK adolescent health counsellors**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park 28e calls the medico-social worker a "paramedical worker", but the same textbook records that the World Health Organization no longer uses the term "paramedical" for the health professions allied to medicine (Park 28e, Ch 2, p 48), so prefer "allied health professional" if you comment on nomenclature; the Revised National Tuberculosis Control Programme is now the National Tuberculosis Elimination Programme, and the schemes for the disabled quoted from Park 28e sit under what is now the Ministry of Social Justice and Empowerment (named "Ministry of Social Welfare" in the text).

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 2 — Concept of Health and Disease; Ch 5 — Epidemiology of Communicable Diseases, leprosy; Ch 7 — Health Programmes in India; Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics; Ch 10 — Health Care of the Community; Ch 12 — Medicine and Social Sciences; Ch 22 — Communication for Health Education); Sunder Lal — Textbook of Community Medicine (Ch 15 — Mental Health; Ch 17 — Healthcare of the Community); Oxford Textbook of Public Health 7e (Ch 47 — Health promotion, health education and the public's health, Ottawa Charter action areas); Jugal Kishore — National Health Programmes of India (tuberculosis section). Coverage note: the corpus describes the medico-social worker's duties chiefly in the hospital/clinic setting and does not lay down a sanctioned community-level job description or staffing norm for the post, so the community role above is built from the worker's stated link function plus the counsellor and social-worker posts explicitly named in the national programmes; no captioned figure of the medico-social worker exists in the indexed corpus, so the diagrams here are structural rather than reproduced plates.



Women empowerment measures in India

Asked in: Paper I 26-May-2025 — RGUHS MD Community Medicine

Definition — Sex is biologically and genetically determined, whereas **gender** is the socially constructed set of roles assigned to women and men; "it is not sex, but gender that has far greater impact on women's health" (Sunder Lal, Ch 10, p 630). **Women's empowerment** is the process by which women gain control over resources, money and decisions — inside the household and outside it — the absence of which is the key determinant of ill health in Indian women. Its goal is **gender equality**, defined in the book as the state in which "women and men experience equal conditions for realizing their full human rights and have the opportunity to contribute to and benefit from national, political and economic, social and cultural development"; the route to it is **gender equity**, "the process of being fair to both men and women" (Sunder Lal, Ch 10, p 630). The empowerment of women is listed among the fundamental conditions and resources for health in the **Jakarta Declaration on Health Promotion, 1997** (Park 28e, Ch 2, p 49), and is the substance of Sustainable Development Goal 5 (Park 28e, Ch 14, p 854).

The gender disadvantage these measures are meant to correct

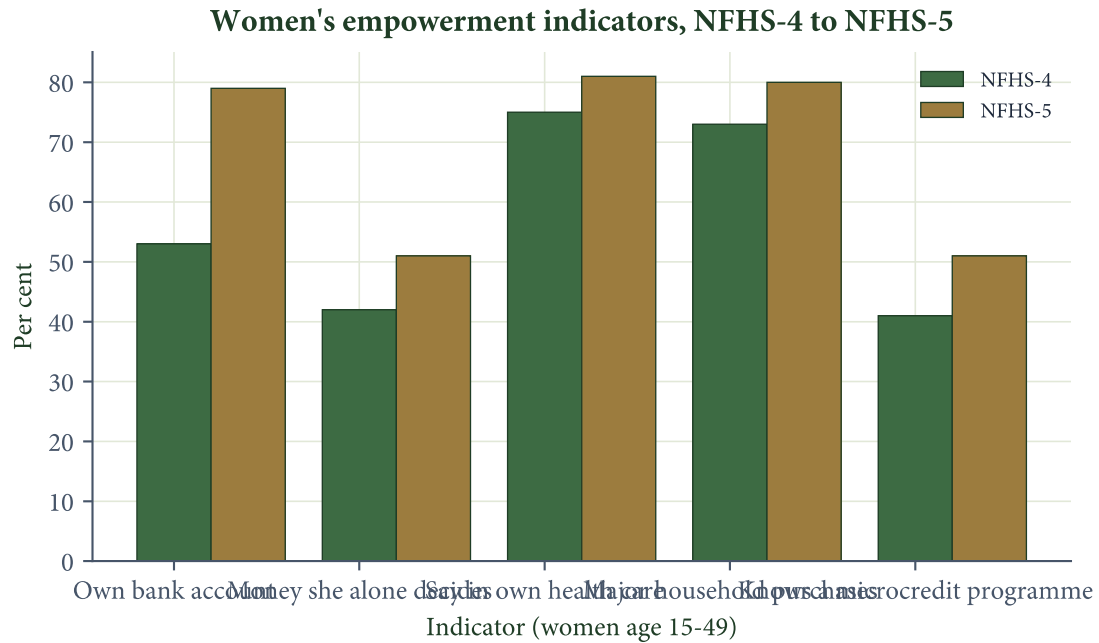
Domain	Current Indian position (with year)
Overall sex ratio	1020 females per 1000 males — favourable for the first time (NFHS-5, 2019–21); women were 48.46% of the population in Census 2011
Sex ratio at birth	929 (NFHS-5) improved from 919 (NFHS-4); natural expected ratio is 105 males per 100 females
Child sex ratio (0–6 y)	914 girls per 1000 boys (Census 2011), fallen 13 points from 927 (Census 2001); worst in Haryana (830) and Punjab (846) , lowest district Jhajjar, Haryana (774) (Sunder Lal, Ch 9, p 546)
Literacy	Overall 74%, male 82.5%, female 65.46% (Census 2011); only 39% of adult women reach secondary education vs 63.5% of men
Child marriage and adolescent fertility	23% of women aged 20–24 married before 18 years; 6.8% of women 15–19 already mothers or pregnant (NFHS-5)
Maternal mortality ratio	97 per 100,000 live births; Sustainable Development Goal target is 70 by 2030
Nutrition	57% of women aged 15–49 anaemic; 18.7% have body mass index below 18.5 (NFHS-5)
Work	Women form 29% of the labour force vs 81% of men; 89.5% of the women's workforce is landless
Contraceptive burden	Of 56.5% current users of modern methods, female sterilisation 37.9% vs male sterilisation 0.3%; condom 9.5% (NFHS-5)
Spousal violence	29.2% of ever-married women aged 18–49 have experienced physical or sexual violence by the husband; 31.9% emotional, physical or sexual (NFHS-5)
Political representation	Women hold 13.5% of seats in Parliament

Domain	Current Indian position (with year)
Composite index	Gender Inequality Index value 0.488, rank 132 of 187 countries (UNDP, 2020); ranked 140th of 162 on the World Economic Forum Global Gender Gap Index, 2021

(Sunder Lal, Ch 10, pp 630–633; NFHS-5 India Report, Ch 7, p 617 and Ch 8, pp 703–704)

How empowerment is measured

Measure	What it captures	Dimensions / components
Gender Empowerment Measure (GEM) , UNDP — the index this question names	Political and economic participation and control wielded by women, computed for individual Indian States; higher GEM = greater empowerment	<i>Political participation</i> — number of women legislators at the Centre, in the Assembly and in Panchayati Raj institutions, number of women candidates fielded by parties, and per cent women voters. <i>Economic participation</i> — share of jobs held by women in the civil services, share of women among medical and engineering graduates, and among High Court and Supreme Court judges (Mahajan & Gupta 4e, Ch 26, p 537)
Gender Development Index (GDI) , UNDP	Gender gap in human development; higher GDI = greater gender equality ; India's GDI improved 0.568 (1996) → 0.633 (2006)	Infant mortality, life expectancy at age one, literacy rate, average years of education and estimated income per capita, compared for girls versus boys. Chandigarh topped GDI and Andhra Pradesh topped GEM in 2006; Bihar was bottom on both GDI and GEM in 1996 and 2006 (Mahajan & Gupta 4e, Ch 26, pp 537–538)
Gender Inequality Index (GII) , UNDP	Loss in human development due to gender inequality	Reproductive health, empowerment, economic (labour-market) activity
Global Gender Gap Index , World Economic Forum	Gap between men and women, ranked across countries	Includes a political-empowerment sub-index, on which India is among the top 20 performers
Human Development Index (HDI) — sex-disaggregated	Aggregate progress in health, education and income, read separately for women	Life expectancy at birth; mean and expected years of schooling; gross national income per capita (Sunder Lal, Ch 1, p 49)
Sex ratio, sex ratio at birth, child sex ratio	Survival and social value of the female — "an indicator of the low status of women in India"	Census and NFHS; sex ratio at birth is the more refined measure, being free of post-birth mortality and neglect (Sunder Lal, Ch 9, p 546)
NFHS women's empowerment module	Individual-level empowerment, the operational Indian measure	Employment and cash earnings, control over own earnings, participation in three household decisions, attitudes to wife beating, house and land ownership, bank account and mobile phone use (NFHS-5 India Report, Ch 2, pp 39–40)
Gender-disaggregated programme data	Whether services actually reach women	All primary health centre and ICDS reports must be gender-wise for morbidity, mortality, nutrition and service use



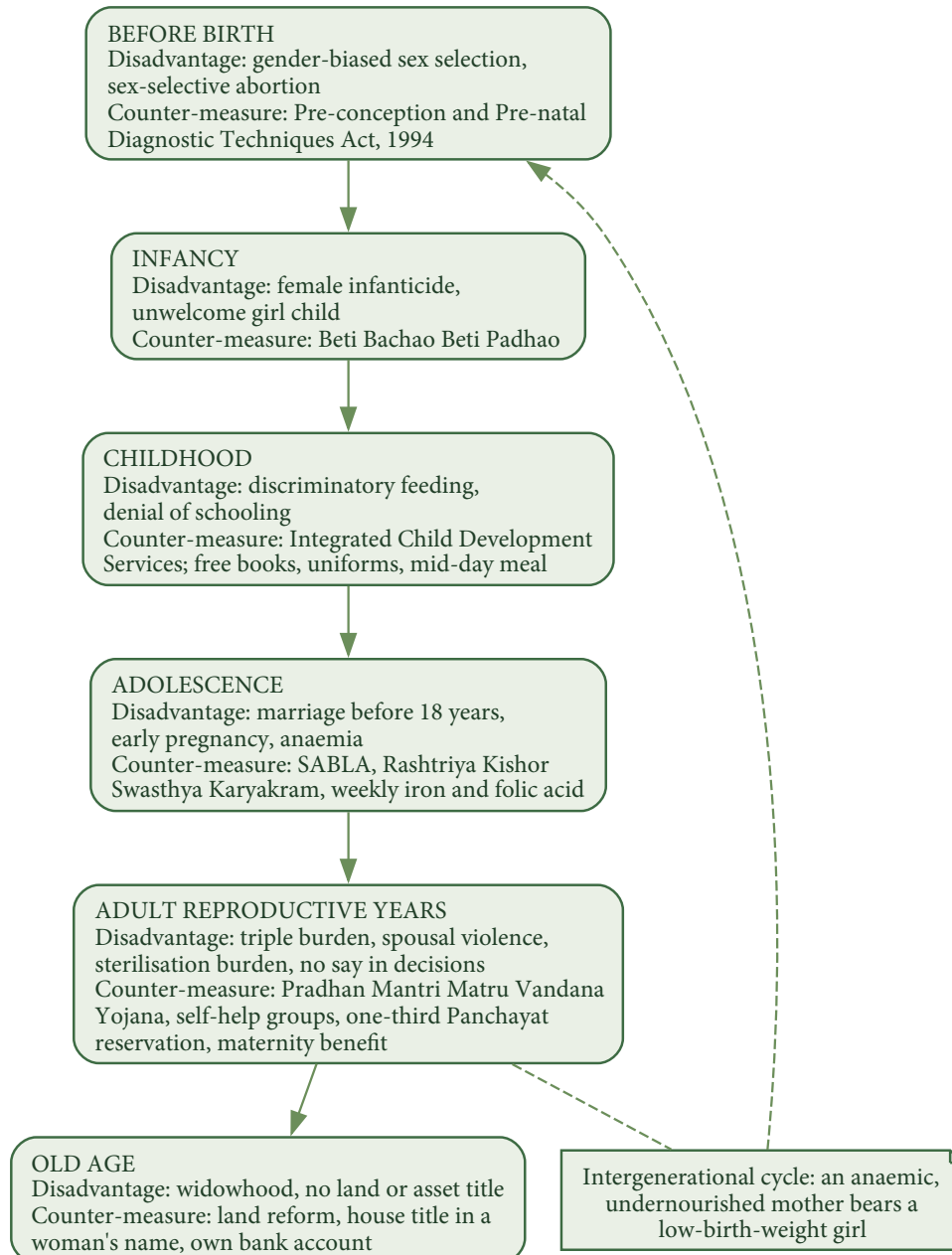
Source: NFHS-5 India Report (2019-21), Ch 7, pp 619-622, with NFHS-4 comparison figures as given in the same report.

Measures taken in India

Domain	Instrument	Key provision
Constitutional	Constitution of India	Grants equality to women and empowers the State to adopt positive discrimination in their favour
Political	73rd and 74th Constitutional Amendment Acts (passed December 1992, law April 1993)	Not less than one-third of total seats and of chairperson posts at every level of Panchayats and Nagar Palikas reserved for women; 158 of 475 Zila Parishads are chaired by women, 17,000 of 51,000 block samiti members are women
Policy	National Policy for the Empowerment of Women , adopted 2001 (Department of Women and Child Development, then under the Ministry of Human Resource Development)	Umbrella policy; the Ministry of Women and Child Development is now the nodal ministry for women's and girl-child schemes (Sunder Lal, Ch 10, pp 633–634)
Planning and finance	Women's Component Plan (Ninth Plan); Gender Budget cells in all departments from 2005	30% of funds earmarked for women-related sectors; gender budgeting mandatory in all central and state health programmes, including RCH-II and the National Rural Health Mission
Legislative — protection	Dowry Prohibition Act 1961; Immoral Traffic (Prevention) Act; Indecent Representation of Women (Prohibition) Act 1986; Commission of Sati (Prevention) Act 1987; Muslim Women (Protection of Rights on Marriage) or "triple talaq" Act, July 2019	Criminalise dowry, trafficking, indecent portrayal, sati and instant talaq

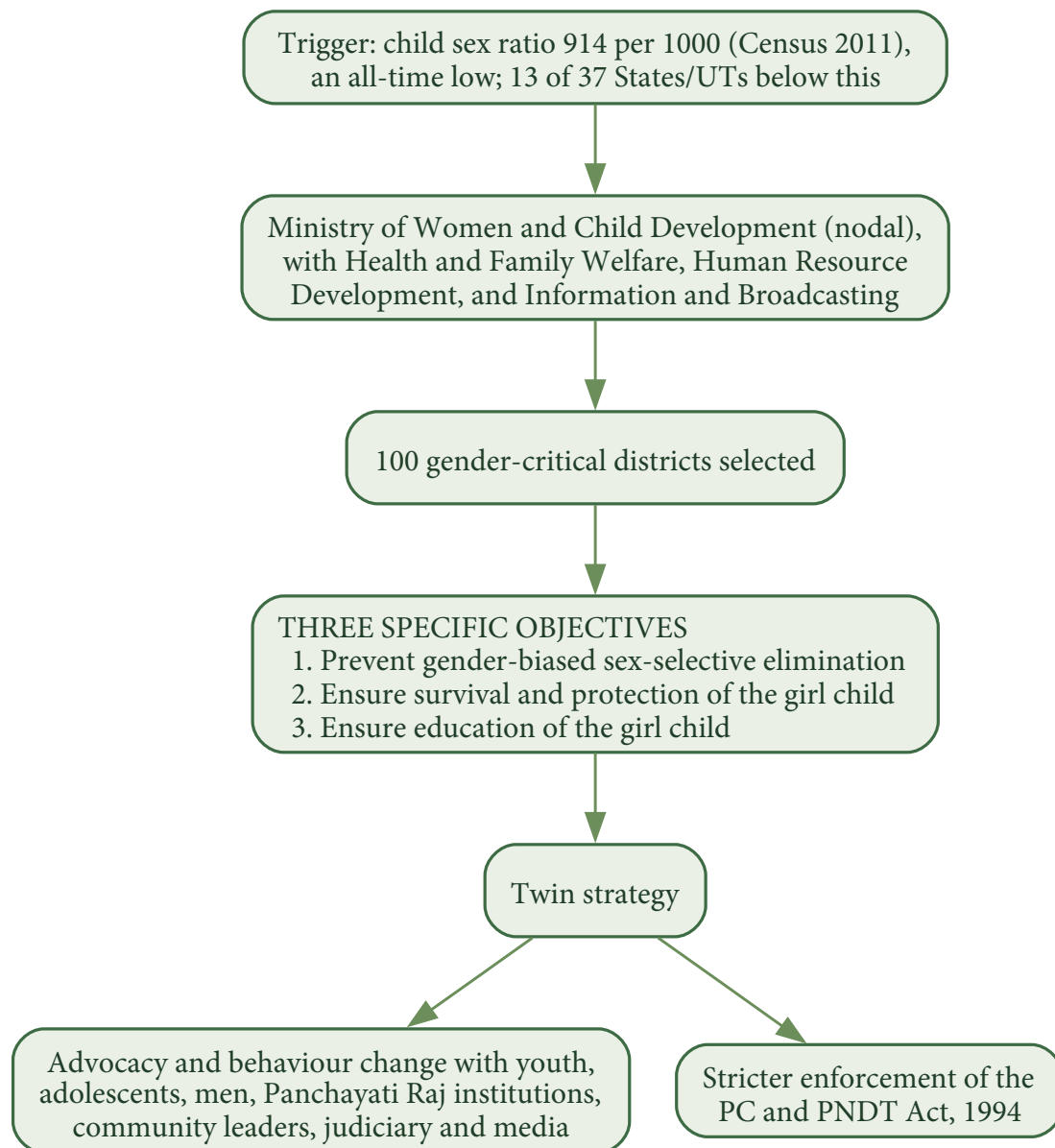
Domain	Instrument	Key provision
Legislative — reproductive	Pre-conception and Pre-natal Diagnostic Techniques (Prohibition of Sex Selection) Act, 1994; Medical Termination of Pregnancy Act	Bans sex determination and sex-selective abortion, to raise the sex ratio at birth
Legislative — workplace	Maternity Benefit (Amendment) Act, 2017; Employees' State Insurance Act 1948; Factories Act; Indian Mines Act 1923	26 weeks paid maternity leave (up to 8 weeks before confinement; 12 weeks after two children); crèche where more than 50 women are employed; no night work between 7 pm and 6 am; no underground work (Park 28e, Ch 18, p 967)
Economic	Swarnajayanti Gram Swarozgar Yojana with DWCRA merged into it; self-help groups ; land reform	40% of benefits earmarked for women; group income generation and distribution of surplus land to self-help group women; 78% of women now have a savings bank account and 54% use a mobile phone themselves (Sunder Lal, Ch 10, p 633)
Housing	Pradhan Mantri Awas Yojana–Urban	House ownership in the name of a female member or in joint name; preference to widows and single women (Park 28e, Ch 7, p 575)
Educational	"Education for All"; National Education Policy 2020; women's Industrial Training Institutes	Free textbooks, scholarships, free uniforms, mid-day meals, flexible school timings, attached hostels and crèches to raise girls' enrolment and retention; gender sensitisation of teachers
Health-sector	National Health Policy 1983, 2002 and 2017; National Population Policy 2000; National Nutrition Policy; Pradhan Mantri Matru Vandana Yojana (₹5,000 in three instalments)	Reduce maternal and infant mortality and address gender issues in health
Adolescent girls	Kishori Shakti Yojana; SABLA (Rajiv Gandhi Scheme for Empowerment of Adolescent Girls), 11–18 years; Rashtriya Kishor Swasthya Karyakram	Nutrition, health, life-skills, vocational and self-development inputs delivered through ICDS
Girl child	Beti Bachao Beti Padhao (see flowchart)	Improve child sex ratio and girls' education
Community platforms	Mahila Swasthya Sangh, Mahila Arogya Samiti (10–12 women per 50–100 urban slum households, annual grant ₹5,000), Mahila Mandals, Village Health Sanitation and Nutrition Committee	Women's collectives as peer educators, demand generators and community-based monitors (Sunder Lal, Ch 17, p 758; Park 28e, Ch 7, p 535)
Media and environment	Portrayal of women consistent with human dignity; removal of gender stereotypes from curricula and textbooks; smokeless <i>chullah</i> , biogas, LPG under Pradhan Mantri Ujjwala Yojana	Reduce drudgery, indoor air pollution and stereotyping

Domain	Instrument	Key provision
International commitment	CEDAW — Convention on the Elimination of All Forms of Discrimination Against Women, adopted by the UN General Assembly 1979, ratified by India on 25 June 1993 ; Beijing Conference on Women 1995; SDG 5	Binding framework for eliminating discrimination



Gender disadvantage across the female life cycle and the Indian counter-measure at each stage

Beti Bachao Beti Padhao — the flagship girl-child measure



Beti Bachao Beti Padhao - structure and objectives

Role of the community physician

- ▶ **Enforce the PC and PNDT Act in your district** — registration and inspection of ultrasound clinics, sealing of machines, and prosecution of sex determination.
- ▶ **Practise gender budgeting and gender-disaggregated reporting** — insist that primary health centre and ICDS returns for morbidity, mortality, nutrition and service utilisation are split by sex, and judge programme performance on that split.
- ▶ **Use women's collectives as the delivery platform** — Mahila Arogya Samiti, Mahila Swasthya Sangh, Mahila Mandals and self-help groups for interpersonal communication, community-based monitoring and demand generation, supported by the accredited social health activist and *anganwadi* worker.

- **Target the adolescent girl** — school retention, weekly iron and folic acid supplementation, delaying marriage beyond 18 years and first birth beyond 19 years, through SABLA and the Rashtriya Kishor Swasthya Karyakram.
- **Involve men** — male participation in contraception (male sterilisation is only 0.3% of modern method use) and in sharing domestic workload, since discrimination "begins and ends at the family level" and changing family attitudes is the core strategy of empowerment.
- **Screen and respond to violence** — antenatal and clinic screening for spousal violence with linkage to support services, since economic-plus-gender empowerment interventions are among the few shown to reduce violence against women (Oxford Textbook of Public Health 7e, Ch 82, pp 3114–3115).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Draw: child sex-ratio trend chart** — x-axis Census years 1991, 2001 and 2011; y-axis girls per 1000 boys aged 0–6 years. Plot All India 945 → 927 → 914, and two contrast lines: Punjab 875 → 798 → 846 and Haryana 879 → 819 → 830. Mark the Pre-conception and Pre-natal Diagnostic Techniques Act, 1994 on the time axis, and annotate the 2001→2011 upturn in Punjab and Haryana as the effect of its enforcement (Sunder Lal, Ch 9, Table 9.5, p 546).

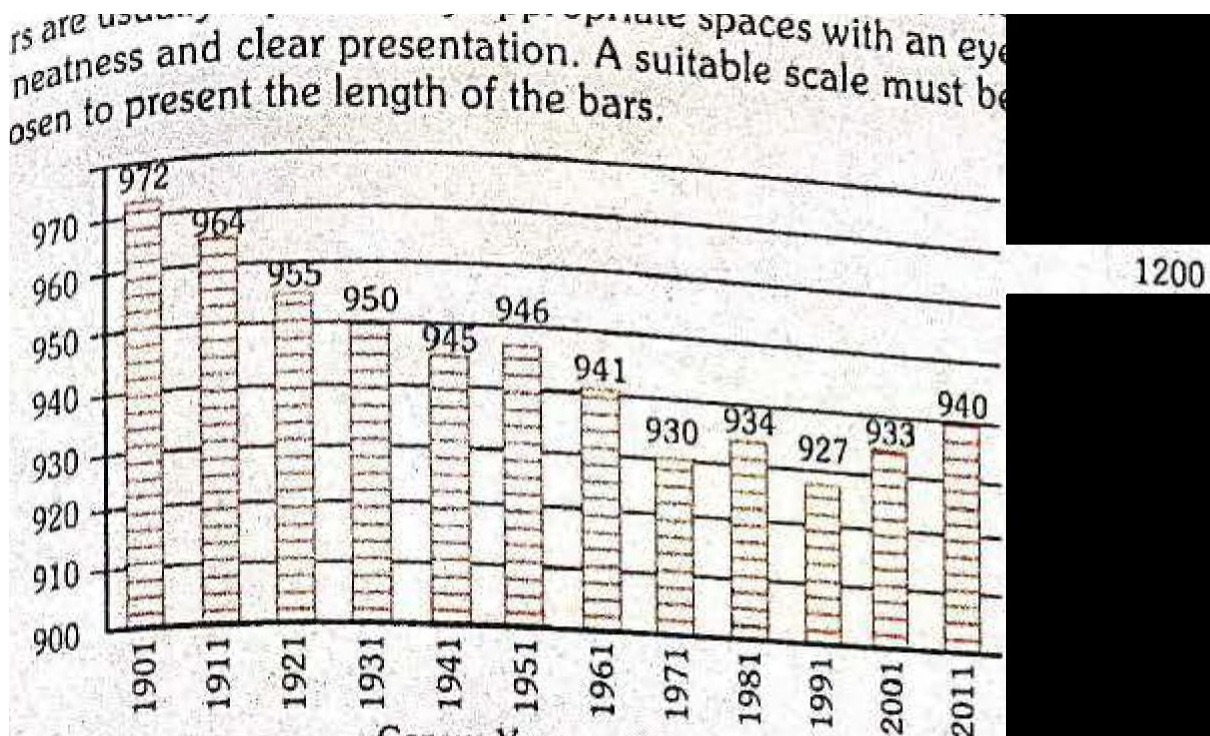


Fig 1 — India; Sex Ratio 1901–2001 (Park's Textbook of Preventive and Social Medicine, 28e, p.1001)

High-yield exam pointers

- **Definitions to open with:** sex is biological and genetic; **gender** is socially constructed and "it is not sex, but gender that has far greater impact on women's health"; gender **equity** (fairness) leads to gender **equality** (equal status).

- ▶ **The named indices:** GEM — political participation (women legislators at Centre, Assembly and Panchayati Raj institutions, women candidates, women voters) plus economic participation (women in civil services, medical and engineering graduates, High Court and Supreme Court judges); GDI — infant mortality, life expectancy at age one, literacy, average years of education and income per capita, girls versus boys; higher value = better on both. **Bihar bottom on both, 1996 and 2006.**
- ▶ **Five-domain mnemonic — "P-L-E-E-H":** Political/constitutional, Legislative, Economic, Educational/social, Health and programmatic.
- ▶ **Numbers that must appear:** child sex ratio 914 (Census 2011) from 927 (2001), worst Haryana 830 and Punjab 846, Jhajjar district 774; sex ratio at birth 929 and overall sex ratio 1020 (NFHS-5, 2019–21); female literacy 65.46% (Census 2011); MMR 97 with SDG target 70 by 2030; anaemia 57% in women 15–49 (NFHS-5); 29.2% spousal physical or sexual violence (NFHS-5); women 13.5% of Parliament; GII 0.488, rank 132 (UNDP 2020); India's GDI 0.633 (2006).
- ▶ **Dates that carry marks:** CEDAW 1979, ratified by India 25 June 1993; 73rd and 74th Amendments 1992, in force April 1993; PC and PNDT Act 1994; National Policy for the Empowerment of Women 2001; Gender Budget cells 2005; Maternity Benefit (Amendment) Act 2017.
- ▶ **BBBP in one line:** Ministry of Women and Child Development, 100 gender-critical districts, three objectives — prevent sex-selective elimination, ensure survival and protection, ensure education of the girl child.
- ▶ **Close with the core strategy:** discrimination begins and ends at the family level, so changing societal attitudes with active participation of both men and women is the core strategy of women's empowerment.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written: the retrieved textbooks predate the **106th Constitutional Amendment (Nari Shakti Vandan Adhiniyam), 2023**, which reserves one-third of Lok Sabha and State Legislative Assembly seats for women, so Sunder Lal's "move for 33% reservation is gaining strength" and the figure of 13.5% women in Parliament are now historical. The Gender Inequality Index, introduced in 2010, replaced the Gender Empowerment Measure in the United Nations Development Programme's Human Development Report series, but the Gender Empowerment Measure is the index this question names and the one on which Indian state rankings are published, so write both. Sunder Lal's Ch 10 (p 631) prints Haryana 820 and Punjab 793 as the Census 2011 child sex ratios; those are the Census 2001 values, and the same book's Ch 9 Table 9.5 (p 546) gives the correct 2011 figures of 830 and 846 — quote 830 and 846. Sunder Lal dates the 26-week maternity leave revision to 2016 and Park to the Maternity Benefit (Amendment) Act, 2017 — quote the 2017 Act with 26 weeks.

SOURCE & COVERAGE

Sunder Lal — Textbook of Community Medicine, Ch 10 (Reproductive, Maternal, Newborn, Child and Adolescent Health, RMNCH+A — sections on Gender Issues and Women Empowerment, and Beti Bachao Beti Padhao), pp 630–634; Ch 1, pp 49–52; Ch 9, pp 545–546; Ch 17, p 758. Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e, Ch 26 (Health Planning, Administration and Management), pp 537–538 (Gender Development Index and Gender Empowerment Measure). Park's Textbook of Preventive and Social Medicine, 28e — Ch 2, p 49 (Jakarta Declaration on Health Promotion); Ch 7, pp 535, 575; Ch 14, p 854 (Sustainable Development Goal 5 targets); Ch 18, p 967 (protection of women workers). National Family Health Survey-5 India Report (2019–21), Ch 2, pp 39–40, Ch 7, pp 617–622, Ch 8, pp 703–705. Oxford Textbook of Public Health 7e, Ch 82, pp 3114–3115. Census of India 2011; National Policy for the Empowerment of Women 2001; CEDAW (ratified by India 1993).



Methods of communication — individual vs group approach

Asked in: Paper I 19-Nov-2020 — RGUHS MD Community Medicine

Definition — Communication is a **two-way process of exchanging or shaping ideas, feelings and information**, and refers broadly "to the countless ways that humans have of keeping in touch with one another". It is more than the mere exchange of information: it is the process that paves the way for desired changes in human behaviour and for informed individual and community participation in achieving predetermined goals. Its components are **sender (source)**, **receiver (audience)**, **message (content)**, **channel (medium)** and **feedback (effect)** (Park 28e, Ch 22, p 1010).

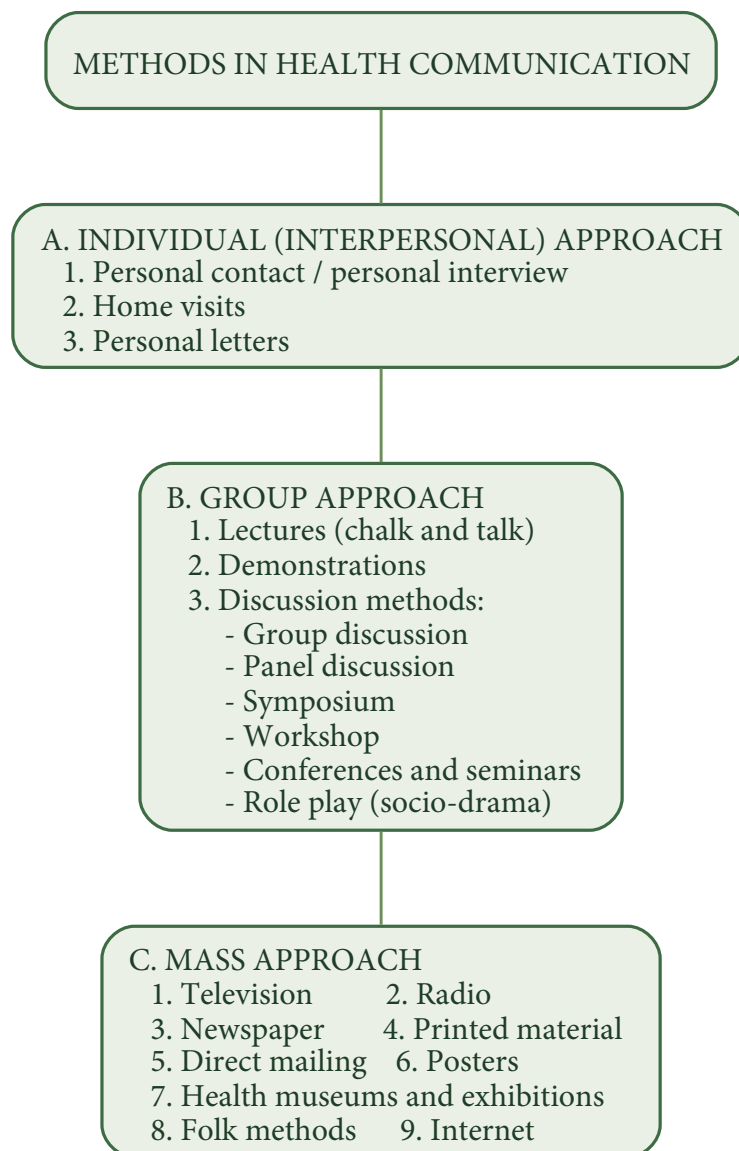
Types of communication

Type	Description	Key points
1. One-way (didactic method)	Flow of communication is one-way from communicator to audience; the familiar example is the lecture method in class rooms	Drawbacks: knowledge is imposed; learning is authoritative; little audience participation; no feedback ; does not influence human behaviour
2. Two-way (Socratic method)	Both communicator and audience take part; audience may raise questions and add their own information, ideas and opinions	Learning is active and democratic ; more likely to influence behaviour
3. Verbal	Communication by word of mouth — the oldest form	May be loaded with hidden meanings; direct verbal communication is persuasive
4. Non-verbal	Communication occurring without words — bodily movements, postures, gestures, facial expressions (smile, raised eyebrow, frown, staring, gazing)	Silence is non-verbal communication; it can speak louder than words

Type	Description	Key points
5. Formal and informal	Formal follows the lines of authority ; informal is the " grape vine " (e.g. gossip circles), present in all organisations	Informal channels become more active if formal channels do not cater to information needs
6. Visual	Charts and graphs, pictograms, tables, maps, posters	Supports and reinforces verbal messages
7. Telecommunication and internet	Communicating over distance using electromagnetic instruments	Radio, television and internet are mass media ; telephone is a point-to-point system, closer to interpersonal communication

The total communication effort rests on three media systems — (a) **interpersonal communication**, (b) **mass media**, and (c) **traditional or folk media** (Park 28e, Ch 22, pp 1011-1012).

Enlist: methods in health communication



Methods in health communication — the individual, group and mass approaches (after Park 28e, Ch 22, Fig 3, p 1020)

Any one or a **combination** of these methods may be used selectively at different times, depending on the objective to be achieved, the behaviour to be influenced and the funds available (Park 28e, Ch 22, p 1020).

Individual (interpersonal) approach

- ▶ **What it is** — one-to-one, face-to-face communication between the health worker and the individual or care-giver. Being personal and direct, it is more persuasive and effective than any other form, and its superiority over mass media for creating a **motivational effect** is well documented (Park 28e, Ch 22, p 1011).
- ▶ **Where the opportunities arise** — personal interviews in the **consultation room of the doctor**, in the **health centre**, or in the **homes of the people**. The individual comes because of illness; the opportunity is taken to educate on diet, the causation, nature and prevention of illness, personal hygiene and environmental hygiene (Park 28e, Ch 22, p 1020).
- ▶ **How it is conducted** — the health educator must first **create an atmosphere of friendship** and allow the individual to talk as much as possible; topics for health counselling are selected according to the relevance of the situation.
- ▶ **In Indian field practice** — it is the core of the **home-visit and beat programme** of health workers, and is the chief strategy for **behaviour change communication (BCC)**. It is strengthened through *Mahila Swasthya Sangh* (MSS), joint home visits by Integrated Child Development Services (ICDS) *Anganwadi* workers with health workers, *Mahila Mandals*, *Mahila Arogya Samitis*, elected women *Panchayat* members and the accredited social health activist (ASHA). The worker's skills must be **polite, friendly, empathic and patient**, and are sharpened by repeated training in group dynamics (Sunder Lal, Ch 4, p 146).
- ▶ **Advantages** — the biggest advantage is that we can **discuss, argue and persuade** the individual to change behaviour; it gives the person the chance to ask questions in terms of his own specific interests; the message passes directly from source to receiver, and is the most interactive and participatory of all methods.
- ▶ **Limitations** — the **numbers reached are small**; education is given only to those who come in contact with us; coverage is minimal and it is **time-consuming**.

Group approach

- ▶ **Basis** — society contains groups of many kinds: school children, mothers, industrial workers, patients. Group teaching is an **effective way of educating the community**.
- ▶ **Choice of subject is critical** — it must relate directly to the interest of the group. We should not broach the subject of tuberculosis control to a mother who has come for delivery; we should talk to her about child-birth and baby care. Similarly, school children are taught oral hygiene, tuberculosis patients about tuberculosis, and industrial workers about accidents (Park 28e, Ch 22, pp 1020-1021).

Methods of group teaching (Park 28e, Ch 22, pp 1021-1023):

Method	Conduct and optimum size	Key merit / limitation
Chalk and talk (lecture)	Carefully prepared oral presentation of facts and organised ideas by a qualified person; group not more than 30 ; talk not to exceed 15-20 minutes	Aids: flipcharts (about 25 x 30 cm), flannelgraph, exhibits, films and charts. Limitation: minimal involvement, learning is passive , does not stimulate problem-solving, health behaviour not necessarily affected
Demonstration	A carefully prepared presentation showing how to perform a skill or procedure , step by step, before the target group	Upholds " seeing is believing " and " learning by doing "; high motivational value ; used for hand-pump installation, sanitary latrine construction, oral rehydration technique, scabies control
Group discussion	A "group" is an aggregation of people interacting in a face-to-face situation ; not less than 6 and not more than 12 members , seated in a circle so each is fully visible to all; a group leader initiates and sums up, a recorder prepares the group report	Permits free exchange of knowledge, ideas and opinions; valuable where long-term compliance is needed (smoking cessation, obesity reduction). Limitations: the shy do not speak, a few dominate , unequal participation, deviation from the subject
Panel discussion	4 to 8 qualified persons plus a chairman/moderator discuss the topic before a large audience; no set agenda, no order of speaking, no set speeches ; audience joins afterwards	Success depends entirely on the chairman keeping the train of thought going
Symposium	A series of speeches on a selected subject, each expert presenting one aspect briefly	No discussion among the members (unlike a panel); audience may raise questions; chairman gives a comprehensive summary
Workshop	A series of meetings, usually four or more , with emphasis on individual work within the group, aided by consultants and resource personnel; divided into small groups each with a chairman and a recorder	Participants leave with a plan of action on the problem; learning in a friendly, democratic atmosphere
Role play (socio-drama)	The group dramatises a situation; group size best at about 25 ; always followed by discussion	Useful for problems of human relationship ; particularly useful for school children ; audience is active, not passive
Conferences and seminars	Held at regional, state or national level; half-a-day to one week in length	Largely continuing education; a single topic in depth or broadly comprehensive

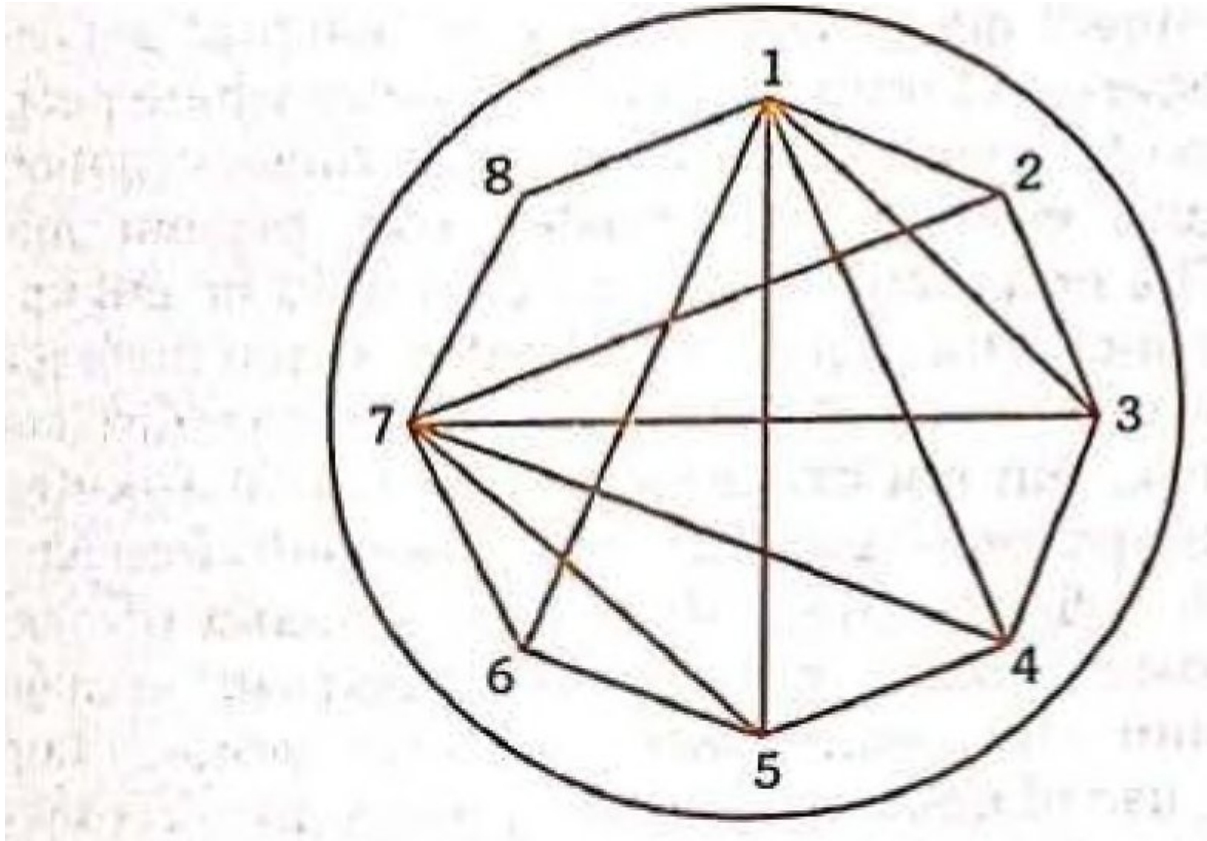


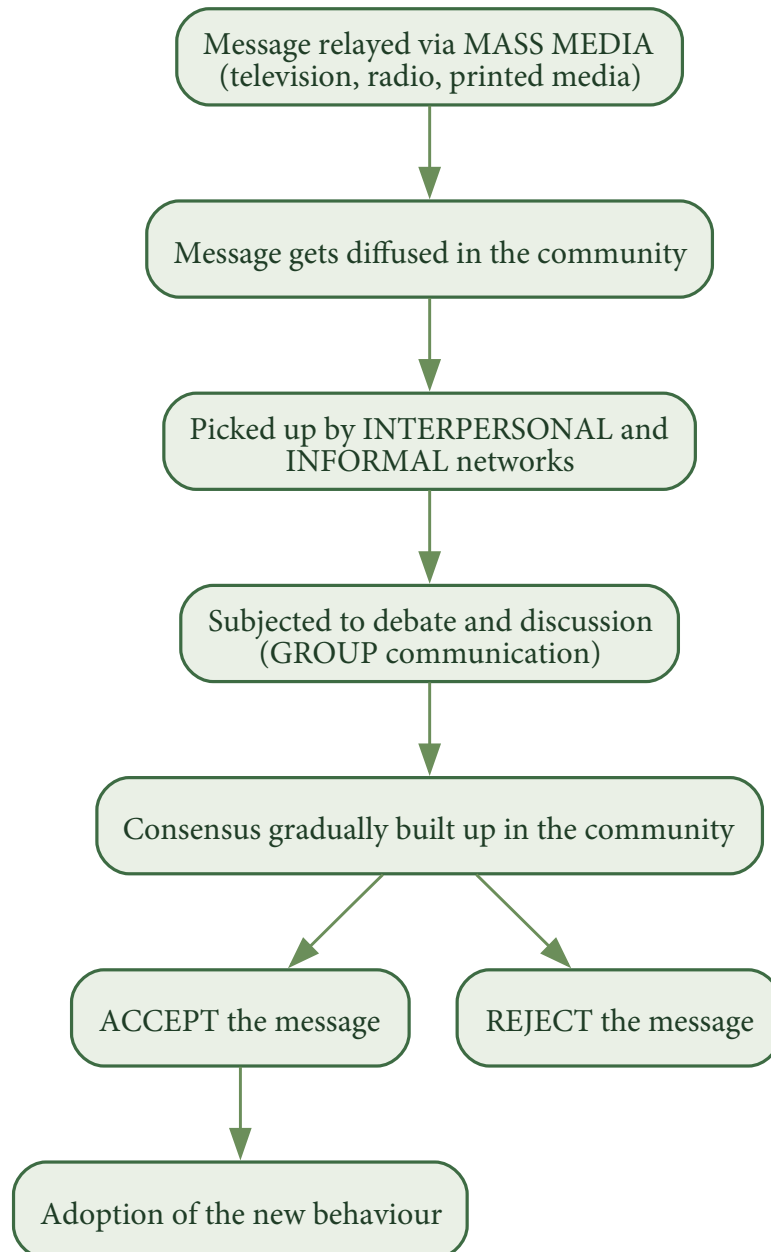
Fig 5 — , A dominated group. No.1 and No.7 dominate the discussion (Park's Textbook of Preventive and Social Medicine, 28e, p.1022)

Individual versus group approach — comparison

Feature	Individual (interpersonal) approach	Group approach
Unit addressed	One person or one family / care-giver	An organised group with a shared interest (mothers, school children, workers, patients)
Setting	Consultation room, health centre, home visit , personal letter	Village meeting, school, factory, <i>Mahila Swasthya Sangh</i> / Village Health Sanitation and Nutrition Committee meeting
Optimum number	One (with the family)	Lecture up to 30; group discussion 6-12; panel 4-8; role play about 25
Direction of flow	Two-way, direct source to receiver	Two-way, but multi-directional — wider interaction among members than any other method
Message	Precise, specific , tailored to that individual's felt need	Common to the group; must relate directly to the group's interest
Persuasive power	Highest — we can discuss, argue and persuade the individual	High — decisions taken by the group have a binding effect on the individual member, so they are adopted more readily than a solitary decision

Feature	Individual (interpersonal) approach	Group approach
Feedback	Immediate, complete, and doubts and fears can be voiced privately	Immediate but shared; may be distorted by dominant members
Coverage per unit time	Very low — only those who come in contact with us	Moderate — many reached in one sitting
Cost and manpower per person reached	High; time-consuming	Lower; more economical
Skill demanded	Interpersonal communication and counselling skills	Group dynamics , leadership and facilitation skills
Chief use	Counselling, behaviour change communication (BCC) in sensitive matters — family planning, HIV, adolescent health, treatment adherence	Changing shared attitudes and practices — breastfeeding, sanitation, anaemia, immunisation, smoking cessation
Main limitation	Numbers reached are small; labour-intensive	Shy members silent, a few dominate, unequal participation, drift from the subject
Suitability	Best where the message is private, individual-specific or emotionally charged	Best where long-term compliance and social reinforcement are required

Personal communication (interpersonal and group methods together) **capitalises on warmth and understanding**, provides the opportunity for involvement, for asking questions and expressing fears, gets people to change personal habits more readily when the discussion presents reasonable explanations, and is **more influential with average and below-average educational levels** — whereas mass media reaches the widest population, gets public attention, supports concentrated campaigns, and is more effective among those with **above-average educational level** (Park 28e, Ch 22, Table 2, p 1021).



The two-step flow — how a mass-media message is converted into behaviour change through interpersonal and group communication (after Park 28e, Ch 22, p 1011)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Methods in health communication (Park, Fig 3)** — three vertical columns under one heading. Labels: *Individual approach* — personal contact, home visits, personal letters; *Group approach* — lectures, demonstrations, discussion methods (group discussion, panel discussion, symposium, workshop, conferences and seminars, role play); *Mass approach* — television, radio, newspaper, printed material, direct mailing, posters, health museums and exhibitions, folk methods, internet.
- ▶ **Sociogram of a group discussion** — a large circle with 8 members numbered 1 to 8 seated around it, and lines drawn between members to show who talks to whom. Labels: *A well-conducted group discussion* (lines spread evenly among all 8) versus *A dominated group* (most lines converging on members No. 1 and No. 7).

High-yield exam pointers

- ▶ **Three approaches = I-G-M:** Individual, Group, Mass (Park, Fig 3) — the enlisting the examiner is looking for.
- ▶ **Numbers to write:** group discussion 6-12 members; panel discussion 4-8 panellists plus a chairman; lecture group **not more than 30**, duration **15-20 minutes**; role play group **about 25**; workshop **four or more** meetings.
- ▶ One-way = **didactic** (lecture; no feedback); two-way = **Socratic** (active, democratic, changes behaviour).
- ▶ Seven types of communication: one-way, two-way, verbal, non-verbal, formal/informal, visual, telecommunication and internet. Three media systems: interpersonal, mass media, traditional/folk.
- ▶ Individual approach — biggest advantage is that we can **discuss, argue and persuade**; biggest limitation is that **numbers reached are small**.
- ▶ Group approach — a group decision has a **binding effect** on the member and is adopted more readily than a solitary decision; demonstration works on "**seeing is believing, learning by doing**".
- ▶ Mass media suits the **above-average** educational level; personal communication suits the **average and below-average** level.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park's Fig. 3 places conferences and seminars under the group approach while listing exhibitions with health museums under the mass approach; some Indian textbooks list exhibitions under group methods, so either placement is accepted provided the three-approach framework is stated correctly.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Communication for Health Education (the communication process, types of communication, Fig. 3 methods in health communication, Table 2 mass media versus personal communication, individual and group approaches, methods of group teaching, Fig. 5 dominated group); Sunder Lal — Textbook of Community Medicine, Principles of Health Promotion and Education (interpersonal and group communication, merits and demerits, Indian field-practice platforms — *Mahila Swasthya Sangh*, *Anganwadi worker*, *ASHA*, Village Health Sanitation and Nutrition Committee).



Social problems and their prevention

Asked in: Paper I 06-Apr-2026 — RGUHS MD Community Medicine

Definition — In a community there are both individual and social problems; **when individual problems come to affect a large number of people, they become social problems** (Park 28e, Ch 12, p 839). Park names **alcoholism, drug dependence, sexually transmitted disease (STD), vagrancy, juvenile delinquency and prostitution**, and several of these carry a direct medical implication (Park 28e, Ch 12, p 839).

Social pathology and what makes a condition a social problem

Social pathology is the term that links disease to adverse social conditions — classically **poverty, crime, delinquency and vagrancy**. It is uncovered by **social surveys**; it was such a survey by *Chadwick* that led to the founding of the **General Board of Health in 1848** in Great Britain (Park 28e, Ch 12, p 816).

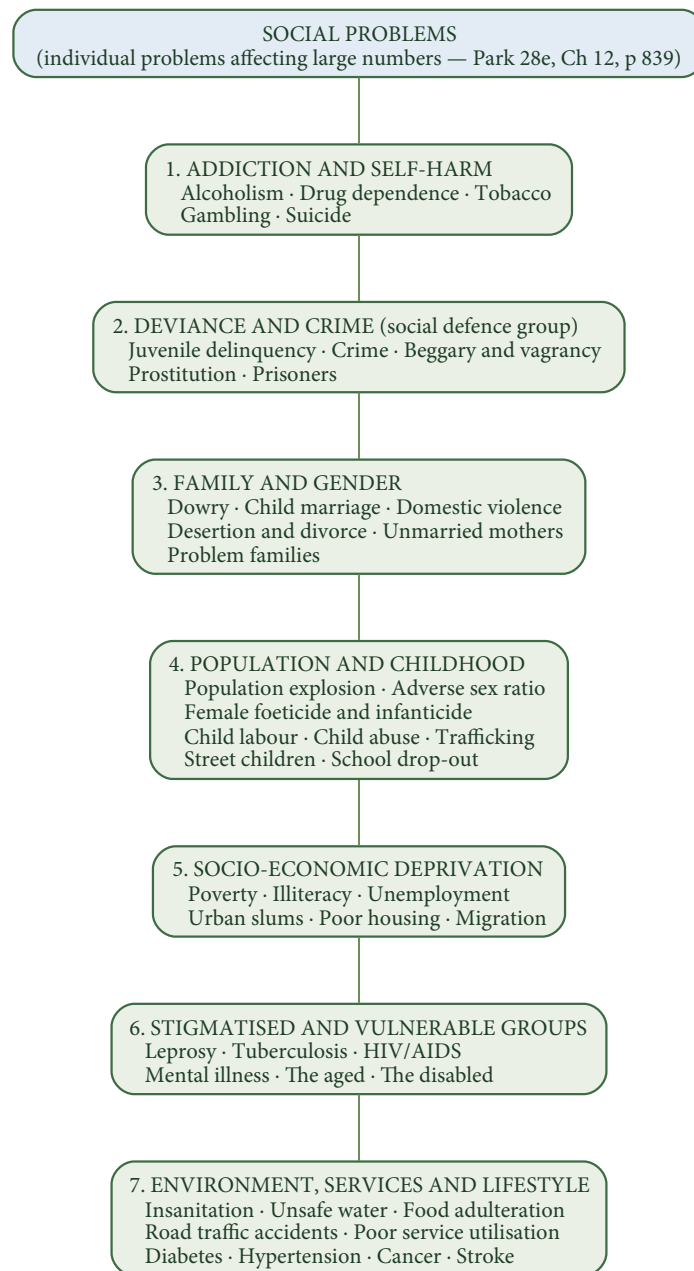
A disease or condition acquires high social relevance and is called a social problem when (Sunder Lal, Ch 2, p 81):

- ▶ It has a **predominant behavioural component in its causation** — for example lifestyle diseases.
- ▶ It is **widespread over a large geographical area and affects many people** (very high incidence and prevalence).
- ▶ It is **severe, with very high mortality**.
- ▶ It leads to **permanent disability or crippling**.
- ▶ It leads to **discrimination and ostracism** (social stigma).
- ▶ It **disrupts trade and affects the economy** of the nation.
- ▶ It affects the **economically productive population** most and leads to poverty.
- ▶ It **threatens the growth and development of young children**.
- ▶ **Treatment is beyond the reach of many**.
- ▶ The disease is **incurable**.

Enumeration of social problems

Group	Social problems	Underlying behaviour or social factor
Addiction and self-harm	Alcoholism; drug dependence / substance use; tobacco use; gambling; suicide	Alcohol and tobacco use in family and peer groups, socially accepted as a status symbol; drug trafficking, illicit use by adolescents and youth, sharing of needles and syringes; stress and inability to cope (Sunder Lal, Ch 2, pp 81-82; Park 28e, Ch 12, p 816)
Deviance and crime	Juvenile delinquency; crime and criminals; beggary and vagrancy; prostitution; prisoner welfare and prison reform	Delinquency arises from social maladjustment, poverty, disturbed home conditions, alcoholism, drug addiction and modern ways of living (Park 28e, Ch 12, p 840); prostitution from change of environment, breakdown of family relations, parental quarrels, want of affection, illegitimate love, easy money, low intelligence quotient (IQ), low moral standards and poverty (Park 28e, Ch 12, p 839)
Family and gender	Dowry system with bride-burning and suicide; child marriage; domestic and marital violence; desertion, polygamy, divorce; unmarried mothers; problem families	Problem families show backwardness, poverty, illness, mental and emotional instability, character defects and marital disharmony; their children become victims of prostitution, crime and vagrancy (Park 28e, Ch 12, pp 828-829); early marriage, dowry harassment and unpaid, unrecognised work of women (Park 28e, Ch 9, p 691). Magnitude (NFHS-5, 2019–21): 32% of ever-married women aged 18–49 have experienced spousal physical, sexual or emotional violence (physical 28%, emotional 14%, sexual 6%) and only 14% of women who experienced violence ever sought help; 23% of women aged 20–24 were married before the legal minimum age of 18 (NFHS-5 India Report, Ch 8, p 677 and Ch 4, p 244)
Population and gender selection	Population explosion; adverse sex ratio and falling child sex ratio	High fertility, early marriage, non-acceptance of or no belief in contraceptives; female foeticide, female infanticide and gender bias (Sunder Lal, Ch 2, p 81)
Childhood	Child labour and child trafficking; child abuse and neglect (battered baby syndrome); street and destitute children; school drop-out	Child labour is rooted in poverty, unemployment and lack of education (Park 28e, Ch 9, p 696)
Socio-economic deprivation	Poverty; illiteracy; unemployment; urban slums; homelessness and poor housing; distress migration	Rising population, large family size, prolonged and recurring illness, addiction in the family, lack of schooling facilities for girls, migration and industrialisation (Sunder Lal, Ch 2, p 82)
Stigmatised and neglected groups	Leprosy, tuberculosis, human immunodeficiency virus (HIV)/acquired immunodeficiency syndrome (AIDS) and mental disorders; the aged and destitute; the disabled	Social stigma, hiding of disease, fear of losing a job, fatalistic attitude that the disease is incurable and disabling, and loss of wages, leading to non-utilisation of treatment (Sunder Lal, Ch 2, p 82)

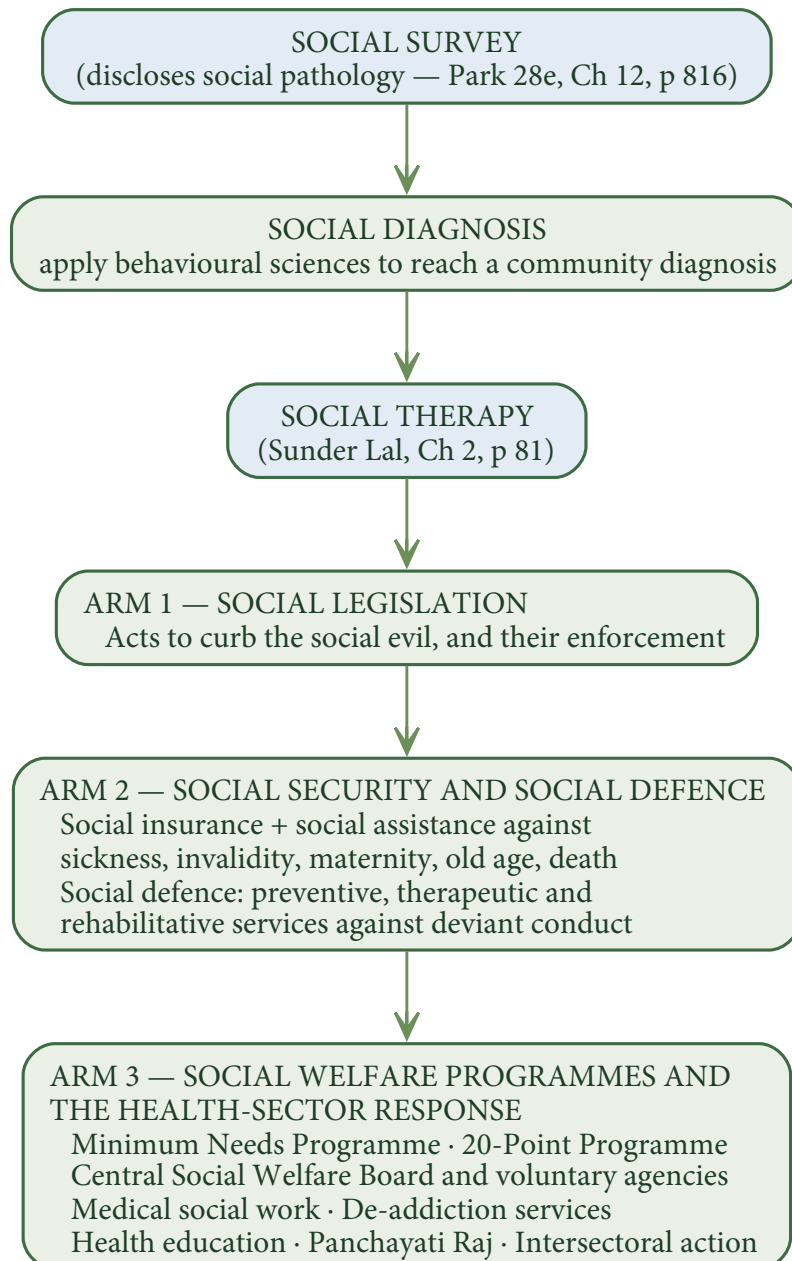
Group	Social problems	Underlying behaviour or social factor
Environment and services	Insanitary environment and open-field defecation; unsafe water ; food adulteration; road traffic accidents ; poor utilisation and poor management of health services	Belief that children's excreta are harmless, non-acceptance of latrines and of chlorination, poor water-storage practice; high-speed driving, disregard of traffic rules, driving under influence; non-accessibility, non-affordability and cultural non-acceptance of services (Sunder Lal, Ch 2, pp 81-82)
Lifestyle-linked disease	Diabetes, hypertension, cancer, cardiovascular disease and stroke	Physical inactivity, faulty diet, tobacco and alcohol (Sunder Lal, Ch 2, p 82)



Classification of social problems for enumeration in the examination

The framework of measures

Park's governing statement is that **social problems** are solved by **social and political action** — that is, by **social welfare programmes, social assistance and social legislation in the community to curb the social evils** (Park 28e, Ch 12, p 839). The working sequence is **social survey** → **social diagnosis** → **social therapy**: social therapy treats the underlying cause or changes behaviour through education, individual and group counselling, motivation, community involvement, and legislation to curb the bad practice or to supply the missing health service, delivered on an equity basis with priority to the economically weaker sections (Sunder Lal, Ch 2, p 81).



From social survey to social therapy — the three arms of action against social problems

Arm 1 — Social legislation

Legislation is the fastest lever but not a sufficient one: well-entrenched social customs cannot be easily erased by an Act of Parliament, and a sustained effort at the root causes is also required (Park 28e, Ch 12, p 840).

Social problem	Legislation and its key provision
Prostitution and trafficking	<i>Suppression of Immoral Traffic in Women and Girls Act, 1956, amended and retitled the Immoral Traffic (Prevention) Act in 1986</i> ; it bans prostitution in its commercialised, organised form and covers all persons, male or female, exploited sexually for commercial purposes (Park 28e, Ch 12, pp 839-840)
Dowry	<i>Dowry Prohibition (Amendment) Act, 1986</i> — minimum punishment for taking or abetting the taking of dowry raised to 5 years' imprisonment and a fine of ₹15,000 (Park 28e, Ch 12, p 840)
Juvenile delinquency	<i>Children Act, 1960</i> — specialised care, protection, maintenance, training and rehabilitation of delinquent children (Park 28e, Ch 12, p 840). <i>Juvenile Justice (Care and Protection of Children) Act, 2015</i> , in force from 15 January 2016 — children aged 16–18 years in conflict with law may be tried as adults for heinous offences and detained in a place of safety until age 21 (Park 28e, Ch 9, p 699)
Drug abuse	<i>Narcotic Drugs and Psychotropic Substances Act, in force 1985</i> (Park 28e, Ch 12, p 840)
Tobacco	<i>Cigarettes and Other Tobacco Products Act (COTPA), 2003</i> — ban on smoking in public places (in force 2 October 2008) , on direct and indirect advertisement , on sale to a person below 18 years , on sale within 100 yards of educational institutions , and on tobacco in gutkha, pan masala and khaini ; mandatory statutory and pictorial warnings on 85% of both sides of the pack (60% picture, 25% text). The <i>Prohibition of Electronic Cigarettes Act, 2019</i> bans electronic nicotine delivery systems (Sunder Lal, Ch 15, p 691)
Food adulteration	<i>Prevention of Food Adulteration Act</i> (Park 28e, Ch 12, p 839)
Unwanted pregnancy and unsafe abortion	<i>Medical Termination of Pregnancy Act</i> (Park 28e, Ch 12, p 839)

Arm 2 — Social security and social defence

Social security is "security that society furnishes through appropriate organization, against certain risks to which its members are exposed"; the risks covered in most countries are **sickness, invalidity, maternity, old age and death**, and social security comprises both **social insurance** and **social assistance** (Park 28e, Ch 12, pp 845-846).

Instrument	Content
Social insurance for industrial workers	<i>Workmen's Compensation Act, 1923; Employees State Insurance (ESI) Act, 1948; Employees Provident Fund and Miscellaneous Provisions Act, 1952; Central Maternity Benefit Act, 1961; Family Pension Scheme, 1971; Payment of Gratuity Act, 1972</i> (Park 28e, Ch 12, pp 845-846)

Instrument	Content
ESI Scheme — the first major landmark social security law in India	Covers non-seasonal factories and establishments employing 10 or more persons for wages (Sunder Lal, Ch 17, p 763). Benefits: medical benefit in kind; sickness benefit — 70% of daily wages for 91 days; maternity benefit up to 26 weeks; temporary and permanent disablement benefit; dependants' benefit — 90% of wages; funeral expenses ₹15,000; rehabilitation and vocational rehabilitation allowance; old-age medical care (Sunder Lal, Ch 11, p 652)
Social assistance	Non-contributory support, funded from general revenues, for the unemployed, disabled, old, widowed, orphaned and handicapped — widow pension, old-age and geriatric care, assistance to persons affected by leprosy, unemployment scheme, family-planning assistance (Suryakantha, Ch 18, p 255)
Social defence	A system developed to defend society against criminality, not merely by treating the offender but by creating conditions in the community conducive to healthy and wholesome growth. It covers the entire gamut of preventive, therapeutic and rehabilitative services for protection of society from antisocial, criminal or deviant conduct: prevention and control of juvenile delinquency, eradication of beggary, social and moral hygiene programmes, welfare of prisoners, prison reforms, elimination of prostitution, and control of alcoholism, drug addiction, gambling and suicide. The Government of India renamed the Central Bureau of Correctional Services as the National Institute of Social Defence in 1975 (Park 28e, Ch 12, p 816)

Arm 3 — Social welfare programmes and the health-sector response

Programme / agency	Content
Minimum Needs Programme (MNP) — Fifth Five Year Plan (1974–78)	Provides certain basic minimum needs to raise living standards of the underprivileged. Eight components: rural health, rural water supply, rural electrification, elementary education, adult education, nutrition, environmental improvement of urban slums, and houses for landless labourers. Two implementation principles: (a) facilities go first to areas that are at present underserved, to remove disparities between areas; (b) facilities are provided as a package to an area through intersectoral area projects for greater impact. Rural health norms: one primary health centre (PHC) per 30,000 population in the plains and per 20,000 in tribal and hilly areas; one sub-centre per 5,000 in the plains and per 3,000 in tribal and hilly areas; one community health centre (CHC)/rural hospital per one lakh population or per one Community Development block. Nutrition objectives: extend the special nutrition programme to 11 million eligible persons; expand it to all Integrated Child Development Services (ICDS) projects; consolidate the mid-day meal programme and link it to health, potable water and sanitation (Park 28e, Ch 7, p 575)

Programme / agency	Content
Twenty Point Programme	Initiated by the Government of India in 1975 as an <i>agenda for national action</i> to promote social justice and economic growth; restructured on 20 August 1986 and again in 2006 . Objectives — eradication of poverty, raising productivity, reducing inequalities, removing social and economic disparities and improving the quality of life . The 2006 points include poverty eradication, power to people, food security, housing for all, clean drinking water, health for all, education for all, welfare of scheduled castes, scheduled tribes, minorities and other backward classes, women's welfare, child welfare, youth development, improvement of slums, environment protection, social security and rural roads (Park 28e, Ch 7, pp 575-576)
Central Social Welfare Board and the voluntary sector	Voluntary-sector activity is coordinated by the Ministry of Social Welfare and Women's Affairs (Park 28e, Ch 12, p 841). The autonomous Central Social Welfare Board (set up August 1953) surveys the needs of voluntary welfare organisations, promotes new ones and gives financial aid to deserving ones; it runs <i>Family and Child Welfare Services</i> projects (initiated 1968) — craft teaching, social education, literacy classes, maternity aid, milk distribution, <i>balwadis</i> and play centres — plus women's hostels, legal aid and support to women exploited by families or employers (Park 28e, Ch 10, p 747)

- **Medical social work at the health facility.** Begun in England in 1895 with the *Almoner* at the Royal Free Hospital, London, it uses **social case work** to uncover the social background of illness, enabling the doctor to reach a **social diagnosis**, treat and estimate prognosis. The **medical social worker** mobilises community resources and assists rehabilitation in chronic disabling disease such as tuberculosis and leprosy (Park 28e, Ch 12, pp 836-837).
- **Management of the addicted individual.** Though addiction is a social problem, the first step is medical care: identification and motivation → **detoxification (requires hospitalisation)** → post-detoxification counselling and follow-up → **rehabilitation**, with a change of environment and a **complete break with the drug-using group, failing which relapse is 100 per cent** (Park 28e, Ch 12, p 840). Delivery is through the **National Drug De-addiction Programme** (begun 1987–88; the Ministry of Health and Family Welfare's role is *demand reduction*), with self-help groups, halfway homes and non-governmental organisations (Sunder Lal, Ch 15, pp 688-689).
- **Health education.** Mass media (television, radio talks, leaflets, posters) create awareness in target groups and the public; for alcoholism, public education and discussion with investigation of public attitudes, **combined with social welfare and health services**, is what yields measurable improvement (Park 28e, Ch 12, p 840).
- **Community participation and democratic decentralisation.** **Panchayati Raj is nothing but democratic decentralisation — a three-tier system of local self-government: Gram Panchayat (village), Panchayat Samiti (block) and Zilla Parishad (district)** (Park 28e, Ch 12, p 841), the structure recommended by the *Balwant Rai Mehta* Committee in 1957 and given constitutional status by the **73rd Amendment Act, 1992**, which devolved 29 subjects including primary health care (Sunder Lal, Ch 2, p 67). It is the platform for intersectoral action across health, education, water and sanitation, rural development and social welfare.
- **Rehabilitation of problem families.** The health visitor, health inspector, midwife, social worker and medical officer of health together rehabilitate problem families — the most effective way of

preventing the next generation from becoming victims of prostitution, crime and vagrancy (Park 28e, Ch 12, pp 828-829).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Draw the **wheel of social security** as one circle divided into three sectors. Labels to write:

- ▶ Outer rim, three arcs: **SOCIAL DEFENCE, SOCIAL INSURANCE, SOCIAL ASSISTANCE**
- ▶ Under **Social defence**: *Prison reforms · Welfare of prisoners · Eradication of beggary · Juvenile delinquency · Antidowry Act · Criminals · Prostitution · Gambling · Suicides · Drug abuse · Alcoholism*
- ▶ Under **Social insurance**: *ESI benefits · Life insurance · Provident fund · Retirement benefit · Family pension · Plantation Act · Port and Dock Act · Mines Act*
- ▶ Under **Social assistance**: *Widow pension · Assistance to persons affected by leprosy · Family planning assistance · Geriatric care · Unemployment scheme · Immoral Traffic Act*

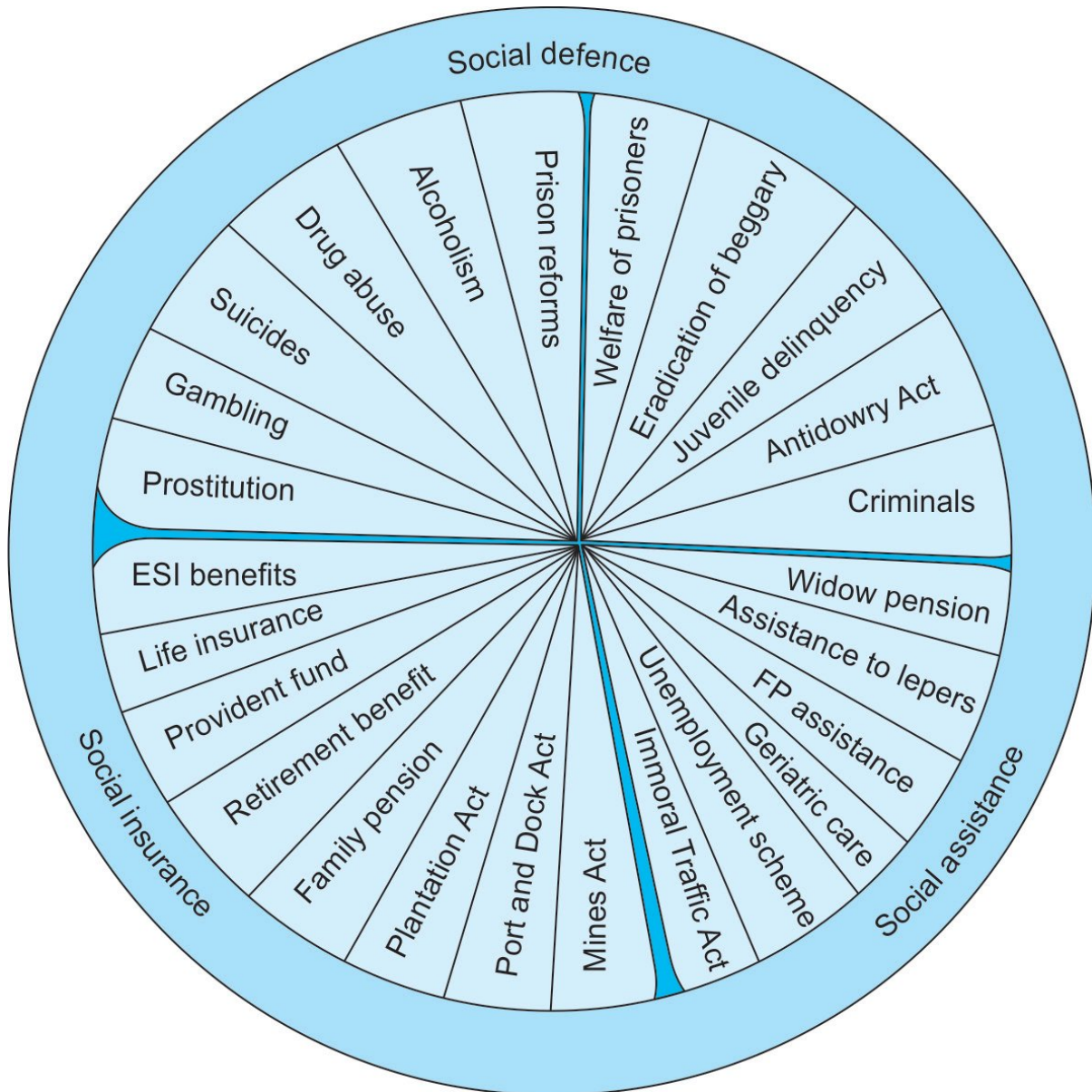


Fig 17.2 — Showing relation between social insurance, social assistance and social defence (Suryakantha Environment Best, p.256)

High-yield exam pointers

- ▶ **Open with Park's definition:** "When individual problems affect a large number of people, they become social problems", then name his six — **alcoholism, drug dependence, sexually transmitted disease, vagrancy, juvenile delinquency, prostitution.**
- ▶ **Three technical terms:** **social pathology** (poverty, crime, delinquency, vagrancy), **social survey** (discloses it — Chadwick's survey led to the General Board of Health, 1848), **social defence** (National Institute of Social Defence, renamed 1975).
- ▶ **The measures line to quote:** social problems are solved by **social and political action** — **social welfare programmes, social assistance and social legislation**; the sequence is **social survey → social diagnosis → social therapy.**
- ▶ **Dated Acts:** Immoral Traffic (Prevention) Act 1956, **retitled 1986**; Dowry Prohibition (Amendment) Act 1986 — **5 years and ₹15,000**; NDPS Act 1985; Children Act 1960 and

Juvenile Justice Act 2015 (in force 15 January 2016); COTPA 2003 (public-place smoking ban in force 2 October 2008, sale banned below 18 years).

- ▶ **Social security = social insurance + social assistance**, against sickness, invalidity, maternity, old age and death; ESI Act 1948 covers establishments with 10 or more workers — sickness benefit 70% of wages for 91 days, maternity 26 weeks, dependants' benefit 90% of wages.
- ▶ **MNP norms are the examiner's favourite line**: PHC per 30,000 (plains) / 20,000 (tribal and hilly); sub-centre per 5,000 / 3,000; CHC per 1 lakh or one Community Development block.
- ▶ **Dated Indian figures (NFHS-5, 2019–21)**: spousal violence 32% (ever-married women 18–49), help sought by only 14%; married before 18 — 23% of women aged 20–24.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the raise of the legal tobacco age from 18 to 21 years, the ban on loose cigarettes and the higher penalty sit in the COTPA (Amendment) Bill, a proposed amendment not yet enacted, so write only the 2003 provisions as being in force; Park's Ministry of Social Welfare and Women's Affairs is today the Ministry of Social Justice and Empowerment with the Ministry of Women and Child Development; and the Minimum Needs Programme belongs to the Five Year Plan era, which ended in 2017.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 12 — Medicine and Social Sciences, pp 816, 828-829, 836-841, 845-846; Ch 9 — Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 691-699; Ch 10 — Health Care of the Community, p 747; Ch 7 — Health Programmes in India, pp 575-576); Sunder Lal — Textbook of Community Medicine (Ch 2 — Relationship of Social and Behavioural Sciences to Health and Disease, pp 67, 81-82; Ch 11 — Occupational Health, p 652; Ch 15 — Mental Health, pp 688-691; Ch 17 — Healthcare of the Community, p 763); Suryakantha — Community Medicine with Recent Advances (Ch 18 — Occupational Health, pp 255-256, Figure 17.2). Named sources cited within: National Family Health Survey (NFHS-5), India Report, 2019–21; Immoral Traffic (Prevention) Act, 1986; Dowry Prohibition (Amendment) Act, 1986; Narcotic Drugs and Psychotropic Substances Act, 1985; Juvenile Justice (Care and Protection of Children) Act, 2015; Cigarettes and Other Tobacco Products Act, 2003; Prohibition of Electronic Cigarettes Act, 2019; Employees State Insurance Act, 1948; 73rd Constitutional Amendment Act, 1992.



Behaviour change communication in disease control

Asked in: Paper I 26-May-2025 — RGUHS MD Community Medicine

Definition — Behaviour change communication (BCC) is a systematic process that motivates individuals, families and communities to change their inappropriate or unhealthy behaviour, or to continue a healthy behaviour (Park 28e, Ch 7, p 499). In disease control it is a **supportive strategy** that converts an available preventive or curative service into an actually used service, acting in the sequence knowledge → attitude → practice (Sunder Lal, Ch 4, p 149). It is the behaviour-specific, audience-segmented and outcome-evaluated arm of information, education and communication (IEC), judged not by activities counted but by how many people actually changed practice (Sunder Lal, Ch 4, pp 153–154). Current national documents increasingly call it **social and behaviour change communication (SBCC)** (Routine Immunization Manual 2024, Ch 11, p 248).

Scope 1 — the behaviours BCC targets along the chain of disease control

Point in the chain	Behaviour to be changed	Suitable example, with the Indian gap that justifies it
Reducing exposure	Sleeping under an insecticide-treated / long-lasting insecticidal net (ITN/LLIN); accepting indoor residual spray (IRS)	Core content of BCC under the National Vector Borne Disease Control Programme (NVBDCP) (Park 28e, Ch 7, p 499)
Reducing susceptibility	Bringing the child for timely and complete immunisation	76.4% of children aged 12–23 months fully immunised, NFHS-5 (Sunder Lal, Ch 4, p 154)
Early recognition	Recognising signs and symptoms of malaria; reporting a hypopigmented anaesthetic patch before disability appears	Malaria BCC objective (Park 28e, Ch 7, p 499); ASHA-led case search under NLEP (Jugal Kishore, p 464)
Prompt care seeking	Early treatment seeking from an appropriate provider rather than an unqualified one	Explicit BCC objective under NVBDCP (Park 28e, Ch 7, p 499)
Treatment adherence	Completing the full antimalarial, anti-tuberculosis or multidrug therapy course	Adherence to treatment regimens (Park 28e, Ch 7, p 499); ASHA follow-up for treatment completion under NLEP (Jugal Kishore, p 464)
Protecting the vulnerable	Ensuring protection of children and pregnant women	Malaria high-risk groups (Park 28e, Ch 7, p 499)
Interrupting transmission	Condom use, avoiding multi-partner sex, sterilised needles and syringes, voluntary blood donation	Stated objectives of the IEC strategy of the National AIDS Control Programme (Jugal Kishore, pp 368–369); comprehensive HIV knowledge only 21.6% in adult women vs 30.7% in men, NFHS-5 (Sunder Lal, Ch 8, p 347)

Point in the chain	Behaviour to be changed	Suitable example, with the Indian gap that justifies it
Environmental and hygiene behaviour	Toilet construction and, more importantly, toilet use; handwashing; safe household water storage	Swachh Bharat Abhiyan generates demand for toilets through awareness and behaviour change with Panchayati Raj Institutions; improved sanitation 71% of Indian households (urban 79%, rural 67%), 2021 (Sunder Lal, Ch 1, pp 40–41)
Risk-factor behaviour (non-communicable disease)	Tobacco and alcohol avoidance, physical activity, reduced saturated fat, sugar and salt	38% of men and 8.9% of women use tobacco, NFHS-5 (Sunder Lal, Ch 4, p 154); BCC is the named tool for these risk factors (Sunder Lal, Ch 2, pp 82–83)
Home management of illness	Correct use of oral rehydration salts and continued feeding in childhood diarrhoea	ORS use rate 60.6%, NFHS-5 (Sunder Lal, Ch 4, p 154)
Stigma and rehabilitation	Non-discrimination, acceptance of the affected person back into family and work	Creating a supportive environment for care and rehabilitation of persons with HIV/AIDS is a stated IEC objective (Jugal Kishore, pp 368–369)

Scope 2 — the three tiers at which BCC operates

Tier	Target audience	Purpose	Example
Advocacy	Policy makers, legislators, Zila Parishad, urban local bodies, media owners, corporate sector	Political and media support, budget, an enabling and non-stigmatising policy environment	Advocacy meetings with Zila Parishad, Mahila Mandals and NGOs under the National Leprosy Eradication Programme (NLEP) (Jugal Kishore, pp 466–467)
Social mobilisation	Community as a whole — Panchayati Raj Institutions, village headmen, school teachers, religious and faith leaders, self-help groups, NGOs	Collective ownership, a supportive social norm, demand for services	Anti-Malaria Month every June, before the monsoon, using mass media plus interpersonal communication and inter-sectoral collaboration (Park 28e, Ch 7, p 499)
Behaviour change communication proper (interpersonal)	Individual, household, care-giver, high-risk group	Precise, negotiated change in one named practice, with counselling and follow-up	Home visits by the auxiliary nurse midwife (ANM) and ASHA; one-to-one counselling of a care-giver on the next due vaccine dose (Routine Immunization Manual 2024, Ch 11, p 252)

Under the National Tuberculosis Elimination Programme these three tiers are used together and are named **advocacy, communication and social mobilisation (ACSM)** (Jugal Kishore, p 304).

Scope 3 — worked examples in national disease-control programmes

Programme	Behaviour targeted	BCC activity actually used
NVBDCP — malaria	Early recognition, early treatment seeking, adherence, protection of children and pregnant women, use of ITN/LLIN, acceptance of IRS	Listed as a supportive intervention alongside integrated vector management; ASHA and Anganwadi worker engagement, involvement of local faith healers and forest department in forest areas, social marketing of bed-nets, Anti-Malaria Month (Park 28e, Ch 7, pp 493, 499)
National Tuberculosis Elimination Programme (NTEP, formerly RNTCP)	Recognising prolonged cough, sputum testing, treatment completion, notification by private practitioners	ACSM with decentralised state- and district-specific communication plans and local innovation to reach all target groups; a multidisciplinary advocacy and IEC unit at the centre (Jugal Kishore, p 304)
National AIDS Control Programme	Safer sex, condom use, safe injection, voluntary blood donation, uptake of testing, reduced stigma	National IEC division for political and media advocacy; state-level decentralisation into local languages; mass media, targeted interventions through peer-led NGO/community-based organisation outreach, Red Ribbon Clubs in schools and colleges, adolescence education programme, integrated counselling and testing centres (Jugal Kishore, pp 368–369; Sunder Lal, Ch 8, p 347)
NLEP	Early self-reporting of skin lesions, completion of multidrug therapy, ending stigma	Decentralised IEC/BCC — mass media (television, radio, local-language press), outdoor media (hoardings, bus panels, wall paintings, posters, rallies), rural media (interpersonal meetings, school talks and quizzes, folk media, exhibitions, health melas), advocacy meetings, women's mobilisation, a toll-free complaint line (Jugal Kishore, pp 464, 466–467)
Universal Immunisation Programme	Timely vaccination; overcoming hesitancy, fear of adverse events following immunisation (AEFI), fear of wage loss and gender-related barriers	SBCC to raise awareness and address myths, and to motivate care-givers and communities to actively seek immunisation; interpersonal communication as the core strategy, "vaccination on demand", <i>Teeka Jaagruk Ghar</i> , engagement of Mahila Arogya Samiti and religious leaders, crisis communication after an AEFI (Routine Immunization Manual 2024, Ch 11, pp 247–249, 252, 257, 261)
Swachh Bharat Abhiyan (formerly Total Sanitation Campaign)	Building and using a household toilet; ending open defecation	Awareness and behaviour change through Panchayati Raj Institutions, <i>Nirmal Gram Puraskar</i> incentive to open-defecation-free villages (launched 24 February 2005); India declared open defecation free in October 2020 (Sunder Lal, Ch 1, pp 40–41)
COVID-19 response	Mask use, physical distancing, hand hygiene	Pan-India "360-degree" strategic IEC campaign using mass media, mid-media, social media and interpersonal communication, including a nationwide pre-caller tune carrying the do's and don'ts (Sunder Lal, Ch 2, p 85)

Programme	Behaviour targeted	BCC activity actually used
Adolescent health	Delayed age at marriage, responsible reproductive behaviour, resisting negative peer pressure	National communication strategy for adolescents with UNICEF — interpersonal communication, mass-media advocacy, decentralised IEC, NGO and private sector involvement, capacity building, and the GATHER approach to counselling (Sunder Lal, Ch 10, p 623)

Scope 4 — channels through which the scope is delivered

Channel	Strength	Examples used in disease control
Mass media	Triggers thought and acts as a "hook"; reaches many people very quickly and repeatedly; reinforces messages sent through other channels	Television and radio spots in prime time, community radio, local-language newspapers (Routine Immunization Manual 2024, Ch 11, pp 250–254)
Mid-media (outdoor and folk)	Reinforces and expands mass-media messages; acts as "reminders" or message take-aways	Wall paintings, <i>nukkad natak</i> , miking, rallies, cycle <i>prachar</i> , health melas, exhibitions (Routine Immunization Manual 2024, Ch 11, p 253; Jugal Kishore, pp 466–467)
Interpersonal communication (IPC)	Direct interaction; allows discussion, dispels myths and misconceptions; encourages, motivates and reinforces action; most precise for one household's need	Home visits, one-to-one counselling by ANM/ASHA, mothers' and fathers' meetings, community meetings (Routine Immunization Manual 2024, Ch 11, pp 252–253; Sunder Lal, Ch 4, p 146)

No single channel is the best channel — multiple mutually reinforcing channels give the greater impact on behaviour (Routine Immunization Manual 2024, Ch 11, p 253).

Scope 5 — planning and evaluating a BCC intervention

- ▶ **Collect information** on the specific problem as the community sees it, and identify the behavioural problem (Park 28e, Ch 22, p 1024).
- ▶ **Decide priorities** and set goals with measurable objectives — the behaviour, the expected level and the target year.
- ▶ **Segment the audience** by economic status, caste, literacy and current practice, because one size does not fit all (Sunder Lal, Ch 4, p 153).
- ▶ **Frame a specific message** that does not merely create awareness but names the action expected, and prepare the education material for it (Sunder Lal, Ch 4, p 153).
- ▶ **Select the right channel mix** for that audience, and assess resources and budget.
- ▶ **Prepare and implement the plan of action** — what will be done, when, and by whom — training front-line workers in interpersonal communication skills.
- ▶ **Monitor** implementation and use monitoring and mentoring visits to modify the plan for local barriers (Routine Immunization Manual 2024, Ch 11, p 261).
- ▶ **Evaluate against behaviour, not activity** — compare baseline and repeat-survey knowledge, attitude and practice; counting clients contacted, camps held, handouts distributed and film

shows held is not evaluation, the real test is how many individuals changed their practice (Sunder Lal, Ch 4, p 154).

- **Observe practice directly** where possible — handwashing, nail cutting, cord cutting, cooking and feeding practices — rather than relying on reported behaviour alone (Sunder Lal, Ch 4, p 154).

The standard BCC evaluation matrix records, for each area of concern: current behaviour with its year → expected behaviour with its target year → target audience → message → media and channel → outcome indicator of change (Sunder Lal, Ch 4, Table 4.1, p 154).

Scope 6 — where the scope ends (limits)

- Education is necessary but **alone insufficient**: the target population must first have access to the proven preventive measure or service being promoted (Park 28e, Ch 22, pp 1014–1015).
- Individual behaviour-change models pay scant attention to the social, economic and cultural environment in which the behaviour occurs, which is why purely individual campaigns have often failed (Oxford Textbook of Public Health 7e, Ch 6, p 309).
- **Barriers to communication** blunt it — physiological (hearing, expression), psychological (emotional disturbance, comprehension), environmental (noise, congestion) and cultural (illiteracy, beliefs) (Park 28e, Ch 22, p 1012).
- Supply-side failure defeats demand generation: weak micro-planning, vacant ANM posts, limited front-line worker skills and poor service experience will nullify any amount of BCC (Routine Immunization Manual 2024, Ch 11, pp 247–248).
- Persuasion deliberately used to manipulate feelings, attitudes and beliefs stops being education and becomes **propaganda** (Park 28e, Ch 22, p 1014).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Labels to reproduce (dynamics of behavioural change — draw as a closed cycle): Satisfied quiet individual → *Need felt* → Tense individual → *Motivation develops* → Tense motivated individual → *Behavioural change occurs* → back to Satisfied quiet individual.

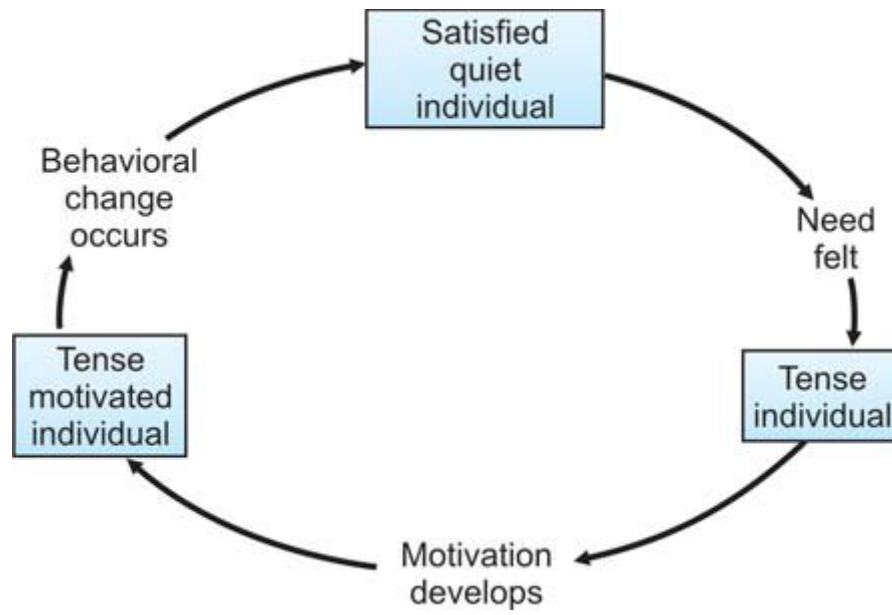
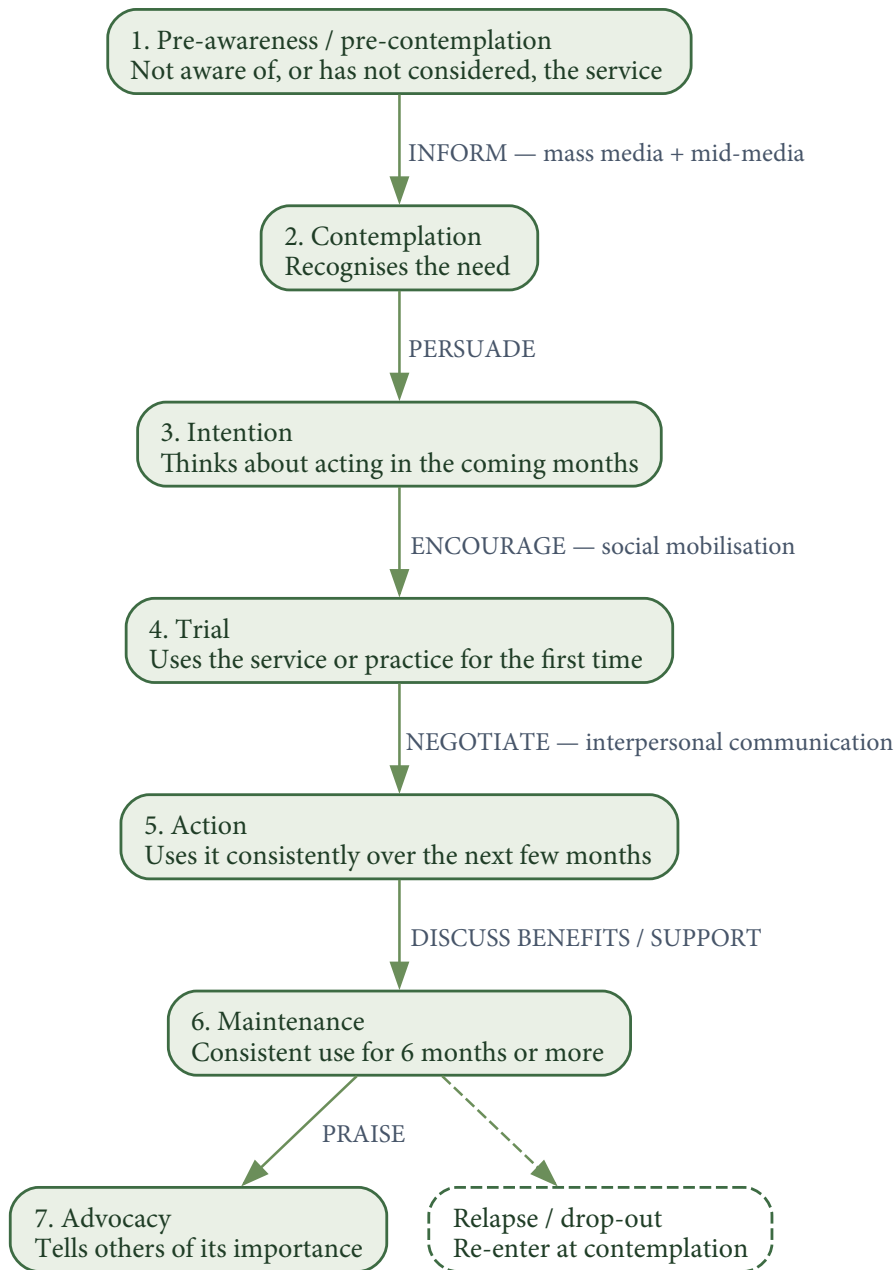
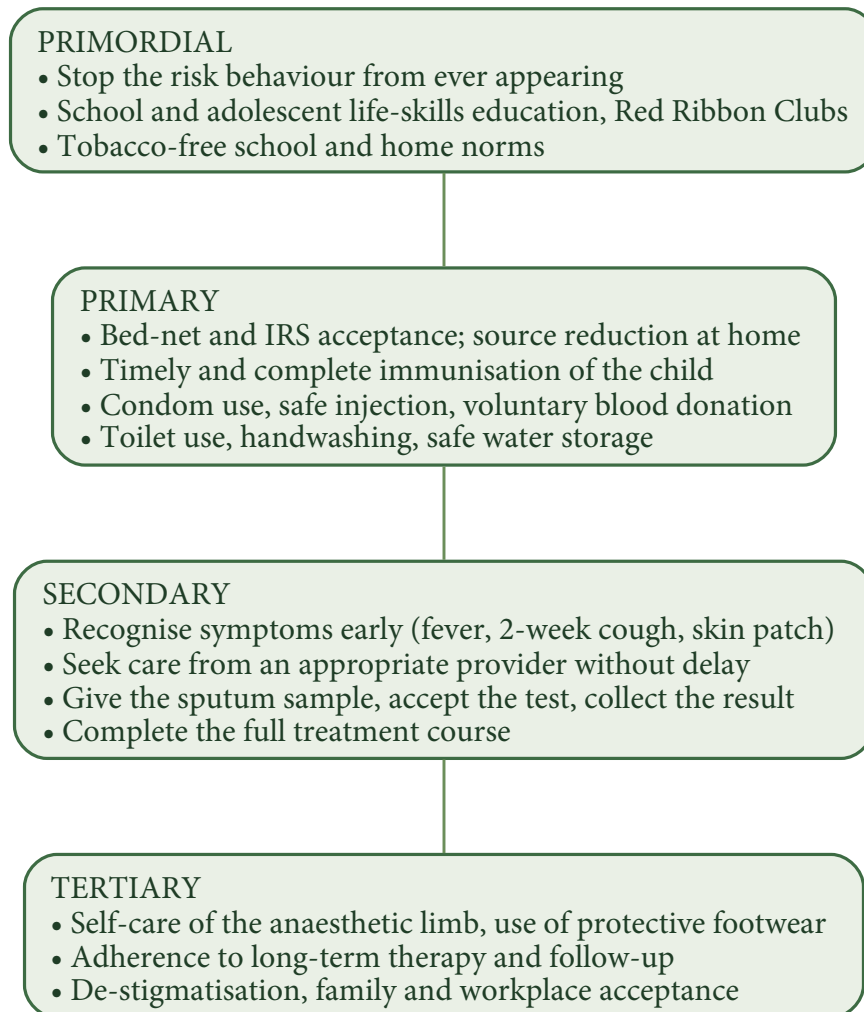


Fig 29.1 — Dynamics of behavioral change* (Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e, p.575)



Behaviour change cycle and the communication task at each stage



Scope of behaviour change communication across the levels of prevention in disease control

High-yield exam pointers

- **Definition to write verbatim:** BCC is a systematic process that motivates individuals, families and communities to change inappropriate or unhealthy behaviour, or to continue a healthy behaviour (Park 28e, Ch 7, p 499).
- **Six malaria BCC objectives (mnemonic RESPIA):** Recognise symptoms early, Early treatment seeking, Stick to the regimen (adherence), Protect children and pregnant women, ITN/LLIN use, Accept IRS.
- **ACSM** = Advocacy, Communication and Social Mobilisation — the name of the three-tier strategy under NTEP.
- **Behaviour change cycle, 7 stages:** pre-awareness → contemplation → intention → trial → action → maintenance → advocacy; inform, persuade, encourage, negotiate, support, praise.
- **Adoption process (diffusion):** awareness → interest → evaluation → trial → adoption (Sunder Lal, Ch 4, p 149); Park's condensed version is awareness → motivation → action.
- **Dated Indian figures for the BCC need:** full immunisation 76.4%, ORS use 60.6%, tobacco use 38% of men and 8.9% of women (all NFHS-5); improved sanitation 71% of households (2021); India declared open defecation free October 2020.

- **Evaluation rule:** count changed behaviour, not activities held — baseline versus repeat survey of knowledge, attitude and practice.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the Routine Immunization Manual for Medical Officers 2024 and other current national documents use the term **social and behaviour change communication (SBCC)** for what the textbooks still print as BCC, and the Revised National Tuberculosis Control Programme was renamed the **National Tuberculosis Elimination Programme** in 2019 (Park 28e, Ch 23, p 1040), so quote the current programme name with the former one in brackets once.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, MoHFW, ICMR, WHO

Programme and policy updates (India)

- **HPV vaccine has entered the national immunisation schedule** — a free nationwide HPV vaccination drive was launched on 28 February 2026 targeting 1.15 crore girls aged 14, placing India among the 160-plus countries carrying HPV vaccine in the national programme; misinformation recycling the 2009 PATH demonstration-project controversy spread on social media within days of launch, so the recommended communication response is pre-bunking rooted in inoculation theory, framing the vaccine as cancer prevention rather than as protection against a sexually transmitted infection, training providers to give a confident recommendation, and school-based sessions with multilingual short films circulated to parents. Two further recommendations are made separately in the paper: testimonials from healthcare professionals, celebrity parents and community leaders on social media, justified by the two-step flow theory of mass communication; and launching culturally tailored digital tools on the MoHFW U-WIN platform so that schools can fold those resources into the health-education curriculum [PMID 42544634].
- **SBCC is now written into the national policy instruments themselves** — a scoping review of central-government instruments found SBCC strategies embedded in Rashtriya Kishor Swasthya Karyakram, Eat Right India, Ayushman Bharat (School Health and Wellness Programme and PM-JAY), Fit India, the FSSAI school-food regulations and the National Programme for Prevention and Control of Non-Communicable Diseases (NP-NCD); NP-NCD stands out, in the authors' words, for its targeted approach to preventing overweight and obesity within the broader context of NCDs, carrying SBCC at the individual, interpersonal, community and organisational levels — but it is the one instrument of the six that lacks policy-level SBCC, where RKSK, Ayushman Bharat, Eat Right India and Fit India are marked present at all five levels; the stated recommendation is to extend the same SBCC architecture into POSHAN 2.0 and RMNCAH+N under the National Health Mission [PMID 39866726].
- **Information and physical access, not money, decide whether an entitlement is actually used** — among 320 enrolled PM-JAY beneficiaries surveyed in Jammu and Kashmir between February and April 2025, only 48.4% had ever used the scheme, and three of the ten barriers tested were independently significant: awareness of scheme benefits (OR 1.253, $p = 0.042$) and availability of the empanelled-hospital list (OR 1.245, $p = 0.027$) raised the odds of use, while poor road quality

lowered them (OR 0.761, $p = 0.033$); out-of-pocket expense, informal payment, wage loss, documentation and claim rejection were all non-significant once the others were controlled, so the authors call for region-specific IEC plus last-mile connectivity rather than more financial cover alone; on the digital side, ABHA registration reached only 23.96% of 537,278 outpatient visits over eight months and 88% of attendees said they needed training and support to use the Ayushman Bharat Digital Mission [PMID 41878109; PMID 41787376].

Current national data

- ▶ **Counselling shifts child feeding more than food or cash does — but only towards the healthy foods it talks about** — in a six-state phone survey of 6,627 Indian mothers in late 2021, 57% of mothers but only 23% of children achieved minimum dietary diversity; infant and young child feeding counselling doubled the odds of a child achieving it (OR 2.14, 95% CI 1.65–2.78) and halved the odds of zero vegetable or fruit intake (OR 0.48, 95% CI 0.39–0.60), and children exposed to counselling plus food plus cash together were 3.5 times as likely to achieve it. Counselling was not associated with any fall in unhealthy-food consumption, and it was associated with **higher** odds of sweet-beverage consumption in children — almost all of it sweetened tea, coffee or milk — with a matching small rise in the NCD-risk score (β 0.13, 95% CI 0.02–0.23); the authors' own reading is that counselling in low- and middle-income settings teaches what to add and never what to avoid, which marks the practical ceiling of message-only work in a changing food environment [PMID 39053606].
- ▶ **The coverage–compliance gap in mass drug administration** — pooled across 34 South-East Asian studies published between January 2019 and March 2025 (28 of them Indian), MDA coverage was 80.16% (95% CI 74.68–85.64) but compliance, meaning the drug actually swallowed, only 67.25% (95% CI 60.17–74.33), both falling in studies done from 2020 onwards; the barriers named are fear of side effects, feeling well and not diseased, too many tablets and bad taste — every one of them a communication target rather than a supply target. Lymphatic filariasis remains endemic in 345 districts against a 2030 elimination goal, with advocacy, social mobilisation and MDA non-compliance the domains flagged for public–private partnership support [PMID 40605551; PMID 41951149].

Elimination, surveillance and disease status

- ▶ **100-Day TB Elimination Campaign, 7 December 2024 to 17 March 2025** — a difference-in-differences analysis of Ni-kshay portal data comparing 53 campaign with 27 non-campaign districts in Maharashtra found notification fell 1% in campaign districts against a 10% fall in non-campaign districts (DID IRR 1.10, 95% CI 1.03–1.17, $p = 0.003$) — a slowed decline, not a rise; the relative gain was confined to rural areas (DID IRR 1.21) and the public sector (DID IRR 1.17), with no effect in urban areas or private care, and the authors conclude only that the campaign helped mitigate a larger decline or stabilise notification — so intensified case-finding with a communication push held the line exactly where interpersonal channels reach, and nowhere else [PMID 42457294].
- ▶ **Kala-azar has moved into a post-elimination surveillance phase** — in endemic villages of Bihar and Jharkhand, rK39 seroprevalence was 20–30% and *Leishmania*-infected sandfly pools were still captured despite indoor residual spraying (1.4% and 1.2% of pools), so low-level transmission persists; the authors conclude that sustaining elimination now rests on combining

epidemiological, serological and entomological surveys, which lets a reported case be converted into a rapid assessment of how much transmission is actually going on and a timely, targeted response [PMID 41324335].

- **Measles–rubella elimination is targeted for December 2026** in India and the rest of the WHO South-East Asia Region; a central-India laboratory series from May 2023 to April 2024 found rubella IgM positivity 7.2% alongside IgG 81.0% overall and 89.6% among pregnant women, so population immunity is broadly good but continuing IgM positivity in young children marks the residual immunity gaps that intensified campaign rounds and their accompanying communication must close [PMID 42159053].

Guideline and legislative updates

- **Persuasion is being outpaced by regulation** — a set of rapid reviews found both size-based nudges (portion, package and tableware size) and price-based measures effective at reducing consumption of unhealthy products, yet at least 118 countries have legislated taxes on sugar-sweetened beverages and 42 on unhealthy foods, against only 20 reports of size-based interventions implemented at sub-national level, of which just one involved regulation; the examinable lesson for the limits of behaviour change communication is that fiscal and structural measures scale where communication alone does not [PMID 40157301].

Global and WHO developments

- **Behaviour-change communication now carries a cost-effectiveness figure** — a behavioural-insights-informed social media campaign layered on Bangladesh's Gavi-supported single-dose HPV rollout raised uptake by 9.5 and 5.3 percentage points beyond the national campaign alone, at an incremental \$6.02 per girl vaccinated and an incremental cost-effectiveness ratio of \$39.57 per DALY averted (sensitivity range \$25–50), far below the 40%-of-GDP-per-capita threshold used in low- and middle-income countries; the mirror image is Colombia, where viral misinformation about mass psychogenic reactions in 2014 drove first-dose HPV coverage from 98% down to 14% — both bear on the WHO goal of cervical cancer elimination by 2030 [PMID 40322786; PMID 42544634].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 7 Health Programmes in India; Ch 22 Communication for Health Education; Ch 23 Health Planning and Management); Sunder Lal — Textbook of Community Medicine (Ch 1, Ch 2, Ch 4 Principles of Health Promotion and Education, Ch 8, Ch 10); Jugal Kishore — National Health Programmes of India (National AIDS Control Programme, NTEP/RNTCP, NLEP sections); Routine Immunization I Manual for Medical Officers 2024 (Ch 11 Communication for increasing vaccine demand for immunization); Oxford Textbook of Public Health 7e (Ch 6 Determinants of health); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e (Fig 29.1); National Family Health Survey-5 (2019–21) figures as quoted in Sunder Lal.

Recent-advances sources: 42544634 — Addressing HPV vaccine hesitancy and misinformation in India through public health communication strategies, *Human Vaccines & Immunotherapeutics* 2026, doi:[10.1080/21645515.2026.2711551](https://doi.org/10.1080/21645515.2026.2711551); 39866726 — A Scoping Review of Existing Policy Instruments to Tackle Overweight and Obesity in India: Recommendations for a Social and Behaviour Change Communication Strategy, *F1000Research* 2024, doi:[10.12688/f1000research.149857.2](https://doi.org/10.12688/f1000research.149857.2); 41878109 — Beyond coverage: why Ayushman Bharat-Pradhan Mantri Jan Arogya Yojana struggles to deliver healthcare in Jammu and Kashmir, *Frontiers in Public Health* 2026, doi:[10.3389/fpubh.2026.1783458](https://doi.org/10.3389/fpubh.2026.1783458); 41787376 — Adoption, digital health literacy, and patient satisfaction of Ayushman Bharat Digital Mission, *BMC Health Services Research* 2026, doi:[10.1186/s12913-026-14314-7](https://doi.org/10.1186/s12913-026-14314-7); 39053606 — Diet Quality Among Mothers and Children in India: Roles of Social and Behavior Change Communication and Nutrition-Sensitive Social Protection Programs, *Journal of Nutrition* 2024, doi:[10.1016/j.tjnut.2024.07.026](https://doi.org/10.1016/j.tjnut.2024.07.026); 40605551 — Coverage and determinants of Mass Drug Administration against lymphatic filariasis in Southeast Asian regional countries during the end-game strategy, *Journal of Vector Borne Diseases* 2026, doi:[10.4103/jvbd.jvbd_58_25](https://doi.org/10.4103/jvbd.jvbd_58_25) (abstract-only); 41951149 — Public-private partnership offers a strategic opportunity for lymphatic filariasis elimination in India by 2030, *Acta Tropica* 2026, doi:[10.1016/j.actatropica.2026.108083](https://doi.org/10.1016/j.actatropica.2026.108083) (abstract-only); 42457294 — Effect of the '100 days TB elimination campaign' on TB notification in Maharashtra, *Indian Journal of Tuberculosis* 2025, doi:[10.1016/j.ijtb.2025.08.014](https://doi.org/10.1016/j.ijtb.2025.08.014) (abstract-only); 41324335 — High Seropositivity to rK39 and Capture of Leishmania-Infected Sand Flies in Bihar and Jharkhand Emphasize the Need for Post-Elimination Surveillance of Visceral Leishmaniasis in India, *Journal of Infectious Diseases* 2026, doi:[10.1093/infdis/jiaf543](https://doi.org/10.1093/infdis/jiaf543) (abstract-only); 42159053 — Monitoring Rubella Immunity and Transmission in Central India Amidst Elimination Efforts, *Microbiology and Immunology* 2026, doi:[10.1111/1348-0421.70065](https://doi.org/10.1111/1348-0421.70065) (abstract-only); 40157301 — Do nudges need a regulatory push? Comparing the effectiveness and implementation of exemplar nudge (size-based) and non-nudge (price-based) dietary interventions, *Social Science & Medicine* 2025, doi:[10.1016/j.socscimed.2025.118004](https://doi.org/10.1016/j.socscimed.2025.118004) (abstract-only); 40322786 — Cost-effectiveness of a behavioral insights-informed digital campaign to increase HPV vaccination in Bangladesh, *Human Vaccines & Immunotherapeutics* 2025, doi:[10.1080/21645515.2025.2500264](https://doi.org/10.1080/21645515.2025.2500264).



Family in health and disease

Asked in: Paper I 21-Jun-2024 — RGUHS MD Community Medicine

Definition — The family is the primary unit in all societies: a group of **biologically related individuals living together and eating from a common kitchen** (Park 28e, Ch 12, p 825); the Bureau of Census (USA) definition is simpler — "a group of two or more persons related by blood, marriage or adoption and residing together" (Sunder Lal, Ch 2, p 69). The family differs from a **household**, in which all members need not be blood relations (e.g. servants): a **household** is defined (NFHS-5) as *a person or group of related or unrelated persons who live together in the same dwelling unit, who acknowledge one adult male or female as the head of the household, who share the same housekeeping arrangements and who are considered a single unit*; the number of households in a house is decided by the **number of kitchens (chulahs)** (Sunder Lal, Ch 2, p 69). The family is at once a **biological unit** (shared pool of genes), a **social unit** (common physical and social environment), a **cultural unit**, an **epidemiological unit**, and the **unit for providing social services as well as comprehensive medical care** (Park 28e, Ch 12, p 825).

Characteristics and types of family

Five essential characteristics: a **mating relationship**; a recognised **form of marriage**; a **common habitation** for child-bearing and rearing; a **system of nomenclature and descent**; and an **economic provision** — some source of income to sustain the economic needs of the family, the head of the family performing a vocation to earn money (Sunder Lal, Ch 2, pp 69-70).

Basics	Type	Defining features	Health significance
Structure	Nuclear (elementary)	Married couple and their dependent children occupying the same dwelling space; universal in all societies; arises with industrialisation, urbanisation and employment away from the parental home (Park 28e, Ch 12, pp 826-827; Sunder Lal, Ch 2, p 70)	Absence of grandparents, uncles, aunts and near relatives throws a greater burden of child rearing on the couple, and the husband–wife relationship is likely to be more intimate (Park 28e, Ch 12, pp 826-827); where both parents work, the child is left with an agency or a servant, and crèches, child-care centres, balwadis or nurseries are used (Sunder Lal, Ch 2, p 70)
Structure	Joint (extended)	Several married couples and their children in one household; all men related by blood, the women are their wives, unmarried girls and widows of the kinsmen; common residence, common purse, common kitchen, common property ; authority vested in the eldest member as head (Park 28e, Ch 12, p 827; Sunder Lal, Ch 2, p 70)	Motto "union is strength" — pooled income and shared responsibility give economic and social security to the old, the helpless and the unemployed ; grandparents look after children while the men and women are away at field work or in a job; prevents fragmentation of land; being lost as the price of education, urbanisation and industrialisation (Park 28e, Ch 12, p 827; Sunder Lal, Ch 2, p 70)

Basis	Type	Defining features	Health significance
Structure	Three-generation	Representatives of three generations related by direct descent living together; occurs usually when young couples are unable to find separate housing accommodation and continue to live with their parents and have their own children (Park 28e, Ch 12, p 827)	Fairly common in the West and is commonly confused with the joint family — the distinction (direct descent only, not several collateral married couples) must be made explicitly (Park 28e, Ch 12, p 827)
Authority	Patriarchal / matriarchal	Male head is the authority in decision-making and controls family affairs, descent reckoned through the male line / authority vested in a female head (Sunder Lal, Ch 2, p 70)	Decides who controls money, food distribution and permission to seek care
Form of marriage	Monogamous / polygamous (polygynous, polyandrous)	One spouse versus more than one wife or several husbands (Sunder Lal, Ch 2, p 70)	Monogamy with one regular trusted partner, both free of HIV , carries no chance of acquiring HIV or other sexually transmitted infection; polygamy (more than one wife or multiple sexual partners) carries a greater risk of HIV and other sexually transmitted infection ; in polyandry a regular partnership carries somewhat less risk of HIV transmission, but an irregular partnership with multiple-partner behaviour carries distinctly more (Sunder Lal, Ch 2, p 70)
Sanction of marriage	Endogamous / exogamous	Marriage only within the in-group versus with an out-group	Consanguineous (cousin) marriage carries higher risk of transmission of genetic disorders, and the incidence of congenital malformation is higher in offspring of consanguineous than non-consanguineous marriages; exogamy carries low risk (Sunder Lal, Ch 2, p 70; Park 28e, Ch 12, p 828)
Residence	Patrilocal / matrilocal	Wife moves to the husband's house / husband to the wife's house (Park 28e, Ch 12, pp 827-828)	Decides who actually gives care during pregnancy, childbirth and illness

Family life cycle and family stress

A normal family cycle is conceived as having **six phases** (Park 28e, Ch 12, p 825).

No.	Phase	Begins with	Ends with
I	Formation	Marriage	Birth of 1st child
II	Extension	Birth of 1st child	Birth of last child
III	Complete extension	Birth of last child	1st child leaves home
IV	Contraction	1st child leaves home	Last child leaves home of parents

No.	Phase	Begins with	Ends with
V	Completed contraction	Last child has left home of parents	1st spouse dies
VI	Dissolution	1st spouse dies	Death of survivor (extinction)

(Park 28e, Ch 12, Table 1, pp 825-826)

- Crises confronting a family are **transitional** — arising within the life cycle (birth of the first child, loss of a spouse) — or **non-transitional** (acts of war, uprooting, mental disorders) (Park 28e, Ch 12, p 826).
- **Stress and health hazards increase when environmental change strikes at critical developmental periods** — adolescence, first pregnancy, menopause and retirement.
- Ability to cope depends on perception of threat, motivation and readiness to respond creatively, available emotional and social supports, and cultural provisions.
- Sociologists have documented the effect of sudden shifts in economic status, migration, uprooting, disasters and physical incapacity of a member at different phases of the cycle.
- Variations that must be allowed for: early death of a child or of a spouse, divorce and childlessness.

Functions of the family relevant to health

Function	Content	Why it matters to the health worker
Residence	To provide a clean and decent home; patrilocal or matrilineal (Park 28e, Ch 12, pp 827-828)	Housing, overcrowding, water storage, indoor air pollution and excreta disposal are family-level exposures (Sunder Lal, Ch 2, p 79)
Division of labour	Rigid gender roles in primitive societies (male earns, female rears); blurred in industrialised and urbanised communities as women take professional careers (Park 28e, Ch 12, pp 826-827)	Determines who is available for child care and for nursing the sick
Child rearing	Physical care of the dependent young so that they survive to adulthood; feeding, hygiene, sleep, clothing, discipline and habit training are handed down between generations (Park 28e, Ch 12, p 827)	Traditional child-care practices are the main obstacle a community health worker meets — e.g. permissive on-demand breastfeeding in the East versus rigid rule-bound care in the West
Socialization / cultural transmission	Family is the bridge between generations and the "transfer point of civilization" ; eating, cleanliness, dress, speech, language, behaviour, attitudes and codes of conduct are transmitted by example and precept (Park 28e, Ch 12, pp 827-828)	Health beliefs, attitudes and practices — and therefore health behaviour — are learnt here
Personality formation	The family acts as a "placenta" — excluding some influences, modifying others, adding its own, laying the foundation of physical, mental and social health of the child (Park 28e, Ch 12, p 828)	Capacity to withstand stress is set by early experience with father, mother and siblings

Function	Content	Why it matters to the health worker
Care of dependent adults	(a) Sick and injured — family gives front-line care, chiefly the mother; the family does more nursing than the hospital , even in highly developed countries. (b) Women in pregnancy and childbirth — maternity leave, diet and nutritional supplements, reduced responsibility in pregnancy and puerperium. (c) Aged and handicapped — long-term care, which in India the joint family provides (Park 28e, Ch 12, p 828)	Societal attitude to pregnancy bears directly on infant deaths, prematurity, stillbirths and maternal morbidity and mortality; without family support no amount of medical care succeeds
Stabilization of adult personality	Family is the " shock absorber " for injury, illness, births, deaths, tension, emotional upset, anxiety and economic insecurity, allowing release of tension and return to mental equilibrium (Park 28e, Ch 12, p 828)	Explains the family's place in community mental health
Status, protection and regulation	Confers name and status; protects members from insult and defamation; regulates marital, political, religious and general social activity, and sex relations through incest taboos (Park 28e, Ch 12, p 827)	Underpins social support and social control of risk behaviour
Biological function	Bearing children and propagating progeny, now to be judged by quality of life rather than number (Sunder Lal, Ch 2, pp 70-71)	Total fertility rate has fallen from 6.0 (1951) to 2.0 (2019-21) , below the replacement level of 2.1 (Sunder Lal, Ch 2, pp 70-71)

The family in disease

Mechanism	Explanation and examples
Shared gene pool	Members share a pool of genes and a common environment, and together these decide susceptibility to disease — haemophilia, colour blindness, diabetes and mental illness run in families; schizophrenia, psychoneurosis and some forms of mental deficiency have a familial incidence (Park 28e, Ch 12, p 828)
Consanguinity	Higher incidence of congenital malformation and of genetic disorders in the offspring of consanguineous marriages (Park 28e, Ch 12, p 828; Sunder Lal, Ch 2, p 70)
Shared environment — communicable disease	The family is the " playground " for communicable disease — tuberculosis, common cold, scabies, diphtheria, measles, mumps, rubella, chickenpox, dysentery, diarrhoea and enteric fever spread rapidly because of the common environment members share (Park 28e, Ch 12, p 828)
Family size	Episodes of infectious diarrhoea and the prevalence of malnutrition increase with the size of the family , malnutrition being commoner in families with more than 4 children ; in short, fewer children would mean better nutrition, better health care, less morbidity and lower infant mortality (Park 28e, Ch 9, p 652). Infant mortality itself increases with family size , severe respiratory infection is commoner, the duration of illness is much longer in families with 3 or more children , and deprivation of maternal care is found in large families (Park 28e, Ch 9, p 674)

Mechanism	Explanation and examples
Stress ("psychosomatic") diseases	Peptic ulcer, colitis, high blood pressure, rheumatism and skin diseases are accepted as stress diseases with a prominent emotional element; alcoholism and narcotic addiction reflect the same trend (Park 28e, Ch 12, p 828)
Lifestyle and non-communicable disease	Lifestyle diseases, chronic non-communicable diseases and behavioural problems all originate at the place of living and in the family , and their natural history can be traced to the family setting (Sunder Lal, Ch 2, p 72)
Gender inequity	Gender roles are determined in the family; food distribution within the family, gender bias in care-seeking, female foeticide and infanticide and the falling child sex ratio are family-level behaviours (Sunder Lal, Ch 2, pp 79, 81-82)
Health-care-seeking behaviour	Household members, particularly mothers, make the primary diagnosis , judge severity and probable outcome, choose treatment (including home treatment of diarrhoea, cough and fast breathing), procure and administer drugs, and decide when to seek formal care for pregnant women and sick children (Sunder Lal, Ch 2, p 82)

Broken families and problem (high-risk) families

- ▶ **Broken family** — one in which the parents have separated, or one or both parents have died (Park 28e, Ch 12, pp 828-829).
- ▶ *Bowlby* brought out **maternal deprivation** as one of the most dangerous pathogenic factors in child development; **paternal separation** and **dual-parental separation** are also important.
- ▶ Children of broken families may later show **psychopathic behaviour, immature personality and retardation of speech and intellect**, and may drift to prostitution, crime and vagrancy.
- ▶ **Problem family** — one that **lags behind the rest of the community**: standards of life far below the accepted minimum, parents unable to meet the physical and emotional needs of their children, home life utterly unsatisfactory (Park 28e, Ch 12, pp 828-829).
- ▶ Underlying factors are usually those of **personality and of relationship** — backwardness, poverty, illness, mental and emotional instability, character defects and marital disharmony; such families are recognised as problems in **social pathology**. They occur in all social classes but are commoner in the lower social classes.
- ▶ **High-risk families to be prioritised in field practice** (Sunder Lal, Ch 2, pp 71-72): large-sized families, families below the poverty line, broken families, addiction in the husband, and prolonged illness of a child or any other member — targeted for health care, social development and poverty alleviation.
- ▶ **Rehabilitation is a team task** — the health visitor, health inspector, midwife, social worker and medical officer of health all render useful service (Park 28e, Ch 12, p 829).

The family as the unit of health care

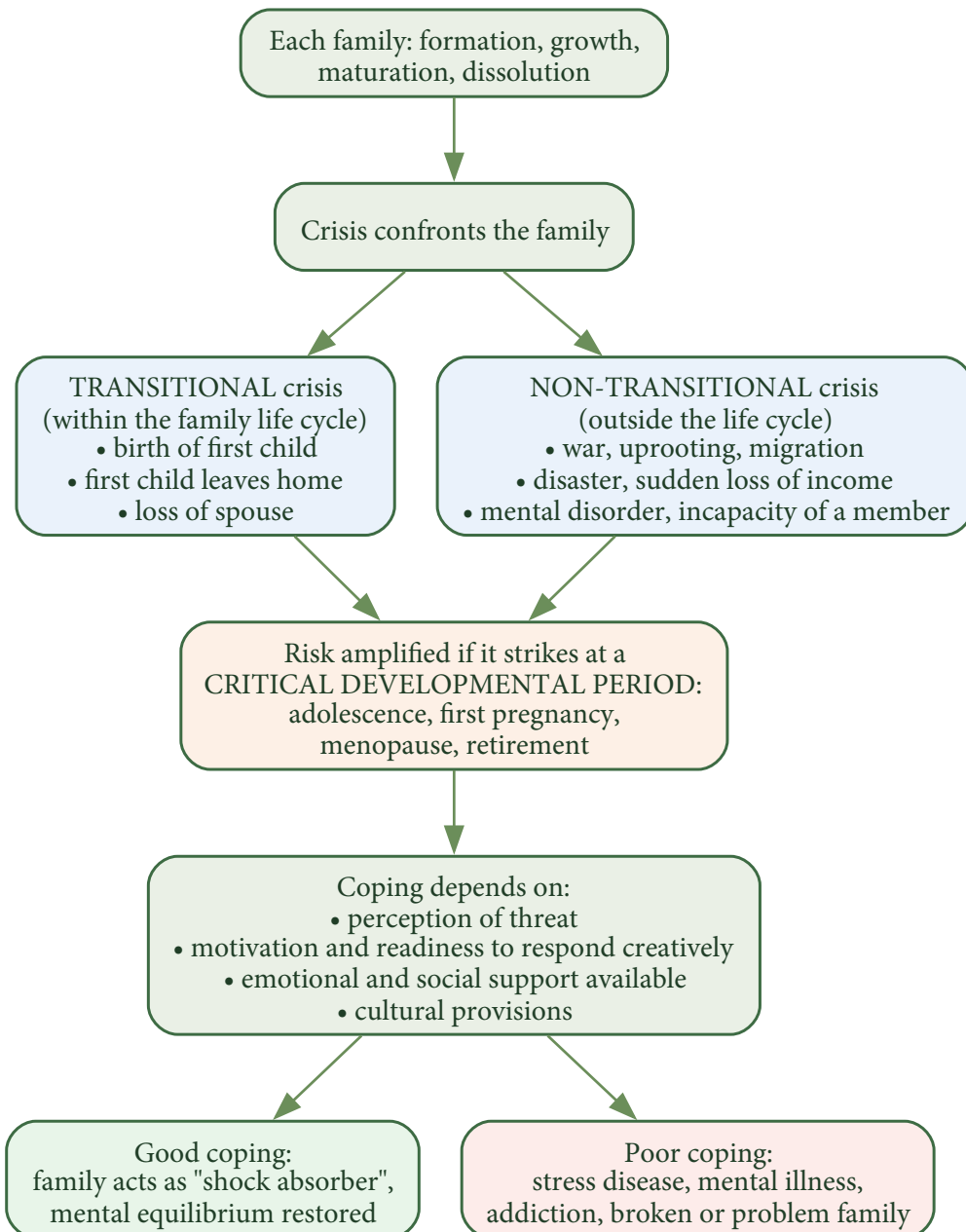
- ▶ **In most societies the family is the fulcrum of health services**; illness is a family event — the individual reacts on the family and the family reacts on the individual (Park 28e, Ch 12, p 829; Sunder Lal, Ch 2, pp 76-78).
- ▶ **Family medicine** is family-oriented medicine, or health care centred on the family as the unit, from first contact to the ongoing care of chronic problems and from prevention to rehabilitation;

applied to patients and their families it becomes the speciality of **family practice**, specially designed to deliver primary care (Park 28e, Ch 1, p 22).

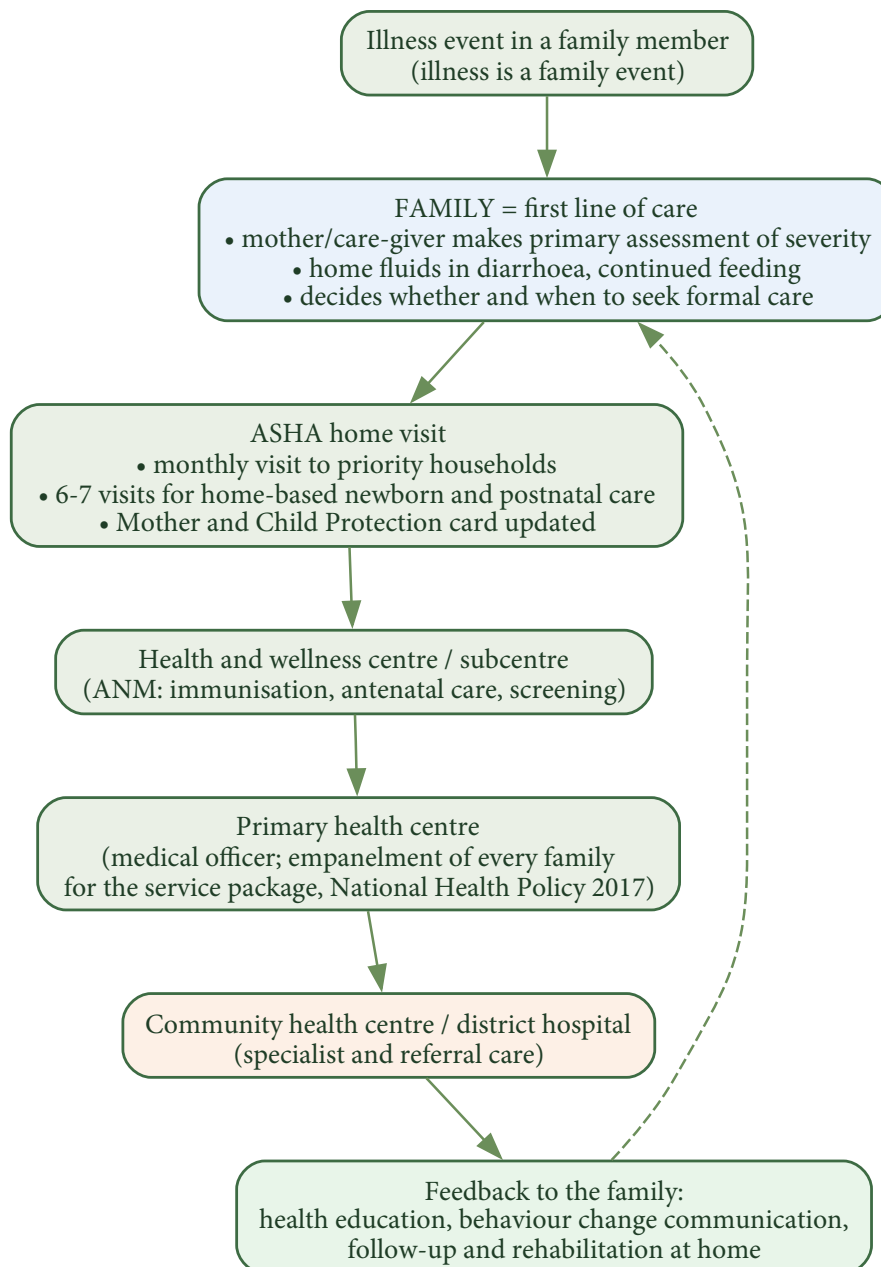
- ▶ **Family-retained records** — the home-based **Mother and Child Protection (MCP) card**, issued jointly by the Ministry of Women and Child Development and the Ministry of Health and Family Welfare, educates and motivates the family to complete all health actions up to 3 years of age (Sunder Lal, Ch 2, pp 76-78).
- ▶ **Outreach through the accredited social health activist (ASHA)** — monthly home visits to priority households, i.e. those with a pregnant woman, a newborn, a child under 2 years, a malnourished child, an eligible couple aged 20–29 years, the poorest families, or a member with tuberculosis or a chronic non-communicable disease; **6–7 visits** for home-based newborn and postnatal care; annual household survey with surveillance of births, deaths and target diseases (Sunder Lal, Ch 2, p 83).
- ▶ **Empanelment of families** — linking every family/household to a primary health-care facility for a defined package of services, under the **National Health Policy 2017** (Sunder Lal, Ch 2, p 83).
- ▶ **Family study** is the core teaching–learning method of community medicine: families are allotted in the first professional year and followed longitudinally, covering socio-economic classification, vital events and their registration, environment, health beliefs and practices, food distribution within the family, infant and young child feeding, growth monitoring, supplementary nutrition, health check-up of members and care of the elderly, immunisation and health-care-seeking behaviour (Sunder Lal, Ch 2, p 79).
- ▶ As *Florence Nightingale* said, "**the secret of national health lies in the homes of the people**" (Park 28e, Ch 12, p 829).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Labelled diagram** — "**The family as a unit**": draw four concentric rings with the health system as the outermost. Labels, innermost outward: **Individual**; **FAMILY — biological unit (common pool of genes)**, **social unit (common physical and social environment)**, **cultural unit**, **epidemiological unit**, **unit of comprehensive medical care and social service**; **Community**; **Health care system (subcentre – primary health centre – community health centre)**. Add two opposing arrows between Individual and Family labelled "*the individual reacts on the family*" and "*the family reacts on the individual*", and label the family ring "*common kitchen, common roof*".



Family cycle and stress — transitional and non-transitional crises acting at critical developmental periods



The family as the unit of health care — from the family's own first-line response to the referral chain

High-yield exam pointers

- ▶ **Family is five units in one:** biological (gene pool), social (common environment), cultural, epidemiological, and the unit of comprehensive medical care and social service.
- ▶ **Household is NOT family:** NFHS-5 household = **related OR unrelated** persons, one dwelling unit, one acknowledged adult head, same housekeeping, counted as one unit; **households per house = number of kitchens (chulahs)**.
- ▶ **Joint family = four "commons" + one head:** common residence, common purse, common kitchen, common property, authority with the eldest member.
- ▶ **Six phases:** Formation → Extension → Complete extension → Contraction → Completed contraction → Dissolution.

- ▶ **Two metaphors examiners look for:** family as a "**placenta**" in personality formation; family as a "**shock absorber**" in stabilization of adult personality.
- ▶ **Broken family** → *Bowlby's maternal deprivation*; **problem family** → "lags behind the rest of the community, standards of life far below the accepted minimum".
- ▶ **Dated Indian figure:** total fertility rate 6.0 (1951) → 2.0 (2019–21), below the replacement level of 2.1; family size — diarrhoea, malnutrition (>4 children) and **infant mortality** all rise with it.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park's six-phase model was constructed for low-mortality Western settings and requires amendment in high-mortality situations (early death of children or of the mother), while "leaving home" is essentially an American and European phenomenon with little relevance in many developing countries of Asia and Africa — quote the model, then add this caveat (Park 28e, Ch 12, p 826).

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 12 (Medicine and Social Sciences, pp 825-829: definition of the family, family life cycle and stress, nuclear/joint/three-generation family, functions of the family, family in health and disease, broken and problem families), Ch 1 (p 22: family medicine and family practice), Ch 9 (p 652: family size, diarrhoea, malnutrition and infant mortality; p 674: family size as a determinant of infant mortality); Sunder Lal — Textbook of Community Medicine, Ch 2 (Relationship of Social and Behavioural Sciences (Humanities) to Health and Disease, pp 69-83: definitions of family including the Bureau of Census (USA) definition, the NFHS-5 household definition and the house/household distinction by kitchens, five characteristics of the family, classification by authority, marriage pattern and sanction of marriage, nuclear and joint family, biological function and total fertility rate 2019-21, high-risk families, Mother and Child Protection card, ASHA home visits and family empanelment under the National Health Policy 2017, family study); National Family Health Survey-5 (household definition); National Health Policy 2017 (family empanelment).



Functions of health communication

Asked in: Paper I 16-Dec-2023 · Paper I 07-May-2022 · Paper I Nov 2017 (3×) — RGUHS MD Community Medicine

Definition — Communication is a **two-way process of exchanging or shaping ideas, feelings and information**, broadly "the countless ways that humans have of keeping in touch with one another" (Park 28e, Ch 22, p 1013). **Health communication** is that process applied to health; the term is **often used synonymously with health education**, which itself suggests "outward and downward" communication of knowledge, and health education is the foundation of a preventive health care system (Park 28e, Ch 22, p 1012). The ultimate goal of all communication is a **change in the desired direction in the person who receives it** — **cognitive** (increase in knowledge), **affective** (change in attitudes and existing patterns of behaviour) and **psychomotor** (acquisition of new skills) — the three **learning objectives** (Park 28e, Ch 22, p 1013).

The eight functions

Park introduces the list with the sentence "**Health communication has to cater to the following needs:**" and prints exactly **eight** items (Park 28e, Ch 22, p 1012).

#	Function	Level of change targeted
1	Information	Cognitive
2	Education	Cognitive + psychomotor
3	Motivation	Affective
4	Persuasion	Affective
5	Counselling	Affective
6	Raising morale	Affective (team)
7	Health development	Cognitive (societal)
8	Health organization	Organisational

Functions 1–4: moving the individual from knowing to doing

- **Information** — the primary function. It provides scientific knowledge about health problems and about how to maintain and promote health. People rarely seek such information although they have a **right to know** the facts about health and disease, so it must be made **easily accessible**. Exposure to the right kind of health information can **eliminate the social and psychological barriers of ignorance, prejudice and misconception**, increase awareness to the point where people **perceive their health needs**, and turn **unfelt needs into felt needs and felt needs into demands**. Government, media and health providers carry the social responsibility to supply factual and balanced information. The assumption that acquiring information by itself changes behaviour and attitudes is **fallacious** — people decide about their health **only after much thought, over a period of time and after several educational contacts**, and their **cultural values, beliefs and norms** govern acceptance. Correct information is the basic part of health education (Park 28e, Ch 22, p 1012).

- ▶ **Education.** Education of the general public is integral to a **prevention-oriented approach**, and the basis of all education is **communication**. It works on the assumption that **knowledge determines attitudes and attitudes determine behaviour**, and it can change lifestyles and risk factors — most of the world's major health problems and premature deaths are **preventable through changes in human behaviour at low cost**. Its stated limit is the examiner's point: **education alone is insufficient to achieve optimum health; the target population must also have access to proven preventive measures or procedures** (Park 28e, Ch 22, p 1012).
- ▶ **Motivation.** Motivation is **the power that drives a person from within to act**. One of the goals of health communication is to **motivate individuals to translate health information into personal behaviour and lifestyle for their own health**. Motivation includes the stages of **interest, evaluation and decision-making**, and health communication assists the individual in passing from the state of **awareness and interest → decision-making → adoption of the new idea or programme**. Motivation is **not long-lasting and may diminish with lapse of time**, so it must be reinforced. The best chance of success lies with people who already have strong motivation — those with **chronic illness or a disability** and those facing an **acute crisis such as surgery or childbirth** — so the **quickest pay-off comes in patient education** (Park 28e, Ch 22, p 1012).
- ▶ **Persuasion.** Park defines it as "a conscious attempt by one individual to change or influence the general beliefs, understanding, values and behaviour of another individual or group of individuals in some desired way" — **the art of winning friends and influencing people, employing neither force nor deliberate manipulation**. Persuasive communication is **more effective than coercion or authoritative communication** and can change lifestyle and modify the risk factors of disease. The ethical boundary must be written: when persuasive communication is deliberately used to **manipulate feelings, attitudes and beliefs it becomes "propaganda" or "brain washing"** (Park 28e, Ch 22, p 1014).

Functions 5–8: supporting the person, the team and the system

- ▶ **Counselling.** A process that helps people **understand and deal better with their problems** and communicate better with those with whom they are emotionally involved. It **improves and reinforces motivation to change behaviour** and **provides support at times of crisis**. It differs from advising: **counselling implies choice, not force**, whereas advising directs people with do's and don'ts. The counsellor must communicate information, gain trust, listen sympathetically to the anxious, distressed and possibly hostile, understand and respond to feelings so the person can express them freely, and help reduce or resolve problems. It therefore relies heavily on **communication and relationship skills**, develops positive attitudes, and is an integral part of **treatment, disease prevention and health promotion** alike (Park 28e, Ch 22, p 1014).
- ▶ **Raising morale.** Morale is "the capacity of a group of people (or team) to pull together persistently or consistently". Communication — **vertical and horizontal, internal and external** — is the **first step in any attempt to raise the morale** of the health team or a group of people (Park 28e, Ch 22, p 1014). This is the function that holds the field workforce — auxiliary nurse midwife (ANM), accredited social health activist (ASHA), multipurpose worker — together as a team.
- ▶ **Health development.** Communication plays a powerful role in health development by **diffusing knowledge about the goals of development and preparing people for the roles expected of**

them. Park is explicit that its own role here is **essentially supportive**, not primary — but **judicious use of communication media does contribute to health development** (Park 28e, Ch 22, pp 1014–1015).

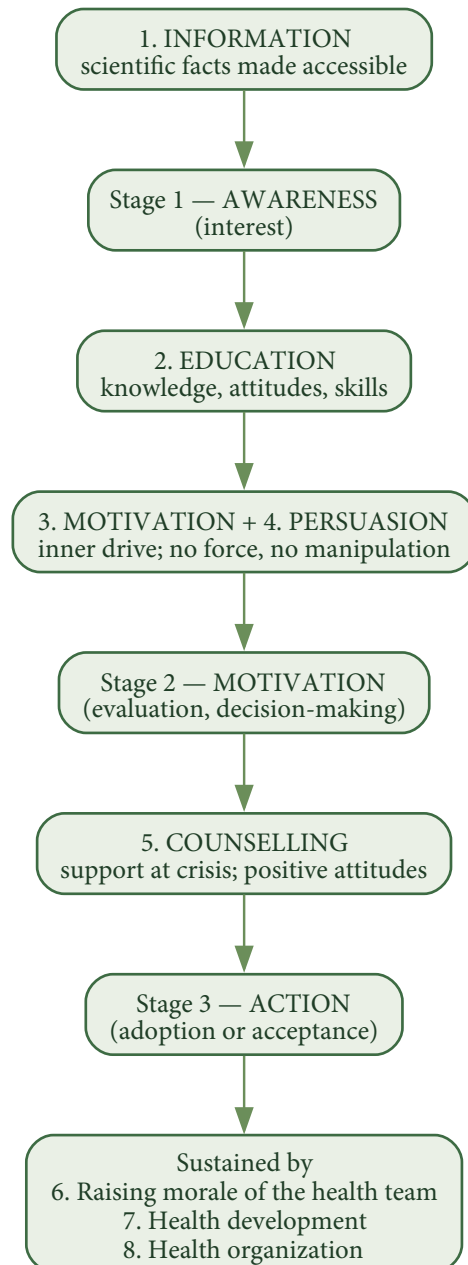
- **Health organization.** Communication is **the life and blood of an organization**. It flows in two major directions: **vertical**, which may be *downward* — from the top administrator through the hierarchy of professionals and non-professionals to the beneficiaries or employees — or *upward*; and **horizontal or cross communication**, usually between equals at any level. The direction in which communication flows **suggests the degree of freedom in the internal communication network**. Communication is therefore an important dimension of health organization and an important means of **intra-sectoral and inter-sectoral coordination** (Park 28e, Ch 22, p 1014).

The functions operationalised in Indian public health practice

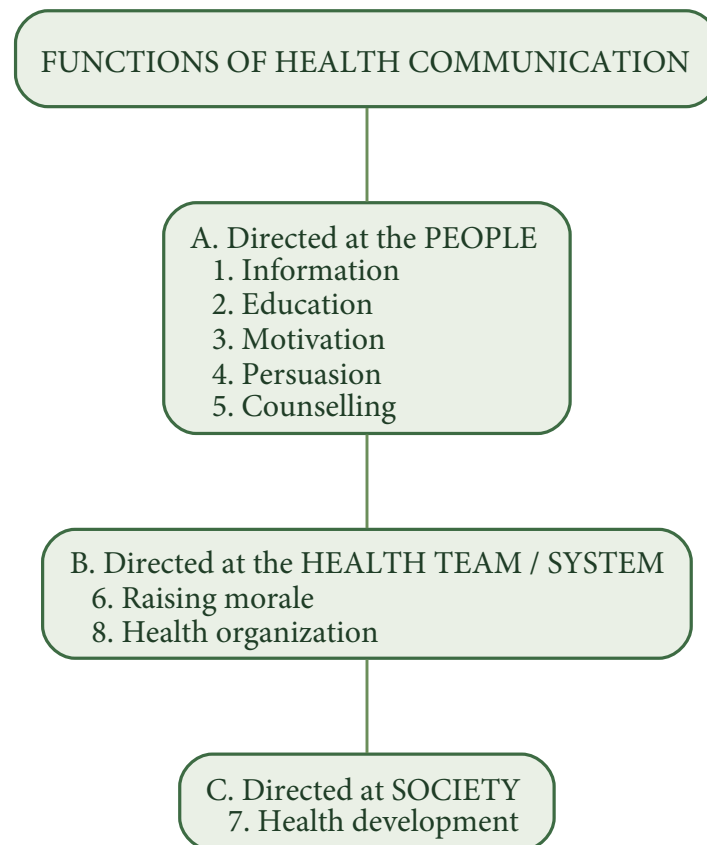
Function in action	Where it is operationalised
Information + education at scale	Strategic Information, Education and Communication (IEC) plan using a 360-degree approach — mass media, mid-media, social media and interpersonal communication — with a year-long plan built round health days and health themes (Sunder Lal, Ch 2, p 85)
Motivation + persuasion for service uptake	National communication strategy under the Reproductive and Child Health (RCH) programme — interpersonal communication for behaviour change at field level, decentralised planning with defined roles for centre, state and district (Sunder Lal, Ch 4, p 155)
Counselling	GATHER approach — Greet, Ask, Tell, Help, Explain, Return for follow-up; for adolescents, information alone is not sufficient — life skills are needed to act on it (Sunder Lal, Ch 4, pp 156–157; Ch 10, p 623)
Health organization	Advocacy through mass media, decentralisation of IEC to states and districts , involvement of non-governmental organisations and the private sector, and capacity building at all levels (Sunder Lal, Ch 10, p 623)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Fig 1 — The communication process (Park's Fig. 1).** Labels: *Sender (source) → Message (content) → Channel (medium) → Receiver (audience)*, with a return arrow labelled *Feedback (effect)* from receiver to sender. Beside it write: feedback is **immediate in interpersonal communication**, delayed in mass communication, and is obtained by **opinion polls, attitude surveys and interviews**.
- **Fig 2 — Adoption model (Park's Fig. 2, "3 stages in the process of change in behaviour").** Labels exactly as printed, top to bottom: *1. Awareness — Interest; 2. Motivation — Evaluation, Decision-making; 3. Action — Adoption or acceptance*. Add: the stages are **not rigid**, stages may be **skipped**, and acceptance that becomes part of the person's own value system is called **internalization** (Park 28e, Ch 22, p 1017).



Derived teaching map — where each function bites along Park's Fig. 2 stages (Park does not itself sequence the functions this way)



The eight functions of health communication grouped by whom they act upon

High-yield exam pointers

- ▶ Mnemonic for the eight functions — I-E-M-P-C-M-D-O: Information, Education, Motivation, Persuasion, Counselling, raising Morale, health Development, health Organization.
- ▶ **Information is the primary function; education alone is insufficient** unless proven preventive measures are accessible.
- ▶ Three learning objectives = **cognitive** (knowledge), **affective** (attitude and behaviour), **psychomotor** (skills).
- ▶ Park's Fig. 2 **Adoption model** — Awareness–Interest → Motivation (evaluation, decision-making) → Action (adoption or acceptance); motivation diminishes with lapse of time; internalization is the end point.
- ▶ Persuasion used to manipulate feelings, attitudes and beliefs = "**propaganda**" or "**brain washing**" — the one-line ethical caveat examiners look for.
- ▶ Communication within a health organization flows **vertically** (downward and upward) and **horizontally** (cross communication between equals); the direction shows the degree of freedom in the network.
- ▶ Five components of the communication process: **sender, receiver, message, channel, feedback**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park prints the list inside a box headed *Functions of health communication* whose lead-in reads "Health communication has to cater to the following needs:", and in that box the sixth and eighth items are printed simply as "Raising morals" and "Organization", while the body sections that follow are headed "6. Raising morale" and "8. Health organization" — write the body headings in the exam. Every "Ch 22, p 1012" citation above refers to the same chapter, *Communication for Health Education*: the scanned copy of Park 28e indexed in this corpus carries the functions box and the sections on information, education and motivation on p 1012, which the index files under Chapter 21 (Health Information and Basic Medical Statistics) because of a chapter-boundary defect in the scan, and the identical text appears in Park 27e, Ch 22, pp 984–985. The "Level of change targeted" column of the first table and the first flowchart (which maps the functions onto Park's Fig. 2) are teaching aids derived here from Park's three learning objectives and adoption model — Park neither tabulates the functions that way nor sequences them against Fig. 2. Park's Fig. 2 places *Interest* against stage 1 (Awareness) while the accompanying text says motivation "includes the stages of interest, evaluation and decision-making"; reproduce the figure exactly as printed.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Communication for Health Education (the functions of health communication, the communication process and its five components, the three learning objectives, and the adoption model), cross-checked against Park's Textbook of Preventive and Social Medicine, 27e, Ch 22 for the passages garbled in the 28e scan. Sunder Lal — Textbook of Community Medicine — Relationship of Social and Behavioural Sciences to Health and Disease (the strategic Information, Education and Communication plan), Principles of Health Promotion and Education (the national communication strategy under the Reproductive and Child Health programme; the GATHER approach to counselling), and Reproductive, Maternal, Newborn, Child and Adolescent Health (the national communication strategy for adolescents).



Gender issues and women's health

Asked in: Paper III 06-Jan-2025 — RGUHS MD Community Medicine

Definition — **Sex** is the biological difference between females and males — anatomical, genetic, biochemical and physiological — and is largely universal and constant (Oxford Textbook of Public Health 7e, Ch 85, p 3178). **Gender** is a social construct: the roles, activities, responsibilities and power relations a society attributes to being female or male, acquired through socialisation and therefore changeable across time and culture (Oxford 7e, Ch 85, p 3178). Only a small proportion of the roles assigned to men and women is attributable to sex; it is **gender, not sex, that has the far greater impact on women's health**, acting as a cross-cutting social determinant alongside class, caste, education and location (Sunder Lal, Ch 10, p 630; Oxford 7e, Ch 85, p 3177).

Sex versus gender; equity versus equality

Attribute	Sex	Gender
Basis	Biological — XX in females, XY in males; the spermatozoon (not the ovum) is the sex determinant	Social, cultural and behavioural construction
Origin	Genetically determined at conception	Learned through socialisation, beginning in the family and school
Variability	Universal, largely constant across societies	Varies between societies and cultures, and across time
Changeability	Not readily changeable	Changeable — hence amenable to public health intervention
Health example	Pregnancy, childbirth, cervical cancer; autoimmune disease commoner in women	Son preference, differential feeding, restricted mobility, spousal violence
Corollary	Women are wrongly blamed for infertility and for the birth of a girl child	Patriarchy defines decision-making, authority and access to resources

Term	Meaning
Gender equity	The process of being fair to both men and women; may require positive discrimination in favour of women. Equity leads to equality (Sunder Lal, Ch 10, p 630)
Gender equality	The end-state — women and men enjoy the same status, with equal conditions for realising their human rights and for contributing to and benefiting from development; not sameness, but equal valuation of similarities and differences (Sunder Lal, Ch 10, p 630)
Gender mainstreaming	Incorporating a gender perspective into policies, strategies, programmes, project activities, administrative functions and organisational cultures, rather than running separate "women's" schemes; popularised from the mid-1990s and adopted by the United Nations Economic and Social Council in 1997, by the World Health Organization (WHO) and by development organisations (Oxford 7e, Ch 85, pp 3182–3183)

Term	Meaning
Critique of mainstreaming	Its health and development impact has been limited — it was interpreted differently in different implementation contexts, which diluted awareness and action, and it depoliticised gender power relations by taking a largely technocratic, "equal representation" approach (Oxford 7e, Ch 85, p 3183)

Gender issues that determine women's health in India

Life domain	Gender issue	Health consequence
Before birth	"Son syndrome" — son preference, the girl child seen as <i>paraya dhan</i> (a liability); misuse of diagnostic technology for sex determination	Gender-biased sex selection, female foeticide, adverse child sex ratio
Infancy and childhood	Discriminatory feeding, rearing and care-seeking during illness; female infanticide	Excess female mortality — female exceeds male mortality by about 10% at 1–11 months, and female child mortality is about 1.5 times male beyond age 1 y, worse in rural India (Sunder Lal, Ch 10, p 632)
Education	Low enrolment, retention and continuation for girls; girls used as caregivers for younger siblings	Illiteracy, lack of health information, a permanent lifelong disadvantage
Marriage and fertility	Early marriage; no say in family size, spacing or method of contraception	Adolescent pregnancy, high-risk and repeated pregnancy, maternal death
Nutrition	Women "eat last and eat least"; discriminatory intra-family food distribution	Anaemia, chronic energy deficiency, low-birth-weight babies — an intergenerational effect (Sunder Lal, Ch 10, p 631)
Work	"Triple burden" — work within the home, work outside it, and child-rearing; unpaid, unrecognised drudgery of fuel, fodder and water collection	Energy-sapping workload continued through pregnancy, making pregnancy unsafe
Resources and autonomy	Less control over money; permission needed to visit the market or relatives; low political representation	Delayed decision to seek care and delayed referral; out-of-pocket expenditure falls hardest on women, whose untreated illness exceeds men's even at higher socio-economic status (Oxford 7e, Ch 85, p 3183)
Violence	Domestic violence, dowry death, honour killing, rape, trafficking, sexual harassment; social acceptance of wife-beating	Injury, depression and anxiety, unintended pregnancy, sexually transmitted infection, loss of reproductive control
Contraception	Female-centric programme with poor male participation; women cannot negotiate condom use	Burden of sterilisation on women; unmet need; vulnerability to HIV acquired from the spouse
Health system	Services organised around reproductive needs alone; gender-insensitive providers; poor mass-media exposure	Health needs beyond the reproductive years neglected; very low cancer-screening uptake

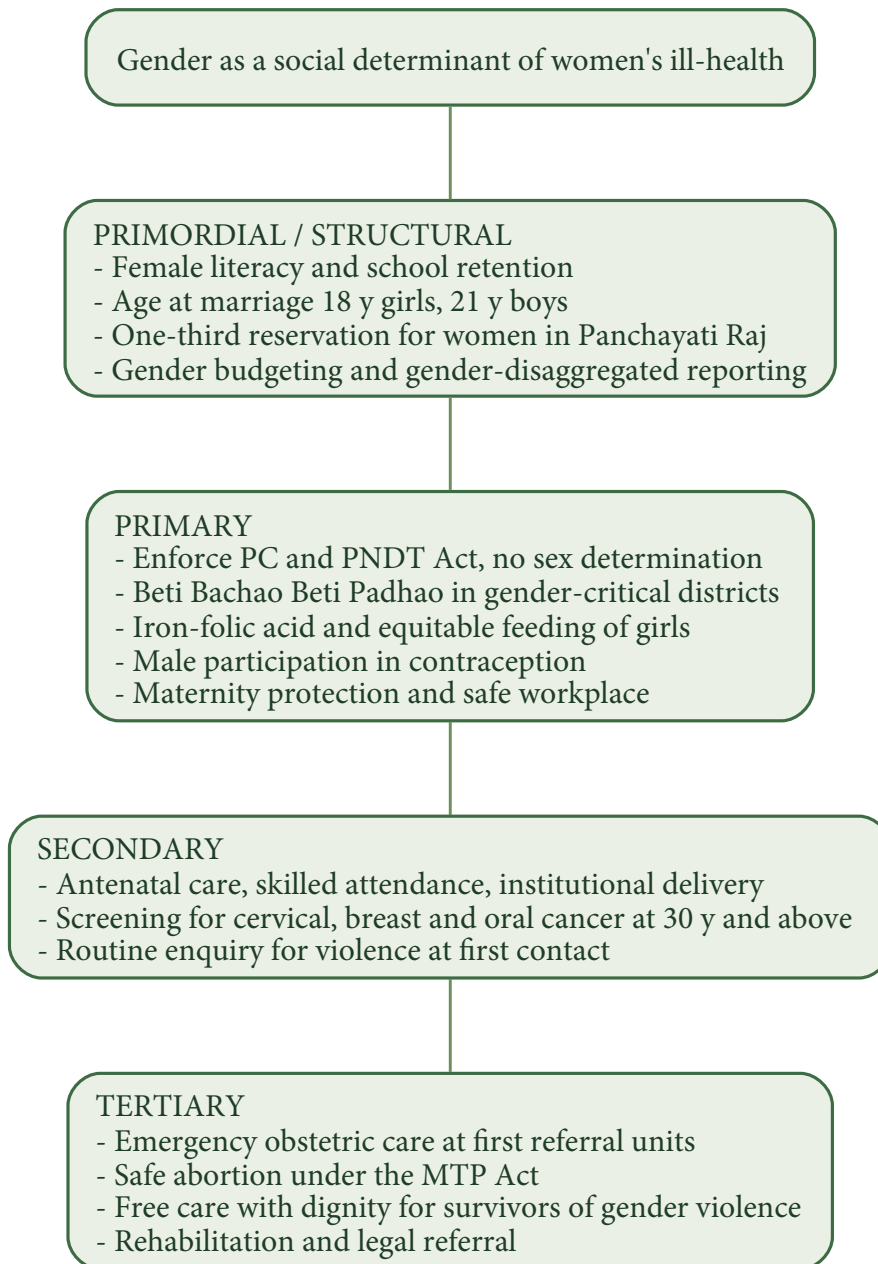
Magnitude — gender-disaggregated Indian indicators

Indicator	Current value	Source (year)
Sex ratio, all ages	1,020 females per 1,000 males — favourable for the first time	NFHS-5 (2019–21)
Sex ratio, age 0–6 y	928 (urban 924, rural 930)	NFHS-5 (2019–21)
"Missing women"	35.7 million females missing in 2011 — 587.5 million females against 623.2 million males in a population of 1,210 million (overall sex ratio 943)	Census 2011 (Sunder Lal, Ch 9, pp 545–546)
Child sex ratio, 0–6 y	914 — an all-time low since 1961 (927 in 2001; rural 919, urban 902); worst — Punjab 793, Haryana 820	Census 2011 (Park 28e, Ch 8)
Sex ratio at birth	907 (rural 907, urban 910); Kerala 974 to Uttarakhand 844	Sample Registration System, 3-year average 2018–20 (Park 28e, Ch 8)
Literacy	Female 65.46% vs male 82.14% (overall 74.04%)	Census 2011 (Park 28e, Ch 8, p 585)
Secondary schooling attained	39% of women vs 63.5% of men	Sunder Lal, Ch 10, p 631
Married before age 18 y (women 20–24 y)	23%	NFHS-5 (2019–21)
Median age at first marriage	19.2 y (women) vs 24.9 y (men)	NFHS-5 (2019–21)
Anaemia, age 15–49 y	57% of women vs 25% of men	NFHS-5 (2019–21)
Body mass index (BMI) below 18.5 kg/m ²	18.7% of women 15–49 y	NFHS-5 (2019–21)
Maternal mortality ratio (MMR)	97 per 100,000 live births; target 70 by 2030	Sunder Lal, Ch 10, p 631
Any spousal violence, ever-married women 18–49 y	32% (physical 28%, emotional 14%, sexual 6%); only 14% of survivors of any physical or sexual violence sought help	NFHS-5 (2019–21), Ch 15
Contraception — modern methods 56.5%	Female sterilisation 37.9% vs male sterilisation 0.3% ; condom 9.5%; unmet need 9.4%	NFHS-5 (2019–21)
Ever screened, women 30–49 y	Cervical cancer 1.9%; breast and oral cavity examination 0.9%	Sunder Lal, Ch 7, p 299
Labour force and politics	Women 29% of the workforce vs men 81%; women 13.5% of Parliament	Sunder Lal, Ch 10, pp 631–632
Life expectancy at birth (2019)	Female 71 y vs male 68.5 y	Sunder Lal, Ch 10, p 631
Gender Inequality Index (GII) — reproductive health, empowerment, economic activity	0.488, rank 132 of 187; Global Gender Gap Index 2021 rank 140	Sunder Lal, Ch 10, p 630

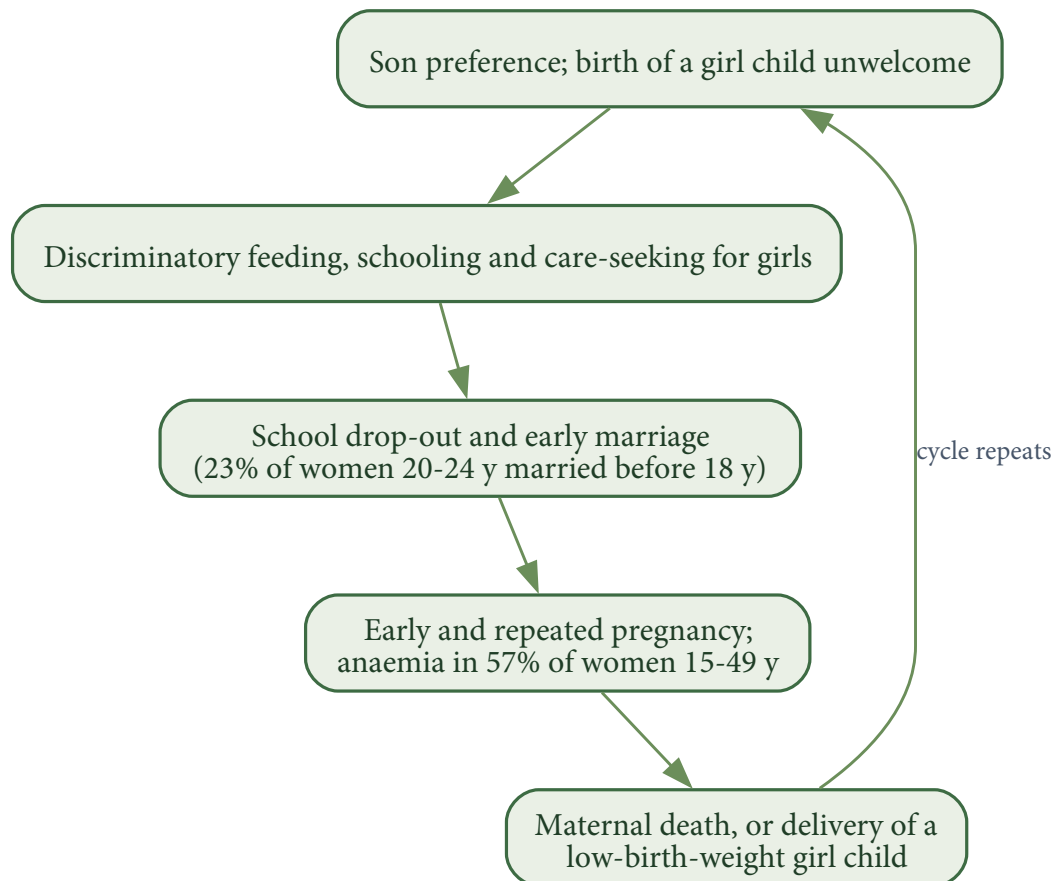
Women's health problems across the life cycle

Life stage	Predominant gender-linked health problem
Pre-conception and foetal	Sex-selective abortion, female foeticide
Infancy and childhood	Excess female mortality, more severe malnutrition in girls, missed care during illness
Adolescence (10–19 y)	Anaemia, school drop-out, child marriage, adolescent pregnancy, sexual abuse
Reproductive years (15–49 y)	Maternal death from haemorrhage, sepsis, abortion, hypertensive disorders, obstructed labour and anaemia (Park 28e, Ch 9, p 663); reproductive tract and sexually transmitted infection; unsafe abortion, causing about 8% of maternal deaths; unmet need for contraception; spousal violence
Peri-menopausal and 40+ y	Cervical, breast and oral cancer; diabetes, hypertension and cardiovascular disease — under-detected because services traditionally stop at the reproductive age (Sunder Lal, Ch 16, p 699)
Elderly women	Widowhood, destitution, absent social security, chronic degenerative disease and neglect

Health-sector, programme and legal response



Levels of prevention applied to gender issues in women's health



The intergenerational cycle by which gender discrimination reproduces ill-health in women

Legislative measures

Instrument	Core provision
Dowry Prohibition Act 1961, Amendment 1986	Minimum punishment for taking or abetting dowry raised to 5 years' imprisonment and a fine of Rs 15,000 (Park 28e, Ch 12, p 840)
Prohibition of Child Marriage Act 2006 (PCMA) — in force 1 November 2007; replaced the Child Marriage Restraint Act 1978 (originally the <i>Sharda Act</i> 1929)	Legal age at marriage 18 y for girls and 21 y for boys. PCMA prohibits child marriage rather than merely restraining it: it makes child marriage an offence and punishes those who conduct, abet, promote or permit the marriage (Park 28e, Ch 9, p 698)
PC & PNDT Act 1994 (Pre-conception and Pre-natal Diagnostic Techniques)	Prohibits sex selection before and after conception. Prenatal diagnostic techniques are permitted for six purposes only — chromosomal abnormality, genetic metabolic disorder, haemoglobinopathy, sex-linked genetic disease, congenital abnormality, and any other abnormality specified by the Central Supervisory Board (Sunder Lal, Ch 9, p 567)
Protection of Women from Domestic Violence Act 2005 (in force 2006)	Civil protection covering physical, emotional, verbal, sexual and economic violence; protection officers appointed; right to reside in the shared household (NFHS-5 India Report, Ch 15)

Instrument	Core provision
Maternity Benefit (Amendment) Act 2017	26 weeks' paid maternity leave, up to 8 weeks before the expected date of confinement; 12 weeks after two children; crèche where more than 50 women are employed (Park 28e, Ch 18, p 967)
Factories Act (Sections 66 and 34); Indian Mines Act 1923	No night work between 7 pm and 6 am; no carrying of excessive weights; no underground work
Immoral Traffic (Prevention) Act; Indecent Representation of Women (Prohibition) Act 1986; Commission of Sati (Prevention) Act 1987; Muslim Women (Protection of Rights on Marriage) Act 2019	Trafficking, obscene portrayal, sati and instant <i>talaq</i> respectively (Sunder Lal, Ch 10, p 634)

Programme and policy measures

- **Beti Bachao Beti Padhao (BBBP)** — the Ministry of Women and Child Development is nodal, supported by the Ministries of Health and Family Welfare, Human Resource Development and Information and Broadcasting; **100 gender-critical districts**; objectives — prevent gender-biased sex-selective elimination, ensure the survival and protection of the girl child, and ensure her education, through advocacy plus stricter PC & PNDT enforcement (Sunder Lal, Ch 10, p 634).
- **National Health Policy 2017** — "women's health and gender mainstreaming" — extend provision to reproductive morbidity and to the health needs of women **beyond the reproductive age (40+)**, orient staff to gender-sensitive and women-friendly services, and provide free services with dignity to victims of gender violence (Sunder Lal, Ch 16, p 699).
- **Gender budgeting** — the Ninth Plan's Women's Component Plan earmarked 30% of funds for women-related sectors; all departments were instructed to open Gender Budget Cells in 2005; programme performance is judged on gender-disaggregated data, with primary health centre (PHC) and Integrated Child Development Services (ICDS) reports rendered sex-wise for morbidity, mortality, nutrition and service use (Sunder Lal, Ch 10, p 633).
- **Reproductive and Child Health / RMNCH+A** — after the International Conference on Population and Development (Cairo, 1994) the family planning programme was repositioned to cover the **total reproductive span**, adding reproductive tract and sexually transmitted infection care, adolescent sexual and reproductive health, and screening for breast, cervical and oral cancer; targets were replaced by community need assessment (Sunder Lal, Ch 10, p 628).
- **Political and economic empowerment** — the 73rd and 74th Constitutional Amendment Acts (passed December 1992, law from April 1993) reserve **not less than one-third** of the total seats and of the offices of chairperson at each level for women in Panchayats and Nagar Palikas (Sunder Lal, Ch 10, p 633); self-help groups, land and inheritance rights, and the Pradhan Mantri Matru Vandana Yojana cash maternity incentive of Rs 5,000 (Sunder Lal, Ch 9, p 552).
- **National Policy for the Empowerment of Women 2001**, following the shift from a **welfare** to a **development** approach to women's issues from the Fifth Plan onwards; the Constitution both grants equality to women and empowers the State to adopt positive discrimination in their favour (Sunder Lal, Ch 10, pp 630, 632).

- **International commitments** — CEDAW (Convention on the Elimination of All Forms of Discrimination Against Women), adopted by the United Nations General Assembly in 1979 and ratified by India on 25 June 1993 (Sunder Lal, Ch 10, p 634); **Sustainable Development Goal (SDG) 5**, targets 5.2 (eliminate all forms of violence against women and girls, including trafficking and exploitation), 5.3 (eliminate harmful practices such as child, early and forced marriage and female genital mutilation) and 5.6 (universal access to sexual and reproductive health and reproductive rights) (Park 28e, Ch 14, p 854).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Sex-versus-gender box pair** — draw two boxes side by side: the left labelled "SEX — biological, XX/XY, universal, unchangeable; pregnancy, childbirth, reproductive cancers"; the right labelled "GENDER — social, learned, varies by culture, changeable; son preference, differential feeding, restricted mobility, violence". Draw a single thick arrow from the GENDER box to a central circle labelled "Women's health status", and a thin arrow from the SEX box, to show which determinant predominates.

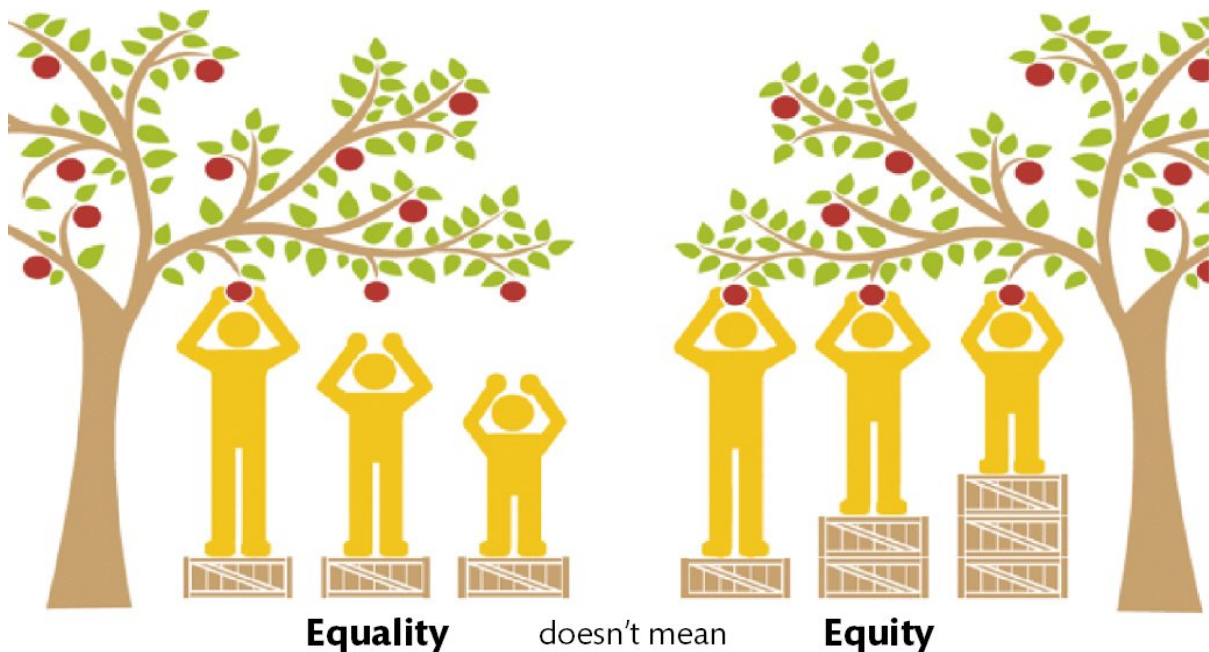
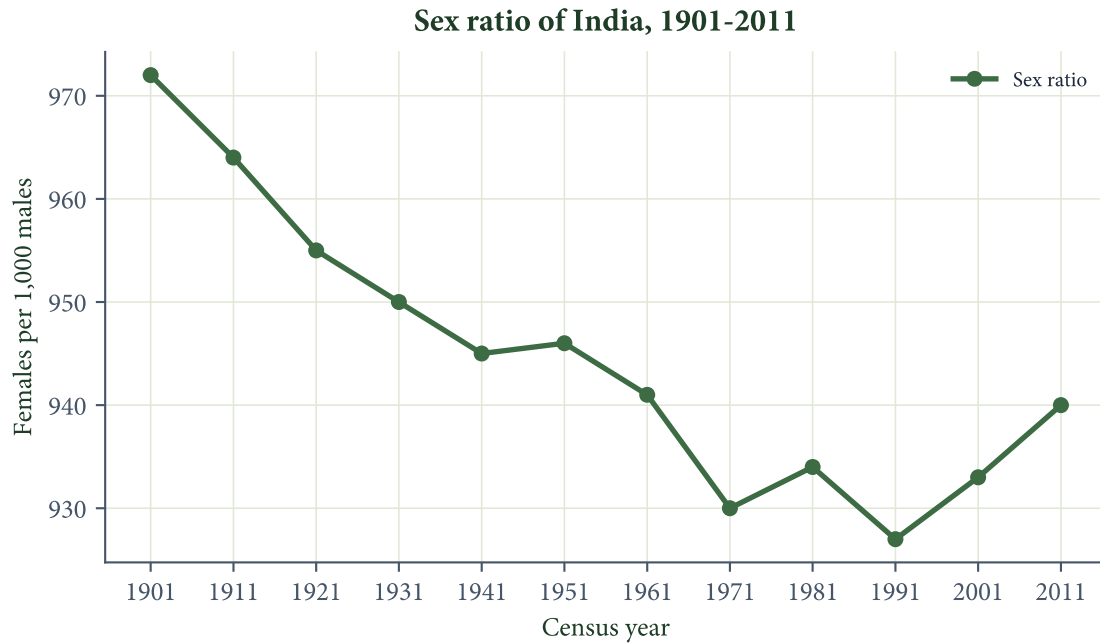


Fig 10.2.1 — The difference between equity and equality (Oxford public health 7th ed, p.3181)



Long-term decline in India's sex ratio since 1901 with partial recovery after 1991 (Park's Textbook of Preventive and Social Medicine 28e, Ch 8, Table 9; Census of India 2011). The 2011 point is the provisional 940 printed by Park; the final Census 2011 figure is 943 (Sunder Lal, Ch 9, p 545).

High-yield exam pointers

- ▶ **Sex is biological and constant; gender is social and changeable** — and it is gender, not sex, that mainly determines women's health. Equity is the *process*, equality is the *outcome*; mainstreaming (UN Economic and Social Council, 1997) is the *method*, and its impact has been limited because it depoliticised gender power relations.
- ▶ Quote every sex ratio **with its source**: all ages **1,020** and age 0–6 y **928** (NFHS-5, 2019–21) against child sex ratio **914** (Census 2011; Punjab 793, Haryana 820); **35.7 million women were "missing" in Census 2011**.
- ▶ The strongest single gender contrast to write: **anaemia 57% in women versus 25% in men, age 15–49** (NFHS-5), and **female sterilisation 37.9% versus male sterilisation 0.3%**.
- ▶ Violence: **32% of ever-married women 18–49 y report any spousal violence** and only **14% of survivors ever sought help** (NFHS-5); the governing law is the **Protection of Women from Domestic Violence Act 2005**.
- ▶ MMR **97 per 100,000 live births** with a target of **70 by 2030**; unsafe abortion causes about **8% of maternal deaths**.
- ▶ **PC & PNDT Act 1994** — remember its **six permitted purposes** for prenatal diagnosis; the current child-marriage law is the **Prohibition of Child Marriage Act 2006** (in force 1 Nov 2007), not the old Restraint Act; BBBP covers **100 gender-critical districts** under the Ministry of Women and Child Development.
- ▶ CEDAW adopted **1979**, ratified by India **25 June 1993**; SDG 5 targets **5.2, 5.3, 5.6**; the 73rd and 74th Amendments reserve **one-third** of local body seats for women.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

A Bill passed in December 2021 to raise the legal age at marriage for girls from 18 to 21 years is still pending in Parliament, so 18 years remains the law (Sunder Lal, Ch 9). The figures used here are from the National Family Health Survey-5 (2019–21) and Census 2011 as carried in the retrieved textbooks; later rounds of the National Family Health Survey and the Sample Registration System may revise the sex ratio, anaemia and maternal mortality values, so always attach the survey round and year to any figure written in the examination.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 8 Demography and Family Planning, Table 9 and Table 13; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics; Ch 12 Medicine and Social Sciences; Ch 14 Sustainable Development Goals; Ch 18 Occupational Health); Sunder Lal — Textbook of Community Medicine (Ch 7 Epidemiology; Ch 9 Demography, Vital Statistics and Population Stabilization; Ch 10 Reproductive, Maternal, Newborn, Child and Adolescent Health; Ch 16 Health Planning and Management); Oxford Textbook of Public Health 7e (Ch 85, "Women, men, and health"); Sociology: The Essentials 6e (Ch 11 Gender); National Family Health Survey-5 India Report (2019–21), Ch 15 Domestic Violence; Census of India 2011; National Health Policy 2017; National Policy for the Empowerment of Women 2001; CEDAW (1979).



Family life education

Asked in: Paper III 29-Sep-2025 — RGUHS MD Community Medicine

Definition — Family life education (FLE) is that component of health education which builds correct knowledge and healthy attitudes towards human reproduction, marriage, parenthood and family relationships (Park 28e, Ch 9, p 685; Suryakantha, Ch 22, p 601). It sits within *family health*, one of the eight contents of health education (Park 28e, Ch 22, p 1018), and overlaps *population education* (Park 28e, Ch 8, pp 615-616).

1. Situation analysis and communication needs assessment

Baseline is drawn from the district fact sheet of the National Family Health Survey (NFHS-5), the Health Management Information System, civil registration, school enrolment records, and a fresh **communication needs assessment (CNA)** — measurement of the prevailing knowledge, attitudes, beliefs and practices of adolescents, newly married couples and parents on the chosen problem (Sunder Lal, Ch 4, p 153). National values below are the benchmark against which the district is compared.

Indicator	Value (source, year)	Implication for the plan
Women aged 20–24 married before 18 y (legal minimum age)	23.3% (NFHS-5, 2019–21)	Age at marriage is the single largest gap
Median age at first marriage	Women 20–49: 19.2 y; men 25–49: 24.9 y (NFHS-5, 2019–21)	Legal minimum 18 y (women), 21 y (men)
Women 15–19 who have begun childbearing	6.8% overall; rural 7.9%, urban 3.8% (NFHS-5, 2019–21)	Rural blocks get priority
The same, by schooling	17.6% with no schooling vs 4.2% with ≥ 12 years of schooling (NFHS-5, 2019–21)	Out-of-school girls are the highest-risk segment
Age-specific fertility rate, 15–19 y	43 per 1,000 women (NFHS-5, 2019–21)	Falling but still a programme target
Comprehensive knowledge of HIV, 15–24 y	Women 20%, men 29% (NFHS-5); was 22% and 32% in NFHS-4	Knowledge is <i>declining</i> — content must be refreshed
Total fertility rate	2.0 (rural 2.1, urban 1.6) (NFHS-5, 2019–21)	Small-family norm largely internalised
Share of men in sterilisation	0.3% of men vs 37.9% of women (NFHS, quoted Sunder Lal, Ch 4, p 154)	Male responsibility is a core message
Sex ratio at birth, last 5 years	929 (NFHS-5; 919 in NFHS-4)	Son preference must be addressed
Adolescent nutrition	24% thin (body mass index for age below -2 SD), 5% overweight or obese, 54% anaemic (Comprehensive National Nutrition Survey 2016–18, quoted Sunder Lal, Ch 10, p 620)	Links FLE to the nutrition package

2. Objectives, target groups and content

Objectives are stated as measurable behaviour change within a stated period — a *single over-riding communication objective* per audience (Sunder Lal, Ch 4, p 153); e.g. "to raise the proportion of girls married at or after 18 years from the district baseline by 15 percentage points in 3 years."

Target group	Where reached	Core content
In-school adolescents (classes VI–XII)	Government and aided schools	Human biology, puberty and menstrual hygiene, right age of marriage, life skills, prevention of substance misuse, gender and consent
Out-of-school adolescents	Village/ward peer-education groups, Anganwadi, youth clubs	Same package, plus vocational and school re-entry linkage
Newly married and eligible couples	Home visits, sub-centre, postpartum clinic	Age at first birth, spacing of 2 years, informed choice of all contraceptive methods, male responsibility

Target group	Where reached	Core content
Parents and guardians	Parent–teacher meetings, Village Health, Sanitation and Nutrition Committee	Sanction for the school curriculum, communication with adolescents, no early marriage, no sex selection
Teachers, ASHAs, Anganwadi workers, auxiliary nurse midwives (ANMs)	Block-level training	Content, counselling skills, referral
Opinion leaders, Panchayat, religious and self-help group leaders	Advocacy meetings	Legitimising the programme within the community

Content is delivered as **life skills** — "the abilities for adaptive and positive behaviour that enable individuals to deal effectively with the demands and challenges of everyday life" (WHO 1993) — the ten core skills being self-awareness, critical thinking, decision-making, effective communication, coping with stress, empathy, creative thinking, problem-solving, interpersonal relationship and coping with emotions (Sunder Lal, Ch 10, p 622).

3. Organisation and delivery plan

The District Health Officer is the nodal officer, and the whole activity is run as a district activity of the **District Health Society** under the National Health Mission; an intersectoral task force under that Society brings in Education, Women and Child Development, Youth Services, Social Welfare, Panchayati Raj and non-governmental organisations. Delivery runs on three existing national platforms rather than a parallel structure.

Stream	Platform	Activity and periodicity
School	School Health Programme under Ayushman Bharat	Two teachers per school (one male, one female) trained as Health and Wellness Ambassadors ; classroom transaction 1 hour every week ; fortnightly/monthly thematic school assembly and question-box responses; quarterly thematic Adolescent Health Days and parent–teacher meetings; bi-annual deworming (Sunder Lal, Ch 10, p 623; Park 28e, Ch 9, p 685)
School (senior)	Adolescence Education Programme, under the National AIDS Control Programme	16-hour sessions during the academic terms of classes IX and XI , on life skills and coping with the physical and psychological changes of growing up (Park 28e, Ch 7, p 526)
Community	Rashtriya Kishor Swasthya Karyakram (RKSK, launched 2014)	Peer educators — 2 male and 2 female per village / per 1,000 population / urban ward , each forming a group of 15–20 same-sex adolescents, weekly participatory sessions of 1–2 hours using the peer education kit; quarterly Adolescent Health Day in every village/ward; Weekly Iron and Folic Acid Supplementation; Menstrual Hygiene Scheme (Sunder Lal, Ch 10, pp 621–622)

Stream	Platform	Activity and periodicity
Facility	Adolescent Reproductive and Sexual Health services	Adolescent Friendly Health Clinics (AFHC) on fixed days at primary health centre, community health centre and district hospital, for married and unmarried, boys and girls — counselling, contraceptives, menstrual problems, reproductive tract and sexually transmitted infection management, antenatal care and anaemia (Park 28e, Ch 7, p 556)
Household	ASHA, Anganwadi worker, ANM home visits	Newly-wed and eligible-couple contact, tracking and repeat messaging

Resources are budgeted from the existing information, education and communication / behaviour change communication head of the District Health Action Plan; no new cadre is created.

4. Methods, media and principles of communication

Approach	Methods	Best use in this plan
Individual	Personal contact, home visits, personal letters	Counselling at AFHC and ANM home visits; permits discussion and persuasion, but reaches few
Group	Lecture ("chalk and talk"), demonstration, group discussion (6–12 members seated in a circle with a group leader and a recorder), panel discussion (4–8 panellists), symposium, workshop, role play (Park 28e, Ch 22, pp 1021–1022)	The core method — peer-education sessions and classroom transactions
Mass	Television, radio, newspaper, printed material, direct mailing, posters, health museums and exhibitions, folk methods, internet	District-wide awareness; reaches widest population but changes behaviour less than personal methods

Personal communication capitalises on warmth and understanding, allows questions and fears to be expressed, and is more influential at average and below-average educational levels; mass media reach the widest population and are more effective among those of above-average educational level (Park 28e, Ch 22, p 1021) — so the plan uses both.

Principles applied (Park 28e, Ch 22, pp 1018–1019): **credibility** (facts compatible with local culture), **interest** (start from felt needs), **participation** (adolescents plan their own sessions), **motivation**, **comprehension** (local language, no jargon), **reinforcement** (repetition at intervals), **learning by doing**, **known to unknown**, **setting an example**, **good human relations**, **feedback**, and **working through leaders**.

5. Anticipated constraints and how the plan meets them

Constraint	Response built into the plan
Teachers shy away from discussing sexuality and reproduction; most of the reproductive content already in the curriculum is not transacted in practical terms (Sunder Lal, Ch 10, pp 623–624)	Structured teacher training with a scripted module; question box for anonymous queries

Constraint	Response built into the plan
Parental and community resistance	Advocacy first with Panchayat, religious and self-help group leaders, then parent–teacher meetings, before classroom roll-out
Earlier programmes reached mainly urban schools, rural schools were not covered	Block-wise phasing that starts with rural and tribal blocks
Only more than half of adolescents are in school, so the school stream reaches barely half the target group (Sunder Lal, Ch 10, p 623); drop-outs are left out	Out-of-school adolescents covered through peer educators, Anganwadi and youth clubs, not schools alone
Communication barriers — physiological, psychological, environmental and cultural, including illiteracy, customs, beliefs, religion and language variation (Park 28e, Ch 21, p 1012)	Local dialect, folk media, pictorial material, small-group face-to-face sessions

6. Monitoring and evaluation

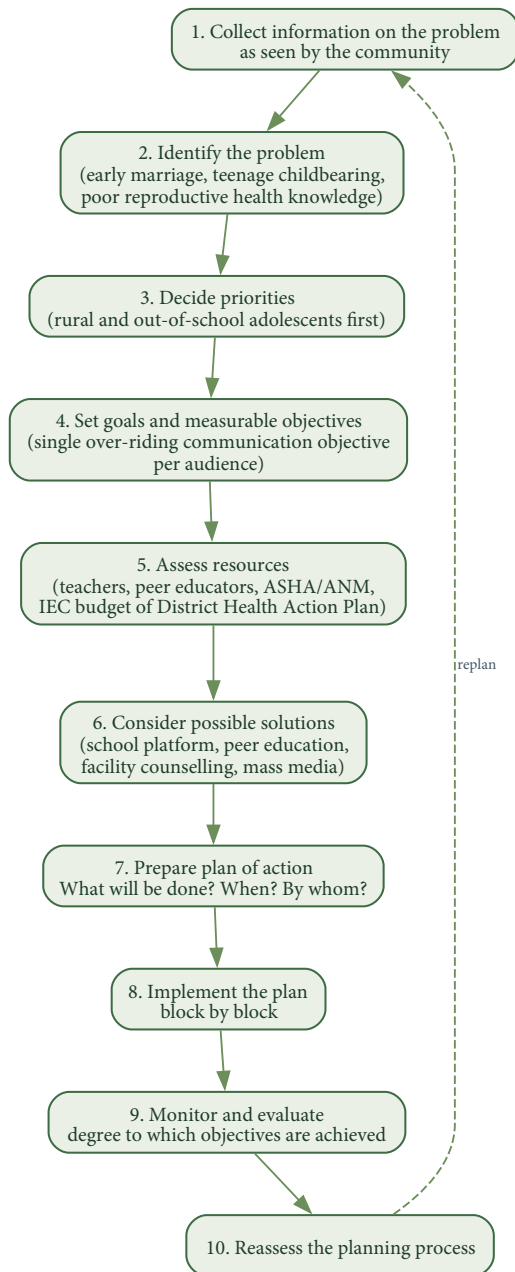
Counting activities alone is not evaluation; the real test is how many individuals changed their behaviour in thinking, feeling and practice, which needs baseline data and a repeat measurement after a stated interval (Sunder Lal, Ch 4, p 154).

Level	Indicator	Data source
Input / process	Schools with two trained Health and Wellness Ambassadors; peer educators trained per 1,000 population; weekly sessions held vs due; Adolescent Health Days held per quarter	Programme registers, HMIS
Output	Adolescents attending peer sessions; AFHC footfall; question-box queries answered; parent–teacher meetings held	AFHC register, school records
Effect (knowledge, attitude, practice)	Adolescents correctly stating the legal age at marriage; comprehensive knowledge of HIV; attitude to spacing measured on a 3-, 5- or 7-point <i>Likert</i> scale (Sunder Lal, Ch 4, pp 153–154); contraceptive use among eligible couples	Repeat KABP survey, exit interviews
Impact	Women aged 20–24 married before 18 y; proportion of women 15–19 who have begun childbearing; age-specific fertility rate 15–19; sex ratio at birth; maternal mortality ratio	NFHS district fact sheet, civil registration, Sample Registration System

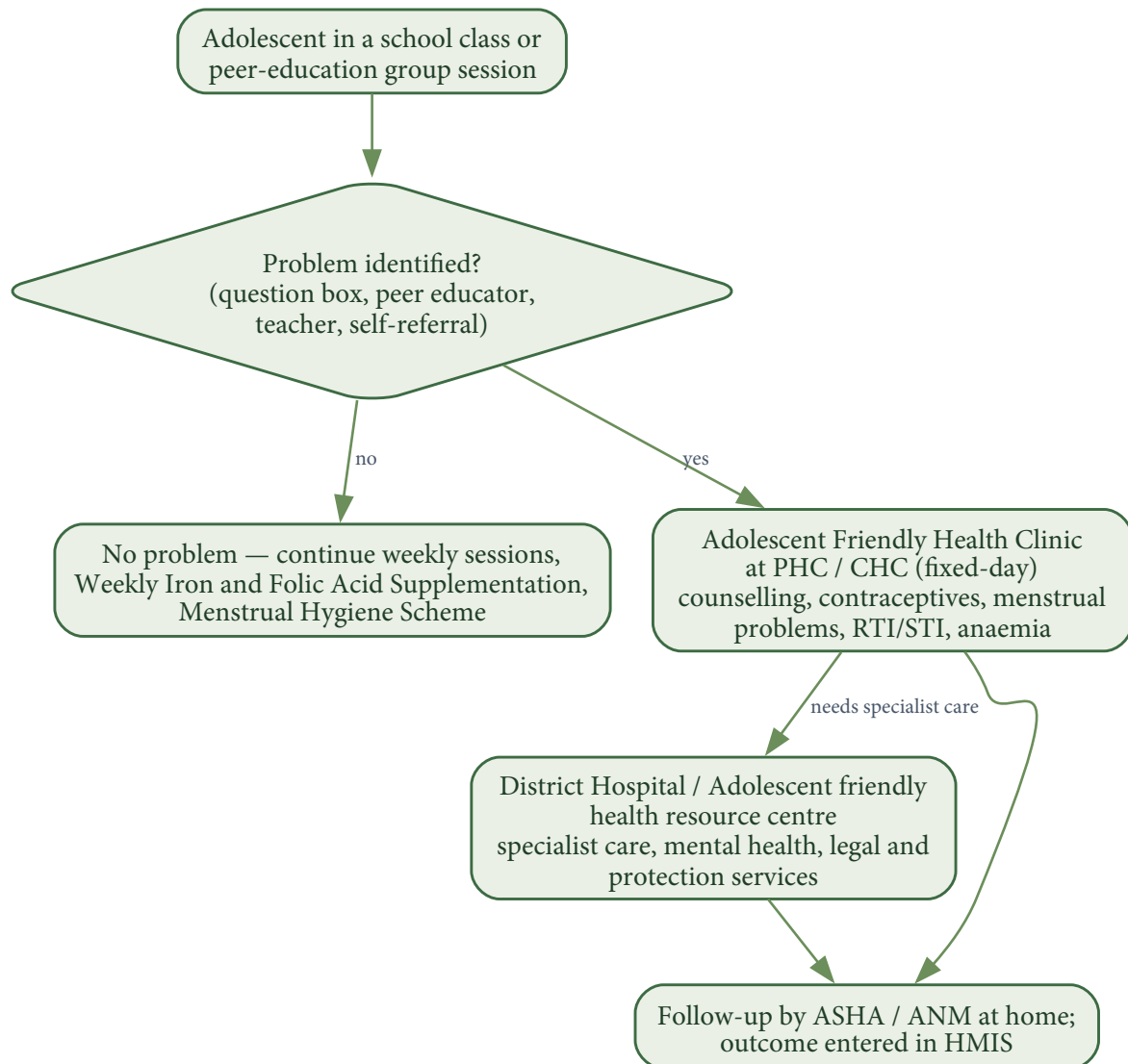
DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Labelled diagram 1 — Adoption model (Park 28e, Ch 22, Fig 2). Draw three stacked boxes with downward arrows and label exactly: Awareness → Motivation (interest → evaluation → decision-making) → Action (adoption or acceptance).

Labelled diagram 2 — District family life education framework. Draw four horizontal bands, bottom to top, labelled: Household — ASHA / Anganwadi worker / ANM home visits; Village or ward — peer educators (2 male + 2 female per 1,000 population), Adolescent Health Day, Village Health, Sanitation and Nutrition Committee; School — 2 Health and Wellness Ambassadors, 1 hour weekly, question box; Facility — Adolescent Friendly Health Clinic at PHC / CHC / District Hospital; with a vertical arrow on the right labelled District Health Society — intersectoral task force (Health, Education, Women and Child Development, Youth Services, Panchayati Raj, NGOs).



Ten steps of health education planning applied to family life education in a district



Referral pathway for an adolescent identified during a family life education session

High-yield exam pointers

- Park's 10 steps of health education planning are the skeleton of any "give a plan" answer: information → identify problem → priorities → objectives → resources → solutions → plan of action (what/when/by whom) → implement → monitor and evaluate → reassess (Park 28e, Ch 22, p 1024).
- **Contents of health education (8):** human biology, nutrition, hygiene, **family health**, disease prevention and control, mental health, prevention of accidents, use of health services — family life education sits in the fourth.
- *Population education* — "an educational programme which provides for a study of the population situation in the community, nation and world with the purpose of developing in the students rational and responsible attitudes and behaviour towards that situation" (Park 28e, Ch 8, pp 615–616) — is the school-curriculum twin of FLE.
- Quantities to reproduce exactly: Health and Wellness Ambassadors **2 per school, 1 hour weekly**; Adolescence Education Programme **16 hours in classes IX and XI**; RKSK peer educators **2 male**

+ 2 female per 1,000 population, groups of 15–20, weekly 1–2 hour sessions; Adolescent Health Day quarterly; group discussion 6–12 members; panel discussion 4–8 panellists.

- ▶ Dated Indian figures: women 20–24 married before 18 y = 23.3% (NFHS-5, 2019–21); women 15–19 who have begun childbearing = 6.8% (NFHS-5); age-specific fertility rate 15–19 = 43/1,000 (NFHS-5); comprehensive HIV knowledge 15–24 y = 20% women, 29% men (NFHS-5); total fertility rate 2.0 (NFHS-5).
- ▶ Legal minimum age at marriage: 18 years for women, 21 years for men.
- ▶ Evaluation must be at three levels — **process** (sessions held), **effect** (knowledge, attitude by *Likert* scale, practice/coverage) and **impact** (age at marriage, teenage childbearing, sex ratio at birth); counting activities alone is the commonest evaluation error.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: the Adolescent Reproductive and Sexual Health (ARSH) scheme and school-based adolescent education were absorbed into the Rashtriya Kishor Swasthya Karyakram (RKSK) in 2014 and into the School Health Programme under Ayushman Bharat, so the current district plan should be described under those two names with the older names mentioned only once in parentheses; the retrieved corpus does not carry a formal one-line textbook definition of "family life education", so the definition above is assembled from Park's school-health and family-health passages and Suryakantha's school health education list.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 7 Health Programmes in India; Ch 8 Demography and Family Planning; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics; Ch 21 Health Information and Basic Medical Statistics; Ch 22 Communication for Health Education); Sunder Lal — Textbook of Community Medicine (Ch 4 Principles of Health Promotion and Education; Ch 10 Reproductive, Maternal, Newborn, Child and Adolescent Health); Suryakantha — Community Medicine with Recent Advances (Ch 22 Maternal and Child Health Services); National Family Health Survey (NFHS-5) India Report, 2019–21; Comprehensive National Nutrition Survey 2016–18 (quoted in Sunder Lal).

♦ ♦ ♦

Health education approach for a primitive society

Asked in: Paper I 21-Jun-2024 — RGUHS MD Community Medicine

Definition — Health education is "a process that informs, motivates and helps people to adopt and maintain healthy practices and lifestyles, advocates environmental changes as needed to facilitate this goal and conducts professional training and research to the same end" (National Conference on Preventive Medicine, USA; Park 28e, Ch 22, p 1014), and after the *Alma-Ata Declaration* (1978) the dynamic definition is "a process aimed at encouraging people to want to be healthy, to know how to stay healthy, to do what they can individually and collectively to maintain health and to seek help when needed" (Park 28e, Ch 22, p 1014). A **primitive society** in public-health usage is a pre-literate, tradition-bound community whose explanation of disease is still supernatural — in India the **scheduled tribes**, including the particularly vulnerable tribal groups. For them health education cannot be a transfer of information: it must be culture-first, interpersonal, in the local dialect, and paired with a service the community can actually reach.

Why a special approach is needed — the profile that dictates the method

Feature of the society	Grounding figure (with year)	Implication for health education
Size and marginality — the target group	Over 104 million tribal people in 705 tribes, 8.6% of India's population; nearly 90% rural, concentrated in ten states and eight North-Eastern states; marginal geographically, socio-economically and politically (Park 28e, Ch 13, p 848)	A national mass campaign will not reach them; the unit of work is the hamlet , not the district
Demographic profile	Sex ratio 990 versus national 943; child (0–6 y) sex ratio 957 versus 914 all-India (Census 2011) (Park 28e, Ch 13, p 848)	The girl child is not the entry point here; do not import an urban message list
Low literacy — the printed word is largely useless	Scheduled tribe literacy 59.0% (male 68.5%, female 49.4%) versus 73.0% all-India (Census 2011) (Park 28e, Ch 13, Table 1, p 848)	Spoken, seen and done media only; posters and pamphlets cannot be the main channel
Supernatural theory of disease	Illness attributed to wrath of gods and goddesses, breach of taboo, past sins, evil eye and spirit intrusion; where disease is held to be divine wrath, giving drugs is thought harmful (Park 28e, Ch 12, p 829). Traditional healers "live close to the people" (Park 28e, Ch 1, p 14)	Do not ridicule the belief; work <i>through</i> the healer as a change agent
Language and comprehension gap	Health messages must be written in local dialects (Sunder Lal, Ch 1, p 54); language, dialect and medical jargon are listed barriers to communication (Sunder Lal, Ch 4, p 155); tribal information, education and communication (IEC) is "not tuned to the tribal vocabulary, beliefs and practices" (Sunder Lal, Ch 17, p 760)	Message built in the local dialect by local people; never words strange and new to them

Feature of the society	Grounding figure (with year)	Implication for health education
Geographic isolation	Inaccessible terrain, forest, hill and riverbed habitation, high mobility, outdoor sleeping (Park 28e, Ch 7, p 498). Relaxed norms — 1 health sub-centre/3,000, 1 primary health centre (PHC)/20,000, 1 community health centre (CHC)/80,000; even so, shortfalls of 27–40% against these norms in about half the states (Park 28e, Ch 13, p 850)	Hamlet-level contact and mobile services; education must not outrun the service
Poverty and low service uptake	Institutional delivery 68.0% (scheduled tribes) versus 88.6% all-India, and deliveries by skilled health personnel 71.5% versus 89.4% (NFHS-5, 2019-21; Park 28e, Ch 13, Table 2, p 849). Total fertility rate 2.09 versus 1.99 ; any contraceptive use 64.4% versus 66.7% (same source). Only about 37% of tribal women received postnatal care within 48 h of a home delivery (Park 28e, Ch 13, p 848)	Education must be <i>paired</i> with an accessible service, or it raises a demand it cannot meet
Immunisation — near-parity overall, gaps in completion	Infants fully immunised 76.8% (scheduled tribes) versus 76.6% all-India, but hepatitis B third dose 81.2% versus 88.6% , pregnant-women tetanus toxoid 79.0% versus 85.5% , measles 86.7% versus 87.9%, BCG 94.3% versus 95.2%, no vaccination at all 4.1% versus 3.6% (NFHS-5, 2019-21; Park 28e, Ch 13, Table 2, p 849)	The message is completing and timing the schedule, not merely starting it; reinforcement and repeat contact matter more than awareness
Child survival gap	Infant mortality rate 41.7 versus 35.2, neonatal 28.8 versus 24.9, post-neonatal 12.9 versus 10.3, under-five 50.3 versus 41.9 and 1–4 year child mortality 9.0 versus 6.9 per 1,000 (NFHS-5, 2019-21; Park 28e, Ch 13, Table 2, p 849)	Home-based newborn care and danger-sign recognition are the highest-yield messages
Undernutrition	Under-five stunting 40.9% versus 35.5%, wasting 23.2% versus 19.3%, underweight 39.5% versus 32.1% (NFHS-5, 2019-21; Park 28e, Ch 13, Table 2, p 849)	Feeding demonstration, not a feeding lecture
Triple burden with an awareness gap	Communicable disease and malnutrition; emerging non-communicable disease — one in four tribal adults is hypertensive, yet two of three do not know the signs and only 5% of men and 9% of women know their status; mental illness and addiction — tobacco use in 72% of tribal men aged 15–54 versus 56% non-tribal, and about 50% consume alcohol (Park 28e, Ch 13, pp 849-850)	The gap is awareness , so screening plus counselling, not information alone
Genetic disorders	Sickle cell disease and thalassaemia together 1–40% across tribal communities; sickle cell anaemia in 1 in 86 births in central India; glucose-6-phosphate dehydrogenase (G6PD) deficiency 0.7–15.6% in 14 primitive tribal populations (Park 28e, Ch 13, p 850)	Counselling on screening, carrier status and marriage must be built into the message
Concentrated malaria risk	Tribes are about 8% of the population but account for about 30% of all malaria cases, over 60% of <i>Plasmodium falciparum</i> and about 50% of malaria mortality (Park 28e, Ch 13, p 849)	Malaria is the natural felt need and the entry point for the whole programme

The four approaches to health education — and which one is chosen

Approach	What it does	Example	Fit for a primitive society
1. Regulatory (managed prevention)	Any governmental intervention, direct or indirect, designed to alter human behaviour by external control or law; forms range from prohibition to imprisonment	<i>Child Marriage Restraint Act</i> , India; compulsory seat-belt use in western countries	Poorest fit — the cause of disease cannot be eradicated by legislation, and coercion in matters of personal choice takes away individual rights; important failures exist (prohibition of alcohol; the 1976 sterilization campaign) (Park 28e, Ch 22, p 1015)
2. Service approach	Provide all the basic health services needed at the people's doorstep, assuming they will then use them	Government water-seal latrines provided free of cost — many rural people did not use them	Failed approach — it was not based on the felt needs of the people (Park 28e, Ch 22, p 1016)
3. Health education approach	Educate people to identify their own problems and make their own informed choice for a healthy life; consistent with a democratic philosophy that does not order the individual; results slow but enduring	Group discussion and demonstration in the hamlet	Necessary but not sufficient — education alone cannot deliver health unless proven preventive measures are also accessible (Park 28e, Ch 22, pp 1015-1016)
4. Primary health care approach	Health education planned and delivered as part of primary health care — community involvement, intersectoral coordination and self-reliance ; described as "health by the people" and "placing people's health in people's hands" (Park 28e, Ch 2, p 49)	Community-selected local tribal workers — hamlet-wise accredited social health activist (ASHA), <i>Anganwadi</i> worker, trained <i>dai</i>	Approach of choice — a radically new approach starting from the people, with full participation and active involvement in planning and delivery; a fundamental shift from the earlier approaches (Park 28e, Ch 22, p 1016)

Since *Alma-Ata* the emphasis has shifted from prevention of disease to promotion of healthy lifestyles, from modifying individual behaviour to modifying the **social environment**, from community participation to community **involvement**, and towards individual and community **self-reliance** (Park 28e, Ch 22, p 1015). What is delivered must also be education and not propaganda: education is behaviour-centred, appeals to reason, disciplines primitive desires and is acquired through self-reliant activity, whereas propaganda is information-centred, spoon-fed, appeals to emotion and discourages thinking by ready-made slogans (Central Health Education Bureau, Government of India; Park 28e, Ch 22, Table 1, p 1016).

Principles of health education applied to a primitive society

Principle (Park 28e, Ch 22, pp 1018-1019)	How it is operationalised here
1. Credibility	The message must be trusted — consistent with scientific knowledge <i>and</i> with local culture; the <i>Expert Committee on Tribal Health</i> (Dr Abhay Bang) holds that the provider should, as far as possible, be a local tribal , trained and deployed in the health workforce, with the centre of gravity placed near the community rather than at the specialist end (Park 28e, Ch 13, p 851)
2. Interest	Start from what the community itself feels — a malaria death, a snake bite, a child's fever — not from a national priority list
3. Participation	The community works <i>with</i> the health team; <i>Alma-Ata</i> holds that people have the right and duty to participate individually and collectively in planning and implementing their health care
4. Motivation	Use secondary motives — praise, reward, recognition — and living examples from within the group; "motivation is contagious"
5. Comprehension	Teach within the level of understanding, education and literacy of the audience; never use words strange and new to the people
6. Reinforcement	Repeat at intervals in different ways — few people learn all of it in a single period; this is what closes the hepatitis B and tetanus toxoid completion gap
7. Learning by doing	"If I hear, I forget; if I see, I remember; if I do, I know" — demonstrate oral rehydration, bed-net use, complementary feeding
8. Known to unknown	Proceed from the concrete to the abstract and simple to complicated, using existing tribal knowledge as the peg
9. Setting an example	The educator must practise what he teaches — he cannot preach against smoking and smoke himself
10. Good human relations	Information and feelings pass most easily between people who have a good relationship; relationship-building goes with communication skill
11. Feedback	A key concept of the systems approach — modify message and channel in the light of audience response
12. Leaders	Penetrate through local leaders — village headman, school teacher, <i>panchayat</i> member, and here also the traditional or faith healer, <i>Anganwadi</i> worker and ASHA; leaders are the agents of change

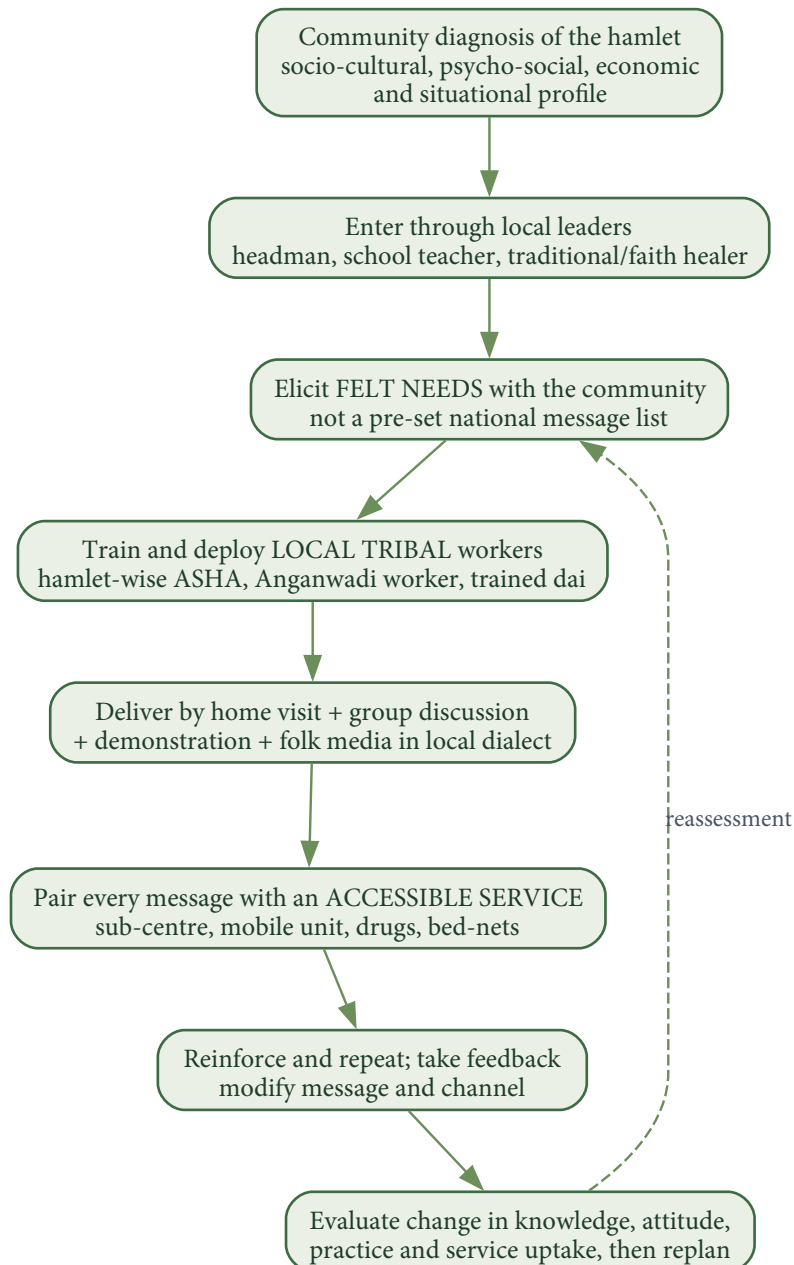
Customs are not uniformly harmful, and a blanket attack on tradition destroys credibility. Classify them as **good, bad, unimportant and uncertain** and attack only the harmful — prolonged breast-feeding, oil bath, massage and exposure to sun are good customs to be actively reinforced (Park 28e, Ch 12, p 830).

Choice of methods and media

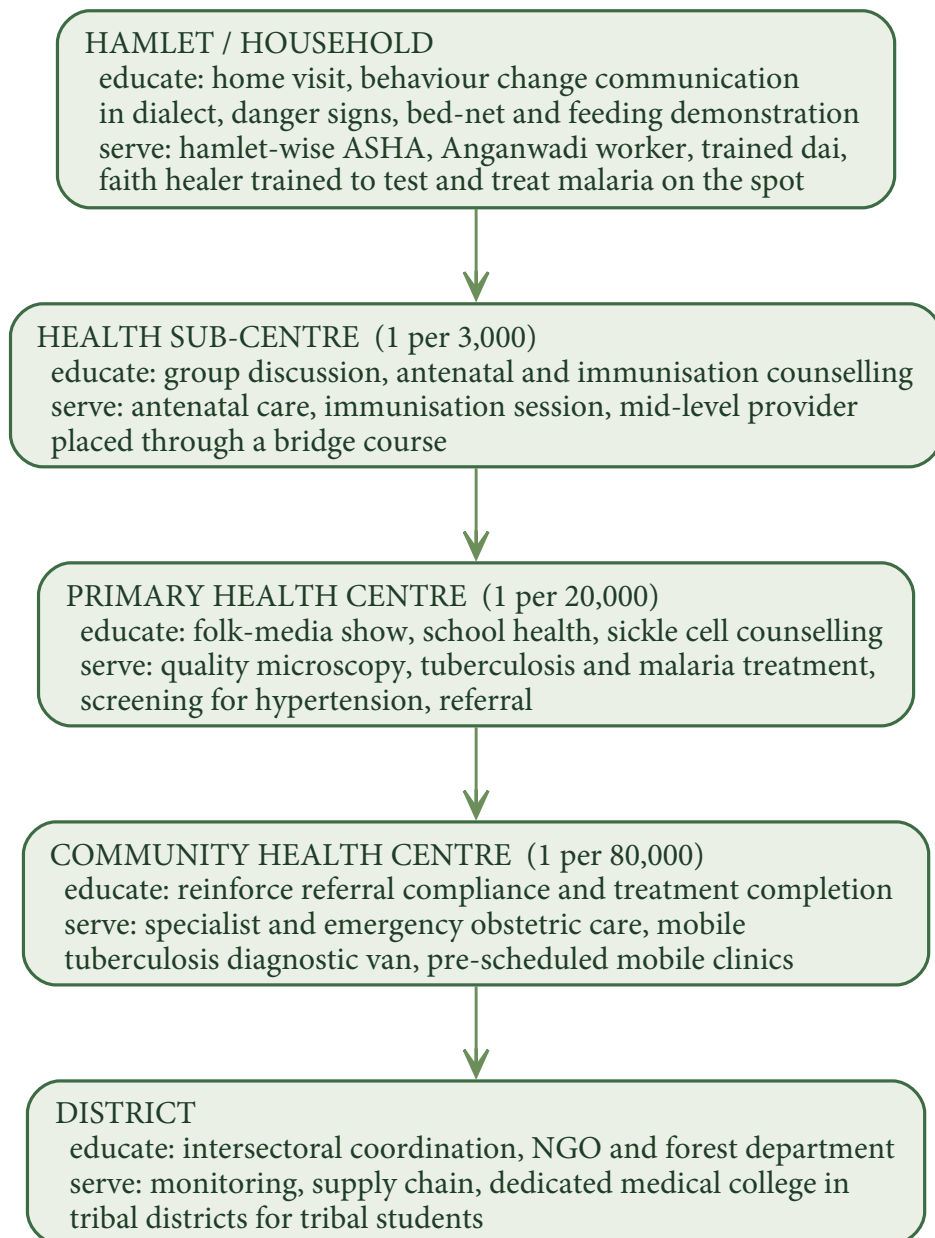
Method	Suitability in a primitive society
Individual — personal contact, home visit, counselling	First choice ; one-to-one, most interactive and participatory, dealing with the precise need of that family, though limited in reach and time-consuming (Sunder Lal, Ch 4, p 146)

Method	Suitability in a primitive society
Group — group discussion, demonstration , role play	Core method ; demonstration dramatises the message and upholds "seeing is believing" and "learning by doing" (Park 28e, Ch 22, p 1021)
Folk media — <i>keerthan</i> , folk song, dance, drama and puppet show, and <i>ghazal</i> and <i>kawali</i> among Muslim groups (Park 28e, Ch 22, p 1024); regionally <i>nautanki</i> in Uttar Pradesh, <i>Burrakatha</i> in Andhra Pradesh and <i>Harikatha</i> (Park 28e, Ch 22, p 1011)	Most powerful mass channel here ; these indigenous forms are close to the culture, preserve the cultural heritage and reach the pre-literate — the medium is neither good nor bad, what matters is the message it carries and the way it is delivered (Park 28e, Ch 22, p 1024)
Mass media — television, radio, newspaper, printed material, internet	Least useful alone ; mass media are more effective among those of above-average educational level, while personal communication is more influential at average and below-average levels, and mass media alone are generally inadequate to change behaviour (Park 28e, Ch 22, Table 2, p 1021; p 1023)

Any one or a combination of these is used selectively, depending on the objectives to be achieved, the behaviour to be influenced and the funds available (Park 28e, Ch 22, p 1020). There is no single way to do public education — health education is **an art rather than a science**, and each community should develop the techniques that meet its own needs (Park 28e, Ch 22, p 1024).



Delivery pathway of health education in a primitive (tribal) community — the primary health care approach



Pairing each health-education activity with the service that must back it, along the relaxed tribal care pathway

Steps in planning the programme

Health education is never planned in a vacuum: it is planned in connection with a specific health programme, and the strategy is fitted to that community's socio-cultural, psycho-social, political, economic and situational characteristics. The ten steps of scientific planning are (Park 28e, Ch 22, p 1024):

- ▶ Collecting information on the specific problems **as seen by the community**.
- ▶ Identification of the problem.
- ▶ Deciding on priorities.
- ▶ Setting goals and measurable objectives.
- ▶ Assessment of resources — local manpower, money and material first.
- ▶ Consideration of possible solutions.

- ▶ Preparation of a plan of action: **what** will be done, **when**, and **by whom**.
- ▶ Implementing the plan.
- ▶ Monitoring and evaluating the degree to which the stated objectives have been achieved.
- ▶ Reassessment of the process of planning.

Evaluation

Level	What is measured
Process (activity count)	Clients contacted, home visits, orientation camps, group discussions, film and folk-media shows, leaders contacted, hamlets covered — the prevailing measure, necessary but not sufficient on its own, though process evaluation itself is held to be most important (Sunder Lal, Ch 4, p 154)
Knowledge	Change measured by standard questions on the specific problem, baseline versus repeat or annual survey (Sunder Lal, Ch 4, pp 153-154)
Attitude	Measured on a <i>Likert</i> scale — 3-, 5- or 7-point agree-to-disagree (Sunder Lal, Ch 4, p 154)
Practice / behaviour	Direct observation of practice — hand-washing, nail cutting, cord care, cooking and feeding practices (Sunder Lal, Ch 4, p 154)
Service outcome	Rise in immunisation completion, institutional delivery, bed-net use and treatment completion in the same population, read against the NFHS-5 baseline above

The real test of effectiveness is not how many activities were generated but **how many individuals actually changed their behaviour** in thinking, feeling and practice, which needs baseline and repeat behaviour data (Sunder Lal, Ch 4, p 154).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Adoption (behaviour-change) model** — Park 28e, Ch 22, Fig 2, p 1017. Labels, top to bottom: 1. **AWARENESS** → 2. **MOTIVATION** (branching to three: **Interest**, **Evaluation**, **Decision-making**) → 3. **ACTION** (**Adoption or acceptance**). Write beside it, as the accompanying text does: "Reliance on one method is likely to lead to failure — an appropriate combination using all methods is needed to change behaviour."
- ▶ **Methods in health communication** — Park 28e, Ch 22, Fig 3, p 1020. A three-branch tree with the stem *Health communication*, branching to: **Individual approach** (personal contact, home visits, personal letters); **Group approach** (lectures, demonstrations, group discussion, panel discussion, symposium, workshop, conference, seminar, role play); **Mass approach** (television, radio, newspaper, printed material, direct mailing, posters, health museums and exhibitions, **folk media**, internet).

High-yield exam pointers

- ▶ **Four approaches (Park):** Regulatory (managed prevention) → Service → Health education → **Primary health care approach** = the answer for a primitive society; primary health care = community involvement + intersectoral coordination + self-reliance.
- ▶ **Adoption model:** Awareness → Motivation (interest, evaluation, decision-making) → Action/Adoption.

- ▶ **Twelve principles, in Park's order:** credibility, interest, participation, motivation, comprehension, reinforcement, learning by doing, known to unknown, setting an example, good human relations, feedback, **leaders**.
- ▶ **Numbers to quote with their year:** 104 million tribal people in 705 tribes, 8.6% of the population; literacy 59.0% versus 73.0% all-India and sex ratio 990 versus 943 (**Census 2011**); institutional delivery **68.0% versus 88.6%**, skilled attendant **71.5% versus 89.4%**, total fertility rate **2.09 versus 1.99**, infants fully immunised **76.8% versus 76.6%** but hepatitis B3 81.2% versus 88.6%, infant mortality rate 41.7 versus 35.2 and under-five 50.3 versus 41.9 per 1,000 (NFHS-5, 2019-21).
- ▶ **Tribal service norms:** 1 sub-centre/3,000, 1 PHC/20,000, 1 CHC/80,000 population — with a 27–40% shortfall against these norms in about half the states.
- ▶ **Two one-liners that fetch marks:** "There is no single way to do public education — health education is an art rather than a science; each community should develop techniques that meet its own needs"; and "If I hear, I forget; if I see, I remember; if I do, I know."
- ▶ **Malaria worked example:** the Tribal Malaria Action Plan provides **hamlet-wise ASHAs instead of village-wise**, and trains *Anganwadi* workers, faith healers, village headmen and school teachers to diagnose and treat malaria wherever an ASHA cannot be placed (Park 28e, Ch 7, pp 498-499).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

"Primitive society" is the older textbook term; current Indian programme language is *scheduled tribes* and *particularly vulnerable tribal groups*, and Park 28e retains "primitive tribal populations" only when quoting genetic-prevalence data, so write the modern term at least once. Park's narrative text on p 848 also carries an older comparator — "the 68.0 per cent rate of institutional delivery is much lower than the national average of 78.9 per cent" and a total fertility rate of 2.48 — which do not match its own NFHS-5 table on p 849; quote the table values (88.6% and 2.09/1.99).

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 22 (Communication for Health Education), Ch 13 (Tribal Health in India), Ch 12 (Medicine and Social Sciences), Ch 10 (Health Care of the Community), Ch 7 (Health Programmes in India), Ch 2 (Concept of Health and Disease), Ch 1 (Man and Medicine: Towards Health for All); Sunder Lal — Textbook of Community Medicine, Ch 1 (Concept of Health and Disease), Ch 4 (Principles of Health Promotion and Education), Ch 17 (Healthcare of the Community). Named sources within these: National Conference on Preventive Medicine, USA (definition of health education); Alma-Ata Declaration (1978); Central Health Education Bureau, Government of India (health education versus propaganda); Census of India 2011 (literacy, sex ratio); National Family Health Survey-5, 2019-21, as tabulated in Park 28e Ch 13 Table 2 (tribal mortality, immunisation, nutrition and maternal indicators); Report of the Expert Committee on Tribal Health, Ministry of Health and Family Welfare and Ministry of Tribal Affairs.



Health education to reduce tobacco hazards

Asked in: Paper I 12-Nov-2019 — RGUHS MD Community Medicine

Definition — Health education is a process that informs, motivates and helps people to adopt and maintain healthy practices, and advocates the environmental and service changes needed to make those practices possible; it aims to increase knowledge and to **reinforce desired behaviour patterns**, but education alone is insufficient unless the population also has access to proven preventive measures (Park 28e, Ch 22, pp 1014-1015). Applied to tobacco, a health education programme is the planned, community-wide use of interpersonal, group and mass communication — backed by legislation, taxation and cessation services — to prevent initiation and promote quitting. Public information and education is the **first** of the four prevention strategies against tobacco-related cancer, and "nothing less than a national anti-smoking campaign" will change tobacco-related behaviour (Park 28e, Ch 6, p 465).

Magnitude of the problem that the programme must address

Indicator	Value (year, source)
Adults (15+) using tobacco in any form	28.6% in GATS-2 (2016-17), down from 34.6% in GATS-1 (2009-10) (Sunder Lal, Ch 15, pp 689-690)
Current smokers / smokeless tobacco users	10.7% / 21.4% (GATS-2, 2016-17) — smokeless exceeds smoking in India
Current bidi smokers / cigarette smokers	7.7% / 4.0% (GATS-2, 2016-17)
Absolute burden	267 million adult users, of whom ~199 million use smokeless tobacco (khaini, gutkha, pan masala) (Sunder Lal, Ch 15, p 690)
School children 13-15 y using tobacco	14.6% (2009) → 8.5% (GYTS-4, 2019) — a 42% fall (Sunder Lal, Ch 15, p 689)

Indicator	Value (year, source)
Deaths and cancers, India	~1 million deaths/year; ~30% of all cancers tobacco-related (Sunder Lal, Ch 15, p 690)
Economic cost, India	₹1,04,500 crore = 1.16% of GDP (2011-12) (Sunder Lal, Ch 15, p 689)
Educational lever already proven	62% of adults thought of quitting cigarettes, 54% bidi and 46% smokeless because of the pack warning (GATS-2)

Content of the message — hazards to be communicated

Theme	Facts to convey
Toxic constituents	Nicotine, carbon monoxide and tar; >4000 chemicals in tobacco smoke, ~60% known or suspected carcinogens (Sunder Lal, Ch 15, p 690)
Mechanism, in plain terms	Nicotine raises heart rate, blood pressure, serum cholesterol, triglycerides and free fatty acids; carbon monoxide forms carboxyhaemoglobin (COHb 5-6% in average smokers, up to 10% in heavy smokers), cutting oxygen supply to heart muscle and tissues
Attributable disease	Smoking causes ~71% of lung cancer, 42% of chronic respiratory disease and ~10% of cardiovascular disease (Park 28e, Ch 6, p 434)
Smokeless tobacco and oral cancer	Oral cancer and precancerous lesions occurred almost solely in those who smoked or chewed; risk 36 times higher if the quid is kept in the mouth during sleep (Park 28e, Ch 6, p 461)
Second-hand smoke	~600,000 deaths a year globally, of which 170,000 are children (Park 28e, Ch 6, p 434); <i>Hirayama</i> first showed excess lung cancer in non-smoking Japanese wives of smokers (Park 28e, Ch 20, pp 990-991)
Positive, gain-framed message	Quitting at any stage is beneficial; techniques exist — social support, cessation clinics, skills training, nicotine replacement therapy (Sunder Lal, Ch 15, pp 690-691)

Target groups, settings and channels

Target group	Setting	Core message	Channel
School children, adolescents	School, <i>anganwadi</i> , college	Never start; refusal skills; teacher as role model	School health programme, child-to-child, debates, posters
General adult population	Community, mass media	Hazards, quitline, 31 May World No Tobacco Day	TV, radio, newspaper, folk media, internet
Current users seeking help	Health and wellness centre, PHC/CHC, tobacco cessation centre	Brief advice at every contact; cessation counselling and relapse support	Interpersonal counselling, quitline
Pregnant women, families	Home visit by ASHA/ANM	Second-hand smoke, child health	Interpersonal communication (IPC)

Target group	Setting	Core message	Channel
Opinion leaders, teachers, doctors, health workers	Village, school, hospital	Advocacy and personal example	Training, orientation camps
Bidi rollers, tobacco farmers	Workplace, farm	Alternate livelihood and cropping	Intersectoral, Central Tobacco Research Institute pilots

Methods of health education and their merits

Approach	Methods (Park 28e, Ch 22, pp 1020-1021)	Merits	Limitations
Individual (interpersonal)	Personal contact, home visits, personal letters, counselling	Two-way; allows argument and persuasion; tailored to precise need; most influential in average and below-average education groups	Numbers reached are small; reaches only those who come in contact
Group	Lectures, demonstrations, group discussion, panel discussion, symposium, workshop, seminar, conference, role play	Group support "legitimises" the new behaviour; participatory; subject can be matched to the group's interest	Needs skilled facilitation; message must relate directly to the group's own concern
Mass	Television, radio, newspaper, printed material, direct mailing, posters, health museums and exhibitions, folk methods, internet	Reaches the widest population; gets public attention; supports concentrated campaigns such as a No Tobacco Week	One-way, no feedback; by itself does not change behaviour; more effective in above-average education groups

Reliance on **one** method alone is likely to lead to failure; a mass-media message becomes effective only when it is picked up, debated and legitimised through interpersonal and group networks (Park 28e, Ch 21, p 1011; Ch 22, p 1017).

Principles of health education applied to tobacco

Principle (Park 28e, Ch 22, p 1019)	Application to a tobacco programme
1. Credibility	Use the doctor, teacher and respected local figure; quote GATS/GYTS data, not scare stories
2. Interest	Begin from a felt need — cost of the daily bidi, a father's cough, a bad-smelling mouth
3. Participation	Village health sanitation and nutrition committee, self-help groups and school clubs plan their own tobacco-free pledge
4. Motivation	Use gain-framed secondary motives — money saved, appearance, family pride — not only fear of death
5. Comprehension	Pitch at the audience's literacy; explain "tar" and "carbon monoxide" in ordinary words

Principle (Park 28e, Ch 22, p 1019)	Application to a tobacco programme
6. Reinforcement	Repeat the message through several channels over months; no single exposure changes a habit
7. Learning by doing	Role play refusing an offered cigarette; demonstrate cleaning a stained tooth
8. Known to unknown	Start from the locally familiar quid or bidi, then move to cancer risk
9. Setting an example	A health educator who smokes cannot teach the hazards of smoking; tobacco-free health facilities and schools
10. Good human relations	Non-judgemental counselling of the user; no blaming
11. Feedback	Modify message and channel from audience response and exit interviews
12. Leaders	Penetrate the community through the village headman, teacher, political worker and faith leader

Park describes four successive **models** of health education — the **medical model** (facts alone; failed to bridge knowledge and behaviour), the **motivation model**, the **social intervention model** (born precisely because motivation alone could not reduce smoking — it is the *social environment*, not only the individual, that must change) and the **primary health care approach** (community participation and intersectoral action) (Park 28e, Ch 22, pp 1016-1017). Tobacco is the textbook example that forced the shift to the last two.

Delivery: the National Tobacco Control Programme (NTCP) and its legal support

NTCP was launched in the 11th Five Year Plan with a pilot in 16 districts of 9 states in 2007-08 (Park 28e, Ch 7, p 567) and now covers **677 districts across 36 States/UTs** (Sunder Lal, Ch 15, p 691). Its nine components are:

- ▶ Public awareness and mass media campaigns for awareness building and **behaviour change**.
- ▶ Establishment of tobacco product testing laboratories, as required under COTPA 2003.
- ▶ Mainstreaming programme components into the health delivery mechanism under the National Health Mission.
- ▶ Research and training on alternate crops and livelihoods with other nodal ministries.
- ▶ Monitoring and evaluation including surveillance — Global Adult Tobacco Survey (GATS) and Global Youth Tobacco Survey (GYTS).
- ▶ Dedicated tobacco control cells at national, state and district level.
- ▶ Training of health and social workers, non-governmental organisations and school teachers.
- ▶ School programme.
- ▶ Provision of tobacco cessation facilities (Tobacco Cessation Centres).

A 24×7 toll-free national helpline (**1800-110-456**) receives reports of violations of the ban on smoking in public places (Sunder Lal, Ch 15, p 692).

Legal / policy instrument	Provisions that education must publicise
COTPA 2003 (Cigarettes and Other Tobacco Products Act)	Ban on smoking in public places (from 2 Oct 2008); ban on direct and indirect advertisement; ban on sale to persons below 18 years; ban on sale within 100 yards of educational institutions; regulation of smoking scenes in films; pictorial warning covering 85% of both principal sides (60% picture + 25% text); mandatory display and testing of tar and nicotine content (Sunder Lal, Ch 15, p 691)
WHO FCTC	India signed 10 Sep 2003, ratified 5 Feb 2004; hosted the 7th Conference of Parties in 2016
WHO MPOWER	Monitor tobacco use and prevention policies; Protect people from tobacco smoke; Offer help to quit; Warn about the dangers; Enforce bans on advertising, promotion and sponsorship; Raise taxes (Oxford Textbook of Public Health 7e, Ch 78, pp 2948-2950)
Prohibition of Electronic Cigarettes Act, 2019	Sale of e-cigarettes banned; Food Safety and Standards Authority of India bans tobacco/nicotine in any food product

Planning and evaluation of the programme

Health education planning follows the ten steps of scientific planning — collect information on the problem as the community sees it; identify the problem; decide priorities; set goals and **measurable objectives**; assess resources; consider possible solutions; prepare a plan of action (what, when, by whom); implement; monitor and evaluate against the stated objectives; reassess the planning process (Park 28e, Ch 22, p 1024). The behaviour change communication matrix operationalises this — area of concern → current practice → expected practice → target audience → message → media and channel → outcome indicator (Sunder Lal, Ch 4, p 154).

Level	Indicator for a community tobacco education programme
Input	Funds released; IEC material produced; staff trained in interpersonal communication
Process	Number of group discussions, film shows, school sessions, home visits, leaders contacted; helpline calls received
Output	Proportion of the target population reached; proportion aware that tobacco causes cancer
Effect (KAP)	Change in knowledge, attitude and practice on repeat survey against baseline
Impact	Fall in prevalence of current tobacco use (GATS/GYTS); quit attempts and quit rate; COTPA prosecutions; fall in tobacco-attributable morbidity

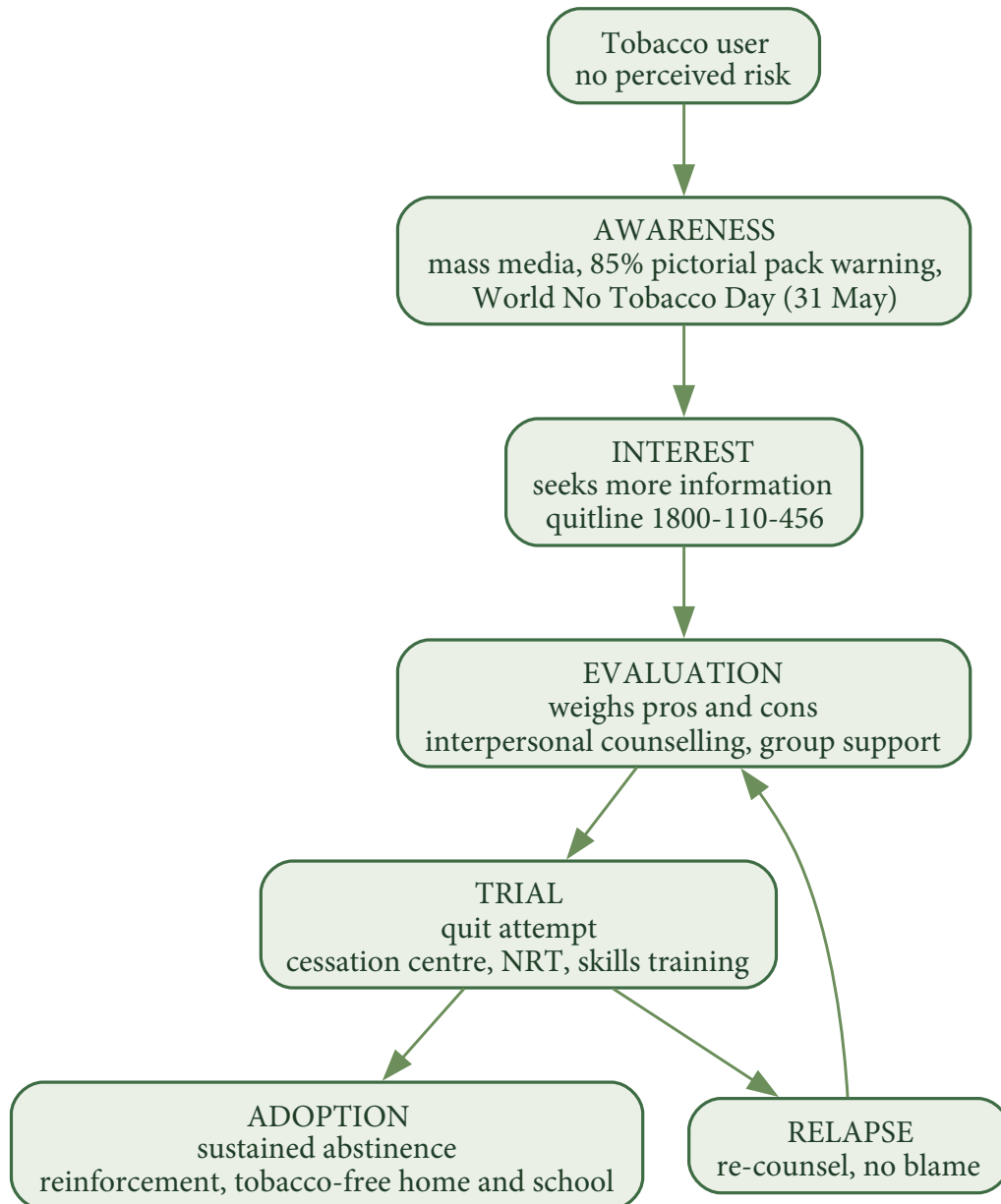
Counting activities alone is the commonest fault — "the real test of effectiveness is how many individuals have changed their behaviour", which requires baseline and repeat survey data (Sunder Lal, Ch 4, p 154). **Process evaluation is the most important component.**

Barriers and challenges

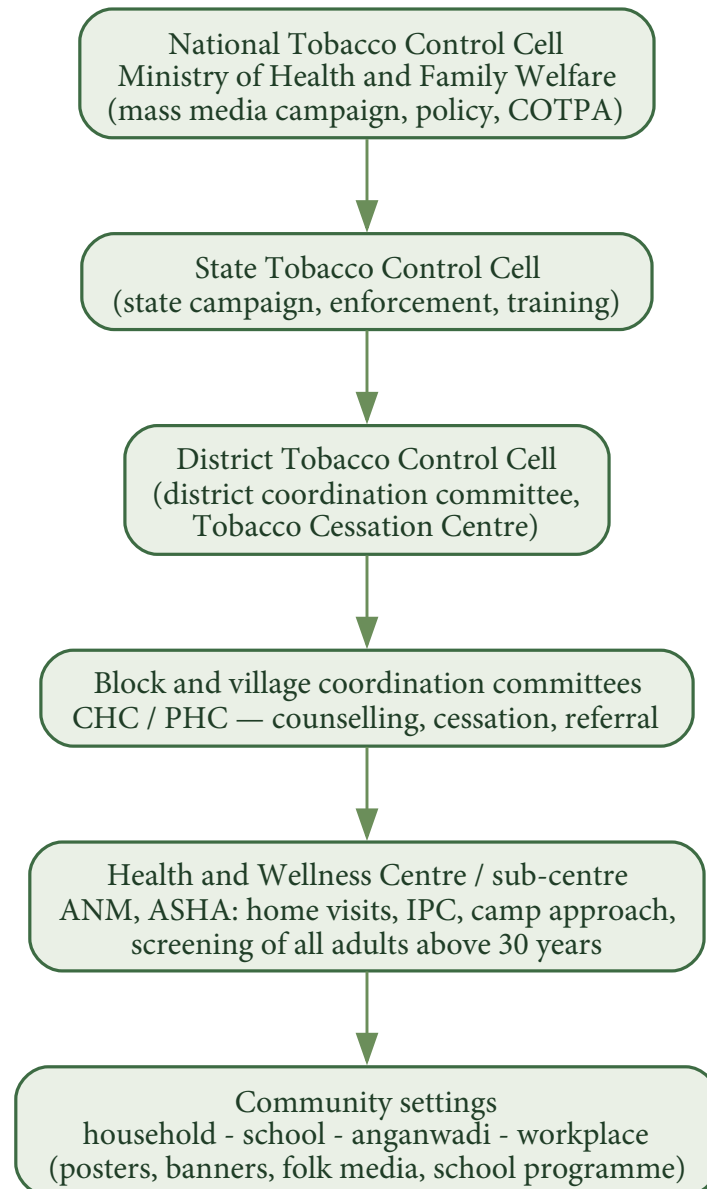
Communication barriers (Sunder Lal, Ch 4, p 155)	Programme challenges (Sunder Lal, Ch 15, p 692)
Talking <i>at</i> people, advising without listening; lack of empathy or trust	Low awareness of harmful effects of tobacco
Physical barriers — noise, interruptions, no eye contact	Very large number and variety of tobacco products
Cultural and social barriers, arrogance of the educator	Low age of initiation of tobacco use
Low credibility of the source	Limited state capacity to enforce the law
Language, dialect and medical jargon	Limited availability of tobacco cessation facilities
Low visual and health literacy	Alternate livelihood for the tobacco workforce; poor intersectoral coordination

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

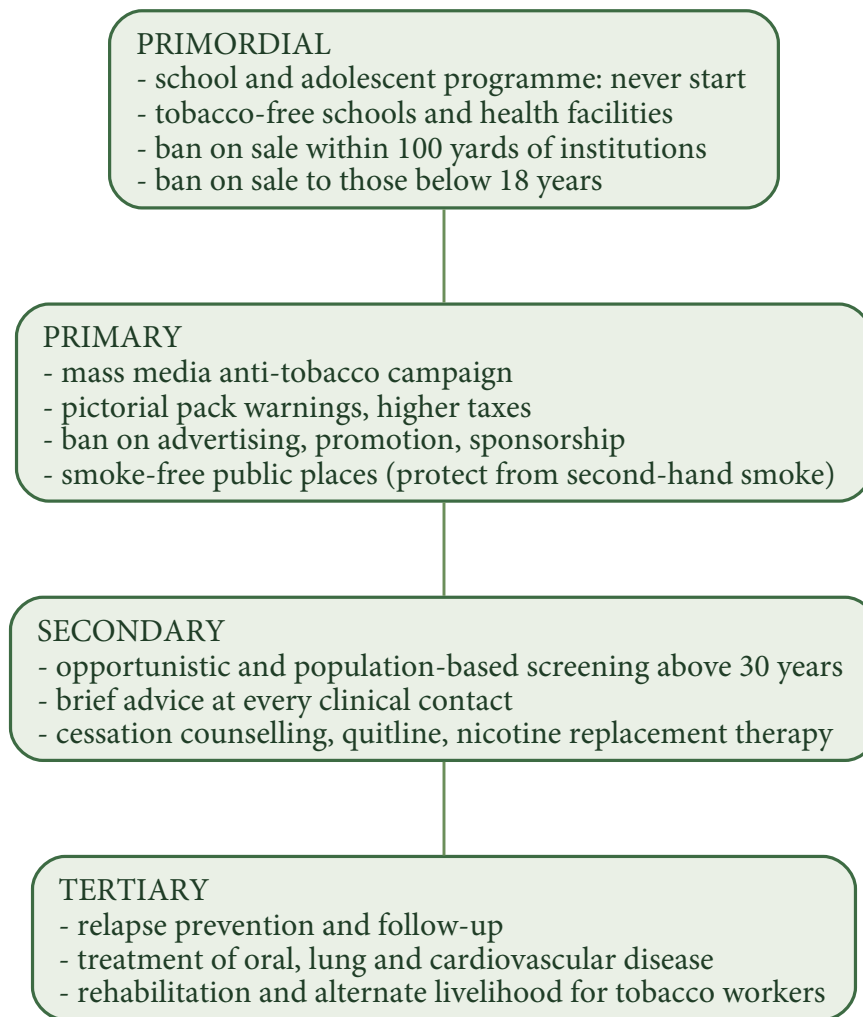
- **DRAWN figure** — "Methods in health communication" (Park 28e, Ch 22, Fig 3). Draw one box at the top labelled *Methods in health communication*, with three branches. Label them exactly: **Individual approach** — 1. Personal contact, 2. Home visits, 3. Personal letters. **Group approach** — 1. Lectures, 2. Demonstrations, 3. Discussion methods (group discussion, panel discussion, symposium, workshop, conferences, seminars, role play). **Mass approach** — 1. Television, 2. Radio, 3. Newspaper, 4. Printed material, 5. Direct mailing, 6. Posters, 7. Health museums and exhibitions, 8. Folk methods, 9. Internet.



Adoption ladder in tobacco cessation and the health-education input needed at each stage



Delivery pathway of tobacco health education under the National Tobacco Control Programme



Levels of prevention delivered through tobacco health education

High-yield exam pointers

- ▶ **MPOWER** = Monitor, Protect, Offer, Warn, Enforce, Raise taxes — the six FCTC-derived policies; India signed the FCTC on 10 Sep 2003 and ratified it on 5 Feb 2004.
- ▶ **GATS-2 (2016-17)**: tobacco use in any form **28.6%** (GATS-1 2009-10: 34.6%); smokeless **21.4%** > smoking 10.7%; **267 million** adult users. **GYTS-4 (2019)**: 8.5% among 13-15 year olds.
- ▶ **COTPA 2003** numbers to write: pictorial warning **85%** of both sides (60% picture + 25% text); no sale within **100 yards** of an educational institution; no sale below **18 years**; public-place smoking ban from **2 Oct 2008**.
- ▶ **NTCP** — 9 components, pilot 2007-08 in 16 districts/9 states, now **677 districts** in 36 States/UTs; helpline **1800-110-456**; endgame target = prevalence <5%.
- ▶ **Park's 12 principles**: credibility, interest, participation, motivation, comprehension, reinforcement, learning by doing, known to unknown, setting an example, good human relations, feedback, leaders.
- ▶ **Adoption sequence**: awareness → interest → evaluation → trial → adoption (Park groups these as awareness → motivation → action).
- ▶ **Evaluation rule**: count behaviour change against a baseline, not the number of film shows held; World No Tobacco Day is **31 May**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: Park 28e still prints the older National Tobacco Control Programme coverage of 108 districts in 31 states while Sunder Lal gives the current 677 districts across 36 States/Union Territories, so quote the larger figure with its source; the proposal to raise the legal age for tobacco purchase from 18 to 21 years, ban loose cigarettes and regulate electronic nicotine delivery systems is described in Sunder Lal as a proposed amendment to COTPA and not as enacted law, whereas the ban on the sale of e-cigarettes under the Prohibition of Electronic Cigarettes Act 2019 is in force.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, MoHFW National Tobacco Control Programme, COTPA Packaging and Labelling Amendment Rules 2024, WHO report on the global tobacco epidemic 2025, Cochrane Library

Programme and policy updates (India)

- The MoHFW tobacco control portal now runs three cessation channels alongside the district Tobacco Cessation Centres: **National Tobacco Quitline Services on toll-free 1800-11-2356**, staffed by counsellors in all major languages, and **mCessation**, an SMS-based text-support service (011-22901701). Quote **1800-11-2356** when asked for the quitline — the 1800-110-456 number in the textbooks is the line for reporting public-place smoking violations under COTPA, not for cessation counselling, and confusing the two is a common loss of marks. [MoHFW/NTCP 2025]
- A national consultation of 40 experts from government, development agencies and academia issued a consensus statement sequencing India's tobacco control priorities against SDG target 3.a. Short-term: anti-tobacco curricula inside health-professional teaching institutions, expanded cessation services, a national task force for tobacco control, and a strengthened Tobacco Control Act. Intermediate: advancing tobacco taxation, creating tobacco-free environments, adopting **FCTC Article 5.3** (protecting policy from tobacco industry interference), building capacity of the national tobacco testing laboratories, and **tobacco vendor licensing**. Long-term: a multisectoral strategy sustaining resources toward a tobacco-free generation. [PMID 40975577]

Current national data

- Smokeless tobacco use among Indian women aged 15-49 fell across three survey rounds — **8.4% (NFHS-3, 2005-06) → 5.6% (NFHS-4, 2015-16) → 3.8% (NFHS-5, 2019-21)** — but the fall is not uniform, and this is the more examinable half: north-eastern women rose from 5.6% to 17.6% between NFHS-4 and NFHS-5, Mizoram recorded 45% and Manipur 41.4%, and NFHS-5 prevalence stayed highest in the poorest quintile (**8.3%**) and among Scheduled Tribes (**10.3%**). Little or no schooling is the strongest educational marker, this time as adjusted odds rather than prevalence: taking women with higher education as reference, the NFHS-5 odds of use were **7.98 (95% CI 5.73-11.11) for primary schooling and 7.56 (5.43-10.52) for no schooling** (the corresponding no-schooling figure in NFHS-4 was 9.05, 6.93-11.83). Gutkha/pan masala is now the commonest product used by women. Household exposure to someone else's smoking independently predicted a woman's own smokeless use (AOR 1.85, 95% CI 1.70-2.02), which is

the epidemiological argument for educating the whole family rather than the user alone. [PMID 42545929]

- Health-worker advice is the weakest link in the education chain and now has a number attached. Among Indian men using cigarettes or bidis who had actually seen a healthcare provider in the previous year, the proportion advised to quit was **54.2% in NFHS-4 and 56.7% in NFHS-5** (AOR 1.27, 95% CI 1.14-1.42) — which works out to only **9.7% of all male smokers surveyed**. Advice was least likely to reach Scheduled Tribe men (AOR 0.60, 95% CI 0.50-0.71) and other backward classes (AOR 0.80, 95% CI 0.72-0.90), and most likely once an NCD had already been diagnosed (AOR 1.80). Cite this when explaining why "brief advice at every clinical contact" reads well on paper and fails in the field. [PMID 42149875]

Elimination, surveillance and disease status

- Current burden estimates split the Indian epidemic by product rather than lumping it: combustible tobacco accounts for about **922,000 deaths and 8.9 million disease events a year** and smokeless tobacco for **165,000 deaths and 400,000 disease events**, with direct plus indirect costs of ₹3.56 trillion and ₹358 billion respectively. A modelled 50% rise in tobacco taxes would avert 1.8 million deaths (~9.8% of tobacco-attributable deaths) and 66.8 million healthy life-years over 10 years — the arithmetic behind the "R" of MPOWER. [PMID 41269237]
- A five-centre Indian case-control study (2,320 buccal mucosa cancers, 2,302 controls, 2010-2022) puts hard numbers on the gain-framed message that quitting always helps. Current chewers had **21.56 times** the odds of buccal mucosa cancer of never-chewers (95% CI 19.51-23.82) and current smokers 6.35 times (5.45-7.40); those who had quit ≥10 years earlier had only **0.39 (0.28-0.54)** the odds of a current smoker and 0.58 (0.43-0.81) of a current chewer. Odds fell per decade quit — 0.68 (0.56-0.81) for smoking, 0.77 (0.65-0.91) for chewing — yet ex-chewers 10 years abstinent were still about 12.5 times above never-users, so primordial prevention still outranks cessation. Quit rates were far lower for chewing (27%) than smoking (39%), which is exactly where Indian health education is under-targeted. [PMID 41657059]

Guideline and legislative updates

- **Cigarettes and Other Tobacco Products (Packaging and Labelling) Amendment Rules, notified 2 December 2024**, replaced the specified pictorial health warning image set carried on all tobacco packs, and separate Amendment Rules 2023 under the advertisement and trade regulation extended controls on tobacco depiction and promotion. The **85%** area of both principal display panels (60% pictorial + 25% text) is unchanged — what rotates is the mandated image set, so write "the current specified health warnings notified in December 2024" rather than describe a particular picture. [MoHFW 2024-Dec]
- The Cochrane review of smokeless tobacco cessation (43 trials, 20,346 people, searched to February 2024) supplies the evidence grade behind the education methods tabulated above: **counselling RR 1.76 (95% CI 1.44-2.16)** and **brief advice RR 1.24 (1.03-1.48)** against minimal or no support, both moderate-certainty; varenicline RR 1.35 (1.08-1.68), moderate-certainty; NRT RR 1.18 (1.05-1.33), low-certainty; bupropion showed no benefit (RR 0.89, 0.54-1.44). Only 5 of the 43 trials were conducted in India although South and South-East Asia holds most of the world's smokeless users — a stated research gap worth one line in a "way forward" paragraph. [PMID 40232040]

Global and WHO developments

- ▶ The WHO report on the global tobacco epidemic 2025, the tenth edition, was released on 23 June 2025 and is themed on the "W" of MPOWER — warning about the dangers of tobacco — which makes it the single most citable global source for this particular question. Over **6 billion people, roughly 7 in 10 worldwide, are now covered by at least one MPOWER measure at best-practice level**. WHO's current headline figures are 1.2 billion tobacco users with about 80% living in low- and middle-income countries, and more than 7 million deaths a year including over 1.6 million non-smokers killed by second-hand smoke. [WHO 2025-Jun]
- ▶ Evidence for the "Offer help" limb has consolidated, with India used as a worked case study: only **31 low- and middle-income countries** meet WHO best practice, defined as cost-covered behavioural support *plus* pharmacotherapy, and the scalable package recommended is quitlines, digital and SMS interventions, and systematic screening with clinician advice delivered through the **Ask-Advise-Connect** model, with cytisine and NRT on the WHO Model List of Essential Medicines. The supporting meta-analyses are quotable: clinic-delivered cessation across South Asia (37 studies, 26 of them Indian) raised continuous abstinence beyond 3 months (RR 1.68, 95% CI 1.14-2.47), with combined behavioural plus drug therapy the strongest arm although that comparison fell just short of significance (RR 1.70, 0.98-2.92); telephone counselling and quitlines raised continuous abstinence by 43% (RR 1.43, 95% CI 1.11-1.84), with better results after ≥3 counselling sessions. [PMID 42585647] [PMID 42394572] [PMID 41747792]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 22 Communication for Health Education; Ch 21 Health Information; Ch 20 Mental Health — tobacco and second-hand smoke; Ch 6 Epidemiology of Chronic Non-communicable Diseases; Ch 7 Health Programmes in India); Sunder Lal — Textbook of Community Medicine (Ch 4 Principles of Health Promotion and Education; Ch 8 Non-communicable Diseases and related programmes; Ch 15 Mental Health — tobacco control legislation and NTCP); Oxford Textbook of Public Health 7e (Ch 78 Tobacco; Ch 100 Strategies and structures for public health interventions). Named surveys and instruments: Global Adult Tobacco Survey India (GATS-1 2009-10, GATS-2 2016-17), Global Youth Tobacco Survey-4 (2019), WHO Framework Convention on Tobacco Control (2003), WHO MPOWER package (2008), COTPA 2003, Prohibition of Electronic Cigarettes Act 2019.

Recent-advances sources: PMID 40975577 — India's consensus statement tobacco control priorities: review of current strategies and the way forward, Indian Journal of Tuberculosis 2024, DOI [10.1016/j.ijtb.2024.11.007](https://doi.org/10.1016/j.ijtb.2024.11.007) (abstract-only); PMID 42545929 — Trends and correlates of smokeless tobacco use among Indian women: evidence from three rounds of the National Family Health Survey 2005-2021, PLoS One 2026, DOI [10.1371/journal.pone.0353704](https://doi.org/10.1371/journal.pone.0353704); PMID 42149875 — Receipt of healthcare provider advice to quit tobacco use among Indian men, PLoS One 2026, DOI [10.1371/journal.pone.0349022](https://doi.org/10.1371/journal.pone.0349022); PMID 41269237 — Estimating the expected health and economic benefits of price increases via taxation to counter the dual combustible and smokeless tobacco epidemics in India, Nicotine & Tobacco Research 2026, DOI [10.1093/ntr/ntaf242](https://doi.org/10.1093/ntr/ntaf242) (abstract-only); PMID 41657059 — The impact of tobacco cessation on buccal mucosa cancer risk: a multi-centre case-control study in India, International Journal of Cancer 2026, DOI [10.1002/ijc.70372](https://doi.org/10.1002/ijc.70372); PMID 40232040 — Interventions for smokeless tobacco use cessation, Cochrane Database of Systematic Reviews 2025, DOI [10.1002/14651858.CD015314.pub2](https://doi.org/10.1002/14651858.CD015314.pub2) (abstract-only); PMID 42585647 — Evidence-based tobacco-cessation strategies for low- and middle-income countries, New England Journal of Medicine 2026, DOI [10.1056/NEJMr2507061](https://doi.org/10.1056/NEJMr2507061) (abstract-only); PMID 42394572 — Effectiveness of tobacco cessation interventions delivered in clinical settings in South Asia: a systematic review and meta-analysis, Global Health Action 2026, DOI [10.1080/16549716.2026.2695701](https://doi.org/10.1080/16549716.2026.2695701) (abstract-only); PMID 41747792 — Effectiveness of telephone counseling and quitlines in tobacco cessation: a systematic review and meta-analysis, Annals of Epidemiology 2026, DOI [10.1016/j.annepidem.2026.110059](https://doi.org/10.1016/j.annepidem.2026.110059) (abstract-only). Non-journal sources: Ministry of Health and Family Welfare, National Tobacco Control Programme portal — National Tobacco Quitline Services and mCessation (accessed August 2026); Cigarettes and Other Tobacco Products (Packaging and Labelling) Amendment Rules, 2 December 2024; Cigarettes and Other Tobacco Products (Prohibition of Advertisement and Regulation of Trade and Commerce, Production, Supply and Distribution) Amendment Rules, 2023; WHO report on the global tobacco epidemic 2025, tenth edition, released 23 June 2025; WHO tobacco fact sheet, updated 2026.

♦ ♦ ♦

Health as a relative concept

Asked in: Paper I Nov 2018 — RGUHS MD Community Medicine

Definition — Health is not an absolute or ideal state but a biologically "normal" state judged against statistical averages derived from a defined reference population (Park 28e, Ch 2, p 28). Because those averages differ with country, culture, social class, age and sex, health standards are not universal: **health in any society must be defined in terms of the prevailing ecological conditions**, and each country decides its own norms for its own set of conditions before working out how to attain them (Park 28e, Ch 2, p 28).

1. Absolute (positive) health versus relative health

Feature	Absolute / "positive health" view	Relative view
Basis	Perfect functioning: every cell and organ at optimum capacity, a sense of perfect well-being and mastery over the environment, optimal capacity for social participation (Park 28e, Ch 2, p 28)	A biologically normal state based on statistical averages of a reference population (Park 28e, Ch 2, p 28)
Reference used	A single universal ideal, the World Health Organization (WHO) standard of complete physical, mental and social well-being (Park 28e, Ch 2, p 26)	A reference group matched for age, sex, community and region (Park 28e, Ch 2, p 26)
Measurability	Not an operational definition — it does not lend itself to direct measurement (Park 28e, Ch 2, p 26)	Operational and measurable — normal values and normal ranges permit a decision to be made (Sunder Lal, Ch 1, pp 38-39)
Attainability	<i>Dubos</i> : "the concept of perfect positive health cannot become a reality because man will never be so perfectly adapted to his environment that his life will not involve struggles, failures and sufferings" — positive health will always remain a mirage (Park 28e, Ch 2, p 28)	Attainable, revisable and locally set
Practical use	An aspiration and a goal towards which nations strive (Park 28e, Ch 2, p 26)	Everyday clinical and public-health decision-making, screening and programme targets

Dubos himself defined health relatively: "health implies the **relative** absence of pain and discomfort and a continuous adaptation and adjustment to the environment to ensure optimal function" (Park 28e, Ch 2, p 25).

2. The axes along which health is relative

Axis of relativity	Why health is relative on this axis	Worked example
Statistical	Normal values are derived from measurements made on a healthy population by statistical reasoning; what occurs most frequently is taken as normal (Sunder Lal, Ch 1, pp 38-39)	"Point normal" values (mean, median, mode) and "normal ranges"; some apparently healthy persons fall outside as outliers (Sunder Lal, Ch 6, p 246)

Axis of relativity	Why health is relative on this axis	Worked example
Geographical / national	Anthropometric standards vary from country to country (Park 28e, Ch 2, p 28)	An Indian newborn weighs 2.8 kg on average against 3.5 kg in developed countries, yet compares favourably in health (Park 28e, Ch 2, p 28)
Socio-economic	Height and weight standards vary between socio-economic groups (Park 28e, Ch 2, p 28); mortality, morbidity and self-assessed health all follow a social gradient (Oxford Textbook of Public Health 7e, Ch 94, p 3535)	The better-off on average live longer, enjoy better health and use more services — in market, social-insurance and tax-funded systems alike
Age and sex	Normal values differ between sexes and vary with age, and the pattern is not the same in all human populations (Sunder Lal, Ch 1, pp 38-39)	"Normal" blood pressure, pulse rate, height, weight, serum cholesterol and haemoglobin each differ by age and sex — a statement of normal values must name the population referred to
Individual biological variation	Deviation from the average is not automatically disease	Many normal people show heart murmurs, enlarged tonsils and chest X-ray shadows and yet show no sign of ill-health (Park 28e, Ch 2, p 28)
Temporal, within a person	Health is dynamic, a process of continuous change subject to frequent subtle variation — what is maximum health today may be minimum tomorrow (Park 28e, Ch 2, p 30)	A person functions at maximum health today and at a diminished level tomorrow; health is never attained once and for all, but ever to be renewed
Cultural and historical	Definitions of illness and wellness are culturally relative and time-dependent — sickness in one culture may be wellness in another (Sociology: The Essentials 6e, Ch 14, p 390)	A body habitus regarded as optimal in one era is defined as sickness in another era in the same culture
Perceptual / behavioural	<i>Illness behaviour</i> — different people react in different ways to the same disease process; some make light of symptoms, others respond in an exaggerated manner to the slightest discomfort (Park 28e, Ch 12, p 812)	<i>Susser's usage</i> : disease is a physiological or psychological dysfunction, illness is the subjective state of feeling unwell, sickness is a state of social dysfunction (Park 28e, Ch 2, p 51) — the three need not coincide

3. Health and disease lie on a continuum — the spectral concept

Health and disease lie along a continuum and **there is no single cut-off point**; the lowest point on the health–sickness spectrum is death and the highest corresponds to the WHO definition of positive health (Park 28e, Ch 2, p 30). Health therefore fluctuates within a range from optimum well-being to various levels of dysfunction, including the total dysfunction that is death. The transition from optimum health to ill-health is gradual, and **where one state ends and the other begins is a matter of judgment** — the single strongest demonstration that health is relative and not categorical. As long as we are alive there is some degree of health in us: there are degrees or "levels of health" just as there are degrees of severity of illness (Park 28e, Ch 2, p 30).

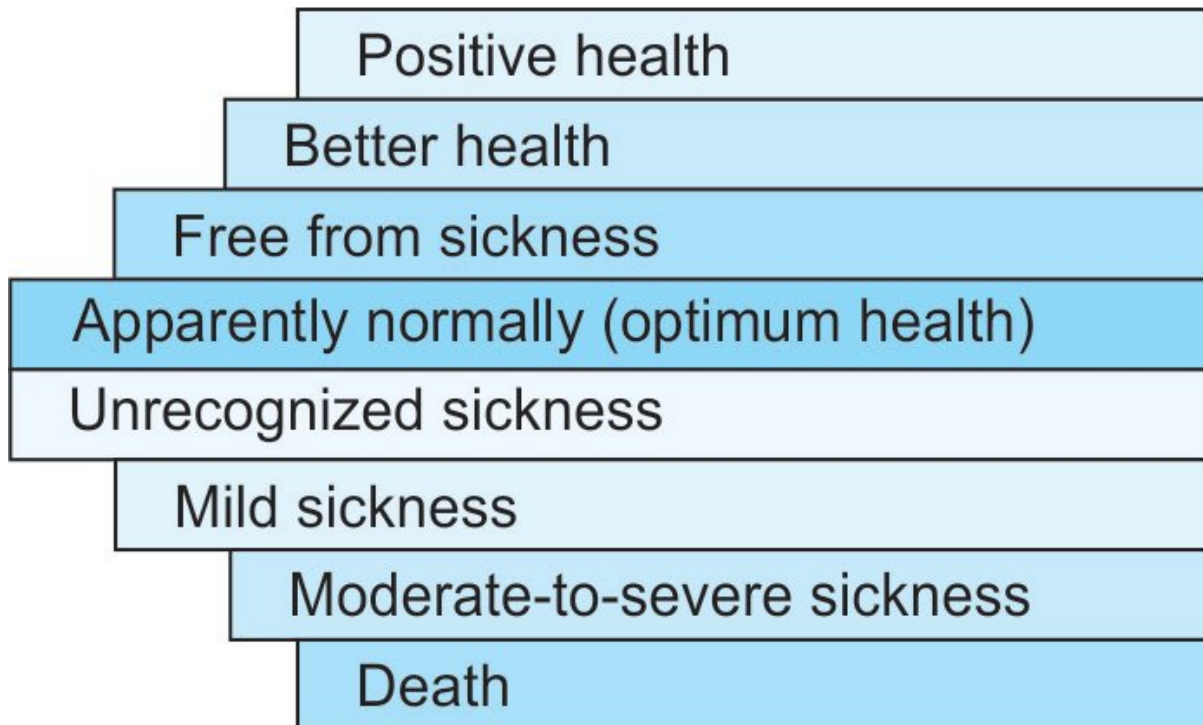


Fig 2.1 — Spectrum of health (Suryakantha Environment Best, p.27)

4. Where the line is drawn — the operational and epidemiological consequence

- ▶ **The operational definition is explicitly relative.** A WHO study group defined health, for measuring purposes, as (a) no obvious evidence of disease, the person functioning normally, that is *conforming within normal limits of variation to the standards of health criteria generally accepted for one's age, sex, community and region*, and (b) the organs functioning adequately in themselves and in relation to one another — a homeostasis that is relatively stable but which varies as human beings adapt to internal and external stimuli (Park 28e, Ch 2, p 26). Age, sex, community and region are written into the definition itself.
- ▶ **The waterline is set by the observer.** In the **iceberg phenomenon of disease**, the floating tip is what the physician sees (clinical cases) and the submerged mass is latent, inapparent, presymptomatic and undiagnosed cases plus carriers; the "waterline" is the demarcation between apparent and inapparent disease (Park 28e, Ch 2, p 58).
- ▶ **How hard you look decides how many are healthy.** For hypertension, diabetes, anaemia, malnutrition and mental illness the unknown morbidity far exceeds the known morbidity (Park 28e, Ch 2, p 58); shift the diagnostic cut-off or apply a screening test and people move across the healthy–diseased boundary without any change in their bodies.
- ▶ **Variability is inherent** in the biomedical measurements on which individual care and community programmes are based, which is precisely why standards ("normal values") have to be established at all (Sunder Lal, Ch 6, p 233).

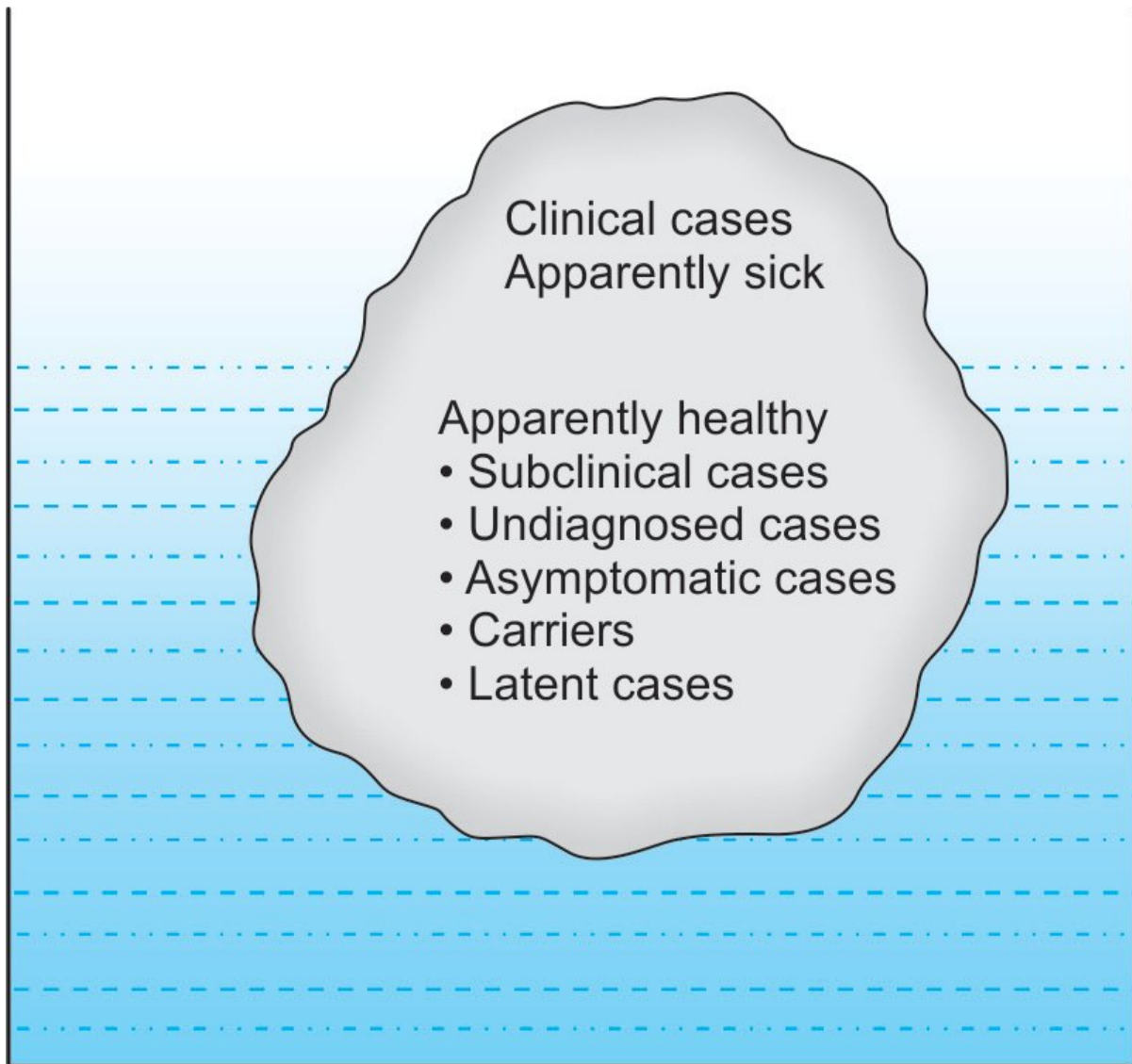


Fig 3.2 — Iceberg of diseases (Suryakantha Environment Best, p.35)

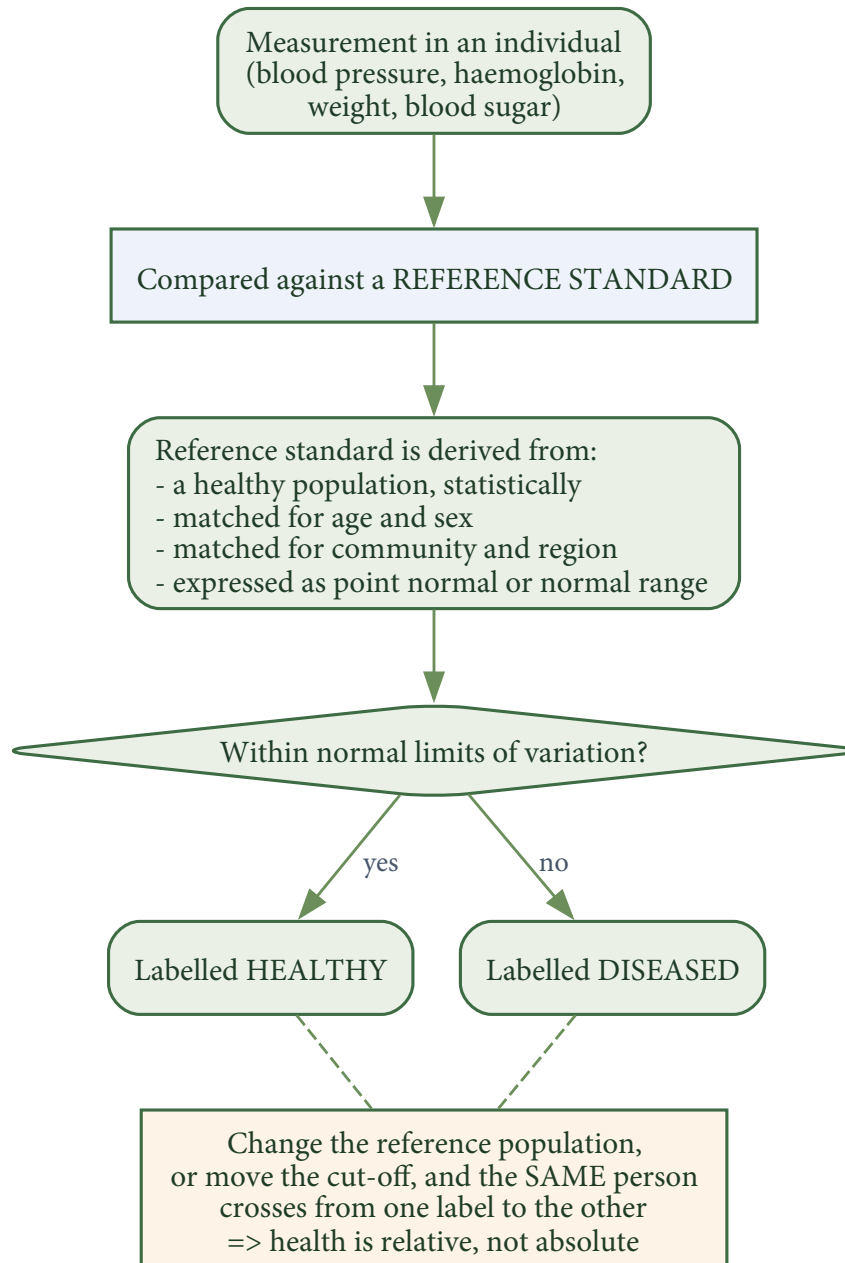
5. Implications for public health practice

- ▶ **Set local, not universal, norms** — instead of setting universal health standards, each country decides on its own norms for a given set of prevailing conditions and then looks into ways to achieve that level (Park 28e, Ch 2, p 28).
- ▶ **Always name the reference population** — a statement of normal values must indicate the population referred to; establishing normal values is what permits the selection of appropriate action in medical practice (Sunder Lal, Ch 1, pp 38-39).
- ▶ **Measure health comparatively** — because health cannot be measured absolutely, it is assessed through indicators compared between groups, between places and across time.
- ▶ **Treat screening cut-offs as conventions** to be justified, not as natural boundaries, and state the criterion used whenever prevalence is reported.
- ▶ **Act on the gradient, not only on the average** — the social gradient in health is one of the most enduring characteristics of contemporary health systems and is a value judgment about justice, not a biological given (Oxford Textbook of Public Health 7e, Ch 94, pp 3535, 3547).

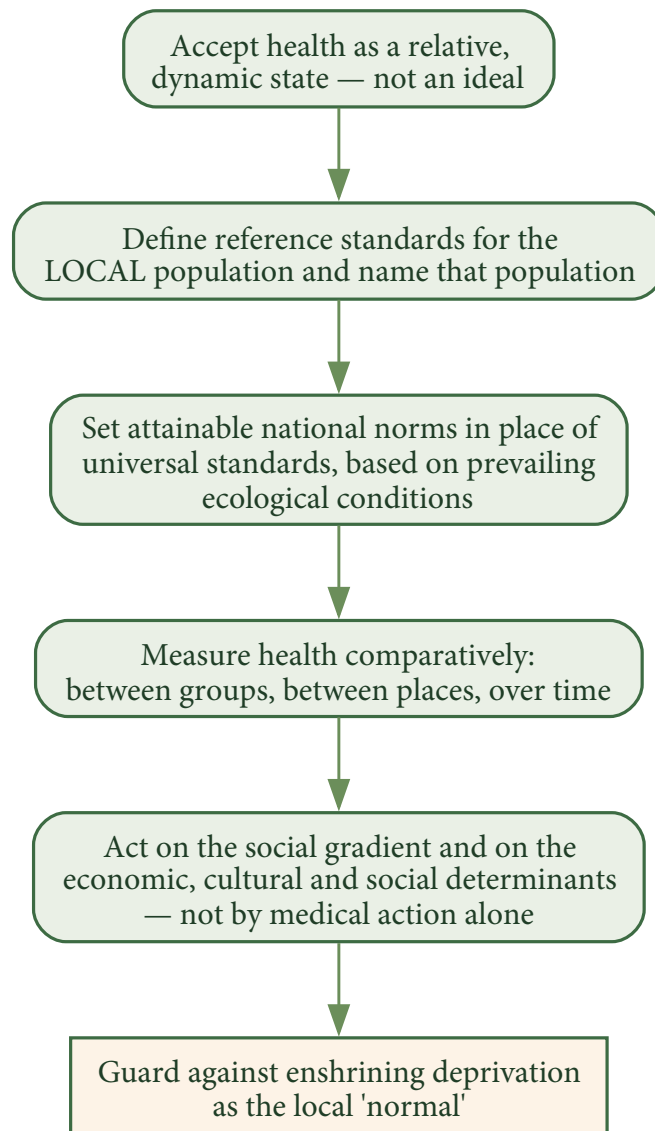
- ▶ **Work beyond the health sector** — positive health depends not only on medical action but on all the other economic, cultural and social factors operating in the community, within the broader concept of improving quality of life (Park 28e, Ch 2, p 28).
- ▶ **Do not enshrine deprivation as the local normal.** The WHO Child Growth Standards released in April 2006 demonstrated for the first time that children born in different regions of the world, given an optimum start in life, have the potential to grow and develop within the same range of height-for-age and weight-for-age (Sunder Lal, Ch 6, p 246) — so a low local average that reflects poverty must be corrected, not adopted as the standard.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Diagram 1 — The health-sickness spectrum** (draw as a vertical ladder of overlapping bands, top to bottom). Labels: *Positive health* → *Better health* → *Freedom from sickness* → *Apparently normal (optimum health)* → *Unrecognized sickness* → *Mild sickness* → *Moderate-to-severe sickness* → *Death*. Mark on the right: "highest point = WHO positive health", "lowest point = death", "no single cut-off — the boundary is a matter of judgment".
- ▶ **Diagram 2 — Iceberg of disease.** Labels: *waterline* (demarcation between apparent and inapparent disease); above the waterline — *clinical cases, apparently sick, what the physician sees*; below the waterline — *subclinical cases, undiagnosed cases, asymptomatic cases, carriers, latent cases, apparently healthy*.



How the same person can be called healthy or diseased — relativity built into the judgement



From the relative concept of health to public health action

High-yield exam pointers

- ▶ Anchor sentence to write first: *"health is a relative concept and health standards vary among cultures, social classes and age-groups; health in any society should be defined in terms of prevailing ecological conditions"* (Park 28e, Ch 2, p 28).
- ▶ The one number that wins marks: **Indian newborn 2.8 kg versus 3.5 kg in developed countries**, yet comparable health.
- ▶ *Dubos* quotation: positive health is a **mirage**; his own definition uses the words "**relative** absence of pain and discomfort".
- ▶ Operational definition carries the four relativity terms — **age, sex, community, region**.
- ▶ Spectrum of health, top to bottom: positive health → better health → freedom from sickness → unrecognized sickness → mild → severe sickness → death; **no single cut-off point**, the boundary is "a matter of judgment".
- ▶ *Susser's* triad: **disease** = physiological/psychological dysfunction; **illness** = subjective state of feeling unwell; **sickness** = social dysfunction.

- Iceberg phenomenon: the **waterline** separates apparent from inapparent disease — the answer to "how many are healthy?" depends on the test used.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the corpus supports every point above from Park's Textbook of Preventive and Social Medicine 28e Chapter 2, Sunder Lal Chapters 1 and 6, the Oxford Textbook of Public Health 7e and Sociology: The Essentials 6e; the two embedded plates (spectrum of health, iceberg of disease) are reproduced from Suryakantha's Community Medicine with Recent Advances because Park's own Figure 2 and Figure 10 exist in the corpus only as unusable scanned page fragments.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 2 — Concept of Health and Disease, pp 25-30, 51, 58; Ch 12 — Medicine and Social Sciences, pp 811-812); Sunder Lal — Textbook of Community Medicine (Ch 1 — Concept of Health and Disease, pp 38-40; Ch 6 — Basic Statistics and its Applications, pp 233, 246); Oxford Textbook of Public Health 7e (Ch 94 — Health Needs Assessment, pp 3534-3535, 3547; Ch 47 — Health Promotion, p 1740); Sociology: The Essentials 6e (Ch 14 — Education and Health Care, p 390); Suryakantha — Community Medicine with Recent Advances (Figs 2.1 and 3.2); WHO Child Growth Standards, 2006.



Methods of health education

Asked in: Paper I 29-Jul-2021 — RGUHS MD Community Medicine

Definition — Health education is the planned use of communication to give people the knowledge, attitudes and skills to make their own choice for a healthy life, so that the desired behaviour is adopted voluntarily and endures (Park 28e, Ch 22, p 1016). The **methods** of health education are the channels that carry the message, classified by the size of the audience reached into the **individual (interpersonal), group and mass approaches** (Park 28e, Ch 22, Fig 3, p 1021). Any one method, or a combination, may be used selectively at different times, depending on the objective to be achieved, the behaviour to be influenced and the funds available (Park 28e, Ch 22, p 1020).

Individual (interpersonal) approach

One-to-one communication between the health worker and one person, dealing with the precise need of that individual — the working vehicle of behaviour change communication (BCC) (Sunder Lal, Ch 4, p 146).

Method	How it is used in practice
Personal contact / personal interview	In the doctor's consultation room, at the health centre, or in the person's home. The individual attends because of illness; the opportunity is taken to educate on matters of interest to him — diet, causation and prevention of the illness, personal hygiene, environmental hygiene. Topics for health counselling are selected according to the relevance of the situation (Park 28e, Ch 22, p 1020)
Home visits	Contact with the family and care-givers by the health worker on her allotted beat; the most precise and specific communication, and the setting in which small group meetings can also be held when women cannot come out (Sunder Lal, Ch 4, pp 146-147)
Personal letters	A written message addressed to a named individual, listed by Park as the third individual-approach method (Park 28e, Ch 22, Fig 3, p 1021)

Merits	Limitations
The message passes directly from source to receiver ; the most interactive and participatory of all approaches (Sunder Lal, Ch 4, p 146)	The numbers reached are small (Park 28e, Ch 22, p 1020)
The biggest advantage — the educator can discuss, argue and persuade the individual to change his behaviour (Park 28e, Ch 22, p 1020)	Education is given only to those who come in contact with the health worker, so the healthy and the non-attender are missed (Park 28e, Ch 22, p 1020)
Provides the opportunity to ask questions on specific interests , in an atmosphere of friendship in which the individual is allowed to talk as much as possible (Park 28e, Ch 22, p 1020)	Depends wholly on the interpersonal communication skill of the worker , which must be sharpened by training and retraining in group dynamics and group discussion (Sunder Lal, Ch 4, p 146)

Group approach

Society contains groups of many kinds — school children, mothers, industrial workers, patients. Group teaching is an effective way of educating the community, but the **choice of subject must relate directly to the interest of the group**: one does not broach tuberculosis control with a mother who has come for delivery — one talks to her about child-birth and baby care (Park 28e, Ch 22, pp 1020-1021).

Method	Essentials — size, duration and conduct	Best suited for
Chalk and talk (lecture)	A carefully prepared oral presentation of facts and organised thoughts by a qualified person; the "chalk" supplies the visual component. Group not more than 30 , talk not exceeding 15–20 minutes , or listeners become bored and restless (Park 28e, Ch 22, p 1021)	Transferring new information to a small, captive, reasonably literate group
Demonstration	A carefully prepared presentation showing how to perform a skill or procedure, carried out step by step before the target group, the demonstrator confirming that the audience has understood and drawing them into discussion (Park 28e, Ch 22, p 1021)	Skills — installing a hand-pump, building a sanitary latrine, preparing oral rehydration solution, disinfection of a well, treatment of scabies (Park 28e, Ch 22, p 1021)

Method	Essentials — size, duration and conduct	Best suited for
Group discussion	6–12 members seated in a circle so each is visible to all, with a group leader who initiates the subject, prevents side-conversations, encourages everyone to participate and sums up, and a recorder who prepares the group report (Park 28e, Ch 22, pp 1021-1022). In Indian field practice a village group of 10–15 meeting for 30–60 minutes at the Anganwadi centre, Mahila Mandal premises or subcentre (Sunder Lal, Ch 4, p 147)	Changing attitudes and behaviour — a group decision has a binding effect on the member and is adopted more readily than a solitary one
Panel discussion	4–8 qualified persons discuss the topic before a large audience, with a chairman/moderator who opens the meeting, introduces the panel and the topic, and keeps the train of thought going. No specific agenda, no order of speaking, no set speeches ; the audience joins in after the panel has explored the subject (Park 28e, Ch 22, p 1022)	Presenting several viewpoints on a contested subject to a large audience
Symposium	A series of speeches on a selected subject, each expert presenting one aspect briefly. No discussion among the members (unlike a panel); the audience may raise questions at the end and the chairman makes a comprehensive summary (Park 28e, Ch 22, p 1022)	Covering a wide subject authoritatively in limited time
Workshop	A series of meetings, usually four or more , divided into small groups each with a chairman and a recorder, working with consultants and resource personnel. Emphasis is on individual work within the group; each participant leaves with a plan of action (Park 28e, Ch 22, p 1022)	Building professional competence and producing an actionable output
Role playing (socio-drama)	The group dramatises a situation, members enacting roles as they have observed or experienced them. Group size best at about 25 ; the audience is not passive but suggests alternative solutions and may take an active part. Always followed by discussion of the problem (Park 28e, Ch 22, pp 1022-1023)	Problems of human relationship ; a particularly useful device for school children
Conferences and seminars	Held at regional, state or national level, half a day to one week long, covering a single topic in depth or a broad field, using formats from self-instruction to multimedia (Park 28e, Ch 22, p 1023)	Continuing education of professionals

Conduct of a group discussion — rules for members (Park 28e, Ch 22, pp 1021-1022):

- ▶ Express ideas clearly and concisely.
- ▶ Listen to what others say.
- ▶ Do not interrupt when others are speaking.
- ▶ Make only relevant remarks.
- ▶ Help the group to reach conclusions.
- ▶ Accept criticism gracefully.

Limitations of group discussion (Park 28e, Ch 22, p 1022):

- ▶ Those who are shy may not take part at all.

- Some members dominate the discussion, producing **unequal participation** unless properly guided.
- Some deviate from the subject and make the discussion irrelevant or unprofitable.
- It succeeds only when members know each other beforehand and can discuss freely — so it is unsuited to a first, anonymous gathering.

Limitations of the lecture (Park 28e, Ch 22, p 1021): students are involved to a minimum extent; learning is passive; it does not stimulate thinking or problem-solving capacity; comprehension varies from listener to listener; and the health behaviour of the listeners is not necessarily affected.

Mass approach

No health worker or health team can mount an effective health education programme for a whole community except through the mass media (Park 28e, Ch 22, pp 1022-1023).

Medium	Strength	Caution / limitation
Television	The most popular of all media ; creates awareness, influences public opinion, introduces new ways of life and familiarises people with what they have not seen (Park 28e, Ch 22, p 1023)	A one-way channel ; only an aid to teaching, and it cannot cover all areas of learning
Radio	Found in nearly every home and has a broadier audience than television in many developing countries ; with television it reaches the illiterate, who are not accessible through the printed word (Park 28e, Ch 22, p 1023)	A purely didactic medium — no feedback
Newspapers	The most widely disseminated of all printed matter; provide more factual and statistical material than radio or television (Park 28e, Ch 22, pp 1022-1023)	News must be newsworthy before it is carried; readers often turn to newspapers seeking entertainment
Printed material and direct mailing	Pamphlets, books, periodicals and mailed leaflets deliver detail that a broadcast cannot (Park 28e, Ch 22, Fig 3, p 1021)	Reaches only the literate
Posters, billboards and signs	Catch the eye and create awareness cheaply, being seen by very large numbers; humour and fear are used to hold attention. Where exposure time is short (streets) the message must be short, simple and direct; where people wait (bus stops, railway stations, hospitals, health centres) more information can be given (Park 28e, Ch 22, p 1023)	Short life — must be changed frequently or they lose effect; have much less effect on behaviour than their users hope; indiscriminate pasting on walls serves no purpose. Content must be topical — display viral hepatitis posters during a hepatitis epidemic, not cholera posters

Medium	Strength	Caution / limitation
Health museums and exhibitions	Attract large numbers and increase knowledge and awareness. Photographic panels attract more people than graphic panels because photographs humanise the message, and three-dimensional models with lighted visuals are more effective still . Workers explain the exhibits and free printed literature is distributed, so an exhibition combines personal and mass methods (Park 28e, Ch 22, pp 1023-1024)	Effective only if properly organised and manned
Folk media	Traditional folk dance, song and drama — <i>Nautanki</i> and <i>Harikatha</i> of Uttar Pradesh, <i>Burrakatha</i> of Andhra Pradesh, <i>keerthan</i> and religious meetings — are close to the cultural values of the rural population and have preserved its cultural heritage, so health messages carried through them are readily accepted (Park 28e, Ch 22, p 1024)	Message must be adapted to the form without distorting either
Internet	A vast and growing amount of health-related literature, including WHO material, reaches a rapidly rising number of users in India, and the numbers are growing every day (Park 28e, Ch 22, pp 1023-1024)	Unregulated content; access still uneven

Key caveat: mass media alone are generally inadequate to change human behaviour and must be combined with other methods. Their real power lies in creating political will in favour of health, raising health consciousness, setting norms, delivering technical messages, popularising health knowledge and fostering community involvement. Messages must be carefully thought out and culturally appropriate before use (Park 28e, Ch 22, p 1023).

Audio-visual aids that carry these methods

Group	Aids
Auditory aids	Radio, tape-recorder, microphones, amplifiers, earphones
Visual aids — not requiring projection	Posters, charts, flannelgraph , blackboard, leaflets, exhibits, models, specimens
Visual aids — requiring projection	Slides, film strips
Combined audio-visual aids	Television, sound films (cinema), slide-tape combination

(Park 28e, Ch 22, p 1020)

Audio-visual aids simplify unfamiliar concepts, clarify where words fail, reinforce learning by appealing to more than one sense, and avoid monotony (Park 28e, Ch 22, p 1020). Two are asked about repeatedly: **flipcharts** — a series of charts about **25 × 30 cm** or larger, each flashed before the group in turn as the talk proceeds, the message on each brief and to the point; and the **flannelgraph** — rough flannel or khadi fixed over a wooden board, on which cut-out pictures backed with sandpaper or rough cloth adhere at once. The flannelgraph is very cheap, easy to transport, keeps a pre-arranged sequence of pictures continuous, and promotes thought and criticism (Park 28e, Ch 22, p 1021).

Mass media versus personal communication — assets compared

Mass media (television, radio, newspaper)	Personal communication (interpersonal and group methods)
Reaches the widest population	Capitalises on warmth, understanding and the communicator's knowledge
Gets public attention	Provides the opportunity for involvement — asking questions, expressing fears, learning more
Gives greater support for concentrated campaigns, such as a health week or health month	Can get people to change personal habits more readily, because discussion supplies reasonable explanations for the change
More effective among those of above-average educational level	More influential with those of average and below-average educational level

(Park 28e, Ch 22, Table 2, p 1021)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Diagram 1 — a well-conducted group discussion** (Park 28e, Fig 4). Draw 6–12 numbered circles arranged in a complete circle; two-headed arrows running from every member to every other member; label one member "Group leader — initiates, guides, prevents side-conversations, sums up" and one "Recorder — prepares the group report". Caption: *Optimum group size 6–12; everyone fully visible to everyone else.*
- **Diagram 2 — a dominated group** (Park 28e, Fig 5). The same circle of numbered members, but with nearly all arrows converging on No. 1 and No. 7, and members 3, 4 and 6 drawn with no arrows at all. Labels: "Dominant members (No. 1, No. 7)", "Silent/shy members — no participation", "Unequal participation". Caption: *The commonest failure of the group discussion method.*

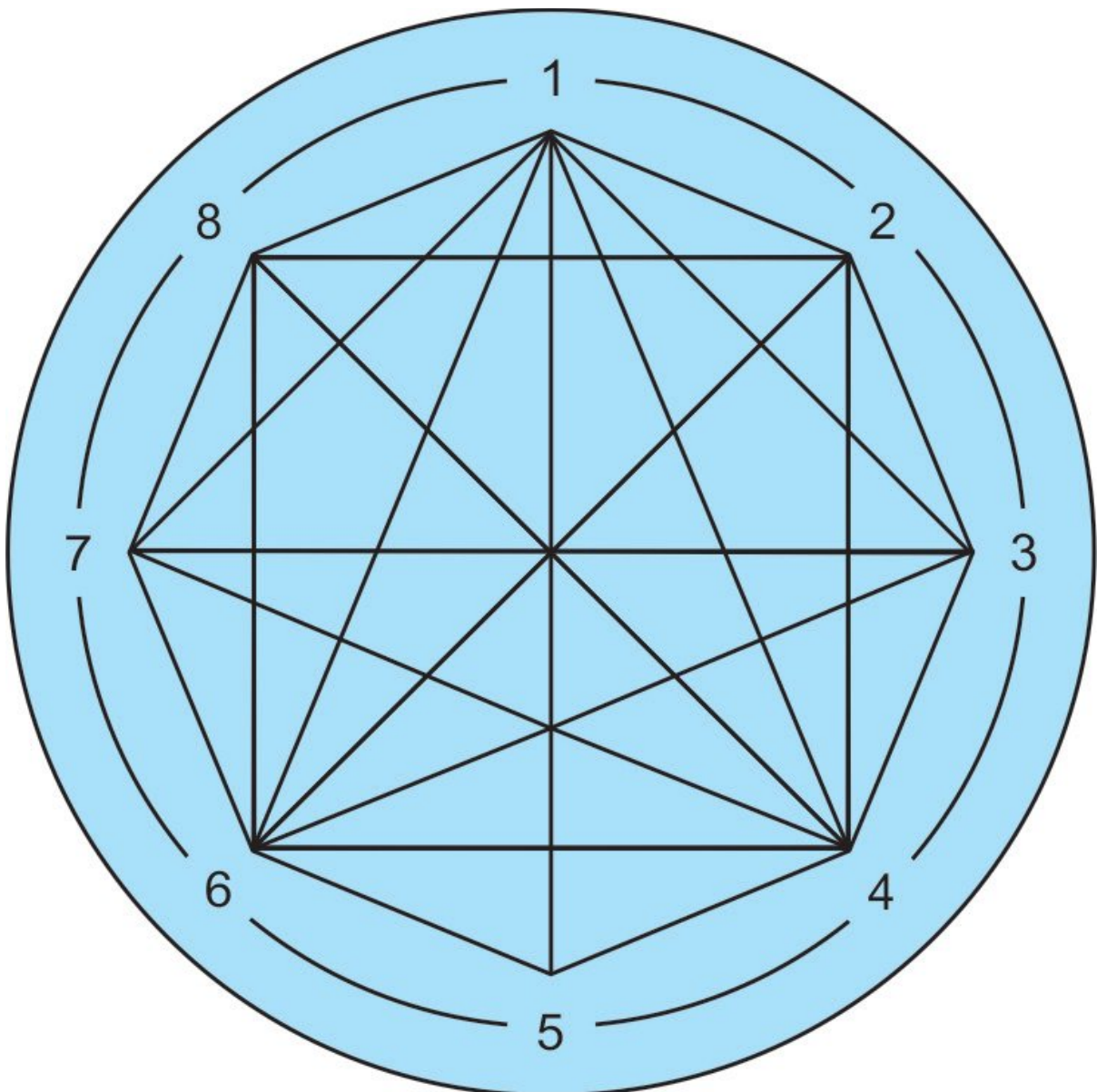


Fig 27.1 — A group discussion (Suryakantha Environment Best, p.744)

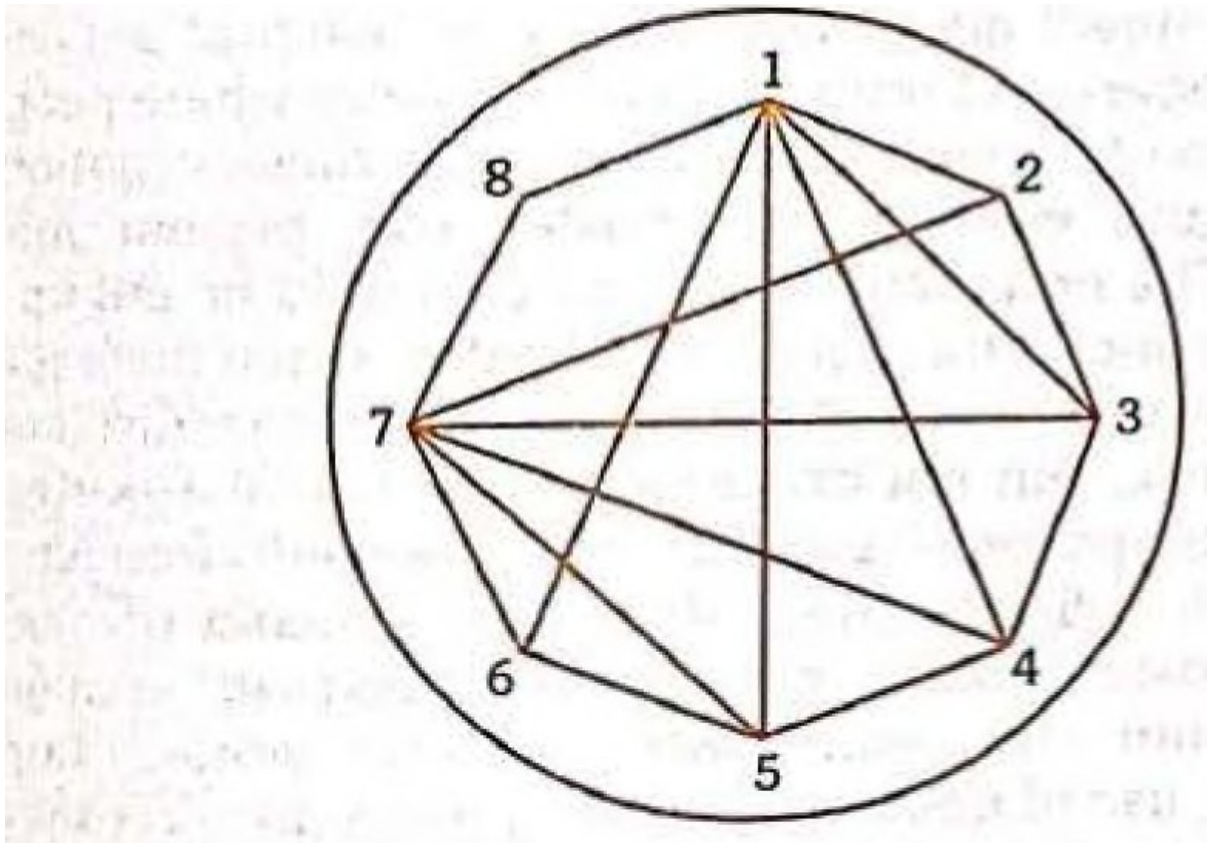
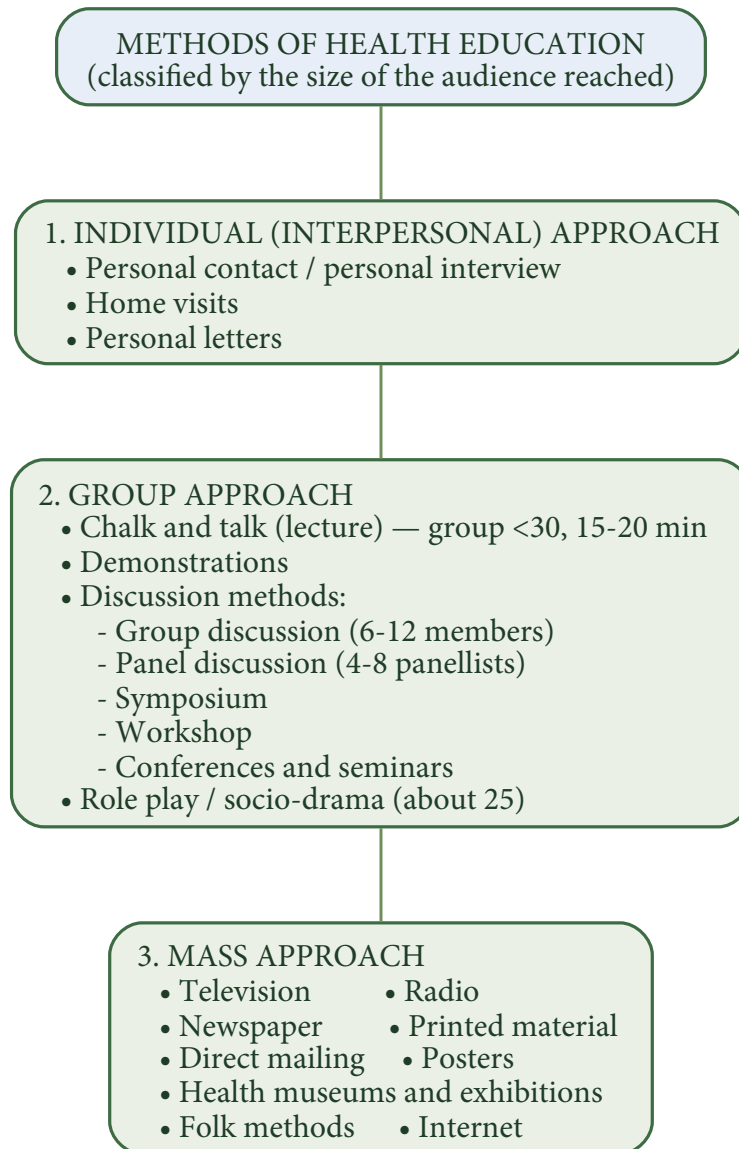
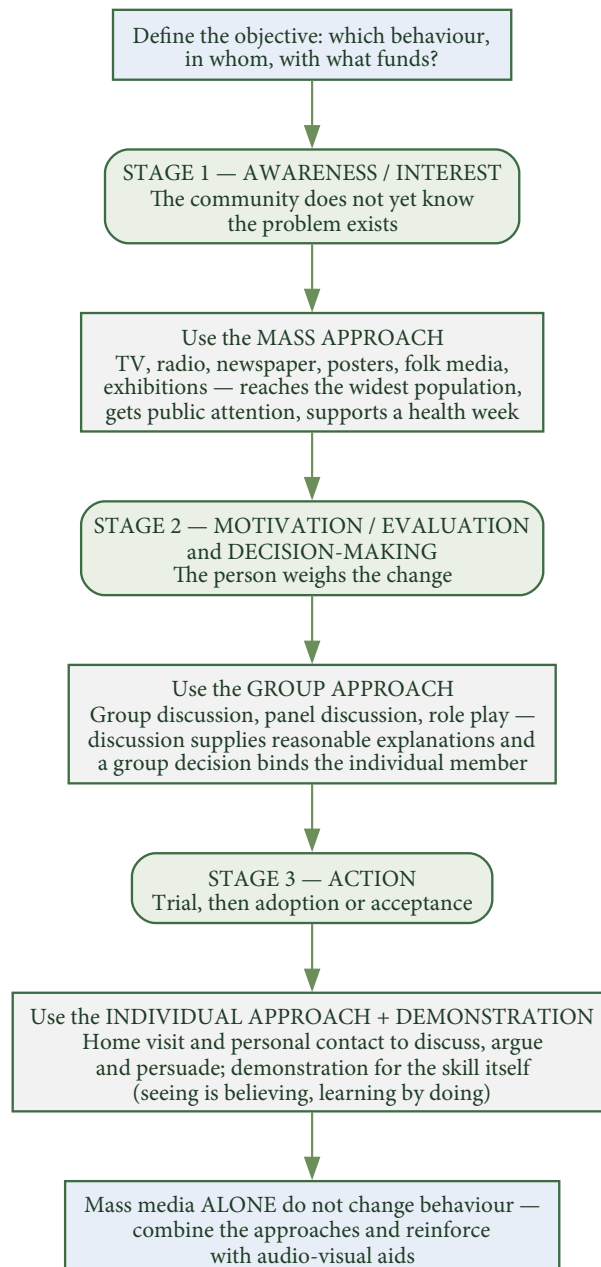


Fig 5 — , A dominated group. No.1 and No.7 dominate the discussion (Park's Textbook of Preventive and Social Medicine, 28e, p.1022)



Classification of the methods of health education by size of audience reached (Park 28e, Ch 22, Fig 3, p 1021)



Matching the method of health education to the stage of behaviour change (Park 28e, Ch 22, pp 1016-1017, 1020-1023)

High-yield exam pointers

- ▶ Park's three approaches = individual (interpersonal), group, mass — classified by size of audience reached (Fig 3, Ch 22).
- ▶ Numbers to write: lecture — group <30, talk 15–20 min; group discussion — 6–12 members in a circle (Indian field group 10–15, meeting 30–60 min); panel — 4–8 panellists plus a chairman; workshop — four or more meetings; role play — about 25; flipchart — 25 × 30 cm; conferences/seminars — half a day to one week.
- ▶ Panel versus symposium — the single commonest one-mark discriminator: a panel has free discussion among the members and no set speeches; a symposium is a series of set speeches with no discussion among the members, the audience questioning only at the end.

- ▶ **Demonstration works on four principles:** dramatises and arouses interest; persuades onlookers to adopt the practice; upholds "*seeing is believing*" and "*learning by doing*"; and brings a desirable change in behaviour.
- ▶ **Audio-visual aids in four groups:** auditory; visual not requiring projection; visual requiring projection; combined.
- ▶ **Mass media are better for those of above-average education; personal communication is more influential at average and below-average educational level** (Park Table 2) — and mass media alone are inadequate to change behaviour.
- ▶ **Group dynamics** — the pattern of interaction between members and with the facilitator; when the interaction arrows are joined up it forms a **sociogram** (Sunder Lal, Ch 4, pp 147-148).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: Park's Fig 3 classifies methods purely by audience size, whereas the National Health Mission's current service vocabulary calls the individual approach **interpersonal communication (IPC)** and places it inside **behaviour change communication (BCC)** under the wider **Information, Education and Communication (IEC)** strategy — the same three methods under newer programme labels. Park gives the optimum group-discussion size as 6–12 while Sunder Lal's Indian field guidance gives 10–15 for a village women's group; both are quotable, and either should be attributed to its source if written together.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 22 (Communication for Health Education), pp 1014–1024, and Ch 21, pp 1010–1011 (folk media); Sunder Lal — Textbook of Community Medicine, Ch 4 (Principles of Health Promotion and Education), pp 146–148; Suryakantha — Community Medicine with Recent Advances, Fig 27.1, p 744 (group discussion plate).



Poverty and its impact on health

Asked in: Paper I 07-Jul-2023 — RGUHS MD Community Medicine

Definition — **Poverty** is the lack of the resources needed for a decent way of life; **absolute poverty** is the lack of the minimum income needed for simple survival, priced as a basket of essential goods and services, while **relative poverty** is the lack of the resources required to participate meaningfully in the society one lives in (Oxford Textbook of Public Health 7e, Ch 8, pp 343, 348–349; Ch 10, pp 413–414). **Deprivation** exists when people lack or are denied the resources to take part in the social and economic activities of their society (Oxford 7e, Ch 8, p 356). In Park's framing poverty is one of the socio-economic host factors that determine health, alongside education, occupation, housing and stress (Park 28e, Ch 2, p 55), and it is described as wielding "its destructive influence at every stage of human life from the moment of conception to the grave" (Park 28e, Ch 12, p 844).

Types of poverty and how each is measured

Concept	Meaning	Measure / cut-off
Absolute (extreme) poverty	Income below that needed for simple survival	World Bank headline measure — living on less than US\$ 1.90 per day per capita (PPP-corrected) ; moderate poverty < US\$ 3.10 a day (Oxford 7e, Ch 8, p 343; Ch 10, pp 413–414)
Relative poverty	Inability to function socially to the normative standard of one's own society; varies by place and time (<i>Peter Townsend</i>)	Income relative to the societal average / "weakly relative lines" (Oxford 7e, Ch 8, pp 348–349)
Deprivation	Being denied resources needed for social and economic participation	Composite deprivation indices; has both absolute and relative forms (Oxford 7e, Ch 8, p 356)
Income poverty, India	Head-count below the national poverty line	Derived from household consumer expenditure data collected by the National Sample Survey Organisation (NSSO) every fifth year (Sunder Lal, Ch 2, pp 85–86)
Multidimensional Poverty Index (MPI)	Deprivation in three dimensions — <i>health</i> (nutrition, child mortality), <i>education</i> (school attendance, mean years of schooling), <i>standard of living</i> (cooking fuel, sanitation, drinking water, electricity, housing, assets)	Head-count of the multidimensionally poor; captures people above the income line who are still deprived (Sunder Lal, Ch 2, p 86)
Human Development Index (HDI)	Aggregate progress in health, education and income	Life expectancy at birth; mean and expected years of schooling; gross national income per capita — a more comprehensive measure than per-capita income alone (Park 28e, Ch 2, p 29; Sunder Lal, Ch 1, p 49)

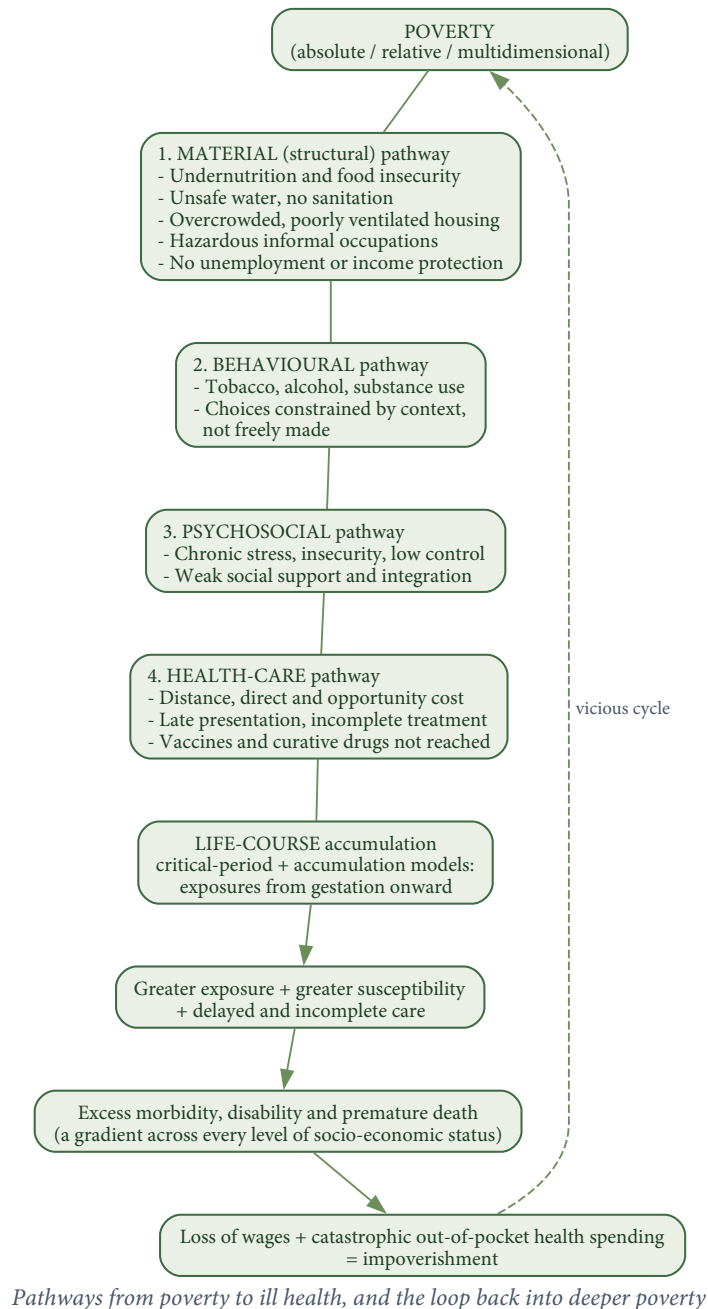
Concept	Meaning	Measure / cut-off
Standard of living / level of living	The objective component of well-being	United Nations "level of living" has nine components — health, food consumption, education, occupation and working conditions, housing, social security, clothing, recreation and leisure, human rights; health is the most important , because its impairment always means impairment of the level of living (Park 28e, Ch 2, p 28)
Global Hunger Index (GHI)	Hunger, tracked yearly by the International Food Policy Research Institute	Proportion of undernourished population + child underweight + child mortality, on a 0 (best) to 100 (worst) scale (Park 28e, Ch 12, p 844)

Magnitude of poverty in India (every figure dated)

Indicator	Value and year	Source
Population below the national poverty line	~22% (270 million), 2011–12 estimate	Park 28e, Ch 12, p 845
Same, disaggregated	25.7% rural, 13.7% urban; 21.9% overall, 2011–12 (NITI Aayog)	Sunder Lal, Ch 2, pp 85–86
Poverty line — old (calorie) definition	Expenditure for a daily intake of 2,400 kcal rural and 2,100 kcal urban ; for the year 2000 fixed at Rs 328/month rural and Rs 454/month urban	Park 28e, Ch 12, p 845
Poverty line — current (Tendulkar) definition	Since 2005 the calorie anchor was replaced by a basket of goods ; per-capita monthly consumption expenditure < Rs 1,640 rural and < Rs 2,500 urban (2014 prices); Tendulkar line for 2009–10 was Rs 672.8 rural and Rs 859.6 urban	Sunder Lal, Ch 2, pp 85–86
Multidimensional poverty	645 million (2005–06) fell to 230 million (2019–21) — about 415 million people lifted out between 2006 and 2021 ; 16.4% multidimensionally poor in 2021	Sunder Lal, Ch 2, p 86 (United Nations Development Programme, 2022)
Deprived rural households	~ 106 million (60%) of 179 million rural households classed as deprived	Socio-Economic and Caste Census 2011, in Sunder Lal, Ch 2, p 85
Global Hunger Index, India	27.3, rank 105 of 127 countries (2024)	Park 28e, Ch 12, p 844
Per-capita gross national income (PPP), India	US\$ 10,030 (2024) , with life expectancy at birth 72 years and infant mortality rate 23 per 1,000 live births	Park 28e, Ch 12, p 844, Table 6

Pathways — how poverty produces ill health

Four complementary explanations are used; no single one is sufficient, and the purely behavioural account is the weakest because it ignores the social contexts that condition and constrain individual choice and fails to explain the excess disease burden that persists after all individual risk factors are accounted for (Oxford 7e, Ch 8, pp 366–367).



Pathways from poverty to ill health, and the loop back into deeper poverty

Pathway	Evidence / mechanism
Materialist or structural	Health is determined by access to income, housing, employment, nutritious food, safe water and sanitation, unemployment protection, health services and education services — factors that reflect embedded economic and power relations (Oxford 7e, Ch 8, pp 366–367)
Behavioural / cultural	Poorer patterns of tobacco, alcohol and diet — but this explanation alone ignores the social context that constrains choice and cannot explain the residual excess disease burden (Oxford 7e, Ch 8, pp 366–367)
Psychosocial	There is a gradient in life expectancy, mortality and morbidity across all levels of the socio-economic hierarchy (<i>Marmot, Whitehall studies</i>); inequalities that persist without material deprivation are best explained by social support, integration and control at work (Oxford 7e, Ch 18, pp 714–715)

Pathway	Evidence / mechanism
Life-course	Disease risk in lower socio-economic groups results from socially patterned exposures during gestation, childhood, adolescence and adult life — the critical-period model and the accumulation model (Oxford 7e, Ch 9, pp 400–401)
Education as the modifier	The world map of illiteracy closely coincides with the maps of poverty, malnutrition, ill health and high infant and child mortality; education, especially female education, partly compensates the effect of poverty on health irrespective of the availability of health facilities — Kerala's infant mortality rate was 6 versus 28 for all-India (2020) , with female literacy 98.3% versus 71.5% all-India (2019–20) (Park 28e, Ch 2, p 32)

Impact of poverty on health and on disease pattern

Domain	Impact
Overall	The poor die younger and suffer more from disability ; they face greater risk from unhealthy conditions at home and at work; malnutrition and the legacy of past illness make them more likely to fall ill and slower to recover, especially with little access to health care (Park 28e, Ch 12, p 844)
Named consequences (Park)	Poverty is the main reason babies are not vaccinated , clean water and sanitation are not provided, and curative drugs are unavailable; it is the main cause of low life expectancy, low birth weight, higher maternal mortality, handicap and disability, mental illness, stress, suicide, family disintegration and substance abuse (Park 28e, Ch 12, p 844)
Maternal and child health	In every region, physical and intellectual development varies with family socio-economic level — under-privileged children of the same age are smaller, lighter and less advanced in psychomotor and intellectual performance ; poverty, illiteracy (especially the mother's) and sickness create a vicious circle spanning one generation to the next (Park 28e, Ch 9, p 652)
Communicable disease	Tuberculosis is mainly a disease of the poor — victims are migrant labourers, slum dwellers, residents of backward areas and tribal pockets; poor living conditions, malnutrition, shanty housing and overcrowding are the main reasons for spread (Park 28e, Ch 5, pp 221–222). Poor families also bear the brunt of diarrhoeal disease, acute respiratory infection and worm infestation from overcrowding and insanitary conditions (Sunder Lal, Ch 2, p 74)
Nutrition	Undernourishment (chronic calorie deficiency, < 1,800 kcal/day), hidden hunger (micronutrient deficiency) and, with the nutrition transition, a "triple burden" of undernutrition, micronutrient deficiency and obesity (Park 28e, Ch 12, pp 844–845)
Non-communicable disease	The poorest are affected most — cases are detected late, people die younger and in their most productive years; at household level, cardiovascular and other non-communicable diseases in turn deepen poverty through catastrophic health spending (Park 28e, Ch 6, p 437). Control of rheumatic fever explicitly requires breaking the poverty–disease–poverty cycle (Park 28e, Ch 6, p 455). Coronary heart disease, diabetes and obesity, by contrast, show higher rates in the upper socio-economic groups (Park 28e, Ch 2, p 32)
Mental health and social pathology	Poverty and low socio-economic status are associated with a greater incidence of mental disorder (Park 28e, Ch 12, p 825). "Social pathology" is classically defined by poverty, crime, delinquency and vagrancy; poverty is listed among the causes of juvenile delinquency, prostitution and problem families (Park 28e, Ch 12, pp 816, 828–829, 839–840)

Domain	Impact
Urban poor	Slum populations face greater hazards from overcrowding, poor sanitation, lack of safe drinking water and pollution; in some areas their health indices are worse than the rural population's — "poverty in plenty" in terms of primary health care (Sunder Lal, Ch 17, p 756)
Occupational	The poor are forced into occupations that harm their health , and lack access to health care and education (Park 28e, Ch 12, p 844)

Ill health as a cause of poverty — the reverse arm of the cycle

"People are poor because of sickness and sick because of poverty" (Sunder Lal, Ch 2, p 85). When a family's breadwinner becomes ill, the household first copes by working harder and by reducing consumption, even of food — both adjustments harm the health of the whole family (Park 28e, Ch 12, p 844).

Element	Indian figure
Out-of-pocket expenditure (OOPE)	Highest in the world; 1.52% of gross domestic product and Rs 2,155 per capita per year; household revenues finance over 60% of current health expenditure (Sunder Lal, Ch 16, pp 727–729)
Impoverishment	About 4.2% of the population — roughly 50 million people — are pushed into poverty every year because of high out-of-pocket health spending (Sunder Lal, Ch 16, pp 728–729)
Catastrophic health expenditure	Household health costs exceeding 10% of total monthly consumption expenditure, or 40% of non-food consumption expenditure (Sunder Lal, Ch 16, pp 728–729)
Global monitoring	Sustainable Development Goal indicator 3.8.2 — fraction of the population protected against catastrophic/impoverishing out-of-pocket health expenditure (Park 28e, Ch 14, p 853)

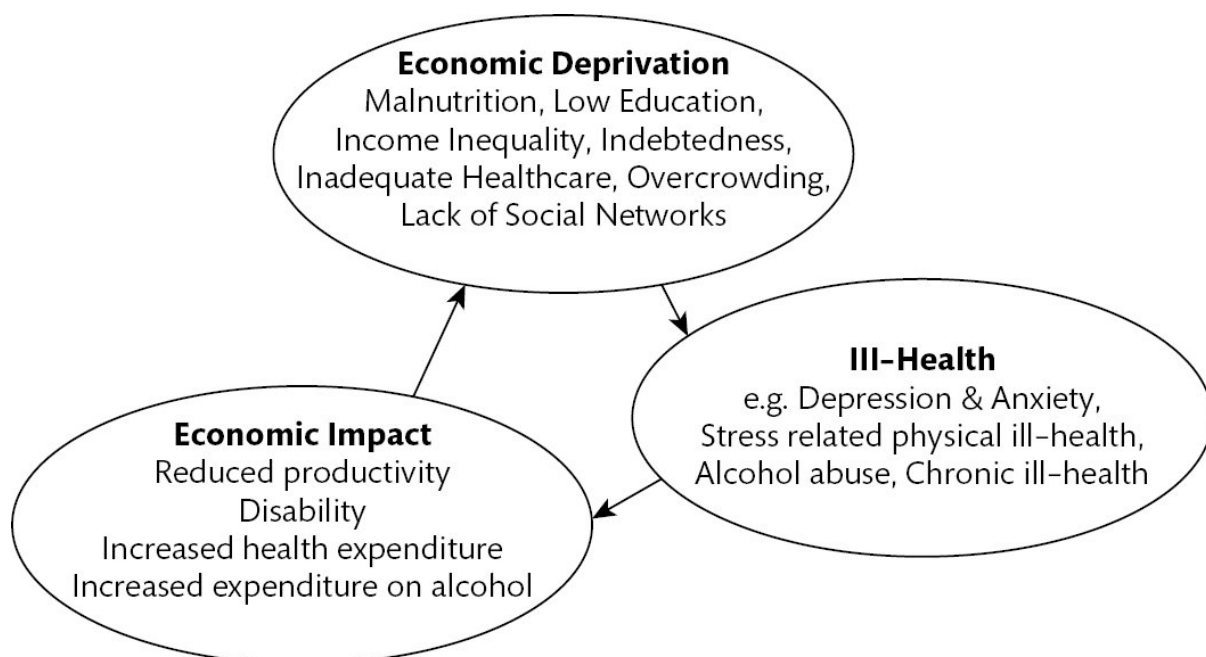
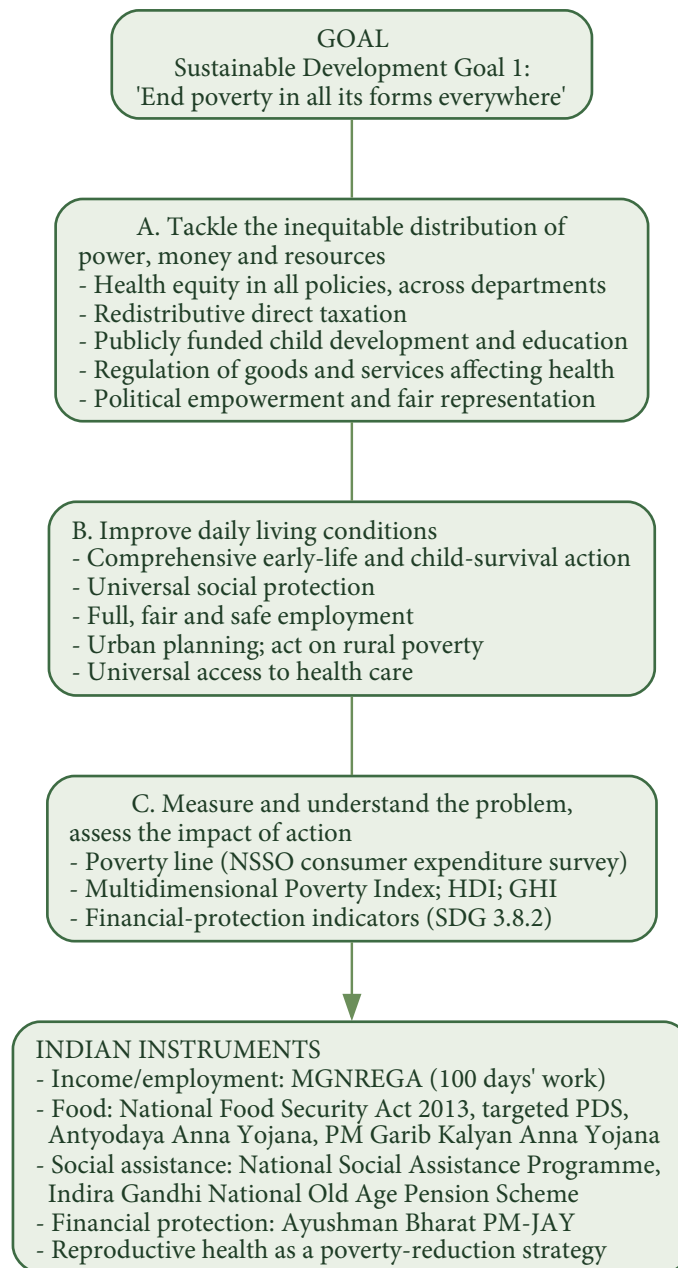


Fig 1.4 — 3 The cycle of poverty and mental illness. From poverty, inequality, a (Oxford public health 7th ed, p.241)

Control — breaking the poverty–disease–poverty cycle



Levels of action against poverty-related ill health, from structural policy down to the household

Level	Measure (with the detail an examiner expects)
Global goal	Sustainable Development Goal 1 — "End poverty in all its forms everywhere." Ending poverty must go hand-in-hand with economic growth, education, health, social protection and a rising basic standard of living; the five Ps of the 2030 Agenda are people, planet, prosperity, peace and partnership (Park 28e, Ch 14, p 852; Sunder Lal, Ch 2, p 85)
Structural / policy	The WHO Commission on Social Determinants of Health (2008) recommends three principles of action — (i) improve daily living conditions at home and at work, for women and men, rural and urban, at all ages; (ii) tackle the inequitable distribution of power, money and resources; (iii) act on knowledge — measure and understand the problem and assess the impact of action (Oxford 7e, Ch 66, pp 2462–2463; Ch 100, pp 3800–3801)

Level	Measure (with the detail an examiner expects)
Employment and income	Mahatma Gandhi National Rural Employment Guarantee Act and Scheme (MGNREGA) — 100 days of employment to able-bodied rural persons (Sunder Lal, Ch 5, p 216)
Food security	Targeted Public Distribution System under the National Food Security Act 2013 — about 4.62 lakh fair price shops reaching roughly 160 million families, 25 kg of rice or wheat per month at 50% of economic cost; Antyodaya Anna Yojana gives 35 kg of foodgrain to the poorest of the poor; Pradhan Mantri Garib Kalyan Anna Yojana gives 5 kg of free foodgrain per person per month; Annapurna Yojana gives 10 kg free to old destitutes without a pension (Sunder Lal, Ch 5, pp 185, 216)
Social security / social assistance	Social security is "security that society furnishes through appropriate organization, against certain risks to which its members are exposed" — the risks covered are sickness, invalidity, maternity, old age and death (Park 28e, Ch 12, p 845). Under the National Social Assistance Programme, the Indira Gandhi National Old Age Pension Scheme (launched 19 November 2007) gives a monthly pension to older persons living below the poverty line, covering about 1.57 crore people, with the central and state governments contributing equal amounts (Park 28e, Ch 9, pp 704–705)
Financial protection in health	Ayushman Bharat – Pradhan Mantri Jan Arogya Yojana (AB PM-JAY), launched 23 September 2018 — cover of Rs 5 lakh per family per year for secondary and tertiary hospitalisation for 10.74 crore poor and vulnerable families (about 50 crore people); entitlement-based on the Socio-Economic and Caste Census database; cashless and paperless at any empanelled public or private hospital, with no family-size limit; the Rashtriya Swasthya Bima Yojana was subsumed into it (Park 28e, Ch 10, p 745; Sunder Lal, Ch 17, pp 763–764)
Health-sector strategy	Investment to reduce health risks among the poor, plus insurance against catastrophic health-care costs, are core elements of a poverty-reduction strategy (Park 28e, Ch 12, p 844). Reproductive health programmes reduce healthy years of life lost, meet unmet need for family planning, relieve the poor of unwanted pregnancies and large families, and promote gender equality — and thereby reduce poverty (Park 28e, Ch 12, p 845)
Education	Female education partly offsets the health effects of poverty independently of health facilities — the Kerala model (Park 28e, Ch 2, p 32)

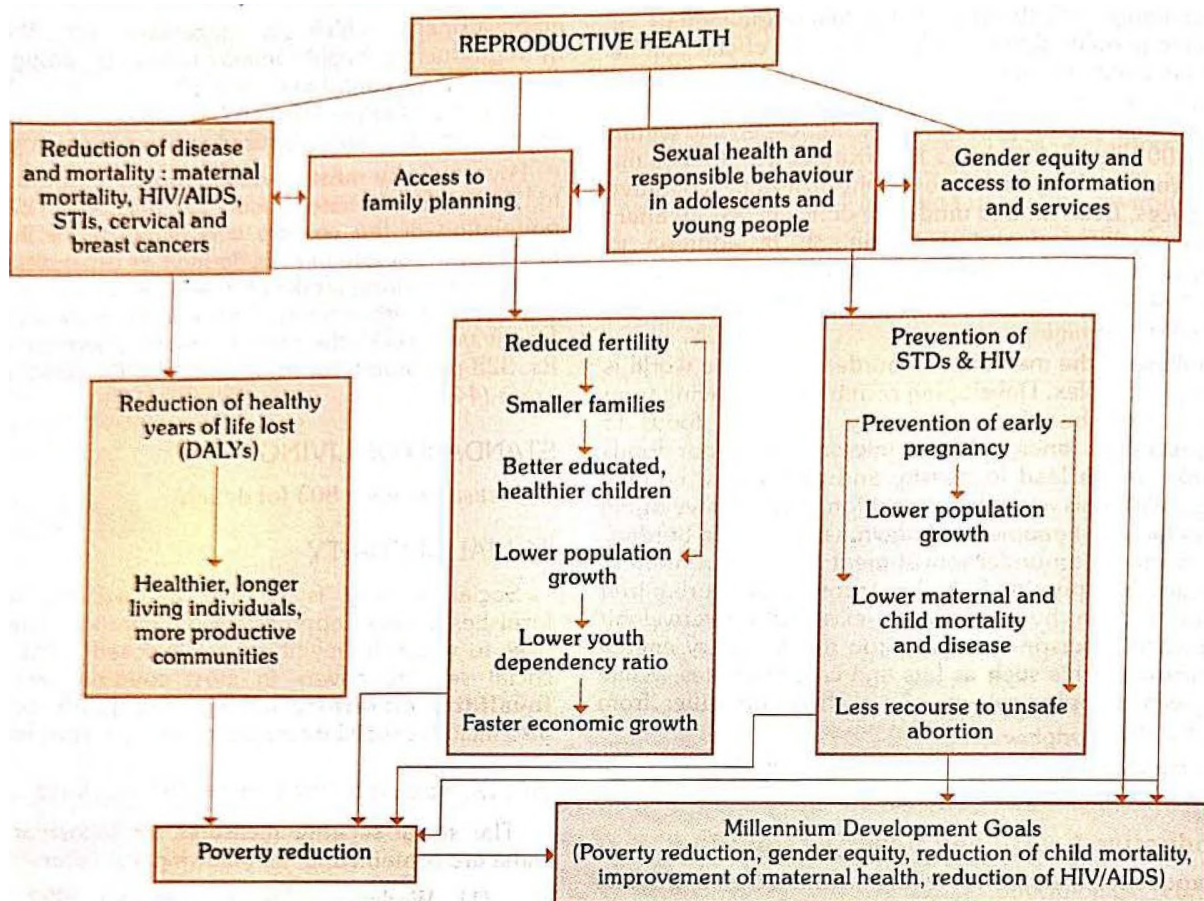


Fig 3 — Reproductive health as a poverty reduction strategy Source : (49) (Park's Textbook of Preventive and Social Medicine, 28e, p.846)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ The poverty–disease–poverty vicious cycle — draw three linked ovals in a clockwise loop and label: *Economic deprivation* (malnutrition, low education, income inequality, indebtedness, inadequate health care, overcrowding, lack of social networks) → *Ill health* (depression and anxiety, stress-related physical ill health, alcohol abuse, chronic ill health) → *Economic impact* (reduced productivity, disability, increased health expenditure, increased expenditure on alcohol) → back to *Economic deprivation*.
- ▶ **Multidimensional Poverty Index tree** — draw one parent box *Multidimensional poverty* with three branches, labelled: *Health* (nutrition; child mortality); *Education* (school attendance; mean years of schooling); *Standard of living* (cooking fuel, sanitation, drinking water, electricity, housing, assets).

High-yield exam pointers

- ▶ **Park's one-line quote to open with** — "Poverty wields its destructive influence at every stage of human life from the moment of conception to the grave" (Park 28e, Ch 12, p 844); and Sunder Lal's aphorism, "people are poor because of sickness and sick because of poverty".
- ▶ **Poverty line, India** — old definition 2,400 kcal rural / 2,100 kcal urban; current Tendulkar definition is a basket of goods, Rs 1,640 rural / Rs 2,500 urban monthly per capita (2014 prices);

data from the NSSO consumer expenditure survey every fifth year; head-count 21.9% (2011–12).

- ▶ **MPI = 3 dimensions, 10 indicators** — health (nutrition, child mortality), education (school attendance, mean years of schooling), standard of living (cooking fuel, sanitation, drinking water, electricity, housing, assets); India 645 → 230 million poor, 415 million lifted out, 2006–2021.
- ▶ **World Bank cut-offs** — extreme poverty < US\$ 1.90/day, moderate < US\$ 3.10/day (PPP).
- ▶ **The financial-protection numbers** — out-of-pocket expenditure 1.52% of GDP, Rs 2,155 per capita/year; ~4.2% (50 million) impoverished annually; catastrophic expenditure = >10% of total or >40% of non-food consumption expenditure; SDG indicator 3.8.2.
- ▶ **CSDH 2008, three actions** — improve daily living conditions; tackle inequitable distribution of power, money and resources; measure, understand and evaluate.
- ▶ **Level of living = 9 components** (health, food, education, occupation and working conditions, housing, social security, clothing, recreation, human rights) — **health is the most important**; HDI = health + education + income.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: the retrieved textbooks stop at the 2011–12 Tendulkar head-count and the 2019–21 multidimensional estimates, so quote each Indian poverty figure with the year given above rather than as a current value; the National Multidimensional Poverty Index reviews published by NITI Aayog after these editions are not carried in the retrieved corpus.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 2 Concept of Health and Disease, pp 28–29, 32, 55; Ch 5 Epidemiology of Communicable Diseases, pp 221–222; Ch 6 Epidemiology of Chronic Non-communicable Diseases, pp 437, 455; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics, pp 652, 704–705; Ch 10 Health Care of the Community, p 745; Ch 12 Medicine and Social Sciences, pp 811, 816, 825, 828–829, 839–840, 843–846; Ch 14 Sustainable Development Goals, pp 852–853); Sunder Lal — Textbook of Community Medicine (Ch 1, p 49; Ch 2 Relationship of Social and Behavioural Sciences to Health and Disease, pp 74, 85–86; Ch 5 Nutrition, pp 185, 216; Ch 16 Health Planning and Management, pp 727–729; Ch 17 Healthcare of the Community, pp 756, 763–764); Oxford Textbook of Public Health 7e (Ch 8 Poverty, justice and health, pp 343, 348–349, 356, 366–367; Ch 9 Socioeconomic inequalities in health in high-income countries, pp 400–401; Ch 10 Reducing health inequalities in developing countries, pp 413–414; Ch 18 Ethical principles and ethical issues in public health, pp 714–715; Ch 66, pp 2462–2465; Ch 100, pp 3800–3801). Named sources: WHO Commission on Social Determinants of Health, 2008; United Nations Development Programme Multidimensional Poverty Index, 2022 report; Socio-Economic and Caste Census 2011; Tendulkar Committee methodology; Global Hunger Index 2024; National Food Security Act 2013; Ayushman Bharat PM-JAY (23 September 2018).



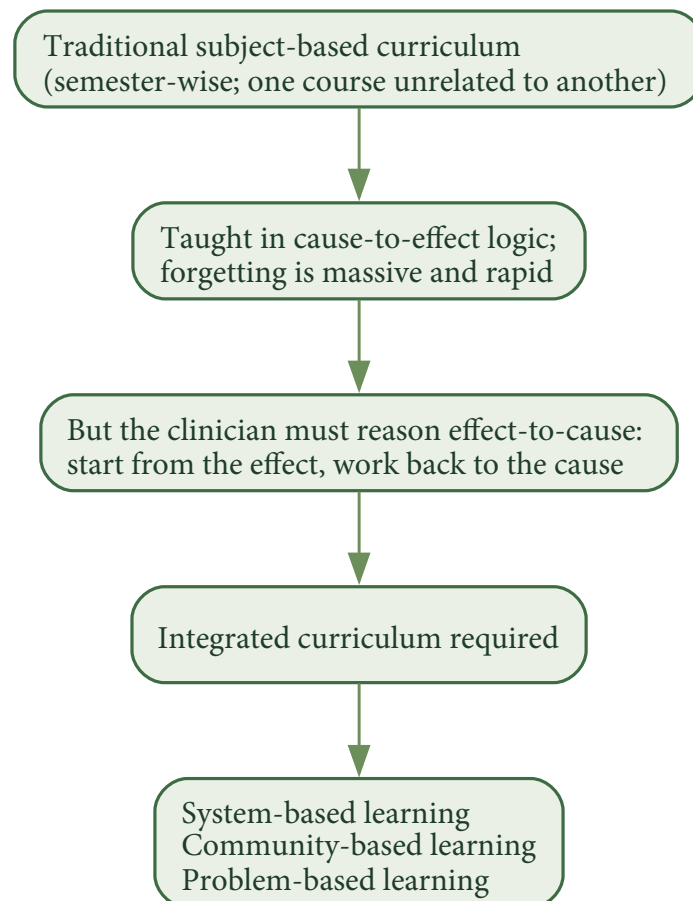
Problem-based learning in medical education

Asked in: Paper I May 2019 — RGUHS MD Community Medicine

Definition — Problem-based learning (PBL) is learning that results from the process of working towards the understanding or evaluation of a community health problem. The problem is encountered **first** and used as a springboard: it stimulates the search for knowledge and the development of **problem-solving skills** needed to understand the mechanisms (root causes) responsible for the problem and its solution. Once the information has been gathered, digested and understood, that knowledge can be used to solve other problems (Sunder Lal, Ch 1, p 34).

Why problem-based learning — the educational rationale

Undergraduate medical education is traditionally subject-based, taught semester-wise, with one course unrelated to the next. It assumes that knowledge gained in the pre-clinical and para-clinical periods will be remembered and will be transferred to illness situations. **Both assumptions are unsupported — forgetting is massive and rapid** (Sunder Lal, Ch 1, p 33). Basic science teachers present material in *cause-to-effect* logic, the logic in which they organise their own knowledge and in which most textbooks are written; but the student, as a clinician, must relearn it in the *effect-to-cause* sequence, because a clinical problem starts with an **effect** of the disease process and the student must work backwards through possible explanations to identify the likely **cause** (Sunder Lal, Ch 1, pp 33–34).



Why an integrated, problem-first curriculum is needed

Feature	Teacher-centred (conventional)	Student-centred (problem-based)
Starting point	The subject/discipline, taught in sequence	A major public health problem, encountered first
Method	Lectures and demonstrations; knowledge imparted	Small tutorial groups; knowledge sought by the learner
Nature of the process	Passive	Active , participatory, self-directed
Role of teacher	Instructor	Facilitator
Direction of reasoning	Cause to effect	Effect to cause (epidemiological reasoning)
Domains addressed	Mainly cognitive	Cognitive, affective and psychomotor together
Outcome sought	Recall of the subject	Self-directed lifelong learning and transferable problem-solving

(Sunder Lal, Ch 1, p 33; domains of learning — Park 28e, Ch 12, p 820)

Key features and the PBL learning climate

Four key features (Sunder Lal, Ch 1, p 34):

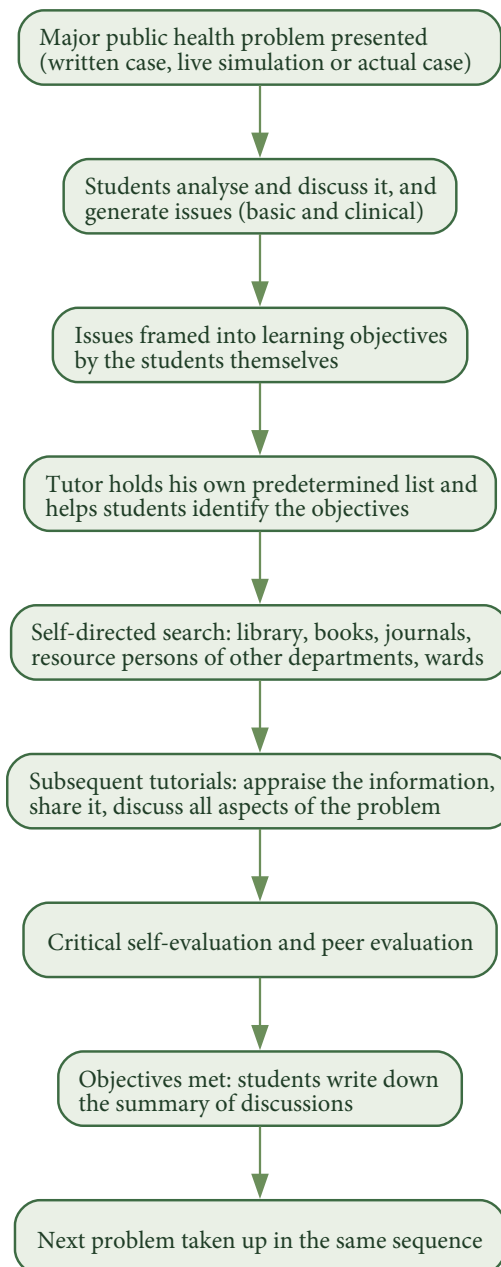
- ▶ **Analysis of health care problems in the community** as the main method of acquiring and applying knowledge.
- ▶ **Self-directed lifelong learning.**
- ▶ **Use of small tutorial groups.**
- ▶ Development of an **epidemiological reasoning process.**

The climate PBL creates — each of these is a marking point:

- ▶ Encourages **active participation** and emphasises the personal nature of learning.
- ▶ **Accepts differences between learners** and recognises the learner's right to commit mistakes; tolerates imperfection.
- ▶ Encourages **openness of mind and trust in self**, and facilitates discovery.
- ▶ Puts emphasis on **self-evaluation** and accepts confrontation of ideas.

The problem, the process and the roles

The problem selected should be a **major public health problem**; the entire subject may be taught on the basis of problems. Problems are presented as **written cases, live simulations or actual cases** (Sunder Lal, Ch 1, p 34).



The problem-based learning cycle, from presentation of the health problem to closure

Actor	Function in the PBL tutorial
Student	Analyses the problem, generates issues, frames own learning objectives, searches for information, shares and appraises it, evaluates own and peers' performance, writes the summary
Tutor (facilitator)	Holds the predetermined list of learning objectives for that problem; helps students arrive at the objectives themselves; does not lecture; certifies with the group that objectives have been met
Resources	Library, books, journals, and resource persons in different disciplines of the college and the wards of the hospital

(Sunder Lal, Ch 1, p 34)

Assessment of the student in problem-based learning

Evaluation is of the **achievement of the programme objectives** and of performance in the PBL exercises, and is deliberately multi-source (Sunder Lal, Ch 1, p 34):

Component	What it assesses
Self-evaluation	The learner's own critical appraisal of performance — built into every tutorial
Peer evaluation	Contribution to the group, sharing of retrieved information
Tutor evaluation	Reasoning, participation, attainment of the agreed objectives
OSCE (objective structured clinical examination)	Clinical competencies, station-wise and objectively scored
OSPE (objective structured practical examination) and special exercises	Practical and psychomotor competencies in competency-based learning

Advantages and limitations

Advantages	Limitations
Knowledge is acquired in the effect-to-cause sequence in which it will actually be used, so transfer to real problems is better	Resource-intensive: needs many small groups running in parallel, and a room and a trained facilitator for each
Counters the massive and rapid forgetting of the subject-based curriculum	Faculty must be retrained as facilitators ; teachers "must reinvent themselves" to practise these competencies (Sunder Lal, Ch 1, p 33)
Builds self-directed lifelong learning and epidemiological reasoning	Difficult to scale to Indian class sizes — over 91,927 MBBS students are admitted every year (Sunder Lal, Ch 2, p 79)
Integrates cognitive, affective and psychomotor domains, and horizontal with vertical content	Systematic coverage of basic sciences may be uneven if problems are poorly chosen
Small-group format permits confrontation of ideas, tolerance of error and self-evaluation	Assessment is labour-intensive and needs OSCE/OSPE infrastructure
The community problem is the "laboratory", so learning is directly relevant to service	Weak without a functioning field practice area and continuous community linkage

Problem-based learning in the Indian medical curriculum

- The Medical Council of India (MCI) 2018 Regulations revised the undergraduate curriculum with structured **horizontal integration** (elements in the same phase) and **vertical integration** (elements from other phases) of teaching and training. **MCI was replaced by the National Medical Commission (NMC) in 2020** (Sunder Lal, Ch 1, p 34).
- The parent document is the **Competency Based Undergraduate Curriculum for the Indian Medical Graduate**; community medicine within it is described as comprising **107 core competencies, of which 18 are skill-based** (Sunder Lal, Ch 20, p 814).

- ▶ **AETCOM (Attitudes, Ethics and Communication)** runs across all four years — **27 modules over 140 hours** — and is taught by a **hybrid PBL** design in which an anchoring lecture is combined with case-based tutorials (Sunder Lal, Ch 1, pp 55-57).
- ▶ Under the NMC curriculum effective **February-March 2022**, MBBS students **adopt a village** (Family Adoption Programme) and study yoga; the family is allotted in the **first professional year** and followed **longitudinally for four years** until the summative assessment — the standing community-based problem on which PBL in community medicine rests (Sunder Lal, Ch 2, p 79).
- ▶ **Clinico-social case review (CSCR)** is the allied method: a clinical review (history, examination, diagnosis, treatment, counselling, follow-up) joined to a social review, used to learn the natural history of disease and the levels of prevention (Sunder Lal, Ch 1, p 34).

Globally, PBL is one instrument of the shift away from purely science-based curricula proposed by the **Global Commission on Health Professionals Education (2010)**, launched on the 100th anniversary of the *Flexner* Report, and endorsed by **World Health Assembly resolution WHA 66.23 (2013)** on transforming health workforce education in support of universal health coverage (Oxford Textbook of Public Health 7e, Ch 103, pp 3921-3922).

Level of learning	Aim	Produces
Informative	Fact memorisation and skill building	<i>Experts</i>
Formative	Critical reasoning and systematic thinking — where PBL principally acts	<i>Professionals</i>
Transformative	Analysis, synthesis and enlightenment	<i>Leaders for change</i>

(Oxford Textbook of Public Health 7e, Ch 103, pp 3927-3928)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Reproduce the **hybrid PBL schedule** as a three-week grid with these exact labels:

- ▶ **Week 1** — "Introduction of group forming"; "Introduction of case 1 — development of learning objectives, assignments for team members"
- ▶ Arrow from case 1 to case 2 labelled "**Self-directed learning**"
- ▶ **Week 2** — "Anchoring lecture or other learning activity"
- ▶ **Week 3** — "Discussion and closure of case 1"; "Introduction of case 2 — development of learning objectives, assignments for team members"

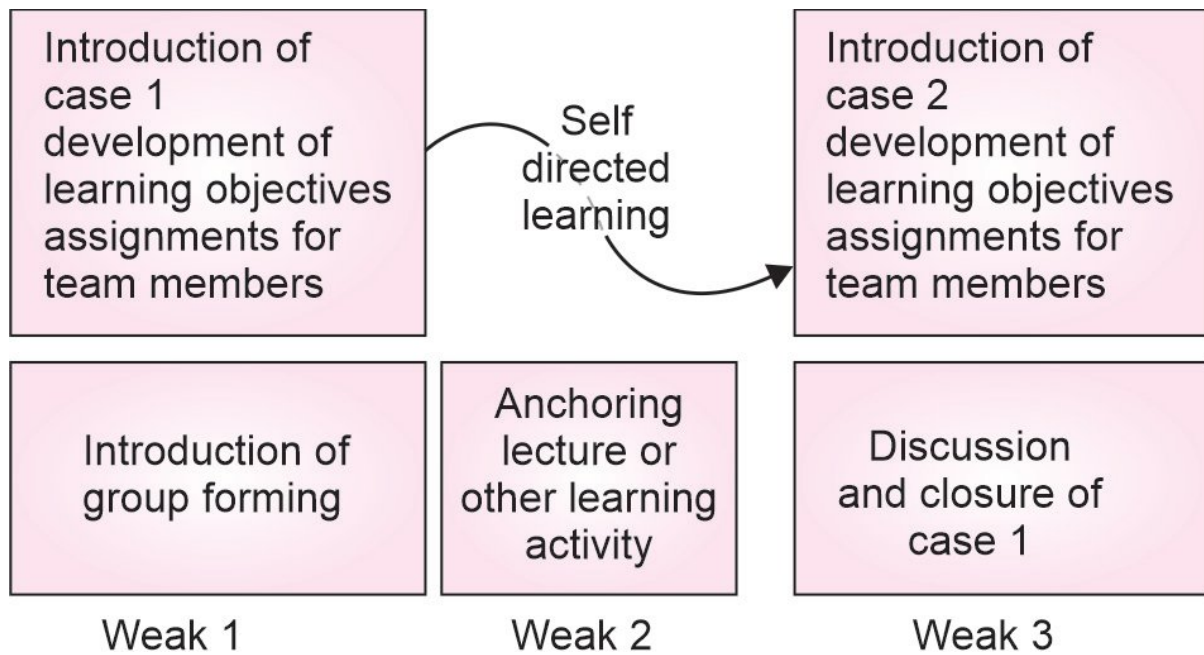


Fig 1.5 — Hybrid problem based learning suggested for AETCOM cases (Sunderlal, p.56)

High-yield exam pointers

- ▶ **Definition to write verbatim:** learning that results from the process of working towards the understanding or evaluation of a community health problem; the problem is the **springboard**, encountered first.
- ▶ **Four key features (mnemonic "CoSSE")** — Community health problem analysis, Self-directed lifelong learning, Small tutorial groups, Epidemiological reasoning.
- ▶ **Cause-to-effect vs effect-to-cause** — the one line that earns the rationale mark; add "forgetting is massive and rapid".
- ▶ **Three integrating techniques** of the integrated curriculum: system-based, community-based, problem-based learning.
- ▶ **Assessment quartet:** self + peer + tutor evaluation, plus OSCE/OSPE.
- ▶ **Indian dates:** MCI competency-based curriculum 2018; MCI replaced by NMC in 2020; Family Adoption Programme and yoga from February-March 2022; AETCOM 27 modules / 140 hours / 4 years.
- ▶ **Three domains:** cognitive (knowledge), affective (attitudes), psychomotor (skills).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: the Medical Council of India (MCI) was restructured as the National Medical Commission (NMC) in 2020, so quote the 2018 competency-based curriculum as an MCI document but name NMC as the current regulator; the retrieved textbooks describe the strengths and the learning climate of problem-based learning but do not enumerate its disadvantages, so the "Limitations" column above states the standard, universally accepted operational constraints (faculty training, small-group logistics, class size) rather than textbook-quoted text, with only the class-size and faculty-retraining points carrying a source.

SOURCE & COVERAGE

Sunder Lal — Textbook of Community Medicine (Ch 1, pp 33-34, 55-57; Ch 2, pp 78-79; Ch 20, p 814); Park's Textbook of Preventive and Social Medicine 28e (Ch 12, pp 819-820; Ch 22, pp 1013, 1021-1022); Oxford Textbook of Public Health 7e (Ch 103, pp 3914, 3921-3922, 3927-3928); Medical Council of India — Competency Based Undergraduate Curriculum for the Indian Medical Graduate, 2018; World Health Assembly resolution WHA 66.23, 2013.

Coverage note: the indexed community medicine corpus carries problem-based learning as a section within Sunder Lal Ch 1 and as curriculum-reform material in Oxford Ch 103; there is no dedicated medical-education textbook in the corpus, so the answer is confined to what these sources state.

♦ ♦ ♦

Role of leaders in information, education and communication

Asked in: Paper I 15-Nov-2022 — RGUHS MD Community Medicine

Definition — Information, education and communication (IEC) is the planned use of communication to supply people with correct health information, to help them acquire knowledge, attitudes and skills, and to carry them from awareness through motivation to sustained health action. A **leader**, in this setting, is any person — formal or informal — whom a community respects and follows, and who can therefore be used as an **agent of change**: psychologists have established that we learn best from people whom we respect and regard, so health education tries to penetrate the community through the local leaders, such as the village headman, the school teacher or the political worker (Park 28e, Ch 22, p 1019). The operating rule is that **if the leaders are convinced first about a given programme, the rest of the task of implementing the programme becomes easy** (Park 28e, Ch 22, p 1019).

Typology of leaders usable in IEC

Category	Who they are	Particular IEC strength
Elected / political	Village headman (sarpanch), Panchayati Raj Institution (PRI) members, ward councillor, political worker	Command over village decisions, untied funds and convergence with other departments
Religious and traditional	Priest, imam, temple/mosque functionary, traditional healer, traditional birth attendant (dai)	Reach resistant families; mosque/temple announcements; legitimise a message against a cultural objection
Educational	School teacher, headmaster, National Service Scheme (NSS) / National Cadet Corps (NCC) officers	A school teacher is a leader to his class; children carry messages home (Sunder Lal, Ch 16, p 712)
Grassroots health leaders	Accredited social health activist (ASHA), Anganwadi worker (AWW), auxiliary nurse midwife (ANM), village health guide	Resident, same language and culture, daily face-to-face contact
Organised community groups	Village Health Sanitation and Nutrition Committee (VHSNC), Mahila Mandal, Mahila Swasthya Sangh, Mahila Arogya Samiti, self-help groups, youth clubs, Rogi Kalyan Samiti	Ready-made platform for group education and community monitoring
Peer / natural opinion leaders	Satisfied acceptors, cured patients, respected elders	Motivation is contagious — men who have already had a vasectomy are among the best advertisements for male sterilisation (Park 28e, Ch 22, p 1019)
Professional and voluntary	Private practitioners, Indian Medical Association, Rotary International, non-governmental organisations	Advocacy, credibility with the better-off, resources for campaigns (Mission Indradhanush Guidelines, pp 56–57)

Attributes of an effective leader

Park lists the qualities that make a leader usable in health education work (Park 28e, Ch 22, pp 1019–1020):

- Understands the **needs and demands** of the community.

- ▶ Provides **proper guidance** and **takes the initiative**.
- ▶ Is **receptive** to the views and suggestions of the people.
- ▶ **Identifies himself with the community** (is one of them, not above them).
- ▶ Is **selfless, honest, impartial, considerate and sincere**.
- ▶ Is **easily accessible** to the people.
- ▶ Is able to **control and compromise the various factions** in the community.
- ▶ Possesses the **skill and knowledge to elicit cooperation** and to achieve **coordination of the various official and non-official organisations**.

Behavioural basis — why working through leaders works

Mechanism	What it states	Practical consequence for IEC
Credibility (principle 1 of health education)	The degree to which the message is perceived as trustworthy by the receiver; unless people trust the communicator no desired action follows (Park 28e, Ch 22, pp 1018–1019)	The leader lends the health worker a credibility an outsider cannot generate
Learning from the respected	We learn best from people whom we respect and regard (Park 28e, Ch 22, p 1019)	Message routed through the leader is retained and acted upon
Two-step flow of communication	A mass-media message diffuses into the community, is picked up by interpersonal and informal networks, is debated and discussed, and on this scrutiny a consensus is gradually built up whether to accept or reject it (Park 28e, Ch 21, p 1011)	Leaders dominate that second step; mass media alone are generally inadequate to change human behaviour (Park 28e, Ch 22, p 1023)
Diffusion of innovation (Rogers)	Innovations are adopted through natural social networks, aided by the community's opinion leaders, because of their perceived trustworthiness (Oxford Textbook of Public Health 7e, Ch 33, p 1228)	Deliberately recruit opinion leaders rather than hope for spontaneous diffusion
Popular Opinion Leader model (Kelly)	Uses the natural leaders found in any social group as agents of change; demonstrated to change behaviour in high-risk groups (Oxford 7e, Ch 1, p 171)	Peer leaders for stigmatised behaviours (HIV, substance use, adolescent health)
Motivation is contagious	One motivated person may spread motivation throughout a group (Park 28e, Ch 22, p 1019)	A single convinced leader converts a whole hamlet
Setting an example (principle 9)	The health educator must practise what he teaches (Park 28e, Ch 22, p 1019)	The leader's own adoption is itself the strongest message
Participation (principle 3)	Alma-Ata: the people have a right and duty to participate individually and collectively in the planning and implementation of their health care; without community participation health programmes are unlikely to succeed (Park 28e, Ch 22, p 1019)	Leaders are the practical vehicle through which participation is organised

Role of leaders across the three components

Component	Functions performed by the leader	Illustration
Information	Legitimises and relays correct information; announces at public gatherings; identifies and corrects rumour and misconception; acts as the addressee of programme material — direct mailing is sent to village leaders, literate persons, panchayats and local bodies considered opinion leaders (Park 28e, Ch 22, p 1023)	Mosque/temple announcements and local miking during immunisation drives (Mission Indradhanush Guidelines, p 99)
Education	Convenes and hosts group education (mothers' meetings, Mahila Mandal meetings, parent–teacher meetings, community and influencer meetings); helps identify felt needs ; translates the message into the local idiom and selects the appropriate folk medium; teaches by demonstration and example	School teacher orientation and parent–teacher meetings built into the campaign communication plan (Mission Indradhanush Guidelines, p 99)
Communication	Converts a one-way message into two-way (Socratic) communication — answers questions, lets fears be expressed, and carries feedback back to the health system; mobilises households to the session site; identifies left-outs, dropouts and resistant families	Social mobilisation brings stakeholders and influencers together to inform, answer questions, clear misconceptions and instil trust (Mission Indradhanush Guidelines, pp 33–34)

Their role also runs through the whole programme cycle: (1) **needs assessment** — the health team learns from the community by talking with community leaders about child-rearing and feeding practices, attitudes, beliefs and customs (Sunder Lal, Ch 16, p 715); (2) **planning** — leaders sit on the village health action plan; (3) **resource mobilisation** — elected representatives, Mahila Swasthya Sangh, Mahila Mandals and voluntary organisations count first among village-level resources (Sunder Lal, Ch 16, p 704); (4) **implementation and service delivery**; (5) **community-based monitoring** — the community monitors government-run programmes and gives feedback; and (6) **evaluation** — the community gives its report card for corrective action (Sunder Lal, Ch 2, pp 68–69).

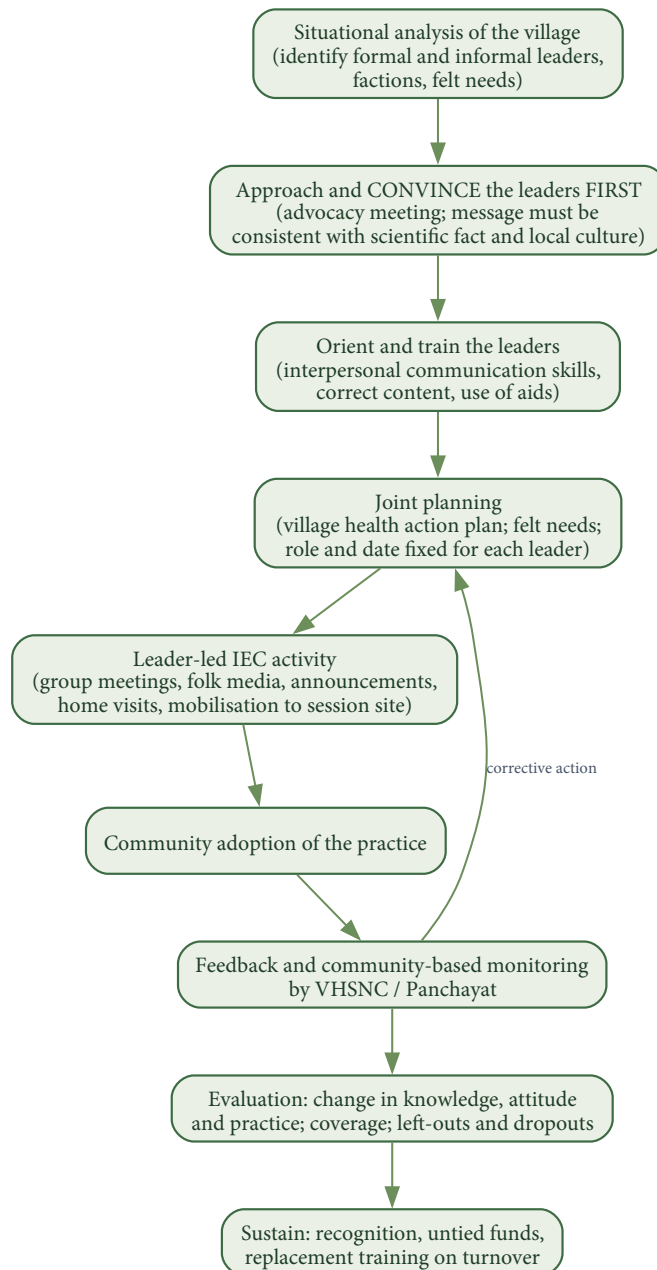
Leaders in Indian community structures and national programmes

Structure / programme	Leadership role built into it
VHSNC	A standing sub-committee of the Panchayat, with PRI members, ANM, AWW, ASHA, teacher and non-governmental organisations, with ASHA as member secretary; prepares and implements the village health action plan with an untied fund of Rs 10,000 per annum; monitors AWW, ASHA and subcentre services; over 5.55 lakh set up by 2021 (Sunder Lal, Ch 17, p 744; Park 28e, Ch 7, p 539)
ASHA and AWW	Resident village-level leaders; ASHA one per 1000 population, over 10.72 lakh working; AWW guides ASHA in organising the monthly health day (Sunder Lal, Ch 17, p 744)
Village Health, Nutrition and Sanitation Day (VHSND)	ASHA with AWW, ANM, self-help groups, community-based organisations and PRIs holds it monthly at the Anganwadi centre for reproductive, maternal, newborn, child and adolescent health services (Sunder Lal, Ch 5, pp 229–230)

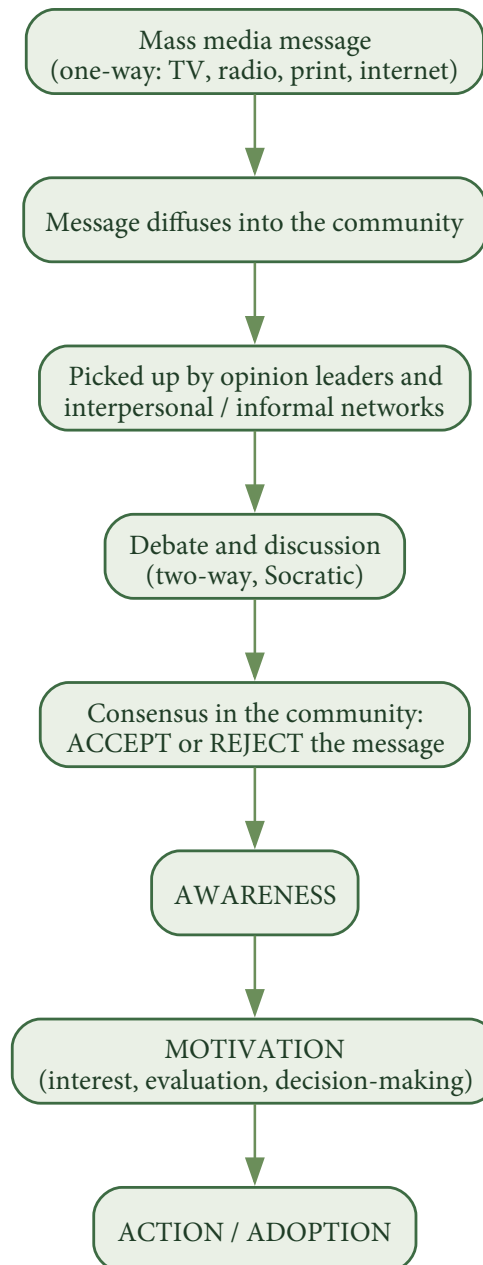
Structure / programme	Leadership role built into it
Health worker (male/female) job description	Identify male community leaders in each village and assist the supervisor in training them; maintain rapport with acceptors, village leaders, ASHA and dais; identify women leaders and help train them (Sunder Lal, Ch 16, pp 718–719; Park 28e, Ch 10, pp 739–740)
Pulse Polio / Intensified Mission Indradhanush	Social mobilisation intensified by involving local influencers, community leaders and religious leaders to improve participation and vaccine acceptance (Park 28e, Ch 5, p 264); block-level meetings with civil-society and religious leaders, interpersonal-communication training of frontline workers and mobilisers, and community meetings with VHSNC, self-help groups and Mahila Mandals (Mission Indradhanush Guidelines, pp 34, 99)
National Population Policy IEC strategy	Coverage and outreach of IEC activities are to be extended by using elected leaders, opinion makers and religious leaders , with convergence across rural development, social welfare, transport, cooperatives and education (Sunder Lal, Ch 9, pp 556–557)
ICDS	PRIs involved in selection of Anganwadi workers and in supervision, monitoring, location, construction and maintenance of centres; VHSNC involved in village-level management and supervision (Sunder Lal, Ch 5, pp 229–230)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Position of the leader in the health communication process** — draw a horizontal chain and label exactly: *Communicator (medical officer / health worker) → Message → Channel (mass media, mid-media, interpersonal) → Local leader (opinion leader / change agent) → Interpersonal and informal networks of the community → Receiver (individual, family, community)*, with a **Feedback** arrow curving back from receiver to communicator through the leader, and a side box labelled **Barriers to communication** — **physiological, psychological, environmental, cultural** (Park 28e, Ch 21, p 1012).



Stepwise engagement of community leaders in an IEC programme



Two-step flow of health communication and the adoption model

Constraints in using leaders, and safeguards

Constraint	Safeguard
Elite capture — the leader may not represent the poorest, women or marginalised castes	Deliberately co-opt women's group leaders, self-help group and adolescent peer leaders; VHSNC reflects the aspirations of poor households (Park 28e, Ch 7, p 539)
Factionalism in the village splitting the message	Choose a leader who can control and compromise the various factions (a listed attribute)
Political turnover ends the engagement	Institutionalise leadership in a body (VHSNC as a Panchayat sub-committee) rather than in an individual

Constraint	Safeguard
The leader's own misconception or resistance	Convince and train the leader before any community activity; base content on fact, consistent with scientific knowledge and local culture
Overload and no incentive for voluntary leaders	Recognition, untied funds, defined and limited tasks, refreshments/mementoes for mobilisers (Mission Indradhanush Guidelines, pp 56–57)
Barriers to communication persisting despite a leader	Address the physiological, psychological, environmental and cultural barriers directly; avoid words the public cannot understand (Park 28e, Ch 21, p 1012; Ch 22, p 1019)

Evaluation of leader-based IEC is by the change in knowledge, attitude and practice (KAP) produced, assessed by workers during home visits and contacts and by exit interviews (Sunder Lal, Ch 4, p 153), together with programme output measures such as session attendance and reduction in left-outs, dropouts and resistant families.

High-yield exam pointers

- ▶ **Park's 12 principles of health education** — Credibility, Interest, Participation, Motivation, Comprehension, Reinforcement, Learning by doing, Known to unknown, Setting an example, Good human relations, Feedback, **Leaders**. Leaders is **principle 12**; credibility is **principle 1**.
- ▶ One-line quotable: *"Leaders are agents of change... if the leaders are convinced first about a given programme, the rest of the task of implementing the programme will be easy"* (Park 28e, Ch 22, p 1019).
- ▶ **Eight attributes of a leader** — write them as a list; the examiner looks for "identifies himself with the community" and "able to control and compromise the various factions".
- ▶ **Two-step flow**: mass media → interpersonal/informal networks and opinion leaders → community consensus. Mass media alone are inadequate to change behaviour.
- ▶ **Adoption model** (Park): Awareness → Motivation (interest, evaluation, decision-making) → Action/Adoption. Sunder Lal's five stages: awareness, interest, evaluation, trial, adoption.
- ▶ **Named models**: Rogers' diffusion of innovation; Kelly's Popular Opinion Leader model.
- ▶ **Indian numbers to quote with year**: VHSNC — over 5.55 lakh set up by 2021, untied fund Rs 10,000 per annum; ASHA — one per 1000 population, over 10.72 lakh working.
- ▶ Alma-Ata (1978): *"the people have a right and duty to participate individually and collectively in the planning and implementation of their health care."*

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only — the corpus describes leadership *styles* (concern for people versus concern for the task; four situational styles matched to follower development levels D1–D4, Sunder Lal, Ch 16, p 712); this belongs to health management and is not required for an IEC answer unless the stem asks for leadership styles. Coverage note: of Park's twelve principles of health education, eleven were confirmed verbatim in the retrieved text, and only principle 10 (good human relations) is quoted from standard Park numbering without a retrieved passage.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 21 — Health Information and Basic Medical Statistics; Ch 22 — Communication for Health Education; Ch 5; Ch 7; Ch 10); Sunder Lal — Textbook of Community Medicine (Ch 2, Ch 4, Ch 5, Ch 9, Ch 16, Ch 17); Oxford Textbook of Public Health 7e (Ch 1 — scope and concerns of public health; Ch 33 — community intervention trials); Intensified Mission Indradhanush operational guidelines (Ministry of Health and Family Welfare); Alma-Ata Declaration 1978; National Population Policy 2000 IEC strategy as summarised in Sunder Lal.

**Socio-cultural factors in health and disease — social control**

Asked in: Paper I 12-Nov-2019 — RGUHS MD Community Medicine

Definition — Culture is "learned behaviour which has been socially acquired", transmitted from one generation to the next by formal and informal learning, and it lays down the norms of behaviour of a society (Park 28e, Ch 12, p 815). **Socio-cultural factors in health** are therefore the customs, habits, beliefs, religion, caste, social class and family patterns through which a community decides what causes disease, what is eaten, how a child is reared and when care is sought. **Social control mechanisms** are the rules of a society, formal and informal, for the maintenance of relationships of authority and subordination; they work largely through **reward and punishment** (Park 28e, Ch 12, p 814).

Sociological concepts underlying health behaviour

Term	Meaning (as defined in the text)	Relevance to health practice
Society	A group of individuals who have organised themselves and follow a given way of life; it controls and regulates individual behaviour both by law and by custom (Park 28e, Ch 12, pp 813-814)	The unit on which community diagnosis is made
Socialisation	The process by which an individual gradually acquires culture and becomes a member of a social group (Park 28e, Ch 12, p 814)	Health behaviour is learnt in childhood; school health is the entry point

Term	Meaning (as defined in the text)	Relevance to health practice
Convention	The practice promoted for the convenience of society or the individual; the starting point of all customs (Park 28e, Ch 12, pp 814-815)	Explains why a habit persists without any reason being offered
Customs → folkways	The "right ways" of doing things in the less vital areas of conduct — greeting, dress, cooking (Park 28e, Ch 12, p 814)	Loosely enforced; easiest to modify by health education
Customs → mores	The more stringent customs, in whose enforcement the public takes an active part (Park 28e, Ch 12, p 814)	Resist education alone; need leaders and time
Laws	Written rules defining right and wrong; laws are generally customs-inspired, i.e. formalised mores (Park 28e, Ch 12, p 814; Sociology: The Essentials 6e, Ch 2, p 60)	The formal arm of social control
Taboo	The strictest norm; its violation brings the most serious sanction (Sociology: The Essentials 6e, Ch 2, p 60)	Food taboos in pregnancy, menstrual seclusion
Ritual	"A set or series of acts, usually involving religion or magic, with the sequence established by tradition" (Park 28e, Ch 12, p 830)	Ceremonial baths, naming ceremony — fixed points to attach services to
Social sanction	The mechanism of social control that enforces a norm; positive (praise, approval, reward) or negative (ridicule, imprisonment, physical coercion) (Sociology: The Essentials 6e, Ch 2, pp 60-61)	The lever every behaviour-change programme actually pulls
Acculturation	"Culture contact" — two-way diffusion of culture through (1) trade and commerce, (2) industrialisation, (3) propagation of religion, (4) education, (5) conquest (Park 28e, Ch 12, p 815)	The route by which scientific medicine itself entered India

Cultural factors in health and disease

Cultural factors are deeply involved in personal hygiene, nutrition, immunisation, seeking early medical care, family planning, child rearing and disposal of refuse and excreta — in short, the whole way of life. Not all customs are bad: some rest on centuries of trial and error and have positive value, while others are useless or positively harmful (Park 28e, Ch 12, pp 815, 829).

Cultural area	Belief / practice prevailing in India	Health consequence
1. Aetiology — supernatural causes	Wrath of gods and goddesses (chickenpox as <i>Chhoti Mata</i>); breach of taboo (venereal disease attributed to illicit contact); past sins (leprosy, tuberculosis); "evil eye", especially in children; spirit or ghost intrusion (hysteria, epilepsy) (Park 28e, Ch 12, p 829)	Cases are not notified, <i>pujas</i> are performed to appease the gods, drugs are considered harmful; the exorcist is consulted before the doctor — delay in diagnosis and treatment

Cultural area	Belief / practice prevailing in India	Health consequence
2. Aetiology — physical causes	Summer heat causing <i>loo</i> (heat stroke), treated with oil and <i>ghee</i> on the soles and <i>mango-phool</i> ; impure water linked to disease; boils and scabies ascribed to "impure blood", treated by eating neem (Park 28e, Ch 12, p 829)	Folk remedy replaces cooling, rehydration and specific treatment
3. Environment and housing	Rural houses <i>katcha</i> , damp, ill-lighted and ill-ventilated; no window or only a small hole; no separate kitchen, latrine, bathroom or drainage; humans and animals infrequently under one roof; interiors kept clean and plastered with mud and cowdung (Park 28e, Ch 12, pp 829-830)	Overcrowding, indoor smoke, zoonoses, vector breeding; open defaecation persists despite facilities being provided
4. Food habits	Vegetarianism honoured in Hindu society; Muslims abhor pork, Hindus beef; hot foods (meat, fish, eggs, jaggery) versus cold foods (curd, milk, vegetables, lemon); fasts in Ramzan and on Hindu occasions; alcohol tabooed among Muslims and high-caste Hindus; men eat first and women last and poorly; milk adulterated partly on the belief that boiling pure milk dries up the donor animal's secretion (Park 28e, Ch 12, p 830)	Food taboos keep nutritious foods away even when available; intra-household discrimination sustains maternal undernutrition and anaemia
5. Mother and child health	Marriage universal; family "incomplete without the birth of a male child"; foods forbidden in pregnancy; delivery by the untrained <i>dai</i> (Park 28e, Ch 12, p 830)	Adverse male : female ratio and female foeticide, high fertility, unsafe delivery, neonatal sepsis and death
6. Personal hygiene	Oral hygiene by neem twigs, ash or charcoal; <i>pan</i> leaf with lime, with or without tobacco; regular and ceremonial baths, oil bath and massage; unsterilised instruments used by the village barber; <i>purdah</i> ; sleeping on the ground; going barefoot; circumcision with religious sanction (Park 28e, Ch 12, p 830)	<i>Pan</i> chewing → oral cancer (Park 28e, Ch 12, p 815); <i>purdah</i> → higher reported tuberculosis and loss of the benefit of sunlight; sleeping on the ground → insect bites; bare feet → hookworm
7. Habits — tobacco and alcohol	Smoking is a socially acceptable practice and a status symbol in family and peer groups (Sunder Lal, Ch 2, pp 81-82); hubble-bubble smoking is a social custom and can spread tuberculosis; the reverse cigar (burning end inside the mouth) in Andhra Pradesh (Park 28e, Ch 12, p 830)	Cancer of lung, mouth, pharynx, larynx, oesophagus and bladder; chronic bronchitis and emphysema; coronary occlusion and angina; pulmonary tuberculosis; fetal growth retardation if the mother smokes (Park 28e, Ch 12, p 830). Cirrhosis of the liver from drinking is the parallel example (Park 28e, Ch 12, p 815)
8. Sex and marriage	Sexual customs vary by religious and ethnic group; menstruation regarded as a time of uncleanness; early marriage held desirable; marriage near-universal. Mean age at marriage in India — 24 years for boys and 21 years for girls , lower among the depressed classes (Park 28e, Ch 12, pp 830-831)	Direct bearing on family planning acceptance and on early childbearing; but also fewer unmarried mothers and illegitimate births than in western countries

Social factors in health and disease

Factor	How it acts	Documented health effect
Caste	Rules and sanctions on endogamy, food taboos, ritual purity, a fixed occupation and social contact; supported by religion; a "closed class" with little social mobility, less rigid in urban and industrial areas (Park 28e, Ch 12, pp 825, 831)	Restricts service utilisation and inter-group contact; consanguinity where cousin marriage is customary; reform meets resistance in a closed-class system
Social class	Bound up with economic status, education, way of life, attitudes and exposure to stress; measured in India by Kuppuswamy's socio-economic status scale (education + occupation + income; total score 26-29 = class I upper, 16-25 II upper-middle, 11-15 III lower-middle, 5-10 IV upper-lower, <5 V lower) (Park 28e, Ch 12, Table 2, p 833)	Upper classes have longer life expectancy, lower mortality and better nutritional status ; coronary heart disease, hypertension and diabetes highest in class I; skin, eye and ear disease, diarrhoea and dysentery higher in lower classes; infant, general and maternal mortality all track social class (Park 28e, Ch 12, p 834)
Wealth index (the survey alternative)	Household assets and housing characteristics reduced by principal-components analysis to five quintiles, used by the National Family Health Survey (NFHS) instead of income (Park 28e, Ch 12, p 833)	NFHS-5 (2019-21): 45.5% of the urban population is in the highest wealth quintile against only 8.1% of the rural population (Park 28e, Ch 12, Table 3, p 833)
Education	The ignorant and the illiterate have much difficulty pursuing measures conducive to good health (Park 28e, Ch 12, p 834)	Illiteracy heads Park's list of cultural barriers to health communication
Occupation	Widely used by social scientists to fix social standing because it determines economic reward, authority, obligations, status, values and life-style; classified internationally by ISCO-08 into 10 major groups; India has no satisfactory occupational class system (Park 28e, Ch 12, pp 831-832)	Occupation-specific risk; but each class is a heterogeneous grouping, people change occupation, and women are hard to classify — every measure of social differentiation has a "snag" and must be used critically
Religion	Bears on age-old habits of cleanliness, eating, clothing and child care (Park 28e, Ch 12, p 832)	Infant mortality in India examined by religion was lowest in Christians, high in Hindus and highest in Muslims
Family	<i>Family of origin</i> (into which one is born) versus <i>family of procreation</i> (which one sets up); it is a biological, social, cultural and epidemiological unit (Park 28e, Ch 12, p 825)	The family is the unit of care; broken and problem families breed psychopathy, delinquency and vagrancy (Park 28e, Ch 12, p 829)
Rural versus urban	Differences in life-style, opinions and values (Park 28e, Ch 12, p 832)	Different health and sickness profiles and different service-utilisation patterns

The six factors through which social class produces differences in health and disease (Park 28e, Ch 12, p 834)

#	Factor	Mechanism
1	Physical environment	Housing, water and access to clean air are better in the upper social hierarchy
2	Differences in services provided	Some areas are relatively "undoctored", and a substantial proportion of people live in them
3	Material resources	Income, wealth and possession of tools that can promote or achieve better health
4	Genetic endowment	People tend to marry within the same social class, so liability to disease differs
5	Educational status	The ignorant and the illiterate cannot pursue measures conducive to good health
6	Attitude to disease	Illness seen as punishment versus natural cause; self-diagnosis versus early medical aid

Lower classes also use hospitals less, consult doctors less often, and take antenatal and postnatal care, check-ups and immunisation less. **The aim of preventive medicine should be to reduce the social class differences in health and disease** (Park 28e, Ch 12, p 834).

Wealth quintile	Neonatal mortality rate	Infant mortality rate	Under-5 mortality rate
Lowest	33.8	48.0	59.0
Middle	23.2	33.7	39.1
Highest	12.2	17.0	20.1

Early childhood mortality per 1000 by wealth index, NFHS-5, India 2019-21 (Park 28e, Ch 12, Table 4, p 834).

Customs classified for programme action

Park classifies customs in the field of maternal and child health as **good, bad, unimportant and uncertain**, and gives worked examples for three of the four classes (Park 28e, Ch 12, p 830).

Class	Examples given in the text	Action by the health worker
Good	Prolonged breast-feeding, oil bath, massage, exposure to sun	Promote and build the programme message on them
Bad	Foods forbidden in pregnancy (eggs, meat, fish, milk, leafy vegetables); delivery by the untrained <i>dai</i> ; branding of the skin; administration of opium and drastic purgatives	Modify by education, leaders and, where justified, legislation
Unimportant	Named as a class; no examples enumerated in the text	Leave alone — do not waste programme credibility

Class	Examples given in the text	Action by the health worker
Uncertain	Application of <i>kajal</i> (black soot in oil) to the eyelids, partly cosmetic and partly to ward off the evil eye; often blamed for transmitting trachoma and other eye infections	Observe and study before pronouncing

Social control mechanisms prevailing in our country

	Formal — organised, written, legal sanction	Informal — unwritten, moral sanction
Agencies named in the text	The laws and enactments of Parliament; the various central, state and local Acts that help maintain the standards of health; sets of formal rules and regulations even in small organisations (Park 28e, Ch 12, p 814)	Powerful groups, individuals or friends who bring informal social pressure to bear on individuals and help construct "norms" of behaviour (Park 28e, Ch 12, p 814)
Institutions through which they act	The state, law and the courts (Sociology: The Essentials 6e, Ch 4, p 101)	The family, the school, the church, the club, the hospital, political parties, professional associations and the <i>panchayats</i> — all listed by Park as social institutions within which rights and duties are defined (Park 28e, Ch 12, p 814); caste, governed by its own rules and sanctions on endogamy, food and ritual purity (Park 28e, Ch 12, p 825); the peer group, through ridicule (Sociology: The Essentials 6e, Ch 4, p 101)
Sanction	Fine, imprisonment, physical coercion (Sociology: The Essentials 6e, Ch 2, p 61)	Positive — praise, approval, reward; negative — ridicule and other subtle mechanisms of control (Sociology: The Essentials 6e, Ch 2, pp 60-61)
Deviance controlled	<i>Formal deviance</i> — behaviour that breaks laws or official rules	<i>Informal deviance</i> — behaviour that violates customary norms (Sociology: The Essentials 6e, Ch 7, p 177)
Indian health example	The central, state and local Acts that maintain the standards of health (Park 28e, Ch 12, p 814)	The small financial reward offered by government for undergoing sterilisation — described by Park as a sort of informal social pressure to further the family planning programme (Park 28e, Ch 12, p 814)

Health and social legislation used as social control in India (the formal arm)

- **Epidemic Diseases Act, 1897** — notification and emergency powers in an outbreak; being replaced by a **Public Health Act** covering epidemics, bioterrorism and disaster management (Sunder Lal, Ch 7, p 286).
- **Factories Act, 1948 (amended 1987)** — applies at 10 or more workers where power is used, 20 or more where it is not; 29 occupational diseases are notifiable (Sunder Lal, Ch 11, p 649).
- **Employees' State Insurance (ESI) Act, 1948** — the first major social security legislation, covering sickness, maternity, temporary and permanent disablement and death due to employment injury; preceded by the **Workmen's Compensation Act, 1923** (Sunder Lal, Ch 11, p 650).

- ▶ **Food Safety and Standards Act, 2006** (consolidating the food laws formerly under the **Prevention of Food Adulteration Act, 1954**) — science-based standards for manufacture, storage, sale and import of food; FSSAI established 2008 (Park 28e, Ch 11, p 802).
- ▶ **Medical Termination of Pregnancy (MTP) Act, 1971**, the **Immoral Traffic (Prevention) Act (1956)** and the **Children Act, 1960**, superseded by the **Juvenile Justice Act, 1986** — Park's own examples of social legislation curbing social evils, under the umbrella of **social defence** (Park 28e, Ch 12, pp 816, 839; Ch 9, p 698).
- ▶ **Cigarettes and Other Tobacco Products Act (COTPA), 2003** — bans smoking in public places, all direct and indirect advertisement, sale to those under 18 years and within 100 yards of an educational institution, and tobacco in *gutkha* and *pan masala*; pictorial warnings over 85% of both faces of the pack (Sunder Lal, Ch 15, p 691).
- ▶ **Pre-conception and Pre-natal Diagnostic Techniques (Prohibition of Sex Selection) Act, 1994**, amended rules in force from February 2003 — prohibits sex selection before and after conception and restricts prenatal diagnostic techniques to chromosomal, metabolic, haemoglobin, sex-linked and congenital disorders (Sunder Lal, Ch 9, p 567).
- ▶ **Prohibition of Child Marriage Act, 2006** (enforced 1 November 2007) — makes child marriage an offence; it replaced the **Child Marriage Restraint Act, 1978**, which raised the legal age of marriage from 15 to 18 years for girls and 18 to 21 years for boys (Park 28e, Ch 9, p 698; Sunder Lal, Ch 15, p 694).
- ▶ **Consumer Protection Act, 1986 (COPRA)** — first gave consumers a forum for speedy redressal against medical services; recognises six rights — **safety, information, choice, being heard, redressal and consumer education** (Park 28e, Ch 12, p 837).

Limits of the formal arm — the cause of a disease, medical or social, cannot be eradicated by legislation, and in areas of personal choice (diet, exercise, smoking) no government can legislate behaviour without taking away individual rights. The **sterilisation campaign of 1976**, which led to the Congress defeat in the 1977 elections, is the standing Indian example: "it is difficult to enforce a law unless majority of people are in favour of it and if it does not interfere with the rights of the individual" (Park 28e, Ch 22, p 1015).

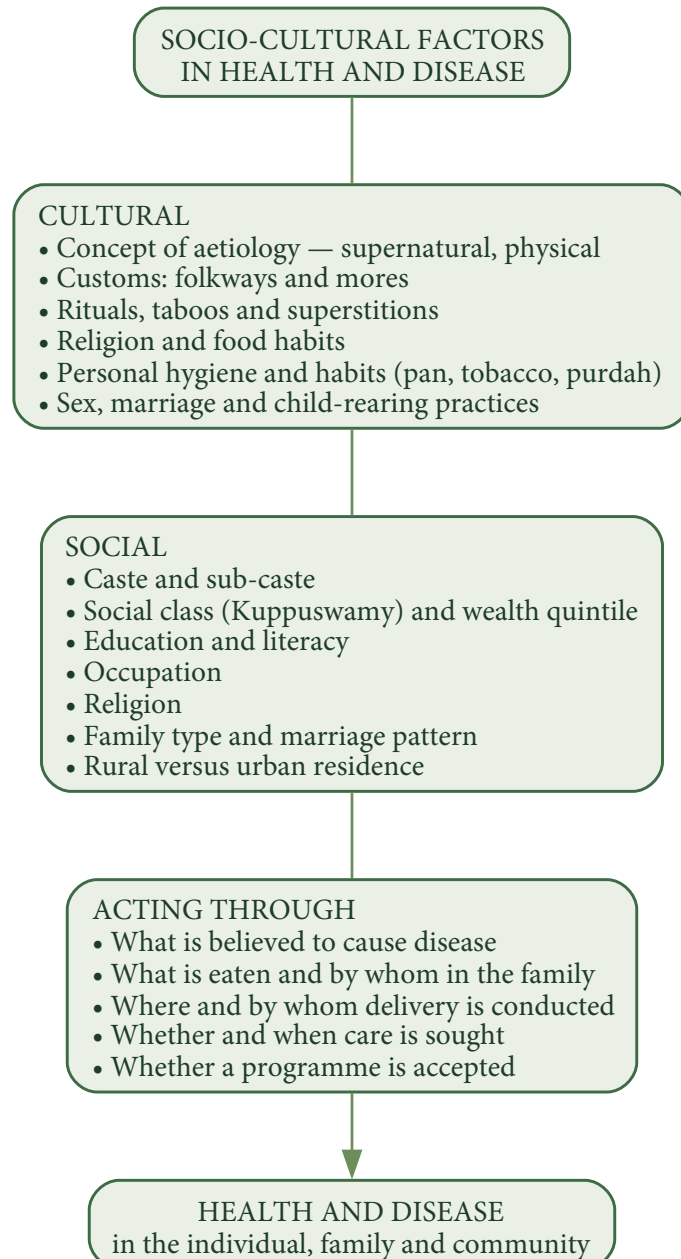
Implications for the community physician

- ▶ **Study the local culture first** — a knowledge of local customs and beliefs is very important for improving the health status of mothers and children, and is part of social paediatrics (Park 28e, Ch 12, p 830).
- ▶ **Work through the informal control system** — under the National Health Mission the **accredited social health activist (ASHA)** is a native woman of 25-45 years, educated to the 8th class, **chosen by and accountable to the village panchayat** as the interface between community and health system (Sunder Lal, Ch 17, p 738).
- ▶ **Remove the four barriers to communication** — **physiological** (hearing, expression), **psychological** (emotional disturbance, intelligence, comprehension), **environmental** (noise, invisibility, congestion) and **cultural** (illiteracy, customs, beliefs, religion, attitudes, class differences, language variation); even when services are readily available these barriers obstruct behaviour change (Park 28e, Ch 22, p 1012).

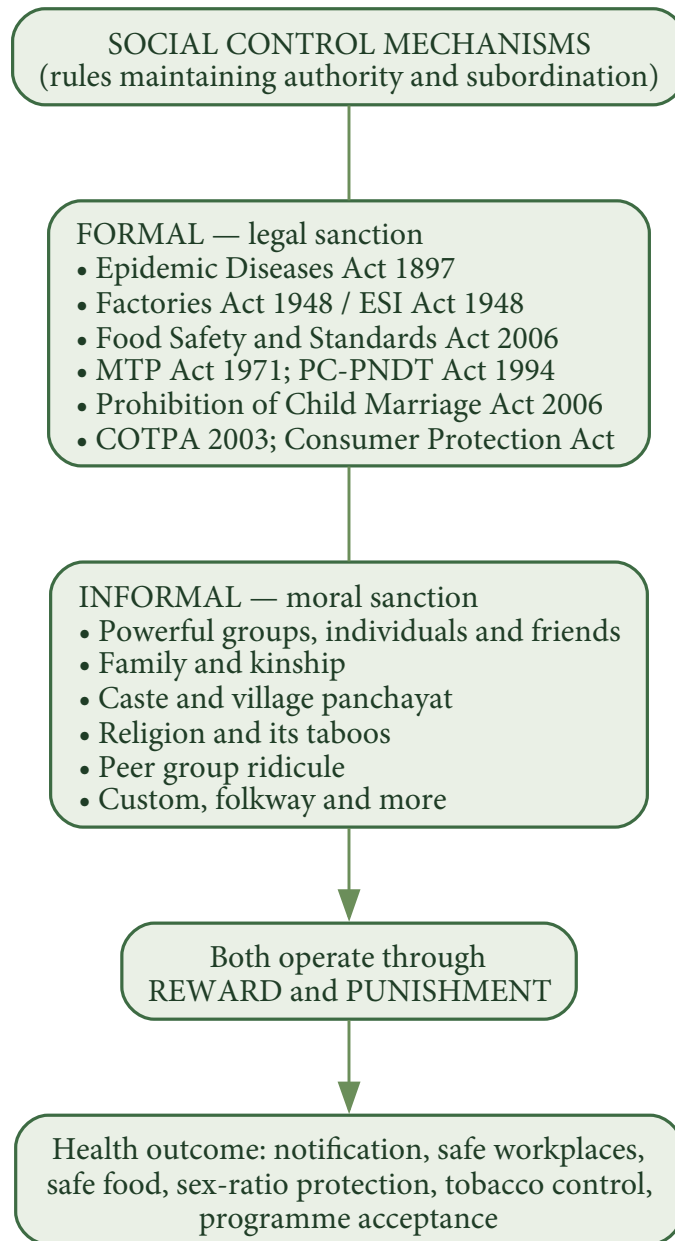
- **Use legislation last and selectively**, only where public opinion already supports it and paired with health education; laws remain useful in emergencies such as epidemic control and the management of fairs and festivals (Park 28e, Ch 22, p 1015).
- **Use social surveys** — they disclose social pathology and have driven public health reform, as Chadwick's survey led to the General Board of Health in 1848; joined to aetiological enquiry they become **social epidemiology** (Park 28e, Ch 12, p 816).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

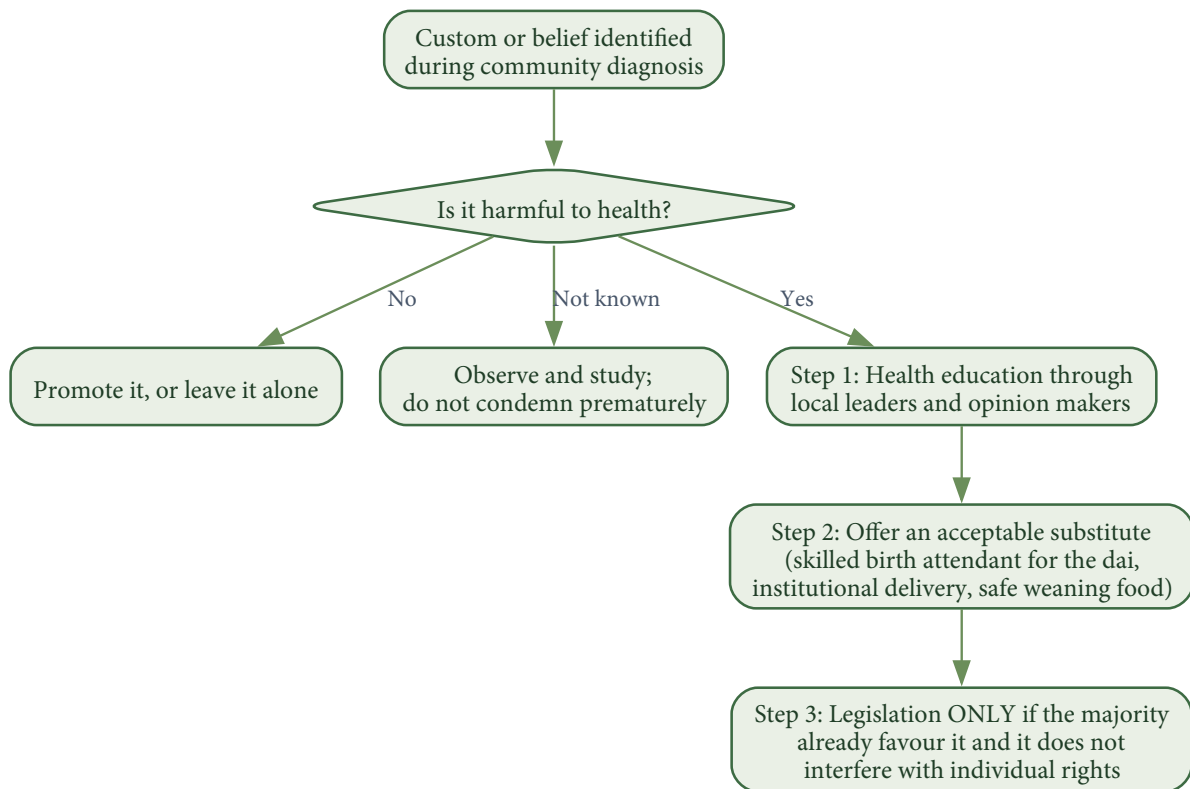
- **Diagram 1 — "Social factors influencing the health of people"** (Park 28e, Ch 12, Fig 1, p 812). Draw three vertical columns headed **SOCIAL/STRUCTURAL FACTORS**, **PREDISPOSING FACTORS** and **HEALTH OUTCOME**. In column 1 write two boxes: (a) **DEMOGRAPHIC, ECONOMIC, POLITICAL, LEGAL, RELIGIOUS, EDUCATIONAL, TECHNOLOGICAL**; (b) **COMMUNICATIONS, MIGRATION, URBANIZATION, LIFE EXPECTANCY, MARRIAGE, FERTILITY, ROLE OF WOMEN**. In column 2 write three boxes: **CURRENT CIRCUMSTANCES** (income, family life, housing, employment, stresses and crises, social support); **INDIVIDUAL PREDISPOSITIONS** (inheritance, perinatal events, previous accidents or illness, upbringing, childhood nutrition); **OPPORTUNITIES/SITUATIONS** (access to health services, exposure to pathogens, availability of education, food, sanitation, alcohol, drugs, marriage partners). Column 3 is a single label, **HEALTH**. Draw arrows from column 1 to column 2 and from column 2 to **HEALTH**, and a long **feedback arrow from HEALTH back to the social/structural factors**.
- **Diagram 2 — "Socialisation as social control"** (Sociology: The Essentials 6e, Ch 4, Fig 4.1, p 101). Draw six concentric circles and label them from the centre outwards: **SELF → FAMILY → PEERS: ridicule → LANGUAGE: cultural symbols → INSTITUTIONS: law, religion, economy, education, state → COERCION: punishment, imprisonment, violence**. Write beneath: control ranges from the most subtle (the expectations of others) to the most overt (physical coercion).



Classification of socio-cultural factors acting on health



Social control mechanisms operating in India and their health instruments



Action algorithm for a locally prevalent custom found on community diagnosis

High-yield exam pointers

- ▶ **Culture** = "learned behaviour which has been socially acquired"; **customs** = folkways (less vital) + mores (stringent); **laws** are formalised mores; all customs start as **convention**. **Acculturation** = **culture contact** by five routes — trade and commerce, industrialisation, propagation of religion, education, conquest.
- ▶ **Five supernatural causes of disease believed in India**: wrath of gods (chickenpox = *Chhoti Mata*), breach of taboo, past sins (leprosy, tuberculosis), evil eye, spirit/ghost intrusion (hysteria, epilepsy).
- ▶ **MCH customs** = good / bad / unimportant / uncertain — *kajal* on the eyelids is the standard "uncertain" example, blamed for trachoma.
- ▶ **Kuppuswamy's scale** = education + occupation + income, class I upper (26-29) to class V lower (<5); NFHS uses **wealth-index quintiles** instead.
- ▶ **NFHS-5 (2019-21)**: infant mortality rate 48.0 per 1000 in the lowest wealth quintile versus 17.0 in the highest (under-5, 59.0 versus 20.1). **Mean age at marriage in India** — 24 years for boys, 21 years for girls.
- ▶ **Six factors in social class differences**: physical environment, differences in services provided (undoctored areas), material resources, genetic endowment, educational status, attitude to disease.
- ▶ **Social control** = formal (laws and enactments of Parliament, central/state/local health Acts) + informal (pressure from powerful groups, individuals or friends), both working through **reward and punishment**; Park's Indian example of *informal* pressure is the cash reward for sterilisation. "It is difficult to enforce a law unless majority of people are in favour of it and if it

does not interfere with the rights of the individual" — the 1976 sterilisation campaign is the stock example of coercion failing.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: three statutes above have moved on since the textbook passages describing them — the Prevention of Food Adulteration Act, 1954 is superseded by the Food Safety and Standards Act, 2006; the six consumer rights quoted are those of the Consumer Protection Act, 1986, and the Consumer Protection Act, 2019 now governs; and the Epidemic Diseases Act, 1897 is being replaced by a comprehensive public health law. Park's cultural illustrations are deliberately dated village examples retained across editions; quote them as the textbook does, not as current survey data.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 12 (Medicine and Social Sciences), pp 811-839, including Fig 1 p 812, Table 2 p 833 (Kuppuswamy's socio-economic status scale), Table 3 p 833 and Table 4 p 834 (wealth quintiles and early childhood mortality, NFHS-5 2019-21), with Ch 9 p 698, Ch 11 p 802 and Ch 22 pp 1012, 1015; Sunder Lal — Textbook of Community Medicine, Ch 2 pp 81-82, Ch 7 p 286, Ch 9 p 567, Ch 11 pp 649-650, Ch 15 pp 691, 694, Ch 17 p 738; Sociology: The Essentials 6e, Ch 2 (Culture and the Media) pp 60-61, Ch 4 p 101 with Fig 4.1, Ch 7 (Deviance and Crime) p 177. National Family Health Survey-5 (2019-21) figures as reproduced in Park 28e.



Ronald Ross

Asked in: Paper I May 2019 — RGUHS MD Community Medicine

Definition — Sir Ronald Ross (1857–1932) was a British physician of the Indian Medical Service who, working in India, demonstrated the development of the malaria parasite inside the *Anopheles* mosquito and so proved that malaria is mosquito-borne (Gilles & Warrell 4e, Ch 1, p 15). For this he received the Nobel Prize for Physiology or Medicine in 1902 (Oxford Textbook of Public Health 7e, Ch 41, pp 1499–1500). He is doubly important to public health: he supplied the missing link in the *mode of transmission* of a major disease, and he founded the mathematical modelling of infectious disease transmission (Nelson & Williams, Ch 6, p 150).

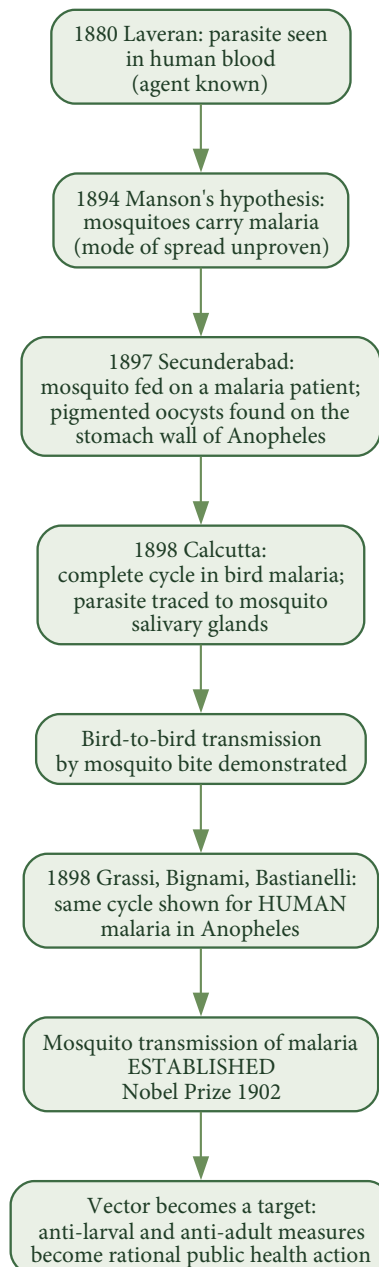
Biographical profile

Item	Detail
Born / died	15 May 1857 at Almora, India; died 16 September 1932 (Suryakantha, Ch 54, p 943)
Medical training	St Bartholomew's Hospital, London; joined the Indian Medical Service in 1881, first posted to Madras (Park Plus, Ch 1, p 14)

Item	Detail
Earlier interests	Music, then literature and novel-writing, then mathematics — the mathematical aptitude later proved decisive (Suryakantha, Ch 54, p 943)
Mentor	Sir Patrick Manson , who in 1894 proposed that malaria is carried from person to person by mosquitoes, and who showed Ross the plasmodia in a blood film (Gilles & Warrell 4e, Ch 1, p 16; Suryakantha, Ch 54, p 943)
Indian postings relevant to the work	Bangalore (1883, Acting Garrison Surgeon — noticed mosquitoes could be controlled by denying them access to water); Ooty; Secunderabad (1897); Calcutta , Presidency General Hospital (1898) (Park Plus, Ch 1, p 14)
Later career	Returned to England; worked at the Liverpool School of Tropical Medicine (Textbook of International Health, Ch 2, p 66); Director of the Ross Institute and Hospital , founded in his honour (Suryakantha, Ch 54, p 943)
Honour	Nobel Prize for Physiology or Medicine, 1902 (Nelson & Williams, Ch 27, p 899)

The discovery: chronology of the proof

Year	Event
1880	Laveran , a French army surgeon in Algeria, discovers and describes the malaria parasite in human blood (Gilles & Warrell 4e, Ch 1, p 16)
1894	Manson puts forward the theory that malaria is transmitted from person to person by mosquitoes — a hypothesis without proof (Gilles & Warrell 4e, Ch 1, p 16)
1895	Ross returns to India and begins his study at Secunderabad; publishes his first observations on the parasite within the mosquito (Suryakantha, Ch 54, p 943; Oxford 7e, Ch 41, p 1537)
1897	At Secunderabad, Ross finds pigmented cysts (oocysts) on the stomach wall of an <i>Anopheles</i> mosquito (probably <i>Anopheles stephensi</i>) that had fed on a malaria patient — the first demonstration of the parasite developing in the vector (Gilles & Warrell 4e, Ch 1, p 16)
1898	At Calcutta, Ross works out the complete cycle of bird malaria in naturally infected sparrows , tracing the parasite to the mosquito's salivary glands, and shows transmission from infected to healthy birds by mosquito bite (Gilles & Warrell 4e, Ch 1, pp 16–17; Park Plus, Ch 1, p 14)
1898	Grassi, Bignami and Bastianelli in Italy describe the cycle of the <i>human</i> malaria parasites in <i>Anopheles</i> and demonstrate sporozoites in the salivary glands (Gilles & Warrell 4e, Ch 1, pp 16–17)
1900	Manson confirms mosquito–malaria transmission by human volunteer experiments in the Roman Campagna and London (Gilles & Warrell 4e, Ch 1, pp 16–17)
1902	Nobel Prize awarded to Ross (Oxford 7e, Ch 41, pp 1499–1500)



The chain of reasoning by which Ronald Ross established mosquito transmission of malaria

Contributions to public health

Contribution	What Ross did	Public health significance
Proof of vector-borne transmission	Showed the malaria parasite develops in <i>Anopheles</i> (1897–98)	Completed the chain of infection for malaria; placed malaria alongside Bruce's tsetse–sleeping sickness (1896) and Reed's <i>Aedes</i> –yellow fever (1900) as the elucidation of modes of transmission that made specific control possible (Park 28e, Ch 1, p 18)

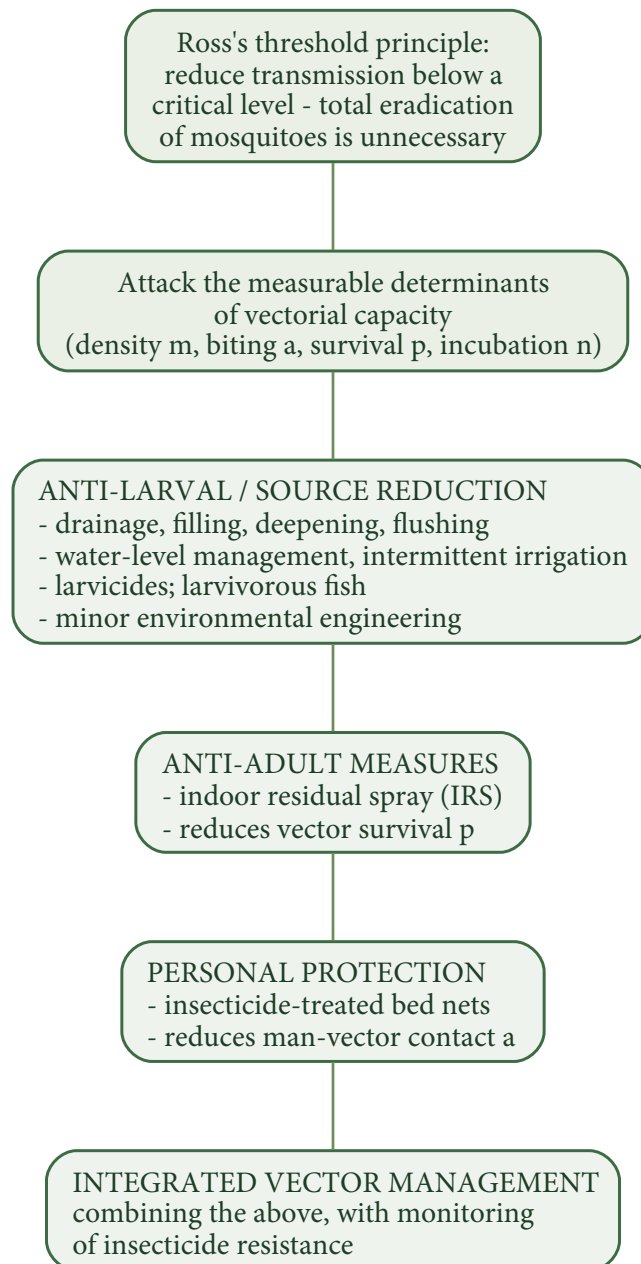
Contribution	What Ross did	Public health significance
Vector control in the field	Initiated anti-larval measures in Sierra Leone (1899); organised an anti-mosquito campaign at Ismailia, Egypt (1901–03) (Gilles & Warrell 4e, Ch 1, pp 18–19)	Converted a laboratory finding into an operational control method — source reduction and larval control
Malaria survey methodology	Contributed to the epidemiology of malaria and to methods of its survey and assessment (Park Plus, Ch 1, p 14)	Forerunner of modern malariometric and entomological surveillance
Mathematical epidemiology	Mauritius survey, 1908 originated his mathematical approach; elaborated in <i>The Prevention of Malaria</i> (Gilles & Warrell 4e, Ch 1, pp 18–19; Oxford 7e, Ch 41, p 1500)	First mathematical model of an infectious disease used in public health practice (Nelson & Williams, Ch 6, p 161)
Organisation building	Founded organisations for malaria prevention in the planting industries of India and Ceylon (Park Plus, Ch 1, p 14)	Early model of workplace/estate-based disease control

Ross's mathematical epidemiology and the threshold principle

- ▶ Ross developed one of the first models of disease transmission based on **differential equations**, to defend — against considerable scepticism — the claim that **malaria could be eradicated by reducing the mosquito population below a critical level, without eradicating mosquitoes entirely** (Oxford 7e, Ch 41, pp 1499–1500).
- ▶ His model made transmission depend on four measurable quantities: **mosquito density**, the **proportion of mosquitoes surviving long enough to transmit**, the **frequency of biting humans**, and the **proportion of humans who are infectious** (carry gametocytes) (Nelson & Williams, Ch 6, p 161).
- ▶ The idea of a **critical threshold below which transmission cannot persist** is the central point of all subsequent transmission modelling, and is embodied today in the **basic reproduction number** (R_0) (Oxford 7e, Ch 41, p 1500).
- ▶ The concept of R_0 traces through **Alfred Lotka and Ronald Ross**; its first modern epidemiological application was by **George Macdonald** in population models of malaria — hence the **Ross–Macdonald model** (Oxford 7e, Ch 41, p 1501). Ross and Macdonald together defined the basic reproduction rate as the number of new malaria cases generated by a single case in a given period (Nelson & Williams, Ch 27, pp 906–907).
- ▶ This line of work yields the still-used index of **vectorial capacity**:

$C = m \cdot a^2 \cdot p^n \div (-\log_e p)$ — where **m** = relative density of female anophelines, **a** = probability that a mosquito takes a human blood meal on a given day, **p** = probability of daily survival of the vector, **n** = days of the parasite's incubation period in the vector (Gilles & Warrell 4e, Ch 5, p 100). Because **p** is raised to the power **n**, **vector longevity is the single most powerful lever** — the justification for adult-mosquito measures such as indoor residual spraying.

- Ross's model, later refined by Macdonald, served as the basis for the global malaria eradication effort launched in the 1950s (Nelson & Williams, Ch 6, p 161).



Ross's threshold principle translated into the levers of malaria vector control

(Control measures as listed in Park 28e, Ch 5, p 333 and Ch 7, p 498.)

Legacy in India

Milestone	Detail
Burden before control	Estimated 75 million cases and 0.8 million deaths annually in India prior to 1953 (Park 28e, Ch 7, p 493)
Programme born of Ross's principle	National Malaria Control Programme (NMCP), 1953, built on residual insecticide spraying of dwellings and cattle sheds, malaria surveys and antimalarial drug supply (Jugal Kishore, Ch 1, p 388)

Milestone	Detail
Peak success	Converted to the National Malaria Eradication Programme in 1958; cases fell to 0.1 million by 1965 (Park 28e, Ch 7, p 493)
Current umbrella	Successively National Anti-Malaria Programme (1999) and then the National Vector Borne Disease Control Programme , which became an integral part of the National Rural Health Mission in 2005 (Park 28e, Ch 7, pp 493–494)
Commemoration	World Mosquito Day, 20 August (Suryakantha, Ch 54, p 944)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Diagram 1 — The cycle Ross demonstrated (sporogony in the mosquito).** Labels to write: *Gametocytes taken up in blood meal* → *Exflagellation / male and female gametes* → *Zygote* → *Ookinete* → *Ookinete penetrates stomach wall* → *Oocyst on stomach (mid-gut) wall* — *Ross, Secunderabad 1897* → *Sporozoites released* → *Salivary glands of female Anopheles* → *Injected into man at next bite*. Mark the mosquito half of the diagram as the **sexual cycle / extrinsic incubation period** and the human half as the **asexual cycle** (pre-erythrocytic then erythrocytic).
- **Diagram 2 — Ross's two-host cycle in one line.** Two boxes, *Man* and *Female Anopheles*, joined by two arrows: *man* → *mosquito (gametocytes ingested)* and *mosquito* → *man (sporozoites injected)*; label the break points where control acts.

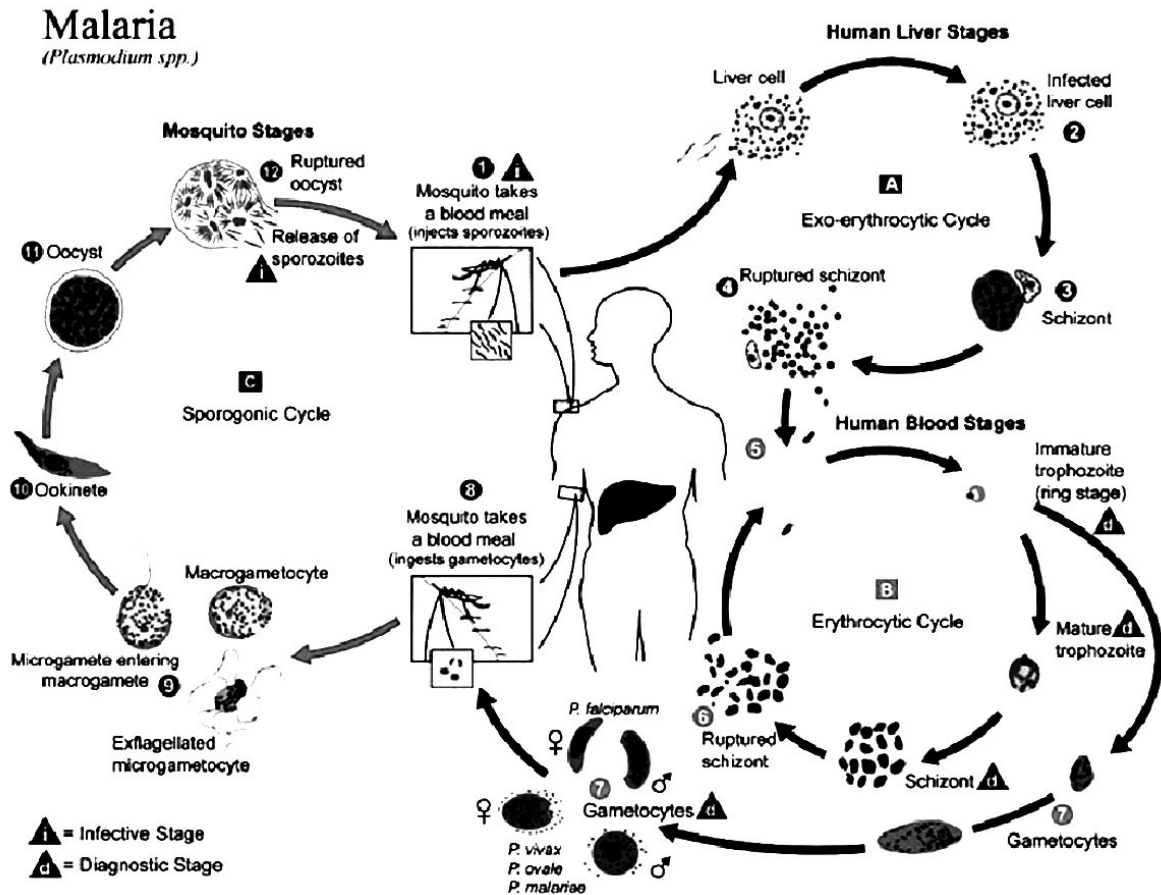


Fig 34.5 — Life Cycle of The Malaria Parasite (AFMC Red Book of Community Medicine 2024, p.1022)

High-yield exam pointers

- ▶ Ross's dates: 1857–1932; Nobel Prize 1902 (Physiology or Medicine); Indian Medical Service, joined 1881.
- ▶ "1897 Secunderabad — oocyst; 1898 Calcutta — bird malaria." That two-line pair is the whole discovery; write both, with the places.
- ▶ Sequence of names to quote: Laveran 1880 (parasite) → Manson 1894 (hypothesis) → Ross 1897–98 (proof in *Anopheles*) → Grassi, Bignami, Bastianelli 1898 (human malaria cycle).
- ▶ Vectorial capacity $C = m \cdot a^2 \cdot p^n \div (-\log_e p)$ — quote it and say why p (daily survival) dominates.
- ▶ Threshold principle: malaria is controlled by pushing the vector below a critical density, not by exterminating mosquitoes — the ancestor of R_0 and of the Ross–Macdonald model.
- ▶ India link: 75 million cases and 0.8 million deaths before 1953 → NMCP 1953 → NMEP 1958 → 0.1 million cases in 1965.
- ▶ World Mosquito Day — 20 August.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Textbooks differ by one year on the discovery because two separate steps are being dated: Park 28e states "in 1898 Ross demonstrated that malaria was transmitted by the *Anopheles*", which refers to the completed bird-malaria cycle at Calcutta, whereas Gilles & Warrell and Suryakantha date the original oocyst finding at Secunderabad to 1897 — writing both years with their respective steps is safest. Similarly, the edition of *The Prevention of Malaria* carrying his transmission model is cited as 1910 (Oxford Textbook of Public Health 7e) and as 1911 (Park Plus; Nelson & Williams call it "Ross's 1911 model"), so quote the book by name rather than by year.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 1 Man and Medicine; Ch 5 Epidemiology of Communicable Diseases; Ch 7 Health Programmes in India); Gilles & Warrell — Essential Malariology 4e (Ch 1 Historical outline; Ch 5 The epidemiology of malaria); Oxford Textbook of Public Health 7e (Ch 41 Mathematical models of transmission and control of infectious agents; Ch 73 Malaria); Nelson & Williams — Infectious Disease Epidemiology (Ch 6 Infectious Disease Dynamics; Ch 27 The Epidemiology and Control of Malaria); Suryakantha — Community Medicine with Recent Advances (Ch 21; Ch 54 Annexures); Park Plus (Amrutha & Vijayalakshmi), Ch 1; Sunder Lal — Textbook of Community Medicine, Ch 8; Jugal Kishore — National Health Programmes of India, Ch 1; Textbook of International Health, Ch 2.

Coverage note: the indexed corpus carries Ross's biography and discovery chronology well, but holds no dedicated captioned portrait or "history of public health" plate, so the embedded figure is the malaria life-cycle plate that his work established.



Sanitary awakening

Asked in: Paper I 12-Nov-2019 — RGUHS MD Community Medicine

Definition — The "great sanitary awakening" is the anti-filth movement of mid-nineteenth-century England in which filth was recognised as man's greatest enemy and the physical environment — housing, drainage, water and refuse — became the object of organised State action. Its central new idea was that **the State has a direct responsibility for the health of the people**, and its legislative expression was the **Public Health Act of 1848** (Park 28e, Ch 1, p 17).

Setting — why filth became the enemy

Determinant (England, 18th–19th century)	Consequence for health
Industrial revolution of the 18th century	Mass migration into towns; creation of slums
Overcrowding of the new industrial towns	All its recognised ill-effects, including tuberculosis

Determinant (England, 18th–19th century)	Consequence for health
Accumulation of filth in cities and towns	No drainage or refuse disposal; the Thames was both the source of drinking water and the depository for sewage
High sickness and death rates, especially in women and children	Health of the people at "the lowest ebb"
Industrial and social problems of factory labour	Poverty and ill-health reinforcing each other
Sharp class gradient in survival	Mean age at death in London, 1842 — 44 years for gentry and professionals vs 22 years for the working class
Frequent visitations of cholera ("the father of public health")	The great cholera epidemic of 1832 was the trigger for enquiry

The prevailing explanation was the **miasma theory** — that dirty environments, unclean drinking water and poor ventilation caused disease. Paradoxically it was this mistaken theory that justified sanitary reform and made the movement possible (Oxford 7e, Ch 53, p 1951).

Chadwick's enquiry and the legislative response

- ▶ **Edwin Chadwick (1800–1890)**, a lawyer and Poor Law Commissioner, was set by the cholera epidemic of 1832 to investigate the health of the inhabitants of the large towns, with a view to improving the conditions under which they lived (Park 28e, Ch 1, p 17).
- ▶ In 1842 he published "**The Sanitary Conditions of the Labouring Population in Great Britain**" — a landmark in the history of public health. Begun as an enquiry into the causes of poverty, it concluded that people often became poor *as a result of* ill-health from an adverse environment (Oxford 7e, Ch 99, pp 3741–42). It focussed the attention of people and government on the urgent need to improve public health.
- ▶ It fed the **Royal Commission on the Health of Towns**, which concluded that physical changes to the city itself were necessary to improve the well-being of the poor (Oxford 7e, Ch 58, pp 2154–55).
- ▶ Filth was recognised as man's greatest enemy, and with this began the **anti-filth crusade** — the "**great sanitary awakening**" (Park 28e, Ch 1, p 17).
- ▶ **Public Health Act 1848** — the first example of governmental involvement in public health affairs (Oxford 7e, Ch 53, p 1951); it recommended a **national board** and **district boards of health** and provided for **Inspectors of Nuisances**, the predecessors of today's environmental health officers (Oxford 7e, Ch 99, pp 3741–42).
- ▶ **Public Health Act 1875** — a comprehensive piece of legislation for the control of man's physical environment.
- ▶ Methodologically, Chadwick's enquiry is the classic **social survey** disclosing social pathology and driving policy; the **General Board of Health** was founded in 1848, and Park notes it was such a survey by Chadwick that led to its foundation (Park 28e, Ch 12, p 816).

Personalities of the sanitary awakening

Person	Year(s)	Contribution
<i>Johann Peter Frank</i> (1745–1821)	Pre-1840	Health philosopher; conceived public health as good health laws enforced by the police, and enunciated that the State is responsible for the health of its people. The Act of 1848 fulfilled his dream
<i>Edwin Chadwick</i> (1800–1890)	1842	Report on the Sanitary Conditions of the Labouring Population in Great Britain; lit the torch of sanitary reform
<i>John Snow</i>	1848–1854	Studied the epidemiology of cholera in London and established the role of polluted drinking water ; spot map of the 1854 Golden Square epidemic focussed attention on the Broad Street pump ; his "grand experiment" is the classic natural experiment (Park 28e, Ch 3, pp 86–87, 106–07)
<i>William Budd</i>	1856	Concluded from a typhoid outbreak in the rural north of England that spread was by drinking water, not by miasma and sewer gas
<i>Sir John Simon</i> (1816–1904)	From 1848	First medical officer of health of London ; more than any other man responsible for sanitary reforms; built a system of public health admired by the rest of the world
<i>Lemuel Shattuck</i> (1793–1859)	1850	Bookseller and publisher; his report on the health conditions in Massachusetts stirred the American conscience, as Chadwick's had the English
<i>Louis Pasteur</i> (1822–1895)	1860, 1873	Demonstrated bacteria in air and disproved spontaneous generation; advanced the germ theory of disease
<i>Robert Koch</i> (1843–1910)	1877	Showed anthrax was caused by a bacterium; then the golden age of bacteriology — tubercle bacillus 1882, cholera vibrio 1883 (Park 28e, Ch 1, p 18)

Snow's and Budd's discoveries are the more remarkable because the causative agents of cholera and typhoid had not yet been identified; together they converted a crusade against dirt into a popular demand for clean water (Park 28e, Ch 1, p 17).

Impact and significance

- **Birth of the public health concept** in England around 1840, and of the State's direct responsibility for health — the single most important legacy.
- **Legislation as a public health tool** — sanitary legislation and sanitary reform became the dominant instrument of the nineteenth century.
- **Creation of public health machinery** — General Board of Health, district boards, Inspectors of Nuisances, and the office of medical officer of health.
- **Sanitary engineering** — piped water supply, sewerage and waste removal, later chlorination of water and pasteurisation of milk; sanitary commissions were created in European and American cities (Oxford 7e, Ch 58, pp 2153–55).
- **Right action for the wrong reason, and slowly** — Chadwick's "Sanitary Idea" rested on the obsolescent miasmatic theory, and his blueprint (constant-pressure clean water in, self-scouring arterial sewers out) had to be paid for from local rates; legal battles over the sanctity of private

property delayed its implementation in many fast-growing British towns by as much as a generation, until the 1870s (Oxford 7e, Ch 2, pp 190–91).

- **Foundations laid worldwide** — France, Spain, Australia, Germany, Italy, Belgium and Scandinavia all developed their public health, so that by the beginning of the 20th century clean water, clean surroundings, wholesome housing and control of offensive trades were established across the western world (Park 28e, Ch 1, pp 17–18).
- **Mortality decline** — over half the gain in life expectancy at birth in England and Wales between 1871 and 1911 came from reduced infectious-disease mortality, about 20% of the total from tuberculosis alone (Oxford 7e, Ch 46, p 1725). *McKeown* attributed the fall mainly to better nutrition, not medical intervention; his critics credit the civic and State changes that made sanitary reform possible (Oxford 7e, Ch 22, p 863).
- **Enduring valuation** — in 2006 *British Medical Journal* readers voted **clean drinking water and waste disposal** the greatest medical advance since 1840, ahead of antibiotics, anaesthesia, vaccines and germ theory (Oxford 7e, Ch 12, p 476).

The sanitary awakening in India

Public health in India followed the English pattern, but progress was extremely slow during the British regime (Mahajan & Gupta 4e, Ch 1, p 17).

Year	Development
1825	Quarantine Act passed (Red Book, Ch 2)
After 1858	A Royal Commission was sent to India to find the reasons for heavy morbidity and mortality among European troops, from malaria and other preventable diseases — the starting point of public health in India (Mahajan & Gupta 4e, Ch 1, p 17)
1859	Commission of Public Health pointed out the need for safe water and environmental sanitation to prevent epidemics (Red Book, Ch 2)
c. 1864	Public Health Departments started as vaccination departments , later as Sanitation Departments , at the Centre and in the Provinces (Mahajan & Gupta 4e, Ch 1, p 17)
Same period	Sanitary Commissioners appointed in all three provinces of Bombay, Madras and Bengal (Red Book, Ch 2)
1866	Births, Deaths and Marriages Registration Act — beginning of voluntary registration of vital events in India; Madras and Mysore later made it compulsory (Mahajan & Gupta 4e, Ch 25, p 479)
1885	Local Self Government Act passed, devolving the sanitary function to local bodies (Red Book, Ch 2)
1904	After ~40 years of tussle over whether sanitation should answer to Government or to the Surgeon General, Public Health Departments were made to function separately ; Sanitary Commissioner and Assistant Sanitary Commissioner were redesignated Director and Assistant Director of Public Health (Mahajan & Gupta 4e, Ch 1, p 17)
1919	Decentralisation of health administration began with the Montague–Chelmsford reforms (Red Book, Ch 2)
1946	Bhore Committee recommended integration of the separated curative (Medical) and preventive (Public Health) departments (Mahajan & Gupta 4e, Ch 1, p 17)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Timeline of the sanitary awakening** (single horizontal line, label above and below): 1832 great cholera epidemic — 1842 Chadwick's Report on the Sanitary Conditions of the Labouring Population — 1848 Public Health Act and General Board of Health — 1848–54 John Snow, Broad Street pump — 1850 Shattuck's Massachusetts report — 1856 William Budd, typhoid — 1860/1873 Pasteur, germ theory — 1875 Public Health Act — 1880–1920 disease control phase.

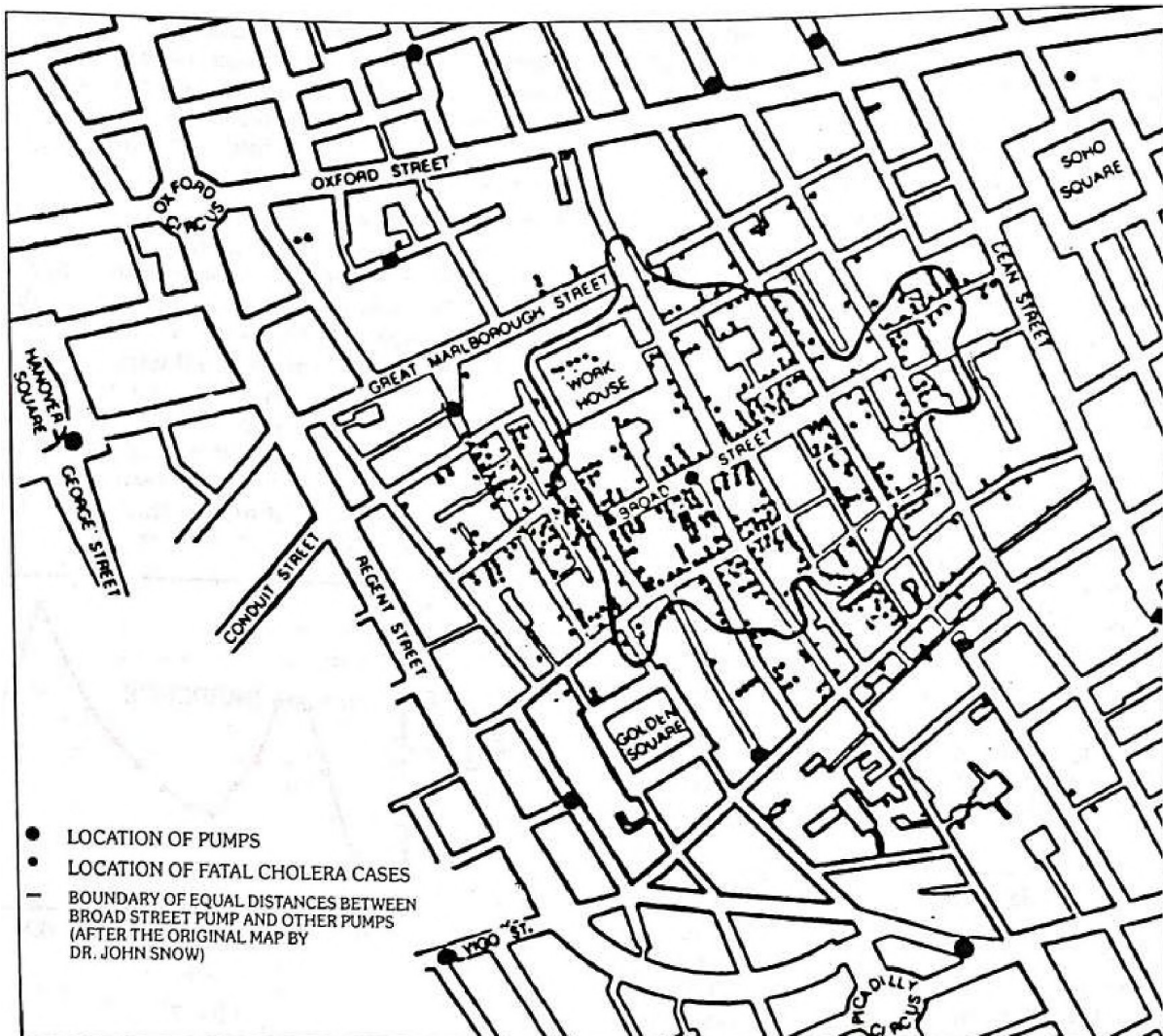
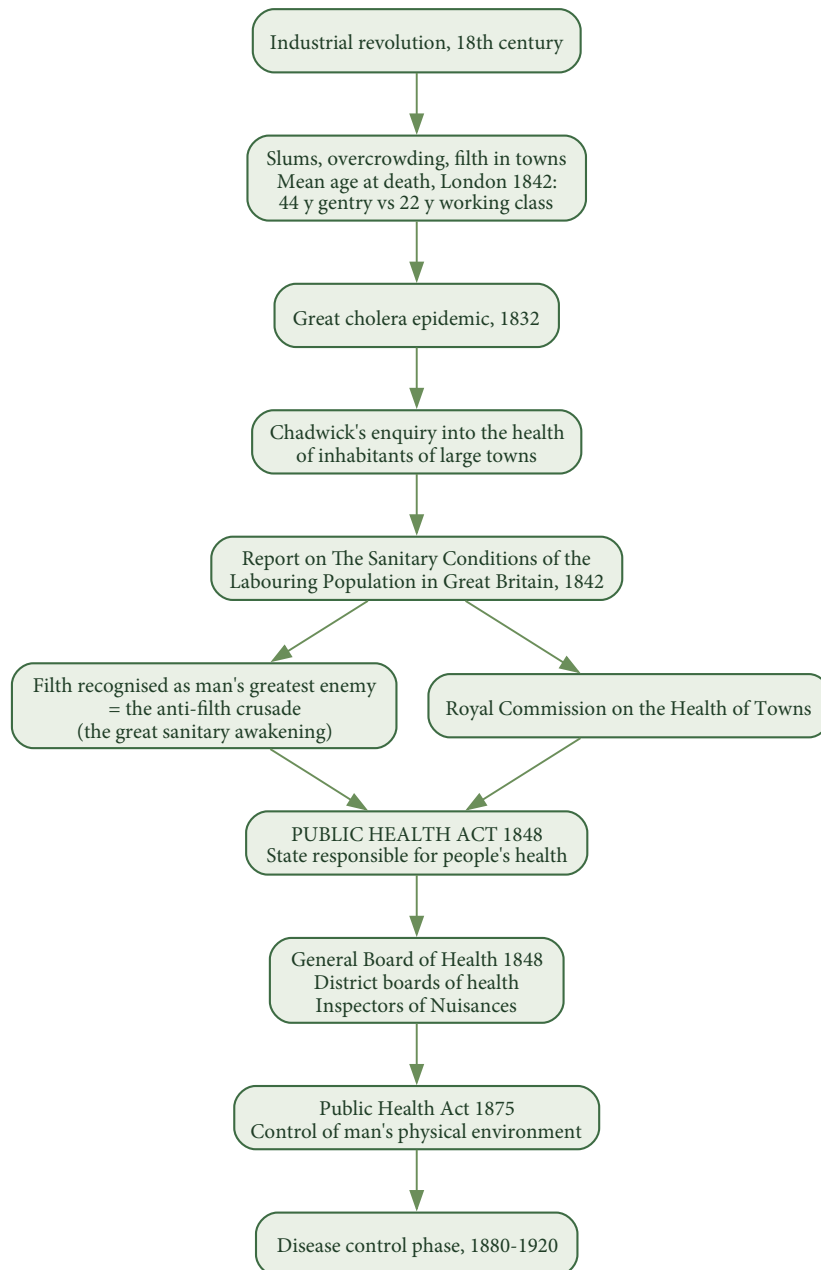
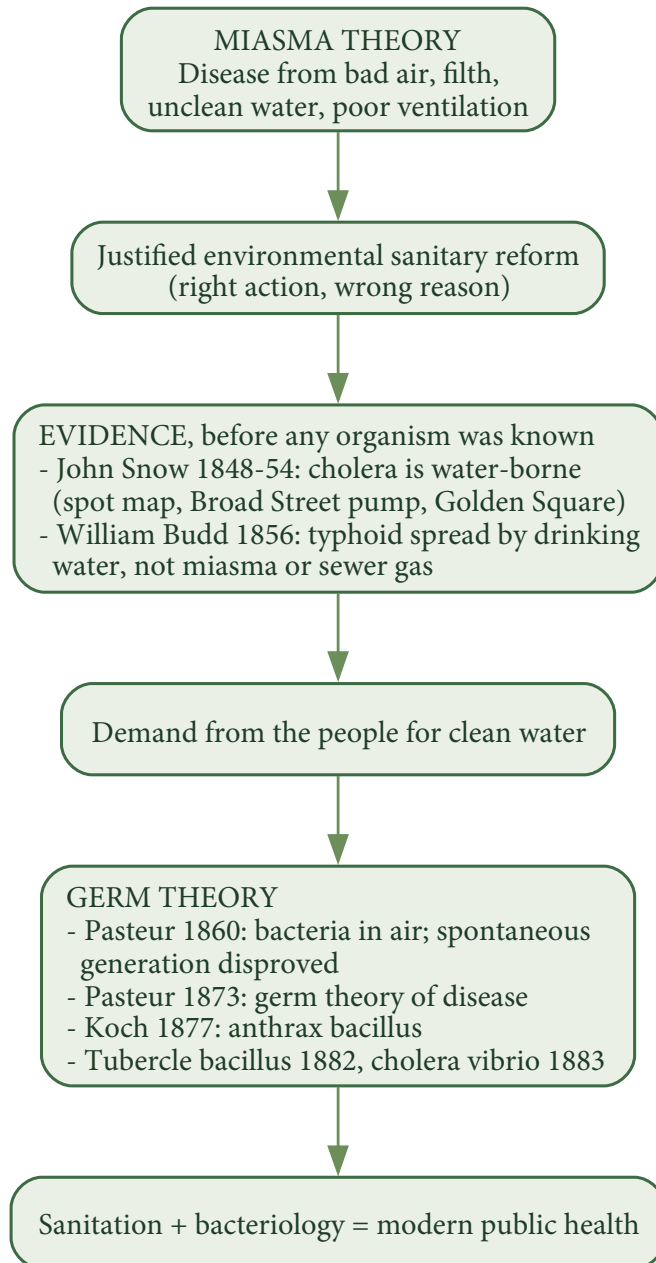


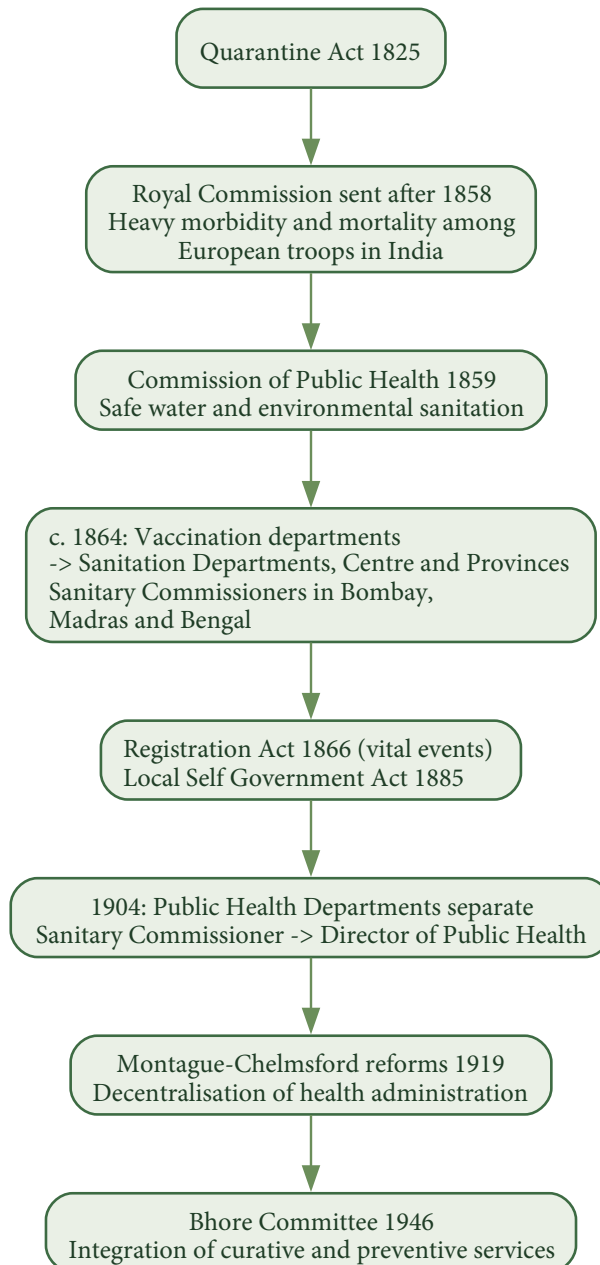
Fig 6 — Spot map of Asiatic cholera in London (Park's Textbook of Preventive and Social Medicine, 28e, p.87)



Causal chain of the great sanitary awakening, England 1832–1875



From miasma to germ theory - the evidence that turned the sanitary idea into a science



The sanitary awakening reaches India - colonial sanitary administration to integration

What followed — the four phases of public health

Phase	Period	Defining feature
Disease control	1880–1920	Sanitary legislation and sanitary reform; general cleanliness, garbage and refuse disposal, quarantine conventions
Health promotional	1920–1960	State assumes direct responsibility for the health of the <i>individual</i> ; personal health services — maternal and child health, school health, industrial health, mental health, rehabilitation; public health nursing. C.E.A. Winslow (1920) defined public health as "the science and art of preventing disease, prolonging life and promoting health and efficiency through organized community effort"
Social engineering	1960–1980	Chronic disease and behavioural problems; social and behavioural aspects given new priority; goals of public health and preventive medicine become identical

Phase	Period	Defining feature
"Health for All"	1981–2000	Glaring contrasts between developed and developing countries addressed through primary health care

(Park 28e, Ch 1, pp 20–21)

Relevance to India today

- ▶ The core principle — that **the environment, not only the patient, must be treated** — survives in the "**sanitation barrier**": segregating faeces by a sanitary latrine and disposal pit so the agent cannot reach a new host (Park 28e, Ch 15, p 911).
- ▶ **Swachh Bharat Mission**, launched 2 October 2014 at Rajghat, New Delhi, is the contemporary anti-filth crusade: **Gramin** makes gram panchayats free of open defecation through solid and liquid waste management; **Urban** adds eradication of manual scavenging, conversion of insanitary latrines and scientific municipal solid waste management (Park 28e, Ch 7, pp 573–74).
- ▶ **Local self-government still carries the sanitary function** — municipal boards and corporations remain responsible for sanitation and drainage, water supply and registration of births and deaths, exactly as the Victorian district boards were (Park 28e, Ch 23, p 1035).

High-yield exam pointers

- ▶ **Four dates to write:** Chadwick's report **1842** → Public Health Act **1848** (+ General Board of Health) → Snow's Broad Street pump **1854** → Public Health Act **1875**.
- ▶ **The one-line significance:** filth recognised as man's greatest enemy → anti-filth crusade → the State has a direct responsibility for the health of the people.
- ▶ **The killer statistic:** mean age at death in London, 1842 — 44 years (gentry/professionals) vs 22 years (working class).
- ▶ **Cholera is called the "father of public health";** the 1832 epidemic triggered Chadwick's enquiry.
- ▶ **Torch metaphor:** "the torch was lit by Chadwick, but **Sir John Simon**, first medical officer of health of London, actually delivered the sanitary reforms."
- ▶ **Indian dates:** Quarantine Act 1825 → Royal Commission after 1858 → Commission of Public Health 1859 → vaccination/sanitation departments c. 1864 → Local Self Government Act 1885 → Public Health Departments separated 1904 → Bhole Committee 1946.
- ▶ **Phase sequence:** disease control (1880–1920) → health promotional (1920–60) → social engineering (1960–80) → Health for All (1981–2000).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the sanitary awakening acted on the **miasma theory**, which was aetiologically wrong but administratively right, and Park's own account notes that Snow's and Budd's water-borne conclusions preceded the identification of *Vibrio cholerae* and *Salmonella typhi* — so do not date the sanitary movement to the germ theory.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 1 (Man and Medicine: Towards Health for All), pp 17–18, 20–21; Ch 3 (Principles of Epidemiology and Epidemiologic Methods), pp 86–87, 106–07; Ch 7 (Health Programmes in India), pp 573–74; Ch 12 (Medicine and Social Sciences), p 816; Ch 15 (Environment and Health), p 911; Ch 23 (Health Planning and Management), p 1035. Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e — Ch 1 (Evolution of Preventive and Social Medicine), pp 16–18; Ch 25 (Demography and Vital Statistics), p 479. Red Book — Public Health and Preventive Medicine — Ch 2, p 299. Oxford Textbook of Public Health 7e — Ch 2 (History and development of public health in developed countries) pp 190–91, Ch 12 (Water and sanitation) p 476, Ch 22 (Public health policy in developed countries) p 863, Ch 46 (Demography and public health) p 1725, Ch 53 (Environmental health methods) p 1951, Ch 58 (Urbanization and health) pp 2153–55, Ch 99 (Practice of environmental health) pp 3741–42. Programme documents: Swachh Bharat Mission (Gramin) and Swachh Bharat Mission (Urban) Guidelines 2014, Government of India.



Social engineering phase

Asked in: Paper I May 2019 — RGUHS MD Community Medicine

Definition — The **social engineering phase (1960–1980)** is the third of the four phases into which the history of modern public health is conventionally divided. It is the period in which public health, having brought most acute communicable disease under control, needed **reorientation more towards social objectives**: social and behavioural aspects of disease and health were given a new priority, and public health moved into the **preventive and rehabilitative aspects of chronic diseases and behavioural problems** (Park 28e, Ch 1, pp 20–21).

The four phases of public health — where the social engineering phase sits

Phase	Period	Dominant health problem	Characteristic approach	Landmarks
Disease control	1880 – 1920	Acute communicable and "filth" diseases	Sanitary legislation and sanitary reform; general cleanliness, garbage and refuse disposal; quarantine conventions to contain disease	Public Health Act 1875, England; <i>Sir John Simon</i> (1816–1904), first medical officer of health of London (Park 28e, Ch 1, p 17)

Phase	Period	Dominant health problem	Characteristic approach	Landmarks
Health promotional	1920 – 1960	Individual ill-health lying outside the reach of a purely environmental sanitation — maternal, child, school, industrial and mental ill-health	State accepted direct responsibility for the individual's health; health promotion of individuals added as a goal; personal health services — maternal and child health, school health, industrial health, mental health and rehabilitation services; public health nursing a direct offshoot	<i>Winslow's</i> definition of public health, 1920; health centre concept mooted by <i>Lord Dawson</i> , 1920; rise of social medicine (chair at Oxford, 1942); Bhore Committee 1946 (India); Community Development Programme (Park 28e, Ch 1, p 20)
Social engineering	1960 – 1980	Chronic non-communicable disease and behavioural problems of affluent societies — cancer, diabetes, cardiovascular disease, alcoholism, drug addiction	Reorientation towards social objectives ; social and behavioural aspects of disease and health given a new priority	Term " community health " comes into preference — EURO symposium 1966; WHO Expert Committee caution, 1973 (Park 28e, Ch 1, pp 20–21)
"Health for All"	1981 – 2000 A.D.	Glaring inequity in health between and within countries	Equity, primary health care, intersectoral coordination, community participation	30th World Health Assembly resolution, May 1977; Alma-Ata Declaration 1978 ; Global Strategy for Health for All adopted by the 34th World Health Assembly, 1981 (Park 28e, Ch 1, pp 21–23)

Why public health entered the social engineering phase — the drivers

- **Change in the pattern of disease in the developed world.** Many acute illnesses were brought under control, and as old problems were solved new ones emerged as chronic disease — cancer, diabetes, cardiovascular disease, alcoholism, drug addiction — especially in affluent societies (Park 28e, Ch 1, p 20).
- **Traditional tools failed.** These problems could not be tackled by isolation, immunization and disinfection, nor explained on the basis of the **germ theory of disease** (Park 28e, Ch 1, pp 20–21; Ch 2, p 52).
- **A chronic burden on society.** Unlike the swift death brought by acute infection, their consequences placed a **chronic burden on the society that created them** (Park 28e, Ch 1, p 21).
- **Multifactorial causation, and with it the risk-factor concept.** Social, economic, cultural, genetic, environmental and psychological factors were recognised as equally important, most linked to **lifestyle and behaviour**. The one-to-one "single cause" idea gave place to **multifactorial causation** — first mooted by *Pettenkofer of Munich* (1819–1901), whose ideas were lost in the bacteriological era — and its revival by epidemiologists carried with it the newer concept of "**risk factors**" in the aetiology of disease (Park 28e, Ch 1, p 18; Ch 2, p 52).
- **Groundwork already laid by the preceding phase.** In the health promotional phase (1920–1960), sociology, psychology and anthropology had rediscovered that man is not only a biological

animal but also a **social being**, and that disease has **social causes, social consequences and social therapy**; the social engineering phase built on that base rather than creating it (Park 28e, Ch 1, p 20).

What "social engineering" meant in practice — defining features

- ▶ **Reorientation of public health towards social objectives** rather than purely technical and environmental ones.
- ▶ **New priority to the social and behavioural aspects** of disease and health — behaviour, habits, beliefs, attitudes and lifestyle practices became legitimate objects of public health action (Park 28e, Ch 1, p 21; Sunder Lal, Ch 4, pp 143–145).
- ▶ **Extension of prevention and rehabilitation to chronic disease.** Public health moved into both the **preventive and the rehabilitative aspects of chronic diseases and behavioural problems**; rehabilitation services already existed as a personal health service of the previous phase, and what was new was their extension to chronic non-communicable disease and behavioural disorders (Park 28e, Ch 1, p 21; Ch 2, pp 65–66).
- ▶ **Social diagnosis followed by social therapy** as the working method (Sunder Lal, Ch 2, p 81).
- ▶ **Convergence of disciplines.** The goals of public health and preventive medicine, which already overlapped considerably, became **identical — prevention of disease, promotion of health and prolongation of life** (Park 28e, Ch 1, p 21).
- ▶ **Terminological shift.** "Public health" kept its name but its original meaning changed, and "**community health**" — "all the personal health and environmental services in any human community, irrespective of whether such services were public or private ones" — was preferred by some leaders, though WHO cautioned that the term is ambiguous without further qualification (Park 28e, Ch 1, p 21; Ch 2, p 66).

Conceptual shifts produced by the phase

Element	Before (up to 1960)	After the social engineering phase
Model of causation	Germ theory; one-to-one relationship — <i>disease agent</i> → <i>man</i> → <i>disease</i>	Multifactorial causation; "agent" replaced by causative factors ; risk factors
Chief target of action	Environment and the micro-organism	Lifestyle, behaviour, and social and economic conditions
Disease of interest	Acute communicable disease	Chronic non-communicable disease and behavioural problems
Service package	Sanitation plus personal preventive services	Preventive, therapeutic and rehabilitative care of chronic disease
Preferred discipline label	Public health / preventive medicine / social medicine	Community health / community medicine
Scientific base	Bacteriology and sanitary science	Epidemiology, biostatistics and the social and behavioural sciences

Limitations, and the transition to the "Health for All" phase

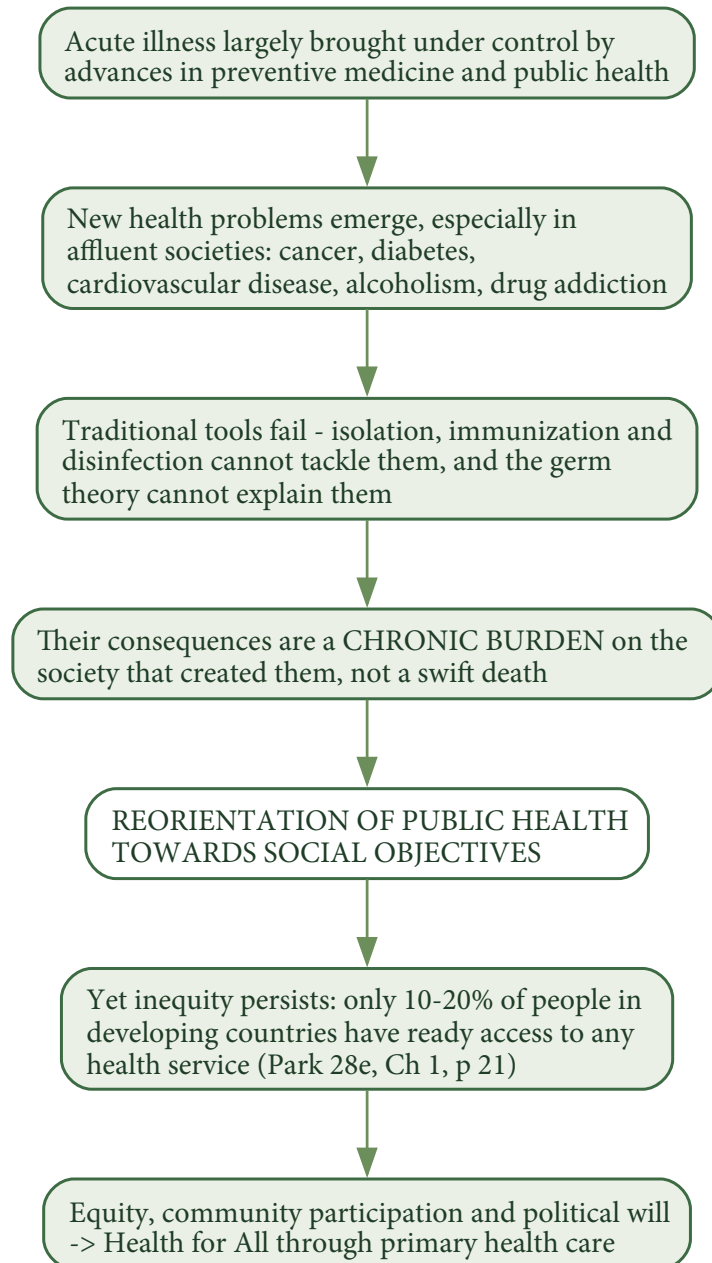
- ▶ The phase was largely a preoccupation of **affluent, industrialised societies**, while the developing world still carried the unfinished agenda of communicable disease, malnutrition and sanitation (Park 28e, Ch 1, pp 20–21).
- ▶ The **glaring contrasts** between developed and developing countries came into sharper focus despite advances in medicine: only **10–20%** of the population in developing countries enjoyed ready access to health services of any kind; death claimed **60–250 of every 1000 live births** in the first year of life; life expectancy was **30% lower** than in developed countries. *John Bryant*, in "**Health and the Developing World**", wrote that perhaps more than half the world's people have no access to health care at all (Park 28e, Ch 1, p 21).
- ▶ The resulting new ideas — **social justice and equity, community participation, changing ideas about health and development, and political will** — culminated in the 30th World Health Assembly resolution of May 1977, the **Alma-Ata Declaration (1978)** naming primary health care as the route, and the **Global Strategy for Health for All adopted by the 34th World Health Assembly in 1981**. The fundamental principle of that strategy is **equity**, Health for All being defined by the member countries of WHO as the "*attainment of a level of health that will enable every individual to lead a socially and economically productive life*" (Park 28e, Ch 1, pp 22–23; Ch 10, p 720).

Legacy

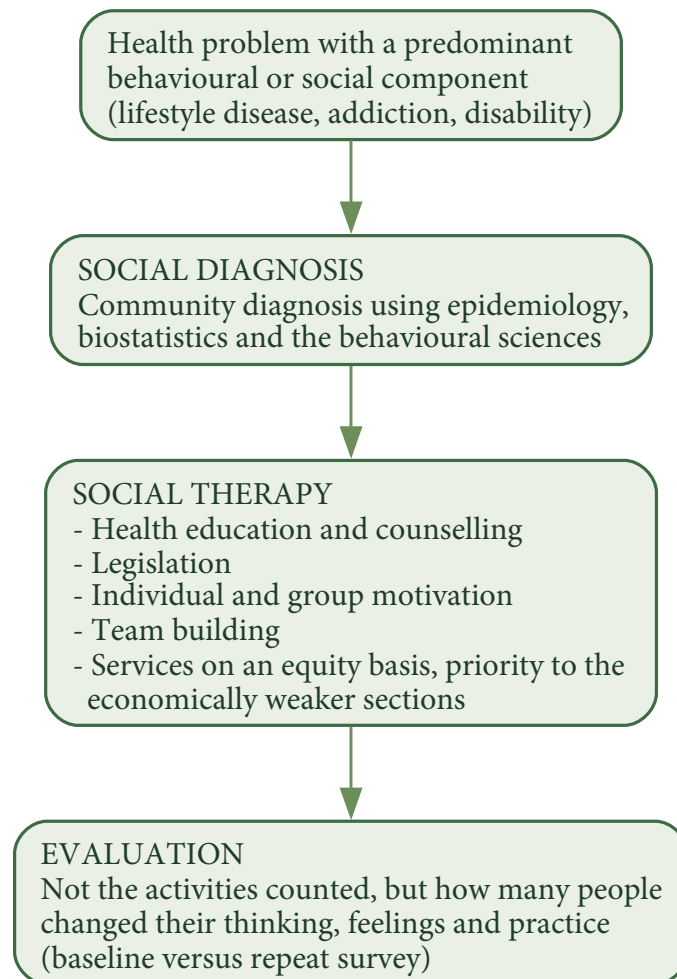
- ▶ The **risk-factor and lifestyle framework** born in this phase is the operating logic of all present non-communicable disease prevention, and **behaviour change communication** a routine programme tool — integrated bio-behavioural surveillance (IBBS) and behavioural surveillance survey data have been used extensively in India's national programme on HIV/AIDS (Sunder Lal, Ch 2, p 81).
- ▶ The health-promotion movement that followed formalised the same logic — health promotion as "the process of enabling people to increase control over, and to improve health" (Park 28e, Ch 2, p 61). The **Jakarta Declaration on Health Promotion**, from the **fourth international conference on health promotion**, July 1997, listed the fundamental conditions and resources for health as **peace, shelter, education, social security, social relations, food, income, the empowerment of women, a stable ecosystem, sustainable resource use, social justice, respect for human rights, and equity** — adding that **above all, poverty is the greatest threat to health** (Park 28e, Ch 2, p 49).
- ▶ It supplies the rationale for **Health in All Policies** and for action on the social determinants of health, since most health problems are as much sociological as biological (Sunder Lal, Ch 2, pp 80–81; Oxford Textbook of Public Health 7e, Ch 47, pp 1733–1734).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Diagram 1 — Timeline: four phases of public health.** Draw a single horizontal time-line and label, in order: 1880–1920 Disease control phase; 1920–1960 Health promotional phase; 1960–1980 SOCIAL ENGINEERING PHASE; 1981–2000 A.D. Health for All phase. Under each, write one key label: Sanitary legislation / Public Health Act 1875; Personal health services / Bhole Committee 1946; Social & behavioural priority / risk factors / rehabilitation of chronic disease; Alma-Ata 1978 / Primary health care / Equity.
- ▶ **Diagram 2 — Germ theory versus multifactorial causation.** Draw two panels side by side. Left panel labelled Germ theory (one-to-one): Disease agent → Man → Disease. Right panel labelled Multifactorial causation: a central box Disease with arrows converging on it from Social, Economic, Cultural, Genetic, Environmental, Psychological and Lifestyle & behaviour.



The causal chain into and out of the social engineering phase



Social diagnosis to social therapy - the working method of the social engineering phase

High-yield exam pointers

- ▶ **Sentence worth reproducing verbatim:** "Social and behavioural aspects of disease and health were given a new priority; public health moved into the preventive and rehabilitative aspects of chronic diseases and behavioural problems."
- ▶ **The three identical goals:** prevention of disease, promotion of health, prolongation of life.
- ▶ **Names:** *Pettenkofer* of Munich (multifactorial causation, 1819–1901); *Winslow* (1920 definition); *Lord Dawson* (health centre, 1920); *Bhore Committee* (1946); *John Bryant* ("Health and the Developing World").
- ▶ **Dated figures for the transition:** 10–20% of people in developing countries had ready access to any health service; infant deaths 60–250 per 1000 live births; life expectancy 30% lower (Park 28e, Ch 1, p 21).
- ▶ **Terminology chain:** public health → community health (EURO symposium 1966; WHO Expert Committee 1973 called the term ambiguous without qualification).
- ▶ **Jakarta Declaration, July 1997** (fourth international conference on health promotion) — thirteen fundamental conditions and resources for health; "above all, poverty is the greatest threat to health".

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park's 28th edition heads the phase "Social engineering phase (1960–1980)" while the running text says public health entered this phase "in the 1960s" — write **1960–1980**; the four-phase scheme is Park's, and other textbooks divide the history of public health differently.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 1 (Man and Medicine: Towards Health for All), pp 17–23; Ch 2 (Concept of Health and Disease), pp 49–66; Ch 10 (Health Care of the Community), p 720. Sunder Lal — Textbook of Community Medicine, Ch 2 (Relationship of Social and Behavioural Sciences (Humanities) to Health and Disease), pp 80–81; Ch 4 (Principles of Health Promotion and Education), pp 143–154. Oxford Textbook of Public Health 7e, Ch 47 (Health promotion, health education, and the public's health), pp 1733–1774. Coverage note: no captioned plate in the indexed figure library matched the phases of public health, so the figures are given as a "diagram to draw" label box and Graphviz flowcharts rather than an embedded plate.



Social factors influencing the health of people

Asked in: Paper I Nov 2017 — RGUHS MD Community Medicine

Definition — **Social factors** are the customs, beliefs, institutions, relationships and material conditions of the society a person belongs to, which act on health directly and by governing exposure to biological, physical and behavioural hazards. The World Health Organization calls them the **social determinants of health** — "the conditions in and under which people are born, grow, live, work and age, and the broader set of forces and systems that shape the conditions of daily life" — and terms them the "**causes of the causes**" (Sunder Lal, Ch 1, pp 41–42). Health cannot be isolated from its social context: over the last few decades social and economic factors have been shown to have as much influence on health as medical interventions, and they bear directly on the incidence, course and outcome of communicable and non-communicable disease alike (Park 28e, Ch 12, p 811).

How social factors act on health

Park's model holds that health is influenced by **four sets of variables** — **individual predispositions, ecological predispositions, current circumstances and opportunities**. These four are themselves driven by the major sources of social change: the **economic, political, educational and other systems** of the society. The relationship is a **loop, not a one-way chain** — the health status of the people feeds back into the social structure, which in turn alters the predisposing variables (Park 28e, Ch 12, pp 811–812). Poverty, malnutrition, poor sanitation, lack of education, inadequate housing, unemployment, poor working conditions and cultural and behavioural factors all predispose to ill-health (Park 28e, Ch 12, p 811).

Socio-economic factors and their health effects

Social factor	How it operates	Health consequence
Social class (composite of income, occupation, education)	Bound up with economic status, way of life, attitudes, expectations and exposure to stress; determines the external resources and internal mechanisms available for dealing with a health problem	Upper classes have longer life expectancy, lower mortality and better nutritional status ; coronary heart disease, hypertension and diabetes are commoner in class I, while skin, eye and ear disease, diarrhoea and dysentery are commoner in lower classes; infant, general and maternal mortality all track social class (Park 28e, Ch 12, pp 831, 834)
Economic status / income	Determines family size, purchasing power, standard of living, pattern of disease, deviant behaviour and the ability to seek health care	Economic progress has been the major factor in reducing morbidity and raising life expectancy in developing countries; ironically affluence itself contributes to coronary heart disease, diabetes and obesity in the upper groups (Park 28e, Ch 2, p 32)
Standard and level of living	"Level of living" (United Nations) has nine components : health, food consumption, education, occupation and working conditions, housing, social security, clothing, recreation and leisure, and human rights	Health is regarded as the most important component, because its impairment always means impairment of the level of living (Park 28e, Ch 2, p 28)
Education and literacy	Measures the inculcation of values, knowledge and achievement; governs the ability to understand and act on health advice	The ignorant and the illiterate have great difficulty in pursuing measures conducive to good health (Park 28e, Ch 12, pp 832, 834)
Occupation	Determines economic rewards, extent of authority, extent of obligations and degree of social status; also fixes the specific hazard faced and may alter sleep, alcohol, smoking and shift patterns	Coal-mine workers suffer silicosis, sedentary workers face heart disease (Park 28e, Ch 3, p 89; Ch 12, p 831)
Housing and environmental sanitation	Rural houses are typically <i>katcha</i> , damp, ill-lighted and ill-ventilated, with no windows and no separate kitchen, latrine, bathroom or drainage; humans and animals may share one roof	Overcrowding and poor lighting favour respiratory infection and tuberculosis; open-field defecation and refuse thrown before the house pollute water and soil and breed flies and mosquitoes (Park 28e, Ch 12, pp 829-830)
Rural / urban residence	Rural and urban people differ in lifestyle, opinions and values, and in access to services; "undoctored" areas are those where a substantial proportion of people live	Health and sickness status differ measurably between rural and urban populations (Park 28e, Ch 12, pp 832, 834)
Social security	"Security that society furnishes through appropriate organisation against certain risks to which its members are exposed" — covering sickness, invalidity, maternity, old age and death , through social insurance and social assistance	Protects against catastrophic loss of income during illness; the Indian industrial worker is covered by the Employees State Insurance Act 1948 and allied laws (Park 28e, Ch 12, pp 845-846)

Social factor	How it operates	Health consequence
Social change, migration and urbanisation	Culture contact through trade and commerce, industrialisation, propagation of religion, education and conquest; rural-to-urban migration diminishes the traditional family support system	Transitional societies show a clash of cultures; young people are caught between the parental and the mass-media culture (Park 28e, Ch 12, pp 815-816)
Social stress	Conflict between new opportunities and frustrations — migration, exposure to culturally alien ideas, tourism, obsolete skills, and commercially encouraged unrealistic aspirations	Psychological stress and poor coping are implicated as contributory factors in virtually all disease, with direct links to hypertension and coronary heart disease (Park 28e, Ch 12, pp 815-816)
Social pathology and social problems	Poverty, crime, delinquency and vagrancy; alcoholism, drug dependence, sexually transmitted disease and prostitution — individual problems become social problems when they affect the welfare or safety of the whole group	Uncovered by social surveys ; solved by social welfare programmes, social assistance and social legislation, not by clinical medicine alone (Park 28e, Ch 12, pp 816, 839)

Socio-cultural factors and their health effects

Cultural domain	Belief or practice	Health consequence
Concept of aetiology	Supernatural causes — wrath of gods and goddesses (chickenpox as <i>Chhoti Mata</i>), breach of taboo, past sins, evil eye, spirit or ghost intrusion; physical folk causes — weather, impure water, "impure blood"	Drugs considered harmful, cases not notified, <i>pujas</i> and exorcists sought instead of treatment — delayed care and failure of notification (Park 28e, Ch 12, p 829)
Environmental sanitation	Open-field defecation regarded as harmless; refuse and cow dung accumulated before the house; the well and tank used for bathing, washing clothes and watering animals; "holy" river water drunk and bottled for distribution	Faecal pollution of water and soil, fly and mosquito breeding — cholera and gastroenteritis epidemics, guinea-worm disease (Park 28e, Ch 12, p 829)
Food habits	Vegetarianism, religious avoidance of pork and beef, the hot/cold food concept, fasting, adulteration of milk, eating from common utensils, women eating last and least	Restricts protective foods; the hot/cold concept is met by the physician during treatment; women eating last and poorly perpetuates female undernutrition and anaemia (Park 28e, Ch 12, pp 829-830)
Mother and child health	Customs classed as good, bad, unimportant and uncertain — good: prolonged breast-feeding, oil bath, massage, sun exposure; bad: food taboos in pregnancy (egg, meat, fish, milk, leafy vegetables), delivery by the untrained <i>dai</i> , branding of skin, opium and drastic purgatives; uncertain: <i>kajal</i> to the eyelids	Bad customs cause maternal and neonatal sepsis and undernutrition; <i>kajal</i> has been blamed for transmitting trachoma and other eye infections (Park 28e, Ch 12, p 830)

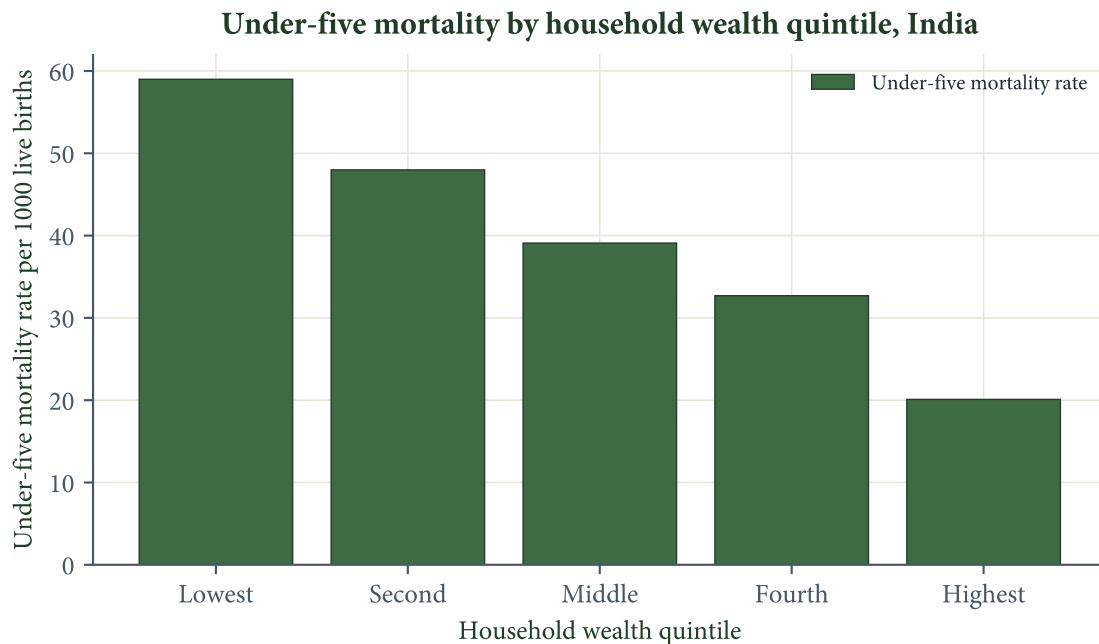
Cultural domain	Belief or practice	Health consequence
Personal hygiene	<i>Pan</i> chewing with lime and tobacco; smoking the hubble-bubble; smoking with the burning end of the cigar inside the mouth (villagers of Andhra Pradesh); shaving by the barber without sterilising instruments; <i>purdah</i> ; sleeping on the ground; not wearing shoes	Oral cancer from <i>pan</i> and reverse smoking; tuberculosis spread by shared hubble-bubble and reported to be high among those observing <i>purdah</i> ; hookworm from bare feet; insect bites from sleeping on the ground (Park 28e, Ch 12, p 830)
Religion and caste	The Indian caste system is a "closed class" — no mobility between classes; each caste governed by rules on endogamy, food taboos and ritual purity, and follows a definite occupation; caste is supported by religion	Little social mobility makes reform difficult without meeting resistance of the people; caste correlates with a host of other socio-cultural factors and with occupation-linked risk (Park 28e, Ch 12, pp 825, 831, 833)
Sex and marriage	Universal and early marriage; the family regarded as incomplete without a male child; menstrual seclusion in some groups	Bears on age at first pregnancy, the sex ratio, the population problem and family-planning acceptance (Park 28e, Ch 12, pp 830-831)
The family	A biological unit (shared pool of genes), a social unit (common physical and social environment), a cultural unit, an epidemiological unit and the unit for social services and comprehensive care; broken and problem families carry backwardness, poverty, illness, emotional instability and marital disharmony	Children reared in problem families become victims of prostitution, crime and vagrancy; the family is the fulcrum of health services — "the secret of national health lies in the homes of the people" (<i>Florence Nightingale</i>) (Park 28e, Ch 12, pp 825-829)

Evidence of the social gradient in India

Social characteristic	Infant mortality rate (per 1000 live births)	Under-five mortality rate (per 1000 live births)
Urban residence	26.6	31.5
Rural residence	38.4	45.7
Mother with no schooling	49.0	60.3
Mother with 12 or more years of schooling	22.0	24.9
Scheduled tribe	41.6	50.3
Scheduled caste	40.7	48.9
Other backward class	34.1	40.5
"Other" (none of the above)	28.0	32.8
India — all	35.2	41.9

Source: National Family Health Survey-5, 2019-21 (NFHS-5 India Report, Tables 7.1-7.2); the wealth-quintile rows are reproduced in Park 28e, Ch 12, p 834.

The gradient extends to the distribution of wealth itself: 45.5 per cent of the urban population but only 8.1 per cent of the rural population falls in the highest wealth quintile (Park 28e, Ch 12, p 833; NFHS-5, 2019-21).



National Family Health Survey-5, 2019-21 (reproduced in Park 28e, Ch 12, p 834). A child in the poorest fifth of households is about three times as likely to die before the fifth birthday as a child in the richest fifth.

Why social class differences in health arise

Park lists six mechanisms behind the class gradient (Park 28e, Ch 12, p 834):

- ▶ **Physical environment** — differences in housing, water supply and access to clean air; the upper social hierarchy enjoys better physical facilities.
- ▶ **Differences in services provided** — availability of services differs between social groups, and the relatively "undoctored" areas are those where a substantial proportion of people live.
- ▶ **Material resources** — income, wealth and possession of the tools that promote or help achieve health.
- ▶ **Genetic endowment** — people of one social class tend to marry within it, so differences in genetic endowment also track class.
- ▶ **Educational status** — the ignorant and illiterate cannot easily pursue health-promoting measures.
- ▶ **Attitude to disease** — some regard illness as punishment and others as natural causation; some self-diagnose and others seek early medical aid.

Measuring social position

Socio-economic status is "the position an individual or family occupies with reference to the prevailing average standards of cultural and material possessions, income, and participation in group activity of the community" (Park 28e, Ch 12, p 832). *Hollingshead* (USA) used education, occupation and residential address; *Kuppuswamy* built an Indian urban scale on **education, occupation and income**;

Pareek and Kulshrestha prepared scales for rural and urban use; and the National Family Health Survey uses a **wealth index** derived by principal component analysis from household assets and housing characteristics (Park 28e, Ch 12, pp 832-833).

Kuppuswamy total score (education 1-7 + occupation 1-10 + monthly family income 1-12)	Socio-economic class
26-29	Upper (I)
16-25	Upper-middle (II)
11-15	Lower-middle (III)
5-10	Upper-lower (IV)
Below 5	Lower (V)

Limitations of classifying by occupation alone (Park 28e, Ch 12, p 832):

- **Heterogeneous grouping** — each class lumps unrelated occupations that also differ in education and family background.
- **Occupational mobility** — people change occupation, and occupations themselves are promoted or demoted in the social hierarchy, invalidating comparison over time.
- **Women** — some occupations are exclusively male or female, and the same job title carries a different standing for a man and a woman.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Diagram to reproduce by hand** — Park's model of social factors influencing health (Park 28e, Ch 12, Fig 1, p 812). Draw a central circle labelled **HEALTH**. Four boxes on the left feeding arrows into it, labelled **Individual predispositions**, **Ecological predispositions**, **Current circumstances**, **Opportunities**. A band above these four labelled **SOURCES OF SOCIAL CHANGE** — **economic system**, **political system**, **educational system**, **other systems**, with arrows down into the four boxes. A long **feedback arrow** running from **HEALTH** back up to the band, labelled *health status feeds back into social structure*.

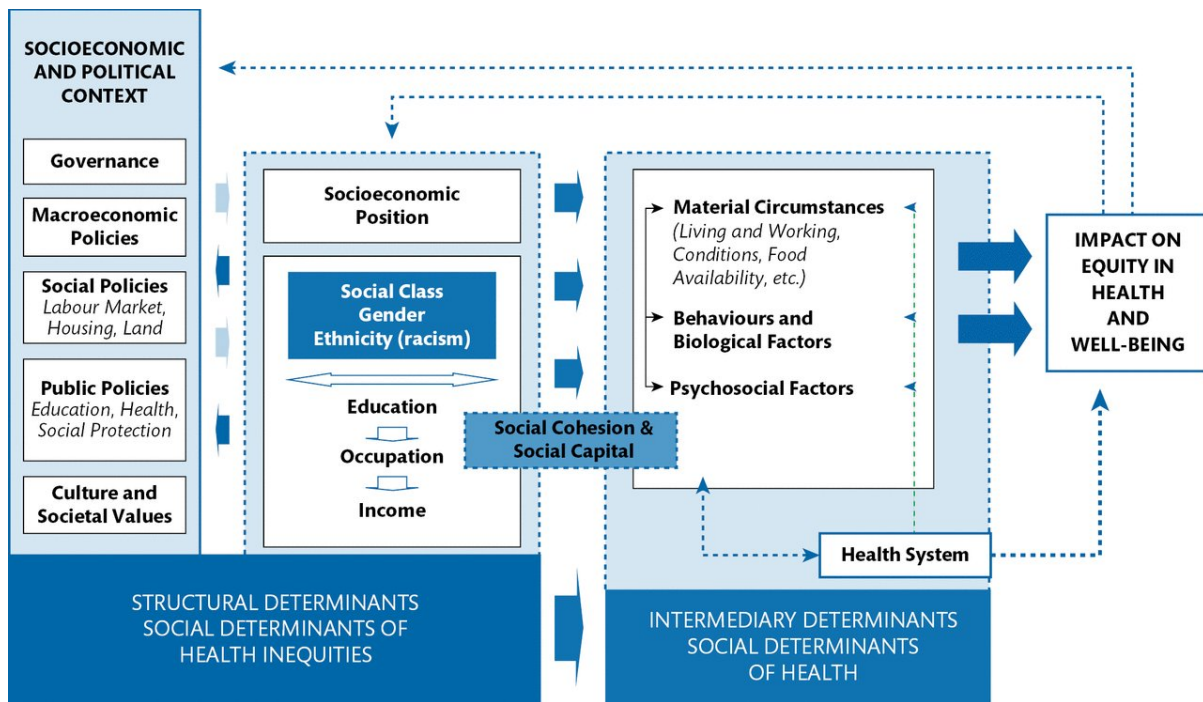
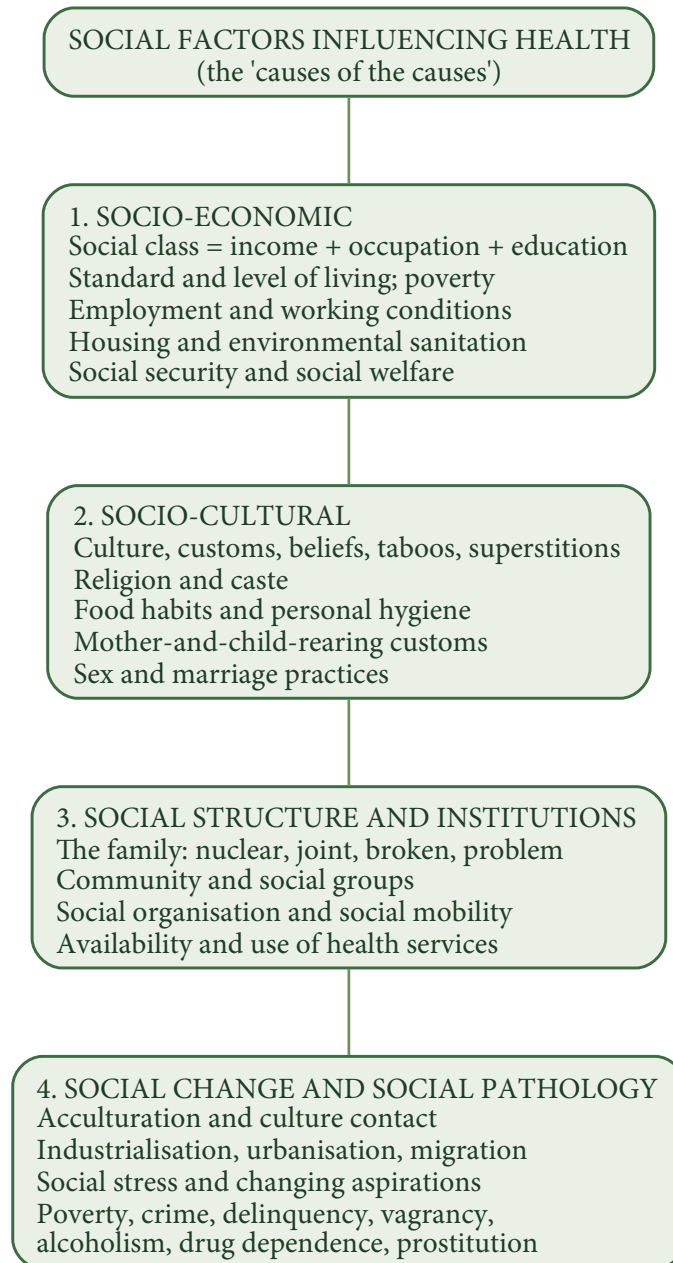
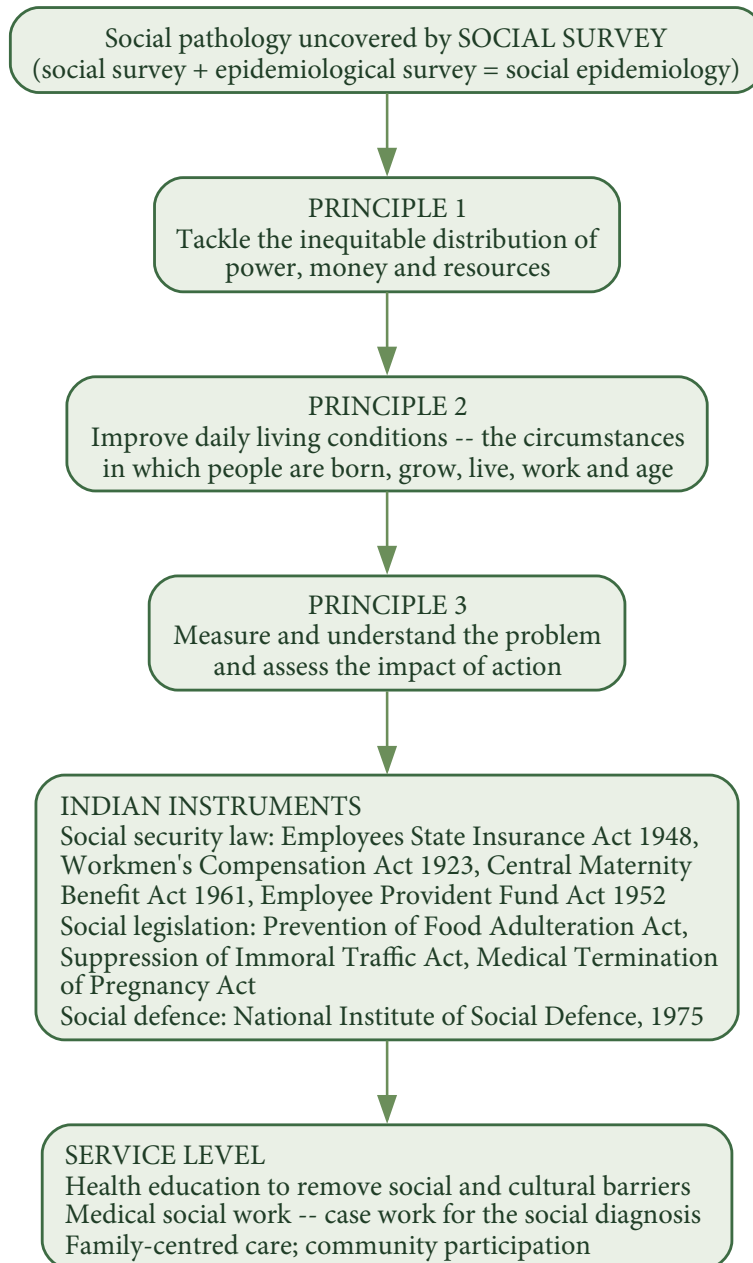


Fig 2.1 — 1 Social determinants of health and health inequities. Adapted with pe (Oxford public health 7th ed, p.299)



Classification of the social factors that influence the health of people



Acting on the social determinants of health — from social survey to service delivery

High-yield exam pointers

- ▶ **Definition to write first:** social determinants of health = the conditions in which people are born, grow, live, work and age, plus the forces shaping them — the "causes of the causes" (WHO).
- ▶ **Park's four sets of variables (Fig 1, p 812):** individual predispositions, ecological predispositions, current circumstances, opportunities — driven by the economic, political and educational systems, with feedback from health.
- ▶ **Nine components of the level of living (United Nations):** health, food consumption, education, occupation and working conditions, housing, social security, clothing, recreation and leisure, human rights.
- ▶ **Six mechanisms of the class gradient:** physical environment, services provided, material resources, genetic endowment, educational status, attitude to disease.

- **Dated Indian figures to quote:** NFHS-5 (2019-21) — under-five mortality **59.0** in the lowest wealth quintile versus **20.1** in the highest; infant mortality **49.0** with an unschooled mother versus **22.0** with 12 or more years of schooling; rural infant mortality **38.4** versus urban **26.6**.
- **Kuppuswamy's scale = education + occupation + income**, total score 3-29, five classes from Upper (I) 26-29 to Lower (V) below 5.
- **Two named quotations that earn marks:** *Florence Nightingale* — "the secret of national health lies in the homes of the people"; *Chadwick's* social survey led to the General Board of Health, 1848.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park 28e states that infant mortality in India was lowest among Christians, higher among Hindus and highest among Muslims; the National Family Health Survey-5 (2019-21) does not bear this out, giving infant mortality of 36.0 among Hindus and 33.3 among Muslims, so quote the dated survey figure rather than the textbook's older statement. The three principles of action are those of the World Health Organization Commission on Social Determinants of Health, *Closing the Gap in a Generation* (2008), as set out in the Oxford Textbook of Public Health 7e; and Kuppuswamy's income slabs must be re-indexed to the Consumer Price Index each year, Park 28e printing the 2012, 2016 and 2024 slabs side by side.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 12 Medicine and Social Sciences, pp 811-846 — social context of medicine, culture and acculturation, social stress, social pathology and social surveys, the family, cultural factors in health and disease, social class, socio-economic status scales, medical social work, social security; Ch 2 Concept of Health and Disease, pp 28-32 and p 56 psychosocial environment; Ch 3 Principles of Epidemiology and Epidemiologic Methods, pp 72, 89). Sunder Lal — Textbook of Community Medicine (Ch 1 Concept of Health and Disease, pp 40-42, social determinants of health; Ch 2 Relationship of Social and Behavioural Sciences to Health and Disease, p 81). Oxford Textbook of Public Health 7e (Ch 2.1 Determinants of health: overview — social determinants framework figure, p 299; Ch 8.8 Dental public health, pp 2462-2465, which sets out the Commission on Social Determinants of Health framework and its three principles of action). National Family Health Survey-5, India Report, 2019-21 (Tables 7.1-7.2, early childhood mortality by background characteristics). World Health Organization Commission on Social Determinants of Health, *Closing the Gap in a Generation*, 2008.



Socio-economic status scale

Asked in: Paper I 15-Nov-2022 — RGUHS MD Community Medicine

Definition — Socio-economic status (SES) is the position that an individual or family occupies with reference to the prevailing average standards of cultural and material possessions, income, and participation in group activity of the community (Park 28e, Ch 12, pp 832-833). A **socio-economic status scale** is a composite weighted instrument that converts a small set of measurable proxies — characteristically the education and occupation of the head of the household and the family income — into a single score and an ordinal social class (I to V). Social status may be inherited, but in modern society it is achieved on the basis of occupation, income, type of housing and neighbourhood, membership of associations and organisations, and material possessions.

Why socio-economic status is measured

- ▶ **A host factor of disease** — Park lists socio-economic status, education, occupation, stress, marital status and housing among the *social and economic* host characteristics (Park 28e, Ch 2, p 55).
- ▶ **A graded determinant, not a threshold one** — the gradient in life expectancy, mortality and morbidity runs across *all* levels of the socio-economic hierarchy and persists where material deprivation is absent (Oxford Textbook of Public Health 7e, Ch 18, pp 714-715).
- ▶ **Service targeting** — the declared objective of SES assessment in Indian practice is to identify deprived households, vulnerable populations and priority groups for health services (Sunder Lal, Ch 2, p 73), and to classify a family as below poverty line (BPL) or above poverty line (APL) (Sunder Lal, Ch 2, p 74).
- ▶ **Research use** — the standard descriptor, matching variable, stratifier and confounder in community-based studies; socio-economic status is one of the eight PROGRESS equity stratifiers (Place of residence, Race/ethnicity, Occupation, Gender, Religion, Education, Socioeconomic status, Social capital) (Oxford 7e, Ch 10, pp 423-424).
- ▶ **Equity monitoring** — national surveys disaggregate every major indicator by SES, exposing the social gradient shown in the chart below.

Scales available — which to use where

Scale	Setting	Variables used	Output
Kuppuswamy's scale	Urban India	Education of head + occupation of head + total monthly family income	Total score 3-29 → class I to V
B.G. Prasad's classification (1961)	Rural and urban	Per capita monthly income only	Five income bands → class I to V
Udai Pareek's scale	Rural India	Caste, occupation, education, land, social participation, family size, house, farm power, material possession	Total score → class I to V
Pareek and Kulshrestha scale	Rural and urban	Multi-item	Social class

Scale	Setting	Variables used	Output
<i>Hollingshead's scale</i> (USA)	Urban, Western	Education, occupation, residential address	Social class
Registrar General's classification (Britain)	Census populations	Occupation alone	Five social classes

Historical anchor — J.H.C. Stevenson of the Registrar General's office arranged the 1911 British census into five occupation-based social classes to display disease prevalence by social level; Kuppuswamy devised his Indian scale in 1959 and published it in 1962 (Park Plus). India has **no satisfactory occupational class system** of its own, which is why composite multi-variable scales are needed here (Park 28e, Ch 12).

Kuppuswamy's socio-economic status scale (urban)

Three components are scored and summed; the income limits alone are recalculated periodically against the All-India Average Consumer Price Index for Industrial Workers (Park 28e, Ch 12, pp 832-833; Sunder Lal, Ch 2, p 74).

(A) Education of head	Score	(B) Occupation of head	Score
Profession or honours	7	Legislators, senior officials and managers	10
Graduate or postgraduate	6	Professionals	9
Intermediate / post-high-school diploma	5	Technicians and associate professionals	8
High school certificate	4	Clerks	7
Middle school certificate	3	Skilled workers, shop and market sales workers	6
Primary school certificate	2	Skilled agricultural and fishery workers	5
Illiterate	1	Craft and related trade workers	4
		Plant and machine operators and assemblers	3
		Elementary occupation	2
		Unemployed	1

(C) Total monthly family income (₹) — 2012	2016	2024	Score
≥ 30,375	≥ 51,646	≥ 2,13,814	12
15,188 – 30,374	25,811 – 51,645	1,06,850 – 2,13,813	10
11,362 – 15,187	19,351 – 25,809	80,110 – 1,06,849	6
7,594 – 11,361	12,890 – 19,350	53,361 – 80,109	4

(C) Total monthly family income (₹) — 2012	2016	2024	Score
4,556 – 7,593	7,725 – 12,889	31,978 – 53,360	3
1,521 – 4,555	2,586 – 7,724	10,703 – 31,977	2
< 1,520	< 2,585	< 10,702	1

Total score (A + B + C)	Socio-economic class
26 – 29	Upper — class I
16 – 25	Upper middle — class II
11 – 15	Lower middle — class III
5 – 10	Upper lower — class IV
Below 5	Lower — class V

(Park 28e, Ch 12, Table 2, pp 832-833; class labels as in Sunder Lal, Ch 2, Table 2.4, p 74.)

Modified Prasad's classification (per capita monthly income)

Uses **per capita** — not total family — monthly income, so it serves rural and urban settings alike and is the scale of choice when only income data exist (Sunder Lal, Ch 2, p 74).

Social class	Prasad 1961 (₹ per capita per month)	Modified for May 2021 (₹)
I	100 and above	≥ 7,863
II	50 – 99	3,931 – 7,862
III	30 – 49	2,359 – 3,930
IV	15 – 29	1,179 – 2,358
V	Below 15	< 1,179

(Sunder Lal, Ch 2, Table 2.6, p 75, after Majhi & Bhatnagar, updated for base year 2016.)

Limits are refreshed by multiplying the 1961 figures by a **multiplication factor** = $4.93 \times 4.63 \times \text{current All-India Consumer Price Index for Industrial Workers} \div 100$ (4.63 being the linking factor between the 1982 and 2001 index series) and rounding to the nearest ₹10 (Park Plus).

Udai Pareek's scale (rural)

Nine components, each weighted; the total is graded into five classes (Sunder Lal, Ch 2, Table 2.5, pp 74-75).

Component	Range of weighted score	Component	Range of weighted score
(A) Caste — scheduled caste 1 → dominant caste 6	1 – 6	(F) Family members — up to 5 = 1; above 5 = 2	1 – 2
(B) Occupation — none 0, labourer 1, caste occupation 2, business 3, independent profession 4, cultivation 5, service 6	0 – 6	(G) House — no house 1, katcha 2, mixed 3, pucca 4, mansion 6	1 – 6
(C) Education — illiterate 0 → graduate and above 6	0 – 6	(H) Farm power — no draught animal 1, 1-2 animals 2, 3-4 animals 3, 5-6 animals or tractor 6	1 – 6
(D) Land — no land 0 → 20 acres and above 6	0 – 6	(I) Material possession — bullock cart / cycle / radio / chairs 1 each, improved agricultural equipment 2, none 0	0 – 6
(E) Social participation — none 0, member of one organisation 1, of more than one 2, office holder 3, wider public leader 6	0 – 6	Grading: > 43 = I; 33-42 = II; 24-32 = III; 13-23 = IV; < 13 = V	—

Asset- and deprivation-based measures used in Indian surveys

Measure	Basis	Categories
Wealth index (National Family Health Survey)	30-odd household assets and housing characteristics — electrification, windows, drinking-water source, toilet, flooring, walls, roofing, cooking fuel, house ownership, persons per sleeping room, bank/post-office account, and ownership of listed durables; weights derived by principal components analysis , standardised to mean 0 and standard deviation 1, households ranked and the population split into quintiles	Lowest, second, middle, fourth, highest — 20% of the <i>national</i> population in each (Park 28e, Ch 12, pp 832-833)
Standard of living index (SLI)	Score 0-40 from drinking water, type of house, lighting, cooking fuel, toilet and ownership of listed items	Low ≤ 9; medium > 9 to ≤ 19; high > 19 — used in District Level Household Surveys and NFHS (Sunder Lal, Ch 2, pp 74-75)
Socio-Economic and Caste Census (SECC) 2011	Door-to-door enumeration across 640 districts ; 14 automatic-exclusion parameters, 5 automatic-inclusion parameters, 7 deprivation criteria (D1-D7) in rural areas; residential, social and occupational vulnerability with 11 occupational categories in urban areas	Rank order for the BPL list runs from the largest to the smallest number of deprivations (Sunder Lal, Ch 2, p 75)
Household consumer expenditure	National Sample Survey Organisation — thin annual sample plus a quinquennial large-sample survey; sums the monetary value of all goods and services consumed by the household in the reference period	Expenditure classes (Sunder Lal, Ch 2, p 75)

Because the wealth index quintiles are **national**, they expose the urban-rural divide directly: at NFHS-5 (2019-21), 45.5% of urban but only 8.1% of rural people fell in the highest quintile, and half the population of Bihar and Jharkhand fell in the lowest (Park 28e, Ch 12, pp 832-833).

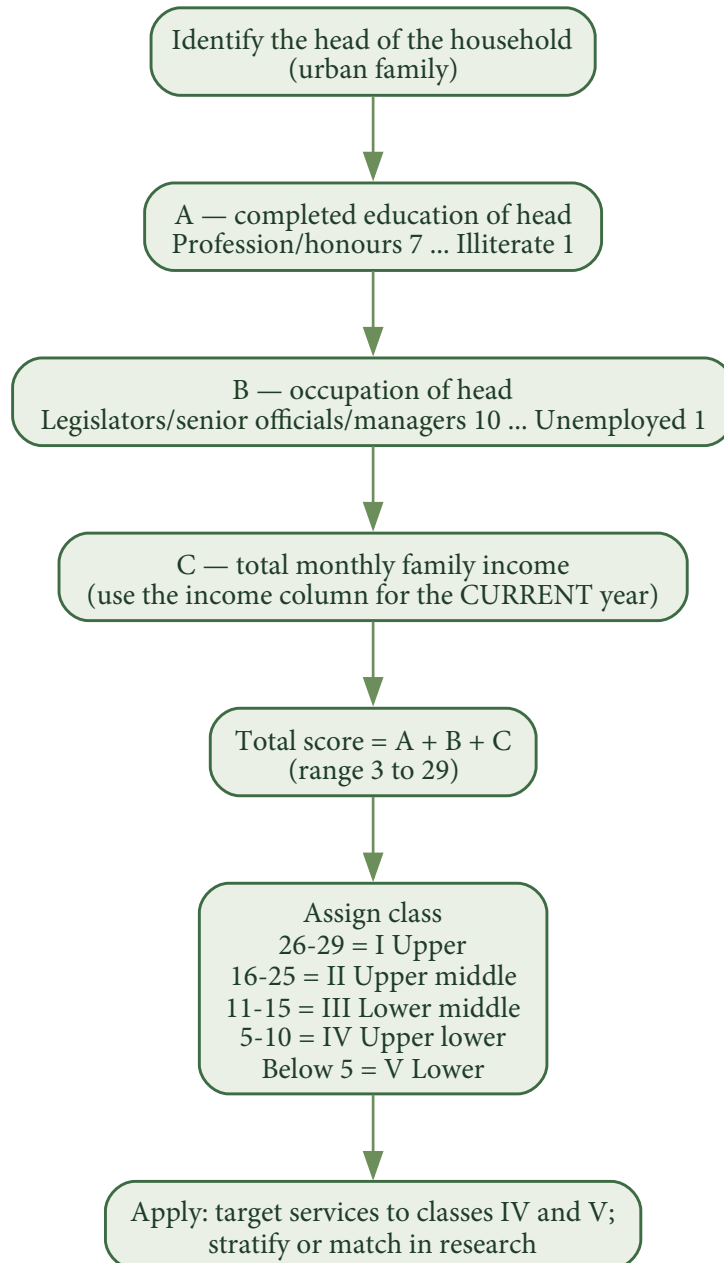
Limitations and cautions

- ▶ **Heterogeneous grouping** — each occupation-defined class lumps together dissimilar occupations whose holders also differ in education and family background; the categorisation is one of convenience only (Park 28e, Ch 12).
- ▶ **Occupational mobility** — people change occupations, and occupations themselves rise or fall in the hierarchy, invalidating comparison of the same class across time (Park 28e, Ch 12).
- ▶ **Classification of women** — the same job title carries different standing for men and women, so assigning a woman a class alongside men is unreliable (Park 28e, Ch 12).
- ▶ **Income is concealed or misreported** and is the hardest of the three variables to obtain truthfully (Park 28e, Ch 12; Sunder Lal, Ch 2, p 74).
- ▶ **Rapid obsolescence** — an income-based scale drifts with inflation and is invalid unless CPI-updated; always quote a scale with its update year.
- ▶ **No single best measure** — education, occupation and income usually point the same way but differ in the strength of their association with health, because the underlying causal mechanisms differ (Oxford 7e, Ch 9, pp 389-390).
- ▶ **Setting mismatch** — Kuppuswamy's income bands and occupational groups fit rural agrarian families poorly, while *Pareek's* caste, land and draught-animal items have no urban meaning.
- ▶ **Wealth is not health** — obesity, hypertension, coronary heart disease and diabetes are emerging in the transitional Indian middle class while poor families still carry malnutrition, diarrhoeal disease, acute respiratory infection, tuberculosis and worm infestation (Sunder Lal, Ch 2, p 74).

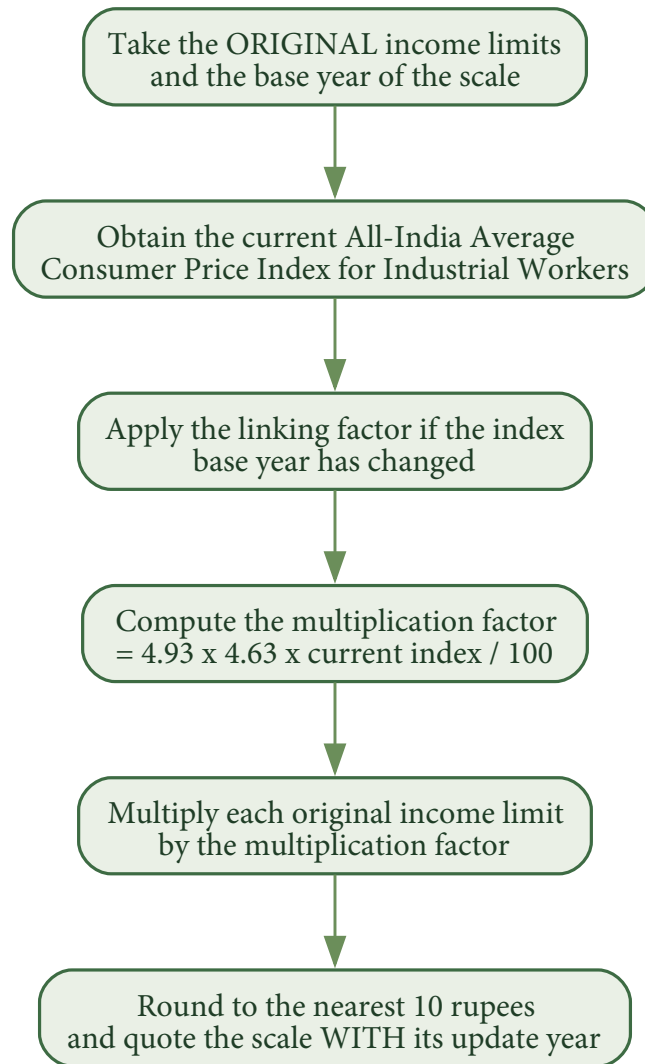
DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Drawn figure — "Three-variable model of socio-economic class (Kuppuswamy)". Three input boxes feed one summing box, which feeds a five-tier pyramid. Labels:

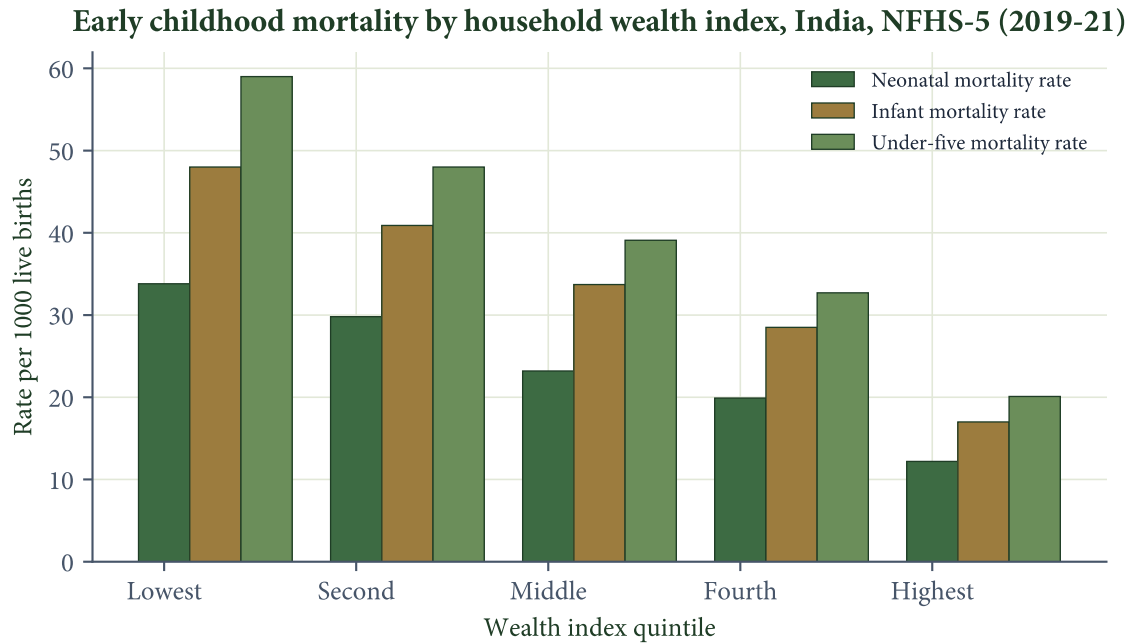
- ▶ Education of head — score 1 to 7
- ▶ Occupation of head — score 1 to 10
- ▶ Total monthly family income — score 1 to 12 (CPI-updated)
- ▶ Total score = A + B + C (range 3 to 29)
- ▶ Pyramid tiers, apex downwards: I Upper (26-29), II Upper middle (16-25), III Lower middle (11-15), IV Upper lower (5-10), V Lower (< 5)



Stepwise assignment of an urban family to a socio-economic class using Kuppuswamy's scale



Keeping the income limits of an Indian socio-economic status scale current with the consumer price index



The social gradient that a socio-economic status scale is meant to capture: under-five mortality is nearly three times higher in the lowest than in the highest wealth quintile. Source: Park 28e, Ch 12, Table 4 (National Family Health Survey-5, 2019-21).

High-yield exam pointers

- ▶ Kuppuswamy = urban; education + occupation + income; total 3-29; classes 26-29 / 16-25 / 11-15 / 5-10 / < 5. Its income is whole-family per month; Prasad's is per capita per month — the wrong denominator is the commonest error.
- ▶ Score ceilings: education 7, occupation 10, income 12; each floor is 1.
- ▶ Park 28e prints Kuppuswamy income limits for 2012, 2016 and 2024 (2024 top band ≥ ₹2,13,814); modified Prasad's for May 2021, class I ≥ ₹7,863 per capita per month.
- ▶ Udai Pareek = rural, nine components including caste, land, farm power; grading > 43 = I down to < 13 = V — score-based, not income-based.
- ▶ Standard of living index 0-40: low ≤ 9, medium 10-19, high > 19. NFHS wealth index: principal components analysis, national quintiles of 20% each.
- ▶ SECC-2011: 640 districts; 14 exclusion, 5 inclusion, 7 deprivation criteria.
- ▶ Dated figure to quote: NFHS-5 (2019-21) — 45.5% urban vs 8.1% rural in the highest wealth quintile.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: only the income bands change at each revision of these scales — the education, occupation and asset scores stay fixed — so always attach the update year to whichever version you write. The caste item in *Udai Pareek's* rural scale reflects 1960s Indian sociology and is increasingly omitted or replaced in current research practice.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, Labour Bureau CPI-IW series, NITI Aayog National Multidimensional Poverty Index 2023, NFHS-6 (2023-24), PM-JAY / National Health Authority

Programme and policy updates (India)

- The SECC-2011 criteria are no longer only a BPL-listing exercise — they are the live eligibility rule for **Ayushman Bharat PM-JAY**: a rural household qualifies on **any one of six** SECC deprivation criteria, an urban household on **any one of 11** SECC occupational categories, with legacy RSBY and state-added beneficiaries additionally included. This covers **107.4 million households (535 million people — the socio-economically bottom 40%)** at ₹5,00,000 per household per year for secondary and tertiary hospitalisation, and was in force in 33 of the then 36 states and union territories as of 2021 (not West Bengal, Odisha or Delhi). It is the answer to quote when asked how an SES classification is actually *used* in Indian programme practice. [PMID 41393311]
- Deprivation-based targeting does not by itself deliver equity: within the same bottom 40%, only 7% of empanelled hospitals in Aspirational Districts offer cardiothoracic and vascular surgery against 17% elsewhere, and men raised 23.6% more surgical claims than women once obstetric and gynaecological procedures were excluded — hence the recommendation to enrol the *lower quintiles within* the bottom 40% rather than treat the eligible group as homogeneous. [PMID 41393311]

Current national data

- **NFHS-6 (2023-24) fact sheets have been released by the MoHFW** — this is the round to quote from now in place of NFHS-5. It reports persistent undernutrition, rising urban overweight, **widening rural-urban and tribal inequities** and falling exclusive breastfeeding, alongside gains in immunisation and antenatal coverage; note that **mortality indicators are absent from this round**, so wealth-quintile mortality gradients must still be cited from NFHS-5. [PMID 42554934]
- **NITI Aayog's National Multidimensional Poverty Index (MPI Progress Review 2023)** is the deprivation-based classification to know alongside the SECC: it is built on the Alkire-Foster method with OPHI and UNDP as technical partners — three dimensions and **12 indicators** (3 health, 2 education, 7 standard of living), each dimension weighted one-third and each indicator equally weighted within its dimension, a household counted poor at a weighted deprivation score of **33.33% or more**, and $MPI = \text{headcount ratio (H)} \times \text{intensity (A)}$. Multidimensionally deprived households fell from **24.85% in 2015-16 to 14.96% in 2019-21**. [PMID 41291549]
- Across five NFHS rounds (1993-2021) the wealth gradient in child death narrowed for the later deaths but **widened for the earliest ones**. Between the lowest and highest wealth quintiles,

absolute inequality fell for late neonatal (12.0 to 2.9), post-neonatal (26.5 to 9.7) and child mortality (39.5 to 8.4), but for **early neonatal mortality both absolute (18.3 to 18.4) and relative (1.9 to 2.9) wealth inequality increased**. The lowest quintile's 2021 early neonatal rate (27.8 per 1000) still exceeded the highest quintile's 1993 rate, and at current rates of decline the SDG early-neonatal target of 7 per 1000 is out of reach for every wealth quintile except the highest. [PMID 40316411]

- The same paper records that the wealth-index asset list is **not identical across NFHS rounds** — mobile phones are counted in the 2021 round but not in 1993 — while stating explicitly that this does not undermine comparability between rounds. If you add the usual caution about comparing quintiles across time, present it as your own gloss and say the authors judged comparability unaffected. [PMID 40316411]

Guideline and legislative updates

- A **third linking factor, 2.88, now sits in the CPI-IW chain, so the older two-factor formula is obsolete**. The Labour Bureau launched the CPI-IW series on base 2016 = 100 with effect from September 2020 and issued the 2001-to-2016 linking factor of 2.88 on 21 October 2020; the correct multiplier is therefore $\text{current CPI-IW} \times 4.93 \times 4.63 \times 2.88 \div 100$, and the 4.93×4.63 version printed in older texts under-states every income limit. Published anchors: CPI-IW 134.2 (April 2023) gives *Prasad's* class I $\geq ₹8,822$; CPI-IW 138.9 (January 2024) gives $\geq ₹9,131$. [PMID 38249702] [PMID 41451071] [PMID 40949512]
- Applying that equation to the latest published index — **All-India CPI-IW 151.9 for June 2026 (base 2016 = 100)** — yields a multiplication factor of about 99.9, putting modified *Prasad's* class I at roughly ₹9,990 per capita per month; multiply the original 1961 limits of 100 / 50 / 30 / 15 by the same factor for classes II to V and round each lower limit to the nearest ₹10. [Labour Bureau 2026-Jun] [PMID 41451071]
- **Centre-wise CPI-IW may be used instead of the All-India index, and the choice shifts the class boundaries materially**. For October 2023, the Coimbatore centre index (138.0) with its own linking factors ($2.6 \times 4.49 \times 4.93$) gives a factor of 79.42 and class I $\geq ₹7,940$, against a factor of 90.98 and $\geq ₹9,100$ on the All-India index for the same month. The published seven-step template also settles the rounding convention: round each lower limit to the nearest ₹10, then set the upper limit of a class at one rupee below the next class's lower limit. [PMID 41451071]
- New instruments are being psychometrically validated against the classical scales. **Solomon's Socio-Economic Status Index (SESI)**, tested in 1,247 households across India, weights nine dimensions — housing 20%, education 15%, occupation 15%, land 10%, vehicle 10%, business 10%, durable assets 10%, locality 5% and **digital access 5%** — with Cronbach's α 0.84, test-retest r 0.89, CFI 0.94 and RMSEA 0.051, correlating 0.72 with the modified *Kuppuswamy* scale and 0.68 with *Prasad's*. Scoring digital access as an SES item is the genuinely new element. [PMID 42362781]
- **Area deprivation, not only household status, is now being indexed in India**. A district-level Punjab Index of Multiple Deprivation, built by maximum-likelihood factor analysis over six domains (economic, infrastructure, health, healthcare, living environment, education) and applied to NFHS-5 data, found diabetes concentrated among richer *individuals* (concentration index 0.1595) and also in *less deprived districts* (0.0526) — but the household-wealth gradient

was significantly the stronger of the two, so the area index supplements a household SES scale rather than replacing it. [PMID 41507894]

Global and WHO developments

- The measurement platform behind the NFHS wealth index is itself now precarious: following the dissolution of USAID, Demographic and Health Survey activity was disrupted in 23 countries and data access was briefly restricted to existing users, with a three-year Gates Foundation emergency grant in July 2025 and a UN-led task force now exploring alternative long-term funding. **NFHS-6 was the first Indian round conducted without DHS technical support and was financed wholly by the Government of India** — and it is exactly the DHS-standardised wealth index, the basis of cross-country and cross-round comparability, that a country-by-country funding model puts at risk. [PMID 41258330]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 12, Medicine and Social Sciences, pp 832-833, 847; Ch 2, p 55); Park Plus (companion volume, historical background and consumer-price-index multiplication factor); Sunder Lal — Textbook of Community Medicine (Ch 2, Relationship of Social and Behavioural Sciences to Health and Disease, pp 73-75); Oxford Textbook of Public Health 7e (Ch 9, pp 389-390; Ch 10, pp 423-424; Ch 18, pp 714-715). Survey and programme sources named in the text: National Family Health Survey-5 (2019-21), Socio-Economic and Caste Census 2011, National Sample Survey Organisation household consumer expenditure surveys.

Recent-advances sources: (PubMed; each PMID below re-checked against the PMC full text except the two marked abstract-only) 41451071 — A Seven-step Process to Update Dr. B.G. Prasad's Socioeconomic Status Scale, Indian J Community Med 2025, doi:[10.4103/ijcm.ijcm.825.23](https://doi.org/10.4103/ijcm.ijcm.825.23); 40949512 — B.G. Prasad Socioeconomic Scale 2024: An Update Using Conventional Approach and Linear Regression Modelling, Indian J Community Med 2025, doi:[10.4103/ijcm.ijcm.153.24](https://doi.org/10.4103/ijcm.ijcm.153.24); 38249702 — Updated BG Prasad's Socioeconomic Status Classification for the Year 2023, Indian J Community Med 2023, doi:[10.4103/ijcm.ijcm.401.23](https://doi.org/10.4103/ijcm.ijcm.401.23); 42362781 — Development and validation of Solomon's socioeconomic status index (SESI): a multidimensional household classification tool for South Asia, Eur J Health Econ 2026, doi:[10.1007/s10198-026-01959-8](https://doi.org/10.1007/s10198-026-01959-8) (abstract-only); 41507894 — Developing a Punjab Index of Multiple Deprivation to investigate regional health inequalities in North-Western India, BMC Public Health 2026, doi:[10.1186/s12889-025-26073-x](https://doi.org/10.1186/s12889-025-26073-x); 42554934 — Trends in Child Health Indicators in India: Insights from National Family Health Survey (NFHS)-6, Indian Pediatr 2026, doi:[10.1007/s13312-026-00412-6](https://doi.org/10.1007/s13312-026-00412-6) (abstract-only); 41291549 — A study on spatial association of poor child health outcomes and multidimensional poverty in India, BMC Public Health 2025, doi:[10.1186/s12889-025-25222-6](https://doi.org/10.1186/s12889-025-25222-6); 40316411 — Trends in socioeconomic inequality in mortality during childhood between 1993 and 2021 in India, BMJ Glob Health 2025, doi:[10.1136/bmjgh-2024-016386](https://doi.org/10.1136/bmjgh-2024-016386) (inequality figures taken from its Table 2, household wealth quintile rows); 41393311 — Assessing Surgical Care under the Government-Funded Health Insurance Scheme — Pradhan Mantri Jan Arogya Yojana in India, Ann Glob Health 2025, doi:[10.5334/aogh.4908](https://doi.org/10.5334/aogh.4908); 41258330 — Global health data precarity: Safeguarding the Demographic and Health Survey program as a global public good, Sci Data 2025, doi:[10.1038/s41597-025-06128-9](https://doi.org/10.1038/s41597-025-06128-9). Labour Bureau, Ministry of Labour and Employment (labourbureau.gov.in) — All-India Consumer Price Index for Industrial Workers (base 2016 = 100), June 2026 index value 151.9 (May 2026, 150.8), and the 2001-to-2016 linking factor 2.88 issued 21 October 2020.

♦ ♦ ♦

Theory of contagion

Asked in: Paper I 14-Jul-2020 — RGUHS MD Community Medicine

Definition — The **theory of contagion** holds that disease passes from a diseased person to a healthy person by **contact**, carried by **minute invisible particles**, and not by divine wrath, disordered body humors or foul air. It was enunciated by *Fracastorius* (Girolamo Fracastoro, 1483–1553), an Italian physician of the revival of medicine, who envisaged the transfer of infection via minute invisible particles, explained the cause of epidemics, recognised that syphilis was transmitted from person to person during sexual relations, and became the **founder of epidemiology** (Park 28e, Ch 1, p 17).

Core propositions of the theory

- ▶ **A living, self-multiplying seed.** Fracastoro is credited with being the first to suggest that "seeds of contagions" caused specific diseases; Girolamo Cardano expanded this in 1557, holding that "seeds of disease were minute animals, capable of reproducing their kind". This is the doctrine of a *contagium vivum* — the cause of an epidemic is a **living organism** able to reproduce its own kind in a new host (Lilienfeld's Foundations of Epidemiology, Ch 2, p 37).
- ▶ **Transfer by minute invisible particles.** In Park's formulation, Fracastorius envisaged the transfer of infection via minute invisible particles and explained the cause of epidemics (Park 28e, Ch 1, p 17).
- ▶ **Specificity — each disease breeds true from its own seed.** This is the direct ancestor of the **one-to-one** relationship between causal agent and disease later embodied in the germ theory, whose model is *disease agent* → *man* → *disease* (Park 28e, Ch 2, p 51).
- ▶ **Transmission requires a source, a route and a recipient.** Both **direct and indirect** contact with sick people or animals might transmit disease (Oxford Textbook of Public Health 7e, Ch 53, p 1951).
- ▶ **The particles were inferred, not demonstrated.** Proof waited for the microscope and finally for Pasteur, who in 1860 demonstrated the presence of bacteria in air (Park 28e, Ch 1, p 18).
- ▶ **Corollary for practice** — if disease travels by contact, preventing contact prevents disease; "hence both isolation and quarantine had been suggested to prevent the spread of diseases" (Oxford Textbook of Public Health 7e, Ch 53, p 1951).

Growth of the contagium vivum doctrine

Who and when	Contribution
Emperor Justinian, 549 CE	Proclaimed the first effective quarantine laws — travellers from territories struck by plague to be isolated and avoided; the earliest contagion-based control measure, alongside the Biblical edict that the leper must "live outside the camp" (Lilienfeld's Foundations of Epidemiology, Ch 2, p 37)
Girolamo Fracastoro (1478–1553)	First to suggest that "seeds of contagions" caused specific diseases; <i>De Contagione et Contagiosis Morbis</i> , 1546 (Lilienfeld, Ch 2, p 37 and reference list p 52)
Girolamo Cardano, 1557	"Seeds of disease were minute animals, capable of reproducing their kind" — the <i>contagium vivum</i> stated explicitly (Lilienfeld, Ch 2, p 37)

Who and when	Contribution
Athanasius Kircher, 1658	<i>Scrutinium Pestis</i> — referred to "true latent germs" and claimed to have seen living microorganisms in the blood of plague victims, 25 years before Leeuwenhoek (Lilienfeld, Ch 2, p 37)
Cotton Mather versus Benjamin Rush (18th century)	Mather, a leading exponent of <i>contagium vivum</i> , proposed that "animalcula" caused smallpox and advocated inoculation; Rush adopted the miasma theory when <i>contagium vivum</i> failed to explain the yellow fever epidemics of North America (Lilienfeld, Ch 2, p 38)
Agostino Bassi	Showed that small organisms within a host could produce disease — traced a fungus as the cause of muscardine in silkworms and recommended disinfectants (Lilienfeld, Ch 2, p 38)
Pierre-Fidèle Bretonneau, 1826	Showed communicable diseases to be specific — distinguished scarlet fever from diphtheria (Lilienfeld, Ch 2, p 38)
Jacob Henle, 1840	<i>Von den Miasmen und Kontagien</i> — the first summary of all scientific evidence for and against the <i>contagium vivum</i> doctrine; widely considered cofounder of the theory of microorganisms as the cause of infective disease. His student was Robert Koch , who later discovered the tubercle bacillus (Lilienfeld, Ch 2, pp 38-39)
By the mid-1800s	<i>Contagium vivum</i> widely espoused by European physicians and scientists; merging with better data and statistical methods, "thus began the development of epidemiology as a distinct discipline" (Lilienfeld, Ch 2, p 39)

Place among the historical theories of disease causation

Up to the time of Louis Pasteur (1822–1895) various concepts of disease causation were in vogue — the supernatural theory, the theory of humors, the concept of contagion, the miasmatic theory and the theory of spontaneous generation (Park 28e, Ch 2, p 51).

Theory	Period / proponent	What disease was held to be	Public-health action it generated
Supernatural theory	Prehistoric (about 5000 B.C.); primitive man	Wrath of gods, invasion by "evil spirits", malevolent influence of stars and planets	Prayers, rituals, sacrifices, witchcraft, charms and amulets (Park 28e, Ch 1, p 13)
Theory of humors	Greeks (Hippocrates) and Indians	Health when the four humors — phlegm, yellow bile, blood, black bile — are in equilibrium; the Ayurvedic tridosha theory is strikingly similar: <i>vata, pitta, kapha</i>	Diet and regimen to restore balance; Hippocrates' <i>Airs, Water and Places</i> tied man to his environment (Park 28e, Ch 1, pp 14-16)
Theory of contagion	<i>Fracastorius</i> (1483–1553), during the revival of medicine (1453–1600 A.D.)	Transfer of infection person to person by minute invisible particles; a living, self-reproducing <i>contagium vivum</i>	Isolation of the sick, quarantine of the exposed (Park 28e, Ch 1, p 17; Lilienfeld, Ch 2, p 37)

Theory	Period / proponent	What disease was held to be	Public-health action it generated
Miasmatic theory	18th–19th century sanitary era	Noxious air and vapours — dirty living environments, unclean drinking water, poor ventilation from overcrowding	The great sanitary awakening: anti-filth crusade, Public Health Act 1848 , safe water supply and sewage disposal (Park 28e, Ch 1, pp 17-18; Oxford 7e, Ch 53, p 1951)
Theory of spontaneous generation	Before 1860	Living organisms arise afresh from non-living matter	Disproved by Pasteur in 1860 , who demonstrated bacteria in air (Park 28e, Ch 1, p 18)
Germ theory of disease	Pasteur 1873; confirmed when Koch showed in 1877 that anthrax was caused by a bacterium	Microbes as the sole cause — a one-to-one relation, <i>disease agent</i> → <i>man</i> → <i>disease</i>	The golden age of bacteriology — gonococcus 1847, typhoid bacillus and pneumococcus 1880, tubercle bacillus 1882, cholera vibrio 1883, diphtheria bacillus 1884 (Park 28e, Ch 1, p 18; Ch 2, p 51)
Multifactorial causation	Pettenkofer of Munich (1819–1901), overshadowed by the germ theory and later revived by epidemiologists	Disease from many interacting factors — agent, host, environment plus social, economic, cultural, genetic and psychological factors	The risk-factor approach, the epidemiological triad, health promotion and lifestyle modification (Park 28e, Ch 1, p 18; Ch 2, p 52)

Contributions of the theory of contagion

Contribution	Substance
Removed the supernatural from epidemics	Epidemics became a natural, transmissible and therefore <i>preventable</i> event rather than a visitation (Park 28e, Ch 1, pp 13, 17)
First clear statement of person-to-person spread	Syphilis identified as transmitted from person to person during sexual relations (Park 28e, Ch 1, p 17)
Founded epidemiology	Park credits Fracastorius as the founder of epidemiology ; Lilienfeld dates the emergence of epidemiology as a distinct discipline to the merging of <i>contagium vivum</i> with statistical method (Park 28e, Ch 1, p 17; Lilienfeld, Ch 2, p 39)
Created the first control tools	Quarantine from the late fourteenth century onwards in Ragusa (Dubrovnik) and then across Italy — 40 days free from disease was the period found necessary to clear infected ships; a law for isolation of smallpox patients and for ship quarantine was enforced from 1701 in Massachusetts (Oxford 7e, Ch 2, p 182; Ch 21, p 804)
Anticipated the germ theory by three centuries	Henle's 1840 synthesis carried the doctrine directly to his student Koch; Pasteur and Koch then supplied the microbiological proof (Lilienfeld, Ch 2, pp 38-39; Park 28e, Ch 1, p 18)
Bequeathed the working vocabulary	<i>Contagious disease</i> , <i>contact</i> and <i>fomite</i> remain formal epidemiological terms in current use (Park 28e, Ch 3, pp 113, 116-117)

Limitations and why it was superseded

- ▶ **The particles were never demonstrated** in Fracastorius' time — the theory stayed a speculation, and Kircher's 1658 "living microorganisms" in plague blood were probably red blood cells (Lilienfeld, Ch 2, p 37).
- ▶ **It could not explain selectivity of epidemics** — with ubiquitous filth, epidemics still occurred only sometimes and in some places, which neither miasma nor contagion could account for (Oxford 7e, Ch 58, p 2155).
- ▶ **Contagion-based control did not stop outbreaks** — the strategies evolved from contagion theory "had not stopped the outbreak of many infectious diseases" (Oxford 7e, Ch 53, p 1951), and belief in the direct passage of poison from one person to another "led to large quarantines of immigrants and justified state-sponsored interventions in the economy" (Oxford 7e, Ch 58, p 2155).
- ▶ **It missed the non-contact routes** — John Snow's study of cholera in London from 1848 to 1854 established polluted drinking water as the vehicle, and William Budd in 1856 concluded that typhoid fever spread by drinking water and not by miasma and sewer gas; vector-borne spread was elucidated later (Ross and Anopheles, 1898). Indeed the idea that a mosquito could transmit a germ from person to person could not yet be imagined, which is why Rush abandoned *contagium vivum* for miasma in yellow fever (Park 28e, Ch 1, pp 17-18; Lilienfeld, Ch 2, p 38).
- ▶ **It ignored host susceptibility and the carrier state** — not everyone exposed to tuberculosis develops tuberculosis; poverty, overcrowding and malnutrition contribute, so the agent is not "sufficient" (Park 28e, Ch 2, pp 51-52).
- ▶ **It has nothing to say about non-communicable disease** — coronary heart disease, cancer, diabetes, mental illness and accidents are neither explained nor prevented by contact-based reasoning, which is why the field moved to multifactorial causation (Park 28e, Ch 1, p 18).

Where contagion survives in present-day practice

Current term	Definition as printed (Park 28e, Ch 3, p 113)	Examples
Contagious disease	A disease that is transmitted through contact	Scabies, trachoma, sexually transmitted disease, leprosy
Infectious disease	A disease due to an infectious agent; some infectious diseases are contagious, others are non-contagious — all infectious diseases and infestations are communicable diseases	Tetanus is the standard illustration of an infectious but non-contagious disease
Communicable disease	An illness due to a specific infectious agent or its toxic products arising through transmission of that agent or its products from an infected person, animal or reservoir to a susceptible host, either directly or indirectly	The wider class within which contagious disease sits

Modes of transmission — contagion is the *direct* half of this classification (Park 28e, Ch 3, pp 116-117):

Direct transmission	Indirect transmission
1. Direct contact (skin-to-skin, mucosa-to-mucosa — touching, kissing, sexual intercourse)	1. Vehicle-borne
2. Droplet infection (spray of saliva and nasopharyngeal secretions on coughing, sneezing, speaking, spitting)	2. Vector-borne — (a) mechanical, (b) biological
3. Contact with soil	3. Air-borne — (a) droplet nuclei, (b) dust
4. Inoculation into skin or mucosa	4. Fomite-borne
5. Transplacental (vertical)	5. Unclean hands and fingers

The two contact-based measures, as currently defined (Park 28e, Ch 3, pp 145-146):

- **Isolation** — separation of infected persons for the **period of communicability**, to prevent transfer of infection from the reservoir to susceptible hosts. Duration is decided by the duration of communicability and the effect of chemotherapy on infectivity.
- **Quarantine** — "the limitation of freedom of movement of such well persons or domestic animals exposed to communicable disease for a period of time **not longer than the longest usual incubation period** of the disease, in such manner as to prevent effective contact with those not so exposed."
- **The present position** — isolation applies to the *sick*, quarantine to the *healthy contacts*; with better early diagnosis and treatment, quarantine has been **replaced by active surveillance**, and isolation is now used only when the risk of transmission is exceptionally serious.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Diagram to reproduce — the epidemiological triad.** Draw a triangle and label the three vertices **AGENT**, **HOST** and **ENVIRONMENT**, with the agent–host side marked *transmission*; write beneath it that the germ theory's single-agent model gave way to this because the same exposure produces disease only in the susceptible.

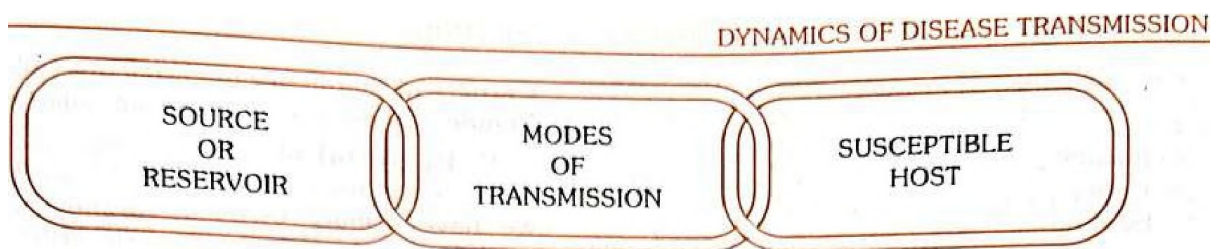
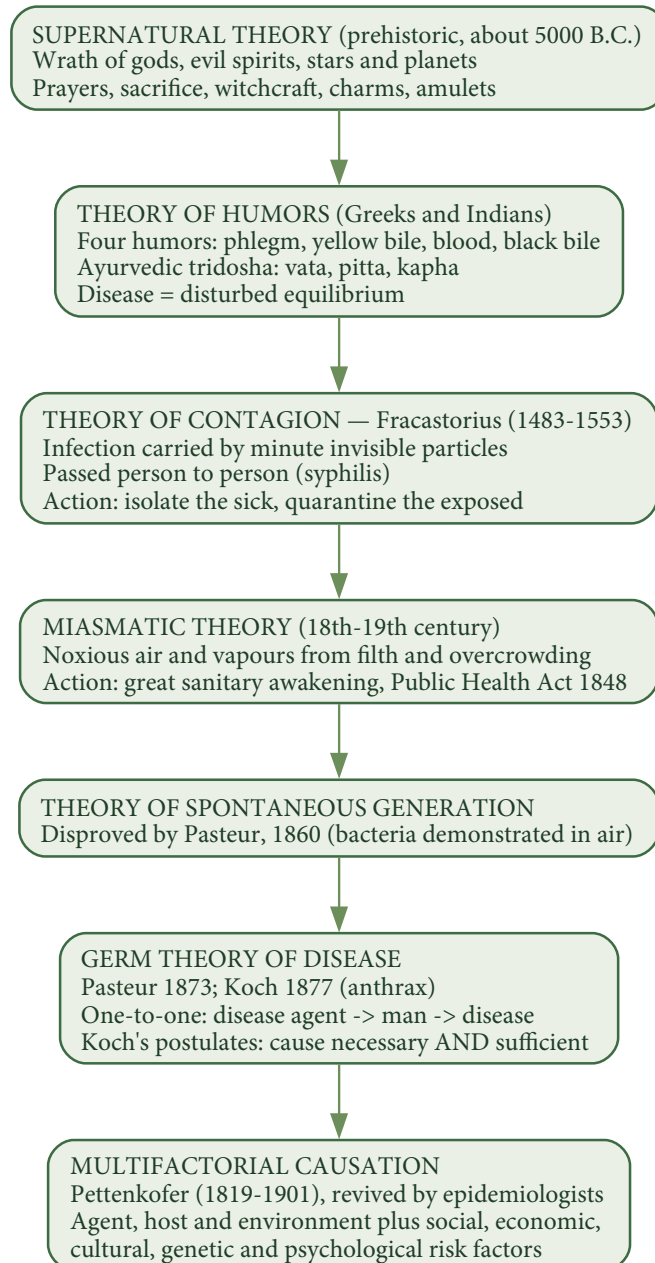
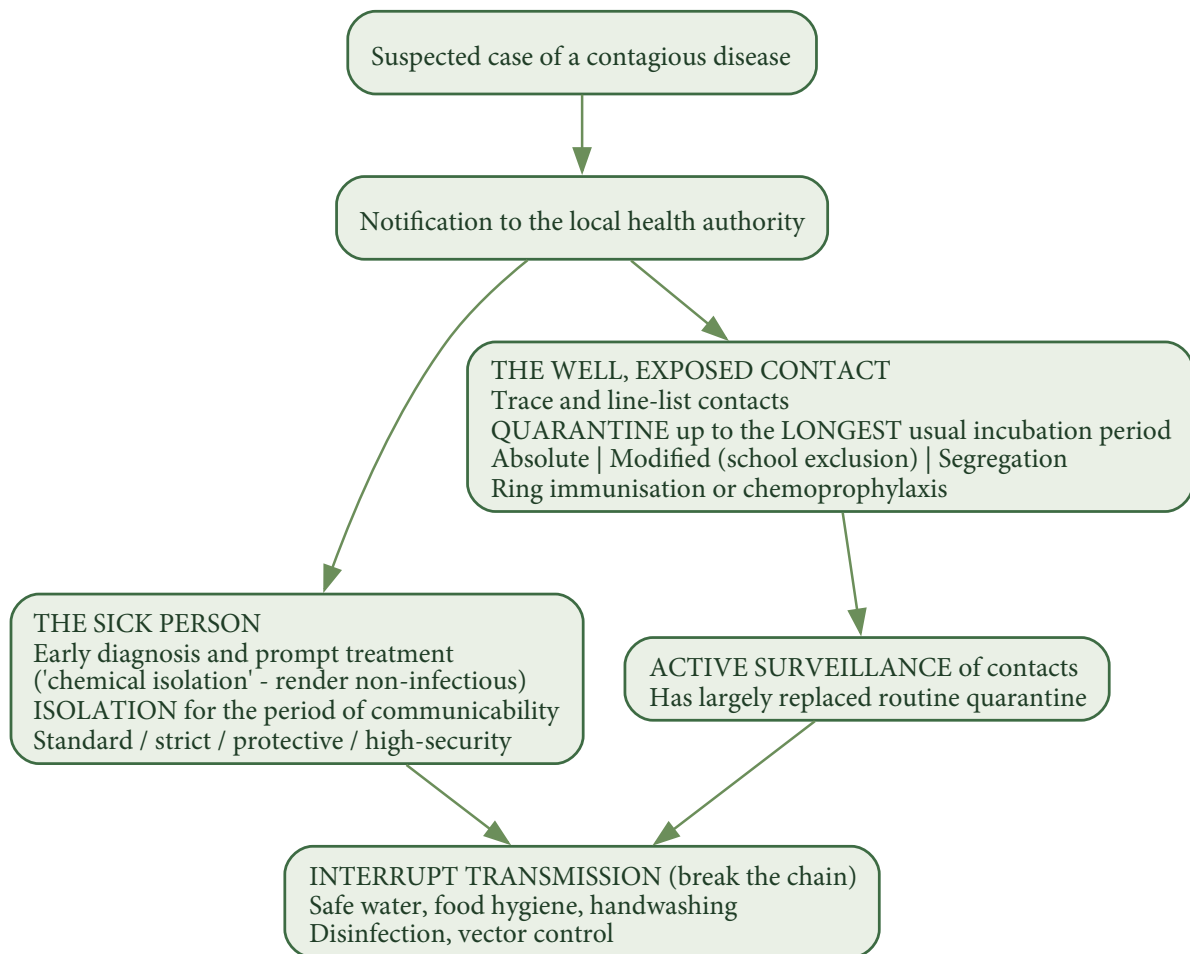


Fig 16 — Chain of Infection (Park's Textbook of Preventive and Social Medicine, 28e, p.115)



Sequence of the theories of disease causation, showing where the theory of contagion sits



Contact-based control of a contagious disease - the operational legacy of the theory of contagion

High-yield exam pointers

- **Open with the name and the dates:** *Fracastorius*, 1483–1553, Italian, "theory of contagion", founder of epidemiology (Park 28e, Ch 1, p 17).
- **The one Latin term to write:** *contagium vivum* — contagion as a living agent, capable of reproducing its kind (Cardano, 1557).
- **Sequence of theories (write in this order):** supernatural → humors/tridosha → **contagion** → miasmatic → spontaneous generation → germ theory → multifactorial causation.
- **Dates that must be exact:** Justinian's quarantine laws 549 CE; Fracastoro's treatise 1546; Henle's *Von den Miasmen und Kontagien* 1840; Pasteur disproved spontaneous generation 1860; germ theory 1873; Koch and anthrax 1877; Snow and cholera 1848–1854; Budd and typhoid 1856.
- **The bridge to the germ theory:** Henle taught Koch — quote this to show the theory was not a dead end but the direct ancestor of bacteriology.
- **Contagious ≠ infectious ≠ communicable:** contagious = transmitted through contact (scabies, trachoma, sexually transmitted disease, leprosy); some infectious diseases are non-contagious; all infectious diseases and infestations are communicable.
- **Isolation versus quarantine:** isolation applies to the sick for the period of communicability; quarantine to the well exposed for the longest usual incubation period, now largely replaced by active surveillance.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the two indexed sources differ on Fracastoro's year of birth — Park 28e gives 1483–1553 while Lilienfeld's Foundations of Epidemiology gives 1478–1553 (and spells the surname "Fracastoro" in its reference list); write Park's dates in an RGUHS answer, since Park is the prescribed spine.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 1 Man and Medicine: Towards Health for All, pp 13-18; Ch 2 Concept of Health and Disease, pp 51-52; Ch 3 Principles of Epidemiology and Epidemiologic Methods, pp 113, 115-117, 145-146); Lilienfeld's Foundations of Epidemiology (Ch 2 Threads of Epidemiologic History, pp 37-39, 52); Oxford Textbook of Public Health 7e (Ch 2 The history and development of public health in developed countries, p 182; Ch 21 Health policy in developing countries, p 804; Ch 53 Environmental health methods, p 1951; Ch 58 Urbanization and health, p 2155). Figure: Park 28e, Fig 16 Chain of Infection, p 115.



Treatment adherence and non-adherence in long-term illnesses

Asked in: Paper II Nov 2016 — RGUHS MD Community Medicine

Definition — Patient **compliance** is "the extent to which patient behaviour (in terms of taking medicines, following diets or executing other lifestyle changes) coincides with clinical prescription" (Park 28e, Ch 6, p 449). **Adherence** describes the same behaviour from the patient's side — the extent to which the patient actually follows a treatment plan he has understood and agreed to — and is the preferred term because long-term treatment works only when the patient is an active partner rather than a passive recipient of orders. **Non-adherence** is any departure from that plan: doses omitted or mistimed, diet and lifestyle advice ignored, follow-up visits missed, or treatment stopped before the course is complete.

Why long-term illness makes adherence the central problem

Aspect	Content
Which illnesses count as "long-term"	There is no international definition of what duration should be considered long-term; many consider chronic conditions to be those of at least 3 months duration, and a practical definition should be established to suit the particular community (Park 28e, Ch 6, p 433). Sunder Lal takes symptoms, or a course of treatment, continuing more than one month as chronic, and notes that diabetes, hypertension and human immunodeficiency virus (HIV) infection are lifelong and need lifelong treatment and follow-up (Sunder Lal, Ch 1, p 43)

Aspect	Content
Why adherence fails here specifically	Treatment is lifelong or runs for many months; it is self-administered and unsupervised at home; the condition is often symptomless (hypertension, early type 2 diabetes), so the benefit is invisible while cost, side-effects and daily routine are felt every day; and the patient must sustain not one act but a daily habit . Financial incentives, which work well for one-off health behaviours, are least effective for exactly this kind of near-daily pill-taking (Oxford PH 7e, Ch 50, pp 1860–1861)
Clinical importance	Behaviour is central to outcome — the prognosis of type 1 (insulin-dependent) diabetes depends on the patient's compliance with the insulin prescription (Oxford PH 7e, Ch 15, pp 595–596); the treatment of high blood pressure must normally be lifelong , and this is precisely where the problem of compliance arises (Park 28e, Ch 6, p 449). Secondary prevention in diabetes rests on self-care — the patient taking major responsibility for adherence to diet and drug regimens, self-monitoring, abstinence from alcohol, weight control and periodic check-ups (Park 28e, Ch 6, p 471)

Types and patterns of non-adherence

Type	What the patient does	Typical long-term example
Non-initiation	Prescription is accepted but never started or never collected	Newly screened hypertensive at a population-based screening camp who never reaches the health facility
Erratic / partial adherence	Doses omitted, halved, doubled or taken at the wrong time; drug "holidays" on festival days, travel or fasting	Missed antihypertensive or oral hypoglycaemic doses
Selective adherence	Takes the drug but ignores the diet, physical-activity, tobacco or alcohol advice — the part of the prescription that is behavioural rather than pharmacological	Diabetic on metformin who does not modify diet or activity
Non-persistence (premature discontinuation)	Stops once symptoms subside, when side-effects appear, or when the drug is out of stock	Tuberculosis (TB) patient who stops after sputum becomes negative
Treatment interruption / default	Treatment interrupted for 1 consecutive month or more — assigned the programme outcome " lost to follow-up " under India's National Tuberculosis Elimination Programme (NTEP) (Sunder Lal, Ch 8, p 364)	TB patient not retrieved after missed doses
Over-adherence	Takes more than prescribed, adds indigenous or over-the-counter drugs, doctor-shops	Elderly patient on multiple prescriptions from several providers

Determinants of non-adherence

Dimension	Factors that reduce adherence
Patient-related	Illiteracy and poor comprehension; forgetfulness; lack of information about the disease and drugs; illness behaviour — people react very differently to the same disease process or regimen by reason of education, religion, social class or occupation (Park 28e, Ch 12, p 812); denial of a symptomless diagnosis; belief that the drug is unnecessary once symptoms remit; depression; substance abuse disorders (Park 28e, Ch 5, pp 249–250); a prolonged sick role deliberately maintained to escape everyday responsibilities (Park 28e, Ch 12, p 836)
Condition-related	Symptomless disease; slow or invisible benefit; long or lifelong duration; stigma (tuberculosis, leprosy, HIV, mental illness); co-morbidities such as diabetes and liver disease that complicate the regimen (Park 28e, Ch 5, pp 232–233)
Therapy-related	Complex regimen, high pill burden, multiple daily doses; adverse effects; malabsorption ; wrong dosages or drug combinations prescribed (Park 28e, Ch 5, pp 249–250)
Provider-related	Absence of guidelines or inappropriate guidelines; non-compliance with guidelines; inadequate training of health staff; no monitoring of treatment (Park 28e, Ch 5, pp 249–250); failure to discuss treatment goals and priorities, to make a plan the patient can actually carry out in daily life, and to give clear instructions on symptoms and when to seek care (Oxford PH 7e, Ch 101, p 3873); the doctor's authoritative role and the " social distance " between a sophisticated practitioner and an illiterate patient, so that advice is issued as an order and neither understood nor accepted (Park 28e, Ch 12, pp 835–836); <i>infantilization</i> — treating the adult patient as a child (Sociology: The Essentials 6e, Ch 14, p 390)
Health-system-related	Non-availability of drugs — stock-outs or delivery disruptions; poor drug quality; poor storage conditions; non-availability of free drugs; poorly organised or poorly funded programmes (Park 28e, Ch 5, pp 249–250); distant or inconvenient facility, long waiting time, inconvenient clinic hours; no defaulter-tracking mechanism
Socio-economic and community	Cost of drugs, investigations and travel; loss of daily wages; migration for work; social and economic barriers (Park 28e, Ch 5, pp 249–250); lack of family support; competing traditional or folk treatment; advice pitched against an existing cultural pattern (hot/cold classification of diseases and medicines) rather than couched within it (Park 28e, Ch 12, p 836)

Consequences of non-adherence

- ▶ **Treatment failure and relapse** — a tuberculosis patient whose sputum smear or culture is positive at month 5 or later during treatment is classified as **treatment failed** and must have therapy modified (Park 28e, Ch 5, p 224; Jugal Kishore, Ch 1, p 310).
- ▶ **Emergence of drug resistance** — poor adherence and poor directly observed treatment are listed among the patient-side causes of inadequate treatment that generate multidrug-resistant tuberculosis (MDR-TB), which takes at least two years with second-line drugs and gives poor results (Park 28e, Ch 5, pp 249–250). Low treatment success with high **death and default rates** under the National Tuberculosis Programme, plus spreading MDR-TB, was why India adopted directly observed treatment, short-course (DOTS) in 1993 (Park 28e, Ch 7, p 507).
- ▶ **Avoidable complications and premature death** — **insufficient follow-up leads to unnecessary treatments and severe complications, often resulting in death** in diabetes (Park 28e, Ch 6, p

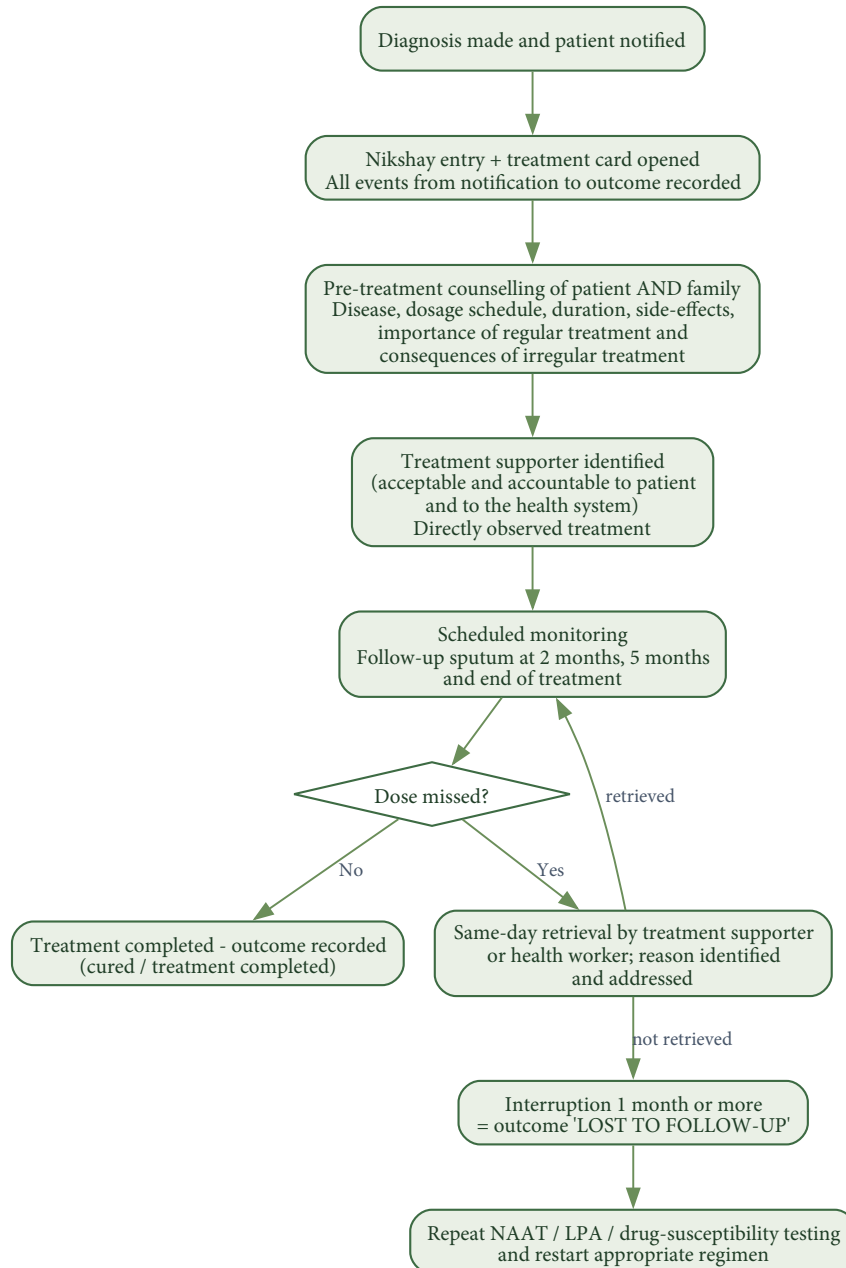
468); uncontrolled hypertension results in stroke, heart failure and renal failure, all of which fall when blood pressure is controlled at community level (Park 28e, Ch 6, p 449).

- **Continued transmission** — an irregularly treated infectious tuberculosis patient stays infectious longer and keeps the chain of transmission going; in HIV, poor adherence prevents viral suppression, and viral load suppression significantly reduces the risk of ongoing transmission (Sunder Lal, Ch 8, p 349; Oxford PH 7e, Ch 50, pp 1865–1866).
- **Programme failure** — treatment success rate, cure rate and viral suppression are the very indicators by which national programmes are judged; India's TB treatment success rate stood at **81% in 2021** with 95% of notified patients put on treatment (Park 28e, Ch 7, p 515).
- **Wasted resources and misleading evaluation** — drugs dispensed but not taken are wasted expenditure, and an effective regimen is wrongly judged ineffective; in pragmatic trials, discontinuation and non-compliance over time are themselves important outcome measurements (Oxford PH 7e, Ch 23, pp 908–909).

Assessment of adherence

Method	How it is done	Limitation
Direct observation of ingestion	Directly observed treatment (DOT) by a treatment supporter acceptable and accountable to both the patient and the health system (Jugal Kishore, Ch 1, p 310)	Labour-intensive; success depends on how acceptable and convenient it is to the patient
Drug-marker / urine colour check	The patient is told that rifampicin will turn the urine red ; this can be used as a test of compliance (Park 28e, Ch 5, p 230)	Proves ingestion of one dose only, not sustained intake
Biological and clinical response	Follow-up sputum microscopy at the end of the initial phase (2 months), at 5 months and at the end of treatment in pulmonary TB (Jugal Kishore, Ch 1, p 310); CD4 count every six months on antiretroviral therapy (ART) (Park 28e, Ch 7, p 523); glycosylated haemoglobin , which reflects mean blood glucose over the previous 2–3 months (Park 28e, Ch 6, p 471); blood pressure and blood sugar at each follow-up visit in non-communicable disease (NCD) care	Response is also affected by drug resistance, malabsorption and disease severity, not by adherence alone
Self-report, pill count and on-time pill pick-up	Under the national AIDS programme, treatment adherence is monitored by self-verified adherence, pill counting and on-time pill pick-up (Sunder Lal, Ch 8, p 349); adherence is assessed by the counsellor at every monthly ART visit, with support provided as required (Park 28e, Ch 7, p 523)	Self-report over-estimates adherence; pills can be discarded; pick-up proves only collection, not ingestion
Programme records	Treatment card and Nikshay entry at every event from notification to treatment outcome (Park 28e, Ch 5, p 234); patient-wise monitoring and tracking, with online referral/transfer in real time to prevent patient loss (Jugal Kishore, Ch 1, p 306); the individual's health card under the national NCD programme (Park 28e, Ch 7, p 565)	Records only what reaches the system; drug collected is not drug swallowed

Method	How it is done	Limitation
Attendance / retention	Regularity of follow-up visits; interruption of treatment for 1 consecutive month or more is recorded as "lost to follow-up" under NTEP (Sunder Lal, Ch 8, p 364)	Detects only gross default, and only after the event



Adherence support and defaulter retrieval in a long-term treatment programme, illustrated by the National Tuberculosis Elimination Programme

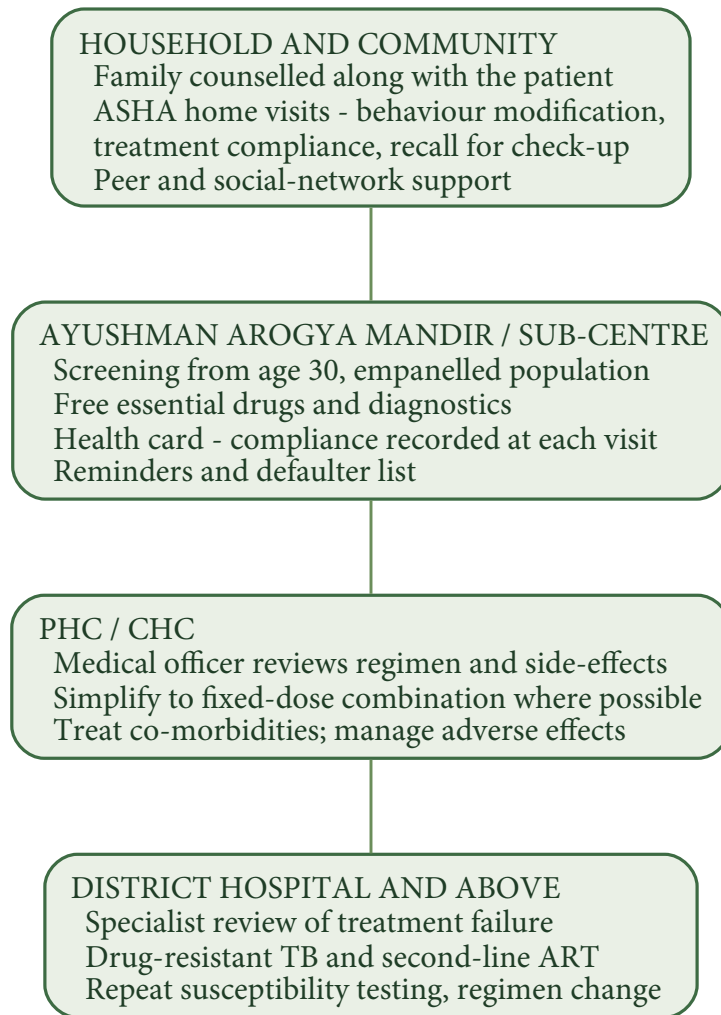
Improving adherence — measures at each level

Level	Measures
Patient and family	Counselling of the patient and family members on the type of disease, mode of spread, duration and dosage schedule, side-effects and how to prevent them, the importance of regular treatment and the consequences of irregular treatment , co-morbidities and substance abuse, and linkage to welfare schemes (Park 28e, Ch 5, pp 232–233); compliance rates improve through education directed at patients, families and the community (Park 28e, Ch 6, p 449); medication adherence improves when others in the patient's social network have themselves learned about the need to control blood pressure and lipids (Oxford PH 7e, Ch 33, p 1225)
Communication and counselling technique	Communicate on the emotional plane (a sympathetic ear, quick rapport), the cultural plane (couch scientific advice within the patient's existing cultural pattern instead of challenging it openly) and the intellectual plane (reduce social distance, use words the patient understands) — the doctor who communicates on all three planes gives maximum psychological satisfaction (Park 28e, Ch 12, pp 835–836). Adapt the six-step GATHER sequence — Greet, Ask, Tell, Help, Explain, Return for follow-up — <i>originally a family-planning counselling sequence</i> (Sunder Lal, Ch 4, pp 156–157)
Health-education principles applied to the regimen	Build on the patient's felt needs and interest; ensure comprehension within his mental capacity and literacy; reinforce by repetition at intervals and in different ways, because an unreinforced message lets the individual slip back to the pre-awareness stage; use motivation and incentives ; learning by doing ; proceed from the known to the unknown ; use feedback , community participation and local leaders as agents of change (Park 28e, Ch 22, p 1019). Behaviour-change interventions are commonly built on the Health Belief Model (Becker 1974), Social Cognitive Theory (Bandura 2004) and the Transtheoretical Model of Stages of Change (Prochaska et al. 2002) (Oxford PH 7e, Ch 15, p 605)
Simplify the therapy	Fixed-dose combinations were developed precisely to improve adherence and thereby reduce treatment failure and the development of drug resistance (Oxford PH 7e, Ch 71, pp 2659–2660); prefer the shortest effective regimen — the shorter all-oral MDR-TB regimen cures drug-resistant TB in six months against up to 20 months of traditional treatment, with fewer side-effects and lower cost (Park 28e, Ch 5, p 240)
Supervise, remind and track	Directly observed treatment by a treatment supporter, with supervision and motivation by a health or non-health worker and monitoring of disease status by the health services (Park 28e, Ch 7, p 511). Reminder text messages improve adherence in long-term therapy — weekly text messages to adults on ART significantly increased the proportion with adherence exceeding 90%, and two-way weekly messages raised both self-reported adherence and viral suppression (Oxford PH 7e, Ch 50, pp 1865–1866). Community follow-up by the accredited social health activist (ASHA) , who prioritises households with treatment defaulters and those with complications; some states have given her a blood-pressure apparatus and glucometer for village-level examination, and the auxiliary nurse midwife (ANM) also makes regular home visits (Park 28e, Ch 7, p 565)

Level	Measures
Remove system barriers	Assured, uninterrupted supply of good-quality free drugs and diagnostics; decentralised treatment near the patient's home; a patient-centred strategy that assesses and promotes adherence and addresses poor adherence when it occurs, with measures tailored to the individual and mutually acceptable to patient and provider (Jugal Kishore, Ch 1, p 310)

Adherence built into Indian national programmes

Programme	Built-in adherence mechanism
National Tuberculosis Elimination Programme (erstwhile Revised National Tuberculosis Control Programme)	DOTS = appropriate medical treatment + supervision and motivation by a health or non-health worker + monitoring of disease status; DOT agents drawn from the community (teachers, anganwadi workers, dais, ex-patients, social workers) with an honorarium of Rs 150 per patient completing treatment ; Nikshay case-based web-based tracking with online transfer; scheduled follow-up sputum examination; defined outcome categories including "lost to follow-up" (Park 28e, Ch 7, p 511; Ch 5, pp 224, 234; Jugal Kishore, Ch 1, p 306)
National AIDS Control Programme	First-line ART free at ART centres; monthly follow-up with clinical evaluation and screening for opportunistic infections at every visit; adherence assessed by the counsellor at every visit with support as required; counselling on treatment adherence, nutrition, positive prevention and positive living; CD4 six-monthly; adherence monitored by self-verified adherence, pill counting and on-time pill pick-up; decentralised Link ART Centres; care-and-support centres for psycho-social support and social benefit schemes (Park 28e, Ch 7, p 523; Sunder Lal, Ch 8, p 349)
National Programme for Prevention and Control of Non-Communicable Diseases (NP-NCD; erstwhile National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Diseases and Stroke, NPCDCS)	Patients already on private treatment may opt to take drugs from the public system after appropriate confirmation; frontline workers visit them regularly and monitor compliance with treatment and lifestyle changes, recorded in the health card ; the stated aim is to minimise treatment defaulters and achieve hypertension and glycaemic control at individual and population level (Park 28e, Ch 7, p 565)
Ayushman Arogya Mandir (erstwhile Ayushman Bharat Health and Wellness Centre, launched 2018) — the delivery platform	Population empanelment and universal screening from age 30 for hypertension, diabetes and three common cancers; "treatment compliance and follow up for all diagnosed cases" listed as an essential service; free essential drugs and diagnostics; two-way referral and continuum of care; 1,72,148 centres operational as on 31 March 2024 (Park 28e, Ch 10, p 735; Ch 7, pp 541, 570)



Tiered adherence support for a patient on long-term treatment, from household to district

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Determinants of non-adherence — the five-dimension wheel.** A central circle *Non-adherence in long-term illness* with five spokes to boxes labelled exactly: *Patient-related* (illiteracy, forgetfulness, illness behaviour, denial, depression, substance abuse), *Condition-related* (symptomless, lifelong, stigma, co-morbidity), *Therapy-related* (complex regimen, pill burden, side-effects, malabsorption), *Health-system and provider* (stock-outs, no free drugs, no monitoring of treatment, social distance), *Socio-economic* (cost, wage loss, migration, no family support).
- ▶ **The three planes of doctor–patient communication.** Three stacked boxes joined by a vertical arrow, labelled exactly: *Emotional plane — sympathetic ear, rapport*; *Cultural plane — advice couched within the patient's existing cultural pattern*; *Intellectual plane — reduce social distance, use words the patient understands*; ending in *Maximum psychological satisfaction → advice followed*.

High-yield exam pointers

- ▶ **Compliance (Park's definition, learn verbatim)** — "the extent to which patient behaviour (in terms of taking medicines, following diets or executing other lifestyle changes) coincides with clinical prescription".
- ▶ **Five dimensions of determinants** — mnemonic P-C-T-H-S: Patient, Condition, Therapy, Health system/provider, Socio-economic. Park's table of causes of inadequate treatment splits the same ground three ways — **providers/programmes, drugs, patients**.
- ▶ **"Lost to follow-up" under India's NTEP** = treatment interrupted for **1 consecutive month or more** (Sunder Lal, Ch 8, p 364). TB follow-up sputum at **2 months, 5 months and end of treatment**; smear or culture positive at **month 5 or later** = **treatment failed**. India's TB treatment success rate **81% (2021)**.
- ▶ **DOTS = three components** — appropriate medical treatment + supervision and motivation by a health or non-health worker + monitoring of disease status. DOT agent honorarium **Rs 150** per patient completing treatment.
- ▶ **Adherence checks worth naming** — rifampicin turning the **urine red**; **glycosylated haemoglobin** for the previous **2–3 months**; self-verified adherence, **pill count** and **on-time pill pick-up**; CD4 every **six months** on ART with counsellor assessment at **every** visit.
- ▶ **GATHER counselling** — Greet, Ask, Tell, Help, Explain, Return for follow-up.
- ▶ **Ayushman Arogya Mandir** (erstwhile Ayushman Bharat Health and Wellness Centre, 2018) — screening from age 30 years; "treatment compliance and follow up for all diagnosed cases" is a listed service; **1,72,148** centres operational as on **31 March 2024**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference, not to be written in the exam. Park uses the older word "compliance" for the definition quoted above, whereas current writing prefers "adherence" (and, in prescribing, "concordance") to stress that the patient agreed to the plan rather than merely obeyed it; the behaviour is the same, and the RGUHS stem uses "adherence". Park's treatment-outcome table (Ch 5, p 224) reproduces the World Health Organization wording of "2 consecutive months or more" for lost to follow-up in drug-susceptible tuberculosis, whereas India's National Tuberculosis Elimination Programme uses **1 consecutive month or more** (Sunder Lal, Ch 8, p 364), and the Indian figure is the one to write; the one-month rule on Park's page 240 belongs to the drug-resistant tuberculosis interruption passage, which is why the flowchart labels that branch as the drug-resistant rule. Two items are not retrievable from the indexed corpus and are given as standard material: the six-type taxonomy of non-adherence (non-initiation, erratic, selective, non-persistence, interruption, over-adherence), which follows the usual adherence literature, and the current programme name **NP-NCD** (National Programme for Prevention and Control of Non-Communicable Diseases), correct today but still printed as NPCDCS throughout Park 28e. Every other figure, schedule and programme component is traceable to the cited pages.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, WHO, NTEP programme data

Programme and policy updates (India)

- ▶ **Ni-kshay Poshan Yojana** — the Rs 500 per month direct benefit transfer paid to every notified tuberculosis patient since 1 April 2018 — has had its first nationwide evaluation, covering 3,712,551 patients notified in nine states between 2018 and 2022: receipt of at least one instalment rose from 56.9% to 76.1%, and the median gap between diagnosis and first payment halved from 200 days to 91 days. Non-receipt stayed far commoner among patients notified by the private sector (adjusted RR 2.10; 95% CI 2.08–2.12) and those HIV-reactive (aRR 1.69; 1.64–1.74); of the 2022 patients with unfavourable outcomes who were paid at all, 65.1% got the first instalment only after the outcome had already been declared — money arriving after the adherence failure it was meant to prevent [PMID 38273246].
- ▶ A patient-cost cohort of 1,482 adults on drug-susceptible treatment across 118 tuberculosis units in 16 districts of four states (2019–2023) found only 31–54% had received any Ni-kshay Poshan benefit by the end of treatment and just 5–11% the full Rs 3,000; 34–60% still met the catastrophic-cost definition ($\geq 20\%$ of pre-tuberculosis annual household income), and the scheme kept only about 2% below that threshold. Raising the benefit to about Rs 10,000 across the six months would clear catastrophic costs for 43% of those affected, and because roughly 60% of the total cost is incurred *before* diagnosis, the authors argue at least half of any revised amount should be paid immediately on treatment registration; the Ni-kshay Mitra community-support scheme is proposed as the route to remote rural and tea-garden patients [PMID 38985653].
- ▶ **IMPACT Diabetes**, a cluster-randomised trial in 16 villages across eight primary health centres in two states, gave direct trial evidence for the accredited social health activist (ASHA)-based compliance monitoring that the national non-communicable disease programme and Ayushman

Arogya Mandir already prescribe on paper: ASHAs screened 1,785 people, and nine months of decision-support-guided ASHA care under physician supervision achieved an HbA1c fall of $\geq 0.5\%$ in 21.8% versus 10.3% of controls ($p < 0.05$), 63.0% versus 43.1% still on glucose-lowering medication at the end of the study, which the trial reports as its medication-adherence outcome ($p < 0.05$), and primary-health-centre physician visits in 85.0% versus 29.8%. Mean HbA1c at the end did not differ (8.6% versus 8.8%), so the gain is in who improves and who reaches care, not in the group average [PMID 41120559].

Current national data

- Pooled Indian figures are worse than textbook accounts imply: a meta-analysis of 12 studies and 3,164 hypertensive patients assessed on the Morisky scale put adherence to antihypertensives at only 15.8% (95% CI 4.4–43.4), with higher age, regimen complexity, low socioeconomic status, low education, uncontrolled blood pressure and comorbidity recurring as correlates of non-adherence [PMID 39710532].
- A systematic review of Indian randomised trials of medication-adherence interventions in chronic non-communicable disease found just 22 trials in all: 12 tested patient education, four used multiple components, two electronic reminders, two psychological approaches, and one each a change of medication regimen and better practitioner competency. Education delivered by community health workers or pharmacists, reinforced by regular follow-up, was the most promising approach, but the primary studies were of mixed methodological quality and 15 of the 18 trials that measured adherence directly relied on self-report or questionnaires rather than pill counts — thin ground for a programme-scale roll-out [PMID 37397765].

Elimination, surveillance and disease status

- **RATIONS** (Jharkhand, 28 tuberculosis units of the National Tuberculosis Elimination Programme) showed that food rations for household contacts of microbiologically confirmed pulmonary tuberculosis patients cut incident tuberculosis by 39% (all forms; adjusted HR 0.59, 95% CI 0.42–0.83) and by 48% for confirmed pulmonary disease, with about 30 households (111 contacts) needing supplementation to prevent one case. In the parallel 2,800-patient cohort — 82.4% underweight at enrolment — nutritional support accompanied 93.7% treatment success with loss to follow-up of only 1.0%, and a $\geq 5\%$ weight gain by two months carried a sharply reduced hazard of death (adjusted HR 0.39, 95% CI 0.18–0.86). Undernutrition is therefore an adherence and survival variable, not background detail [PMID 37567200; PMID 37567210].
- A meta-synthesis of 24 qualitative studies from the Indian subcontinent (12 from India) sorted adherence barriers into patient-related, provider-related and structural themes, with gender and stigma cutting across all three — women faced harsher stigma and poorer health-seeking, and marriage prospects both drove and sabotaged adherence. Its central warning is that the classical failures of directly observed treatment (wage loss, travel, loss of confidentiality) and the newer reminder and 99DOTS interventions share the same structural roots, so technology alone cannot fix them; more flexible arrangements such as family-observed treatment are needed [PMID 37142319].

Guideline and legislative updates

- The largest synthesis of digital adherence technologies — 102 studies mapped to the reach–effectiveness–adoption–implementation–maintenance framework — reports weighted mean

engagement across dose-days of 87% for smart pillboxes, 85% for phone-based systems, 82% for video-supported therapy, 81% for text messages and 61% for 99DOTS; mobile access ranged from 50% to 100% between settings and 2–31% of patients were excluded outright by technology barriers. Provider-side fidelity was the weak link, with low engagement often failing to trigger intensified patient management, and in Himachal Pradesh's 99DOTS roll-out across 74 tuberculosis units uptake between units ranged from 0.0% to 97.5% [PMID 39947715].

- ▶ Trial evidence for these technologies is genuinely split, which is the examinable point. A multicentre randomised trial in Tibet of an electronic medication monitor package cut poor-adherence months from 37% to 10% (adjusted risk difference –29.2 percentage points, 95% CI –35.3 to –22.2) and raised treatment success from 73% to 94% (+21 points, 12.4–29.4). A larger Chinese cluster-randomised trial of 2,686 patients improved pill-taking just as markedly (missing more than 20% of a month's doses, adjusted RR 0.36, 95% CI 0.27–0.50) yet left its primary composite unchanged (adjusted RR 1.01, 95% CI 0.73–1.40) — that composite counting death, loss to follow-up, treatment failure, a switch to multidrug-resistant therapy **or tuberculosis recurrence**, all measured to 18 months from the start of treatment. The narrower secondary composite, the same list minus recurrence, was equally flat (adjusted RR 0.97, 95% CI 0.69–1.35), and loss to follow-up alone accounted for most events. The reason is instructive — intensive management was triggered for only 16% of the intervention group and directly observed treatment for 8%, and reported doctor visits did not actually rise afterwards. Monitoring alone changes behaviour; outcomes move only when the monitor sits inside differentiated care that is genuinely delivered [PMID 38309280; PMID 37061308].

Global and WHO developments

- ▶ WHO's current figures put 10.7 million people falling ill with tuberculosis and 1.23 million deaths in 2024, with only about two in five drug-resistant patients treated and roughly 34,000 started on the six-month all-oral regimens (BPaLM, BDLLfxC) — regimen shortening is now the principal global adherence lever, the same logic that made **SHINE** practice-changing: in 1,204 children in Uganda, Zambia, South Africa and India, four months of treatment was non-inferior to six for non-severe, smear-negative, drug-susceptible tuberculosis (adjusted difference –0.4 percentage points, 95% CI –2.2 to 1.5) with 94% adherence, and it now underpins the shortened paediatric regimen [WHO 2025; PMID 35263517].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 5 (Epidemiology of Communicable Diseases), pp 224, 230, 232–234, 240, 249–250; Ch 6 (Epidemiology of Chronic Non-communicable Diseases and Conditions), pp 433, 449, 468, 471; Ch 7 (Health Programmes in India), pp 507, 511, 515, 523, 541, 565, 570; Ch 10 (Health Care of the Community), p 735; Ch 12 (Medicine and Social Sciences), pp 812, 835–836; Ch 22 (Communication for Health Education), p 1019. Sunder Lal — Textbook of Community Medicine, Ch 1 (Concept of Health and Disease), p 43; Ch 4 (Principles of Health Promotion and Education), pp 156–157; Ch 8 (Epidemiology of Communicable and Non-Communicable Diseases), pp 349, 364. Oxford Textbook of Public Health 7e — Ch 15 (Behavioural determinants), pp 595–596, 605; Ch 23 (Information systems), pp 908–909; Ch 33 (Community intervention trials), p 1225; Ch 50 (Behavioural economics and health), pp 1860–1861, 1865–1866; Ch 71 (Acquired immunodeficiency syndrome), pp 2659–2660; Ch 101 (Strategies for health services), p 3873. Sociology: The Essentials 6e, Ch 14 (Education and Health Care), p 390. Jugal Kishore — National Health Programmes of India, Ch 1, pp 306, 310 (International Standards for Tuberculosis Care, Standards 9 and 10).

Recent-advances sources: 38273246 — Direct benefit transfer for nutritional support of patients with TB in India—analysis of national TB program data of 3.7 million patients, 2018–2022, BMC Public Health 2024, doi:[10.1186/s12889-024-17777-7](https://doi.org/10.1186/s12889-024-17777-7); 38985653 — Estimation of potential social support requirement for tuberculosis patients in India, Health Policy and Planning 2024, doi:[10.1093/heapol/czae065](https://doi.org/10.1093/heapol/czae065); 41120559 — Preliminary effectiveness and feasibility of ASHA-led mobile health intervention for diabetes care in Indian primary health care settings, Scientific Reports 2025, doi:[10.1038/s41598-025-20728-w](https://doi.org/10.1038/s41598-025-20728-w); 39710532 — Prevalence of antihypertensive medication adherence and associated factors in India: a systematic review and meta-analysis, Hipertensión y Riesgo Vascular 2024, doi:[10.1016/j.hipert.2024.11.005](https://doi.org/10.1016/j.hipert.2024.11.005) (abstract-only); 37397765 — Interventions to promote medication adherence for chronic diseases in India: a systematic review, Frontiers in Public Health 2023, doi:[10.3389/fpubh.2023.1194919](https://doi.org/10.3389/fpubh.2023.1194919); 37567200 — Nutritional supplementation to prevent tuberculosis incidence in household contacts of patients with pulmonary tuberculosis in India (RATIONS), Lancet 2023, doi:[10.1016/S0140-6736\(23\)01231-X](https://doi.org/10.1016/S0140-6736(23)01231-X) (abstract-only); 37567210 — Nutritional support for adult patients with microbiologically confirmed pulmonary tuberculosis: outcomes in a programmatic cohort nested within the RATIONS trial in Jharkhand, India, Lancet Global Health 2023, doi:[10.1016/S2214-109X\(23\)00324-8](https://doi.org/10.1016/S2214-109X(23)00324-8) (abstract-only); 37142319 — Patient adherence to tuberculosis treatment in the Indian subcontinent: systematic review and meta-synthesis of qualitative research, BMJ Open 2023, doi:[10.1136/bmjopen-2022-063926](https://doi.org/10.1136/bmjopen-2022-063926); 39947715 — Implementation outcomes of tuberculosis digital adherence technologies: a scoping review using the RE-AIM framework, BMJ Global Health 2025, doi:[10.1136/bmjgh-2024-016535](https://doi.org/10.1136/bmjgh-2024-016535); 38309280 — Effectiveness of a comprehensive package based on electronic medication monitors at improving treatment outcomes among tuberculosis patients in Tibet: a multicentre randomised controlled trial, Lancet 2024, doi:[10.1016/S0140-6736\(23\)02270-5](https://doi.org/10.1016/S0140-6736(23)02270-5) (abstract-only); 37061308 — Digital adherence technologies to improve tuberculosis treatment outcomes in China: a cluster-randomised superiority trial, Lancet Global Health 2023, doi:[10.1016/S2214-109X\(23\)00068-2](https://doi.org/10.1016/S2214-109X(23)00068-2); 35263517 — Shorter Treatment

for Nonsevere Tuberculosis in African and Indian Children (SHINE), New England Journal of Medicine 2022, doi:[10.1056/NEJMoa2104535](https://doi.org/10.1056/NEJMoa2104535). Plus WHO tuberculosis fact sheet, 2025 update (2024 global estimates).



Changing concepts in public health

Asked in: Paper I 29-Jul-2021 — RGUHS MD Community Medicine

Definition — The term "public health" came into general use around 1840 and first appeared in law in the Public Health Act of England, 1848. C.E.A. Winslow (1920), as adapted by the WHO Expert Committee on Public Health Administration, defined it as "the science and art of preventing disease, prolonging life, and promoting health and efficiency through organized community efforts for the sanitation of the environment, the control of communicable infections, the education of the individual in personal hygiene, the organization of medical and nursing services for early diagnosis and preventive treatment of disease, and the development of social machinery to ensure for every individual a standard of living adequate for the maintenance of health" (Park 28e, Ch 2, p 65). The term survives but its content, target and goals have shifted repeatedly, through **four distinct phases** (Park 28e, Ch 1, pp 20–21).

The four phases in the evolution of public health

1. Disease control phase (1880–1920)

- ▶ Public health in the 19th century was largely **sanitary legislation and sanitary reform** aimed at control of man's physical environment — water supply, sewage disposal, garbage and refuse disposal, general cleanliness; **quarantine conventions** were held to contain disease (Park 28e, Ch 1, pp 17, 20).
- ▶ These measures were **not aimed at any specific disease** and were not based on technical knowledge, yet they improved health through "**death control**".
- ▶ Landmarks: the **Public Health Act of 1875** in England, and **Sir John Simon (1816–1904)**, first medical officer of health of London, who built the English public health system (Park 28e, Ch 1, p 17).

2. Health promotional phase (1920–1960)

- ▶ It was realised that public health had neglected the individual and that **the State had a direct responsibility for the health of the individual**; **health promotion of individuals** was therefore added as a second goal alongside disease control (Park 28e, Ch 1, p 20).
- ▶ **Personal health services** were organised — maternal and child health, school health, industrial health, mental health and rehabilitation services; **public health nursing** was a direct offshoot. Winslow's 1920 definition belongs to this phase.
- ▶ Two great movements followed: (a) **basic health services** through health centres and subcentres — concept mooted by **Lord Dawson (England, 1920)**, urged by the **League of Nations Health Organization (1931)**, recommended in India by the **Bhore Committee (1946)**; (b) the

Community Development Programme, which attempted too much too quickly with inadequate resources and failed to survive, but left behind the rural health infrastructure (Park 28e, Ch 1, p 20).

- **Social medicine** matured in parallel — seeded by **Neumann (1847)** and **Virchow (1848)**, with **Alfred Grotjahn** naming the role of social factors in aetiology "*social pathology*" (Park 28e, Ch 1, p 20).

3. Social engineering phase (1960–1980)

- **Acute illness problems were largely brought under control** in the developed world, and as old problems were solved **new ones emerged** — cancer, diabetes, cardiovascular disease, alcoholism and drug addiction, especially in affluent societies.
- These could **neither be tackled by the traditional tools** (isolation, immunization, disinfection) **nor explained on the basis of the germ theory**; hence the new concept of "**risk factors**" as **determinants of disease** (Park 28e, Ch 1, pp 20–21).
- Unlike the swift death of acute infection, they placed a **chronic burden on the society that created them**, forcing reorientation towards **social objectives**; social and behavioural aspects were given new priority and public health moved into the **preventive and rehabilitative aspects of chronic disease and behavioural problems**.
- The goals of public health and preventive medicine thereby became **identical** — prevention of disease, promotion of health, prolongation of life — and some leaders preferred the term "**community health**" (Park 28e, Ch 1, p 21).

4. "Health for All" phase (1981–2000 A.D.)

- Glaring contrasts between developed and developing countries came into sharp focus: only **10–20 per cent** of people in developing countries had ready access to health services of any kind, **60–250 of every 1000 live births** died within the first year, and life expectancy was **30 per cent lower** (Park 28e, Ch 1, p 21). *John Bryant (Health and the Developing World)* wrote that "large numbers of the world's people... have no access to health care at all".
- The **30th World Health Assembly (May 1977)** resolved that the main social target should be a level of health permitting all citizens by 2000 "to lead a socially and economically productive life" (Park 28e, Ch 1, p 21).
- The **Joint WHO–UNICEF conference at Alma-Ata (1978)**, of 134 countries, proclaimed **primary health care** as the way to Health for All; the **Global Strategy for Health for All** was adopted by the **34th World Health Assembly in 1981** (Park 28e, Ch 1, p 23).
- **Equity in health** — that all people should have an opportunity to enjoy good health — became the guiding principle (Park 28e, Ch 2, p 49).

The contemporary phase (2000 onwards): from Health for All to universal health coverage

- **Health promotion charters** — the First International Conference on Health Promotion (**Ottawa, 21 November 1986**) proclaimed the **Ottawa Charter**, with five action areas: build healthy public policy; create supportive environments; strengthen community action; develop personal skills;

re-orient health services. Further conferences followed (Jakarta, 1997) (Park 28e, Ch 2, pp 34, 49–50).

- **Millennium Development Goals** — at the Millennium Summit, New York, **September 2000**, representatives of **189 countries** adopted the UN Millennium Declaration with targets for **2015**; **3 of the 8 goals, 8 of the 18 targets and 18 of the 48 indicators** were health-related (Park 28e, Ch 1, p 23).
- **Sustainable Development Goals** — the UN General Assembly adopted the 2030 Agenda in **September 2015: 17 goals and 169 targets**, with **SDG 3** ("Ensure healthy lives and promote well-being for all at all ages") carrying **13 targets**; the **universal health coverage** target is key to achieving all the others (Park 28e, Ch 1, p 23; Ch 2, p 50).
- **Renewal of primary health care** — the importance of primary health care was reinforced by WHO in the **Astana Declaration (2018)**, and universal health coverage was adopted by the United Nations within the SDGs (Oxford Textbook of Public Health 7e, Ch 47, pp 1751–1752; Ch 101, pp 3857–3858).
- **Deprofessionalization and community participation** — a new cadre of health workers (community health workers, anganwadi workers, multipurpose workers) joined the health team, and health came to require "**self care**" plus community participation in planning and management — "Health by the People" (Park 28e, Ch 1, pp 21–23; Ch 2, p 34).

Parallel shifts in the underlying concepts

Axis	Earlier concept	Current concept
Concept of health	<i>Biomedical</i> — absence of disease; germ theory; body as a machine and the doctor its repairer	<i>Holistic</i> — a synthesis of the biomedical, ecological (health as a dynamic equilibrium between man and environment — <i>Dubos</i>) and psychosocial concepts; a sound mind in a sound body in a sound family in a sound environment (Park 28e, Ch 2, pp 25–26)
Explanation of causation	Single specific agent (germ theory)	Multifactorial causation; risk factors ; social, psychological, cultural, economic and political determinants (Park 28e, Ch 1, p 20; Ch 2, p 25)
Target of action	Man's physical environment, then the individual	The population as a whole, both sick and well, and the whole quality of life
Scope of public health	Sanitation and control of communicable disease	Also includes the organization of medical care services — proposed by a EURO symposium in 1966 , endorsed by a WHO Expert Committee in 1973 (Park 28e, Ch 2, p 66)
Preferred terminology	Public health → preventive and social medicine	Community health / community medicine — services to the population at large (Park 28e, Ch 1, p 21)
Levels of prevention	Primary, secondary, tertiary	Primordial and community prevention added; new dimensions of preventive paediatrics, preventive geriatrics and preventive cardiology (Park 28e, Ch 2, pp 65–66)
Locus of responsibility	The State alone	Individual (self care) + community participation + State + international responsibility (Park 28e, Ch 2, pp 26, 34)

Axis	Earlier concept	Current concept
Global goal	Health for All by 2000 A.D. (pledged 1981)	SDG 3 and universal health coverage by 2030 (Park 28e, Ch 1, p 23)

Changing goal of disease programmes: control → elimination → eradication

Goal	Definition	Example / status
Disease control	The disease agent is permitted to persist in the community at a level where it ceases to be a public health problem, according to the tolerance of the local population; an equilibrium is established between agent, host and environment	Malaria control — distinct from malaria eradication (Park 28e, Ch 2, p 58)
Disease elimination (regional elimination)	Interruption of transmission of a disease from a large geographic region or area; an intermediate goal between control and eradication, now seen as an important precursor of eradication	Elimination of measles, polio and diphtheria from large regions (Park 28e, Ch 2, p 58)
Disease eradication	Literally "to tear out by roots" — termination of all transmission of infection by extermination of the infectious agent ; an absolute, "all or none" process, reserved for cessation of infection and disease from the whole world	Smallpox is the only disease eradicated globally ; polio, measles and dracunculiasis are the candidates advanced for global eradication; polio remains endemic only in Afghanistan and Pakistan (Park 28e, Ch 2, p 58)

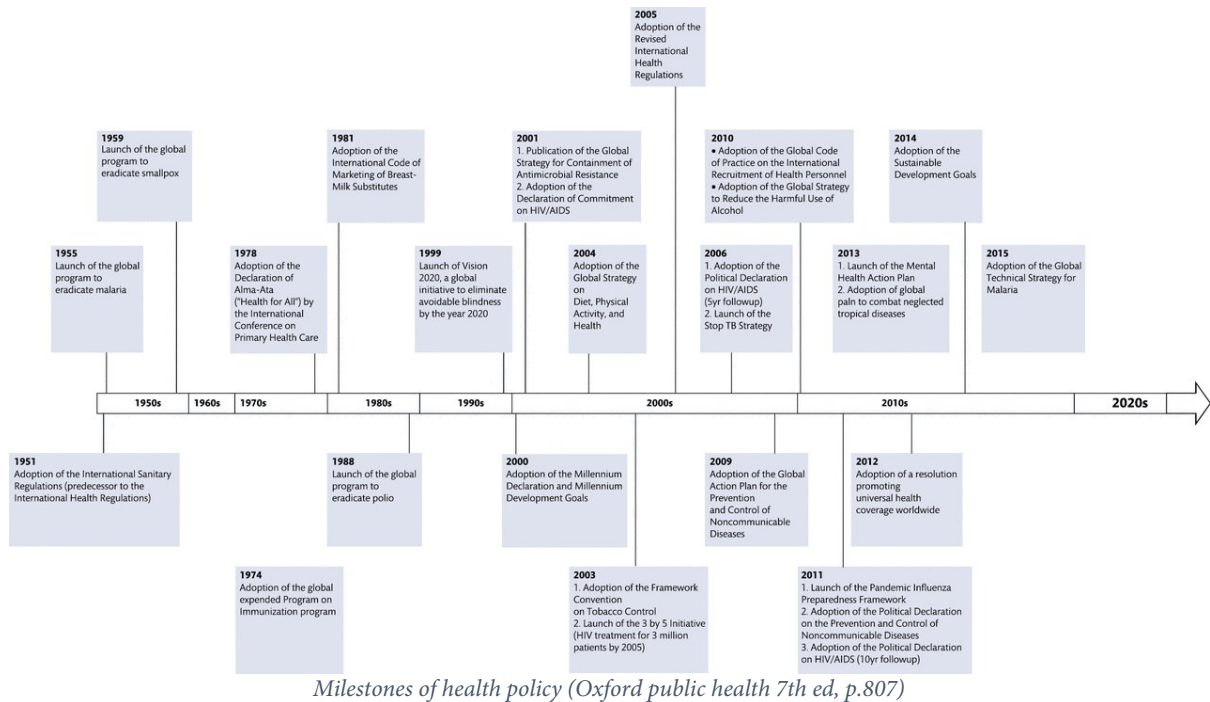
The resulting "new philosophy of health"

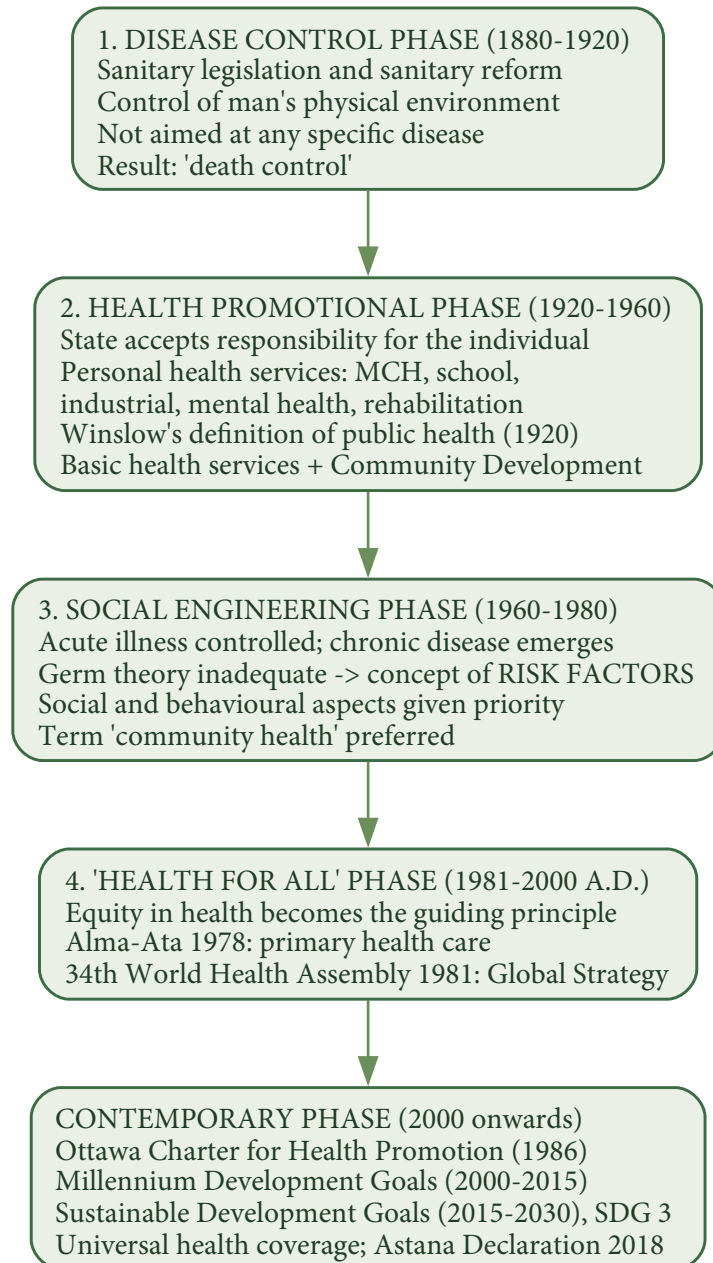
Park summarises the changed concept as eight propositions (Park 28e, Ch 2, p 26):

- ▶ Health is a **fundamental human right**.
- ▶ Health is the **essence of productive life**.
- ▶ Health is **intersectoral**.
- ▶ Health is an **integral part of development**.
- ▶ Health is **central to the concept of quality of life**.
- ▶ Health involves **individual, State and international responsibility**.
- ▶ Health and its maintenance is a **major social investment**.
- ▶ Health is a **worldwide social goal**.

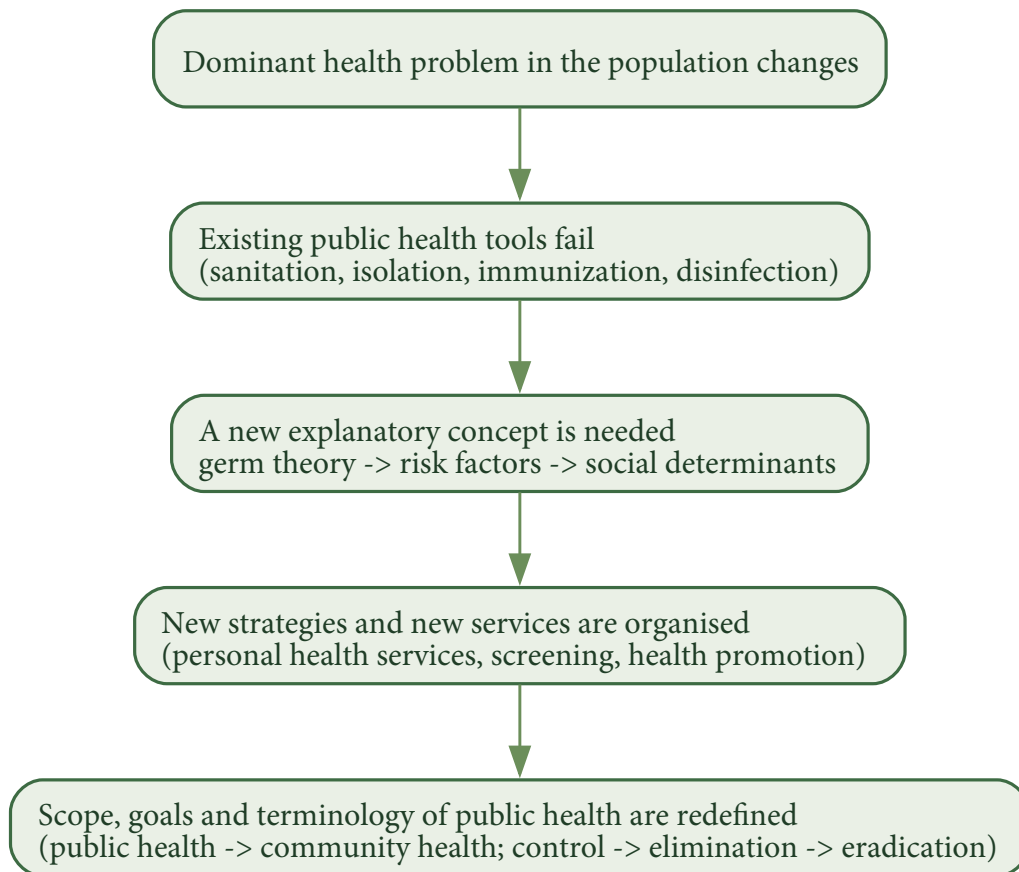
DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Hand-drawn figure — "Changing concepts of health" (four overlapping circles, one inside the next): label the circles, innermost outwards — Biomedical (absence of disease; germ theory) → Ecological (man–environment equilibrium; Dubos) → Psychosocial (social, psychological, cultural, economic, political factors) → Holistic (synthesis; all sectors of society affect health). Label the outer margin "health = a worldwide social goal".





The four phases in the evolution of public health, with the contemporary phase (Park 28e, Ch 1, pp 20-21)



How a changing disease pattern forces a changing concept of public health

High-yield exam pointers

- ▶ **Mnemonic for the four phases** — "D-H-S-H": Disease control (1880–1920) → Health promotional (1920–1960) → Social engineering (1960–1980) → Health for All (1981–2000 A.D.).
- ▶ Quote Winslow's 1920 definition verbatim and add that the WHO Expert Committee on Public Health Administration adapted it; the term "public health" itself came into general use around 1840, the Public Health Act of England was 1848, and the Act for control of the physical environment was 1875.
- ▶ Dated milestones to write: Alma-Ata 1978 (134 countries, WHO–UNICEF); Global Strategy for HFA, 34th World Health Assembly, 1981; Ottawa Charter, 21 Nov 1986 (5 action areas); MDGs, Millennium Summit 2000, 189 countries; SDGs, September 2015 — 17 goals, 169 targets, SDG 3 with 13 targets.
- ▶ Health for All inequity figures: only 10–20 % of people in developing countries had ready access to health services; 60–250 per 1000 live births died in the first year; life expectancy 30 % lower.
- ▶ **Control vs elimination vs eradication**: control = agent persists below the level of a public health problem; elimination = interruption of transmission in a **region**; eradication = **worldwide, all-or-none**, only **smallpox** achieved.
- ▶ Changing concepts of **health** (a favourite twin question): **biomedical** → **ecological** → **psychosocial** → **holistic**. Two eponyms worth naming: Sir John Simon and Alfred Grotjahn ("social pathology").

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park's 28th edition ends its formal list at the "Health for All" phase (1981–2000 A.D.) and names no fifth phase; the contemporary phase above is assembled from the Millennium Development Goals, Sustainable Development Goals and universal health coverage material in the same chapter, and the Astana Declaration of 2018 comes from the Oxford Textbook of Public Health 7e, not from Park.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 1, pp 17–23; Ch 2, pp 25–26, 33–34, 49–50, 58, 65–66); Oxford Textbook of Public Health, 7e (Ch 2, pp 196–201; Ch 47, pp 1751–1752; Ch 101, pp 3857–3858); Sunder Lal — Textbook of Community Medicine (Ch 1, p 38; Ch 2, Table 2.1 — historical development of medicine as a social science, "empirical health era" through the "health for/in all era, universal coverage"); WHO Ottawa Charter for Health Promotion, 1986; Declaration of Alma-Ata, 1978; UN Millennium Declaration, 2000; UN 2030 Agenda for Sustainable Development, 2015; WHO Astana Declaration on Primary Health Care, 2018.



Determinants and indicators of health status

Asked in: Paper I 12-Nov-2019 — RGUHS MD Community Medicine

Definition — Health is **multifactorial**: what a person is, and what diseases he or she may fall victim to, depends on the interaction of two sets of factors — the **genetic constitution** and the **environment** to which the person is exposed; these interactions may be health-promoting or deleterious, so the health of individuals and of whole communities is the result of many interactions (Park 28e, Ch 2, pp 30–31). A **health indicator** is a variable that helps measure change, directly or indirectly (Sunder Lal, Ch 16, p 725); in relation to health trends the term *indicator* is preferred to *index*, a **health index** being an amalgamation of several health indicators (Park 28e, Ch 2, p 36). Because health is multidimensional, health status can never be described by a single indicator — it must be conceived as a **profile employing many indicators** (Park 28e, Ch 2, p 36).

Determinants of health

#	Determinant	What it covers	Illustrative example
1	Biological (genetic)	Genetic make-up is unique and cannot be altered after conception; state of health depends partly on it	Chromosomal anomalies, inborn errors of metabolism, mental retardation, some types of diabetes; addressed today by genetic screening and gene therapy

#	Determinant	What it covers	Illustrative example
2	Behavioural and socio-cultural conditions (lifestyle)	"The way people live" — cultural and behavioural patterns and lifelong personal habits learnt through socialisation (parents, peers, school, mass media)	Harmful: smoking, alcoholism → coronary heart disease, obesity, lung cancer, drug addiction. In India traditional lifestyles add risks from poor sanitation, poor nutrition and personal hygiene. Health-promoting lifestyles also exist — adequate nutrition, enough sleep, sufficient physical activity
3	Environment	<i>Internal</i> environment (tissues, organs, organ systems — the domain of internal medicine) and <i>external/macro-environment</i> — "all that which is external to the individual human host", divided into physical, biological and psychosocial components; the <i>micro- or domestic environment</i> is the individual's personal way of living	Housing, water supply, psychosocial stress, family structure, social and economic support systems; Hippocrates first related disease to climate, water and air, revived later by <i>Pettenkofer</i> in Germany
4	Socio-economic conditions	(i) Economic status — gross national product (GNP), which determines purchasing power, standard of living, family size and pattern of disease; (ii) education , especially female education; (iii) occupation — being employed in productive work itself promotes health; (iv) political system — resource allocation, manpower policy, choice of technology	Female literacy → low infant mortality (Kerala, Table below). Affluence may itself cause illness — high rates of coronary heart disease, diabetes and obesity in upper socio-economic groups. WHO target: at least 5 % of GNP on health care; India spends about 1 % of GNP on health and family welfare
5	Health services	Availability, accessibility, affordability and acceptability of promotive, preventive, curative and rehabilitative care; to be effective, services must reach the social periphery , be equitably distributed, accessible and affordable	Immunisation of children and provision of safe water prevent morbidity and mortality; but "health care does not produce good health" — there is a strong correlation between GNP and expectation of life at birth, yet no significant correlation between medical density and expectation of life at birth
6	Ageing of the population	By 2020 the world had over 1.4 billion people aged 60 and over , more than two-thirds of them in developing countries	Increased prevalence of chronic diseases and disabilities accompanying ageing
7	Gender	Women's health placed on the development agenda; the Global Commission on Women's Health was established in 1993	Its agenda covered nutrition, reproductive health, health consequences of violence, ageing, lifestyle-related conditions and the occupational environment

#	Determinant	What it covers	Illustrative example
8	Other factors	The information and communication revolution; contributions from systems outside the formal health care system	Food and agriculture, education, industry, social welfare, rural development — medicine is not the sole contributor to the health of a population

(Park 28e, Ch 2, pp 30–32.) Sunder Lal condenses the same idea into **four inputs converging on the health goal** — **environment, behaviour of the host, health services and heredity** (Sunder Lal, Ch 1, pp 39–42).

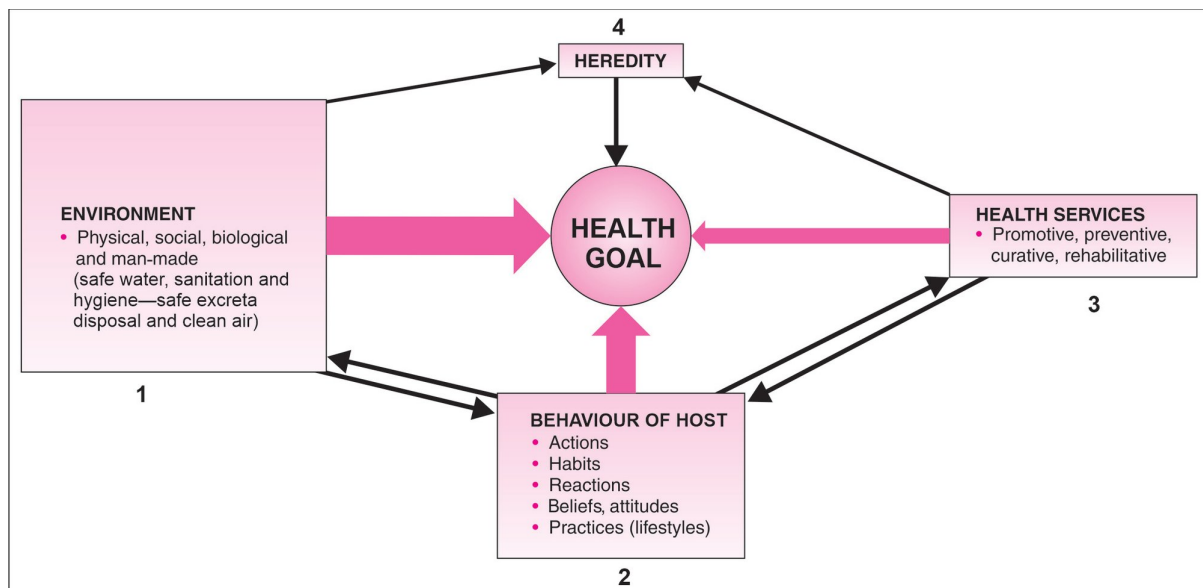
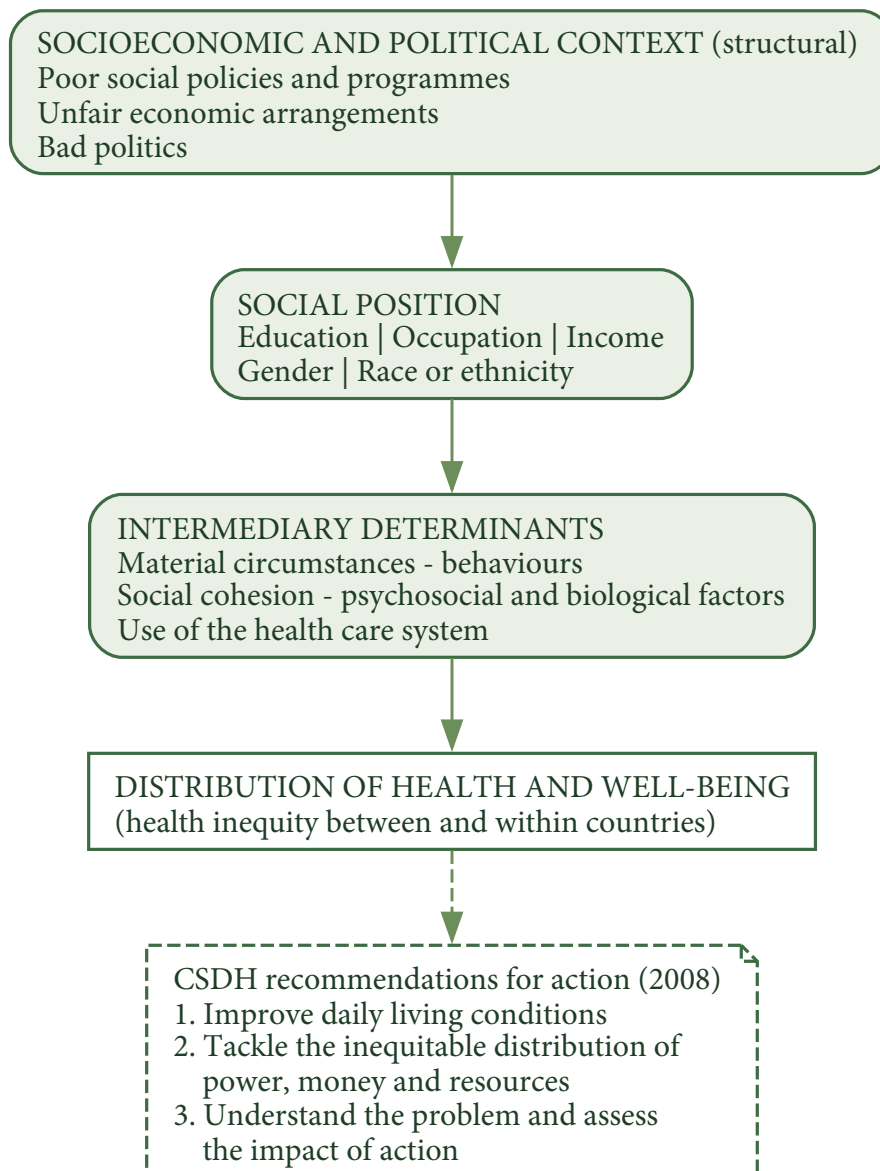


Fig 1.1 — Determinants of health (WHO*) (Sunderlal, p.40)

Social determinants of health are the "causes of the causes" — the conditions in and under which people are born, grow, live, work and age, and the broader set of forces and systems that shape the conditions of daily life (WHO). They include early years' experiences, education, economic status, employment and decent work, housing and environment (Sunder Lal, Ch 1, pp 41–42).



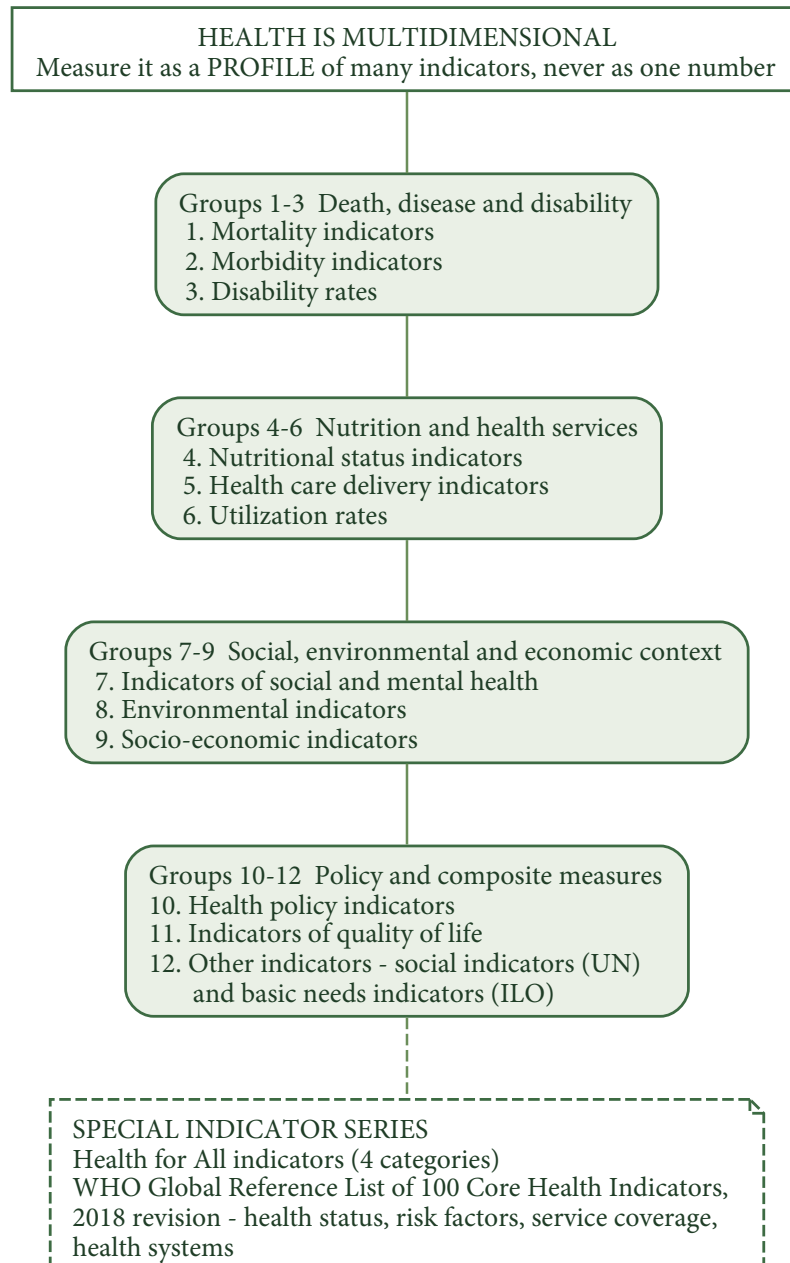
Social determinants of health - the Commission on Social Determinants of Health (CSDH) cascade from structural context to health outcome

Characteristics of a good indicator, and how indicators are classified

Characteristic	Meaning
Valid	It actually measures what it is supposed to measure
Reliable and objective	The answer is the same if measured by different people in similar circumstances
Sensitive	It is sensitive to — and measures — changes in the situation concerned as soon as they occur
Specific	It reflects changes only in the situation or phenomenon concerned
Feasible	The data needed can actually be obtained, and are simple to collect and analyse
Relevant	It contributes to the understanding of the phenomenon of interest

Characteristic	Meaning
Precise (Sunder Lal)	It carries sufficient detail — age, sex, geographical area

(Park 28e, Ch 2, p 36; Sunder Lal, Ch 16, p 725.) In real life **few indicators comply with all these criteria** (Park 28e, Ch 2, p 36). For programme evaluation, keep the list **short and specific (KISS)** (Sunder Lal, Ch 16, p 725).



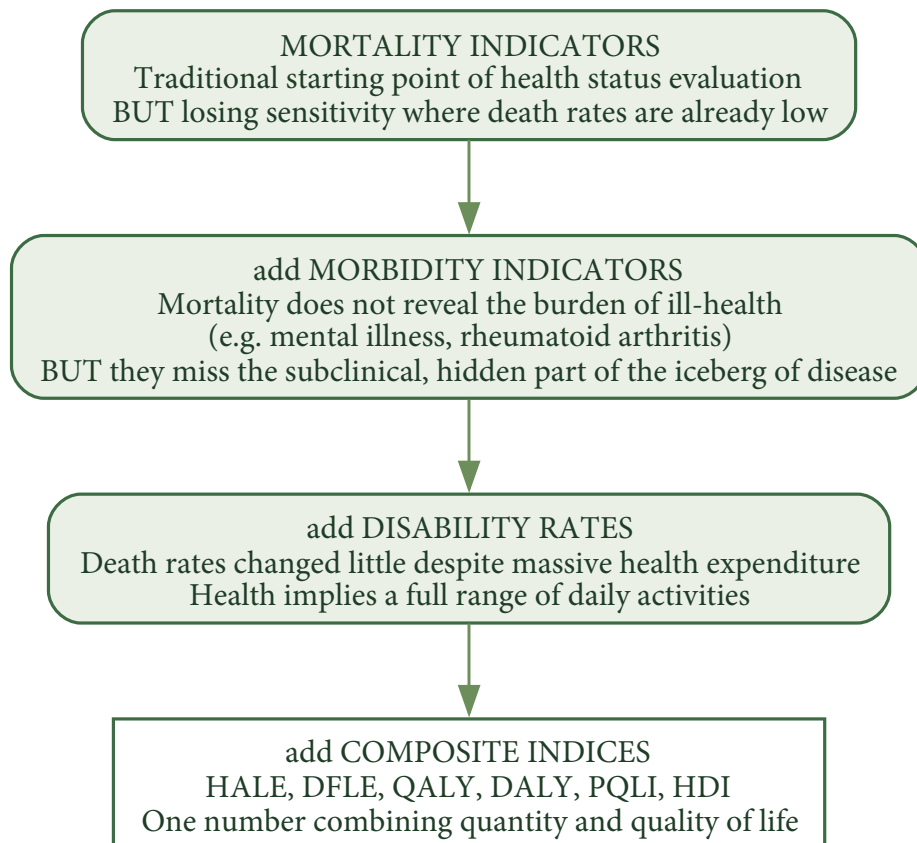
Park's twelve groups of health indicators, stacked for recall, plus the special indicator series

Mortality indicators — definition, denominator and current Indian value

Indicator	Definition (note the denominator)	Current Indian value (year)
Crude death rate (CDR)	Number of deaths per 1000 population in a given year	6.0 (SRS 2020)

Indicator	Definition (note the denominator)	Current Indian value (year)
Expectation of life at birth	Average number of years that will be lived by those born alive into a population if current age-specific mortality rates persist; a positive health indicator and a global health indicator	69.6 years (2014–17)
Age-specific death rate	Deaths in a specific age group in a defined area and period, per 1000 estimated population of the same age group, area and period	—
Infant mortality rate (IMR)	Deaths under 1 year of age in a given year per 1000 live births in the same year	28 (SRS 2020); one in every 36 infants dies within the first year
Child death rate (1–4 years)	Deaths at ages 1–4 years in a given year per 1000 children in that age group at the mid-point of the year ; it excludes infant deaths	—
Under-5 proportionate mortality rate	Proportion of total deaths occurring in the under-5 age group; reflects both infant and child mortality	Under-five mortality rate 35 (2019)
Adult mortality rate	Probability of dying between the ages of 15 and 60 years per 1000 population	—
Maternal mortality ratio (MMR)	Maternal (puerperal) deaths per 100,000 live births; accounts for the greatest proportion of deaths among women of reproductive age in the developing world	103 (SRS 2017–19)
Disease-specific mortality rate	Deaths from a named disease in the population	Deaths from cancer, cardiovascular disease, accidents, diabetes
Proportional mortality rate	Proportion of all deaths currently attributed to one disease — the simplest measure of a disease's burden	Coronary heart disease causes 25–30 % of all deaths in most western countries; proportional mortality from communicable disease indicates the magnitude of preventable mortality
Case fatality rate	Deaths from a specific disease in a period ÷ cases of that disease in the same period × 100; links mortality to morbidity and measures pathogenicity, severity or virulence	—
Years of potential life lost (YPLL)	Years of life lost through premature death, i.e. death before an arbitrarily determined age, usually 75 years	A 30-year-old dying in a road accident loses 45 years

(Park 28e, Ch 2, pp 36–38; Indian rates from Park 28e, Ch 2, Table 1, p 35 and SRS Bulletin 2022, Vol 55 No 1, pp 3–4.)



Why one group of indicators is not enough - the escalation from mortality to composite indices

Morbidity, disability and composite (summary) indices

Group	Indicators used
Morbidity indicators	(a) Incidence and prevalence; (b) notification rates; (c) attendance rates at out-patient departments and health centres; (d) admission, readmission and discharge rates; (e) duration of stay in hospital; (f) spells of sickness or absence from work or school
Disability — event-type	Number of days of restricted activity; bed disability days; work-loss (or school-loss) days within a specified period
Disability — person-type	<i>Limitation of mobility</i> (confined to bed, confined to house, needing a special aid to get around); <i>limitation of activity</i> (basic activities of daily living — eating, washing, dressing, going to the toilet, moving about; and limitation in a major activity such as ability to work at a job or do housework)
HALE (health-adjusted life expectancy; renamed from DALE)	Life expectancy at birth adjusted for time spent in poor health — the equivalent number of years in full health a newborn can expect to live at current rates of ill-health and mortality
DFLE (disability-free life expectancy; syn. active life expectancy)	Average number of years an individual is expected to live free of disability if current patterns of mortality and disability continue
QALY (quality-adjusted life year)	Combines quality and quantity of life; used to assess value for money of an intervention. 1 QALY = one year lived in perfect health; one year lived at a utility value of 0.5 = 0.5 QALY

Group	Indicators used
DALY (disability-adjusted life year)	DALY = YLL + YLD. YLL = deaths at each age × expected remaining years by a global standard life expectancy (Japanese life expectancy is used as the standard). YLD = incident cases × average duration × a disability weight from 0 (perfect health) to 1 (dead) . One DALY = one year of healthy life lost. Developed by Harvard University for the World Bank in 1990; adopted by WHO in 2000
PQLI (physical quality of life index)	Consolidates three indicators — infant mortality, life expectancy at age one, and literacy ; each scaled 0–100, averaged with equal weight ; deliberately excludes per capita GNP ("money is not everything"); the ultimate objective is a PQLI of 100
HDI (human development index)	Composite of three dimensions — a long and healthy life (life expectancy at birth), knowledge (mean years of schooling + expected years of schooling) and a decent standard of living (gross national income per capita, PPP US\$). Value 0–1: low < 0.550, medium 0.550–0.699, high 0.700–0.799, very high ≥ 0.800. Minimum goalposts: 20 years life expectancy, 0 years schooling, US\$ 100 GNI

(Park 28e, Ch 2, pp 38–39; PQLI and HDI — Park 28e, Ch 2, pp 29–30.)

The remaining indicator groups, with examples

Group	Indicators	Worked example
4. Nutritional status	(a) Anthropometry of preschool children — weight, height, mid-arm circumference; (b) heights (and sometimes weights) of children at school entry; (c) prevalence of low birth weight (< 2.5 kg)	NFHS-5 (2019–21): 36 % of under-fives stunted, 19 % wasted, 32 % underweight; 67 % of children 6–59 months anaemic
5. Health care delivery	Doctor–population ratio; doctor–nurse ratio; population–bed ratio; population per health centre/subcentre; population per trained birth attendant. They reflect the equity of distribution of health resources	Doctor–population ratio per 10,000 (2010–18): 2.5 in least developed countries, 11.5 in developing countries, 30.4 in high-HDI countries (Park 28e, Ch 2, Table 7)
6. Utilization rates	Proportion of people in need of a service who actually receive it in a given period, usually a year — e.g. proportion of infants fully immunised; proportion of pregnant women receiving antenatal care or delivering under a trained birth attendant; percentage using family planning methods; bed-occupancy rate; average length of stay; bed turnover ratio	NFHS-5 (2019–21): 76.6 % of children 12–23 months received all basic vaccinations; 3.6 % received none
7. Social and mental health	Positive indicators are scarce, so indicators of social and mental pathology are used: suicide, homicide, other violence and crime, road traffic accidents, juvenile delinquency, alcohol and drug abuse, smoking, consumption of tranquillisers, obesity, family violence, battered-baby and battered-wife syndromes, neglected and abandoned youth	These indicators are a guide to social action , not merely a measurement

Group	Indicators	Worked example
8. Environmental	Pollution of air and water, radiation, solid wastes, noise, exposure to toxic substances in food or drink; the most useful are proportion of population with access to safe water and sanitation	Households with safe water in the home or within 15 minutes' walking distance of a standpoint or protected well; adequate sanitary facilities in or near the home
9. Socio-economic	Rate of population increase; per capita GNP; level of unemployment; dependency ratio; literacy rates, especially female literacy; family size; housing (persons per room); per capita calorie availability. They do not directly measure health but are essential to interpret health indicators	India (SRS 2020): crude birth rate 19.5, natural growth rate 13.5 per 1000
10. Health policy	The single most important indicator of political commitment is allocation of adequate resources — proportion of GNP on health services; proportion of GNP on health-related activities (water supply, sanitation, housing, nutrition, community development); proportion of total health resources devoted to primary health care	—
11. Quality of life	Composite indices (PQLI above); WHO defines quality of life as the product of the interplay between social, health, economic and environmental conditions affecting human and social development	—
12. Other indicators	Social indicators (UN Statistical Office) — 12 categories including population, family formation, learning and educational services, earning activities, distribution of income and consumption, social security and welfare services, health services and nutrition, housing and environment, public order and safety, time use, social stratification and mobility. Basic needs indicators (ILO) — calorie consumption, access to water, life expectancy, deaths due to disease, illiteracy, doctors and nurses per population, rooms per person, GNP per capita	—

(Park 28e, Ch 2, pp 39–40; NFHS-5 India Report, 2019–21.)

Illustration — Kerala as a yardstick: determinants translated into indicators

Indicator (year)	Kerala	All India	The determinant it reflects
Crude death rate per 1000 (2020)	7.0	6.0	Higher in Kerala — a reminder that the crude death rate is influenced by the age–sex composition of the population and so is restricted for comparison
Rural birth rate (2020)	13.1	21.1	Education, age at marriage, family planning utilisation
Infant mortality rate (2020)	6.0	28	Female literacy, antenatal and perinatal care

Indicator (year)	Kerala	All India	The determinant it reflects
Under-five mortality rate (2019)	9.0	35	Nutrition, immunisation, environment
Maternal mortality ratio (2017–19)	30	103	Skilled birth attendance, health service accessibility
Life expectancy at birth (2014–17)	75.2	69.6	Cumulative effect of all determinants
Literacy rate, % (2021)	96.2	77.7	Education as a determinant
Female literacy rate, % (2019–20)	98.3	71.5	The single strongest correlate of Kerala's low IMR
Mean age at marriage, females (2019)	23.2	22.3	Socio-cultural determinant
Per capita income (2021–22)	₹2,36,093	₹1,28,829	Economic status

Kerala demonstrates that in a democratic system with **strong political commitment to equitable socio-economic development**, **high levels of health can be achieved even on modest levels of income**; it can therefore be considered a **yardstick** for judging health status in the country (Park 28e, Ch 2, Table 1, p 35). The converse illustration is that the world map of **illiteracy closely coincides** with the maps of poverty, malnutrition, ill-health and high infant and child mortality (Park 28e, Ch 2, p 32).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Determinants of health (Sunder Lal Fig 1.1)** — a central circle labelled *HEALTH GOAL* with four boxes converging on it by bold arrows, labelled exactly: **1 ENVIRONMENT** (physical, social, biological and man-made — safe water, sanitation and hygiene, safe excreta disposal, clean air); **2 BEHAVIOUR OF HOST** (actions, habits, reactions, beliefs and attitudes, practices/lifestyles); **3 HEALTH SERVICES** (promotive, preventive, curative, rehabilitative); **4 HEREDITY**. Add double-headed arrows between environment ↔ behaviour, behaviour ↔ health services, environment ↔ heredity and health services ↔ heredity to show that the determinants also act on one another.
- **Construction of the human development index** — three columns labelled *DIMENSIONS* (long and healthy life / knowledge / decent standard of living), *INDICATORS* (life expectancy at birth / mean years of schooling + expected years of schooling / GNI per capita in PPP US\$) and *DIMENSION INDEX* (life expectancy index / education index / income index), converging into one box labelled *HUMAN DEVELOPMENT INDEX* (geometric mean, value 0–1).

High-yield exam pointers

- **Park's 8 determinants** — *B-B-E-S-H-A-G-O*: Biological, Behavioural/socio-cultural, Environment, Socio-economic (economic status, education, occupation, political system), Health

services, Ageing of the population, Gender, Other factors. Sunder Lal's four inputs: environment, behaviour of host, health services, heredity.

- ▶ Social determinants = "**causes of the causes**" — conditions in which people are **born, grow, live, work and age** (WHO).
- ▶ **Characteristics of an indicator** — *valid, reliable, sensitive, specific, feasible, relevant* (+ precise); few indicators satisfy all six.
- ▶ **12 indicator groups** — mortality, morbidity, disability, nutritional status, health care delivery, utilization, social and mental health, environmental, socio-economic, health policy, quality of life, other (social and basic needs indicators).
- ▶ **Denominators that are commonly written wrong:** IMR = infant deaths ÷ **live births** × 1000; child death rate (1–4 y) = deaths ÷ **mid-year population aged 1–4** × 1000; MMR = maternal deaths ÷ **100,000 live births**; case fatality rate = deaths ÷ **cases** × 100; proportional mortality rate = deaths from a disease ÷ **all deaths**.
- ▶ **DALY = YLL + YLD**; disability weight 0 (perfect health) to 1 (dead); 1 DALY = 1 year of healthy life lost. **PQLI = IMR + life expectancy at age 1 + literacy** (excludes GNP, scaled 0–100). **HDI = life expectancy + schooling + GNI per capita PPP** (scaled 0–1; very high ≥ 0.800).
- ▶ **Dated Indian figures to quote:** CDR 6.0, crude birth rate 19.5, IMR 28, natural growth rate 13.5 (SRS 2020); under-five mortality 35 (2019); MMR 103 (SRS 2017–19); life expectancy 69.6 years (2014–17); TFR 2.0, stunting 36 %, all basic vaccinations 76.6 % (NFHS-5, 2019–21).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only — Park 28e still quotes the Millennium Development Goal indicators alongside the Sustainable Development Goal indicators; the current international indicator set is the **WHO Global Reference List of 100 Core Health Indicators, 2018 revision**, covering four domains (health status, risk factors, service coverage, health systems), and the maternal mortality ratio and infant/under-five mortality figures above are the latest carried in the retrieved corpus (Sample Registration System up to 2020 and 2017–19), which are superseded periodically by newer Sample Registration System bulletins.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 2 (Concept of Health and Disease), pp 29–41 (determinants of health pp 30–32; indicators of health pp 36–41; quality of life, PQLI and HDI pp 29–30; Lessons from Kerala State and Table 1, p 35). Sunder Lal — Textbook of Community Medicine, Ch 1 (Concept of Health and Disease, Ethics in Medicine and Doctor–Patient Relationship), pp 39–42 and Fig 1.1, p 40; Ch 16 (Health Planning and Management), p 725. Oxford Textbook of Public Health 7e — Ch 66, pp 2462–2463 (Commission on Social Determinants of Health framework and its three recommendations, CSDH 2008). Sample Registration System Bulletin 2022, Vol 55 No 1 (rates for 2020). National Family Health Survey (NFHS-5) India Report, 2019–21.



Counselling in HIV/AIDS

Asked in: Paper IV 13-May-2022 — RGUHS MD Community Medicine

Definition — Counselling is **face-to-face** communication by which one person helps another make a decision or solve a problem; a helping process aimed at problem-solving, it helps people understand themselves better in terms of their own needs, strengths, limitations and available resources, and brings change through a supportive relationship that aims to make the client **independent** (Sunder Lal, Ch 4, p 155). Park adds that it helps people deal with their problems, communicate better with those they are emotionally involved with, reinforce motivation to change behaviour, and provides support at times of crisis (Park 28e, Ch 22, p 1014).

Concept — what counselling is, and what it is not

Feature	Counselling	Advising	Health education
Basic transaction	Two-way, face-to-face helping relationship (Sunder Lal, Ch 4, p 155)	One-way direction — telling do's and don'ts (Park 28e, Ch 22, p 1014)	Educational process translating health knowledge into desirable behaviour (Park 28e, Ch 22, p 1014)
Element of choice	Implies choice, not force ; ends in an informed choice by the client (Park 28e, Ch 22, p 1014)	Amounts to directing and cautioning people	Persuasion without force or deliberate manipulation
Unit addressed	Individual, couple or family in a private setting	Individual	Individual, group or whole community
Intended outcome	Client resolves his own problem and becomes independent	Client complies with the adviser	Adoption and reinforcement of healthy behaviour patterns

Abilities required of a counsellor (Park 28e, Ch 22, p 1014) — in different circumstances different people may counsel, but the counsellor must be able:

- ▶ To communicate information.
- ▶ To gain the trust of the people.
- ▶ To listen sympathetically to people who are anxious, distressed and possibly hostile.
- ▶ To understand the other person's feelings and respond so that the other person feels free to express them.
- ▶ To help people reduce or resolve their problems.

Counselling thus relies on communication and relationship skills, develops positive attitudes, and is integral to treatment, disease prevention and health promotion in all health care programmes (Park 28e, Ch 22, p 1014). Its core is the "**holy trinity**" of counselling — **respect, empathy and genuineness** (Oxford Textbook of Public Health 7e, Ch 47, p 1754).

Steps of counselling — the GATHER approach

There are **six steps** to counselling; **GATHER** stands for Greet, Ask, Tell, Help, Explain and Return for follow-up (Sunder Lal, Ch 4, pp 156–157). Framed for contraceptive counselling, it is the standard sequence applied across programmes.

Step	What the counsellor does	Purpose
G — Greet	Greet the client politely, make them comfortable and give them full attention	Builds the connection and the trust on which clients rely
A — Ask	Ask clients about themselves — their needs, situation, existing knowledge and concerns	Obtains information that lets the provider advise each client individually
T — Tell	List all available options and clearly describe those that interest the client — how each works, advantages, disadvantages and possible side-effects	Even a client with a fixed preference should know the other options for future reference and to inform family and friends
H — Help	Help the client choose a method suited to her needs and plans; if a method is unsuitable, explain why and help select another; compare risks and benefits	The final decision must be an informed choice made by the client ; involving the husband/partner is desirable
E — Explain	Explain how to use the chosen method — when, where and how it is given, common side-effects, warning signs, and reasons to return	Prevents unnecessary discontinuation; supported by printed take-home cards, flip charts, cue cards and checklists
R — Return	Plan the follow-up visit, and encourage the client to return at any time with questions, problems or warning signs	Ensures continuation, and that the client knows when and where to seek care

Role of counselling in HIV/AIDS

Counselling here is concerned with preventing HIV infection and its transmission from persons living with HIV/AIDS (Sunder Lal, Ch 4, p 157). It operates at every level of prevention and is the mandatory gateway to testing — under the National AIDS Prevention and Control Policy **all HIV tests are voluntary, based on the client's consent, accompanied by counselling and confidentiality of the results** (Sunder Lal, Ch 4, p 160).

A. Main steps of preventive counselling (Sunder Lal, Ch 4, p 157)

- ▶ Determine whether the individual or group has a high-risk behaviour pattern.
- ▶ Help people understand and acknowledge their high-risk behaviour patterns.
- ▶ Define with them how their lifestyle and self-image are linked to their behaviour.
- ▶ Help individuals define their own potential for changing behaviour.
- ▶ Work with individuals to introduce and sustain the modified behaviour.

B. Role by level of prevention (Sunder Lal, Ch 4, p 157)

Level	Role of counselling
Health promotion	Counselling individuals who have high-risk behaviour but are unaware of the magnitude and nature of the risk; focuses on behaviour that presents a risk of HIV infection and reviews ways of managing individual change
Specific protection	Instructing those already infected on preventing onward spread — do not donate blood, use condoms during sexual intercourse, do not share needles and syringes — with an attitude of understanding, since a positive result is itself traumatic

Level	Role of counselling
Psychosocial support	Emotional and practical support for anxiety over hospitalisation and depression from stigma; enables people living with HIV to take charge of their lives, decide for themselves and remain active in work and education; involving family and friends reduces dependence on health and social services

C. Pre-test counselling — gives the person being tested simple, up-to-date information on the technical aspects of screening and the personal, medical, social, psychological and legal implications of a positive or negative result, with testing organised so as to minimise the possibility of disclosure. It has two parts (Sunder Lal, Ch 4, p 158):

- ▶ **Personal history and assessment of risk** — sexual behaviour (multiple partners, commercial sex, unprotected sex, homosexuality, bisexuality), injecting drug use, blood transfusion, organ transplant.
- ▶ **Assessment of factors and knowledge** — why the test is requested; what behaviour or symptoms concern the client; what he knows about the test and its uses; what he will do if positive or negative; beliefs about HIV transmission; the role the family will play.

D. Post-test counselling (Sunder Lal, Ch 4, p 158)

Result	Content of counselling
Negative	The client may feel relief, but caution is needed — there is a window period of 6–12 weeks between infection and detectable antibodies; a test in this period may be falsely negative, so retest after 12 weeks , and a negative result becomes a certainty only if 6 months have elapsed since the last exposure. Advocate safe sex, avoidance of high-risk behaviour and no needle-sharing
Positive	Give the result privately and in confidence ; a single positive test does not establish infection — three antibody tests based on different antigens must be positive. Allow time to absorb the news, then give a clear factual explanation — not speculation about prognosis or time left to live — and provide support, hope of achievable solutions, information on where resources and treatment are available, and the efficacy of antiretroviral treatment. Address the psychological sequelae: anxiety, anger, stigma, fear of death and even suicidal thinking

E. Requirements constant to all HIV counselling (Sunder Lal, Ch 4, pp 157–158) — **rappport** (do not rush the client with the news; the number of sessions depends on his baseline understanding); **acceptance** (treat with respect whatever the background or behaviour); **accessibility** (available whenever doubts arise); **consistency and accuracy** (keep abreast of current knowledge, never pretend to know); **confidentiality**. Since HIV infection is progressive, counselling must be **repeated regularly** and must tell the client where medical support and community resources lie.

F. Programme delivery — HIV counselling and testing services — services began in 1997; the former voluntary counselling and testing centres (VCTCs) were remodelled as **integrated counselling and testing centres (ICTCs)**, and after the National Guidelines of December 2016 the umbrella term is **HIV counselling and testing services (HCTS)** = ICTC + prevention of parent-to-child transmission (PPTCT) + HIV/tuberculosis collaborative activities (Sunder Lal, Ch 4, p 159; Park 28e, Ch 7, p 520). ICTC functions: early detection of HIV; basic information on transmission and prevention to promote behavioural change and reduce vulnerability; **linking people living with HIV to prevention, care and**

treatment services such as antiretroviral therapy (ART) centres; and outreach visits to positive clients in severe crisis (Park 28e, Ch 7, p 520).

Programme element	Detail
Types of ICTC	Fixed facility — standalone (SA-ICTC) with full-time counsellor and laboratory technician in medical colleges, district and sub-district hospitals and community health centres; facility-integrated (F-ICTC) below block level at 24×7 primary health centres using trained existing staff; plus mobile and public-private partnership ICTCs (Park 28e, Ch 7, pp 520–521; Sunder Lal, Ch 4, p 160)
Core staff	ICTC in-charge/manager (often a microbiologist), two trained counsellors (one male, one female; about 8–10 counselling sessions per counsellor per day), one trained laboratory technician (Sunder Lal, Ch 4, p 160)
Opt-out — provider-initiated	Client referred by a provider — symptoms suggestive of HIV (pneumonia, tuberculosis, persistent diarrhoea), sexually transmitted or reproductive tract infection, or a pregnant woman registering at the antenatal clinic; the client is informed and asked, and is tested unless she opts out (Sunder Lal, Ch 4, p 160)
Opt-in — client-initiated	Direct walk-in or self-referred client acting on his own risk perception or peer-educator advice; counselled, then actively opts in — written consent is taken before testing (Sunder Lal, Ch 4, p 160)
PPTCT	Started 2002 ; aims at universal HIV testing of every pregnant woman to eliminate vertical transmission; community-based HIV screening is done by auxiliary nurse midwives (ANMs) at sub-centre level (Park 28e, Ch 7, p 521)
Peer counselling	HIV-positive men and women specially trained to hear the concerns of other clients (Sunder Lal, Ch 4, p 160)

G. Contribution to national targets — counselling drives the first cascade target, since a person must be counselled and tested before he can be treated: against a target of 90%, **78% of people living with HIV in India know their status** (Park 28e, Ch 7, p 526, national achievement). NACP Phase-V (2021–2026) aims to cut annual new HIV infections and AIDS-related mortalities by **80% by 2025–26** from the 2010 baseline, attain dual elimination of vertical transmission, and **eliminate HIV/AIDS-related stigma** — achievable only through counselling and communication (Park 28e, Ch 7, p 517).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- DRAWN figure — reproduce the **model of HIV counselling and testing services by level of care** as a pyramid, labelled: *Medical college / District hospital — doctor, counsellor, laboratory technician — standalone ICTC; Community health centre — standalone or facility-integrated ICTC, designated microscopy centre; Primary health centre and 24×7 PHC — doctor, nurse/ANM, laboratory technician — facility-integrated ICTC; Sub-centre and anganwadi — ASHA, ANM, multipurpose worker — community-based HIV screening; with a referral arrow running upward on the left and the service package (ICTC, PPTCT, sexually transmitted infection and opportunistic infection care, ART, condom promotion, information-education-communication) on the right.*

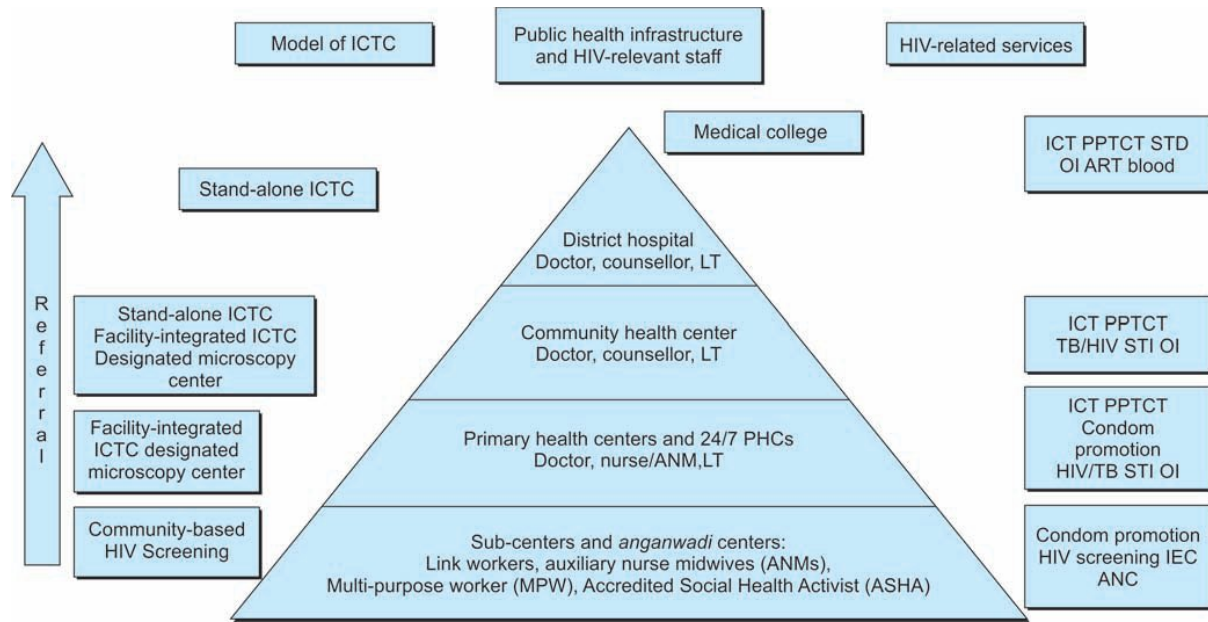
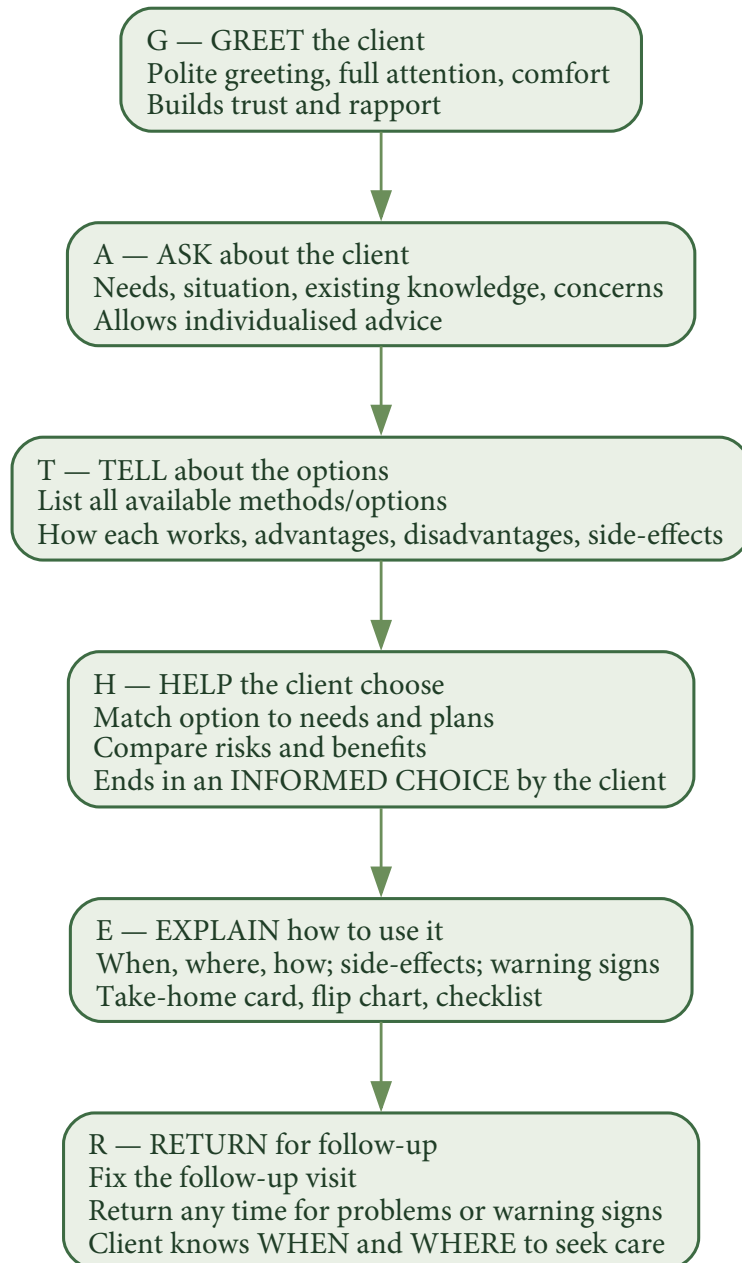
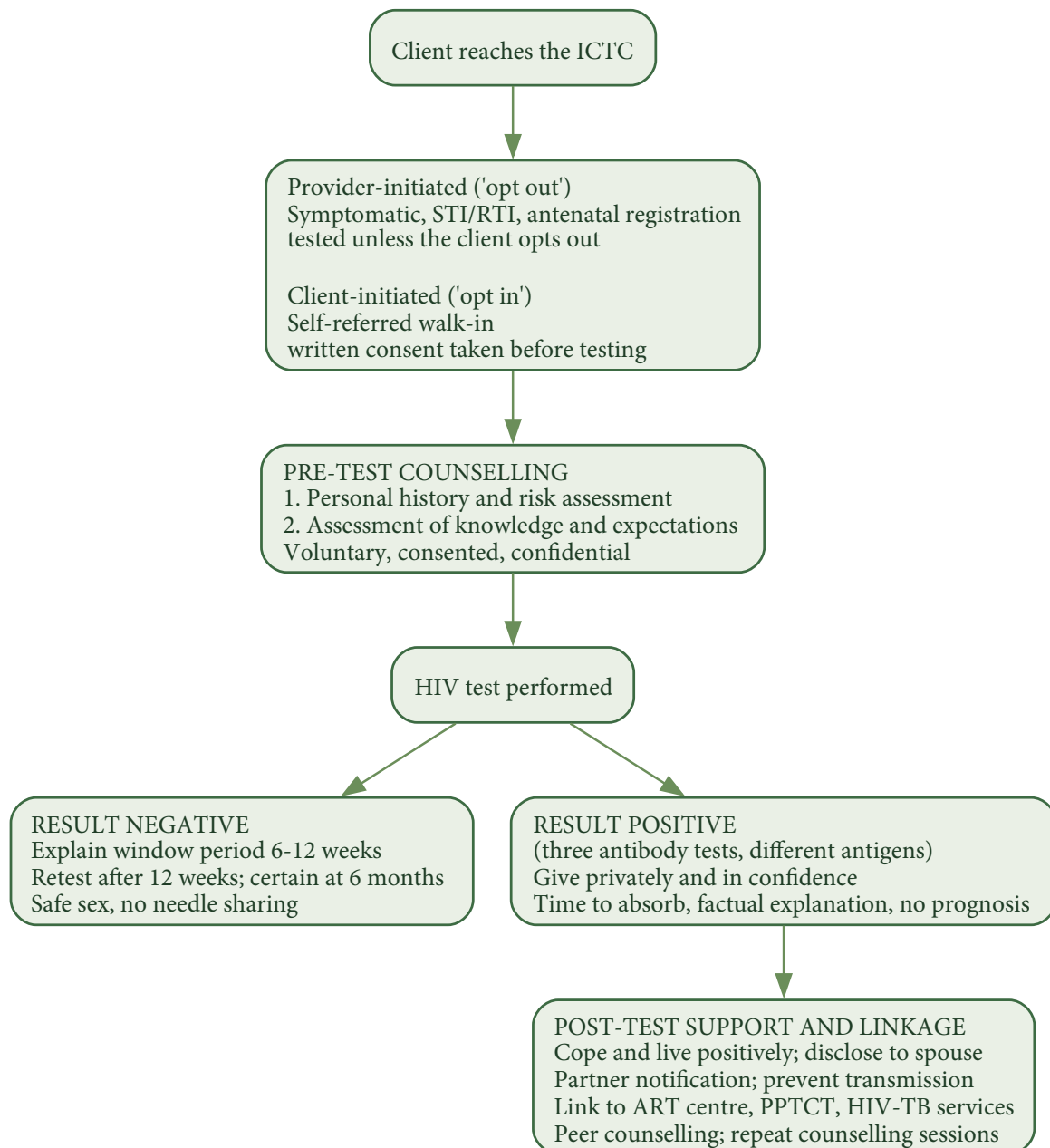


Fig 43.5 — Model of ICTC (Suryakantha Environment Best, p.863)



The six steps of counselling — the GATHER approach



HIV counselling and testing pathway at an integrated counselling and testing centre (ICTC)

High-yield exam pointers

- ▶ **GATHER = Greet, Ask, Tell, Help, Explain, Return for follow-up** — six steps, the standard answer to "enumerate the steps of counselling".
- ▶ Counselling **implies choice, not force**; advising directs people to do's and don't's (Park 28e, Ch 22, p 1014). The "holy trinity" is **respect, empathy, genuineness**.
- ▶ **Three C's of HIV testing** — **Consent, Counselling, Confidentiality**: all HIV tests are voluntary, consented, counselled and confidential under the National AIDS Prevention and Control Policy.
- ▶ **Window period 6–12 weeks**; retest at 12 weeks; negative certain at 6 months after last exposure. **Three antibody tests on different antigens** are needed to declare a person HIV positive.
- ▶ Five requirements of HIV counselling — **rapport, acceptance, accessibility, consistency/accuracy, confidentiality**.

- ▶ VCTC → ICTC → HCTS (National guidelines, December 2016); HCTS = ICTC + PPTCT + HIV/TB collaborative activities. Services began 1997; PPTCT began 2002.
- ▶ 78% of people living with HIV in India know their status against a 90% target (Park 28e, Ch 7, p 526); NACP Phase-V (2021–2026) targets an 80% reduction in new infections and AIDS-related deaths by 2025–26 from the 2010 baseline.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only — the counts of counselling and testing centres differ between texts (Sunder Lal gives more than 34,500 centres, Park 28e more than 15,000 ICTCs offering PPTCT) and are revised periodically by the National AIDS Control Organisation, so quote the structure rather than an exact count unless the current NACO India HIV Estimates report is at hand.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, WHO, UNAIDS, FDA, HIV and AIDS (Prevention and Control) Act 2017

Programme and policy updates (India)

- ▶ **Counselling has moved onto the phone — "virtual counsellors" now carry HIV self-testing.** The USAID-supported *Safe Zindagi* web platform distributed a Drug Controller General of India-approved oral-fluid self-test kit free of cost; between 30 June 2021 and 30 September 2022, 5,324 clients ordered a kit, 94% of those who received one uploaded a result, 6% screened positive (about 27 times the general-population prevalence), 72% of them completed confirmatory testing and 85% of confirmed cases reached antiretroviral therapy. Three-quarters completed the test with counsellor help, and client interviews showed the counsellor supplied exactly what GATHER supplies at a centre — trust, privacy reassurance, emotional support at the moment of a reactive result, and linkage; 53% of those who answered had never tested before [PMID 38861458].
- ▶ **STAR III showed the "Ask" and "Tell" steps can be generated virtually and at scale.** Run by PATH with the National AIDS Control Organisation and WHO across 50 districts in 14 states (April 2022 to February 2023), social-media advertising, influencers and virtual peer navigators reached over 1.4 million people; 712 of 1,087 eligible registrants ordered a self-test kit, 98.9% were followed up, 1.8% were reactive and 80% of those started antiretroviral therapy; 95.9% found the kit easy to use, 93.6% easy to interpret, 87.1% would choose self-testing again and 61% were first-time testers [PMID 41195967].
- ▶ **Decentralised antiretroviral therapy is now offered by counselling, not assigned — and counselling must handle the privacy trade-off.** In Andhra Pradesh, 20,761 of 213,375 people in care (9.7%) were travelling more than 50 km for refills; 18,418 were contacted with a standard counselling script and 48.2% chose a facility nearer home. Twelve-month retention was 96.5% versus 95.2% with lower loss to follow-up (2.3% versus 3.6%) and identical mortality (1.2%). Among those who declined, 29% cited loss of anonymity and 5% self-perceived stigma — bringing services closer to home can threaten confidentiality, which is now an explicit counselling task [PMID 42231306].

Current national data

- **The "first 95" figure has moved past the textbook's 78%.** India has about 2.5 million people living with HIV, adult prevalence 0.21%, and on 2022–23 estimates 79% know their status, 86% of those are on antiretroviral therapy and 93% of those tested for viral load are suppressed — so quote **79–86–93** for the cascade. The epidemic stays concentrated: prevalence 9.03% in people who inject drugs, 3.78% in transgender people, 3.26% in men who have sex with men and 1.85% in female sex workers, which is why counselling is targeted at key populations rather than the general population [PMID 38861458].
- **NFHS-5 (2019–21) measures the knowledge gap counselling exists to close.** Among 84,482 men aged 15–54, only **36.4%** had comprehensive knowledge of HIV; it was 47.4% in men ever tested versus 36.2% in never-tested men (adjusted OR 1.26; 95% CI 1.20–1.33), and full mass-media exposure raised the odds by two-thirds (adjusted OR 1.67; 95% CI 1.56–1.78). Pre-test and post-test counselling is thus itself a health-education intervention, not merely a gateway to a test [PMID 40700410].

Elimination, surveillance and disease status

- **Diagnosis, not treatment, is the weak link in this region.** South and South-East Asia carries 3.5 million people living with HIV, and as of 2022 the Asia-Pacific cascade stood at **78–84–95** — the first 95 lagging furthest. As of August 2022 India, Indonesia and Pakistan had no national self-testing policy while Cambodia, Sri Lanka, Thailand and Vietnam were already implementing one; secondary distribution of kits through social networks in India's STAR III reached 68.6% first-time testers with 96% acceptance, making self-testing the main lever on the first target [PMID 39397317].

Guideline and legislative updates

- **The HIV and AIDS (Prevention and Control) Act, 2017 puts the "three C's" into statute.** Informed consent is mandatory before an HIV test, and **pre-test and post-test counselling are written into the consent requirement itself**, with narrow exceptions for epidemiological work, blood-bank screening, court directions and research. No one may be compelled to disclose status except by court order; only a physician or a counsellor may inform a partner, only where non-disclosure poses a serious risk, and only after telling the person concerned of that intention. Disclosure is not required of a woman where it could bring violence or abandonment, and a person living with HIV carries a duty — **after counselling** — to take reasonable precautions against transmitting to sexual or needle-sharing partners [PMID 40983475].
- **What post-exposure and pre-exposure counselling must now contain in India.** First-line therapy is the fixed-dose combination tenofovir disoproxil fumarate/lamivudine/dolutegravir, with "Treat All" and routine viral-load monitoring as policy. The national post-exposure prophylaxis document keeps a 28-day three-drug course begun as early as possible — preferably within 2 hours and always within 72 hours — and several institutions have shifted to a dolutegravir-based course in line with WHO's 2024 advice favouring three drugs. Tenofovir/emtricitabine is approved for pre-exposure prophylaxis with national technical guidance in place, but rollout remains patchy, so eligibility counselling matters [PMID 42543939].

Global and WHO developments

- ▶ **WHO now recommends twice-yearly injectable lenacapavir for pre-exposure prophylaxis — and simplified testing to deliver it.** The guidelines issued on 14 July 2025 add long-acting lenacapavir as an extra prevention option and, critically for counselling services, endorse a public-health approach using **HIV rapid tests** rather than complex algorithms to support long-acting injectable prophylaxis, opening delivery through pharmacies, community sites and telehealth. The US regulator approved the agent for prevention in June 2025 [WHO 2025-Jul] [FDA 2025-Jun] [PMID 42543939].
- ▶ **Latest global figures to quote alongside the Indian ones.** For 2025: 41.0 million people living with HIV, 1.2 million new infections and 570,000 AIDS-related deaths; 88% of people with HIV knew their status, 89% of those were on treatment and 95% of those treated were virally suppressed — which works out to 88–78–74 across everyone living with HIV [WHO 2026] [UNAIDS 2026].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 22, p 1014; Ch 7, pp 517, 520–521, 526. Sunder Lal — Textbook of Community Medicine, Ch 4, pp 155–160. Oxford Textbook of Public Health 7e, Ch 47, p 1754. Figure: Suryakantha — Community Medicine with Recent Advances, Fig 43.5, p 863. Programme documents cited by these texts: National AIDS Prevention and Control Policy; National HIV Counselling and Testing Services Guidelines, December 2016; National AIDS Control Programme Phase-V (2021–2026).

Recent-advances sources: 38861458 — HIV self-testing in India: implementation and qualitative evaluation of a web-based programme with virtual counsellor support, *Journal of the International AIDS Society* 2024, DOI [10.1002/jia2.26302](https://doi.org/10.1002/jia2.26302); 41195967 — Virtual model for HIV self-testing demand generation and kit distribution: results from the STAR III study in India, *Journal of Acquired Immune Deficiency Syndromes* 2026, DOI [10.1097/QAI.0000000000003792](https://doi.org/10.1097/QAI.0000000000003792) (abstract-only); 42231306 — Factors associated with opting care from closer location to residence among PLHIV at Andhra Pradesh, India between 2023 and 2024, *BMC Health Services Research* 2026, DOI [10.1186/s12913-026-14523-0](https://doi.org/10.1186/s12913-026-14523-0); 40700410 — Factors influencing HIV knowledge among Indian men: a cross-sectional study, *PLoS One* 2025, DOI [10.1371/journal.pone.0327411](https://doi.org/10.1371/journal.pone.0327411); 39397317 — Leveraging HIV self-testing to achieve the UNAIDS 2025 targets in the South and Southeast Asia region, *Journal of the International AIDS Society* 2024, DOI [10.1002/jia2.26357](https://doi.org/10.1002/jia2.26357); 40983475 — Critiquing the HIV and AIDS Act, 2017 in India through a reproductive justice framework, *Medical Law Review* 2025, DOI [10.1093/medlaw/fwaf034](https://doi.org/10.1093/medlaw/fwaf034); 42543939 — Antiretroviral therapy in India (2025): drugs, indications, contraindications, mechanisms, and prophylaxis (PEP/PrEP), *Journal of the Association of Physicians of India* 2026, DOI [10.59556/japi.74.1559](https://doi.org/10.59556/japi.74.1559) (abstract-only). WHO guideline recommendation on twice-yearly injectable lenacapavir for pre-exposure prophylaxis, 14 July 2025; US FDA approval of lenacapavir for pre-exposure prophylaxis, June 2025; WHO HIV fact sheet and UNAIDS global fact sheet (2025 estimates).

♦ ♦ ♦

Acculturation

Asked in: Paper I 12-Nov-2021 — RGUHS MD Community Medicine

Definition — Acculturation means "**culture contact**": when two peoples with different types of culture come into contact, there is **diffusion of culture both ways** (Park 28e, Ch 12, p 815). It is therefore the diffusion of culture between groups and communities, occurring through migration, urbanization, industrialization and social mobility such as education, trade, globalization and information technology (Sunder Lal, Ch 2, pp 83-84). Because cultural factors are deeply involved in personal hygiene, nutrition, immunization, seeking early medical care, family planning, child rearing and disposal of refuse and excreta — in short the whole way of life — culture contact "**has both good and bad aspects**" for health (Park 28e, Ch 12, p 815).

Related sociological terms

Term	Meaning
Cultural diffusion	Transmission of cultural elements from one society or cultural group to another — the mechanism underlying acculturation (Sociology: The Essentials 6e, Ch 2, p 75)
Assimilation	A minority becomes socially, economically and culturally absorbed within the dominant society, giving up much of its own culture (Sociology 6e, Ch 10, p 270)
Pluralism	The opposite pole — maintenance and persistence of one's own culture, language, practices and art despite contact (Sociology 6e, Ch 10, p 270)
Culture lag	Delay in cultural adjustment to changing social conditions; material culture changes first, non-material culture (habits, mores) lags behind — term coined by <i>William F. Ogburn</i> , 1922 (Sociology 6e, Ch 2, p 75; Ch 16, p 444)
Culture shock	Feeling of disorientation on encountering a new or rapidly changed cultural situation (Sociology 6e, Ch 2, p 75)

Channels through which culture contact takes place

Park lists five classical channels; Sunder Lal adds the contemporary drivers.

Channel	Illustration / health-relevant consequence
1. Trade and commerce	Entry of commercial products — marketed breast-milk substitutes, tobacco, processed and fast foods
2. Industrialization	Mechanized work, sedentary occupations, factory and urban living
3. Propagation of religion	Missionary hospitals and dispensaries; changes in food and ritual practice
4. Education	Literacy and adoption of scientific explanations of disease in place of supernatural ones
5. Conquest	The British brought their culture into India through conquest — "an Indian is said to be the next best Englishman" (Park 28e, Ch 12, p 815)
Contemporary drivers	Migration, urbanization, industrialization and social mobility — education, trade, globalization, information technology (Sunder Lal, Ch 2, pp 83-84)

Channel	Illustration / health-relevant consequence
Mass media as amplifier	"The radio, [television] and the cinema have been important factors in shaping cultural-behaviour patterns of people" (Park 28e, Ch 12, p 815)

Positive aspects of acculturation

Positive aspect	Substance
Introduction of scientific medicine	"The introduction of scientific medicine is through culture contact" (Park 28e, Ch 12, p 815) — the first benefit Park names
Displacement of harmful customs	Along with scientific medicine go the customs Park classifies as <i>bad</i> — delivery by the untrained <i>dai</i> , branding of the skin, administration of opium and drastic purgatives, and food taboos in pregnancy on eggs, meat, fish, milk and leafy vegetables (Park 28e, Ch 12, p 830)
Beneficial change in food habits	"The changes in food habits of people is brought about through culture contact; many orthodox brahmins in India today eat meat" (Park 28e, Ch 12, p 815)
Diffusion genuinely runs both ways	The traditional society is not only a receiver — Western people may adopt Indian culture, become vegetarian and take up the Indian diet, <i>yoga</i> and meditation (Sunder Lal, Ch 2, pp 83-84)
Lowering of the cultural barrier to services	Poor acceptability arising from wrong beliefs, attitudes and practices is listed as a standing cultural barrier to health-seeking behaviour; understanding the culture of the people is what makes a programme or service acceptable — acceptance of family planning needs thorough understanding of the community's reproductive knowledge, beliefs, attitudes and practices (Sunder Lal, Ch 2, p 84)
Makes migrant studies possible	Comparing migrants with kin who stayed at home, or with the host population, separates genetic from environmental causation — men of Japanese ancestry in the USA have a higher coronary heart disease rate than Japanese in Japan, and third-generation Japanese-Americans reach US stomach and colon cancer rates (Park 28e, Ch 3, p 88)

Negative aspects of acculturation

Negative aspect	Substance and dated Indian figure
Spread of tobacco	"The widespread use of tobacco in the world is because of culture contact" (Park 28e, Ch 12, p 815) — culture contact is what carried tobacco worldwide in the first place, even though Indian prevalence is now falling under tobacco control. India: 28.6% of adults use tobacco in any form (Global Adult Tobacco Survey-2, 2016-17, down from 34.6% in GATS-1, 2009-10), and 8.5% of minors aged 13-15 years (Global Youth Tobacco Survey-4, 2019, down 42% from 14.6% in 2009) (Sunder Lal, Ch 15, p 689; Ch 8, pp 501-502)
Tobacco's disease and money cost	Nearly 30% of cancers in India are related to tobacco use; economic cost Rs 1,04,500 crore = 1.16% of GDP (2011-12) (Sunder Lal, Ch 15, p 689)
Alcohol and pan	Cirrhosis of liver from drinking and lung cancer from smoking in developed countries are "the result of the abuse of widely proclaimed social habits"; in India chewing <i>pan</i> is associated with oral cancer (Park 28e, Ch 12, p 815). Dependence on or harmful use of alcohol is prevalent in 4.6% of Indian adults (Sunder Lal, Ch 8, pp 501-502)

Negative aspect	Substance and dated Indian figure
Dietary transition to junk food	Globalization changed the food habits of children and adolescents through the availability, accessibility and affordability of fast and junk foods (Sunder Lal, Ch 8, p 483). Under-5 overweight/obesity rose from 2.1% (NFHS-4, 2015-16) to 3.4% (NFHS-5, 2019-21), attributed to junk-food consumption (Sunder Lal, Ch 5, pp 179-180)
Physical inactivity	Urbanization, industrialization and mechanization cut energy expenditure — 41% of Indian adults fail to meet minimal levels of physical exercise (WHO-ICMR study, 2020) (Sunder Lal, Ch 8, p 485)
The migrant's own evidence	People of Indian origin living in the USA, UK and Malaysia have higher incidence and prevalence of obesity than the local people — the direct demonstration that acquiring a host culture damages health (Sunder Lal, Ch 8, p 485)
Shift in the national disease burden	As lifestyles changed and urbanization and globalization altered living patterns, disability-adjusted life years (DALYs) lost to communicable, maternal, neonatal and nutritional causes fell from 61% (1990) to 33% (2016), non-communicable disease rose from 30% to 55% and injuries from 9% to 12% (ICMR, PHFI, IHME 2017, <i>India: Health of the Nation's States — The India State-Level Disease Burden Initiative</i> , quoted in Sunder Lal, Ch 8, p 498)
Erosion of good traditional practice	Commercial promotion of breast-milk substitutes threatened India's universal breastfeeding, requiring the WHO International Code of Marketing of Breast-milk Substitutes (1981) and the Indian National Code, later reinforced by the Infant Milk Substitutes Act 1992 (amended 2003); non-nutritive liquids reduce breast-milk intake (Sunder Lal, Ch 5, pp 176-177)
Sociological cost	See <i>culture lag</i> and <i>culture shock</i> above

Social stress — Park's heading for the clash of two cultures

Park sets this out under **SOCIAL STRESS** (Ch 12, p 815). A typical feature of traditional societies is a sense of continuity and immutability in patterns of social life, whereas transitional societies "may be better able to cope with change" and modern societies "are perhaps best adapted to assimilate rapid changes". In transitional societies young people are therefore simultaneously involved in **two cultures** — the traditional one their parents grew up in and still value, and the modern one portrayed in the mass media; a similar clash occurs in families who have migrated to another country or from a rural to an urban area. The stress-inducing conditions Park lists are:

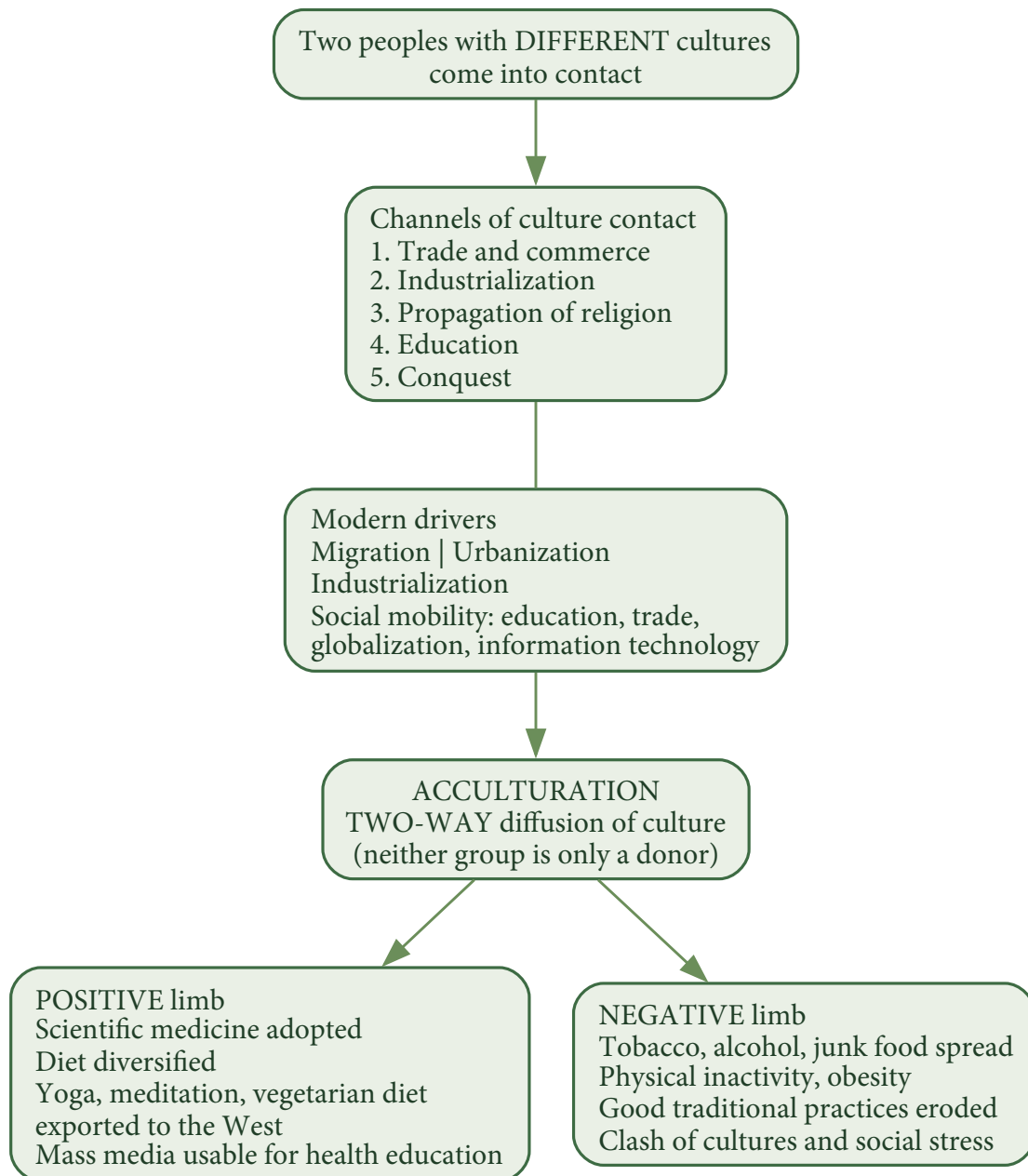
- ▶ **Rural-to-urban migration** and the consequent diminution of the traditional family support system.
- ▶ **Greater exposure through mass media** to ideas that had previously been culturally alien.
- ▶ **Tourism**.
- ▶ **Changing technological needs** demanding skills different from the previous generation's, for which available training or education may be inadequate.
- ▶ **Commercially encouraged economic aspirations** that are often unrealistic.

Pressure is greatest where young people have little control over their own destinies, where rapid population expansion means greater competition in younger age groups, and where resources are inadequate to meet their needs. Psychological stress with inadequate coping ability has been implicated

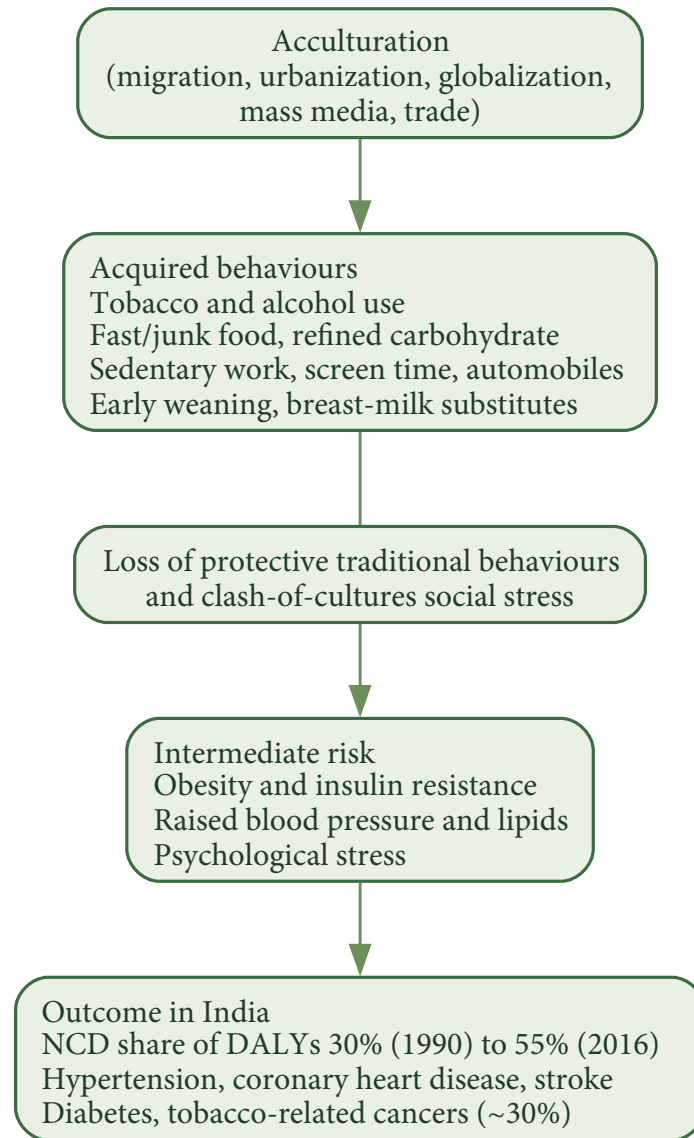
as a contributory factor in virtually all diseases, with **direct links between stress and hypertension and coronary heart disease** (Park 28e, Ch 12, p 815).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Two-way culture contact (acculturation) diagram** — draw two overlapping circles labelled "Culture A (host/dominant)" and "Culture B (migrant/traditional)", with two arrows crossing in opposite directions in the overlap labelled "diffusion of culture BOTH ways". Around the circles write the five channels: **trade and commerce, industrialization, propagation of religion, education, conquest**. From the overlap draw two arrows down to two boxes labelled "Good aspects" and "Bad aspects".



Acculturation — channels of culture contact and its two-sided outcome



Mechanism linking acculturation to the non-communicable disease burden in India

High-yield exam pointers

- ▶ **One-line definition to write first:** acculturation = "culture contact"; contact between two peoples of different culture produces **diffusion of culture both ways** (Park).
- ▶ **Five channels of culture contact (mnemonic "TIPEC")** — Trade and commerce, Industrialization, Propagation of religion, Education, Conquest; contemporary drivers are migration, urbanization, industrialization and social mobility.
- ▶ **Park's three named examples:** scientific medicine introduced by culture contact; food habits changed (orthodox brahmins now eat meat); worldwide spread of tobacco.
- ▶ **Dated Indian figures to quote:** adult tobacco use **28.6%** (GATS-2, 2016-17, from 34.6% in GATS-1, 2009-10); tobacco use in 13-15-year-olds **8.5%** (GYTS-4, 2019, from 14.6% in 2009); alcohol dependence/harmful use **4.6%** of adults; under-5 overweight **3.4%** (NFHS-5, 2019-21) vs 2.1% (NFHS-4, 2015-16); physical inactivity **41%** of adults (WHO-ICMR, 2020); NCD share of DALYs **30% (1990) → 55% (2016)** (ICMR-PHFI-IHME 2017).

- **Write Park's own heading "SOCIAL STRESS"** for the clash of two cultures — linked directly to hypertension and coronary heart disease.
- **Balance line to close with:** acculturation is neither good nor bad in itself — the community physician's task is to promote the beneficial diffusion (scientific medicine, health education) and to blunt the harmful (tobacco, alcohol, junk food) by legislation and behaviour change communication.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park's textbook treats acculturation in a single short paragraph, so the positive and negative aspects above are built by combining Park's own examples with the culture-contact drivers and the dated Indian lifestyle-disease data from Sunder Lal's Textbook of Community Medicine; the sociological terms (cultural diffusion, assimilation, pluralism, culture lag, culture shock) come from a standard sociology text and are worth one line each, not more, in an examination answer.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 12, Medicine and Social Sciences — Culture, Acculturation, Dynamics of Social Change, Social Stress, pp 813-830; Ch 3, Principles of Epidemiology and Epidemiologic Methods, p 88); Sunder Lal — Textbook of Community Medicine (Ch 2, Relationship of Social and Behavioural Sciences to Health and Disease, pp 81-85; Ch 5, Nutrition, pp 176-180; Ch 8, Epidemiology of Communicable and Non-Communicable Diseases, pp 483-502; Ch 15, Mental Health, pp 689-690); Sociology: The Essentials 6e (Ch 2, Culture and the Media, pp 64-78; Ch 10, Race and Ethnicity, p 270; Ch 16, Population, the Environment and Social Change, p 444). Named survey, guideline and report sources: Global Adult Tobacco Survey (GATS-1, 2009-10; GATS-2, 2016-17); Global Youth Tobacco Survey-4 (2019); National Family Health Survey-4 (2015-16) and NFHS-5 (2019-21); WHO-ICMR physical activity study 2020; ICMR, PHFI, IHME 2017 — *India: Health of the Nation's States, The India State-Level Disease Burden Initiative*; WHO International Code of Marketing of Breast-milk Substitutes 1981 and the Indian Infant Milk Substitutes Act 1992 (amended 2003). **Coverage note:** no usable captioned figure on culture or acculturation exists in the indexed corpus (the one candidate plate, Park 28e Fig 1 p 812, has its labels stripped in the scan), so the answer carries hand-drawn and flowchart figures only.



Social marketing

Asked in: Paper III 19-Nov-2022 — RGUHS MD Community Medicine

Definition — Social marketing applies commercial marketing skills and technologies to the analysis, planning, execution and evaluation of programmes designed to influence the **voluntary behaviour** of target groups, in order to improve their personal well-being and that of their society (Sunder Lal, Ch 4, p 161). *Kotler and Zaltman*, who introduced the term in 1971, defined it as "the design, implementation and control of programmes aimed at increasing the acceptability of a social idea or practice in one or more groups of target adopters" (Mahajan & Gupta 4e, Ch 29). Operationally it is the use of commercial marketing concepts, tools and resources to encourage socially beneficial behaviour among population segments **not served, or not adequately served, by existing public or private systems** (Sathe & Sathe, Ch 13, p 149).

Concepts, evolution and scope

- ▶ **Evolution** — *Kotler* and *Levy* broadened the concept of marketing beyond commerce in 1969 ("Broadening the concept of marketing"); *Kotler* and *Zaltman* coined the term "social marketing" in 1971 (Park Plus, Ch 21, pp 290-291).
- ▶ **Exchange theory is the basis** — individuals and organisations exchange resources for perceived benefits. Exchange may be by coercion, by threat, or voluntary; marketing deals only with **voluntary exchange** (Park Plus, Ch 21, p 290).
- ▶ **Behaviour targeted** — the audience is influenced to *accept* a new behaviour, *reject* a potentially undesirable one, *modify* a current one, or *abandon* an established undesirable one (Park Plus, Ch 21, p 290).
- ▶ **The consumer is "king"** — the programme first analyses the client's felt needs, likes, dislikes and satisfaction using focus group discussions, interviews and surveys. This early consumer consultation is called **feed-forward** (*Manoff*). Two audiences are defined: the **primary target audience**, whose behaviour is to change, and the **secondary target audience**, who influence them (Sunder Lal, Ch 4, p 161; Park Plus, Ch 21, p 292).
- ▶ **Core tactic** — the *precise* delivery of a *precise* product, service or idea, to the *precise* client, at the *precise* time of their *precise* need (Sunder Lal, Ch 4, p 161).
- ▶ **Relation to information, education and communication (IEC)** — IEC *creates* health-seeking behaviour; social marketing *facilitates the practice* of that behaviour by promoting, distributing and selling socially beneficial products, services and ideas at affordable cost to low-income groups through a wide range of private outlets (Sunder Lal, Ch 4, p 161).
- ▶ **Scope** — family welfare products and services first; also attitude and behaviour change in smoking, seat-belt use, drug abuse, heart disease and other lifestyle-related chronic disease; and in public health generally, health promotion, injury prevention, environmental protection and community mobilisation (Sunder Lal, Ch 4, p 161; Park Plus, Ch 21, p 293).

Feature	Social marketing	Commercial marketing
Focus	On the consumer	On the product
What is sold	Behaviours, ideas and practices	Products and services

Feature	Social marketing	Commercial marketing
Purpose	Social good / population well-being	Financial gain
Time frame	Long	Short
Emphasis	Non-tangible goods (with a tangible base added, e.g. condoms)	Tangible products
Task	Create demand for the <i>idea</i> (family planning) and for the <i>tool</i> (the condom)	Steer an existing pattern of demand towards one brand

(Park Plus, Ch 21, p 294; Mahajan & Gupta 4e, Ch 29)

The marketing mix — the six Ps

Element	What it means	Indian illustration
Product	The tangible product, service, idea or practice offered, once the client perceives the need; it carries a distinctive label or logo	Condoms, oral pills, sanitary napkins; services such as medical termination of pregnancy; practices such as breastfeeding and oral rehydration therapy
Price	Must be affordable to the poor in both rural and urban settings; price also includes non-financial cost — time, effort, discomfort, embarrassment and social censure	Deluxe Nirodh at Rs 3 per packet of 5; Mala-D at Rs 3 per cycle
Place	Outlets must be identified, accessible and known to people	<i>Mahila Swasthya Sangh</i> and women's groups, grocery shops, pharmacies and chemists, bus-stand stalls
Promotion	Mass media advertisement, publicity and interpersonal communication, to create sustained demand	Nirodh advertised on radio, cinema, press, matchboxes, hoardings and point-of-purchase material
Partnership	No single agency can cover a country as diverse as India; government, private sector and non-governmental organisations must act together	Hindustan Latex Ltd, DKT India, Population Services International as social marketing organisations
Policy environment	Government policy must encourage and sustain the behaviour the programme initiates	National Population Policy 2000 mandated social marketing schemes for products and services

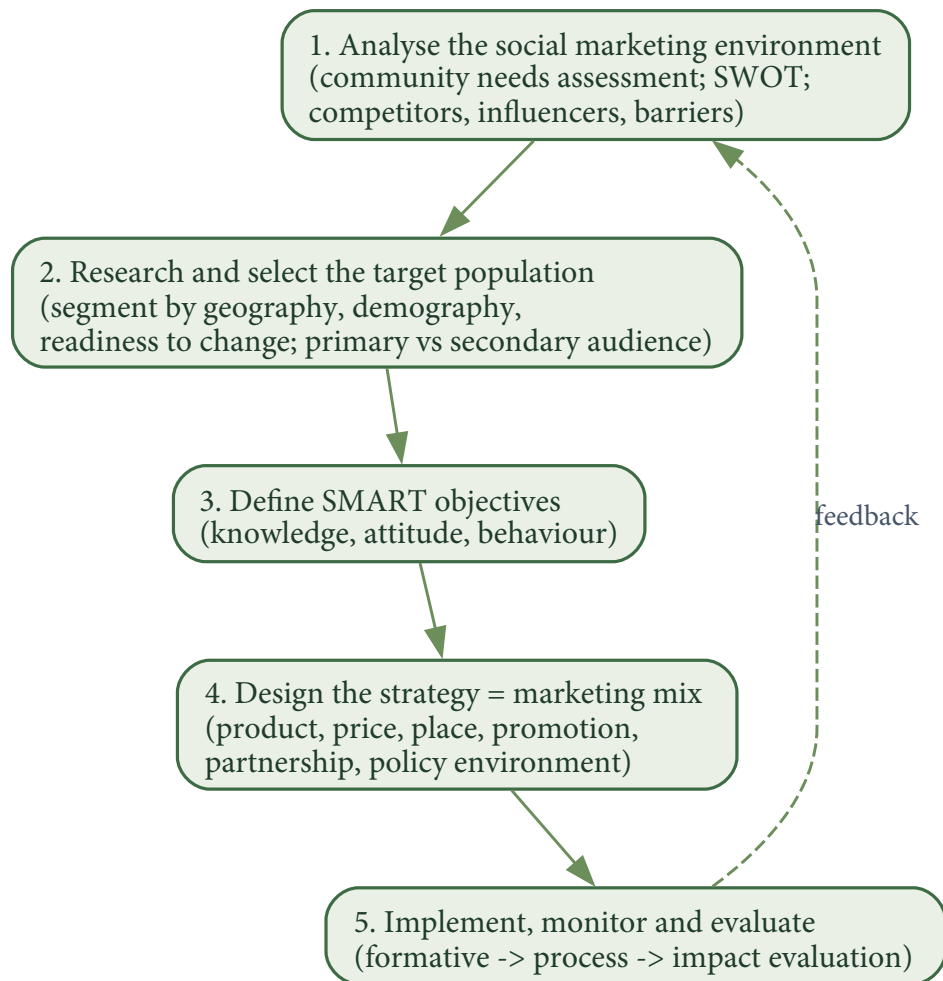
(Sunder Lal, Ch 4, pp 161-162; Park Plus, Ch 21, pp 292-293)

Steps in planning a social marketing programme

- **Analyse the social marketing environment** — community needs assessment for felt needs, and a SWOT analysis (strengths, weaknesses, opportunities, threats) to identify competitors, influencers and barriers; the plan must respect all principles of health education.
- **Research and select the target population** — segment the audience by geography, demography and readiness to change; segments must be homogeneous enough to plan an intervention and large enough to justify resources. Direct methods are focus group discussions, questionnaires, interviews and random surveys; the indirect method uses secondary data such as the census. Failure to segment misses hard-to-reach groups.

- **Define the objective** — specific, measurable, achievable, relevant and time-bound (SMART) objectives for change in knowledge, attitude and behaviour.
- **Design the strategy** — harmonise audience needs with organisational resources through the marketing mix (the Ps above).
- **Implement, monitor and evaluate** — formative evaluation (needs analysis, message design, pre-testing) precedes and continues through implementation; messages should raise awareness, then knowledge, then motivation, and finally build skills.

(Park Plus, Ch 21, pp 291-292; Oxford Textbook of Public Health 7e, Ch 33.8, pp 1222-1223)



The social marketing planning cycle

Social marketing in India

India was the **first country to adopt social marketing for its family welfare programme**, and it is one of the earliest examples of public-private partnership in health (Sathe & Sathe, Ch 13, p 149). The operational sequence used is eight-step: establish management and operating procedures → select the product → identify the consumer population → decide brand name and packaging → fix the price → identify the distribution system → recruit sales outlets → implement sales promotion (Sathe & Sathe, Ch 13, p 149).

Milestone	Detail
1968	Social marketing of condoms begins under the National Family Welfare Programme (the <i>Nirodh</i> programme, run with Ford Foundation assistance, is described as the first national social marketing programme)
1987	Social marketing of oral contraceptive pills begins (brand <i>Mala-D</i>)
1995	Centchroman (<i>Saheli</i>) introduced under social marketing through Hindustan Latex Ltd with product and promotional subsidy
1996–97	All socially marketed oral pill brands made available without prescription
Subsidy	70–85% of cost, depending on the year's procurement price
Prices	Social marketing organisations receive Deluxe Nirodh at Rs 2 per packet of 5 and sell at Rs 3; one cycle of oral pills at Rs 1.60, sold at Rs 3 per cycle as Mala-D (a 28-pill pack: 21 pills of levonorgestrel 0.15 mg + ethinylestradiol 0.03 mg, with 7 tablets of ferrous fumarate 60 mg). Mala-N is the free-supply brand through primary health centres
Brands	3 government and 13 social-marketing-organisation brands of condoms; 1 government and 8 organisation brands of pills

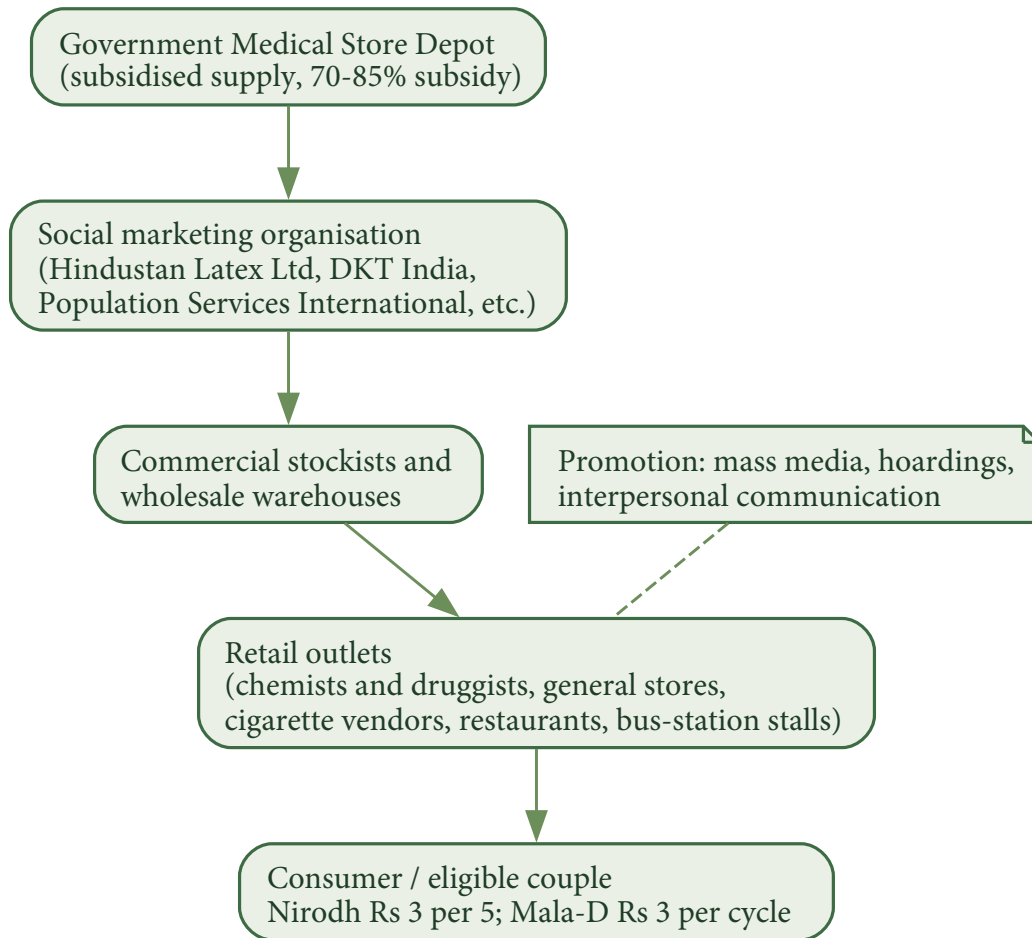
(Jugal Kishore, Ch 1, p 238; Park 28e, Ch 8, p 599; Park Plus, Ch 21, p 293)

Product	Baseline	Latest reported
Condoms	16 million pieces (1968–69)	483.21 million pieces (2017–18); 459.51 million pieces (2023–24)
Oral contraceptive pills	7.24 lakh cycles (1987–88)	206.31 lakh cycles (2017–18)
Saheli (centchroman)	—	77.52 lakh tablets (2023–24)

Socially marketed condoms and oral pills each account for about **one-third** of all such contraceptives distributed in India annually; the contraceptive prevalence rate rose from 10.4% (1951) to **56.5% (2019–21)**, and *NIRODH*, a Government of India brand, has become the generic word for a condom in the country (Sunder Lal, Ch 4, pp 161–162).

Other applications — condom promotion for high-risk groups and bridge populations under the National AIDS Control Programme (Park 28e, Ch 7, p 524); social marketing to increase the use of bed nets in malaria control (Park 28e, Ch 7, p 499); reduced-osmolarity oral rehydration salts — a campaign by ICICI Bank with United States Agency for International Development funding and Indian Academy of Pediatrics support in urban areas of 8 Hindi-speaking states; iron–folic acid tablets; and sanitary napkins (Park Plus, Ch 21, p 293). Social marketing is also a recognised support strategy for food fortification programmes (WHO/FAO Guidelines on Food Fortification with Micronutrients, Ch 10).

National strategy for strengthening it — maximise use of public health infrastructure (sub-centres, primary and community health centres, Indian systems of medicine outlets) since the programme is still largely urban; expand the basket of products; deepen private–non-governmental–government partnership; introduce **social franchising** of services so the private sector delivers preventive care cheaply; and promote IEC — all to accelerate the socio-demographic goals of the National Population Policy 2000 (Sunder Lal, Ch 4, p 162).



Distribution channel of the contraceptive social marketing programme in India

Advantages, limitations and challenges

Advantages	Limitations and challenges
Reaches hard-to-reach groups that classical health education and government outlets miss	The payback to the individual is rarely immediate or explicit
Uses an existing commercial network — cheaper and more effective than building a parallel government one	Strong competition from rival behaviours and products (e.g. bottle feeding versus breastfeeding)
A subsidised price attaches value to a product that is not valued when given free	Criticised for " blaming the victim " — it targets individual behaviour rather than the underlying social and environmental causes
Brand name and packaging aid recognition and consumer loyalty, and permit communication where the subject is taboo	Market analysis is often inaccurate, and there is no control over intangible prices such as embarrassment
Expands choice of method and improves quality and access to family welfare services	Audience segmentation may itself become discriminatory

Advantages	Limitations and challenges
Engages end-users, influencers and stakeholders in programme design and evaluation	For contraceptives specifically: poor access to outlets for the poorest, low retailer profit margins, out-of-date pharmacist knowledge, competition from free government supply, client embarrassment, and failure to reach the lowest income groups

(*Park Plus*, Ch 21, pp 291, 293-294; *Sathe & Sathe*, Ch 13, p 149)

Prerequisites for success — a nationwide communication network, reasonably high literacy, a good commercial distribution infrastructure, a supportive government policy, and a large proportion of married couples with purchasing power (*Sathe & Sathe*, Ch 13, p 149).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **The six-P social marketing mix wheel.** Draw a central circle labelled "Consumer (target audience)"; six spokes to boxes labelled, clockwise: **Product** (condom / oral pill / idea), **Price** (subsidised, affordable), **Place** (outlets — chemist, grocery, *Mahila Swasthya Sangh*), **Promotion** (mass media + interpersonal), **Partnership** (government + non-governmental organisation + private sector), **Policy environment** (National Population Policy 2000). Add an outer arrow labelled "feed-forward — consumer consultation before design" entering the centre.

High-yield exam pointers

- Definition to write: *Kotler and Zaltman* (1971) — "design, implementation and control of programmes aimed at increasing the acceptability of a social idea or practice in one or more groups of target adopters".
- Mnemonic for the mix — **6 Ps: Product, Price, Place, Promotion, Partnership, Policy environment.** (Some texts give only the classical 4 Ps.)
- Key dates: 1969 Kotler & Levy broaden marketing; 1971 term coined; 1968 condom social marketing in India; 1987 oral pills; 1995 Saheli.
- Prices to quote: Deluxe Nirodh Rs 3 per packet of 5; Mala-D Rs 3 per cycle; subsidy 70–85%.
- Dated Indian figures: condoms 459.51 million pieces (2023–24); oral pills 206.31 lakh cycles (2017–18); Saheli 77.52 lakh tablets (2023–24); contraceptive prevalence rate 56.5% (2019–21); socially marketed condoms and pills ≈ **one-third** of national distribution.
- One-line distinction: IEC *creates* health-seeking behaviour, social marketing *enables its practice* by selling the product cheaply through private outlets.
- Two named criticisms: "**blames the victim**" and **discriminatory segmentation**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park's textbook does not treat social marketing as a separate concept chapter — it appears there only as the contraceptive distribution scheme (Ch 8) and within the National AIDS Control and malaria programmes, so the conceptual content above is grounded on Sunder Lal, Mahajan & Gupta, Sathe & Sathe and Park Plus; the launch year of the Nirodh programme is given as 1968 by the family welfare programme documents and as 1969 (Ford Foundation-assisted) by Sathe, and either may be written.

SOURCE & COVERAGE

Sunder Lal — Textbook of Community Medicine, Ch 4 (Principles of Health Promotion and Education), pp 160-162; Park's Textbook of Preventive and Social Medicine 28e, Ch 8 (Demography and Family Planning), pp 599, 612; Jugal Kishore — National Health Programmes of India, p 238; Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e, Ch 29 (Information, Education, Communication and Training in Health); Sathe & Sathe — Epidemiology and Management for Health Care, Ch 13, p 149; Park Plus (Amrutha & Vijayalakshmi), Ch 21 (Health Planning and Management), pp 290-294; Oxford Textbook of Public Health 7e, Ch 33.8, pp 1222-1223. Programme documents named: National Population Policy 2000; National Family Welfare Programme.

**Women abuse — a medico-social problem**

Asked in: Paper III 30-May-2025 — RGUHS MD Community Medicine

Definition — Gender-based violence is defined by the United Nations as **any act of violence that results in physical, sexual or psychological harm or suffering to women, girls, men and boys, as well as threats of such acts, coercion, or the arbitrary deprivation of liberty**; domestic violence is its commonest form directed against women (NFHS-5 India Report 2019-21, Ch 15, p 677). The *Declaration on the Elimination of Violence against Women* (United Nations, 1993) adds that such acts count "whether occurring in public or in private life". Women abuse is a **medico-social problem** because its roots lie in social conditions — patriarchy, son preference, dowry custom, economic dependence and weak legal literacy — while its consequences present to the health system as injury, unwanted pregnancy, sexually transmitted infection, depression and death. Neither medical treatment nor social action alone can control it; the two must run together.

Types of abuse across the woman's life cycle

Violence can occur in any phase of a woman's life and many women experience multiple episodes across their lifetime (Jugal Kishore, Ch 2, p 891, after WHO 1996).

Phase of life	Types of violence
Pre-birth	Female foeticide; battering during pregnancy (physical and emotional harm to the woman, adverse birth outcomes); coerced pregnancy following rape
Infancy	Female infanticide; malnutrition
Girls under 5 years	Differential access to food and medical care; neglect of education; sexual abuse; physical abuse; emotional violence through a sense of rejection
Adolescent and adult women	Early, forced marriage and early pregnancy; sexual violence; domestic violence; dowry harassment; violence for infertility or failure to bear a son; witch-hunting; honour killing; desertion; sexual harassment at the workplace; acid burns; denial of entitlements; mental torture and false accusations of infidelity
Elderly	Desertion; general and medical neglect of old women and widows

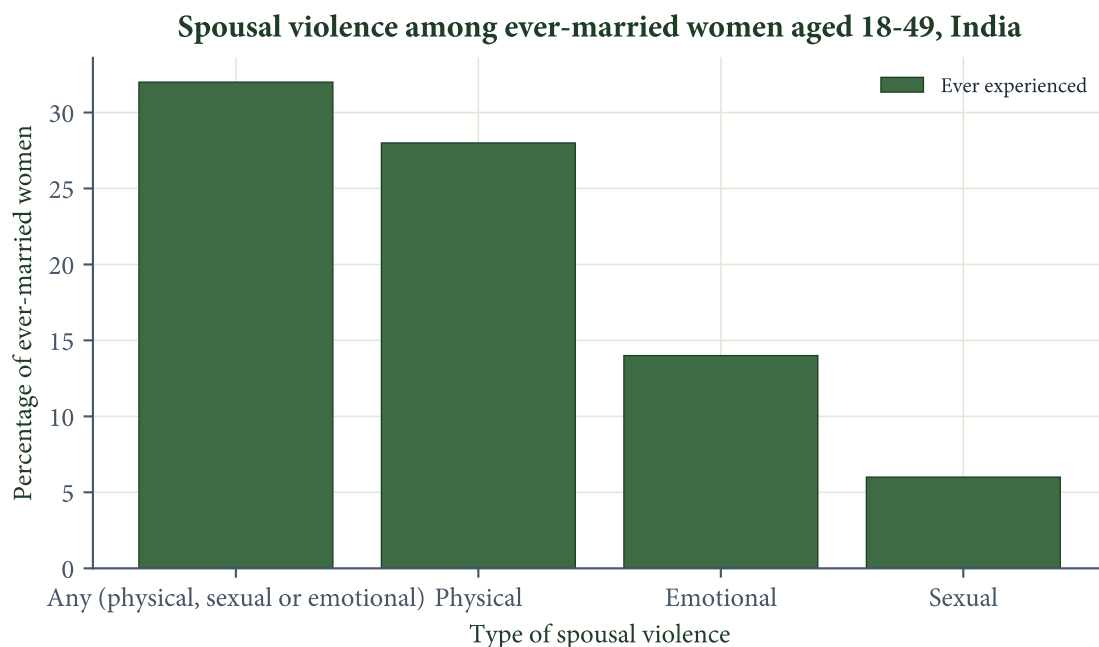
By form, abuse is classified as physical, sexual, emotional/psychological and economic. Economic abuse — depriving a wife of money or selling her jewellery and assets — is explicitly recognised as violence under Indian law (Jugal Kishore, Ch 2, p 893).

Magnitude of the problem

Indicator	Value	Source and year
Ever-married women 18–49 who have ever experienced spousal violence (physical and/or sexual), India	22.3% (urban 17.5%, rural 24.4%)	NFHS-6 fact sheet, 2023-24, India Key Indicators 95
Ever-married women 18–49 who experienced physical violence during any pregnancy	2.7% (urban 2.3%, rural 2.9%)	NFHS-6 fact sheet, 2023-24, India Key Indicators 96
Young women 18–29 who experienced sexual violence by age 18	0.7% (urban 0.4%, rural 0.8%)	NFHS-6 fact sheet, 2023-24, India Key Indicators 97
Trend in spousal physical or sexual violence	31% (NFHS-4) → 29% (NFHS-5, 2019-21) → 22.3% (NFHS-6, 2023-24)	NFHS-5 India Report; NFHS-6 fact sheet
Any spousal physical, sexual or emotional violence, ever-married women 18–49 (physical 28%, emotional 14%, sexual 6%)	32%	NFHS-5, 2019-21 (Table 15.9)
Ever experienced physical violence since age 15, Indian women 18–49	30%	NFHS-5, 2019-21 (Table 15.1)
State range for any spousal violence	2% (Lakshadweep) to 48% (Karnataka)	NFHS-5, 2019-21 (Table 15.13)
Among women 18–49 who ever experienced physical or sexual violence: sought help to stop it	14% (77% never sought help nor told anyone)	NFHS-5, 2019-21 (Table 15.17)
Women (and men) 15–49 who believe wife-beating is justified for at least one of seven stated reasons	45% of women, 44% of men (52% of women in NFHS-4)	NFHS-5, 2019-21 (Tables 14.14.1 and 14.14.2)

Indicator	Value	Source and year
Ever experienced physical or sexual intimate partner violence, women aged 15+ (global)	30%	Devries 2013 (Oxford Public Health 7e, Ch 82, p 3088)
Lifetime intimate partner violence — Eastern Mediterranean, Africa and South-East Asia vs high-income countries, Western Pacific and European regions	37% vs 23–25%	WHO 2013 (Oxford Public Health 7e, Ch 82, pp 3088–3089)
Non-partner sexual violence (global)	7.2% (95% CI 5.3–9.1%)	Devries 2013 (Oxford Public Health 7e, Ch 82, p 3090)
Share of all female homicides attributable to intimate partner violence	36%	WHO 2013 (Oxford Public Health 7e, Ch 82, p 3089)
Range of intimate partner violence across the WHO multi-country study	15% (urban Japan) to 71% (rural Ethiopia)	WHO/Lancet 2006 (Jugal Kishore, Ch 2, p 891)

Two features make these figures a public-health emergency rather than a legal curiosity. First, onset is **early in marriage** — 11% of women married only once report their first episode within 2 years of marriage and 21% within 5 years (NFHS-5, Table 15.14). Second, the problem is **hidden**: dowry deaths reach the media, but routine domestic violence is seldom reported or notified (Sunder Lal, Ch 10, p 632), so the falling survey prevalence must be read alongside the fact that three-quarters of affected women tell no one.



Sources: NFHS-5 India Report, Chapter 15 (Domestic Violence), for NFHS-4 and NFHS-5; NFHS-6 Fact Sheet, India Key Indicators, indicator 95, 2023-24 (provisional).

Why it is a medico-social problem — the social determinants

Domestic violence in India is deeply embedded in patriarchal norms about gender relations, and these are reinforced by weak government attention — both direct (failure to protect victims and prosecute offenders) and indirect (failure to enforce laws on minimum age at marriage and on inheritance) (Jugal Kishore, Ch 2, pp 891–892).

Domain	Factors that perpetuate abuse (UNICEF 2000, in Jugal Kishore, Ch 2, p 892)
Cultural	Gender-specific socialisation; cultural definitions of appropriate sex roles; belief in the inherent superiority of the male; values giving men proprietary rights over women and girls; the family seen as a private sphere under male control; marriage customs of bride price and dowry; acceptance of violence as a way of settling conflict
Economic	Women's economic dependence on men; limited access to cash and credit; discriminatory law on inheritance, property and maintenance after divorce or widowhood; limited access to employment, education and training
Legal	Lesser legal status of women in written law or in practice; restrictive legal definitions of rape and domestic abuse; low legal literacy among women; insensitive handling by police and judiciary
Political	Under-representation of women in power, politics, the media, and the legal and medical professions; domestic violence not taken seriously; the family held to be beyond the reach of the state; limited organisation of women as a political force

The same determinants are usefully arranged by **ecological level** — individual, relationship, community and societal — because each level offers a different entry point for prevention (Oxford Public Health 7e, Ch 82). Indian correlates confirmed by survey data include rural residence, illiteracy and poverty (Sunder Lal, Ch 10, p 632), plus three steep gradients:

Correlate (NFHS-5, 2019-21)	Exposed group	Comparison group
Number of marital control behaviours shown by the husband	76% where 5–6 behaviours shown	18% where none shown (Table 15.12)
Husband's drunkenness (spousal <i>physical or sexual</i> violence)	70% where husband often gets drunk	23% where husband does not drink (Figure 15.4)
Intergenerational transmission — respondent's father beat her mother	57% where her father beat her mother	25% where he did not (Table 15.11)

Health consequences

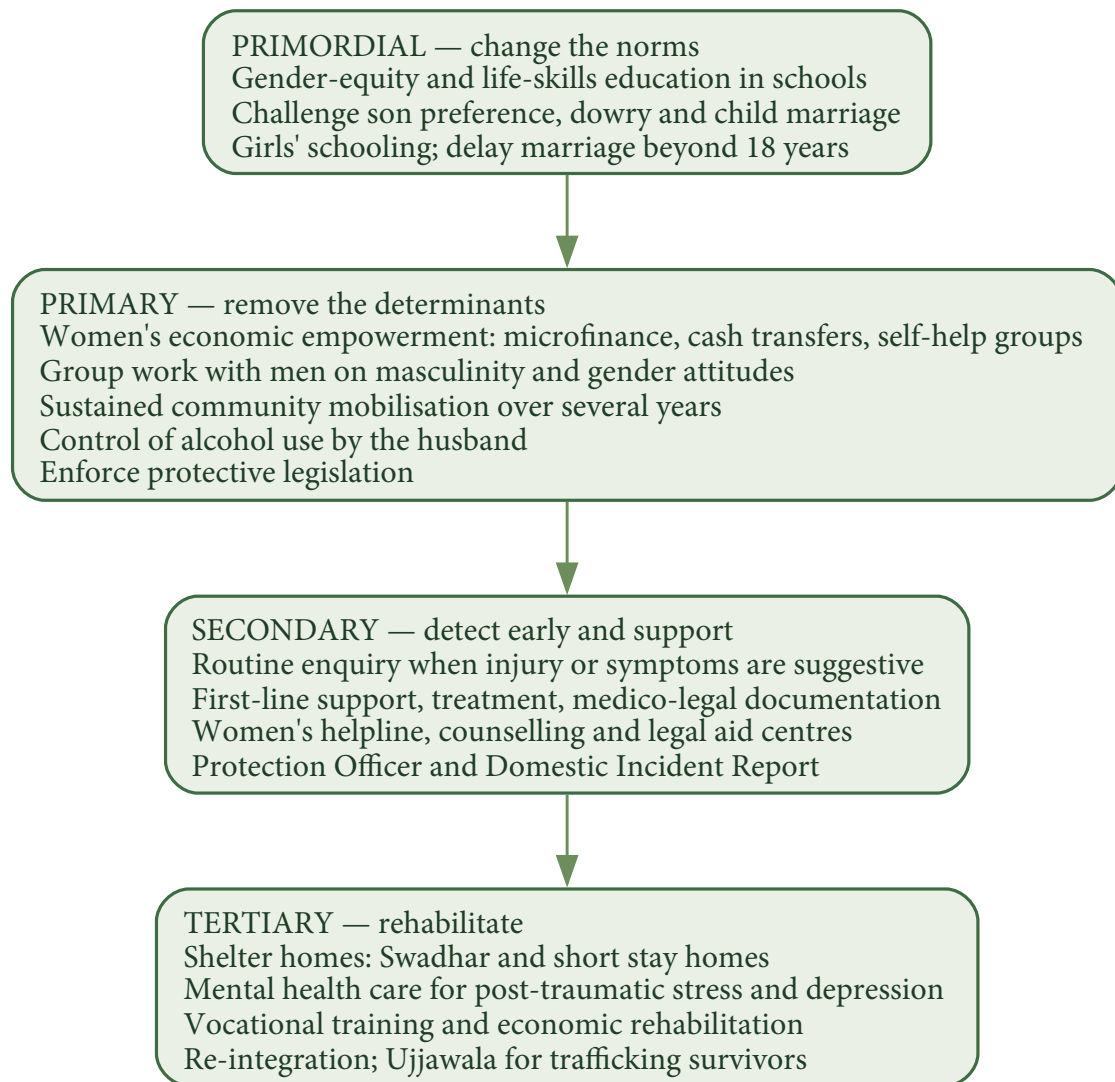
Category	Consequences
Non-fatal — physical	Injury; unwanted pregnancy; gynaecological problems; sexually transmitted diseases including HIV; miscarriage; pelvic inflammatory disease; chronic pelvic pain; headache; asthma; irritable bowel syndrome; self-injurious behaviour such as smoking and unprotected sex (Jugal Kishore, Ch 2, p 892)

Category	Consequences
Non-fatal — mental	Depression; fear and anxiety; low self-esteem; sexual dysfunction; eating problems; obsessive-compulsive disorder; post-traumatic stress disorder; post-partum depression (Jugal Kishore, Ch 2, p 892). Exposed women carry more than twice the risk of depression (WHO 2013, in Oxford Public Health 7e, Ch 82); rape causes post-traumatic stress disorder more often than any other violence exposure (Tolin and Foa 2006, in Oxford Public Health 7e, Ch 82)
Fatal	Suicide; homicide; maternal death; death from HIV/AIDS (Jugal Kishore, Ch 2, p 892)
Injury pattern (India)	One-fourth of ever-married women who experienced spousal physical or sexual violence report injuries — 7% eye injuries, sprains, dislocations or burns and 6% deep wounds, broken bones or broken teeth (NFHS-5, 2019-21, Table 15.15)
Reproductive and HIV	Violence limits a woman's sexual and reproductive control and health-seeking, producing unintended pregnancy, poorer maternal and perinatal outcomes and sexually transmitted infection; it also increases HIV acquisition and undermines adherence to antiretroviral therapy (Oxford Public Health 7e, Ch 85 and Ch 82)
Intergenerational and economic	Children who witness violence carry lifelong risk of mental illness, substance use and further violence; women exposed to violence were estimated to earn 35% less than unexposed women (Duvvury 2012; Oxford Public Health 7e, Ch 82, p 3088)

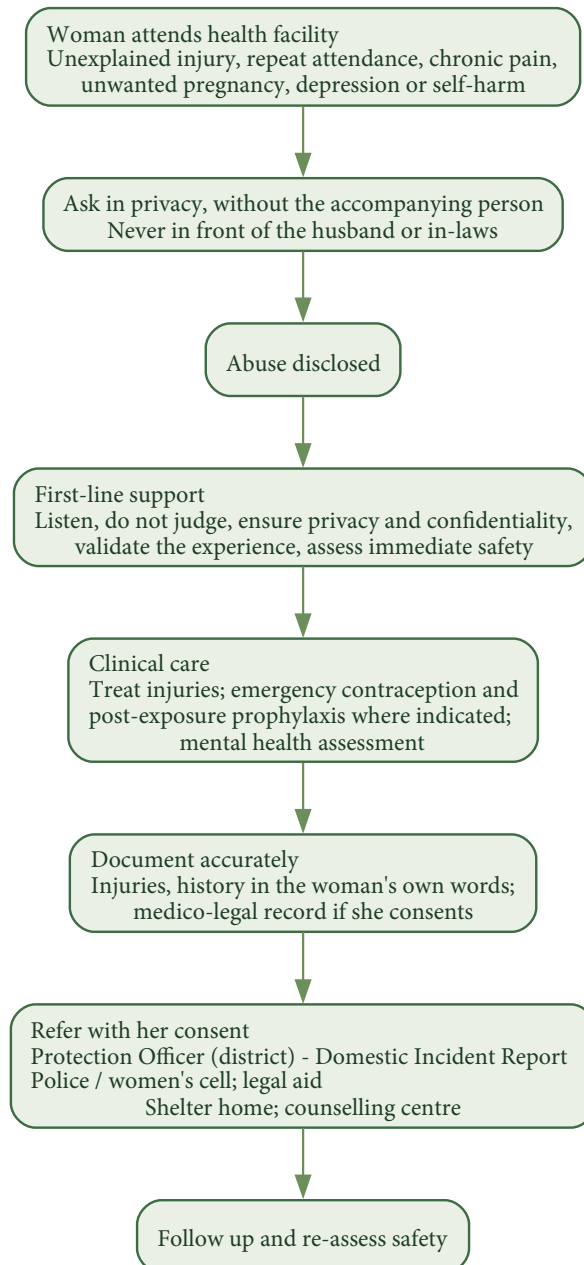
The health system is largely bypassed: among women who sought help, the commonest source was her own family (58%), then the husband's family (27%) and a friend (18%); only 9% approached the police and only 2% a **doctor or other medical personnel** (NFHS-5, Ch 15, Table 15.18). Health services are frequently unprepared to respond (Oxford Public Health 7e, Ch 85).

Prevention and control

The current framework is **RESPECT** (WHO) for violence against women, with **INSPIRE** its counterpart for violence against children; together they give a practical basis for understanding the prevention challenge (Oxford Public Health 7e, Ch 82, p 3117). Mapped onto the levels of prevention:



Levels of prevention applied to abuse of women



Health system response to a woman presenting with suspected abuse (clinical package as in the RMNCH+A service list, Park 28e, Ch 10, p 734)

1. **Legal safeguards** (a medico-social problem needs a legal arm; every postgraduate should know these by name):

Legislation	Key provision
Protection of Women from Domestic Violence Act, 2005	Covers relationships of consanguinity, marriage, live-in relationships, adoption and joint family — not the marital tie alone; recognises physical, sexual, verbal, emotional and economic abuse; Protection Officers in every district file the <i>Domestic Incident Report</i> and arrange legal aid, medical services and shelter; courts pass protection orders ; the woman has the right to secure housing in the shared household; breach of a protection order is cognizable and non-bailable, punishable with imprisonment up to 1 year and/or fine up to Rs 20,000 (Jugal Kishore, Ch 2, pp 892–894)

Legislation	Key provision
Sexual Harassment of Women at Workplace (Prevention, Prohibition and Redressal) Act, 2013 ("POSH Act")	Defines sexual harassment at the workplace and provides a time-bound complaints procedure; mandates an Internal Complaints Committee in every workplace with 10 or more employees and a Local Complaints Committee in each district for the unorganised sector and small establishments (see Note)
Dowry Prohibition Act, 1961; Amendment Act, 1986	Minimum punishment for taking or abetting dowry raised to 5 years' imprisonment and a fine of Rs 15,000 (Park 28e, Ch 12, p 840)
Indian Penal Code Section 498A	Cruelty to a woman within the matrimonial home (Jugal Kishore, Ch 2, p 886)
Indian Penal Code Sections 375, 376, 228-A, 509; Section 114A, Indian Evidence Act 1872	Rape and other sexual offences, and the evidentiary provision that supports a rape prosecution (Jugal Kishore, Ch 2, p 886)
Immoral Traffic (Prevention) Act, 1956 (retitled 1986)	Bans commercialised prostitution; covers all persons sexually exploited for commercial purposes (Park 28e, Ch 12, pp 839–840)
Commission of Sati (Prevention) Act, 1987	Prevention and punishment of sati (Sunder Lal, Ch 10, p 634)
Indecent Representation of Women (Prohibition) Act, 1986	Bans indecent portrayal of women (Sunder Lal, Ch 10, p 634)
Pre-Conception and Pre-Natal Diagnostic Techniques Act, 1994 (amended 2003)	Prohibits sex determination and sex-selective abortion (Sunder Lal, Ch 9, p 569)
Medical Termination of Pregnancy Act, 1971 (amended 2002, Rules 2021)	Legal, safe abortion including after rape (Sunder Lal, Ch 9, p 569)
National Commission for Women Act, 1990	Statutory watchdog for women's rights (Jugal Kishore, Ch 2, p 781)
Muslim Women (Protection of Rights on Marriage) Act, 2019	Bans instant triple <i>talaq</i> (Sunder Lal, Ch 10, p 634)
73rd and 74th Constitutional Amendments	Reservation of seats for women in <i>Panchayats</i> and municipalities (Jugal Kishore, Ch 2, p 781)
Factories Act Sections 66 and 34; Employees' State Insurance Act 1948; Maternity Benefit (Amendment) Act 2017	Prohibit night work 7 pm–6 am and carrying excessive weights; maternity benefit; 26 weeks' maternity leave and creches where more than 50 women are employed (Park 28e, Ch 18, p 967)

2. Support services and schemes (Ministry of Women and Child Development) —

- **Swadhar homes and short stay homes** — shelter, food, clothing, counselling, clinical, medical and legal support, training, economic rehabilitation and helplines (Jugal Kishore, Ch 2, pp 890–891).
- **Ujjawala (2007)** — prevention of trafficking, and rescue, rehabilitation and re-integration of victims of commercial sexual exploitation (Jugal Kishore, Ch 2, p 901).
- **Working women's hostels with creches** (Jugal Kishore, Ch 2, pp 890–891).

- ▶ **Beti Bachao Beti Padhao** — mass campaign with multisectoral action in **100 gender-critical districts** to reverse the declining child sex ratio (Jugal Kishore, Ch 2, pp 780–781).
- ▶ **National Policy for Empowerment of Women, 2001 structures** — women's cells in police stations, women police stations, family courts, *Mahila* courts, counselling and legal aid centres, *Nyaya Panchayats* (Jugal Kishore, Ch 2, pp 780–781).
- ▶ **One Stop Centre (*Sakhi*) and women's helpline 181** — integrated medical, police, legal, psychosocial and shelter support under one roof, one centre per district (see Note).

3. Interventions with proven effect on violence itself (Oxford Public Health 7e, Ch 82) —

- ▶ **Combined economic and gender empowerment** — the *IMAGE* study, South Africa: microfinance plus a gender-empowerment intervention and mobilisation of women into community action.
- ▶ **Sustained community mobilisation** over several years at sufficient intensity — the *SASA!* cluster-randomised trial, Kampala.
- ▶ **Workshop series with men** to reflect critically on masculinity and use of violence — *Stepping Stones* (South Africa), *Program H* (Brazil).
- ▶ **Couples' interventions**, effective even in complex settings, notably Rwanda.
- ▶ **Psychotherapy with a substance-use module** — the Common Elements Transdiagnostic Approach (CETA), Zambia, where alcohol and drug use drives perpetration.

Awareness-raising alone raises knowledge but does **not** change behaviour; it must sit inside a larger package that also strengthens health-sector and police response.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Ecological model of violence against women** — four concentric rings, innermost to outermost: *Individual* (witnessing abuse in childhood, alcohol use, low education); *Relationship / family* (marital conflict, male control of decisions, dowry demand); *Community* (poverty, weak sanctions, tolerant social norms); *Societal* (patriarchy, lesser legal status of women, acceptance of violence). Label an arrow from each ring to a matching prevention entry point.
- ▶ **Iceberg of women abuse** — waterline labelled "reported"; above the water: dowry deaths, homicide, rape reported to police; below the water: routine wife-beating, marital rape, emotional and economic abuse, sexual harassment — annotate the submerged part "77% of women who experienced physical or sexual violence never sought help nor told anyone (NFHS-5, 2019-21)".

High-yield exam pointers

- ▶ **Current national figure** — NFHS-6 (2023-24): ever-married women 18–49 who have ever experienced spousal violence (physical and/or sexual) = **22.3%** (urban 17.5%, rural 24.4%); physical violence during any pregnancy 2.7%; sexual violence by age 18 among young women 18–29 = 0.7%. Trend: 31% (NFHS-4) → 29% (NFHS-5, 2019-21) → **22.3%** (NFHS-6, 2023-24).
- ▶ **NFHS-5 (2019-21)** — the latest round carrying the detailed domestic-violence chapter: any spousal violence 32%; physical violence since age 15 in 30% of all women 18–49; **only 14%**

sought help, 77% never sought help nor told anyone; one-fourth report injuries; Karnataka highest at 48%; 76% vs 18% by marital-control behaviours, 70% vs 23% by husband's drunkenness.

- ▶ **Attitudes (NFHS-5, 2019-21)** — 45% of women and 44% of men aged 15–49 justify wife-beating for at least one of seven reasons; down 7 percentage points for women from 52% in NFHS-4, but up 2 points for men.
- ▶ **WHO/global set** — 30% of women aged 15+ ever experienced physical or sexual intimate partner violence; 37% in the Eastern Mediterranean, Africa and South-East Asia versus 23–25% in high-income countries, the Western Pacific and European regions; non-partner sexual violence 7.2%; intimate partner violence causes 36% of female homicides.
- ▶ **Life-cycle table** — pre-birth (foeticide) → infancy (infanticide) → under-5 girl (neglect, abuse) → adolescent/adult (dowry, domestic, sexual, workplace) → elderly (desertion, medical neglect). Write this table first; it earns the "medico-social" marks.
- ▶ **Four determinant domains, UNICEF 2000** — C-E-L-P: Cultural, Economic, Legal, Political.
- ▶ **PWDVA 2005 essentials** — Protection Officer in every district, Domestic Incident Report, protection order, right to shared household, economic abuse recognised, breach punishable up to 1 year and/or Rs 20,000. **POSH Act 2013** — Internal Complaints Committee at 10 or more employees, Local Complaints Committee per district.
- ▶ **Frameworks and evidence-based prevention** — **RESPECT** (women) and **INSPIRE** (children), both WHO; **IMAGE** (economic plus gender empowerment), **SASA!** (community mobilisation), *Stepping Stones / Program H* (work with men).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written as fact in the exam. The first-line-support sequence in the health-system flowchart (enquire in privacy, listen, do not judge, validate, assess immediate safety) follows standard World Health Organization clinical guidance and was not retrievable from the indexed corpus, though the clinical and referral package it feeds into is corpus-grounded (Park 28e, Ch 10, p 734). The Sexual Harassment of Women at Workplace Act 2013 and the One Stop Centre (*Sakhi*) and helpline 181 services appear in the corpus only as a title in a contents listing, so they are included here as standard current Indian public health rather than as retrieved text. The older textbook figure that 56% of women justify wife-beating refers to six stated reasons and carries no survey round in its source (Sunder Lal, Ch 10, p 632), so it must not be quoted alongside the 2019-21 seven-reason data. The NFHS-6 fact sheet (2023-24) results are provisional and carry only three violence indicators, so the detailed NFHS-5 (2019-21) Chapter 15 findings are still quoted for help-seeking, injuries, state variation and attitudes.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, Ministry of Women and Child Development (Mission Shakti, SHe-Box), National Crime Records Bureau, Global Burden of Disease 2023, WHO global estimates 2025, RESPECT Women framework, PRS Legislative Research

Programme and policy updates (India)

- **Mission Shakti (effective 1 April 2022) has renamed and merged the women's welfare schemes that the textbooks still list separately.** It runs as two verticals — *Sambal* (safety and security: One Stop Centre, Women Helpline, *Beti Bachao Beti Padhao*, *Nari Adalat*) and *Samarthya* (empowerment: *Palna*, PMMVY, *Sakhi Niwas*, *Shakti Sadan*, SANKALP: Hub for Empowerment of Women). *Swadhar Greh* and *Ujjawala* no longer exist as stand-alone schemes: both are subsumed into *Shakti Sadan*, an integrated relief and rehabilitation home for women in distress, while the working women's hostel is now *Sakhi Niwas*. *Nari Adalat* adds *gram panchayat*-level alternative dispute resolution by negotiation, mediation and reconciliation for lesser grievances of harassment or denial of rights, piloted in 50 *gram panchayats* each in Assam and Jammu & Kashmir. A renamed or merged scheme is the most examinable single update in this answer. [MWCD Mission Shakti 2022]
- **Health-sector integration in India remains thin, and this is a defensible criticism to write.** A systematic review found only 11 evaluated interventions in India that combine domestic-violence work with reproductive health care, just five of them cluster-randomised and most confined to Maharashtra; the effective ones shared psychoeducation, skill-building and counselling, and several deliberately engaged husbands. India still has no national policy mandating enquiry about domestic violence in health facilities — consistent with only 2% of affected women reaching a doctor. [PMID 38951870]

Current national data

- **District-level violence estimates now exist for all 640 districts,** derived by small-area estimation linking NFHS-4 to Census 2011: past-12-month physical intimate partner violence 22.5% (95% CI 21.9–23.2), emotional 11.4% (11.0–11.9) and sexual 5.2% (4.9–5.5). State averages hide the real spread — within Bihar, physical violence ranged from 23.5% (Siwan) to 42.7% (Purbi Champaran), and across India from 1.1% (East Sikkim) to 47.8% (Viluppuram, Tamil Nadu). Hotspots overlapped geographically with male alcohol use and child marriage, and clustered across state borders rather than within them — the argument for district-level rather than state-level targeting. [PMID 37734802]
- **Reported crime against women is rising even as survey prevalence falls — hold both figures together.** National Crime Records Bureau cases rose from 371,503 (2020) to 445,256 (2022), a rate of 57 → 67 per 100,000 women. *Cruelty by husband or his relatives* stayed the single largest head (30.0% of cases in 2020, 31.4% in 2022). Highest rate Delhi 145, lowest Nagaland 5; significant hotspots in Rajasthan, Madhya Pradesh, Haryana, Telangana and Odisha. On district-level spatial regression only sex ratio and population density were significantly associated with risk — female literacy and alcohol use were not, which the authors attribute to working with aggregated data. The rise may partly reflect improved reporting and law-enforcement practice rather than more violence, but the paper offers that as only one candidate — it lists population size and socioeconomic factors alongside it, notes that crimes reported to official agencies are far fewer than crimes actually committed, and concludes that the crime rate is still increasing and calls this alarming. Do not present the reporting explanation as settled. [PMID 39081347]

Elimination, surveillance and disease status

- **The Global Burden of Disease Study 2023 moved intimate partner violence into the front rank of risk factors.** Globally in 2023, **608 million women aged 15 years and over (95% UI 518–724 million) had ever been exposed**, with **18.5 million DALYs (8.74–30.0) attributable**; among women aged 15–49 years it ranked **fourth** of all risk factors for DALYs, with sexual violence against children fifth. Anxiety disorders (5.43 million DALYs) and major depressive disorder (3.96 million) were the leading attributed causes — the burden is predominantly mental, not surgical. [PMID 41386261]
- **Sustainable Development Goal target 5.2 — eliminate all forms of violence against women and girls by 2030 — is judged increasingly unrealistic.** The number of women exposed has stayed largely unchanged for more than two decades; the UN Office on Drugs and Crime reported that the number of women and girls intentionally killed in 2022 was the highest recorded in two decades; and on the most optimistic rate of decline consistent with the published confidence interval for low- and middle-income countries (0.4% a year), a recalculation puts it at roughly **173 years merely to halve intimate partner violence**. Prevention is failing the youngest cohorts hardest — lifetime prevalence in the 2024 EU survey was **higher in women aged 18–29 (35%) than in women of all ages (30%)** — and technology-facilitated gender-based violence (image-based abuse, sextortion, deepfakes) is an emerging category that prevalence surveys do not yet capture. [PMID 42303635]

Guideline and legislative updates

- **The Indian Penal Code no longer exists — sections 498A, 375/376 and 509 must now be cited under their successor law.** Three Acts passed in December 2023 replace the colonial trio: the **Bharatiya Nyaya Sanhita 2023** (for the Indian Penal Code 1860), the **Bharatiya Nagarik Suraksha Sanhita 2023** (for the Code of Criminal Procedure 1973) and the **Bharatiya Sakshya Adhiniyam 2023** (for the Indian Evidence Act 1872). For women, the age at which a gangrape victim is treated as a major was raised from 16 to 18 years, and sexual intercourse obtained by deceit or a false promise of marriage was made a distinct offence; the marital-rape exception was retained, and the Justice Verma Committee's recommendation of a gender-neutral rape definition was not adopted. [PRS 2023]
- **SHe-Box gives the POSH Act 2013 a national single-window complaint route.** Any woman, in any sector and irrespective of her employment status, can lodge a workplace sexual-harassment complaint online; it is forwarded to the authority holding jurisdiction and tracked through Central, State and District nodal officers. The portal currently carries **89,395 registered Internal Committees**, which is the practical answer to "how is the POSH Act actually monitored?" [MWCD SHe-Box portal]

Global and WHO developments

- **WHO's 2025 modelled estimates supersede the 2013 figures quoted in the textbooks.** Lifetime physical and/or sexual intimate partner violence or non-partner sexual violence among women aged 15 years and over is now put at **30.4% (uncertainty interval 28.0–33.7%)**, **about 840 million women**, and **23.3% (21.5–25.7%) of ever-partnered adolescent girls aged 15–19 years have already experienced it**. The **RESPECT Women** framework has been reissued in a second edition, its seven strategies — Relationship skills, Empowerment of women, Services for

survivors, Poverty reduction, Environments made safe, Child and adolescent abuse prevented, Transformed attitudes and norms — now evidence-graded separately for high-income and for low- and middle-income settings. [PMID 42303635]

- **Climate has entered the list of determinants of violence against women in India.** The study used two definitions of drought, and they do not give the same answer. On the *government* (socioeconomic) definition — analysed in NFHS-4 only (**n = 62,464 married women**), because district drought declarations were unavailable after 2016 — living in a district officially declared drought-affected raised the odds of past-year physical (adjusted OR 1.34; 95% CI 1.21–1.48), sexual (1.49; 1.27–1.74) and emotional (1.41; 1.22–1.63) intimate partner violence. On the *rainfall* (precipitation-based) definition, analysed across both NFHS-4 and NFHS-5 (n = 122,696), effects were much weaker and significant only for physical (1.05; 1.001–1.10) and emotional violence (1.10; 1.02–1.16), not sexual (1.07; 0.97–1.16) — the authors state that associations for sexual violence appeared only with the socioeconomic definition. The socioeconomic-drought association held only in districts *not* historically classified as drought-prone — that is, where no relief or coping machinery was already in place. Use this if the question asks for emerging or climate-related determinants. [PMID 37968380]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 10 Health Care of the Community — RMNCH+A service package; Ch 12 Medicine and Social Sciences; Ch 18 Occupational Health); Sunder Lal — Textbook of Community Medicine (Ch 9 and Ch 10, Reproductive, Maternal, Newborn, Child and Adolescent Health); Jugal Kishore — National Health Programmes of India (Ch 2, Legislations Related to Health, and women's welfare schemes); Oxford Textbook of Public Health 7e (Ch 82 Interpersonal Violence, pp 3088–3090 and p 3117; Ch 85 Women, Men and Health); National Family Health Survey (NFHS-6) India Fact Sheet, 2023-24 (India Key Indicators 95–97, provisional); National Family Health Survey (NFHS-5) India Report 2019-21, Chapter 14 (section 14.8 Attitudes toward Wife Beating, pp 623–624) and Chapter 15 (Domestic Violence, pp 675–715); WHO Global and Regional Estimates of Violence Against Women, 2013; Tolin and Foa 2006 (rape and post-traumatic stress disorder); UNICEF 2000 (factors perpetuating domestic violence).

Recent-advances sources: 37734802 — Spatial heterogeneity in intimate partner violence across the 640 districts of India: a secondary analysis of a cross-sectional, population-based survey by use of model-based small-area estimation, *Lancet Global Health* 2023, doi:[10.1016/S2214-109X\(23\)00377-7](https://doi.org/10.1016/S2214-109X(23)00377-7); 39081347 — Crime against women in India: district-level risk estimation using the small area estimation approach, *Frontiers in Public Health* 2024, doi:[10.3389/fpubh.2024.1362406](https://doi.org/10.3389/fpubh.2024.1362406); 41386261 — Disease burden attributable to intimate partner violence against females and sexual violence against children in 204 countries and territories, 1990-2023: a systematic analysis for the Global Burden of Disease Study 2023, *Lancet* 2025, doi:[10.1016/S0140-6736\(25\)02503-6](https://doi.org/10.1016/S0140-6736(25)02503-6); 42303635 — Is the elimination of violence against women a realistic goal for the near future?, *Nature Communications* 2026, doi:[10.1038/s41467-026-74656-y](https://doi.org/10.1038/s41467-026-74656-y); 38951870 — Integrated domestic violence and reproductive health interventions in India: a systematic review, *Reproductive Health* 2024, doi:[10.1186/s12978-024-01830-0](https://doi.org/10.1186/s12978-024-01830-0); 37968380 — Climate and Gender: Association Between Droughts and Intimate Partner Violence in India, *American Journal of Epidemiology* 2024, doi:[10.1093/aje/kwad222](https://doi.org/10.1093/aje/kwad222). Ministry of Women and Child Development — Mission Shakti scheme portal (Sambal and Samarthya sub-schemes; effective 1 April 2022) and the SHe-Box (Sexual Harassment electronic Box) portal. PRS Legislative Research — Bharatiya Nyaya (Second) Sanhita 2023 and companion Bills (passed Lok Sabha 20 December 2023, Rajya Sabha 21 December 2023). WHO global estimates of violence against women 2025 and the RESPECT Women framework, second edition (both as reported in *Nature Communications* 2026, doi:[10.1038/s41467-026-74656-y](https://doi.org/10.1038/s41467-026-74656-y)).

♦ ♦ ♦

Epidemiology

♦ ♦ ♦

Epidemiological triad for non-communicable disease

Asked in: Paper I 29-Jul-2021 — RGUHS MD Community Medicine

Definition — The **epidemiological triad** (triangle of epidemiology) is the model of disease causation that regards disease as the product of an **interaction between agent, host and environment**, with **time** as the fourth element accounting for latency and duration (Park 28e, Ch 2, pp 51-52). These three causative factors, operating in the **prepathogenesis phase**, determine both the onset of disease in an individual and its distribution in the community; mere presence of the three is not enough — interaction is required (Park 28e, Ch 2, p 54). Applied to **non-communicable disease (NCD)**, the classical "agent" is replaced by **multiple causative factors (risk factors)**, giving the *advanced model of the triangle of epidemiology* (Park 28e, Ch 2, p 52).

1. The three limbs of the triad, and how each is read in non-communicable disease

Limb	Classical (communicable) meaning	Reading in non-communicable disease
Agent	The single living cause of disease — "the seed" (Park 28e, Ch 2, p 55)	Usually no single known agent ; the term becomes causative factors — multiple aetiological factors of disease, disability, injury and death (Park 28e, Ch 2, p 52)
Host	The individual who harbours the agent — "the soil"	Groups or populations and their characteristics — demographic, biological, socio-economic and lifestyle attributes that raise or lower risk
Environment	External surroundings permitting transmission	Widened to behaviour, culture, physiological factors and ecological elements — the total life situation of the patient and the community (Park 28e, Ch 2, p 52)
Time	Incubation period, duration of illness	Long latent period and the "lag-time" effect — today's cardiovascular mortality reflects behavioural exposures of decades ago (Park 28e, Ch 6, p 438)

2. Agent (causative factors) in non-communicable disease

Park's classification of disease agents is deliberately broad and includes both living and non-living agents (Park 28e, Ch 2, p 55).

Class of agent	Non-communicable disease example
Biological — viruses, rickettsiae, fungi, bacteria, protozoa, metazoa	Human papillomavirus, hepatitis B and C virus, <i>Helicobacter pylori</i> — about 18% of the global cancer burden is attributable to specific chronic infections (Park 28e, Ch 6, p 435)
Nutrient — proteins, fats, carbohydrates, vitamins, minerals, water	Excess saturated fat, salt and refined carbohydrate; low fruit and vegetable intake — obesity, hypertension, coronary heart disease (Park 28e, Ch 6, p 438)
Physical — heat, cold, humidity, pressure, radiation, electricity, sound	Solar and ionizing radiation in cancer; noise-induced hearing loss (Park 28e, Ch 2, p 57)

Class of agent	Non-communicable disease example
Chemical — endogenous (urea, bilirubin, ketones, uric acid) and exogenous (allergens, metals, fumes, dust, gases, insecticides)	Gout from uric acid; benzene, arsenic, cadmium, chromium, vinyl chloride, asbestos, polycyclic hydrocarbons as occupational carcinogens (Park 28e, Ch 6, pp 458-459)
Mechanical — chronic friction and other mechanical forces	Crushing, sprains, dislocations and death in road traffic injury
Absence, insufficiency or excess of a factor necessary to health — chemical, nutrient, lack of structure or part of structure, chromosomal, immunological	Insulin deficiency in diabetes; congenital cardiac defects; chromosomal disorders (Park 28e, Ch 2, p 55)
Social agents	Poverty, smoking, abuse of drugs and alcohol, unhealthy lifestyles, social isolation (Park 28e, Ch 2, p 55)

The four shared **behavioural risk factors** that the World Health Organization (WHO) targets across the major NCDs are **tobacco use, unhealthy diet, physical inactivity and harmful use of alcohol**; they act through the **physiological (metabolic) risk factors** — raised blood pressure, raised blood glucose, dyslipidaemia and overweight/obesity — to produce diabetes, heart disease, stroke, cancer and chronic respiratory disease (Park 28e, Ch 6, p 437; NCD Combined Manual for Medical Officers, p 8).

3. Host factors (intrinsic)

Group of host factors	Content	Non-communicable disease relevance
Demographic	Age, sex, ethnicity	Blood pressure rises with age in both sexes; age, sex and family history are non-modifiable risk factors for coronary heart disease (Park 28e, Ch 6, pp 440, 447)
Biological	Genetic factors; biochemical levels (e.g. cholesterol); blood groups and enzymes; cellular constituents of blood; immunological factors; physiological function (blood pressure, forced expiratory volume)	Serum cholesterol, blood glucose, body mass index and blood pressure are the measurable intermediate host states through which behaviour acts
Social and economic	Socio-economic status, education, occupation, stress, marital status, housing	Poverty and low literacy cluster with tobacco use and late presentation
Lifestyle	Personality traits, living habits, nutrition, physical exercise, use of alcohol, drugs and smoking, behavioural patterns	The dominant modifiable limb in NCD; type A personality is listed among risk factors for heart disease (Park 28e, Ch 2, pp 55, 57)

Modifiable versus non-modifiable: modifiable risk factors include smoking, hypertension, elevated serum cholesterol, physical inactivity and obesity — they are amenable to intervention. Non-modifiable (immutable) factors — **age, sex, race, family history and genetic factors** — cannot be changed but act as **signals** identifying persons who need closer surveillance (Park 28e, Ch 2, p 57).

4. Environmental factors (extrinsic)

The **macro-environment** is "all that which is external to the individual human host, living and non-living, and with which he is in constant interaction"; the **micro- (domestic or personal) environment** covers the individual's own way of living — eating habits, smoking, drinking, drug use (Park 28e, Ch 2, pp 31, 55).

Component	Content	Non-communicable disease example
Physical	Air, water, housing, radiation, workplace, roads	Ambient and indoor air pollution and occupational hazards are the named environmental risk factors for NCD, along with possession of destructive weapons in injuries (Park 28e, Ch 6, p 435)
Biological	Living things external to the host	Oncogenic infections and their reservoirs; food-borne contaminants and adulterants
Psychosocial	Cultural values, customs, habits, beliefs, attitudes, morals, religion, education, lifestyles, community life, health services, social and political organization	Occupational and domestic stress; tobacco and alcohol advertising; urbanisation. Disease is viewed as maladjustment of the human organism to its psychosocial environment (Park 28e, Ch 2, p 56)

5. Worked application of the triad to major non-communicable diseases

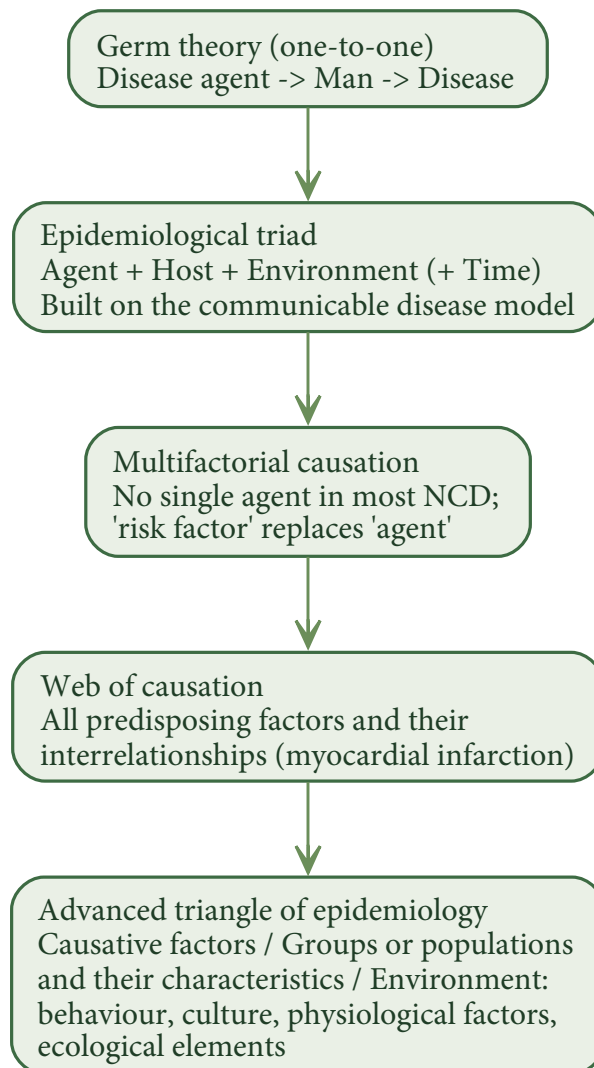
Disease	Causative factors (agent limb)	Host limb	Environment limb
Coronary heart disease	Cigarette smoking, high blood pressure, elevated serum cholesterol, diabetes, obesity, lack of exercise, type A personality	Age, sex, family history, genetic factors (non-modifiable)	Sedentary urban living, dietary transition to saturated fat and salt (Park 28e, Ch 2, p 57; Ch 6, p 440)
Cancer	Tobacco (smoked and chewed), alcohol, dietary factors, solar and ionizing radiation, occupational carcinogens, oncogenic infections	Age, sex, genetic susceptibility, immune status	Environmental factors account for 80-90% of all human cancers ; at least one-third of cancers are preventable (Park 28e, Ch 6, p 458)
Type 2 diabetes	Obesity, diet	Genetic predisposition, age, physical inactivity	Urbanisation, food environment, lack of space for physical activity (Park 28e, Ch 2, p 57; Ch 6, p 471)
Hypertension	High sodium intake (populations ingesting >400 mmol/day have high incidence; <60 mmol/day virtually none), alcohol	Age, genetic abnormality of renal salt excretion, obesity	Salt-heavy food supply, psychosocial stress (Park 28e, Ch 6, p 447)
Road traffic injury	Mechanical force, alcohol, speed	Age, sex, non-use of seat belts	Automobile design, roadway design (Park 28e, Ch 2, p 57)

6. Why the classical triad needs modification in non-communicable disease

Four gaps in the natural history of chronic disease limit the simple triad (Park 28e, Ch 6, pp 435-436):

- ▶ **Absence of a known agent** — the cause is known in only a few (silica in silicosis, asbestos in mesothelioma), which makes both diagnosis and specific prevention difficult.
- ▶ **Multifactorial causation** — rarely a one-to-one cause-effect relationship; chronic disease results from the **cumulative effect of multiple risk factors**, environmental, behavioural or constitutional.
- ▶ **Long latent period** between first exposure and manifest disease (e.g. cancer cervix), which breaks the link between antecedent and outcome and forces a search for **precursor lesions**.
- ▶ **Indefinite onset** — slow, insidious development so that the diseased and non-diseased states cannot be sharply separated (diabetes, hypertension); by the time advice is sought, damage may be irreversible.

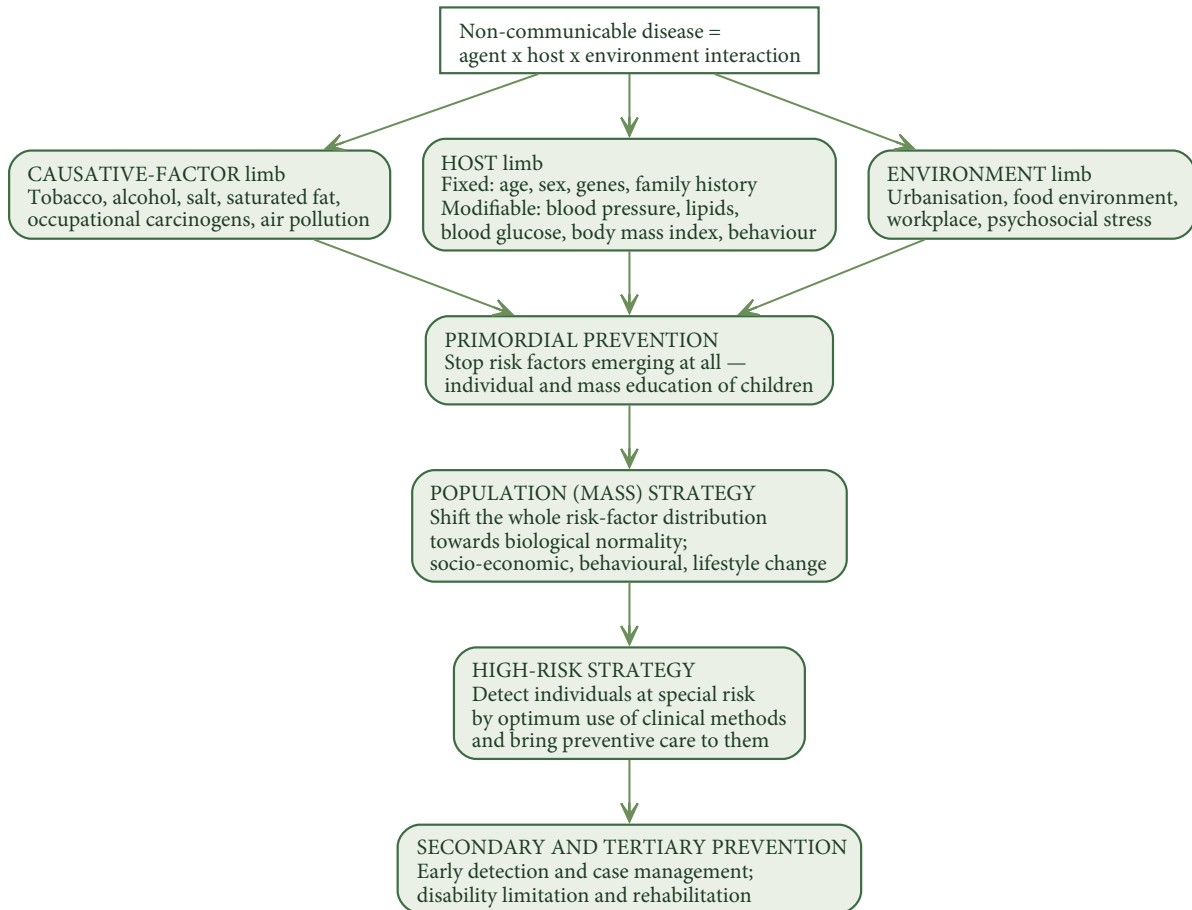
Two extensions therefore replace the rigid triad: the **web of causation** (proposed in *Epidemiology: Principles and Methods*), which considers all predisposing factors of any type and their complex interrelationships and is ideally suited to chronic disease where the agent is unknown; and the **advanced triangle of epidemiology**, in which *agent* becomes *causative factors* and *environment* absorbs behaviour, culture, physiological factors and ecological elements (Park 28e, Ch 2, pp 52-53). Importantly, the web does **not** require every link to be cut — **removal of one sufficiently important link may control the disease**, and the relative weight of each factor is expressed as its **relative risk** (Park 28e, Ch 2, p 53).



Evolution of causation models — from the germ theory to the advanced triangle used for non-communicable disease

7. Using the triad for prevention and control

The primary mission of epidemiology is to provide information that breaks one of the legs of the triangle, disrupting the connection between agent, host and environment (Park 28e, Ch 2, p 52). Because NCD aetiology is multifactorial, the preventive attack must also be multifactorial and is delivered through three complementary strategies — **primordial prevention, the population (mass) strategy and the high-risk strategy** (Park 28e, Ch 2, p 60; Ch 6, p 442).



Breaking a leg of the epidemiological triad — points of attack for non-communicable disease control

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **The epidemiological triad** — draw an equilateral triangle. Label the apex ENVIRONMENT, the left base angle AGENT (causative factors in NCD), the right base angle HOST. Write TIME alongside as the fourth element (latent period, lag-time effect, duration of illness). Add double-headed arrows along each side to show **interaction**, and mark one side with a cross labelled "*break a leg of the triangle = control*".
- **The advanced triangle of epidemiology** — the same triangle relabelled: apex ENVIRONMENT — behaviour, culture, physiological factors, ecological elements; left base CAUSATIVE FACTORS; right base GROUPS OR POPULATIONS AND THEIR CHARACTERISTICS.

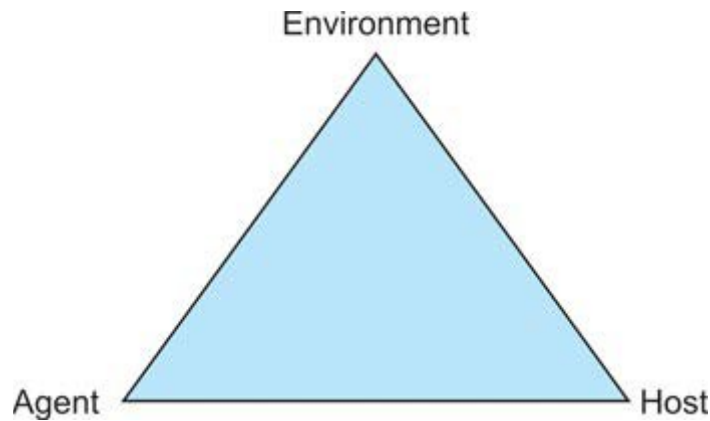


Fig 3.1 — The epidemiological triad (Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e, p.27)

High-yield exam pointers

- ▶ **Mnemonic for the triad limbs:** A-H-E-T — Agent, Host, Environment, Time. Host = "soil", agent = "seed".
- ▶ **Seven classes of agent (Park):** biological, nutrient, physical, chemical, mechanical, absence/insufficiency/excess of a factor necessary to health, social.
- ▶ **Four host groups:** demographic, biological, social and economic, lifestyle. **Three environmental components:** physical, biological, psychosocial.
- ▶ **Four gaps in the natural history of chronic disease:** absence of a known agent, multifactorial causation, long latent period, indefinite onset.
- ▶ **WHO's four shared behavioural risk factors:** tobacco use, unhealthy diet, physical inactivity, harmful use of alcohol — monitored by the WHO STEPS framework for surveillance of NCD risk factors (WHO, 2011).
- ▶ **Indian and global figures with dates:** non-communicable disease accounted for 63% of all deaths in India in 2016 (Park 28e, Ch 10, p 723); an estimated 2.78 million people died of cardiovascular disease in India in 2021 (Park 28e, Ch 6, pp 437-438); global target — one-third relative reduction in premature mortality from cardiovascular disease, cancer, diabetes and chronic respiratory disease (WHO Global Action Plan, extended to 2030).
- ▶ **Three complementary preventive strategies:** primordial prevention, population (mass) strategy, high-risk strategy — all three must be applied together.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park's 28th edition still prints the India non-communicable-disease mortality share as 63% of all deaths (2016) alongside its 2021 cardiovascular estimates, so quote each figure with its own year rather than merging them; the WHO Global Action Plan for the Prevention and Control of Non-communicable Diseases (2013-2020) has been extended to 2030, and India's national programme is now the **National Programme for Prevention and Control of Non-Communicable Diseases (NP-NCD)** (formerly NPCDCS).

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 2 (Concept of Health and Disease, pp 51-60), Ch 6 (Epidemiology of Chronic Non-communicable Diseases and Conditions, pp 433-471), Ch 10 (Health Care of the Community, p 723); Gordis Epidemiology 7e, Ch 2 (The dynamics of disease transmission, pp 35-36); NCD Combined Manual for Medical Officers (Government of India), Sections 1 and 3; Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e, Fig 3.1, p 27; WHO Global Action Plan for the Prevention and Control of NCDs (2013-2020, extended to 2030) and WHO STEPS Framework for Surveillance of NCD Risk Factors (2011) as cited in Park 28e.



Study design for breast self-examination training

Asked in: Paper I 12-Nov-2021 — RGUHS MD Community Medicine

Definition — The officer proposes to *introduce* breast self-examination (BSE) training into a healthy community and then measure what it achieves, which makes this an **experimental (intervention) study**. Park classifies experimental studies as randomized controlled trials (patients as the unit of study), **field trials** (healthy people as the unit) and **community trials or community intervention studies** (communities as the unit) (Park 28e, Ch 3, p 82). Because BSE training is delivered to a whole village through the sub-centre team, the intervention is defined at group level, so the design of choice is a **cluster-randomised community intervention trial** — village/ward as the unit of randomisation, the individual woman as the unit of analysis.

1. Fix the research question, objectives and hypothesis

Element (PICOT)	Specification to be written into the protocol
P — Population	All women aged 30 years and above, usual residents (≥ 6 months) of the villages served by the PHC; women aged 30 and above are the group already covered by the national screening package (Park 28e, Ch 10, p 735)
I — Intervention	Structured BSE training by the auxiliary nurse midwife (ANM), community health officer (CHO) and accredited social health activist (ASHA) — demonstration on a breast model, return demonstration, printed pictorial card, monthly reinforcement
C — Comparison	Existing routine services alone (no BSE training) — the control clusters continue to receive standard Health and Wellness Centre care
O — Outcome	Primary: breast-cancer mortality rate; feasible primary for a PHC-level study: proportion of incident breast cancers detected at stage I–II. Secondary: correct BSE practice, tumour size at presentation
T — Time	Intervention 12 months, behaviour outcomes at 6 and 12 months, stage-shift and mortality over 5–10 years

- **Null hypothesis (H_0)** — the proportion of breast cancers detected at an early stage is the same in trained and untrained clusters.

- **Alternative hypothesis (H_1)** — BSE training increases the proportion detected early (two-sided test retained, because screening can also harm).
- The **primary outcome must be the clinical disease to be prevented or controlled** — it provides the central justification for the trial and determines the study size; everything else is secondary (Handbook of Epidemiology, Ch I.8, p 369).

2. Justify the choice of design

Design considered	What it would yield	Verdict
Knowledge–attitude–practice cross-sectional survey	Baseline awareness only	Rejected as the main design — no comparison, cannot attribute change
Before-and-after study in one village (no control)	Change over time in one group	Rejected — confounded by secular trends , the classic flaw of non-randomised community trials (Handbook of Epidemiology, Ch I.8, p 368)
Non-randomised community trial (one village trained, a neighbouring one not)	A comparison, but groups not comparable	Rejected — comparability is low and the chance of a spurious result higher (Park 28e, Ch 3, p 106)
Individually randomised trial within a village	Highest internal validity	Rejected — the intervention is delivered at group level and neighbours would teach each other (contamination)
Cluster-randomised community intervention trial	Unbiased comparison, avoids contamination, matches how the service is actually delivered	Design of choice — clusters may be geographic (villages) or otherwise (workplaces, schools); village randomisation also improves long follow-up where personal identifiers are unreliable but community links are strong (Handbook of Epidemiology, Ch I.8, p 368)
Case-control study of BSE practice among diagnosed cases	Cheap, quick	Only a fallback if a trial is unaffordable — recall bias is severe

3. Build the trial: Park's six steps applied to this study

The six basic steps of an intervention trial (Park 28e, Ch 3, p 101), applied here:

- **Drawing up the protocol** — written *before* the first woman is enrolled and unaltered thereafter. It fixes the objectives, the operational definitions ("trained" = attended the session and gave a correct return demonstration; "early stage" = stage I or II), the training curriculum, inclusion/exclusion criteria, sample size, the outcome measures with their denominators, the analysis plan and the referral pathway.
- **Reference and experimental populations** — the *reference (target) population* is women aged 30 and above in the PHC area, to whom findings will be generalised; the *experimental (study) population* is the subset actually taking part after exclusions and consent. Participants must (a) give informed consent, (b) be representative of the reference population and (c) be eligible for the trial (Park 28e, Ch 3, pp 101–102). The community must be large enough for age-specific rates to be meaningful and **stable, without migration into or out of the area**, with clear

membership; community participation is the most essential ingredient (Park 28e, Ch 3, pp 82–83).

- ▶ **Randomization and sample size** — randomise **whole villages/wards**, not women. Compute the size for the difference in the primary proportion at alpha (two-sided) = 0.05 and beta = 0.20, i.e. power 80 per cent, from the anticipated proportion in the control arm and the smallest worthwhile difference (Browner 5e, Ch 6, pp 130–135). Then inflate for clustering by the **design effect** = $1 + (m - 1)r$, where m is the number of women per cluster and r the intra-cluster correlation coefficient; with $m = 50$ and $r = 0.02$ this is $1 + (49 \times 0.02) \approx 2$, i.e. roughly double the individually-randomised number (Handbook of Epidemiology, Ch I.8, p 369). Include **as many clusters as possible** — a large number improves the chance that randomisation balances the arms; with few villages, use **restricted randomisation after matching or stratifying** them on socio-demographic factors (as in the eleven matched community pairs of the *Community Intervention Trial for Smoking Cessation, COMMIT*), matching only on variables strongly related to the outcome (Handbook of Epidemiology, Ch I.8, p 369). Generate the allocation sequence centrally and conceal it until the villages are assigned (Ch I.8, p 354).
- ▶ **Manipulation (the intervention)** — a standardised BSE training package: what to look for, when in the cycle to examine, and what to do on finding a lump. The rationale is Park's — the basic methods of early detection of breast cancer are BSE by the patient, palpation by a physician, thermography and mammography, and **all women should be encouraged to perform BSE** because breast cancers are more frequently found by women themselves than by a physician at routine examination (Park 28e, Ch 6, p 460).
- ▶ **Follow-up** — active, equal and complete in **both** arms, since unequal ascertainment would itself create a difference. Keep a line list of enrolled women and a monthly ANM contact register, cross-linked to the population-based cancer registry and death records; report losses arm-wise in a CONSORT participant-flow diagram (Handbook of Epidemiology, Ch I.8, p 354).
- ▶ **Assessment of outcome** — neither the woman nor the trainer can be masked to health education, so a double- or triple-blind trial is impossible here; **blind the outcome assessor and the data analyst** instead, because observer bias and bias in evaluation are prevented neither by randomisation nor by sample size (Park 28e, Ch 3, p 103).

4. Outcome measures, indicators and analysis

Level	Indicator	Numerator / denominator
Process	Training coverage	Women trained in BSE / eligible women enumerated in intervention clusters $\times 100$
Process (quality)	Correct technique	Women completing all steps on a standard observation checklist / women trained $\times 100$
Behavioural (intermediate)	BSE practice	Women reporting BSE at least once a month over the previous 6 months / women surveyed $\times 100$

Level	Indicator	Numerator / denominator
Service response	Referral completion	Women with a self-detected lump examined at the PHC within one month / women reporting a lump $\times 100$
Test performance	Sensitivity, specificity, predictive value of BSE against clinical breast examination and confirmatory diagnosis	Sensitivity = $a/(a+c) \times 100$; specificity = $d/(b+d) \times 100$; predictive value of a positive test = $a/(a+b) \times 100$ (Park 28e, Ch 4, p 167)
Early impact	Stage shift	New breast cancers in stage I–II / all new breast cancers in the cluster $\times 100$; also median tumour size at diagnosis
Definitive impact	Breast-cancer mortality rate	Deaths from breast cancer / women-years of follow-up in the whole allocated cluster
Harm	Benign biopsy rate, anxiety from false positives	Benign biopsies / women referred $\times 100$
Public-health outcome	All-cause mortality, reported alongside disease-specific mortality — a screening strategy can cut disease-specific mortality yet leave overall mortality unchanged (Gordis 7e, Ch 10, p 224)	Deaths from all causes / women-years

- Analyse by **intention to treat**, i.e. by the cluster to which the village was allocated, counting *all* outcomes regardless of whether the woman attended the training.
- **The analysis must account for clustering** — outcomes of women in the same village are not independent; if clustering is ignored, p-values are artificially small and confidence intervals over-narrow (Handbook of Epidemiology, Ch I.8, pp 368–369). Use a random-effects model or generalised estimating equations.
- Check **baseline (prognostic) comparability** of the arms on age, parity, education and family history before attributing any difference to the intervention (Gordis 7e, Ch 10, p 224).

5. Biases peculiar to a screening trial, and how this design defeats them

Bias	How it would mislead here	Safeguard built into the design
Lead-time bias	BSE moves the date of diagnosis earlier, so survival <i>appears</i> longer while only "disease time" is lengthened	Use mortality rates, not survival rates , as the endpoint (Fletcher, <i>Clinical Epidemiology</i> 5e, Ch 10, Table 10.2, p 178)
Length-time bias	Self-examination preferentially finds slow-growing, good-prognosis lumps, flattering the trained arm	Randomised control arm; count all outcomes regardless of how the cancer was detected (Fletcher, Ch 10, Table 10.2, p 178)

Bias	How it would mislead here	Safeguard built into the design
Referral (volunteer) bias	Women who take up training are healthier and more adherent than those who do not, so they would do better anyway — in the <i>Health Insurance Plan</i> trial, mortality from causes other than breast cancer differed between women who accepted and refused screening (Gordis 7e, Ch 18, p 381)	Randomise, and compare the entire allocated groups , not screened versus unscreened (Gordis 7e, Ch 18, pp 373, 391)
Compliance bias	The trained-and-compliant women look better because they are compliant, not because of BSE	Count all outcomes regardless of compliance (Fletcher, Ch 10, Table 10.2, p 178)
Overdiagnosis bias	Detection of lesions that would never have caused symptoms inflates the "cancers found" in the trained arm	Long follow-up; a persistent excess of total cancers in the intervention arm signals overdiagnosis
Contamination	Control-village women learn BSE from neighbours, radio or the same ANM, diluting the difference	Randomise clusters , choose geographically separated villages, use different field staff (Handbook of Epidemiology, Ch I.8, p 368)
Secular trend / confounding	Awareness rises everywhere over the study period	Concurrent randomised control arm, not a before–after comparison (Handbook of Epidemiology, Ch I.8, p 368)
Observer bias in outcome assessment	The trainer, knowing allocation, reports outcomes more favourably	Independent, allocation-blind outcome assessor and analyst (Park 28e, Ch 3, p 103)

6. Ethics, feasibility and programme fit

- **Ethics committee clearance before enrolment**, with a participant information sheet and **written informed consent** in the local language; in a community-based study this is additionally preceded by **group/community consent from the community head or culturally appropriate authority**, which never replaces individual consent (ICMR National Ethical Guidelines 2017, section 10.4.6).
- **Guarantee the diagnostic and treatment pathway before starting**. The "cost" of a screening test is not the test but the entire follow-up it sets in motion, even when the result proves false-positive (Gordis 7e, Ch 18, p 391); a trial that finds lumps it cannot get diagnosed is unethical.
- **Ancillary care and post-research benefit sharing** — communicate the findings back to the community and plan to extend the BSE training to the control clusters at the end, so the control arm also benefits (ICMR 2017, sections 2.11 and 6.2.13).
- **Register the trial** with the Clinical Trials Registry–India (ICMR 2017, section 3.7) and report it against the CONSORT checklist — sequence generation, allocation concealment, implementation, masking and statistical methods (Handbook of Epidemiology, Ch I.8, p 354).
- **Be honest about feasibility**. Where disease frequency is low and the natural history long, a mortality endpoint needs thousands of people followed for a decade or more, and cost and

logistics are often prohibitive (Park 28e, Ch 3, p 106). Advise the officer to **power the study on the stage-shift and behavioural endpoints** and declare mortality a long-term secondary objective, escalating to a district-wide collaboration if mortality must be primary.

- **Nest the trial inside the existing service, not beside it.** Universal screening of the population aged 30 years and above for hypertension, diabetes and three common cancers — oral, breast and cervix — is already part of the Health and Wellness Centre package (Park 28e, Ch 10, p 735), operationalised by the 2016 *Operational Guidelines: Prevention, Screening and Control of Common Non-Communicable Diseases* (Park 28e, Ch 7, p 577). The trial therefore tests an **add-on** against that standard package.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Diagram 1 (to sketch)** — *Design of a cluster-randomised trial of BSE training*: one box "Reference population: women 30 y and above in the PHC area" → "Eligible villages/wards enumerated; exclusions made; consent obtained" → a box labelled **RANDOMIZE (unit = village)** splitting into two vertical arms, "Intervention clusters — offered BSE training" and "Control clusters — routine services", each arm carrying the sub-labels "cancers detected", "stage at diagnosis", "deaths from breast cancer", with the comparison arrow at the bottom labelled "compare mortality from breast cancer in the whole allocated group".
- **Diagram 2 (to sketch)** — *2 × 2 table for validating BSE as a detection test*: columns "Disease present / Disease absent" (by confirmatory diagnosis), rows "BSE positive / BSE negative", cells labelled **a = true positive**, **b = false positive**, **c = false negative**, **d = true negative**, with margins labelled "Sensitivity = $a/(a+c) \times 100$ ", "Specificity = $d/(b+d) \times 100$ ", "Predictive value of a positive test = $a/(a+b) \times 100$ ".

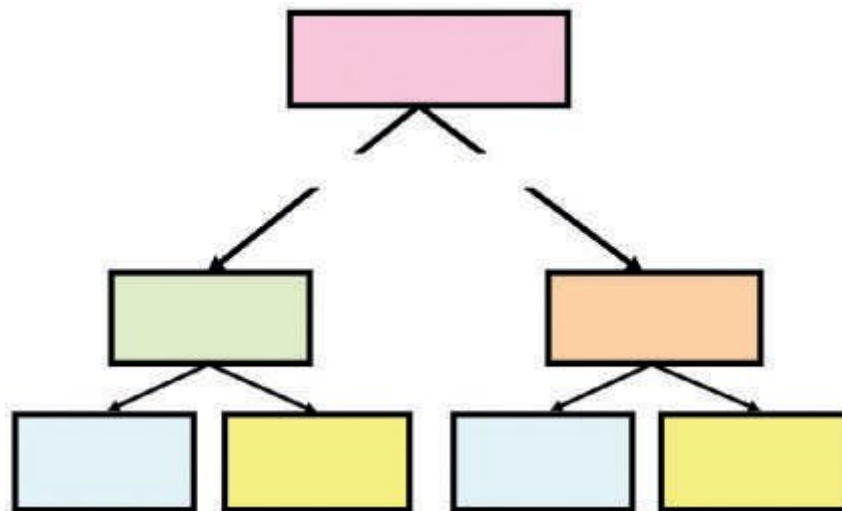
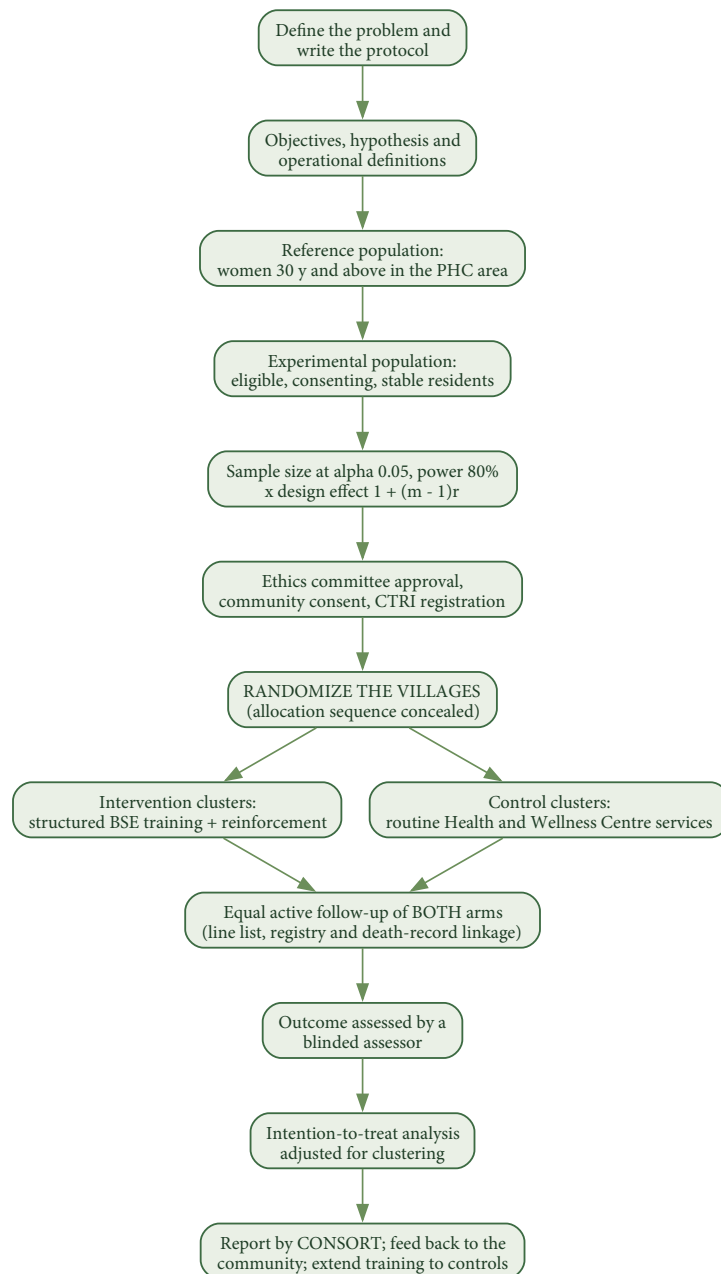
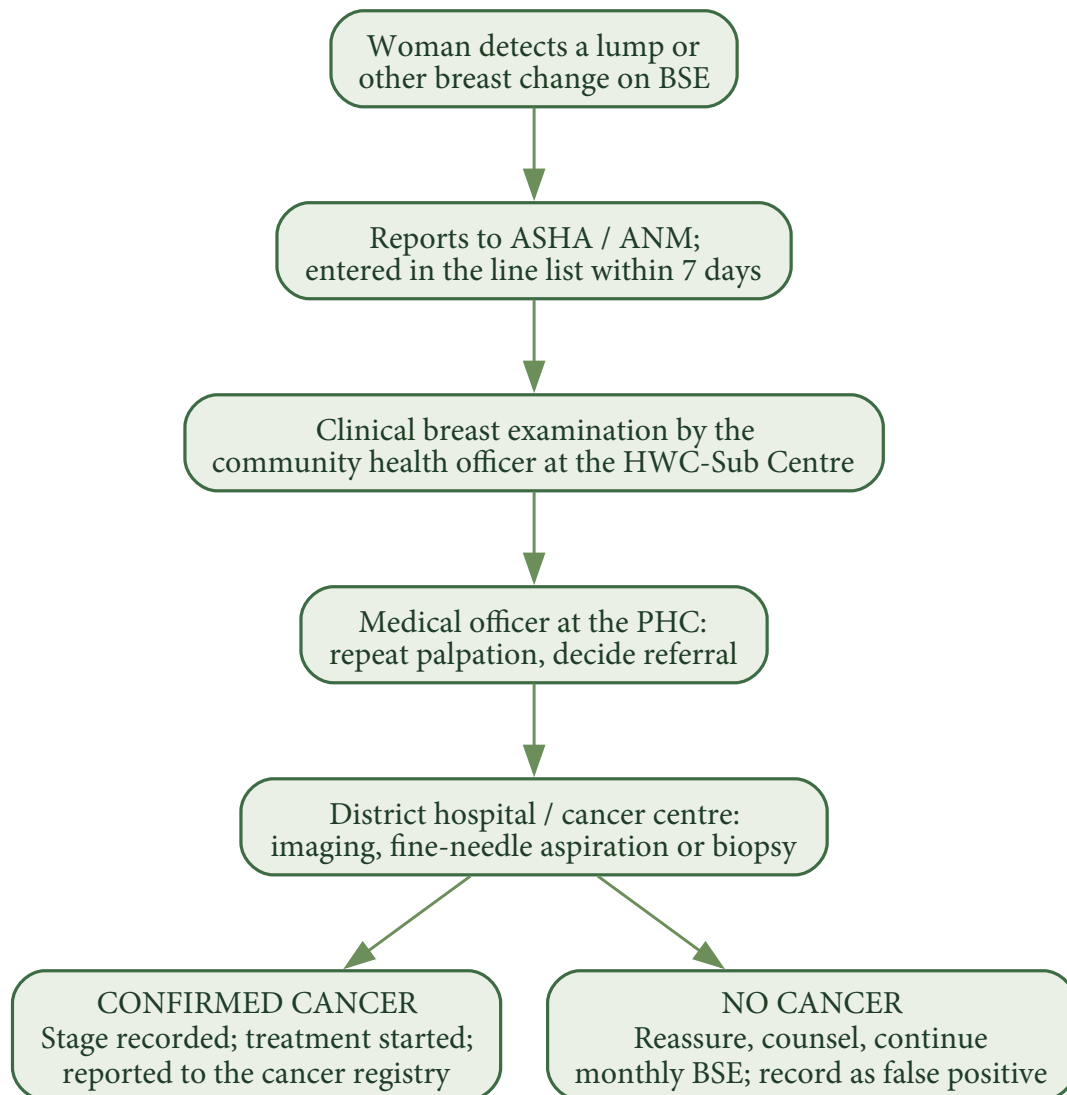


Fig 18.5 — Design of a randomized trial of the benefits of screening. (Gordis Epidemiology 7e, p.374)



Sequence for designing and running a cluster-randomised community trial of breast self-examination training



Response pathway to be guaranteed before the trial begins, for a woman who detects a lump on self-examination

High-yield exam pointers

- **Name the design in the first line:** a *cluster-randomised community intervention trial (field trial)* — unit of randomisation = village/ward, unit of analysis = individual woman.
- **Park's six steps of a trial:** protocol → reference and experimental populations → randomization → manipulation/intervention → follow-up → assessment of outcome (Park 28e, Ch 3, p 101).
Experimental studies = clinical trials (patients), field trials (healthy people), community trials (communities) (Park 28e, Ch 3, p 82).
- **Design effect** = $1 + (m - 1)r$; ignoring clustering makes p-values artificially small and confidence intervals over-narrow (Handbook of Epidemiology, Ch I.8, pp 368–369).
- **The endpoint of any screening trial is disease-specific MORTALITY in the whole allocated group** — never 5-year survival (lead-time bias); length-time and compliance bias are defeated by counting all outcomes regardless of detection method or compliance (Fletcher, Ch 10, Table 10.2, p 178).
- **Screening-test validity:** sensitivity = $a/(a+c) \times 100$, specificity = $d/(b+d) \times 100$, predictive value of a positive test = $a/(a+b) \times 100$ (Park 28e, Ch 4, p 167).

- **Blinding:** single = participant blind, double = participant and doctor, triple = participant, investigator and analyst (Park 28e, Ch 3, p 103) — none is possible for health education, so blind only the outcome assessor.
- **Dated Indian figures:** breast cancer tops the list of cancers in India — age-standardised incidence 26.6 and mortality 13.7 per 100,000 in 2022; mean age of occurrence about 42 years in India versus 53 in white women (Park 28e, Ch 6, pp 457, 463–464). Screening for oral, breast and cervical cancer for all aged 30 years and above is part of the Health and Wellness Centre package (Park 28e, Ch 10, p 735).
- **Park's sentence on BSE, worth quoting:** its effectiveness "has not been adequately quantified", but it is a useful adjuvant to early case detection and in many countries the only feasible approach to wide population coverage (Park 28e, Ch 6, p 460).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only — Park 28e records that the effectiveness of breast self-examination has not been adequately quantified, so the trial must be framed as a test of a strategy whose mortality benefit is still unproven; because screening of women aged 30 years and above for breast cancer is already part of the Health and Wellness Centre service package, the control clusters receive that standard care and are not denied any established service. Coverage note: the indexed corpus yielded the study-design, screening-bias and ethics content in full, but no Indian trial of breast self-examination with its own effect estimate, so no numerical effect size for BSE is quoted here.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, WHO Global Breast Cancer Initiative, MoHFW, PIB India, NFHS-6 (2023-24), CONSORT 2025 statement, SPIRIT 2025 statement

Programme and policy updates (India)

- **Ayushman Bharat Health and Wellness Centres are now "Ayushman Arogya Mandir"** — the Union Health Ministry directed states on 25 November 2023 to rebrand all AB-HWCs, with the tagline *Arogyam Parmam Dhanam*. The sub-centre and PHC that will deliver the trial (and the standard package the control clusters receive) must be named Ayushman Arogya Mandir in the protocol, not "Health and Wellness Centre". [MoHFW 2023-Nov] [PMID 40302063]
- **NPCDCS has been renamed NP-NCD** — the change was communicated to the states on 3 May 2023, revised operational guidelines were issued as NP-NCD 2023-30, and the CPHC NCD IT system became the **National NCD Portal**. Under the current protocol ASHAs enumerate men and women aged 30–65 years with the **Community Based Assessment Checklist (CBAC)** and clinical breast examination is repeated only once in five years; the 2025 review of the programme points out that neither the starting age of 30 nor the five-year interval is backed by evidence, which is precisely the kind of gap a PHC-level trial can address. [MoHFW 2023-May] [PMID 40302063]
- **Union Budget 2025-26 promised Day Care Cancer Centres in every district hospital within three years (200 in 2025-26)**, alongside full basic-customs-duty exemption on 36 cancer and rare-disease medicines. This is the treatment leg of the diagnostic-and-treatment pathway that

the answer insists must be guaranteed before the first woman is enrolled — name the district DCCC in the referral pathway. [PIB 2025-Feb]

Current national data

- ▶ **NFHS-6 (2023-24) was released in May 2026** — about 6.79 lakh households across 715 districts, with IIPS Mumbai as nodal agency; household coverage by any health insurance or financing scheme rose from 41.0% to 60.2%. Quote NFHS-6 for the general health indicators, but quote NFHS-5 (2019-21) for cancer-screening coverage. [PIB 2026-May]
- ▶ **Only 0.9% of Indian women aged 30-49 had ever been screened for breast cancer (NFHS-5)** — against 1.9% for cervical and 1.2% for oral cancer in men of the same age, with Tamil Nadu and Kerala well ahead of the rest; an analysis of 399,039 eligible women confirmed uptake below 1%, with higher age, education, urban residence, employment and access to internet or mass media predicting screening. Two consequences for the design: contamination of control clusters by routine services is negligible, and the anticipated control-arm proportion used in the sample-size calculation must be set from these near-zero figures, not from textbook assumptions. [PMID 40302063] [PMID 38347257]

Elimination, surveillance and disease status

- ▶ **The Mumbai cluster-randomised trial of clinical breast examination (Mittra, BMJ 2021) is the design template for this very question** — 20 geographically distinct clusters, 151,538 women aged 35-64, allocated by draw of lots to four biennial rounds of CBE by trained female primary health workers plus cancer awareness, or awareness alone, with 20 years of follow-up and analysis by intention to screen. Stage III or IV disease at diagnosis fell from 47% to 37% ($P=0.001$); breast-cancer mortality rate ratio was 0.85 (95% CI 0.71 to 1.01, $P=0.07$) overall and 0.71 (0.54 to 0.94) in women aged 50 and above; all-cause mortality rate ratio 0.95 (0.81 to 1.10). Incidence was raised while screening was active (rate ratio 1.34, 1.05 to 1.71) but had equalised by 20 years (0.97, 0.87 to 1.09) — the cleanest available demonstration that the excess was earlier detection and not overdiagnosis, and the strongest argument for the long follow-up the answer recommends. [PMID 33627312]
- ▶ **The bottleneck is the screening-to-diagnosis-to-treatment cascade, not coverage** — in the Uttar Pradesh Breast Health Initiative only 41% of women found to have an abnormal CBE at primary level reached a diagnostic facility, the barriers being fear and stigma around cancer, family responsibilities taking priority over the woman's own health, discomfort at being examined by male providers, and inability to navigate the district hospital; referring providers had no way of learning the outcome. A 2026 review argues India should judge screening by **stage shift, time to diagnosis and time to treatment** rather than numbers screened, and that organised early diagnosis using CBE with a streamlined diagnostic pathway is more feasible and equitable than population-wide mammography. Build a patient navigator, a female examiner and a tracking system into the protocol rather than assuming referral will complete itself. [PMID 39388655] [PMID 42594327]

Guideline and legislative updates

- ▶ **CONSORT 2025 supersedes CONSORT 2010 (BMJ and JAMA, April 2025)** — a 30-item checklist with seven new items, three revised, one deleted, content absorbed from the Harms, Outcomes, Non-pharmacological Treatment and TIDieR guidance, a new **open science** section

(trial registration, where the protocol and statistical analysis plan can be accessed, sharing of de-identified participant-level data, funding and conflicts of interest) and a dedicated **patient and public involvement** item. Write "reported against CONSORT 2025, with the CONSORT extension for cluster trials", not simply "CONSORT". [PMID 40228833]

- **SPIRIT 2025 supersedes SPIRIT 2013 for the protocol itself** (BMJ, April 2025) — 34 minimum items, with two new items, five revised, three deleted and two merged, its own open-science section, and a new item on how patients and the public will be involved in the design, conduct and reporting of the trial. Park's step 1, "drawing up the protocol", should now be written against SPIRIT 2025 and lodged with the Clinical Trials Registry-India before enrolment. [PMID 40294953]

Global and WHO developments

- **The WHO Global Breast Cancer Initiative (launched 2021) supplies ready-made outcome targets** — reduce breast-cancer mortality by **2.5% per year**, saving 2.5 million lives over 20 years, through three pillars (health promotion for early detection, timely diagnosis, comprehensive management) measured by the **60-60-80 benchmarks**: at least 60% diagnosed at stage I or II, diagnosis completed within 60 days of the woman or a health worker noticing the abnormality, and at least 80% completing recommended treatment. WHO does not endorse breast self-examination as a screening test and promotes **breast awareness** instead, so the intervention is best framed as awareness plus a guaranteed CBE and referral pathway. Globally there were 2.3 million new cases and 670,000 deaths in 2022, with cases projected to rise 38% and deaths 68% by 2050 and only seven countries currently achieving the 2.5% annual decline. [WHO 2021] [PMID 39388655] [PMID 39994475]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 3 Principles of Epidemiology and Epidemiologic Methods, pp 82–83, 101–103, 106; Ch 4 Screening for Disease, p 167; Ch 6 Epidemiology of Chronic Non-communicable Diseases, pp 457, 460, 463–464; Ch 7 Health Programmes in India, p 577; Ch 10 Health Care of the Community, p 735); Gordis Epidemiology 7e (Ch 10 Assessing Preventive and Therapeutic Interventions: Randomized Trials, pp 214, 224; Ch 18 Epidemiologic Approach to Evaluating Screening Programs, pp 368, 373–374, 381, 391); Handbook of Epidemiology, Ahrens & Pigeot (Ch I.8 Intervention Trials, pp 354, 368–369); Fletcher — Clinical Epidemiology: The Essentials (Ch 10 Prevention, pp 178–179); Browner — Designing Clinical Research 5e (Ch 6 Estimating Sample Size, pp 130–135); ICMR National Ethical Guidelines for Biomedical and Health Research Involving Human Participants, 2017 (sections 2.11, 3.7, 6.2.13, 10.4.6); Government of India, Operational Guidelines — Prevention, Screening and Control of Common Non-Communicable Diseases: Hypertension, Diabetes and Common Cancers (Oral, Breast, Cervix), MoHFW, 2016.

Recent-advances sources: 33627312 — Effect of screening by clinical breast examination on breast cancer incidence and mortality after 20 years: prospective, cluster randomised controlled trial in Mumbai, BMJ 2021, doi:[10.1136/bmj.n256](https://doi.org/10.1136/bmj.n256); 40302063 — Current Status of Implementation of Cancer Screening Programme in India: A Review of Policies and Practice, Asian Pacific Journal of Cancer Prevention 2025, doi:[10.31557/APJCP.2025.26.4.1117](https://doi.org/10.31557/APJCP.2025.26.4.1117); 38347257 — Socio-cultural and financial issues against breast cancer screening behaviour among eligible Indian women: evidence for action, Breast Cancer Research and Treatment 2024, doi:[10.1007/s10549-024-07244-7](https://doi.org/10.1007/s10549-024-07244-7) (abstract-only); 39388655 — Barriers to Follow-Up of an Abnormal Clinical Breast Examination in Uttar Pradesh, India: A Qualitative Study, JCO Global Oncology 2024, doi:[10.1200/GO.24.00001](https://doi.org/10.1200/GO.24.00001); 42594327 — Cancer Screening in India: Lessons, Limitations, and Pragmatic Solutions for the Next Decade, JCO Global Oncology 2026, doi:[10.1200/GO-26-00018](https://doi.org/10.1200/GO-26-00018) (abstract-only); 40228833 — CONSORT 2025 statement: updated guideline for reporting randomised trials, BMJ 2025, doi:[10.1136/bmj-2024-081123](https://doi.org/10.1136/bmj-2024-081123); 40294953 — SPIRIT 2025 statement: updated guideline for protocols of randomised trials, BMJ 2025, doi:[10.1136/bmj-2024-081477](https://doi.org/10.1136/bmj-2024-081477); 39994475 — Global patterns and trends in breast cancer incidence and mortality across 185 countries, Nature Medicine 2025, doi:[10.1038/s41591-025-03502-3](https://doi.org/10.1038/s41591-025-03502-3) (abstract-only). Also: WHO Global Breast Cancer Initiative, launched 2021 (three pillars and the 60-60-80 benchmarks); MoHFW letter to States and Union Territories renaming Ayushman Bharat Health and Wellness Centres as Ayushman Arogya Mandir, 25 November 2023; MoHFW communication to States renaming NPCDCS as the National Programme for Prevention and Control of Non-Communicable Diseases (NP-NCD), 3 May 2023, with revised operational guidelines NP-NCD 2023-30; PIB India, Union Budget 2025-26, February 2025 (Day Care Cancer Centres in all district hospitals); PIB India, release of the National Family Health Survey-6 (2023-24), May 2026.



Study design for prevalence of obesity

Asked in: Paper I 06-Apr-2026 — RGUHS MD Community Medicine

Definition — Prevalence is the total number of all individuals (old and new cases) who have an attribute or disease at a particular point in time, divided by the population at risk of having that attribute at the same point in time; though called a "rate" it is really a ratio (Park 28e, Ch 3, p 81). A question that asks only *how much obesity exists now* is answered by a **community-based descriptive cross-sectional (prevalence) study** — a single examination of a cross-section of the population at one point in time, whose result can be projected to the whole population provided sampling is done correctly (Park 28e, Ch 3, p 89). Exposure and outcome are ascertained simultaneously, so a cross-sectional study identifies prevalent, not incident, cases (Gordis 7e, Ch 7, p 172).

Choice of study design and its justification

Design	What it yields	Suitability here
Cross-sectional (prevalence) survey	Point prevalence + simultaneous data on age, sex, physical exercise, body weight, diet (Park 28e, Ch 3, p 89)	Design of choice — cheap, quick, single contact; more useful for chronic than short-lived conditions, and obesity is chronic and long-lasting
Cohort study	Incidence and risk	Not required; costly, long follow-up
Case-control study	Odds ratio for determinants	Cannot generate a prevalence at all
Review of the physician's clinic records	Counts among care-seekers only	Rejected — selection bias, denominator (population at risk) unknown

Cross-sectional data suffer **prevalence-incidence (survival) bias** and cannot establish the temporal sequence of exposure and outcome; their findings therefore generate hypotheses for later analytical study (Gordis 7e, Ch 7, p 172).

Objectives and operational definition (case definition)

- **Primary objective** — to estimate the prevalence of overweight and obesity among adults aged 18 years and above residing in the physician's field practice area, with 95% confidence limits.
- **Secondary objectives** — to describe the prevalence by age, sex, socio-economic status and occupation, and to estimate the prevalence of **abdominal (central) obesity**.

Obesity is defined as an abnormal growth of adipose tissue due to enlargement of fat cell size (hypertrophic), an increase in fat cell number (hyperplastic), or both (Park 28e, Ch 6, p 472). The **operational** case definition must be fixed in the protocol before fieldwork, because prevalence is entirely determined by the cut-off chosen:

Criterion	WHO cut-off (kg/m ² , BMI = weight in kg / height in m ²)	Cut-off to be used in India
Underweight	< 18.50	< 18.5
Normal range	18.50–24.99	18.5–22.9

Criterion	WHO cut-off (kg/m ² , BMI = weight in kg / height in m ²)	Cut-off to be used in India
Overweight (pre-obese)	25.00–29.99	23.0–24.9
Obese class I	30.00–34.99	Obesity ≥ 25.0
Obese class II	35.00–39.99	—
Obese class III	≥ 40.00	—
Abdominal obesity — waist circumference	High waist-hip ratio (WHR) > 0.9 in men, > 0.85 in women indicates abdominal fat accumulation	Waist circumference should be < 90 cm in men and < 80 cm in women; normal WHR < 0.95 men, < 0.85 women

WHO classification: Park 28e, Ch 6, Table 2, p 474. Indian cut-offs and waist criteria: National Programme for Prevention and Control of NCDs — Combined Manual for Medical Officers, pp 16–17. WHR: Park 28e, Ch 6, p 476. Waist circumference is measured at the **mid-point between the lower border of the rib cage and the iliac crest**; it is unrelated to height and is an approximate index of intra-abdominal fat mass (Park 28e, Ch 6, pp 475–476).

Study population, sampling and sample size

Element	Specification for this study
Reference population	All adults ≥ 18 years in the physician's practice area (a defined urban ward / group of villages)
Sampling frame	Household list from the ward/panchayat register, ASHA or auxiliary nurse midwife (ANM) family register, or a fresh house-to-house enumeration; its accuracy and completeness decide the quality of the sample (Park 28e, Ch 21, p 1006)
Inclusion	Usual residents ≥ 18 years, resident ≥ 6 months, consenting
Exclusion	Pregnant women and women who have had a birth in the two months preceding the survey (as in NFHS-5); persons with oedema, ascites or spinal deformity in whom height/weight are invalid
Study period	Fixed calendar window (e.g. 3 months), so the estimate is a point prevalence

Sampling technique — a probability sample is mandatory; only then can chance error be calculated (Mahajan, *Methods in Biostatistics*, Sampling, p 121).

Method	Procedure	Use here
Simple random	Number every unit in the frame, select by random-number table	Feasible if the frame is small and complete
Systematic random	k = total population / sample size desired; pick a random start < k, then every kth house (Park 28e, Ch 21, p 1006)	Preferred for a house-to-house survey — simple, field-friendly
Stratified random	Divide into homogeneous strata (urban/rural, socio-economic group, sex), sample each in proportion to size	Add as strata to give representation to all groups; gives greater accuracy (Mahajan, Sampling, p 137)

Method	Procedure	Use here
Cluster / 30-cluster	Population divided by 30 gives the sampling interval; clusters picked by cumulative population	Only if the frame is unavailable; gives a higher standard error but is cheaper and quicker (Mahajan, Sampling, p 138)
Multistage	Successive random selection of ward → street → household	For a large practice area

Sample size — for estimating a single proportion with absolute precision, $n = 4pq / L^2$ (95% confidence, where L is the allowable absolute error) (Mahajan, Sampling, pp 130–132).

- ▶ Anticipated prevalence **p = 24 per cent**, taken from NFHS-5 (2019–21), in which 24 per cent of women and 23 per cent of men aged 15–49 years were overweight or obese (body mass index ≥ 25.0). $q = 76$.
- ▶ Absolute allowable error $L = 5$ per cent; $n = (4 \times 24 \times 76) / 5^2 = 7296 / 25 = 292$.
- ▶ If cluster sampling is used, multiply by the **design effect** — the ratio of the standard error under the actual design to that under simple random sampling; a value of 2 is a sensible ad hoc choice, i.e. $292 \times 2 = 584$ (Handbook of Epidemiology, Ch II.1, pp 570, 594).
- ▶ Inflate for expected non-response/absentees, say 10 per cent: $584 / 0.9 \approx 650$ adults.

Data collection, tools and quality control

- ▶ **Pre-test every form before the first participant is recruited**; any entry involving subjective judgement needs an explicit operational definition written on the form itself and in an operations manual (Browner, *Designing Clinical Research* 5e, Ch 18, pp 570–571).
- ▶ Adopt the WHO **STEPS** structure for a non-communicable disease risk-factor survey — Step 1 questionnaire (demographic, health status, health behaviours), Step 2 physical measurements, Step 3 biochemical measurements (Park 28e, Ch 6, pp 436–437). Steps 1 and 2 suffice for a prevalence-of-obesity survey; Step 3 only if metabolic correlates are an objective.
- ▶ **Anthropometry** — height on a stadiometer to 0.1 cm without footwear; weight on a calibrated digital scale to 0.1 kg in light clothing; waist and hip circumference with a non-stretchable tape. Calibrate the scale daily against a standard weight.
- ▶ **Minimise non-sampling error**, which arises from inadequately calibrated instruments and observer variation (Park 28e, Ch 21, p 1006): train all observers together, take duplicate readings, and measure inter-observer agreement in a pilot on ~30 subjects.
- ▶ Conduct a **pilot study / dress rehearsal** to test feasibility, recruitment rate, reproducibility of measurement and cost (Browner 5e, Ch 18, p 573).
- ▶ Handle **non-response** by up to three repeat visits including an evening or holiday visit, and report the response rate; compare responders with non-responders on age and sex.

Analysis, presentation and ethics

- ▶ Prevalence = (number of persons with body mass index ≥ 25.0 at the survey point / number of persons examined) $\times 100$.
- ▶ Report the **95 per cent confidence interval** as $p \pm 1.96 \times \text{standard error of the proportion}$, $SE = \sqrt{pq/n}$ (Park 28e, Ch 21, standard error of proportion).

- Give **age- and sex-specific prevalence**, since point prevalence can be made specific for age, sex and other attributes (Park 28e, Ch 3, p 81); age-standardise before comparing with NFHS-5 or a neighbouring area.
- Test subgroup differences by chi-square; report prevalence of abdominal obesity separately.
- **Ethics** — written informed consent with a participant information sheet in the local language, approval of the institutional ethics committee, confidentiality and privacy of data, and referral of every person found obese to the physician / Health and Wellness Centre for lifestyle counselling (ICMR National Ethical Guidelines 2017, Box 4.4).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Diagram 1 (to sketch)** — *Cross-sectional study design*: one box "Defined population of the practice area", an arrow labelled "single examination at one point in time", splitting into four cells of a 2×2 table labelled *Exposed–Disease (a)*, *Exposed–No disease (b)*, *Not exposed–Disease (c)*, *Not exposed–No disease (d)*, with the analysis arm labelled "prevalence of disease in exposed = $a/(a+b)$ ".
- **Diagram 2 (to sketch)** — *Anthropometric landmarks*: an outline figure marked with "height — stadiometer, no footwear", "waist — midpoint between lower border of rib cage and iliac crest", "hip — widest point over greater trochanters".

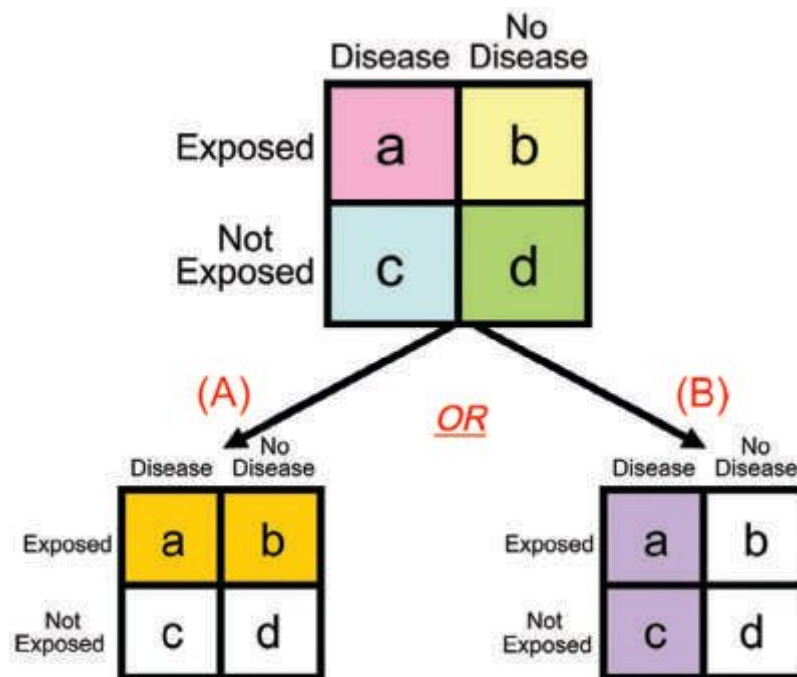
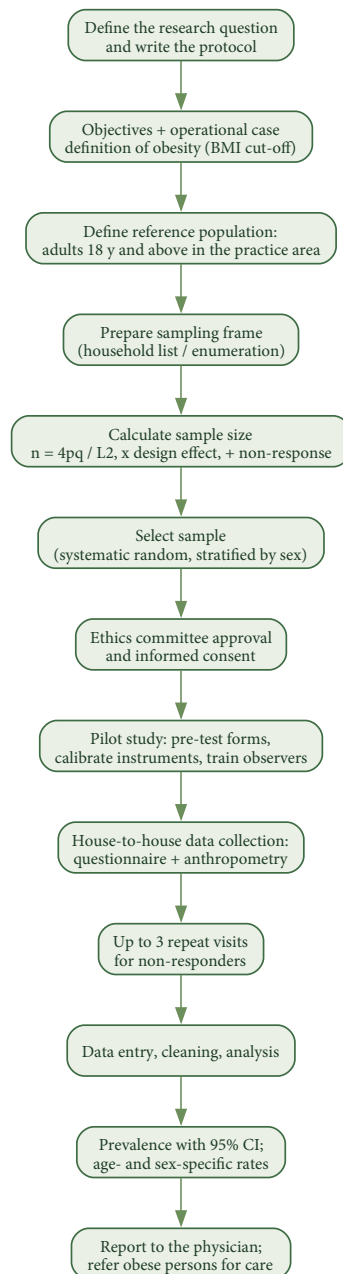
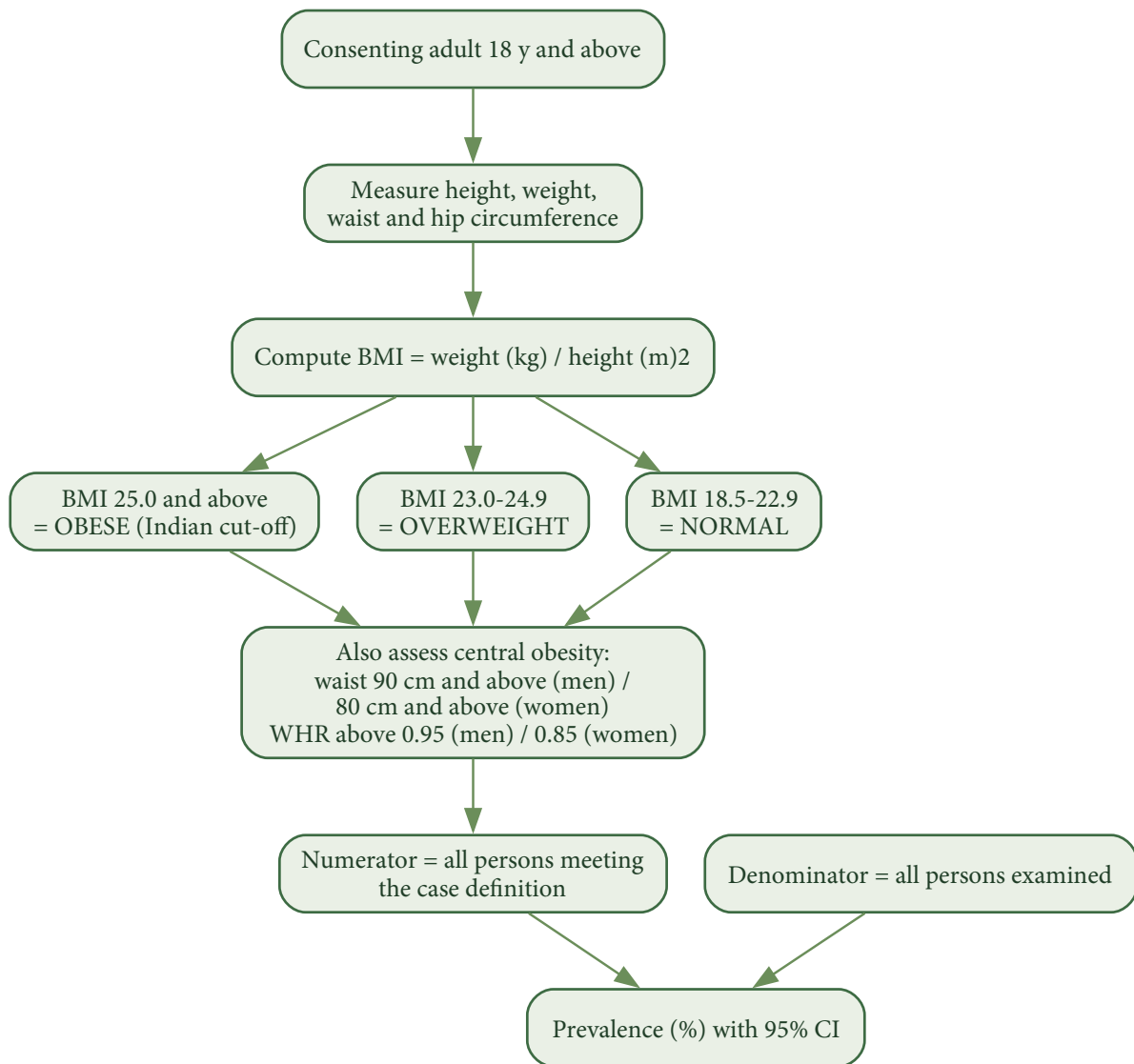


Fig 7.5 — Design of a hypothetical cross-sectional study. (top) A 2 x 2 table of (Gordis Epidemiology 7e, p.173)



Operational sequence of a community-based cross-sectional survey to estimate the prevalence of obesity



Field algorithm for classifying a surveyed adult using Indian anthropometric cut-offs

High-yield exam pointers

- **Design word to write first:** "community-based descriptive **cross-sectional** study" — also called a **prevalence study** (Park 28e, Ch 3, p 89; Gordis 7e, Table 7.11).
- **Point prevalence** = (all current cases, old and new, at a point in time / estimated population at the same point in time) × 100 (Park 28e, Ch 3, p 81).
- **Sample size for a proportion:** $n = 4pq / L^2$ at 95 per cent confidence; then × design effect (take 2 for cluster sampling) and + 10 per cent for non-response.
- **Indian body mass index cut-offs:** normal 18.5–22.9, overweight 23–24.9, obesity ≥ 25 kg/m²; waist circumference < 90 cm (men) and < 80 cm (women) (NCD Combined Manual for Medical Officers). WHO obesity is BMI ≥ 30.
- **Dated Indian figure:** NFHS-5 (2019–21) — 24 per cent of women and 23 per cent of men aged 15–49 years are overweight or obese; the figure for women rose from 21 per cent in NFHS-4 (2015–16) to 24 per cent; mean BMI 22.4 in both sexes; 57 per cent of women and 48 per cent of men have a waist-to-hip ratio placing them at substantially increased risk of metabolic complications.

- ▶ **Chief limitation to name:** prevalence-incidence (survival) bias and inability to establish temporal sequence — a cross-sectional study generates, not tests, aetiological hypotheses (Gordis 7e, Ch 7, p 172).
- ▶ **WHO STEPS** for a non-communicable disease risk-factor survey = questionnaire → physical measurements → biochemical measurements (Park 28e, Ch 6, pp 436–437).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park's 28th edition still prints the WHO classification (obesity = body mass index $\geq 30 \text{ kg/m}^2$), while the Government of India non-communicable disease programme manual uses the lower Asian Indian cut-off (obesity = body mass index $\geq 25 \text{ kg/m}^2$); state which cut-off you have adopted in the protocol, because the prevalence obtained differs greatly between the two.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 3 — Principles of Epidemiology and Epidemiologic Methods, pp 80–81, 89; Ch 6 — Epidemiology of Chronic Non-communicable Diseases, pp 436–437, 472–476; Ch 21 — Health Information and Basic Medical Statistics, p 1006); Gordis Epidemiology 7e (Ch 7 — Observational studies, pp 172–174, 192); Mahajan's Methods in Biostatistics for Medical Students and Research Workers (Sampling, pp 121–138); Handbook of Epidemiology, Ahrens & Pigeot (Ch II.1 — Sample Size Determination in Epidemiologic Studies, pp 570, 594); Browner — Designing Clinical Research 5e (Ch 18 — Study Implementation and Quality Control, pp 570–573); ICMR National Ethical Guidelines for Biomedical and Health Research 2017 (Box 4.4); National Programme for Prevention and Control of Non-Communicable Diseases — Combined Manual for Medical Officers (pp 16–17); National Family Health Survey (NFHS-5), India Report 2019–21 (Ch 10 — Nutritional status of adults).

♦ ♦ ♦

Nipah virus outbreak management

Asked in: Paper III 11-May-2022 — RGUHS MD Community Medicine

Definition — Nipah virus infection is an emerging **zoonotic** disease caused by a virus of the family **Paramyxoviridae**, genus **Henipavirus**; it takes its name from the village in Malaysia where the person from whom the virus was first isolated died in **1999** (Park 28e, Ch 5, p 349). An **epidemic (outbreak)** exists when the observed number of cases is in excess of the expected frequency for that population, based on past experience (Park 28e, Ch 3, p 159). As **district medical and health officer (DMHO)** I am the district's public health authority and the officer accountable for detection, investigation, containment and closure.

1. Epidemiological facts that drive every decision

Feature	Fact	Operational consequence for the district
Agent	Paramyxoviridae, genus <i>Henipavirus</i> (Park 28e, Ch 5, p 349)	Handle as a high-consequence pathogen; no field culture
Natural reservoir	Fruit bats of family Pteropodidae , genus <i>Pteropus</i> ; no apparent disease in the bat (Park 28e, Ch 5, p 349)	Reservoir cannot be eliminated — control must target exposure, not the bat
Modes of transmission	Direct contact with infected bats , with infected pigs , or person-to-person from an infected patient — the pattern reported in Bangladesh and India, seen in family members and caregivers ; consumption of raw date palm sap contaminated by bat secretions and excreta (Park 28e, Ch 5, p 349); respiratory secretions and other body fluids are the main drivers of person-to-person spread (Sunder Lal, Ch 8, p 475)	Contact tracing of household and health-care contacts is the single highest-yield activity
Incubation period	4 to 14 days (Park 28e, Ch 5, p 349); 4 days to 2 weeks (Sunder Lal, Ch 8, p 475)	Fixes the contact-surveillance period and the date the outbreak can be closed
Clinical spectrum	Asymptomatic infection → acute respiratory illness → fatal encephalitis. Fever, headache, myalgia, vomiting, sore throat; then dizziness, drowsiness, altered consciousness and neurological signs; atypical pneumonia and acute respiratory distress; seizures and coma within 24–48 hours in severe cases (Park 28e, Ch 5, p 350)	Acute encephalitis syndrome plus respiratory distress is the alert syndrome
Sequelae	~20% left with residual neurological damage (seizure disorder, personality change); a few relapse or develop delayed-onset encephalitis (Park 28e, Ch 5, p 350)	Survivors need follow-up; do not close the register at discharge
Case fatality rate	40–75% , varying with local surveillance and clinical-management capacity (Park 28e, Ch 5, p 350); may go up to 100% (Sunder Lal, Ch 8, p 475)	Justifies emergency-mode response from case one
Diagnosis	Real-time RT-PCR on body fluids and antibody detection by ELISA ; also PCR assay and virus isolation in cell culture (Park 28e, Ch 5, p 350)	Specimens must be referred; no district-level confirmation

Feature	Fact	Operational consequence for the district
Treatment / vaccine	No specific drug and no vaccine; intensive supportive care for respiratory and neurological complications (Park 28e, Ch 5, p 350)	Prevention is the <i>only</i> tool — there is no immunisation campaign to fall back on
Indian experience	India affected since 2001; West Bengal (Siliguri, Nadia); Kerala, May 2018 — Kozhikode and Malappuram, 16 deaths among 18 confirmed cases (CFR 88.8%) in 20 days, including a 31-year-old nurse; again 2019 and 2021; most recent Kerala outbreak September 2023 (Sunder Lal, Ch 8, pp 475-476; Park 28e, Ch 5, p 349)	Nosocomial amplification is the documented Indian pattern — protect health workers first

2. First 24 hours — verification, notification and activation

- ▶ **Verify the diagnosis myself, on the spot.** Reports may be spurious and arise from lay misinterpretation of signs and symptoms. It is not necessary to examine every case; clinical examination of a **sample of cases** suffices, and epidemiological investigation **must not be delayed until laboratory results arrive** (Park 28e, Ch 3, pp 158-159).
- ▶ **Confirm that an outbreak exists** by comparing current acute encephalitis syndrome frequency with the same period in previous years, and by checking whether cases are epidemiologically linked or clustered (Park 28e, Ch 3, p 159). Under the **Integrated Disease Surveillance Programme (IDSP)** the encephalitis trigger is set as low as a **single probable case of Japanese encephalitis, or two cases of fever with altered consciousness or seizures in a village of 1,000 population** (Sunder Lal, Ch 7, p 310) — so a single suspected Nipah case already crosses the alert threshold.
- ▶ **File the First Information Report (Form 'C')** from the reporting unit to the **district surveillance officer (DSO)** as soon as the suspected outbreak is verified (Sunder Lal, Ch 7, p 310), and report upward under IDSP. This is a **trigger-2/3 event** — district response by the DSO and rapid response team, escalating to state level once the outbreak is established (Jugal Kishore, Ch 1, p 491).
- ▶ **Activate the district Rapid Response Team (RRT)** — a multi-speciality team of an epidemiologist, a clinician, a microbiologist and other specialists as required (Park 28e, Ch 7, p 568).
- ▶ **Collect and despatch specimens from the first 10 cases** — enough to confirm the aetiology (Sunder Lal, Ch 7, pp 310-311) — through the cold chain to the designated referral laboratory, that is, the same national laboratories used for other emerging viral diseases: the **National Institute of Virology (NIV), Pune** and the **National Centre for Disease Control (NCDC), Delhi** (Jugal Kishore, Ch 1, p 669).
- ▶ **Isolate every suspected case immediately** and start barrier nursing before confirmation.
- ▶ **Inform the District Collector** as chairperson of the **District Disaster Management Authority (DDMA)** under the Disaster Management Act, 2005 (Sunder Lal, Ch 13, p 669), and open a district control room with a 24-hour helpline; the IDSP 24×7 toll-free number **1075** and the

Media Scanning and Verification Cell capture rumours and additional alerts (Sunder Lal, Ch 7, p 307).

3. Epidemiological investigation

Objectives: define the magnitude of the outbreak in time, place and person; determine the conditions responsible; identify the cause, source and mode of transmission so that control measures can be chosen; and recommend measures to prevent recurrence (Park 28e, Ch 3, p 159).

Step (Park 28e, Ch 3, pp 159-160)	What I actually do for Nipah
1. Verification of diagnosis	Bedside review + RT-PCR/ELISA from NIV
2. Confirmation of the existence of an epidemic	Compare with previous years' acute encephalitis syndrome data; any confirmed case counts
3. Defining the population at risk	Obtain/prepare a map of the affected area with landmarks, roads and numbered dwellings; house-to-house census by age and sex to fix the denominator, without which attack rates cannot be computed
4. Rapid search for all cases	Active house-to-house medical survey ; structured epidemiological case sheet recording name, age, sex, occupation, travel, previous exposure, date of onset, signs and symptoms, personal contacts at home, work and school; ask each patient about other ill persons in the household, neighbourhood, school and workplace with onset within one incubation period of the index case; daily search for secondary cases
5. Data analysis by time, place, person	Epidemic curve by date of onset — distinguishes a common-source from a propagated epidemic (successive waves one incubation period apart indicate person-to-person spread); spot map of cases against bat roosts, palm-sap collection points, piggeries and the index household; attack rate and case fatality rate by age, sex, occupation (especially health workers) and exposure
6. Formulation of hypothesis	Explain the outbreak in terms of source, agent, mode of spread and enabling environmental factors, ranked by likelihood
7. Testing of hypothesis	Compare attack rates in exposed and unexposed; a case-control study yielding an odds ratio for exposures such as raw sap/toddy, fallen or partly-eaten fruit, bat-roost proximity and caregiving contact (Sunder Lal, Ch 7, p 311)
8. Evaluation of ecological factors	Survey bat roosts, date-palm and toddy collection practice, piggeries and animal deaths jointly with the animal husbandry and forest departments
9. Further investigation of the population at risk	Serological survey of contacts and health workers to detect inapparent infection; sampling of bats and domestic animals
10. Writing the report	Background (geography, climate, socio-economic and demographic profile, organisation of health services, surveillance and early-warning system, normal disease prevalence) and historical data on previous epidemics (Park 28e, Ch 3, Table 47)

Working case definition used for the search (framed from the clinical and laboratory description above; the operative definition is issued by the state/NCDC):

Category	Definition
Suspect	Acute fever with headache/myalgia/vomiting and altered sensorium, seizures or acute respiratory distress, in a resident of the affected area during the outbreak period
Probable	A suspect case with an epidemiological link — contact with a confirmed case, a caregiver or health worker, or history of raw date-palm sap/toddy consumption or bat exposure
Confirmed	A suspect or probable case with real-time RT-PCR positivity on body fluids or Nipah-specific antibody by ELISA

4. Control measures — breaking the chain of transmission

Communicable disease control acts on three links: the **reservoir/source**, the **modes of transmission**, and the **susceptible host** (Park 28e, Ch 3, pp 144-146).

Link	Measures in a Nipah outbreak
Controlling the reservoir and source	Early diagnosis, notification, isolation, treatment, quarantine, surveillance and disinfection — the general measures for a human reservoir (Park 28e, Ch 3, p 144). Isolate every case in a designated facility for the period of communicability; line-list every contact (household, caregivers, health-care staff, ambulance crew, laboratory staff, funeral attendants) and keep them under active surveillance for the maximum incubation period from last exposure, with restriction of movement. Quarantine as a stand-alone tool has largely been replaced by active surveillance (Park 28e, Ch 3, p 146)
Animal component (zoonoses control)	Park's three components for zoonoses: control in animals (diagnosis, treatment, destruction, quarantine, immunisation); control of vehicles of transmission (food hygiene, safe handling of animal products, proper disposal of animal carcasses and wastes, disinfection); prevention and treatment in man (protection of high-risk groups, monitoring of health status including occupational health, early diagnosis and treatment, health education, prevention of environmental and food contamination, improved diagnostic facilities) — Park 28e, Ch 15, p 942. Culling of infected pigs stopped the Malaysian and Singapore outbreaks (Nelson & Williams, Ch 13, p 376). Bats are not culled; the reservoir is managed by removing exposure
Interrupting transmission	Ban sale and consumption of raw date-palm sap and toddy ; discard fallen and partly-eaten fruit; wash all fruit; hand hygiene with soap and water after any contact with a patient or animal (Sunder Lal, Ch 8, p 475). Environmental disinfection with freshly prepared 1% sodium hypochlorite , and 70% alcohol for metal surfaces where bleach is unsuitable; never spray disinfectant on heavily contaminated surfaces, as splashes spread the agent (Park 28e, Ch 3, pp 157-158)
Protecting the susceptible host	No vaccine and no chemoprophylaxis exist ; protection is therefore entirely non-specific — full personal protective equipment and correct mask technique for all health-care and burial staff (Park 28e, Ch 3, pp 158-159), restriction of visitors, one designated attendant per patient, and safe dead-body handling. In the absence of a vaccine the only way to reduce infection in people is by raising awareness of risk factors and educating them on reducing exposure (Park 28e, Ch 5, p 350)

5. Service delivery, coordination and legal powers

- Designate a Nipah treatment centre at the district hospital or medical college with a physically separate isolation ward, a triage area at the entrance, an intensive care facility for respiratory and

neurological support, and a dedicated ambulance; restrict entry to a named, trained, rostered team.

- ▶ **Protect health workers first** — the Indian outbreak pattern is nosocomial and caregiver-centred; maintain a named register of exposed staff with daily temperature monitoring, and remove any symptomatic worker from duty at once.
- ▶ **Strengthen surveillance across the district** — sensitise all government and private hospitals, sentinel private practitioners and laboratories to notify every suspected acute encephalitis syndrome case **immediately** to the district surveillance officer, by telephone and on the outbreak line list, over and above the routine **weekly** IDSP formats — 'S' (syndromic), 'P' (probable) and 'L' (laboratory-confirmed), filled by health workers, clinicians and laboratory staff respectively and collected **every Monday** (Sunder Lal, Ch 7, p 307). Private providers treat over **70% of spells of sickness**, so they must be brought into notification and given the national guidelines on outbreak prevention and control (Sunder Lal, Ch 20, p 787).
- ▶ **Intersectoral and One Health coordination** through the DDMA — animal husbandry and veterinary services (pigs, animal deaths), forest and wildlife (bat roosts), revenue and police (movement restriction), panchayati raj institutions, education (school closure if indicated), and the district information officer.
- ▶ **Use statutory powers** — the **Epidemic Diseases Act, 1897** confers on health authorities the power to take special measures during an epidemic; this 125-year-old Act is being replaced by a Public Health (Prevention, Control and Management of Epidemics, Bioterrorism and Disasters) legislation (Sunder Lal, Ch 8, p 410).
- ▶ **Risk communication** — daily bulletins to the press to pre-empt rumour and panic, community meetings, and messages in the local language on sap and fruit avoidance, hand hygiene, early reporting of fever with altered sensorium, and safe funeral practice.

6. Closing the outbreak and evaluating the response

Active **daily** search for secondary cases continues until the area is declared free of the epidemic, a period usually taken as **twice the incubation period of the disease since the occurrence of the last case** (Park 28e, Ch 3, p 160) — for Nipah, **28 days (2 × 14 days)**. I then submit a complete outbreak report with recommendations to prevent recurrence, revise the district epidemic preparedness plan, restock personal protective equipment, repeat RRT training, and score the response on these indicators:

Indicator	What it measures	Target
Onset-to-notification interval (index case)	Sensitivity of surveillance and provider awareness	Shortest achievable — every day of delay adds one generation of cases
Onset-to-specimen-despatch interval	Field readiness of the RRT and the cold chain	Same day as case verification
Onset-to-laboratory-confirmation interval	Referral-laboratory turnaround and transport	Minimise; response must not wait on it
Proportion of listed contacts traced and followed for the full incubation period	Completeness of containment	100%

Indicator	What it measures	Target
Case fatality rate	Quality of clinical and critical care	Below the 88.8% recorded in Kerala, May 2018
Number of secondary cases and health-care-worker cases	Effectiveness of isolation and infection control	Zero
Completeness of the line list and outbreak report	Quality of the record for future preparedness	100%, with Table 47 background items filled

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Diagram 1 — Spot map of the affected area.** Labels to write: *index household; secondary cases in household contacts; health-care-worker cases marked separately; district hospital / designated isolation facility; bat roost (Pteropus colony); date-palm sap collection points; piggery; water body and village boundary; scale and north arrow; legend — confirmed / probable / suspect / death.*
- **Diagram 2 — Propagated epidemic curve.** Labels to write: *X-axis — date of onset; Y-axis — number of cases; index case; first generation of cases; second generation; interval between peaks = one incubation period (4–14 days); last case; outbreak declared over at $2 \times$ incubation period (28 days) after last case.*

exposure.

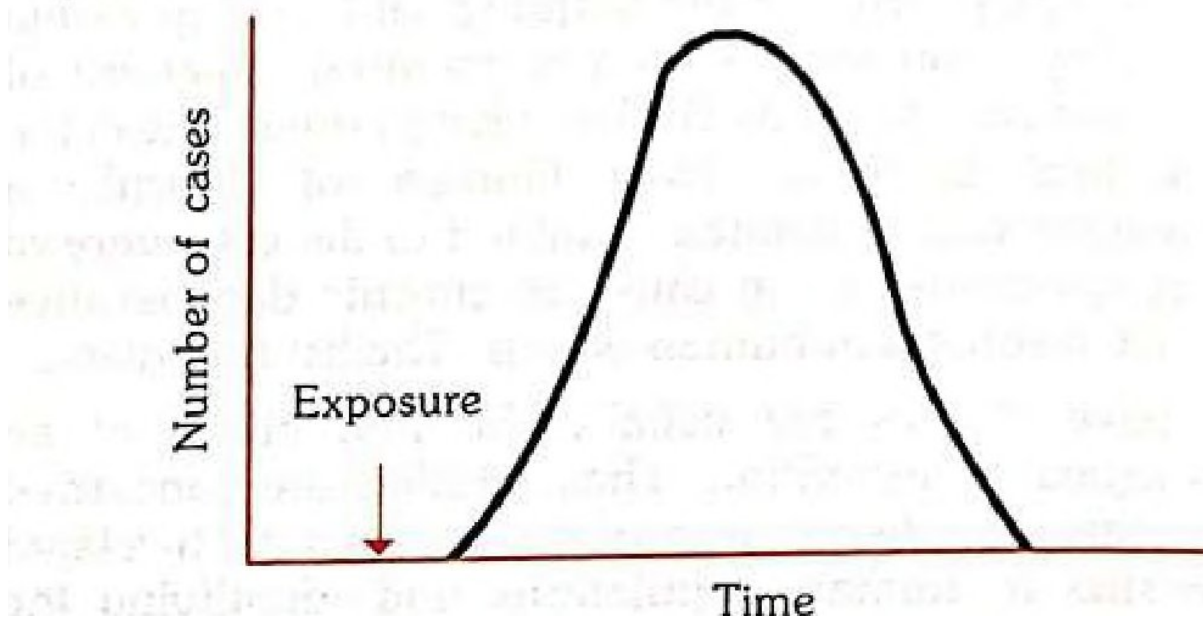
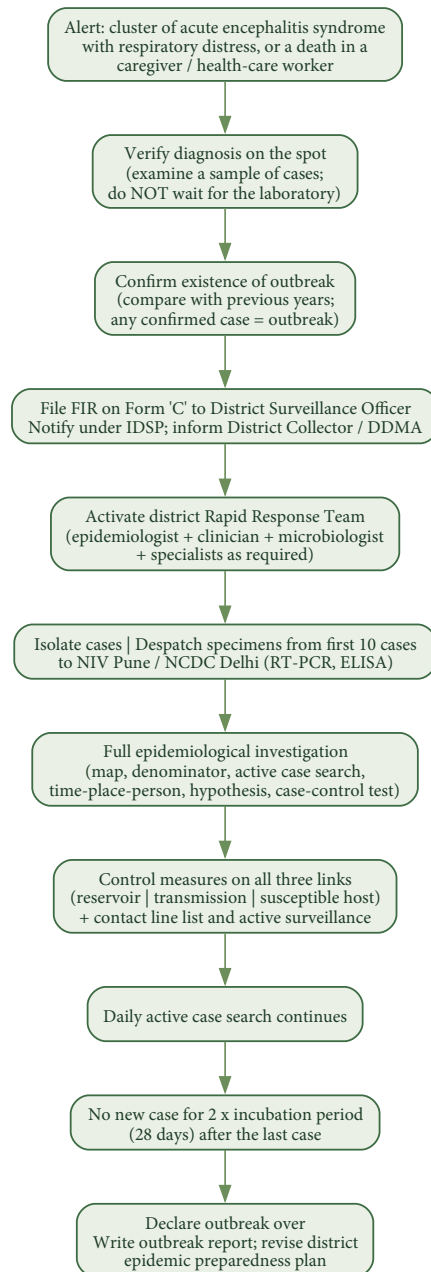
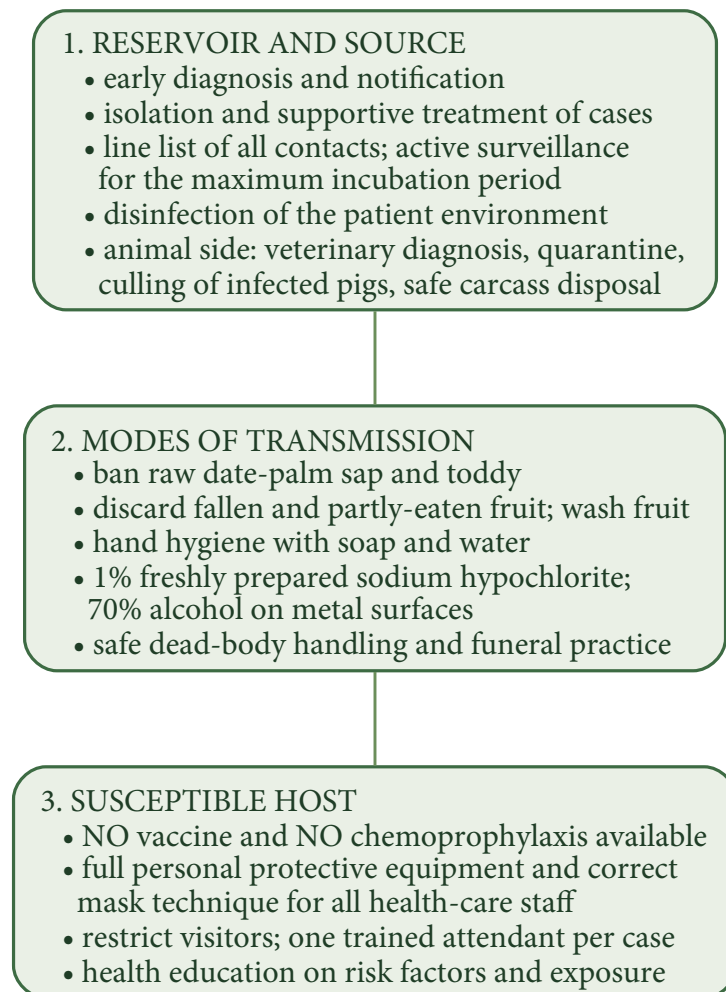


Fig 4 — Epidemic curve (Park's Textbook of Preventive and Social Medicine, 28e, p.84)



District response algorithm for a suspected Nipah virus outbreak



Nipah control measures arranged along the chain of transmission

High-yield exam pointers

- ▶ **Park's ten steps, as one string:** Verify → Confirm → Define population at risk → Rapid case search → Analyse time-place-person → Formulate hypothesis → Test hypothesis → Evaluate ecological factors → Further investigate population at risk → Write report.
- ▶ **Closure arithmetic:** $2 \times \text{maximum incubation period after the last case} = 2 \times 14 = 28 \text{ days}$ for Nipah.
- ▶ **IDSP ladder to quote:** trigger 1 local, 2 district (DSO + RRT), 3 state, 4 national, 5 international; FIR on Form 'C'; weekly 'S' / 'P' / 'L' formats every **Monday**; helpline 1075.
- ▶ **Why one case is an outbreak:** a single case of a communicable disease long absent from a population, or a first invasion by a disease not previously recognised there, requires immediate reporting and full field investigation (Park 28e, Ch 3, p 113).
- ▶ **Indian benchmark to date:** Kerala, **May 2018** — Kozhikode and Malappuram, **16 deaths among 18 confirmed cases, CFR 88.8%, in 20 days**, including a 31-year-old nurse; most recent Kerala outbreak **September 2023**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam. The working case definition given above is composed from the clinical and laboratory description in Park 28e (Ch 5, pp 349-350); the definition actually in force in an Indian outbreak is the one issued by the state health department and the National Centre for Disease Control, and that is the one a district officer must use. Park 28e dates the first isolation of the virus to 1999, while Sunder Lal (Ch 8, p 475) records the first report as 1998 in Nipah village at Kampung, Malaysia — quote Park's 1999. Park 28e gives the incubation period as 4–14 days, and this is what fixes the 28-day closure rule; operational contact-surveillance periods used in recent Indian Nipah outbreaks, and any fixed hour-targets for notification and specimen despatch, are set by the state protocol in force rather than by the textbooks. Sunder Lal (Ch 8, p 475) records that ribavirin may reduce mortality in Nipah encephalitis and that a monoclonal antibody developed in Australia has been used to save life; Park 28e states flatly that no specific drug or vaccine is available, so the safe examination answer is intensive supportive care with these two named only as investigational.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, WHO Disease Outbreak News, International Health Regulations (2005) as amended, ICMR, NCDC, National One Health Mission, PIB India

Programme and policy updates (India)

- ▶ **IDSP reporting now runs on the Integrated Health Information Platform (IHIP).** IHIP was launched as the next-generation IDSP digital platform in April 2021 and reporting migrated from the weekly IDSP portal to the real-time IHIP portal from 1 October 2021, making notification case-based and real-time rather than weekly and aggregate. The paper 'S', 'P' and 'L' formats survive at field level and remain the right thing to describe in the exam, but the First Information Report, the outbreak line list and the whole outbreak investigation are now transacted electronically on IHIP, and that is the platform a serving DMHO actually uses. [PIB 2021-Apr]
- ▶ **The National One Health Mission is the standing structure for a zoonotic outbreak of exactly this kind.** Approved by the Prime Minister's Science, Technology and Innovation Advisory Council and steered by the Office of the Principal Scientific Adviser with ICMR, it runs on four verticals — pandemic preparedness, integrated disease surveillance, outbreak investigation, and federated cross-sector data — and operates a geographically distributed network of 22 BSL-3 and BSL-4 laboratories accessible for human, livestock and wildlife samples alike. The Union Cabinet approved the National Institute of One Health, Nagpur as the mission's anchor institution in February 2024. This is the named national framework behind the intersectoral coordination the answer describes. [ICMR 2024]
- ▶ **Laboratory confirmation no longer has to leave the district.** ICMR deployed an indigenously developed, bus-mounted mobile BSL-3 laboratory to Kozhikode during the September 2023 Kerala outbreak (6 confirmed cases, 2 deaths), providing on-site Nipah diagnosis and cutting turnaround time to about 4 hours [PMID 41178748]. A network of 17 Virus Research and

Diagnostic Laboratories has since been established for severe acute respiratory illness surveillance in Kerala and West Bengal [PMID 40253778]. The textbook position that there is no district-level confirmation still holds for routine practice, but a mobile high-containment laboratory can now be brought to the epicentre.

Current national data

- **The current run to quote is Kerala 2025 → West Bengal January 2026 → Kozhikode June 2026.** Kerala reported 4 confirmed cases with 2 deaths in 2025 — 2 in Malappuram and 2 in Palakkad, the first Palakkad cases — with 723 contacts listed across five districts as of 17 July 2025 (Palakkad 394, Malappuram 212, Kozhikode 114, Ernakulam 2, Thrissur 1) [WHO 2025-Aug]. West Bengal then reported 2 laboratory-confirmed cases with no deaths at Barasat, North 24 Parganas in January 2026; over 190 contacts including health and care workers were traced, monitored and tested, all negative, and WHO graded the risk moderate at sub-national level and low nationally [WHO 2026-Jan]. Kerala was back within months: a further laboratory-confirmed case in Kozhikode district — an adult male, symptom onset 30 May 2026, hospitalised 10 June 2026 and on ventilatory support — with 104 contacts identified as of 18 June 2026, risk-stratified as 4 very-high-risk, 14 high-risk and 86 low-risk, 45 of them health and care workers, and no secondary case at the time of reporting [WHO 2026-Jun]. India's Nipah map is therefore no longer only Kerala, while Kozhikode remains the most repeatedly affected single district.
- **The Indian case-fatality ladder, and the syndrome that predicts nosocomial spread.** Indian outbreak CFRs are usually quoted as a 33–100% band — Siliguri 2001 (66 cases, 68%), Nadia 2007 (5 cases, 100%), Kozhikode 2023 (6 cases, 33%, the lowest of any Indian outbreak with more than one case), Malappuram 2024 (2 cases, 100%) and Malappuram–Palakkad 2025 (4 cases, 50%) [PMID 42160364]. Quote the band with its exclusion attached: the same review's own outbreak table records Ernakulam 2019 as a single case with a CFR of 0%, a patient who survived on supportive care alone, so 33% is the floor across multi-case outbreaks and not the lowest figure India has recorded. Encephalitis-predominant outbreaks (2019, 2021, 2024) produced no human-to-human transmission at all, whereas the ARDS-predominant outbreaks (2001, 2007, 2018, 2023) did [PMID 40253778] — so a respiratory rather than purely neurological presentation is the signal to escalate infection control immediately.

Elimination, surveillance and disease status

- **Most Nipah cases infect nobody, which is the epidemiological case for tracing rather than blanket restriction.** Modelling of 33 primary and 152 secondary laboratory-confirmed cases drawn from 23 studies in Bangladesh and India found that 92.0% (95% CrI 87.0–95.4) of cases generate no secondary case, and that 5 seeded cases carry only a 1.1% (95% CrI 0.3–2.3) chance of producing an outbreak above 50 cases [PMID 41935612]. The reproduction number in the 2018 Kerala outbreak was about 0.4, and Indian estimates generally sit below 1 but can cross the epidemic threshold in high-contact hospital settings if isolation is delayed [PMID 42160364]. Rapid contact tracing and immediate isolation, not movement bans, are what the numbers support.
- **The reservoir is wider than the two hotspot states, and the spillover route into the index case remains unsettled.** An ICMR-NIV cross-sectional survey of 533 *Pteropus medius* bats from 21 roosts across 8 non-outbreak states detected Nipah RNA in a bat from Uttar Pradesh whose

genome was closely homologous to Kerala sequences, plus anti-Nipah IgG in bats from Goa, Maharashtra and Uttar Pradesh — an argument for extending surveillance well beyond Kerala and West Bengal [PMID 42202672]. In Kerala the index cases have had no traceable bat or sap contact, viral RNA has been recovered from bats 42–55 km from the 2023 index household, proximity to a roost has not tracked risk, and seroprevalence among close contacts was only about 1%, which has led ICMR investigators to propose rare airborne spillover as the missing route [PMID 40844108]. The same evidence is why roost destruction is counterproductive and must be discouraged in risk communication.

Guideline and legislative updates

- **The Indian clinical and containment protocol is now explicit, and it is 21 days at the contact end.** NCDC guidance and the Kerala State Medical Board recommend early compassionate-use remdesivir and favipiravir, with ribavirin for severe disease — all still investigational, none licensed. Containment in the 2023 Kerala outbreak combined Truenat point-of-care testing, ring containment, risk-stratified 21-day quarantine and 1,260–1,288 contacts under surveillance, while the 2018 response monitored over 2,600 contacts for 21 days; designated Nipah intensive care units, powered air-purifying respirators and chemical disinfection protocols were the operational core of infection control [PMID 42160364]. Write the textbook closure rule of twice the maximum incubation period after the last case, but know that the working contact-surveillance period used in Indian outbreaks is 21 days.

Global and WHO developments

- **The amended International Health Regulations (2005) are in force and add a tier above PHEIC.** The Seventy-seventh World Health Assembly adopted the amendment package on 1 June 2024 and the amended text became current on 19 September 2025. It creates a new highest level of global alert, the **pandemic emergency** — a PHEIC caused by a communicable disease that is spreading, or is at high risk of spreading, widely across several countries, is overwhelming or close to overwhelming their health systems, is causing or risks causing major social and economic disruption, and therefore needs fast, fair and stepped-up international action. [WHO 2024-Jun]
- **The first Nipah vaccine efficacy pathway has opened, but monoclonal antibodies may matter more for India.** In a CEPI-funded phase 1 trial, 192 adults were randomised 5:1 to the Hendra soluble-G subunit vaccine HeV-sG-V or placebo; two 100 µg doses 28 days apart produced neutralising geometric mean titres of 1485.6 against a NiV-Bangladesh and 2581.9 against a NiV-Malaysia reporter virus seven days after the second dose, with no serious adverse events, hospitalisations or deaths, while a single dose was not sufficiently immunogenic [PMID 41248666]. Serum Institute of India is manufacturing ChAdOx1 NipahB with the University of Oxford and CEPI, supporting the first phase 2 Nipah vaccine trial in Bangladesh and a reserve of roughly 100,000 doses planned at Pune for rapid South Asian deployment — a stated intention, not a stock already held [PMID 42160364]. CEPI modelling nonetheless finds that under the sporadic, small-outbreak epidemiology India actually faces, a stockpile of 10,000 therapeutic monoclonal-antibody regimens would avert more deaths (49; 95% CI 23–87 over 30 years) than either proactive or reactive vaccination, whose benefit only appears in livestock-amplified or extended scenarios [PMID 42119335]. Park's statement that no vaccine exists remains correct for

the exam; what has changed is that candidates are now in human trials and stockpiling policy is being written.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 3 (Principles of Epidemiology and Epidemiologic Methods), p 113 (definition of epidemic; single case as an outbreak), pp 144-146 (control of communicable disease along the three links; quarantine replaced by active surveillance), pp 157-160 (disinfection, personal protective equipment and mask technique; investigation of an epidemic; Table 47) and Fig 4, p 84; Ch 5 (Epidemiology of Communicable Diseases), pp 349-350 (Nipah virus infection); Ch 7 (Health Programmes in India), p 568 (IDSP and rapid response team); Ch 15 (Environment and Health), p 942 (control of zoonoses). Sunder Lal — Textbook of Community Medicine, Ch 7 (Epidemiology), pp 307-311 (IDSP weekly 'S', 'P', 'L' formats, outbreak triggers, principles and measures to control an epidemic); Ch 8, p 410 (Epidemic Diseases Act 1897 and its proposed replacement) and pp 475-476 (Nipah, including the Kerala 2018 outbreak); Ch 13 (Disaster Preparedness and Management), p 669 (District Disaster Management Authority); Ch 20 (Recent Advances in Community Medicine), p 787 (weak district epidemiological units, involvement of private providers). Jugal Kishore — National Health Programmes of India, Ch 1, p 491 (IDSP surveillance trigger levels) and p 669 (national laboratory capacity at NIV Pune and NCDC Delhi for emerging viral disease). Nelson & Williams — Infectious Disease Epidemiology, Ch 13 (Emerging and New Infectious Diseases), pp 375-376 (Malaysian and Bangladeshi outbreaks, pig culling, date-palm sap). Gordis Epidemiology 7e, Ch 2, p 53 (Box 2.2, steps in investigating an acute outbreak).

Recent-advances sources: 42160364 — The Nipah virus threat in India: epidemiological trends, risk drivers, and lessons for pandemic preparedness, PLoS Pathog 2026;22(5):e1014226, doi:[10.1371/journal.ppat.1014226](https://doi.org/10.1371/journal.ppat.1014226); 40844108 — Nipah virus infection in humans in Kerala, India: hypothesis of air-borne transmission, Indian J Med Res 2025;161(5):567-571, doi:[10.25259/IJMR_2180_2024](https://doi.org/10.25259/IJMR_2180_2024); 41248666 — Safety and immunogenicity of a Nipah virus vaccine (HeV-sG-V) in adults: a single-centre, randomised, observer-blind, placebo-controlled, phase 1 study, Lancet 2025;406(10521):2792-2803, doi:[10.1016/S0140-6736\(25\)01390-X](https://doi.org/10.1016/S0140-6736(25)01390-X) (abstract-only); 41935612 — Superspreading potential of Nipah virus disease despite self-limited human-to-human transmission in community, Int J Infect Dis 2026;168:108683, doi:[10.1016/j.ijid.2026.108683](https://doi.org/10.1016/j.ijid.2026.108683) (abstract-only); 42202672 — Survey on the prevalence of Nipah virus in Pteropus medius bats from non-outbreak regions of India, J Infect Public Health 2026;19(7):103273, doi:[10.1016/j.jiph.2026.103273](https://doi.org/10.1016/j.jiph.2026.103273) (abstract-only); 42119335 — Modeling optimal deployment strategies for Nipah vaccines and monoclonal antibodies, Vaccine 2026;84:128688, doi:[10.1016/j.vaccine.2026.128688](https://doi.org/10.1016/j.vaccine.2026.128688) (abstract-only); 41178748 — Rapidly deployable mobile BSL-3 laboratory: a response to the Nipah virus outbreak in Kozhikode, Kerala, India, 2023, Pathog Glob Health 2025;120(1):18-27, doi:[10.1080/20477724.2025.2582036](https://doi.org/10.1080/20477724.2025.2582036) (abstract-only); 40253778 — Encephalitis-predominant Nipah virus outbreaks in Kerala, India during 2024, J Infect Public Health 2025;18(7):102782, doi:[10.1016/j.jiph.2025.102782](https://doi.org/10.1016/j.jiph.2025.102782) (abstract-only). Non-journal sources: WHO Disease Outbreak News, Nipah virus infection — India (Kerala), 6 August 2025; WHO Disease Outbreak News, Nipah virus disease — India (West Bengal), 30 January 2026; WHO Disease Outbreak News 2026-DON609, Nipah virus infection — India (Kerala), 25 June 2026 (Kozhikode case, symptom onset 30 May 2026; 104 contacts risk-

stratified as of 18 June 2026); WHO, International Health Regulations (2005) amendments adopted by the Seventy-seventh World Health Assembly on 1 June 2024, amended text current from 19 September 2025; National One Health Mission, Office of the Principal Scientific Adviser and ICMR — National Institute of One Health, Nagpur approved February 2024; PIB India, launch of the Integrated Health Information Platform as the next-generation IDSP digital platform, April 2021.



Cervical cancer screening programme

Asked in: Paper IV 04-Aug-2021 — RGUHS MD Community Medicine

Definition — Screening is "the search for unrecognized disease or defect by means of rapidly applied tests, examinations or other procedures in apparently healthy individuals" (Park 28e, Ch 4, p 163). Planning such a programme is a full run of the **planning cycle** — situation analysis, objectives, resources, priorities, written plan, implementation, monitoring and evaluation (Park 28e, Ch 23, pp 1026-1027) — with a guaranteed diagnosis-and-treatment linkage for every woman who screens positive.

1. Justifying screening for cancer cervix in this village

Pre-planning needs political will translated into legislation, an organisation for planning, and administrative capacity (Park 28e, Ch 23, p 1026) — all three already exist through the National Programme for Prevention and Control of Non-Communicable Diseases. The first step is to decide whether screening is worthwhile at all, which needs **ethical, scientific and financial justification** (Park 28e, Ch 4, p 165):

Criterion for the disease (Park 28e, Ch 4, p 165)	Position for cancer cervix in a village
1. Important health problem, prevalence high	Second commonest cancer in Indian women, after breast — the top three by cases in females are breast, cervix uteri and ovary, while lip and oral cavity leads in men and is only 4th in women (Park 28e, Ch 6, p 457). In India cancer cervix is 9.0% of all cancer incidence among women , age-standardised incidence rate 17.7 per 100,000 with 79,979 estimated deaths, 2022 (Park 28e, Ch 6, pp 457, 461). World context: 4th commonest cancer in women, ~6,60,000 new cases and ~3,50,000 deaths in 2022, 94% of deaths in low- and middle-income countries (Park 28e, Ch 6, p 461)
2. Recognisable latent or early asymptomatic stage	Prolonged pre-invasive phase — carcinoma-in-situ detectable long before symptoms (Park 28e, Ch 6, p 460)
3. Natural history adequately understood	Human papillomavirus (HPV) found in >95% of cancers and accepted as a necessary cause ; takes 15-20 years to progress in a normal immune system, 5-10 years if immunity is weakened, e.g. untreated HIV (Park 28e, Ch 6, p 462)
4. A test that detects disease before signs and symptoms	Pap (Papanicolaou) smear, visual inspection with acetic acid (VIA), HPV-DNA / HPV-mRNA test (Park 28e, Ch 6, pp 460-461)

Criterion for the disease (Park 28e, Ch 4, p 165)	Position for cancer cervix in a village
5. Facilities for confirmation of diagnosis	Colposcopy, Pap smear and biopsy at the district hospital (Park 28e, Ch 6, p 460) — must be verified to exist before the programme starts
6. Effective treatment available	Ablative/excisional treatment of pre-cancer; surgery, radiotherapy, chemotherapy and palliative care for invasive cancer (Park 28e, Ch 6, p 462)
7. Agreed policy on whom to treat	Written cut-off for a positive test and a written treatment protocol issued before screening begins
8. Evidence that early detection reduces morbidity and mortality	Organised population-based programmes have reduced incidence and mortality in many developed countries (Park 28e, Ch 6, p 460)
9. Benefits exceed risks and costs	Judged against the cost of false positives, referral load and the district hospital's capacity

2. Situation analysis and setting objectives (planning cycle steps 1-2)

- **Collect the minimum essential planning data** for the PHC area (Park 28e, Ch 23, p 1027): (a) population with age and sex structure; (b) morbidity and mortality statistics; (c) epidemiology and geographical distribution of disease; (d) medical care facilities, public and private; (e) technical manpower of each category; (f) training facilities available; (g) **attitudes and beliefs of the population towards the disease, its cure and prevention** — decisive here, because tests involving vaginal examination are by themselves poorly acceptable in mass campaigns (Park 28e, Ch 4, pp 165-166).
- **Build the denominator by enumeration, not by assumption.** A rural PHC serves 30,000 in the plains and 20,000 in hilly/tribal areas; each sub centre covers 5,000 (3,000 in hilly, tribal or desert areas) and each accredited social health activist (ASHA) 1,000 (Park 28e, Ch 10, pp 725, 731-732; tribal norms Ch 13, p 850). The first community-level task is **active enumeration of the eligible population and registration of families** — the ASHA line-lists every woman aged 30 years and above, habitation by habitation, on an individual health card (AFMC Red Book 2024, Ch 38, Table 38.13, p 1204; Sunder Lal, Ch 8, p 503); about a third of the population is aged 30 years and above (Sunder Lal, Ch 16, p 703).
- **Risk-assess separately, using the Community Based Assessment Checklist (CBAC)** — a distinct second community-level task (AFMC Red Book 2024, Ch 38, Table 38.13, p 1204). CBAC carries **six variables** — **age, family history of non-communicable disease, waist circumference, physical inactivity, tobacco use and alcohol use**; a score above 4 marks the person **high-risk** and gives priority for screening (Sunder Lal, Ch 16, p 703; Ch 8, p 503).
- **Decide the type of screening** — mass, high-risk (selective) or multiphasic (Park 28e, Ch 4, p 165). Since cancer cervix occurs relatively less often in upper social groups, **selective screening of the lower socio-economic groups raises the yield of new cases**.
- **Write measurable objectives** with a numerator, a denominator and a deadline — line-listing, first-round coverage, and time to complete diagnostic evaluation of screen positives — since evaluation measures the degree to which objectives and targets are fulfilled (Park 28e, Ch 23, p 1027).

3. Choosing the screening test and the screening policy

The test must satisfy **acceptability, repeatability and validity**, besides yield, simplicity, safety, rapidity, ease of administration and cost; tests that best fulfil one condition are often least likely to fulfil another, so the choice is always a compromise (Park 28e, Ch 4, pp 165-166).

Test	Strengths	Limitations	Fit for a village PHC
Pap smear (cytology)	Detects prolonged pre-invasive phase; long track record	Needs excessive resources — laboratories, equipment and trained personnel; repeatability poor (a smear from the same woman may be normal one day and abnormal on another; Park 28e, Ch 4, p 166)	Only where a cytology laboratory and cytotechnicians genuinely exist
VIA (5% acetic acid)	Sensitivity similar to cytology ; easy to carry out; health workers easy to train; result immediate; 5 ml glacial acetic acid in 95 ml distilled water (Park 28e, Ch 6, p 460)	Subjective — subject to inter-observer variation; needs re-training and supervision	The workable field test ; ASHA, ANM and mid-level provider are trained in it under the national programme (Sunder Lal, Ch 8, p 503)
HPV-DNA / HPV-mRNA test	Objective ; simpler; prevents more pre-cancers and cancers; saves more lives; more cost-effective than visual inspection or cytology (Park 28e, Ch 6, pp 460-461)	Needs sample transport and laboratory linkage; HPV-positive women may need a triage test	The preferred test where the laboratory link can be assured

Screening policy fixed in the plan (Park 28e, Ch 6, pp 459-462):

- ▶ **Age of entry** — 30 years in the general population of women; women living with HIV from 25 years.
- ▶ **Interval** — a validated HPV test every 5 to 10 years; women living with HIV every 3 to 5 years; the older cytology policy is a Pap test at 30 years and every 3 years thereafter.
- ▶ **Screening must be linked to treatment** — a screening test without an assured treatment pathway is not a programme.
- ▶ HPV vaccination of girls aged 9-14 years, before sexual debut, runs alongside but **does not** replace screening.

4. Assessment of resources, priorities and the written plan (planning cycle steps 3-5)

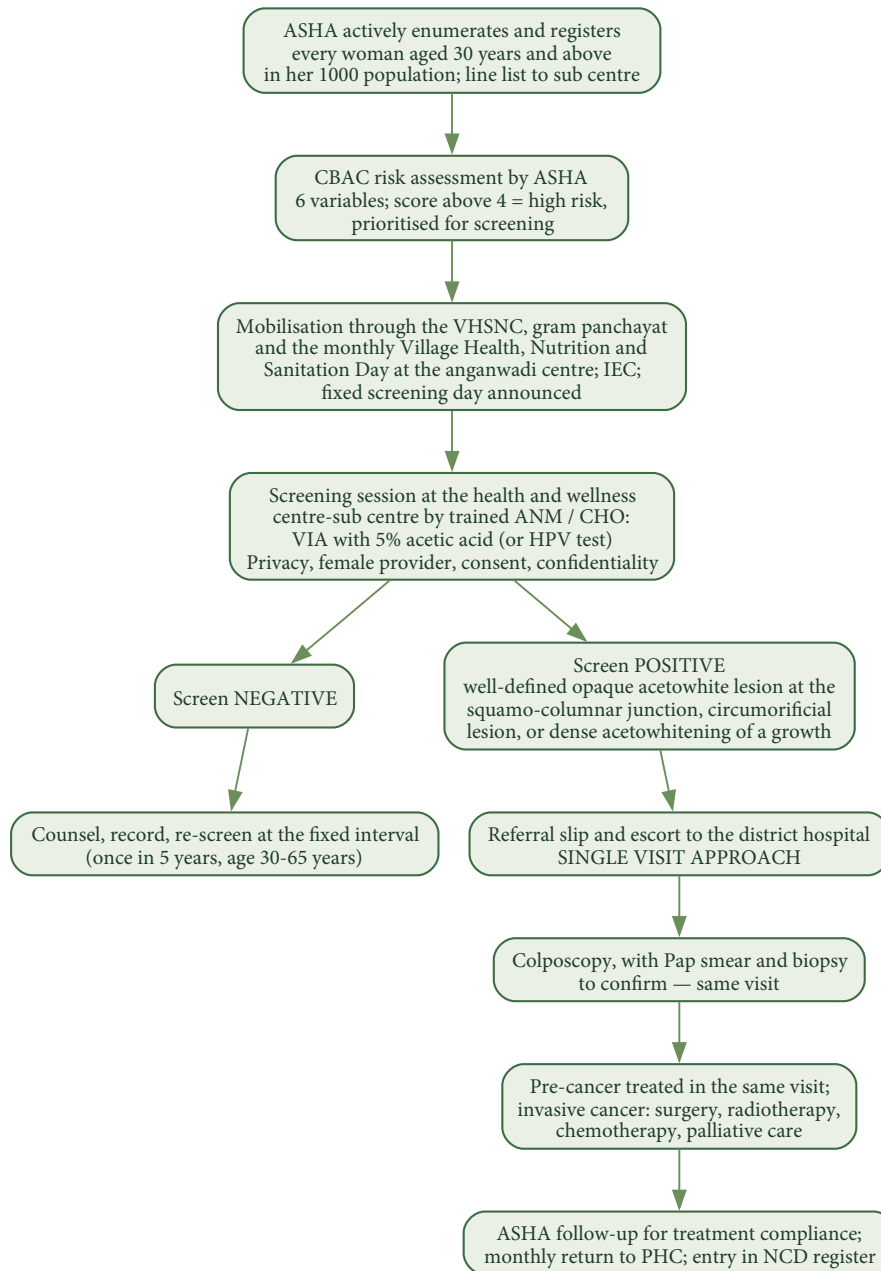
Resources means the **manpower, money, materials, skills, knowledge and techniques** needed, balanced against what is actually available (Park 28e, Ch 23, p 1027).

Resource head	What the medical officer must arrange
Manpower	ASHA (enumeration, CBAC, mobilisation, follow-up); auxiliary nurse midwife (ANM) and community health officer (CHO) as screeners; staff nurse and medical officer at the PHC; named gynaecologist/pathologist at the district hospital

Resource head	What the medical officer must arrange
Training	Hands-on VIA training with a written positivity definition; observer error is reduced by standardisation of procedures, intensive training of observers, and independent assessment by two or more observers (Park 28e, Ch 4, pp 165-166)
Materials	Speculum sets and sterilisation, examination couch with privacy screen and good light source, freshly prepared 5% acetic acid, gloves, referral slips, register and NCD/electronic health record entry
Money and logistics	Budget for consumables, transport of screen-positive women, camp costs, printing of information-education-communication (IEC) material
Community mobilisation	The village health, sanitation and nutrition committee (VHSNC) — a standing subcommittee of the panchayat, minimum 15 members, chaired by a <i>mahila panch</i> with the ASHA as member secretary, and including 5 elected gram panchayat members, the ANM, the anganwadi worker, a school teacher and self-help group volunteers (Sunder Lal, Ch 1, p 42); the monthly Village Health, Nutrition and Sanitation Day at the anganwadi centre as the fixed community platform (Sunder Lal, Ch 17, p 744); plus intensive IEC to sensitise the community to the disease and to early detection (Park 28e, Ch 6, p 460)
Fixing priorities	Where resources fall short, priority follows financial constraints, mortality and morbidity data, diseases preventable at low cost, and community and political interest (Park 28e, Ch 23, p 1027)

The **written plan** then states what will be done, when, and by whom, area by area, with a **built-in system of evaluation** (Park 28e, Ch 23, p 1027).

5. Implementation — the village screening pathway



Village-level cervical cancer screening pathway planned from a primary health centre

The plan must sit inside the existing **National Programme for Prevention and Control of Non-Communicable Diseases (NP-NCD)**, whose package of services already assigns the tasks:

Level	Assigned task (AFMC Red Book 2024, Ch 38, Table 38.13, pp 1203-1205; Sunder Lal, Ch 8, pp 502-503, 506)
Community (ASHA)	(a) Active enumeration of the eligible population and registration of families; (b) NCD risk assessment using CBAC; (c) mobilisation for screening at the health and wellness centre; (d) follow-up for treatment compliance

Level	Assigned task (AFMC Red Book 2024, Ch 38, Table 38.13, pp 1203-1205; Sunder Lal, Ch 8, pp 502-503, 506)
Sub centre / health and wellness centre	Health education; screening of all persons aged 30 years and above for hypertension, diabetes and the three common cancers — oral, breast and cervical. Frequency: once a year for diabetes and hypertension; once in five years for the three cancers, in persons aged 30-65 years (Sunder Lal, Ch 8, pp 502-503; Ch 16, p 703). Referral of suspects; electronic health record entry
PHC	Screening of outpatient attendees, including cervix by VIA; confirmation of diagnosis , treatment initiation, bi-directional referral linkage (Park 28e, Ch 7, p 565)
CHC / district hospital	Diagnosis and management; specialist confirmation and treatment of cancer and pre-cancer; chemotherapy follow-up and palliative care

6. Monitoring and evaluation

Indicator	Formula	What it detects
Coverage (participation) rate	$\text{Women screened} \div \text{eligible women enumerated} \times 100$	Whether mobilisation and acceptability worked
Screen-positivity rate	$\text{Screen positives} \div \text{women screened} \times 100$	Drift in VIA reading; training need
Referral compliance rate	$\text{Screen positives who reached the district hospital} \div \text{all screen positives} \times 100$	The weakest link in most village programmes
Positive predictive value	$\text{True positives (pre-cancer or cancer)} \div \text{all screen positives} \times 100$	Falls as disease prevalence falls, so a low-prevalence village generates mostly false positives (Park 28e, Ch 4, p 168; Gordis 7e, Ch 5, pp 123-124)
Yield	Amount of previously unrecognised disease diagnosed as a result of the screening effort (Park 28e, Ch 4, p 168)	Depends on sensitivity, prevalence and participation
Treatment completion rate	$\text{Treated} \div \text{diagnosed} \times 100$	Whether screening was truly linked to treatment
Impact indicators	Stage distribution at diagnosis; age-standardised incidence and mortality from cancer cervix (India 2022 baseline 17.7 per 100,000)	Needs a population-based cancer registry — registration is a <i>sine qua non</i> for any cancer control programme and the base for assessing magnitude and planning services (Park 28e, Ch 6, pp 459-460)

Guard against these when claiming the programme worked (Gordis 7e, Ch 18, pp 368, 373, 391):

- **Referral (volunteer) bias** — volunteers are healthier and more adherent than non-participants.
- **Length-biased sampling** — screening preferentially picks up slow-growing, better-prognosis lesions.

- **Lead time bias** — earlier diagnosis lengthens measured survival without postponing death, so judge on disease-specific mortality, not 5-year survival.
- **Overdiagnosis bias** — lesions detected that would never have surfaced clinically.
- Only a **randomised controlled trial** settles these selection biases (Oxford Textbook of Public Health 7e, Ch 98, p 3720); false positives burden diagnostic facilities and discredit screening (Park 28e, Ch 4, p 168).

Diagrams

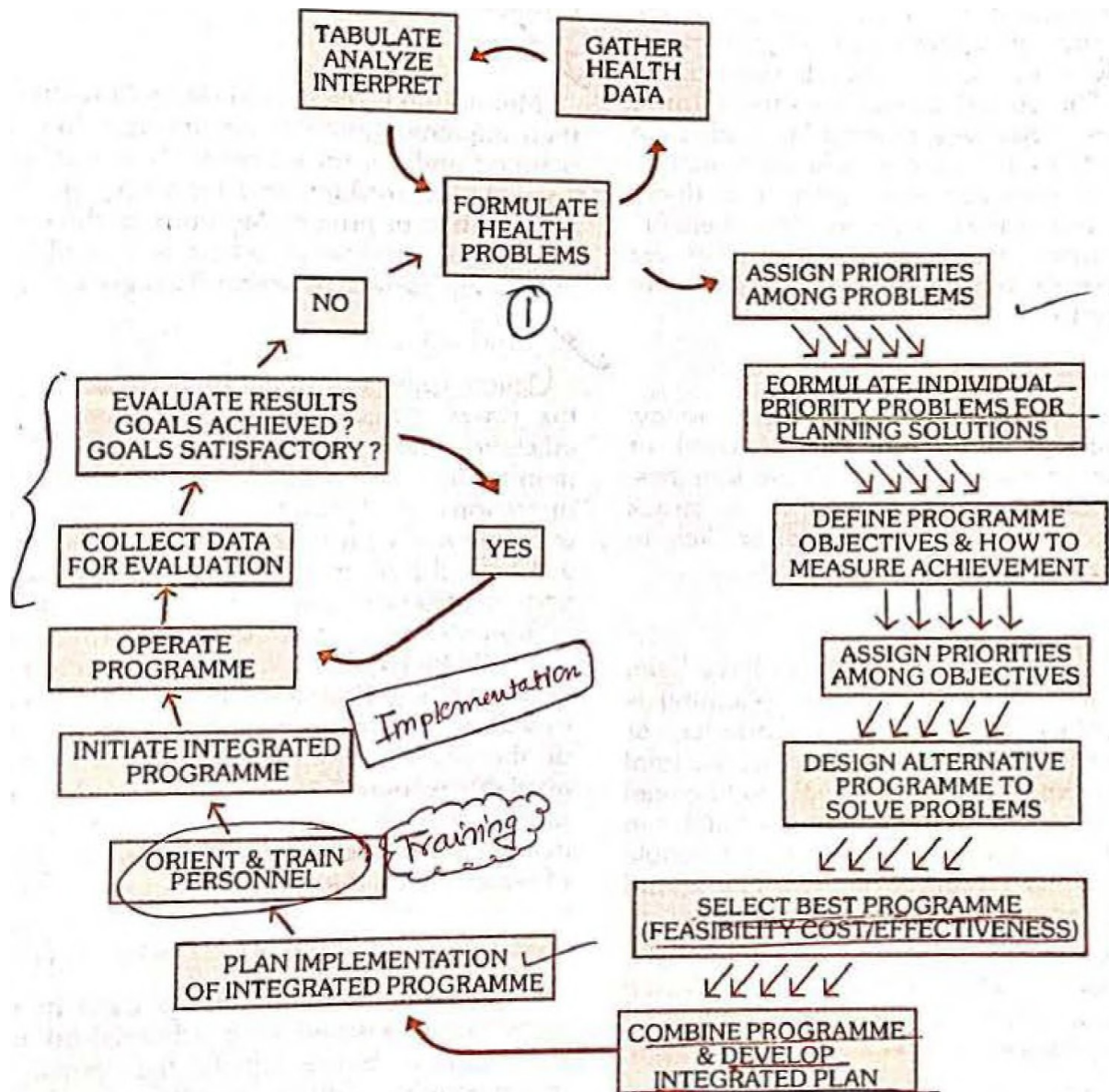


Fig 1 — The planning cycle (Park's Textbook of Preventive and Social Medicine, 28e, p.1026)

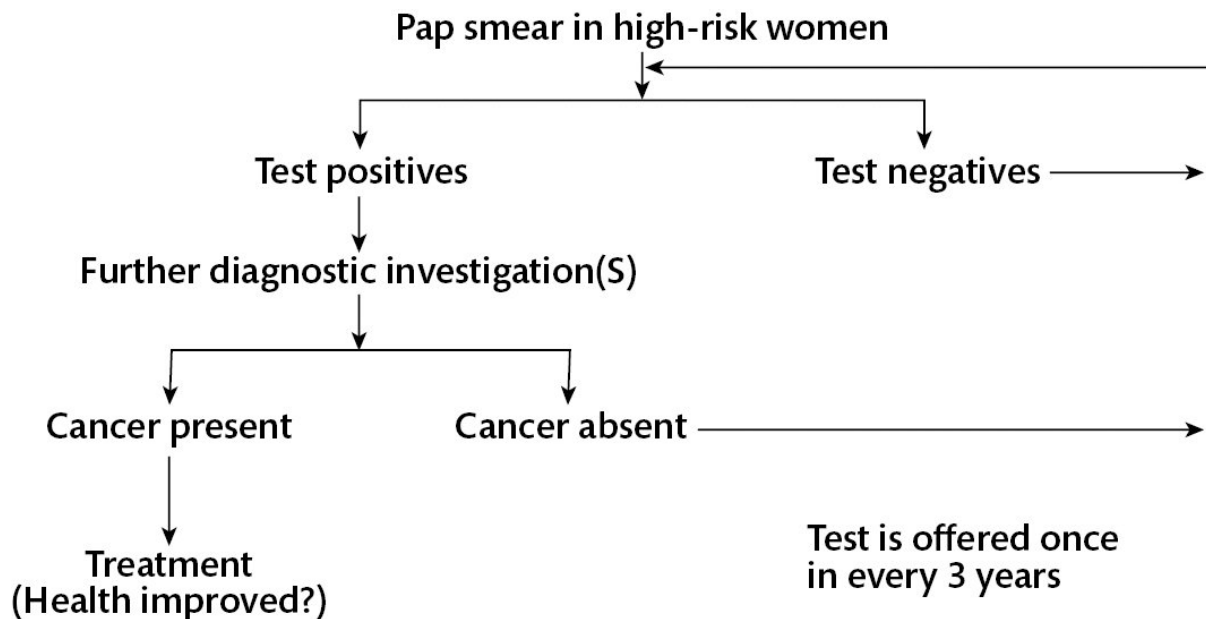


Fig 11.5 — 1 Participants flow and components of a screening programme: Pap smear for screening cervical cancer (Oxford public health 7th ed, p.3699)

High-yield exam pointers

- ▶ Criteria for screening = DISEASE (9 points) + TEST (acceptability, repeatability, validity, plus yield, simplicity, safety, rapidity, ease, cost); repeatability fails from observer variation (intra- and inter-observer), biological/subject variation and technical error.
- ▶ Indian burden to quote: cancer cervix = second commonest cancer in Indian women after breast; 9.0% of all cancer incidence in women, age-standardised incidence rate 17.7 per 100,000, 79,979 deaths (2022); globally 94% of cervical cancer deaths are in low- and middle-income countries.
- ▶ Policy numbers: entry at 30 years (25 if living with HIV); HPV test every 5-10 years (3-5 years if living with HIV); Pap every 3 years; HPV vaccine at 9-14 years; NP-NCD field frequency — cancers once in 5 years, ages 30-65; diabetes and hypertension yearly.
- ▶ CBAC = 6 variables (age, family history of NCD, waist circumference, physical inactivity, tobacco, alcohol); score >4 = high risk, prioritised for screening.
- ▶ VIA = 5 ml glacial acetic acid + 95 ml distilled water (5%); positive = well-defined opaque acetowhite lesion near the squamo-columnar junction; then the single visit approach.
- ▶ Yield = previously unrecognised disease diagnosed by the screening effort; depends on sensitivity, prevalence, participation. Positive predictive value falls as prevalence falls — so screen a high-risk group. Four screening biases: referral/volunteer, length, lead time, overdiagnosis — judge on mortality, never 5-year survival.
- ▶ Planning cycle: situation analysis → objectives and goals → resources → priorities → written plan → implementation → monitoring → evaluation.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Pap 28e prints the older cytology policy (Pap smear at 30 years, then every 3 years) beside the updated World Health Organization position of a validated human papillomavirus test every 5 to 10 years from 30 years, while the test actually deployed in the field under the National Programme for Prevention and Control of Non-Communicable Diseases remains visual inspection with acetic acid — state both and name the one your district uses.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, ICMR, NFHS-5, IARC, WHO Cervical Cancer Elimination Initiative

Programme and policy updates (India)

- ▶ **HPV vaccination has entered the national immunisation programme.** A free nationwide campaign was launched on 28 February 2026, targeting about **1.15-1.2 crore girls aged 14 years**, delivered through government facilities and schools as a **voluntary, free single dose of the quadrivalent vaccine**; girls who turned 15 within three months of launch are also eligible. Before this, only two states — **Punjab and Sikkim** — ran school-based HPV vaccination, from 2016, reaching roughly **97% of target girls** [PMID 42544634] [PMID 42518306]. For the answer this means vaccination and screening are now two arms of one national plan, not a vaccine "running alongside".
- ▶ **The policy agenda now names two specific asks** — putting HPV vaccination formally inside the Universal Immunization Programme and adding the **HPV nucleic acid amplification test to the National Essential Diagnostics List** — while repurposing the COVID-era RT-PCR laboratory network for HPV molecular testing. The starting point quoted is stark: India records roughly **127,000 new cases and 80,000 deaths a year, about one-fifth of the global burden**, with **under 2% of women ever screened and under 1% vaccinated**; the indigenous quadrivalent vaccine Cervavac, introduced in 2023, is the new lever [PMID 41645272] (abstract-only).
- ▶ **HPV testing is becoming deliverable at village level.** In a four-setting Maharashtra study (urban non-slum, urban slum, rural, tribal), self-collected sampling was accepted by **98.2% of enrolled women** after art-based health education, though initial participation was only 66%; oncogenic HPV prevalence was **4.4-11.5% (overall 5.6%), highest in tribal women**, and 87.6% of women took the sample at home. Referral compliance, however, split sharply by distance from the tertiary centre — 81.8% urban non-slum and 89.3% urban slum, but 50% rural (63 km) and **0% tribal (139 km)**; running community triage camps with Pap and VIA lifted these to 66.7% and 88.9%, and treatment compliance among women with proven pre-cancer was 100% [PMID 41715026].
- ▶ **An Indian point-of-care HPV assay has now passed formal validation.** Truenat HR-HPV-Plus (8 types) reached sensitivity **80.4% (95% CI 71.1-87.8)** and specificity **91.5% (95% CI 89.5-93.2)** for CIN2+ and met IARC validation criteria, whereas the 7-type PathoDetect assay (sensitivity 68.1%) and two 14-type assays did not — a near-patient molecular option for a PHC that already runs Truenat for TB [PMID 42252568] (abstract-only).

Current national data

- **Cancer screening coverage in India remains below 2%.** Participation among women is **1.9% for cervical cancer**, 0.9% for breast and 0.9% for oral cavity (oral cavity in men 1.2%) — this after more than a decade of NP-NCD. NFHS-5 (2019-21) figures for seven purposively selected districts (Bathinda, Shimla, South Delhi, Dibrugarh, Palghar, Dakshina Kannada, Tiruvannamalai) show ever-screening for the three cancers ranging from **0% to 8%**, and an ICMR-led implementation study running to 2027 is testing a package there to raise coverage, screening quality and speed of treatment initiation. The system barriers named are a shortage of trained staff, multitasking, weak data management, interrupted supply of logistics and inadequate referral linkage [PMID 42285578].
- **The Tamil Nadu STEPS survey (Dec 2023-Mar 2024; 4,184 women aged 30-69) quantifies exactly where a village programme leaks.** Lifetime screening coverage was 10% — **the highest of any Indian state**, against a national ~2%. Of 423 women ever screened, 108 (26%) were screen-positive, but only 62 (15%) were referred for colposcopy, 56 (13%) actually underwent it, 38 (9%) had a biopsy and 1 (0.2%) was diagnosed with cancer. Step by step, only **57% of screen-positive women were referred**, while 94% of those referred did attend — so **referral by the provider, not default by the patient, is the single largest bottleneck** [PMID 41888339] — direct evidence for the referral-compliance indicator in the monitoring table.

Elimination, surveillance and disease status

- **The elimination target is a rate, not a date: incidence below 4 cases per 100,000 women-years.** The WHO Cervical Cancer Elimination Initiative baseline recorded **604,127 cases and 341,831 deaths globally in 2020** (age-standardised incidence 13.3, mortality 7.2 per 100,000 women-years), with **India carrying 21% of cases and 23% of deaths** and more than 58% of all cases occurring in Asia. India's own age-standardised incidence, **18.0 per 100,000 women-years**, sits far above the elimination threshold [PMID 36528031].
- **Girls-only vaccination alone may not get India below the threshold.** The EpiMetHeos agent-based model, calibrated to Indian data, projected that girls-only vaccination at 60% coverage would cut age-standardised incidence from 11 to **4.7 per 100,000 woman-years — still above the threshold** — whereas gender-neutral vaccination at the same 60% coverage reached **2.8, below it** (2.4 at 90% coverage). Gender-neutral cover was also far more resilient to interruption: after a 5-year disruption, cases still prevented per 100,000 girls born were 302 versus 107, a 2.8-fold gain [PMID 37486822].

Guideline and legislative updates

- **The single-dose schedule is now evidence-backed from Indian data.** The IARC multicentre India cohort, at a median 12 years after vaccination, found efficacy against persistent HPV 16/18 infection of **92.0% (95% CI 87.0-95.0) after one dose** in 3,022 recipients, statistically indistinguishable from two doses (94.8%) and three doses (95.3%). Crucially there was **no HPV 16/18-associated high-grade precancer in vaccinated girls** — 4 CIN2/CIN3 lesions did occur in vaccinated women, none of them HPV 16/18-related, versus **12 CIN2/CIN3 lesions plus 1 invasive squamous cell carcinoma in unvaccinated women** (8 of those lesions HPV 16/18-related) [PMID 39529521]. This is the evidence base for WHO's 2022 position permitting a one- or two-dose schedule [PMID 39529520] (abstract-only).

- **Single-dose vaccination is cost-effective for India at state level, not just nationally.** Modelling on Indian demographic, epidemiological and cost data put the incremental cost-effectiveness ratio for single-dose vaccination of 10-year-old girls at 90% coverage at **US\$406 per DALY averted** versus no vaccination (state range **US\$67-593**), below an opportunity-cost threshold of 30% of GDP per capita in every state; two-dose vaccination cost US\$1,404 per DALY averted, and the annual outlay for the single-dose programme is under a tenth of the existing universal childhood immunisation programme [PMID 37931940].

Global and WHO developments

- **Ten of the eleven WHO South-East Asia Region countries have now introduced HPV vaccination nationally** — all except DPR Korea — with Bhutan (first in the region, 2010), Sri Lanka and Thailand above 90% coverage, while India, launching only in 2026, is still at the start. The recurring barriers are policy and financing constraints, sociocultural resistance, weak health-system capacity and vaccine hesitancy fed by misinformation; the WHO Global Strategy targets remain **90% vaccination, 70% screening and 90% treatment coverage** [PMID 42518306].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 4 (Screening for Disease), Ch 6, Ch 7, Ch 10, Ch 13, Ch 23 (Health Planning and Management); Gordis Epidemiology 7e — Ch 5, Ch 18 (Evaluating Screening Programs); Oxford Textbook of Public Health 7e — Ch 98 (Medical screening), including the participant-flow figure; Sunder Lal — Textbook of Community Medicine, Ch 1, Ch 8, Ch 16, Ch 17; AFMC Red Book of Community Medicine 2024 — Ch 38, Table 38.13.

Recent-advances sources: 42544634 — Addressing HPV vaccine hesitancy and misinformation in India through public health communication strategies, Hum Vaccin Immunother 2026, doi:[10.1080/21645515.2026.2711551](https://doi.org/10.1080/21645515.2026.2711551) (full text verified); 42518306 — HPV vaccination in the WHO South-East Asia Region: a scoping review of uptake, implementation challenges, and pathways to acceleration, Hum Vaccin Immunother 2026, doi:[10.1080/21645515.2026.2693359](https://doi.org/10.1080/21645515.2026.2693359) (full text verified); 41645272 — Cervical cancer elimination in India: repurposing diagnostics, vaccination, and accelerating policy for the 2030 target, Cancer 2026, doi:[10.1002/cncr.70292](https://doi.org/10.1002/cncr.70292) (abstract-only, no PMC record); 41715026 — Experiences with HPV self-sampling in India: advancing cervical cancer prevention and screening, BMC Public Health 2026, doi:[10.1186/s12889-026-26212-y](https://doi.org/10.1186/s12889-026-26212-y) (full text verified; the HPV prevalence range is taken from the Results text and Table 3, 4.4-11.5%, in preference to the article's own abstract, which prints 11.2%); 42252568 — Clinical validation of four point-of-care high-risk HPV assays, including two reduced-valency assays, for cervical cancer screening in low-resource settings, Int J Cancer 2026, doi:[10.1002/ijc.70534](https://doi.org/10.1002/ijc.70534) (abstract-only, no PMC record); 42285578 — Enhancing screening, early diagnosis and treatment initiation of oral, breast and cervical cancer in selected districts of India: an implementation research protocol, BMJ Open 2026, doi:[10.1136/bmjopen-2025-112428](https://doi.org/10.1136/bmjopen-2025-112428) (full text verified); 41888339 — Cervical cancer screening coverage and care cascade in Tamil Nadu, India, based on a community-based cross-sectional survey, Sci Rep 2026, doi:[10.1038/s41598-026-45005-2](https://doi.org/10.1038/s41598-026-45005-2) (full text verified); 36528031 — Global estimates of incidence and mortality of cervical cancer in 2020: a baseline analysis of the WHO Global Cervical Cancer Elimination Initiative, Lancet Glob Health 2022, doi:[10.1016/S2214-109X\(22\)00501-0](https://doi.org/10.1016/S2214-109X(22)00501-0) (full text verified); 37486822 — Building resilient cervical cancer prevention through gender-neutral HPV vaccination, eLife 2023, doi:[10.7554/eLife.85735](https://doi.org/10.7554/eLife.85735) (full text verified); 39529521 — A prospective cohort study comparing efficacy of 1 dose of quadrivalent human papillomavirus vaccine to 2 and 3 doses at an average follow up of 12 years postvaccination, J Natl Cancer Inst Monogr 2024, doi:[10.1093/jncimonographs/lgae042](https://doi.org/10.1093/jncimonographs/lgae042) (full text verified, including Table 6 and the CIN 2/CIN 3 counts); 39529520 — Single-dose human papillomavirus vaccination: an update, J Natl Cancer Inst Monogr 2024, doi:[10.1093/jncimonographs/lgae030](https://doi.org/10.1093/jncimonographs/lgae030) (abstract-only, PMC record carries no full text); 37931940 — Health and economic effects of introducing single-dose or two-dose human papillomavirus vaccination in India, BMJ Glob Health 2023, doi:[10.1136/bmjgh-2023-012580](https://doi.org/10.1136/bmjgh-2023-012580) (full text verified).

♦ ♦ ♦

Cancer screening

Asked in: Paper II 26-Sep-2025 — RGUHS MD Community Medicine

Definition — Screening is "the search for unrecognized disease or defect by means of rapidly applied tests, examinations or other procedures in apparently healthy individuals" (Park 28e, Ch 4, p 163).

Cancer screening is correspondingly the "search for unrecognized malignancy by means of rapidly applied tests", aimed at detecting cancer at a pre-invasive (in situ) or pre-malignant stage (Park 28e, Ch 6, pp 459-460). It is an act of **secondary prevention** and is the main weapon for early detection of cancer; it is an initial examination only, and every positive must be referred for diagnostic work-up.

Why cancer is amenable to screening

- ▶ In many instances malignant disease is preceded for **months or years by a pre-malignant lesion**, whose removal prevents subsequent development of cancer (Park 28e, Ch 6, pp 459-460).
- ▶ **Most cancers begin as localized lesions**, and if found at this stage a high rate of cure is obtainable.
- ▶ A substantial proportion of cancers arise at **body sites accessible to clinical examination** — oral cavity, breast and uterine cervix — so effective screening programmes have been developed for exactly these three (Park 28e, Ch 6, p 459).
- ▶ There is a usable **lead time** — the advantage gained by screening, i.e. the interval between diagnosis by early detection and diagnosis by other means (Park 28e, Ch 4, p 164). In oral cancer, pre-cancerous lesions are detectable for **up to 15 years** before invasive change; in cervical cancer it takes **15-20 years** for cancer to develop in an immunocompetent woman, and only **5-10 years** in the immunosuppressed (Park 28e, Ch 6, pp 461-462).

Criteria to be satisfied before starting a cancer screening programme

Criteria rest on two considerations — the **disease** and the **test** (Park 28e, Ch 4, pp 165-166).

#	Disease criteria	Applied to cancer
1	Important health problem (prevalence high)	Cancer of cervix, breast, oral cavity — leading cancers in India
2	Recognizable latent or early asymptomatic stage	Carcinoma in situ; leukoplakia/erythroplakia
3	Natural history adequately understood	Point at which the process ceases to be reversible must be known
4	A test can detect it before signs and symptoms appear	Cytology, human papillomavirus (HPV) test, mammography
5	Facilities available to confirm the diagnosis	Colposcopy, biopsy, histopathology
6	Effective treatment exists	Surgery, radiotherapy, cryotherapy
7	Agreed policy on whom to treat as a patient	Which grade of pre-cancer is treated
8	Good evidence that early detection and treatment reduce morbidity and mortality	Established for cervix; absent for lung
9	Expected benefits exceed the risks and costs	Radiation, false positives, anxiety

Test criteria	What it means
Acceptability	High cooperation is essential; painful, discomforting or embarrassing tests (rectal, vaginal examination) are poorly accepted in mass campaigns
Repeatability (reliability)	Same result on repeat testing — observer variation, biological variation and test error must be small
Validity (accuracy)	Ability of the test to separate those who have the disease from those who do not — sensitivity and specificity
Others	Yield, simplicity, safety, rapidity, ease of administration and cost; the choice is usually a compromise, since the most accurate test is often the most expensive

Types and methods of cancer screening

Type (Park 28e, Ch 4, p 165)	Meaning	Cancer example
Mass screening	Screening of a whole population or a sub-group, offered to all irrespective of individual risk	Universal screening of everyone aged 30 years and above
High-risk / selective screening	Applied selectively to groups defined by epidemiological research; raises the yield	Oral cavity examination of tobacco chewers and bidi smokers; cervical screening of lower social groups, in whom cancer cervix is commoner
Multiphasic screening	Two or more screening tests applied in combination to a large number of people at one time, with a questionnaire, clinical examination and investigations	Randomized studies in the UK and USA showed no mortality or morbidity benefit and increased cost — utility doubtful

Methods used in cancer control are (a) mass screening by **comprehensive cancer examination** — a rapid clinical examination of one or more body sites by the physician; (b) mass screening **at single sites** — uterine cervix, breast or lung; and (c) **selective screening** of defined high-risk groups (Park 28e, Ch 6, p 460).

Screening tests for individual cancers

Cancer	Test and schedule	Key points
Cervix	Papanicolaou (Pap) smear from age 30 years, then every 3 years, with periodic pelvic examination (Park 28e, Ch 6, p 460)	Organized population-based programmes have reduced incidence and mortality in developed countries; Pap needs laboratories, equipment and trained cytologists
Cervix — low-resource alternatives	Visual inspection with 5 per cent acetic acid (VIA), VIA with magnification (VIAM), visual inspection after Lugol's iodine (VILI)	Cheap, single-visit, feasible at primary care

Cancer	Test and schedule	Key points
Cervix — WHO current preference	HPV DNA or HPV mRNA test from age 30, repeated every 5-10 years; women living with HIV screened from 25 years, every 3-5 years (Park 28e, Ch 6, pp 459-461)	Objective, simpler, prevents more pre-cancer and cancer, more cost-effective than cytology or visual inspection; must be linked to treatment
Breast	Breast self-examination (BSE), palpation by a physician, thermography, mammography (Park 28e, Ch 6, p 460)	Mammography is the most sensitive and specific; thermography is not sensitive; palpation is unreliable in large fatty breasts. Mammography delivers about 500 milliroentgen against 30-40 milliroentgen for a chest X-ray, needs high-standard equipment and very experienced radiologists, and biopsy of a suspicious lesion ends in a false positive in 5-10 cases for every cancer detected. Not advised under 35 years unless symptomatic or with a family history of early-onset breast cancer
Oral cavity	Visual and digital examination of the oral cavity; case-finding by primary health care workers during home visits (Park 28e, Ch 6, p 461)	Site is easily accessible; leukoplakia can be cured by tobacco cessation; over 50 per cent of oral cancers in developing countries are still detected at an advanced stage
Lung	Chest radiograph and sputum cytology	Not recommended — evidence unconvincing, and it is doubtful whether the disease satisfies the criteria for screening (Park 28e, Ch 6, p 460)

Validity, predictive value and yield

Measure	Formula (a = true positive, b = false positive, c = false negative, d = true negative)	Interpretation
Sensitivity	$a / (a + c) \times 100$	Ability to correctly identify all who have the disease
Specificity	$d / (b + d) \times 100$	Ability to correctly identify those free of disease
Predictive value of a positive test	$a / (a + b) \times 100$	Probability that a test positive truly has cancer
Predictive value of a negative test	$d / (c + d) \times 100$	Probability that a test negative is truly free of cancer
False negative rate	$c / (a + c) \times 100$	Cases missed — a missed cancer
False positive rate	$b / (b + d) \times 100$	Healthy people sent for further diagnostic tests, at inconvenience, anxiety and expense

Sensitivity and specificity are **inversely related** — one is raised only at the expense of the other, and no test is 100 per cent sensitive and 100 per cent specific (Park 28e, Ch 4, pp 167-168). Predictive value of a positive test **falls as disease prevalence declines**, which is precisely why cancer screening is restricted to older or high-risk groups. **Yield** is the amount of previously unrecognized disease diagnosed as a

result of the screening effort, and depends on the sensitivity of the test, the prevalence of the disease and the participation of individuals in the programme (Park 28e, Ch 4, p 168).

Cancer screening in India

- ▶ The **National Cancer Control Programme** was launched in 1975-76, revised in 1984-85 and again in **December 2004**, and in **2010** was integrated with the National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Diseases and Stroke (Park 28e, Ch 7, p 566).
- ▶ Under the comprehensive primary health care package, **universal population-based screening of everyone aged 30 years and above** is done for hypertension, diabetes and **three common cancers — oral, breast and cervical** (Park 28e, Ch 10, p 735).
- ▶ The **decentralized non-governmental organization scheme** funds information, education and communication activities plus early detection of cancer of the cervix, mouth and breast by screening and self-examination; **7 November** is observed as National Cancer Awareness Day (Park 28e, Ch 7, p 566).
- ▶ **HPV vaccination of girls aged 9-14 years does not replace cervical screening** — population-based screening programmes are still needed to find and treat pre-cancer (Park 28e, Ch 6, p 462).

Evaluating a cancer screening programme

Bias / pitfall	How it misleads
Referral (volunteer) bias	Those who come for screening differ from those who do not; volunteers are generally healthier and more adherent, so the screened group may do better regardless of screening (Gordis 7e, Ch 18)
Length-biased sampling (prognostic selection)	Screening preferentially detects slow-growing tumours with a long pre-clinical phase and hence a better prognosis; fast, aggressive tumours surface as interval cases
Lead time bias	Survival is measured from an earlier point, so survival appears longer even if death occurs at the same time
Overdiagnosis bias	Detection of lesions that would never have caused illness or death — incidence rises while mortality does not
Misclassification bias	Errors in assigning cause of death between screened and unscreened groups
5-year survival rate	An invalid measure of screening benefit; the valid endpoint is cause-specific mortality in a randomized controlled trial

The definitive design is a **randomized controlled trial**, which equalizes both selection and prognostic differences — the *Health Insurance Plan (HIP)* trial begun in **1963** established mammography screening this way (Gordis 7e, Ch 18, p 380). Conversely, **neuroblastoma screening** in infants detected many more cases but did not lower mortality in Quebec (476,654 children) or Germany (2,581,188 children), showing that the ability to detect a disease by screening cannot be equated with benefit (Gordis 7e, Ch 18, p 389).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Lead time in cancer screening** (reproduce the figure above): a horizontal time arrow labelled, left to right — *No disease* → *Biologic onset of disease* → *Disease detectable by screening* → *Symptoms* → *Therapy given* → *Clinical outcome*; draw a brace between "disease detectable by screening" and "symptoms" and label it **LEAD TIME**.
- **Sensitivity-specificity 2 × 2 table**: a four-cell box with columns *Diseased* / *Not diseased* and rows *Test positive* / *Test negative*; label cells **a true positive**, **b false positive**, **c false negative**, **d true negative**, with row totals (a+b), (c+d) and column totals (a+c), (b+d).

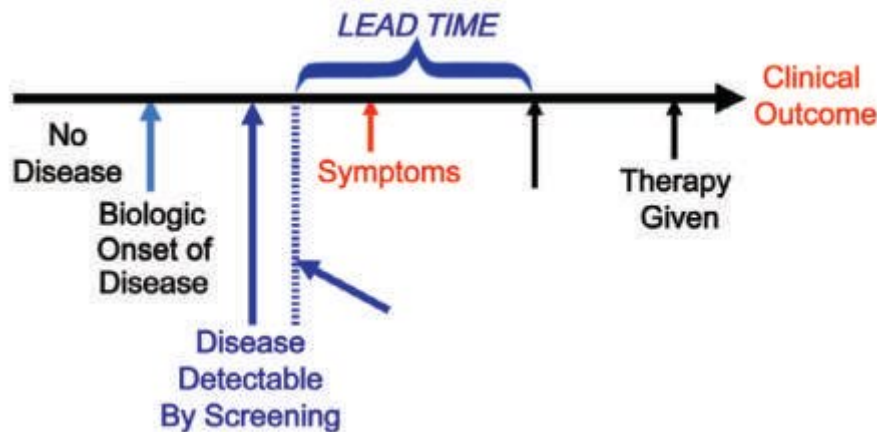
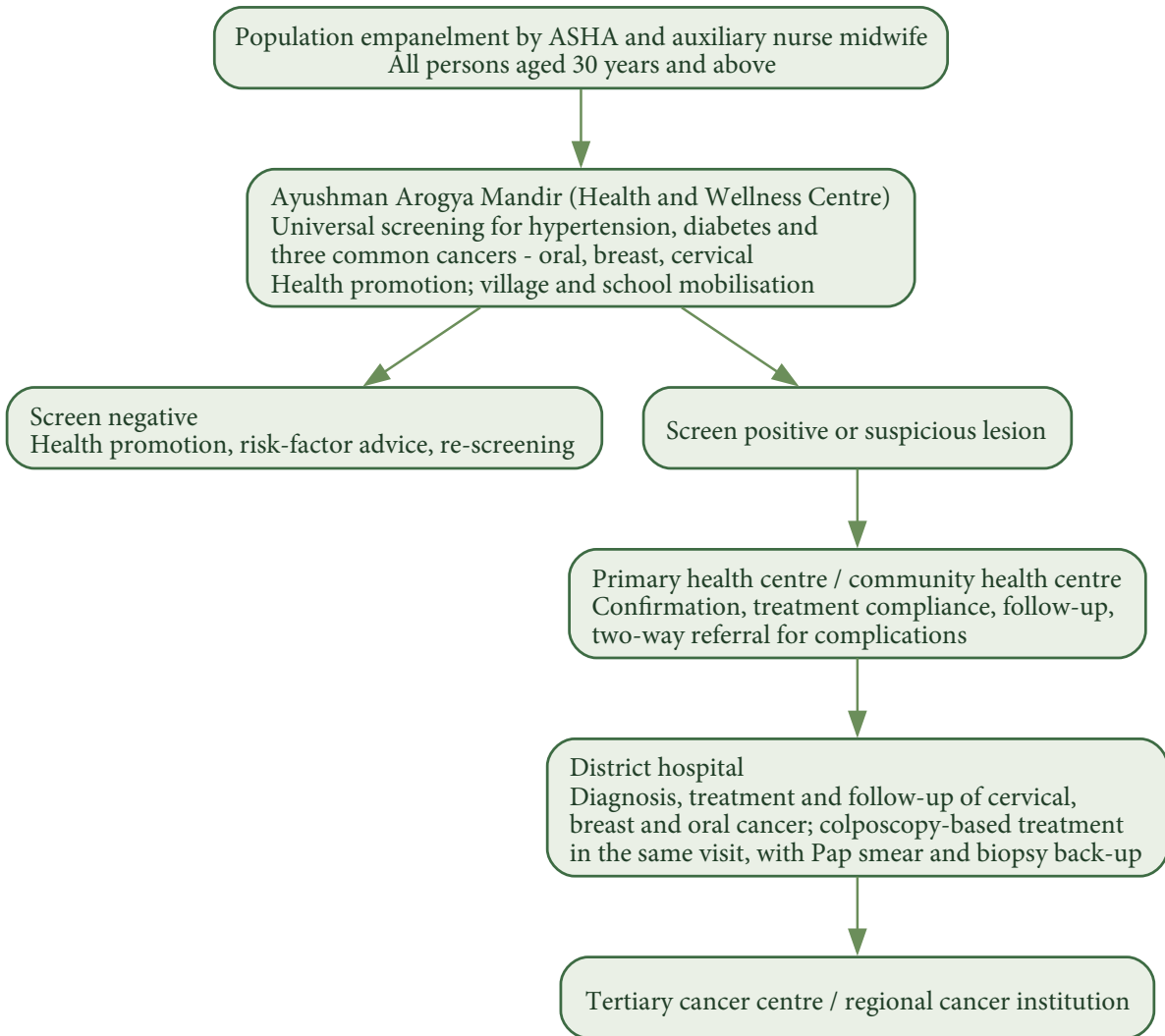
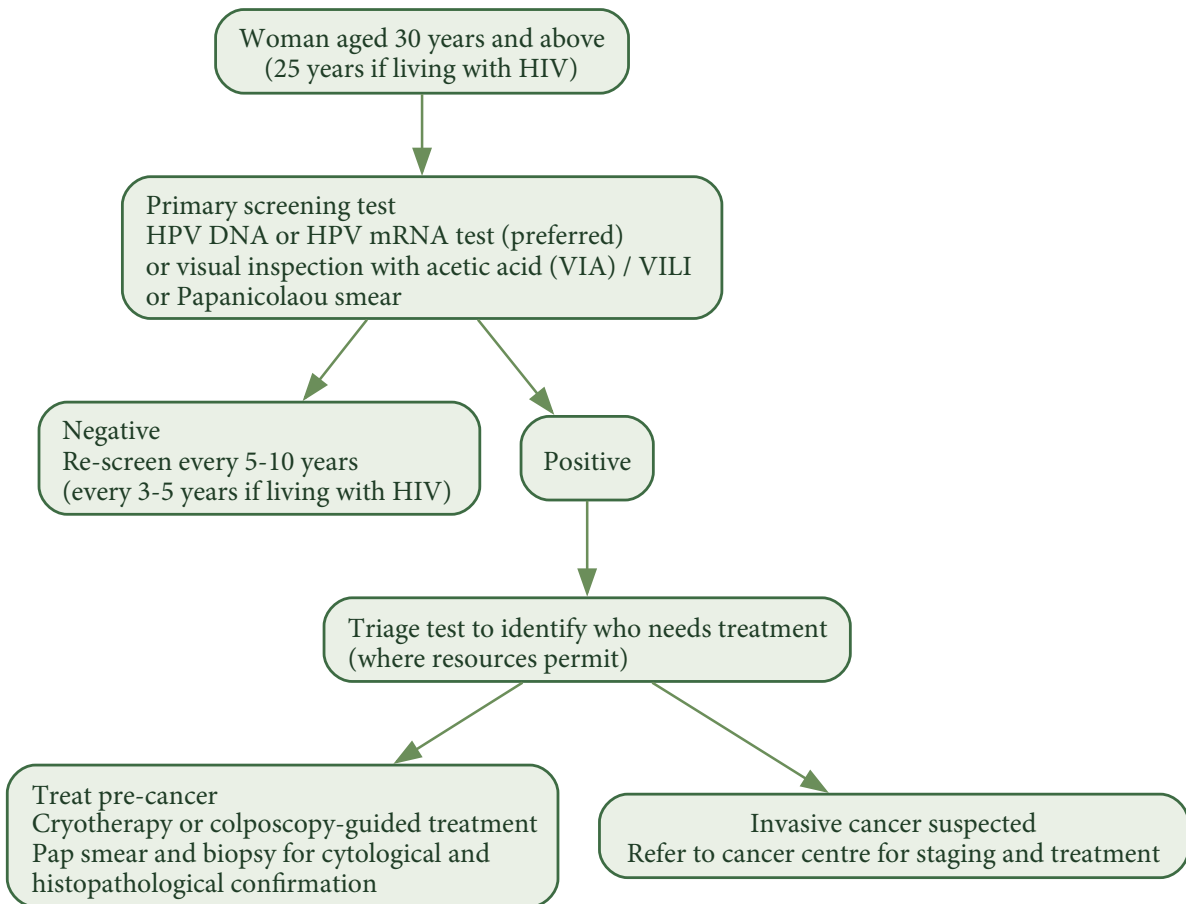


Fig 18.2 — Preclinical phase of the disease. (A) Natural history with point at wh (Gordis Epidemiology 7e, p.371)



Population-based cancer screening pathway for oral, breast and cervical cancer, India



WHO screen-and-treat algorithm for cervical cancer, women aged 30 years and above

High-yield exam pointers

- ▶ Definition to write verbatim: cancer screening = "search for unrecognized malignancy by means of rapidly applied tests"; screening ≠ diagnosis (screening test is done on apparently healthy people, applied to groups, based on one cut-off, less accurate, less expensive, not a basis for treatment, initiative from the agency).
- ▶ Three cancers with effective screening programmes: **cervix, breast, oral cavity**. Lung cancer screening is **not** recommended.
- ▶ Cervix: Pap from **30 years, every 3 years**; WHO preferred HPV test from **30 years every 5-10 years**; **25 years, every 3-5 years** if living with HIV; HPV vaccine at **9-14 years** does not replace screening.
- ▶ Mammography radiation dose **500 milliroentgen** versus **30-40 milliroentgen** for chest X-ray; false-positive biopsy in **5-10 cases per cancer detected**; avoid under **35 years**.
- ▶ Indian delivery: universal screening of **30 years and above** for oral, breast and cervical cancer; National Cancer Control Programme **1975-76**, revised **1984-85** and **2004**, integrated **2010**; National Cancer Awareness Day **7 November**.
- ▶ Four biases when evaluating screening — **referral (volunteer)**, **length**, **lead time**, **overdiagnosis**; the valid endpoint is **cause-specific mortality in a randomized controlled trial**, never the 5-year survival rate.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park 28e still prints the classical Pap-smear policy (from 30 years, every 3 years) alongside the newer World Health Organization preference for human papillomavirus testing; write both, giving the human papillomavirus test as the current recommendation. The Health and Wellness Centre has since been renamed Ayushman Arogya Mandir, and the National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Diseases and Stroke has been restructured as the National Programme for Prevention and Control of Non-Communicable Diseases; the screening service package quoted above is unchanged. Coverage note — the retrieved corpus did not carry the Community Based Assessment Checklist or the screening-interval circular for oral and breast cancer under the current programme, so no interval is quoted for those two cancers.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, WHO Global Strategy for Cervical Cancer Elimination, American Cancer Society 2026 cervical cancer screening guideline, AOGIN India 2026 policy statement, NFHS-5

Programme and policy updates (India)

- ▶ **Indigenous HPV vaccine and the push into routine immunisation** — Cervavac, India's own quadrivalent HPV vaccine, was introduced in 2023, but fewer than 1 per cent of Indian women have received any HPV vaccine and fewer than 2 per cent are ever screened. The stated policy priorities are to bring HPV vaccination into the Universal Immunization Programme, add the HPV nucleic acid amplification test to the National Essential Diagnostics List, and reuse the COVID-era reverse transcription polymerase chain reaction laboratory network for HPV testing. [PMID 41645272]
- ▶ **AOGIN India policy statement (2026)** on HPV vaccination for cervical cancer elimination sets out dose, schedule and delivery guidance for Indian conditions, and states that the World Health Organization benchmark of 90 per cent of girls vaccinated by age 15 cannot be reached through private purchase alone — it needs integration into the Universal Immunization Programme. [PMID 42169564]

Current national data

- ▶ **NFHS-5 (2019-21) shows the screening package is barely operating.** National oral cancer screening coverage was **4.4 per 1,000 people**, and residents of high-tobacco-prevalence states were 42 per cent *less* likely to be screened (adjusted odds ratio 0.58) — the reverse of what a risk-targeted programme should produce [PMID 40857966]. Coverage among women aged 30-49 years was **0.77 per cent** (men 1.06 per cent) against a programmatic expectation of 50 per cent [PMID 39562702], and among tobacco users only **1.07 per cent of men and 0.53 per cent of women** had ever had an oral examination [PMID 40964758].
- ▶ **Cervical cancer burden** — roughly **127,000 new cases and 80,000 deaths a year** in India, about one-fifth of the world total, against fewer than 2 per cent of women ever screened. Always attach the survey round and year (NFHS-5, 2019-21) when quoting any of these coverage figures in the answer. [PMID 41645272]

Elimination, surveillance and disease status

- ▶ **WHO cervical cancer elimination target** (global strategy launched November 2020) — elimination is defined as an incidence below **4 cases per 100,000 women per year**, to be approached through the **90-70-90 targets for 2030**: 90 per cent of girls fully HPV-vaccinated by age 15, 70 per cent of women screened with a high-performance test at ages 35 and 45, and 90 per cent of women with pre-cancer or cancer treated [PMID 40794406]. Globally, **64 per cent of women aged 30-49 years (662 million) have never been screened**, and coverage is at least seven times higher in high-income than in low- and lower-middle-income countries [PMID 35839811].
- ▶ **Oral cancer screening now has mortality evidence, not merely feasibility.** The Indian cluster-randomised trial followed for 15 years recorded an **81 per cent reduction in oral cancer mortality** among tobacco and/or alcohol users who completed four rounds of visual screening; observational programmes report 21-22 per cent fewer advanced oral cancers and 24-26 per cent lower oral cancer mortality in high-risk groups. [PMID 36599217]

Guideline and legislative updates

- ▶ **Self-collected HPV samples have entered guidelines.** The American Cancer Society's 2026 update accepts a **self-collected vaginal sample for primary HPV testing in average-risk women aged 25-65 years** (a clinician-collected cervical sample remains preferred), with repeat testing at **3 years** after a negative self-collected test against **5 years** after a negative clinician-collected one. The same update tightens the rules for stopping screening — negative HPV tests at ages **60 and 65**, the last no earlier than 65. [PMID 41342729]
- ▶ **Self-collection performs well and roughly doubles uptake.** Pooled sensitivity for cervical intraepithelial neoplasia grade 2 or worse was 90 per cent for HPV testing on self-collected samples, 93 per cent on clinician-collected samples and 87 per cent (95% CI 76-93) for cytology [PMID 40643571]. In the PRESTIS randomised trial, mailing self-collection kits raised participation from **17.4 to 41.1 per cent** (relative participation 2.36; 95% CI 1.99-2.80), and to 46.6 per cent when a patient navigator followed up [PMID 40478588] — directly relevant in India, where embarrassment at pelvic examination is a documented barrier to acceptability.

Global and WHO developments

- ▶ **Single-dose HPV vaccination.** WHO changed its advice in **2022** to allow one or two doses for girls and young women up to age 20. A meta-analysis of 902,368 vaccinated women found one dose comparable to two or three doses for pre-cancer (CIN2/3 risk ratio 1.54; 95% CI 0.91-2.62) but slightly weaker against high-risk HPV infection (risk ratio 1.27; 95% CI 1.02-1.57) [PMID 38180991]; the PRIMAVERA immunobridging trial met non-inferiority for HPV-18 but not HPV-16 (geometric mean concentration ratio 0.50; 96% CI 0.44-0.57) [PMID 40683285], so definitive efficacy trials are still running [PMID 39529520].
- ▶ **Lung cancer screening has changed abroad, though not in India.** Low-dose computed tomography lowers lung cancer mortality in high-risk smokers (NLST, NELSON) [PMID 36563698], and the SUMMIT cohort of 12,773 Londoners screened in 2019-21 found lung cancer in 2.0 per cent, with 79.3 per cent of screen-detected cancers at stage I or II, sensitivity 97.0 per cent and specificity 95.2 per cent [PMID 40154514]. India has no low-dose computed tomography screening programme, and chest radiography with sputum cytology remains

ineffective — so "not recommended" still stands for the Indian answer, but add that it is the modality, not the screening principle, that has moved.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 4 (Screening for Disease, pp 163-169), Ch 6 (Epidemiology of Chronic Non-communicable Diseases and Conditions, pp 457-462), Ch 7 (Health Programmes in India, pp 564-566), Ch 10 (Health Care of the Community, p 735); Gordis Epidemiology 7e — Ch 18 (Epidemiologic Approach to Evaluating Screening Programs, pp 368-391); Jugal Kishore — National Health Programmes of India (National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Diseases and Stroke, objectives).

Recent-advances sources: 41645272 — Cervical cancer elimination in India: repurposing diagnostics, vaccination, and accelerating policy for the 2030 target, *Cancer* 2026, doi:[10.1002/cncr.70292](https://doi.org/10.1002/cncr.70292) (abstract-only); 42169564 — AOGIN India policy statement on the use of HPV vaccination for cervical cancer elimination, *Asian Pac J Cancer Prev* 2026, doi:[10.31557/APJCP.2026.27.5.1603](https://doi.org/10.31557/APJCP.2026.27.5.1603) (abstract-only); 40857966 — Oral cancer screening in India: a secondary analysis of disparities using National Family Health Survey data, *Oral Oncol* 2025, doi:[10.1016/j.oraloncology.2025.107632](https://doi.org/10.1016/j.oraloncology.2025.107632) (abstract-only); 39562702 — Effect of socioeconomic status and women empowerment status on coverage of oral cancer screening among Indian women within reproductive age group, *Sci Rep* 2024, doi:[10.1038/s41598-024-80346-w](https://doi.org/10.1038/s41598-024-80346-w); 40964758 — Prevalence and factors associated with oral cancer screening among tobacco users in India: findings from the National Family Health Survey-5, *Indian J Public Health* 2025, doi:[10.4103/ijph.ijph_534_24](https://doi.org/10.4103/ijph.ijph_534_24) (abstract-only); 40794406 — Cervical cancer screening, HPV vaccination, and cervical cancer elimination, *JAMA Netw Open* 2025, doi:[10.1001/jamanetworkopen.2025.26683](https://doi.org/10.1001/jamanetworkopen.2025.26683); 35839811 — Cervical cancer screening programmes and age-specific coverage estimates for 202 countries and territories worldwide, *Lancet Glob Health* 2022, doi:[10.1016/S2214-109X\(22\)00241-8](https://doi.org/10.1016/S2214-109X(22)00241-8); 36599217 — Perspective on oral cancer screening: time for implementation research and beyond, *J Cancer Policy* 2023, doi:[10.1016/j.jcpo.2022.100381](https://doi.org/10.1016/j.jcpo.2022.100381) (abstract-only); 41342729 — Self-collected vaginal specimens for human papillomavirus testing and guidance on screening exit: an update to the American Cancer Society cervical cancer screening guideline, *CA Cancer J Clin* 2026, doi:[10.3322/caac.70041](https://doi.org/10.3322/caac.70041); 40643571 — Accuracy of HPV self-collection compared with clinician-collected HPV testing and cytology: a meta-analysis, *Cancer Epidemiol Biomarkers Prev* 2025, doi:[10.1158/1055-9965.EPI-25-0362](https://doi.org/10.1158/1055-9965.EPI-25-0362) (abstract-only); 40478588 — Self-collection for cervical cancer screening in a safety-net setting: the PRESTIS randomized clinical trial, *JAMA Intern Med* 2025, doi:[10.1001/jamainternmed.2025.2971](https://doi.org/10.1001/jamainternmed.2025.2971); 38180991 — The clinical effectiveness of one-dose vaccination with an HPV vaccine: a meta-analysis of 902,368 vaccinated women, *PLoS One* 2024, doi:[10.1371/journal.pone.0290808](https://doi.org/10.1371/journal.pone.0290808); 40683285 — HPV type 16 and type 18 antibody concentrations after a single dose of bivalent HPV vaccine in girls aged 9-14 years (PRIMAVERA), *Lancet Infect Dis* 2025, doi:[10.1016/S1473-3099\(25\)00284-1](https://doi.org/10.1016/S1473-3099(25)00284-1); 39529520 — Single-dose human papillomavirus vaccination: an update, *J Natl Cancer Inst Monogr* 2024, doi:[10.1093/jncimonographs/lgae030](https://doi.org/10.1093/jncimonographs/lgae030) (abstract-only); 36563698 — Lung cancer screening, *Lancet* 2022, doi:[10.1016/S0140-6736\(22\)01694-4](https://doi.org/10.1016/S0140-6736(22)01694-4) (abstract-only); 40154514 — Low-dose CT for lung cancer screening in a high-risk population (SUMMIT), *Lancet Oncol* 2025, doi:[10.1016/S1470-2045\(25\)00082-8](https://doi.org/10.1016/S1470-2045(25)00082-8) (abstract-only). WHO Global Strategy to Accelerate the

Elimination of Cervical Cancer, launched November 2020 (90-70-90 targets for 2030); WHO HPV vaccine schedule revision 2022 (one or two doses); American Cancer Society cervical cancer screening guideline update 2026.



Sero-epidemiology

Asked in: Paper I May 2017 — RGUHS MD Community Medicine

Definition — Sero-epidemiology (serological epidemiology) is the branch of epidemiology that studies the distribution, determinants and trends of **infection and immunity in populations by measuring specific antibodies or antigens in serum**, instead of counting only clinically diagnosed cases. Park lists it among the modern off-shoots of epidemiology, and states that herd immunity in a population is determined by such serological surveys (Park 28e, Ch 3, pp 72, 123). Its unique value is that a serum sample reports not only the infection a person has now but also infections recovered from years or decades earlier, so it measures the **whole iceberg of infection and not merely its clinical tip** (Giesecke, Ch 16).

Immunological basis — what a serum marker tells the epidemiologist

Serum marker / measure	Behaviour	Epidemiological meaning
Latent period after first antigenic exposure	3–10 days before any antibody appears in blood (Park 28e, Ch 3, p 121)	Serology is negative very early in infection; a single early sample cannot exclude infection
Immunoglobulin M (IgM)	Antibody elicited first ; rises over the next 2–3 days, peaks, then declines almost as fast as it developed	Marker of recent or current infection → estimates incidence
Immunoglobulin G (IgG)	Appears a few days after IgM, peaks in 7–10 days, then falls gradually over weeks to months and persists long	Marker of past infection or vaccination → estimates cumulative prevalence and immunity
Titre	Highest serum dilution still testing positive (e.g. 1:32) in dilution-based tests — haemagglutination, haemagglutination inhibition, complement fixation	Semi-quantitative measure of antibody; 1:1000 is a <i>higher</i> titre than 1:10
Geometric mean titre (GMT)	n th root of the product of all titres (or antilog of the mean of the log titres)	Correct group average on an exponential dilution scale; cannot be computed if any sample is seronegative
Paired sera (acute sample + second sample ~2 weeks later)	A four-fold or greater rise in titre = seroconversion	Proof of recent infection; built into case definitions — e.g. laboratory-confirmed mumps requires positive IgM or a four-fold rise in IgG titre (Park 28e, Ch 5, p 184)

Serum marker / measure	Behaviour	Epidemiological meaning
Cut-off point	Separates true (specific) reactivity from non-specific reactivity, whose distributions overlap	Raising the cut-off gains specificity and loses sensitivity , and vice versa; every good serological study must state how the cut-off was set

The current standard assay is the **enzyme-linked immunosorbent assay (ELISA / EIA)**, which gives a directly quantitative antibody value; neutralisation tests such as the plaque-reduction test are used where functional (protective) antibody must be shown (Giesecke, Ch 16).

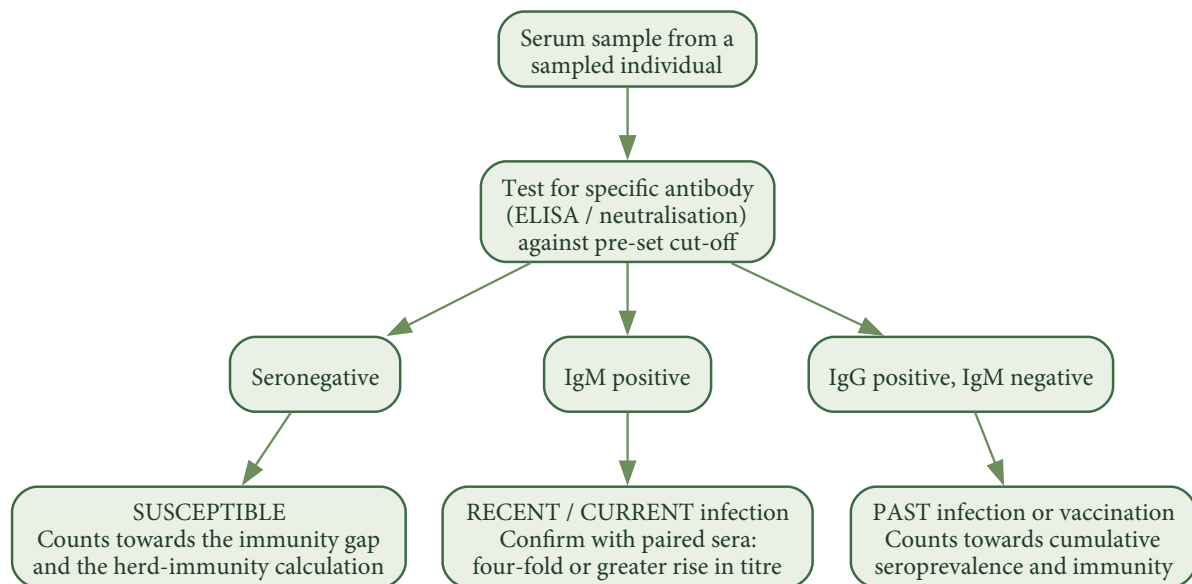
Types of sero-epidemiological study

Design	Serum sampled	What it yields	Chief caveat
Cross-sectional seroprevalence survey	One random population sample, one bleed	Seroprevalence by age, sex, place, occupation, behaviour; risk factors by odds ratio	Gives prevalence, not incidence; vulnerable to the age cohort effect
Serial (repeated) cross-sections	Fresh samples at two or more time points	Incidence estimated from the change in prevalence between rounds, without continuous case surveillance	Rounds must be comparable in sampling and assay
Longitudinal cohort with repeated / paired sera	Same individuals bled repeatedly	True incidence and exact timing of seroconversion; best for identifying transmission routes	Costly, attrition, repeated venepuncture
Stored sera / serum-bank study	Frozen sera collected earlier for other reasons	Rapid, cheap, allows retrospective study of a new agent	Strong selection bias — sera were taken for clinical reasons
Sentinel sero-surveillance	Fixed sentinel sites, sampled annually	Trend of infection over time in defined groups	Sentinel groups are not the general population

Steps in conducting a sero-epidemiological survey

- ▶ **Define the objective** — prevalence of past infection, immunity gap, incidence, vaccine impact or aetiological hypothesis.
- ▶ **Define seropositivity** — the antigen tested, the assay, and the pre-declared cut-off titre.
- ▶ **Design the sample** — probability sampling (commonly stratified cluster sampling of blocks → households → individuals); fix sample size for the required precision.
- ▶ **Obtain ethics clearance and consent** — institutional review board approval, and pre-decided rules for interpreting a positive result and for follow-up of the participant, submitted with the protocol (Handbook of Epidemiology, Ch 33, p 1343).
- ▶ **Pilot the field procedures** — questionnaire, specimen collection, laboratory test — for feasibility, acceptability and reliability before launch.
- ▶ **Collect, preserve and transport specimens** with an assured cold chain to the reference laboratory.

- **Test in a single quality-controlled laboratory** to keep the cut-off uniform across sites.
- **Analyse with sampling weights** for the design (block, household and individual selection), and report **weighted seroprevalence with 95 per cent confidence intervals**, stratified by age and place, then risk factors by univariate and logistic regression analysis (Handbook of Epidemiology, Ch 33, pp 1341–1343).
- **Act on the result** — identify susceptible groups and the priority area for immunisation or other control.



Interpreting a serum sample in a sero-epidemiological study

Uses of sero-epidemiology

Use	How it works	Example from the corpus
Measuring the true extent of infection (the submerged iceberg)	Subclinical infection can be detected only by laboratory tests such as antibody response	Poliovirus: inapparent infection in 91–96 per cent of infections; about 1000 subclinical cases per clinical case in children and 75 in adults (Park 28e, Ch 5, p 259)
Detecting silent transmission	Infection may circulate with no clinical picture	Influenza "may smoulder in a community, being manifest only by serological surveys" (Park 28e, Ch 5, p 184)
Estimating herd immunity and immunity gaps	Proportion seropositive is compared with the herd-immunity threshold for that agent	"Herd immunity may be determined by serological surveys (serological epidemiology)" (Park 28e, Ch 3, p 123); the European Sero-Epidemiology Network (1996–2014, 18 states) exposed immunity gaps for diphtheria, measles and rubella (Plotkin 7e, Ch 74, p 2088)
Estimating force of infection and basic reproduction number (R_0)	Age-stratified seroprevalence: the extra seroprevalence in age class i over class $i-1$ estimates one year's hazard	Nelson & Williams, Ch 6, p 170

Use	How it works	Example from the corpus
Evaluating vaccines and immunisation programmes	Seroconversion and titre rise as immune correlates; vaccine effectiveness from a seroprevalence cross-section as $VE^* = 1 - \text{prevalence ratio}$	Nelson & Williams, Ch 11, pp 301, 305
Demonstrating natural boosting	Rise in titre after exposure in already-immune persons	Measles blood-drive study: 11 students with a pre-exposure titre below 1000 showed a four-fold or greater boost in titre, whereas none of the 7 with a pre-exposure titre above 1000 did (Giesecke, Ch 16)
Serological surveillance	Continuous serological surveillance through serum banks , listed by Park as a core data source of surveillance; it "identifies patterns of current and past infection"	Park 28e, Ch 2, p 59 and Ch 3, pp 114, 151
Outbreak investigation	In the step of "further investigation of the population at risk", serological study reveals clinically inapparent cases and clarifies pathogenesis	Park 28e, Ch 3, p 160
Measuring transmission intensity in elimination settings	Detects residual low-level transmission by seroprevalence in new entrants to the population, when parasite rate and spleen rate have become insensitive	Malaria sero-epidemiology (Gilles & Warrell 4e, Ch 3, p 68)
Reconstructing past epidemics	Age-seroprevalence curves act as "seroepidemiological archaeology"	Sero-epidemiological studies suggest the causative subtypes of the influenza epidemics of 1890 (H2N8) and 1900 (H3N8), which predate virus isolation (Jawetz 28e, Ch 39)
Generating aetiological hypotheses, including for chronic disease	Seropositivity related to a chronic outcome	<i>Chlamydia pneumoniae</i> and atherosclerotic coronary and cerebrovascular disease (association remains controversial); HTLV-1 and HTLV-1-associated myelopathy / tropical spastic paraparesis (Jawetz 28e, Ch 27, Ch 43)

Indian applications

Application	Method and finding
ICMR national COVID-19 sero-surveys	Four national rounds; antibody prevalence 0.7 per cent (May 2020), 7.1 per cent (Aug–Sep 2020), 21.4 per cent (Dec 2020–Jan 2021) and 67.6 per cent (Jun–Jul 2021). Round 4 covered 28,975 persons across 70 districts of 21 states, and the 67.6 per cent figure is for persons above 6 years of age; seropositivity 62.3 per cent in the unvaccinated, 80 per cent after one dose, 89.8 per cent after two doses — showing that one-third of the population was still susceptible (Sunder Lal, Ch 20, pp 803–804)

Application	Method and finding
HIV Sentinel Surveillance	Annual cross-sectional survey of the same sites by unlinked anonymous serological testing (specimen tested without identifying the person); pregnant women attending antenatal clinics are taken as a proxy for the general population . Begun in 1994 at 55 sentinel sites, raised to 180 in 1998; used to grade states as high, moderate and low prevalence and to categorise districts (Park 28e, Ch 7, pp 518–520)
HIV sero-surveillance under NACO	Listed separately from sentinel surveillance as sero-surveillance for estimating HIV prevalence in general and high-risk populations (Jugal Kishore, National Health Programmes of India)

Limitations and sources of error

- ▶ **Non-specific reactions are unavoidable** — the seropositive and seronegative titre distributions overlap, so sensitivity and specificity can only be traded against each other by moving the cut-off, never both maximised.
- ▶ **Antibody is not the same as protection** — waning titres mean that absence of detectable antibody does not necessarily indicate susceptibility (Nelson & Williams, Ch 11, p 305).
- ▶ **Serology usually cannot separate natural infection from vaccine-induced immunity**, unless an antigen absent from the vaccine is used.
- ▶ **Serum-bank (selection) bias** — stored sera were nearly always taken for clinical reasons, so donors are ill, and the age-sex mix is skewed; blood-donor sera are equally unrepresentative (Giesecke, Ch 16).
- ▶ **Non-response bias** — the infected may be absent, may not open the door, or may refuse to donate blood, biasing infectious-disease surveys systematically (Handbook of Epidemiology, Ch 33, p 1341).
- ▶ **The age cohort effect** — a rising cross-sectional seroprevalence with age reflects the *higher past exposure of older cohorts*, not the current age-specific incidence; predicting the future of today's children from such a curve is a fallacy.
- ▶ **Cellular immunity is missed** — the humoral response alone is measured; cell-mediated immunity (as tested by the tuberculin reaction) is not captured.
- ▶ **Operational limits** — surveys are expensive and cannot easily be repeated; they need venepuncture, cold chain, a reference laboratory and ethics clearance.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **The primary immune response (basis of sero-epidemiological interpretation)** — x-axis *Time after antigenic stimulus (days)*; y-axis *Antibody concentration*; label the *latent period of 3–10 days*, the early short **IgM** curve, the later, taller, longer **IgG** curve, and mark **IgM = recent infection** and **IgG = past infection or vaccination**.
- ▶ **Age-seroprevalence curve** — x-axis *Age in years (0–70)*; y-axis *Seroprevalence (%)*; a rising curve flattening at older ages, annotated "*steep rise = high past force of infection*" and "*age cohort effect — reflects past, not present, transmission*".

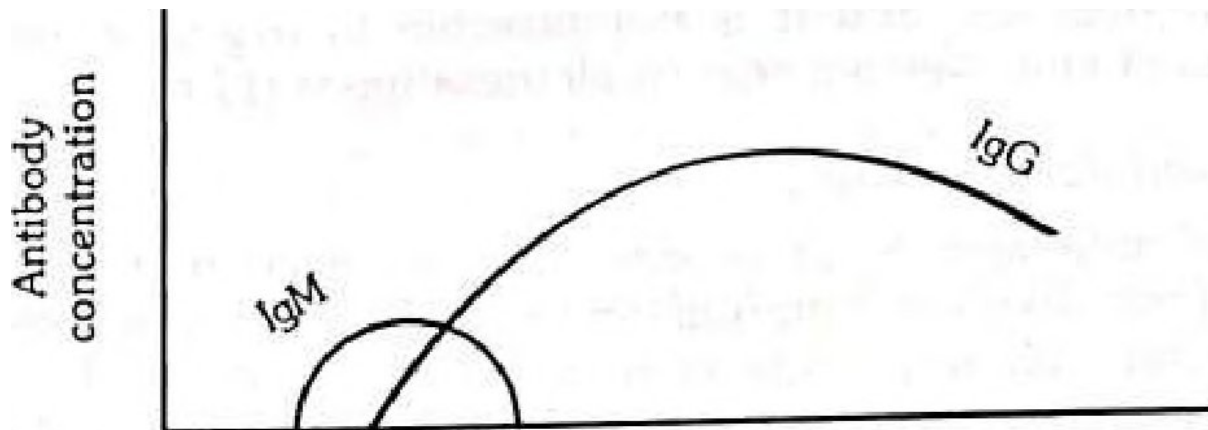
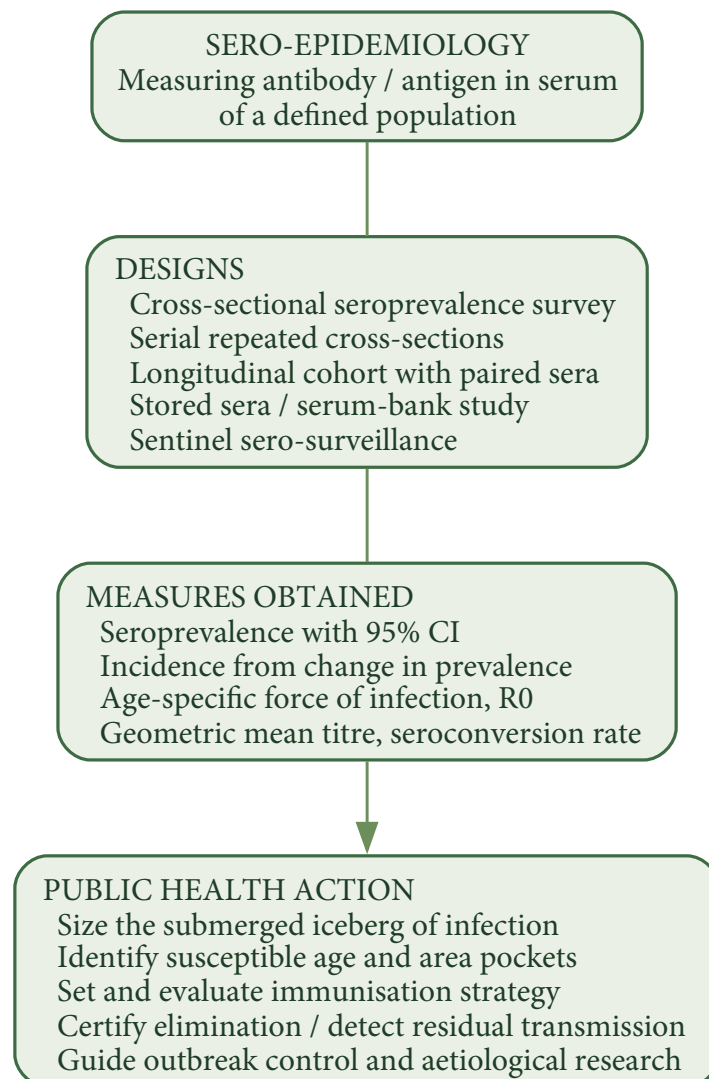


Fig 19 — The primary response (Park's Textbook of Preventive and Social Medicine, 28e, p.121)



Sero-epidemiology — study designs and the public health outputs each yields

High-yield exam pointers

- **One-line definition to write first:** epidemiological study of infection and immunity in populations using serum antibody/antigen, not clinical case counts — Park's own phrase is "*herd immunity may be determined by serological surveys (serological epidemiology)*".
- **IgM = recent, IgG = past;** latent period 3–10 days; IgG peaks 7–10 days (Park 28e, Ch 3, p 121).

- ▶ **Seroconversion** = four-fold or greater rise in titre in paired sera taken about 2 weeks apart.
- ▶ **Geometric mean titre**, not arithmetic mean, for group titres; cannot be calculated if any sample is seronegative.
- ▶ **Iceberg figure**: polio — 91–96 per cent inapparent infection, ~1000 subclinical per clinical case in children and 75 in adults.
- ▶ **Indian data with year**: ICMR national COVID-19 sero-survey round 4 (Jun–Jul 2021) — 67.6 per cent seroprevalence in those over 6 years, up from 21.4 per cent in round 3 (Dec 2020–Jan 2021).
- ▶ **HIV Sentinel Surveillance**: unlinked anonymous testing, antenatal clinic attendees as proxy for the general population, 55 sites (1994) → 180 sites (1998).
- ▶ **Two named biases to quote**: serum-bank selection bias and the **age cohort effect**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only — the ICMR national sero-survey figures above are from the four COVID-19 rounds up to June–July 2021 as reported in the retrieved text; any later round should be quoted with its own date. The corpus contains no single textbook paragraph headed "sero-epidemiology" in Park 28e; the definition here is assembled from Park's own uses of the term (off-shoots of epidemiology, herd immunity by serological survey, serological surveillance through serum banks) together with the dedicated seroepidemiology chapter in Giesecke.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 2 Concept of Health and Disease; Ch 3 Principles of Epidemiology and Epidemiologic Methods; Ch 5 Epidemiology of Communicable Diseases; Ch 7 Health Programmes in India); Giesecke — Modern Infectious Disease Epidemiology (Ch 16, Seroepidemiology); Handbook of Epidemiology, Ahrens & Pigeot (Ch IV.1 Infectious Disease Epidemiology); Nelson & Williams — Infectious Disease Epidemiology (Ch 6 Infectious Disease Dynamics; Ch 11 Vaccines); Plotkin's Vaccines 7ed (Ch 74 Immunization in Europe); Gilles & Warrell — Essential Malariology 4e (Ch 3 Diagnostic methods in malaria); Jawetz Medical Microbiology 28e (Ch 27, Ch 39, Ch 43); Sunder Lal — Textbook of Community Medicine (Ch 20 Recent Advances, ICMR national sero-surveys); Jugal Kishore — National Health Programmes of India (NACO surveillance).



Cancer registry

Asked in: Paper II Nov 2017 — RGUHS MD Community Medicine

Definition — Cancer registration is the process of systematically and continuously collecting information on every malignant neoplasm occurring in a defined patient group or population (Jugal Kishore, National Health Programmes of India, p 584). Registration implies something more than notification: a **register** requires that a permanent record be established, that the cases be followed up, and that basic statistical tabulations be prepared both on frequency and on survival, the patients on a register also being available for special studies (Park 28e, Ch 21, pp 997-998). In the terminology of the International Epidemiological Association dictionary, the *register* is the file of data that can be related to a population base, while the *registry* is the system of ongoing registration (Handbook of Epidemiology, Ch 4, pp 242-243). Cancer registration is a *sine qua non* for any cancer control programme: it provides the base for assessing the magnitude of the problem and for planning the necessary services, and Park places it as the first measure of **secondary prevention** of cancer (Park 28e, Ch 6, p 459).

Registration, register and registry

Term	What it means
Notification	A single one-way report of a diagnosed case to the health authority; the transaction ends there
Registration	Permanent record + follow-up of the case + tabulation of frequency and survival + availability for special studies (Park 28e, Ch 21, pp 997-998)
Register	The file of data itself, relatable to a population base (Handbook of Epidemiology, Ch 4, p 243)
Registry	The organisation and system that carries out ongoing registration (Handbook of Epidemiology, Ch 4, p 243)

Types of cancer registry

Cancer registries are basically of two types — hospital-based and population-based (Park 28e, Ch 6, pp 459-460).

Feature	Hospital-based cancer registry (HBCR)	Population-based cancer registry (PBCR)
Unit covered	All cancer patients treated by one particular institution, whether in-patients or out-patients (Park 28e, Ch 6, pp 459-460)	The complete cancer situation in a given geographic area (Park 28e, Ch 6, p 459)
Base population	Not defined — the hospital population is always a selected population	Defined resident population; optimum size 2–7 million (Park 28e, Ch 6, p 459)
Denominator	Absent; incidence rate cannot be derived (Sunder Lal, Ch 8, p 524)	Present; incidence rate can be computed (Park 28e, Ch 6, p 459)
Strongest data	Cancer extent, stage, therapy and survival are more reliable (Sunder Lal, Ch 8, p 524)	Incidence, mortality, time trends, geographic and occupational variation

Feature	Hospital-based cancer registry (HBCR)	Population-based cancer registry (PBCR)
Principal use	Evaluation of diagnostic and treatment programmes, provided there is long-term follow-up (Park 28e, Ch 6, p 459)	Initiating epidemiological enquiries into causes of cancer, surveillance of time trends, and planning and evaluation of operational activities in all main areas of cancer control (Park 28e, Ch 6, p 459)
Epidemiological value	Limited , because the hospital population is selected (Park 28e, Ch 6, p 459)	The only source that can give a true community incidence rate (Lilienfeld's Foundations of Epidemiology, Ch 7, p 136)
Indian examples	Chandigarh, Dibrugarh, Thiruvananthapuram (the original three, 1982); later Chennai, Mumbai, Bengaluru (Jugal Kishore, p 585; Sunder Lal, Ch 8, p 524)	Bengaluru, Chennai, Mumbai (the original three, 1982); Delhi, Bhopal, Ahmedabad and others; rural site at Barshi , Maharashtra (Jugal Kishore, p 585; Sunder Lal, Ch 8, p 524)

A hospital tumour registry is a **case-based** registry: it describes the characteristics of persons developing the disease well, but it cannot measure how often the disease occurs, because it has no reference population. A patient visiting from another state who falls ill and is registered where treated is assigned to the wrong population at risk. Only a population-based registry, or a population-based survey, supplies the correct reference population (Lilienfeld, Ch 7, p 136).

Registries may also be classified by their legal basis. **Mandatory** registries require every physician, laboratory and hospital to report each diagnosed case to a central agency; **voluntary** registries are cheaper to run but tend to under-report cases (Lilienfeld, Ch 7, p 157). In the Nordic countries cancer registration rests on compulsory reporting by all physicians and all pathologists, in both public and private practice (Handbook of Epidemiology, Ch 4, p 245).

Organisation, sources of case finding, and core data items

A registry aims to collect as much information as is practical on **all newly recognised cases** in the population, and to achieve completeness it draws on several independent sources simultaneously (Lilienfeld, Ch 7, p 136).

- ▶ **Hospitals and nursing homes**, public and private — in-patient and out-patient records, medical record departments.
- ▶ **Pathology, cytology and haematology laboratories** — the histological confirmation that anchors the diagnosis.
- ▶ **Practising physicians and clinicians**, including private practitioners and radiotherapy and oncology units.
- ▶ **Official death certificates** — deaths certified as cancer are traced back to identify cases never registered in life.
- ▶ **Imaging, endoscopy and other diagnostic departments** for cases with no tissue diagnosis.

Registries must be organised on a statutory footing with secured core funding and permanent scientific as well as clerical staff; short-term project funding cannot sustain the required expertise (Handbook of Epidemiology, Ch 4, pp 243-244). A **unique personal identification number** is the single technical necessity in any individual-level register; where only names and addresses are available, biases and

follow-up problems follow (Handbook of Epidemiology, Ch 4, p 244). Registries should collect the uniform minimum set of data recommended in the **WHO Handbook for Standardized Cancer Registries** (Park 28e, Ch 6, p 459).

Domain	Core items in a cancer register
Identification	Name, personal identification number, age, sex
Place	Place of residence (fixes the denominator population)
Tumour	Site of tumour, histological type
Diagnosis	Basis of diagnosis, date of diagnosis
Source	Reporting hospital and department, reporting pathology or cytology department

(Handbook of Epidemiology, Ch 4, Table 4.1, p 244)

Uses of a cancer registry

Level of use	What the registry delivers
Descriptive epidemiology	Crude incidence rate for all ages; age-specific rates; age-adjusted (world population) standardised incidence rates; truncated incidence rate — the age-adjusted rate based on age-specific rates in the 35–64 year age group, which holds more than 60% of registered cancer cases (Sunder Lal, Ch 8, p 524)
Aetiological research	Initiating epidemiological enquiries into the causes of cancer; geographical and temporal differences in incidence, mortality and prevalence generate hypotheses (Park 28e, Ch 6, p 459; Handbook of Epidemiology, Ch 35, p 1418)
Record-linkage studies	Large-scale, high-quality and relatively low-cost studies — cancer risk in migrants by region of origin and length of stay; linkage of census with registry data to assess risk from specific occupations; second primary neoplasms; cancer risk after a non-neoplastic diagnosis (Handbook of Epidemiology, Ch 35, p 1418; Ch 4, pp 246–247)
Study base	A sampling frame for case-control studies — cases are drawn from the register and controls from the general population; also the follow-up mechanism for cohort studies (Handbook of Epidemiology, Ch 4, p 247)
Survival and outcome	Linkage of the cancer register with the cause-of-death register permits survival analysis of cancer patients; morbidity registers give duration of illness, case fatality and survival (Handbook of Epidemiology, Ch 4, pp 246–247; Park 28e, Ch 21, p 998)
Programme planning and evaluation	Assessing the magnitude of the problem, planning services, surveillance of time trends, and planning and evaluation of operational activities in all main areas of cancer control (Park 28e, Ch 6, pp 459–460)
International comparison	Registries meeting defined criteria of completeness and validity are published at five-year intervals in <i>Cancer Incidence in Five Continents</i> (Handbook of Epidemiology, Ch 3, p 169)

Quality, completeness and limitations

Cancer diagnoses are in general of good quality among routinely collected data, and are better in younger than in older patients, who often carry several diseases (Handbook of Epidemiology, Ch 4, p 245).

Aspect	Detail
Completeness achievable	A Swedish study estimated the deficit in cancer registration at 4.5%, falling to under 2% when the diagnosis was histologically verified (Handbook of Epidemiology, Ch 4, p 245)
Where completeness fails	A Finnish study found good coverage for solid tumours but roughly 10% under-registration of benign neoplasms of the central nervous system, chronic lymphatic leukaemia and multiple myeloma (Handbook of Epidemiology, Ch 4, p 245)
Routine internal checks	Computer detection of invalid codes, inconsistent code combinations, duplicate registrations and illogical time sequences — for example testis cancer coded in a female, or distant metastasis with carcinoma <i>in situ</i> (Handbook of Epidemiology, Ch 4, p 245)
What checks cannot fix	Missing data and false primary diagnoses; in the Finnish Cancer Register about 2% of cancers are coded to an unknown primary site (Handbook of Epidemiology, Ch 4, p 245)
Inherent strengths	Data are already collected and collected objectively with regard to any specific study; observations are numerous and often nationwide, covering a whole population (Handbook of Epidemiology, Ch 4, p 247)
Inherent limitations	A restricted set of variables, not specified in advance for the research question in focus; data quality can differ between registries (Handbook of Epidemiology, Ch 4, p 247)
Cost	Developing and maintaining a registry is costly; where cost is prohibitive, surveillance systems, cross-sectional studies and population-based morbidity surveys are alternatives for common outcomes (Lilienfeld, Ch 7, p 138)
Ethics	Registration is justified only if the registered data are used; the balance between the individual's right to privacy and the right to benefit from register-based knowledge must be explicitly maintained (Handbook of Epidemiology, Ch 4, pp 254-255)

National Cancer Registry Programme (NCRP), India

The NCRP was launched by the **Indian Council of Medical Research** in **1981–82**, with registration commencing on **1 January 1982**, to provide true information on cancer incidence and prevalence and to support the planning and evaluation of the National Cancer Control Programme (Jugal Kishore, pp 584-585; Sunder Lal, Ch 8, pp 524-525).

Objectives

- ▶ To generate reliable data on the magnitude and patterns of cancer, and on trends over time.
- ▶ To undertake epidemiological studies based on the results of registry data, including the geographical variation of cancer incidence and its associated risk factors, so that control measures can be made region-specific.

- To help in designing, planning, monitoring and evaluation of cancer control activities under the National Cancer Control Programme, and to evaluate the effectiveness of control strategies.
- To develop training programmes in cancer registration and epidemiology.

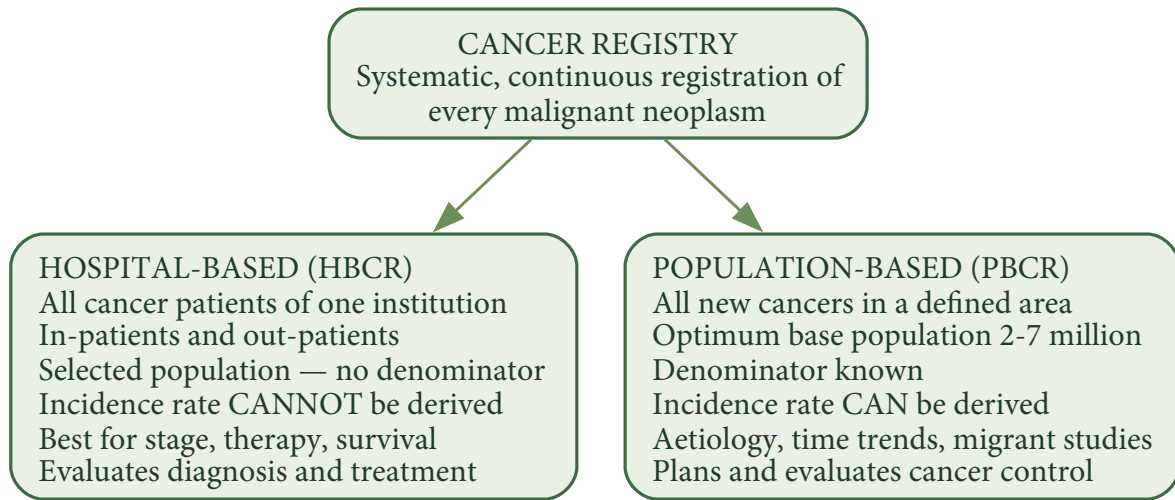
(Jugal Kishore, pp 584-585; Sunder Lal, Ch 8, p 524)

Structure and current position

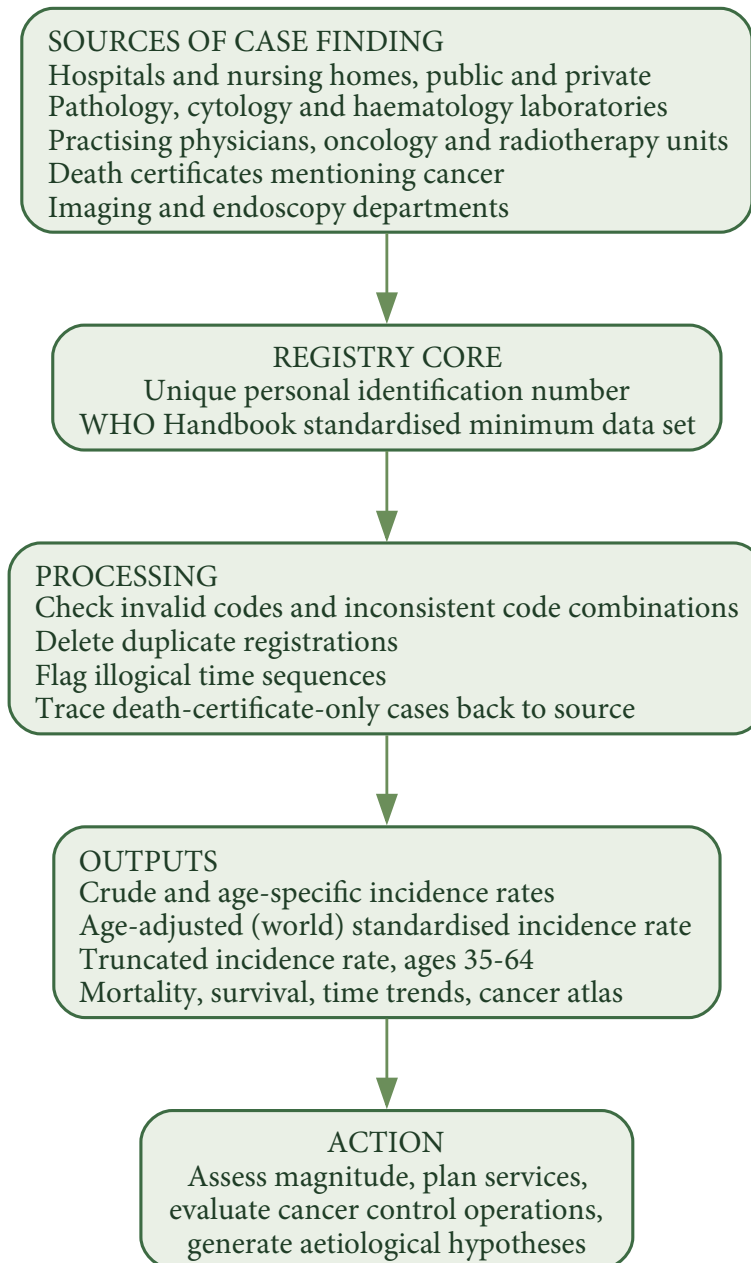
Item	Position
Started with	3 population-based registries (Bengaluru, Chennai, Mumbai) and 3 hospital-based registries (Chandigarh, Dibrugarh, Thiruvananthapuram), from 1 January 1982 (Jugal Kishore, p 585)
Present network	28 population-based and 5 hospital-based registries , supplying data on incidence, mortality and distribution of cancer (Park 28e, Ch 6, p 457)
Coordination	Operational wing at ICMR headquarters, Delhi, with technical wings at Delhi and Mumbai, which coordinate, review and monitor the registries (Jugal Kishore, p 585)
Population coverage	Low — quoted as 7.45% of the country's population by Jugal Kishore (p 585) and as 3.3% overall (urban 12.8%, rural 0.06%) by Sunder Lal (Ch 8, p 524)
Rural representation	Essentially one expanded rural registry at Barshi , Maharashtra (Sunder Lal, Ch 8, p 524)
Cancer Atlas of India	Developed precisely because extensive areas remain uncovered and the cancer picture in most rural and many urban areas is unknown (Jugal Kishore, p 585)
Burden it reports (2022)	Five-year prevalent cases about 3,258,518 ; new cases 1,413,318 ; deaths 916,827 (Park 28e, Ch 6, p 457)
Leading sites (2022)	Breast, cervix uteri and lip/oral cavity lead the age-standardised incidence; together breast, cervix and oral cavity account for about 34% of all cancers in India (Park 28e, Ch 6, p 457; Sunder Lal, Ch 8, p 503)
Programme it serves	National Cancer Control Programme, launched 1975-76, revised 1984-85 and again in December 2004, integrated in 2010 with the National Programme for Prevention and Control of Diabetes, Cardiovascular Disease and Stroke (NPCDCS); National Cancer Awareness Day is 7 November (Park 28e, Ch 7, p 566; Jugal Kishore, p 584)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

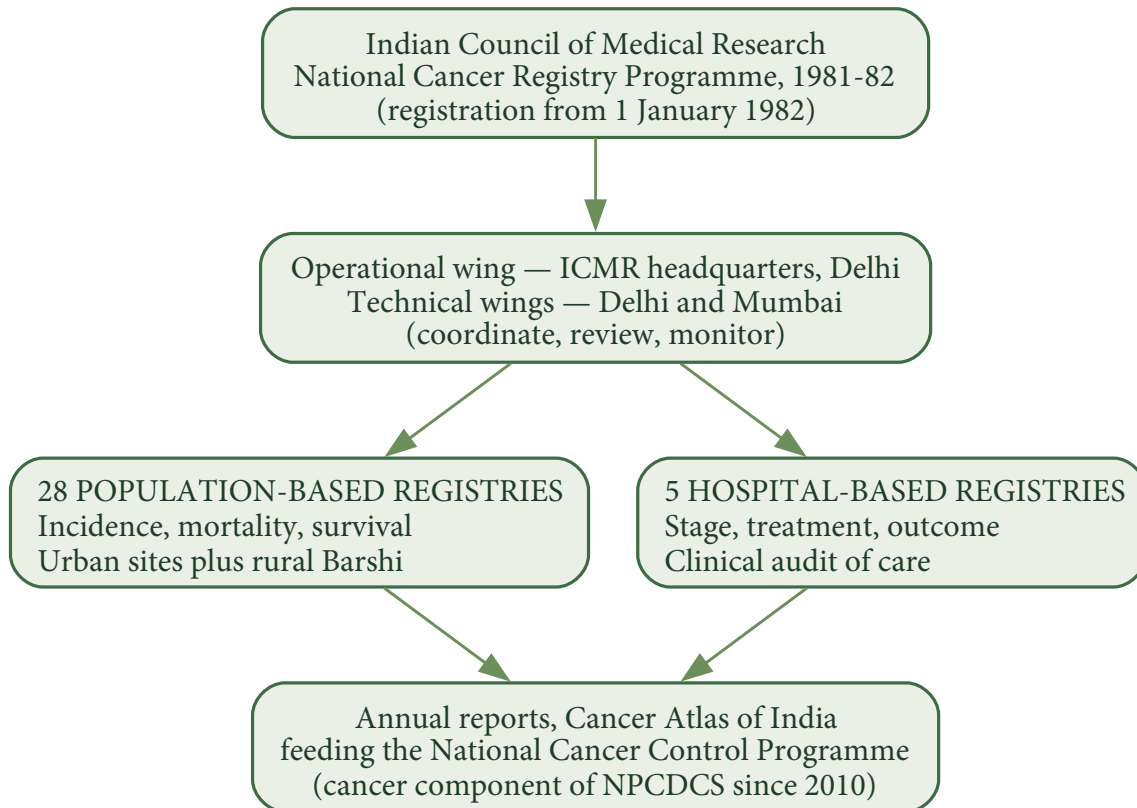
Diagram 1 — Hospital-based versus population-based registry, drawn as numerator and denominator. Labels to reproduce: a large outer rectangle labelled "Defined geographic population, 2–7 million = DENOMINATOR"; inside it a shaded circle labelled "All new cancers in the area = NUMERATOR → incidence rate obtainable (PBCR)"; overlapping it, a smaller circle drawn partly outside the rectangle labelled "Cancers treated at one hospital, in-patient + out-patient (HBCR)"; the part of the small circle lying outside the rectangle labelled "Patients from outside the area — assigned to the wrong population at risk"; a caption arrow to the small circle reading "No denominator → no incidence rate; but best data on stage, therapy and survival".



Classification of cancer registries and what each type can and cannot yield



Flow of data through a population-based cancer registry, from case finding to programme action



Organisation of the National Cancer Registry Programme of the Indian Council of Medical Research

High-yield exam pointers

- ▶ **Definition line to write first:** registration = permanent record + follow-up + tabulation of frequency and survival + availability for special studies; and cancer registration is a *sine qua non* for any cancer control programme (Park 28e).
- ▶ **The one number examiners look for:** optimum base population for a population-based cancer registry = 2–7 million (Park 28e, Ch 6, p 459).
- ▶ **The one-line discriminator:** HBCR has **no denominator**, so no incidence rate — it gives stage, therapy and survival; PBCR has a denominator, so it alone gives incidence.
- ▶ **Truncated incidence rate** = age-adjusted rate computed on age-specific rates in the 35–64 year band, adjusted to the world population; over 60% of registered cases fall there.
- ▶ **NCRP:** ICMR, 1981–82, registration from 1 Jan 1982; started with 3 PBCRs (Bengaluru, Chennai, Mumbai) + 3 HBCRs (Chandigarh, Dibrugarh, Thiruvananthapuram); now 28 PBCRs + 5 HBCRs; rural coverage rests essentially on **Barshi**.
- ▶ **India 2022 burden to quote with its year:** new cases 1,413,318; deaths 916,827; five-year prevalence about 3,258,518 (Park 28e, Ch 6, p 457).
- ▶ **Standard data set** = WHO Handbook for Standardized Cancer Registries; **international benchmark publication** = *Cancer Incidence in Five Continents*, issued every five years for registries meeting completeness and validity criteria.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the number of registries and their population coverage differ between the retrieved textbooks because each quotes a different year — Park 28e gives 28 population-based and 5 hospital-based registries, Sunder Lal gives 29 population-based registries with 3.3% total population coverage, and Jugal Kishore gives 25 with 7.45% coverage; write Park 28e's figures as the current position and, if the coverage percentage is asked, quote it with the source and year attached. The National Centre for Disease Informatics and Research (NCDIR), Bengaluru, which now runs the National Cancer Registry Programme for the Indian Council of Medical Research, is not named in the retrieved passages, so the answer keeps to the operational and technical wings described in the corpus. The retrieved text lists hospitals, pathology laboratories, practising physicians and death certificates as registry sources; imaging, endoscopy and radiotherapy departments are added as standard registry practice rather than as a quoted list.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, ICMR-NCDIR National Cancer Registry Programme, IARC GLOBOCAN 2022, MoHFW NP-NCD 2023-30 operational guidelines, WHO 90-70-90 cervical cancer elimination targets

Programme and policy updates (India)

- ▶ **NPCDCS has been renamed and re-scoped as NP-NCD** — the programme the textbooks call the National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Diseases and Stroke now runs as the **National Programme for Prevention and Control of Non-Communicable Diseases**, under revised operational guidelines issued in 2023 and covering 2023-30; the Ayushman Bharat Health and Wellness Centres that deliver the screening were renamed **Ayushman Arogya Mandir**. Under it the ASHA enumerates the eligible population aged 30-65 years using the Community-Based Assessment Checklist and mobilises them for screening once in five years by oral visual inspection, clinical breast examination and VIA. Use the new name in any answer on cancer control [PMID 40302063]
- ▶ **Cancer-care infrastructure attached to the registry-driven programme has expanded** — nationwide screening under NP-NCD, treatment financing through Ayushman Bharat PM-JAY, 19 State Cancer Institutes and 20 tertiary care cancer centres, and 200 district day-care cancer centres planned for 2025-26 [PMID 40833697]
- ▶ **Registries are being wired into the Ayushman Bharat Digital Mission** — the Ministry of Health has adopted HL7 FHIR R4 as the base exchange standard under ABDM, and an Indian oncology data model of 25 FHIR profiles and 50 terminology value sets (mapped to SNOMED CT and LOINC) has been built for breast, cervical, oesophageal, lung and oral cancer screening in the Meghalaya pilot (East Khasi Hills district), expressly to link screening programmes, hospitals and cancer registries into one interoperable network; existing records and registries are not yet FHIR-enabled and still need bridging software [PMID 41329957]

Current national data

- ▶ **The NCRP network is now 43 population-based registries, not 28** — 33 are run by ICMR's National Centre for Disease Informatics and Research (NCDIR), Bengaluru, which now operates

the programme, 9 by the Tata Memorial Centre, Mumbai, and 1 by the Tamil Nadu Cancer Registry Programme; together they cover about **18% of India's population** and reported 708,223 incident cases and 206,457 deaths for **2015-19**. In that observed 2015-19 data the lifetime risk of developing cancer in India was **11.0%**, Mizoram reached 21.1% in men and 18.9% in women, and Aizawl district recorded the country's highest age-adjusted incidence (males 256.1, females 217.2 per 100,000, world standard population). **Cancer Incidence in Five Continents Volume XII** carries data from 24 Indian registries [PMID 40833697]

- **India's estimated burden for 2024 is 1,562,099 new cases and 874,404 deaths** (crude incidence 110.3, crude mortality 61.7 per 100,000), superseding the 2022 figures of 1,413,318 cases and 916,827 deaths given in the body above; the 2024 estimate carries a cumulative risk of developing cancer up to age 74 of **11.3%** for both sexes (11.6% men, 11.1% women). Leading sites are mouth, lung and prostate in men and breast, cervix and ovary in women [PMID 40833697]
- **NFHS-5 (2019-21) quantifies how little screening actually happens** — among women aged 30-49 only **1.9%** had ever been screened for cervical cancer and **0.9%** for breast cancer, and among men of the same age only **1.2%** had ever been screened for oral cancer; Tamil Nadu and Kerala are well ahead of every other state [PMID 40302063]

Elimination, surveillance and disease status

- **Population-based survival is now a routine NCRP output, not just incidence** — the multi-registry analysis puts 5-year age-standardised relative survival for oral cancer at **37.2%** across 10 PBCRs (covering 2.9% of India's population), ranging from 58.4% in Ahmedabad urban down to 20.9% in Manipur, and 34.1% in rural against 48.5% in urban residents. **In pooled Kollam and Thiruvananthapuram data**, the only two registries judged complete enough for stage analysis, survival was 70.6% for localised, 37.4% for locoregional and 9.3% for metastatic disease, and in the same pooled data distant metastasis carried a hazard ratio of 3.95 (95% CI 2.78-5.60) and age 65 years or more a hazard ratio of 1.76 (95% CI 1.44-2.14) [PMID 40198071]
- **A surveillance blind spot worth naming in a registry answer** — a state-wide WHO STEPS survey in Tamil Nadu (2023-24, 4,184 women aged 30-69) found 10% ever screened for cervical cancer against a national figure of about 2%, but **59% of that screening was done in private facilities that NP-NCD routine reporting does not capture**, so programme data understate true coverage; referral for colposcopy was the main leak, with only 57% of screen-positive women referred. WHO's 90-70-90 elimination targets require 70% of women screened with a high-performance test by ages 35 and 45 and 90% of those with precancer or invasive cancer treated [PMID 41888339]

Guideline and legislative updates

- **Modelling favours HPV testing over VIA, but VIA remains the standardised NP-NCD test** — a Markov cost-utility analysis from ICMR-NCDIR comparing VIA, provider-collected HPV sampling and HPV self-sampling in women aged 30-65 found provider-collected HPV testing gave the most quality-adjusted life-years (5.97 on three-yearly screening against 4.71 for self-sampling and 4.01 for VIA); VIA was cost-saving but delivered the least health gain, and the authors conclude only that policy analysis is needed to reach the 70% coverage target — no programme-level switch has been made [PMID 40975493]. On the ground the national programme still screens all women aged 30 and above by VIA performed by a trained staff nurse

in government facilities, and it is private facilities that use Pap, HPV DNA or liquid-based cytology [PMID 41888339]

Global and WHO developments

- **GLOBOCAN 2022 (IARC) is the current global reference** — close to **20 million** new cancer cases and **9.7 million** cancer deaths worldwide in 2022, with lung cancer both the most frequently diagnosed cancer (12.4% of all cancers) and the leading cause of cancer death; roughly one in five people develop cancer in a lifetime, and **35 million** new cases are projected for 2050 [PMID 38572751]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 6 (Epidemiology of Chronic Non-communicable Diseases and Conditions), pp 457, 459-460; Ch 21 (Health Information and Basic Medical Statistics), pp 997-998; Ch 7 (Health Programmes in India), p 566. Handbook of Epidemiology (Ahrens & Pigeot) — Ch 4 (Use of Disease Registers), pp 240-247, 254-255; Ch 3 (Descriptive Studies), pp 168-169; Ch 35 (Cancer Epidemiology), pp 1392, 1418. Lilienfeld's Foundations of Epidemiology — Ch 7 (Morbidity Statistics), pp 134-138, 157. Sunder Lal — Textbook of Community Medicine, Ch 8, pp 503, 524-525. Jugal Kishore — National Health Programmes of India, pp 577, 584-585. Named documents: WHO Handbook for Standardized Cancer Registries; *Cancer Incidence in Five Continents* (IARC); International Epidemiological Association Dictionary of Epidemiology (Last), for the register-versus-registry distinction.

Recent-advances sources: PMID 40833697 — Cancer Incidence and Mortality Across 43 Cancer Registries in India, *JAMA Network Open* 2025, doi:[10.1001/jamanetworkopen.2025.27805](https://doi.org/10.1001/jamanetworkopen.2025.27805). PMID 40302063 — Current Status of Implementation of Cancer Screening Programme in India: A Review of Policies and Practice, *Asian Pacific Journal of Cancer Prevention* 2025, doi:[10.31557/APJCP.2025.26.4.1117](https://doi.org/10.31557/APJCP.2025.26.4.1117). PMID 40198071 — Geographic Disparities in Oral Cancer Survival From 10 Population-Based Cancer Registries in India, *JAMA Network Open* 2025, doi:[10.1001/jamanetworkopen.2025.3910](https://doi.org/10.1001/jamanetworkopen.2025.3910). PMID 41888339 — Cervical cancer screening coverage and care cascade in Tamil Nadu, India, based on a community-based cross-sectional survey, *Scientific Reports* 2026, doi:[10.1038/s41598-026-45005-2](https://doi.org/10.1038/s41598-026-45005-2). PMID 41329957 — FHIR Standard-Based Oncology Data Model for Cancer Screening: Design and Implementation Study, *JMIR Cancer* 2025, doi:[10.2196/79011](https://doi.org/10.2196/79011). PMID 40975493 — A comparison of the cost-effectiveness of HPV (self-sampling and health care provider sampling) versus VIA for cervical cancer screening in India, *Journal of Cancer Policy* 2025, doi:[10.1016/j.jcpo.2025.100642](https://doi.org/10.1016/j.jcpo.2025.100642) (Markov cost-utility **modelling** study, not a programme report; abstract-only, and the bullet is written to the abstract). PMID 38572751 — Global cancer statistics 2022: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries, *CA: A Cancer Journal for Clinicians* 2024, doi:[10.3322/caac.21834](https://doi.org/10.3322/caac.21834) (abstract-only). Named documents: MoHFW operational guidelines, National Programme for Prevention and Control of Non-Communicable Diseases (NP-NCD) 2023-30; NFHS-5 (2019-21); WHO 90-70-90 cervical cancer elimination targets; *Cancer Incidence in Five Continents* Volume XII (IARC).



Monitoring versus surveillance

Asked in: Paper I 21-Jun-2024 — RGUHS MD Community Medicine

Definition — In ordinary dictionaries the words *monitoring* and *surveillance* are almost synonymous, but in public health practice they have taken on rather specific and somewhat different meanings (Park 28e, Ch 2, pp 58-59). **Monitoring** is the performance and analysis of routine measurements aimed at detecting changes in the environment or in the health status of a population. **Surveillance** is the continuous, systematic collection, analysis, interpretation and feedback of health data, closely integrated with timely dissemination to those who have the right to know, so that control action can be taken.

Monitoring — meaning, scope and essential features

- ▶ **Epidemiological sense** — routine, repeated measurement of a defined variable to detect change in the environment or in the health status of a population; hence monitoring of **air pollution, water quality, growth and nutritional status** (Park 28e, Ch 2, pp 58-59).
- ▶ **Service sense** — on-going measurement of the performance of a **health service or a health professional**, or of the extent to which patients comply with or adhere to advice from health professionals.
- ▶ **Management sense** — "the **episodic oversight** of the implementation of an activity, seeking to ensure that input deliveries, work schedules, targeted outputs and other required actions are proceeding according to plan" (Park 28e, Ch 2, pp 58-59).
- ▶ It keeps track of **achievements, staff movements and utilisation, supplies and equipment, and the money spent in relation to the resources available**, so that if anything goes wrong **immediate corrective measures** can be taken.
- ▶ It requires **careful planning and the use of standardised procedures and methods of data collection**, and can then be carried out over extended periods by **technicians and automated instrumentation** (Park 28e, Ch 2, p 59).

Surveillance — meaning, elements and uses

- ▶ **Definition 1** — continuous analysis, interpretation and feedback of systematically collected data, generally using methods distinguished by their **practicality, uniformity and rapidity rather than by accuracy or completeness**; by observing trends in **time, place and persons**, changes can be observed or anticipated and appropriate action, including investigative or control measures, taken (Park 28e, Ch 2, p 59).
- ▶ **Definition 2** — systematic and continuous collection, analysis and interpretation of data, closely integrated with the **timely and coherent dissemination of the results** to those who have the right to know so that action can be taken; the final phase in the surveillance chain is the **application of information to health promotion and to disease prevention and control** (Park 28e, Ch 2, p 59).
- ▶ **Definition 3 (disease-control usage)** — "the **continuous scrutiny of all aspects of occurrence and spread of disease that are pertinent to effective control**"; surveillance goes beyond the passive reporting of cases and includes **laboratory confirmation of a presumptive diagnosis**, finding out the source of infection and routes of transmission, and identification of **all cases, susceptible contacts and others at risk** (Park 28e, Ch 3, p 151).

- **Components of a surveillance activity** — (a) collection of data; (b) compilation of data; (c) analysis and interpretation; (d) follow-up action; (e) **feedback** (Park 28e, Ch 7, p 569).
- **Prerequisites of effective surveillance** — use of a **standard case definition**, ensuring **regularity of reports**, and **action on reports** (Park 28e, Ch 7, p 569).
- **Sources of surveillance data** — mortality and morbidity reports based on death certificates, hospital records, general-practice sentinels or notifications; laboratory diagnosis; outbreak reports; **vaccine uptake and side effects**; sickness-absence records; changes in disease agents, vectors or reservoirs; and serological surveillance through **serum banks**, which can also be seen as an example of biological monitoring (Park 28e, Ch 2, p 59).
- Surveillance programmes can assume any character and dimension — hence **epidemiological, demographic and nutritional** surveillance (Park 28e, Ch 2, p 59).

Monitoring versus surveillance — point-by-point comparison

Point of difference	Monitoring	Surveillance
Core definition	Performance and analysis of routine measurements to detect change in environment or health status (Park 28e, Ch 2, pp 58-59)	Continuous collection, analysis, interpretation and feedback of systematically collected data so that action can be taken (Park 28e, Ch 2, p 59)
Periodicity	Intermittent or episodic (Park 28e, Ch 2, p 59)	Continuous and on-going (Park 28e, Ch 2, p 59)
Conceptual relation	Monitoring is one specific and essential part of the broader concept (Park 28e, Ch 2, p 59)	The broader concept , which embraces monitoring
What is watched	A defined input, process or measurement — air, water, growth, supplies, achievements, expenditure against resources	The occurrence and spread of disease or health event in time, place and persons
Personnel and skill	Carried out over extended periods by technicians and automated instrumentation using standardised procedures (Park 28e, Ch 2, p 59)	Requires professional analysis and sophisticated judgement of data leading to recommendations for control activities (Park 28e, Ch 2, p 59)
Data quality emphasis	Standardised, repeatable measurement to a defined procedure	Methods distinguished by practicality, uniformity and rapidity rather than accuracy or completeness
Immediate purpose	Detect deviation from plan so that immediate corrective measures can be taken (Park 28e, Ch 2, p 59)	Detect or anticipate changes and trends , and take investigative or control measures
Where the information goes	Corrective action is taken within the activity itself — the manager acts on what the measurement shows (Park 28e, Ch 2, p 59)	Timely and coherent dissemination to those who have the right to know is intrinsic to the definition (Park 28e, Ch 2, p 59)

Point of difference	Monitoring	Surveillance
Unit observed, in the worked paediatric example	All infants enrolled, starting before 6 months and continued monthly , in small groups of 10-20 (Park 28e, Ch 11, Table 39, p 793)	A representative sample , at representative ages at longer intervals , any size group, 50-100 most efficient (Park 28e, Ch 11, Table 39, p 793)
End product	Course correction of an on-going activity	Public health action — investigation, prevention and control

Suitable examples

Field	Example of monitoring	Example of surveillance
Child nutrition (the classic paired example — Park 28e, Ch 11, p 793)	Growth monitoring — oriented to the individual child , a dynamic measure of health from month to month; monthly weight plotted on the growth chart, weighed by the mother guided by the worker on a simple card emphasising growth; response is early home intervention based on local knowledge	Nutritional surveillance — "keeping watch over nutrition in order to make decisions that will lead to improvement in nutrition in population"; done on a representative sample of children of a village or area by a trained worker on a precise card; response is nutritional rehabilitation , often with supplements
Communicable disease	Weekly monitoring of case detection, drug and test-kit supplies and reporting completeness of a district programme against plan	Integrated Disease Surveillance Programme (IDSP) — weekly S, P and L reporting from sub-centres, primary health centres (PHCs), community health centres (CHCs), hospitals and medical colleges to detect early warning signals of impending outbreaks (Park 28e, Ch 7, pp 568-569)
HIV/AIDS	Monitoring the programme at facility level — service delivery, supplies and adherence at the antiretroviral therapy centre	HIV sentinel surveillance — annual cross-sectional unlinked anonymous testing to determine the level, trend and geographical spread of HIV infection and to estimate HIV prevalence and burden (Park 28e, Ch 7, pp 519-520)
Malaria	Monitoring of indoor residual spray rounds, bed-net distribution and rapid diagnostic test kit supply	Epidemiological surveillance of malaria using annual parasite incidence (API), annual blood examination rate (ABER), annual <i>falciparum</i> incidence (AFI), slide positivity rate (SPR) and slide <i>falciparum</i> rate (SFR) (Park 28e, Ch 7, p 498)
Environment	Air pollution and water quality monitoring by standardised instruments at fixed stations (Park 28e, Ch 2, p 58)	Surveillance for changes in disease agents, vectors or reservoirs as a source of data on factors influencing disease (Park 28e, Ch 2, p 59)
Immunisation	Monitoring of session-wise vaccine utilisation and of coverage achieved against the microplan	Surveillance of vaccine uptake and side effects (adverse events following immunisation) (Park 28e, Ch 2, p 59)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- Reproduce this plate bottom-up as **Diagram 1**: community informants and *Anganwadi* workers/ASHAs → multipurpose health worker at the sub-centre → health supervisor → PHC → CHC → district (hospitals reporting monthly by messenger or online) → state bureau of health intelligence (medical colleges reporting monthly by post or online) → Central Bureau of Health Intelligence (Sunder Lal, Ch 7, Fig 7.7, p 304).

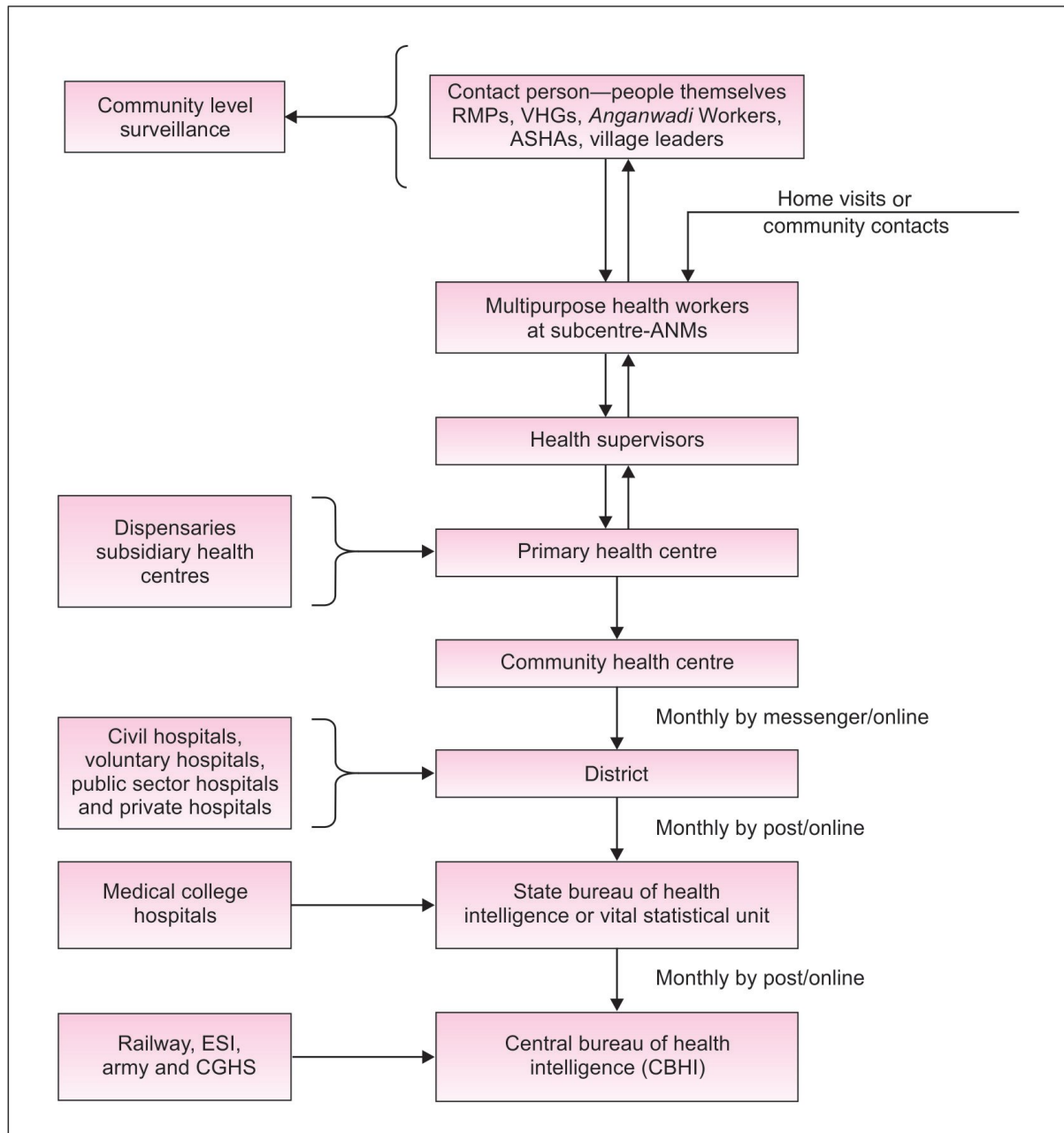
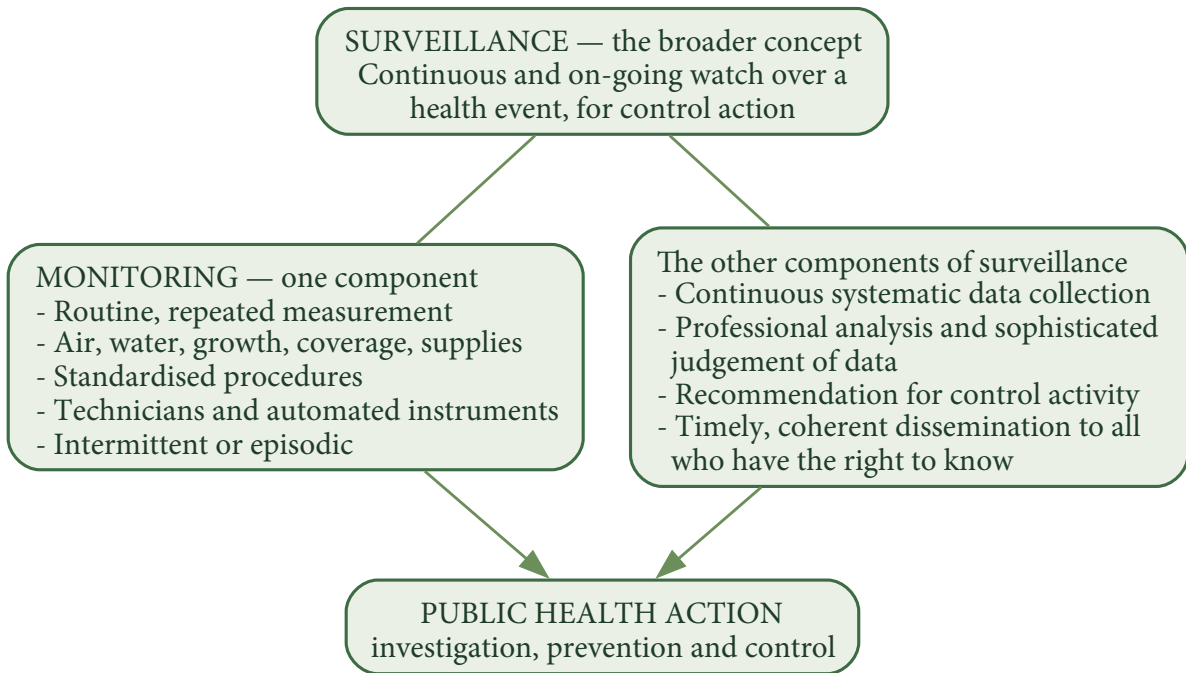
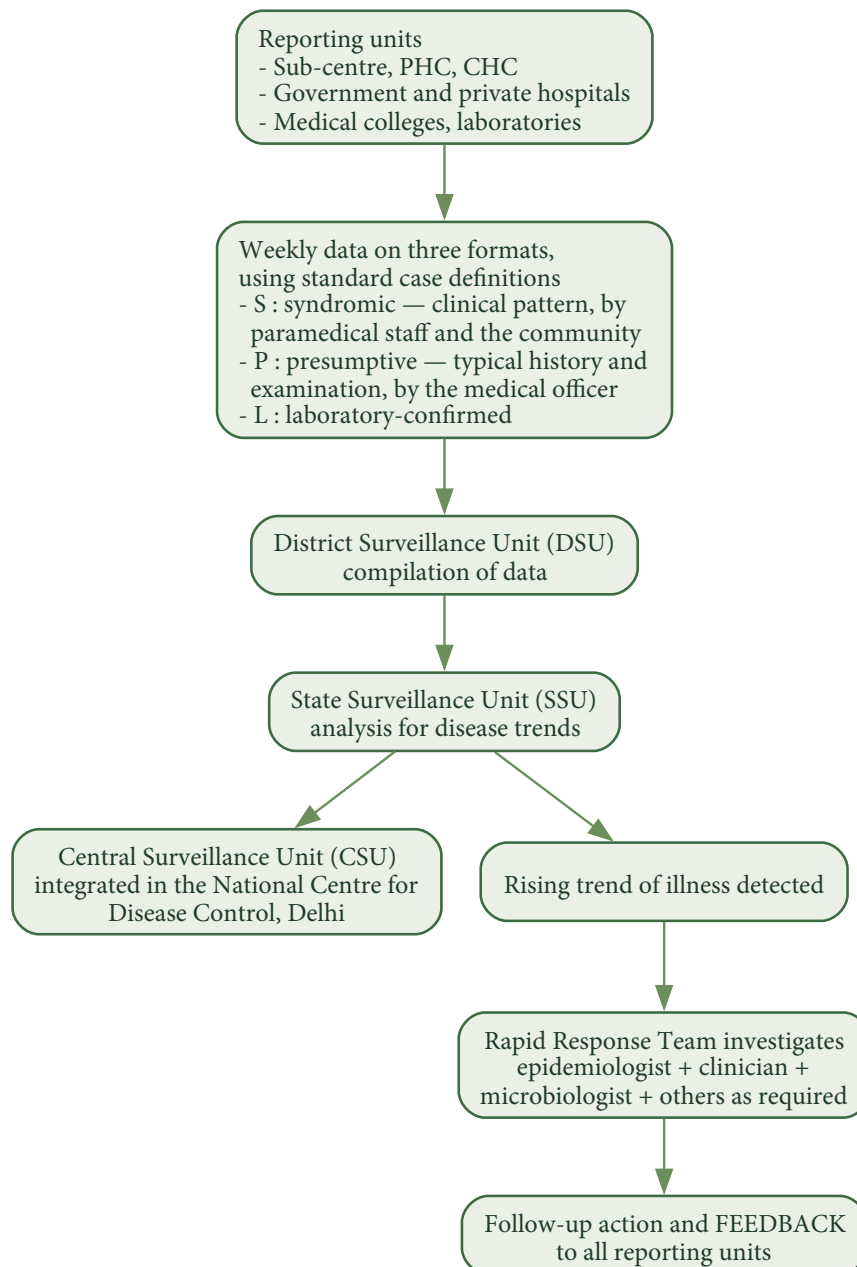


Fig 7.7 — Channel of information flow in community and institutional level surveillance (Sunderlal, p.304)



Monitoring is one specific and essential part of the broader concept embraced by surveillance



The surveillance chain in practice - Integrated Disease Surveillance Programme

High-yield exam pointers

- Open with the one-line discriminator: "**monitoring is intermittent or episodic, surveillance is continuous and on-going, and monitoring is one specific and essential part of the broader concept embraced by surveillance**" (Park 28e, Ch 2, p 59).
- Quote verbatim: surveillance uses methods distinguished by their **practicality, uniformity and rapidity rather than by accuracy or completeness** — the commonest single mark lost.
- Monitoring is done by **technicians and automated instrumentation**; surveillance needs professional analysis and sophisticated judgement leading to recommendations for control activities.

- Five components of surveillance: **collection** → **compilation** → **analysis and interpretation** → **follow-up action** → **feedback**. Three prerequisites: **standard case definition**, **regularity of reports**, **action on reports**.
- The paired example the examiner wants: **growth monitoring** (all infants, from before 6 months, monthly, individual child) versus **nutritional surveillance** (representative sample, representative ages, longer intervals) — Park 28e, Ch 11, Table 39, p 793.
- Three objectives of nutritional surveillance: aid **long-term planning** in health and development, provide input for **programme management and evaluation**, and give **timely warning and intervention to prevent short-term food consumption crises**.
- IDSP: launched **November 2004** as a 5-year project to March 2010, restructured and extended to March 2012; **CSU integrated in the National Centre for Disease Control, Delhi**, SSU at every state/union territory headquarters, DSU at every district; weekly **S, P and L** reports.
- Malaria surveillance parameters to write in full: **API, ABER, AFI, SPR, SFR**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park 28e records that the Integrated Disease Surveillance **Project** continues in the 12th Five Year Plan with a domestic budget as the Integrated Disease Surveillance **Programme** under the National Health Mission for all states (Park 28e, Ch 7, p 568), so write the current name first and the former name in brackets once; separately, the **Integrated Health Information Portal (IHIP)** — the step forward from the Health Management Information System — is being expanded to all states and union territories to connect all public health laboratories (Sunder Lal, Ch 20, p 805; Ch 8, p 532).

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, WHO International Health Regulations (2024 amendments), WHO Pandemic Agreement 2025, WHO South-East Asia Regional Office

Programme and policy updates (India)

- IDSP moved in **2018** from aggregate, paper-based weekly reporting to **real-time case-based reporting on the Integrated Health Information Platform**, and the surveillance arm is now formally written as **IHIP-IDSP**; a 2023-25 field study in two high-burden districts found dengue still flowing through **two parallel channels** with different outbreak criteria — **IHIP-IDSP** (threshold derived from the preceding 3-5 years of district-level data) and the vector-borne disease programme **NVBDCP (five or more positives in a village, or one dengue death)** — with only **13% of sanctioned IDSP district public health laboratories functional** and 4 of 36 districts lacking sentinel-hospital cover; in practice the NVBDCP criterion was the one applied by both programmes [PMID 42245358].
- Under the **Pradhan Mantri Ayushman Bharat Health Infrastructure Mission (PM-ABHIM, announced 25 October 2021)** four zonal ICMR National Institutes of Virology are being set up at **Bengaluru, Jammu, Dibrugarh and Jabalpur**; a 2024 four-zone prioritisation exercise with 186 stakeholders (28% of them from IDSP) ranked **dengue, influenza, measles, Japanese encephalitis and hepatitis A** as the top priority viral diseases for outbreak response [PMID 42522023].

Current national data

- ▶ Secondary analysis of the IDSP portal, **2010-2023: 1158 outbreaks** of ten selected zoonoses, **Japanese encephalitis/AES the commonest**; anthrax, JE/AES, Crimean-Congo haemorrhagic fever, leptospirosis and scrub typhus were reported through the year, while most other zoonoses clustered in the **July-September monsoon** — a worked illustration of surveillance data driving seasonal preparedness [PMID 41738949].
- ▶ For non-communicable disease, an integrated **surveillance-and-monitoring framework** has been proposed for India that fuses National Programme for Prevention and Control of Non-Communicable Diseases data, national surveys and electronic health records, using model-based **small-area estimation** to produce district-level estimates of the diabetes care continuum (awareness, treatment, control) [PMID 38815594].

Elimination, surveillance and disease status

- ▶ **Measles-rubella, WHO South-East Asia Region:** the 2013 goal of "measles elimination and rubella control by 2020" was revised in 2019 to **measles and rubella elimination by 2023**; as of 2023 five of eleven countries (Bhutan, DPR Korea, Maldives, Sri Lanka, Timor-Leste) were verified as having eliminated both. The Region completed the switch from **outbreak-based to laboratory-supported case-based fever-and-rash surveillance by 2022**, WHO-proficient laboratories rose from 23 (2013) to 58 (2023), reported measles incidence rose from 16.2 to 40.9 per million while rubella fell from 5.5 to 1.4 per million [PMID 39460260]; India's rubella elimination is now targeted for **December 2026**, and central-India serosurveillance during 2023-24 found rubella **IgM 7.2%** against **IgG 81.0%** (89.6% in pregnant women), i.e. continuing transmission despite good population immunity [PMID 42159053].
- ▶ **Kala-azar:** elimination is defined as fewer than **one VL case per 10,000 population at block level** in India. Malda district, West Bengal held that target throughout 2018-2022 with VL cases falling from 19 to 7, but **PKDL rose from 40 to 55** and formed the majority of reported cases by 2022 — post-elimination surveillance must now chase the **PKDL human reservoir**, not VL alone [PMID 42591813].
- ▶ **Malaria:** India carries close to half the global *Plasmodium vivax* burden; the Integrated Health Information Platform, **Tribal Malaria Action Plans** and the **Urban Malaria Scheme** are named as central to elimination, but molecular surveillance still cannot map relapse for want of a hypnozoite biomarker, and private-sector reporting and cross-border coordination remain the weak links [PMID 42307231].

Guideline and legislative updates

- ▶ The **2024 amendments to the International Health Regulations**, in force from **19 September 2025**, require every State Party to **establish or designate a national IHR authority** (new Article 4(1)bis) to coordinate implementation across sectors — in addition to the existing national IHR focal point — explicitly to remedy fragmented authority, thin legal mandate and weak intersectoral coordination [PMID 40953844].

Global and WHO developments

- ▶ The **77th World Health Assembly agreed the IHR amendments on 1 June 2024**, creating the new category of "**pandemic emergency**" that the Director-General may declare alongside a PHEIC, with equity provisions on access to health products (Article 13) and sustainable

financing for developing countries (Article 44); the **78th World Health Assembly adopted the WHO Pandemic Agreement (35 Articles) on 20 May 2025**, its Pathogen Access and Benefit-Sharing annex going to the 2026 Assembly [PMID 40589223].

- **Wastewater-based surveillance** is consolidating as a One Health early-warning complement to case reporting: a systematic review of 16 studies found pathogens in wastewater linked positively to clinical cases and gave early warning of outbreaks [PMID 42515017]; 2.25 years of Delhi monitoring (819 samples, 59.6% positive) showed wastewater SARS-CoV-2 titres leading clinical case trends by **one week** in January 2022 ($r = 0.867$, $p = 0.002$) and by **two weeks** in April 2023 ($r = 0.879$, $p = 0.001$) [PMID 41998553]; and in Vellore, sampling of **40 open-drainage sites** detected *Salmonella* Typhi at **26 of them (65%; overall sample positivity 11.4%)**, while blood culture captured only **11 cases** against an estimated seroincidence of **10.4 per 100 person-years** [PMID 40029872].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 2 (Concept of Health and Disease, pp 58-59, definitions of monitoring and surveillance, sources of surveillance data), Ch 3 (Principles of Epidemiology and Epidemiologic Methods, p 151, surveillance in disease control), Ch 7 (Health Programmes in India, p 498 malaria surveillance parameters, pp 519-520 HIV sentinel surveillance, pp 568-569 IDSP structure, formats, components and prerequisites of surveillance), Ch 11 (Nutrition and Health, p 793 and Table 39, nutritional surveillance versus growth monitoring); Sunder Lal — Textbook of Community Medicine — Ch 7 (Epidemiology, Fig 7.7, p 304, channel of information flow in community and institutional level surveillance), Ch 8 (p 532, Health Management Information System and the move to IHIP), Ch 20 (Recent Advances in Community Medicine, p 805, expansion of IHIP to all states/union territories).

Recent-advances sources: 42245358 — Informational challenges of dengue surveillance and response. A case study from India, *Frontiers in Public Health* 2026, doi:[10.3389/fpubh.2026.1809584](https://doi.org/10.3389/fpubh.2026.1809584); 42522023 — Strengthening India's pandemic preparedness with the four zonal institutes of virology, *Infectious Diseases of Poverty* 2026, doi:[10.1186/s40249-026-01484-z](https://doi.org/10.1186/s40249-026-01484-z); 41738949 — Trend of zoonotic diseases outbreaks in India: a secondary data analysis of Integrated Disease Surveillance Programme data, 2010-2023, *Transactions of the Royal Society of Tropical Medicine and Hygiene* 2026, doi:[10.1093/trstmh/trag011](https://doi.org/10.1093/trstmh/trag011) (abstract-only); 38815594 — Advancing diabetes surveillance ecosystems: a case study of India, *The Lancet Diabetes & Endocrinology* 2024, doi:[10.1016/S2213-8587\(24\)00124-4](https://doi.org/10.1016/S2213-8587(24)00124-4) (abstract-only); 39460260 — Progress towards Measles and Rubella Elimination in the South-East Asia Region 2013-2023, *Vaccines* 2024, doi:[10.3390/vaccines12101094](https://doi.org/10.3390/vaccines12101094); 42159053 — Monitoring Rubella Immunity and Transmission in Central India Amidst Elimination Efforts, *Microbiology and Immunology* 2026, doi:[10.1111/1348-0421.70065](https://doi.org/10.1111/1348-0421.70065) (abstract-only); 42591813 — Sustaining the Elimination of Visceral Leishmaniasis: A Secondary Data Analysis of the National Kala-Azar Elimination Programme in Malda District, West Bengal, India (2018-2022), *Cureus* 2026, doi:[10.7759/cureus.112553](https://doi.org/10.7759/cureus.112553); 42307231 — *Plasmodium vivax* malaria in India: microbiological barriers to diagnosis, treatment, and elimination, *Clinical Microbiology Reviews* 2026, doi:[10.1128/cmr.00025-26](https://doi.org/10.1128/cmr.00025-26) (abstract-only); 40953844 — National IHR authorities in the 2024 amendments to WHO's International Health Regulations, *BMJ Global Health* 2025, doi:[10.1136/bmjgh-2025-019284](https://doi.org/10.1136/bmjgh-2025-019284); 40589223 — The 2025 World Health Assembly Pandemic Agreement and the 2024 Amendments to the International Health Regulations, *Medical Science Monitor* 2025, doi:[10.12659/MSM.950411](https://doi.org/10.12659/MSM.950411); 42515017 — Wastewater Surveillance for Early Warning of Infectious Disease Outbreaks: A Systematic Review of Evidence and Implications for One Health Surveillance, *Pathogens* 2026, doi:[10.3390/pathogens15070690](https://doi.org/10.3390/pathogens15070690); 41998553 — Designing effective SARS-CoV-2 wastewater surveillance system: lessons learnt from sustained monitoring in Delhi, India, *BMC Infectious Diseases* 2026, doi:[10.1186/s12879-026-13330-6](https://doi.org/10.1186/s12879-026-13330-6); 40029872 — Wastewater surveillance for Salmonella Typhi and its association with seroincidence of enteric fever in Vellore, India, *PLOS Neglected Tropical Diseases* 2025, doi:[10.1371/journal.pntd.0012373](https://doi.org/10.1371/journal.pntd.0012373). Also: WHO International Health Regulations amendments (adopted 1 June 2024; in force 19 September 2025) and the WHO Pandemic Agreement (adopted 20 May 2025).



Bias in research studies

Asked in: Paper I May 2019 — RGUHS MD Community Medicine

Definition — Bias is any systematic error in the determination of the association between the exposure and the disease, so that the estimate of relative risk is pushed up or down, and it reflects some non-comparability between the study and the control groups (Park 28e, Ch 3, p 93). Gordis puts it as "any systematic error in the design, conduct or analysis of a study that results in a mistaken estimate of an exposure's effect on the risk of disease" (Gordis 7e, Ch 15, p 302). It is therefore a threat to the **internal validity** of the study first, and through internal validity to its external validity.

1. Bias, chance and confounding — the three rival explanations for any observed association

Bias is an error of *method*, not of chance: it is a fault built into the study, and it persists however large the study becomes.

Feature	Random error (chance)	Bias (systematic error)	Confounding
Nature	On average as likely to put the observed value on one side of the truth as the other (Fletcher, Ch 11, p 193)	Distorts the finding in one direction (Browner 5e, Ch 1, p 30)	Mixing or confusion of the effect of another factor with the effect of the study exposure (Handbook of Epidemiology, Ch I.9, p 389)
Where it enters	Sampling variation, measurement variation	Design, conduct or analysis	Non-comparability of subgroups within the source population (Handbook, Ch I.9, p 388)
Effect of enlarging the sample	Reduced — precision improves	No effect whatever (Browner 5e, Ch 1, p 30)	No effect
Handled by	Inferential statistics, confidence interval, power	Prevention at design and conduct stage	Design (randomisation, restriction, matching) or analysis (stratification, adjustment)
Measured by	<i>P</i> value, 95% confidence interval	Cannot be quantified from the study's own data	Difference between crude and adjusted estimate

- A *P* value answers only the question of chance. **Statistical association does not imply causation, and a *P* value is an inadequate although common end-point of case-control studies** — systematic error must be ruled out first (Park 28e, Ch 3, pp 92–93).
- Once confounding is set aside, the remaining biases are categorised by **whether they arise from the selection of study subjects (selection bias) or from their classification (information bias)**; most observational studies suffer from more than one form, and the effects of multiple biases compound the error (Handbook of Epidemiology, Ch I.9, p 389).

2. Selection bias — arises in choosing the participants

Selection bias concerns the way study participants are selected from the source population, so that an association is observed which does not exist in the source population (Handbook of Epidemiology, Ch I.9, p 388).

Type	How it arises	Classic example	Influence on the estimate
Selection bias proper	Cases and controls are not representative of cases and controls in the general population; systematic differences in characteristics between them (Park 28e, Ch 3, p 93)	Hospital controls used for a community disease	Relative risk or odds ratio raised or lowered unpredictably
<i>Berksonian bias</i> (Joseph Berkson)	Different rates of admission to hospital for people with different diseases, so hospital cases and hospital controls differ from their community counterparts (Park 28e, Ch 3, pp 93–94)	Any hospital-based case-control study	Spurious association between two conditions
Non-response bias	Data missing from eligible persons who are not invited or decline to participate; relevant when non-participants share a characteristic linked to their non-response (Gordis 7e, Ch 20, p 530)	Smokers refusing to join a cohort	If refusers are also at higher disease risk, association is diluted toward the null (Gordis 7e, Ch 8, pp 202–203)
Self-selection / volunteer bias, the <i>healthy worker effect</i>	Employment status is correlated with health status — the fit obtain and keep jobs, so workers show fewer adverse outcomes than the general population (Gordis 7e, Ch 20, p 500)	Occupational cohorts compared with the general population	Occupational hazard is underestimated
Loss to follow-up (withdrawal) bias	Differential attrition in a cohort study; if diseased persons are selectively lost, incidence in that arm falls (Gordis 7e, Ch 8, pp 202–203)	Long-term cohorts, migration, change of occupation	Either direction; Park advises a system built into the design to reach as close to 95 per cent follow-up as possible (Park 28e, Ch 3, pp 97–98)
Screening-evaluation biases — referral bias, length-biased sampling, lead-time bias, overdiagnosis bias	Screened and unscreened populations differ, and early detection lengthens measured survival without postponing death (Gordis 7e, Ch 18, p 368)	Comparing survival of screen-detected with symptom-detected cancers	Screening falsely appears beneficial

3. Information (measurement) bias — arises in classifying the participants

Information bias is the result of **misclassification of study participants with respect to disease or exposure status**, and refers to the people actually included in the study (Handbook of Epidemiology, Ch I.9, p 388). Gordis lists its sources as bias in abstracting records, bias in interviewing, bias from surrogate interviews, surveillance bias, recall bias and reporting bias (Gordis 7e, Ch 15, Box 15.1, pp 305–306).

Type	How it arises	Influence on the estimate
Memory or recall bias	Cases, having the disease, recall past exposures better than healthy controls — a mother of a malformed baby recalls a mild pregnancy infection that a control mother forgets (Park 28e, Ch 3, p 93; Gordis 7e, Ch 15, p 305)	Differential — creates a spurious association, or hides a real one
Interviewer's bias	The interviewer knows the hypothesis and who the cases are, and questions them more thoroughly (Park 28e, Ch 3, p 94)	Exposure over-ascertained in cases; odds ratio inflated
Reporting bias, including <i>wish bias</i>	The subject is reluctant to report a known exposure because of attitudes, beliefs and perceptions; the diseased subject answering "why me?" denies lifestyle exposures or over-reports workplace exposures (Gordis 7e, Ch 15, pp 306–307)	Differential; direction depends on the subject's motive
Surveillance bias	Exposed persons are watched more closely, so their outcomes are detected more often — physicians examined oral-contraceptive users more thoroughly for phlebitis (Gordis 7e, Ch 15, p 306)	Apparent excess risk in the exposed
Bias from surrogate interviews and record abstraction	The case is dead or too ill, so exposure is supplied by a spouse, close relative or friend who may not know it accurately; school, employment and medical records may be incomplete (Lilienfeld's Foundations, Ch 10, pp 219–220; Gordis 7e, Ch 15, p 306)	Differential whenever surrogates are needed more often for cases than for controls — the usual situation in a high-fatality case-control study such as pancreatic cancer, where a widow tends to raise her husband's occupational level and to convert him posthumously into a non-drinker and non-smoker (Gordis 7e, Ch 15, p 306). Non-differential only if surrogate reporting and record abstraction are applied equally to both groups and blind to case status

Type	How it arises	Influence on the estimate
Non-differential misclassification	Errors in classification occur equally in the groups compared, an inherent property of the data-collection method — most people under-report their own smoking, abortion or drug-use history (Gordis 7e, Ch 15, p 305; Lilienfeld's Foundations, Ch 10, p 220)	Predictable — bias toward the null ; the true effect is diluted and may be missed (Gordis 7e, Ch 20, p 495)
Differential misclassification	Error frequency differs between cases and controls, or between exposed and unexposed — spouses of cirrhosis patients over-report alcohol consumption while spouses of controls under-report it, inflating the odds ratio from both ends (Lilienfeld's Foundations, Ch 10, p 220)	Unpredictable — can manufacture an association or abolish one (Gordis 7e, Ch 15, p 305)
Subject variation, observer bias, bias in evaluation (in trials)	Participants feel better knowing they receive the new treatment; the investigator measuring outcome knows the allocation; the investigator subconsciously reports the outcome favourably (Park 28e, Ch 3, p 103)	Treatment effect exaggerated

4. Confounding, and the biases of analysis and reporting

Confounding — a confounding factor is one which is associated both with the exposure and with the disease, is distributed unequally in the study and control groups, and is — independently of its association with the exposure — itself a risk factor for the disease (Park 28e, Ch 3, p 92).

Example	Exposure studied	Confounder	Why it confounds
Alcohol and oesophageal cancer	Alcohol	Smoking	Associated with alcohol consumption <i>and</i> an independent risk factor for oesophageal cancer; the effect of alcohol emerges only when smoking is neutralised (Park 28e, Ch 3, pp 92–93)
Altitude and endemic goitre	High altitude	Iodine deficiency	Related both to altitude and to goitre, so it explains the statistical association wholly or in part — an <i>indirect association</i> (Park 28e, Ch 3, p 108)

Confounding differs from other bias in one practical respect: it can be dealt with **in the analysis**, provided the confounder was measured. Selection bias arising from selection on factors affected by the exposure cannot be repaired this way and should not be treated as confounding (Handbook of Epidemiology, Ch I.9, p 388). If the confounder is itself misclassified, attempts to control it will not fully control it and **may actually increase the bias** (Handbook, Ch I.9, p 389).

Bias of analysis and reporting — a fourth family, and a bias of **dissemination**, not of participant classification. **Reporting biases arise when the dissemination of research findings is influenced by the nature and direction of the results** (Cochrane Handbook, Ch 10, p 322). They act after the study is

finished, on which results and which analyses reach the reader, so they distort the pooled evidence rather than the single study's own estimate.

Type	What the nature and direction of the results influences (Cochrane Handbook, Ch 10, pp 323–324)
Publication bias	Whether the findings are published at all — Rosenthal's <i>file-drawer problem</i> : statistically significant "positive" results are more likely to be published
Time lag bias	How rapidly they are published
Multiple (duplicate) publication bias	Whether they are published once or several times over
Location bias	Whether they appear in a journal that is easily accessible and well indexed in standard databases
Citation bias	Whether others cite them
Language bias	Whether they are published in a particular language
Outcome reporting bias	Which of the study's outcomes are selectively reported and which are suppressed
Post-hoc hypotheses and data dredging	Which analyses are run and presented, hypotheses generated after seeing the data being passed off as <i>a priori</i> (Handbook of Epidemiology, Ch I.12, pp 473–474)

5. Influence of bias on the study and on public health practice

- **It destroys internal validity first.** If the effect measured within the study is wrong, no amount of generalisation is safe — bias threatens internal validity and, through it, external validity (Gordis 7e, Ch 15, p 302).
- **The direction of distortion is often predictable and should be stated in the discussion.** Non-differential misclassification biases toward the null and hides real effects; differential misclassification, selection bias and surveillance bias can move the estimate either way (Gordis 7e, Ch 15, pp 305–306).
- **It is not fixed by size, statistics or significance testing.** Increasing the sample size has no effect on systematic error (Browner 5e, Ch 1, p 30), and randomisation and sample size guard against neither observer bias nor bias in evaluation (Park 28e, Ch 3, p 103).
- **It is quantitatively large.** Trials with inadequate or unclearly reported allocation concealment gave effect estimates on average **18 per cent more "beneficial"** than trials with adequate concealment (95 per cent confidence interval 5 to 29 per cent), the exaggeration being confined to subjective outcomes (Cochrane Handbook, Ch 8, p 239).
- **It propagates.** A biased primary study enters systematic reviews and meta-analyses, and the reporting biases add a second layer on top, so that pooled evidence over-estimates benefit and the guidelines built on it are distorted (Cochrane Handbook, Ch 10, pp 322–323; Handbook of Epidemiology, Ch III.8, p 1187).

- **It misdirects public health action** — a falsely positive association triggers needless control measures and expenditure, while a falsely null one lets a genuine hazard continue; either way scarce resources and participants' goodwill are wasted.
- **Bias is a result of an error in the design or conduct of a study**, so the effort must go into reducing or eliminating it or, at the very least, recognising it and taking it into account when interpreting the findings (Gordis 7e, Ch 15, pp 306–307).

6. How bias can be reduced — stage by stage

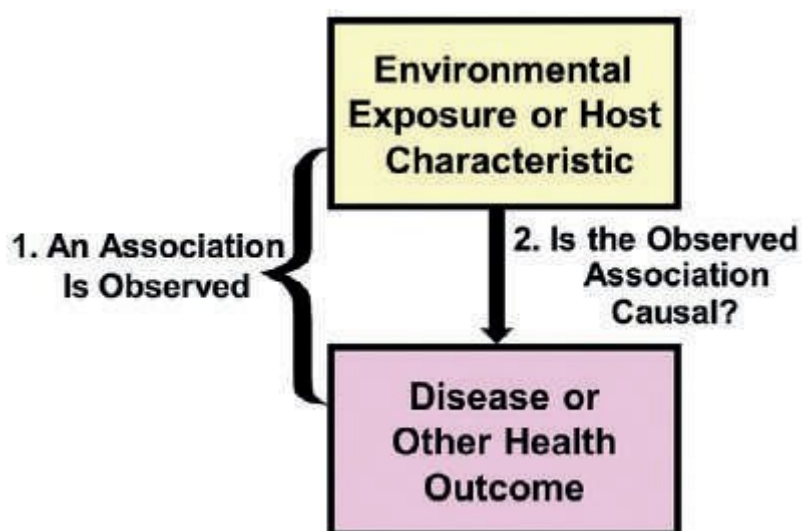
Stage	Measure	Bias controlled
Protocol	Write the protocol before enrolment: <i>a priori</i> hypotheses, precise operational definitions of case, exposure and outcome, and a pre-specified analysis plan; mark and report post-hoc hypotheses as such (Handbook of Epidemiology, Ch I.12, pp 473–474, 494–495)	Analysis and reporting bias
Design — sampling	Prevent selection bias rather than correct it; select controls comparable to cases, prefer population-based rather than hospital-based controls , and repeat the study in more than one geographic area — consistency raises validity (Park 28e, Ch 3, pp 92–93)	Selection and Berksonian bias
Design — allocation	Randomisation with concealment of the allocation sequence ; an unpredictable sequence plus concealment is sufficient to prevent selection bias in a trial (Cochrane Handbook, Ch 8, pp 218, 236)	Selection bias, confounding by known <i>and</i> unknown factors
Design — confounder control	Restriction (narrow the eligibility, e.g. to one age-sex group) or matching on variables known to influence the outcome (Park 28e, Ch 3, p 92; Handbook, Ch I.9, p 383)	Confounding
Conduct — measurement	Obtain exposure information in precisely the same manner for cases and controls (Park 28e, Ch 3, p 92); use validated instruments and objective sources; self-administered computer-assisted interviewing for sensitive questions (Gordis 7e, Ch 15, p 306)	Recall, interviewer and reporting bias
Conduct — masking	Blinding — <i>single</i> (participant unaware), <i>double</i> (participant and doctor unaware), <i>triple</i> (participant, investigator and analyst unaware); double blinding is the method most frequently used, and blinding is less essential when the outcome is death (Park 28e, Ch 3, p 103)	Subject variation, observer bias, bias in evaluation
Conduct — outcome ascertainment	Examine all members of the cohort equally , keep assessors unaware of group, and set careful written rules for deciding that an outcome event has occurred (Fletcher, Ch 7, p 122)	Surveillance and measurement bias
Conduct — follow-up	Build a system for obtaining basic outcome information on those who cannot be followed in full; aim for ≥ 95 per cent follow-up (Park 28e, Ch 3, pp 97–98)	Loss-to-follow-up bias

Stage	Measure	Bias controlled
Analysis	Stratification by categories of the confounder, and adjustment by regression modelling; analyse trials by intention to treat ; collect confounder data so they can be controlled later (Gordis 7e, Ch 20, p 530; Handbook, Ch I.12, pp 494–495)	Confounding
Reporting	Prospective trial registration , comprehensive searching including grey literature, and funnel plots where the number of studies allows — though an asymmetric funnel plot must not be equated with publication bias (Cochrane Handbook, Ch 10, p 322)	Publication and selective outcome reporting bias
Interpretation	Where bias is unavoidable, state its likely direction and magnitude ; seek a second sample from another setting to assess its importance (Browner 5e, Ch 1, p 30)	Residual bias

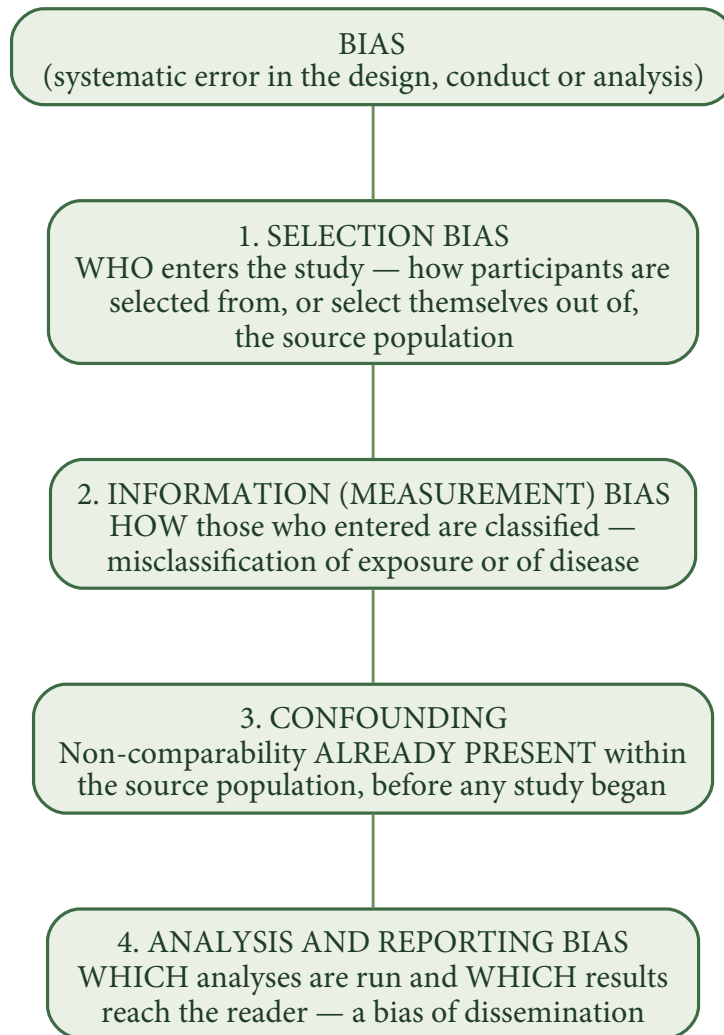
Two cautions on over-correction. Controlling a confounder at the design stage is **irreversible** and forbids the exploration of later hypotheses (Handbook, Ch I.12, pp 494–495). **Overmatching** — matching on a factor that is not itself a risk factor for the disease — needlessly makes the exposure distributions of cases and controls alike and reduces the efficiency of the estimate (Handbook, Ch I.6, pp 307–308), while **restriction** narrows generalisability, since a study limited to persons under 50 can say nothing about older people (Gordis 7e, Ch 20, p 495).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

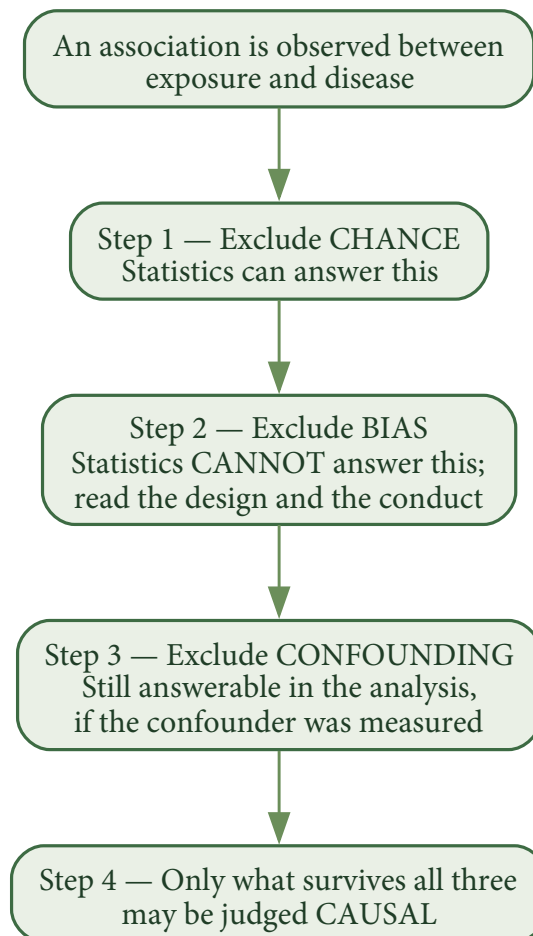
- **Diagram to sketch** — *The confounding triangle*: draw EXPOSURE and DISEASE as two boxes joined by a horizontal arrow labelled "association under study", and a third box CONFOUNDER above them with one arrow down to EXPOSURE labelled "associated with the exposure" and a second arrow down to DISEASE labelled "independent risk factor for the disease"; write beside it "and unequally distributed between the two groups".



An association is observed — is it causal? (Gordis Epidemiology 7e, p.196)



The four families of bias, named by the stage of the study at which each enters



Order in which the four explanations for an observed association are excluded

High-yield exam pointers

- ▶ **Definition to write first:** bias = any *systematic* error in the design, conduct or analysis that gives a mistaken estimate of the exposure's effect on disease risk (Park 28e, Ch 3, p 93; Gordis 7e, Ch 15, p 302).
- ▶ **The one-line discriminator:** chance is reduced by a bigger sample, bias is not — **increasing the sample size has no effect on systematic error** (Browner 5e, Ch 1, p 30).
- ▶ **Park's five biases of case-control studies** (write them in this order): confounding — memory/recall — selection — Berksonian — interviewer's (Park 28e, Ch 3, pp 93–94).
- ▶ **Direction rule:** *non-differential* misclassification → always **toward the null**; *differential* misclassification → either direction (Gordis 7e, Ch 15, p 305).
- ▶ **Trials — three biases, one antidote:** subject variation, observer, evaluation → **blinding**.
- ▶ **Confounding — five words:** randomise, restrict, match (design); stratify, adjust (analysis).
- ▶ **Quotable figures:** inadequate allocation concealment exaggerates the treatment effect by about 18 per cent (95% CI 5–29%) (Cochrane Handbook, Ch 8, p 239); aim for **≥ 95 per cent follow-up** in a cohort study (Park 28e, Ch 3, pp 97–98).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference, not to be written in the exam: some authors treat confounding as a form of bias and others keep it separate, because unlike selection and information bias a measured confounder can still be dealt with in the analysis — the RGUHS answer is safest if it presents confounding as the third of the three threats and says so explicitly.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 3 Principles of Epidemiology and Epidemiologic Methods, pp 92–94, 97–98, 103, 108); Gordis Epidemiology 7e (Ch 8 Cohort Studies, pp 202–203; Ch 15 More on Causal Inference: Bias, Confounding and Interaction, pp 302–307, incl. Box 15.1 and Fig 8.1 reproduced from p 196; Ch 18 Epidemiologic Approach to Evaluating Screening Programs, p 368; Ch 20 and the answers to the review questions, pp 495, 500, 530); Handbook of Epidemiology, Ahrens & Pigeot (Ch I.6 Case-Control Studies, pp 307–308; Ch I.9 Confounding and Interaction, pp 383, 388–389; Ch I.12 Design and Planning of Epidemiological Studies, pp 473–474, 494–495; Ch III.8 Clinical Epidemiology, p 1187); Lilienfeld's Foundations of Epidemiology (Ch 10 Observational Studies: Case-Control Studies, pp 219–220); Fletcher — Clinical Epidemiology: The Essentials (Ch 7 Prognosis, p 122; Ch 11 Chance, p 193); Browner — Designing Clinical Research, 5e, 2022 (Ch 1 Getting Started: The Anatomy and Physiology of Clinical Research, p 30); Cochrane Handbook for Systematic Reviews of Interventions (Ch 8 Assessing Risk of Bias in Included Studies, pp 218, 236, 239; Ch 10 Addressing Reporting Biases, pp 322–324).

♦ ♦ ♦

Mixed study designs

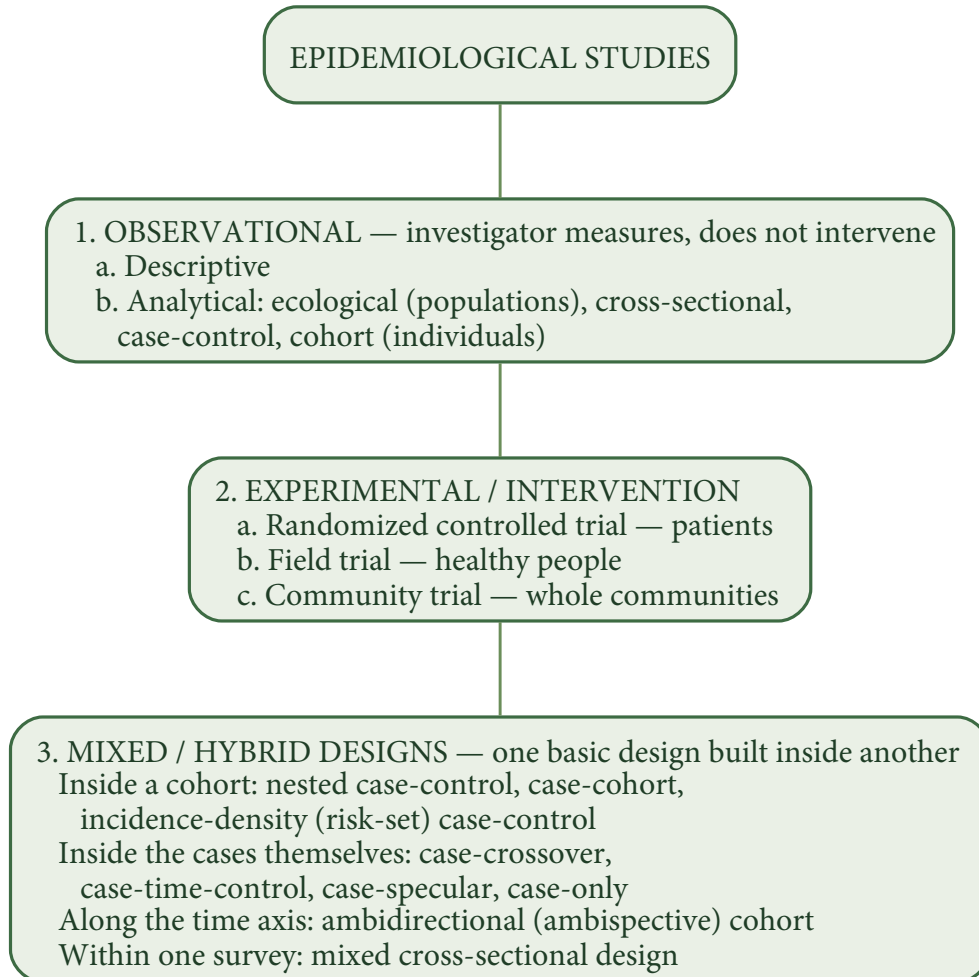
Asked in: Paper I 24-Sep-2025 — RGUHS MD Community Medicine

Definition — A **mixed (hybrid) study design** is an epidemiological design that deliberately combines the elements of two basic designs — most often the **cohort** and the **case-control** design — inside a single study, so as to keep the validity of the cohort (a fully enumerated study base, exposure recorded before the outcome occurs, calculable incidence) while buying the speed and economy of the case-control approach. Such designs "usually combine elements of the case-control and the cohort designs", by measuring the expensive exposure on only a sampled fraction of an enumerated cohort instead of on every member (Handbook of Epidemiology, Ch 12, p 484). The term is also used when a single **cross-sectional** study simultaneously carries current, retrospective and prospective components (Oxford Textbook of Public Health 7e, Ch 27, p 1003).

Where mixed designs sit in the classification of epidemiological studies

Park classifies epidemiological studies as **observational** (descriptive; and analytical — ecological, cross-sectional, case-control, cohort) and **experimental or intervention** studies (randomized controlled trial with patients, field trial with healthy people, community trial with communities) as the units of study

(Park 28e, Ch 3, p 82). Park adds the caveat that these "cannot be regarded as watertight compartments; they complement one another" (Park 28e, Ch 3, p 82). Mixed designs are the formal expression of that overlap: they are not a separate universe of studies but **basic designs nested one inside another** — a case-control study built inside a cohort, or a comparison built inside the case series itself.



Where mixed (hybrid) designs sit in the classification of epidemiological studies

Types of mixed study designs with suitable examples

Design	What it mixes	Comparison group	Effect measure	Suitable example
Nested case-control	Case-control study built inside an enumerated cohort	Controls = random sample of those still at risk at the moment each case is diagnosed (risk-set or incidence-density sampling), matchable to the case (Handbook of Epidemiology, Ch 7, p 333)	Odds ratio, an unbiased estimate of the incidence rate ratio (Handbook of Epidemiology, Ch 7, p 333)	<i>Levin et al.</i> — Epstein-Barr virus antibodies and multiple sclerosis in a cohort of over 3 million US military personnel who gave sera from 1988 onwards; 83 cases with a stored sample, 2 controls each matched on age, sex, race/ethnicity and date of blood collection; only these sera were assayed (Lilienfeld's Foundations of Epidemiology, Ch 10, pp 221-222)

Design	What it mixes	Comparison group	Effect measure	Suitable example
Case-cohort (case-base)	Cases from the whole cohort compared against a random subcohort drawn at baseline	Controls = random sample of the cohort at t_0 (sampling fraction p), chosen without regard to future outcome (Handbook of Epidemiology, Ch 7, pp 329-330)	Cumulative incidence / risk ratio; also allows absolute risk (Hunter's Diseases of Occupations, Ch 85, p 1132)	Baseline serum vitamin D in cohort members who later died of prostate cancer compared with levels in a random sample of the whole cohort (Browner, Designing Clinical Research 5e, Ch 20, pp 675-676)
Incidence-density (risk-set) case-control	Cohort follow-up time + case-control sampling	Controls sampled in proportion to person-time at risk	Incidence rate ratio	Used when risk factors change over time and there is loss to follow-up during a long cohort (Browner 5e, Ch 9, pp 242-243)
Case-crossover	Crossover trial + case-control, but observational — each case is its own control	The same case at an earlier "control" time period; the participant, not the investigator, determines exposure (Handbook of Epidemiology, Ch 7, p 336)	Matched-pair odds ratio (conditional logistic regression)	<i>McEvoy et al.</i> — mobile phone use in the 10 minutes before a car crash compared with the same driver's use at the same time of day 24 h, 72 h and 7 days earlier; phone use was about four times more likely just before the crash (Browner 5e, Ch 9, p 263)
Case-time-control	Case-crossover + a control series	Adds control subjects to remove time trends in exposure (Suissa 1995), used in pharmacoepidemiology where confounding by indication is a problem (Handbook of Epidemiology, Ch 30, p 1238)	Ratio of case and control odds ratios	Study of an acute adverse drug event where prescribing of the drug is itself changing over calendar time
Case-specular	Case series + a <i>counterfactual</i> (imaginary) control	A "specular" residence created by reflecting the case's house across the centre of its street, so the neighbourhood is matched perfectly (Handbook of Epidemiology, Ch 7, pp 342-343)	Matched-pair odds ratio over wire-code categories	Power-line wire codes as a surrogate for electromagnetic fields and childhood cancer — removes confounding by neighbourhood characteristics (Handbook of Epidemiology, Ch 7, pp 342-344)

Design	What it mixes	Comparison group	Effect measure	Suitable example
Case-only (genetic)	Cases alone, using the independence of genotype and environment	No external controls at all; gene-environment interaction estimated within cases (Handbook of Epidemiology, Ch 7, §7.7, p 344)	Interaction odds ratio	Genetic epidemiology studies of gene-environment interaction
Ambidirectional (ambispective) cohort	Retrospective and prospective cohort in one study	The same cohort, assembled from past records, assessed to date and then followed forward (Park 28e, Ch 3, p 96)	Relative risk	<i>Court-Brown and Doll (1957)</i> — a cohort assembled in 1955 of 13,352 patients given large doses of radiotherapy for ankylosing spondylitis between 1934 and 1954; deaths from leukaemia and aplastic anaemia in 1935-54 were far above the general population, and the same cohort was then followed prospectively from 1955 (Park 28e, Ch 3, p 96)
Mixed cross-sectional design	One survey carrying current, past and future measurements	Within the same survey	Prevalence odds ratio; becomes a relative risk if the survey is followed up	A survey that measures a current biomarker (vitamin E), asks about health-service use in the past year, identifies old and new asthma cases, and then follows the participants up — converting itself into a cohort (Oxford Textbook of Public Health 7e, Ch 27, p 1003)

Nested case-control study — the prototype mixed design

- ▶ **Assemble and enumerate a cohort.** Record baseline exposure data and bank the biological specimens (frozen sera, urine, images, records) on everyone at entry (Gordis 7e, Ch 8, p 207).
- ▶ **Follow the cohort forward** until a sufficient number of incident cases of the outcome accrue.
- ▶ **Define the cases** — all cohort members who develop the outcome during follow-up.
- ▶ **Sample the controls from the same cohort**, at the time each case is diagnosed, from those still at risk (the risk set); one or several controls per case, matchable on age, sex and date of specimen collection (Handbook of Epidemiology, Ch 12, pp 482-483).
- ▶ **Thaw and assay only the sampled specimens** — laboratory work is done on a few hundred rather than the whole cohort, and the cost falls dramatically (Gordis 7e, Ch 8, p 207).
- ▶ **Analyse as a matched case-control study**; the odds ratio estimates the incidence rate ratio.

Why it works. Recall bias is eliminated because exposure was recorded before the disease appeared; the exposure measurement reflects the true **pre-illness state** rather than an effect of subclinical disease, which matters for cancers with long asymptomatic phases such as prostate cancer and chronic lymphocytic leukaemia; and because cases and controls come from the same cohort, comparability is

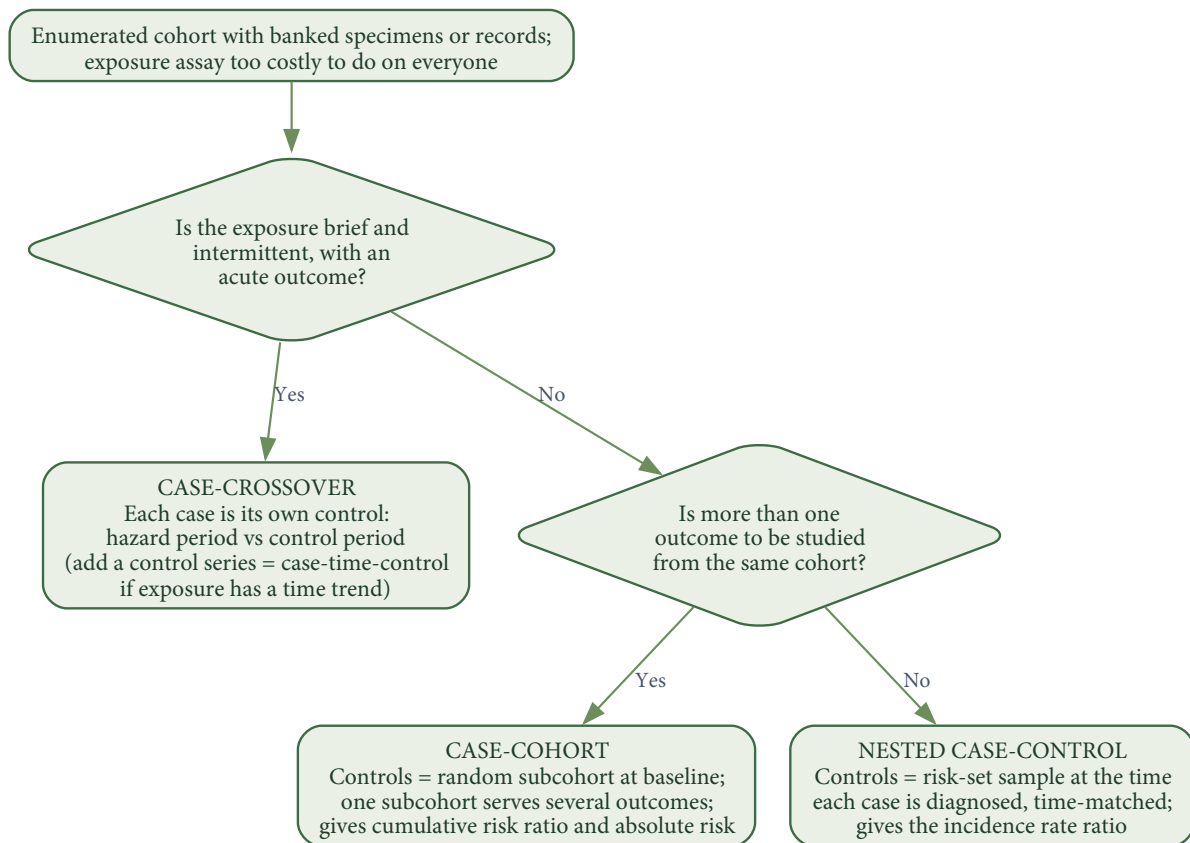
far better than in a hospital-based case-control study (Gordis 7e, Ch 8, p 207; Ch 9, p 211). Selection bias is minimised for the same reason (Lilienfeld's Foundations, Ch 10, pp 221-222).

Nested case-control versus case-cohort — the examiner's discriminator

Feature	Nested case-control	Case-cohort
When controls are chosen	At the time of diagnosis of each case	Once, at baseline (t_0), before any case occurs
Matching on follow-up time	Yes — controls matched to the case on time at risk	No — a case-cohort study is essentially an unmatched nested case-control study (Handbook of Epidemiology, Ch 7, pp 329-330)
Can a control later become a case?	Yes — controls may develop the disease after serving as controls	Yes — the subcohort contains future cases
Number of outcomes studied	Usually one; a new control set is needed for each outcome	The same subcohort serves several different outcomes — a major efficiency (Handbook of Epidemiology, Ch 6, p 303)
Estimating population exposure/genotype frequency	Not suitable	Suitable — the subcohort is a simple random sample of the full cohort (Handbook of Epidemiology, Ch 6, p 303)
Effect measure	Incidence rate ratio (density sampling)	Cumulative risk ratio; absolute risk also obtainable

Advantages and limitations of mixed designs

Advantages	Limitations
Exposure measured before the outcome — no recall bias, correct temporal sequence (Gordis 7e, Ch 9, p 211)	Exposure data or specimens must still be collected on the entire cohort at baseline, so a cohort must already exist (Handbook of Epidemiology, Ch 7, p 334)
Cases and controls come from the same study base — selection bias minimised, comparability high (Gordis 7e, Ch 8, p 207)	Shares the cohort's limitations: long follow-up, differential loss to follow-up can still bias the result (Handbook of Epidemiology, Ch 7, p 334)
Cost-efficient — costly assays, record abstraction or database linkage are done only on cases and a sample of non-cases (Fletcher, Clinical Epidemiology, Ch 6, p 102)	Analysis is more complex — conditional logistic regression, risk-set sampling, weighted estimators
Yields the full set of risk measures — absolute risk, relative risk, attributable risk and population risk (Fletcher, Ch 6, p 102)	Case-crossover-type designs assume no period or carry-over effect , so they suit only brief, intermittent exposures with acute outcomes (Handbook of Epidemiology, Ch 7, p 336)
Self-controlled designs abolish confounding by all time-invariant host factors — genes, past nutrition — because each case is its own stratum (Handbook of Epidemiology, Ch 7, p 342)	Case-specular and case-only designs rest on untestable assumptions (symmetry; gene-environment independence)



Choosing a mixed design once an enumerated cohort with banked specimens exists

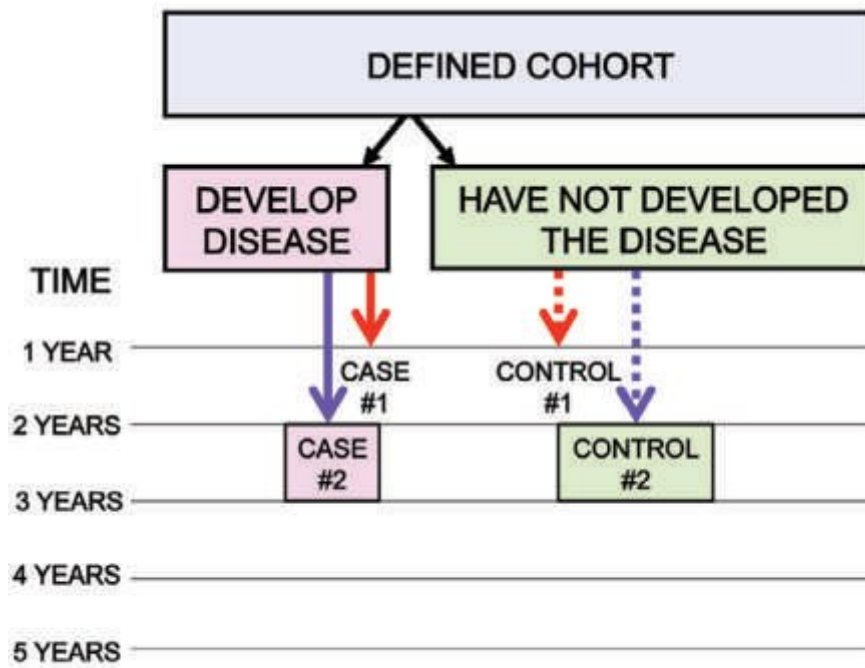
Mixed-methods designs — the second reading of the term

If "mixed" is taken to mean **mixed methods** rather than mixed epidemiological designs, it denotes designs that "collect, analyse and merge both quantitative and qualitative data" in one study (Oxford Textbook of Public Health 7e, Ch 52, pp 1937-1939).

- ▶ **Role of each strand** — qualitative methods identify strategies for delivering an intervention, or explain *why* implementation succeeded or failed; quantitative methods test the hypothesis and measure the predictors of success (Oxford Textbook of Public Health 7e, Ch 52, pp 1937-1939).
- ▶ **Rationale** — *triangulation*, described by Denzin (1978) in four forms — data-source, researcher, theory and methodological triangulation — with *interdisciplinary* triangulation added by Janesick (1994); triangulation increases the **credibility** of the findings (Oxford Textbook of Public Health 7e, Ch 31, pp 1158-1159).
- ▶ **Suitable example** — an implementation-research evaluation of a national programme in which coverage and outcome indicators are measured quantitatively, while focus-group discussions and in-depth interviews with beneficiaries, providers and administrators assess acceptability and appropriateness of the service (Oxford Textbook of Public Health 7e, Ch 52, pp 1937-1939).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Nested case-control design** — draw one box at the top labelled *DEFINED COHORT*, splitting into two boxes: *DEVELOP DISEASE* and *HAVE NOT DEVELOPED THE DISEASE*. Down the left margin mark a time axis: *1 YEAR, 2 YEARS, 3 YEARS, 4 YEARS, 5 YEARS*. At year 1 mark *CASE #1* on the left with an arrow across to *CONTROL #1* on the right; at year 2 mark *CASE #2* with an arrow across to *CONTROL #2*. Add the caption "*controls sampled from those still at risk at the time each case occurs*".
- **Case-crossover design** — draw a single horizontal time line for one case. Mark the event at the right end labelled *OUTCOME* (e.g. *road-traffic crash / myocardial infarction*). Immediately before it shade a short block labelled *HAZARD (case) PERIOD* — *exposure in the 10 min before the event*. Further to the left shade one or more equal blocks labelled *CONTROL PERIOD(S)* — *same driver, same time of day, 24 h / 72 h / 7 days earlier*. Add the caption "*each case is its own control*".



Design of a hypothetical nested case-control study: steps in selecting cases and controls (Gordis Epidemiology 7e, Ch 8, p 205)

High-yield exam pointers

- Definition line to write first: a mixed/hybrid design combines elements of the cohort and the case-control design in one study — cohort validity at case-control cost (Handbook of Epidemiology, Ch 12, p 484).
- The single discriminator between the two core hybrids: nested case-control = controls sampled at the time of each case's diagnosis (time-matched, risk-set sampling → incidence rate ratio); case-cohort = controls are a random subcohort at baseline (unmatched → cumulative risk ratio, and one subcohort serves several outcomes).

- ▶ Three named examples worth memorising: **EBV and multiple sclerosis** in 3 million US military personnel (nested case-control); **mobile phone use and car crash**, about 4-fold, comparing the 10 min before the crash with 24 h/72 h/7 days earlier (case-crossover); **Court-Brown and Doll 1957**, 13,352 ankylosing spondylitis patients irradiated 1934-54 and followed on from 1955 (ambidirectional cohort).
- ▶ Why the mixed design beats a plain case-control study: **no recall bias** (exposure recorded before disease), exposure reflects the **pre-illness state**, and cases and controls share one study base.
- ▶ Case-crossover is valid only for a **transient, intermittent exposure with an acute outcome** and assumes no period or carry-over effect; its extension for exposures with a secular time trend is the **case-time-control** design (Suisse 1995).
- ▶ Park's own words for the classification: the study types "cannot be regarded as watertight compartments; they complement one another" (Park 28e, Ch 3, p 82) — this sentence justifies the whole answer.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park's Textbook 28e does not use the heading "mixed study designs" — it covers only the combined retrospective-plus-prospective (ambidirectional) cohort under types of cohort studies, so the remaining hybrid designs in this answer are grounded on Gordis 7e, the Handbook of Epidemiology, Lilienfeld's Foundations, Fletcher and Browner; the phrase "mixed designs" itself is used in the Oxford Textbook of Public Health 7e (Ch 27, p 1003).

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 3 — Principles of Epidemiology and Epidemiologic Methods); Gordis Epidemiology 7e (Ch 7 Observational studies, Ch 8 Cohort studies, Ch 9 Comparing cohort and case-control studies); Handbook of Epidemiology, Ahrens & Pigeot (Ch 6 Case-Control Studies, Ch 7 Modern Epidemiologic Study Designs, Ch 12 Design and Planning of Epidemiological Studies, Ch 30 Pharmacoepidemiology); Lilienfeld's Foundations of Epidemiology (Ch 10 — Observational Studies: Case-Control Studies); Fletcher — Clinical Epidemiology: The Essentials (Ch 6 — Risk: From Disease to Exposure); Browner — Designing Clinical Research 5e, 2022 (Ch 9 — Designing Case-Control Studies); Hunter's Diseases of Occupations (Ch 85); Oxford Textbook of Public Health 7e (Ch 27 Cross-sectional studies, Ch 31 Qualitative research imagination, Ch 52 Implementation science and translational public health).



Adverse events following immunization surveillance

Asked in: Paper I 26-May-2025 — RGUHS MD Community Medicine

Definition — An **adverse event following immunization (AEFI)** is any untoward medical occurrence which follows immunization and which does not necessarily have a causal relationship with the usage of the vaccine; it may be any unfavourable or unintended sign, abnormal laboratory finding, symptom or disease (Park 28e, Ch 3, p 135). **AEFI surveillance** is the continuous and systematic detection, reporting, investigation, causality assessment and response to such events. In India it runs inside the Universal Immunization Programme (UIP) under the *AEFI Surveillance and Response Operational Guidelines 2024* of the Ministry of Health and Family Welfare; sustaining robust AEFI surveillance is a stated UIP objective (Routine Immunization Manual for Medical Officers 2024, Ch 1, p 29; Ch 7, pp 167–169).

Types of AEFI used for reporting

Type	Defining features	Examples	Action required
Minor	Common, self-limiting; occur within a few hours of injection; resolve over a short period; pose little danger	Pain, swelling, redness at injection site (about 10% of vaccinees, up to 50% with pentavalent or tetanus–adult diphtheria booster); fever (about 10%, about 50% with pentavalent); irritability, malaise; fever, rash and/or conjunctivitis in 5–15% after measles–rubella	Reassure, paracetamol; enter in AEFI register, SAFE-VAC and HMIS. No Case Reporting Form, no case investigation
Severe	Clinically more severe than minor, may or may not be life-threatening; must not have been admitted (treated in outpatient/day care) and must have recovered completely without sequelae ; most leave no long-term problem	Seizures, hypotonic hyporesponsive episode (HHE), persistent screaming, thrombocytopenia, high-grade fever (102°F) treated without admission, injection-site abscess drained in day care, vasovagal syncope managed at the session site	Report immediately , investigate, causality assessment (State/National AEFI committee, by Day 100)
Serious	Death; inpatient hospitalization (any admission, whatever the clinical severity); persistent or significant disability (including a non-hospitalized case that recovered with sequelae); AEFI cluster; congenital anomaly/birth defect; parental, community or media concern	Anaphylaxis needing admission; abscess admitted for treatment; death after vaccination; a cluster of fainting among schoolchildren in a measles–rubella campaign	Report immediately , investigate, causality assessment (State/National AEFI committee, by Day 100)

The severe-versus-serious split decides the reporting category and the record kept, **not** whether the case is worked up: both are reported immediately without waiting to categorise them, both are investigated, and cause-specific categorization after causality assessment applies to serious *and* severe AEFIs (Routine Immunization Manual 2024, Ch 7, pp 169–173, 181).

Cause-specific categorization of AEFIs

Assigned **after** causality assessment, using the classification revised by the Council for International Organizations of Medical Sciences (CIOMS) and WHO in 2012 and updated in 2018.

#	Category	Definition	Example
1	Vaccine product-related reaction	Caused or precipitated by a vaccine due to one or more inherent properties of the vaccine product, even when correctly prepared, handled and administered	Extensive limb swelling after DTP (diphtheria–tetanus–pertussis) vaccination
2	Vaccine quality defect-related reaction	Caused by one or more quality defects of the vaccine product, including its administration device as supplied by the manufacturer; now very rare with good manufacturing practices	Failure by the manufacturer to fully inactivate a lot of inactivated polio vaccine, causing paralytic polio (<i>Cutter incident</i>)
3	Immunization error-related reaction	Caused by inappropriate vaccine handling, prescribing or administration; preventable by nature	Transmission of infection by a contaminated multidose vial; local reactions after vaccine frozen in transport; failure to protect due to loss of potency
4	Immunization stress-related response (formerly immunization anxiety-related reaction)	Arises from anxiety about the injection, unrelated to the content of the vaccine; common in adults, adolescents and children over 5 years and in mass campaigns	Fainting, breath-holding, light-headedness, tingling of hands and around the mouth
5	Coincidental event	Caused by something other than the vaccine product, immunization error or immunization stress; only a temporal association	Fever at the time of vaccination that is in fact due to malaria

(Park 28e, Ch 3, pp 135–136, 141; Routine Immunization Manual 2024, Ch 7, pp 181–184)

Reporting

- **Every AEFI after any vaccination must be reported, and there is no upper time limit for reporting** — an AEFI is reportable whatever the interval between vaccination and the event, or between the event and the report, and whether the vaccine is in the UIP schedule or not, given in a government or a private setting, to a child or an adult.
- **The digital rail is U-WIN.** All AEFIs are entered into **SAFE-VAC** (Surveillance and Action For Events following Vaccination) through **U-WIN**, the national immunization platform: the vaccinator or District Immunization Officer (DIO) uses the "Report AEFI" button against the beneficiary, which opens the **Case Notification Form (CNF)** — the first form in the chain. The DIO categorises the case (minor/severe/serious) and verifies it; the **Case Reporting Form (CRF)** is then initiated, the **Case Investigation Form (CIF)** is generated automatically for the DIO to

complete, and the **Causality Assessment Form** is submitted by the State user (Routine Immunization Manual 2024, Ch 7, Fig 7.4, p 175 and Unit 7 summary, pp 192–193; Ch 8, p 202).

- ▶ A **severe or serious** AEFI is notified immediately, by telephone or in person, to the medical officer or nearest government health facility. The medical officer visits, completely fills **Section A of the CRF** and submits it to the DIO **within 24 hours** of notification; the DIO countersigns it and uploads the scanned form. Events after COVID-19 vaccines go through the Co-WIN-linked SAFE-VAC.
- ▶ The auxiliary nurse midwife (ANM) reports all AEFIs in her area **weekly** for the block AEFI register; a weekly **NIL report** is valid only after she has actively looked for events among recently vaccinated children.

Categories reported	Reporting format	Frequency
Serious, severe and minor (all vaccines except COVID-19)	SAFE-VAC (via U-WIN)	Immediate
Serious, severe and minor	AEFI registers	Weekly
Serious, severe and minor (numbers only)	HMIS (Health Management Information System)	Monthly

(Routine Immunization Manual 2024, Ch 7, Table 7.2, pp 173–177)

Trigger events and their expected onset window (the case-finding prompt in AEFI surveillance):

Onset after immunization	Events to be reported
Within 24–48 hours	Anaphylaxis; anaphylactoid (acute hypersensitivity) reaction; persistent inconsolable screaming (more than 3 hours); hypotonic hyporesponsive episode; toxic shock syndrome
Within 7 days	Severe local reaction; sepsis; injection-site abscess (bacterial or sterile)
Within 14 days	Seizures including febrile seizures, and encephalopathy (6–12 days for measles/MMR; 0–2 days for DTP)
2–28 days	Brachial neuritis after a tetanus toxoid-containing vaccine
15–35 days	Thrombocytopenia after measles/MMR
Within 3 months	Acute flaccid paralysis (4–30 days in an oral polio vaccine recipient; 4–75 days in a contact)
1–12 months after BCG	Lymphadenitis; disseminated BCG infection; osteitis/osteomyelitis
After rotavirus vaccine	Intussusception
No time limit	Any death, hospitalization, disability or other severe and unusual event thought by health workers or the public to be related to immunization

(Park 28e, Ch 3, pp 141–142)

Investigation of a severe or serious AEFI

The **District AEFI Committee** investigates, coordinated by the DIO, with the medical officer, ANM, ASHA and anganwadi worker gathering information — by interview, observation and inspection of

records at **four sites**: the treating hospital or clinic, the vaccination session site, the cold chain point, and the community.

- ▶ **Confirm the information in the report** — obtain the patient's medical file or clinical record, verify the details of patient and event, and identify any other cases to be included.
- ▶ **Investigate and collect data** — about the *patient* (immunization history, previous medical history including earlier similar reactions or allergies, family history of similar events); about the *event* (history, clinical description, laboratory results, diagnosis, treatment, whether hospitalized, outcome); about the *suspected vaccine* (conditions of shipment, present storage, vaccine vial monitor status, refrigerator temperature record, storage before arrival at the facility).
- ▶ **Assess the immunization service** — vaccination technique, diluent and reconstitution practice, use of auto-disable syringes, multidose vial handling and cold chain condition at the site.
- ▶ **Formulate a working hypothesis** on the likely cause(s) of the event.
- ▶ **Test the working hypothesis** — does the case distribution fit it? The vaccine may be tested for sterility and adjuvant (for example aluminium content), the diluent for sterility and chemical composition, and needles and syringes for sterility. Testing is requested **only on clear suspicion, never routinely, and never before the working hypothesis has been formulated**.
- ▶ **Conclude the investigation** — reach a conclusion on the cause, complete the CIF, take corrective action and recommend further action. The signed CIF is entered into SAFE-VAC with a scanned upload.
- ▶ **In an AEFI death**, the medical officer persuades the family to allow a postmortem, which must be videographed and shared with the investigating committee; if postmortem is not possible, a thorough **verbal autopsy** in the prescribed format is done. Postmortem, histopathology, toxicology and final cause-of-death reports are tracked and submitted to the DIO.
- ▶ **An AEFI cluster** is *two or more cases of the same adverse event related in time, place or vaccine administration*. Investigation starts by framing a case definition and identifying every case meeting it; a cluster of similar events most often arises from **programme (immunization) error** (Park 28e, Ch 3, pp 142–143).

Timelines, structure and responsibilities

Timeline	Activity	Instrument
Day 0	Case notification of a severe or serious AEFI	—
24 hours	Medical officer visits the case and reports to the district	Case Reporting Form (CRF)
48 hours	District decides on investigation and informs the State	—
Day 21	Case investigation completed; district reports to the State	Case Investigation Form (CIF)
Day 100	Causality assessment	State causality assessment report

Level	Functionary / body	Core role in AEFI surveillance
Community	ASHA, anganwadi worker, frontline workers	Inform every AEFI encountered to the ANM/medical officer; assist referral and the investigating team; build community confidence
Sub-centre	ANM	Before the session record manufacturer, expiry date, batch number, vaccine vial monitor status and date/time of opening; keep an in-date anaphylaxis kit with adrenaline and paracetamol syrup 125 mg/5 mL; observe recipients for 30 minutes; give the vaccinated list to ASHA/anganwadi worker; treat minor AEFIs; weekly reporting
PHC / block	Medical Officer	Build the reporting network; notify and submit Section A of the CRF within 24 hours; support the investigation; collect hospital, laboratory and postmortem records; maintain and analyse the AEFI register; certify anaphylaxis kits quarterly; train staff; refer media queries to the district
District	District Immunization Officer	Overall coordinator — expands reporting, verifies and countersigns CRFs, enters SAFE-VAC, convenes the committee, handles media and community communication
District	District AEFI Committee (DIO is member secretary)	Investigates all severe and serious AEFIs; meets at least once a quarter ; block medical officer as special invitee; gives feedback to the medical officer. Investigates but never assigns causality
State / National	State and National AEFI Committees	Causality assessment of an AEFI case is done by state or national AEFI committees only ; the cause-specific category follows from it
National	National AEFI Secretariat	Established to strengthen AEFI surveillance, with collaboration of medical colleges for technical and research assistance and support of the WHO–National Public Health Support Network (Routine Immunization Manual 2024, Ch 1, p 29); Central Drug Laboratory, Kasauli , tests vaccine samples in AEFI cases (Jugal Kishore, Ch 2, p 772)
Global	Global Advisory Committee on Vaccine Safety (GACVS) (Plotkin 7e, Ch 76, p 2154); Brighton Collaboration (Plotkin 7e, Ch 82, pp 2254–2255)	GACVS reviews vaccine safety signals of global importance; the Brighton Collaboration develops and implements globally accepted standard case definitions for assessing adverse events following immunization, graded by level of evidence

(Routine Immunization Manual 2024, Ch 7, pp 177–184)

Diagrams

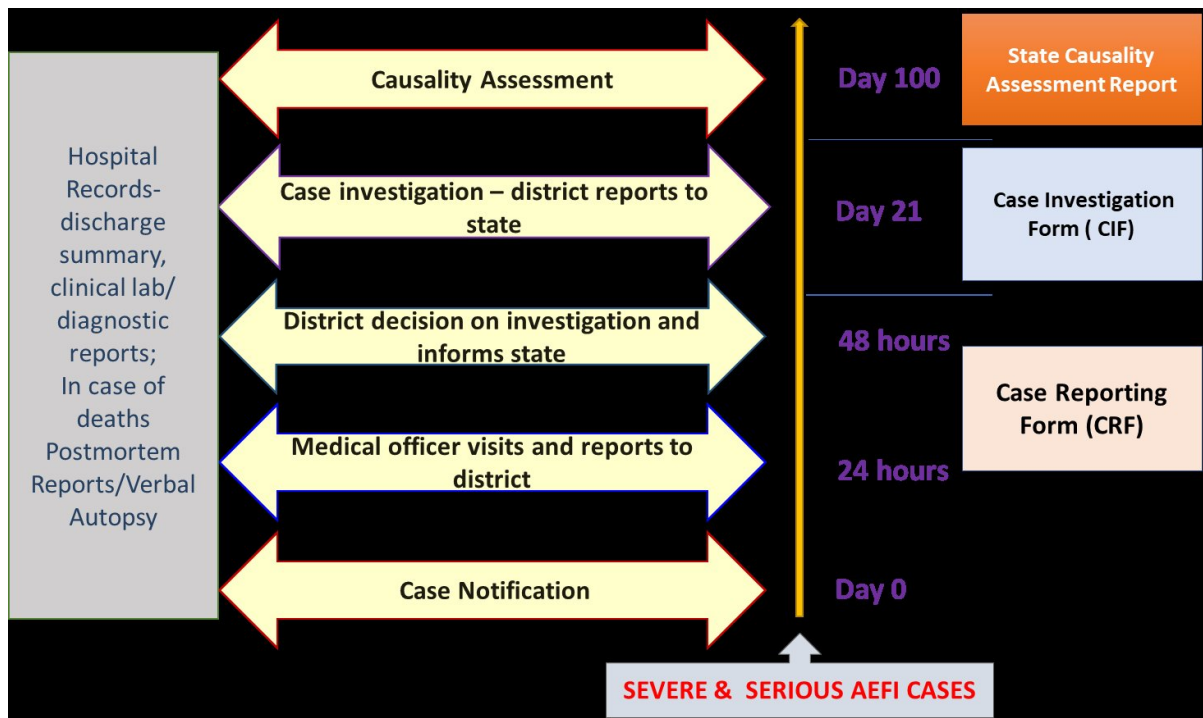


Fig 7.6 — Timelines in AEFI Surveillance (Routine Immunization I Manual For Medical Officers 2024, p.178)

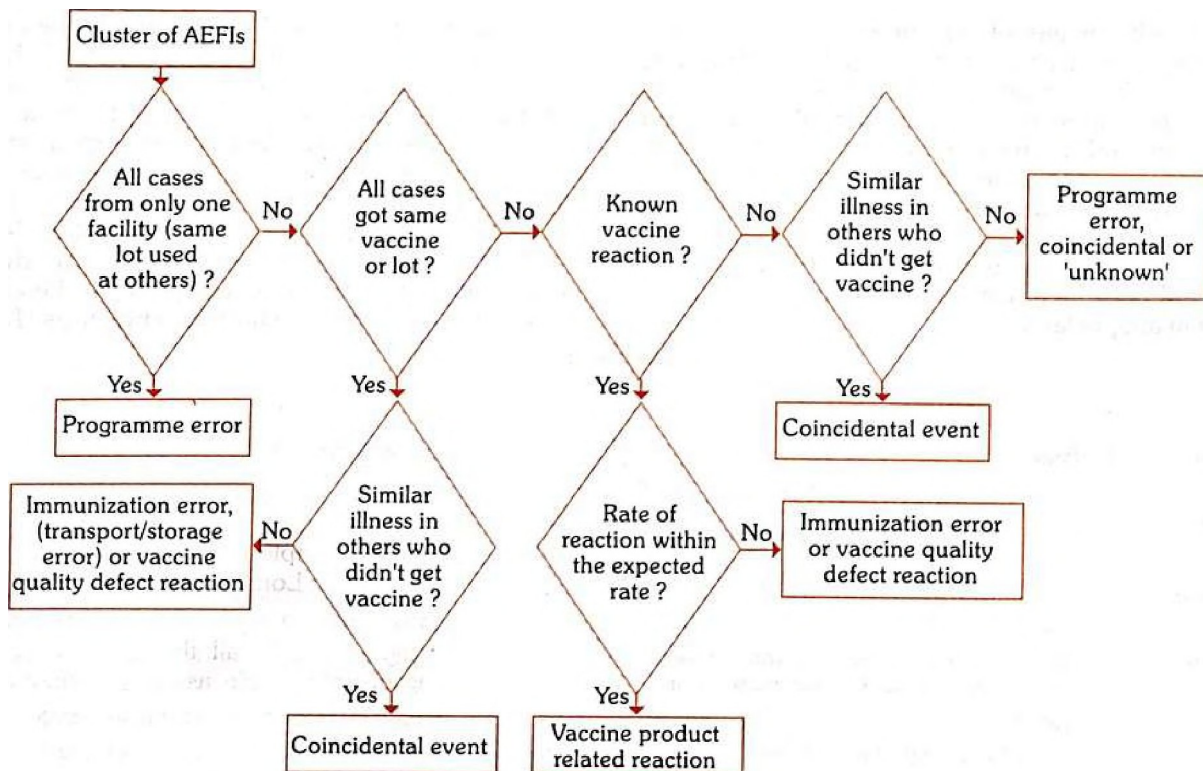
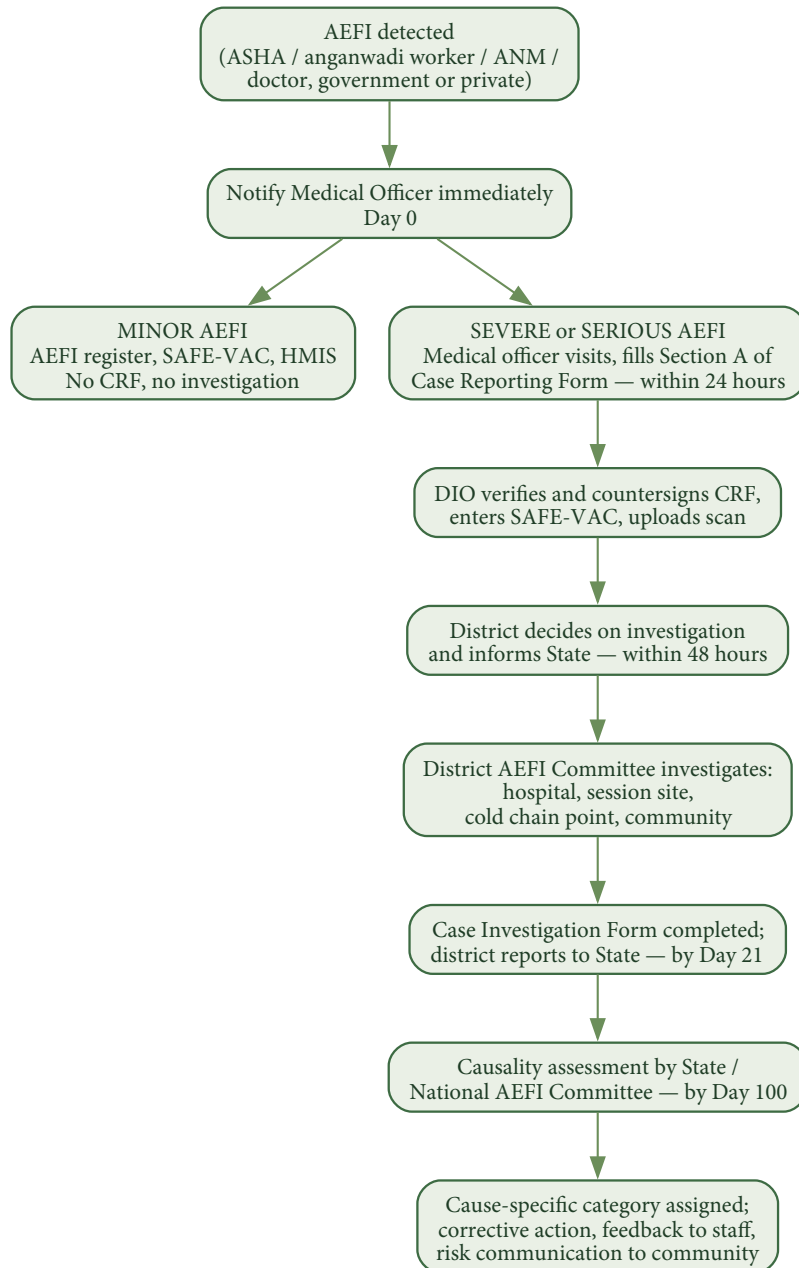
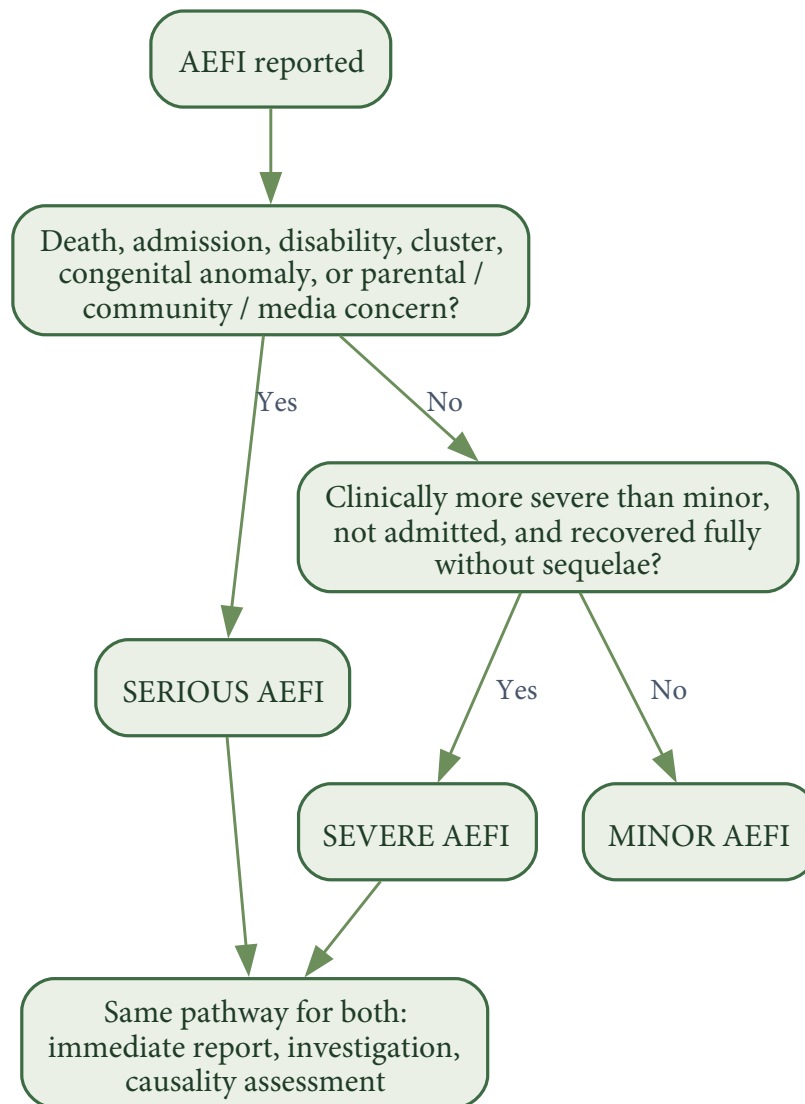


Fig 27 — Identifying cause of AEFI cluster (Park's Textbook of Preventive and Social Medicine, 28e, p.142)



Flow of an AEFI from detection to causality assessment and corrective action



Decision rule for triaging a reported AEFI into minor, severe or serious

High-yield exam pointers

- ▶ AEFI = any untoward medical occurrence following immunization that **does not necessarily have a causal relationship** with the vaccine.
- ▶ Serious AEFI mnemonic — **D-H-D-C-C-C**: Death, Hospitalization, Disability, Cluster, Congenital anomaly, Concern (parental/community/media). Any admission, whatever its severity, is serious.
- ▶ Five CIOMS/WHO cause-specific categories: vaccine **product**-related, vaccine **quality defect**-related, immunization **error**-related, immunization **stress**-related response, **coincidental**.
- ▶ Timeline to write verbatim: **Day 0** notification → **24 h** CRF to DIO → **48 h** district decision → **Day 21** CIF to State → **Day 100** State causality assessment report.
- ▶ Channels: **SAFE-VAC** through **U-WIN** **immediately**, AEFI register **weekly**, HMIS **monthly**. Form chain: CNF → CRF → CIF → **Causality Assessment Form**. No CRF and no investigation for minor AEFIs. **No upper time limit** for reporting.

- ▶ **Both severe and serious AEFIs** are reported immediately, investigated and sent for causality assessment; **causality assessment is done only by State or National AEFI committees**, the district investigates but does not assign causality.
- ▶ AEFI cluster = **two or more** cases of the same event related in time, place or vaccine administration, and usually signals a **programme error**; laboratory testing of vaccine, diluent or syringe only on clear suspicion and never before a working hypothesis is formed.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park's 28th edition still carries the older WHO position that common minor reactions need not be reported, whereas India's Routine Immunization Manual (2024) requires every AEFI including minor ones to be recorded in the AEFI register, SAFE-VAC and the Health Management Information System — write the 2024 Indian position.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, NFHS-5, WHO Global Vaccine Safety Blueprint, Brighton Collaboration case definitions, NTAGI

Programme and policy updates (India)

- ▶ **Active AEFI surveillance now sits alongside the passive SAFE-VAC system.** The Ministry of Health and Family Welfare tasked the International Clinical Epidemiology Network (INCLEN) in 2016 with building the **Multicentre Active AEFI Surveillance System (MAASS)** — a hospital-based sentinel network of 22 tertiary care hospitals (15 public, 7 private) across 17 states and 1 union territory, guided by a technical advisory group that included the Immunization Division and the National AEFI Committee. Over 29 months (November 2017 to March 2020) it covered 90,147 admissions of children aged 1–24 months, from which 7,616 confirmed participants contributed 8,362 confirmed outcomes across 12 pre-specified conditions — 8 case conditions (anaphylaxis, seizure, intussusception, Guillain–Barré syndrome, acute demyelinating encephalomyelitis, aseptic meningitis, thrombocytopenia, Kawasaki disease) and 4 control conditions with no known vaccine link (sepsis, dengue, malaria, urinary tract infection) — using Brighton Collaboration case definitions with case-control, self-controlled case series and self-controlled risk interval designs. The rationale is examinable: spontaneous passive reporting cannot reliably detect events rarer than 1 per 10,000 doses, and has no unvaccinated comparison group [PMID 40779307].
- ▶ **Digital case notification is being pushed down to treating doctors in teaching hospitals.** A pilot across four government medical college hospitals in Chennai (August 2023 to March 2024) added nodal officers, skill workshops, visual job aids and a smartphone Case Notification Form app, working through the paediatrics, community medicine and pharmacology departments: 179 doctors registered as notifiers, 27 of them submitted 56 severe and serious paediatric AEFI notifications, and after 2 cross-notifications to other states and 1 discard all 53 remaining cases (100%) were carried through to the national SAFE-VAC portal with case ID, case reporting and investigation forms; 26% were notified the same day and 64.2% within four days. Under-reporting from tertiary teaching hospitals is the acknowledged weak link in Indian AEFI surveillance, and 85% of the notifiers were postgraduate scholars [PMID 40845796].

- ▶ **HPV vaccine has entered the national programme, and its Indian history is itself an AEFI case study.** The 2009–10 PATH–ICMR demonstration projects in Andhra Pradesh and Gujarat were suspended in April 2010 after four deaths among enrolled girls; an AIIMS committee subsequently attributed the deaths to poisoning, drowning and fever, unrelated to the vaccine, but the scare kept HPV vaccine out of routine use for more than a decade. Serum Institute launched **Cervavac** in September 2022; NTAGI recommended countrywide introduction for girls aged 9–14 years in December 2022, mainly through school-located immunization; and in **February 2026 India launched a free national HPV vaccination programme for girls aged 14 years**, restricted to facilities with a medical officer and internet connectivity so every dose enters the national immunization registry, with a 90-day promotional campaign [PMID 42210649].

Current national data

- ▶ **Zero-dose children (no DTP1), NFHS-5 (2019–21): 6.67% of children aged 12–23 months**, on a sample of 41,132 — down from 33.4% in NFHS-1 (1992–93). Highest in Nagaland (17.02%), Meghalaya (16.76%) and Mizoram (14.61%). Prevalence fell from 8.85% in the poorest wealth quintile to 5.72% in the richest — a 3.1 percentage-point gap — with a Wagstaff concentration index of –0.12 (SE 0.011), confirming that zero-dose status is concentrated among the poor. Decomposition of that index attributed 24.35% of the inequality to household wealth, 12.06% to maternal education, 10.88% to media exposure, 10.61% to antenatal care visits and 9.79% to place of delivery; the measured covariates explained 46.31% of the inequality, the rest residual [PMID 41372226].
- ▶ **Full immunisation coverage, NFHS-5: 77.08% of children aged 12–23 months** (BCG, 3 DPT, 3 polio, 1 measles), with a clear birth-order gradient — crude OR 0.91, 0.74, 0.64, 0.56 and 0.41 for the 2nd to 6th-or-later birth order against first-borns, and odds still 34% lower (aOR 0.66) for the sixth or later after adjustment. Coverage was highest in the south (82.57%) and lowest in the north-east (67.50%), the north-east carrying aOR 0.52 against the northern region as reference; four or more ANC visits (aOR 1.62) and public-facility delivery (aOR 1.64) were the strongest positive predictors. Useful for the "who gets missed" half of an immunisation answer [PMID 41637385].

Elimination, surveillance and disease status

- ▶ **Vaccine-preventable disease resurgence despite the widest vaccine basket India has ever offered.** Recent years show spikes in reported measles across Indian states, a notable rise in mumps particularly among older children, and re-emergence of diphtheria in states such as Kerala and Assam with the burden shifting towards adolescents and adults; overall rubella incidence has fallen but under-immunised pockets continue to impede elimination. The stated remedies are real-time tracking through Co-WIN and U-WIN, regular field assessment of geographical and income inequity, and moving the UIP towards **life-course vaccination** rather than childhood-only cover [PMID 40278999].

Guideline and legislative updates

- ▶ **New Brighton Collaboration case definition (2026) for acute intraocular inflammation** — covering anterior, intermediate and posterior uveitis, panuveitis, retinal vasculitis and neuroretinitis — produced under the CEPI-funded SPEAC project with graded levels of diagnostic certainty for use throughout the vaccine life cycle, in background-incidence work,

causality-hypothesis testing and post-licensure pharmacovigilance. It is explicitly not a clinical management guide. Standard case definitions are what make AEFI data comparable across countries, and the catalogue keeps expanding [PMID 42481266].

- **The WHO AEFI causality tool remains the only vaccine-specific instrument of its kind.** A scoping review of case-level causality assessment tools developed or updated between 2008 and 2023 identified 18 tools — 12 algorithmic, 4 hybrid, 1 global-introspection and 1 probabilistic — of which only the WHO AEFI tool was designed for vaccines. The authors argue that future causality assessment should also weigh product quality, potential medication or immunization error, and adherence to risk-minimisation measures [PMID 41910704].

Global and WHO developments

- **WHO prequalification and national regulators now expect active vaccine safety surveillance, not spontaneous reporting alone.** Because low- and middle-income country systems rely on passive reporting that under-detects events, risk management plans submitted to WHO prequalification and national regulatory authorities increasingly carry commitments to active surveillance at sentinel sites or through formal observational studies — a shift accelerated by the rapid deployment of COVID-19 vaccines with limited safety data. The Developing Countries Vaccine Manufacturers Network ran a "learn-by-doing" project to build this capacity [PMID 39813974].
- **Coded surveillance data need chart-level validation before a signal is believed.** In the Global COVID Vaccine Safety project, Guillain–Barré syndrome diagnostic codes in three Spanish real-world data sources were validated against Brighton Collaboration levels of diagnostic certainty: across 180 identified cases, positive predictive values for definite or probable cases (levels 1–2) ranged from 4.8% to 90%, the highest where in-hospital medical records were accessible. The difference tracked the type of data source; the effect of the 2021 European regulatory alert on GBS was inconsistent [PMID 41806493].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Chapter 3, Principles of Epidemiology and Epidemiologic Methods); Routine Immunization — I Manual for Medical Officers, Ministry of Health and Family Welfare, Government of India, 2024 (Unit 1 Introduction, Unit 7 AEFI Surveillance, Unit 8 Sources and Use of Data); AEFI Surveillance and Response Operational Guidelines, Ministry of Health and Family Welfare, 2024; Plotkin's Vaccines, 7e (Chapter 76, Immunization in Developing Countries; Chapter 82, Vaccine Safety); Jugal Kishore — National Health Programmes of India; CIOMS/WHO cause-specific categorization of AEFIs (2012, updated 2018).

Recent-advances sources: PMID 40779307 — Establishing a Multicenter Active Adverse Events Following Immunization Sentinel Surveillance Network Across 22 Tertiary Care Hospitals in India: Protocol for a Prospective Observational Study, *JMIR Res Protoc* 2025, doi:[10.2196/64050](https://doi.org/10.2196/64050); PMID 40845796 — Implementation of smartphone application for reporting adverse events following immunization by the treating doctors in tertiary care teaching hospitals in Chennai, India: Pilot and feasibility study, *Vaccine* 2025, doi:[10.1016/j.vaccine.2025.127644](https://doi.org/10.1016/j.vaccine.2025.127644) (abstract-only); PMID 42210649 — Resilience to HPV vaccine safety scares in seven countries, *J Glob Health* 2026, doi:[10.7189/jogh.16.04148](https://doi.org/10.7189/jogh.16.04148); PMID 41372226 — Decomposition of socioeconomic inequalities in zero-dose children aged 12–23 months in India, *Sci Rep* 2025, doi:[10.1038/s41598-025-27394-y](https://doi.org/10.1038/s41598-025-27394-y); PMID 41637385 — How does birth order influence full immunization coverage among children aged 12–23 months in India? Evidence from the National Family Health Survey, *PLoS One* 2026, doi:[10.1371/journal.pone.0342300](https://doi.org/10.1371/journal.pone.0342300); PMID 40278999 — Vaccine-Preventable Diseases in Pediatric Age Group in India: Recent Resurgence, Implications and Solutions, *Indian J Pediatr* 2025, doi:[10.1007/s12098-025-05531-9](https://doi.org/10.1007/s12098-025-05531-9) (abstract-only); PMID 42481266 — Acute intraocular inflammation: Brighton Collaboration case definition and guidelines for data collection, analysis, and presentation of immunization safety data, *Vaccine* 2026, doi:[10.1016/j.vaccine.2026.128919](https://doi.org/10.1016/j.vaccine.2026.128919) (abstract-only); PMID 41910704 — A Scoping Review of Case-Level Causality Assessment Tools Developed Between 2008 and 2023; Strengths, Weaknesses and Potential Future Improvements, *Drug Saf* 2026, doi:[10.1007/s40264-026-01665-7](https://doi.org/10.1007/s40264-026-01665-7) (abstract-only); PMID 39813974 — Active vaccine safety surveillance in low- and middle-income countries: Challenges for vaccine manufacturers from emerging countries, *Vaccine* 2025, doi:[10.1016/j.vaccine.2025.126727](https://doi.org/10.1016/j.vaccine.2025.126727) (abstract-only); PMID 41806493 — Validation of Guillain-Barré syndrome case identification in three heterogeneous VAC4EU real-world data sources in Spain using the Brighton Collaboration criteria, *Vaccine* 2026, doi:[10.1016/j.vaccine.2026.128397](https://doi.org/10.1016/j.vaccine.2026.128397) (abstract-only).



Criteria for judging causality

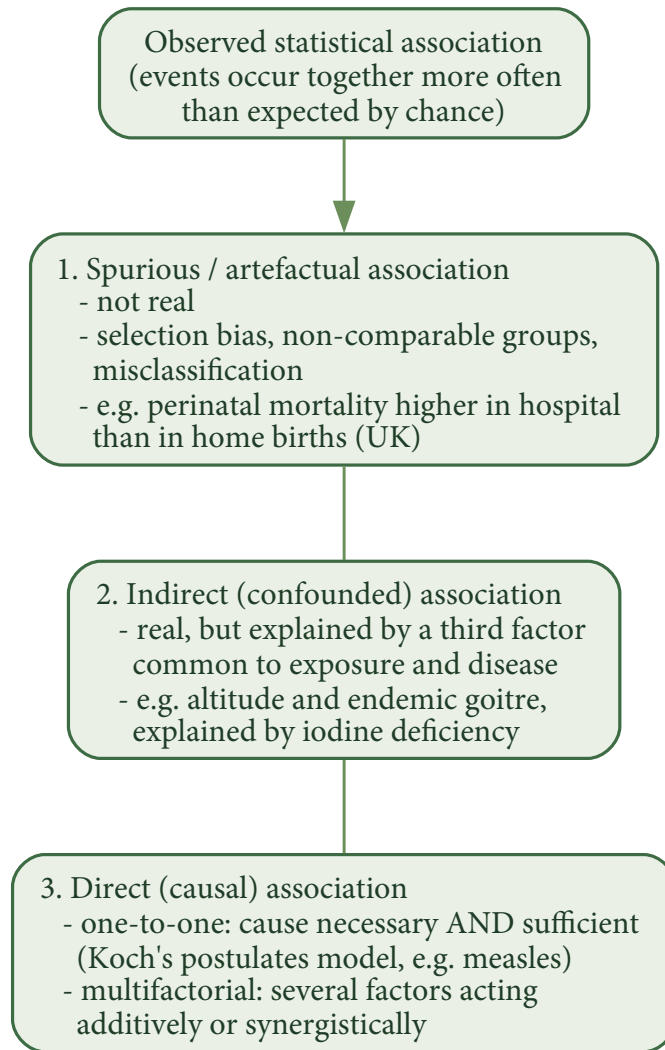
Asked in: Paper I 07-Jul-2023 — RGUHS MD Community Medicine

Definition — Two events are **associated** when they occur together more often than would be expected by chance; association does not by itself imply a causal relationship (Park 28e, Ch 3, p 107). The **criteria for judging causality** are a set of judgemental guidelines — set out in the US Surgeon-General's report *Smoking and Health* (1964) and restated by Sir Austin **Bradford Hill** in 1965 — applied to decide, in the absence of a controlled human experiment, whether an observed statistical association is likely to be causal (Park 28e, Ch 3, p 109). Correlation alone cannot settle this, because a correlation coefficient (range -1.0 to $+1.0$) neither establishes that exposure preceded disease nor measures risk: causation implies correlation, but correlation does not imply causation (Park 28e, Ch 3, p 107).

Step 1 — First classify the observed association

Type of association	What it means	Classical example	What to do
a. Spurious (artefactual)	The observed association is not real — it arises from selection bias, non-comparable study and control groups, non-representative sampling or misclassification	UK study of 5,174 home births and 11,156 hospital births: perinatal mortality 5.4 per 1,000 (home) versus 27.8 per 1,000 (hospital), because hospitals attract high-risk women (Park 28e, Ch 3, p 107)	Discard; prevent by design
b. Indirect (confounded)	A real statistical association produced by a third (confounding) factor common to both the characteristic and the disease — e.g. age, sex, social class	Altitude and endemic goitre, wholly explained by environmental iodine deficiency ; higher sucrose intake in myocardial infarction (<i>Yudkin and Roddy</i>), confounded by cigarette smoking (Park 28e, Ch 3, p 108)	Pursue — indirect associations often yield aetiological clues
c. Direct (causal) — one-to-one	Factor A is both necessary and sufficient : when A is present B follows, and when B is present A must have been present — the Koch's-postulates model	Measles (Park 28e, Ch 3, p 108)	Accept as causal
c. Direct (causal) — multifactorial	Several factors act together, additively or synergistically , at cellular level to produce disease; a one-to-one relationship is an oversimplification in biology	Most chronic disease (Park 28e, Ch 3, pp 108–109)	Accept as causal for the set of factors

Only after **chance** (judged by the test of significance and the confidence interval), **bias** and **confounding** have been excluded is it worth applying the causal criteria (Gordis 7e, Ch 14, pp 290–291; Lilienfeld's Foundations, Ch 4, p 75).



Classification of an observed statistical association (Park 28e, Ch 3, pp 107-109)

Step 2 — The criteria themselves

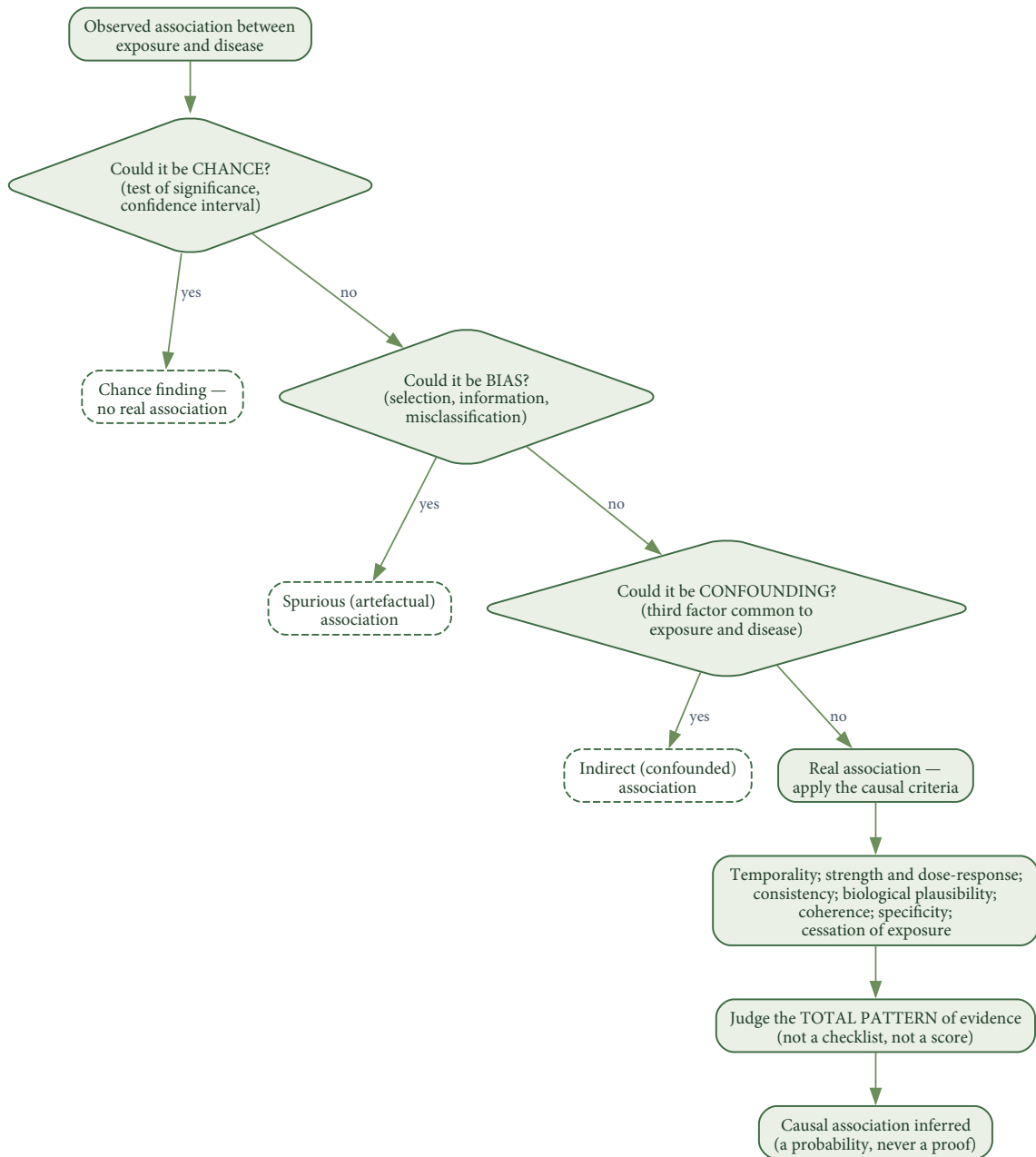
Park lists six criteria; Bradford Hill's 1965 address gave **nine** "viewpoints"; Gordis 7e gives **nine** guidelines. The table sets out all of them with the question each one answers.

Criterion	Core question it answers	What strengthens the causal inference	Park 28e	Hill 1965	Gordis 7e
Temporal association (temporality)	Does the suspected cause precede the observed effect?	Exposure documented before disease onset, with a latent period compatible with the disease. London smog, December 1952: a rise in airborne particle concentration was followed by a rise in daily deaths, and their fall by a fall in mortality	Yes	Yes	Yes

Criterion	Core question it answers	What strengthens the causal inference	Park 28e	Hill 1965	Gordis 7e
Strength of association	(a) Is the relative risk large? (b) Is there a dose-response or duration-response relationship?	The larger the relative risk — the greater its departure from unity — the greater the likelihood of causality; absence of a dose-response or duration-response argues against causality	Yes (dose-response included here)	Yes	Yes
Biological gradient (dose-response)	Does risk rise steadily with increasing level of exposure?	Rising incidence with rising exposure; may legitimately be absent when the biological relationship is dichotomous or has a threshold	folded into strength	Yes	Yes (listed separately)
Specificity of association	Does one cause produce one disease, and one disease follow one cause?	The weakest criterion — one factor can cause several diseases and most diseases are multifactorial, so apparent lack of specificity does not refute causality	Yes	Yes	Yes
Consistency / replication of findings	Are the findings reproduced in different populations, in different settings, by different methods and different investigators?	Evidence from a single study is seldom sufficient; consistency is also expected within population subgroups, and unexplained inconsistency weakens the case	Yes	Yes	Yes (as two separate items)
Biological plausibility	Is the association coherent with current biological knowledge , and is a mechanism conceivable?	A plausible mechanism supports causality, but epidemiological observation may legitimately precede biology (<i>Gregg's</i> rubella and congenital cataract; high oxygen concentration and retrolental fibroplasia)	Yes	Yes	Yes
Coherence of the association	Does the association fit the other known and relevant facts — secular trends, sex and time differences, occupational data?	Rising tobacco consumption paralleled by rising lung cancer incidence, with the rise appearing later in women in line with their later adoption of smoking	Yes	Yes	Yes (as "consistency with other knowledge")
Consideration of alternate explanations	Have confounding and other explanations been actively sought and ruled out?	Explicit matching, restriction or statistical adjustment for confounders, and their documented exclusion by the investigators	—	—	Yes

Criterion	Core question it answers	What strengthens the causal inference	Park 28e	Hill 1965	Gordis 7e
Cessation of exposure / experimental evidence	Does removing the exposure reduce risk?	A fall in risk after withdrawal of exposure is strong support; it may fail when the pathogenic process is already irreversible — emphysema is not reversed by stopping smoking, although its progression is slowed	—	Yes (as experiment)	Yes
Analogy	Is there an accepted analogous cause-and-effect relationship?	An established analogy (for example one drug already proved teratogenic) lowers the weight of evidence demanded for a similar exposure	—	Yes	—

(Park 28e, Ch 3, pp 109–111; Gordis 7e, Ch 14, pp 289–296; Handbook of Epidemiology, Ch I.1, p 67)



Sequence from an observed association to a causal inference

Step 3 — Applying the criteria: cigarette smoking and lung cancer

This is the worked example on which the criteria were first elaborated, in *Smoking and Health*, the 1964 report of the Advisory Committee to the US Surgeon General (Park 28e, Ch 3, pp 109–111).

Criterion	Evidence in the smoking–lung cancer hypothesis
Temporal association	Lung cancer occurs in smokers of long standing; the rise in cigarette consumption preceded the rise in lung cancer death rates by about 30 years, compatible with the long latent period of carcinogenesis
Strength of association	Large relative risk, with a clear dose-response and duration-response gradient — risk rises with the number of cigarettes smoked per day and with the duration of smoking

Criterion	Evidence in the smoking–lung cancer hypothesis
Specificity	Apparently poor, since smoking is also linked to coronary heart disease, chronic bronchitis, emphysema and cancer cervix; yet about 80–90 per cent of lung cancer is attributable to cigarette smoking . Not every smoker develops cancer (multifactorial disease) and not every case has smoked (chromates, asbestos, nickel, uranium, air pollution)
Consistency	Replicated in many retrospective studies and in at least nine prospective studies in different countries
Biological plausibility	Carcinogenic substances have been isolated from cigarette smoke — a complex of nicotine, carbon monoxide, benzpyrene and particulate matter with additive and synergistic action — although the precise carcinogens responsible for human lung cancer are unknown
Coherence	The historical rise in cigarette consumption is coherent with the rise in lung cancer; death rates rose first in men and later, more rapidly, in women; relative risk falls after cessation; risk is greatly increased when asbestos- or uranium-exposed workers also smoke

Step 4 — How the criteria must be used, and their limitations

- ▶ **No criterion is self-sufficient.** The Surgeon-General's Report (1964) states that the causal significance of an association is a matter of **judgement which goes beyond any statement of statistical probability**; no single criterion is by itself *sine qua non*, but each adds to the quantum of evidence and all together give a probability of the association being causal (Park 28e, Ch 3, p 109).
- ▶ **Not a checklist and not a score.** What counts is not how many criteria are ticked but an assessment of the **total pattern of evidence** drawn from several different types of study (Gordis 7e, Ch 14, pp 292–293, 299).
- ▶ **Temporality is the one requirement that cannot be waived** — an exposure that occurred after disease onset cannot have caused it (Gordis 7e, Ch 14, p 289).
- ▶ **Weighting.** Gordis 7e presents a modified two-tier scheme in which temporal relationship, biological plausibility, consistency and the exclusion of alternative explanations are the primary criteria, with dose-response, strength of association and cessation effects as further considerations (Gordis 7e, Ch 14, pp 296–297).
- ▶ **Causal inference is inductive, never deductive.** There is no necessary or sufficient criterion for deciding whether an association is causal, and causal inferences cannot attain the certainty of a logical deduction (Handbook of Epidemiology, Ch I.1, p 67); epidemiology **infers** causation rather than proving it (Lilienfeld's Foundations, Ch 4, pp 76–77).
- ▶ **Infectious disease is judged differently.** Causation of a communicable disease is still demonstrated by the **Henle–Koch postulates** — the organism is found in all cases of the disease, can be isolated from patients and grown in pure culture, and reproduces the disease when the pure culture is inoculated into a susceptible animal or human — updated by *Fredricks and Relman* (1996) to incorporate DNA typing of the organism (Lilienfeld's Foundations, Ch 4, pp 75, 83–84).

- ▶ **The criteria exist to permit action.** A causal hypothesis that satisfies them, even without experimental proof, becomes sufficiently probable to provide a reasonable basis for **preventive and public health action** — the route by which tobacco control was launched (Lilienfeld's Foundations, Ch 4, pp 83–84).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Draw these two small model diagrams alongside the criteria table.

1. Model of an indirect association (Park 28e, Ch 3, Fig 12, p 108) — exact labels:

- ▶ Box A: *Altitude*
- ▶ Box B: *Endemic goitre*
- ▶ Box C: *Iodine deficiency* — the common (confounding) factor
- ▶ Solid arrows from C to A and from C to B
- ▶ Dotted line joining A and B, labelled *apparent (indirect) association*

2. Model of multifactorial causation showing synergism (Park 28e, Ch 3, Fig 15, p 109) — exact labels:

- ▶ *Factor 1, Factor 2, Factor 3* — three boxes stacked on the left
- ▶ All three arrows converging on *Reaction at cellular level*
- ▶ A single arrow from there to *Disease*

High-yield exam pointers

- ▶ **Park's six criteria, in order:** Temporal association, Strength of association, Specificity, Consistency, Biological plausibility, Coherence.
- ▶ **Hill's nine viewpoints (1965):** strength, consistency, specificity, temporality, biological gradient, plausibility, coherence, experimental evidence, analogy.
- ▶ **Gordis 7e's nine guidelines:** temporal relationship, strength of association, dose-response relationship, replication of findings, biologic plausibility, consideration of alternate explanations, cessation of exposure, consistency with other knowledge, specificity of association.
- ▶ **Three types of association (Park):** spurious (artefactual), indirect (confounded), direct or causal (one-to-one, or multifactorial).
- ▶ **Confounding variable:** a factor associated with the exposure under study which is itself, independently, a risk factor for the disease, and is distributed unequally between study and control groups.
- ▶ **Quotable line:** "The causal significance of an association is a matter of judgement which goes beyond any statement of statistical probability" — Surgeon-General's Report, 1964.
- ▶ **Smoking–lung cancer figures to write:** about a 30-year lag between rising cigarette consumption and rising lung cancer deaths; 80–90 per cent of lung cancer attributable to smoking; at least nine prospective studies supporting consistency.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park's 28th edition lists six criteria while Bradford Hill's 1965 address gave nine viewpoints and Gordis 7e gives nine guidelines; the lists overlap rather than conflict, since Park folds the biological gradient into "strength of association" and omits experiment or cessation of exposure and analogy — so writing Park's six as the core and naming the additional Hill and Gordis items separately, as above, is the safe form for this answer.

SOURCE & COVERAGE

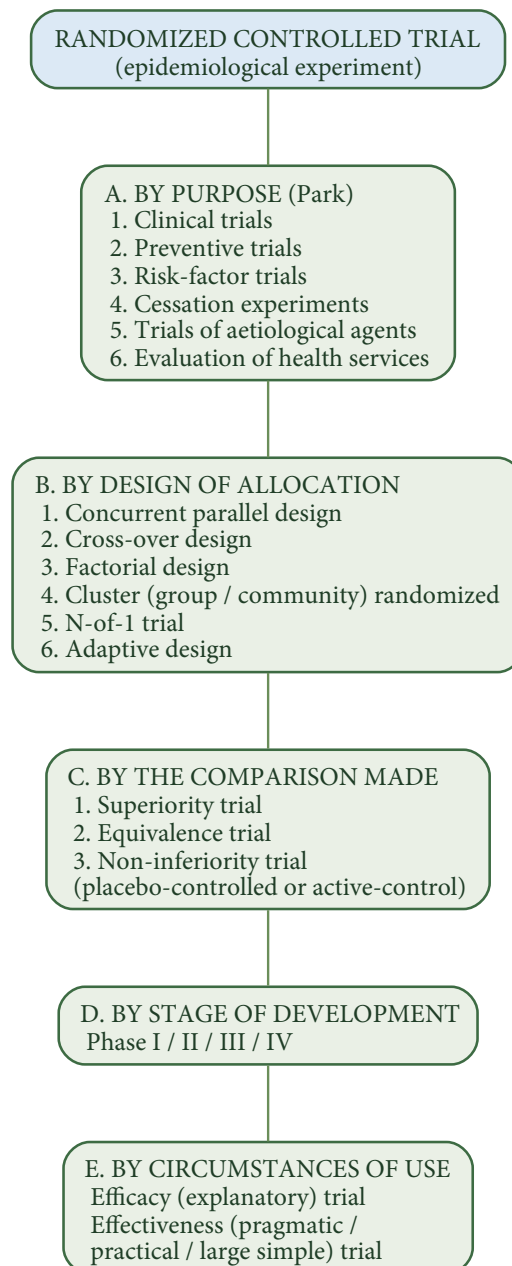
Park's Textbook of Preventive and Social Medicine, 28e (Ch 3, Principles of Epidemiology and Epidemiologic Methods, pp 107–111); Gordis Epidemiology 7e (Ch 14, From Association to Causation: Deriving Inferences From Epidemiologic Studies, pp 282–299); Lilienfeld's Foundations of Epidemiology (Ch 4, Inferring Causal Relationships, pp 74–84); Handbook of Epidemiology, Ahrens & Pigeot (Ch I.1, Basic Concepts, p 67); *Smoking and Health* — Report of the Advisory Committee to the Surgeon General of the Public Health Service, US, 1964.

◆ ◆ ◆

Randomised controlled trials

Asked in: Paper I Nov 2017 — RGUHS MD Community Medicine

Definition — A randomized controlled trial (RCT) is an epidemiological experiment in which participants drawn from a defined reference population are allocated **by chance alone** to an experimental (intervention) arm or a control arm, followed identically, and compared for a pre-defined outcome (Park 28e, Ch 3, p 101). Because allocation is random, known and unknown confounders are distributed alike between the arms, so for a new programme or a new therapy the RCT is the number-one method of evaluation (Park 28e, Ch 3, p 101). RCTs are classified along **five axes** — **purpose, design of allocation, nature of the comparison, stage of product development, and circumstances of use** (efficacy versus effectiveness).



Classification of randomized controlled trials along five axes

A. Types by purpose — the classical epidemiological classification

#	Type	What it evaluates	Classic example from the corpus
1	Clinical trials	Therapeutic agents, mainly drugs, in patients who already have the disease	Beta-blockers to reduce cardiovascular mortality in survivors of the acute phase of myocardial infarction; tonsillectomy for recurrent throat infection; coronary bypass surgery to prevent myocardial infarction (Park 28e, Ch 3, p 104)
2	Preventive trials	Measures of primary prevention in healthy people — chiefly vaccines and chemoprophylactic drugs	Medical Research Council (UK) whooping-cough vaccine trial, 1946: attack rate 1.45 per 1000 child-months in 3,801 vaccinated children versus 6.72 per 1000 child-months in 3,759 unvaccinated controls (Park 28e, Ch 3, p 104)
3	Risk-factor trials	Modification of a suspected risk factor in people who carry the factor but not yet the disease; may be single-factor or multi-factor	World Health Organization clofibrate trial — the largest preventive trial, >15,000 men (one-third clofibrate, two-thirds olive oil), 3 centres (Edinburgh, Prague, Budapest), double-blind, mean observation 9.6 years: fewer non-fatal infarcts but 25 per cent more deaths in the clofibrate arm (Park 28e, Ch 3, pp 104–105)
4	Cessation experiments	Effect of stopping a habit or removing a suspected agent believed to be causally related to the disease; a reduction in disease greatly strengthens the causal hypothesis	Randomized trial of smoking cessation in the primary prevention of coronary heart disease; smoking cessation and lung cancer incidence (Park 28e, Ch 3, p 105)
5	Trials of aetiological agents	Confirm or refute an aetiological hypothesis by deliberately allocating the suspected agent	Retrolental fibroplasia (<i>Kinsey and Hemphill</i>): premature babies of birth weight 1500 g or less randomized to 50 per cent oxygen for 28 days versus a curtailed-oxygen control group (Park 28e, Ch 3, p 105)
6	Evaluation of health services ("health services research")	Effectiveness and efficiency of alternative policies of health-care delivery, so that limited resources can be prioritised	Indian controlled trials of tuberculosis chemotherapy showing domiciliary treatment as effective as sanatorium treatment ; RCT of multiphasic screening in South-East London; substitution of nurses and paramedical workers for physicians (Park 28e, Ch 3, p 105)

Other named risk-factor intervention trials in coronary heart disease are the **Stanford Three-Community Study**, the **North Karelia Project** (Finland), the **Oslo Study**, and the **Multiple Risk Factor Intervention Trial (MRFIT)**, United States (Park 28e, Ch 3, p 105).

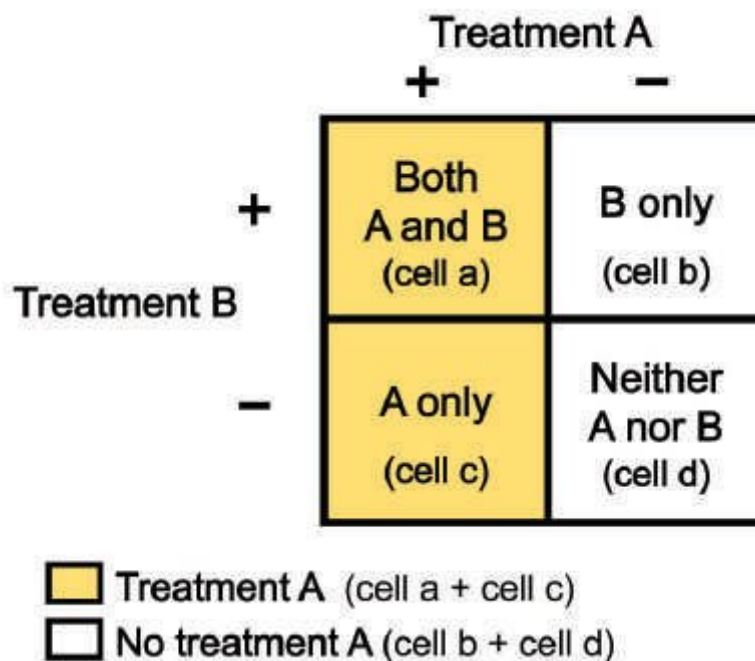
The analysis of a **preventive trial** must end in a clear statement about three things (Park 28e, Ch 3, p 104):

- The **benefit** the community will derive from the measure.
- The **risks** involved.
- The **cost** to the health service in money, men and material resources.

B. Types by design of allocation

Design	How subjects are allocated	Chief merit	Chief limitation
Concurrent parallel	Two randomly assigned groups, one exposed to the treatment and one not; each patient stays in the same arm for the whole investigation (Park 28e, Ch 3, p 103)	Simplest, most widely used, suits any intervention including surgery	Between-person variation is not eliminated, so it needs more subjects than a cross-over of the same power — the corollary of Park's statement that cross-over studies "economize on the total number of patients required" (Park 28e, Ch 3, pp 103–104)
Cross-over	Each patient serves as his own control — randomly assigned to treatment or control, then after a wash-out period (its length set by the pharmacologic properties of the drug, to allow elimination and the disappearance of carry-over effects) the two arms are switched	Every patient is assured of receiving the new therapy at some point; economises on the total number of patients	Costs time; order effects and the <i>Hawthorne effect</i> may operate (Park 28e, Ch 3, pp 103–104; Gordis 7e, Ch 10, p 227)
Factorial	Two (or more) interventions tested simultaneously in the same trial; each subject falls into one of four cells — both, either alone, or neither	Two questions answered for nearly the price of one; permits study of interaction	Effect modification of one intervention by the other reduces power and complicates interpretation; one arm may need to be stopped early — in the <i>Physicians' Health Study</i> the aspirin component was terminated while the beta-carotene component continued (Gordis 7e, Ch 11, pp 247–248; Browner 5e, Ch 12, p 399)
Cluster (group / community) randomized	Naturally occurring groups are randomized — geographical (villages, towns) or otherwise (schools, workplaces, clinical practices) — and outcomes counted in the individuals within them	The only valid design when the intervention is delivered at group level; avoids contamination between arms; improves long follow-up where personal identifiers are unreliable	Outcomes within a cluster are not independent — analysis must allow for clustering, else <i>p</i> values are artificially small and confidence intervals too narrow; sample size must be inflated by the design effect $1 + (m - 1)r$, where <i>m</i> = subjects per cluster and <i>r</i> = intra-cluster correlation coefficient (Handbook of Epidemiology, Ch 8, pp 368–369)

Design	How subjects are allocated	Chief merit	Chief limitation
N-of-1 trial	A single patient receives active treatment or placebo in random order for successive brief periods, patient and physician blinded, and the repeated responses are analysed statistically	Tailors the decision to the individual patient	Only usable when disease activity is unpredictable, response is prompt and there is no carry-over — e.g. migraine, asthma, fibromyalgia; rarely done and even more rarely published (Fletcher, Ch 9, pp 164–165)
Adaptive design	The protocol is allowed to change on the basis of pre-specified interim analyses (e.g. dropping an ineffective dose arm, or modifying sample size or duration)	Efficient; fewer patients exposed to inferior arms	Feasible only when outcomes are measured and analysed quickly; requires strict pre-specification and statistical safeguards (Browner 5e, Ch 12, pp 382–383)



Factorial design: effects of treatment A versus no treatment A (Gordis Epidemiology 7e, Ch 10, p 229)

A **cross-over design** is unsuitable in three situations (Park 28e, Ch 3, pp 103–104):

- ▶ When the drug of interest **cures** the disease.
- ▶ When the drug is effective only during a **certain stage** of the disease.
- ▶ When the disease itself **changes radically** during the time required for the study.

Note the distinction between a **planned cross-over** (a design, as above) and an **unplanned cross-over** — subjects randomized to one arm actually receiving the other, also called **contamination of treatment arms**, which is dealt with by intention-to-treat analysis, not by design (Gordis 7e, Ch 10, p 228).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Concurrent parallel and cross-over controlled therapeutic trials** (after Park 28e, Ch 3, Fig 10, p 104). Draw two panels, one above the other, with an arrow marked *Time* running left to right beneath each.
 - **Panel (a) Concurrent parallel:** box "Patients" → "Random assignment" → upper branch "Exposed to specific treatment R" and lower branch "Unexposed to specific treatment R" → both converge on "COMPARE OUTCOME".
 - **Panel (b) Cross-over:** box "Patients" → "Random assignment" → upper branch "Exposed to R", lower branch "Unexposed to R" → "COMPARE OUTCOME" → **wash-out period** (draw as diagonal crossing lines) → arms switched, upper now "Unexposed to R" and lower now "Exposed to R" → "COMPARE OUTCOME — EXPOSED AND UNEXPOSED TO R".

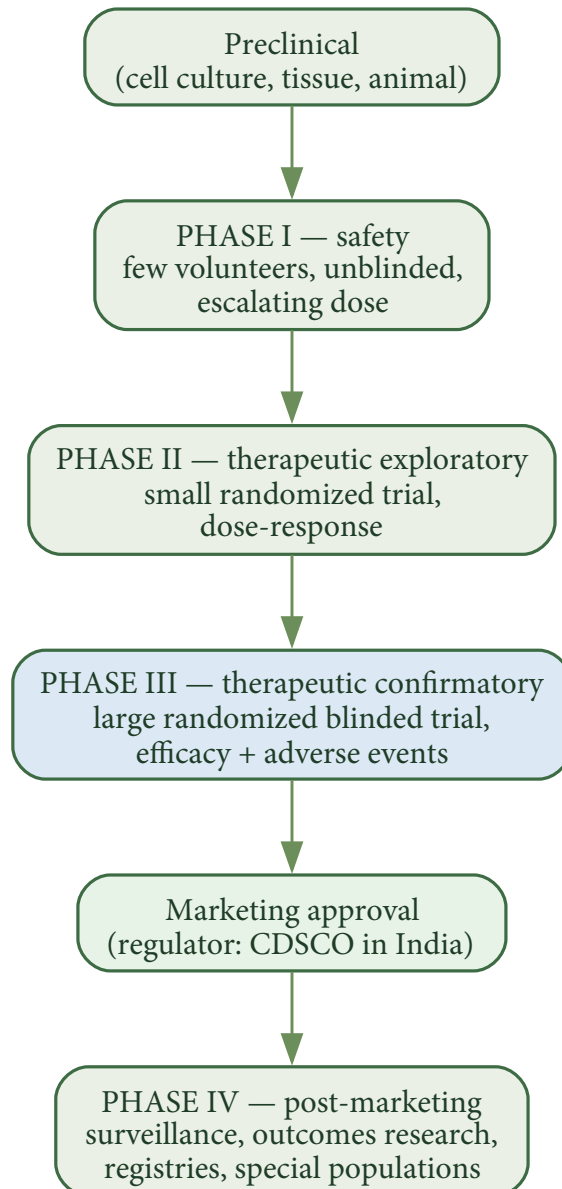
C. Types by the nature of the comparison

Type	Question asked	Statistical criterion	Practical note
Superiority trial	Is the new treatment better than the comparator?	Reject the null hypothesis of no difference	The default design; comparator may be placebo or an active drug (Fletcher, Ch 9, p 163)
Equivalence trial	Is the new treatment neither better nor worse to a clinically important extent?	Confidence interval for the difference lies within a two-sided equivalence margin	Used when a new therapy is preferred for being easier to use, less invasive or safer, at similar efficacy (Browner 5e, Ch 12, p 379)
Non-inferiority trial	Is the new treatment unlikely to be worse than the standard by more than a clinically important amount?	One-directional: the confidence interval must exclude a loss greater than the pre-specified inferiority (non-inferiority) margin	Chosen when the new treatment is safer, cheaper or easier to give; needs a larger sample size than a comparable superiority trial. Example: single-dose oral azithromycin versus injectable penicillin for yaws in Papua New Guinea, margin 10 per cent — cure 96 per cent versus 93 per cent (Fletcher, Ch 9, p 163)
Active-control (comparative effectiveness) trial	Which of two treatments in current use is preferable?	As for superiority, equivalence or non-inferiority, as pre-specified	The control group also receives an active intervention; optimal when a "standard of care" exists and placebo would be unethical (Browner 5e, Ch 12, p 379)

Anything that blunts the difference between arms — heavy drop-out, cross-over, imprecise outcome measurement, or an intention-to-treat analysis in such a trial — makes it **easier to declare non-inferiority falsely**; a non-inferiority trial is therefore a weak test if poorly conducted (Fletcher, Ch 9, pp 163–164).

D. Types by phase of drug or vaccine development

Stage	Subjects and design	Objective
Preclinical	Studies in cell cultures, tissues and animals (Browner 5e, Ch 11, Table 11.2, pp 351–352)	Toxicology and biological activity before any human exposure
Phase I	A few volunteers; unblinded, uncontrolled, escalating doses (Browner 5e, Ch 11, pp 351–352)	Safety and tolerability, pharmacokinetics and pharmacodynamics; for vaccines, dose and route of administration with immunogenicity, and the study must be low risk (ICMR National Ethical Guidelines 2017, pp 86, 89)
Phase II — <i>therapeutic exploratory trial</i> (Mahajan & Gupta 4e, Ch 4, p 57)	Relatively small randomized or time-series trials in patients selected by relatively narrow criteria and closely monitored (Browner 5e, Ch 11, pp 351–352; ICMR 2017, p 86)	Explore therapeutic efficacy — estimate then confirm the dose response, and test effect on biomarkers or clinical outcomes; immunogenicity for vaccines (ICMR 2017, p 86)
Phase III — <i>therapeutic confirmatory trial</i> (Mahajan & Gupta 4e, Ch 4, p 57)	Relatively large randomized, blinded (preferably) trials; planned as bridging and pivotal trials (Browner 5e, Ch 11, pp 351–352; ICMR 2017, p 87)	Demonstrate or confirm therapeutic benefit (protection rate for vaccines) conclusively on clinical outcomes and adverse events; provides the basis for marketing approval (ICMR 2017, p 87)
Phase IV	Large trials or observational studies conducted after the product is approved (Browner 5e, Ch 11, pp 351–352)	Post-marketing surveillance, outcome research and registries; detects uncommon serious adverse effects and suggests additional therapeutic uses (ICMR 2017, pp 87, 89)



Sequence of trial phases in the development of a new drug or vaccine

E. Efficacy (explanatory) versus effectiveness (pragmatic) trials

Feature	Efficacy / explanatory trial	Effectiveness / pragmatic trial
Question	Can the treatment work under ideal circumstances?	Does offering the treatment work under ordinary circumstances?
Patients	Narrow eligibility, highly selected, adherent	Broad eligibility — the kinds of patients seen in ordinary care
Care given	Best possible care, tightly protocolised	Usual care, decided by the treating clinician
Analysis emphasised	Per-protocol / explanatory analysis	Intention-to-treat analysis
Chief strength	Internal validity	Generalizability

Feature	Efficacy / explanatory trial	Effectiveness / pragmatic trial
Related forms	—	Large simple trial — simplified entry criteria, minimal extra testing, one simple clinically important outcome (e.g. discharged from hospital alive)

Efficacy trials usually precede effectiveness trials, because if a treatment does not work under the best circumstances it cannot work under ordinary ones; the gap between the two is the "**implementation gap**" between ideal and ordinary care, and is itself a target for improvement (Fletcher, Ch 9, pp 154, 161–162).

High-yield exam pointers

- ▶ **Park's six basic steps of an RCT:** drawing up a protocol → selecting reference and experimental populations → randomization → manipulation or intervention → follow-up → assessment of outcome (Park 28e, Ch 3, p 101).
- ▶ **Park's six types by purpose (mnemonic "C-P-R-C-A-E"):** Clinical, Preventive, Risk-factor, Cessation, Aetiological-agent, Evaluation of health services.
- ▶ **Park's two named study designs:** concurrent parallel and cross-over — draw Fig 10 with the wash-out shown as diagonal lines.
- ▶ **Cluster-trial design effect** = $1 + (m - 1)r$ (m = subjects per cluster, r = intra-cluster correlation coefficient) — quote this whenever a community-based or school-based trial is described.
- ▶ **WHO clofibrate trial:** >15,000 men, 3 centres, mean 9.6 years, fewer non-fatal infarcts but **25 per cent more deaths** on clofibrate — the standard example of a risk-factor trial that protected the public. **MRC whooping-cough vaccine trial 1946:** attack rate 1.45 versus 6.72 per 1000 child-months — the standard preventive-trial example.
- ▶ **Retrolental fibroplasia / 50 per cent oxygen for 28 days in babies ≤ 1500 g** — the standard trial-of-an-aetiological-agent example; **domiciliary versus sanatorium treatment of tuberculosis in India** — the standard health-services-evaluation example.
- ▶ **Phase III = pivotal/confirmatory; Phase IV = post-marketing.** Non-inferiority needs a **pre-specified margin** and a **larger sample size** than superiority.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: under the Indian Council of Medical Research National Ethical Guidelines 2017, a Phase IV clinical trial on a drug with market authorisation of less than four years needs regulatory approval from the Central Drugs Standard Control Organisation in addition to ethics committee approval, whereas routine post-marketing surveillance may not require ethics committee approval.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 3, Principles of Epidemiology and Epidemiologic Methods, pp 101–105 (the spine for the basic steps, the six types by purpose, and the concurrent parallel and cross-over designs); Gordis Epidemiology 7e — Ch 10, pp 227–229 and Ch 11, pp 247–248 (planned and unplanned cross-over, factorial design, Physicians' Health Study); Handbook of Epidemiology (Ahrens & Pigeot) — Ch 8, Intervention Trials, pp 368–369 (cluster randomization and the design effect); Fletcher — Clinical Epidemiology: The Essentials, Ch 9, pp 154, 161–165 (efficacy versus effectiveness, large simple and pragmatic trials, superiority/equivalence/non-inferiority, N-of-1 trials); Browner — Designing Clinical Research 5e, Ch 11 (Table 11.2, pp 351–352) and Ch 12 (pp 379–383, 399) (phases of testing, active-control, equivalence, adaptive and factorial designs); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e, Ch 4 (General Epidemiology), p 57 (the "therapeutic exploratory" and "therapeutic confirmatory" labels for Phase II and Phase III); Indian Council of Medical Research National Ethical Guidelines for Biomedical and Health Research Involving Human Participants, 2017, pp 86–89 (phase definitions for drugs and vaccines, Phase IV scope).

**Disease surveillance**

Asked in: Paper I Nov 2018 — RGUHS MD Community Medicine

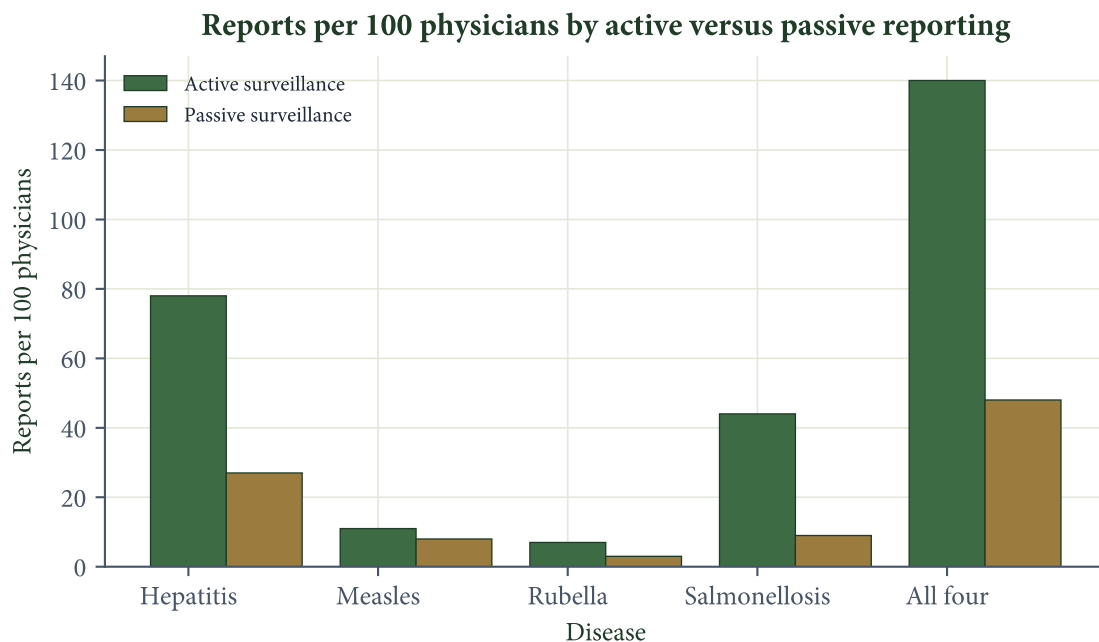
Definition — Surveillance is the "continuous scrutiny of all aspects of occurrence and spread of disease that are pertinent to effective control"; it goes beyond passive reporting of cases and its ultimate objective is prevention (Park 28e, Ch 3, p 151). The Centers for Disease Control and Prevention (CDC) define epidemiological surveillance as the ongoing systematic collection, analysis and interpretation of health data essential to the planning, implementation and evaluation of public health practice, closely integrated with the timely dissemination of these data to those who need to know (Gordis 7e, Ch 3, p 57). Surveillance is continuous and ongoing and needs professional analysis and judgement leading to control recommendations, whereas monitoring is intermittent or episodic (Park 28e, Ch 2, p 59).

Surveillance is classified along four axes — **who initiates the data, how much of the population is covered, the content and diagnostic certainty of the data, and the method or source that generates it.** The same system is usually described by more than one of these labels at once.

1. By who initiates the data — passive, active and sentinel

Type	What it is	Strengths	Limitations	Example
Passive (routine reporting, institution-based, provider-initiated)	Relies on the physician, laboratory or hospital staff to take the initiative to report; reporting is mandated or requested by government or the local health authority (Oxford Public Health 7e, Ch 42, p 1553; Gordis 7e, Ch 3, p 58)	Simplest, cheapest and quickest method; easy to set up; allows international comparison across countries with notification systems	Under-reporting and incomplete, variable-quality data; local outbreaks missed because few cases are diluted in a large denominator; only those who attend a facility are counted	Monthly institutional reporting of the enlisted 20 communicable diseases from sub-centre/PHC/CHC/hospital → district → State Bureau of Health Intelligence → Central Bureau of Health Intelligence (CBHI) (Sunder Lal, Ch 7, p 302)
Stimulated or enhanced passive	During an outbreak the investigator writes to providers describing the situation and asks for reports of similar cases (Sunder Lal, Ch 7, p 303)	Rapidly improves case ascertainment without a new system	Temporary; depends on provider goodwill	Letter to practitioners during a suspected measles or hepatitis outbreak
Active (active case detection)	The public health agency itself contacts providers at fixed intervals, or dedicated staff make periodic field and house visits for case finding (Oxford Public Health 7e, Ch 42, p 1553; Gordis 7e, Ch 3, p 58)	Most complete and uniform reporting; direct link with providers allows definitions to be clarified; best approach in an outbreak and in the final phase of an eradication programme	Costly in staff and money, so must be restricted to a few high-priority, epidemic-prone diseases	Fortnightly fixed-beat house visits with thick and thin blood smears in malaria; photo-flash-card searches in smallpox and guinea-worm eradication; acute flaccid paralysis (AFP) active search, including the pre-certification phase where zero incidence must hold for 3 years (Sunder Lal, Ch 7, p 303)
Sentinel	A small number of selected sites or providers report; the sentinel data are extrapolated to the whole population to estimate prevalence (Park 28e, Ch 2, p 59)	Reporting bias minimised and feedback to providers simplified; high-quality, timely data at low cost because staff are specially trained and supervised; combines well with active surveillance	Not effective for rare disease or disease outside the catchment area; representativeness must be designed in	HIV sentinel surveillance (55 sites from 1994, raised to 180 sites in 1998); 1–3 sentinel sites in a large hospital per high-endemic district for severe malaria and malaria deaths; sentinel influenza-like illness (ILI) and severe acute respiratory infection (SARI) sites (Park 28e, Ch 7, pp 498, 520)

Type	What it is	Strengths	Limitations	Example
Comprehensive	Includes all cases occurring in the whole population of the geographical area covered (national or regional)	Complete denominator; no extrapolation needed	Expensive; often unachievable	Notifiable-disease reporting across a whole state
Compulsory vs voluntary	Data submission made mandatory by law, professional edict or policy, versus a voluntary approach (Oxford Public Health 7e, Ch 42, p 1553)	A legal or professional obligation improves completeness and timeliness	Does not by itself improve representativeness or validity	Notification of cholera, plague and yellow fever under the International Health Regulations



Active surveillance yielded 2.9 times as many reports overall as passive surveillance (ratios: hepatitis 2.9, measles 1.4, rubella 2.3, salmonellosis 4.9), at a cost of \$30,861 against \$11,053 for the passive system. Source: Handbook of Epidemiology (Ahrens & Pigeot), Ch 33, Tables 1.2 and 1.3, pp 1323-1324 — Monroe County, New York, 1980-81 and Vermont, 1981.

2. By the population covered

Type	Scope	Example (Park 28e, Ch 3, pp 151-152)
Individual surveillance	Surveillance of an infected person until they are no longer a significant risk to other individuals	Contact of a case kept under watch through the incubation period
Local population surveillance	A defined local community or district	Surveillance of malaria

Type	Scope	Example (Park 28e, Ch 3, pp 151-152)
National population surveillance	The whole country	Surveillance of smallpox after eradication
International surveillance	The World Health Organization maintains surveillance of important diseases and gives timely warning to all national governments	Influenza, malaria, polio; diseases subject to the International Health Regulations — cholera, plague, yellow fever (Park 28e, Ch 3, p 144)

3. By the content and diagnostic certainty of the data

Type	Definition	Trigger for action
Indicator-based	Routine reporting of disease cases — notifiable-disease systems, sentinel systems and laboratory systems — collected weekly or monthly in a structured, standardised way to build historical trends and baselines (Oxford Public Health 7e, Ch 42, p 1555)	Public health response is triggered when a pre-defined threshold is crossed
Event-based	Rapid detection, reporting, confirmation and assessment of public health events — clusters of disease and rumours of unexpected deaths — using broader, more sensitive definitions and flexible reporting structures (Oxford Public Health 7e, Ch 42, pp 1555-1556)	Any rare, new or high-impact event, detected while still geographically localised
Syndromic (S)	Diagnosis made on the basis of the clinical pattern , by paramedical personnel and members of the community (Park 28e, Ch 7, p 569)	Broad syndromes: fever with rash, fever with altered sensorium or convulsions, cough >3 weeks, acute flaccid paralysis, diarrhoea, jaundice, unusual events causing death or hospitalisation
Presumptive (P)	Diagnosis made on typical history and clinical examination by a medical officer	Probable case counted pending laboratory work
Confirmed / laboratory (L)	Clinical diagnosis by a medical officer and/or positive laboratory identification	Confirmed case; feeds outbreak confirmation

Indicator-based and event-based systems have different functions but are complementary components of a single national surveillance system, not alternatives.

4. By the method and source of the data

Type	What it collects
Community-level surveillance	People themselves, registered medical practitioners, village health guides, <i>Anganwadi</i> workers, ASHAs and village leaders report births, deaths, outbreaks and unusual events; informants are contacted regularly by health workers (Sunder Lal, Ch 7, p 302)
Routine institutional reporting	Facility records of notifiable cases and deaths, transmitted up the reporting chain — the commonest method because it is inexpensive

Type	What it collects
Surveys and special studies	Sample or disease surveys repeated at intervals to show change in incidence — blindness surveys, the National Oral Health Survey and fluoride mapping, non-communicable disease risk-factor surveys at 1–5 year intervals , and multipurpose designs such as the National Family Health Survey (Sunder Lal, Ch 7, p 303)
Case and outbreak investigation	Age-specific attack rates, case fatality rates, rate of serious disability and estimates of vaccine efficacy — information not obtainable from routine systems. Measles surveillance in India is based on outbreak investigation; case-based surveillance and focus investigation began under the malaria elimination programme in 2017
Verbal autopsy	A trained investigator interviews the mother or a witness on a standard format to ascertain the most probable cause of a maternal, infant or other death and whether it was avoidable
Laboratory surveillance	Confirms presumptive syndromes; also supports molecular subtyping, strain characterisation and antimicrobial drug-sensitivity monitoring (Oxford Public Health 7e, Ch 42, p 1562)
Serological surveillance	Patterns of current and past infection, through serum banks — identifies immunity gaps (Park 28e, Ch 3, p 151; Ch 2, p 59)
Entomological surveillance	Vector density and insecticide susceptibility under the vector-borne disease control programme
Environmental surveillance	Sewage samples for wild poliovirus , and air, water, noise and chemical monitoring (Sunder Lal, Ch 7, p 303)
Disease registries	Comprehensive longitudinal listings of people with a condition, with diagnosis, treatment and outcome; cut across severity levels and give information over time on individuals — cancer and birth-defect registries
Behavioural and second-generation surveillance	Repeat cross-sectional behavioural surveys combined with biological (sentinel sero-) surveillance and case, death and tuberculosis data, to explain an epidemic rather than only count it — the World Health Organization/UNAIDS model for HIV (Oxford Public Health 7e, Ch 42, p 1566)
Digital, internet-based and big-data surveillance	ProMED (from 1994 , which carried the earliest public notice of SARS in early 2003), the Global Public Health Intelligence Network scanning online media in six languages in near real time, search-trend monitoring and participatory "crowd-sourced" reporting (Oxford Public Health 7e, Ch 42, pp 1566-1567)

By subject matter these same systems are named **epidemiological, demographic and nutritional surveillance** — nutritional surveillance being "keeping watch over nutrition, in order to make decisions that will lead to improvement in nutrition in population" (Park 28e, Ch 2, p 59; Ch 11, p 793).

5. Types of surveillance in Indian practice

The **Integrated Disease Surveillance Programme (IDSP)** — launched as the Integrated Disease Surveillance Project in **November 2004**, with a Central Surveillance Unit at the National Centre for Disease Control, Delhi, State Surveillance Units at every State/Union Territory headquarters and

District Surveillance Units in every district — runs four surveillance streams together (Park 28e, Ch 7, pp 568-569):

- ▶ **Regular (passive) surveillance** — malaria; acute diarrhoeal disease (cholera) and typhoid; tuberculosis; measles; polio; road traffic accidents (linked with police records); plague; and unusual clinical syndromes such as meningoencephalitis, respiratory distress and haemorrhagic fevers.
- ▶ **Sentinel surveillance** — sexually transmitted and blood-borne infection (HIV, hepatitis B virus, hepatitis C virus), water quality and outdoor air quality in large urban centres.
- ▶ **Regular periodic surveys** — non-communicable disease risk factors: anthropometry, physical activity, blood pressure, tobacco and nutrition.
- ▶ **Additional state priorities** — each state may add up to five further conditions.

Vertical national surveillance also runs for AFP (polio eradication), HIV/AIDS, vector-borne diseases, vaccine-preventable diseases, acute respiratory infection and diarrhoea in children, nutrition, and blindness (Sunder Lal, Ch 7, p 302).

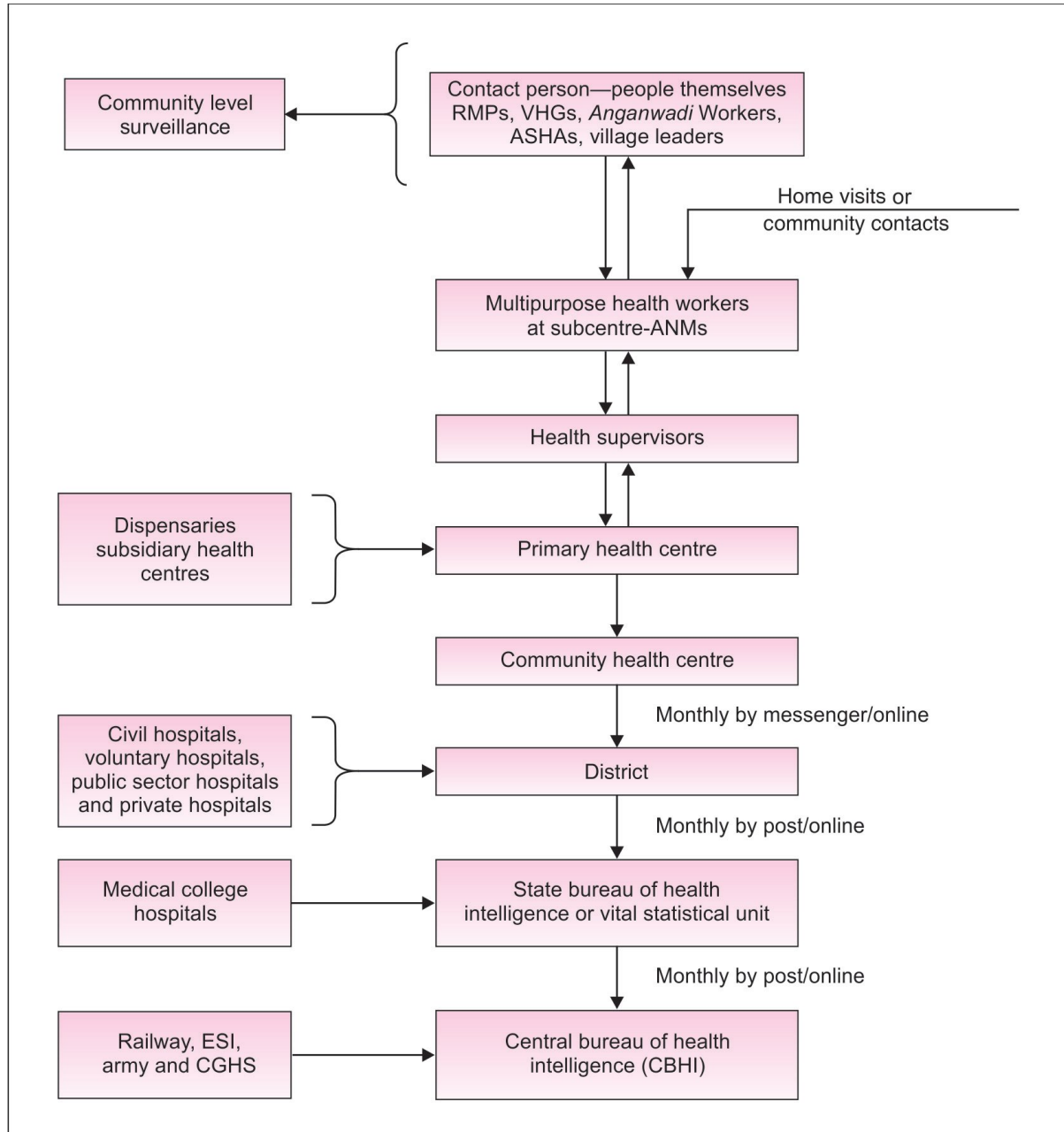
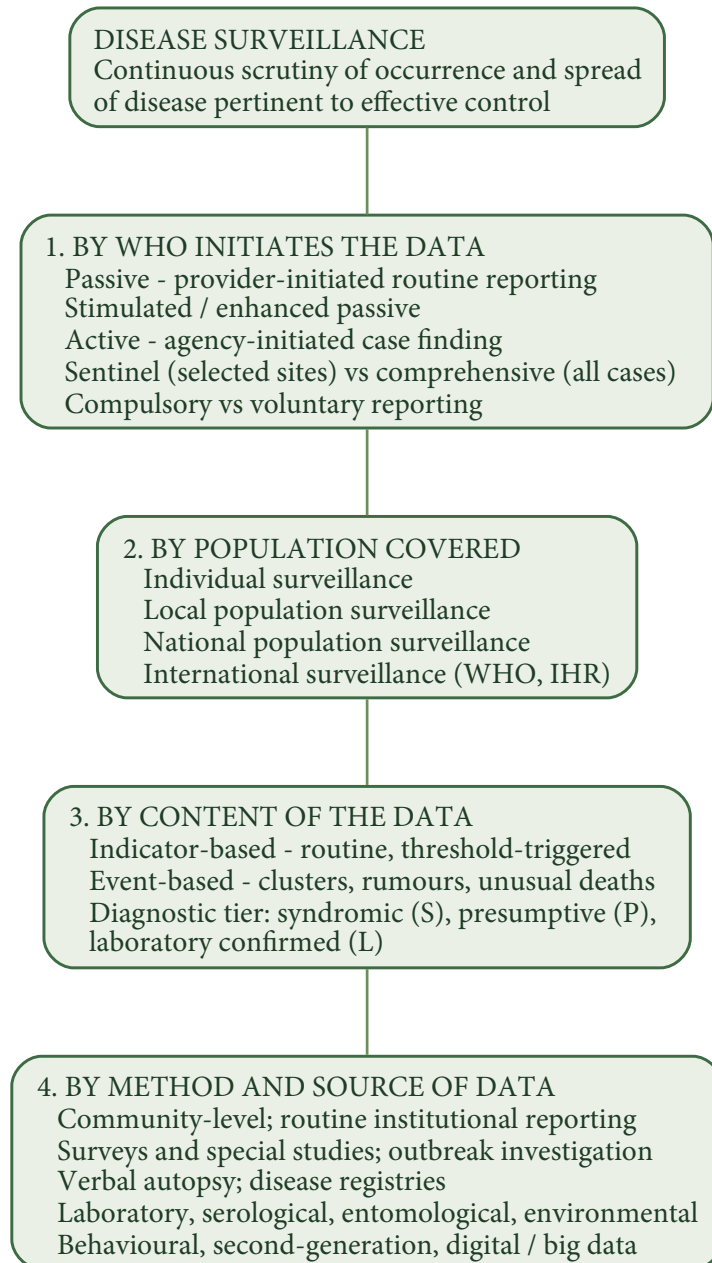


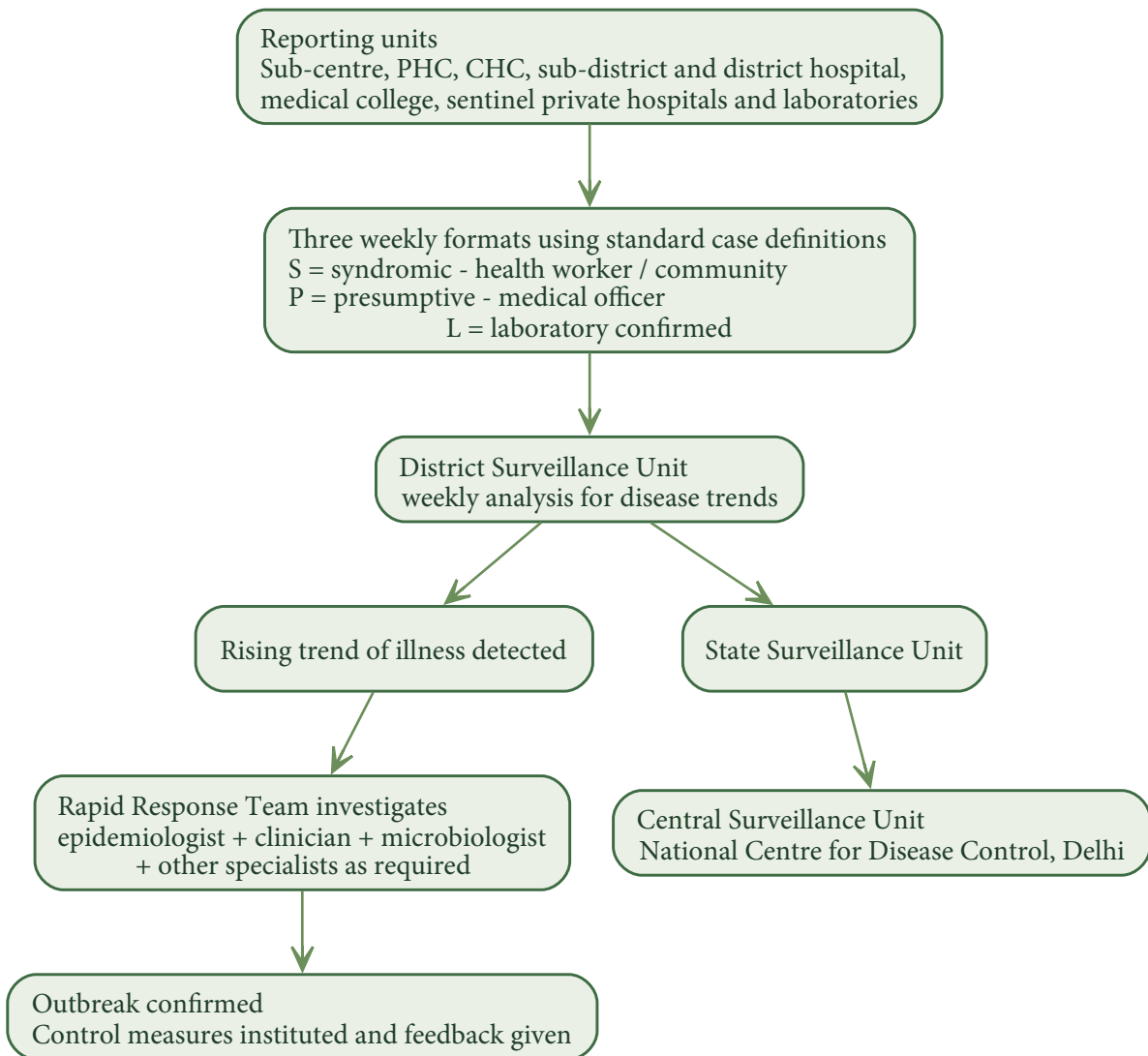
Fig 7.7 — Channel of information flow in community and institutional level surveillance (Sunderlal, p.304)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Labelled diagram to sketch by hand — the surveillance cycle.** Draw a closed loop of five boxes and label them exactly: (a) Collection of data → (b) Compilation and transmission of data → (c) Analysis and interpretation → (d) Follow-up action / decision making → (e) Feedback, with the arrow from (e) returning to (a). Write alongside the loop the three prerequisites of effective surveillance: **use of a standard case definition, regularity of reports, action on reports** (Park 28e, Ch 7, p 569; Sunder Lal, Ch 7, p 302).



Classification of disease surveillance along four axes



Weekly reporting and response pathway under the Integrated Disease Surveillance Programme

High-yield exam pointers

- **Four axes mnemonic "I-P-C-M"** — Initiator (passive/active/sentinel), Population covered (individual/local/national/international), Content (indicator-based/event-based; S-P-L), Method and source.
- **Park's four types by population:** individual → local (malaria) → national (smallpox after eradication) → international (WHO — influenza, malaria, polio).
- **IDSP letters S, P, L** = syndromic by paramedical staff and community; presumptive by the medical officer; laboratory-confirmed. IDSP launched **November 2004**; Central Surveillance Unit at the National Centre for Disease Control, Delhi; weekly reporting; a rising trend triggers the **Rapid Response Team**.
- **Five essential elements of surveillance:** collection → compilation and transmission → analysis and interpretation → action → feedback.
- **Sentinel surveillance** exists because no routine notification system identifies all cases; sentinel data are **extrapolated** to the whole population; reporting bias is minimised and feedback simplified.

- ▶ **Active surveillance is reserved for eradication endgames and outbreaks** — smallpox, polio (AFP), guinea worm, malaria; it gave **2.9 times** as many reports as passive surveillance in the classic Monroe County comparison, at nearly three times the cost.
- ▶ Surveillance versus monitoring: **surveillance is continuous and needs professional judgement; monitoring is episodic and can be delegated to technicians and instruments.**

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park 28e still describes the system as the Integrated Disease Surveillance *Project* and notes that more than 90 per cent of districts were sending weekly data by e-mail or portal; weekly S, P and L reporting has since moved to the web-enabled Integrated Health Information Platform (IHIP), which Park 28e mentions only in passing in the leprosy chapter, so quote the platform name with that caveat rather than as a Park-sourced fact.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, WHO International Health Regulations (2005) 2024 amendments, WHO Pandemic Agreement 2025, Africa CDC and US CDC event-based surveillance indicator framework, ICMR Antimicrobial Resistance Surveillance Network

Programme and policy updates (India)

- ▶ **Weekly S, P and L reporting has moved onto the web-enabled Integrated Health Information Platform, but IHIP-IDSP now runs in parallel with the vertical vector-borne programme rather than replacing it.** A 2023–25 field study in two high-burden districts of a western state found IHIP-IDSP capturing case-based, real-time S, P and L forms while the National Vector Borne Disease Control Programme (NVBDCP) aggregated the same dengue caseload weekly on paper, each with a different outbreak trigger — NVBDCP declaring an outbreak on five cases or one death, IHIP-IDSP on a threshold derived from the previous three to five years of district data. Entomological data and platelet counts stayed outside both systems; district public health laboratories set up under IHIP-IDSP were working in only three of 36 districts, and four districts had no sentinel hospital at all. [PMID 42245358]
- ▶ **One Health surveillance in India is now being audited with a formal instrument.** A policy content analysis in Gujarat applied the One Health Surveillance Matrix to the two parallel national systems — IDSP-IHIP for human health and the **National Animal Disease Reporting System (NADRS)** for animal health — and found the human-health side weak on data sharing, sampling, outreach to decision-makers and external communication, and the animal-health side weak on data storage, interpretation and dissemination to decision-makers. Cross-sectoral surveillance for zoonoses is therefore structurally, not just operationally, incomplete. [PMID 39453563]

Current national data

- ▶ **Cholera, IDSP 2015–19 against the 2017–18 serosurvey.** IDSP recorded 29,400 cases from 22 of 36 States/Union Territories and 124 of 641 districts, with Maharashtra (8,040) and West Bengal (5,291) heaviest; the national serosurvey of 7,882 sera aged 9–45 years found 11.7 per cent (95% CI 10.4–13.3) with a vibriocidal titre of 1:320 or above. The gap is the cleanest available Indian

illustration of how outbreak-triggered passive reporting under-ascertains a notifiable disease. [PMID 39226336]

- **Measles–rubella campaign coverage, 2019–20 post-campaign evaluation in five districts.** Documented-or-recall coverage in children aged 9 months to 5 years ranged from 74.2 per cent (95% CI 64.1–82.2) in Kanpur Nagar, Uttar Pradesh to 90.4 per cent (95% CI 83.7–94.6) in Dibrugarh, Assam — no district reached the 95 per cent elimination target, despite caregiver awareness of 88–98 per cent. [PMID 38551928]
- **Antimicrobial resistance surveillance.** Within the ICMR Antimicrobial Resistance Surveillance Network (ICMR-AMRSN), a north Indian tertiary centre's 2018–2024 series reported 20,509 of 130,077 specimens culture-positive (15.8 per cent), 80.3 per cent Gram-negative, *Escherichia coli* commonest (32.3 per cent) and *Acinetobacter baumannii* the leading intensive-care isolate (41.9 per cent), with carbapenem-resistant *A. baumannii* and MRSA rising. [PMID 40536523]

Elimination, surveillance and disease status

- **Kala-azar — target met, but only provisionally.** India crossed the WHO threshold for elimination as a public health problem (under 1 case per 10,000 population at sub-district/block level) in **December 2023**: annual cases fell from 77,102 in 1992 to 599 in 2023, deaths from 1,419 to four, and endemic blocks still above the threshold from 633 in 2005 to 136 in 2015, eight in 2021 and none in 2023. India has entered a **consolidation phase and must hold the threshold for three consecutive years** before submitting a dossier for WHO validation — Bangladesh was validated first, in 2023. Surveillance rests on the NCVBDC-run **Kala-azar Management Information System (KAMIS)** register, and independent programme assessments in 2019, 2021 and 2023 flagged surveillance, data-use and vector-control bottlenecks; post-kala-azar dermal leishmaniasis (9,108 cases over 2014–23) and VL–HIV co-infection (5–7 per cent of Indian VL patients) remain the human reservoirs requiring continued active case detection. [PMID 40037366] [PMID 40828749]
- **Residual-specimen sentinel serosurveillance is emerging as a cheap alternative to household serosurveys.** Anonymised leftover diagnostic sera from public and private laboratories in Kanpur Nagar and Palghar districts were tested for measles and rubella IgG before and after the 2018–19 supplementary immunisation activity, reproducing the community-serosurvey trend for rubella; selection bias and thin metadata are the trade-offs, but head-to-head work elsewhere put the cost at roughly one-seventh that of a school-based cluster sample. This extends classical serological surveillance from serum banks to routine laboratory left-overs. [PMID 39552145]

Guideline and legislative updates

- **Event-based surveillance finally has a standard evaluation framework.** Africa CDC with the US CDC published an adaptable input–activity–output–outcome–impact indicator framework in 2024, the first comprehensive guide for monitoring and evaluating EBS systems; the two attributes that define EBS — timeliness and sensitivity — had until then been measured by only a small minority of published evaluations, even though EBS had been rolled out in communities and health facilities across 33 African countries and WHO guidance dates back to 2008. A parallel AMSTAR-appraised evidence map of low- and middle-income countries (15 systematic reviews, 82 primary articles, India among the implementing countries) concluded that EBS improves early outbreak detection and alert response. [PMID 38364834] [PMID 40388686]

Global and WHO developments

- **The legal architecture of international surveillance has been rewritten twice in two years.** The 77th World Health Assembly agreed amendments to the International Health Regulations (2005) on **1 June 2024**, adding a "**pandemic emergency**" tier above the public health emergency of international concern and a revised Article 44 committing to sustainable financing for developing countries through a new coordinating financial mechanism under the Health Assembly; the amendments entered into force in **September 2025**. On **20 May 2025** the 78th World Health Assembly adopted the **WHO Pandemic Agreement** (35 Articles, 124 governments), which layers legally binding commitments on surveillance, pathogen data sharing, health-worker protection and equitable access to countermeasures on top of the IHR — with enforceability, the definition of "pandemic" and unequal national capacity flagged as its weak points. [PMID 40589223] [PMID 40443067]
- **Wastewater and environmental surveillance has broken out of the polio programme.** An Asia-wide assessment counted 89 wastewater surveillance projects across 45 institutions in 19 countries, most still research-driven with only about one-fifth embedded in national surveillance systems, and SARS-CoV-2, antimicrobial-resistant bacteria, enterovirus, influenza and poliovirus as the five commonest targets. In Mumbai, sampling of slum open drains and two sewage treatment plants from August 2022 to June 2023 detected rising SARS-CoV-2 viral load about three weeks before the city's clinical case surge and sequenced the shifting variants — a working demonstration that environmental surveillance can serve as a population-level early-warning system beyond sewage testing for wild poliovirus. [PMID 40828164] [PMID 39913514]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 2 (Concept of Health and Disease, p 59), Ch 3 (Principles of Epidemiology and Epidemiologic Methods, pp 144, 151-152), Ch 7 (Health Programmes in India, pp 498, 519-520, 568-569), Ch 11 (Nutrition and Health, p 793). Gordis Epidemiology 7e — Ch 3 (The Occurrence of Disease: I. Disease Surveillance and Measures of Morbidity, pp 57-58). Oxford Textbook of Public Health 7e — Ch 42 (Public Health Surveillance, pp 1553-1567). Handbook of Epidemiology (Ahrens & Pigeot) — Ch 33 (Infectious Disease Epidemiology, pp 1322-1326). Sunder Lal — Textbook of Community Medicine, Ch 7 (Epidemiology, pp 302-307), including Fig 7.7, p 304.

Recent-advances sources: 42245358 — Informational challenges of dengue surveillance and response: a case study from India, *Frontiers in Public Health* 2026, doi:[10.3389/fpubh.2026.1809584](https://doi.org/10.3389/fpubh.2026.1809584); 39453563 — Application of the One Health Surveillance (OHS) Matrix to evaluate the disease surveillance systems in Gujarat, India: a policy content analysis, *Journal of Epidemiology and Global Health* 2024, doi:[10.1007/s44197-024-00317-2](https://doi.org/10.1007/s44197-024-00317-2); 39226336 — Analysis of cholera risk in India: insights from 2017-18 serosurvey data integrated with epidemiologic data and societal determinants from 2015-2019, *PLoS Neglected Tropical Diseases* 2024, doi:[10.1371/journal.pntd.0012450](https://doi.org/10.1371/journal.pntd.0012450); 38551928 — Post-campaign coverage evaluation of a measles and rubella supplementary immunization activity in five districts in India, 2019-2020, *PLoS One* 2024, doi:[10.1371/journal.pone.0297385](https://doi.org/10.1371/journal.pone.0297385); 40536523 — Superbug surge: a tale of seven-year battle against escalating antimicrobial resistance in North India, *Current Microbiology* 2025, doi:[10.1007/s00284-025-04321-y](https://doi.org/10.1007/s00284-025-04321-y) (abstract-only); 40037366 — Kala-azar elimination in India: reflections on success and sustainability, *International Health* 2025, doi:[10.1093/inthealth/ihaf013](https://doi.org/10.1093/inthealth/ihaf013); 40828749 — The story of elimination of visceral leishmaniasis (kala-azar) in India: challenges towards sustainment, *PLoS Neglected Tropical Diseases* 2025, doi:[10.1371/journal.pntd.0013321](https://doi.org/10.1371/journal.pntd.0013321); 39552145 — Increased measles and rubella seroprevalence in children using residual blood samples from health facilities and household serosurveys after supplementary immunization activities in two districts in India, *Epidemiology and Infection* 2024, doi:[10.1017/S0950268824001353](https://doi.org/10.1017/S0950268824001353); 38364834 — An indicator framework for the monitoring and evaluation of event-based surveillance systems, *The Lancet Global Health* 2024, doi:[10.1016/S2214-109X\(24\)00034-2](https://doi.org/10.1016/S2214-109X(24)00034-2); 40388686 — Event-based surveillance in middle- and low-income countries: an evidence map, *Medwave* 2025, doi:[10.5867/medwave.2025.04.3031](https://doi.org/10.5867/medwave.2025.04.3031) (abstract-only); 40589223 — The 2025 World Health Assembly Pandemic Agreement and the 2024 amendments to the International Health Regulations combine for pandemic preparedness and response, *Medical Science Monitor* 2025, doi:[10.12659/MSM.950411](https://doi.org/10.12659/MSM.950411); 40443067 — The Pandemic Agreement: what's next?, *International Journal of Health Planning and Management* 2025, doi:[10.1002/hpm.70000](https://doi.org/10.1002/hpm.70000) (abstract-only); 40828164 — Wastewater surveillance for early pathogen detection in Asia, *International Journal of Environmental Health Research* 2025, doi:[10.1080/09603123.2025.2544736](https://doi.org/10.1080/09603123.2025.2544736) (abstract-only); 39913514 — Wastewater-based genomic surveillance of SARS-CoV-2 in vulnerable communities in Mumbai, *Indian Journal of Medical Research* 2024, doi:[10.25259/ijmr.299.24](https://doi.org/10.25259/ijmr.299.24). Plus: WHO — amendments to the International Health Regulations (2005), 77th World Health Assembly, 1 June 2024, in force September 2025;

WHO Pandemic Agreement, 78th World Health Assembly, 20 May 2025.

♦ ♦ ♦

Broad Street cholera outbreak

Asked in: Paper I 02-Jan-2025 — RGUHS MD Community Medicine

Definition — The Broad Street (Golden Square, Soho) outbreak of 1854 is the investigation in which *John Snow*, an English physician-epidemiologist, used a spot map of cholera deaths and a comparison of death rates by water supplier to show that cholera was transmitted by faecally polluted drinking water, and had the handle of the incriminated Broad Street pump removed. Cholera is often called the "**father of public health**", and Snow's work between 1848 and 1854 established the role of polluted drinking water in its spread long before *Vibrio cholerae* was known (Park 28e, Ch 1, p 17).

The outbreak — the facts an examiner expects

Item	Fact
Place, year	Golden Square district, Soho, London; 1854 epidemic (Park 28e, Ch 3, pp 86–87)
Magnitude	About 600 deaths in the first week of September 1854 among people living within a few blocks of the Broad Street pump (Gordis 7e, Ch 1, pp 29–30)
Investigator	John Snow — better known in his own day as the anaesthetist who gave chloroform to Queen Victoria in childbirth (Gordis 7e, Ch 1, pp 29–30)
Data source	Death registration of the Registrar General, William Farr , who initially favoured the miasma theory and was later convinced (Gordis 7e, Ch 1, pp 29–30; Nelson & Williams, Ch 1, p 26)
Prevailing theory	Miasma — disease from foul or malodorous air; no successful control measure had ever come from it (Nelson & Williams, Ch 2, p 39)
State of knowledge	<i>Vibrio cholerae</i> unknown; the conclusion rested entirely on observational data (Gordis 7e, Ch 1, p 31)
Action taken	Snow lobbied local politicians to shut the Broad Street pump (now Broadwick Street); the epidemic abated (Lilienfeld's Foundations, Ch 2, pp 43–44)

What Snow actually did, and the modern step each prefigures

Snow's step	Present-day equivalent in outbreak investigation
Counted deaths house to house and recorded each house's water supplier — "shoe-leather epidemiology" (Gordis 7e, Ch 1, pp 29–31)	Case finding and line listing against a case definition, with active daily search for secondary cases (Park 28e, Ch 3, p 160)
Plotted deaths on a spot map , showing clustering around one pump (Park 28e, Ch 3, pp 86–87)	Analysis by place using spot or shaded maps; the hand-drawn map already meets the definition of a geographic information system (GIS) (Nelson & Williams, Ch 6, p 185)

Snow's step	Present-day equivalent in outbreak investigation
Dated the deaths and followed the rise and fall of the epidemic	Analysis by time — the epidemic curve separates a common-source from a propagated epidemic; water-borne cholera begins as common-source, peaks sharply, then tails off as it continues person to person (Park 28e, Ch 3, pp 84–85)
Reasoned that the mouth was the portal of entry and water the vehicle (Lilienfeld's Foundations, Ch 2, p 43)	Formulation of hypothesis covering possible source, causative agent, mode of spread and the environmental factors that permitted it (Park 28e, Ch 3, p 160)
Ascertained the houses supplied by each company, computed cholera deaths per 10,000 houses for the first seven weeks, and compared these with the rest of London (Lilienfeld's Foundations, Ch 2, p 45)	Testing of hypothesis by comparing attack rates in those exposed and those not exposed (Park 28e, Ch 3, p 160) — in modern terms a cohort study (Handbook of Epidemiology, Ch 5, pp 262–263)
Had the pump handle removed (Park 28e, Ch 3, pp 143–144)	Immediate interruption of the vehicle of transmission, without waiting for laboratory confirmation

The "grand experiment" — Snow's natural experiment on water supply, London 1854

The Lambeth Company had moved its Thames intake **upstream** of the sewage discharge point while the Southwark and Vauxhall Company continued to draw water **downstream**; because the two companies' mains were interspersed, adjacent houses in the same street received water from different sources, so exposure was allocated independently of locality, occupation and social class (Lilienfeld's Foundations, Ch 2, pp 43–45; Handbook of Epidemiology, Ch 5, pp 262–263).

Water supply	Number of houses	Deaths from cholera	Deaths per 10,000 houses
Southwark and Vauxhall Company	40,046	1,263	315
Lambeth Company	26,107	98	37
Rest of London	256,423	1,422	59

Mortality in Southwark and Vauxhall houses was **eight to nine times** that in Lambeth houses (Snow, *On Cholera*, reproduced in Lilienfeld's Foundations of Epidemiology, Ch 2, p 45).

Public health lessons learned

- **Descriptive epidemiology can locate a source before the agent is known.** The clustering of deaths on the spot map around one pump was enough to incriminate the common water supply, and let Snow hypothesise a water-borne disease "long before the birth of bacteriology" (Park 28e, Ch 3, pp 86–87).
- **Do not delay control for aetiological certainty.** Park states the principle explicitly — control measures should not be delayed because of incomplete or absent knowledge of the aetiological agent — and cites this very act of removing the handle of the incriminated pump (Park 28e, Ch 3, pp 143–144).

- ▶ **Denominators, not counts.** Snow first ascertained how many houses each company supplied and only then expressed deaths **per 10,000 houses** (Lilienfeld's Foundations, Ch 2, p 45). A raw count of deaths would have been uninterpretable; the rate carries the comparison.
- ▶ **Recognise and use a natural experiment.** Snow's achievement lay in the logical organisation of observations, the recognition of a natural experiment in the 1854 outbreak, and a quantitative approach to disease occurrence (Lilienfeld's Foundations, Ch 2, p 45) — the ancestor of the exposed-versus-unexposed cohort comparison.
- ▶ **Field work is irreplaceable.** The evidence came from walking the district and enumerating households, not from a desk (Gordis 7e, Ch 1, pp 29–31) — the enduring case for sending a rapid response team to the field.
- ▶ **Vital registration and surveillance are the raw material of public health.** Farr's death registration made the study possible, and Snow's data were so convincing that Farr then required every registrar in south London to record which water company supplied each house in which a person died of cholera (Gordis 7e, Ch 1, p 31) — in effect the first exposure-specific surveillance variable.
- ▶ **Evidence must be converted into law.** Snow's report led to legislation requiring all London water companies to filter their water by **1857** (Lilienfeld's Foundations, Ch 2, p 45); the wider sanitary movement produced the Public Health Act of 1848 and the comprehensive Public Health Act of 1875 (Park 28e, Ch 1, p 17).
- ▶ **Environmental engineering outperforms exhortation.** At that time the Thames was simultaneously the source of drinking water and the depository for sewage (Park 28e, Ch 1, p 17); separating drinking water from excreta — a structural intervention, not advice to individuals — was what ended the problem.
- ▶ **Cholera is a marker of inequity and of the State's duty.** In 1842 the mean age at death in London was 44 years for the gentry and professionals but 22 years for the working class (Park 28e, Ch 1, p 17); cholera remains "a key indicator of lack of social development" and mainly a disease of the poor (Park 28e, Ch 5, pp 287, 292). Chadwick's report and the Public Health Act of 1848 that followed it established that the State is directly responsible for the health of its people (Park 28e, Ch 1, p 17).
- ▶ **Advocacy is part of the epidemiologist's role, but attribution must stay honest.** Snow had to lobby politicians to get the pump closed (Lilienfeld's Foundations, Ch 2, pp 43–44); yet it remains unclear whether removal of the handle actually aided the epidemic's decline, since the outbreak may already have passed its peak (Gordis 7e, Ch 20, p 418). A measure applied to a falling epidemic curve cannot claim credit without a comparison.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- Snow's spot map of Asiatic cholera, Golden Square, London 1854 — labels to reproduce: Broad Street; the **Broad Street pump** (large dot at the centre); the neighbouring pumps that were not implicated (Warwick Street, Great Marlborough Street, Rupert Street, Bridle Street); black bars, one per death, stacked against each house; **Workhouse** (few deaths — it had its own well); **Brewery** (no deaths among the workers); Golden Square. Caption: deaths cluster tightly around a single pump (Park 28e, Ch 3, p 87, Fig 6).

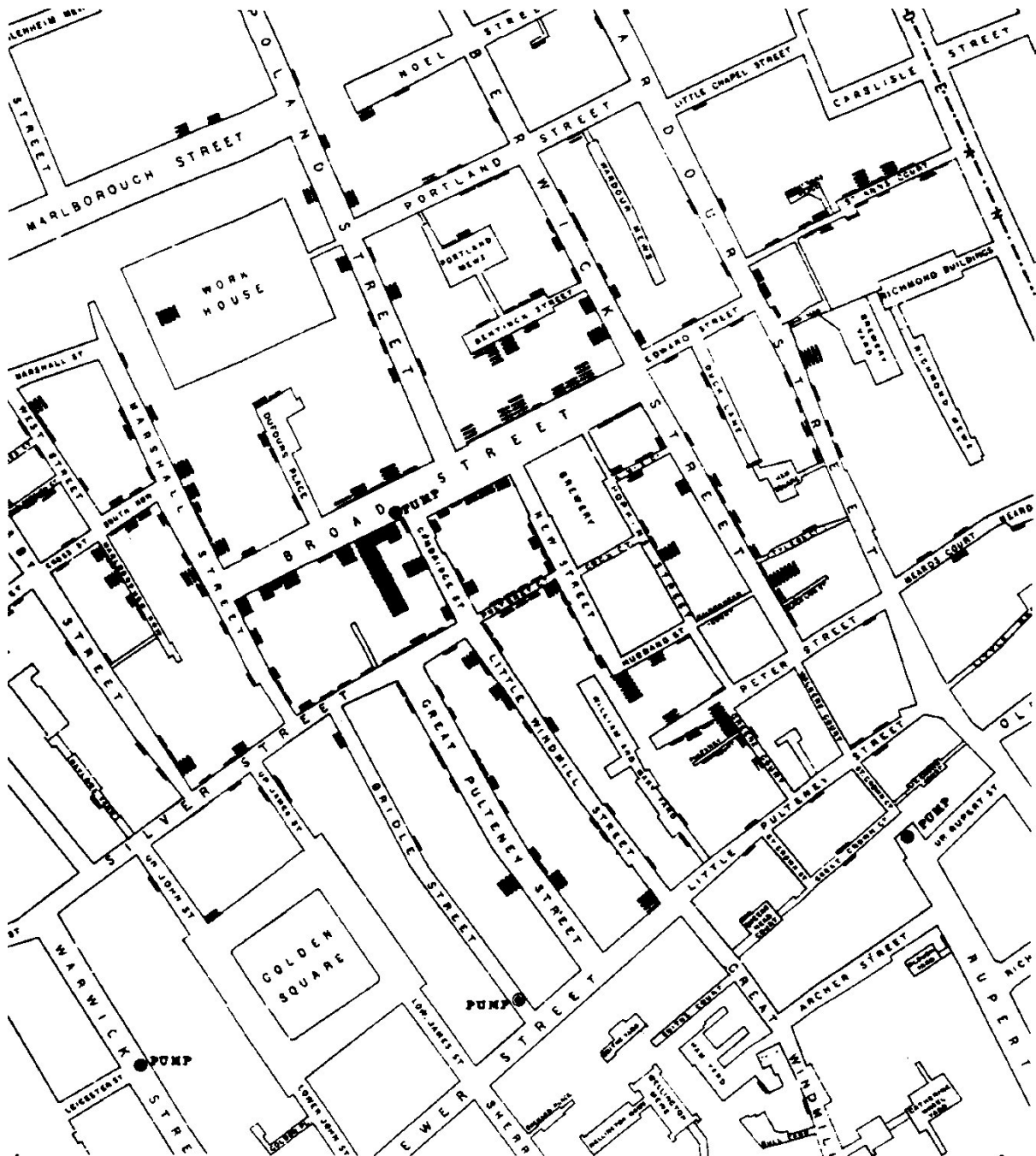
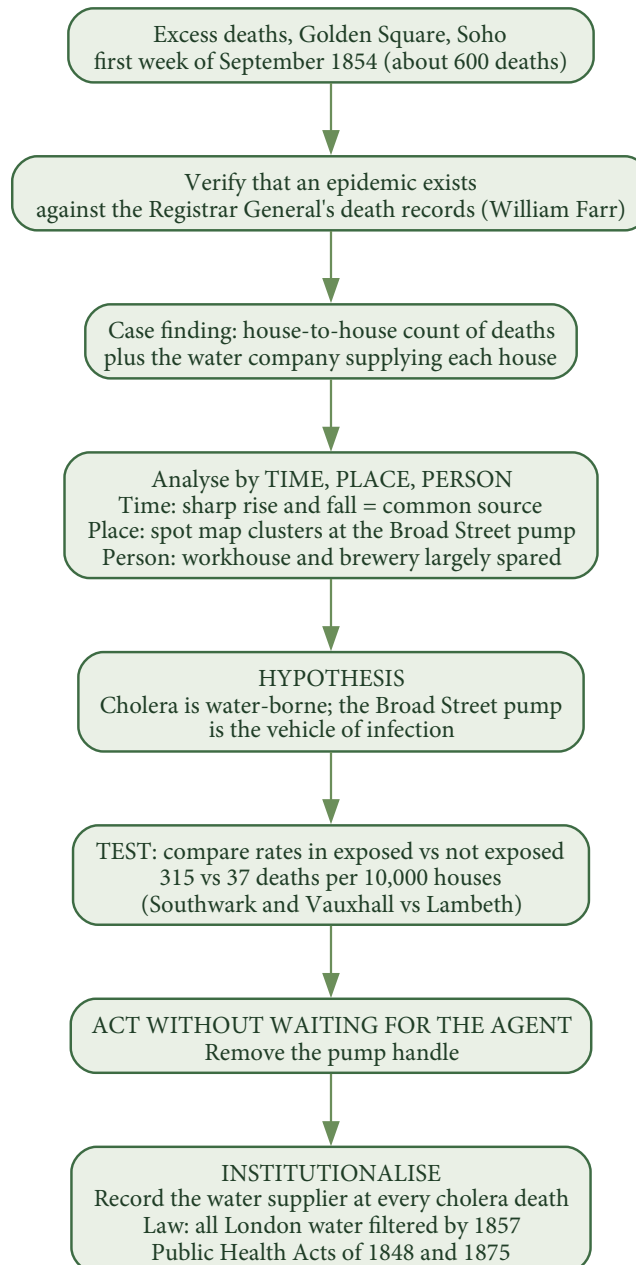


Fig 7.1 — Map of the cholera outbreak in London around Golden Square and the Broad Street pump, indicating deaths, the pumps in the vicinity, and various land use features as drafted by John Snow (Nelson & Williams — *Infectious Disease Epidemiology*, p.186)

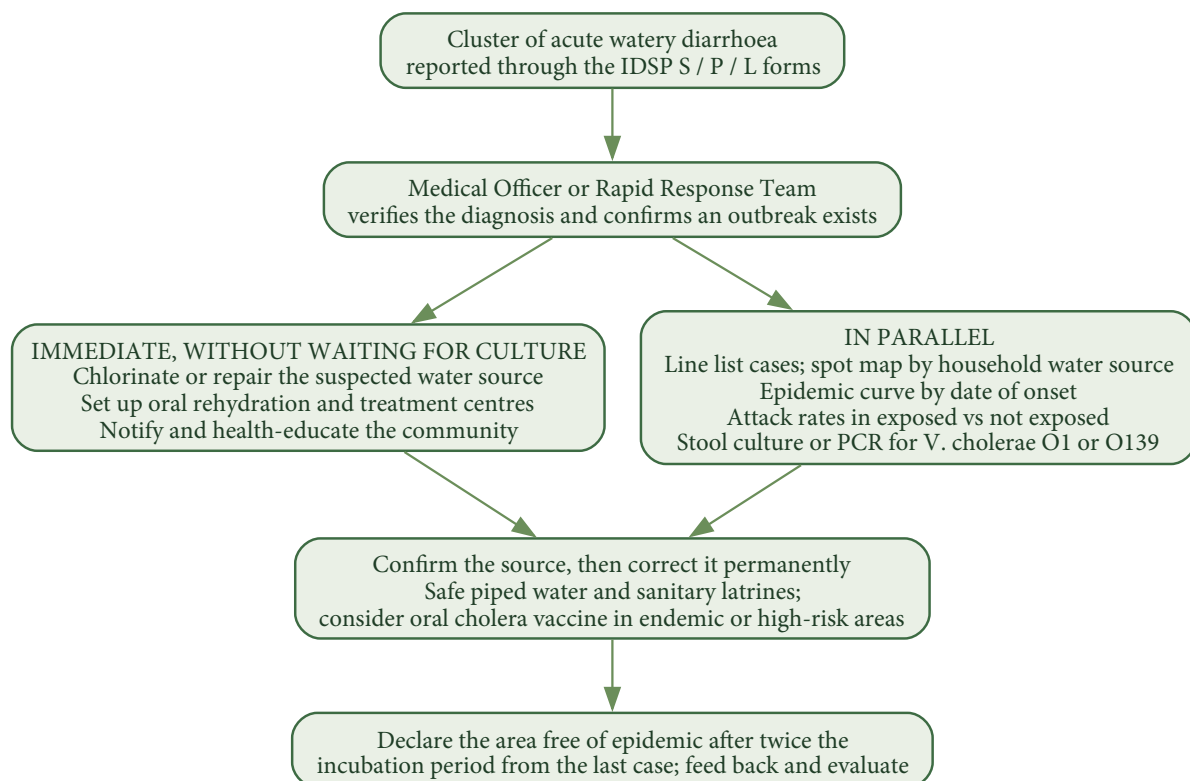


Snow's line of reasoning at Broad Street mapped onto the modern steps of outbreak investigation

Applying the lessons to a suspected water-borne outbreak in India today

Broad Street lesson	Current practice in India
Cases must be counted and reported before they can be acted on	Integrated Disease Surveillance Programme (IDSP) , launched November 2004 — case-based surveillance to detect early warning signals of impending outbreaks, with weekly S (health worker), P (clinician) and L (laboratory) reporting formats; a rising trend is investigated by the Medical Officer or the Rapid Response Team (Jugal Kishore, National Health Programmes of India, pp 486–488; Park 28e, Ch 7, p 568)

Broad Street lesson	Current practice in India
Confirm the outbreak fast	Clinical suspicion of cholera in a person aged 2 years or older with acute watery diarrhoea and severe dehydration; rapid diagnostic tests to detect an outbreak, with stool culture or polymerase chain reaction for <i>V. cholerae</i> O1 or O139 to confirm (Park 28e, Ch 5, pp 290, 292)
Cut the vehicle of transmission first	Treated drinking water carrying free residual chlorine supplied to every family and stored in narrow-mouthed covered containers; boiling or chlorination in rural areas; simple, cheap sanitary excreta disposal (Park 28e, Ch 5, p 291)
Treat where the people are	Home oral rehydration for the mildly dehydrated, who are over 90 per cent of cases; intravenous fluids at the nearest treatment centre for the severely dehydrated; avoid transporting cases long distances, which has itself spread disease (Park 28e, Ch 5, p 291)
Sustain it beyond the epidemic	Cholera surveillance as part of an integrated disease surveillance system with local feedback and global information sharing (Park 28e, Ch 5, p 292); health education on oral rehydration therapy, early reporting, food hygiene, handwashing after defecation and safe water (Park 28e, Ch 5, p 292)



Present-day response to a cluster of acute watery diarrhoea, following the Broad Street principles

High-yield exam pointers

- **Numbers:** 1854, Golden Square (Soho), London; about 600 deaths in the first week of September 1854; Southwark and Vauxhall 315 versus Lambeth 37 deaths per 10,000 houses (rest of London 59) — an eight- to nine-fold difference; London water compulsorily filtered by 1857.
- **Names:** John Snow (investigator, and Queen Victoria's chloroform anaesthetist), William Farr (Registrar General), Edwin Chadwick (report on the sanitary condition of the labouring

population, the "great sanitary awakening"), Sir John Simon (first Medical Officer of Health of London); Public Health Acts of 1848 and 1875.

- ▶ **Principle to quote verbatim:** control measures should not be delayed because of incomplete or absent knowledge of the aetiological agent (Park 28e, Ch 3, pp 143–144).
- ▶ **Method labels to name:** spot map (place), epidemic curve (time), natural experiment, cohort comparison, "shoe-leather epidemiology", attack rates in the exposed versus the not exposed.
- ▶ **The honest caveat:** it is uncertain whether removing the pump handle actually shortened the epidemic — writing this earns marks rather than losing them.
- ▶ **Indian link:** IDSP (2004), its weekly S/P/L reporting and the Rapid Response Team are the direct descendants of Farr's registration plus Snow's field visit.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: Gordis 7e quotes the Lambeth rate as 38 deaths per 10,000 houses whereas Lilienfeld's Foundations, reproducing Snow's own table, gives 37 — write the figures from Snow's table (315 versus 37) and the eight- to nine-fold ratio, on which both texts agree.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 1, 3, 5 and 7); Gordis Epidemiology 7e (Ch 1 and 20); Lilienfeld's Foundations of Epidemiology (Ch 2, Threads of Epidemiologic History); Handbook of Epidemiology, Ahrens & Pigeot (Ch 5 and 24); Nelson & Williams — Infectious Disease Epidemiology (Ch 1, 2 and 6–7); Jugal Kishore — National Health Programmes of India (Integrated Disease Surveillance Programme); WHO Guidelines for Cholera Control as summarised in Park 28e.



Uses of epidemiology

Asked in: Paper I 26-May-2025 — RGUHS MD Community Medicine

Definition — Epidemiology is "the study of the occurrence and distribution of health-related events, states and processes in specified populations, including the study of the determinants influencing such processes, and the application of this knowledge to control relevant health problems" (Park 28e, Ch 3, p 72). Its distinguishing feature is that the unit of study is a **defined population** or population at risk, not a case, so it is concerned with the sick and the healthy alike (Park 28e, Ch 3, p 73). The three aims set by the *International Epidemiological Association* — to describe the distribution and magnitude of health problems, to identify aetiological factors, and to provide the data essential to the planning, implementation and evaluation of services and to the setting of priorities among them — are served respectively by descriptive, analytical and experimental studies, the ultimate aim being effective action to reduce the health problem and promote well-being (Park 28e, Ch 3, p 73). *J.N. Morris* set out **seven distinct uses** of epidemiology, five of which carry the subject beyond the search for causes and bring it close to the day-to-day concerns of modern medicine (Park 28e, Ch 3, p 111).

The seven uses of epidemiology (Morris)

#	Use	What epidemiology contributes	Suitable example
1	To study historically the rise and fall of disease in the population	Health and disease patterns in a community are never constant; epidemiology charts disease profiles and secular (time) trends , allows projection into the future and identifies emerging health problems and their correlates	The first contribution of epidemiology to coronary heart disease was the recognition that it was an epidemic ; accidents, cancer and diabetes were later shown to be "epidemic". As old diseases were conquered (smallpox), new ones were identified (Legionnaires' disease, Lassa fever, AIDS) (Park 28e, Ch 3, p 111)
2	Community diagnosis	Identification and quantification of health problems in a community by mortality and morbidity rates and ratios, plus identification of their correlates, to define individuals or groups at risk or in need of care ; it lays down priorities in disease control, gives a benchmark for later evaluation of services, and yields new knowledge on disease distribution, causation and prevention. Epidemiology is therefore the " diagnostic tool " of community medicine (Park 28e, Ch 3, p 111)	A health situation analysis of a defined population — e.g. the field-practice area of a primary health centre — assembling (i) morbidity and mortality statistics, (ii) demographic conditions, (iii) environmental conditions bearing on health, (iv) socio-economic factors, (v) cultural background, attitudes, beliefs and practices, (vi) medical and health services available and (vii) other services available; the problems that emerge are then ranked by priority or urgency for allocation of resources (Park 28e, Ch 10, pp 721-722)

#	Use	What epidemiology contributes	Suitable example
3	Planning and evaluation	Provides the fundamental basis for the rational allocation of limited resources , for developing needed services, and for assessing their impact on people's problems; applying epidemiological principles to problems of health care constitutes the " new epidemiology "	Planning facilities for medical care — the number of hospital beds required for patients with a specific disease and health manpower planning . In developing countries too many hospitals have been built and equipped without knowledge of the community's particular disease problems (Park 28e, Ch 3, p 111)
4	Evaluation of individual's risks and chances	Beyond the incidence rate and specific rates (measures of absolute risk), epidemiology states the degree of risk carried by an individual with a given exposure, using relative risk, attributable risk and population attributable risk	Lung cancer death rates in UK physicians: 224 per 100,000 person-years in heavy smokers, 10 per 100,000 person-years in non-smokers and 74 per 100,000 person-years in the total population — individual relative risk 22.40 and population attributable risk per cent 86 per cent , i.e. 86 per cent of lung cancer deaths could be avoided if cigarette smoking were eliminated (Park 28e, Ch 3, Table 19, pp 98-99). The rising risk of bearing a child with Down syndrome with advancing maternal age, and of some hereditary disorders, is the other classic example (Park 28e, Ch 3, p 112)
5	Syndrome identification	Medical syndromes are identified by observing frequently associated findings in individual patients; epidemiological investigation defines and refines syndromes by observing whole groups, and corrects clinical misconceptions	Of roughly 3000 so-called syndromes described in the contemporary paediatric literature, a primary defect is known in only about 20 per cent . The poverty gradient in the certification of gastric ulcer, absent in duodenal ulcer , led to the differentiation of the two main types of peptic ulcer (little appreciated until 1920). Conversely, the <i>Patterson-Kelly</i> association of dysphagia with iron-deficiency anaemia was not confirmed when tested by epidemiological methods (Park 28e, Ch 3, p 112)
6	Completing the natural history of disease	Epidemiology is concerned with the entire spectrum of disease in a population, whereas the picture built from hospital patients is quite different from that in the community; by studying agent, host and environmental factors in the community the epidemiologist fills the gaps the clinician cannot	One-third to two-thirds of all deaths from ischaemic heart disease are sudden , occurring in less than one hour — a conclusion hospital studies could never reach because most victims never reach hospital; this finding gave tremendous impetus to the development of intensive coronary care units (Park 28e, Ch 3, p 112). The iceberg of disease expresses the same principle

#	Use	What epidemiology contributes	Suitable example
7	Searching for causes and risk factors	By relating disease to inter-population differences and other attributes of the populations examined, epidemiology identifies causes and risk factors ; for chronic disease the hope of a single cause is unfulfilled, and the "risk factor" concept replaced it	Rubella as a cause of congenital defects in the newborn; thalidomide as a teratogenic agent; cigarette smoking as a cause of lung cancer; exposure of premature babies to oxygen as the cause of retrolental fibroplasia (Park 28e, Ch 3, p 112)

Quantifying risk — the indices used in Use 4

Index	Formula, with Park's worked value	Interpretation
Relative risk (risk ratio)	Incidence of disease (or death) among exposed ÷ incidence among non-exposed; UK physicians = $224 \div 10 = 22.40$	Direct measure of the strength of association ; relative risk 1 = no association, >1 = positive association; relative risk 2 = incidence twice that of the unexposed (a 100 per cent increase in risk); 0.25 = a 75 per cent reduction. Quote its 95 per cent confidence interval (Park 28e, Ch 3, p 98)
Attributable risk (risk difference)	Incidence among exposed – incidence among non-exposed; Park's hypothetical cohort = 10 per 1000 – 1 per 1000 = 9 per 1000	The excess incidence in the exposed that is attributable to the exposure; "some authors use the term risk difference" (Park 28e, Ch 3, p 98)
Attributable risk per cent	(Incidence among exposed – incidence among non-exposed) ÷ incidence among exposed × 100 = $9/10 \times 100 = 90$ per cent	The proportion of disease in the exposed due to the exposure — 90 per cent of lung cancer among smokers was due to their smoking; the measure of preventive potential in the exposed group
Population attributable risk	Incidence in the total population – incidence among the non-exposed; UK physicians = $74 - 10 = 64$ per 100,000 person-years	The excess incidence in the whole community attributable to the exposure (Park 28e, Ch 3, p 98)
Population attributable risk per cent	(Incidence in total population – incidence in non-exposed) ÷ incidence in total population × 100 = $(74 - 10)/74 \times 100 = 86$ per cent	The amount by which the disease could be reduced in that population if the factor were eliminated or modified — 86 per cent of lung cancer deaths avoidable if cigarettes were eliminated; the measure that guides public health priority setting (Park 28e, Ch 3, pp 98-99)
Which to quote	—	Relative risk is the better index of the aetiological role of a factor; attributable risk better reflects the public health importance , i.e. the impact a successful preventive programme would have (Park 28e, Ch 3, pp 98-99)

Use 3 expanded — planning and evaluating health services

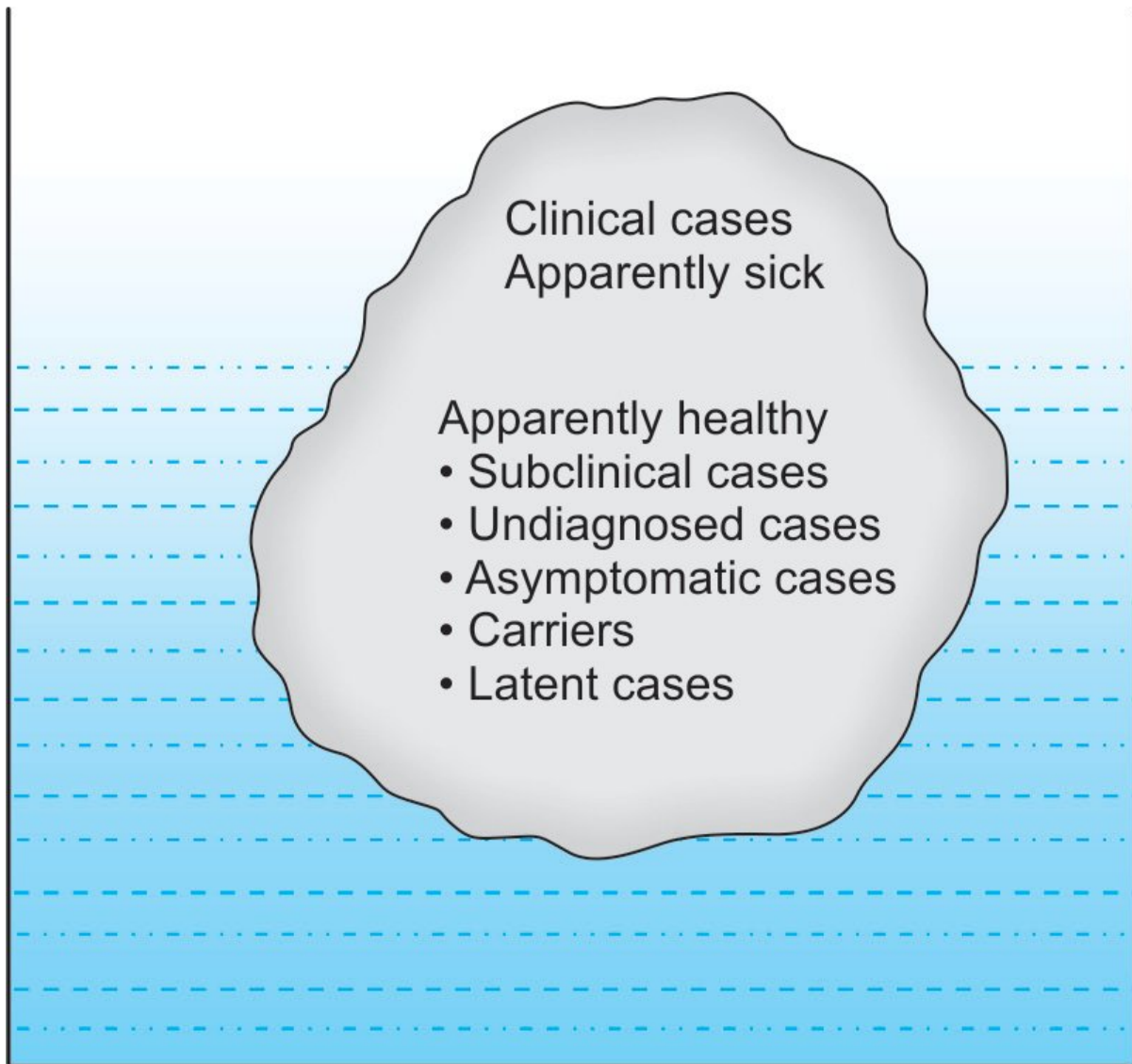
- **Planning facilities for preventive services** — screening programmes, immunisation campaigns and the provision of sanitary services (Park 28e, Ch 3, p 111).

- **Planning for research** — setting the research agenda from the burden actually identified.
- **Evaluation of any control measure** — any measure taken to control or prevent a disease must be followed by evaluation to find out whether it actually reduced the frequency of the disease. Evaluating a **hepatitis vaccine**, for example, requires more than a demonstration of its effectiveness: the **cost of large-scale application** — vaccine, trained personnel, storage, transport — must be measured, and its value against alternative methods assessed by **cost-effectiveness studies** (Park 28e, Ch 3, pp 111-112).
- **The randomised controlled trial** made it possible to evaluate treatment modalities on a firm scientific basis, and such trials raised doubts about **multiphasic screening**, tonsillectomy, varicose-vein stripping and prolonged hospitalisation of patients with myocardial infarction. It is not enough to know that a programme provides some benefit — we need to know **how much benefit, at what risk and at what cost** (Park 28e, Ch 3, p 112).
- **Process versus outcome measures** — *outcome* denotes whether the patient or the community at large actually benefits from the care provided; traditionally mortality and morbidity, now extended to **patient satisfaction and well-being, quality of life, and degree of dependence and disability** (Gordis 7e, Ch 17, p 350).

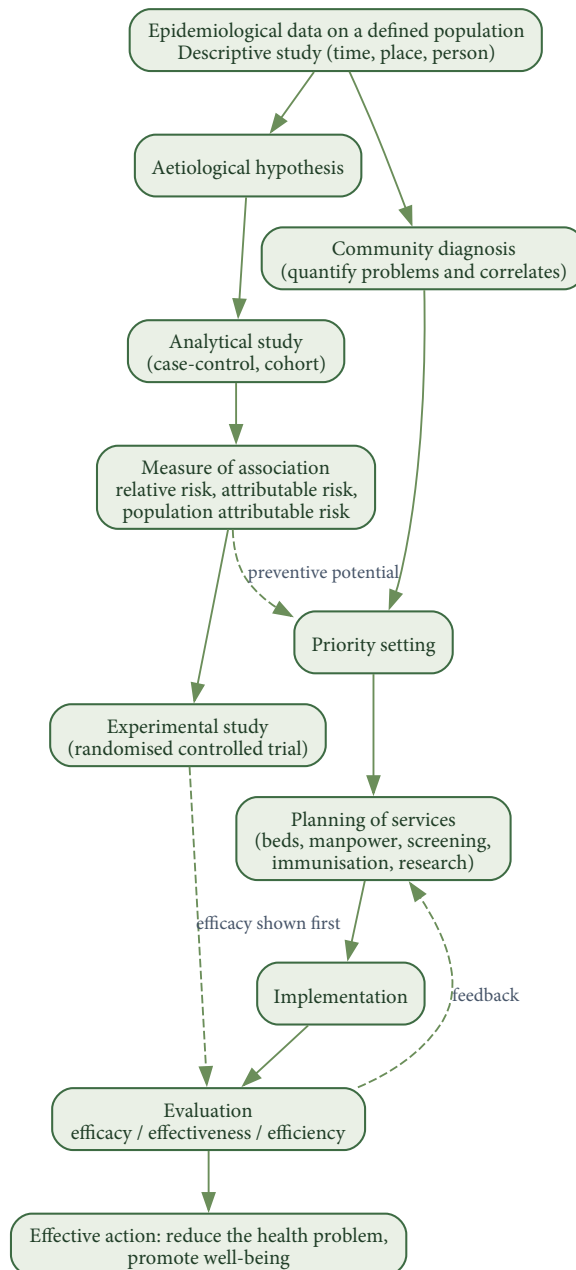
Term	Question it answers	Comment
Efficacy	Does the intervention work under ideal, controlled ("laboratory") conditions ?	Established first, usually by a randomised controlled trial
Effectiveness	Does it do what it is intended to do in the community-dwelling population at large ?	An efficacious measure may still fail under field conditions
Efficiency	What is the cost-benefit ratio — can the goal be reached less expensively and better?	Cost includes money plus discomfort, pain, absenteeism, disability and social stigma; if a measure is not effective there is little point examining efficiency (Gordis 7e, Ch 17, p 350)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Iceberg of disease** (draw for Use 6). Labels: *floating tip above the waterline* — "Clinical cases / apparently sick (what the physician sees)"; *the waterline* — "demarcation between apparent and inapparent disease"; *submerged mass* — "Apparently healthy: subclinical cases, undiagnosed cases, asymptomatic cases, carriers, latent cases"; caption "hidden mass of disease — the undiagnosed reservoir in the community" (Park 28e, Ch 2, p 58).



Iceberg of disease — the community picture that epidemiology completes (Suryakantha — Community Medicine with Recent Advances, Fig 3.2, p 35)



How epidemiological information becomes public health action — the aetiological limb and the health-services limb

High-yield exam pointers

- Mnemonic for Morris's seven uses — **H C P I S N C**: Historical study, Community diagnosis, Planning and evaluation, Individual risks and chances, Syndrome identification, Natural history completed, Causes and risk factors. Write "*Morris — seven uses, five of which extend beyond the search for causes*".
- Write the three **International Epidemiological Association** aims verbatim: describe distribution and magnitude; identify aetiological factors; provide data for planning, implementation and evaluation of services and for setting priorities among them.
- Formulas: $\text{relative risk} = \text{incidence in exposed} \div \text{incidence in non-exposed}$; $\text{attributable risk} = \text{incidence in exposed} - \text{incidence in non-exposed}$; $\text{attributable risk per cent} = (\text{incidence exposed} - \text{incidence non-exposed}) \div \text{incidence exposed} \times 100$; **population attributable risk = incidence**

in total population – incidence in non-exposed; population attributable risk per cent =
 (incidence in total population – incidence in non-exposed) ÷ incidence in total population ×
 100.

- ▶ Numbers that earn marks: UK physicians — 224 / 10 / 74 per 100,000 person-years, giving relative risk **22.40**, population attributable risk **64 per 100,000 person-years** and population attributable risk per cent **86 per cent**; attributable risk per cent **90 per cent** in Park's hypothetical smoking cohort; ~3000 paediatric syndromes with a primary defect known in only about **20 per cent**; **one-third to two-thirds** of ischaemic heart disease deaths sudden (within one hour).
- ▶ Name the classic causal examples in one line: rubella, thalidomide, cigarette smoking, oxygen and retrolental fibroplasia.
- ▶ Define **efficacy** / **effectiveness** / **efficiency** as ideal conditions / community conditions / cost–benefit — a routine follow-up question on Use 3.
- ▶ Epidemiology is the "**diagnostic tool**" of **community medicine**; applying it to problems of health care is the "**new epidemiology**".

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park 28e uses the older term "mongol child" for the maternal-age example under Use 4; the current term is Down syndrome, and the fact itself is unchanged.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 3, Principles of Epidemiology and Epidemiologic Methods, pp 72-73, 98-99 including Tables 18 and 19, and pp 111-112; Ch 2, Concept of Health and Disease, p 58; Ch 10, Health Care of the Community, pp 721-722); Gordis Epidemiology 7e (Ch 17, Using Epidemiology to Evaluate Health Services, p 350); Suryakantha — Community Medicine with Recent Advances (Fig 3.2, p 35, iceberg of disease).



Endemic index

Asked in: Paper I 15-Nov-2022 — RGUHS MD Community Medicine

Definition — *Endemic* (en = in, demos = people) refers to the constant presence of a disease or infectious agent within a given geographic area or population group without importation from outside, and also to the "usual" or expected frequency of that disease in that area or population (Park 28e, Ch 3, p 113). The **endemic index** is the quantitative measure by which this habitual level is expressed and graded, so that the endemicity of one community can be compared with another and any excess over the expected level can be recognised. In classical malariology the term is used specifically for the **spleen rate**, which is widely used for measuring the endemicity of malaria in a community (Park 28e, Ch 5, pp 323–324).

The two senses in which "endemic index" is used

Sense	What it measures	How it is obtained	Typical use
Disease-specific endemic index (classical, malarimetric)	The prevalence of infection or of its cardinal sign in an indicator age group of a community	A cross-sectional malarimetric survey — spleen palpation and blood films in children (Park 28e, Ch 5, pp 323–324)	Grading a locality as hypo-, meso-, hyper- or holoendemic; stratifying areas for control
Generic endemic index / endemic level (surveillance)	The expected number of cases of a notifiable disease in a defined population and a defined week or month	Mean or median number of cases in the corresponding period of the previous 3, 5 or 7 years, from continuous notification (Sathe — Health Management, Ch 1)	Early detection of an outbreak; an arbitrary limit of two standard errors above the endemic occurrence defines the epidemic threshold, e.g. for influenza (Park 28e, Ch 3, p 159)

The examiner's expected core is the first sense; the second is added because it is what "endemic index" means outside malaria and is the basis of the Integrated Disease Surveillance Programme's endemic channel.

Classical malarimetric indices (the endemic index and its refinements)

Index	Definition (verbatim sense)	Denominator / age group	What it indicates
Spleen rate (<i>the endemic index proper</i>)	Percentage of children 2–10 years of age showing enlargement of the spleen	Children 2–10 y examined	The standard measure of endemicity ; adults are excluded because causes other than malaria frequently produce splenomegaly in them
Average enlarged spleen (AES)	The average size of the enlarged spleen	Only children with palpable splenomegaly	A refinement of the spleen rate — grades intensity, not just presence
Parasite rate	Percentage of children 2–10 years showing malaria parasites in their blood films	Children 2–10 y with slides examined	Parasitological counterpart of the spleen rate; grades endemicity

Index	Definition (verbatim sense)	Denominator / age group	What it indicates
Parasite density index	Average degree of parasitaemia in a sample of a well-defined group of the population	Only positive slides are included in the denominator	Intensity of infection
Infant parasite rate	Percentage of infants below one year showing malaria parasites in their blood films	Infants < 1 y	The most sensitive index of recent transmission; zero for 3 consecutive years indicates absence of transmission even if anopheline vectors persist
Proportional case rate	Number of cases diagnosed as clinical malaria for every 100 patients attending hospitals and dispensaries	Total attendances	A crude index — cases are not related to their time/space distribution

(All definitions: Park 28e, Ch 5, pp 323–324.)

Grading of endemicity by the endemic index

Level of endemicity	Spleen rate (children 2–9 y)	Parasite rate	Epidemiological description
Hypoendemic	Not exceeding 10%	Not exceeding 10% in children 2–9 y (may be higher for part of the year)	Little transmission; effects on the general population unimportant in an average year
Mesoendemic	11–50%	11–50% in children 2–9 y	Typically rural communities of subtropical zones with wide geographical variation in transmission risk
Hyperendemic	Constantly over 50%; also high in adults (over 25%)	Constantly over 50% in children 2–9 y	Intense but seasonal transmission; immunity insufficient in all age groups
Holoendemic	Constantly over 75%, but low in adults	Constantly over 75% among infants 0–11 months	Perennial, intense transmission producing considerable immunity beyond early childhood

(Gilles & Warrell — *Essential Malariology* 4e, Ch 5, pp 101–102, Table 5.2; concordant with Maxcy-Rosenau-Last 15e, Table 15-11.)

Park's descriptive definitions match these cut-offs: **hyperendemic** means the disease is constantly present at a high incidence and/or prevalence rate and affects **all age groups equally**, while **holoendemic** means a high level of infection beginning early in life and affecting most of the child population, leading to a state of equilibrium such that the **adult population shows evidence of the disease much less commonly than do the children**, as in malaria (Park 28e, Ch 3, p 113).

Endemic indices in other diseases

Disease	Index	Definition
Lymphatic filariasis	Microfilaria rate	Percentage of persons showing microfilariae in peripheral blood (20 cu. mm), one slide from each person in the sample population
	Filarial endemicity rate	Percentage of persons examined showing microfilariae, or disease manifestation, or both
	Microfilarial density	Number of microfilariae per unit volume (20 cu. mm) of blood — indicates intensity of infection
	Average infestation rate	Average number of microfilariae per positive slide, each slide from 20 cu. mm of blood

(Park 28e, Ch 5, p 338.)

Uses and limitations

Uses

- **Stratification** — classifies localities as hypo-, meso-, hyper- or holoendemic, so that control inputs (indoor residual spraying, long-lasting insecticidal nets, mass drug administration) are directed where endemicity is highest.
- **Baseline and impact assessment** — a fall in the spleen and parasite rate on repeat surveys is direct evidence of the effect of a control programme.
- **Independent of case reporting** — in a control programme where the case-detection machinery is weak, the classical malariometric measures may still provide the needed information, namely the trend of the disease (Park 28e, Ch 5, p 323).
- **Sensitive detection of residual transmission** — the infant parasite rate detects fresh, locally acquired transmission when all other indices have fallen.
- **Epidemic detection** (generic sense) — the endemic level plotted as a channel over 52 weeks allows the current week's incidence to be judged against expectancy.

Limitations

- The spleen and parasite rate **cut-offs are arbitrary** and, being cross-sectional, do not capture the seasonal nature of transmission or the annual change in the rates (Gilles & Warrell 4e, Ch 5, p 102).
- Two areas with very different infection pressure — for example 20 versus over 200 new infections per person per year — can both be classified holoendemic on a single cross-sectional survey.
- **Spleen palpation is observer-dependent**, and splenomegaly in a community may be caused by kala-azar, haemoglobinopathies and other conditions.
- The proportional case rate is crude, and a hospital-based denominator is not the population at risk.
- For the generic endemic index, reliable calculation requires **complete and continuous notification** and a fairly large population; irregular reporting produces distortions and artefacts unrelated to true incidence (Sathe — Health Management, Ch 1).

What is used in India today

The malariometric endemic indices belong to the pre-eradication era. Under current epidemiological surveillance for malaria in India the parameters in use are the **annual parasite incidence (API)**, **annual blood examination rate (ABER)**, **annual falciparum incidence (AFI)**, **slide positivity rate (SPR)** and **slide falciparum rate (SFR)** (Park 28e, Ch 5, p 324; Ch 7, p 498).

Parameter	Formula	Interpretation
Annual parasite incidence (API)	$(\text{Confirmed cases during one year} \div \text{Population under surveillance}) \times 1000$	A sophisticated measure of malaria incidence in a community; the operational stratification unit
Annual blood examination rate (ABER)	$(\text{Number of slides examined} \div \text{Population}) \times 100$	An index of operational efficiency ; the aim is to screen 10% of the population
Slide positivity rate (SPR)	Percentage of slides found positive for malarial parasite, irrespective of species	Trend of transmission
Slide falciparum rate (SFR)	Percentage of slides positive for <i>P. falciparum</i>	Trend of falciparum transmission

(Park 28e, Ch 5, p 324.)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Endemic channel (endemic index curve)** — draw x-axis "week of the year (1–52)", y-axis "number of cases"; a central solid line labelled "**expected endemic level (median of previous 5 years)**", two dashed lines above it labelled "**+1 standard error**" and "**+2 standard error = epidemic threshold**", and a jagged solid line for "**observed cases, current year**" crossing above the +2 SE line at one point, marked "**epidemic — investigate**".

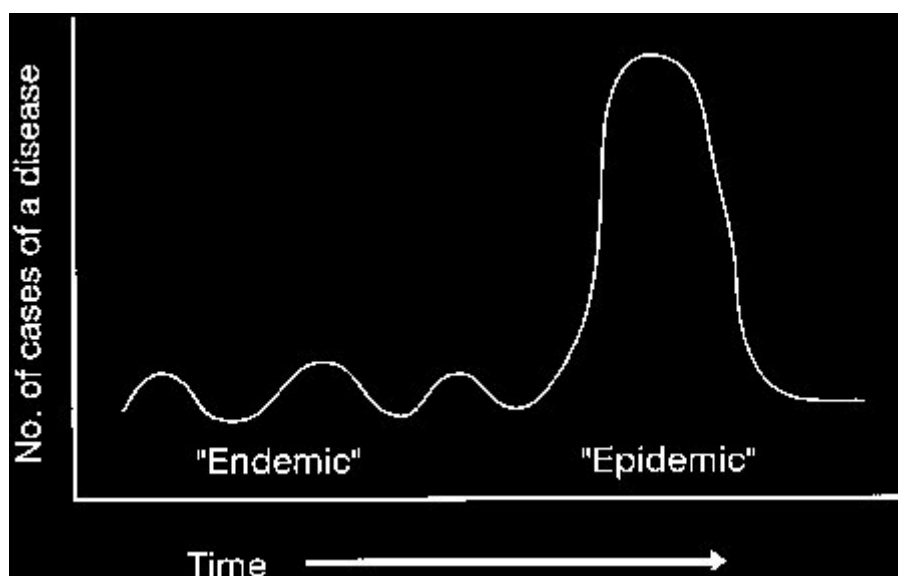
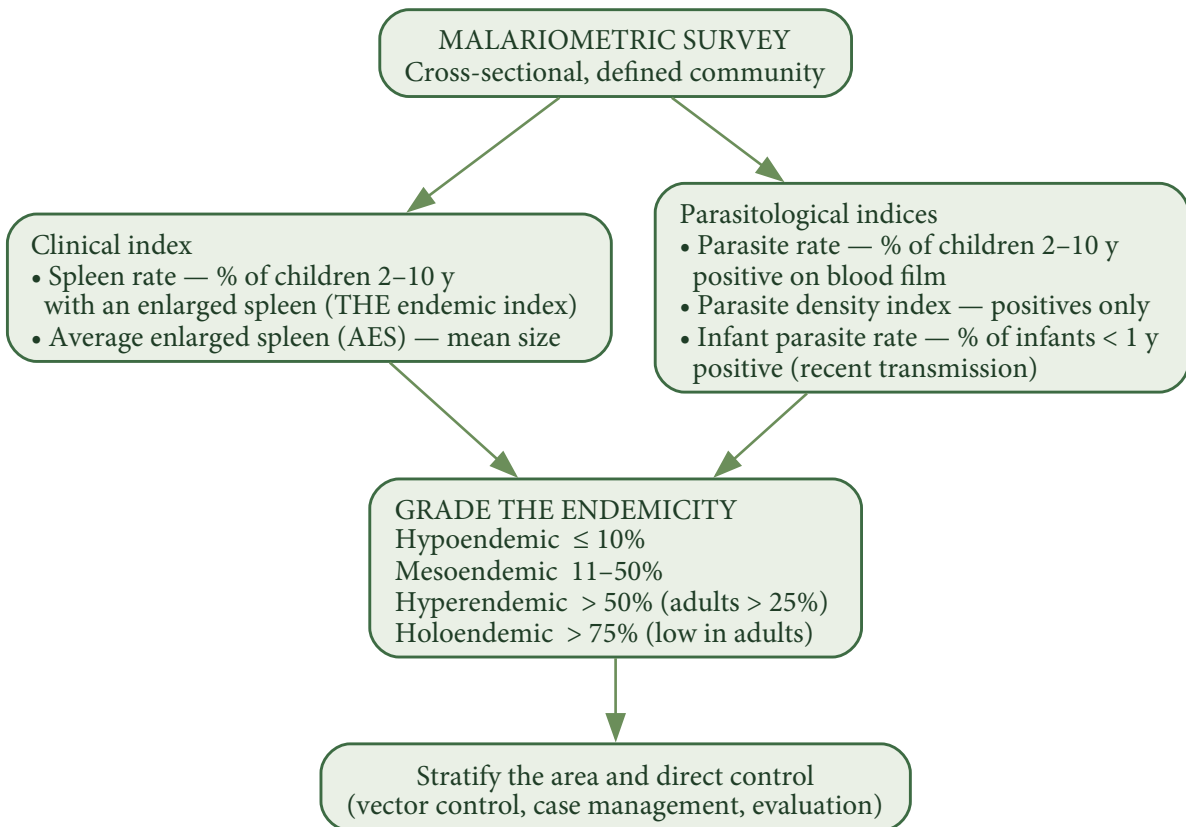
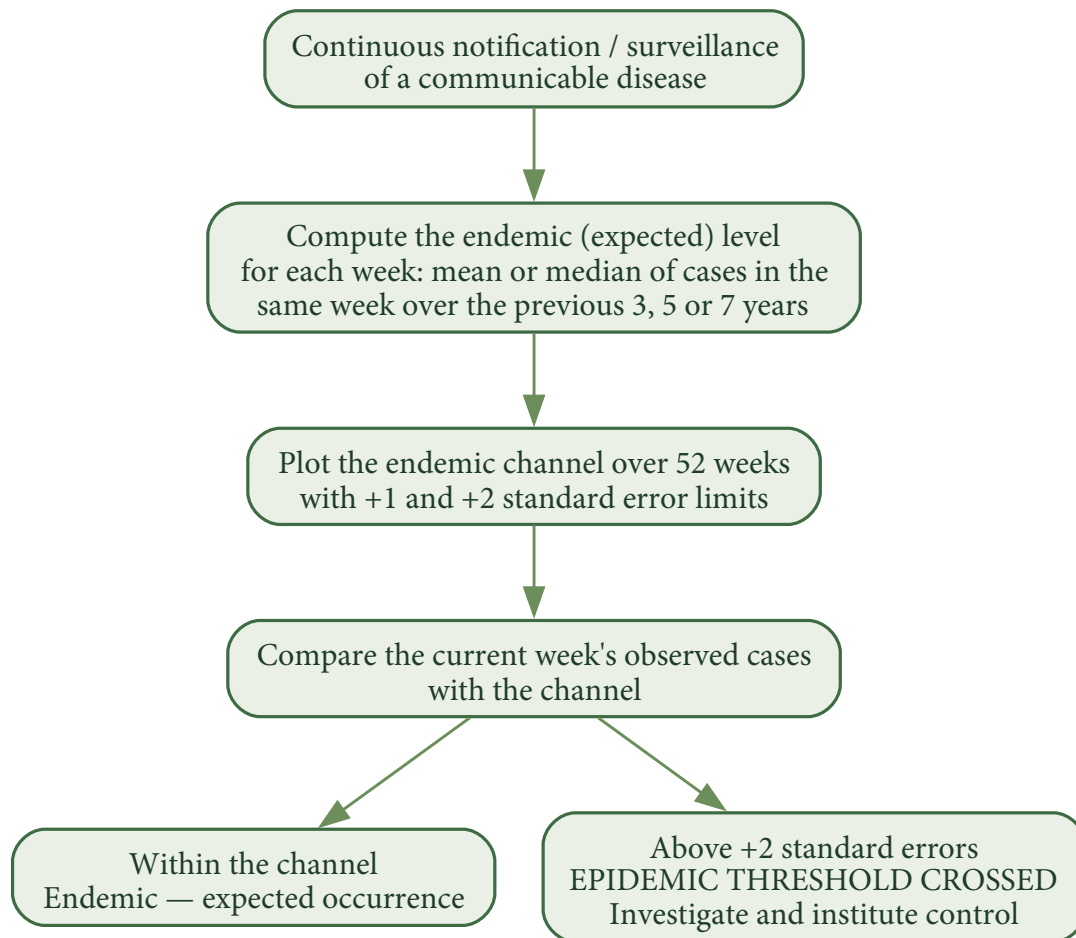


Fig 2.6 — Endemic versus epidemic disease. (Gordis Epidemiology 7e, p.39)



Endemic indices obtained from a malarionetric survey and their use in grading endemicity



The generic endemic index — computing the expected level and detecting an epidemic

High-yield exam pointers

- ▶ **Endemic index = spleen rate** = % of children 2–10 years (Park) with splenomegaly; WHO/Gilles tables use 2–9 years. Adults excluded — other causes of splenomegaly.
- ▶ Endemicity ladder by spleen rate: ≤10% **hypo**, 11–50% **meso**, >50% **hyper** (adults >25%), >75% **holo** (low in adults).
- ▶ **Infant parasite rate** (< 1 year) = **most sensitive index of recent transmission**; zero for 3 consecutive years = no transmission.
- ▶ **Parasite density index** — only positive slides in the denominator; **average infestation rate** in filariasis — microfilariae **per positive slide** (20 cu. mm).
- ▶ **API** = confirmed cases in a year ÷ population under surveillance × 1000; **ABER** = slides examined ÷ population × 100 (operational efficiency; target 10%).
- ▶ **Epidemic threshold** = 2 standard errors above the endemic occurrence (Park 28e, Ch 3, p 159).
- ▶ **Filarial endemicity rate** = % examined showing microfilariae, or disease manifestation, or both — wider than the microfilaria rate.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the four-level endemicity classification (holo-, hyper-, meso-, hypoendemic) is now much less used — neither the 2018 World Malaria Report nor the World Health Organization manual on malaria surveillance refers to this approach, and current programme stratification in India is by annual parasite incidence (Oxford Textbook of Public Health 7e); Park gives the spleen and parasite rate age band as 2–10 years while the World Health Organization endemicity table uses 2–9 years, so quote whichever band you quote consistently.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 3 — Principles of Epidemiology and Epidemiologic Methods, pp 113, 159; Ch 5 — Epidemiology of Communicable Diseases, pp 323–324, 338; Ch 7 — Health Programmes in India, p 498); Gilles & Warrell — Essential Malariology 4e (Ch 5, The epidemiology of malaria, pp 101–102, Table 5.2 Classifications of endemicity); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Table 15-11, Levels of endemicity by spleen and parasite rates in children 2–9 years of age); Gordis Epidemiology 7e (Ch 2, The dynamics of disease transmission, p 39, Fig 2.6); Sathe — Health Management (Ch 1, General Concepts — determination of endemic/epidemic levels); Oxford Textbook of Public Health 7e (malaria endemicity classification, current status).

Coverage note: the exact phrase "endemic index" appears in the indexed corpus only in Sathe's general-epidemiology sense; the malariometric sense is grounded on Park's statement that the spleen rate "is widely used for measuring the endemicity of malaria in a community" together with the World Health Organization endemicity table reproduced in Gilles & Warrell and Maxcy-Rosenau-Last.



Case-control vs cohort studies — risk estimation

Asked in: Paper I 21-Jun-2024 — RGUHS MD Community Medicine

Definition — A **case control study** is an analytical observational design in which both exposure and outcome have already occurred when the study is initiated; it starts with people who have the disease (**cases**) and a comparable group without it (**controls**), and proceeds backwards from "effect to cause" to compare exposure rates (Park 28e, Ch 3, pp 90-91). A **cohort study** starts with a group exposed to a putative aetiological factor (**study cohort**) and a group not so exposed (**control cohort**), both free of the disease at entry, and works forward from "cause to effect" to compare the **incidence** of disease (Park 28e, Ch 3, p 95). Calendar time is *not* the distinguishing feature — a retrospective (historical) cohort study also uses data recorded in the past, yet still moves from cause to effect (Gordis 7e, Ch 7).

Differences between case control and cohort studies

#	Point of difference	Case control study	Cohort study
1	Direction of enquiry	Proceeds from "effect to cause"	Proceeds from "cause to effect"
2	Starting point	Starts with the disease	Starts with people exposed to the risk factor or suspected cause
3	Question tested	Whether the suspected cause occurs more frequently in those with the disease than among those without it	Whether the disease occurs more frequently in those exposed than in those not similarly exposed
4	Groups compared	Cases versus controls — in Park's Table 11 letters, cases = a + c and controls = b + d	Exposed versus non-exposed persons — in Park's Table 16 letters, exposed = a + b and non-exposed = c + d
5	Outcome measured	Proportion of cases exposed versus proportion of controls exposed	Incidence of disease in the exposed versus in the non-exposed
6	Role in hypothesis testing	Usually the first approach to testing a causal hypothesis; also useful for exploratory studies	Reserved for testing a precisely formulated hypothesis
7	Number of subjects	Involves fewer subjects	Involves a larger number of subjects
8	Time to result	Yields relatively quick results	Long follow-up period often needed, with delayed results
9	Rare disease	Suitable — the design of choice	Inappropriate when the disease under investigation is rare (see Note)
10	Rare exposure	Unlikely to identify a sufficient number of exposed persons	Desirable — the design of choice when the exposure of interest is rare
11	Measure of risk obtained	Generally yields only an estimate of relative risk, i.e. the odds ratio (= ad/bc , cross-product of Park's Table 11)	Yields incidence rates , relative risk and attributable risk directly

#	Point of difference	Case control study	Cohort study
12	Number of outcomes studied	Cannot yield information about diseases other than the one selected	Can yield information about more than one disease outcome for the same exposure
13	Temporal relationship	Harder to establish, as exposure and disease have both already occurred	Prospective design gives better data on whether the exposure preceded the outcome
14	Main potential bias	Bias in the assessment of exposure — recall bias where past exposure is self-reported, plus difficulty in selecting an appropriate control group	Bias in the assessment of outcome , which is greater than in a case control study; recall bias is minimised prospectively, and misclassification into exposed and non-exposed is reduced because the groups are formed before disease develops
15	Attrition	Not applicable	Loss to follow-up is a major problem; the safest course is to achieve as close to a 95 per cent follow-up as possible
16	Cost	Relatively inexpensive	Expensive

(Rows 1-3, 6-9, 11-12, 16 follow Park 28e, Ch 3, Table 22, p 99; rows 4-5, 10, 13-14 follow Gordis 7e, Ch 9, Table 9.1 and pp 210-212; row 15, Park 28e, Ch 3, p 98.)

The two 2 × 2 frameworks — do not confuse the letters

Framework of a case control study (Park 28e, Ch 3, Table 11, p 91) — the columns are the two groups the investigator assembles:

Suspected or risk factor	Cases (disease present)	Controls (disease absent)
Present	a	b
Absent	c	d
Total	a + c	b + d

Exposure rate among cases = $a/(a + c)$; among controls = $b/(b + d)$; odds ratio = ad/bc .

Framework of a cohort study (Park 28e, Ch 3, Table 16, p 95) — here the rows are the two groups the investigator assembles:

Cohort	Disease: yes	Disease: no	Total
Exposed to putative aetiological factor	a	b	a + b
Not exposed to putative aetiological factor	c	d	c + d

Incidence among exposed = $a/(a + b)$; among non-exposed = $c/(c + d)$.

Estimation of risk in cohort studies

Analysis is the fifth and final step of a cohort study, and the data are analysed in two stages — first the incidence rates, then the estimation of risk (Park 28e, Ch 3, p 98).

- **Incidence rates.** Because a defined denominator — the population at risk — exists in both arms, incidence can be calculated **directly** in those exposed and those not exposed, which no case control study permits. Where subjects are observed for different lengths of time, the denominator becomes the total **person-time (person-years) of observation** instead of persons, so that each subject contributes only the time for which they were actually followed and free of the outcome (Gordis 7e, Ch 3, p 65).
- **Estimation of risk.** Having calculated the incidence rates, the next step is to estimate the risk of the outcome — disease or death — in the exposed and non-exposed cohorts. This is done in terms of two well-known indices, **relative risk** and **attributable risk** (Park 28e, Ch 3, p 98). A third index, the **population attributable risk**, extends attributable risk from the exposed group to the whole community.
- **Tests of statistical significance and precision.** The difference between the two incidence rates is tested for statistical significance ($P < 0.001$ in Park's worked example below), and the **95 per cent confidence interval** of the relative risk is quoted, since it indicates both the likely and the maximum level of risk (Park 28e, Ch 3, p 98).

Measures of risk

Measure	Formula	What it expresses
Relative risk (RR), or risk ratio	Incidence of disease (or death) among exposed ÷ Incidence among non-exposed = $[a/(a + b)] \div [c/(c + d)]$	Direct measure of the strength of association between suspected cause and effect; the key index in aetiological enquiry
Attributable risk (AR), or risk difference	Incidence among exposed – Incidence among non-exposed	The excess incidence of disease in the exposed that is attributable to the exposure
Attributable risk per cent (AR%)	$[(\text{Incidence among exposed} - \text{Incidence among non-exposed}) \div \text{Incidence among exposed}] \times 100$	The proportion of disease among the exposed attributable to the exposure — the amount that would be eliminated if the factor were controlled
Population attributable risk (PAR)	Incidence in the total population – Incidence among non-exposed	Excess disease in the whole community attributable to the exposure
Population attributable risk per cent	$[(\text{Incidence in total population} - \text{Incidence among non-exposed}) \div \text{Incidence in total population}] \times 100$	The amount by which disease in that population could be reduced if the suspected factor were eliminated or modified

(Park 28e, Ch 3, pp 98-99)

Interpretation of the relative risk

Value	Interpretation
RR = 1	Risk in exposed equal to risk in non-exposed — no association
RR > 1	Risk in exposed greater than in non-exposed — positive association, which may be causal

Value	Interpretation
RR < 1	Risk in exposed less than in non-exposed — negative association, possibly indicating a protective effect, as with an effective vaccine

(Gordis 7e, Ch 12, Table 12.4, pp 256-257)

Worked example — cigarette smoking and lung cancer

Cigarette smoking	Developed lung cancer	Did not develop lung cancer	Total
Yes	70 (a)	6,930 (b)	7,000 (a + b)
No	3 (c)	2,997 (d)	3,000 (c + d)

- Incidence among smokers = $70/7,000 = 10$ per 1,000; among non-smokers = $3/3,000 = 1$ per 1,000 ($P < 0.001$).
- **Relative risk** = $10 \div 1 = 10$ — smokers are 10 times at greater risk of developing lung cancer than non-smokers.
- **Attributable risk per cent** = $[(10 - 1) \div 10] \times 100 = 90$ per cent — if the association is causal, 90 per cent of the lung cancer among smokers was due to their smoking.

(Park 28e, Ch 3, Table 18, p 98)

For the **population attributable risk**, Park uses lung cancer death rates per 100,000 person-years among United Kingdom physicians: heavy smokers 224 (a), non-smokers 10 (b), and deaths in the total population 74 (c). Individual relative risk = $a/b = 22.40$. The absolute population attributable risk = $c - b = 74 - 10 = 64$ per 100,000 person-years, and the **population attributable risk per cent** = $(c - b)/c = 86$ per cent, meaning 86 per cent of the lung cancer deaths in that population could be avoided if cigarette smoking were eliminated. Park's Table 19 prints this last line simply as "Population AR" (Park 28e, Ch 3, Table 19, pp 98-99).

Relative risk versus attributable risk

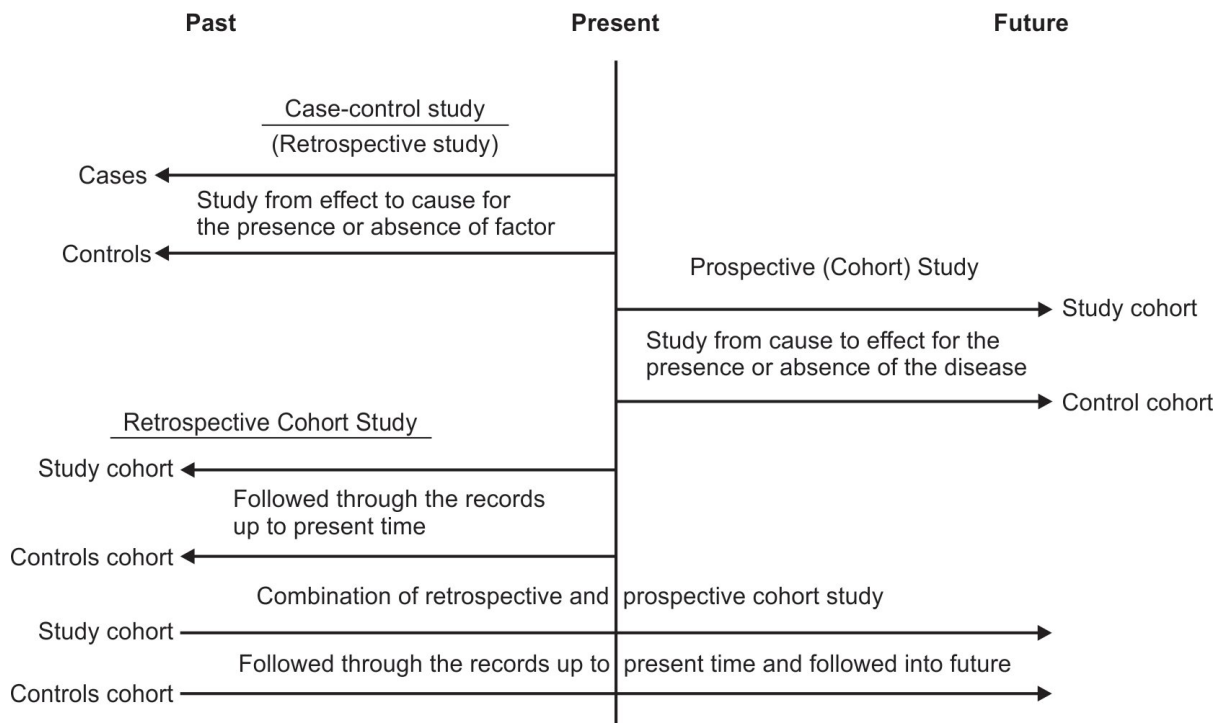
Relative risk is the better index for assessing the **aetiological** role of a factor — the larger it is, the stronger the association between cause and effect. But a large relative risk does not by itself indicate **public health importance**; attributable risk gives the better idea of how much of the problem a successful preventive programme would actually remove (Park 28e, Ch 3, p 99). The contrast is sharp when the same exposure is set against two different outcomes.

Cause of death	Death rate/1,000 — smokers	Death rate/1,000 — non-smokers	Relative risk	Attributable risk %
Lung cancer	0.90	0.07	12.86	92.2
Coronary heart disease	4.87	4.22	1.15	13.3

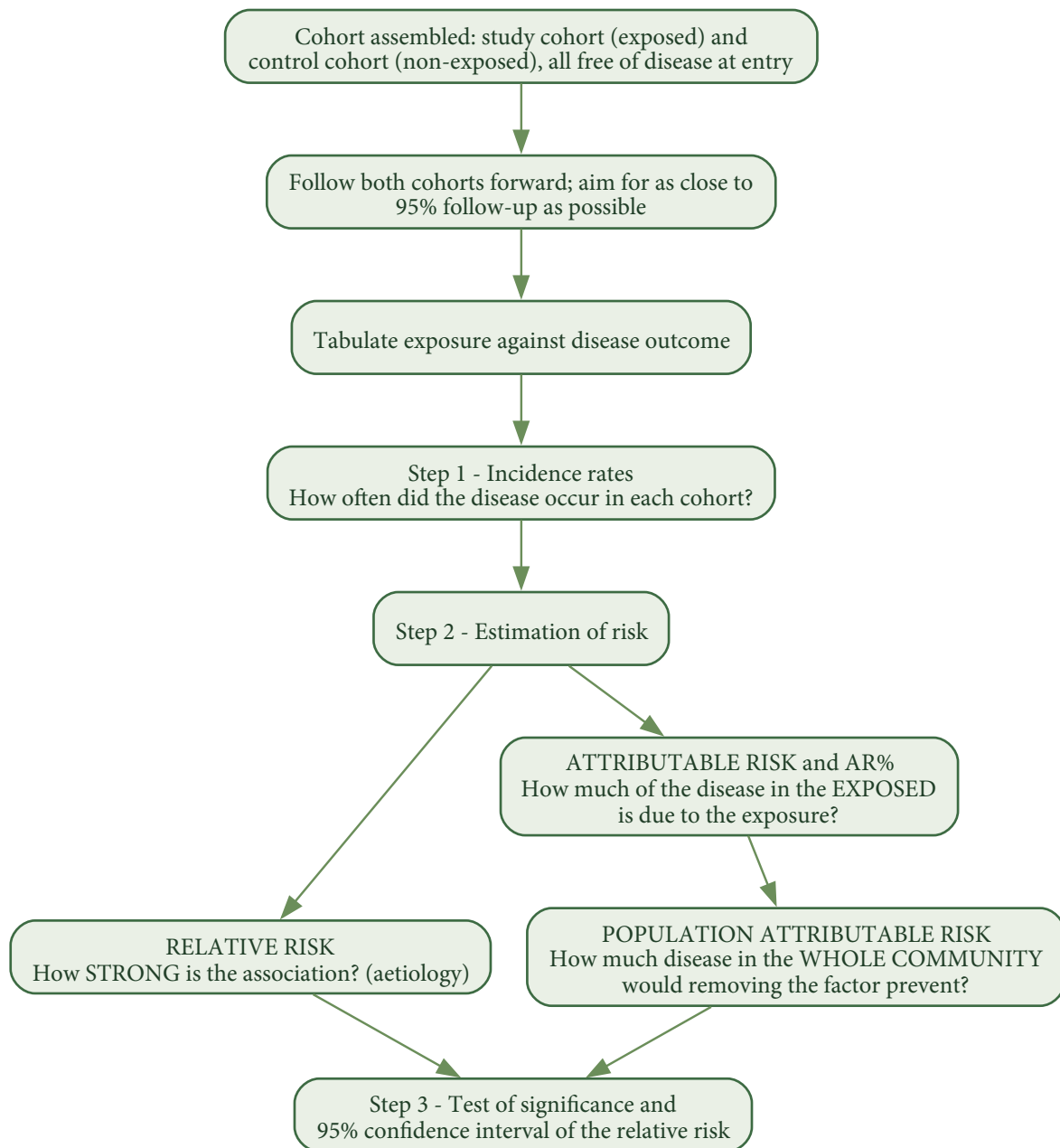
(Park 28e, Ch 3, Table 21, p 99) — smoking accounts for 92.2 per cent of lung cancer but only 13.3 per cent of coronary heart disease, so far less coronary disease than lung cancer could be prevented by removing the factor.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- Schematic diagram of the design of case control and cohort studies (Park 28e, Ch 3, Fig. 8, p 90). Draw **two side-by-side panels**, each with a horizontal **TIME** axis and a "**Direction of inquiry**" arrow. Label the left panel *Design of a Case Control Study* — "Start with: **Cases** (people with the disease) and **Controls** (people without the disease)", with the direction-of-inquiry arrow running **backwards** in time to *exposed / not exposed*. Label the right panel *Design of a Cohort Study* — "Start with: **Exposed** and **Not exposed**", with the direction-of-inquiry arrow running **forwards** in time to *disease / no disease*.



Schematic representation of case-control study and Cohort studies (Suryakantha, Community Medicine with Recent Advances, Fig 18.8, p 274)



Sequence of analysis and estimation of risk in a cohort study

High-yield exam pointers

- Mnemonic for the two directions: **case control** = "effect to cause", **cohort** = "cause to effect"; a cohort study is the only observational design that gives **incidence** directly, and therefore the only one that gives relative risk **and** attributable risk (Park Table 22).
- Odds ratio = ad/bc , valid as an estimate of relative risk only on three assumptions — the disease is relatively rare, the cases are representative of those with the disease, and the controls are representative of those without the disease (Park 28e, Ch 3, p 93).
- Attributable risk per cent = $(\text{incidence exposed} - \text{incidence non-exposed}) \div \text{incidence exposed} \times 100$ — the denominator is the incidence in the **exposed**, not the total.

- ▶ Population attributable risk per cent = $(\text{incidence in total population} - \text{incidence in non-exposed}) \div \text{incidence in total population} \times 100$ — the denominator is the incidence in the **whole population**.
- ▶ Park's numbers worth reproducing: smoking and lung cancer, incidence 10 versus 1 per 1,000, relative risk **10**, attributable risk **90 per cent**; United Kingdom physicians, relative risk **22.40**, population attributable risk **86 per cent**; lung cancer relative risk **12.86** with attributable risk **92.2 per cent** versus coronary heart disease relative risk **1.15** with attributable risk **13.3 per cent**.
- ▶ One line that earns marks: *relative risk answers "how strong is the association?" (aetiology); attributable risk answers "how much disease would prevention actually remove?" (public health impact)*.
- ▶ Calendar time does **not** distinguish the two designs — a retrospective (historical) cohort study also looks at past data, but it still moves from cause to effect.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference, not to be written in the exam: Park's printed Table 22 lists ten points of difference, and its seventh row combines the two situations into one line reading "inappropriate when the disease **or exposure** under investigation is rare" — rows 9 and 10 above split that line because Park's own "Indications for cohort studies" (p 95) and Gordis Table 9.1 both make the cohort study the design of choice when the **exposure** is rare. Rows 4-5, 10 and 13-15 are likewise drawn from Gordis 7e (Table 9.1 and Chapter 9) and from Park's text on cohort follow-up rather than from the ten printed rows, because RGUHS examiners expect a fuller enumeration than Table 22 alone.

SOURCE & COVERAGE

- ▶ Park's Textbook of Preventive and Social Medicine, 28e — Chapter 3 (Principles of Epidemiology and Epidemiologic Methods), pp 90-99, including Tables 11, 16, 18, 19, 21 and 22 and Fig. 8.
- ▶ Gordis Epidemiology 7e — Chapter 3 (Disease surveillance and measures of morbidity), pp 61-65 on person-time denominators; Chapter 7 (Observational studies); Chapter 9 (Comparing cohort and case-control studies), pp 210-212 with Table 9.1; Chapter 12 (Estimating risk: Is there an association?), pp 256-257 with Table 12.4.
- ▶ Suryakantha — Community Medicine with Recent Advances, Fig 18.8, p 274 (schematic representation of case-control and cohort studies).

♦ ♦ ♦

Observational study designs

Asked in: Paper I May 2017 — RGUHS MD Community Medicine

Definition — In an observational study the investigator allows nature to take its own course: exposure and outcome are measured, but the investigator never allocates the exposure and introduces no intervention (Park 28e, Ch 3, p 82). A descriptive study is limited to describing the occurrence of disease in a population; an analytical study goes further and analyses the relationship between the health outcome and other variables. Cohort, case-control, cross-sectional and ecological studies are collectively observational, in contrast to randomised controlled trials (Gordis 7e, Ch 8, p 195).

Types of observational studies

Design	Alternative name	Unit of study	Direction of enquiry	Principal yield
Case report / case series	Clinical observation	Individual patient(s)	None	Earliest origin of a research question (Gordis 7e, Ch 7, p 166)
Descriptive study	Time–place–person study	Individual or population	None	Disease load; formulation of an aetiological hypothesis (Park 28e, Ch 3, p 83)
Ecological study	Correlational study	Populations as unit of study	Group-level exposure vs group-level outcome	Hypothesis generation; liable to the <i>ecological fallacy</i> (Park 28e, Ch 3, p 82; Gordis 7e, Ch 7, p 194)
Cross-sectional study	Prevalence study	Individuals	Exposure and disease measured simultaneously	Prevalence and its correlates; simplest observational design (Park 28e, Ch 3, p 89)
Case-control study	Case-reference / retrospective study	Individuals	Effect to cause (backwards)	Odds ratio; tests a causal hypothesis (Park 28e, Ch 3, p 91)
Cohort study	Follow-up / longitudinal / prospective study	Individuals	Cause to effect (forwards)	Incidence in exposed and unexposed; relative risk (Park 28e, Ch 3, p 95)

Sub-types worth naming. Cohort studies are *prospective (concurrent)*, *retrospective (historical or non-concurrent)*, or a combination of the two (Gordis 7e, Ch 7, p 192). Hybrid designs derived from a cohort include the *nested case-control* study and the *case-crossover* study, in which the subject serves as their own control (Gordis 7e, Ch 7, p 192; Ch 8, p 205). Descriptive epidemiology may itself use a cross-sectional or a longitudinal design — a cross-sectional study has been likened to a photograph and a longitudinal study to a cine film (Park 28e, Ch 3, pp 89–90).

Design selected to test the alcohol–cirrhosis association

Alcohol is already listed as the prominent risk factor for cirrhosis of the liver (Park 28e, Ch 2, p 57, Table 8), so the task here is **hypothesis testing**, not hypothesis generation. A **case-control study** is selected.

Option	Why rejected / selected
Descriptive or ecological study	Describes distribution, or correlates group-level alcohol consumption with group-level cirrhosis mortality; cannot establish an individual-level association
Cross-sectional study	Exposure and disease measured at one point in time; tells us about the distribution of disease rather than its aetiology, and cannot establish that drinking preceded cirrhosis (Park 28e, Ch 3, p 89)
Cohort study	Valid but needs a very large cohort followed for decades, is costly and suffers attrition; suitable only when follow-up is easy, the cohort is stable and ample funds are available (Park 28e, Ch 3, p 95)
Randomised controlled trial	Ethically impossible — a harmful exposure cannot be allocated
Case-control study — SELECTED	Cirrhosis is a chronic disease whose causal pathway may span decades, which is precisely where the case-control method has its major use (Park 28e, Ch 3, p 91); it is rapid, inexpensive, needs few subjects and has no attrition

The case-control method has three distinct features: both exposure and outcome have already occurred before the study starts; the study proceeds **backwards from effect to cause**; and it uses a control or comparison group to support or refute the inference (Park 28e, Ch 3, p 91).

Conducting the study — the four basic steps

There are four basic steps in conducting a case-control study (Park 28e, Ch 3, p 91).

1. Selection of cases and controls

Element	What is done in this study
Diagnostic criteria	Fix, <i>before</i> the study begins, exactly what constitutes a case of cirrhosis (the agreed clinical, imaging and/or histological criteria, and the stage to be included). Once established, the criteria must not be altered until the study is over
Eligibility criteria	Take only newly diagnosed (incident) cases arising within a specified period, in preference to old cases or cases in advanced stages (prevalent cases)
Sources of cases	(i) <i>Hospital</i> — a single hospital or a network of hospitals over a specified period, taking the entire case series or a random sample of it; (ii) <i>General population</i> — all cases of cirrhosis occurring in a defined geographic area during a specified period, ascertained through a survey, a disease registry or a hospital network. Cases must be fairly representative of all cases in the community
Controls	Must be free of cirrhosis and as similar to the cases as possible except for the absence of the disease under study
Sources of controls	Hospital patients with unrelated illnesses; relatives (spouses, siblings); neighbourhood controls (same locality, factory or school); general-population controls taken as a random sample of a defined geographic area
Safeguard	Draw cases from one source and controls from more than one source, and repeat the study in a different geographic area — consistent findings increase the validity of the inference (Park 28e, Ch 3, p 92)

Hospital controls are a well-recognised source of selection bias, because many hospital patients have diseases that are themselves influenced by the factor under study — here, patients admitted with

pancreatitis, alcoholic gastritis or a road traffic injury would be unsuitable comparison patients (Park 28e, Ch 3, pp 91–92).

2. Matching

Matching is the process by which controls are selected so that they are similar to the cases in respect of certain pertinent variables (e.g., age) which are known to influence the outcome of disease and which, if not adequately matched, could distort or confound the result. A **confounding factor** is one that is associated both with the exposure and with the disease, is distributed unequally between the study and control groups, and is independently a risk factor for the disease (Park 28e, Ch 3, p 92). Here **smoking** is the classic confounder — Park's own worked illustration is alcohol and oesophageal cancer, where smoking confounds because it is associated with the consumption of alcohol *and* is an independent risk factor for the disease, so the effect of alcohol can be determined only if the influence of smoking is neutralised by matching (Park 28e, Ch 3, pp 92–93). Chronic hepatitis B and C infection must likewise be recorded and allowed for, since they cause progressive liver fibrosis leading to cirrhosis in their own right (Park 28e, Ch 5, p 269).

Two rules are examinable. **Never match on alcohol itself** — the suspected aetiological factor must not be matched, because by matching, its aetiological role is eliminated from the study. And do not over-match: if matching is overdone it becomes difficult to find controls, and excess zeal in matching tends to reduce the odds ratio (Park 28e, Ch 3, p 92).

3. Measurement of exposure

Definitions and criteria about exposure are just as important as those used to define cases and controls, and the information must be obtained in **precisely the same manner** in both groups — by interviews, by questionnaires, or by studying past records of cases such as hospital records and employment records (Park 28e, Ch 3, p 92). Alcohol exposure is therefore defined in advance as ever/never use, average quantity consumed per day, duration of drinking in years, and type of beverage, so that a **dose–response gradient** can be examined. The interviewer is kept blind to case or control status.

4. Analysis and interpretation

(a) *Exposure rates* — a case-control study provides a direct estimation of the frequency of exposure to the suspected factor in the disease and non-disease groups: exposure rate among cases = $a/(a+c)$, and among controls = $b/(b+d)$.

(b) *Estimation of risk* — a typical case-control study provides **no incidence rates**, because there is no appropriate denominator or population at risk; relative risk can be determined exactly only from a cohort study. Risk is therefore estimated by the **odds ratio (OR)**, or **cross-product ratio** = ad/bc , whose derivation rests on three assumptions: the disease being investigated is relatively rare, the cases are representative of those with the disease, and the controls are representative of those without it (Park 28e, Ch 3, p 93).

Alcohol exposure	Cases (cirrhosis present)	Controls (cirrhosis absent)
Present	150 (a)	120 (b)
Absent	50 (c)	280 (d)

Alcohol exposure	Cases (cirrhosis present)	Controls (cirrhosis absent)
Total	200 (a+c)	400 (b+d)

Exposure rate in cases = $150/200 = 75$ per cent; in controls = $120/400 = 30$ per cent. Odds ratio = $ad/bc = (150 \times 280) / (120 \times 50) = 7.0$, i.e., the odds of cirrhosis among drinkers are 7 times those among non-drinkers. (*Figures illustrative, to demonstrate the computation.*)

(c) *Test of significance* — for discrete variables such as exposure and disease, the standard error of the difference between two proportions or the **chi-square test**; for continuous variables (age, quantity consumed) the standard error of the difference between two means, or the t test. By convention $P \leq 0.05$ is regarded as statistically significant, but **statistical association does not imply causation**, and a statement of the P value alone is an inadequate end-point of a case-control study (Park 28e, Ch 3, p 93).

(d) *Matched analysis* — where each control has been individually matched to a case, the matched-pairs odds ratio is the ratio of the discordant pairs (pairs in which the case was exposed and the control was not, to pairs in which the control was exposed and the case was not); concordant pairs are ignored (Gordis 7e, Ch 12, p 266).

(e) *Bias to be excluded before concluding causation* — bias is any systematic error in the determination of the association between exposure and disease (Park 28e, Ch 3, p 93).

Bias	How it would operate here	Control
Bias due to confounding	Smoking and hepatitis B/C are associated both with drinking and with cirrhosis	Matching; stratified or multivariable analysis
Memory (recall) bias	Cirrhotic patients recall past drinking more readily than healthy controls; drinkers also under-report intake	Structured questionnaire; corroboration from records and family
Selection bias	Cases and controls not representative of the general population	Best controlled by prevention at the design stage; multiple control sources
Berksonian bias	Different rates of admission to hospital for people with different diseases	Prefer population-based cases and controls to hospital-based ones
Interviewer's bias	An interviewer who knows the hypothesis and who the cases are probes them more thoroughly	Double-blinding; check the average interview time for cases against controls

Causation is finally argued from a strong odds ratio, a dose–response gradient with quantity and duration of alcohol, biological plausibility, and consistency of the finding when the study is repeated in different geographic areas.

Advantages and disadvantages of the case-control design

Advantages (Park 28e, Ch 3, p 94, Table 13)	Disadvantages (Park 28e, Ch 3, p 94, Table 13)
1. Relatively easy to carry out	1. Problems of bias — relies on memory or past records, the accuracy of which may be uncertain and validation of which is difficult or sometimes impossible

Advantages (Park 28e, Ch 3, p 94, Table 13)	Disadvantages (Park 28e, Ch 3, p 94, Table 13)
2. Rapid and comparatively inexpensive (compared with cohort studies)	2. Selection of an appropriate control group may be difficult
3. Requires comparatively few subjects	3. Incidence cannot be measured ; relative risk can only be estimated
4. Particularly suitable for rare diseases, or diseases about which little is known	4. Does not distinguish between causes and associated factors
5. No risk to subjects	5. Not suited to the evaluation of therapy or prophylaxis of disease
6. Allows the study of several different aetiological factors at once	6. Representativeness of cases and controls is a major concern
7. Risk factors can be identified, so rational prevention and control programmes can be established	
8. No attrition problems, because no follow-up into the future is required	
9. Ethical problems are minimal	

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Labels to reproduce by hand (design of a case-control study): "Disease occurrence over the time" (arrow pointing right, along the top); "Direction of inquiry" (arrow pointing left); "Cause" at the left end, "Effect" at the right end; box "Cases (people with disease)" branching to "Exposed — find exposure rate among cases" and "Not exposed — find non-exposure rate among cases"; box "Controls (people without disease)" branching to "Exposed — find exposure rate among controls" and "Not exposed — find non-exposure rate among controls".

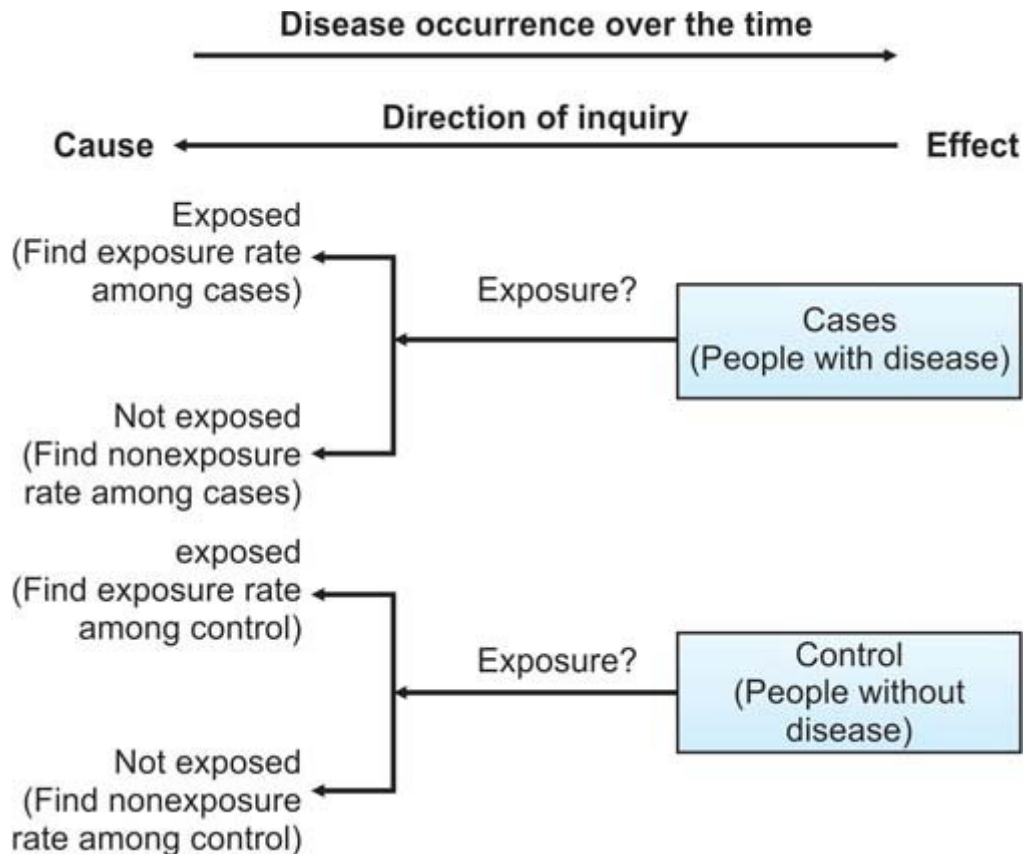
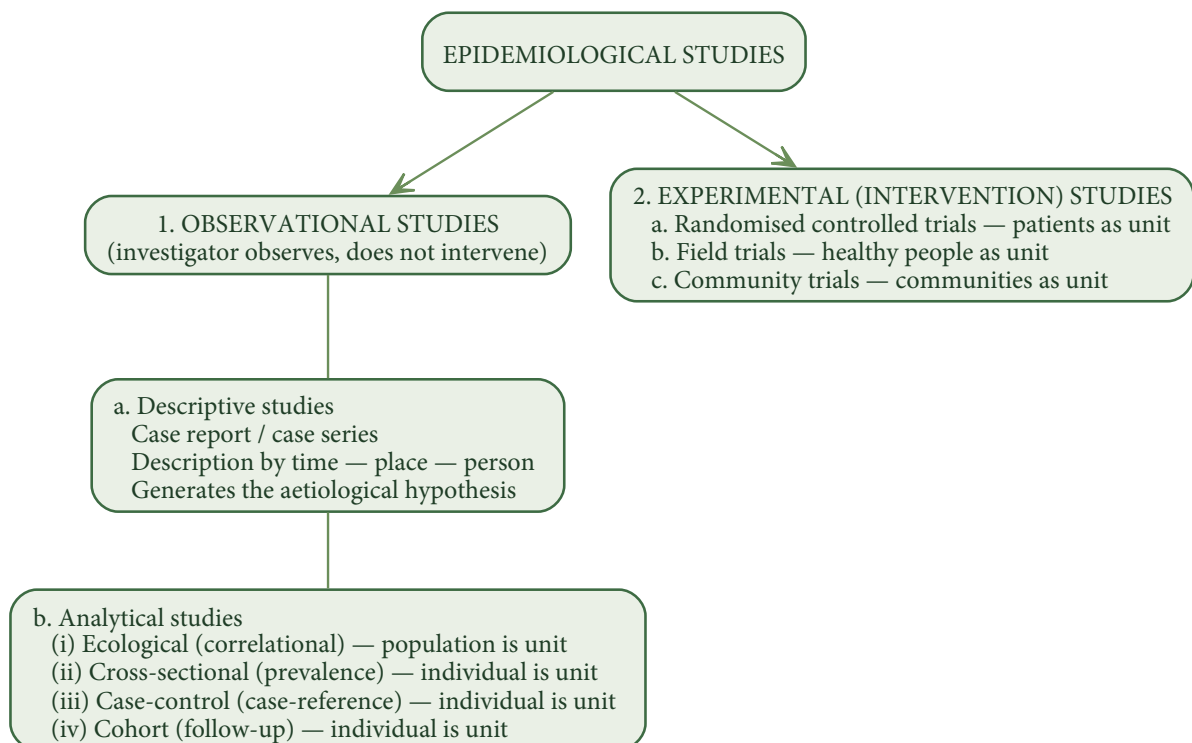
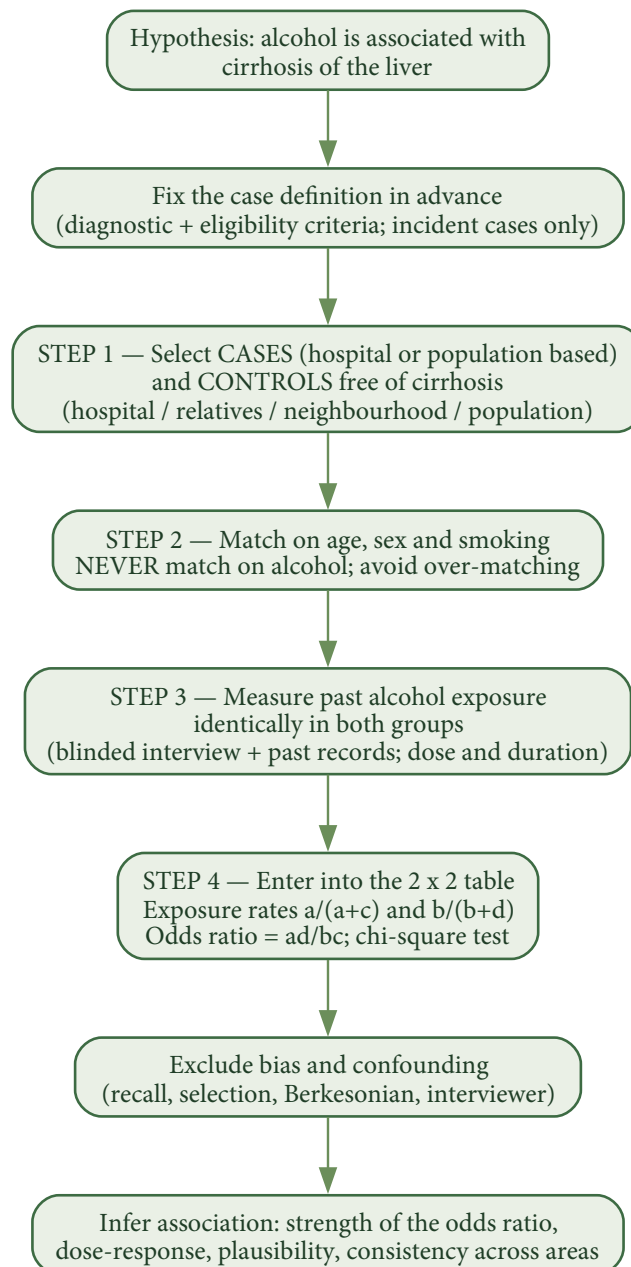


Fig 4.2 — Case control study design (Effect to cause study) (Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e, p.47)



Classification of epidemiological studies, with the observational branch expanded



Operational plan of a case-control study to test the alcohol and cirrhosis association

High-yield exam pointers

- ▶ **Park's classification (Ch 3, p 82):** Observational = descriptive + analytical (ecological, cross-sectional, case-control, cohort); Experimental = randomised controlled trial, field trial, community trial.
- ▶ **Three features of a case-control study:** exposure and outcome have both already occurred; the enquiry runs backwards from effect to cause; a control group is used.
- ▶ **Four basic steps:** selection of cases and controls, matching, measurement of exposure, analysis and interpretation.
- ▶ **Odds ratio = ad/bc** (cross-product ratio). Its three assumptions: the disease is relatively rare; cases represent those with the disease; controls represent those without. Matched-pairs odds ratio = ratio of the two discordant pairs.

- ▶ A case-control study yields no incidence, hence no direct relative risk — relative risk is determined exactly only from a cohort study.
- ▶ Never match on the suspected aetiological factor (alcohol), or its aetiological role is eliminated; over-matching reduces the odds ratio.
- ▶ Bias mnemonic "C-R-S-B-I": Confounding, Recall (memory), Selection, Berksonian, Interviewer's bias.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the sub-types named above (nested case-control, case-crossover, ambispective cohort) come from Gordis 7e and are worth naming in the enumeration, but the RGUHS answer is marked against Park's four-way observational classification; the 2 x 2 figures in the worked example are illustrative arithmetic to show the computation, not data from any published Indian study.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 3 (Principles of Epidemiology and Epidemiologic Methods, pp 82–95: classification of epidemiological studies, cross-sectional studies, the case-control method, its four basic steps, odds ratio, bias, and Table 13 advantages and disadvantages), Ch 2 (p 57, Table 8: prominent risk factors — cirrhosis of liver and alcohol), Ch 5 (p 269: hepatitis B and progression to cirrhosis). Gordis Epidemiology 7e — Ch 7 (Observational studies, pp 166, 192), Ch 8 (Cohort studies, pp 195, 205), Ch 12 (Estimating risk: Is there an association?, pp 262–266: odds ratio and matched-pairs odds ratio). Figure: Mahajan & Gupta, Textbook of Preventive and Social Medicine 4e, Fig 4.2, p 47.



Validity and reliability in epidemiological research

Asked in: Paper I Nov 2018 — RGUHS MD Community Medicine

Definition — Validity (accuracy) is the extent to which a measurement, or a whole study, actually measures what it purports to measure; it is the absence of **systematic error (bias)** and is judged against an external true value or gold standard (Park 28e, Ch 4, pp 166-167; Handbook of Epidemiology, Ch 13, p 521). **Reliability (repeatability, reproducibility, precision, consistency)** is the extent to which the test gives the **same result when repeated more than once on the same individual or material under the same conditions**; it is the absence of **random error** and needs no external standard, only repeat readings (Park 28e, Ch 4, p 165). Validity is closeness to the truth; reliability is closeness of repeat readings to one another.

Validity contrasted with reliability

Feature	Validity (accuracy)	Reliability (repeatability, precision)
Question answered	Is the measurement close to the truth ?	Do repeated measurements agree with each other?
Type of error opposed	Systematic error — bias	Random error — chance variability
External reference needed	Yes — a gold standard or true value	No — only replicate readings
Assessed by	Sensitivity, specificity, predictive value; mean difference from the gold standard	Percent agreement, kappa, intraclass correlation coefficient, coefficient of variation
Consequence in a study	Wrong (biased) estimate of effect — invalid conclusions	Loss of statistical power at a given sample size; wider confidence intervals
Improved by	Calibrating instruments, unobtrusive measurement, blinding (plus all reliability strategies)	Standardised operations manual, training and certifying observers, refining and automating instruments, repetition and use of the mean
Direction of dependence	A test can be highly reliable yet invalid — consistently wrong in the same direction	If validity is low, reliability becomes meaningless (Gordis 7e, Ch 5, p 127)

(Browner, *Designing Clinical Research 5e*, Ch 4, pp 83-96.)

The three levels at which validity operates

Level	Meaning	Example	Chiefly threatened by
Measurement (test/instrument) validity	The ability of a test to separate those who have the disease from those who do not	Glycosuria is a useful screening test for diabetes, but the glucose tolerance test is more valid (Park 28e, Ch 4, p 166)	Misclassification of exposure, outcome or confounder

Level	Meaning	Example	Chiefly threatened by
Internal validity	The extent to which the comparison made within the study is correct for the subjects actually studied — the legitimacy of the inference about exposure and outcome	The relative risk observed among the study participants is the true relative risk in them	Selection bias, information bias, confounding (Gordis 7e, Ch 15, p 303)
External validity (generalisability)	The extent to which the findings can be applied beyond the study population — to the defined population and, presumably, to the total population	A trial in urban tertiary-care patients applied to a rural district	Non-representative sampling, volunteer participation, restrictive eligibility criteria

Gordis draws a distinction the examiner looks for: **selecting** study subjects from a larger target population affects **generalisability** but does not by itself damage internal validity; **selection bias**, a *systematic error* in choosing one or more of the groups being compared, damages **internal validity** (Gordis 7e, Ch 15, p 303).

Measuring the validity of a test against a gold standard

Screening test result by diagnosis (Park 28e, Ch 4, Table 3-A):

Test result	Diseased	Not diseased	Total
Positive	a (true positive)	b (false positive)	a + b
Negative	c (false negative)	d (true negative)	c + d
Total	a + c	b + d	a + b + c + d

Index	Formula	Interpretation
Sensitivity	$a / (a + c) \times 100$	Ability of the test to correctly identify all those who have the disease (true positives); term introduced by <i>Yerushalmy</i> in the 1940s (Park 28e, Ch 4, p 167)
Specificity	$d / (b + d) \times 100$	Ability of the test to correctly identify those who do not have the disease
False-negative rate	$c / (a + c) \times 100$	Diseased persons wrongly reassured
False-positive rate	$b / (b + d) \times 100$	Healthy persons wrongly alarmed and subjected to further diagnostic tests, at inconvenience, anxiety and expense
Positive predictive value	$a / (a + b) \times 100$	Probability that a test-positive person truly has the disease
Negative predictive value	$d / (c + d) \times 100$	Probability that a test-negative person is truly free of disease

Three properties of these indices carry marks:

- ▶ Sensitivity and specificity are **inherent properties of the test**; predictive value is **not** — it depends on sensitivity, specificity **and the prevalence of the disease** in the population screened (Park 28e, Ch 4, p 168).
- ▶ Sensitivity and specificity are **inversely related** — sensitivity can be raised only at the expense of specificity and vice versa; an ideal test would be 100 per cent sensitive and 100 per cent specific, which in practice seldom occurs (Park 28e, Ch 4, p 167).
- ▶ **Specificity has a greater effect than sensitivity on positive predictive value**, because in a low-prevalence population most of the people tested are non-diseased, so a small fall in specificity generates a large number of false positives (Gordis 7e, Ch 5, pp 126-127).

Validity of a measurement when there is no gold standard

For abstract and subjective constructs — pain, quality of life, health beliefs, social isolation — no gold standard exists, and validity is established by these constructs instead (Browner 5e, Ch 4, pp 91-92; Fletcher, Clinical Epidemiology, Ch 3, pp 51-52; Handbook of Epidemiology, Ch 10, p 421):

Type of validity	Meaning	How it is established
Face validity	The measurement <i>seems</i> inherently reasonable on inspection, e.g. assessing frailty by muscle weakness, exhaustion, weight loss and slow walking speed	Inspection by investigators and experts; the least rigorous method
Content validity	The instrument covers all dimensions of the construct intended and nothing more , e.g. a pain scale asking about aching, throbbing, pressure, burning and stinging but not itching or nausea	Literature review, expert panel, and asking prospective respondents
Construct validity	The instrument measures the theoretical trait it claims to and not something else	Established experimentally — administered to groups known by experts to differ on the construct
Criterion validity — concurrent	The new measure agrees with an already established, validated measure applied at the same time	Correlate the new instrument with the criterion measure; valuable only if the criterion is convincing
Criterion validity — predictive	The measure predicts a future directly observable outcome, e.g. a frailty score predicting later long-term-care admission	Correlate baseline scores with the later observed outcome

Reliability — sources of variation and how each is minimised

Repeatability depends on three groups of factors: **observer variation**, **biological (subject) variation** and **errors relating to technical methods**; blood pressure is poorly reproducible because it is subject to all three (Park 28e, Ch 4, pp 165-166).

Source	What it is	Example	Minimised by
Intra-observer (within-observer) variation	Variation between repeated observations by the same observer on the same subject or material at the same time	One clinician records two different blood pressures in the same subject at the same sitting	Taking the average of several replicate measurements at the same time
Inter-observer (between-observer) variation	Variation between different observers on the same subject or material	One observer reads a blood smear as malaria while a second reads the same slide as normal; among 14,867 chest X-ray films read independently by eight radiologists, wide observer variation was demonstrated	Standardisation of procedures for measurement and classification, intensive training of observers, use of two or more observers making independent assessments
Biological (subject) variation	(a) Variation in the parameter itself over time; (b) variation in the way subjects perceive and report symptoms (recall error); (c) regression to the mean — extreme values tend to move towards the average on repeat measurement	A cervical smear from the same woman may be normal one day and abnormal on another; myocardial infarction may occur with or without pain	Repeat measurement over time on a subsample; awareness of regression to the mean when evaluating therapy
Technical / instrument errors	Defective instruments, faulty calibration, faulty reagents, or a test that is itself unreliable	A manometer out of adjustment giving consistently high readings	Refining and automating instruments; periodic calibration against a gold standard

Observer error can never be eliminated absolutely; the purpose of quality control is not to abolish measurement error but to reduce it to an acceptable level (Park 28e, Ch 3, p 74; Ch 4, p 165).

Statistics used to quantify reliability

Statistic	Data type	Interpretation
Overall percent agreement	Paired categorical	Proportion of all readings classified the same way by two observers; falsely inflated when most readings are negative, because agreement on normals dominates
Percent positive agreement	Paired categorical	Uses only pairs in which at least one reading was positive (denominator $a + b + c$), discarding the concordant-negative cell d (Gordis 7e, Ch 5, p 130)
Kappa (κ), proposed by Cohen in 1960	Paired categorical	(Percent agreement observed – percent agreement expected by chance) $\div$ (100% – percent agreement expected by chance). <i>Landis and Koch</i> : $\kappa > 0.75$ excellent , 0.40–0.75 intermediate to good , < 0.40 poor agreement beyond chance (Gordis 7e, Ch 5, pp 131–132)
Weighted kappa	Paired ordered categories	As for kappa, but gives partial credit for near-agreement

Statistic	Data type	Interpretation
Intraclass correlation coefficient (ICC)	Continuous	Between-person variance ÷ total (between + within-person) variance; equals 1.0 only on exact agreement; the continuous analogue of kappa (Handbook of Epidemiology, Ch 13, pp 526-527)
Pearson correlation coefficient (r)	Continuous	Measures correlation, not agreement — r may be 0.9 even when one observer consistently records twice the other's value; also very sensitive to outliers and to the range of values
Coefficient of variation (CV)	Replicate continuous readings	Standard deviation expressed as a percentage of the mean of replicate observations
Test-retest and equivalence reliability	Questionnaire / scale	Same instrument on two occasions, or two equivalent forms; the correlation coefficient should be at least 0.70 (Handbook of Epidemiology, Ch 10, pp 418-419)
Internal consistency — Cronbach's alpha	Multi-item scale	Ranges 0 to 1.0; alpha below 0.70 indicates the items measure different characteristics and should probably not be combined into one scale (Browner 5e, Ch 17, p 540)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Panel A (external validity)** encloses Total population → Defined population → Study population; **Panel B (internal validity)** encloses Study population → RANDOMIZED → New treatment / Current treatment.
- ▶ **Target-and-arrows diagram (draw four targets in a row):** label them *Precise + Unbiased* (tight cluster on the bullseye — reliable and valid), *Imprecise + Unbiased* (scattered around the bullseye — valid on average, unreliable), *Precise + Biased* (tight cluster off-centre — reliable but invalid), *Imprecise + Biased* (scattered and off-centre — neither).

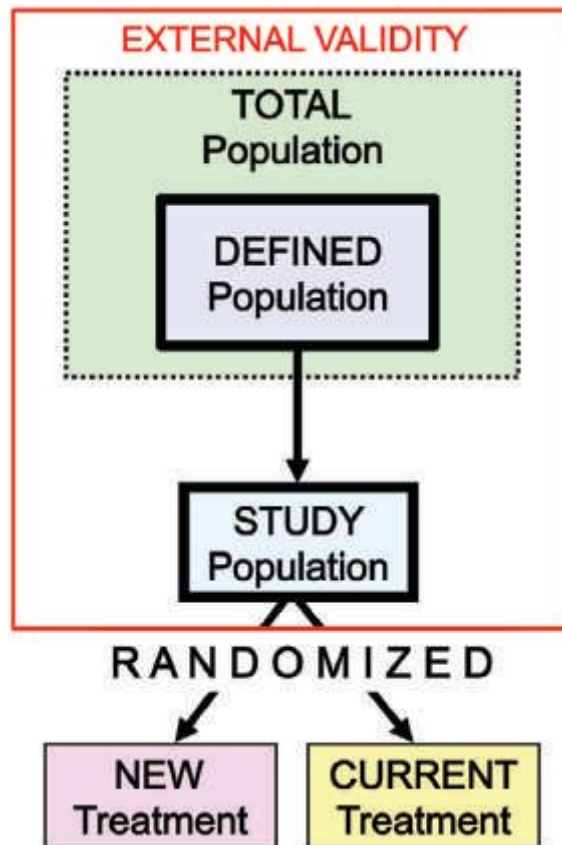


Fig 11.8 — (A) External validity (generalizability) in a randomized trial. Findin (Gordis Epidemiology 7e, p.241)

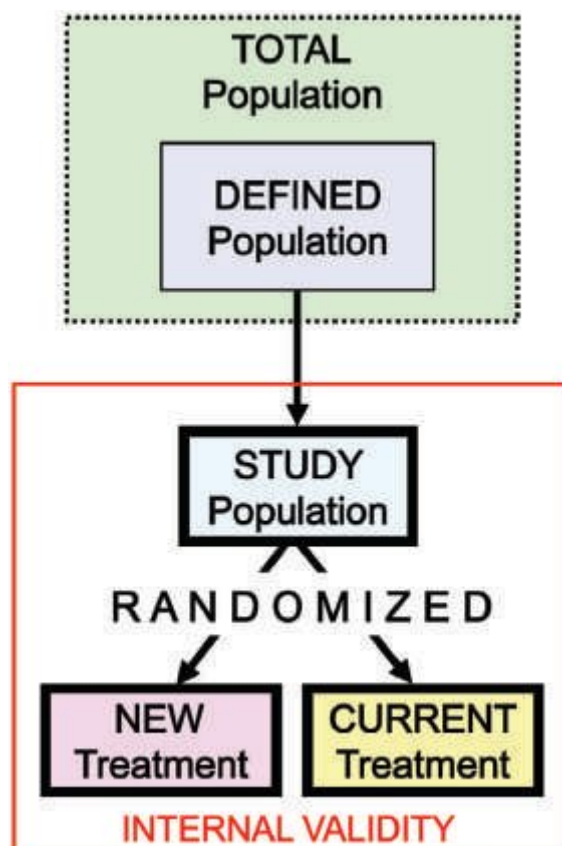
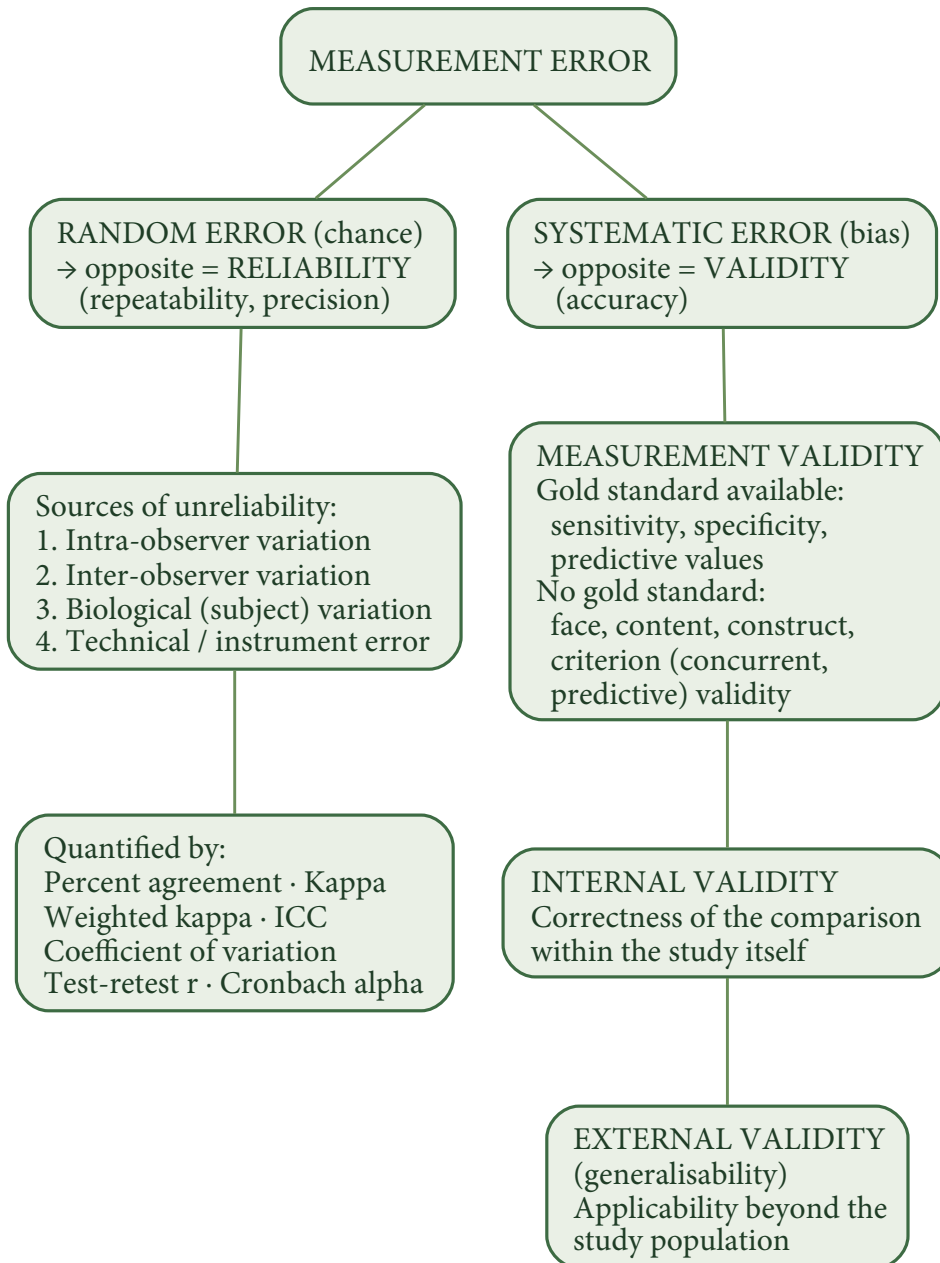
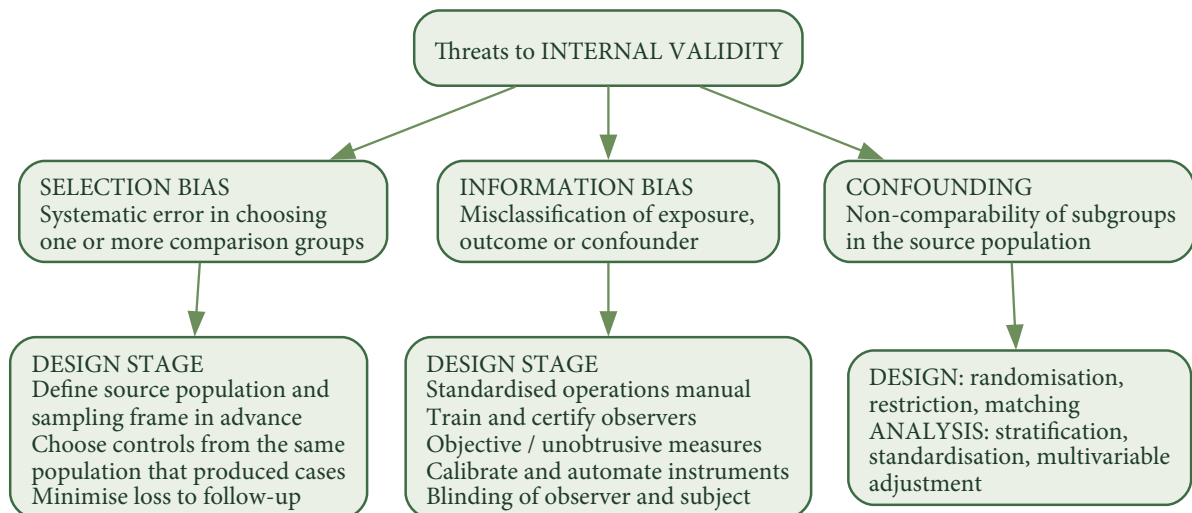


Fig 11.8 — (A) External validity (generalizability) in a randomized trial. Findin (Gordis Epidemiology 7e, p.241)



Validity and reliability in epidemiological research — where each concept sits



Threats to internal validity and the stage at which each is controlled

High-yield exam pointers

- ▶ One-line contrast to open with: **validity = freedom from bias (are we hitting the bullseye?); reliability = freedom from random error (are the shots clustered?)** — a test can be reliable but invalid, but if validity is low, reliability is meaningless.
- ▶ Write the 2×2 table first, then all six formulas: sensitivity $a/(a+c)$, specificity $d/(b+d)$, positive predictive value $a/(a+b)$, negative predictive value $d/(c+d)$, false negative $c/(a+c)$, false positive $b/(b+d)$ — all $\times 100$.
- ▶ Sensitivity and specificity are **inherent** to the test; predictive value additionally depends on **prevalence**; **specificity** influences positive predictive value more than sensitivity does.
- ▶ Kappa cut-offs (*Landis and Koch*): **> 0.75 excellent, 0.40–0.75 intermediate to good, < 0.40 poor**; kappa proposed by *Cohen* (1960); the term sensitivity was introduced by *Yerushalmy* (1940s).
- ▶ Reliability thresholds to quote: test–retest and equivalence correlation ≥ 0.70 ; Cronbach's alpha < 0.70 means the items do not belong on one scale; intraclass correlation coefficient = 1.0 only on exact agreement.
- ▶ Park's triad of causes of poor repeatability: **observer variation (intra- and inter-observer) + biological (subject) variation including regression to the mean + technical/instrument errors**.
- ▶ Remember Gordis's distinction: **choosing** a study population limits *generalisability*; **selection bias** destroys *internal validity*.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only — Pearson's correlation coefficient is still frequently reported as a "reliability" statistic in Indian dissertations, but it measures correlation rather than agreement and can be near-perfect even when one observer systematically reads twice as high as the other; the intraclass correlation coefficient (for continuous data) or kappa (for categorical data) is the correct choice.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 3, Principles of Epidemiology and Epidemiologic Methods; Ch 4, Screening for Disease); Gordis Epidemiology 7e (Ch 5, Assessing the Validity and Reliability of Diagnostic and Screening Tests; Ch 11, Randomized Trials: Some Further Issues; Ch 15, More on Causal Inference: Bias, Confounding and Interaction); Handbook of Epidemiology, Ahrens and Pigeot (Ch 10, Epidemiological Field Work in Population-Based Studies; Ch 13, Quality Control and Good Epidemiological Practice); Browner — Designing Clinical Research 5e, 2022 (Ch 4, Planning the Measurements: Precision, Accuracy and Validity; Ch 17, Designing, Selecting and Administering Self-Reported Measures); Fletcher — Clinical Epidemiology: The Essentials (Ch 3, Abnormality).

**Sensitivity and specificity of screening tests**

Asked in: Paper I May 2018 — RGUHS MD Community Medicine

Definition — A screening test must satisfy the criteria of acceptability, repeatability (reliability) and **validity (accuracy)**, and validity has exactly two components — sensitivity and specificity (Park 28e, Ch 4, pp 165-167). **Sensitivity** — a term introduced by *Yerushalmy* in the 1940s as a statistical index of diagnostic accuracy — is the ability of a test to identify correctly all those who have the disease, that is, the true-positives; a 90 per cent sensitivity means 90 per cent of diseased persons screened give a true-positive result and the remaining 10 per cent a false-negative result. **Specificity** is the ability of a test to identify correctly those who do not have the disease, that is, the true-negatives; a 90 per cent specificity means 10 per cent of non-diseased persons are wrongly classified as diseased (Park 28e, Ch 4, p 167).

The 2 × 2 table from which both are computed

Both are measured by applying the test to one group of persons **having** the disease and to a reference group **not having** the disease, the disease status being fixed independently by a definitive (gold-standard) diagnosis; both are expressed as percentages (Park 28e, Ch 4, pp 166-167).

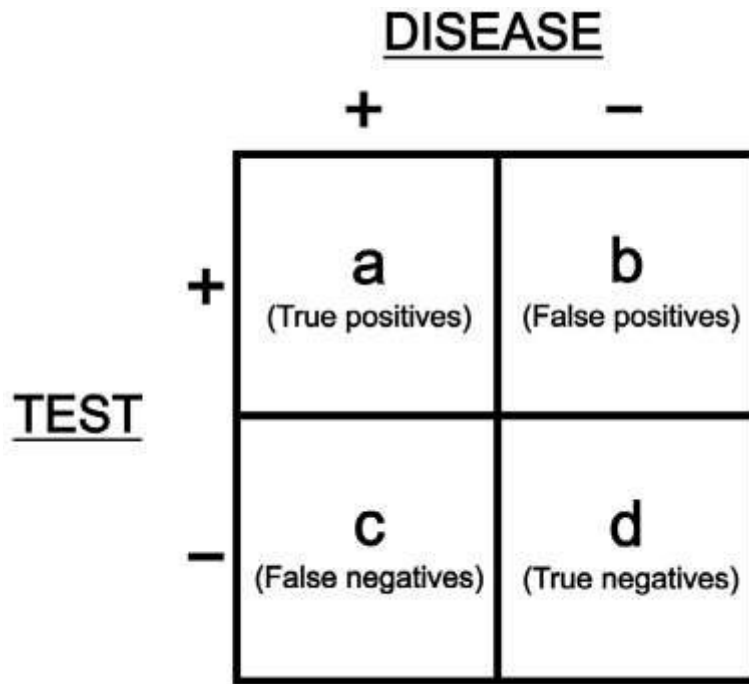


Fig 5.5 — Diagram showing four possible groups resulting from screening with a dichotomous test. (Gordis Epidemiology 7e, p.117)

Cell	Test result	True disease status	Name
a	Positive	Diseased	True-positive
b	Positive	Not diseased	False-positive
c	Negative	Diseased	False-negative
d	Negative	Not diseased	True-negative

Measure	Formula	Denominator is read	Question it answers
Sensitivity	$a/(a + c) \times 100$	Down the diseased column	Of all who truly have the disease, what per cent does the test call positive?
Specificity	$d/(b + d) \times 100$	Down the non-diseased column	Of all who truly do not have the disease, what per cent does the test call negative?
Percentage of false-negatives	$c/(a + c) \times 100$	Diseased column	= 100 – sensitivity
Percentage of false-positives	$b/(b + d) \times 100$	Non-diseased column	= 100 – specificity
Predictive value of a positive test	$a/(a + b) \times 100$	Across the positive-test row	Given a positive result, what is the probability of disease?
Predictive value of a negative test	$d/(c + d) \times 100$	Across the negative-test row	Given a negative result, what is the probability of being free of disease?

(All six formulas: Park 28e, Ch 4, p 167.) The single commonest error is to interchange the denominators: sensitivity and specificity are computed **vertically** (by true disease status), predictive values **horizontally** (by test result).

Worked example (Park's Table 3-B)

Screening test result	Diseased	Not diseased	Total
Positive	40 (a)	20 (b)	60
Negative	100 (c)	9,840 (d)	9,940
Total	140	9,860	10,000

Measure	Working	Value
Sensitivity	$40/140 \times 100$	28.57 %
Specificity	$9,840/9,860 \times 100$	99.79 %
False-negatives	$100/140 \times 100$	71.4 %
False-positives	$20/9,860 \times 100$	0.20 %
Predictive value of a positive test	$40/60 \times 100$	66.66 %
Predictive value of a negative test	$9,840/9,940 \times 100$	98.9 %

(Park 28e, Ch 4, Table 3-B, p 167.) Two tests may be compared on both properties simultaneously — in the diagnosis of brain tumours, electroencephalography (EEG) gives a sensitivity of $36/40 = 90$ per cent and a specificity of $306,000/360,000 = 85$ per cent, whereas computer-assisted axial tomography (CAT) gives $39/40 = 97.5$ per cent and $342,000/360,000 = 95$ per cent; CAT is therefore both more sensitive and more specific than EEG (Park 28e, Ch 4, Tables 4 and 5, pp 167-168).

Sensitivity contrasted with specificity

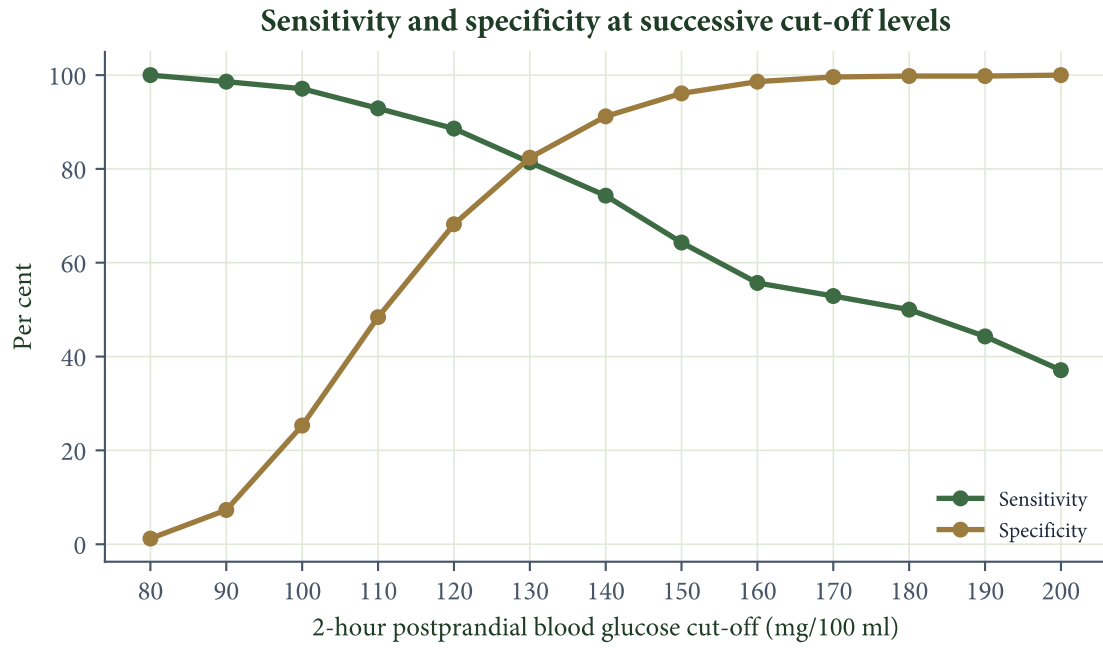
Attribute	Sensitivity	Specificity
Identifies correctly	Those with the disease (true-positives)	Those without the disease (true-negatives)
Formula	$a/(a + c) \times 100$	$d/(b + d) \times 100$
Reference group	Persons known to have the disease	Persons known not to have the disease
Error it minimises	False-negatives (missed cases)	False-positives (false alarms)
A high value means	Few cases are missed; a negative result largely excludes disease	Few healthy persons are labelled; a positive result largely confirms disease
Preferred when	The disease is serious, treatable and missing a case is dangerous; disease prevalence is high	Confirmatory work-up is costly, invasive or hazardous; a false label causes harm; disease is rare
Effect of lowering the cut-off	Increases	Decreases

Attribute	Sensitivity	Specificity
Affected by disease prevalence?	No — an inherent property of the test	No — an inherent property of the test

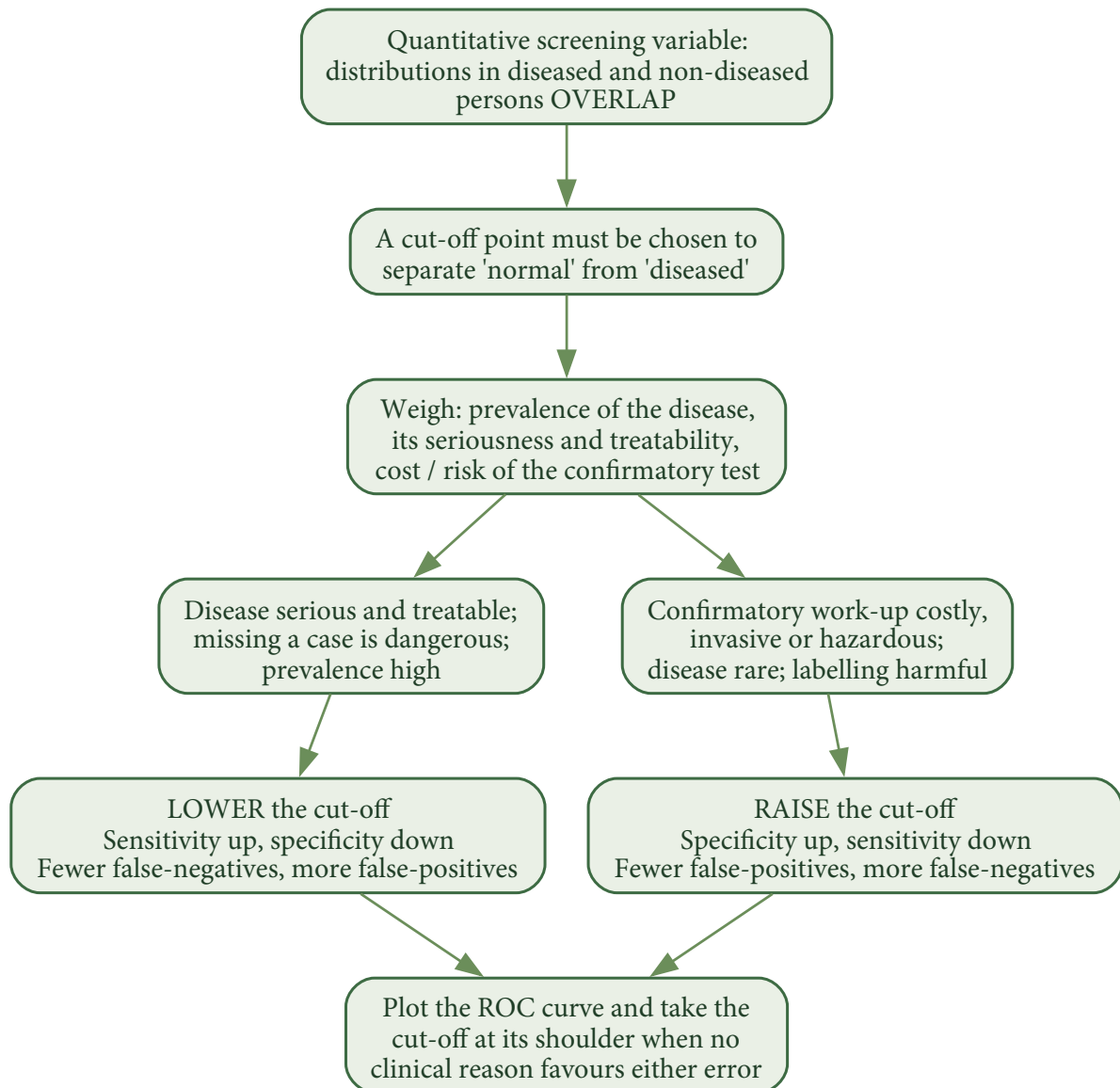
(Park 28e, Ch 4, pp 167-169; Gordis 7e, Ch 5, p 123.)

The inverse relationship and the problem of the borderline

- ▶ With a **quantitative** test (blood sugar, blood pressure, serum cholesterol) the distributions of the attribute in diseased and non-diseased persons **overlap**; false-positives and false-negatives constitute the area of overlap, and no single value separates the two groups completely (Park 28e, Ch 4, p 167).
- ▶ Consequently **sensitivity and specificity are inversely related** — sensitivity can be increased only at the expense of specificity and vice versa. An ideal test would be 100 per cent sensitive and 100 per cent specific; in practice this seldom occurs (Park 28e, Ch 4, p 167).
- ▶ Where the distribution is **bimodal** (as in a genetically transmitted condition such as phenylketonuria), the point at which the two distributions intersect is conventionally used as the cut-off, since it generally minimises false-positives and false-negatives together (Park 28e, Ch 4, p 169).
- ▶ Where the distribution is **unimodal**, there is no natural dividing line at all; a lower cut-off renders the test highly sensitive (few cases missed, many false-positives) and a higher cut-off renders it highly specific (Park 28e, Ch 4, p 169).
- ▶ Factors weighed while fixing the cut-off: **disease prevalence** (when prevalence is high the level is set lower, increasing sensitivity), the **nature of the disease** (seriousness and treatability), and the **cost, risk and availability** of the confirmatory diagnostic work-up (Park 28e, Ch 4, p 169).
- ▶ The trade-off is displayed graphically by the **receiver operating characteristic (ROC) curve**, sensitivity plotted against 1 – specificity over a range of cut-off values; the best cut-off usually lies at or near the "shoulder" of the curve, and the **area under the curve** expresses the overall accuracy of the test (Fletcher, Clinical Epidemiology, Ch 8, p 132; Handbook of Epidemiology, Ch 31, pp 1263-1264).



The inverse relationship in real data: a 2-hour postprandial blood glucose test in 70 true diabetics and 510 true non-diabetics. At 180 mg/100 ml sensitivity is 50 per cent but specificity 99.8 per cent; the two curves cross near 130 mg/100 ml (81.4 and 82.4 per cent). Source: Park's Textbook of Preventive and Social Medicine 28e, Ch 4, Table 6.



Fixing the cut-off point of a screening test — the sensitivity-specificity trade-off

False-negatives and false-positives — the clinical face of the two indices

The epidemiologist thinks in terms of sensitivity and specificity; the clinician thinks in terms of false-negatives and false-positives (Park 28e, Ch 4, p 168).

	False-negative	False-positive
Meaning	A person who actually has the disease is told he does not	A person without the disease is told he has it
Consequence	"False reassurance" — signs and symptoms may be ignored and treatment postponed; serious if the disease is grave and the test unlikely to be repeated soon	Healthy persons are put through further diagnostic tests with inconvenience, discomfort, anxiety and expense until freedom from disease is established
Programme consequence	The impact of the detection programme on disease transmission is greatly reduced	Diagnostic facilities are burdened and the screening programme is brought into discredit

False-negative		False-positive
Reduced by	High sensitivity — the lower the sensitivity, the more the false-negatives	High specificity

Why sensitivity and specificity alone do not answer the clinician's question

Sensitivity and specificity are inherent properties of the test; **predictive accuracy** additionally depends on disease prevalence in the population screened, and the predictive value of a positive result falls as prevalence declines (Park 28e, Ch 4, p 168; Gordis 7e, Ch 5, p 123).

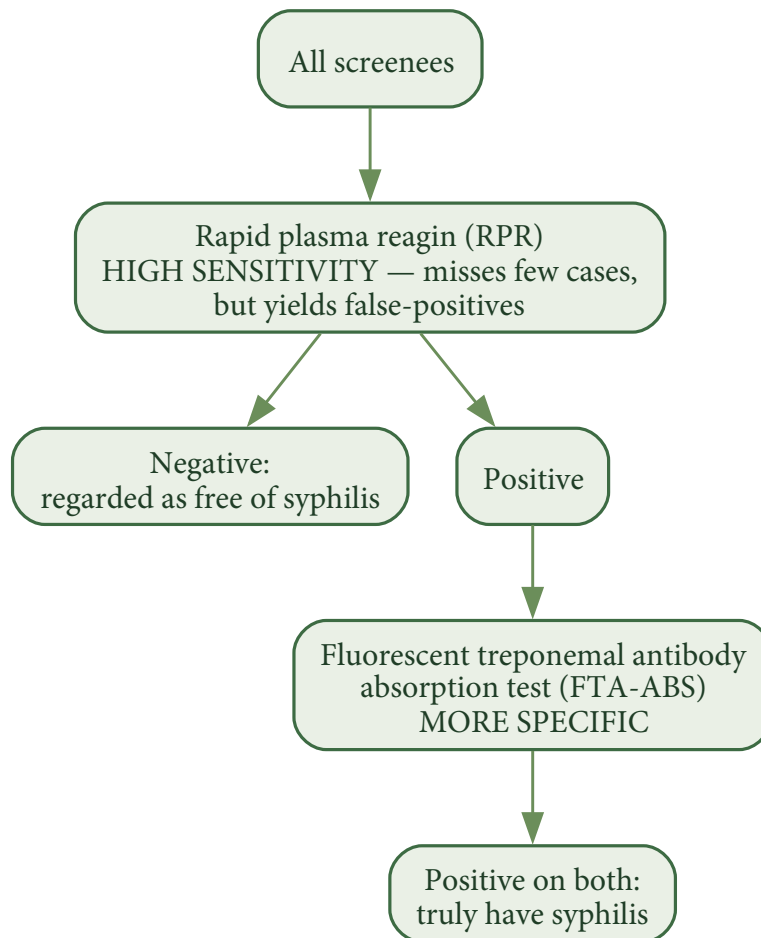
Illustration	Sensitivity	Specificity	Prevalence	Predictive value of a positive test
Gram-stained cervical smear for gonorrhoea (Park 28e, Ch 4, Table 7)	50 %	—	5 %	21 %
same test	50 %	—	15 %	47 %
same test	50 %	—	25 %	63 %
Hypothetical test (Gordis 7e, Ch 5, Table 5.8)	99 %	95 %	1 %	17 %
same test	99 %	95 %	5 %	51 %

Hence **yield** — the amount of previously unrecognised disease diagnosed as a result of the screening effort — depends on the sensitivity and specificity of the test, the prevalence of the disease and participation; restricting a diabetes screening programme to persons over 40 years raises the yield, and high-risk groups are therefore usually selected for screening (Park 28e, Ch 4, p 168).

Combining tests to raise sensitivity or specificity

Mode	How a person is called positive	Net effect
Sequential (two-stage) — a sensitive test first, a specific test on those positive	Positive on both tests	Net loss of sensitivity, net gain of specificity
Simultaneous (parallel) — both tests applied together	Positive on any one test	Net gain of sensitivity, net loss of specificity

(Gordis 7e, Ch 5, p 121.)



Sequential (two-stage) testing in syphilis screening — a sensitive test first, a specific test to confirm

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Overlap of distributions and the cut-off point (Park's Fig. 3)** — labels: x -axis "Value of the screening variable (e.g. blood glucose, mg/100 ml)"; y -axis "Number of persons"; left curve "Non-diseased"; right curve "Diseased"; vertical line "Cut-off point"; shaded zone of overlap "Borderline group"; area of the non-diseased curve to the right of the cut-off "False-positives (b)"; area of the diseased curve to the left of the cut-off "False-negatives (c)"; annotation "Point of intersection of the two distributions — minimises both errors".
- **Receiver operating characteristic (ROC) curve** — labels: x -axis "1 – specificity (false-positive rate), 0 to 1"; y -axis "Sensitivity (true-positive rate), 0 to 1"; diagonal "Line of no discrimination — useless test"; curve bowing towards the upper-left corner "Test with good discrimination"; marked point on the curve "Shoulder = best cut-off"; annotation "Area under the curve = overall accuracy of the test".

High-yield exam pointers

- Sensitivity = $a/(a + c) \times 100$ and specificity = $d/(b + d) \times 100$ — both read **down the columns** (true disease status); predictive values read **across the rows** (test result). Sensitivity + false-negative rate = 100; specificity + false-positive rate = 100.

- ▶ *Yerushalmy*, 1940s, introduced the term "sensitivity"; validity (accuracy) = sensitivity + specificity; the other two screening-test criteria are acceptability and repeatability.
- ▶ **SnNout / SpPin**: a highly Sensitive test, when Negative, rules the disease **out**; a highly Specific test, when Positive, rules it **in**.
- ▶ Anchor figures for the trade-off (Park Table 6): 2-hour postprandial glucose at 180 mg/100 ml → sensitivity 50 %, specificity 99.8 %; at 120 mg/100 ml → 88.6 % and 68.2 %.
- ▶ Sensitivity and specificity do **not** change with prevalence; predictive value of a positive test does — 17 % at 1 per cent prevalence versus 51 % at 5 per cent prevalence for a test of 99 per cent sensitivity and 95 per cent specificity (Gordis Table 5.8).
- ▶ Sequential testing → net loss of sensitivity, net gain of specificity; simultaneous testing → net gain of sensitivity, net loss of specificity.
- ▶ ROC curve = sensitivity (y) against 1 – specificity (x); best cut-off at the shoulder; area under the curve = overall accuracy.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park does not develop the receiver operating characteristic curve or likelihood ratios, so those two points are grounded on Fletcher's Clinical Epidemiology and the Handbook of Epidemiology instead; the SnNout/SpPin mnemonic is standard teaching shorthand for what Park states in words about false-negatives and false-positives, not a phrase used in Park itself.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Chapter 4, Screening for Disease, pp 165-169 (Tables 3-A, 3-B, 4, 5, 6, 7); Gordis Epidemiology, 7e — Chapter 5, Assessing the Validity and Reliability of Diagnostic and Screening Tests, pp 115-127 (Fig. 5.5, Tables 5.7-5.9); Fletcher — Clinical Epidemiology: The Essentials, Chapter 8 (Diagnosis), p 132; Handbook of Epidemiology (Ahrens & Pigeot), Chapter 31 (Screening), pp 1263-1264.



Randomisation procedures in research

Asked in: Paper I 24-Sep-2025 — RGUHS MD Community Medicine

Definition — Randomization (random allocation) is the allocation of eligible, consenting participants to the study and control groups purely by chance, so that every participant has an equal chance of being allocated to either group and, critically, the next assignment is unpredictable (Park 28e, Ch 3, p 102; Gordis 7e, Ch 10, p 219). It is the "heart" of a randomized controlled trial (RCT), and it is the single step that separates an experimental study from an analytical observational study, in which comparability can only be attempted by matching (Park 28e, Ch 3, pp 102-103).

Bradford Hill used randomization as a means of preventing bias from entering the experimental trial. After the Second World War both streptomycin for tuberculosis and a pertussis vaccine were tested using Hill's randomization approach; the pertussis vaccine trial began earlier, but the streptomycin trial

published its results first, and the design became a standard of the field (Lilienfeld's Foundations, Ch 2, pp 50-51).

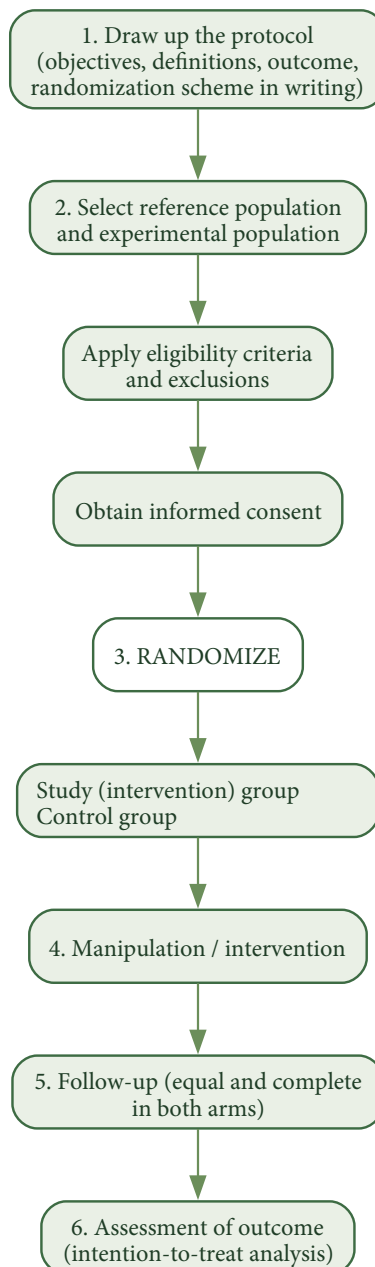
Why randomize — the purposes served

Purpose	How randomization achieves it
Removes selection (allocation) bias	The investigator has no control over which group a participant enters; assignment is by chance alone (Park 28e, Ch 3, p 102)
Balances known confounders	Age, sex, disease severity and similar prognostic variables tend to be distributed equally between the arms
Balances unknown and unmeasured confounders	Matching can only balance factors already known to be important; randomization also distributes factors whose effect is unrecognised or cannot be measured (Park 28e, Ch 3, p 102; Fletcher, Ch 5, p 91)
Prevents confounding by indication	The clinician's choice of treatment based on patient characteristics is taken out of the allocation process (Handbook of Epidemiology, Ch 9, p 379)
Provides a valid basis for statistical inference	Any residual difference between arms is attributable to chance, which the significance test is designed to quantify
Preserves intention-to-treat analysis	Analysis in the group to which the participant was randomized keeps the benefit of randomization despite non-compliance and loss to follow-up (Handbook of Epidemiology, Ch 9, p 379)

Randomization increases the *likelihood* of comparability but does not *guarantee* it; by chance, one arm may still collect most of the high-risk participants, which is why restricted procedures are used (Gordis 7e, Ch 10, p 222).

Where randomization sits in a randomized controlled trial

Randomization is performed **only after** the volunteer has been screened as eligible and has given informed consent — never before (Park 28e, Ch 3, pp 102-103). Allocating first and consenting later reintroduces the very bias randomization is meant to remove.



Position of randomization within the six basic steps of a randomized controlled trial (Park 28e, Ch 3, pp 101-103)

Randomization procedures, with suitable examples

Procedure	How it is carried out	Suitable example	Comment
Simple randomization	Coin toss, lottery method (numbered cards shuffled and drawn), a table of random numbers, or a computer-generated list of random assignments, with no restriction of any kind	The model low-risk description quoted by Cochrane: "We generated the two comparison groups using simple randomization, with an equal allocation ratio, by referring to a table of random numbers" (Cochrane Handbook, Ch 8, p 237)	Simplest and unpredictable; yields groups of similar size only in large trials (usually meaning at least 100 per randomized group) and in small trials can give groups differing substantially in size or in case-mix (Cochrane Handbook, Ch 8, p 237)

Generating and concealing the allocation sequence

- ▶ **Decide and write down the scheme before recruitment begins** — the method of sequence generation and any restriction (blocking, stratification) must be specified *in writing* before the first allocation (Gordis 7e, Ch 10, p 220; CONSORT item 8, Handbook of Epidemiology, Ch 8, p 354).
- ▶ **Generate the sequence** — computer random-number generation is the usual method; a table of random numbers is the manual alternative used as a backup or where access to a computer is limited (Gordis 7e, Ch 10, p 219; Park 28e, Ch 3, pp 102-103).
- ▶ **Worked example using a random number table** — decide that odd digits denote treatment A and even digits treatment B; enter the table at a randomly chosen row and column, note the starting point and the direction of reading, and read consecutively. A run 5, 3, 8 ... allocates the first two patients to A, the third to B, and so on. The next assignment is not predictable — unlike strict alternation, which is (Gordis 7e, Ch 10, pp 219-220).
- ▶ **Report who did what** — CONSORT requires reporting who generated the allocation sequence, who enrolled participants and who assigned participants to their groups; keeping these three roles separate is the standard safeguard (CONSORT item 10, Handbook of Epidemiology, Ch 8, p 354).
- ▶ **Conceal the allocation** until the participant is irreversibly entered.
- ▶ **Blind after allocation** where feasible, and analyse by intention to treat.

Allocation concealment means preventing anyone who enrolls a participant from foreseeing the forthcoming assignment; **blinding (masking)** means keeping participants, care providers and outcome assessors unaware of the allocation *after* it has been made (Handbook of Epidemiology, Ch 29, p 1179).

Adequate method of concealment	How it works
Central randomization by a third party (telephone, web-based, pharmacy-controlled) — the most desirable method	Participant details are transmitted to a randomization office remote from the recruitment centres, and the allocation is released only after the participant is irreversibly registered (Cochrane Handbook, Ch 8, pp 223-224, 241)
Sequentially numbered drug containers of identical appearance	Prepared by an independent pharmacy, sequentially numbered and opened in sequence; identical in appearance, tamper-proof and equal in weight (Cochrane Handbook, Ch 8, p 241)
Sequentially numbered, opaque, sealed envelopes	Opened in strict sequence only after the participant's details are written on the envelope; carbon or pressure-sensitive paper transfers the details inside, cardboard or aluminium foil makes the envelope impermeable to intense light, and tamper-proof security tape seals it. Envelope methods are more open to manipulation than the other two (Cochrane Handbook, Ch 8, pp 240-241)

Adequate method of concealment	How it works
Inadequate (high risk of bias) — an open random allocation schedule (e.g. a list of random numbers on display); assignment envelopes used without safeguards (unsealed, non-opaque or not sequentially numbered); alternation or rotation; date of birth; case record number	Those enrolling participants can foresee the assignment and can delay or divert a participant to the "appropriate" arm, so selection bias re-enters (Cochrane Handbook, Ch 8, pp 224, 239)

What is not randomization

- **Systematic assignment by date of admission.** In a post-war trial of anticoagulant therapy, patients admitted on odd-numbered days went to one group and those admitted on even-numbered days to the other; more patients than expected then arrived on odd days, because physicians who had observed the benefit "speeded up, where feasible, the hospitalization of those patients ... in order to bring as many as possible under the odd-day deadline" (Gordis 7e, Ch 10, p 218).
- **Alternate allocation (quasi-randomization).** In the New York City trial of bacille Calmette-Guérin (BCG) vaccine, physicians themselves chose who was vaccinated, and the children of cooperative, health-conscious parents were preferentially vaccinated, producing a spuriously large benefit. Changing to vaccinating alternate children removed the difference — an improvement, but *this does not constitute randomization*, because alternation is predictable (Gordis 7e, Ch 10, pp 218-219).
- **Matching.** Used in analytical (case-control and cohort) studies where groups are already defined by disease or exposure; it can balance only recognised factors (Park 28e, Ch 3, pp 102-103).
- **Random sampling is not random allocation.** Random *sampling* — simple random, systematic, stratified, cluster and multistage (Mahajan's Methods in Biostatistics, Ch 7 (Sampling), pp 135-138) — decides **who enters** the study and governs external validity or generalisability. Random *allocation* decides **which arm** they enter and governs internal validity. A trial may be randomized without being randomly sampled, and a survey randomly sampled without any allocation.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Draw the stratified randomization scheme with these exact labels: **1,000 patients** → *Step 1, stratify by sex: 600 males, 400 females* → *Step 2, stratify by age: 360 younger males, 240 older males, 300 younger females, 100 older females* → *Step 3, randomize each subgroup* → **500 NEW treatment** and **500 CURRENT treatment**.

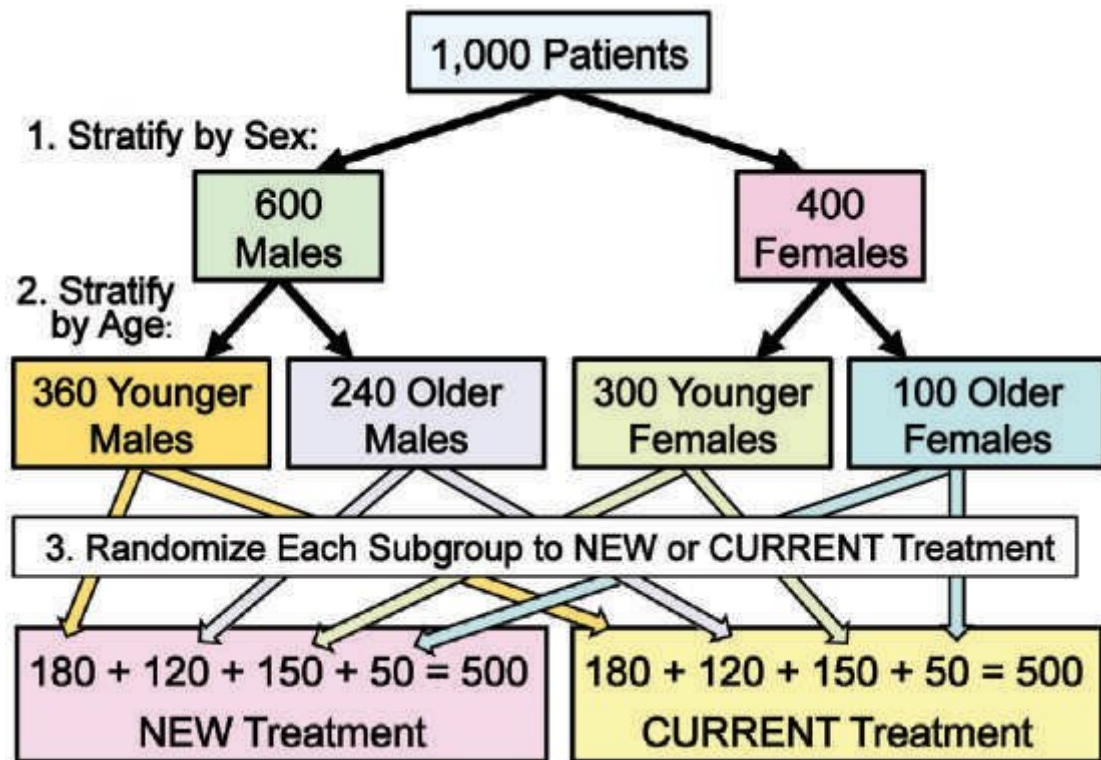
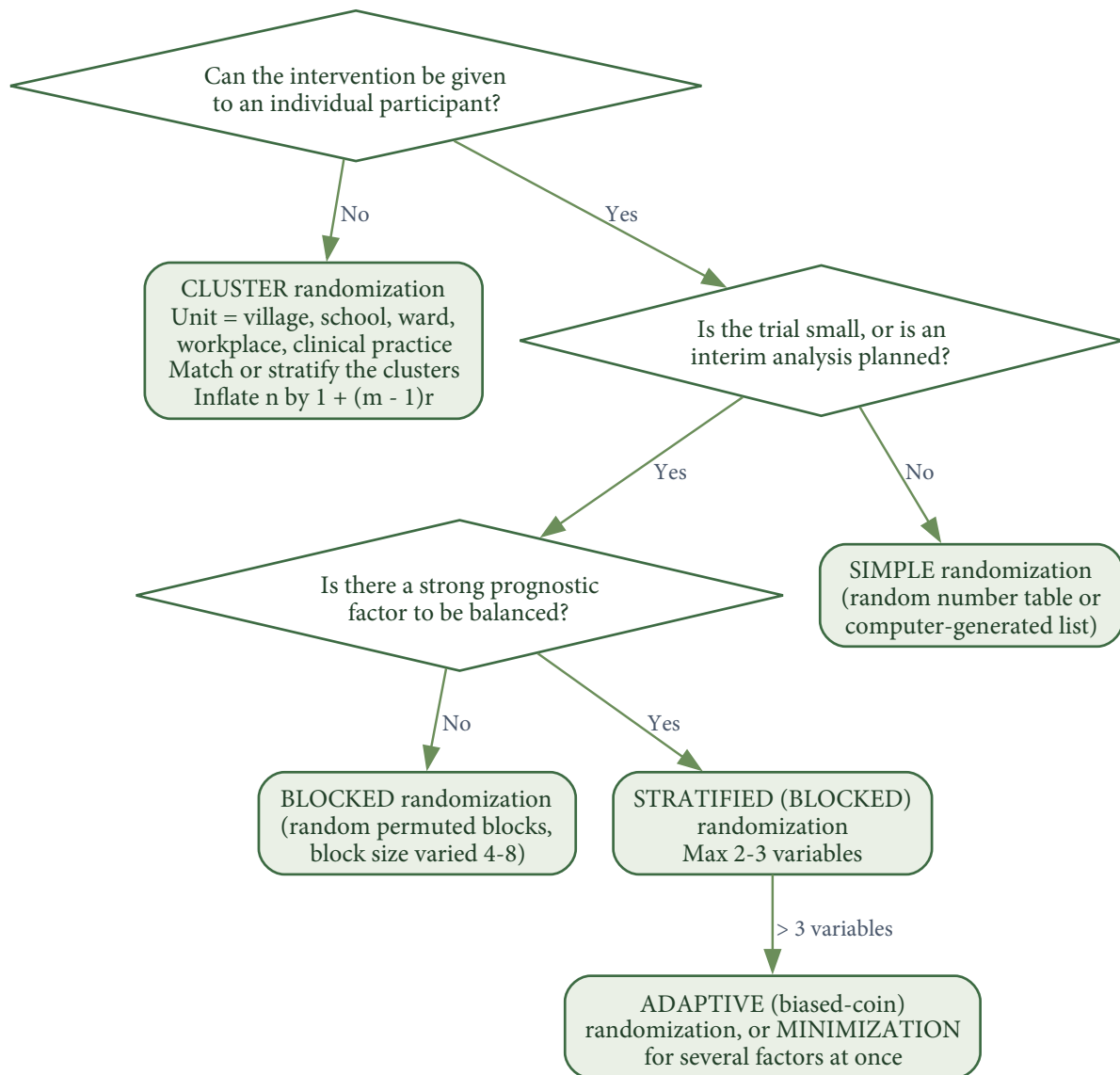


Fig 10.3 — Example of stratified randomization. See discussion in text. (Gordis Epidemiology 7e, Ch 10 (Assessing preventive and therapeutic interventions: Randomized trials), p 223)



Choosing the randomization procedure for a given research question

High-yield exam pointers

- ▶ Randomization is done **after** eligibility screening and **after** informed consent, never before (Park 28e).
- ▶ The critical element is **unpredictability of the next assignment** — alternation and day-of-admission assignment are not randomization (Gordis 7e).
- ▶ Seven procedures to name: **simple**, **blocked (permuted)**, **stratified**, **stratified blocked**, **adaptive (biased coin)**, **minimization**, **cluster (matched or stratified)**.
- ▶ Adaptive randomization alters the *probability* of the next assignment; minimization allocates the next participant to the arm that *minimises total imbalance* across several factors.
- ▶ Blocked randomization with a **fixed** block size in an unblinded trial is predictable — vary the block size (4 to 8).
- ▶ Stratified blocked randomization balances only about **two or three** baseline variables; simple randomization needs about **100 per arm** to give similar group sizes.

- ▶ Cluster trial sample-size design effect = $1 + (m - 1)r$ (m = subjects per cluster, r = intra-cluster correlation coefficient).
- ▶ Three adequate concealment methods: **central randomization, sequentially numbered identical drug containers, sequentially numbered opaque sealed envelopes**. Concealment is *before* allocation; blinding is *after*.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the reporting items quoted here (sequence generation, allocation concealment, implementation, blinding) are the randomization items of the CONSORT statement as reproduced in the Handbook of Epidemiology, and the concealment grading is the Cochrane risk-of-bias tool; both are cited from the indexed textbooks rather than from the original guideline documents.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 3, Principles of Epidemiology and Epidemiologic Methods); Gordis Epidemiology 7e (Ch 10, Assessing Preventive and Therapeutic Interventions: Randomized Trials); Browner — Designing Clinical Research 5e (Ch 11, Designing Randomized Blinded Trials; Ch 12, Alternate Interventional Study Designs); Handbook of Epidemiology, Ahrens & Pigeot (Ch 8, Intervention Trials; Ch 9, Confounding and Interaction; Ch 29, Clinical Epidemiology); Fletcher — Clinical Epidemiology: The Essentials (Ch 5, Risk; Ch 9, Treatment); Cochrane Handbook for Systematic Reviews of Interventions (Ch 8, Assessing Risk of Bias in Included Studies); Lilienfeld's Foundations of Epidemiology (Ch 2, Threads of Epidemiologic History); Mahajan's Methods in Biostatistics for Medical Students and Research Workers (Ch 7, Sampling).



Experimental epidemiology — steps in RCTs

Asked in: Paper I Nov 2016 — RGUHS MD Community Medicine

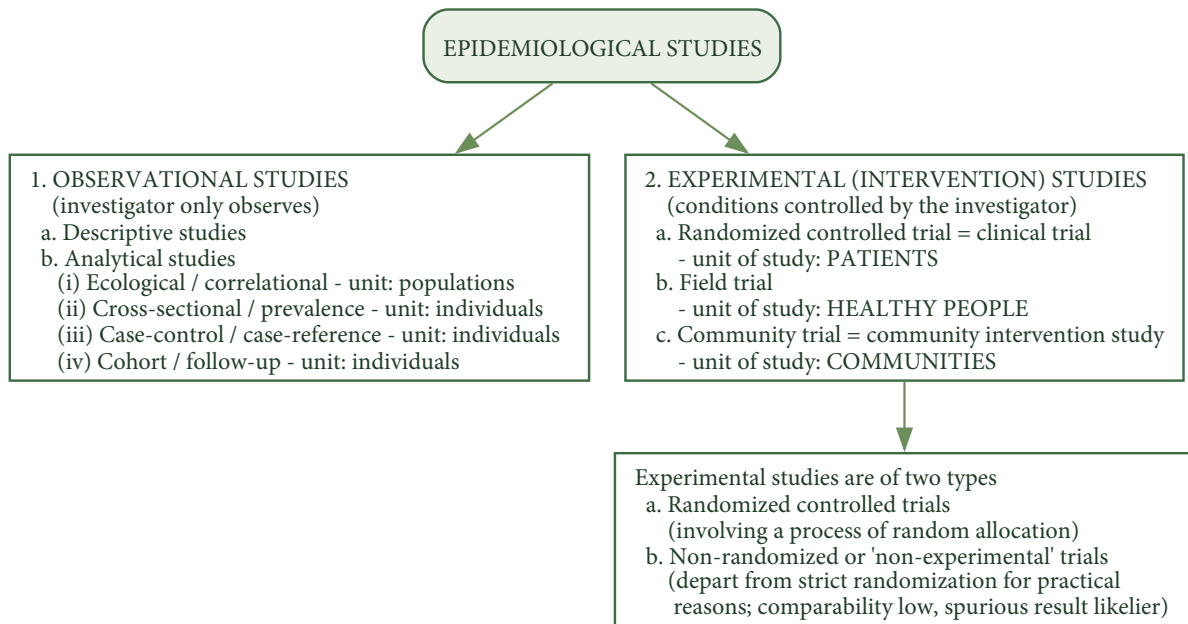
Definition — In the 1920s "experimental epidemiology" meant the study of epidemics among colonies of experimental animals such as rats and mice; in modern usage it is often equated with **randomized controlled trials (RCT)** (Park 28e, Ch 3, p 100). Experimental or intervention studies resemble cohort studies except that the conditions under which the study is carried out are under the **direct control of the investigator** — the suspected cause is deliberately applied, withdrawn or altered in the experimental group, no change is made in the control group, and the outcome is compared in both. This contrasts sharply with observational studies, in which the epidemiologist takes no action but only observes the natural course of events (Park 28e, Ch 3, p 100).

Aims, features and place of experimental studies among epidemiological designs

Item	Content
Aim (a)	To provide "scientific proof" of aetiological (or risk) factors, which may then permit modification or control of those diseases (Park 28e, Ch 3, p 100)
Aim (b)	To provide a method of measuring the effectiveness and efficiency of health services for the prevention, control and treatment of disease and for improving the health of the community (Park 28e, Ch 3, p 100)
Defining feature	Investigator-controlled intervention/manipulation + a comparison (control) group + randomization
Merits	All the advantages of a prospective cohort study; the randomized trial is generally considered the gold standard of study designs , and the randomized, double-masked, placebo-controlled trial is the highest level of evidence for new-drug approval (Gordis 7e, Ch 10, p 231)
Extra problems	All the disadvantages of a cohort study plus three additional problems — cost, ethics and feasibility (Park 28e, Ch 3, p 100)
Settings	May be conducted in animals or in human beings (Park 28e, Ch 3, p 100)

Setting	Uses	Limitations / landmarks (Park 28e, Ch 3, pp 100–101)
Animal experiments	(a) Experimental reproduction of human disease in animals to confirm an aetiological hypothesis; (b) testing efficacy of vaccines and drugs; (c) completing the natural history of disease	Animals breed and multiply rapidly, so genetic experiments needing generations in man are feasible. But not all human diseases can be reproduced in animals, and conclusions may not hold in man — the WHO typhoid vaccine trial in Yugoslavia (mid-1950s) found the alcohol-killed vaccine superior in animals yet less than half as effective as the heat-killed phenol-preserved vaccine in human trials

Setting	Uses	Limitations / landmarks (Park 28e, Ch 3, pp 100–101)
Human experiments	Indispensable to investigate aetiology and evaluate preventive and therapeutic measures, especially for diseases not reproducible in animals	Lind's scurvy experiment in 12 seamen (1747); Jenner's cowpox experiments (1796); Finlay and Reed on yellow fever; Goldberger's pellagra experiments (1915) proving a nicotinic-acid deficiency, not an infection. Ethical and logistic considerations often prevent their use; WHO laid down a strict code of practice for human trials in 1980



Place of experimental (intervention) studies in the classification of epidemiological studies

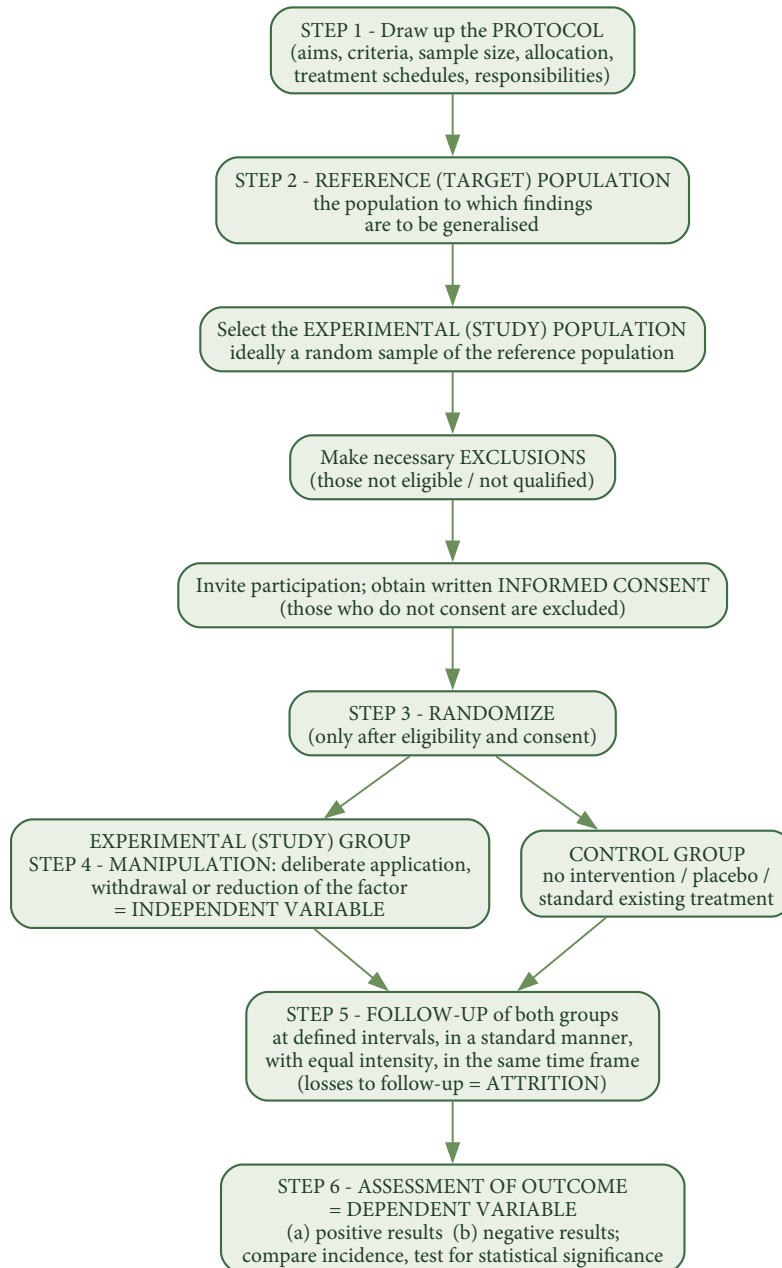
Types of randomized controlled trial

Type	Purpose	Classical example (Park 28e, Ch 3, pp 104–106)
1. Clinical trials	Evaluate therapeutic agents, mainly drugs, in patients	Beta-blockers to reduce cardiovascular mortality in acute myocardial infarction; folate supplementation before conception to prevent recurrence of neural tube defects; tonsillectomy for recurrent throat infection
2. Preventive trials	Trials of primary preventive measures — vaccines and chemoprophylactic drugs; may be applied to groups rather than individuals	Medical Research Council (UK) whooping-cough vaccine trial, 1946 — attack rate 1.45 per 1000 child-months in the vaccinated group versus 6.72 per 1000 child-months in the control group
3. Risk factor trials	Investigator intervenes to interrupt the usual sequence of disease development in those with a risk factor; may be single-factor or multi-factor	WHO clofibrate trial for primary prevention of coronary heart disease (>15,000 men, 3 European centres, double-blind, mean observation 9.6 years) — fewer non-fatal infarctions but 25% more deaths in the clofibrate group; also the Stanford Three Community Study, North Karelia Project, Oslo Study and MRFIT

Type	Purpose	Classical example (Park 28e, Ch 3, pp 104–106)
4. Cessation experiments	Evaluate termination of a habit or removal of a suspected agent; a fall in disease strengthens the causal hypothesis	Cigarette-smoking cessation and lung cancer / coronary heart disease
5. Trial of aetiological agents	Confirm or refute an aetiological hypothesis — rarely possible in man for ethical reasons	Retrolental fibroplasia — Kinsey and Hemphill randomized babies ≤ 1500 g to 50% oxygen for 28 days versus a "curtailed oxygen group"; peak incidence 1952–53 fell sharply after 1953 with reduced oxygen use
6. Evaluation of health services	Assess effectiveness and efficiency of alternative health-care delivery policies ("health services research")	Indian controlled trials of tuberculosis chemotherapy showing domiciliary treatment as effective as costlier sanatorium treatment ; South-East London trial of multiphasic screening; substitution of nurses and paramedical workers for physician tasks

Steps in conducting a randomized controlled trial

The RCT is really an **epidemiologic experiment** and is the number-one method of evaluating a new programme or therapy; its basic steps are six (Park 28e, Ch 3, p 101).



Design and operational sequence of a randomized controlled trial (after Park 28e, Ch 3, Fig 9)

1. Drawing up a protocol. An essential feature of an RCT is that it is conducted under a **strict protocol**, whose purpose is to **prevent bias and reduce sources of error**. Once evolved it must be **strictly adhered to**; a **pilot study** beforehand tests feasibility, operational efficiency of procedures, unknown effects and acceptability, and the final version must be **agreed upon by all concerned before the trial begins** (Park 28e, Ch 3, p 102). In India the protocol must also be cleared by a properly constituted **ethics committee** and the trial **registered with the Clinical Trials Registry–India (CTRI)** before enrolment (ICMR National Ethical Guidelines 2017, sections 3.7, 7.1).

The protocol specifies (Park 28e, Ch 3, p 102):

- ▶ Aims and objectives of the study, and the questions to be answered
- ▶ Criteria for selection of the study and control groups
- ▶ Size of the sample

- ▶ Procedure for allocation of subjects into study and control groups
- ▶ Treatments to be applied — when, where, how and to what kind of patients
- ▶ Standardization of working procedures and schedules
- ▶ Responsibilities of all parties involved, up to evaluation of outcome

2. Selecting reference and experimental populations.

Population	Meaning
Reference (target) population	The population to which the findings of the trial, if found successful, are expected to be applicable/generalised . It may be as broad as mankind, or geographically limited, or limited to persons of specific age, sex or occupation — a whole city, school children, industrial workers, an obstetric population (Park 28e, Ch 3, pp 101–102)
Experimental (study) population	The actual population that participates , derived from the reference population; ideally randomly chosen from it so that it has the same characteristics. If it is not derived from the reference population, the findings cannot be generalised to it. Choose a stable population whose cooperation is assured, to avoid losses to follow-up

Volunteers must fulfil **three criteria**: (a) **informed consent** — agreement to participate after being informed of the purpose, procedures and possible dangers of the trial; (b) **representative** of the reference population; and (c) **qualified or eligible**, i.e. fully susceptible to the disease under study — subjects already immune to whooping cough, or non-anaemic subjects in a trial of haematinics, are not eligible (Park 28e, Ch 3, p 102).

3. Randomization. A statistical procedure by which participants are allocated into "study" and "control" groups, to receive or not to receive the experimental procedure, manoeuvre or intervention. It is the "heart" of a **controlled trial**, and is an attempt to **eliminate bias and allow for comparability** — matching can equalise only factors already known to be important, whereas randomization distributes **known and unknown factors equally** between the groups, so that "like can be compared with like". It ensures the **investigator has no control over allocation**, thus eliminating **selection bias**; every individual gets an equal chance of being allocated to either group. Where a variable may itself influence outcome (e.g. age, sex), the study population is first divided into sub-groups by that variable and individuals within each sub-group randomly allocated (**stratified randomization**). Randomization is done **only after** the participant has qualified for the trial and given informed consent, and is **best done using a table of random numbers**. The essential difference from an analytical study is that there no randomization is possible, because differentiation into diseased/non-diseased or exposed/non-exposed has already occurred, leaving matching as the only option (Park 28e, Ch 3, pp 102–103).

4. Manipulation or intervention. The study (experimental) group is intervened upon by the **deliberate application, withdrawal or reduction of the suspected causal factor** — drug, vaccine, dietary component, habit — as laid down in the protocol. This creates an **independent variable**, whose effect is determined by measuring the final outcome, the **dependent variable** (incidence of disease, survival time, recovery period) (Park 28e, Ch 3, p 103).

5. Follow-up. Examination of experimental and control subjects at **defined intervals, in a standard manner, with equal intensity, under the same circumstances and in the same time frame**, until final

assessment. Duration is fixed by the expectation that a significant difference (e.g. in mortality) will be demonstrable at a given point after the start, and may run into years. Losses through death, migration and loss of interest are inevitable — this is **attrition**; if substantial, results cannot be generalised to the reference population, so it must be minimised (Park 28e, Ch 3, p 103).

6. **Assessment of outcome.** The final step, in terms of:

Component	Content
(a) Positive results	Benefits of the experimental measure — reduced incidence or severity of disease, cost to the health service, or other appropriate outcome, in study versus control group
(b) Negative results	Severity and frequency of side-effects and complications, including death ; adverse effects are missed if they are not actively sought
Analysis	Incidence of positive/negative results rigorously compared in both groups and differences tested for statistical significance ; sequential analysis is possible as data accrue, but analysis at the end of the trial is more useful (Park 28e, Ch 3, p 103)
Expression of result	Efficacy = (rate of disease in the control/placebo group – rate in the intervention group) ÷ rate in the control/placebo group, giving the reduction in risk ; rates are often calculated per person-years (Gordis 7e, Ch 11, pp 239–240)
Analysis principle	Primary analysis is by intention to treat — according to the original randomized assignment, whatever treatment was actually received; an "as-treated" analysis breaks the randomization and reduces the trial to a cohort study (Gordis 7e, Ch 10, p 227)

Bias in assessment and blinding. Errors of assessment from the human element arise in three ways — **subject variation** (participants feel better if they know they are on the new treatment), **observer bias** (the investigator measuring outcome knows the allocation), and **bias in evaluation** (the investigator subconsciously gives a favourable report). Neither randomization nor sample size guards against these; **blinding** is the remedy (Park 28e, Ch 3, p 103).

Type of blinding	Who is kept unaware
Single blind trial	The participant is not aware whether he belongs to the study or the control group
Double blind trial	Neither the doctor nor the participant is aware of the group allocation and the treatment received — the most frequently used method
Triple blind trial	The participant, the investigator and the person analyzing the data are all blind — ideally this should be used

Blinding of the subject is achieved by a **placebo**, of identical appearance to the active agent (Gordis 7e, Ch 10, p 224). When the outcome measured is a hard end-point such as **death**, blinding is not so essential (Park 28e, Ch 3, p 103).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Diagram (to sketch)** — *Design of a randomized controlled trial* (Park 28e, Ch 3, Fig 9): a vertical chart with the boxes labelled, from top down — "Reference or target population" → "Experimental or study population" → side-arrows out for "Those not eligible" and "Those who do not wish to give consent" → "RANDOMIZE" → splitting into two boxes "Experimental group" and "Control group" → both converging on "FOLLOW-UP" → "ASSESSMENT".

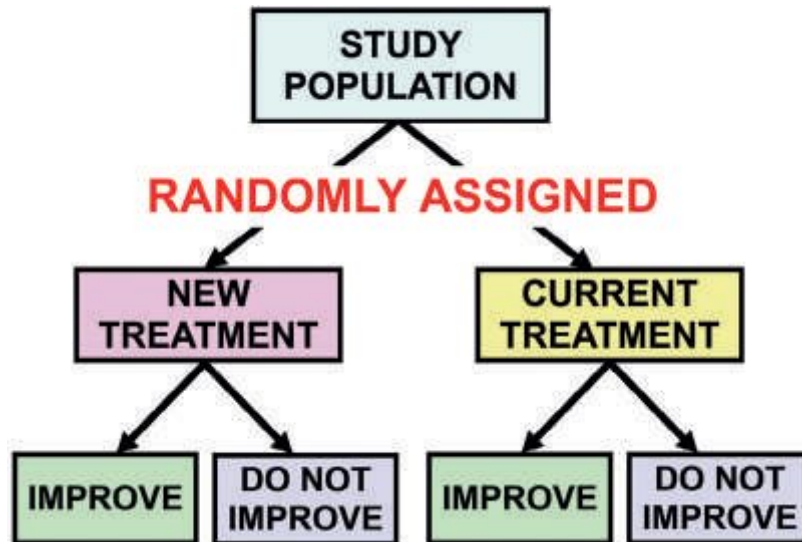


Fig 12.1 — Design of a randomized clinical trial. (Gordis Epidemiology 7e, p.255)

High-yield exam pointers

- **Definition line:** 1920s = epidemics in colonies of experimental animals (rats, mice); modern usage = **randomized controlled trials** (Park 28e, Ch 3, p 100).
- **Six steps, in order** — P-R-R-M-F-A: Protocol → Reference and experimental populations → Randomization → Manipulation → Follow-up → Assessment of outcome.
- **Three criteria for a volunteer:** informed consent; representative of the reference population; qualified/eligible (fully susceptible to the disease studied).
- **Randomization is the "heart" of a controlled trial** — done only after eligibility and consent, by a table of random numbers; distributes **known and unknown** confounders equally and eliminates selection bias.
- **Three sources of assessment bias:** subject variation, observer bias, bias in evaluation → countered by **blinding** (double commonest, triple ideal).
- **Independent variable** = intervention; **dependent variable** = outcome; **attrition** = losses to follow-up; primary analysis = **intention to treat**.
- **Six types of RCT:** clinical, preventive, risk-factor, cessation, aetiological agents, evaluation of health services. Indian example — **domiciliary versus sanatorium chemotherapy for tuberculosis**.

- **Indian regulatory step:** ethics committee approval + **CTRI registration** (launched 20 July 2007 by ICMR, WHO-linked; made mandatory by CDSCO on 15 June 2009 under the Drugs and Cosmetics Act); serious adverse events reported **within 24 hours** (ICMR 2017).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park's six-step list dates from the classical era of trial methodology, so add allocation concealment, prospective registration with the Clinical Trials Registry–India and CONSORT-style reporting (ICMR National Ethical Guidelines 2017) as one line after the six steps, not in place of them.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 3 — Principles of Epidemiology and Epidemiologic Methods: classification of studies p 82; experimental epidemiology, aims, animal and human experiments pp 100–101; the six steps of a randomized controlled trial pp 101–103; blinding p 103; types of randomized controlled trial and non-randomized trials pp 104–106); Gordis Epidemiology 7e (Ch 10 — Assessing Preventive and Therapeutic Interventions: Randomized Trials, masking and placebo p 224, intention-to-treat analysis p 227; Ch 11 — Randomized Trials: Some Further Issues, efficacy pp 239–240); ICMR National Ethical Guidelines for Biomedical and Health Research Involving Human Participants, 2017 (section 3.7 — Registration with Clinical Trials Registry–India; section 7.1 — clinical trials).



Framingham Heart Study

Asked in: Paper I 12-Nov-2021 — RGUHS MD Community Medicine

Definition — The Framingham Heart Study is the classical **prospective (concurrent) cohort study** of cardiovascular disease, initiated in **1948 by the United States Public Health Service** in the town of Framingham, Massachusetts (population 28,000 in 1948), to relate a set of suspected factors — serum cholesterol, blood pressure, body weight and cigarette smoking — to the subsequent development of cardiovascular disease (Park 28e, Ch 3, p 100). It is the study in which the term "**risk factor**" was introduced (Handbook of Epidemiology, Ch 34, p 1372), and it became the prototype of similar studies in the United States and other countries (Park 28e, Ch 3, p 100).

Objectives and the hypotheses tested

Framingham assembled its cohort from a **defined general population** identified by place of residence rather than by exposure (Park 28e, Ch 3, p 96) — Gordis's second method of selecting a cohort (Gordis 7e, Ch 8, p 199) — so that many exposures and many outcomes could be studied in the same cohort. The hypotheses set at inception, none of them established at the time, were (Gordis 7e, Ch 8, p 199):

- The incidence of coronary heart disease (CHD) increases with age, and occurs earlier and more frequently in males.
- Persons with hypertension develop CHD at a greater rate than normotensives.
- Elevated blood cholesterol is associated with an increased risk of CHD.

- ▶ Tobacco smoking and habitual use of alcohol are associated with increased incidence of CHD.
- ▶ Increased physical activity is associated with a decrease in the development of CHD.
- ▶ An increase in body weight predisposes to the development of CHD.
- ▶ CHD develops at an increased rate in patients with diabetes mellitus.

Study design and methodology

Element	Framingham Heart Study
Design	Prospective (concurrent) cohort study — "cause to effect", the epidemiologist observing only, without intervention (Park 28e, Ch 3, pp 96, 99-100)
Choice of community	A stable community of adequate size with little out-migration, willing to participate, not overtly different from other communities of the region, and with a health facility close enough for easy access — Framingham satisfied all these criteria (Park 28e, Ch 3, p 83; Gordis 7e, Ch 8, p 198); it also had a local community hospital and lay close to a large medical centre (Lilienfeld, Ch 9, pp 196-197)
Sampling frame	About 10,000 residents aged 30–59 years; a random sample of 6,507 of both sexes was invited (Park 28e, Ch 3, p 100; Gordis 7e, Ch 8, p 199)
Age limits and rationale	30–59 years — those below 30 would seldom manifest cardiovascular end-points within the planned follow-up, and many above this range would already have established coronary disease (Park 28e, Ch 3, p 100; Gordis 7e, Ch 8, p 198)
Exclusion at baseline	Persons with clinically evident CHD at the initial examination — the cohort must be free of the disease under study (Park 28e, Ch 3, pp 95, 100)
Exposures measured	Serum cholesterol, blood pressure, body weight, cigarette smoking; also age, sex, physical activity and diabetes (Park 28e, Ch 3, p 100; Gordis 7e, Ch 8, p 199)
Follow-up	Biennial (every 2 years) re-examination , planned for 20 years because heart disease develops slowly; continued for about 30 years and then into the offspring generation (Park 28e, Ch 3, p 100; Fletcher, Ch 5, p 81)
Outcome ascertainment	Biennial examination was the main source of follow-up information, supplemented by daily surveillance of hospital admissions at the town's only hospital (Gordis 7e, Ch 8, p 199) and by death certificate records (Park 28e, Ch 3, p 100)
Analysis	Incidence in exposed and non-exposed, relative risk and attributable risk; later, multivariable (multiple logistic) analysis to study several exposures and their interaction simultaneously (Park 28e, Ch 3, pp 98-99; Gordis 7e, Ch 8, p 199)

Principal findings

Finding	Detail
Major risk factors of CHD established	Elevated serum cholesterol, cigarette smoking, hypertension and sedentary habits — hence the four main possibilities of intervention: lowering serum cholesterol, cessation of smoking, control of hypertension, promotion of physical activity (Park 28e, Ch 6, p 443)
Predictors confirmed on repeated follow-up	The predictive value of serum lipids, blood pressure and cigarette smoking was repeatedly demonstrated; male sex, advancing age, diabetes mellitus, obesity, low vital capacity and certain electrocardiographic abnormalities were also identified as risk factors (Park 28e, Ch 3, p 100)

Finding	Detail
Serum cholesterol in middle age	Increasing risk of CHD with increasing serum cholesterol levels in the 45–54 year age group (Park 28e, Ch 3, p 100)
Smoking varies with the manifestation of disease	The association between smoking and CHD varied with the manifestation of the disease — smoking was more strongly associated with sudden death from CHD than with less fatal forms (Park 28e, Ch 3, p 100)
First landmark report (6 years of follow-up)	Serum cholesterol concentration, blood pressure and electrocardiographic evidence of left ventricular hypertrophy predicted the development of CHD (Handbook of Epidemiology, Ch 34, pp 1371-1372)
Gradient of risk with cholesterol	In men the relation was dose-related in both age groups over the first 12 years — relative risk rising from 1.0 to 4.1 across the cholesterol bands in men aged 30–49 years. In women the gradient was not monotonic over the same period (Gordis 7e, Ch 12, Table 12.7, p 258 — see chart)
Systolic versus diastolic blood pressure	Systolic blood pressure is the better predictor of disease outcome, although clinicians long favoured diastolic elevation (Handbook of Epidemiology, Ch 13, pp 513-514)
Obesity	A dramatic increase in sudden death among men more than 20 per cent overweight compared with those of normal weight (Park 28e, Ch 6, p 476)
Interaction of risk factors	Risk factors combine more than additively — the effect of smoking is greater in the presence of high total and low high-density lipoprotein (HDL) cholesterol, so that adding five risk factors to a 60-year-old man raises 10-year cardiovascular risk almost ten-fold (Park 28e, Ch 2, p 57; Fletcher, Ch 12, pp 213-214)

Contributions to epidemiology and public health

- ▶ **The concept and the term "risk factor."** The Framingham investigators coined "risk factor" to describe such predictive characteristics (Handbook of Epidemiology, Ch 34, p 1372); the concept gave a new dimension to epidemiological research (Park 28e, Ch 3, p 104).
- ▶ **Separation of modifiable from unmodifiable risk factors** — the modifiable are amenable to intervention, the unmodifiable act only as signals alerting the health professional (Park 28e, Ch 2, p 57); see the second flowchart for the two classes.
- ▶ **Multivariable risk prediction.** Age, sex, systolic blood pressure, total cholesterol, cigarette use and relative weight were entered into a multiple logistic regression model to predict the **12-year** future probability of a coronary event — myocardial infarction, sudden death or angina pectoris (Handbook of Epidemiology, Ch 32, pp 1302-1303). The later **Framingham Risk Score**, derived from the original and offspring cohorts, estimates risk from age, sex, HDL, low-density lipoprotein (LDL) and total cholesterol levels, smoking status, blood pressure and diabetes history (Handbook of Epidemiology, Ch 2, p 107), and is among the most-used risk-prediction tools in medicine (Fletcher, Ch 4, p 72).
- ▶ **Basis for the two preventive strategies** — the population (mass) strategy and the high-risk strategy for primary prevention of chronic disease (Park 28e, Ch 2, p 60), and the rationale for the risk-factor intervention trials (Park 28e, Ch 6, p 443); see the second flowchart.
- ▶ **A methodological prototype.** Its intensive biennial examination schedule, long-term continuity of follow-up and investigator involvement, and incorporation of new design components over its

decades-long history made it a uniquely rich source of data on individual cardiovascular risk and a reference example for later cohort studies (Handbook of Epidemiology, Ch 1, p 23); it became a prototype of similar studies in the United States and other countries (Park 28e, Ch 3, p 100), and its experience was combined with four other cohorts (Albany civil servants, Chicago Gas Company, Chicago Western Electric Company, Tecumseh) into the **United States National Cooperative Pooling Project** (Handbook of Epidemiology, Ch 34, p 1372).

Critical appraisal

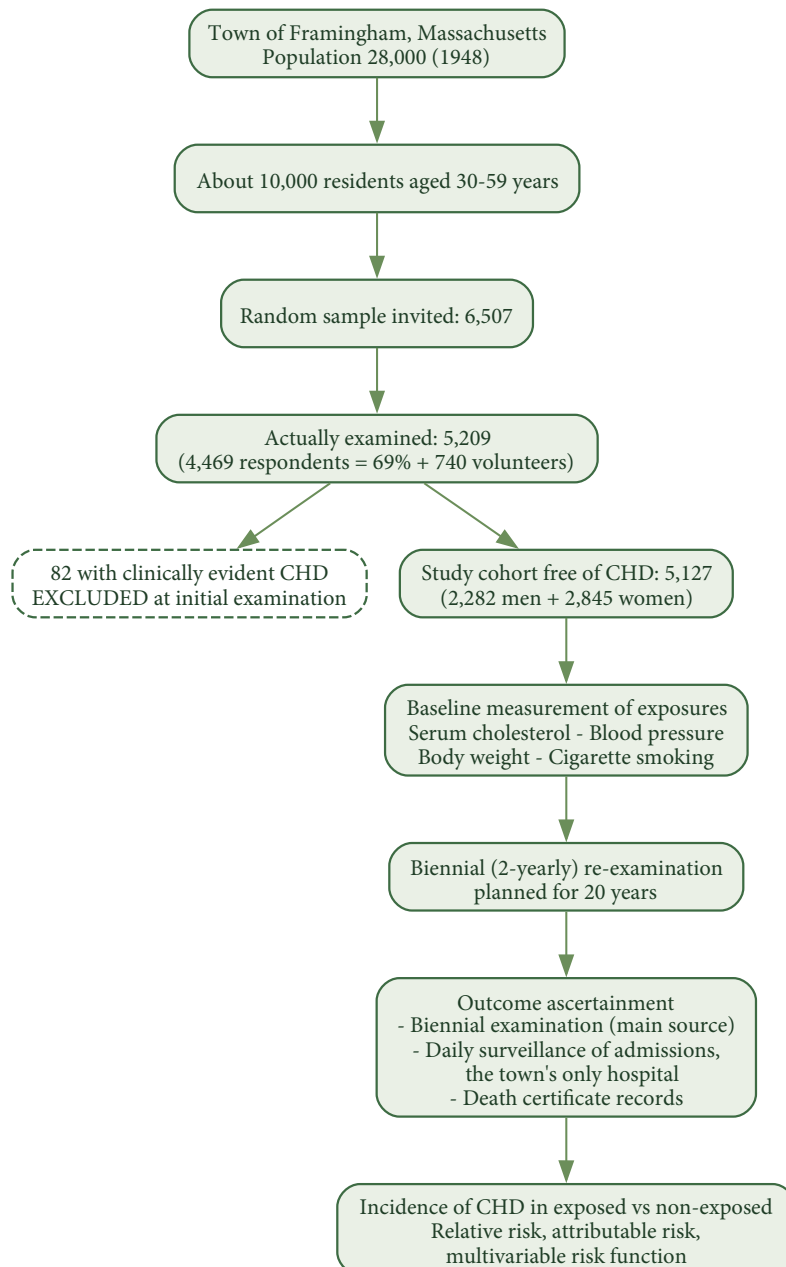
Strengths	Limitations
Incidence can be measured directly in exposed and non-exposed, giving relative risk and attributable risk (Park 28e, Ch 3, p 99)	Needs a very large number of people followed for a very long time; only 5 per cent of the Framingham cohort had a coronary event in the first 8 years (Park 28e, Ch 3, p 99; Fletcher, Ch 5, p 85)
A defined population permits study of many exposures and many outcomes , and of their interaction, in one cohort (Gordis 7e, Ch 8, p 199; Ch 9, p 210)	Very expensive, and results are delayed by decades (Park 28e, Ch 3, p 99; Fletcher, Ch 5, p 85)
Exposure recorded before disease occurs — recall bias and misclassification into exposed/non-exposed are minimised (Park 28e, Ch 3, p 99)	Loss to follow-up; as close to 95 per cent follow-up as possible is the recommended safeguard (Park 28e, Ch 3, p 98)
Standardised, repeated measurement improves validity of exposure data (Fletcher, Ch 5, pp 81-82)	Measurement error attenuates risk estimates — in Framingham data the logistic coefficient for systolic blood pressure was smaller with a single reading than with the average of two, and larger still after bias correction (Handbook of Epidemiology, Ch 18, pp 742-743)
Temporal sequence (cause before effect) is directly demonstrable (Park 28e, Ch 3, p 99)	The study itself may alter people's behaviour (Park 28e, Ch 3, p 99)
Long multigenerational follow-up, continued as the <i>Framingham Offspring Study</i> (Fletcher, Ch 5, p 81)	Generalisability limited — one small community in the north-eastern United States, and volunteers were added to the probability sample (Handbook of Epidemiology, Ch 34, pp 1371-1372; Gordis 7e, Ch 8, p 199)
Desirable when the exposure of interest is rare — a case-control design would not yield enough exposed persons (Gordis 7e, Ch 9, p 210)	Inappropriate for rare diseases , or diseases of low incidence in the population (Park 28e, Ch 3, p 99; Gordis 7e, Ch 9, p 210)

Application in India

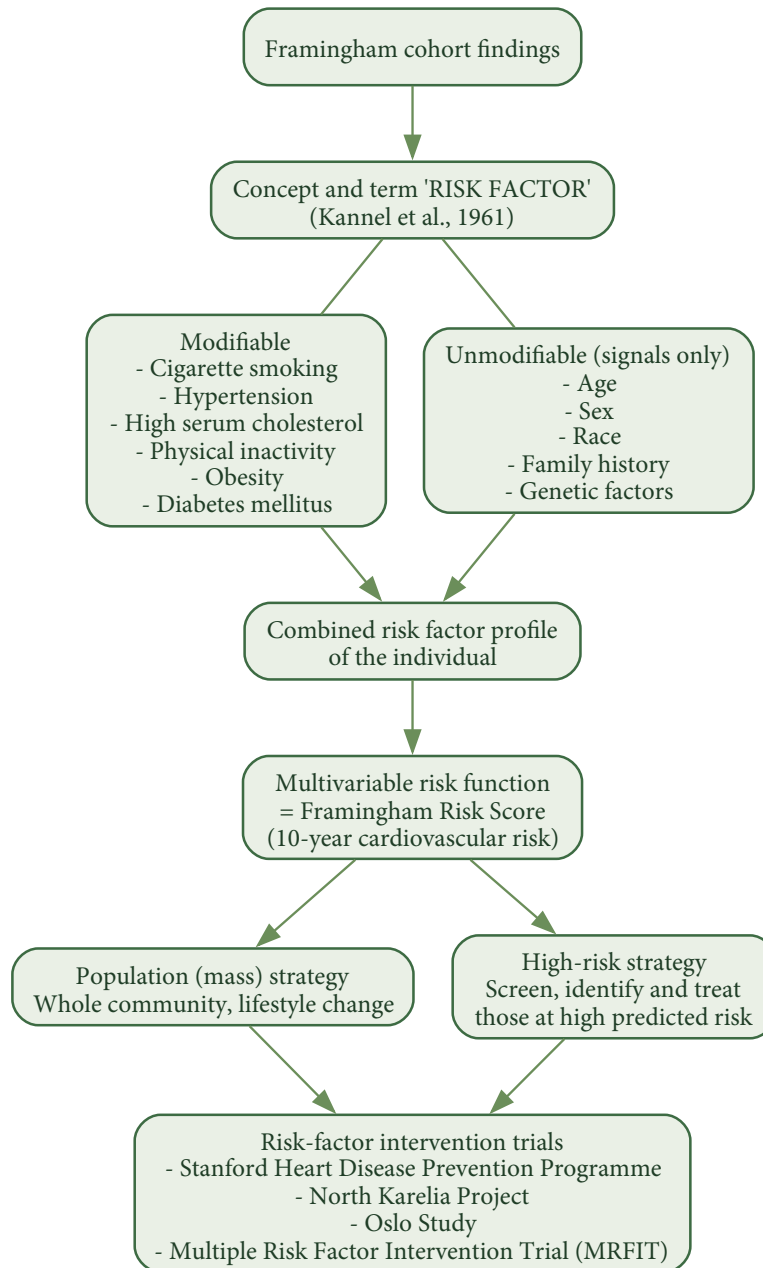
India's non-communicable disease programme applies the same Framingham logic through the **World Health Organization / International Society of Hypertension (WHO/ISH) risk prediction charts**, of which a set applicable to the Indian population is used. The charts give the 10-year risk of a fatal or non-fatal major cardiovascular event (myocardial infarction or stroke) from age, sex, systolic blood pressure, smoking status, total blood cholesterol and presence of diabetes, with a second set for settings where cholesterol cannot be measured (NCD Combined Manual for Medical Officers, pp 10-13).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- Draw: the cohort study design — a single box "Defined population of Framingham, free of CHD at entry" splitting into "Exposed (e.g. high serum cholesterol)" and "Not exposed", each followed forward in time to "Develops CHD" and "Does not develop CHD", with the arrow of time and the arrow "direction of enquiry" both pointing forward; mark the 2×2 cells a, b, c, d and write **incidence in exposed** = $a/(a+b)$, **incidence in non-exposed** = $c/(c+d)$.

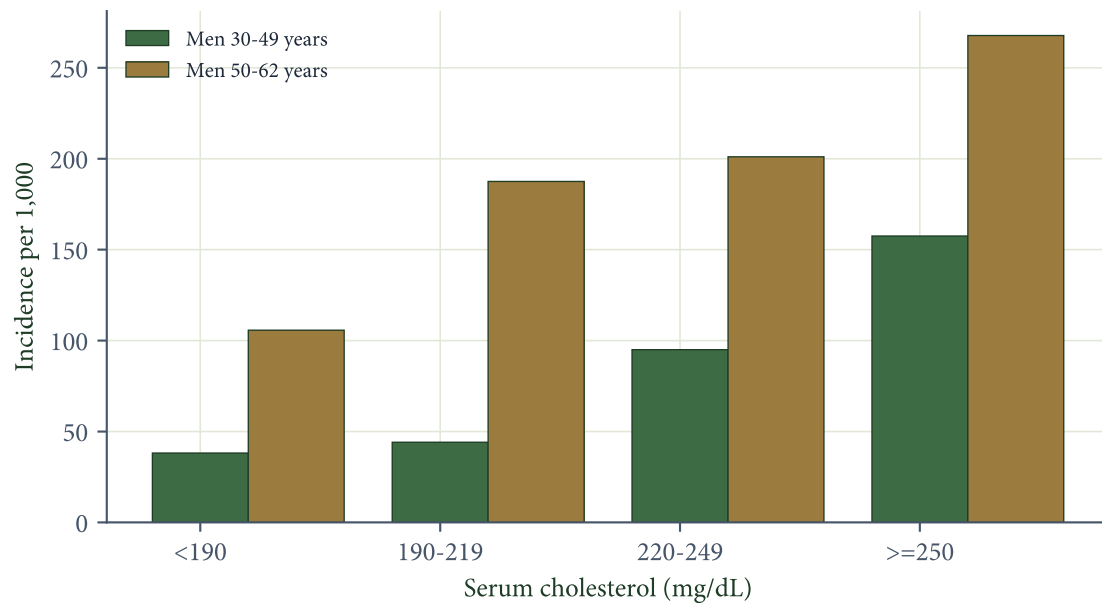


Derivation and follow-up of the Framingham Heart Study cohort, 1948



From the Framingham cohort to cardiovascular disease prevention

Serum cholesterol and 12-year incidence of coronary heart disease, Framingham m



Framingham Study, first 12 years; relative risk rises from 1.0 to 4.1 across cholesterol bands in men aged 30-49 years. The gradient was not consistent in women (Gordis Epidemiology 7e, Ch 12, Table 12.7, p 258, after Truett, Cornfield and Kannel, 1967).

High-yield exam pointers

- **Numbers to write:** started 1948, United States Public Health Service; town population 28,000; age 30–59 years; 6,507 invited; 4,469 respondents (69%) plus 740 volunteers = 5,209 examined; 82 with CHD excluded, cohort 5,127 (2,282 men + 2,845 women); **biennial** examination; planned for 20 years.
- **Relative risk** = incidence in exposed ÷ incidence in non-exposed; **attributable risk** = (incidence in exposed – incidence in non-exposed) ÷ incidence in exposed × 100.
- **The four intervention targets from Framingham:** lower serum cholesterol, stop smoking, control hypertension, promote physical activity (Park 28e).
- **Framingham gave the world the word "risk factor"** — and the Framingham Risk Score (age, sex, HDL, LDL and total cholesterol, smoking, blood pressure, diabetes → 10-year cardiovascular risk).
- **Systolic blood pressure** predicts outcome better than diastolic; smoking is more strongly linked to **sudden death** from CHD than to less fatal forms.
- **Indian counterpart in programme use:** WHO/ISH 10-year cardiovascular risk prediction charts for the Indian population, under the national non-communicable disease programme.
- **One-line criticism:** a single small north-eastern United States town plus volunteers — externally valid only with caution.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not written in the exam: the corpus differs slightly on two starting details — Park's Textbook 28e and Gordis give the year of initiation as 1948 while Fletcher and the Handbook of Epidemiology give 1949, and Park, Fletcher and the Handbook give the entry age band as 30–59 years while Gordis and Lilienfeld give 30–62 years; quote Park's figures (1948, 30–59 years) as the spine, since Park is the prescribed text (Lilienfeld, Ch 9, p 186). The Framingham Offspring Study is documented in the corpus only as the continuation into the children of the original cohort, without a start year, so do not quote one.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 2, pp 57, 60; Ch 3, pp 83, 95-100, 104; Ch 6, pp 443, 476); Gordis Epidemiology 7e (Ch 8, pp 196-199; Ch 9, p 210; Ch 12, p 258); Handbook of Epidemiology, Ahrens & Pigeot (Ch 1, p 23; Ch 2, p 107; Ch 13, pp 513-514; Ch 18, pp 742-743; Ch 32, pp 1302-1303; Ch 34, pp 1371-1372); Fletcher — Clinical Epidemiology: The Essentials (Ch 4, p 72; Ch 5, pp 80-85; Ch 12, pp 213-214); Lilienfeld's Foundations of Epidemiology (Ch 9, pp 186, 196-197); NCD Combined Manual for Medical Officers, Government of India (WHO/ISH risk prediction charts, pp 10-13).



Burden of coronary heart disease

Asked in: Paper II 14-Nov-2019 — RGUHS MD Community Medicine

Definition — Coronary heart disease (CHD, syn. ischaemic heart disease) is impairment of heart function due to inadequate blood flow to the heart compared with its needs, caused by changes in the coronary circulation; the World Health Organization (WHO) regards it as a modern "epidemic" affecting populations, not an unavoidable attribute of ageing (Park 28e, Ch 6, p 438). "Burden" means the total impact of the disease on a population — deaths, sickness, disability, lost healthy life-years and preventable fraction — and no single index captures it. Park therefore lists eight complementary measures, "each illustrating a different aspect of the picture" (Park 28e, Ch 6, pp 439-440).

1. Fix the case definition first — CHD is hard to count

Burden estimates are only as good as the case definition, because CHD presents in five ways of very unequal specificity, and two other cardiac diseases mimic it.

Presentation	Specific to CHD?	How it is captured in a burden study
Angina pectoris of effort	No	Standardised chest-pain questionnaire in a community survey
Myocardial infarction	Yes	Electrocardiogram (ECG) evidence of infarction; hospital record; death certificate

Presentation	Specific to CHD?	How it is captured in a burden study
Irregularities of the heart (arrhythmia)	No	Clinical record, ECG
Cardiac failure	No	Clinical record; often mis-assigned
Sudden death	No	Defined by the <i>International Society and Federation of Cardiology</i> as death "occurring instantly or within an estimated 24 hours of the onset of acute symptoms or signs"

- **Sources of diagnostic confusion:** rheumatic heart disease and cardiomyopathy (Park 28e, Ch 6, p 438).
- **Standard survey instrument:** prevalence is estimated from cross-sectional surveys using ECG for evidence of infarction plus a history of prolonged chest pain; the reference manual is *Cardiovascular Survey Methods* by Rose and Blackburn (Park 28e, Ch 6, p 439).
- **Natural history is variable** — death may occur in the first episode or after a long history — so incidence, prevalence and mortality can each move independently.

2. The eight epidemiological measures of CHD burden

#	Measure	Definition / formula	What it adds	Dated illustrative value
a	Proportional mortality ratio	Deaths from CHD ÷ all deaths in the same period × 100	Simplest measure; needs only cause-of-death data, no denominator population	CHD causes ~30% of deaths in men and ~25% in women in most western countries (Park 28e, Ch 6, p 439)
b	Loss of life expectancy	Average gain in life expectation if all cardiovascular deaths were eliminated, other mortality rates unchanged	Expresses burden as time lost, not as counts	Gain of 3.4 to 9.4 years for men, greater for women (Park 28e, Ch 6, p 439)
c	CHD incidence rate	Sum of fatal + non-fatal attack rates per 1,000 population at risk per year	The true measure of new disease and of causal factors	Accurate rates are difficult to compute because of the varied manifestations; mortality rates are used as a crude indicator of incidence
d	Age-specific death rates	Deaths from CHD in an age group ÷ mid-year population of that age group × 1,00,000	Essential when analysis is aimed at aetiology; a rise here suggests a true increase in incidence rather than better certification	India 2021: crude death rate from ischaemic heart disease 110.9 per 1,00,000 (Park 28e, Ch 6, pp 437-438)

#	Measure	Definition / formula	What it adds	Dated illustrative value
e	Prevalence rate	All current cases (old + new) ÷ population at risk at that point in time × 1,000	Measures the existing case-load for service planning; obtained by cross-sectional survey (ECG + prolonged chest pain history)	India: ischaemic heart disease 96.7 per 1,000 urban vs 27.1 per 1,000 rural adults on clinical + ECG criteria (Park 28e, Ch 6, pp 437-438)
f	Case fatality rate	Proportion of attacks that are fatal within 28 days of onset , × 100	Links mortality to morbidity; measures severity and quality of acute care	25% of heart-attack patients die suddenly; in ~55% of all cardiac deaths, death occurs within the first hour (Park 28e, Ch 6, p 439)
g	Levels of risk factors in the community	Population mean / prevalence of cigarette smoking, blood pressure, alcohol consumption and serum cholesterol	Measures the <i>future</i> burden already "banked" in the population (lag-time effect)	India 2019: tobacco use 24% (aged 15+), hypertension 31% (aged 30-79), mean salt 10 g/day, physical inactivity 49% (Park 28e, Ch 6, p 437)
h	Levels of medical care	Availability, coverage and quality of coronary care, secondary-prevention drugs (aspirin, beta-blockers, angiotensin-converting enzyme inhibitors, statins) and revascularisation in the community	Converts burden into a service-gap statement	WHO global target: at least 50% of eligible people to receive drug therapy and counselling (Park 28e, Ch 6, p 437)

Applying (a) to (f) to India in 2021 gives the scale of the problem: an estimated **2.78 million cardiovascular deaths**, of which **15,67,700 were from ischaemic heart disease**; globally CHD caused **9.02 million** of the **19.20 million cardiovascular deaths**, about **13.22 per cent** of all deaths worldwide (Park 28e, Ch 6, pp 437, 440). India also suffers the **highest loss of potentially productive years of life** of any country from cardiovascular deaths in the 35-64 year age band.

3. Composite (health-gap) measures — combining death with disability

Mortality alone understates CHD because survivors live with angina and heart failure; composite indices weight both.

Index	Formula / basis	Interpretation
Disability-adjusted life year (DALY)	DALY = YLL + YLD; developed by Harvard for the World Bank in 1990, adopted by WHO in 2000	One DALY = one lost year of healthy life
Years of life lost (YLL)	Deaths at each age × expected remaining years of life by a global standard life expectancy (Japanese life expectancy used as the standard)	Premature-mortality component

Index	Formula / basis	Interpretation
Years lost to disability (YLD)	Incident cases × average duration of disease × disability weight, on a scale from 0 (perfect health) to 1 (dead)	Non-fatal component
Health-adjusted life expectancy (HALE)	Life expectancy at birth adjusted for time spent in poor health	Equivalent years of full health a newborn can expect
Quality-adjusted life year (QALY)	Years of life added by an intervention × utility value (1 year in perfect health = 1 QALY; 1 year at utility 0.5 = 0.5 QALY)	Used to judge value-for-money of coronary interventions
Disability-free life expectancy (DFLE)	Average years expected to be lived free of disability at current mortality and disability patterns	Positive-health framing

All indices above are defined in Park 28e, Ch 2, p 38. Ischaemic heart disease was the **2nd leading global cause of DALYs in 2019** (180,847 thousand DALYs; 7.1% of all DALYs; 2,346 DALYs per 1,00,000) and again ranked **2nd in 2021, behind COVID-19** (Gordis 7e, Ch 4, Table 4.14, pp 105-106; Park 28e, Ch 2, p 65).

4. Attributable-risk approach — how much of the burden is preventable

Measure	Formula	Worked CHD example (smoking)
Attributable risk in the exposed	Incidence in exposed – incidence in unexposed	10.6 of the 28 per 1,000 incident cases in smokers
Attributable risk per cent (exposed)	(Above ÷ incidence in exposed) × 100	37.9% of CHD incidence among smokers
Population attributable risk (PAR)	Incidence in total population – incidence in unexposed	Excess CHD incidence in the whole community
PAR per cent	(PAR ÷ incidence in total population) × 100	21.3% of CHD incidence in the total population — the maximum reduction an ideal cessation programme could achieve
Levin's formula	PAR% = $p(r - 1) \div [p(r - 1) + 1]$, where p = proportion of population exposed and r = relative risk (or odds ratio)	Allows PAR to be estimated from case-control data

All formulas and both worked figures are from Gordis 7e, Ch 13, pp 275-277 and the chapter appendix, pp 280-281.

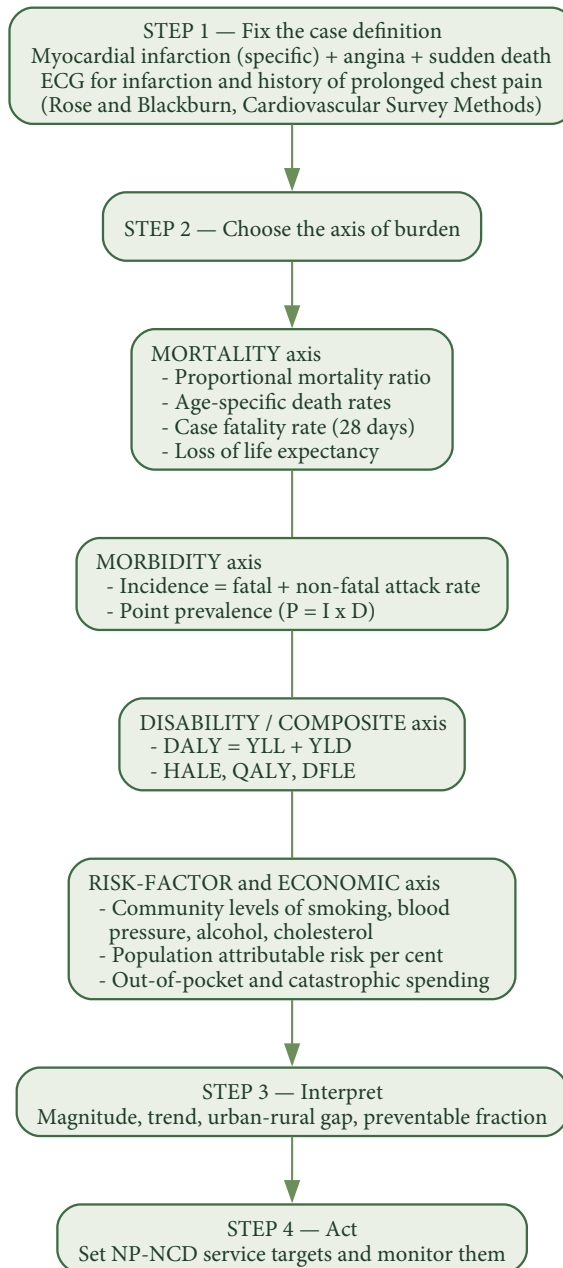
5. Sources of data for measuring CHD burden in India

Source	What it yields for CHD	Main limitation
Civil Registration System with Medical Certification of Cause of Death (MCCD) , ICD-10 coded	Cause-specific mortality; proportional mortality ratio. Urban MCCD attributes 25.1% of total deaths to diseases of the circulatory system (Park 28e, Ch 6, p 440)	Only ~22% of deaths were medically certified in 2020, and >75% of India's ~95 lakh annual deaths occur at home (Sunder Lal, Ch 8, p 535)
Sample Registration System (SRS) with verbal autopsy (RHIME) — the Million Death Study, 1998-2014	Nationally representative cause-of-death fractions where certification is absent	Verbal autopsy misclassifies sudden and unwitnessed cardiac deaths (Park 28e, Ch 9, p 663)
Community cross-sectional surveys (ECG + prolonged chest-pain history, per Rose and Blackburn)	Point prevalence; silent infarction; urban-rural differentials	Costly; single ECG has limited sensitivity
Hospital records	Acute coronary events, case fatality, duration of stay, cost of care	Only the "tip of the iceberg" — mild and subclinical cases never attend; admission policy varies (Park 28e, Ch 21, p 997)
Disease registers and record linkage	Duration of illness, case fatality, survival, follow-up; associations between diseases	Registers for CHD are not national in India; record linkage is data-heavy (Park 28e, Ch 21, pp 997-998)
WHO STEPS non-communicable disease risk-factor surveillance — Step 1 questionnaire (tobacco, alcohol, diet, physical activity), Step 2 physical measurements (blood pressure, anthropometry), Step 3 biochemical measurements (blood sugar, cholesterol)	Community levels of every modifiable CHD risk factor, comparable over time and place; integrated with the Integrated Disease Surveillance Programme	Measures risk, not disease (Park 28e, Ch 6, pp 436-437; Jugal Kishore, p 574)
National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Diseases and Stroke (NPCDCS / NP-NCD) population-based screening of those aged ≥ 30 years	Programme-generated prevalence of hypertension and diabetes; NCD clinic case-loads	Screened, not surveyed — selection bias (Park 28e, Ch 7, p 565)
Modelled estimates — Global Burden of Disease study and the ICMR India State-Level Disease Burden Initiative (2017)	State-wise DALYs, epidemiological transition level	Modelled from sparse primary data
Cohort and multicentre studies — the <i>Framingham Study</i> (since 1951) and the WHO <i>MONICA</i> project (multinational monitoring of trends and determinants in cardiovascular diseases; 41 centres in 26 countries, ending 1994)	True incidence, attack rates, trends and their determinants	Expensive, long, few Indian equivalents (Park 28e, Ch 6, pp 440, 443)

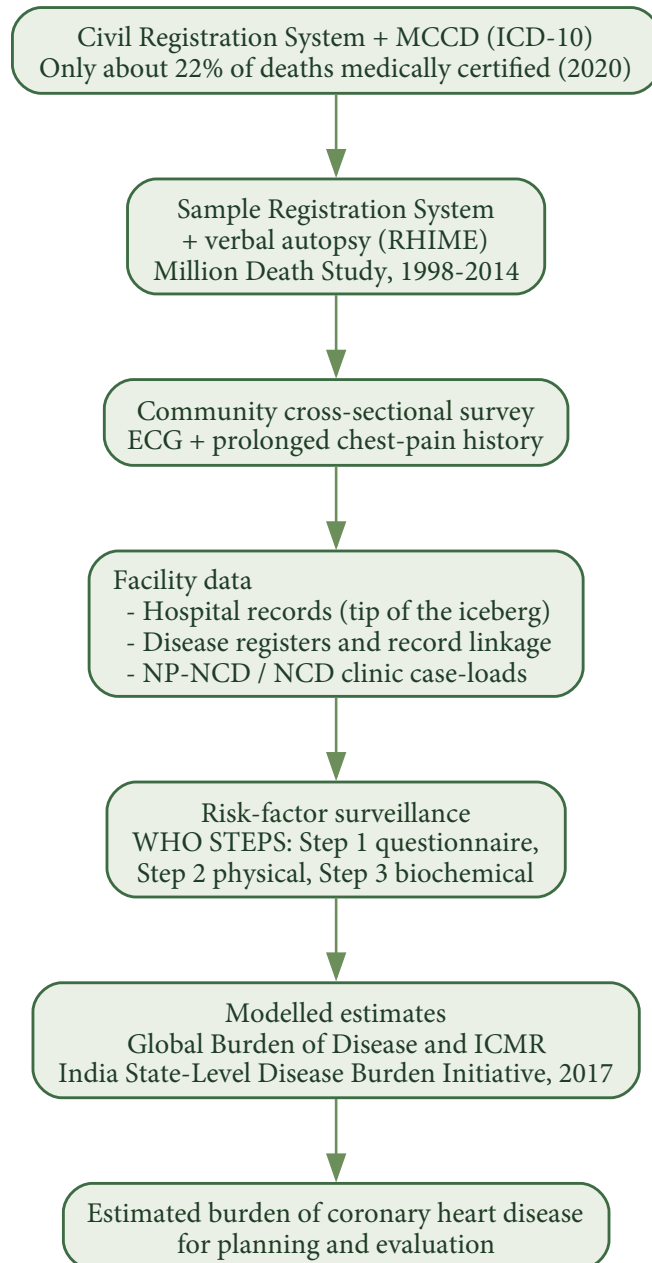
DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Figure — "Iceberg of coronary heart disease" (draw a waterline across an iceberg and label as below):

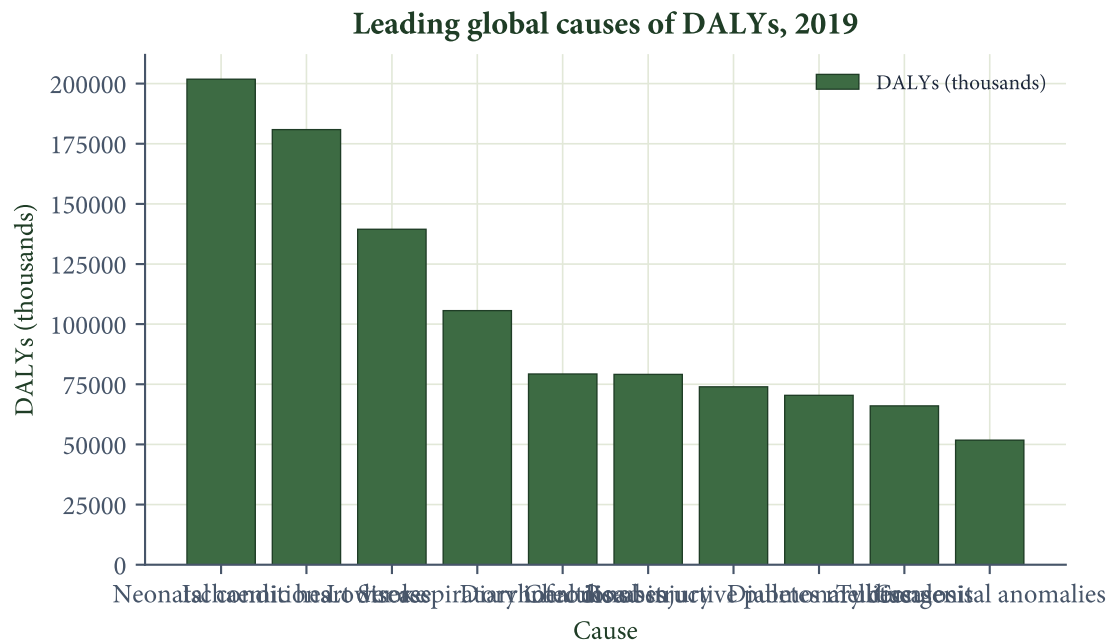
- ▶ Above the waterline (what routine data capture): CHD deaths certified under MCCD/ICD-10; hospitalised acute coronary events
- ▶ The waterline: "routine mortality and hospital data stop here"
- ▶ Below the waterline (only surveys reach these): symptomatic angina in the community; silent / unrecognised myocardial infarction (ECG-positive only); asymptomatic coronary atherosclerosis; population carrying risk factors — smoking, raised blood pressure, raised serum cholesterol, diabetes, obesity, physical inactivity
- ▶ Side arrows: label the top "proportional mortality ratio, age-specific death rates, case fatality rate" and the bottom "prevalence survey, WHO STEPS risk-factor survey"



Sequence for measuring the burden of coronary heart disease in a population



Ladder of data sources for coronary heart disease burden in India, from routine records to modelled estimates



Ischaemic heart disease ranks second, contributing 7.1 per cent of all disability-adjusted life years (2,346 DALYs per 1,00,000 population). Source: WHO Global Health Estimates 2019, reproduced in Gordis Epidemiology 7e, Ch 4, Table 4.14, pp 105-106.

High-yield exam pointers

- ▶ **Mnemonic for Park's eight measures, in his order — "P-L-I-A-P-C-R-M":** Proportional mortality ratio, Loss of life expectancy, Incidence rate, Age-specific death rates, Prevalence rate, Case fatality rate, Risk-factor levels, Medical care levels.
- ▶ **Write the two definitional cut-offs exactly:** case fatality rate = attacks fatal **within 28 days** of onset; sudden death = death **instantly or within an estimated 24 hours** of onset of acute symptoms or signs.
- ▶ **DALY = YLL + YLD;** one DALY = one lost year of healthy life; disability weight runs 0 (perfect health) to 1 (dead).
- ▶ **$P = I \times D$** — prevalence rises with duration, so improved survival after myocardial infarction *raises* CHD prevalence while incidence falls.
- ▶ **$PAR\% = \frac{p(r-1)}{[p(r-1)+1]}$** (*Levin's formula*); smoking accounts for 21.3% of CHD in the total population and 37.9% among smokers (Gordis 7e, Ch 13).
- ▶ **Dated Indian figures:** ischaemic heart disease deaths 15,67,700 in 2021, crude death rate 110.9 per 1,00,000; prevalence 96.7 per 1,000 urban vs 27.1 per 1,000 rural adults; only ~22% of deaths medically certified (2020).
- ▶ **Incidence is the hard one:** accurate CHD incidence rates cannot be computed directly, so mortality rates are used as a crude proxy.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The programme is now written as the National Programme for Prevention and Control of Non-Communicable Diseases (NP-NCD), formerly the National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Diseases and Stroke (NPCDCS); the electrocardiogram coding scheme used in the Rose and Blackburn survey manual is conventionally the Minnesota code, which the retrieved Park text does not name explicitly, so write it only as "standardised ECG coding" if you wish to stay strictly with the textbook.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, NFHS-5, ICMR-INDIAB, India Civil Registration System, WHO HEARTS, Global Burden of Disease 2023

Programme and policy updates (India)

- **The programme has been renamed — write the new name.** NPCDCS (launched 2010) became the **National Programme for Prevention and Control of Non-Communicable Diseases (NP-NCD) in 2023**, and hypertension screening now runs largely through Health and Wellness Centres created under the National Health Policy 2017. Repeat India Hypertension Control Initiative (IHCI) surveys in nine districts of five states show care-seeking moving to the public sector between 2018-19 and 2023-24 (20% to 32%), with blood-pressure control among those treated rising 37% to 48% in government facilities, drug stocks reported as always available rising 72% to 81%, and the proportion paying for medicines falling 47% to 9% [PMID 41146104].
- **Programme cohorts now supply a national figure for Park's measure (h), levels of medical care.** The IHCI cohort covered 12,35,453 registered adults aged ≥30 years across 89 districts in 15 states, of whom 44% under care had controlled blood pressure. The measurement lesson matters more than the number: control fell from **68% on a single recent reading to 52% when two consecutive visits were required** — the case definition, not the disease, moved the estimate [PMID 41314840].

Current national data

- **ICMR-INDIAB-25 is the current national risk-factor survey for lipids** — stratified multistage cluster sample of 1,21,078 adults aged ≥20 years across all states and union territories, with fasting venous lipids in 23,665. Weighted dyslipidaemia prevalence was 87.3%, driven mainly by low HDL-cholesterol (66.8%), and it was present in 40% of people with no diabetes, hypertension or obesity. Quote this in place of older lipid figures under "levels of risk factors in the community" [PMID 42547322].
- **The composite measures now have Indian sex-specific estimates.** Combining the Sample Registration System with the Longitudinal Ageing Study in India (LASI Wave 1), **multimorbidity-free life expectancy at ages 45-49 was 25.50 years in women vs 23.30 years in men**, yet women spent far longer with complex multimorbidity (10.77 vs 8.93 years) and their complex-multimorbidity-free life expectancy was barely better (20.83 vs 20.22 years). Cardiovascular disease was the **strongest single negative contributor across all ages to the female-male gap in disease-free life expectancy**, most pronounced at ages 45-49 (-0.36 years) — a direct illustration of why mortality alone understates CHD burden [PMID 41409690].

Elimination, surveillance and disease status

- ▶ **India's 2021 Civil Registration System report, released only in May 2025, resets the mortality denominator.** It recorded 1.02 crore registered deaths in 2021, a 26% rise over 2020, against 3,35,004 reported COVID-19 deaths — about 24 lakh excess deaths, a 7.2-fold excess-to-COVID-death ratio. Reporting fidelity varied enormously by state — only six states had ratios below 2-fold, while Gujarat exceeded 40-fold (about 2,51,000 excess deaths against 5,800 reported COVID-19 deaths). The same report notes 47.3% of the deceased received no medical attention at death (34.5% in 2019), which is precisely why cause-specific CHD mortality stays weakly certified; the Sample Registration System has historically found the civil system capturing only 75-80% of deaths [PMID 41232123].
- ▶ **Under-registration of deaths is sex-patterned, so CHD mortality by sex is biased at source.** NFHS-5 recorded death registration of 73% for males vs 64% for females, widening to 57% vs 44% in the poorest wealth quintile and narrowing to 89% vs 85% in the richest. Female sex independently reduced the odds of registration (OR 0.65 at 15-59 years; OR 0.58 at ≥60 years), as did rural residence (OR 0.78 and 0.65) [PMID 41296742].

Guideline and legislative updates

- ▶ **Registration of Births and Deaths (Amendment) Act, 2023** made the birth certificate mandatory for a host of services, and the states that designate community and primary health centres as statutory *Registrars* of births and deaths — Delhi, Uttar Pradesh, Manipur, Uttarakhand, Bihar — are largely the same states that recorded the biggest gains in registration. Registration of children delivered in CHCs/PHCs rose from 80.8% to 90.2% between NFHS-4 and NFHS-5 (the largest rise of any delivery setting), and such births had 38% higher adjusted odds of being registered than births in private hospitals [PMID 41752339].
- ▶ **Attributable-risk arithmetic is now used to cost legislation.** An India-specific microsimulation attributes about 9,22,000 deaths a year to combustible tobacco and 1,65,000 to smokeless tobacco, and estimates that a 50% tobacco tax rise would avert 1.8 million deaths (~9.8% of tobacco-attributable deaths) and 66.8 million healthy life-years lost over ten years. This is the population-attributable-risk method of section 4 applied to a real policy instrument [PMID 41269237].

Global and WHO developments

- ▶ **Global Burden of Disease 2023 supersedes the GBD 2019 and 2021 figures carried in textbooks.** Across the G20 in 2023 there were 8.52 million incident ischaemic heart disease cases and 5.81 million deaths; age-standardised incidence and mortality fell (estimated annual percentage change -1.61 and -1.49), but India and China carry the largest absolute burden, incidence is rising in women aged 15-49, and high systolic blood pressure remains the dominant modifiable risk, with particulate matter and lead as leading environmental contributors [PMID 42552754].
- ▶ **WHO HEARTS supplies the current service-coverage benchmark for measure (h).** By December 2024 about 34 million people had been started on antihypertensive treatment at more than 2,20,000 primary-care facilities across 38 countries, yet only about 1 in 5 people with hypertension worldwide have controlled blood pressure. The programme names the blood-pressure control rate as the critical metric and treats 50% community-level control as

achievable — cite this alongside the WHO 50% drug-therapy-and-counselling target [PMID 41338844].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 2 (indicators of health, DALY/HALE/QALY, pp 37-38, 65), Ch 3 (measurement of mortality and morbidity, incidence and prevalence, pp 76-82), Ch 6 (cardiovascular diseases and coronary heart disease, pp 433-443), Ch 7 (NPCDCS and NCD targets, pp 565-566), Ch 9 (verbal autopsy and RHIME, p 663), Ch 21 (hospital records, disease registers, record linkage, pp 996-998); Gordis Epidemiology 7e — Ch 4 (mortality and other measures of disease impact, Global Health Estimates 2019 DALY table, pp 105-107) and Ch 13 (attributable risk, population attributable risk and Levin's formula, pp 275-281); Sunder Lal — Textbook of Community Medicine (medical certification of cause of death and the Million Death Study, Ch 8, p 535; sources of vital statistics, Ch 9, p 557); Jugal Kishore — National Health Programmes of India (assessment of the burden of non-communicable diseases by the WHO STEPS approach, p 574); WHO Global Health Estimates 2019 and 2021; ICMR India State-Level Disease Burden Initiative, 2017.

Recent-advances sources: 41146104 — Changing patterns in health-seeking behaviour and prescription practices for treatment of hypertension in nine districts: insights from India Hypertension Control Initiative, 2018-19 and 2023-24, BMC Public Health 2025, doi:[10.1186/s12889-025-24898-0](https://doi.org/10.1186/s12889-025-24898-0); 41314840 — Scaling India Hypertension Control Initiative strategies to 15 states: treatment outcomes and risk factors for uncontrolled blood pressure, India: a cohort study, 2018-2021, BMJ Open 2025, doi:[10.1136/bmjopen-2025-106372](https://doi.org/10.1136/bmjopen-2025-106372); 42547322 — Prevalence of dyslipidemia in India: The ICMR-INDIAB National Study (ICMR-INDIAB-25), Journal of Clinical Lipidology 2026, doi:[10.1016/j.jacl.2026.07.011](https://doi.org/10.1016/j.jacl.2026.07.011) (abstract-only); 41409690 — Outliving and outlasting: explaining sex inequities in healthy life expectancy through the lens of multimorbidity and within morbidities in older populations in India, Frontiers in Public Health 2025, doi:[10.3389/fpubh.2025.1613516](https://doi.org/10.3389/fpubh.2025.1613516); 41232123 — India's death toll in 2021 during the COVID-19 pandemic: insights from delayed official civil registration data, Journal of Global Health 2025, doi:[10.7189/jogh.15.03045](https://doi.org/10.7189/jogh.15.03045); 41296742 — Who gets counted? Understanding low female death registration in India, PLoS One 2025, doi:[10.1371/journal.pone.0337224](https://doi.org/10.1371/journal.pone.0337224); 41752339 — Strengthening Civil Registration Through Grassroots Health Institutions in India, International Journal of Environmental Research and Public Health 2026, doi:[10.3390/ijerph23020257](https://doi.org/10.3390/ijerph23020257); 41269237 — Estimating the Expected Health and Economic Benefits of Price Increases Via Taxation to Counter the Dual Combustible and Smokeless Tobacco Epidemics in India, Nicotine & Tobacco Research 2026, doi:[10.1093/ntr/ntaf242](https://doi.org/10.1093/ntr/ntaf242) (abstract-only); 42552754 — Ischemic Heart Disease in G20 Countries: Forecasted Burden and Regional Disparities Based on GBD 2023 Estimates, Catheterization and Cardiovascular Interventions 2026, doi:[10.1002/ccd.70783](https://doi.org/10.1002/ccd.70783) (abstract-only); 41338844 — Lessons Learned From Treating 34 Million People With Hypertension: The Global HEARTS Initiative, Journal of the American College of Cardiology 2025, doi:[10.1016/j.jacc.2025.09.324](https://doi.org/10.1016/j.jacc.2025.09.324) (abstract-only). Also cited: Registration of Births and Deaths (Amendment) Act, 2023; Civil Registration System of India, 2021 report (released May 2025); National Family Health Survey 5 (2019-21); WHO HEARTS technical package; Global Burden of Disease Study 2023.



Investigation of a waterborne outbreak

Asked in: Paper II 28-May-2025 — RGUHS MD Community Medicine

Definition — An **epidemic** is "the occurrence in a community or region of cases of an illness or other health-related events clearly in excess of normal expectancy", the community, region and time period being specified precisely (Park 28e, Ch 3, p 84). A **waterborne outbreak** is such an excess in which the common vehicle of transmission is water — a piped supply, well, tank, tanker or bottled water. It is investigated by the standard epidemic-investigation sequence, with the **water supply itself added as the object of environmental study**. Investigation calls for inference as well as description; teams are often summoned after the peak has passed, so the exercise is largely retrospective, and while no "cook-book" fits every situation, the orderly sequence below applies to almost any epidemic, several steps running concurrently (Park 28e, Ch 3, p 159).

1. Objectives of the investigation

- ▶ To **define the magnitude** of the outbreak in terms of time, place and person.
- ▶ To determine the **particular conditions and factors** responsible for its occurrence.
- ▶ To identify the **cause, source(s) of infection and modes of transmission**, so as to decide the measures necessary to control the epidemic.
- ▶ To make **recommendations to prevent recurrence** (Park 28e, Ch 3, p 159).

2. The ten steps, applied to water

#	Step (Park 28e, Ch 3, pp 159-161)	What is done	Waterborne application
1	Verification of diagnosis	The first step — the report may be false or based on a mistaken diagnosis; confirm clinically and in the laboratory	Stool or rectal swab in Venkatraman–Ramakrishnan medium, alkaline peptone water or Cary–Blair; <i>V. cholerae</i> O1/O139 by culture or polymerase chain reaction (PCR); an immunochromatographic dipstick rapid diagnostic test (RDT) helps at the bedside, but confirmation is by culture or PCR (Park 28e, Ch 5, pp 290, 292)
2	Confirmation of the existence of an epidemic	Compare disease frequency with the same period of previous years; an epidemic exists when observed frequency exceeds expected frequency. The epidemic threshold is arbitrarily set at two standard errors from the endemic occurrence	Often self-evident: common-source epidemics of cholera, food poisoning and hepatitis A are easily recognised without such comparison

#	Step (Park 28e, Ch 3, pp 159-161)	What is done	Waterborne application
3	Defining the population at risk	(a) <i>Map</i> — obtain or prepare a detailed current map with landmarks, roads and every dwelling unit; divide into segments by natural boundaries, number the houses. (b) <i>Count</i> — house-to-house census by age and sex; without a denominator of "population at risk", attack rates cannot be calculated	Overlay each household's water source (tap, well, tank, tanker, bottled) plus the mains, valves, overhead tanks and sewer alignment on the same map
4	Rapid search for all cases and their characteristics	(a) <i>Medical survey</i> of the defined area to pick up cases who never sought care and those exposed to risk. (b) <i>Epidemiological case sheet</i> ("case interview form") designed after a rapid preliminary inquiry — name, age, sex, occupation, social class, travel, previous exposure, time of onset, signs and symptoms; lay health workers may be trained to administer it. (c) <i>Searching for more cases</i> — ask each patient about cases in home, family, neighbourhood, school and workplace with onset within one incubation period of the index case; include hospital admissions; repeat daily until the area is declared free, i.e. twice the incubation period after the last case	The case sheet must additionally record drinking-water source, storage vessel, household treatment (boiling, filtering, chlorine tablet), meals eaten outside, and any known pipe leak or sewage mixing
5	Data analysis	Ongoing analysis by time, place and person ; if the agent is known, rearrange into the agent–host–environment model (see section 3)	Water-source-specific attack rates are the equivalent of the food-specific attack rates used in foodborne outbreaks
6	Formulation of hypothesis	From the time–place–person distribution, frame hypotheses on (a) possible source, (b) causative agent, (c) possible modes of spread, (d) environmental factors that enabled the epidemic; rank them in order of likelihood	e.g. "the epidemic is due to <i>V. cholerae</i> O1 entering the distribution main through a leaking joint adjacent to a sewer in ward X"
7	Testing of hypothesis	Weigh all reasonable hypotheses by comparing attack rates in those exposed and not exposed to each suspected factor, to find the one consistent with all known facts	Compare attack rates between households on the suspect main and on other mains, and between users and non-users of the suspect well

#	Step (Park 28e, Ch 3, pp 159-161)	What is done	Waterborne application
8	Evaluation of ecological factors	Study circumstances that made the epidemic possible — sanitary status of eating establishments, water and milk supply, breakdown in the water supply system , population movements, temperature, humidity, air pollution, insect and animal reservoir dynamics; the outbreak may be studied in a case-control fashion	This is where the sanitary survey and water sampling of section 4 belong
9	Further investigation of the population at risk	Study the population at risk or a sample — medical examination, screening tests, examination of suspected food, faeces or blood samples, biochemical studies, assessment of immunity status; serological study reveals clinically inapparent cases; healthy persons from the same universe are studied in case-control fashion, classifying everyone by exposure to specific potential vehicles and whether ill or not	Stool survey of contacts and food handlers; serological survey for hepatitis A/E where these are suspected
10	Writing the report	The report must be complete and convincing ; sections include background (geographical location, climatic conditions), interpretation of data (comprehensive picture of the outbreak, hypotheses as to cause(s), formulation and testing of hypotheses by statistical analysis), control measures (strategies and methodology of implementation, constraints, results) and evaluation (significance of results) (<i>Park 28e, Ch 3, Table 47, pp 160-161</i>)	Submit through the district surveillance officer under the Integrated Disease Surveillance Programme (IDSP)

3. Analysis by time, place and person

Axis	What is prepared	What it yields
Time	Epidemic curve — chronological distribution of dates of onset	Time relationship with exposure to the suspected source; whether common-source or propagated; seasonal or cyclic pattern suggestive of a particular infection
Place	Spot map (geographic distribution) of cases and their relation to possible sources — water supply, air pollution, foods eaten, occupation	Clustering indicates a common source; this was how <i>John Snow</i> incriminated the Broad Street pump in the Golden Square district of London, 1854 (<i>Park 28e, Ch 3, pp 86-87</i>)
Person	Analysis by age, sex, occupation and other risk factors; attack rates and case fatality rates for exposed and non-exposed	Identifies the exposure common to cases and the group at highest risk

Rates to compute (*Park 28e, Ch 3, p 81*):

Rate	Formula
Attack rate	(Number of new cases of a specified disease during a specified time interval ÷ Total population at risk during the same interval) × 100
Secondary attack rate	Number of exposed persons developing the disease within the range of the incubation period following exposure to a primary case
Source-specific attack rate	Attack rate computed separately for users and non-users of each water source, and compared

Interpreting the curve — the features of a **point-source (common-source, single-exposure) epidemic** are: (i) the epidemic curve **rises and falls rapidly with no secondary waves**; (ii) the epidemic is **explosive**, with clustering of cases within a narrow interval of time; and (iii) **all cases develop within one incubation period** of the disease. A useful landmark is the **median incubation period**, the time for 50% of cases to occur after exposure (*Park 28e, Ch 3, p 84*). Waterborne cholera classically shows a **hybrid curve** — it begins from a common source, reaches a sharp peak, then tails off gradually over a longer period as it continues by propagated person-to-person spread (*Park 28e, Ch 3, pp 84-85*).

4. Investigating the water supply itself

Component	Action	Standard / interpretation
Sanitary survey	Trace the entire system — source, treatment works, storage tanks, mains, service pipes, household taps and vessels	Look for corrosion of pipe lines, leaky joints and cross connections between water supply pipes and sewage drainage pipes ; low water pressure increases infiltration of pollutants. Surveillance must be exercised at <i>every point</i> in the distribution system (<i>Park 28e, Ch 15, p 865; Ch 17, p 953</i>)
Sampling for bacteriology	Sterilised neutral-glass bottle of 200–250 ml with ground-glass stopper; add sodium thiosulphate (0.1 ml of 3% solution, or a small crystal) before sterilisation if the water may contain chlorine; run a tap in regular use to waste for 2 minutes ; sterilise taps not in regular use	Begin examination as soon as possible; otherwise keep in ice and analyse within 48 hours — samples not so preserved must be rejected. Send with date and time of collection and despatch, source of water, recent rainfall and sanitary survey findings (<i>Park 28e, Ch 15, p 884</i>)
Sampling for chemistry	Clean glass-stoppered neutral-glass ("Winchester Quart") bottles of not less than 2 litres , rinsed three times with the sample water	(<i>Park 28e, Ch 15, p 884</i>)
Sampling for cholera	1–3 litres of suspect water in sterile bottles (for the filter method), or 9 volumes of sample water added to 1 volume of 10% peptone water , despatched by the quickest transport	(<i>Park 28e, Ch 5, p 290</i>)

Component	Action	Standard / interpretation
Bacteriological tests	Presumptive coliform test (multiple tube method, giving the most probable number, MPN); confirmatory test by subculture in brilliant green bile broth at 37°C for up to 48 hours (needed for chlorinated water); tests for faecal streptococci and <i>C. perfringens</i> ; colony (plate) count	(Park 28e, Ch 15, p 881)
Chemical surveillance	pH, colour, turbidity, chlorides, ammonia, chlorine demand and residual chlorine	Regular measurement of chlorine residuals in the supply may in part replace bacteriological surveillance (Park 28e, Ch 15, p 881)

Indicator organisms and what each means (Park 28e, Ch 15, p 876):

Indicator	Significance
Coliform group (faecal — <i>E. coli</i> ; non-faecal — <i>Klebsiella aerogenes</i>)	Coliforms have greater resistance to natural purification than waterborne pathogens, so their presence implies the probable presence of intestinal pathogens
<i>E. coli</i> / thermotolerant coliform	The precise indicator of faecal pollution — must not be detectable in any 100 ml sample (Park 28e, Ch 15, p 877)
Faecal streptococci	Occur in faeces in much smaller numbers than <i>E. coli</i> ; in doubtful cases their finding is important confirmatory evidence of recent faecal pollution ; useful after laying new mains or repairs
<i>Clostridium perfringens</i>	Spores survive longer than coliforms and resist chlorination at usual doses ; presence in the absence of coliforms suggests faecal contamination at some remote time , and in filtered supplies indicates deficient filtration

5. Control measures instituted alongside the investigation

- **Notification** — cholera is notifiable locally and nationally; under the International Health Regulations 2005 it is no longer mandatorily notifiable to the World Health Organization, but the event is assessed against the IHR Annex 2 decision instrument. Cases and deaths are reported daily and weekly until the area is declared free (Park 28e, Ch 5, pp 290-292).
- **Early case-finding and treatment** — aggressive search for mild, moderate and severe cases; establish **treatment centres** without delay, rehydration being the mainstay and antibiotics adjuncts (Park 28e, Ch 5, p 291).
- **Interrupt the vehicle immediately** — stop the suspect supply, issue a boil-water advisory, arrange safe tankered water, repair the leak or cross-connection.
- **Disinfection** — chlorinate to a **free residual chlorine of 0.5 mg/litre at the end of one hour's contact**; the correct dose is the **chlorine demand of that water plus 0.5 mg/L free residual**. Chlorine does **not act on spores, protozoal cysts or helminthic ova** except in higher doses, and turbid or highly polluted water is unsuited to direct chlorination (Park 28e, Ch 15, pp 870-871).
- **Disinfection of a well** — volume = $3.14 \times d^2 \times h \div 4$ (m³; 1 m³ = 1,000 litres); chlorine demand by *Horrock's apparatus*; roughly **2.5 g of good-quality bleaching powder per 1,000 litres**, an applied dose of about **0.7 mg chlorine per litre**. Paste not more than 100 g per bucket, let the

lime settle and use only the supernatant chlorine solution (lime raises hardness), then lower the bucket below the water surface and agitate (*Park 28e, Ch 15, pp 872-873*).

- ▶ **Sanitation and health education** — excreta disposal, hand hygiene, food safety, safe household storage, community engagement.
- ▶ **Declaring the outbreak over** — an area is declared free of cholera when **twice the incubation period (10 days)** has elapsed since the death, recovery or isolation of the last case (*Park 28e, Ch 5, p 291*).
- ▶ **Reporting channel in India** — under IDSP, weekly data flow on the '**S**' form (health worker), '**P**' form (clinician) and '**L**' form (clinical laboratory); a rising trend is investigated by the medical officer or the **rapid response team (RRT)** constituted and deputed by the district surveillance unit. Response is graded by trigger: 1 local, 2 district (district surveillance officer and RRT), 3 state, 4 national, 5 international (*Jugal Kishore, National Health Programmes of India, pp 488-491*).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Diagram 1 — Epidemic curve of a point-source waterborne outbreak.** Labels to write: *x-axis — date/hour of onset; y-axis — number of cases; arrow marking single exposure; arrow marking the minimum, median and maximum incubation period; single sharp peak, rapid rise and fall, no secondary waves; all cases within one incubation period; dotted tail marked "propagated (secondary person-to-person) spread"*.
- ▶ **Diagram 2 — Spot map of the affected locality.** Labels to write: *streets and house numbers; one dot per case; symbol for each water source — piped tap / well / overhead tank / tanker; the suspect source ringed; sewer line; boundary of equal distance between the suspect source and other sources; legend*.

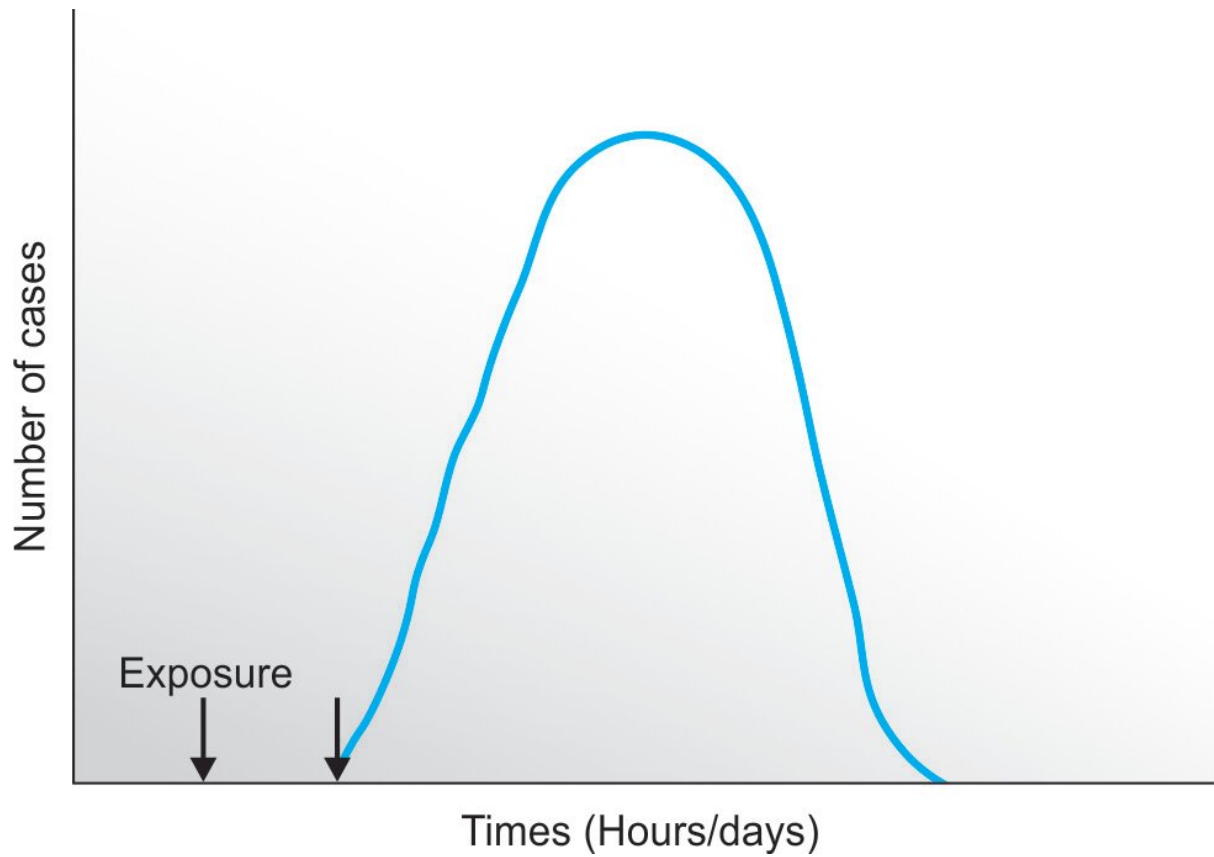


Fig 18.3 — Epidemic curve (Point source) (Suryakantha Environment Best, p.268)

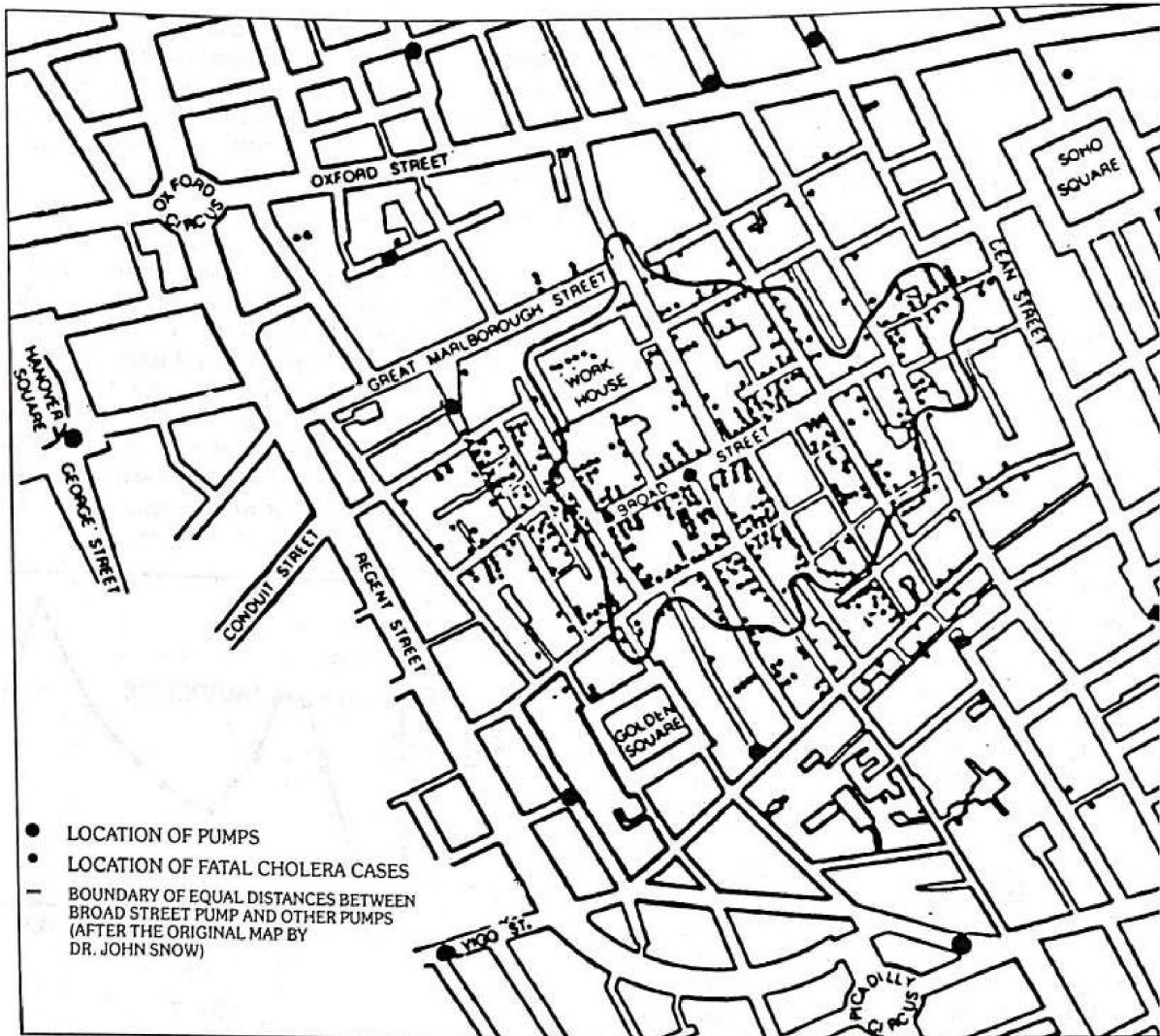
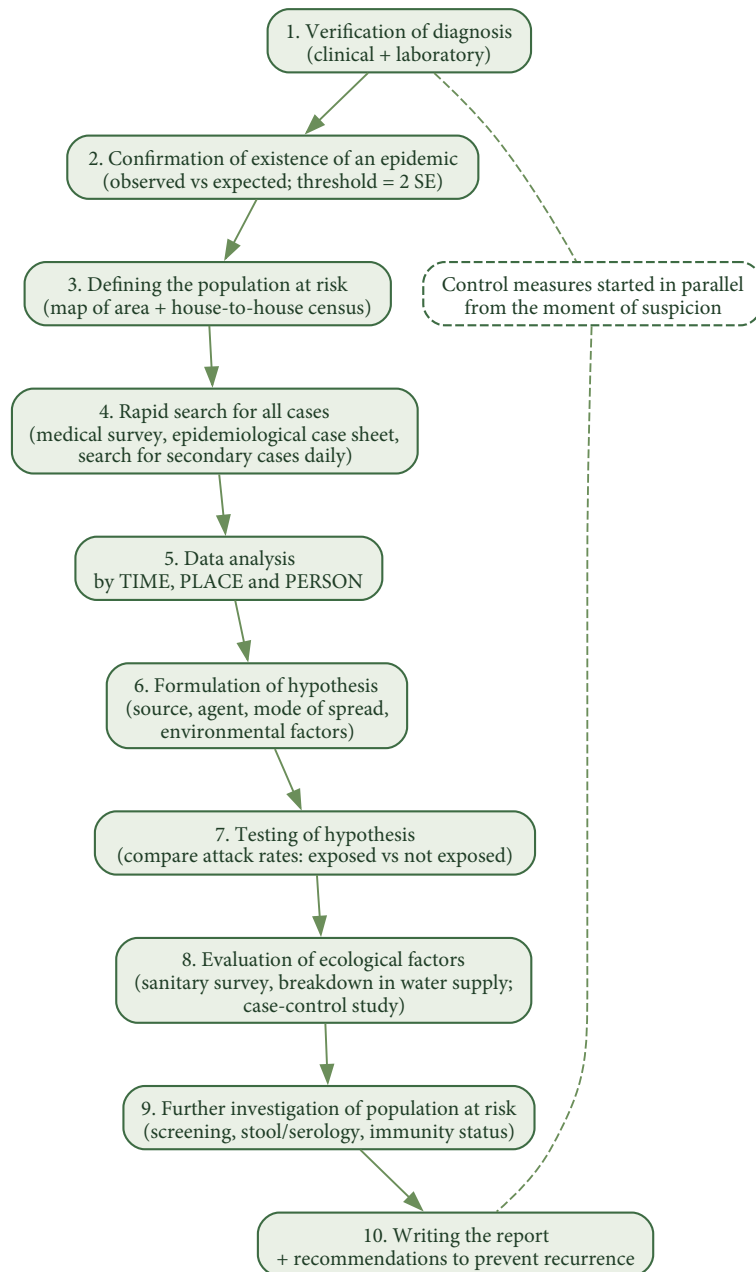
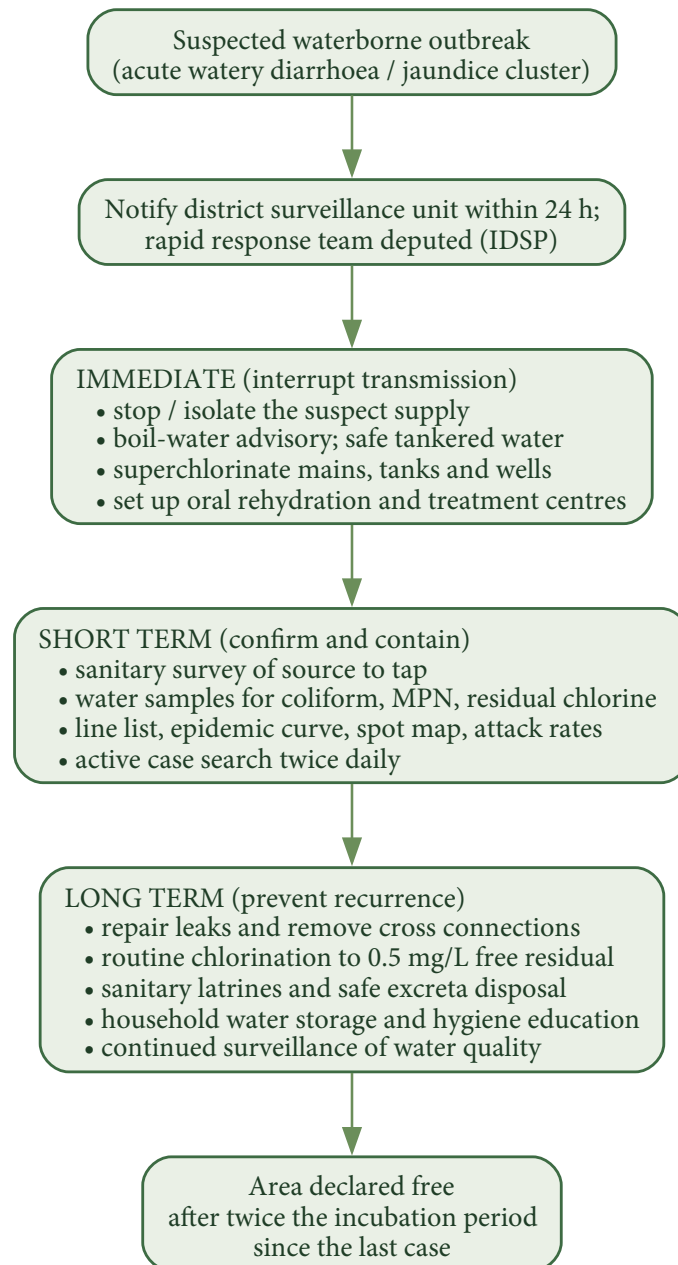


Fig 6 — Spot map of Asiatic cholera in London (Park's Textbook of Preventive and Social Medicine, 28e, p.87)



Sequence of steps in investigating a waterborne outbreak (Park 28e, Ch 3, pp 159-161)



Control pathway once a water supply is incriminated

High-yield exam pointers

- ▶ **Ten steps mnemonic:** Verify → Confirm → Define population → Search cases → Analyse (time, place, person) → Formulate → Test → Ecological factors → Further investigation → Report.
- ▶ **Attack rate** = new cases in the interval ÷ population at risk in the same interval × 100. **Secondary attack rate** = exposed persons developing disease within the incubation period after a primary case.
- ▶ **Point-source epidemic:** rapid rise and fall, no secondary waves, explosive clustering, **all cases within one incubation period**; epidemic threshold = **two standard errors** above endemic occurrence.
- ▶ **Water bacteriology:** *E. coli* / thermotolerant coliform **not detectable in any 100 ml sample**; faecal streptococci = recent pollution; *C. perfringens* without coliforms = remote pollution.

- ▶ **Sampling:** 200–250 ml sterile bottle, sodium thiosulphate 0.1 ml of 3% solution if chlorinated, tap flushed 2 minutes, **on ice and analysed within 48 hours**.
- ▶ **Chlorination:** dose = chlorine demand + **0.5 mg/L free residual after 1 hour**; **2.5 g bleaching powder per 1,000 litres** (≈ 0.7 mg/L applied chlorine); chlorine demand measured by **Horrock's apparatus**; no action on spores, cysts or ova.
- ▶ **Declared free:** twice the incubation period (10 days for cholera) since death, recovery or isolation of the last case.
- ▶ **Waterborne diseases to list** (*Park 28e, Ch 15, p 865*): viral — hepatitis A, hepatitis E, poliomyelitis, rotavirus diarrhoea of infants; bacterial — typhoid and paratyphoid, bacillary dysentery, *E. coli* diarrhoea, cholera; protozoal — amoebiasis, giardiasis; helminthic — roundworm, threadworm, hydatid disease; leptospiral — Weil's disease.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: in disaster and emergency settings Park advises raising the residual chlorine level in a public supply to about **0.2–0.5 mg/litre** as the first priority of ensuring water quality (*Park 28e, Ch 17, p 953*); the 0.5 mg/litre after one hour's contact quoted in the answer body is the routine chlorination standard. Coverage note — the scanned text of Park's Table 47 (contents of the final epidemic report) is only partly legible in the indexed corpus, so only its legible sections have been reproduced above.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 3 (Principles of Epidemiology and Epidemiologic Methods), p 81 (attack rates), pp 84–87 (types of epidemics, epidemic curve, spot map, John Snow), pp 159–161 (Investigation of an Epidemic, the ten steps and Table 47); Ch 5 (Epidemiology of Communicable Diseases), pp 290–293 (laboratory diagnosis, water and stool sampling, and control of cholera); Ch 15 (Environment and Health), p 865 (water-related diseases, pollution of distribution systems), pp 869–873 (chlorination, break-point, well disinfection, Horrock's apparatus), pp 876–877 (indicator organisms and bacteriological guideline values), p 881 (presumptive and confirmatory coliform tests, chemical surveillance), p 884 (sampling, transport and storage); Ch 17 (Disaster Management), p 953 (survey of public water supplies in an emergency). Suryakantha — Community Medicine with Recent Advances, Ch 5 (Environment and Water), pp 62–63 (bacteriological indicators) and Ch 18, p 268 (point-source epidemic curve). Jugal Kishore — National Health Programmes of India, pp 487–491 (Integrated Disease Surveillance Programme: S, P and L forms, rapid response teams, trigger levels).

♦ ♦ ♦

Investigation of a food poisoning outbreak

Asked in: Paper I 07-May-2022 — RGUHS MD Community Medicine

Definition — Food poisoning is an acute gastroenteritis caused by ingestion of food or drink contaminated with either living bacteria or their toxins, or with inorganic chemical substances and poisons derived from plants and animals (Park 28e, Ch 5, p 297). It is recognised as an outbreak by a triad: a history of ingestion of a **common food**, attack of **many persons at the same time**, and **similarity of signs and symptoms** in the majority of cases. Investigation is therefore a classic common-source, point-exposure outbreak enquiry in which the vehicle is a specific food item and the denominator is the set of persons who shared the meal.

Hallmarks of the outbreak and objectives of the investigation

Feature	Food poisoning outbreak	Cholera outbreak (chief differential)
Epidemiology	Usually a single group who shared a common meal; no secondary cases	Epidemic form, cases in the neighbourhood; secondary cases occur
Incubation period	1–24 hours	A few hours up to 5 days
Onset	With vomiting	With purging
Nausea and retching	Present	None
Vomiting	Often a single severe vomit	Projectile, effortless, watery, continuous
Stools	Contain mucus	Copious rice-watery, inoffensive

(Park 28e, Ch 5, p 299. Food poisoning may also be mistaken for bacillary dysentery and for chemical — arsenic — poisoning.)

Objectives (Park 28e, Ch 3, p 159):

- ▶ To define the magnitude of the outbreak in terms of **time, place and person**.
- ▶ To determine the particular conditions and factors responsible for its occurrence.
- ▶ To identify the **cause, source(s) of infection and mode of transmission**, so that control measures can be determined.
- ▶ To make **recommendations to prevent recurrence**.

Steps in the investigation

Some steps run concurrently; no rigid "cook-book" sequence applies to every outbreak (Park 28e, Ch 3, p 159).

- ▶ **Verification of the diagnosis** — examine a sample of cases clinically and exclude cholera, bacillary dysentery and chemical poisoning. Field work begins without waiting for laboratory results.
- ▶ **Confirmation that an outbreak exists** — compare the observed frequency with that expected for the population from past experience; an arbitrary limit of **two standard errors** above endemic

occurrence defines the epidemic threshold. A common-source event such as food poisoning is usually self-evident and needs no such comparison.

- ▶ **Defining the population at risk** — (a) obtain or prepare a **map** with landmarks, roads and numbered dwelling units; (b) **count the population**: secure a complete list of everyone who shared any part of the meal, including kitchen and dining-hall staff. Without this denominator attack rates cannot be calculated.
- ▶ **Rapid search for all cases** — a **medical survey** of the defined group picks up cases who never sought care. Data go on a pre-designed **epidemiological case sheet**, administered by trained lay health workers, recording age, sex, occupation, residence, all foods eaten in the previous **2 days** and place of consumption, **time of onset**, and symptoms (nausea, vomiting, diarrhoea, abdominal pain, headache, fever, prostration) in order of occurrence; a line list is built from these sheets.
- ▶ **Data analysis by time, place and person** — construct an **epidemic curve** from hours of onset, a **spot map** of cases against kitchens and water sources, and **food-specific attack rates** and case fatality rates by age, sex and occupation.
- ▶ **Formulation of hypothesis** — on the agent–host–environment model, state the probable source, agent, mode of spread and permitting environmental factors, in order of likelihood.
- ▶ **Testing of hypothesis** — compare attack rates in exposed and unexposed for each suspected food; cross-tabulate to separate the true vehicle from a confounded one; a **case-control study** may be mounted to establish the association between illness and a particular food.
- ▶ **Evaluation of ecological factors** — inspect sanitation of the eating establishment, kitchen, water and milk supply, storage and refrigeration, and note temperature and humidity.
- ▶ **Further investigation of the population at risk** — screen or sample a subset (stool, blood, suspected food) to detect inapparent infection and classify every member by exposure and by illness.
- ▶ **Institute control measures and write the report** — the report must be complete and convincing and must carry recommendations to prevent recurrence.

Specimens and laboratory investigation

Specimen / method	Purpose
Stool or rectal swab from cases	Isolate the causative organism; examine aerobically and anaerobically ; phage typing completes the work-up
Vomit	Isolate organism or demonstrate preformed toxin
Remnants of the incriminated food	Isolate the organism and determine the total number of bacteria and the relative numbers of each kind — this gives a better indication of the organism actually involved
Stool of kitchen employees and food handlers	Detect carriers and the source of contamination
Blood for antibodies	Retrospective (serological) diagnosis

Specimen / method	Purpose
Animal experiments	Feeding remnants of food to rhesus monkeys; for botulism , the protection test — a saline filtrate of the foodstuff injected subcutaneously into mice protected with antitoxic sera, with suitable controls
Environmental study	Inspection of the eating place(s) and kitchen(s), and questioning of food handlers about food preparation

(Park 28e, Ch 5, p 299)

Analysis: the food-specific attack rate

The attack rate is the incidence rate as used in epidemic situations (Gordis 7e, Ch 12, p 256). For every food item served, tabulate:

Food item	Ate the food: ill / total (attack rate %)	Did not eat: ill / total (attack rate %)	Risk ratio and risk difference
Item A	a / (a+b)	c / (c+d)	ratio of the two rates; difference of the two rates

The incriminated food is the one with a **high attack rate among those who ate it and a low attack rate among those who did not**, with the largest ratio and difference. Where two items are correlated, **cross-tabulation** separates them, as in the streptococcal foodborne outbreak at Dade County Jail, Miami, 16 August 1974:

Stratum	Ate egg salad — ill / total (attack rate)	Did not eat egg salad — ill / total (attack rate)
Drank beverage	152 / 201 (75.6%)	19 / 72 (26.4%)
Did not drink beverage	12 / 15 (80.0%)	7 / 28 (25.0%)

Egg salad raised the attack rate within **both** beverage strata, while the beverage raised it in **neither** egg-salad stratum — so the egg salad was the vehicle and the beverage association was spurious (Gordis 7e, Ch 2, p 54).

Identifying the agent from incubation period and clinical picture

The median incubation period read off the epidemic curve, with the dominant symptom, narrows the agent before the laboratory reports.

Agent	Incubation period	Foods usually implicated	Distinguishing features
<i>Staphylococcus aureus</i> enterotoxin	1–8 h	Salads, custards, milk and milk products	Sudden vomiting, cramps, diarrhoea; fever rare ; toxin heat-stable, resists boiling 30 min; death uncommon
<i>Bacillus cereus</i> — emetic form	1–6 h	Raw, dried and processed foods	Predominantly upper gastrointestinal symptoms; mimics staphylococcal poisoning; $\geq 10^5$ organisms per gram of food confirms

Agent	Incubation period	Foods usually implicated	Distinguishing features
<i>Bacillus cereus</i> — diarrhoeal form	12–24 h	Same	Predominantly lower gastrointestinal symptoms; mimics <i>C. perfringens</i> ; lasts about 24 h
<i>Salmonella</i> (<i>S. typhimurium</i> , <i>S. cholerae-suis</i> , <i>S. enteritidis</i>)	12–24 h	Meat, milk and milk products, sausages, custards, egg and egg products	Acute enteritis and colitis: chills, fever , nausea, vomiting, profuse watery diarrhoea for 2–3 days; mortality about 1%; convalescent carriers occur
<i>Clostridium perfringens</i> (<i>welchii</i>)	6–24 h (peak 10–14 h)	Meat, meat dishes and poultry cooked ≥ 24 h earlier, cooled slowly, reheated before serving	Diarrhoea and cramps, little or no fever, vomiting rare , illness ≤ 1 day, no deaths reported
<i>Clostridium botulinum</i> exotoxin (types A, B, E)	18–36 h	Home-preserved foods — home-canned vegetables, smoked or pickled fish	Gastrointestinal symptoms slight; dysphagia, diplopia, ptosis, dysarthria, blurred vision, muscle weakness; afebrile, conscious; kills two-thirds of victims, death in 4–8 days from respiratory or cardiac failure
Non-bacterial (arsenic, cadmium, mercury, pesticides, plant and sea-food toxins)	Minutes to hours	Any contaminated or adulterated food	No fever; requires toxicological analysis rather than culture

(Park 28e, Ch 5, pp 297–299)

Control measures and the report

- ▶ **Withdraw and destroy** the incriminated food; suspend and disinfect the kitchen or establishment.
- ▶ **Treat cases** — oral rehydration and symptomatic care; for botulism, antitoxin **50,000–100,000 units intravenously** to everyone who ate the food, since it is useless once toxin is fixed to nervous tissue.
- ▶ **Exclude food handlers** with infected wounds, boils, diarrhoea, dysentery or throat infection. Medical inspection of handlers is of limited value in detecting carriers, though it removes some sources of infection.
- ▶ **Correct food-handling technique** — minimise bare-hand handling of ready-to-eat food, keep the interval between preparation and consumption short, pasteurise milk, milk products and egg products, cook so heat penetrates the centre leaving no cool spots (most food-poisoning organisms die above 60°C), and separate raw meat, poultry, seafood and eggs from ready-to-eat food.
- ▶ **Temperature control** — "cook and eat the same day" is the golden rule; the **danger zone** is 10°C to 49°C , and refrigeration must not exceed 4°C , below which cold is bacteriostatic.
- ▶ **Notify and report** — report under the Integrated Disease Surveillance Programme (IDSP), whose weekly **S, P and L** forms are filled by health worker, clinician and laboratory; a rising trend triggers the medical officer or **rapid response team**. Send food samples for analysis under the

Food Safety and Standards Act 2006, administered by the Food Safety and Standards Authority of India (FSSAI, established 2008), and maintain continuing periodic surveillance of establishments.

- **Write the final report** — background and geography, methods, findings, incriminated food and agent, control measures taken, and recommendations.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Draw the **point-source epidemic curve** and label: x -axis "Time (hours/days) of onset"; y -axis "Number of cases"; arrow marked "Exposure (common meal)"; arrow marked "first case"; the interval between them labelled "**minimum incubation period**"; the peak labelled "**median incubation period** → identifies the agent"; the single sharp rise and fall labelled "**common source, single exposure — no secondary cases**".

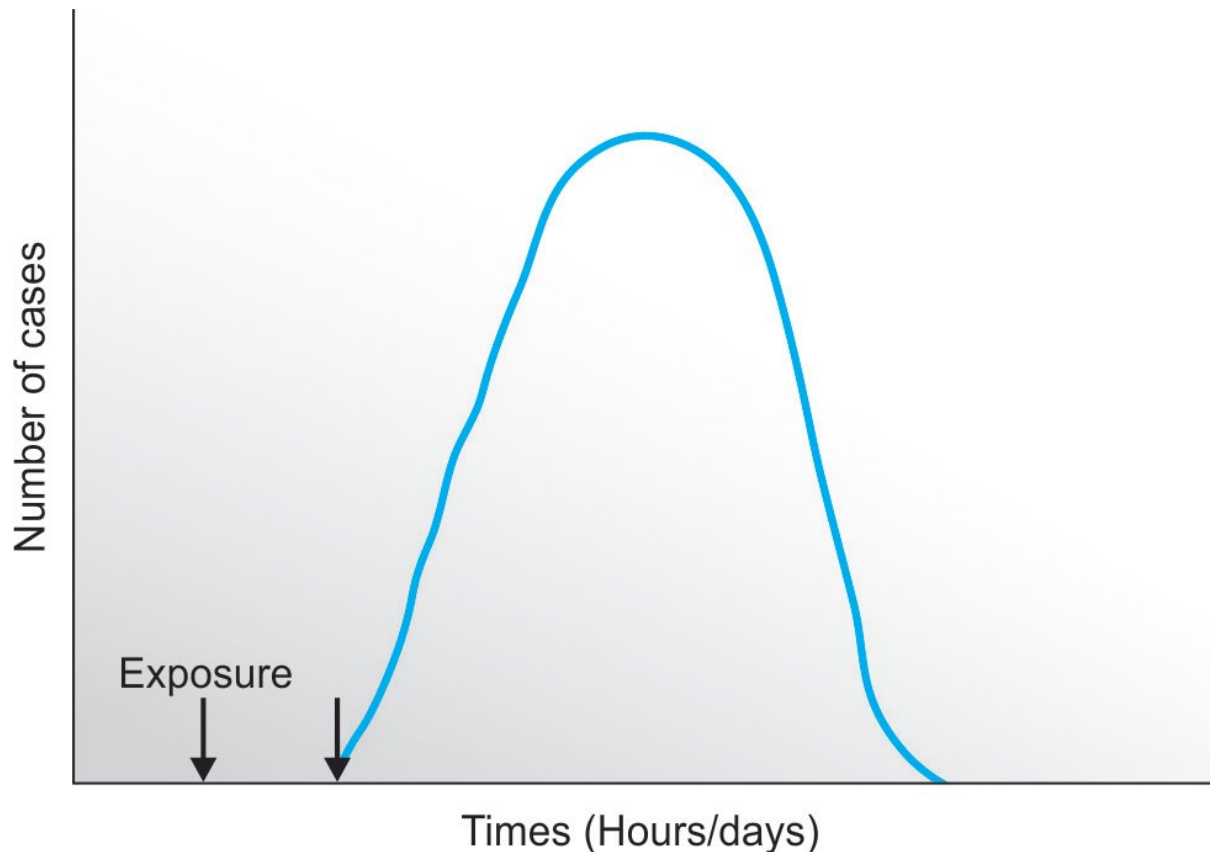
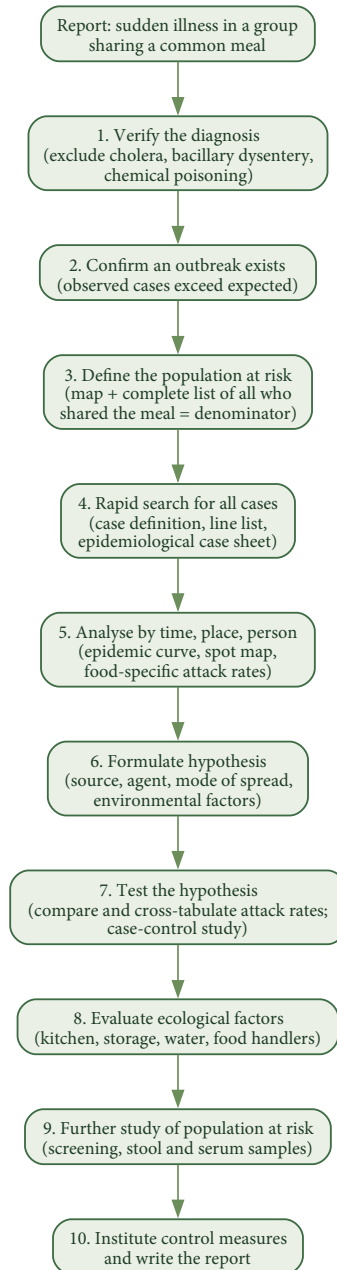
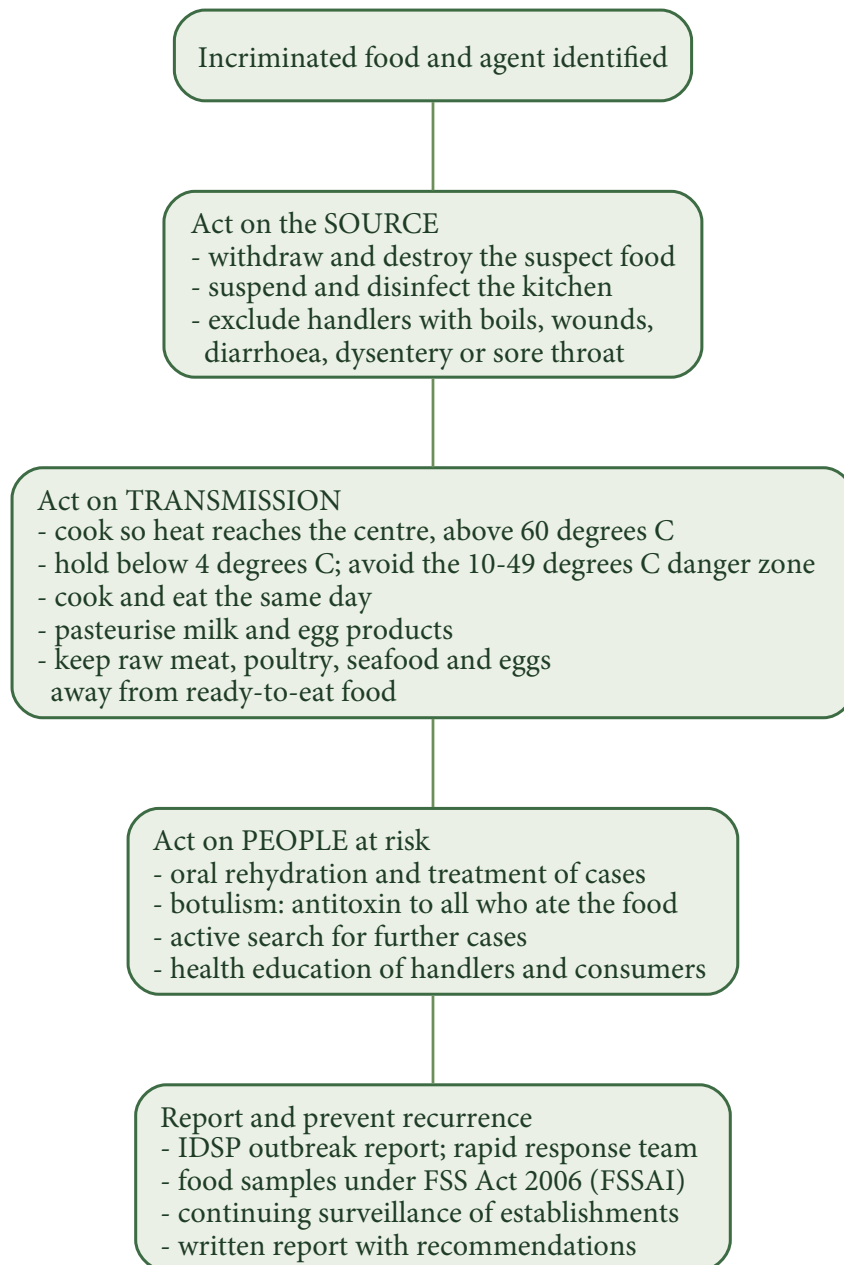


Fig 18.3 — Epidemic curve (Point source) (Suryakantha Environment Best, p.268)



Sequence of steps in the investigation of a food poisoning outbreak



Control measures arising from the investigation of a food poisoning outbreak

High-yield exam pointers

- ▶ **Triad of food poisoning:** common food + many persons at the same time + similar symptoms; **no secondary cases** (this alone separates it from cholera).
- ▶ **Food-specific attack rate** = (ill among those who ate the food) ÷ (total who ate the food); incriminate the food with a high rate in eaters, low rate in non-eaters, and the widest ratio and difference. Attack rate = incidence rate in an epidemic.
- ▶ **Incubation period mnemonic** — 1–8 h *Staphylococcus*; 1–6 h emetic *B. cereus*; 6–24 h *C. perfringens*; 12–24 h *Salmonella* and diarrhoeal *B. cereus*; 18–36 h botulism.
- ▶ **Epidemic threshold** = 2 standard errors above the endemic occurrence.
- ▶ **Temperatures:** kill above 60°C; danger zone 10–49°C; refrigerate below 4°C; staphylococcal toxin survives boiling for 30 minutes; botulinum toxin is destroyed at 100°C in a few minutes.

- **Ten steps** in order: verify → confirm → define population → search cases → analyse (time, place, person) → hypothesis → test → ecological factors → further study → report.
- **Botulism antitoxin** 50,000–100,000 units intravenously to all who ate the food.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park describes the epidemic-investigation sequence as guidelines rather than a rigid order, and states that several steps are done concurrently — an examiner accepts the ten steps in the order given above, but control measures are begun as soon as the vehicle is suspected and are not deferred to step 10.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 3 (Principles of Epidemiology and Epidemiologic Methods, pp 159–161: investigation of an epidemic) and Ch 5 (Epidemiology of Communicable Diseases, pp 297–300: food poisoning, its investigation, prevention and control); Ch 11, p 802 (Food Safety and Standards Act 2006, FSSAI). Gordis Epidemiology 7e — Ch 2, p 54 (food-specific attack rates and cross-tabulation, Dade County Jail outbreak 1974) and Ch 12, p 256 (attack rate as incidence rate; risk ratio and risk difference). Jugal Kishore — National Health Programmes of India, pp 488–491 (Integrated Disease Surveillance Programme: S, P and L forms, rapid response teams, epidemic response). Suryakantha — Community Medicine with Recent Advances, Fig 18.3, p 268 (point-source epidemic curve).



Disease surveillance — role of the microbiology laboratory

Asked in: Paper I 02-Jan-2025 — RGUHS MD Community Medicine

Definition — Surveillance is the systematic and continuous collection, analysis and interpretation of health data, closely integrated with the timely dissemination of results to those who have the right to know so that action can be taken; its listed data sources explicitly include **laboratory diagnosis** and **serological surveillance through serum banks** (Park 28e, Ch 2, p 59). Surveillance "goes beyond the passive reporting of cases" — it includes laboratory confirmation of a presumptive diagnosis, finding the source of infection and routes of transmission, and identifying all cases and susceptible contacts (Park 28e, Ch 3, p 151). The microbiology laboratory is therefore a constitutive element of infectious disease surveillance, not an optional support service.

Justification 1 — what the microbiology laboratory contributes

Contribution	What the laboratory actually does	Why surveillance fails without it
Aetiological confirmation	Isolation and identification of the agent from clinical specimens	Clinical diagnosis alone is unreliable — "the diagnosis of cholera can never be made with certainty on clinical grounds" (Park 28e, Ch 5, p 290)

Contribution	What the laboratory actually does	Why surveillance fails without it
Case classification	Supplies the laboratory criterion separating <i>suspected</i> → <i>probable</i> → <i>confirmed</i> cases	Without it every reported case stays presumptive; a case can be called "epidemiologically linked" only if at least one case in the chain of transmission is laboratory confirmed (Handbook of Epidemiology, Ch 33, p 1327)
Uncovering the submerged iceberg	Serological surveillance identifies patterns of current and past infection	Inapparent and subclinical infection, carriers and immunity gaps are invisible to clinical notification (Park 28e, Ch 3, p 151)
Detecting outbreaks against a constant background	Serotyping, phage typing and whole-genome sequencing of routine isolates	Serotyping of salmonellae discovers outbreaks and eliminates a source even when overall incidence is almost constant (Giesecke, Ch 20, p 234)
Source attribution	Matching the patient isolate with food, water, environmental or carrier isolates	In legionellosis it is essential to obtain <i>Legionella</i> from the patient and from the implicated water source and show the isolates are identical (Nelson & Williams, Ch 5, p 138)
Antimicrobial resistance surveillance	Characterises antimicrobial susceptibility; monitors emergence of multidrug-resistant pathogens	Drug-resistance surveillance requires "appropriate laboratory support for early detection, confirmation and communication" (Park 28e, Ch 5, p 430)
Monitoring the agent itself	Genotyping and strain characterisation	Intratype differentiation classifies poliovirus as wild, Sabin-derived (<1% genomic divergence from the Sabin parent) or vaccine-derived (>1% divergence) (Plotkin 7e, Ch 49, p 1310)
Certifying elimination and eradication	Certification-level laboratory surveillance and laboratory containment	Polio-free certification rests on laboratory evidence that wild poliovirus transmission has been interrupted (Plotkin 7e, Ch 74, pp 2103–2104)
Programme evaluation	Laboratory-confirmed endpoints for vaccine effectiveness studies	Laboratory confirmation increases the precision of the point estimate of vaccine effectiveness (Nelson & Williams, Ch 11, p 304)
Hospital infection surveillance	Culture reports, isolate storage, environmental microbiological studies, molecular fingerprinting	Culture reports "are among the most important pieces of data included in surveillance" of healthcare-associated infection (Nelson & Williams, Ch 14, p 420)
Global health security	National capacity to detect and assess events for notification	The International Health Regulations (2005) require states to detect, assess and notify the World Health Organization through the National IHR Focal Point within 24 hours (Oxford Textbook of Public Health 7e, Ch 105, pp 3996–3997)

Justification 2 — the laboratory decides the case-definition tier

Tier	Basis of diagnosis	Reported by (IDSP form)	Weakness if used alone
Syndromic	Clinical pattern recognised by paramedical personnel and the community	Health worker — form "S"	Sensitive but poorly specific
Presumptive / probable	Typical history and clinical examination by a medical officer	Clinician — form "P"	Laboratory results consistent with, but not meeting, confirmation criteria
Confirmed	Clinical diagnosis plus positive laboratory identification	Laboratory staff — form "L"	The only tier that permits specific, agent-directed action

(Tiers: Park 28e, Ch 7, p 569. The "S", "P" and "L" weekly formats, filled respectively by health workers, clinicians and laboratory staff: Sunder Lal, Ch 7, p 307.)

The decisive argument is that the **predictive value of a clinical case definition falls as incidence falls**. A clinical measles definition of fever, rash and cough has a high positive predictive value in a high-incidence country but a very low one where vaccination coverage is high, because most cases meeting it will be other viral infections or adverse drug reactions (Giesecke, Ch 13, p 153). Precisely where a programme approaches elimination, clinical surveillance becomes least trustworthy and laboratory confirmation becomes mandatory.

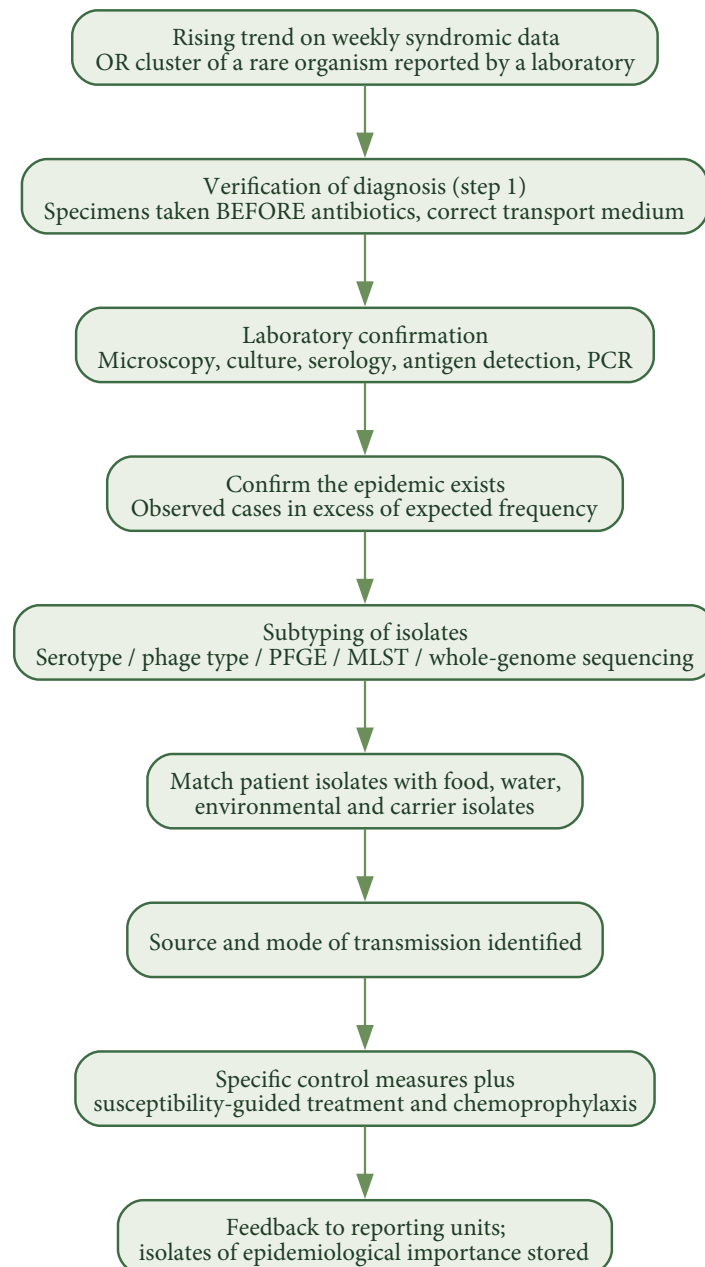
Justification 3 — the laboratory improves every attribute of a surveillance system

Attribute (standard evaluation criteria)	How the microbiology laboratory improves it
Sensitivity — ability to identify every case; critical for outbreak and eradication programmes	Environmental (sewage) sampling supplements acute flaccid paralysis surveillance and raises sensitivity because most poliovirus infections are inapparent; sustained wild poliovirus type 1 circulation in Israel in 2013–14 was detected by environmental surveillance with no clinical cases (Plotkin 7e, Ch 49, pp 1272–1273)
Predictive value positive — proportion of reported cases truly having the event	Low values mean non-cases are investigated, outbreaks are exaggerated and pseudo-outbreaks declared; laboratory confirmation removes this misclassification (Handbook of Epidemiology, Ch 33, p 1325)
Representativeness	Requiring laboratories to submit all, or a defined sample of, their isolates gives an unbiased picture of circulating strains and resistance (Handbook of Epidemiology, Ch 33, p 1331)
Timeliness	Global Polio Laboratory Network standards: results reported on ≥80% of acute flaccid paralysis specimens within 14 days , and ≥80% of suspected isolates forwarded for intratypic differentiation within 7 days (Plotkin 7e, Ch 49, p 1312)
Flexibility	A system must adapt to changes in laboratory methodology; a new assay lets an existing system take on a new pathogen without redesign (Handbook of Epidemiology, Ch 33, pp 1325–1326)

Attribute (standard evaluation criteria)	How the microbiology laboratory improves it
Usefulness	Susceptibility data directly inform treatment protocols, infection prevention and control, and water, sanitation and hygiene action (Park 28e, Ch 14, p 858)

Justification 4 — the laboratory in outbreak detection and investigation

- **Detection** — outbreaks reach the health department from a doctor's office, hospital, nursing home or a **laboratory**, or through analysis of routine reports and specialised aberration-detection routines applied to subtyped isolates (Nelson & Williams, Ch 5, p 124).
- **Verification of diagnosis** — the first step of any epidemic investigation (Park 28e, Ch 3, p 159); specimens must be collected before antibiotics are given and in the correct transport medium (for cholera, Venkatraman–Ramakrishnan medium or alkaline peptone water — Park 28e, Ch 5, p 290).
- **Confirmation that an epidemic exists** — observed frequency in excess of the expected frequency for that population (Park 28e, Ch 3, p 159).
- **Subtyping** — pulsed-field gel electrophoresis, restriction fragment length polymorphism, multi-locus variable-number tandem repeat analysis, multi-locus sequence typing or whole-genome sequencing link cases to one another and to the hypothesised source (Nelson & Williams, Ch 5, p 138).
- **Source and mode of transmission** — identified by matching human isolates with food, water, environmental and carrier isolates (Jugal Kishore, Ch 1, p 491).
- **Specific control and feedback** — susceptibility results guide treatment and chemoprophylaxis, and isolates of epidemiological importance are stored for later comparison (Nelson & Williams, Ch 14, p 420).

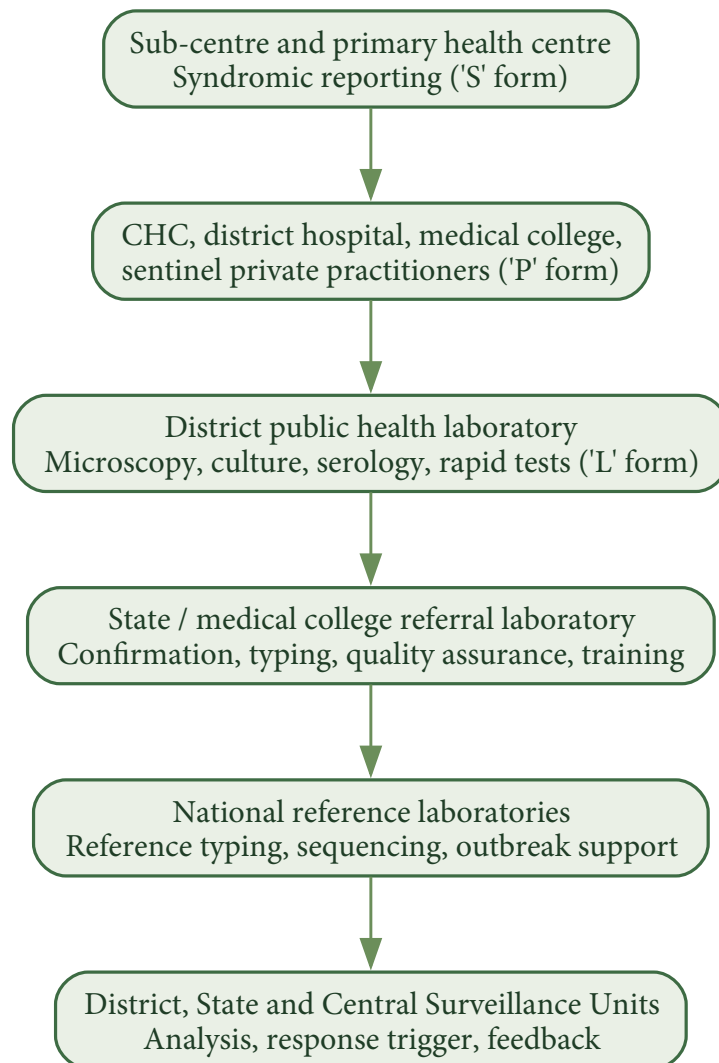


The microbiology laboratory in the outbreak detection-to-control sequence

Justification 5 — the laboratory in India's surveillance architecture

The Integrated Disease Surveillance Programme (IDSP), launched in 2004 with World Bank assistance, carries "to improve laboratory support for disease surveillance" as a specific objective and "upgradation of laboratories" as a project activity (Sunder Lal, Ch 7, p 306). Weekly reporting is done every Monday on the S, P and L formats using standard case definitions, and more than 90 per cent of districts report weekly (Park 28e, Ch 7, p 568). The laboratory network component supplies incremental infrastructure, equipment, reagents, communication and networking, and training and human resource development (Jugal Kishore, Ch 1, p 491). A referral laboratory network was established in 9 states — Gujarat, Punjab, Rajasthan, Uttarakhand, Karnataka, Tamil Nadu, Maharashtra, Andhra Pradesh and West Bengal — using existing functional laboratories in medical colleges and other major centres and linking them with adjoining districts for diagnostic services during outbreaks of epidemic-prone

disease, with 23 further medical college laboratories added in Bihar, Assam, Odisha, Tripura, Kerala, Haryana, Jammu & Kashmir and Manipur (Jugal Kishore, Ch 1, pp 492–493). Medical colleges are integrated into the system as **reference laboratories** and for **quality assurance**, training and epidemic investigation (Jugal Kishore, Ch 1, pp 490–491). When a rising trend is investigated, the Rapid Response Team is a multi-speciality team of an **epidemiologist**, a **clinician** and a **microbiologist**, with other specialists as required (Park 28e, Ch 7, p 568) — the microbiologist is written into the response structure itself.



Tiered laboratory referral pathway supporting disease surveillance in India

Programme-specific laboratory networks — each is a surveillance system that could not exist without a laboratory:

Network	Laboratory role	Retrieved benchmark
Global Polio Laboratory Network	Virus isolation, intratypic differentiation and sequencing of all acute flaccid paralysis stool specimens	145 formally accredited laboratories; in India, stool specimen collection completeness rose from 59% in 1998 to 82% by end-September 2020 (Plotkin 7e, Ch 49, p 1310; Park 28e, Ch 5, p 264)

Network	Laboratory role	Retrieved benchmark
WHO Global Measles and Rubella Laboratory Network	Standardised case confirmation and virus characterisation	Established 2000; more than 730 laboratories across all six WHO regions; 13 of the 24 known measles genotypes detected during 2005–14 (Plotkin 7e, Ch 37, p 863)
HIV sentinel surveillance, India	Unlinked anonymous serological testing of general population and high-risk groups	Begun 1994 with 55 sentinel sites, increased to 180 by 1998 (Park 28e, Ch 7, pp 519–520)
WHO Global Antimicrobial Resistance Surveillance System (GLASS)	Frequency of antimicrobial-resistant pathogens in common acute bacterial infections including bloodstream infection	The 2019 data call gathered data from 66 countries and territories (Park 28e, Ch 14, p 858)
National Leprosy Eradication Programme	Antimicrobial resistance surveillance for leprosy	Guidelines launched and rollout planned under the National Strategic Plan and Roadmap for Leprosy 2023–2027 (Park 28e, Ch 7, p 506)

Prerequisites and limitations to acknowledge

- The result is only as good as the **specimen**: correct site, collection before antimicrobials, and correct transport medium (Park 28e, Ch 5, p 290).
- **Molecular typing criteria mislead without epidemiological data** — for pulsed-field gel electrophoresis, isolates differing by up to three bands are treated as subtypes of one strain, and differences between epidemiologically unrelated strains occur randomly, so interpretive guidelines applied in a larger context without epidemiological information give misleading information (Nelson & Williams, Ch 9, p 254).
- Sustained **quality assurance, trained manpower, reagent supply, equipment and networking** are required; the IDSP laboratory component funds exactly these inputs (Jugal Kishore, Ch 1, p 491).
- Laboratory confirmation must not delay action — for practical epidemic response a confirmed aetiological diagnosis is not always required before control measures begin (Jugal Kishore, Ch 1, p 487).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Weekly surveillance information flow** — draw the vertical spine and label: Subcentres → PHC → District Surveillance Unit → State Surveillance Unit → Central Surveillance Unit; entering the District Surveillance Unit from the left, label CHC, District Hospital, Medical College, Public Health Laboratory; from the right, label Private Practitioner, Private Health Facilities, Other Hospitals, Private Laboratory.

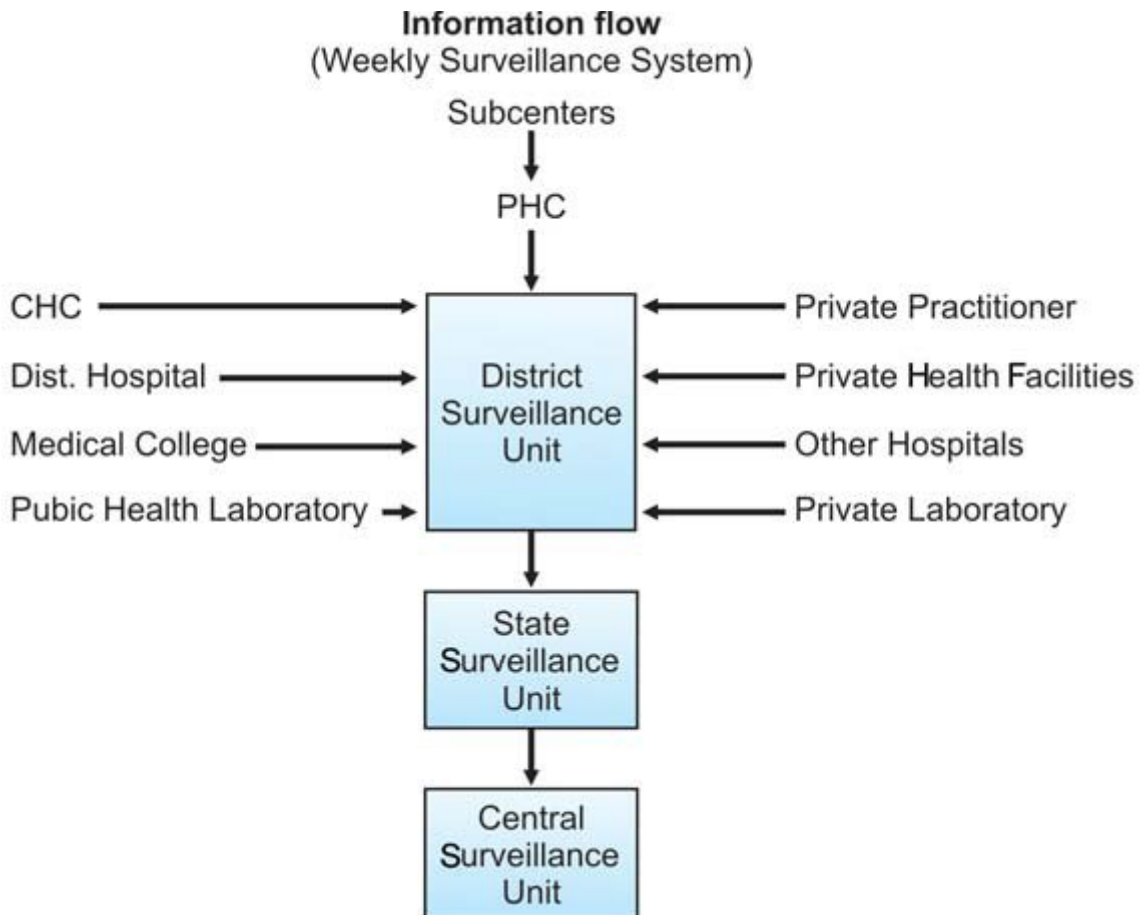


Fig 21.1 — Information flow (Weekly Surveillance System) (Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e, p.398)

High-yield exam pointers

- ▶ **IDSP weekly forms — S / P / L:** "S" syndromic (health worker), "P" probable (clinician), "L" laboratory-confirmed (laboratory staff); reported every Monday; more than 90% of districts reporting weekly.
- ▶ **Case tiers:** suspected → probable → confirmed; a case is "epidemiologically linked" only if at least one case in the chain of transmission is laboratory confirmed.
- ▶ **Surveillance evaluation attributes, in order:** usefulness, sensitivity, predictive value positive, representativeness, simplicity, flexibility, data quality, acceptability, timeliness, stability.
- ▶ **Polio laboratory benchmarks:** 145 accredited Global Polio Laboratory Network laboratories; results on ≥80% of specimens within 14 days; intratypic differentiation referral within 7 days; wild versus Sabin-derived (<1%) versus vaccine-derived (>1% genomic divergence).
- ▶ **Quotable line:** "The diagnosis of cholera can never be made with certainty on clinical grounds" (Park 28e, Ch 5).
- ▶ **Rapid Response Team = epidemiologist + clinician + microbiologist** (plus others as required) — cite this whenever asked to justify the laboratory's place in surveillance.
- ▶ **International Health Regulations (2005):** notify the World Health Organization through the National IHR Focal Point within **24 hours**; core capacity to detect, assess, notify, report and contain.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Coverage note — the counts given for the Integrated Disease Surveillance Programme's referral laboratory network and for the proportion of districts reporting weekly are those printed in the cited editions and change with each programme review, so they must always be quoted with their source and year as done above; the retrieved corpus carried no updated post-2024 figure for the network, and no retrieved passage described India's national antimicrobial resistance laboratory network, so it has not been detailed here.

Recent advances [RA]

Current as of 2026-08-19 · depth: elaborate · sources: PubMed, WHO, ICMR, NCVBDC (MoHFW)

Programme and policy updates (India)

- ▶ **Four zonal National Institutes of Virology are being built under the Pradhan Mantri Ayushman Bharat Health Infrastructure Mission (PM-ABHIM)**, decentralising virology capacity beyond NIV Pune. A four-zone prioritisation exercise in March 2024 drew 186 participants from the Virus Research and Diagnostic Laboratory (VRDL) network, the Integrated Disease Surveillance Programme and ICMR, and ranked dengue, influenza, measles, Japanese encephalitis and hepatitis A as the top five priority viral diseases, with clear zone-to-zone variation in the lists [PMID 42522023].
- ▶ **IDSP reporting has migrated from paper returns to the Integrated Health Information Platform (IHIP)** — a digital, case-based rather than aggregate system carrying the same S, P and L formats. Odisha went live in 2021; analysis of IHIP P-form data for Jagatsinghpur and Ganjam districts (April 2021 to October 2022) put acute respiratory illness at the highest notification rate in both districts. Use the IHIP name in the answer; the weekly-Monday S/P/L logic in the body is unchanged [PMID 41629233].
- ▶ **Laboratory access at primary-care level now rests on the Free Diagnostics Service Initiative (2015) and Comprehensive Primary Health Care (2018) under Ayushman Bharat**, which take an essential-diagnostics package down to Ayushman Arogya Mandir level. Standardisation, capacity building, infrastructure and robust monitoring remain the binding constraints — the same prerequisites the textbook lists for the IDSP laboratory component [PMID 40214332].

Current national data

- ▶ **Kala-azar, quoted with its year:** 436 cases and 27 deaths in 2025, and 198 cases with no deaths up to July 2026 (provisional), against 9,241 cases in 2014; the residual burden is in Bihar, Jharkhand, West Bengal and Uttar Pradesh [NCVBDC 2026-Jul].
- ▶ **Laboratory-confirmed antimicrobial resistance in *Salmonella* Typhi, India 1977–2024** (meta-analysis of 188 studies): classical multidrug resistance to chloramphenicol, ampicillin and co-trimoxazole fell from 65% (95% CI 63–67) in 1990–94 to near 0% (95% CI 0–1) from 2020 onward, while fluoroquinolone resistance rose from 2% to 85% (95% CI 84–86) by 2015–19; third-generation cephalosporin and azithromycin resistance stayed near 3%. This is exactly the isolate-based trend that only a laboratory network can generate, and it now drives typhoid conjugate vaccine and treatment policy [PMID 41990105].

Elimination, surveillance and disease status

- ▶ **Measles-rubella elimination for the WHO South-East Asia Region has been pushed to 2026, and India remains endemic for both.** India's reported measles incidence rose from 4 to 45.3 cases per million and rubella from 1.2 to 2.1 per million between 2021 and 2023; Nepal became the sixth regional country to eliminate rubella (August 2025). India is also absorbing the WHO polio surveillance network — which had been carrying measles-rubella surveillance as well — into its own national surveillance programme, a laboratory-network transition worth naming in any elimination answer [PMID 41635705].
- ▶ **Kala-azar has entered a post-elimination phase in which serological and entomological laboratory surveillance replaces case counting.** In endemic villages of Bihar and Jharkhand, rK39 seroprevalence was 20–30% and qPCR found *Leishmania* in 1.4% and 1.2% of sand fly pools despite indoor residual spraying — residual transmission wholly invisible to clinical notification, and the clearest current illustration of the submerged-iceberg argument [PMID 41324335].
- ▶ **pflhrp2/pflhrp3 gene deletions are quietly degrading malaria rapid diagnostic tests.** Reported deletion prevalence ranges from 2.4% to 57.8% (3.8% from Chhattisgarh), and WHO's 2024 response plan requires a programme to switch to non-HRP2 diagnostics once ≥5% of symptomatic infections give false-negative HRP2 results, with PCR, digital droplet PCR and whole-genome sequencing used for molecular surveillance of the deletions [PMID 42024904; WHO 2024].

Guideline and legislative updates

- ▶ **WHO issued its first recommendations on near-point-of-care molecular TB tests in February 2026,** alongside tongue-swab specimens and a pooled-testing strategy to cut cost and waiting time. These near-point-of-care NAATs return a result in under an hour on battery-operable instruments needing no specialised laboratory infrastructure, but they do not detect rifampicin resistance and so require reflex testing (Xpert MTB/RIF Ultra, Truenat MTB Rif-Dx and equivalents). Fewer than half of people newly diagnosed with TB in 2024 had a molecular test as their first test; India's Truenat platform is WHO-recommended and the Huwel UniAMP card is ICMR-approved for national use [PMID 42024898; WHO 2026-Feb].

Global and WHO developments

- ▶ **The 2024 amendments to the International Health Regulations entered into force on 19 September 2025 and add a *national IHR authority* to the familiar National IHR Focal Point.** Amended Article 4 requires each state to establish or designate an entity to coordinate IHR implementation nationally, targeting the fragmented authority, thin legal mandate and weak intersectoral coordination that COVID-19 exposed. The 24-hour notification duty through the focal point, as given in the body above, is unchanged [PMID 40953844].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 2, 3, 5, 7, 14) — spine. Nelson & Williams, Infectious Disease Epidemiology (Ch 5, 9, 11, 14). Giesecke, Modern Infectious Disease Epidemiology (Ch 12, 13, 20). Handbook of Epidemiology, Ahrens & Pigeot (Ch 33 — surveillance evaluation criteria). Plotkin's Vaccines, 7ed (Ch 37, 49, 74). Jugal Kishore, National Health Programmes of India (IDSP objectives, laboratory network, epidemic response). Sunder Lal, Textbook of Community Medicine (Ch 7 — IDSP objectives and the S/P/L weekly formats). Oxford Textbook of Public Health 7e (Ch 75, 105 — International Health Regulations 2005). Mahajan & Gupta, Textbook of Preventive and Social Medicine 4e (Fig 21.1, p 398). National Strategic Plan and Roadmap for Leprosy 2023–2027; WHO Global Antimicrobial Resistance Surveillance System (GLASS), 2019 data call.

Recent-advances sources: 42522023 — Strengthening India's pandemic preparedness with the four zonal institutes of virology: stakeholder consultations for outbreak response and collaborative research priorities, *Infectious Diseases of Poverty* 2026, doi:[10.1186/s40249-026-01484-z](https://doi.org/10.1186/s40249-026-01484-z) (abstract-only); 41629233 — Epidemiological Profile of Diseases Reported in the P Form of Integrated Health Information Platform in Two Districts of Odisha, *Indian Journal of Public Health* 2026, doi:[10.4103/ijph.ijph.660.25](https://doi.org/10.4103/ijph.ijph.660.25) (abstract-only); 40214332 — Transforming Health-Care System: Free Diagnostic Service Initiative and Comprehensive Primary Health Care, *Indian Journal of Public Health* 2025, doi:[10.4103/ijph.ijph.1474.23](https://doi.org/10.4103/ijph.ijph.1474.23) (abstract-only); 41990105 — Burden of antimicrobial resistance in culture-confirmed *Salmonella* Typhi isolates in India from 1977 to 2024: a systematic review and meta-analysis, *PLoS Neglected Tropical Diseases* 2026, doi:[10.1371/journal.pntd.0014206](https://doi.org/10.1371/journal.pntd.0014206); 41635705 — Accelerating measles and rubella elimination in southeast Asia, *The Lancet Regional Health — Southeast Asia* 2025, doi:[10.1016/j.lansea.2025.100682](https://doi.org/10.1016/j.lansea.2025.100682); 41324335 — High Seropositivity to rK39 and Capture of *Leishmania*-Infected Sand Flies in Bihar and Jharkhand Emphasize the Need for Post-Elimination Surveillance of Visceral Leishmaniasis in India, *The Journal of Infectious Diseases* 2026, doi:[10.1093/infdis/jiaf543](https://doi.org/10.1093/infdis/jiaf543) (abstract-only); 42024904 — Are HRP2/3 deletions silently crippling malaria rapid diagnostic tests?, *Indian Journal of Medical Research* 2026, doi:[10.25259/ijmr.2856.2025](https://doi.org/10.25259/ijmr.2856.2025); 42024898 — New tools and policies herald a major advance in tuberculosis diagnosis, *Indian Journal of Medical Research* 2026, doi:[10.25259/ijmr.923.2026](https://doi.org/10.25259/ijmr.923.2026); 40953844 — National IHR authorities in the 2024 amendments to WHO's International Health Regulations: transforming the opportunity into reality, *BMJ Global Health* 2025, doi:[10.1136/bmjgh-2025-019284](https://doi.org/10.1136/bmjgh-2025-019284). Plus: WHO recommendations on near-point-of-care molecular TB testing, tongue-swab specimens and pooled testing, February 2026; WHO pfhrp2/3 response plan and global surveillance protocol, 2024; International Health Regulations (2005), 2024 amendments in force 19 September 2025; National Center for Vector Borne Diseases Control (MoHFW) national kala-azar situation, figures to July 2026 (provisional).

♦ ♦ ♦

Screening for diabetes mellitus

Asked in: Paper I May 2017 — RGUHS MD Community Medicine

Definition — Screening is "the search for unrecognized disease or defect by means of rapidly applied tests, examinations or other procedures in apparently healthy individuals" (Park 28e, Ch 4, p 163). Whether a condition deserves a screening programme is judged on two separate sets of criteria — those relating to the **disease** and those relating to the **test** (Park 28e, Ch 4, p 165). Diabetes mellitus is named by Park among the diseases for which case-detection ("prescriptive") screening is justified (Park 28e, Ch 4, p 164), and India operationalises this as universal population-based screening from **30 years of age** (Park 28e, Ch 10, p 735).

Reasons for selecting diabetes as a disease suitable for screening

Each of the nine disease criteria is satisfied, and this is precisely why diabetes was chosen.

#	Criterion for a screenable disease (Park 28e, Ch 4, p 165)	How diabetes mellitus satisfies it
1	The condition sought should be an important health problem — prevalence should be high	Global prevalence 8.5% in adults aged 18 years and over (WHO 2014, in Park 28e, Ch 6, p 468). In India prevalence is 10.9–14.9% in urban and 3.0–7.8% in rural populations aged 20 years and above (Park 28e, Ch 6, pp 468-469). NFHS-5 (2019-21) found random blood glucose above 140 mg/dL or on glucose-lowering medication in 12% of women and 14% of men aged 15 years and over (NFHS-5 India Report, Ch 6, pp 525-526)
2	There should be a recognizable latent or early asymptomatic stage	Type 2 diabetes is typically gradual in onset and "often discovered by chance"; impaired glucose tolerance is a defined intermediate at-risk state detectable only by the oral glucose tolerance test (Park 28e, Ch 6, pp 467-468). This detectable preclinical phase gives a long lead time in which to intervene
3	The natural history must be adequately understood	Agent (insulin deficiency — absolute in type 1, partial in type 2), host factors (age, genetic susceptibility, obesity, sedentary habit) and environmental factors (diet, alcohol, viral infection, early malnutrition, stress, chemical agents), and the progression to microvascular and macrovascular complications, are all well characterised (Park 28e, Ch 6, pp 468-470)
4	There is a test that detects the disease before signs and symptoms appear	Fasting and 2-hour post-load plasma glucose, random plasma glucose and glycated haemoglobin (HbA1c), with defined WHO 2019 cut-offs; automated biochemistry now makes it possible to screen thousands of samples for glucose (Park 28e, Ch 6, p 470)
5	Facilities for confirmation of diagnosis must be available	The standard oral glucose tolerance test remains the cornerstone of diagnosis and is available at primary health centre level and above; cases suspected on strip testing at sub-centre level are referred upward for confirmation (Park 28e, Ch 6, p 470; Ch 7, p 564)

#	Criterion for a screenable disease (Park 28e, Ch 4, p 165)	How diabetes mellitus satisfies it
6	There must be an effective treatment	Diet alone; diet plus oral antidiabetic drugs; or diet plus insulin. Good control of blood glucose protects against the development of complications (Park 28e, Ch 6, p 471)
7	There should be an agreed policy on whom to treat — Park itself cites "border-line diabetes" as the example of the problem	WHO 2019 cut-offs define diabetes unambiguously, while impaired fasting glucose and impaired glucose tolerance define the at-risk group for lifestyle intervention rather than drug treatment (Park 28e, Ch 4, p 165; Ch 6, pp 467-470)
8	There must be good evidence that early detection and treatment reduce morbidity and mortality	Documented in Park's own efficacy table — 35–58% reduction in incidence with lifestyle intervention in high-risk people, and 60–70% reduction in serious vision loss with annual eye examination (Park 28e, Ch 6, p 471; detailed below)
9	The expected benefits must exceed the risks and costs	Diabetes is a major cause of disability through blindness, kidney failure, coronary thrombosis and gangrene of the lower extremities (Park 28e, Ch 6, p 460, tertiary prevention). Screening costs one glucose strip applied by an existing health worker, which is small against the lifetime cost of these complications

Evidence base for criterion 8 — interventions of proven efficacy

Intervention (Park 28e, Ch 6, p 471, Table 4)	Benefit
Lifestyle intervention to prevent type 2 diabetes in high-risk people	35–58% reduction in incidence
Metformin to prevent type 2 diabetes in high-risk people	25–31% reduction in incidence
Glycaemic control where HbA1c exceeds 9%	30% reduction in microvascular disease per 1% drop in HbA1c
Blood pressure control where pressure exceeds 130/80 mmHg	Reduction in macrovascular and microvascular disease per 10 mmHg drop in blood pressure
Annual eye examination	60–70% reduction in serious vision loss
Foot care in people at high risk of ulcers	50–60% reduction in serious foot disease
Angiotensin-converting-enzyme (ACE) inhibitor in all people with diabetes	42% reduction in nephropathy; 22% drop in cardiovascular disease

Why the age cut-off is 30 years and above

- **Age is itself the qualifying risk factor.** A person is classified as at high risk for diabetes if he or she is above the age of 30 years — the first item on the national risk list, ahead of overweight (body mass index above 23 kg/m²), physical inactivity, raised blood pressure, and impaired fasting glucose or impaired glucose tolerance (NCD Combined Manual for Medical Officers, pp 19-20).

- ▶ **Indians develop type 2 diabetes a decade earlier than Western populations.** Type 2 diabetes "usually occurs after the age of forty years but occurs frequently even at lower age among Indians" (NCD Combined Manual for Medical Officers, pp 19-20); migrant studies from Singapore, South Africa, the United Kingdom and the United States demonstrate this ethnic susceptibility (Park 28e, Ch 6, p 468).
- ▶ **The prevalence gradient rises steeply from this age.** For both women and men the prevalence of high and very high blood glucose increases sharply with age (NFHS-5 India Report, Ch 6, pp 525-526), so screening from 30 years captures the steep part of the curve and keeps the yield of the programme high.
- ▶ **It converts an unrewarding mass screen into a productive selective screen.** Screening the whole population for diabetes is not considered rewarding; screening of high-risk groups is considered more appropriate (Park 28e, Ch 6, p 471). Setting the entry point at 30 years applies that high-risk principle at population scale instead of screening children and young adults in whom the yield would be negligible.
- ▶ **It matches the national service package already in place.** Health and Wellness Centres deliver universal screening of the population aged 30 years and above for hypertension, diabetes and three common cancers — oral, breast and cervical (Park 28e, Ch 10, p 735); the sub-centre carries out opportunistic screening of the population above 30 years with blood pressure measurement and blood glucose by strip (Park 28e, Ch 7, p 564).
- ▶ **One contact yields several screens.** Because hypertension, diabetes and the three cancers share the same 30-plus target population, a single household or clinic contact screens for all of them, which improves cost-effectiveness and acceptability far beyond what a diabetes-only screen could achieve.

The screening test that makes this feasible (criterion 4 in detail)

Test	WHO 2019 diagnostic cut-off (Park 28e, Ch 6, p 470, Table 3)	Suitability for field screening
Fasting venous or capillary plasma glucose (after an 8–14 h fast)	≥7.0 mmol/L (126 mg/dL)	Simple and cheap, but true fasting cannot be assured in the field, so spurious diagnosis may occur
2-hour post-load venous plasma glucose (after 75 g oral glucose)	≥11.1 mmol/L (200 mg/dL)	Preferred for epidemiological purposes, alone or with the fasting value
2-hour post-load capillary plasma glucose	≥12.2 mmol/L (220 mg/dL)	Point-of-care capillary testing where laboratory measurement is unavailable
Random plasma glucose	≥11.1 mmol/L (200 mg/dL)	Unsatisfactory for epidemiological use; gives only a crude estimate of frequency in a population
HbA1c	≥6.5% (48 mmol/mol)	No fasting needed; reflects mean blood glucose over the preceding 2–3 months

Test	WHO 2019 diagnostic cut-off (Park 28e, Ch 6, p 470, Table 3)	Suitability for field screening
Urine test for glycosuria	—	Largely abandoned: sensitivity only 10–50% (milder cases missed), although specificity exceeds 90% (Park 28e, Ch 6, pp 469-470)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- Iceberg of disease applied to diabetes — draw an iceberg with the waterline labelled *clinical horizon*. Above the water label **diagnosed, symptomatic diabetes** (what the physician sees). Below the water, in descending order, label **undiagnosed diabetes**, **impaired glucose tolerance / impaired fasting glucose**, and **persons carrying risk factors** — **obesity, sedentary life, family history**. Add an upward arrow crossing the waterline labelled **screening**.

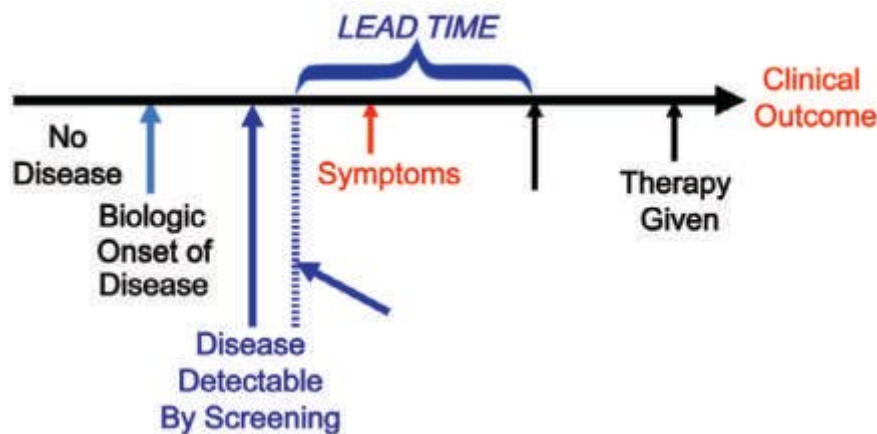
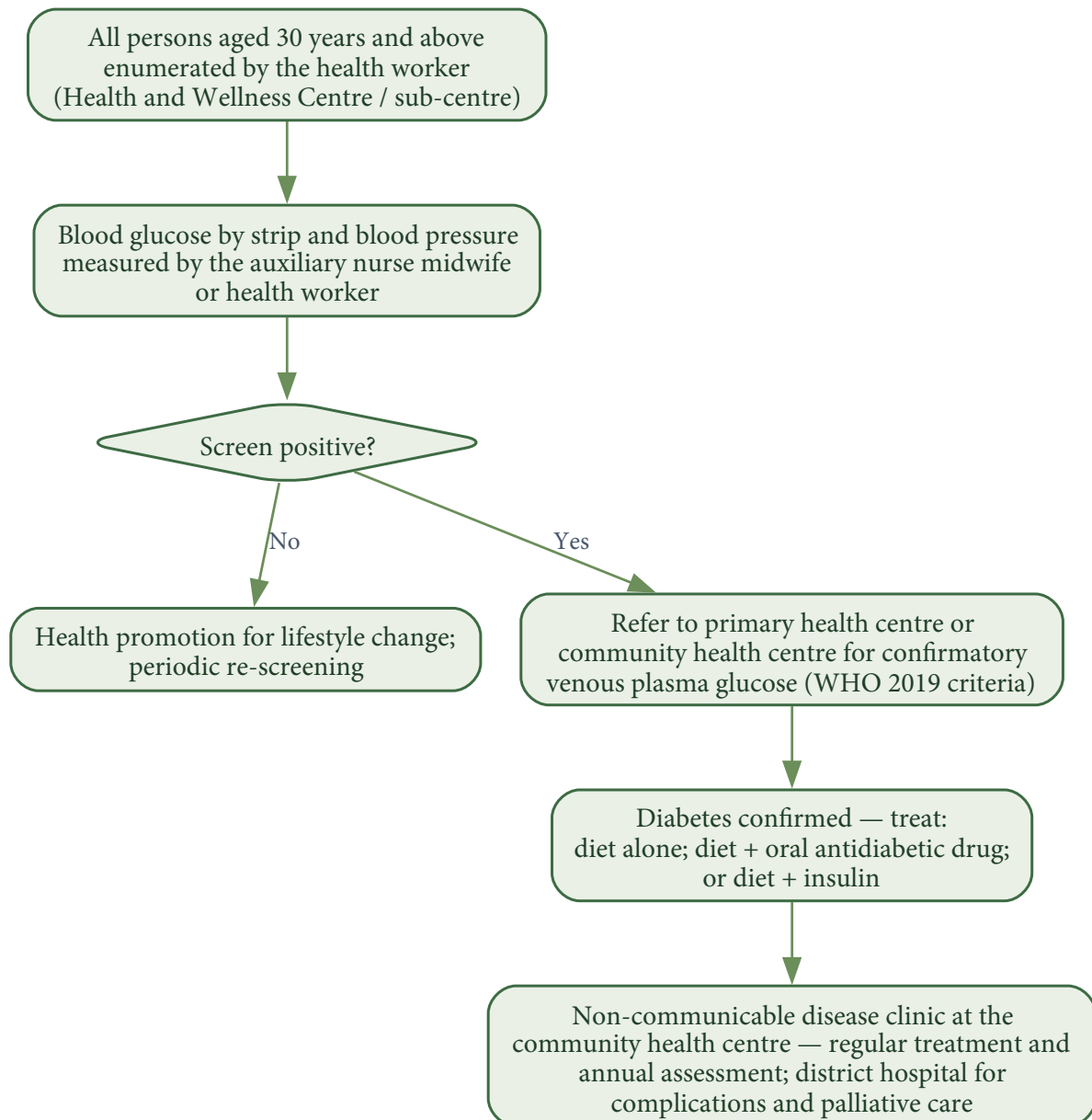


Fig 18.2 — Preclinical phase of the disease. (A) Natural history with point at which disease is detectable by screening. (B) Natural history with detectable preclinical phase. (C) Natural history with lead time. (Gordis Epidemiology 7e, p.371)



Population-based screening pathway for diabetes in persons aged 30 years and above

High-yield exam pointers

- Screening criteria split into **DISEASE (9 points)** and **TEST (acceptability, repeatability, validity; plus yield, simplicity, safety, rapidity, ease of administration and cost)** — name both headings, then answer the disease side, which is what this stem asks.
- Indian prevalence to quote with its source: **urban 10.9–14.9%, rural 3.0–7.8%** in those aged 20 years and above (Park 28e); **12% of women and 14% of men** aged 15 years and over above 140 mg/dL or on medication (NFHS-5, 2019-21).
- WHO 2019 cut-offs: fasting ≥ 126 mg/dL; 2-hour post-load venous ≥ 200 mg/dL; 2-hour post-load capillary ≥ 220 mg/dL; random ≥ 200 mg/dL; HbA1c $\geq 6.5\%$. The capillary 2-hour value of 220 mg/dL is the one most often written wrongly.
- Glycosuria as a screening test: **sensitivity 10–50%, specificity above 90%** — the classic "too many false negatives" example in Park.

- ▶ Park's own high-risk groups for diabetes screening: age **40 years and over**, family history of diabetes, the obese, women who have delivered a baby weighing more than **4.5 kg** (3.5 kg in constitutionally small populations), women with excess weight gain in pregnancy, and patients with premature atherosclerosis.
- ▶ Efficacy figures worth one line each: lifestyle intervention **35–58%** and metformin **25–31%** reduction in incidence; annual eye examination **60–70%** reduction in serious vision loss.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park 28e places the textbook high-risk screening threshold at 40 years and calls whole-population screening "not rewarding", while the Indian service package under Ayushman Bharat sets universal screening at 30 years and above; both statements sit in the same book, and the 30-year figure is the programme answer expected by the examiner. The programme delivering this screening is named in Park 28e as the National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Diseases and Stroke (NPCDCS) and has since been renamed the National Programme for Prevention and Control of Non-Communicable Diseases (NP-NCD) — that rename is not carried in the retrieved corpus, so the older name is safe to write.

Recent advances [RA]

Current as of 2026-08-19 · depth: elaborate · sources: PubMed, MoHFW NP-NCD Operational Guidelines 2023-2030, ICMR-INDIAB, LASI, NFHS-6, WHO global diabetes coverage targets, IDF Diabetes Atlas 2025

Programme and policy updates (India)

- ▶ **Both programme names in this answer have changed, and the programme now carries a numerical target.** NPCDCS is now the National Programme for Prevention and Control of Non-Communicable Diseases (NP-NCD), governed by Ministry of Health and Family Welfare operational guidelines covering 2023–2030; under it the "75/25" initiative commits India to placing 75 million people with hypertension and diabetes on standard care by 2025. In November 2023 the Ayushman Bharat Health and Wellness Centres were renamed Ayushman Arogya Mandirs, with the tagline *Arogyam Paramam Dhanam* — health is the supreme wealth — and NCD care, which includes population-based and opportunistic screening of everyone aged 30 years and above for hypertension, diabetes and oral, breast and cervical cancer, is one of the twelve comprehensive primary health care service packages delivered there. Write NP-NCD and Ayushman Arogya Mandir; about 1.78 lakh sub-health centres and primary health centres had been transformed as of the 2025 ICMR report. [PMID 40878336] [PMID 42257165] [PMID 40991221] [MoHFW 2023]
- ▶ **Criterion 5 — facilities for confirmation of diagnosis — is now a measured quantity rather than an assumption.** An ICMR facility survey of 561 sub-health-centre Ayushman Arogya Mandirs across nine districts in the second half of 2024 found a community health officer or mid-level provider in 94.1%, a glucometer in 83.8%, a blood pressure apparatus in 85% and at least two essential antidiabetic drugs in 78.4%. A parallel readiness assessment of 415 public and private facilities in 19 districts of seven States, surveyed in 2021 and 2023 and scored by WHO's service availability and readiness method with 70% as the pass mark, put public primary health

centres at 73% but sub-centres at 61% and community health centres at 59% — the weakest link sits precisely where the 30-plus screen is performed. [PMID 40991221] [PMID 40536370]

Current national data

- **Quote ICMR-INDIAB, not the older urban–rural split, for national prevalence.** The ICMR-INDIAB national cross-sectional study surveyed 113,043 adults aged 20 years and above drawn from all 31 States and union territories, with fieldwork running from October 2008 to December 2020, and reported weighted prevalences of diabetes 11.4% (95% CI 10.2–12.5), prediabetes 15.3% (13.9–16.6), hypertension 35.5%, generalised obesity 28.6% and abdominal obesity 39.5% — considerably higher than previously estimated, still increasing in most States while stabilising in the more developed ones; commentary on these data puts the headline at over 100 million Indians living with diabetes. Among the 1,032 pregnant women of the same national survey who had a glucose test and no overt diabetes, gestational diabetes by NICE criteria was 22.4% (16.7–28.0), already 19.2% in those tested before 20 weeks of gestation and 23.4% at or after 20 weeks — which is the argument for screening every pregnant woman from early pregnancy rather than only in the second half. [PMID 37301218] [PMID 42344079] [PMID 41454808]
- **The submerged part of the iceberg has been measured.** In the Longitudinal Ageing Study in India, a nationally and State-representative survey of 57,810 adults aged 45 years and older with data collected in 2017–19, diabetes prevalence by self-report or HbA1c $\geq 6.5\%$ was 19.8% (95% CI 19.4–20.2), about 50.4 million people, urban 30.0% against rural 15.0%; only 60.1% were aware of the diagnosis, and among those diagnosed 45.7% had glycaemic control (HbA1c below 7.0%), 58.9% blood pressure control and just 6.4% were taking a lipid-lowering drug. This is the strongest recent Indian number for criterion 2 — a large detectable but undetected pool — and simultaneously the warning that screening without a working treatment cascade buys little. [PMID 40784358]
- **Date every survey figure, because the round has moved on.** NFHS-6 fact sheets covering 2023–24 have been released; the indicators published so far are the maternal, child nutrition, immunisation and childhood-illness set, with mortality indicators absent. The adult blood-glucose figures still citable are therefore NFHS-5, 2019–21 — 12% of women and 14% of men aged 15 years and over above 140 mg/dL or on medication — and they must be written with the round and its years attached. [PMID 42554934]

Elimination, surveillance and disease status

- **Screening is tracked on a national digital register, and coverage no longer equals control.** The National NCD Portal is the information system for population-based screening of the 30-plus population for diabetes, hypertension and oral, breast and cervical cancer; a qualitative study of 19 users in Dakshina Kannada found they valued the accessibility of patient data but were held back by heavy workload, digital incompetence and lack of accountability, and asked for a dedicated data-entry operator. In Tamil Nadu, the doorstep *Makkalai Thedi Maruthuvam* programme was evaluated in a household survey of 6,856 adults aged 30 and above across all 38 districts and showed diabetes screening coverage of 79.4% and hypertension 82.6% with more than 95% of those diagnosed on treatment — yet only 9.8% of the diabetes cases were controlled against 35.4% of the hypertension cases, and cancer screening coverage lagged well behind, which qualifies the "one contact, several screens" argument. [PMID 41555328] [PMID 41398228]

- **The efficacy figure behind criterion 8 is smaller in Indian community conditions than the textbook table suggests.** The Kerala Diabetes Prevention Program, a cluster-randomised trial in 1,007 adults aged 30–60 years with an Indian Diabetes Risk Score of 60 or more, has reported its 9-year follow-up: cumulative diabetes incidence 125 of 419 (30%) with the peer-led lifestyle intervention against 142 of 422 (34%) in controls, relative risk 0.88 (95% CI 0.73–1.08, $p=0.24$) — not statistically significant. Judged on fasting plasma glucose alone the reduction was 28% (RR 0.72, 0.53–0.98, $p=0.036$). Park's 35–58% remains the figure to write as the textbook value, but the honest examination line is that long-term community effect sizes in India are more modest. [PMID 41997166]

Guideline and legislative updates

- **The risk-stratification checklist used in the field performs poorly, and point-of-care HbA1c now has Indian validation.** The Community Based Assessment Checklist, the paper tool applied before testing, was compared against measured values in 32,686 residents aged 30 and above in Mysuru city (data collected 2019–20): the area under the curve was only 0.62 (95% CI 0.61–0.63) for identifying those at risk of diabetes on random capillary blood glucose ≥ 140 mg/dL, with a best Youden index of 0.17 at a checklist cut-off above 4 — the authors recommend that trained frontline staff measure capillary glucose and blood pressure alongside the checklist rather than triage on the score alone. Against that, a point-of-care HbA1c analyser compared with HPLC in 200 outpatients at a tertiary centre in eastern India gave sensitivity 96.7% and specificity 97.1% on capillary blood, with Lin's concordance coefficient 0.96 on capillary blood (0.95 on venous). Note the cut-off tested there was HbA1c $\geq 7\%$ for detecting hyperglycaemia in a clinic population, not the $\geq 6.5\%$ diagnostic threshold in the table above, so it validates the device rather than a screening cut-off — but it does make a non-fasting field test realistic where the classical objection to fasting glucose is that a true fast cannot be assured. [PMID 42555696] [PMID 42344074]
- **The 30-year threshold is under pressure from below, not above.** A national white paper built on 25 expert consultations argues that nearly 25% of Indian adults aged 30–39 are already hypertensive or pre-hypertensive and proposes that a first cardiovascular assessment — blood pressure, glucose, lipids and obesity metrics — be institutionalised at age 30, noting that national screening programmes still predominantly target those over 40. From the diabetes side, the Indian Diabetes Risk Score, which weights higher age heavily, is argued to underestimate risk in young-onset type 2 diabetes, a phenotype that is younger and leaner yet more insulin-resistant in Indians and is linked to maternal undernutrition, low birth weight and rapid childhood weight gain. Both support the answer the examiner wants: 30 years, not Park's textbook 40, is the Indian entry point. [PMID 42033036] [PMID 41550251]

Global and WHO developments

- **There are now numbered global diabetes targets, and the world is short of every one of them.** After the Global Diabetes Compact (April 2021) and resolution WHA74.4, Member States endorsed the first global diabetes coverage targets in 2022: by 2030, 80% of people with diabetes diagnosed; among those diagnosed, 80% with good glycaemic control (HbA1c below 8.0%) and 80% with blood pressure control (below 140/90 mmHg); 60% of those older than 40 years on a statin; and access to affordable insulin and blood glucose self-monitoring for people with type 1

diabetes. A pooled analysis of individual participant data from national surveys in 100 low-, middle- and high-income countries set the 2021 baseline at 63.2% (95% CI 61.8–64.6) diagnosed against that 80% target, met by only 8 of the 100 countries; among those diagnosed, 63.2% (62.1–64.4) achieved glycaemic control (7 countries), 70.8% (69.8–71.9) blood pressure control (15 countries), and only 31.8% (30.4–33.2) were on a statin (8 countries) — statin use is the metric furthest from its target. Diagnosis fell from 69.9% in high-income countries to 35.3% in low-income countries, so the gap that a screening programme has to close is widest in the poorest countries. [PMID 41386244]

- **The diagnostic gap is the specific global argument for a population screen.** WHO puts 828 million adults aged 18 years and over living with diabetes in 2022, 14% of adults against 7% in 1990, with 445 million adults aged 30 and above — about 59% — on no glucose-lowering drug treatment, and undiagnosed cases accounting for 84–97% of that untreated group. The World Health Assembly adopted resolution WHA76.5 on strengthening diagnostics capacity in May 2023, promoting national diagnostics strategies, national essential diagnostics lists and point-of-care and self-testing; glucose and HbA1c both sit on the WHO model list of essential in vitro diagnostics. WHO's package of essential NCD interventions still triggers testing at symptoms or at age over 40 with overweight or obesity, so India's 30-plus universal screen is ahead of the global package. [PMID 41409098]
- **The projection is the closing line to write.** The 11th edition of the IDF Diabetes Atlas (2025) estimates 589 million adults with diabetes worldwide in 2024, projected to reach 853 million by 2050; the accompanying Indian commentary argues that NP-NCD implementation remains fragmented and episodic, and that the response now needs health-systems thinking, user-centred design and digital integration through the Ayushman Bharat Digital Mission rather than programmatic expansion alone. [PMID 42344079]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 4 (Screening for Disease, pp 163-167), Ch 6 (Epidemiology of Chronic Non-communicable Diseases and Conditions, pp 467-471), Ch 7 (Health Programmes in India, pp 564-565), Ch 10 (Health Care of the Community, p 735); Gordis Epidemiology 7e — Ch 18 (Epidemiologic approach to evaluating screening programs, p 371); NCD Combined Manual for Medical Officers, Ministry of Health and Family Welfare, pp 11-20; National Family Health Survey-5 India Report, 2019-21, Adult Health chapter, pp 525-526; WHO recommendations for the diagnostic criteria for diabetes, 2019, as reproduced in Park 28e Table 3. **Recent-advances sources:** 40878336 — Assessing India's NCD preparedness in health systems: methodological considerations, *Indian Journal of Medical Research* 2025;161:748-749, DOI [10.25259/ijmr.1468.2025](https://doi.org/10.25259/ijmr.1468.2025); 42257165 — Renaming of Ayushman Bharat Health and Wellness Centres as Ayushman Arogya Mandirs with a tagline of 'Arogyam Paramam Dhanam' — a step towards holistic care, *Journal of Family Medicine and Primary Care* 2026;15:1459-1460, DOI [10.4103/jfmprc.jfmprc.1966.24](https://doi.org/10.4103/jfmprc.jfmprc.1966.24); 40991221 — Functionality of Ayushman Arogya Mandirs for hypertension and diabetes care in nine districts of India, *Indian Journal of Medical Research* 2025;162:50-52, DOI [10.25259/ijmr.1693.2025](https://doi.org/10.25259/ijmr.1693.2025); 40536370 — Preparedness of public and private health facilities for management of diabetes and hypertension in 19 districts in India, *Indian Journal of Medical Research* 2025;161:327-335, DOI [10.25259/ijmr.755.2024](https://doi.org/10.25259/ijmr.755.2024); 37301218 — Metabolic non-communicable disease health report of India: the ICMR-INDIAB national cross-sectional study (ICMR-INDIAB-17), *The Lancet Diabetes & Endocrinology* 2023;11:474-489, DOI [10.1016/S2213-8587\(23\)00119-5](https://doi.org/10.1016/S2213-8587(23)00119-5) (abstract-only); 41454808 — Prevalence of gestational diabetes mellitus in India: the ICMR-INDIAB national study (ICMR-INDIAB-24), *Indian Journal of Medical Research* 2025;162:460-469, DOI [10.25259/ijmr.1874.2025](https://doi.org/10.25259/ijmr.1874.2025); 40784358 — Prevalence, awareness, treatment, and control of diabetes in India: a nationally representative survey of adults aged 45 years and older, *The Lancet Global Health* 2025;13:e1543-e1552, DOI [10.1016/S2214-109X\(23\)00502-8](https://doi.org/10.1016/S2214-109X(23)00502-8); 42554934 — Trends in child health indicators in India: insights from National Family Health Survey (NFHS)-6, *Indian Pediatrics* 2026, DOI [10.1007/s13312-026-00412-6](https://doi.org/10.1007/s13312-026-00412-6) (abstract-only); 41555328 — A qualitative study on users' perspective on the functioning of the National Non-Communicable Disease Portal in Dakshina Kannada district, Karnataka, *BMC Health Services Research* 2026;26:104, DOI [10.1186/s12913-026-14008-0](https://doi.org/10.1186/s12913-026-14008-0); 41398228 — Makkalai Thedi Maruthuvam: evaluating equity in non-communicable disease care through a doorstep primary health program in Tamil Nadu, India, *BMC Primary Care* 2025;27:17, DOI [10.1186/s12875-025-03064-w](https://doi.org/10.1186/s12875-025-03064-w); 41997166 — Effects of a peer-support lifestyle intervention on diabetes incidence in India (K-DPP): 9-year follow-up of a cluster-randomised, controlled trial, *The Lancet Global Health* 2026;14:103927, DOI [10.1016/S2214-109X\(26\)00057-4](https://doi.org/10.1016/S2214-109X(26)00057-4) (abstract-only); 42555696 — Screening performance of community-based assessment checklist compared with point of care screening tests to identify high-risk individuals for diabetes and hypertension in urban Mysore, India, *Indian Journal of Medical Research* 2026;164:227-232, DOI [10.25259/ijmr.78.2026](https://doi.org/10.25259/ijmr.78.2026) (abstract-only); 42344074 — Validation of the HemoCue HbA1c 501 System for point-of-care HbA1c estimation using venous and capillary blood: a comparative study with HPLC in eastern India, *Indian Journal of Community Medicine* 2026;51:497-502, DOI

[10.4103/ijcm.ijcm_432_25](#); 42033036 — A national hypertension awareness and early screening initiative: "Act at 30 — It's Time for Heart Check", *Journal of Clinical Hypertension* 2026;28:e70203, DOI [10.1111/jch.70203](#); 41550251 — Re-evaluating higher age as a risk factor for diabetes in the Indian population: the emergence of young-onset type 2 diabetes, *Journal of Diabetes and Metabolic Disorders* 2026;25:36, DOI [10.1007/s40200-026-01856-0](#) (abstract-only); 41386244 — Attainment of global diabetes targets in 2021: a pooled analysis of individual-level data from national surveys in 100 low-income, middle-income, and high-income countries, *The Lancet Global Health* 2026;14:e21-e32, DOI [10.1016/S2214-109X\(25\)00393-6](#); 41409098 — Bridging the diagnostic gap in diabetes, *Bulletin of the World Health Organization* 2026;104:50-52, DOI [10.2471/BLT.25.293828](#); 42344079 — A global wake-up call: the 2025 International Diabetes Federation Diabetes Atlas and what it means for India, *Indian Journal of Community Medicine* 2026;51:441-442, DOI [10.4103/ijcm.ijcm_307_25](#). Plus the Ministry of Health and Family Welfare Operational Guidelines, National Programme for Prevention and Control of Non-Communicable Diseases (NP-NCD), 2023-2030.



Mathematical models of infectious diseases

Asked in: Paper II 28-May-2025 — RGUHS MD Community Medicine

Definition — A mathematical model of an infectious disease is a set of equations describing the **population dynamics of transmission** — the time evolution of incidence or prevalence — in terms of measurable parameters such as contact frequency between infectious and susceptible persons and rate of recovery. Because one person's risk depends on whether others are infected, infectious disease epidemiology cannot use the independence assumption of chronic-disease epidemiology; **modelling and prevention by immunisation** are therefore its two most distinctive features (Handbook of Epidemiology, Ch I.1, p 42; Ch II.8, p 863).

Models are used to **predict** the future course of an outbreak — the original motive, prompted by the regular biennial measles peaks of the pre-vaccine era; to **understand** which determinants matter most and so deserve better measurement; to **quantify transmissibility** as a reproduction number and generation time; to **set programme targets** such as the critical vaccination coverage for herd immunity; to **compare interventions before deployment** and support policy in real time during an outbreak; to **estimate burden where surveillance is insufficient**, as for rubella and hepatitis B; and to **teach** infectious disease epidemiology (Giesecke, Ch 11, pp 120-121; Nelson & Williams, Ch 6, pp 170-171; Plotkin's Vaccines 7e, Ch 76, p 2154). Their chief contribution is how they change thinking about the epidemic process, not any one quantitative result (Nelson & Williams, Ch 6, p 170).

Types of models

Type	Structure	Best suited to
Static (Bernoulli-type)	Constant hazard of infection, no transmission mechanism; Bernoulli's 1766 smallpox/variola analysis	Long-term risk from an endemic disease; cannot explain epidemic ebb and flow
Chain binomial	Discrete generations; En'ko, later Reed–Frost, who were also responsible for what may be the first epidemic simulator (a pre-computer mechanical device using small beads)	Households, schools and other tightly interacting units
Compartmental (mass action)	SIR, SIS, SIRS, SEIR, host–vector; ordinary differential equations	Population-level dynamics of immunising infections
Network- and agent-based	Each individual is a node/agent with its own contacts and disease state	Sexually transmitted infections (STIs), pandemic preparedness; needs very large data inputs
Deterministic vs stochastic	Fixed outcome vs repeated computer simulation with chance built in	Stochastic needed early in an epidemic and in small populations

(Nelson & Williams, Ch 6, pp 149-151, 154-159, 167)

Core parameters, and how R_0 is measured

Parameter	Symbol	Meaning	Public health measure acting on it
Transmission risk per contact	β	Probability that one contact transmits; essentially the attack rate per contact	Condoms, face masks, hand washing
Contact rate	κ	Number of such contacts an average person has per unit time	Isolation, quarantine, school closure, fewer sexual partners
Duration of infectivity	D	Mean time a case remains infectious	Early treatment (antibiotics); ineffective for <i>Salmonella</i> carriage
Basic reproduction number	$R_0 = \beta \times \kappa \times D$	Average number of persons directly infected by one case during her entire infectious period , when she enters a totally susceptible population	—
Effective (current) reproduction number	R, also R_t or R_e ; $R = R_0 \times s$	Secondary cases per case at time t, where s = proportion still susceptible	Vaccination and natural infection reduce s
Generation time (serial interval)	—	Mean time from infection of the infector to infection of the infectee	—
Latent period	—	Infection → onset of infectiousness (differs from incubation period: infection → symptom onset)	—

Threshold rule: $R_0 < 1$ — each wave is smaller than the last and the disease dies out; $R_0 = 1$ — endemic state; $R_0 > 1$ — an ever-increasing number of infections, i.e. an epidemic (Giesecke, Ch 11, pp 121-123; Nelson & Williams, Ch 6, pp 151-152).

Nelson & Williams give **at least four methods** of estimating R_0 (Ch 6, pp 169-170):

- ▶ **From the constituent processes** — multiply the measured β , κ and D (in a vector-borne disease, the biting rate, vector survival, gametocyte rate and human recovery rate).
- ▶ **From the initial growth rate of cases** — early infections grow exponentially with the force of infection; comparing the **doubling time** of incidence with the mean **generation time** gives R_0 (if both are 10 days, each case infected two others, so $R_0 = 2$) (Giesecke, Ch 11, pp 125-126).
- ▶ **From the final size of the outbreak** in a closed population.
- ▶ **From equilibrium age-specific prevalence** — at equilibrium $R = 1$, so $R_0 = 1/s$, s being the susceptible fraction on **serological survey** (more robust than symptom-based surveillance); the gross average-age approximation is $R_0 = L/A + 1$ (L = life span, A = mean age at acquisition), so $L = 70$ y and $A = 7$ y give $R_0 = 11$.

For populations with several types of infectious individual (age strata, sexual-activity classes, or latent/active/treated tuberculosis), R_0 is defined as the **spectral radius (largest eigenvalue) of the next-generation matrix** (Nelson & Williams, Ch 6, p 170).

The simple SIR model

The population N is split into three proportions — S susceptible, I infected and infectious, R removed (immune); before the first case $S = 1$, $I = R = 0$. The three differential equations are (Giesecke, Ch 11, pp 126-127):

- ▶ $dS/dt = -\beta \kappa S I$
- ▶ $dI/dt = +\beta \kappa S I - I/D$
- ▶ $dR/dt = + I/D$

Assumptions: a fixed population (nobody enters, leaves or dies); zero latent period, so a person is infectious as soon as infected; infectivity lasting exactly as long as the clinical illness; and homogeneous mixing, so susceptible-with-infectious contacts form the fraction $S \times I$ — the **law of mass action**, borrowed from chemistry (Giesecke, Ch 11, pp 126-128).

Worked example (Giesecke, Ch 11, pp 128-129): $\beta = 0.15$, $\kappa = 12$ contacts per week, $D = 1$ week gives $R_0 = 1.8$. One case in a closed population of 1000 peaks at about 140 new cases per week around week 11 and ends by week 16-17; about 800 become immune and **about 200 escape infection entirely**, because as immunity accumulates more exposures are "thrown away" on non-susceptibles — the mechanism of herd immunity.

Extensions: **SEIR** adds an exposed (latent) compartment, as for measles; **SIS** suits non-immunising carriage; **SIRS** waning immunity; **host-vector** models pair human and mosquito populations for malaria and dengue; adding births and deaths generates the multiannual measles cycles (Nelson & Williams, Ch 6, pp 154-156, 166).

Herd immunity threshold — the key programme application

If proportion p is already immune, one case infects $R_0 - p \cdot R_0$ others; requiring this to be under 1 gives the fundamental vaccination formula:

$$p > 1 - 1/R_0$$

(Giesecke, Ch 11, p 124; Ch 19, pp 230-231; Nelson & Williams, Ch 16, p 518). This level prevents epidemics, though sporadic secondary cases still occur. Distinguish **vaccinated** from **immunised**: the threshold is a proportion of the population *protected*, not a coverage figure, so with a vaccine of imperfect efficacy — measles vaccine given at 9 months has a primary failure rate of about 15% — coverage must exceed the threshold (Nelson & Williams, Ch 16, p 518; Giesecke, Ch 19, p 231).

Pathogen	R_0 (Table 6-1)	Generation time (Table 6-1)	Herd immunity threshold
Measles	7.7; 7.1–29.3; 11–18	9–17 days; 12 days	93–95% for R_0 12–18 — <i>quoted</i> (Ch 16, p 518)
Smallpox	3.2; 3.5–6; 6.87 (95% CI 4.52–10.1)	14–16 days; 16 days (s.d. 4 days); 14–20 days	80–85% for R_0 5–7 — <i>quoted</i> (Ch 16, p 518)
Poliovirus	5–7 (Ch 16, p 518; not in Table 6-1)	—	80–85% — <i>quoted</i> , given together with smallpox (Ch 16, p 518)
Rubella	2.9–7.8; 3.4–5.6	22 days; 15–23 days	~66–87% (derived)
Cholera	2.6; 4–15 (also 5.0, flagged as a reproductive-number, not an R_0 , estimate)	7.1–9.3 days; 7–10 days	~62–93% (derived)
Severe acute respiratory syndrome (SARS)	2.7; 1.2; 2.2–3.6	8.4 days	~17–72% (derived)
Dengue	1.3–6.3	19–22 days; 24 days	~23–84% (derived)
Influenza	1.5–2	3.6 days (s.d. 1.6, seasonal); 2.3 days (H1N1, range 1.5–2.7); 3.1 days (H3N2, range 2.2–4.0); 2.7 days (2009 pandemic H1N1)	~33–50% (derived)
Malaria	1–10 (low-transmission areas) to 100–1000 (high-transmission areas); also 1–3000	~60–120 days; > 200 days	—

(R_0 and generation time: Nelson & Williams, Ch 6, Table 6-1, pp 151-152. Estimates vary considerably by pathogen and from setting to setting.)

Park adds that it is **neither possible nor necessary to reach 100% herd immunity** to halt an epidemic; that herd immunity rests on clinical and subclinical infection, immunisation of the herd, and herd structure, which shifts constantly with births, deaths and migration; that smallpox was eradicated not

by herd immunity but by eliminating the source through surveillance and containment; and that for tetanus herd immunity does not protect the individual at all (Park 28e, Ch 3, pp 122-123).

Applications to public health problems

Problem	Model used	Effect on policy
Malaria eradication	Ross–MacDonald host–vector model	Showed the <i>Anopheles</i> vector need not be eliminated, only substantially reduced — MacDonald held that the worst African conditions could be overcome by raising daily vector mortality from about 5% to about 45%. Underpinned the World Health Organization (WHO) campaign focused on dichlorodiphenyltrichloroethane (DDT), which eliminated malaria transmission among 500 million people
Rubella vaccination	Age-structured dynamic model	Vaccination raises the average age of infection ; partial coverage in a high-birth-rate setting can push infection into the childbearing years and increase congenital rubella syndrome (CRS) . Raising coverage to 80% reduced CRS in every setting examined; such models are used by WHO to advise which countries should add rubella vaccine and what the risks are
Foot-and-mouth disease, United Kingdom 2001	Spatially explicit deterministic and stochastic models, used in real time	Policy changed from culling only on infected farms to culling on farms within a radius of infected farms; the decision remains controversial
Pandemic influenza preparedness	Agent-based models	"Targeted layered containment" (antivirals, school closure, quarantine, social distancing) could cut the overall attack rate by 75% or more if $R_0 \approx 2$ or less; Ferguson et al. estimated containment was still possible if detected after 20 confirmed cases with $R_0 < 1.8$, needing a 3-million-dose antiviral stockpile
2009 H1N1 response	Real-time parameter estimation	By May 2009 gave R_0 1.2–1.6, mean generation time 1.9 days and case fatality below 0.5% — later analyses were largely confirmatory
Human immunodeficiency virus (HIV) burden	Back-calculation from the AIDS incubation-period distribution	Reconstructed United States HIV incidence 1978–1990; projections of AIDS incidence for 1991–1995 were broadly consistent with what was observed
Gonorrhoea and STI control	Core-group and network models	Hethcote and Yorke estimated that screening targeted at the high-activity core group is four times more effective in reducing gonorrhoea incidence than broadly targeted screening
Human papillomavirus (HPV) vaccination	Age-structured compartmental model (17 age groups, 3 sexual-activity levels, both sexes)	Vaccinating both boys and girls at 12 years gives the greatest reduction in HPV incidence and is the most cost-effective strategy

Problem	Model used	Effect on policy
Measles periodicity	Mass-action model (Hamer), later with stochastic and seasonal terms	Explains 2–5 year epidemic cycles; coverage above 80% lengthens the interepidemic period from 2–5 to 4–8 years

(Nelson & Williams, Ch 6, pp 149-151, 170-176; Ch 16, p 512)

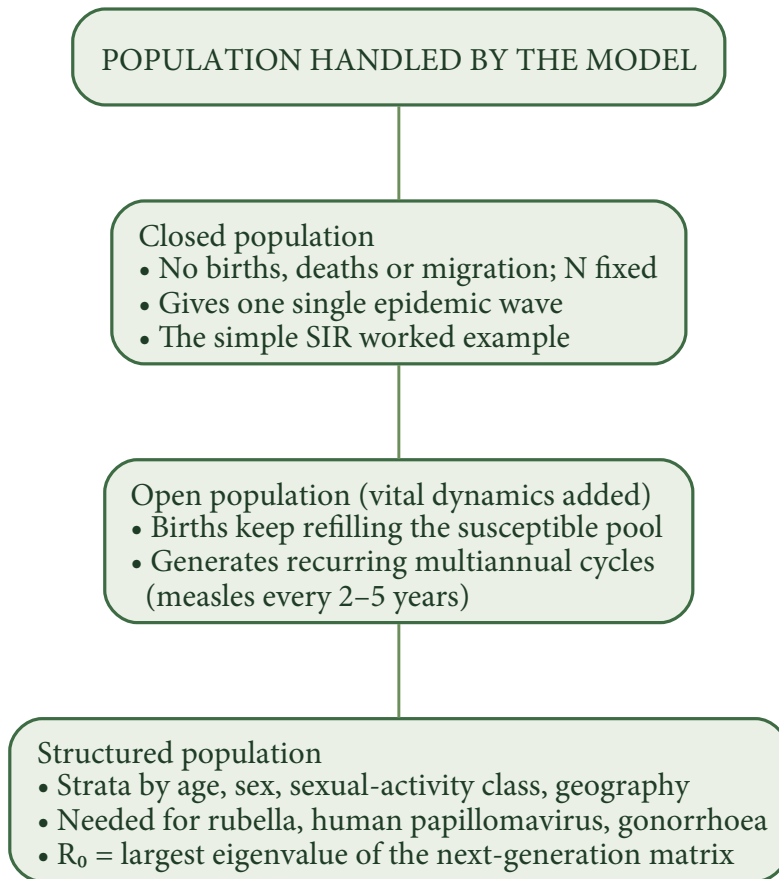
Assumptions violated and limitations

- ▶ **Homogeneous mixing is unrealistic** — family, school, workplace, age and geography decide who meets whom, so most infections spread **more slowly** than the model predicts.
- ▶ **A single contact type with one β is unrealistic** — in HIV, anal intercourse and needle sharing carry a far higher β than vaginal intercourse, so one introduction gives very different epidemics in different subgroups.
- ▶ **The epidemic itself changes the contact pattern** — the hardest thing of all to model.
- ▶ **Deterministic models ignore chance**, which dominates early and in small populations: an index case who meets only immune people starts no epidemic at all.
- ▶ **R_0 is an average** — meaningless when subgroups differ greatly in rate of spread, and it varies from setting to setting.
- ▶ **Data, not equations, are the bottleneck** — "it is much easier to write down the formulae of a model than to collect good data to put into it"; contact-rate data are especially scarce and predictions very sensitive to exact values.
- ▶ **Greater realism costs more assumptions** — agent-based models embed structural assumptions that are hard to communicate, replicate or test by sensitivity analysis. No model is better than its assumptions, which mathematical phrasing makes look more exact than they are.

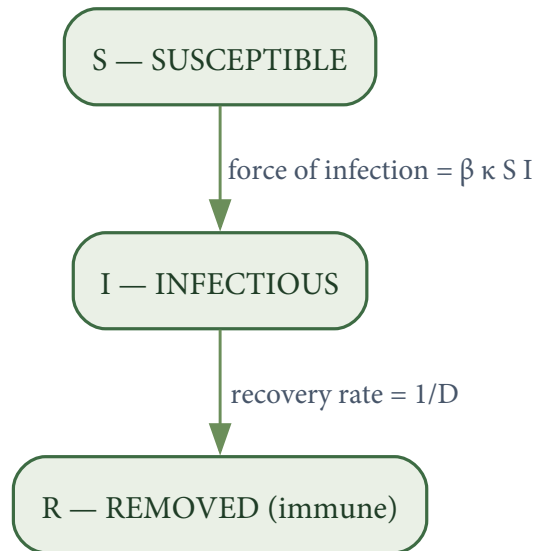
(Giesecke, Ch 11, pp 129-131; Nelson & Williams, Ch 6, pp 158-159, 172-173)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **The SIR epidemic curve in a closed population** (after Giesecke Fig 11.4). Draw three curves on one graph. Labels: y -axis "Number of persons (population 1000)"; x -axis "Weeks"; a falling curve labelled "**Susceptible (S)**" starting at 1000 and levelling at about 200; a rising mirror-image curve labelled "**Removed / immune (R)**" ending near 800; a low hump labelled "**Infectious (I)**" peaking around week 11; an arrow at the peak labelled "**R falls below 1 — epidemic begins to tail off**"; a bracket on the right labelled "**~200 escape infection (herd immunity)**".
- ▶ **Time periods of one infected individual** (after Nelson & Williams Fig 6-3). A single horizontal time line with labels: "**Infection**", "**Latent period**" (infection $\rightarrow$ infectiousness), "**Infectiousness**" (shaded block), "**Incubation period**" (infection $\rightarrow$ symptom onset), "**Symptom onset**", and "**Generation time**" (infection of infector $\rightarrow$ infection of infectee).



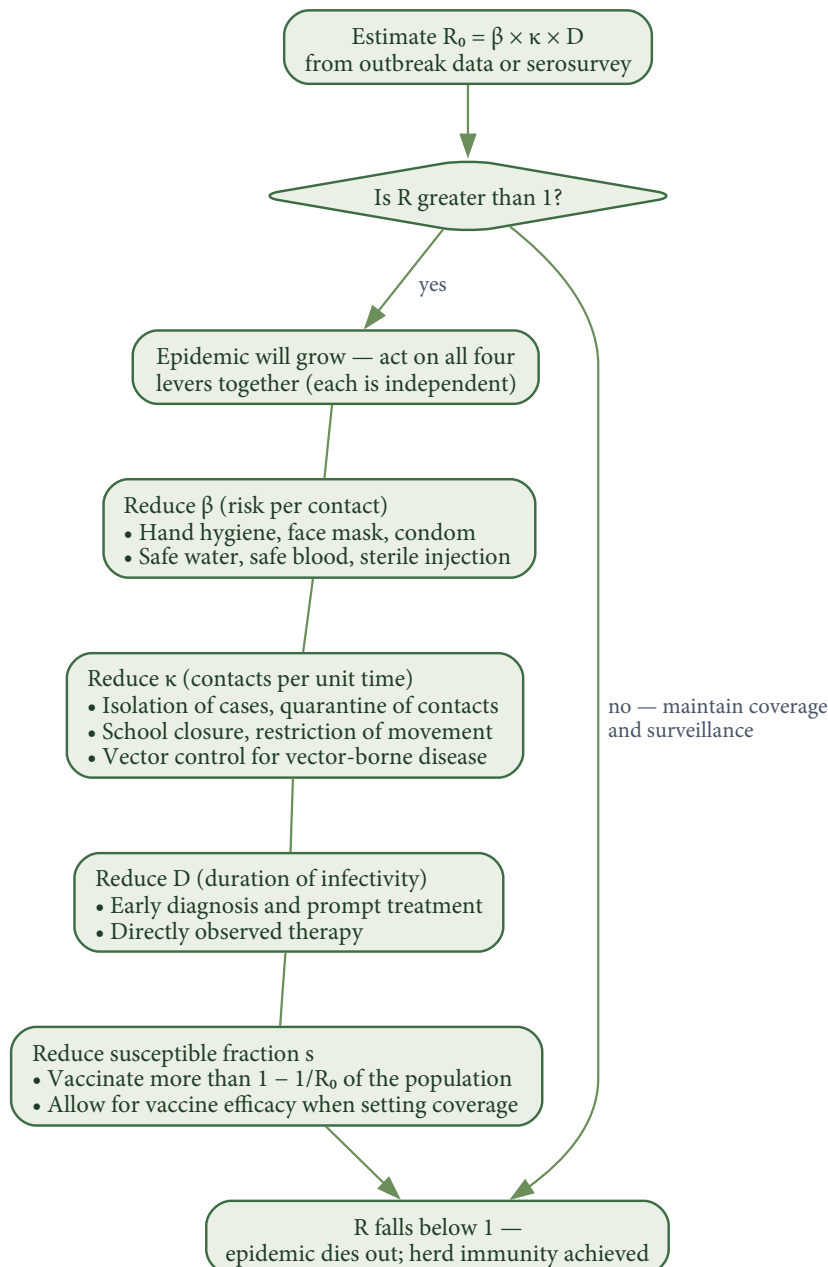
How a model handles the population — the choice that decides which dynamics it can reproduce



VARIANT 1 — SEIR
 Insert E (exposed / latent) BETWEEN S and I,
 i.e. $S \rightarrow E \rightarrow I \rightarrow R$ (e.g. measles)

- OTHER VARIANTS (not part of the chain above)
- SIS — no immunity; I returns to S
 - SIRS — waning immunity; R returns to S
 - Host–vector — paired human and mosquito models (malaria, dengue)

The SIR compartmental model, with its standard variants listed separately



Translating a model into epidemic control — driving R below 1

High-yield exam pointers

- ▶ $R_0 = \beta \times \kappa \times D$; $R = R_0 \times s$; epidemic if $R_0 > 1$, endemic if $R_0 = 1$, dies out if $R_0 < 1$.
- ▶ **Herd immunity threshold** $p > 1 - 1/R_0$; measles R_0 12–18 → 93–95%; smallpox and poliovirus R_0 5–7 → 80–85%. Vaccinated $\neq$ immunised.
- ▶ $R_0 = L/A + 1$; $L = 70$ y, $A = 7$ y → $R_0 = 11$.
- ▶ SIR: $dS/dt = -\beta\kappa SI$; $dI/dt = \beta\kappa SI - I/D$; $dR/dt = I/D$. Variants SIS, SIRS, SEIR, host-vector.
- ▶ Names: **Bernoulli** 1766 (first model, smallpox), **Hamer** (law of mass action), **Ross** (malaria, threshold principle), **En'ko and Reed-Frost** (chain binomial), **Kermack & McKendrick** (critical density of susceptibles), **MacDonald** (Ross-MacDonald vector model).
- ▶ Four ways to estimate R_0 : constituent processes; initial growth rate; final outbreak size; equilibrium age-specific seroprevalence. Structured populations → next-generation matrix.

- Biggest limitation: the contact pattern is unknown, complex and changes during the epidemic.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park 28e covers herd immunity and epidemic curves but not the reproduction-number formulation, so the quantitative content here is grounded on Giesecke Ch 11 and Nelson & Williams Ch 6. In the R_0 table the thresholds for measles (93–95%) and for smallpox and poliovirus (80–85%) are quoted directly from Nelson & Williams Ch 16, p 518, whereas the rows marked "derived" are simple arithmetic from the textbook formula $1 - 1/R_0$ applied to the R_0 ranges in Table 6-1 and are not separately quoted figures.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed

Programme and policy updates (India)

- **BharatSim** is an open-source agent-based framework built specifically for India: it generates a synthetic population from the Census of India, the India Human Development Survey, the National Sample Survey and the Gridded Population of the World, and scales to roughly 50 million agents — the size of an average Indian state. It was used to ask when schools could safely reopen at given levels of hybrid immunity, how new variants behave against a background of prior infection and vaccination, and how non-pharmaceutical interventions perform in a model Indian city — a direct answer to the standard criticism that agent-based models demand data India did not have [PMID 39775067].
- An age-stratified compartmental tuberculosis transmission model for India, extended with body-mass-index strata linked to progression and treatment outcome, estimated that nutritional support for tuberculosis-affected households at 50% coverage would avert 361,200 deaths (95% uncertainty interval 318,000–437,700) and 880,700 disease episodes (802,700–974,900) between 2023 and 2035, at an incremental cost-effectiveness ratio of US\$167 per disability-adjusted life-year averted, with about 24 households needing the intervention to prevent one death — modelling used to argue the nutrition component of the national tuberculosis programme [PMID 39824201].

Current national data

- The first nationwide community-based rabies estimate in two decades: a multistage cluster survey of 337,808 people in 78,807 households across 60 districts in 15 states (March 2022 to August 2023) found an animal-bite incidence of 6.6 per 1000 per year (95% CI 5.7–7.6), about 9.1 million bites nationally, and a decision-tree probability model gave 5726 human rabies deaths per year (95% uncertainty interval 3967–7350). One in five dog-bite victims received no anti-rabies vaccine and about half of those who started one did not finish the course — the arithmetic underlying India's 2030 dog-mediated rabies elimination target [PMID 39362224].
- Paired facility-based (residual blood specimens from diagnostic laboratories) and household serosurveys in Kanpur Nagar, Uttar Pradesh and Palghar, Maharashtra measured measles and rubella IgG before and after a measles–rubella supplementary immunisation activity. Rubella seroprevalence rose significantly in both specimen sources and both districts, but measles trends were inconsistent between the two sources: in Kanpur Nagar the residual specimens showed a

rise in younger children where the community survey showed none, and in Palghar the reverse — the community survey rose in both age groups while the residual specimens did not change. This is the serological route to the susceptible fraction s — the fourth method of estimating R_0 listed above — obtainable cheaply from specimens already in the laboratory, but the discordance makes residual specimens a qualified rather than a clean substitute for a household survey [PMID 39552145].

Elimination, surveillance and disease status

- COMET-LF, a new compartmental model of lymphatic filariasis calibrated to microfilaria and disease prevalence in ten endemic districts of Bihar, projects that six rounds of mass drug administration at 85% coverage suffice for elimination where baseline microfilaria prevalence is up to 10% — agreeing with the older EPIFIL and LYMFASIM models — but that lymphoedema and hydrocele prevalence falls only 8–11.5% over 20 years after five rounds, because chronic disease outlives transmission. Run as a two-patch metapopulation model, migration from a high-prevalence neighbouring area can require up to 4–5 extra rounds of the two-drug diethylcarbamazine–albendazole regimen (2–3 extra with the three-drug ivermectin-containing regimen), and a disease-free block adjacent to a region above 5% microfilaria prevalence carries a significant resurgence risk [PMID 41199190].
- India's reported kala-azar incidence reached the elimination threshold of under 1 case per 10,000 population at subdistrict (block) level in 2023; independent programme assessments in 2019, 2021 and 2023 credited political commitment and intensified implementation, while flagging weak surveillance, poor data use and patchy vector control [PMID 40037366]. Post-elimination surveillance in two endemic villages of Bihar and Jharkhand nonetheless found rK39 seroprevalence of 20–30% and *Leishmania*-infected sand-fly pools (1.2–1.4%) despite indoor residual spraying, which the authors read as persisting low-level transmission — in modelling terms, R pushed just below 1 rather than to zero [PMID 41324335].

Guideline and legislative updates

- An age-stratified compartmental transmission model calibrated to Indian tuberculosis epidemiology projected 2025–2050 vaccination scenarios: M72/AS01E averted about 40% more cases and deaths than BCG revaccination, at an average incremental cost of US\$190 million versus US\$23 million per year, and nearly all scenarios were cost-effective. The dominant uncertainties were the vaccine's mechanism of effect and whether efficacy requires prior infection at the time of vaccination [PMID 37542319].
- Simulation models fitted to country-specific data, using single-dose protection scenarios derived from the International Agency for Research on Cancer India vaccine trial, found that — assuming lifelong single-dose protection — switching from two-dose to one-dose human papillomavirus vaccination and reallocating the freed doses to a female catch-up campaign could prevent an additional 467–1336 cervical cancers per 100,000 women born in cohorts aged 11–30 years in India; reallocating instead to improve routine coverage was enough to reach elimination in India [PMID 39529528].

Global and WHO developments

- A multiple-strain metapopulation model of global SARS-CoV-2 transmission, calibrated to epidemiological, phylogenetic and high-resolution air-travel data, showed that adding traveller-

focused genomic surveillance at a small number of highly connected hubs shortens the delay from a variant's emergence to its first sequencing while using less total surveillance effort. The practical version — enhancing traveller surveillance at the two largest-volume hubs without moving resources away from anywhere else — cut the mean detection lag from 14.1 to 11.5 days in forward simulations of a novel variant; the authors stress it complements rather than replaces local sequencing capacity [PMID 41832161].

- A PRISMA systematic review of 67 studies (Semantic Scholar, 2018 onward) of artificial intelligence in early-warning systems for infectious disease surveillance found machine learning, deep learning and natural language processing combining epidemiological, web, climate and wastewater data streams, with earlier outbreak detection and better prediction accuracy as the main gains; data quality and bias, the "black box" transparency problem and integration into existing public health systems remain the limiting factors [PMID 40626156].

SOURCE & COVERAGE

Giesecke — *Modern Infectious Disease Epidemiology* (CRC Press), Ch 11 "Mathematical Models for Epidemics" and Ch 19 "The Epidemiology of Vaccination"; Nelson & Williams — *Infectious Disease Epidemiology*, Ch 6 "Infectious Disease Dynamics" (incl. Table 6-1 and Box 6-2) and Ch 16 "Measles"; Park's Textbook of Preventive and Social Medicine 28e, Ch 3 "Principles of Epidemiology and Epidemiologic Methods"; Handbook of Epidemiology (Ahrens & Pigeot), Ch I.1 "Basic Concepts" and Ch II.8 "Geographical Epidemiology"; Plotkin's Vaccines 7e, Ch 76 "Immunization in Developing Countries".

Recent-advances sources: 39775067 — BharatSim: an agent-based modelling framework for India, *PLoS Comput Biol* 2024, doi:[10.1371/journal.pcbi.1012682](https://doi.org/10.1371/journal.pcbi.1012682); 39824201 — Estimating the epidemiological and economic impact of providing nutritional care for tuberculosis-affected households across India: a modelling study, *Lancet Glob Health* 2025, doi:[10.1016/S2214-109X\(24\)00505-9](https://doi.org/10.1016/S2214-109X(24)00505-9); 39362224 — Estimates of the burden of human rabies deaths and animal bites in India, 2022-23, *Lancet Infect Dis* 2024, doi:[10.1016/S1473-3099\(24\)00490-0](https://doi.org/10.1016/S1473-3099(24)00490-0) (abstract-only); 39552145 — Increased measles and rubella seroprevalence in children using residual blood samples from health facilities and household serosurveys after supplementary immunization activities in two districts in India, *Epidemiol Infect* 2024, doi:[10.1017/S0950268824001353](https://doi.org/10.1017/S0950268824001353); 41199190 — COMET-LF: a compartmental model of dynamics of infection, disease, and elimination strategies for lymphatic filariasis, *BMC Infect Dis* 2025, doi:[10.1186/s12879-025-11691-y](https://doi.org/10.1186/s12879-025-11691-y); 40037366 — Kala-azar elimination in India: reflections on success and sustainability, *Int Health* 2025, doi:[10.1093/inthealth/ihaf013](https://doi.org/10.1093/inthealth/ihaf013); 41324335 — High seropositivity to rK39 and capture of *Leishmania*-infected sand flies in Bihar and Jharkhand emphasize the need for post-elimination surveillance of visceral leishmaniasis in India, *J Infect Dis* 2026, doi:[10.1093/infdis/jiaf543](https://doi.org/10.1093/infdis/jiaf543) (abstract-only — the PMC record carries no full text); 37542319 — New tuberculosis vaccines in India: modelling the potential health and economic impacts of adolescent/adult vaccination with M72/AS01E and BCG-revaccination, *BMC Med* 2023, doi:[10.1186/s12916-023-02992-7](https://doi.org/10.1186/s12916-023-02992-7); 39529528 — Leveraging single-dose human papillomavirus vaccination dose-efficiency to attain cervical cancer elimination in resource-constrained settings, *J Natl Cancer Inst Monogr* 2024, doi:[10.1093/jncimonographs/lgae035](https://doi.org/10.1093/jncimonographs/lgae035); 41832161 — Optimizing global genomic surveillance for early detection of emerging SARS-CoV-2 variants, *Nat Commun* 2026, doi:[10.1038/s41467-026-70664-0](https://doi.org/10.1038/s41467-026-70664-0); 40626156 — Artificial intelligence in early warning systems for infectious disease surveillance: a systematic review, *Front Public Health* 2025, doi:[10.3389/fpubh.2025.1609615](https://doi.org/10.3389/fpubh.2025.1609615).

♦ ♦ ♦

Cohort study — design, merits and demerits

Asked in: Paper I 12-Nov-2019 — RGUHS MD Community Medicine

Definition — A cohort study is an analytical (observational) study in which a *cohort* — a group of people who share a common characteristic or experience within a defined time period — is identified **before** the disease under study appears, classified according to exposure to a suspected causal factor, and then observed forward in time to compare the frequency of the outcome in the exposed and non-exposed groups (Park 28e, Ch 3, p 95). It therefore proceeds from **cause to effect**, the reverse of the case control study, and is also called a prospective, longitudinal, incidence or forward-looking study.

Design framework, prerequisites and indications

Framework of a cohort study (Park 28e, Ch 3, Table 16, p 95):

Cohort	Disease: yes	Disease: no	Total
Exposed to putative aetiologic factor (<i>study cohort</i>)	a	b	a + b
Not exposed to putative aetiologic factor (<i>control cohort</i>)	c	d	c + d

Conditions to be satisfied while assembling the cohorts (Park 28e, Ch 3, p 95):

- ▶ Both cohorts must be **free of the disease under study** at entry — members already showing evidence of it are excluded after examination.
- ▶ Study and control cohorts should be **equally susceptible** to the disease, or efficiently reflect any difference in its occurrence (e.g. males over 35 years for lung cancer).
- ▶ The two groups should be **comparable for all other variables** influencing the frequency of the disease.
- ▶ **Diagnostic and eligibility criteria** must be defined beforehand, and depend on a reliable method of recognising the disease when it develops.

A cohort design is indicated when (a) there is already good evidence of an association from clinical observation, descriptive and case control studies; (b) the **exposure is rare but incidence among the exposed is high** (industrial workers, those exposed to X-rays); (c) attrition can be minimised — follow-up easy, cohort stable and accessible; and (d) ample funds are available (Park 28e, Ch 3, p 95).

Types of cohort study by time frame

Type	Position of outcome when study begins	Direction of work	Classical example
Prospective (current/concurrent) cohort	Outcome has not yet occurred ; study begins in the present and continues into the future	Assemble now, follow forward	Framingham Heart Study; Doll and Hill's study of smoking and lung cancer; Royal College of General Practitioners' study of oral contraceptives (Park 28e, Ch 3, p 96)

Type	Position of outcome when study begins	Direction of work	Classical example
Retrospective (historical / non-concurrent prospective) cohort	All outcomes have already occurred before the investigation starts	Go back 10–30 years into existing employment or medical records, trace forward to the present	Records-based occupational cohorts (Park 28e, Ch 3, p 96)
Combined (ambispective) cohort	Cohort identified from past records and assessed for outcome to date, then followed prospectively for further outcome	Both backward and forward	<i>Court-Brown and Doll</i> (1957): 13,352 patients given radiation therapy for ankylosing spondylitis between 1934 and 1954, deaths from leukaemia or aplastic anaemia 1935–1954, with a prospective component added from 1955 (Park 28e, Ch 3, p 96)

Steps (elements) in the design of a cohort study

The elements of a cohort study are five (Park 28e, Ch 3, pp 96–98):

- ▶ **Selection of study subjects.** Subjects are assembled in one of two ways —
 - *General population:* used when the exposure or cause of death is fairly frequent; the cohort is drawn from a well-defined geographical, political or administrative area (Framingham Heart Study). If the population is very large an appropriate **sample** is taken, so results can be generalised.
 - *Special groups:* (i) **select groups** — doctors, nurses, lawyers, teachers, civil servants, insured persons, an obstetric population, college alumni, government employees, volunteers; homogeneous and easy to follow for a protracted period (Doll used British doctors). (ii) **exposure groups** — chosen for special exposure to a physical or chemical agent, e.g. industrial workers, radiologists exposed to X-rays; more economical when the exposure is rare and it permits grading by degree and duration of exposure.
- ▶ **Obtaining data on exposure.** Sources are — (a) **the cohort members themselves**, by personal interview or mailed questionnaire (simple and economical for the large numbers involved; Doll and Hill used them for British doctors); (b) **review of records**, for dose of radiation, type of surgery or details of treatment; (c) **medical examination or special tests** — blood pressure, serum cholesterol, electrocardiogram; (d) **environmental surveys**, the best source of exposure levels where the cohort lived or worked. Data must allow classification of members (i) as exposed or not exposed and (ii) by degree of exposure in broad classes; demographic variables affecting disease frequency are also recorded for later analysis.
- ▶ **Selection of comparison groups.** Three options —
 - *Internal comparison:* no outside group is needed; a single cohort enters the study and its members are classified into comparison groups by level of exposure (smoking, blood pressure, serum cholesterol) **before** the disease develops, then compared for subsequent morbidity and mortality.

- *External comparison*: used when information on degree of exposure is not available — e.g. smokers versus non-smokers, a cohort of radiologists versus a cohort of ophthalmologists. Study and control cohorts must be similar in demographic and other important variables.
 - *Comparison with general population rates*: the mortality experience of the exposed group is compared with that of the general population of the same geographical area (lung cancer in uranium miners versus the general population where they resided; cancer in asbestos workers versus local rates). Limiting factors are non-availability of population rates for the required outcome, and difficulty in ensuring that the study and comparison groups represent the exposed and non-exposed segments of the general population.
- **Follow-up.** Procedures are decided at the start and depend on the outcome — (a) **periodic medical examination** of each cohort member, (b) **review of physician and hospital records**, and (c) **routine surveillance of death records**. Losses are inevitable through death, change of residence, migration or loss of interest, and such losses may bias the result; the design must therefore build in a system for obtaining basic outcome information on those who cannot be followed in detail. The safest course is to achieve a follow-up as close as possible to **95 per cent**.
- **Analysis.** The data are analysed in terms of (a) incidence rates of the outcome among the exposed and the non-exposed, and (b) estimation of risk — relative risk and attributable risk (detailed below).

Measures computed at analysis

Measure	Formula	Worked value in Park's example	Interpretation
Incidence rate in each group	$a/(a+b)$ among exposed; $c/(c+d)$ among non-exposed	$70/7000 = 10$ per 1000 in smokers; $3/3000 = 1$ per 1000 in non-smokers ($P < 0.001$)	Cohort design alone yields incidence directly in both groups (Park 28e, Ch 3, Table 18, p 98)
Relative risk (RR)	Incidence among exposed $\div$ incidence among non-exposed	$10 \div 1 = 10$	Direct index of the strength of association. $RR = 1$ no association; $RR > 1$ positive association; RR of 2 means twice the incidence; RR of 0.25 means a 75 per cent reduction. A high RR does not by itself imply causation; quote the 95 per cent confidence interval (Park 28e, Ch 3, p 98)
Attributable risk (AR), per cent	(Incidence among exposed – incidence among non-exposed) $\div$ incidence among exposed $\times 100$	$(10 - 1)/10 \times 100 = 90$ per cent	The proportion of disease in the exposed attributable to the exposure — i.e. how much could be eliminated if the factor were removed (Park 28e, Ch 3, p 98)

Measure	Formula	Worked value in Park's example	Interpretation
Population attributable risk (PAR), per cent	$(\text{Incidence in total population} - \text{incidence among non-exposed}) \div \text{incidence in total population} \times 100$	UK physicians, lung cancer deaths per 100,000: heavy smokers 224, non-smokers 10, total population 74 $\rightarrow$ RR 22.4 and PAR $(74 - 10)/74 = 86$ per cent	The reduction achievable in the whole community if the factor were eliminated — the measure of public-health importance (Park 28e, Ch 3, Table 19, p 99)

Merits

#	Merit	Basis in the design
1	Incidence can be calculated directly in exposed and non-exposed	Disease-free groups are followed forward, so new cases are counted (Park 28e, Ch 3, p 99)
2	Several outcomes can be studied simultaneously for one exposure	Cohorts set up for smoking and lung cancer also demonstrated associations with coronary heart disease, peptic ulcer and cancer of the oesophagus (Park 28e, Ch 3, p 99)
3	Provides a direct estimate of relative risk	Both incidence rates are measurable, so no surrogate such as the odds ratio is needed (Park 28e, Ch 3, p 99)
4	Dose-response relationships can be calculated	Members can be graded by level and duration of exposure at entry (Park 28e, Ch 3, p 99)
5	Certain biases are minimised — notably misclassification of individuals as exposed or non-exposed, and recall bias	Comparison groups are formed before the disease develops, so exposure is recorded independently of the outcome (Park 28e, Ch 3, p 99; Gordis 7e, Ch 9, p 210)
6	The design of choice when the exposure is rare	A case control study is unlikely to capture enough exposed persons (Gordis 7e, Ch 9, p 210)
7	Best observational evidence of the temporal sequence (exposure precedes outcome)	Exposure is ascertained while the cohort is still disease-free (Gordis 7e, Ch 9, p 210)

Demerits

#	Demerit	Practical consequence
1	Requires a large number of subjects; unsuitable for uncommon diseases or those of low incidence	Too few events accrue for a valid analysis; a case control design is preferred for rare disease (Park 28e, Ch 3, p 99; Gordis 7e, Ch 9, p 210)
2	Long duration — 20–30 years or more in cancer studies	Investigators may die and participants may change their exposure classification before results emerge; even for common chronic disease such as coronary heart disease it is hard to keep large numbers under indefinite surveillance (Park 28e, Ch 3, p 99)
3	Administrative problems are inevitable — loss of experienced staff, loss of funding, extensive record keeping	Continuity of the study is threatened (Park 28e, Ch 3, p 99)

#	Demerit	Practical consequence
4	Loss to follow-up of a substantial proportion of the original cohort — migration, loss of interest, refusal to give information	Differential loss biases the incidence estimates; hence the 95 per cent follow-up target (Park 28e, Ch 3, pp 98–99)
5	Selection of a representative comparison group is a limiting factor; volunteers may not represent all persons with the characteristic of interest	Non-comparability of study and control cohorts confounds the result (Park 28e, Ch 3, p 99)
6	Diagnostic criteria and standard methods may change during prolonged follow-up	Once the protocol is fixed, new knowledge or new tests are difficult to introduce (Park 28e, Ch 3, p 99)
7	Expensive , because of the infrastructure needed to follow a large population over time	Cost is the commonest reason a cohort study is not undertaken (Park 28e, Ch 3, p 99; Gordis 7e, Ch 9, p 210)
8	The study itself may alter behaviour	Being observed for smoking and lung cancer may induce subjects to stop or reduce smoking, diluting the association (Park 28e, Ch 3, p 99)
9	Ethical problems arise as evidence accumulates	Once the implicating factor is reasonably established, the investigator is obliged to intervene and reduce or eliminate the exposure (Park 28e, Ch 3, p 99)
10	Practical constraints force concentration on a limited number of factors	Other potentially relevant exposures go unmeasured (Park 28e, Ch 3, p 99)
11	Greater potential for bias in ascertaining the outcome than in a case control study	Special surveillance of cohort members may detect more disease than would be found in the general population — valid for internal comparison but it invalidates external comparison with general-population rates (Gordis 7e, Ch 9, p 210; Handbook of Epidemiology, Ch 5, pp 285–286)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Reproduce the same figure by hand with these exact labels:

- ▶ Top line: "Start with:" — two boxes side by side, **EXPOSED** and **NOT EXPOSED**, joined by the word **AND**.
- ▶ Left margin below: "Then, follow up for:"
- ▶ Four boxes in the lower row, two under each upper box: **DISEASE**, **NO DISEASE** (under **EXPOSED**); **DISEASE**, **NO DISEASE** (under **NOT EXPOSED**).
- ▶ Arrow of time drawn left to right beneath, labelled "**cause → effect**", with the 2 × 2 cell letters a, b, c, d written into the four lower boxes.

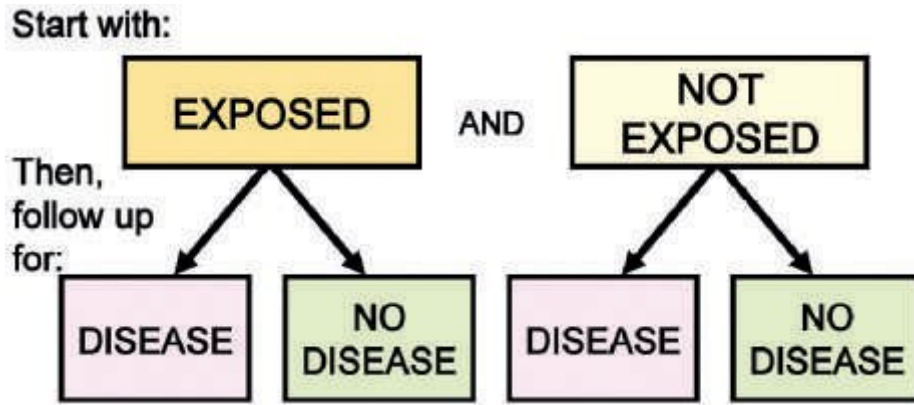


Fig 8.3 — Design of a cohort study beginning with exposed and unexposed groups. (Gordis Epidemiology 7e, p.196)

1. SELECTION OF STUDY SUBJECTS

- General population of a defined area (e.g. Framingham)
- Select groups: doctors, nurses, insured, alumni
- Special exposure groups: industry, radiologists
- All must be free of the disease at entry

2. OBTAINING DATA ON EXPOSURE

- Interview or mailed questionnaire
 - Review of employment / medical records
 - Medical examination or special tests
 - Environmental survey
- Classify: exposed vs not, and by degree of exposure

3. SELECTION OF COMPARISON GROUP

- Internal comparison (levels within one cohort)
- External comparison (separate unexposed cohort)
- Comparison with general population rates

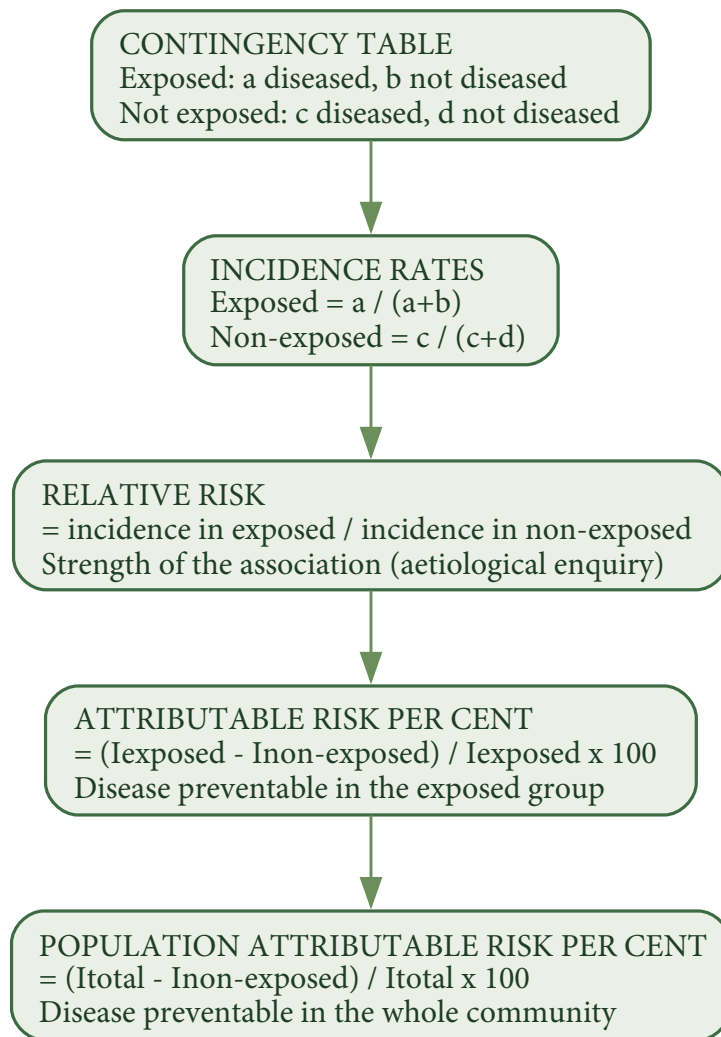
4. FOLLOW-UP

- Periodic medical examination
 - Review of physician and hospital records
 - Routine surveillance of death records
- Target: follow-up as close to 95 per cent as possible

5. ANALYSIS

- Incidence in exposed and in non-exposed
- Relative risk, attributable risk, population attributable risk

The five elements (steps) in the design of a cohort study



Analysis pathway of a cohort study, from the contingency table to the measures of risk

High-yield exam pointers

- **Five elements, in order** — Subjects, Exposure data, Comparison group, Follow-up, Analysis (mnemonic: S-E-C-F-A).
- **Three types** — prospective (current), retrospective (historical / non-concurrent prospective), and combined (Court-Brown and Doll, 13,352 ankylosing spondylitis patients irradiated 1934–1954).
- Follow-up target: as close to **95 per cent** as possible; losses are the chief threat to validity.
- **Relative risk = incidence in exposed ÷ incidence in non-exposed; attributable risk % = (I_e – I_o)/I_e × 100; population attributable risk % = (I_{total} – I_o)/I_{total} × 100.** Park's worked figures: RR = 10 and AR = 90 per cent for smoking and lung cancer; PAR = 86 per cent for lung cancer deaths in UK physicians.
- Relative risk answers "how strong is the association?"; attributable risk answers "how much disease can we prevent?" — the latter is the public-health measure.
- Three ways to build a comparison group: **internal, external, general population rates.**
- Cohort study is the design of choice for a **rare exposure**; the case control study for a **rare disease**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park lists the five steps under the heading "Elements of a cohort study" — writing them as "steps in the design" is the same list, and the RGUHS stem uses the word "design"; Park's worked example for relative risk and attributable risk (Table 18) is explicitly hypothetical, whereas the population attributable risk example (Table 19, UK physicians) uses observed data.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e, Ch 3 (Principles of Epidemiology and Epidemiologic Methods), pp 95–99; Gordis Epidemiology 7e, Ch 8 (Cohort studies) and Ch 9 (Comparing cohort and case-control studies), pp 196–211; Handbook of Epidemiology (Ahrens & Pigeot), Ch 5 (Cohort Studies), pp 285–286.

**Nested case-control study**

Asked in: Paper I 07-Jul-2023 — RGUHS MD Community Medicine

Definition — A **nested case-control study** is a hybrid analytical design in which a case-control study is "nested" inside a defined, enumerated cohort: all incident cases arising in the cohort during follow-up become the **cases**, and the **controls** are a random sample of cohort members who were still free of the disease and at risk at the moment each case was diagnosed (Handbook of Epidemiology, Ch 7, p 333). Because cases and controls are both drawn from the same cohort — the same study base — the design is itself a form of cohort study, and the expensive exposure measurement is made only on the sampled subset instead of on the whole cohort (Gordis 7e, Ch 8, p 207).

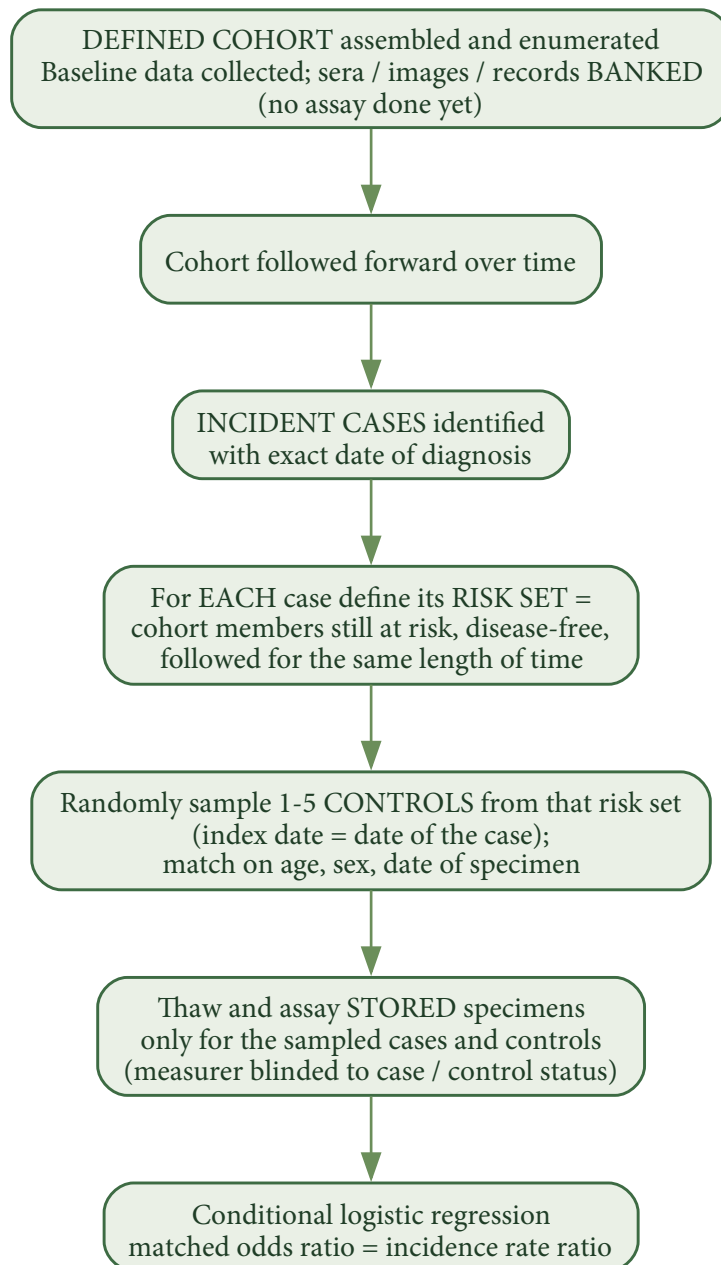
Design features

Feature	Nested case-control study
Study base	A single enumerated cohort with entry and exit times known, in which baseline data, sera, images or records have already been banked (Handbook of Epidemiology, Ch 12, pp 482–483)
Cases	All (or a sample of) incident cases occurring in the cohort between entry into the cohort and the end of the defined follow-up period (Browner, Designing Clinical Research 5e, Ch 9)
Controls	A random sample of those in the cohort at risk at the time of diagnosis of each case , optionally matched to that case on age, sex, date of specimen collection and other factors (Handbook of Epidemiology, Ch 7, p 333)
Sampling method	Risk-set (incidence-density) sampling — one risk set per case, comprising cohort members followed for the same length of time who had not yet become cases (Browner 5e, Ch 9)
Exposure measurement	Made after the cases have occurred, but on material collected before disease onset — stored specimens are thawed and assayed only for the selected cases and controls (Gordis 7e, Ch 8, p 207)

Feature	Nested case-control study
Effect measure	Exposure odds ratio , an unbiased estimate of the incidence rate ratio when controls are sampled by risk set (Handbook of Epidemiology, Ch 7, p 333)
Analysis	Conditional logistic regression , since controls are matched to cases on follow-up time (Browner 5e, Ch 9)

Steps in conducting a nested case-control study

- ▶ **Assemble and enumerate the cohort** and record the date of entry for every member; obtain baseline exposure data and **bank** biological specimens, images or records for later analysis (Browner 5e, Ch 9).
- ▶ **Define the outcome and the follow-up period** — the outcome must occur after the date of entry into the cohort and before the end of follow-up.
- ▶ **Follow the cohort forward** until sufficient incident cases accrue, with outcome ascertainment by staff blinded to exposure status.
- ▶ **Identify all incident cases together with their exact dates of occurrence** during follow-up.
- ▶ **Sample the controls from the risk set of each case** — those alive, still under follow-up and disease-free on the date that case occurred (the *index date* for the control); commonly one to five controls per case, matched on region, age and sex (Browner 5e, Ch 9).
- ▶ **Measure the exposure in the cases and controls only** — thaw and assay the stored specimens, or abstract the records — with the measurer blinded to case-control status (Gordis 7e, Ch 8, p 207).
- ▶ **Analyse** as a matched case-control study by conditional logistic regression, and report the adjusted odds ratio with its 95% confidence interval.



Sequence of a nested case-control study within a defined cohort

Control selection and the measure of effect

- ▶ A control is selected **each time a case develops**, so cases and controls are in effect **matched on calendar time and on length of follow-up**, that is, on the opportunity for the disease to occur (Gordis 7e, Ch 8, p 204; Handbook of Epidemiology, Ch 7, p 333).
- ▶ A person **selected as a control early in follow-up may later become a case** in the same study, and a case may have served as a control in an earlier risk set — this is correct, not an error (Gordis 7e, Ch 8, p 204).
- ▶ Because sampling is by incidence density, the exposure odds ratio estimates the **incidence rate ratio directly, without invoking the rare-disease assumption** (Handbook of Epidemiology, Ch 7, p 333).

- If instead controls are chosen so that they can never become cases, the analysis reverts to that of a **cumulative-incidence case-control study**, and the odds ratio estimates the risk ratio only under the rare-disease assumption (Handbook of Epidemiology, Ch 7, p 333).
- The design accommodates **time-varying exposure and incomplete or variable follow-up**, which a single baseline measurement in a plain cohort analysis cannot handle; this variant is the **incidence-density nested case-control study** (Browner 5e, Ch 9).

Advantages and limitations

Advantages	Limitations
No recall bias — exposure histories are determined before the cases are diagnosed (Handbook of Epidemiology, Ch 7, p 328)	Exposure data and/or specimens must still be collected on the entire cohort at baseline , so data-collection costs exceed those of a traditional case-control study (Handbook of Epidemiology, Ch 7, p 328)
Correct temporal sequence — exposure is known to precede disease, which is appropriate for causal inference (Handbook of Epidemiology, Ch 7, p 328)	Needs a pre-existing enumerated cohort , and the foresight to bank the right specimens under storage conditions that preserve them for years (Browner 5e, Ch 9)
Selection bias minimised — cases and controls come from one study base, giving greater comparability than a hospital-based case-control study (Gordis 7e, Ch 8, p 207)	Total study time, measured from assembly of the cohort, is long , so it is less suitable than a traditional case-control study for very rare diseases or very long latent periods (Handbook of Epidemiology, Ch 7, p 328)
Fatal cases are not excluded , and non-participation bias is reduced, because exposure data already exist for everyone (Handbook of Epidemiology, Ch 7, p 328)	Residual confounding — the parent cohort may not have collected the covariates the nested question needs; collecting them afterwards reintroduces recall bias (Handbook of Epidemiology, Ch 7, pp 328-329)
Cost-efficient — costly assays, record abstraction or database linkage are done on a few hundred people instead of the whole cohort (Gordis 7e, Ch 8, p 207; Fletcher, Clinical Epidemiology, Ch 6, p 102)	Silent preclinical disease at baseline can distort the baseline measurement (effect-cause rather than cause-effect); minimised by excluding cases diagnosed within a short defined period after baseline (Handbook of Epidemiology, Ch 7, p 329)
Statistically more efficient than a full cohort analysis — detects the same difference as significant with a smaller sample (Handbook of Epidemiology, Ch 7, p 328)	Differential loss of cases and controls from the cohort can still introduce selection bias, exactly as differential loss to follow-up does in a cohort study (Handbook of Epidemiology, Ch 7, p 328)
Yields the whole set of risk measures — absolute risk and crude incidence from the cohort, plus a covariate-adjusted relative risk from the nested analysis (Fletcher, Ch 6, p 102)	A separate matched risk set is required for each outcome studied (Handbook of Epidemiology, Ch 7, p 330)

Nested case-control compared with the case-cohort and the conventional case-control design

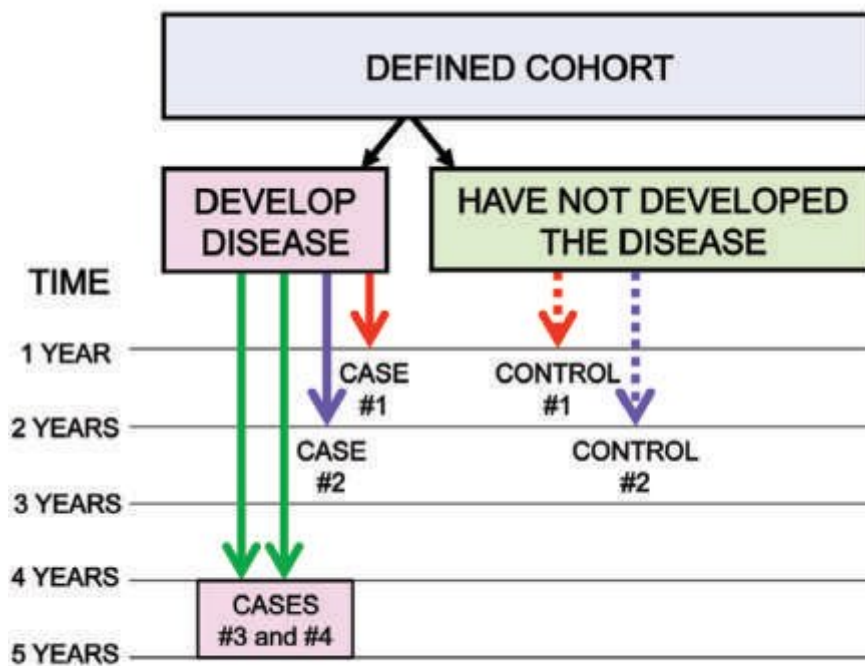
Feature	Nested case-control	Case-cohort	Conventional case-control
Source of controls	Risk set at the time each case is diagnosed	Random sample of the cohort at baseline , chosen without regard to future outcome (Handbook of Epidemiology, Ch 7, p 329)	Hospital or community, outside any enumerated cohort
Matched on follow-up time	Yes	No — it is essentially an unmatched nested case-control study (Handbook of Epidemiology, Ch 7, pp 329-330)	No
One control group for several outcomes	No — a fresh risk set is needed per outcome	Yes — the same subcohort serves studies of multiple outcomes (Handbook of Epidemiology, Ch 7, p 330)	No
Recall bias	Absent	Absent	Present
Effect measure	Odds ratio, equal to the incidence rate ratio	Cumulative-incidence (risk) ratio	Odds ratio, approximating relative risk only when the disease is rare
Incidence obtainable	Yes, from the parent cohort	Yes	No

Classical examples and applications

- **Epstein-Barr virus and multiple sclerosis** — nested in a cohort of over **3 million** active and reserve US military personnel who gave blood on induction from 1988 and at 2-yearly intervals; the **83** subjects meeting diagnostic criteria for multiple sclerosis who had at least one stored sample were the cases, with **2 controls per case** matched on age, sex, race/ethnicity and date of blood collection; only these stored sera were assayed, and strong positive associations between Epstein-Barr virus antibodies and multiple sclerosis were found (Lilienfeld's Foundations of Epidemiology, Ch 10, pp 221-222).
- **Opiate agonist treatment and fatal overdose** — an incidence-density nested case-control study inside the Italian *VEdeTTE* cohort of **4,444** heroin users followed from 1998 to 2005; the **95** deaths coded as overdose were the cases, with **4 controls per case** alive on the date of the case's death, matched on region, age (± 5 years) and sex; conditional logistic regression gave an odds ratio of **0.09** (95% CI 0.03-0.24) for opiate agonist treatment (Browner 5e, Ch 9).
- **Typical applications** — nutritional, hormonal, infectious and molecular or genetic markers measured on banked sera within large chronic-disease cohorts; drug-safety questions where the exposure changes over time; and any question in which the assay, record abstraction or database linkage is too costly to perform on the entire cohort (Browner 5e, Ch 9, pp 262-263).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Nested case-control design (Gordis's schematic — reproduce this).** Draw one box at the top labelled *DEFINED COHORT*, with two arrows down to two boxes: *DEVELOP DISEASE* (left) and *HAVE NOT DEVELOPED THE DISEASE* (right). Down the left margin rule a time axis labelled *TIME*, with horizontal lines at *1 YEAR*, *2 YEARS*, *3 YEARS*, *4 YEARS*, *5 YEARS*. On the left mark *CASE #1* at 1 year, *CASE #2* at 2 years, *CASES #3 and #4* at 4 years and *CASE #5* at 5 years; directly opposite each, on the right, mark *CONTROL #1*, *CONTROL #2*, *CONTROLS #3 and #4* and *CONTROL #5* at the same time level, joined by dotted arrows. Write beneath: "5 cases, 5 controls — one control is drawn from those still at risk each time a case occurs."
- **Risk sets on a person-time diagram.** Draw one horizontal line per cohort member from a common left margin; end some lines with a filled circle labelled *incident case*, some with a cross labelled *lost to follow-up*, and let the rest run to the right margin labelled *not a case by end of follow-up*. Drop a vertical dashed line through each case and label the resulting vertical bands *RISK SET 1*, *RISK SET 2*, *RISK SET 3*.



Design of a hypothetical nested case-control study: steps in selecting cases and controls (Gordis Epidemiology 7e, Ch 8, p 205)

High-yield exam pointers

- Definition to write first: **a case-control study nested inside an enumerated cohort, with controls sampled from those still at risk at the moment each case is diagnosed.**
- Key phrase for control selection: **risk-set = incidence-density sampling**; therefore the odds ratio equals the **incidence rate ratio without the rare-disease assumption**, and the analysis is **conditional logistic regression**.

- ▶ The discriminator to write against case-cohort: **nested case-control samples controls at the time of each case (time-matched); case-cohort samples a random subcohort at baseline (unmatched), and one subcohort can serve several outcomes.**
- ▶ Efficiency rule of thumb: **four to five controls per case gets close to maximum efficiency** (Browner 5e, Ch 20, p 653).
- ▶ Three biases abolished: **recall bias, selection bias from sampling two different populations, and exclusion of fatal cases** — all because exposure was recorded before the disease appeared and both groups came from one cohort.
- ▶ Named example worth quoting: **Epstein-Barr virus and multiple sclerosis** — over 3 million US military personnel, 83 cases, 2 matched controls each, stored sera assayed only for the sampled subset.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park's Textbook of Preventive and Social Medicine 28e carries no section on the nested case-control design (Chapter 3 describes only the conventional case-control and cohort designs), so this answer is grounded on Gordis Epidemiology 7e, the Handbook of Epidemiology, Browner's Designing Clinical Research 5e, Lilienfeld's Foundations of Epidemiology and Fletcher's Clinical Epidemiology; a documented Indian example of the design was not retrievable from the indexed corpus.

SOURCE & COVERAGE

Gordis Epidemiology 7e (Ch 8 — Cohort studies, pp 204-207; Ch 9 — Comparing cohort and case-control studies, p 211); Handbook of Epidemiology, Ahrens & Pigeot (Ch 7 — Modern Epidemiologic Study Designs, pp 329-334; Ch 12 — Design and Planning of Epidemiological Studies, pp 479-483); Browner — Designing Clinical Research 5e, 2022 (Ch 9 — Designing Case-Control Studies, pp 242-263; Ch 20 — Writing a Proposal for Funding a Research Study, p 653); Lilienfeld's Foundations of Epidemiology (Ch 10 — Observational Studies: Case-Control Studies, pp 221-222); Fletcher — Clinical Epidemiology: The Essentials (Ch 6 — Risk: From Disease to Exposure, p 102); Park's Textbook of Preventive and Social Medicine, 28e (Ch 3 — Principles of Epidemiology and Epidemiologic Methods, pp 90-96, for the parent case-control and cohort designs).



Quasi-experimental studies

Asked in: Paper I 16-Dec-2023 — RGUHS MD Community Medicine

Definition — A quasi-experimental study is an interventional study in which the investigator applies and controls the intervention but allocates it **without randomisation** — the study and comparison groups are formed by geography, time, policy or convenience instead of by chance (Browner, Designing Clinical Research 5e, Ch 12, p 392). Abramson defines it as a study in which observations are made before and after an intervention, ideally with parallel observations in an unexposed group, but where allocation is not random (Abramson, Research Methods in Community Medicine, Ch 2, pp 13-14). Park classifies these under **non-randomized (non-experimental) trials**, i.e. designs "departing from strict randomization for practical purposes, but in such a manner that non-randomization does not seriously affect the theoretical basis of conclusions" (Park 28e, Ch 3, p 101). Because allocation is not random, comparability of groups is low and the validity of causal inference "remains largely a matter of extra-statistical judgement" (Park 28e, Ch 3, p 106).

When a quasi-experimental design is chosen

- ▶ **Randomisation is unethical** — a preventive measure of established benefit cannot be withheld (screening for cervical cancer was introduced before trials were thought necessary, so mortality before and after the programme had to be compared instead) (Handbook of Epidemiology, Ch 31, p 1277).
- ▶ **The intervention can only be applied to a whole group** — water fluoridation, seat-belt legislation, a health-education campaign or a school/factory programme is delivered community-wide and cannot be split within a community (Park 28e, Ch 3, p 106).
- ▶ **Too few allocation units to randomise** — when only a handful of communities, schools or factories are involved, true random allocation is impossible and the term "quasi-experimental study" is used, while systematic comparison of similar communities with and without the intervention is still implied (Handbook of Epidemiology, Ch 34, p 1379).
- ▶ **Rare disease with a long natural history** — a randomised trial would need thousands of subjects followed for a decade or more, and cost and logistics become prohibitive (Park 28e, Ch 3, p 106).
- ▶ **Political, social, administrative or legal constraints** — evaluation of a policy change, a health-service reorganisation or an implementation programme, where the investigator cannot control who receives it (Browner 5e, Ch 12, p 392).
- ▶ **Health services and programme evaluation** — "most studies of public health interventions are nonexperimental or quasi-experimental", the investigator having no power to decide allocation (Abramson, Ch 33, pp 337, 345); sound experimental designs are extremely rare and even a quasi-experimental design with a control group may be hard to mount (Handbook of Epidemiology, Ch 37, p 1490).

Classification of quasi-experimental (non-randomised) designs

Using **Campbell and Stanley notation** — O = observation/measurement, X = intervention (Handbook of Epidemiology, Ch 37, p 1490):

Design	Notation	Structure	Classical/worked example	Principal weakness
Uncontrolled trial	X O	Intervention given, outcome measured, no comparison group (implied "historical controls")	Papanicolaou test for cervical cancer, introduced in the 1920s without any randomised study (Park 28e, Ch 3, p 106)	No comparison at all; any secular change is read as effect
Before-and-after study without control (single-group pre-test/post-test)	O X O	Incidence compared before and after a preventive measure; the group serves as its own control, which eliminates virtually all group differences	James Lind and fresh fruit for scurvy, 1750; Salk and Sabin polio vaccines; compulsory seat-belts in Victoria, Australia, 1971 — deaths 564 to 464 (–17.7%) and injuries 14,620 to 12,454 (–14.8%) (Park 28e, Ch 3, Table 24, pp 106-107)	Secular trend, regression to the mean, maturation
Before-and-after study with control (non-equivalent control group)	O X O / O — O	A second, "natural" comparison community as similar as possible is chosen; effect = difference-in-differences , i.e. the difference in the change over time in outcomes between the intervention and control groups	Victoria compared with other Australian states without seat-belt legislation (Park 28e, Ch 3, Table 25, p 107); Newburgh (fluoridated) vs Kingston, New York (Gordis 7e, Ch 1, pp 23-25); analysis principle from Browner 5e, Ch 12, pp 398-400	Groups not equivalent at baseline; regression to the mean persists
Interrupted time series	O O O X O O O	Several serial measurements before and after the intervention; estimates both a change in level and a change in slope (trend)	The commonest quasi-experimental design for aggregated annual programme data (Handbook of Epidemiology, Ch 37, p 1490); ≥3 measurements each side are needed to estimate the trend (Browner 5e, Ch 12, pp 397-398)	Seasonality, coincident policy change, altered data-collection or classification methods
Controlled interrupted time series	O O O X O O O / O O O — O O O	Adds a non-randomised comparison series; outcome = between-group difference in the change in level or trend	Premium-rebate study in a German sickness fund, aggregated series plus a control series (Handbook of Epidemiology, Ch 37, p 1492)	Requires a genuinely comparable control series

Design	Notation	Structure	Classical/worked example	Principal weakness
Natural experiment	—	Nature or circumstance, not the investigator, separates the population into exposed and unexposed	Smokers vs non-smokers; migrants; religious/social groups; atomic bombing of Japan; famines; the 1981 Athens earthquake and cardiovascular mortality; Snow's "grand experiment" — Southwark and Vauxhall Co. 315 cholera deaths per 10,000 houses vs Lambeth Co. 37 (Park 28e, Ch 3, Table 23 and pp 105-107)	Investigator controls nothing; exposure may be confounded
Pseudo-randomised (quasi-randomised) allocation	—	Allocation by an alternating or predictable rule, e.g. odd/even hospital record number	Offers logistic advantage only	Allocation is predictable, so staff can manipulate sequence or eligibility; baseline comparability is not guaranteed (Browner 5e, Ch 12, p 393)
Regression discontinuity	—	Exposure is determined by whether a continuous "running variable" crosses a threshold; subjects just above and just below the cut-off are compared	Early vs delayed antiretroviral therapy in a large South African cohort, CD4 count <200 cells/μL as the running variable; birth weight and blood lead are other typical running variables (Browner 5e, Ch 10, pp 294-295)	Assumes nothing else differs across the threshold; needs large numbers near the cut-off
Instrumental variable / Mendelian randomisation	—	An external factor unrelated to outcome determines exposure and acts as the "instrument"	Regional proportion of coiling used as the instrument to compare coiling with clipping of cerebral aneurysms (Browner 5e, Ch 10, p 294)	Validity of the instrument cannot be fully tested

Conditions required for a valid before-and-after study

For a before-and-after comparison to constitute evidence, all six of the following are needed (Park 28e, Ch 3, p 106):

- ▶ Data on the incidence of the disease must be available both before and after introduction of the preventive measure.
- ▶ Only one factor is introduced or manipulated, all other factors remaining the same — for example fluoridation of drinking water to prevent dental caries.
- ▶ Diagnostic criteria of the disease must remain unchanged throughout.
- ▶ Adoption of the preventive measure should be over a wide area.
- ▶ The reduction in incidence must be large, because there is no control group against which a small change could be judged.

- Several trials may be needed before the evaluation is regarded as conclusive.

Threats to internal validity and their remedies

Threat	What it does to the result	Remedy
Secular / temporal and seasonal trend	Disease frequency or clinical care changes over time regardless of the intervention (Browner 5e, Ch 12, p 395)	Add a concurrent control group; use interrupted time series so the pre-intervention trend is estimated and allowed for
Regression to the mean	Groups selected because of an outbreak or an extreme baseline value drift back toward the mean on re-measurement; remains a major problem even in controlled before-after studies (Browner 5e, Ch 12, pp 398-399)	Take several stable baseline measurements before intervening; avoid selecting the group on an extreme value
Maturation effect	People, staff and clinicians learn and improve with time and repeated testing without any intervention (Browner 5e, Ch 12, p 396)	Concurrent control group; longer follow-up to check the effect is sustained
Confounding by indication / non-comparable groups	Clinicians choose who gets the intervention, so recipients differ systematically — the apparent benefit of intervention is often greater in non-randomised than in randomised studies of the same question (Browner 5e, Ch 12, p 393)	Propensity score (a modelled probability of receiving the treatment, used for stratification or matching), stratification or multivariable adjustment (Browner 5e, Ch 10, p 301)
Change in data collection or case classification	An apparent step in the series is an artefact of measurement, not of the intervention (Browner 5e, Ch 12, pp 397-398)	Keep case definition and ascertainment identical across the whole series; prespecify a falsification test
Residual (unmeasured) confounding	Statistical adjustment handles only measured baseline differences (Browner 5e, Ch 12, p 393)	Cannot be removed — acknowledge as a limitation
Non-comparable baseline in the control area	Difficulty in analysis if baseline mortality differs between intervention and comparison areas (Handbook of Epidemiology, Ch 31, p 1277)	Choose control areas matched on disease frequency and characteristics (Park 28e, Ch 3, p 107)

Quasi-experimental study versus randomised controlled trial

Feature	Quasi-experimental study	Randomised controlled trial
Allocation	Non-random — by area, time, policy or convenience	Random allocation after eligibility and informed consent
Comparability of groups	Low; achieved only by matching or statistical adjustment	Groups equal in all relevant characteristics except the intervention
Unit of allocation	Usually community, clinic, district or time period	Usually the individual
Confounding	Measured confounders adjustable; unmeasured confounding remains	Both measured and unmeasured confounders balanced

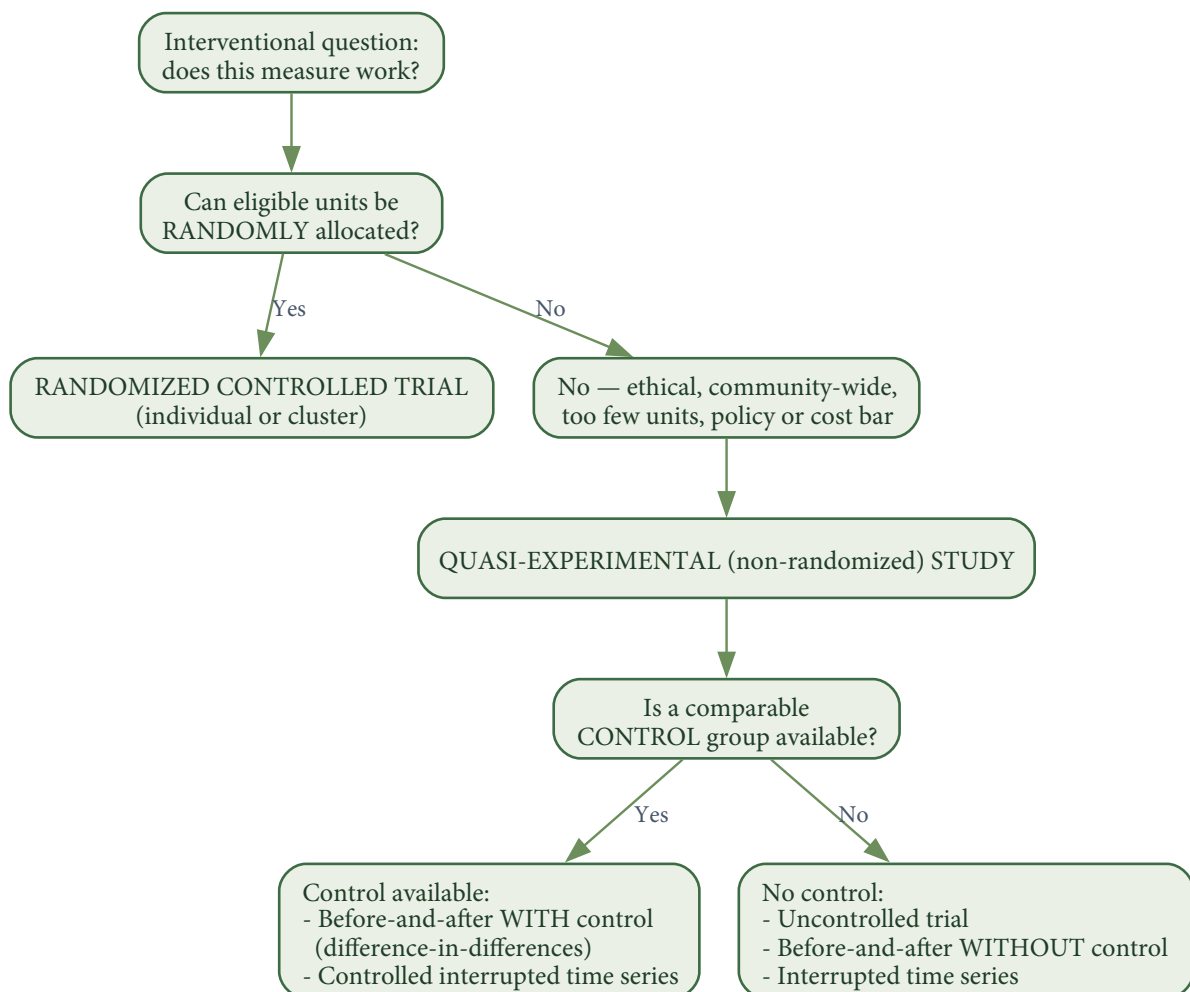
Feature	Quasi-experimental study	Randomised controlled trial
Blinding	Usually impossible	Single/double blinding feasible
Strength of causal inference	Weaker — "extra-statistical judgement" (Park 28e, Ch 3, p 106)	Strongest; the gold standard for evaluating health care (Handbook of Epidemiology, Ch 37, pp 1461-1462)
Feasibility, cost, ethics	High feasibility; usable where randomisation is impossible or unethical	Recruitment difficult and expensive; may be ethically or logistically impossible
Typical analysis	Difference-in-differences; segmented (piecemeal) regression of the time series; propensity-score methods	Intention-to-treat comparison of randomised groups
Level of evidence	Below the randomised trial	Highest for an intervention question

Advantages and limitations

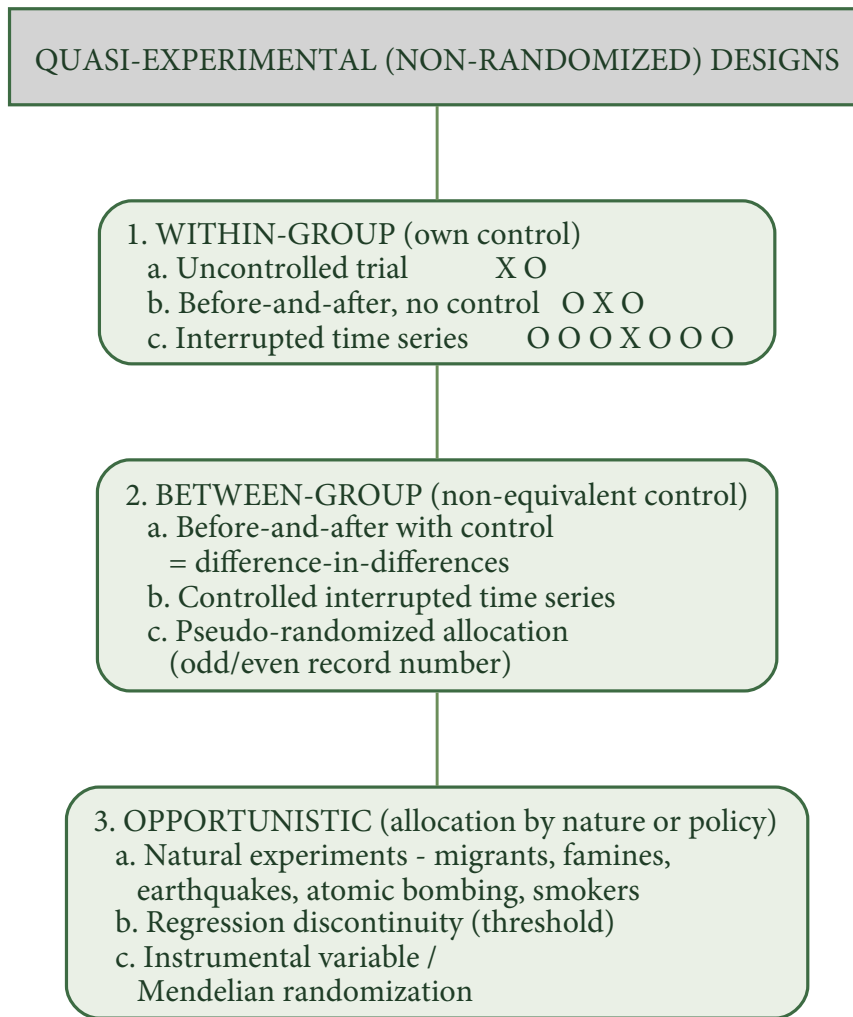
Advantages	Limitations
Applicable where randomisation is impossible, unethical or politically unacceptable	No randomisation, so degree of comparability is low and chance of a spurious result higher (Park 28e, Ch 3, p 106)
Suited to community-wide and policy interventions, and to programme and health-service evaluation	Effect sizes tend to be exaggerated relative to randomised studies of the same question (Browner 5e, Ch 12, p 393)
In the before-and-after design each group serves as its own control, so time-invariant confounders are eliminated, not merely balanced (Browner 5e, Ch 12, p 395)	Vulnerable to secular trend, regression to the mean and maturation
Cheaper and quicker; can use routine and administrative data (electronic records, registers, vital statistics)	If different people are measured before and after, ordinary confounders such as age and disease severity re-enter (Browner 5e, Ch 12, p 395)
Designs with a control group and with pre-intervention measurements are far more convincing than those with neither (Abramson, Ch 33, p 337)	Demands the same methodological rigour as a randomised trial and, needing much larger populations, may cost more (Handbook of Epidemiology, Ch 31, p 1277)
Historically decisive for public health — Lind, Snow, seat-belt legislation, fluoridation	Blinding usually impossible; residual confounding is never excluded

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Interrupted time series plot** — draw and label: x-axis *Time (serial measurements)*; y-axis *Outcome rate*; a vertical dashed line marked **Intervention**; at least three data points before and three after; label the **pre-intervention trend (slope)**, the **change in level (step at the intervention)**, the **change in slope (post-intervention trend)**, and the dotted **counterfactual line** (the pre-intervention trend projected forward).
- **Design notation strip** — draw the Campbell and Stanley notation for the four core designs one under the other, with O = observation and X = intervention: X O (uncontrolled), O X (before-after without control), O X O above O — O (before-after with control), O O O X O O O (interrupted time series).

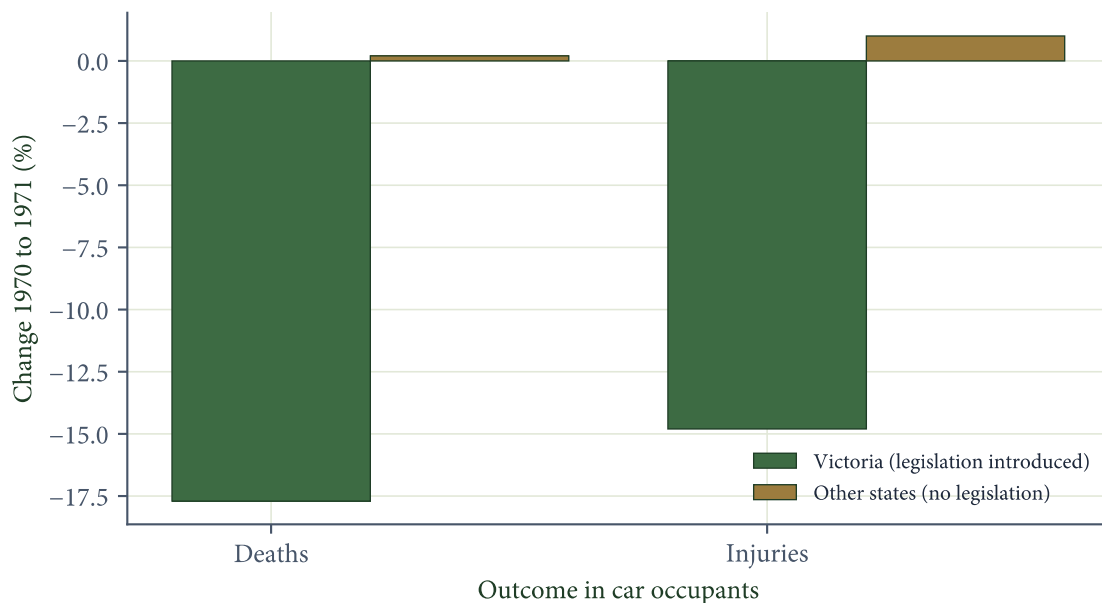


Choosing between a randomised and a quasi-experimental design



Classification of quasi-experimental (non-randomized) designs

Before-and-after WITH control: compulsory seat-belt legislation, Victoria (Australia)



Deaths fell 564 to 464 and injuries 14,620 to 12,454 in Victoria, while the other Australian states changed by +0.2% and +1.0% — the concurrent control converts an ambiguous before-after fall into evidence of effect (Park 28e, Ch 3, Table 25, p 107).

High-yield exam pointers

- ▶ **Definition trigger:** intervention applied by the investigator but allocated without randomisation; Park's term is **non-randomized (non-experimental) trial** (Park 28e, Ch 3, p 101).
- ▶ **Park's three headings, in order:** (1) uncontrolled trials, (2) natural experiments, (3) before-and-after comparison studies — **A without control, B with control** (Park 28e, Ch 3, pp 106-107).
- ▶ **Six requirements for a before-and-after study** (mnemonic **D-O-D-W-L-S**): **D**ata before and after; **O**ne factor manipulated; **D**agnostic criteria unchanged; **W**ide area of adoption; **L**arge reduction in incidence; **S**everal trials.
- ▶ **Notation:** O = observation, X = intervention (Campbell and Stanley, 1966). Interrupted time series needs ≥ 3 measurements before and ≥ 3 after.
- ▶ **Effect measure with a non-equivalent control** = **difference-in-differences**; analysis of a time series = **segmented (piecemeal) regression**; **confounding by indication** = **propensity score**.
- ▶ **Three named threats to internal validity** to write every time: **secular/temporal trend, regression to the mean, maturation effect**.
- ▶ **Park's three tables — quote the numbers:** **Table 23** — Snow's "grand experiment" (1854), cholera deaths per 10,000 houses: **Southwark and Vauxhall Company 315** (sewage-polluted Thames intake) vs **Lambeth Company 37** (intake above London) (Park 28e, Ch 3, pp 105-106); **Table 24** — Victoria seat-belt legislation 1971, before-and-after **without** control, deaths 564→464 (−17.7%), injuries 14,620→12,454 (−14.8%); **Table 25** — the same data **with** control, other Australian states +0.2% deaths and +1.0% injuries (Park 28e, Ch 3, pp 106-107).
- ▶ **Other classic examples:** Lind and scurvy (1750); Newburgh vs Kingston fluoridation (Gordis 7e, Ch 1, pp 23-25); Bhopal 1984 and Hiroshima-Nagasaki as natural experiments (Gordis 7e, Ch 14, p 283).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park lists natural experiments among non-randomized trials, whereas Designing Clinical Research 5e calls them opportunistic observational studies and keeps "quasi-experimental" for an intervention the investigator applies but does not randomise. The stepped-wedge design is randomised and is therefore not quasi-experimental.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 3, Principles of Epidemiology and Epidemiologic Methods, pp 101-107, Tables 23, 24 and 25); Browner — Designing Clinical Research, 5e, 2022 (Ch 10, Estimating Causal Effects Using Observational Studies, pp 294-301; Ch 12, Alternate Interventional Study Designs, pp 392-401); Handbook of Epidemiology (Ahrens & Pigeot) (Ch 31 Screening, p 1277; Ch 34 Cardiovascular Diseases, p 1379; Ch 37 Health Services Research, pp 1461-1492); Gordis Epidemiology 7e (Ch 1, pp 23-25; Ch 14, p 283); Abramson — Research Methods in Community Medicine (Ch 2 Types of investigation, pp 13-14; Ch 33 Programme trials, pp 337-345).



Role of surveillance in public health

Asked in: Paper I 16-Dec-2023 — RGUHS MD Community Medicine

Definition — Surveillance is the continuous, systematic collection, analysis and interpretation of health data, closely integrated with the timely and coherent dissemination of the results to those who have the right to know, so that action can be taken (Park 28e, Ch 2, p 59). The Centers for Disease Control and Prevention (CDC) definition adds that the data must be essential to the planning, implementation and evaluation of public health practice (Gordis 7e, Ch 3, pp 57-58). It is distinguished from **monitoring**, which is intermittent and episodic and can be run by technicians, whereas surveillance is continuous and demands professional analysis and judgement leading to recommendations for control; the ultimate objective of surveillance is **prevention** (Park 28e, Ch 2, p 59; Ch 3, pp 151-152).

Objectives and roles served by surveillance

#	Role	What surveillance delivers	Illustration
1	Early warning and outbreak detection	Detects early warning signals of impending outbreaks so that an effective response is initiated in time	Weekly fever alerts under the Integrated Disease Surveillance Programme give early warning of malaria outbreaks (Park 28e, Ch 7, p 499)
2	Measuring the burden of disease	Systematic morbidity and mortality data by time, place and person; most of what is known about disease frequency comes from established surveillance programmes	Mortality registration and morbidity reporting are the oldest surveillance elements (Gordis 7e, Ch 3, pp 57-58)
3	Detecting change over time	By observing trends in time, place and person, changes can be observed or anticipated and control measures taken	Changes in disease agents, vectors or reservoirs; serological surveillance through serum banks identifies patterns of current and past infection (Park 28e, Ch 2, p 59; Ch 3, p 151)

#	Role	What surveillance delivers	Illustration
4	Identifying high-risk groups and geographic spread	Levels and trends of infection in general and high-risk populations, and identification of emerging pockets	Objectives of human immunodeficiency virus (HIV) sentinel surveillance in India (Park 28e, Ch 7, pp 519-520)
5	Directing case-based control action	Goes beyond passive reporting — laboratory confirmation of a presumptive diagnosis, tracing the source of infection and routes of transmission, identification of all cases, susceptible contacts and others at risk	Communicable disease control (Park 28e, Ch 3, p 151)
6	Planning and resource allocation	Essential data to monitor progress of ongoing disease control programmes and to allocate health resources more efficiently	Stated objective of the Integrated Disease Surveillance Programme (Park 28e, Ch 7, p 568)
7	Evaluating control measures	Once control measures are instituted their effectiveness is evaluated; if unsuccessful, the reasons for failure are identified, measures modified and evaluation continued	Surveillance is an essential pre-requisite to the rational design and evaluation of any disease control programme (Park 28e, Ch 3, pp 151-152)
8	Sustaining elimination and eradication	Proves absence of transmission and detects re-introduction	National population surveillance of smallpox after eradication; malaria Category 0 — detect any re-introduced case, notify immediately, maintain malaria-free status (Park 28e, Ch 3, p 152; Ch 7, p 497)
9	Meeting international obligations	Cholera, plague and yellow fever notifiable to the World Health Organization (WHO) under the International Health Regulations (IHR); louse-borne typhus, relapsing fever, paralytic polio, malaria, influenza-A and severe acute respiratory syndrome are under WHO surveillance, with timely warning given to national governments	Park 28e, Ch 3, p 144; Ch 21, p 997; Ch 3, p 152
10	Extending beyond infection	Increasingly used for non-communicable diseases and their risk factors, genetic traits, environmental toxins, vaccination coverage, drug-resistant organisms, and injuries after natural disasters	Gordis 7e, Ch 3, pp 57-58

Components, prerequisites and data sources

Components of surveillance activity (Park 28e, Ch 7, p 569):

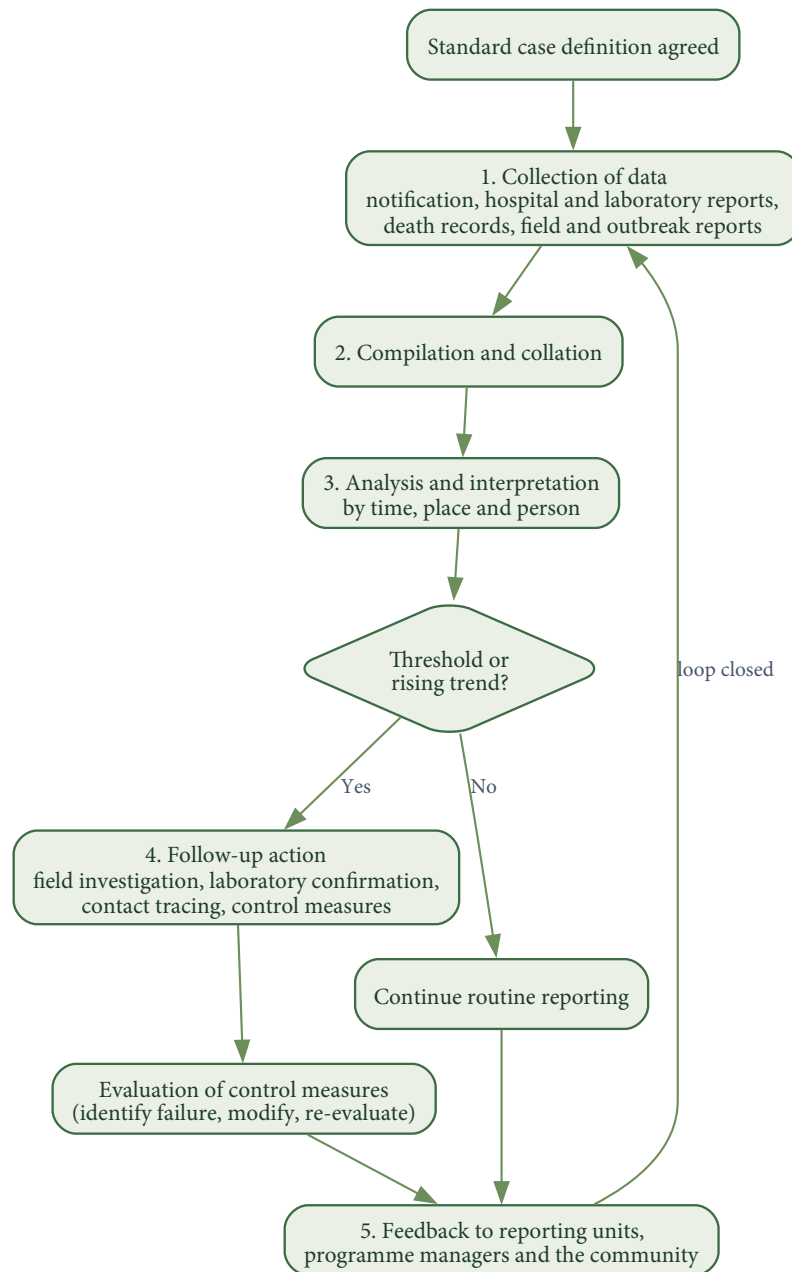
- Collection of data
- Compilation of data
- Analysis and interpretation

- ▶ Follow-up action
- ▶ Feedback

Prerequisites of effective surveillance — use of a **standard case definition**, regularity of reports, and action on reports (Park 28e, Ch 7, p 569). A surveillance case definition is "*a set of uniform criteria used to define a disease for public health*", and is deliberately different from the clinical definition used to treat an individual patient (Gordis 7e, Ch 3, p 58).

The five questions surveillance must answer — who gets the disease, how many get the disease, where did they get the disease, why did they get the disease, and what is to be done as public health response (Park 28e, Ch 7, p 569).

Sources and elements of the data — mortality reports based on death certificates; morbidity reports from hospital records, general practice sentinels and notifications; laboratory diagnosis; outbreak reports; vaccine uptake and side-effects; sickness absence records; changes in disease agents, vectors or reservoirs; serological surveillance through serum banks (Park 28e, Ch 2, p 59). The WHO list of elements adds individual case investigations, epidemic field investigations, surveys, animal reservoir and vector distribution studies, biologics and drug utilisation, and knowledge of the population and the environment (Handbook of Epidemiology, Ch IV.1, p 1326).



The surveillance-to-action cycle — data are of public health value only when they return as feedback and control

Types and levels of surveillance

Type	How it works	Strength	Limitation
Passive	The agency puts a reporting system in place and waits for the care provider to report; reporting is mandated by the health authority	Near-universal, inexpensive, easy to set up initially, allows international comparison	Compliance irregular and incomplete; under-reporting; local outbreaks diluted in a large population denominator (Gordis 7e, Ch 3, p 58; Handbook of Epidemiology, Ch IV.1, p 1322)

Type	How it works	Strength	Limitation
Active	Specifically recruited and trained staff make periodic field visits to clinics, primary health centres and hospitals, interview physicians and patients, review records and search villages for cases	More accurate, more uniform and more complete data; outbreaks more easily identified	Much more expensive, harder to develop and maintain (Gordis 7e, Ch 3, pp 58-59; Handbook of Epidemiology, Ch IV.1, p 1323)
Sentinel	Selected competent physicians or institutions in selected areas report cases; sentinel data are extrapolated to the whole population	Identifies the cases that no routine notification system can capture; minimises reporting bias; simplifies feedback; may be cheaper than a full notification system	Not a complete count; representativeness depends on site selection (Park 28e, Ch 2, p 59)
Serologic al (sero-surveillance)	Testing of sera, often through serum banks or unlinked anonymous testing of a risk group repeated annually at the same site	Identifies patterns of current and past infection and true prevalence	Needs laboratory capacity; ethical safeguards for anonymity (Park 28e, Ch 3, p 151; Ch 7, pp 519-520)
Syndromic	Pre-diagnostic indicators (calls to poison centres, over-the-counter sales of anti-diarrhoeals, animal deaths) monitored centrally in real time	Shifts the epidemic curve to the left — detects outbreaks in their pre-diagnostic phase	Low specificity; needs verification (Lilienfeld's Foundations of Epidemiology, Ch 14, p 293)
Behavioural	Periodic survey of behaviours that determine risk	Tracks the determinant before the disease appears	Self-report bias; one of the six surveillance types used in the national HIV programme (Park 28e, Ch 7, p 519)

Levels at which surveillance operates (Park 28e, Ch 3, p 152):

- **Individual surveillance** — of infected persons until they are no longer a significant risk to others.
- **Local population surveillance** — e.g. surveillance of malaria.
- **National population surveillance** — e.g. surveillance of smallpox after eradication.
- **International surveillance** — WHO maintains surveillance of important diseases (influenza, malaria, polio) and gives timely warning to all national governments.

Surveillance in Indian public health practice — the Integrated Disease Surveillance Programme

Element	Detail
Launch and status	Launched November 2004 as the World Bank–assisted Integrated Disease Surveillance Project; continues from the 12th Five Year Plan on domestic budget as the Integrated Disease Surveillance Programme under the National Health Mission (Park 28e, Ch 7, p 568; Sunder Lal, Ch 7, p 306)

Element	Detail
Aim	Detect early warning signals of impending outbreaks and initiate an effective response in a timely manner, in urban and rural areas (Park 28e, Ch 7, p 568)
Objectives	A decentralised, state- and district-based surveillance system for communicable and non-communicable diseases; integration of the surveillance activities of the vertical disease control programmes; laboratory strengthening; human resource development including rapid response teams; involvement of the private sector, non-governmental organisations and medical colleges (Sunder Lal, Ch 7, p 306; Jugal Kishore, pp 487-491)
Structure	Central Surveillance Unit at the National Centre for Disease Control, Delhi; State Surveillance Units at every state/union territory headquarters; District Surveillance Units in all districts, the district committee chaired by the District Collector (Park 28e, Ch 7, p 568; Jugal Kishore, pp 489-490)
Data flow	Weekly reporting, Monday to Sunday, on three formats — S (syndromic, health worker), P (presumptive, medical officer), L (laboratory confirmed, laboratory staff) — using standard case definitions; more than 90 per cent of districts send weekly data by e-mail or portal (Park 28e, Ch 7, pp 568-569; Jugal Kishore, p 488)
Response	A rising trend in any area triggers investigation by the medical officer or rapid response team; graded triggers define who responds (Jugal Kishore, pp 490-491)
Medical college role	Reference laboratories, quality assurance, training, epidemic investigation and non-communicable disease surveillance (Jugal Kishore, p 490)

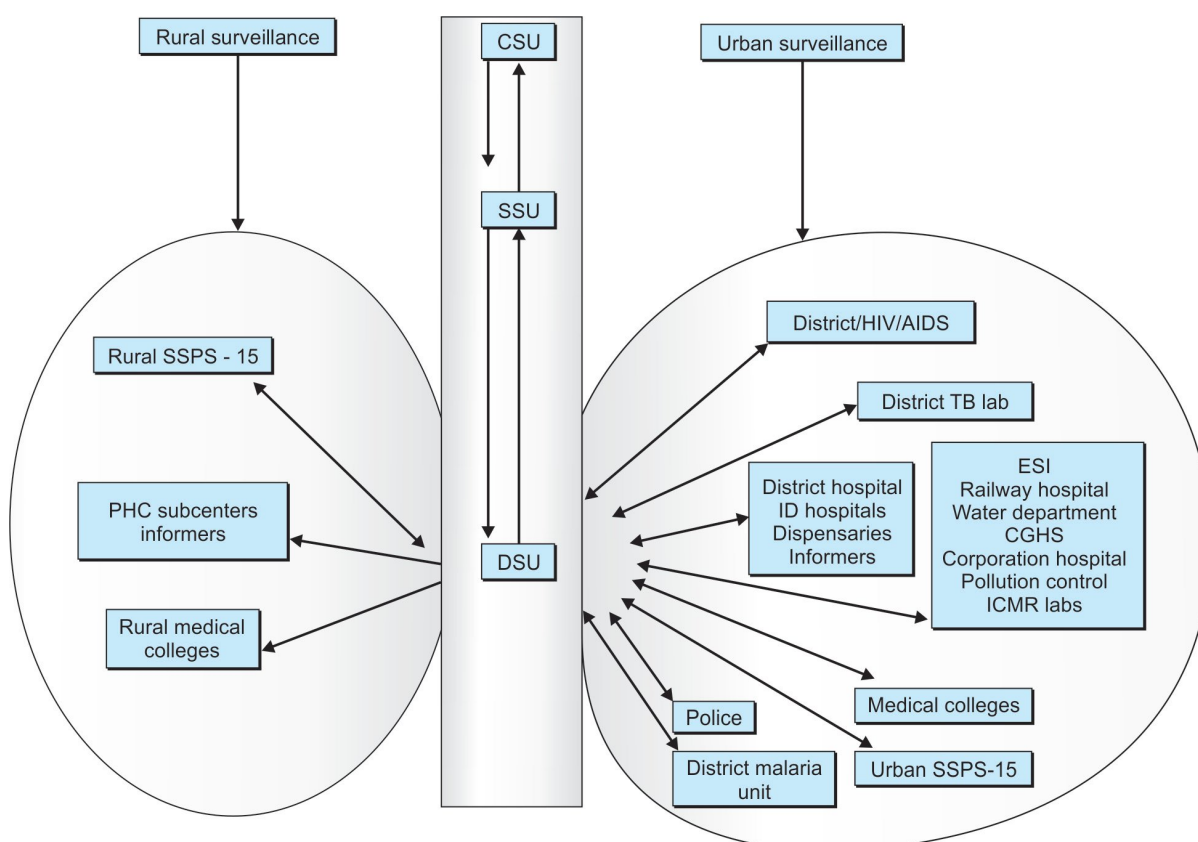
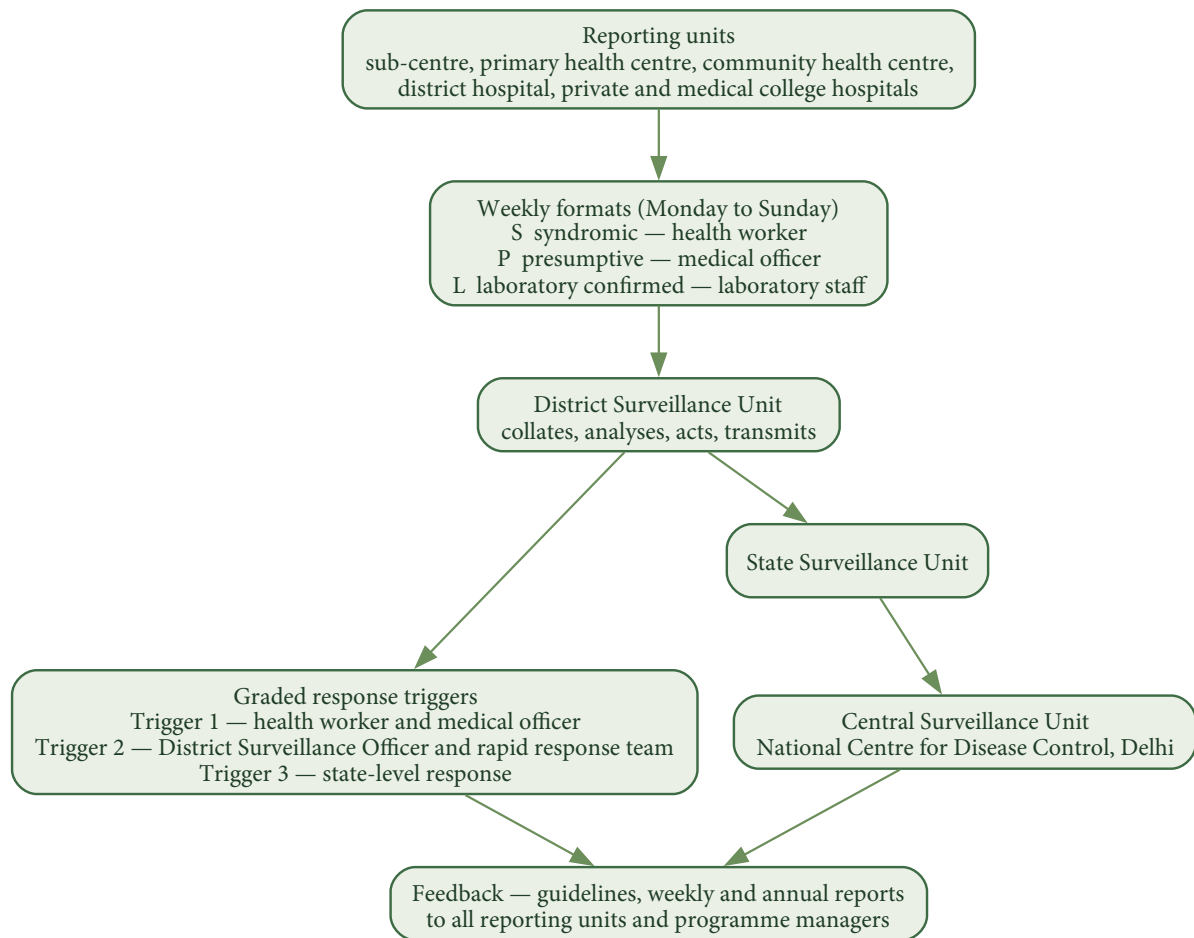


Fig 46.1 — Structural framework of integrated disease surveillance project Abbrev (Suryakantha Environment Best, p.905)



Integrated Disease Surveillance Programme — weekly reporting flow and graded response triggers

Evaluating a surveillance system

Attribute	What is judged (CDC criteria)
Usefulness	Whether conclusions are reached and recommendations made — a system that only collects and publishes without preventive use has failed
Sensitivity	Ability to identify every single case; critical for outbreak investigation and eradication programmes
Predictive value positive	Proportion of reported cases that truly have the event; a low value means non-cases investigated, exaggerated or pseudo-outbreaks, and misclassification that corrupts aetiological enquiry
Representativeness	Whether the occurrence and distribution of reported cases reflect the real situation in the population
Simplicity	Essential to gain acceptance, especially when relying on outside reporters
Flexibility	Ability to adapt to changed epidemiological patterns, laboratory methods, operating conditions, funding or reporting sources
Data quality	Completeness (blank or unknown values) and validity of recorded data
Acceptability	Shown by the participation of providers in the system
Timeliness	Most important in surveillance of epidemics

Attribute	What is judged (CDC criteria)
Stability	Reliability of data handling without failure, and availability when needed

(All rows: *Handbook of Epidemiology*, Ch IV.1, pp 1325-1326.)

Limitations

- ▶ **The morbidity iceberg** — reported cases are only the tip; some have symptoms but never seek care, some seek care but are not diagnosed or are misdiagnosed, so reported cases may not represent true endemicity (*Handbook of Epidemiology*, Ch IV.1, pp 1327-1328).
- ▶ **Under-reporting** — notification covers only a small part of the total sickness in the community and suffers a great deal of under-reporting (Park 28e, Ch 21, p 997).
- ▶ **No uniform legal basis in India** — the list of notifiable diseases varies between states and between urban and rural areas; some states invoke only the Epidemic Diseases Act, 1897, and a uniform Act on the model of the Registration of Births and Deaths Act, 1969 is still considered necessary (Park 28e, Ch 21, p 997).
- ▶ **Cost and feasibility** — active surveillance gives far better data but is much more expensive, and the areas most in need may be physically difficult to reach (Gordis 7e, Ch 3, pp 58-59; *Handbook of Epidemiology*, Ch IV.1, p 1323).
- ▶ **Reporter fatigue** — passive reporting burdens clinicians who see it as peripheral to patient care, so instruments must be short and feedback must be given (Gordis 7e, Ch 3, p 58; *Handbook of Epidemiology*, Ch IV.1, pp 1322-1323).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Reproduce the structural framework of the Integrated Disease Surveillance Programme (embedded above) with these labels:

- ▶ Two vertical columns: **Rural surveillance** and **Urban surveillance**
- ▶ Rural column, bottom upward: sub-centre → primary health centre → community health centre → district hospital
- ▶ Urban column: urban health posts and hospitals; **selected sentinel private sites** (sentinel private practitioners and sentinel hospitals)
- ▶ Other reporting units feeding the district: district tuberculosis centre, district HIV/AIDS unit, Employees' State Insurance and railway hospitals, Central Government Health Scheme, corporation hospitals, water department, informers
- ▶ **District Surveillance Unit (DSU) → State Surveillance Unit (SSU) → Central Surveillance Unit (CSU)** at the National Centre for Disease Control
- ▶ Medical colleges shown as reference laboratories and training/epidemic-investigation support
- ▶ **Feedback arrows** drawn downward from CSU → SSU → DSU → reporting units

High-yield exam pointers

- ▶ **Definition to write first:** continuous, systematic collection, analysis and interpretation of data with timely dissemination to those who need to know so that action can be taken; *"the ultimate objective of surveillance is prevention"* (Park 28e, Ch 3, pp 151-152).
- ▶ **Five components** — collection, compilation, analysis and interpretation, follow-up action, feedback. **Three prerequisites** — standard case definition, regularity of reports, action on reports (Park 28e, Ch 7, p 569).
- ▶ **Four levels** — individual, local population, national, international (Park 28e, Ch 3, p 152).
- ▶ **Surveillance vs monitoring** — surveillance is continuous and needs professional judgement; monitoring is intermittent/episodic and can be done by technicians with standardised procedures (Park 28e, Ch 2, p 59).
- ▶ **Integrated Disease Surveillance Programme numbers** — launched November 2004; Central Surveillance Unit at the National Centre for Disease Control, Delhi; weekly S/P/L formats; more than 90 per cent of districts reporting weekly (Park 28e, Ch 7, pp 568-569).
- ▶ **International Health Regulations** — cholera, plague and yellow fever notifiable to WHO (Park 28e, Ch 3, p 144).
- ▶ **Evaluation attributes** — usefulness, sensitivity, predictive value positive, representativeness, simplicity, flexibility, data quality, acceptability, timeliness, stability (Handbook of Epidemiology, Ch IV.1, pp 1325-1326).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Under the current International Health Regulations, notification of every case of cholera is no longer mandatory — public health events involving cholera are instead assessed against the Annex 2 decision instrument to decide whether notification to the World Health Organization is required (Park 28e, Ch 5, pp 292-293). Coverage note: the retrieved corpus describes Integrated Disease Surveillance Programme reporting by e-mail and web portal but does not describe the later migration of that reporting to the Integrated Health Information Platform in any detail, so no figures for that platform are quoted here.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, WHO, World Health Assembly resolutions

Programme and policy updates (India)

- ▶ **Integrated Disease Surveillance Programme reporting has migrated from weekly aggregate returns to the case-based Integrated Health Information Platform.** The platform was introduced in Odisha in 2021 and its P form now carries individual patient records rather than a weekly count — two Odisha districts registered 255,071 and 63,895 patients between April 2021 and October 2022, acute respiratory illness having the highest notification rate in both. Vector-borne disease reporting is being moved onto the same platform to give near-real-time malaria surveillance. [PMID 41629233] [PMID 41629232]
- ▶ **Sensitivity of the new platform is the present weak point, and it responds to ordinary quality-improvement methods.** In Raigarh, Chhattisgarh (September 2023) the platform captured only 44 per cent of the laboratory-tested malaria cases held in primary health centre registers; refresher training of laboratory technicians plus weekly feedback from the district surveillance unit was used to drive sensitivity towards an 80 per cent target. The reporter's side of the same problem: at Chandigarh health and wellness centres the surveillance portal scored only 51–70 per cent on implementation, and 89.2 per cent of health workers said duplicate paper-plus-digital entry had increased their workload — reporter fatigue in its digital form. [PMID 41629241] [PMID 40854229]

Current national data

- ▶ **Zoonoses on the surveillance portal, 2010–2023** — 1,158 outbreaks were reported for ten selected zoonotic diseases, Japanese encephalitis and acute encephalitis syndrome being the commonest. Anthrax, Crimean-Congo haemorrhagic fever, leptospirosis and scrub typhus were reported throughout the year, while most other diseases clustered in the July–September monsoon; the authors caution that the apparent rising trend partly reflects improving reporting rather than rising incidence. [PMID 41738949]
- ▶ **Antimicrobial resistance surveillance networks now generate trend data of programme quality.** The Delhi network (24 laboratories, 2020–2024) analysed 10,689 typhoidal *Salmonella* isolates, annual isolates rising from 522 to 3,679; multidrug resistance in *S. Typhi* fell to 1 per cent by 2024 and ampicillin and co-trimoxazole resistance declined, but ciprofloxacin non-susceptibility persisted at roughly 48–60 per cent and fluoroquinolone resistance in *S. Paratyphi* A ran at 88–98 per cent. [PMID 41643898]

Elimination, surveillance and disease status

- ▶ **Measles and rubella — the elimination deadline has moved twice, and the textbook target is superseded.** The South-East Asia Region's 2013 goal of measles elimination and rubella control by 2020 was revised in September 2019 to measles *and* rubella elimination by 2023; as of 2023 only five of the eleven countries (Bhutan, DPR Korea, Maldives, Sri Lanka, Timor-Leste) had been verified, regional measles incidence had risen from 16.2 to 40.9 per million with significant increases in India and Indonesia, and **2026** was set as the next feasible target. Laboratory-supported case-based fever-and-rash surveillance is the core strategy, and the accredited laboratory network grew from 23 to 58 laboratories over the decade. [PMID 39460260]
- ▶ **Kala-azar — elimination is being held, but post-kala-azar dermal leishmaniasis is now the surveillance target.** In Malda district, West Bengal, visceral leishmaniasis fell from 19 cases (2018) to 7 (2022) while post-kala-azar dermal leishmaniasis rose from 40 to 55 and became the majority of reported cases; all 15 endemic blocks stayed under the elimination threshold, with treatment completion above 90 per cent and indoor residual spraying coverage above 91 per cent. Because the dermal form is a long-term human reservoir, active case detection and follow-up of treated patients — not case counts alone — decide whether elimination holds. [PMID 42591813]
- ▶ **Emerging disease picked up through the routine stream — Chandipura virus, 2024:** the outbreak yielded 245 confirmed cases with 82 deaths, a reminder that this sandfly-borne neurotropic virus remains an active differential within acute encephalitis syndrome surveillance in children. [PMID 41769747]

Guideline and legislative updates

- ▶ **International Health Regulations (2005) — the 2024 amendments entered into force on 19 September 2025.** Adopted by consensus at the Seventy-seventh World Health Assembly on 1 June 2024 (resolution WHA77.17), they insert a **pandemic emergency** as a tier above a public health emergency of international concern — a communicable-disease event with wide or threatened spread across multiple States that is exceeding health-system capacity, causing substantial social or economic disruption, and requiring rapid, equitable and coordinated international action. Revised Articles 13 and 44 add equity and sustainable-financing obligations, and amended Article 4 requires every State Party to **establish or designate a National International Health Regulations Authority** above the long-standing National Focal Point, to coordinate implementation across sectors and at subnational level — a reform aimed squarely at fragmented authority, thin legal mandates and weak intersectoral coordination, and one that obliges corresponding change in domestic law. [PMID 40589223] [PMID 40953844] [PMID 40223587] [WHO 2025-Sep]

Global and WHO developments

- ▶ **WHO Pandemic Agreement adopted 20 May 2025** at the Seventy-eighth World Health Assembly, where 124 governments approved its 35 articles after three years of negotiation; it is only the second instrument negotiated under Article 19 of the WHO Constitution and sits alongside the amended Regulations rather than replacing them. Separately, **mpox was declared a public health emergency of international concern on 14 August 2024** after clade Ib spread

beyond Africa — the working example of the notification-and-assessment machinery this answer describes. [PMID 40589223] [PMID 39921615]

► **The streams that supplement notification are now evidence-based rather than aspirational.**

Environmental (wastewater) surveillance detects circulation independently of health-seeking behaviour — multiplex amplicon sequencing of wastewater from south India returned a *S. Typhi* genotype and/or resistance profile in 8 of 110 samples, and the genotypes matched those in contemporaneous clinical cases. Digital epidemiology gave a modest but free gain: adding Google Trends to seasonal ARIMA models of surveillance-reported acute febrile illness in Haryana lifted adjusted R^2 across the four disease models from 20–49 per cent to 46–52 per cent. Formal One Health evaluation of the Indian human-health and animal-health surveillance systems in Gujarat found the human side weakest on data sharing, sampling, outreach to decision-makers and external communication. [PMID 40627640] [PMID 40155818] [PMID 39453563]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 2 Concept of Health and Disease; Ch 3 Principles of Epidemiology and Epidemiologic Methods; Ch 5 Epidemiology of Communicable Diseases; Ch 7 Health Programmes in India; Ch 21 Health Information and Basic Medical Statistics); Gordis Epidemiology 7e (Ch 3 Disease Surveillance and Measures of Morbidity); Handbook of Epidemiology, Ahrens & Pigeot (Ch IV.1 Infectious Disease Epidemiology); Lilienfeld's Foundations of Epidemiology (Ch 14 Field Epidemiology); Sunder Lal — Textbook of Community Medicine (Ch 7 Epidemiology); Jugal Kishore — National Health Programmes of India (Integrated Disease Surveillance chapter); Suryakantha — Community Medicine with Recent Advances (figure, p 905). Surveillance evaluation attributes follow the CDC 2001 updated guidelines as reproduced in the Handbook of Epidemiology.

Recent-advances sources: 41629233 — Epidemiological Profile of Diseases Reported in the P Form of Integrated Health Information Platform in Two Districts of Odisha, Indian J Public Health 2026, doi:[10.4103/ijph.ijph.660.25](https://doi.org/10.4103/ijph.ijph.660.25) (abstract-only); 41629232 — Epidemiology of Malaria in Koraput District, Odisha, 2019–2023: A Descriptive Analysis of National Vector Borne Disease Control Programme Surveillance Data, Indian J Public Health 2026, doi:[10.4103/ijph.ijph.653.25](https://doi.org/10.4103/ijph.ijph.653.25) (abstract-only); 41629241 — Quality Improvement Initiative to Improve the Sensitivity of Malaria Reporting under Integrated Health Information Platform, Raigarh, Chhattisgarh, 2023, Indian J Public Health 2026, doi:[10.4103/ijph.ijph.648.25](https://doi.org/10.4103/ijph.ijph.648.25) (abstract-only); 40854229 — Implementation Status and Usability of Digital Health Interventions Among Health Care Workers and End Users at the Primary Health Care Level in Chandigarh, North India, JMIR Form Res 2025, doi:[10.2196/69824](https://doi.org/10.2196/69824); 41738949 — Trend of zoonotic diseases outbreaks in India: a secondary data analysis of Integrated Disease Surveillance Programme data, 2010–2023, Trans R Soc Trop Med Hyg 2026, doi:[10.1093/trstmh/trag011](https://doi.org/10.1093/trstmh/trag011) (abstract-only); 41643898 — Tracking antimicrobial resistance in typhoidal Salmonella, 2020–2024: output from an antimicrobial resistance surveillance network in New Delhi, India, Indian J Med Microbiol 2026, doi:[10.1016/j.ijmmb.2026.101074](https://doi.org/10.1016/j.ijmmb.2026.101074) (abstract-only); 39460260 — Progress towards Measles and Rubella Elimination in the South-East Asia Region 2013–2023, Vaccines 2024, doi:[10.3390/vaccines12101094](https://doi.org/10.3390/vaccines12101094); 42591813 — Sustaining the Elimination of Visceral Leishmaniasis: A Secondary Data Analysis of the National Kala-Azar Elimination Programme in Malda District, West Bengal, India (2018–2022), Cureus 2026, doi:[10.7759/cureus.112553](https://doi.org/10.7759/cureus.112553); 41769747 — Chandipura Virus: A Neglected Sandfly-Borne Neurotropic Virus in Children — Insights From Seven Decades of Evidence, Rev Med Virol 2026, doi:[10.1002/rmv.70119](https://doi.org/10.1002/rmv.70119) (abstract-only); 40589223 — Editorial: The 2025 World Health Assembly Pandemic Agreement and the 2024 Amendments to the International Health Regulations Combine for Pandemic Preparedness and Response, Med Sci Monit 2025, doi:[10.12659/MSM.950411](https://doi.org/10.12659/MSM.950411); 40953844 — National IHR authorities in the 2024 amendments to WHO's International Health Regulations: transforming the opportunity into reality, BMJ Glob Health 2025, doi:[10.1136/bmjgh-2025-019284](https://doi.org/10.1136/bmjgh-2025-019284); 40223587 — The 2024 Amendments to the International Health Regulations: A New Era for Global Health Law in Pandemic Preparedness and Response?, J Law Med Ethics 2025, doi:[10.1017/jme.2025.16](https://doi.org/10.1017/jme.2025.16) (abstract-only); 39921615 — Mpox: Global epidemic situation and

countermeasures, Virulence 2025, doi:[10.1080/21505594.2025.2457958](https://doi.org/10.1080/21505594.2025.2457958); 40627640 — Determining genotype and antimicrobial resistance of Salmonella Typhi in environmental samples by amplicon sequencing, PLoS Negl Trop Dis 2025, doi:[10.1371/journal.pntd.0013211](https://doi.org/10.1371/journal.pntd.0013211); 40155818 — Investigating Google Trends to forecast acute febrile illness outbreaks in North India reported through the Integrated Disease Surveillance Program, BMC Infect Dis 2025, doi:[10.1186/s12879-025-10801-0](https://doi.org/10.1186/s12879-025-10801-0); 39453563 — Application of the One Health Surveillance (OHS) Matrix to Evaluate the Disease Surveillance Systems in Gujarat, India: A Policy Content Analysis, J Epidemiol Glob Health 2024, doi:[10.1007/s44197-024-00317-2](https://doi.org/10.1007/s44197-024-00317-2). Also: World Health Organization — International Health Regulations (2005), 2024 amendments adopted 1 June 2024 by resolution WHA77.17 and entered into force 19 September 2025 [WHO 2025-Sep]; WHO Pandemic Agreement adopted by the Seventy-eighth World Health Assembly on 20 May 2025.



Screening for cardiovascular disease in the community

Asked in: Paper II 16-Jul-2020 — RGUHS MD Community Medicine

Definition — Screening is "the search for unrecognized disease or defect by means of rapidly applied tests, examinations or other procedures in apparently healthy individuals" (Park 28e, Ch 4, p 163). Applied to cardiovascular disease (CVD) in the community it is an organised, active search — among people who are **not** seeking care — for the *risk factors* that antedate CVD and for *unrecognised early disease*. Epidemiologists deliberately extended screening from disease to risk factors because factors of a pathophysiological nature such as serum cholesterol and blood pressure antedate disease and are amenable to effective intervention, so prevention can be applied before disease occurs (Park 28e, Ch 4, p 165).

Justification — the criteria for screening applied to cardiovascular disease

Criteria rest on two considerations, the **disease** to be screened and the **test** to be applied (Park 28e, Ch 4, pp 165–166).

Criterion (the disease)	Position for CVD in India
1. Important health problem, high prevalence	Per the NCD Country Profile 2016, overall mortality due to non-communicable diseases (NCDs) was 60%, of which CVD alone accounted for 26% (Sunder Lal, Ch 8, p 477). Coronary heart disease (CHD) prevalence is 1.6–7.4% in rural and 6.5–13.2% in urban adults ; 25% of urban and 10% of rural adults are hypertensive (Sunder Lal, Ch 8, p 477). Rheumatic heart disease (RHD) prevalence is 0.9 per 1,000 in the 5–15 year age group and it forms 20–30% of hospital admissions for CVD (Sunder Lal, Ch 8, p 477). Compared with all other countries, India suffers the highest loss in potentially productive years of life from CVD deaths at ages 35–64 (Park 28e, Ch 6, pp 437–438)

Criterion (the disease)	Position for CVD in India
2. Recognisable latent or early asymptomatic stage	Raised blood pressure, dysglycaemia, dyslipidaemia and atherosclerosis are silent for years — the ICMR-NCDIR study 2021 found 28.0% of adults hypertensive and 72% unaware of their status (Sunder Lal, Ch 8, p 491)
3. Natural history adequately understood	Established by long prospective cohorts, chiefly the <i>Framingham Study</i> running since 1951 (Park 28e, Ch 6, p 443)
4. A test detects disease before signs and symptoms	Sphygmomanometry, capillary/venous blood glucose, serum cholesterol, waist circumference, electrocardiogram (ECG), cardiac auscultation
5. Facilities for confirmation of diagnosis	NCD clinic at community health centre (CHC) with blood sugar, lipid profile, X-ray and ECG; district-hospital NCD clinic (Park 28e, Ch 7, pp 564–565)
6. Effective treatment exists	Antihypertensives, statins, antiplatelets, oral hypoglycaemics; benzathine penicillin prophylaxis for RHD
7. Agreed policy on whom to treat	Blood pressure $\geq 140/90$ mmHg, fasting plasma glucose ≥ 126 mg/dL, and the WHO/ISH 10-year risk bands (below)
8. Early detection and treatment reduces morbidity and mortality	Control of risk factors has produced a 50–80% decline in CVD incidence in high-income countries (Jugal Kishore, p 593)
9. Benefits exceed risks and costs	Screening for risk factors in a whole population is costly and must be supported by proven interventions and services close to the community (Sunder Lal, Ch 8, pp 502–503)

Criterion (the test)	Requirement	How CVD screening tools score
Acceptability	Not painful, discomfiting or embarrassing, since a high rate of cooperation is essential (Park 28e, Ch 4, pp 165–166)	Blood pressure and waist measurement are highly acceptable; venepuncture less so
Repeatability (reliability, precision, reproducibility)	Consistent result on repetition under the same conditions; depends on observer variation, biological (subject) variation and errors of the method (Park 28e, Ch 4, pp 165–166)	Blood pressure has large biological variation — hence two consecutive readings at least a minute apart, seated, twice daily (morning and evening) over several days, all values averaged , before labelling hypertension (Sunder Lal, Ch 8, p 492)
Validity	Sensitivity and specificity — sensitivity, the term introduced by <i>Yerushalmy</i> in the 1940s, is the ability to correctly identify all those who have the disease (Park 28e, Ch 4, p 167)	Auscultation alone is a reasonable RHD screen but needs echocardiographic confirmation (Sunder Lal, Ch 8, p 497)

Criterion (the test)	Requirement	How CVD screening tools score
Yield	"The amount of previously unrecognized disease that is diagnosed as a result of the screening effort"; raised by restricting screening to older/high-risk groups (Park 28e, Ch 4, p 168)	Restricting to age ≥ 30 years and prioritising a Community Based Assessment Checklist score >4 is precisely a yield-raising device
Others	Simplicity, safety, rapidity, ease of administration, cost — the choice is always a compromise, since more accurate tests are more expensive and time-consuming (Park 28e, Ch 4, pp 165–166)	Sensitivity and specificity move in opposite directions: at a 2-hour post-prandial glucose of 180 mg/dL sensitivity is 50% and specificity 99.8% (Park 28e, Ch 4, p 167)

What is screened, with which test and at which cut-off

Target	Screening tool	Cut-off / action threshold
Raised blood pressure	Mercury-free electronic sphygmomanometer	Normal $<120/ <80$; pre-hypertension 120–139/80–89; stage 1 140–159/90–99; stage 2 $\geq 160/ \geq 100$ mmHg (JNC VII; NCD Manual, pp 29–30; Sunder Lal, Ch 8, pp 491–492)
Diabetes / dysglycaemia	Fasting or random capillary or venous plasma glucose	Diabetes: fasting ≥ 126 mg/dL (WHO 2019; Park 28e, Ch 6, p 470) or 2-hour post-load / casual with symptoms ≥ 200 mg/dL; impaired fasting glucose 110 to <126 ; impaired glucose tolerance 2-hour >140 to <200 mg/dL (NCD Manual, pp 20–21). With a glucometer in the fed state the >200 mg/dL cut-off may be revised to >220 mg/dL (Sunder Lal, Ch 8, p 487)
Central obesity	Waist circumference	Scores 0 if male <90 cm / female <80 cm, 1 if 90–100 / 80–90 cm, 2 if >100 / >90 cm (Sunder Lal, Ch 8, p 504)
Tobacco and alcohol use, physical inactivity	Structured history	Physical activity <150 minutes/week scores 1; daily alcohol scores 1; daily smoking or smokeless tobacco scores 2 (Sunder Lal, Ch 8, p 504)
Dyslipidaemia	Total cholesterol ($\pm$ HDL)	Total cholesterol ≥ 320 mg/dL, low-density lipoprotein (LDL) cholesterol ≥ 240 mg/dL, or total-cholesterol/HDL ratio >8 places the person directly in the high-risk group (NCD Manual, pp 10–11); safe level of total cholesterol is taken as 200 mg/dL (Sunder Lal, Ch 6, p 246)
Established coronary disease	ECG at CHC/district NCD clinic	Used for confirmation, not for mass screening (Park 28e, Ch 7, pp 564–565)
Rheumatic heart disease (school children)	Cardiac auscultation	"Reliable auscultation is sufficient for screening of RHD in most cases"; confirm by echocardiography , supported by ECG, ASO titre, C-reactive protein, ESR and chest X-ray (Sunder Lal, Ch 8, p 497)

Types and strategies of screening used for cardiovascular disease

Type / strategy	Content	Place in CVD control
Mass screening	Screening of a whole population or a sub-group such as all adults, offered irrespective of the individual's particular risk (Park 28e, Ch 4, pp 164–165)	Basis of the national population-based screening of everyone aged 30 years and above
High-risk (selective) screening	Applied selectively to groups defined by epidemiological research; the family is one productive sub-group, since diabetes, hypertension and breast cancer tend to aggregate in families, so screening other members and close relatives detects additional cases (Park 28e, Ch 4, p 165)	Operationalised by the Community Based Assessment Checklist (CBAC) family-history question and score
Multiphasic screening	Two or more tests applied at one time	Randomised controlled studies in the UK and USA showed no benefit in mortality or morbidity reduction while increasing the cost of health services, casting doubt on its utility (Park 28e, Ch 4, p 165)
Opportunistic screening	Screening persons above 30 years at the point of primary contact with any health facility — village, sub-centre, PHC, CHC, district or tertiary hospital (Sunder Lal, Ch 8, p 502)	Cheap and continuous; misses those who never attend
Population strategy (Rose)	Directed at the whole population irrespective of individual risk; a small reduction in mean blood pressure or serum cholesterol produces a large reduction in CVD incidence (Park 28e, Ch 2, p 60)	Greatest potential; more than half of CHD cases occur in people not apparently at special risk — the central limitation of the high-risk strategy (Park 28e, Ch 6, p 443)
High-risk strategy	Brings preventive care to individuals at special risk, detected by optimum use of clinical methods (Park 28e, Ch 2, p 60)	This is what community screening delivers; complementary to, not a substitute for, the population strategy

Delivery in the community — population-based screening under the national NCD programme

Screening of hypertension, diabetes and three common cancers with fixed periodicity is an integral part of comprehensive primary health care at the sub-centre health and wellness centre (Sunder Lal, Ch 8, pp 502–503).

- **Target group and frequency** — all men and women **30–65 years**; **once a year** for diabetes, hypertension and obesity in persons above 30 years, and once in five years for oral, breast and cervical cancer (Sunder Lal, Ch 8, pp 502–503).
- **Who screens** — ASHAs and the ANM, under supervision of the lady health visitor, the medical officer and the mid-level provider, trained in oral visual examination, clinical breast examination and visual inspection with acetic acid; **about 30 persons can be screened in a day** (Sunder Lal, Ch 8, p 503).

- ▶ **Enumeration** — the first step is active enumeration by the ASHA of all persons over 30 years in each household on an individual health card; about 32–37% of the population is above 30 years (Sunder Lal, Ch 8, p 503; Ch 16, p 703).
- ▶ **Risk assessment** — the ASHA completes **Part A of the Community Based Assessment Checklist (CBAC)**, a six-item score covering age, tobacco use, daily alcohol use, waist circumference, physical activity and family history of hypertension, diabetes, heart disease/stroke or cancer (Sunder Lal, Ch 8, pp 503–504).
- ▶ **Early-detection questions (CBAC Part B)** — B1 for both sexes, **seven items**: shortness of breath; coughing more than 2 weeks; blood in sputum; history of fits; difficulty in opening the mouth; ulcer/patch/growth in the mouth not healed in two weeks; **any change in the tone of your voice**. B2 for women additionally: lump in the breast, blood-stained nipple discharge, change in shape and size of breast, bleeding between periods, bleeding after menopause, bleeding after intercourse, foul-smelling vaginal discharge. A **yes** to **any one** symptom means immediate referral to the nearest facility with a medical officer (Sunder Lal, Ch 8, pp 504–505).
- ▶ **Referral thresholds** — under the Government of India operational guidelines (2016), a systolic blood pressure over 140 and diastolic over 90 mmHg, or a random blood sugar of 140 mg/dL and above, is referred to the medical officer at the nearest facility for confirmation, laboratory investigation and initiation of treatment (Park 28e, Ch 7, p 565).
- ▶ **Treatment and continuity of care** — once hypertension or diabetes is established the patient receives at least a **month's supply of drugs from the PHC**; when stable, a three-month supply may be given with the ANM/ASHA visiting monthly to check compliance, diet, lifestyle and blood pressure/glucose. First follow-up at the PHC is at the **end of three months**, with an annual specialist consultation at the nearest nodal CHC NCD clinic (Park 28e, Ch 7, p 565).
- ▶ **Health promotion alongside** — camps, interpersonal communication, posters and banners in the community, household, schools, *anganwadis* and workplaces on healthy food, physical activity, avoiding tobacco and alcohol, and warning signs (Sunder Lal, Ch 8, p 530).

Total cardiovascular risk approach — WHO/ISH risk prediction charts

Charts indicate the **10-year risk of a fatal or non-fatal major cardiovascular event (myocardial infarction or stroke)** from age, gender, blood pressure, smoking status, total blood cholesterol and presence or absence of diabetes; one set is used where cholesterol can be measured and another where it cannot (NCD Manual, pp 10–11). A "smoker" includes all current smokers and those who quit less than one year before assessment, and a person with diabetes is one on insulin or an oral hypoglycaemic, or with fasting venous plasma glucose ≥ 126 mg/dL or 2-hour post-prandial ≥ 200 mg/dL on two separate occasions (NCD Manual, p 11).

10-year risk of a cardiovascular event	Risk class	Action (NCD Manual, section 4.8, p 15)
<10%	Low risk	Low risk is not <i>no</i> risk — conservative lifestyle management; reassess after 5 years unless health status changes significantly
10% to <20%	Moderate risk	Monitor risk profile every 2 years

10-year risk of a cardiovascular event	Risk class	Action (NCD Manual, section 4.8, p 15)
20% to <30%	High risk	Monitor risk profile yearly ; first-line antihypertensive may be considered (NCD Manual, p 18)
≥30%	Very high risk	Very high risk of fatal or non-fatal vascular events — monitor risk profile every 3–6 months ; first-line antihypertensive may be considered (NCD Manual, p 18)

Risk stratification is **unnecessary** — the person is high risk by definition — for those with established CVD, total cholesterol ≥320 mg/dL or LDL ≥240 mg/dL or total-cholesterol/HDL ratio >8, persistent blood pressure ≥160/≥100 mmHg, or renal failure/impairment (NCD Manual, pp 10–11). Risk factors **not** captured by the charts but which raise the estimate include premature CHD or stroke in a first-degree relative (male <55, female <65 years), triglycerides ≥150 mg/dL, HDL ≤40 mg/dL in men or ≤50 mg/dL in women, impaired fasting glycaemia or glucose tolerance, microalbuminuria and socioeconomic deprivation (NCD Manual, p 15).

Evaluation of a community CVD screening programme

Domain	Measure / issue
Validity of the test	Sensitivity and specificity; a test with high specificity has few false positives, and false positives both burden diagnostic facilities and bring discredit to the programme (Park 28e, Ch 4, p 168)
Predictive accuracy	Positive predictive value depends on sensitivity, specificity and disease prevalence — as prevalence falls the positive predictive value falls (Park 28e, Ch 4, p 168); specificity has a greater effect than sensitivity on predictive value (Gordis 7e, Ch 5, pp 126–127)
Yield	Previously unrecognised disease diagnosed per 1,000 screened; raised by targeting age ≥30 years and prioritising CBAC score >4 (Park 28e, Ch 4, p 168)
Coverage and compliance	Proportion of the enumerated ≥30-year population actually screened; proportion of screen-positives who complete referral and stay on treatment
Biases that inflate apparent benefit	Lead-time bias (survival appears longer only because diagnosis was earlier), length-biased sampling (slowly progressive disease is preferentially detected), overdiagnosis bias , referral bias and misclassification bias (Gordis 7e, Ch 18, p 368)
Definitive design	A randomised controlled trial in which one randomly selected group receives the screening test and the control group does not — ideally in the setting where the programme will run; case-control studies and comparison of trends between areas of differing coverage are the alternatives (Park 28e, Ch 4, p 170)
Cost	The cost of a test is not only the test procedure but the whole follow-up process set in motion by a positive result, even a false-positive one (Gordis 7e, Ch 18, p 391)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **DRAWN figure** — the screening-test 2×2 table with its formulas (draw as a four-cell grid with the four marginal totals). Column heading: **Diagnosis** — split into *Diseased* and *Not diseased*. Row heading: **Screening test result** — split into *Positive* and *Negative*. Fill the cells a = true-positive, b = false-positive, c = false-negative, d = true-negative; write the row totals a+b and c+d and the column totals a+c and b+d. Beneath the grid write the five formulas exactly (Park 28e, Ch 4, p 167): Sensitivity = $a/(a+c) \times 100$; Specificity = $d/(b+d) \times 100$; Predictive value of a positive test = $a/(a+b) \times 100$; Predictive value of a negative test = $d/(c+d) \times 100$; False-negatives = $c/(a+c) \times 100$ and false-positives = $b/(b+d) \times 100$. Add the one-line legend: *sensitivity and specificity are inversely related — for a 2-hour post-prandial glucose of 180 mg/dL they are 50% and 99.8%, so the cut-off chosen for a community CVD screen is always a compromise.*

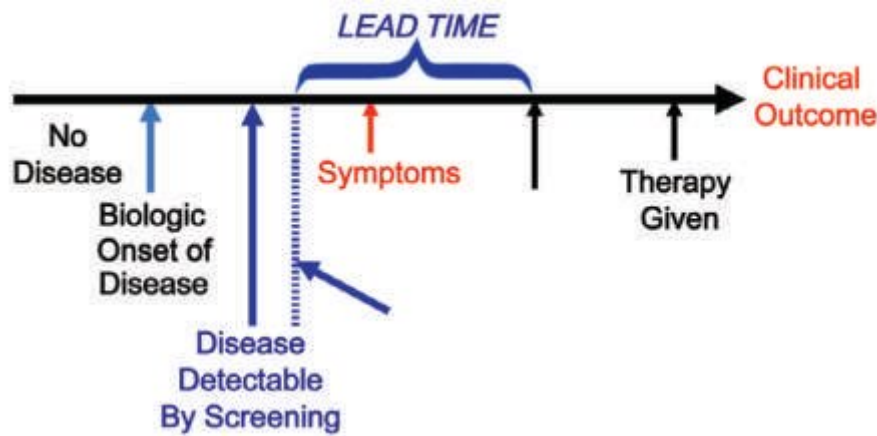
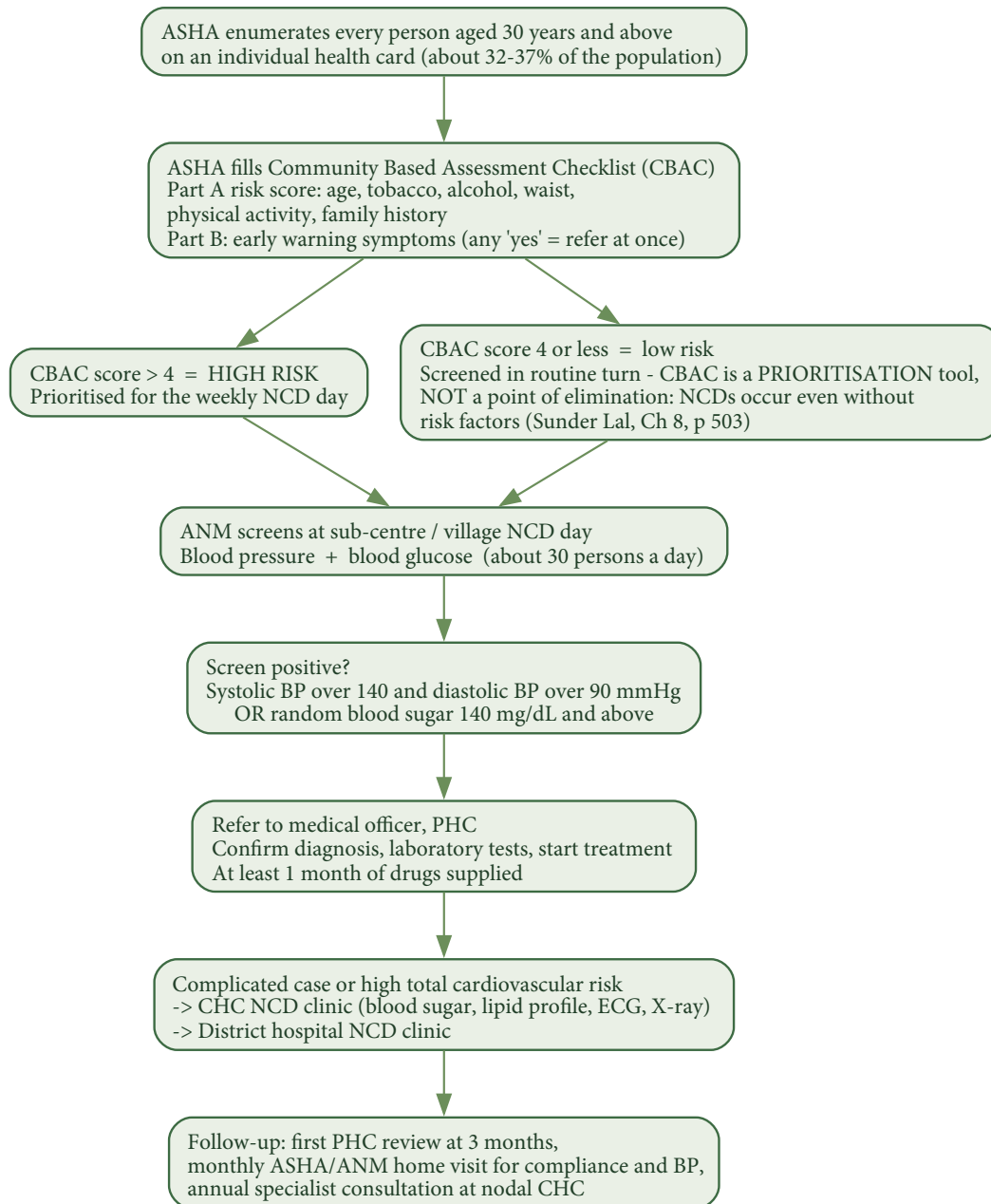
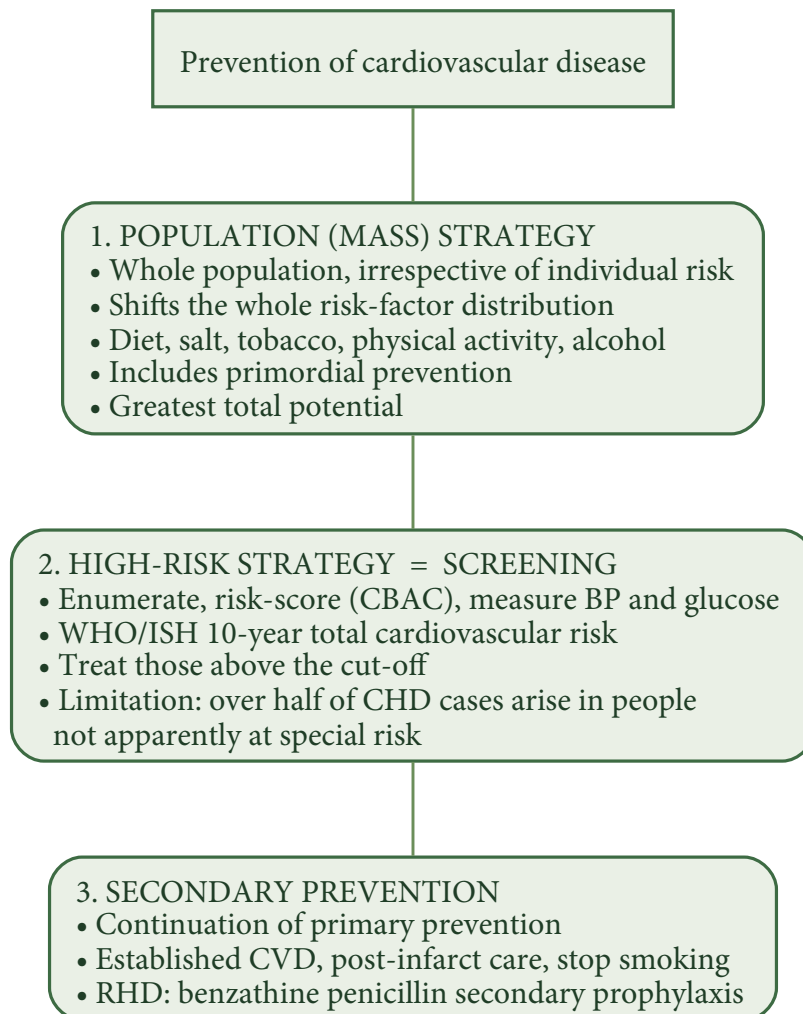


Fig 18.2 — Preclinical phase of the disease; natural history and lead time (Gordis Epidemiology 7e, p.371)



Community pathway for cardiovascular screening at a health and wellness centre, from enumeration to lifelong follow-up



Where community screening sits among the three complementary strategies of cardiovascular disease prevention

High-yield exam pointers

- ▶ **Definition to write verbatim:** screening is "the search for unrecognized disease or defect by means of rapidly applied tests, examinations or other procedures in apparently healthy individuals" (Park 28e).
- ▶ **Criteria split into DISEASE (9 points) and TEST (acceptability, repeatability, validity + yield, simplicity, safety, rapidity, ease of administration, cost) — write both headings; do not merge them.**
- ▶ **Programme numbers:** target 30–65 years; hypertension/diabetes **annually**, three cancers **once in 5 years**; CBAC score >4 = **prioritised for the weekly NCD day** (a low score never excludes anyone from screening); ASHA/ANM screen ~30 persons a day.
- ▶ **Referral cut-offs (GoI operational guidelines 2016):** systolic BP over 140 and diastolic over 90 mmHg, or random blood sugar 140 mg/dL and above → medical officer. Diagnostic cut-offs: fasting plasma glucose ≥126 mg/dL (WHO 2019), hypertension ≥140/90 mmHg.
- ▶ **WHO/ISH risk bands:** <10% low (reassess 5 y), 10–<20% moderate (2 y), 20–<30% high (yearly), ≥30% very high (3–6 monthly). Skip the chart altogether if established CVD, total cholesterol ≥320 mg/dL, BP ≥160/100 mmHg, or renal impairment.

- **Dated Indian figures:** NCD Country Profile 2016 — NCDs 60% of deaths, CVD 26%; ICMR-NCDIR 2021 — hypertension 28.0% of adults, **72% unaware**, urban 34% vs rural 25.7%; NFHS-5 (2019–21) hypertension 24% men, 21.3% women aged ≥ 15 y; RHD 0.9/1,000 at 5–15 years.
- **Four biases to name when asked to evaluate:** lead-time, length-biased sampling, overdiagnosis, referral bias — and the answer to all of them is a randomised controlled trial.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam — the programme under which this screening runs was launched as the National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Diseases and Stroke (NPCDCS) and is now the National Programme for Prevention and Control of Non-Communicable Diseases (NP-NCD), delivered through health and wellness centres now branded Ayushman Arogya Mandir, so a recent paper may use either name; the blood pressure classification quoted is the JNC VII scheme reproduced in the Indian NCD manual and in Sunder Lal, and newer international guidelines use a lower hypertension threshold, but the Indian programme cut-off for referral remains 140/90 mmHg; the retrieved corpus gives the proportion of the population aged over 30 years as 37% in one place and 32% in a worked sub-centre example, so quote the range rather than a single figure.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, ICMR, WHO, India Hypertension Control Initiative, World Heart Federation 2023 rheumatic heart disease screening criteria, 2025 AHA/ACC hypertension guideline, UN High-Level Meeting on NCDs 2025

Programme and policy updates (India)

- The Community Based Assessment Checklist has now been checked against actual measurements in the community — on the authors' own account probably the first large-scale community-based validation, the earlier evidence having been hospital-based — and it performed poorly. Among 32,686 adults aged ≥ 30 years in Mysuru (2019–20) the area under the ROC curve was only 0.62 (95% CI 0.61–0.63) against a random capillary glucose ≥ 140 mg/dL and 0.61 (95% CI 0.60–0.61) against blood pressure $\geq 140/90$ mmHg; at the programme cut-off of >4 the sensitivity was 63.0% with specificity 54.3% for dysglycaemia and 58.6% with specificity 56.9% for raised blood pressure. The authors' recommendation is that frontline workers measure glucose and blood pressure *alongside* the checklist rather than triage on the score alone — the exact reason the checklist is a prioritisation tool and never a point of elimination [PMID 42555696]
- The India Hypertension Control Initiative, built on the WHO HEARTS package, has scaled from 26 districts to 89 districts across 15 states: of 1,235,453 people registered between 2018 and 2021, 1,046,512 remained in care and 44% had controlled blood pressure, ranging 26–57% by facility type. Control fell from 68% when defined on a single recent reading to 52% when defined on two-visit readings — a direct field demonstration of the repeatability problem that forces repeat measurement before labelling hypertension [PMID 41314840]
- A qualitative study in West Garo Hills, Meghalaya (52 participants across health workers, village health councils and community members) mapped why uptake stays low despite an ASHA-

delivered programme: fear of diagnosis and of lifelong treatment, loss of a working day, elders deciding whether the household participates, preference for traditional medicine, a high village-to-health-and-wellness-centre ratio and unfilled posts. Hypertension screening coverage in that state was 42.5% against a national figure of 70.5% [PMID 41933310]

Current national data

- ICMR-INDIAB-17 (2023) supplies the current national denominators from 113,043 adults aged ≥ 20 years across all 31 states and union territories: diabetes 11.4% (95% CI 10.2–12.5), prediabetes 15.3%, hypertension 35.5%, generalised obesity 28.6%, abdominal obesity 39.5% and dyslipidaemia 81.2% — substantially higher than the older NCD Country Profile and ICMR-NCDIR figures quoted above, and worth citing as the newest nationally representative estimate [PMID 37301218]
- The India Hypertension Control Initiative baseline population survey in nine districts (2018–19; 7,047 people with hypertension aged 18–69) gives the cascade to quote when asked what community screening has to fix: 52.4% aware, 40.8% on treatment, only 14.5% controlled, with control below 5% in two of the nine districts [PMID 40121324]
- ICMR-INDIAB-22 (2025) found impaired kidney function ($\text{eGFR} < 60 \text{ mL/min/1.73 m}^2$) in 3.2% (95% CI 2.9–3.5) of 25,408 adults, rising to 8.1% in those with both hypertension and diabetes (adjusted OR 6.2, 95% CI 4.5–8.6 versus neither). Diabetes alone carried more risk than hypertension alone (OR 3.2 versus 2.4). The authors propose adding serum creatinine to the existing NP-NCD screening package — which would give the "renal impairment = automatically high risk" rule of the risk charts a measurable basis at primary care level [PMID 41100657]

Elimination, surveillance and disease status

- The 2023 World Heart Federation echocardiographic criteria for rheumatic heart disease were written specifically to permit task-shifting to handheld devices and non-specialist operators. A large real-world programmatic test — 11,198 schoolchildren aged 5–17 years in Lira District, northern Uganda, screened by nurse-performed handheld echocardiography on a three-view protocol — gave, after correction for partial verification bias, sensitivity 60.0% (95% CI 46.6–73.5) and specificity 96.3% (95% CI 96.0–96.7), with positive predictive value 19.5% and negative predictive value 99.4%; missed cases were mostly earliest-stage (stage A) or isolated aortic regurgitation, and no stage C or D case was missed among those who had confirmatory echocardiography. It is not the first such field evaluation — the same report sets its numbers beside an earlier northern Uganda programme run by primary-care nurses across more than 30 providers, which reached sensitivity 55% and specificity 97%. High specificity with a moderate sensitivity is exactly the profile of a frontline triage test, so the auscultation-then-confirmation model in the table above still holds, but the confirmation step is moving from the cardiologist to a trained nurse with a handheld probe [PMID 42419874]

Guideline and legislative updates

- DANCavas (2022) is the definitive randomised answer to "does population cardiovascular screening work": 46,611 Danish men aged 65–74 randomised 1:2 to invitation for non-contrast ECG-gated CT (coronary calcium, aneurysm, atrial fibrillation), ankle-brachial pressures and blood tests, or to usual care. Uptake was 62.6%, and at a median 5.6 years all-cause mortality — the primary outcome — was not significantly reduced (HR 0.95, 95% CI 0.90–1.00, $p = 0.06$); the

stroke hazard ratio was 0.93 (95% CI 0.86–0.99). The companion trial-based economic analysis put comprehensive screening at €10,812 per life-year and €9,075 per QALY gained — cost-effective at conventional thresholds, with the target group left unsettled. Cite it when asked to evaluate a screening programme: it is the randomised controlled trial design the textbook calls for, and it returned a negative primary result [PMID 36027560]; [PMID 36029019]

- The August 2025 AHA/ACC hypertension guideline moved drug treatment to a blood pressure of $\geq 130/80$ mmHg for adults with diabetes, chronic kidney disease, or a PREVENT-estimated 10-year cardiovascular risk $\geq 7.5\%$ — a materially lower trigger than the 140/90 mmHg referral cut-off the Indian programme still uses, and the discrepancy to flag rather than to blur in an answer. Modelled on NHANES, about 22.8 million US adults (28% of those with hypertension) became eligible, of whom 43% were untreated; treating all of them was projected to prevent roughly 200,900 all-cause and 162,600 cardiovascular deaths over ten years [PMID 42522939]

Global and WHO developments

- The Fourth UN High-Level Meeting on the Prevention and Control of NCDs and Promotion of Mental Health, held during the UN General Assembly in September 2025, produced a Political Declaration since adopted by 175 member states, which frames early detection, integrated care and community-driven strategies as essential rather than optional components of progress in reducing premature NCD deaths — SDG target 3.4; the Fifth such meeting is scheduled for 2031. India is among the few countries to have written meaningful involvement of people living with NCDs into its national NCD programme, with state-level guidelines in Punjab and Maharashtra and community advisory boards in Rajasthan and Andhra Pradesh [PMID 42144248]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 2 Concept of Health and Disease, p 60; Ch 4 Screening for Disease, pp 163–170; Ch 6 Epidemiology of Chronic Non-communicable Diseases and Conditions, pp 437–438, 443, 470; Ch 7 Health Programmes in India, pp 564–565); Gordis Epidemiology 7e (Ch 5 Assessing the Validity and Reliability of Diagnostic and Screening Tests, pp 126–127; Ch 18 The Epidemiologic Approach to Evaluating Screening Programs, pp 368, 371, 391); Sunder Lal — Textbook of Community Medicine (Ch 6, p 246; Ch 8 Epidemiology of Communicable and Non-communicable Diseases, pp 477, 487, 491–492, 497, 502–505, 530; Ch 16 Health Planning and Management, p 703); NCD Combined Manual for Medical Officers, Ministry of Health and Family Welfare (pp 10–11, 15, 18, 20–21, 29–30) incorporating the WHO/ISH Pocket Guidelines for Assessment and Management of Cardiovascular Risk; Jugal Kishore — National Health Programmes of India (p 593); Government of India Operational Guidelines for Prevention, Screening and Control of Common Non-Communicable Diseases, 2016 (as reproduced in Park 28e); WHO diagnostic criteria for diabetes, 2019.

Recent-advances sources: 42555696 — Screening performance of community-based assessment checklist compared with point of care screening tests to identify high-risk individuals for diabetes and hypertension in urban Mysore, India, Indian Journal of Medical Research 2026, doi:[10.25259/IJMR.78.2026](https://doi.org/10.25259/IJMR.78.2026); 41314840 — Scaling India Hypertension Control Initiative strategies to 15 states — treatment outcomes and risk factors for uncontrolled blood pressure, India: a cohort study, 2018–2021, BMJ Open 2025, doi:[10.1136/bmjopen-2025-106372](https://doi.org/10.1136/bmjopen-2025-106372); 41933310 — Community and health workers perspectives on barriers to diabetes and hypertension screening in North Eastern India: a qualitative study, BMC Health Services Research 2026, doi:[10.1186/s12913-026-14435-z](https://doi.org/10.1186/s12913-026-14435-z); 37301218 — Metabolic non-communicable disease health report of India: the ICMR-INDIAB national cross-sectional study (ICMR-INDIAB-17), Lancet Diabetes & Endocrinology 2023, doi:[10.1016/S2213-8587\(23\)00119-5](https://doi.org/10.1016/S2213-8587(23)00119-5) (abstract-only); 40121324 — Population-based surveillance for hypertension awareness, treatment, and control in nine districts — India Hypertension Control Initiative, 2018–19, Journal of Human Hypertension 2025, doi:[10.1038/s41371-025-01005-9](https://doi.org/10.1038/s41371-025-01005-9); 41100657 — Prevalence of impaired kidney function and its association with diabetes and hypertension in India: the ICMR-INDIAB study (ICMR-INDIAB-22), Indian Journal of Medical Research 2025, doi:[10.25259/IJMR.1027.2025](https://doi.org/10.25259/IJMR.1027.2025); 42419874 — Operational performance of the 2023 World Heart Federation screening criteria for rheumatic heart disease in a high-risk, resource-limited population, Open Heart 2026, doi:[10.1136/openhrt-2026-004196](https://doi.org/10.1136/openhrt-2026-004196); 36027560 — Five-Year Outcomes of the Danish Cardiovascular Screening (DANCAVAS) Trial, New England Journal of Medicine 2022, doi:[10.1056/NEJMoa2208681](https://doi.org/10.1056/NEJMoa2208681) (abstract-only); 36029019 — Cost effectiveness of population screening vs. no screening for cardiovascular disease: the Danish Cardiovascular Screening trial (DANCAVAS), European Heart Journal 2022, doi:[10.1093/eurheartj/ehac488](https://doi.org/10.1093/eurheartj/ehac488) (abstract-only); 42522939 — Population-Level 10-Year Implications of the New 2025 American Heart Association/American College of Cardiology Hypertension Guideline Recommendations for Primary Prevention in the United States: A NHANES-Based Cohort Study (2009–2018), Journal of the American Heart Association 2026, doi:[10.1161/JAHA.125.048616](https://doi.org/10.1161/JAHA.125.048616) (abstract-only);

42144248 — Integrating meaningful involvement of people living with non-communicable diseases in India's National NCD Programme, BMJ Global Health 2026, doi:10.1136/bmjgh-2025-020189. Also used: World Heart Federation 2023 echocardiographic criteria for rheumatic heart disease; AHA/ACC hypertension guideline, August 2025; Political Declaration of the Fourth UN High-Level Meeting on NCDs and Mental Health, September 2025.



Sensitivity and specificity issues in HIV screening

Asked in: Paper I 21-Jun-2024 — RGUHS MD Community Medicine

Definition — **Sensitivity** is the ability of a test to identify correctly all those who have the disease, that is the "true positives"; the term was introduced by *Yerushalmy* as a statistical index of diagnostic accuracy. **Specificity** is the ability of the test to identify correctly those who do not have the disease, that is the "true negatives". Both, with predictive accuracy, are **inherent properties of a screening test** and constitute its **validity** (Park 28e, Ch 4, p 167). In human immunodeficiency virus (HIV) screening these two properties pull against each other, and because adult HIV prevalence in India is very low, the operational problem is not the sensitivity or specificity of the kit but the **predictive value** of the result it gives.

Measures used to evaluate an HIV screening test

Cell of the 2 × 2 table	Meaning
a	True positive — infected, test reactive
b	False positive — not infected, test reactive
c	False negative — infected, test non-reactive
d	True negative — not infected, test non-reactive

Measure	Formula	What it answers in HIV screening
Sensitivity	$a / (a + c) \times 100$	Of all infected persons, what proportion does the kit pick up?
Specificity	$d / (b + d) \times 100$	Of all uninfected persons, what proportion does the kit correctly clear?
Predictive value of a positive test	$a / (a + b) \times 100$	Given a reactive result, what is the probability the person is truly infected?
Predictive value of a negative test	$d / (c + d) \times 100$	Given a non-reactive result, what is the probability the person is truly uninfected?
False-negative rate	$c / (a + c) \times 100$	Infected persons wrongly told they are free of infection
False-positive rate	$b / (b + d) \times 100$	Healthy persons wrongly labelled infected

(Formulas: Park 28e, Ch 4, p 167.) Predictive value depends on sensitivity, specificity and **most importantly on the prevalence** of infection in the population tested; because prevalence varies widely between States and regions, predictive value is a more important determinant of the usefulness of an HIV test than sensitivity and specificity alone (Sunder Lal, Ch 7, pp 299-300).

Validity of the tests actually used for HIV

Test	Role	Reported validity	Limitation that creates the "issue"
Enzyme-linked immunosorbent assay (ELISA)	Sensitive screening test	Sensitivity > 99.9 %	Repeatedly reactive results must be confirmed by Western blot, otherwise false positives are reported (Park 28e, Ch 5, p 421)
Western blot	Confirmatory test (antibody to core protein p24 and envelope glycoprotein gp41)	Specificity, when combined with ELISA, > 99.99 %	Indeterminate results in early HIV infection, HIV-2 infection, autoimmune disease, pregnancy and recent tetanus toxoid administration (Park 28e, Ch 5, pp 420-421); technically demanding, needs trained readers
Rapid / simple tests (used in Integrated Counselling and Testing Centres, ICTC)	Point-of-care screening, result within 30 minutes	National AIDS Control Organisation (NACO) recommends kits detecting > 99.5 % of infected persons with false positives in < 2 % of those tested	User-friendly but a single reactive rapid test is never a diagnosis (Jugal Kishore, Ch 1, p 347)
First-generation ELISA (historical)	Screening	Sensitivity only 95 %	Second-generation and current ELISA / rapid / simple tests reach about 99 % for HIV-1 (Sunder Lal, Ch 7, p 300)

The issues in HIV screening

Issue	Mechanism	Public health consequence
1. Sensitivity and specificity are inversely related	Raising sensitivity by lowering the cut-off is possible only at the expense of specificity, and vice versa; no test is 100 % sensitive and 100 % specific (Park 28e, Ch 4, pp 167-168)	A single kit cannot simultaneously protect the blood supply and avoid mislabelling healthy people
2. The window period — a <i>biological</i> false negative	HIV antibodies take 2-12 weeks to appear (Park 28e, Ch 5, p 417); the window is conventionally taken as 6-12 weeks, and a negative becomes certain only 6 months after last exposure (Sunder Lal, Ch 4, pp 158-159)	The person is most infectious in this phase because of high viraemia, yet tests negative. This false negative is a property of the host, not of the kit — a "100 % sensitive" assay cannot remove it

Issue	Mechanism	Public health consequence
3. Very low prevalence in India	Adult (15-49 years) HIV prevalence 0.20 % (0.17-0.25) , India HIV Estimates 2023 ; prevalence among antenatal clinic attendees 0.22 % (HIV Sentinel Surveillance 2021) (Park 28e, Ch 5, p 414)	Even a highly specific test yields mostly false positives on mass screening — the predictive value of a positive test collapses (see next section)
4. Indeterminate results on the confirmatory test	Cross-reacting antibody producing indeterminate Western blot results in early HIV infection, HIV-2 infection, autoimmune disease, pregnancy and recent tetanus toxoid administration (Park 28e, Ch 5, p 421)	Neither a positive nor a negative report can be issued; repeat testing after an interval and referral of the sample to a reference laboratory are needed
5. Consequences of a false positive are unusually severe for HIV	A wrong positive is not merely an "extra diagnostic work-up" as in other screening	Stigma, discrimination, psychological trauma, marital and employment consequences; counselling must strengthen coping with stigma and social impact (Sunder Lal, Ch 4, pp 159-160). False positives also discredit screening programmes and needlessly discard blood units
6. Consequences of a false negative	Missed infection	Transfusion-transmitted HIV, false reassurance with continued high-risk behaviour, missed opportunity for prevention of parent-to-child transmission and for "test and treat"
7. Repeatability and laboratory error	Repeatability depends on observer variation, biological variation and errors of the test itself (Park 28e, Ch 4, pp 165-166); laboratory results differ between laboratories	Requires External Quality Assessment Scheme — coded panels sent from the State Reference Laboratory twice a year, and each ICTC sends 20 % of all positive and 5 % of all negative samples collected in the first week of every quarter for cross-checking to the State Reference Laboratory once every quarter (Jugal Kishore, Ch 1, pp 347-348)

Effect of prevalence on the predictive value of a reactive test

Using Sunder Lal's illustrative ELISA figures — sensitivity **98 %**, specificity **99 %**, the values of his worked example rather than the **> 99.9 %** sensitivity quoted for current ELISA above — applied to 1,00,000 persons tested at different prevalences:

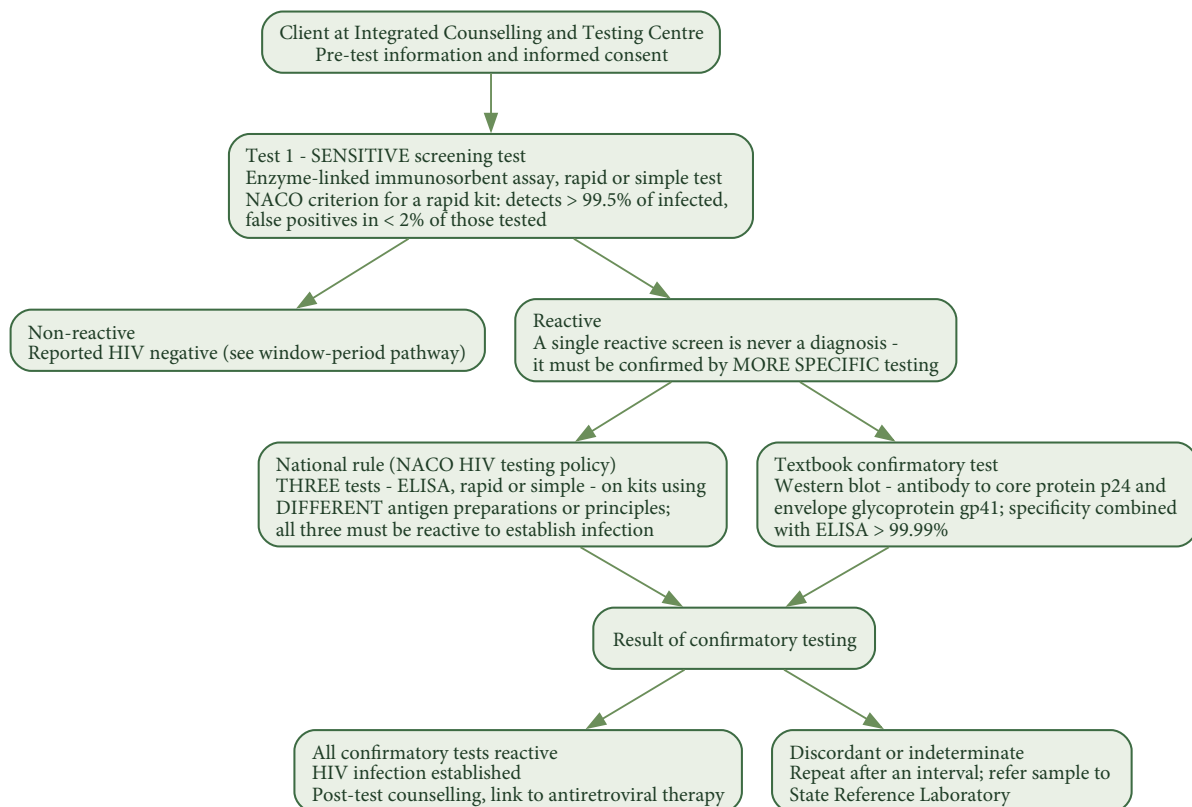
Prevalence of HIV in the population screened	True positives	False positives	Predictive value of a positive test
5 % (Sunder Lal's worked example — hypothetical 5 % prevalence)	4,900	950	83.8 %
1 %	980	990	49.7 %
0.22 % (antenatal attendees, HIV Sentinel Surveillance 2021)	216	998	17.8 %

Prevalence of HIV in the population screened	True positives	False positives	Predictive value of a positive test
0.20 % (Indian adults, India HIV Estimates 2023)	196	998	16.4 %

(5 % row: Sunder Lal, Ch 7, p 300, Table 7.11. Lower rows: the same sensitivity and specificity arithmetically re-applied at Indian prevalence figures from Park 28e, Ch 5, p 414.)

Two corollaries the examiner expects:

- **Specificity has a greater effect than sensitivity on predictive value**, because in a low-prevalence population the non-diseased are overwhelmingly the larger group, so even a 1 % false-positive rate generates more false positives than there are true cases (Gordis 7e, Ch 5, pp 126-127).
- The **more sensitive** a test, the better its **negative** predictive value; the **more specific** a test, the better its **positive** predictive value (Fletcher, Ch 8, p 136).



Sensitivity 98% and specificity 99% (Sunder Lal, Ch 7, p 300, Table 7.11) applied to 1,00,000 persons at Indian prevalences from Park 28e, Ch 5, p 414.

How the issue is resolved — the serial (multi-test) strategy

Because no single test satisfies both requirements, **two or three different tests are applied in sequence**: first a **sensitive** test to detect HIV antibody, then a **specific confirmatory** test to weed out the false positives (Park 28e, Ch 5, p 420). To establish HIV infection, **three tests based on different antigen preparations must be reactive** (Sunder Lal, Ch 4, p 159; Jugal Kishore, Ch 2, p 763); NACO uses a **two-test protocol** for sentinel surveillance (Park 28e, Ch 7, pp 520-521).

The gain is shown by Sunder Lal's two-stage data (Ch 7, p 300, Tables 7.11 and 7.12): of 1,00,000 persons at 5 % prevalence, ELISA gave **5,850** reactives — 4,900 true positives and 950 false positives — a

predictive value of only 83.8 %; Western blot applied to those 5,850 gave 4,656 reactivities of whom 4,655 were truly infected — **predictive value 99.9 %**. The price is a loss of sensitivity: **net sensitivity falls to 4,655/5,000 = 93.1 %** — 100 infected persons missed by the ELISA and a further 245 by the Western blot, 345 in all. This is the general rule — **serial testing gains specificity and loses sensitivity**, whereas **parallel testing gains sensitivity and loses specificity and positive predictive value** (Gordis 7e, Ch 5, p 121; Fletcher, Ch 8, pp 144-145).

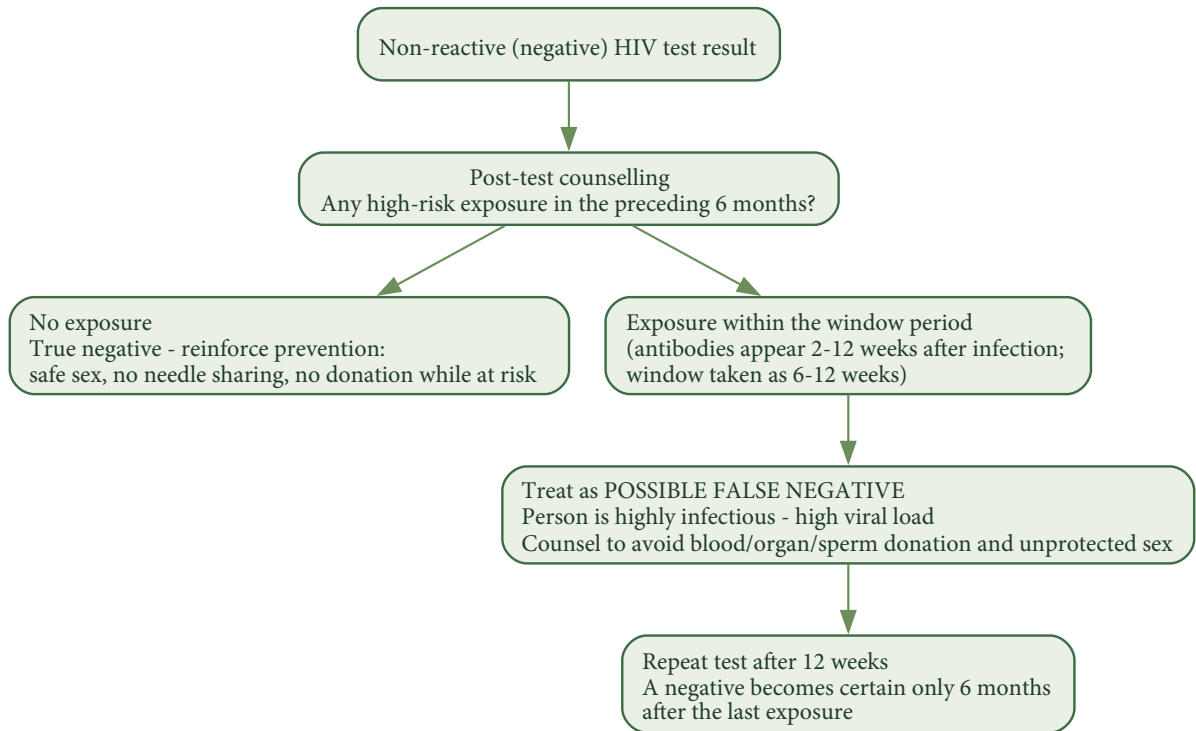
Choosing the strategy according to the purpose of testing

Purpose	Which property is maximised	Why	Practice
Blood and blood-product safety	Sensitivity — a low false-negative rate is the priority	One missed unit transmits a lethal, incurable infection	Mandatory testing of every unit of blood for HIV, hepatitis B and C, malaria and syphilis (Sunder Lal, Ch 8, p 349); a single ELISA / rapid / simple test, unlinked and anonymous, sero-reactive units discarded by incineration (Jugal Kishore, Ch 1, pp 351-352)
Diagnosis / case identification in an individual	Specificity and positive predictive value — a low false-positive rate is the priority	A wrong positive causes stigma, discrimination and lifelong therapy	Serial algorithm: reactive screen → confirmatory test(s) on different antigens; result given only after post-test counselling (Park 28e, Ch 5, p 420; Jugal Kishore, Ch 2, p 763)
Sentinel surveillance (measuring prevalence)	Balance — errors in both directions must be small and constant over time	The estimate, not the individual, is the output	Unlinked anonymous testing with informed consent, two-test protocol , dried blood spot or serum, sample size 400 at antenatal sites and 250 at high-risk-group sites, once in 2 years (Park 28e, Ch 7, pp 520-521)

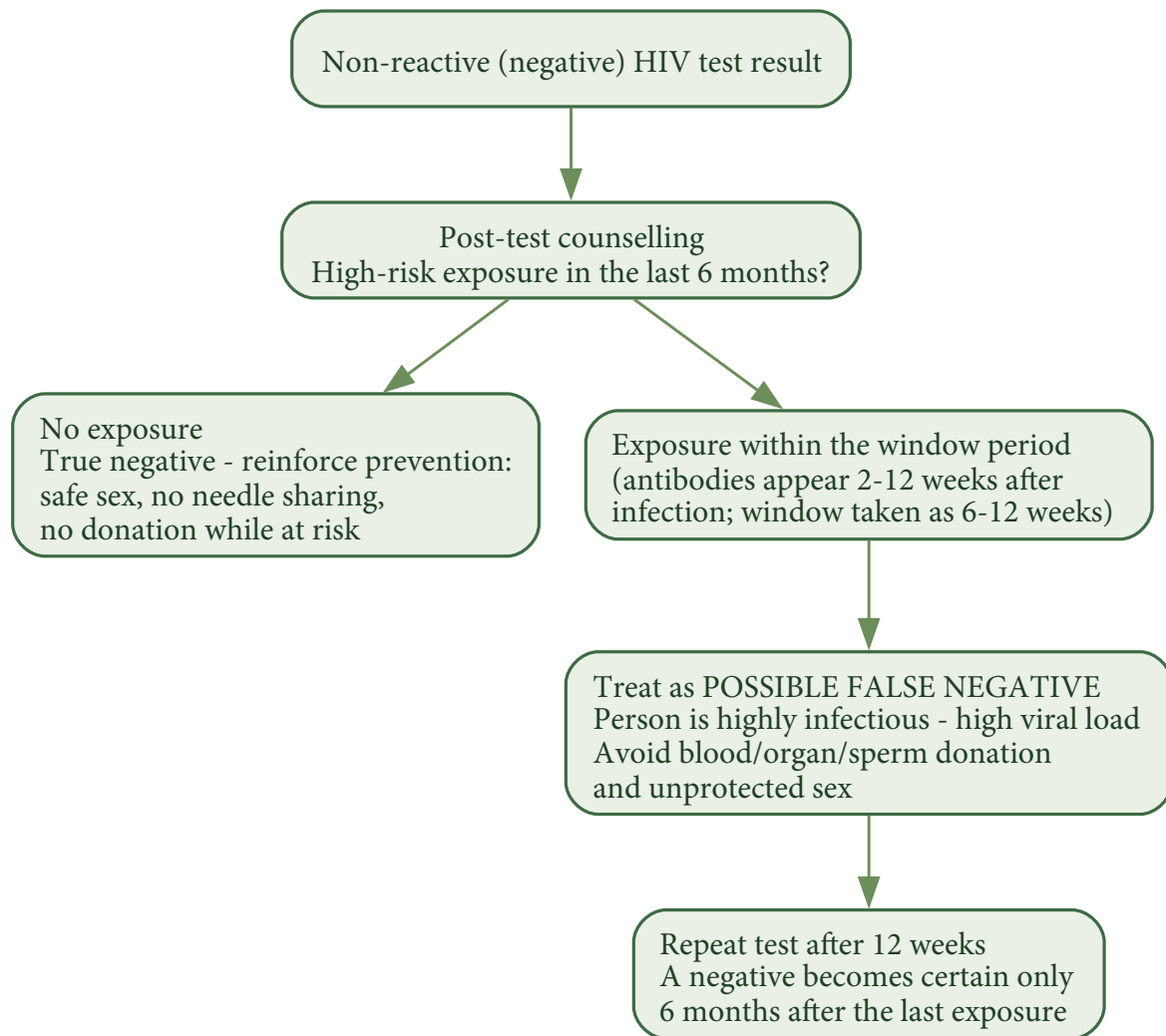
The five WHO principles that must accompany every HIV testing service are the **five C's** — **consent, confidentiality, counselling, correct test results and connection to care and treatment** (Park 28e, Ch 5, p 421).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Sensitivity–specificity 2 × 2 table:** four-cell box, columns *Infected / Not infected*, rows *Test reactive / Test non-reactive*; label cells **a true positive, b false positive, c false negative, d true negative**; add row totals (a+b), (c+d) and column totals (a+c), (b+d); write the six formulas alongside.
- **Overlapping distributions and the cut-off:** two overlapping bell curves labelled *Non-diseased* and *Diseased* on a horizontal axis of test value; mark a vertical cut-off line; shade the overlap right of the line **false positives**, left of it **false negatives**; arrow showing that moving the cut-off left raises sensitivity and lowers specificity, and right does the reverse.



Serial HIV testing algorithm — a sensitive screen followed by specific confirmation



Interpreting a non-reactive HIV result — the window period as a source of false negatives

High-yield exam pointers

- **Formulas to write first:** sensitivity = $a/(a+c)$; specificity = $d/(b+d)$; positive predictive value = $a/(a+b)$; negative predictive value = $d/(c+d)$; false-negative rate = $c/(a+c)$; false-positive rate = $b/(b+d)$.
- **Numbers to quote:** ELISA sensitivity > 99.9 %; Western blot with ELISA specificity > 99.99 %; NACO rapid-kit recommendation > 99.5 % detection with < 2 % false positives; first-generation ELISA sensitivity only 95 %.
- **Indian denominators, dated:** adult HIV prevalence 0.20 % (India HIV Estimates 2023); antenatal clinic prevalence 0.22 % (HIV Sentinel Surveillance 2021) — quote these to explain why the predictive value of a positive test is low.
- **Window period 2-12 weeks (Park);** retest at 12 weeks, negative certain at 6 months — the false negative that no amount of test sensitivity can abolish.
- **Serial testing** → gains specificity, loses sensitivity; **parallel testing** → gains sensitivity, loses specificity and positive predictive value. Specificity affects positive predictive value more than sensitivity does.

- ▶ **Sunder Lal's two-stage worked example:** 1,00,000 tested at 5 % prevalence → ELISA 5,850 reactive, positive predictive value 83.8 %; Western blot → positive predictive value 99.9 % but net sensitivity only **93.1 %** (345 of 5,000 infected missed).
- ▶ **Purpose decides the property:** blood safety → maximise sensitivity (single test); individual diagnosis → maximise specificity (three tests, different antigens); surveillance → unlinked anonymous two-test protocol.
- ▶ **Five C's of HIV testing services:** consent, confidentiality, counselling, correct results, connection to care.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park 28e still describes Western blot as the standard confirmatory test, whereas the national HIV testing policy requires three tests on kits with different antigen preparations for diagnosis in an Integrated Counselling and Testing Centre; the retrieved corpus supports the three-test rule (Sunder Lal, Jugal Kishore) without printing the current kit-wise algorithm, so no kit names are quoted here. Coverage note — the corpus did not contain a fourth-generation antigen-antibody combination assay or nucleic acid testing algorithm for Indian HIV screening, so the shortening of the window period by those tests is not quantified in this answer.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 4 (Screening for Disease, pp 165-169), Ch 5 (Epidemiology of Communicable Diseases, pp 414, 417, 420-421), Ch 7 (Health Programmes in India, pp 518-521); Sunder Lal — Textbook of Community Medicine, Ch 4 (Principles of Health Promotion and Education, pp 158-160), Ch 7 (Epidemiology, pp 299-301), Ch 8 (Epidemiology of Communicable and Non-Communicable Diseases and Related Disease-Specific National Health Programmes, p 349); Gordis Epidemiology 7e — Ch 5 (Assessing the Validity and Reliability of Diagnostic and Screening Tests, pp 121, 126-127); Fletcher — Clinical Epidemiology: The Essentials, Ch 8 (Diagnosis, pp 136, 144-145); Jugal Kishore — National Health Programmes of India, Ch 1 (National AIDS Control Programme, pp 341-352) and Ch 2 (national HIV testing policy and safe blood, pp 762-763); India HIV Estimates 2023 (NACO/ICMR) and HIV Sentinel Surveillance 2021, as quoted in Park 28e.



Sentinel surveillance

Asked in: Paper II 16-Jul-2020 — RGUHS MD Community Medicine

Definition — Sentinel surveillance is the systematic and routine collection of data on a health event from a small, deliberately selected set of reporting units — hospitals, clinics, laboratories or individual practitioners — instead of from every provider, the findings being extrapolated to the whole population to estimate disease burden. It exists because no routine notification system can identify all cases of an infection or disease, so a method of picking up the missing cases and thereby supplementing the notified cases is required (Park 28e, Ch 2, p 59). It is typically an indicator-based system and the most efficient way of collecting high-quality data in a timely way (Oxford 7e, Ch 42, p 1563).

Origin — The approach has its roots in occupational health: birds detecting odourless lethal gases such as carbon monoxide in mines may be the earliest form, sentinel animal flocks later detected arboviruses heralding human encephalitis epidemics, and *Rutstein* and colleagues extended the concept to the health-care system generally (Maxcy-Rosenau-Last 15e, Ch 2, p 39).

Rationale, objectives and uses

- ▶ **Complete the case count and estimate the burden.** Routine (passive) notification is incomplete and biased; sentinel data supplement the notified cases and are extrapolated to estimate symptom frequency, prevalence, incidence and geographical distribution, for the whole population or for a defined high-risk subpopulation (Park 28e, Ch 2, p 59; Oxford 7e, Ch 24, pp 945–946).
- ▶ **Signal trends, detect outbreaks and monitor indicators over time,** as a rapid economical alternative to whole-population surveillance; a fixed network — a hybrid between a health information system and a sample survey — also tracks demographic and epidemiological indicators including intervention impact (Oxford 7e, Ch 42, p 1563; Handbook of Epidemiology, Ch 38, pp 1540–1541).
- ▶ **Capture events the regular system cannot capture** — severe malaria, malaria deaths and antimalarial effectiveness, which routine malaria reporting misses because it is poorly articulated with hospitals (Park 28e, Ch 7, p 498; Sunder Lal, Ch 8, p 377).
- ▶ **Establish the seasonal pattern** so preventive programmes can be timed to it, and trigger immediate local investigation whenever an abnormal rise in cases is reported (Sunder Lal, Ch 7, pp 303–305).
- ▶ **Act as a "watch dog" of preventive service quality** — whether children still develop a vaccine-preventable disease after completing immunisation — and reveal a shift of disease to higher age groups or clustering in particular localities (Sunder Lal, Ch 7, pp 304–305).
- ▶ **Sustain laboratory work that is impossible nationally** — a few sites can collect specimens for viral culture or other diagnostics alongside case counts (Handbook of Epidemiology, Ch 33, p 1325).

Design and operation

Design element	What is done
Choice of unit	A limited number of reporting units chosen for their high likelihood of accurately identifying cases and their commitment to prompt reporting (Oxford 7e, Ch 24, p 946); providers most likely to serve the affected population — e.g. child health clinics and paediatricians for childhood disease (Handbook, Ch 33, p 1325)
Matching the unit to disease severity	For influenza, sentinel influenza-like illness (ILI) sites in out-patient settings monitor mild illness while sentinel severe acute respiratory infection (SARI) sites in hospitals monitor admitted severe illness (Oxford 7e, Ch 42, p 1563); in the World Health Organization COVID-19 framework the sentinel ILI/acute respiratory infection/SARI site's assigned role is immediate case notification plus virological surveillance (Park 28e, Ch 5, pp 216–217, Table 3)
Typical sentinel sites	Infectious disease / tuberculosis / paediatric hospital, health centre, antenatal clinic, sexually transmitted infection clinic, laboratory, rehabilitation centre — any unit seeing a large number of the relevant cases (Sunder Lal, Ch 7, p 303)
Site-selection criteria	Representativeness of the population of interest (climate zone, demographic and socioeconomic characteristics); feasibility and sustainability — staff and leadership motivation, specimen collection/storage/transport, high-quality diagnostics (Oxford 7e, Ch 42, p 1563)
Case definition	One standard surveillance case definition — "a set of uniform criteria used to define a disease for public health" — applied at every site, essential for comparison between areas within a country and between countries (Gordis 7e, Ch 3, p 58; Sunder Lal, Ch 8, p 472)
Data items	Cases and deaths diagnosed at the facility, plus additional variables such as age and immunisation status (Sunder Lal, Ch 7, p 303)
Site strengthening	Special training and supervision of staff to ensure complete and accurate reporting (Sunder Lal, Ch 7, p 303); laboratory equipment, extra paid trained staff and better communications where the network is fixed (Handbook, Ch 38, p 1540)
Periodicity	Continuous, or fixed periodic rounds — e.g. general practitioners telephoned every week for influenza counts (Handbook, Ch 33, p 1325); a fixed 3-month round conducted once in 2 years in HIV sentinel surveillance (Park 28e, Ch 7, pp 520–521)
Analysis and use	Extrapolation to the total population; simplified feedback to the reporting providers, which also minimises reporting bias (Park 28e, Ch 2, p 59)

Sentinel surveillance compared with other surveillance methods

Feature	Passive (routine notification)	Active	Sentinel
Who reports	All providers; reporting mandated or requested by government/local health authority (Gordis 7e, Ch 3, p 58)	Project staff specifically recruited for that disease, making periodic field visits to clinics, primary health centres and hospitals for case finding (Gordis 7e, Ch 3, p 58)	A selected group of physicians, hospitals, laboratories or institutions (Oxford 7e, Ch 42, pp 1554–1555)

Feature	Passive (routine notification)	Active	Sentinel
Population coverage	Whole population of the area ("comprehensive")	Facilities the recruited staff can visit periodically, extending in rural settings to surveying villages and towns (Gordis 7e, Ch 3, p 58)	Catchment of the chosen sites only; data extrapolated
Data quality	Under-reporting and incompleteness likely	High — staff are specifically employed and trained for the task (Gordis 7e, Ch 3, p 58)	High — trained staff and specialists handle the cases (Sunder Lal, Ch 7, p 303)
Cost	Low	Very high	Intermediate; may be less costly than building and maintaining full notification (Park 28e, Ch 2, p 59)
Typical use	Notifiable diseases, long-term trends	Final phase of eradication (smallpox, polio acute flaccid paralysis, Guinea worm), outbreak investigation	Trends, burden estimation, laboratory-supported surveillance of high-volume conditions

Choice between a comprehensive and a sentinel system rests on the surveillance objective, disease characteristics and resources available; a sentinel system may be combined with active surveillance (Oxford 7e, Ch 42, pp 1554–1555; Handbook, Ch 33, p 1325).

Sentinel surveillance systems in India

System	Sentinel unit	What is collected
HIV Sentinel Surveillance, National AIDS Control Programme	1,323 sites in 2016–17 — antenatal clinic 829, female sex worker 245, men who have sex with men 89, injecting drug user 87, truckers 28, migrant 27, transgender 18 (Park 28e, Ch 7, p 520)	HIV prevalence by unlinked anonymous testing; pregnant women attending antenatal clinics act as proxy for the general population
Dengue and chikungunya, National Vector Borne Disease Control Programme	521 sentinel surveillance hospitals with laboratory support in endemic states, backed by 14 Apex Referral Laboratories; kits supplied through National Institute of Virology, Pune (Park 28e, Ch 7, p 502)	Serological confirmation of dengue and chikungunya
Malaria	1–3 sentinel sites per high-endemic district in large hospitals (Park 28e, Ch 7, p 498)	All out-patient and in-patient malaria cases and malaria-related deaths; a Sentinel Site Medical Officer and trained laboratory technicians are posted (Sunder Lal, Ch 8, p 377)
Japanese encephalitis	Sentinel sites in tertiary care institutions in endemic and adjoining areas (Jugal Kishore, p 434)	Clinical and serological surveillance

System	Sentinel unit	What is collected
Integrated Disease Surveillance Programme (IDSP)	Sentinel private practitioners and sentinel hospitals/nursing homes in the private sector, alongside public units (Park 28e, Ch 7, p 570)	Weekly syndromic (S), presumptive (P) and laboratory (L) reports
National influenza surveillance (ILI and SARI)	Surveillance of influenza-like illness and severe acute respiratory infection is in place under IDSP (Sunder Lal, Ch 7, p 307), supported by a network of 12 laboratories in medical colleges (Sunder Lal, Ch 8, p 472)	Standard-case-definition ILI counts with virological typing of influenza viruses
National Programme for Control of Blindness	Designated sentinel units (Jugal Kishore, pp 486–487)	Blindness surveillance

HIV Sentinel Surveillance — objectives (Park 28e, Ch 7, pp 519–520): (1) determine the level of HIV infection in the general population and in high-risk groups in each state; (2) understand trends of the epidemic in both; (3) understand geographical spread and identify emerging pockets; (4) inform prioritisation of programme resources and evaluation of programme impact; (5) estimate HIV prevalence and burden in the country. It began in 1994 with 55 sites, rose to 180 by 1998, and is run as a repeated cross-sectional survey of the same risk group at the same place, by unlinked anonymous serological testing.

HIV Sentinel Surveillance round 2016–17 design (Park 28e, Ch 7, pp 520–521):

Parameter	High-risk groups	Bridge population	General population
Sentinel site	Targeted intervention projects	Sexually transmitted disease clinic, targeted intervention projects	Antenatal clinic
Sample size	250	250	400
Duration	3 months	3 months	3 months
Frequency	Once in 2 years	Once in 2 years	Once in 2 years
Sampling method	Consecutive / random	Consecutive	Consecutive
Age group	15–49 years	15–49 years	15–49 years
Testing strategy	Unlinked anonymous with informed consent	Unlinked anonymous at sexually transmitted disease clinic, with informed consent at targeted intervention sites	Unlinked anonymous
Blood specimen	Dried blood spot	Serum at sexually transmitted disease clinic, dried blood spot at targeted intervention sites	Serum
Testing protocol	Two-test protocol	Two-test protocol	Two-test protocol

Advantages and limitations

Advantages	Limitations
Most efficient way to obtain high-quality data in a timely way; rapid and economical alternative to whole-population surveillance (Oxford 7e, Ch 42, p 1563)	Conducted only in selected locations, so it is less effective for rare diseases or for disease occurring outside the catchment of the sentinel sites (Oxford 7e, Ch 42, p 1563)
More reliable and accurate data — trained staff and specialists handle the cases (Sunder Lal, Ch 7, p 303)	Extrapolation to the whole population is valid only if the sites are genuinely representative; representativeness must be built in at the design stage (Oxford 7e, Ch 42, pp 1554–1555)
Reporting bias is minimised and feedback to providers is simplified (Park 28e, Ch 2, p 59)	Upgrading the site alters the health situation there — better facilities attract more patients, better treatment changes transmission dynamics, better health education lowers incidence, biasing the very indicators being measured (Handbook, Ch 38, p 1541)
Yields more valuable and detailed information than traditional notification, and sites can later be developed into a notification system (Park 28e, Ch 2, p 59)	Rests on a few participating units, so output degrades if they lose motivation or drop out — staff and leadership motivation is an explicit sustainability criterion at site selection (Oxford 7e, Ch 42, p 1563)
Sustains laboratory confirmation and specimen collection that could not be maintained nationally (Handbook, Ch 33, p 1325)	Historically fragmented — separate sentinel systems ran under different vertical programmes with duplication and no national consolidation, one reason IDSP was created (Jugal Kishore, pp 486–487, p 772)
Cost reduction; can be combined with active surveillance (Handbook, Ch 33, p 1325)	Facility-based, so it misses cases that never reach a health facility

Evaluating a sentinel surveillance system

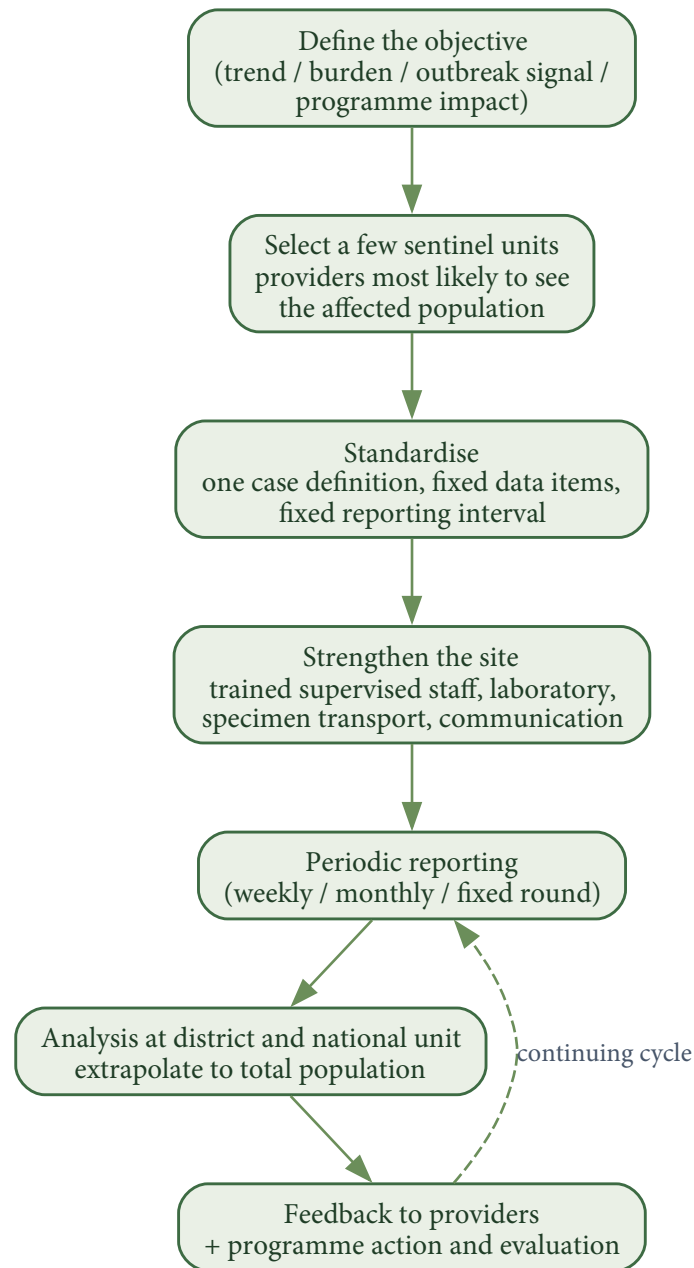
Assessed on the standard surveillance-system attributes (Handbook of Epidemiology, Ch 33, pp 1325–1326; Oxford 7e, Ch 42, p 1575). Four are decisive here:

- **Representativeness** — whether the occurrence and distribution of reported cases reflect the real situation in the population; the key attribute, since the system rests on extrapolation.
- **Sensitivity** — ability to identify every case; the attribute most obviously traded away by sampling a few units.
- **Timeliness** — interval between event, report and action; the main reason a small network is preferred to a national one.
- **Cost-effectiveness** — outcomes against direct, indirect and societal costs, cost reduction being the purpose of sampling providers.

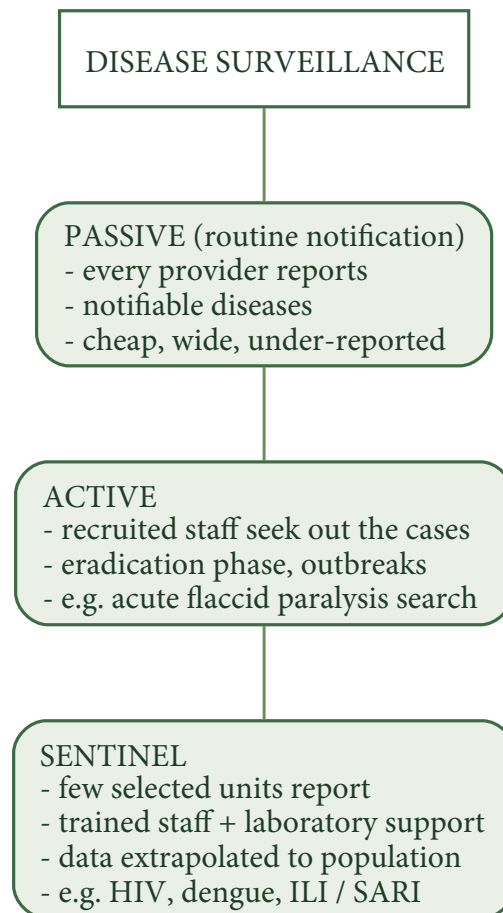
The remaining attributes apply as to any system: **usefulness, predictive value positive, simplicity, flexibility, data quality, acceptability and stability.**

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- Schematic: "Sentinel surveillance — sampling a few units to estimate the whole" — labels to write: *Total population; All cases occurring (many never reach a facility); Cases reaching health facilities; Selected sentinel units (antenatal clinic, sexually transmitted infection clinic, large-OPD hospital, laboratory); Standard case definition + trained staff + laboratory confirmation; Periodic report; District / national surveillance unit; upward arrow Extrapolation to total population; downward arrow Feedback and programme action.*



Operating cycle of a sentinel surveillance system



Where sentinel surveillance sits among surveillance methods

High-yield exam pointers

- ▶ **Definition to write first:** no routine notification system identifies all cases, so selected competent physicians or institutions report in detail and their data are extrapolated to the whole population (Park 28e, Ch 2, p 59).
- ▶ **Indian numbers with their year:** HIV Sentinel Surveillance 55 sites (1994) → 180 (1998) → 1,323 (2016–17), of which 829 antenatal clinic sites; dengue 521 sentinel surveillance hospitals + 14 Apex Referral Laboratories; malaria 1–3 sites per high-endemic district.
- ▶ **HIV round design to reproduce:** antenatal clinic n = 400, high-risk group and bridge population n = 250 each; 3-month round, **once in 2 years**; age 15–49 years; unlinked anonymous, two-test protocol; antenatal attendees are the proxy for the general population.
- ▶ **The killer limitation to quote:** upgrading a sentinel site changes the health situation there, so the network can bias its own indicators (Handbook of Epidemiology, Ch 38, p 1541).
- ▶ **Evaluation mnemonic — "US-PRoF-DATS"** covers nine attributes: Usefulness, Sensitivity, Predictive value positive, Representativeness, Flexibility, Data quality, Acceptability, Timeliness, Stability. **Simplicity and cost-effectiveness fall outside it — add them separately.**

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park's narrative on p 520 describes HIV sentinel surveillance as an "annual cross-sectional survey of the risk group", while Table 6 of the same chapter (pp 520–521) specifies a 3-month round once in 2 years — write the table figure. Site counts are revised periodically and must carry their year: Park 28e gives 521 dengue sentinel surveillance hospitals with 14 Apex Referral Laboratories, Sunder Lal over 646, Jugal Kishore 527 with 15 laboratories for 2015; Park gives 1–3 malaria sites per high-endemic district, Sunder Lal a minimum of two.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 2 Concept of Health and Disease; Ch 5 Epidemiology of Communicable Diseases; Ch 7 Health Programmes in India); Oxford Textbook of Public Health 7e (Ch 24 Community diagnosis and health information systems in low- and middle-income countries; Ch 42 Public health surveillance); Handbook of Epidemiology, Ahrens & Pigeot (Ch 33 Infectious Disease Epidemiology; Ch 38 Epidemiology in Developing Countries); Gordis Epidemiology 7e (Ch 3 The occurrence of disease: I. Disease surveillance and measures of morbidity); Sunder Lal — Textbook of Community Medicine (Ch 7 Epidemiology; Ch 8 Epidemiology of Communicable and Non-Communicable Diseases and Related Disease-Specific National Health Programmes); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 2 Epidemiology and Public Health); Jugal Kishore — National Health Programmes of India. Programme figures: HIV Sentinel Surveillance round 2016–17 (National AIDS Control Organisation).



Environmental surveillance in reducing morbidity and mortality

Asked in: Paper I May 2019 — RGUHS MD Community Medicine

Definition — Surveillance is the **continuous and systematic collection, analysis and interpretation of data, closely integrated with timely dissemination of the results to those who have the right to know, so that action can be taken** (Park 28e, Ch 2, p 59). Surveillance of environmental factors applies this cycle not to cases of disease but to the **agent and its vehicle** — the physical, chemical and biological quality of water, air, food, soil, housing, the workplace and the vector population — so that the hazard is detected and corrected *before* it produces disease. Park expressly allows this, since sources of surveillance data "may relate directly to disease or to factors influencing disease" (Park 28e, Ch 2, p 59); the ultimate objective, as with all surveillance, is **prevention** (Park 28e, Ch 3, pp 151–152).

Surveillance and monitoring — the distinction the examiner expects

Feature	Monitoring	Surveillance
Timing	Intermittent, episodic	Continuous, ongoing

Feature	Monitoring	Surveillance
Who performs it	Technicians and automated instrumentation, using standardised procedures	Requires professional analysis and sophisticated judgement of data
Output	A measurement (e.g. today's residual chlorine)	Recommendations for control activities
Relationship	Monitoring is one specific and essential part of the broader concept of surveillance	Surveillance subsumes monitoring

(Park 28e, Ch 2, p 59)

Components of the surveillance activity (Park 28e, Ch 7, p 569): (a) collection of data, (b) compilation, (c) analysis and interpretation, (d) **follow-up action**, (e) **feedback**. **Prerequisites of effective surveillance**: a standard case (or standard breach) definition, regularity of reports, and action on the reports. Surveillance may be **passive** (data reported by the existing system — cheap, but under-reported and incomplete) or **active** (dedicated staff make periodic field visits and undertake case finding — more accurate but resource-intensive) (Gordis 7e, Ch 3, p 58).

What is surveilled: domains, indicators and the standard applied

Domain	What is measured	Standard / cut-off applied	Morbidity–mortality averted
Drinking water — bacteriological	<i>E. coli</i> or thermotolerant coliforms; total coliforms; <i>C. perfringens</i> ; colony count	Must not be detectable in any 100 ml sample, in the supply, entering distribution and in distribution; immediate investigative action if detected (Park 28e, Ch 15, pp 876–877)	Cholera, typhoid, hepatitis A and E, dysentery, diarrhoeal deaths
Drinking water — chemical	pH, colour, turbidity, chlorides, ammonia, chlorine demand, free residual chlorine	Bacteriological rule — free residual chlorine 0.5 mg/L for a contact period of at least 1 hour (Park 28e, Ch 15, p 870). Virological rule — 0.5 mg/L free chlorine residual after a contact period of at least 30 minutes at pH 8.0, insisted upon in all disinfected supplies in areas suspected of hepatitis A endemicity; 0.2 mg/L for half an hour elsewhere, and turbidity must be 1 NTU or less before disinfection (Park 28e, Ch 15, p 876). Regular residual measurement may in part replace bacteriological surveillance (Park 28e, Ch 15, p 881)	Cholera, typhoid, dysentery and diarrhoeal deaths by the bacteriological rule; hepatitis A and E by the virological rule

Domain	What is measured	Standard / cut-off applied	Morbidity–mortality averted
Drinking water — toxic chemicals	Arsenic, lead, cadmium, mercury, selenium, cyanide	Arsenic 0.01, lead 0.01, cadmium 0.003, mercury 0.001, selenium 0.01, cyanide 0.07 mg/L (Suryakantha, Ch 5, p 61)	Arsenicosis, plumbism, itai-itai, Minamata disease
Fluoride in water	Fluoride concentration by area	Optimum 0.5–0.8 ppm for India; endemic areas carry 3–12 mg/L (Park 28e, Ch 11, p 765; Ch 15, p 882)	Dental and crippling skeletal fluorosis at excess; dental caries at deficiency
Ambient air	Sulphur dioxide, smoke and suspended particles are the best indicators, monitored daily at several sites, results collated by a central agency (Park 28e, Ch 15, p 891). India — under the National Air Quality Monitoring Programme (NAMP) the Central Pollution Control Board (CPCB), with the State Pollution Control Boards and the National Environmental Engineering Research Institute (NEERI), Nagpur, monitors SO ₂ , NO ₂ , PM ₁₀ and PM _{2.5} through 793 stations covering 344 cities and towns in 28 states and 8 union territories (Sunder Lal, Ch 3, p 123). In its earlier configuration, sponsored by CPCB from 1990 and reported by Park as a 14-year database from 10 major cities, the programme also covered inhalable dust, lead, hydrogen sulphide, ammonia and polycyclic aromatic hydrocarbons; suspended particulate matter (SPM) exceeded CPCB standards in all cities most of the year and the respirable (<PM ₁₀) fraction was 30–60% of SPM (Park 28e, Ch 15, p 892)	National Ambient Air Quality Standards 2009 (legally enforceable, under revision) — PM _{2.5} 40 µg/m ³ annual and 60 µg/m ³ 24-hour; PM ₁₀ 60 and 100; NO ₂ 40 annual and 80 24-hour; SO ₂ 80 24-hour; O ₃ 100 µg/m ³ 8-hour. International comparator, WHO Air Quality Guideline levels 2021 — PM _{2.5} 5 annual / 15 24-hour, PM ₁₀ 15 / 45, NO ₂ 10 annual (WHO 2005 values were PM _{2.5} 10 / 25 and PM ₁₀ 20 / 50) (Sunder Lal, Ch 3, p 123; Park 28e, Ch 15, p 890). A breach is communicated through the National Air Quality Index (AQI), which transforms complex multi-pollutant data into a single index value with a name and a colour — six bands by health breakpoint: good 0–50, satisfactory 51–100, moderate 101–200, poor 201–300, very poor 301–400, severe >400 (Sunder Lal, Ch 3, pp 123–124)	Chronic obstructive pulmonary disease, ischaemic heart disease, stroke, lung cancer, impaired neuropsychological development from lead

Domain	What is measured	Standard / cut-off applied	Morbidity–mortality averted
Household air	Use of solid fuels; indoor pollutant levels	—	4.3 million premature deaths a year from household air pollution: 34% stroke, 26% ischaemic heart disease, 22% chronic obstructive pulmonary disease, 12% pneumonia, 6% lung cancer; exposure almost doubles the risk of childhood pneumonia (WHO Household Air Pollution and Health, Fact Sheet No. 292, 2016, quoted in Park 28e, Ch 15, pp 891, 893)
Work place	Environmental (air) monitoring → biological monitoring (substance or metabolite in blood/urine/expired air) → biological effect monitoring (early biochemical change) → health surveillance (Oxford Handbook of Occupational Health 2e, Ch 31, pp 652–653)	Threshold limit values and biological exposure indices of the ACGIH; biological tolerance values of the DFG; for lead, statutory levels triggering health surveillance, an action level and a suspension level (Hunter's, Ch 5, p 85; Oxford OH 2e, Ch 2, p 99)	Lead, mercury, solvent and pesticide poisoning; organophosphate exposure flagged by a fall in plasma cholinesterase $\geq 15\%$ or erythrocyte $\geq 12\%$ (Hunter's, Ch 43, p 442)
Vector / entomological	<i>Aedes aegypti</i> index (house index) — percentage of houses in a defined area with actual breeding, of houses examined (Park 28e, Ch 15, p 925). <i>Anopheles</i> indices — human blood index (proportion of freshly-fed female anophelines whose stomach contains human blood), sporozoite rate (percentage of female anophelines with sporozoites in the salivary glands), mosquito density (number per man-hour catch), man-biting rate (average incidence of anopheline bites per day per person) and inoculation rate (man-biting rate $\times$ infective sporozoite rate) (Park 28e, Ch 5, p 324)	Aedes index kept at zero at all ports, with a 400-metre mosquito-free perimeter under the International Health Regulations; below 1 per cent in towns and seaports for yellow fever control (Park 28e, Ch 15, p 925; Ch 5, pp 347–348)	Dengue, yellow fever, chikungunya, malaria; prevents international spread
Sanitation coverage	Safely managed water and sanitation, handwashing facility	2 billion people lack safely managed water; only 54% have safely managed sanitation; 2.3 billion (29%) lack a basic handwashing facility, 2020 (Park 28e, Ch 15, p 861)	Diarrhoea and intestinal worm infestation, together 10% of the total burden of disease in developing countries (Park 28e, Ch 15, p 860)

Elements of an environmental surveillance programme — drinking water as the model

- ▶ **Sanitary survey** — an on-the-spot inspection and evaluation, by a qualified person, of the entire water supply system, to detect and correct faults and deficiencies; it is **essential for adequate interpretation of the laboratory results** (Park 28e, Ch 15, p 880).
- ▶ **Sampling** — carried out with the thoroughness and aseptic precaution of a surgical operation, by trained personnel, strictly following the method and **frequency prescribed in the WHO Guidelines for Drinking-water Quality (2011) or the ICMR Manual of Standards for Drinking Water Supplies** (Park 28e, Ch 15, pp 880–881).
- ▶ **Bacteriological examination** — presumptive coliform test (multiple tube, most probable number), confirmatory tests, tests for faecal streptococci and *C. perfringens*, and colony count (Park 28e, Ch 15, p 881).
- ▶ **Biological examination** — the type and number of organisms prevailing in the water.
- ▶ **Chemical surveillance** — of rising importance because of industrial and agricultural pollutants entering raw water sources (Park 28e, Ch 15, p 881).
- ▶ **Supporting activities** — periodic flushing and cleaning of storage, certification of operators, cross-connection and back-flow control, leak detection, central and regional analytical laboratory services, and quality control of bottled water and ice (Park 28e, Ch 15, p 880).

Institutional framework in India

Level	Agency / instrument	Surveillance function
Statute	Water (Prevention and Control of Pollution) Act 1974; Air (Prevention and Control of Pollution) Act 1981; Environment (Protection) Act 1986	Constitute Central and State Water Boards with wide powers to control pollution; provide the legal deterrent and the enforcement teeth for monitoring findings (Park 28e, Ch 15, p 866; Ch 23, pp 1038–1039)
National — air	National Air Quality Monitoring Programme (NAMP), Central Pollution Control Board with the State Boards and NEERI Nagpur; National Clean Air Programme (NCAP), launched 2019	Determines the status and trends of ambient air quality and long-term trend analysis for criteria pollutants; NCAP carries a revised target of a 40% reduction in PM_{2.5} concentration by 2026 (Sunder Lal, Ch 3, pp 122–123)
Health system	Integrated Disease Surveillance Programme — Central, State and District Surveillance Units; weekly (Monday–Sunday) data on S (health worker), P (clinician) and L (laboratory) forms; classified as syndromic, presumptive and confirmed	Detects the rising trend that a water or air failure produces; a Rapid Response Team investigates; the water department is an explicit reporting partner in the urban surveillance network (Jugal Kishore, pp 488–490; Park 28e, Ch 7, p 569)
Global	WHO Environmental Health Criteria Programme; the WHO international network of laboratories for monitoring and study of air pollution; the Global Environment Monitoring System (GEMS)	Evaluates the health effects of environmental pollutants; the network laboratories issue warnings where and when air pollution is high (Park 28e, Ch 15, pp 866, 893; Ch 25, p 1056; Sunder Lal, Ch 3, p 123)

How it reduces morbidity and mortality

- ▶ **It intercepts the causal chain before disease occurs.** Disease surveillance counts cases already ill; environmental surveillance measures **exposure**, one step earlier. Occupational practice makes this explicit — keep the worker to the **left of the exposure–disease spectrum**, because early detection and control of exposure is more desirable than diagnosing occupational disease after it is fully established (Hunter's, Ch 5, pp 81–82).
- ▶ **It gives early warning of an impending outbreak.** A failed sanitary survey, a coliform breach or a loss of free residual chlorine identifies a supply at risk days before the first case of cholera or hepatitis is notified — the exposure is measured and corrected while the population is still well.
- ▶ **It localises the source and permits a single decisive act.** *John Snow's* spot map of the 1854 Golden Square cholera epidemic focused attention on the Broad Street pump and let him conclude that cholera was water-borne **long before the birth of bacteriology** (Park 28e, Ch 3, pp 86–87) — the archetype of environmental data converting directly into mortality reduction.
- ▶ **It targets scarce interventions geographically.** Fluoride mapping identifies the endemic belts (Andhra Pradesh — Nellore, Nalgonda, Prakasam; Punjab, Haryana, Karnataka, Kerala, Tamil Nadu) where the alternative-source or the **Nalgonda defluoridation technique** of the National Environmental Engineering Research Institute, Nagpur, must be deployed; non-skeletal fluorosis is **easily reversed** by safe drinking water at that stage (Park 28e, Ch 11, pp 786–787; Suryakantha, Ch 16, pp 186–187).
- ▶ **It generates and enforces standards.** Guideline values are expressed as a maximum upper limit with an **averaging time**, since the time of exposure is critical in determining toxicity (Park 28e, Ch 15, p 892); the measured breach is then what the Pollution Control Board acts upon in law.
- ▶ **It evaluates control measures.** "Surveillance must follow control measures... once control measures have been instituted, their effectiveness should be evaluated. If they have not been successful, the reason(s) for failure should be identified, the existing measures modified and evaluation continued" (Park 28e, Ch 3, pp 151–152).
- ▶ **It protects at the international frontier.** Entomological surveillance at ports and airports under the International Health Regulations holds vector density at the prescribed level and so prevents the export and import of yellow fever and dengue.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Label box (draw and label): the surveillance cycle** — five boxes in a closed ring: *Collection of data* → *Compilation* → *Analysis and interpretation* → *Follow-up action* → *Feedback*, with the feedback arrow returning to collection; annotate the ring "prerequisites: standard case definition, regularity of reports, action on reports".

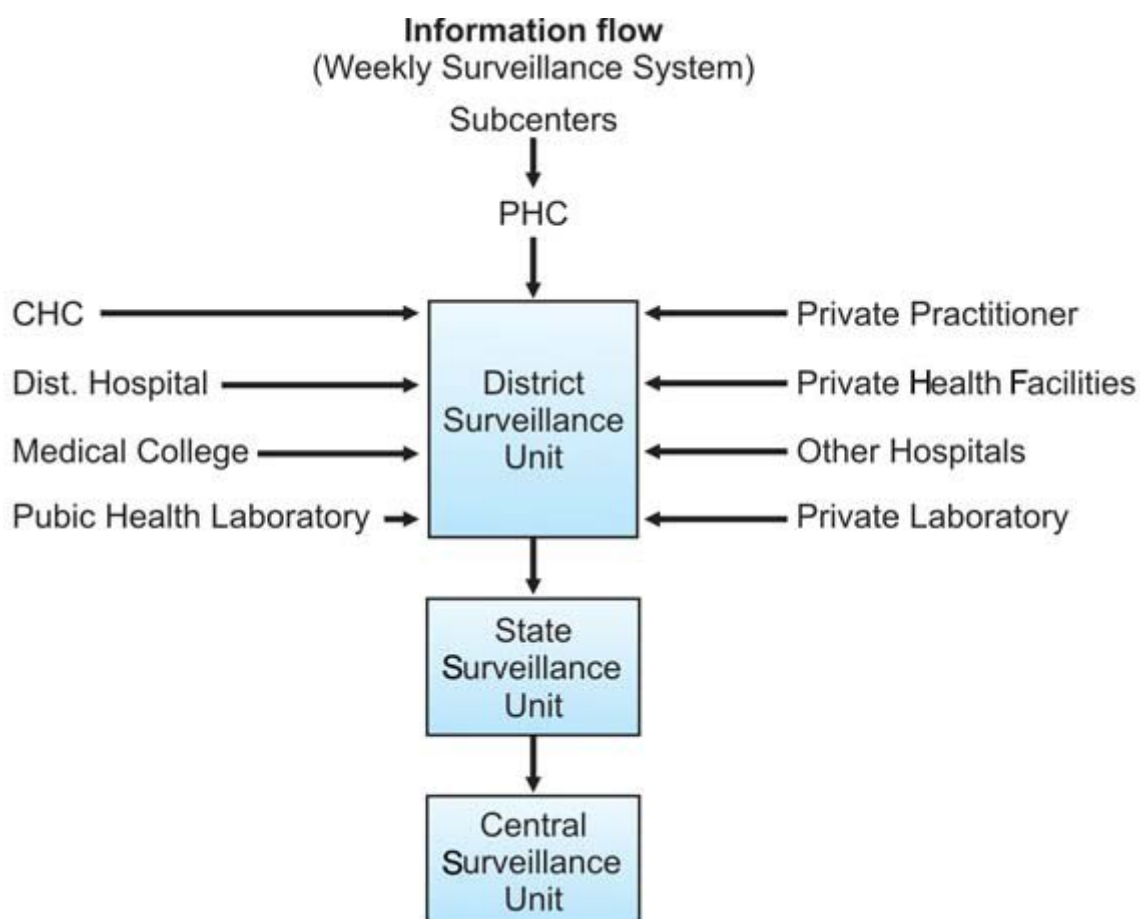
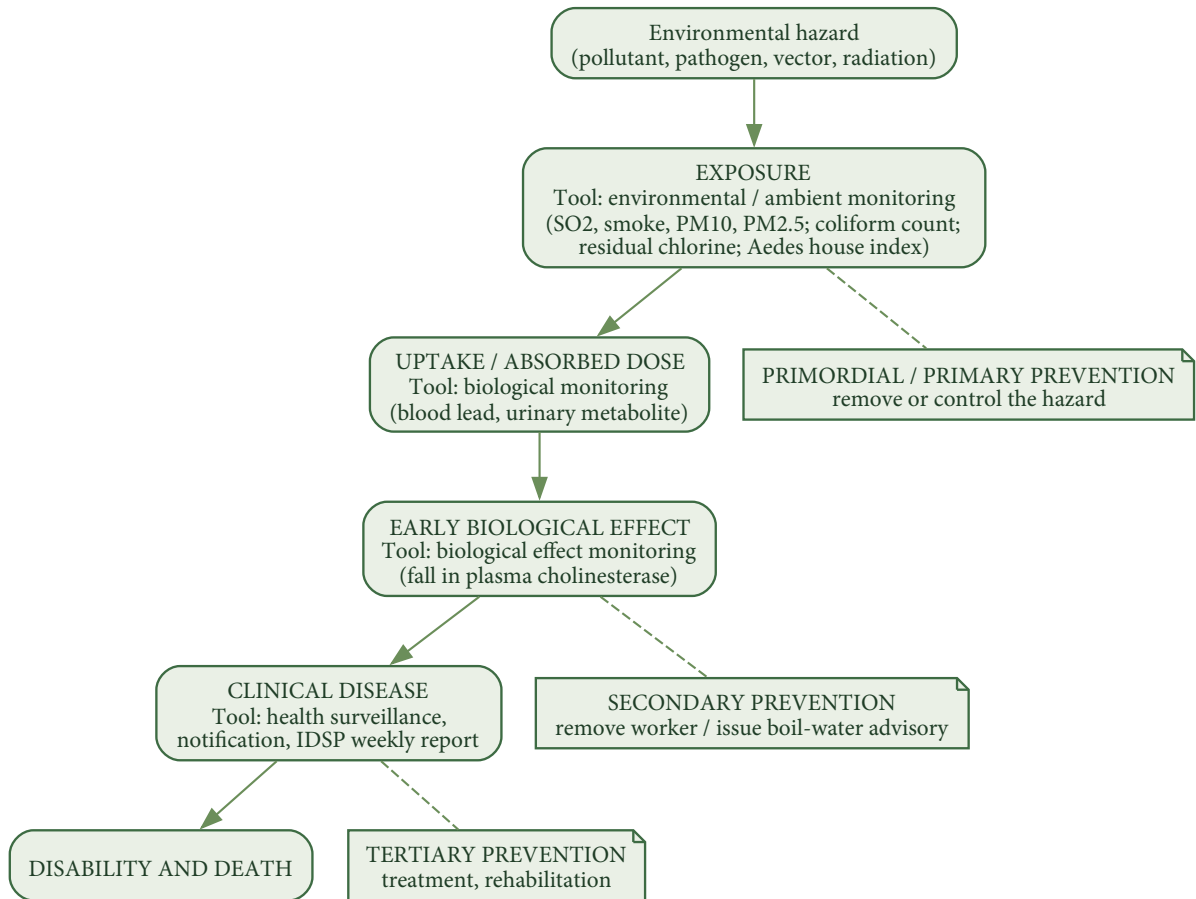
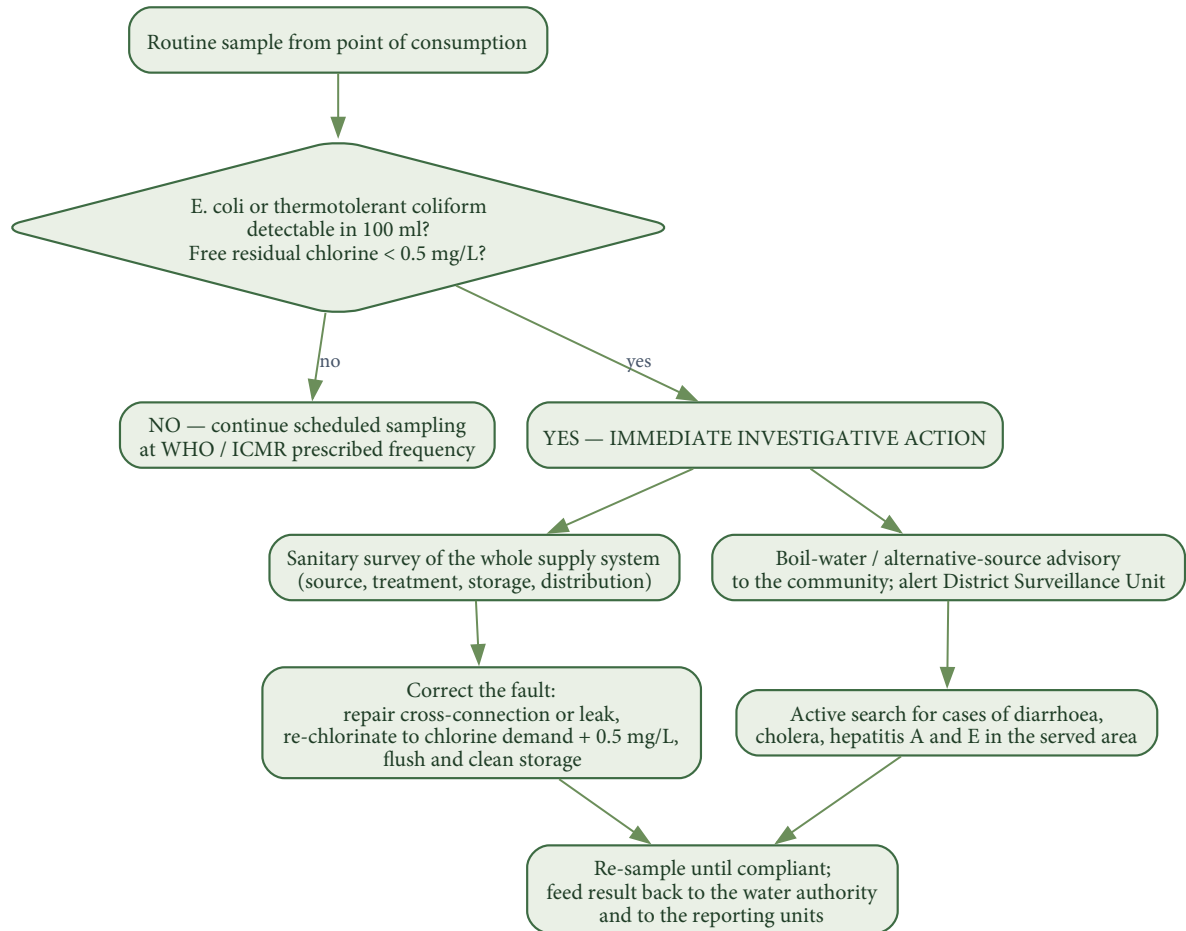


Fig 21.1 — Information flow (Weekly Surveillance System) (Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e, p.398)



The exposure-disease paradigm — the point at which each environmental surveillance tool intercepts, and the level of prevention it delivers



Action algorithm when routine drinking-water surveillance fails

High-yield exam pointers

- **Surveillance vs monitoring:** surveillance is *continuous* and needs professional judgement; monitoring is *intermittent* and can be automated — **monitoring is a part of surveillance**, not a synonym.
- **Five components** — Collect, Compile, Analyse, **Follow-up action**, **Feedback**. Three prerequisites — standard case definition, regularity of reports, action on reports. Ultimate objective = **prevention**.
- **Water numbers to write:** *E. coli* must not be detectable in any 100 ml sample; free residual chlorine **0.5 mg/L for 1 hour** (bacteriological) and **0.5 mg/L for 30 minutes at pH 8.0** (virological — insisted upon where hepatitis A is endemic; **0.2 mg/L for half an hour** elsewhere, turbidity ≤ 1 NTU before disinfection); fluoride optimum **0.5–0.8 ppm** in India (endemic areas 3–12 mg/L); arsenic and lead **0.01 mg/L**, mercury **0.001 mg/L**.
- **Air numbers to write:** best indicators = **SO₂, smoke and suspended particles**; **National Ambient Air Quality Standards 2009** — PM_{2.5} **40 µg/m³ annual / 60 µg/m³ 24-hour**, PM₁₀ **60 / 100**, NO₂ **40 annual / 80 24-hour**, SO₂ **80 24-hour**, O₃ **100 µg/m³ 8-hour**; **WHO AQG 2021** — PM_{2.5} **5 / 15**, PM₁₀ **15 / 45**, NO₂ **10 annual**; **AQI** six bands, severe **>400**; **NAMP network 793 stations, 344 cities, 28 states and 8 union territories**; **NCAP 2019, target 40% PM_{2.5} reduction by 2026**; household air pollution kills **4.3 million a year** (WHO Fact Sheet No. 292, 2016) — 34% stroke, 26% ischaemic heart disease, 22% COPD, 12% pneumonia, 6% lung cancer.

- ▶ **Vector:** *Aedes aegypti* index = % of houses with actual breeding among houses examined; **zero at ports**, <1% in towns/seaports; **400 m** mosquito-free perimeter under the International Health Regulations. Anopheline: **inoculation rate** = **man-biting rate** × **infective sporozoite rate**.
- ▶ **Legislation triad:** Water Act 1974, Air Act 1981, Environment (Protection) Act 1986.
- ▶ **Named example for the "role" limb:** *John Snow*, Broad Street pump, Golden Square, 1854 — environmental spot map that reduced mortality before bacteriology existed.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Suryakantha's Public Health Acts chapter prints the Air (Prevention and Control of Pollution) Act as 1974, which is a misprint — Park 28e dates it correctly to 1981, and 1981 is the answer to write. The National Ambient Air Quality Standards 2009 are recorded as being under revision.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 2, 3, 5, 7, 11, 15, 23, 25); Sunder Lal — Textbook of Community Medicine (Ch 3, Environmental Health Problems — Table 3.3 national standards and WHO AQG, Table 3.4 national air quality index, NAMP and NCAP); Gordis Epidemiology 7e (Ch 3, Disease surveillance); Suryakantha — Community Medicine with Recent Advances (Ch 5, 16, 52); Hunter's Diseases of Occupations (Ch 5 Biological monitoring; Ch 43 Pesticides); Oxford Handbook of Occupational Health 2e (Ch 2, 31); Jugal Kishore — National Health Programmes of India (Integrated Disease Surveillance Programme); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e (Fig 21.1); National Ambient Air Quality Standards 2009 (Central Pollution Control Board); WHO Air Quality Guidelines 2005 and 2021; WHO Household Air Pollution and Health, Fact Sheet No. 292 (2016); National Clean Air Programme 2019; WHO Guidelines for Drinking-water Quality 2011; ICMR Manual of Standards of Drinking Water Supplies.



Time trends in epidemiology

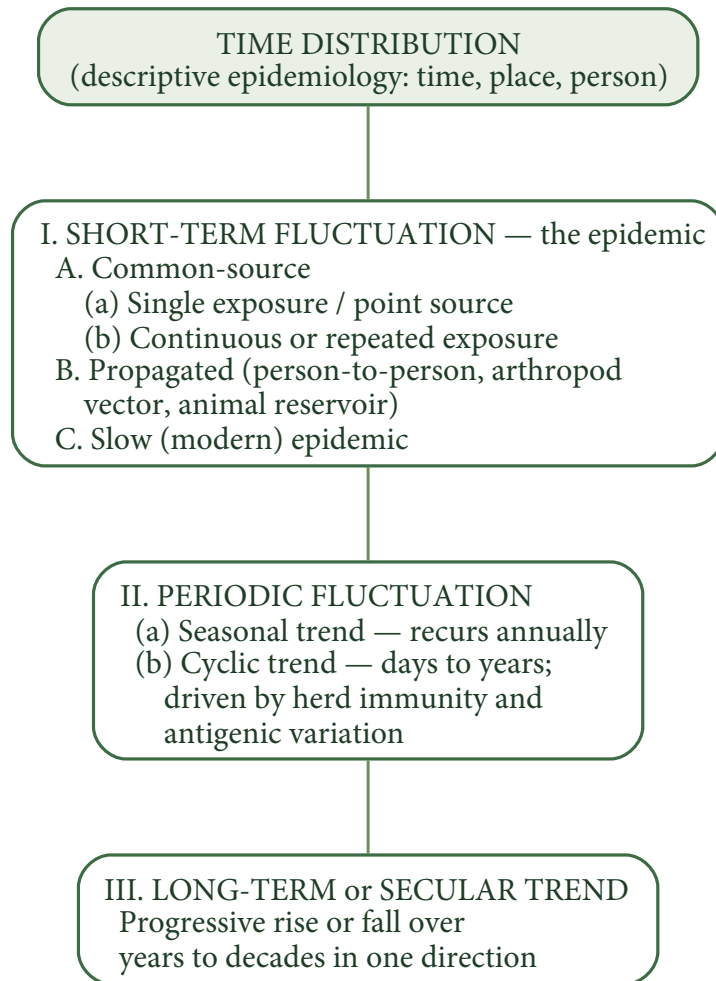
Asked in: Paper I 16-Dec-2023 — RGUHS MD Community Medicine

Definition — In descriptive epidemiology, the pattern of a disease is described under the three heads of **time, place and person**; the *time distribution* describes disease occurrence by year, season, month, week, day or hour of onset (Park 28e, Ch 3, p 84). A **time trend** is the systematic change in the frequency of a disease in a defined population over a specified period. Epidemiologists recognise **three kinds of time trend** — short-term fluctuations, periodic fluctuations and long-term (secular) trends — and studying them yields clues to the source and aetiology of disease, thereby suggesting preventive measures (Park 28e, Ch 3, p 84).

Types of time trend

Type	Time frame	Underlying mechanism	Classical examples
I. Short-term fluctuation	Hours to weeks (rarely months)	Sudden introduction of, or common exposure to, an agent; person-to-person spread in a susceptible population	An epidemic — food poisoning, cholera, hepatitis A, the Bhopal gas tragedy, Minamata disease (Park 28e, Ch 3, p 84)
II. Periodic fluctuation — (a) seasonal trend	Recurrs at the same time each year	Temperature, humidity, rainfall, overcrowding, vector life cycle	Measles and varicella peak in early spring; upper respiratory infection in winter; bacterial gastrointestinal infection in summer; dengue peaks September–November (Park 28e, Ch 3, p 85)
II. Periodic fluctuation — (b) cyclic trend	Days, weeks, months or years (cycles longer than one year)	Naturally occurring variation in herd immunity — a fresh pool of susceptibles must build up; antigenic variation	Measles every 2–3 years and rubella every 6–9 years in the pre-vaccination era; influenza pandemics at 7–10 year intervals (Park 28e, Ch 3, p 85)
III. Long-term or secular trend	Several years to decades	Real change in aetiological agent, host or environment; also changes in diagnosis and reporting	Consistent rise of coronary heart disease, lung cancer and diabetes in developed countries over ~50 years, with a decline of tuberculosis, typhoid fever, diphtheria and polio (Park 28e, Ch 3, pp 85–86)

Seasonal trend is regarded by some epidemiologists as a form of cyclic trend (Park 28e, Ch 3, p 85).



Classification of time trends in disease occurrence (Park 28e, Ch 3, pp 84-86)

I. Short-term fluctuations — the epidemic

An **epidemic** is "the occurrence in a community or region of cases of an illness or other health-related events **clearly in excess of normal expectancy**", with the community, the population and the time period specified precisely; epidemicity is therefore *relative* to the usual frequency of that disease, in that area, in that population, at that season of the year (Park 28e, Ch 3, p 84). Three major types are distinguished (Park 28e, Ch 3, p 84):

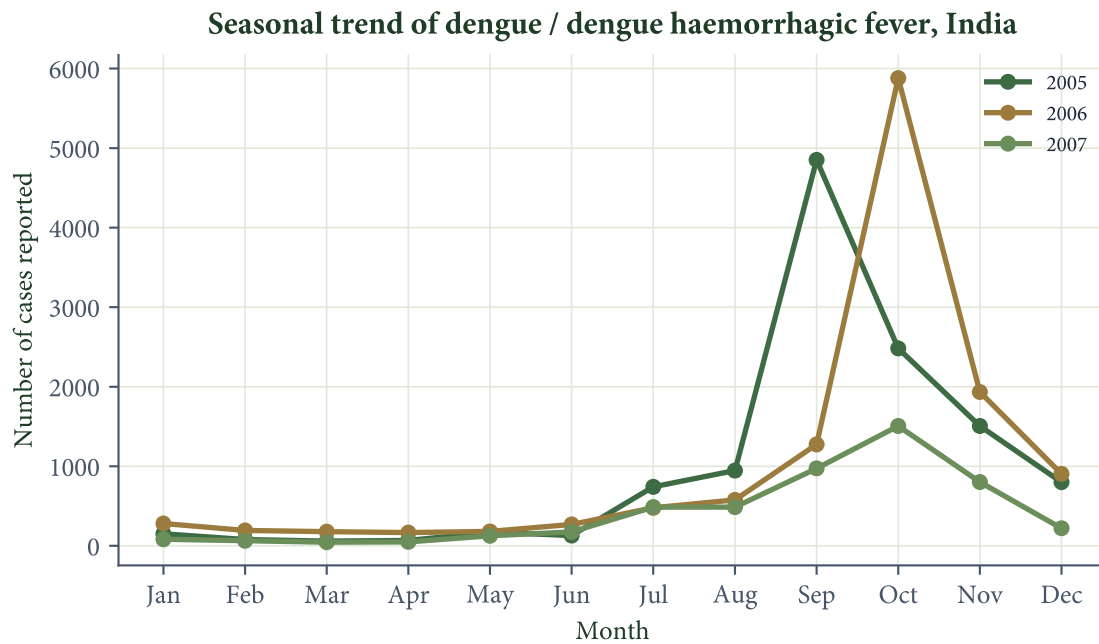
Type of epidemic	Nature of exposure	Shape of the epidemic curve	Example
A(a) Common-source, single exposure ("point-source")	Brief and essentially simultaneous	Rises and falls rapidly , explosive, one peak , no secondary waves ; all cases fall within one incubation period	Food-poisoning outbreak; Bhopal gas tragedy; Minamata disease
A(b) Common-source, continuous or repeated exposure	Prolonged, repeated or intermittent, not necessarily at the same time or place	No explosive rise; extends beyond one incubation period ; irregular; no secondary cases among contacts	Gonorrhoea from a single source; contaminated well; a nationally distributed contaminated vaccine or food; Legionnaires' disease, Philadelphia

Type of epidemic	Nature of exposure	Shape of the epidemic curve	Example
B. Propagated	Person-to-person, arthropod vector or animal reservoir	Gradual rise, tails off over a much longer period; successive waves one incubation period apart	Hepatitis A, poliomyelitis
C. Slow (modern) epidemic	Chronic exposure to lifestyle and environmental determinants	Rise measured over decades	Coronary heart disease, lung cancer

- ▶ **Median incubation period** — on a point-source curve, the time required for 50 per cent of the cases to occur after exposure (Park 28e, Ch 3, p 84).
- ▶ A **mixed pattern** occurs when an epidemic begins from a common source and then continues as a propagated one — water-borne cholera reaches a sharp peak but tails off gradually (Park 28e, Ch 3, p 85).
- ▶ Speed of propagated spread depends on **herd immunity, opportunities for contact and the secondary attack rate**; transmission continues until susceptibles are depleted, so such epidemics favour settings with a regular supply of new susceptibles through birth or immigration (Park 28e, Ch 3, p 85).
- ▶ The **epidemic curve** — the distribution of times of onset of disease — is described by three factors: the distribution of times of onset, the incubation period, and the time of exposure; **if any two are known the third can be deduced** (Gordis 7e, Ch 2, pp 44–45; Lilienfeld's Foundations, Ch 3, pp 66–67).

II. Periodic fluctuations

- ▶ **Seasonal trend** — variation related to environmental conditions (temperature, humidity, rainfall, overcrowding, vector life cycle) that directly or indirectly favour transmission. For many infections such as poliomyelitis the basis of the seasonal variation remains unknown. Non-infectious conditions also show it — sunstroke, hay fever, snakebite (Park 28e, Ch 3, p 85).
- ▶ **Cyclic trend** — cycles arising from the periodic accumulation of susceptibles, so that herd immunity falls below the threshold needed to block transmission; effective immunisation abolishes the cycle. Non-infectious events also cycle, e.g. automobile accidents in the United States are commoner at week-ends, especially Saturdays. A knowledge of cyclicity is useful because it enables communities to **anticipate and defend themselves** (Park 28e, Ch 3, pp 85–86).



Outbreaks begin in July and peak in September-October-November, coinciding with late summer and rain; annual totals 11,990 (2005), 12,317 (2006) and 5,023 (2007) (Park's Textbook of Preventive and Social Medicine 28e, Ch 3, Table 10, p 85).

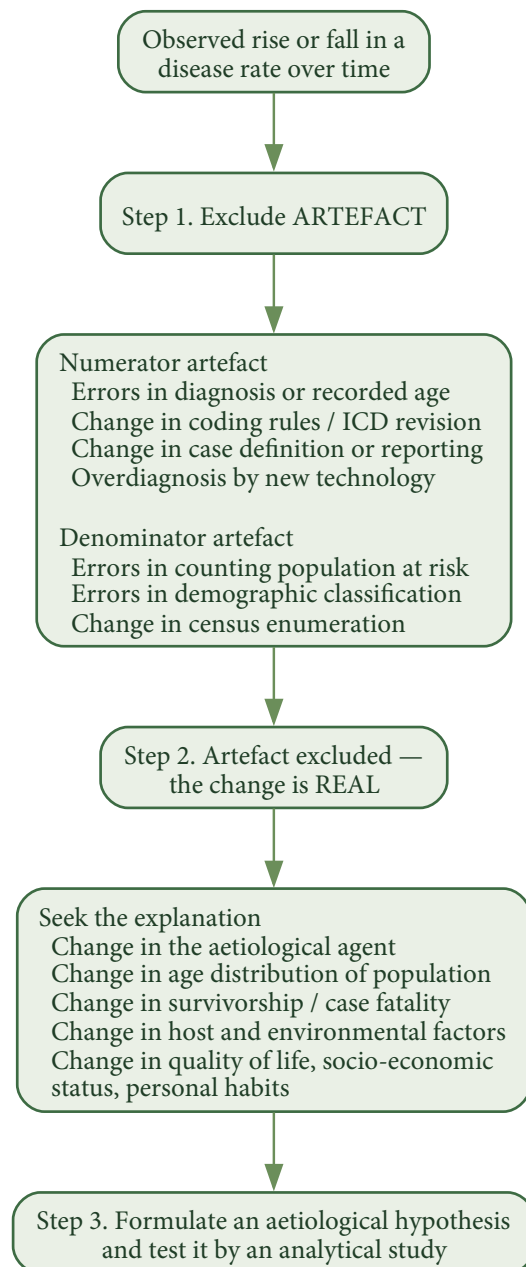
III. Long-term or secular trends

- ▶ A **secular trend** means a progressive increase or decrease in the occurrence of a disease over a long period, **generally several years or decades** (Park 28e, Ch 3, pp 85–86); trends over time in vital statistics are likewise termed secular trends (Lilienfeld's Foundations, Ch 6, p 116).
- ▶ Short-term fluctuations may be **superimposed** on it, but a secular trend implies a **consistent tendency to change in one direction** (Park 28e, Ch 3, p 86).
- ▶ **Birth-cohort analysis** is the tool that separates a secular trend from a generational one: successive birth cohorts are followed longitudinally instead of comparing cross-sections. *William Farr* used the life-table technique on successive cohorts in the mid-1800s, and cohort analysis in the 1920s–1930s established the importance of **reactivated tuberculosis infection later in life** (Lilienfeld's Foundations, Ch 6, p 127).
- ▶ A **cohort effect** is a change confined to persons born in the same period who share a common experience; a cross-sectional view of the same data can show an apparent decline while the cohort itself is still worsening, as with obesity prevalence in the United States (Gordis 7e, Ch 4, p 103).

Interpretation of an observed time trend

Before a trend is accepted as real it must be shown not to be an **artefact** of the numerator or the denominator (Gordis 7e, Ch 4, pp 103–104).

Level	Artefactual explanations	Real explanations
Numerator (cases/deaths)	Errors in diagnosis; errors in recorded age; changes in the coding rules ; changes in classification (successive ICD revisions); changed clinical recognition of the outcome; changed rules and procedures for reporting (Gordis 7e, Ch 4, p 104; Lilienfeld's Foundations, Ch 6, pp 117–118)	Change in the incidence of the outcome due to host factors or environmental factors (Lilienfeld's Foundations, Ch 6, p 117)
Denominator (population at risk)	Errors in counting the population; errors in classifying it by demographic characteristics such as age, sex or race; changed enumeration of the population at risk (Gordis 7e, Ch 4, p 104)	Change in the age distribution of the population (Lilienfeld's Foundations, Ch 6, p 117)
Case ascertainment / survival	Overdiagnosis from a new technology — the rise and fall of prostate cancer mortality in the United States after prostate-specific antigen (PSA) screening entered routine use in 1987; new imaging improving diagnostic acumen (Lilienfeld's Foundations, Ch 6, pp 117–118)	Change in survivorship ; change in case fatality (Lilienfeld's Foundations, Ch 6, p 117; Park 28e, Ch 3, p 86)
Comparability over time	Data not gathered in the same way across time — e.g. education and prenatal-care items from the 2003 revision of the United States birth certificate are not comparable with earlier ones (Lilienfeld's Foundations, Ch 6, p 117)	Changes in specific and non-specific determinants — quality of life, socio-economic status, personal habits (Park 28e, Ch 3, p 86)



Stepwise interpretation of an observed change in a disease rate over time

Uses of time-trend analysis

- ▶ **Surveillance and monitoring** — identifies which diseases are increasing, which are decreasing, which are the **emerging health problems**, and how effective control measures against old ones have been (Park 28e, Ch 3, p 86).
- ▶ **Hypothesis generation** — *time clustering* of cases of adenocarcinoma of the vagina in young women incriminated *in utero* exposure to **diethylstilbestrol** (Park 28e, Ch 3, p 86).
- ▶ **Outbreak investigation** — a chronological distribution of dates of onset is plotted as an epidemic curve, which suggests a time relationship with a suspected source, whether the epidemic is common-source or propagated, and whether the pattern is seasonal or cyclic and so suggestive of a particular infection (Park 28e, Ch 3, p 160).

- **Projection and planning** — study of disease profiles and time trends allows useful projections into the future and identification of emerging health problems and their correlates; epidemiology first showed coronary heart disease to be an *epidemic*, and later accidents, cancer and diabetes (Park 28e, Ch 3, p 111).
- **Programme evaluation** — a secular fall following an intervention (e.g. in tuberculosis, diphtheria, poliomyelitis) is evidence of impact, once artefact is excluded.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Epidemic curve, common-source single exposure ("point-source")** — label: x-axis *Time (days/hours of onset)*, y-axis *Number of cases*; a single sharp peak; bracket marking *one incubation period*; arrow marking the *median incubation period* (time to 50 per cent of cases); arrow at the start marking *time of exposure*; note "no secondary waves".
- **Epidemic curve, propagated epidemic** — same axes; successive waves of increasing then decreasing size, each *one incubation period* apart; label *gradual rise, long tail, transmission stops when susceptibles are depleted*.

exposure.

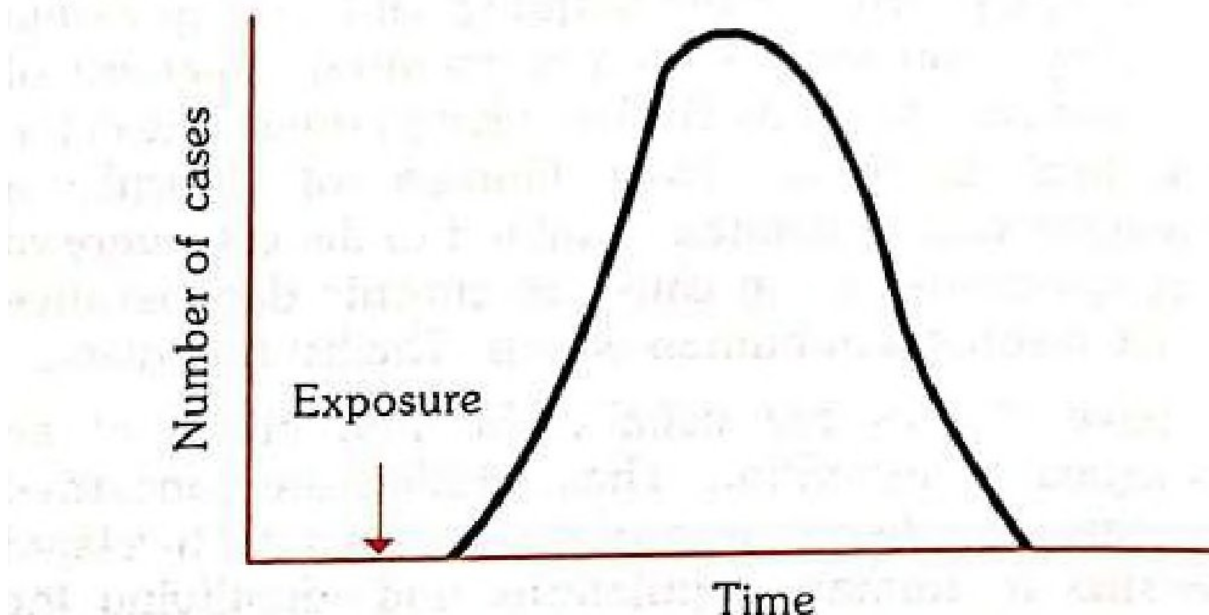


Fig 4 — Epidemic curve (Park's Textbook of Preventive and Social Medicine, 28e, p.84)

High-yield exam pointers

- Three time trends = Short-term fluctuation, Periodic fluctuation (seasonal + cyclic), Secular (long-term) trend — quote them in this order (Park 28e, Ch 3, p 84).
- Cycle lengths to write exactly: measles 2–3 years, rubella 6–9 years, influenza pandemics 7–10 years.
- Point-source epidemic triad: explosive rise and fall, no secondary waves, all cases within one incubation period; the median incubation period is the time to 50 per cent of cases.
- Epidemic curve identity: onset distribution + incubation period + time of exposure — know any two, deduce the third.

- Interpretation checklist: **artefact (numerator vs denominator) before reality** — coding/ICD change, case-definition change, overdiagnosis, census error.
- Named examples worth one line each: **diethylstilbestrol and vaginal adenocarcinoma** (time clustering), **PSA screening and prostate cancer mortality** (overdiagnosis), **dengue in India peaking September–November** (seasonal).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the dengue figures quoted are Park's Table 10 for **2005–2007** and are used here to illustrate the seasonal *pattern*, not the current dengue burden; the seasonal shape, not the absolute counts, is what the examiner wants.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 3, Principles of Epidemiology and Epidemiologic Methods, pp 84–86, 111, 160 — time distribution, types of epidemics, periodic and secular trends, interpretation of time trends, epidemic curve in outbreak investigation); Gordis Epidemiology 7e (Ch 2, pp 44–45 — epidemic curve; Ch 4, pp 103–104 — cohort effect and Table 4.13, artefactual versus real explanations of trends in mortality); Lilienfeld's Foundations of Epidemiology (Ch 3, pp 65–70 — epidemic curve and its three factors; Ch 6, pp 116–118, 127 — secular trends, Table 6.1 possible explanations for trends in epidemiologic rates, birth-cohort analysis).



Vaccine efficacy and its impact on disease control

Asked in: Paper I 16-Dec-2023 — RGUHS MD Community Medicine

Definition — **Vaccine efficacy (VE)** is the percentage reduction in the incidence (attack rate) of disease among the vaccinated compared with the unvaccinated, measured prospectively under the ideal conditions of a randomised, double-blind, placebo-controlled trial (Giesecke, Ch 19, p 232; Plotkin 7e, Ch 73, pp 2065–2066). It measures only the **direct** protection conferred on the individual vaccinee. The additional benefit to unvaccinated persons, arising from reduced circulation of the agent, is **indirect protection** or **herd immunity** — the protection that occurs when vaccination of a portion of the population protects unprotected individuals (Park 28e, Ch 3, p 122).

1. Efficacy versus effectiveness

Feature	Vaccine efficacy	Vaccine effectiveness
Setting	Ideal — pre-licensure randomised controlled trial (RCT)	Real world — post-licensure programme conditions
Design	Randomised, double-blind, placebo-controlled	Observational: cohort, case-control, household secondary attack rate, screening method (Plotkin 7e, Ch 73, pp 2065–2066)

Feature	Vaccine efficacy	Vaccine effectiveness
Comparison group	Placebo recipients, equal exposure guaranteed by randomisation	Unvaccinated in the community; equal exposure must be <i>demonstrated</i> , not assumed (Giesecke, Ch 19, p 232)
End-point	Narrow, laboratory-confirmed disease	Broader clinical/socioeconomic events (Plotkin 7e, Ch 32, p 745)
Captures indirect protection?	No — measures direct effect only	Yes, partly; cluster-randomised designs give the overall benefit of the programme (Plotkin 7e, Ch 77, p 2183)
Usual value	Higher	Equal to or lower than efficacy

2. Measurement — formulae and a worked example

Measure	Formula	Note
Vaccine efficacy / effectiveness	$VE = (ARU - ARV) / ARU \times 100$	ARV = attack rate in vaccinated; ARU = attack rate in unvaccinated (Plotkin 7e, Ch 73, p 2066)
Same, as relative risk	$VE = (1 - RR) \times 100$	RR = relative risk of disease, vaccinated vs unvaccinated (Giesecke, Ch 19, pp 220-221)
Case-control setting	$VE = (1 - OR) \times 100$	Odds ratio (OR) substitutes for RR to approximate ARV/ARU (Plotkin 7e, Ch 73, p 2066)
Screening method	$VE = (PPV - PCV) / [PPV \times (1 - PCV)]$	PPV = proportion of the population vaccinated; PCV = proportion of cases vaccinated (Giesecke, Ch 19, p 222)

Worked example (Park's pertussis field trial). Of 3,801 vaccinated children 149 developed whooping cough; of 3,759 unvaccinated controls 687 did (Park 28e, Ch 3, p 104).

- ▶ $ARV = 149 / 3,801 = 3.9\%$
- ▶ $ARU = 687 / 3,759 = 18.3\%$
- ▶ $VE = (18.3 - 3.9) / 18.3 \times 100 = \approx 79\%$

Screening-method example (measles outbreak, Niamey, Niger, 1990-91, infants 9-11 months). Pre-outbreak coverage PPV = 0.63; proportion of cases vaccinated PCV = 0.187. $VE = (0.63 - 0.187) / [0.63 \times (1 - 0.187)] = 87\%$ (Giesecke, Ch 19, p 222).

3. Study designs, in descending order of validity

Design	Where used	Chief limitation
Randomised controlled trial	Pre-licensure; the reference standard	Large, long, costly; unethical once vaccine is in the programme
Prospective/retrospective cohort	Post-licensure field evaluation	Confounding by indication; requires reliable vaccination records

Design	Where used	Chief limitation
Case-control	Rare disease, rapid answer	Assumes equal risk of exposure apart from vaccination; unreliable if >80% are vaccinated, as unvaccinated cases and controls become too few (Nelson & Williams, Ch 18, p 564)
Household secondary attack rate	Intense, uniform exposure within families	Small numbers; not generalisable
Indirect-cohort (Broome) method	Serotype-specific vaccines, e.g. pneumococcal	Needs similar vaccine-type/non-vaccine-type distribution in the unvaccinated (Plotkin 7e, Ch 47, pp 1209-1210)
Screening method	Rapid triage in an outbreak, using routine coverage data	Crude; only indicates whether a fuller investigation is warranted

4. Determinants of observed efficacy — why field values fall short

Determinant	Effect on measured efficacy
Type of vaccine	Live vaccines need fewer doses, give longer and greater protection than killed vaccines; killed cholera vaccine offers only 50% protection (Park 28e, Ch 3, p 124)
Number and spacing of doses	Three doses of pertussis vaccine give about 80% protection in the first three years, almost nil by 12 years after immunisation (Park 28e, Ch 3, p 124)
Primary vaccine failure	Failure to seroconvert at all — 16-24% of children after one dose of varicella vaccine (Plotkin 7e, Ch 62, pp 1688-1689)
Secondary vaccine failure	Initial seroconversion followed by waning to non-protective levels; measles in 5% of seroconverters within 10 years in a Canadian study (Plotkin 7e, Ch 37, p 879)
Cold chain break	Loss of potency, especially of live vaccines; hepatitis B loses potency on freezing as HBsAg dissociates from the alum adjuvant (Park 28e, Ch 3, p 130; Plotkin 7e, Ch 25, p 525)
Case definition used	A loose clinical definition underestimates, a laboratory-confirmed definition overestimates effectiveness (Plotkin 7e, Ch 62, p 1689)
Clustering of vaccinated persons	Vaccinated and unvaccinated are then unequally exposed, and crude efficacy overestimates the true immunising capacity (Plotkin 7e, Ch 77, p 2183)

5. Impact on disease control — from efficacy to herd immunity

Efficacy translates into population impact only through **coverage**: to a first approximation incidence falls by the product of coverage and efficacy, though real-world impact also depends on how vaccine and risk are distributed, on the *type* of immunity induced, and on indirect protection (Plotkin 7e, Ch 77, p 2166).

Quantity	Formula	Meaning
Basic reproduction number	R_0	Average number of secondary cases from one case in a wholly susceptible population
Herd immunity threshold	$H = 1 - 1/R_0$	Proportion immune above which large outbreaks are prevented (Plotkin 7e, Ch 77, p 2168)
Coverage needed	$CH = H / P$	P = protective efficacy; coverage must exceed the threshold in inverse proportion to efficacy (Plotkin 7e, Ch 77, p 2173)
Coverage needed, transmission-blocking vaccine	$(R_0 - 1) / [R_0 \times (P_i + P_t - P_i P_t)]$	P_i = efficacy against infection, P_t = efficacy against transmissibility (Plotkin 7e, Ch 77, p 2173)

A vaccination coverage of 100% yields an *immunisation* coverage only equal to the vaccine's efficacy — so a low-efficacy vaccine can never reach the threshold, however complete the coverage (Giesecke, Ch 19, pp 230-231).

Disease	R_0 (developed countries)	Herd immunity threshold H (%)
Measles	12-18	92-94
Pertussis	5-17	80-94
Polio	2-20	50-95
Diphtheria	6-7	83-85
Rubella	6-7	83-85
Smallpox	5-7	80-85
Mumps	4-7	75-86
Influenza	1.4-4	30-75
Tetanus	Not applicable	Not applicable — herd immunity does not protect the individual

(Plotkin 7e, Ch 77, p 2169; tetanus qualifier, Park 28e, Ch 3, p 123)

Levels of disease control achieved. In **control**, the agent is permitted to persist at a level at which it ceases to be a public health problem, an equilibrium between agent, host and environment. **Elimination** is interruption of transmission in a large geographic region or political jurisdiction, and is the precursor of eradication. **Eradication** is termination of all transmission worldwide by extermination of the agent — an absolute, "all-or-none" process achieved so far only for smallpox (Park 28e, Ch 2, p 58).

Two cautions. Smallpox eradication owed less to herd immunity than to surveillance-and-containment removal of the source of infection (Park 28e, Ch 3, p 123). And where coverage is partial, mass vaccination raises the average age at infection — for rubella this can paradoxically *increase* congenital rubella syndrome, avoided by rapid high coverage in prepubertal children or by vaccinating postpubertal females (Plotkin 7e, Ch 53, p 1445; Ch 77, p 2175).

6. Demonstrated impact in India

Disease	Burden before vaccination	Present status
Smallpox	1.88 lakh cases (1974)	Eradicated; zero cases
Poliomyelitis	2.00 lakh cases (1978, at start of the Expanded Programme on Immunization)	Polio-free in 2014; zero cases
Neonatal tetanus	45,948 cases (1980)	Eliminated in 2015 (<1 case per 1,000 live births); 81 cases in 2021

(Routine Immunization Manual for Medical Officers 2024, Ch 1, pp 31-32)

Coverage indicator	Value
Full immunisation, children 12-23 months, NFHS-5 (2019-21)	76.4% (NFHS-4, 2015-16: 62%)
Diphtheria-pertussis-tetanus 3rd dose (Penta-3), 2019	>90%; dipped 6 percentage points to 85% in 2020-21 with COVID-19 disruption
Universal Immunization Programme (UIP) target	>90% full immunisation, year on year
UIP annual beneficiaries	~2.6 crore newborns and 3.0 crore pregnant women, against 12 vaccine-preventable diseases

(Routine Immunization Manual for Medical Officers 2024, Ch 1, pp 29, 32-34; NFHS-5 India Report, Ch 5, pp 361-362, 382-383)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Labels to reproduce by hand** on the above graph: y-axis "Thousands of paralytic poliomyelitis cases"; x-axis "Years 1958-1961"; upper curve "**Expected** cases if the vaccine's effect were limited to those vaccinated"; lower curve "Cases actually **observed**"; the shaded gap between them "**Reduced by herd immunity**" (Gordis 7e, Ch 2, p 43).
- **Second diagram (draw freehand)**: herd immunity threshold curve — x-axis "Basic reproduction number R_0 " (0 to 30), y-axis "Herd immunity threshold H (%)" (0 to 100), rising curve labelled " $H = 1 - 1/R_0$ ", with dropped markers at "Measles R_0 12-18, H 92-94%" and "Influenza R_0 1.4-4, H 30-75%" (Plotkin 7e, Ch 77, p 2171).

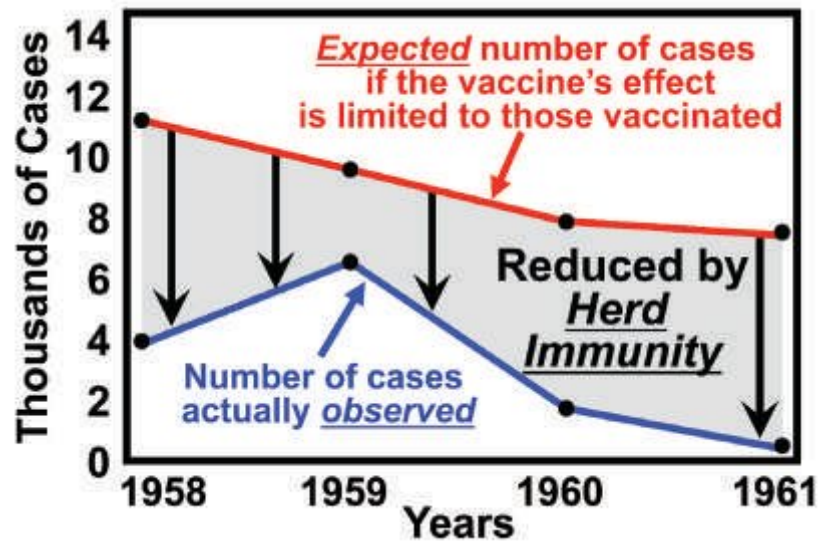
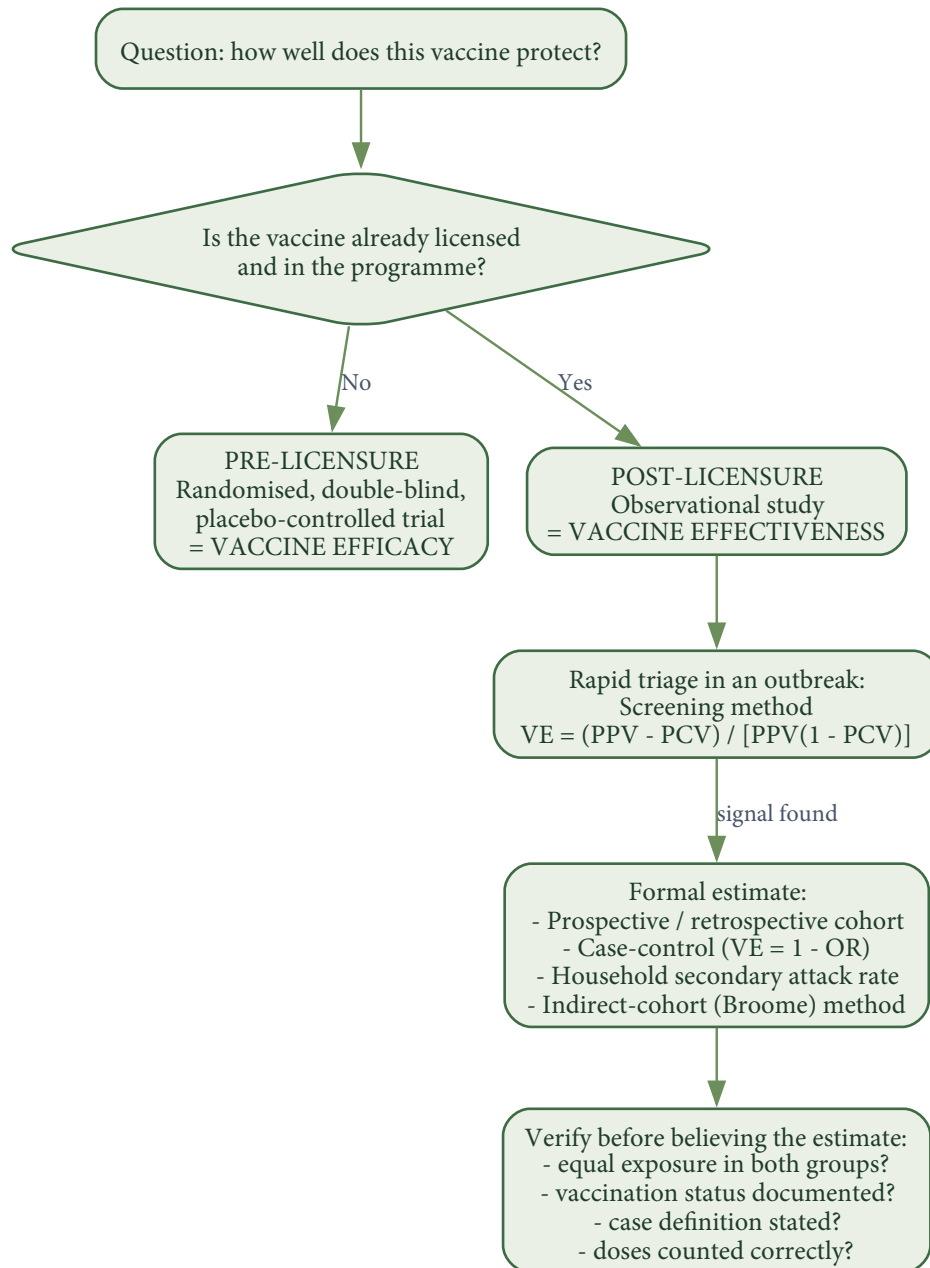
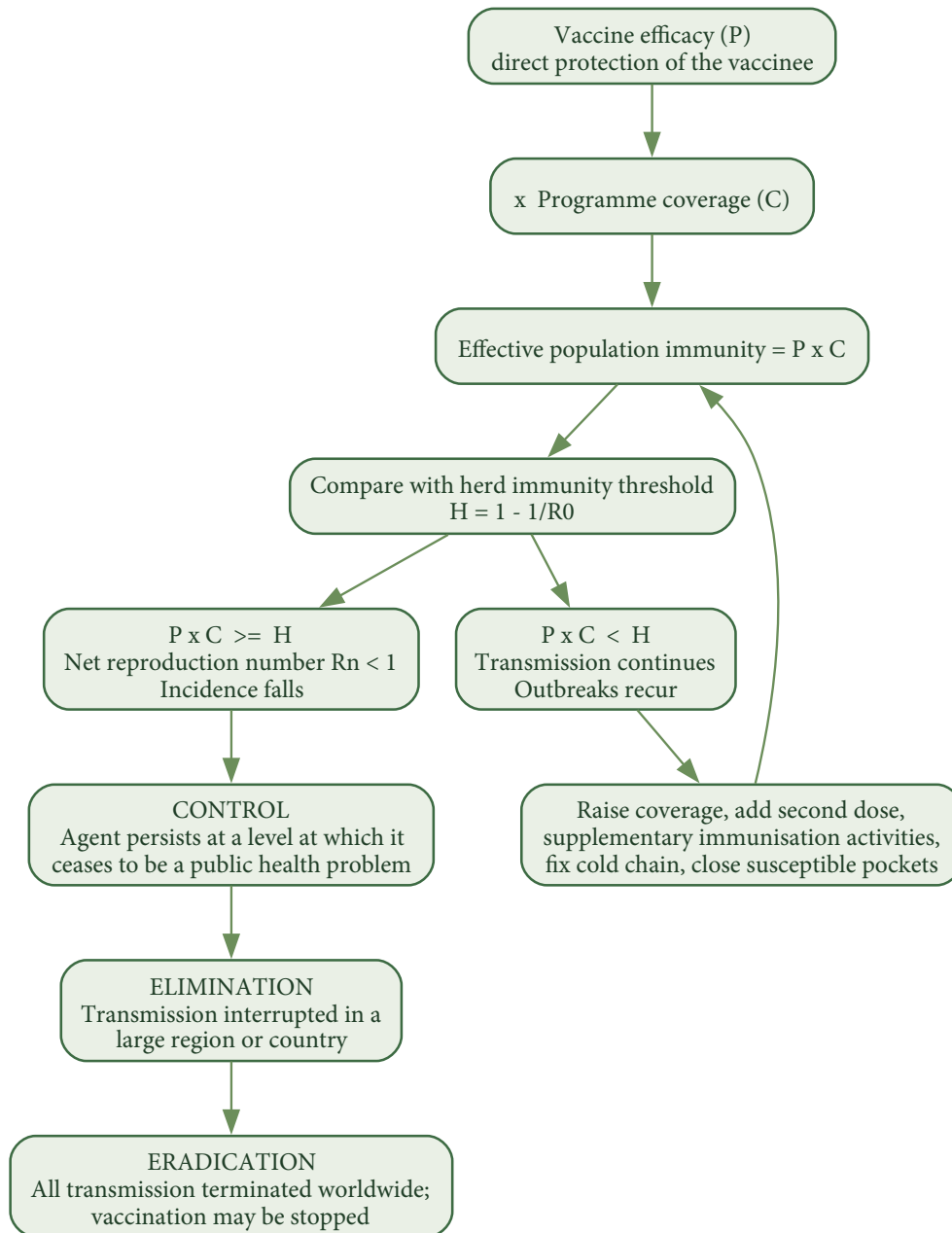


Fig 2.9 — Effect of herd immunity, United States, 1958–61. (A) Expected number o (Gordis Epidemiology 7e, p.43)



Choosing the design to measure vaccine efficacy or effectiveness



From vaccine efficacy to control, elimination and eradication

High-yield exam pointers

- ▶ $VE = (ARU - ARV) / ARU \times 100 = (1 - RR) \times 100$; in case-control substitute the odds ratio for RR.
- ▶ **Screening method:** $VE = (PPV - PCV) / [PPV(1 - PCV)]$ — PPV is the proportion of the *population* vaccinated, PCV the proportion of *cases* vaccinated.
- ▶ $H = 1 - 1/R_0$; coverage needed $CH = H / P$. Measles R_0 12-18, H 92-94% — the highest threshold in the table.
- ▶ Efficacy = ideal (trial) conditions; effectiveness = field conditions. **Primary** vaccine failure = never seroconverted; **secondary** = seroconverted then waned.
- ▶ India: **polio-free 2014, maternal and neonatal tetanus eliminated 2015**; full immunisation 76.4% (NFHS-5, 2019-21) against a UIP target of >90%.

- Tetanus is the exception — herd immunity does not protect the individual, because the reservoir is the soil, not other people.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: the herd immunity thresholds in the table are crude theoretical estimates whose validity depends on assumptions about maternal immunity, age at vaccination and mixing patterns, and Plotkin's authors caution that they should not be accepted uncritically as eradication targets (Plotkin 7e, Ch 77, pp 2172, 2184). India's Universal Immunization Programme currently protects against 12 vaccine-preventable diseases and is committed to measles-rubella elimination following the 2013 South-East Asia Regional Committee resolution.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed

Programme and policy updates (India)

- **Intensified Mission Indradhanush (IMI)** — run October 2017 to January 2018 across 121 districts and 17 urban areas — is the reference "periodic intensification of routine immunisation" strategy for closing coverage pockets: full immunisation coverage in IMI districts rose about 18.5% over pre-IMI levels, and a controlled interrupted time-series put the median gain in delivery of routine UIP antigens at 10.6% (95% CI 5.1–16.5) during implementation. A national drug-sales interrupted time series, however, found the knock-on effect on paediatric antibiotic demand small and heterogeneous — fluoroquinolone sales fell 12.6% in the first month (IRR 0.874, 95% CI 0.777–0.983) while broad-spectrum penicillins, cephalosporins and macrolides continued to rise — a useful caution that programme "impact" must be measured on the outcome actually claimed, not on a convenient proxy [PMID 40413400].
- **Resurgence of vaccine-preventable disease is now an explicit programme concern**, despite the widest vaccine basket and highest coverage India has ever had: spikes in reported measles cases across several states, a marked rise in mumps particularly in older children, and re-emergent diphtheria in Kerala and Assam with the burden shifting to adolescents and adults, all against persisting geographic and income inequity in coverage. The stated remedies are real-time electronic beneficiary tracking (Co-WIN and U-WIN) and converting the UIP from an infant programme into a **life-course immunisation** programme [PMID 40278999].

Current national data

- **NFHS-5 (2019–21), re-analysed for equity**: among more than 43,000 children aged 12–23 months, 77.1% were fully immunised, but coverage fell stepwise with birth order — sixth-or-later children had 34% lower adjusted odds of full immunisation than first-borns (aOR 0.66) — and ranged from 82.6% in the south to 67.5% in the north-east (north-east aOR 0.52). Four or more antenatal visits (aOR 1.62) and public-facility delivery (aOR 1.64) were the strongest positive predictors; girls remained slightly less likely to be fully immunised (aOR 0.95). This is the *coverage* half of the efficacy × coverage product, and it names the pockets that keep $P \times C$ below the herd immunity threshold [PMID 41637385].

- **Pooled national coverage gap:** a 2026 meta-analysis of 45 Indian cross-sectional studies (>428,000 children, 2015–2025) estimated incomplete immunisation in under-fives at **26% (95% CI 21–32)** — about one child in four. The dominant determinants were absence of an immunisation card (OR 9.46, 95% CI 3.28–27.28), non-institutional delivery (OR 3.59), maternal illiteracy (OR 3.25), poverty (OR 2.03), higher birth order (OR 1.88) and female sex (OR 1.34) — that is, documentation and contact with the health system rather than outright refusal, and the main obstacle to India's Immunization Agenda 2030 targets [PMID 42320383].

Elimination, surveillance and disease status

- **Rotavirus — the cleanest recent Indian demonstration of efficacy converting into impact.** India was the first country in the WHO South-East Asia Region to add rotavirus vaccine to the UIP (Rotavac, 26 March 2016; Rotasiil added 2018; nationwide by 2019). Eight years of multicentre sentinel surveillance (2016–2023; 27,862 stool samples) found rotavirus positivity of 24.2% against 37% before rollout, the steepest fall in infants (29.4% in 2016 to 15.1% in 2023, a 48.6% reduction) with smaller falls in older children consistent with indirect protection. G3P[8] became the dominant genotype (44.7%), but the same shift occurred in unvaccinated countries, arguing against vaccine-driven strain replacement [PMID 41832515].
- **Diphtheria illustrates failure of boosters rather than of primary doses.** A 15-year Delhi tertiary-hospital series (2007–2021) isolated *Corynebacterium diphtheriae* from 2,131 of 5,493 suspected cases (38.8%); all of the 278-isolate subset tested were toxigenic. The case-fatality ratio was 19.1%; only 1.1% of confirmed cases were fully vaccinated for age, and partially vaccinated (aOR 4.5, 95% CI 2.8–7.4) or unvaccinated/unknown-status children (aOR 4.2, 95% CI 2.7–6.5) had roughly four times the odds of confirmed disease. Median age rose from 5 to 7 years across the period, and a 2017–18 national serosurvey of 6–17 year-olds found 10.5% non-immune and 59.8% only partially immune. India accounted for close to a quarter of the roughly 25,000 diphtheria cases reported globally in 2024 [PMID 42555698].

Guideline and legislative updates

- **Typhoid conjugate vaccine (TCV)** — WHO-recommended for endemic countries, with four prequalified products (Typbar-TCV, TYPHIBEV, SKYTyphoid, ZyVac TCV) — now has pooled real-world effectiveness data: a 2025 meta-analysis of four post-licensure studies (24,180 children and adolescents in India, Pakistan and Zimbabwe) found single-dose effectiveness of **87% (95% CI 57–94)** against culture-confirmed typhoid and **97% (95–98)** against extensively drug-resistant *S. Typhi*, with protection declining 4–5 years after vaccination. Only ten countries had introduced TCV into national schedules at the time of review — a live illustration of the gap between demonstrated efficacy and programmatic adoption [PMID 41125003].

Global and WHO developments

- **First national-level mortality evidence for HPV vaccination.** In England (programme from 2008 for girls aged 12–13, 80–90% coverage), no cervical cancer deaths occurred in women aged 20–24 during 2020–2024 against 23.1 expected — a reduction of **100% (95% CI 84–100)**; the reduction was 69% (55–79) at ages 25–29 in the same period, and about 199.6 deaths (95% CI 125.0–274.2) were averted up to end-2024. This is the strongest available proof that a high-efficacy vaccine given at high coverage *before exposure* converts into mortality reduction, and it underpins the WHO cervical cancer elimination goal [PMID 42309117].

- ▶ **Coverage and timing, not efficacy alone, decide population impact.** A 2026 global systematic review of 63 population-level HPV studies reported declines of 58–100% in HPV infection, 30–88% in CIN2+, 60–90% in genital warts and 70–88% in invasive cervical cancer — with the largest reductions confined to settings reaching at least 70% coverage and vaccinating before sexual debut, usually through school-based delivery [PMID 42053091].
- ▶ **Gender-neutral vaccination and the difficulty of separating direct from indirect protection.** A systematic review of 17 studies from high-income countries found consistent falls in anogenital warts among adolescent and young adult males after male HPV vaccination began (2011 onwards), with oropharyngeal cancer incidence down 11–19% in US men aged 20–44 years and penile cancer down 6–38% at ages 15–34 years; the authors caution that part of this benefit is herd immunity carried over from earlier girls-only programmes — exactly the confounding that makes field *effectiveness* differ from trial *efficacy* [PMID 42217245].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 2 — Concept of Health and Disease; Ch 3 — Principles of Epidemiology and Epidemiologic Methods); Gordis Epidemiology 7e (Ch 2 — The dynamics of disease transmission; Ch 11 — Randomized trials: some further issues); Plotkin's Vaccines, 7ed (Ch 37 — Measles Vaccines; Ch 47 — Pneumococcal Polysaccharide Vaccines; Ch 53 — Rubella Vaccines; Ch 62 — Varicella Vaccines; Ch 73 — Immunization in the United States; Ch 77 — Community Protection); Giesecke — Modern Infectious Disease Epidemiology (Ch 19 — The Epidemiology of Vaccination); Nelson & Williams — Infectious Disease Epidemiology (Ch 18 — Tuberculosis); Routine Immunization I Manual for Medical Officers, Government of India, 2024; National Family Health Survey-5 India Report (2019-21).

Recent-advances sources: PMID 40413400 — Did the periodic intensification of routine immunisation strategy (Intensified Mission Indradhanush) reduce the demand for pediatric antibiotic formulations in India?, BMC Infectious Diseases 2025, doi:[10.1186/s12879-025-11082-3](https://doi.org/10.1186/s12879-025-11082-3); PMID 40278999 — Vaccine-Preventable Diseases in Pediatric Age Group in India: Recent Resurgence, Implications and Solutions, Indian Journal of Pediatrics 2025, doi:[10.1007/s12098-025-05531-9](https://doi.org/10.1007/s12098-025-05531-9) (abstract-only); PMID 41637385 — How does birth order influence full immunization coverage among children aged 12-23 months in India? Evidence from the National Family Health Survey, PLOS ONE 2026, doi:[10.1371/journal.pone.0342300](https://doi.org/10.1371/journal.pone.0342300); PMID 42320383 — Burden of incomplete immunization and its determinants among children under five years in India: a systematic review and meta-analysis (2015-2025), Vaccine 2026, doi:[10.1016/j.vaccine.2026.128850](https://doi.org/10.1016/j.vaccine.2026.128850) (abstract-only); PMID 41832515 — Molecular epidemiology of rotavirus causing diarrhea among under-five children after the introduction of rotavirus vaccines, Rotavac and Rotasil, into the national immunization program of India, Virology Journal 2026, doi:[10.1186/s12985-026-03126-0](https://doi.org/10.1186/s12985-026-03126-0); PMID 42555698 — Persistent high diphtheria positivity in North India: evidence from a tertiary hospital-based study, 2007-2021, Indian Journal of Medical Research 2026, doi:[10.25259/IJMR.249.2026](https://doi.org/10.25259/IJMR.249.2026); PMID 41125003 — Real-world vaccine effectiveness of typhoid conjugate vaccine in children and adolescents: a systematic review and meta-analysis, Vaccine 2025, doi:[10.1016/j.vaccine.2025.127872](https://doi.org/10.1016/j.vaccine.2025.127872) (abstract-only); PMID 42309117 — Cervical cancer mortality trends following HPV vaccination in England, 2001-24: an analysis of population-based mortality data, The Lancet 2026, doi:[10.1016/S0140-6736\(26\)00918-9](https://doi.org/10.1016/S0140-6736(26)00918-9) (abstract-only); PMID 42053091 — Population-level impact of HPV vaccination: a global systematic review of ecological, cross-sectional, and cohort studies, Expert Review of Vaccines 2026, doi:[10.1080/14760584.2026.2667734](https://doi.org/10.1080/14760584.2026.2667734) (abstract-only); PMID 42217245 — Impact of Male Human Papillomavirus (HPV) Vaccination on HPV Infection and HPV-Related Diseases: A Systematic Review, Reviews in Medical Virology 2026, doi:[10.1002/rmv.70171](https://doi.org/10.1002/rmv.70171) (abstract-only).

♦ ♦ ♦

Web of causation

Asked in: Paper I Nov 2017 — RGUHS MD Community Medicine

Definition — The **web of causation** is a model of disease causation suggested by *MacMahon* and *Pugh* in their book *Epidemiology: Principles and Methods* (Boston, Little Brown, 1970), which considers **all the predisposing factors of any type and their complex interrelationship with each other** (Park 28e, Ch 2, p 53; reference 87, p 71). Disease is pictured not as the end of a single causal chain but as the end-point of a network of many interlocking chains, each strand of which is itself the product of earlier causes. The model is **ideally suited to chronic disease, where the disease agent is often not known** and the illness is the outcome of interaction of multiple factors (Park 28e, Ch 2, p 53).

Place of the web among models of disease causation

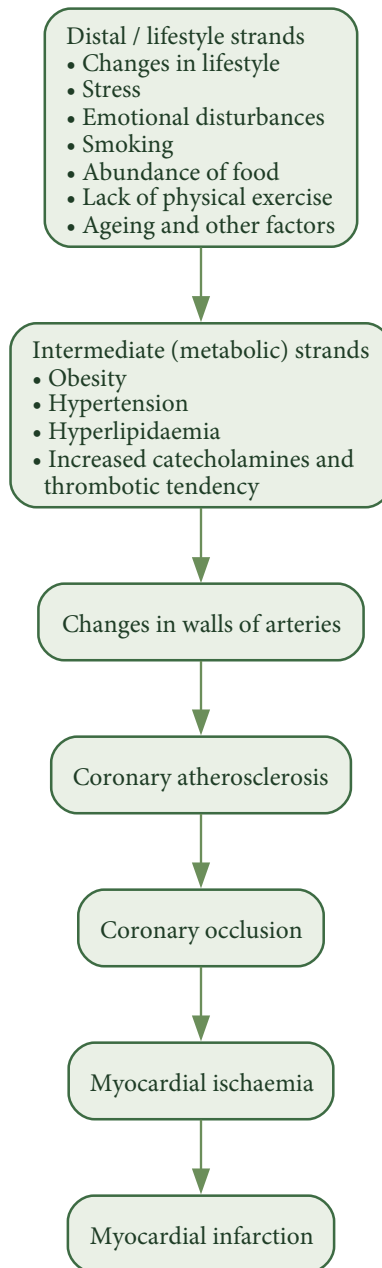
Model	Attributed to / shown as	Core idea	Fits best
Germ theory (one-to-one)	19th–early 20th century bacteriology	Disease agent → man → disease; one agent both necessary and sufficient	Classical infections; fails even for tuberculosis and leprosy (Park 28e, Ch 2, pp 51–52)
Epidemiological triad	Park Fig 4	Agent–host–environment interact; break one leg of the triangle to stop the outbreak	Communicable disease (Park 28e, Ch 2, pp 51–52)
Advanced triangle of epidemiology	Park Fig 6	"Agent" is replaced by causative factors ; adds behaviour, lifestyle, ecological, physical factors and the characteristics of groups or populations	Present-day disease, disability, injury and death (Park 28e, Ch 2, p 52)
Multifactorial causation — alternative factors	Park Fig 14	Several factors, each able to produce the disease independently — smoking, air pollution and asbestos in lung cancer	Non-communicable disease with several separate routes (Park 28e, Ch 3, p 108)
Multifactorial causation — synergism	Park Fig 15	Factors 1 + 2 + 3 act together at cellular level to produce disease	Interaction / effect modification (Park 28e, Ch 3, p 109)
Web of causation	<i>MacMahon</i> and <i>Pugh</i> , 1970; Park Fig 7	The whole nexus of predisposing factors and their interrelations — many chains, many entry points for intervention	Chronic disease where the agent is unknown (Park 28e, Ch 2, p 53)
Sufficient-component cause ("causal pies")	<i>Rothman</i> , 1976	A sufficient cause is a minimal set of conditions that inevitably produces the disease; no component acts alone, but each component within a set is necessary	Explaining why the same exposure does not produce disease in everyone (Lilienfeld's Foundations, Ch 4, pp 77–78; Abramson RCM, Ch 28, p 273)

Structure and essential features of the web

- ▶ **Many chains, not one chain.** In a single chain of causation $A \rightarrow X \rightarrow Y \rightarrow Z \rightarrow B$, interest lies not only in the initiating factor A but in the intervening causes X, Y, Z . The web is the **total nexus** — that chain, its modifying factors, every other chain that ends in the same disease, and the interactions between chains (Abramson RMCM, Ch 28, pp 272–273).
- ▶ **Distal and proximal causes lie on a continuum.** Remote social and economic causes (poverty, crowding, undernutrition) feed into proximal biological ones (exposure, infection, tissue change); clinicians preoccupied with pathogenesis regularly underestimate the distal end (Fletcher, Clinical Epidemiology, Ch 12, p 215).
- ▶ **Causes act at different levels** — molecular, cellular, personal, familial, community and societal — so preventive action is possible at each level (Abramson RMCM, Ch 28, p 273).
- ▶ **Where the agent is unknown, aetiology is written in terms of risk factors** — an attribute or exposure significantly associated with the disease, or a determinant modifiable by intervention. Risk factors may be truly causative (smoking for lung cancer), merely contributory (lack of physical exercise for coronary heart disease), or predictive only in a statistical sense (illiteracy for perinatal mortality) (Park 28e, Ch 2, p 57).
- ▶ **The strands are not of equal weight.** In a multifactorial event individual factors are by no means all equally important, and their relative importance is expressed as the **relative risk** (Park 28e, Ch 2, p 53).
- ▶ **Strands interact.** Where several factors act together the resulting risk may exceed the simple sum of their separate effects — that is, effect modification (Fletcher, Ch 12, pp 213–214).
- ▶ **The web is dynamic.** It changes even for old diseases: tuberculosis case numbers in the United States, falling for a century, rose again between 1985 and 1992 as immigration from high-incidence countries and the acquired immunodeficiency syndrome (AIDS) epidemic added new strands (Fletcher, Ch 12, p 215).
- ▶ **It applies to infectious disease as well.** A causal web is best understood in chronic degenerative disease, but it also underlies infection, where the microbe is a **necessary but not sufficient** cause — exposure to human immunodeficiency virus (HIV) does not invariably produce disease, and not everyone exposed to tuberculosis becomes infected (Fletcher, Ch 12, pp 213–214).
- ▶ **The whole web need not be dismantled.** The model does not imply that disease cannot be controlled unless all the multiple causes or chains are removed; **elimination of a single link may suffice, provided that link is sufficiently important** in the pathogenetic process (Park 28e, Ch 2, p 53).

Example — web of causation for myocardial infarction

Park's own worked example (Fig 7) is the causal web of **myocardial infarction**, chosen because the "agent" of coronary heart disease is not identifiable and the event is the convergence of lifestyle, metabolic and vascular strands (Park 28e, Ch 2, p 53).



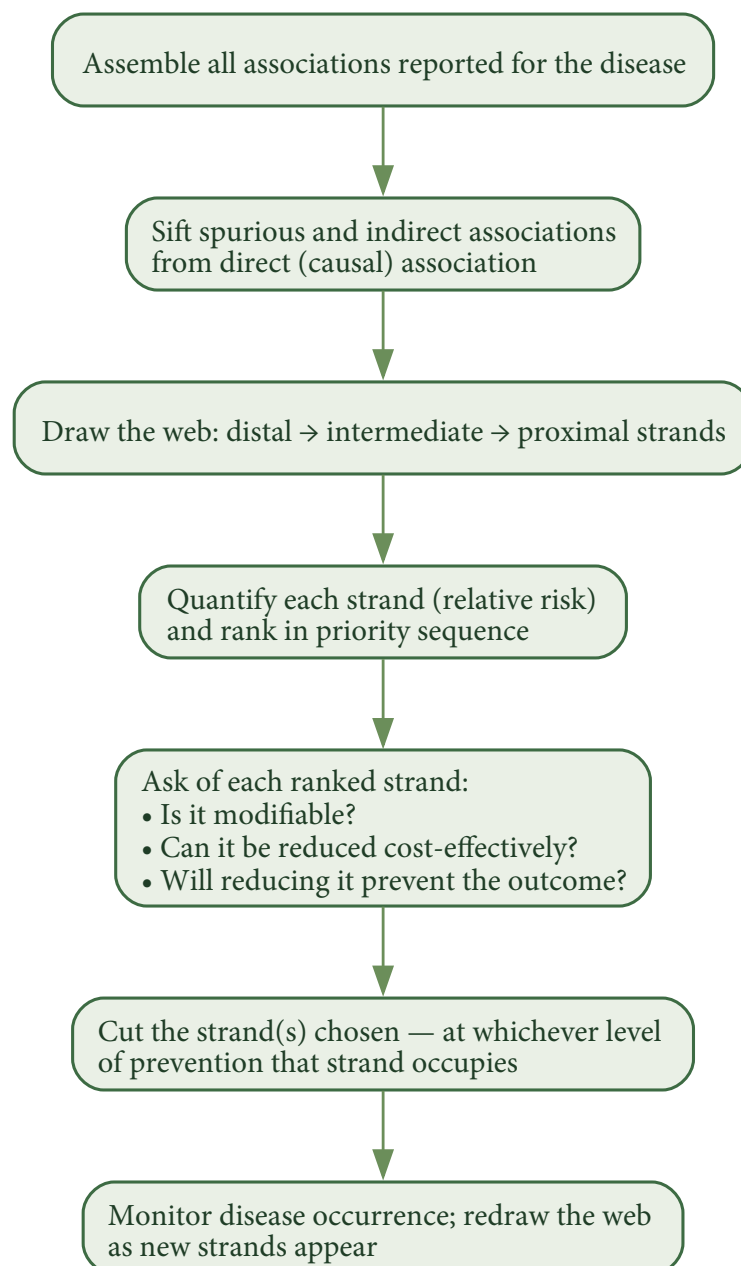
Web of causation for myocardial infarction (after Park's Textbook of Preventive and Social Medicine 28e, Ch 2, Fig 7, p 53)

Level of the web	Factors shown in Park Fig 7	Where intervention can cut the strand
Distal (social and behavioural)	Changes in lifestyle, stress, emotional disturbances, ageing and other factors	Primordial prevention — food, tobacco and physical-activity policy; workplace and community stress reduction
Behavioural	Smoking, abundance of food, lack of physical exercise	Primary prevention — tobacco cessation, dietary fat and salt reduction, exercise promotion
Intermediate (metabolic)	Obesity, hypertension, hyperlipidaemia, raised catecholamines and thrombotic tendency	Primary and secondary prevention — weight reduction, detection and treatment of hypertension, lipid lowering, antiplatelet therapy

Level of the web	Factors shown in Park Fig 7	Where intervention can cut the strand
Proximal (vascular)	Changes in arterial walls → coronary atherosclerosis → coronary occlusion	Secondary prevention — screening of high-risk adults, treatment of established coronary disease
Terminal event	Myocardial ischaemia → myocardial infarction	Tertiary prevention — emergency care, revascularisation, cardiac rehabilitation

The value of drawing the web is precisely this: it shows a variety of possible interventions, any of which might reduce the occurrence of myocardial infarction, rather than one "magic bullet" (Park 28e, Ch 2, p 53).

Using the web to plan prevention and control



Stepwise use of a web of causation to plan disease control

- ▶ **The purpose of identifying multiple factors** is to quantify them and arrange them in **priority sequence** for modification or amelioration; the multifactorial concept thereby offers **several alternative approaches** to prevention and control (Park 28e, Ch 2, p 52).
- ▶ **Detection of a risk factor is a prelude to intervention, not proof of cause.** For each factor ascertained, ask whether it can be reduced cost-effectively and whether its reduction will actually prevent or delay the unwanted outcome (Park 28e, Ch 2, p 57).
- ▶ **A strand can be cut before the agent is known.** Before the cholera vibrio was discovered, elimination of certain water supplies produced a marked fall in new cases — an indirect association exploited for prevention (Park 28e, Ch 3, p 108).
- ▶ **Look for the strand that is always or frequently necessary** — a factor featuring in every, or in most, of the operating constellations of causes. Removing or blocking such a factor acts as a "preventive broom" that sweeps away much of the web; immunisation works this way, by abolishing an always-necessary cause (low level of immunity) (Abramson RMCM, Ch 28, p 273).

Strengths and limitations

Strengths	Limitations
Handles chronic non-communicable disease, where no single agent exists (Park 28e, Ch 2, p 53)	Becomes unwieldy — the more associations, the more investigation needed to "disentangle the web", and the notion of <i>cause</i> becomes confused (Park 28e, Ch 3, p 107)
Displays many alternative points of intervention instead of one	Does not by itself distinguish association from causation; Bradford Hill-type criteria are still required
Accommodates causes at every level from molecular to societal, so it supports both clinical and social action	Criticised for describing strands while leaving the generating social structure unexamined — <i>Krieger's</i> "has anyone seen the spider?" (Krieger 1994, cited in Abramson RMCM, Ch 28, reference 36)
Predicts that removal of one important link can control disease	Gives no rule for which link matters most; relative risk must be measured separately (Park 28e, Ch 2, p 53)
Directly generates aetiological hypotheses for analytical study	Superseded in part by the sufficient-component and counterfactual models, and by eco-epidemiology / "Chinese boxes", which specify the levels explicitly (Lilienfeld's Foundations, Ch 4, p 78; Abramson RMCM, Ch 28, p 280)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Web of causation for myocardial infarction** (Park 28e, Ch 2, Fig 7) — draw many arrows converging downward. **Exact labels:** Changes in lifestyle; Stress; Emotional disturbances; Smoking; Abundance of food; Lack of physical exercise; Ageing and other factors → Obesity; Hypertension; Hyperlipidaemia; Increased catecholamines, thrombotic tendency → Changes in walls of arteries → Coronary atherosclerosis → Coronary occlusion → Myocardial ischaemia → **Myocardial infarction**.
- ▶ **A community-medicine web — deaths from gastroenteritis** (Abramson RMCM, Ch 28, pp 272–273). **Exact labels:** Poverty; Lack of clinics; Treatment by traditional practitioners; Early weaning; Dirty feeding bottles; Various microorganisms; Respiratory infections; Malnutrition → **Death from gastroenteritis**.

High-yield exam pointers

- ▶ **Attribution:** web of causation = *MacMahon and Pugh, Epidemiology: Principles and Methods*, 1970 — write the names, the book and the year.
- ▶ **One-line definition to reproduce:** "considers all the predisposing factors of any type and their complex interrelationship with each other."
- ▶ **Park's own example is myocardial infarction** (Fig 7) — reproduce the chain ending "...coronary atherosclerosis → coronary occlusion → myocardial ischaemia → myocardial infarction".
- ▶ **Key examination sentence:** removal of only one link may control the disease, provided that link is sufficiently important; strands are not of equal weight, and their relative importance is expressed as **relative risk**.
- ▶ **Sequence of causal models to name:** germ theory (one-to-one) → epidemiological triad → multifactorial causation → web of causation → sufficient-component ("causal pie", *Rothman* 1976).
- ▶ **Contrast to state in one line:** the triad is broken by removing one leg (agent, host or environment) in communicable disease; the web is broken by cutting the heaviest strand in chronic disease.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park's Fig 7 is unchanged between the 27th and 28th editions; the node labels listed above are transcribed from the printed figure, since the 28e figure caption alone does not carry them.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 2 — Concept of Health and Disease, pp 51–53, 57, reference list p 71; Ch 3 — Principles of Epidemiology and Epidemiologic Methods, pp 107–109); Fletcher — Clinical Epidemiology: The Essentials (Ch 12, Cause, pp 213–215); Abramson — Research Methods in Community Medicine (Ch 28, Making Sense of Associations, pp 272–273); Lilienfeld's Foundations of Epidemiology (Ch 4, Inferring Causal Relationships, pp 77–78); MacMahon B and Pugh TF (1970), *Epidemiology: Principles and Methods*, Boston, Little Brown (as cited by Park 28e).



Community diagnosis

Asked in: Paper I 24-Sep-2025 — RGUHS MD Community Medicine

Definition — The diagnosis of disease in an individual patient rests on signs, symptoms and the inferences drawn from them; when the same idea is applied to a population it is called **community diagnosis**, defined as *the pattern of disease in a community described in terms of the important factors which influence this pattern* (Park 28e, Ch 2, p 68). The World Health Organization (WHO, 1995) describes it as *"a quantitative and qualitative description of the health of citizens and the factors which influence their health; it identifies problems, proposes areas for improvement and stimulates action"*, and it is also called a **community health profile** or a **health needs assessment** (Oxford Textbook of Public Health 7e, Ch 4.2, p 922).

Community diagnosis contrasted with clinical diagnosis

Element	Clinical diagnosis	Community diagnosis
Unit of concern	The individual patient, through the patient–physician interaction	A defined population sharing at least one common characteristic — village, ward, occupation, ethnicity, exposure or risk factor (Oxford 7e, Ch 4.2, p 922)
"History"	Presenting complaints, past and family history	Felt needs and "chief complaints" of the community, elicited by listening to people, elected and informal leaders, the village or urban health sanitation and nutrition committee (VHSNC), <i>Mahila Arogya Samiti</i> , anganwadi workers and school teachers (Sunder Lal, Ch 1, p 53)
"Examination"	Physical examination of the patient	Direct observation and examination of the community — environment, water and sanitation, and the records kept at the subcentre, primary health centre (PHC) or community health centre (CHC) (Sunder Lal, Ch 6, pp 268–269)
"Investigations"	Laboratory tests, imaging	Household surveys, facility surveys, interviews, focus group discussions, mapping and records review — quantitative and qualitative (Oxford 7e, Ch 4.2, p 924)
Expressed as	A disease label in one person	Mortality and morbidity rates and ratios and their correlates, defining those at risk or in need of care (Park 28e, Ch 3, p 111)

Element	Clinical diagnosis	Community diagnosis
Diagnosis includes	The presenting illness	The presenting illness plus its underlying determinants — unsafe water, open defecation, unhealthy behaviour, diet, poor knowledge and attitudes, poor access to comprehensive primary health care, poverty and illiteracy (Sunder Lal, Ch 17, pp 734–736)
"Treatment"	Prescription for the patient	Community treatment / community health action — the sum of steps decided upon to meet the health needs of the community, taking account of available resources and the wishes of the people (Park 28e, Ch 2, p 68)
Follow-up	Review of the patient	Follow-up of the community, evaluation of interventions and replanning of services (Sunder Lal, Ch 17, pp 734–737)

Data required for a community diagnosis, and where they come from in India

Park lists seven categories of data required to analyse the health situation and define the health problems (Park 28e, Ch 10, pp 721–722); each is drawn from a defined source in the national health information system (Park 28e, Ch 21, pp 996–998).

Data required (Park's seven)	Principal Indian source
1. Morbidity and mortality statistics	Notification of diseases, hospital records, disease registers, epidemiological surveillance under national disease control programmes, other health service records
2. Demographic conditions of the population	Census (last held March 2011, under the Census Act 1948); Civil Registration System under the Registration of Births and Deaths Act 1969; Sample Registration System (SRS), begun in the mid-1960s as a dual-record system and now covering the whole country
3. Environmental conditions bearing on health	Environmental health statistics — air, water, food additives, industrial toxicants, noise, waste disposal
4. Socio-economic factors with a direct effect on health	Census (income, occupation, housing, working conditions); economic and social-security statistics
5. Cultural background, attitudes, beliefs and practices	Non-quantifiable information and qualitative enquiry — health interview surveys, focus group discussions
6. Medical and health services available	Health service records, health manpower statistics, hospital indices (bed-occupancy rate, duration of stay)
7. Other services available	Other routine statistics related to health (demographic, economic, social-security schemes)

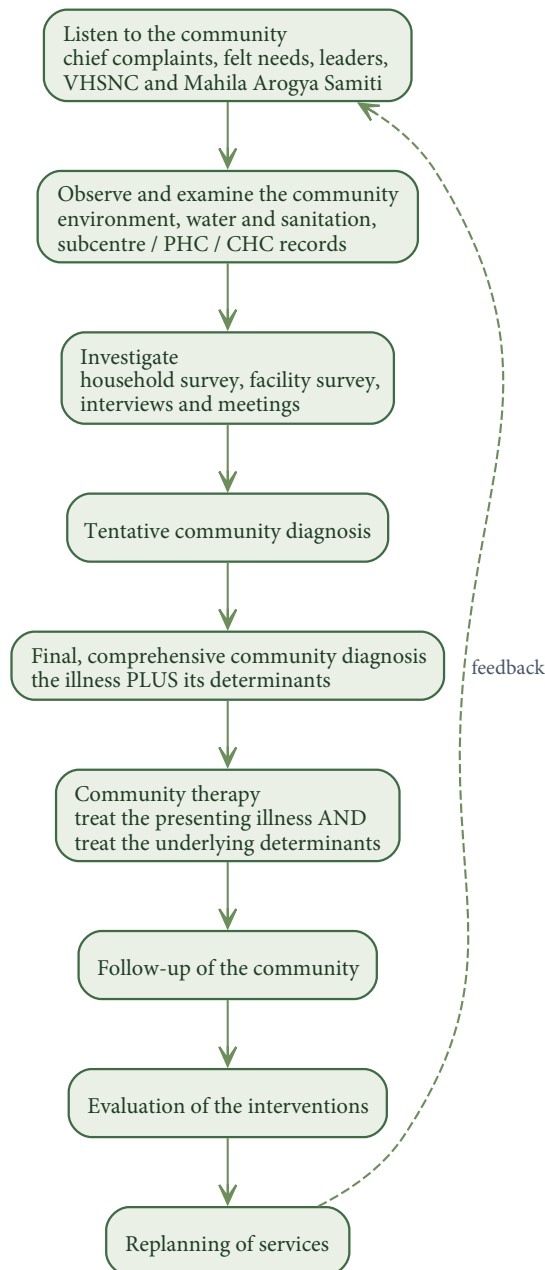
Population surveys are needed because routine statistics never give the whole picture; Park explicitly classes "*surveys for evaluating the health status of a population*" as community diagnosis (Park 28e, Ch 21, p 998). The National Family Health Survey-5 (NFHS-5, 2019–21) supplies key estimates for all 28 states and 8 union territories and for **707 districts** (as on 31 March 2017), continuing the district-level estimation begun in NFHS-4 (2015–16), so a district-level community diagnosis can be anchored on survey data (NFHS-5 India Report, 2022).

Choosing the indicators. Each metric selected should be **feasible, valid, reliable, relevant, sensitive, specific, actionable and accessible**, and should be capable of being disaggregated by geography, age, sex and ethnicity; no single metric suffices, so a complementary *group* of metrics is chosen (Oxford 7e, Ch 4.2, pp 947–948).

Steps in making a community diagnosis

Community diagnosis follows a customisable five-step process (Oxford 7e, Ch 4.2, pp 924–927):

- ▶ **Initiation** — constitute a cross-functional team from government departments, health bureaux, non-governmental organisations and community representatives; refine the scope, budget, timeline, data-collection and analysis methods, the community-engagement strategy, and the indicators to be used; anticipate obstacles and risks.
- ▶ **Collection** — gather both primary data (surveys, interviews, focus group discussions, mapping, observation) and secondary data (records review), and both quantitative and qualitative data, since neither alone gives a complete picture.
- ▶ **Analysis** — compile, tabulate and summarise with descriptive statistics; compute rates and ratios; examine correlates and determinants; compare with historical trends, with similar communities and with the district, state and national figures.
- ▶ **Diagnosis** — interpret the analyses into a statement of the current state of health, the determinants of health, and the health problems prioritised for intervention. The diagnosis must be **informative** (it improves understanding) and **actionable** (the actions to be taken follow easily from it).
- ▶ **Dissemination** — produce a community diagnosis report setting out scope, methods, results and recommendations, written so that policymakers, health workers *and* community members can all understand it.



Community diagnosis and community treatment — the clinical method applied to a population

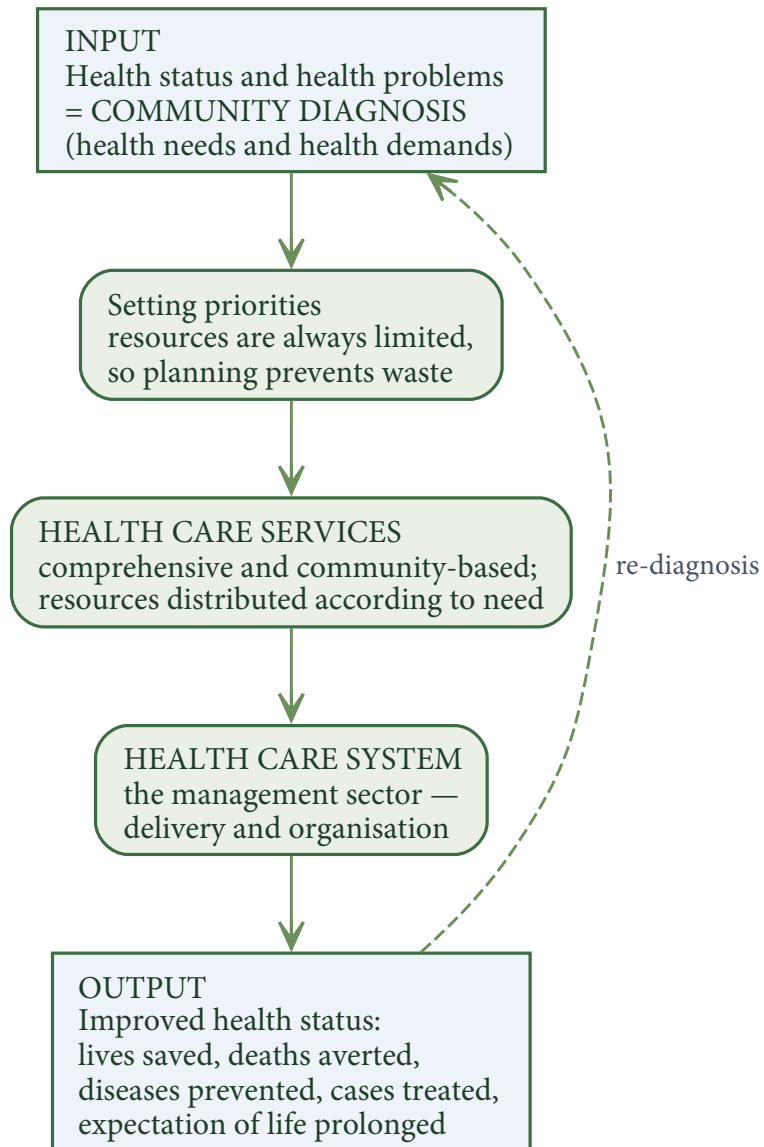
How community diagnosis improves health care of the community

Community diagnosis is the second of Morris's seven uses of epidemiology, which is why epidemiology is described as the "diagnostic tool" of community medicine (Park 28e, Ch 3, p 111). Its contributions to health care are these:

Contribution	What it delivers
1. It is the first requisite of any planned effort	An assessment of health status and health problems must precede any attempt to develop health care services; it forms the INPUT of the health care model, representing the health needs and health demands of the community (Park 28e, Ch 10, pp 721–722)

Contribution	What it delivers
2. Sets priorities	Quantification of health problems allows priorities in disease control and prevention to be laid down (Park 28e, Ch 3, p 111). Priority is fixed by weighing financial constraints, mortality and morbidity data, diseases preventable at low cost, saving the lives of the young in whom there has been considerable social investment, and community and political interest (Park 28e, Ch 23, p 1027)
3. Rationalises resource allocation	Resources always fall short of total need. Park's example: in developing countries, <i>too many hospitals have been built and equipped without knowledge of the particular disease problems in the community</i> ; epidemiological information on the distribution of health problems over time and place is the fundamental basis for planning and developing the needed health services (Park 28e, Ch 3, p 111)
4. Designs the right service package	It tells the planner how many hospital beds are needed for specific diseases, what health manpower is required, and which preventive services to mount — screening programmes, immunisation campaigns, provision of sanitary services (Park 28e, Ch 3, p 111)
5. Identifies groups at risk and unmet needs	It defines those individuals or groups at risk or in need of health care (Park 28e, Ch 3, p 111); health surveys evaluate population health needs, unmet needs and the use of health services (Park 28e, Ch 21, p 998)
6. Treats determinants, not merely disease	Because a comprehensive community diagnosis names the underlying determinant, the therapy can be directed at it — water treatment and chlorination, safe excreta and waste disposal, universal salt iodation, immunisation, condom promotion, mass drug administration, behaviour-change communication, and legislation for tobacco, helmet and seat-belt use (Sunder Lal, Ch 17, p 736)
7. Provides the baseline for evaluation	Quantified morbidity and mortality serve as a benchmark against which health services are evaluated at a later date (Park 28e, Ch 3, p 111); re-measuring the same indicators after the intervention is what establishes effectiveness (Oxford 7e, Ch 4.2, p 926)
8. Generates new knowledge	Quantification of health problems is itself a source of new knowledge about disease distribution, causation and prevention (Park 28e, Ch 3, p 111)
9. Secures community participation	Community involvement is essential in community diagnosis, in community therapy and in replanning of services (Sunder Lal, Ch 17, pp 734–737); authentic partnership makes the community feel heard and improves the yield of future projects (Oxford 7e, Ch 4.2, pp 928–929)
10. Closes the planning loop	Evaluation results are fed back into the system to fill gaps and improve organised health services (Sunder Lal, Ch 17, pp 734–737) — community diagnosis is repeated, not done once

The measurable **OUTPUT** of this chain is a changed, improved health status of the community, expressed as lives saved, deaths averted, diseases prevented, cases treated and expectation of life prolonged (Park 28e, Ch 10, p 721).



Where community diagnosis sits in the health care delivery model (after Park 28e, Ch 10, Fig 1)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Community diagnosis is the entry arc of the planning cycle — reproduce it with these exact labels:

- ▶ Gather health data → Tabulate, analyse, interpret → Formulate health problems (this arc is the community diagnosis)
- ▶ Assign priorities among problems → Formulate individual priority problems for planning solutions → Define programme objectives and how to measure achievement → Assign priorities among objectives
- ▶ Design alternative programmes to solve problems → Select best programme (feasibility, cost/effectiveness) → Combine programmes and develop integrated plan
- ▶ Plan implementation of integrated programme → Orient and train personnel → Initiate integrated programme → Operate programme
- ▶ Collect data for evaluation → Evaluate results: goals achieved? goals satisfactory? → YES back to *Operate programme*; NO back to *Formulate health problems*

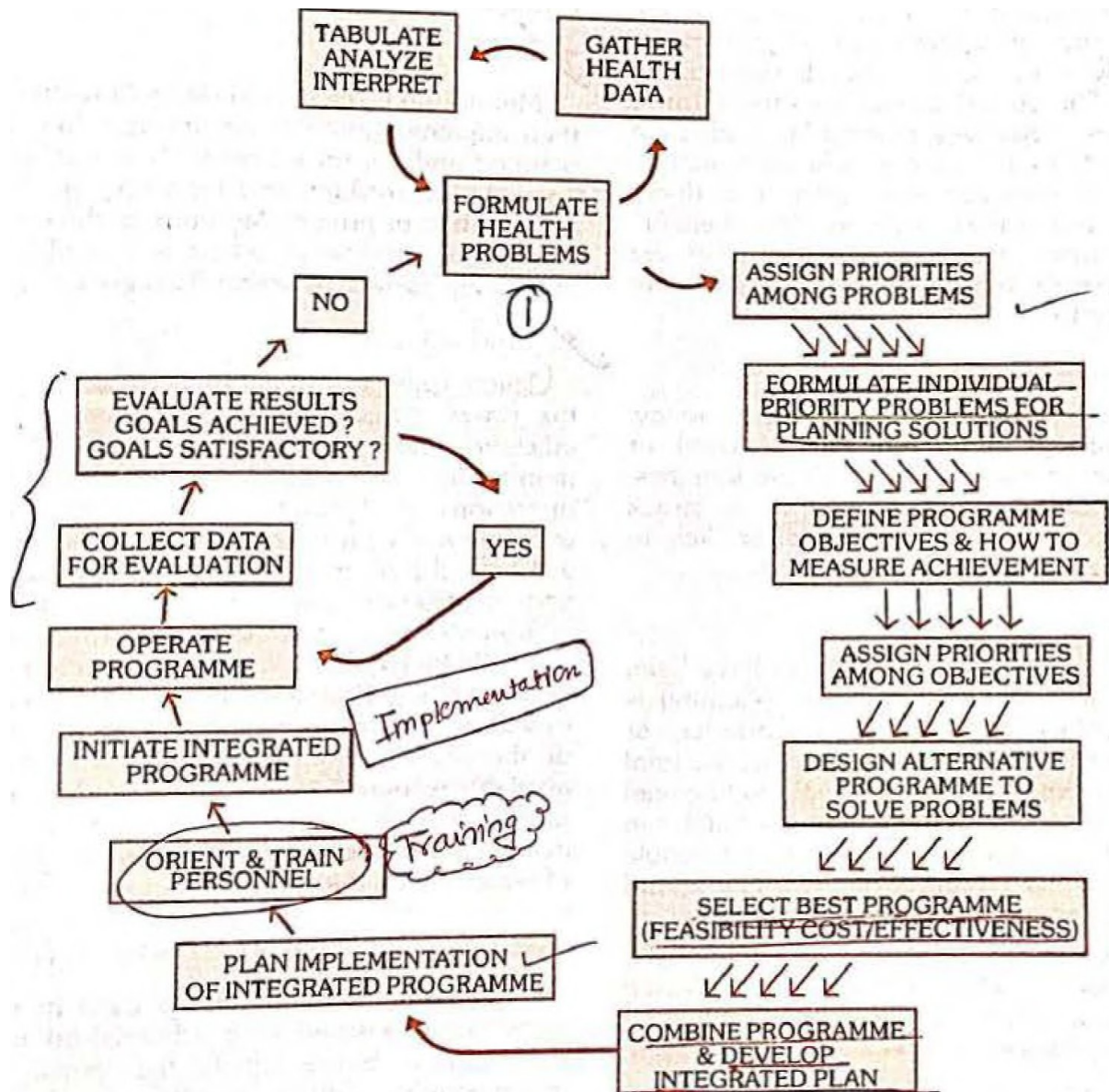


Fig 1 — The planning cycle (Park's Textbook of Preventive and Social Medicine, 28e, p.1026)

High-yield exam pointers

- ▶ Two definitions to quote: Park — "the pattern of disease in a community..."; WHO 1995 — "a quantitative and qualitative description of the health of citizens...".
- ▶ Morris's seven uses of epidemiology (community diagnosis is No. 2): historical study of the rise and fall of disease; **community diagnosis**; planning and evaluation; evaluation of individual risks and chances; syndrome identification; completing the natural history of disease; searching for causes and risk factors.
- ▶ Park's seven data items — mnemonic "M-D-E-S-C-M-O": Morbidity and mortality, Demography, Environment, Socio-economic, Cultural, Medical and health services, Other services.
- ▶ Oxford's five steps — I-C-A-D-D: Initiation, Collection, Analysis, Diagnosis, Dissemination.
- ▶ Indicator criteria — feasible, valid, reliable, relevant, sensitive, specific, actionable, accessible.

- **Current Indian data anchors:** Census 2011 (last completed census); Sample Registration System (continuous, whole country); NFHS-5, 2019–21 (all 28 states and 8 union territories, and 707 districts as on 31 March 2017; district-level estimation began with NFHS-4, 2015–16).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park's data list and health care model are written around the older term "health situation analysis"; in current Indian practice the same exercise is performed at district level using the Health Management Information System and district fact sheets, but the retrieved textbook corpus does not describe those platforms, so they are not quoted as facts here.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 2 — Concept of Health and Disease; Ch 3 — Principles of Epidemiology and Epidemiologic Methods; Ch 10 — Health Care of the Community; Ch 21 — Health Information and Basic Medical Statistics; Ch 23 — Health Planning and Management, Fig 1 "The planning cycle"); Oxford Textbook of Public Health 7e (Ch 4.2 — Community Diagnosis and Health Information Systems in Low- and Middle-Income Countries); Sunder Lal — Textbook of Community Medicine (Ch 1, Ch 6, Ch 17 — Healthcare of the Community); National Family Health Survey-5 (NFHS-5) India Report, 2019–21 (published 2022).



Likelihood ratio

Asked in: Paper I 26-May-2025 — RGUHS MD Community Medicine

Definition — The **likelihood ratio (LR)** of a given test result is the probability of that result in people **with** the disease divided by the probability of the same result in people **without** the disease (Fletcher — Clinical Epidemiology, Ch 8, p 140). It expresses how many times more (or less) likely a particular test result is in a diseased than in a non-diseased person, and so states by how much that result raises or lowers the pre-test probability of the target disorder (Handbook of Epidemiology, Ch 29, p 1173). It is derived entirely from sensitivity and specificity, and is the single test property that converts what was known about the patient before the test into what is known after it.

Derivation from the 2 × 2 table

The standard screening table (Park 28e, Ch 4, p 167) is: **a** = true positive, **b** = false positive, **c** = false negative, **d** = true negative.

Screening test result	Diseased	Not diseased	Total
Positive	a (true positive)	b (false positive)	a + b
Negative	c (false negative)	d (true negative)	c + d
Total	a + c	b + d	a + b + c + d

Quantity	Formula	What it answers
Sensitivity	$a / (a + c)$	Proportion of diseased correctly called positive (Park 28e, Ch 4, p 167)
Specificity	$d / (b + d)$	Proportion of non-diseased correctly called negative (Park 28e, Ch 4, p 167)
Positive likelihood ratio (LR+)	$[a/(a+c)] \div [b/(b+d)] = \text{sensitivity} \div (1 - \text{specificity})$	How much a positive result raises the odds of disease (Sackett & Haynes, Ch 10, pp 297-298)
Negative likelihood ratio (LR-)	$[c/(a+c)] \div [d/(b+d)] = (1 - \text{sensitivity}) \div \text{specificity}$	How much a negative result lowers the odds of disease (Sackett & Haynes, Ch 10, pp 297-298)
Pre-test odds	$\text{prevalence} \div (1 - \text{prevalence})$	Odds form of the pre-test (prior) probability (Fletcher, Ch 8, p 140)
Bayes' step	Pre-test odds $\times$ LR = post-test odds	The central working equation (Fletcher, Ch 8, p 140)
Post-test probability	$\text{post-test odds} \div (1 + \text{post-test odds})$	Equals the predictive value of that result (Fletcher, Ch 8, pp 142-143)
Diagnostic odds ratio	$\text{LR+} \div \text{LR-}$	Single-number summary of discrimination; e.g. LR+ 5.1 and LR- 0.4 give a diagnostic odds ratio of 13 (Sackett & Haynes, Ch 10, pp 304-305)

Pre-test odds carry the same information as prevalence, the likelihood ratios the same information as sensitivity and specificity, and the post-test odds the same information as predictive value (Fletcher, Ch 8, pp 140-141). The likelihood ratio thus answers the question the clinician actually asks — "given this result, what is the likelihood that this patient has the disease?" (Gordis 7e, Ch 5, pp 122-123).

Interpreting the size of a likelihood ratio

An LR of exactly 1 makes the result useless — post-test probability equals pre-test probability. Values above 1 raise and values below 1 lower the probability of disease, and the further from 1, the greater the shift (Handbook of Epidemiology, Ch 29, p 1173).

LR of a positive result	LR of a negative result	Shift from pre- to post-test probability
> 10	< 0.1	Large, often conclusive change
5 – 10	0.1 – 0.2	Moderate shift
2 – 5	0.2 – 0.5	Small , but sometimes important
1 – 2	0.5 – 1	Small and rarely important
1	1	No change — test adds nothing

Bands from Handbook of Epidemiology, Ch 29, p 1173. Tests with LR_s far from 1.0 (LR₊ > 10, LR_– < 0.1) produce few false positives and few false negatives (Fletcher, Ch 8, p 141).

Bedside rule of thumb (McGee's multiples of 15), used when no nomogram or calculator is at hand (Fletcher, Ch 8, Table 8.3, p 143):

Likelihood ratio	Approximate change in disease probability
10	+ 45 %
9	+ 40 %
6	+ 35 %
5	+ 30 %
4	+ 25 %
3	+ 20 %
2	+ 15 %
1	No change
0.5	– 15 %
0.4	– 20 %
0.3	– 25 %
0.2	– 30 %
0.1	– 45 %

Applications

- **Bayesian revision of probability in an individual patient.** Two patients with chest pain — an elderly woman at high pre-test probability and a young man with anxiety at low pre-test probability — may share the same borderline troponin I level, yet the same likelihood ratio moves them to very different post-test probabilities and therefore to different management (Handbook of Epidemiology, Ch 29, p 1172).
- **Using the *degree* of abnormality, not just "positive or negative" (multilevel likelihood ratios)** — the chief advantage over sensitivity and specificity, which are locked to one cut-off. An LR can be computed for **each stratum** of a continuous result (Fletcher, Ch 8, pp 140-141). In the *Hill* troponin I study a value ≥ 10.0 $\mu\text{g/L}$ occurred in 3/20 (15 %) of those with an adverse cardiac outcome and 4/332 (1.2 %) of those without, giving an LR of 12.5, whereas a value < 0.5 $\mu\text{g/L}$ gave an LR of 0.7 (Handbook of Epidemiology, Ch 29, pp 1172-1173).
- **Serial (sequential) testing.** The post-test odds after one test become the pre-test odds for the next, so successive likelihood ratios multiply to give an overall probability accounting for every test in the series (Fletcher, Ch 8, pp 146-147). Predictive values cannot be chained in this way.
- **Quantifying history and physical examination findings**, turning "soft" bedside items into numbers. In the *CARE-COAD1* study of obstructive airway disease: the patient's own report of the disease LR+ 7.3 / LR– 0.5; smoking for more than 40 years LR+ 8.3 / LR– 0.8; age 45 years or more LR+ 1.3 / LR– 0.4; maximum laryngeal height 4 cm or less LR+ 2.8 / LR– 0.8 (Sackett & Haynes, Ch 10, p 304).
- **Ruling a diagnosis in or out.** A very high LR+ on a positive result rules the target disorder **in** (*SpPin* — high Specificity, Positive result, rules **in**); a very low LR– on a negative result rules it **out** (*SnNout* — high Sensitivity, Negative result, rules **out**) (Sackett & Haynes, Ch 10, p 299).

- ▶ **Comparing tests and appraising the literature.** Performance at each level of result condenses into one number with its 95 % confidence interval, making the LR the natural currency for appraising and pooling diagnostic-accuracy studies; the diagnostic odds ratio ($LR+ \div LR-$) condenses it further (Sackett & Haynes, Ch 10, pp 304-305).
- ▶ **Transporting test performance to a new population — the screening application.** Predictive value falls as prevalence declines (Park 28e, Ch 4, p 168), and one quoted from a study with prevalence artificially set at 50 % means little (Fletcher, Ch 8, p 140). The LR, built only from sensitivity and specificity — inherent properties of the test (Park 28e, Ch 4, p 167) — carries across populations, so a published LR can be applied to the **local** prevalence to obtain the predictive value a community screening programme will actually see.
- ▶ **Deciding when to stop testing.** If the post-test probability crosses the treatment threshold, testing stops; if it lands in the indeterminate zone (roughly 10-90 %), a further test is needed — a positive D-dimer reaching only 22 % means another test is required to confirm deep vein thrombosis (Fletcher, Ch 8, pp 142-143).

Worked example — D-dimer assay for deep vein thrombosis

Pre-test probability (prevalence) 10 %; $LR+$ of the D-dimer assay 2.5; $LR-$ 0.03 (Fletcher, Ch 8, pp 140, 142-143).

- ▶ **Pre-test odds** = $0.10 / (1 - 0.10) = 0.111$
- ▶ **Post-test odds** = $0.111 \times 2.5 = 0.278$
- ▶ **Post-test probability** = $0.278 / (1 + 0.278) = 0.22$, i.e. 22 % — identical to the positive predictive value.
- ▶ **Nomogram check** — a straight edge joining 10 % on the pre-test scale to 2.5 on the likelihood-ratio scale crosses the post-test scale at about 22 %.
- ▶ **Rule-of-thumb check** — an LR of about 2.5 adds roughly 15 %, giving about 25 %; close enough to reach the same clinical conclusion.

Advantages and limitations

Advantages	Limitations
Applies to multiple levels of a test result, not a single cut-off (Fletcher, Ch 8, pp 140-141)	Must be used with odds, not probabilities ; conversion to and from odds needs arithmetic, a nomogram or a calculator (Fletcher, Ch 8, p 142)
One number summarises the information contributed by each level of result (Fletcher, Ch 8, pp 141-142)	Most clinicians find thinking in odds harder than thinking in probabilities (Fletcher, Ch 8, p 142)
Well suited to serial testing — likelihood ratios chain, predictive values do not (Fletcher, Ch 8, pp 141-142, 146-147)	Chaining assumes the tests are independent ; if they are not, the combined result is over-estimated (Fletcher, Ch 8, pp 146-147)
Independent of prevalence, being built from sensitivity and specificity, which are inherent properties of the test (Park 28e, Ch 4, p 167)	Needs a valid pre-test probability ; a wrong prior gives a wrong posterior however good the likelihood ratio

Advantages	Limitations
Directly quantifies history and examination findings, not only laboratory tests (Sackett & Haynes, Ch 10, p 304)	An LR near 1 means the test is uninformative — a result that is easily missed if only sensitivity is quoted (Handbook of Epidemiology, Ch 29, p 1173)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Fagan nomogram — labels to reproduce by hand:

- ▶ Left vertical axis: **Pre-test probability** (0.001 at top to 0.999 at bottom)
- ▶ Central vertical axis: **Likelihood ratio** (1000 at top through 1 in the middle to 0.001 at bottom)
- ▶ Right vertical axis: **Post-test probability** (0.999 at top to 0.001 at bottom)
- ▶ A **straight line** drawn from the pre-test probability through the likelihood ratio, extended to cut the post-test probability axis
- ▶ Annotate: *Fagan TJ, Nomogram for Bayes' theorem, 1975*

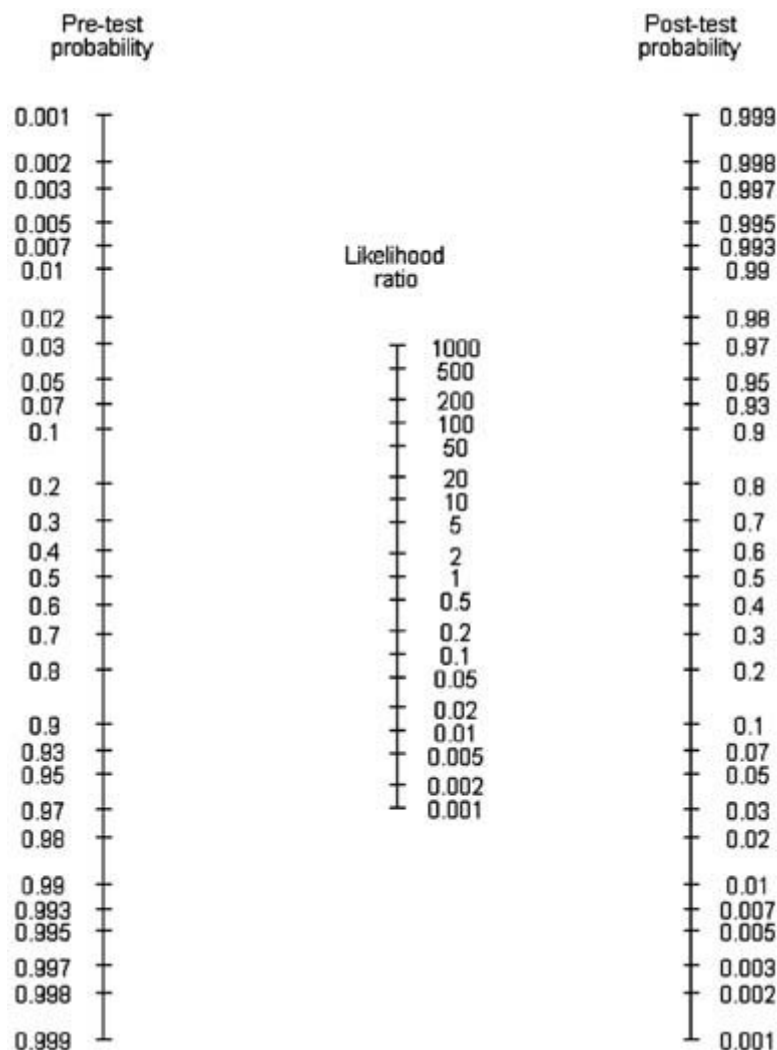
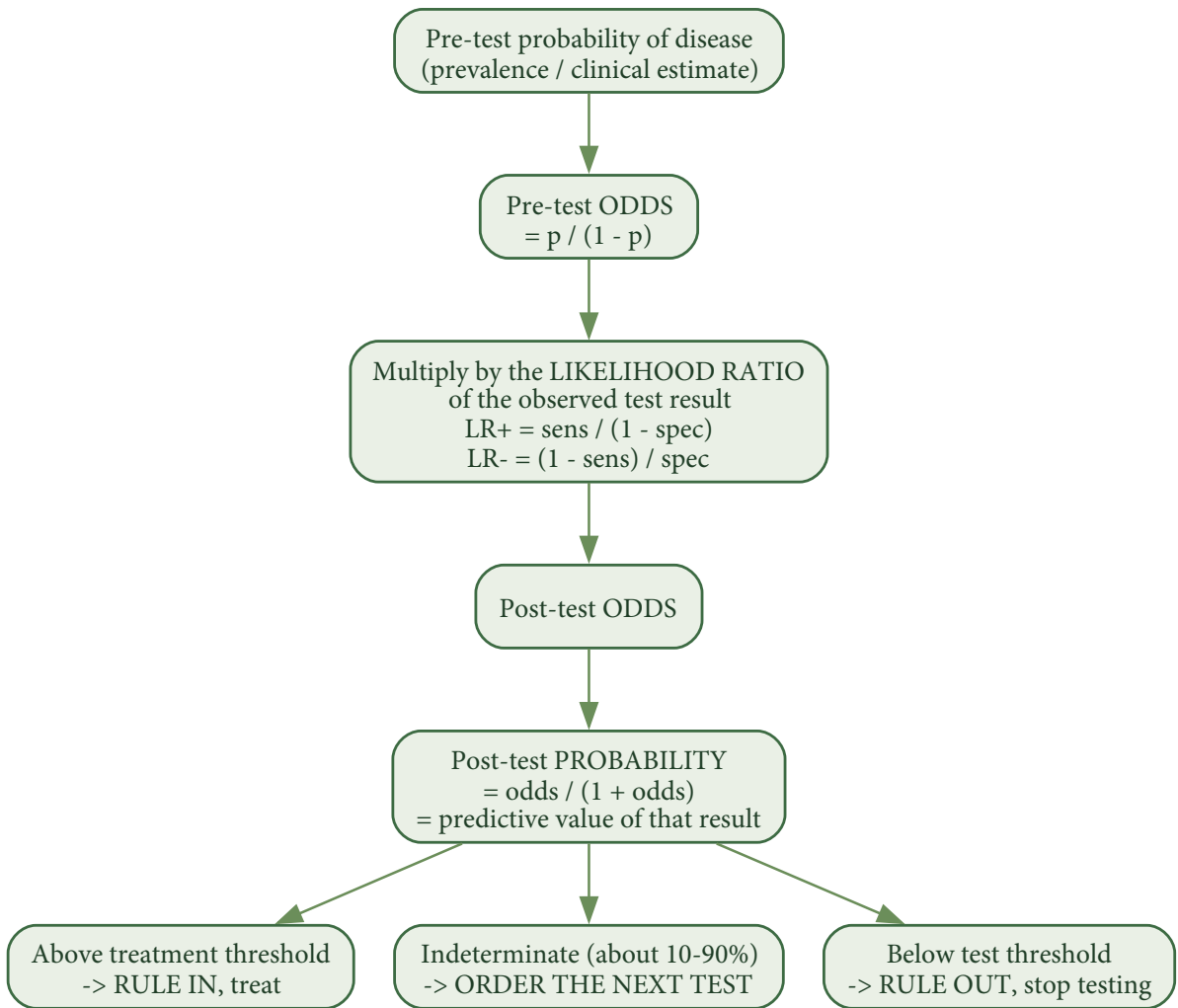
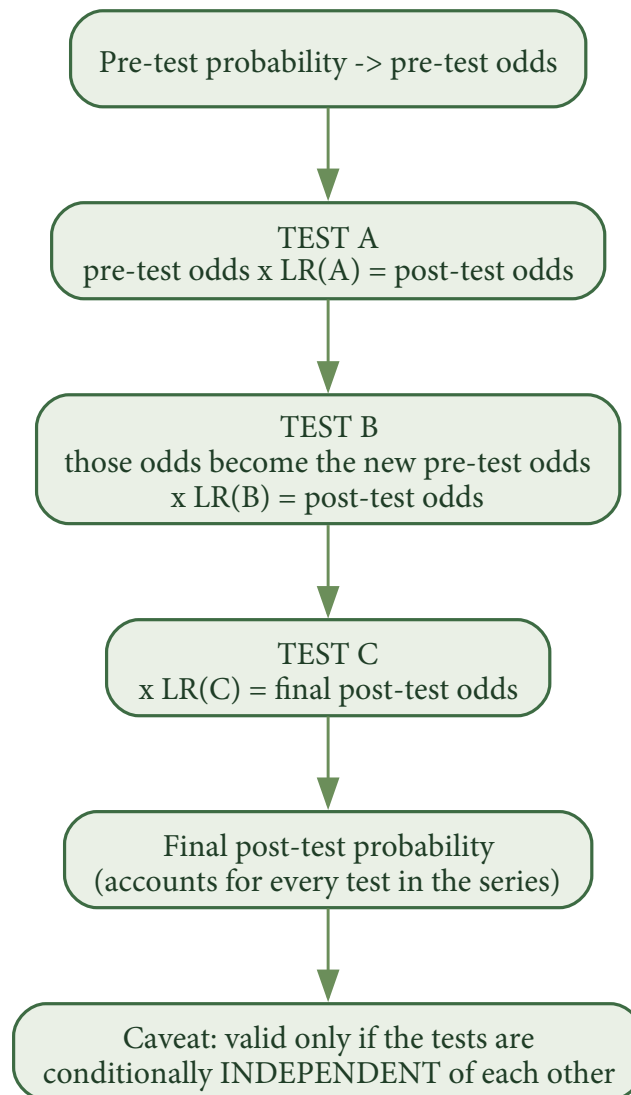


Fig 8.3 — Likelihood ratio (Fagan) nomogram (Copyright 1975 Massachusetts Medica (Handbook of Epidemiology (Ahrens & Pigeot), p.1174)



Using the likelihood ratio to revise disease probability in a single patient



Serial testing - the post-test odds of one test become the pre-test odds of the next

High-yield exam pointers

- ▶ **Both formulae first:** $LR+ = \text{sensitivity} \div (1 - \text{specificity})$; $LR- = (1 - \text{sensitivity}) \div \text{specificity}$.
- ▶ **Bayes' working line:** $\text{pre-test odds} \times LR = \text{post-test odds}$; $\text{odds} = p/(1 - p)$; $p = \text{odds}/(1 + \text{odds})$.
- ▶ **Bands:** $LR+ > 10$ or $LR- < 0.1 \rightarrow$ large, often conclusive; 5-10 and 0.1-0.2 $\rightarrow$ moderate; 2-5 and 0.2-0.5 $\rightarrow$ small; $LR = 1 \rightarrow$ useless.
- ▶ **McGee's multiples of 15:** LR 2, 5, 10 raise probability by about 15 %, 30 %, 45 %; LR 0.5, 0.2, 0.1 lower it by the same amounts.
- ▶ **Mnemonics:** *SpPin* (high Specificity, Positive rules in) and *SnNout* (high sensitivity, Negative rules out).
- ▶ **The discriminator:** predictive value changes with prevalence; sensitivity, specificity and hence the LR do not — so the LR is quoted in the literature and the predictive value computed locally.
- ▶ **Name the nomogram:** *Fagan* (1975); diagnostic odds ratio = $LR+ \div LR-$.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not written in the exam: Park's Textbook 28e covers sensitivity, specificity, predictive value and yield under "Screening for Disease" but does **not** treat the likelihood ratio, so the likelihood-ratio content above is grounded in the clinical-epidemiology texts of the corpus, with Park used only for the 2×2 table, sensitivity, specificity and the prevalence-dependence of predictive value.

SOURCE & COVERAGE

Fletcher — Clinical Epidemiology: The Essentials, Ch 8 (Diagnosis), pp 126-147; Sackett, Haynes, Guyatt & Tugwell — Clinical Epidemiology: How to Do Clinical Practice Research, Evaluating Diagnostic Tests, pp 297-305; Handbook of Epidemiology (Ahrens & Pigeot), Ch 29 (Clinical Epidemiology), pp 1172-1177; Park's Textbook of Preventive and Social Medicine 28e, Ch 4 (Screening for Disease), pp 166-168; Gordis Epidemiology 7e, Ch 5 (Assessing the Validity and Reliability of Diagnostic and Screening Tests), pp 118-123. Named within these: *Fagan* nomogram (1975); *McGee*, Simplifying likelihood ratios (2002); *Hill* troponin I study (2003); *CARE-COAD1* study (2000).



WHO STEPwise approach to NCD risk factor surveillance (STEPS)

Asked in: Paper II 10-Jul-2023 — RGUHS MD Community Medicine

Definition — The WHO STEPwise approach to Surveillance (STEPS) is a standardised survey methodology developed by the World Health Organization to help countries establish a **non-communicable disease (NCD) risk factor surveillance system**. It gathers a *hierarchy of information* through standardised instruments and protocols so that findings are comparable over time and across locations, and prescribes **three sequential steps — questionnaire, physical measurements and biochemical measurements** (Park 28e, Ch 6, pp 436-437; Jugal Kishore, p 574). Surveys are run at country or sub-national level, allowing flexibility and integration at each step.

Rationale and aims of NCD risk factor surveillance

Surveillance is the systematic and continuous collection, analysis and interpretation of data, closely integrated with timely dissemination to those who must act; it is distinguished from **monitoring**, which is intermittent or episodic (Park 28e, Ch 2, p 59). NCD mortality reflects exposure acquired years earlier — the long latent period of chronic disease (Park 28e, Ch 6, p 436) — so waiting to count cases is too late; the object of surveillance must be the *risk factor* itself.

#	Aim of NCD risk factor surveillance
1	Monitor trends of important NCD risk factors in the community over a period of time

#	Aim of NCD risk factor surveillance
2	Evolve strategies for intervention on these risk factors so as to reduce the burden of disease due to NCDs
3	Strengthen NCD surveillance at district level
4	Integrate NCD risk factor surveillance with the Integrated Disease Surveillance Programme (IDSP)

(Jugal Kishore, p 491)

The risk factors chosen for STEPS are those that are **measurable and modifiable**, and for which the available evidence indicates that reduction is feasible and would have a major impact on disease outcomes (Jugal Kishore, p 574). The four behavioural risk factors targeted globally are **tobacco use, unhealthy diet, physical inactivity and harmful use of alcohol** (Park 28e, Ch 6, p 437).

The three STEPS and the grading within each step

Step	Method	What is collected	Field requirement
STEP 1	Information by interview alone	Demographic and socio-economic data; health status and history; tobacco use; alcohol use; dietary habits; physical activity pattern	Trained interviewer; standard questionnaire
STEP 2	STEP 1 plus physical measurements	Height, weight (for body mass index), waist circumference, pulse rate, blood pressure	Calibrated stadiometer, weighing scale, tape, sphygmomanometer
STEP 3	STEPS 1 and 2 plus biochemical measurements	Blood collection for blood glucose and blood (total) cholesterol	Venepuncture, laboratory or point-of-care assay
At every step, items are graded	Core → expanded → optional	Core is <i>mandatory</i> for all users and forms the minimum comparable data set (demographic data, health status and health behaviours, yielding the socio-economic and metabolic risk factor estimates); expanded items add detail by choice; optional modules are country-specific	Grading decides questionnaire length, field time and cost

(Jugal Kishore, p 574 and p 491; Park 28e, Ch 6, pp 436-437)

Risk factors measured and the cut-offs applied in India

Risk factor	Step	Definition / cut-off used
Current tobacco use	1	Self-reported current use; the global target refers to persons aged 15 years and over (Park 28e, Ch 6, p 437)
Harmful use of alcohol	1	Self-reported quantity, frequency and pattern of drinking

Risk factor	Step	Definition / cut-off used
Insufficient physical activity	1	Below the recommended 150 minutes of moderate or vigorous aerobic activity per week, spread over at least 3 days (WHO HEARTS-D guidance, p 14)
Unhealthy diet — salt	1	Recommended intake is less than 5 g of salt per person per day (WHO SHAKE package, p 23); adult sodium chloride requirement is about 5 g/day and intake above 10 g/day has a definite tendency to raise blood pressure (Park 28e, Ch 11, p 763)
Unhealthy diet — fruit and vegetable	1	Servings consumed on a typical day
Overweight / obesity	2	Body mass index = weight (kg) / height (m) ² ; Indian cut-offs — 18.5–22.9 normal, 23–24.9 overweight, ≥25 obesity (NCD Manual, p 16)
Central obesity	2	Central obesity when waist circumference is ≥90 cm in men or ≥80 cm in women (desirable is <i>below</i> these values); waist-hip ratio ≥0.95 in men or ≥0.85 in women (normal is below these) (NCD Manual, p 16)
Raised blood pressure	2	Systolic ≥140 mmHg or diastolic ≥90 mmHg or currently taking antihypertensive medication (Park 28e, Ch 6, p 446). <i>JNC VII</i> grading: pre-hypertension 120–139/80–89; stage 1 140–159/90–99; stage 2 ≥160/≥100 (NCD Manual, p 29)
Raised blood glucose	3	Fasting venous plasma glucose ≥126 mg/dl , or post-prandial (about 2 hours after a main meal) ≥200 mg/dl , on two separate occasions (NCD Manual, p 11)
Blood cholesterol	3	Total cholesterol is measured ; the corpus gives no STEPS survey cut-off for "raised cholesterol". It supplies only a <i>treatment</i> threshold — a person without established cardiovascular disease whose total cholesterol is ≥320 mg/dl (or low-density lipoprotein cholesterol ≥240 mg/dl, or total/high-density lipoprotein ratio >8) belongs to the high-risk category without any risk charting, and needs intensive lifestyle intervention and appropriate drug therapy (NCD Manual, p 10)

Conduct of a STEPS survey

- **Define scope and objectives** — decide whether the survey is national or sub-national, and which risk factor indicators must be reported.
- **Adapt the instrument** — take all core items, then add expanded and optional modules according to local requirement and available resources.
- **Sample** — draw a population-based sample of adults in the defined age band; the Indian rounds have reported on ages **15–64 years** (Jugal Kishore, p 575).
- **Standardise** — train interviewers, calibrate instruments and follow the standard protocol; this is what makes results comparable over time and across locations.
- **Collect data sequentially** — Step 1, then Step 2, then Step 3 on the same respondent at the same site; a country short of resources may legitimately stop at Step 1 or Step 2 and still report core indicators.
- **Analyse** — age-standardised prevalence of each risk factor, disaggregated by sex, age group and urban/rural residence.

- **Disseminate and act** — fact sheet and report to programme managers and policy makers; the final phase of the surveillance chain is the application of information to prevention and control (Park 28e, Ch 2, p 59).
- **Repeat the round** to generate trends against the agreed targets.

Use of STEPS data — global targets and STEPS in India

WHO's comprehensive **global monitoring framework** for the prevention and control of NCDs, endorsed by the World Health Assembly in May 2013, carries **25 indicators and 9 voluntary global targets** under the **Global Action Plan for the Prevention and Control of NCDs 2013–2020, extended to 2030**; India is committed to its implementation (Jugal Kishore, p 597; Park 28e, Ch 6, p 437 and Ch 7, p 565). Most of these are risk factor targets that only a STEPS-type survey can measure. Sustainable Development Goal **target 3.4** — a one-third reduction in premature mortality from NCDs by 2030 — extends the global NCD mortality target, and STEPS supplies its risk factor baselines (Park 28e, Ch 6, p 437).

#	Voluntary global target (as of August 2023)	Measured by
1	One-third relative reduction in premature mortality from cardiovascular disease, cancer, diabetes and chronic respiratory disease	Mortality registration (not STEPS)
2	At least 20% relative reduction in the harmful use of alcohol	STEP 1
3	15% relative reduction in the prevalence of insufficient physical activity	STEP 1
4	30% relative reduction in mean population intake of salt/sodium	STEP 1 dietary questions; mean population intake is best estimated from urinary sodium excretion (WHO SHAKE package, p 24)
5	30% relative reduction in prevalence of current tobacco use in persons aged 15+ years	STEP 1
6	25% relative reduction in prevalence of raised blood pressure	STEP 2
7	Halt the rise of diabetes and obesity	STEPS 2 and 3
8	At least 50% of eligible people receive drug therapy and counselling (including glycaemic control) to prevent heart attacks and strokes	Health-facility data
9	80% availability of affordable basic technology and essential medicines, including generics, required to treat major NCDs in public and private facilities	Facility survey

(Park 28e, Ch 6, p 437)

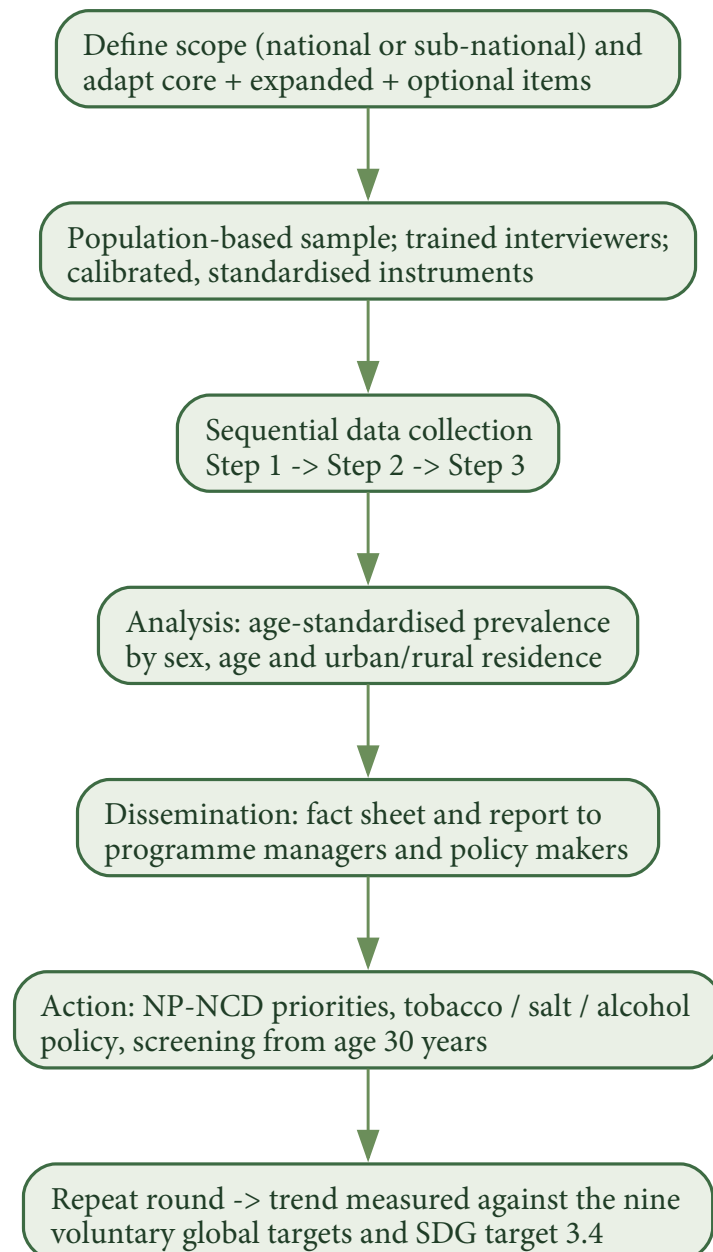
STEPS in India — element	Detail
Rounds conducted	Survey conducted from April 2003 ; in 6 states in 2005 and again in 7 states in 2007 (Park 28e, Ch 6, p 437)
Illustrative finding	WHO STEPwise surveillance in 5 Indian states — only 50% of those aged 15–64 years consumed vegetables daily and 60–80% led a sedentary lifestyle (Jugal Kishore, p 575)
System integration	NCD risk factor surveillance is to be integrated with IDSP , whose four aims of NCD risk factor surveillance are listed above (Jugal Kishore, p 491); each medical college is integrated into the surveillance system , providing reference laboratories, quality assurance, training, epidemic investigation and non-communicable disease surveillance (Jugal Kishore, p 490)
Programme link	NP-NCD (formerly NPCDCS) — its second immediate objective is " <i>surveillance of NCDs and their risk factors in the population</i> " (Jugal Kishore, p 595); NCD cells at central, state and district levels implement and monitor it (Park 28e, Ch 7, p 565)
Service-level counterpart	Opportunistic screening of persons above 30 years for blood glucose and blood pressure (Park 28e, Ch 7, p 564); blood sugar and blood pressure are checked for those 30 years and above and all pregnant women (Park 28e, Ch 7, p 565)
Complementary national data	NFHS-5 (2019-21) — hypertension in 21% of women and 24% of men aged 15 years and over (Park 28e, Ch 6, p 446)

Strengths and limitations

Strengths	Limitations
Standardised instruments and protocols give comparability over time and across locations	Cross-sectional — yields prevalence only; no incidence and no causal inference
Resource-graded and flexible — a country may stop at Step 1 or Step 2 and still report core indicators	Step 1 is self-reported , so recall bias and social-desirability bias affect tobacco, alcohol and diet data
Targets factors that are measurable and modifiable , where reduction is feasible and high-impact	Step 3 needs venepuncture, fasting, laboratory support and cold chain — often omitted, leaving the metabolic picture incomplete
Sequential integration — each step builds on the previous one in the same respondent	A single blood pressure or glucose reading may misclassify, since the standard definitions require two occasions
Directly feeds the nine voluntary global targets and SDG target 3.4	Working men and urban migrants are under-represented when non-response is high
Serves as a host platform onto which other modules can be added — population salt-intake, knowledge and dietary-source data are recommended to be integrated into a planned STEPS survey (WHO SHAKE package, p 23)	Covers risk factors only — must be combined with mortality registration and disease registries for the full NCD picture

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **The STEPS pyramid** — three horizontal bands, widest at the base. Labels to write: base = *STEP 1 — Questionnaire (interview alone): demographic, health status, tobacco, alcohol, diet, physical activity*; middle band = *STEP 2 — Physical measurements: height, weight, waist circumference, pulse, blood pressure*; apex = *STEP 3 — Biochemical measurements: blood glucose, blood cholesterol*. Along the right margin write *Core → Expanded → Optional at every step*; along the left margin write *increasing cost, complexity and information yield*.



NCD risk-factor surveillance cycle, from STEPS survey to programme action

High-yield exam pointers

- STEPS = STEPwise approach to Surveillance; source document is WHO (2011), STEPS: A Framework for Surveillance of NCD Risk Factors (Park 28e, Ch 6, reference 18).

- ▶ Mnemonic for the three steps — "Ask, Measure, Test": Step 1 questionnaire, Step 2 physical measurements, Step 3 biochemical measurements; every step graded **core** → **expanded** → **optional**.
- ▶ Four behavioural risk factors: **tobacco, unhealthy diet, physical inactivity, harmful alcohol**; metabolic factors measured in Steps 2–3: **raised blood pressure, raised blood glucose, blood cholesterol, overweight/obesity**.
- ▶ Indian body mass index cut-offs: **≥23 overweight, ≥25 obesity**; **waist circumference ≥90 cm in men and ≥80 cm in women defines central obesity** (waist–hip ratio ≥0.95 men, ≥0.85 women).
- ▶ Hypertension = **≥140/≥90 mmHg or on antihypertensive medication**; diabetes = **fasting ≥126 mg/dl or post-prandial ≥200 mg/dl on two separate occasions**; **total cholesterol ≥320 mg/dl = high-risk category without risk charting**.
- ▶ Global monitoring framework = **25 indicators + 9 voluntary global targets** (World Health Assembly, May 2013); headline targets **30% tobacco, 30% salt, 25% raised blood pressure, 20% alcohol, 15% physical inactivity, halt the rise in diabetes and obesity**.
- ▶ India: STEPS from **April 2003**, 6 states in 2005 and 7 states in 2007; risk factor surveillance to be **integrated with IDSP** and delivered through the **NCD cells** of NP-NCD.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Diseases and Stroke (NPCDCS) is now the National Programme for Prevention and Control of Non-Communicable Diseases (NP-NCD), and Park 28e still prints the older name in places. The Indian waist and waist–hip-ratio cut-offs used in the national NCD programme are deliberately lower than the international World Health Organization cut-offs printed in Park 28e, Ch 6, p 476, where increased metabolic risk is placed at a waist of about 102 cm in men and 88 cm in women and a high waist–hip ratio at above 0.9 in men and 0.85 in women. Coverage note: the retrieved corpus describes the three steps, the core/expanded/optional grading, the aims and the Indian rounds, but it does not give a STEPS survey cut-off defining "raised total cholesterol" (only the ≥320 mg/dl treatment threshold quoted above), nor the standard STEPS adult age band of 18–69 years, nor a recommended repeat interval, so none of these is quoted here; the Indian rounds are reported for ages 15–64 years.

Recent advances [RA]

Current as of 2026-08-19 · depth: elaborate · sources: PubMed, WHO, MoHFW, ICMR, FSSAI, NFHS-5

Programme and policy updates (India)

- ▶ NPCDCS was renamed the **National Programme for Prevention and Control of Non-Communicable Diseases (NP-NCD)** in 2023, with screening delivered through sub-centres and primary health centres upgraded to Health and Wellness Centres since 2018. Repeat representative surveys of adults aged 18–69 years in nine India Hypertension Control Initiative districts across five states show, between 2018–19 and 2023–24, a rise in care-seeking at government facilities from 20% to 32% (aPR 1.54; 95% CI 1.4–1.7), blood pressure control from 37% to 48% in government facilities, always-available antihypertensive stock from 72% to 81%,

and a fall in the proportion paying for medicines from 47% to 9% — a working demonstration that a repeated STEPS-type survey, not only facility data, is what measures programme effect [PMID 41146104].

- **Ayushman Bharat Health and Wellness Centres were renamed Ayushman Arogya Mandir in November 2023**, with the tagline *Arogyam Paramam Dhanam*, an AYUSH-informed holistic-wellness brief covering physical, mental and social wellness, lifestyle counselling, yoga and elderly care, and a renaming outlay of about ₹48 crore with photographic verification uploaded to the programme portal [PMID 42257165].

Current national data

- **The National NCD Monitoring Survey (NNMS), 2017-18, is India's first comprehensive national round** and the operational successor to the early state STEPS rounds: 10,659 adults aged 18–69 years from 600 primary sampling units, with instruments adapted from WHO STEPS, the IDSP NCD risk factor survey and the Global Adult Tobacco Survey. Prevalences — tobacco use 32.8% (95% CI 30.8–35.0), alcohol use 15.9% (14.2–17.7), insufficient physical activity 41.3% (39.4–43.3), fewer than five servings of fruit or vegetables daily 98.4% (97.8–98.8), raised blood pressure 28.5% (27.0–30.1) and raised blood glucose 9.3% (8.3–10.5); 12.8% of those aged 40–69 years had a ten-year cardiovascular risk of $\geq 30\%$ or established disease [PMID 33651825].
- **Mean population salt intake in India is 8.0 g/day** (men 8.9, women 7.1), measured for the first time nationally from **spot urinary sodium** within NNMS — well above the WHO recommendation of under 5 g/day, and fewer than one-third of adults were aware that high salt intake harms health. This is the measurement that the 30% salt-reduction global target actually requires [PMID 37741845].
- **ICMR-INDIAB-17 (2023)** — 113,043 adults aged ≥ 20 years across 31 states and union territories, surveyed 2008–2020: diabetes 11.4% (95% CI 10.2–12.5), prediabetes 15.3% (13.9–16.6), hypertension 35.5% (33.8–37.3), generalised obesity 28.6% (26.9–30.3), abdominal obesity 39.5% (37.7–41.4) and dyslipidaemia 81.2% (77.9–84.5); quote this alongside, not instead of, the NFHS-5 hypertension figures, because the age band and cut-offs differ [PMID 37301218].

Elimination, surveillance and disease status

- **Industrially produced trans fat is the first NCD risk factor with a global elimination programme** — WHO REPLACE, with a Director-General target of elimination by 2023; 43 countries have adopted best-practice regulation, covering an additional 3.2 billion people and about 66% of annual trans-fat-attributable deaths. India's route: FSSAI capped trans fat at 10% by weight in 2013 and 5% in 2015, ran the *India@75 — Freedom from Trans-fats* campaign from 2018, gazetted a 3% limit effective 1 January 2021 and 2% effective 1 January 2022, and in February 2021 extended the 2% limit to **all food products** [PMID 39111974].
- **District-level NCD surveillance now exists for the whole country** — hypertension and diabetes were modelled for adults aged 15–49 years in all **720 districts** using the 2016 (NFHS-4) and 2021 (NFHS-5) rounds, showing that prevalence rose in hundreds of districts and that districts starting low gained fastest; this is the sub-national granularity the STEPS aim of "strengthening surveillance at district level" was written for [PMID 41133252].

Guideline and legislative updates

- The WHO non-laboratory cardiovascular risk chart lets Step 1 and Step 2 data alone yield a risk estimate, without the venepuncture of Step 3 — applied to 4,480 NNMS adults aged 40–69 years it classified 84.9% as very low-to-low, 14.4% moderate and 0.7% high-to-very-high ten-year risk, with unemployment (aOR 5.12; 95% CI 3.63–7.24) and raised blood glucose (aOR 1.81; 1.39–2.34) predicting risk $\geq 10\%$. This is the practical answer to the resource constraint that makes countries stop at Step 2 [PMID 39382425].

Global and WHO developments

- WHO's updated NCD position (September 2025) — at least 43 million NCD deaths in 2021, about 75% of non-pandemic deaths worldwide; 18 million occurred before age 70, of which 82% were in low- and middle-income countries; cardiovascular disease alone accounted for at least 19 million. The Global Action Plan was extended to **2030** and now carries a 2023–2030 implementation roadmap, so the "2013–2020" framing in older texts is superseded [WHO 2025-Sep].
- **India is one of the few large countries where NCD mortality worsened.** Benchmarking WHO Global Health Estimates for 185 countries, the probability of dying from an NCD before age 80 fell between 2010 and 2019 in 152 countries for women and 147 for men — but rose for **both sexes in India**, and the decadal change for Indian women deteriorated against 2001–2010. On present trajectory SDG target 3.4 is off track [PMID 40945529].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 2 — surveillance and monitoring; Ch 6 — epidemiology of chronic non-communicable diseases, STEPS methodology, voluntary global targets, obesity and waist measurement, hypertension definition and NFHS-5 prevalence; Ch 7 — NPCDCS/NP-NCD, NCD cells and opportunistic screening; Ch 11 — sodium requirement); Jugal Kishore — National Health Programmes of India (STEPwise approach, IDSP structure and the role of medical colleges, aims of NCD risk factor surveillance, NPCDCS objectives, WHO Global Action Plan 2013-2020); NCD Combined Manual for Medical Officers, Ministry of Health and Family Welfare (Indian body mass index, waist and waist-hip-ratio values, JNC VII blood pressure grading, diabetes diagnostic criteria, WHO/ISH risk assessment and the ≥ 320 mg/dl very-high-risk threshold); WHO HEARTS-D non-communicable disease management guidance (physical activity recommendation); WHO SHAKE Technical Package for Salt Reduction (STEPS as a host survey for salt-intake monitoring; urinary sodium); WHO (2011), STEPS: A Framework for Surveillance of NCD Risk Factors; WHO Global Action Plan for the Prevention and Control of NCDs 2013-2020, extended to 2030; National Family Health Survey-5, 2019-21.

Recent-advances sources: 41146104 — Changing patterns in health-seeking behaviour and prescription practices for treatment of hypertension in nine districts: insights from India Hypertension Control Initiative, 2018-19 and 2023-24, BMC Public Health 2025, doi:[10.1186/s12889-025-24898-0](https://doi.org/10.1186/s12889-025-24898-0); 42257165 — Renaming of Ayushman Bharat Health and Wellness centres as Ayushman Arogya Mandirs, Journal of Family Medicine and Primary Care 2026, doi:[10.4103/jfmpc.jfmpc.1966.24](https://doi.org/10.4103/jfmpc.jfmpc.1966.24); 33651825 — National noncommunicable disease monitoring survey (NNMS) in India: estimating risk factor prevalence in adult population, PLoS One 2021, doi:[10.1371/journal.pone.0246712](https://doi.org/10.1371/journal.pone.0246712); 37741845 — Awareness, behavior, and determinants of dietary salt intake in adults: results from the National NCD Monitoring Survey, India, Scientific Reports 2023, doi:[10.1038/s41598-023-42694-x](https://doi.org/10.1038/s41598-023-42694-x); 37301218 — Metabolic non-communicable disease health report of India: the ICMR-INDIAB national cross-sectional study (ICMR-INDIAB-17), Lancet Diabetes & Endocrinology 2023, doi:[10.1016/S2213-8587\(23\)00119-5](https://doi.org/10.1016/S2213-8587(23)00119-5) (abstract-only); 39111974 — Ending trans fat: the first-ever global elimination program for a noncommunicable disease risk factor, Journal of the American College of Cardiology 2024, doi:[10.1016/j.jacc.2024.04.067](https://doi.org/10.1016/j.jacc.2024.04.067); 41133252 — Measuring changes in prevalence of hypertension and diabetes across 720 districts in India using cross-sectional data from 2016 to 2021, BMJ Public Health 2025, doi:[10.1136/bmjph-2025-002653](https://doi.org/10.1136/bmjph-2025-002653); 39382425 — Ten-year risk assessment for cardiovascular disease & associated factors among adult Indians (aged 40-69 yr): insights from the National Noncommunicable Disease Monitoring Survey, Indian Journal of Medical Research 2024, doi:[10.25259/ijmr.1748.23](https://doi.org/10.25259/ijmr.1748.23); 40945529 — Benchmarking progress in non-communicable diseases: a global analysis of cause-specific mortality from 2001 to 2019, Lancet 2025, doi:[10.1016/S0140-6736\(25\)01388-1](https://doi.org/10.1016/S0140-6736(25)01388-1). Plus WHO noncommunicable diseases fact sheet, updated September 2025; FSSAI trans-fat regulation gazetted December 2020 (3% from 1 January 2021, 2% from 1 January 2022) and extended to all food products February 2021.

♦ ♦ ♦

Biostatistics



Cluster sampling technique

Asked in: Paper I 14-Jul-2020 — RGUHS MD Community Medicine

Definition — Cluster sampling is a probability sampling method in which the **sampling unit is a group (cluster) of individuals, not an individual**, and the sampling frame is a list of these clusters rather than a list of persons (Abramson, *Research Methods in Community Medicine*, Ch 8, p 80). A cluster is a randomly selected naturally occurring group — a village, ward, block, slum, school, factory, household or family — and clusters may also be created artificially (Mahajan, *Methods in Biostatistics*, Ch 7 Sampling, pp 137–138; Sunder Rao & Richard, pp 79–80). Its role in sampling is to make **large-area community surveys feasible when no list of individuals exists**, at the price of a higher standard error than a simple random sample of the same size (Mahajan, p 138; Sunder Lal, Ch 6, p 252).

Place of cluster sampling among the sampling methods

Method	Sampling unit	Frame needed	Distinguishing feature	Relative sampling error
Simple random	Individual	Complete list of every individual	Each unit has an equal chance; lottery or random-number table (Sunder Rao, pp 77–78)	Reference standard (lowest for a given n)
Systematic random	Individual	Serial list / numbered houses	Random start, then every <i>k</i> th unit; each unit has the same chance but far fewer possible samples (Park 28e, Ch 21, p 1006)	≈ simple random
Stratified random	Individual within stratum	Separate frame for each stratum	Population first divided into homogeneous strata, then sampled independently; guarantees representation of minority subgroups (Sunder Lal, Ch 6, p 250)	Lower than simple random
Cluster	Group	Only a list of clusters with their populations	Groups selected, then all or a sub-sample of members studied; probability proportional to size (PPS) (Sunder Lal, Ch 6, p 250)	Higher than simple random
Multistage	Successively smaller units	Frame built only for the units actually selected	Cluster sampling extended — state → district → taluka → village → household (Mahajan, p 137; Sunder Rao, pp 80–81)	Higher, accumulates at each stage

The five recognised methods of drawing an unbiased (probability) sample are simple random, stratified random, systematic, cluster and multistage sampling; only with a probability method can the chance error be calculated and the findings safely extended from sample to population (Sunder Rao, pp 77–78, 80–81).

Types of cluster sample

Type	Procedure	Typical use
One-stage (whole-cluster)	Clusters selected at random; every member of each chosen cluster is surveyed	Small clusters, e.g. all persons in 30 selected villages (Mahajan, p 138)
Two-stage	Clusters (primary units) selected first; a sub-sample of secondary units is then drawn within each — the process is called sub-sampling	The standard community survey design; households selected within selected villages (Sunder Rao, pp 80–81; Sunder Lal, Ch 6, p 251)
Multistage cluster	Sub-sampling continued through progressively lower levels	National surveys over a whole country (Sunder Rao, p 81)
Stratified cluster	Population first stratified, then clusters selected within strata	<i>Pathfinder</i> surveys, designed to include the important population subgroups likely to differ in disease (Sunder Lal, Ch 6, p 252)

Clusters are almost always selected with **probability proportional to size (PPS)**, so that a village of twice the population has twice the chance of selection and, if its population exceeds the sampling interval, may contribute two or more clusters; this makes the final sample self-weighted (Sunder Lal, Ch 6, pp 250–251; Sunder Rao, p 80).

The WHO 30 × 7 cluster survey — the classical technique

Steps in selecting the 30 clusters (Mahajan, pp 138–140; Sunder Rao, p 80):

- ▶ List every city, town, village and ward falling in the target area, with its population.
- ▶ Cumulate the populations down the list.
- ▶ **Sampling interval = total cumulative population ÷ 30.**
- ▶ Choose a **random number less than or equal to the sampling interval**, with the same number of digits; the community against which it falls in the cumulative column contains **cluster 1**.
- ▶ **Cluster 2 = random number + sampling interval**; cluster 3 = cluster 2 + interval, and so on, until the **30th cluster = 29th cluster + sampling interval**.
- ▶ Within each selected community, choose the **starting household at random** — go to a central point, spin a pen or bottle to fix a direction, count the households from the centre to the edge in that direction and pick one at random (the *random walk* method). Thereafter take the nearest unvisited household, in one direction, until the required number of eligible subjects is found (Abramson, Ch 31, p 307; Sunder Lal, Ch 6, p 251).
- ▶ Define the **age group and the item to be studied before starting** — e.g. children aged 12–23 months and whether they are fully immunised (Mahajan, p 138).

Sample size convention — the original Expanded Programme on Immunization (EPI) recommendation for surveys of vaccination status was **30 clusters × 7 children = 210 children** aged 12–23 months; nutritional surveys use **30 clusters × 30 individuals** (Mahajan, p 138; Abramson, Ch 31, p 306; Sunder Rao, pp 79–80). This size delivers an estimate valid at **95 per cent confidence limits with about ±10 per cent variation** (Mahajan, pp 137–138). Thirty is regarded as the practical minimum — the more clusters, the more valid the result (Abramson, Ch 31, p 306).

Statistical consequence of clustering — the design effect

Concept	Statement / formula	Source
Why precision falls	People within a natural cluster are more alike than people in the population at large (a city block tends to have one socio-economic class), so each extra person in a cluster adds less new information	Browner, <i>Designing Clinical Research</i> 5e, Ch 3, pp 69–70
Intraclass correlation, ρ or ICC	Between-cluster variance $\div$ (between-cluster + within-cluster variance); interpretable as the proportion of total variation in the response explained by between-cluster variation	Machin, Ch 6, pp 58–59; <i>Methods and Applications of Statistics in Clinical Trials</i> , Ch 18, pp 220–221
Design effect (variance inflation factor)	$DE = 1 + (M - 1) \times ICC$, where M is the average cluster size	Cochrane Handbook, Ch 16, pp 521–522
Effective sample size	Actual sample size $\div$ design effect	Cochrane Handbook, Ch 16, pp 521–522
Sample size for a cluster design	Sample size calculated for individual sampling is multiplied by the design effect	Machin, Ch 6, pp 58–59
Typical magnitude of ICC	Values below 0.05 are usual; larger clusters carry smaller ICCs, but even a small ICC widens confidence intervals appreciably when clusters are large	Cochrane Handbook, Ch 16, p 521
Worked example	ICC 0.02, average cluster size 29.8 $\rightarrow DE = 1 + (29.8 - 1) \times 0.02 = 1.576$; 295 children become an effective 187	Cochrane Handbook, Ch 16, pp 521–522
Design rule	Other things equal, many small clusters beat few large clusters ; power responds far more to the number of clusters than to the number sampled per cluster	Abramson, Ch 8, p 81; <i>Statistics in Clinical Trials</i> , Ch 18, p 222

Advantages and limitations

Advantages	Limitations
Cuts the cost of preparing a sampling frame — only clusters, not individuals, need listing (Sunder Lal, Ch 6, p 251)	Sampling error (standard error) is higher than for a simple random sample of the same size (Sunder Lal, Ch 6, p 252; Mahajan, p 138)
Cuts the cost of travel between selected units, since subjects live close together (Sunder Lal, Ch 6, p 251; Abramson, Ch 31, p 306)	Cluster members are homogeneous (high intraclass correlation) , so precision cannot be estimated without special methods of analysis (Abramson, Ch 8, pp 80–81)
Simple field procedure usable by health workers with minimal technical support (Abramson, Ch 31, p 306)	The EPI second stage uses a random walk, not true random selection, so it falls short of a true probability sample (Abramson, Ch 31, p 306)

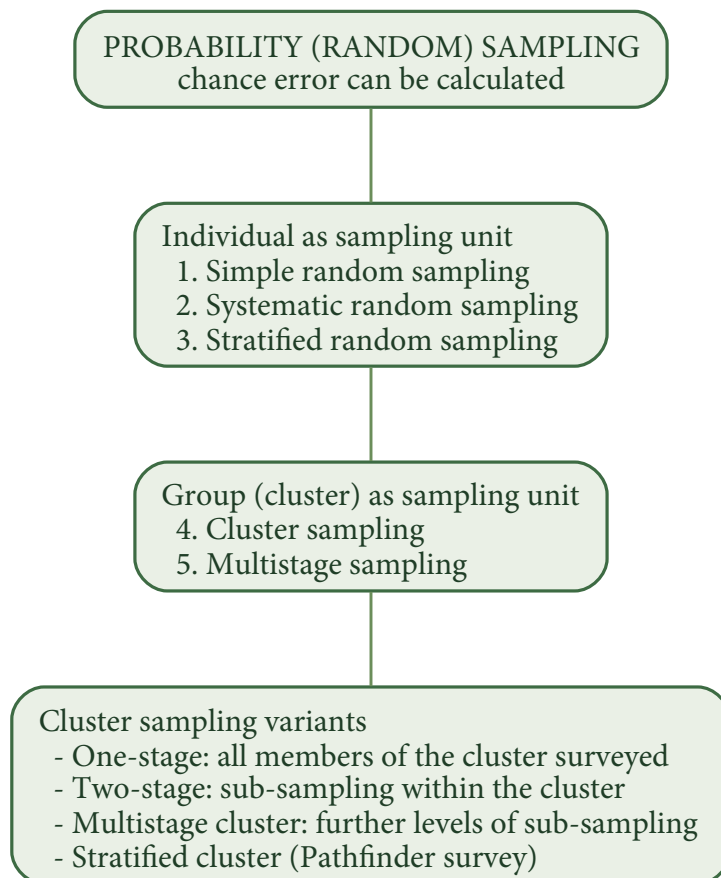
Advantages	Limitations
Rapid — suited to emergencies and to any situation demanding quick results (Abramson, Ch 31, p 306)	Sample size and analysis must be inflated/adjusted for the design effect , or confidence intervals are falsely narrow (Browner 5e, Ch 3, p 70; Cochrane, Ch 16, p 521)
Eliminates " packing " — neighbours moving into a selected house to receive survey benefits (Sunder Lal, Ch 6, p 251)	The random walk over-samples households near the centre; remedies include taking the fifth-nearest household or compact segment sampling (Abramson, Ch 31, pp 308–309)
Introduces flexibility and exploits existing administrative divisions, saving extra labour (Mahajan, p 137)	Requires reasonably accurate population estimates for each community for the PPS step (Abramson, Ch 31, p 306)

Role of cluster sampling in community medicine practice

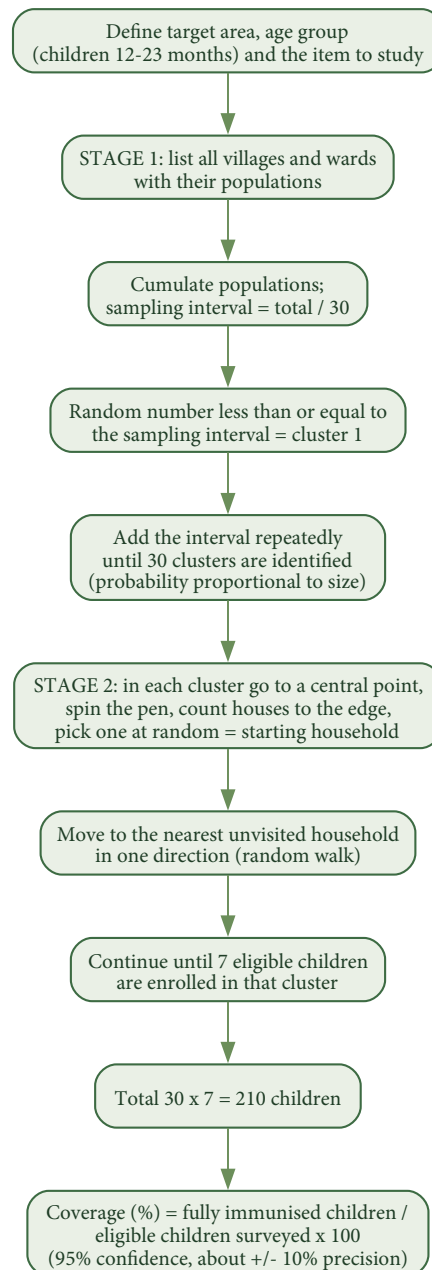
Application	How cluster sampling is used
Immunisation coverage evaluation	The WHO-approved module for evaluating coverage under EPI and the Universal Immunization Programme — 30 clusters × 7 children aged 12–23 months (Mahajan, p 138; Sunder Rao, pp 79–80)
National household surveys	NFHS-5 (2019–21) used a stratified two-stage design: the 2011 Census as frame, primary sampling units = villages (rural) and Census Enumeration Blocks (urban) chosen by PPS systematic sampling , then 22 households per cluster by systematic selection; 30,456 PSUs selected across 707 districts, fieldwork completed in 30,198 (NFHS-5 India Report, Ch 1, pp 38–39)
Programme monitoring and evaluation	Two-stage cluster sampling is widely used in diarrhoeal disease surveys, leprosy elimination monitoring surveys and evaluation of the national AIDS programme (Sunder Lal, Ch 6, p 251)
Rapid epidemiological assessment	Cluster surveys are indicated whenever resources are limited or rapid results are needed, especially in emergency and disaster settings (Abramson, Ch 31, p 306)
Prevalence surveys where no register exists	Blindness, specific-disease prevalence, service coverage and health-service needs surveys (Abramson, Ch 31, p 306)
Nutrition surveys	30 clusters × 30 individuals (Abramson, Ch 31, p 306)
Community-based interventional research	Cluster (community) randomised trials, where the community — not the individual — is allocated; analysis must use the intracluster correlation (<i>Statistics in Clinical Trials</i> , Ch 18, pp 220–222)
Large national disease surveys	Multistage design — random districts within states, then talukas, then villages, then houses; e.g. a hookworm survey taking 10 per cent of villages and every 10th house (Mahajan, p 137)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Diagram 1 (to sketch)** — *Sampling unit in four methods*: draw a rectangle "Population" filled with small dots; in panel (a) **Simple random** circle scattered individual dots; in panel (b) **Stratified** draw two horizontal bands labelled "Stratum A", "Stratum B" with dots circled in each; in panel (c) **Cluster** draw nine squares labelled "villages", shade three whole squares and label "all members surveyed"; in panel (d) **Two-stage cluster** shade the same three squares but circle only a few dots inside each, labelled "sub-sampling of households". Label the axis line beneath: "*sampling unit = individual → sampling unit = group*".
- **Diagram 2 (to sketch)** — *PPS cumulative-population line*: a vertical column of village names with a second column "cumulative population", a right-hand ruler marked "*sampling interval = total ÷ 30*", arrows from the random start and successive additions striking villages, with a bracket showing one large village struck twice = two clusters.



Classification of probability sampling methods, showing where cluster sampling sits



Two-stage 30 x 7 cluster survey for evaluating immunisation coverage

High-yield exam pointers

- ▶ **One-line definition to write first:** any method of sampling in which a **group** is taken as the sampling unit is cluster sampling (Sunder Rao, p 80).
- ▶ **Sampling interval = total cumulative population ÷ 30;** first cluster = a random number ≤ the interval; each subsequent cluster = previous + interval.
- ▶ **30 × 7 = 210 children aged 12–23 months** for immunisation coverage; **30 × 30** for nutrition surveys; precision ±10 per cent at 95 per cent confidence.
- ▶ **Design effect DE = 1 + (M – 1) × ICC;** effective sample size = n ÷ DE; multiply the calculated sample size by DE. ICC is usually < 0.05.

- ▶ **Dated Indian example:** NFHS-5 (2019–21) — stratified two-stage cluster sample, PSU = village / Census Enumeration Block selected by PPS from the 2011 Census frame, 22 households per cluster, 30,456 PSUs, 707 districts.
- ▶ **Single greatest drawback to state:** higher sampling error (higher standard error) than a simple random sample of equal size, because of intracluster homogeneity.
- ▶ **Design rule:** many small clusters are better than a few large ones; ≥ 30 clusters is the minimum.
- ▶ Cluster sampling $\neq$ stratified sampling — in **stratified** sampling you sample **from every** stratum and precision improves; in **cluster** sampling you sample **only some** clusters and precision worsens.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park's 28th edition covers only simple random, systematic and stratified random sampling in its Health Information and Basic Medical Statistics chapter, so the cluster-sampling content above is grounded on the biostatistics and research-methods texts of the corpus rather than on the spine.

SOURCE & COVERAGE

Mahajan's Methods in Biostatistics for Medical Students and Research Workers (Ch 7 — Sampling, pp 137–140); Sunder Rao & Richard — Introduction to Biostatistics and Research Methods (Sampling, pp 77–81); Abramson — Research Methods in Community Medicine (Ch 8 — Sampling, pp 80–81; Ch 31 — Rapid Epidemiological Methods, pp 306–309); Sunder Lal — Textbook of Community Medicine (Ch 6 — Basic Statistics and its Applications and Research Methodology, pp 250–252); Browner — Designing Clinical Research 5e (Ch 3 — Choosing the Study Participants, pp 69–70); Cochrane Handbook for Systematic Reviews of Interventions (Ch 16 — Special Topics in Statistics, pp 521–522); Machin et al. — Sample Size Tables for Clinical Studies (Ch 6 — Cluster Designs, pp 58–59); Methods and Applications of Statistics in Clinical Trials (Ch 18 — Cluster Randomization, pp 220–222); Park's Textbook of Preventive and Social Medicine 28e (Ch 21 — Health Information and Basic Medical Statistics, p 1006); National Family Health Survey (NFHS-5) India Report 2019–21 (Ch 1 — Introduction, Sample Design, pp 38–39).



Cluster sampling

Asked in: Paper I 29-Jul-2021 — RGUHS MD Community Medicine

Definition — Cluster sampling is a probability sampling method in which the sampling unit is a **group (cluster) of population members** rather than an individual: any method of sampling wherein a group is taken as the sampling unit is cluster sampling (Sunder Rao & Richard, pp 79-80). A random sample of clusters is drawn from a list of clusters, and all members — or a random sub-sample of members — within each selected cluster are studied (Park Plus, Ch 19, p 237). Clusters are usually **natural groups**: villages, wards, blocks, slums of a town, factories, workshops, schools, households or hostels; artificial clusters may also be constructed (Mahajan's Methods in Biostatistics, Ch 7, p 138; Sunder Rao & Richard, pp 79-80).

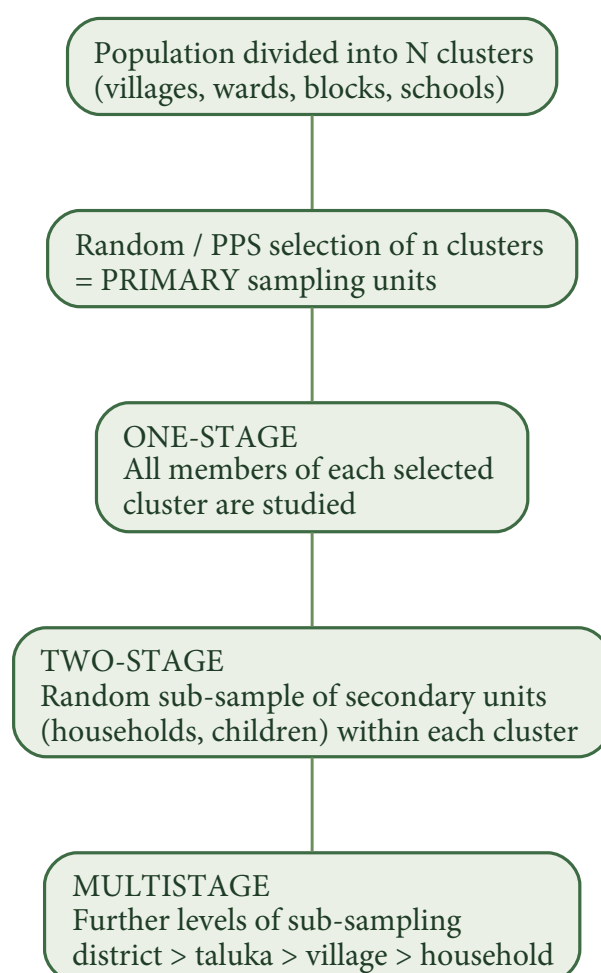
When cluster sampling is chosen

- ▶ When it is **difficult or costly to prepare a complete list (sampling frame) of every individual** in the population, so simple random sampling is not feasible — only a list of clusters is needed (Park Plus, Ch 19, p 237; Sunder Lal, Ch 6, pp 251-252).
- ▶ When the **population is widely dispersed geographically** and travel between scattered individuals would be prohibitive (Park Plus, Ch 19, p 237).
- ▶ When the population already exists in **natural administrative groupings** — villages for rural and wards for urban areas — whose population sizes are known from the census (Sunder Lal, Ch 6, p 250).
- ▶ When a **rapid, field-practicable estimate** of coverage or prevalence is needed for an area as a whole, accepting ± 10 per cent precision (Sunder Rao & Richard, p 80).
- ▶ When the intervention itself must be delivered to a group — a village, school or health facility — making the cluster the unit of allocation in a **cluster randomised trial** (Browner, Designing Clinical Research 5e, Ch 12, pp 389-390).

Types of cluster sampling

Type	What is done	Where used
One-stage (single-stage) cluster sampling	A simple random sample of n clusters is drawn from the list of N clusters; all elements in each selected cluster are studied (Park Plus, Ch 19, p 237)	Classical 30-cluster survey, in which the entire eligible population of each chosen cluster is surveyed (Mahajan, Ch 7, p 138)
Two-stage cluster sampling (sub-sampling)	Clusters (primary units) are selected first; within each selected cluster a random sub-sample of secondary units is then drawn, so $n_i < N_i$ (Sunder Rao & Richard, pp 80-81)	Immunisation coverage evaluation — villages/wards first, then households and children (Sunder Lal, Ch 6, pp 250-251)
Multistage cluster sampling	Sub-sampling is carried on through progressively lower levels — district $\rightarrow$ taluka $\rightarrow$ village $\rightarrow$ household (Sunder Rao & Richard, pp 80-81; Mahajan, Ch 7, p 137)	Large national and state surveys

Type	What is done	Where used
Probability proportional to size (PPS) cluster sampling	Clusters are selected with a chance proportional to their population, so a large village may contain two, three or four clusters (Sunder Lal, Ch 6, pp 250-251); cluster sampling routinely uses PPS (Sunder Rao & Richard, p 80)	Coverage evaluation surveys; National Family Health Survey (NFHS)
Stratified cluster sampling	The population is first stratified, then clusters are drawn within strata; the <i>Pathfinder</i> method uses this to include the most important population subgroups likely to differ in disease levels (Sunder Lal, Ch 6, p 252)	World Health Organization (WHO) Pathfinder oral health surveys
Cluster randomisation	The cluster, not the individual, is the unit of allocation to intervention and control (Browner 5e, Ch 12, pp 389-390)	Community and programme trials



Cluster sampling — one-stage, two-stage and multistage designs

Procedure — the WHO 30-cluster (30 × 7) coverage evaluation survey

Cluster sampling is best known in community medicine as the WHO-approved module for evaluating vaccination coverage under the Expanded Programme on Immunization (EPI) and India's Universal

Immunization Programme (UIP), in which **7 children aged 12-23 months are surveyed in each of 30 clusters, giving 210 children** (Mahajan, Ch 7, p 138; Sunder Rao & Richard, pp 79-80).

First stage — locating the 30 clusters (probability proportional to size)

- ▶ **List** every village, town and ward of a city in the area whose service coverage is to be evaluated, with its population (Mahajan, Ch 7, p 138; Sunder Lal, Ch 6, p 250).
- ▶ **Cumulate** the populations down the list to obtain the cumulative population.
- ▶ **Sampling interval (SI) = total cumulative population ÷ 30 clusters** (Sunder Rao & Richard, p 80).
- ▶ Choose a **random number (R)**, less than or equal to the sampling interval and with the same number of digits, from a random number table or a currency note. The first cumulative entry that equals or exceeds R locates **cluster 1** (Sunder Lal, Ch 6, p 250).
- ▶ **R + SI** locates cluster 2; add the sampling interval repeatedly (cluster 2 + SI = cluster 3, and so on) until all **30 clusters** are identified (Mahajan, Ch 7, p 138).
- ▶ Villages with a large population acquire **more than one start number and therefore more than one cluster** — this is what makes selection probability proportional to size (Sunder Lal, Ch 6, p 251).

Worked example (Sunder Lal, Ch 6, pp 250-251)

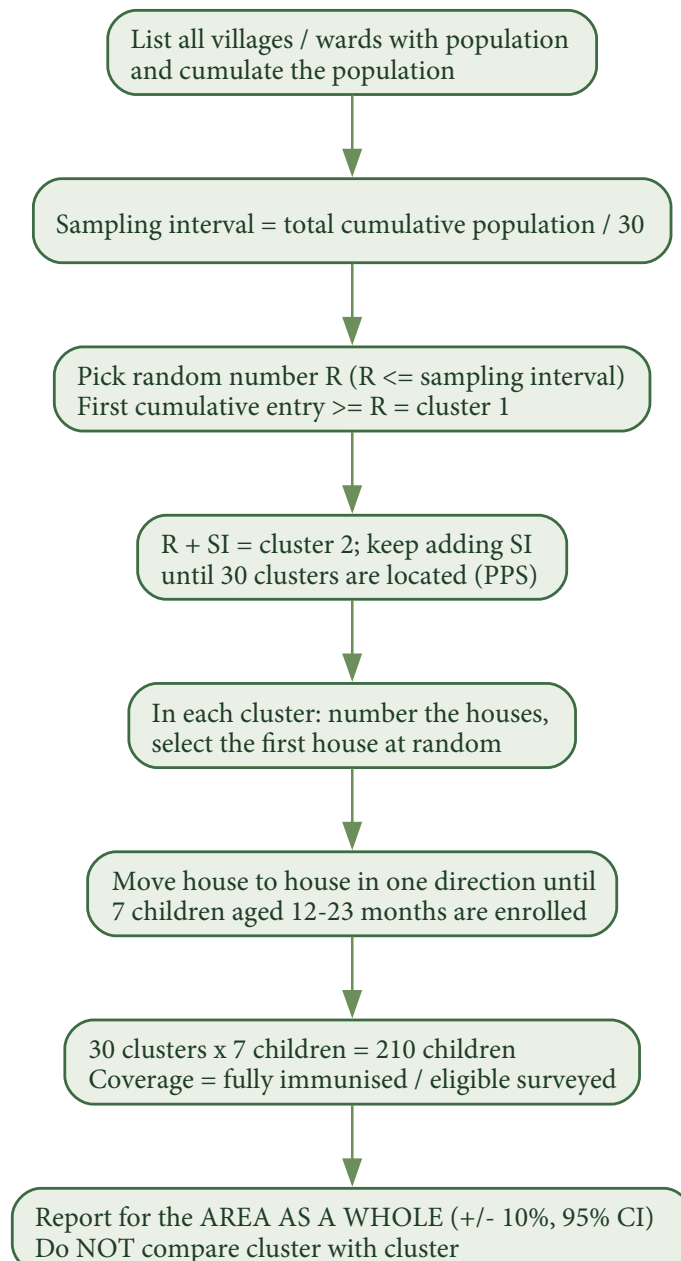
Step	Value
Total cumulative population	23,455
Number of clusters desired	30
Sampling interval (23,455 ÷ 30)	782
Random start number R (< SI)	690
First cumulative entry ≥ 690 → cluster 1	802 = village A
690 + 782 = 1,472 → cluster 2	2,170 = village C
1,472 + 782 = 2,254 → cluster 3	village D, and so on to cluster 30

Second stage — selecting subjects within a cluster

- ▶ Number all houses in the selected cluster and pick the **first house at random** using a random number table or the serial number on a currency note (Mahajan, Ch 7, p 138).
- ▶ Define the **age group and the item being studied in advance** — for example, children aged 12-23 months who are fully vaccinated (3 doses of diphtheria-pertussis-tetanus vaccine, 3 doses of polio vaccine, 1 dose of Bacillus Calmette-Guérin vaccine and 1 dose of measles vaccine) (Mahajan, Ch 7, p 138).
- ▶ Survey **house to house moving in one direction** until 7 eligible children aged 12-23 months are completed; households may alternatively be randomly selected using the Anganwadi worker's or accredited social health activist's (ASHA) household survey register, which is a complete frame of families in the village (Sunder Lal, Ch 6, p 251).
- ▶ Compute **coverage = fully immunised children ÷ eligible children surveyed**, expressed as a percentage.

Reliability the 30 × 7 design guarantees (Sunder Rao & Richard, p 80)

Standard	Statement
Precision	Accuracy of ± 10 per cent — a survey result of 70 per cent means true coverage lies between 60 and 80 per cent
Confidence	95 per cent probability that the true value lies within that range
Reference period	Reflects coverage of the 1 year preceding the survey
Level of inference	Valid for the area as a whole; comparisons between individual clusters are not valid



The 30-cluster (30 x 7) immunisation coverage evaluation survey

Statistical consequence — design effect and intraclass correlation

Concept	Meaning	Practical rule
Intraclass correlation coefficient (ICC)	Degree of similarity between members of the same cluster; a positive ICC means the cluster is more homogeneous than chance would produce (Handbook of Epidemiology, Ch 37, p 1472; Abramson, Ch 33, p 343)	ICC values below 0.05 are typical , but even small values widen confidence intervals substantially when clusters are large; larger clusters tend to have smaller ICCs (Cochrane Handbook, Ch 16, p 521)
Cluster effect	Ratio of cluster-sampling variance to simple-random-sampling variance; because ICC is positive in most real populations, simple-sampling variance grossly underestimates the true variance (Handbook of Epidemiology, Ch 37, p 1472)	Never analyse a cluster sample as though it were a simple random sample
Design effect (DEFF)	Ratio of the standard error of the estimate under the actual design to that under simple random sampling (Handbook of Epidemiology, Ch 14, p 570); equivalently, the ratio of the sample sizes required for a cluster and a random sample (Abramson, Ch 8, p 88)	$DEFF = 1 + (M - 1) \times ICC$, where M is the average cluster size (Cochrane Handbook, Ch 16, p 521)
Sample size inflation	Compute the simple-random-sample size, then multiply it by the design effect (Abramson, Ch 8, p 88; Handbook of Epidemiology, Ch 14, p 570)	Design effects of 2 or more are not uncommon ; a DEFF of 2 doubles the required sample size. The larger the cluster size, the larger the design effect ; if several studies give different values, use the highest (Abramson, Ch 8, pp 88-89)
Effective sample size	Actual number studied ÷ design effect (Cochrane Handbook, Ch 16, pp 521-522)	In a cluster randomised trial the effective sample size lies between the number of clusters and the total number of participants (Browner 5e, Ch 12, pp 389-390)
Number of clusters	Power depends more on the number of clusters than on their size	Prefer many small clusters to a few large ones ; at least 4 clusters in every arm of a cluster randomised trial (Browner 5e, Ch 12, pp 389-390; Abramson, Ch 33, p 343)

Advantages and limitations

Advantages	Limitations
Cuts down the cost of preparing a sampling frame — only a list of clusters is required (Sunder Lal, Ch 6, pp 251-252)	Sampling error / standard error is higher than for a simple random sample of the same size (Sunder Lal, Ch 6, p 252; Mahajan, Ch 7, p 138)

Advantages	Limitations
Cuts down the cost of travelling between selected units, giving significant gains in data-collection cost (Sunder Lal, Ch 6, pp 251-252; Park Plus, Ch 19, p 238)	Less precise , because units within a cluster tend to be homogeneous and a cluster gives a poorer cross-section of the population (Park Plus, Ch 19, p 238)
Only a sample of clusters need be covered, so fieldwork is simpler and quicker (Park Plus, Ch 19, p 238; Mahajan, Ch 7, p 138)	Sample size must be inflated by the design effect , offsetting part of the cost saving (Abramson, Ch 8, p 88)
Eliminates the problem of " packing " — neighbouring households temporarily transferring members into a selected house during case-finding or treatment surveys (Sunder Lal, Ch 6, pp 251-252)	Sampling efficiency falls as cluster size decreases or the number of clusters increases beyond the optimum (Park Plus, Ch 19, p 238)
Statistically valid results at 95 per cent confidence with 10 per cent variation from only 210 children (Mahajan, Ch 7, p 138)	Results are valid for the area as a whole only ; between-cluster comparison is invalid (Sunder Rao & Richard, p 80)
Uses existing administrative divisions , saving extra labour (Mahajan, Ch 7, p 137)	In cluster randomised trials, baseline imbalance between arms is common because few units are randomised, and analysis is complex (Browner 5e, Ch 12, pp 389-390)

Cluster sampling compared with simple random and stratified random sampling

Feature	Simple random sampling	Stratified random sampling	Cluster sampling
Sampling unit	Individual	Individual, within each stratum	Group / cluster
Sampling frame needed	Complete list of all individuals	Complete list plus stratum information for each individual	List of clusters only
Desired internal composition	—	Units within a stratum homogeneous, between strata heterogeneous (Park Plus, Ch 19, pp 236-237)	Units within a cluster ideally heterogeneous — cluster sampling beats stratified sampling when cluster elements are heterogeneous (Park Plus, Ch 19, pp 237-238)
Precision for the same sample size	Reference standard	More precise and more representative (Park Plus, Ch 19, p 237)	Least precise — higher standard error (Mahajan, Ch 7, p 138)
Design effect	1 by definition	Usually gains precision	Greater than 1 whenever ICC is positive (Handbook of Epidemiology, Ch 37, p 1472)
Field cost and feasibility	High — subjects scattered	High — needs detailed knowledge of the distribution of the characteristic (Park Plus, Ch 19, p 237)	Lowest cost, most field-practicable (Park Plus, Ch 19, p 238)

Applications in community medicine

Application	Design used
Coverage evaluation of immunisation under EPI and UIP	WHO 30 × 7 two-stage PPS cluster survey, 210 children aged 12-23 months (Mahajan, Ch 7, p 138)
Diarrhoeal disease surveys, leprosy elimination monitoring surveys, evaluation of the national AIDS programme and other programmes	Two-stage cluster sampling (Sunder Lal, Ch 6, pp 251-252)
National Family Health Survey-5 (2019-21)	Stratified two-stage sample; Census 2011 as frame; primary sampling units = villages (rural) and Census Enumeration Blocks (urban), selected by PPS systematic sampling; a cluster is a primary sampling unit or a segment of one; 22 households per cluster; 30,456 primary sampling units selected across 707 districts, fieldwork completed in 30,198 (NFHS-5 India Report, Ch 2, pp 38-39)
Oral health assessment and planning (WHO Pathfinder surveys)	Stratified cluster sampling; index ages 5, 12, 15, 35-44 and 65-74 years; 25-50 subjects per index age group per cluster (Sunder Lal, Ch 6, p 252)
Community / programme (cluster randomised) trials	Villages, schools or health facilities randomised as whole units (Browner 5e, Ch 12, pp 389-390)
Area sampling for large surveys	Clusters are city blocks or other well-defined geographic sections (Park Plus, Ch 19, p 238)
Large country-wide surveys (e.g. hookworm survey in a district)	Multistage: districts → talukas → villages → every 10th house (Mahajan, Ch 7, p 137)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Diagram 1 — "Simple random vs stratified vs cluster sampling"**: three identical squares representing one population, each divided into 16 small blocks. Labels: *Square 1 — Simple random sampling: scattered shaded individuals across all blocks; Square 2 — Stratified random sampling: population divided into 4 horizontal strata (Stratum I-IV), a few shaded individuals in each stratum; Square 3 — Cluster sampling: 4 whole blocks shaded completely (selected clusters), all members within them studied.* Caption labels to write: **sampling unit = individual / individual within stratum / whole group; frame = list of individuals / list by strata / list of clusters.**
- ▶ **Diagram 2 — "Probability proportional to size selection of 30 clusters"**: a vertical bar/ladder representing the cumulative population from 0 to 23,455. Labels: *Sampling interval SI = $23,455 \div 30 = 782$; Random start R = 690 $\rightarrow$ cluster 1 (village A, cumulative 802); R + SI = 1,472 $\rightarrow$ cluster 2 (village C, cumulative 2,170); 1,472 + SI = 2,254 $\rightarrow$ cluster 3 (village D); mark a large village receiving two or three start numbers = two or three clusters (PPS); end label cluster 30 = cluster 29 + SI.*

High-yield exam pointers

- ▶ **Definition trigger**: sampling unit is a **group, not an individual**; clusters = villages, wards, blocks, slums, factories, workshops, schools.
- ▶ **$30 \times 7 = 210$** : 30 clusters $\times$ 7 children aged 12-23 months; **sampling interval = total cumulative population $\div$ 30**; selection is **probability proportional to size**; precision ± 10 per cent at 95 per cent confidence; valid for the area as a whole, **never cluster vs cluster.**
- ▶ **Design effect DEFF** = $1 + (M - 1) \times ICC$; required size = simple-random-sample size $\times$ DEFF; DEFF of 2 or more is common; effective sample size = observed size $\div$ DEFF.
- ▶ **Rule of thumb for trials**: many small clusters beat few large ones; **minimum 4 clusters per arm**; ignoring clustering gives spuriously significant results and unduly narrow confidence intervals.
- ▶ **One-line trade-off**: cluster sampling buys **cheapness and field feasibility** at the price of **precision** — lowest cost, highest standard error.
- ▶ **Indian example to quote**: NFHS-5 (2019-21) — stratified two-stage design, Census 2011 frame, 22 households per cluster, 30,456 primary sampling units across 707 districts.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park's Textbook of Preventive and Social Medicine 28e describes only simple random, systematic and stratified random sampling in its statistics chapter, so the cluster-sampling content above is grounded on the biostatistics texts and Park Plus rather than on the Park spine; the ± 10 per cent, 95 per cent confidence, 30×7 specification is the original WHO Expanded Programme on Immunization cluster-survey module as reproduced in the Indian textbooks, and WHO has since issued an updated vaccination-coverage cluster-survey reference manual that is not held in this corpus.

SOURCE & COVERAGE

Mahajan's Methods in Biostatistics for Medical Students and Research Workers (Ch 7, Sampling); Sunder Rao & Richard — An Introduction to Biostatistics and Research Methods (Sampling, pp 77-81); Sunder Lal — Textbook of Community Medicine (Ch 6, Basic Statistics and its Applications Including Research Methodology); Park Plus (Amrutha & Vijayalakshmi), Ch 19 — Health Information and Basic Medical Statistics; Abramson — Research Methods in Community Medicine (Ch 8 Sampling; Ch 33 Programme trials); Handbook of Epidemiology, Ahrens & Pigeot (Ch 14 Sample Size Determination; Ch 37 Health Services Research); Cochrane Handbook for Systematic Reviews of Interventions (Ch 16, Cluster-randomized trials); Browner — Designing Clinical Research 5e, 2022 (Ch 12, Alternate Interventional Study Designs); National Family Health Survey-5 India Report, 2019-21 (Ch 2, Sample Design).

**Coefficient of variation**

Asked in: Paper I 14-Jul-2020 — RGUHS MD Community Medicine

Definition — The **coefficient of variation (CV)** is the standard deviation expressed as a **percentage of the arithmetic mean** (Sunder Rao & Richard, §7.7, pp 54–55). It is therefore a **relative, unit-free** measure of dispersion, adopted when the variability of groups of observations depends on the average size of those groups, and/or when the observations are in **different units of measurement**, so that the relative variability of different groups or of different measurements can be compared (Sunder Rao & Richard, §7.7; AFMC Red Book 2024, §26.27). It measures the spread of the data **relative to the size of the mean** and is independent of the unit of measurement (AFMC Red Book 2024, §26.27).

Formula and steps of computation

CV (per cent) = (Standard deviation ÷ Arithmetic mean) × 100

- ▶ Calculate the **arithmetic mean** ($\bar{x}$) of the series.
- ▶ Find the **deviation of each observation from the mean**, $(x - \bar{x})$.
- ▶ **Square** each deviation, $(x - \bar{x})^2$.
- ▶ **Add** the squared deviations to obtain the sum of squares, $\Sigma(x - \bar{x})^2$.
- ▶ Divide by the **number of observations minus one** to get the mean squared deviation, i.e. the **variance** (σ^2).
- ▶ Take the **square root** of the variance to get the **standard deviation (SD)** — the root-mean-square deviation. (*Steps 1–6: Mahajan, Ch 4, p 78.*)
- ▶ Divide the SD by the mean and multiply by 100 to express it as a percentage — the **coefficient of variation** (Mahajan, Ch 4, pp 86–88).

Where the coefficient of variation sits among the measures of variability

Measure	Basis	Unit	Chief merit / limitation
Range	Difference between the two extreme observations, $L - S$	Same as the variable	Simplest and crudest; based on only two extreme values that are themselves prone to chance variation, so it gives only a rough idea of spread (AFMC 2024, §26.27; Sunder Lal, Ch 6, p 246)
Interquartile range	Middle 50 per cent of observations, $Q_3 - Q_1$	Same as the variable	Unaffected by rare or extreme values, hence a good measure of dispersion for skewed data (Sunder Lal, Ch 6, p 247)
Mean (average) deviation	Mean of the absolute deviations from the mean	Same as the variable	Listed among the measures of dispersion (Park 28e, Ch 21, p 1004; Mahajan, Ch 4, p 75), but absolute deviations are algebraically awkward, so indices built on them are seldom used (Feinstein, Ch 5, §5.5.2, p 73)
Standard deviation	Root-mean-square deviation from the mean	Same as the variable	Best and most widely used; based on all observations and the basis of most inference procedures — but it is an absolute measure and cannot be compared across different units or across widely different means (Sunder Lal, Ch 6, p 247; AFMC 2024, §26.27)
Variance	Sum of squared deviations from the mean $\div n$ (or $n - 1$)	Square of the original unit	Not expressed in the units of the original observations, so not directly interpretable (Sunder Lal, Ch 6, p 247)
Coefficient of variation	$(SD \div \text{mean}) \times 100$	None — a percentage	The only relative measure in the list; permits comparison across units and across unequal means, but presupposes a meaningful, non-trivial mean (Sunder Rao & Richard, §7.7; AFMC 2024, §26.27)

Why the standard deviation alone will not do

Situation	Illustration (Mahajan, Ch 4, pp 86–88)	What the SD says	What the CV says
Same character, two groups with widely different means	Height of adults, mean 160 cm, SD 10 cm $\rightarrow$ CV 6.25 %; height of children, mean 60 cm, SD 5 cm $\rightarrow$ CV 8.33 %	Adults' heights vary twice as much (SD 10 vs 5)	Children's heights are relatively more variable — the conclusion is reversed
Two characters measured in different units in the same group	Blood pressure (mmHg) versus height (cm) in the same adults — CV of blood pressure 8.3 % versus CV of height 3.1 %	One SD is in mmHg and the other in cm, so they are not comparable at all	Blood pressure is about 2.7 times as variable as height

Situation	Illustration (Mahajan, Ch 4, pp 86–88)	What the SD says	What the CV says
Reproducibility of a measurement or assay	Repeated assays of a single specimen across technicians, laboratories or machines	An SD that itself rises as the level of the measurement rises	CV summarises inter-assay, inter-observer and inter-instrument variability; the assay with the smallest CV is the most precise (Browner 5e, Ch 13, p 421)

Uses and applications

- ▶ To compare the variability of the **same characteristic between two or more series or data sets** — for example, the same anthropometric measurement in two populations or two survey rounds (AFMC 2024, §26.27).
- ▶ To compare the variability of **two or more different characteristics within the same series**, which is possible only because CV carries no unit (AFMC 2024, §26.27; Sunder Rao & Richard, §7.7).
- ▶ To compare series whose observations are in **different units of measurement**, or whose averages **differ widely** (Sunder Rao & Richard, §7.7).
- ▶ To study the **reliability and precision of an instrument or method** (AFMC 2024, §26.27). Reproducibility of a continuous measurement is expressed either as the **within-participant standard deviation** or as the coefficient of variation, i.e. the within-participant SD divided by the mean (Browner 5e, Ch 4, p 84).
- ▶ In **study implementation and quality control**: commercial laboratories used by a study must supply reproducibility data for their measurements, such as coefficients of variation (Browner 5e, Ch 18, pp 581–582).
- ▶ To judge how **effectively the mean represents the individual items** of a data set — if the coefficient is too large, the mean is not an effective summary (Feinstein, Ch 5, §5.5.1, p 72).

Interpretation

Point	Rule of thumb from the corpus
Direction	A compact distribution — tall and narrow — has a relatively small SD and therefore a small CV; a widely scattered distribution has a large CV (Feinstein, Ch 5, p 72)
Working boundary	Results are held to be of doubtful validity when the coefficient exceeds 0.10–0.15 (10–15 %) , though Feinstein cautions that this may be too strict for the kind of variation actually found in medical data (Feinstein, Ch 5, §5.5.1, p 73)
Quick mental check	In a symmetrical, relatively compact data set the SD is substantially smaller than the mean; if SD exceeds about 25 % of the mean , the distribution is either eccentric or highly dispersed (Feinstein, Ch 5, §5.8.1.1, p 77)
Comparing two assays	The one with the smallest CV is the most precise — though it may not be the most accurate (Browner 5e, Ch 13, p 421)

Limitations and the related relative indices

Index	Formula	When used instead of the CV
Quartile coefficient of variation	$(Q_3 - Q_1) \div (Q_3 + Q_1)$	When the distribution is skewed or eccentric , so that neither the mean nor the SD represents the data well; $(Q_3 - Q_1)/2$ corresponds to an SD and $(Q_3 + Q_1)/2$ to a mean (Feinstein, Ch 5, §5.5.3, p 73)
Coefficient of dispersion (CD)	Average absolute deviation from the median $\div$ median	Directly analogous to the CV but built on absolute deviations from the median; seldom used because absolute deviations are algebraically awkward (Feinstein, Ch 5, §5.5.2, p 73)
Coefficient of stability (c.s.)	$(SD \div \sqrt{n}) \div \text{mean}$, i.e. standard error $\div$ mean	To judge the stability of the mean itself across repeated sampling rather than the scatter of the raw data; about 0.05 is a rough guide to intrinsic stability (Feinstein, Ch 7, §7.4.1, pp 118–119)

Two further cautions: the coefficient of variation is a **descriptive** summary and does not by itself test whether a difference in variability between two groups is statistically significant; and because the mean sits in the denominator, the index becomes unstable and uninformative as the mean approaches zero.

Worked illustration — physiological variables in normal persons

Observation	Range	Mean	SD	CV
Age at menarche (years)	9–18	14.16	1.13	7.98 %
Menstrual cycle (days)	21–35	28	2.00	7.14 %
Pulse rate (beats/min)	55–90	72	3.50	4.86 %
Weight at birth (kg)	1.8–4.5	3.05	0.39	12.79 %
Systolic blood pressure (mmHg)	90–150	115	12.00	10.43 %

(Mahajan, Ch 5 — Normal Distribution and Normal Curve, p 111. Pulse rate is the most consistent and birth weight the most variable of these five, a comparison the standard deviations alone — 3.50 beats/min against 0.39 kg — could never have supported.)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- Draw two superimposed frequency curves with an identical peak position. Labels to write exactly: common vertical line at the centre — "Same mean ($\bar{x}$)"; the tall narrow curve — "Small SD $\rightarrow$ small CV $\rightarrow$ compact, consistent, precise"; the low broad curve — "Large SD $\rightarrow$ large CV $\rightarrow$ widely scattered"; x-axis — "Value of the variable"; y-axis — "Frequency"; a side box — "CV % = $(SD \div \text{mean}) \times 100$ ".

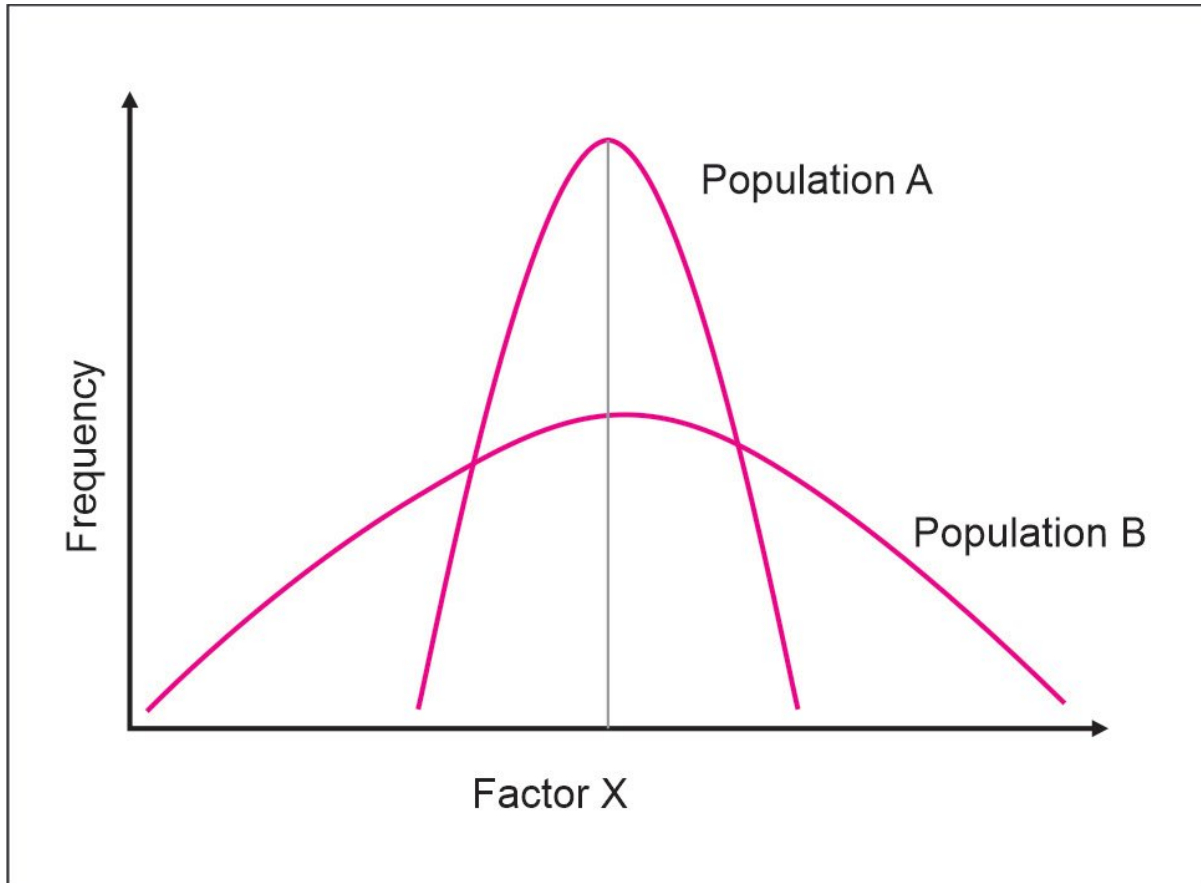
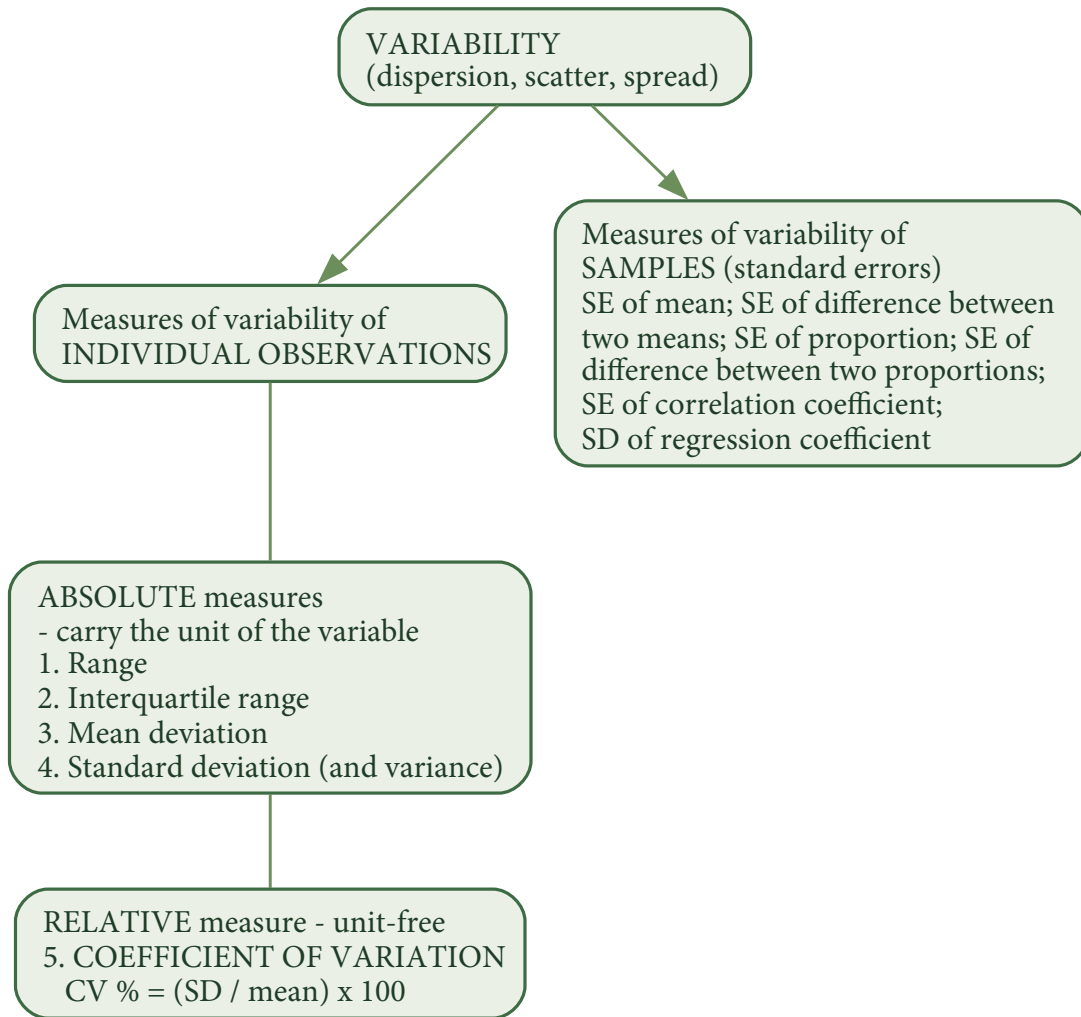
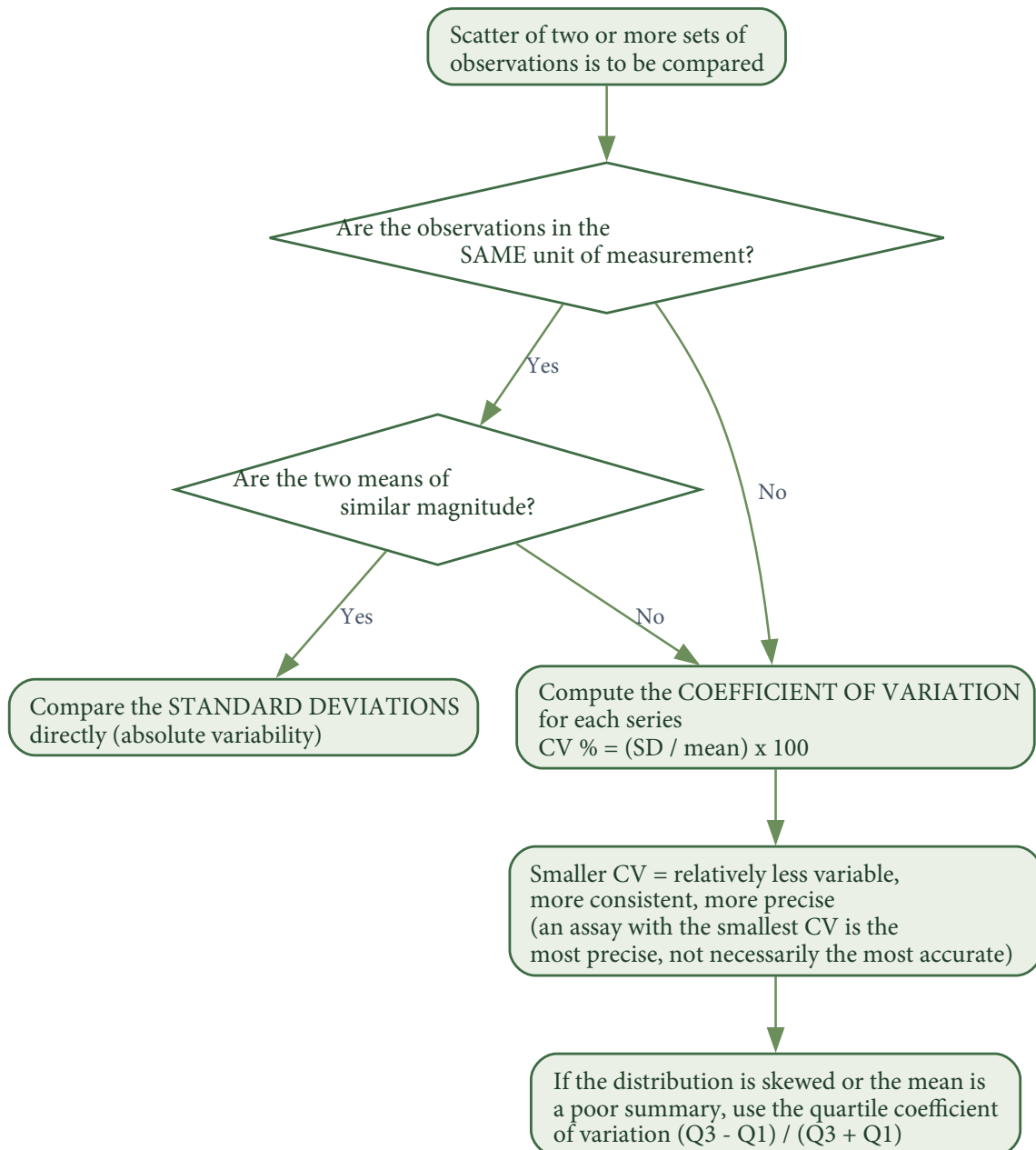


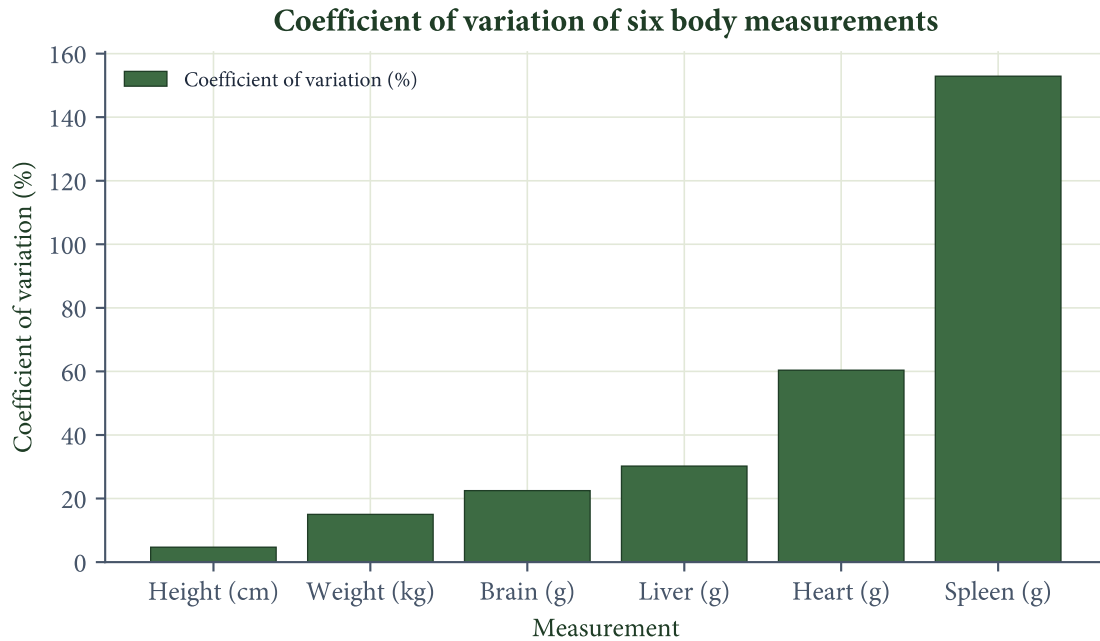
Fig 6.8 — Two distributions with same central location but different spread or σ (Sunderlal, p.243)



Measures of variability — where the coefficient of variation fits



Deciding between the standard deviation and the coefficient of variation



South Indian males aged 21-30 years: relative variability rises steeply from height (mean 164.6 cm, SD 7.64) to spleen weight (mean 367.2 g, SD 561.4), although all six are body measurements of the same men. Source: autopsy statistics, Department of Pathology, Christian Medical College Hospital, Vellore, reproduced as Table 7.9 in Sunder Rao & Richard, pp 54-55.

High-yield exam pointers

- ▶ $CV\% = (SD \div \text{mean}) \times 100$ — the only **relative, unit-free** measure of dispersion; write the formula first.
- ▶ **Two indications:** observations in **different units**, or series with **widely different averages**. Mnemonic — "**different units, different means**".
- ▶ Reversal example to quote: adult height mean 160 cm SD 10 → CV 6.25 %; child height mean 60 cm SD 5 → CV 8.33 % — the larger SD is the smaller relative variation.
- ▶ **Precision rule:** among two assays or instruments, the one with the **smallest CV is the most precise, though not necessarily the most accurate**.
- ▶ **Boundaries:** CV above 10–15 % raises doubt about adequacy (Feinstein); if SD exceeds 25 % of the mean, suspect an eccentric or highly dispersed distribution.
- ▶ For **skewed data** use the **quartile coefficient of variation** = $(Q_3 - Q_1) / (Q_3 + Q_1)$; for the stability of the **mean** use the **coefficient of stability** = $\text{standard error} \div \text{mean}$ (guide value 0.05).
- ▶ Classic illustration: CV of **spleen weight** 152.9 % versus **height** 4.7 % in the same men — organ weights are far more variable than linear body measurements.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park's Textbook of Preventive and Social Medicine 28e carries the measures of dispersion in Chapter 21 (Health Information and Basic Medical Statistics, p 1004), but the scanned text of that chapter does not yield a legible coefficient-of-variation passage, so the definition, formula, worked examples and applications above are grounded in Mahajan, Sunder Rao & Richard, the AFMC Red Book of Community Medicine 2024, Feinstein and Browner rather than in the spine text; the single statement that a difference between two coefficients of variation is descriptive and not itself a test of significance, and that the index becomes unstable as the mean approaches zero, is standard statistics not carried verbatim in the retrieved passages.

SOURCE & COVERAGE

Mahajan's Methods in Biostatistics for Medical Students and Research Workers (Ch 4 — Variability and its Measures, pp 70–88; Ch 5 — Normal Distribution and Normal Curve, p 111); Sunder Rao & Richard — Biostatistics and Research Methods (Ch 7 — Measures of Dispersion, §7.7 Coefficient of Variation and Table 7.9, pp 54–55); AFMC Red Book of Community Medicine 2024 (Ch 26 — Epidemiology and Biostatistics, §26.27 Measures of Variability, pp 781–782); Feinstein — Principles of Medical Statistics (Ch 5 — Inner Zones and Spreads, §§5.5, 5.8, pp 72–77; Ch 7 — Confidence Intervals and Stability: Means and Medians, pp 118–119); Browner et al. — Designing Clinical Research, 5e, 2022 (Ch 4 — Planning the Measurements, p 84; Ch 13 — Designing Studies of Medical Tests, p 421; Ch 18 — Study Implementation and Quality Control, pp 581–582); Sunder Lal — Textbook of Community Medicine (Ch 6 — Basic Statistics and its Applications and Research Methodology, pp 243–247); Park's Textbook of Preventive and Social Medicine, 28e (Ch 21 — Health Information and Basic Medical Statistics, p 1004).



Correlation and regression coefficients

Asked in: Paper I 12-Nov-2019 — RGUHS MD Community Medicine

Definition — Correlation is the statistical technique that measures the **degree and direction of the linear relationship between two quantitative variables**, neither of which is assumed to depend on the other; it is summarised by the **correlation coefficient (r)**, which ranges from -1.0 to $+1.0$ (Park 28e, Ch 21, p 1009). **Regression** describes the **dependence of one variable on another** so that the value of the dependent variable can be **estimated or predicted** from the independent variable; it is summarised by the **regression coefficient (b)**, the slope of the fitted straight line (Park 28e, Ch 21, p 1009; Sunder Rao & Richard, Ch 13, p 104). Correlation answers "*how strongly do they move together?*"; regression answers "*by how much does Y change per unit of X?*"

Scatter diagram and types of correlation

The first step in any bivariate analysis is to **plot the scatter (X–Y) diagram** — one variable on the X-axis, the other on the Y-axis — because data sets with identical means, identical r and identical

regression equations can have completely different underlying patterns that only the plot reveals (Red Book — Public Health and Preventive Medicine, Ch on Biostatistics, pp 117-118).

Type of correlation	Value of r	Pattern on scatter diagram	Example
Perfect positive	$r = +1$	All points on one straight upward line	Not found in nature; height and weight, age and height approach it up to a certain age
Perfect negative	$r = -1$	All points on one straight downward line	Both variables move in exactly opposite proportion
Partial (moderately) positive	$0 < r < +1$	Elliptical upward cloud about the mean line	Height and weight of adults; haemoglobin and packed cell volume
Partial (moderately) negative	$-1 < r < 0$	Elliptical downward cloud	Mean weekly temperature and number of deaths registered
Absolutely no correlation	$r = 0$	Points scattered at random; the two regression lines lie at right angles	Two unrelated characters

(Mahajan's Methods in Biostatistics, Ch 12 — Correlation and Regression, pp 236-241)

The correlation coefficient (r)

Also called Karl Pearson's product-moment coefficient of linear correlation (Sunder Rao & Richard, Ch 13, p 104).

Item	Statement
Formula	$r = \Sigma xy \div \sqrt{(\Sigma x^2 \times \Sigma y^2)}$, where $x = (X - \bar{X})$ and $y = (Y - \bar{Y})$
Range	-1.0 to +1.0; the sign gives direction, the absolute value gives strength
Unit	Dimensionless (a pure number) — unaffected by the units in which X and Y are measured
Relation to b	$r = \sqrt{(b_{yx} \times b_{xy})}$ — r is the geometric mean of the two regression coefficients
Square of r	r^2 = coefficient of determination = proportion of variation in Y explained by X

Interpretation of the magnitude of r (Red Book — Public Health and Preventive Medicine, Ch on Biostatistics, p 137):

Value of r	Interpretation
0 to ± 0.25	Little or no linear relationship
± 0.25 to ± 0.50	Fair degree of linear relationship
± 0.50 to ± 0.75	Moderate to good linear relationship
± 0.75 and above	Very good linear relationship

Testing whether an observed r could have arisen by chance:

- **Null hypothesis** — the population correlation coefficient (ρ) is zero, i.e. there is no linear relationship.
- Compute r from the sample.
- **Degrees of freedom** = $n - 2$, where n is the number of *pairs* of observations.
- Refer the calculated r to the **table of the correlation coefficient**; e.g. at 10 degrees of freedom (a sample of 12 pairs) an r exceeding 0.576 in absolute value occurs by chance in fewer than 5 in 100 ($P < 0.05$) and is significant at the 5% level (Mahajan's Methods in Biostatistics, Appendix V, pp 417-420).
- Equivalently, apply Student's t -test: $t = r\sqrt{(n - 2) \div (1 - r^2)}$, with $n - 2$ degrees of freedom.

The regression coefficient (b) and the regression equation

The regression coefficient b is the change in the dependent variable (Y) for one unit change in the independent variable (X) (Mahajan's Methods in Biostatistics, Ch 12, p 244). Because either variable may be treated as dependent, **two regression lines exist** for every data set (Sunder Rao & Richard, Ch 13, pp 107-108).

Quantity	Formula	Meaning
Regression of Y on X	$b_{yx} = \Sigma xy \div \Sigma x^2 = r \times (\text{SD of Y} \div \text{SD of X})$	Change in Y per unit rise in X
Regression of X on Y	$b_{xy} = \Sigma xy \div \Sigma y^2 = r \times (\text{SD of X} \div \text{SD of Y})$	Change in X per unit rise in Y
Regression equation	$Y_c = \bar{Y} + b(X - \bar{X}) = a + bX$	Predicted (calculated) value of Y for any given X
Intercept (a)	$a = \bar{Y} - b\bar{X}$	Value of Y when X = 0
Standard error of estimate	$S = \sqrt{[\Sigma(Y - Y_c)^2 \div (n - 2)]}$	Scatter of observed points about the regression line, like an SD around it
Test of significance of b	$t = b \div \text{standard error of } b, \text{ df} = n - 2$	Tests whether X is a significant predictor of Y

Key properties:

- The coefficients are estimated by the **method of least squares**, which minimises the sum of squared vertical deviations of the observed points from the line.
- The regression line **always passes through the point** ($\bar{X}, \bar{Y}$), the mean of both variables.
- Unlike r , **b carries units** (e.g. kg per cm) and therefore changes if the measurement scale changes; r does not.
- About half the observed points lie on either side of the fitted line, and roughly 95% lie within ± 2 standard errors of estimate of it, giving the 95% confidence band.
- Worked illustration: for weight regressed on height, $b = 1.16$ means a **1.16 kg rise in weight for every 1 cm rise in height** (Jekel's Epidemiology, Biostatistics and Preventive Medicine, Ch 11, pp 154-155).

Coefficient of determination (r^2) — the proportion of the total variation in the dependent variable explained by the regression, also called *explained variation*. If r between age and blood pressure is 0.8, then $r^2 = 0.64$, i.e. **64% of the variability in blood pressure is accounted for by age** and the remaining 36% by other factors such as weight, diet and exercise (Red Book, Ch on Biostatistics, p 137; Sunder Rao & Richard, Ch 13, pp 104-105).

Correlation versus regression

Feature	Correlation coefficient (r)	Regression coefficient (b)
Purpose	Measures strength and direction of association	Predicts/estimates one variable from another
Variables	Both mutual; no dependent-independent distinction	One dependent (Y), one independent (X)
Symbol and range	r ; -1 to +1	b ; any value, positive or negative, unbounded
Units	None (pure number)	Units of Y per unit of X
Number of values	One value for a pair of variables	Two (b_{yx} and b_{xy})
Effect of changing scale	Unchanged	Changes with units of measurement
Graphical form	Scatter diagram	Regression (mean correlation) line
Typical question	Are haemoglobin and packed cell volume related?	What is the expected weight at a given height?

Extensions used in public health research

Method	When used
Spearman's rank correlation (r_{rank} , ρ)	Variables not normally distributed, or measured on an ordinal/ranked scale; ranges -1 to +1 and is read exactly like r
Kendall's tau (τ)	Alternative rank correlation for ordered observations
Multiple linear regression	One continuous outcome predicted from two or more independent (explanatory) variables
Coefficient of multiple correlation (R) and R^2	Strength of the joint relationship; R^2 is the variation in Y explained by all predictors together
Partial correlation coefficient	Relationship of Y with one predictor while holding the others constant — the basis of adjusting for confounders
Multiple logistic regression	Outcome variable is binary (disease present/absent)

(Sunder Rao & Richard, Ch 13, pp 109-110 and Ch 16, p 132; Red Book, Ch on Biostatistics, p 137)

Uses in community medicine and research

Use	What the coefficient does	Grounded example
Dose-response / exposure-effect	r quantifies how closely an exposure tracks its effect	Fluoride in drinking water (mg/litre) vs community fluorosis index; drug dose vs measured response (Mahajan, Ch 17 Exercises, pp 363-364)
Causal inference	Helps disentangle the chain of direct and indirect causes; supplies "strength of association"	Correlation coefficient tests in aetiological studies (Mahajan, Ch 12, pp 236-237)
Prediction and reference standards	Regression equation estimates a hard-to-measure variable from an easy one	Crown-heel length (cm) = $42.97 + 0.120 \times$ gestational age (weeks); at 40 weeks, estimated length 47.77 cm — the basis of growth charts (Sunder Rao & Richard, Ch 13, pp 102-108)
Adjusting for confounding	Partial/multiple coefficients give each predictor's effect with the others held constant	Diastolic blood pressure predicted from body mass index and waist-hip ratio together (Red Book, Biostatistics)
Comparing two measurement methods	Warns that high r is not agreement	Even $r = 0.94$ may conceal poor agreement between two methods of measuring the same substance (Principles of Medical Statistics, Ch 19, p 379)
Paired biological variables with no cause-effect direction	r describes co-variation when both are effects of a common cause	Biparietal diameter and femur length in foetuses; birth weight and maternal height (Sunder Rao & Richard, Ch 13, p 104)

Assumptions, restrictions and common errors

- **Correlation does not prove causation.** Causation implies correlation, but correlation does not imply causation — the temporal sequence of exposure preceding disease cannot be assumed, and correlation does not measure risk (Park 28e, Ch 3, p 107).
- **Spurious or "nonsense" correlation** arises when two variables share a common time trend. Yule found $r = 0.951$ between the proportion of marriages celebrated in Anglican churches in England and Wales (1866–1911) and the standardised mortality rate over the same period; Charlier found $r = 0.86$ between the stork population of Oslo and the number of babies born there (Sunder Rao & Richard, Ch 13, p 109).
- **r measures only linear relationships.** A curvilinear relationship may give a low r despite a strong association; $r = 0$ means uncorrelated, **not** independent.
- **Data must be homogeneous.** If the scatter clusters into two or more groups, a spuriously high r results from the heterogeneity alone.
- **Do not extrapolate** the regression line beyond the range of the observed data — a line of height on age fitted to school-age children cannot predict the heights of adults.

- ▶ **Pearson's r is strongly influenced by outliers** and is trustworthy only when each variable is approximately normally distributed (Jekel, Ch 11, p 153); otherwise use Spearman's rank correlation.
- ▶ Regression modelling further assumes **linearity, independence of observations and homoscedasticity** (equal variance of Y across all levels of X) (Jekel, Ch 13, pp 180-181).
- ▶ A third common factor may produce the association: a tall brother is not the cause of a tall sister (Sunder Rao & Richard, Ch 13, pp 108-109).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Draw the five scatter diagrams of the types of correlation as one strip, labelling each panel exactly:

- ▶ Panel A — X-axis "X", Y-axis "Y", all points on a rising straight line; label *Perfect positive correlation, $r = +1$*
- ▶ Panel B — all points on a falling straight line; label *Perfect negative correlation, $r = -1$*
- ▶ Panel C — upward elliptical cloud with two lines crossing at the mean, labelled *Y on X* and *X on Y*; label *Partial positive correlation*
- ▶ Panel D — downward elliptical cloud with the same two lines; label *Partial negative correlation*
- ▶ Panel E — random round cloud with *Y on X* and *X on Y* lines at right angles; label *Absolutely no correlation, $r = 0$*

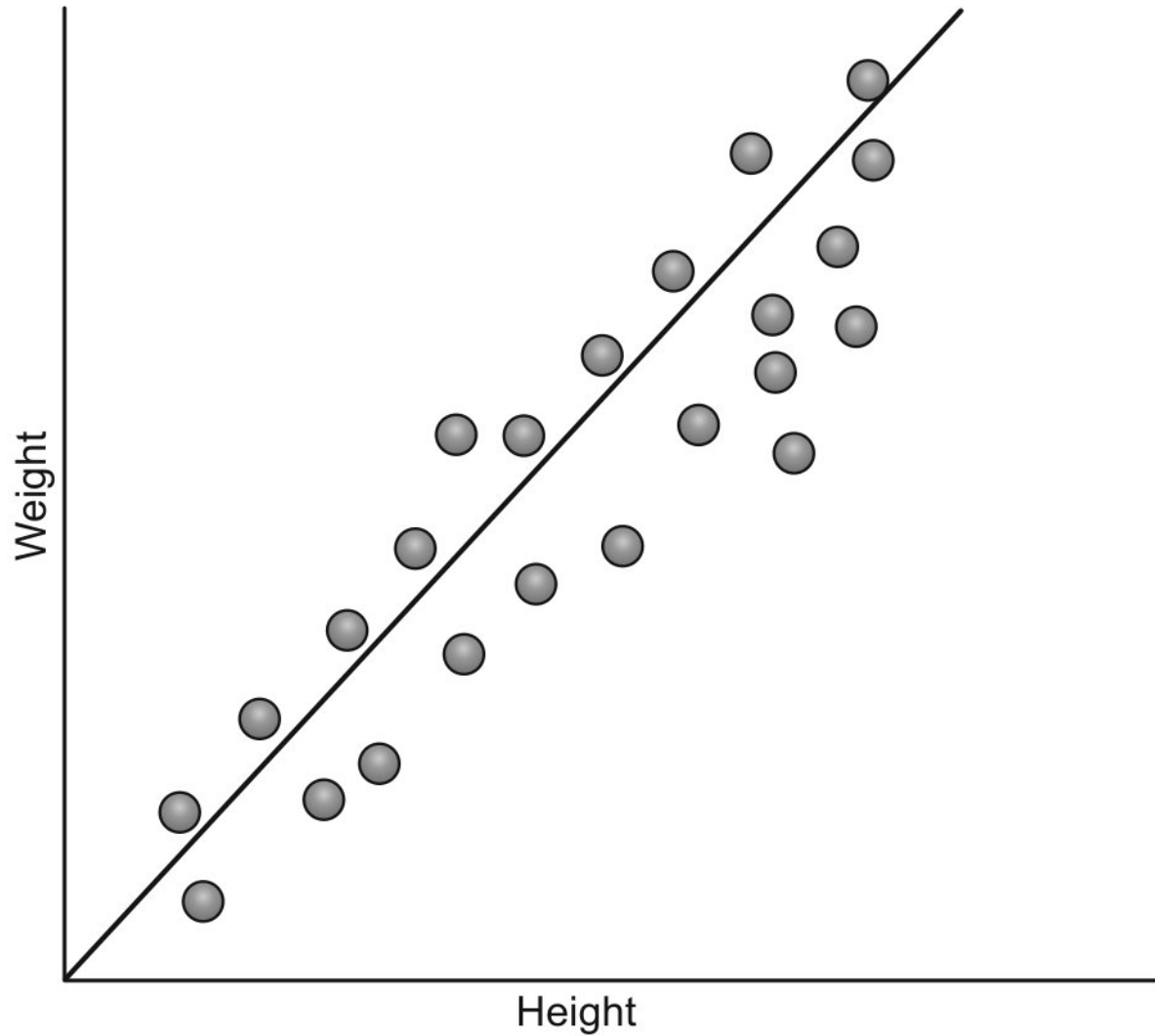


Fig 2.9 — Scatter diagram showing positive correlation (Mahajan's Methods in Biostatistics for Medical Students and Research Workers (PDFDrive), p.42)

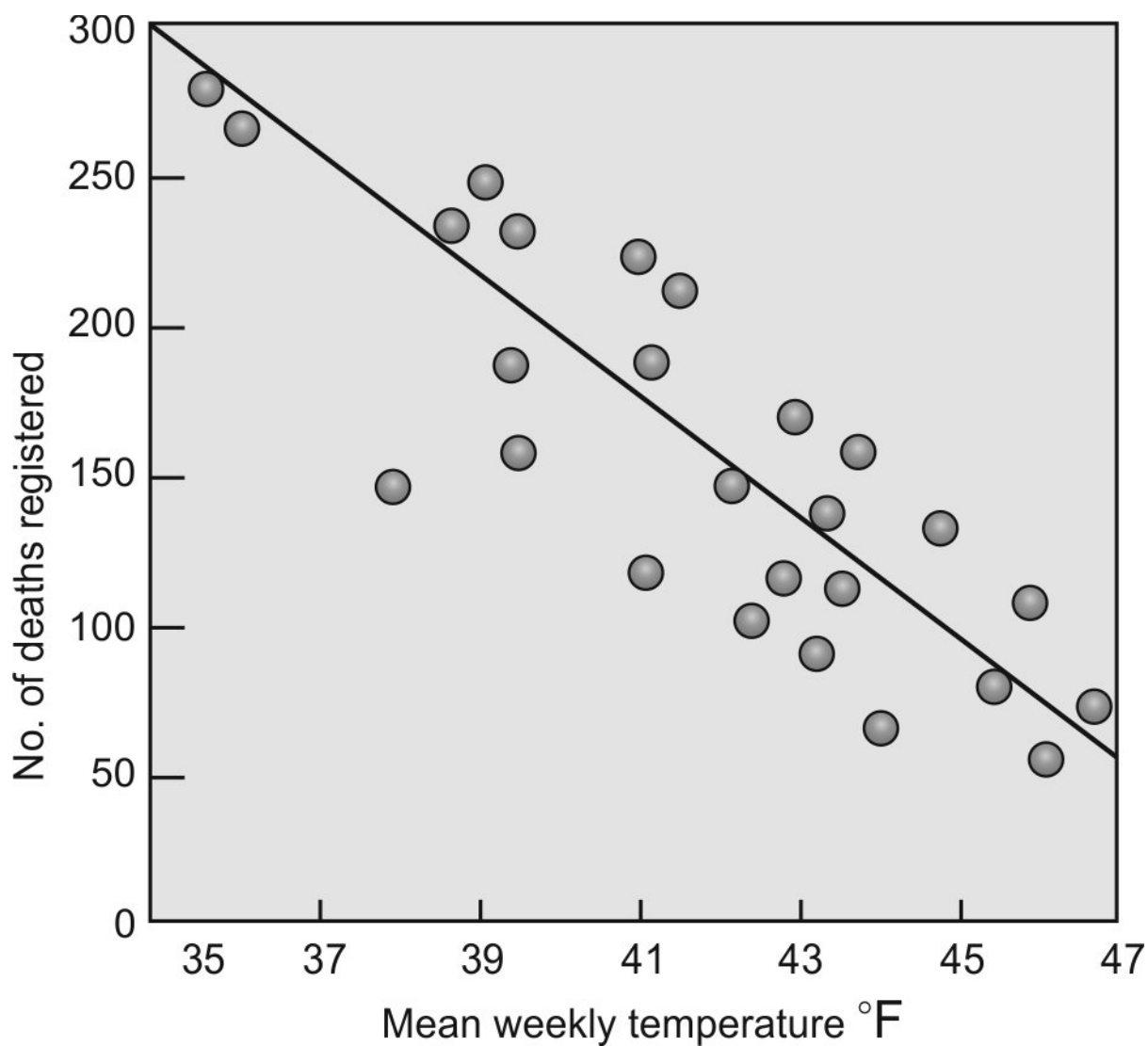


Fig 2.10 — Scatter diagram showing negative correlation (Mahajan's Methods in Biostatistics for Medical Students and Research Workers (PDFDrive), p.42)

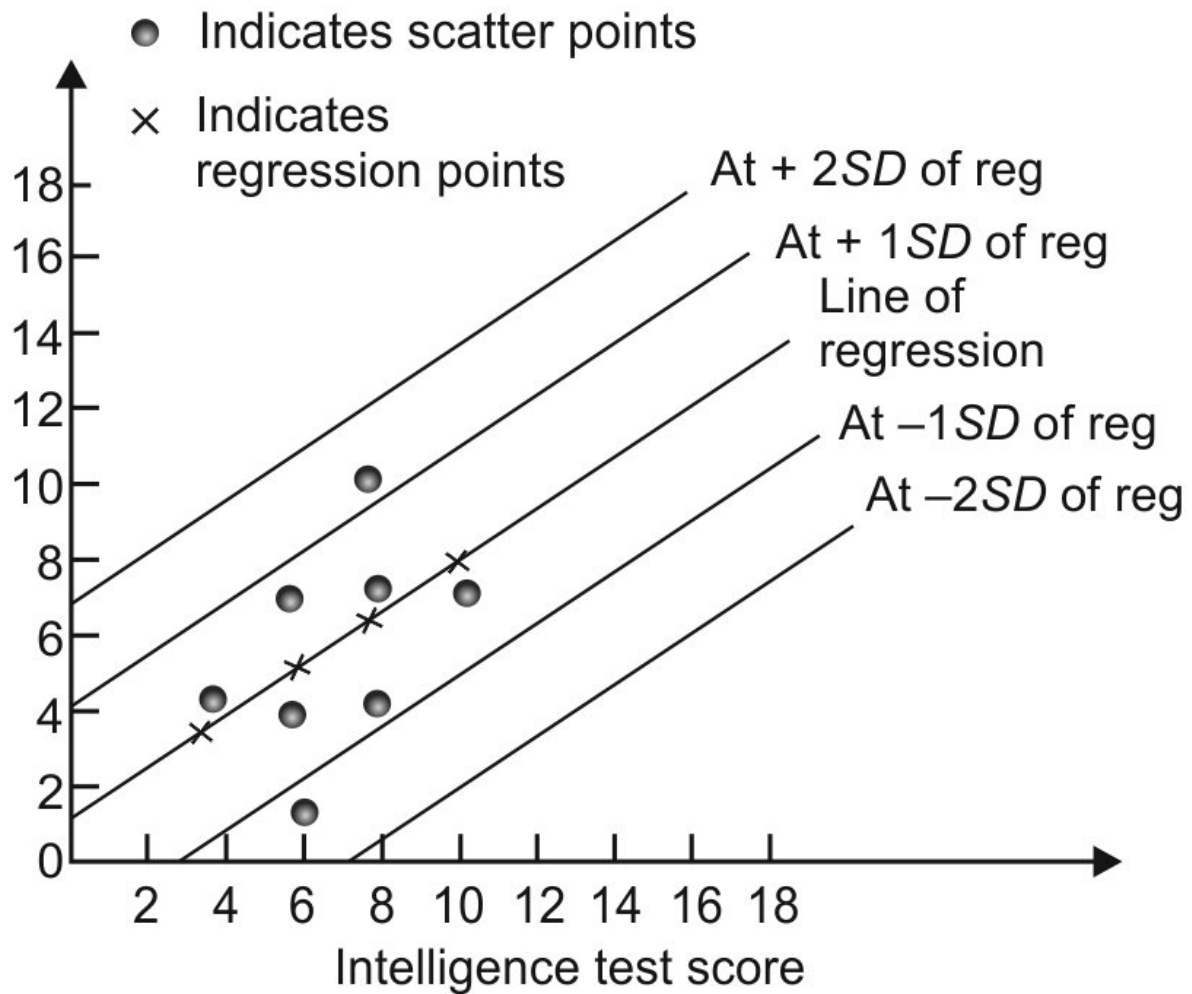
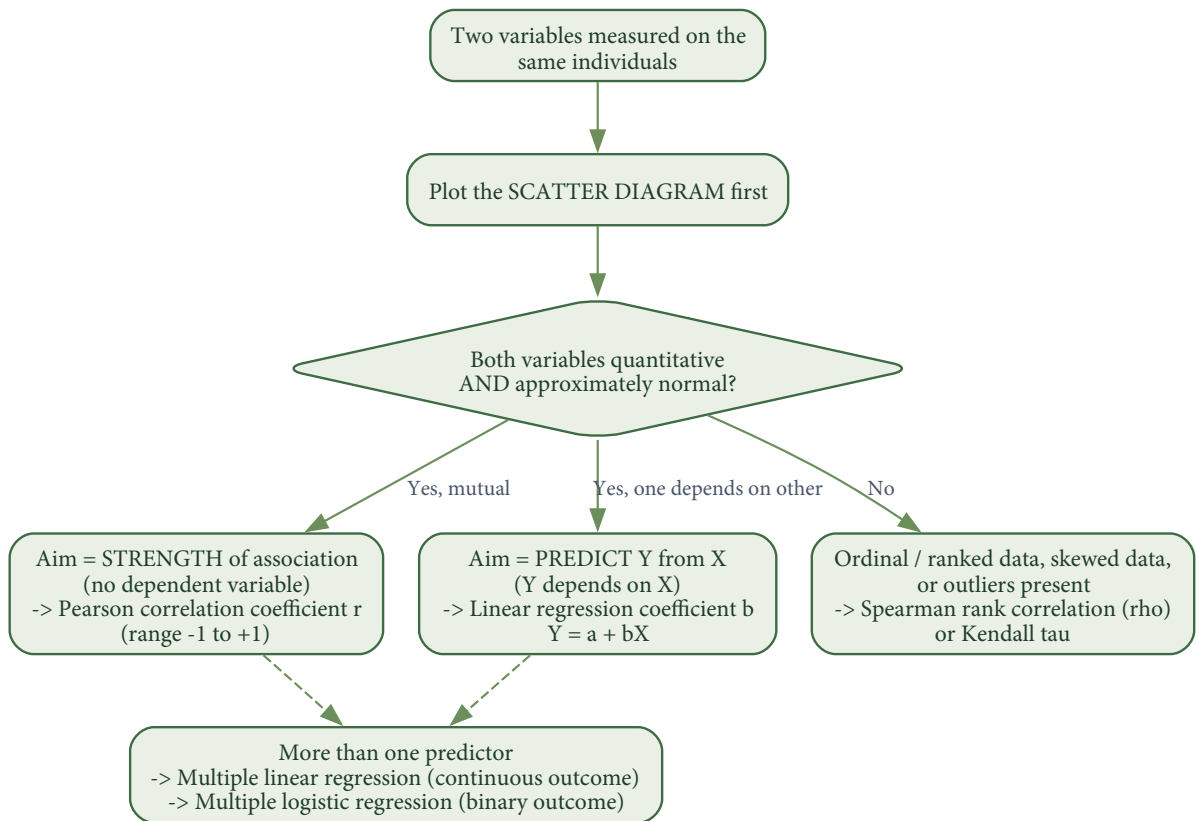
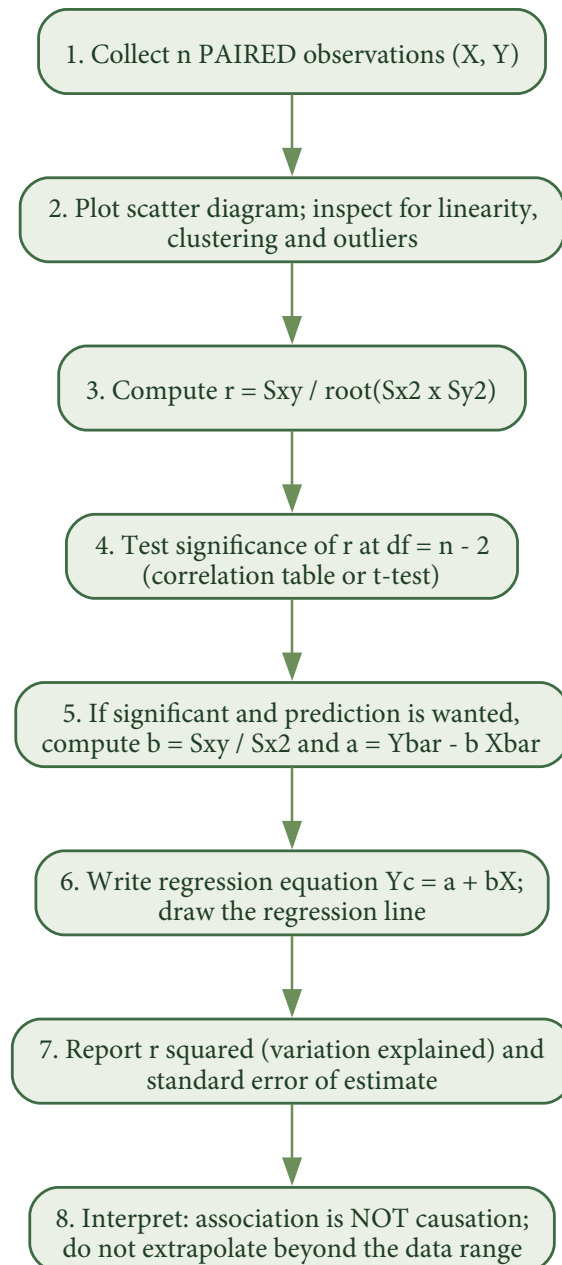


Fig 12.2 — Regression line and standard deviation of the Y measurements (Mahajan's Methods in Biostatistics for Medical Students and Research Workers (PDFDrive), p.241)



Choosing the correct measure of association between two variables



Sequence of a correlation-and-regression analysis of paired data

High-yield exam pointers

- ▶ $r = \Sigma xy \div \sqrt{(\Sigma x^2 \times \Sigma y^2)}$; range -1 to $+1$; dimensionless; **df for testing $r = n - 2$** (n = number of pairs).
- ▶ $b_{yx} = \Sigma xy \div \Sigma x^2 = r \times (\text{SD of } Y \div \text{SD of } X)$; regression equation $Y_c = a + bX$ with $a = \bar{Y} - b\bar{X}$; the line always passes through $(\bar{X}, \bar{Y})$.
- ▶ $r = \sqrt{(b_{yx} \times b_{xy})}$ — r is the geometric mean of the two regression coefficients.
- ▶ r^2 = **coefficient of determination** = per cent of variation in Y explained by X ($r = 0.8 \rightarrow 64\%$ explained).
- ▶ Grading of r : $0-0.25$ little/none, $0.25-0.50$ fair, $0.50-0.75$ moderate-to-good, ≥ 0.75 very good.

- Eponyms to name: **Karl Pearson** (product-moment r), **Spearman** (rank correlation, ρ), **Kendall** (τ), **Yule** (nonsense correlation, $r = 0.951$, Anglican marriages and mortality).
- One-liner to close with: "**Causation implies correlation, but correlation does not imply causation**" (Park 28e, Ch 3, p 107).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: the printed algebraic form of Spearman's rank correlation coefficient, $r_s = 1 - [6\sum d^2 \div n(n^2 - 1)]$ where d is the difference between the paired ranks, appears in the indexed corpus as an unreadable scanned image, so only its structure (rank the two series, take the differences d , square and sum them) is directly traceable to the retrieved text; the formula itself is standard and is safe to quote.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 3, Principles of Epidemiology; Ch 21, Health Information and Basic Medical Statistics); Mahajan's Methods in Biostatistics for Medical Students and Research Workers (Ch 12, Correlation and Regression; Ch 2, Sources and Presentation of Data; Appendix V, Correlation Coefficient Table); Sunder Rao & Richard — Biostatistics and Research Methods (Ch 13, Linear Regression and Correlation; Ch 16, Non-parametric Tests); Jekel's Epidemiology, Biostatistics and Preventive Medicine (Ch 11, Bivariate Analysis; Ch 13, Multivariable Analysis); Red Book — Public Health and Preventive Medicine (Biostatistics); Principles of Medical Statistics (Ch 18, Principles of Associations; Ch 19, Evaluating Trends).



Null hypothesis and tests of significance

Asked in: Paper I 07-May-2022 — RGUHS MD Community Medicine

Definition — The **null hypothesis (H_0)** is the assumption of **no difference** between the populations from which the samples were drawn, or of **no association** between the two factors studied — the whole of the difference observed in the samples being attributed to sampling variation, i.e. to chance (Sunder Rao & Richard, p 85). The observed difference is compared with the sampling distribution expected under H_0 , and H_0 is rejected only if such a difference is too large to be credited to chance. Park illustrates it identically for two whooping cough vaccines — "we set up a hypothesis, called the *Null Hypothesis*, that there was no difference between the effect of the two vaccines" (Park 28e, Ch 21, p 1008). **Testing the null hypothesis is fundamental to all tests of significance.**

Null hypothesis and the terms that go with it

Term	Sym bol	Statement	Worked example (Mahajan, p 175)
Null hypothesis	H_0	There is no real difference / no association; the observed difference is due to chance	"Vitamins A and D make no difference to growth"
Alternative hypothesis	H_1	A real difference or association exists in the population	"Vitamins A and D play a positive, significant role in promoting growth"
Two-tailed test	—	Direction <i>not</i> specified ("is it different? "); rejection region split 2.5% at each tail	Is the mean intelligence quotient (IQ) of malnourished children <i>different</i> from the well-nourished?
One-tailed test	—	Direction specified ("is it better / higher? "); rejection region at one tail only	Are malnourished children of <i>lower</i> mean IQ? Is drug A <i>better</i> than drug B?
Level of significance	α	Pre-fixed probability at which H_0 will be rejected; conventionally 5%, sometimes 1%	$P \leq 0.05$ taken as "rare" (Sunder Rao & Richard, p 85)
P value	P	Probability that the observed difference, or a greater one, arises by chance alone	$P = 0.01185$ for $t = 2.863$ at 15 degrees of freedom (Sunder Rao & Richard, p 100)

Rule of interpretation (Park 28e, Ch 21, pp 1007–1008; Sunder Rao & Richard, pp 100–101):

Result	Normal deviate	P value	Decision on H_0
Outside 95% confidence limits	> 2	$P < 0.05$	Reject H_0 — difference significant at 5% level
Exactly at 95% confidence limits	$= 2$	$P = 0.05$	Just significant at 5% level
Within 95% confidence limits	< 2	$P > 0.05$	Do not reject H_0 — difference not significant
Far outside limits	—	$P < 0.01$	Reject H_0 — significant at 1% level

In a one-tailed test the tabulated value is read against $P = 0.10$ to obtain the 5% level of significance, because the probability tables for Z, t and χ^2 are printed for both tails together (Mahajan, pp 178–179).

Steps in testing the null hypothesis

- ▶ **State H_0 and H_1** — H_0 of no difference or chance difference, with its alternative (Mahajan, p 175).
- ▶ **Fix the level of significance (α)** in advance — 5% or 1% — and decide whether the test is one-tailed or two-tailed from the way the research question is worded.
- ▶ **Verify the assumptions** — random sampling, and homogeneity of variances tested by Fisher's F test (variance ratio test); if variances are heterogeneous, Z and t must not be used as such (Mahajan, p 180).
- ▶ **Select the test** by type of data, number of groups, paired or independent observations, and normality of the variable.
- ▶ **Compute the test statistic**, which in Z and t tests is the **critical ratio = observed difference ÷ standard error of that difference** (Mahajan, p 175; Sunder Rao & Richard, pp 96–97).
- ▶ **Find the degrees of freedom (df)** — unpaired t: $n_1 + n_2 - 2$; paired t: $n - 1$; chi-square: $(r - 1)(c - 1)$, where r is the number of rows and c the number of columns (Mahajan, pp 191–192, 223).
- ▶ **Compare the calculated value with the table value** at that df and level of significance, and read off P.
- ▶ **Draw the inference** — if the calculated value exceeds the table value, reject H_0 and accept H_1 and call the difference significant; if not, do not reject H_0 and call the difference not significant, i.e. not large enough to exclude chance (Sunder Rao & Richard, pp 85–86).

Errors in deciding on the null hypothesis

Decision taken	H_0 is actually true	H_0 is actually false
H_0 rejected	Type I error (α)	No error
H_0 not rejected	No error	Type II error (β)

(Sunder Rao & Richard, p 86)

- ▶ **Type I error (α)** — calling a difference significant when no real difference exists; its magnitude is the level of significance chosen.
- ▶ **Type II error (β)** — failing to detect a difference that really exists. In Sunder Rao's illustration, two samples of 30 each comparing a 30% against a 55% cure rate reached significance in only 40% of repeated studies — a Type II error of 60% (p 86). **Power = $1 - \beta$** .
- ▶ Type II error is the less serious because it only calls for confirmation — **increase the sample size and repeat** (Mahajan, pp 173–174).
- ▶ Two-tailed normal deviates for these risks: $\alpha = 0.05 \rightarrow Z = 1.96$; $\alpha = 0.01 \rightarrow Z = 2.575$; $\alpha = 0.001 \rightarrow Z = 3.29$ (Sunder Rao & Richard, pp 214–215).

Parametric tests for the null hypothesis

Test	Null hypothesis tested	Conditions for use	Test statistic	Degrees of freedom
Standard error of proportion / normal deviate	Sample proportion equals the expected population proportion	Population divides into only 2 classes — male/female, sick/healthy, success/failure	$(\text{observed} - \text{expected}) \div \text{SE}$, where $\text{SE} = \sqrt{pq/n}$	— (compare with 2)
Standard error of difference between two proportions	Two proportions are equal	Qualitative data, two groups	$\text{difference} \div \text{SE of difference}$	— (compare with 2)
Z test	Two means (or a mean and the population mean) are equal	Random samples, quantitative data, normal distribution, sample size > 30	$(\bar{X}_1 - \bar{X}_2) \div \text{SE of difference}$	— (compare with 1.96)
Unpaired (independent) t test	Means of two independent groups are equal	Random samples, quantitative data, normal distribution, sample size < 30 , homogeneous variances	$(\bar{X}_1 - \bar{X}_2) \div \text{SE of difference}$	$n_1 + n_2 - 2$
Paired t test	Mean of the before–after differences is zero	Single sample, correlated observations on the same individuals; eliminates individual sampling variation	$\text{mean difference} \div \text{SE of the differences}$	$n - 1$
Analysis of variance (ANOVA) — F test	Means of three or more groups are all equal, i.e. all groups come from the same population	Quantitative data, more than two samples	$\text{between-group variance} \div \text{within-group variance}$	Two df — between groups and within groups
Fisher's F (variance ratio) test	The two variances are homogeneous	Applied <i>before</i> Z or t	$\text{larger variance} \div \text{smaller variance}$	—
Chi-square (χ^2) test	The two attributes are independent — no association	Qualitative data expressed as actual frequencies , not percentages	$\Sigma (O - E)^2 / E$	$(r - 1)(c - 1)$

Criteria for applying the t test (Mahajan, pp 192–193): random samples; quantitative data; normally distributed variable; sample size less than 30. The criteria for Z differ only in the last condition.

Chi-square is the alternative to the standard error of difference between two proportions, with the advantage that **more than two groups can be compared** (Park 28e, Ch 21, p 1008). It answers three kinds of question: (i) **independence or association** between two variables, by far the commonest use; (ii) **homogeneity** of several samples; (iii) **goodness of fit** of an observed to a theoretical distribution (Sunder Rao & Richard, p 113). Precautions:

- Calculate on the **actual observed numbers** — never on proportions or percentages, since for the same percentage χ^2 can range from near zero to a very large value (Sunder Rao & Richard, p 118).
- Apply **Yates' correction for continuity** — reduce $|O - E|$ by 0.5 — when $df = 1$ and any expected cell value is less than 5 (Mahajan, pp 223–224).
- The test is valid only if the expected number in each cell is not less than two (Park 28e, Ch 21, p 1008).
- χ^2 is only a **test of significance**, not a measure of the strength of association (Sunder Rao & Richard, p 118).

Non-parametric (distribution-free) tests for the null hypothesis

Used when parametric assumptions cannot be met — data are ranks, scores or simple classifications, the population distribution is unknown, or the sample is very small (Sunder Rao & Richard, pp 126–128).

Situation in which H_0 is to be tested	Non-parametric test	Parametric counterpart
Two categories, small sample	Binomial test	—
Discrete categories (2 or more), expected frequencies large	Chi-square test	—
Fit of a continuous distribution	Kolmogorov–Smirnov test	—
Two independent samples, weaker (ranked) measurements, large samples	Mann–Whitney U test	Unpaired t test
Two independent samples that can only be dichotomised above/below the combined median	Median test (with Fisher's test if combined n is small)	Unpaired t test
Independent samples falling into mutually exclusive classes, small numbers	Fisher's exact probability test	Chi-square
Matched pairs, data are frequencies in categories	McNemar test	—
Matched pairs, variable continuous but measurable only grossly	Sign test	Paired t test
Matched pairs whose differences can be meaningfully ranked	Wilcoxon test	Paired t test
k (more than two) independent samples	Kruskal–Wallis one-way analysis of variance by ranks — H is distributed as χ^2 with $k - 1$ df	One-way ANOVA (F test)
Correlation between ranked observations	Spearman's rho (ρ) or Kendall's tau (τ)	Pearson correlation coefficient

- **Advantages** — probabilities obtained are exact; usable when samples are as small as 6, where there is no parametric alternative unless the population distribution is precisely known; suitable for ranked and for simple classificatory data; easier to learn and apply.

- **Disadvantages** — wasteful of data when parametric assumptions are in fact satisfied; **no non-parametric method exists for testing interactions in analysis of variance**; tables of critical values may not be easily available.
- Kruskal–Wallis is the most efficient non-parametric test for k independent samples — **power efficiency 95.5%** against the F test (Sunder Rao & Richard, pp 131–132).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Labels for hand reproduction — "Zones of acceptance and rejection of the null hypothesis (two-tailed test)":

- Bell-shaped normal curve, horizontal axis labelled $(\bar{X}_1 - \bar{X}_2)$
- Central shaded area: **Acceptance region of null hypothesis (H_0)**, marked $Z = \text{or} < 1.96 \text{ SE}$
- Left tail: **Rejection region 2.5%**
- Right tail: **Rejection region 2.5%**
- Cut-off lines at $Z = \pm 1.96 \text{ SE}$, with $Z > 1.96 \text{ SE}$ spanning both rejection regions
- Formula alongside: $SE = \sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$

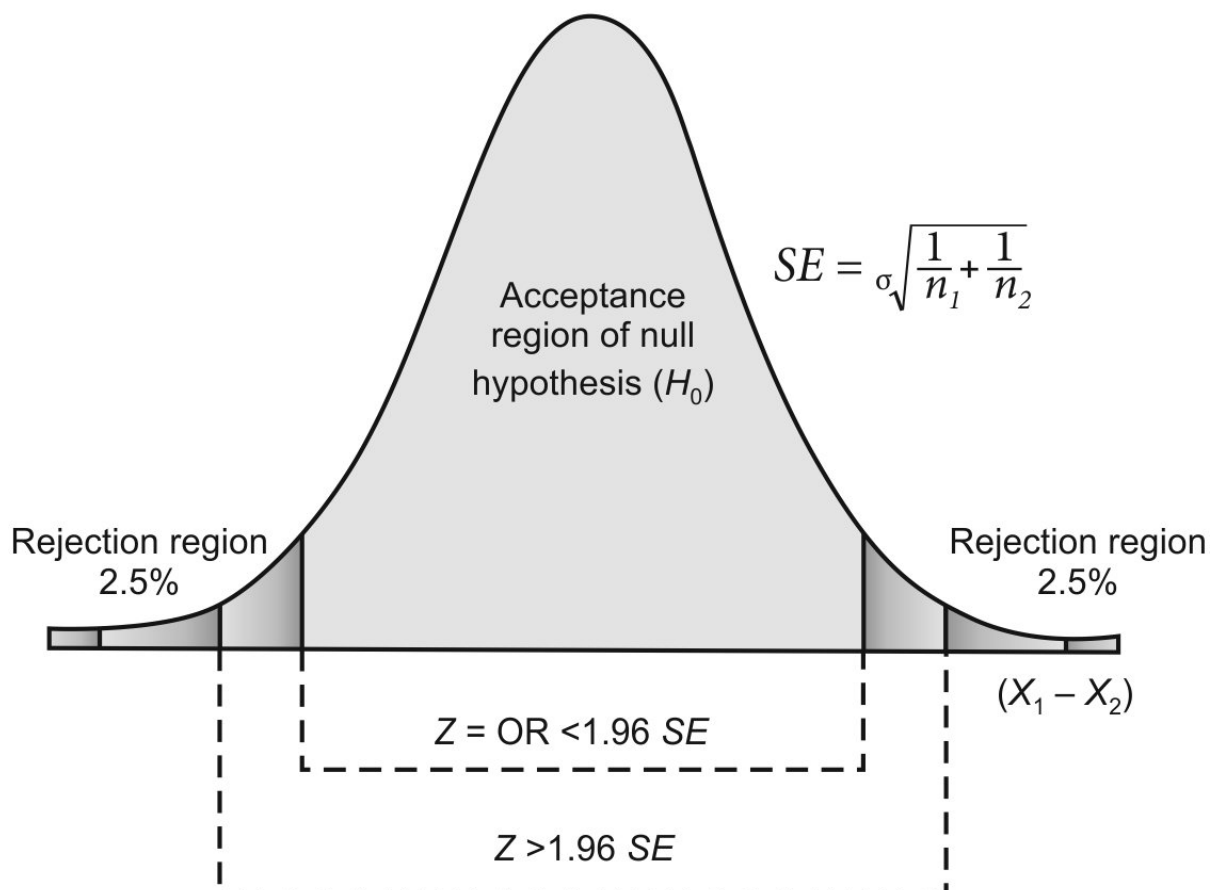
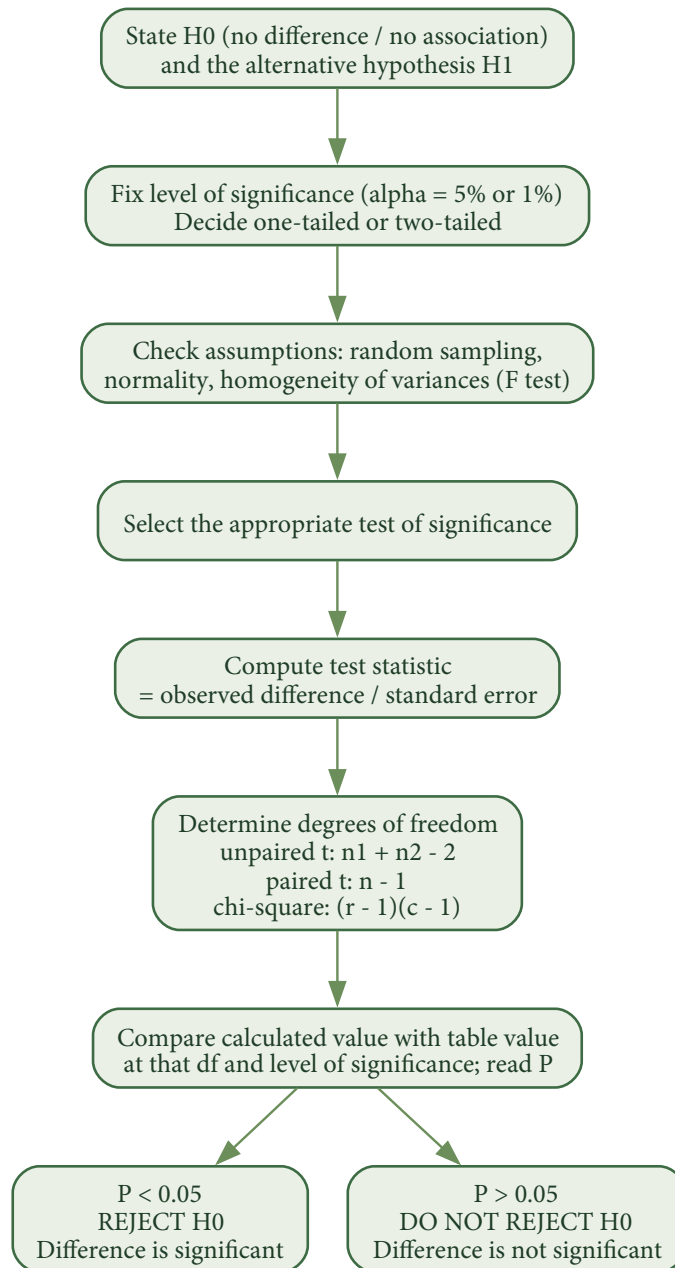
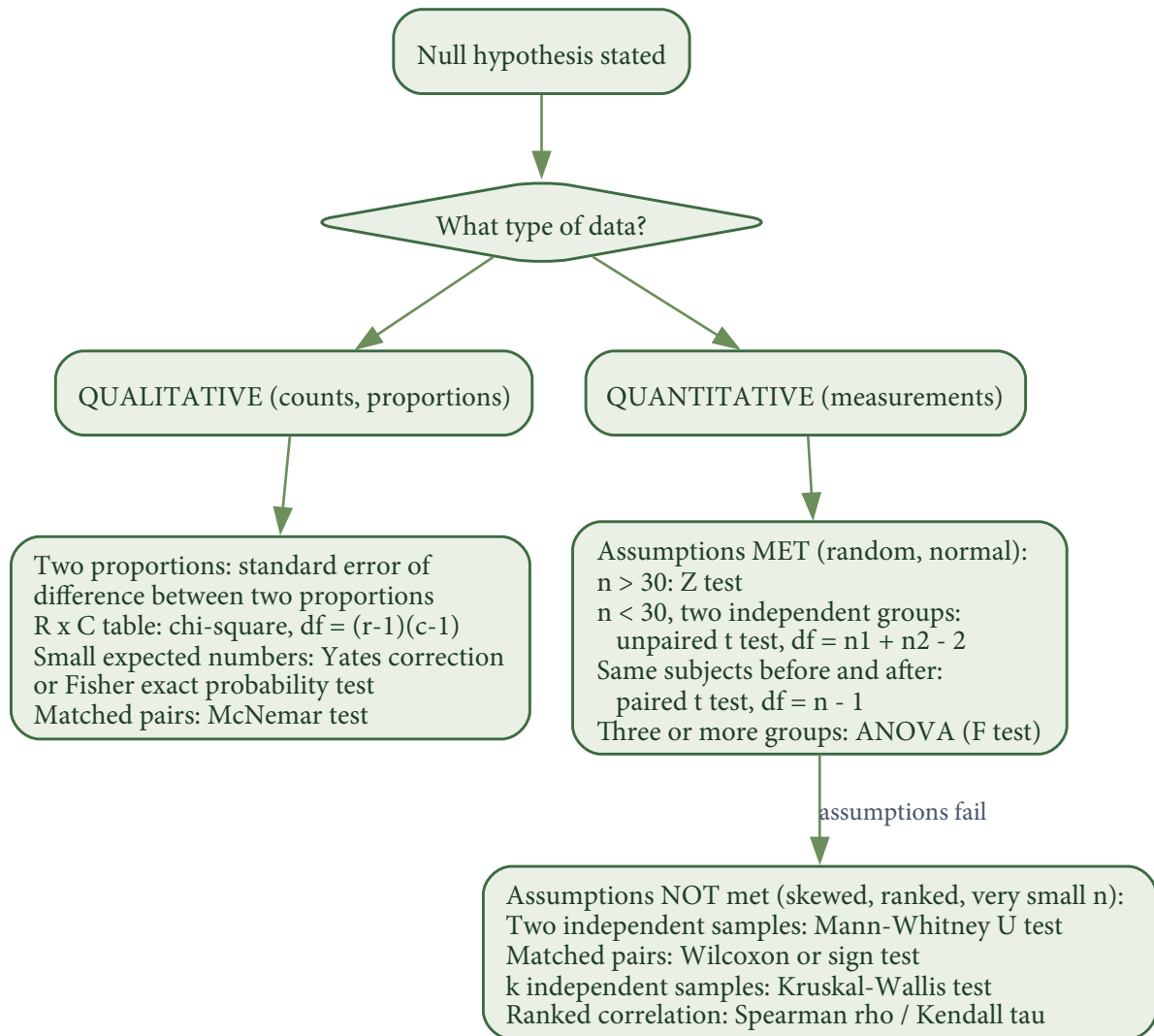


Fig 8.6 — Zones of acceptance and rejection of null hypothesis. Rejection (Mahajan's Methods in Biostatistics for Medical Students and Research Workers (PDFDrive), p.171)



Sequence followed in testing a null hypothesis



Choosing the test that will decide the null hypothesis

High-yield exam pointers

- ▶ H_0 = "no difference, no association"; a test of significance can only **reject** it or **fail to reject** it — H_0 is never proved.
- ▶ **Critical ratio** = **observed difference** ÷ **standard error of the difference**; this single ratio underlies the Z test and both t tests.
- ▶ **Degrees of freedom**: unpaired $t = n_1 + n_2 - 2$; paired $t = n - 1$; chi-square = $(r - 1)(c - 1)$; Kruskal-Wallis $H = k - 1$.
- ▶ **Two-tailed normal deviates**: $\alpha = 0.05 \rightarrow 1.96$; $\alpha = 0.01 \rightarrow 2.575$; $\alpha = 0.001 \rightarrow 3.29$. χ^2 at 1 df, 5% level = 3.84.
- ▶ **Type I error** = α (rejecting a true H_0 , equals the level of significance); **Type II error** = β (missing a real difference); **power** = $1 - \beta$; remedy for β is a **larger sample**.
- ▶ **Yates' correction**: subtract 0.5 from $|O - E|$ when $df = 1$ and any expected cell value is < 5 .
- ▶ Discriminator between t and Z is **sample size** (< 30 vs > 30); between parametric and non-parametric is **normality and level of measurement**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference, not to be written in the exam: Park 28e and Mahajan word the decision as "accept the null hypothesis", whereas the retrieved text of Sunder Rao & Richard words it as "do not reject the null hypothesis" — the latter is the phrasing current statistical practice prefers, and either is accepted by RGUHS examiners.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 21, Health Information and Basic Medical Statistics, pp 1006–1009; Ch 3, Principles of Epidemiology, p 93); Mahajan's Methods in Biostatistics for Medical Students and Research Workers (Ch 8, Sampling Variability and Significance, pp 169–179; Ch 9, Significance of Difference in Means, pp 180–199; Ch 11, The Chi-square Test, pp 220–224); Sunder Rao & Richard — Introduction to Biostatistics and Research Methods (Tests of Significance pp 85–101; The Chi-square Test pp 113–118; Analysis of Variance pp 119–123; Non-parametric or Distribution-free Statistical Tests pp 126–132; Sample Size Determination pp 214–215).



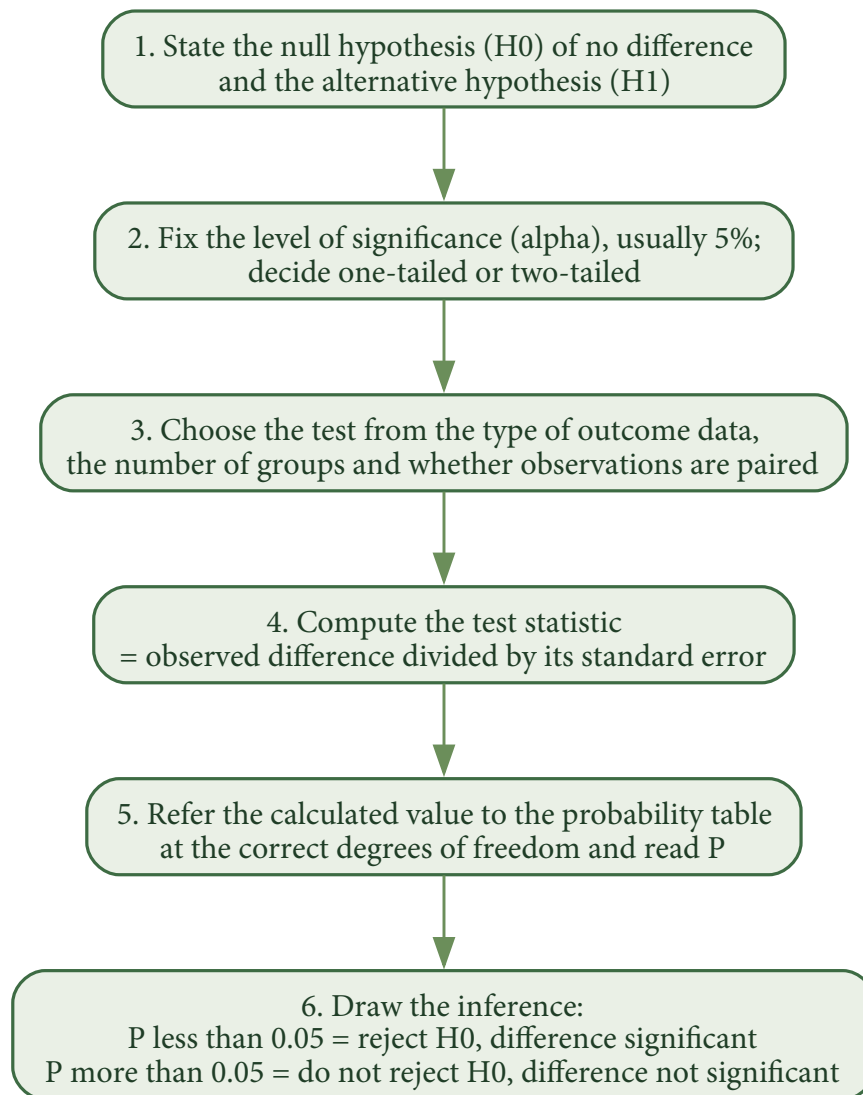
Tests of significance in health research

Asked in: Paper I Nov 2017 — RGUHS MD Community Medicine

Definition — Tests of significance are mathematical methods by which the probability (P), or relative frequency, of an observed difference occurring by chance is found; the difference may be between the mean or proportion of a sample and that of the universe, or between the estimates of an experimental and a control group (Mahajan, Ch 8, p 175). The **Z-test** and **t-test** express the observed difference in terms of its **standard error (SE)**, which is the measure of variation in sample estimates that occurs by chance; the **chi-square test** instead compares observed with expected frequencies. A test of significance quantifies only the play of sampling variability — it does not by itself establish causation.

Logic and steps common to every test

Four problems in health research call for a test of significance (Sunder Rao & Richard, Ch 12, pp 86–88): comparison of a sample mean with a population mean; comparison of two sample means; comparison of a sample proportion with a population proportion; and comparison of two sample proportions. The basis and the sequence of steps are the same for all of them, only the test statistic changes.



The six steps common to every test of significance

Errors in the decision

	Null hypothesis is TRUE	Null hypothesis is FALSE
H0 rejected	Type I error (alpha) — a difference declared where none exists	Correct decision
H0 not rejected	Correct decision	Type II error (beta) — a real difference missed

Type I error is fixed in advance by the level of significance chosen — at the 5% level, 5% of all true null hypotheses will be rejected on average — and in medical studies it is the more serious of the two (Mahajan, Ch 8, pp 174-175). **Power** = **1 – beta** is the numerical value indicating the sensitivity of the test. For a given sample size the two errors cannot be reduced together, so Type I error is fixed at a tolerable limit and Type II error is then minimised by enlarging the sample (Sunder Rao & Richard, Ch 11, pp 86-87).

Critical values of the normal deviate (Z) (Sunder Rao & Richard, Appendix 5, p 259)

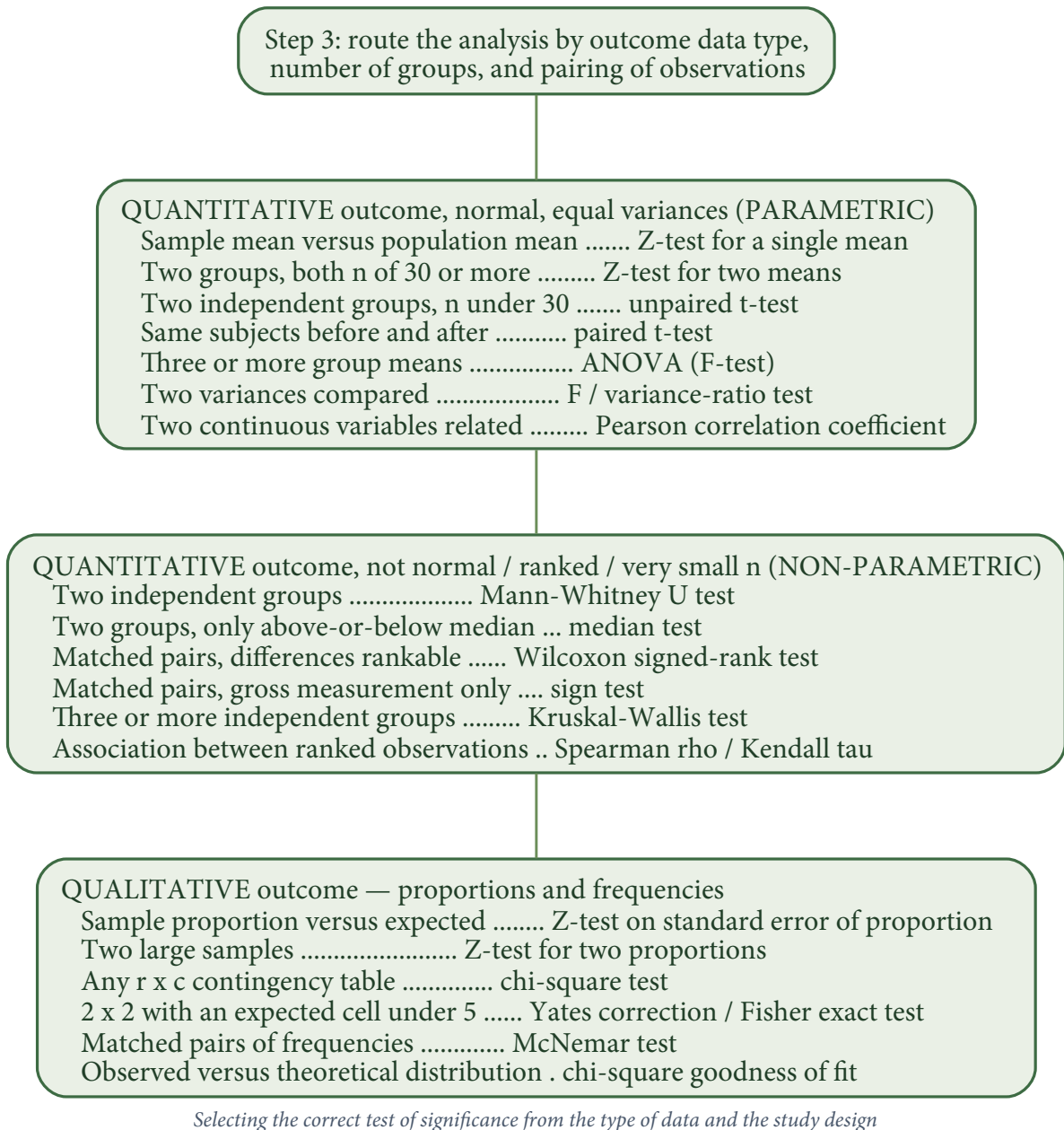
Level of significance (alpha)	Two-tailed Z	One-tailed Z
0.10	1.645	1.282
0.05	1.96	1.645
0.02	2.326	2.054
0.01	2.576	2.326
0.001	3.291	3.090

Choosing between a parametric and a non-parametric test

A parametric test may be applied only when all of the following hold (Mahajan, Ch 9, pp 180, 192-193):

- ▶ The sample is drawn **at random** from the corresponding population.
- ▶ The data are **quantitative** (measured on a numerical scale).
- ▶ The variable is **normally distributed** in the population.
- ▶ There is **homogeneity of variances** — tested by Fisher's *F* or variance-ratio test; if the ratio is significant the samples are heterogeneous and the t-test must not be used.
- ▶ Sample size decides Z versus t: **n of 30 or more** for Z, **n under 30** for Student's t. This is the only criterion on which the two differ.

When the distribution is unknown or not normal, when the data are qualitative or in ranks, or when samples are as small as six, a **non-parametric (distribution-free)** test is used instead; in many of these the measurements are converted to ranks or even to signs (Sunder Rao & Richard, Ch 16, pp 126-127).



Parametric tests, with worked examples from the corpus

Test	Applied to test	Test statistic and degrees of freedom	Example
Z-test on SE of mean	A sample mean against a known population mean, n of 30 or more	$Z = (\text{sample mean} - \text{population mean}) / \text{SE of mean}$; SE = $SD/\sqrt{n}$; significant if Z exceeds 1.96	Mean height of 100 schoolboys compared with the state mean (Mahajan, Ch 8, pp 175-176)

Test	Applied to test	Test statistic and degrees of freedom	Example
Z-test on SE of difference between two means	Two large-sample means	$SE(d) = \sqrt{(SD_1^2/n_1 + SD_2^2/n_2)}$; difference more than twice $SE(d)$ is significant	Kidney weight of 12 control mice (mean 318 mg, SD 10.2) versus 12 drug-treated mice (mean 370 mg, SD 24.1): $SE(d) = 7.5$, observed difference 52 mg, i.e. more than twice SE — significant (Park 28e, Ch 21, p 1007)
Z-test on SE of proportion	A sample proportion against an expected proportion (data in only two classes: male/female, sick/healthy, success/failure)	$SE(p) = \sqrt{(pq/n)}$; relative deviate = (expected – observed) / SE	Expected proportion of males 52%; a sample of 100 gave 40%. $SE = \sqrt{(52 \times 48/100)} = 5.0$, 95% limits 42–62, relative deviate = $12/5 = 2.4$ — exceeds 2, so significant (Park 28e, Ch 21, p 1007)
Z-test on SE of difference between two proportions	Two attack rates, cure rates or prevalences from large samples	$SE(p_1 - p_2) = \sqrt{(p_1q_1/n_1 + p_2q_2/n_2)}$	Failure rates of drug A (36/120) versus drug B (16/80) in toothache (Mahajan, Ch 17, p 357); prevalence of diabetes in rural versus urban Ahmedabad (Mahajan, Ch 17, pp 356-357)
Student's t-test — unpaired (W.S. Gossett, pen-name <i>Student</i>)	Two independent small samples drawn from two populations	$t = \text{observed difference} / SE \text{ of difference}$; $df = n_1 + n_2 - 2$	Mean chest circumference of two groups: $t = 6.738$ at 18 df; the largest tabled t at 18 df is 3.92 at $\alpha 0.001$, so $P < 0.001$ (Mahajan, Ch 19, p 394)
Student's t-test — paired	Correlated observations on the <i>same</i> individuals, before and after, or the same character at two sites	$t = \text{mean of differences} / SE \text{ of mean difference}$; $df = n - 1$	Blood pressure of every patient measured before and again after treatment; pairing removes individual sampling variation (Mahajan, Ch 9, pp 198-199; Ch 19, p 394)
F-test (variance-ratio test)	Homogeneity of two variances — the pre-condition for t	$F = \text{larger variance} / \text{smaller variance}$; $df = n_1 - 1$ and $n_2 - 1$	Two height samples of 9 and 10 observations, sums of squares 36 and 42: F far below the tabled 5% value, so variances homogeneous and the t -test may proceed (Mahajan, Ch 9, pp 202-203)
ANOVA — one-way	Simultaneous comparison of three or more means	$F = \text{between-group variance} / \text{within-group variance}$; $df = (k - 1)$ and $(n - k)$	Haemoglobin of 33 children on three different diets (means 11.27, 8.96, 10.61 g%): $F = 16.54$ with $df 2$ and 30, against tabled 3.32 at 5% and 5.39 at 1% — groups differ, $P < 0.01$ (Sunder Rao & Richard, Ch 15, pp 120-123)

Test	Applied to test	Test statistic and degrees of freedom	Example
ANOVA — two-way	Effect of two factors simultaneously	Separate F for each factor from the ANOVA table	Days of hospitalisation by age group and by sex: age groups $F = 9.182$, $P = 0.027$ (significant); sex $F = 0.416$, $P = 0.554$ (not significant) (Sunder Rao & Richard, Ch 15, pp 123-124)

ANOVA acts as a **screening test** — a significant F says that at least one group differs from the others, and further examination is then needed to identify which group is the distinctive one (Principles of Medical Statistics, Ch 29, pp 624-625).

The chi-square test — the workhorse for qualitative data

Devised by *Karl Pearson* in 1889 as an index of dispersion, $\chi^2 = \sum (O - E)^2 / E$, where O is the observed and E the expected frequency in each cell, summed over all cells; $df = (c - 1)(r - 1)$, where c is the number of columns and r the number of rows (Sunder Rao & Richard, Ch 14, p 112; Mahajan, Ch 11, p 223; Park 28e, Ch 21, p 1008). It has the advantage over the Z-test for proportions that it can be used when there are **more than two groups or more than two classes** (Park 28e, Ch 21, p 1008).

Application	What is tested	Example
Test of association (independence) — the commonest use	Whether two attributes occur together more often than by chance, in a fourfold (2×2) contingency table	Severity of disease by blood group: 9.4% of group 'O' severe versus 6.1% of others; $\chi^2 = 5.536$ at 1 df, $0.01 < P < 0.02$ — the two are associated (Sunder Rao & Richard, Ch 14, pp 113-115). Diabetes by residence in 7,011 subjects: $\chi^2 = 25.472$ at 1 df, $P < 0.001$ (Mahajan, Ch 19, pp 399-400)
Test of proportions / homogeneity — larger than fourfold table	Whether several proportions are similar to one another	Microfilaria positivity across social classes I–IV (positivity 5%, 10%, 12%, 20%): a 2-column $\times$ 4-row table, $df = (2 - 1)(4 - 1) = 3$ (Mahajan, Ch 11, pp 220-223)
Test of goodness of fit	Whether an observed frequency distribution fits a hypothetical or theoretical distribution; k = number of classes	Seasonal variation in disease onset over 12 months against an even expectation of 6 per month: $\chi^2 = 22.02$ at 11 df, $0.02 < P < 0.05$ — seasonality present (Sunder Rao & Richard, Ch 14, pp 116-117)

Restrictions and corrections

- In a fourfold table with 1 df, χ^2 is unreliable if the **expected value in any cell is less than 5**; Yates's **correction for continuity** is then applied — reduce $|O - E|$ by 0.5 before squaring (Mahajan, Ch 11, pp 223-224; Sunder Rao & Richard, Ch 14, p 117). In the blood-group example the corrected χ^2 fell from 5.536 to 5.059 — still significant at 5% but no longer at 2%.
- **Fisher's exact probability test** replaces χ^2 when the sample is small and the scores fall into two mutually exclusive classes (Sunder Rao & Richard, Ch 16, pp 127-128).
- χ^2 must be calculated on the **actual numbers observed in each category** — **never on a proportion or a percentage**; for a given percentage abnormal the value of χ^2 can be anything

from almost zero to a very large number depending on the sample size (Sunder Rao & Richard, Ch 14, p 118). The tabled 5% value at 1 df is **3.84** (Sunder Rao & Richard, Ch 14, p 113).

- ▶ χ^2 is **only a test of significance, not a measure of the strength of association**: a large χ^2 means only that such a result is unlikely to have arisen by chance, not that the association is stronger (Sunder Rao & Richard, Ch 14, pp 118-119).
- ▶ It is a **large-sample approximation**; if more than one cell has an expected number of less than 1, χ^2 should not be applied at all, and combining neighbouring classes to meet the expected-count rule weakens the sensitivity of the test (Sunder Rao & Richard, Ch 14, pp 118-119).

Non-parametric (distribution-free) tests

Non-parametric test	Parametric counterpart	Used when	Example
Mann-Whitney U test	Unpaired t-test	Two independent samples, weaker (ordinal) measurements	Pain scores of two treatment groups
Median test	Unpaired t-test	Data cannot even be meaningfully ranked; only dichotomised as above or below the combined median (use Fisher's test if the combined n is small)	Duration of stay above/below median in two wards
Wilcoxon signed-rank test	Paired t-test	Matched pairs whose differences can be meaningfully ranked	Drug effect in 11 paired mice: smaller rank sum $T = 14$ against a critical T of 11 at $P = 0.05$ for $n = 11$ — H_0 not rejected (Sunder Rao & Richard, Ch 16, p 130)
Sign test	Paired t-test	Matched pairs on a variable with underlying continuity but measured only grossly (only the direction of change is known)	Better/worse after a health-education session
McNemar test	—	Matched pairs where the data are frequencies in categories (concordance)	Ranks given to two food preparations by 8 judges (Sunder Rao & Richard, Ch 16, p 128)
Kruskal-Wallis one-way ANOVA by ranks	One-way ANOVA	Three or more independent samples; H is distributed as χ^2 with $k - 1$ df; power efficiency 95.5% relative to the F-test	Three groups ranked jointly: $H = 1.2$ at 2 df, $P > 0.5$ — H_0 not rejected (Sunder Rao & Richard, Ch 16, pp 130-132)
Kolmogorov-Smirnov test	χ^2 goodness of fit	Goodness of fit when the variable has a continuous distribution	Fit of birth weights to an assumed distribution
Spearman's rank correlation (rho), Kendall's tau	Pearson's correlation coefficient	Association when the observations are in ranks	Agreement of two nutritionists ranking 10 children: $r_s = 0.855$, $P = 0.002$ — significant agreement (Sunder Rao & Richard, Ch 16, p 275)

Advantages (Sunder Rao & Richard, Ch 16, pp 126-127): the probability statements are exact; with samples as small as six there is no alternative unless the population distribution is precisely known; tests exist for data inherently in ranks and for simply classificatory data; and they are easier to learn and apply. **Limitation** — where the parametric assumptions do hold, the non-parametric test is the less powerful of the two, so a real difference is more likely to be missed.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Reproduce the normal curve of the sampling distribution with these exact labels:

- ▶ Bell-shaped curve, x-axis labelled $(X_1 - X_2)$, the observed difference between two means
- ▶ Central shaded area: **Acceptance region of the null hypothesis (H_0)**
- ▶ Left tail: **Rejection region 2.5%**; right tail: **Rejection region 2.5%** (together the 5% level, two-tailed)
- ▶ Vertical cut-offs marked $Z = 1.96 SE$ on either side of the centre
- ▶ Bracket under the central zone: **Z equal to or less than 1.96 SE — difference not significant**
- ▶ Bracket under the whole span: **Z more than 1.96 SE — difference significant, H_0 rejected**
- ▶ Formula written alongside: $SE = \sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$

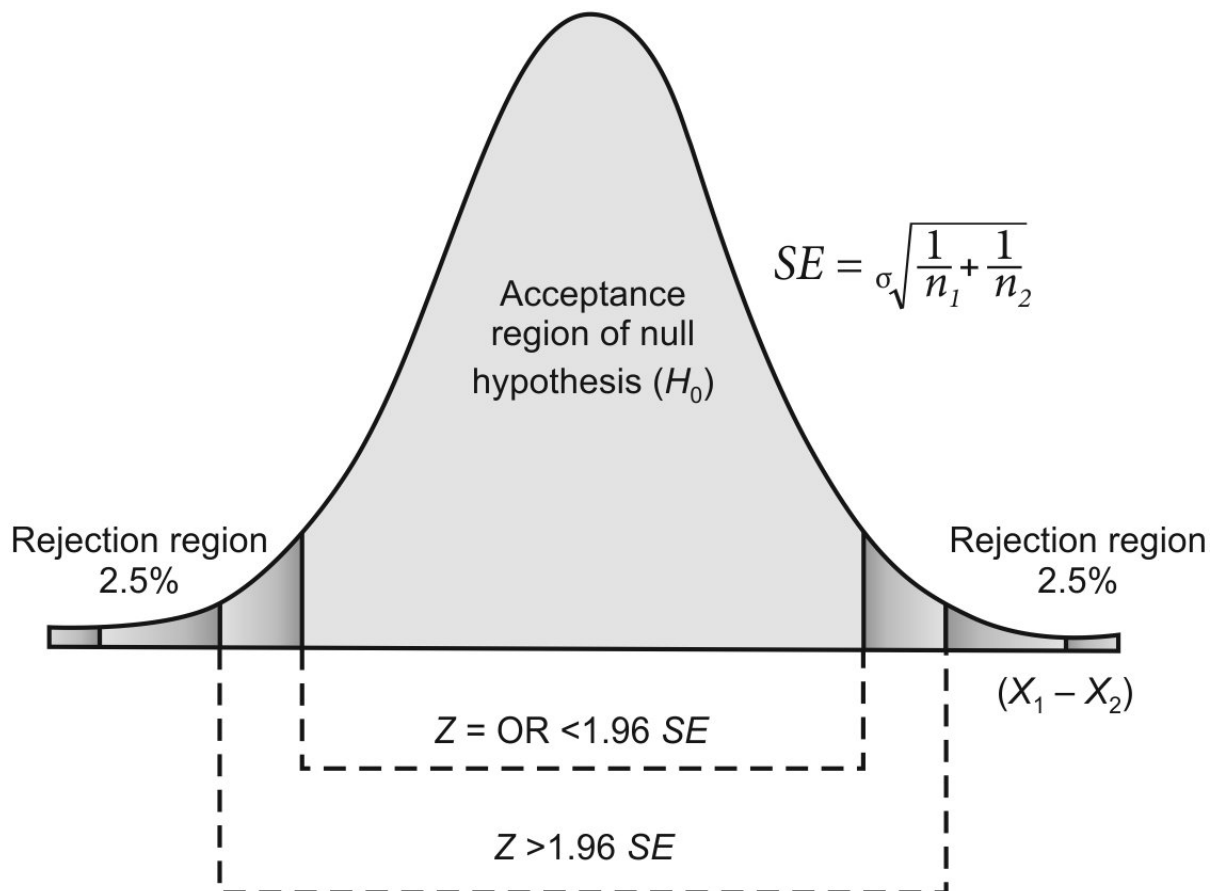


Fig 8.6 — Zones of acceptance and rejection of null hypothesis. Rejection (Mahajan's Methods in Biostatistics for Medical Students and Research Workers (PDFDrive), p.171)

High-yield exam pointers

- ▶ Mnemonic for the sequence — **H-A-T-S-P-I**: Hypothesis, Alpha, Test chosen, Statistic computed, P read from table, Inference drawn.
- ▶ **Test statistic = observed difference ÷ standard error** — this single sentence covers Z, t and F; χ^2 alone works on O and E frequencies.
- ▶ Degrees of freedom to write without hesitation: unpaired t = $n_1 + n_2 - 2$; paired t = $n - 1$; $\chi^2 = (r - 1)(c - 1)$; Kruskal-Wallis H = $k - 1$; F = $n_1 - 1$ and $n_2 - 1$.
- ▶ Two-tailed critical values: Z = 1.96 at 5% and 2.576 at 1%; $\chi^2 = 3.84$ at 1 df, 5%.
- ▶ Z versus t differ on one criterion only — **sample size 30 or more versus less than 30**; both need random sampling, quantitative normally distributed data and homogeneous variances (F-test).
- ▶ **Yates's correction** — subtract 0.5 from $|O - E|$ in a 2×2 table when any expected cell is under 5; use **Fisher's exact test** if the sample is smaller still.
- ▶ **Type I error (alpha)** is fixed by the level of significance and is the more serious in medical research; **Type II error (beta)** is reduced by enlarging the sample; **power = 1 - beta**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Some journals accept the reporting convention $P < 0.05$ or $P < 0.01$, while others require the exact probability to three or four decimal places, and a computer output printed as $P = 0.000000$ should be read as a very small probability with the null hypothesis rejected (Sunder Rao & Richard, Ch 12, pp 100-101).

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e, Ch 21 (Health Information and Basic Medical Statistics), pp 994-1008; Mahajan's Methods in Biostatistics for Medical Students and Research Workers, Ch 8 (Sampling Variability and Significance), Ch 9 (Significance of Difference in Means), Ch 10 (Significance of Difference in Proportions), Ch 11 (The Chi-square Test) and Ch 19 (SPSS Manual); Sunder Rao & Richard — Introduction to Biostatistics and Research Methods, Ch 11 (Basis of Statistical Inference), Ch 12 (Tests of Significance and Estimation), Ch 14 (The Chi-square Test), Ch 15 (Analysis of Variance) and Ch 16 (Non-parametric or Distribution-free Statistical Tests); Principles of Medical Statistics, Ch 13 (Parametric Sampling: Z and t Tests) and Ch 29 (Analysis of Variance).



Sampling methods

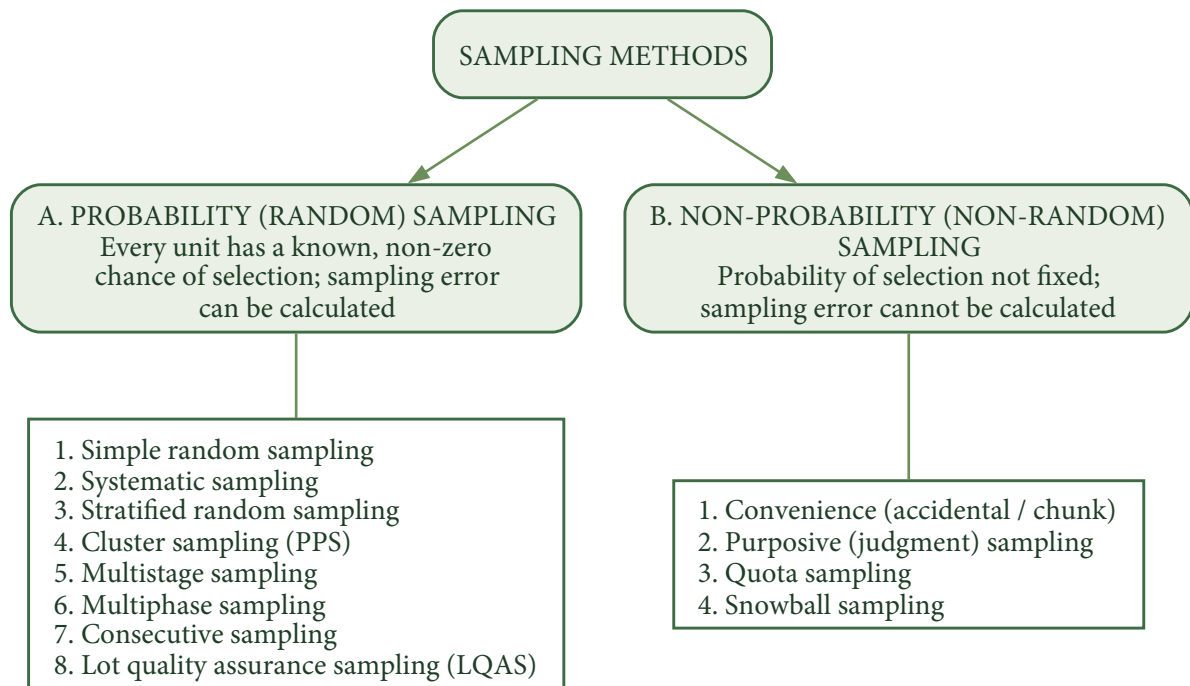
Asked in: Paper I May 2019 — RGUHS MD Community Medicine

Definition — Sampling is the process of choosing a small group (the **sample**) from the totality of individuals carrying a specified characteristic (the **population** or **universe**), so that the characteristics of the population can be estimated from the sample (Sunder Rao & Richard, Ch 10, p 62). Because studying every member is costly, slow and often less accurate — a large field force cannot be kept uniform — an appropriately drawn sample is preferred to complete enumeration (Mahajan, Ch 6, p 113). A value calculated from the population is a **parameter**; the corresponding value calculated from the sample is a **statistic**, and the two objectives of sampling are estimation of parameters from statistics and testing of hypotheses about the population (Mahajan, Ch 6, pp 113–114).

Basic terms that must be defined before any sample is drawn

Term	Meaning
Population / universe	Totality of all individuals with the specified characteristic (Sunder Rao, Ch 10, p 62). It must be further qualified before sampling — which doctors, those in service or in general practice; which women, married or unmarried (Mahajan, Ch 6, p 113); for a patient population, state the type of patients and the stage of disease (Sunder Rao, Ch 10, p 63). Types — finite, infinite, hypothetical (Sunder Rao, Ch 10, p 63)
Sampling unit	Each member of the population — a person, family, house, street or ward. Units must not overlap and together must constitute the whole population (Sunder Rao, Ch 10, p 69)
Sampling frame	The listing of all members of the universe from which the sample is drawn; its accuracy and completeness decide the quality of the sample (Park 28e, Ch 21, p 994)
Sample size (n)	Number of individuals/observations in the sample (Sunder Rao, Ch 10, p 62). Ordinarily should not be below 30 (Mahajan, Ch 6, pp 114–115)
Sampling interval (k)	$k = N/n$, where N = population size and n = sample size; Sunder Lal calls this the <i>sampling interval</i> or <i>sampling ratio</i> . For 1000 families with $n = 100$, $k = 10$ — every 10th family is taken (Sunder Lal, Ch 6, p 219)
Sampling fraction (f)	$f = n/N$ — the fraction of the whole that is included in the sample (Fletcher, <i>Clinical Epidemiology: The Essentials</i> , Ch 3); here $100/1000 = 1/10 = 10\%$. Proportionate stratified sampling applies the <i>same</i> sampling fraction to every stratum (Sunder Lal, Ch 6, p 219). Do not confuse f with k — they are reciprocals
Precision	Depends on sample size; precision is proportional to $\sqrt{n}$ and inversely proportional to the standard deviation — a 4-fold rise in n doubles precision (Mahajan, Ch 6, pp 115–116)

Classification of sampling methods



Classification of sampling methods

Probability sampling is the only design that permits descriptive statistics and tests of significance to be legitimately applied to the population, because the chance error can itself be calculated (Mahajan, Ch 6, p 120; Browner, DCR 5e, Ch 3, pp 31–33). Non-probability methods may be convenient and cheap, but their real defect is that there is no yardstick against which representativeness or precision can be measured, so the results are unconvincing (Sunder Rao, Ch 10, p 65).

Probability (random) sampling methods

Method	Procedure	Worked example	Merits	Limitations
Simple random sampling (unrestricted random sampling)	Every unit of the population has an equal chance of selection. Number all units in the frame, then select by lottery method or a table of random numbers (Mahajan, Ch 6, p 121; Park 28e, Ch 21, p 994)	To put 10 of 100 admitted patients on trial: number the 100 cards, shuffle, draw with replacement, rejecting repeats (Mahajan, Ch 6, p 121)	Simplest; eliminates conscious or unconscious personal selection; provides the greatest number of possible samples (Park 28e, Ch 21, p 994)	Needs a complete frame; impracticable for large scattered populations; sub-groups may be missed

Meth od	Procedure	Worked example	Merits	Limitations
System atic sampli ng	Number all units serially, compute the sampling interval k , choose a random start between 1 and k , then take every k -th unit thereafter (Sunder Lal, Ch 6, p 219)	1000 families, $n = 100 \rightarrow k = 10$; random start 7 $\rightarrow$ families 7, 17, 27, 37... (Sunder Lal, Ch 6, p 219). Filaria survey: number the houses, random start between 1 and 10, then every 10th house (Park 28e, Ch 21, p 994)	Convenient and simple; a complete frame need not be prepared in advance; all sections of the list are represented; accurate when the population is large, homogeneous and numbered (Sunder Rao, Ch 10, p 67; Mahajan, Ch 6, p 124)	Number of possible samples is greatly reduced (Park 28e, Ch 21, p 994); periodicity in the list biases the estimate; locked/vacant houses and non-residential premises fall in the series (Sunder Rao, Ch 10, p 67)
Stratifi ed rando m sampli ng	Used when the population is not homogeneous . Divide it into internally homogeneous groups (strata) by the characteristic of interest, prepare a frame for each stratum, then draw a simple random sample from every stratum, usually with the same sampling fraction (Mahajan, Ch 6, p 124; Sunder Lal, Ch 6, p 219)	Village split into landowning and landless strata for a malnutrition survey (Sunder Lal, Ch 6, p 219); population divided into five socio-economic strata, 10% from large strata and 50% or more from small strata so that no sub-sample is below 30 (Mahajan, Ch 6, p 124)	Proportionate representation of every stratum; greater accuracy than simple random sampling; minority sub-groups can be assured adequate numbers by varying the sampling fraction (Mahajan, Ch 6, p 124; Sunder Lal, Ch 6, p 219)	A separate frame is needed for each stratum; varying the sampling fraction between strata destroys the proportional representativeness of the sample as a whole (Sunder Lal, Ch 6, p 219)
Cluster sampli ng (with probabi lity proport ional to size, PPS)	The sampling unit is a natural group — village, ward, block, slum, school, factory — not an individual. Clusters are selected with probability proportional to their population, then all (or a fixed number of) units within the chosen cluster are studied (Sunder Rao, Ch 10, p 67; Sunder Lal, Ch 6, p 219)	Immunisation-status survey: select schools rather than children, then study all class-1 children of the selected schools (Sunder Rao, Ch 10, p 67)	No frame of individuals required, cutting the cost of frame preparation and of travel; prevents "packing" of neighbouring households into a selected house (Sunder Lal, Ch 6, p 220); field work is simpler, quicker and cheaper (Mahajan, Ch 6, p 125)	Higher standard error / sampling error than a simple random sample of the same size (Mahajan, Ch 6, p 125; Sunder Lal, Ch 6, p 221); comparisons <i>between</i> individual clusters are not valid (Sunder Rao, Ch 10, p 68)

Meth od	Procedure	Worked example	Merits	Limitations
Multistage sampling	Sampling is carried out in successive stages, a random technique being applied at each stage: state → district → taluka → village → household (Mahajan, Ch 6, p 124). Selected clusters are themselves sub-sampled instead of fully inspected (sub-sampling), giving two-stage and higher-order designs (Sunder Rao, Ch 10, pp 68–69)	Hookworm survey in a district — choose 10% of the villages in the talukas, then examine stools of all persons in every 10th house (Mahajan, Ch 6, p 124)	Introduces flexibility not available in other techniques; uses the existing administrative divisions and sub-divisions, saving labour (Mahajan, Ch 6, p 124)	Sampling error accumulates across stages; estimation of error is more complex
Multiphase sampling	Part of the information is collected from the whole sample and the remaining information from a subsample , the numbers falling at every succeeding phase so that the complicated and costly procedure is reserved for the last and smallest phase (Park Plus, Ch 19, pp 238–239)	Cross-sectional nutrition survey — phase 1: knowledge, attitude and practice (KAP) study on all families; phase 2: dietary intake on a subsample; phase 3: anthropometry on a subsample of phase 2; phase 4: biochemical estimation of nutrients on a subsample of phase 3 (Park Plus, Ch 19, p 239). Tuberculosis survey — Mantoux test in all, sputum examination only in the Mantoux-positive (AFMC Red Book 2024, Ch 26, p 739)	Makes the study less expensive, less time-consuming, less laborious and more purposeful, because the costly test is applied to a shrinking subsample (Park Plus, Ch 19, p 239; AFMC Red Book 2024, Ch 26, p 739)	Only the first phase carries the full sample; every later phase has a smaller n, so the costliest estimates are the least precise, precision varying with $\sqrt{n}$ (Mahajan, Ch 6, p 115)
Consecutive sampling	Random sampling in which the sampling interval is zero — every successive eligible attender meeting the selection criteria is recruited in the order of attendance until the required sample size or the survey period is completed (Sunder Lal, Ch 6, p 219)	Annual HIV Sentinel Surveillance among antenatal clinic attenders, used specifically to remove selection bias (Sunder Lal, Ch 6, p 219)	No prior enumeration of the accessible population needed; best when recruitment runs long enough to cover seasonal variation (Browner, DCR 5e, Ch 3, p 33)	Generalisability is limited — early attenders may differ from later ones (Browner, DCR 5e, Ch 3, p 33)

Method	Procedure	Worked example	Merits	Limitations
Lot quality assurance sampling (LQAS)	Borrowed from industrial quality control; small samples are surveyed in two or more geographical "lots" with the same objective, and the weighted results of each lot are compared against a pre-set acceptable threshold (Sunder Lal, Ch 6, p 221)	Used to identify the low-vaccination-coverage areas among five cities in Bangladesh (Sunder Lal, Ch 6, p 221)	Rapidly separates the sets of population giving acceptable results from those giving unfavourable results and requiring urgent intervention, so effort can be targeted (Sunder Lal, Ch 6, p 221)	Identifies priority areas against a benchmark rather than yielding a precise coverage estimate for a single lot (Sunder Lal, Ch 6, p 221; Park Plus, Ch 19, p 239)

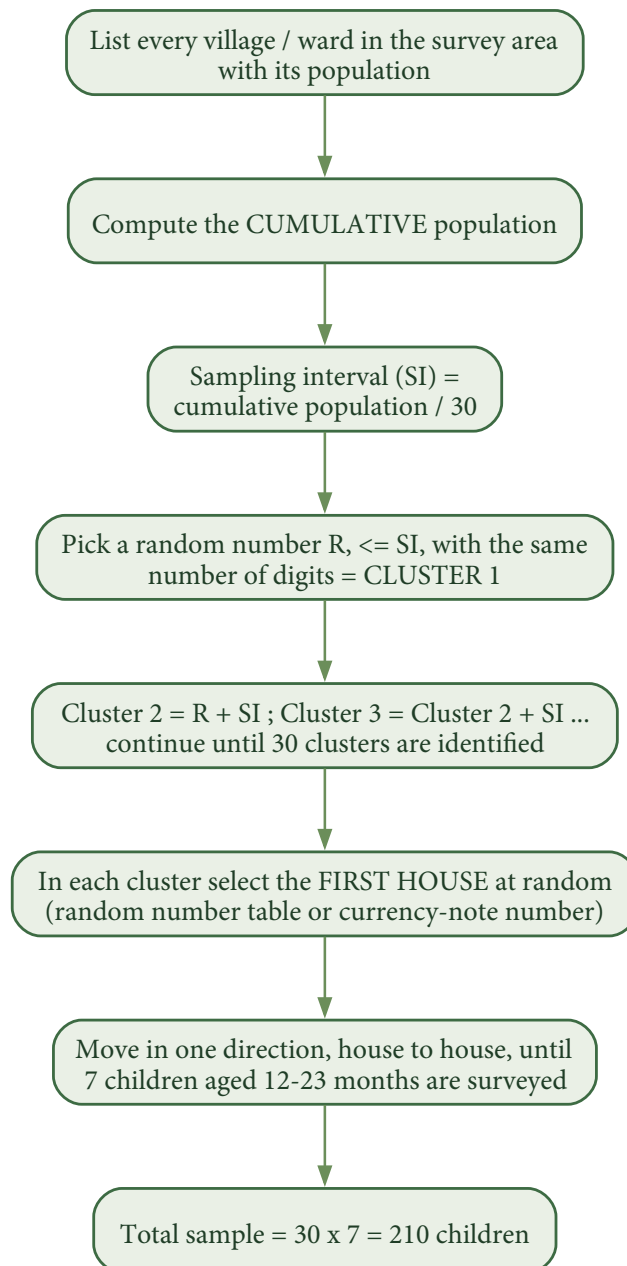
Non-probability (non-random) sampling methods

Method	How the sample is chosen	Where used	Chief defect
Convenience sampling (also <i>accidental, accessibility, haphazard, chunk</i>)	Whoever is easiest for the investigator to reach is enrolled (Sunder Rao, Ch 10, p 65; Sunder Lal, Ch 6, p 221)	Pilot studies, to gain a first idea of the study variable (Sunder Lal, Ch 6, p 221)	Maximum bias; representativeness is disregarded; conclusions usually unsatisfactory
Purposive / judgment sampling	The investigator deliberately exercises subjective choice to obtain what he regards as a "representative" sample (Sunder Rao, Ch 10, p 65)	When the universe is small, or an urgent public-health/policy problem will not wait for a probability design (Sunder Lal, Ch 6, p 221)	Distortion from personal prejudice or ignorance of key features of the population structure
Quota sampling	The universe is divided into quotas by age, sex, socio-economic status, religion; the number in each quota may be statistically fixed, but the individuals within it are picked by the interviewer's judgment (Sunder Rao, Ch 10, p 65; Sunder Lal, Ch 6, p 221)	Public-opinion and market surveys; the commonest non-random method	Judgment plus convenience — subjectivity within the quota cannot be removed even with careful instruction
Snowball sampling	A few known participants are recruited, each of whom names further eligible participants, until the sample size is reached (Sunder Lal, Ch 6, p 221)	Hard-to-reach populations with no sampling frame — female sex workers, people who inject drugs (Sunder Lal, Ch 6, p 221)	Sample is confined to networks of the initial contacts; no basis for estimating precision

WHO 30 × 7 cluster survey — the standard field application

Cluster sampling with PPS is the design used for immunisation coverage evaluation under the Expanded Programme on Immunization (EPI) and the Universal Immunization Programme (UIP), and the same

two-stage method has been applied to diarrhoeal disease surveys, leprosy elimination monitoring and evaluation of the national AIDS programme (Mahajan, Ch 6, p 125; Sunder Lal, Ch 6, p 220).



Steps in the WHO 30-cluster (30 x 7) immunisation coverage evaluation survey

Reliability standards accepted for this design: accuracy ± 10 percentage points (a survey figure of 70% means true coverage lies between 60% and 80%), confidence level 95%, results reflecting coverage of the one year preceding the survey, and representing the **area as a whole** (Sunder Rao, Ch 10, p 68; Mahajan, Ch 6, p 125).

Errors arising in a sample survey

Feature	Sampling error	Non-sampling error
Origin	Data were gathered from a sample rather than from the entire population; inherent chance variation from one sample to another (Park 28e, Ch 21, p 994; Sunder Rao, Ch 10, p 64)	Inadequately calibrated instruments, observer variation, incomplete coverage of the selected subjects, conceptual errors, faulty definitions, non-response, careless recording (Park 28e, Ch 21, p 994; Sunder Rao, Ch 10, p 64)
Determinants	(a) Size of the sample and (b) natural variability of the individual readings (Park 28e, Ch 21, p 994)	Quality of planning, tools, training and supervision
Behaviour	Decreases as sample size increases; cannot be abolished but is minimised by a suitable sampling method (Park 28e, Ch 21, p 994; Sunder Rao, Ch 10, p 64)	Independent of sample size; often more important than the sampling error (Park 28e, Ch 21, p 994)
Measurable?	Yes — quantifiable only in probability designs	No
Remedy	Probability sampling; adequate n	Standardised definitions and tools, training, blinding, supervision, monitoring, control of non-response (Sunder Rao, Ch 10, p 64; Mahajan, Ch 6, pp 119–120)

Bias in sampling is separate from sampling error: it drives the sample statistic systematically away from the population parameter. The classical example is the mean birth weight of 50 babies born in a private maternity home, which validly estimates the mean weight in the affluent class but seriously **overestimates** it for a mixed population; likewise cure rates from private practice cannot be compared with hospital cases, who present late and come from lower socio-economic strata (Mahajan, Ch 6, p 119).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Population, sampling frame and sample** — label: *Universe (N) → Sampling frame (complete listing of units) → Sample (n) → Statistic ($\bar{X}$, s , p) estimating Parameter (μ , σ , P)*, with a two-headed arrow marked *sampling error* between statistic and parameter.
- **The probability designs, shown as grids of dots** — label each: (a) *Simple random* — scattered shaded dots; (b) *Systematic* — every k -th dot shaded, random start marked; (c) *Stratified* — grid split into strata I, II, III with shaded dots in each; (d) *Cluster* — two or three whole blocks fully shaded; (e) *Multistage* — district → taluka → village → household, one shaded at each stage; (f) *Multiphase* — the same grid drawn four times, the shaded area shrinking from phase 1 to phase 4.

High-yield exam pointers

- Two objectives of sampling — **estimation of parameters** and **testing of hypotheses**; two characteristics of a representative sample — **precision** (size, ordinarily ≥ 30) and **unbiased character** (Mahajan, Ch 6, pp 114–115).

- ▶ Do not swap the two ratios: sampling **interval** $k = N/n$, sampling **fraction** $f = n/N$; the random start must lie between 1 and k .
- ▶ Precision $\propto \sqrt{n} / \text{SD}$ — increasing n four-fold **doubles** precision; nine-fold trebles it (Mahajan, Ch 6, p 116).
- ▶ WHO cluster survey = 30 clusters $\times$ 7 children aged 12–23 months = 210 children; SI = cumulative population $\div$ 30; cluster selection is PPS.
- ▶ Cluster sampling gives a **higher standard error** than simple random sampling of the same size — the price paid for cheaper, faster field work.
- ▶ Multiphase = whole sample for the cheap test, shrinking subsamples for the costly ones (KAP $\rightarrow$ diet $\rightarrow$ anthropometry $\rightarrow$ biochemistry). Multistage $\neq$ multiphase: multistage narrows the *sampling units*, multiphase narrows the *investigations*.
- ▶ Park lists **three** commonest methods (simple random, systematic, stratified random); write all eight probability methods plus the four non-probability methods for a 10-mark answer.
- ▶ Sampling error falls as n rises; **non-sampling error does not**, and is often the larger of the two.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Sunder Lal's textbook classifies **consecutive sampling** under random (probability) methods because its sampling interval is zero, whereas Designing Clinical Research 5e classifies it as non-probability, and the same book (Chapter 20 answer key) also classifies **systematic sampling** as non-probability, while Park, Mahajan, Sunder Rao and Sunder Lal all treat systematic sampling as a random method — write the Indian-textbook position in the exam and give this caveat only if asked to justify.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Chapter 21, Health Information and Basic Medical Statistics (Sampling; Sampling Errors; Non-Sampling Errors), pp 993–994. Park Plus — Chapter 19, Health Information and Basic Medical Statistics, pp 234–239. Mahajan's Methods in Biostatistics for Medical Students and Research Workers — Chapter 6, Sampling, pp 113–127. Sunder Rao & Richard, Introduction to Biostatistics and Research Methods — Chapter 10, Sampling, pp 62–70. Sunder Lal, Textbook of Community Medicine — Chapter 6, Basic Statistics and its Applications Including Research Methodology, pp 217–221. AFMC Red Book of Community Medicine 2024 — Chapter 26, Epidemiology and Statistical Methods, p 739. Browner et al., Designing Clinical Research, 5e (2022) — Chapter 3, Choosing the Study Participants: Specification, Sampling and Recruitment, pp 31–33, and Chapter 20 answer key. Fletcher, Clinical Epidemiology: The Essentials — Chapter 3, Abnormality.



Standard normal curve in health research

Asked in: Paper I 06-Apr-2026 — RGUHS MD Community Medicine

Definition — When a large random sample of a continuous biological variable (height, birth weight, blood pressure, haemoglobin) is plotted with a small class interval, the frequency polygon becomes a smooth, symmetrical, bell-shaped curve with a single peak at the mean; this is the **normal** or **Gaussian distribution**, and its curve is the **normal curve** (Mahajan's Methods in Biostatistics, Ch 5, pp 99-101). Because every mean and standard deviation (SD) generates a different normal curve, the variable is **standardised** by subtracting the mean and dividing by the SD. The resulting single reference curve, with mean 0, standard deviation 1 and total area 1, is the **standard normal curve** (Park 28e, Ch 21, p 1005).

Characteristics of the standard normal curve

Property	Statement
Shape	Smooth, bell-shaped, perfectly symmetrical about the mean, with a single peak (Park 28e, Ch 21, p 1005)
Central values	Mean, median and mode all coincide at the centre (Sunder Rao, Ch 9, p 59)
Parameters	Mean = 0 and standard deviation = 1 after standardisation; the original curve is described completely by its mean and SD (Park 28e, Ch 21, p 1005; Sunder Rao, Ch 9, p 60)
Total area	Equals 1 (i.e. 100% of observations); area is therefore read directly as a probability (Park 28e, Ch 21, p 1005)
Symmetry of frequencies	Frequencies at equal distances on either side of the mean are equal; roughly half the observations lie above and half below the mean (Sunder Rao, Ch 9, p 59)
Range covered	Six standard deviations, three on either side of the mean, cover almost the whole range of a variable; the SD divides the range into 6 equal subranges (Mahajan, Ch 5, p 100)
Basis of inference	It is the parent distribution from which the <i>t</i> , chi-square and <i>F</i> sampling distributions are derived, and it is the basis of the tests of significance (Sunder Rao, Ch 9, p 61; Mahajan, Ch 5, p 104)

Areas under the curve and the standard normal deviate (Z)

The **standard normal deviate**, also called the **relative deviate** or **standard normal variate (Z)**, is the distance of any observation from the mean expressed in units of standard deviation (Park 28e, Ch 21, p 1005):

$Z = (\text{Observation} - \text{Mean}) \div \text{SD}$, i.e. $Z = (X - \bar{x})/\text{SD}$ for a sample and $Z = (X - \mu)/\sigma$ for a population.

Z is a pure number without units, so any normally distributed variable in any unit can be converted to the one standard curve and its probability read from the **unit normal distribution table**.

Limits about the mean	Observations included	Observations outside
Mean $\pm$ 1 SD	68.27% (roughly two-thirds)	~32%
Mean $\pm$ 1.96 SD	95%	5%
Mean $\pm$ 2 SD	95.45%	4.55% (about 1 in 20)

Limits about the mean	Observations included	Observations outside
Mean $\pm$ 2.58 SD	99%	1%
Mean $\pm$ 3 SD	99.73%	0.27% (about 1 in 370)

(Mahajan, Ch 5, p 100; Sunder Rao, Ch 9, p 59)

Relative deviate (Z)	Area from the middle of the curve to Z
0.00	0.0000
0.50	0.1915
1.00	0.3413
1.50	0.4332
2.00	0.4772
3.00	0.4987
4.00	0.49997
5.00	0.4999998

(Park 28e, Ch 21, Table 4, p 1005. The area beyond $Z = 0.5$ – the tabulated value; doubling the tabulated value gives the central area.)

Applications in health research

Application	How the standard normal curve is used	Grounded example
Defining normal or critical limits of a clinical or laboratory variable	Mean $\pm$ 1.96 SD of a healthy reference population is taken as the range of normality; values outside are unusual and possibly pathological	Birth weight mean 3.05 kg, SD 0.39 kg $\rightarrow$ normal limits 2.29–3.81 kg; 4 kg is abnormal, 2.5 kg is normal (Mahajan, Ch 5, p 104)
Estimating the proportion or probability of the population beyond a value	Compute Z and read the corresponding tail area from the unit normal table	$Z = 1.5 \rightarrow$ area 0.0668, i.e. 6.68% of individuals exceed that observation and 93.32% do not (Mahajan, Ch 5, p 104)
Setting screening and case-definition cut-offs	Cut-offs are fixed at a stated number of SDs from a reference median	WHO Child Growth Standards: stunting = height-for-age Z below $-2SD$, wasting = weight-for-height Z below $-2SD$, underweight = weight-for-age Z below $-2SD$, severe acute malnutrition = weight-for-height/length Z below $-3SD$ (Park 28e, Ch 11, p 783)
Confidence interval estimation	Sample mean $\pm$ 1.96 standard error (SE) gives the 95% confidence limits within which the population mean lies; the odds of the population mean lying outside are 1:19	Systolic blood pressure of 566 males, mean 128.8 mm Hg, SD 13.05, SE 0.55 $\rightarrow$ limits taken at ± 2 SE = 127.7–129.9 mm Hg (Mahajan, Ch 9, pp 169-170)

Application	How the standard normal curve is used	Grounded example
Tests of significance in large samples (Z test)	Observed difference ÷ standard error is referred to the standard normal curve; a ratio beyond ± 1.96 is significant at 5%, beyond ± 2.58 at 1%	A ratio much less than 1.96 means the difference is probably due to chance and is not significant (Mahajan, Ch 9, p 178)
Obtaining the p value	The p value is the area under the curve beyond +Z plus the area beyond -Z	Ethnic subgroup birth weight: population mean 2.9 kg, SD 0.5, n = 100, sample mean 2.8 kg $\rightarrow Z = -2$; area = 0.023 in each tail $\rightarrow p = 0.046$, so only 4.6% of such samples would arise by chance (Sunder Lal, Ch 6, p 258)
Sample size calculation	The Z value corresponding to the desired confidence enters the sample-size formula, given the SD of the population	Used to fix the size of sample needed for the desired confidence limits (Mahajan, Ch 9, p 169)
Data summarisation and comparison	When a distribution is normal, the mean and SD describe it completely, so no essential detail is lost; Z is a unit-free number, so variables in different units can be compared on one scale	Anthropometric indices are all expressed in SD (Z-score) units from the reference median (Sunder Rao, Ch 9, p 60; Park 28e, Ch 11, p 783)
Justifying the use of large-sample methods	Many other theoretical distributions approximate the normal curve when the sample is large; <i>t</i> closely approximates the normal distribution as sample size rises, so the Z test replaces the <i>t</i> test in large samples	Small samples and their Z values do not follow the normal distribution, hence Student's <i>t</i> test is used instead (Mahajan, Ch 9, p 178; Sunder Rao, Ch 9, pp 60-61)

Worked example (write one in the exam)

- ▶ **Question** — Average birth weight is 3.05 kg with SD 0.39 kg. Is 4 kg abnormal?
- ▶ **Normal limits** = mean ± 1.96 SD = $3.05 \pm (1.96 \times 0.39) = 2.29$ kg to 3.81 kg.
- ▶ **Interpretation** — 4 kg lies outside these limits, so it is abnormal; a weight of 2.5 kg lies within them and is normal (Mahajan, Ch 5, p 104).
- ▶ **By the Z route** — pulse rate mean 72/min, SD 2/min; probability of a randomly chosen normal male having a pulse of 80 or more: $Z = (80 - 72)/2 = 4$; area from the mean to $Z = 4$ is 0.49997, so the probability beyond it = $0.5 - 0.49997 = 0.00003$, i.e. about 3 in 100,000 (Park 28e, Ch 21, p 1005).

Limitations — when the curve cannot be applied

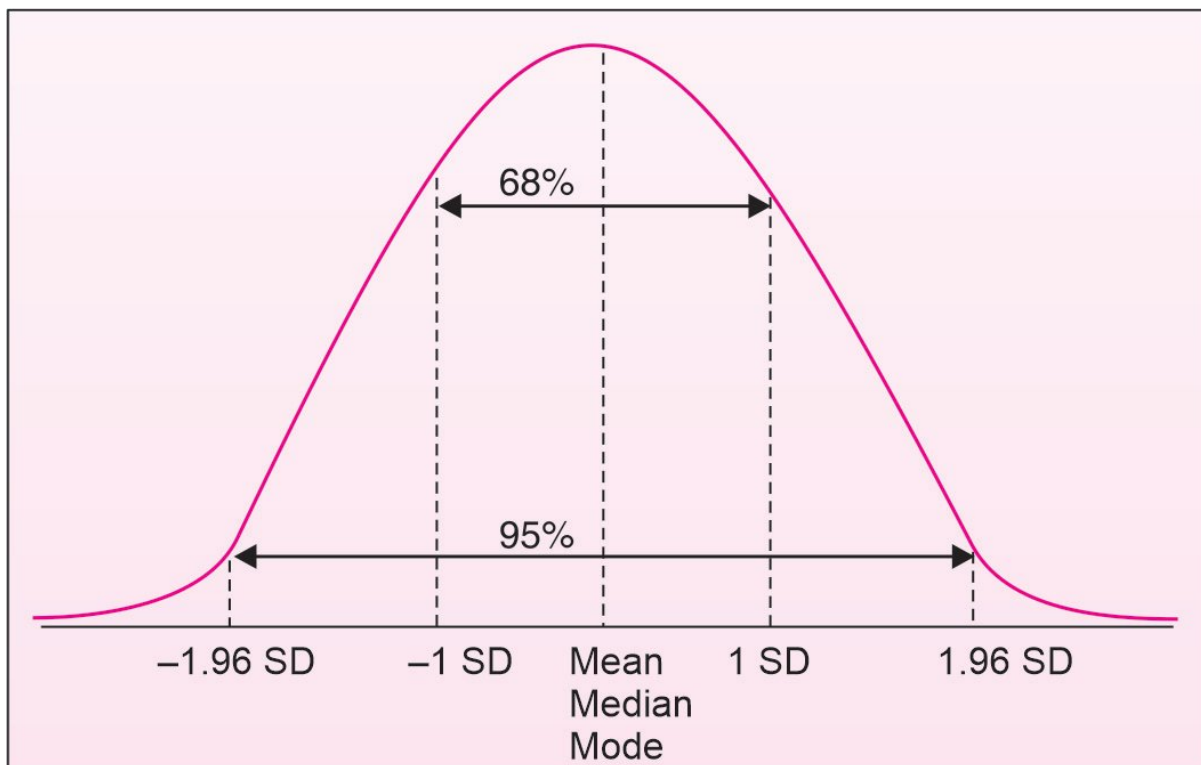
Situation	Consequence for the researcher
Distribution is skewed (positively or negatively) — mean, median and mode no longer coincide	Mean $\pm$ SD limits are misleading; use the median and percentiles, transform the data, or use a non-parametric test (Sunder Rao, Ch 6, pp 29-30)

Situation	Consequence for the researcher
Small sample (roughly under 30)	Z values do not follow the normal distribution; use Student's <i>t</i> test, whose shape varies with degrees of freedom (Mahajan, Ch 9, p 178)
Qualitative or ordinal data	The normal curve applies to continuous quantitative variables only
Sample not large and random	Normality is a virtue of a large random sample; a biased sample gives a distorted curve (Mahajan, Ch 5, p 99)

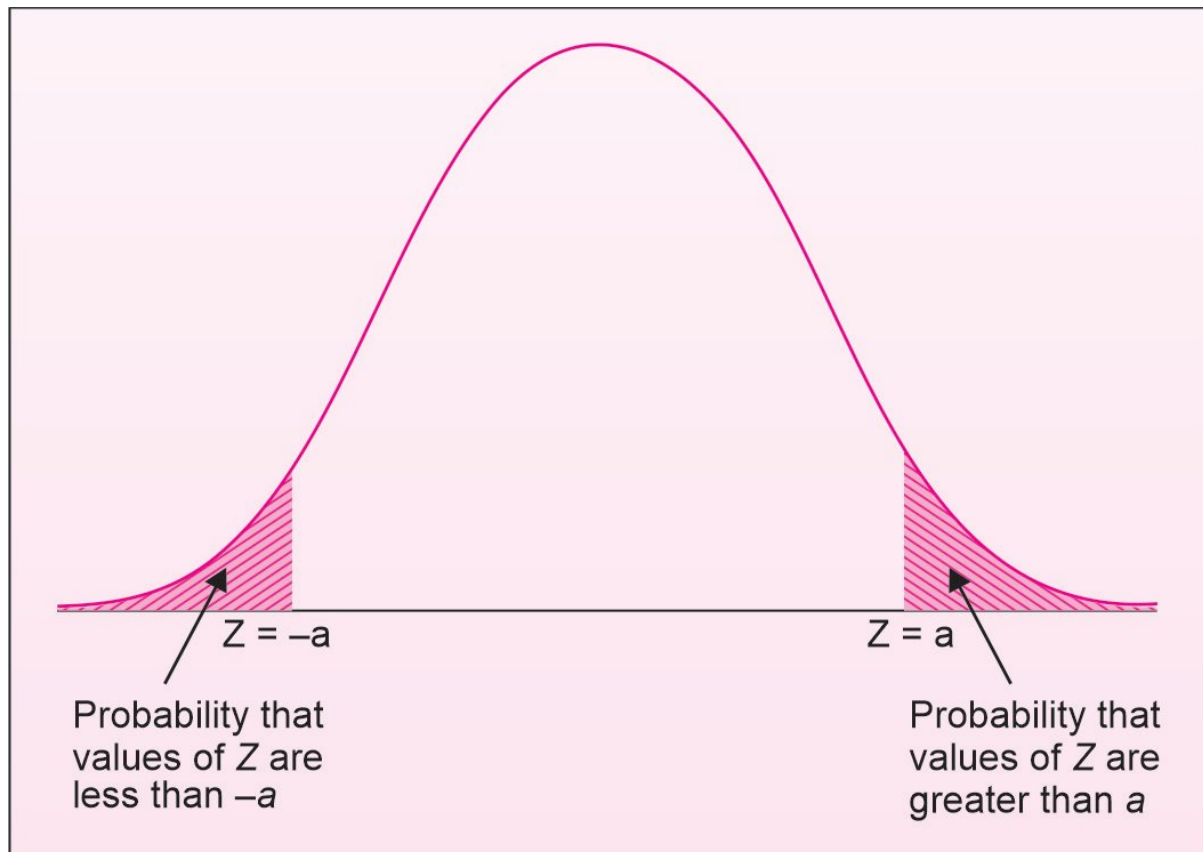
DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Labels to reproduce by hand (one bell curve, X-axis in SD units):

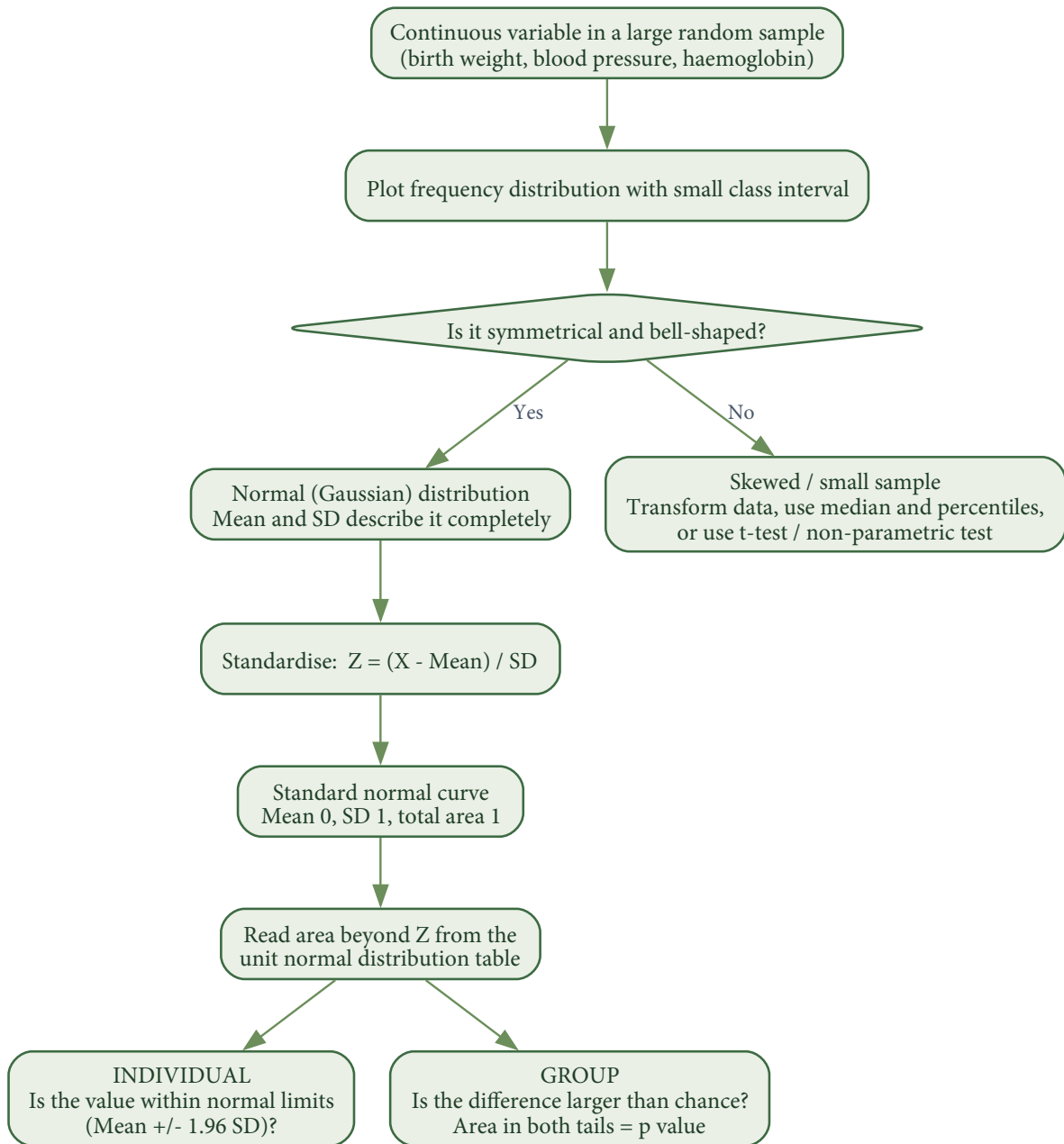
- ▶ Peak marked Mean = Median = Mode ($Z = 0$)
- ▶ X-axis ticks at $-3, -2.58, -1.96, -1, 0, +1, +1.96, +2.58, +3$ SD
- ▶ Arrowed brackets: ± 1 SD = 68.27%, ± 1.96 SD = 95%, ± 2.58 SD = 99%, ± 3 SD = 99.73%
- ▶ Shade both tails beyond ± 1.96 SD and label 2.5% each tail = 5% level of significance
- ▶ Write on the figure: $Z = (X - \text{Mean}) \div \text{SD}$; Total area = 1



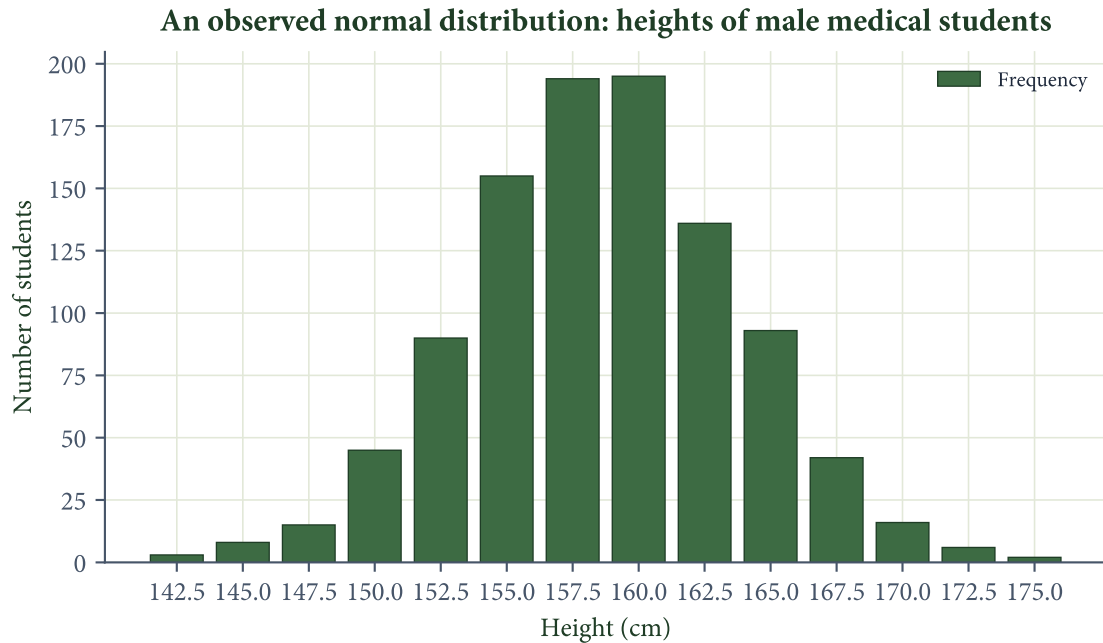
The standard normal distribution curve (Sunder Lal — Textbook of Community Medicine, Ch 6, p 244)



Area under the normal curve (Sunder Lal — Textbook of Community Medicine, Ch 6, p 248)



Converting a health observation into a probability using the standard normal curve



$n = 1000$; mean 160 cm, SD 5 cm. Mean ± 1 SD holds 680 (68%), ± 2 SD holds 950 (95%), ± 3 SD holds 995 (99%). Source: Mahajan's Methods in Biostatistics, Ch 5, Table 5.1, p 101.

High-yield exam pointers

- ▶ $Z = (\text{Observation} - \text{Mean}) \div \text{SD}$; standard normal curve has mean 0, SD 1, total area 1; it is also called the *Gaussian* curve.
- ▶ Memorise the ladder: $\pm 1 \text{ SD} = 68.27\%$, $\pm 1.96 \text{ SD} = 95\%$, $\pm 2 \text{ SD} = 95.45\%$, $\pm 2.58 \text{ SD} = 99\%$, $\pm 3 \text{ SD} = 99.73\%$ — and that 6 SDs cover almost the entire range.
- ▶ Normal limits of a variable = **mean $\pm 1.96 \text{ SD}$** ; confidence limits of a mean = **sample mean $\pm 1.96 \text{ SE}$** . Do not interchange SD (individual variation) with SE (sample-to-sample variation).
- ▶ p value = area beyond +Z **plus** area beyond -Z (two-tailed); significant if Z beyond ± 1.96 (5%) or ± 2.58 (1%).
- ▶ WHO Child Growth Standards cut-offs, all Z-score based: **stunting, wasting and underweight at below -2SD ; severe acute malnutrition at below -3SD** weight-for-height/length, or mid-upper arm circumference under 115 mm, or nutritional oedema (Park 28e).
- ▶ Values beyond $\pm 2 \text{ SD}$ occur in about **1 in 20**, beyond $\pm 3 \text{ SD}$ in about **1 in 370** observations.
- ▶ Large sample ($n \geq 30$) $\rightarrow$ **Z test**; small sample $\rightarrow$ **Student's *t* test** (W.S. Gosset), which approaches the normal curve as n rises.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: statistical writing sometimes prefers the term *Gaussian distribution* to *normal distribution* to avoid confusion with the clinical meaning of "normal", since a value inside mean $\pm 1.96 \text{ SD}$ is statistically usual but not necessarily healthy.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 21, Health Information and Basic Medical Statistics; Ch 11, Nutrition and Health); Mahajan's Methods in Biostatistics for Medical Students and Research Workers (Ch 5, Normal Distribution and Normal Curve; Ch 8, Sampling Variability and Significance; Ch 9, Significance of Difference in Means; Appendix I, Table of Unit Normal Distribution); Sunder Rao & Richard — Biostatistics and Research Methods (Ch 6, Measures of Central Tendency; Ch 9, Probability Distributions); Sunder Lal — Textbook of Community Medicine (Ch 6, Basic Statistics and its Applications and Research Methodology); WHO Child Growth Standards, as quoted in Park 28e.



Sampling error and standard error

Asked in: Paper I May 2017 — RGUHS MD Community Medicine

Definition — **Sampling error** is the discrepancy between a sample estimate (statistic) and the true population value (parameter) that arises purely because data were gathered from a sample and not from the entire population of concern; it is the variation seen from one sample to another when repeated samples are drawn from the same universe (Park 28e, Ch 21, p 1006; Sunder Rao, p 83). **Standard error** (SE) is the standard deviation of the sampling distribution of a statistic — the single number that *measures* that chance variation from sample to sample, or from sample to universe (Mahajan, Ch 8, p 164). In short, sampling error is the phenomenon; the standard error is its measurement.

Sampling error — nature, causes and control

Feature	Explanation
Synonyms	Sampling variability; statistical error; chance variability (Mahajan, Ch 4, p 72)
Why it occurs	Only a fraction of the universe is examined; no two random samples are alike, so each sample estimate differs from the parameter and from the estimates of other samples (Park 28e, Ch 21, p 1006)
Direction	Random, not systematic — estimates scatter symmetrically around the parameter , so sample means are distributed normally about the population mean μ (Mahajan, Ch 8, p 164)
Determinant 1 — size of the sample (n)	As the size of the sample increases, sampling error decreases (Park 28e, Ch 21, p 1006)
Determinant 2 — natural variability of the individual readings	The more widely individual readings vary from one another, the more variability there is from one sample to another (Park 28e, Ch 21, p 1006)
Determinant 3 — sampling method	The inherent chance variation is minimised by a suitable probability sampling design — simple random, systematic or stratified (Sunder Rao, p 76)

Feature	Explanation
Can it be abolished?	No. It is natural and inevitable; it can be <i>reduced</i> by enlarging the sample but never totally eliminated (Mahajan, Ch 4, p 72)
How it is measured	By calculating the standard error (Mahajan, Ch 4, p 72)

Sampling error versus non-sampling error

Point	Sampling error	Non-sampling error
Source	Studying a sample instead of the whole population	Faults in the conduct of the survey itself (Park 28e, Ch 21, p 1006)
Examples	Chance difference between the sample mean and the population mean	Inadequately calibrated instruments, observer variation, incomplete coverage of the subjects selected, non-response, inconsistent definitions, conceptual and recording errors (Park 28e, Ch 21, p 1006; Sunder Rao, p 76)
Nature	Random, quantifiable, predictable in size	Systematic, i.e. bias; usually not quantifiable
Effect of increasing n	Falls, in proportion to 1/square-root of n	Unchanged, and may even worsen
Remedy	Larger sample, probability sampling	Adequate planning, standardisation of tools, training, supervision, monitoring, complete follow-up (Sunder Rao, p 76)
Relative importance	Measurable and allowed for statistically	Often more important than the sampling error (Park 28e, Ch 21, p 1006)

Standard error — meaning and formulae

Because in practice only one sample is taken, the standard error is the yardstick that says how reliable an estimate is likely to be, and it allows confidence limits to be set around that estimate (Park 28e, Ch 21, p 1006). Its value falls as the square root of n rises, so quadrupling the sample halves the standard error. The standard error of proportion is defined in the same way for qualitative data — the unit that measures the variation occurring by chance in the proportion of a character from sample to sample or from sample to population (Mahajan, Ch 10, p 196).

Statistic	Standard error	Symbols and source
Mean of one sample	$SE(\bar{x}) = \sigma / \text{square-root of } n$ (use the sample standard deviation s when σ is unknown)	σ = population standard deviation, n = sample size (Park 28e, Ch 21, p 1006)
Proportion	$SE(p) = \text{square-root of } (pq/n)$	p = proportion with the attribute, $q = 1 - p$ (or, in per cent, $p + q = 100$) (Park 28e, Ch 21, p 1007)
Difference between two means	$SE(d) = \text{square-root of } (\sigma_1^2/n_1 + \sigma_2^2/n_2)$	Two independent large samples (Park 28e, Ch 21, p 1007)
Difference between two proportions	$SE(p_1 - p_2) = \text{square-root of } [pq(1/n_1 + 1/n_2)]$	p = pooled proportion of the two samples (Mahajan, Ch 10, p 201)

Statistic	Standard error	Symbols and source
Other statistics	Standard error of the correlation coefficient; standard deviation of the regression coefficient; standard error of estimate	All listed among the measures of variability of <i>samples</i> (Mahajan, Ch 4, p 75)

Reading it on the normal curve. Sample means are normally distributed about μ in standard-error units: 95 per cent of them lie within $\mu \pm 1.96$ SE and 99 per cent within $\mu \pm 2.58$ SE. Park uses the working approximation of two standard errors for 95 per cent confidence limits (Park 28e, Ch 21, p 1006; Mahajan, Ch 8, pp 166–168).

Applications and uses of the standard error

- **To set confidence limits for the population parameter** — the population mean μ lies within sample mean ± 1.96 SE in 95 estimates out of 100; the odds of it lying outside these limits are 1 : 19 (Mahajan, Ch 9, p 182).
- **To decide whether a sample has been drawn from a known population** — if the sample mean lies beyond the known population mean ± 1.96 SE, there is a 95 per cent chance that the sample does not belong to that universe, or that it is under the influence of some external factor (Mahajan, Ch 9, p 182).
- **To test the significance of an observed difference** — the Z test and Student's t test both express the observed difference in standard-error units, as $Z = \text{observed difference} / \text{standard error of that difference}$. A difference greater than 1.96 SE is significant at the 5 per cent level; a difference greater than 3 SE is real in more than 99 per cent of cases (Mahajan, Ch 9, pp 185–186).
- **To calculate the required size of the sample** for a desired width of confidence limits, when the population standard deviation is known (Mahajan, Ch 9, p 182).
- **To express the precision of a published estimate** — the standard error underlies the error bars and confidence intervals reported with means, proportions, rates and regression coefficients.

Standard deviation contrasted with standard error

Point	Standard deviation (SD)	Standard error (SE)
What it measures	Variation of <i>individual observations</i> within one sample	Variation of <i>sample estimates</i> from sample to sample, about the parameter (Mahajan, Ch 17, answer to exercise 35)
Belongs to	Descriptive statistics — spread of data	Inferential statistics — precision of an estimate
Formula for the mean	$s = \text{square-root of } [\text{sum of } (x - \bar{x})^2 / (n - 1)]$	s divided by the square root of n
Effect of increasing n	Approaches the true population sigma; does not shrink	Steadily becomes smaller

Point	Standard deviation (SD)	Standard error (SE)
Used for	Defining normal limits — mean $\pm$ 2 SD covers 95.45 per cent and mean $\pm$ 1.96 SD covers 95 per cent of observations (Mahajan, Ch 5, pp 112–113)	Confidence limits and tests of significance

Worked illustrations from the spine text

- **Standard error of the mean.** A random sample of 25 males aged 20–24 years had a mean temperature of 98.14 °F with a standard deviation of 0.6. $SE = 0.6 / \text{square-root of } 25 = 0.12$. The 95 per cent confidence limits for the population mean are $98.14 \pm (2 \times 0.12)$, that is **97.90 °F to 98.38 °F**; the chance that the true mean lies outside these limits is only 1 in 20 ($P = 0.05$) (Park 28e, Ch 21, p 1006).
- **Standard error of a proportion.** Males formed 52 per cent of a village population. In a random sample of 100 people only 40 per cent were male. $SE(p) = \text{square-root of } (52 \times 48 / 100) = 5.0$, so a truly representative sample could by chance yield $52 \pm 2(5)$, i.e. 42 to 62 per cent. The observed 40 per cent lies outside; the relative deviate is $(52 - 40)/5 = 2.4$, which exceeds 2, so the difference is significant and unlikely to have arisen by chance (Park 28e, Ch 21, p 1007).
- **Standard error of the difference between two means.** Twenty-four comparable mice were randomly split into a control group ($n = 12$, mean kidney weight 318 mg, SD 10.2) and an experimental group ($n = 12$, mean 370 mg, SD 24.1). $SE(d) = \text{square-root of } (10.2^2/12 + 24.1^2/12) = \text{square-root of } 57.07 = 7.5$. The observed difference of 52 mg is more than twice the standard error of the difference, and is therefore significant — the treatment affected kidney weight (Park 28e, Ch 21, p 1007).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Sampling distribution of sample means in standard-error units** — draw one bell-shaped curve carrying these exact labels: y-axis *Frequency*; x-axis *Standard error units*; the centre ordinate *mu (population mean)*; a formula box $SE \text{ of mean} = \text{sigma} / \text{square-root of } n$; ordinates at -1 and $+1$ enclosing 34% and 34%; the next bands 13.5% on either side; further ordinates at $+1.96$ and $+2.58$ with the two tails marked 2.5% each; and brackets beneath reading 95% *confidence limits* ($\pm 1.96 SE$) and 99% *confidence limits* ($\pm 2.58 SE$).

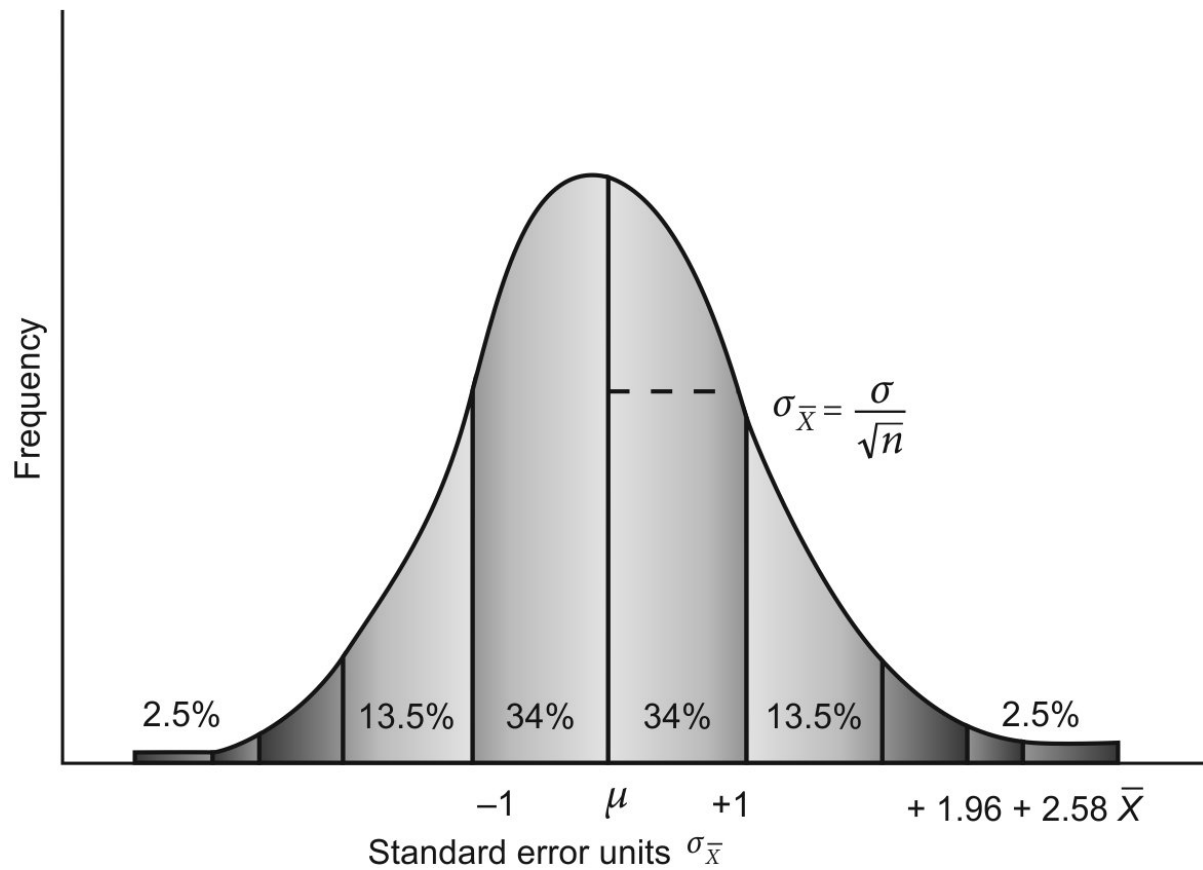
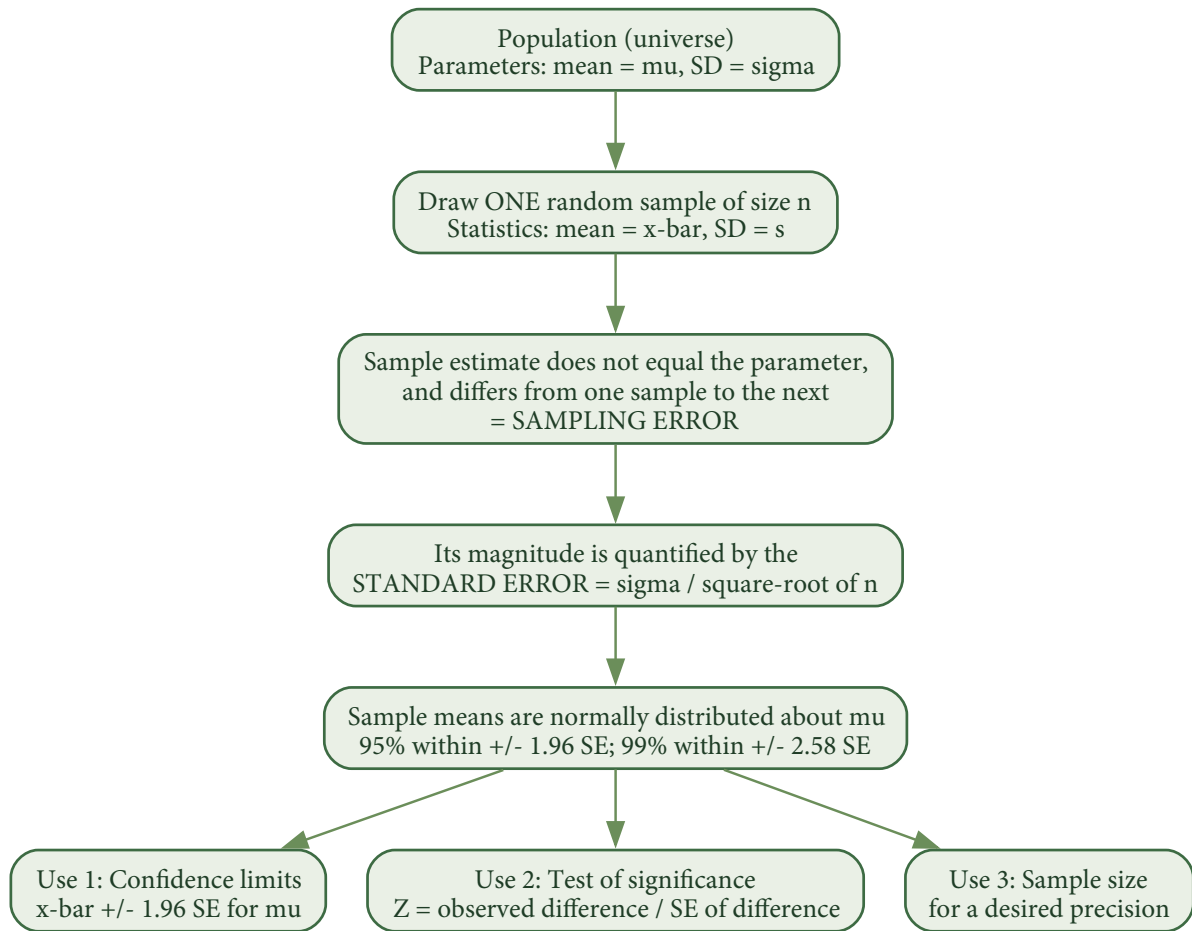
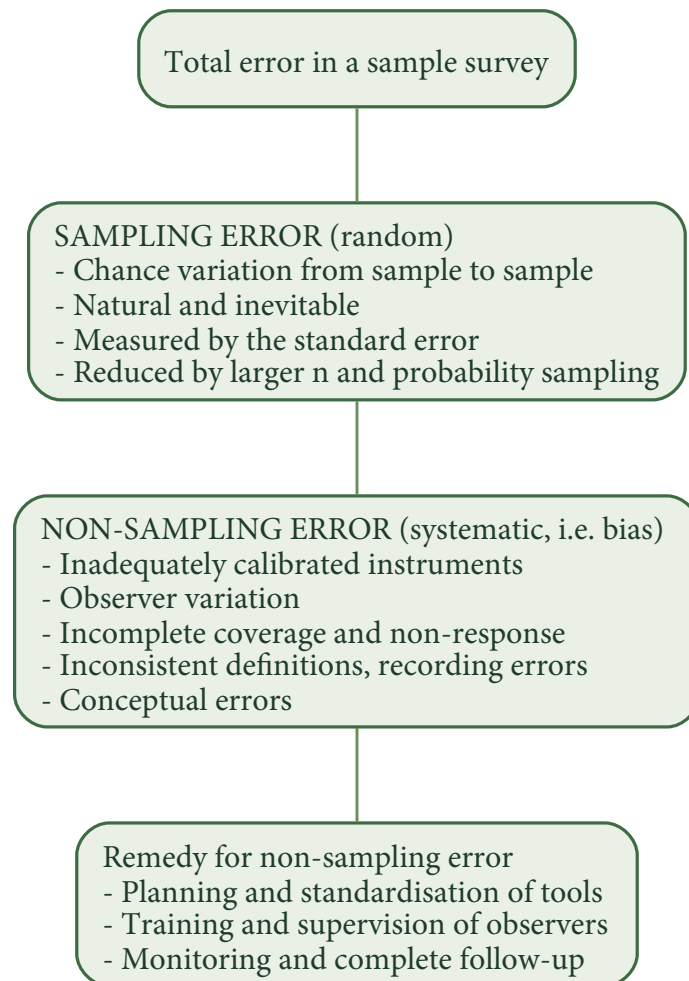


Fig 8.1 — Normal distribution of sample values (means) (Mahajan's Methods in Biostatistics for Medical Students and Research Workers (PDFDrive), p.166)



From population to inference - how sampling error arises and how the standard error measures it



Errors in a sample survey - sampling error versus non-sampling error, and their remedies

High-yield exam pointers

- ▶ **One-line distinction to open with:** the standard deviation measures variation *within* a sample; the standard error measures variation *between* samples. Both carry the same units as the original observations.
- ▶ **Four standard errors to write out:** $SE(\text{mean}) = \sigma / \sqrt{n}$; $SE(\text{proportion}) = \sqrt{pq/n}$; $SE(\text{difference between two means}) = \sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}$; $SE(\text{difference between two proportions}) = \sqrt{pq(1/n_1 + 1/n_2)}$.
- ▶ **Two determinants of sampling error (Park 28e):** the size of the sample and the natural variability of the individual readings.
- ▶ **Confidence limits:** $\text{mean} \pm 1.96 SE = 95 \text{ per cent}$; $\text{mean} \pm 2.58 SE = 99 \text{ per cent}$; Park's working rule uses 2 SE for 95 per cent.
- ▶ **Significance rule of thumb:** an observed difference greater than $1.96 \times SE$ of the difference is significant at $P < 0.05$; greater than $3 \times SE$ it is real in more than 99 per cent of cases.
- ▶ **Examiner's trap:** sampling error can be reduced but never eliminated, while non-sampling error is not reduced by enlarging the sample and is often the more important of the two.
- ▶ **Park's numbers worth memorising:** $n = 25$, mean 98.14°F , SD 0.6 , SE 0.12 , 95 per cent limits $97.90\text{--}98.38^\circ\text{F}$.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only — Park writes the working formulae with the population standard deviation sigma, whereas in practice the sample standard deviation s is substituted when sigma is unknown; and for small samples (n below about 30) the t distribution replaces the normal deviate, so the multiplier becomes the tabulated t value at $n - 1$ degrees of freedom rather than 1.96.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 21, Health Information and Basic Medical Statistics, pp 1006–1007); Mahajan's Methods in Biostatistics for Medical Students and Research Workers (Ch 4 Variability and its Measures; Ch 8 Sampling Variability and Significance; Ch 9 Significance of Difference in Means; Ch 10 Significance of Difference in Proportions of Large Samples); Sunder Rao & Richard — Biostatistics and Research Methods (Sampling and Statistical Inference, pp 76, 83). Figure reproduced from Mahajan's Methods in Biostatistics, Fig 8.1, p 166.



Lot quality assurance sampling (LQAS)

Asked in: Paper I 14-Jul-2020 — RGUHS MD Community Medicine

Definition — The lot quality technique, or **lot quality assurance sampling (LQAS)**, is a rapid small-sample method in which a defined administrative unit (the "lot") is **classified** as acceptable or unacceptable against a pre-set standard, rather than having its coverage estimated as a percentage. A small sample is drawn from the lot and the number of "defectives" — sampled persons who did not receive the service, or who have the condition under study — is counted; if that number exceeds a pre-decided **decision rule**, the lot is rejected as under-performing (Park Plus, Ch 19, p 239; Sunder Lal, Ch 6, p 252). What the manager receives is therefore a verdict on a supervision area, delivered quickly and cheaply enough to be repeated at every monitoring round (Plotkin's Vaccines 7e, Ch 76, pp 2151–2152).

Origin and underlying principle

Element	Content
Origin	Originally developed in the 1920s to control the quality of output in industrial production processes — a small random sample of a manufactured batch (lot) is tested for quality (Park Plus, Ch 19, p 239)
Industrial rule	If the number of defective items in the sample exceeds a predetermined criterion (decision rule), the whole lot is rejected without examining every item (Park Plus, Ch 19, p 239)
Basis of the decision rule	The desired production standard plus a statistically determined sample size (Park Plus, Ch 19, p 239); in health use the sample size and the cut-off number of cases are calculated from cumulative binomial probabilities (Abramson, Ch 31, p 309)

Element	Content
Transfer to public health	Statisticians and epidemiologists found the industrial technique suitable for assessing a variety of primary health care services, e.g. measuring immunisation coverage (Sunder Lal, Ch 6, p 252); proposed as a methodology for monitoring programme performance at health centre or community level (Park Plus, Ch 19, p 239)
Unit of assessment	Used locally at the level of a " supervision area " to identify priority areas, or indicators, that are not reaching average coverage or an established benchmark (Park Plus, Ch 19, p 239)
What it identifies	Communities with a high (or low) prevalence of the characteristic under study (Abramson, Ch 31, p 309)
Output	A classification — acceptable or unacceptable — against predefined upper and lower cut-offs (Plotkin's Vaccines 7e, Ch 76, pp 2151–2152)

Key terms

Term	Meaning in a lot quality survey
Lot / supervision area	The smallest unit for which a separate management decision is to be taken (district, block, urban ward, sub-centre area, facility catchment)
Defective	A sampled individual who did NOT receive the service (e.g. an unimmunised child), or who HAS the condition (e.g. a child with trachoma)
Upper threshold	The benchmark coverage the programme has set; a lot at or above this is classified "adequate"
Lower threshold	The coverage below which performance is frankly unacceptable and demands intervention
Grey area	The central range of coverage that is excluded by neither the "adequate" nor the "inadequate" classification (Plotkin's Vaccines 7e, Ch 76, p 2152)
Decision value (d)	The maximum number of defectives permitted in the sample; more than d defectives rejects the lot
Alpha error	Wrongly rejecting a lot that is truly adequate
Beta error	Wrongly accepting a lot that is truly inadequate

Steps in conducting a lot quality survey

- **Define the lot.** List the supervision areas — blocks, wards, villages, sub-centre areas — for which separate managerial decisions will actually be made. A unit that no supervisor can act on separately should not be made a lot, because the whole value of the method lies in naming an area someone is accountable for.
- **Define the indicator and the "defective".** The indicator must be a single, clear, dichotomous one so that every sampled subject falls unambiguously into one of two boxes, for example a child aged 12–23 months who is either fully immunised or not.
- **Fix the two thresholds.** State in advance the benchmark coverage the programme demands (upper threshold) and the coverage that will be regarded as outright failure (lower threshold). Both are managerial decisions taken before any data are collected, never after seeing the results.

- ▶ **Fix the acceptable alpha and beta errors**, then read off the required **sample size (n)** and the **decision value (d)** (Abramson, Ch 31, p 309). The sample needed is far smaller than for a coverage survey because the question being answered is a yes/no classification against a fixed benchmark, not the estimation of a percentage with confidence limits.
- ▶ **Select the sample within the lot**. In its classical form LQAS uses **simple random sampling**; where this is not feasible for a field survey, a systematic or *random-walk* sample of houses is used, or a **clustered LQAS** design in which clusters are selected within each lot with probability proportional to estimated size, from a list of all villages with their populations (Abramson, Ch 31, p 309; Plotkin's Vaccines 7e, Ch 76, pp 2151–2152).
- ▶ **Collect the data and count the defectives**. Classical LQAS examines the full fixed sample of n subjects. In the rapid field variant, examination may instead be curtailed as soon as the predetermined number of cases has been found, which is what makes that version so quick (Abramson, Ch 31, p 309).
- ▶ **Apply the decision rule**. Defectives $\leq d$ means the lot is accepted; defectives $> d$ means the lot is rejected and flagged as a priority area. No further calculation and no significance test are needed — the arithmetic ends with a count.
- ▶ **Act on the verdict and re-survey**. Supervise, retrain and redeploy supplies and staff to the rejected lots, then repeat the survey at the next monitoring round to see whether the classification has changed. A lot quality survey that is not followed by supervisory action has wasted the fieldwork.
- ▶ **Pool only if an overall figure is needed**. Under certain circumstances results from several subunits can be pooled to derive an overall estimate for the larger area (Plotkin's Vaccines 7e, Ch 76, p 2152).

Worked field example — trachoma

Step	Content
Sample	Systematic or <i>random-walk</i> sample of houses in a village or other community; a maximum of 50 children in the sample are examined
Stopping rule	Stop as soon as the predetermined number of cases has been found
Statistical property	The stopping number is the one that correctly identifies about 95% of communities having the sought prevalence
Decision	9 cases found (then stop) → highly accurate pointer to a prevalence of 20% or more ; no case or only one → prevalence less than 5%
Use	The two prevalence bands determine which treatment strategy the community receives

(Abramson, Ch 31, pp 309–310)

Applications in public health and national programmes

Application	Detail and source
Immunisation coverage	The classic use; identifies the areas of low vaccination coverage that need urgent intervention (Sunder Lal, Ch 6, p 252)

Application	Detail and source
Campaign (supplementary immunisation activity) monitoring	Clustered LQAS: within each lot, e.g. a district, clusters are selected with probability proportional to estimated size and a fixed number of children is sampled per cluster (Plotkin's Vaccines 7e, Ch 76, pp 2151–2152)
Quality of immunisation services	Immunisation coverage among clinic attendees, cold-chain maintenance, safe injection technique, and use of all opportunities to vaccinate (Plotkin's Vaccines 7e, Ch 76, p 2152)
Validation of elimination targets	The WHO Technical Advisory Group suggested pilot testing of validation of elimination of leprosy by the lot quality assurance sampling technique (Suryakantha, Ch 43 National Health Programs, p 851)
Other primary health care services	Family planning, antenatal care, use of oral rehydration therapy and health-worker performance, especially in developing countries (Abramson, Ch 31, p 311)
Nutrition programme monitoring	Listed with 30-cluster surveys and sentinel-site monitoring as a simple routine system that a fortification programme can "piggy-back" on; such systems are widely used to monitor universal salt iodisation and immunisation coverage, at least annually (WHO/FAO Guidelines on Food Fortification with Micronutrients, Ch 8 Monitoring and evaluation, pp 225–226)
Community diagnosis	Counted among the other sources of information that contribute to community diagnosis, alongside community key informants and risk-factor surveillance (Oxford Textbook of Public Health 7e, Community diagnosis and health information systems in low- and middle-income countries, p 946)
Published field use	LQAS used to determine the low vaccination coverage areas in five cities of Bangladesh (<i>Bulletin of the World Health Organization</i> 2001;79:6, cited in Sunder Lal, Ch 6, p 252)

Lot quality survey versus the 30-cluster (EPI) coverage survey

Feature	Lot quality assurance sampling	30-cluster survey
Question answered	Is THIS supervision area meeting the benchmark?	What is the coverage of the area as a whole?
Sample	Small; a fixed number of subjects, or clusters, per lot	210 children — 7 from each of 30 clusters, aged 12–23 months (Mahajan, Ch 6 Sampling, p 138)
Precision	Classification only, against the two pre-set cut-offs	Accuracy of ±10% at 95% confidence (Sunder Rao & Richard, p 80)
Selection	Simple random, systematic/random-walk, or clustered LQAS	Clusters by probability proportional to size; sampling interval = cumulative population ÷ 30
Subunit comparison	The whole purpose — the failing areas are named	Not valid; comparisons between clusters are not permitted (Sunder Rao & Richard, p 80)
Reference period	Current programme performance, repeated at each round	Coverage of the 1 year preceding the survey (Sunder Rao & Richard, p 80)

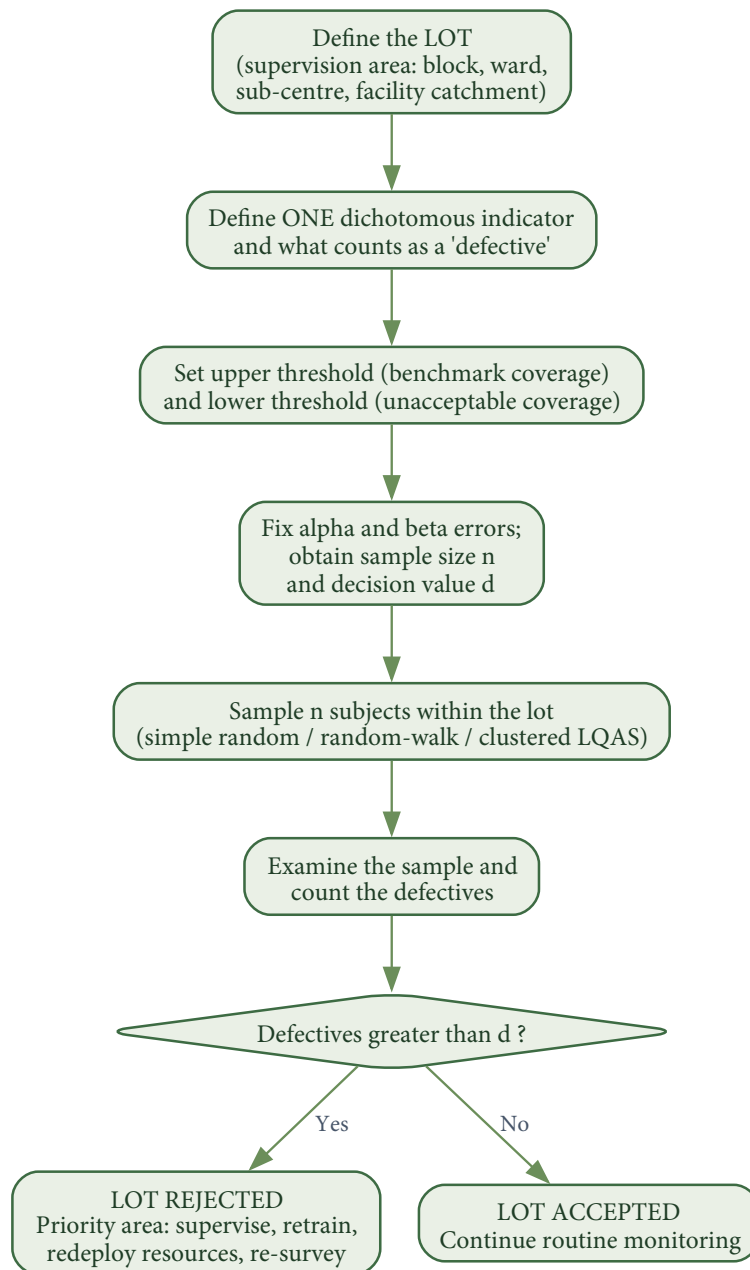
Advantages and limitations

The method earns its place in routine supervision because it converts a statistical survey into a management decision at a cost a district programme officer can actually meet; its weaknesses all follow from the same small sample that makes it cheap.

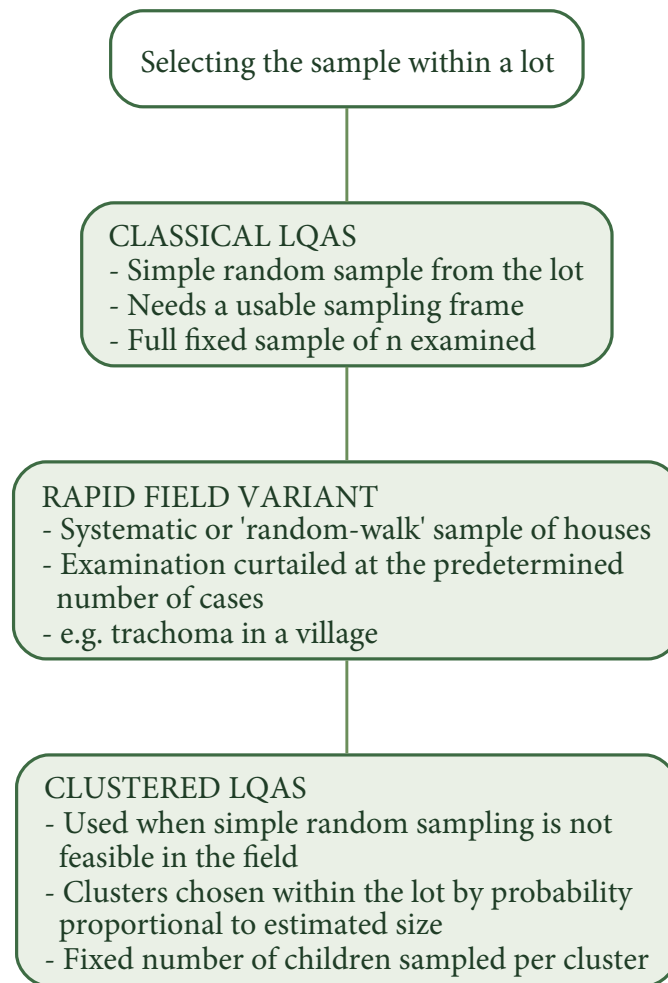
Advantages	Limitations
Very small sample — rapid, cheap and feasible as routine supervision rather than as a research exercise	As a consequence of the small sample sizes, quantitative estimates for each subunit are not available (Plotkin 7e, Ch 76, p 2152)
Usable locally , at supervision-area level, by the programme manager who must act on the result (Park Plus, Ch 19, p 239)	An unresolved central band of coverage remains between the two thresholds, so a lot in the middle is not classified either way (Plotkin 7e, Ch 76, p 2152)
Names the priority areas and the indicators that are not reaching the benchmark (Park Plus, Ch 19, p 239)	In clustered LQAS, children within a cluster are most often chosen by non-probability methods (Plotkin 7e, Ch 76, pp 2151–2152)
In the rapid field variant, examination stops at the cut-off, saving fieldwork (Abramson, Ch 31, p 309)	Ideally more clusters would be used and children selected by random sampling within each cluster than common practice allows (Plotkin 7e, Ch 76, p 2152)
Under certain circumstances , results from multiple subunits can be pooled to derive an overall estimate (Plotkin 7e, Ch 76, p 2152)	Depends wholly on thresholds and a decision rule fixed correctly in advance; the sample size and cut-off are only as good as the standard set for them (Park Plus, Ch 19, p 239)
Simple enough to repeat at short intervals, so a rejected lot can be re-classified after corrective action	Applied to a service already at, or near, the benchmark, the verdict adds little that the routine reporting system did not already show

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **The lot quality decision line** — a single horizontal coverage scale from 0% to 100% carrying the following labels: *Lot (supervision area)*; *lower threshold — unacceptable coverage*; *grey area — not resolved by either classification*; *upper threshold — benchmark/desired coverage*; *sample of n subjects*; *decision value d* ; *defectives $\leq d \rightarrow$ lot ACCEPTED*; *defectives $> d \rightarrow$ lot REJECTED*; *alpha error = adequate lot wrongly rejected*; *beta error = inadequate lot wrongly accepted*.



Decision algorithm of a lot quality assurance sampling survey



Sampling designs used within a lot

High-yield exam pointers

- ▶ **The four quantities fixed before fieldwork** — upper threshold, lower threshold, alpha error, beta error. These four together fix the sample size n and the decision value d ; nothing about the survey is decided after the data are seen.
- ▶ **Polio campaign figure to quote: commonly 6 clusters of 10 children each are sampled per lot in clustered LQAS** (Plotkin 7e, Ch 76, pp 2151–2152).
- ▶ **The two errors, stated in programme language** — alpha error rejects an adequate lot and wastes supervisory effort; beta error accepts an inadequate lot and leaves a failing area untouched, which is the more damaging error in programme monitoring.
- ▶ **The commonest mistake to avoid** — reporting a lot as "72% covered". A lot is reported as "adequate" or "inadequate"; writing a percentage for it shows the candidate has confused LQAS with a coverage survey.
- ▶ **Indian programme hooks worth naming:** pilot testing of validation of leprosy elimination by the LQAS technique (WHO Technical Advisory Group) and universal salt iodisation monitoring.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park's Textbook of Preventive and Social Medicine 28e carries no section on the lot quality technique, so this answer is grounded on Sunder Lal, Park Plus, Abramson, Plotkin's Vaccines 7e, Suryakantha and the WHO/FAO fortification guidelines. Two points often taught with this topic — that each survey handles only one dichotomous indicator so that several indicators multiply the fieldwork, and that the method suits neither rare conditions nor the measurement of small changes over time — are general statistical statements that the retrieved corpus does not carry, and they are recorded here rather than in the body for that reason. The corpus also contains no ready-made table of LQAS sample sizes and decision values, so none has been reproduced.

SOURCE & COVERAGE

Sunder Lal — Textbook of Community Medicine, Ch 6 (Basic Statistics and its Applications and Research Methodology); Park Plus (Amrutha & Vijayalakshmi), Ch 19 (Health Information and Basic Medical Statistics); Abramson — Research Methods in Community Medicine, Ch 31 (Rapid Epidemiological Methods); Plotkin's Vaccines 7ed, Ch 76 (Immunization in Developing Countries); Mahajan's Methods in Biostatistics for Medical Students and Research Workers, Ch 6 (Sampling); Sunder Rao & Richard — Biostatistics and Research Methods; Suryakantha — Community Medicine with Recent Advances, Ch 43 (National Health Programs); WHO/FAO Guidelines on Food Fortification with Micronutrients, Ch 8 (Monitoring and Evaluation), 2006; Oxford Textbook of Public Health 7e (Community Diagnosis and Health Information Systems in Low- and Middle-Income Countries).

**Non-parametric tests**

Asked in: Paper I 07-Jul-2023 — RGUHS MD Community Medicine

Definition — Non-parametric or **distribution-free** tests are tests of significance that make no assumption about the form of the parent population and use no population constant (parameter such as the mean or variance) in their computation (Mahajan's Methods in Biostatistics, Ch 1, pp 20-21). In place of the original measurements they analyse **ranks, signs or simple counts in categories**, so they remain valid when the data are not truly numerical or the distribution is unknown (Sunder Rao & Richard, Ch 16, pp 114-116; Principles of Medical Statistics, Ch 15, pp 271-272). Because most of them replace values by ranks they are also called **rank tests**.

Indications — when a non-parametric test is chosen

- **The population distribution is unknown or non-normal.** Parametric tests assume the populations are normally distributed and have the same variance; only when every assumption is true is the resulting probability statement meaningful (Sunder Rao & Richard, Ch 16, p 114).

- ▶ **The sample is too small to verify the assumptions.** With samples as small as 6 there is no alternative to a non-parametric test unless the nature of the population distribution is precisely known (Sunder Rao & Richard, Ch 16, p 115).
- ▶ **The measurement scale is ordinal.** Grades of severity of an illness, or ranks given to anaesthetic agents for effectiveness, are expressed numerically but do not possess the properties needed for addition, division and multiplication; applying arithmetic to them distorts the data (Sunder Rao & Richard, Ch 16, p 114).
- ▶ **The data are nominal (simple classificatory)** — present or absent, cured or not cured — where a mean cannot be computed at all.
- ▶ **The distribution is markedly skewed or contains outliers**, so that the median rather than the mean is the honest central index and rank methods are the matching test (Principles of Medical Statistics, Ch 3, pp 39-40).
- ▶ **Several different populations are being compared**, or the observations are inherently in ranks (Sunder Rao & Richard, Ch 16, p 115).
- ▶ **A rapid, crude answer is wanted.** The sign and median tests compress the data into a binary scale and can settle significance without more complex work; if significance is reached at this crude level, the refined tests are almost never necessary (Principles of Medical Statistics, Ch 15, p 290).

Parametric versus non-parametric tests

Feature	Parametric test	Non-parametric test
Assumption about distribution	Data follow an assumed or established distribution — normal, binomial, Poisson	None; distribution-free (Mahajan, Ch 1, pp 20-21)
Population constants used	Mean, standard deviation, variance	None — ranks, signs and counts only
Scale of measurement required	Interval or ratio (truly numerical)	Nominal or ordinal upwards
Central index summarised	Mean	Median and rank order
Minimum sample size	Larger; assumptions must be testable	Usable with n as small as 6
Probability obtained	Approximate, from a theoretical distribution	Exact probabilities in most tests (Sunder Rao & Richard, Ch 16, p 114)
Efficiency when assumptions truly hold	Maximum	Reduced — relative power-efficiency about 0.64 to 0.96 of the parametric test (Principles of Medical Statistics, Ch 27)
Examples	Z test, Student's t test, F test (analysis of variance), Pearson correlation coefficient r	Chi-square, Mann-Whitney U, Wilcoxon, Kruskal-Wallis, Spearman's rho

Classification — which test answers which question

Question being asked	Data situation	Non-parametric test	Parametric counterpart
Does one sample fit an assumed distribution?	Two categories, small sample	Binomial test	Z test for a proportion
"	Discrete categories (2 or more), expected frequencies adequate	Chi-square test	—
"	Variable has a continuous distribution	Kolmogorov-Smirnov test	—
Do two independent samples come from the same population?	Weaker (ordinal) measurement, larger samples	Mann-Whitney U test (Wilcoxon rank-sum)	Unpaired Student's t test
"	Measurements on a numerical scale, small sample	Randomisation test	Unpaired t test
"	Data cannot even be ranked, only dichotomised above or below the combined median	Median test (Fisher's test if the combined sample is small)	Unpaired t test
"	Scores fall into mutually exclusive classes, small numbers	Fisher's exact probability test	Z test for two proportions
Do more than two independent samples differ?	Ranked or ordinal data, at least 5 per group	Kruskal-Wallis one-way analysis of variance by ranks	One-way analysis of variance (F test)
Do more than two related samples differ?	Two-way layout of ranked data	Friedman two-way analysis by ranks	Two-way / repeated-measures analysis of variance
Do matched pairs differ?	Frequencies in categories	McNemar test	Test of paired proportions
"	Underlying continuity but only gross measurement possible	Sign test	Paired t test
"	Differences between pairs can be meaningfully ranked	Wilcoxon signed-rank test	Paired t test
Are two ranked variables associated?	Both variables available as ranks	Spearman's rho (rs) or Kendall's tau	Pearson correlation coefficient r

Test-selection rules from Sunder Rao & Richard, Ch 16, pp 115-116; Kruskal-Wallis and Friedman as the rank analogues of one-way and two-way analysis of variance from Principles of Medical Statistics, Ch 29, p 638.

Test statistics and decision rules

Test	Statistic	Distribution and decision rule
McNemar	chi-square = $(\text{mod}(a - b) - 1)^2 / (a + b)$, where a and b are the numbers favouring each of the two conditions	Chi-square with 1 degree of freedom (Sunder Rao & Richard, Ch 16, pp 116-117)
Sign test	Count of + and – signs of the paired differences; null hypothesis $P(+) = P(-) = 0.5$	Binomial probability; e.g. 8 negative and 2 positive signs with $N = 10$ gives $P = 0.055$, i.e. not significant (Sunder Rao & Richard, Ch 16, p 117)
Wilcoxon signed-rank	Rank the paired differences ignoring sign, then re-attach the signs; T = the smaller of the sums of the positive and of the negative ranks	Reject the null hypothesis only if T is less than or equal to the critical value for n pairs — greater values of T carry higher probabilities (Sunder Rao & Richard, Ch 16, pp 117-118)
Mann-Whitney U	Pool and rank all observations; $U1 = R1 - n1(n1 + 1)/2$, and $U + U' = n1 \times n2$	Refer the smaller of U and U' to the U/P table; when both groups exceed about 20, the sampling distribution of U approaches normal and Z is used (Principles of Medical Statistics, Ch 15, pp 275-277)
Kruskal-Wallis	$H = [12 / n(n + 1)] \times \sum(R_i^2 / n_i) - 3(n + 1)$, where R_i is the rank sum of the i-th group and n the total observations	Distributed approximately as chi-square with k – 1 degrees of freedom provided each group has at least 5 observations; a correction is applied for tied ranks (Sunder Rao & Richard, Ch 16, pp 118-120)
Spearman's rank correlation	$r_s = 1 - [6 \times \sum(d^2)] / [n(n^2 - 1)]$, where d is the difference between the paired ranks	Compare with critical values for n pairs; e.g. $r_s = 0.661$ with $n = 10$ gives $0.02 < P < 0.05$, so the null hypothesis of no correlation is rejected (Sunder Rao & Richard, Ch 16, pp 120-121)
Median test	Split both groups at the combined median to form a 2×2 table	Analyse the table by chi-square, or by Fisher's exact test when the numbers are small (Principles of Medical Statistics, Ch 15, p 290)
Chi-square	chi-square = $\sum[(O - E)^2 / E]$	Expected value must not be less than 5 in any cell; Yates' correction is applied for small samples in a 2×2 table (Mahajan, Ch 11, pp 218-219)

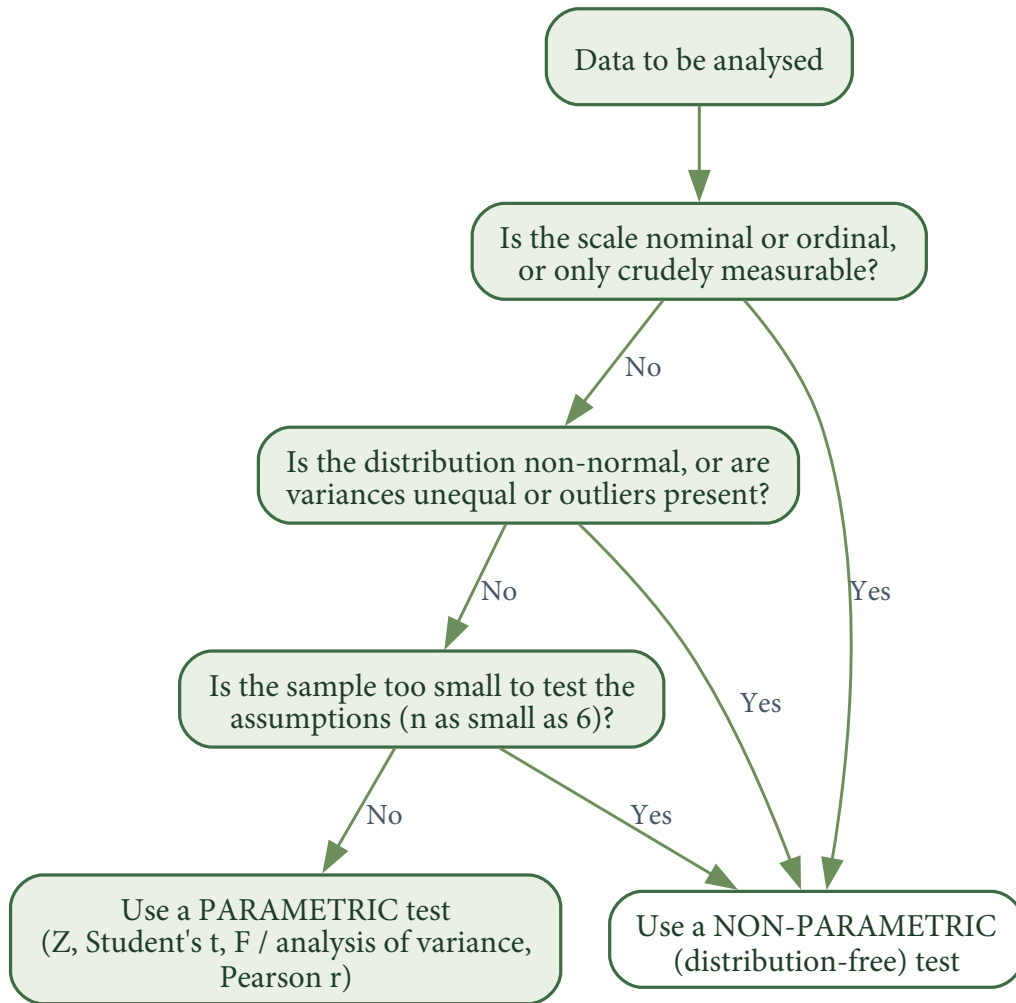
Advantages and limitations

Advantages (Sunder Rao & Richard, Ch 16, pp 114-115)	Limitations (Sunder Rao & Richard, Ch 16, p 115)
Probability statements from most non-parametric tests are exact probabilities	If all the assumptions of the parametric test are in fact met and the measurement is of the required strength, non-parametric tests are wasteful of data
Usable when samples are as small as 6, where no parametric alternative exists	There are no non-parametric methods for testing interactions in the analysis of variance
Suitable tests exist for observations drawn from several different populations	Tables of critical values may not be easily available

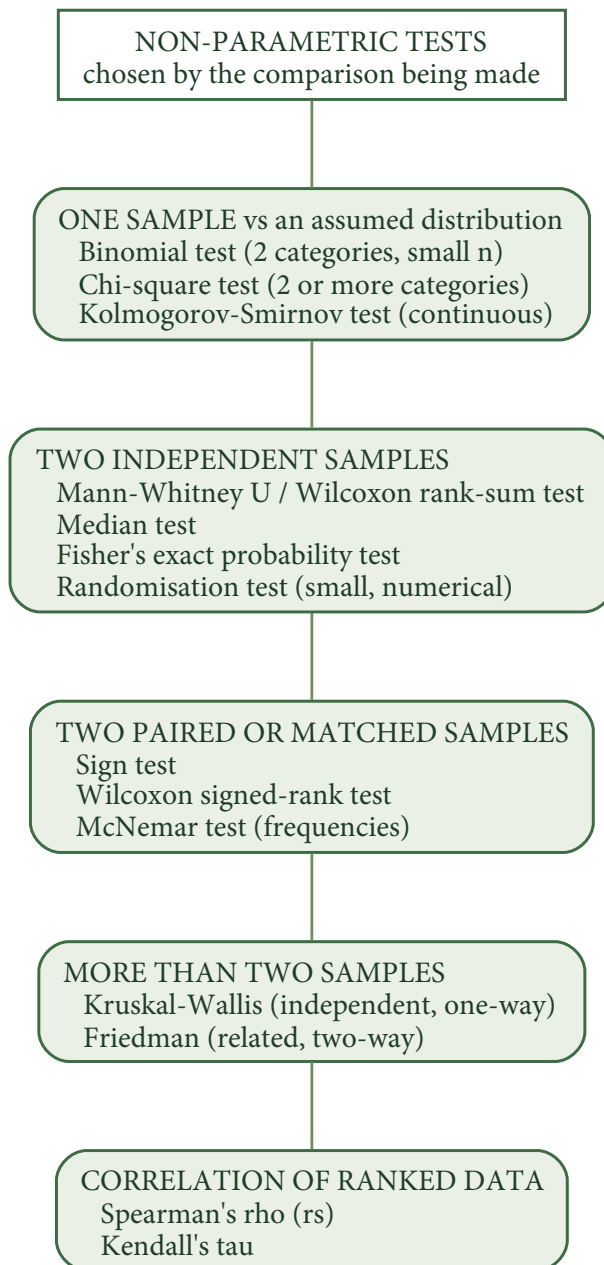
Advantages (Sunder Rao & Richard, Ch 16, pp 114-115)	Limitations (Sunder Rao & Richard, Ch 16, p 115)
Can treat data inherently in ranks, and scores that have only the strength of ranks	Reduced power-efficiency, about 0.64 to 0.96 of the parametric test, when the data are truly normal (Principles of Medical Statistics, Ch 27)
Methods available for simple classificatory (nominal) data	Information is lost when measured data are reduced to ranks
Much easier to learn, compute and apply than parametric tests	Confidence intervals are not readily obtained for ranked data (Principles of Medical Statistics, Ch 15)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Right-skewed distribution that justifies rank methods** — draw a single-peaked curve with a long tail to the right; label the horizontal axis *Values in data* and the vertical axis *Relative frequency*; drop three vertical lines from the curve, labelled from the crest outwards as **Mode**, then **Median**, then **Mean**, and mark that the mean is dragged towards the tail by outlier values while the median stays central (Principles of Medical Statistics, Ch 3, Fig 3.5).



Deciding between a parametric and a non-parametric test



Choice of non-parametric test by the type of comparison

High-yield exam pointers

- **Pairing to memorise:** sign and Wilcoxon signed-rank tests ↔ paired t test; Mann-Whitney U ↔ unpaired t test; Kruskal-Wallis ↔ one-way analysis of variance; Friedman ↔ two-way analysis of variance; Spearman's rho ↔ Pearson's r.
- **Spearman:** $rs = 1 - 6 \cdot \text{sum}(d^2) / [n(n^2 - 1)]$.
- **Kruskal-Wallis:** $H = [12 / n(n + 1)] \cdot \text{sum}(R_i^2/n_i) - 3(n + 1)$; degrees of freedom = $k - 1$; needs at least 5 per group; power efficiency $3/\pi = 95.5$ per cent against the F test (Sunder Rao & Richard, Ch 16, p 120).
- **Mann-Whitney:** $U1 = R1 - n1(n1 + 1)/2$; $U + U' = n1 \times n2$; always refer the **smaller** value of U.
- **Wilcoxon signed-rank:** T = the smaller of the positive and negative rank sums; unusually, the null hypothesis is rejected when T is **small**.

- **McNemar:** $\chi^2 = (\text{mod}(a - b) - 1)^2 / (a + b)$, with 1 degree of freedom.
- **Arithmetic check:** in any two-group rank test the two rank sums must add up to $N(N + 1)/2$.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only — Park's Textbook of Preventive and Social Medicine 28e (Ch 21, Health Information and Basic Medical Statistics) covers the parametric tests and chi-square but carries no separate section on rank-based non-parametric tests, so the test statistics and decision rules above are grounded on the dedicated biostatistics texts listed below.

SOURCE & COVERAGE

Sunder Rao & Richard — Introduction to Biostatistics and Research Methods (Ch 16, Non-parametric or Distribution-free Statistical Tests, pp 114-121); Principles of Medical Statistics (Ch 15, Non-Parametric Rank Tests, pp 271-292; Ch 27, Indexes of Categorical Association; Ch 29, Analysis of Variance, p 638; Ch 3, Central Index of a Group, pp 39-42); Mahajan's Methods in Biostatistics for Medical Students and Research Workers (Ch 1, Introduction, pp 20-21; Ch 11, The Chi-square Test, pp 218-219). Coverage note: the Friedman test appears in the retrieved corpus only as the rank analogue of two-way analysis of variance, so no worked formula for it is stated here.



Nonparametric tests of significance in statistics

Asked in: Paper I 15-Nov-2022 — RGUHS MD Community Medicine

Definition — Non-parametric or distribution-free tests are tests of significance that make **no assumption about the form of the parent population distribution** and use **no population constant (parameter)** such as the mean or the variance (Mahajan's Methods in Biostatistics, Ch 1, pp 8-9). In most of them the data are converted from measurements or scores into **ranks, signs or simple classifications**, and the test is applied to these rather than to the original values (Sunder Rao & Richard, Ch 16, p 114; Principles of Medical Statistics, Ch 15, p 271). Because the arithmetic of addition and division that parametric tests require would distort observations that are not truly numerical, these tests are the correct choice whenever the strength of measurement is weak.

When a non-parametric test is indicated

- **The distribution of the population is unknown** or is known to be non-normal, and the number of observations is too few to test the assumptions of normality and equal variance (Sunder Rao & Richard, Ch 16, p 114).
- **Very small samples** — with samples as small as 6, there is **no alternative** to a non-parametric test unless the nature of the population distribution is precisely known (Sunder Rao & Richard, Ch 16, p 114).

- ▶ **Ordinal (ranked) data** — clinical grading scales such as absent / mild / moderate / severe, or improvement scales, where numbers are only labels of order and arithmetic on them is meaningless.
- ▶ **Nominal or classificatory data** — counts falling into mutually exclusive categories, tested by chi-square or the Fisher exact probability test.
- ▶ **Data with extreme outliers.** In a worked two-group comparison, one outlier so inflated the variance that the t test gave $t = 0.67$ (not significant) although every other value in group A was exceeded by the lowest value in group B; the rank test detected the difference (Principles of Medical Statistics, Ch 15, p 285).
- ▶ **Variables with underlying continuity that can be measured only in a gross way** — the direction of change is reliable but its magnitude is not (Sunder Rao & Richard, Ch 16, p 116).

Parametric versus non-parametric tests

Feature	Parametric test	Non-parametric test
Assumption about population distribution	Data follow an assumed/established distribution — normal, binomial, Poisson	None — hence "distribution-free"
Population constants used	Mean, standard deviation, variance	None
Scale of measurement required	Interval / ratio (dimensional)	Nominal or ordinal will do
What is analysed	The measurements themselves	Ranks, signs or category counts
Small samples ($n < 10$)	Unreliable	Applicable; often the only option
Effect of outliers	Variance inflated, test may fail	Little effect — value replaced by its rank
Probability statement	Derived from mathematical theory	Exact probability in most such tests
Statistical power when assumptions ARE met	Higher	Lower ("wasteful of data")
Power when distribution is non-Gaussian	May fail	Can be the more powerful of the two
Computation	More laborious	Much easier to learn and apply
Examples	Z test, Student's t test, F test (ANOVA), Pearson correlation coefficient r	Chi-square, sign, Wilcoxon, Mann-Whitney U, Kruskal-Wallis, Spearman rho

(Mahajan's Methods in Biostatistics, Ch 1, pp 8-9; Sunder Rao & Richard, Ch 16, pp 114-115; Principles of Medical Statistics, Ch 15, pp 285-287)

The individual tests, their statistic and their parametric counterpart

Study situation	Non-parametric test	Test statistic and decision rule	Parametric counterpart
One sample, fit to an assumed distribution; two categories, small n	Binomial test	Exact binomial probability	—
One sample, discrete categories (2 or more) with adequately large expected frequencies	Chi-square goodness of fit	$\chi^2 = \Sigma(O - E)^2/E$; k - 1 degrees of freedom	—
One sample, variable with a continuous distribution	Kolmogorov-Smirnov test	Maximum deviation between observed and expected cumulative distributions	—
Matched pairs, data as frequencies in categories (before/after concordance)	McNemar test	$\chi^2 = (n - a - b) / 4$	a - b
Matched pairs, only the direction of the difference is trustworthy	Sign test	Counts of + and - signs against the null hypothesis $P(+) = P(-) = 0.5$, tested by binomial expansion	Paired t test
Matched pairs, the differences themselves can be ranked	Wilcoxon signed-rank test	Rank the differences ignoring sign, re-attach the sign; T = the smaller of the sums of the positive and the negative ranks; compare with the critical T for n pairs	Paired t test
Two independent samples, weaker (ordinal) measurement	Mann-Whitney U test (Wilcoxon rank-sum)	Pool and rank both groups; $U = R - n(n + 1)/2$ for each group; refer the smaller of U and U' to the U/P table; for n > 20 convert U to a Z value	Unpaired t test
Two independent samples that cannot even be meaningfully ranked	Median test	Dichotomise all values as above/below the combined median → 2 × 2 table → chi-square (Fisher test if total is small)	Unpaired t test
Two independent samples, all scores in mutually exclusive classes, small numbers	Fisher's exact probability test	Exact probability of the observed 2 × 2 table	Z test for two proportions
More than two independent samples	Kruskal-Wallis one-way analysis of variance by ranks	$H = [12 / \{N(N + 1)\}] \times \Sigma(R_j^2/n_j) - 3(N + 1)$; H is distributed as χ^2 with k - 1 degrees of freedom	One-way ANOVA (F test)
More than two related samples (repeated measures)	Friedman two-way analysis of ranked data	Ranks within each block, tested as χ^2	Two-way / repeated-measures ANOVA

Study situation	Non-parametric test	Test statistic and decision rule	Parametric counterpart
Correlation between two ranked variables	Spearman's rank correlation coefficient	$r_s = 1 - [6 \sum d^2 / (n^3 - n)]$, where d = difference between the paired ranks	Pearson's correlation coefficient r
Correlation between two ranked variables (concordant/discordant pairs)	Kendall's rank correlation coefficient (tau)	tau = +1 when all $n(n-1)/2$ pairs are concordant, -1 when all discordant, 0 = no association	Pearson's correlation coefficient r
Association between two qualitative attributes in a contingency table	Chi-square test of association	$\chi^2 = \sum (O - E)^2 / E$; degrees of freedom = (rows - 1)(columns - 1)	—

(Sunder Rao & Richard, Ch 16, pp 115-121; Principles of Medical Statistics, Ch 15, pp 276-283 and Ch 27, p 590 and Ch 29, p 638; Mahajan's Methods in Biostatistics, Ch 11, pp 207-211)

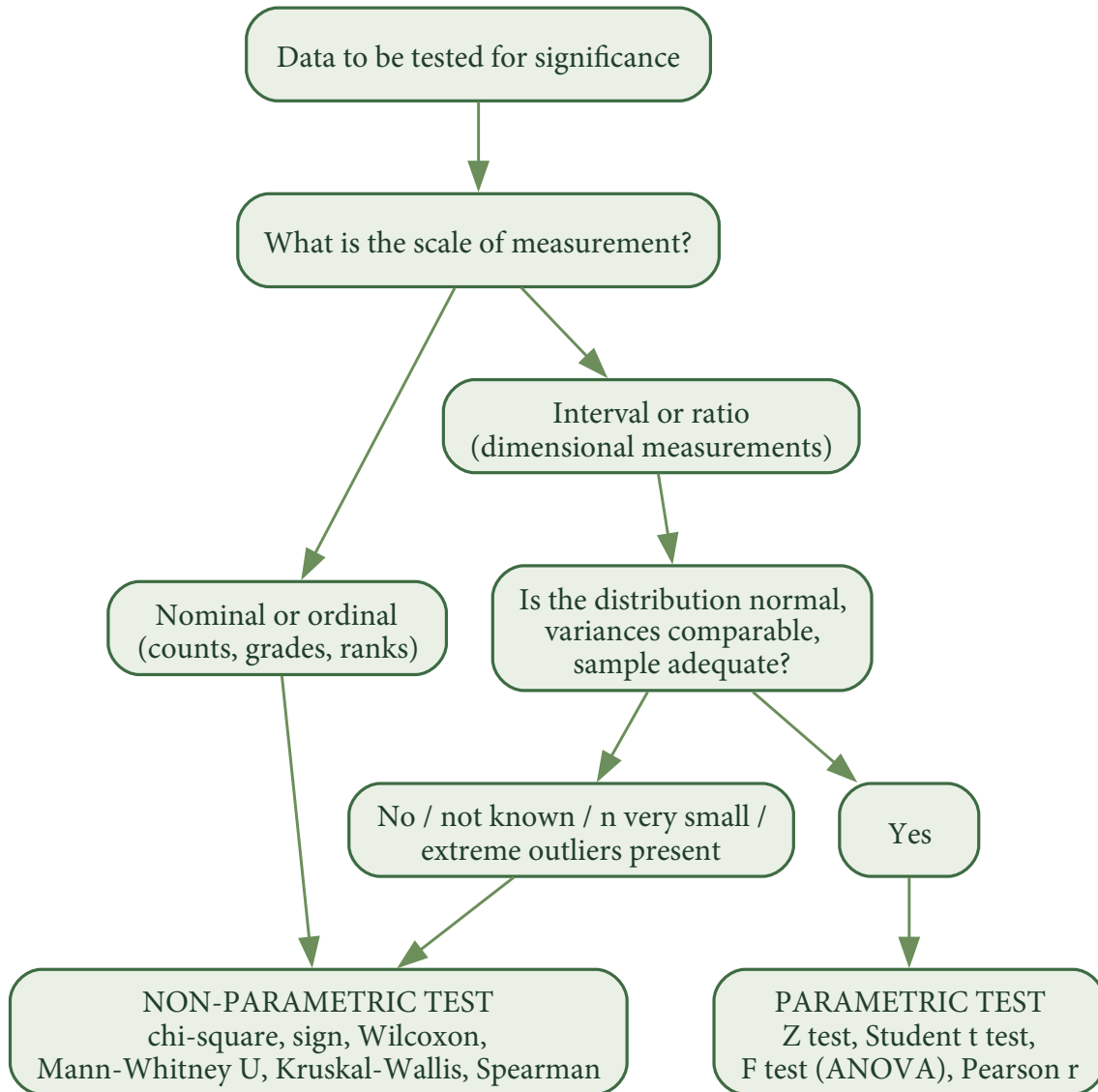
Advantages and limitations

Advantages (Sunder Rao & Richard, Ch 16, p 114)	Limitations (Sunder Rao & Richard, Ch 16, p 115)
Probability statements obtained from most non-parametric tests are exact probabilities	When all the assumptions of the parametric test are in fact met and the measurement is of the required strength, they are wasteful of data — a larger sample is needed for the same power
Usable with samples as small as 6, where no parametric alternative exists	No non-parametric method exists for testing interactions in analysis of variance
Tests exist for observations drawn from several different populations	Tables of critical values may not be easily available
Handle data that are inherently ranks , and numerical scores that in truth have only the strength of ranks	Ties are frequent when ordinal scales have few grades; they complicate ranking and need adjustment (Principles of Medical Statistics, Ch 15, p 278)
Handle simple classificatory (nominal) data	Descriptive interpretation is awkward — the result is a probability statement about ranks, and confidence intervals for medians need special formulas (Principles of Medical Statistics, Ch 15, pp 279, 287)
Much easier to learn and apply ; robust to outliers	Sensitive only to the alternative for which they were designed; a poorly chosen test loses information

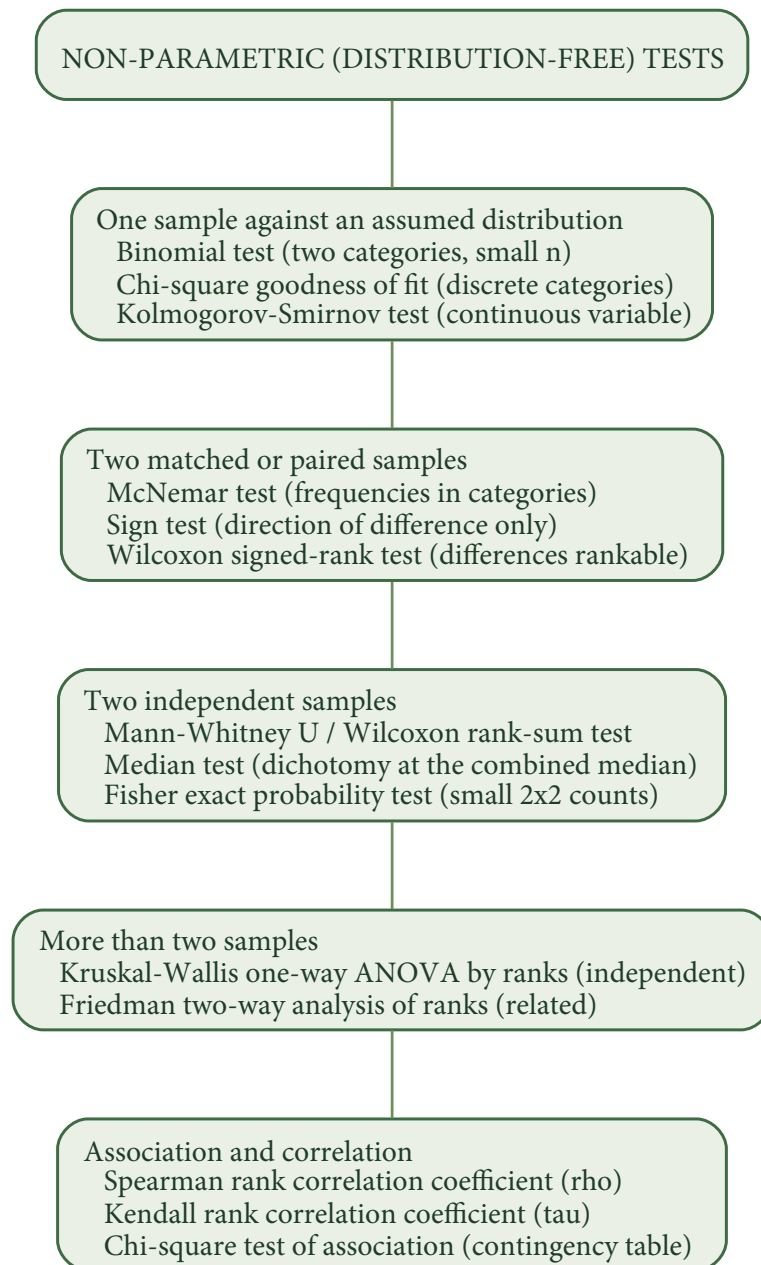
Power efficiency — the loss of power is small: the Kruskal–Wallis test has a **power efficiency of 95.5 per cent** compared with the F test, the most powerful parametric test for k independent samples (Sunder Rao & Richard, Ch 16, p 120). Where the distribution is non-Gaussian, a rank test can be **more powerful** than the corresponding parametric test (Principles of Medical Statistics, Ch 15, p 286).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Skewed distribution with an outlier** — why a rank test is used. Draw one frequency curve with a long right tail. Labels: x-axis *value of the variable*; y-axis *frequency*; mark, from left to right, **mode**, **median**, **mean**; mark one isolated point far out on the right tail as **outlier**; annotate *variance inflated* → *t test fails to reach significance*, and *ranks unaffected* → *rank test still detects the difference*.



Deciding between a parametric and a non-parametric test of significance



Non-parametric tests classified by the study situation

High-yield exam pointers

- **Definition trigger:** no assumption about the distribution + no population parameter used + works on ranks, signs or counts.
- **Pairing rule (write the counterpart every time):** paired $t \rightarrow$ sign / Wilcoxon signed-rank; unpaired $t \rightarrow$ Mann-Whitney U; one-way ANOVA $\rightarrow$ Kruskal-Wallis; two-way ANOVA $\rightarrow$ Friedman; Pearson $r \rightarrow$ Spearman rho / Kendall tau.
- **Formulas to reproduce:** Spearman $r_s = 1 - 6\sum d^2 / (n^3 - n)$; Kruskal-Wallis $H = [12 / \{N(N + 1)\}] \sum (R_j^2 / n_j) - 3(N + 1)$ at $k - 1$ degrees of freedom; McNemar $\chi^2 = (|a - b| - 1)^2 / (a + b)$ at 1 degree of freedom.
- **Decision rules:** Wilcoxon signed-rank uses the **smaller** sum of signed ranks (T); Mann-Whitney uses the **smaller** of U and U', and converts to Z when n exceeds 20.

- **Chi-square caution:** in a fourfold (2×2) table with 1 degree of freedom, apply **Yates' correction** (reduce $|O - E|$ by 0.5) when any expected cell frequency is less than 5; use **Fisher's exact test** when numbers are very small (Mahajan's Methods in Biostatistics, Ch 11, p 211).
- **Numbers to quote:** usable with samples as small as 6; Kruskal–Wallis power efficiency **95.5 per cent** against the F test.
- **Three limitations to name:** wasteful of data when parametric assumptions hold; no non-parametric test for ANOVA interactions; critical-value tables not easily available.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only — Park's Textbook of Preventive and Social Medicine 28e (Ch 21, Health Information and Basic Medical Statistics) covers the Z, *t* and chi-square tests but carries no separate section on non-parametric tests, so this answer is grounded on the dedicated biostatistics texts listed below; Park is consistent with it in directing chi-square to discrete variables and the *t* test to continuous variables (Park 28e, Ch 3, p 93).

SOURCE & COVERAGE

Sunder Rao & Richard — Introduction to Biostatistics and Research Methods, Ch 16 "Non-parametric or Distribution-free Statistical Tests", pp 114-121; Principles of Medical Statistics (Feinstein), Ch 15 "Non-Parametric Rank Tests", pp 271-292, Ch 27 "Indexes of Categorical Association", p 590, and Ch 29 "Analysis of Variance", p 638; Mahajan's Methods in Biostatistics for Medical Students and Research Workers, Ch 1 "Introduction", pp 8-9 and Ch 11 "The Chi-square Test", pp 207-211; Park's Textbook of Preventive and Social Medicine 28e, Ch 3 p 93 and Ch 21.



Poisson distribution

Asked in: Paper I May 2018 — RGUHS MD Community Medicine

Definition — The **Poisson distribution** is a discrete probability distribution describing the number of times an uncommon event occurs independently and at random within a fixed unit of time, space or population, when the average rate of occurrence stays constant (Machin, Sample Size Tables, Ch 2, pp 19-20). It applies to those situations in which the occurrences of an event can be counted but the non-occurrences cannot be counted, and counting them would be meaningless (Sunder Rao & Richard, Ch 9.3, pp 68-69). It is fixed by a single parameter — its mean, written λ (or **m**) — and its defining property is that the variance equals the mean (Jekel's Epidemiology, Ch 7, p 134).

Conditions in which counts follow a Poisson distribution

- **The variable is a discrete count**, taking only the values 0, 1, 2, 3 ... with no fixed upper limit, recorded over a defined unit of time, area, volume or person-time (Machin, Ch 2, p 20).
- **Only occurrences are countable.** The number of spells of diarrhoea in a group of infants over a set period, the radioactive fallout after a nuclear explosion, or the goals scored in a series of

matches can each be counted, but the spells, fallout or goals that did *not* occur cannot be (Sunder Rao & Richard, Ch 9.3, pp 68-69). This is the single feature that separates a Poisson situation from a binomial one.

- ▶ **The event is rare** in relation to the opportunity for it to happen — that is, the probability of the event at any one instant is very small while the population or period of observation is large.
- ▶ **Events occur independently of one another**; one occurrence neither raises nor lowers the chance of the next. Contagion, familial aggregation or epidemic spread violates this and produces counts more variable than Poisson.
- ▶ **The underlying rate is constant** over the units being compared. If the rate genuinely differs between age-groups, seasons or places, the pooled counts will not be Poisson.
- ▶ **Only the mean need be known**. Once the observed mean is available, the whole theoretical frequency distribution can be generated from it (Sunder Rao & Richard, Ch 9.3, p 70) — no second parameter is estimated, unlike the normal distribution which needs both mean and standard deviation.

Probability function and properties

Item	Statement
Probability function	$P(r) = e^{-\lambda} \cdot \lambda^r / r!$, where r = number of times the event occurs (0, 1, 2 ...), λ = mean number of events per unit, $r! = 1 \times 2 \times 3 \dots \times r$, and $e = 2.71828$ (Machin, Ch 2, p 20; Sunder Rao & Richard, Ch 9.3, p 69)
Parameter	One — the mean λ ; the distribution is completely specified by it
Mean	λ (the rate per unit time, space or person-time)
Variance	λ — variance equals the mean; this equality is the diagnostic property of the distribution (Machin, Ch 2, p 20; Jekel's Epidemiology, Ch 7, p 134)
Standard deviation	$\sqrt{\lambda}$
Range	0 to infinity (discrete, unbounded above); always positive
Shape at small λ	$\lambda = 1 \rightarrow$ markedly right-skewed (positively skewed), mode at 0 and 1 (Machin, Ch 2, p 20)
Shape at large λ	$\lambda = 4 \rightarrow$ skewness much reduced; $\lambda = 10$ or $15 \rightarrow$ nearly symmetrical and approaching the normal distribution (Machin, Ch 2, p 20)
Additive property	The sum of independent Poisson counts is itself Poisson, with mean equal to the sum of the individual means
Test of fit	Chi-square test of goodness of fit, comparing observed with theoretical Poisson frequencies (Mahajan's Methods in Biostatistics, Ch 11, p 221)

Poisson compared with the binomial and normal distributions

Feature	Binomial	Poisson	Normal (Gaussian)
Type of data	Discrete, dichotomous (present / absent)	Discrete counts of a rare event	Continuous measurement

Feature	Binomial	Poisson	Normal (Gaussian)
Non-occurrences	Countable ($n - r$)	Not countable / meaningless	Not applicable
Parameters	Two: n and p	One: λ	Two: mean and standard deviation
Mean	np	λ	μ
Variance	npq	λ (= mean)	σ^2 (independent of the mean)
Denominator	Fixed number of persons (n)	Unit of time, space or person-time	Not applicable
Typical example	Proportion of sick persons in a sample of 4 from a village with 40% sickness (Sunder Rao & Richard, Ch 9.2, p 68)	Deaths in a town from a particular disease per day; hospital admissions per day (Machin, Ch 2, p 20)	Height, weight, body temperature, intelligence quotient (Sunder Rao & Richard, Ch 9.4, p 70)
Relation	Limiting form when n is large and p is very small, with $\lambda = np$	Approaches the normal distribution as λ becomes large (Machin, Ch 2, p 20)	Common large-sample limit of both

Applications in community medicine

Application	What is counted	How the Poisson model is used
Routine vital statistics	Deaths in a town from a particular disease per day; admissions to a hospital per day	Provides the probability of observing any given daily count (Machin, Ch 2, p 20)
Laboratory counts	Yeast cells per haemocytometer square, bacterial colonies per plate, blood cells per counting chamber	Tests whether the counting technique is behaving randomly (Sunder Rao & Richard, Ch 9.3, pp 69-70)
Investigation of disease clusters	Cases of a rare disease in one locality or workplace	Poisson is "especially useful in evaluating the clustering of rare events, such as suspected cancer clusters" — observed cases are compared with the number expected under a Poisson model (Jekel's Epidemiology, Ch 7, p 134)
Space-time clustering	Pairs of cases close in both place and time (<i>Knox</i> test statistic Z)	Z is tested as a Poisson observation to obtain its p value (Handbook of Epidemiology, Ch 21 — II.8 Geographical Epidemiology, p 880)
Standardised mortality ratio (SMR)	Observed number of deaths in the study group	The variance and the 95% confidence interval of the SMR are derived assuming the observed death count is Poisson (Principles of Medical Statistics, Ch 26, p 566; Handbook of Epidemiology, Ch 2 — I.2 Rates and Risks, p 153)

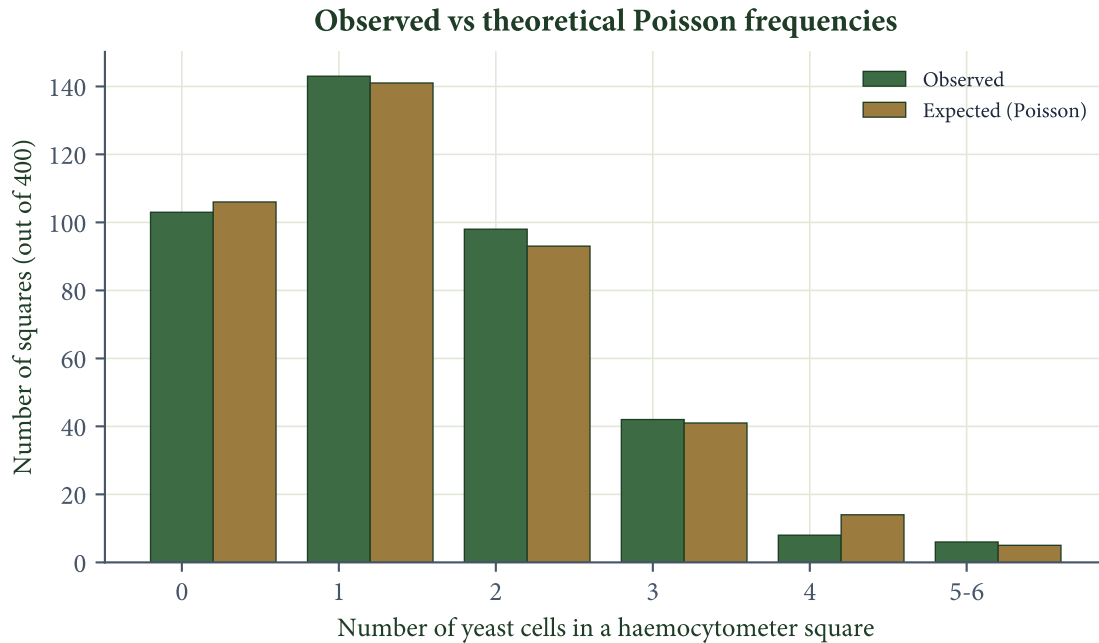
Application	What is counted	How the Poisson model is used
Cohort and surveillance analysis	Events per person-year of follow-up	Poisson regression models incidence rates and yields adjusted rate ratios (Oxford Textbook of Public Health 7e, Ch 5.5 Cohort studies, pp 1121-1122)
Post-marketing surveillance	Rare adverse drug reactions among N patients monitored	Sample size to observe a given number of reactions at an anticipated incidence λ assumes the reactions follow a Poisson distribution (Machin, Ch 11, p 143)
Interpreting a zero numerator	Serious adverse events, none observed	<i>Rule of three</i> (Hanley) — see below (Cochrane Handbook, Ch 16, p 547)
Goodness of fit	Any observed frequency distribution of counts	Chi-square test determines whether the observed distribution departs from the theoretical Poisson by more than chance (Mahajan's Methods in Biostatistics, Ch 11, p 221)

Worked example — fitting a Poisson distribution to observed counts

Yeast cells were counted in each of 400 tiny squares of a haemocytometer. From the observed mean alone, the theoretical frequencies and probabilities were generated by the Poisson formula and compared with what was actually seen (Sunder Rao & Richard, Table 9.2, pp 69-70).

Yeast cells in the square	Squares observed	Squares expected (Poisson)	Poisson probability
0	103	106	0.2650
1	143	141	0.3525
2	98	93	0.2325
3	42	41	0.1025
4	8	14	0.0350
5 and 6	6	5	0.0125
Total	400	400	1.0000

The close agreement between columns 2 and 3 is what "the counts follow a Poisson distribution" means in practice; the departure is formally tested by chi-square goodness of fit.



Poisson fit to yeast-cell counts in 400 haemocytometer squares (Sunder Rao & Richard, Biostatistics and Research Methods, Table 9.2, pp 69-70)

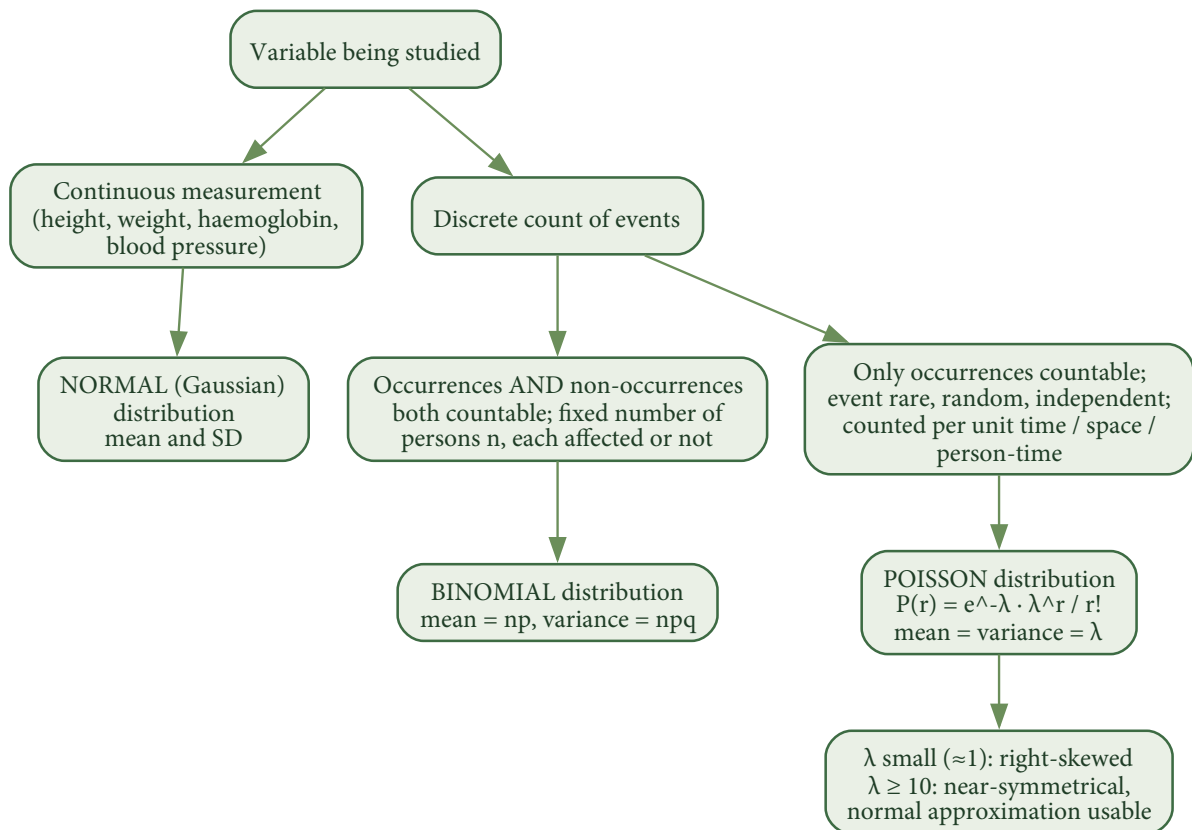
Inference and analysis using the Poisson model

- **Standard error of a rate.** If r events occur in m units of observation, the rate is $R = r/m$ and, because r is small relative to m , the standard error simplifies to $SE(R) = \sqrt{R/m}$ (Machin, Ch 2, p 25). The approximate 95% confidence interval is $R \pm 1.96 \times SE(R)$; this normal approximation must not be used when the rate is very low or the number observed is small.
- **Worked confidence interval.** Organ donations recorded over 731 days gave $R = 1.82$ per day, $SE(R) = \sqrt{(1.82/731)} = 0.05$, and a 95% confidence interval of **1.72 to 1.92 donations per day** (Machin, Ch 2, p 25).
- **Rule of three (zero numerator).** If **no** events are observed in a group of size N , the upper 95% confidence limit for the *number* of events is 3 and for the *risk* is $3/N$ — a limit taken directly from the Poisson distribution (Cochrane Handbook, Ch 16, p 547). Put the other way, about 300x patients must be observed to be 95% sure of seeing at least one event when the true incidence is $x\%$ — which is why reactions occurring in fewer than 1 in 1000 patients are routinely missed by a 2000–5000 patient development programme and need phase IV surveillance (Methods and Applications of Statistics in Clinical Trials, Ch 59, p 713). Using 3.7 instead of 3 gives the more conventional two-tailed 95% limit (Cochrane Handbook, Ch 16, pp 547-548).
- **Poisson regression.** For events accrued over person-time, $\log(\text{number of events})$ is modelled as a linear function of the exposures with $\log(\text{person-time at risk})$ as an offset, so that $\log(\text{event rate}) = a + b_1X_1 + b_2X_2 \dots$ (Oxford Textbook of Public Health 7e, Ch 5.14 Statistical methods, pp 1450-1452). A **Poisson regression coefficient is the log of a rate ratio**, just as a logistic coefficient is the log of an odds ratio (Abramson, Research Methods in Community Medicine, Ch 28).
- **Overdispersion (extra-Poisson variation)** is the commonest practical failure of the model: when events cluster in time or place, or unmeasured characteristics mix several different rates together, the variance exceeds the mean. It is handled by a negative binomial model, or by robust

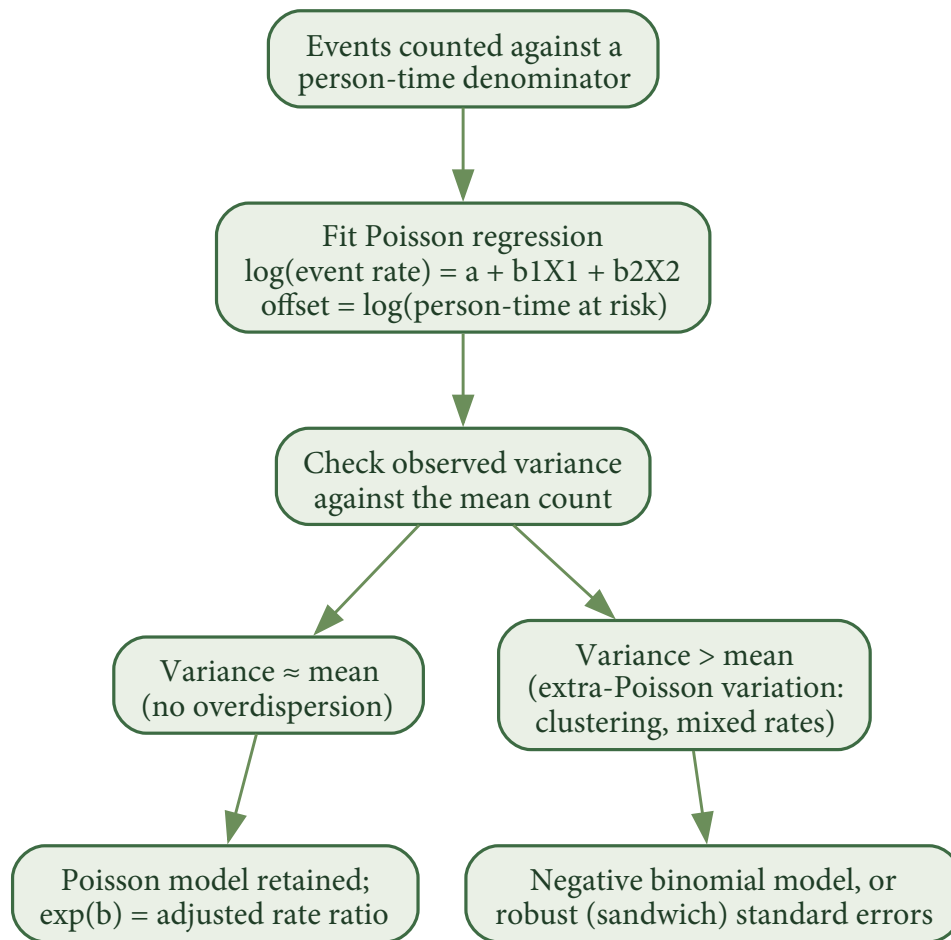
(sandwich) standard errors within a generalised estimating equation framework (Handbook of Epidemiology, Ch 2 — I.2 Rates and Risks; Ch 16 — II.3 Regression Methods).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Diagram 1 — Shape of the Poisson distribution as λ increases** (four small bar charts side by side, redrawn from Machin, Fig 2.4, Ch 2, p 20). Labels: x -axis "Number of events (r)"; y -axis "Probability"; panel titles " $\lambda = 1$ ", " $\lambda = 4$ ", " $\lambda = 10$ ", " $\lambda = 15$ "; annotate panel 1 "markedly right-skewed, mode at 0–1", panel 2 "skewness reduced", panels 3 and 4 "nearly symmetrical — approaches the normal distribution".



Choosing the correct probability distribution for a public-health variable



Analytic pathway for count and rate data in a cohort or surveillance dataset

High-yield exam pointers

- ▶ $P(r) = \frac{e^{-\lambda} \lambda^r}{r!}$ — write the formula first; define r , λ , $r!$ and $e = 2.71828$.
- ▶ **Mean = variance = λ** ; $SD = \sqrt{\lambda}$. One parameter only. This single line earns the most marks.
- ▶ The operational test of a Poisson situation: *the non-occurrences cannot be counted* (diarrhoeal spells, radioactive fallout, goals scored) — contrast with binomial, where $n - r$ is countable.
- ▶ Shape sequence: $\lambda = 1$ right-skewed $\rightarrow \lambda = 4$ less skewed $\rightarrow \lambda = 10-15$ nearly symmetrical $\rightarrow$ normal.
- ▶ **SE of a rate** = $\sqrt{R/m}$; 95% CI = $R \pm 1.96\sqrt{R/m}$.
- ▶ **Rule of three**: no events in N observations $\rightarrow$ upper 95% limit of risk = $3/N$.
- ▶ Public-health uses to name: rare-disease death counts, cell/colony counts, cancer-cluster investigation, *Knox* space-time clustering, standardised mortality ratio confidence intervals, Poisson regression of person-time rates, post-marketing surveillance of rare adverse drug reactions.
- ▶ Goodness of fit to a Poisson distribution is tested by the **chi-square test**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only — Park's Textbook of Preventive and Social Medicine 28e does not deal with probability distributions in its Health Information and Basic Medical Statistics chapter, so this answer is grounded on the biostatistics and epidemiology texts in the corpus rather than on the spine book; the standard deviation as $\sqrt{\lambda}$, the additive property of independent Poisson counts, and the statement that the Poisson is the limiting form of the binomial when n is large and p is very small ($\lambda = np$) are universally accepted textbook statistics stated here without a page citation from the retrieved passages.

SOURCE & COVERAGE

Sunder Rao & Richard — An Introduction to Biostatistics and Research Methods (Ch 9, Probability Distributions); Machin, Campbell, Tan & Tan — Sample Size Tables for Clinical Studies 3e (Ch 2, Distributions and confidence intervals; Ch 11, Post-marketing surveillance); Jekel's Epidemiology, Biostatistics and Preventive Medicine (Ch 7); Mahajan's Methods in Biostatistics for Medical Students and Research Workers (Ch 11, The Chi-square Test); Handbook of Epidemiology, Ahrens & Pigeot (I.2 Rates, Risks, Measures of Association and Impact; II.3 Regression Methods for Epidemiologic Analysis; II.8 Geographical Epidemiology); Oxford Textbook of Public Health 7e (5.5 Cohort studies; 5.14 Statistical methods); Principles of Medical Statistics, Feinstein (Ch 26); Cochrane Handbook for Systematic Reviews of Interventions (Ch 16, Special topics in statistics); Abramson — Research Methods in Community Medicine (Ch 28); Methods and Applications of Statistics in Clinical Trials (Ch 59, Phase IV Trials).



Sampling techniques suitable for rapid survey

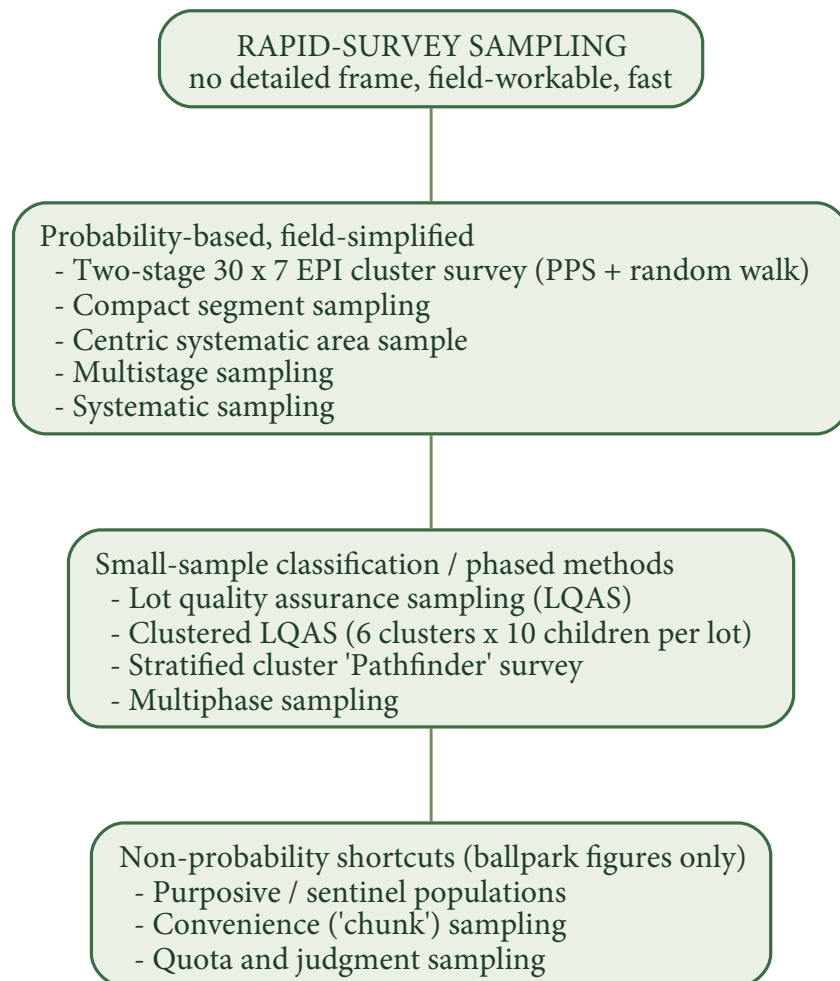
Asked in: Paper I Nov 2018 — RGUHS MD Community Medicine

Definition — A **rapid survey** (rapid epidemiological method, rapid appraisal, rapid assessment) is a simplified survey procedure that speeds up a study so that real-time results are available as a basis for programme decisions and action (Abramson 6e, Ch 31, p 305). It is simplified along four axes — sampling, data collection, processing, and evaluation of care — of which the sampling method is the decisive one (Abramson 6e, Ch 31, pp 305-306). A rapid-survey sampling technique must therefore work **without a detailed sampling frame**, must be executable in the field **by health workers with minimal technical support**, and must still give data precise enough for a programme decision.

Sampling techniques suitable for a rapid survey

Technique	Principle	Typical rapid-survey use	Main limitation
Two-stage EPI cluster survey (30 x 7)	Stage 1: communities chosen with probability proportional to size (PPS) ; stage 2: a "self-weighted" cluster picked in the field by random walk	Immunization coverage, disease prevalence, service coverage, blindness, health-service needs (Abramson 6e, Ch 31, p 306)	Second stage is not random — it is effectively quota sampling; higher standard error than simple random sampling (Abramson 6e, Ch 31, p 308; Mahajan, Ch 6, p 138)
Compact segment sampling	Selected community is sketch-mapped and split into equal segments of a preselected target segment size ; one segment chosen at random and all eligible people in it surveyed	Preferred cleaner alternative to the random walk; suits surveys with several measurements (Abramson 6e, Ch 31, p 309)	Needs a rough sketch map — slightly slower
Centric systematic area sample	Survey area divided into equal squares on a map; the community nearest each square's centre is taken and every household in it investigated	Map-based rapid appraisal where no population list exists (Abramson 6e, Ch 31, pp 307-308)	Purely geographic, ignores population size
Multistage sampling	Successive random stages — state → district → taluka → village → household	Large country/region surveys; the very large area is reached in steps (Mahajan, Ch 6, p 137; Abramson 6e, Ch 31, p 307)	Sampling error accumulates at each stage
Systematic sampling	Every <i>k</i> th unit from a random start	Field-convenient where no list exists (houses along a street) (Sunder Rao, Ch 10, p 79)	Periodicity in the list produces bias
Lot quality assurance sampling (LQAS)	A very small sample classifies a "lot" (supervision area) as acceptable or unacceptable against a decision rule	Identifying priority districts/blocks with low coverage or high prevalence (Sunder Lal, Ch 6, p 252; Park Plus, Ch 19, p 239)	Gives a verdict, not a precise coverage estimate

Technique	Principle	Typical rapid-survey use	Main limitation
Stratified cluster ("Pathfinder") survey	Stratified clusters chosen to include the population subgroups most likely to differ	Oral-health assessment and planning; 25-50 subjects per index age group per cluster (Sunder Lal, Ch 6, p 252)	Deliberate, not representative, stratification
Multiphase sampling	Sample shrinks at each successive phase, so the costly test is done on the last, smallest subsample	Nutrition surveys — interview → clinical → anthropometry → biochemistry (Park Plus, Ch 19, p 239)	Later phases too small for subgroup analysis
Purposive / sentinel sampling	Deliberately chosen "sentinel" groups stand in for the population	Influenza surveillance by sick absenteeism in selected schools or workplaces; antenatal attenders for sexually transmitted infection prevalence in refugee camps (Abramson 6e, Ch 31, pp 305-306)	Not a probability sample; no yardstick of representativeness (Sunder Rao, Ch 10, p 77)
Convenience ("chunk"), quota, judgment sampling	Units taken for ease of access or by the investigator's judgment	Only when rough "ballpark figures" suffice, or for qualitative appraisal (Abramson 6e, Ch 31, p 305)	Maximum bias; results unconvincing (Sunder Lal, Ch 6, p 252)



Sampling techniques usable in a rapid survey, grouped by how they cut time and cost

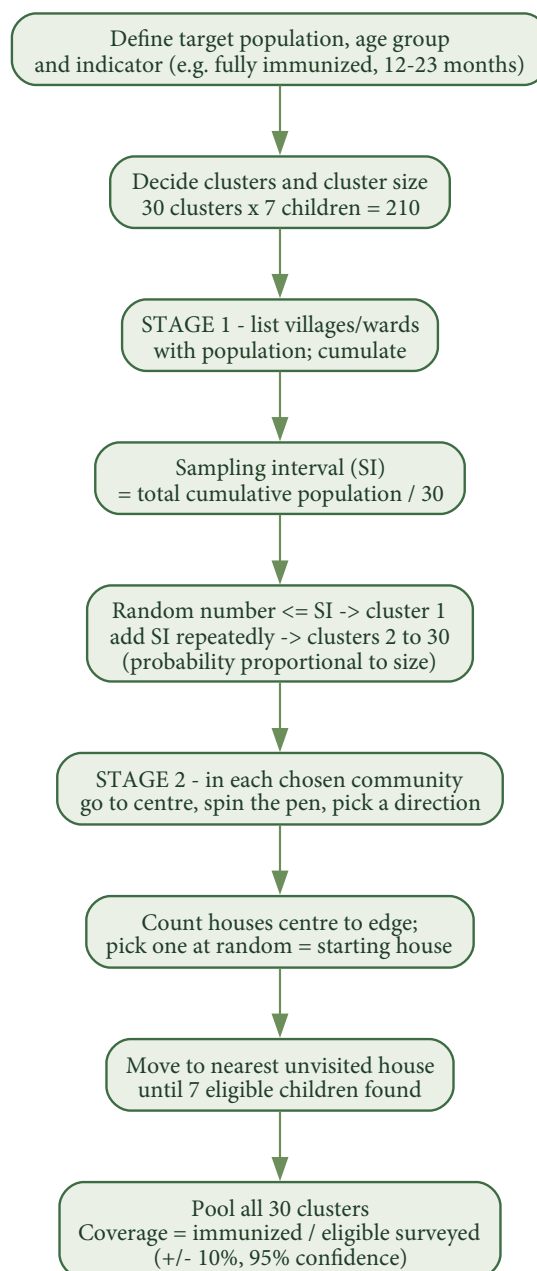
The WHO EPI 30 x 7 cluster survey — the standard rapid survey

Advocated by the World Health Organization's **Expanded Programme on Immunization (EPI)** as a rapid, cheap and reasonably accurate basis for coverage surveys, and since extended to many other topics (Abramson 6e, Ch 31, p 306). It has four advantages: **no detailed sampling frame is needed**, subjects live close together, field selection is simple enough for peripheral health workers, and the data are reasonably representative (Abramson 6e, Ch 31, p 306).

- **Fix the number and size of clusters.** More clusters give more valid results; **30 is the practical minimum**. Required sample size ÷ number of clusters = subjects per cluster. The original EPI recommendation for vaccination status was **30 clusters x 7 children aged 12-23 months = 210 children**; **30 clusters of 30** are suggested for nutrition surveys (Abramson 6e, Ch 31, pp 306-307; Mahajan, Ch 6, p 138).
- **Select the 30 communities by PPS.** List all villages/wards with their populations, cumulate the populations, and compute **sampling interval (SI) = total cumulative population ÷ 30**. Draw a random number ≤ SI with the same number of digits; the first cumulative entry equalling or exceeding it is cluster 1. Add SI repeatedly for clusters 2, 3 ... 30. A large village may thus contribute more than one cluster — this is probability proportional to size (Sunder Lal, Ch 6, pp 250-251; Abramson 6e, Ch 31, p 307).

- **Select the cluster within the community (random walk).** Go to a central point, choose a direction at random by spinning a bottle or throwing a pen ("spin-the-pen"), count the households from centre to edge in that direction and pick one at random as the starting house; thereafter visit the **unvisited household whose door is nearest** the last one, until the required number of eligible subjects is found (Abramson 6e, Ch 31, p 307). In India the anganwadi or ASHA household survey register can serve as a ready frame for this stage (Sunder Lal, Ch 6, p 251).
- **Collect and analyse.** Coverage = eligible children found immunized ÷ eligible children surveyed, pooled over all 30 clusters.

Reliability standards of the 30 x 7 design: accuracy ± 10 percentage points, at 95% confidence; results reflect the **area as a whole** and **comparisons between individual clusters are not valid**; the estimate refers to the year preceding the survey (Sunder Rao, Ch 10, p 80; Mahajan, Ch 6, p 138).

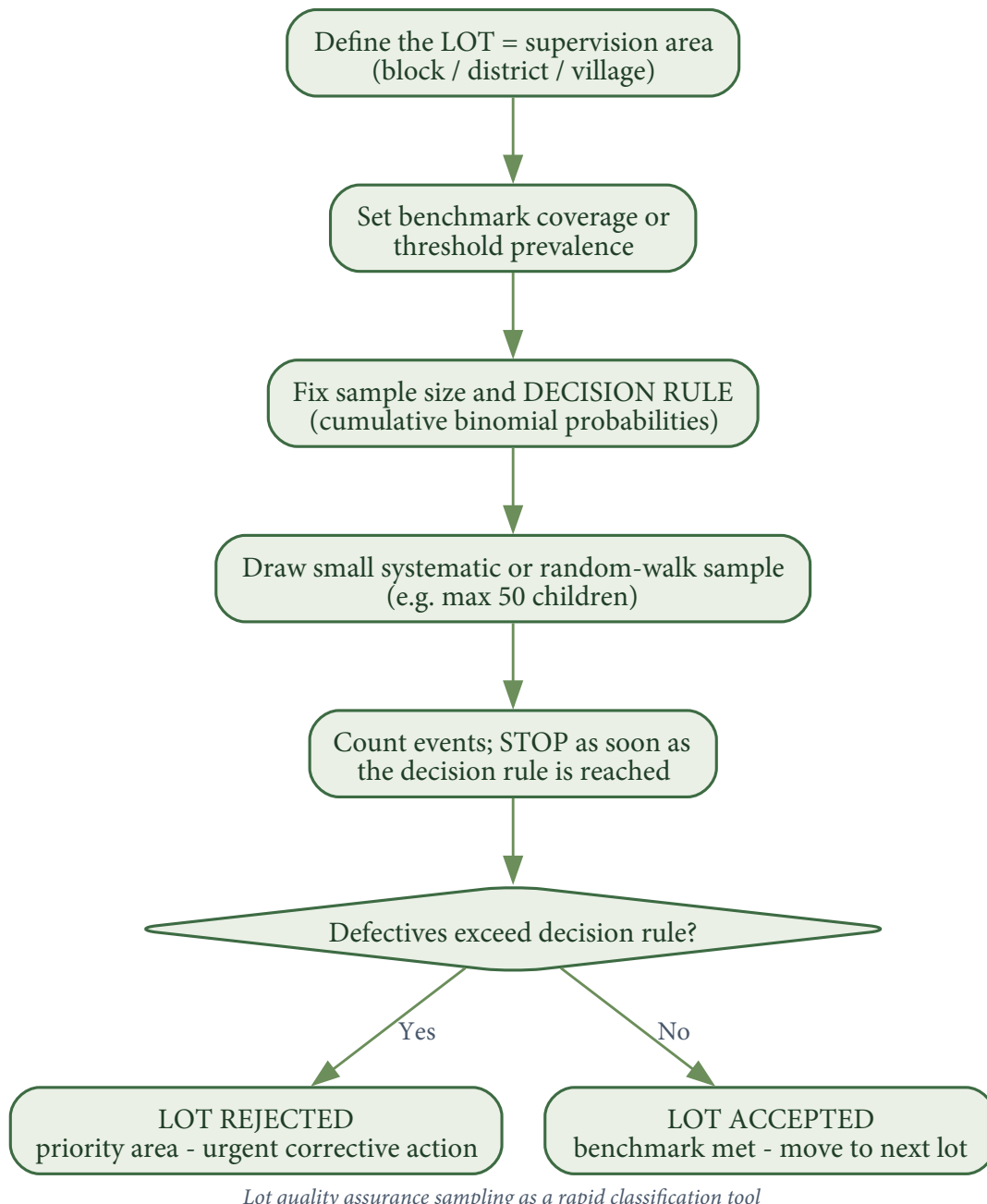


Sequence of a WHO EPI 30 x 7 two-stage cluster survey

Lot quality assurance sampling (LQAS)

Borrowed from **industrial quality control** of the 1920s, where a small random sample of a manufactured batch (**lot**) is tested and the lot is **rejected if defectives exceed a predetermined decision rule** set by the desired standard and a statistically determined sample size (Park Plus, Ch 19, p 239). In public health the "lot" is a **supervision area** — a block, district or sub-district — and LQAS identifies which areas fail to reach an established benchmark and need urgent intervention (Park Plus, Ch 19, pp 239-241; Sunder Lal, Ch 6, p 252).

Element	Rapid-survey application
Sample per lot	Very small — a maximum of 50 children examined in a village, stopping the moment a predetermined number of cases is reached (Abramson 6e, Ch 31, pp 309-310)
Decision rule	The count that correctly classifies about 95% of communities at the sought prevalence, from cumulative binomial probabilities; e.g. for trachoma, 9 cases points to prevalence $\geq 20\%$, while 0 or 1 case means $< 5\%$ (Abramson 6e, Ch 31, pp 309-310)
Clustered LQAS	Within each lot, clusters chosen by PPS from a village list; for polio campaign evaluation commonly 6 clusters of 10 children per lot (Plotkin's Vaccines 7e, Ch 76, pp 2151-2152)
Field example	Used to identify the low-vaccination-coverage areas among five cities of Bangladesh (Sunder Lal, Ch 6, p 252)



Merits and limitations of rapid-survey sampling

Merits	Limitations
No detailed sampling frame or census required (Abramson 6e, Ch 31, p 306)	The EPI second stage is not random — it is quota sampling; results mislead if similar people live close together (Abramson 6e, Ch 31, p 308)
Cuts the cost of frame preparation and of travel between units; removes "packing" of neighbouring households into a selected house (Sunder Lal, Ch 6, pp 251-252)	Sampling error is higher than a simple random sample of the same size (Sunder Lal, Ch 6, p 252; Mahajan, Ch 6, p 138)
Simple enough for peripheral health workers; less time and cost (Mahajan, Ch 6, p 138)	Comparisons between clusters are invalid unless samples are large (Sunder Rao, Ch 10, p 80)

Merits	Limitations
Real-time results — a district report in about 10 days (Abramson 6e, Ch 31, p 305)	Unsuitable when the survey covers many topics needing different cluster sizes, or when associations are being studied (Abramson 6e, Ch 31, p 308)
Special value in emergencies — cluster surveys with morbidity surveillance guided the Rwandan refugee response at Goma, 1994 (Abramson 6e, Ch 31, pp 313-314)	Bias remedies (5th-nearest household, quadrant split, random selection from a listed frame) make the method less "rapid" (Abramson 6e, Ch 31, pp 308-309)

Rapid qualitative appraisal complements these: informal interviews with the public and **key informants**, focus-group discussions, the nominal group technique and unstructured observation, with **triangulation** across methods to confirm validity (Abramson 6e, Ch 31, pp 310-311).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Diagram 1 — PPS selection line for a 30-cluster survey.** Labels: horizontal line of villages A, B, C ... with *estimated population*; *cumulative population* below each; *total cumulative population*; *sampling interval (SI) = total ÷ 30*; *random start R (≤ SI)*; tick marks at R, R+SI, R+2SI ...; arrow to *selected cluster*; note *large village = 2 or more clusters (PPS)*.
- **Diagram 2 — Random walk within a village.** Labels: *village boundary*; *central point*; *spin-the-pen direction arrow*; *houses counted centre → edge*; *randomly chosen starting house*; *nearest unvisited house chain of arrows*; *stop at 7th eligible child (12-23 months)*.

High-yield exam pointers

- **30 x 7 = 210 children aged 12-23 months** is the classical WHO EPI immunization coverage cluster survey; **30 x 30** for nutrition surveys.
- **Sampling interval = total cumulative population ÷ 30**; random start ≤ SI with the same number of digits; clusters selected with **probability proportional to size**.
- Precision of the 30 x 7 design: **±10 percentage points at 95% confidence**; valid for the area as a whole, **not** for cluster-to-cluster comparison.
- Stage 2 field methods: **spin-the-pen random walk** (classical) vs **compact segment sampling** (cleaner — sketch map, equal segments, all eligibles in one random segment).
- LQAS classifies a lot as acceptable/unacceptable against a **decision rule**; it identifies priority areas, it does not estimate coverage precisely. Clustered LQAS for polio campaigns: **6 clusters x 10 children per lot**.
- Cluster sampling has a **higher standard error** than simple random sampling of the same size — the price paid for speed and cheapness.
- Non-probability shortcuts (convenience, quota, purposive/sentinel) are acceptable only for **ballpark figures** or qualitative appraisal.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the classical EPI random walk has been criticised because its second stage is non-random, and current practice favours listing or mapping the households in each selected cluster and choosing them randomly, or using compact segment sampling instead (Abramson 6e, Ch 31, pp 308-309; Plotkin's Vaccines 7e, Ch 76, pp 2151-2152).

SOURCE & COVERAGE

Abramson & Abramson — Research Methods in Community Medicine, 6e, Ch 31 (Rapid Epidemiological Methods) and Ch 8 (Sampling); Mahajan's Methods in Biostatistics for Medical Students and Research Workers, Ch 6 (Sampling); Sunder Rao & Richard — Introduction to Biostatistics and Research Methods, Ch 10 (Sampling); Sunder Lal — Textbook of Community Medicine, Ch 6 (Basic Statistics and its Applications Including Research Methodology); Plotkin's Vaccines, 7e, Ch 76 (Immunization in Developing Countries); Park Plus (Amrutha & Vijayalakshmi), Ch 19 (Health Information and Basic Medical Statistics). Coverage note: Park's Textbook of Preventive and Social Medicine 28e (Ch 21) describes only simple random, systematic and stratified random sampling and carries no dedicated rapid-survey section, so the rapid-survey content above is grounded in the biostatistics and research-methods texts of the corpus.



Tests of significance

Asked in: Paper I 12-Nov-2021 · Paper I 19-Nov-2020 (2×) — RGUHS MD Community Medicine

Definition — A test of significance decides whether an observed difference — between a sample estimate and a population value, or between two sample estimates — is a **real** difference or merely the play of **chance** (**sampling variability**). It expresses the observed difference in units of its **standard error** (SE), giving the critical ratio, and reads off the probability of that or a larger ratio arising by chance from the appropriate sampling distribution (Sunder Rao, Ch 12, pp 88-89). A probability of 5% or less is conventionally treated as rare, and the null hypothesis is then rejected (Sunder Rao, Ch 11, pp 85-86).

Basic concepts and terminology

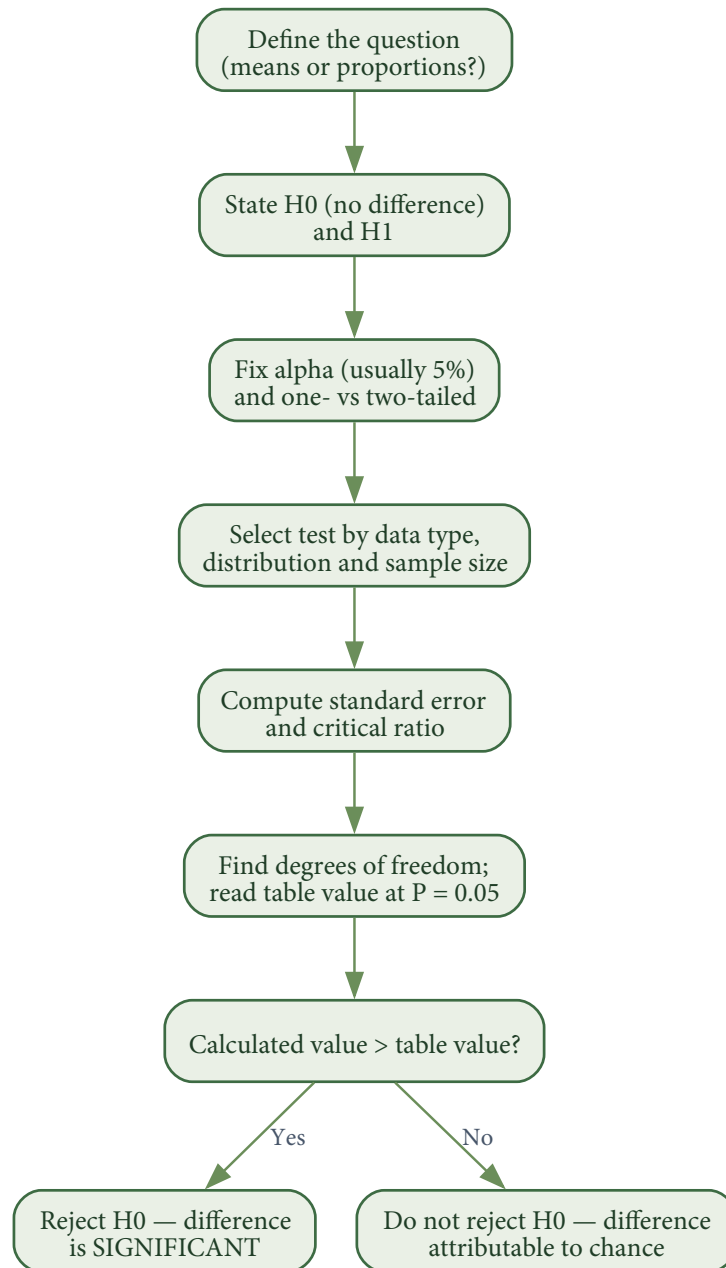
Term	Meaning
Null hypothesis (H_0)	Assumption of <i>no</i> real difference; any observed difference is due to chance alone (Mahajan, Ch 8, pp 175-176)
Alternative hypothesis (H_1)	The factor under study does exert a real effect — accepted only when H_0 is rejected (Mahajan, Ch 9, pp 191-192)
Standard error (SE)	Measure of the chance variation of a sample estimate; the yardstick against which the observed difference is measured (Mahajan, Ch 9, pp 180-182)

Term	Meaning
Critical ratio	Observed difference between the statistics ÷ standard error of that difference (Sunder Rao, Ch 12, pp 88-89)
Level of significance (α)	Pre-fixed probability at which H_0 is rejected — usually 5%, 1% or 0.5% (Mahajan, Ch 8, pp 174-175)
P value	Probability that the observed (or a larger) difference could have arisen by chance; $P \leq 0.05$ = significant (Sunder Rao, Ch 11, pp 85-86)
Degrees of freedom (df)	Number of observations/cells free to vary given a fixed total; the row of the table against which the calculated value is read (Mahajan, Ch 9, pp 191-192)
One- vs two-tailed test	Rejection region placed in one tail (direction of difference specified) or split between both tails (Mahajan, Ch 8, pp 173-175)

Standard normal deviates for two-tailed tests: $Z = 1.96$ at $\alpha = 0.05$, 2.575 at $\alpha = 0.01$, 3.29 at $\alpha = 0.001$ (Sunder Rao, Ch 23, pp 214-215). An observed difference greater than 1.96 SE is significant at 5%; greater than 3 SE it is real in more than 99% of cases (Mahajan, Ch 9, pp 185-186).

Steps in performing a test of significance

- **Define the problem** — comparison of a sample mean with a population mean, of two sample means, of a sample proportion with a population proportion, or of two sample proportions (Sunder Rao, Ch 12, pp 88-89).
- **State the null hypothesis** (no difference) and the alternative hypothesis (Mahajan, Ch 8, pp 175-176).
- **Fix the level of significance** (usually 5%) and decide one- or two-tailed testing, before the analysis (Mahajan, Ch 8, pp 174-175).
- **Choose the sampling distribution** appropriate to the data type and sample size, and compute the **standard error** of the statistic (Sunder Rao, Ch 12, pp 88-89).
- **Calculate the critical ratio** (Z , t , χ^2 , F) = difference ÷ standard error.
- **Find the degrees of freedom** and refer the calculated value to the relevant table at the chosen P (Mahajan, Ch 9, pp 191-192).
- **Draw the inference** — value exceeding the table value at $P = 0.05$ means H_0 is rejected and H_1 accepted; otherwise the difference is attributed to sampling variability (Mahajan, Ch 9, pp 191-192).



Sequence of steps in performing a test of significance

Type I and Type II errors, and power

	H ₀ is actually true	H ₀ is actually false
H ₀ rejected	Type I error (α)	Correct decision
H ₀ not rejected	Correct decision	Type II error (β)

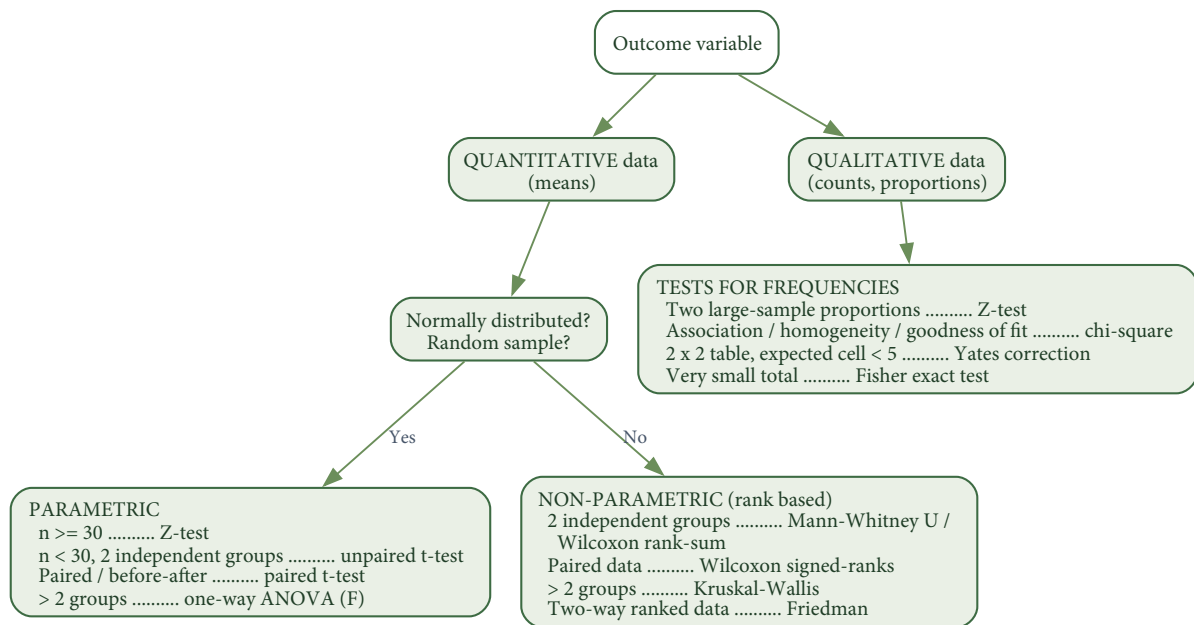
- **Type I error (α)** — rejecting a true null hypothesis, i.e. declaring a difference that does not exist. Its size is fixed in advance by the level of significance: at 5% we shall wrongly reject 5% of all true hypotheses (Mahajan, Ch 8, pp 174-175; Sunder Rao, Ch 11, p 86).
- **Type II error (β)** — failing to reject a false null hypothesis, i.e. missing a real difference. It measures the risk of failure to detect a difference that truly exists (Sunder Rao, Ch 11, p 86).

- **Power** = $1 - \beta$ — the numerical value indicating the sensitivity of the test, i.e. its ability to detect a real difference (Sunder Rao, Ch 11, pp 86-87).
- In medical studies **Type I error is the more serious** of the two; Type II error is less serious because it only calls for confirmation with a larger sample (Mahajan, Ch 8, pp 173-175).
- For a fixed sample size the two errors move in opposite directions — **both cannot be minimised simultaneously**. The accepted practice is to fix α at a tolerable limit and then reduce β by **increasing the sample size** (Sunder Rao, Ch 11, pp 86-87).

Classification and choice of the test

Test	Type of data	Criteria / when applied	Degrees of freedom
Z (standard normal deviate) test	Quantitative, normally distributed	Random sample, $n \geq 30$; also for proportions in large samples	Not required (Z table)
Unpaired (independent) Student's t-test	Quantitative, normally distributed	Two independent groups, $n < 30$	$n_1 + n_2 - 2$
Paired t-test	Quantitative, normally distributed	Two observations on the <i>same</i> individuals (before/after, two sites)	$n - 1$
Variance ratio (F) test	Quantitative	Tests whether two samples have homogeneous variances	$n_1 - 1$ and $n_2 - 1$
Analysis of variance (ANOVA), one-way	Quantitative	Simultaneous comparison of means of more than two groups	Between and within group df
Two-way ANOVA	Quantitative	Two factors acting simultaneously on the response	As per factors and residual
Chi-square (χ^2) test	Qualitative (frequencies)	Association, homogeneity, goodness of fit	$(r - 1)(c - 1)$

Criteria for the t-test: random samples, quantitative data, normally distributed variable, sample size less than 30 — the Z-test criteria differ only in the last (Mahajan, Ch 9, pp 192-193). ANOVA uses the **F-ratio = between-group variance ÷ within-group variance**; a significant F means the groups are not all from populations with identical means, after which further analysis shows which groups differ (Sunder Rao, Ch 15, pp 120-123).



Chi-square (χ^2) test — applications and restrictions

Devised by *Karl Pearson*, $\chi^2 = \sum (O - E)^2 / E$, where O and E are the observed and expected frequencies (Sunder Rao, Ch 14, p 112). At 1 df the 5% table value is **3.84** (Sunder Rao, Ch 14, p 113).

Applications — (1) test of **independence or association** between two attributes in a contingency table; (2) test of **homogeneity** (similarity of two or more samples); (3) test of **goodness of fit** between an observed and a theoretical distribution (Sunder Rao, Ch 14, p 113).

Restrictions (Mahajan, Ch 11, pp 223-224; Sunder Rao, Ch 14, pp 118-119)

- ▶ Must be applied to **actual frequencies**, **never to percentages** (Mahajan, Ch 11, pp 219-220).
- ▶ In a fourfold (2×2) table with 1 df, if any **expected value is less than 5**, apply **Yates's correction for continuity** — subtract 0.5 from the absolute difference $|O - E|$ before squaring.
- ▶ Even after Yates's correction the test is misleading if an expected frequency is much below 5; if more than one cell has an expected number below 1, χ^2 should not be used at all.
- ▶ In tables larger than 2×2 , Yates's correction cannot be applied — **pool** the small class with the adjacent one.
- ▶ Interpret with caution when the grand total is **less than 50**.
- ▶ χ^2 indicates the **presence or absence** of an association but is **not a measure of its strength**, and a statistical association does not establish cause and effect.

Non-parametric (distribution-free) tests

Used when the assumption of a normal distribution cannot be made, when samples are very small, or when data are only ranks or categories; measurements are converted to **ranks or signs** before analysis (Sunder Rao, Ch 16, pp 126-127).

Parametric test	Non-parametric counterpart
Unpaired t-test	Mann-Whitney U test / Wilcoxon two-group rank-sum test
Paired t-test	Wilcoxon signed-ranks test
One-way ANOVA	Kruskal-Wallis H test (H follows χ^2 with $df = k - 1$)
Two-way ANOVA	Friedman test (two-way analysis of ranked data)
χ^2 with very small numbers	Fisher's exact test
Comparison of two continuous distributions	Kolmogorov-Smirnov test

Advantages — probabilities obtained are exact; usable with samples as small as 6; applicable to inherently ranked and to purely classificatory data; easier to learn and apply (Sunder Rao, Ch 16, pp 126-127). **Limitation** — they discard information contained in the actual measurements and are therefore less powerful when the parametric assumptions do hold.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Normal curve showing zones of acceptance and rejection (two-tailed test) — label exactly:

- ▶ Central shaded area: Acceptance region of null hypothesis (H_0)
- ▶ Left tail: Rejection region 2.5%; right tail: Rejection region 2.5%
- ▶ Baseline: $(X_1 - X_2)$, difference between two means
- ▶ Boundary markers: $Z = 1.96 \text{ SE}$ on either side of the mean
- ▶ Inner bracket: $Z \leq 1.96 \text{ SE}$ — difference not significant; outer bracket: $Z > 1.96 \text{ SE}$ — difference significant at 5%
- ▶ Formula on the curve: $\text{SE} = \sigma \sqrt{(1/n_1 + 1/n_2)}$

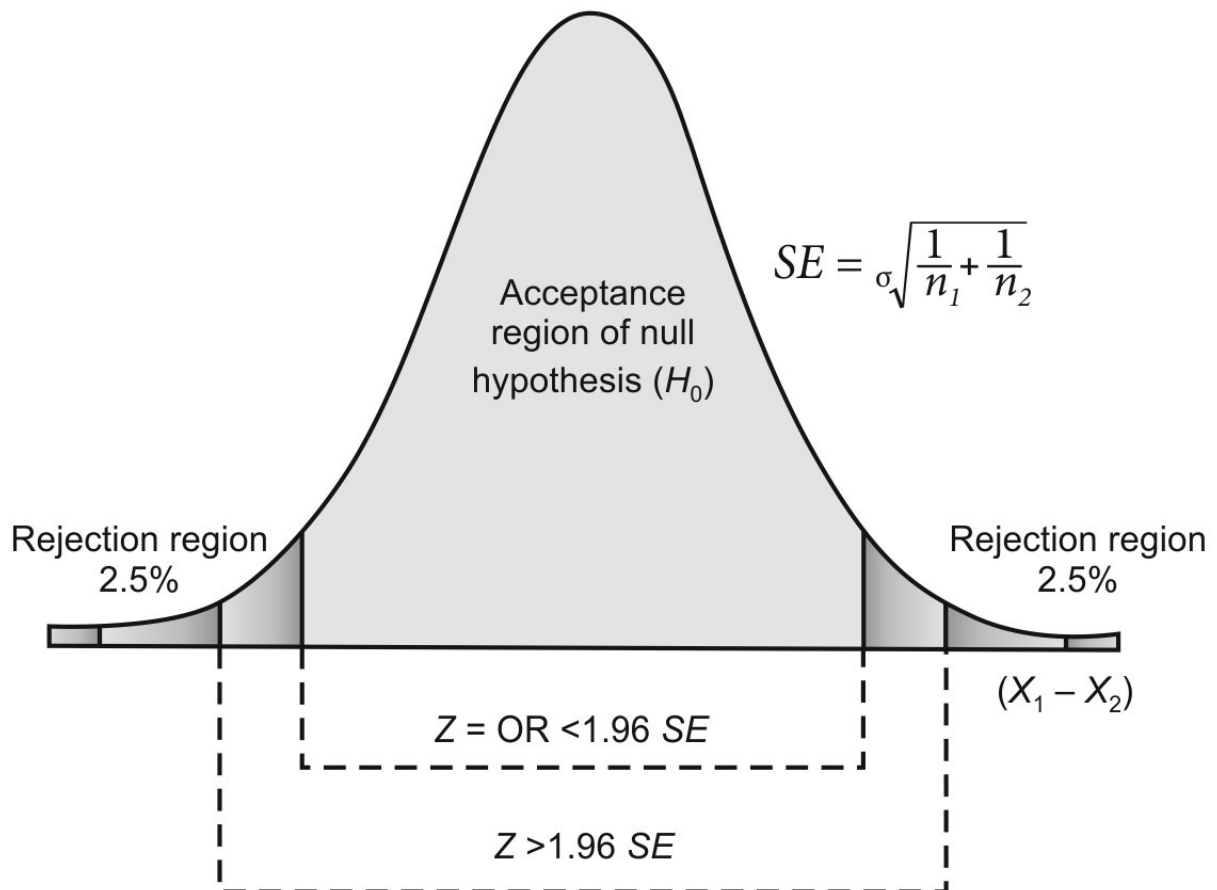


Fig 8.6 — Zones of acceptance and rejection of null hypothesis. Rejection (Mahajan's Methods in Biostatistics for Medical Students and Research Workers (PDFDrive), p.171)

High-yield exam pointers

- ▶ Critical ratio = **observed difference ÷ standard error of the difference**; $Z = 1.96$ ($P = 0.05$), 2.575 ($P = 0.01$), 3.29 ($P = 0.001$), two-tailed.
- ▶ df: unpaired $t = n_1 + n_2 - 2$; paired $t = n - 1$; $\chi^2 = (r - 1)(c - 1)$; Kruskal-Wallis = $k - 1$.
- ▶ $\chi^2 = \Sigma (O - E)^2 / E$; table value at 1 df, 5% = 3.84; Yates's correction = subtract 0.5 from $|O - E|$.
- ▶ t-test criteria (mnemonic R-Q-N-S): Random sample, Quantitative data, Normal distribution, Sample size < 30 ($\geq 30 \rightarrow Z$ -test).
- ▶ α = **Type I** (reject a true H_0 , "false alarm"); β = **Type II** (miss a real difference); **power** = $1 - \beta$. Fix α , cut β by enlarging the sample.
- ▶ Three uses of χ^2 : **independence/association, homogeneity, goodness of fit**.
- ▶ More than two group means $\rightarrow$ ANOVA (F test), never repeated t-tests.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Modern reporting practice increasingly asks for a **confidence interval alongside (or instead of) the P value**, since the two are mathematically reciprocal but the interval also shows how large the difference might really be, and a very large sample can yield an impressive P value for a difference of no clinical importance (Principles of Medical Statistics, Ch 13, pp 235-237; Ch 25, pp 546-547).

SOURCE & COVERAGE

Mahajan's Methods in Biostatistics for Medical Students and Research Workers (Ch 8 Sampling Variability and Significance; Ch 9 Significance of Difference in Means; Ch 11 The Chi-square Test); Sunder Rao & Richard — An Introduction to Biostatistics and Research Methods (Ch 11 Basis of Statistical Inference; Ch 12 Tests of Significance and Estimation; Ch 14 The Chi-square Test; Ch 15 Analysis of Variance; Ch 16 Non-parametric Tests; Ch 23 Sample Size Determination); Principles of Medical Statistics (Ch 13 Parametric Sampling: Z and t Tests; Ch 15 Non-Parametric Rank Tests; Ch 29 Analysis of Variance); Park's Textbook of Preventive and Social Medicine 28e (Ch 21 Health Information and Basic Medical Statistics).



Box plots in data presentation

Asked in: Paper IV 18-Nov-2019 — RGUHS MD Community Medicine

Definition — A **box plot** (Tukey's "**box-and-whiskers plot**", proposed 1977; antecedent the *range bar* of Mary Spear, 1952) is a graphical summary of a continuous (dimensional) variable built entirely on **quantiles** rather than on parametric indexes such as the mean and standard deviation. A rectangular box spans the interquartile range with a line at the median, and two whiskers summarise the rest of the distribution, so that the central value, the middle-50% zone, the symmetry, the spread and the outliers of a distribution are all read off a single figure (Principles of Medical Statistics, Ch 5, pp 75-76).

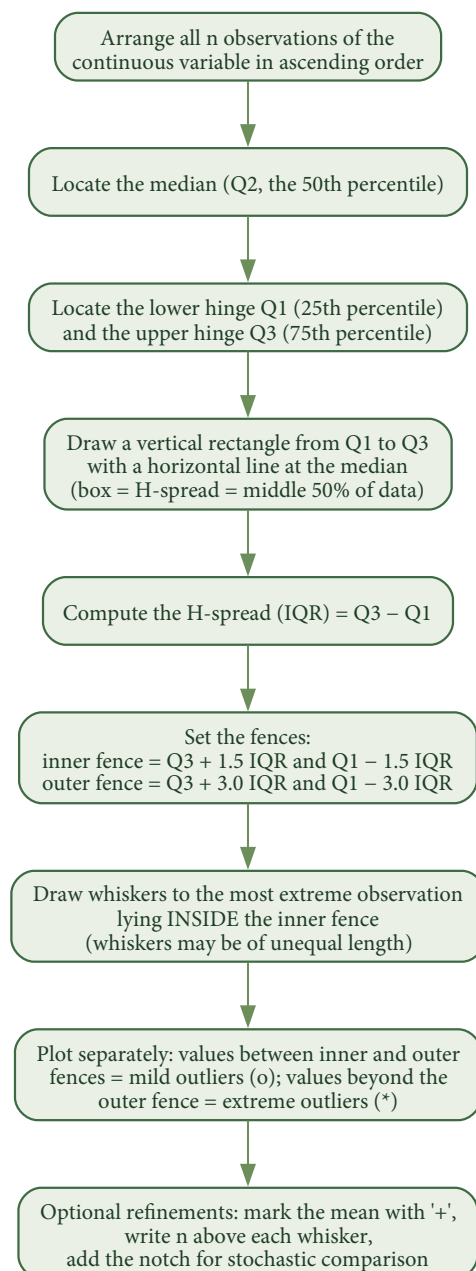
Anatomy — what each element of a box plot represents

Element of the plot	Statistic it marks	What the examiner expects you to say it conveys
Line across the box	Median (Q2 = D5 = P50)	Central index; a quartile, decile and percentile at once (Mahajan's Methods in Biostatistics, Ch 4, p 62)
Bottom of box	Lower quartile Q1 (P25) — Tukey's <i>lower hinge</i>	Value below which 25% of observations fall (Sunder Rao & Richard, §26.9.4)
Top of box	Upper quartile Q3 (P75) — <i>upper hinge</i>	Value below which 75% of observations fall
Height of box	Interquartile range (IQR) = Q3 – Q1, Tukey's H-spread	The middle 50% of the data — the "bulk" of the group
Position of median within box	Ratio of the two half-boxes	Symmetry vs skewness of the distribution
Whiskers	Depends on convention (see below)	Spread of the non-outlying data
Points beyond the whiskers	Outliers (o = mild, * = extreme)	Extreme/aberrant values and data-entry errors
Optional + sign	Arithmetic mean	Mean-vs-median discrepancy = a direct read-out of skew

Element of the plot	Statistic it marks	What the examiner expects you to say it conveys
Optional figure above whisker	n for that group	Group size, which a box plot does not otherwise show

The five values minimum, Q1, median, Q3 and maximum constitute the **five-number summary**, and the box plot is essentially its picture. In Mahajan's worked height data, $Q1 = 166.84$ cm, $Q2 = 171.46$ cm, $Q3 = 174.93$ cm, giving $IQR = 8.09$ cm and semi-interquartile range $Q = 4.04$ cm (Mahajan's Methods in Biostatistics, Ch 5, pp 76-77).

Construction



Steps in constructing a Tukey box-and-whiskers plot

Conventions for the whiskers — state which one you used

Convention	Whisker extends to	Comment
Tukey's 1.5 H-spread rule (standard)	Most extreme value within 1.5 IQR of the hinge; everything beyond is an outlier	The usual default; used by SAS, where o = outliers between 1.5 and 3 IQR and * = beyond 3 IQR (Principles of Medical Statistics, Ch 5, p 76)
1.0 H-spread rule	Most extreme value within 1 IQR of the hinge	Declares more points as outliers
ipr95 (percentile) whiskers	2.5th and 97.5th percentiles	Whiskers become unequal because the percentile boundaries are not symmetrical about the quartiles
10th–90th percentile whiskers	10th percentile below, 90th percentile above	The convention given in Indian PSM biostatistics texts (Sunder Rao & Richard, §26.9.4, p 239)
Range whiskers	Absolute minimum and maximum	Simplest; no outlier category (Spear's original range bar)
SPSS convention	Inner outlier 1.0–1.5 IQR marked X; extreme outlier beyond 1.5 IQR marked E	Software conventions differ — always label the figure legend

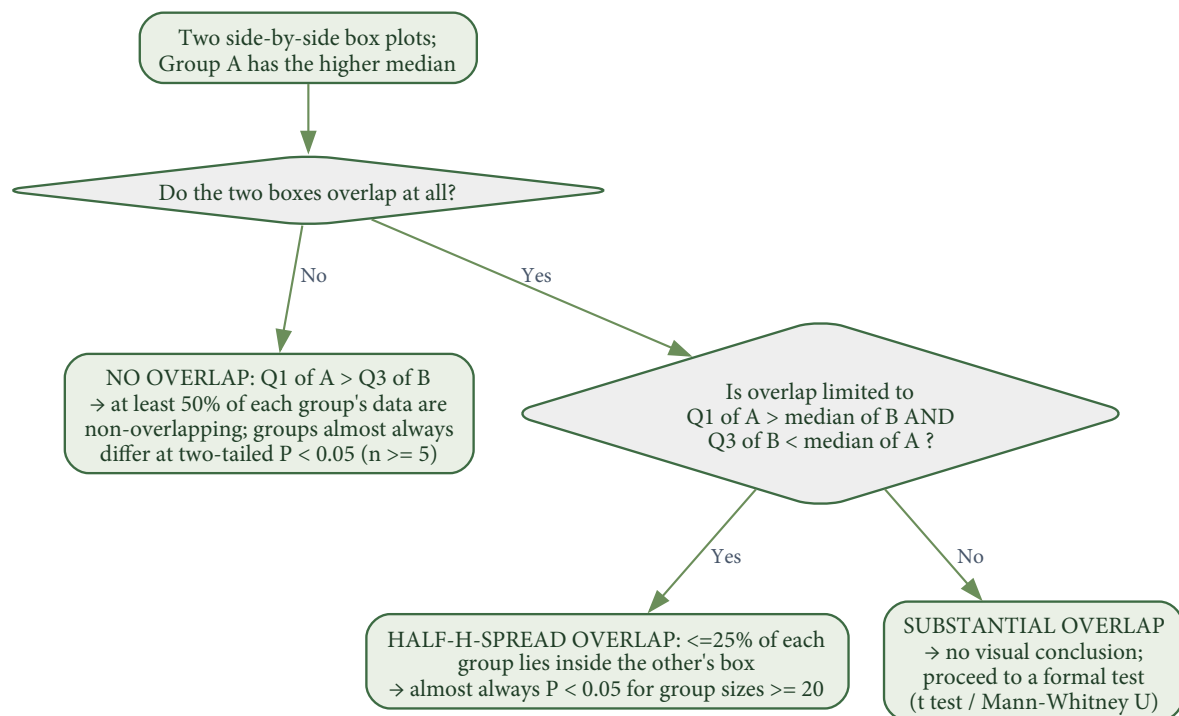
Uses of the box plot in data presentation

#	Use	How the box plot serves it
1	Compact summary of a continuous variable	Displays the whole five-number summary of a large data set in one figure; superior to a histogram, whose bin widths are arbitrary
2	Screening a distribution for normality before choosing a test	Direct inspection is the simplest method of diagnosing an "eccentric" (non-Gaussian) distribution and therefore of deciding parametric vs non-parametric analysis (Principles of Medical Statistics, Ch 5, §5.8.2)
3	Showing skewness and its direction	A "top-heavy" box with the mean above the median = right (positive) skew, as in income, length of stay, parasite counts, serum triglycerides
4	Detecting outliers and data-quality errors	Outliers are plotted individually, which flags transcription and measurement errors during data cleaning
5	Comparing two or more groups side by side	The most valuable use — spreads and overlaps of the "bulk" of data are compared directly, e.g. birth weight of babies of illiterate vs higher-secondary-educated mothers (Sunder Rao & Richard, Fig 26.13); CD4 lymphocyte counts across subgroups of men (Principles of Medical Statistics, Fig 16.13)
6	Rapid visual screening for statistical significance	Degree of overlap of two boxes translates into a stochastic conclusion (see flowchart), used as a quick screening test before the formal test
7	Data with open-ended class intervals or unrecorded extremes	Median and interquartile range remain computable and are unaffected by rare extreme values, unlike mean and standard deviation (Sunder Rao & Richard, measures of dispersion)

#	Use	How the box plot serves it
8	Ordinal data	Legitimate because the plot rests on ranks/quantiles; but it becomes uninformative when many categories are tied at the same percentile
9	Censored (survival) data	<i>Truncated</i> or censored box plots built from median and quartile survival times taken off the survival curve compare two treatments better than overlaid survival curves (Principles of Medical Statistics, Ch 22)
10	Journal, thesis and report presentation	The natural companion figure whenever results are reported as median (IQR) rather than mean $\pm$ SD — the norm for skewed community-medicine data

Interpretation

Reading a single box plot — three features indicate an approximately Gaussian distribution: the mean and median lie close together; the two halves of the box are roughly symmetrical about the median; and there are no egregious asymmetrical outliers. Violation of any one indicates an eccentric distribution (Principles of Medical Statistics, Ch 5, §5.8.2). Because plotting software may round odd values, always check the actual quartile values before declaring a box asymmetrical.



Visual screening of two side-by-side box plots for a difference between groups

Notched box plot — a notch is cut around the median covering $\text{median} \pm 1.57 \times \text{IQR} / \sqrt{n}$; the difference between two medians is taken as significant if the notch of one box does not overlap the median of the other. The half-H-spread rule is easier and needs no extra measurement (Principles of Medical Statistics, Ch 16, §16.1.2.5).

Advantages and limitations

Advantages	Limitations
Shows central index, spread, symmetry and outliers together	Group size (n) is not shown unless deliberately added at the top of each whisker
Distribution-free — valid for skewed and ordinal data	Conventions for whiskers and outliers differ between texts and software, so an unlabelled plot is ambiguous
Resistant to extreme values (based on quantiles, not mean/SD)	Hides multimodality and the actual pattern of individual points — a stem-and-leaf plot or dot plot is better for pattern
Permits rapid visual comparison of several groups on one axis	Uninformative for ordinal scales with many tied percentile values
No arbitrary class intervals, unlike a histogram	Not suitable for qualitative/nominal data (use bar or pie diagram)
Allows a quick screening judgement about significance	Screening rules assume roughly equal group sizes and variances; they never replace the formal test

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Draw a single vertical box-and-whiskers plot with these exact labels:

- ▶ Y-axis: "Birth weight (g)" with a scale
- ▶ Horizontal line at the top of the box — **Q3 (upper hinge, 75th percentile)**
- ▶ Horizontal line inside the box — **Median (Q2, 50th percentile)**
- ▶ Horizontal line at the bottom of the box — **Q1 (lower hinge, 25th percentile)**
- ▶ Vertical brace beside the box — **Interquartile range (H-spread) = $Q3 - Q1$ = middle 50% of data**
- ▶ + sign inside the box — **Mean**
- ▶ Upper whisker — **Largest value within $Q3 + 1.5 \text{ IQR}$ (inner fence)**
- ▶ Lower whisker — **Smallest value within $Q1 - 1.5 \text{ IQR}$ (inner fence)**
- ▶ Dotted lines above and below — **Inner fence ($\pm 1.5 \text{ IQR}$) and Outer fence ($\pm 3.0 \text{ IQR}$)**
- ▶ Small circle beyond the inner fence — **Mild outlier (o); asterisk beyond outer fence — Extreme outlier (*)**
- ▶ **n = ...** written above the upper whisker

Second diagram — two side-by-side box plots labelled "Illiterate mothers" and "Higher secondary and above", on a common "Birth weight (g)" axis, with the non-overlap of the two boxes marked.

High-yield exam pointers

- ▶ Tukey 1977, "box-and-whiskers plot"; quartiles = **hinges**; IQR = **H-spread**; five-number summary = min, Q1, median, Q3, max.
- ▶ **IQR = $Q3 - Q1$; semi-interquartile range $Q = (Q3 - Q1)/2$; median = $Q2 = D5 = P50$.**

- ▶ **Fences:** inner = ± 1.5 IQR from the hinges (whisker ends); outer = ± 3.0 IQR. Between the two = mild outlier; beyond the outer = extreme outlier.
- ▶ **Notch** = $\text{median} \pm 1.57 \times \text{IQR}/\sqrt{n}$; non-overlapping notches $\Rightarrow$ medians differ.
- ▶ Two-group screening: **no overlap of boxes** $\Rightarrow P < 0.05$ for $n \geq 5$; **half-H-spread overlap** $\Rightarrow P < 0.05$ for $n \geq 20$.
- ▶ Box plot is the figure of choice whenever the summary reported is **median (IQR)**, i.e. skewed data; mean $\pm$ SD data go with a bar or error-bar chart.
- ▶ Box plot $\Rightarrow$ quantitative/continuous data. Qualitative data $\Rightarrow$ bar, pie, pictogram or spot map (Park 28e, Ch 21, pp 1000-1003).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park's Textbook of Preventive and Social Medicine 28e (Ch 21, Health Information and Basic Medical Statistics, pp 1000-1003) lists bar chart, histogram, frequency polygon, line diagram, pie chart, pictogram, statistical map and scatter diagram, but does not describe the box plot, so the construction, fence and interpretation detail above is taken from the dedicated biostatistics texts cited; whisker conventions differ between them (1.5 IQR in Tukey/Feinstein, 10th–90th percentile in Sunder Rao), and an answer should state which convention is being used.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, CONSORT 2025 statement, SAMPL guidelines, EQUATOR Network, NFHS-5 (2019-21)

Guideline and legislative updates

- ▶ **CONSORT 2025** replaced CONSORT 2010 as the reporting standard for randomised trials: a 30-item checklist with seven new items, three revised, one deleted, and a restructured layout carrying a new **open science** section that groups trial registration (item 2), where the protocol and statistical analysis plan can be accessed (item 3), sharing of de-identified participant-level data (item 4), and funding and conflicts of interest (item 5). The implication for figures is direct: a box plot published in a trial report now sits alongside the deposited participant-level data, so it should be reproducible from that dataset [PMID 40228833].
- ▶ Adherence to the **SAMPL** (Statistical Analyses and Methods in the Published Literature) guidelines was audited across 150 statistical reviews done for clinical journals between 2020 and 2025: 99 of 150 manuscripts (66%) were accepted after SAMPL-based revision — 87 after one round and 12 after two ($p < 0.001$) — while the 51 rejections (34%) were driven mainly by inappropriate test use or unjustified methods. "Presentation of descriptive data" is one of the SAMPL principles scored, so reporting skewed results as median (IQR) with the matching box plot rather than mean $\pm$ SD is a point on which manuscripts are actually revised [PMID 40845506].
- ▶ An AI-versus-editor audit of 100 oncology papers scored against ten SAMPL items compared Gemini Advanced 2.0 Flash with a statistical editor and found substantial overall agreement (weighted kappa 0.77; 95% CI 0.6-0.94), with full agreement (kappa 1) on four items including

naming the statistical package and reporting confidence intervals. Agreement was only **moderate** for "summarising non-normally distributed data" (kappa 0.42) — the median/IQR-versus-mean/SD decision that settles whether a box plot or a bar chart is the right figure — and for test directionality (0.43), while the **lowest agreement of all, kappa 0.37, was on multiple-comparison adjustments**, where justification for post hoc tests was missing. Read the kappa for what it is: a measure of how closely the AI and the human grader scored the same paper, not of how consistently authors in the literature apply the rule [PMID 41004884].

- Contemporary guidance restates the rule the box plot embodies: health data are commonly skewed, mean $\pm$ SD is distorted by extreme values, and the correct pair for such data is the **median with the interquartile range (Q3 – Q1)**, with the distribution first inspected by histogram or Q-Q plot and, if tested, by Kolmogorov-Smirnov or Shapiro-Wilk (both over-sensitive in very large samples). The companion point for exams: SD measures dispersion of the observations, whereas the standard error underlies the 95% CI of the population mean — the two must not be interchanged in a figure legend [PMID 38841386] [PMID 33201265].

Current national data

- NFHS-5 (2019-21) illustrates why a national mean can mislead where a box plot would not: 70.8% of 84,390 reported deaths were registered nationally, yet district-level coverage ranged from **5% to 100% across 707 districts**, and 122 districts registered under 40% of female deaths (female vs male aOR 0.61; 95% CI 0.59-0.64). A district-level indicator with that spread is exactly the skewed, outlier-rich distribution that the five-number summary and the box plot are built for [PMID 36733620].
- A district-level NFHS-5 spatial analysis of 232,920 pregnancies put the national unintended-pregnancy rate at **9.1%**, mapped state and district rates as **choropleth maps**, and confirmed clustering across the 707 districts with Global Moran's I 0.47 ($p < 0.01$), Anselin Local Moran's I locating high-high clusters in Arunachal Pradesh, northern West Bengal, Bihar, western Uttar Pradesh, Haryana and Delhi [PMID 39402475]. That published analysis stops at the map; adding a box plot is this answer's own recommendation, not an established convention — a choropleth shows *which* districts are high but not *how* the district values are spread, so pairing the map with a box plot of those values is the sensible addition in a thesis or a report.

Global and WHO developments

- The **raincloud plot** is the current successor to the plain box plot for continuous data from factorial designs: it superimposes a box-and-whiskers plot (the summary statistics), a density estimate (the "cloud") and the individual observations (the "raindrops") in one figure, and so shows the bimodality, skew and outliers that a box plot alone hides. A "raincloud quartet" of four datasets built to share an identical mean, SD and one-sample t test makes the point for group comparisons the way Anscombe's quartet did for correlation. Uptake is measurable — in an audit of brief reports in the Psychonomic Society's flagship journal, raincloud plots were 9% of plots in 2023 (none in 2013) and figures showing individual data points rose from 2% (2013) to 29% (2023); the tool is now built into the free software JASP [PMID 40825889].
- Figure-quality auditing has become a research field of its own. Across 641 phase III oncology trials enrolling 518,235 patients, **18% of Kaplan-Meier plots carried missing visual elements** — a truncated survival axis (which exaggerates between-group separation) or an absent number-at-

risk table — and only 3% of surrogate-endpoint plots were fully interpretable; the defect rate fell from 45% (2002-07) to 8% (2016-19). The identical discipline is demanded of a box plot: state the whisker convention and write n on the figure, or the plot is not interpretable [PMID 38824414].

- Because a box plot's five-number summary is often all that a primary paper reports, new estimators now **recover the mean and SD from reported medians, quartiles and extremes** so such studies are not excluded from meta-analysis: weighted (inverse-variance and inverse-variance-covariance) estimators accommodate any set of reported quantiles and non-normal underlying distributions, and Box-Cox/maximum-likelihood approaches serve when the distribution is unknown. This turns the box plot from a display-only device into a usable input for evidence synthesis [PMID 41983277] [PMID 34668458].
- The **Banksia plot** is a newer display for method-comparison work rather than a box-plot variant, but it answers the same need for a compact visual comparison: it plots point estimates and confidence intervals from different analysis methods across datasets whose outcome scales differ, centring the reference estimate at zero and scaling its confidence limits to -0.5 to 0.5 so that all comparisons become visually commensurable [PMID 39515488].

SOURCE & COVERAGE

Principles of Medical Statistics (Feinstein, 2002) — Ch 5 "Inner Zones and Spreads" pp 75-78, Ch 9, Ch 16 "Interpretations and Displays for Two-Group Contrasts" pp 294-304, Ch 22; Sunder Rao & Richard — Biostatistics and Research Methods, §26.9.4 Box Plot (p 239) and the chapter on measures of dispersion; Mahajan's Methods in Biostatistics for Medical Students and Research Workers — Ch 3 "Measures of Location — Averages and Percentiles" and Ch 4 "Variability and its Measures"; Park's Textbook of Preventive and Social Medicine 28e — Ch 21, Health Information and Basic Medical Statistics.

Recent-advances sources: 40228833 — CONSORT 2025 statement: updated guideline for reporting randomised trials, BMJ 2025, doi:[10.1136/bmj-2024-081123](https://doi.org/10.1136/bmj-2024-081123); 40845506 — Could adherence to SAMPL Guidelines improve statistical reporting in clinical manuscripts? Insights from 150 editorial reviews for specialty journals, Current Research in Translational Medicine 2025, doi:[10.1016/j.retram.2025.103536](https://doi.org/10.1016/j.retram.2025.103536) (abstract-only); 41004884 — AI-assisted statistical review of 100 oncology research articles: compliance with SAMPL guidelines, Current Research in Translational Medicine 2025, doi:[10.1016/j.retram.2025.103544](https://doi.org/10.1016/j.retram.2025.103544) (abstract-only); 38841386 — It is normal to be "not-normal": reporting of correct descriptive statistics in dental research, Restorative Dentistry & Endodontics 2024, doi:[10.5395/rde.2024.49.e19](https://doi.org/10.5395/rde.2024.49.e19); 33201265 — Statistical data presentation: a primer for rheumatology researchers, Rheumatology International 2021, doi:[10.1007/s00296-020-04740-z](https://doi.org/10.1007/s00296-020-04740-z) (abstract-only); 36733620 — Death registration coverage 2019-2021, India, Bulletin of the World Health Organization 2023, doi:[10.2471/BLT.22.288889](https://doi.org/10.2471/BLT.22.288889); 39402475 — Spatial heterogeneity in unintended pregnancy and its determinants in India, BMC Pregnancy and Childbirth 2024, doi:[10.1186/s12884-024-06850-z](https://doi.org/10.1186/s12884-024-06850-z); 40825889 — Informative data visualization with raincloud plots in JASP, Behavior Research Methods 2025, doi:[10.3758/s13428-025-02773-5](https://doi.org/10.3758/s13428-025-02773-5); 38824414 — Lost in the plot: missing visual elements in Kaplan-Meier plots of phase III oncology trials, The Oncologist 2024, doi:[10.1093/oncolo/oyae067](https://doi.org/10.1093/oncolo/oyae067); 41983277 — Minimum distance estimation of mean and standard deviation from reported quantiles, Research Synthesis Methods 2026, doi:[10.1017/rsm.2026.10090](https://doi.org/10.1017/rsm.2026.10090) (abstract-only); 34668458 — Estimating the sample mean and standard deviation from order statistics and sample size in meta-analysis, Statistical Methods in Medical Research 2021, doi:[10.1177/09622802211047348](https://doi.org/10.1177/09622802211047348) (abstract-only); 39515488 — The Banksia plot: a method for visually comparing point estimates and confidence intervals across datasets, Journal of Clinical Epidemiology 2024, doi:[10.1016/j.jclinepi.2024.111591](https://doi.org/10.1016/j.jclinepi.2024.111591) (abstract-only). Reporting standards used: CONSORT 2025 (April 2025); SAMPL guidelines; EQUATOR Network. Indian national data: NFHS-5 (2019-21).

♦ ♦ ♦

Histogram in data presentation

Asked in: Paper IV May 2019 — RGUHS MD Community Medicine

Definition — A **histogram** is a pictorial diagram of a **frequency distribution** of a characteristic measured on a **continuous scale**. Class intervals are marked on the horizontal axis (*abscissa*) and frequencies on the vertical axis (*ordinate*), and a series of adjoining rectangles (blocks) is erected over the class intervals such that the **area of each block is proportional to the frequency** of that class (Park 28e, Ch 21, p 1001; Sunder Rao & Richard, Ch 1, pp 33-34). It is therefore a special kind of bar diagram reserved for quantitative measured data.

Where the histogram sits among methods of data presentation

Nature of data	Correct mode of presentation	Examples
Quantitative, continuous, measured	Graphs	Histogram, frequency polygon, frequency curve, line chart, cumulative frequency diagram (ogive), scatter or dot diagram (Mahajan, Ch 3, pp 32-33)
Qualitative, discrete, counted	Diagrams	Bar diagram (simple, multiple, component, proportional), pie or sector diagram, pictogram, map or spot map (Mahajan, Ch 3, pp 32-33)
Data tied to a geographic or administrative area	Statistical maps	Shaded maps, dot maps (Park 28e, Ch 21, p 1002)

The histogram is thus the **first-line graph for a single continuous variable**. Graphs make a more lasting impression than detailed numbers and give a grasp of the overall picture rather than the details; the price is loss of detail, so the original data must be returned to for exact study (Sunder Rao & Richard, Ch 1, pp 33-34; Park 28e, Ch 21, p 1000).

Uses of the histogram in data presentation

#	Use	What the histogram shows	Applied example
1	Display a frequency distribution	Converts a bulky frequency table into one picture — at a glance, how many observations fall in each group and where the main concentration lies (Park 28e, Ch 21, p 1000)	Diastolic blood pressure in women aged 45-64 years (Park 28e, Ch 21, Fig 5, p 1001)
2	Show the range and the shape of the distribution	Whether the data are symmetrical convex (Gaussian), uniform, skewed, concave or multi-modal; only symmetrical distributions are well represented by the mean and standard deviation, all others being <i>eccentric</i> (Principles of Medical Statistics, Ch 3, p 36)	Heights of adults form a near-symmetrical bell shape

#	Use	What the histogram shows	Applied example
3	Locate the mode and judge central tendency	The crest of the histogram is the modal class; a visibly skewed histogram warns that the mean is pulled away from the crest and misrepresents the group (Principles of Medical Statistics, Ch 3, pp 36-38; Sunder Rao & Richard, Ch 1, p 41)	Duration of hospital stay, parity, income — all right-skewed
4	Screen the normality assumption before analysis	Univariate distributions are inspected to see whether the data are so eccentric that non-parametric rather than parametric procedures are required, and whether the mean and standard deviation are fit to summarise them (Principles of Medical Statistics, Ch 5, pp 76, 80)	Deciding Student's <i>t</i> -test versus Mann-Whitney U test
5	Detect a mixture of two populations (bimodality) and read off a cut-off	Two modes with an antimode (trough) between them indicate two underlying groups; the point at which the two distributions intersect is frequently used as the cut-off between the "normal" and the "diseased", because it generally minimises false positives and false negatives (Park 28e, Ch 4, p 169)	In a tuberculin infection survey the size of induration is plotted as a histogram, and reactors can be told from non-reactors because the induration measures cluster around two modes separated by an antimode (Handbook of Epidemiology, Ch 38, p 1557). The bimodal age-incidence curve of Hodgkin's disease — an early peak at 15-35 years and a later peak from about 50 years — is the classic epidemiological example (Park 28e, Ch 3, p 88)
6	Detect outliers and errors	Isolated blocks separated by a large gap from the main bulk of the data flag either an error of observation or transfer, or genuinely extreme observations (Principles of Medical Statistics, Ch 3, p 36; Ch 5, p 76)	A recorded haemoglobin of 25 g/dL in a survey
7	Construct the epidemic curve in outbreak investigation	Cases plotted by date or time of onset ; it suggests the time relationship with a suspected source, whether the epidemic is common-source or propagated , and whether there is a seasonal or cyclic pattern suggestive of a particular infection (Park 28e, Ch 3, p 160)	Cases of acute hepatitis E by date of onset after a break in water treatment
8	Estimate the incubation period of an outbreak	The time from the presumed exposure to the peak of the curve is the hypothesised median incubation time , which helps identify the organism (Nelson & Williams, Ch 5, p 137)	Food-poisoning outbreak after a common meal

#	Use	What the histogram shows	Applied example
9	Judge whether transmission is continuing	It indicates whether transmission is continuing or has ended — vital to knowing whether the outbreak needs further control measures or whether the existing measures are containing it (Nelson & Williams, Ch 5, p 137)	Ongoing cholera transmission from an unrepaired source
10	Base for derived graphs	Joining the mid-points of the tops of the histogram blocks gives the frequency polygon ; with a large total frequency and narrow class intervals the polygon smooths into the frequency curve (Park 28e, Ch 21, pp 1001-1002; Sunder Rao & Richard, Ch 1, pp 34-36)	Superimposing a frequency curve of heights on its histogram
11	Compare two subgroups back to back	Two age frequency distributions plotted on either side of a central axis give the population (age and sex) pyramid , which compares the male with the female population and shows the number of dependents and the general structure of the population (Park 28e, Ch 8, p 581)	India's pyramid has a broad base and a tapering top , typical of a developing country, set against that of Switzerland (Park 28e, Ch 8, Fig 2, p 581)

Construction of a histogram — steps and rules

- **Tabulate first.** Glance through the data, pick out the lowest and the highest observation, select a suitable class width and divide the whole range into **10 to 15 classes**; make a tally mark for each observation against its class, then count the tallies (Sunder Rao & Richard, Ch 1, p 33). Park's practical rule is a **maximum of 20 groups when the data are large and a minimum of 5 groups when there is not much data** (Park 28e, Ch 21, p 1000); Mahajan puts the number **ordinarily between 6 and 16**, depending on the detail needed and the size of the sample (Mahajan, Ch 3, pp 31-32).
- **Keep class intervals equal throughout** — for example 10 mm intervals for systolic blood pressure from 80 to 160 mmHg — so that class frequencies are directly comparable; the interval must be neither too broad (detail is lost) nor too narrow (the purpose of condensing the data is defeated) (Mahajan, Ch 3, pp 31-32; Park 28e, Ch 21, p 1000; Sunder Rao & Richard, Ch 1, pp 27-28).
- **Arrange groups in ascending order** from the lowest value upwards (80-89, 90-99, ..., 150-159), and define the limits so that there is no overlap: the **upper limit means "up to but not including"**, so a height of exactly 162 cm is counted in the next group, 162-164 cm (Mahajan, Ch 3, pp 30-32).
- **Plot the class intervals on the x-axis (abscissa) and the frequency on the y-axis (ordinate)**, choosing a scale that is easy on the eye (Mahajan, Ch 3, pp 32-33; Park 28e, Ch 21, p 1000).
- **Erect one rectangle over each class interval**, its area proportional to that class frequency (Park 28e, Ch 21, p 1001).

- **Bars must touch one another.** In a histogram the bars always touch, because the horizontal categories are the ordered intervals of a continuous scale; the bars at the two ends may be omitted if the extreme intervals are unbounded (open-ended) (Principles of Medical Statistics, Ch 9, pp 154-155).
- **Correct for unequal class intervals.** Where the width of a class differs from the rest, the **height** of that bar must be adjusted to represent the average value for the whole interval, i.e. plot frequency per unit class width. Plotting raw frequency over an unequal interval is the commonest mistake in histogram drawing (Principles of Medical Statistics, Ch 9, pp 154-155; Sunder Rao & Richard, Ch 1, pp 34-35).
- **Title and label completely** — a title that explains in exact terms what the data represent, both axes labelled with their units, and the source and year given (Sunder Rao & Richard, Ch 1, p 33).

Histogram compared with bar diagram

Feature	Histogram	Bar diagram
Type of data	Quantitative, continuous, measured	Qualitative, discrete, counted
x-axis	A continuous numerical scale divided into class intervals; bars contiguous	Unrelated nominal categories, separated by spaces
What represents the frequency	The area of the block	The length of the bar
Order of bars	Fixed — ascending order of the variable	Interchangeable at will
Width of bars	Fixed by the class interval	Arbitrary, only kept equal for appearance
Derived graph possible	Frequency polygon, frequency curve, ogive	Multiple, component and proportional bar charts

(Park 28e, Ch 21, pp 1000-1002; Mahajan, Ch 3, pp 32-33; Principles of Medical Statistics, Ch 9, pp 154-155)

Limitations and errors to avoid

- **Detail is lost** — simplicity is obtained at the expense of accuracy, and the original table must be consulted for exact values (Park 28e, Ch 21, p 1000).
- **Only one continuous variable at a time** — two groups cannot be compared on the same histogram; use two frequency polygons on one pair of axes instead (Sunder Rao & Richard, Ch 1, pp 35-36).
- **Shape depends on the class interval chosen** — too large a group omits the details and too small a group defeats the purpose of condensing the data, so the same observations can be made to look different (Mahajan, Ch 3, Figs 2.1 and 2.2, pp 34-35).
- **Bimodality must be interpreted with caution** — small numbers of observations are a frequent source of bimodality, and the absence of bimodality does **not** signify that the data have come from a homogeneous source (Park 28e, Ch 3, p 88).
- **Not for qualitative, nominal or rate data** — a histogram of blood groups or of sex distribution is wrong; that is a bar diagram.

- ▶ A truncated or non-zero y-axis exaggerates differences; the frequency axis must start at zero, with a break mark shown if the x-axis is interrupted.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Labels for the plain histogram the candidate must sketch (tuberculin reaction in 206 persons, Mahajan, Ch 3, Table 2.8, p 33):

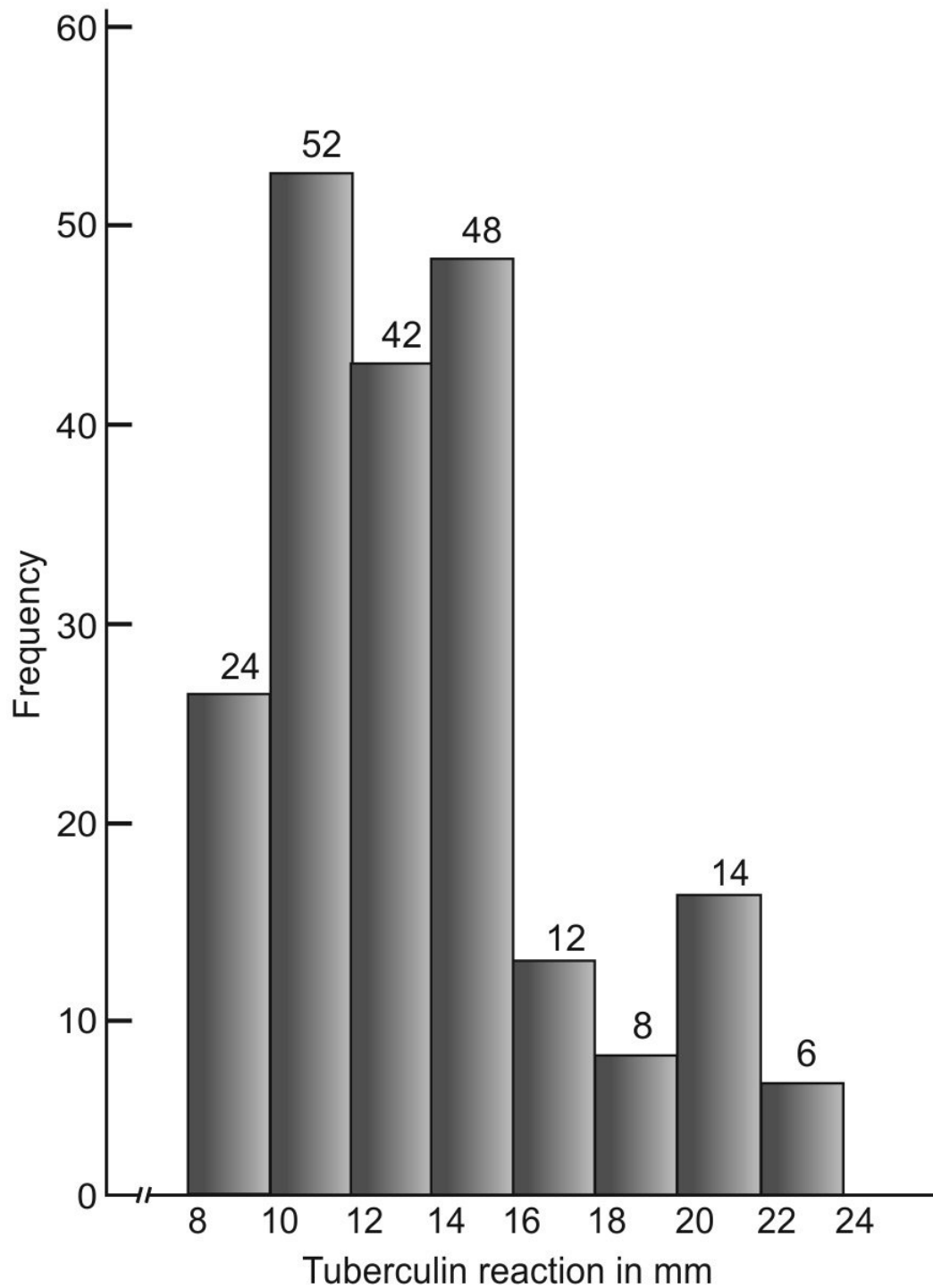
- ▶ Title: "Histogram — tuberculin (Mantoux) reaction in 206 persons"
- ▶ x-axis (abscissa): "Tuberculin reaction in mm"; marks at 8, 10, 12, 14, 16, 18, 20, 22, 24
- ▶ y-axis (ordinate): "Frequency (number of persons)"; scale 0 to 60
- ▶ Adjoining rectangles, no gaps, heights: 8-10 mm = 24; 10-12 = 52; 12-14 = 42; 14-16 = 48; 16-18 = 12; 18-20 = 8; 20-22 = 14; 22-24 = 6 (total 206)
- ▶ Annotate only "frequency distribution of a continuous variable; modal class = 10-12 mm" — do **not** label any millimetre value on this plate as a diagnostic cut-off
- ▶ Optional overlay: dotted line joining the mid-points of the tops of the blocks, labelled "frequency polygon"

Labels for the schematic bimodal histogram (the two-population use):

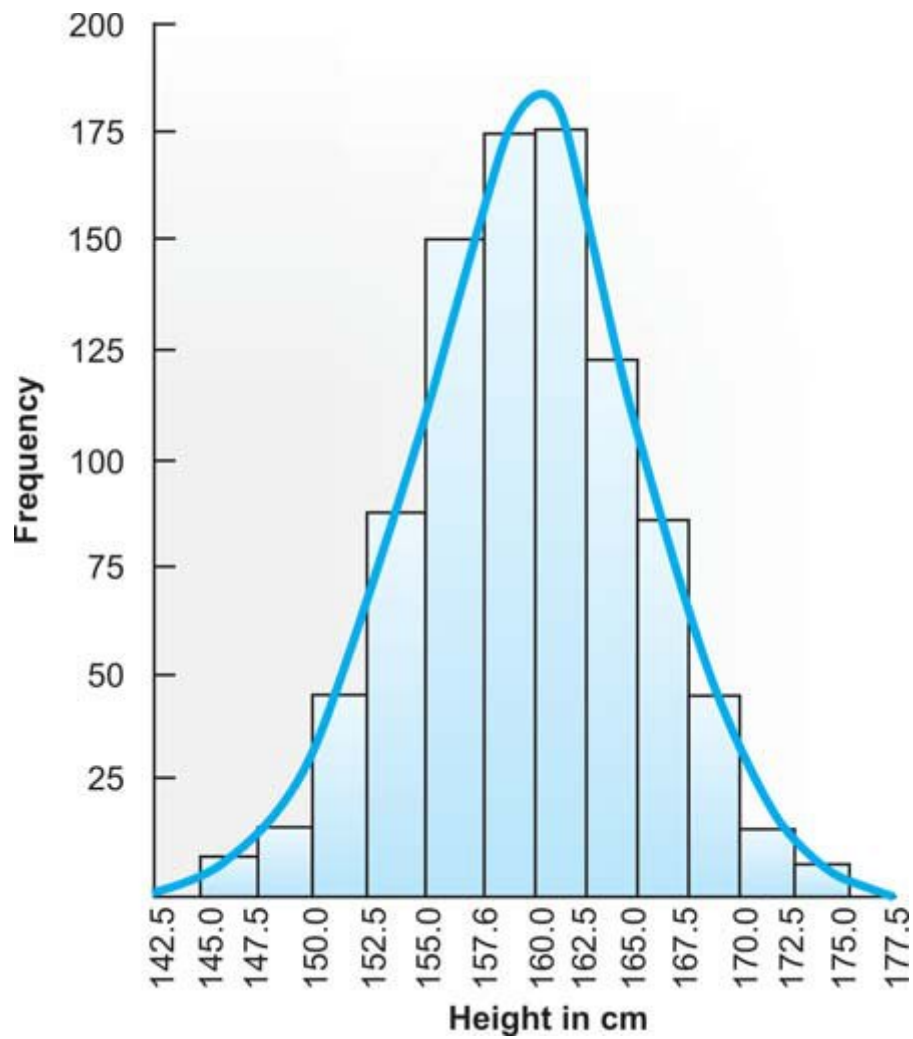
- ▶ x-axis: "Size of induration (mm)"; y-axis: "Number of persons"
- ▶ Two crests with a trough between them, labelled "first mode — non-reactors", "**antimode**", "second mode — reactors" (Handbook of Epidemiology, Ch 38, p 1557)
- ▶ Vertical dotted line dropped at the antimode, labelled "point at which the two distributions intersect = cut-off between 'normal' and 'diseased'" (Park 28e, Ch 4, p 169)

Labels for the epidemic curve the candidate must sketch:

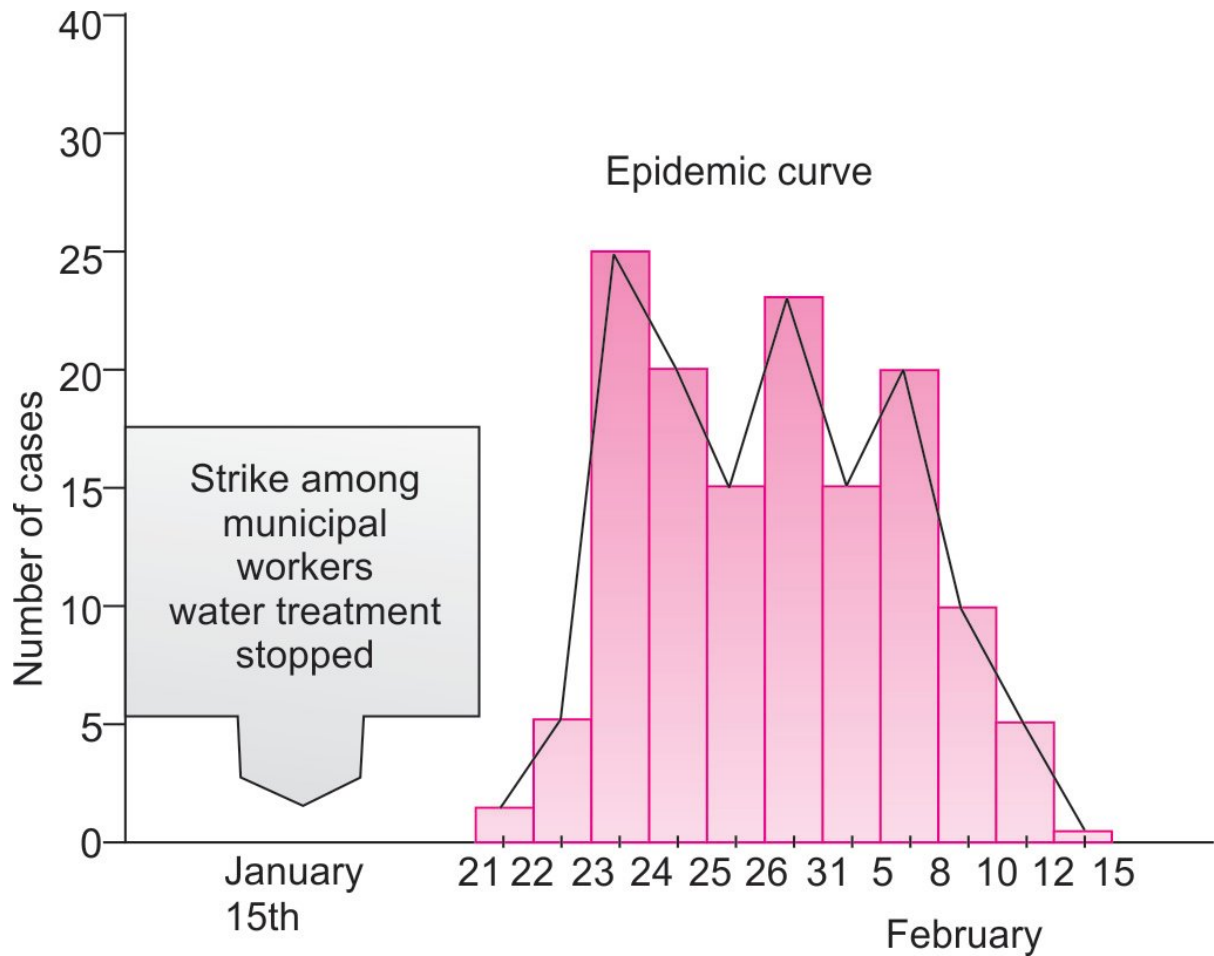
- ▶ x-axis: "Date of onset of illness"; y-axis: "Number of cases"
- ▶ Arrow on the x-axis marking "Exposure / break in water treatment"
- ▶ Contiguous bars forming a single sharp peak, labelled "common-source (point) epidemic"
- ▶ Horizontal bracket from the exposure arrow to the peak, labelled "median incubation period"



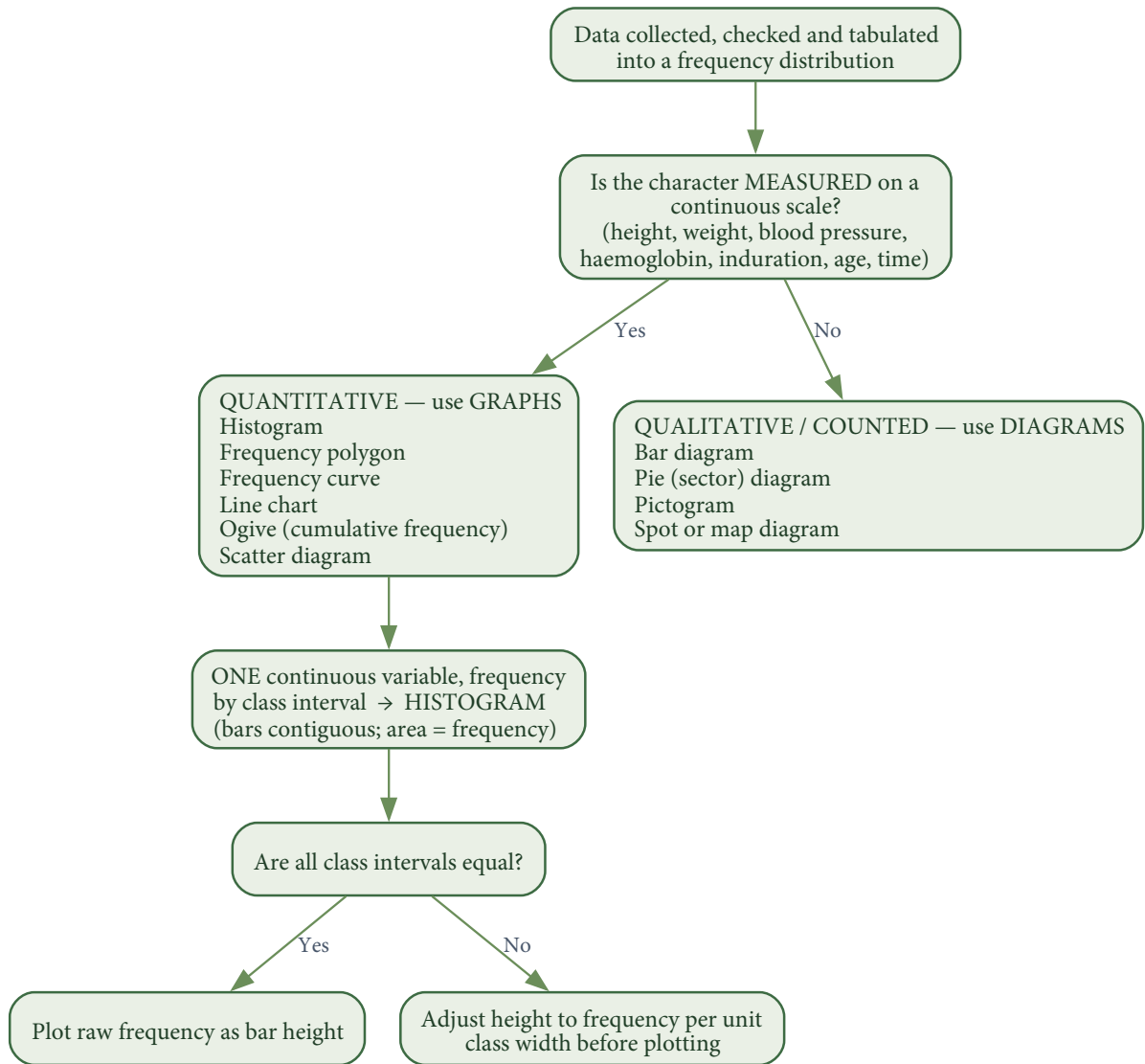
Histogram showing tuberculin reaction in 206 persons (Mahajan's Methods in Biostatistics for Medical Students and Research Workers, p.34)



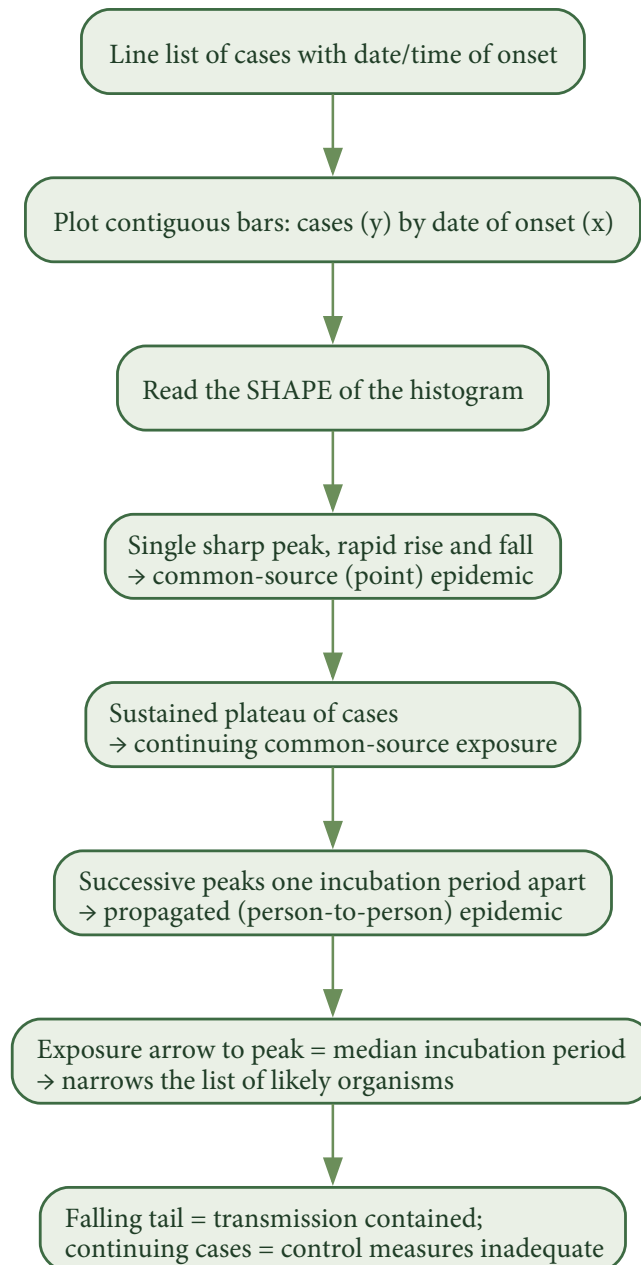
Frequency curve of heights superimposed over histogram (Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e, p.452)



Cases of acute hepatitis (E) by date of onset — epidemic curve drawn as a histogram (Sunder Lal — Textbook of Community Medicine, p.308)



Choosing the correct graphic for the type of data — the histogram is the graph for a single continuous variable



Reading an epidemic curve drawn as a histogram of cases by date of onset

High-yield exam pointers

- ▶ **One-line definition to write first:** "A histogram is a pictorial diagram of a frequency distribution of a **continuous** variable, in which the **area** of each block is proportional to the frequency of its class interval" (Park 28e, Ch 21, p 1001).
- ▶ **Two rules that earn marks:** bars **touch** (no gaps); with **unequal class intervals** the height is adjusted to frequency per unit width.
- ▶ **Class number rule:** Sunder Rao 10-15 classes; Park maximum 20 (large data) and minimum 5 (small data); Mahajan ordinarily 6-16 — intervals **equal** throughout.
- ▶ **Mnemonic** — "Measured = Graph, Counted = Diagram."
- ▶ **Frequency polygon** = join the **mid-points of the tops** of the histogram blocks; smoothed = frequency curve.

- **Outbreak use:** epidemic curve = cases by **date of onset**; exposure-to-peak = **median incubation time**; single sharp peak = point source, a second rise one incubation period later = propagated.
- **Bimodality trigger words:** two modes separated by an **antimode**; the intersection of the two distributions is the **cut-off point**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not written in the exam: the tuberculin plate is Mahajan's Table 2.8 and Figure 2.1 and is used here purely to show what a histogram of continuous data looks like — the diagnostic Mantoux cut-offs are Park's (induration exceeding 10 mm positive, less than 6 mm negative, 6 to 9 mm doubtful; Park 28e, Ch 5, pp 228-229) and are not read off this plate, while the separation of reactors from non-reactors by two modes and an antimode is a finding of tuberculin infection surveys in children without a BCG scar (Handbook of Epidemiology, Ch 38, p 1557).

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed

Programme and policy updates (India)

- **The IDSP epidemic curve is now drawn from a digital line list, not a weekly paper form.** The Integrated Health Information Platform (IHIP) was soft-launched pan-India on 1 April 2021 under IDSP and carries 33 epidemic-prone diseases and syndromes. It replaces the weekly S, P and L forms with near-real-time, case-based web entry, adds GIS-based mapping and geo-tagging of symptomatic cases and health facilities, and raises event and outbreak alerts automatically — so the histogram of cases by date of onset is regenerated from the line list rather than tallied by hand at the district surveillance unit. The 2015 Joint Monitoring Mission of IDSP had recommended exactly this redesign [PMID 37082408].
- **What the digital line list actually yields at district level.** Secondary analysis of the IHIP P (presumptive) form in Ganjam and Jagatsinghpur districts of Odisha (April 2021 to October 2022) registered 255,071 and 63,895 patients — notification rates 6,618 and 5,146 per lakh population. Acute respiratory illness carried the highest notification rate in both districts, and rates were higher in males and in those above 60 years; block-to-block variation was wide enough that the authors called for follow-up analysis of why some blocks notify so little [PMID 41629233].
- **Under-notification flattens the histogram before anyone draws it.** In Raigarh district, Chhattisgarh, IHIP captured only 44% of laboratory-tested malaria cases entered in primary health centre registers in September 2023. Refresher training of laboratory technicians plus a weekly feedback loop from the district surveillance unit raised that sensitivity to 100% within five weeks. The exam point: the height of every block in an epidemic curve is bounded by reporting completeness, so data quality must be stated before the curve is interpreted [PMID 41629241].

Current national data

- **One condition, several national numbers — and why district-level distributions are being built.** A review of India's diabetes surveillance shows that NFHS, the National NCD Monitoring Survey and ICMR-INDIAB differ in sampling and measurement and therefore return different awareness, treatment and control estimates for the same care continuum. The proposed fix is an integrated framework fusing survey data with the electronic records of the **National Programme for Prevention and Control of Non-Communicable Diseases**, using model-based **small-area estimation**, so that state and district administrators see a distribution across districts rather than one national summary figure [PMID 38815594].

Elimination, surveillance and disease status

- **"Backfilling" — the modern caution about reading a live epidemic curve.** In England's 2022 mpox outbreak, delays from symptom onset to specimen date to reporting meant the most recent bars of the curve filled in only over subsequent weeks, so the real-time curve falsely appeared to be turning down. A hierarchical **generalised additive model** fitted to individual-level line-list data corrected the right-hand tail and recovered the true shape of the curve in real time, and its outputs went into the national technical briefings [PMID 37721951].
- **Measles, England 2023-24 — the largest outbreak there in a decade.** Laboratory confirmation lags made case data inherently retrospective rather than real-time, and the reporting delay itself varied with measles prevalence and with the purpose of testing. A generalised additive nowcast allowing for delay, location and day-of-week reduced the daily log weighted interval score by 42% against baseline and improved four-week trend prediction (ranked probability score down 69%) [PMID 40782889].

Guideline and legislative updates

- **Reporting guidance is actively pushing the bar chart out and the histogram back in.** An audit of 270 figure panels showing continuous univariate data across four ecology and conservation journals found bar charts (31%) and dot-and-whisker plots (29%) together outnumbered every other plot type, and only 19% of panels showed the underlying distribution or the raw data at all. The proposed replacement — the **sea stack plot** — is a vertical histogram with mean and dispersion overlaid, recommended for continuous data with $n \geq 20$, on the argument that a histogram treats every observation equally and, unlike a smoothed density curve, cannot invent observations in a bin that is actually empty [PMID 38633526].
- **Raincloud plots — the histogram idea carried into group comparisons.** A raincloud plot shows a box plot, a density estimate (the "cloud") and every individual observation (the "raindrops") in one panel. A **raincloud quartet** built so that mean, SD, n and the t -test inference are identical across four panels resolves on inspection into normal, bimodal, right-skewed and outlier-driven data — Anscombe's lesson restated for a single group, and precisely the argument for inspecting a histogram before quoting a mean. Raincloud plots made up 9% of plots in one journal in 2023, and panels showing individual data points rose from 2% (2013) to 29% (2023) [PMID 40825889].

Global and WHO developments

- **Not every epidemic curve carries the same information.** An analytical framework now quantifies how much usable information under-reporting and reporting latency strip out of an

incidence time series, and ranks surveillance streams by informativeness for estimating the instantaneous reproduction number. It directly challenges the routine assumption that the death curve is the more trustworthy curve for acute infections such as COVID-19 and influenza [PMID 38177483].

- **Nowcasting buys about a week of situational awareness.** Retrospective validation across 18 months of Italian COVID-19 surveillance found that adjusting the epidemic curve for right-truncation cut the median lag in estimating the time-varying reproduction number from **13 days to 8 days**, and detected periods of epidemic growth **6 to 23 days** earlier than the uncorrected curve [PMID 39234082].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 21, Health Information and Basic Medical Statistics, pp 1000-1002; Ch 3, Principles of Epidemiology and Epidemiologic Methods, pp 84-88, 160; Ch 4, Screening for Disease, p 169; Ch 5, Epidemiology of Communicable Diseases, pp 228-229; Ch 8, Demography and Family Planning, p 581); Mahajan's Methods in Biostatistics for Medical Students and Research Workers (Ch 3, Sources and Presentation of Data, pp 27-37); Sunder Rao & Richard — Biostatistics and Research Methods (Ch 1, Frequency distribution and its presentation, pp 27-41); Principles of Medical Statistics (Ch 3, Central Index of a Group, pp 33-38; Ch 5, Inner Zones and Spreads, pp 76-80; Ch 9, Communication and Display of Univariate Data, pp 154-155); Handbook of Epidemiology (Ahrens & Pigeot) (Ch 38, Epidemiology in Developing Countries, pp 1556-1557); Nelson & Williams — Infectious Disease Epidemiology (Ch 5, Outbreak Epidemiology, pp 137-138); Sunder Lal — Textbook of Community Medicine (epidemic curve, p 308); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e (Ch 23, p 452).

Recent-advances sources: 37082408 — IHIP: A Leap into India's Dream of Digitalizing Healthcare, Indian Journal of Community Medicine 2023, DOI [10.4103/ijcm.ijcm.739.22](https://doi.org/10.4103/ijcm.ijcm.739.22); 41629233 — Epidemiological Profile of Diseases Reported in the P Form of Integrated Health Information Platform in Two Districts of Odisha, Indian Journal of Public Health 2026, DOI [10.4103/ijph.ijph.660.25](https://doi.org/10.4103/ijph.ijph.660.25) (abstract-only); 41629241 — Quality Improvement Initiative to Improve the Sensitivity of Malaria Reporting under Integrated Health Information Platform, Raigarh, Chhattisgarh, 2023, Indian Journal of Public Health 2026, DOI [10.4103/ijph.ijph.648.25](https://doi.org/10.4103/ijph.ijph.648.25) (abstract-only); 38815594 — Advancing diabetes surveillance ecosystems: a case study of India, Lancet Diabetes & Endocrinology 2024, DOI [10.1016/S2213-8587\(24\)00124-4](https://doi.org/10.1016/S2213-8587(24)00124-4) (abstract-only); 37721951 — Nowcasting the 2022 mpox outbreak in England, PLoS Computational Biology 2023, DOI [10.1371/journal.pcbi.1011463](https://doi.org/10.1371/journal.pcbi.1011463); 40782889 — Nowcasting cases and trends during the measles 2023/24 outbreak in England, Journal of Infection 2025, DOI [10.1016/j.jinf.2025.106569](https://doi.org/10.1016/j.jinf.2025.106569) (abstract-only); 38633526 — Sea stack plots: replacing bar charts with histograms, Ecology and Evolution 2024, DOI [10.1002/ece3.11237](https://doi.org/10.1002/ece3.11237); 40825889 — Informative data visualization with raincloud plots in JASP, Behavior Research Methods 2025, DOI [10.3758/s13428-025-02773-5](https://doi.org/10.3758/s13428-025-02773-5); 38177483 — Quantifying the information in noisy epidemic curves, Nature Computational Science 2022, DOI [10.1038/s43588-022-00313-1](https://doi.org/10.1038/s43588-022-00313-1) (abstract-only); 39234082 — Increasing situational awareness through nowcasting of the reproduction number, Frontiers in Public Health 2024, DOI [10.3389/fpubh.2024.1430920](https://doi.org/10.3389/fpubh.2024.1430920).



Scatter diagram in data presentation

Asked in: Paper IV 19-Nov-2021 — RGUHS MD Community Medicine

Definition — A **scatter diagram** (also called scattergram, correlation diagram, XY diagram or dot diagram) is a graphic method of presenting **bivariate quantitative data**, in which the paired values of two variables X and Y measured on the *same individual or the same group* are plotted as one dot each on a rectangular coordinate system. It is prepared after tabulation in which the frequencies of at least two variables have been cross-classified, and it displays the **nature, direction and closeness of the relationship** between the two variable characters — for example height and weight in men aged 20 years (Mahajan's Methods in Biostatistics, Ch 2, pp 40–42). Park states the rule of reading it: if the dots cluster round a straight line it shows evidence of a relationship of a linear nature, and if there is no such cluster it is probable that there is no relationship between the variables (Park 28e, Ch 21, pp 1002–1003).

Place of the scatter diagram among methods of data presentation

Method	Type of data suited	What it displays	No. of variables
Histogram	Quantitative continuous, grouped	Shape of the frequency distribution	One
Frequency polygon / frequency curve	Quantitative continuous	Smoothed frequency distribution	One
Line chart (line diagram)	Quantitative, ordered in time	Trend of events with the passage of time	One, over time
Cumulative frequency diagram (ogive)	Quantitative continuous	Percentiles and cumulative frequency	One
Scatter or XY diagram	Quantitative, paired (bivariate)	Relationship / correlation between two variables	Two
Bar diagram (simple, multiple, proportional)	Qualitative, discrete, counted	Comparison of magnitudes	One or two categories
Pie or sector diagram	Qualitative	Proportional composition of a whole	One
Pictogram	Qualitative, for lay audiences	Magnitude by picture symbols	One
Statistical map (shaded map, spot map)	Data by geographic or administrative area	Spatial distribution	One, by place

(Mahajan's Methods in Biostatistics, Ch 2, pp 32–33; Park 28e, Ch 21, pp 1000–1003)

Its unique place is that it is the **only routine method of presentation that displays two quantitative variables simultaneously**, and it is therefore the compulsory first step of every correlation and regression analysis.

Construction — steps

- Tabulate the paired observations so that both characteristics are cross-classified for each individual; n individuals give n pairs.

- Take the **independent, causal or predictor variable on the horizontal axis (X, abscissa)** and the **dependent or predicted variable on the vertical axis (Y, ordinate)**.
- Choose a scale for each axis that covers the observed range; label each axis with the variable name and its unit of measurement.
- Plot **one dot for each individual** at the meeting point of its X and Y values.
- **Do not join the dots** — joining them converts the figure into a line diagram, which shows trend over time and not correlation.
- Inspect the configuration of the dots. If they cluster along a straight line, superimpose the **line of best fit (regression line, $Y = a + bX$)** computed by least squares; it passes through the scatter with about half the points on either side and is symmetrically distributed if the correlation is significant (Mahajan's Methods in Biostatistics, Ch 12, pp 247–249).
- Caption the figure with both variables, their units, the number of observations and the data source.

What the scatter diagram reveals — the five patterns of correlation

Pattern	Correlation coefficient (r)	Appearance of the dots	Example
Perfect positive	$r = +1$	All dots on one straight line rising left to right; X varies directly and proportionately with Y	Not found in nature; approached by height and weight, age and height, age and weight up to a certain age
Perfect negative	$r = -1$	All dots on one straight line falling left to right	Not found in nature
Moderately (partial) positive	$0 < r < +1$	Dots scattered about a rising line	Height and weight of adults; gestational age and crown–heel length of newborns
Moderately (partial) negative	$-1 < r < 0$	Dots scattered about a falling line	Mean weekly temperature and number of deaths registered
Absolutely no correlation	$r \approx 0$	Dots scattered at random over the whole field, no line discernible	Two unrelated characters

The extent of correlation always lies between -1 and $+1$ (Mahajan's Methods in Biostatistics, Ch 12, pp 236–237). Pearson's coefficient r is unit-free, independent of change of origin and scale, and lies between -1 and $+1$ (Sunder Lal, Ch 6, p 232). If the dots instead cluster along a **curve**, a non-linear relationship is present — growth curves are typically logistic or exponential and seasonal curves are cyclic (Sunder Rao & Richard, Ch 13, pp 90–91).

Uses of the scatter diagram

- **Obligatory first step of bivariate analysis** — the analysis of a relationship is started by representing the two characteristics of each individual as a point on a graph, and the

configuration of the points indicates the nature of the relationship before any coefficient is calculated (Sunder Rao & Richard, Ch 13, p 90).

- ▶ **Graphic test of correlation.** Of the two methods of deciding whether two variable characters are correlated, one is mathematical (compute r and its standard error) and the other is **graphic** — plot the meeting points of the two variables and note the trend in the scatter diagram (Mahajan's Methods in Biostatistics, Ch 12, pp 250–251).
- ▶ **Display of the regression line for prediction** — the expected value of Y for any given value of X can be read straight off the fitted line without further calculation (Mahajan's Methods in Biostatistics, Ch 12, pp 247–249).
- ▶ **Checking the assumption of linearity before applying Pearson's r or linear regression.** In simple bivariate analysis, inspecting the graph of the original points is easier than examining residual plots; if the points show a clearly non-rectilinear pattern the straight line is an unsatisfactory model (Principles of Medical Statistics, Ch 19, pp 390–391). Correlation gives the intensity of relationship only when the relationship is linear (Sunder Rao & Richard, Ch 13, p 97).
- ▶ **Detection of outliers and influential points.** A single outlier can greatly alter the covariance and variances and so distort r and the regression line — as demonstrated by *Anscombe's quartet* — four sets of bivariate data ($n = 11$ each) with identical summary values, identical $r^2 = 0.667$ and the identical regression line $Y = 3 + 0.5X$, yet four completely different scatter patterns (Principles of Medical Statistics, Ch 19, pp 391–392).
- ▶ **Detection of heterogeneity in the data.** If the dots tend to cluster in two or more groups, a spuriously high value of r may result merely from the heterogeneity of the data (Sunder Rao & Richard, Ch 13, p 97).
- ▶ **Ecological (correlational) studies in epidemiology**, where the unit of analysis is the population and group-level exposure is plotted against group-level outcome — for example bacillus Calmette-Guérin (BCG) vaccine coverage against COVID-19-related mortality per million population across 173 countries (Gordis 7e, Ch 7, pp 168–169). Such plots generate hypotheses only, and are open to the **ecologic fallacy**.
- ▶ **Presentation of nutritional and public-health correlations**, for example the positive correlation between intake of fat and intake of sugar in the average diets of 41 countries (Park 28e, Ch 21, pp 1002–1003, after Yudkin J, Lancet 1964).
- ▶ **Health-care quality improvement.** The scatter diagram is one of the **seven basic tools** of statistical quality control used by quality-improvement committees, alongside the cause-and-effect (Ishikawa or fishbone) diagram, check sheet, histogram, Pareto chart, control chart and flowchart; it shows the relationship between pairs of numerical data and makes the direction of that relationship easy to see (Health Program Planning and Evaluation, Ch 10, pp 277–280).
- ▶ **Publication and reporting** — the bivariate graph of points should always be drawn, inspected, and preferably displayed in the published report, because for a two-dimensional relationship a single graphic picture is worth more than a thousand words of statistical calculation (Principles of Medical Statistics, Ch 19, p 391).

Advantages and limitations

Advantages	Limitations
Simple, needs no calculation, and gives an immediate visual impression of direction and closeness of relationship	Purely qualitative — it shows the trend but does not quantify it; the correlation coefficient must still be computed and tested
Retains every individual observation, so no information is lost by grouping	Becomes crowded and unreadable with very large numbers of observations, and identical pairs overlap
Reveals non-linear patterns, outliers and clustered subgroups that a single summary coefficient conceals	Reading is subjective; two observers may judge the same scatter differently
Guides the correct choice of subsequent analysis (linear regression, data transformation, or non-parametric method)	Restricted to two quantitative variables measured on the same individual; not usable for qualitative data
Displays the regression line and allows prediction within the observed range	Regression lines must not be extended beyond the range of the data on which they are based (Sunder Rao & Richard, Ch 13, p 96)
Useful for hypothesis generation in ecological studies and quality improvement	Shows association only — a correlation does not prove that one variable causes the change in the other, and time-related <i>nonsense correlations</i> may appear (Sunder Rao & Richard, Ch 13, p 97)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Diagram 1 — Panel of the five correlation patterns** (draw five small square boxes side by side). Labels: *X axis — independent variable; Y axis — dependent variable*; box A **Perfect positive correlation, $r = +1$** ; box B **Perfect negative correlation, $r = -1$** ; box C **Moderately (partial) positive correlation, $0 < r < +1$** ; box D **Moderately (partial) negative correlation, $-1 < r < 0$** ; box E **Absolutely no correlation, $r \approx 0$** . In boxes C and D also draw and label the two regression lines, *Y on X* and *X on Y*.
- **Diagram 2 — Scatter diagram with the line of best fit**. Labels: *X axis — Height (cm); Y axis — Weight (kg)*; *each dot = one individual (paired X, Y value)*; *line of regression of Y on X, $Y = a + bX$* ; *about half the dots on each side of the line*; *outlier point*.

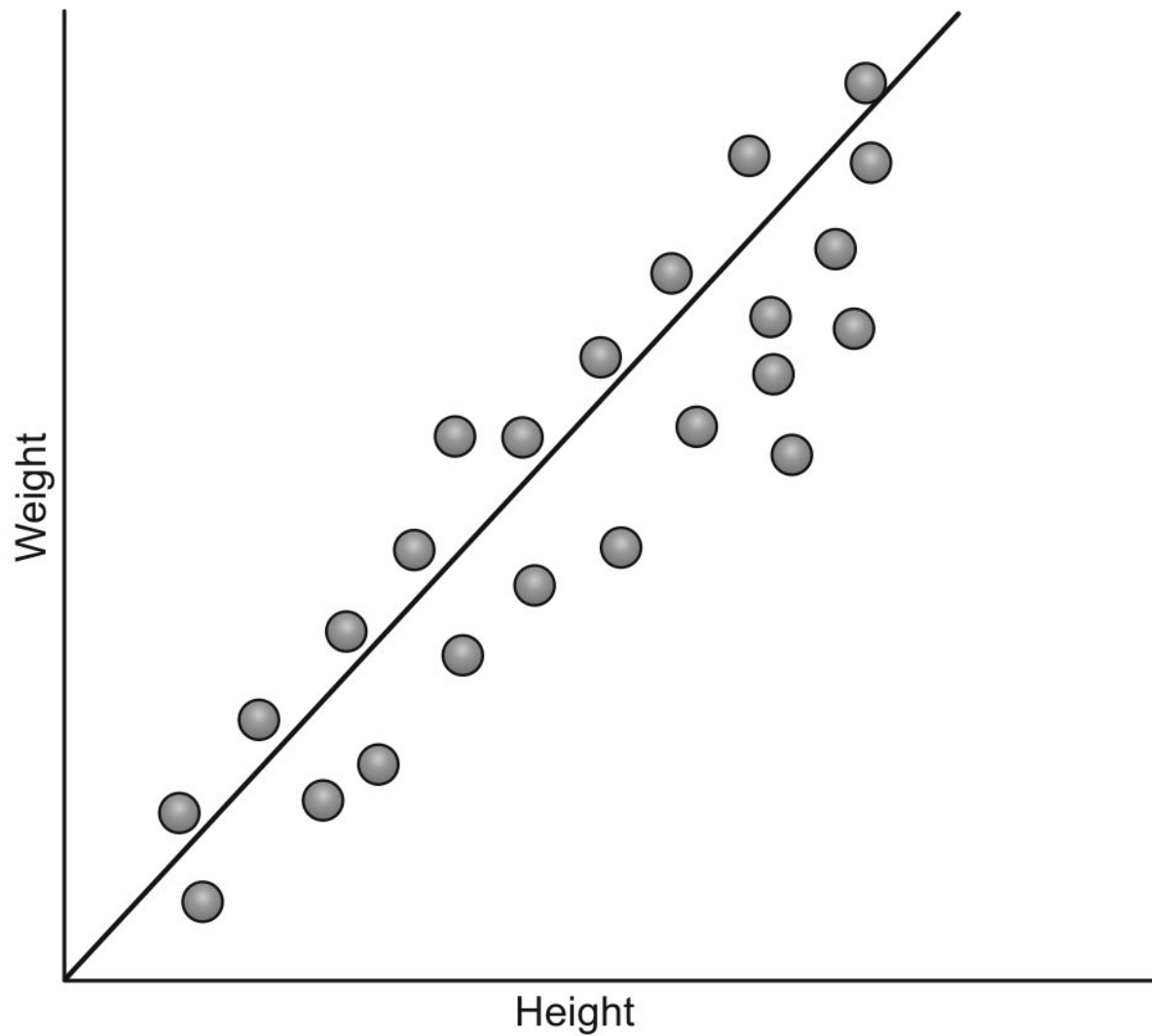


Fig 2.9 — Scatter diagram showing positive correlation (Mahajan's Methods in Biostatistics for Medical Students and Research Workers (PDFDrive), p.42)

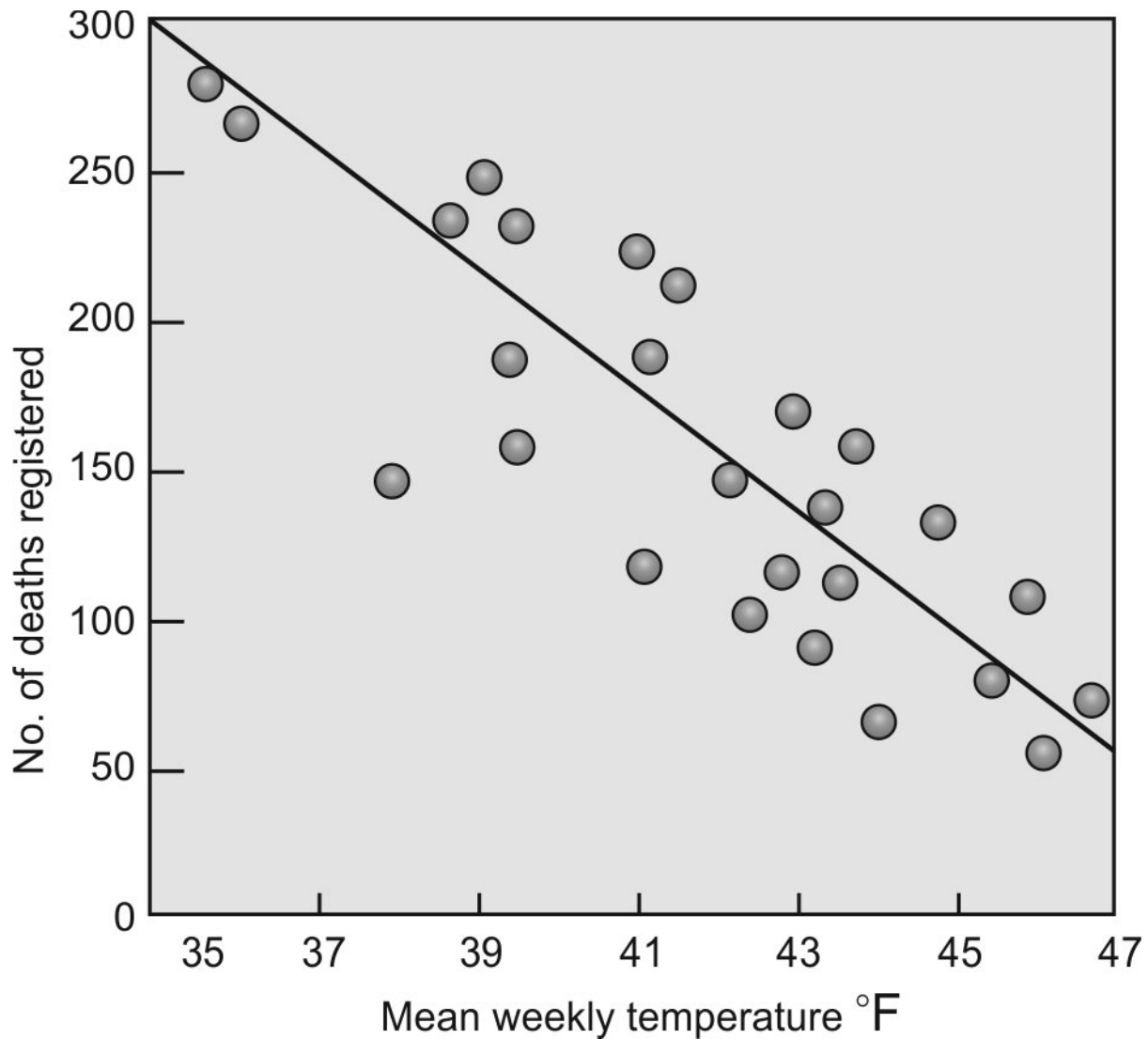
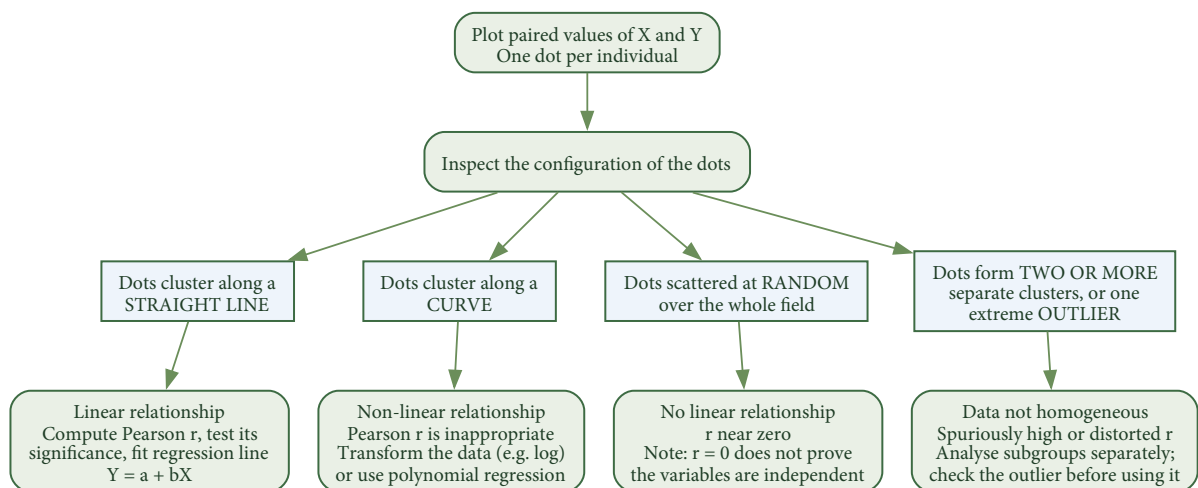
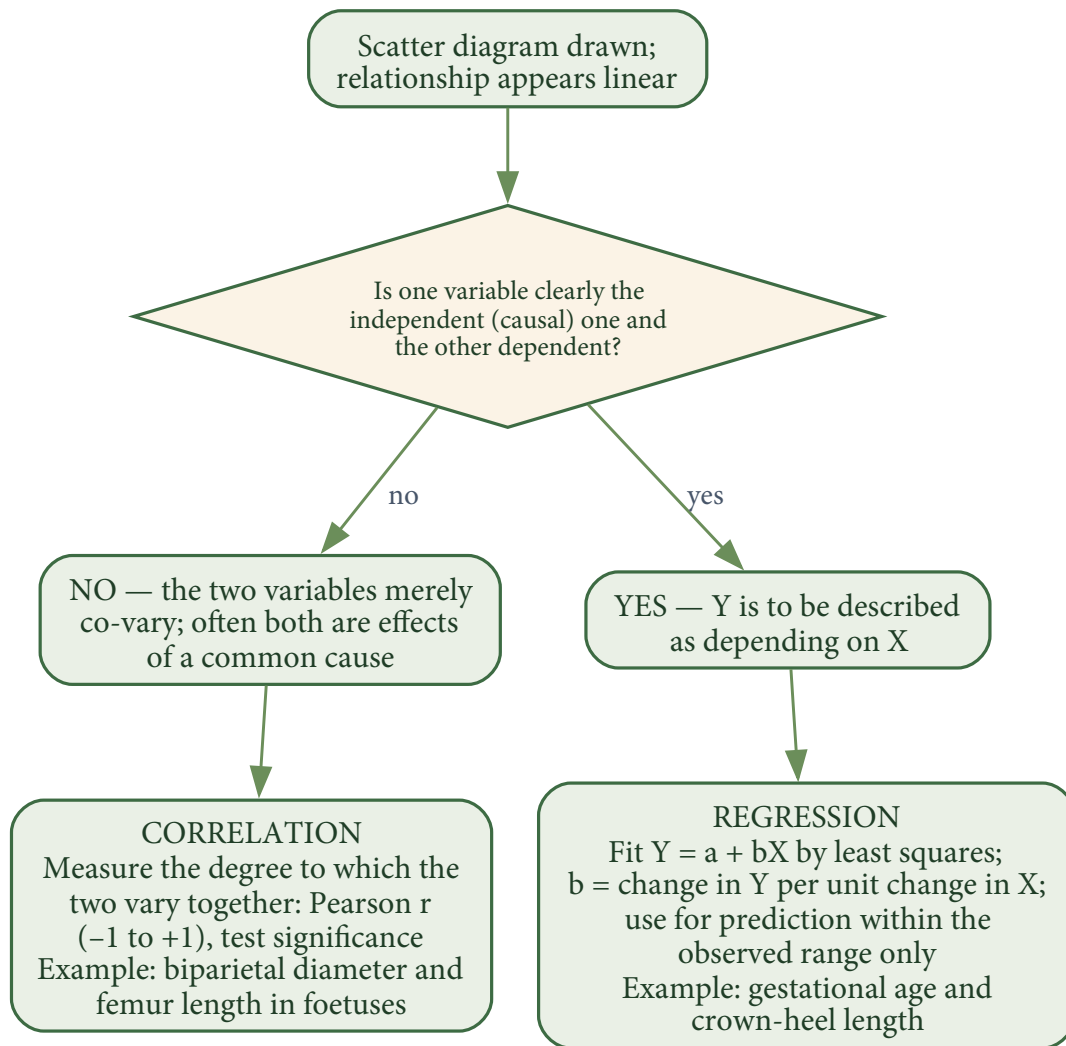


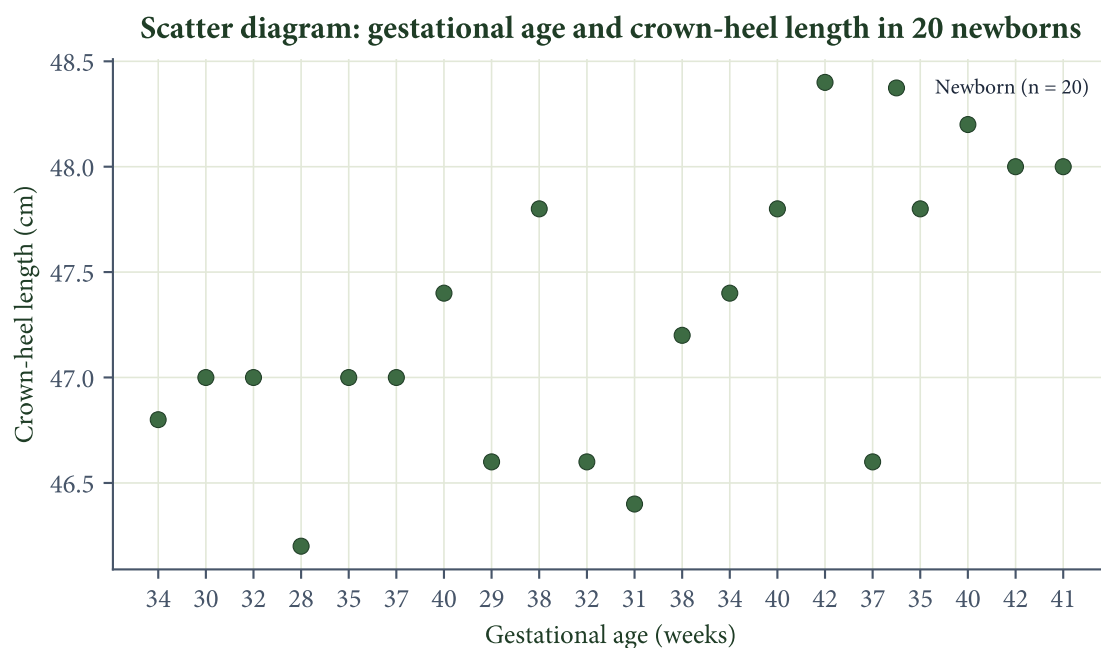
Fig 2.10 — Scatter diagram showing negative correlation (Mahajan's Methods in Biostatistics for Medical Students and Research Workers (PDFDrive), p.42)



Reading a scatter diagram — pattern seen and the correct next analytic step



Using the scatter diagram to decide between correlation and regression



High-yield exam pointers

- ▶ Synonyms to write: **scattergram, correlation diagram, XY diagram, dot diagram**; it is the graph of **paired quantitative (bivariate)** data, one dot per individual, dots **never joined**.
- ▶ Axis rule: **independent variable on X (abscissa), dependent variable on Y (ordinate)**; regression line $Y = a + bX$, where b (regression coefficient) = change in Y per unit change in X .
- ▶ Five patterns with their r : perfect positive **+1**, perfect negative **-1**, moderately positive **0 to +1**, moderately negative **-1 to 0**, no correlation ≈ 0 ; extent of correlation always **-1 to +1**.
- ▶ Park's reading rule verbatim: dots clustering round a straight line = evidence of a **linear relationship**; no cluster = probably **no relationship** (Park 28e, Ch 21).
- ▶ Two methods of testing correlation: **mathematical** (r and its standard error) and **graphic** (the scatter diagram) — Mahajan.
- ▶ Three classic cautions: $r = 0$ **does not mean independence**; **heterogeneous clusters give a spuriously high r** ; **Anscombe's quartet** — identical coefficients, entirely different scatters, so always plot before you compute.
- ▶ Applied uses to name: **ecological (correlational) study** (BCG coverage vs COVID-19 mortality, Gordis 7e) and **one of the seven basic quality-improvement tools** with the fishbone diagram, check sheet, histogram, Pareto chart, control chart and flowchart.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: the Sunder Rao & Richard and Sunder Lal page numbers cited above are the printed book pages; the retrieval index also carries a slightly different internal PDF pagination for these two titles.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, NFHS-5 (2019–21), CONSORT 2025 statement, WHO GLASS, Ayushman Bharat Digital Mission dashboard

Programme and policy updates (India)

- ▶ The paired quantitative indicators that a scatter diagram plots are now produced continuously by programme dashboards rather than by one-off surveys: an assessment of the **Ayushman Bharat Digital Mission** (launched 2021) reported 400 million beneficiary registrations and 273 million linked health records as of 3 June 2023, with about 208,000 health facilities and 190,000 health professionals verified — but with wide inter-state variation, which is precisely the kind of two-variable spread best shown as a state-wise XY plot rather than a table [PMID 39437234].
- ▶ **Civil registration is now analysed district-wise as a bivariate spatial problem.** Using NFHS-5 (2019–21) data on 79,449 deaths across 707 districts, only 71% of deaths in India were registered; district-level death registration showed strong positive spatial autocorrelation (Moran's $I = 0.69$, $p < 0.01$), with two "cold spot" clusters totalling 152 districts (Arunachal Pradesh–Nagaland–Manipur, and Jharkhand–Bihar–Uttar Pradesh). Spatial-error regression linked lower registration to higher proportions of uneducated, poor, rural, Muslim and Scheduled Tribe households; the authors explicitly caution that these district-level correlations are subject to the **ecological fallacy** [PMID 40348997].

Current national data

- ▶ The classical scatter diagram has an explicit spatial descendant, the **Moran's scatter plot**, in which each state's own value is plotted on X against the mean value of its neighbouring states on Y. An ecological study of NFHS-5 state data found total fertility rate ranging 1.1–3.0 (mean 1.81), weak global clustering (Moran's $I = 0.020$), LISA high-fertility clusters in Bihar, Jharkhand and Assam against low-fertility clusters in Tamil Nadu and Kerala, and the multidimensional poverty index as the only significant predictor of TFR (OLS coefficient 9.443, $p < 0.001$) [PMID 42266520].
- ▶ When the outcome is **qualitative**, the bivariate first step is a cross-tabulation and chi-square, not a scatter diagram — a distinction worth stating in the exam. NFHS-5 analysis of 673,094 non-pregnant, non-amenorrhoeic women aged 15–49 found anaemia prevalence 57%, used a bivariate percentage distribution with Pearson's chi-square before logistic regression, and reported higher odds of anaemia with traditional contraceptive methods (OR 1.08; 95% CI 1.05–1.11) [PMID 38938050].

Elimination, surveillance and disease status

- ▶ National surveillance now reports **rate against denominator-utilisation as a paired measurement per unit**, the exact bivariate structure a scatter diagram displays for quality-improvement benchmarking. India's health-care-associated infection surveillance network (about 200 ICUs in 54 hospitals, 1 May 2017 – 30 April 2024) recorded 8,629 central line-associated bloodstream infections over 977,052 central-line days: pooled CLABSI rate 8.83 per 1000 central-line days with a central-line utilisation ratio of 0.32, highest in neonatal ICUs (13.86 per 1000, CLUR 0.11); carbapenem resistance reached 87.1% in *Acinetobacter baumannii* [PMID 40845882].

Guideline and legislative updates

- ▶ **CONSORT 2025 supersedes CONSORT 2010** and should no longer be cited in its old form: a 30-item checklist plus the participant flow diagram, developed through a three-round Delphi of 317 participants and a 30-expert consensus meeting. Seven items were added, three revised and one deleted, with a new open-science section (registration, access to protocol and statistical analysis plan, sharing of de-identified participant data, funding and conflicts of interest); item 26 now requires both absolute and relative effect sizes for binary outcomes [PMID 40228833].
- ▶ A statistical editors' catalogue of common errors restates the rule that governs the scatter diagram's place among presentation methods — **use tables when specific values matter and graphs when general patterns and relationships matter** — and flags pie charts, "dynamite" bar-and-whisker plots, double-axis graphs and stacked bar charts as poor choices, recommends showing interactions graphically, and lists failure to check model assumptions (linearity among them) as a recurring fault [PMID 39284617].
- ▶ When the dots cluster along a **curve** rather than a line, the current recommended treatment is **restricted cubic splines** rather than categorising the continuous variable: an *International Journal of Epidemiology* tutorial presents RCS as a flexible extension of categorisation, sets out its two steps as model fitting and *graphical presentation* of the exposure–outcome curve, and supplies R, Stata and SAS code [PMID 40527479].

Global and WHO developments

- ▶ Genomic surveillance is now delivered to programme managers as an interactive plotting dashboard: **TyphiNET** v1.6 (February 2024) carried 11,836 *Salmonella* Typhi genomes from 101 countries, India contributing 2,327. It reports a national prevalence only when $N \geq 10$ (the same floor used by **WHO GLASS**) and maps it only at $N \geq 20$, and is designed to inform typhoid conjugate vaccine introduction — TCVs having been WHO-prequalified in 2018 and made Gavi-eligible [PMID 40346656].
- ▶ Purpose-built spatio-temporal visualisation software now formalises the textbook rule of *plotting before computing*: the Epid Data Explorer supports exploration without a pre-formulated hypothesis, and was demonstrated on COVID-19 data to compare indicators across areas and time and to assess correlation between viral spread and weather conditions [PMID 39224960].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 21, Health Information and Basic Medical Statistics); Mahajan's Methods in Biostatistics for Medical Students and Research Workers (Ch 2, Sources and Presentation of Data; Ch 12, Correlation and Regression); Sunder Rao & Richard — Introduction to Biostatistics and Research Methods (Ch 13, Linear Regression and Correlation); Principles of Medical Statistics — Feinstein (Ch 18, Principles of Associations; Ch 19, Evaluating Trends); Gordis Epidemiology 7e (Ch 7, Observational Studies); Sunder Lal — Textbook of Community Medicine (Ch 6, Basic Statistics and its Applications); Health Program Planning and Evaluation (Ch 10, Program Quality and Fidelity). Figures reproduced from Mahajan's Methods in Biostatistics, p 42.

Recent-advances sources: 39437234 — The Ayushman Bharat Digital Mission of India: An Assessment, *Health Systems and Reform* 2024, doi:[10.1080/23288604.2024.2392290](https://doi.org/10.1080/23288604.2024.2392290) (abstract-only); 40348997 — Spatial disparities in death registration across states and districts of India, 2019-21, *BMC Public Health* 2025, doi:[10.1186/s12889-025-22707-2](https://doi.org/10.1186/s12889-025-22707-2); 42266520 — Spatial Distribution and Determinants of Fertility in India, *Indian Journal of Community Medicine* 2026, doi:[10.4103/ijcm.ijcm_555_25](https://doi.org/10.4103/ijcm.ijcm_555_25); 38938050 — Modern Contraception and Anaemia Among Reproductive-age Women in India: Results From a Household Survey, *Journal of Preventive Medicine and Public Health* 2024, doi:[10.3961/jpmpmh.23.504](https://doi.org/10.3961/jpmpmh.23.504); 40845882 — Profile of central line-associated bloodstream infections in adult, paediatric, and neonatal intensive care units of hospitals participating in a health-care-associated infection surveillance network in India: a 7-year multicentric study, *Lancet Global Health* 2025, doi:[10.1016/S2214-109X\(25\)00221-9](https://doi.org/10.1016/S2214-109X(25)00221-9) (abstract-only); 40228833 — CONSORT 2025 statement: updated guideline for reporting randomised trials, *BMJ* 2025, doi:[10.1136/bmj-2024-081123](https://doi.org/10.1136/bmj-2024-081123); 39284617 — Common errors in statistics and methods, *BMJ Paediatrics Open* 2024, doi:[10.1136/bmjpo-2024-002755](https://doi.org/10.1136/bmjpo-2024-002755); 40527479 — Estimating and presenting non-linear associations with restricted cubic splines, *International Journal of Epidemiology* 2025, doi:[10.1093/ije/dyaf088](https://doi.org/10.1093/ije/dyaf088) (abstract-only); 40346656 — The TyphiNET data visualisation dashboard: unlocking *Salmonella* Typhi genomics data to support public health, *Genome Medicine* 2025, doi:[10.1186/s13073-025-01470-4](https://doi.org/10.1186/s13073-025-01470-4); 39224960 — Epid data explorer: A visualization tool for exploring and comparing spatio-temporal epidemiological data, *Health Informatics Journal* 2024, doi:[10.1177/14604582241279720](https://doi.org/10.1177/14604582241279720) (abstract-only).



Life table

Asked in: Paper I 26-May-2025 — RGUHS MD Community Medicine

Definition — A **life table** is a particular way of expressing the death rates experienced by a population during a particular period, set out as the life history of a **hypothetical cohort** (conventionally 1,00,000 births) that is subjected to the current age-specific death rates and decreases gradually by death until every member has died (Mahajan's Methods in Biostatistics, Ch 16 — Life Table, p 324). It is a special type of cohort analysis, and is a device not only for mortality but also for other vital events — natality, reproduction and chances of survival. *William Farr* called the life table the "**biometer of population**" (Sunder Rao & Richard, Ch 19, p 171). Its principal output, **expectation of life** (e_x), is "the average number of years which a person of that age may expect to live, according to the mortality pattern prevalent in that country" (Park 28e, Ch 8, pp 585-586).

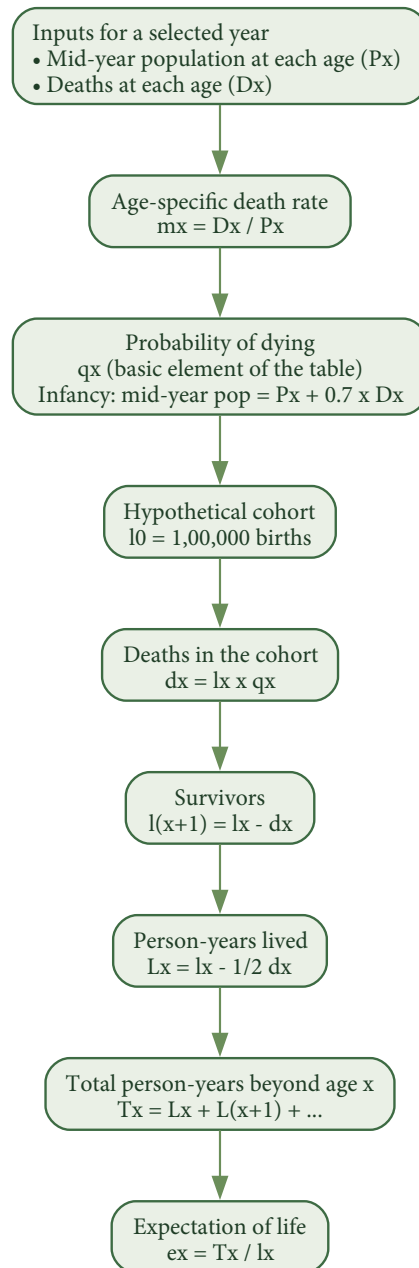
Columns of a life table and what each means

Column	Meaning	Relation / formula
x	Exact age in completed years (or age interval x to $x+1$)	—
P_x	Mid-year population between ages x and $x+1$ (observed)	Denominator for m_x
D_x	Deaths observed between ages x and $x+1$ (observed)	Numerator for m_x
m_x	Age-specific death rate	$m_x = D_x / P_x$
a_x	Average fraction of the last year of life lived by those dying between x and $x+1$	Taken as $\frac{1}{2}$ after age 5; not $\frac{1}{2}$ in infancy
q_x	Probability of dying — proportion of those alive at exact age x who die before $x+1$; the <i>basic element</i> of the life table	$q_x = D_x / \text{mid-year population of that age}$
p_x	Probability of surviving from x to $x+1$	$p_x = 1 - q_x$
l_x	Number of the cohort surviving to exact age x	$l_0 = 1,00,000$; $l_{x+1} = l_x - d_x$
d_x	Number of the cohort dying between x and $x+1$	$d_x = l_x \times q_x$
L_x	Person-years lived by the cohort between x and $x+1$	$L_x = l_x - \frac{1}{2}d_x (= l_{x+1} + \frac{1}{2}d_x)$
T_x	Total person-years lived by the cohort beyond age x	$T_x = L_x + L_{x+1} + \dots$
e_x	Expectation of life at age x — average further years a person aged x may expect to live	$e_x = T_x / l_x$

(Sunder Rao & Richard, Ch 19, pp 171-173; Mahajan, Ch 16, pp 326-330)

Data required and how it is built

- ▶ **Population living at every individual age** in a selected year — the selected year being the most recent one for which accurate statistics are available.
- ▶ **Number of deaths occurring at those ages** during that same year (Mahajan, Ch 16, p 325).
- ▶ Life tables are usually worked out **after each decennial census**, and are prepared **separately for males and females**, because male and female mortality experience differ; separate tables may also be made for geographical subdivisions, ethnic or occupational groups (Sunder Rao & Richard, Ch 19, p 171).
- ▶ **Infancy needs special handling** — deaths are not distributed evenly across the first two years of life, so the "half the deaths by 30th June" assumption does not hold; in the first year roughly 7/10 of infant deaths are taken to have occurred by mid-year. From the third year onwards deaths are more or less evenly distributed and the $\frac{1}{2}$ assumption is used (Mahajan, Ch 16, p 326).



Sequence of construction of a life table from vital statistics to expectation of life

Uses of the life table

#	Use	What the life table supplies	Concrete application
1	Estimating survivors at any age (service planning)	l_x column	Age 5 — children likely to enter primary school; age 15 — women entering the fertile period and adolescents entering the community; age 18 — persons eligible to vote; age 45+ — women reaching menopause; age 55/58/60 — persons becoming eligible for pension; 65+ — size of the geriatric problem, and of concessions such as income-tax benefit and railway concession (Mahajan, Ch 16, pp 324-325)

#	Use	What the life table supplies	Concrete application
2	Measuring longevity	e_x , especially e_0	Expectation of life at birth is a positive health indicator and one of the best indicators of a country's development and overall health status; adopted as a global health indicator (Park 28e, Ch 2, p 37; Ch 8, pp 585-586)
3	Comparing mortality across groups	e_x and life-table death rates	Comparison of two places, periods, professions or social groups — and such comparison is independent of age and sex composition , so no separate standardisation is needed (Mahajan, Ch 16, pp 325, 330)
4	Analysis by cause of death	Multiple decrement table	Splits life-table deaths by cause; gives the chance of ultimately dying of a stated cause, and the gain in expectation of life if that cause were eliminated (Sunder Rao & Richard, Ch 19, p 175)
5	Fertility and family welfare	Cohort analysis by the life-table method	Total fertility and the net reproduction rate (NRR) are computed by the life-table method; the technique has been used to evaluate the effectiveness of population-control programmes (Mahajan, Ch 16, p 330; Sunder Rao & Richard, Ch 19, p 175)
6	Actuarial and administrative use (its original purpose)	q_x and l_x	Number likely to die after joining service and before retirement — hence budgeting of insurance premium, risk cover and pension liability (Mahajan, Ch 16, p 325; Sunder Rao & Richard, Ch 19, p 171)
7	Morbidity and disability analysis	Multiple decrement and select tables	Estimating the lifetime chance of becoming diabetic, or of acquiring a cardiovascular-renal condition, from a general-population life table plus prevalence and mortality of the condition (Sunder Rao & Richard, Ch 19, p 175)
8	Hospital and health-service studies	'Hospital stay' tables built on the life-table pattern	Expectation of mental disease computed from first admissions; expected length of hospital stay for specific conditions; multiple decrement table for patients separated by discharge, death or withdrawal (Sunder Rao & Richard, Ch 19, p 175)
9	Clinical survival analysis	Modified (clinical) life table	Survival after treatment of a chronic disease such as tuberculosis or cancer, or after cardiac surgery such as bypass or transplantation; adapted by <i>Kaplan and Meier</i> to handle censored observations — subjects lost to follow-up or still alive when the study closes (Mahajan, Ch 16, p 325; Sunder Rao & Richard, Ch 19, pp 175-176)
10	Population projection and population studies	l_x and L_x by age and sex	Projection of population estimates by age and sex; studies of marriage, population growth, the labour force and the money value of a person to dependants (Mahajan, Ch 16, p 325; Sunder Rao & Richard, Ch 19, p 175)

In survival analysis, two time-dependent functions are derived: the **survival function** $S(t)$, the probability of surviving at least to time t , and the **hazard function** $h(t)$, the conditional probability of

dying at time t having survived to that time; the plot of $S(t)$ against t is the **survival curve** (Sunder Rao & Richard, Ch 19, p 176).

Suitable example 1 — constructing a short life table

Problem: Population and deaths for ages 0-5 are as below. If 10,000 children start life, construct a life table and find the number surviving at each successive birthday up to the 5th (Mahajan, Ch 16, pp 330-332).

Age (x)	Population	Deaths	Mid-year population	q_x	dx (per 10,000)	lx = survivors at exact age
0-1	4,151	70	$4,151 + (7/10 \times 70) = 4,200$	$70/4,200 = 0.0167$	167	10,000 (at birth)
1-2	4,792	16	$4,792 + (\frac{1}{2} \times 16) = 4,800$	$16/4,800 = 0.0033$	32	$10,000 - 167 = 9,833$ at age 1
2-3	4,797	6	$4,797 + 3 = 4,800$	$6/4,800 = 0.0013$	13	$9,833 - 32 = 9,801$ at age 2
3-4	4,998	4	$4,998 + 2 = 5,000$	$4/5,000 = 0.0008$	8	$9,801 - 13 = 9,788$ at age 3
4-5	4,798	4	$4,798 + 2 = 4,800$	$4/4,800 = 0.0008$	8	$9,788 - 8 = 9,780$ at age 4
—	—	—	—	—	—	$9,780 - 8 = 9,772$ at age 5

Interpretation: of 10,000 children starting life under this mortality experience, **9,772 survive to their fifth birthday**; the first year alone accounts for 167 of the 228 deaths, so any programme gain must be sought in infancy.

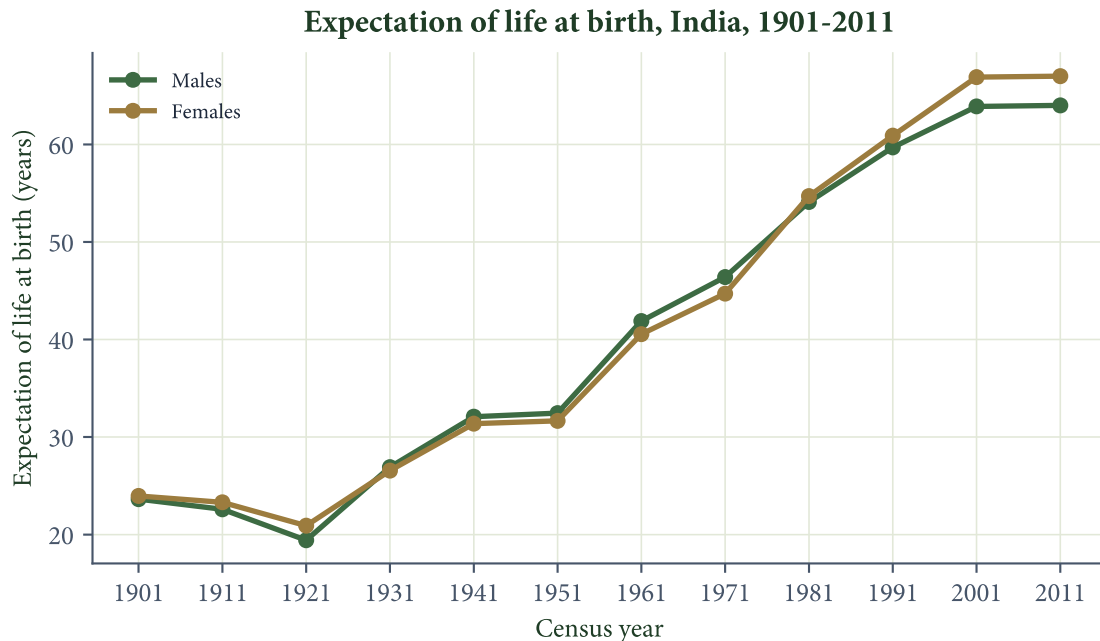
Suitable example 2 — reading a national life table

Extract from the life table for females in India, 1961-70 (Sunder Rao & Richard, Ch 19, Table 19.6, pp 172-173):

Age x	lx (survivors)	dx	q_x	L_x	T_x	ex (years)
0	1,00,000	12,837	0.12837	91,014	44,65,447	44.7
1	87,163	2,969	0.03406	85,382	43,74,437	50.2
2	84,194	1,348	0.01601	83,520	42,89,051	50.9
3	82,846	1,198	0.01446	82,367	42,05,531	50.8
4	81,648	1,072	0.01313	81,326	41,23,164	50.5
5	80,576	971	0.01205	80,091	40,41,838	50.2

Interpretation: expectation of life at birth was 44.7 years, but at age 1 it was 50.2 years and at age 2 it peaked at 50.9 years — a newborn's figure is dragged down by the very high infant mortality ($q_0 = 0.128$, i.e. 12,837 of every 1,00,000 girls died before their first birthday). This is exactly why life expectancy at age 1 excludes the influence of infant mortality and life expectancy at age 5 excludes the influence of

child mortality (Park 28e, Ch 2, p 37). For contrast, expectation of life at birth in India in 2024 was 71 years for males and 74 years for females (Park 28e, Ch 8, Table 15, p 586).



Life-table output over a century: expectation of life at birth falls in 1921, then rises steadily, with the female curve overtaking the male curve from 1981 (Park's Textbook of Preventive and Social Medicine 28e, Ch 8, Table 14, p 586)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Life-table survivorship curve (l_x curve)** — plot survivors (l_x) on the Y-axis against age (x) on the X-axis. Labels to mark: Y-axis "Number of survivors out of a cohort of 1,00,000 (l_x)"; X-axis "Age in completed years (x)"; starting point " $l_0 = 1,00,000$ births"; steep initial drop "infant and child mortality"; flat middle segment "low mortality of childhood and youth"; terminal fall "senescent mortality"; shaded area under the curve = " T_0 , total person-years lived"; $T_0/l_0 = e_0$.
- ▶ **Clinical (modified) life-table survival curve** — plot the survival function $S(t)$ on the Y-axis (0 to 1) against time since entry on the X-axis; label "censored observation (tick mark)", "5-year survival rate", and "Kaplan-Meier stepped curve".

High-yield exam pointers

- ▶ **Farr's phrase:** the life table is the "biometer of population"; $e_x = T_x/l_x$, $d_x = l_x \times q_x$, $L_x = l_x - \frac{1}{2}d_x$, $T_x = \sum L_x$, $p_x = 1 - q_x$.
- ▶ **q_x is the basic element;** radix of the cohort $l_0 = 1,00,000$; two data inputs only — mid-year population by age and deaths by age for the selected year.
- ▶ **Infancy exception:** the $\frac{1}{2}$ -year assumption fails in the first two years; take about 7/10 of infant deaths as occurring before mid-year.
- ▶ **e_x rises after birth** wherever infant mortality is high (India females 1961-70: e_0 44.7 vs e_1 50.2, e_2 50.9) — a favourite viva point.

- **Dated Indian figures:** expectation of life at birth, India 2024 — males 71 years, females 74 years (Park 28e); world combined e_0 rose from 46.5 years in 1950-55 to 73 years in 2024.
- **Uses mnemonic — "SLIM-FACE":** Survivors at any age, Longevity (e_x), Insurance/pension actuarial work, Mortality comparison free of age-sex structure, Fertility (NRR by life-table method), Analysis by cause (multiple decrement), Clinical survival (Kaplan-Meier), Estimation/projection of population.
- Comparison of communities by e_0 alone places **undue emphasis on mortality** and is distorted by selective migration — a valid criticism worth one line.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the RGUHS syllabus treats the life table both as a biostatistical tool (survival analysis) and as a demographic tool; the Sample Registration System abridged life tables of the Office of the Registrar General of India are the routine Indian source for expectation of life, but the retrieved corpus carried only the older 1961-70 and 1986-90 tables, so the worked figures above are quoted with their own periods rather than being updated.

SOURCE & COVERAGE

Mahajan's Methods in Biostatistics for Medical Students and Research Workers (Chapter 16 — Life Table); Sunder Rao & Richard — An Introduction to Biostatistics and Research Methods (Chapter 19 — Standardised Death Rates and Life Tables); Park's Textbook of Preventive and Social Medicine, 28e (Chapter 2 — Concept of Health and Disease; Chapter 8 — Demography and Family Planning, Tables 14 and 15). Coverage note: the corpus yielded no captioned life-table plate suitable for embedding, so the survivorship curve is given as a "Diagram to draw" box instead.



Regression analysis

Asked in: Paper I 24-Sep-2025 — RGUHS MD Community Medicine

Definition — Regression analysis is the statistical method that describes and quantifies the dependence of one variable, the **dependent (outcome) variable Y**, on one or more other variables, the **independent (explanatory or predictor) variables X**, so that the value of Y can be estimated from a known value of X. For the simple linear case the dependence is expressed as the equation of a straight line, $Y = a + bX$, whose constants are fitted to the observed data by the **method of least squares** (Sunder Rao & Richard, Ch 13). Its function is to provide a means of estimating the value of one variable from the value of another (Park 28e, Ch 21, p 1009).

Terms of the regression equation

Term	Symbol / formula	Meaning	How it is read
Independent variate	X	The variable that is known or set by the investigator	Customarily denoted x (Park 28e, Ch 21, p 1009)
Dependent variate	Y	The variable to be estimated or predicted	Denoted y
Regression line	$Y_c = a + bX$	Straight line of best fit through the scatter diagram	Mean line — about half the points lie on either side (Mahajan, Ch 12, p 234)
Intercept	a	Value of Y when X = 0	Anchors the line
Regression coefficient (slope)	b	Change expected in Y for one unit change in X	The single most examinable quantity (Sunder Rao & Richard, Ch 13)
b from the correlation coefficient	$b_{yx} = r \times (\text{SD of Y}) / (\text{SD of X})$	Links regression to correlation	Sunder Rao & Richard, Ch 13
Two regression lines	b_{yx} and b_{xy}	Y on X, and X on Y	Every dataset yields two lines, not one
Standard error of estimate	$S = \sqrt{[\Sigma(Y - Y_c)^2 / (n - 2)]}$	Scatter of observed points about the fitted line	Behaves like a standard deviation (SD) around the regression line (Mahajan, Ch 12)
95% limits of a predicted value	Predicted value $\pm 2 \times$ SD of regression	Confidence limits for an individual estimate	Mahajan, Ch 12 (Summary)
Coefficient of determination	r^2	Proportion of total variation in Y explained by the regression line	"Explained variation"; the remainder is due to unmeasured factors (Sunder Rao & Richard, Ch 13)
Multiple correlation coefficient	R (and R^2)	Strength of relation of Y with all explanatory variables together	R^2 = proportion of variation in Y explained by the whole model

Correlation and regression contrasted

Feature	Correlation	Regression
Question asked	Do the two variables co-vary?	How does Y depend on X?
Status of variables	No distinction — relationship is mutual; often both are effects of a common cause	Distinct dependent and independent variables
Measure obtained	Correlation coefficient r, ranging –1.0 to +1.0	Regression coefficient b, in units of Y per unit of X
Number of results	One value of r	Two lines/coefficients (Y on X and X on Y)
Purpose	Degree (strength) of association	Estimation and prediction; support for hypotheses of causation; statistical control of a variable
Typical example	Biparietal diameter and femur length in foetuses; birth weight and maternal height	Effect of age on blood pressure; protein intake on child growth rate; insecticide dose on insect mortality

(Both rows grounded in Sunder Rao & Richard, Ch 13; Park 28e, Ch 3, p 107 — correlation does not imply causation.)

Types of regression model

Model	Outcome (dependent) variable	Effect measure produced	Typical community-medicine use
Simple linear regression	Continuous, one predictor	Slope b (units of Y per unit X)	Estimating crown–heel length from gestational age
Multiple (multivariable) linear regression	Continuous, ≥ 2 predictors	Partial regression coefficients, each adjusted for the others	Birthweight on maternal weight and parity (Oxford Public Health 7e, Ch 39, pp 1446–1447)
Non-linear regression (log-transformation, polynomial)	Continuous, curvilinear relation	Coefficients after transformation, e.g. $\log Y = \log b + 2 \log X$	Growth and exponential dose–response curves (Sunder Rao & Richard, Ch 13)
Logistic / multiple logistic regression	Binary (diseased / not diseased, died / survived)	Adjusted odds ratio (OR)	Analysis of case-control studies (Handbook of Epidemiology, Ch I.3, p 183; Browner DCR 5e, Ch 8, p 236)
Poisson (or negative binomial) regression	Counts, or events per person-time	Incidence rate ratio (IRR)	Admissions per year; trends in incidence over calendar time (Oxford Public Health 7e, Ch 30 Cohort Studies, pp 1117–1118)
Cox proportional hazards regression	Time to event, with censoring	Hazard ratio (HR)	Survival analysis in cohort studies (Browner DCR 5e, Ch 8, p 224)

Model	Outcome (dependent) variable	Effect measure produced	Typical community-medicine use
Analysis of covariance (ANCOVA)	Continuous outcome with a group factor plus a covariate	Group means adjusted to a common covariate value	Comparing groups after removing a confounding covariate (Principles of Medical Statistics, Ch 29)

Rule of thumb: linear regression for quantitative outcomes, logistic regression for binary outcomes, Poisson regression for rates (Oxford Public Health 7e, Ch 32, p 1198).

Applications of regression analysis

Application	What regression contributes	Grounded example
1. Estimation and prediction	Supplies the expected value of Y for any observed X, mathematically or by reading off the fitted line	Crown–heel length (CHL) from gestational age (GA): $CHL = 42.97 + 0.120 \times GA$; for GA 40 weeks the estimated CHL is 47.77 cm (Sunder Rao & Richard, Ch 13)
2. Quantifying a dose–response or exposure–outcome effect	The slope b is itself the effect size, reported with its 95% confidence interval (CI) and p-value	Birthweight (g) = $2,832 + 11.3 \times \text{maternal pre-pregnancy weight (kg)}$: birthweight rises 11.3 g per kg (Oxford Public Health 7e, Ch 39, p 1446)
3. Control of confounding (multivariable adjustment / risk adjustment)	Regression procedures adjust the dependent variable for one or more confounding covariates, replacing stratification when confounders are many	Adding parity to the model above: birthweight = $2,830 + 11.0 \times \text{maternal weight} + 16.9 \times \text{parity}$ — the parity effect halves and becomes non-significant, exposing it as confounding by maternal weight (Oxford Public Health 7e, Ch 39, pp 1446–1447; Sunder Rao & Richard, Ch 24)
4. Testing effect modification (interaction)	Provides a uniform framework to test whether an effect is uniform across population subgroups	Handbook of Epidemiology, Ch I.3, pp 181–182
5. Analysis of analytic study designs	Yields the design-appropriate adjusted measure of association	Case-control → adjusted OR by logistic regression; cohort with person-time → IRR by Poisson regression; cohort with time-to-event → HR by Cox regression (Browner DCR 5e, Ch 8, p 224)
6. Identifying independent risk factors	Each coefficient reflects association with the outcome after adjusting for every other predictor in the model	Building a set of predictors of birthweight — income, tobacco, alcohol (Oxford Public Health 7e, Ch 39, pp 1445–1446)
7. Trend analysis of rates over time	Models change in incidence or mortality by calendar period, age and birth cohort	Poisson methods are apt for trends in incidence over calendar time (Oxford Public Health 7e, Ch 30, pp 1117–1118)

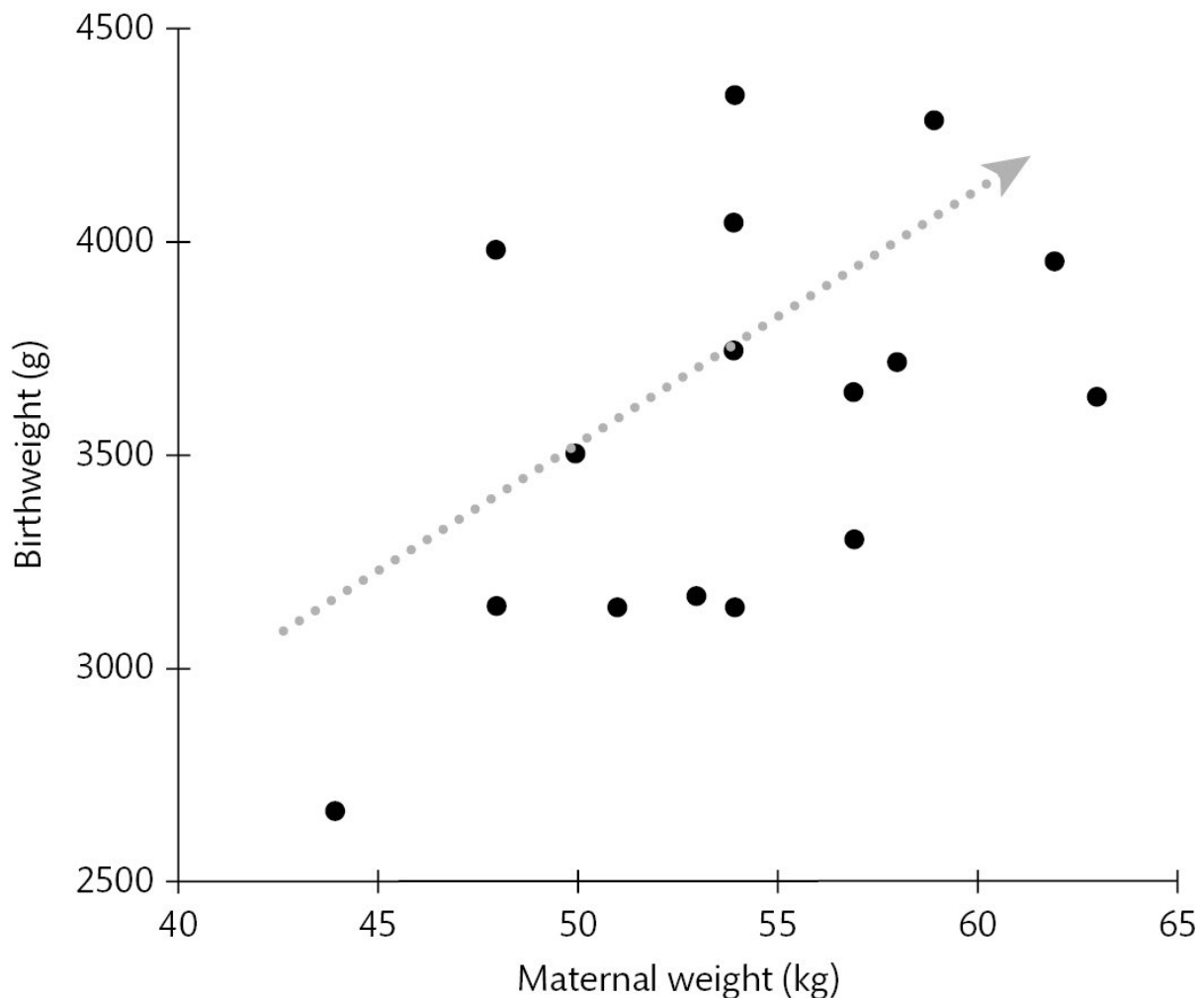
Application	What regression contributes	Grounded example
8. Health services research and health economics	Risk adjustment of outcomes across hospitals and districts; modelling of costs and utilisation	Ordinary least squares (OLS) regression on the logarithm of costs; Poisson regression for number of hospital admissions per year (Handbook of Epidemiology, Ch IV.5, p 1473)
9. Analysis of complex trial designs	Two-stage cluster-level analysis and random-effects models are built on regression	Cluster randomised trials (Oxford Public Health 7e, Ch 32, pp 1198, 1200)
10. Model parsimony	Tests whether a given model gives a parsimonious yet adequate description of the observed data	Handbook of Epidemiology, Ch I.3, pp 181–182

Assumptions, restrictions and pitfalls

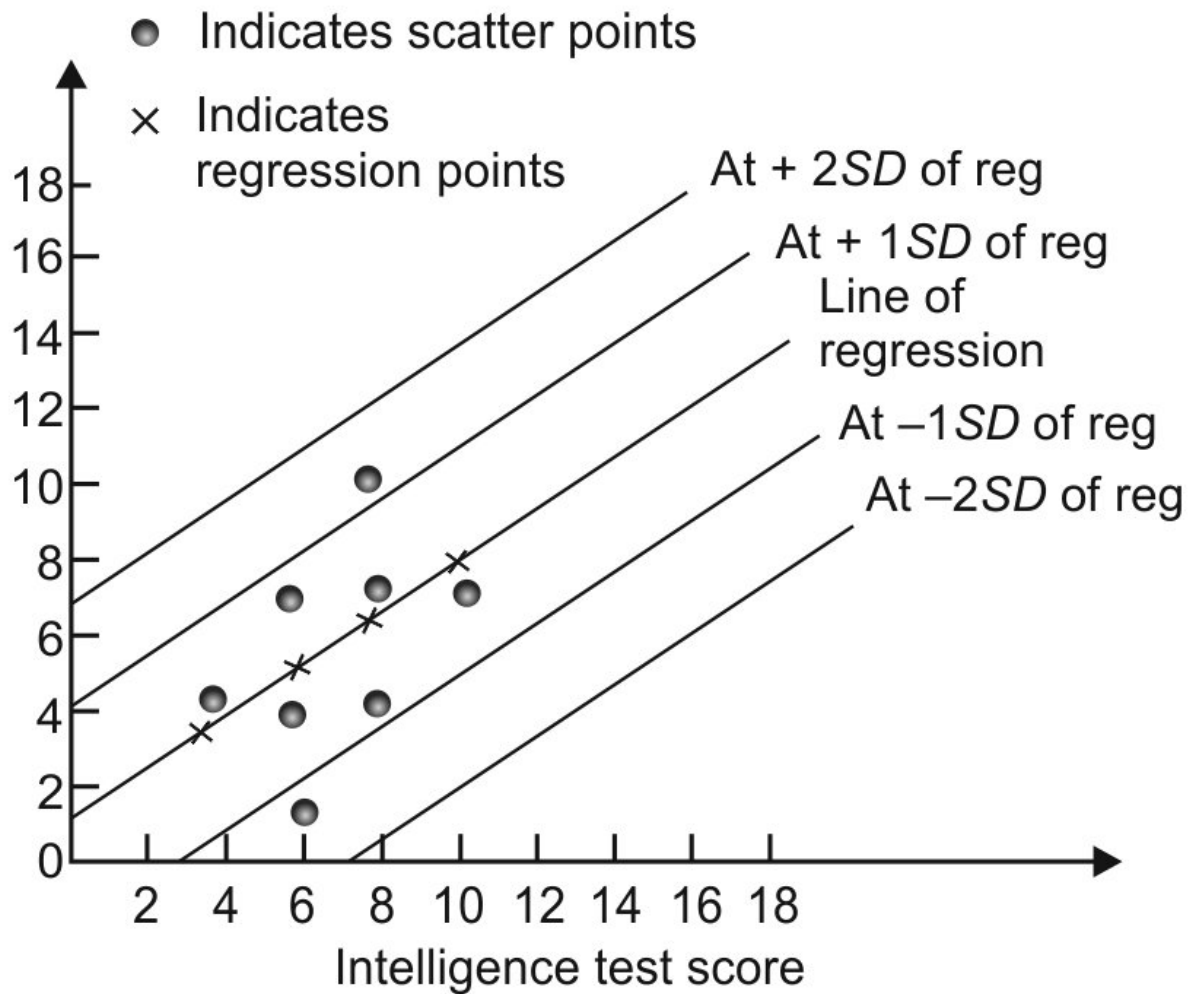
- ▶ **Linearity** — least-squares fitting assumes the data are suitably fitted by a straight line; where the scattergram is curved, transform the data (logarithm) or use polynomial regression (Principles of Medical Statistics, Ch 19, p 373; Sunder Rao & Richard, Ch 13).
- ▶ **No extrapolation** — the regression line must not be extended beyond the range of the data on which it was based. The regression of height on age of school-age children cannot be used to predict the heights of adults.
- ▶ **Uniform trend on interpolation** — predicting Y for an X lying between two observed values assumes the trend is uniform across that interval.
- ▶ **Homogeneous data** — if the scatter clusters in two or more groups, a spuriously high r may result from the heterogeneity alone.
- ▶ **r = 0 does not mean independence** — the two variables are merely uncorrelated, i.e. have no *linear* relationship.
- ▶ **Spurious ("nonsense") correlations** — two variables that merely share a time trend can correlate highly. *Yule* found $r = 0.951$ between the proportion of marriages celebrated in Anglican churches (England and Wales, 1866–1911) and the standardised mortality rate; *Charlier* found $r = 0.86$ between the stork population of Oslo and the number of babies born there (Sunder Rao & Richard, Ch 13).
- ▶ **Direction of causation** — it is not always clear which variable is cause and which is effect; regression supports but never proves causation.
- ▶ **Multicollinearity** — when two independent variables are highly correlated, one masks the effect of the other and may be dropped with a small, non-significant partial regression coefficient although it truly matters. Variables must be chosen on prior theoretical grounds, not by feeding in every available variable (Sunder Rao & Richard, Ch 13).
- ▶ **Variable selection** — forward (stepwise) inclusion and backward elimination are used to arrive at the "best" set of predictors; multiple regression is performed on a computer.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

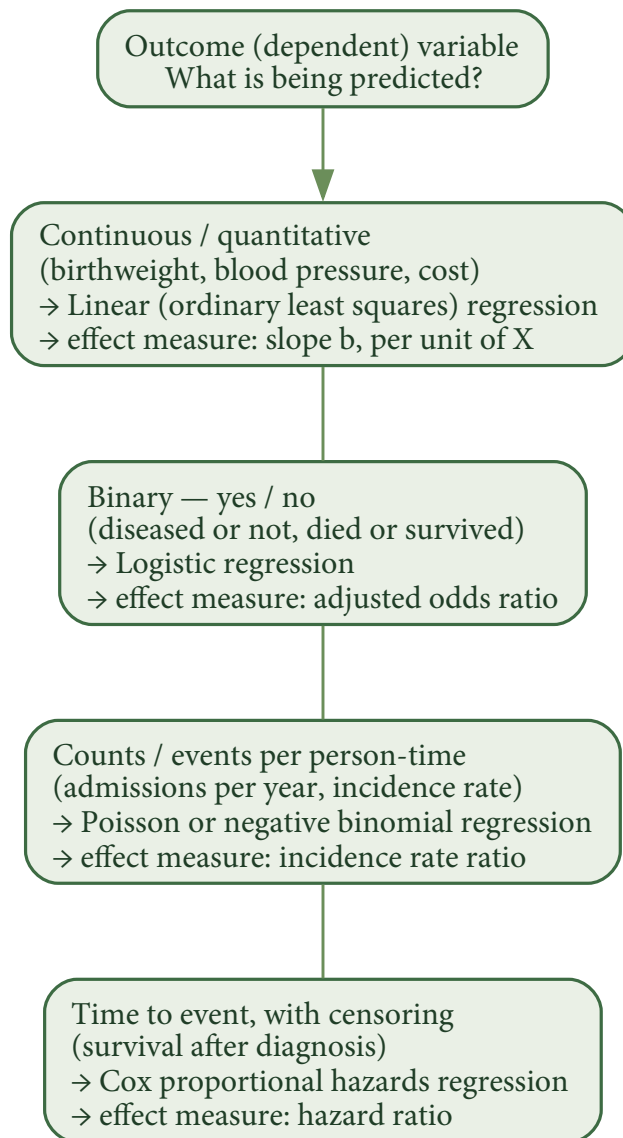
- **Draw: scatter diagram with the regression line.** Labels — X axis "Independent variable (X), e.g. maternal weight (kg)"; Y axis "Dependent variable (Y), e.g. birthweight (g)"; scatter points; the fitted straight line labelled "Regression line $Y = a + bX$ "; mark the intercept a on the Y axis; mark a right-angled step on the line showing "1 unit of X" and "b units of Y"; add the flanking lines "+1 SD of regression", "-1 SD of regression", "+2 SD of regression", "-2 SD of regression" and annotate "95% limits of a predicted value".



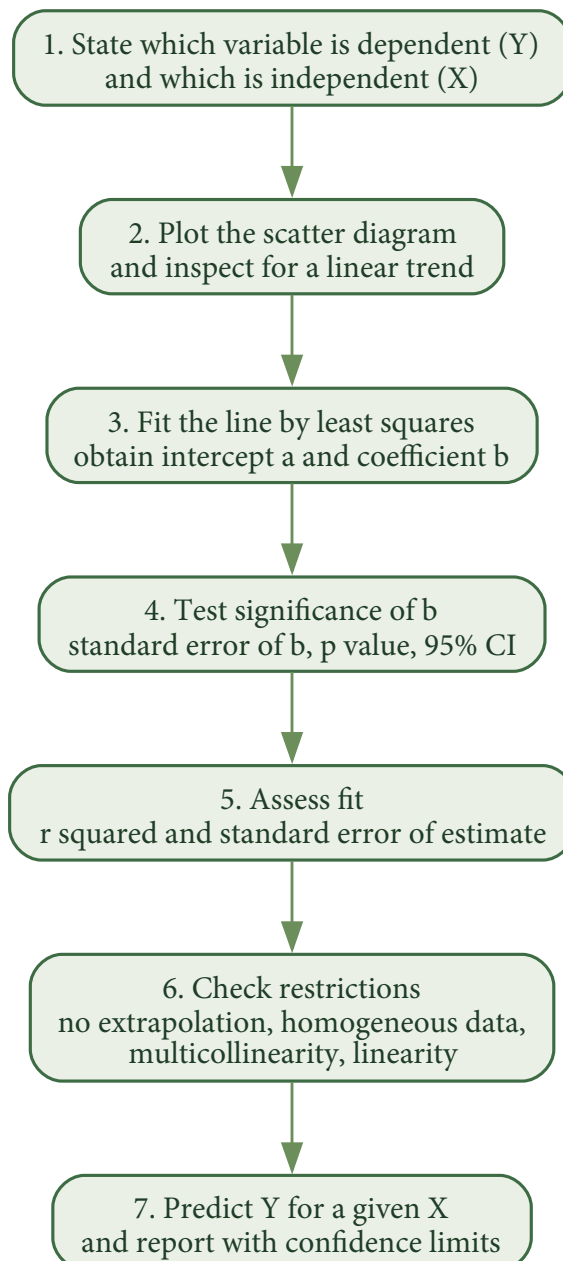
Scatter plot and regression line of birthweight against maternal pre-pregnancy weight (Oxford public health 7th ed, p.1438)



Regression line with ± 1 SD and ± 2 SD limits of regression (Mahajan's Methods in Biostatistics, p.241)



Choosing the regression model from the type of outcome variable



Steps in performing a regression analysis

High-yield exam pointers

- ▶ **Write the equation first:** $Y = a + bX$; a = intercept (value of Y when $X = 0$), b = regression coefficient = change in Y per **one unit** change in X .
- ▶ **Link to correlation:** $b_{yx} = r \times (\text{SD of } Y) / (\text{SD of } X)$; r^2 = coefficient of determination = proportion of variation in Y explained by the regression.
- ▶ **Two lines per dataset** — Y on X and X on Y ; correlation gives only one r .
- ▶ **Model-by-outcome grid** (guaranteed marks): continuous $\rightarrow$ linear; binary $\rightarrow$ logistic (odds ratio); rate/count $\rightarrow$ Poisson (rate ratio); time-to-event $\rightarrow$ Cox (hazard ratio).
- ▶ **The public-health application to name:** multivariable regression (multivariate methods) as the fifth method of **controlling confounding**, after restriction, matching, stratification and standardisation of rates.

- **Two restrictions always worth writing:** never extrapolate beyond the range of the data; a high r may be a spurious/"nonsense" correlation from a shared time trend (Yule's $r = 0.951$ marriages–mortality example).
- **Standard error of estimate** = $\sqrt{[\Sigma(Y - Y_c)^2/(n - 2)]}$; a predicted value ± 2 SD of regression gives the 95% limits.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park's Textbook covers regression very briefly under "Correlation and Regression" in the Health Information and Basic Medical Statistics chapter, so the depth expected of an MD Community Medicine answer — model families, adjusted odds ratios and hazard ratios, and confounder adjustment — comes from the biostatistics and epidemiology texts rather than from Park alone.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 21 Health Information and Basic Medical Statistics; Ch 3 Principles of Epidemiology); Sunder Rao & Richard — Introduction to Biostatistics and Research Methods (Ch 13 Linear Regression and Correlation; Ch 24 Confounding and Causality); Mahajan's Methods in Biostatistics for Medical Students and Research Workers (Ch 12 Correlation and Regression; Ch 19 Computer Application in Biostatistical Analysis); Oxford Textbook of Public Health 7e (Statistical Methods; Cohort Studies; Cluster Randomized Trials); Browner — Designing Clinical Research 5e, 2022 (Ch 8 Designing Cross-Sectional and Cohort Studies); Handbook of Epidemiology, Ahrens & Pigeot (Ch I.3 Descriptive Studies; Ch II.2 General Principles of Data Analysis; Ch IV.5 Health Services Research); Principles of Medical Statistics (Ch 19 Evaluating Trends; Ch 29 Analysis of Variance).



Research methodology and ethics



Focus group discussion

Asked in: Paper I 02-Jan-2025 — RGUHS MD Community Medicine

Definition — A **focus group discussion (FGD)** is a form of group interview that deliberately capitalises on the interaction between members of the group in responding to a question (Mahajan & Gupta 4e, Ch 24, p 469). It is a **thematic discussion among a small, homogeneous group of potential programme beneficiaries**, designed to learn directly from the "clients", in the clients' own words, what they think of a product or practice and why they hold it — a technique adapted from applied psychology to obtain "top of the mind" responses (Sunder Lal, Ch 6, p 270). It is a **qualitative method: the group, not the number of individuals, is the unit of analysis**, and it is chosen to find out how a group discusses and makes sense of an issue, never as a device to enrol more participants cheaply (Browner 5e, Ch 14, p 470; Ch 20, p 663).

Essential requirements of an FGD

Element	Specification
Participants	6–12 persons willing to talk about the issue (Park Plus, Ch 19, p 232); Indian PSM texts prescribe 6–8 per group (Mahajan & Gupta 4e, Ch 24, p 469; Sunder Lal, Ch 6, p 270). Krueger and Casey advise 5–10, with 4 as the minimum and 12 the maximum — small enough for everyone to speak, large enough to give a range of perceptions (Oxford Textbook of Public Health 7e, Ch 31, pp 1151–1152)
Composition	As homogeneous as possible for background characteristics — sex, age, social status. Too much disparity kills interaction; if men are present women may not speak, especially on sensitive topics (Mahajan & Gupta 4e, Ch 24, pp 469–470; Park Plus, Ch 19, p 232)
Prior acquaintance	Members should preferably not know each other beforehand (Mahajan & Gupta 4e, Ch 24, p 469; Oxford 7e, Ch 44, p 1636)
Moderator	One researcher conducts but does not lead the discussion, using a predetermined, pretested line of open-ended questioning; sets the group up with instructions and then moves to the side, intervening only if it goes badly off track (Park Plus, Ch 19, p 232; Oxford 7e, Ch 31, p 1152)
Recorder (note-taker)	A second researcher takes detailed notes and draws the sociogram — a pictorial representation of how interactions occurred between participants (Park Plus, Ch 19, p 232)
FGD guide	A predetermined guide of open-ended questions, i.e. questions needing an in-depth response, not a "yes/no" (Park Plus, Ch 19, p 232)
Recording	Audio recording with a backup ; verbatim transcription where detailed content analysis is required (Park Plus, Ch 19, p 232; Oxford 7e, Ch 31, p 1152)
Venue and seating	Comfortable, relaxed, quiet enough for a good recording (Oxford 7e, Ch 31, p 1152); participants seated in a circle so each is fully visible to all others (Park 28e, Ch 22, p 1021)
Duration	Not more than 1–1½ hours (Mahajan & Gupta 4e, Ch 24, p 469); up to 2 hours in Western practice (Oxford 7e, Ch 31, p 1150)

Element	Specification
Number of groups	Decided by the research goal and resources, and often cannot be fixed in advance; several FGDs are run for one problem, stratified where a variable matters (separate groups for men and women, or for providers and users) (Mahajan & Gupta 4e, Ch 24, p 469; Oxford 7e, Ch 31, p 1152)
Sampling	Purposive, non-probability sampling; qualitative research is not concerned with sample size in the statistical sense (Park Plus, Ch 19, p 233; Mahajan & Gupta 4e, Ch 24, p 466)

Conduct of an FGD

The three specific components are **preparation** (recruitment of participants, physical arrangements, preparation of the FGD guide), **conduct of the session**, and **facilitating the group discussion** (Park Plus, Ch 19, p 233), expanded below.

- ▶ **Select the theme** from what field workers observe on home visits, at mother-and-child protection sessions, at the sub-centre or the *anganwadi* — for example, use of oral rehydration solution (ORS) or of home-available fluids to prevent dehydration in diarrhoea (Sunder Lal, Ch 6, p 270).
- ▶ **Prepare the guide** — a short pretested list of open-ended, probing questions in the local dialect.
- ▶ **Recruit** 6–12 homogeneous participants purposively; recruiters bring them together; obtain informed consent.
- ▶ **Arrange the setting** — circular seating, quiet private venue, recorder placed so as not to distract the group.
- ▶ **Open the session** — moderator explains the purpose, sets ground rules (one speaker at a time, no right or wrong answer), builds rapport.
- ▶ **Facilitate** — probe, seek clarification, actively draw out the quiet members and restrain the dominant ones, follow up interesting leads, keep to the theme.
- ▶ **Record** — audio recording plus written notes plus the sociogram; **even non-verbal communication is recorded** (Mahajan & Gupta 4e, Ch 24, p 470).
- ▶ **Transcribe and translate** the recording verbatim.
- ▶ **Analyse** by content and thematic analysis — the **frequency with which an issue is mentioned and the exact expressions used** are analysed, and themes are drawn out (Mahajan & Gupta 4e, Ch 24, p 470; Oxford 7e, Ch 31, p 1152).
- ▶ **Repeat FGDs** until no new theme emerges, and establish rigour by **triangulation** (using more than one qualitative method to see whether they reach common conclusions) and by **independent analysis of the data by more than one investigator** (Abramson, Ch 15, p 145).

Uses in community medicine

Situation	Example
Exploratory study of a health issue	Understanding why men visit commercial sex workers without a condom despite knowing the risk (Mahajan & Gupta 4e, Ch 24, p 466)

Situation	Example
Testing acceptance of a new programme; solving specific programme problems; evaluating health programmes	Standard FGD applications listed for public health practice (Park Plus, Ch 19, p 233)
Community diagnosis / participatory rural appraisal (PRA)	Thematic discussion with beneficiaries on their perception and practices about a disease such as diarrhoea (Sunder Lal, Ch 6, p 269)
Behaviour change communication (BCC) and social marketing	Popularising contraceptives, ORS and immunisation; FGD tells us the "why" behind a problem, while a close-ended survey tells us only "what" the problem is (Sunder Lal, Ch 6, p 270)
Building a quantitative instrument	Securing background information, generating project ideas, learning the community's own vocabulary for an education programme, formulating questions for the formal interview schedule, and interpreting data already collected (Mahajan & Gupta 4e, Ch 29, p 580)
National programme monitoring (India)	Assessment of behavioural and social drivers of immunisation in low-coverage areas and households with missed children — done by questionnaire and <i>supplemented with focus group discussion</i> for specific questions (Routine Immunization Manual for Medical Officers 2024, Ch 10, p 240)
Health needs assessment and health promotion	Widely used, e.g. 14 FGDs to explore views on health inequalities across socio-economic groups (Oxford 7e, Ch 31, p 1151)

FGD compared with the individual in-depth interview

Feature	Focus group discussion	In-depth interview
Unit of analysis	The group	The individual
What it reveals	Reactions and responses of the group to one another and to the facilitator; group norms and the range of views within a subgroup	Individual perspectives, beliefs and experiences
Choose it when	You want to know how a community discusses and makes sense of an issue	You want the perspectives of individuals from different communities
Typical size	6–12 per group, several groups	One respondent per sitting, commonly 20–30 interviews
Chief hazard	Frequent talkers shape the collective view	Interviewer bias; costly and time-consuming

(Browner 5e, Ch 14, pp 469–470; Oxford 7e, Ch 31, p 1150)

Merits

- **Group interaction is the yield** — an answer by one member prompts another to respond more fully and more emotionally; this is the strength of a focus group and cannot be captured in a one-to-one interview (Mahajan & Gupta 4e, Ch 24, pp 469–470).

- ▶ Focus groups are **not merely a way of pooling individual knowledge** — they give rise synergistically to insights and solutions that would not come about without them (Oxford 7e, Ch 31, p 1151).
- ▶ Explores **not only what people think, but how they think and why they think that way** (Mahajan & Gupta 4e, Ch 24, pp 469–470).
- ▶ Yields a **large amount of information in a relatively short time** and is inexpensive for the quantity and type of data obtained (Park Plus, Ch 19, p 232; Health Program Planning and Evaluation, Ch 15, p 391).
- ▶ Participants can **control the agenda and formulate opinions** in a way not possible in an individual interview, and topics can be explored in an open format at the start of research (Oxford 7e, Ch 31, p 1151).
- ▶ Gives the **collective view and the social norms** of a community or subgroup, along with the range of perspectives within it (Park Plus, Ch 19, p 232).
- ▶ **Built-in validation** — members confirm or disconfirm each other's experiences, so the "instant feedback" makes the method more efficient than in-depth interviews, which need a separate interview to validate interpretations; a group setting is also less intimidating for some people (Health Program Planning and Evaluation, Ch 15, p 392).
- ▶ Produces the **clients' own words and vocabulary**, which is exactly what is needed to frame questionnaire items and IEC messages (Sunder Lal, Ch 6, p 270; Mahajan & Gupta 4e, Ch 29, p 580).

Demerits

Limitation	Explanation
Unequal participation	The shy may not take part at all and one or two members may dominate, so participation is unequal unless properly guided; some deviate from the subject and make the discussion unprofitable (Park 28e, Ch 22, p 1022)
Dominant individuals	Frequent talkers shape the collective view; skilful moderation is vital to minimise this (Oxford 7e, Ch 31, p 1151; Browner 5e, Ch 14, p 470)
"Group think"	Members are discouraged from voicing opinions at odds with the majority; participants are influenced by the social pressure of small-group interaction (Oxford 7e, Ch 31, p 1151; Ch 44, p 1636)
Confidentiality cannot be guaranteed	The conversation takes place in public, in front of strangers, yet confidentiality is an essential component of informed consent in research on human subjects (Oxford 7e, Ch 44, p 1636)
Low validity and reliability	The data are of low or uncertain validity and reliability; the method should be used with caution and largely for exploratory work (Oxford 7e, Ch 44, p 1636)
Wrong tool for individual experience	Focus groups reveal group interactions, not individual perspectives, and are not a strategy for increasing the number of research participants (Browner 5e, Ch 14, p 470; Ch 20, p 663)
Heterogeneity silences people	Mixed or disparate groups interact poorly; women may not speak on sensitive topics if men are present (Mahajan & Gupta 4e, Ch 24, pp 469–470)

Limitation	Explanation
Analysis is cumbersome	A complete record of what each person said, including non-verbal communication, has to be kept, transcribed and content-analysed; participants talk over each other, voices are hard to identify on the recording and the note-taker inevitably misses comments (Mahajan & Gupta 4e, Ch 24, p 470; Health Program Planning and Evaluation, Ch 15, pp 392–393)
Less researcher control	Researchers have less control over the data generated, and recording itself is a challenge and can be intrusive (Oxford 7e, Ch 31, p 1151; Health Program Planning and Evaluation, Ch 15, p 392)
Skill-dependent	Data quality depends wholly on how the group is run — facilitation skill, knowledge of group dynamics and command of the subject are essential; training is needed in managing the group process (Oxford 7e, Ch 31, p 1152; Health Program Planning and Evaluation, Ch 15, p 391)
Logistically heavy	Recruiting and scheduling 8–12 people, travel, transport, meals and compensation, then recording, transcribing, translating and analysing a large volume of text (Browner 5e, Ch 14, p 463; Health Program Planning and Evaluation, Ch 15, p 392)
Not for outcome evaluation	Better suited to community assessment and programme planning; it may be inappropriate to ask individuals to reveal in a group what difference a programme made to them (Health Program Planning and Evaluation, Ch 15, p 392)
No quantification	Purposive, non-probability sampling means the findings describe perceptions and explain behaviour; they do not estimate prevalence or permit generalisation (Park Plus, Ch 19, p 233; Mahajan & Gupta 4e, Ch 24, p 466)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Sociogram of a focus group discussion — draw two circles, each with 8–10 participant seats around the rim and the moderator at the top; inside each, arrows for who spoke to whom.

- ▶ Left circle — *a poor discussion*: nearly all arrows converge on one or two participants and on the moderator; two seats have no arrow at all (dominated group, silent members).
- ▶ Right circle — *a good discussion*: arrows criss-cross between all seats, few arrows to the moderator (participant-to-participant interaction).
- ▶ Label: moderator, participants 1–8, direction of interaction, "dominant participant", "silent participant".

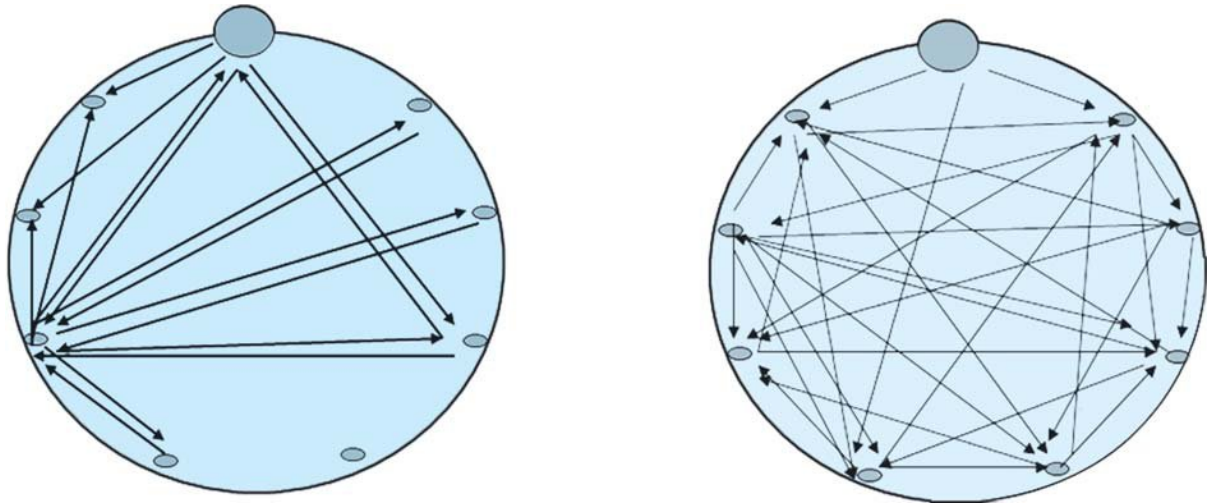
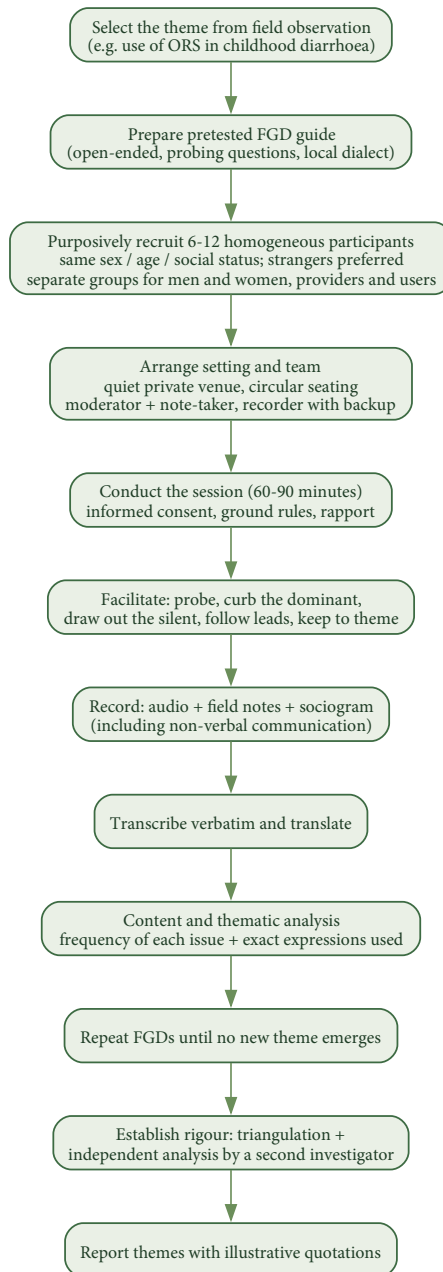
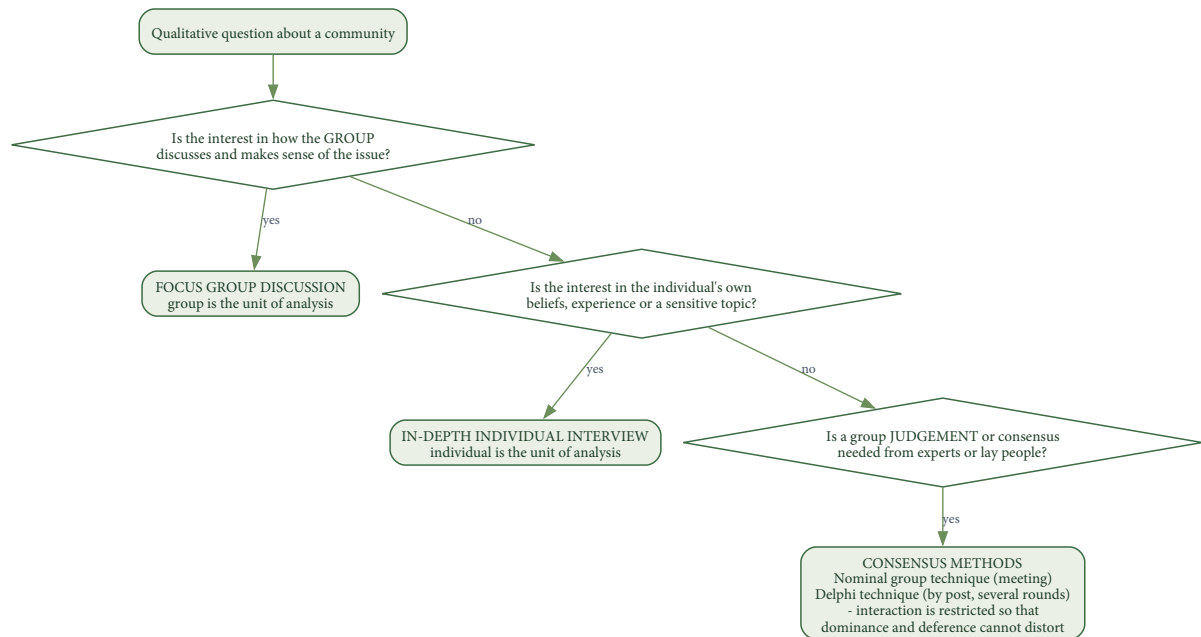


Fig 29.2 — Sociogram showing examples of Not a good discussion and Good discussion (Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e, p.581)



Steps in conducting a focus group discussion



Choosing between a focus group discussion and other group or individual methods

High-yield exam pointers

- ▶ **Numbers to write:** 6–12 participants (6–8 in Indian texts), 1–1½ hours, several FGDs per problem until no new theme emerges; purposive non-probability sampling.
- ▶ **The team:** moderator (guides, does not lead) + note-taker/recorder who draws the **sociogram**; pretested FGD **guide**; audio recorder **with backup**.
- ▶ **Three composition rules:** homogeneous, strangers to each other, separate groups for men and women on sensitive topics.
- ▶ **One-line strength:** interaction gives insights that would not arise otherwise — it shows *what, how and why* people think; survey tells "what", FGD tells "why".
- ▶ **One-line weakness:** the group is the unit of analysis — confidentiality cannot be assured, group think and dominant talkers distort, and the findings cannot be quantified or generalised.
- ▶ **Analysis:** verbatim transcription → content and thematic analysis of frequency and exact expressions → triangulation and independent second analyst for rigour.
- ▶ **Indian application to quote:** FGD supplements the questionnaire in assessing behavioural and social drivers of immunisation in low-coverage areas (Routine Immunization Manual for Medical Officers 2024).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park's Textbook of Preventive and Social Medicine 28e carries no separate section on focus group discussion — its Chapter 22 describes **group discussion** (6–12 members, circular seating, a group leader and a recorder) and it is from there that the limitations of unequal participation and dominance are quoted here; the FGD-specific detail has been taken from Park Plus, Mahajan & Gupta 4e, Sunder Lal and the Oxford Textbook of Public Health 7e.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 22, Communication for Health Education); Park Plus (Ch 19, Health information and basic medical statistics); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e (Ch 24, Research Methodology; Ch 29, Information, Education, Communication and Training in Health); Sunder Lal — Textbook of Community Medicine (Ch 6, Basic Statistics and its Applications and Research Methodology); Oxford Textbook of Public Health 7e (Ch 31, Qualitative research imagination; Ch 44, Sociology and psychology in public health); Browner — Designing Clinical Research 5e, 2022 (Ch 14, Qualitative Approaches in Clinical Research); Abramson — Research Methods in Community Medicine (Ch 15, Methods of collecting data; Ch 20, Consensus methods); Health Program Planning and Evaluation (Ch 15, Qualitative Methods for Planning and Evaluation); Routine Immunization Manual for Medical Officers, Government of India, 2024 (Ch 10, Supervision and monitoring).



Evidence-based health care

Asked in: Paper IV 21-Nov-2022 — RGUHS MD Community Medicine

Definition — Evidence-based health care is the practice of basing decisions about health care on the best available, current, valid and relevant evidence, the decisions being made by those receiving care, informed by the tacit and explicit knowledge of those providing care, within the context of available resources (*Sicily Statement on Evidence-Based Practice*, 2003; Lilienfeld's Foundations of Epidemiology, Ch 15). Applied to the individual patient it is called **evidence-based medicine (EBM)**; applied to populations it is **evidence-based public health (EBPH)**. It rejects intuition, unsystematic clinical experience and pathophysiological reasoning as *sufficient* grounds for decisions, while still valuing expertise (Handbook of Epidemiology, Ch 29).

1. The four components of an evidence-based decision

Component	What it contributes	Consequence if ignored
Research evidence	Quantified estimates of benefit, harm, cost from valid studies	Practice drifts to eminence-, vehemence- and eloquence-based medicine (Fletcher, Ch 1, p 22)
Clinical state and circumstances	Comorbidity, severity, feasibility, resources available locally	Evidence generated elsewhere applied to a population it does not fit
Patient / community preferences and actions	Values placed on the trade-off between benefit, harm, inconvenience and cost	Technically correct but unacceptable or non-adhered-to care
Clinical (or public-health) expertise	The integrating skill that weighs the other three	Guidelines used as rules rather than as a starting point (Fletcher, Ch 14, p 171)

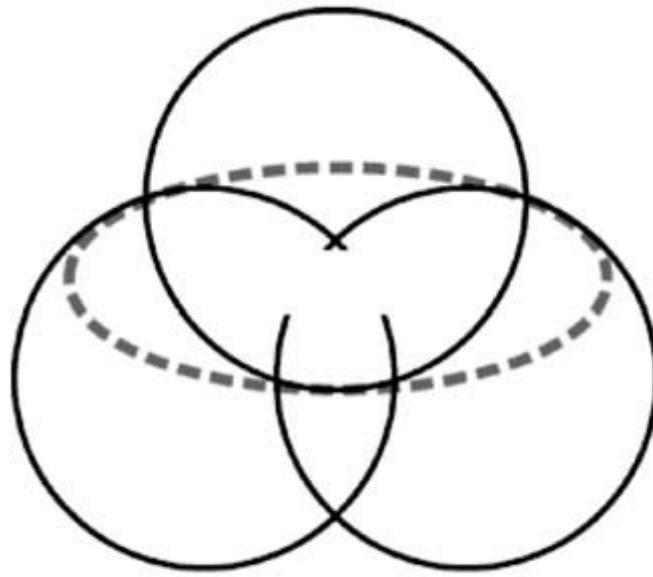


Fig 8.4 — Model for Evidence-Based Decision Making. Reproduced with permission f (Handbook of Epidemiology (Ahrens & Pigeot), p.1199)

2. Steps in practising evidence-based health care (the five A's)

- ▶ **Ask** — convert the problem into a specific, answerable question, structured as **PICO**: Population, Intervention, Comparator, Outcome (extended to PICOTS with Timeframe and Setting) (CONSORT 2025, item on objectives).
- ▶ **Acquire** — search the best available research evidence: the *Cochrane Library* for systematic reviews, MEDLINE through PubMed, EMBASE, and pre-appraised point-of-care resources (Fletcher, Ch 14, pp 229–230).
- ▶ **Appraise** — judge the retrieved evidence for **validity** (control of bias and chance), **importance** (size and precision of effect) and **applicability** to the patient or population (Fletcher, Ch 1, p 22).
- ▶ **Apply** — integrate the appraisal with expertise, with the clinical or epidemiological circumstances, and with the values of the patient or community.
- ▶ **Assess** — audit one's own performance and outcomes, and repeat the cycle as new evidence appears.

For information to be usable at the **point of care** it must be rapid (available within minutes), current (hence electronic), tailored to the specific question, sorted by scientific strength, and available where the clinician actually works (Fletcher, Ch 14, Table 14.2).

3. Hierarchy of evidence

Rank	Study design	Basis for the rank
1	Meta-analysis (quantitative systematic review)	Summarises multiple studies into one estimate — the "platinum standard"
2	Experimental designs — randomized controlled trial, large simple trial, randomized cluster trial	Randomisation controls both known and unknown confounders; the RCT is the "gold standard" and the No. 1 method for evaluating a new programme or therapy (Park 28e, Ch 3, p 101)
3	Observational analytical designs — cohort, case-control	Have a comparison group, so association can be assessed; confounding less well controlled
4	Descriptive studies — prevalence, cross-sectional, ecological, case series, case reports	Case series and case reports have no comparison group; useful only for generating hypotheses

Source: Lilienfeld's *Foundations of Epidemiology*, Ch 15, Table 15.1.

4. Grading the body of evidence — the GRADE approach

The Grades of Recommendation, Assessment, Development and Evaluation (GRADE) system, adopted by the Cochrane Collaboration and by over 20 bodies including the World Health Organization and the National Institute for Health and Care Excellence, rates the **quality (certainty) of evidence separately for each outcome**, and then rates the **strength of the recommendation** separately again (Cochrane Handbook, Ch 12, pp 385–392).

Quality level	Underlying methodology
High	Randomized trials; or double-upgraded observational studies
Moderate	Downgraded randomized trials; or upgraded observational studies
Low	Double-downgraded randomized trials; or observational studies
Very low	Triple-downgraded randomized trials; downgraded observational studies; case series / case reports

Five factors that downgrade quality	Three factors that upgrade quality
1. Limitations in design and implementation (risk of bias)	1. Large magnitude of effect
2. Indirectness of evidence (population, intervention, control, outcome)	2. All plausible confounding would reduce the demonstrated effect, or suggest a spurious effect where none is shown
3. Unexplained heterogeneity or inconsistency of results	3. Dose–response gradient
4. Imprecision (wide confidence intervals)	
5. High probability of publication bias	

Recommendations are then labelled **strong** (1) or **weak** (2) and paired with the evidence grade (A high, B moderate, C low) — so **1A** is a strong recommendation on high-quality evidence that applies to most patients without reservation, while **2C** is a very weak recommendation where other options are equally reasonable (Fletcher, Ch 14, Table 14.1). Strength depends not only on the evidence but on the balance of benefits against harms.

5. Evidence synthesis — systematic review, meta-analysis and guidelines

Key characteristics of a systematic review (Cochrane Handbook, Ch 1, p 31):

- ▶ A clearly stated set of objectives with **pre-defined eligibility criteria** for studies.
- ▶ An **explicit, reproducible methodology**.
- ▶ A **systematic search** that attempts to identify all studies meeting the eligibility criteria.
- ▶ An **assessment of validity** of the included studies (risk of bias).
- ▶ A **systematic presentation and synthesis** of characteristics and findings.

Purposes of a meta-analysis — to resolve conflicting conclusions; to increase statistical power for major endpoints and subgroups; to sharpen the boundaries of effect size; to answer new questions; and, for trial data, to generate the **number needed to treat (NNT)** (Chalmers, cited in Lilienfeld, Ch 15). Individual study results with their confidence intervals are displayed against the null value in a **forest plot**; pooling uses a **fixed-effects model** (assumes one true effect, weights larger studies) or a **random-effects model** (assumes a distribution of effects, so no study dominates).

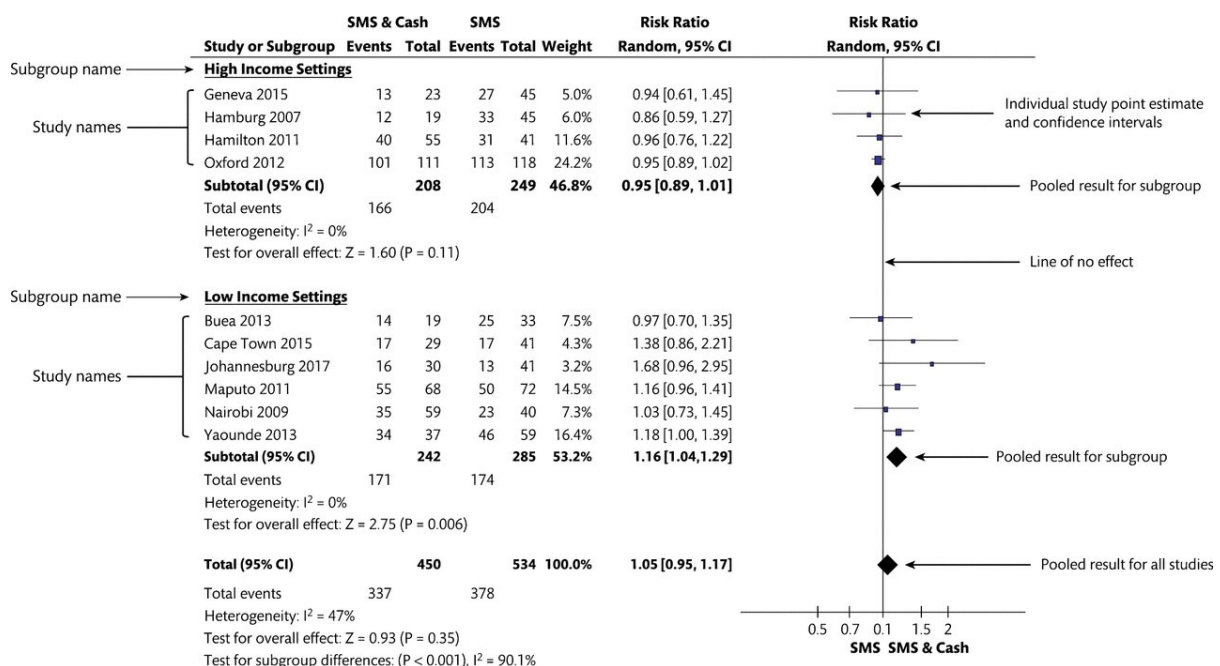


Fig 5.13 — 1 Forest plot of invented data, from a meta-analysis of ten studies in (Oxford public health 7th ed, p.1383)

Element	Detail
Reporting standard for reviews of trials	QUOROM (1996), updated in 2009 as PRISMA — 27 items

Element	Detail
Reporting standard for reviews of observational studies	MOOSE (2000); additionally requires each study to be rated for quality
Methodological standard	<i>Cochrane Handbook for Systematic Reviews of Interventions</i>
Synthesising bodies	Cochrane Collaboration (named for Archie Cochrane; >50 groups; health-care interventions) and the Campbell Collaboration (named for Donald Campbell; social interventions)
Guideline bodies	US Preventive Services Task Force, National Institute for Health and Care Excellence

Clinical practice guidelines are advice about the care of patients with a specific condition; a trustworthy guideline shows transparency of development and funding, declared conflicts of interest, a balanced multidisciplinary group, a basis in high-quality systematic review, an explicit evidence level and recommendation strength for each statement, precise wording, external review, and a stated updating plan (Fletcher, Ch 14, Table 14.3, after the Institute of Medicine). Guidelines are guidelines, not rules.

6. Evidence-based public health — the community medicine application

EBPH differs from EBM in that it addresses **populations**, and therefore draws on several kinds of evidence, not epidemiological studies alone. Its characteristics are (Lilienfeld's Foundations of Epidemiology, Ch 15):

- ▶ Making decisions using the **best available peer-reviewed evidence**, both quantitative and qualitative.
- ▶ Using **data and information systems** systematically.
- ▶ Applying **programme-planning frameworks**, often grounded in behavioural science theory.
- ▶ **Engaging the community** in assessment and in decision making.
- ▶ Conducting **sound evaluation**.
- ▶ **Disseminating** what is learned to key stakeholders and decision makers.

Because populations are complex, the public-health practitioner must know not merely *whether* an intervention works but *under which circumstances* it works — results of randomized controlled trials cannot simply be generalised to whole populations, so multiple methodologies are needed for global problems such as preventable maternal and child mortality.

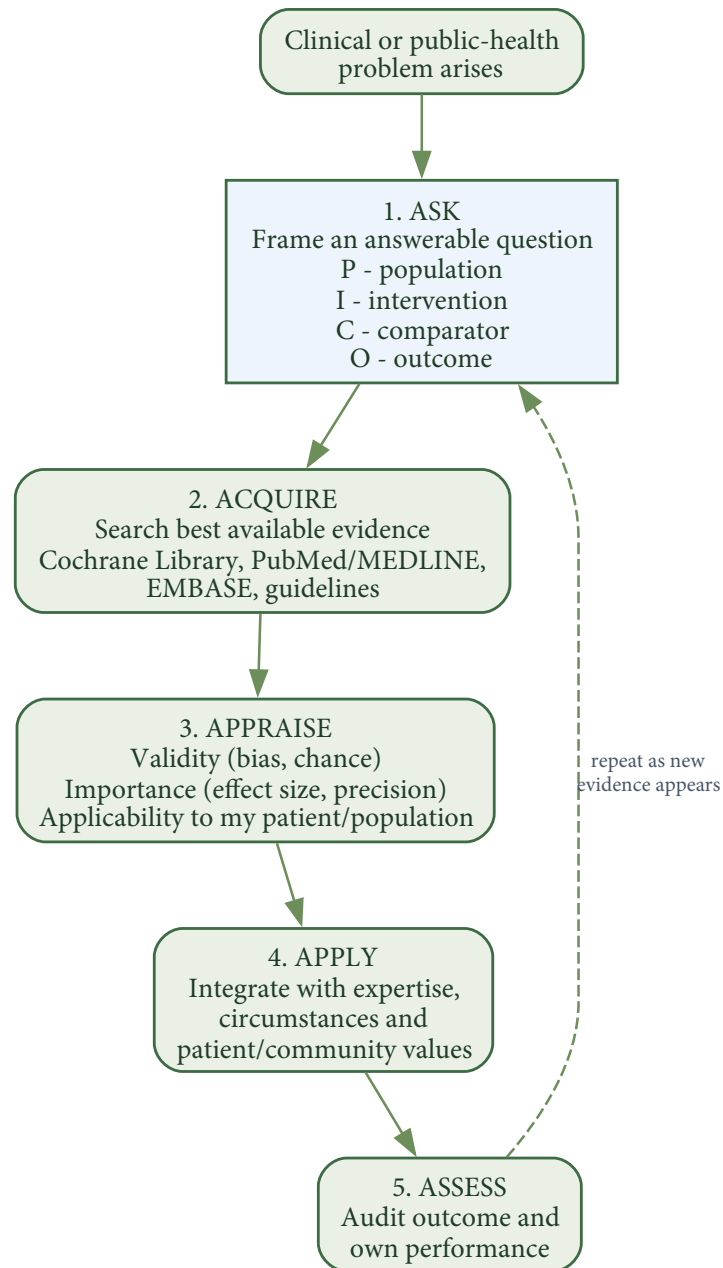
7. Limitations and barriers

Barrier	Explanation
Publication bias	Positive results are preferentially submitted and published, so the retrievable literature is unrepresentative; must be considered when framing eligibility criteria and reported per PRISMA
Heterogeneity / failed homogeneity	Pooling is valid only if studies are sufficiently similar in subject selection, exposure and outcome measurement and follow-up

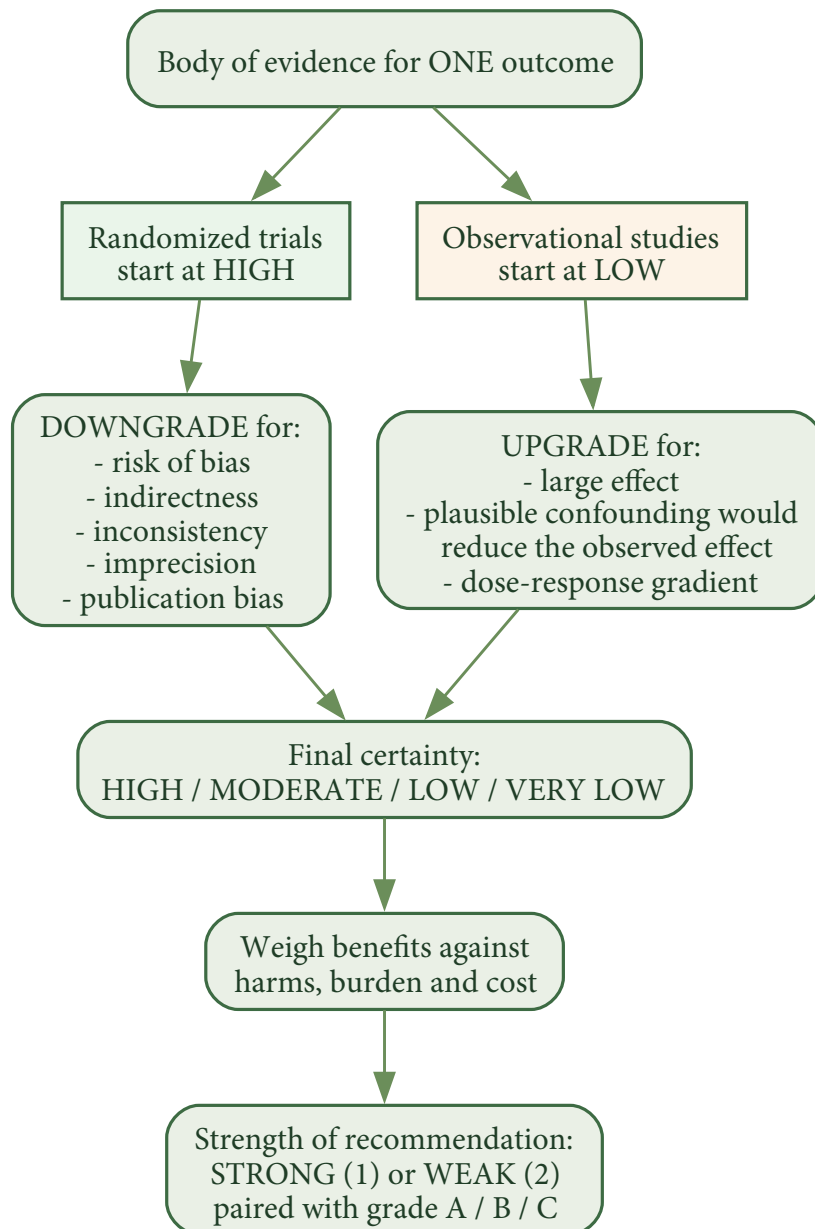
Barrier	Explanation
Conflict of interest and scientific misconduct	Financial interest, personal relationships, intellectual passion, institutional loyalty, career advancement; expressed as fraud, selective reporting, or "spin" (Fletcher, Ch 14, pp 226–228)
Search inefficiency	PubMed searches miss relevant articles and return many irrelevant ones — in one study nephrologists retrieved only 46% of relevant articles, with 1 relevant per 16 retrieved
Evidence gaps for populations	Little evidence on the <i>effectiveness of interventions</i> to reduce inequalities, as against evidence on their causes (Oxford Textbook of Public Health 7e, Ch 9.4)
Political and commercial pressure	Adoption of evidence-based policy is weakened by economic manipulation, delay, harassment and rhetoric; leaders may reject evidence for ideological, electoral or partisan reasons (Oxford Textbook of Public Health 7e, Ch 7.2, pp 319–321)
Time and skill at the point of care	Clinicians cannot appraise every question themselves and must rely on trustworthy pre-appraised sources

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Hierarchy of evidence pyramid** — draw an inverted-triangle pyramid, apex at the top, and label from apex downwards: *Meta-analysis / systematic review; Randomized controlled trial, large simple trial, randomized cluster trial; Cohort study; Case-control study; Cross-sectional / prevalence / ecological study; Case series and case reports*; base label *Expert opinion, unsystematic clinical experience*. Mark the right-hand margin with two arrows: "increasing freedom from bias / increasing confidence" pointing up, and "increasing volume of available literature" pointing down.



The five-step cycle of evidence-based health care practice



Rating certainty of evidence and strength of recommendation using GRADE

High-yield exam pointers

- ▶ **Definition to write verbatim** — Sicily Statement 2003: "decisions about health care should be based on the best available, current, valid and relevant evidence."
- ▶ **Five A's:** Ask (PICO) → Acquire → Appraise → Apply → Assess.
- ▶ **Hierarchy:** meta-analysis > RCT / large simple trial / cluster RCT > cohort > case-control > cross-sectional and ecological > case series and case reports.
- ▶ **GRADE = 4 levels** (high, moderate, low, very low); **5 downgrading factors** (bias, indirectness, inconsistency, imprecision, publication bias) and **3 upgrading factors** (large effect, plausible confounding reducing effect, dose-response).
- ▶ **Five key characteristics of a systematic review:** pre-defined objectives and eligibility criteria; explicit reproducible methodology; systematic search; validity assessment; systematic synthesis.

- **Reporting standards:** PRISMA (27 items, 2009, from QUOROM 1996) for trials; MOOSE (2000) for observational studies.
- **Named organisations:** Cochrane Collaboration (Archie Cochrane, health-care interventions), Campbell Collaboration (Donald Campbell, social interventions), USPSTF, NICE.
- **Two threats to any meta-analysis:** publication bias and failure of the homogeneity assumption.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park's Textbook of Preventive and Social Medicine 28e carries no dedicated chapter on evidence-based health care; the answer above is therefore grounded chiefly in the epidemiology and research-methods texts of the corpus, with Park 28e supplying only the standing of the randomized controlled trial as the primary method of evaluation. The corpus also contains no Indian institutional material on this topic, so India's Health Technology Assessment body and any Indian evidence-to-policy mechanism have deliberately not been described.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO, HTAIn (Department of Health Research), CONSORT 2025 statement, GRADE Working Group

Programme and policy updates (India)

- **India now has an institutional home for evidence-to-policy work — Health Technology Assessment in India (HTAIn)**, committed to by the National Health Policy 2017 and housed in the Department of Health Research, Ministry of Health and Family Welfare. It runs on a hub-and-spoke model — a Secretariat, a Board, a multidisciplinary Technical Appraisal Committee, and Regional Resource Centres that grew from 1 in 2017 to 30 by 2024 (about 72 scientific and technical staff), with six centres upgraded to HTA resource hubs covering the north, west and central, south, union territory and north-east zones. Method is governed by an Indian Reference Case for economic evaluation plus an HTA Quality Appraisal Checklist, and the One Nation One Subscription initiative — nationwide access to over 13,000 e-journals from 30 publishers, with article-processing charges funded — is aimed squarely at the literature-access barrier that cripples appraisal in Indian institutions. [PMID 40913689]
- **HTA is itself cost-effective in India:** an evaluation of three assessments commissioned by HTAIn between 2018 and 2020, using the University of Glasgow return-on-investment framework and netting out the counterfactual, found a return of 9:1, rising to a potential 71:1 if recommendations were fully implemented, and varying from 5:1 to 40:1 between interventions — the variation itself an argument for careful topic selection and for planning implementation from the outset. [PMID 40113231]
- **Ayushman Bharat PM-JAY has acquired its own appraisal arm — the Health Financing and Technology Assessment (HeFTA) unit within the National Health Authority — which applies HTA evidence to inclusions and exclusions in the health benefit package, to package rates and to the framing of standard treatment guidelines, with reported savings to the scheme; the recommended next step is value-based pricing integrated with strategic purchasing.** [PMID 38715207]

Current national data

- **What HTAIn had actually evaluated up to October 2023:** 39 studies published, of which 33 were analysable — 24.2% addressed maternal and neonatal disorders, 33.3% of interventions were devices, 45.5% were for screening, and none addressed rehabilitation or palliation; 84.8% of the interventions came out cost-effective at a $1 \times$ GDP-per-capita threshold. Rated against the Indian Reference Case, equity (25.7% adherence) and heterogeneity or subgroup analysis (24.3%) were the weakest domains, and only 18.2% of reports scored above 8 out of 10 — the Indian evidence base for public-health decisions is growing fast but is still thin exactly where inequality would show up. A separate mixed-methods appraisal of 26 reports from ten Regional Resource Centres (2018–2023) graded 14% excellent, 50% good, 32% average and 4% poor. [PMID 41316276] [PMID 40913689]

Elimination, surveillance and disease status

- **The National TB Elimination Programme adopted the 6-month all-oral BPaLM regimen (bedaquiline, pretomanid, linezolid, moxifloxacin) in September 2024,** following the WHO recommendation of December 2022 — a clean worked example of global trial evidence entering an Indian national programme against the 2025 elimination target. Locally generated evidence immediately qualified it: in a northern-India tertiary cohort, fluoroquinolone resistance reached 78.7% (74/94), so only 24.4% (23/94) of drug-resistant cases were eligible for the regimen. Evidence valid elsewhere still has to be tested against local drug-susceptibility data before it is generalised to a population. [PMID 40225357]
- **The economic evidence points the same way:** a Markov model across four high-burden countries put empirical BPaLM, replacing BPaL where fluoroquinolone susceptibility testing is unavailable, at \$32 per DALY averted in India (IQR \$23–53) against an Indian cost-effectiveness threshold of \$422 per DALY, and the conclusion held across a fluoroquinolone-resistance prevalence of 0–70%. Global drug-resistance surveillance data are thereby feeding directly into a national treatment-policy choice. [PMID 41100518]

Guideline and legislative updates

- **CONSORT 2025 (April 2025) supersedes CONSORT 2010,** which its authors state should no longer be used: a 30-item checklist with 7 new items, 3 revised and 1 deleted, produced through a three-round international Delphi survey of 317 participants and a two-day consensus meeting of 30 experts. The checklist is restructured around a new **open science** section (trial registration; where the protocol and statistical analysis plan can be accessed; sharing of de-identified participant-level data; funding and conflicts of interest), and adds items on patient and public involvement, how harms were defined and assessed, how missing data were handled, and how the intervention and comparator were actually delivered. SPIRIT 2025, the matching protocol standard, was published alongside it — so the PICO-framed question, the protocol and the final report are now harmonised end to end. [PMID 40228833]
- **GRADE is being extended to planetary health:** the GRADE Working Group is developing guidance to bring environmental impact into guideline development as an explicit criterion within the Evidence-to-Decision framework, using sources such as life-cycle assessment of medical products where direct quantitative evidence is absent; the ARIA 2024–2025 allergic rhinitis guideline is the worked example. This is the first structural addition to the GRADE

Evidence-to-Decision architecture that a community-medicine candidate is likely to be asked about. [PMID 40505857]

Global and WHO developments

- ▶ **Living evidence syntheses have moved from pandemic improvisation to standing method:** living systematic reviews, living guidelines, living evidence maps and living policy briefs are continuously surveilled rather than updated on a fixed cycle — COVID-19 treatment guidelines were refreshed weekly — and are now produced by WHO and Cochrane as global public goods, reaching governments through intermediaries such as the WHO Evidence-informed Policy Network, the McMaster Health Forum and PAHO. Unsettled questions are when a living approach is warranted, how policy built on shifting evidence is implemented and defended, and how resource-constrained countries get equitable access to the syntheses. [PMID 39318895]
- ▶ **Who writes global guidance is itself an equity question:** of the 33 countries designated high-burden for multidrug- or rifampicin-resistant TB, 72.1% are low- or lower-middle-income, yet only 30.3% of members of the WHO Guideline Development Groups for 2016–2022 and 10.4% of authors of the individual-patient-data meta-analyses feeding those guidelines were affiliated to institutions in such countries. Because multilateral funding (World Bank, Global Fund) is often conditioned on compliance with WHO guidance, a recommendation poorly matched to local epidemiology forces a country to choose between fitting guidance and external money — a concrete illustration of the political and structural barriers to evidence-based policy. [PMID 39616361]

SOURCE & COVERAGE

Lilienfeld's Foundations of Epidemiology (Schneider & Lilienfeld), Ch 15 — Evidence-Based Practices; Fletcher — Clinical Epidemiology: The Essentials, Ch 1 (Introduction) and Ch 14 (Knowledge Management); Cochrane Handbook for Systematic Reviews of Interventions, Ch 1 (Introduction), Ch 12 (Interpreting results and drawing conclusions) and Ch 13 (Including non-randomized studies); Handbook of Epidemiology (Ahrens & Pigeot), Ch 29 — Clinical Epidemiology; Park's Textbook of Preventive and Social Medicine 28e, Ch 3 — Principles of Epidemiology and Epidemiologic Methods; Oxford Textbook of Public Health 7e, Ch 7.2 (Politics of public health) and Ch 9.4; Jekel's Epidemiology, Biostatistics and Preventive Medicine, Ch 8 and Ch 18 (US Preventive Services Task Force); CONSORT 2025 Statement (PICO framing of the research question). Named documents: Sicily Statement on Evidence-Based Practice (2003), PRISMA (2009), MOOSE (2000), GRADE Working Group.

Recent-advances sources: 40913689 — A Mixed-Methods Assessment of India's Health Technology Assessment Ecosystem, Appl Health Econ Health Policy 2025, doi:[10.1007/s40258-025-00995-4](https://doi.org/10.1007/s40258-025-00995-4); 40113231 — Is health technology assessment value for money? Estimating the return on investment of health technology assessment in India (HTAIn), BMJ Evid Based Med 2025, doi:[10.1136/bmjebm-2023-112487](https://doi.org/10.1136/bmjebm-2023-112487) (abstract-only); 38715207 — Establishing a Health Technology Assessment Evidence Ecosystem in India's Pradhan Mantri Jan Arogya Yojana, Health Syst Reform 2024, doi:[10.1080/23288604.2024.2327097](https://doi.org/10.1080/23288604.2024.2327097) (abstract-only); 41316276 — Insights and implications: an analysis of the studies undertaken by Health Technology Assessment in India (HTAIn), Health Res Policy Syst 2025, doi:[10.1186/s12961-025-01422-9](https://doi.org/10.1186/s12961-025-01422-9); 40225357 — Detection of extensive drug resistance by the Xpert MTB/XDR assay in multidrug resistant tuberculosis cases at a tertiary care centre in northern India, and therapeutic decision making for the six-month BPALM regimen, J Clin Tuberc Other Mycobact Dis 2025, doi:[10.1016/j.jctube.2025.100520](https://doi.org/10.1016/j.jctube.2025.100520); 41100518 — Impact of fluoroquinolone resistance on the cost-effectiveness of empiric treatment for multidrug- or rifampicin-resistant tuberculosis, PLOS Glob Public Health 2025, doi:[10.1371/journal.pgph.0005275](https://doi.org/10.1371/journal.pgph.0005275); 40228833 — CONSORT 2025 statement: updated guideline for reporting randomised trials, BMJ 2025, doi:[10.1136/bmj-2024-081123](https://doi.org/10.1136/bmj-2024-081123); 40505857 — A Novel Approach to Consider Planetary Health in Guideline Development: A GRADE Approach Using the ARIA 2024-2025 Guidelines as a Case Study, J Allergy Clin Immunol Pract 2025, doi:[10.1016/j.jaip.2025.04.060](https://doi.org/10.1016/j.jaip.2025.04.060) (abstract-only); 39318895 — Implications of living evidence syntheses in health policy, Bull World Health Organ 2024, doi:[10.2471/BLT.23.290540](https://doi.org/10.2471/BLT.23.290540); 39616361 — Revealing disparities in representation in knowledge generation and guideline development, BMC Health Serv Res 2024, doi:[10.1186/s12913-024-11958-1](https://doi.org/10.1186/s12913-024-11958-1). Non-journal sources: WHO recommendation on the 6-month BPALM regimen (December 2022); Government of India / National TB Elimination Programme adoption of BPALM (September 2024); CONSORT 2025 and SPIRIT 2025 statements (April 2025); GRADE Working Group planetary-health guidance (in development, 2025).

♦ ♦ ♦

Indian Council of Medical Research

Asked in: Paper IV Oct 2018 — RGUHS MD Community Medicine

Definition — The Indian Council of Medical Research (ICMR) is the apex national body in India for the formulation, coordination and promotion of biomedical and health research, founded in 1911 with headquarters in New Delhi (Park 28e, Ch 23, p 1034; ICMR Ethics 2017, p 10). It is one of the oldest medical research bodies in the world, functions under the Department of Health Research, Ministry of Health and Family Welfare, and is funded wholly from the Union Health Ministry budget (ICMR Ethics 2017, p 11; Park 28e, Ch 23, p 1034). Beyond conducting research, it sets the country's ethical standards for research on human participants and runs the national trial registry.

Identity, evolution and mandate

Attribute	Detail
Year founded	1911, New Delhi (Park 28e, Ch 23, p 1034)
Present headquarters	V. Ramalingaswami Bhawan, Ansari Nagar, New Delhi 110 029 (ICMR Ethics 2017, p 11)
Parent administrative arm	Department of Health Research, Ministry of Health and Family Welfare (ICMR Ethics 2017, p 11)
Executive head	Director-General, ICMR (ICMR Ethics 2017, pp 1-3)
Source of funds	Wholly from the budget of the Union Ministry of Health (Park 28e, Ch 23, p 1034)
Core role	Aiding, promoting and coordinating scientific research on human diseases — their causation, prevention and cure (Park 28e, Ch 23, p 1034)

Evolution of its ethical guidance — the strand of ICMR's work most often examined (ICMR Ethics 2017, pp 12-14):

Year	Document	Chaired by
1980	Policy Statement on Ethical Considerations Involved in Research on Human Subjects	Justice <i>H.R. Khanna</i>
2000	Ethical Guidelines for Biomedical Research on Human Subjects	Justice <i>M.N. Venkatachaliah</i>
2006	Ethical Guidelines for Biomedical Research on Human Participants	—
2017	National Ethical Guidelines for Biomedical and Health Research Involving Human Participants (current; released October 2017)	Advisory Group chaired by Dr <i>Vasantha Muthuswamy</i>

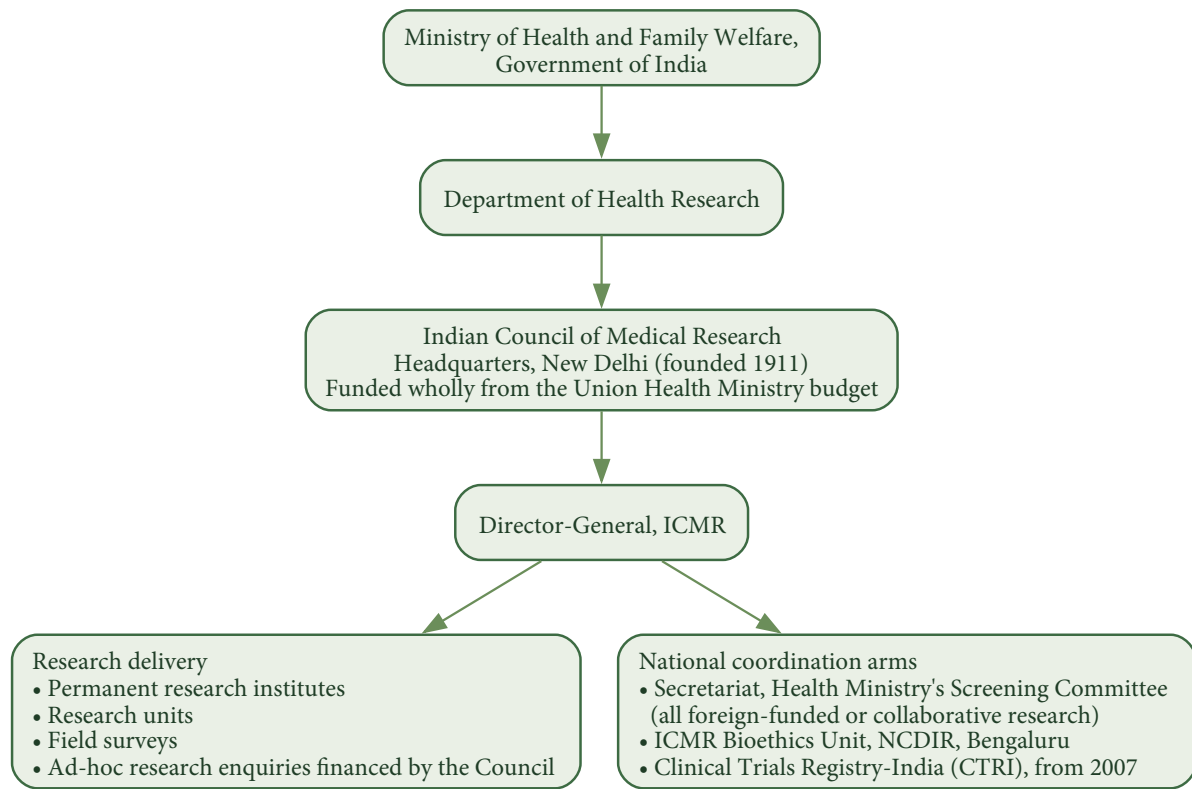
Functions

- **Research generation** — conducted through the Council's permanent research institutes, research units, field surveys, and a large number of ad-hoc research enquiries financed by the Council (Park 28e, Ch 23, p 1034).

- ▶ **Coordination of national medical research** — medical research in the country is organised largely through the Council (Park 28e, Ch 23, p 1034).
- ▶ **Setting national research ethics standards** — the 2017 guidelines apply to all biomedical, social and behavioural science research for health involving human participants, their biological material and data (ICMR Ethics 2017, p 18).
- ▶ **Regulating international collaboration** — all biomedical and health research proposals involving foreign assistance and/or collaboration must be cleared by the Health Ministry's Screening Committee (HMSC) before initiation; the HMSC secretariat sits at ICMR Headquarters, New Delhi (ICMR Ethics 2017, pp 38-39).
- ▶ **Running the Clinical Trials Registry–India (CTRI)** — see below.
- ▶ **Setting technical norms for practice** — for example Recommended Dietary Intakes for Indians (ICMR, 1990) and Dietary Guidelines for Indians (National Institute of Nutrition, Hyderabad, 2011) (Park 28e, Ch 11, p 810), and Guidelines for Good Clinical Laboratory Practices (ICMR Ethics 2017, p 160).
- ▶ **National surveys and task-force studies** — the National Nutrition Monitoring Bureau surveys (1997–2012) and the ICMR task-force study on micronutrient deficiency disorders in 16 districts of India, 2001 (Sunder Lal, Ch 5, p 210).
- ▶ **Health-policy contribution** — the joint ICSSR–ICMR report *Health for All: An Alternative Strategy*, 1981 (Sunder Lal, Ch 6, pp 271-272).

Institutes and centres

Institute / centre	Location
National Institute of Nutrition	Hyderabad (Park 28e, Ch 23, p 1034)
Virus Research Centre (established with Rockefeller Foundation aid)	Pune (Park 28e, Ch 23, p 1034; Sunder Lal, Ch 18, p 777)
Tuberculosis Chemotherapy Centre — now the National Institute for Research in Tuberculosis (NIRT), a designated National Reference Laboratory of the National Tuberculosis Elimination Programme	Chennai (Park 28e, Ch 23, p 1034; Ch 7, pp 509-510)
Cancer Research Centre	(Park 28e, Ch 23, p 1034)
Blood Group Reference Centre	Mumbai (Park 28e, Ch 23, p 1034)
National Centre for Disease Informatics and Research (NCDIR), which houses the ICMR Bioethics Unit	Bengaluru (ICMR Ethics 2017, pp 185-187)



Governance and research-delivery structure of the Indian Council of Medical Research

The 2017 National Ethical Guidelines: principles and scope

Four basic principles — **respect for persons (autonomy), beneficence, non-maleficence and justice** — are expanded into 12 general principles (ICMR Ethics 2017, pp 18-19):

#	Principle	Core requirement
1	Essentiality	Use of human participants is essential after considering all alternatives; vetted by an independent ethics committee
2	Voluntariness	Right to agree, refuse or withdraw at any time; safeguarded by the informed consent process
3	Non-exploitation	Equitable selection; benefits and burdens distributed fairly; safeguards for vulnerable groups
4	Social responsibility	Research must not create or deepen social divisions or disturb community harmony
5	Privacy and confidentiality	Identity and records kept confidential; breach only with ethics committee consultation for valid scientific or legal reasons (e.g. suicidal ideation, homicidal tendency, court order)
6	Risk minimisation	Risks minimised at every stage; care and compensation if harm occurs
7	Professional competence	Research planned, conducted, evaluated and monitored by appropriately qualified persons
8	Maximisation of benefit	Design maximises direct or indirect benefit to participants and society

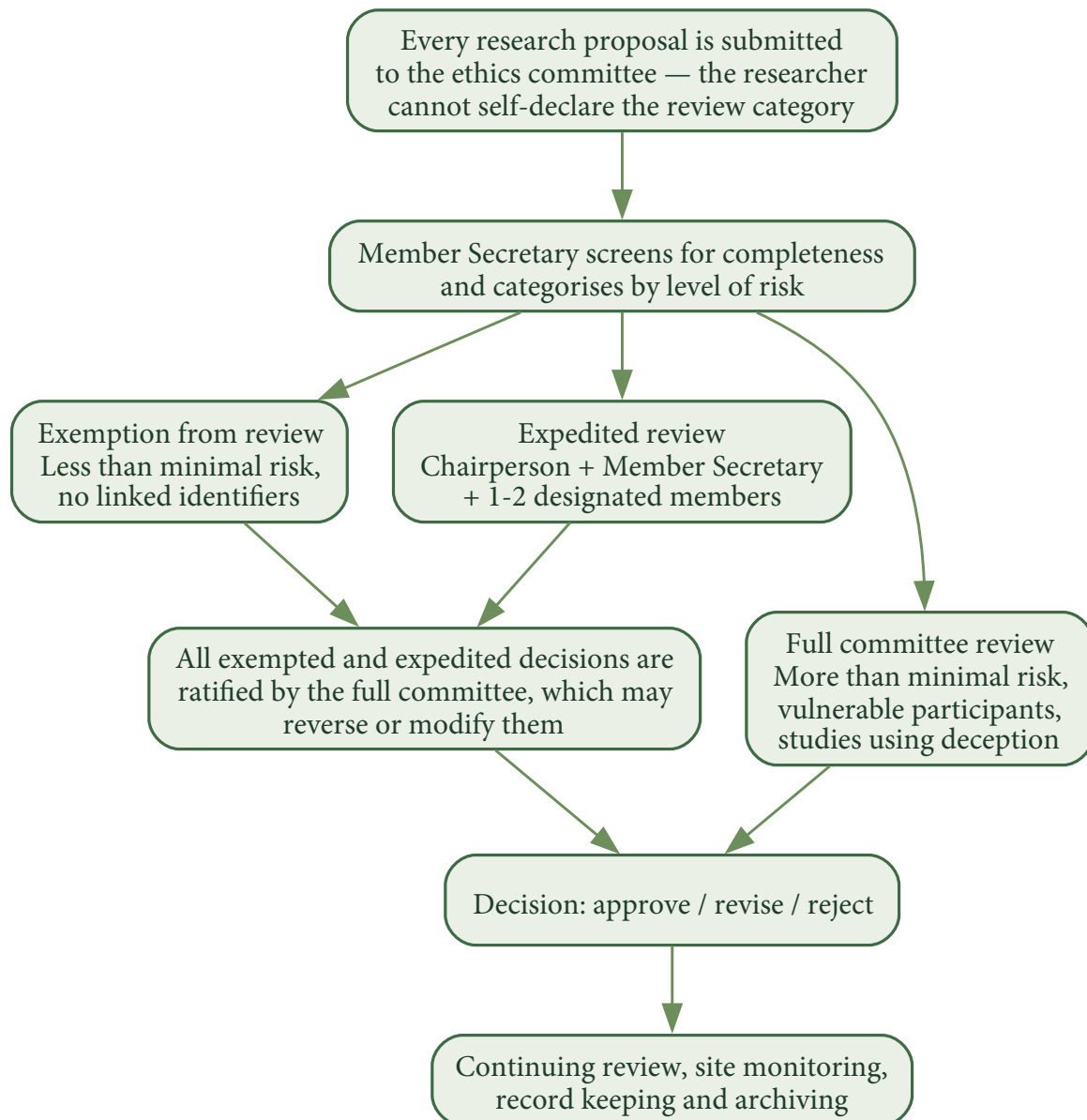
#	Principle	Core requirement
9	Institutional arrangements	Institution provides research-governance policy, infrastructure, manpower, funds, training
10	Transparency and accountability	Plan and outcomes in the public domain via registries and publications; conflict of interest disclosed; records retained for audit
11	Totality of responsibility	All stakeholders answerable for their actions
12	Environmental protection	Protection of environment and resources at all stages

Twelve sections of the document (ICMR Ethics 2017, pp 4-9): 1 General principles; 2 General ethical issues; 3 Responsible conduct of research; 4 Ethical review procedures; 5 Informed consent process; 6 Vulnerability; 7 Clinical trials of drugs and other interventions; 8 **Public health research**; 9 Social and behavioural sciences research for health; 10 Human genetics testing and research; 11 Biological materials, biobanking and datasets; 12 Research during humanitarian emergencies and disasters.

Ethical review machinery

Element	ICMR requirement (ICMR Ethics 2017, pp 42-46)
Character	Multi-disciplinary and multi-sectoral; adequate age and gender representation
Size	Preferably 7 to 15 members; minimum 5 present for quorum
Non-affiliated members	Preferably 50% from outside the institution; at least one must be in the quorum, and preferably the lay person too
Composition	Chairperson (non-affiliated) ± Vice-Chairperson, Member Secretary (affiliated), basic medical scientist(s), clinician(s), legal expert(s), social scientist/philosopher/ethicist, lay person
Balance	Between medical and non-medical, technical and non-technical members
Head of institution	Not a member; acts as appellate authority to appoint the committee and handle disputes
Validity	No decision is valid without the quorum being met

Risk categories that decide the review route (ICMR Ethics 2017, p 21): *less than minimal risk* (harm nil or not expected — anonymous data, public-domain data, meta-analysis); *minimal risk* (harm no greater than in routine daily life — history taking, physical examination, chest X-ray, non-invasive body fluids); *minor increase over minimal risk (low risk)* — e.g. routine research on children, on persons unable to consent.



Ethical review pathway for a research proposal under the ICMR National Ethical Guidelines 2017

Clinical Trials Registry–India and ethics-committee registration

Item	Detail (ICMR Ethics 2017, pp 35, 62-64)
CTRI launch	20 July 2007 by ICMR; linked to the WHO registry
Nature	Free, online, public record system for clinical trials, postgraduate theses and other biomedical research
Mandatory registration	Made mandatory by the Central Drugs Standard Control Organization (CDSCO) on 15 June 2009 for trials registered under the Drugs and Cosmetics Act and its Rules; voluntary for other biomedical and health research
Publication link	Indian biomedical journal editors consider only trials on a public database for publication
Ethics committee registration	Committees must register with the relevant authority as per regulatory requirements

Item	Detail (ICMR Ethics 2017, pp 35, 62-64)
Accreditation (voluntary)	SIDCER, AAHRPP, CDSCO, or Quality Council of India through NABH

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **ICMR organogram** — label: Ministry of Health and Family Welfare; Department of Health Research; ICMR Headquarters, New Delhi (1911); Director-General; permanent research institutes; research units; field surveys; ad-hoc financed enquiries; HMSC secretariat; ICMR Bioethics Unit (NCDIR, Bengaluru); CTRI.
- ▶ **Risk ladder governing ethics review** — label, bottom upward: less than minimal risk → exemption from review; minimal risk → expedited review; minor increase over minimal risk / more than minimal risk → full committee review; with the side note "all routes ratified by the full committee".

High-yield exam pointers

- ▶ **1911 New Delhi**; apex body for formulation, coordination and promotion of biomedical research; under the Department of Health Research, Ministry of Health and Family Welfare; funded wholly from the Union Health Ministry budget.
- ▶ **Four ethical principles → 12 general principles**. Mnemonic for the 12: E-V-N-S-P-R-P-M-I-T-T-E — Essentiality, Voluntariness, Non-exploitation, Social responsibility, Privacy and confidentiality, Risk minimisation, Professional competence, Maximisation of benefit, Institutional arrangements, Transparency and accountability, Totality of responsibility, Environmental protection.
- ▶ **Ethics committee numbers**: 7–15 members, quorum 5, ≥1 non-affiliated in quorum, preferably 50% non-affiliated; head of institution excluded and acts as appellate authority.
- ▶ **Three review types**: exemption from review, expedited review, full committee review — the ethics committee, never the researcher, decides which.
- ▶ **CTRI**: launched 20 July 2007 by ICMR; mandatory by CDSCO from 15 June 2009 for trials under the Drugs and Cosmetics Act; voluntary otherwise.
- ▶ **Guideline lineage**: 1980 (Khanna) → 2000 (Venkatachaliah) → 2006 → 2017, the 2017 version carrying 12 sections including a dedicated section on public health research.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the Council is widely described as having begun as the Indian Research Fund Association and been redesignated the Indian Council of Medical Research in 1949, and as being governed by a Governing Body chaired by the Union Health Minister with a Scientific Advisory Board — these two administrative details, and the fact that ICMR now maintains far more than the five institutes Park lists (which retains several older institute names), are standard knowledge but were not found in the retrieved corpus passages, so they are flagged rather than presented as cited fact.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, New Drugs and Clinical Trials Rules 2019, ICMR National Ethical Guidelines 2017, Department of Health Research (NECRBHR / Naitik portal), CDSCO, NABH, WHO South-East Asia Regional Office

Programme and policy updates (India)

- ▶ **Ethics-committee registration has shifted to a second, separate authority.** Under the New Drugs and Clinical Trials Rules 2019, every committee reviewing biomedical and health research — not only regulatory drug trials — must register with the **National Ethics Committee Registry for Biomedical and Health Research (NECRBHR)** at the Department of Health Research, applying in form CT-01 through the online **Naitik** portal; provisional registration runs 2 years and final registration 5 years. The requirement covers medical, dental, pharmacy, nursing, AYUSH, hospital and independent committees, irrespective of any existing CDSCO registration. Uptake lagged badly: about 2,100 committees had opened portal accounts and more than 1,560 had applied by July 2023, against an expected ~4,000, with roughly 40% of medical and dental colleges still unregistered. [PMID 37737078]
- ▶ **India now runs two parallel ethics-committee registries, and accreditation remains weak.** A national audit found 1,400 committees registered with CDSCO (90.9% institutional, 9.1% independent) against only 952 with the Department of Health Research, of which just 14% held final registration; only 22% of CDSCO-registered committees had even applied for NABH accreditation and only 11% published their standard operating procedures. Registration validity is now 5 years under the 2019 Rules, replacing the 3 years allowed under the older Rule 122DD of Schedule Y. [PMID 40322477]
- ▶ **Site-by-site review of multi-centre trials is being reconsidered.** In a survey run by the ICMR Bioethics Unit with the Clinical Studies and Trials Unit, only 21% of 106 researchers and 31% of 136 ethics-committee members wanted to retain separate approval at every participating site; researchers leaned towards a designated committee and members towards a joint review, and about four-fifths of both groups wanted the approving committee to share its decisions with the other sites' committees. [PMID 42337749]

Current national data

- ▶ **ICMR-INDIAB-17 is the current national benchmark for metabolic non-communicable disease** (113,043 adults aged ≥ 20 years, 31 states and union territories, surveyed 2008–2020): weighted prevalence of diabetes 11.4% (95% CI 10.2–12.5), prediabetes 15.3%, hypertension

35.5%, generalised obesity 28.6%, abdominal obesity 39.5% and dyslipidaemia 81.2% — considerably above earlier estimates, with prediabetes outnumbering diabetes in most states of lower human development index. [PMID 37301218]

- **ICMR-INDIAB-23 reframes obesity screening for Indian practice.** The commonest phenotype is metabolically obese but non-obese by BMI (43.3%), more frequent in rural areas, carrying the highest adjusted risk of chronic kidney disease (OR 1.81, 95% CI 1.21–2.70); the metabolically obese obese group carries the highest risk of type 2 diabetes (OR 12.89, 9.54–17.43). ICMR-INDIAB also counts 254 million adults with generalised and 351 million with abdominal obesity — the case against BMI-only screening. [PMID 40844097]

Elimination, surveillance and disease status

- **Dog-mediated rabies: elimination target 2030.** India's National Action Plan for Dog-Mediated Rabies Elimination was released in 2020 and became operational in 2021 with the endorsement of six ministries. It turns on more than 70% dog vaccination coverage (oral rabies vaccine still awaiting central licensure and bait development), universal access to post-exposure prophylaxis and rabies immunoglobulin at block and municipal level, and molecular epidemiology through the newly created Integrated District Public Health Laboratories. [PMID 37970176]
- **AMR surveillance is being pushed onto a One Health, district-level footing,** taking the district under the Collector or Deputy Commissioner as the administrative unit where the human, animal and environmental sectors actually intersect, so that joint multisectoral surveillance generates locally actionable resistance data rather than sentinel-laboratory data alone. [PMID 37651765]

Guideline and legislative updates

- **The 2019 Rules and the 2017 ICMR guidelines do not align on post-trial access.** Both address continued supply of an investigational drug after a trial closes, but their provisions conflict on interpretation, and Indian drug law still carries no express compassionate-use or expanded-access route — a live gap between statute and ICMR guidance that falls hardest on participants with poor access to care. [PMID 37616571]
- **CTRI data quality is itself now a research finding.** Retrospective registration has been disallowed since 1 April 2018, yet records remain mislabelled and first-enrolment dates have been edited after registration; separately, among 473 multinational phase 2 or 3 trials registered 2013–2020, 35% of the records listed as not completed were overdue for updating, and a number of trials recruited 60% or more of their participants from India, mostly under foreign sponsorship. [PMID 35699281; PMID 37729309]

Global and WHO developments

- **A single national digital ethics-review platform is the emerging model.** Replacing isolated committees with one access-controlled national system carrying common ICMR templates would harmonise review, allow one submission to serve every site of a multi-centre study, and generate both the ethics-committee registry and the clinical trial registry as automatic transactional by-products — dissolving the need for separate registration steps. India's e-EC application (built on the ICMR 2017 guidelines and the 2019 Rules) and the WHO South-East Asia Regional Office research portal are the working proofs of concept. [PMID 39304929]

SOURCE & COVERAGE

ICMR National Ethical Guidelines for Biomedical and Health Research Involving Human Participants, 2017; Park's Textbook of Preventive and Social Medicine 28e (Ch 23 Health Planning and Management, Ch 7 Health Programmes in India, Ch 11 Nutrition and Health); Sunder Lal — Textbook of Community Medicine (Ch 5 Nutrition, Ch 6 Basic Statistics and Research Methodology, Ch 18 International Health).

Coverage note: the spine textbook devotes only a single paragraph to ICMR, so the organisational and functional detail here is drawn mainly from the ICMR 2017 guideline document itself; no retrieved passage described the Council's Governing Body or its current full institute list.

Recent-advances sources: 37737078 — Overview of the National Ethics Committee Registry for Biomedical and Health Research in India: stepping up to safeguard the ethical aspect of research involving human participants, Indian J Pharmacol 2023, doi:[10.4103/ijp.ijp_216_22](https://doi.org/10.4103/ijp.ijp_216_22); 40322477 — Evaluation of the current status of ethics committees in India using the NABH, CDSCO and Department of Health Research databases, Perspect Clin Res 2024, doi:[10.4103/picr.picr_40_24](https://doi.org/10.4103/picr.picr_40_24); 42337749 — Ethics review of multi-centre trials in India: a survey of researchers and ethics committee members on perspectives, challenges, and opportunities, BMC Med Ethics 2026, doi:[10.1186/s12910-026-01489-1](https://doi.org/10.1186/s12910-026-01489-1); 37301218 — Metabolic non-communicable disease health report of India: the ICMR-INDIAB national cross-sectional study (ICMR-INDIAB-17), Lancet Diabetes Endocrinol 2023, doi:[10.1016/S2213-8587\(23\)00119-5](https://doi.org/10.1016/S2213-8587(23)00119-5) (abstract-only); 40844097 — High prevalence of metabolic obesity in India: the ICMR-INDIAB national study (ICMR-INDIAB-23), Indian J Med Res 2025, doi:[10.25259/IJMR_328_2025](https://doi.org/10.25259/IJMR_328_2025); 37970176 — National Action Plan for Dog-Mediated Rabies Elimination: a One Health approach to catalytic response is key to India achieving the target by 2030, Indian J Community Med 2023, doi:[10.4103/ijcm.ijcm_554_23](https://doi.org/10.4103/ijcm.ijcm_554_23); 37651765 — One Health based joint surveillance for AMR containment at district and national level in India: a conceptual model, Indian J Med Microbiol 2023, doi:[10.1016/j.ijmmb.2023.100470](https://doi.org/10.1016/j.ijmmb.2023.100470) (abstract-only); 37616571 — Post-trial access to investigational drugs in India: addressing challenges in the regulatory framework, Med Law Rev 2024, doi:[10.1093/medlaw/fwad028](https://doi.org/10.1093/medlaw/fwad028) (abstract-only); 35699281 — CTRI requirement of prospective trial registration: not always consistent, Indian J Med Ethics 2022, doi:[10.20529/IJME.2022.033](https://doi.org/10.20529/IJME.2022.033) (abstract-only); 37729309 — Representation from India in multinational, interventional, phase 2 or 3 trials registered in Clinical Trials Registry-India: a cross-sectional study, PLoS One 2023, doi:[10.1371/journal.pone.0284434](https://doi.org/10.1371/journal.pone.0284434); 39304929 — A transformative solution to build effective, transparent, and resilient "fit-for-purpose" national health research ethics systems, Health Res Policy Syst 2024, doi:[10.1186/s12961-024-01219-2](https://doi.org/10.1186/s12961-024-01219-2). Regulatory and guideline instruments cited: New Drugs and Clinical Trials Rules 2019 (GSR 227(E), 19 March 2019); ICMR National Ethical Guidelines 2017; Department of Health Research NECRBHR / Naitik portal; CDSCO ethics-committee registry; NABH accreditation scheme.

♦ ♦ ♦

Meta-analysis

Asked in: Paper I Nov 2018 — RGUHS MD Community Medicine

Definition — Meta-analysis is the statistical combination of results from two or more separate studies (Cochrane Handbook, Ch 9, p 268) — the use of statistical methods to summarise the results of independent studies, a definition credited to *Glass* (1976) (Cochrane Handbook, Ch 1, p 31). Operationally it identifies a research question, specifies the sources from which studies answering it will be found, transforms the findings to a **common metric**, and analyses them statistically (Lilienfeld's Foundations of Epidemiology, Ch 15, p 305). It is the quantitative step **inside** a systematic review — which is itself defined by pre-defined eligibility criteria, an explicit reproducible methodology, a systematic search, an assessment of risk of bias and a systematic synthesis; many, but not all, systematic reviews contain a meta-analysis (Cochrane Handbook, Ch 1, p 31).

1. Purposes — why a meta-analysis is done, and when it must not be

Reason to pool	What it buys the reviewer
To increase power	Power is the chance of detecting a real effect as statistically significant if it exists; many individual studies are too small to detect small effects, but combined they have a higher chance (Cochrane Handbook, Ch 9, p 271)
To improve precision	The estimate of the intervention effect improves when based on more information — narrower confidence interval than any single study can give (Cochrane Handbook, Ch 9, p 271; Browner 5e, Ch 16, p 519)
To answer questions not posed by the individual studies	Primary studies use a specific patient type and defined interventions; studies differing in these allow the consistency of effect, and reasons for differences, to be investigated (Cochrane Handbook, Ch 9, p 271)
To settle controversies from apparently conflicting studies, or to generate new hypotheses	The degree of conflict is formally assessed and reasons for different results quantified (Cochrane Handbook, Ch 9, p 271)
To sharpen the boundaries of effect size, and for trials, to derive the number needed to treat (NNT)	<i>Chalmers'</i> stated purposes of meta-analysis (Lilienfeld's Foundations of Epidemiology, Ch 15, p 305)

Situation in which meta-analysis should NOT be done	Why
Clinically diverse studies — the "apples and oranges" criticism	If studies are clinically diverse a meta-analysis may be meaningless and genuine differences in effect may be obscured; it is often nonsensical to combine a mix of different treatments and comparators in one analysis (Cochrane Handbook, Ch 9, pp 271–272)
Included studies at risk of bias	Meta-analysis simply compounds the errors and produces a wrong result that is then read as having more credibility (Cochrane Handbook, Ch 9, p 272)

Situation in which meta-analysis should NOT be done	Why
Serious publication or reporting bias	Likely to produce an inappropriate summary (Cochrane Handbook, Ch 9, p 272)
Non-randomised studies (NRS) pooled with, or across, dissimilar designs	Effect estimates from NRS should not be combined with those from randomised trials, or across NRS using dissimilar design features; pooling NRS is expected to be the exception, not the rule (Cochrane Handbook, Ch 13, p 449)

2. Steps in conducting a meta-analysis

- ▶ **Define the question and aim** — as a pair-wise comparison, experimental versus control intervention (Lilienfeld's Foundations of Epidemiology, Ch 15, p 305; Cochrane Handbook, Ch 9, p 268).
- ▶ **Write the protocol first** — objectives; eligibility criteria (types of studies, participants, interventions, outcome measures); search methods; data collection and analysis plan (Cochrane Handbook, Ch 2, pp 38–39). The analysis plan must pre-specify the effect measure, how heterogeneity will be quantified, and whether a fixed-effect or random-effects model will be used (Cochrane Handbook, Ch 9, Section 9.1.7).
- ▶ **Systematic search** for all completed studies — MEDLINE/PubMed, Embase, CINAHL, PsycINFO, AMED and the Cochrane Controlled Trials Register, plus hand search of articles and reviews; record the decision on unpublished, non-peer-reviewed material, since excluding it invites **publication bias** (Lilienfeld's Foundations of Epidemiology, Ch 15, pp 305–306).
- ▶ **Apply the pre-set inclusion and exclusion criteria** *before* examining the articles, recording numbers screened, excluded and included (Lilienfeld's Foundations of Epidemiology, Ch 15, p 306).
- ▶ **Assess risk of bias** in each included study — sequence generation, allocation concealment, blinding, incomplete outcome data, selective reporting (Cochrane Handbook, Ch 8).
- ▶ **Extract data on a piloted form, independently by two reviewers**, with a pre-specified rule for resolving disagreement (discussion, then arbitration); this minimises information bias (Cochrane Handbook, Ch 7, pp 194–195).
- ▶ **Convert results to a common effect measure** — odds ratio, risk ratio or risk difference for dichotomous data; mean or standardised mean difference for continuous data; hazard ratio for time-to-event data. Designs yielding different outcome measures need separate meta-analyses (Cochrane Handbook, Ch 9, p 300; Lilienfeld's Foundations of Epidemiology, Ch 15).
- ▶ **Pool by weighted averaging** under a fixed-effect or random-effects model (Section 3); then **assess heterogeneity** and explore it by subgroup analysis or meta-regression (Section 4), and **assess reporting bias** by funnel plot (Section 5).
- ▶ **Sensitivity analyses** — repeat the analysis under the alternative choices made at each arbitrary decision node, to show the finding is robust (Cochrane Handbook, Ch 9, Section 9.7).
- ▶ **Present and report** — forest plot, summary-of-findings table, **PRISMA** checklist (Section 6).

3. Statistical methods of pooling

Every method is a variation on a **weighted average of the effect estimates from the different studies** (Cochrane Handbook, Ch 9, p 268). The **inverse-variance** weight is one divided by the square of the standard error, so larger studies with smaller standard errors carry more weight — a weighting that minimises the imprecision of the pooled estimate; ratio measures are combined on the **natural logarithm scale** (Cochrane Handbook, Ch 9, pp 288–289).

Type of data	Effect measure	Fixed-effect methods	Random-effects method
Dichotomous	Odds ratio (OR)	Mantel-Haenszel (M-H), inverse variance (IV), Peto	M-H, IV
Dichotomous	Risk ratio (RR); risk difference (RD)	M-H, IV	M-H, IV
Continuous	Mean difference (MD); standardised mean difference (SMD)	Inverse variance	Inverse variance
Time-to-event	"O–E and Variance" from the log-rank test	Peto	None available
Time-to-event	Log hazard ratio and its standard error (Cox model)	Generic inverse variance	Generic inverse variance

All random-effects options above use the DerSimonian and Laird weighting (Cochrane Handbook, Ch 9, Table 9.4.a, p 300).

- **Mantel-Haenszel** is the default fixed-effect method and is **generally preferable to inverse variance**, because when data are sparse (low event rates, small studies) inverse-variance standard errors are poor (Cochrane Handbook, Ch 9, p 291).
- **Peto's method** pools **only odds ratios**, as a sum of "O – E" statistics (O = observed, E = expected events in the experimental arm) (Cochrane Handbook, Ch 9, p 291). At event rates **below 1 per cent** it is the least biased and most powerful method, provided arm sizes are not badly unbalanced and the effect is not exceptionally large (Cochrane Handbook, Ch 16, p 548).
- **DerSimonian and Laird (1986)** is the standard random-effects method — inverse variance with weights adjusted for between-study variation (Cochrane Handbook, Ch 9, pp 289, 291).
- **Zero cells** (no events in an arm) break the computation; software adds **0.5 to each cell of the 2 × 2 table** for such studies (Cochrane Handbook, Ch 9, pp 290–291; Ch 16, p 546).

	Fixed-effect model	Random-effects model
Assumption	The true effect is the same value in every study ; observed differences are due to chance alone, i.e. no statistical heterogeneity	The effects being estimated in the different studies are not identical but follow a distribution (conventionally normal)
What the pooled estimate means	"What is the best estimate of the intervention effect?"	"What is the average intervention effect?" — the centre of the distribution

	Fixed-effect model	Random-effects model
Weighting	By precision alone	Precision plus the between-study variance, so small studies gain relative weight
Confidence interval where heterogeneity exists	Narrower; P value more significant	Wider ; claims of significance more conservative
When both agree	Identical results when there is no heterogeneity	Identical results when there is no heterogeneity
Caution	Ignores heterogeneity; the pooled effect may not exist in any population and its CI can be meaninglessly narrow	Does not "take account of" heterogeneity — causes must still be explored; the CI describes uncertainty about the mean, not the degree of heterogeneity

(Cochrane Handbook, Ch 9, pp 291–292, 305–306)

4. Heterogeneity — types, measurement and management

Type	What varies
Clinical diversity (clinical heterogeneity)	Participants, interventions and outcomes studied
Methodological diversity (methodological heterogeneity)	Study design and risk of bias
Statistical heterogeneity (conventionally called simply "heterogeneity")	The intervention effects themselves — a consequence of clinical or methodological diversity, or both; it shows as observed effects differing more from each other than random error alone would explain

(Cochrane Handbook, Ch 9, Section 9.5.1)

Measure	What it is	Interpretation
Poor overlap of the confidence intervals on the forest plot	Informal visual check	Generally indicates statistical heterogeneity (Cochrane Handbook, Ch 9, p 302)
Cochran's Q — chi-squared (χ^2) test of heterogeneity	Tests whether observed differences are compatible with chance alone	A low P value, or a large χ^2 relative to its degrees of freedom, is evidence of heterogeneity. Low power when studies are few or small, so a non-significant result is not evidence of no heterogeneity — hence P < 0.10 is often used instead of 0.05. With many studies it may detect trivial heterogeneity (Cochrane Handbook, Ch 9, pp 302–303)

Measure	What it is	Interpretation
I^2 statistic — calculated from Q and its degrees of freedom, $I^2 = (Q - df)/Q \times 100$	The percentage of the variability in effect estimates that is due to heterogeneity rather than sampling error (chance)	Rough guide: 0–40 % might not be important; 30–60 % may represent moderate heterogeneity; 50–90 % substantial; 75–100 % considerable — importance also depends on the magnitude and direction of effects and on the strength of evidence for heterogeneity (Cochrane Handbook, Ch 9, p 303)
tau-squared (τ^2)	The estimated between-study variance in a random-effects meta-analysis; τ (its square root) is the standard deviation of the underlying effects across studies	For absolute effect measures, an approximate 95 % range of underlying effects runs from 2τ below to 2τ above the random-effects pooled estimate (Cochrane Handbook, Ch 9, p 306)

Seven strategies when heterogeneity is found (Cochrane Handbook, Ch 9, pp 303–305):

- ▶ **Check the data are correct** — severe heterogeneity often means data were mis-extracted or mis-entered (e.g. standard errors entered as standard deviations).
- ▶ **Do not do a meta-analysis** — where results vary widely, especially in the *direction* of effect, quoting an average is misleading.
- ▶ **Explore it** — by **subgroup analysis** or **meta-regression**, ideally pre-specified in the protocol; explanations devised after seeing the heterogeneity are hypothesis-generating only.
- ▶ **Ignore it** — a fixed-effect analysis; its P value still tests the null hypothesis of no effect in every study.
- ▶ **Use a random-effects model** — for heterogeneity that cannot be explained; not a substitute for investigating it.
- ▶ **Change the effect measure** — heterogeneity may be an artefact of the wrong measure (mean difference where the standardised mean difference is needed; homogeneous odds ratios necessarily give heterogeneous risk differences when control-group risks vary).
- ▶ **Exclude outlying studies** — unwise on results alone as it introduces bias; do it as a **sensitivity analysis**, with and without the outliers.

5. Publication and reporting bias

Publication bias is a selection bias occurring when the published studies are not representative of all research done — typically because positive studies are more likely to be published than negative ones (Lilienfeld's Foundations of Epidemiology, Ch 15, pp 305–306; Browner 5e, Ch 16, pp 519–520).

Detection / handling	Detail
Funnel plot	Plots each study's effect estimate against a measure of its size (usually the standard error). It is a generic means of displaying small-study effects — a tendency for effects estimated in smaller studies to differ from those in larger ones — and asymmetry must not be equated automatically with publication bias (Cochrane Handbook, Ch 10, pp 337–338)

Detection / handling	Detail
Sources of funnel-plot asymmetry	(1) Selection biases — publication bias, delayed ("time-lag" or "pipeline") publication bias, location biases (language bias, citation bias, multiple publication bias), selective outcome reporting; (2) poor methodological quality inflating effects in smaller studies — poor design, inadequate analysis, fraud; (3) true heterogeneity (effect differs by study size); (4) artefactual — sampling variation linking the effect to its standard error; (5) chance (Cochrane Handbook, Ch 10, Table 10.4.a, pp 337–338)
Statistical tests of asymmetry	<i>Egger's</i> test regresses the intervention effect estimates on their standard errors, weighted by 1/variance; under no small-study effects the line is vertical (Cochrane Handbook, Ch 10, p 339)
Rule for using the tests	Use only when there are at least 10 studies — with fewer, power is too low to distinguish chance from real asymmetry; do not use if all studies are of similar size; interpret alongside visual inspection. A negative test does not exclude bias (Cochrane Handbook, Ch 10, pp 342–343)
Caveat for odds ratios	The standard error of the log odds ratio is mathematically linked to the size of the odds ratio, so funnel plots of odds ratios can look asymmetric and Egger P values can be too small (Cochrane Handbook, Ch 10, p 348)
What to do about it	Comprehensive searching including unpublished sources; sensitivity analyses comparing fixed- and random-effects estimates; funnel plots and tests alert to the problem but do not solve it (Cochrane Handbook, Ch 10, pp 342–344, 348)

6. Presenting, interpreting and reporting the result

- **Forest plot** — effect estimates and confidence intervals for both the individual studies and the meta-analysis. Each study is a **block at its point estimate whose area is its weight**, with a **horizontal line for the 95 per cent confidence interval**; each subgroup and overall pooled result is a **diamond**; the vertical line of no effect sits at 1 for ratio measures and 0 for differences (Cochrane Handbook, Ch 11, pp 362–365).
- It also carries the raw 2×2 data or means for every study, and — **only when the data are actually pooled** — the heterogeneity statistics (τ^2 , χ^2 , I^2 , test for subgroup differences), the test for overall effect and each study's **per cent weight** (Cochrane Handbook, Ch 11, Box 11.3.a, p 363).
- **Label the axis** to show which side favours which intervention, rather than relying on convention (Cochrane Handbook, Ch 11, p 365).
- **Report against PRISMA** — *Preferred Reporting Items for Systematic Reviews and Meta-Analyses*, 2009, a 27-item checklist that updated the 1996 QUOROM statement; **MOOSE** (2000) covers meta-analyses of observational studies and additionally requires each study to be rated for quality (Lilienfeld's Foundations of Epidemiology, Ch 15).

Variant of meta-analysis	Definition and value
Aggregate (published) data meta-analysis	The usual form — only the published summary results of each study are combined

Variant of meta-analysis	Definition and value
Individual patient/participant data (IPD) meta-analysis	Original research data for each participant are sought directly from the researchers, allowing data checking, thorough exploration and consistent re-analysis; the gold standard , but slower and costlier, and usually run by a collaborative group (Cochrane Handbook, Ch 18, p 572; Ch 7, p 189)
Prospective meta-analysis (PMA)	Studies are identified, evaluated and determined eligible before the results of any of them are known , so hypotheses, selection criteria and analyses are all stated <i>a priori</i> ; usually collaborative and IPD-based (Cochrane Handbook, Ch 19, p 584)
Meta-regression	Relates study-level characteristics to the size of the intervention effect; identifies determinants of heterogeneity even where studies are too heterogeneous to pool (Cochrane Handbook, Ch 9, p 309; Ch 13, p 448)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Diagram 1 (to sketch)** — *Forest plot*: vertical line of no effect at RR/OR = 1; per study a square at the point estimate, area = weight, with a horizontal line = 95 % CI; a diamond = pooled estimate, its width = the pooled CI. Left margin "Study or Subgroup / Events / Total / Weight %"; right margin "Risk Ratio (95 % CI)"; foot "Heterogeneity: τ^2 , χ^2 , df, P, I^2 = %" and "Test for overall effect: Z, P"; axis "Favours experimental $\leftarrow \rightarrow$ Favours control".
- **Diagram 2 (to sketch)** — *Funnel plot*: y-axis standard error of the effect estimate, inverted (most precise at the top); x-axis the effect estimate on a log scale; a symmetrical inverted funnel of dots widening downwards about a vertical pooled-estimate line; mark the missing lower-left corner as "small negative studies unpublished = publication bias / small-study effects".

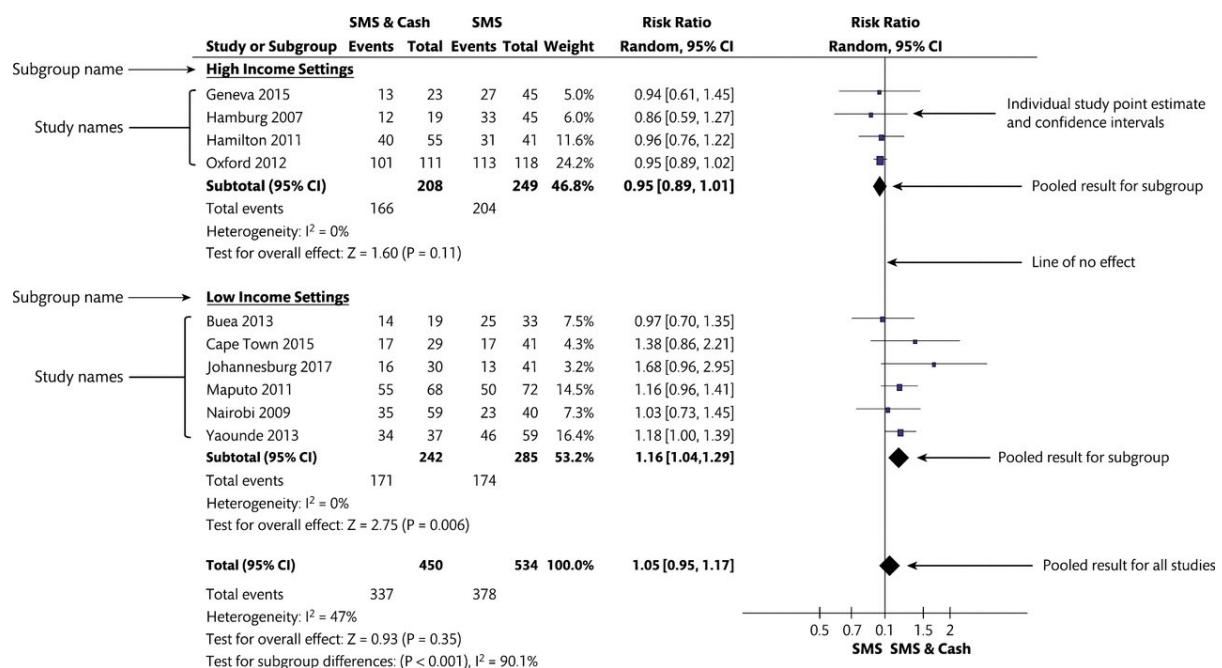
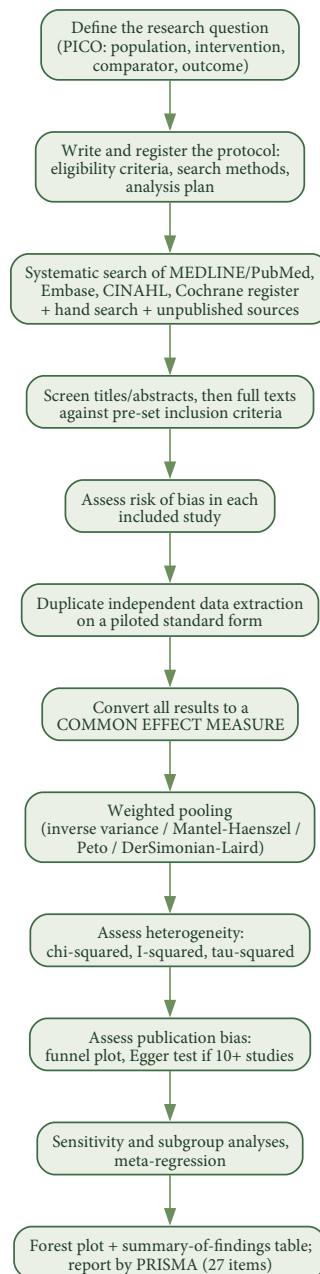
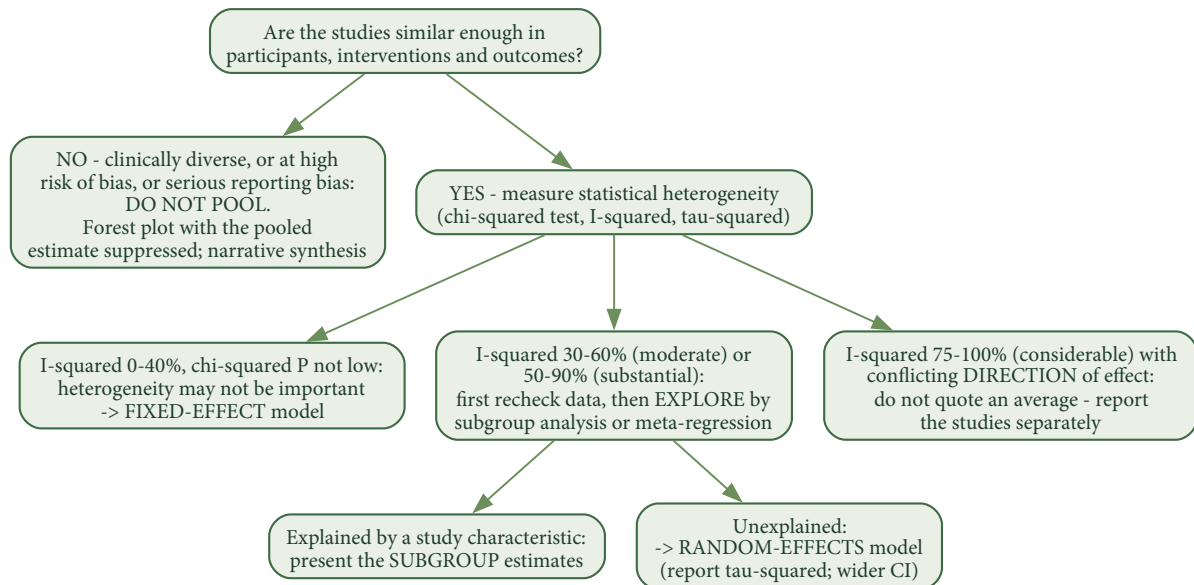


Fig 5.13 — 1 Forest plot of invented data, from a meta-analysis of ten studies in (Oxford public health 7th ed, p.1383)



Sequence of a systematic review with meta-analysis, from question to reported pooled estimate



Decision algorithm for pooling once heterogeneity has been measured

High-yield exam pointers

- **Open with the definition:** the statistical combination of results from two or more separate studies (Cochrane Handbook, Ch 9, p 268) — systematic review = the method, meta-analysis = its quantitative step.
- **Four advantages:** increase power; improve precision; answer questions not posed by the individual studies; settle controversies / generate hypotheses (Cochrane Handbook, Ch 9, p 271).
- $I^2 = (Q - df) / Q \times 100$ = per cent of variability due to heterogeneity rather than chance. Bands: 0–40 % may not matter; 30–60 % moderate; 50–90 % substantial; 75–100 % considerable (Cochrane Handbook, Ch 9, p 303). The heterogeneity χ^2 test uses $P < 0.10$, not 0.05, because of low power.
- **Fixed effect = one true effect** ("best estimate"); **random effects = a distribution of effects** ("average effect"), wider CI; identical when there is no heterogeneity (Cochrane Handbook, Ch 9, pp 291–292, 305–306).
- **Methods:** Mantel-Haenszel is the default fixed-effect method, preferred when events are few; **Peto pools only odds ratios**, least biased at event rates $< 1\%$; DerSimonian and Laird is the random-effects method (Cochrane Handbook, Ch 9, pp 291, 300; Ch 16, p 548).
- **Funnel plot asymmetry $\neq$ publication bias** — it shows **small-study effects**; five causes: selection biases, poor methodological quality, true heterogeneity, artefact, chance. Test asymmetry **only if ≥ 10 studies** (Cochrane Handbook, Ch 10, Table 10.4.a, pp 342–343).
- **Forest plot:** block area = study weight, horizontal line = 95 % CI, **diamond = pooled estimate**, vertical line = line of no effect (Cochrane Handbook, Ch 11, pp 362–363). **Reporting:** PRISMA 2009, 27 items (updated the 1996 QUOROM statement); MOOSE 2000 for observational meta-analyses (Lilienfeld's Foundations of Epidemiology, Ch 15).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not written in the exam: the indexed corpus holds version 5.x of the Cochrane Handbook, whose reference software is RevMan and whose risk-of-bias instrument is the original Cochrane tool, whereas current practice uses the revised tools and an updated PRISMA statement — the statistics quoted here (weighting, fixed versus random effects, I^2 , funnel plots) are unchanged. Coverage note: Park's Textbook of Preventive and Social Medicine 28e carries no section on meta-analysis, so this answer rests on the Cochrane Handbook, Lilienfeld's Foundations of Epidemiology and Designing Clinical Research, and the retrieved corpus offered no Indian meta-analysis example.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, PRISMA 2020, PRIOR 2022, ROBINS-E 2024, GRADE Guidance 41, AMSTAR-2, ROBIS, HTAIn (Department of Health Research, MoHFW)

Programme and policy updates (India)

- **Meta-analysis now feeds Indian policy formally through HTAIn** — Health Technology Assessment in India sits under the **Department of Health Research, Ministry of Health and Family Welfare**, and commissions evidence syntheses to guide allocation of finite health-system resources; an evaluation of three HTAs commissioned between 2018 and 2020 put the return on investment at **9:1**, rising to **71:1** if the recommendations were fully implemented, with the ratio varying from 5:1 to 40:1 across interventions and diseases [PMID 40113231].

Current national data

- **The pooled Indian estimate of tuberculosis infection exceeds the national survey figure** — 77 studies from 10,521 records; DerSimonian–Laird random-effects pooling (MetaXL) of 15 community-based cohorts gave a TB-infection prevalence of **41 % (95 % CI 29.5–52.6)**, $I^2 = 99 %$, and **36 % (95 % CI 28–45)**, $I^2 = 83 %$ in the general population once risk groups were excluded, against the crude **31.3 % (95 % CI 30.8–31.9)** in those above 15 years recorded by the **National TB Prevalence Survey 2019–21**; a sensitivity analysis dropping the heterogeneity-driving studies gave **38 % (29–46)**, $I^2 = 72 %$. Reporting bias was judged not by a funnel plot but by a **Doi plot with the LFK index** (2.00 = minor asymmetry), a newer alternative to the funnel plot [PMID 37202933].

Elimination, surveillance and disease status

- **Risk-group meta-analysis is being used to reprioritise TB preventive treatment under NTEP** — 68 Indian studies pooled by random-effects model with Cochran's Q and I^2 gave TB-infection prevalence of **36 % overall**, **59 % in smokers**, **53 % in people with diabetes** and **in those with alcohol use**, **48 % in the malnourished**, **47 % in household contacts**, **44 % in people with HIV** and **31 % in healthcare workers**; WHO does not at present recommend preventive treatment for smokers, diabetics, alcohol users or the malnourished, and the review is offered as the missing evidence for widening eligibility under the National Strategic Plan to end TB in India [PMID 39421504].

Guideline and legislative updates

- ▶ **PRISMA 2020 replaces PRISMA 2009** — quote this version in the exam, not the 2009 one — the checklist still has 27 items but is restructured into seven sections with sub-items, and the authors state that the 2009 statement should no longer be used. New content includes **assessment of certainty (confidence) in the body of evidence as both a methods item (15) and a results item (22)**, a competing-interests item (26) and an item on public availability of data, analytic code and other materials (27); registration and protocol move to a new "Other information" section (24a–24c); and the synthesis item is split into 13a–13f — which studies are eligible for each synthesis, data preparation, tabulation and visual display, the synthesis model with heterogeneity methods and software, exploration of heterogeneity, and sensitivity analysis. A **12-item PRISMA 2020 for Abstracts checklist** and revised flow diagrams for original and updated reviews are included, and the statement explicitly covers continually updated (**living**) systematic reviews [PMID 33782057].
- ▶ **PRIOR (2022) is the reporting guideline for overviews of reviews / umbrella reviews** — developed by the EQUATOR four-stage process with a modified Delphi requiring $\geq 70\%$ agreement; the final checklist carries **27 main items with 19 sub-items**, filling the gap PRISMA leaves for syntheses whose unit of inclusion is a systematic review rather than a primary study [PMID 35944924].
- ▶ **ROBINS-E (2024) extends domain-based risk-of-bias assessment to exposure studies** — built on ROBINS-I (non-randomised studies of interventions) and RoB 2 (randomised trials), it appraises a single exposure-effect estimate across **seven bias domains** through signalling questions, yielding judgements of **low risk / some concerns / high risk / very high risk of bias** together with a predicted **direction of bias**. The present version covers follow-up (cohort) studies only, with case-control variants under development — directly relevant to the environmental, occupational and behavioural exposure reviews that inform public-health policy, and to the Handbook's caution against pooling non-randomised studies uncritically [PMID 38555664].
- ▶ **AMSTAR-2 and ROBIS are complementary, not interchangeable, when appraising a published review** — applied to 200 systematic reviews (AMSTAR-2 has 16 items, ROBIS 24): **73 % were rated low or critically low methodological quality by AMSTAR-2 and 81 % high risk of bias by ROBIS**, 9 % received opposing ratings, and median completion time was 51 versus 64 minutes [PMID 41626887].
- ▶ **Report a prediction interval alongside the confidence interval for a random-effects pooled estimate** — across 1500 screened meta-analyses (866 using a random-effects model), a prediction interval was given in only 1.7 % (95 % CI 0.9–3.3) of sports-medicine and 3.9 % (2.4–6.6) of general medical reports; where one could be calculated, **60 % of studies disagreed in conclusion with the reported confidence interval**, and prediction intervals were on average 3.4 times wider — the practical expression of the $\tau^2 / 2\tau$ range already described above [PMID 38501202].

Global and WHO developments

- ▶ **GRADE Guidance 41 (2025) drops the "levels of contextualisation"** — certainty of evidence remains defined as the certainty that the true effect lies on one side of a **threshold** or within a **particular range**, but the GRADE working group has abandoned the categorisation of those thresholds into levels of contextualisation, which proved internally inconsistent and confusing,

and now simply names the threshold or range chosen as the target of the certainty rating [PMID 40587852].

- **Large-language-model screening is now measurably accurate and far cheaper than manual screening** — optimised GPT-4 prompts tested on 48,425 citations across 10 systematic reviews achieved weighted **sensitivity 97.7 % and specificity 85.2 %** at abstract screening and **96.5 % / 91.2 %** at full-text screening, whereas unoptimised zero-shot prompts retrieved under half the eligible records (sensitivity $\approx$ 49 %); screening 10,000 citations cost roughly **\$157 and under one day** against more than **83 hours and \$1,667** for a single human screener. PRISMA 2020 items 8 and 9 already require any such automation tools to be declared [PMID 39993313].

SOURCE & COVERAGE

Cochrane Handbook for Systematic Reviews of Interventions (Ch 1, p 31; Ch 2, pp 38–39; Ch 7, pp 189, 194–195; Ch 8; Ch 9 Analysing Data and Undertaking Meta-analyses, pp 268–272, 288–306, 309 and Sections 9.5.1–9.5.4, 9.7; Ch 10 Addressing Reporting Biases, pp 327–348 incl. Table 10.4.a; Ch 11 Presenting Results, pp 362–365 incl. Box 11.3.a; Ch 13, pp 442–449; Ch 16 Special Topics in Statistics, pp 546–548; Ch 18 Reviews of Individual Patient Data, p 572; Ch 19 Prospective Meta-analysis, p 584); Lilienfeld's Foundations of Epidemiology (Ch 15 Evidence-Based Practices, pp 305–309); Browner — Designing Clinical Research 5e (Ch 16, pp 518–520, 525; Ch 20 glossary, p 738); Oxford Textbook of Public Health 7e (Fig 5.13, p 1383 — annotated forest plot).

Recent-advances sources: 33782057 — The PRISMA 2020 statement: an updated guideline for reporting systematic reviews, *BMJ* 2021, doi:[10.1136/bmj.n71](https://doi.org/10.1136/bmj.n71); 35944924 — Reporting guideline for overviews of reviews of healthcare interventions: development of the PRIOR statement, *BMJ* 2022, doi:[10.1136/bmj-2022-070849](https://doi.org/10.1136/bmj-2022-070849); 38555664 — A tool to assess risk of bias in non-randomized follow-up studies of exposure effects (ROBINS-E), *Environment International* 2024, doi:[10.1016/j.envint.2024.108602](https://doi.org/10.1016/j.envint.2024.108602); 41626887 — Exploring the methodological quality and risk of bias in 200 systematic reviews: a comparative study of ROBIS and AMSTAR-2 tools, *Research Synthesis Methods* 2025, doi:[10.1017/rsm.2025.10032](https://doi.org/10.1017/rsm.2025.10032); 38501202 — Meta-analysis prediction intervals are under reported in sport and exercise medicine, *Scandinavian Journal of Medicine & Science in Sports* 2024, doi:[10.1111/sms.14603](https://doi.org/10.1111/sms.14603) (abstract-only); 40587852 — GRADE certainty ratings: thresholds rather than categories of contextualization (GRADE Guidance 41), *Annals of Internal Medicine* 2025, doi:[10.7326/ANNALS-25-00548](https://doi.org/10.7326/ANNALS-25-00548) (abstract-only); 39993313 — Development of prompt templates for large language model-driven screening in systematic reviews, *Annals of Internal Medicine* 2025, doi:[10.7326/ANNALS-24-02189](https://doi.org/10.7326/ANNALS-24-02189) (abstract-only); 40113231 — Is health technology assessment value for money? Estimating the return on investment of health technology assessment in India (HTAIn), *BMJ Evidence-Based Medicine* 2025, doi:[10.1136/bmjebm-2023-112487](https://doi.org/10.1136/bmjebm-2023-112487) (abstract-only); 37202933 — The prevalence of tuberculosis infection in India: a systematic review and meta-analysis, *Indian Journal of Medical Research* 2023, doi:[10.4103/ijmr.ijmr_382_23](https://doi.org/10.4103/ijmr.ijmr_382_23); 39421504 — Prevalence of tuberculosis infection among various risk groups in India: a systematic review and meta-analysis, *Indian Journal of Community Medicine* 2024, doi:[10.4103/ijcm.ijcm_36_24](https://doi.org/10.4103/ijcm.ijcm_36_24).



Qualitative research methods

Asked in: Paper I May 2019 · Paper I Nov 2016 (2×) — RGUHS MD Community Medicine

Definition — Qualitative research provides insight into experiences, meanings and culture **without using numeric measures, predesigned tests or closed-ended surveys**; the investigator engages with participants and interprets their narrative responses (Browner, Designing Clinical Research 5e, Ch 14, pp 454-455). Its findings are described **in words rather than numbers** (Abramson, Ch 15, p 144). It uses an **interpretive, naturalistic** approach that penetrates to the deeper significance the participant attaches to the topic (Cochrane Handbook, Ch 20, p 598).

When qualitative methods are used in community medicine

- ▶ To study **beliefs, perceptions and practices** about health and disease, prevention, treatment and the utilisation of traditional and other health care (Abramson, Ch 15, pp 144-145).
- ▶ To answer the "why" behind a problem, where a closed-ended survey only tells us "what" the problem is (Sunder Lal, Ch 6, p 270).
- ▶ To **develop or improve measurements** — semi-structured interviews identify what matters to participants, reveal how items may be misunderstood, and refine closed-ended questionnaire items (Browner, Ch 14, pp 464-465).
- ▶ To **capture complex, context-dependent or sensitive topics holistically**, and to **explore a new phenomenon** on which nothing is known (Browner, Ch 14, pp 480-481).
- ▶ As a **formative arm before a survey** — barriers to preventive service use were identified qualitatively in Haiti and then measured in a quantitative survey; equally as a **follow-up** to explain and expand quantitative findings, and as a source of hypotheses (Abramson, Ch 15, p 145).
- ▶ Within a **mixed-methods study**, providing individual-level data that make sense of group-level findings and improve the validity of the parent quantitative study (Browner, Ch 14, pp 464-465).
- ▶ In **social marketing and behaviour change communication** — focus groups are routinely used to position products and services such as contraceptives, oral rehydration solution (ORS) and immunisation (Sunder Lal, Ch 6, p 270).

Qualitative versus quantitative research

Feature	Quantitative study	Qualitative study
Research focus	Question established a priori	Question may change, or even be discovered, during the study
Participants	Specification by inclusion/exclusion criteria; sampling representative of target population	Flexible, inclusive criteria that may change; purposeful or snowball , may be non-representative
Designs	Cross-sectional, cohort, case-control, randomised trial	Observational, exploratory, descriptive, comparative
Logic	Predominantly deduction	Induction as well as deduction; hypotheses formed from observations

Feature	Quantitative study	Qualitative study
Sample size	Based on estimated effect size and power	Based on how many are needed to understand the concept; saturation
Data	Numbers, standard data format, coding scheme	Rich narrative description ; text, field notes, transcripts
Analysis	Pre-planned hypothesis testing by statistical inference	Iterative review of coded output and memos, seeking findings and counter-examples
How rigour is achieved	Representativeness; set procedures to eliminate bias	Purposefulness of sampling and active participation of the investigator in data collection

(Browner, Ch 14, Tables 14.1 and 14.3, pp 456-457, 472-473)

Qualitative study designs and approaches

Design / approach	Origin and aim	Principal data
Ethnography	Anthropology; immerse in the setting to understand the "way of life" of a group or institution over a prolonged period, producing <i>thick description</i>	Participant observation , field diaries, interviews
Interview-based study	Individual or group; explores understanding, experience and the meanings attached to them	In-depth interviews, focus groups
Case study	Detailed study of a bounded case; may combine qualitative and quantitative methods	In-depth and/or focus group interviews
Document-based research and content analysis	Analysis of records, policy documents, diaries, media	Written/textual excerpts
Participative (participatory) action research	Links research to action , manages power relations, values experiential knowledge	Mixed methods with community as partners
Phenomenology	Philosophy/psychology; describes the lived experience and its essence	Interviews (3-10 participants)
Grounded theory	Sociology; derives an abstract theory of a process grounded in participants' views	Multiple stages of data collection (20-30 participants)
Narrative research	Humanities; life stories retold as a chronology	1-2 individuals

(Oxford Textbook of Public Health 7e, Ch 31, pp 1147-1148; Creswell, Research Design, Ch 1 p 46, Ch 9 p 248)

Methods of collecting qualitative data

Method	How it is done	Best used for
In-depth interview (IDI)	Unstructured (oral/life history) or guided by a topic guide used flexibly; tape-recorded and transcribed; typically 20-30 interviewees	Individual perspectives, beliefs and experiences
Focus group discussion (FGD)	Thematic discussion in a small homogeneous group of 6-8 beneficiaries (Sunder Lal); Oxford quotes 5-10, range 4-12 , lasting up to 2 hours; run by a moderator with a separate note-taker , participants brought in by recruiters	Group interaction and collective reasoning — not individual perspectives
Participant observation	Observer lives with the community and becomes part of it; central method of ethnography	Actual (not reported) behaviour and practice
Non-participant observation	Observer keeps distance, has no formal role in the activity	Clinic and facility processes
Key informant interview	Conversation with community members holding special knowledge	Rapid orientation to a community
Document analysis	Records, diaries, official publications, open-ended written responses	Policy and institutional processes
Group discussion / <i>Mahila Swasthya Sangh</i> meetings, exit interviews, opinion polls	Rapid group methods conducted by anganwadi worker (AWW) and auxiliary nurse midwife (ANM)	Community and behaviour-change communication needs
Delphi method, projective techniques, transect walk	Iterative expert consensus; pictorial elicitation; a structured walk through a village or slum observing facilities and practices	Consensus, deeper perceptions, spatial data

(Oxford 7e, Ch 31, pp 1149-1152; Sunder Lal, Ch 6, pp 268-270; Abramson, Ch 15, p 145)

Sampling and sample size

- ▶ Sampling is **purposive, not random** — cases are chosen to give the **richest and most relevant** data (Oxford 7e, Ch 31, p 1154).
- ▶ **Theoretical sampling** (Glaser and Strauss) — the sample is adjusted as the theory develops, to cover as many angles as possible.
- ▶ **Snowball (chain-referral) sampling** — a few members of the group help find others; useful for hard-to-reach groups such as people who use illicit drugs.
- ▶ **Negative or deviant cases** are deliberately sought, to test and strengthen emerging explanations.
- ▶ **Saturation** — sampling continues serially until fresh data yield **no new themes or conceptual insight** (Browner, Ch 20, p 739).
- ▶ Rules of thumb: **6-8 sampling units** suffice for a homogeneous group, **up to 20** when seeking disconfirming evidence or maximum variation (Oxford 7e, p 1155); a grounded-theory study needs **20-25 interviews**, and **30-40 participants** is a sensible start for a modest single-investigator study (Browner, Ch 14, pp 468-469).
- ▶ The **logic** of the sampling strategy — not its size — is the criterion on which it is judged.

Analysis of qualitative data

- **Framework analysis** (Ritchie and Spencer) is the most widely used applied method, in five stages: familiarisation → identifying a thematic framework → indexing (coding) → charting → mapping and interpretation (Oxford 7e, Ch 31, pp 1156-1158).
- **Thematic coding** — marking sections of data with codes compiled in a **codebook** to link them to research concepts (Browner, Ch 20, p 739).
- Other purposes of analysis: **content analysis**, **narrative analysis**, **conversation analysis** (turn-taking, intonation) and **discourse analysis** (how knowledge is produced through language).
- Analysis is **iterative**, running alongside data collection, with **analytical memos** guiding further sampling.
- Software such as **NVivo** manages, retrieves and visualises data, but **the intellectual work of description and interpretation remains the researcher's** — software cannot conduct the analysis.

Ensuring rigour (quality) in qualitative research

Criterion (Lincoln and Guba)	Quantitative parallel	How it is achieved
Credibility	Internal validity	Rigour of technique, credibility and training of the researcher, stated philosophical assumptions; triangulation ; analysis of negative cases
Transferability	External validity / generalisability	Report enough detail on methods, setting and parameters for others to judge; theory carries the findings across settings
Dependability	Reliability	Consistency over time, researchers and methods; two independent coders
Confirmability	Objectivity	Confirmation by a source outside the research team; an audit trail of sampling, recording, coding and theme construction

Types of triangulation (Denzin, plus Janesick): **data source**, **researcher**, **theory**, **methodological**, and **interdisciplinary** (researchers from different professions, or non-researchers with a stake in the findings) (Oxford 7e, Ch 31, pp 1158-1159). Because the researcher inevitably influences the findings, the aim is not to eliminate bias but to **make it visible** through **reflexivity and transparency** (Oxford 7e, pp 1140-1141).

Ethical issues (ICMR National Ethical Guidelines 2017)

- A hypothesis may not be stateable at the outset and instruments **keep changing as the research progresses**; the ethics committee (EC) must be informed of changes and **appropriate re-consent** taken (Section 9, Box 9.3).
- **Informed consent is dynamic and negotiable**; where written consent is not possible, other means may be used and documented.

- ▶ **Community/gatekeeper consent** must be obtained from community heads or culturally appropriate local authorities, but **community tradition does not substitute for individual consent** unless a waiver has been granted (Box 9.4).
- ▶ **Prior EC permission with justification is required for audio or video recording** of participants' interviews.
- ▶ Participants may face **non-physical harm** — social, psychological, economic — which the researcher and EC must recognise in the cultural context.
- ▶ **Privacy and confidentiality** must be safeguarded; preliminary observation for note-making need not be submitted for review, but any ethical issue arising in that phase must be included in the proposal.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ Reproduce the above as a labelled ladder with these EXACT labels, bottom to top: Raw data (transcripts, field notes, images) → Organizing and preparing data for analysis → Reading through all data → Coding the data (hand or computer) → branching to Themes and Description → Interrelating themes/description (e.g. grounded theory, case study) → Interpreting the meaning of themes/descriptions; with a side box Validating the accuracy of the information arrowed back to the coding and preparation steps.

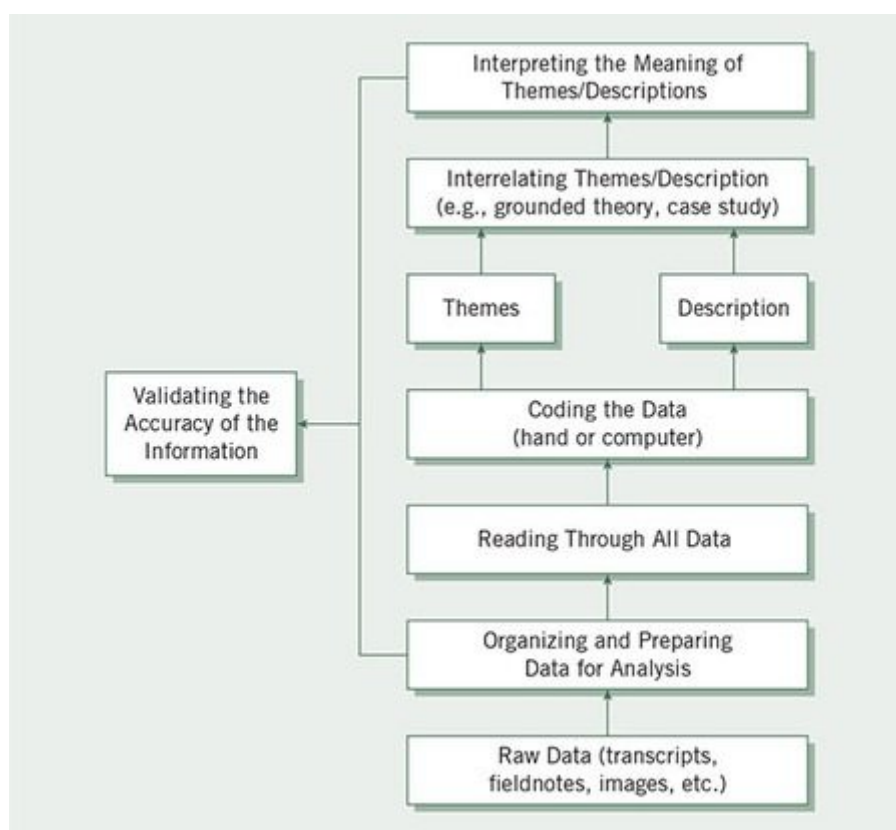
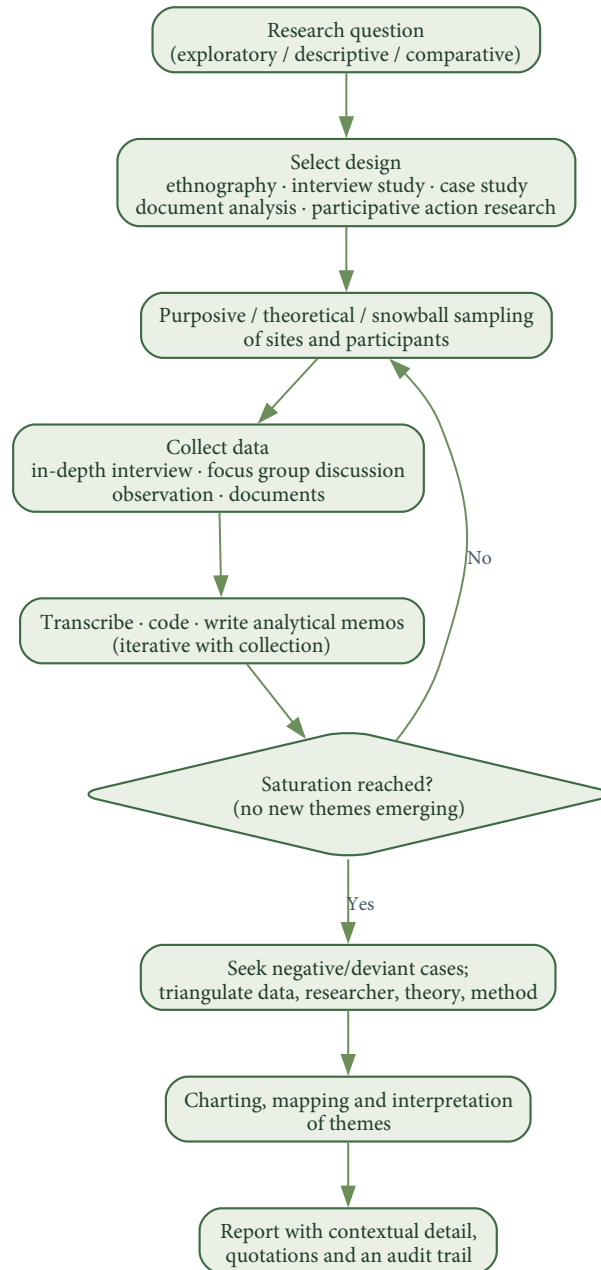
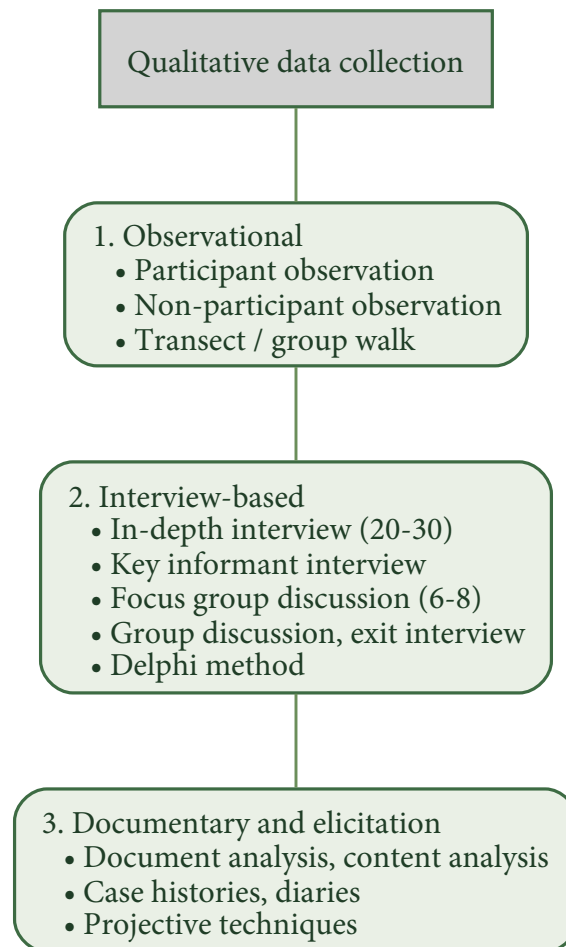


Fig 9.2 — Data Analysis in Qualitative Research (Creswell — Research Design, p.255)



Sequence of conducting a qualitative study, from research question to reporting



Classification of qualitative data collection methods

High-yield exam pointers

- ▶ **Definition line to write:** findings described **in words, not numbers**; answers "**why**", whereas a closed-ended survey answers "**what**".
- ▶ **FGD numbers:** 6-8 homogeneous participants (Sunder Lal); 5-10, range 4-12, up to 2 hours (Oxford 7e); one **moderator** + one **note-taker**. In-depth interviews typically 20-30; grounded theory 20-25 interviews (Browner) or 20-30 participants (Creswell); a modest single-investigator study 30-40 participants.
- ▶ **Sampling mnemonic** — PTSN-S: Purposive, Theoretical, Snowball, Negative case, until Saturation.
- ▶ **Rigour mnemonic** — "C-T-D-C": Credibility, Transferability, Dependability, Confirmability (Lincoln and Guba), with **triangulation** (data, researcher, theory, method, interdisciplinary) and an **audit trail**.
- ▶ **Framework analysis five stages:** familiarisation, thematic framework, indexing, charting, mapping and interpretation.
- ▶ **ICMR 2017 points:** consent is **dynamic and re-negotiable**; **gatekeeper consent does not replace individual consent**; **prior EC permission for audio/video recording**.
- ▶ **Interview vs focus group:** an interview reveals the **individual's** perspective; a focus group reveals **group interaction** — choose by research question, not by efficiency.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park's Textbook of Preventive and Social Medicine 28e, the spine for this bank, carries no dedicated section on qualitative research methods, so this answer is grounded on the Oxford Textbook of Public Health 7e, Designing Clinical Research 5e, Abramson, Sunder Lal, Creswell and the ICMR National Ethical Guidelines 2017; the corpus also contains no Indian textbook statement of a single agreed focus group size, which is why two ranges (6-8 and 5-10) are quoted with their sources.

SOURCE & COVERAGE

Oxford Textbook of Public Health 7e (Ch 31, Qualitative research imagination, pp 1139-1163); Browner — Designing Clinical Research 5e, 2022 (Ch 14, Qualitative Approaches in Clinical Research, pp 454-481; glossary Ch 20); Abramson — Research Methods in Community Medicine (Ch 15, Methods of Collecting Data, pp 144-146); Sunder Lal — Textbook of Community Medicine (Ch 6, Basic Statistics and its Applications and Research Methodology, pp 268-270); Creswell — Research Design (Ch 1, p 46; Ch 9, Qualitative Methods, pp 245-255); Cochrane Handbook for Systematic Reviews of Interventions (Ch 20, Qualitative research and Cochrane reviews, p 598); ICMR National Ethical Guidelines for Biomedical and Health Research Involving Human Participants, 2017 (Section 9, Social and Behavioural Sciences Research for Health, pp 121-126).



Social mapping

Asked in: Paper IV 04-Aug-2021 — RGUHS MD Community Medicine

Definition — Social mapping is a **participatory rural appraisal (PRA)** method in which the members of a community themselves draw a map of their own village or urban slum — on the ground or on paper — to depict its layout, households, infrastructure and social characteristics. In Indian community medicine practice, the spot map of a village or slum area prepared after a walk-through, showing the various facilities of that area, is what is called social mapping (Sunder Lal, Ch 6, p 269). Mapping is described as the "**mother of PRA methods**", because it is usually the first exercise done and it feeds every method that follows (Narayanasamy PRA, Ch 3, p 76).

What a social map shows

Element	Content
Who draws it	Community members themselves — the outsider only convenes and facilitates; PRA calls this " handing over the stick " and " they do it " (Narayanasamy PRA, Ch 2, pp 57-58)

Element	Content
Who participates	Key informants from every section of the community; men and women of all age groups, including children — some gather people, some gather materials, some draw, some give information, some copy the map to paper (Narayanasamy PRA, Ch 3, pp 63-64)
Medium	Preferably the ground ; alternatively paper
Materials	Chalk, stones, <i>rangoli</i> powder, pebbles, seeds, twigs, flowers, leaves, goat drops — locally available, so material or its size itself carries meaning
Unit depicted	The household — each house given a number and a symbol chosen by the participants
Information carried	Village layout, infrastructure, population, social stratification, chronic health cases, the handicapped, malnourished children, family planning and vaccination status, widows, the destitute; it can also expose social issues such as female infanticide and untouchability (Narayanasamy PRA, Ch 3, p 63)
Two map families in PRA	Social map (people and social characteristics) and resource map (land, water, trees, soil, cropping pattern); each has topical sub-maps

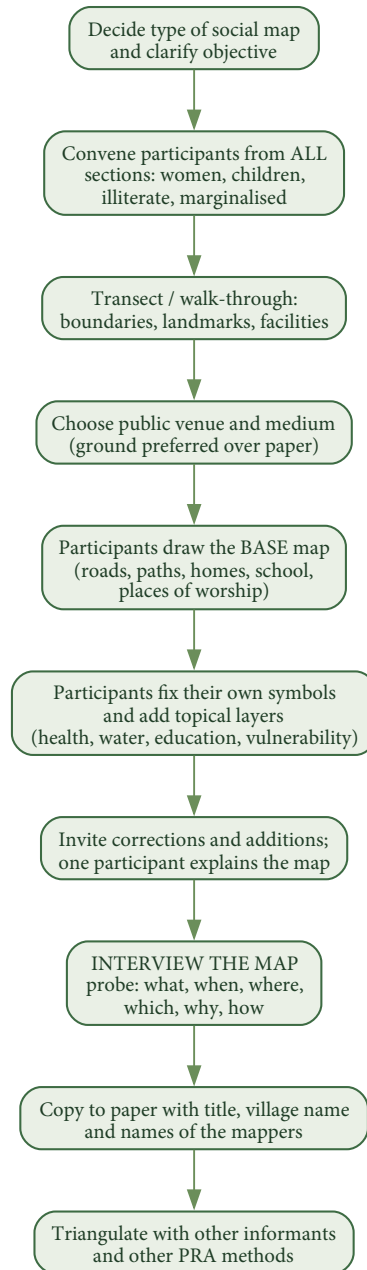
Types of topical social map

A single map overloaded with symbols becomes clumsy and hard to "interview". A plain **base map** of the village's physical structure is therefore drawn first, and a series of topical maps is built on it (Narayanasamy PRA, Ch 3, p 66).

Topical map	What it depicts
Base map	Roads, paths, lands, homes; school, temple, church, mosque, forest, hills — the physical skeleton
Health map	Chronic illness and disability; the sick classified by type of disease, location and social group; pregnant women with month of pregnancy; childless couples; family planning status; users and non-users of contraceptives; women with severe anaemia; deaths by category; children affected by polio; malnourished children; hazards such as pollution and zones of open defecation
Well-being / ill-being map	Local categories of prosperity and deprivation; forms the basis for well-being ranking
Water and sanitation map	Wells, ponds, piped supply, taps, latrines, wastewater and refuse disposal
Education map	School drop-outs with their socio-economic background; age-wise and sex-wise children in and out of school
Leadership map	Where influence sits — leaders, <i>panchayat</i> members, gatekeepers
Indigenous skills map	Local skills and traditional practitioners
Vulnerability map	Vulnerability = exposure to risk + inability to cope : female-headed households, child labourers, street children, orphans, commercial sex workers, people living with HIV/AIDS, the aged, widows, bonded labourers, malnourished children, migrant families, families living near pollution zones, families that eat only once a day (Narayanasamy PRA, Ch 3, pp 71-72)

Methods — how a social mapping exercise is conducted

- ▶ **Decide the type of map and clarify the objective.** Fix in advance what information is wanted and why it is wanted; PRA's principle of **optimal ignorance** says collect only what is sufficient for the stated objective and no more (Narayanasamy PRA, Ch 2, p 53).
- ▶ **Convene the participants.** Find community members who know the area and are willing to share; deliberately include the illiterate, women, the marginalised and children, so that a literate male elite does not speak for the village.
- ▶ **Do a transect (walk-through) first.** Walk the village or slum with the participants, fix compass directions and boundaries, get a spatial overview from a high vantage point, and observe access, schools, *anganwadis*, health facility, water sources, refuse and wastewater disposal, housing, worship places, community centres, cattle sheds and pig rearing while interviewing as you walk (Sunder Lal, Ch 6, p 269).
- ▶ **Choose the place and the medium.** A public, easily accessible venue that can hold a large group; the ground is preferred because it is user-friendly to local people.
- ▶ **Help people get started, then withdraw.** Facilitate a simple outline base map of key features and landmarks; once confidence is gained, ask for further layers of information. Do not impose your own way of drawing.
- ▶ **Let the participants fix the symbols.** They discuss among themselves and decide the symbol for each variable; the facilitator gives them adequate time to do so and supplies no ready-made legend.
- ▶ **Encourage corrections and additions.** Ask one participant to explain the map aloud so that omissions and errors are added or altered by the group itself.
- ▶ **"Interview the map."** Use the map as the visual stimulus and probe with the PRA helpers — **what, when, where, which, why, how** — to draw out narrative on the life of the village.
- ▶ **Add a household census if needed.** Two methods exist — the **card method** (a 3 × 4 inch card per household, numbered as on the social map, symbols filled in and compiled later; first innovated at Kethanaickenpatti village, Tamil Nadu) and the **social map method** (census data shown on the map itself) (Narayanasamy PRA, Ch 3, pp 77-80).
- ▶ **Transfer the ground map to paper** with a title, the name of the site or village, and the **names of the participants** — this credits the mappers and creates a permanent record.
- ▶ **Triangulate** the information with others in the locality, then thank the participants.



Steps in conducting a social mapping exercise (PRA)

Ground versus paper as the medium

Criterion	Ground	Paper
Nature of participation	Participatory, inclusive, democratic	A few dominate
Type of participants	All sections — the illiterate, marginalised, women, children	The literate
Number of participants	Quite large	Two or three
Time	Less, as work is divided	More, as one person draws
Cost	Almost none	Paper, pens, pencils, stickers
Scope for triangulation	Very high — many observers point out omissions	Limited, medium unfamiliar

Criterion	Ground	Paper
Extent of information	A great deal	Less
Scope for modification	Large, easily altered	Small
Main problem	Lack of permanency	Low participation, rigidity

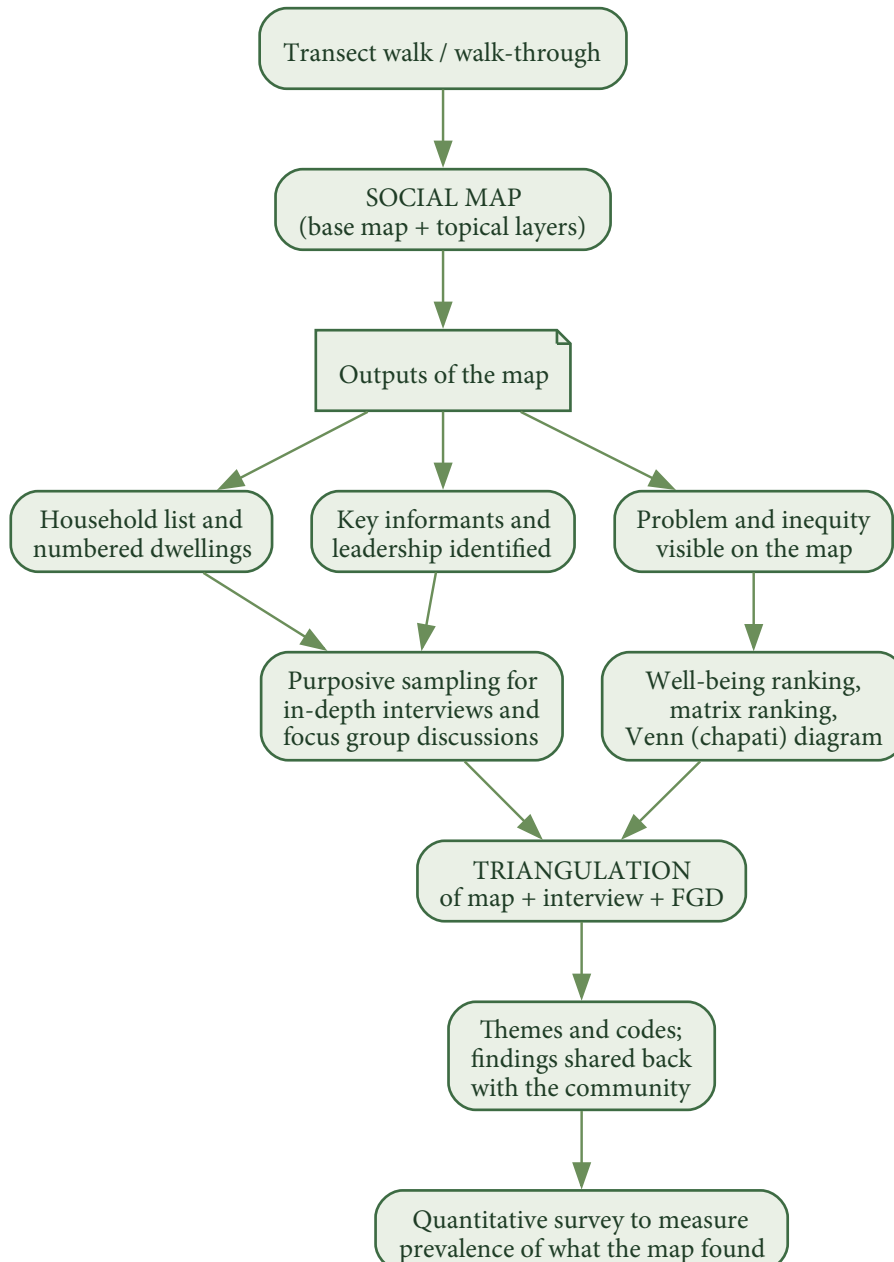
(Narayanasamy PRA, Ch 3, Table 3.1, p 75)

Use in qualitative studies

Use	How the social map serves the qualitative researcher
Icebreaker and rapport	Creates a convivial atmosphere in which outsider and community interact as long-standing friends — the usual entry point of fieldwork
Sampling frame for purposive selection	The numbered household list generated by the map is used to select information-rich informants for in-depth interview and focus group discussion ; qualitative studies use purposeful, not random, selection (Browner DCR 5e, Ch 14, pp 454-455)
Visual elicitation	"Interviewing the map" yields narrative on attitudes, perceptions, motivations, feelings and behaviour that a closed-ended questionnaire cannot reach (Abramson, Ch 15, p 145)
Triangulation	The map is cross-checked against the transect walk, key informant interview and focus group discussion — the use of more than one qualitative method to see whether they reach a common conclusion (Abramson, Ch 15, p 145)
Sequencing of methods	Social map → well-being / ill-being analysis → Venn (<i>chapati</i>) diagram of institutions → semi-structured interview; each step cross-checks and enriches the previous one (Narayanasamy PRA, Ch 22, pp 341-342)
Surfacing inequity	Differential access becomes visible on the map — in Pillaianatham village a Harijan colony of 40 households had not one water tap, while streets of 25-35 higher-caste households had two or three taps each (Narayanasamy PRA, Ch 3, p 72)
Identifying the individual case	While a reproductive and child health map was being drawn, the women identified a severely anaemic pregnant woman from a very poor household who was then taken for treatment — the map produced not only the problem but its solution
Monitoring and evaluation	With a baseline map, the community itself checks whether every child has been vaccinated, whether every poor family holds a ration card, and whether welfare schemes are reaching beyond a few families
Complementing quantitative work	Qualitative work gives " culture-specific maps that help improve the fit of programmes to people" — it shows the presence of beliefs and behaviours, not their numerical prevalence , which a subsequent quantitative survey must measure (Abramson, Ch 15, pp 144-145)

Distinguish from a spot map. A *spot map* is drawn by the investigator by plotting cases on an existing map to show clustering — as John Snow did for cholera in the Golden Square district of London in 1854, which pointed to the Broad Street pump (Park 28e, Ch 3, pp 86-87). A *social map* is drawn by the community itself and depicts the social structure it lives in.

Programme use in India. Health workers prepare simple local area working maps for mass campaigns on pulse polio immunisation, malaria and other national health programmes, showing facility locations, high-risk areas and hot spots (Sunder Lal, Ch 6, p 269). In routine immunisation microplanning this is formalised: master listing of all areas and **area mapping (Form SC-2)** precedes the house-to-house **head count survey (Form SC-3)** and the **session due list (Form SC-5)**; the map need not be to scale but must carry demarcation of areas for the ASHA, anganwadi worker and mobilisers and of **high-risk areas**. The head count survey is repeated **every six months** and high-risk-area validation is checked for the **last six months** (Routine Immunization Manual for Medical Officers 2024, Ch 4).



Place of the social map in a community-based qualitative study

Merits and limitations

Merits	Dangers and limitations
Easy to understand and construct; participants are comfortable starting with a map before moving to complex diagrams or sensitive issues	Outsiders overemphasise the product at the expense of the process and the discussion
Inclusive — the illiterate, the marginalised, women and children can all take part	Outsiders tend to intervene in the process and to impose their own way of drawing
Rapid, almost cost-free, uses locally available material	Too much information on one map makes it clumsy and hard to use, analyse and interpret
Owned and retained by the community for its own monitoring and evaluation	A ground map has no permanency unless copied to paper
Very high scope for triangulation because many people watch and correct	A large village with a single team takes long; participants' interest wanes
Generates both the sampling frame and the qualitative narrative in one exercise	Mapping areas where land tenure and boundary rights are uncertain can bring existing conflicts to the surface
Findings can be fed straight back to the <i>panchayat</i> , e.g. to cross-check the government's list of poor households	Mapping stigmatising attributes (HIV status, caste, defecation zones) risks breach of confidentiality and harm; gatekeeper/community consent must precede individual consent , and community tradition does not substitute for individual consent unless a waiver is granted by the ethics committee (ICMR National Ethical Guidelines 2017, section 9)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Social map of a village (label exactly these):

- ▶ Village boundary with **compass direction** (N)
- ▶ **Approach road**, internal lanes and paths
- ▶ **Households**, each numbered, with the participants' own symbols (e.g. different symbol for a household with a chronic illness, a malnourished child, a pregnant woman, a widow or a female-headed household)
- ▶ **Sub-centre / primary health centre, anganwadi, school**
- ▶ **Places of worship** (temple / church / mosque), *chaupal* or community centre
- ▶ **Water sources** — well, pond, hand pump, piped taps, overhead tank
- ▶ **Sanitation** — latrines, zones of open defecation, refuse and wastewater disposal points
- ▶ **Cattle sheds / pig rearing areas**
- ▶ **High-risk area (HRA)** shaded, migrant settlement marked
- ▶ **Legend / key of symbols** decided by the community, plus **title, village name, date and names of the mappers**

High-yield exam pointers

- ▶ Sunder Lal's definition: a **spot map of the village/slum depicting the various facilities**, prepared after a walk-through, is social mapping; mapping is the "**mother of PRA methods**".
- ▶ PRA methods (name them): **village profile map (resource and social mapping), time line / trend analysis, seasonality studies, matrix ranking, transect walk, Venn (*chapati*) diagram, focus group discussion** (Sunder Lal, Ch 6, p 269).
- ▶ PRA principles: reversal of learning, learning rapidly and progressively, offsetting biases, optimising trade-off (**optimal ignorance**, appropriate imprecision), triangulation, seeking complexity and diversity, sequencing, **they do it** ("**handing over the stick**"), self-critical awareness, sharing.
- ▶ Topical social maps: **health, well-being/ill-being, water and sanitation, education, leadership, indigenous skills, vulnerability**.
- ▶ **Vulnerability = exposure to risk + inability to cope**.
- ▶ Ground beats paper on participation, cost, time, triangulation and information; paper wins only on permanency.
- ▶ Qualitative maps show **presence, not prevalence** — always follow with a quantitative survey for numbers.
- ▶ Programme link: RI microplanning — **area mapping (SC-2) → head count survey (SC-3) → due list (SC-5)**; head count survey and high-risk-area validation **every 6 months** (Routine Immunization Manual 2024).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The RGUHS corpus carries no single "social mapping" chapter — the definition and the Indian programme usage come from Sunder Lal and the Routine Immunization Manual 2024, while the detailed procedure, topical map types and merits/limitations come from Narayanasamy's *Participatory Rural Appraisal: Principles, Methods and Application*; the figure plates of that book are blank in the scanned corpus, so a "Diagram to draw" box replaces a real plate.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, NFHS-5 (2019-21), e-RaktKosh 2024, WHO Ending Cholera roadmap 2030, WHO NTD road map

Programme and policy updates (India)

- ▶ **Household enumeration is now the front end of Indian campaign microplanning — the social map's numbered household list has become a digital line list.** A pre-emptive two-dose oral cholera vaccine campaign in Bishnupur-II block, South 24 Parganas, West Bengal (two phases, October and November 2024) opened with a household census that enumerated 45,073 people across three gram panchayats and generated a vaccination line list carrying unique household identifiers; door-to-door delivery by ASHAs and anganwadi workers on that census-based microplan reached 66.4% for one dose and 63.1% for both doses, with 95.1% second-dose retention, only mild AEFI (0.3% of 58,390 doses), and a delivery cost of INR 13.24 per person (INR 416.44 per fully vaccinated person including vaccine) [PMID 42092212]
- ▶ **Facility mapping has moved from village level to national infrastructure planning.** A pan-India geospatial model of blood banking facilities across 735 districts, using 2024 e-RaktKosh data, counted 5,679 facilities (4.13 per million people); 93.86% of the population lay within 60 minutes of one by motorised transport but only 16.21% within a 30-minute walk, with north-eastern states and island union territories worst served [PMID 42019942]

Current national data

- ▶ **NFHS-5 (2019-21) is the round to quote, and it is now being analysed district-wise rather than state-wise.** NFHS-5 puts elevated random blood glucose (≥ 140 mg/dL or on diabetes medication) at 13.5% of women and 15.6% of men aged 15 years and over; a biomarker analysis covering all 707 NFHS-5 districts found the district figure for women ranging from about 6% in parts of the north-east to over 30% in Punjab, Kerala and Himachal Pradesh, and multiscale geographically weighted regression fitted far better than ordinary least squares (adjusted R^2 0.834 vs 0.650), meaning the determinants themselves change with place — obesity dominating in southern districts, early marriage in central and northern ones [PMID 42057028]
- ▶ **Composite district indices built on NFHS-5 are the new sub-state planning tool.** A maternal and child health index for all 75 districts of Uttar Pradesh weighted nine NFHS-5 (2019-21) indicators by analytic hierarchy process (skilled birth attendance and antenatal visits at 22% each), ranked districts by TOPSIS and mapped five performance bands; Barabanki, Mahamaya Nagar and Unnao ranked highest, eastern UP and Bundelkhand clustered in the lower bands, and the lowest-ranked districts were the urbanised Gautam Buddha Nagar, Kanpur Nagar and

Ghaziabad — their poor scores read as slum and peri-urban inequality hidden inside a district average, which is exactly what a village or slum social map exists to expose [PMID 41479241]

Elimination, surveillance and disease status

- ▶ **Tuberculosis (NTEP):** a Mysuru district study of 8,459 notifications (2017-19) combined the analytic hierarchy process with GIS to rank priority groups rather than merely map cases — urban men aged 40-59 and 20-39 years emerged as "very high" priority, with 64.8% of cases urban and Mysuru city alone contributing 36.5%; the national burden is cited as 2.69 million cases a year [PMID 42295326]
- ▶ **Malaria — community mapping as larval-source surveillance:** in an isolated village in Nhamatanda district, Mozambique, ten focus groups with 94 residents sketched maps of their community and marked permanent and temporary mosquito breeding sites, which were then visited and GPS-referenced, yielding 125 potential sites (36 permanent, 89 temporary); who sat in the group changed what reached the map — boys aged 10-13 named the most sites and men spoke more freely than the women — and malaria cases in under-fives (96 of 156 children PCR-positive) were georeferenced onto the same map, the authors proposing the method for national malaria programmes building larval surveillance systems [PMID 39210393]
- ▶ **Mass drug administration for neglected tropical diseases:** 14 transect walks with community leaders, followed by 14 social mapping sessions of six to eight members each — deliberately separated by sex, age and community role so that leaders did not speak over members — in Kaduna and Ogun states, Nigeria, surfaced medicine-delivery venues beyond those the programme already used: Qur'anic and *almajiri* schools, football fields, mechanics' shops, film-viewing centres, a shrine and the village head's palace; locally available sticks, stones and leaves let residents map whatever their literacy level, and the disaggregated groups exposed that custom keeps women out of the palace and largely out of the mosque [PMID 36960807]

Guideline and legislative updates

- ▶ **Reporting standards for participatory mapping have now been proposed.** A scoping review following PRISMA-ScR (39 empirical studies retained from 2,240 records across seven databases) found that what counts as "participation" and the naming of methods remain unsettled, and set out a three-part remedy — a harmonised participatory-mapping framework, a checklist of mapping components, and reporting standards for broader impacts such as community-level health effects. It also fixes the modern vocabulary an examiner may use: participatory/community mapping, public participation GIS (PPGIS), participatory GIS (PGIS), volunteered geographic information, photomapping, activity-space mapping, body mapping and sketch mapping [PMID 41812314]

Global and WHO developments

- ▶ **Community engagement now has a graded vocabulary** — community-oriented, community-based, community-managed and community-owned — applied to the recurrent Nipah virus outbreaks of Bangladesh and India, where no approved vaccine or therapeutic exists and prevention rests on ecological and behavioural pathways such as contaminated date-palm sap; participatory, locally grounded methods are argued to improve acceptability, early detection and outbreak response [PMID 42606920]

- **Participatory dialogue reaches national policy only when it is timed to the policy cycle.** The Wellcome Trust's Responsive Dialogues framework (groundwork → community conversations → post-conversation sharing) was evaluated in Malawi and Thailand: Malawi's community-generated recommendations, including tighter regulation of antibiotic sales, entered the revised National Action Plan on AMR, while Thailand's locally specific proposals did not survive into the final national strategy. Entrenched hierarchy — village heads, deference to seniority — had to be actively managed with pre-event briefings, co-developed ground rules and rotating leadership roles, which is the same power problem PRA answers with "handing over the stick" [PMID 42246464]

SOURCE & COVERAGE

Sunder Lal — Textbook of Community Medicine (Ch 6, Basic Statistics and its Applications and Research Methodology); N. Narayanasamy — Participatory Rural Appraisal: Principles, Methods and Application (Ch 2 Concept and Principles of PRA, Ch 3 Participatory Mapping, Ch 22 Application of PRA Methodology); Abramson — Research Methods in Community Medicine (Ch 15 Methods of Collecting Data, Ch 34 Community-Oriented Primary Care); Browner — Designing Clinical Research 5e, 2022 (Ch 14 Qualitative Approaches in Clinical Research); Park's Textbook of Preventive and Social Medicine 28e (Ch 3 Principles of Epidemiology and Epidemiologic Methods); Routine Immunization I Manual for Medical Officers, 2024 (Ch 4 Routine Immunization Microplanning); ICMR National Ethical Guidelines for Biomedical and Health Research Involving Human Participants, 2017 (Section 9, Social and Behavioural Sciences Research for Health).

Recent-advances sources: 42092212 — Mass vaccination with two-dose oral cholera vaccine in vulnerable populations, using existing public health infrastructure from planning to delivery through multi-sectoral collaboration in West Bengal, India, *J Infect Public Health* 2026, doi:[10.1016/j.jiph.2026.103237](https://doi.org/10.1016/j.jiph.2026.103237) (abstract-only); 42019942 — Timely geographic access to blood banking facilities: a pan-India modelling study, *Transfus Med* 2026, doi:[10.1111/tme.70075](https://doi.org/10.1111/tme.70075) (abstract-only); 42057028 — Subnational inequalities and spatially varying associations of elevated blood glucose among women in India: a national district-level biomarkers analysis, *BMC Public Health* 2026, doi:[10.1186/s12889-026-27482-2](https://doi.org/10.1186/s12889-026-27482-2); 41479241 — District-level maternal and child health index in Uttar Pradesh: a GIS-based analysis using AHP-TOPSIS on NFHS-5 data, *Prim Health Care Res Dev* 2026, doi:[10.1017/S146342362510073X](https://doi.org/10.1017/S146342362510073X); 42295326 — Prioritizing the tuberculosis burden by the analytic hierarchy process for effective targeted intervention, *Geospat Health* 2026, doi:[10.4081/gh.2026.1454](https://doi.org/10.4081/gh.2026.1454) (abstract-only); 39210393 — Community participatory mapping of malaria mosquito breeding sites in Mozambique, *Malar J* 2024, doi:[10.1186/s12936-024-05084-w](https://doi.org/10.1186/s12936-024-05084-w); 36960807 — Lessons from participatory community mapping to inform neglected tropical disease programmes in Nigeria, *Int Health* 2023, doi:[10.1093/inthealth/ihac074](https://doi.org/10.1093/inthealth/ihac074); 41812314 — Synergies of participatory mapping and health geography: a scoping review, *Soc Sci Med* 2026, doi:[10.1016/j.socscimed.2026.119104](https://doi.org/10.1016/j.socscimed.2026.119104) (abstract-only); 42606920 — Community engagement strategies for preventing recurrent Nipah virus outbreaks in Bangladesh and India: adapting a framework for outbreak preparedness and response, *J Glob Health* 2026, doi:[10.7189/jogh.16.03024](https://doi.org/10.7189/jogh.16.03024) (abstract-only); 42246464 — 'Responsive Dialogues', an engagement framework for antimicrobial resistance: a cross-country qualitative evaluation in Thailand and Malawi, *Glob Health Action* 2026, doi:[10.1080/16549716.2026.2676473](https://doi.org/10.1080/16549716.2026.2676473).

♦ ♦ ♦

Good clinical practice — guidelines and informed consent

Asked in: Paper I 07-May-2022 — RGUHS MD Community Medicine

Definition — Good Clinical Practice (GCP) is "an international ethical and scientific quality standard for designing, conducting, recording, and reporting trials that involve the participation of human subjects" (Browner, *Designing Clinical Research* 5e, Ch 18, pp 576–577). Compliance provides public assurance that the rights, safety and well-being of trial participants are protected and that the reported data are credible — GCP therefore has two inseparable arms, an **ethical** arm (protection of the participant) and a **scientific/quality** arm (credibility of the data).

Why GCP was needed, and the guidelines that constitute it

The need for these protections arose largely from abuses — the medical experiments conducted by the Nazis and studies conducted elsewhere without the full knowledge of the subjects (Barr, *Good Clinical Practice*, Ch 35 in *Methods and Applications of Statistics in Clinical Trials*, 2014, p 438). The regulations and guidances promulgated under national drug law, European Union rules and the International Conference on Harmonisation (ICH) are **collectively termed** "Good Clinical Practices" (GCP Ch 35, p 439).

Instrument	Year	What it contributes to GCP
Nuremberg Code	1947	The first international treatise on research ethics: voluntary consent of the human subject is <i>absolutely essential</i> — free, without force, fraud, deceit or duress, and given by a person with legal capacity (ICMR 2017, p 16; Sunder Lal, Ch 1, p 58)
WMA Declaration of Helsinki	1964; seven revisions, latest Fortaleza Oct 2013	Ethical principles for medical research involving human subjects; consent preferably written; every study registered in a publicly accessible database before recruitment of the first subject (ICMR 2017, pp 16, 35)
Belmont Report	1979	For the first time enunciated the three basic ethical principles for research involving human subjects — respect for persons, beneficence and justice (ICMR 2017, p 16)
ICH E6(R1) Good Clinical Practice	1996	The consolidated international GCP guideline; quality-control standard used by regulators in Europe, the United States and Japan (ICMR 2017, p 16; Browner 5e, Ch 18, p 577)
ICH E6(R2) Integrated Addendum	2016	Revision of E6(R1) — risk-based quality management, data integrity and electronic records (ICMR 2017, p 16 and reference 12, p 158)

Instrument	Year	What it contributes to GCP
US Code of Federal Regulations Title 21 (Parts 11, 50, 58, 312, 812)	—	Statutory GCP detail: protection of human subjects, Investigational New Drug (IND) application, Good Laboratory Practice, electronic records and signatures (GCP Ch 35, pp 440–441)
Indian GCP Guidelines, Central Drugs Standard Control Organization (CDSCO)	2001 (ICMR's own reference list dates the CDSCO text 2004)	The national GCP standard binding on Indian trials (ICMR 2017, p 16 and reference 4, p 158)
New Drugs and Clinical Trials Rules — notified by the Ministry of Health and Family Welfare, G.S.R. 227(E) dated 19 March 2019, administered through CDSCO	2019	The current Indian regulatory framework for clinical trials — ethics-committee registration, compensation for trial-related injury, and fixed approval timelines; supersedes Schedule Y
Schedule Y, Drugs and Cosmetics Act 1940 and Rules 1945	amended June 2005; Rules further amended 2013	The former Indian regulatory requirement for clinical trials, as stated in ICMR 2017 (p 16) — now replaced by the 2019 Rules
ICMR National Ethical Guidelines for Biomedical and Health Research Involving Human Participants	2017	12 general principles; ethics committee, consent, compensation and public-health-research ethics
Clinical Trials Registry–India (CTRI)	Launched by ICMR 20 July 2007; registration made mandatory by CDSCO 15 June 2009	Free online public record linked to the WHO registry; prospective registration before enrolment — mandatory for trials under the Drugs and Cosmetics Act, voluntary for other biomedical and health research (ICMR 2017, section 3.7, pp 35–36)

The binding Indian rule is explicit: **all clinical trials must be conducted in accordance with the Indian GCP guidelines, the Declaration of Helsinki (2013 or later), the National Ethical Guidelines 2017, and the prevailing rules under the Drugs and Cosmetics Act 1940 (ICMR 2017, section 7.1.3, p 85).**

The 12 general ethical principles that GCP operationalises

The four basic principles — respect for persons (autonomy), beneficence, non-maleficence and justice — are expanded by ICMR into 12 principles applicable to all biomedical, social and behavioural health research (ICMR 2017, section 1.1, pp 18–19).

#	Principle	Core requirement
1	Essentiality	Use of human participants is essential after considering all alternatives; vetted by an independent ethics committee (EC)
2	Voluntariness	Right to agree, decline or withdraw at any time; safeguarded by the informed consent process
3	Non-exploitation	Equitable selection; benefits and burdens distributed fairly; safeguards for vulnerable groups

#	Principle	Core requirement
4	Social responsibility	Research must not create or deepen social divisions
5	Privacy and confidentiality	Identity and records confidential, access restricted; breach only for valid scientific or legal reason in consultation with the EC
6	Risk minimisation	Risks minimised at all stages; care and compensation if harm occurs
7	Professional competence	Planned, conducted, evaluated and monitored by appropriately qualified, experienced, trained persons
8	Maximisation of benefit	Design to maximise benefit to participants and/or society
9	Institutional arrangements	Institution provides research governance, infrastructure, manpower, funds, training
10	Transparency and accountability	Registries and publication; conflict of interest declared; records retained for external scrutiny/audit
11	Totality of responsibility	All stakeholders answerable for their actions
12	Environmental protection	Protection of environment and resources at all stages

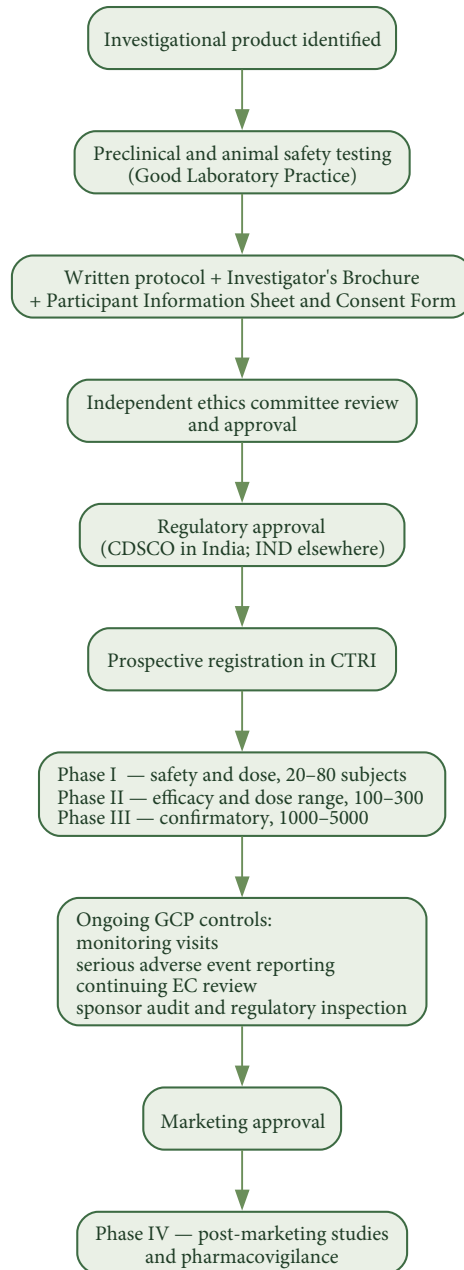
Scope — what GCP actually governs

Domain	GCP requirement
Scientific basis	Design supported by preclinical, animal and other data; adequate preclinical data must indicate the intervention is acceptably safe before humans are exposed (Browner 5e, Table 18.2; ICMR 2017, pp 87–88)
Ethical review	Protocol, amendments and all consent materials reviewed by an independent EC/institutional review board of medical and non-medical members, with continuing review (GCP Ch 35, p 439)
Protocol	Written protocol carefully followed — objectives, design, endpoints, measures to minimise bias, selection and withdrawal criteria, treatment, discontinuation criteria, statistics, quality control, data handling (GCP Ch 35, pp 439–440)
Investigator's Brochure (IB)	Compilation of clinical and non-clinical data on the product, supplied to the EC to judge the risk–benefit balance and to each investigator (GCP Ch 35, p 440)
Regulatory application	Investigational New Drug (IND)/Investigational Device Exemption approval before any human testing; in India, CDSCO permission (GCP Ch 35, pp 440–441; ICMR 2017, pp 87–88)
Investigational product	Manufactured, handled and stored per current Good Manufacturing Practice (GMP) and used per protocol (GCP Ch 35, pp 440–441)
Personnel	Investigators and clinical staff trained and qualified; research conducted only by scientifically qualified persons under a clinically competent physician (Browner 5e, Table 18.2; GCP Ch 35, p 441)
Procedures	All clinical and laboratory procedures meet quality standards; standard operating procedures (SOPs) cover every aspect of study conduct (Browner 5e, Ch 18, p 577)

Domain	GCP requirement
Data and documentation	Case Report Forms (CRFs) complete and supported by source documents; records legible; corrections by cross-out, dated and initialled, never obliterating original data; retained ≥ 2 years after approval of the marketing application (GCP Ch 35, pp 444–445)
Monitoring	Sponsor ensures monitoring to verify participant protection, data accuracy and verifiability from source documents, and protocol/GCP compliance (GCP Ch 35, p 444)
Safety	Adverse experiences evaluated and reported; regulator and all investigators promptly informed of significant new adverse effects; all adverse events reported to the EC on a pre-planned timetable (GCP Ch 35, p 443; ICMR 2017, section 2.6.3, pp 24–25)
Audit and inspection	Sponsor's quality-assurance audits during the study; regulatory inspections of sites, sponsors, monitors and ECs; a clinical hold stops dosing when subjects face unreasonable risk (GCP Ch 35, pp 445–446)
Reporting	Statistical methods prespecified and followed; results clearly and fairly reported (Browner 5e, Table 18.2)

Responsibilities of the stakeholders

Stakeholder	Key GCP obligations
Sponsor	Ultimate responsibility for legal and regulatory compliance; obtains regulatory approval; manufactures and labels investigational product; selects qualified investigators and monitors; provides the IB and training; ensures monitoring; evaluates and reports adverse experiences; initiates, withholds or discontinues the trial; refrains from commercialising the investigational product; brings a deviating site into compliance or stops the study (GCP Ch 35, pp 443–444)
Investigator	Conducts the trial per regulations and the investigational plan; protects participants' rights, safety and welfare; controls the investigational drug; keeps CRFs and source documents current (GCP Ch 35, p 444). In an investigator-initiated academic trial the investigator carries the dual role of investigator and sponsor (ICMR 2017, section 7.16.1, pp 101–102)
Monitor	Oversees progress before, during and after the trial; verifies participant protection, data accuracy and protocol/GCP compliance; must be trained and familiar with the product, protocol and consent form (GCP Ch 35, p 444)
Ethics committee (EC)	7–15 members, multidisciplinary and multisectoral, preferably 50% non-affiliated; quorum minimum five including medical/non-medical and at least one non-affiliated member; initial and continuing review, review of amendments, serious adverse events and consent documents; may halt a study for serious protocol violations (ICMR 2017, sections 4.3 and 4.11, pp 42–46, 60–61)
Contract Research Organization (CRO)	Assumes, in writing, one or more transferred sponsor obligations (GCP Ch 35, p 444)



Regulatory and ethical pathway of a clinical trial under Good Clinical Practice

Informed consent — process, elements and documentation

Informed consent is the process by which a competent person, having received and understood adequate information in a language and manner she or he understands, **voluntarily** agrees to take part in research. The **informed consent document (ICD)** has two parts — the **participant information sheet (PIS)** and the **informed consent form (ICF)**; both must be approved by the EC before any enrolment, and it is the researcher's primary responsibility to obtain written consent from the participant or the **legally acceptable/authorised representative (LAR)** (ICMR 2017, section 2.2.1, p 21).

Essential elements of the informed consent document (ICMR 2017, Box 5.1, pp 64–65):

#	Element
1	Statement that it is research

#	Element
2	Purpose and methods in simple language
3	Expected duration and frequency of contact, number of participants, types and methods of data collection
4	Benefits reasonably expected to the participant, community or others
5	Foreseeable risks, discomfort or inconvenience
6	Extent to which confidentiality can be maintained, its limits and the consequences of a breach
7	Payment/reimbursement for participation and incidental expenses
8	Free treatment and/or compensation for research-related injury or harm
9	Freedom to participate or withdraw at any time without penalty or loss of entitled benefits
10	Identity of the research team and contact persons — principal investigator for study queries, and EC Chairperson/Member Secretary or helpline for appeals against ethical violations

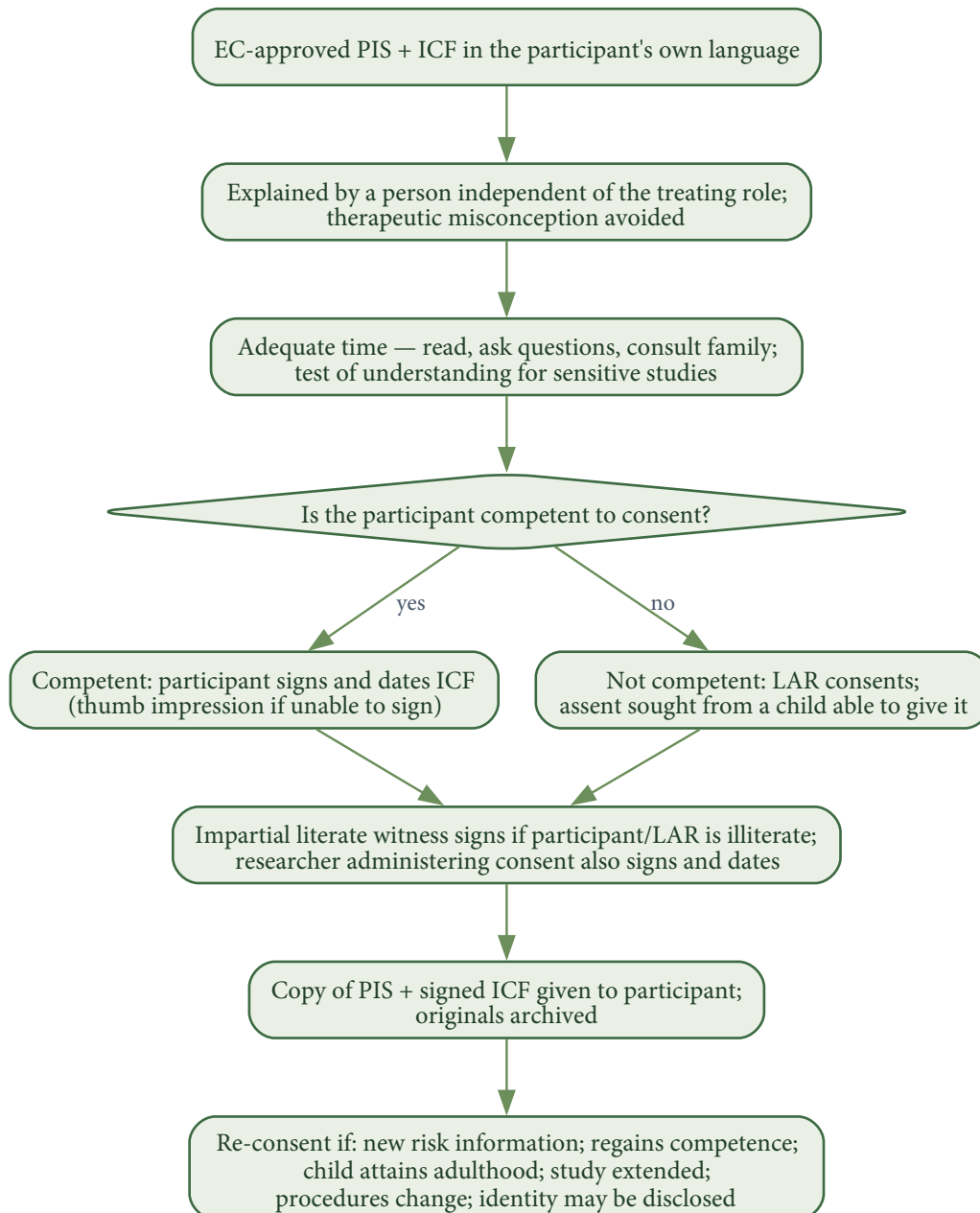
Additional elements added when the study type demands (ICMR 2017, Box 5.1, pp 64–65):

- ▶ Alternative treatments available.
- ▶ Pre- and post-test counselling where a stigmatising condition may be revealed (for example human immunodeficiency virus infection, genetic disorders).
- ▶ Insurance coverage for the participant.
- ▶ For stored biological material — period of storage, secondary use, sharing, right to withdraw the sample, risk of biologically sensitive findings, benefit sharing on commercialisation, and the publication plan including photographs and pedigree charts.

Rules that make the consent valid (ICMR 2017, sections 5.3, 5.4 and 5.9, pp 66–70):

- ▶ Only the **EC-approved version** and its local translations may be used.
- ▶ Information must be given in a language and manner easily understood; braille or other adapted methods for differently abled participants.
- ▶ Adequate time must be given to read, to consult family and friends and to ask questions; no unjustifiable assurance, influence or intimidation.
- ▶ Therapeutic misconception must be avoided — for example by having a co-investigator who is not the treating physician administer the consent (ICMR 2017, section 7.1.6, p 85).
- ▶ A **test of understanding** should be administered wherever possible for sensitive studies and repeated until understood.
- ▶ The participant signs and dates the ICF; if **unable to sign**, a **thumb impression** is taken; an **impartial literate witness**, not a relative and unconnected with the research, must witness the process for an illiterate participant or LAR.
- ▶ The **researcher who administers the consent must also sign and date** the form.
- ▶ Refusal must not affect the patient–clinician relationship, routine care or any entitled benefit; routine care continues even after withdrawal.
- ▶ A copy of the PIS and the signed ICF is given to the participant (reluctance to take them is recorded); the originals are archived.

Situation	What ICMR 2017 requires
Waiver of consent — permitted only when the research carries less than minimal risk and the waiver will not adversely affect participants' rights and welfare	Six EC-grantable situations: research not practicable without the waiver and scientifically justified; retrospective studies with de-identified or uncontactable participants; anonymised biological samples/data; certain public health studies, surveillance or programme evaluation; data already in the public domain; humanitarian emergencies and disasters, where consent is still sought at the earliest (section 5.7, Box 5.2, pp 68–69)
Re-consent (fresh consent)	New information changes the benefit–risk ratio; an unconscious participant regains consciousness or mental competence; a child becomes an adult during the study ; long-term follow-up or extension; change in treatment modality, procedures, visits, data-collection methods or duration; identity may be disclosed through data or photographs in a publication (section 5.8, p 69)
Child participant	Assent in addition to LAR consent — not documented below 7 years, oral assent recorded 7–12 years, written assent 12–18 years also signed by parent/LAR; dissent respected (Box 6.6, p 79)
Gatekeeper and community consent	Permission of the head/leader of the group or culturally appropriate local authority, or of a representative community body meeting its own quorum, before individual consent — individual consent is still required (section 5.10, pp 69–70)
Electronic consent	Must carry all Box 5.1 elements in the participant's language, be EC-approved in advance, be supervised by the principal investigator, and not be used where the participant is uncomfortable with electronic media; a copy is still archived and given (section 5.5, p 68)
Studies using deception	Two-step consent — an initial consent plus debriefing after participation ; unjustified deception, undue influence and intimidation avoided at all costs (section 5.11, pp 70–71)



The informed consent process from ethics committee approval to archiving and re-consent

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **The two arms of GCP** — a single box "Good Clinical Practice" splitting into two labelled columns: *Ethical arm* (labels: independent ethics committee review, informed consent, risk minimisation, privacy and confidentiality, compensation for research-related injury) and *Scientific/quality arm* (labels: written protocol, Investigator's Brochure, standard operating procedures, Case Report Form and source documents, monitoring, audit and inspection, prespecified statistics, fair reporting).
- ▶ **Structure of the informed consent document** — one rectangle labelled "Informed Consent Document (ICD)" divided into two labelled halves: "Participant Information Sheet (PIS)" and "Informed Consent Form (ICF)", with the signature block labelled: participant signature/thumb impression, date; impartial witness (if illiterate); researcher administering consent, date.

High-yield exam pointers

- ▶ **GCP definition to reproduce verbatim:** "an international ethical and scientific quality standard for designing, conducting, recording, and reporting trials that involve the participation of human subjects."
- ▶ **Guideline chain to name in order:** Nuremberg Code 1947 → Declaration of Helsinki 1964 (revised Fortaleza 2013) → Belmont Report 1979 (respect for persons, beneficence, justice) → ICH E6(R1) 1996 and E6(R2) 2016 → Indian GCP Guidelines, CDSCO 2001 → Schedule Y (amended June 2005) → ICMR National Ethical Guidelines 2017 → **New Drugs and Clinical Trials Rules 2019 (current Indian regulation)**.
- ▶ **ICMR's 12 principles mnemonic** — "E-V-N-S-P-R-P-M-I-T-T-E": Essentiality, Voluntariness, Non-exploitation, Social responsibility, Privacy/confidentiality, Risk minimisation, Professional competence, Maximisation of benefit, Institutional arrangements, Transparency and accountability, Totality of responsibility, Environmental protection.
- ▶ **Ethics committee numbers:** 7–15 members, preferably 50% non-affiliated, quorum of **five** including medical and non-medical members with at least one non-affiliated member; head of institution must NOT be a member (acts as appellate authority).
- ▶ **Consent document = PIS + ICF, with 10 essential elements** (Box 5.1); illiterate participant → **thumb impression + impartial literate witness**; consent administered by someone other than the treating physician to avoid therapeutic misconception.
- ▶ **Waiver only for less-than-minimal-risk research** — six situations in Box 5.2; re-consent triggers include a **child attaining adulthood** during the study; assent written from 12–18 years, oral 7–12 years, none below 7 years.
- ▶ **Dates to quote:** CTRI launched **20 July 2007**, registration mandatory from **15 June 2009**; study records kept at least **2 years** after approval of the marketing application.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the local corpus copy of the New Drugs and Clinical Trials Rules 2019 is an unsearchable scanned image, so no clause of those Rules is quoted above, and the Schedule Y position is reproduced as it is stated in the Indian Council of Medical Research National Ethical Guidelines 2017, which predates the 2019 Rules.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, ICH E6(R3), EMA, WHO, WMA Declaration of Helsinki 2024

Programme and policy updates (India)

- ▶ **Ethics committees must now register with the National Ethics Committee Registry for Biomedical and Health Research (NECRBHR), Department of Health Research, MoHFW** — the New Drugs and Clinical Trials Rules 2019 extended registration beyond CDSCO-registered clinical-trial ECs to *every* EC that reviews biomedical and health research, including postgraduate thesis work; application is on form **CT-01** through the online **Naitik portal**, processed in 30 days, with provisional registration valid 2 years and final registration 5 years. Uptake lags badly — against an expected ~4,000 ECs, only about 2,100 had created portal accounts and roughly 30% had submitted applications up to July 2023, and about 40% of medical and dental colleges were still unregistered [PMID 37737078].
- ▶ **Site-by-site ethics review of multi-centre trials is under active reform debate** — an all-India survey (August–October 2024) of 106 researchers and 136 EC members experienced in multi-centre trials found only 22% and 27% respectively rated their EC as efficient at avoiding delays, and fewer than one-third favoured retaining independent review at every participating site; joint, designated and national single-review models were canvassed without a clear consensus [PMID 42337749].
- ▶ **Early-phase capacity building under the ICMR Phase I Network** — a November 2024 PGIMER Chandigarh workshop with CDSCO, ICMR, academia and industry worked through Investigational New Drug and pre-IND submissions, mock Subject Expert Committee meetings, on-site monitoring and consent processes, and debated the phase-lag policy that governs first-in-human studies in India [PMID 42189405].

Current national data

- ▶ **CTRI is now the third-largest trial registry in the world** — 71,444 records on 29 July 2024, behind ClinicalTrials.gov (503,188) and ChiCTR (85,723); yet despite registration being mandatory since 2009 and prospective since 1 April 2018, an audit of CDSCO permission letters found 46 of 381 regulatory trials (12%) with no matching CTRI record, with no penalty for non-registration and no mechanism cross-linking CDSCO and CTRI records [PMID 39966939].
- ▶ **Methodological quality of Indian phase III trials improved after the mid-2010s reforms** — among 1,385 phase III trials registered with CTRI from 2007 to June 2025, randomised designs rose from 88.7% (2007–2014) to 94.8% (2015–2025; $p < 0.001$), industry sponsorship fell from 92.1% to 81.6% while educational institutions' share rose nearly five-fold (2% → 9.7%), and

vaccine (5.7% → 7%) and device/biologic/cell-and-gene trials (5.2% → 12.3%) increased [PMID 42437223].

Elimination, surveillance and disease status

- **Consent and confidentiality have become programme-design problems in TB elimination work** — as active case finding, household contact tracing and adherence monitoring move out of clinics into homes, workplaces and informal settlements, obtaining meaningful individual consent and protecting confidentiality against stigma directly determine community participation, trust and long-term programme effectiveness; Indian and South African experience is used to argue for explicit consent and confidentiality standards in community-level TB interventions [PMID 42362239].

Guideline and legislative updates

- **ICH E6(R3) has replaced E6(R2)** — the single most examinable change to GCP since 2016 — the Principles plus **Annex 1** package took effect on 23 July 2025 and is built on **quality by design**, critical-to-quality factors and **risk proportionality**, defining trial quality as fitness for purpose and scaling sponsor and investigator oversight and monitoring to risk instead of uniform source-data verification; **Annex 2**, covering pragmatic and decentralised trials and real-world data sources, reached Step 4 on 3 June 2026 and applies from 15 January 2027 [ICH 2025] [EMA 2026-Jun] [PMID 41194530].
- **India's framework is broadly consistent with the 2024 Declaration of Helsinki, with named gaps** — the ICMR 2017 National Ethical Guidelines align well on safeguards for vulnerable populations and on ethical review, but the outstanding harmonisation work is registration of *all* medical research rather than only trials under the Drugs and Cosmetics Act, empowering ethics committees, and ensuring equitable access and inclusion [PMID 41195926].

Global and WHO developments

- **Declaration of Helsinki 2024 revision (adopted 19 October 2024, 75th WMA General Assembly) supersedes Fortaleza 2013** — "subjects" is replaced throughout by "**participants**"; responsible inclusivity of under-represented and marginalised groups is required with appropriate safeguards; the role of research ethics committees in approval, monitoring and implementation is strengthened; and new material covers community engagement, avoidance of research waste and of ecological impact, data privacy and artificial-intelligence governance, with dual ethical review for cross-border studies [PMID 39616449] [PMID 39671397].
- **WHO Guidance for best practices for clinical trials, 25 September 2024** — issued in response to World Health Assembly resolution **WHA75.8 (2022)** on strengthening clinical trials, it sets out measures to strengthen the global clinical-trial ecosystem, improve research efficiency and reduce research waste, and to sustain trial capacity for endemic conditions that can be adapted rapidly during emergencies and pandemics [WHO 2024-Sep].

SOURCE & COVERAGE

Browner — *Designing Clinical Research*, 5e (2022), Ch 18 and Ch 7; Barr — *Good Clinical Practice*, Ch 35 in *Methods and Applications of Statistics in Clinical Trials* (2014), pp 438–446; Indian Council of Medical Research — *National Ethical Guidelines for Biomedical and Health Research Involving Human Participants*, 2017 (Introduction; sections 1.1, 2.2, 3.7, 4.3, 4.11, 5.1–5.11, 6.5, 7.1, 7.16; Boxes 5.1, 5.2, 6.6; List of References); World Medical Association *Declaration of Helsinki*, 2013 (Fortaleza) revision; Sunder Lal — *Textbook of Community Medicine*, Ch 1. Coverage note: Park's Textbook of Preventive and Social Medicine 28e carries no section on Good Clinical Practice, so this answer is grounded on the dedicated GCP chapter, Browner 5e and ICMR 2017.

Recent-advances sources: 37737078 — Overview of the National Ethics Committee Registry for Biomedical and Health Research in India, *Indian J Pharmacol* 2023, [DOI](#); 42337749 — Ethics review of multi-centre trials in India: a survey of researchers and ethics committee members, *BMC Med Ethics* 2026, [DOI](#); 42189405 — Regulatory and Ethical Frameworks for First-in-Human Clinical Trials in India, *Clin Drug Investig* 2026, [DOI](#) (abstract-only); 39966939 — Clinical trial registration in India: 12% of drug regulatory trials are not registered, as required by law, *Health Res Policy Syst* 2025, [DOI](#); 42437223 — Phase III Trials in India: A Comprehensive Analysis of the Clinical Trials Registry-India, *Cureus* 2026, [DOI](#); 42362239 — Informed consent and confidentiality in community-level tuberculosis interventions, *Indian J Tuberc* 2026, [DOI](#) (abstract-only); 41194530 — Evolving Standards: Good Clinical Practice Insights from US FDA, MHRA UK, and Health Canada, *Clin Pharmacol Ther* 2025, [DOI](#); 41195926 — 2024 Revision of Declaration of Helsinki: policy perspectives from India, *Indian J Med Ethics* 2025, [DOI](#) (abstract-only); 39616449 — The 2024 Revision of the Declaration of Helsinki and its Continued Role as a Code of Ethics to Guide Medical Research, *Med Sci Monit* 2024, [DOI](#); 39671397 — The 2024 revision of the Declaration of Helsinki: a modern ethical framework for medical research, *Postgrad Med J* 2025, [DOI](#) (abstract-only). Also ICH E6(R3) Guideline for Good Clinical Practice — Principles and Annex 1, in effect 23 July 2025, and Annex 2 adopted at Step 4 on 3 June 2026 (effective 15 January 2027); WHO *Guidance for best practices for clinical trials*, 25 September 2024, responding to WHA75.8 (2022); WMA *Declaration of Helsinki*, 2024 revision adopted 19 October 2024.



Medical entomology and zoonoses



Aedes aegypti index – interpretation and control

Asked in: Paper II May 2019 — RGUHS MD Community Medicine

Definition — Under the International Health Regulations (IHR), the *Aedes aegypti* index is "the ratio, expressed as a percentage, between the number of houses in a limited well-defined area on the premises of which actual breeding of *Aedes aegypti* is found, and the total number of houses examined in that area" — a **house (premises) index of larval breeding**, not a count of adult mosquitoes. The same Regulations require **airports and seaports to be kept free of *Aedes* breeding for 400 metres all round their perimeter**, with the index kept at zero at all ports (Park 28e, Ch 15, p 925); the yellow fever chapter states the working ceiling as the index kept below 1, and less than 1 per cent in towns and seaports as the index of freedom from yellow fever (Park 28e, Ch 5, pp 347-348).

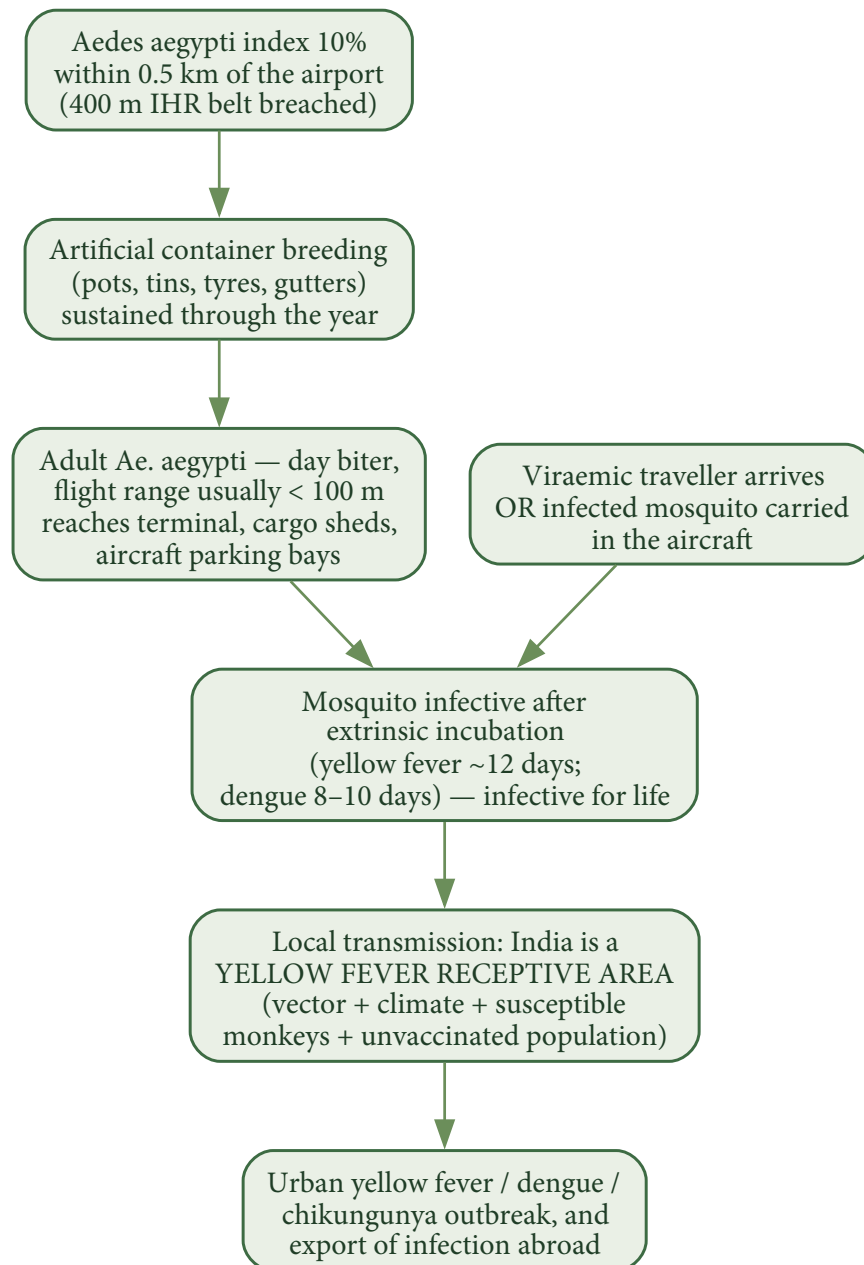
1. Inference on a value of 10 per cent

Yardstick	Prescribed standard	Observed here	Inference
IHR standard inside the 400 m perimeter belt of an international port	<i>Aedes aegypti</i> index = zero (Park 28e, Ch 15, p 925)	10%	Gross failure; the mandatory <i>Aedes</i> -free zone does not exist
Park's operational ceiling for ports and towns	< 1 per cent (Park 28e, Ch 5, pp 347-348)	10%	Ten times the permitted ceiling
Larval house index risk band for dengue transmission	High risk > 10%; low risk < 1% (Sunder Lal, Ch 8, p 396)	10%	Sits at the high-risk transmission threshold — not a low-risk area by any reading
Flight range of <i>Ae. aegypti</i>	usually less than 100 metres from the breeding place (Park 28e, Ch 15, pp 925-926)	breeding at ≤ 0.5 km	Every positive house is within easy flight reach of the terminal, cargo sheds and aircraft parking bays
Status of India for yellow fever	" Receptive area " — the vector is abundant, the climate favourable, <i>Macacus</i> monkeys susceptible and the population unvaccinated; only the virus is missing (Park 28e, Ch 5, p 348)	vector breeding at the point of entry	The one missing link can arrive by a single aircraft

What the value tells the district health officer, in points:

- **One in every ten premises surveyed is actively breeding the vector** — the survey area of 0.5 km entirely contains and exceeds the **400 metre statutory belt**, so the airport is in breach of a binding international obligation, not merely of good practice.
- It reflects **failure of the airport health organisation's routine anti-larval work**, not a chance event: sustained peridomestic breeding in water-storage pots, discarded tins, tyres, coconut shells, broken bottles, flower pots and roof gutters — the artificial containers *Ae. aegypti* prefers (Park 28e, Ch 15, pp 925-926).

- ▶ **Risk of importing yellow fever is real.** The mosquito becomes infective after an extrinsic incubation period of about **12 days** and remains infective for life; a viraemic traveller (blood infective for the first 3-4 days of illness) or an infected mosquito carried in an aircraft could seed urban yellow fever (Park 28e, Ch 5, pp 347-348).
- ▶ **Risk of dengue, chikungunya and Zika in both directions.** *Ae. aegypti* is highly domesticated, strongly anthropophilic, a **nervous feeder** and a **discordant species** — it bites more than one person per blood meal, which clusters cases; dengue extrinsic incubation is **8-10 days** and transovarial transmission occurs (Park 28e, Ch 5, p 307).
- ▶ **Aircraft amplify dispersal.** Aircraft and ships "have created fresh problems of public health" by moving mosquitoes from country to country (Park 28e, Ch 15, pp 925-926); a high index at a point of entry is an international, not a local, hazard.
- ▶ **Corrective action is mandatory and urgent**, and the target is not "reduction" but **zero breeding within 400 m**, with the wider 0.5 km zone held below 1%.



Why a 10 per cent Aedes aegypti index within 0.5 km of an international airport is a point-of-entry hazard

2. Immediate response (first 72 hours to 1 week)

- **Confirm and map the survey.** Repeat the larval survey with a trained entomological team over the whole 0.5 km belt, house by house, recording **house index, container index, Breteau index and adult landing/biting rate**; spot-map every positive premises and every positive container type.
- **Notify and convene.** Report to the **Airport Health Officer / Port Health Organisation**, the district health officer and the state programme officer under the **National Vector Borne Disease Control Programme (NVBDCP)**, and convene the airport's inter-sectoral committee (airport authority, municipal body, airlines, ground handlers, cargo operators, construction contractors).

- **Emergency source reduction sweep** of all positive containers — empty, dry, scrub and invert; remove and dispose of scrap tyres, tins, bottles and coconut shells; clear roof gutters and unused water tanks.
- **Emergency adulticiding** — space spray to knock down the existing infective adult population, repeated as the programme directs, recognising that its effect is transient because aerosol droplets do not reach indoor microhabitats (Jugal Kishore, National Health Programmes of India, p 441).
- **Larvicide every container that cannot be emptied** (see the package below).
- **Fever surveillance and case search** in the belt, with dengue notification (dengue has been a notifiable disease since 2016 — Park 28e, Ch 7, p 502) and reporting through the Integrated Disease Surveillance Programme.

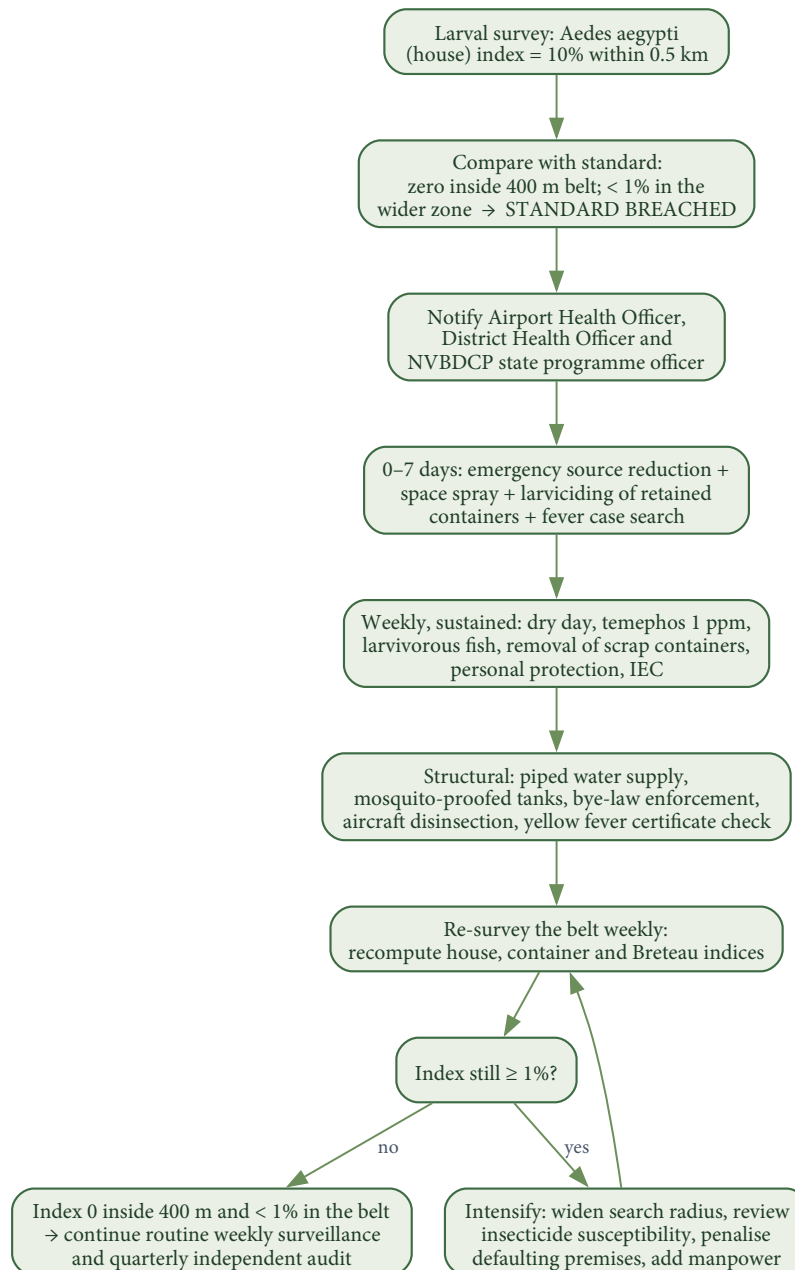
3. Sustained vector-control package to bring the index to zero

Component	Specific action in the 0.5 km belt	Agent / dose / frequency
Anti-larval — environmental modification (permanent)	Piped, reliable, 24-hour water supply so that water need not be stored; mosquito-proofing of overhead tanks, cisterns and underground reservoirs; filling and levelling; correct drainage of sewage and waste water	Engineering works; permanent results (Park 28e, Ch 15, pp 925-926; Jugal Kishore, p 442)
Anti-larval — environmental manipulation (recurrent)	Weekly "dry day": empty, dry and scrub all storage containers; do not store water beyond a week; strain stored water through a clean cloth weekly and sun-dry the cloth; remove all non-essential containers, scrap tyres and construction-site water	Weekly cycle, because the mosquito life cycle occupies about 8 days (Jugal Kishore, pp 447-448; Park 28e, Ch 15, p 926)
Anti-larval — chemical	Larvicide only where water must be conserved	Temephos (Abate) 50 EC at 1 ppm = 1 mg per litre — odourless, colourless, low mammalian toxicity, safe in potable water; once weekly (Jugal Kishore, p 442; Park 28e, Ch 15, p 938). Field dosage 56-112 g/ha; fenthion and chlorpyrifos are alternatives (Park 28e, Ch 15, p 927)
Anti-larval — biological	Ornamental tanks, fountains, fire buckets, large permanent containers	Larvivorous fish — <i>Gambusia</i> and guppy; <i>Bacillus thuringiensis</i> serotype H-14 (Bt H-14) (Jugal Kishore, pp 442, 447-448; Park 28e, Ch 15, p 922)
Anti-adult — space spray	Indoor knock-down of resting adults in terminal, staff quarters and adjacent houses	Pyrethrum extract 0.1-0.2% , 30-60 ml per 1,000 cubic feet; doors and windows kept closed half an hour; effect temporary, reinfestation from outside is rapid (Jugal Kishore, p 442; Park 28e, Ch 15, p 927)
Anti-adult — outdoor fogging	Outdoor thermal fogging of the belt during an outbreak	Malathion — the insecticide advised for outdoor fogging in the national dengue control strategy (Park 28e, Ch 7, p 502)

Component	Specific action in the 0.5 km belt	Agent / dose / frequency
Personal protection	Airport and cargo staff, and residents of the belt	Repellents, protective clothing, coils, mats and aerosols; screening of doors and windows . Insecticide-treated nets have limited value because the vector bites by day , but are useful for infants and night-shift workers sleeping in daytime (Jugal Kishore, p 442)
Integrated vector management (IVM)	Combine the above rather than rely on one method	Optimal use of biological, chemical and environmental interventions, with safe use of insecticides and monitoring of insecticide resistance (Park 28e, Ch 7, p 498)

4. Statutory, administrative and intersectoral measures at the point of entry

- ▶ **Enforce the 400 metre *Aedes*-free belt** under the IHR as the airport health authority's standing duty, with a permanent, funded anti-larval squad and a fixed weekly search-and-destroy round.
- ▶ **Disinsection of conveyances** — aircraft and ships arriving from endemic areas are subjected to **aerosol spraying with prescribed insecticides on arrival** for destruction of insect vectors (Park 28e, Ch 5, p 348).
- ▶ **Traveller measures** — enforce the **valid international certificate of vaccination against yellow fever** (17D vaccine, 0.5 ml subcutaneously, immunity from the 7th day, possibly lifelong); a traveller from an infected area without a valid certificate is **quarantined in a mosquito-proof ward for 6 days** from the date of leaving the infected area (Park 28e, Ch 5, p 348).
- ▶ **Municipal bye-laws and penalties** against water-holding scrap, uncovered storage and stagnant water on airport, cargo, hotel and construction premises — cooperation in reducing peridomestic habitats is difficult "unless local legislation is strictly enforced" (Service, *Medical Entomology for Students*, Ch 1, p 91).
- ▶ **Intersectoral coordination** — health, water and sanitation, local body and panchayati raj, town planning, builders and construction contractors, transport and education sectors (Sunder Lal, Ch 8, p 397; Park 28e, Ch 7, p 502).
- ▶ **Information, education and communication (IEC) and community participation** — long-term source reduction succeeds only when supported by health education securing community participation (Park 28e, Ch 5, p 348); use mass media, airport staff briefings and the *India Fights Dengue* mobile application (Jugal Kishore, p 441).
- ▶ **Training** of airport health staff, insect collectors and municipal workers in larval survey technique, index calculation, larviciding and outbreak response (Sunder Lal, Ch 8, p 397).



Escalating response algorithm for an *Aedes aegypti* index above the permitted level at an international airport

5. Monitoring and evaluation — the indicators to report

Indicator	Formula	Target after corrective action
<i>Aedes aegypti</i> / house index	(Houses positive for <i>Ae. aegypti</i> larvae or pupae ÷ houses inspected) × 100	0 inside the 400 m belt; < 1% in the 0.5 km zone (Park 28e, Ch 15, p 925; Ch 5, p 348)
Container index	(Containers positive for <i>Aedes</i> breeding ÷ water-holding containers inspected) × 100	No numerical cut-off; a single positive container warrants immediate source reduction (Sunder Lal, Ch 8, pp 395–396)
Breteau index	Number of containers positive for <i>Ae. aegypti</i> larvae per 100 houses inspected	< 5 (low risk); > 50 is high risk (Sunder Lal, Ch 8, p 396)

Indicator	Formula	Target after corrective action
Adult landing / biting rate	$Aedes$ collected on human bait $\div$ man-hours	< 2 per man-hour (Sunder Lal, Ch 8, p 396)
Process indicators	Premises covered per round; containers treated; kg of scrap removed; disinsection certificates issued; yellow fever certificates verified	100% coverage of the 0.5 km belt every week

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Diagram 1 — Plan of the *Aedes*-free perimeter zone of an international airport.** Labels to write: runway and terminal building; aircraft parking bay; cargo shed; airport perimeter fence; 400 metre statutory *Aedes*-free belt (International Health Regulations); 0.5 km survey zone — houses surveyed, 10% positive; positive premises marked (water pots, discarded tyres, tins, roof gutters); flight range of *Aedes aegypti* usually < 100 metres (arrow from a positive house to the terminal); isolation / mosquito-proof ward for unvaccinated travellers (6 days); aircraft disinsection point.
- **Diagram 2 — Larval survey record sheet.** Labels: houses inspected; houses positive; containers inspected; containers positive; and the three derived indices — house index, container index, Breteau index.

High-yield exam pointers

- **The two numbers the examiner wants:** 400 metres *Aedes*-free belt around every international airport and seaport; index zero at ports (Park's yellow fever chapter: below 1, and < 1% for towns and seaports). 10% is ten-fold above the ceiling and infinitely above the port target.
- **Three larval indices and one adult index:** house index (% houses positive) — high risk > 10%, low risk < 1%; container index (% containers positive); Breteau index = positive containers per 100 houses — high risk > 50, low risk < 5; landing/biting rate — high risk > 2 per man-hour.
- **Vector facts:** first proved vector of a virus disease (yellow fever); day biter; artificial-container breeder; eggs laid singly, cigar-shaped, drought-resistant; flight range usually < 100 m; survives best at 16-30°C, 60-80% relative humidity; extrinsic incubation ~12 days (yellow fever) and 8-10 days (dengue), infective for life.
- **Park's classification of mosquito control** — the skeleton of the control answer: 1. Anti-larval (environmental, chemical, biological); 2. Anti-adult (residual sprays, space sprays, genetic control); 3. Protection against mosquito bites (mosquito net, screening, repellents).
- **Doses:** temephos 1 ppm (1 mg/litre) weekly; pyrethrum 0.1-0.2%, 30-60 ml per 1,000 cubic feet; malathion for outdoor fogging; *Bacillus thuringiensis* H-14 and *Gambusia*/guppy for biological control.
- **India is a yellow fever "receptive area"** — vector, climate and susceptible monkeys present, population unvaccinated; only the virus is missing. Unvaccinated traveller from an infected area → quarantine in a mosquito-proof ward for 6 days.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference, not to be written as a caveat in the exam: Park's 28th edition states the port standard in two places and in two forms — Chapter 15 (page 925) says the index "is kept at zero at all ports", while Chapter 5 (page 348) says it "is kept below 1" and elsewhere "less than 1 per cent in towns and seaports"; quote **zero for the 400 metre port belt and under 1 per cent for the surrounding town**, which reconciles both. The National Vector Borne Disease Control Programme has since been restructured as the National Center for Vector Borne Diseases Control, but the retrieved corpus uses the former name throughout, so the older name is used here.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 15 (Environment and Health), pp 922, 925-927, 938 (*Aedes aegypti* index and the 400 m port belt, *Aedes* bionomics and dispersal, classification of mosquito control measures, larvicide and space-spray dosages); Ch 5 (Epidemiology of Communicable Diseases), pp 307, 316, 339, 347-348 (dengue vector biology, yellow fever control, India as a receptive area, international measures at ports); Ch 7 (Health Programmes in India), pp 493, 498, 502 (National Vector Borne Disease Control Programme, integrated vector management, dengue guidelines and outdoor fogging). Sunder Lal — Textbook of Community Medicine, Ch 8, pp 395-397 (entomological surveillance, larval indices and their epidemiological interpretation, behaviour change communication, intersectoral coordination). Jugal Kishore — National Health Programmes of India, pp 441-442, 447-448 (dengue vector management: environmental, chemical, biological and personal protection measures with doses). Service — Medical Entomology for Students, Ch 1, pp 39, 91-92 (*Aedes aegypti* container breeding, source reduction, temephos and *Bacillus thuringiensis israelensis*, need for legislative enforcement).



Arthropod-borne diseases and transmission

Asked in: Paper I 19-Nov-2020 — RGUHS MD Community Medicine

Definition — An **arthropod-borne disease** is a communicable disease in which an arthropod acts as the **vector**, defined as "an arthropod or other invertebrate which transmits infection by inoculation into or through the skin or mucous membrane by biting, or by deposit of infective materials on the skin or on food or other objects" (Park 28e, Ch 15, p 922). Transmission by such a vector may be **mechanical** or **biological** (Park 28e, Ch 3, pp 117-118). The study of these arthropods is **medical entomology**.

A. Arthropods of medical importance — the basis of any vector classification

Arthropod-borne diseases constitute a major health problem in India: malaria remains the important vector-borne disease with an **annual morbidity of 4 to 5 million cases**, filaria has an estimated **236 million people living in filaria-endemic areas**, scabies is widespread especially in rural areas, and

dengue, haemorrhagic fever, Japanese encephalitis and Kyasanur Forest Disease (KFD) are the important arthropod-borne virus diseases (Park 28e, Ch 15, p 921).

Class Insecta	Class Arachnida	Class Crustacea
1. Mosquitoes — anophelines, culicines	1. Ticks — hard ticks, soft ticks	1. Cyclops
2. Flies — houseflies, sandflies, tsetse flies, blackflies	2. Mites (chiggers) — <i>Leptotrombidium</i> and trombiculid mites; itch mite	
3. Human lice — head and body lice; crab lice		
4. Fleas — rat fleas, sand fleas		
5. Reduviid bugs		

(Park 28e, Ch 15, p 921, Table 1)

Separating the three classes at sight is the standard viva follow-up:

Character	Insecta	Arachnida	Crustacea
Body divisions	Head, thorax and abdomen	Cephalothorax and abdomen (no division in some cases)	Cephalothorax and abdomen
Legs	3 pairs	4 pairs	5 pairs
Antennae	1 pair	None	2 pairs
Wings	One or two pairs; some are wingless	None	None
Where found	On land	On land	In water

(Park 28e, Ch 15, p 921, Table 2)

B. Classification of arthropod-borne diseases

1. By the arthropod vector and the disease it transmits — the single most examinable table:

Arthropod	Diseases transmitted
1. Mosquito	Malaria, filaria, viral encephalitis (e.g. Japanese encephalitis), viral fevers (e.g. dengue, West Nile), viral haemorrhagic fevers (e.g. yellow fever, dengue haemorrhagic fever)
2. Housefly	Typhoid and paratyphoid fever, diarrhoea, dysentery, cholera, gastro-enteritis, amoebiasis, helminthic infestations, poliomyelitis, conjunctivitis, trachoma, anthrax, yaws
3. Sandfly	Kala-azar, oriental sore, sandfly fever, oroya fever
4. Tsetse fly	Sleeping sickness
5. Louse	Epidemic typhus, relapsing fever, trench fever, pediculosis
6. Rat flea	Bubonic plague, endemic typhus, chiggerosis, <i>Hymenolepis diminuta</i>

Arthropod	Diseases transmitted
7. Blackfly	Onchocerciasis
8. Reduviid bug	Chagas disease
9. Hard tick	Tick typhus, viral encephalitis, viral fevers, viral haemorrhagic fever (e.g. Kyasanur Forest Disease), tularaemia, tick paralysis, human babesiosis
10. Soft tick	Q fever, relapsing fever
11. Trombiculid mite	Scrub typhus, rickettsial-pox
12. Itch-mite	Scabies
13. Cyclops	Guinea-worm disease, fish tapeworm (<i>Diphyllobothrium latum</i>)
14. Cockroaches	Enteric pathogens

(Park 28e, Ch 15, p 922, Table 3)

2. Epidemiological classification of vector-borne diseases (Park 28e, Ch 3, pp 117-118) — four axes:

Axis	Categories
I. By vector	(a) Invertebrate type — arthropod vectors fall largely into seven orders: (1) <i>Diptera</i> — flies and mosquitoes; (2) <i>Siphonaptera</i> — fleas; (3) <i>Orthoptera</i> — cockroaches; (4) <i>Anoplura</i> — sucking lice; (5) <i>Hemiptera</i> — bugs, including kissing bugs; (6) <i>Acarina</i> — ticks and mites; (7) <i>Copepoda</i> — cyclops. (b) Vertebrate type — mice, rodents, bats
II. By transmission chain (heterogeneous infection chain, three principal patterns)	(a) Man and a non-vertebrate host — man-arthropod-man (malaria); man-snail-man (schistosomiasis). (b) Man, another vertebrate host and a non-vertebrate host — mammal-arthropod-man (plague); bird-arthropod-man (encephalitis). (c) Man and two intermediate hosts — man-cyclops-fish-man (fish tapeworm); man-snail-fish-man (<i>Clonorchis sinensis</i>); man-snail-crab-man (paragonimiasis)
III. By the method by which the vector transmits the agent	(a) Biting; (b) regurgitation; (c) scratching-in of infective faeces; (d) contamination of the host with body fluids of the vector
IV. By the method by which the vector is involved in transmission and propagation of the parasite	(a) Mechanical transmission; (b) biological transmission — propagative, cyclo-propagative, cyclo-developmental

3. **By reservoir** — where the chain is **man-arthropod-man** the reservoir is human (malaria, bancroftian filariasis); where a lower vertebrate is the reservoir the disease is a **zoonosis**, classed by Park as **anthropozoonoses** (transmitted to man from lower vertebrate animals, e.g. plague), **zooanthropoonoses** (man to lower vertebrate animals) and **amphixenoses** (maintained in both and transmissible in either direction). In the life-cycle classification of zoonoses the arthropod-borne ones are the **meta-zoonoses** — "transmitted biologically by invertebrate vectors; in the invertebrate the agent multiplies or develops, or both, and there is always an extrinsic incubation (prepatent) period before transmission to another vertebrate host is possible", e.g. arbovirus infections and plague (Park 28e, Ch 5, p 341).

4. **By causative agent**

Agent class	Diseases
Viral — arboviruses, defined as viruses "maintained in nature principally, or to an important extent, through biological transmission between susceptible vertebrate hosts by haematophagous arthropods; they multiply in the tissues of arthropods, and are passed on to new vertebrates by the bites of arthropod after a period of extrinsic incubation" (Park 28e, Ch 5, p 350)	Dengue, Japanese encephalitis, yellow fever, Kyasanur Forest Disease, chikungunya, Zika
Rickettsial	Four groups — see the group-by-group table below
Bacterial	Plague, tularaemia
Protozoal	Malaria, kala-azar, sleeping sickness, Chagas disease
Helminthic	Lymphatic filariasis, onchocerciasis, dracunculiasis

Rickettsial diseases are grouped on the basis of clinical features and epidemiological aspects, and all are transmitted to man by arthropod vectors, **Q fever excepted** (Park 28e, Ch 5, p 365). The vector is **not** the same across the four groups, and two groups carry more than one vector — the commonest error is to assume a one-to-one mapping:

Group	Organism	Disease	Vector
Typhus group	<i>Rickettsia prowazekii</i>	Epidemic typhus (louse-borne typhus), Brill-Zinsser disease	Louse
Typhus group	<i>Rickettsia typhi</i>	Murine typhus, endemic typhus (flea-borne typhus)	Flea
Scrub typhus group	<i>Orientia tsutsugamushi</i>	Scrub typhus	Mite
Spotted fever group	<i>Rickettsia rickettsii</i>	Rocky Mountain spotted fever	Tick
Spotted fever group	<i>Rickettsia conorii</i>	Fiebre boutonneuse, Mediterranean and Israeli spotted fever, South African tick fever, African (Kenya) tick typhus, Indian tick typhus	Tick
Spotted fever group	<i>Rickettsia sibirica</i>	Siberian (North Asian) tick typhus	Tick
Transitional group	<i>Rickettsia akari</i>	Rickettsial pox	Mite
Transitional group	<i>Rickettsia australis</i>	Queensland tick typhus	Tick
(Q fever — listed in the same table but not arthropod-transmitted to man)	<i>Coxiella burnetii</i>	Q fever	Airborne, fomites, tick

(Park 28e, Ch 5, p 365, Table 1)

C. Transmission of arthropod-borne diseases

Three types of transmission cycle are involved in the spread of arthropod-borne disease (Park 28e, Ch 15, p 922):

Type of transmission	What happens to the agent in the arthropod	Examples
1. Direct contact	The arthropod itself is transferred directly from man to man by close contact; no agent-vector interaction	Scabies, pediculosis
2. Mechanical transmission	The agent is mechanically transported by a crawling or flying arthropod through soiling of its feet or proboscis, or passes through its gastro-intestinal tract and is passively excreted. No development and no multiplication in or on the vector	Housefly transmitting diarrhoea, dysentery, typhoid, food poisoning and trachoma
3. Biological transmission — (a) Propagative	The agent merely multiplies , with no change in form	Plague bacilli in the rat flea
3. Biological transmission — (b) Cyclo-propagative	The agent changes in form and multiplies (cyclical change plus multiplication)	Malaria parasite in the anopheline mosquito
3. Biological transmission — (c) Cyclo-developmental	The agent undergoes only development, without multiplication	Microfilaria in <i>Culex</i> ; guineaworm embryo in cyclops

(Park 28e, Ch 3, p 118 and Ch 15, pp 922-923)

Vertical transmission within the vector (Park 28e, Ch 3, p 118):

- **Transovarial transmission** — the infectious agent is transmitted vertically from the infected female to her progeny through the egg. Examples: dengue virus enters fully developed eggs at the time of oviposition (Park 28e, Ch 5, p 307); scrub typhus rickettsiae are maintained transovarially from one generation of trombiculid mite to the next (Park 28e, Ch 5, p 366); rickettsial pox in the mite (Park 28e, Ch 5, p 368); and in Indian tick typhus infection in nature is maintained by **transovarian and transstadial passage** in the tick (Park 28e, Ch 5, p 367).
- **Transstadial transmission** — survival of the agent through successive life-cycle stages, larva to nymph to adult, as in ticks, each haematophagous stage being able to transmit (Service, *Medical Entomology for Students*, pp 300-301).
- **Venereal transmission** — pathogens passed from congenitally infected males to females during mating, described in some vectors of yellow fever and Crimean-Congo haemorrhagic fever (Service, pp 300-301).

D. Terms the examiner expects to be defined alongside the classification

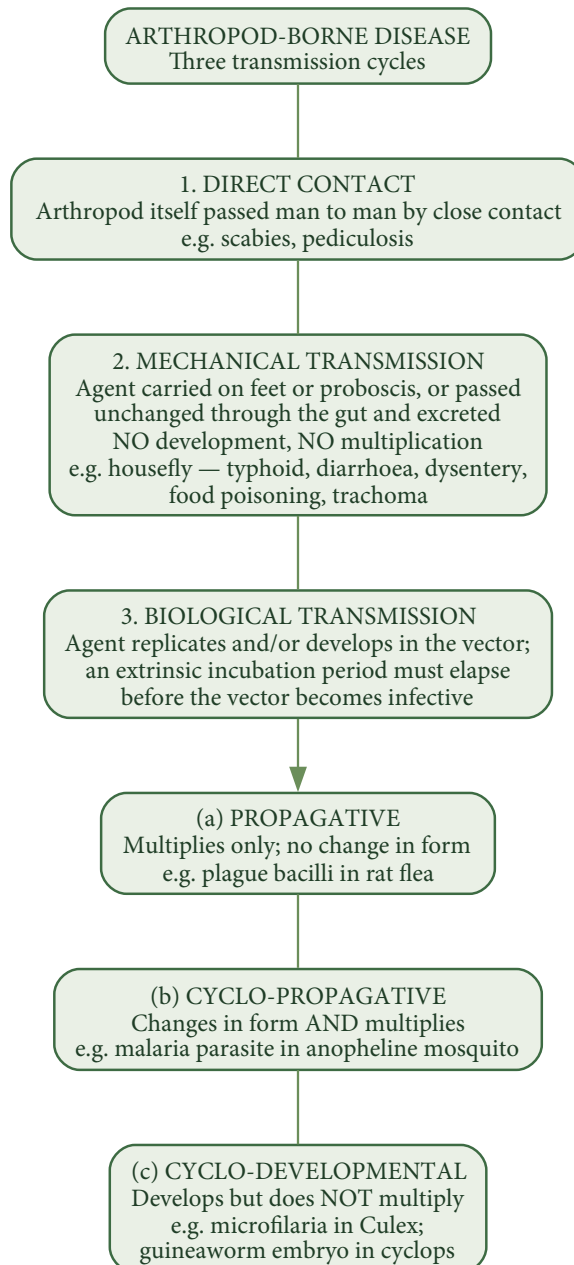
Term	Definition
Extrinsic incubation period	The period of time necessary for the development of the disease agent in the arthropod host — the interval between the vector becoming infected and its becoming infective. In malaria and filaria it is 10 to 14 days or longer, depending on environmental temperature (Park 28e, Ch 15, p 922)
Definitive host	The host in which the sexual cycle of the agent occurs, e.g. the mosquito is the definitive host in malaria
Intermediate host	The host in which the asexual cycle of the agent occurs, e.g. the mosquito in filaria and cyclops in guinea-worm disease
Infestation	The lodgement, development and reproduction of arthropods on the surface of the body or in the clothing , e.g. louse infestation

Factors influencing the ability of a vector to transmit disease (Park 28e, Ch 3, p 118):

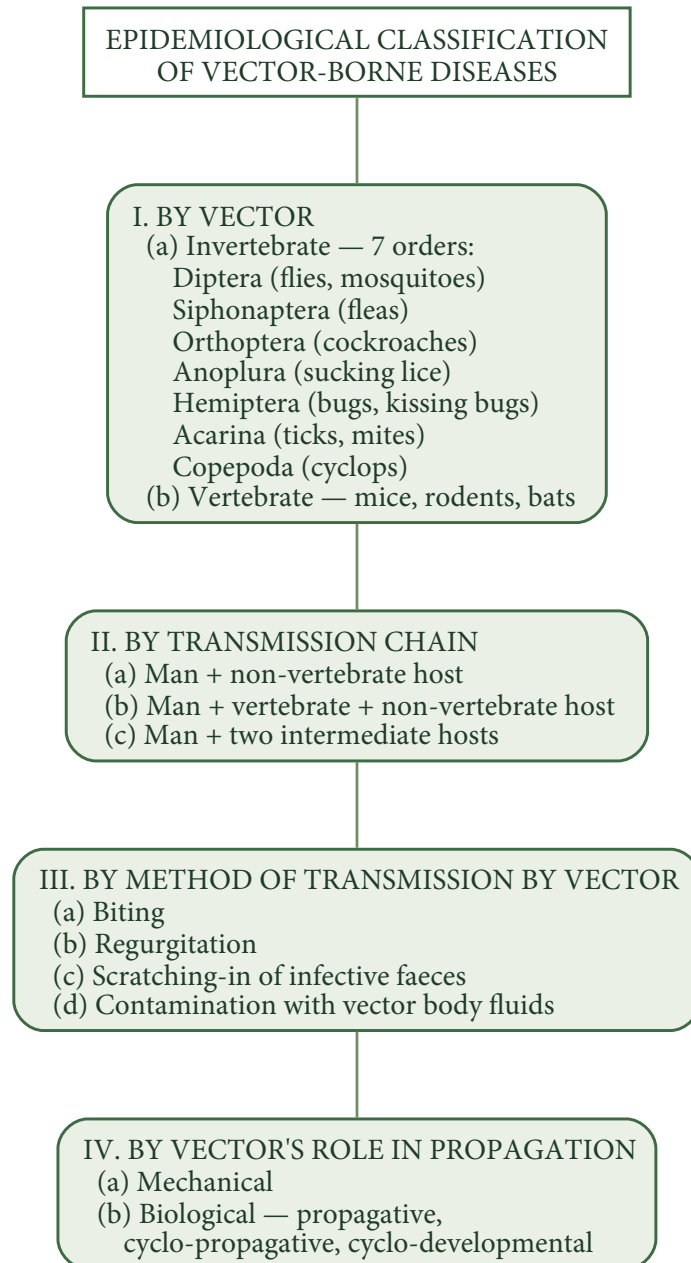
Factor	What it means
(a) Host feeding preferences	Whether the vector takes human or animal blood — some mosquitoes prefer human blood, some animal blood, and some vary widely in feeding habit (Park 28e, Ch 5, p 322)
(b) Infectivity	The ability to transmit the disease agent
(c) Susceptibility	The ability to become infected
(d) Survival rate of the vector	How long the vector survives in the environment
(e) Domesticity	The degree of association with man
(f) Suitable environmental factors	Season, temperature and breeding conditions — seasonal occurrence of a disease such as malaria may be related to intense breeding and thereby greater density of the vector during certain periods of the year

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

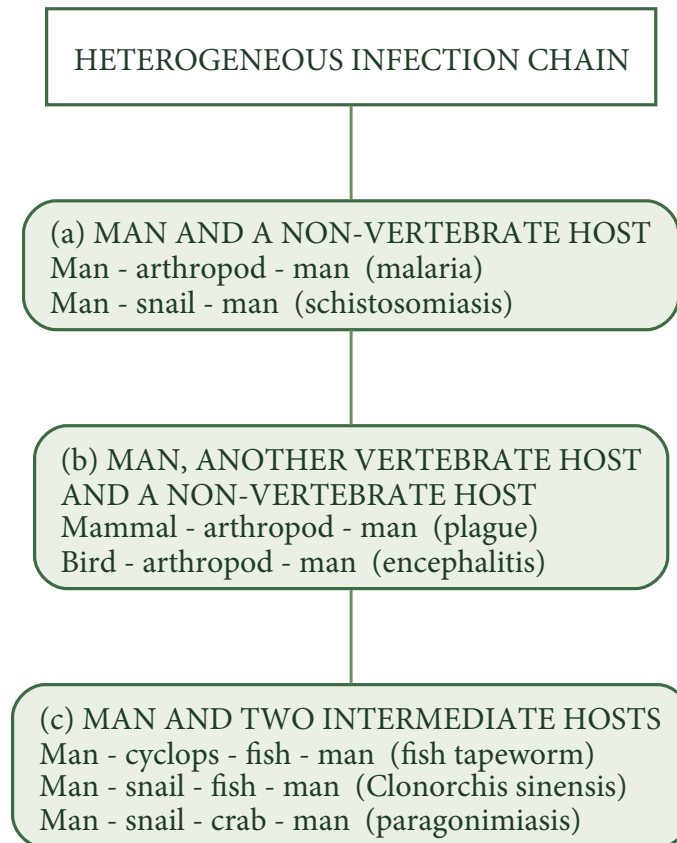
- **Hand-drawn figure** — draw the **three classes of arthropods of medical importance** side by side, labelling on each: *body divisions* (head + thorax + abdomen for Insecta; cephalothorax + abdomen for Arachnida and Crustacea), *number of pairs of legs* (3 / 4 / 5), *antennae* (1 pair / none / 2 pairs), *wings* (present in Insecta only), *habitat* (land / land / water); annotate representative members — mosquito, housefly, sandfly, louse, flea, reduviid bug (Insecta); hard tick, soft tick, trombiculid mite, itch mite (Arachnida); cyclops (Crustacea).



Transmission cycles in arthropod-borne disease (Park 28e, Ch 15, p 922)



Epidemiological classification of vector-borne diseases — the four axes (Park 28e, Ch 3, pp 117-118)



Transmission chains in vector-borne disease — the three principal patterns (Park 28e, Ch 3, p 118)

High-yield exam pointers

- ▶ **Three transmission cycles = D-M-B:** Direct contact (scabies, pediculosis) → Mechanical (housefly: typhoid, diarrhoea, dysentery, food poisoning, trachoma) → Biological (three sub-types).
- ▶ **Biological transmission sub-types, one word each:** Propagative = **multiply only** (plague bacilli in rat flea); Cyclo-propagative = **change + multiply** (malaria in anopheline); Cyclo-developmental = **change only, no multiplication** (microfilaria in *Culex*, guineaworm embryo in cyclops).
- ▶ **Seven orders of arthropod vectors** — Diptera, Siphonaptera, Orthoptera, Anoplura, Hemiptera, Acarina, Copepoda.
- ▶ **Rickettsial groups do NOT map one vector each:** typhus group = **louse** (epidemic typhus) *and* **flea** (murine typhus); scrub typhus group = **mite**; spotted fever group = **tick**; transitional group = **mite** (*R. akari*, rickettsial pox) *and* **tick** (*R. australis*, Queensland tick typhus); Q fever is the arthropod-transmission exception (Park 28e, Ch 5, p 365).
- ▶ **Extrinsic incubation period in malaria and filaria = 10 to 14 days or longer**, depending on environmental temperature (Park 28e, Ch 15, p 922) — quote the figure with its qualifier.
- ▶ **Definitive host = sexual cycle** (mosquito in malaria); **intermediate host = asexual cycle** (mosquito in filaria, cyclops in guinea-worm). Reversing these is the commonest error.
- ▶ **Transovarial = female to progeny through the egg** (dengue, scrub typhus, rickettsial pox; probable but not proven in Kyasanur Forest Disease ticks — Service, p 256); **transstadial = larva**

to nymph to adult (ticks). Indian burden to quote: malaria 4-5 million cases annually, 236 million people in filaria-endemic areas (Park 28e, Ch 15, p 921).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: the question asks solely for classification and transmission, so the principles of arthropod control (environmental, chemical, biological and genetic) and the National Vector Borne Disease Control Programme have been deliberately left out; add them only if a stem explicitly asks for control. The Indian malaria and filaria burden figures quoted above are Park 28e's own text figures and are not dated by the book to a specific year, so state them as Park's figures rather than as current surveillance data.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 3 (Principles of Epidemiology and Epidemiologic Methods, pp 117-118), Ch 5 (Epidemiology of Communicable Diseases, pp 307, 322, 341, 350, 365-368), Ch 15 (Environment and Health — Medical Entomology, pp 921-923); Service — Medical Entomology for Students (Kyasanur Forest Disease, p 256; glossary: transovarial, transstadial, venereal and vertical transmission, pp 300-301). Coverage note: Park 28e Tables 1-3 were cross-checked against the 27th edition where the 28e scan was optically degraded.



Zoonoses — classification and control

Asked in: Paper I Nov 2017 — RGUHS MD Community Medicine

Definition — Zoonoses are "those diseases and infections [the agents of] which are naturally transmitted between [other] vertebrate animals and man" (Park 28e, Ch 5, p 341). Control rests on three components — in animals, on the vehicles of transmission, and in man (Park 28e, Ch 15, p 942).

Classification of zoonoses

- ▶ More than **150 zoonoses** are recognised, several emerging recently (Kyasanur forest disease, monkeypox); besides morbidity and mortality they cause **great economic loss** in animals and animal products, developing countries losing most (Park 28e, Ch 5, p 341).
- ▶ India has the **largest animal population in the world, nearly 11% of the world total** (Park 28e, Ch 15, p 941), and **68.86%** of Indians are rural, in close contact with domestic and often wild animals (Park 28e, Ch 5, p 341).
- ▶ G.S. Nelson stressed that the **direction** of transmission must be stated: what matters is the relative significance of man versus animals as the **maintenance host** (Park 28e, Ch 5, p 341).

A. By reservoir host and direction of transmission

Category	Direction of transmission	Examples
Anthropo-zoonoses	Lower vertebrate animal → man	Rabies, plague, hydatid disease, anthrax, trichinosis (Park 28e, Ch 3, pp 113–114)
Zoo-anthropozoonoses	Man → lower vertebrate animal	Human tuberculosis in cattle (Park 28e, Ch 3, p 114)
Amphixenoses	Maintained in both man and lower vertebrates; transmissible in either direction	<i>Trypanosoma cruzi</i> , <i>Schistosoma japonicum</i> (Park 28e, Ch 3, p 114)

B. By life cycle of the infecting agent (four categories, of considerable teaching value)

Each category shares epidemiological features, which is why this axis drives the choice of control strategy (Park 28e, Ch 5, p 341).

Category	Life cycle requirement	Key epidemiological feature	Examples
1. Direct zoonoses	Infected vertebrate → susceptible vertebrate by direct contact, fomite or mechanical vector	Agent undergoes little or no propagative change and no essential developmental change during transmission	Rabies, trichinosis, brucellosis
2. Cyclo-zoonoses	More than one vertebrate host species, but no invertebrate host	Developmental cycle completed entirely among vertebrates	Human taeniasis, echinococcosis, pentastomid infections
3. Meta-zoonoses	Transmitted biologically by invertebrate vectors	Agent multiplies and/or develops in the invertebrate; there is always an extrinsic incubation (prepatent) period before onward transmission	Arbovirus infections, plague, schistosomiasis
4. Sapro-zoonoses	A vertebrate host plus a non-animal developmental site or reservoir	Organic matter (including food), soil and plants are the non-animal site	Larva migrans, some of the mycoses

C. By aetiological agent, with the animal principally involved

Agents may be **viral, bacterial, rickettsial, fungal, helminthic, protozoal or arthropod**; Park's Table 1, "Partial list of zoonoses" (WHO 1967), prints **four** groups — bacterial, viral, protozoan, helminthic (Park 28e, Ch 15, pp 941–942); rickettsial zoonoses appear separately (Ch 5, Table 1, p 365).

Group	Disease in man → animal principally involved
A. Bacterial	Anthrax → cattle, sheep, goats, camels, pigs; brucellosis → herbivores, pigs, dogs, horses, buffaloes; ornithosis → wild and domestic birds; Q fever → cattle, sheep, goats; leptospirosis → rodents, mammals, wild animals; tuberculosis → cattle, sheep, goats, pigs, cats; plague → rodents
B. Viral	Rabies → dog, fox, skunk, mongoose, jackal and bat ; cowpox → cattle; monkeypox → monkeys, rodents; Eastern equine encephalitis → horses, rodents; Ross river fever → horses, cattle, goats, sheep, rats, bats, pigs; yellow fever → monkeys; Japanese encephalitis → wild birds; Lassa fever → multi-mammate rat

Group	Disease in man → animal principally involved
C. Protozoan	Leishmaniasis → dogs, cats, swine; toxoplasmosis → cats, mammals, birds; trypanosomiasis → game animals, cattle; babesiosis → cattle. Indian kala-azar is an exception — it is considered non-zoonotic , with man as the sole reservoir (Park 28e, Ch 5, pp 370–371)
D. Helminthic	Taeniasis → cattle (<i>Taenia saginata</i>); the pig is the main intermediate host of <i>T. solium</i> (Park 28e, Ch 5, p 368); echinococcosis → dogs, wild carnivores, domestic and wild ungulates; trichinellosis → swine, rodents, wild carnivores, marine mammals; clonorchiasis → dogs, cats, swine, wild mammals, fish; fasciolopsis → swine, dogs; schistosomiasis → rodents
Rickettsial (Park 28e, Ch 5, p 365)	Scrub typhus → rodents; murine (endemic) typhus → rodents; Rocky Mountain spotted fever and Indian tick typhus → rodents, dogs; Siberian tick typhus → rodents; rickettsial pox → mice; Queensland tick typhus → rodents, marsupials; Q fever → sheep, cattle, goats

Control measures of zoonoses

Component	Measures
(a) Control in animals	Diagnosis of the zoonotic condition in the animal; treatment; destruction; quarantine; immunisation
(b) Control of the vehicles of transmission	Establishment of food hygiene practices; ensuring safety of animal products — wool, hides, horn, bones, fat; proper disposal of animal carcasses and wastes ; disinfection procedures
(c) Prevention and treatment in man	Protection of high-risk groups by immunisation and chemoprophylaxis ; monitoring of health status including occupational health programmes ; prevention of spread by man; early diagnosis and treatment; health education ; prevention of environmental and food contamination; improvement of diagnostic facilities

These three components are operationalised as follows (Park 28e, Ch 15, p 942).

- **Controlling the reservoir** is the most desirable measure where possible and is comparatively easy with an **animal** reservoir (bovine tuberculosis, brucellosis); in man only early diagnosis, notification, isolation, treatment, quarantine, surveillance and disinfection are possible (Park 28e, Ch 3, p 144). For brucellosis, **test-and-slaughter with full compensation to the farmer** is the only satisfactory solution aimed at eradication (Park 28e, Ch 5, p 357).
- **Mass immunisation of the animal reservoir.** Urban rabies — elimination of stray and ownerless dogs plus immunisation, in the shortest possible time, of **at least 80% of the dog population** (Park 28e, Ch 5, p 346). Brucellosis — *B. abortus* **strain 19** vaccine for young animals, compulsory yearly for all heifers; sustained **7 to 10 years** it may eliminate the disease (Park 28e, Ch 5, p 357).
- **Vector and rodent control** for meta-zoonoses and rodent-borne zoonoses: measures aimed at the vector and its breeding places, with control of cattle and pets (Park 28e, Ch 3, p 146). Anti-rodent work rests on **environmental sanitation** — proper garbage storage and disposal, rat-proof godowns, burrows blocked with concrete. **Sanitation is essential to the permanent control of rats; trapping, poisoning and fumigation are supplementary** (Park 28e, Ch 15, pp 940–941).

- **Food-chain and product safety:** at farm level, hygienic feed and a sanitary environment, aiming at "**salmonella-free**" animals; then hygienic slaughtering, pasteurisation of milk and eggs, solid-waste disposal and cold storage (Park 28e, Ch 5, pp 364–365).
- **Protection of occupationally exposed groups** — Park names **farmers, shepherds, milkmen and abattoir workers**: personal hygiene, care in handling placenta, discharges and aborted fetuses, protective clothing for carcasses, **pasteurisation or boiling of milk**, and early treatment in man (Park 28e, Ch 5, p 357). For Kyasanur forest disease — protective clothing, the repellents dimethyl phthalate (DMP) and diethyltoluamide (DEET), daily tick self-examination and removal, and **killed KFD vaccine** for the exposed (Park 28e, Ch 5, pp 354–355).
- **Surveillance and notification:** notification enables early detection of outbreaks and immediate action by the health authority (Park 28e, Ch 3, p 144). The One Health agenda calls for **joint collaborative activities between the animal and human health sectors in surveillance and response and in risk assessment** (Oxford Textbook of Public Health 7e, Ch 75, p 2861); in India these field data flow through the Integrated Disease Surveillance Programme / Integrated Health Information Platform (IDSP/IHIP) (One Health Booklet, AIIMS, Ch 1, pp 5–7).
- **Legislation and health education** for rabies: registration and licensing of domestic dogs, restraint in public places, destruction of dogs and cats bitten by rabid animals, quarantine of imported dogs for **about 6 months** (Park 28e, Ch 5, p 346).

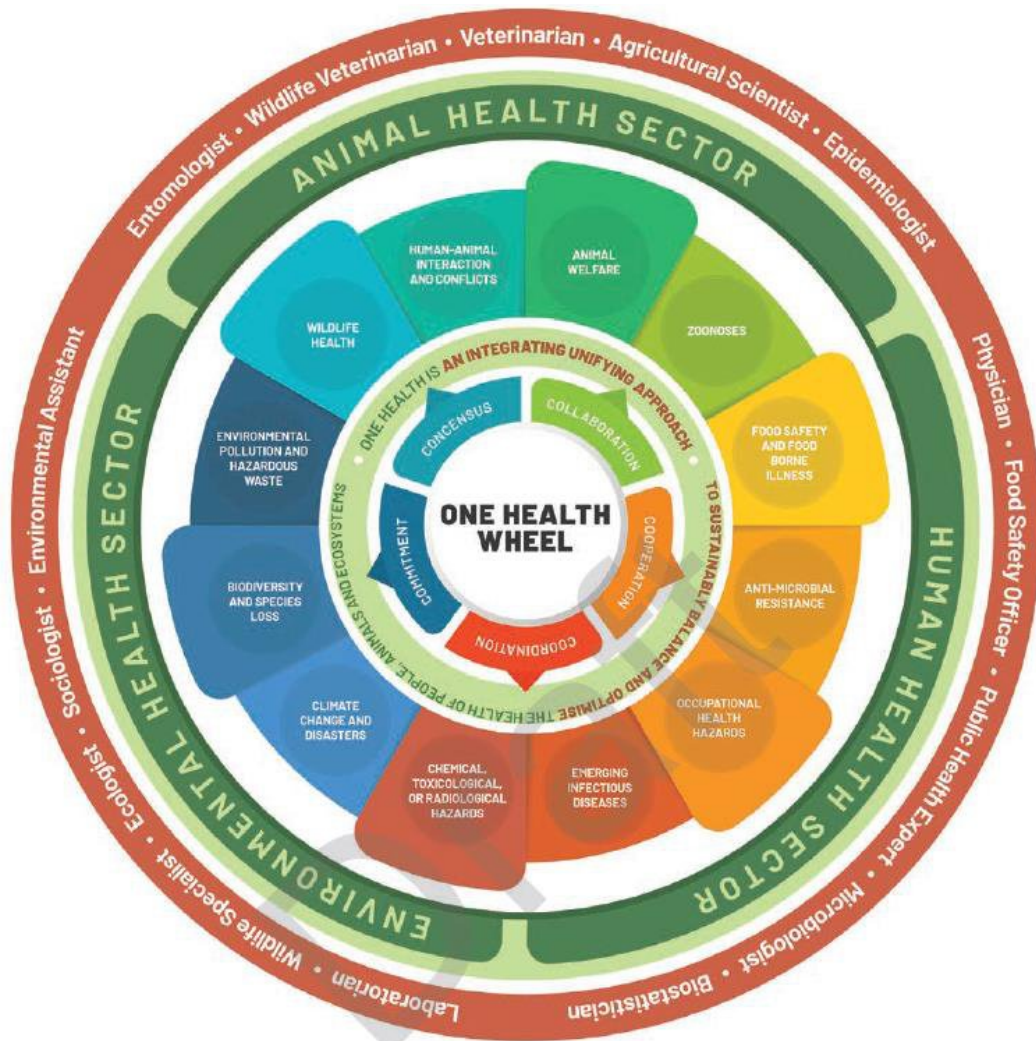
Intersectoral action, One Health and the Indian programmes

- Park makes the point for **salmonellosis** — the health sector alone cannot solve it, so responsibility may fall to agriculture, veterinary and other ministries outside the health sector (Park 28e, Ch 5, p 365) — and it generalises to every zoonosis; community participation, political support and intersectoral coordination are prerequisites (Park 28e, Ch 3, p 144).
- The "**One Medicine**" concept, blending human medicine and veterinary science, evolved into "**One World – One Health**" (conceived 2004) — an **integrative, multi-disciplinary** strategy joining multisector expertise for the health of man, animals and ecosystems (One Health Approach, Ch 1, p 2).

Programme / initiative	Nodal agency and launch	Core content
National One Health Programme for Prevention and Control of Zoonoses (NOHP-PCZ)	National Centre for Disease Control (NCDC), Ministry of Health and Family Welfare; the first programme on strengthening intersectoral coordination for zoonoses came in the 12th Five Year Plan	A network of 15 Regional Coordinators and 33 sentinel / surveillance sites across India; NCDC is establishing a National Centre on One Health ; field data are collected and analysed through IDSP/IHIP to keep away the threat of emerging and re-emerging diseases; NCDC also runs the National Programme on Climate Change and Human Health (One Health Booklet, AIIMS, Ch 1, pp 3–7)
National One Health Mission	Office of the Principal Scientific Adviser to the Government of India	Coordinates India's efforts with the international quadripartite (WHO–FAO–WOAH–UNEP); under PM-ABHIM a biosafety- and biosecurity-based laboratory network is being created for the wildlife–domestic animal–human interface (One Health Booklet, AIIMS, Ch 1, p 5)

Programme / initiative	Nodal agency and launch	Core content
National Rabies Control Programme (NRCP)	Launched in the 12th Five Year Plan; NCDC nodal for the human component, Ministry of Environment and Forests nodal for the animal component	Capacity building of health and veterinary professionals; intradermal route of vaccine; surveillance of animal bites; dog population survey; mass vaccination of dogs; dog population management; IEC; strengthening laboratory diagnosis; collaboration with veterinary and animal husbandry (Sunder Lal, Ch 8, p 429)
National Action Plan for Dog-mediated Rabies Elimination from India by 2030	Released September 2021 by NCDC	Three elements — (a) epidemiological surveillance : rabies made notifiable in both the national health and the veterinary systems , with reporting of clinical and laboratory-confirmed cases and of persons receiving post-exposure prophylaxis; (b) mass canine parenteral vaccination of at least 70% of the dog population in endemic areas, with registration and permanent identification of vaccinated dogs (oral vaccine as a supplementary measure for inaccessible dogs); (c) dog population management (Sunder Lal, Ch 8, p 430)

- **Dog population management in India.** Until 1998 the stray-dog population was controlled by impounding and euthanising, which was against animal rights; the **Animal Birth Control (ABC)** policy — sterilise and release in the same area — was therefore adopted, and **culling of stray dogs has been banned since 2001**, the programme now running under the ABC Rules (latest revision 2023). To succeed it must sterilise **70% of stray dogs in a given area within 6 months**, before the next reproductive cycle (Jugal Kishore, pp 482–483; Sunder Lal, Ch 8, p 430).

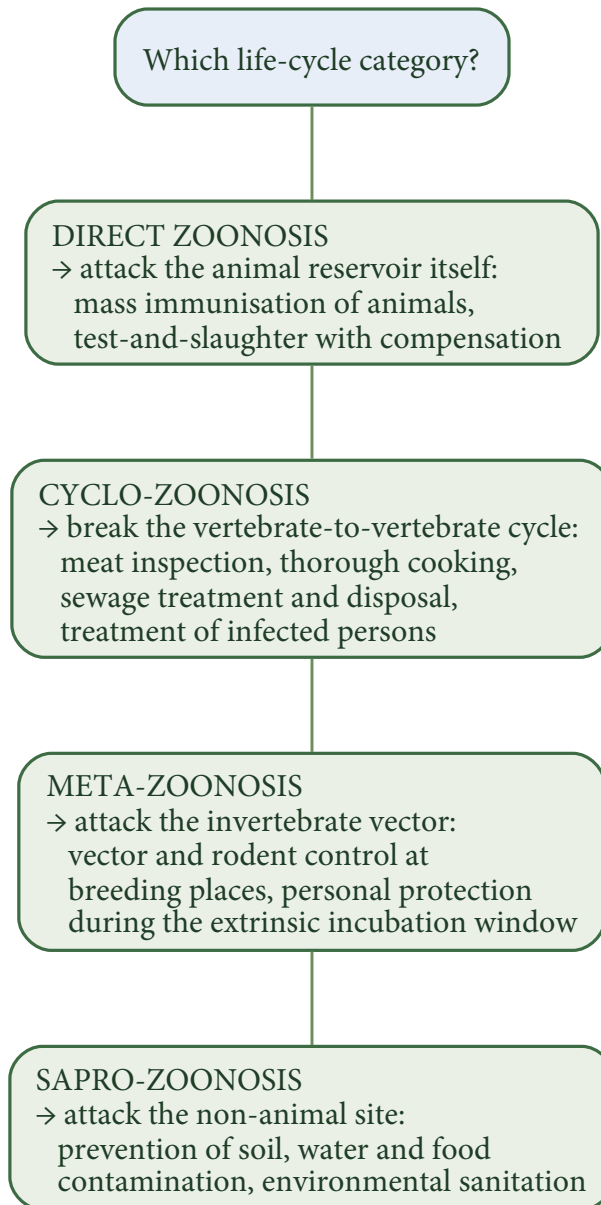


The One Health wheel — the sectors and disciplines involved (One Health Booklet (AIIMS), p.10)

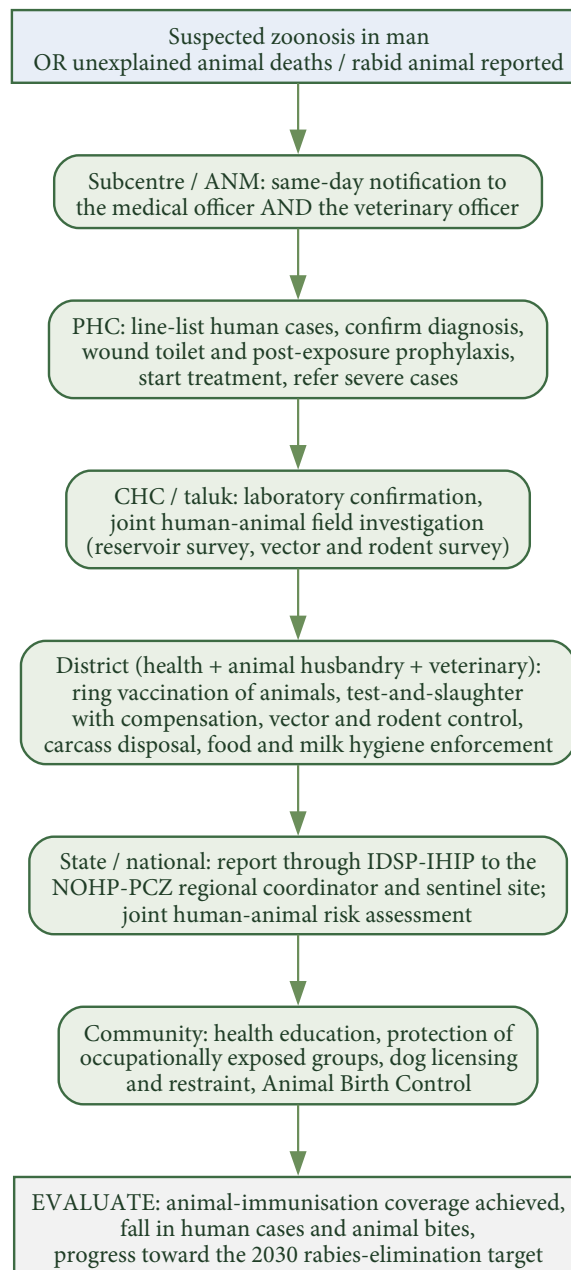
DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Chain of transmission of a zoonosis — five boxes in a line, with the three points of attack drawn as arrows striking the chain. Exact labels:

- ▶ Animal reservoir (domestic / wild vertebrate)
- ▶ Portal of exit (saliva, urine, milk, faeces, carcass)
- ▶ Mode of transmission (direct contact / vehicle: food, milk, hides / biological vector)
- ▶ Portal of entry in man (bite, ingestion, inhalation, skin abrasion)
- ▶ Susceptible human host
- ▶ Attack point (a) — Control in animals
- ▶ Attack point (b) — Control of the vehicles of transmission
- ▶ Attack point (c) — Prevention and treatment in man



The life-cycle category of a zoonosis decides which control lever is the dominant one



Operational response to a suspected zoonosis, by level of care

High-yield exam pointers

- ▶ **Definition verbatim:** diseases and infections naturally transmitted between vertebrate animals and man; > 150 recognised.
- ▶ **Life-cycle mnemonic "D-C-M-S":** Direct, Cyclo, Meta, Sapro. Only meta-zoonoses have an **extrinsic incubation period**; only sapro-zoonoses have a **non-animal** site.
- ▶ **Three components of control:** in animals → vehicles of transmission → prevention and treatment in man.
- ▶ **Rabies numbers:** Park — ≥ 80% of dogs immunised, imported dogs quarantined **6 months**; National Action Plan 2021 — ≥ 70% canine parenteral vaccination; ABC — 70% of strays sterilised within **6 months**; culling banned **2001**.

- ▶ **NOHP-PCZ:** NCDC, 15 **Regional Coordinators**, 33 **sentinel sites**, data via IDSP/IHIP. NRC — NCDC nodal for man, Ministry of Environment and Forests for animals.
- ▶ **Brucellosis:** strain 19 vaccine for heifers, sustained 7–10 years; test-and-slaughter with **full compensation**.
- ▶ **Rats:** sanitation alone is **permanent control**. **India:** 68.86% rural, about 11% of the world's animals.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: Park prints the 68.86% rural proportion without a year. Q fever appears in both the bacterial and the rickettsial groups because *Coxiella burnetii* was historically classified with the rickettsiae; either placement is accepted. Park's "elimination of stray and ownerless dogs" is now stale for India — culling of stray dogs has been banned since 2001, and dog-population management is by Animal Birth Control sterilise-and-release plus mass vaccination (Sunder Lal, Ch 8, p 430).

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, WHO, National Action Plan for Rabies Elimination 2021, National Guidelines for Rabies Prophylaxis 2019, Animal Birth Control Rules 2023, WHO Pandemic Agreement

Programme and policy updates (India)

- ▶ **NAPRE is the operative rabies document and it has widened its toolkit.** The National Action Plan for Rabies Elimination (2021) commits India to zero dog-mediated human rabies deaths by 2030 and now endorses **catch-neuter-vaccinate-release (CNVR) together with parenteral and oral dog vaccination**; animal-bite reporting has moved from IDSP paper returns onto the real-time IHIP digital platform. **Sikkim and Goa have reached rabies-controlled status.** The national block is not absent policy but non-interoperable human–veterinary–municipal data and episodic, NGO-dependent vaccination drives. [PMID 42494549]
- ▶ **A second national One Health architecture has grown up beside NOHP-PCZ.** The Department of Biotechnology launched India's **first One Health Consortium** — 27 (now 28) veterinary, medical and wildlife centres mapping nationwide prevalence of **10 priority zoonoses and 5 transboundary animal diseases**; ICMR with ICAR are establishing a **National Institute of One Health at Nagpur**; **One Health Support Units** were piloted in Karnataka and Uttarakhand (2022) and then expanded; and the **Principal Scientific Adviser** carries the National One Health Mission. Two older anchors worth quoting: the **Division of Zoonoses at NICD (now NCDC) in 1964** and the **National Committee on Zoonoses in 1978**. [PMID 39897352]
- ▶ **Oral vaccination of dogs is the emerging answer to the coverage problem.** With roughly **6.2 crore free-roaming dogs**, a purely catch-vaccinate-release national campaign would need about **1.1 million vaccinators**; oral bait handout (SAG2, SPBN GASGAS) needs no restraint and reaches shy and aggressive dogs, and cost optimisation for an Indian city shows baits stay the cheapest option even priced up to **10 times parenteral vaccine**. Proposed first as pilots in hotspot districts, alongside — not instead of — parenteral campaigns. [PMID 40511438] [PMID 37606523]

Current national data

- ▶ **Dog-bite and PEP-quality figures now exist at community level.** A 2022–23 survey across 15 states gave **6.6 animal bites and 5.6 dog bites per 1000 per year**, yet **20.5%** of victims received no anti-rabies vaccine, **49.1%** of those who started did not complete the course, and only **11.2%** of eligible victims received immunoglobulin. A Delhi survey (2023–24) found annual dog-bite incidence **25.5 per 1000**, ranging from **15.9** in planned colonies to **32.1** in Jhuggi-Jhopri clusters; **75.5%** of bites were by stray dogs, **65.7%** were Category III, and only **26.9%** of Category III bites received immunoglobulin or monoclonal antibody. [PMID 41354795]
- ▶ **Scrub typhus has been quantified in the community for the first time.** A two-year cohort of 32,279 people in 37 Tamil Nadu villages gave a clinical incidence of **6.0 per 1000 person-years** (95% CI 4.8–7.5), seroconversion **81.2 per 1000 person-years**, baseline IgG seroprevalence **42.8%**, scrub typhus accounting for **29.3% of all fever hospitalisations**, and case fatality **1.5%**; diabetes carried a 2.4-fold higher risk of severe disease. [PMID 40073309]

Elimination, surveillance and disease status

- ▶ **"Zero by 30" and India's share of it.** Rabies causes about **59,000 deaths, 3.7 million DALYs and USD 8.6 billion** of losses worldwide each year, of which India carries roughly **one-third (35%)**; the WHO–FAO–WOAH–GARC "United Against Rabies" target is zero human deaths from dog-mediated rabies by 2030, and the operational benchmark is **≥70% canine vaccination coverage annually for three consecutive years** — quote this figure alongside Park's **≥80%**, not instead of it. [PMID 40511438] [PMID 39713832]
- ▶ **Emerging zoonoses and formal priority-setting.** Nipah remains a WHO R&D Blueprint priority pathogen requiring BSL-4 containment, case fatality **38–75%**; the 2023 Kerala outbreak had six cases and two deaths, following a larger Bangladesh season, and **no vaccine or drug is licensed** — current evidence supports trialling the monoclonal antibodies **1F5 and m102.4** and the **small molecule remdesivir**, while **chloroquine is explicitly not recommended**. Structured expert prioritisation in Punjab ranked **brucellosis (0.70) > rabies (0.69) > anthrax (0.64) > leptospirosis = toxoplasmosis = highly pathogenic avian influenza (0.62)**, placing endemic zoonoses above epidemic-potential ones — a directly writable illustration of how a state sets its control priorities. [PMID 39549708] [PMID 40437297] [PMID 37989991]

Guideline and legislative updates

- ▶ **Rabies monoclonal antibody is displacing equine immunoglobulin.** A phase 4 randomised trial in 3,994 Category III exposure patients across 15 Indian tertiary hospitals compared rabies monoclonal antibody (3.33 IU/kg, infiltrated into and around the wound on day 0) plus purified vero-cell vaccine against equine immunoglobulin (40 IU/kg) plus the same vaccine: day-14 geometric mean neutralising antibody **16.05 IU/mL (95% CI 13.25–19.44)** with the monoclonal versus **13.48 IU/mL (9.51–19.11)** with ERIG, **no rabies in either arm over one year**, and fewer causally related adverse events with the monoclonal. [PMID 40783290]
- ▶ **The legal frame around stray dogs has shifted, and it is examinable as a tension rather than a rule.** Dog-bite management now rests on the Prevention of Cruelty to Animals Act 1960, the National Guidelines for Rabies Prophylaxis 2019, the Animal Birth Control Rules 2023 and the Bharatiya Nyaya Sanhita 2023. Reported burden for 2024 was **over 37 lakh dog bites with 54 human rabies deaths**, against modelled estimates of **17.4 million animal bites and 20,565 rabies**

deaths a year — quote both and label which is reported and which is estimated. **August 2025 Supreme Court proceedings** permitting forcible removal of street dogs in defined circumstances are widely read as conflicting with the ABC sterilise-and-release mandate. [PMID 41696785] [PMID 42494549]

Global and WHO developments

- The WHO Pandemic Agreement is the new global instrument to name. It was adopted unanimously by the 78th World Health Assembly, and an Intergovernmental Working Group is drafting the Pathogen Access and Benefit-Sharing (PABS) annex under Article 12, covering the sharing of biological materials and data and of the vaccines and therapeutics derived from them. WHO's rabies position (fact sheet updated **June 2024**) keeps the 59,000-deaths estimate, prefers the intradermal over the intramuscular post-exposure route, and now names monoclonal antibodies alongside immunoglobulin. Separately, WHO and Africa CDC declared mpox a public health emergency in August 2024, with India reporting its first clade Ib case the following month. [PMID 41407491] [WHO 2024-Jun] [PMID 39361760]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 3, Ch 5, Ch 15 (pp 113–371, 940–942); Sunder Lal — Textbook of Community Medicine, Ch 8 (pp 429–430); One Health Booklet (AIIMS), Ch 1 (pp 3–10); Jugal Kishore — National Health Programmes of India (pp 481–483); Oxford Textbook of Public Health 7e, Ch 75 (p 2861); One Health Approach, Ch 1 (p 2). Named documents: National Action Plan for Dog-mediated Rabies Elimination from India by 2030 (NCDC, September 2021); Animal Birth Control Rules (2023).

Recent-advances sources: 42494549 — One Health implementation for rabies elimination in India by 2030: progress, governance, challenges, and policy alignment, *One Health* 2026, doi:[10.1016/j.onehlt.2026.101520](https://doi.org/10.1016/j.onehlt.2026.101520); 39897352 — One Health initiative in India: genesis and hurdles in establishing the first consortium, *Veterinary World* 2024, doi:[10.14202/vetworld.2024.2925-2931](https://doi.org/10.14202/vetworld.2024.2925-2931); 40511438 — Oral vaccination of dogs in India as a strategy for elimination of rabies: scope of implementation, *Indian Journal of Community Medicine* 2025, doi:[10.4103/ijcm.ijcm_34_24](https://doi.org/10.4103/ijcm.ijcm_34_24); 37606523 — Optimising rabies vaccination of dogs in India, *Epidemiology and Infection* 2023, doi:[10.1017/S0950268823001334](https://doi.org/10.1017/S0950268823001334); 41354795 — Burden of dog bites and associated healthcare seeking behaviour for rabies post-exposure prophylaxis in North and South Shahdara municipal zones of Delhi, India, *BMC Public Health* 2025, doi:[10.1186/s12889-025-25860-w](https://doi.org/10.1186/s12889-025-25860-w); 40073309 — Incidence of scrub typhus in rural South India, *New England Journal of Medicine* 2025, doi:[10.1056/NEJMoa2408645](https://doi.org/10.1056/NEJMoa2408645); 39713832 — Elimination of dog-mediated human rabies: scientific tools, One Health and partnerships, *Revue Scientifique et Technique (WOAH)* 2024, doi:[10.20506/rst.SE.3560](https://doi.org/10.20506/rst.SE.3560) (abstract-only); 39549708 — Therapeutics for Nipah virus disease: a systematic review to support prioritisation of drug candidates for clinical trials, *The Lancet Microbe* 2024, doi:[10.1016/j.lanmic.2024.101002](https://doi.org/10.1016/j.lanmic.2024.101002); 40437297 — A One Health approach to understanding and managing Nipah virus outbreaks, *Nature Microbiology* 2025, doi:[10.1038/s41564-025-02020-9](https://doi.org/10.1038/s41564-025-02020-9) (abstract-only); 37989991 — Assessing and prioritizing zoonotic diseases in Punjab, India: a One Health approach, *EcoHealth* 2023, doi:[10.1007/s10393-023-01654-7](https://doi.org/10.1007/s10393-023-01654-7) (abstract-only); 40783290 — Post-exposure prophylaxis regimen of rabies monoclonal antibody and vaccine in category 3 potential exposure patients: a phase 4, open-label, randomised, active-controlled trial, *The Lancet* 2025, doi:[10.1016/S0140-6736\(25\)00735-4](https://doi.org/10.1016/S0140-6736(25)00735-4) (abstract-only); 41696785 — Stray dog bites and rabies in India: a silent epidemic, *Medico-Legal Journal* 2026, doi:[10.1177/00258172251376869](https://doi.org/10.1177/00258172251376869) (abstract-only); 41407491 — Achieving equity to fully realise the pandemic agreement, *BMJ Global Health* 2025, doi:[10.1136/bmjgh-2025-020691](https://doi.org/10.1136/bmjgh-2025-020691); 39361760 — Africa aims to avert an mpox pandemic, *Science* 2024, doi:[10.1126/science.adt4793](https://doi.org/10.1126/science.adt4793) (abstract-only). Also used: WHO rabies fact sheet, updated 5 June 2024.

♦ ♦ ♦

One Health approach

Asked in: Paper II 26-Sep-2025 — RGUHS MD Community Medicine

Definition — **One Health** is an integrated, unifying approach that aims to sustainably balance and optimise the health of human beings, companion animals, wild life, plants and ecosystems (One Health Booklet, AIIMS Bibinagar–NCDC, p 5). It is a holistic, systems-based approach that recognises the interconnection between the four major systems of the ecosphere — humans, animals, plants and the environment — and works through multisectoral expertise with intersectoral coordination rather than through single-sector silos (One Health Approach, Ch 1, p 2).

Evolution of the concept

Stage	Content
One Medicine	Blending of human medicine and veterinary science as a reaction to zoonotic diseases (One Health Approach, Ch 1, p 2)
Definition of zoonoses	"Those diseases and infections which are naturally transmitted between vertebrate animals and man"; more than 150 zoonoses are recognised, and WHO (1967) classified them into groups (Park 28e, Ch 5, p 341; Ch 15, p 941)
"One World – One Health", conceived 2004	Added the health of the entire ecosystem, including wild flora and fauna, to the human–animal pairing (One Health Approach, Ch 1, p 2)
One Health (current)	A worldwide strategy that is integrative and multidisciplinary, covering the human–animal–plant–environment interface, including plant health, since infected plants can act as vehicles for infections of humans (One Health Approach, Ch 1, p 2)
Quadripartite governance, 2022	World Health Organization (WHO), Food and Agriculture Organization (FAO), World Organisation for Animal Health (WOAH) and United Nations Environment Programme (UNEP) issued the One Health Joint Plan of Action (2022–2026) (One Health Booklet, pp 5, 7)

Rationale — why a single-sector approach fails

Evidence	Value (source, year)
Existing human infections of animal origin	60% (One Health Booklet, p 7)
Emerging human infections of animal origin	at least 75% ; more than 60% of emerging infectious diseases are zoonotic, and about 70% of these originate in wildlife (One Health Booklet, p 7, pp 15-16)
New human diseases appearing each year	3 of every 5 are of zoonotic origin (One Health Booklet, p 7)
Agents with bioterrorism potential	80% are zoonotic pathogens (One Health Booklet, p 7)
Human disease burden in low-income countries from infectious disease	more than 60% (One Health Booklet, pp 15-16)
Global burden of unsafe food	600 million illnesses and more than 400,000 deaths a year (WHO, 2015) (One Health Booklet, pp 20-21)

Evidence	Value (source, year)
Economic loss from foodborne disease	USD 95 billion productivity loss a year in low- and middle-income countries; USD 15 billion a year to treat (World Bank, 2018) (One Health Booklet, p 21)
India-specific vulnerability	High human population density, a very large livestock population and diverse wildlife habitats; nearly 68.86% of India's population is rural and lives in close contact with domestic and often wild animals (Park 28e, Ch 5, p 341; One Health Booklet, p 7)

Drivers that push pathogens across the interface — population growth and expansion into new geographical areas, land-use change, intensive food production, unplanned urbanisation, poor waste management and disposal, pollution, biodiversity and climate crises, and increased trade and travel, which convert a local spillover into an epidemic or pandemic (One Health Booklet, p 7, p 11).

Scope — domains, guiding principles and the sectors involved

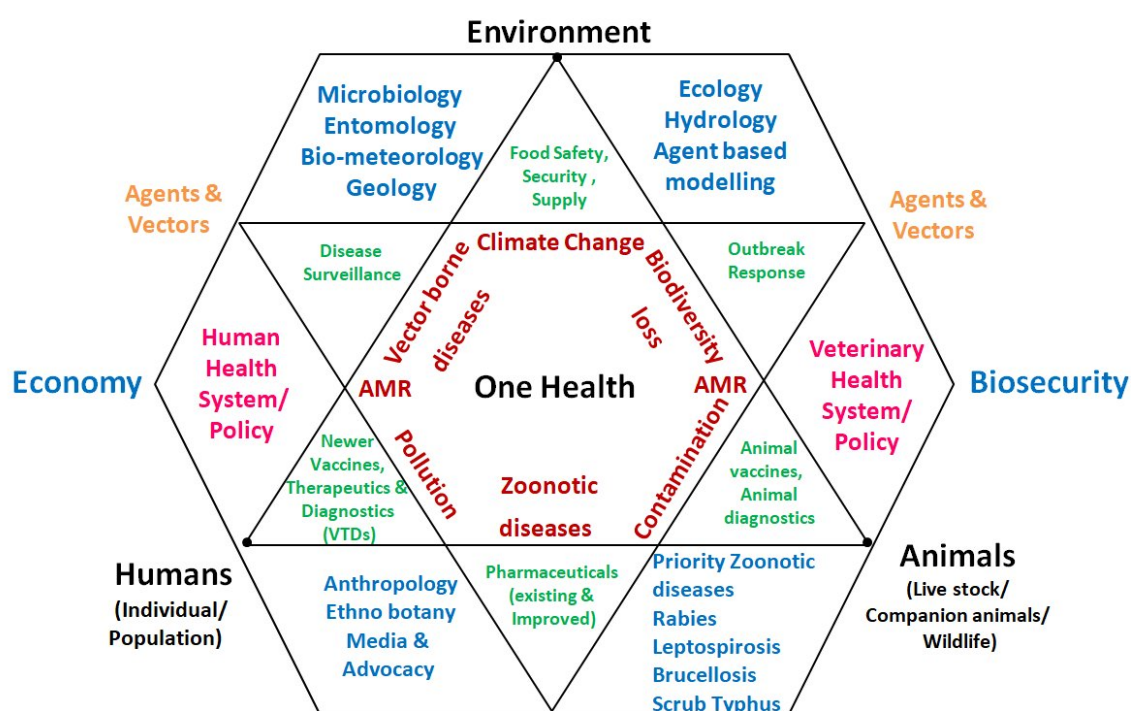


Fig 8 — The One Health Constellation Model (One Health Booklet (AIIMS), p.17)

Domain	What it covers
Zoonotic and emerging infectious diseases	Priority zoonoses (rabies, leptospirosis, brucellosis, scrub typhus), spillover from wildlife, outbreak response, epidemic and pandemic preparedness (One Health Booklet, Fig 8, p 17)
Vector-borne diseases	Defining the topographical and temporal distribution of the various vector species, their lifecycle, and transmission to humans (One Health Approach, Ch 1, p 2)
Food and water safety, security and supply	Zoonotic and non-zoonotic pathogens, chemical contaminants, veterinary drug residues, pesticides and heavy metals entering the food chain from before harvest to consumption (One Health Booklet, pp 20-21; Fig 8, p 17)

Domain	What it covers
Antimicrobial resistance (AMR)	Resistance generated at the human–animal–plant–environment interface by therapeutic, prophylactic and growth-promoter antibiotic use, and by pharmaceutical and industrial effluent (One Health Booklet, pp 13-14, 17)
Environment, climate and biodiversity	Pollution and contamination of air, water and soil, waste disposal, climate change, biodiversity loss (One Health Booklet, p 11, Fig 8, p 17)
Other domains in the One Health wheel	Wildlife health; human–animal interaction and conflicts; animal welfare; occupational health hazards; chemical, toxicological or radiological hazards; climate change and disasters; environmental pollution and hazardous waste (One Health Booklet, Fig 3, p 10)

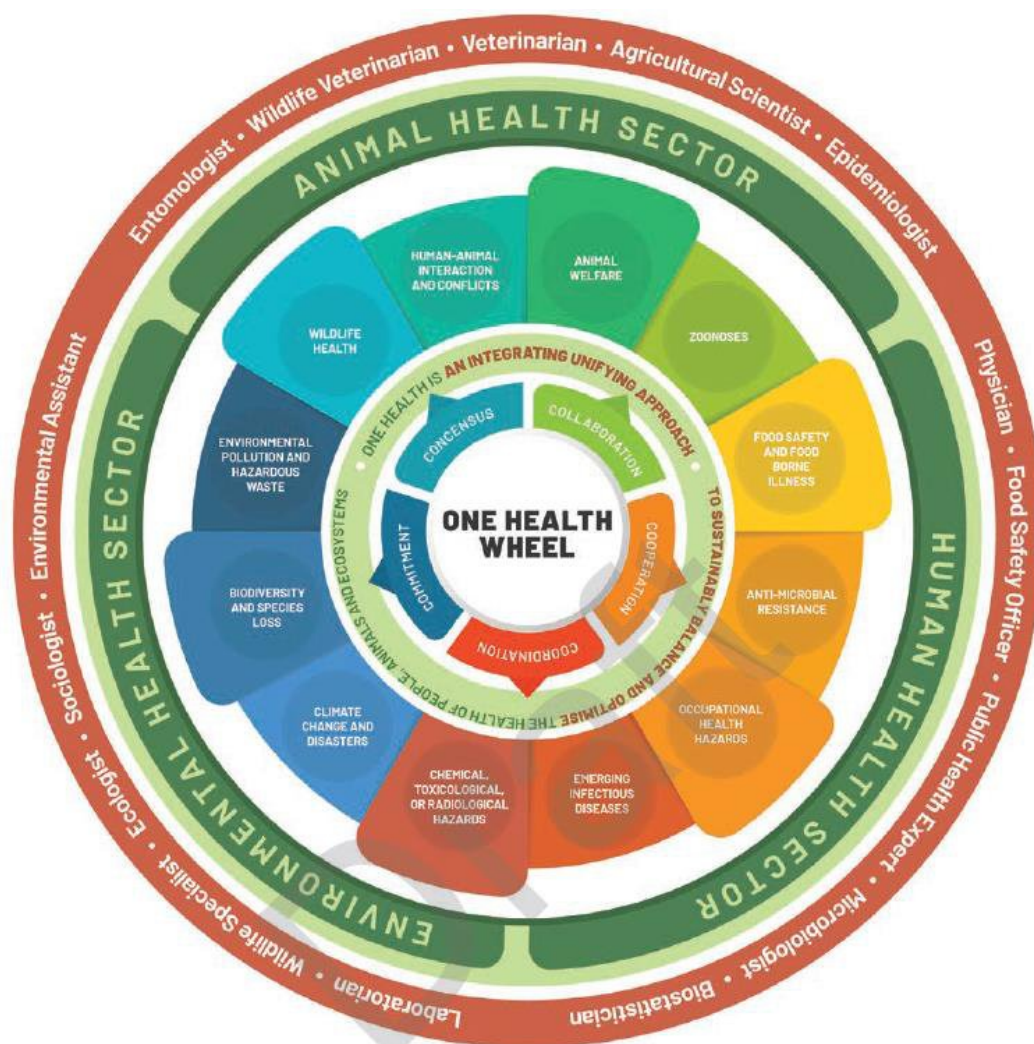


Fig 3 — The One Health wheel – representing the various sectors and discipline (One Health Booklet (AIIMS), p.10)

Five "C" principles at the hub of the One Health wheel — CONSENSUS, COLLABORATION, COOPERATION, COORDINATION and COMMITMENT, around the legend that One Health is an integrating, unifying approach to sustainably balance and optimise the health of people, animals and ecosystems (One Health Booklet, Fig 3, p 10).

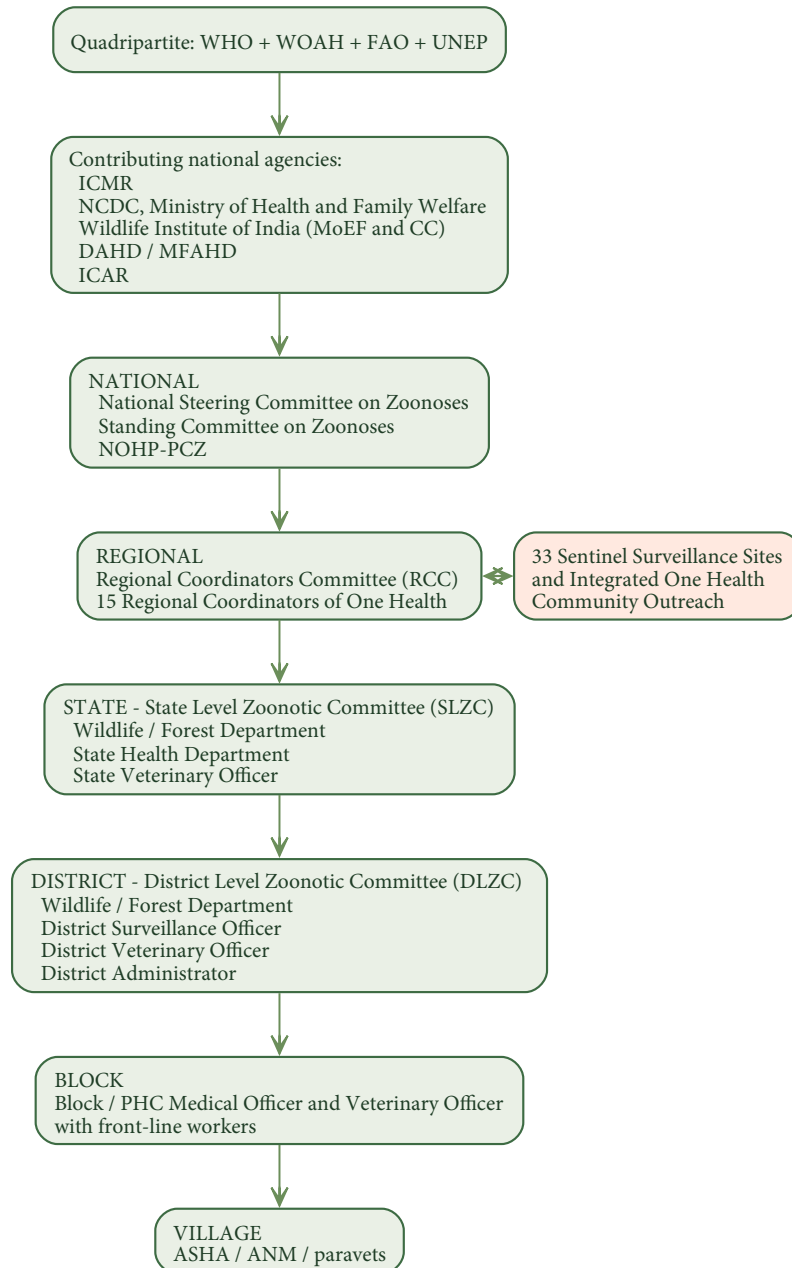
Sector at the table (One Health Booklet, Fig 3, p 10)	Professionals in the ring
Animal health sector	Entomologist, wildlife veterinarian, veterinarian, agricultural scientist, epidemiologist
Human health sector	Physician, food safety officer, public health expert, microbiologist, biostatistician
Environmental health sector	Laboratorian, ecologist, wildlife specialist, sociologist, environmental assistant

Implementation — global framework and the Indian programme

- ▶ The quadripartite **One Health Joint Plan of Action (2022–2026)** combats health risks at the human–animal–environment interface, aiming at public health, animal health, food safety and food security in addition to ecosystem health (One Health Booklet, p 7).
- ▶ In India the **National One Health Mission** is coordinated by the **Office of the Principal Scientific Adviser (PSA) to the Government of India**, which links national efforts with the international agencies (One Health Booklet, p 5).
- ▶ The **National Centre for Disease Control (NCDC)**, under the Ministry of Health and Family Welfare, is the operational lead. It runs the **National One Health Programme for Prevention and Control of Zoonoses (NOHP-PCZ)** and the **National Programme on Climate Change and Human Health (NPCCHH)**, addressing zoonoses, antimicrobial resistance and climate change. NCDC is establishing a **National Centre on One Health** and has already put in place **15 Regional Coordinators and 33 surveillance/sentinel sites** across India (One Health Booklet, pp 3, 5).
- ▶ The first programme on *strengthening intersectoral coordination for prevention and control of zoonoses* was launched in the **12th Five Year Plan**; NCDC has since started a field epidemiology programme, "**Sector Connect**", for an integrated response to emerging public health threats (One Health Booklet, p 3).
- ▶ Under the **Prime Minister's Ayushman Bharat Health Infrastructure Development Mission (PM-ABHIM)** a network of laboratories built on **biosafety and biosecurity** principles is being created for diseases arising at the wildlife–domestic animal–human interface (One Health Booklet, p 5). The national PM-ABHIM One Health package comprises a **national institute for One Health**; 4 new national institutes of virology with a WHO–South-East Asia Region research platform; **9 biosafety level 3 (BSL-3) laboratories**; strengthening of NCDC with 5 new regional Centres for Disease Control and 20 metropolitan surveillance units; integrated public health laboratories in all districts; block public health units at community health centres; expansion of the Integrated Health Information Platform to all states and union territories; 17 new and 33 strengthened public health units at points of entry (32 airports, 11 seaports, 7 land crossings); and 15 health emergency operation centres (Sunder Lal, Ch 20, p 805).
- ▶ Field data flow through the **Integrated Disease Surveillance Programme (IDSP)** and its **Integrated Health Information Platform (IHIP)** (One Health Booklet, p 5). **Activities at every level** — joint capacity building, laboratory strengthening, joint risk assessment and communication plan, joint operational research, joint monitoring with periodic reviews (One Health Booklet, Fig 16, p 29).

Year	Milestone in India (One Health Booklet, p 7)
2009	One Health Alliance of South Asia (OHASA) launched
2010	National Centre for Disease Control (NCDC) established
2014	COHEART initiated in Kerala
2014–2017	ICMR–ICAR joint project on zoonoses in Central India
2018	IDSP's Integrated Health Information Platform (IHIP) commenced
2019	One Health Initiative declared
2021	PM-ABHIM initiated; One Health Consortium formed
Nov 2022	AIIMS Bibinagar designated Regional Coordinator for One Health under NOHP-PCZ; 1200 sq m earmarked there for a BSL-3 laboratory under PM-ABHIM (One Health Booklet, p 28)

Top 10 priority zoonotic diseases for India under NOHP-PCZ (2020) — zoonotic influenza (zoonotic influenza A viruses), anthrax, Japanese encephalitis, leptospirosis, brucellosis, dengue, rabies, scrub typhus, plague, and Crimean-Congo haemorrhagic fever (One Health Booklet, p 19).



Institutional mechanism on One Health in India, from the global quadripartite to the village worker (One Health Booklet, Fig 16, p 29)

Operational strategy — prevention, preparedness and response



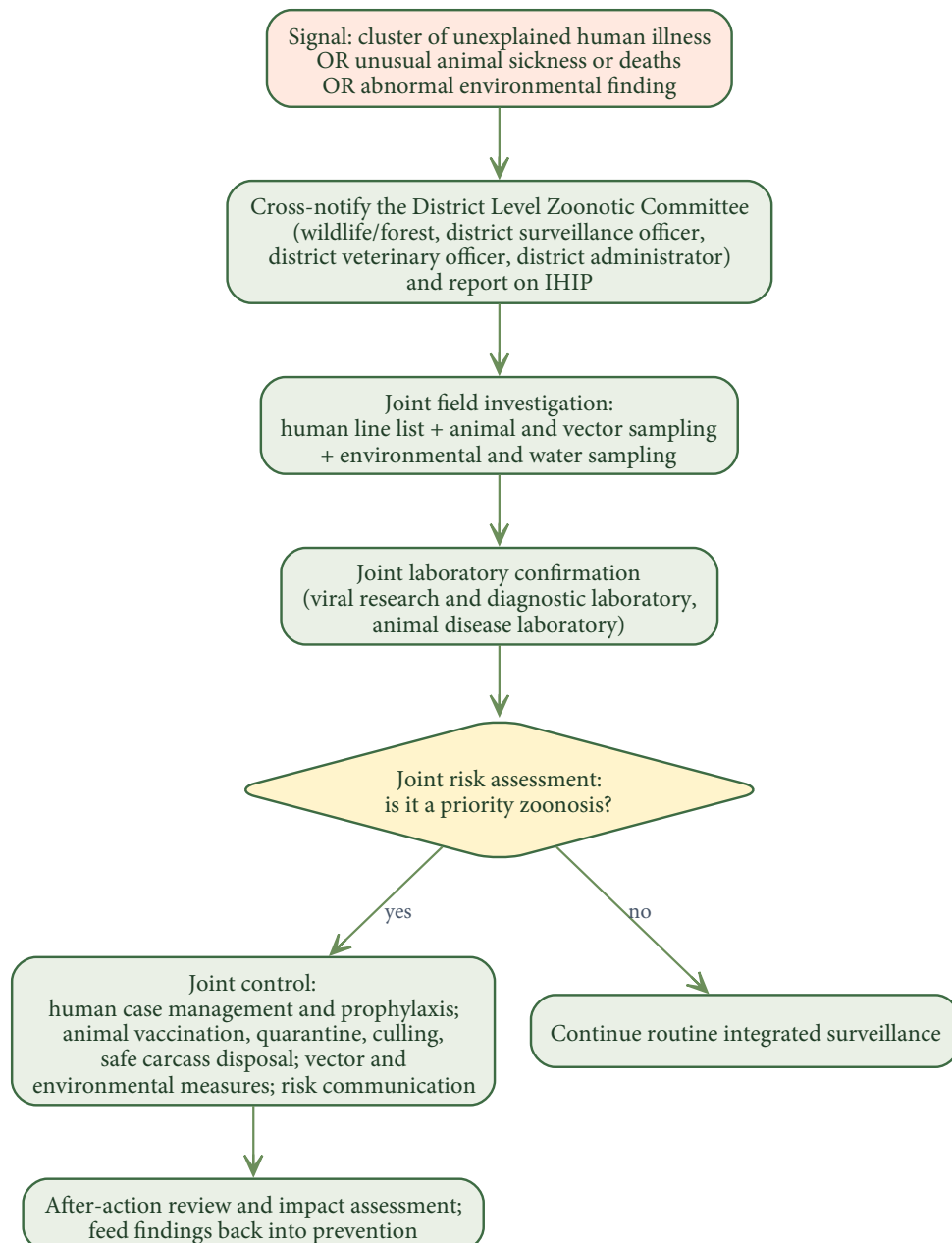
Fig 10 — P-P-R action strategies under One Health (One Health Booklet (AIIMS), p.20)

Pillar	Action strategies (One Health Booklet, p 20)
Prevention (4)	1. Understanding infection dynamics in the natural host · 2. Addressing the drivers of disease emergence · 3. Risk-reduction activities · 4. Multisectoral One Health coordination
Preparedness (7)	1. Strategic planning, financing, leadership and governance · 2. Collaborative surveillance focused on One Health · 3. Early-warning systems · 4. Workforce development · 5. Laboratory preparedness · 6. Hospital preparedness and managing surge capacity · 7. Operational research
Response (4)	1. Integrated joint outbreak investigation · 2. Implementing joint control measures, recovery and rehabilitation · 3. Risk communication and community engagement · 4. After-action report and impact assessment

Cross-reporting between sectors is the practical core of the response pillar: it identifies zoonotic outbreaks sooner, so fewer cases occur, a smaller area is affected and recovery is quicker, whereas the

absence of shared preparedness and agreed procedures delays response. India's experience with avian influenza, Kyasanur forest disease, Nipah and COVID-19 reinforced this need (One Health Booklet, p 19).

Sector-wise control package for a zoonosis (Park 28e, Ch 15, p 942) — (a) *control in animals*: diagnosis of the zoonotic condition, treatment, destruction, quarantine and immunisation; (b) *control of the vehicles of transmission*: food hygiene practices, ensuring the safety of animal products such as wool, hides, horn, bones and fat, proper disposal of animal carcasses and wastes, and disinfection; (c) *prevention and treatment in man*: protection of high-risk groups.



Joint One Health response to a suspected priority zoonotic event

Barriers to implementation and the way forward

Barriers (One Health Booklet, p 26)

Way forward (One Health Booklet, pp 26-27)

Barriers (One Health Booklet, p 26)	Way forward (One Health Booklet, pp 26-27)
The environmental sector (natural resource and wildlife management, biodiversity conservation, pollution and waste management) is not routinely incorporated; environmental determinants poorly understood by other sectors	Consistent integration of environmental considerations into One Health policy and practice
Disjointed legislative schemes; lack of robust regulatory frameworks, legal support, mandates and enabling policies	Effective governance rooted in transdisciplinary and multisectoral principles, backed by appropriate legislation
Lack of data sharing and transparency; absence of multisectoral coordination mechanisms; siloed budgets and decision-making	Institutionalised intersectoral knowledge sharing, joint alert protocols and joint decision processes
Costs and benefits of One Health interventions not sufficiently defined	Operational research and economic evaluation; use the established cross-disciplinary progress against antimicrobial resistance as the building block for other areas
Low awareness among stakeholders and the community	Stakeholder and community engagement; integration of One Health into undergraduate and postgraduate education in related disciplines

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- The One Health triad (three overlapping circles). Label the three circles **HUMANS** (individual and population), **ANIMALS** (livestock, companion animals, wildlife) and **ENVIRONMENT** (plants, ecosystems, water, air, soil). Label the three pairwise overlaps **zoonotic and vector-borne diseases**, **food and water safety**, and **environmental contamination**. Label the central overlap **ONE HEALTH**, and write inside it the four shared threats — **emerging infections**, **antimicrobial resistance**, **climate change**, **biodiversity loss**. Ring the diagram with the three collaborating sectors — **animal health**, **human health** and **environmental health**.

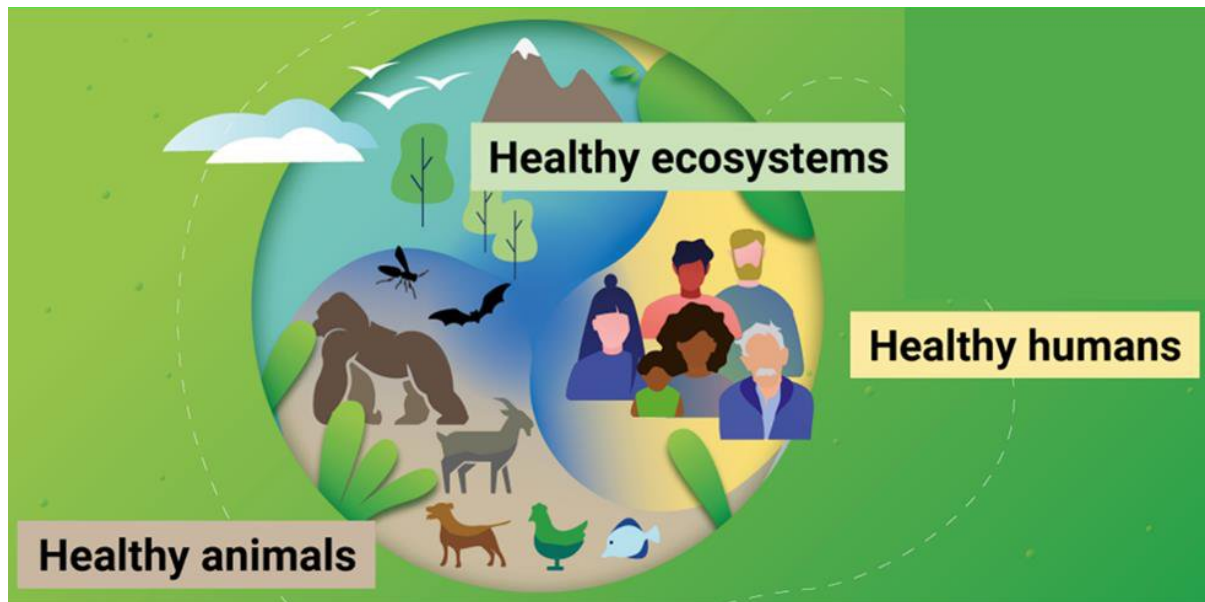


Fig 1 — Concept of One Health for Healthy humans, Healthy animals and (One Health Booklet (AIIMS), p.8)

High-yield exam pointers

- ▶ **Definition to write verbatim:** an integrated, unifying approach that aims to sustainably balance and optimise the health of human beings, companion animals, wild life, plants and ecosystems.
- ▶ **Five C's of the One Health wheel** — Consensus, Collaboration, Cooperation, Coordination, Commitment; **three sectors in the ring** — animal health, human health, environmental health.
- ▶ **Quadrupartite** = WHO + FAO + WOAHA + UNEP; instrument = **One Health Joint Plan of Action 2022–2026**. Concept named "One World – One Health" in 2004; ancestor concept = "One Medicine".
- ▶ **Numbers:** 60% of existing human infections and $\geq 75\%$ of emerging infections are of animal origin; $\sim 70\%$ of emerging zoonoses come from wildlife; 3 of every 5 new human diseases a year are zoonotic; 80% of bioterrorism-capable agents are zoonotic; unsafe food causes 600 million illnesses and $> 400,000$ deaths a year (WHO, 2015); > 150 zoonoses recognised (Park 28e).
- ▶ **India:** National One Health Mission under the **Office of the Principal Scientific Adviser**; NCDC runs NOHP-PCZ and NPCCHH, with **15 Regional Coordinators and 33 sentinel sites**; committee chain = National Steering and Standing Committees on Zoonoses, then RCC, SLZC, DLZC, block, ASHA/ANM/paravets; PM-ABHIM package includes **9 BSL-3 laboratories**.
- ▶ **Top 10 priority zoonoses for India (2020)** — group them to recall: *four bacterial* (anthrax, leptospirosis, brucellosis, plague), *one rickettsial* (scrub typhus), and *five viral* (zoonotic influenza, Japanese encephalitis, dengue, rabies, Crimean-Congo haemorrhagic fever).
- ▶ **Three pillars of the action strategy:** Prevention (4) — Preparedness (7) — Response (4). **Park's zoonosis control triad:** control in animals; control of the vehicles of transmission; prevention and treatment in man.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only — Park's Textbook of Preventive and Social Medicine 28e does not carry a separate "One Health" chapter and treats this material under zoonoses and veterinary public health, so the programme detail, governance structure and current Indian figures in this answer are taken from the NCDC–AIIMS Bibinagar One Health booklet (2023), Sunder Lal's Textbook of Community Medicine and the quadripartite One Health Joint Plan of Action 2022–2026, and Park does not attach a year to the rural-population proportion it quotes.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, WHO, UN General Assembly political declaration on antimicrobial resistance, Office of the Principal Scientific Adviser (Government of India), Quadripartite One Health Joint Plan of Action

Programme and policy updates (India)

- **The National One Health Mission now has a published architecture.** It was approved by the Prime Minister's Science, Technology and Innovation Advisory Council (PM-STIAC) at its 21st meeting, is coordinated by the Office of the Principal Scientific Adviser with **ICMR as the implementing agency**, and rests on a framework built by **more than 13 ministries and departments**. The Executive Committee is chaired by the Union Minister of Health and Family Welfare with the PSA as vice-chair, and a Scientific Steering Committee is chaired by the PSA; four advisory and review committees cover the BSL-3/BSL-4 laboratory network, technology-enabled integrated surveillance and outbreak investigation, research on medical countermeasures, and integration of databases and data sharing. The **National Institute of One Health, Nagpur** is the anchor institute (Director post approved by the Union Cabinet in February 2024), and the BSL-3/BSL-4 network now stands at **22 laboratories** — well beyond the 9 BSL-3 laboratories listed in the older PM-ABHIM package. A **One Health Governance Model Framework for States and Union Territories** was issued in December 2025. [PSA 2025-Dec]
- **Awareness is not the bottleneck — practice is.** Among 400 frontline workers drawn from the Health, AYUSH, Animal Husbandry, Food Safety and Environment departments of Rajkot district, Gujarat (surveyed January–June 2025), 72% were aware of One Health and 71% held positive attitudes, but only **45% reached the threshold for good practice**; just 28% had written standard operating procedures defining their One Health role and only 32% attended joint intersectoral meetings quarterly or more often. Prior One Health training was the strongest predictor of good practice (aOR 3.1, 95% CI 2.0–4.9), and Environment-department staff were the least likely to practise it (aOR 0.5, 95% CI 0.2–0.9). [PMID 42110247]

Current national data

- **Rabies burden re-measured (ICMR, 2022–23).** A nationwide community survey of 78,807 households across 60 districts in 15 states (2 March 2022 to 26 August 2023) put annual animal-bite incidence at **6.6 per 1000** (95% CI 5.7–7.6) — about **9.1 million bites a year** — with dog bites at 5.6 per 1000. **20.5%** of dog-bite victims received no anti-rabies vaccine and **49.1%** of those given a first dose never completed the course; decision-tree modelling estimated **5,726**

human rabies deaths a year (95% uncertainty interval 3,967–7,350). Quote this in preference to older textbook rabies mortality figures. [PMID 39362224]

- **Ticks are India's unmeasured interface.** Annual economic loss from ticks and tick-borne diseases is now estimated at **USD 787.63 million** (an earlier estimate put it at USD 498.7 million), yet India still has no national tick monitoring and surveillance network. Kyasanur forest disease has spread from Karnataka into several states, Crimean-Congo haemorrhagic fever has caused outbreaks in Gujarat, Rajasthan and Uttar Pradesh, and the human-side economic burden has never been estimated for India at all. [PMID 42165723]

Elimination, surveillance and disease status

- **The top 10 priority zoonoses list is now peer-reviewed and ranked.** The national multisectoral workshop behind the NOHP-PCZ list scored **40 zoonoses** using a modified US CDC One Health Zoonotic Disease Prioritization process with the analytic hierarchy process. The five weighted criteria were severity in humans (weight 0.45), capacity for prevention and control (0.22), potential for introduction or increased transmission in India (0.195), economic burden (0.188) and pandemic or epidemic potential (0.187). **Zoonotic influenza ranked first**, followed by anthrax, Japanese encephalitis, leptospirosis, brucellosis, dengue, rabies, scrub typhus, plague and Crimean-Congo haemorrhagic fever; sensitivity analysis left the top 10 essentially unchanged. [PMID 41429420]
- **Nipah** carries a mean case-fatality of about **70%**, with no approved vaccine or treatment. Humans are infected directly from *Pteropus* bats through shared food or through bridging hosts such as pigs and horses, and the spillover pathway for the **2023 Kerala outbreak remains unexplained** — the standard Indian illustration of why laboratory and field One Health work have to run together. [PMID 40437297]

Guideline and legislative updates

- **UN General Assembly political declaration on antimicrobial resistance (26 September 2024)** set the first global AMR targets: a **10% reduction by 2030** in the 4.95 million deaths a year associated with bacterial AMR; at least **70% of human antibiotic use to come from the WHO AWaRe Access group**; a meaningful reduction in antimicrobial use in agri-food systems by 2030; **USD 100 million** in catalytic funding so that at least **60% of countries have funded national AMR action plans by 2030**; and a Quadripartite-established independent panel for evidence for action against AMR from 2025. [UNGA 2024-Sep]
- **The Joint Plan of Action is being implemented lopsidedly.** A scoping review mapping 77 One Health programmes against the quadripartite JPA theory of change found 42 (54.5%) involved only the human and animal sectors, and none addressed the environment sector's own interfaces. Action track 1 (collaborative capacity and high-level coordination) featured in 83.1% of programmes, but the environment action track in only **20.8%** and the AMR action track in **31.2%** — the hard evidence behind the "environmental sector not routinely incorporated" barrier. [PMID 40160937]

Global and WHO developments

- **The WHO Pandemic Agreement was adopted on 20 May 2025** at the 78th World Health Assembly, only the second treaty concluded under Article 19 of the WHO Constitution after the Framework Convention on Tobacco Control. It carries a One Health approach and prevention at

source alongside a commitment that participating manufacturers give WHO real-time access to **20% of their production of pandemic vaccines, therapeutics and diagnostics**. The Pathogen Access and Benefit-Sharing (PABS) annex is still being negotiated by an intergovernmental working group and must be adopted before the agreement opens for signature; **60 ratifications** are needed for entry into force. [WHO 2025-May] [PMID 41423281]

- **The amended International Health Regulations (2005) have been in force since 19 September 2025.** Adopted on 1 June 2024 at the 77th World Health Assembly, they create a new top alert tier — the "**pandemic emergency**" — defined as a public health emergency of international concern caused by a communicable disease that has or risks wide spread across multiple States, exceeds or risks exceeding health-system response capacity, causes substantial social and economic disruption, and requires rapid, equitable and coordinated international action. Temporary Recommendations remain legally non-binding. [WHO 2024-Jun]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 5, Epidemiology of Communicable Diseases; Ch 15, Environment and Health); Sunder Lal — Textbook of Community Medicine (Ch 20, Recent Advances in Community Medicine); One Health Booklet, AIIMS Bibinagar in collaboration with the National Centre for Disease Control, 2023 (Figures 1, 3, 8, 10 and 16); "One Health Approach" (Journal of Public Health and Primary Care, 2022); FAO, UNEP, WHO and WOA — One Health Joint Plan of Action 2022–2026 (Global Plan of Action on One Health, Rome, 2022); NCDC Technical Guidance on Zoonotic Disease Prevention, Preparedness and Response (draft), Directorate General of Health Services, Ministry of Health and Family Welfare, Government of India.

Recent-advances sources: 42110247 — From awareness to public health policy action: operational gaps in One Health implementation among frontline workers in Western India, *Frontiers in Public Health* 2026, DOI [10.3389/fpubh.2026.1814319](https://doi.org/10.3389/fpubh.2026.1814319); 39362224 — Estimates of the burden of human rabies deaths and animal bites in India, 2022–23, *The Lancet Infectious Diseases* 2025, DOI [10.1016/S1473-3099\(24\)00490-0](https://doi.org/10.1016/S1473-3099(24)00490-0) (abstract-only); 42165723 — Ticks and tick-borne diseases: a call for comprehensive tick surveillance in India under One Health approach, *Indian Journal of Medical Research* 2026, DOI [10.25259/IJMR_2011_2025](https://doi.org/10.25259/IJMR_2011_2025); 41429420 — Transforming One Health in India: national multisectoral mixed method study on prioritization of zoonotic diseases, *JMIR Public Health and Surveillance* 2025, DOI [10.2196/77850](https://doi.org/10.2196/77850); 40437297 — A One Health approach to understanding and managing Nipah virus outbreaks, *Nature Microbiology* 2025, DOI [10.1038/s41564-025-02020-9](https://doi.org/10.1038/s41564-025-02020-9) (abstract-only); 40160937 — One Health implementation: a systematic scoping review using the Quadripartite One Health Joint Plan of Action, *One Health* 2025, DOI [10.1016/j.onehlt.2025.101008](https://doi.org/10.1016/j.onehlt.2025.101008); 41423281 — Evidence-based opportunities to address pandemic drivers via the Pandemic Agreement: lessons from the Framework Convention on Tobacco Control, *BMJ Global Health* 2025, DOI [10.1136/bmjgh-2025-021304](https://doi.org/10.1136/bmjgh-2025-021304). Also: UN General Assembly political declaration on antimicrobial resistance, 26 September 2024; WHO Pandemic Agreement adopted by the Seventy-eighth World Health Assembly, 20 May 2025; International Health Regulations (2005) amendments adopted 1 June 2024 and in force 19 September 2025; Office of the Principal Scientific Adviser, Government of India — National One Health Mission, including the One Health Governance Model Framework for States and Union Territories, December 2025.



Itch mite life cycle and scabies

Asked in: Paper I 07-May-2022 — RGUHS MD Community Medicine

Definition — The **itch mite** (*Sarcoptes scabiei*, also called *Acarus scabiei*) is an extremely small globular arthropod of the order **Acarina**, just visible to the naked eye, whose fertilised female burrows into the epidermis, where it breeds and causes the condition known as **scabies** or **itch** (Park 28e, Ch 15, p 935). Its discovery in **1687** marks scabies as the first disease of man with a known cause (Park 28e, Ch 15, p 935). Scabies is a contagious, intensely pruritic infestation transmitted by close personal contact, and because it spreads among those sharing a bed or a room it is classically called a **familial or household infection** (Park 28e, Ch 15, p 936).

Identification and morphology of the itch mite

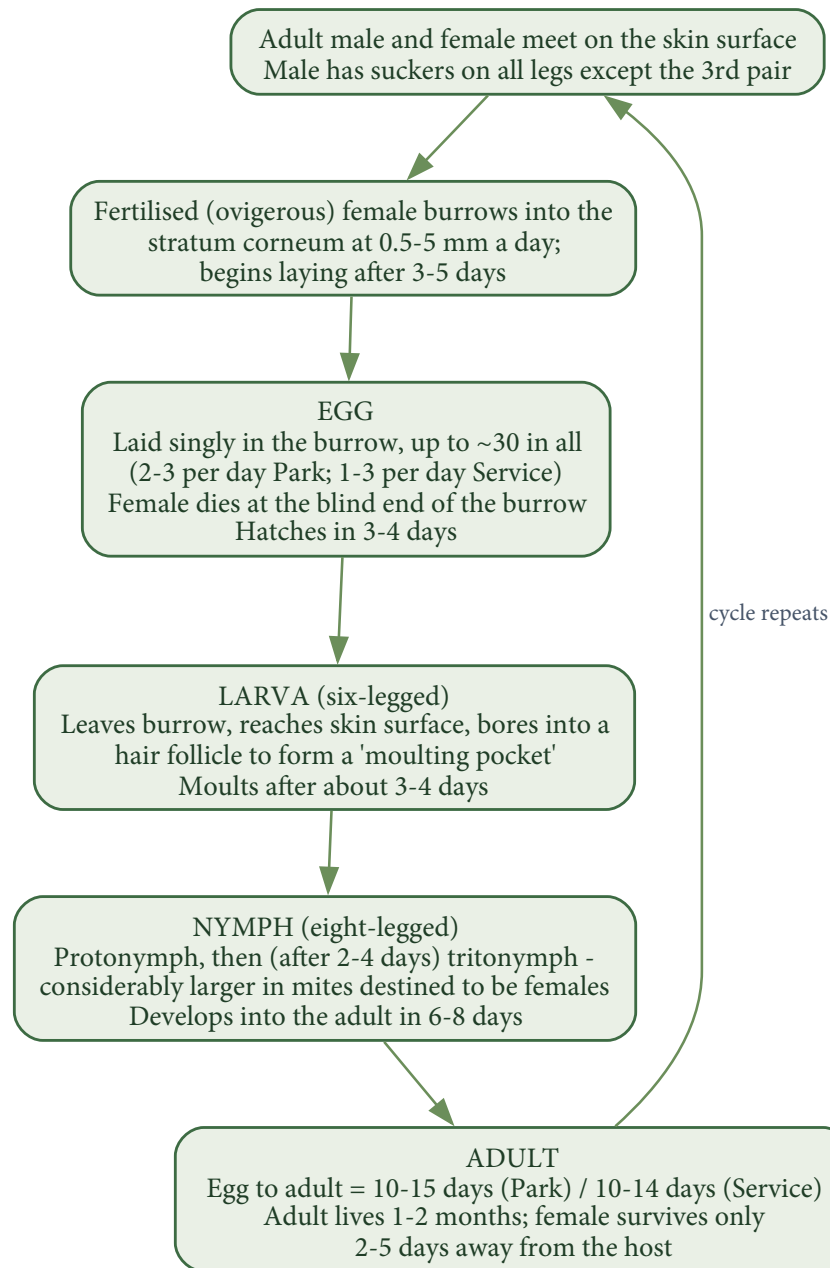
Mites resemble ticks in general appearance, having **four pairs of legs** in the adult and a body **not demarcated into head, thorax and abdomen**. From the public health standpoint two mites matter: the **trombiculid mite** (vector of scrub typhus) and the **itch mite** (Park 28e, Ch 15, p 935).

Feature	Description
Species	<i>Sarcoptes scabiei</i> (<i>Acarus scabiei</i>) — human variety (Park 28e, Ch 15, p 935)
Size	About 0.4 mm , just visible to the naked eye (Park 28e, Ch 15, p 936); Service gives the female as 0.30-0.45 mm (Service, Ch 18, p 264)
Shape	Globular; rounded above and flattened below, like a tortoise (Park 28e, Ch 15, p 936); pale and disc-shaped, with wavy striations dorsally and ventrally (Service, Ch 18, p 264)
Body	No demarcation into cephalothorax and abdomen; surface thrown into folds and covered with short bristles. No distinct head — short fat palps and pincer-like chelicerae protrude anteriorly (Park 28e, Ch 15, p 936; Service, Ch 18, p 264)
Legs (female)	Two pairs in front and two pairs behind; the front two pairs end in long tubular processes known as suckers (pedicels ending in thin-walled roundish "suckers"), while in the female the posterior two pairs have no suckers and end in long, very conspicuous bristles (Park 28e, Ch 15, p 936; Service, Ch 18, p 264)
Sexing	The male has suckers on all the legs excepting the third pair — this distinguishes it from the female (Park 28e, Ch 15, p 936); Service recognises the male by suckers on the last (fourth) pair of legs , absent in the female. The male measures 0.20-0.25 mm , about half the size of the egg-bearing (ovigerous) female (Service, Ch 18, pp 264-266)
Host specificity	<i>Sarcoptes</i> also infests dog, cattle and horse, causing mange . The human acarus is morphologically indistinguishable from the animal variety but is physiologically quite distinct , so animal scabies cannot flourish on human skin; if animal mites do infect man the infestation persists only a few weeks (Park 28e, Ch 15, p 935; Service, Ch 18, pp 263-264)
Survival off the host	The female lives 4-6 weeks on man but only 2-5 days away from the host (Service, Ch 18, p 266); Park gives the adult life span as 1-2 months (Park 28e, Ch 15, p 936)

Life cycle of the itch mite

There are **four stages** — **egg, larva, nymph and adult** (Park 28e, Ch 15, p 936). The entire cycle is completed **on and within human skin**; there is no external or intermediate host.

- ▶ **Mating and burrowing** — the male wanders on the skin surface seeking an unfertilised female in a short pocket or tunnel. **Only after fertilisation** does the female commence burrowing, and she starts laying **3-5 days** later (Service, Ch 18, p 266).
- ▶ **EGG** — she burrows **within the stratum corneum**, tunnelling **0.5-5 mm a day**, laying eggs singly **up to about 30 in all**, and finally **dies at the blind end of the burrow**. Eggs hatch in **3-4 days** (Park 28e, Ch 15, p 936; Service, Ch 18, p 266). *Rate conflict:* Park gives **2-3 eggs per day** (Park 28e, Ch 15, p 936), Service **1-3 eggs per day** (Service, Ch 18, p 266).
- ▶ **LARVA** — **three pairs of legs (six-legged)**, unlike the eight-legged nymph and adult. It leaves the burrow, reaches the surface and bores into a hair follicle to form a small "**moulting pocket**" in which a vesicle develops; most larvae die on the surface. Survivors moult in the pocket — Park: **mature into nymphs in about 3 days**; Service: **the larva moults after 3-4 days** (Park 28e, Ch 15, p 936; Service, Ch 18, p 266).
- ▶ **NYMPH** — eight-legged; the larva moults to a **protonymph**, which after **2-4 days** moults to a second nymph, the **tritonymph** — which in mites destined to become females is considerably **larger than the nymph of a male** — and this after a further **2-3 days** moults into an adult male or female. Park summarises the whole nymphal phase as **nymphs developing into adults in 6-8 days** (Park 28e, Ch 15, p 936; Service, Ch 18, p 266).
- ▶ **ADULT** — adults feed on liquid oozing from dermal cells they have chewed and live **1-2 months** (Park 28e, Ch 15, p 936). *Duration conflict:* egg to adult takes **10-15 days** (Park 28e, Ch 15, p 936) or **10-14 days** (Service, Ch 18, p 266).



Life cycle of the itch mite (Sarcoptes scabiei) — completed entirely in human skin in 10-15 days

Magnitude and epidemiology of scabies

Dimension	Detail
Global burden	Global prevalence is estimated at at least 300 million cases annually (Service, Ch 18, p 267); Harrison puts the figure at about 250 million persons worldwide (Harrison 22e, Ch 472, p 3771)
Secular trend	Increases in incidence appear to run in 15- to 20-year cycles , probably from fluctuating levels of immunity in the human population, though this explanation is disputed (Service, Ch 18, p 267)
Contact needed for transfer	Actual transfer of mites from person to person takes about 15-20 minutes of close skin-to-skin contact — hence spread between bed-sharers and between courting couples who habitually hold hands, while indirect fomite transmission is uncommon (Service, Ch 18, pp 266-267)

Dimension	Detail
Epidemic and disaster settings	Incidence rises in wars and disasters — earthquakes, floods, famines — when people sleep and live in very overcrowded conditions (Service, Ch 18, p 266). Outbreaks occur in preschools, hospitals, nursing homes, prisons and other congregate residences (Harrison 22e, Ch 472, p 3771)
Social distribution	Long associated with poverty, overcrowding and poor hygiene, but people of all socioeconomic backgrounds can become infested (Service, Ch 18, p 264)
Mite load per person	A person with a healthy immune system carries only about 5-15 female mites (children 19-20); only about 3% carry more than 50 mites (Service, Ch 18, p 267)

Mode of spread, sites of lesions and rash distribution

Aspect	Detail
Close contact (chief mode)	Sleeping in the same bed, children playing with each other, nursing an infested person; hence spread through the family — a familial or household infection (Park 28e, Ch 15, p 936)
Contaminated fomites	Occasionally acquired from contaminated clothes and bed linen ; dislodged healthy females can be found on bedding and clothing, and mites may stay alive on towels and discarded clothing for a few days (Park 28e, Ch 15, p 936; Service, Ch 18, p 266)
Sites of burrows / lesions	Hands and wrist — 63%; extensor aspect of the elbows — 10.9% (next commonest); then axillae, buttocks, lower abdomen, feet and ankles, and the palms in infants ; the breasts in women and the genitals in men (Park 28e, Ch 15, p 936; Service, Ch 18, p 264)
Skin that is soft and folded	Anterior folds of the axillae (but not their hollows), umbilical region, prepuce, glans penis, scrotum and lower gluteal regions; genital papules may be mistaken for venereal sores (AFMC Red Book 2024, Ch 32, p 944)
Distribution of the allergic rash	Occurs largely on areas not occupied by burrowing mites — buttocks, waist, shoulders, arms, calves, ankles — and saves the head, the centre of the chest and back, and the palms and soles in adults (Service, Ch 18, p 267)

Clinical features, complications and diagnosis

Feature	Detail
Sensitisation / incubation	The rash appears only 4-6 weeks after a <i>first</i> infestation (exceptionally within 2 weeks) because it is an allergic response to the mite; on re-infestation it appears within 2-4 days (Service, Ch 18, pp 267-268)
Itching	Severe pruritus, characteristically worse at night and after a hot bath, provoking vigorous scratching (Park 28e, Ch 15, p 936; Service, Ch 18, p 267)
Burrow	The pathognomonic lesion — a narrow twisting tunnel, with mite faeces visible through the skin as dark pepper-like spots ; becomes obscure once scratched, ulcerated or secondarily infected (Service, Ch 18, p 267; AFMC Red Book 2024, Ch 32, p 944)

Feature	Detail
Complications	Scratching causes secondary bacterial infection — boils, pustules, ecthyma, eczema and impetigo contagiosa. Secondary impetigo over the gluteal folds and elbows should itself arouse suspicion of underlying scabies . Severity is not proportional to the number of mites (Service, Ch 18, pp 267-268; AFMC Red Book 2024, Ch 32, p 944)
Crusted (Norwegian) scabies	Rare but highly contagious ; vast mite numbers, up to 4,700 mites per gram of skin (about 2 million on one person). Skin becomes scaly and hyperkeratotic with crusts over hands and feet and scaling of head, neck and nails, but itching is much less pronounced . Follows loss of immunity — corticosteroid use, human immunodeficiency virus (HIV) infection, mental impairment — in whom reduced scratching also spares the mites (Service, Ch 18, pp 267-268)
Post-scabetic itch	Rash and itch persist 2-3 weeks after every mite has been killed ; this is not treatment failure — give antihistamines, not further scabicide (Service, Ch 18, pp 268-269)

The main diagnostic features of scabies are five (Park 28e, Ch 15, p 936):

- ▶ The patient **complains of itching which is worse at night**.
- ▶ Examination **reveals follicular lesions at the affected site**.
- ▶ **Secondary infection** leads to **crusted papules and pustules**.
- ▶ The **diagnosis is probable if the other members of the household are affected** — the criterion that links diagnosis directly to the simultaneous-treatment rule below.
- ▶ **Confirmation** is by **searching for the parasite in the skin debris under the microscope**.

Three bedside aids reveal the burrow (Service, Ch 18, p 267):

- ▶ **Ink test** — a drop of ink smeared over suspect skin seeps into the tunnel within minutes and outlines it once the excess is wiped off.
- ▶ **Tetracycline fluorescence** — topical liquid tetracycline makes tunnels fluoresce bright yellow-green under ultraviolet light.
- ▶ **Skin scraping** — the end of the tunnel is scratched with a needle or scraped with a scalpel blade and examined at about ×50 magnification.

With a typical rash and a history of itching, do not hesitate to diagnose scabies even without a demonstrable burrow (AFMC Red Book 2024, Ch 32, p 944).

Treatment and control of scabies

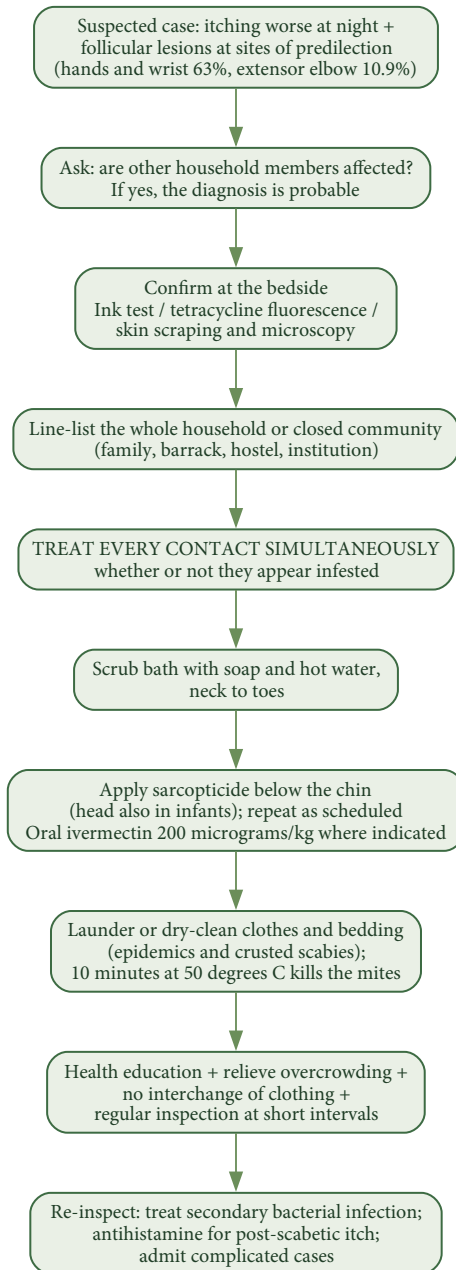
The cardinal public health rule is that **all members of the affected household or closed community must be treated simultaneously, whether or not they appear to be infested** — otherwise reinfection is inevitable (Park 28e, Ch 15, p 936; Service, Ch 18, p 268). Before treatment the patient is given a **good scrub with soap and hot water**, scrubbing the whole body from the neck to the toes (Park 28e, Ch 15, p 936; AFMC Red Book 2024, Ch 32, p 944).

Sarcopti cide	Strength and formulation	How applied
Benzyl benzoate	25% emulsion — an effective sarcopticide	Applied with a paint brush or shaving brush to every inch of the body below the chin including the soles of the feet and allowed to dry; in babies the head must also be treated. Repeat after 12 hours; after a further 12 hours a bath is given and all underclothes, clothes and sheets changed and washed. Not more than two applications per week , as excessive use causes an irritant dermatitis (Park 28e, Ch 15, p 936). <i>Schedule conflict</i> : the AFMC Red Book advises one thorough application is usually adequate, a second only after an interval of 10 days if needed, with the bath after 24 hours (AFMC Red Book 2024, Ch 32, p 944); Service puts the one-dose cure rate at only about 50% and advises a repeat on the third day (Service, Ch 18, p 268)
Permethri n	5% cream — usually the first choice	Applied from the neck downwards and washed off after 8-14 hours; a second application a week later is sometimes advised. Caution in pregnancy and in children aged 2 months to 2 years (Service, Ch 18, p 268; AFMC Red Book 2024, Ch 34, p 1081)
Ivermectin	Oral (or, more recently, topical)	Proving effective as a single oral dose or as a topical application, but resistance to both permethrin and ivermectin has been documented (Service, Ch 18, p 268). Dose — 200 micrograms/kg orally as a single dose for common (non-crusted) scabies; crusted scabies needs 3-7 doses over 8-30 days with repeated topical permethrin and a keratolytic such as 6% salicylic acid to debulk the crusts (Harrison 22e, Ch 472, p 3772)
Gamma- HCH (lindane)	0.5-1.0% in coconut or any vegetable oil, or in vanishing cream	Rubbed on the affected parts of the skin on one or two occasions separated by an interval of 2-3 days (Park 28e, Ch 15, p 936)
Tetmosol	5% solution	An efficient sarcopticide; three daily applications recommended (Park 28e, Ch 15, p 936)
Sulphur ointment	2.5-10%	Daily for 4 days — a cheap remedy (Park 28e, Ch 15, p 936); Service advises daily for at least a week, and it is safe in infants and young children (Service, Ch 18, p 268)
Malathion	Lotion	Second line if permethrin fails; washed off only after 24 hours ; caution in children aged 6 years or younger (Service, Ch 18, p 268)
Crotamito n	Cream or lotion	Very safe, but needs 2-5 daily applications ; also relieves itching (Service, Ch 18, p 268)
Antihista mine	Oral	For the post-scabetic itch that persists after the mites are killed (Service, Ch 18, pp 268-269)

Community and institutional control measures (AFMC Red Book 2024, Ch 32, p 944; Service, Ch 18, pp 267-268):

- **Health education** on the nature of the disease, its mode of spread and precautions — particularly before men go home on leave.
- **Relieve overcrowding** — proper spacing of beds, and verandahs and tents for an influx of recruits, since infestation spreads rapidly in crowded barracks.

- ▶ **Forbid interchange of clothing** and sleeping huddled together.
- ▶ **Personal hygiene** at a high standard, with adequate bathing and laundering facilities.
- ▶ **Early detection by regular inspection at short intervals** — patients report sick only at the late stage of intolerable itching; also inspect everyone returning from leave.
- ▶ **Radical treatment of every case**, thorough and supervised — failure to cure is commonly due to slipshod, poorly supervised application; **mass treatment as a supervised "drill"** where infestation is widespread, as in training and recruiting depots.
- ▶ **Launder or dry-clean clothes and bedding** — in epidemics as well as in crusted scabies; **ten minutes at 50°C will kill the mites**. Because transfer needs 15-20 minutes of close contact and the mite survives only 2-5 days off the host, routine elaborate disinfestation is unnecessary — ordinary laundering suffices.
- ▶ **Admit to hospital** cases complicated by secondary bacterial infection or eczematous change.



Case detection to community control pathway for scabies

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Itch mite, dorsal view** — draw a tortoise-shaped globular body and label: *body 0.4 mm, rounded above and flattened below; no demarcation into cephalothorax and abdomen; body surface thrown into folds with short bristles; two pairs of legs in front ending in long tubular processes (suckers); two pairs of legs behind ending in long bristles in the female; third pair of legs — without suckers in the male.*
- ▶ **Stages of the mite** — draw the **six-legged larva** beside the **eight-legged adult** and label: *larva — three pairs of legs; nymph and adult — four pairs of legs.*
- ▶ **Burrow in the skin** — draw a section of skin and label: *stratum corneum; the burrow (tunnel) made by the fertilised female; ovigerous female at the blind end of the burrow, where she dies; eggs laid singly in the burrow, up to about 30; larva leaving the burrow for the skin surface; moulting pocket in a hair follicle, with the vesicle over it.*

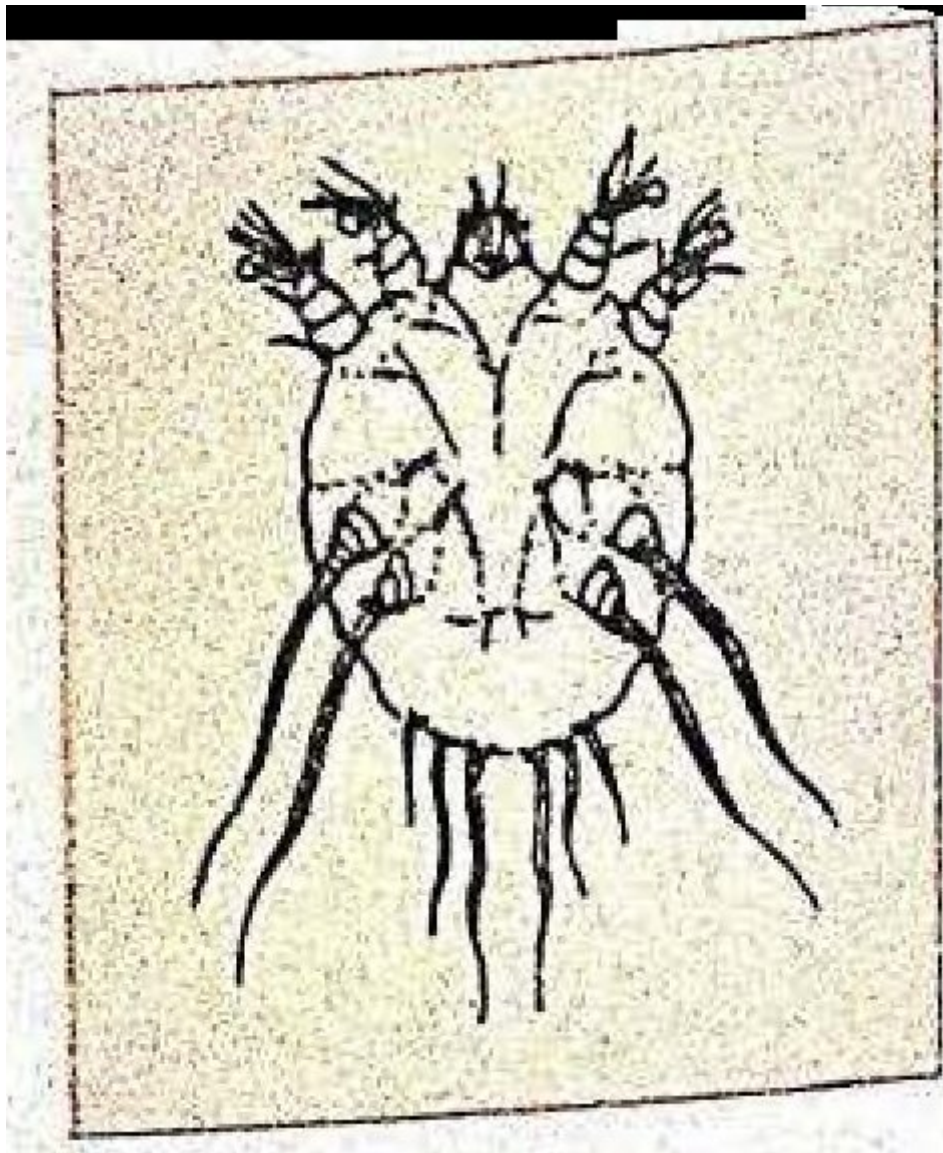


Fig 15 — Itch mite (Park's Textbook of Preventive and Social Medicine, 28e, p.936)

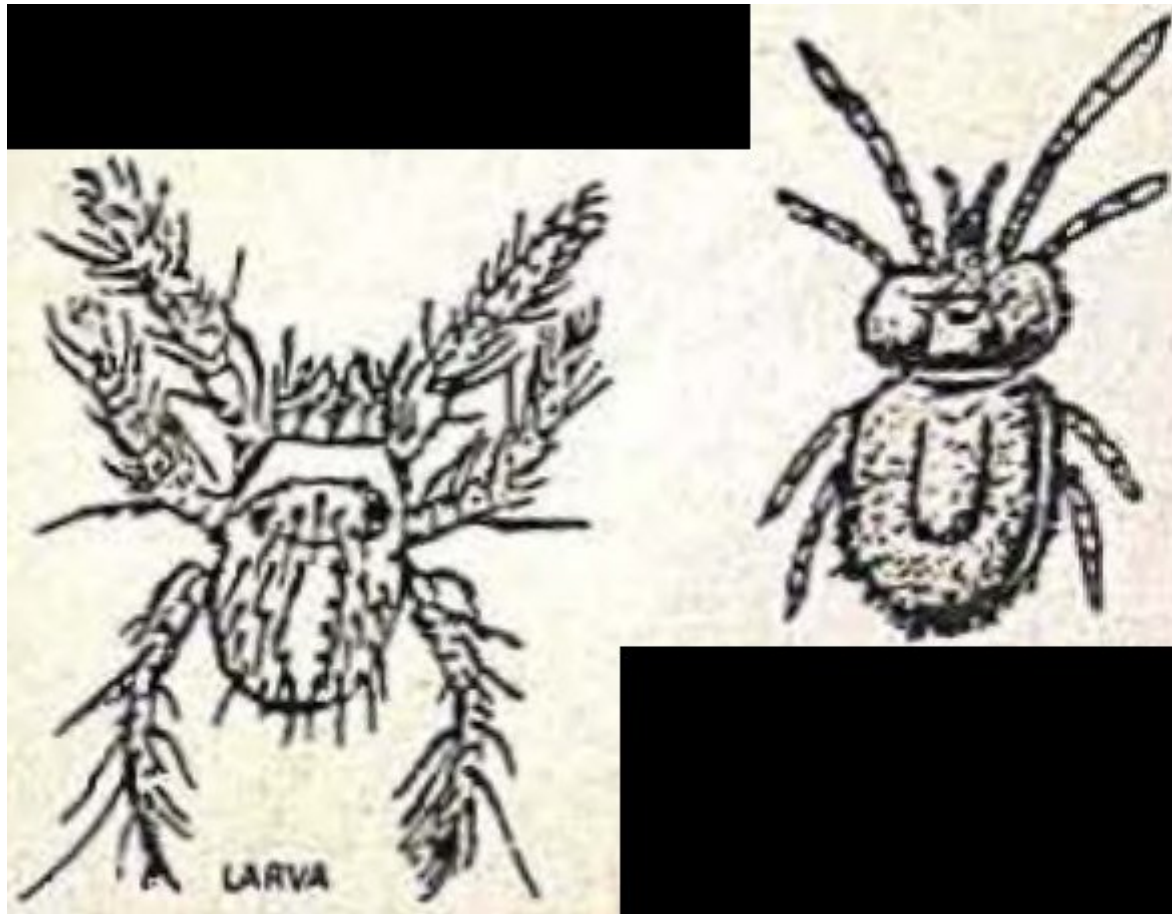


Fig 14 — Life cycle of a mite (Park's Textbook of Preventive and Social Medicine, 28e, p.935)

High-yield exam pointers

- ▶ **Landmark** — the itch mite's discovery in 1687 made scabies the first disease of man with a known cause.
- ▶ **Morphology one-liner** — *Sarcoptes scabiei*, 0.4 mm, tortoise-shaped; in the female the hind two pairs of legs end in long bristles without suckers, while the male has suckers on all legs except the third pair.
- ▶ **Life-cycle timings** — egg hatches 3-4 days → larva moults ~3-4 days → nymph becomes adult 6-8 days; egg to adult 10-15 days (Park) / 10-14 days (Service); adult lives 1-2 months; female lays 2-3 eggs/day (Park), 1-3 (Service), up to ~30, dies at the end of the burrow; survives only 2-5 days off the host.
- ▶ **Burden** — global prevalence at least 300 million cases a year; incidence runs in 15-20 year cycles; transfer needs 15-20 minutes of skin-to-skin contact, so fomite spread is uncommon; rises in wars, earthquakes, floods and famines and affects all socioeconomic groups.
- ▶ **Site percentages** — hands and wrist 63%, extensor aspect of elbows 10.9%; palms in infants, breasts in women, genitals in men.
- ▶ **Five diagnostic features (Park)** — night-worse itching; follicular lesions at the site; crusted papules and pustules on secondary infection; **other household members affected**; parasite in skin debris under the microscope.

- **Rash timing** — 4-6 weeks after first infestation, 2-4 days on re-infestation; persists 2-3 weeks after cure (post-scabetic itch — do not re-treat). **Crusted (Norwegian) scabies** — up to 4,700 mites per gram of skin, *less* itching.
- **Treatment rule** — **treat the whole household simultaneously**; permethrin 5% cream is the usual first choice; benzyl benzoate 25% below the chin; ivermectin 200 micrograms/kg orally single dose (3-7 doses over 8-30 days in crusted scabies); **10 minutes at 50°C** kills mites on linen.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park's chapter still lists gamma-HCH (lindane) and tetmosol among the sarcoptocides, whereas current practice has moved to permethrin 5% cream as first-line topical treatment with oral ivermectin for crusted scabies and for mass treatment of closed communities; the retrieved corpus carries no Indian national scabies treatment schedule and no Indian prevalence estimate, so quote the global burden, mechanisms, timings and drug schedules above rather than any Indian programme target.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, WHO, IACS 2020 consensus criteria, BASHH 2025 guideline

Programme and policy updates (India)

- India still has **no specific guideline for the diagnosis, reporting or treatment of scabies**, and ASHAs posted at sub-centres and health and wellness centres are not routinely trained to identify and manage neglected tropical diseases like scabies. A West Bengal field study showed that a **two-day training** raised ASHA knowledge scores on a ten-question instrument from a mean of 2.44 to 7.24 out of 10 (Wilcoxon signed rank; the paper reports the P-value as 0.00 and its abstract states $p < 0.05$) and that their case-finding matched independent re-screening by the study investigator, putting suspected scabies at **1% of under-fives** (21 of 2,119) — the practical model for building scabies detection into the existing frontline workforce [PMID 41270104].
- **Mass screening and treatment works under Indian field conditions** — a cluster randomised trial in tribal Gadchiroli, Maharashtra, treated every case plus household contacts with **directly observed oral ivermectin** (permethrin 5% in the 18.6% with a contraindication): prevalence fell to **2.8% versus 8.8%** in the usual-care arm at two months (adjusted RR 0.21, 95% CI 0.11-0.38), but rebounded to **7.3% versus 14.1%** at twelve months (adjusted RR 0.49, 95% CI 0.25-0.98) — one round buys about a year, so rounds must be repeated [PMID 33861741].

Current national data

- Quote the burden **with its source and year**: the Global Burden of Disease 2021 estimates put global point prevalence at **206.6 million**, annual incidence at **622.5 million** cases across 204 countries, and the burden at about **5.3 million DALYs**; age-standardised incidence is falling while absolute case numbers rise with population growth, and modelling projects a temporary rebound in incidence around **2027** with burden persisting to 2050 [PMID 42430442]. These supersede the older "at least 300 million" and "about 250 million" figures in the textbooks.

- ▶ India has **no national scabies prevalence survey**. Community studies report **13-59%** across settings [PMID 41270104] and **6.9-20.6%** in tribal populations [PMID 33861741], while the pooled WHO South-East Asia Region estimate is **11.4% (95% CI 6.4-19.5)** [PMID 39575757] — always name the setting rather than quoting one "Indian figure".

Elimination, surveillance and disease status

- ▶ Scabies has been a **WHO neglected tropical disease since 2017**, and the NTD road map 2021-2030 target is **25 countries implementing mass drug administration (MDA) by 2030**, recommended wherever community prevalence is **10% or more**, given as **two doses 7-14 days apart** — the first dose kills mites but not eggs, so the second is aimed at the nymphs hatching after it — with **permethrin 5%** instead for pregnant women, women lactating in the first week after birth, and children under 15 kg or 90 cm [PMID 41642892]; the dose is **oral ivermectin 200 micrograms/kg** [PMID 42384720]. **Fiji and Solomon Islands became the first two countries to run national ivermectin-based MDA** (Solomon Islands July 2022, Fiji October 2022); integration with another programme succeeded only in Fiji, with lymphatic filariasis, while the intended Solomon Islands yaws integration failed when supplies arrived late. First-dose coverage on the MDA registers was **49% in Solomon Islands and 26% in Fiji**, but independent post-MDA coverage surveys estimated far higher first-dose uptake — **78% in Solomon Islands and 45% in Fiji** — so the registers under-count, and the residual shortfall was mainly a failure of **reach rather than refusal** (of Fijians who missed treatment, 62% said no team came to their home and 30% were out when it did) [PMID 41642892].
- ▶ Outbreak surveillance now drives the response: the Rohingya camps at Cox's Bazar recorded **384,852 cases in four years** (attack rate 5,562 per 10,000, highest in under-fives, 5.8% with secondary bacterial infection), and the MDA campaign of 29 November 2023 to 1 February 2024 cut about **1,885 cases a week**, held for **six months**, then rebounded (+72 cases per week, $p=0.002$) — the same lesson as Gadchiroli, that MDA without relieving overcrowding is temporary [PMID 42384720].

Guideline and legislative updates

- ▶ **Diagnosis is now standardised** by the 2020 International Alliance for the Control of Scabies consensus criteria — three levels of certainty with eight subcategories: **A confirmed** (mite, eggs or faeces seen on microscopy, high-powered imaging or dermoscopy), **B clinical** (burrows, typical male genital lesions, or typical lesions in typical distribution with two history features), **C suspected**; the two history features are **itch** and **positive contact history**. Surveys must state the level used, because IACS-based surveys return roughly double the prevalence of the older burrow-and-pruritus approach (**19.5% versus 9.9%**) [PMID 32034956; PMID 39575757]. The BASHH national guideline for adults was updated in **2025** [PMID 40103364].
- ▶ **Permethrin 5%, not oral ivermectin, gives the better individual cure**: the **SCRATCH** cluster randomised trial (294 households, 23 centres) found day-28 cure of **88.5% with permethrin versus 71.8% with ivermectin** (difference -16.7 percentage points, 95% CI -26.3 to -7.1), so ivermectin failed non-inferiority and permethrin was statistically superior, although patients found ivermectin more acceptable [PMID 41494769]; an updated meta-analysis of seven trials agrees (cure RR 0.93, 95% CI 0.86-0.99; treatment failure RR 1.52, 95% CI 1.06-2.20) [PMID

42098632]. Ivermectin's place is population-level MDA and closed communities, not a routine substitute for permethrin.

- **Treatment failure is common and increasing** — pooled failure across 147 studies was **15.2%** (95% CI 12.9-17.6), and **two doses of oral ivermectin failed in 7.1% against 15.2% for a single dose** ($p=0.021$), so the second dose belongs in every schedule; failure rose by 0.27% a year overall and 0.58% a year for permethrin between 1983 and 2021, consistent with drifting mite susceptibility [PMID 37625798]. For severe or crusted disease, a randomised trial found **ivermectin 400 micrograms/kg was not superior to the standard 200 micrograms/kg** (days 0, 7 and 14) when combined with permethrin 5% — cure 75% versus 82%, odds ratio 0.64 (95% CI 0.25-1.67) — so the existing dose stands [PMID 42090793].

Global and WHO developments

- WHO's current framing is **at least 200 million people affected at any one time and more than 400 million cumulatively each year**, with ivermectin on the Essential Medicines List and MDA recommended above the 10% threshold [WHO 2023]. A meta-analysis of 70 studies puts pooled global prevalence at **11.9% (95% CI 9.6-14.7)**, highest in Oceania at 17.9%, and quantifies the household transmission the textbook describes — contact with an itchy household member **OR 11.3 (95% CI 4.82-26.51)**, non-use of soap OR 3.41, bed-sharing OR 2.64, sharing clothes OR 2.52, infrequent bathing OR 2.13 [PMID 39575757].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 15, Environment and Health — "Ticks and mites", Itch mite, pp 935-936, including Fig 14 and Fig 15), with the same passage in the 27e (Ch 15, pp 911-912) used to resolve the scanned text; Service — Medical Entomology for Students (Ch 18, Scabies mites (Sarcoptidae) — external morphology, life cycle, epidemiology, recognition and treatment of scabies, pp 263-269); Harrison's Principles of Internal Medicine 22e (Ch 472, Ectoparasite Infestations and Arthropod Injuries — global burden and scabicide dosing, pp 3771-3772); AFMC Red Book of Community Medicine 2024 (Ch 32, sites of predilection, prevention and control measures and radical treatment, p 944; Ch 34, permethrin 5% cream, p 1081).

Recent-advances sources: 41270104 — The impact of training front-line health care workers in improving the yield of scabies diagnosis among children under five in a rural community of West Bengal, PLoS Negl Trop Dis 2025, doi:[10.1371/journal.pntd.0013739](https://doi.org/10.1371/journal.pntd.0013739); 33861741 — Control of scabies in a tribal community using mass screening and treatment with oral ivermectin: a cluster randomized controlled trial in Gadchiroli, India, PLoS Negl Trop Dis 2021, doi:[10.1371/journal.pntd.0009330](https://doi.org/10.1371/journal.pntd.0009330); 42430442 — Time trends in the burden of scabies from 1990 to 2021, and projections to 2050: insights based on the Global Burden of Disease Study 2021, PLoS Negl Trop Dis 2026, doi:[10.1371/journal.pntd.0014237](https://doi.org/10.1371/journal.pntd.0014237); 39575757 — Prevalence and determinants of scabies: a global systematic review and meta-analysis, Trop Med Int Health 2024, doi:[10.1111/tmi.14058](https://doi.org/10.1111/tmi.14058); 41642892 — National control programs for scabies: experiences from Fiji and Solomon Islands, PLoS Negl Trop Dis 2026, doi:[10.1371/journal.pntd.0013874](https://doi.org/10.1371/journal.pntd.0013874); 42384720 — Massive scabies outbreak in Rohingya refugee camps, Cox's Bazar: epidemiology and impact of a mass drug administration campaign, PLoS Negl Trop Dis 2026, doi:[10.1371/journal.pntd.0013832](https://doi.org/10.1371/journal.pntd.0013832); 32034956 — The 2020 International Alliance for the Control of Scabies Consensus Criteria for the Diagnosis of Scabies, Br J Dermatol 2020, doi:[10.1111/bjd.18943](https://doi.org/10.1111/bjd.18943); 40103364 — British Association for Sexual Health and HIV National Guideline on the Management of Scabies in adults 2025, Int J STD AIDS 2025, doi:[10.1177/09564624251321264](https://doi.org/10.1177/09564624251321264) (abstract-only); 41494769 — Oral ivermectin versus 5% permethrin cream to treat children and adults with classic scabies (SCRATCH): multicentre, assessor blinded, cluster randomised clinical trial, BMJ 2026, doi:[10.1136/bmj-2025-086277](https://doi.org/10.1136/bmj-2025-086277); 42098632 — Comparative effectiveness of ivermectin versus permethrin for the treatment of scabies: an updated systematic review and meta-analysis, BMC Infect Dis 2026, doi:[10.1186/s12879-026-13451-y](https://doi.org/10.1186/s12879-026-13451-y); 37625798 — Failure of scabies treatment: a systematic review and meta-analysis, Br J Dermatol 2024, doi:[10.1093/bjd/ljad308](https://doi.org/10.1093/bjd/ljad308) (abstract-only); 42090793 — Combined Oral Ivermectin and 5% Permethrin Cream to Treat Severe Scabies, N Engl J Med 2026, doi:[10.1056/NEJMoa2411721](https://doi.org/10.1056/NEJMoa2411721) (abstract-only). WHO scabies fact sheet, 31 May 2023.

♦ ♦ ♦

Post-exposure prophylaxis for animal bites

Asked in: Paper II 04-Jan-2025 — RGUHS MD Community Medicine

Definition — Post-exposure prophylaxis (PEP) is the emergency treatment given after a bite, scratch or lick by a suspected rabid mammal, aimed at **neutralising the inoculated virus before it enters the nervous system** (Park 28e, Ch 5, p 343). Every human exposure is a **medical emergency**. PEP is a **three-pronged approach** — wound management, passive immunisation with rabies immunoglobulin (RIG) and active immunisation with anti-rabies vaccine — all three of equal importance and given simultaneously according to the category of exposure (Sunder Lal, Ch 8, p 424).

1. Deciding whom to treat, and the category of exposure

In a rabies-endemic country such as India, where dog-to-dog transmission is sustained, **every animal bite is presumed to be a potentially rabid animal bite** and treatment is begun immediately without waiting for laboratory results (Sunder Lal, Ch 8, p 425).

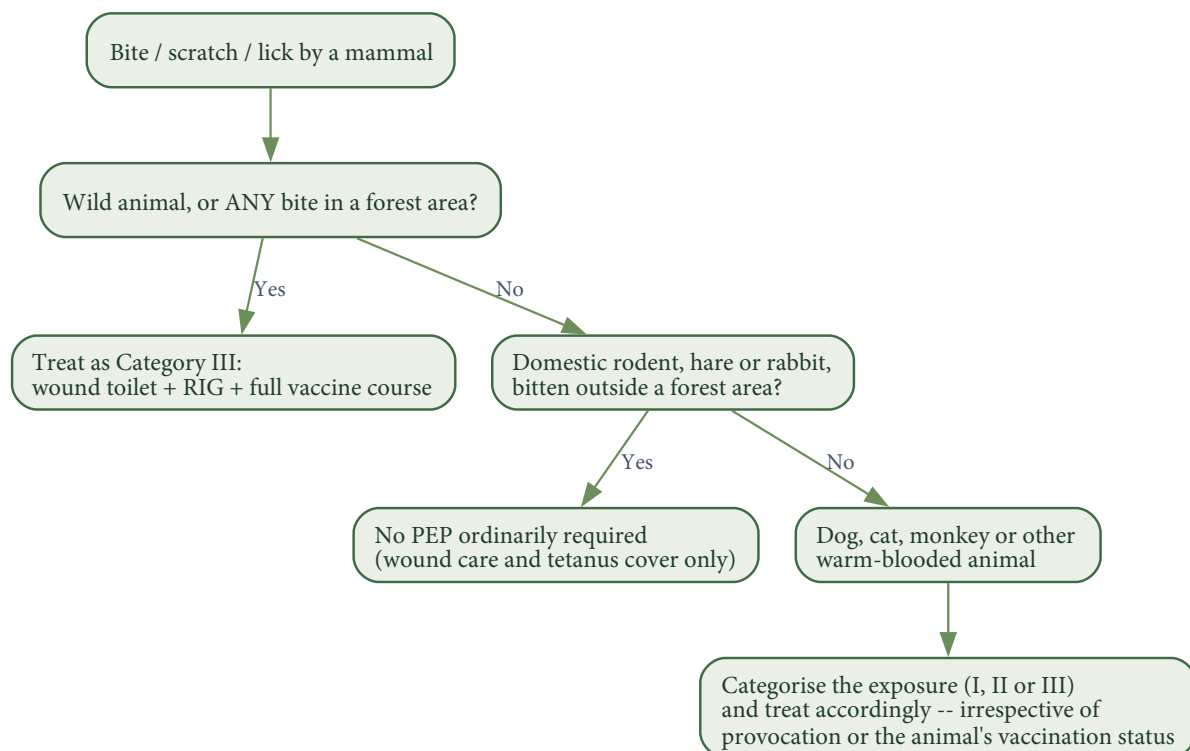
Circumstance	Decision (National Guidelines for Rabies Prophylaxis, India, 2019)
Bites by all warm-blooded animals	PEP indicated as per category
Exposure to wild animals; any bite sustained in a forest area	Treat as Category III
Domestic rodents, hare and rabbits	Ordinarily no PEP ; rodent bites in forest areas do need PEP
Bats	Bat rabies not conclusively proven in India — at present does not warrant PEP
Human-to-human	Risk minimal; PEP may be offered to those closely exposed to a rabies patient's secretions. Cornea/organs must not be taken from a suspected or confirmed case
Provoked bite	A provoked bite does not mean the animal is not rabid — treat as exposure
Biting animal already vaccinated	Animal vaccine failure occurs; PEP is given irrespective of the animal's vaccination status
Late presentation (months or years later)	Evaluate and treat as if the exposure had just occurred
Features that raise the risk (World Health Organization, WHO)	Biting mammal is a known rabies reservoir or vector species; animal looks sick or behaves abnormally; saliva contaminated a wound or mucous membrane; bite unprovoked; animal unvaccinated; animal cannot be traced (<i>Park 28e, Ch 5, p 344</i>)

(Sunder Lal, Ch 8, pp 425–426)

When in doubt, start PEP and refer the patient to an anti-rabies clinic (Sunder Lal, Ch 8, p 426).

Category	Type of exposure	Recommended PEP
I	Touching or feeding animals; licks on intact skin ; contact of intact skin with secretions or excretions of a rabid animal or human case	No PEP if the history is reliable; wash the exposed area with soap and water and apply an antiseptic
II	Nibbling of uncovered skin; minor scratches or abrasions without bleeding	Wound management + rabies vaccine
III	Single or multiple transdermal bites or scratches; licks on broken skin ; contamination of mucous membrane with saliva (<i>Park 28e reproduces the WHO table, which also places contact with bats here</i>)	Wound management + rabies immunoglobulin + rabies vaccine

(Sunder Lal, Ch 8, Table 8.23, p 426 — National Guidelines for Rabies Prophylaxis 2019; Park 28e, Ch 5, Table 1, p 344)



Does this mammal bite need post-exposure prophylaxis?

2. First prong — management of the bite wound

Local treatment removes virus before it is absorbed on nerve endings; animal experiments show it alone reduces the chance of developing rabies by **up to 80%** (Park 28e, Ch 5, p 343). It is of maximal value within minutes, but **must still be done however late the patient reports**, because the virus persists and multiplies at the bite site.

- **Wash** — prompt, gentle, thorough washing of the wound, scratches and adjoining area with soap or detergent and running water for **at least 15 minutes** (Park 28e, Ch 5, p 343) / **approximately 15 minutes** (National Guidelines 2019, in Sunder Lal, Ch 8, p 424); if soap is unavailable flush with copious water, irrigate punctured wounds with a catheter, and never touch the wound with bare hands.

- **Chemical (virucidal) treatment** — after washing and drying, apply alcohol (400–700 ml/litre), tincture or 0.01% aqueous iodine, or povidone-iodine (Park 28e, Ch 5, p 343).
- **Do not apply mud, soil, chillies or oil**; if already applied, wash them off gently and flush again (Sunder Lal, Ch 8, p 424).
- **Do not suture primarily** — place **minimum loose sutures 24–48 hours later** if unavoidable, under cover of local RIG infiltration (Park 28e, Ch 5, p 343; Sunder Lal, Ch 8, p 424).
- **Cauterisation is no longer recommended** — it leaves a bad scar and adds nothing to soap-and-water washing (Sunder Lal, Ch 8, p 424).



Fig 20.69 — In cases of an animal bite/scratch follow these simple first aid steps (Suryakantha Environment Best, p.512)

3. Second prong — passive immunisation with rabies immunoglobulin

RIG supplies ready-made neutralising antibody at the wound before the patient can make his own, and local infiltration in Category III bites is life-saving (Sunder Lal, Ch 8, p 424).

Item	Specification
Indications	All Category III exposures; all wild-animal exposures; in the immunocompromised (HIV/AIDS, steroids or cancer chemotherapy, congenital agammaglobulinaemia) give RIG for Category II as well as III
Human RIG (HRIG)	20 IU/kg body weight
Equine RIG (ERIG) / F(ab') ₂	40 IU/kg body weight
Timing	Given only once , at or as soon as possible after the first vaccine dose; not indicated beyond day 7 after the first dose, as an active antibody response has begun
Site	The entire dose, or as much as anatomically possible , infiltrated into and around the wound(s), avoiding compartment syndrome; may be diluted to infiltrate all wounds. Park adds that any remainder is given intramuscularly distant from the vaccine site; the Indian 2019 guideline holds that this adds little or no protection over infiltration alone, and reserves it for likely additional or occult wounds, mucosal exposure or aerosol exposure
Skin test	Not required before ERIG — it does not predict reactions, and ERIG is given whatever the result; be prepared to treat anaphylaxis
Cautions	Never intravenously ; never in the same syringe or anatomical site as the vaccine; observe the patient 15–20 minutes afterwards
Not indicated	Persons with documented complete previous pre- or post-exposure prophylaxis

(Sunder Lal, Ch 8, pp 424–425; Park 28e, Ch 5, p 345)

Anaphylaxis with modern highly purified ERIG is rare — **1 in 45,000** in Park 28e (Ch 5, p 345) and **1 in 1,50,000** in the 2019 Indian guidelines (Sunder Lal, Ch 8, p 424). **Serum sickness** may appear 7–10 days after ERIG but has not been reported after HRIG (Sunder Lal, Ch 8, p 425).

4. Third prong — active immunisation, and when PEP may be modified or stopped

Only **cell-culture and embryonated egg-based vaccines (CCEEVs)** are used — human diploid cell, purified chick embryo cell, purified Vero cell and purified duck embryo vaccines. **The Government of India stopped producing nervous tissue vaccine in 2004** (Park 28e, Ch 5, pp 344–345).

Regimen	Route and dose	Schedule	Vials / visits
Essen (standard)	One intramuscular (IM) dose (1.0 or 0.5 ml by vaccine type) into deltoid , or anterolateral thigh in children under 2 years — never gluteal	Days 0, 3, 7, 14, 28	5 vials, 5 visits
4-dose IM alternative	Same IM dose; only for healthy, fully immunocompetent persons who receive wound care plus high-quality RIG plus a WHO-prequalified vaccine	Days 0, 3, 7, 14	4 visits
Zagreb (2-1-1) abbreviated multisite IM	One IM dose per site; 2 doses on day 0, one in each deltoid or thigh	Days 0 (×2), 7, 21	4 vials, 3 visits
Updated Thai Red Cross, 2-site intradermal (ID) (2–2–2–0–2)	0.1 ml per ID site = one-fifth of the IM dose , 2 sites per visit (one over each deltoid)	Days 0, 3, 7, 28 (the day-14 dose is omitted)	~2 vials, 4 visits

(Park 28e, Ch 5, p 344; Sunder Lal, Ch 8, pp 427–428)

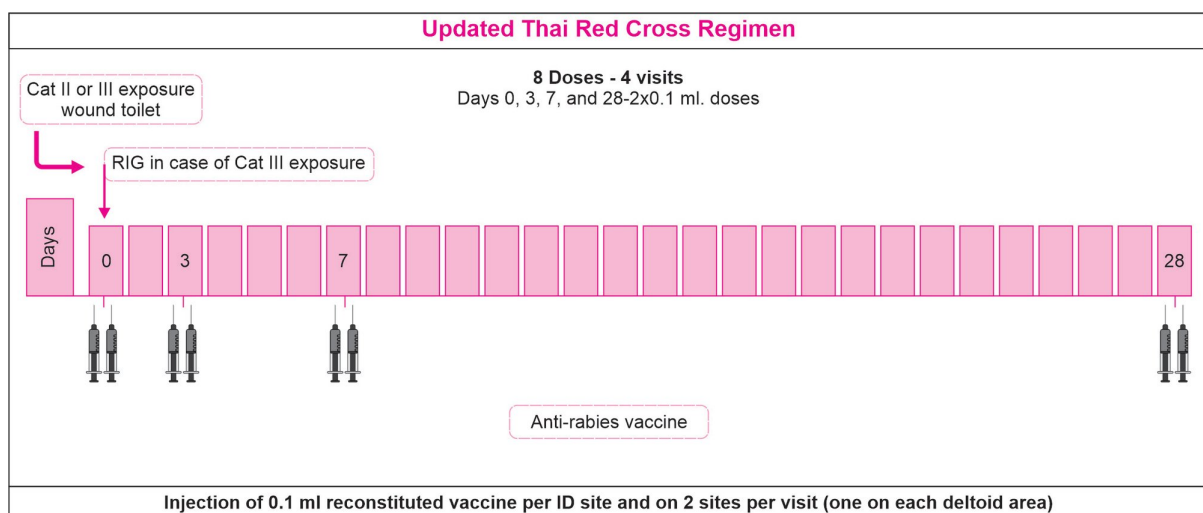


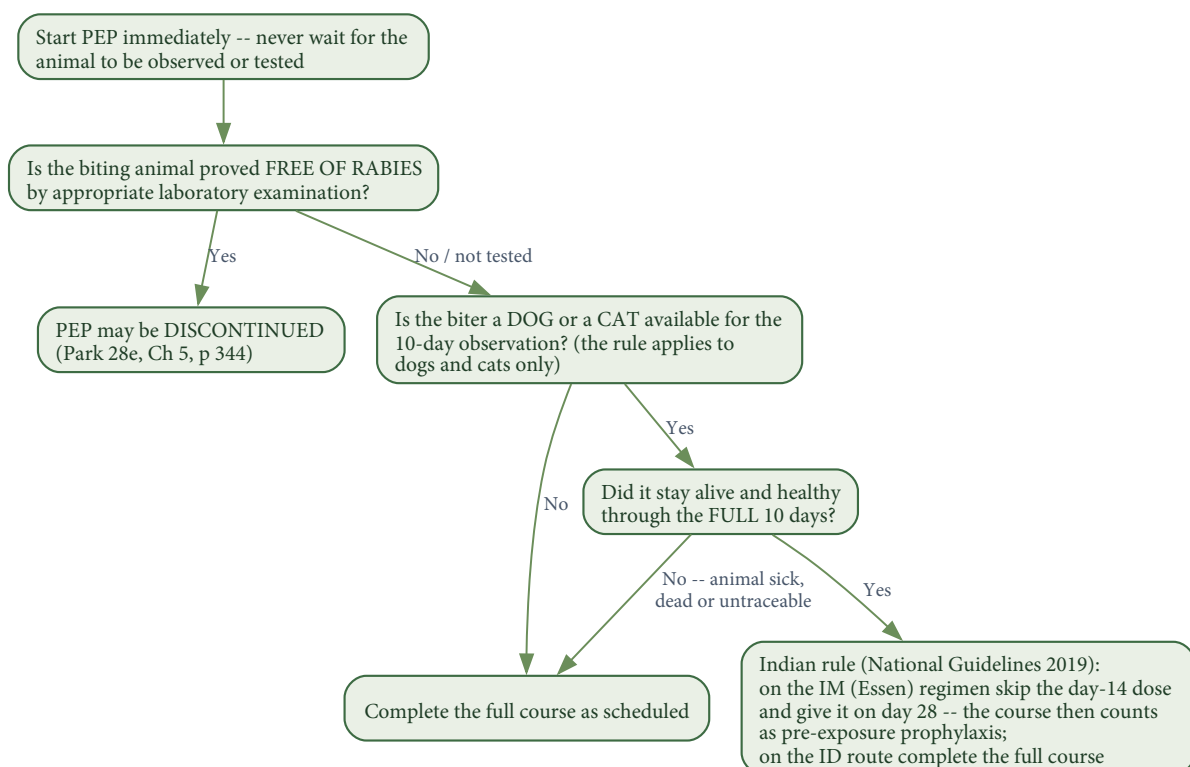
Fig 8.13B — Updated Thai red cross regimen (2–2–2–0–2) (Sunderlal, p.428)

Vaccine handling and response	Specification
Potency	≥2.5 IU per intramuscular dose for every CCEEV

Vaccine handling and response	Specification
Preservative	WHO-prequalified rabies vaccines contain no thimerosal
Storage	+2°C to +8°C, protected from sunlight; shelf-life 2–3 years
After reconstitution	Use immediately, or within 6–8 hours at the correct temperature — batch intradermal sessions so that an opened vial is used up within that time
Protective titre	Neutralising antibody ≥ 0.5 IU/ml, reached in most healthy people by day 14
Changing brands	Discouraged mid-course, but permissible if unavoidable

(Park 28e, Ch 5, p 344; Sunder Lal, Ch 8, p 429)

Modifying or discontinuing PEP. PEP is started at once but need not always be finished. It may be discontinued if the suspected animal is proved by appropriate laboratory examination to be free of rabies, or if a domestic dog, cat or ferret remains healthy throughout the observation period (Park 28e, Ch 5, p 344). The **10-day observation applies to dogs and cats only** — the natural history of rabies in other mammals is not fully understood (Sunder Lal, Ch 8, p 426).



Modifying or discontinuing PEP after observing or testing the biting animal

(Park 28e, Ch 5, p 344; Sunder Lal, Ch 8, p 426)

5. PEP in special groups

Group	Management
Previously fully vaccinated — documented complete pre-exposure prophylaxis (PrEP) of 3 doses, or PEP of 5 doses, with a cell-culture vaccine at any time in the past	Only 2 booster doses on days 0 and 3, IM or ID; no RIG — an anamnestic response occurs. Also applies to a documented titre ≥ 0.5 IU/ml. Alternative: a single-visit 4-site intradermal regimen, 0.1 ml over both deltoids or thighs
Re-exposure within 3 months of completing PEP	Wound toilet alone — neither vaccine nor RIG (WHO 2018)
Immunocompromised including HIV/AIDS	Full 5-dose intramuscular CCEEV series + RIG infiltration + thorough wound management for Category II and III; check neutralising antibody at 2–4 weeks to decide on an extra dose
Pregnancy, lactation, infancy, old age, concurrent illness	Not contraindications — rabies is nearly 100% fatal, so there is no contraindication to PEP; the vaccine has no adverse effect on the fetus, the pregnancy or the mother
On chloroquine (malaria treatment or prophylaxis)	Reduced response to intradermal vaccination — give the vaccine by the intramuscular route
Previous severe reaction to a vaccine component	Contraindicates that same vaccine; a different CCEEV is used

(Park 28e, Ch 5, p 345; Sunder Lal, Ch 8, pp 426, 429)

6. Tetanus and bacterial infection — the non-rabies half of bite PEP

Tetanus toxoid, and a suitable antibiotic to prevent sepsis, follow the local wound treatment in every bite victim (Sunder Lal, Ch 8, p 424; Park 28e, Ch 5, p 343). A mammal bite is penetrating and contaminated, never "clean and minor", so after surgical toilet it is managed under "other wounds" in Park's scheme (Park 28e, Ch 5, Fig 1, p 381):

Immunity category of the victim	Treatment for a bite wound
A — complete toxoid course or booster within the past 5 years	Nothing more required
B — complete course or booster >5 and <10 years ago	Tetanus toxoid, 1 dose
C — complete course or booster >10 years ago	Tetanus toxoid 1 dose + human tetanus immunoglobulin
D — no complete course, or immunity status unknown	Complete course of toxoid + human tetanus immunoglobulin

Adjunct	Regimen
Simultaneous active-passive cover in a non-immune person	250 units of human tetanus immunoglobulin (or 1,500 units of antitetanus serum) in one arm and 0.5 ml adsorbed tetanus toxoid in the other, toxoid repeated at 6 weeks and 1 year

Adjunct	Regimen
Antibiotic	Single IM dose of 1.2 mega units of benzathine penicillin, or erythromycin estolate 500 mg 6-hourly for 7 days if penicillin-sensitive; give as early as possible, within 6 hours of injury — an adjunct, never a substitute for immunisation

(Park 28e, Ch 5, p 380)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Three-pronged post-exposure prophylaxis after a mammal bite.** Draw a single bitten forearm at the centre with three labelled arrows converging on it: (1) *Wound toilet* — soap and running water 15 minutes → virucidal agent (povidone-iodine/alcohol) → no primary suturing, no cauterisation; (2) *Passive immunisation* — rabies immunoglobulin infiltrated **into and around the wound**, human RIG 20 IU/kg or equine RIG 40 IU/kg, once only, not beyond day 7; (3) *Active immunisation* — anti-rabies vaccine, deltoid or anterolateral thigh, **never gluteal**. Add a fourth side-arrow labelled *Tetanus toxoid ± tetanus immunoglobulin, antibiotic*. Label the three categories of exposure (I, II, III) beside the arrows to show that Category I needs none, Category II needs (1) + (3), and Category III needs (1) + (2) + (3).

High-yield exam pointers

- ▶ **Three prongs, given together:** wound toilet — RIG — vaccine. Washing alone cuts risk **up to 80%**; wash at least **15 minutes** with soap and running water.
- ▶ **Categories:** I = intact skin (no PEP); II = nibble or scratch **without bleeding** (vaccine); III = transdermal bite, lick on broken skin, mucosal contamination (RIG + vaccine). **All wild-animal and all forest-area bites = Category III.**
- ▶ **RIG:** HRIG 20 IU/kg, ERIG 40 IU/kg — once only, **not after day 7**, infiltrated into the wound, **no skin test**, never intravenously.
- ▶ **Regimens:** Essen IM 0-3-7-14-28; Zagreb 2-1-1 IM 0(×2)-7-21; updated Thai Red Cross ID 2-2-2-0-2 at 0-3-7-28, **0.1 ml per site = one-fifth of the IM dose**. Deltoid or anterolateral thigh, **never gluteal**. Potency ≥ 2.5 IU/IM dose; reconstituted vaccine within **6–8 hours**; protective titre ≥ 0.5 IU/ml by day 14.
- ▶ **Previously vaccinated:** 2 doses on days 0 and 3, no RIG; re-exposure **within 3 months** of PEP needs wound toilet only.
- ▶ **Stopping PEP:** discontinue if the animal is **laboratory-proven free of rabies**; the **10-day observation is for dogs and cats only** — if such an animal stays healthy for 10 days an Essen course skips day 14 and gives it on day 28 (becoming pre-exposure prophylaxis), but an ID course is always completed.
- ▶ **No contraindication to PEP** — pregnancy, lactation, infancy, old age included; **chloroquine users take the vaccine intramuscularly**. Nervous tissue vaccine production stopped in India in 2004.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park 28e reproduces the World Health Organization table, which places contact with **bats** in Category III, whereas India's National Guidelines for Rabies Prophylaxis 2019 hold that bat rabies is not conclusively proven in India and that exposure to bats does not at present warrant post-exposure prophylaxis — write the Indian position in an RGUHS answer and note the World Health Organization one in a line.

Recent advances [RA]

Current as of 2026-08-19 · depth: elaborate · sources: PubMed, WHO rabies fact sheet 2024

Programme and policy updates (India)

- ▶ India's rabies work is now framed by the **National Action Plan for dog Mediated Rabies Elimination (NAPRE)**, aligned to the global "**Zero by 30**" target within a One Health framework; nationally representative **2022–23** data show that dog-bite exposure and preventable deaths persist, and the stated correctives are integrated human–animal surveillance, sustained mass dog vaccination, **universal intradermal PEP access**, and targeted pre-exposure prophylaxis where PEP access is poorest [PMID 42494549].
- ▶ Alongside the **National Rabies Control Programme**, the Ministry of Health and Family Welfare has pressed states to make rabies a **notifiable disease** — **20 states have already done so** — and NAPRE provides for a dedicated GIS-enabled surveillance portal; India carries roughly **20,000 of the ~59,000 global rabies deaths** a year, and the dose-sparing **intradermal route has been approved in India since 2006** [PMID 41487407].

Current national data

- ▶ State-wise analysis of open government data puts reported dog-bite cases at **7.57 million in 2018**, falling to **2.18 million in 2022**, then rising again to **2.76 million in 2023**, with Uttar Pradesh, Bihar and Maharashtra consistently the highest-burden states — always quote the year with the figure [PMID 41116782].
- ▶ Completion, not initiation, is the weak link: among 735 bite victims at a Bengaluru anti-rabies clinic (2021–22), **80.4% finished the five-dose Essen course**, with stepwise attrition (93.7% → 86.4% → 82.3% → 80.4% for doses 2 to 5); the commonest reason for dropping out was negligence (27.7%) [PMID 41749175].

Elimination, surveillance and disease status

- ▶ **Goa** remains India's demonstration of state-level elimination achieved through a One Health model of mass dog vaccination and community education, but **sporadic cases since 2023**, mostly in border areas, show that the status has to be actively held by continued surveillance [PMID 41487407].
- ▶ Apparent "PEP failure" is usually a delivery failure: of 89 laboratory-confirmed rabies cases (2018–2024) who had received some PEP, only **3 were true breakthroughs**; PEP began within 2 days in 57.3%, wound care was documented in 56.18%, and only **26.39% of Category III cases received immunoglobulin without any deviation** [PMID 41350240].

Guideline and legislative updates

- ▶ The WHO-endorsed abridged **one-week, two-site intradermal regimen on days 0, 3 and 7** (Institut Pasteur du Cambodge, IPC) is the major schedule change beyond the updated Thai Red Cross regimen in the tables above: **RESIST-2** found the day-28 session added nothing (titres plateaued between days 28 and 42; all patients alive at one year); a side-by-side study found all 210 recipients of either ID IPC or four-dose IM Essen reached **≥0.5 IU/ml by day 28**; and the **Resist-3** study showed that 181 people boosted 2, 5 or 10 years after a 3- or 4-session ID course all reached protective titres **within 7 days** [PMID 31570311; PMID 37337899; PMID 42296188].
- ▶ **Rabies monoclonal antibodies are an accepted substitute for RIG** where immunoglobulin is short. An Indian comparison of 159 patients each given the single antibody Rabishield or the two-antibody cocktail TwinRab (docaravimab + miromavimab) for Category III bites found both safe and feasible to infiltrate with no serious systemic events, Rabishield needing more vials per patient (2.28 vs 2.00; $p = 0.001$) [PMID 42490664]; post-marketing surveillance of 200 TwinRab recipients at **40 IU/kg** recorded solicited local reactions in 1.5% [PMID 39629371].
- ▶ Diluting the biological to achieve **full-dose wound infiltration** now has real-world support: across 223 Bengaluru bite cases (2022–24) dilution reached **1:1 for HRIG, 1:5.4 for ERIG, 1:20.7 for Rabishield and 1:8.2 for TwinRab**, with **100% survival** over 6–42 months of follow-up [PMID 41576704].

Global and WHO developments

- ▶ Gavi's **Vaccine Investment Strategy 2021–2025** added human rabies vaccine, funding PEP scale-up in Gavi-eligible countries [PMID 41487407]; supply-chain modelling of bite-patient data from Tanzania and Kenya found that **intradermal vaccination cuts vial use by more than 55%** and lowers stockout risk even in low-throughput facilities [PMID 40274463], and WHO puts the global toll at about **59,000 deaths a year**, with the switch from intramuscular to intradermal administration reducing cost by **60–80%** [WHO 2024-Jun].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 5, pp 342–346 rabies; pp 380–381 prevention of tetanus in the wounded); Sunder Lal — Textbook of Community Medicine (Ch 8, pp 423–430, reproducing the Government of India National Guidelines for Rabies Prophylaxis 2019, National Rabies Control Programme, National Centre for Disease Control); Suryakantha — Community Medicine with Recent Advances (p 512, animal-bite first aid). Underlying document cited by these texts: WHO Expert Consultation on Rabies, third report, WHO Technical Report Series No. 1012, 2018.

Recent-advances sources: 42494549 — One Health implementation for rabies elimination in India by 2030: progress, governance, challenges, and policy alignment, *One Health* 2026, doi:[10.1016/j.onehlt.2026.101520](https://doi.org/10.1016/j.onehlt.2026.101520) (abstract-only); 41487407 — Rabies post-exposure prophylaxis in India: a SWOT analysis, *Therapeutic Advances in Vaccines and Immunotherapy* 2026, doi:[10.1177/25151355251410780](https://doi.org/10.1177/25151355251410780); 41116782 — State-wise geospatial analysis of dog bite burden in India from 2018 to 2023, *Porto Biomedical Journal* 2025, doi:[10.1097/j.pbj.0000000000000305](https://doi.org/10.1097/j.pbj.0000000000000305); 41749175 — Compliance with anti-rabies post-exposure prophylaxis among animal bite victims attending an anti-rabies clinic of a tertiary care centre in urban Bengaluru, *BMC Infectious Diseases* 2026, doi:[10.1186/s12879-026-12879-6](https://doi.org/10.1186/s12879-026-12879-6); 41350240 — Rabies despite postexposure prophylaxis in high-burden settings: vaccine failure or gaps in healthcare delivery?, *Clinical Infectious Diseases* 2026, doi:[10.1093/cid/ciaf662](https://doi.org/10.1093/cid/ciaf662) (abstract-only); 31570311 — A 1-week intradermal dose-sparing regimen for rabies post-exposure prophylaxis (RESIST-2): an observational cohort study, *Lancet Infectious Diseases* 2019, doi:[10.1016/S1473-3099\(19\)30311-1](https://doi.org/10.1016/S1473-3099(19)30311-1) (abstract-only); 37337899 — Side-by-side comparative study of the immunogenicity of the intramuscular and intradermal rabies post-exposure prophylaxis regimens in a cohort of suspected rabies virus exposed individuals, *Clinical Infectious Diseases* 2023, doi:[10.1093/cid/ciad304](https://doi.org/10.1093/cid/ciad304); 42296188 — Comparative long-term immunogenicity and boosting of 3- versus 4-session intradermal rabies post-exposure prophylaxis regimens, Cambodia and Madagascar: the Resist-3 study, *Clinical Infectious Diseases* 2026, doi:[10.1093/cid/ciag368](https://doi.org/10.1093/cid/ciag368) (abstract-only); 42490664 — Safety and operational feasibility of single versus cocktail rabies monoclonal antibodies for post-exposure prophylaxis in a resource-limited setting, *PLoS Neglected Tropical Diseases* 2026, doi:[10.1371/journal.pntd.0014274](https://doi.org/10.1371/journal.pntd.0014274) (abstract-only); 39629371 — Safety assessment of the world's first novel cocktail of two monoclonal antibodies in WHO category-III animal-bite patients, *Journal of Family Medicine and Primary Care* 2024, doi:[10.4103/jfmmpc.jfmmpc_377_24](https://doi.org/10.4103/jfmmpc.jfmmpc_377_24); 41576704 — Effectiveness of rabies immunoglobulins and monoclonal antibodies infiltrated into animal bite wounds after dilution: real-world evidence from an anti-rabies clinic in India, *Vaccine* 2026, doi:[10.1016/j.vaccine.2026.128233](https://doi.org/10.1016/j.vaccine.2026.128233) (abstract-only); 40274463 — Optimising human rabies vaccine supply chains: a modelling study, *Vaccine* 2025, doi:[10.1016/j.vaccine.2025.127108](https://doi.org/10.1016/j.vaccine.2025.127108). Plus the World Health Organization rabies fact sheet, updated 5 June 2024.

♦ ♦ ♦

Challenges in control of arthropod-borne diseases

Asked in: Paper I 02-Jan-2025 — RGUHS MD Community Medicine

Definition — An arthropod-borne (vector-borne) disease is one in which the infectious agent reaches a susceptible person through a **vector**, defined as an arthropod or any living carrier (e.g., snail) that transports an infectious agent to a susceptible individual; transmission may be **mechanical** or **biological** (Park 28e, Ch 3, pp 117-118). Control must therefore break a three-link chain — agent, vector and human host — and its difficulty arises because every link is defended by biological, ecological, social and programmatic obstacles at the same time. In India six such diseases (malaria, lymphatic filariasis, Japanese encephalitis, dengue and chikungunya by mosquitoes; kala-azar by sandflies) are handled by the National Vector Borne Disease Control Programme (NVBDCP), which itself states that prevention and control are complex because transmission depends on the interaction of numerous ecological, biological, social and economic factors including migration (Park 28e, Ch 7, p 493).

Vector-related (entomological) challenges

Challenge	Mechanism	Suitable example
Physiological insecticide resistance	Enzymatic detoxification, altered target site, or reduced cuticular penetration, selected by insecticide pressure (Gilles & Warrell 4e, Ch 6, pp 172-173)	Resistance has appeared in over 100 species of arthropods of public health importance , so vector control by insecticides alone is no longer fully effective (Park 28e, Ch 15, p 922)
Behaviouristic resistance (insecticide avoidance)	The vector changes its resting and feeding behaviour to avoid contact with the treated surface, defeating indoor residual spray (IRS) even while still susceptible (Gilles & Warrell 4e, Ch 6, p 172)	Efficacy of IRS depends on the feeding and resting habits of the vector; outdoor-resting anophelines escape it
Cross-resistance	One selected gene covers a second chemical class, so simple substitution fails	The <i>kdr</i> gene giving DDT resistance also confers cross-resistance to pyrethroids , and pyrethroid selection enhances DDT resistance; an insensitive acetylcholinesterase gives broad organophosphate–carbamate resistance (Gilles & Warrell 4e, Ch 6, pp 173-174)
Scale of vector resistance	Multi-species and multi-compound	55 anopheline species resistant to one or more insecticides — 53 to DDT, 27 to organophosphates, 17 to carbamates, 10 to pyrethroids; 21 of them are important malaria vectors, including the Indian vectors <i>A. culicifacies</i> , <i>A. fluviatilis</i> and <i>A. annularis</i> (Gilles & Warrell 4e, Ch 6, p 177)
Cost escalation from resistance	Forced switch to costlier compounds shrinks the area covered by the same budget	Vector resistance to insecticides necessitated the more expensive pyrethroids, thereby limiting coverage (Park 28e, Ch 5, p 333); DDT and synthetic pyrethroids need 2 rounds and malathion 3 rounds per transmission season (Park 28e, Ch 7, p 498)

Challenge	Mechanism	Suitable example
Agricultural selection pressure	Insecticides sprayed on crops select vector populations outside the health sector's control	Substitute compounds do not stay useful for long where insecticides are widely applied in agriculture, as happened in cotton-growing areas (Gilles & Warrell 4e, Ch 6, p 179)
Vector diversity defeats a single method	Each vector has its own bionomics, so one tool cannot serve all six diseases	Filariasis vectors became resistant to DDT, HCH and dieldrin, so IRS for filariasis was discontinued (Park 28e, Ch 5, p 339); <i>Culex quinquefasciatus</i> is resistant to a wide range of insecticides, limiting chemical choice (Service, Ch 3)
Container and peri-domestic breeding	Breeding sites lie inside private premises, are innumerable and refill after every rain	<i>Aedes aegypti</i> breeds in containers in and around houses and is endophagic and endophilic (Park 28e, Ch 5, p 307); rooftop water tanks are important breeding places of <i>Anopheles stephensi</i> , the urban malaria vector (Service, Ch 3)
Transovarial transmission	The virus persists in the eggs, so the infected vector population is re-seeded without any human case	Transovarial transmission of dengue virus occurs when virus enters fully developed eggs at the time of oviposition (Park 28e, Ch 5, p 307)

Agent- and reservoir-related challenges

- **Antiparasitic drug resistance.** Resistance of *Plasmodium falciparum* to the 4-aminoquinolines and to sulfadoxine–pyrimethamine is widespread; mefloquine resistance is reported from Myanmar and Thailand and reduced quinine susceptibility from Thailand, which forced all endemic countries to adopt **artemisinin-based combination therapy (ACT)** (Park 28e, Ch 5, p 318).
- **Counterfeit and substandard drugs.** Drug-resistant parasites *and the presence of counterfeit antimalarial drugs* further hinder control efforts (Park 28e, Ch 5, p 318).
- **Sylvatic cycles and animal reservoirs.** Where the agent maintains itself in animals, human-directed control cannot interrupt transmission — jungle yellow fever continues to be essentially uncontrollable because the virus maintains itself in the animal population and mosquito control is possible only over restricted areas (Park 28e, Ch 5, pp 347-348). In **Kyasanur forest disease (KFD)** of Karnataka, monkeys are amplifying hosts, cattle provide the blood meals that sustain the *Haemaphysalis* tick population without maintaining the virus, and **man is an incidental, dead-end host** (Park 28e, Ch 5, p 354).
- **Zoonotic amplification.** Japanese encephalitis virus is maintained in wild birds (Park 28e, Ch 15, p 942), so vector work alone cannot stop it and the programme had to add **JE vaccination** (Park 28e, Ch 7, p 493).
- **Hidden human reservoirs.** Post-kala-azar dermal leishmaniasis (PKDL) appears one to several years after apparent cure as nodular skin infiltrations that are usually non-ulcerated yet swarming with parasites, so it escapes notice while sustaining transmission (Park 28e, Ch 5, p 372).

- **Multiplicity of agents.** Dengue circulates as several serotypes, so infection with one does not protect the population against the next, and each new serotype re-opens the community to epidemics.

Environmental, social and behavioural challenges

Challenge	Explanation and example
Unplanned urbanisation and civic failure	Unplanned and under-planned urbanisation, overcrowding, poor sanitation and inadequate public health infrastructure head the list of factors behind emerging and re-emerging infections (Park 28e, Ch 5, p 426); failure of urban authorities to provide civil amenities lets the vector breed at high levels and makes the environment conducive to transmission (Park 28e, Ch 5, p 307)
Water-storage behaviour	Even where piped water exists people keep storing water as insurance against an unreliable supply; covering storage pots is simple but unpopular and is soon neglected, and screens fitted on rooftop tanks become torn or are removed (Service, Ch 3)
Climate and warming	<i>Ae. aegypti</i> survives best between 16 °C and 30 °C at 60–80 % relative humidity; a 2 °C rise shortens the extrinsic incubation period of dengue virus, leaves more infected mosquitoes available for longer, and raises biting frequency through dehydration (Park 28e, Ch 5, p 307). WHO has predicted that temperature shifts with global warming will encourage the spread of malaria and dengue (One Health Approach)
Migration and mobility	Transmission depends on ecological, biological, social and economic factors including migration (Park 28e, Ch 7, p 493); rapid and intense international travel is a recognised driver of re-emergence (Park 28e, Ch 5, p 426)
Hard-to-reach ecotypes	Malaria persists in tribal populations living in foothill, forested or conflict-affected areas of Andhra Pradesh, Chhattisgarh, Jharkhand, Madhya Pradesh, Maharashtra, Meghalaya, Mizoram, Odisha, Telangana and Tripura (Park 28e, Ch 7, p 496)
Occupational and forest exposure	KFD strikes those entering the forest for work; protection depends on adequate clothing, repellents (dimethyl phthalate, DEET), daily body examination for ticks and avoidance of sitting or lying on forest ground — all dependent on individual compliance (Park 28e, Ch 5, pp 354-355)
Community non-participation	Long-term source reduction works only when supported by health education aimed at securing community participation (Park 28e, Ch 5, p 348); environmental control has intensive public health education and political support as essential prerequisites (Park 28e, Ch 15, p 922)

Programmatic, technical and health-system challenges

- **No single method suffices.** There is no single and simple method that would ensure control of transmission, which is why integrated vector control combining bio-environmental and personal protection measures replaced dependence on residual insecticides (Park 28e, Ch 5, p 333).
- **Coverage thresholds are hard to reach.** High-risk villages must be protected by IRS and insecticide-treated nets with coverage of **more than 80 per cent**, whatever the intervention, and high-risk areas and populations must be re-defined at least annually — a heavy recurring operational demand (Park 28e, Ch 7, p 498).

- ▶ **Resurgence when spraying stops.** Discontinuation of spraying has often led to the **resurgence of malaria**; once applied, IRS may need to be continued for an indefinite period and needs exhaustive coverage to work at all (Park 28e, Ch 5, p 332).
- ▶ **Limits of the alternatives.** Biological control with larvivorous fish such as *Gambusia*, and agents such as *Coelomomyces* fungi, is effective **only when used in conjunction with other methods** (Park 28e, Ch 15, p 927); genetic and transgenic approaches are much more difficult to implement than conventional insecticidal methods (Service, Ch 3); and chemical control itself carries environmental contamination, forcing replacement of persistent compounds such as DDT with biodegradable ones (Park 28e, Ch 15, p 922).
- ▶ **Weak entomological surveillance.** Sustaining the human blood index, sporozoite rate, mosquito density per man-hour catch, man-biting rate and inoculation rate (Park 28e, Ch 5, p 324), together with periodic insecticide-susceptibility testing, needs entomologists and insectaries that most districts do not have; early detection of resistance and monitoring of its progress are essential to any programme based on wide insecticide use (Gilles & Warrell 4e, Ch 6, p 177).
- ▶ **Operational load of the elimination phase.** Kala-azar elimination raised active case search from a single annual round to **quarterly searches** during the "Kala-azar Fortnight", backed by ASHA incentives of Rs 300 for each case identified, Rs 100 for one and Rs 200 for two rounds of insecticide spraying, and wage compensation of Rs 500 for kala-azar and Rs 2,000 for PKDL (Park 28e, Ch 7, p 501) — an incentive-driven intensity that is difficult to sustain year on year.
- ▶ **Mass drug administration (MDA) compliance.** Lymphatic filariasis elimination rests on annual MDA with DEC plus albendazole (DA), or ivermectin plus DEC plus albendazole (IDA), with districts released only after successive Transmission Assessment Surveys (Park 28e, Ch 7, p 501); the whole at-risk population must swallow the drug every year for transmission to stop.
- ▶ **International obligations at ports.** Under the International Health Regulations the *Aedes aegypti* index must be kept at **zero at all ports**, and airports and seaports kept free of *Aedes* mosquitoes for a distance of **400 metres around the perimeter** (Park 28e, Ch 15, p 925) — a standard demanding continuous port health vigilance.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Diagram — chain of transmission of an arthropod-borne disease, with the points at which control is attempted.** Labels to write: *Reservoir / infected human*, *Susceptible vector takes blood meal*, *Extrinsic incubation period in vector*, *Infective vector*, *Susceptible host*, *Attack point 1 — early diagnosis and radical treatment*, *Attack point 2 — anti-adult measures (indoor residual spray)*, *Attack point 3 — anti-larval and source-reduction measures*, *Attack point 4 — personal protection (long-lasting insecticidal net, repellent)*, *Attack point 5 — vaccination where available (Japanese encephalitis, Kyasanur forest disease)*.

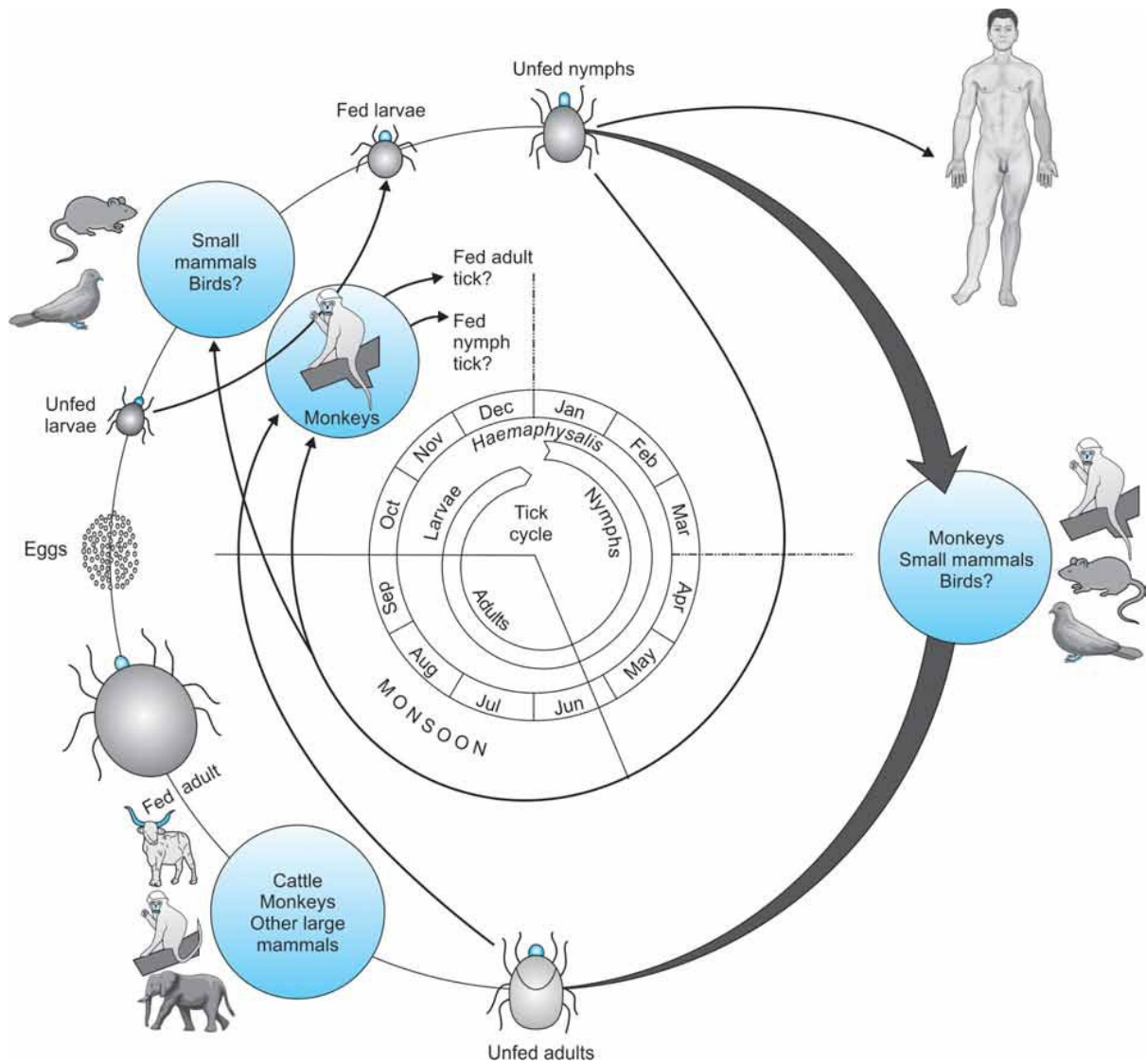
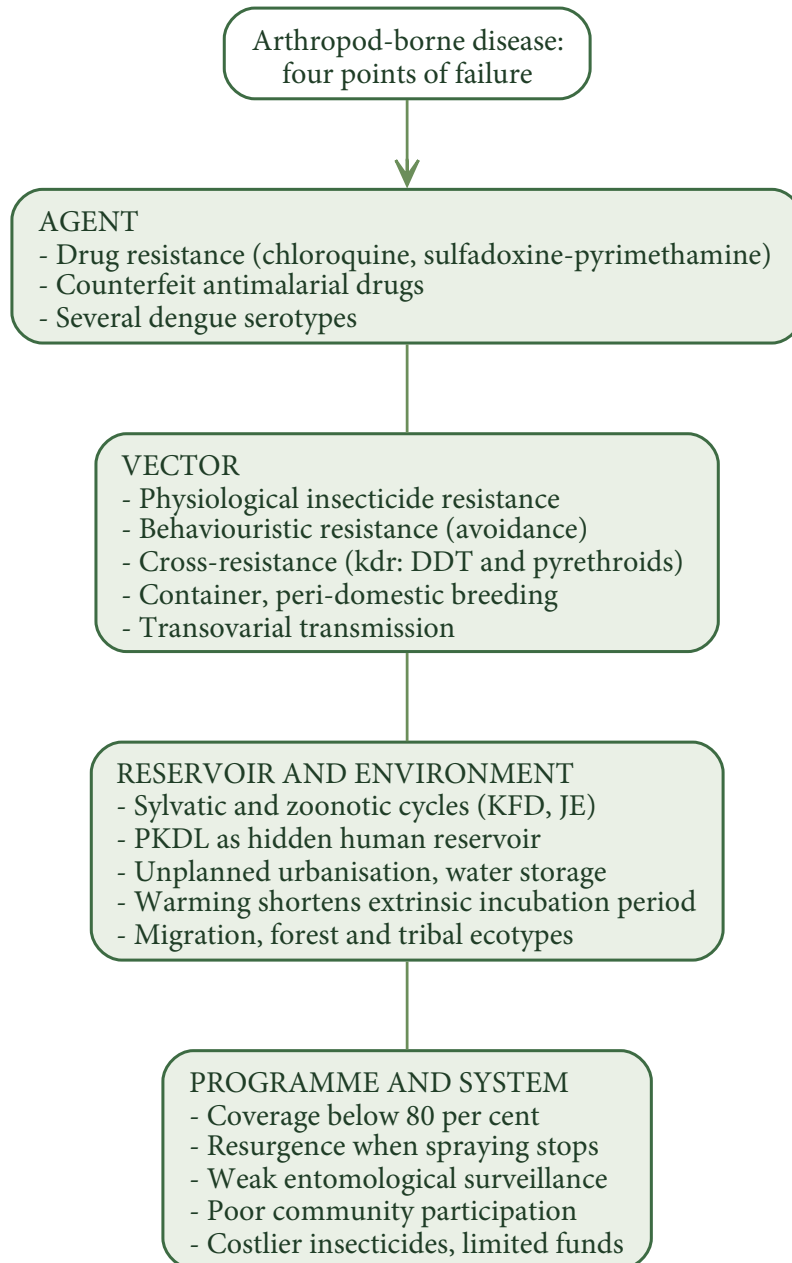
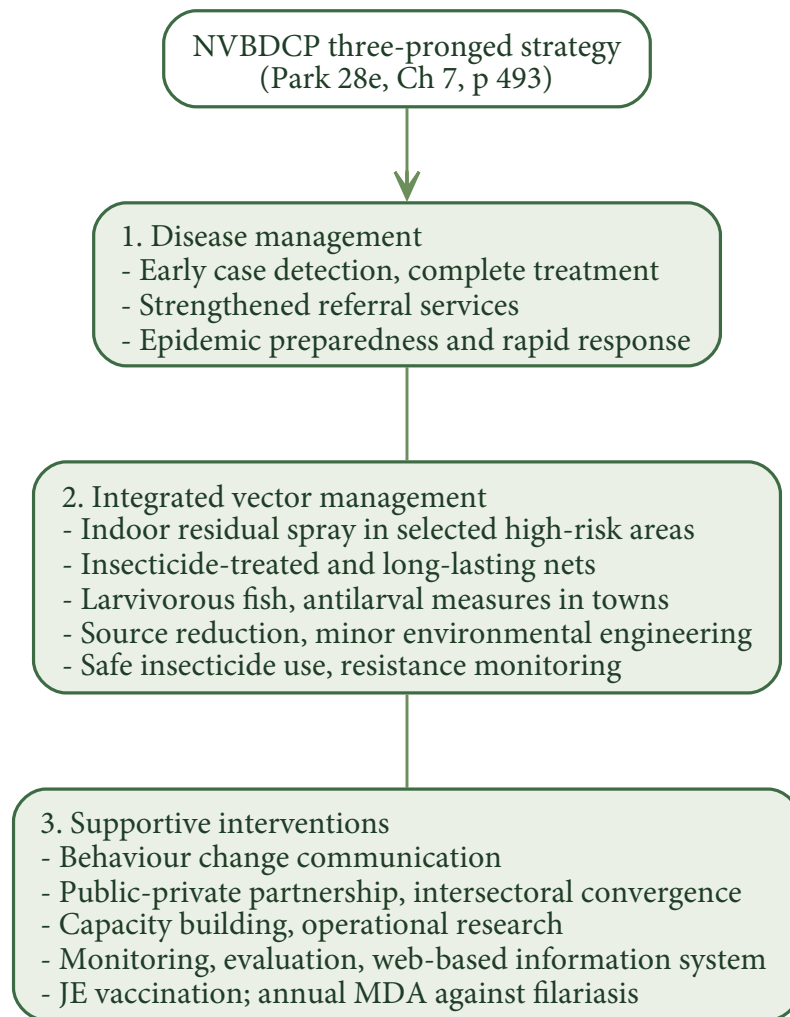


Fig 20.53 — Natural cycle of Kyasanur forest disease (Suryakantha Environment Best, p.473)



Where control of an arthropod-borne disease breaks down - the four defended links



Integrated vector management under NVBDCP - the programmatic answer to these challenges

High-yield exam pointers

- **Four principles of arthropod control** (Park 28e, Ch 15, p 922): environmental, chemical, biological, genetic — and **environmental control is the best approach** because its results are likely to be permanent.
- **Two resistances to name:** *physiological* resistance and *behaviouristic* resistance (insecticide avoidance); add *cross-resistance* through the *kdr* gene linking DDT and pyrethroids.
- **Quote the scale:** resistance in **more than 100 arthropod species** of public health importance (Park 28e); **55 anopheline species** resistant — 53 to DDT, 27 to organophosphates, 17 to carbamates, 10 to pyrethroids (Gilles & Warrell 4e).
- **Annual parasite incidence (API) = confirmed malaria cases during one year × 1000 ÷ population under surveillance** (Park 28e, Ch 5, p 324). Entomological indices to list: human blood index, sporozoite rate, man-hour density, man-biting rate, inoculation rate.
- **IRS rounds:** DDT and synthetic pyrethroids 2 rounds, malathion 3 rounds per transmission season; coverage of high-risk villages must exceed **80 per cent** (Park 28e, Ch 7, p 498).
- **International Health Regulations:** *Aedes aegypti* index kept at **zero at all ports**, and ports kept free of *Aedes* for **400 metres** around the perimeter (Park 28e, Ch 15, p 925).

- **Karnataka example worth writing:** Kyasanur forest disease — first recognised in 1957 in Shimoga district; now four districts (Shimoga, North Kannada, South Kannada, Chikamagaluru) over 6,000 sq km; 400–500 cases a year with case-fatality 3–5 per cent; monkey deaths herald an outbreak (Park 28e, Ch 5, p 354).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference, not to be written in the exam: the National Vector Borne Disease Control Programme has since been re-designated the National Centre for Vector Borne Diseases Control, and elimination target years and insecticide policy are revised periodically, so any programme figure should be quoted with the year of the source it comes from; the retrieved corpus describes the programme under its former NVBDCP name.

Recent advances [RA]

Current as of 2026-08-19 · depth: elaborate · sources: PubMed, NCVBDC (MoHFW), WHO World Malaria Report 2024, WHO Global Arbovirus Initiative

Programme and policy updates (India)

- **The programme has been renamed and its tool box is openly under review.** The former NVBDCP now runs as the **National Center for Vector Borne Diseases Control (NCVBDC)** under the Ministry of Health and Family Welfare [NCVBDC 2026]. A national review lists the tools India must add because the existing ones are losing effect — nets carrying piperonyl butoxide or the pyrrole chlorfenapyr, clothianidin and other neonicotinoids for IRS, *Bacillus sphaericus* larvicides, attractive toxic sugar baits for outdoor biting, endectocides such as ivermectin, insecticidal paints, spatial repellents, insecticide-treated wearables, and sterile-insect, incompatible-insect and *Wolbachia* techniques; the national programme still deploys pyrethroid-only LLINs, with PBO nets only recently cleared for public health use in India [PMID 40018116].
- **Elimination deadlines the programme now carries.** India has pledged elimination of lymphatic filariasis as a public health problem by 2027, three years ahead of the global 2030 target; MDA with the triple-drug IDA regimen covered 63 districts in 2023, effective coverage must exceed 65 per cent, and the 2023 revised strategy adds digital platforms for real-time monitoring [PMID 40639855]. Malaria elimination is targeted for 2030 under the National Framework [PMID 39182925]. An ICMR-journal editorial warns that delivery is thinning at the same time — a sparse new-product pipeline, patchy resistance monitoring, and field operations resting on contractual multipurpose workers with little entomological training [PMID 41100645].

Current national data

- **Malaria figures, each with its year.** The World Malaria Report 2024 put global cases at 263 million in 2023 (252 million in 2022), incidence 60.4 per 1,000 at risk; India's **estimated** burden was **2.04 million cases in 2023**, close to half of the WHO South-East Asia Region. NCVBDC reported 227,564 confirmed cases in 2023 and 255,500 in 2024, *P. vivax* accounting for about 39 and 40 per cent, and 88 per cent of 2024 cases came from eight states — Odisha, Chhattisgarh, Uttar Pradesh, Jharkhand, Maharashtra, Tripura, Mizoram and West Bengal [PMID 41036407].
- **Reported and estimated burden are diverging, so never mix them.** India's *reported* cases rose 29 per cent in 2023 over 2022 while WHO's *estimate* for the same period fell by roughly 1.3 million,

because the estimate applies statistical adjustments to programme returns and one adjustment factor was dropped for India in the 2024 report. Quote a figure with the report and year it came from [PMID 40987489].

Elimination, surveillance and disease status

- **Kala-azar — threshold met, WHO validation still ahead.** By December 2023 every implementation unit in India was below the elimination threshold of fewer than 1 case per 10,000 population at block level; the country is now in the consolidation phase and must sustain that for three consecutive years before a dossier can go to WHO for validation (Bangladesh was validated first). PKDL, HIV–VL co-infection and undiagnosed cases remain the human reservoirs that can restart an epidemic [PMID 40828749].
- **Resistance is now documented in Indian vectors and tracked in the parasites.** In sprayed villages of Muzaffarpur, Bihar, the sandfly *Phlebotomus argentipes* showed raised glutathione S-transferase, esterase and cytochrome P450 activity with the *kdr* mutation **L1014S at 48.5 per cent** — early alpha-cypermethrin resistance — while *ace-1* stayed wild-type, so organophosphates remain available and insecticide rotation is the recommendation [PMID 41918020]. In three Tamil Nadu cities *Aedes aegypti* larvae were widely temephos-resistant (corrected mortality 13.3–72.7 per cent) whereas *Anopheles stephensi* stayed susceptible (97–98 per cent) [PMID 41961265]. Molecular surveillance of *P. falciparum* across Indian states found **no WHO-validated Pfk13 artemisinin-resistance mutation**, supporting continued ACT efficacy, but *Pfcr* K76T persists a decade after chloroquine was withdrawn and *Pfaat1* S258L — linked to reduced lumefantrine susceptibility and previously reported only from Africa — has now been detected [PMID 41846600].

Guideline and legislative updates

- **Kyasanur forest disease — the vaccine has been withdrawn.** The formalin-inactivated KFD vaccine, whose field effectiveness in Karnataka was 62.4 per cent after two doses and 82.9 per cent with a booster, was **discontinued by 2022** on grounds of regulatory non-compliance, painful administration and falling acceptance; a beta-propiolactone-inactivated whole-virion candidate is at the preclinical stage [PMID 42004966] and no antiviral is approved [PMID 41202828]. Tick-borne disease also sits outside NCVBDC — KFD and Crimean-Congo haemorrhagic fever fall to NCDC and the IDSP, and India still has no coordinated national tick surveillance programme under One Health [PMID 42165723].
- **Malaria vaccines exist but do not yet fit India.** WHO recommended RTS,S/AS01 in 2021 and **R21/Matrix-M on 2 October 2023**, R21 being the first to reach WHO's 75 per cent efficacy goal. Both are pre-erythrocytic vaccines directed at the *P. falciparum* circumsporozoite protein, so they leave *P. vivax* — about 40 per cent of India's cases — untouched; India's elimination goal needs a vaccine covering both species across all age groups [PMID 39182925].

Global and WHO developments

- **Global Arbovirus Initiative, and the worst dengue year on record.** WHO launched the Global Arbovirus Initiative in **March 2022** on six pillars — monitoring risk, reducing epidemic risk, strengthening vector control, preparedness, innovation in diagnostics and interventions, and coordination. **2024 saw a record 14 million dengue cases from 134 countries and territories, more than 52,000 severe cases and more than 10,000 deaths**, with a nine-fold rise in African

dengue since 2019 and locally acquired cases now appearing in temperate France and Italy [PMID 41308660].

- **New tools that have crossed the trial threshold.** Singapore's cluster-randomised test-negative trial of male *Wolbachia* (wAlbB) releases suppressed *Ae. aegypti* and gave **71–72 per cent protective efficacy against dengue** (odds ratios 0.28–0.29) [PMID 41671481]; a meta-analysis of nine studies covering over 8 million people pooled an incidence rate ratio of **0.15 (95% CI 0.05–0.39)** for *Wolbachia* deployment [PMID 42212411]. Ivermectin mass drug administration (BOHEMIA, coastal Kenya), 400 µg/kg monthly for three months, cut malaria infection in children aged 5–15 by **26 per cent (IRR 0.74; 95% CI 0.58–0.95)** even where net coverage was already high [PMID 40700688]. Genomes of 645 *Anopheles stephensi* trace the African urban-malaria invasion to a South Asian introduction that seeded a Djibouti bridgehead, carrying metabolic-detoxification resistance haplotypes with it — India's own urban vector now threatens 126 million African city dwellers [PMID 42348676].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 3 Principles of Epidemiology and Epidemiologic Methods; Ch 5 Epidemiology of Communicable Diseases; Ch 7 Health Programmes in India; Ch 15 Environment and Health); Gilles & Warrell — Essential Malariology 4e (Ch 6, Rationale and technique of malaria control); Service — Medical Entomology for Students 5e (Ch 3, mosquitoes and their control); Suryakantha — Community Medicine with Recent Advances (Fig 20.53, natural cycle of Kyasanur forest disease); One Health Approach (climate change and the spread of vector-borne disease). Coverage note: the corpus supports the entomological, parasitological, ecological and NVBDCP programmatic content quoted above; post-2024 programme renaming and current elimination target years were not retrievable from it and are flagged in the Note rather than stated as fact.

Recent-advances sources: 40018116 — Addressing challenges in vector control: current strategies and the imperative for novel tools in India, *BMJ Public Health* 2024, doi:[10.1136/bmjph-2023-000342](https://doi.org/10.1136/bmjph-2023-000342); 40639855 — Global insights can accelerate India's journey towards elimination of lymphatic filariasis, *BMJ Global Health* 2025, doi:[10.1136/bmjgh-2025-018851](https://doi.org/10.1136/bmjgh-2025-018851); 41100645 — Vector control at a crossroad: grappling with multiple challenges, *Indian Journal of Medical Research* 2025, doi:[10.25259/IJMR_1953_2025](https://doi.org/10.25259/IJMR_1953_2025); 41036407 — Hotspot districts of persistent *Plasmodium vivax* malaria in India, *Lancet Regional Health Southeast Asia* 2025, doi:[10.1016/j.lansea.2025.100662](https://doi.org/10.1016/j.lansea.2025.100662); 40987489 — Time to address the drawbacks in estimation and reporting of malaria cases in India and beyond, *BMJ Global Health* 2025, doi:[10.1136/bmjgh-2025-019170](https://doi.org/10.1136/bmjgh-2025-019170); 40828749 — The story of elimination of visceral leishmaniasis (kala-azar) in India: challenges towards sustainment, *PLoS Neglected Tropical Diseases* 2025, doi:[10.1371/journal.pntd.0013321](https://doi.org/10.1371/journal.pntd.0013321); 41918020 — Elevated metabolic enzymes and kdr mutations indicate early alpha-cypermethrin resistance in *Phlebotomus argentipes*, Bihar, *Parasites & Vectors* 2026, doi:[10.1186/s13071-026-07339-8](https://doi.org/10.1186/s13071-026-07339-8); 41961265 — Assessment of temephos resistance in *Aedes aegypti* and *Anopheles stephensi*, Tamil Nadu, *Journal of Vector Borne Diseases* 2026, doi:[10.4103/jvbd.jvbd.324_25](https://doi.org/10.4103/jvbd.jvbd.324_25) (abstract-only); 41846600 — Targeted genomic surveillance of malaria drug resistance in India, *Open Forum Infectious Diseases* 2026, doi:[10.1093/ofid/ofag106](https://doi.org/10.1093/ofid/ofag106); 42004966 — Whole virion inactivated Kyasanur Forest Disease vaccine candidate: preclinical safety and efficacy, *Frontiers in Immunology* 2026, doi:[10.3389/fimmu.2026.1786057](https://doi.org/10.3389/fimmu.2026.1786057); 41202828 — Kyasanur Forest disease: an emerging arboviral threat, *Lancet Infectious Diseases* 2026, doi:[10.1016/S1473-3099\(25\)00589-4](https://doi.org/10.1016/S1473-3099(25)00589-4) (abstract-only); 42165723 — Ticks and tick-borne diseases: a call for comprehensive tick surveillance in India under One Health, *Indian Journal of Medical Research* 2026, doi:[10.25259/IJMR_2011_2025](https://doi.org/10.25259/IJMR_2011_2025); 39182925 — Rethinking malaria vaccines: currently approved vaccines in India's path to elimination, *BMJ Global Health* 2024, doi:[10.1136/bmjgh-2024-016019](https://doi.org/10.1136/bmjgh-2024-016019); 41308660 — Strengthening global preparedness and response to arboviral disease threats, *Lancet Infectious Diseases* 2026, doi:[10.1016/S1473-3099\(25\)00686-3](https://doi.org/10.1016/S1473-3099(25)00686-3); 41671481 — Dengue suppression by male *Wolbachia*-infected mosquitoes, *New England Journal of Medicine* 2026, doi:[10.1056/NEJMoa2503304](https://doi.org/10.1056/NEJMoa2503304) (abstract-only); 42212411 — Impact of *Wolbachia*-infected *Aedes aegypti* on dengue incidence: systematic review and meta-analysis, *Transactions of the Royal*

Society of Tropical Medicine and Hygiene 2026, doi:[10.1093/trstmh/trag063](https://doi.org/10.1093/trstmh/trag063) (abstract-only); 40700688 — Ivermectin to control malaria: a cluster-randomized trial (BOHEMIA), New England Journal of Medicine 2025, doi:[10.1056/NEJMoa2411262](https://doi.org/10.1056/NEJMoa2411262) (abstract-only); 42348676 — The origin, history and resistance architecture of an invasive urban malaria mosquito in Africa, Science 2026, doi:[10.1126/science.adx6925](https://doi.org/10.1126/science.adx6925) (abstract-only). Programme name and current status page: NCVBDC, Ministry of Health and Family Welfare, portal updated August 2026.



Kyasanur Forest disease (KFD)

Asked in: Paper II 08-Apr-2026 — RGUHS MD Community Medicine

Definition — Kyasanur Forest Disease (KFD) is a febrile disease associated with haemorrhages, caused by an arbovirus of the flavivirus group and transmitted to man by the bite of infective hard ticks (Park 28e, Ch 5, p 354). It is a **zoonosis** — an anthroponosis, i.e. an infection transmitted to man from vertebrate animals (Park 28e, Ch 3, pp 113-114) — in which the natural cycle runs between forest ticks, small mammals and monkeys, and **man is an incidental, dead-end host who plays no part in virus transmission** (Park 28e, Ch 5, p 354).

Problem statement — history and distribution

Item	Fact (with source)
Recognition	First recognised in 1957 in Shimoga district, Karnataka , when a febrile illness appeared in villages after unusually heavy monkey mortality in the forest. Local inhabitants called it " monkey disease "; it was later named after the locality — Kyasanur Forest — from where the virus was first isolated (Park 28e, Ch 5, p 354; Mahajan & Gupta 4e, Ch 19, p 353)
Original focus	About 800 sq. km around the Sagar and Sorab taluks of Shimoga district (Park 28e, Ch 5, p 354; Suryakantha, Ch 21, p 471)
Present endemic area	Now over 6,000 sq. km across four Karnataka districts — Shimoga, North Kannada, South Kannada and Chikamagaluru (Park 28e, Ch 5, p 354); Sunder Lal adds Udupi , with Shimoga and Chikamagaluru worst affected (Sunder Lal, Ch 8, p 401)
Geographical extension	Now reported from adjoining Tamil Nadu, Maharashtra and Goa (Sunder Lal, Ch 8, p 401); has spread as far north as Sindhudurg district (Maharashtra) and as far south as Palakkad district (Kerala) (AFMC Red Book 2024, Ch 34, p 1067)
Silent seropositivity	Antibodies to KFD or a closely related virus found in humans and animals — particularly cattle in Kutch and Saurashtra (Gujarat) (Park 28e, Ch 5, p 354), and in parts of West Bengal and the Andaman & Nicobar Islands (AFMC Red Book 2024, Ch 34, p 1067); in 2002 about 22% of Andaman & Nicobar inhabitants were seropositive (Service, Ch 1, pp 255-256)
Annual burden	About 400–500 cases every year , case fatality rate about 3–5 per cent (Park 28e, Ch 5, p 354). Of 823 confirmed cases in 2003–2012 , 28 were fatal (Plotkin's Vaccines 7e, Ch 75, p 2114)
Cumulative burden	An estimated 9,594 cases in 16 districts of India between 1957 and 2017 (AFMC Red Book 2024, Ch 34, p 1067)

Item	Fact (with source)
Major epidemics	1957–58: 681 cases; 1983–84: 2,589 cases; 2002–03: 1,562 cases; 2016–17: 809 cases (AFMC Red Book 2024, Ch 34, p 1067). The 1983 Karnataka epidemic alone reported 2,167 cases (Mahajan & Gupta 4e, Ch 19, p 353) and followed large-scale felling of a virgin forest in September 1982 , which drove monkeys and their tick parasites into neighbouring forests (Mahajan & Gupta 4e, Ch 19, p 353)

Epidemiological determinants

Determinant	Detail
Agent	KFD virus — a group B togavirus (flavivirus) , an RNA virus 25–40 nm in diameter, antigenically related to the other tick-borne flaviviruses, particularly Far Eastern tick-borne encephalitis and Omsk haemorrhagic fever (Park 28e, Ch 5, p 354). Destroyed by heat, ethyl ether and sodium desoxycholate; evokes neutralising (N), haemagglutination-inhibiting (HI) and complement-fixing (CF) antibodies, of which N and HI persist about 2 years and N antibodies are the most specific (Suryakantha, Ch 21, p 471). A closely related virus, Alkhurma , occurs in Saudi Arabia (Service, Ch 1, pp 255-256)
Agent peculiarity	Unlike most other arbovirus infections, KFD has a prolonged viraemia in man of about 10 days or more (Park 28e, Ch 5, p 354) — yet this still does not make man infective to ticks in practice, and the viraemia is too brief to sustain a cycle (Suryakantha, Ch 21, p 472)
Reservoir	Small mammals — particularly rats, squirrels and shrews — are the main reservoirs ; birds and bats are less important hosts (Park 28e, Ch 5, p 354)
Amplifying host	Monkeys — the bonnet monkey <i>Macaca radiata</i> (white-faced) and the langur <i>Presbytis entellus</i> (black-faced), mortality being higher in langurs (Suryakantha, Ch 21, p 472). They amplify the virus but are not effective maintenance hosts because most of them die of the infection (Park 28e, Ch 5, p 354)
Cattle	Provide <i>Haemaphysalis</i> ticks with a plentiful source of blood meals, causing a population explosion of ticks . Cattle are therefore very important in maintaining tick populations but play no part in virus maintenance (Park 28e, Ch 5, p 354)
Man	An incidental or dead-end host who plays no part in virus transmission (Park 28e, Ch 5, p 354)
Vector	Hard ticks (family Ixodidae) of the genus <i>Haemaphysalis</i> . At least 15 species are known to transmit the disease, particularly <i>H. spinigera</i> and <i>H. turturis</i> (Park 28e, Ch 5, p 354); <i>H. spinigera</i> transmits the virus to humans while <i>H. turturis</i> maintains animal transmission (Service, Ch 1, pp 255-256). The virus has also been isolated from soft ticks (Park 28e, Ch 5, p 354)
Vector biology	Larvae feed on birds and small forest rodents; nymphs feed mainly on monkeys and humans ; adults feed on monkeys, cattle and other large mammals. Transstadial and probably transovarial transmission occur in the tick (Service, Ch 1, pp 255-256). Extrinsic incubation period 10–14 days ; the resting habitat is the lantana thicket at the forest–village interface (Suryakantha, Ch 21, p 472)
Age	Majority of cases 20–40 years (Park 28e, Ch 5, p 354)
Sex	Attack rate greater in males (Park 28e, Ch 5, p 354)

Determinant	Detail
Occupation	Mostly cultivators who enter the forest accompanying their cattle or cutting wood; also wood-cutters, hunters and cattle grazers (Park 28e, Ch 5, p 354; AFMC Red Book 2024, Ch 34, p 1067)
Season	Highest human and monkey infection in the drier months, January to June , coinciding with peak nymphal tick activity and with the period of greatest human activity in the forest, i.e. from January until the onset of the rains in June (Park 28e, Ch 5, p 354)
Environmental / man-made	Deforestation, free roaming of cattle for grazing, expansion of rice cultivation and hunting (Suryakantha, Ch 21, p 472). KFD is the classic demonstration of how changes in human behaviour — deforestation and agricultural development — alter ecology and produce outbreaks in the human population (Service, Ch 1, p 256)

Mode of transmission, incubation and case detection

- ▶ **Transmission is entirely by the bite of infective ticks, especially the nymphal stages** (Park 28e, Ch 5, p 354); monkey-to-monkey spread is by infective adult ticks and monkey-to-man by nymphs (Suryakantha, Ch 21, p 472).
- ▶ **There is no evidence of man-to-man transmission** (Park 28e, Ch 5, p 354). An infected person cannot start an urban cycle because the nymphs cannot survive the higher temperatures of human dwellings and live only in cool, shady forest — hence KFD is a **dead-end infection** (Suryakantha, Ch 21, p 472).
- ▶ **Incubation period 3 to 8 days** (Park 28e, Ch 5, p 354).
- ▶ **Clinical picture** — sudden onset of fever, headache and severe myalgia with prostration; the acute phase lasts about **2 weeks**, with gastrointestinal disturbance and haemorrhages from nose, gums, stomach and intestine in severe cases. In a number of cases a **second phase of mild meningoencephalitis follows an afebrile period of 7 to 21 days**, with return of fever, severe headache, neck stiffness, coarse tremors, abnormal reflexes and mental disturbance. **Case fatality 5 to 10 per cent** (Park 28e, Ch 5, p 354).
- ▶ **Diagnosis is established only after detecting the virus in blood and/or by serological evidence** (Park 28e, Ch 5, p 354) — serology using enzyme-linked immunosorbent assay (ELISA) and neutralising antigens, plus virus isolation (AFMC Red Book 2024, Ch 34, p 1067). Samples are referred to the **National Institute of Virology (NIV), Pune**, which is the reference laboratory for KFD virus (Mahajan & Gupta 4e, Ch 19, p 353).

Prevention and control

1. SURVEILLANCE (Park 28e, Ch 5, p 354; Mahajan & Gupta 4e, Ch 19, p 353)

- Karnataka Government system monitors KFD in humans AND monkey mortality in known epidemic and neighbouring areas
- Monkey deaths are 'heralders' of human disease
- Regular monitoring of all suspect cases and deaths

2. VECTOR (TICK) CONTROL (Park 28e, Ch 5, p 354)

- Carbaryl, fenthion, naled or propoxur at 2.24 kg active ingredient per hectare, by power or aircraft-mounted equipment
- Spray 'hot spots': within 50 metres around the spot of a monkey death, besides the endemic foci
- Acaricide dusting of forest tracks used by man: 5% DDT, 0.5% lindane, 0.25-0.48% bendiocarb, 5% carbaryl, 0.5% diazinon, 2% malathion (Mahajan & Gupta 4e, Ch 19, p 353)
- RESTRICT CATTLE MOVEMENT into forest to shrink tick population

3. PERSONAL PROTECTION AND HEALTH EDUCATION (Park 28e, Ch 5, p 354)

- Adequate protective clothing, impregnated if possible
- Repellents: dimethyl phthalate (DMP), diethyltoluamide (DEET); also permethrin, butopyronoxyl (Mahajan & Gupta 4e, Ch 19, p 353)
- Examine the body for ticks at the END OF EACH DAY and remove attached ticks promptly
- Discourage sitting or lying on the forest ground

4. VACCINATION OF THE POPULATION AT RISK (Park 28e, Ch 5, p 354)

- Formalin-inactivated, chick-embryo-fibroblast KFD vaccine
- Two doses one month apart, ages 7-65 years, in endemic taluks
- Booster at 6-9 months, then periodic boosters

Prevention and control of Kyasanur Forest Disease — the four-pronged Karnataka package

Levels of prevention

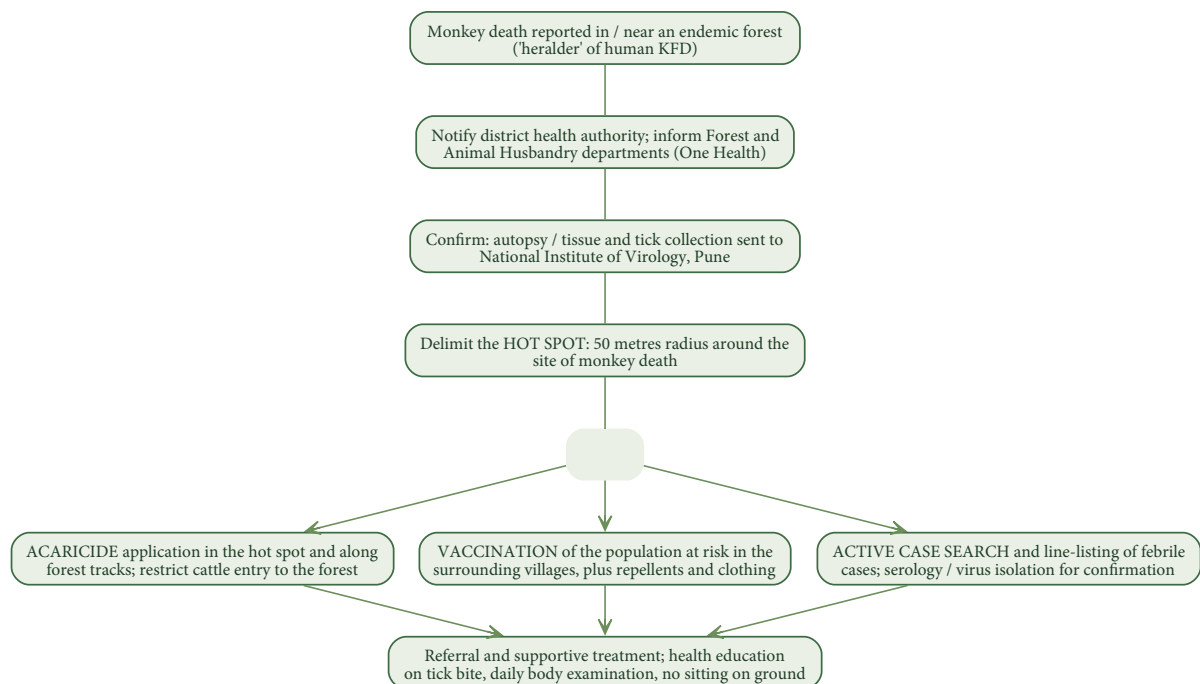
Level	Measures
Primordial / ecological	Prevent the man-made ecological disturbance that creates the disease: avoid indiscriminate deforestation and forest clearing and regulate the expansion of cultivation and grazing into forest — the lantana thicket that colonises the forest-village interface is the vector's resting habitat (Suryakantha, Ch 21, p 472; Service, Ch 1, p 256). NIV Pune recommends that before any forest clearing, forest ticks be examined for KFD virus and forest animals for serological evidence, to establish that the area is free of KFD (Mahajan & Gupta 4e, Ch 19, p 353)
Primary — health promotion	Health education of forest-going cultivators, wood-cutters, hunters and graziers on tick bite, daily body examination and prompt tick removal, and on not sitting or lying on the ground (Park 28e, Ch 5, p 354)

Level	Measures
Primary — specific protection	KFD vaccination of the population at risk; repellents (DMP, DEET) and protective clothing (Park 28e, Ch 5, p 354)
Primary — vector and reservoir attack	Acaricide application to hot spots and forest tracks; restriction of cattle grazing in the forest (Park 28e, Ch 5, p 354; Suryakantha, Ch 21, p 474)
Secondary	Early case detection through surveillance, serological/virological confirmation via NIV Pune, and prompt supportive hospital care — there is no specific antiviral treatment (Sunder Lal, Ch 8, p 401; AFMC Red Book 2024, Ch 34, p 1067)
Tertiary	Management of the second (meningoencephalitic) phase and of prolonged convalescence — recovery is almost complete in 90 per cent of cases but the patient remains asthenic for a long time (Suryakantha, Ch 21, p 472)

The KFD vaccine

Attribute	Detail
Type	Formalin-inactivated (killed), tissue-culture vaccine grown in chick embryo fibroblasts; it replaced the original formalin-inactivated mouse-brain vaccine in about 1990 (Plotkin's Vaccines 7e, Ch 59, p 1599)
Manufacture	Institute of Animal Health and Veterinary Biologicals, Hebbal, Bangalore (Plotkin's Vaccines 7e, Ch 75, pp 2114-2115); Virus Diagnostic Laboratory, Shimoga (Mahajan & Gupta 4e, Ch 19, p 353)
Schedule (Karnataka programme)	Two doses one month apart in persons aged 7–65 years, given annually since 1990 in Shimoga and adjacent districts, with a booster after 6–9 months (Plotkin's Vaccines 7e, Ch 75, pp 2114-2115) and subsequent boosters (Plotkin's Vaccines 7e, Ch 59, p 1599)
Dose and storage	1 mL for adults, 0.5 mL for children, intramuscular; stored at 2–8°C (Suryakantha, Ch 21, p 474)
Efficacy (1990–92 field trial, 72 affected villages)	Attack rate 0.870% (325/37,373) unvaccinated vs 0.15% (14/9,072) after one dose and 0.047% (10/21,083) after two doses → vaccine efficacy 82.4% (one dose) and 94.8% (two doses) (Plotkin's Vaccines 7e, Ch 75, pp 2114-2115)
Field effectiveness (2005–2010)	Much lower — 62.4% (95% CI 26.1–80.8) after two primary doses, rising to 82.9% (95% CI 71.3–89.8) with two doses plus a booster; vaccination in April–May 2011 did not protect in the Shimoga outbreak of December 2011–March 2012, and overall coverage has been low (Plotkin's Vaccines 7e, Ch 75, pp 2118-2119)

Surveillance and outbreak response



Outbreak response algorithm triggered by monkey deaths in a KFD-endemic area

- **Monkey mortality is the sentinel event** — the Karnataka Government surveillance system monitors both KFD in humans and mortality in monkeys in known epidemic areas *as well as neighbouring areas*, because **deaths of monkeys are considered heralders of the disease in endemic areas** (Park 28e, Ch 5, p 354).
- **Regular monitoring of all cases and deaths suspected to be due to KFD** is undertaken (Mahajan & Gupta 4e, Ch 19, p 353).
- **Insecticidal spraying is targeted at "hot spots"** — areas where monkey deaths have been reported, within **50 metres around the spot of the monkey death**, besides the endemic foci (Park 28e, Ch 5, p 354).
- **One Health coordination** is essential: KFD is one of the newly emerged zoonoses (Park 28e, Ch 5, p 341), and successful One Health action requires **inter-sectoral coordination between human, animal and ecosystem health** — here the health, forest, veterinary and revenue departments together (One Health Approach, Ch 1, p 2).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Natural cycle of KFD — labels to reproduce by hand: Eggs → *Unfed larvae* → (blood meal on small mammals — rats, squirrels, shrews; birds) → *Fed larvae* → *Unfed nymphs* → (blood meal on monkeys and MAN) → *Fed nymphs* → *Unfed adults* → (blood meal on cattle, monkeys, other large mammals) → *Fed adult tick* → Eggs. Mark on the drawing: "MAN = dead-end host" beside the nymphal arm; "Monkeys = amplifying host (they die)"; "Small mammals = true reservoir"; "Cattle = maintain tick population, not the virus"; and "transstadial and transovarial transmission" across the egg–larva–nymph–adult arrows.

High-yield exam pointers

- 1957, Shimoga, Karnataka — "monkey disease"; agent a group B togavirus (flavivirus) related to Far Eastern tick-borne encephalitis and Omsk haemorrhagic fever; vector *Haemaphysalis spinigera* and *H. turturis* (≥ 15 *Haemaphysalis* species implicated).
- Host hierarchy mnemonic — R-A-M-C: Reservoir = small mammals (rats, squirrels, shrews); Amplifier = monkeys (*Macaca radiata*, *Presbytis entellus*) which die; Man = dead-end host; Cattle = tick multipliers only.
- Incubation period 3–8 days; prolonged viraemia ~10 days (unusual for an arbovirus); biphasic illness with meningoencephalitis after an afebrile gap of 7–21 days; case fatality 3–5% (up to 5–10% in older accounts).
- Season January–June (peak nymphal activity + peak human forest activity); 400–500 cases a year; endemic in Shimoga, North Kannada, South Kannada, Chikamagaluru over >6,000 sq. km.

- ▶ **Control** = surveillance (monkey deaths) + acaricide in a 50-metre hot spot + restriction of cattle grazing + DMP/DEET and daily body examination + killed vaccine. Insecticide dose: carbaryl / fenthion / naled / propoxur at 2.24 kg active ingredient per hectare.
- ▶ **Vaccine**: formalin-inactivated chick embryo fibroblast vaccine, 2 doses 1 month apart, ages 7–65 years, booster at 6–9 months; trial efficacy 94.8% for two doses, but field effectiveness only 62.4% (two doses) rising to 82.9% with a booster.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference, not to be written in the exam: sources differ on two numbers — Park 28e gives the case fatality as 3–5% in its problem statement but 5–10% under clinical features (the AFMC Red Book 2024 supports 3–5%, Suryakantha supports 5–10%), and the vaccine booster interval is given as **annual** by Suryakantha but as **6–9 months then every 5 years** by Plotkin's Vaccines 7e; quote either with its source attached. KFD has no dedicated national health programme — it is managed by the Karnataka State Department of Health together with the Virus Diagnostic Laboratory, Shimoga and the National Institute of Virology, Pune.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, ICMR-NIV Pune, NCDC, DCGI, Operational Manual for KFD (Government of Karnataka, 2020)

Programme and policy updates (India)

- ▶ **The killed chick-embryo-fibroblast KFD vaccine described in the textbooks is no longer in use.** It was discontinued by 2022 for non-compliance with current regulatory standards and reduced public acceptance, and the Thirthahalli programme audit records no vaccine available for the 2022–23 season. Drift between the vaccine strain and circulating virus is one proposed explanation for its weak field performance: field-forward nanopore sequencing of 19 KFDV genomes from Shivamogga (2018–2020) found 28 non-synonymous variants, with more than ten envelope-protein changes in 80% of samples [PMID 42004966; PMID 39494093; PMID 40168937].
- ▶ **A next-generation whole-virion inactivated KFD vaccine is in the pipeline** — developed by Indian Immunologicals Ltd with ICMR-NIV Pune from a Vero-cell-adapted lineage-2 strain (NIV-164187, isolated in the 2016 Belgaum outbreak), inactivated with β -propiolactone rather than formalin and adjuvanted with aluminium hydroxide. Two 18 μ g doses gave complete protection against a 100 LD₅₀ challenge in mice with significantly lower brain viral load, immune serum passively protected naive mice, and acute plus repeat-dose toxicity studies were clean across mice, rats, hamsters and rabbits. The candidate has DCGI approval to advance to a Phase 1 human trial — it is not yet a licensed vaccine [PMID 42004966].
- ▶ **A programme audit of the Karnataka KFD package** in Thirthahalli taluk (2021–22 season) rated vaccination, monkey necropsy and tick control *unsatisfactory* against the state's own operational manual: 21,586 doses (40%) for an eligible population of 53,264, geolocation plus necropsy for only 2 of 11 monkey deaths, a 2–3 month delay between case reporting and acaricide procurement, and no vaccine at all for the 2022–23 season — while surveillance, diagnosis, treatment and IEC were satisfactory [PMID 39494093].

Current national data

- **Shivamogga 2022 (NCDC field investigation, 35 laboratory-confirmed cases, one death, CFR 3%, median age 46 years):** residence within 500 m of a monkey-death site (aOR 8.6, 95% CI 1.8–41.9) and household tick exposure — tick sightings plus stored dry leaves or firewood (aOR 3.7, 95% CI 1.3–10.7) — were the independent risk factors, whereas **two vaccine doses with or without a booster conferred no measurable protection (aOR 1.5, 95% CI 0.5–4.6)**. Seven cases (28%) reported no forest or plantation activity but handled dry leaves, firewood and cattle at home; the authors hypothesise household transport of infected ticks, though these exposures were not statistically significant in their own analysis (household handling of forest products OR 0.7, 95% CI 0.2–1.8; cattle in the house OR 1.2, 0.4–3.4, both excluded from the multivariate model). No case knew that monkey-death sites should be avoided [PMID 39494093].
- **Goa 2023 (Caranzol village, Sattari, North Goa):** village-wide search of 484 residents found 13 presumptive and 5 confirmed cases, attack rate 3.1%, highest in women (4.6%) and at 45–59 years (5.1%). Forest entry without protective clothing, non-use of repellent and poor post-exposure hygiene were significantly associated with illness; six cases had been vaccinated five years earlier; tick pools were negative and no monkey deaths were recorded — an outbreak without the classic sentinel event [PMID 41629231].

Elimination, surveillance and disease status

- **Transovarial transmission has now been demonstrated in the wild, not merely inferred from models.** Questing (unfed, blood-meal-negative) larvae from 7 of 49 sites in Shivamogga and Wayanad were KFDV-positive, in *Rhipicephalus* as well as *Haemaphysalis* species. Crucially, positive larvae came from around houses and gardens, harvested paddy and areca plantation as well as forest — suggesting ticks themselves may act as a reservoir, and that the standing advice that risk is confined to forest and to the nymphal stage needs revision [PMID 39825388].
- **Vector range extends far beyond the reported case map.** A five-state forest-fringe survey (44 villages across Goa, Maharashtra, Karnataka, Kerala and Tamil Nadu, sampled December–February at peak nymphal activity) collected 10,350 ticks — 24 species across four genera, including 12 recognised KFD vectors. *H. spinigera* and *H. turturis* were the most abundant and widespread species in historically unaffected districts as well as endemic ones, with no significant difference in tick abundance or diversity between the two, arguing for anticipatory ecological surveillance and landscape-based risk forecasting rather than surveillance keyed only to reported monkey deaths and human cases [PMID 41699064].
- **Point-of-care confirmation at the district level has been field-validated.** ICMR-NIV Pune evaluated a portable real-time RT-PCR (Truenat) assay against the NS5 gene on 247 stored 2024-season sera at the Virus Diagnostic Laboratory, Shivamogga and North Goa district hospital, benchmarked against laboratory real-time RT-PCR as the gold standard: sensitivity 97.8% (95% CI 88.4–99.6) and 100% (78.5–100), specificity 98.1% (93.3–99.5) and 95.2% (88.3–98.1), with turnaround falling from about 24 hours to about 2.5 hours. Throughput of four samples per run is the stated bottleneck in an outbreak. This was a retrospective evaluation and the authors conclude only that the assay could be incorporated into existing KFD surveillance — NIV Pune remains the reference laboratory [PMID 42295717].

Guideline and legislative updates

- **Consensus reviews now treat the existing vaccine as inadequate rather than merely imperfect** — field protection of only about 62%, no approved antiviral, and no agreed serological or cellular correlate of protection; the proposed roadmap is better preclinical models that reproduce human disease, defined correlates of protection, and standardised immunogenicity and neutralisation assays so that candidate vaccines can be benchmarked for regulatory evaluation. Monkey mortality is retained as a probable early indicator of forthcoming human outbreaks [PMID 41202828; PMID 42314066].

Global and WHO developments

- **KFD control has been mapped explicitly onto the Sustainable Development Goals** — the first review to align KFD research with SDG 3 (health), SDG 9 (innovation, i.e. rapid diagnostic kits and a mobile app for data collection enabling real-time outbreak monitoring), SDG 13 (climate action), SDG 15 (life on land) and SDG 17 (partnerships), and it calls for cost-effective diagnostics and new vaccines alongside better characterisation of the biphasic illness (about 20% of cases) [PMID 41297079].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 5, Epidemiology of Communicable Diseases, p 354; Ch 3, pp 113-114; Ch 5, p 341); Service — Medical Entomology for Students (Ch 1, pp 252-256); Suryakantha — Community Medicine with Recent Advances (Ch 21, pp 471-474); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e (Ch 19, Arthropod-borne Diseases, p 353); Sunder Lal — Textbook of Community Medicine (Ch 8, p 401); Plotkin's Vaccines, 7ed (Ch 59, p 1599; Ch 75, pp 2114-2119); AFMC Red Book of Community Medicine 2024 (Ch 34, p 1067); One Health Approach (Ch 1, p 2).

Recent-advances sources: 42004966 — Development of whole virion inactivated Kyasanur Forest Disease vaccine candidate and its preclinical safety and efficacy evaluation, *Frontiers in Immunology* 2026, doi:[10.3389/fimmu.2026.1786057](https://doi.org/10.3389/fimmu.2026.1786057); 42295717 — Field evaluation of Kyasanur forest disease real-time PCR point-of-care assay, *Indian Journal of Medical Research* 2026, doi:[10.25259/IJMR_3018_2025](https://doi.org/10.25259/IJMR_3018_2025); 39494093 — Kyasanur Forest Disease: an epidemiological investigation and case-control study in Shivamogga, Karnataka, India — 2022, *International Journal of Public Health* 2024, doi:[10.3389/ijph.2024.1606715](https://doi.org/10.3389/ijph.2024.1606715); 39825388 — First evidence of transovarial transmission of Kyasanur Forest disease virus in *Haemaphysalis* and *Rhipicephalus* ticks in the wild, *Parasites & Vectors* 2025, doi:[10.1186/s13071-024-06643-5](https://doi.org/10.1186/s13071-024-06643-5); 41699064 — Ixodid tick diversity and distribution across forest-fringe landscapes of the Western Ghats, India, with emphasis on Kyasanur Forest Disease vectors, *Scientific Reports* 2026, doi:[10.1038/s41598-026-38517-4](https://doi.org/10.1038/s41598-026-38517-4); 41629231 — Outbreak investigation of Kyasanur Forest Disease in North Goa district, Goa, India 2023, *Indian Journal of Public Health* 2026, doi:[10.4103/ijph.ijph_661_25](https://doi.org/10.4103/ijph.ijph_661_25) (abstract-only); 40168937 — Molecular epidemiology of Kyasanur forest disease employing ONT-NGS, a field-forward sequencing approach, *Journal of Clinical Virology* 2025, doi:[10.1016/j.jcv.2025.105783](https://doi.org/10.1016/j.jcv.2025.105783) (abstract-only); 41202828 — Kyasanur Forest disease: an emerging arboviral threat, *The Lancet Infectious Diseases* 2025, doi:[10.1016/S1473-3099\(25\)00589-4](https://doi.org/10.1016/S1473-3099(25)00589-4) (abstract-only); 42314066 — Kyasanur Forest Disease: advances in experimental animal models, immune assays, and challenges in vaccine development, *Reviews in Medical Virology* 2026, doi:[10.1002/rmv.70177](https://doi.org/10.1002/rmv.70177) (abstract-only); 41297079 — Integrated approach for sustainable management and control of Kyasanur forest disease in alignment with sustainable development goals, *Virology* 2025, doi:[10.1016/j.virol.2025.110749](https://doi.org/10.1016/j.virol.2025.110749) (abstract-only). Programme-implementation benchmarks are those of the Operational Manual for KFD, Department of Health and Family Welfare, Government of Karnataka, 2020, as audited in the NCDC field investigation cited above.



Chemical control of mosquito larvae

Asked in: Paper I 07-May-2022 — RGUHS MD Community Medicine

Definition — Chemical control of mosquito larvae (**larviciding**) is the planned application of chemical agents to aquatic breeding habitats so as to kill the immature aquatic stages before they emerge as adults. Insecticides are classified by their action on the life cycle of mosquitoes into **larvicides** and **imagicides** (Gilles & Warrell 4e, Ch 6, p 146). In Park's scheme it is the second of the three anti-larval measures — environmental control, chemical control and biological control (Park 28e, Ch 15, p 926).

Classification of chemical larvicides

Group	Agents	Mode of action	Chiefly effective against
Surface films — mineral oils	Diesel oil, fuel oil, kerosene, crude-oil fractions, Mosquito Larvicidal Oil (MLO), <i>malariol</i>	Physical: spreads as a thin film, cuts off the air supply to larvae and pupae; also a direct toxic action on these stages (Park 28e, Ch 15, p 926)	All pre-adult stages (larvae and pupae) of surface-breathing mosquitoes
Surface films — monomolecular films	Ethoxylated isostearyl alcohol from plant oils (<i>Arosurf</i> , <i>Agnique</i>)	Alter the air–water interface so that larvae, pupae, emerging and ovipositing adults drown (Service, Ch 1, p 38)	Larvae, pupae and adults at the water surface
Stomach poison	Paris green (copper acetoarsenite), containing over 50 per cent arsenious oxide (Park 28e, Ch 15, p 939)	Must be ingested ; arsenical poisoning of the gut	Mainly Anopheles larvae, because they are surface feeders; culicine bottom feeders only with a granular formulation (Park 28e, Ch 15, p 926)
Contact organophosphates	Temephos (Abate), fenthion (Baytex), chlorpyrifos, malathion, pirimiphos-methyl	Inhibit cholinesterase, so acetylcholine accumulates at the synapse (Park 28e, Ch 15, p 939)	Larvae of all genera; the currently recommended larvicides (Service, Ch 1, p 39)
Pyrethrum-based	Pyrosene oil-E, emulsifiable concentrate containing 0.1–0.2 per cent pyrethrins by weight, diluted 1:4 with water before use (Park 28e, Ch 5, p 339)	Contact nerve poison	<i>Culex quinquefasciatus</i> in urban anti-larval work in India
Insect growth regulators (IGRs)	Methoprene, pyriproxyfen, fenoxycarb — arrest larval development; diflubenzuron, novaluron — inhibit chitin formation in immature stages (Service, Ch 1, p 40)	Interfere with moulting and metamorphosis	Larvae only; adults already formed are unaffected

Group	Agents	Mode of action	Chiefly effective against
Not recommended	DDT and other organochlorines	—	No longer recommended as larvicides, because of environmental persistence and accumulation in food chains (Service, Ch 1, p 39)

Dosage, formulation and frequency

Larvicide	Dosage	Duration / frequency
Mineral oils	40–90 litres per hectare (Park 28e, Ch 15, p 926); 9–27 L/ha for modern high-spreading formulated oils (Service, Ch 1, p 38); 142–190 L in open ponds to cover the water surface (Gilles & Warrell 4e, Ch 6, p 155)	Once weekly; 1–2 weeks of effective action
Paris green	2 per cent dust = 2 kg Paris green + 98 kg inert diluent (soapstone powder or slaked lime), particle size 20–25 microns (Park 28e, Ch 15, p 926)	Once a week over breeding places (Park 28e, Ch 15, p 939)
Temephos (Abate)	56–112 g/ha (Park 28e, Ch 15, p 927); not more than 1.0 ppm in wells and domestic water containers (Park 28e, Ch 15, p 938); 1 per cent sand granules give 1 mg active ingredient per litre in potable water (Service, Ch 1, p 39)	Weekly to fortnightly
Fenthion	22–112 g/ha (Park 28e, Ch 15, p 927); 2 per cent granules; 1 ppm highly effective against <i>C. fatigans</i> larvae (Park 28e, Ch 15, p 938)	Weekly
Chlorpyrifos	11–16 g/ha (Park 28e, Ch 15, p 927)	Weekly; preferred in organically polluted water (Service, Ch 1, p 39)
Malathion	224–672 g/ha (Park 28e, Ch 15, p 927)	Weekly
Methoprene (IGR)	Briquettes of 8 per cent methoprene (<i>Altosid</i>) on small water bodies (Gilles & Warrell 4e, Ch 6, p 155)	30 days; some IGR briquettes give over 100 days (Service, Ch 1, p 40)

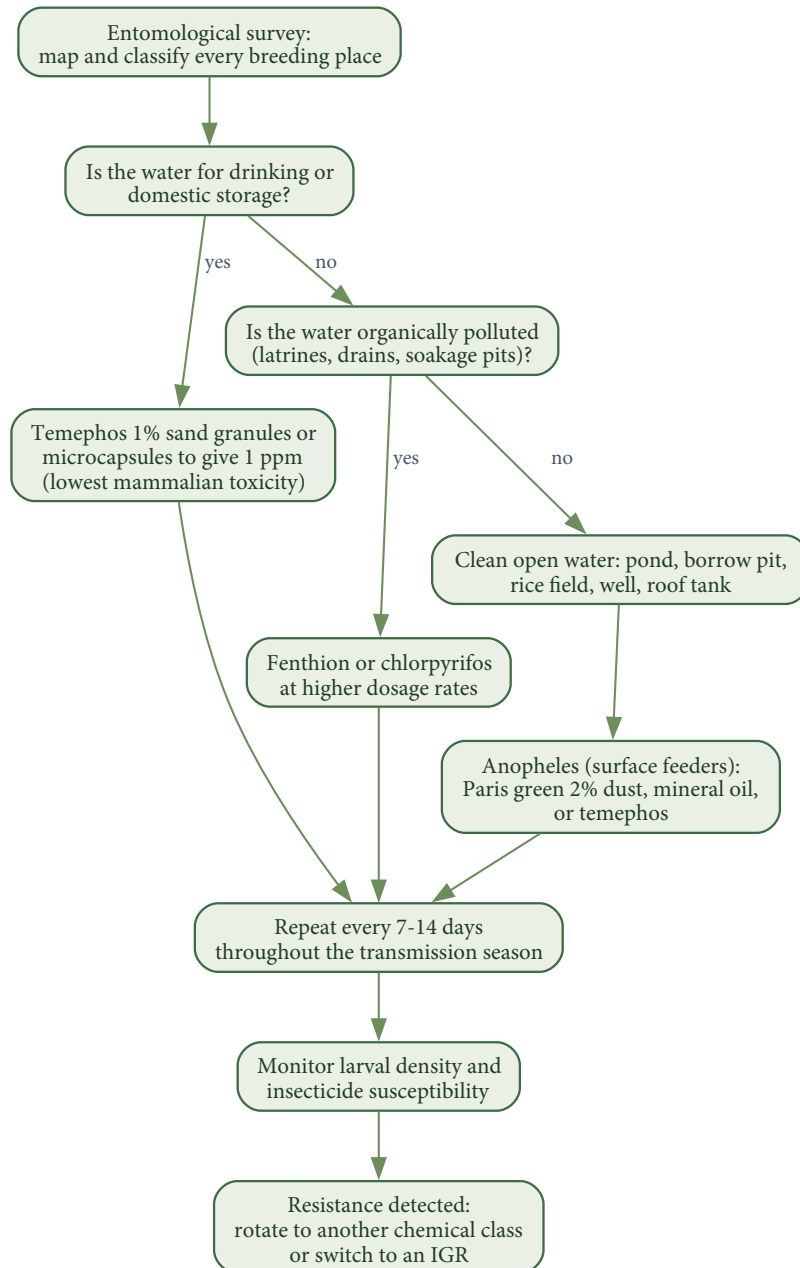
- **Why weekly** — the life cycle of a mosquito occupies about 8 days, so oiling and dusting are repeated once a week on all breeding places (Park 28e, Ch 15, p 926). Oils are re-sprayed about every 7–10 days and the other larvicidal insecticides every 10–14 days in the tropics, more frequently on highly polluted water (Service, Ch 1, p 39); the interval must match the development cycle of the vector, generally 7–14 days (Gilles & Warrell 4e, Ch 6, p 156).
- **Formulations** — emulsions and oil solutions for open water; granules, pellets and gelatine capsules to penetrate dense aquatic vegetation; slow-release sand granules, microcapsules and briquettes for containers (Service, Ch 1, p 39).
- **Pre-flood and post-flood granules** — slow-release granules scattered on dry or marshy ground stay inactive until flooding, then kill larvae as they hatch; this overcomes the installment hatching of drought-resistant *Aedes* eggs (Service, Ch 1, p 39).

- ▶ **Equipment** — hand blowers and rotary blowers for dusts, knapsack sprayers for liquids, vehicle-mounted machines for larger areas, and fixed-wing aircraft or helicopters for large or inaccessible breeding areas (Park 28e, Ch 15, p 926; Service, Ch 1, pp 39–40).

When larviciding is the appropriate choice

Antilarval measures are of greatest value alongside environmental management, and the use of larvicides should be governed by technical, operational and economic considerations (Gilles & Warrell 4e, Ch 6, p 132):

- ▶ **Technical** — very brief seasonal activity of the vector; **exophily** (outdoor resting behaviour) that makes indoor residual spraying ineffective; cultural or other objections to house spraying; and where the same operation also controls the vector of a second endemic vector-borne disease (comprehensive vector control).
- ▶ **Operational** — where the breeding places are **few, fixed and findable**, particularly in urban areas.
- ▶ **Economic** — where two vector-borne diseases co-exist and one operation covers both; operational convenience is usually also the more economical option.
- ▶ **Choice of compound** turns on effective toxicity to the target (allowing for resistance), duration of action at the given dosage, ease of application in local topography and climate, and toxicity to humans, domestic animals and non-target aquatic life (Gilles & Warrell 4e, Ch 6, p 146).



Habitat-based choice of chemical larvicide after an entomological survey of breeding places

Chemical larviciding in Indian programme practice

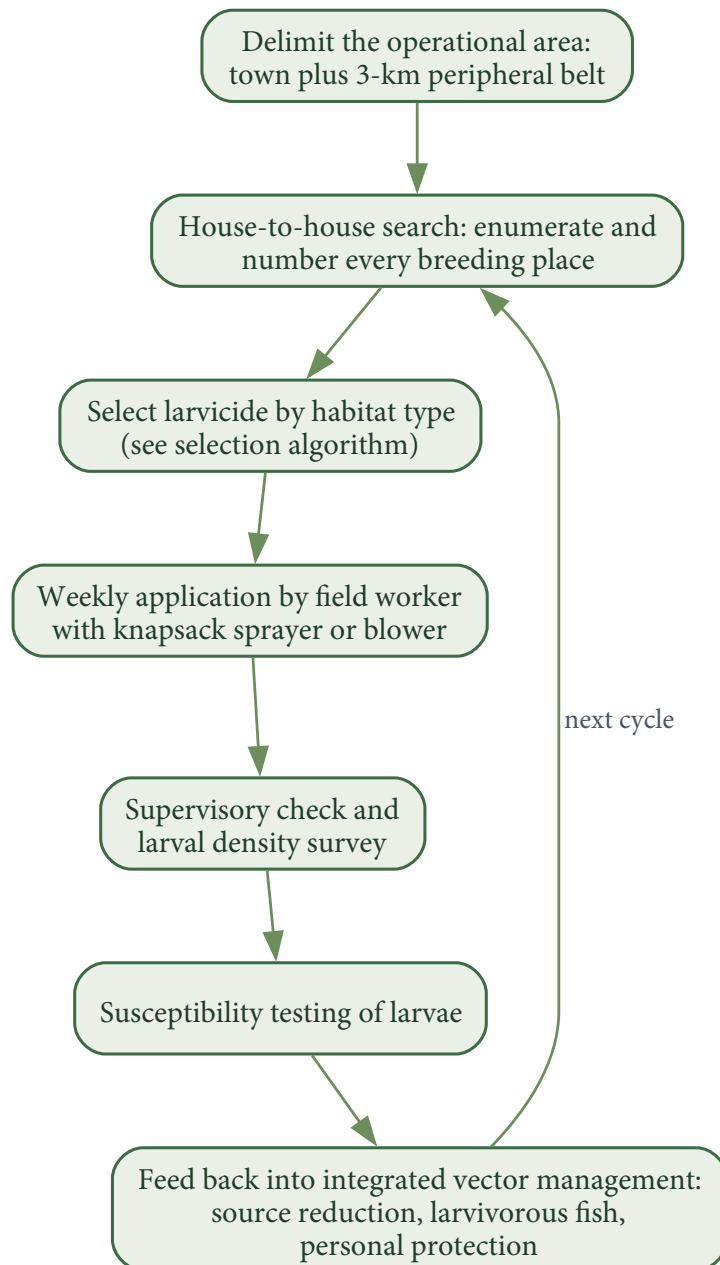
- Under the **National Vector Borne Disease Control Programme (NVBDCP)** — whose directorate is now the **National Centre for Vector Borne Disease Control (NCVBDC)**, the body that issued the *National Strategic Plan: Malaria Elimination 2023–2027* (Park 28e, Ch 5, p 334) — integrated vector management uses indoor residual spray as the primary method in **rural** areas and **anti-larval measures** in **urban** areas, together with larvivorous fish, source reduction and minor environmental engineering (Park 28e, Ch 7, pp 493, 498). The central government supplies DDT and larvicides; states procure spray insecticides and equipment (Park 28e, Ch 7, p 494).
- The **Urban Malaria Scheme**, launched in **1971**, works by intensive anti-larval measures plus drug treatment, because the urban vector *Anopheles stephensi* breeds in man-made containers —

overhead and underground tanks, cisterns, water coolers, roof gutters, flower vases, ornamental ponds and old tyres (Park 28e, Ch 7, p 494).

- ▶ Anti-larval work is deliberately **restricted to urban areas plus a 3-km peripheral belt**, because the flight range of *Culex quinquefasciatus* is about 3 km and covering rural India is operationally and financially impossible (Park 28e, Ch 5, p 339).
- ▶ For *C. quinquefasciatus*, **mosquito larvicidal oil** was the mainstay but is being replaced by pyrethrum oil, temephos and fenthion; organophosphorus larvicides are applied **once weekly on all breeding places** (Park 28e, Ch 5, p 339).
- ▶ **Larval source management** is a named intervention of the *National Framework for Malaria Elimination in India 2016–2030* (Park 28e, Ch 7, p 496), which was launched in **February 2016** (Park 28e, Ch 7, p 495).
- ▶ **Safe use of insecticides and monitoring of insecticide resistance** are formal components of integrated vector management (Park 28e, Ch 7, p 498).

Limitations and precautions

Problem	Detail
Recurrent and costly	Larviciding must be repeated at frequent intervals, which makes it a comparatively costly operation (Park 28e, Ch 5, p 333)
Insecticide resistance	The vector has developed resistance to many organophosphorus larvicides (Park 28e, Ch 5, p 339); there is cross-resistance between most carbamates and most organophosphorus compounds (Park 28e, Ch 15, p 939)
Non-target toxicity	Oils kill fish and injure vegetation; pyrethroid larvicides kill aquatic insects, crustaceans and fish, so are used only in special circumstances; chlorpyrifos causes high fish mortality (Park 28e, Ch 15, p 939; Service, Ch 1, p 39)
Water made unpotable	Oil renders water unfit for drinking (Park 28e, Ch 15, p 926)
Community refusal	Householders may refuse temephos in water pots because of odour or objection to any insecticide in drinking water (Service, Ch 1, p 39)
Stage and species gaps	Paris green does not act on pupae (a non-feeding resting stage) or on culicine bottom feeders; <i>Mansonia</i> larvae take oxygen from the roots of aquatic plants and never surface, so oiling fails and the <i>Pistia</i> plant must be removed or destroyed with herbicide (Suryakantha, Ch 15, p 142; Park 28e, Ch 15, p 925)
Reduced efficacy	Organophosphorus compounds hydrolyse quickly in water (Park 28e, Ch 15, p 927) and are less effective in organically polluted water (Service, Ch 1, p 39); wind breaks the oil film (Suryakantha, Ch 15, p 142)



Operational cycle of an urban anti-larval round under the Urban Malaria Scheme

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Life cycle of the mosquito with the larvicidal window marked** — label: *Egg (1–2 days) → Larva, four instars (5–7 days) → Pupa (1–2 days) → Adult* (Park 28e, Ch 15, p 923); bracket the larval stage and write "larvicides act here"; write against the whole aquatic phase "**life cycle about 8 days — hence weekly application**" (Park 28e, Ch 15, p 926), and beside the adult "**egg to adult 7–10 days under favourable conditions**" (Park 28e, Ch 15, p 924).
- **Anopheles versus Culex larva at the water surface** — label: *Anopheles* larva lying horizontal, flush with the surface, no siphon, **surface feeder** → killed by Paris green; *Culex* larva hanging at an angle by a long respiratory siphon, **bottom feeder** → not killed by Paris green dust; *Mansonia* larva with siphon **pierced into the rootlet of the Pistia plant** → not reached by surface films.

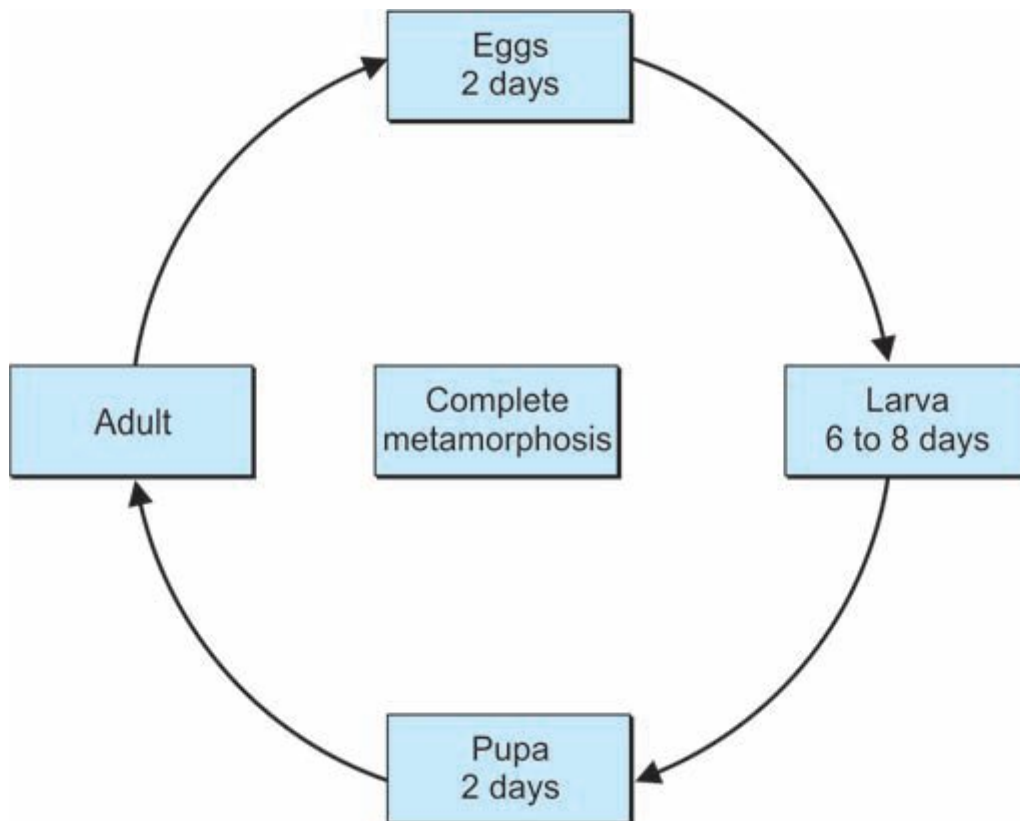


Fig 15.11 — Life-history of mosquito (Suryakantha Environment Best, p.137)

High-yield exam pointers

- **Park's three chemical larvicides:** mineral oils, Paris green, synthetic insecticides — write them in this order.
- **Paris green** = copper acetoarsenite, **over 50 per cent arsenious oxide**, a **stomach poison**, applied as **2 per cent dust** (2 kg + 98 kg diluent), particles **20–25 microns**; historically responsible for the eradication of *A. gambiae* from **Brazil in 1940** and its near-eradication in **Egypt in 1944** (Park 28e, Ch 15, p 939).
- **Temephos (Abate)** is the **least toxic of the organophosphorus larvicides** — at 1 ppm it is a most effective larvicide and also the least toxic (Park 28e, Ch 15, p 927); rat oral LD₅₀ **8600 mg/kg**

versus fenthion 250 and chlorpyrifos 135 (Gilles & Warrell 4e, Ch 6, p 147); dose 56–112 g/ha, or 1 ppm in potable water.

- ▶ Oil dose 40–90 L/ha, applied weekly; 142–190 L in open ponds; life cycle about 8 days; general repeat interval 7–14 days.
- ▶ Pyrosene oil-E = pyrethrum emulsifiable concentrate, 0.1–0.2 per cent pyrethrins, diluted 1:4.
- ▶ Urban Malaria Scheme 1971; anti-larval work confined to the town plus a 3-km belt (flight range of *Culex quinquefasciatus*).
- ▶ Fenthion at 1 ppm was found highly effective against *C. fatigans* larvae by the WHO Filariasis Research Unit, Rangoon — the classic filaria-vector larviciding figure (Park 28e, Ch 15, p 938).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: *Bacillus thuringiensis* var. *israelensis* (Bti) and *Bacillus sphaericus*, although often supplied alongside larvicides, are microbial agents and belong under **biological** control rather than chemical control, so they were not elaborated here; DDT and other organochlorines should not now be used as larvicides, although DDT remains the insecticide of choice for indoor residual spray against adult mosquitoes in India (Park 28e, Ch 7, p 498).

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, NCVBDC (MoHFW), WHO

Programme and policy updates (India)

- ▶ Temephos is now failing at the dose the programme actually applies, not at the test dose. F1 *Anopheles stephensi* larvae from Thiruvananthapuram and Adyar in Chennai were fully susceptible to the WHO discriminatory dosage of 0.25 ppm, yet both strains were resistant to the 1 ppm operational field dosage of temephos 50 per cent EC; complete kill needed 40 ppm in Thiruvananthapuram (LC99 37.47–45.06 ppm) and 50 ppm still gave only 98.7 per cent mortality in Adyar (LC99 59.79 ppm). The operational implication for urban anti-larval work is that discriminatory-dose susceptibility testing alone is no longer sufficient — routine larvicide efficacy monitoring at field dosage is needed [PMID 40829581].
- ▶ Adding larval control to filaria MDA did not pay for itself in Tamil Nadu. In a three-arm cluster-randomised trial across 36 villages, expanded polystyrene beads floated on wells and cesspits against *Culex quinquefasciatus* cost an extra US\$491 per 1,000 population, and beads plus pyrethroid-impregnated curtains an extra US\$1,294 per 1,000 population, over MDA alone (2010 US\$); neither arm produced a statistically significant additional benefit on microfilaria or antigen prevalence from 2010 to 2013, because the well-functioning MDA programme had already reduced transmission so far that little scope for improvement remained. Supplementary vector control is therefore best reserved for blocks where MDA is under-performing rather than added routinely [PMID 39630851].

Current national data

- ▶ National figures (NCVBDC, provisional to 28 February 2026). Malaria: 176,522 cases with 83 deaths (2022), 227,564 cases with 83 deaths (2023) and 255,500 cases with 86 deaths (2024); the annual parasite incidence stood at 0.18 per 1,000 in 2024, against 3.29 per 1,000 in 1995. Dengue:

289,235 cases with 485 deaths (2023), 233,519 cases with 297 deaths (2024) and 121,824 cases with 131 deaths (2025). Quote any of these only with the year attached [NCVBDC 2026].

- **Pan-India dengue serotype surveillance, 2023–2025.** Passive molecular surveillance through 45 ICMR Virus Research and Diagnostic Laboratories across 25 states and union territories tested 6,889 dengue-positive sera: every region showed co-circulation of multiple serotypes, eight states carried all four, DENV-2 predominated, and concurrent multi-serotype infection was associated with more thrombocytopenia, arthralgia and haemorrhagic manifestations than mono-serotypic infection — the evidence base being used to guide dengue vaccine introduction strategy [PMID 42323159].

Elimination, surveillance and disease status

- **The urban container-breeder the Urban Malaria Scheme was designed around has gone global, and a minority of habitats does most of the breeding.** *An. stephensi*, which vectors an estimated 12–15 per cent of India's urban malaria, has invaded Djibouti, Ethiopia, Sudan, Somaliland, Kenya, Nigeria and Ghana, and WHO now reports it across eight African countries with resistance to several public-health insecticides [WHO 2025]. In Kebri Dehar, Ethiopia, 23 per cent of container habitats yielded 77 per cent of all larvae — overwhelmingly ground-level cement cisterns holding water year-round — and a two-patch metapopulation model predicted local vector elimination when only these "superproductive" stable habitats were treated at coverage above 60 per cent, provided a long-residual larvicide such as pyriproxyfen or spinosad (up to six months) was used rather than Bti with its roughly two-week residual. This is the modern, quantified version of Park's "few, fixed and findable" rule [PMID 41126240].
- **Pyrethroid susceptibility is still holding in the dengue vector but is fraying locally.** Across nine localities in Jaipur, diagnostic doses of 0.005 µg/mL deltamethrin, 0.01 µg/mL permethrin and 0.05 µg/mL lambda-cyhalothrin each produced 100 per cent mortality in *Aedes aegypti*, with deltamethrin the most potent; three localities (Sanganer, Shastri Nagar, Shyam Nagar) nonetheless returned resistance ratios of about 3.7–4.5, which is the early-warning signal that triggers a class rotation [PMID 41307257].

Guideline and legislative updates

- **Larviciding still carries only a conditional WHO recommendation.** A full WHO recommendation for large-scale deployment is held by insecticide-treated nets and indoor residual spraying with prequalified products alone; larviciding holds a conditional positive recommendation resting on low-certainty evidence, supported by a Cochrane review that found just four eligible studies of which only one was a cluster-randomised trial. The guideline's "few, fixed and findable" feasibility criterion has confined larviciding largely to urban and arid settings. Although the Global Fund may finance conditionally recommended interventions, budget is rarely offered for larval source management at the end of the grant process, and only 14 of the 35 countries in a recent global review of national malaria programmes were implementing it, many at small scale [PMID 39915825].
- **Non-target toxicity now has a quantified evidence base that applies to programme design.** A systematic review of 95 articles covering 282 case studies confirmed that Bti at recommended rates kills 90–100 per cent of target larvae with far lower non-target mortality than chemical insecticides, but meta-analysis showed consistent reductions in the abundance of Chironomidae

and Crustacea and in chironomid emergence. The authors advise that ecological harm should not be assumed absent, and that a monitoring and assessment programme should accompany sustained larviciding — a caution that applies with greater force to the oils, organophosphates and pyrethroids in the table above [PMID 39294726].

Global and WHO developments

- **Global malaria burden and elimination status.** WHO estimated 282 million malaria cases in 2024, about 3 per cent more than in 2023, and 610,000 deaths (up from 598,000); the African Region carried about 95 per cent of both. **Fourteen** countries have been certified malaria-free **since 2017** — Uzbekistan and Paraguay (2018), Argentina and Algeria (2019), China and El Salvador (2021), Azerbaijan, Tajikistan and Belize (2023), Cabo Verde and Egypt (2024), Georgia, Suriname and Timor-Leste (2025); the figure becomes seventeen only if counted from 2015, which adds Maldives, Sri Lanka and Kyrgyzstan [WHO 2025]. The Global Technical Strategy for Malaria 2016–2030 target remains a 90 per cent reduction in incidence and mortality by 2030 [PMID 41637317].
- **Larviciding is being retested at national scale, and its delivery is being re-engineered.** Tanzania's national programme endorsed countrywide larviciding in its National Strategy for Vector Control 2019–2024 and National Malaria Strategic Plan 2021–2025; the Tanga Region pilot (June 2022 to April 2024, six rounds of eight weeks each, Bti for the first six weeks of a round then *Bacillus sphaericus* for the last two) cut **late-stage anopheline larvae across all three malaria risk strata**, the effect being largest in the high-risk stratum (IRR 0.20, 95% CI 0.12–0.35), but reduced **adult *An. gambiae* s.l. only in the moderate- and low-risk strata** (IRR 0.09 and 0.29; no effect in the high-risk stratum), with no significant association for *An. funestus* in any stratum; densities rebounded rapidly whenever application was paused for the heavy rains [PMID 41309908]. A systematic review of 54 African studies found larval density fell 50–100 per cent, yet only 14 studies measured clinical impact, where malaria incidence fell 31–62 per cent [PMID 40662945]. In Ghana, replacing manual scouting with drone mapping plus AI prioritisation of high-risk water bodies detected on average 3.6 times more potential breeding sites, roughly halved field labour (median person-days per km² 3.92 versus 7.23; $p = 0.029$) and treated about 2.8 times more sites per pack of larvicide, with malaria outcomes no worse despite the reduced larvicide and labour inputs [PMID 41637317].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 5, Ch 7, Ch 15); Service — Medical Entomology for Students (Ch 1); Gilles & Warrell — Essential Malariology 4e (Ch 6); Suryakantha — Community Medicine with Recent Advances (Ch 15, Medical Entomology). Programme content from the National Vector Borne Disease Control Programme and the Urban Malaria Scheme (1971) as described in Park 28e Ch 7, the National Framework for Malaria Elimination in India 2016–2030, and the National Strategic Plan: Malaria Elimination 2023–2027 (National Centre for Vector Borne Disease Control).

Recent-advances sources: 40829581 — Emerging Resistance in *Anopheles stephensi* Larvae to the Larvicide Temephos in Chennai, India, *Am J Trop Med Hyg* 2025, doi:[10.4269/ajtmh.25-0232](https://doi.org/10.4269/ajtmh.25-0232) (abstract-only); 39630851 — Cost-effectiveness of vector control for supplementing mass drug administration for eliminating lymphatic filariasis in India, *PLoS Negl Trop Dis* 2024, doi:[10.1371/journal.pntd.0011835](https://doi.org/10.1371/journal.pntd.0011835); 42323159 — Pan-India molecular surveillance of dengue serotypes, 2023–2025, *Int J Infect Dis* 2026, doi:[10.1016/j.ijid.2026.108897](https://doi.org/10.1016/j.ijid.2026.108897) (abstract-only); 41126240 — *Anopheles stephensi* larval habitat superproductivity and its relevance for larval source management in Ethiopia, *Malar J* 2025, doi:[10.1186/s12936-025-05589-y](https://doi.org/10.1186/s12936-025-05589-y); 41307257 — Evaluation of diagnostic doses and susceptibility status against pyrethroids in *Aedes aegypti* from Jaipur, Rajasthan, *J Vector Borne Dis* 2025, doi:[10.4103/jvbd.jvbd_145_25](https://doi.org/10.4103/jvbd.jvbd_145_25) (abstract-only); 39915825 — Elevating larval source management as a key strategy for controlling malaria and other vector-borne diseases in Africa, *Parasit Vectors* 2025, doi:[10.1186/s13071-024-06621-x](https://doi.org/10.1186/s13071-024-06621-x); 39294726 — Effects of mosquito control using *Bacillus thuringiensis israelensis* (Bti) on aquatic and terrestrial ecosystems: a systematic review, *Environ Evid* 2023, doi:[10.1186/s13750-023-00319-w](https://doi.org/10.1186/s13750-023-00319-w); 41309908 — A large-scale mosquito larviciding in Tanga Region, Tanzania, reduced mosquito densities to varying degrees across malaria transmission risk strata, *Sci Rep* 2025, doi:[10.1038/s41598-025-26675-w](https://doi.org/10.1038/s41598-025-26675-w); 40662945 — Systematic literature review of bacterial larvicides (*Bti* and *B. sphaericus*) for malaria control in Africa, *Int J Environ Health Res* 2025, doi:[10.1080/09603123.2025.2533346](https://doi.org/10.1080/09603123.2025.2533346) (abstract-only); 41637317 — Field evaluation of drone and AI assisted larval source management in Ghana, *PLoS One* 2026, doi:[10.1371/journal.pone.0340690](https://doi.org/10.1371/journal.pone.0340690). Plus National Centre for Vector Borne Diseases Control (MoHFW) national malaria and dengue statistics, provisional to 28 February 2026; WHO malaria fact sheet 2025 (World Malaria Report 2025 estimates for 2024); WHO lymphatic filariasis fact sheet 2025.

♦ ♦ ♦

Rodent-borne diseases and control measures

Asked in: Paper I 12-Nov-2021 — RGUHS MD Community Medicine

Definition — Rodents are mammals of the order Rodentia that live in and around human dwellings and act as reservoirs or sources of over a score of communicable diseases, chiefly plague and typhus fever. Their numbers often exceed the human population, a single female rat producing about 100 offspring a year, and besides disease they destroy or contaminate roughly 36 per cent of stored foodstuffs (Park 28e, Ch 15, p 940). **Deratization** — the destruction of rats and the elimination of their habitat — is therefore a core environmental health measure.

Rodents of public health importance

Group	Species	Habits of public health relevance
Domestic (commensal)	<i>Rattus rattus</i> — black or roof rat	Restricted area of movement; good climber, nests in roofs; readily infests ships, hence great public health importance
	<i>Rattus norvegicus</i> — Norway, brown or sewer rat	Semi-domestic; frequents sewers, drains and houses; burrows
	<i>Mus musculus</i> — house mouse	Lives entirely within dwellings; seldom lives in fields
Wild (field) rodents — India	<i>Tatera indica</i> , <i>Bandicota bengalensis varius</i> (<i>Gunomys kok</i>), <i>B. indica</i> , <i>Millardia meltada</i> , <i>M. gleadowi</i> , <i>Mus booduga</i>	Reservoirs of plague in nature; <i>Tatera indica</i> is the natural reservoir of plague in India , not <i>R. rattus</i> as once thought

(Park 28e, Ch 15, p 940; Park 28e, Ch 5, pp 359–360)

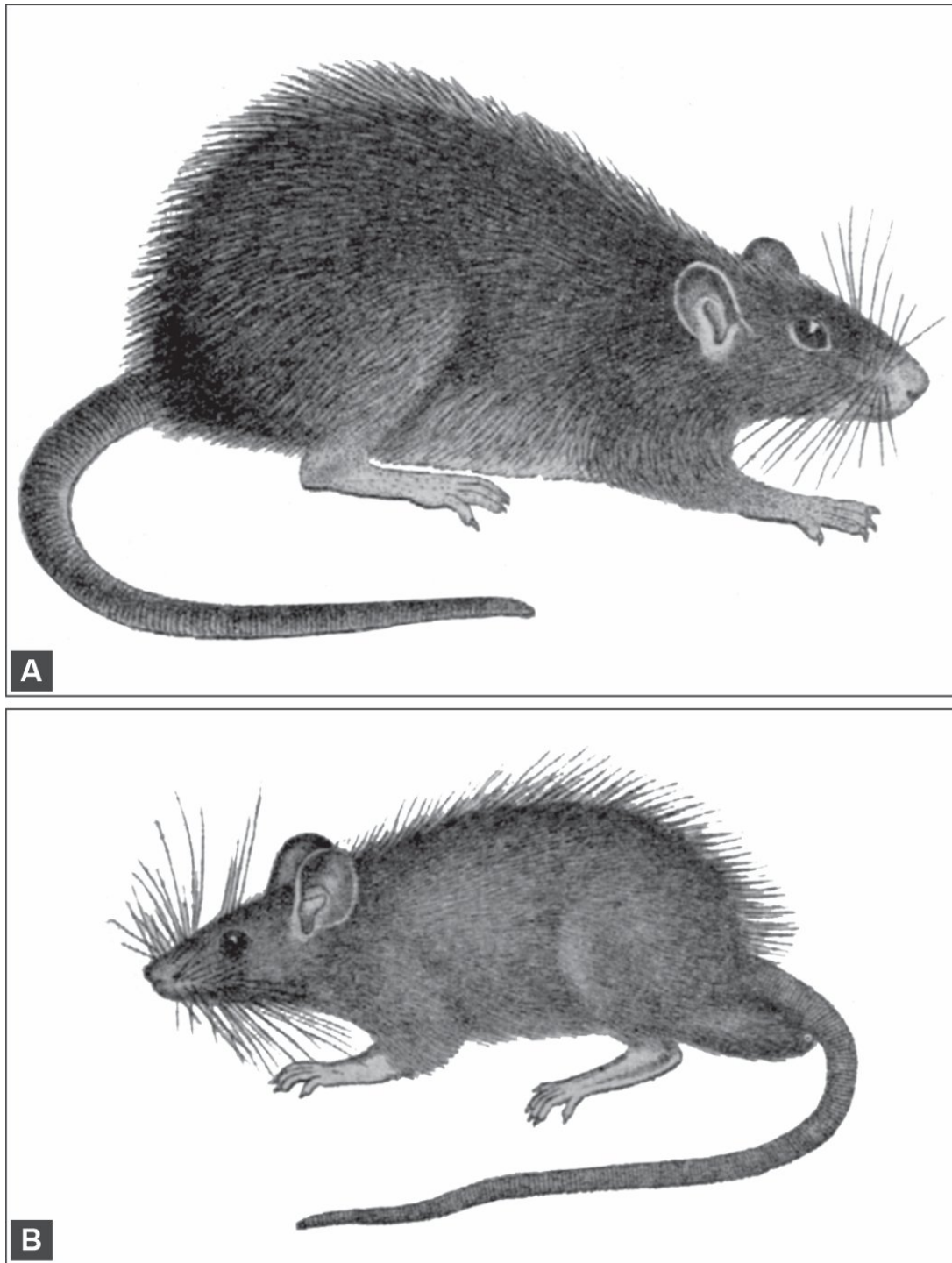


Fig 3.19 — (A) *Rattus norvegicus*, (B) *Rattus rattus* (Sunderlal, p.140)

Identifying feature	<i>Rattus rattus</i>	<i>Rattus norvegicus</i>
Tail	Longer than head + body	Shorter than head + body
Body	Light, slender	Heavy
Ear	Large	Small, thick
Nose	Pointed	Blunt
Eye	Large	Small

(Park 28e, Ch 15, p 940, Fig 1)

Diseases associated with rodents

Park groups rodent-associated diseases into five aetiological classes (Park 28e, Ch 15, p 940; Park 27e, p 912).

Class	Diseases	Causative agent / note
1. Bacterial	Plague	<i>Yersinia pestis</i> — the most important rodent-associated pathogen
	Tularaemia	<i>Francisella (Pasteurella) tularensis</i>
	Salmonellosis	<i>Salmonella</i> spp.; spread mechanically by rat droppings contaminating food
	Melioidosis	<i>Burkholderia pseudomallei</i> ; by ingesting rat-contaminated food (AFMC Red Book 2024, Ch 28, p 829)
2. Viral	Lassa fever	Multimammate rat (<i>Mastomys</i>) reservoir
	Viral haemorrhagic fevers	Includes hantaviral haemorrhagic fever with renal syndrome (Asia) and hantavirus pulmonary syndrome (deer mice, Americas), case fatality over 50 per cent (Park 28e, Ch 5, p 427)
	Viral encephalitis	Rodents act as amplifying hosts
	Lymphocytic choriomeningitis	Mice (AFMC Red Book 2024, Ch 28, p 829)
3. Rickettsial	Scrub typhus	<i>Orientia tsutsugamushi</i> ; mite-borne (<i>Leptotrombidium</i>), rodents and mites are reservoirs
	Murine (endemic, flea-borne) typhus	<i>Rickettsia typhi</i> ; rats are the reservoir, rat flea <i>Xenopsylla cheopis</i> the vector
	Rickettsial pox / tick typhus	<i>R. akari</i> / <i>R. conorii</i> ; mite- and tick-borne, rodent reservoir
4. Parasitic	<i>Hymenolepis diminuta</i>	Rat tapeworm
	Trichinosis	Rat is intermediary host of <i>Trichinella spiralis</i>
	Leishmaniasis, amoebiasis, Chagas disease	Rodents as reservoir or contaminating host
5. Others (spirochaetal, fungal)	Rat-bite fever	<i>Spirillum minus</i> (sodoku); transmitted directly by rat bite
	Leptospirosis (Weil's disease)	<i>Leptospira icterohaemorrhagiae</i> ; <i>R. norvegicus</i> and <i>Mus musculus</i> are the most important reservoirs, excreting leptospire in urine lifelong (Park 28e, Ch 5, pp 357–358)
	Relapsing fever	<i>Borrelia recurrentis</i> , <i>B. duttoni</i>

Class	Diseases	Causative agent / note
	Histoplasmosis, ringworm	Fungal

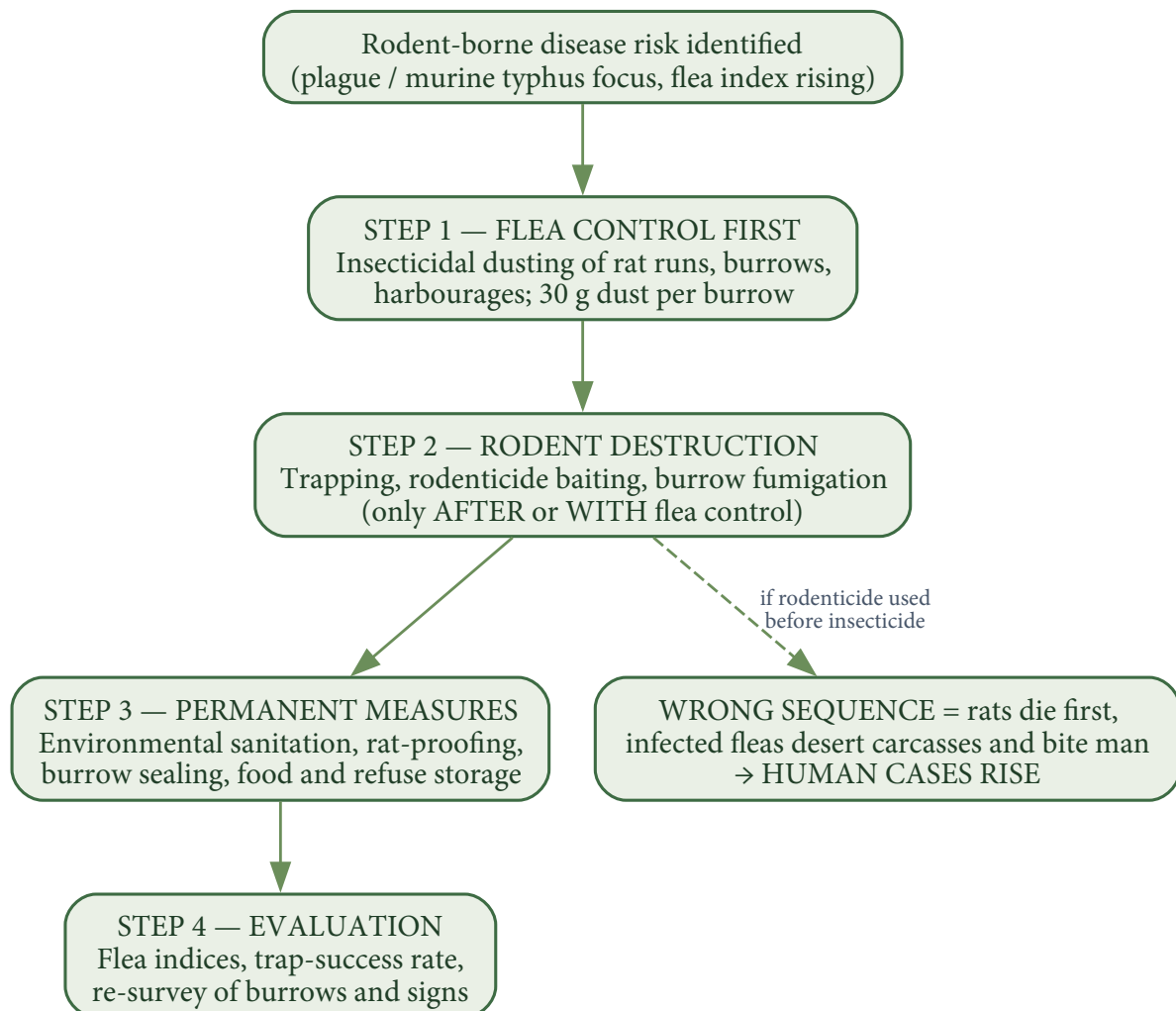
Modes of transmission from rodent to man

Route	Example diseases
Directly through rat bite	Rat-bite fever
Contamination of food or water by rodent urine, faeces or carcasses	Salmonellosis, leptospirosis, tularaemia, melioidosis
Through rat ectoparasites — fleas, mites, ticks, lice	Plague and murine typhus (rat flea <i>X. cheopis</i>); scrub typhus (trombiculid mite)

(Park 28e, Ch 15, p 940)

A dead rat is as dangerous as a living rat, because the ectoparasites leave the cooling body and attack other animals and man (AFMC Red Book 2024, Ch 28, p 829). This single fact governs the sequence of every anti-rodent operation.

Rodent control measures



Correct operational sequence of an anti-rodent campaign — fleas first, then rodents, then permanent environmental measures

The most effective way to break the rodent–flea–man chain is destruction of rat fleas by proper insecticide application; **flea control must precede or coincide with anti-rodent measures** (Park 28e, Ch 5, p 362; Sunder Lal, Ch 8, p 434).

1. Environmental sanitation (permanent measures) — the mainstay. Rats need three things: food, water and shelter; if these are denied, rat density falls naturally. All other measures are supplementary to sanitation (Park 28e, Ch 15, p 940). The four components are: (i) proper storage, collection and disposal of garbage; (ii) proper storage of foodstuffs in rat-proof containers; (iii) construction of rat-proof buildings, godowns and warehouses; and (iv) elimination of rat burrows by blocking them with concrete. Operationally these are grouped as **anti-rodent engineering** (design and construction that deny ingress and harbourage), **anti-rodent hygiene** (hygienic solid waste disposal, incineration of slaughterhouse waste), **anti-rodent housekeeping**, and **health education** to build civic sense — permanent control in urban areas is achievable only by these methods, since destruction alone is followed by rapid rebound to the original density (AFMC Red Book 2024, Ch 28, p 831).

2. Flea control (insecticidal). Ten per cent DDT dust is the cheapest and most widely used formulation; rodents passing over it pick it up on the fur, which kills the fleas. Where rat fleas are resistant to DDT, gamma-HCH or dieldrin, use **carbaryl 2 per cent or malathion 5 per cent**. Sprays are applied to floors

and walls up to about one foot height; about 2–3 g of formulation per square metre of surface is needed, and dust is blown into rodent burrows with a dust-blower at about 30 g per burrow (Park 28e, Ch 15, p 933; Ch 5, p 362).

3. Trapping. A simple operation, but it causes only a temporary reduction in commensal rodent numbers. The number of traps laid should be **at least 5 per cent of the human population**; the *wonder trap* developed by the Haffkine Institute, Mumbai, can trap as many as 25 rats at a time. Traps are baited with indigenous foods of the locality and captured rats destroyed by drowning. Rats are suspicious animals and soon become "**trap-wise**", so trapping is only supplementary to other methods (Park 28e, Ch 15, pp 940–941).

4. Rodenticides (poison baiting).

Type	Mode of action	Examples	Time to death
Single-dose (acute)	Lethal after one feeding	Red squill, norbromide, zinc phosphide, barium carbonate	Hours
Multiple-dose (cumulative)	Anticoagulants causing internal haemorrhage; need repeated feeds over 3 or more days	Warfarin, diphacinone, coumafuryl, pindone	4–10 days

WHO's Expert Committee grouped the acute rodenticides by the care their handling demands (Park 28e, Ch 15, p 941):

Safety group	Compounds
Requiring ordinary care	Red squill, norbromide, zinc phosphide
Requiring maximal precaution	Sodium fluoroacetate, fluoroacetamide, strychnine
Too dangerous for use	Arsenic trioxide, phosphorus, thallium sulphate, ANTU, gophacide

The two poisons commonly used in India are: **barium carbonate**, a cheap white tasteless powder mixed with wheat or rice flour in a ratio of 1 : 4, moistened and made into marbles placed near burrows, killing in 2–24 hours — a weak rodenticide of uneven performance, easily detected in bait, and no longer recommended; and **zinc phosphide**, an efficient rodenticide used at **1 part to 10 parts of wheat or rice flour** with a few drops of edible oil to improve palatability, killing in about three hours. Rubber gloves must be worn and special bait boxes used, as it is highly poisonous to man and domestic animals. Because of its good safety record, low cost and reasonably high effectiveness, **zinc phosphide is recommended for large-scale use against rats in India** (Park 28e, Ch 15, p 941). Continued use of anticoagulants has produced warfarin-resistant Norway rat populations in several European countries; in such settings bromadiolone or coumatetralyl are used (AFMC Red Book 2024, Ch 9, p 233). All rodent poisons are toxic to mammals including man and call for the utmost care.

5. Fumigation. An effective method of destroying **both rats and rat fleas** simultaneously — its unique advantage. Fumigants used in anti-rat campaigns are **calcium cyanide** (cyanogas or cymag), carbon disulphide, methyl bromide and sulphur dioxide. Cyanogas powder is pumped into rat burrows by a **cyanogas foot pump**; about 2 ounces are pumped into each burrow after closing the exit openings, and

the burrow is promptly sealed with wet mud. On contact with moisture it releases hydrogen cyanide, lethal to rats and their fleas. Only trained personnel may fumigate. For eradication of rats from **ships**, either cyanogas or sulphur dioxide is used (Park 28e, Ch 15, p 941).

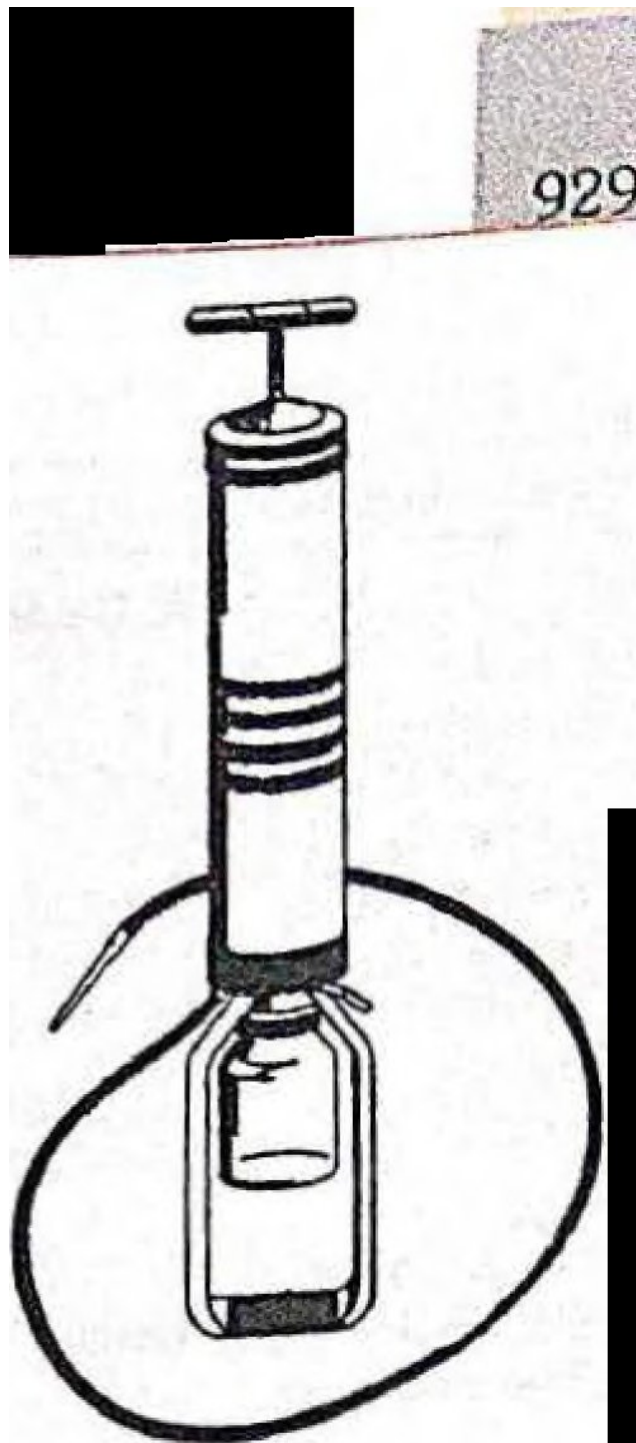


Fig 2 — Cyanogas foot pump (Park's Textbook of Preventive and Social Medicine, 28e, p.941)

6. Chemosterilants. Chemicals causing temporary or permanent sterility in either or both sexes; still at the **experimental stage** (Park 28e, Ch 15, p 941).

7. Biological control. Natural enemies — cats, hawks, eagles, vultures, owls, snakes, mongoose, dogs and ferrets — keep numbers down to some extent in nature but are **of no practicable value** in a rat

control programme, and several are themselves carriers of *Salmonella typhimurium* and *S. enteritidis* (AFMC Red Book 2024, Ch 28, p 834).

8. Ports and ships. Rat-proofing is a requirement in the building of new vessels; storerooms are sealed around pipes and conduits with metal sheeting, food is kept in rat-proof containers, and deratting by trapping and poisoning is carried out at least once every three to six months (AFMC Red Book 2024, Ch 9, pp 232–233).

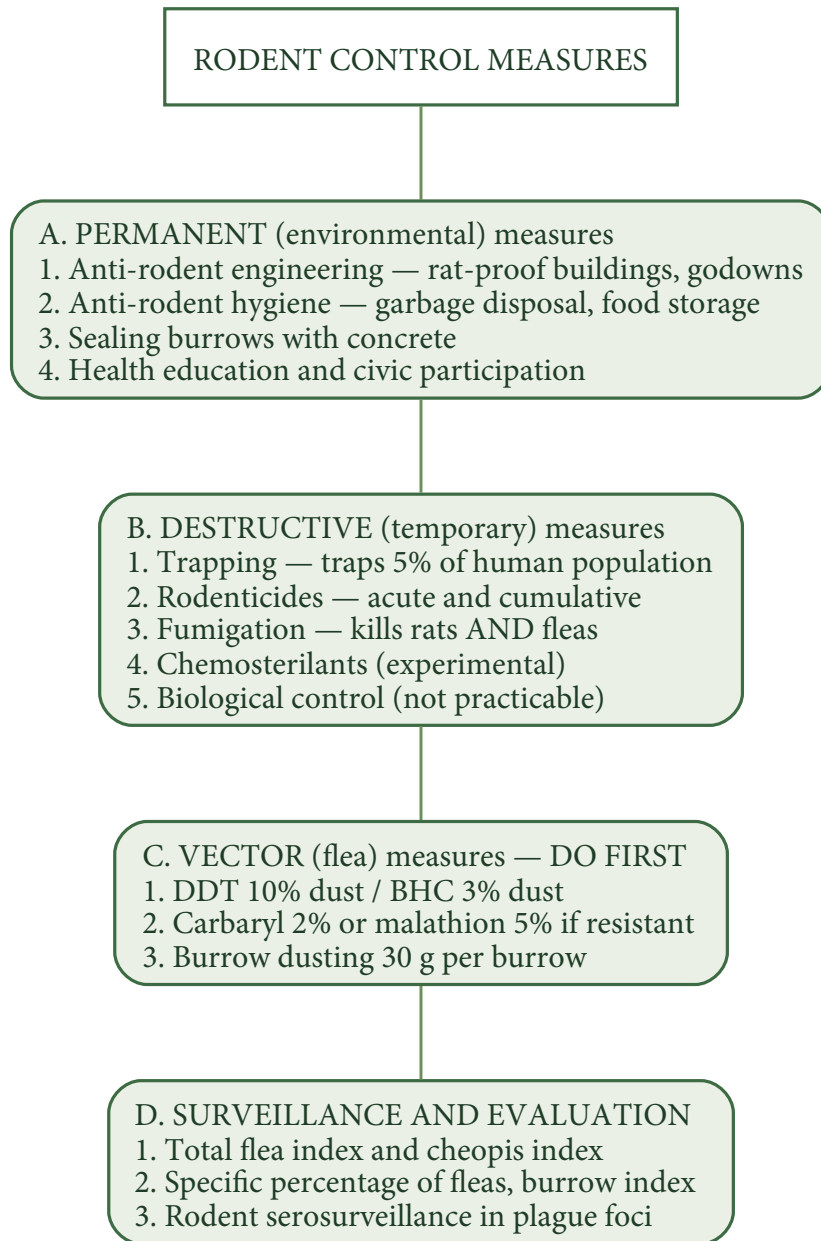
Evaluation and surveillance of rodent control

Index	Definition	Interpretation
Total flea index	Average number of fleas of all species per rat	General measure of flea density
Cheopis index	Average number of <i>X. cheopis</i> per rat (a specific flea index)	More significant than the total flea index; a value greater than 1 indicates potential explosiveness should a plague outbreak occur
Specific percentage of fleas	Percentage of each flea species found on rats	Identifies the vector species present
Burrow index	Average number of free-living fleas per species per rodent burrow	Measures residual burrow infestation

Flea indices do not by themselves predict an imminent epidemic; they serve as a **warning that more stringent control measures are needed** and as the yardstick for evaluating a spraying programme (Park 28e, Ch 5, p 360). Rodent serosurveillance for plague antibody in wild rodents, with dogs as sentinel animals in villages at risk, complements this in known plague foci (Sunder Lal, Ch 8, p 434).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Real plates embedded above:** identifying features of the two commensal rats, and the cyanogas foot pump.
- **Diagram 1 — Comparison of *Rattus rattus* and *Rattus norvegicus*.** Draw two rats side by side. Labels for *R. rattus*: TAIL LONGER THAN HEAD + BODY, LIGHT SLENDER BODY, LARGE EAR, POINTED NOSE, LARGE EYE. Labels for *R. norvegicus*: TAIL SHORTER THAN HEAD + BODY, HEAVY BODY, SMALL THICK EAR, BLUNT NOSE, SMALL EYE.
- **Diagram 2 — Chain of transmission of plague.** Wild rodent (*Tatera indica*) → wild rodent flea → commensal rat (*R. rattus*) → rat flea (*Xenopsylla cheopis*, blocked flea) → MAN. Mark the two points of attack: "insecticide here (first)" on the flea arrow and "rodenticide here (second)" on the rat.



Classification of rodent control measures

High-yield exam pointers

- **Park's five classes of rodent-associated disease (mnemonic B-V-R-P-O):** Bacterial (plague, tularaemia, salmonellosis) — Viral (Lassa, haemorrhagic fever, encephalitis) — Rickettsial (scrub typhus, murine typhus, rickettsial pox) — Parasitic (*Hymenolepis diminuta*, leishmaniasis, amoebiasis, trichinosis, Chagas) — Others (rat-bite fever, leptospirosis, histoplasmosis, ringworm).
- **Three modes of transmission:** rat bite (rat-bite fever) — contaminated food/water (salmonellosis, leptospirosis) — rat fleas (plague, typhus).
- **Cheopis index > 1** = potential explosiveness of a plague outbreak. Learn all four flea indices with their denominators.
- **Golden rule:** insecticide (flea) before rodenticide (rat) — a dead rat is as dangerous as a living rat.

- ▶ **Numbers to write:** traps = 5 per cent of human population; zinc phosphide 1 : 10 with flour (barium carbonate 1 : 4); anticoagulant death in 4–10 days; cyanogas ~2 ounces per burrow; insecticidal dust ~30 g per burrow, 2–3 g per square metre.
- ▶ *Tatera indica* is the natural reservoir of plague in India; *R. norvegicus* and *Mus musculus* are the chief reservoirs of leptospirosis.
- ▶ Sanitation is the permanent control; everything else is supplementary.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: DDT and BHC are named here because Park's Textbook of Preventive and Social Medicine 28e still lists them as the standard flea-control dusts, and the same edition retains barium carbonate and chemosterilants in its account; in current Indian field practice barium carbonate is no longer recommended, chemosterilants remain experimental, and organochlorine use is restricted, so carbaryl 2 per cent or malathion 5 per cent is the practical choice for flea dusting.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 15 (Environment and Health), pp 933, 940–941 (rodents, antirodent measures, rodenticides, fumigation) and Ch 5 (Epidemiology of Communicable Diseases), pp 357–362 (leptospirosis, plague, flea indices, control of fleas); Park's Textbook of Preventive and Social Medicine, 27e, p 912 (clean text of the "Rodents and disease" enumeration); Sunder Lal — Textbook of Community Medicine, Ch 8, pp 434, 439 (flea control before rodent killing, plague serosurveillance, leptospirosis rodent control); AFMC Red Book of Community Medicine 2024, Ch 28, pp 829–834 and Ch 9, pp 232–233 (rodents and disease, permanent versus temporary measures, biological control, ship deratting); Service — Medical Entomology for Students, pp 193–199 (plague and murine typhus cycles, rodent flea control); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e, Ch 11, p 124 (fumigation and poison baiting).



Epidemiology and control of leptospirosis

Asked in: Paper I 12-Nov-2019 — RGUHS MD Community Medicine

Definition — Leptospirosis is essentially an animal infection caused by several serotypes of *Leptospira* (spirochaetes) and transmitted to man under certain environmental conditions (Park 28e, Ch 5, p 357). Manifestations range from a mild febrile illness to severe, sometimes fatal disease with liver and kidney involvement; *Weil's disease* — combined renal and liver failure — is only one of its many presentations (Park 28e, Ch 5, p 357-358). Man is an **accidental (dead-end) host** and does not contribute significantly to transmission (Maxcy-Rosenau-Last 15e, Ch 16, p 462).

Problem statement — global and Indian burden

Item	Figure (with source and year)
Rank among zoonoses	Considered the most widespread disease transmissible from animal to man (Park 28e, Ch 5, p 357)
Global annual burden	>1 million cases and almost 60,000 deaths each year (Park 28e, Ch 5, p 357)
Peak incidence in hyperendemic pockets	Up to 975 cases per lakh population (Park 28e, Ch 5, p 357)
Endemic incidence, humid tropics	0.1–1 per 1,00,000 ; during outbreaks may exceed 100 per 1,00,000 (Sunder Lal, Ch 8, p 438)
Ecological setting	High prevalence in warm humid tropical countries; outbreaks mostly follow heavy rainfall and consequent flooding — links with extreme weather events documented in Guyana, India, the Philippines and Thailand (Park 28e, Ch 5, p 357)
Endemic areas in India	Coastal districts of Gujarat, Maharashtra, Kerala, Tamil Nadu, Andhra Pradesh, Karnataka, Goa, Odisha and the Andamans (Sunder Lal, Ch 8, p 438, citing NICD and WHO 2006, 2018)
Outbreaks reported in India, 2019	15 outbreaks (Sunder Lal, Ch 8, p 438)
Case fatality rate	5–15 per cent among patients with severe illness; >50 per cent in severe pulmonary haemorrhagic syndrome (Park 28e, Ch 5, p 358)
Seasonality	Tropics — rainy season; temperate zones — late summer and early autumn (Maxcy-Rosenau-Last 15e, Ch 16, p 462)

Indian outbreak data (Gujarat and Kerala) — the case fatality of a neglected, late-diagnosed zoonosis (Sunder Lal, Ch 8, Table 8.25, p 438):

Year	Cases	Deaths	Case fatality rate (%)
1994	158	12	7.6
1997	657	76	11.6
1998	537	42	7.8
2000	112	12	10.7
2011	600	104	17.0

Year	Cases	Deaths	Case fatality rate (%)
2018 (Kerala, Jan–Aug)	520	28	5.4

Epidemiological determinants

Determinant	Detail
Agent	<i>Leptospira</i> — thin, motile spirochaetes 0.1–0.2 µm wide and 5–15 µm long with hooked ends; only strains of <i>L. interrogans</i> are pathogenic; seen by dark-field illumination and silver staining (Park 28e, Ch 5, p 357)
Antigenic diversity	23 sero-groups and 200 serovars recognised worldwide, serologically related with cross-reactivity (Park 28e, Ch 5, p 357)
Serovar–host adaptation	Serovar <i>icterohaemorrhagiae</i> with rats and mice; serovar <i>pomona</i> with cattle and swine — infection in these maintenance hosts is endemic and asymptomatic, with chronic renal-tubular carriage (Maxcy-Rosenau-Last 15e, Ch 16, p 462)
Source of infection	Urine of infected animals, excreted for a long time — often for the entire lifetime in rodents (Park 28e, Ch 5, p 357)
Animal reservoirs	Rodents (rats, mice, voles) — <i>Rattus norvegicus</i> and <i>Mus musculus</i> most important globally; in India <i>Rattus rattus</i> , <i>R. norvegicus</i> , the lesser and the larger bandicoot. Domestic animals — cattle, sheep, goats, water buffalo, pigs, horses — infected by grazing on urine-contaminated pasture; dogs among pets (Park 28e, Ch 5, p 357; Sunder Lal, Ch 8, p 439)
Carrier state	Domestic animals are temporary carriers (several months); rodents are permanent carriers (lifelong shedding) (Sunder Lal, Ch 8, p 438)
Age and sex	Males affected more than females owing to greater occupational exposure; commonest in 20–45 years; rare in infants and young children because exposure is minimal (Park 28e, Ch 5, pp 357–358)
Occupation	Essentially an occupational disease — agricultural and livestock farmers, workers in rice and sugarcane fields, underground sewer workers, abattoir workers, meat and animal handlers, veterinarians; also miners, forestry and fish-farm workers and military personnel on field exercises (Park 28e, Ch 5, p 358; Sunder Lal, Ch 8, p 438)
Leisure exposure	Swimming, fishing, rafting and adventure travel in contaminated surface water (Park 28e, Ch 5, p 358; Maxcy-Rosenau-Last 15e, Ch 16, p 462)
Immunity	A solid but serovar-specific immunity follows infection — no cross-protection against other serovars (Park 28e, Ch 5, p 358)
Environment	The infection is unique in being acquired from an environment contaminated by urine and faeces of carrier animals; leptospires shed in urine survive for weeks in soil and water, so contamination reaches high levels after flooding, water-logging and in saline, moist soils (Park 28e, Ch 5, p 358; Sunder Lal, Ch 8, p 438)

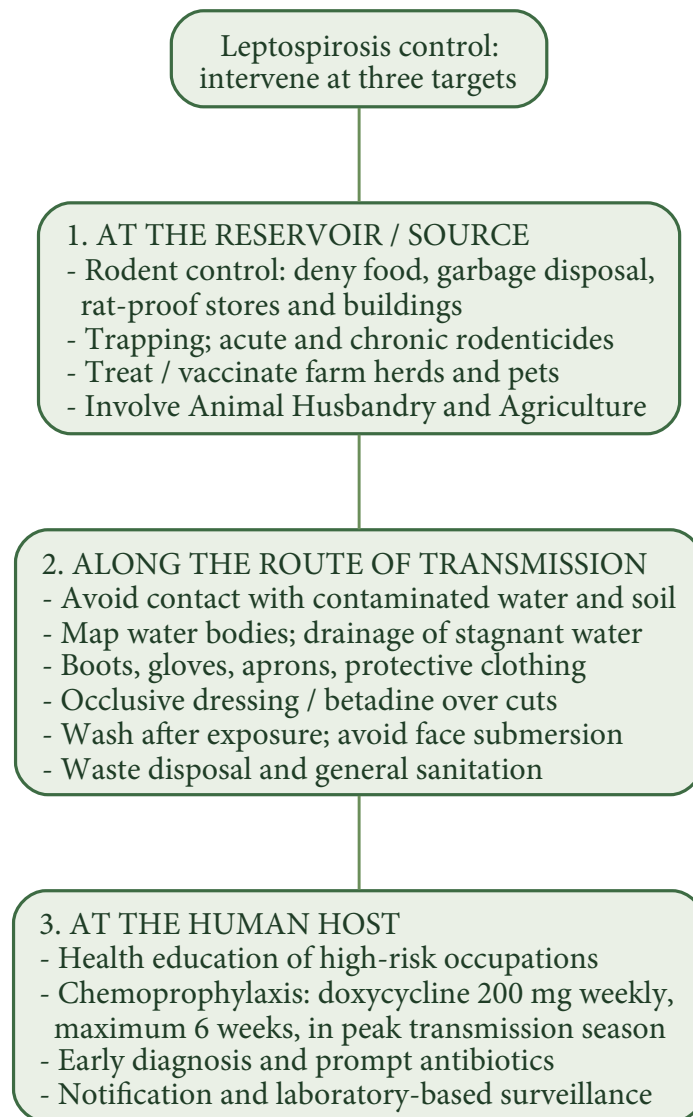
Mode of transmission, incubation and communicability

Route	Mechanism
Direct contact	Entry through skin abrasions or intact mucous membrane on contact with urine or tissue of an infected animal (Park 28e, Ch 5, p 358)

Route	Mechanism
Indirect contact	Broken skin in contact with soil, water or vegetation contaminated by animal urine, or ingestion of contaminated food or water (Park 28e, Ch 5, p 358)
Droplet / inhalation	Breathing air polluted with droplets of urine, e.g. while milking infected cows or goats (Park 28e, Ch 5, p 358)
Conjunctival	Through the conjunctiva and buccal or nasal mucosa (Sunder Lal, Ch 8, p 438)
Man-to-man	Rare or absent ; rare congenital infection reported; leptospirosis in pregnancy can cause foetal death or abortion (Park 28e, Ch 5, pp 357-358; Sunder Lal, Ch 8, p 438)
Incubation period	Usually 10 days , range 4–20 days (Park 28e, Ch 5, p 358)
Period of communicability	Leptospire excreted in human urine for months after illness — recorded up to 11 months ; excretion is lifelong in reservoir species (Sunder Lal, Ch 8, p 438)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Transmission cycle of leptospirosis** — draw a circle with these exact labels: *Maintenance hosts (rats, bandicoots, cattle, pigs, dogs) → chronic renal tubular carriage → leptospire shed in urine → contaminated soil, surface water, mud, paddy field, sewer → entry through abraded skin / intact mucosa / conjunctiva / inhaled urine droplets → Man — accidental dead-end host*. Mark "**no onward man-to-man spread**" as a broken arrow leaving "Man", and label the three control cut-points on the circle as (1) **at the reservoir**, (2) **along the route**, (3) **at the human host**.



Three-point control strategy for leptospirosis

Laboratory diagnosis (supports both case management and surveillance)

- ▶ **Dark-ground illumination** of clinical samples — demonstrates motile leptospires; useful only in experienced hands (Sunder Lal, Ch 8, p 439).
- ▶ **IgM ELISA** — the workhorse for early diagnosis; positive as early as **2 days into illness** (Park 28e, Ch 5, p 358).
- ▶ **Rapid tests** — latex agglutination and the **Leptodipstick** test for field and peripheral use (Park 28e, Ch 5, p 358).
- ▶ **Microscopic agglutination test (MAT)** — the **gold standard** of serodiagnosis because of unsurpassed specificity, and it identifies the serovar; available only in a reference laboratory (Sunder Lal, Ch 8, p 439).
- ▶ **Paired sera** — seroconversion or a **four-fold or greater rise in titre** in the presence of compatible illness is the major criterion for definitive diagnosis (Sunder Lal, Ch 8, p 439).
- ▶ **Isolation** — from blood or cerebrospinal fluid in the **first week**, and from urine **after the 10th day of illness** (Sunder Lal, Ch 8, p 439).

- Polymerase chain reaction (PCR) (Sunder Lal, Ch 8, p 439).

Control measures by level of prevention

Level	Measures
Primordial / health promotion	Improved housing and general sanitation, proper storage of food-grain and disposal of garbage to remove rodents from the domestic environment; health impact assessment of irrigation, canal and other development projects (Sunder Lal, Ch 8, p 439; Jugal Kishore, p 451)
Specific protection	Protective clothing, boots, gloves and aprons for all high-risk work; betadine ointment on cuts and abrasions before entering fields and after exit ; occlusive dressings over open wounds if water contact is unavoidable; washing or showering after exposure; avoid submerging the face (Sunder Lal, Ch 8, p 439; Maxcy-Rosenau-Last 15e, Ch 16, p 462)
Chemoprophylaxis	Doxycycline 200 mg orally once a week for not more than 6 weeks during the period of high exposure, for high-risk groups where clustering of cases is reported. Prolonged and mass chemoprophylaxis is not recommended (Sunder Lal, Ch 8, p 439; Jugal Kishore, p 451)
Reservoir control	Rodent control as above, plus treatment or vaccination of farm herds and pets ; animal vaccine immunity is serovar-specific, needs frequent revaccination, and prevents illness but not necessarily infection or urinary shedding (Sunder Lal, Ch 8, p 439; Maxcy-Rosenau-Last 15e, Ch 16, p 462)
Environmental control	Mapping of water bodies for a proper drainage system; recognise and drain potentially contaminated water and soil; proper waste disposal (Sunder Lal, Ch 8, p 439; Park 28e, Ch 5, p 358)
Early diagnosis and treatment (secondary prevention)	Penicillin is the drug of choice — 6 million units daily intravenously ; tetracycline, amoxycillin, ampicillin and doxycycline are also effective. Antibiotics must be started on clinical suspicion, without waiting for confirmation (Park 28e, Ch 5, p 358; Maxcy-Rosenau-Last 15e, Ch 16, p 462)
Human vaccination	No safe human vaccine effective in field conditions is available for wider use. Occupational-group immunisation against specific serovars has been used in Japan, China, Italy, Spain and France; any such vaccine must contain the serovars predominant in that area, since immunity to one <i>Leptospira</i> type may not protect against another (Sunder Lal, Ch 8, p 439; Park 28e, Ch 5, p 358)
Tertiary prevention	Intensive care for severe icteric disease — dialysis for renal failure (which may last several weeks) and cardiac monitoring (Maxcy-Rosenau-Last 15e, Ch 16, p 462)
Surveillance	Laboratory-based surveillance supplemented by sero-surveys in man and animals ; leptospirosis should be made a reportable disease in all endemic states (Sunder Lal, Ch 8, p 439; Jugal Kishore, p 451)

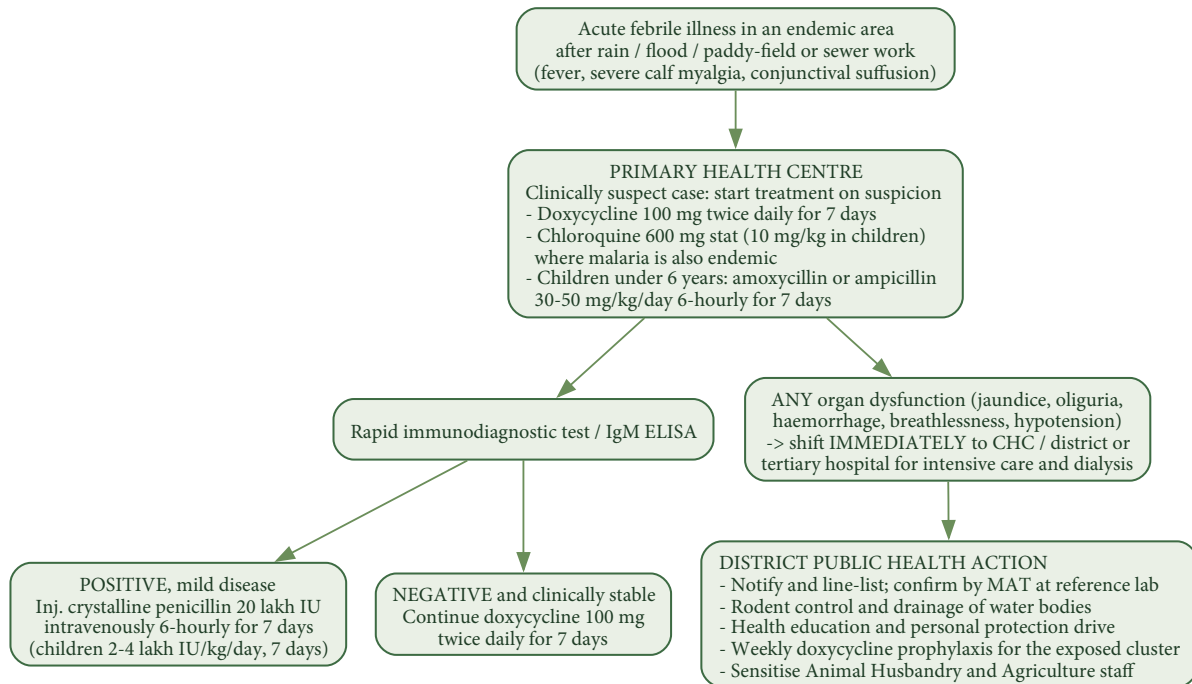
Programme for Prevention and Control of Leptospirosis, India

Launched by the Government of India; current guidance is the **National Guidelines 2015** of the National Centre for Disease Control (NCDC), Directorate General of Health Services (Sunder Lal, Ch 8, p 437).

Objectives (Sunder Lal, Ch 8, p 437):

- Reduction of morbidity and mortality due to leptospirosis.

- ▶ Strengthening diagnostic laboratories.
- ▶ Strengthening patient care management.
- ▶ Development of trained manpower.
- ▶ Creating awareness.
- ▶ Sensitisation of animal husbandry and agriculture personnel.



Case management and referral pathway for suspected leptospirosis in an endemic district

Intersectoral and One Health linkage — leptospirosis control cannot be delivered by the health sector alone; rodent control and animal vaccination require the Departments of Animal Husbandry and Agriculture (Sunder Lal, Ch 8, p 439). NCDC runs the **National One Health Programme for Prevention and Control of Zoonoses (NOHP-PCZ)**, draws field data through the Integrated Disease Surveillance Programme and its Integrated Health Information Platform, and has established **15 regional coordinators and 33 surveillance sites** across India (One Health Booklet, AIIMS, Ch 1, pp 5-7).

High-yield exam pointers

- ▶ **Agent and serotype numbers:** *Leptospira*, 0.1–0.2 μm $\times$ 5–15 μm , hooked ends; only *L. interrogans* strains pathogenic; **23 sero-groups, 200 serovars**; dark-field illumination.
- ▶ **Numbers to write:** incubation **10 days (4–20)**; global **>1 million cases, ~60,000 deaths/year**; CFR **5–15 %** in severe illness, **>50 %** with pulmonary haemorrhage; India 2019 — **15 outbreaks**.
- ▶ **Two prescriptions:** treatment — **penicillin 6 million units IV daily** (or doxycycline 100 mg twice daily $\times$ 7 days); prophylaxis — **doxycycline 200 mg once weekly, maximum 6 weeks**.
- ▶ **Serovar–host pairs:** *icterohaemorrhagiae* $\rightarrow$ rats and mice; *pomona* $\rightarrow$ cattle and swine. Rodents are lifelong carriers, domestic animals temporary carriers, **man an accidental dead-end host**.
- ▶ **Diagnosis one-liner:** IgM ELISA for early diagnosis (positive from day 2), **MAT is the gold standard**, blood/CSF culture in week 1 and urine culture after day 10.

- **Programme:** Programme for Prevention and Control of Leptospirosis, NCDC National Guidelines 2015, six objectives; endemic in the coastal belt — Gujarat, Maharashtra, Kerala, Tamil Nadu, Andhra Pradesh, Karnataka, Goa, Odisha and the Andamans.
- **Mnemonic for the answer plan:** RAT-MOP — Reservoir, Agent and serovars, Transmission routes; then Man-host factors, Occupation, Prevention at three targets.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference, not to be written in the exam: Park's 28th edition gives the incubation period as 10 days (range 4–20 days) while Sunder Lal quotes 2–30 days with an average of 5–14 days — quote Park's figure as the spine answer. Park also states that only *Leptospira interrogans* strains are pathogenic, whereas Sunder Lal's text names both *L. interrogans* and *L. biflexa*; *L. biflexa* is the free-living saprophytic species, so follow Park. Leptospirosis is not nationally notifiable across India — the national guidelines recommend that endemic states make it reportable, and outbreak reporting currently flows through the Integrated Disease Surveillance Programme.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, Cochrane Library, ICMR

Programme and policy updates (India)

- **Post-flood chemoprophylaxis reaches only a minority of those it is meant for.** After the 2018 Kerala floods an ICMR-National Institute of Epidemiology rapid survey in Kozhikode district interviewed 1,573 people; 152 (10%) had flood-water contact and 119 of them were eligible for weekly doxycycline, but only 58 — 38.2% (95% CI 30.8–46.1) — actually consumed it. The weak link is delivery and uptake rather than the policy itself, which is the sharper point to make when writing about chemoprophylaxis. [PMID 38164080]
- **The endemic map runs well beyond the coastal belt.** ICMR-National Institute of Research in Tribal Health tested 518 hepatitis-negative febrile samples from Madhya Pradesh (2018–19) through the viral research and diagnostic laboratory network: 68 (13.1%) were IgM-positive for leptospirosis, with jaundice in 42.3% and renal involvement in 30.9%, clustering in the 31–60 year age group and in the monsoon and post-monsoon months. Central Indian states with no recognised leptospirosis programme should still carry it in the acute febrile illness screening panel. [PMID 39067858]

Current national data

- **Leptospirosis is the most lethal of the monsoon fevers in a coastal Indian hospital series.** Among 314 adults admitted with acute febrile illness at Manipal, Karnataka, leptospirosis made up 7.6% of cases yet carried the highest median SOFA score (7, IQR 2.5–10, against 3 for dengue), acute kidney injury in 52.4%, haemodialysis in 28.6%, inotropes in 38.1% and ICU admission in 57.1%; it caused 4 of the 10 deaths in the cohort, a within-disease fatality of 19%. [PMID 42311109]
- **Species and fatality from a Kerala reference series (2025).** Gene sequencing of 70 lipL32 PCR-positive bloods identified *Leptospira interrogans* in 66 (91%), the remainder being *L. kirschneri* or *L. noguchii*; 14 patients died (reported case fatality 19%) and 13 of those deaths were in *L.*

interrogans infection. Serovars named included *canicola*, *hardjo*, *copenhageni*, *icterohaemorrhagiae* and *manilae* — a current supplement to the textbook serovar–host list. [PMID 40602749]

Elimination, surveillance and disease status

- **Flood duration, not merely the occurrence of a flood, drives the post-flood surge.** A Kerala analysis comparing a non-flood year (2017) with the heavy 2018 and moderate 2019 floods found a post-flood rise in cases in both flood years, much stronger after the heavy flood, with flood duration emerging as the most important severity indicator. Surveillance intensification and prophylaxis rounds should therefore be pegged to how long water stands, not simply to the fact that a flood happened. [PMID 38741103]
- **Man remains a dead-end host, but vertical transmission is now documented.** A global systematic review (27 publications, 34 suspected reports) found strong evidence for transplacental and breast-milk transmission (12 reports), while sexual transmission (3 reports) and transfusion (1 report) were not well proven and no sustained person-to-person chains were recorded. The textbook line "rare or absent" should now be qualified — congenital and lactational spread is real, but the disease still does not propagate human to human. [PMID 42395288]

Guideline and legislative updates

- **The trial evidence behind weekly doxycycline prophylaxis is weak.** The 2024 Cochrane update pooled 5 randomised trials (2,593 participants, including Indian data): laboratory-confirmed leptospirosis RR 0.56 (95% CI 0.25–1.26) and clinically diagnosed disease RR 0.76 (95% CI 0.53–1.08), neither significant and both low to very low certainty, while non-serious adverse events rose steeply (RR 10.13, 95% CI 2.40–42.71). This supports rather than overturns the Indian rule — short courses for a defined high-risk cluster, never mass or prolonged prophylaxis. [PMID 38483067]
- **Antibiotic treatment and adjuvant steroids also rest on very-low-certainty evidence.** Cochrane 2024 on treatment (9 trials, 1,019 participants) found penicillin versus placebo all-cause mortality RR 1.57 (95% CI 0.65–3.79), and Cochrane 2025 on corticosteroids (4 trials, 253 participants) found mortality RR 1.04 (95% CI 0.38–2.80), with a possible fall in Jarisch-Herxheimer reactions (RR 0.13, 95% CI 0.04–0.41, single trial). The public-health instruction is unchanged — start antibiotics early on clinical suspicion — but an answer should not claim a proven mortality benefit. [PMID 38483092] [PMID 40704556]

Global and WHO developments

- **Leptospirosis is still not on the WHO neglected tropical disease list.** A multi-institutional expert group argued in 2025 that its burden — 1.03 million cases, 58,900 deaths and 2.9 million DALYs a year — exceeds that of several listed NTDs including rabies, visceral leishmaniasis and lymphatic filariasis, yet it sits outside both the NTD list and the WHO Research and Development Blueprint, whose road maps have so far covered only viral diseases. That exclusion keeps funding, diagnostics and surveillance under-resourced. [PMID 39847571]
- **Climate-driven flooding is now the dominant epidemic trigger.** In the weeks after the 2024 Rio Grande do Sul floods in Brazil, 7,818 confirmed leptospirosis cases were reported alongside 10 diarrhoeal disease outbreaks, with health networks disrupted, diagnosis delayed and vaccination

campaigns impaired — the order of magnitude a district should plan for when an extreme rainfall event strikes an endemic belt. [PMID 41334923]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 5, Epidemiology of Communicable Diseases, pp 357-358; Ch 15, rodent control, pp 940-941); Sunder Lal — Textbook of Community Medicine (Ch 8, pp 437-439, including Table 8.25 and the NCDC Programme for Prevention and Control of Leptospirosis — National Guidelines 2015); Jugal Kishore — National Health Programmes of India (p 451, level-wise case management, after the NICD and WHO Guidelines for Prevention and Control of Leptospirosis, 2006); Maxcy-Rosenau-Last Public Health and Preventive Medicine, 15e (Ch 16, Diseases Transmitted Primarily from Animals to Humans, pp 460-463); One Health Booklet, AIIMS (National One Health Programme for Prevention and Control of Zoonoses, NCDC). **Coverage note:** the indexed figure catalogue holds no leptospirosis transmission-cycle plate, so the transmission cycle is given as a labelled diagram-to-draw box rather than an embedded figure.

Recent-advances sources: 38164080 — Rapid assessment of coverage of doxycycline/azithromycin chemoprophylaxis against leptospirosis following floods, Kozhikode district, Kerala, 2018, *Trans R Soc Trop Med Hyg* 2024, doi:[10.1093/trstmh/trad092](https://doi.org/10.1093/trstmh/trad092) (abstract-only); 39067858 — Leptospirosis in central India: a retrospective study to explore burden of tropical illness, *Indian J Med Microbiol* 2024, doi:[10.1016/j.ijmmb.2024.100689](https://doi.org/10.1016/j.ijmmb.2024.100689) (abstract-only); 42311109 — Leptospirosis is associated with severity, increased resource utilization, and mortality among acute febrile illnesses from Karnataka, Southwestern India, *Ann Med* 2026, doi:[10.1080/07853890.2026.2687164](https://doi.org/10.1080/07853890.2026.2687164); 40602749 — Molecular and genomic characterization of *Leptospira* isolates in humans and its relation to disease severity, *Indian J Med Microbiol* 2025, doi:[10.1016/j.ijmmb.2025.100910](https://doi.org/10.1016/j.ijmmb.2025.100910) (abstract-only); 38741103 — Analysing the outbreaks of leptospirosis after floods in Kerala, India, *Int J Health Geogr* 2024, doi:[10.1186/s12942-024-00372-9](https://doi.org/10.1186/s12942-024-00372-9); 42395288 — Human-to-human transmission of leptospirosis: a global systematic review, *Front Public Health* 2026, doi:[10.3389/fpubh.2026.1782163](https://doi.org/10.3389/fpubh.2026.1782163); 38483067 — Antibiotic prophylaxis for leptospirosis, *Cochrane Database Syst Rev* 2024, doi:[10.1002/14651858.CD014959.pub2](https://doi.org/10.1002/14651858.CD014959.pub2) (abstract-only); 38483092 — Antibiotics for treatment of leptospirosis, *Cochrane Database Syst Rev* 2024, doi:[10.1002/14651858.CD014960.pub2](https://doi.org/10.1002/14651858.CD014960.pub2) (abstract-only); 40704556 — Corticosteroids for treatment of leptospirosis, *Cochrane Database Syst Rev* 2025, doi:[10.1002/14651858.CD014935.pub2](https://doi.org/10.1002/14651858.CD014935.pub2) (abstract-only); 39847571 — Leptospirosis, melioidosis, and rickettsioses in the vicious circle of neglect, *PLoS Negl Trop Dis* 2025, doi:[10.1371/journal.pntd.0012796](https://doi.org/10.1371/journal.pntd.0012796); 41334923 — Floods and infectious diseases: public health lessons from the 2024 southern Brazil disaster, *Clin Microbiol Rev* 2025, doi:[10.1128/cmr.00165-24](https://doi.org/10.1128/cmr.00165-24) (abstract-only).

♦ ♦ ♦

Epidemiology and control of avian influenza

Asked in: Paper II 18-Dec-2023 — RGUHS MD Community Medicine

Definition — Avian influenza ("bird flu") is a highly contagious viral disease of birds caused by type A strains of the influenza virus; the viruses are highly species specific but have, on rare occasions, crossed the species barrier to infect pigs and humans (Sunder Lal, Ch 8, p 469). It occurs in a **low pathogenic** (LPAI) and a **highly pathogenic avian influenza** (HPAI) form, the latter killing up to 100 per cent of an infected flock within 48 hours (Sunder Lal, Ch 8, p 469). Human infection with avian influenza A (H5N1) is a zoonosis of pandemic potential, because the virus might ultimately adapt into a strain that is contagious among humans (Park 28e, Ch 5, p 185).

Epidemiological determinants

Determinant	Facts
Agent	Influenza virus type A. The types are antigenically distinct with no cross-immunity; only influenza A causes pandemics (Sunder Lal, Ch 8, pp 468-469)
Subtypes	16 haemagglutinin (H) and 9 neuraminidase (N) subtypes. The H subtype governs binding to and entry into the host cell; the N subtype governs release of newly formed virus from the cell (Sunder Lal, Ch 8, p 469; Jugal Kishore, p 663)
Pathogenicity	Of the 16 H subtypes only H5 and H7 cause highly pathogenic disease. They enter flocks as low-pathogenic viruses and, given time to circulate, mutate to high pathogenicity — hence any H5/H7 in poultry is a cause for concern even with mild initial signs (Mahajan & Gupta 4e, Ch 16, p 187)
Reservoir	Migratory waterfowl, most notably wild ducks — most resistant to infection, excrete virus in droppings and carry it over great distances with only mild, short-lived illness (Red Book, Ch 2, pp 885-889). Domestic ducks excrete large quantities of highly pathogenic virus without signs of illness and act as a <i>silent</i> reservoir (Sunder Lal, Ch 8, p 470)
Susceptible hosts	All birds, pigs and human beings. The pig is the classical " mixing vessel ", being in contact with both birds and humans (Jugal Kishore, p 663; Nelson & Williams, Ch 15, p 492)
High-risk humans	Poultry farm and live-bird-market workers, cullers (catching, bagging, transporting and disposing of birds), poultry processing and defeathering workers, veterinary staff, health-care workers — and their family members (Red Book, Ch 2, pp 885-889)
Age and outcome	Most cases in previously healthy children and adults; majority aged under 40 years, mortality highest at 10–19 years; overall case fatality of reported H5N1 cases about 60 per cent (Jugal Kishore, p 662)
Environmental	Backyard flocks roaming freely into homes and children's play areas; live bird markets; virus survives long periods in the environment, especially at low temperature (Sunder Lal, Ch 8, p 470; Jugal Kishore, p 663)

Distribution — time, place and person

Event	Facts
Origin	First identified in Italy more than 100 years ago; occurs worldwide (Red Book, Ch 2, pp 885-889)

Event	Facts
1997, Hong Kong	First documented human infection with an avian influenza virus — 18 severe respiratory cases, 6 deaths; outbreak curbed by prompt culling of about 1.5 million chickens (Red Book, Ch 2, pp 885-889; Maxcy-Rosenau-Last 15e, Ch 9, p 150)
January 2004, Asia	An H5N1 strain caused poultry outbreaks in several Asian countries, leading to the culling of more than 100 million birds ; unprecedented in geographic scale and impact (Maxcy-Rosenau-Last 15e, Ch 8, p 107)
Mid-2003 onward, South-East Asia	Largest and most severe HPAI outbreaks on record; an estimated 150 million birds died or were destroyed; virus now endemic in parts of Indonesia and Vietnam and parts of Cambodia, China, Thailand and Lao PDR (Sunder Lal, Ch 8, p 469)
Cumulative human burden	Since 1997, 478 cases with 286 deaths reported to WHO — Cambodia, Indonesia, Thailand, Vietnam (Park 28e, Ch 5, p 427). A more recent count is about 900 human infections since 2003 with case fatality ~50 per cent (Harrison 22e, Ch 147, p 1192)
India — poultry	Affected India in Feb–March 2006; the West Bengal outbreak of 2008 was more serious than the earlier Maharashtra and Manipur outbreaks; the 2021 outbreak occurred in nine states (Rajasthan, Gujarat, Uttar Pradesh, Kerala, Himachal Pradesh, Madhya Pradesh, Haryana, Maharashtra, Delhi); outbreaks in India were investigated by the National Institute of Communicable Diseases (NICD), now the National Centre for Disease Control (NCDC) (Sunder Lal, Ch 8, pp 469-470)
India — first human case	An 11-year-old child from Gurugram, Haryana , died at AIIMS Delhi on 12 July 2021 , confirmed by the National Institute of Virology (NIV), Pune (Sunder Lal, Ch 8, p 470)
Expanding host range	H5N1 has infected and killed mammalian species previously thought resistant (Sunder Lal, Ch 8, p 470); it began circulating in North American wild birds and poultry in late 2021, with subsequent detection in US dairy cattle herds and barn cats and numerous animal-to-human transmissions (Harrison 22e, Ch 147, p 1192)

Mode of transmission

Route	Detail
Bird to bird	Virus shed in oculonasal discharges and faeces; contaminated drinking water is commonly implicated; airborne spread when birds are in close proximity (Red Book, Ch 2, pp 885-889)
Farm to farm	Mechanical carriage on equipment, feed, cages, vehicles and clothing — especially shoes ; rodents and flies act as mechanical vectors (Jugal Kishore, p 663; Suryakantha, Ch 20, p 353)
Bird to human (main route)	Direct contact with infected live or dead poultry or with contaminated environments such as live bird markets ; exposure most likely during slaughter, defeathering, butchering and preparation of poultry for cooking (Park 28e, Ch 5, pp 185-186; Red Book, Ch 2, pp 885-889)
Human to human	Not sustained. Very limited transmission documented in health-care workers, family members, poultry workers and cullers in the 1997 Hong Kong outbreak; in no instance has the virus spread beyond a first generation of close contacts (Jugal Kishore, p 663; Sunder Lal, Ch 8, p 470)
Country to country	International trade in live poultry, and migratory waterfowl, sea birds and shore birds (Red Book, Ch 2, pp 885-889)

Route	Detail
NOT transmitted by	Properly cooked poultry, eggs or poultry products — the virus is heat sensitive and is killed when food reaches 70 °C in all parts (Red Book, Ch 2, pp 885-889; Jugal Kishore, p 665)

Clinical course and diagnosis

Feature	Detail
Incubation period	2–5 days (Park 28e, Ch 5, p 186)
Onset	Abrupt, with high fever above 38 °C, cough and breathlessness (Red Book, Ch 2, pp 885-889)
Lower respiratory involvement	Develops early — dyspnoea and auscultatory signs; clinically apparent pneumonia with chest X-ray changes in virtually all patients (Red Book, Ch 2, pp 885-889)
Progression	Respiratory distress and respiratory failure within 1 week of onset; most cases die despite ventilatory support (Red Book, Ch 2, pp 885-889)
Laboratory findings	Lymphopenia below $1 \times 10^9/\text{litre}$; slightly to moderately raised alanine and aspartate transaminases (Red Book, Ch 2, pp 885-889)
Confirmatory test	Reverse transcription polymerase chain reaction (RT-PCR) ; rapid influenza diagnostic tests have lower sensitivity (Park 28e, Ch 5, p 186)
Optimal specimen	Nasopharyngeal aspirate within 3 days of onset , then acute and convalescent serum (Red Book, Ch 2, pp 885-889)
Referral laboratory	Any influenza-A-positive specimen suspected of avian influenza must be verified by a designated WHO H5 reference laboratory — in India, human specimens to NIV Pune and animal specimens to the High Security Animal Disease Laboratory, Bhopal (Red Book, Ch 2, pp 885-889)

Control in the animal sector

The World Organisation for Animal Health (OIE), with the Food and Agriculture Organization (FAO) and WHO, builds control on three pillars (Jugal Kishore, pp 663-665):

- **Early detection and warning** — strengthened veterinary services, animal-health surveillance tailored to the species at risk, the biosecurity level and the frequency of poultry contact with wild birds, feeding an OIE early warning system.
- **Transparent notification** — member countries report to OIE; there is no official disease-free status for avian influenza, and countries self-declaring freedom must comply with the Terrestrial Animal Health Code and report zero annually.
- **Rapid response** — contingency planning, **zoning and compartmentalisation, culling, strategic vaccination, decontamination and disposal**, wildlife measures, food safety, import/export controls and **compensation** to owners.

Operational specifics to reproduce:

- **Culling** — all poultry of the infected farm and of every farm within a **3 km radius**; all farms within a **10 km radius** must be tested for the virus (Suryakantha, Ch 20, p 353). In India, culling

is done within a 3 km radius of infected villages, with mopping-up and sanitisation by the Animal Husbandry Department (Jugal Kishore, p 663).

- ▶ **Disposal** — carcasses of suspected and confirmed poultry cases should preferably be incinerated, or buried deep using **lime and soil in a 1 : 3 ratio** (Red Book, Ch 2, pp 885-889).
- ▶ **Biosecurity** — movement control, natural and artificial boundaries, control of human traffic, and no introduction of birds of unknown disease status into the flock (Red Book, Ch 2, pp 885-889).
- ▶ **Laboratory confirmation in birds** — two-step serology: agar gel immunodiffusion or ELISA for the type A group antigen, then the **haemagglutination inhibition test** for the subtype, using OIE-certified reagents (Jugal Kishore, pp 664-665).
- ▶ **Vaccination of birds is strategic, not blanket** — widespread poultry vaccination is inadvisable because vaccinated birds carry antibody and an epidemic will then not be suspected until humans start dying (Suryakantha, Ch 20, p 353). Available options are inactivated monovalent (H5 or H7), bivalent (H5 + H7), combination and live recombinant fowlpox-H5 vaccines (Jugal Kishore, p 665).
- ▶ **Market reform** — discourage direct marketing of live poultry to consumers in areas experiencing outbreaks (Jugal Kishore, p 665).

Control in humans and in the health system

Target group	Measures
Cullers and poultry workers	Protective clothing and equipment against infection, plus antiviral drugs as prophylaxis (Jugal Kishore, p 665). Chemoprophylaxis — oseltamivir 75 mg once daily for at least 7 days, begun immediately or at least within 2 days of exposure, and continued for up to 6 weeks (Red Book, Ch 2, pp 885-889)
General public	Regular hand washing with proper drying, respiratory hygiene, early self-isolation, avoiding close contact with sick people and avoiding touching eyes, mouth and nose (Park 28e, Ch 5, p 186); avoid contact with poultry in affected areas (Suryakantha, Ch 20, p 352)
Food safety	Cook poultry and eggs thoroughly to 70 °C throughout; prevent cross-contamination; care in handling shell eggs and raw egg products (Jugal Kishore, p 665)
Case management	Oseltamivir 75 mg twice daily for 5 days in adults, started within 48 hours of onset; paediatric weight bands <15 kg 30 mg, 15–23 kg 45 mg, 23–40 kg 60 mg, >40 kg 75 mg, all twice daily for 5 days; zanamivir 10 mg (2 inhalations) twice daily for 5 days if resistant (Suryakantha, Ch 20, pp 351-352). M2 inhibitors (amantadine, rimantadine) are not relied upon — currently circulating H5N1 strains are fully resistant, though a reassortant pandemic virus might again be susceptible (Red Book, Ch 2, pp 885-889)
Health facility	Hand hygiene as the cornerstone, standard precautions, and personal protective equipment — gloves, gown, mask or respirator, boots and eye protection (Suryakantha, Ch 20, pp 353-354)
Discharge policy (avian influenza)	Adults discharged when afebrile for 7 days with a normal chest X-ray; children under 12 years discharged 21 days after onset (Suryakantha, Ch 20, pp 351-352)

Target group	Measures
Human vaccine	An inactivated H5N1 vaccine licensed in 2007 for ages 18–64 years, two doses 28 days apart (Park 28e, Ch 5, p 186; Jugal Kishore, p 666). Seasonal influenza vaccine does not protect against avian influenza (Sunder Lal, Ch 8, p 470)

Surveillance is the cornerstone of preparedness and response (Red Book, Ch 2, pp 885-889):

Surveillance element	What must be reported / assessed
Priority reportable events	Individuals with, and clusters of, acute respiratory illness on or during admission; unexplained deaths from acute respiratory illness in the community; the same in health-care facilities; rising sales of antivirals, antimicrobials, decongestants or antitussives
Risk assessment for testing	Clinical presentation including unexplained death from acute respiratory illness; occupational exposure (farm, processing plant, culler, live-animal market, laboratory, health worker); residence where poultry deaths are rumoured; travel in the 7 days before onset to a country reporting HPAI outbreaks plus contact within 1 metre of live or dead domestic fowl, wild birds or swine, or contact with a confirmed case

India's response architecture (Jugal Kishore, pp 666-667; Sunder Lal, Ch 8, pp 470-471):

- ▶ Any suspected case is reported **immediately to the National Centre for Disease Control (NCDC), New Delhi**, which deputs an epidemic investigation team.
- ▶ An **Inter-Ministerial Task Force** and a **Joint Monitoring Group** monitor the situation across the health and animal husbandry sectors.
- ▶ Enhanced surveillance at all international airports, travel advisories, and tracking of travellers from affected countries.
- ▶ **Integrated Disease Surveillance Programme (IDSP)** focal points in states report clusters of influenza-like illness (ILI) and of pneumonia; central and state **rapid response teams** investigate.
- ▶ Identified laboratories at national and state level; central supply of oseltamivir, N-95 masks, personal protective equipment and ventilators to states. India has a contingency plan for human cases and has stockpiled **10 million doses of oseltamivir**.
- ▶ Regional commitment through the **Delhi Declaration on Prevention and Control of Avian Influenza and Pandemic Preparedness in Asia** (New Delhi, 28 July 2006).
- ▶ The governing framework is **One World–One Health**, developed by FAO, OIE, WHO and UNICEF for emerging infections at the animal–human interface (Jugal Kishore, p 664) — the multisectoral **One Health** approach linking human, animal and ecosystem health (One Health Approach, Ch 1, p 2).

The impending pandemic threat

- ▶ **The three preconditions for a pandemic** are that a new influenza virus subtype emerges, that it infects humans and causes serious illness, and that it spreads easily and in sustained fashion among humans. H5N1 amply meets the first two; **all prerequisites have been met save one** — efficient and sustained human-to-human transmission (Sunder Lal, Ch 8, p 470).
- ▶ **Two routes to that last step** — *reassortment*, in which genetic material is exchanged between human and avian viruses during co-infection of a human or a pig, and the slower *adaptive*

mutation, increasing the virus's capacity to bind human cells (Sunder Lal, Ch 8, p 470; Mahajan & Gupta 4e, Ch 16, p 187).

- ▶ **No population immunity** — H5N1 has never circulated widely among people, so no one would have immunity should such a pandemic virus emerge, and once a fully contagious virus emerges its global spread is considered **inevitable** (Sunder Lal, Ch 8, p 470; Park 28e, Ch 5, pp 185-186).
- ▶ **The counter-measures are not ready** — vaccines effective against a pandemic virus are not yet available, seasonal vaccine will not protect, and at least **four months** are needed to produce a new-subtype vaccine in significant quantity; antivirals have limited production capacity and prohibitive cost for many countries (Jugal Kishore, p 663; Sunder Lal, Ch 8, p 470).
- ▶ **The virus is entrenched and spreading in new hosts** — silent excretion by domestic ducks perpetuates transmission, and the host range has widened to mammals including dairy cattle (Sunder Lal, Ch 8, p 470; Harrison 22e, Ch 147, p 1192). Avian H7N7 and H7N9 have also caused zoonotic outbreaks (Harrison 22e, Ch 204, p 1576).
- ▶ **The historical analogy is disturbing** — the 1918 "Spanish influenza", believed to have begun as an avian virus that acquired by stepwise adaptive mutation the ability to sustain human-to-human transmission, killed an estimated 40–50 million people, against about 2 million in 1957 and 1 million in 1968. H5N1 resembles it in severity, in its concentration among the young and previously healthy, and in causing primary viral pneumonia without secondary bacterial infection (Red Book, Ch 2, pp 885-889; Sunder Lal, Ch 8, p 468).
- ▶ **In resource-poor settings, non-medical measures are the main line of defence** through the first wave — early-warning surveillance, health education of small poultry farmers, biosecurity and social distancing (Red Book, Ch 2, pp 885-889; Sunder Lal, Ch 8, p 469).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

WHO pandemic phases — labels to write (Sunder Lal, Ch 8, p 471): Phase 1 — predominantly animal-to-animal transmission, few human cases; Phase 2 — an animal influenza virus capable of human infection; Phase 3 — small outbreaks among close populations; Phase 4 — sustained human-to-human transmission; Phase 5 — spread across two or more countries in one WHO region; Phase 6 — plus spread to another WHO region; then post-peak period, possible new wave, and post-pandemic period.

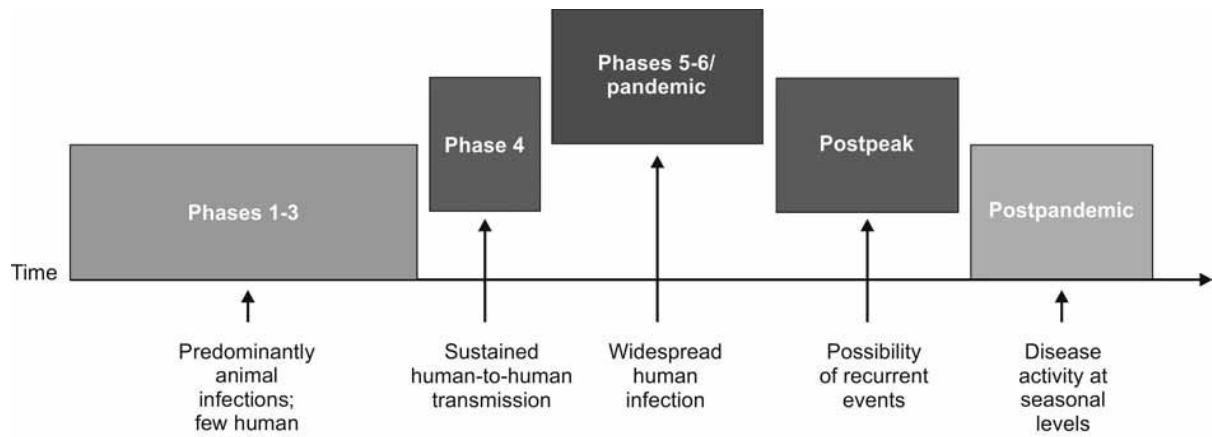
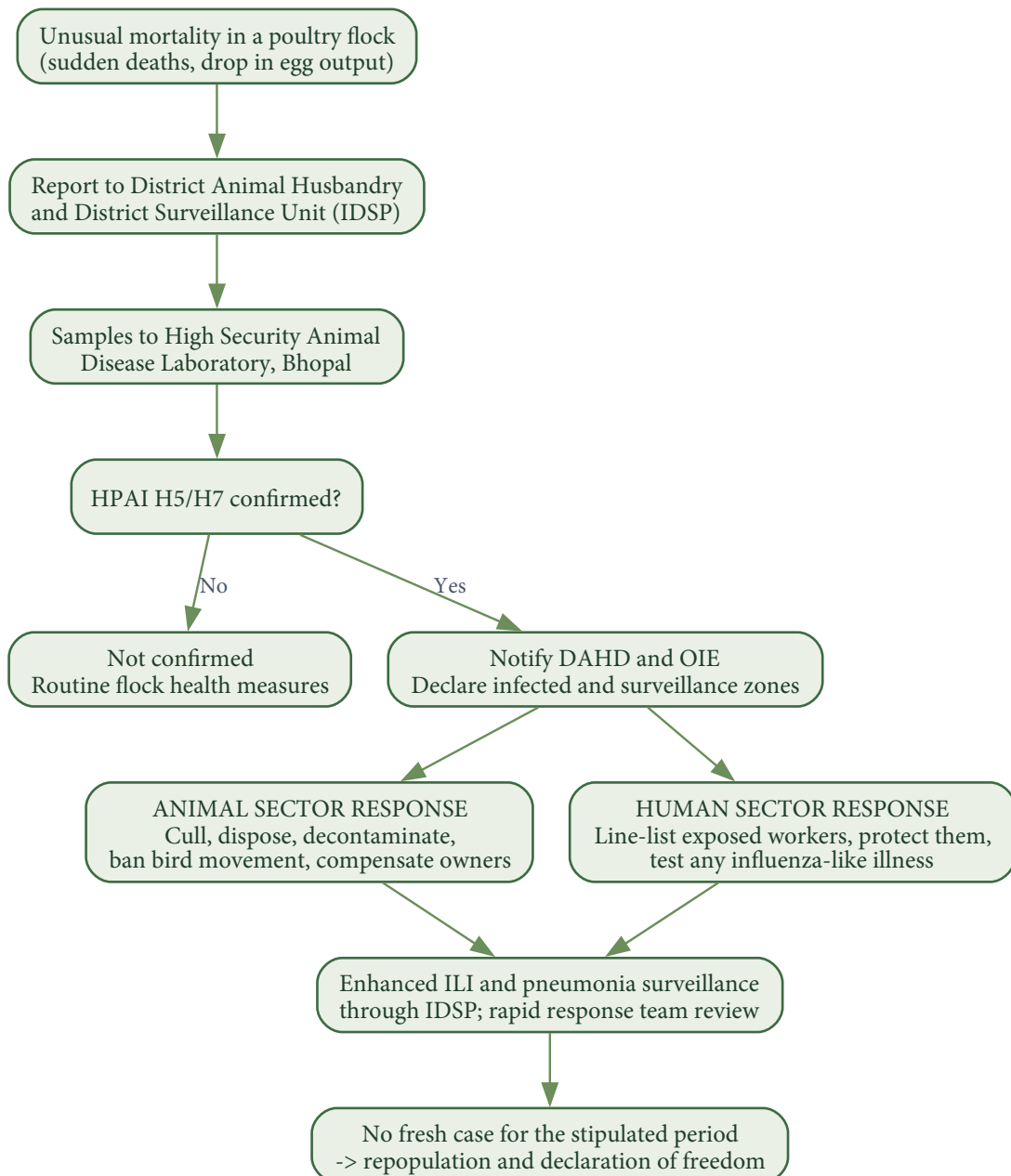
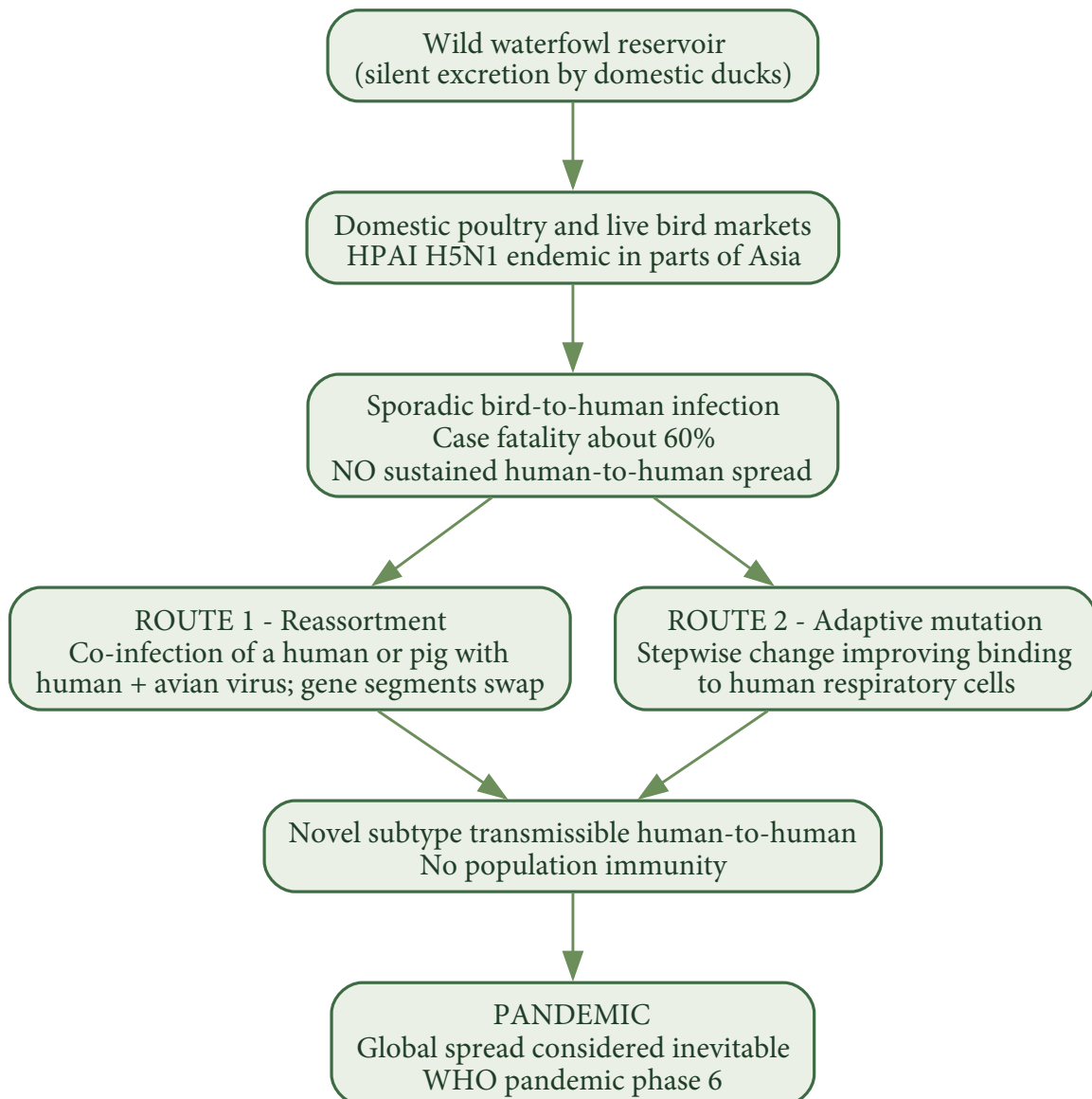


Fig 20.16 — Pandemic influenza phases (2009) (Suryakantha Environment Best, p.355)



Outbreak response to suspected highly pathogenic avian influenza in poultry, India



How avian influenza A (H5N1) could become a pandemic strain

High-yield exam pointers

- ▶ 16 H and 9 N subtypes; only H5 and H7 turn highly pathogenic; only influenza A causes pandemics.
- ▶ Numbers to quote: HPAI kills up to 100% of a flock in 48 h; incubation 2–5 days; case fatality of reported H5N1 about 60%; ~150 million birds destroyed since 2003 and >100 million culled in the January 2004 Asian outbreaks; 478 cases/286 deaths reported to WHO since 1997 (Park 28e).
- ▶ Control radii: cull within 3 km, test within 10 km; carcass burial with lime : soil 1 : 3; cook to 70 °C.
- ▶ Drugs: oseltamivir 75 mg twice daily × 5 days for treatment within 48 h; 75 mg once daily for ≥ 7 days for post-exposure prophylaxis; circulating H5N1 is fully resistant to the M2 inhibitors, so they are not relied upon.
- ▶ India: first human death 11-year-old, Gurugram, 12 July 2021 (NIV Pune); poultry outbreaks first Feb–March 2006, the severe 2008 West Bengal outbreak, and 2021 in nine states; 10 million doses of oseltamivir stockpiled; Delhi Declaration, 28 July 2006.

- ▶ **Three conditions for a pandemic:** new subtype + infects humans with serious illness + spreads easily and sustainably — H5N1 meets the first two only.
- ▶ **Mnemonic for control: CULL** — Cull and compensate, Under-take biosecurity and disposal, Laboratory confirmation and notification, Limit human exposure (PPE, prophylaxis, food safety).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam. Park 28e lists **four** antigenically distinct influenza types — A, B, C and D — and 18 currently identified haemagglutinin subtypes, whereas the Indian texts used for the body of this answer print three types and 16 haemagglutinin with 9 neuraminidase subtypes (Park 28e, Ch 5, pp 184-185). The global H5N1 counts differ by edition: Park 28e quotes 478 cases and 286 deaths since 1997, whereas Harrison 22e (2025) puts the total at about 900 human infections since 2003 with a case fatality near 50 per cent and records the 2024 spread of H5N1 into United States dairy cattle. Sunder Lal's Recent Advances chapter adds two later Indian poultry events — Alappuzha in Kerala in December 2021 and Maharashtra in February 2022 (Sunder Lal, Ch 20, p 786) — and the retrieved corpus carries no Indian outbreak after that date.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO, WOA, EFSA-ECDC avian influenza overviews

Programme and policy updates (India)

- ▶ **Kerala 2024 — clade 2.3.4.4b H5N1 in Indian poultry, and the reason Indian H5N1 is now discussed by clade.** Molecular-clock and phylogeography work at ICMR-NIV Pune traces the clade 2.3.4.4b H5N1 outbreaks reported from Kerala in June 2024 to an introduction into India from East Asia around 2022; that virus is a novel reassortant whose internal genes come from low-pathogenic viruses already reported in Kerala together with some of European origin. India's earliest clade 2.3.4.4b detections were the H5N8 outbreaks in zoological parks in 2016, and these were two separate introductions from East Asia — one traced to China, one to Japan — around 2015. This event postdates every textbook edition used for the body above. [PMID 40930910]
- ▶ **A national HPAI risk map now exists to target surveillance and resource allocation.** Machine-learning models over India's outbreak record — random forest and classification tree the two best performers among eleven algorithms tested — place the highest predicted risk along the coastal areas and wetlands running from the western coast of southern India to the north-eastern states, the belt where paddy cultivation and traditional paddy-duck farming put semi-domestic ducks, wild ducks and migratory birds into the same water bodies. District-level basic reproduction numbers ranged from 0.75 to 3.30, with 52 districts above 1 — the highest Ahmednagar in Maharashtra at 3.30, followed by several districts of Assam. India recorded 312 HPAI outbreaks between 2006 and 2024, the most in any single district being Alappuzha in Kerala (39) and then Jalgaon in Maharashtra (26); they cluster in winter and spring, January alone carrying 32.6 per cent of them and March 17.0, December 13.8 and February 11.5 per cent,

against under 1 per cent each in July and August. The same analysis dates India's first H5N1 outbreak to January 2006 at Nawapur, Maharashtra. [PMID 41102300]

Current national data

- **India has contributed one human A(H5N1) case to each of two EFSA–ECDC reporting rounds, with no Indian case in the two rounds between them.** For 7 June to 8 September 2025, 19 human avian-influenza infections with three deaths were reported from four countries — Cambodia 11 A(H5N1), Bangladesh one A(H5N1), China five A(H9N2) and one A(H10N3), and India one A(H5N1) [PMID 41158675]. The next two rounds carried no Indian case at all: 9 September to 28 November 2025 gave 19 infections with two deaths in four countries — Cambodia, China, Mexico and the United States — in a period when European bird detections were unprecedented for the time of year and HPAI outbreaks were once again reported in vaccinated ducks [PMID 41425765]; and 29 November 2025 to 27 February 2026 gave only 10 infections with no deaths in two countries, Cambodia and China, alongside the first suggested spillover from wild birds to dairy cattle in the Netherlands [PMID 42016297]. The most recent round, 28 February to 4 June 2026, returned to 19 infections with three deaths, now over six countries and territories, with India again contributing one A(H5N1) case, alongside the first detection of A(H9N2) clade G5.5 in European poultry and of A(H5N5) in a polar bear and a walrus in Norway [PMID 42421857]. Every one of these rounds put the risk as low for the general public and low-to-moderate for the occupationally exposed. Quote such counts with the reporting round attached; an undated case number is a fault, and a run of rounds is not an unbroken run of Indian cases.
- **South Asia's twenty-year human denominator is small but recurrent.** Eighteen confirmed human H5N1 cases were reported across the region between 2005 and 2025, from only four countries — Bangladesh, India, Nepal and Pakistan. Bangladesh carries the largest share at 12, spread over 2007, 2010, 2011, 2012, 2014 and four in 2024; India three, one in 2020 and two in 2024; Nepal a single case in 2018; and Pakistan three in 2006, the region's first and so far its last. Quote either the regional total or the country pattern, not both as an arithmetic — the per-country figures as printed add up to 19, one more than the total the same paper states. Bangladesh is also the regional centre of clade 2.3.4.4b persistence, ducks being by far the dominant sampling host there, and genetic clustering linking Bangladesh, India, Sri Lanka and the Maldives points to transboundary movement by migratory birds and poultry trade. Bird losses to avian influenza viruses worldwide since 2005 are put at about 422 million, against roughly 2,634 confirmed human cases and over 100 deaths. [PMID 41857673]

Elimination, surveillance and disease status

- **Two H5N1 clades now co-circulate in the subcontinent, and clade 2.3.2.1a is the one linked to Indian human infection.** A Nepalese turkey isolate from 2023 proved to be a 2.3.2.1a/2.3.4.4b reassortant, four of its eight segments from each clade, with HA, NA and NP coming from the 2.3.2.1a parent — write the four-and-four split rather than a full gene list, because the paper's abstract and its own Results section disagree over whether the fourth 2.3.2.1a segment is M or NS. Its HA shared 99.99 per cent nucleotide identity with A/Victoria/149/2024, the virus recovered from a two-and-a-half-year-old child who returned from Kolkata to Melbourne in February 2024 with severe influenza and respiratory failure needing mechanical ventilation, and

who then recovered fully. Near-identical viruses came from two Indian human cases in 2025, both of them fatal, from two Bangladeshi cases the same year and from two cats in India, so the older clade has not been displaced by 2.3.4.4b; the companion chicken isolate from the same Nepalese outbreaks was pure 2.3.4.4b. [PMID 41305503]

- **The American dairy-cattle episode has rewritten the severity picture, and the textbook 60 per cent case fatality must now be qualified by clade and exposure.** Of 46 laboratory-confirmed human A(H5N1) cases in the United States from March to October 2024, 25 had dairy-cattle exposure, 20 poultry exposure and one no identified exposure; all 45 with an animal exposure had mild illness, none was hospitalised and none died, 93 per cent had conjunctivitis and 33 per cent had conjunctivitis alone, median illness lasted four days, 87 per cent received oseltamivir started a median of two days after onset, and none of 97 household contacts became infected — but protective equipment use was poor (gloves 71 per cent, eye protection 60 per cent, face masks 47 per cent) [PMID 39740051]. The wider panzootic has reached South America and Antarctica for the first time, with sustained mammal-to-mammal transmission on European fur farms, among South American marine mammals and in US dairy cattle [PMID 39317240].

Guideline and legislative updates

- **Oseltamivir resistance in highly pathogenic H5N1 is no longer theoretical.** The H275Y neuraminidase substitution, which confers high-level resistance to oseltamivir and peramivir, was carried by viruses from 8 of 45 sampled poultry farms in the British Columbia outbreak of autumn 2024, with no evidence of oseltamivir or peramivir use on those farms to explain it; the genotype involved, clade 2.3.4.4b D1.1, is the one behind the severe human cases, including a fatal infection in a 65-year-old man in Louisiana and a 13-year-old Canadian admitted to paediatric intensive care with respiratory failure. The resistant virus remained susceptible to zanamivir and to baloxavir and was less fit than the wild type both in cell culture and in mice, so the practical answer is a resistance-testing pathway and a stocked alternative agent rather than an oseltamivir-only stockpile. [PMID 42419257]
- **India's occupational and food-safety law still leaves the exposed workforce short of protection.** Zoonoses are compensable under the Code on Social Security 2020, yet the industries generating the exposure sit outside the "hazardous process" classification of the Occupational Safety, Health and Working Conditions Code 2020 that would trigger mandatory prevention — a compensation-without-prevention gap in a workforce whose seroprevalence includes H9N2 at 6.2 per cent among poultry workers, brucellosis at 15.8 per cent among animal handlers and leptospirosis at 46.3 per cent among high-risk agricultural workers [PMID 42448630]. A Punjab survey of 60 wet-market stalls scored none as excellent on hygiene and biosecurity and 21.67 per cent as poor, and routes the remedy through FSSAI and municipal licensing [PMID 40802022].

Global and WHO developments

- **The amended International Health Regulations are in force and add a tier above the PHEIC.** The amendments adopted by the Seventy-seventh World Health Assembly on 1 June 2024 entered into force on 19 September 2025 and create a "pandemic emergency" — a public health emergency of international concern caused by a communicable disease that has or risks wide spread to and within multiple States, is exceeding or risks exceeding health-system capacity, is causing substantial social or economic disruption, and requires rapid, equitable and enhanced

coordinated international action. An H5N1 event with sustained human transmission would be notified and graded under this framework, not under the 2009 six-phase table drawn above.

[WHO 2024-Jun; WHO 2025-Sep]

- **A pandemic treaty, a replacement risk-assessment tool, and the animal-health body's current name.** The Seventy-eighth World Health Assembly adopted the WHO Pandemic Agreement on 20 May 2025 (124 in favour, none against, 11 abstentions), obliging participating manufacturers to make 20 per cent of real-time production of pandemic vaccines, therapeutics and diagnostics available to WHO, with a Pathogen Access and Benefit-Sharing annex to be adopted before signature opens and 60 ratifications required for entry into force [WHO 2025-May]. WHO now grades candidate viruses with the Tool for Influenza Pandemic Risk Assessment (TIPRA), which scores both the likelihood of a virus acquiring sustained human-to-human transmission and the impact should a pandemic follow — write this alongside, not instead of, the three classical preconditions [WHO 2026]. The body the textbooks call the OIE now publishes as WOAHA, the World Organisation for Animal Health [WOAHA 2026]; and poultry vaccination remains no guarantee of freedom, HPAI outbreaks having once again been reported in vaccinated ducks in Europe during the autumn 2025 reporting round [PMID 41425765].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 5, pp 184-188, 427); Sunder Lal — Textbook of Community Medicine (Ch 8, pp 468-471; Ch 20, p 786); Jugal Kishore — National Health Programmes of India (pp 662-667); Red Book — Public Health and Preventive Medicine, AFMC (Ch 2, pp 885-889; printed as Ch 119, Avian Influenza, pp 853-857); Suryakantha — Community Medicine with Recent Advances (Ch 20, pp 351-355); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e (Ch 16, p 187); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 8, p 107; Ch 9, p 150); Nelson & Williams — Infectious Disease Epidemiology (Ch 15, pp 492-494); Harrison's Principles of Internal Medicine 22e (Ch 147, p 1192; Ch 204, p 1576); One Health Approach (Ch 1, p 2); FAO-OIE-WHO Global Strategy for the Progressive Control of Highly Pathogenic Avian Influenza, 2005; Delhi Declaration on Prevention and Control of Avian Influenza and Pandemic Preparedness in Asia, 2006.

Recent-advances sources: 40930910 — Tracing the possible origins of the clade 2.3.4.4b highly pathogenic avian influenza H5Nx viruses from India, *Virology* 2025, doi:[10.1016/j.virol.2025.110654](https://doi.org/10.1016/j.virol.2025.110654) (abstract-only); 41102300 — Investigating environmental determinants and spatiotemporal dynamics of highly pathogenic avian influenza H5N1 outbreaks in India through machine learning, *Scientific Reports* 2025, doi:[10.1038/s41598-025-96765-2](https://doi.org/10.1038/s41598-025-96765-2); 41158675 — Avian influenza overview June–September 2025, *EFSA Journal* 2025, doi:[10.2903/j.efsa.2025.9702](https://doi.org/10.2903/j.efsa.2025.9702) (abstract-only); 41425765 — Avian influenza overview September–November 2025, *EFSA Journal* 2025, doi:[10.2903/j.efsa.2025.9834](https://doi.org/10.2903/j.efsa.2025.9834) (abstract-only); 42016297 — Avian influenza overview December 2025–February 2026, *EFSA Journal* 2026, doi:[10.2903/j.efsa.2026.10015](https://doi.org/10.2903/j.efsa.2026.10015) (abstract-only); 42421857 — Avian influenza overview March–May 2026, *EFSA Journal* 2026, doi:[10.2903/j.efsa.2026.10196](https://doi.org/10.2903/j.efsa.2026.10196) (abstract-only); 41857673 — Two decades (2005 to 2025) of avian influenza research trends, knowledge gaps and distribution of emerging H5N1 2.3.4.4b clade in South Asia, *One Health Outlook* 2026, doi:[10.1186/s42522-026-00206-7](https://doi.org/10.1186/s42522-026-00206-7) (full text read; its per-country human-case figures sum to 19 against a stated regional total of 18, and the bullet flags that); 41305503 — Zoonotic implications of the co-circulation of clade 2.3.4.4b and 2.3.2.1a H5N1 avian influenza viruses in Nepal in 2023, *Viruses* 2025, doi:[10.3390/v17111481](https://doi.org/10.3390/v17111481) (full text read; the abstract and the Results section disagree on whether M or NS is the 2.3.2.1a-derived segment of the turkey reassortant, so only the four-and-four split and the HA/NA/NP assignment on which both agree are stated); 39740051 — Highly pathogenic avian influenza A(H5N1) virus infections in humans, *New England Journal of Medicine* 2024, doi:[10.1056/NEJMoa2414610](https://doi.org/10.1056/NEJMoa2414610) (abstract-only); 39317240 — The global H5N1 influenza panzootic in mammals, *Nature* 2024, doi:[10.1038/s41586-024-08054-z](https://doi.org/10.1038/s41586-024-08054-z) (abstract-only); 42419257 — Characterization of oseltamivir-resistant A(H5N1) clade 2.3.4.4b, genotype D1.1 variants identified in poultry farms of British Columbia, Canada, *Emerging Microbes & Infections* 2026, doi:[10.1080/22221751.2026.2686474](https://doi.org/10.1080/22221751.2026.2686474); 42448630 — Occupational zoonotic risk at India's human-animal interface, *American Journal of Industrial Medicine* 2026, doi:[10.1002/ajim.70115](https://doi.org/10.1002/ajim.70115) (abstract-only); 40802022 — Assessment of hygiene and biosecurity practices of wet market stalls in Punjab (India), *EcoHealth* 2025, doi:[10.1007/s10393-025-01747-5](https://doi.org/10.1007/s10393-025-01747-5) (abstract-only). WHO — International Health Regulations (2005) amendments adopted 1 June 2024 (WHA77.17), entered

into force 19 September 2025; WHO Pandemic Agreement adopted by the Seventy-eighth World Health Assembly, 20 May 2025; WHO Tool for Influenza Pandemic Risk Assessment (TIPRA), 2026. WOA — organisational naming only (the earlier claim about its poultry-vaccination position has been replaced by the PubMed-grounded vaccinated-duck finding in PMID 41425765).



Genetically modified vectors in malaria control

Asked in: Paper I May 2018 — RGUHS MD Community Medicine

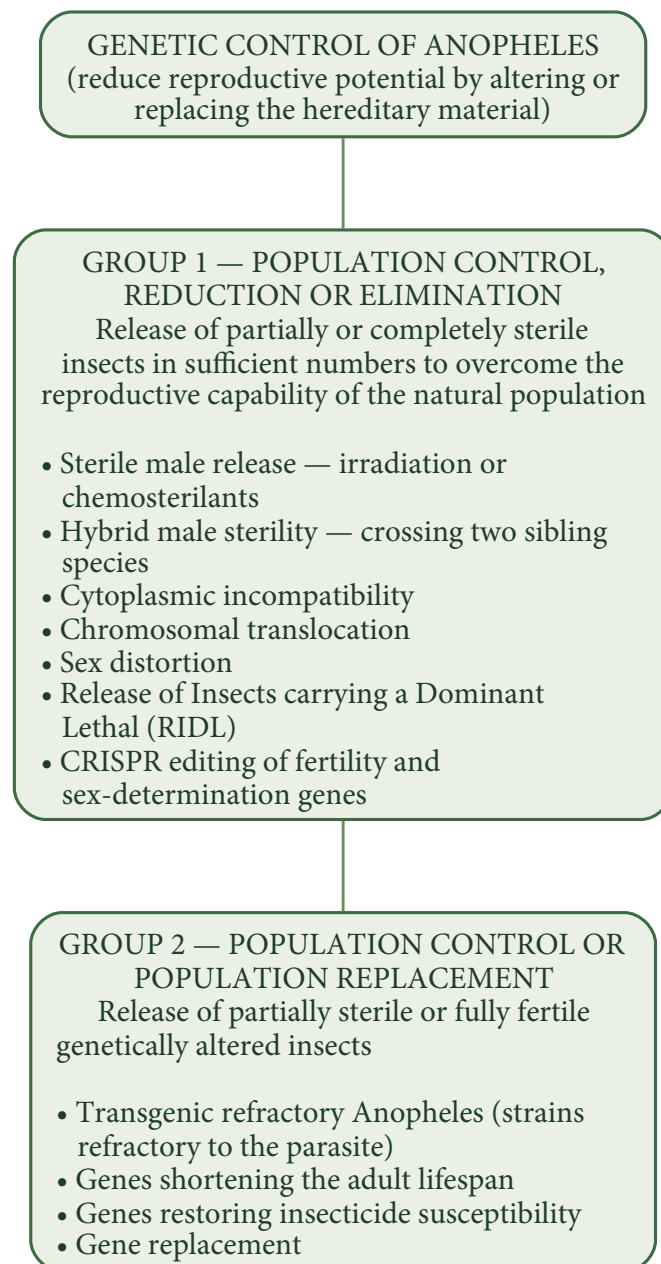
Definition — Genetic control is defined as "the use of any condition or treatment that can reduce the reproductive potential of noxious forms by altering or replacing the hereditary material" (AFMC Red Book 2024, Ch 34, p 1100). A **genetically modified vector (GMV)** is an *Anopheles* mosquito whose genome has been deliberately altered by recombinant DNA or gene-editing technology so that it either fails to reproduce (population suppression) or fails to support development of *Plasmodium* (population replacement). It is a special type of biological control, directed at the adult mosquito (Service — Medical Entomology, Ch 1, p 36; Gilles & Warrell 4e, Ch 6, pp 144-145).

Why this approach is being pursued in malaria

Driver	Grounding
Available vaccines do not interrupt transmission — RTS,S/AS01 and R21/Matrix-M are WHO-prequalified and recommended; both are pre-erythrocytic vaccines that prevent <i>P. falciparum</i> infection and subsequent illness and death in children, and both are not designed to interrupt malaria transmission . First dose from 5 months of age, 3 primary doses at one-month intervals, plus a 4th dose 18 months after the 3rd for RTS,S/AS01 and 12 months after the 3rd for R21/Matrix-M. Neither is in India's routine immunisation schedule. Interrupting transmission therefore still rests on the vector — Park's rationale for vector control being "the only practical approach to malaria control"	Park 28e, Ch 5, pp 333-334; rationale sentence Park 28e, Ch 5, p 322
Insecticide resistance — <i>An. culicifacies</i> (the major rural vector in India) is resistant to DDT and HCH across the Indian subcontinent and to malathion, especially in Gujarat and Maharashtra; of 55 anopheline species showing resistance, 53 are resistant to DDT and 21 are very important malaria vectors	Gilles & Warrell 4e, Ch 6, pp 177-178
Larviciding is logistically impossible over large rural areas — spraying must be repeated every 7-14 days	Service, Ch 1, p 59
New insecticides are expensive and slow to develop — organophosphates and carbamates are unaffordable for most developing tropical countries without external help, and research on new compounds is expensive and increasingly difficult	Gilles & Warrell 4e, Ch 6, p 173

Driver	Grounding
Elimination target — the National Framework for Malaria Elimination in India 2016-2030 (launched February 2016) and the current National Strategic Plan: Malaria Elimination 2023-2027 (NCVBDC, Ministry of Health and Family Welfare, 2023) target interruption of indigenous transmission — all States/UTs in Category 0 — by 2027 and elimination by 2030, which demands tools that outlast insecticide resistance	Park 28e, Ch 7, pp 495-496; Park 28e, Ch 5, pp 324-325
Intrinsic advantages of genetic methods over chemical methods — cheaper, potentially more efficient, and above all not subject to vector resistance	Park 28e, Ch 15, p 927

Classification of genetic control strategies



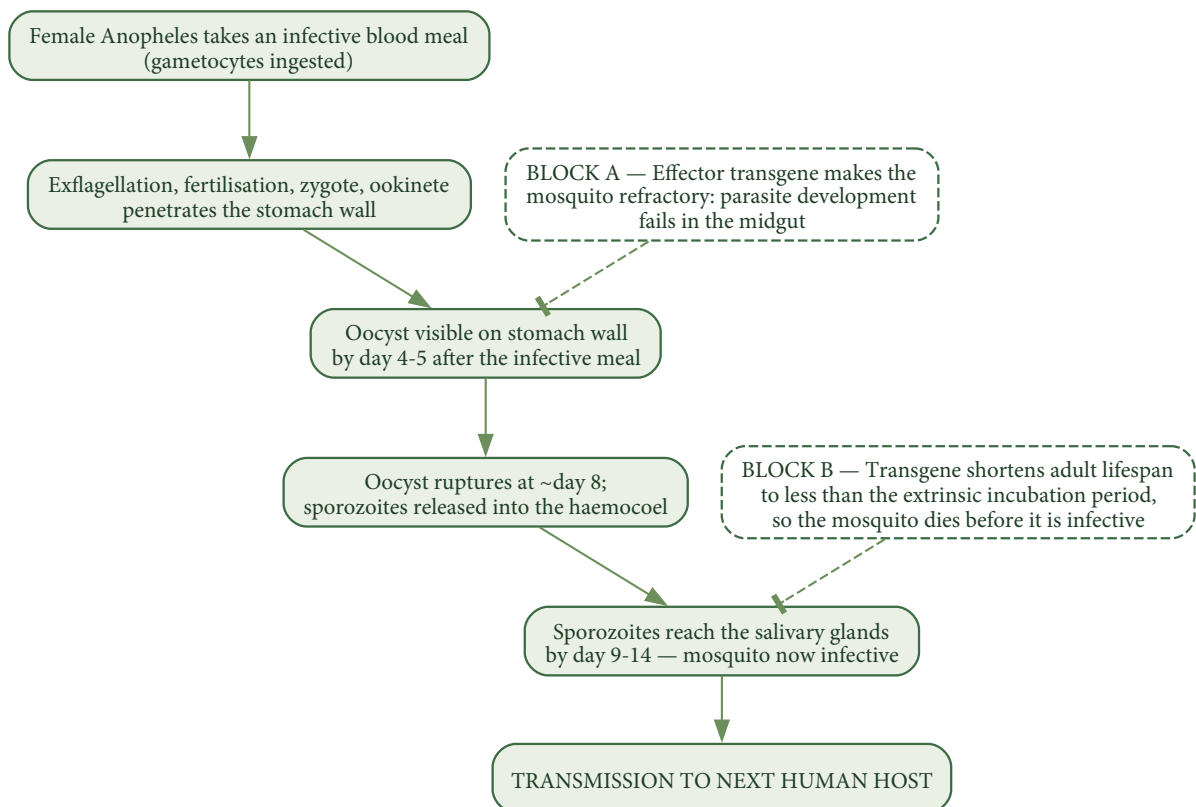
The two general groups of genetic vector control (AFMC Red Book 2024, Ch 34, p 1100) and the techniques under each

Techniques, mechanism and field evidence

Technique	What is altered	Mechanism	Evidence / status
Sterile male release (sterile insect technique, SIT)	No transgene — males sterilised by ionizing radiation or by chemosterilants fed in the insectary	Released males out-compete wild fertile males for mates; inseminated females lay eggs that fail to hatch	<i>An. albimanus</i> in El Salvador, 1970s: ~ 4.36 million chemosterilised males over ~ 15 km² gave >97% reduction in the biting population , but rearing effort per unit area was prohibitive (Service, Ch 1, p 36)
Hybrid male sterility (interspecific crossing)	Hereditary material of a sibling species	Crossing two sibling species of <i>Anopheles</i> gives sterile hybrid males; mass release drives the next generation below the threshold at which transmission is possible	Tried against the <i>A. gambiae</i> complex in Burkina Faso — failed , little mating occurred between sterile males and wild females (Gilles & Warrell 4e, Ch 6, pp 144-145)
RIDL — Release of Insects carrying a Dominant Lethal	Genetically engineered insects homozygous for a dominant lethal construct (instead of irradiation)	Construct passes to offspring, which die before maturing; late-acting lethality is preferred for mosquitoes, which have a density-dependent mortality phase, so killing after that phase maximises suppression	AFMC Red Book 2024, Ch 34, pp 1103-1104
CRISPR gene editing	Key genes regulating fertility and sex determination	Described as safe, self-limiting and scalable genetic control for a specific species	Stated as usable in the field to suppress and even destroy a target species (AFMC Red Book 2024, Ch 34, p 1100)
Transgenic refractory <i>Anopheles</i>	Effector genes blocking parasite development	Mosquito remains alive but cannot support sporogony; may also carry a shortened lifespan	A transgenic <i>An. stephensi</i> created in 2010 had both a reduced life span and complete resistance to infection with <i>P. falciparum</i> (Service, Ch 1, p 36). Still at an experimental phase; malaria-resistant transgenic mosquitoes with higher survival rates could eventually replace parasite-carrying mosquitoes (Oxford Public Health 7e, Ch 97)
Classical genetic methods (Park)	Cytoplasmic incompatibility, chromosomal translocation, sex distortion (suppression); gene replacement (replacement)	Reduce or distort reproductive output, or substitute the wild genotype	Explored in small field trials only; Park places all of them in the " Research Phase " (Park 28e, Ch 15, pp 923, 927)

Technique	What is altered	Mechanism	Evidence / status
Symbiont-based (<i>Wolbachia</i>) — related, not transgenic	Introduction of an intracellular bacterium (family Rickettsiaceae)	Reduces adult longevity and pathogen replication; vectors of dengue and malaria are not naturally infected by <i>Wolbachia</i> . Used either as population alteration or as population reduction	Chiefly validated for <i>Aedes</i> /arboviruses; for malaria, midgut bacteria including <i>Wolbachia</i> inhibit <i>P. falciparum</i> oocyst development (AFMC Red Book 2024, Ch 34, p 1103; Nelson & Williams, Ch 27, p 906)

How a refractory transgenic mosquito interrupts transmission



Point of interruption of the sporogonic cycle by a genetically modified refractory *Anopheles*

Requirements, limitations and unresolved problems

- **Mass rearing on an enormous scale** — the main and most difficult requirement for the success of all genetic methods is production of very large numbers of **healthy, competitive** (though genetically different) mosquitoes, released in the right place and at the right time to mate successfully with wild insects (Gilles & Warrell 4e, Ch 6, pp 144-145).
- **Poor area coverage per unit effort** — in the El Salvador trial enormous numbers had to be reared to control a very small area, which is why interest later lapsed (Service, Ch 1, p 36).
- **Re-invasion from surrounding untreated areas** — in the same El Salvador trial the results seemed promising at first, but the **rapid influx of normal anophelines from the periphery of the area was a setback** (Gilles & Warrell 4e, Ch 6, p 144). Genetic control therefore needs an isolated or fully covered operational unit.

- ▶ **No proven gene-drive mechanism for *Anopheles* in the field** — mechanisms have still to be found to drive the introduced genes through wild populations once transgenic mosquitoes are released (Service, Ch 1, p 36).
- ▶ **Fitness and mating competitiveness** — the Burkina Faso hybrid-sterility attempt failed precisely because released males did not mate with wild females.
- ▶ **Toxicological and mutagenic concerns of chemosterilants** — alkylating agents and other compounds carry implications for mammalian species including humans (Gilles & Warrell 4e, Ch 6, p 144).
- ▶ **Operationally harder than insecticides** — neither genetic approach is simple, and both will be much more difficult to implement than conventional insecticidal methods (Service, Ch 1, pp 36-37).
- ▶ **Not yet ready for scale** — Park concludes these methods are "nowhere near the stage where they can be used large-scale in an effective way" (Park 28e, Ch 15, p 923). Full-scale programmatic implementation of the new tools is **currently not recommended in the absence of strong data on epidemiological impact**, even though entomological effect is promising (AFMC Red Book 2024, Ch 34, p 1103).

Place in the malaria control programme

- ▶ Genetic control is a **supplementary component of integrated vector management**, never a stand-alone measure — no single method controls arthropods, so two or more methods are combined (Park 28e, Ch 15, p 923; Suryakantha, Ch 16 [Medical Entomology], p 133).
- ▶ It is deployed alongside **anti-larval measures** (source reduction, larvicides, insect growth regulators, *Bacillus thuringiensis* var. *israelensis*, larvivorous fish), **anti-adult measures** (indoor residual spraying, space spraying) and **personal protection** (insecticide-treated nets, repellents, protective clothing).
- ▶ Stated **future policy** is to reduce the intensity of chemical selection pressure by cutting the frequency and coverage of insecticide spraying, to strengthen surveillance using remote sensing, geographical information systems and global positioning systems, and to pursue **safer alternatives coupled with intensive research using molecular biology tools for production of Genetically Modified Vectors** (AFMC Red Book 2024, Ch 34, pp 1104-1105).
- ▶ In the Indian setting the target vectors would be **An. culicifacies** (rural) and **An. stephensi** (urban), the two vectors of major importance (Park 28e, Ch 5, p 322).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Diagram 1 — Two groups of genetic vector control. Draw two vertical columns from a common heading "Genetically modified *Anopheles*". Label:

- ▶ Left column: **Group 1 — population control/reduction** → sterile male release → hybrid male sterility, cytoplasmic incompatibility, chromosomal translocation, sex distortion → RIDL (dominant lethal construct) → CRISPR fertility/sex-determination editing → **wild population density falls.**
- ▶ Right column: **Group 2 — population replacement** → transgenic refractory strain → gene replacement, lifespan-shortening and insecticide-susceptibility genes → **population survives but cannot transmit *Plasmodium*.**
- ▶ Common footer box: **Reduced vectorial capacity → interrupted transmission.**

Diagram 2 — Point of block in the mosquito (sporogonic) cycle. Draw the *Anopheles* gut in outline and label, in sequence: gametocytes ingested → ookinete → oocyst on stomach wall (day 4-5) → sporozoites released (day 8) → salivary glands (day 9-14, infective). Mark two crosses: one on the midgut labelled "refractory transgene — parasite development blocked", one before the salivary-gland stage labelled "life-span-shortening transgene — dies before the extrinsic incubation period is completed".

High-yield exam pointers

- ▶ **Definition to reproduce verbatim** — genetic control = "use of any condition or treatment that can reduce the reproductive potential of noxious forms by altering or replacing the hereditary material".
- ▶ **Two groups:** Group 1 = **population control/reduction/elimination** by releasing partially or completely sterile insects; Group 2 = **population control or replacement** by releasing partially sterile or **fully fertile genetically altered** insects. Every technique fits one of the two.
- ▶ **Park's five classical genetic methods:** sterile male technique, cytoplasmic incompatibility, chromosomal translocation, sex distortion, gene replacement — all still in the "**Research Phase**".
- ▶ **Landmark numbers:** El Salvador *An. albimanus*, 1970s — **4.36 million** chemosterilised males over ~15 km², >97% reduction in biting population, defeated by re-invasion from the periphery. Transgenic *An. stephensi*, 2010 — reduced life span plus complete resistance to *P. falciparum*. Cayman Islands 2009 — **3.3 million** sterile transgenic male *Ae. aegypti*, ~80% population reduction (the dengue proof-of-concept for RIDL).
- ▶ **Three advantages over insecticides (Park):** cheaper, potentially more efficient, and **not subject to vector resistance.**
- ▶ **The single limiting requirement:** mass production of very large numbers of healthy, **competitive** mosquitoes released at the right place and right time.
- ▶ **Malaria vaccines do not replace vector control:** RTS,S/AS01 and R21/Matrix-M are WHO-prequalified pre-erythrocytic vaccines from **5 months of age, 3 primary doses one month apart + a 4th dose**, and are **not designed to interrupt transmission.**

- **Indian frame:** *An. culicifacies* (rural), *An. stephensi* (urban); National Framework for Malaria Elimination 2016-2030 and the **National Strategic Plan: Malaria Elimination 2023-2027** target **zero indigenous cases (Category 0) in all States/UTs by 2027** and elimination by 2030.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: the indexed corpus carries the classical genetic-control methods, RIDL, CRISPR-based suppression and transgenic refractory strains, but has no detailed account of engineered CRISPR **gene-drive** constructs (for example doublesex-targeting drives in *An. gambiae*), of the WHO Guidance Framework for testing genetically modified mosquitoes, or of any Indian regulatory pathway (Genetic Engineering Appraisal Committee approvals) for releasing genetically modified vectors — so those are deliberately not stated above; likewise, the statement that the two malaria vaccines are not part of India's routine immunisation schedule is standard current practice rather than a sentence quoted from the retrieved text.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO World Malaria Report 2024, WHO initiative on *Anopheles stephensi* in Africa, NCVBDC National Strategic Plan for Malaria Elimination 2023-2027, ICMR-NIMR

Programme and policy updates (India)

- The **National Strategic Plan for Malaria Elimination 2023-2027** now runs alongside the NFME 2016-2030, the Intensified Malaria Control Programme and the Intensified Malaria Elimination Project, with Global Fund support. An ICMR-NIMR/NCVBDC subnational survey of **37,849 households (177,644 people) across 47 districts and 375 villages in 10 endemic States** found LLIN distribution 85.2% but **regular LLIN use only 73.6%**, and only **57.4%** of febrile persons reaching a government provider within 24-48 hours; lack of formal education and tribal residence were the strongest risk markers. The current programmatic bottleneck is converting coverage into utilisation, not absence of new tools [PMID 42096166].
- Elimination is now driven at **sub-district (micro-level) scale**. In Sonbhadra district, Uttar Pradesh, **18,217 cases** were reported 2017-2024, **API fell from 2.7 (2017) to 0.1 (2024)** with a 95% fall in cases, yet Kulldorff spatial scanning identified a persistent high-risk cluster in **Chopan block** and annual falciparum incidence rose again to 0.09 in 2024 — the residual-focus pattern that makes a resistance-proof tool such as a genetically modified vector attractive [PMID 41871706].

Current national data

- **World Malaria Report 2024** — India accounted for close to **half of estimated cases in the WHO South-East Asia Region in 2023**; India's own programme reporting for 2024 was **255,500 cases and 86 deaths** [PMID 42096166].
- **Species shift in the North-East (NCVBDC data)** — in Mizoram *P. vivax* rose from **23% of cases in 2021 to 47% in 2024 and 42% in 2025**; in Mamit district it was **68% of 2025 cases and 86% of infections in under-fives**. *P. vivax* transmission was perennial while *P. falciparum* stayed monsoon-linked, and chloroquine-resistant *P. vivax* importation from bordering Myanmar is suspected [PMID 42310537].

Elimination, surveillance and disease status

- *An. stephensi*, India's urban vector, is now an invasive vector in Africa — it has spread from East to West Africa within a few years, has been implicated in urban malaria outbreaks and in the spread of drug- and diagnosis-evading *P. falciparum*, and WHO launched an initiative to stop its spread in Africa in 2023. The vector India would target with a GMV is simultaneously a global surveillance priority [PMID 40251402].
- First homing suppression gene drive built in *An. stephensi* — HSDdsx targets the female-specific exon of *doublesex* using two gRNAs with *nanos*-Cas9; drive homozygotes were recessive-sterile in both sexes with intersex phenotypes, resistance-allele formation was very low, and adding a *vasa*-Cas9 line raised drive inheritance to 100% (95% CI 99.5-100%), supporting continuous release at a lower release rate than SIT or fsRIDL [PMID 39856077]. In a parallel *An. stephensi* study all four RNA Pol III promoters tested were active in both germlines with mean inheritance up to 99.8%, but maternally deposited Cas9 generated resistant alleles — hence the move to multiplexed gRNAs [PMID 40251402].

Guideline and legislative updates

- No confined field trial of gene-drive mosquitoes has yet been authorised anywhere. Preparatory sites operate in Tanzania (Transmission Zero), Burkina Faso, Ghana and Uganda (Target Malaria) and São Tomé and Príncipe. Developers identify five roadblocks — an overload of overlapping soft-governance guidance, conflation of small confined trials with large-scale deployment, reliance on advisers lacking technical expertise, uncertainty over which body holds regulatory authority, and questions about financing — and argue small confined trials should be regulated by the host country's own national authority [PMID 40795863].
- Biosafety law versus social licence — Burkina Faso hosted Africa's first release of genetically modified sterile male mosquitoes (non-gene-drive) in 2019, and in 2025 its government suspended Target Malaria's activities citing national interest. A 2025 survey of 874 household heads in Mali (where GMO release falls under Law 08-042/AN-RM of 2008) found only 10.8% aware of GM mosquitoes, 12.2% aware of the biosafety law, 47.7% perceiving risk (chiefly transmission of unknown disease, 95.9%), yet 77% conditional acceptance provided confined testing precedes wide release — the operational meaning of prior community consent [PMID 42376620].

Global and WHO developments

- Gene-drive-capable *An. gambiae* engineered inside an endemic country (Ifakara Health Institute, Tanzania) carried the MM-CP modification — the antimicrobial peptides magainin 2 and melittin expressed from the zinc carboxypeptidase A1 locus. Against *P. falciparum* isolates taken from naturally infected children, oocysts were stunted (median diameter 22.2 µm vs 57.3 µm in wild type) and salivary-gland invasion was nearly abolished (0/42 vs 35% of wild-type in one replicate, 7% vs 55% in another). With a companion Cas9 strain, transgene inheritance was 94.2 ± 4.9% vs 48.3 ± 4.1% in Mendelian controls — non-autonomous drive, deliberately kept separable for staged risk assessment [PMID 41372414].
- Modelled programme impact — across sixteen ~12,000 km² areas of West Africa a female-fertility suppression drive cut vector abundance 71.6-98.4%; added on top of RTS,S vaccination and pyrethroid-PBO nets it averted at least 60% more clinical cases (western Burkina Faso) and

up to **170% more** (Liberia), rising further when all major vector species were targeted. This is the quantitative case that genetic control **supplements** integrated vector management rather than replacing it [PMID 39419965].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 5 — Epidemiology of Communicable Diseases; Ch 7 — Health Programmes in India; Ch 15 — Environment and Health); Service — Medical Entomology for Students (Ch 1); Gilles & Warrell — Essential Malariology 4e (Ch 6 — Rationale and technique of malaria control); AFMC Red Book of Community Medicine 2024 (Ch 34 — Arthropod-borne diseases and medical entomology); Nelson & Williams — Infectious Disease Epidemiology (Ch 27 — The Epidemiology and Control of Malaria); Oxford Textbook of Public Health 7e (Ch 97 — Principles of infectious disease control); Suryakantha — Community Medicine with Recent Advances, Ch 16 (Chapter-15_Medical Entomology), p 133. Programme documents: National Framework for Malaria Elimination in India 2016-2030 (launched February 2016); National Strategic Plan: Malaria Elimination 2023-2027, National Centre for Vector Borne Diseases Control, Ministry of Health and Family Welfare, 2023.

Recent-advances sources: 42096166 — Comprehensive Analysis of Health Service Utilization for Febrile Illness and Malaria in Malaria-Endemic States in India: LLINs as a Pillar of Control Strategies, *Infectious Diseases and Therapy* 2026, DOI [10.1007/s40121-026-01359-8](https://doi.org/10.1007/s40121-026-01359-8); 41871706 — Identification of micro-level malaria hotspots and its implications for achieving malaria elimination target in a remote district of India, *Acta Tropica* 2026, DOI [10.1016/j.actatropica.2026.108066](https://doi.org/10.1016/j.actatropica.2026.108066) (abstract-only); 42310537 — Rising burden of Plasmodium vivax in Mizoram: an emerging threat to malaria elimination in Northeast India, *BMC Infectious Diseases* 2026, DOI [10.1186/s12879-026-13756-y](https://doi.org/10.1186/s12879-026-13756-y) (abstract-only); 40251402 — Optimization of SgRNA expression with RNA pol III regulatory elements in Anopheles stephensi, *Scientific Reports* 2025, DOI [10.1038/s41598-025-98557-0](https://doi.org/10.1038/s41598-025-98557-0); 39856077 — Gene drive-based population suppression in the malaria vector Anopheles stephensi, *Nature Communications* 2025, DOI [10.1038/s41467-025-56290-2](https://doi.org/10.1038/s41467-025-56290-2); 40795863 — Is Gene Drive Research Losing Traction?, *American Journal of Tropical Medicine and Hygiene* 2025, DOI [10.4269/ajtmh.25-0242](https://doi.org/10.4269/ajtmh.25-0242); 42376620 — Risk Perceptions and Acceptability of Genetically Modified Mosquitoes for Malaria Control in Mali, 2025, *Open Research Africa* 2026, DOI [10.12688/openresafrika.16627.1](https://doi.org/10.12688/openresafrika.16627.1); 41372414 — Gene-drive-capable mosquitoes suppress patient-derived malaria in Tanzania, *Nature* 2025, DOI [10.1038/s41586-025-09685-6](https://doi.org/10.1038/s41586-025-09685-6); 39419965 — The potential of gene drives in malaria vector species to control malaria in African environments, *Nature Communications* 2024, DOI [10.1038/s41467-024-53065-z](https://doi.org/10.1038/s41467-024-53065-z). Programme and global sources: WHO World Malaria Report 2024; WHO initiative to stop the spread of *Anopheles stephensi* in Africa, 2023; National Strategic Plan for Malaria Elimination 2023-2027, NCVBDC, Ministry of Health and Family Welfare.

♦ ♦ ♦

Vector bionomics in vector control

Asked in: Paper I 26-May-2025 — RGUHS MD Community Medicine

Definition — Vector bionomics is the study of the life habits of a vector species in relation to its environment: where and in what water it breeds, whose blood it prefers, when it bites, where it rests after feeding, how far it flies, how long it lives, and how susceptible it still is to insecticides. Park lists exactly these as the factors that determine the vectorial importance of a mosquito — density, life span, choice of host, resting habits, breeding habits, time of biting, vectorial capacity and resistance to insecticides (Park 28e, Ch 5, p 322). Vector control becomes rational only when every measure chosen attacks a known bionomic trait of the *local* vector; the same disease therefore needs different measures in different places.

Each bionomic attribute dictates one control decision

Bionomic attribute	What is observed or measured	Control decision it dictates	Illustration
Density (abundance against the critical density below which transmission cannot be sustained)	mosquito density per man-hour catch	whether density reduction alone can interrupt transmission	<i>An. culicifacies</i> needs a high density to propagate malaria, whereas the very efficient <i>An. fluviatilis</i> transmits at a much lower density (Park 28e, Ch 5, p 322)
Life span (longevity)	daily survival probability; parous rate	the single strongest lever — kill the female before sporogony is complete	the vector must survive 10–12 days after an infective feed to become infective; the eradication strategy is to shorten the life span to under 10 days with insecticides (Park 28e, Ch 5, p 322)
Choice of host (anthropophily)	human blood index — the proportion of freshly-fed females whose stomach contains human blood	the value of insecticide-treated nets and personal protection	human blood feeds in <i>An. culicifacies</i> vary from 2–80 per cent; <i>An. fluviatilis</i> is highly anthropophilic (Park 28e, Ch 5, pp 322, 324)
Resting habit — indoor (endophily) or outdoor (exophily)	indoor and outdoor resting collections, kept under constant surveillance	whether indoor residual spraying (IRS) can work at all	the whole concept of malaria eradication rested on the indoor resting habit of most Indian malaria vectors (Park 28e, Ch 5, p 322)
Breeding habit (larval habitat)	larval survey by habitat type	which anti-larval operation to run, and where	<i>An. fluviatilis</i> breeds in moving water, <i>An. sundaicus</i> in brackish water, <i>An. stephensi</i> in wells, cisterns, fountains and overhead tanks (Park 28e, Ch 5, p 322)
Larval feeding behaviour (surface versus bottom feeder)	larval habit in the habitat	choice of larvicide and its formulation	Paris green, a stomach poison, kills mainly <i>Anopheles</i> larvae because they are surface feeders; bottom-feeding larvae are killed only by a special granular formulation (Park 28e, Ch 15, p 926)

Bionomic attribute	What is observed or measured	Control decision it dictates	Illustration
Time of biting	night landing and biting collections	whether bed-nets protect, and when a space spray is timed	Indian anophelines feed nocturnally between dusk and dawn; <i>Aedes</i> is the exception and bites by day (Park 28e, Ch 5, p 322)
Dispersal / flight range	breeding-site mapping around cases	the radius over which control must be applied	flight range varies with species and may reach 11 km, but <i>Ae. aegypti</i> usually flies less than 100 metres, a factor that facilitates its eradication (Park 28e, Ch 15, pp 925–926); <i>Culex quinquefasciatus</i> flies about 3 km, so urban anti-larval work covers an extra 3-km peripheral belt (Park 28e, Ch 5, p 339)
Susceptibility / resistance to insecticides	susceptibility tests and resistance monitoring under integrated vector management	the insecticide chosen and when it is rotated	where DDT resistance has appeared, malathion, propoxur or a synthetic pyrethroid replaces it; resistance has appeared in over 100 arthropod species of public health importance (Park 28e, Ch 15, pp 922, 927; Ch 7, p 498)

Vectorial capacity — bionomics expressed as a number

Vectorial capacity is the combined effect of the density of the vector population, its susceptibility to infection, its life span and the probability of its feeding on man; it is distinct from the physiological capacity to transmit infection (Park 28e, Ch 5, p 322). It is written as $C = ma^2p^n \div (-\log_e p)$ (Gilles & Warrell 4e, Ch 5, p 100). Because the human-biting habit is squared and daily survival is raised to the power of the incubation period, behaviour and longevity dominate the equation — which is why an intervention that shortens vector life expectancy outperforms one that merely reduces numbers.

Term	Bionomic meaning	Control measure that attacks it
m	relative density of female anophelines	source reduction and minor environmental engineering, larvicides, larvivorous fish
a	probability that the mosquito takes a human blood meal on a given day (human-biting habit)	long-lasting insecticidal nets (LLINs), repellents, screening of houses
p	probability of surviving one day (mean life expectancy of human malaria vectors is 20–25 days, average daily death rate 4–5 per cent)	indoor residual spraying and LLINs, both of which shorten adult life
n	days of sporogony (extrinsic incubation period)	not modifiable by a programme; fixed by temperature — optimum 25–30°C, sporogony ceases below 16°C, and <i>P. falciparum</i> takes 9 days at 30°C but 23 days at 20°C (Gilles & Warrell 4e, Ch 5, p 104; Service, p 49)

Davidson's stability index, $a \div (-\log_e p)$, is built from the same two bionomic traits: values above 2.5 indicate stable transmission and below 0.5 unstable transmission (Gilles & Warrell 4e, Ch 5, p 100).

Worked examples — bionomics choosing the measure

- ▶ ***Anopheles stephensi* and urban malaria.** It breeds in wells, cisterns, fountains and overhead tanks (Park 28e, Ch 5, p 322); rooftop tanks are important breeding places in the Indian subcontinent and screening them would prevent breeding (Service, p 37). Because the habitats are few, man-made and fixed, control is anti-larval and civic — larviciding, lidding and screening of tanks, and enforcement of building bye-laws — not household spraying.
- ▶ ***Anopheles culicifacies* and *An. fluviatilis* in rural malaria.** Both rest indoors after feeding and bite at night, so the programme's primary rural measure is indoor residual spraying with DDT 1–2 g of pure DDT per square metre, supplemented by insecticide-treated nets (Park 28e, Ch 15, p 927). Two rounds of spraying are given for DDT and synthetic pyrethroids and three for malathion, timed to cover the whole transmission season (Park 28e, Ch 7, p 498). *An. fluviatilis* breeds in moving water, where larviciding is impractical — a further reason the anti-adult arm carries the load in the foothills.
- ▶ ***Aedes aegypti* and dengue.** It is a domestic container breeder whose population fluctuates with rainfall and water storage, survives best between 16°C and 30°C at 60–80 per cent relative humidity, and is endophagic and endophilic (Park 28e, Ch 5, p 307); it bites by day and rarely flies beyond 100 metres. Control is therefore source reduction in and immediately around the house, temephos in containers holding drinking water at 1 mg of active ingredient per litre (Service, pp 91–92), and ultra-low-volume space spraying in an outbreak to kill infected adults fast. Bed-nets protect only day sleepers, and transovarial transmission of dengue virus into fully developed eggs (Park 28e, Ch 5, p 307) means containers must be removed, not merely emptied. The same short flight range underlies the International Health Regulations requirement that the 400-metre perimeter of ports and airports be kept free of *Aedes* and that the *Aedes aegypti* index be held at zero at ports (Park 28e, Ch 15, p 925).
- ▶ ***Culex quinquefasciatus* and bancroftian filariasis.** It breeds in organically polluted water, so the definitive measure is sanitation and modern sewage disposal (Service, p 92), backed by mosquito larvicidal oil, pyrethrum oil, temephos or fenthion, and minor engineering such as filling ditches and cesspools and maintaining septic tanks and soakage pits (Park 28e, Ch 5, p 339). Its 3-km flight range fixes the operational boundary of an urban anti-larval unit. Because the vector became resistant to DDT, HCH and dieldrin, indoor residual spraying was discontinued for filariasis and personal protection by mosquito nets became the mainstay of prophylaxis (Park 28e, Ch 5, p 339).
- ▶ ***Phlebotomus argentipes* and kala-azar.** The proven Indian vector breeds in cracks and crevices in soil and buildings, in tree holes and caves; only the females bite and their habits are primarily nocturnal, and overcrowding, ill-ventilation and accumulated organic matter favour transmission (Park 28e, Ch 5, p 372). Larval control is effectively impossible because the breeding sites of most sand fly species are unknown (Service, p 117), so control is entirely anti-adult: residual spraying of human dwellings, cattle sheds and other resting places from floor level up to about 6 feet (2 metres), the height at which the flies rest (Park 28e, Ch 5, pp 374–375). Where sand flies rest and bite outdoors, house spraying has little effect (Service, p 117). Under the kala-

azar elimination strategy, DDT — long the first choice because *P. argentipes* was susceptible — is being replaced by a synthetic pyrethroid as resistance emerges (Park 28e, Ch 7, pp 501–502).

Why control fails when bionomics is ignored

- ▶ **Spraying an exophilic vector.** Indoor residual spraying as practised for *Anopheles* is not usually effective against culicines, because most of them do not rest in houses (Service, p 91).
- ▶ **Nets against a day-biter.** Nocturnal protection is worthless where the vector is *Aedes*, which bites during the day (Park 28e, Ch 5, p 322).
- ▶ **Unmonitored resistance.** Insecticidal control alone is no longer fully effective, resistance having appeared in over 100 arthropod species of public health importance (Park 28e, Ch 15, p 922).
- ▶ **Unknown larval habitat.** Anti-larval work cannot be planned for sand flies at all, forcing complete reliance on adult control (Service, p 117).
- ▶ **Man-made breeding.** Borrow pits, garden pools, irrigation channels, dams, roads and bridges create new breeding places — "man-made malaria" (Park 28e, Ch 5, p 322) — so environmental impact appraisal is itself a bionomics-led control measure.

Entomological indicators that keep bionomic knowledge current

Indicator	Definition	Bionomic trait it tracks
Human blood index	proportion of freshly-fed female anophelines whose stomach contains human blood	degree of anthropophilism (Park 28e, Ch 5, p 324)
Sporozoite rate	percentage of female anophelines with sporozoites in their salivary glands	vector infectivity and effective longevity
Mosquito density	number of mosquitoes per man-hour catch	vector abundance against critical density
Man-biting rate (biting density)	average incidence of anopheline bites per day per person, from standardised catches on human bait	man–vector contact
Inoculation rate	man-biting rate multiplied by the infective sporozoite rate	net transmission pressure (Park 28e, Ch 5, p 324)
<i>Aedes aegypti</i> index	percentage of houses in a limited well-defined area on whose premises actual breeding of <i>Ae. aegypti</i> is found, out of houses examined	container-breeding intensity; kept at zero at ports under the International Health Regulations (Park 28e, Ch 15, p 925)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Diagram 1 — Differentiating the three genera (the basis of any bionomics survey).**
Labels: *Anopheles* — eggs laid singly with lateral floats, larva rests parallel to the water surface with a rudimentary breathing tube, adult with maxillary palps as long as the proboscis, proboscis and body in the same straight line, wings spotted. *Aedes* — eggs laid singly without floats, larva at an angle to the surface with a short stout siphon bearing one pair of hair tufts, palps shorter than proboscis, tip of female abdomen pointed. *Culex* — eggs laid in rafts, larva at an angle with a long slender siphon bearing several pairs of hair tufts, palps shorter than proboscis, tip of female abdomen blunt.
- ▶ **Diagram 2 — The mosquito life cycle with its control points.** Labels: egg on water surface, four larval instars, comma-shaped non-feeding pupa, emerging adult, blood meal on host, indoor resting (endophily) or outdoor resting (exophily), oviposition back to the habitat; mark "anti-larval measures" against the egg-larva-pupa limb and "anti-adult measures" against the resting adult.

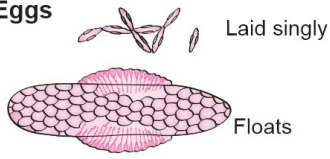
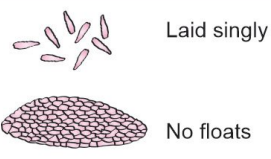
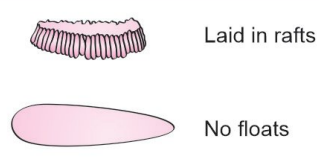
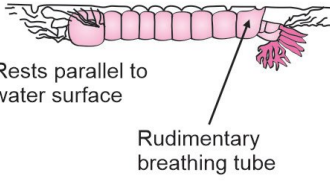
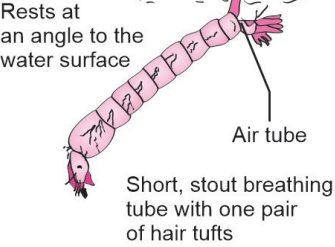
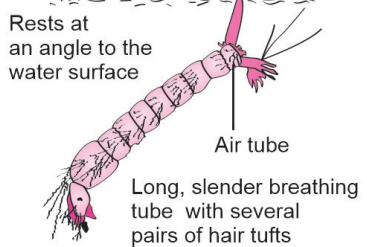
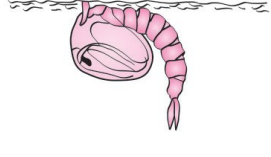
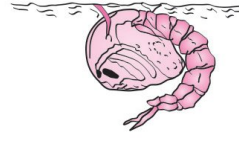
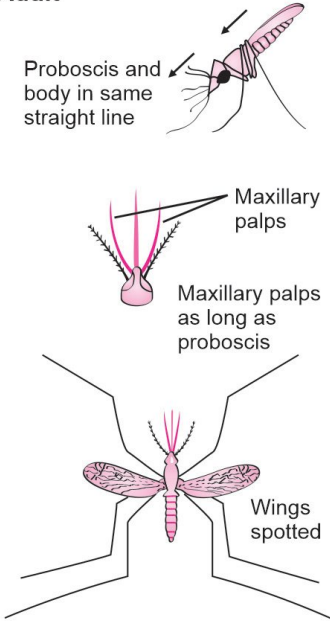
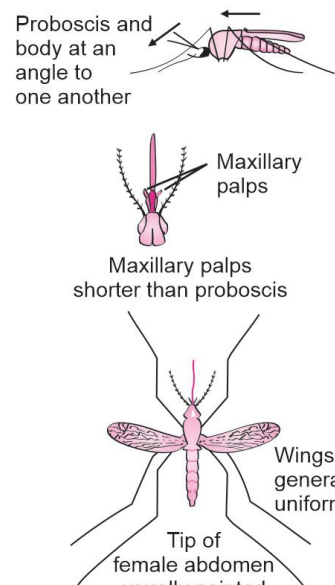
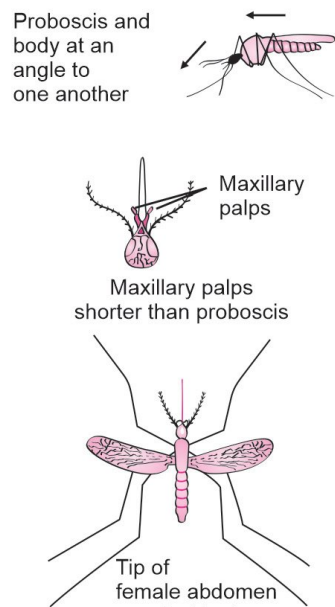
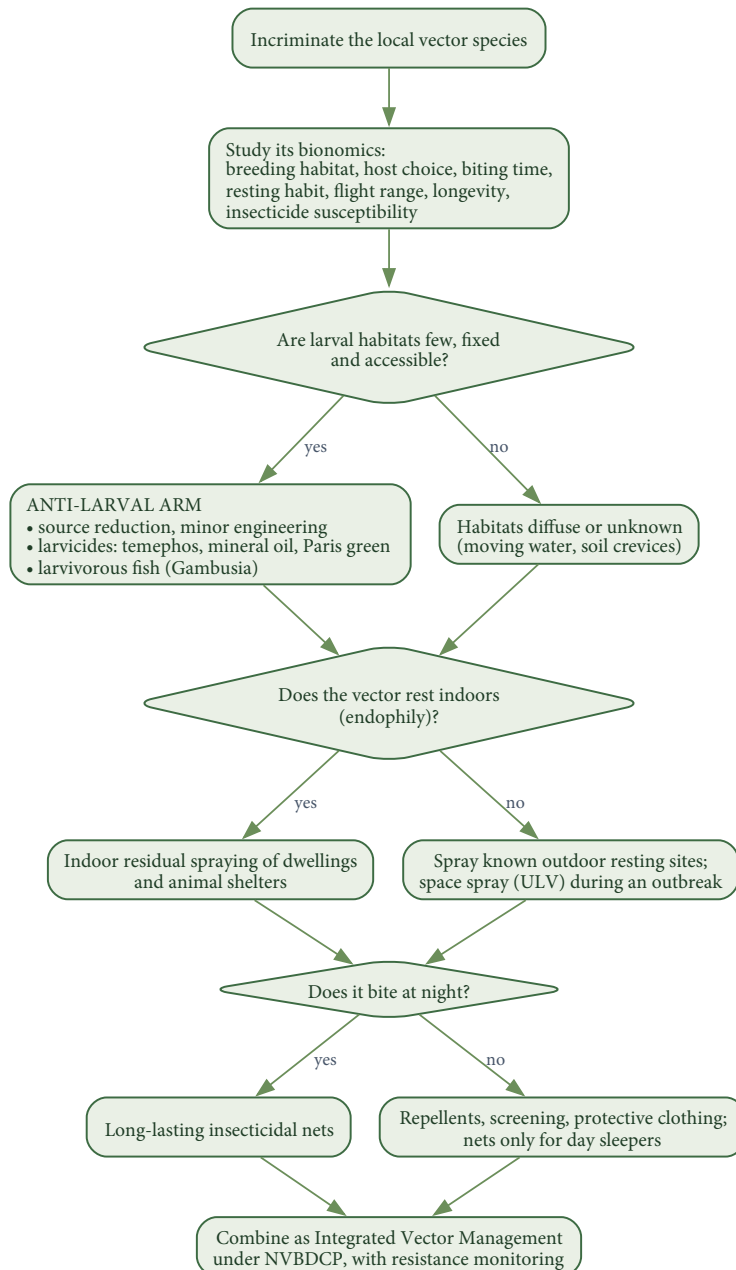
Anopheles	Aedes	Culex
Eggs  Laid singly Floats	Eggs  Laid singly No floats	Eggs  Laid in rafts No floats
Larvae  Rests parallel to water surface Rudimentary breathing tube	Larvae  Rests at an angle to the water surface Air tube Short, stout breathing tube with one pair of hair tufts	Larvae  Rests at an angle to the water surface Air tube Long, slender breathing tube with several pairs of hair tufts
Pupae (differ only slightly) 		
Adult  Proboscis and body in same straight line Maxillary palps Maxillary palps as long as proboscis Wings spotted	Adult  Proboscis and body at an angle to one another Maxillary palps Maxillary palps shorter than proboscis Wings generally uniform Tip of female abdomen usually pointed	Adult  Proboscis and body at an angle to one another Maxillary palps Maxillary palps shorter than proboscis Tip of female abdomen usually blunt

Fig 3.7 — Main features for identification and differentiating *Anopheles*, *Aedes* and *Culex* (Sunderlal, p.126)



Choosing the arm of vector control from the vector's bionomics

High-yield exam pointers

- ▶ The eight bionomic determinants of vectorial importance (Park): **density, life span, choice of host, resting habits, breeding habits, time of biting, vectorial capacity, insecticide resistance.**
- ▶ Vectorial capacity $C = ma^2p^n \div (-\log_e p)$; stability index = $a \div (-\log_e p)$, stable above 2.5, unstable below 0.5.
- ▶ The vector must live **10–12 days** after an infective feed; eradication aims to cut its life span to **under 10 days**.
- ▶ Sporogony optimum **25–30°C**, ceases below **16°C**; parasite development in the vector is optimal at **20–30°C** and stops below **16°C**.
- ▶ Flight ranges that set operational radii: *Ae. aegypti* **<100 m** (and a 400-m insect-free perimeter at ports/airports), *Cx. quinquefasciatus* **~3 km** peripheral belt.

- ▶ Dosages: DDT 1–2 g/m² for indoor residual spraying, 2 rounds/year (3 for malathion); temephos 1 mg active ingredient per litre in potable containers.
- ▶ NVBDCP integrated vector management = indoor residual spraying in high-risk areas + insecticide-treated nets + larvivorous fish + urban anti-larval measures + source reduction and minor environmental engineering.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The retrieved corpus supplies the components of bionomics (breeding, feeding, resting, biting, dispersal, longevity, susceptibility) and their operational consequences in detail, but no textbook in it gives a single verbatim one-line definition of the word "bionomics"; the opening definition therefore states standard entomological usage. The malaria vector control figures quoted are those of the National Vector Borne Disease Control Programme as printed in Park 28e, and the current programme, now working towards malaria elimination, may have revised area-wise spray and net criteria since that edition.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 5 — Epidemiology of Communicable Diseases; Ch 7 — Health Programmes in India; Ch 15 — Environment and Health); Service — Medical Entomology for Students, 5e; Gilles & Warrell — Essential Malariology, 4e (Ch 5 — The Epidemiology of Malaria); Sunder Lal — Textbook of Community Medicine (Fig 3.7, differentiation of *Anopheles*, *Aedes* and *Culex*); National Vector Borne Disease Control Programme integrated vector management strategy and the kala-azar elimination strategy as described in Park 28e.



Principles of arthropod control

Asked in: Paper I 16-Dec-2023 — RGUHS MD Community Medicine

Definition — A **vector** is an arthropod or any other living carrier (e.g., a snail) that transports an infectious agent to a susceptible individual (Park 28e, Ch 3, pp 117-118). The study of arthropods of medical importance is **medical entomology**, and the arthropods concerned fall into three classes — **Insecta** (mosquitoes, houseflies, sandflies, tsetse flies, lice, fleas, blackflies, reduviid bugs, cockroaches), **Arachnida** (hard and soft ticks, trombiculid mites, itch mite) and **Crustacea** (Cyclops) (Park 28e, Ch 15, p 921). **Arthropod control** is the planned reduction of vector density, longevity and man-vector contact below the level at which transmission can be sustained, using environmental, chemical, biological, genetic and newer methods, singly or in combination.

1. The general principles — classification

Park lists **four** general principles, to which a fifth group of newer methods and an overarching integrated approach are added (Park 28e, Ch 15, pp 922-923).

#	Principle	Core content	Chief merit	Chief limitation
1	Environmental control	Source reduction, filling and drainage, water management, waste disposal	Results are likely to be permanent ; the best approach to arthropod control	Costly, slow; needs political support and intersectoral action
2	Chemical control	Organochlorine, organophosphorus, carbamate (and pyrethroid) insecticides	Rapid, large-scale impact; still the operational mainstay in India	Resistance in over 100 species of arthropods of public health importance; environmental contamination
3	Biological control	Larvivorous fish (<i>Gambusia</i>), fungi (<i>Coelomomyces</i>), bacterial larvicides	No environmental pollution; no vector resistance to predators	Slow (days to weeks); species-specific; predators also eat harmless insects
4	Genetic control	Sterile male technique, cytoplasmic incompatibility, chromosomal translocations	Cheap in principle, not subject to insecticide resistance	Tested only in small field trials; nowhere near large-scale effective use
5	Newer methods	Insect growth regulators, chemosterilants, sex attractants (pheromones)	Highly selective, low mammalian toxicity	Still largely at the research phase

(Park 28e, Ch 15, pp 922-923)

2. Environmental control

The best approach, because the results are likely to be permanent (Park 28e, Ch 15, p 922). It works by **environmental manipulation** — denying the arthropod the water, filth or shelter it needs to breed.

- **Elimination of breeding places (source reduction)** — the single most efficient approach to controlling mosquito breeding (Park 28e, Ch 5, p 339).
- **Filling and drainage operations**, and minor engineering works — filling ditches and cesspools, drainage of stagnant water, maintenance of septic tanks and soakage pits (Park 28e, Ch 5, p 339).
- **Carefully planned water management and provision of piped water supply** — a reliable piped supply greatly reduces *Aedes aegypti* breeding in stored water (Service, Ch 1, p 91).
- **Proper disposal of refuse and other wastes**, sanitary disposal of human excreta and animal dung, and general environmental sanitation — the basis of housefly control (Park 28e, Ch 15, pp 928-929).
- **Cleanliness in and around houses**; removal of aquatic weeds (for *Mansonia*, whose larvae depend on plant roots for oxygen) by mechanical removal or herbicides (Park 28e, Ch 5, p 339).
- **Prerequisites** — intensive **health education** of the public and **political support** are essential; a fly-control campaign succeeds only as an organised community effort (Park 28e, Ch 15, p 922).

3. Chemical control

(a) Classification of insecticides by mode of entry (Park 28e, Ch 15, p 937):

Group	Mechanism	Examples
Contact poisons	Kill primarily by contact, penetrating the cuticle	Pyrethrum, DDT, HCH, dieldrin
Stomach poisons	Must be ingested by the insect	Paris green (copper aceto-arsenite), sodium fluoride
Fumigants	Give off vapours with a lethal effect	Hydrogen cyanide, methyl bromide, sulphur dioxide, carbon disulphide

(b) **Classification by chemical family** — organochlorines (DDT, HCH/lindane, dieldrin, methoxychlor), organophosphorus compounds (malathion, fenitrothion, temephos/Abate, fenthion, chlorpyrifos/dursban, dichlorvos), carbamates (propoxur, bendiocarb) and synthetic pyrethroids (deltamethrin, permethrin, lambda-cyhalothrin), some of which are up to ten times as effective as natural pyrethrins (Park 28e, Ch 15, pp 922, 939).

(c) **Methods of application:**

Method	Target	Practical detail
Indoor residual spray (IRS)	Resting adults	DDT 1–2 g of pure DDT per square metre, 1–3 times a year, to walls and resting surfaces; effective for about 6–12 months; where DDT resistance exists, malathion, propoxur (OMS-33) or lindane are used (Park 28e, Ch 15, p 927)
Space spray / fogging / ultra-low-volume spray	Flying adults	Pyrethrum extract as space spray; malathion as ULV to interrupt dengue and encephalitis epidemics; recommended with targeted residual spraying during an outbreak to rapidly reduce adult density (Park 28e, Ch 5, p 355; Ch 15, p 938)
Larviciding — mineral oils	Larvae and pupae	Film cuts off the air supply; 40–90 litres per hectare, applied weekly (the mosquito life cycle is about 8 days); renders water unfit for drinking and kills fish (Park 28e, Ch 15, p 926)
Larviciding — chemicals	Larvae	Paris green as 2 per cent dust (a stomach poison, killing chiefly surface-feeding anopheline larvae); temephos (Abate), fenthion, pyrethrum oil; temephos at 1 mg/litre is also used against <i>Cyclops</i> in guinea-worm control (Park 28e, Ch 15, pp 926, 937; Ch 5, p 339)
Dusts	Lice, fleas, ticks, bugs	DDT 5–10 per cent dust; insecticidal dusts blown into rodent burrows and runways for flea control (Park 28e, Ch 15, p 938; Service, Ch 1, p 198)

(d) **Constraints** — insecticides alone are no longer fully effective because of resistance and environmental contamination; persistent compounds such as DDT are to be replaced progressively by readily **biodegradable** and less toxic compounds (methoxychlor, dursban), although developing countries will depend on organochlorines for some time on grounds of cost (Park 28e, Ch 15, p 922).

4. Biological, genetic and newer methods

Method	Agent / technique	Evidence and caveat
Predators	Larvivorous fish — <i>Gambusia affinis</i> (top minnow), introduced into some 60 countries	Most widely used biological agent; effective only in conjunction with other methods ; will also eat harmless insects (Park 28e, Ch 15, p 927; Service, Ch 1, p 34)
Fungi	<i>Coelomomyces</i> species, <i>Lagenidium</i> , <i>Culicinomyces</i>	Pathogenic to mosquito larvae but none has given satisfactory field control (Park 28e, Ch 15, p 922; Service, Ch 1, p 35)
Bacterial larvicides	<i>Bacillus thuringiensis</i> var. <i>israelensis</i> (Bti); <i>B. sphaericus</i>	Bti is the most effective pathogen — mass-producible, safe to humans and wildlife, largely specific to mosquito larvae; slow-release briquettes act up to about 30 days . <i>B. sphaericus</i> recycles in polluted water and is especially effective against <i>Culex</i> . Strictly microbial insecticides , not true biocontrol agents, as they do not persist by recycling (Service, Ch 1, pp 35, 290)
Other agents	Nematodes (<i>Romanomermis culicivorax</i>), protozoa, viruses	Under study; a fear exists that introducing biological agents may itself pose a hazard to man (Park 28e, Ch 15, p 922)
Genetic control	Sterile male technique (ionising radiation, sterile hybrid males, chemosterilants such as apholate and tepa), cytoplasmic incompatibility, chromosomal translocations	Sterile males compete for mates, giving infertile inseminations and non-hatching eggs; about 4.36 million chemosterilised <i>Anopheles albimanus</i> males were released in El Salvador in the 1970s. Tested only in small field trials (Park 28e, Ch 15, p 923; Service, Ch 1, p 36)
Insect growth regulators (IGRs)	Juvenile hormone analogues (methoprene, pyriproxyfen); chitin synthesis inhibitors (diflubenzuron, novaluron)	Prevent larva-to-pupa or pupa-to-adult development, or kill larvae as they moult (Service, Ch 1, pp 91, 294)
Sex attractants	Pheromone traps	Research phase; potentially cheaper and not subject to vector resistance (Park 28e, Ch 15, pp 923, 927)

5. Operational framework — the three-pronged attack

Whatever the arthropod, control operations are grouped as follows (Park 28e, Ch 15, p 926):

Prong	Components	Key specifications
1. Anti-larval measures	(a) Environmental control (b) Chemical control (c) Biological control	In India, anti-larval work is restricted to urban areas plus a 3-km peripheral belt , because the flight range of <i>Culex quinquefasciatus</i> is about 3 km (Park 28e, Ch 5, p 339)
2. Anti-adult measures	(a) Residual sprays (b) Space sprays (c) Genetic control	IRS reduces vector longevity ; once begun it must be continued, as discontinuation has repeatedly led to resurgence, and it needs exhaustive coverage (Park 28e, Ch 5, p 332)

Prong	Components	Key specifications
3. Protection against bites	(a) Mosquito net (b) Screening (c) Repellents	Net: white material for easy detection of mosquitoes, rectangular pattern, mesh aperture not exceeding 0.0475 inch (about 150 holes per square inch). Screening: copper or bronze gauze of 16 meshes to the inch , aperture not larger than 0.0475 inch — costly but excellent results. Repellents: diethyltoluamide (DEET) , active against <i>Culex fatigans</i> for 18–20 hours ; also indalone, dimethyl phthalate, dimethyl carbate, ethyl hexanediol — chief drawback is the short duration of protection (Park 28e, Ch 15, pp 927-928)

Additional community-level measures are **zooprophylaxis**, protective clothing and mosquito coils/mats; against ticks and mites the best repellents are indalone, diethyltoluamide and benzyl benzoate, with training of exposed workers in self-examination and prompt tick removal (Park 28e, Ch 5, p 332; Ch 15, p 935).

6. Insecticide resistance, integrated vector management and evaluation

Resistance is the ability of arthropods to tolerate doses of insecticide that would be lethal to the majority of normal individuals (Service, Ch 1, p 294). It has appeared in **over 100 species of arthropods of public health importance** (Park 28e, Ch 15, p 922). Among anophelines, WHO recorded **55 species** resistant to one or more insecticides — 53 to DDT, 27 to organophosphorus compounds, 17 to carbamates and 10 to pyrethroids, with 16 species resistant to three or four chemical groups and 21 of the 55 being important malaria vectors (Gilles & Warrell 4e, Ch 6, p 177). **Double resistance** (DDT plus HCH) and **cross-resistance** between most carbamates and most organophosphorus compounds occur, and house-spraying may select **exophilic** populations that rest outdoors (Park 28e, Ch 15, p 939; Service, Ch 1, p 60). Periodic susceptibility testing is therefore essential so that only effective insecticides are applied.

Integrated vector management (IVM) is the utilisation of **all appropriate technological and management techniques to bring about an effective degree of vector suppression in a cost-effective manner** (Gilles & Warrell 4e, Ch 6, pp 179-180). Its essential requirements are:

- ▶ Each method must **favour, or at least not interfere with**, the others (e.g., pond larviciding combined sequentially with source reduction and larvivorous fish).
- ▶ Adequate knowledge of the **biology, ecology and behaviour** of the vector, of local agricultural practice and of aquatic life, to avoid costly mistakes.
- ▶ **Human safety** in the choice and handling of insecticides.
- ▶ **Community involvement** through well-planned, sustained measures related to local socio-economic conditions (Gilles & Warrell 4e, Ch 6, p 180).

In India, the National Vector Borne Disease Control Programme (NVBDCP) delivers IVM as biological, chemical and environmental interventions of proven efficacy, separately or in combination, with optimal use of resources; it explicitly includes safe use of insecticides and **monitoring of insecticide resistance**. Its three components are **indoor residual spray** against adults, **anti-larval measures** (chemical, biological, environmental), and **personal protection** with bed-nets including insecticide-treated nets. IRS is the primary method in rural areas and anti-larval measures in urban areas; high-risk

villages receive IRS and insecticide-treated nets with **coverage above 80 per cent**, and high-risk areas are re-defined **at least annually** (Park 28e, Ch 7, p 498).

Evaluation of an arthropod control programme uses entomological indices rather than case counts alone:

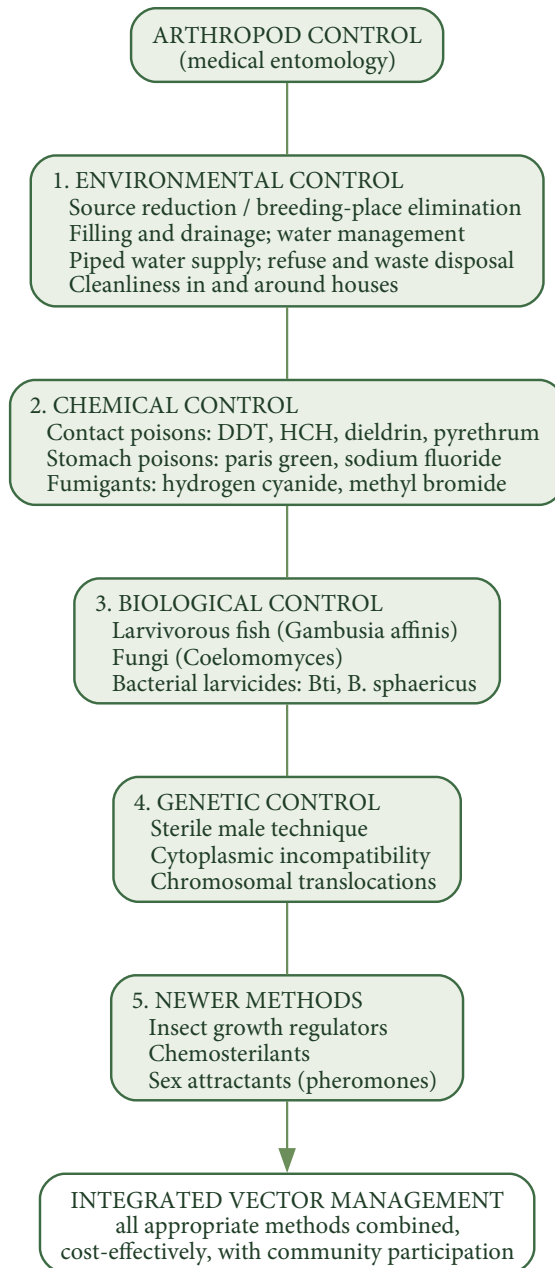
Index	Definition
Mosquito density	Number of mosquitoes per man-hour catch
Man-biting rate	Average incidence of anopheline bites per day per person, from standardised catches on human bait
Human blood index	Proportion of freshly-fed female anophelines whose stomach contains human blood (degree of anthropophilism)
Sporozoite rate	Percentage of female anophelines with sporozoites in the salivary glands
Inoculation rate	Man-biting rate $\times$ infective sporozoite rate
<i>Aedes aegypti</i> index	Percentage of houses, in a limited well-defined area, on the premises of which actual breeding of <i>Aedes aegypti</i> is found; kept at zero at ports , which under the International Health Regulations are kept free of vector breeding for 400 metres around their perimeters
Cheopis index	Average number of <i>Xenopsylla cheopis</i> per rat; a value greater than 1 indicates potential explosiveness should a plague outbreak occur

(Park 28e, Ch 5, pp 324, 360; Ch 15, p 925)

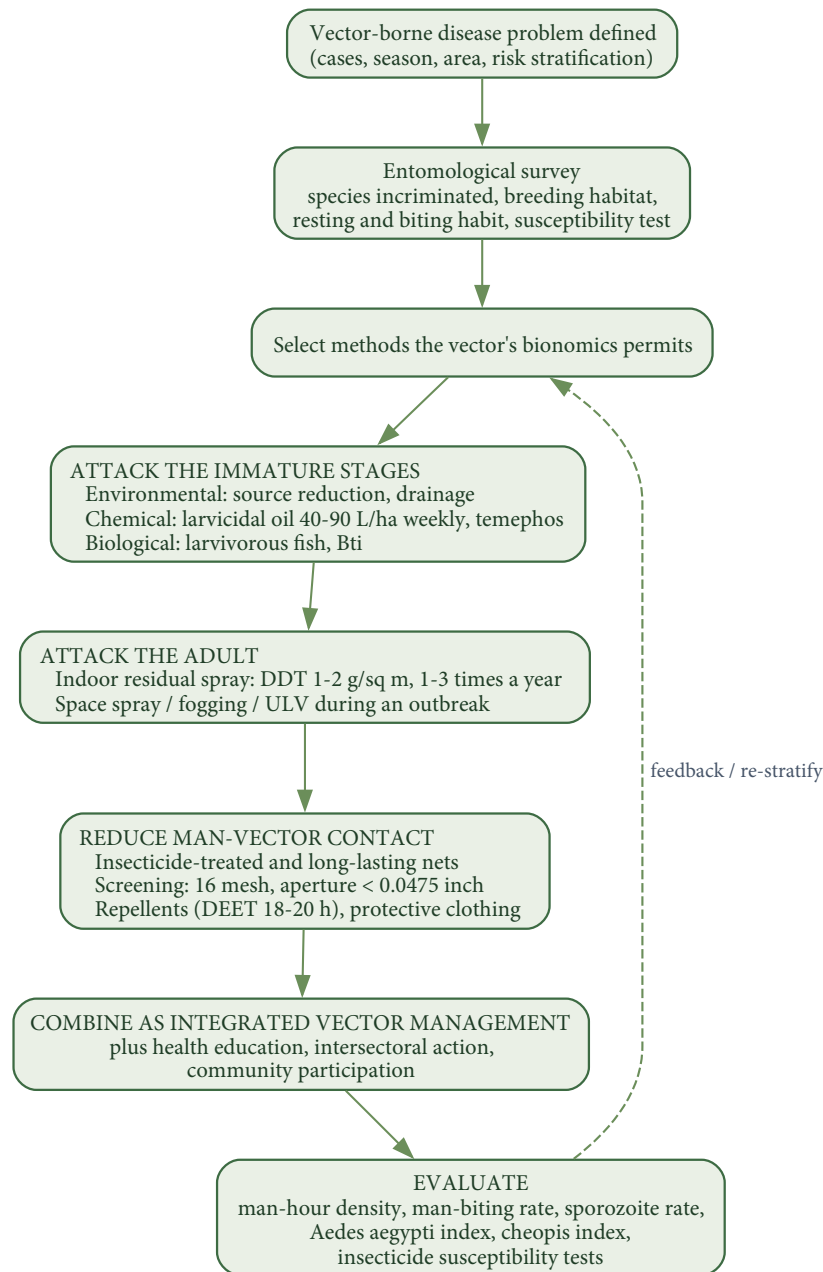
DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Diagram 1 — The three-pronged attack on the vector life cycle (draw the aquatic cycle as a circle and label the three arrows striking it):

- ▶ Egg → Larva → Pupa (aquatic stages, in water) — arrow labelled **ANTI-LARVAL MEASURES: environmental / chemical / biological**
- ▶ Adult mosquito (resting on wall, flying) — arrow labelled **ANTI-ADULT MEASURES: residual spray, space spray, genetic control**
- ▶ Man ↔ Adult vector contact line — arrow labelled **PERSONAL PROTECTION: bed-net, screening, repellent**
- ▶ Side label: **Source reduction = permanent; insecticide = temporary**



The five general principles of arthropod control and their operational content



Operational pathway of a vector control programme, from vector incrimination to evaluation

High-yield exam pointers

- **Park's four principles, in order:** Environmental → Chemical → Biological → Genetic control; plus newer methods (insect growth regulators, chemosterilants, sex attractants) and the **integrated approach**. Environmental control is the **best** because results are **permanent**.
- **Three-pronged operational grouping:** anti-larval (environmental / chemical / biological) — anti-adult (residual spray / space spray / genetic) — protection against bites (net / screening / repellents).
- **Numbers to write:** DDT residual spray 1–2 g/m², 1–3 times a year (effective 6–12 months); mosquito larvicidal oil 40–90 litres/hectare weekly; net and screen aperture ≤ 0.0475 inch, screen gauze 16 mesh/inch; DEET active 18–20 hours; Bti briquettes act up to 30 days; *Culex quinquefasciatus* flight range 3 km.

- **Resistance:** over 100 species of arthropods of public health importance are resistant (Park 28e); among anophelines, 55 species resistant to one or more insecticides, 53 of them to DDT.
- **IVM definition (write verbatim):** "utilisation of all appropriate technological and management techniques to bring about an effective degree of vector suppression in a cost-effective manner."
- **NVBDCP IVM** = indoor residual spray + anti-larval measures + insecticide-treated nets; IRS is primary in **rural** areas, anti-larval measures in **urban** areas; high-risk villages need **>80 per cent** coverage, re-defined at least **annually**.
- **Evaluation indices:** man-hour density, man-biting rate, human blood index, sporozoite rate, inoculation rate, *Aedes aegypti* index (zero at ports; ports vector-free for 400 m), cheopis index (>1 = potential explosiveness in plague).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park 28e retains DDT-based figures and the "over 100 resistant species" statement; the National Vector Borne Disease Control Programme has since been renamed the **National Center for Vector Borne Diseases Control (NCVBDC)**, and long-lasting insecticidal nets have largely replaced conventionally treated nets in the field, but the textbook principles, dosages and indices reproduced above are the ones the RGUHS examiner expects.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, Cochrane Library

Programme and policy updates (India)

- **Triple-drug MDA (IDA) is now the accelerator regimen for lymphatic filariasis.** After WHO endorsed ivermectin + DEC + albendazole in 2017, India committed to an accelerated elimination plan in June 2018 and rolled IDA out from December 2018 in five districts — Arwal (Bihar), Simdega (Jharkhand), Nagpur (Maharashtra), Varanasi (Uttar Pradesh) and Yadgir (Karnataka) — completing all five by November 2019, the first country in South-East Asia to do so. The first round was well tolerated with no serious adverse events, but coverage exposed the urban weak point: **81 per cent in the rural districts against 39 per cent in the urban ones**. India carries about **40 per cent of the global LF burden**, with **272 endemic districts across 16 states and 5 union territories** and roughly **670 million people at risk**; the national deadline slipped from 2015 to 2017 and now aligns with the global **2030 target** [PMID 35292580]. The Indian community trial behind the policy (Yadgir, Karnataka; 4,758 receiving IDA versus 4,160 receiving DEC + albendazole) showed IDA cleared microfilaraemia better, with mild adverse events in 8.3 versus 6.4 per cent and no serious events [PMID 33591979].
- **Countrywide vector bionomics and susceptibility surveillance (13 districts, 8 states, 2017–2020)** is the current operational evidence base for IVM as India targets **malaria elimination by 2030**. *An. culicifacies* showed man-hour density 3.1–15.9, human blood index 0.001–0.220 and sporozoite rate 0.06–4.24 (no infected vectors at all in Madhya Pradesh). It was resistant to **DDT in every district tested (<39 per cent mortality)** and to **malathion (49–78 per cent mortality) in every district except Angul (Odisha)**, which alone stayed malathion-susceptible; **deltamethrin resistance appeared in Maharashtra, Gujarat, Haryana and Karnataka**, susceptibility being retained only in Sidhi (Madhya Pradesh) and Angul — pyrethroid resistance is rising in the

primary vector, and a possible shift towards outdoor resting was seen in Madhya Pradesh [PMID 38885497].

Current national data

- **Pan-India dengue serotype surveillance, 2023–2025** — 6,889 dengue-positive sera tested by real-time RT-PCR through 45 of the 165 Virus Research and Diagnostic Laboratories (VRDLs) across 25 states and union territories. Every region showed co-circulation of multiple serotypes, **eight states carried all four**, and **DENV-2 predominated**; concurrent multi-serotype infection was associated with more thrombocytopenia, arthralgia and haemorrhagic manifestations than mono-serotypic infection. Directly relevant to dengue vaccine introduction strategy [PMID 42323159].
- **Nationwide sand fly molecular xenomonitoring (January 2022 – December 2023, five biogeographical zones)** aligned the National Kala-Azar Elimination Programme with the WHO NTD roadmap 2021–2030. *Phlebotomus argentipes* from Bihar, West Bengal and Kerala carried high *Leishmania donovani* kDNA loads, and **parasite DNA was also found beyond the established vector** — *P. sergenti* and *P. papatasi* in Rajasthan, *P. longiductus*, *P. major* and *P. bruneyi* in Himachal Pradesh — arguing for xenomonitoring as a routine elimination-phase entomological index alongside the classical ones [PMID 39557939].

Elimination, surveillance and disease status

- **A unified "triple surveillance" system for Aedes-borne arboviruses is being argued for in India** — dengue is hyperendemic across much of the country, chikungunya has re-emerged markedly since the mid-2000s and Zika has appeared sporadically in several states. Because all three share the same *Aedes* vectors, co-circulation and co-infection confound the present disease-by-disease vertical surveillance and complicate diagnosis, so integrated dengue–chikungunya–Zika surveillance is proposed [PMID 42314658].
- ***Anopheles stephensi*, India's urban malaria vector, has become an invasive threat abroad.** A case-control investigation of the Dire Dawa outbreak in Ethiopia (April–July 2022) found *P. falciparum* infections clustered spatially around index cases in strong association with *An. stephensi* near the household, with sporozoite-positive mosquitoes recovered. The outbreak involved clonal parasite propagation carrying the **kelch13 R622I** artemisinin-resistance marker (14 of 20 lineage-2 infections) together with ***pfhrp2/pfhrp3* deletions** that cause false-negative rapid diagnostic tests — the strongest evidence yet that this container-breeding urban vector can drive urban malaria outbreaks [PMID 37884028].

Guideline and legislative updates

- **Dual-active-ingredient nets have displaced pyrethroid-only LLINs as the recommended net wherever pyrethroid resistance exists.** In the Benin cluster-randomised superiority trial (60 clusters, 216,289 inhabitants), **chlorfenapyr–pyrethroid** nets cut malaria incidence in children over 2 years against pyrethroid-only nets (0.56 versus 1.03 cases per child-year; **HR 0.54, 95% CI 0.42–0.70**), whereas pyriproxyfen–pyrethroid nets did not (HR 0.86, 95% CI 0.65–1.14) [PMID 36706778]. The four-arm Tanzania trial's third-year results confirmed the benefit persists over a 3-year net lifespan: malaria prevalence **22.8 per cent with chlorfenapyr–pyrethroid versus 37.4 per cent with pyrethroid-only** (OR 0.57, 95% CI 0.38–0.86), while PBO–pyrethroid (OR 0.95) and pyriproxyfen–pyrethroid (OR 0.82) did not differ significantly [PMID 37776879]. The

principle to write: mixing a **second, non-pyrethroid mode of action** into the net, not simply re-treating with the same class.

- **Adding IRS on top of ITNs helps only when the spray insecticide has a different mode of action from the net.** Cochrane review of eight studies (six cluster-RCTs plus two quasi-experimental) in sub-Saharan Africa: adding "**non-pyrethroid-like**" IRS (bendiocarb, pirimiphos-methyl, propoxur) reduced parasite prevalence (RR 0.61, 95% CI 0.42–0.88; 4 cluster-RCTs, 16,394 participants; high-certainty evidence), with a smaller and inconsistent effect on incidence (rate ratio 0.86, 95% CI 0.61–1.23; low-certainty). Adding "**pyrethroid-like**" IRS — a group that includes DDT, which shares the sodium-channel target of pyrethroids — gave **no detectable additional benefit** (prevalence RR 1.11, 95% CI 0.86–1.44). This is the direct evidential basis for rotating insecticide classes rather than doubling up on one [PMID 35038163].

Global and WHO developments

- ***Wolbachia* has become an evidence-backed *Aedes* method, in two distinct modes.** Replacement (**wMel**): Cochrane found wMel-infected *Ae. aegypti* deployments probably reduce the odds of virologically confirmed dengue by **77 per cent** (OR 0.23, 95% CI 0.15–0.35; one cluster-RCT, 6,306 participants, moderate-certainty evidence), with wMel prevalence sustained near 95.8 per cent over 27 months [PMID 38597256]. **Suppression (wAlbB males)**: repeated release of *Wolbachia*-infected males renders wild females' offspring non-viable through **cytoplasmic incompatibility** — the textbook's sterile-male principle delivered biologically. A 15-cluster (8 intervention, 7 control) test-negative cluster-randomised trial in Singapore cut adult wild-type *Ae. aegypti* in the intervention arm from a baseline of **0.18 to 0.041 per trap** over months 3–24, against **0.277 per trap** in the control clusters, and achieved **71–72 per cent protective efficacy** against symptomatic dengue (OR 0.28–0.29) [PMID 41671481].
- **Among the newer methods, one clear win, one null and one still pre-field.** A transfluthrin **spatial repellent** in Busia County, Kenya (58 clusters, 24 months, on top of universal ITN coverage and pyrethroid-resistant vectors) reduced the hazard of first malaria infection by **33.4 per cent** (95% CI 11.1–50.1) and of all new infections by 32.1 per cent (15.9–45.2) — the first epidemiological evidence for spatial repellents against malaria in Africa [PMID 39709979]. **Attractive targeted sugar baits** (dinotefuran stations, 70 clusters, Western Zambia, two transmission seasons at ~90–98 per cent coverage) did **not** reduce clinical malaria incidence (IRR 0.91, 95% CI 0.72–1.15) or *P. falciparum* prevalence (OR 0.89) [PMID 39548456]. A homing **gene drive** targeting the female-specific *doublesex* exon in *An. stephensi* reached **100 per cent drive conversion** when paired with a vasa-Cas9 line, with very low resistance-allele formation — still laboratory-stage, but the modern successor to the chromosomal translocations and sterile-male releases in the textbook's genetic-control row [PMID 39856077].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 15 Environment and Health, pp 921-928, 935, 937-939; Ch 5 Epidemiology of Communicable Diseases, pp 324, 332-333, 339, 355, 360; Ch 7 Health Programmes in India, pp 498-499; Ch 3, pp 117-118); Service — Medical Entomology for Students (Ch 1, pp 33-39, 58-60, 91, 159, 198, 286-294); Gilles & Warrell — Essential Malariology 4e (Ch 6 Rationale and technique of malaria control, pp 167, 177-180).

Recent-advances sources: 35292580 — Introduction of Triple-Drug Therapy for Accelerating Lymphatic Filariasis Elimination in India: Lessons Learned, *Am J Trop Med Hyg* 2022, doi:[10.4269/ajtmh.21-0964](https://doi.org/10.4269/ajtmh.21-0964); 33591979 — An open label, block randomized, community study of the safety and efficacy of co-administered ivermectin, diethylcarbamazine plus albendazole vs. diethylcarbamazine plus albendazole for lymphatic filariasis in India, *PLoS Negl Trop Dis* 2021, doi:[10.1371/journal.pntd.0009069](https://doi.org/10.1371/journal.pntd.0009069); 38885497 — Malaria Vector Bionomics: Countrywide Surveillance Study on Implications for Malaria Elimination in India, *JMIR Public Health Surveill* 2024, doi:[10.2196/42050](https://doi.org/10.2196/42050); 42323159 — Pan-India molecular surveillance of dengue serotypes, 2023-2025, *Int J Infect Dis* 2026, doi:[10.1016/j.ijid.2026.108897](https://doi.org/10.1016/j.ijid.2026.108897) (abstract-only); 39557939 — Nationwide cross-sectional surveillance of *Leishmania donovani* in phlebotomine sand flies and its impact on national kala-azar elimination in India, *Sci Rep* 2024, doi:[10.1038/s41598-024-78915-0](https://doi.org/10.1038/s41598-024-78915-0); 42314658 — Coinfections of Chikungunya, Dengue, and Zika Viruses in India: Rationale for a Unified Triple Surveillance Approach, *Am J Trop Med Hyg* 2026, doi:[10.4269/ajtmh.25-0743](https://doi.org/10.4269/ajtmh.25-0743) (abstract-only); 37884028 — Evidence for a role of *Anopheles stephensi* in the spread of drug- and diagnosis-resistant malaria in Africa, *Nat Med* 2023, doi:[10.1038/s41591-023-02641-9](https://doi.org/10.1038/s41591-023-02641-9); 36706778 — Efficacy of pyriproxyfen-pyrethroid LLINs and chlorfenapyr-pyrethroid LLINs compared with pyrethroid-only LLINs for malaria control in Benin: a cluster-randomised, superiority trial, *Lancet* 2023, doi:[10.1016/S0140-6736\(22\)02319-4](https://doi.org/10.1016/S0140-6736(22)02319-4) (abstract-only); 37776879 — Effectiveness of LLINs with pyriproxyfen-pyrethroid, chlorfenapyr-pyrethroid, or piperonyl butoxide-pyrethroid versus pyrethroid only against malaria in Tanzania: final-year results of a four-arm, single-blind, cluster-randomised trial, *Lancet Infect Dis* 2024, doi:[10.1016/S1473-3099\(23\)00420-6](https://doi.org/10.1016/S1473-3099(23)00420-6) (abstract-only); 35038163 — Indoor residual spraying for preventing malaria in communities using insecticide-treated nets, *Cochrane Database Syst Rev* 2022, doi:[10.1002/14651858.CD012688.pub3](https://doi.org/10.1002/14651858.CD012688.pub3); 38597256 — Wolbachia-carrying *Aedes* mosquitoes for preventing dengue infection, *Cochrane Database Syst Rev* 2024, doi:[10.1002/14651858.CD015636.pub2](https://doi.org/10.1002/14651858.CD015636.pub2); 41671481 — Dengue Suppression by Male Wolbachia-Infected Mosquitoes, *N Engl J Med* 2026, doi:[10.1056/NEJMoa2503304](https://doi.org/10.1056/NEJMoa2503304) (abstract-only); 39709979 — Effect of a spatial repellent on malaria incidence in an area of western Kenya characterised by high malaria transmission, insecticide resistance, and universal coverage of insecticide treated nets (AEGIS Consortium): a cluster-randomised, controlled trial, *Lancet* 2025, doi:[10.1016/S0140-6736\(24\)02253-0](https://doi.org/10.1016/S0140-6736(24)02253-0) (abstract-only); 39548456 — Efficacy of attractive targeted sugar bait stations against malaria in Western Province Zambia: epidemiological findings from a two-arm cluster randomized phase III trial, *Malar J* 2024, doi:[10.1186/s12936-024-05175-8](https://doi.org/10.1186/s12936-024-05175-8); 39856077 — Gene drive-based population suppression in the malaria vector *Anopheles stephensi*, *Nat Commun* 2025, doi:[10.1038/s41467-025-56290-2](https://doi.org/10.1038/s41467-025-56290-2).



Emerging and re-emerging zoonotic diseases

Asked in: Paper II 18-Dec-2023 — RGUHS MD Community Medicine

Definition — Zoonoses are "those diseases and infections [the agents of] which are naturally transmitted between [other] vertebrate animals and man" (Park 28e, Ch 5, p 341). More than **150 zoonoses** have been recognised, and in recent years several new zoonotic diseases have emerged, e.g. Kyasanur Forest disease and mpox (Park 28e, Ch 5, p 341). An **emerging zoonosis** is therefore an animal-derived infection newly recognised in man, or one whose incidence, geographical range or host range is expanding; a **re-emerging** disease is one previously controlled easily by chemotherapy and antibiotics that has now developed antimicrobial resistance and often appears in epidemic form (Park 28e, Ch 5, p 427).

Zoonoses matter particularly in India because nearly **68.86% of the population is rural** and lives in close contact with domestic animals, and often not far from wild ones; developing countries also suffer heavier economic losses in animals, meat and milk because of less-developed public health and veterinary services (Park 28e, Ch 5, p 341).

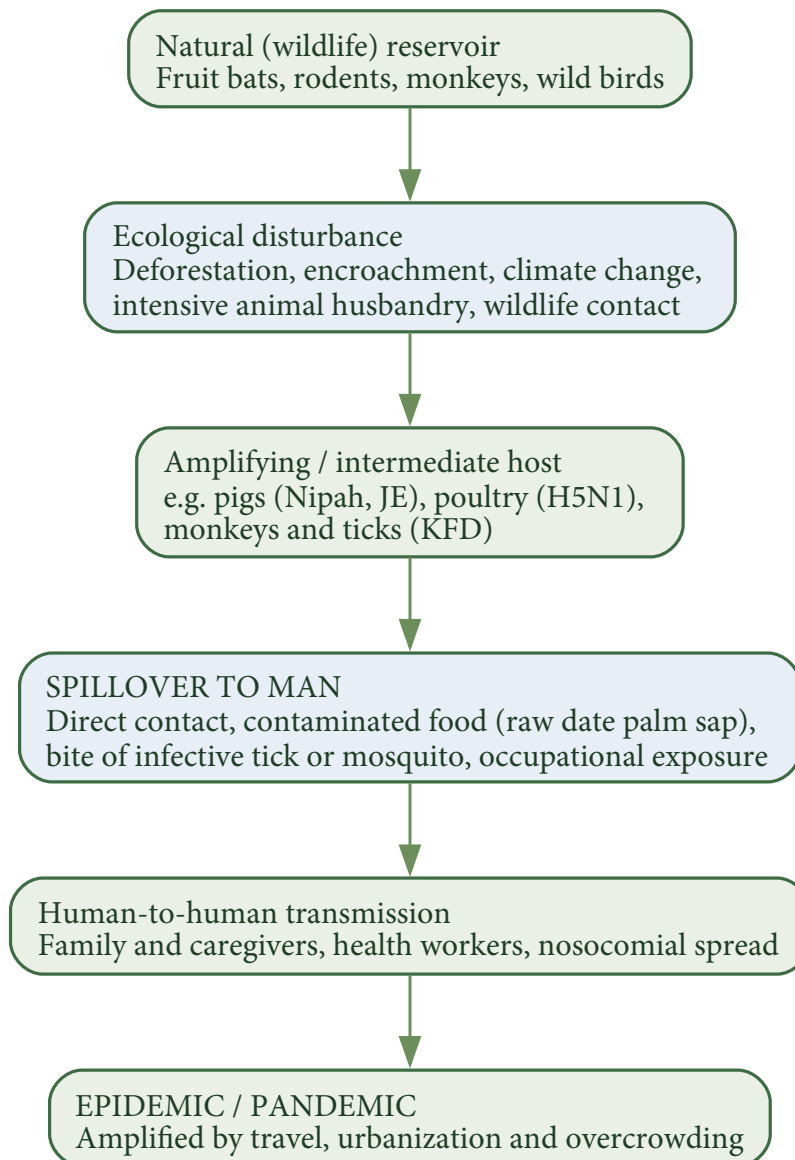
Terms and classification (the frame for naming them)

Term / category	Meaning	Examples
Anthropozoonoses	Infections transmitted to man from lower vertebrate animals	Rabies, brucellosis (Park 28e, Ch 5, p 341)
Zooanthropozoonoses	Infections transmitted from man to lower vertebrate animals	Human tuberculosis in cattle (Park 28e, Ch 3, p 114)
Amphixenoses	Infections maintained in both man and lower vertebrates, transmitted in either direction	<i>T. cruzi</i> , <i>S. japonicum</i> (Park 28e, Ch 3, p 114)
1. Direct zoonoses	Infected vertebrate → susceptible vertebrate by direct contact, fomite or mechanical vector; agent undergoes little or no propagative or developmental change	Rabies, trichinosis, brucellosis (Park 28e, Ch 5, p 341)
2. Cyclo-zoonoses	Need more than one vertebrate host but no invertebrate host to complete the agent's developmental cycle	Human taeniasis, echinococcosis, pentastomida infections (Park 28e, Ch 5, p 341)
3. Meta-zoonoses	Transmitted biologically by invertebrate vectors ; agent multiplies and/or develops in the vector, with an obligatory extrinsic incubation (prepatent) period	Arbovirus infections, plague, schistosomiasis (Park 28e, Ch 5, p 341)
4. Sapro-zoonoses	Have a vertebrate host plus a non-animal developmental site or reservoir — organic matter including food, soil, plants	Larva migrans, some mycoses (Park 28e, Ch 5, p 341)

Most newly emerging zoonoses fall in the **direct** and **meta-zoonotic** categories — that is, bat-, rodent- and tick-associated infections.

Factors driving emergence and re-emergence

Factor	How it operates
Unplanned and under-planned urbanization	Human settlement encroaches on animal reservoirs (Park 28e, Ch 5, p 426)
Overcrowding and rapid population growth	Sustains chains of secondary human-to-human transmission (Park 28e, Ch 5, p 426)
Poor sanitation	Rodent and vector proliferation, contamination of water and food (Park 28e, Ch 5, p 426)
Inadequate public health infrastructure	Delayed detection, weak veterinary and human surveillance (Park 28e, Ch 5, p 426)
Resistance to antibiotics / antimicrobial resistance	Converts controllable zoonoses (e.g. salmonellosis) into re-emerging epidemic disease (Park 28e, Ch 5, p 426; One Health Approach, Ch 1, p 2)
Increased human exposure to vectors and natural reservoirs	Forest work, plantation labour, animal husbandry, bush-meat and raw animal products (Park 28e, Ch 5, p 426)
Rapid and intense international travel	Exports a local spillover into a global event within days (Park 28e, Ch 5, p 426)
Microbial genetic mutation and recombination	New strains and subtypes with pandemic potential (Park 28e, Ch 5, p 426; One Health Approach, Ch 1, p 2)
Global warming and climatic change	Shifts vector distribution; floods, droughts and cyclones disturb the ecosystem (One Health Approach, Ch 1, p 2)
Occupational exposure; bioterrorism	Abattoir, veterinary, forestry and laboratory work; deliberate release (One Health Approach, Ch 1, p 2)



Pathway of emergence of a zoonotic infection, with the point at which each control measure acts

Emerging zoonotic agents recognised since 1973

Selected zoonotic entries from Park's list of infectious agents and diseases in humans and/or animals recognised since 1973 (Park 28e, Ch 5, pp 427-428):

Year	Agent	Type	Disease
1976	<i>Cryptosporidium parvum</i>	Parasite	Acute and chronic diarrhoea
1977	Ebola virus	Virus	Ebola haemorrhagic fever
1977	Hantaan virus	Virus	Haemorrhagic fever with renal syndrome
1977	<i>Campylobacter jejuni</i>	Bacterium	Enteric pathogen distributed globally
1982	<i>Escherichia coli</i> O157:H7	Bacterium	Haemorrhagic colitis; haemolytic uraemic syndrome

Year	Agent	Type	Disease
1982	<i>Borrelia burgdorferi</i>	Bacterium	Lyme disease
1986	BSE agent	Non-conventional agent	Bovine spongiform encephalopathy ("mad cow disease")
1991	Guanarito virus	Virus	Venezuelan haemorrhagic fever
1991	New species of <i>Babesia</i>	Parasite	Atypical babesiosis
1992	<i>Bartonella henselae</i>	Bacterium	Cat-scratch disease; bacillary angiomatosis
1993	Sin Nombre virus	Virus	Hantavirus pulmonary syndrome
1994	Sabia virus	Virus	Brazilian haemorrhagic fever
1996	New-variant CJD; Australian bat lyssavirus	Virus / prion	—
1997	Influenza A H5N1	Virus	Avian influenza (bird flu)
1999	Nipah virus	Virus	Nipah encephalitis
2003	Coronavirus	Virus	Severe acute respiratory syndrome (SARS)
2009	H1N1	Virus	Pandemic A (H1N1) 2009 influenza
2019	Coronavirus (SARS-CoV-2)	Virus	COVID-19

Emerging zoonoses of current importance in India

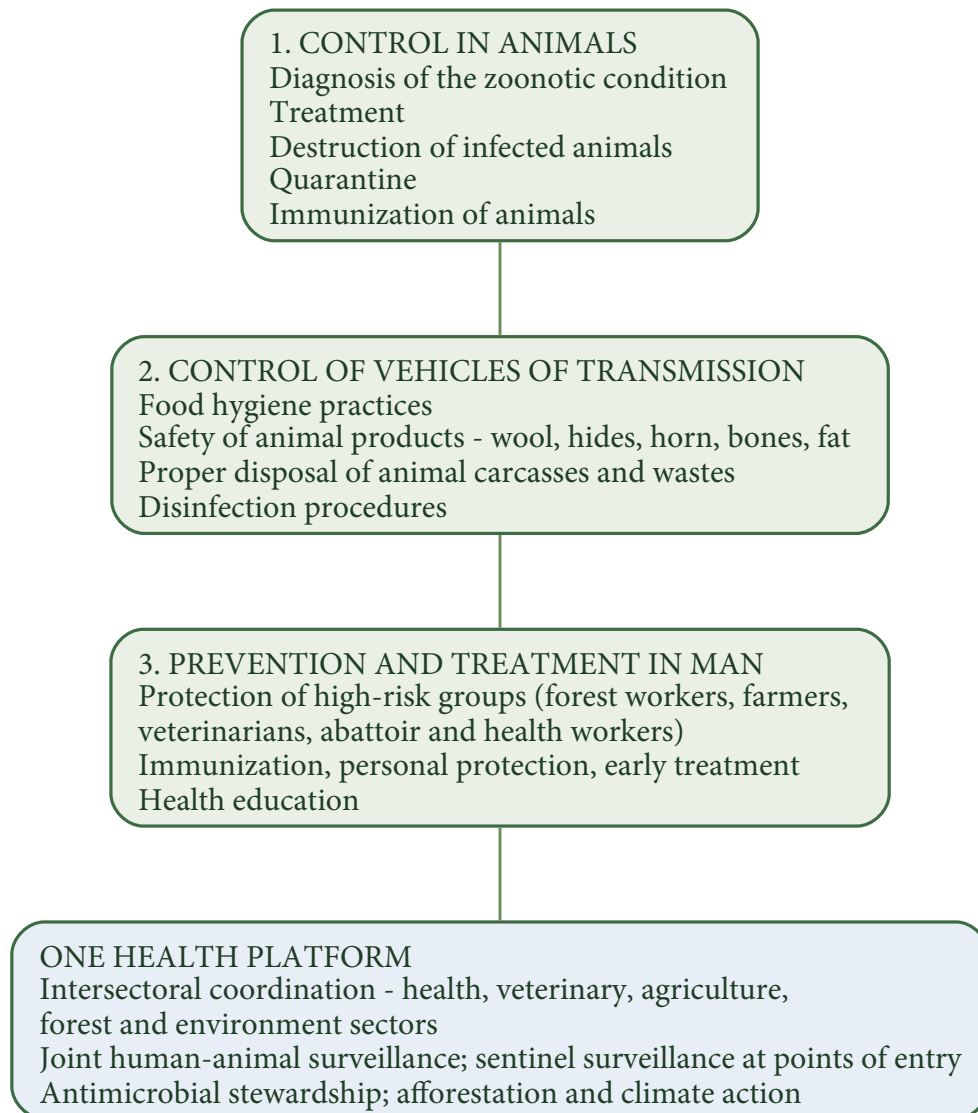
Disease	Agent and reservoir	Transmission	Key facts
Nipah virus infection	Henipavirus (Paramyxoviridae); natural host fruit bats of family Pteropodidae, genus <i>Pteropus</i> , in which there is no apparent disease	Direct contact with infected bats or pigs; raw date palm sap contaminated with bat secretions; person-to-person among family and caregivers	First identified 1999 among pig farmers in Malaysia and Singapore; in India and Bangladesh since 2001, most recently Kerala, September 2023 . Incubation 4-14 days; case fatality 40-75% ; ~20% of survivors left with residual neurological sequelae; diagnosis by RT-PCR and ELISA; no specific drug or vaccine — supportive care and risk-factor education only (Park 28e, Ch 5, pp 349-350)

Disease	Agent and reservoir	Transmission	Key facts
Kyasanur Forest disease (KFD)	Arbo-flavivirus; monkeys and ticks (<i>Haemaphysalis spinigera</i> and ≥ 15 other tick species)	Bite of infective ticks	Febrile disease with haemorrhages; recognised 1957 in Shimoga district, Karnataka and called "monkey disease" locally. Now in four Karnataka districts (Shimoga, North Kannada, South Kannada, Chikkamagaluru) over 6,000 sq km; 400-500 cases per year, CFR 3-5% ; peak January to June; monkey deaths herald outbreaks and are monitored by the Karnataka surveillance system (Park 28e, Ch 5, p 354)
Mpox (formerly monkeypox)	Monkeypox virus (MPXV) , Orthopoxvirus; DNA virus with Clade I (former Congo Basin) and Clade II (former West African), subclades Ia/Ib and IIa/IIb; rope and tree squirrels, Gambian pouched rats, dormice, non-human primates	Animal-to-human by contact with blood, body fluids or cutaneous/mucosal lesions, and by eating inadequately cooked meat; person-to-person by close contact	Has emerged as the most important orthopoxvirus for public health after smallpox eradication in 1980 and cessation of smallpox vaccination. Over 97,000 laboratory-confirmed cases and 186 deaths from 117 countries in all 6 WHO regions ; India's first case 9 September 2024 (Clade II) in a returned traveller. Control rests on surveillance, rapid identification and isolation of cases and standard infection-control precautions (Park 28e, Ch 5, pp 171-173)
Avian influenza (bird flu)	Influenza viruses primarily of birds; H5N1	Rarely crosses to pigs and humans	The vast majority of avian influenza viruses do not infect humans, but H5N1 has pandemic potential should it adapt to contagious human-to-human spread; suspected cases are reported at once to the National Centre for Disease Control, New Delhi (Park 28e, Ch 5, p 185; Jugal Kishore, p 666)
COVID-19	Novel coronavirus (SARS-CoV-2), distinct from but closely related to SARS and MERS coronaviruses; animal origin speculated	Respiratory, person-to-person	Cluster of pneumonia of unknown origin, Wuhan, late December 2019 ; declared a pandemic on 11 March 2020 ; as of 20 September 2022 , 618,141,427 cases and 6,533,467 deaths worldwide (Park 28e, Ch 5, pp 205, 427)
Ebola virus disease	Ebolavirus species (Zaire, Sudan, Tai Forest, Bundibugyo); chimpanzees, gorillas, monkeys, fruit bats	First contact with blood, organs or body fluids of infected animals, then human-to-human through body fluids or contaminated objects such as needles; not air-, water- or food-borne	Incubation 2-21 days , not infective during this period; CFR up to 70% ; the epidemic that began December 2013 had reported 25,515 cases and over 10,000 deaths by 8 April 2015 (Park 28e, Ch 5, p 427)

Disease	Agent and reservoir	Transmission	Key facts
Scrub typhus	<i>Orientia tsutsugamushi</i> ; zoonosis between trombiculid mites and small mammals — field mice, rats, shrews	Bite of infected larval mite (chigger)	The most widespread rickettsiosis; endemic in India and South-East Asia, >1 million cases annually ; one of the commonly reported rickettsial diseases of India. Control by mite control in vegetation, miticide-impregnated clothing and repellents; no vaccine exists at present (Park 28e, Ch 5, pp 365-366)
Leptospirosis	<i>Leptospira</i> — 23 serogroups and 200 serovars ; rodents (<i>R. norvegicus</i> , <i>Mus musculus</i>) are the most important reservoirs, excreting leptospire in urine lifelong	Contact with water, soil or food contaminated by animal urine	Considered the most widespread disease transmissible from animal to man ; high prevalence in warm humid tropical countries; outbreaks follow heavy rainfall and flooding , e.g. after the 1999 Odisha super-cyclone. Males affected more often because of greater occupational exposure (Park 28e, Ch 5, pp 357-358; Ch 17, p 952)
Japanese encephalitis (JE)	Group B arbovirus (Flavivirus); pig as amplifier and ardeid birds as natural hosts	Culicine mosquitoes; man is an incidental "dead-end" host	Explicitly a zoonotic disease "infecting mainly animals and incidentally man"; cycles are pig → mosquito → pig and ardeid bird → mosquito → ardeid bird (Park 28e, Ch 5, pp 350-351)

Re-emerging zoonoses to name alongside these: **plague** (rodent-borne, *Tatera indica* the natural reservoir in India), **chikungunya fever**, which re-emerged as epidemic outbreaks after more than three decades, and **antimicrobial-resistant salmonellosis** (Park 28e, Ch 15, p 940; Ch 7, p 493; Ch 5, p 365).

Prevention and control — the One Health response



The three principal components of control of zoonoses, delivered through a One Health platform

The "One World – One Health" concept, conceived in 2004 as an evolution of the older "One Medicine" idea, amalgamates the health of the entire ecosystem including wild flora and fauna. It is an integrative, multidisciplinary global strategy combining multisector expertise with **intersectoral coordination** across human, animal and ecosystem health, and is particularly important for emerging and re-emerging diseases, which are largely cross-border in nature and so require national **and** international surveillance (One Health Approach, Ch 1, p 2). Because the health sector alone cannot solve a zoonosis, responsibility for prevention and control also falls to agriculture, veterinary and other ministries outside the health sector (Park 28e, Ch 5, p 365).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ World map of selected emerging and re-emerging infectious diseases — outline the continents and place symbols with a key for Nipah/Hendra (South and South-East Asia, Australia), avian influenza H5N1 (East and South-East Asia), SARS coronavirus / novel coronavirus (East Asia, Middle East), Ebola and Crimean–Congo haemorrhagic fever (Africa), mpox (Central and West Africa), hantavirus (the Americas), Lyme borreliosis (North America, Europe), leptospirosis and plague (tropics).
- ▶ Map of Kyasanur Forest disease in Karnataka — draw the outline of Karnataka, shade the four endemic districts Shimoga, North Kannada, South Kannada and Chikkamagaluru, and label the original 1957 focus at Kyasanur Forest, Shimoga, with arrows for monkey → tick (*Haemaphysalis spinigera*) → man.

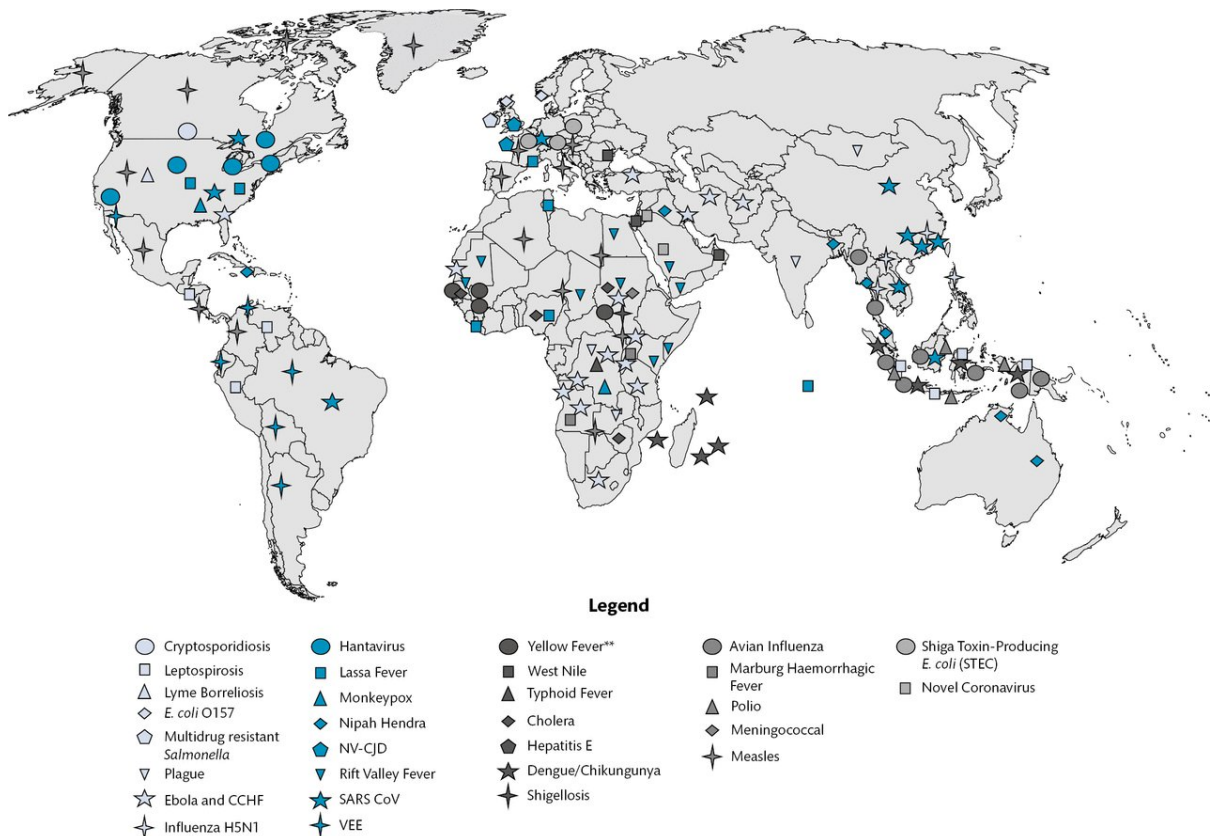


Fig 8.17 — 1 Selected emerging and re-emerging infectious diseases, 1996–2012. So (Oxford public health 7th ed, p.2833)

High-yield exam pointers

- ▶ **Definition verbatim** — zoonoses are "those diseases and infections [the agents of] which are naturally transmitted between [other] vertebrate animals and man"; >150 zoonoses recognised; 68.86% of India's population is rural.
- ▶ **Four-category life-cycle classification** — **direct** (rabies, trichinosis, brucellosis), **cyclo-** (taeniasis, echinococcosis), **meta-** (arbovirus infections, plague, schistosomiasis — obligatory extrinsic incubation period), **sapro-** (larva migrans, mycoses).

- **Eight Park factors for emergence** — unplanned urbanization, overcrowding and rapid population growth, poor sanitation, inadequate public health infrastructure, antibiotic resistance, increased exposure to vectors and reservoirs, rapid international travel, microbial genetic mutation.
- **Nipah numbers** — reservoir *Pteropus* fruit bats; incubation 4-14 days; CFR 40-75%; ~20% residual neurological sequelae; RT-PCR and ELISA; no drug or vaccine; last Indian outbreak Kerala September 2023.
- **KFD numbers** — recognised 1957, Shimoga; 4 Karnataka districts, >6,000 sq km; 400-500 cases/year, CFR 3-5%; January to June peak; vector *Haemaphysalis spinigera*; monkey deaths are the herald sign.
- **Mpox** — Clade I and Clade II (with subclades Ia/Ib, IIa/IIb); India's first case 9 September 2024, Clade II; the most important orthopoxvirus since smallpox eradication (1980).
- **Three components of control** — control in animals; control of vehicles of transmission; prevention and treatment in man — delivered on the **One Health** ("One World – One Health", 2004) platform.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the retrieved corpus does not carry a formal WHO definition of an "emerging zoonosis" or Indian incidence figures for scrub typhus, leptospirosis or Nipah, so the answer names the diseases with only the figures the texts state; India's National One Health Mission and the National Centre for Disease Control's zoonosis programme are the current administrative vehicles for the intersectoral coordination described above, but the retrieved passages do not detail their components.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed — every bullet checked against the article's full text where PMC carries it; the five abstract-only articles are flagged in the source list below

Programme and policy updates (India)

- **India now has a ranked national list of priority zoonoses.** A national multisectoral workshop held at Jaipur in February 2020 and published in 2025 used the US CDC's One Health Zoonotic Disease Prioritization tool: 50 experts drawn from the health, animal husbandry, agriculture and environment ministries scored 40 zoonoses against five weighted criteria — severity in humans carried the highest weight (0.45), followed by capacity for prevention and control (0.22). The top ten in rank order were zoonotic influenza, anthrax, Japanese encephalitis, leptospirosis, brucellosis, dengue, rabies, scrub typhus, plague and Crimean-Congo haemorrhagic fever; zoonotic tuberculosis and Nipah tied on the next rank, with KFD immediately below. Sensitivity analysis barely shifted the top ten, and the list is meant to drive state-level prioritisation exercises and One Health action plans [PMID 41429420].
- **The institutional machinery has been renamed and expanded.** The NCDC has established a **Centre for One Health**, which is building a countrywide network of sentinel surveillance sites for zoonotic diseases; the **National One Health Programme for Prevention and Control of Zoonoses (NOHP-PCZ)** was launched in 2017 to force cross-sectoral coordination; and the

ICMR has set up the **National Institute of One Health (NIOH), Nagpur**. Bringing human tick-borne diseases into notification under **IDSP–IHIP**, supported by the National One Health Mission, is the proposed next step [PMID 41429420; PMID 42165723].

Current national data

- ▶ **Rabies — first nationwide community-based estimate (ICMR, March 2022 to August 2023).** A survey of 337,808 people in 78,807 households across 60 districts in 15 states found an annual animal-bite incidence of **6.6 per 1,000** (95% CI 5.7–7.6), that is roughly **9.1 million bites a year**, with dog bites at 5.6 per 1,000. Among dog-bite victims **20.5% received no anti-rabies vaccine** and only 66.2% received three or more doses; nearly half of those who took one dose never finished the course. Decision-tree modelling estimated **5,726 human rabies deaths annually** (95% uncertainty interval 3,967–7,350) [PMID 39362224].
- ▶ **Scrub typhus — first population-based incidence from India.** A two-year cohort of 32,279 people in 37 Tamil Nadu villages (54,588 person-years) measured clinical scrub typhus at **6.0 per 1,000 person-years** (95% CI 4.8–7.5), while silent seroconversion ran far higher at **81.2 per 1,000 person-years** (95% CI 70.8–91.6), with a baseline IgG seroprevalence of **42.8%**. Severe disease occurred in 8.8% of cases, community case fatality was **1.5%** (5 of 328 — well below hospital series, which miss mild cases), and scrub typhus accounted for **29.3% of all fever hospitalisations**. Pre-existing IgG did not protect against a further clinical attack, though it was associated with milder illness [PMID 40073309].
- ▶ **India's tick burden has been costed, but nobody is counting the ticks.** The country carries **106 tick species** across 15 agro-climatic zones; combined losses from ticks and tick-borne disease are put at about **₹46,199 million (US\$595 million) a year**, and acaricide resistance in Indian tick populations is pooled at **68–78%**. KFD and CCHF are notifiable, yet India still has **no national tick surveillance system** — the existing programmes (NVBDCP, IDSP, NCDC) are built around mosquito-borne disease [PMID 42165723].

Elimination, surveillance and disease status

- ▶ **Rabies: "Zero by 30" is the operative target.** India launched its **National Action Plan for Dog Mediated Rabies Elimination in 2021**, aligned to the WHO goal of zero dog-mediated human rabies deaths by 2030, and **declared rabies a notifiable disease**. Laboratory confirmation remains the weak link — at a national referral laboratory only **41.9% of 317 suspected cases** could be confirmed by real-time PCR or RFFIT, and **children under 10 years made up 42.1%** of confirmed cases [PMID 41061569].
- ▶ **KFD has outgrown its textbook geography.** Outbreaks once confined to Karnataka have now spread to **several other Indian states**, driven by deforestation, forest fragmentation and human incursion into wildlife habitat, and the case-fatality range is now cited as **3–10%**. The formalin-inactivated KFD vaccine gave only about **62% protection** in field studies and was **withdrawn in 2023**, so at present there is neither a vaccine nor an approved antiviral; monkey deaths remain the earliest warning of a human outbreak [PMID 41202828; PMID 42314066; PMID 42165723].
- ▶ **Nipah has been a WHO R&D Blueprint priority pathogen since 2018, still with no licensed vaccine or drug.** Since 2001 Bangladesh has recorded 335 cases and 237 deaths (CFR 70.7%), and the **2023 Kerala outbreak produced 6 cases and 2 deaths**. Only the Bangladesh strain (NiV-B)

has documented person-to-person spread, and the spillover route of that Kerala outbreak remains speculative [PMID 39549708; PMID 40437297].

Guideline and legislative updates

- **A treatment shortlist for Nipah now exists.** A systematic review of 59 studies prioritised the monoclonal antibodies **1F5** and **m102.4** together with **remdesivir**, alone or in combination, for prophylaxis and early treatment; against NiV-B in monkeys 1F5 still gave complete protection when started 5 days after challenge, against 3 days for m102.4. **Ribavirin** is retained in India's Nipah treatment guidelines (though not in Bangladesh's) on equivocal evidence, and **chloroquine should not be used** — it failed to protect ferrets or hamsters and was lethal at higher doses [PMID 39549708].

Global and WHO developments

- **Mpox has been declared a PHEIC twice.** WHO first declared it in **July 2022** (clade IIb), reinstated the designation on **14 August 2024** after a clade I surge across Central Africa, then stood the PHEIC down once African cases fell by about 80% — while the **Africa CDC kept its own continental public-health emergency in force**. Between 1 January 2022 and 28 February 2026, **181,164 laboratory-confirmed cases and 492 deaths were reported from 144 countries**. Overall case fatality is 8.7%, but **clade I reaches 10.6% against 3.6% for clade II**, and the 2024 clade Ib outbreak spread through sex-work networks and household contact, children included. Tecovirimat is still the only drug approved for mpox, and **MVA-BN (JYNNEOS)** the most widely used of the three vaccines in world use (with LC16-KMB and ACAM2000) [PMID 42410978].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 3, Principles of Epidemiology — zoonoses terminology; Ch 5, Epidemiology of Communicable Diseases — Zoonoses, Nipah virus infection, Japanese encephalitis, Kyasanur Forest disease, Leptospirosis, Rickettsial zoonoses and scrub typhus, Mpox, Avian influenza, COVID-19, and "Emerging and re-emerging infectious diseases" with Table 1 of agents recognised since 1973; Ch 7, Health Programmes in India; Ch 15, Environment and Health — rodents and control of zoonoses; Ch 17, Disaster Management); One Health Approach (factors associated with emergence and re-emergence of diseases; "One World – One Health"); Jugal Kishore — National Health Programmes of India (avian influenza epidemic containment and IDSP). Coverage note: the corpus is rich on individual emerging zoonoses and on the factors of emergence, but thin on India-specific incidence data and on the structure of the current national zoonoses programme, so no such figures are quoted.

Recent-advances sources: all from PubMed. Full text read and checked line by line (PMC): 41429420 — Transforming One Health in India: National Multisectoral Mixed Method Study on Prioritization of Zoonotic Diseases, JMIR Public Health Surveill 2025, doi:[10.2196/77850](https://doi.org/10.2196/77850) (PMC12770923; source of both the priority ranking and the NCDC Centre for One Health sentinel-site network); 42165723 — Ticks and tick-borne diseases: a call for comprehensive tick surveillance in India under One Health approach, Indian J Med Res 2026, doi:[10.25259/IJMR_2011_2025](https://doi.org/10.25259/IJMR_2011_2025) (PMC13224214; source of NOHP-PCZ 2017, NIOH Nagpur, IDSP-IHIP notification proposal, the tick figures and the 2023 KFD vaccine withdrawal); 40073309 — Incidence of Scrub Typhus in Rural South India, N Engl J Med 2025, doi:[10.1056/NEJMoa2408645](https://doi.org/10.1056/NEJMoa2408645) (PMC7617504); 39549708 — Therapeutics for Nipah virus disease: a systematic review to support prioritisation of drug candidates for clinical trials, Lancet Microbe 2024, doi:[10.1016/j.lanmic.2024.101002](https://doi.org/10.1016/j.lanmic.2024.101002) (PMC12062192); 42410978 — An integrated review of monkeypox: from pathogen and epidemiology to diagnostics, control, and challenges, Emerg Microbes Infect 2026, doi:[10.1080/22221751.2026.2695529](https://doi.org/10.1080/22221751.2026.2695529) (PMC13347851). Not in PMC — bullets rest on the PubMed abstract only, and claim nothing beyond it: 39362224 — Estimates of the burden of human rabies deaths and animal bites in India, 2022-23, Lancet Infect Dis 2025, doi:[10.1016/S1473-3099\(24\)00490-0](https://doi.org/10.1016/S1473-3099(24)00490-0); 41061569 — Laboratory diagnosis of rabies in an endemic country: progress and challenges on the path to elimination, Diagn Microbiol Infect Dis 2025, doi:[10.1016/j.diagmicrobio.2025.117123](https://doi.org/10.1016/j.diagmicrobio.2025.117123); 41202828 — Kyasanur Forest disease: an emerging arboviral threat, Lancet Infect Dis 2025, doi:[10.1016/S1473-3099\(25\)00589-4](https://doi.org/10.1016/S1473-3099(25)00589-4); 42314066 — Kyasanur Forest Disease: advances in experimental animal models, immune assays, and challenges in vaccine development, Rev Med Virol 2026, doi:[10.1002/rmv.70177](https://doi.org/10.1002/rmv.70177); 40437297 — A One Health approach to understanding and managing Nipah virus outbreaks, Nat Microbiol 2025, doi:[10.1038/s41564-025-02020-9](https://doi.org/10.1038/s41564-025-02020-9).

♦ ♦ ♦

Mosquito control under the Urban Malaria Scheme

Asked in: Paper I May 2019 — RGUHS MD Community Medicine

Definition — The Urban Malaria Scheme (UMS) is a centrally sponsored component of the national vector borne disease control programme, **launched in 1971** to reduce or interrupt malaria transmission in towns and cities; its methodology comprises vector control by **intensive anti-larval measures** and **drug treatment** (Park 28e, Ch 7, p 494). It was sanctioned during **November 1971** under the Modified Plan of Operation (Jugal Kishore, p 397). Anti-larval work — not indoor residual spraying — is the urban mainstay because the urban vector *Anopheles stephensi* breeds in man-made water containers in and around houses, which residual wall spray cannot reach.

1. Why the urban strategy is anti-larval — vector and breeding sources

Vector	Habit	Principal urban breeding sources	Control implication
<i>Anopheles stephensi</i> (main urban malaria vector)	Domestic/peri-domestic container breeder; bites indoors or outdoors, rests mainly indoors — essentially endophilic, though found resting outdoors in southern India and southern Iran	Overhead rooftop and underground water-storage tanks, cisterns, wells, desert coolers, roof gutters, flower vases, bottles, ornamental ponds, old tyres, water stored at construction sites	Weekly larviciding + source reduction + mosquito-proofing of tanks (Park 28e, Ch 7, p 494; Service, Medical Entomology, p 51; Gilles & Warrell 4e, Ch 4, p 83 and Ch 6, p 172)
<i>Aedes aegypti</i>	Domestic species, breeds in the same containers alongside <i>An. stephensi</i> ; outdoors prefers old tyres and drums	Same domestic containers	Same anti-larval package serves dengue control (Jugal Kishore, p 398)
<i>Culex quinquefasciatus</i>	Pest and filaria vector; flight range about 3 km	Sullage collections, drains, septic tanks, soakage pits, cesspools	Anti-larval area of a town is extended by a 3 km peripheral belt , a rule set for the <i>Culex</i> /filaria anti-larval programme whose operational component was merged with the UMS in June 1978 (Park 28e, Ch 5, p 339; Ch 7, p 500)

Two urban forces continually re-create these sources: **large construction activity**, providing fresh breeding sites, and **influx of migrant labour from malarious zones**, importing the parasite (Park 28e, Ch 7, pp 494–495).

2. Scheme profile and inclusion norms

Item	Detail
Coverage now	About 130 million population in 131 towns of 19 states/UTs (Park 28e, Ch 7, pp 494–495; Jugal Kishore, p 398)
Urban share of burden	7.4% of all malaria cases and 10.9% of malaria deaths are reported from urban areas; maximum cases from Chennai, Vadodara, Visakhapatnam, Ahmedabad, Kolkata, New Mumbai, Vijayawada (Park 27e, Ch 7, p 477)

Item	Detail
Norm — size	Town population $\geq 50,000$ (Jugal Kishore, p 398)
Norm — endemicity	Annual parasite incidence (API) ≥ 2 per 1000 ; the Expert Committee on Malaria additionally recommended inclusion of urban areas reporting a slide positivity rate (SPR) of 5% and above , with active surveillance under the scheme (Jugal Kishore, p 398; Park 28e, Ch 7, pp 494–495)
Norm — legal	Town must promulgate and strictly implement civic bye-laws to prevent or eliminate domestic and peri-domestic breeding places (Jugal Kishore, p 398)
Objectives	(a) To control malaria by reducing the vector population in the urban areas; (b) reduce morbidity and mortality through early diagnosis and prompt treatment (EDPT) (Jugal Kishore, p 398)
Implementing agency	Local administrative authorities/ municipalities , under active supervision of state health authorities — town level: Biologist ; state level: Additional/Joint/Deputy Director (Malaria & Filaria) (Jugal Kishore, p 399)

3. Strategies for mosquito control under UMS

A. Source reduction and minor engineering (the first and most efficient line). Filling of ditches, pits and low-lying areas; channelisation, de-silting, de-weeding and trimming of drains; drainage of stagnant water; repair and proper maintenance of septic tanks and soakage pits so that egg-laying females cannot gain access; and emptying, scrubbing and drying every domestic water container **once a week** (Jugal Kishore, pp 398–399; Park 28e, Ch 5, p 339). It is the most efficient line because it removes the habitat permanently instead of treating it repeatedly. Rooftop and storage tanks, the classic *An. stephensi* habitat in the Indian subcontinent, are mosquito-proofed by tight covers and screening — which needs supervision, since such covers are commonly torn or removed (Service, Medical Entomology, p 37; Gilles & Warrell 4e, Ch 4, p 83).

B. Chemical control — recurrent anti-larval measures. Larvicides are applied **once weekly** on all breeding places, because the aquatic life cycle of the mosquito occupies about **8 days** (Park 28e, Ch 15, p 926).

Larvicide	Use and dose
Temephos (Abate) — organophosphate, the UMS workhorse	≤ 1.0 ppm in wells and potable-water containers; least toxic effective larvicide; field dose 56–112 g/ha (Park 28e, Ch 15, p 938 and Table 6, p 927)
Fenthion	22–112 g/ha (Park 28e, Ch 15, Table 6 "Toxicants employed as larvicides in mosquito control", p 927)
Malathion	224–672 g/ha (Park 28e, Ch 15, Table 6, p 927)
Chlorpyrifos	11–16 g/ha ; with temephos, one of the more effective larvicides in organically polluted water (Park 28e, Ch 15, Table 6, p 927; Service, p 92)
Mosquito larvicidal oil (MLO)	Film cuts off air supply to larvae and pupae; 40–90 litres/hectare ; being replaced by pyrethrum oil, temephos and fenthion (Park 28e, Ch 5, p 339; Ch 15, p 926)

Larvicide	Use and dose
Pyrosene oil-E	Pyrethrum-based emulsifiable larvicide, 0.1–0.2% pyrethrins, diluted 1:4 before use (Park 28e, Ch 5, p 339)

Organochlorines (DDT, HCH) are **not** used as larvicides because of long residual effect, water contamination and risk of resistance (Park 28e, Ch 15, p 927).

C. Biological control. Larvivorous fish are stocked in water bodies that are not to be treated chemically — slow-moving streams, ornamental ponds, cisterns, farm ponds, burrow pits and sewage oxidation ponds; *Gambusia affinis* and *Lebistes reticulatus* (Barbados millions) are the standard species, and larvicides are reserved for water bodies unsuitable for fish (Park 28e, Ch 7, pp 494–495; Ch 15, p 927). The microbial larvicides *Bacillus thuringiensis* var. *israelensis* (Bti) — supplied as floating slow-release briquettes giving up to 30 days of control — and *B. sphaericus*, which recycles in the habitat and is especially effective against *Culex* in polluted water, are the other proven agents (Service, pp 35, 93). For *Culex* breeding in pit latrines, cesspits and soakage pits, a floating 1–2 cm layer of 2–3 mm expanded polystyrene beads suffocates larvae and pupae and blocks egg-laying, and a single application can last several years (Service, pp 37, 92).

D. Anti-adult measures — focal space spray. Pyrethrum extract 2% is space-sprayed in 50 houses in and around every malaria-positive case to kill the infective mosquitoes (Jugal Kishore, p 399). Pyrethrum kills on contact but has no residual action, so re-infestation from outside occurs quickly; doors and windows are kept shut for half an hour after spraying. In outbreaks, **ultra-low-volume (ULV) fogging** with malathion or fenitrothion brings the adult population down rapidly (Park 28e, Ch 15, p 927). Indoor residual spraying remains the primary method of vector control in **rural** areas, anti-larval measures in the **urban** areas (Park 28e, Ch 7, p 498).

E. Legislative measures — civic and building bye-laws. Control of urban malaria lies **primarily** in the implementation of civil bye-laws to prevent mosquito breeding in the domestic and peri-domestic areas, enforced by the municipality against defaulting householders and builders. Bye-laws have been enacted and implemented in Delhi, Mumbai, Kolkata, Chandigarh, Bengaluru, Chennai, Ahmedabad and Goa (Park 28e, Ch 7, pp 494–495; Park 27e, Ch 7, p 477).

F. Personal protection. Insecticide-treated and long-lasting insecticidal nets, screening of houses, repellents, protective clothing and mosquito coils reduce man–vector contact (Park 28e, Ch 5, p 333; Ch 7, p 498).

G. Anti-parasitic measures and IEC. Early case detection and prompt treatment through passive agencies — hospitals, dispensaries, malaria clinics and private practitioners — reduces the reservoir of infection, supported by weekly **active surveillance in slum areas** (one worker per 20,000 population, one malaria clinic per 50,000); **information, education and communication (IEC)** campaigns secure the community participation that is indispensable when breeding sites lie inside people's homes (Jugal Kishore, pp 391, 398–399).

All of these are delivered as **integrated vector management (IVM)**, which combines biological, chemical and environmental interventions for maximum result with minimum input, includes safe use of insecticides and monitoring of insecticide resistance, and avoids dependence on any single tool (Park 28e, Ch 7, p 498; Ch 15, p 926).

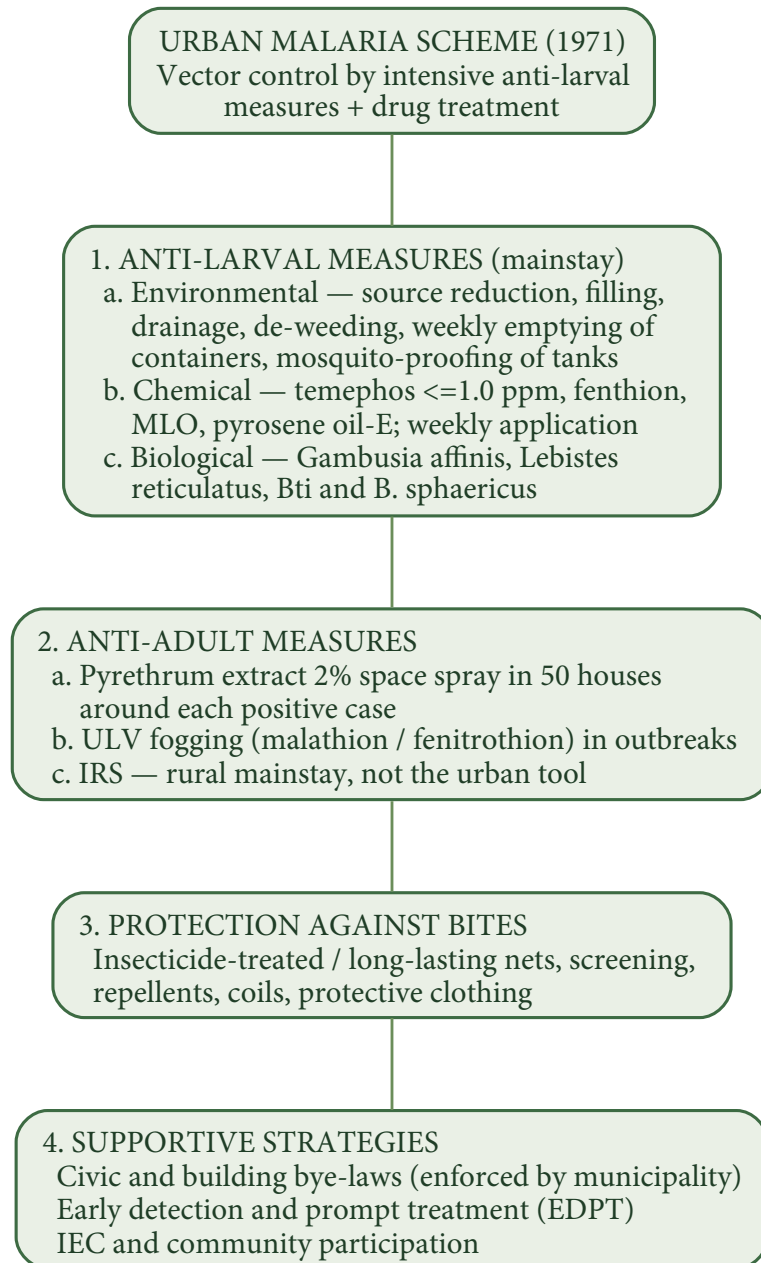
4. Monitoring and evaluation of the strategy

Index	Definition
Annual parasite incidence (API)	Confirmed cases during one year ÷ population under surveillance × 1000 (Park 28e, Ch 5, p 324)
Annual blood examination rate (ABER)	Blood slides examined ÷ population × 100 — the operational check on surveillance effort
Slide positivity rate (SPR)	Percentage of slides positive for malaria parasite of any species
Slide falciparum rate (SFR)	Percentage of slides positive for <i>P. falciparum</i>
Mosquito density	Number of mosquitoes per man-hour catch (Park 28e, Ch 5, p 324)
Human blood index	Proportion of freshly fed female anophelines whose stomach contains human blood — degree of anthropophilism
Sporozoite rate	Percentage of female anophelines with sporozoites in salivary glands
Man-biting rate × sporozoite rate	Gives the inoculation rate

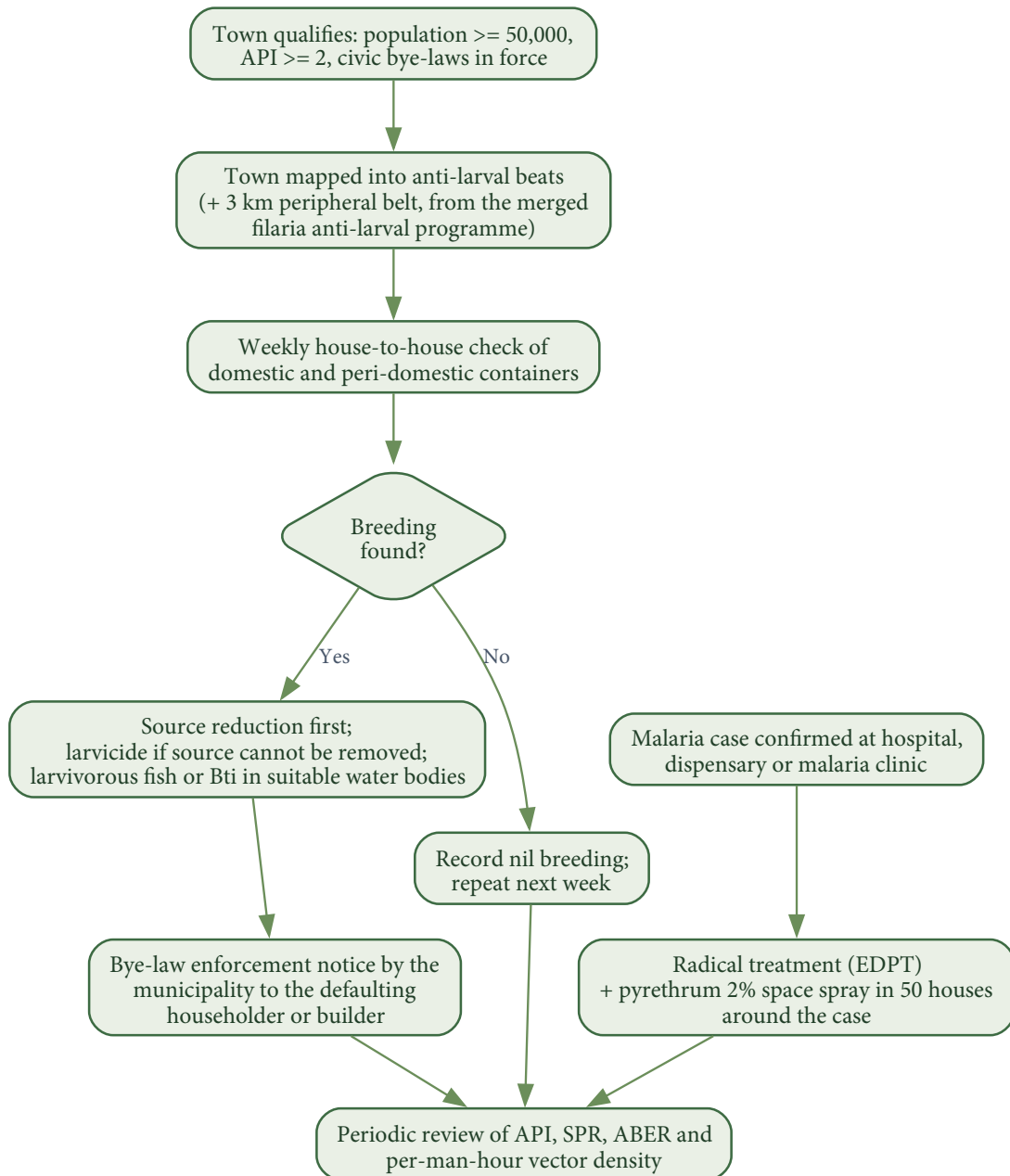
The vector has developed resistance to many of the organophosphorus larvicides, so knowledge of the resistance status decides which insecticide is used, and monitoring of insecticide resistance is a built-in component of IVM (Park 28e, Ch 5, p 339; Ch 5, p 322; Ch 7, p 498).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Urban breeding sources of *Anopheles stephensi* — a labelled section through a typical urban house. Draw a two-storey house and label: (1) overhead rooftop water-storage tank (uncovered), (2) underground sump/water-storage tank, (3) cistern, (4) roof gutter with stagnant water, (5) desert cooler, (6) flower vases and discarded bottles in the courtyard, (7) ornamental pond, (8) discarded old tyres, (9) open well, (10) water drums at an adjacent construction site, (11) blocked surface drain / septic tank with sullage (*Culex quinquefasciatus*), (12) the 3 km peripheral belt drawn as a dotted ring around the town boundary. Against each source write the action — cover / empty weekly / temephos ≤ 1.0 ppm / larvivorous fish / source reduction.



Classification of mosquito control strategies delivered under the Urban Malaria Scheme



Operational anti-larval cycle in a UMS town, with the case-triggered response

High-yield exam pointers

- ▶ **UMS in one line:** 1971, anti-larval measures + drug treatment, 131 towns, 19 states/UTs, ~130 million protected; urban areas contribute 7.4% of cases and 10.9% of deaths.
- ▶ **Inclusion norms mnemonic — "50-2-Bye-law":** population ≥ 50,000, API ≥ 2 (Expert Committee also: SPR ≥ 5%), and enforced civic bye-laws.
- ▶ **Temephos ≤ 1.0 ppm** is the only larvicide dose to remember for wells and potable water; field doses (Park Table 6) — Abate 56–112, fenthion 22–112, malathion 224–672, chlorpyrifos 11–16 g/ha.
- ▶ **Pyrethrum extract 2% in 50 houses around every positive case;** larviciding repeated weekly (aquatic cycle ≈ 8 days).

- ▶ Larvivorous fish = *Gambusia affinis* and *Lebistes reticulatus*; microbial larvicides = Bti and *B. sphaericus*.
- ▶ Anti-larval = urban, IRS = rural — write this contrast; implemented by the municipality, town-level in-charge is the Biologist; the 3 km peripheral belt comes from the *Culex* filaria programme merged with UMS in June 1978.
- ▶ API formula denominator is population under surveillance, not total population.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the programme directorate has been renamed the **National Center for Vector Borne Diseases Control (NCVBDC)**, formerly the National Vector Borne Disease Control Programme (NVBDCP); Jugal Kishore prints the urban share of cases as 7.8 per cent against Park's 7.4 per cent, a difference of reporting year rather than of substance, so quote Park's 7.4 per cent; the town-count, population and case-share figures are as printed and are not annually revised.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, NCVBDC, ICMR, National Framework for Malaria Elimination 2016–2030, National Strategic Plan for Malaria Elimination 2023–2027, WHO World Malaria Report 2024

Programme and policy updates (India)

- ▶ Every Indian district is now stratified by API into four programme categories — on 2024 NCVBDC data, **103 districts in Category 0 (API = 0), 634 in Category 1 (API < 1), 9 in Category 2 (API 1–2) and 28 high-burden districts in Category 3 (API > 2)** — under the **National Framework for Malaria Elimination (NFME) 2016–2030** and the **National Strategic Plan for Malaria Elimination (2017–2022, then 2023–2027)**. [PMID 41663159] Quote the UMS "API ≥ 2" inclusion norm alongside this category ladder, since any UMS town today sits inside a district carrying one of these four labels — that pairing is the answer's own framing, not a claim of the cited paper.
- ▶ A **12-district, 10-state multicentric longitudinal study** by ICMR-VCRC with NCVBDC is profiling malaria over 12 months across urban, peri-urban, rural, forest-foothill and coastal settings, with WHO tube bioassays plus **5× and 10× discriminating-concentration resistance-intensity testing**, and larval dipping deliberately restricted to the urban and peri-urban clusters where the programme routinely larvicides — **Bengaluru Urban and Tiruchirappalli are the urban and peri-urban sites respectively** — with confirmed breeding-site maps handed to local health authorities. This is a study protocol with no results yet; its stated purpose is standardised, setting-specific evidence. [PMID 41663159]
- ▶ The **Malaria Elimination Demonstration Project (Mandla, Madhya Pradesh)** enumerated **1,143,126 people in 248,825 households** and ran **fortnightly household visits with on-the-spot RDT diagnosis and treatment**; incidence fell from **26.13 (2018) to 11.18 (2020) per 100,000**. The authors propose this constant-contact longitudinal design for disease-elimination programmes generally, precisely because the **unit of observation is the household and not the individual** — a household-unit surveillance model that could be adapted to the UMS slum active-surveillance

component, though the paper itself says nothing about the UMS or urban slums. [PMID 39456035]

Current national data

- **Reported cases have risen since 2021** — 161,753 (2021) → 176,522 (2022) → 227,564 (2023) → 255,500 (2024) — even though WHO-estimated cases fell 69% (6.4 million → 2 million) and deaths 68% between 2017 and 2023, a gain that took India out of the "High Burden to High Impact" grouping in 2024; India still carries about **half the WHO South-East Asia Region burden**. [PMID 41663159] Set against this, **402 district outbreak-years occurred during 2011–2020**, and seven states (Mizoram, Tripura, Jharkhand, Odisha, Madhya Pradesh, Uttar Pradesh, Meghalaya) had at least one outbreak year in **over 70% of their districts**. [PMID 41882635] Quote the rebound against the older textbook narrative of uninterrupted decline.

Elimination, surveillance and disease status

- A practical caution on the ABER in the monitoring table above: **10% is the adequacy benchmark**, but Gujarat villages recorded 1.6–78% because mass-survey slides, contact smears and migrant slides were pushed into the numerator while the migrant population was left out of the denominator — an inflated ABER therefore signals bad recording, not good surveillance. [PMID 39618956]
- Sub-national heterogeneity is now the rate-limiting problem: Chhattisgarh (2015–2023) logged 391,387 cases with **API 0.4–3.2 and a 25% annual decline (IRR 0.75; 95% CI 0.71–0.79)**, yet Bastar ran **12.5 times** the central region (95% CI 3.7–43.0); a 2023 evaluation found IDSP malaria surveillance there captured only **5.8% of presumptive and 29% of laboratory-confirmed cases**; and in Sonbhadra (UP) API fell **2.7 (2017) → 0.1 (2024)** with a 95% case reduction while Kulldorff scan statistics still isolated a persistent hotspot at Chopan block — the argument for beat-level rather than town-level targeting. [PMID 40600283; PMID 41629229; PMID 41871706]

Vector-control evidence and resistance management

- Larvicide choice for *An. stephensi* now has systematic-review backing that validates the UMS list: in a quantitative systematic review, **temephos and fenthion**, together with **Bti and Beauveria bassiana**, were the most effective larvicides **in the vector's native range**, though the review states their relative effectiveness in the invaded range was less clear [PMID 39427695]; Indian field trials of Bti (Bactivec SC, 1 ml per 50 L) produced **80–96% falls in larval density and 81–100% in pupal density**, with residual activity of **14–17 days at 1 ml/50 L (7–14 days at 0.5 ml/50 L)** [PMID 39844918].
- **Insecticide-resistance management is now an explicit design criterion rather than an afterthought**. In a deterministic population-genetics simulation (PopGEM model, outcome = vectorial capacity — modelling, not a field trial), next-generation **mixture** ITNs outperformed pyrethroid-only nets for both transmission control and resistance management even when deployed at up to **30% lower coverage**, supporting the mixture-net strategy proposed under WHO's Global Plan for Insecticide Resistance Management (GPIRM) [PMID 42546043]. The mechanisms to monitor are metabolic detoxification (cytochrome P450s, glutathione S-transferases, carboxylesterases), *kdr* target-site mutation, cuticle thickening and behavioural avoidance, with pyrethroid resistance commonest and cross-resistance increasingly reported [PMID 41894285].

Global and WHO developments

- ▶ *An. stephensi* — the UMS vector — has colonised eight sub-Saharan African countries since 2013 [PMID 39054167, abstract-only], with detections in Nigeria (2020), Kenya (2022) and Ghana (2022) [PMID 39844918], and continued spread could place an additional ~126 million people a year at risk; WHO has urged affected countries to run targeted surveillance, share information, issue public-health guidance and prioritise control research [PMID 39427695]. No field trial of any intervention against this vector has yet been done in Africa, so the evidence base being drawn on is its native range — largely India — where the systematic-review verdict is that LLINs and IRS are the effective tools, with repellents as the complement; larviciding and larvivorous fish appear in that same native-range evidence (mostly Indian trials) but the review does not put them forward as the template [PMID 39844918]. Close an answer on that: the UMS is the standing reference experience for urban *An. stephensi* control, not a validated African template.
- ▶ Two forward-looking developments worth a sentence each: climate-suitability modelling projects *An. stephensi* from ~13% of the land surface today (≈40% of the world's population) to over 30% by 2100, exposing up to 56%; and spatial repellent emanators have entered the evidence base as a new vector-control class — in a pragmatic trial in displacement camps at Maiduguri, Nigeria they cut malaria incidence (IRR 0.713; 95% CI 0.584–0.869) with 22.5% protective efficacy against first infection. [PMID 40595312; PMID 41520670]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 5 — Epidemiology of Communicable Diseases; Ch 7 — Health Programmes in India; Ch 15 — Environment and Health); Park's Textbook of Preventive and Social Medicine, 27e (Ch 7); Jugal Kishore — National Health Programmes of India (National Vector Borne Diseases Control Programme chapter, Urban Malaria Scheme section); Service — Medical Entomology for Students; Gilles & Warrell — Essential Malariology 4e. Coverage note: the Urban Malaria Scheme passage in the scanned Park 28e text is badly degraded, so the burden figures (7.4% of cases and 10.9% of deaths) and the figures of 131 towns, 19 states/UTs, 130 million population covered and the $\geq 50,000$ population / SPR $\geq 5\%$ norms were all taken from the identical passage in Park 27e (Ch 7, pp 477–478), where the 28e scan drops the leading digit of the deaths figure and garbles the surrounding sentence; the cross-check against Jugal Kishore agrees.

Recent-advances sources: 41663159 — Multicentric longitudinal study on malaria burden, vector bionomics and health system assessment in diverse eco-epidemiological settings in the context of malaria elimination in India: study protocol, *BMJ Open* 2026, doi:[10.1136/bmjopen-2025-111645](https://doi.org/10.1136/bmjopen-2025-111645); 39456035 — A constant contact community-based epidemiological investigation (C3EI) as part of malaria elimination demonstration project, Mandla district, Madhya Pradesh, India, *Malaria Journal* 2024, doi:[10.1186/s12936-024-05143-2](https://doi.org/10.1186/s12936-024-05143-2); 41882635 — Epidemiology of malaria outbreak years in India provides key insights into malaria elimination, *Malaria Journal* 2026, doi:[10.1186/s12936-026-05794-3](https://doi.org/10.1186/s12936-026-05794-3); 39618956 — Assessment of the quality of malaria surveillance and laboratory services for diagnosis in three districts of Gujarat state, India, *Frontiers in Public Health* 2024, doi:[10.3389/fpubh.2024.1465228](https://doi.org/10.3389/fpubh.2024.1465228); 40600283 — Epidemiology of malaria in Chhattisgarh, India: a surveillance data analysis, 2015–2023, *Transactions of the Royal Society of Tropical Medicine and Hygiene* 2025, doi:[10.1093/trstmh/traf072](https://doi.org/10.1093/trstmh/traf072); 41629229 — Evaluation of Malaria Surveillance System in Government Health Facilities in Six Districts of State Chhattisgarh, India, 2023, *Indian Journal of Public Health* 2026, doi:[10.4103/ijph.ijph_647_25](https://doi.org/10.4103/ijph.ijph_647_25) (abstract-only); 41871706 — Identification of micro-level malaria hotspots and its implications for achieving malaria elimination target in a remote district of India, *Acta Tropica* 2026, doi:[10.1016/j.actatropica.2026.108066](https://doi.org/10.1016/j.actatropica.2026.108066) (abstract-only); 39427695 — A quantitative and systematic analysis of *Anopheles stephensi* bionomics and control approaches, *Acta Tropica* 2024, doi:[10.1016/j.actatropica.2024.107431](https://doi.org/10.1016/j.actatropica.2024.107431); 39844918 — A systematic review of interventions targeting *Anopheles stephensi*, *Wellcome Open Research* 2024, doi:[10.12688/wellcomeopenres.23480.1](https://doi.org/10.12688/wellcomeopenres.23480.1); 42546043 — Impact of insecticide resistance evolution on malaria vector control, *PLoS Computational Biology* 2026, doi:[10.1371/journal.pcbi.1014612](https://doi.org/10.1371/journal.pcbi.1014612); 41894285 — Mechanisms of insecticide resistance in mosquitoes: a systematic review of biochemical and physiological perspectives for sustainable vector control, *Medicine (Baltimore)* 2026, doi:[10.1097/MD.00000000000048068](https://doi.org/10.1097/MD.00000000000048068); 39054167 — Invasive *Anopheles stephensi* in Africa: insights from Asia, *Trends in Parasitology* 2024, doi:[10.1016/j.pt.2024.06.008](https://doi.org/10.1016/j.pt.2024.06.008) (abstract-only); 40595312 — Future global distribution and climatic suitability of *Anopheles stephensi*, *Scientific Reports* 2025, doi:[10.1038/s41598-025-07653-8](https://doi.org/10.1038/s41598-025-07653-8); 41520670 — The effectiveness of long-lasting spatial repellent emanators against malaria in humanitarian crisis settings in northern Nigeria: a

two-arm pragmatic, open-label, controlled trial, *The Lancet Infectious Diseases* 2026, doi:10.1016/S1473-3099(25)00684-X (abstract-only). Programme figures — National Framework for Malaria Elimination 2016–2030, National Strategic Plan for Malaria Elimination 2017–2022 and 2023–2027, the NCVBDC district API categorisation for 2024, and the WHO World Malaria Report 2024 estimates — are all quoted here as reported in PMID 41663159, not from the primary programme documents.



Toxicity of insecticides

Asked in: Paper I 14-Jul-2020 — RGUHS MD Community Medicine

Definition — Insecticides are pesticides used to kill insects of public health or agricultural importance. Because they act on nervous tissue and the target sites are largely shared between insects and mammals, they are not selective and readily injure non-target species including man (Hunter's Diseases of Occupations, Ch 43, pp 420, 428). Pesticide poisoning is one of the commonest causes of chemical poisoning worldwide, and insecticides account for the largest share of it.

Magnitude of the problem

Measure	Figure	Source
Agricultural workers suffering ≥ 1 acute pesticide poisoning episode in a lifetime	1–3% worldwide	Hunter's Diseases of Occupations, Ch 43, p 441
Pesticide-related deaths per year	~300,000	Hunter's Diseases of Occupations, Ch 43, p 441
Acute poisonings needing hospitalisation per year	not less than 1 million	Hunter's Diseases of Occupations, Ch 43, p 428
Share of acute occupational pesticide illness caused by insecticides (SENSOR surveillance, USA)	49% of all cases; 69.7% mild, 29.6% moderate, 0.4% severe	Hunter's Diseases of Occupations, Ch 43, p 428
Mortality in severe organophosphorus poisoning (50–90% cholinesterase inhibition)	~10%	Hunter's Diseases of Occupations, Ch 43, p 435
Population covered by indoor residual spray in India (DDT the insecticide of choice; malathion or synthetic pyrethroids where resistance exists)	~80 million	Park 28e, Ch 7, p 498

Classification by chemical group and relative toxicity to man

Group	Public health examples	Mechanism of human toxicity	Relative toxicity
Organochlorines	DDT, HCH (gammexane, lindane), dieldrin	Nerve poisons — reduce potassium transport, DDT also slows sodium-channel closure, inhibits ATPase and disturbs calcium transport → repetitive neuronal discharge (Hunter's Diseases of Occupations, Ch 43, p 424)	DDT least toxic of the three; median lethal dose in man ~250 mg/kg body weight. Gamma HCH ~2× as toxic as DDT; dieldrin 5–8× as toxic as DDT (Park 28e, Ch 15, p 939)
Organophosphorus compounds	Malathion, fenitrothion, chlorpyrifos, fenthion, temephos	Inhibit cholinesterase, the enzyme degrading acetylcholine at the synapse → acetylcholine accumulates at all cholinergic endings (Park 28e, Ch 15, p 939)	Most important cause of severe insecticide poisoning. Malathion has the least toxicity of all organophosphorus compounds (Park 28e, Ch 15, p 938)
Carbamates	Propoxur, bendiocarb	Same anticholinesterase mechanism, but the carbamyl-enzyme bond dissociates readily (Hunter's Diseases of Occupations, Ch 43, p 429)	Poisoning shorter and less serious; rarely life-threatening even after deliberate ingestion (Hunter's Diseases of Occupations, Ch 43, p 436)
Pyrethrum and synthetic pyrethroids	Pyrethrins, permethrin, deltamethrin, lambda-cyhalothrin	Alter sodium-channel gating in the nerve axon; mammals absorb them poorly and degrade them rapidly in the liver (Hunter's Diseases of Occupations, Ch 43, pp 424, 430)	Lowest systemic toxicity of the four; lambda-cyhalothrin classed moderately hazardous, LD50 56 mg/kg (Gilles & Warrell 4e, Ch 6, p 149)

Toxicity is graded by the WHO/IPCS classification of pesticides by hazard, which uses the oral LD50 in the rat: an LD50 of 5 mg/kg for a solid is **extremely hazardous**, and more than 500 mg/kg is **slightly hazardous** (Hunter's Diseases of Occupations, Ch 43, p 422). On this scale DDT (LD50 113 mg/kg) is moderately hazardous, while dieldrin (LD50 37 mg/kg) is highly hazardous and has virtually ceased to be used in public health after poisoning spray operators and domestic animals (Gilles & Warrell 4e, Ch 6, pp 150–151).

Routes of exposure and groups at risk

- **Dermal** — the dominant occupational route. Dieldrin poisoning is met with more often than DDT or HCH poisoning precisely because dieldrin is absorbed through the skin, whereas DDT and HCH are not absorbed through intact skin in the solid state (Park 28e, Ch 15, p 939).
- **Inhalation** — during space spraying, fogging and fumigation.
- **Ingestion** — deliberate self-harm, accidental ingestion by children from unlabelled containers, and contaminated food or water.
- **Groups at risk, in descending order of dose** — manufacturing and formulation plant workers; mixers and loaders (exposure is greatest at the concentrate-to-dilution stage because of splashing and inadequate protective equipment); field applicators, especially with backpack sprayers; re-

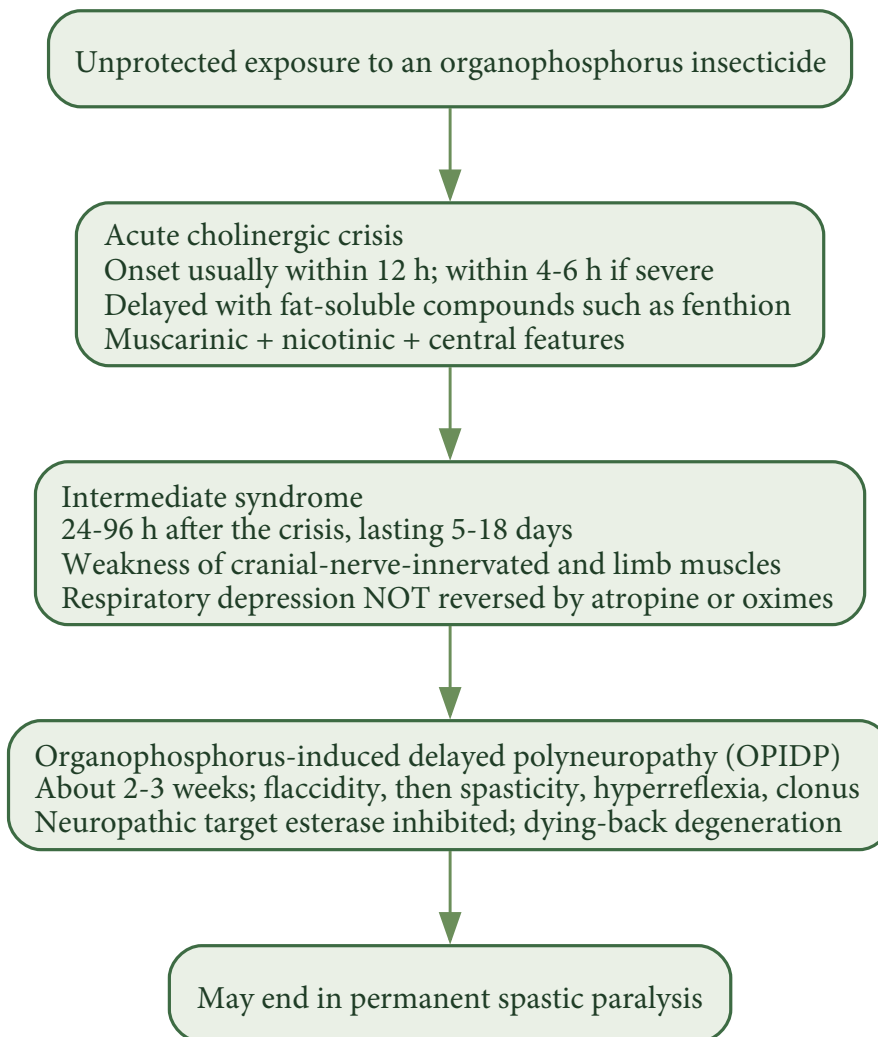
entry workers handling treated crops; and the general population exposed to residues in food, water and the environment (Hunter's Diseases of Occupations, Ch 43, pp 439–440).

Acute clinical effects

Acute effects are divided into **local** — dermal and mucosal irritation, corrosion or ulceration of skin, eye, respiratory and gastric mucosa — and **systemic** effects following absorption of an effective dose (Hunter's Diseases of Occupations, Ch 43, pp 428–429).

Group	Acute clinical picture	Specific treatment
Organochlorines	Increased nervous excitability, tremors, motor seizures, respiratory failure (Park 28e, Ch 15, p 939; Hunter's Diseases of Occupations, Ch 43, p 429)	Supportive; barbiturates, especially phenobarbitone , and diazepam for irritability and convulsions. Avoid sympathomimetic drugs — organochlorines sensitise the myocardium to catecholamines (Park 28e, Ch 15, p 939; Hunter's Diseases of Occupations, Ch 43, p 436)
Organophosphorus compounds	Headache, giddiness, apprehension, restlessness, cold sweating, salivation, uncontrolled urination and defecation, unconsciousness, and in extreme cases ataxia and paralysis of the respiratory centre (Park 28e, Ch 15, p 939). Muscarinic features are secretory-excretory — diarrhoea, bronchorrhoea, emesis, lacrimation, salivation, with miosis, bradycardia and bronchoconstriction; nicotinic features are fasciculation and weakness (Hunter's Diseases of Occupations, Ch 43, pp 435–436)	Atropine is the specific antidote — 1–2 mg intramuscularly, repeated if necessary at 30-minute intervals; also 2-PAM iodide, 2-PAM chloride and P2S (Park 28e, Ch 15, p 939)
Carbamates	Symptoms similar to organophosphorus poisoning but briefer and milder	Atropine titrated to muscarinic symptoms; pralidoxime is not recommended (Hunter's Diseases of Occupations, Ch 43, p 436)
Pyrethrum and pyrethroids	Paraesthesia of face, hands and arms without inflammatory signs, and upper respiratory sensory irritation; severe exposure gives dizziness, headache, fasciculation, convulsions (Hunter's Diseases of Occupations, Ch 43, p 430; Ch 87, p 1186)	None specific — paraesthesiae settle within 24 hours (Hunter's Diseases of Occupations, Ch 87, p 1186)

Severity in anticholinesterase poisoning is directly proportional to enzyme inhibition: symptoms may appear when **50% of cholinesterase activity is inhibited**, and the most life-threatening state occurs at **90% or more** inhibition (Hunter's Diseases of Occupations, Ch 43, p 435).



Time course of human response to organophosphorus insecticide exposure

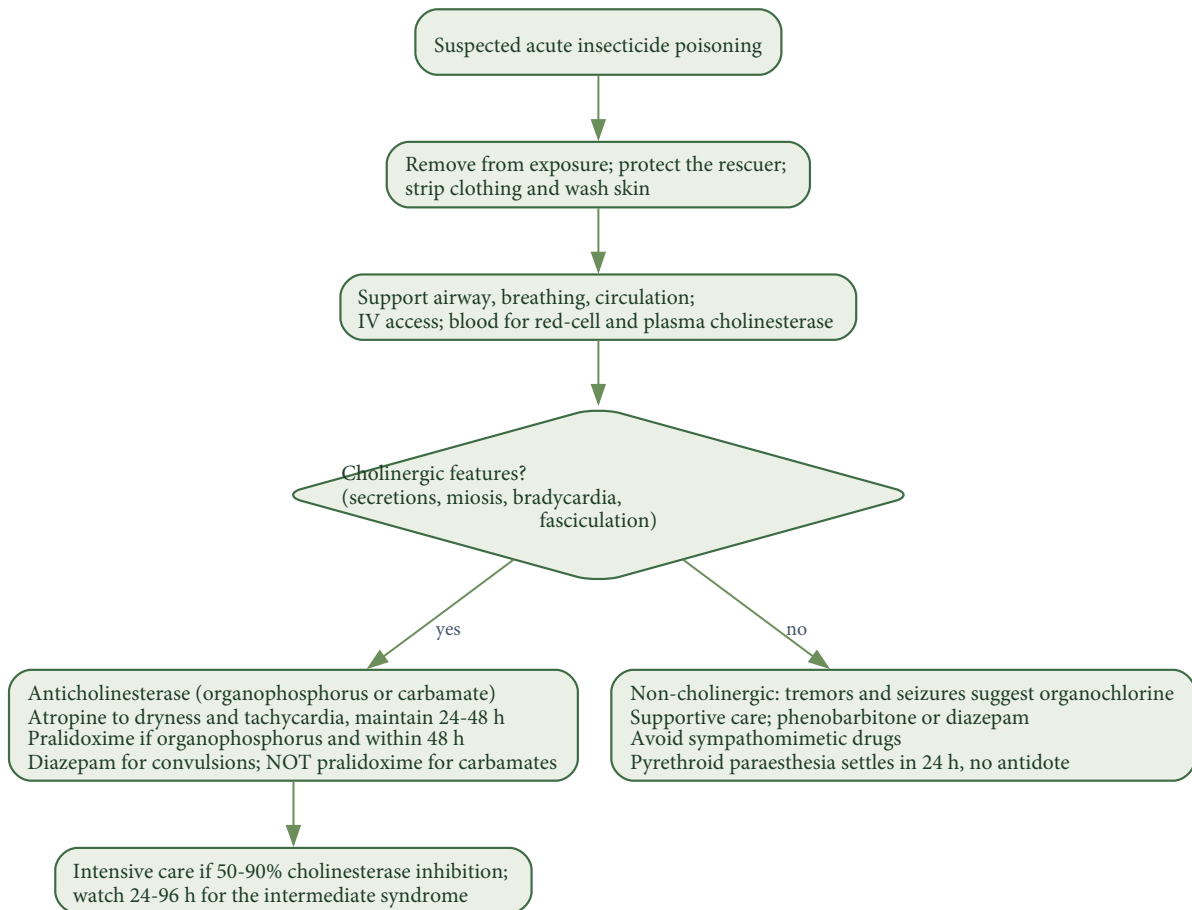
Chronic and long-term effects

Outcome	Strength of evidence
Nervous system	The only fully described peripheral effect is organophosphorus-induced delayed neuropathy, and that is a sequel of acute poisoning rather than a true chronic effect; parkinsonism and neurobehavioural change are suspected but not established (Hunter's Diseases of Occupations, Ch 43, pp 431–432; Ch 87, p 1186)
Cancer	Non-Hodgkin lymphoma in farm workers is the most consistently reported excess, linked to phenoxyacetic acid herbicides and to insecticides; the picture is confounded by multiple concurrent exposures, so a clear risk estimate is still unavailable (Hunter's Diseases of Occupations, Ch 43, p 434; Ch 90, p 1250)
Immune system	Only minor changes in cell populations without overt disease; direct immunotoxicity at low chronic doses is very low or undetectable (Hunter's Diseases of Occupations, Ch 43, p 432)
Reproduction and endocrine disruption	Variably attributed to pesticide exposure; evidence inconclusive (Hunter's Diseases of Occupations, Ch 43, p 431)

Outcome	Strength of evidence
Environmental	Persistence and bioaccumulation of organochlorines in food chains, with adverse effects on cats, poultry, predatory birds and fish; guideline value for DDT in drinking water is 2 µg/litre and for aldrin/dieldrin 0.03 µg/litre (Park 28e, Ch 15, p 879; Gilles & Warrell 4e, Ch 6, p 150)

Diagnosis and management of acute insecticide poisoning

- ▶ **Treat as a medical emergency** and hospitalise; support airway, breathing and circulation first.
- ▶ **Decontaminate safely** — protect yourself, wear protective clothing, remove contaminated clothes and wash the skin in a well-ventilated area; do not allow smoking if a solvent is involved (Oxford Handbook of Occupational Health 2e, Ch 39, pp 831–832).
- ▶ **Confirm the diagnosis** — blood in an EDTA tube for plasma pseudocholinesterase, preferably red-cell acetylcholinesterase; in remote rural areas a finger-prick "paper test" kit gives a quick estimate of inhibition. Organochlorines and their metabolites are measurable in blood by gas-liquid chromatography within a few days (Hunter's Diseases of Occupations, Ch 43, pp 429, 435).
- ▶ **Atropinise** — repeat until secretions are minimal and the skin is dry with a sinus tachycardia, then maintain for 24–48 hours; continuous infusion may be needed in severe cases.
- ▶ **Give an oxime** in moderate-to-severe organophosphorus poisoning — pralidoxime reactivates acetylcholinesterase and is effective **within 48 hours**; it is far less effective later once "ageing" of the phosphorylated enzyme has occurred (Hunter's Diseases of Occupations, Ch 43, p 435).
- ▶ **Diazepam 5–10 mg intravenously** for apprehension, agitation, fasciculation and convulsions.
- ▶ **Do not rely on activated charcoal** — of little benefit, as the pesticide is absorbed rapidly. Monitor blood pressure for hypertensive crisis. Manage moderate-to-severe cases in an intensive care unit.



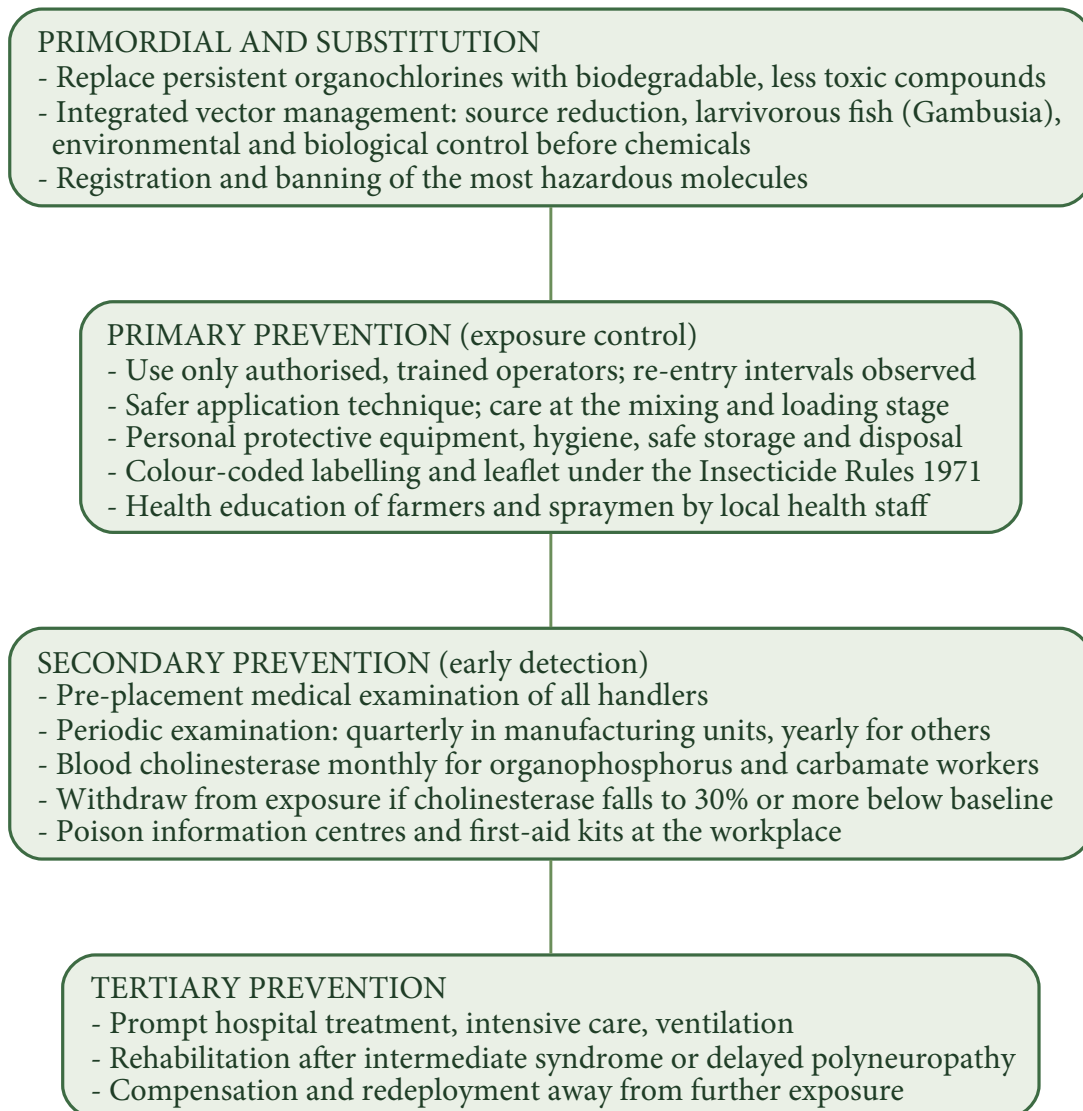
Emergency algorithm for suspected acute insecticide poisoning

Prevention and control, including Indian legislation

The **Insecticides Act, 1968** (Act 46 of 1968) extends to the whole of India, with the Insecticide Rules 1971 framed for its implementation. It was passed after outbreaks of food poisoning from insecticides in Kerala, Madras, West Bengal, Assam, Mumbai and Agra. It created two statutory bodies — the **Central Insecticides Board** (advisory) and the **Registration Committee** (implementing) — to regulate import, manufacture, sale, transport, distribution and use of insecticides so as to prevent risk to human beings and animals. Section 17 forbids import or manufacture of a misbranded insecticide, Section 18 forbids sale, stocking, distribution or transport without registration, and Sections 19 and 20 provide for insecticide analysts and inspectors. Labelling follows FAO guidelines and classification follows WHO (Jugal Kishore, National Health Programmes of India, Ch 3, pp 961–962).

Category under the Insecticide Rules	Label colour	Warning statement
Extremely toxic	Bright red	"Poison"; keep out of the reach of children; if swallowed or if symptoms occur call physician immediately
Highly toxic	Bright yellow	"Poison"; keep out of the reach of children
Moderately toxic	Bright blue	"Danger"; keep out of the reach of children
Slightly toxic	Bright green	"Caution"

Every pack must also carry a leaflet naming the pests it is meant for, its harmfulness to humans and animals with symptoms, safety measures, emergency first aid and the antidote (Jugal Kishore, Ch 3, pp 961–962).



Levels of prevention of insecticide toxicity

Biological monitoring is the backbone of secondary prevention. Baseline plasma and red-cell cholinesterase should be established before each spraying season, or at least 60 days after the last exposure. A fall in activity of **30% or more below the pre-exposure level** requires an immediate repeat test and consideration of removal from exposure; a drop between successive estimations of $\geq 15\%$ in plasma or $\geq 12\%$ in erythrocytes indicates significant absorption (Hunter's Diseases of Occupations, Ch 43, pp 440, 442). Where laboratory support is lacking, as in much of rural India, the emphasis must instead be on training, safe systems of work and care in preparation, use and disposal (Hunter's Diseases of Occupations, Ch 43, p 441).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **The cholinergic synapse in organophosphorus poisoning.** Label: presynaptic neurone; synaptic vesicles containing acetylcholine; synaptic cleft; postsynaptic **muscarinic receptor** (parasympathetic effector organs) and **nicotinic receptor** (neuromuscular junction, autonomic ganglia); **acetylcholinesterase** on the postsynaptic membrane; site of **organophosphorus/carbamate binding to acetylcholinesterase** (irreversible for organophosphorus, reversible for carbamate); accumulated acetylcholine in the cleft; **atropine blocking the muscarinic receptor**; **pralidoxime reactivating the phosphorylated enzyme** (only before "ageing").

High-yield exam pointers

- ▶ Order of human toxicity among organochlorines: **dieldrin (5–8×) > gamma HCH (2×) > DDT**; human median lethal dose of DDT $\approx$ 250 mg/kg body weight; dieldrin poisoning is commonest because it alone is absorbed through intact skin.
- ▶ **Malathion is the least toxic organophosphorus compound**; pyrethroids are the least toxic group overall to mammals.
- ▶ Atropine dose in Park: **1–2 mg intramuscularly, repeated at 30-minute intervals**; oximes are 2-PAM iodide, 2-PAM chloride and P2S, effective only **within 48 hours** (before "ageing").
- ▶ Cholinesterase thresholds: symptoms at **50% inhibition**, life-threatening at $\geq 90\%$; remove from work if activity falls $\geq 30\%$ below baseline.
- ▶ Three phases of organophosphorus poisoning: acute cholinergic crisis (<12 h) $\rightarrow$ intermediate syndrome (24–96 h) $\rightarrow$ delayed polyneuropathy (2–3 weeks).
- ▶ Indian legislation: **Insecticides Act 1968 + Insecticide Rules 1971**; Central Insecticides Board (advisory) and Registration Committee (implementing); label colours **red–yellow–blue–green** for extremely, highly, moderately and slightly toxic.
- ▶ Pyrethroid paraesthesia of face and hands is the commonest field complaint; it is a local nerve-ending effect, needs no antidote, and clears in 24 hours.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park 28e still quotes the WHO Expert Committee position that no toxic effects were reported among 200,000 spraymen and 600–1,000 million people living in repeatedly sprayed houses, and that DDT's safety for human use "remains unchallenged" (Park 28e, Ch 15, p 938); read alongside this, the Stockholm Convention on Persistent Organic Pollutants (2001) restricts DDT to selected public health uses such as indoor residual spraying and net treatment, which remains India's practice (Hunter's Diseases of Occupations, Ch 43, p 421).

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 15 "Toxicity of Insecticides", pp 937–939; Ch 15, p 879; Ch 7, p 498); Hunter's Diseases of Occupations (Ch 43 "Pesticides and other agrochemicals", pp 420–442; Ch 87, pp 1184–1186; Ch 90, p 1250); Oxford Handbook of Occupational Health 2e (Ch 39, pp 831–832); Gilles & Warrell, Essential Malariology 4e (Ch 6, pp 149–151, 183); Jugal Kishore, National Health Programmes of India (Ch 3, pp 961–962 — Insecticides Act 1968 and Insecticide Rules 1971); Service, Medical Entomology for Students (Ch 1, pp 39, 60); WHO/IPCS Recommended Classification of Pesticides by Hazard, 2005.



Trombiculid mites

Asked in: Paper I 15-Nov-2022 — RGUHS MD Community Medicine

Definition — Trombiculid mites (family **Trombiculidae**, class Arachnida, order Acarina) are minute spider-like arthropods with four pairs of legs in the adult and a body not divided into head, thorax and abdomen (Park 28e, Ch 15, p 935). Uniquely among medically important arthropods, **only the six-legged larval stage — the "chigger" — is parasitic on vertebrates**, while the nymphal and adult stages are free-living predators in the soil and never feed on a vertebrate host (Park 28e, Ch 5, p 366). More than 2,000 species exist, of which about 20 commonly attack people; species of the genus *Leptotrombidium* are both the vector and the true reservoir of **scrub typhus** (*Orientia tsutsugamushi*) (Service, Medical Entomology, Ch 19, pp 271–272).

Species of public health importance

Species	Distribution	Public health role
<i>Leptotrombidium deliense</i> , <i>L. akamushi</i>	Asia, South Pacific	Principal vectors of scrub typhus; the two species named in the Indian curriculum (Park 28e, Ch 15, p 935)
<i>L. fletcheri</i> , <i>L. pallidum</i>	South-East Asia, Japan	Additional important vectors (Service, Ch 19, pp 271–272, 276–277)
<i>Neotrombicula autumnalis</i> (harvest mite)	Northern Europe	Nuisance biting only — "harvest-bug itch"; transmits no infection (Service, Ch 19, p 276)
<i>Eutrombicula alfreddugesi</i> (red bug)	USA, Central and South America	Nuisance biting only — "scrub itch", trombidiosis (Service, Ch 19, pp 271–272)

More than 40 mite species in 13 genera are known or suspected vectors of scrub typhus, but **only about seven species are epidemiologically important** (Service, Ch 19, pp 276–277).

External morphology

Feature	Larva ("chigger") — the only parasitic stage	Nymph and adult — free-living
Size	0.15–0.3 mm, just visible to the unaided eye; increases about sixfold after feeding	Adults 1–2 mm
Colour	Reddish or orange, sometimes pale yellowish	Usually reddish
Legs	Three pairs (six legs) , each ending in a pair of large claws	Four pairs (eight legs)
Body covering	Fine feathered hairs on legs and body	Feathered hairs dorsally and ventrally
Mouthparts	Large conspicuous five-segmented palps giving a "false head" appearance	Not adapted to vertebrate feeding
Scutum	Rectangular or pentagonal, weakly sclerotised, on the anterior dorsum; bears five feathered setae plus one pair of sensillae in medically important species	Absent as a diagnostic plate
Eyes	One pair on either side of the scutum	Present

(Service, *Medical Entomology*, Ch 19, pp 271–273)

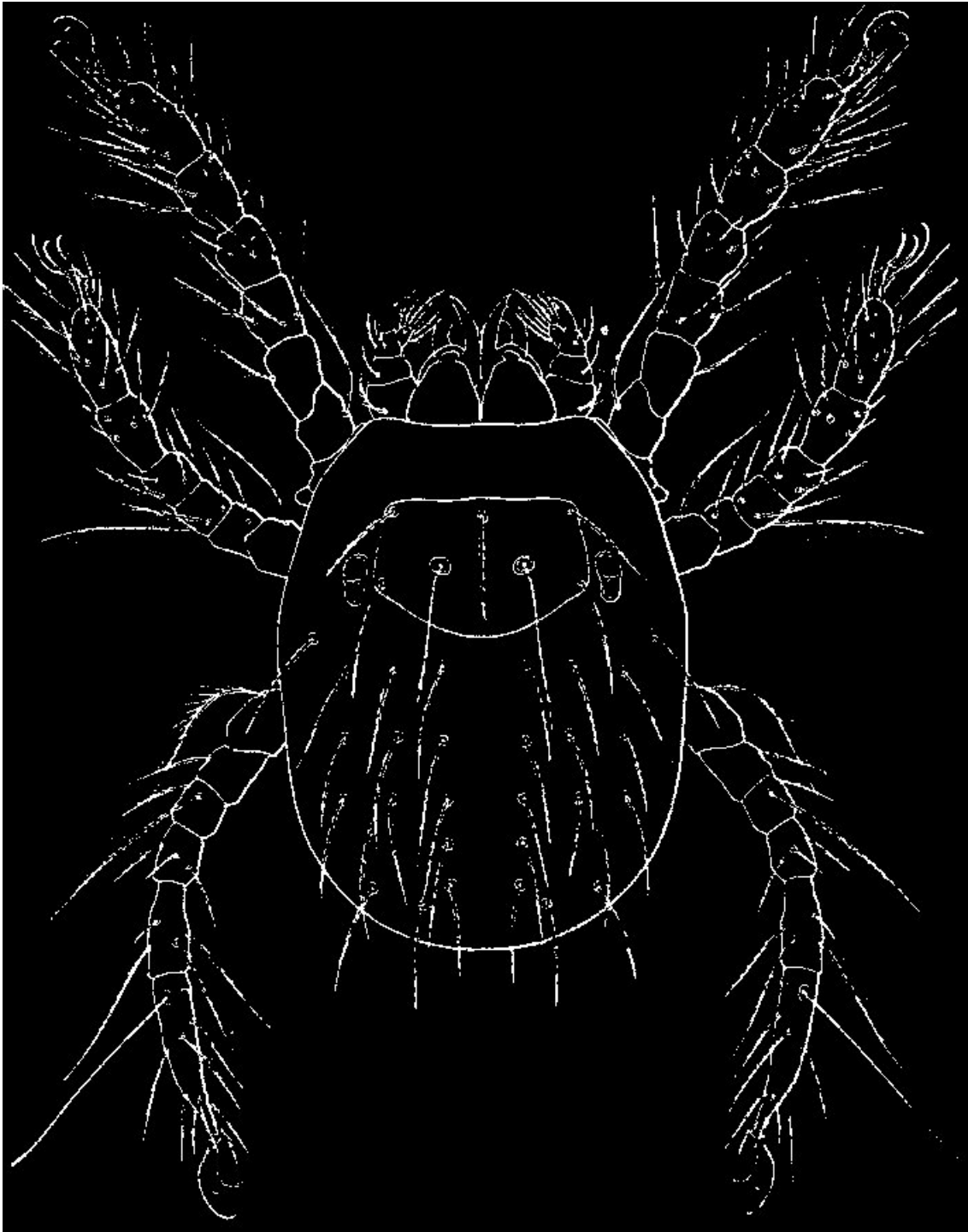


Fig 19.2 — Dorsal view of a larval *Leptotrombidium* mite. Note that the scutum has (Service — *Medical Entomology for Students*, p.273)

Life cycle

The full sequence, with the inactive (non-feeding) stages bracketed, is:

(Egg) → (Deutovum, prelarva) → Larva → (Protonymph) → Deutonymph → (Tritonymph) → Adult
(Service, Ch 19, pp 275–276).

- **Egg and deutovum** — eggs are laid in the soil; the enclosed prelarva (deutovum) is inactive.

- **Larva (chigger)** — hatches, becomes very active, swarms over the ground and climbs grasses and low vegetation seeking a host. This is the **only stage that feeds on a vertebrate**, and it feeds **once only in its lifetime**.
- **Feeding** — the larva pierces the skin and injects saliva that disintegrates host cells. It does **not suck blood**; it feeds on lymph, tissue fluid and semi-digested tissue drawn up a vertical tube-like structure formed in the host's skin called the **stylostome (histiosiphon)**. *Leptotrombidium* larvae remain attached to people for only about **3–10 days**, whereas some other trombiculids stay up to a month.
- **Protonymph** — the engorged larva drops to the ground and buries itself just below the soil surface; it becomes inactive for **7–10 days**.
- **Deutonymph** — an eight-legged, reddish, feather-haired nymph that is **not parasitic**; it is active and predaceous on soil-inhabiting arthropods for a few days to about two weeks.
- **Tritonymph** — inactive for about **14 days**, then moults.
- **Adult** — resembles the nymph but larger; free-living and predaceous on small soil animals; the fertilised female lays the next generation of eggs.

Total duration: usually 40–75 days, but may extend to 12 months (Service, Ch 19, p 275).

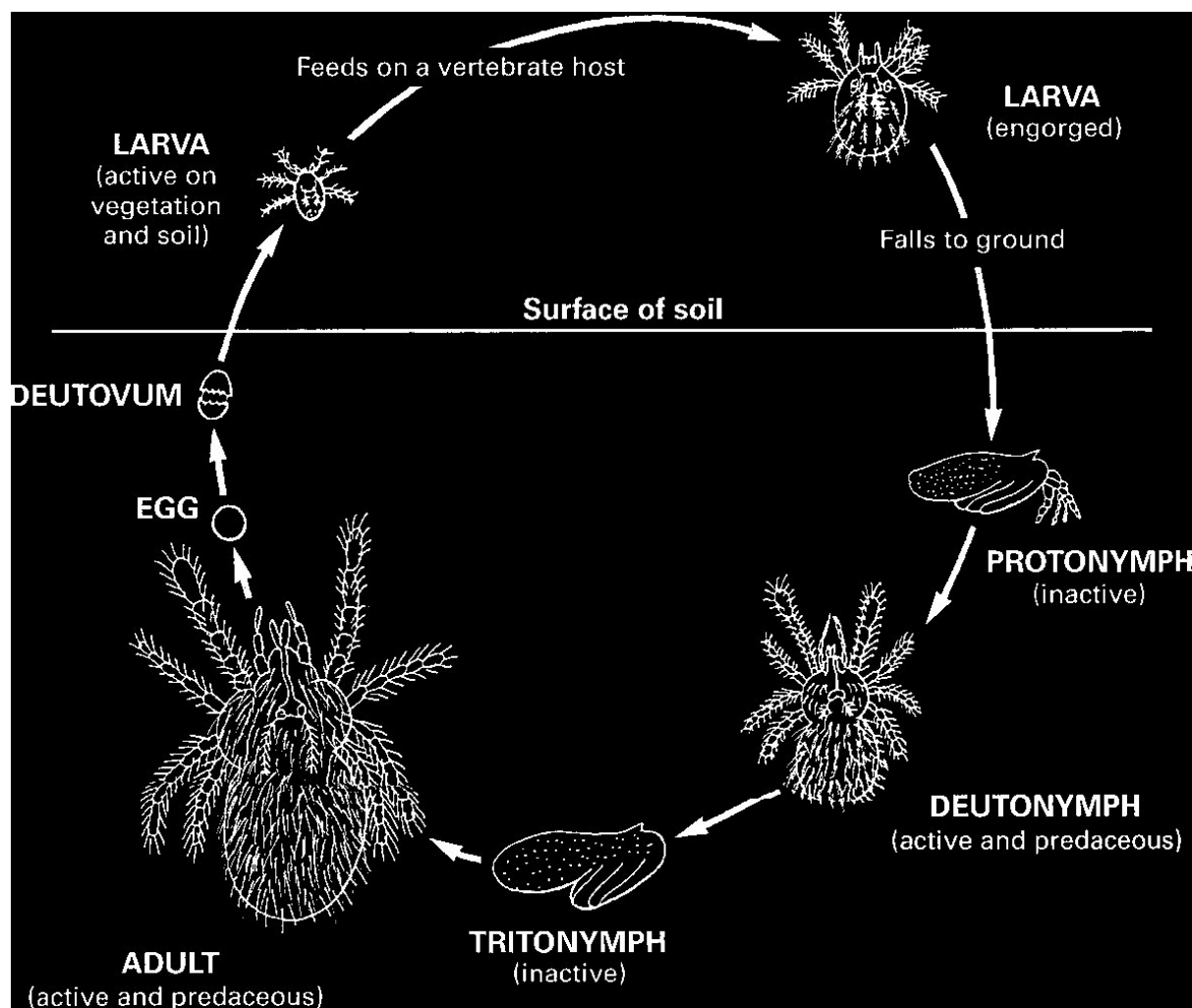


Fig 19.3 — Life cycle of a *Leptotrombidium* mite. (Courtesy of the Natural History (Service — Medical Entomology for Students, p.274)

Bionomics and ecology

- ▶ **Host-seeking behaviour** — larvae ascend grasses and low vegetation and attach to birds and mammals, and to people walking through infested vegetation.
- ▶ **Site of attachment** — on animals the larvae congregate where skin is soft and moist (ears, genitalia, perianal region); **on people they seek sites where clothing is tight against the skin — the waist and the ankles**, which is the anatomical basis for the classic eschar sites.
- ▶ **Natural larval hosts** — rodents (*Rattus*, *Apodemus*, *Microtus*) and insectivores such as shrews (*Suncus*) and tree shrews (*Tupaia*). **Domestic rats appear to play little or no part** in the ecology of scrub typhus. Birds and other incidental hosts help disperse larvae to new areas.
- ▶ **"Mite islands"** — discrete patches of scrub vegetation harbouring dense populations of host-seeking larvae. They range from a few square metres to several square kilometres, and are typically created when people clear vegetation for cultivation or firewood, producing the secondary scrub from which both mite and disease take their name. Their patchy, focal distribution explains the sharply localised, occupationally clustered epidemiology of scrub typhus and makes control difficult.
- ▶ **"Fringe habitats"** — the transition zone between forest and cleared land, where mite islands and rodent hosts overlap, carries the highest risk (Service, Ch 19, pp 275–277).

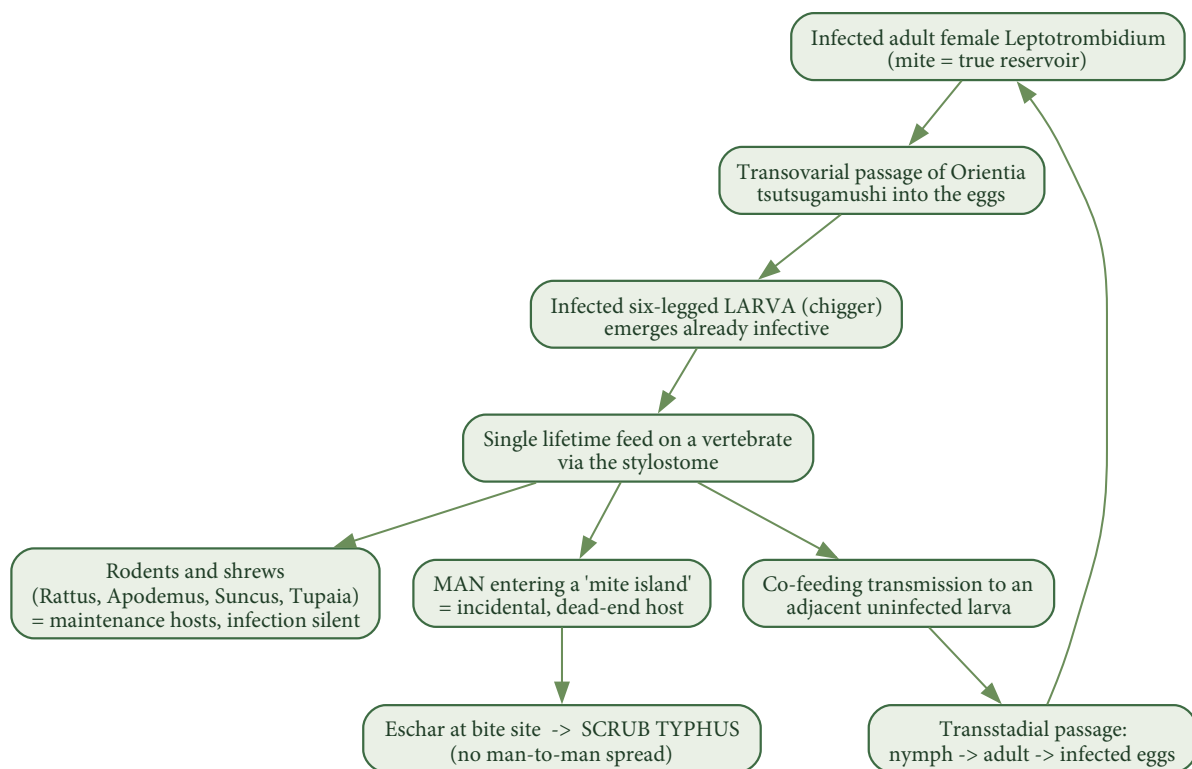
Medical importance

Effect	Mechanism / agent	Comment
Direct nuisance and dermatitis — "scrub itch", "autumnal itch", "harvest-bug itch", trombidiosis	Salivary reaction and the stylostome	Intense itching, usually on the legs; forcible removal leaves mouthparts embedded and worsens irritation (Service, Ch 19, p 276)
Scrub typhus (mite-borne typhus, tsutsugamushi disease, Japanese river fever, chigger-borne rickettsiosis)	<i>Orientia tsutsugamushi</i> — several serologically distinct strains	The most widespread rickettsiosis in man; >1 million cases annually ; endemic in India, Sri Lanka, Japan, South-East Asia, Western Pacific islands, eastern Australia, China and South-Central Russia (Park 28e, Ch 5, p 366)
Historical burden	—	In the Asia-Pacific theatre of World War II (1939–45) scrub typhus incidence in troops was second only to malaria (Service, Ch 19, pp 276–277)

Role of the mite in scrub typhus transmission

Element	Fact
Reservoir	The trombiculid mite itself is the true reservoir; infection is maintained transovarially from one mite generation to the next. Rodents and shrews are amplifying/maintenance hosts (Park 28e, Ch 5, p 366)
Vector stage	The larva (chigger) only — it acquires and transmits at its single lifetime feed

Element	Fact
Additional route	Co-feeding transmission — an uninfected larva feeding beside an infected larva acquires rickettsiae, which then pass transstadially to the adult and thence to the eggs (Service, Ch 19, p 277)
Mode of infection in man	Inoculation of rickettsiae during the bite of the infected larva; not transmitted person to person (Park 28e, Ch 5, p 366)
Incubation period	Usually 10–12 days , range 6–21 days
Bite-site lesion	Eschar — a punched-out ulcer covered with a blackened scab at the site of the mite bite; it localises the mite bite and is the single most useful clinical clue
Other features	Chills, fever 104.5–105°F, headache, prostration, macular rash about the 5th day , generalised lymphadenopathy and lymphocytosis; in untreated cases pyrexia falls by lysis in the 3rd week
Laboratory diagnosis	Weil–Felix reaction strongly positive with <i>Proteus</i> strain OX-K ; also indirect fluorescent antibody test, complement fixation test, ELISA, isolation of rickettsiae (Park 28e, Ch 5, p 366)
Treatment	Tetracycline is the drug of choice ; long-acting doxycycline or minocycline make single-dose treatment possible. With proper therapy mortality is nil
Vaccine	None available (Park 28e, Ch 5, p 366)

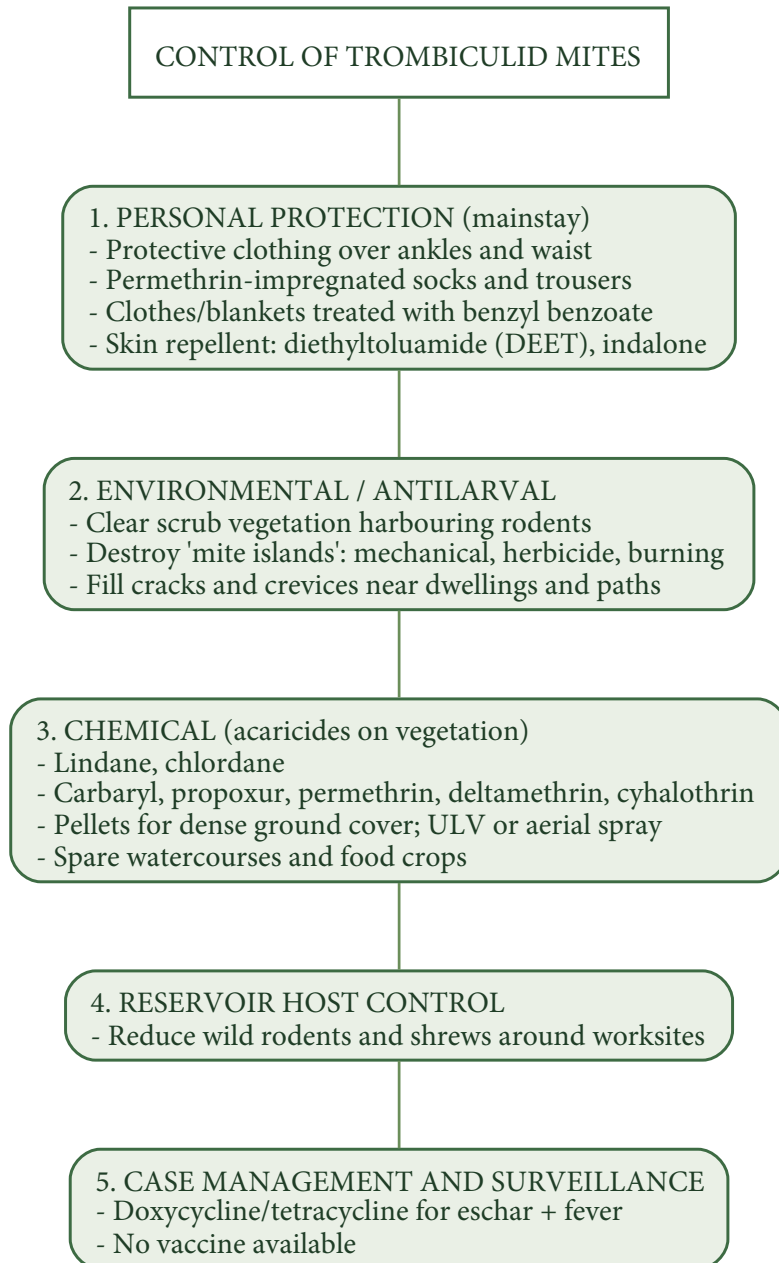


Maintenance and transmission cycle of Orientia tsutsugamushi by trombiculid mites

Control

Approach	Measures
Environmental / antilarval	Clear the scrub vegetation harbouring rats and mice; destroy identified mite islands mechanically, with herbicides or by controlled burning; fill cracks and crevices near buildings and paths (Park 28e, Ch 5, p 366; Ch 15, p 935; Service, Ch 19, pp 277–278)
Chemical (acaricides)	Ground or vegetation application of lindane or chlordane (Park 28e, Ch 5, p 366); carbaryl , propoxur , permethrin , deltamethrin or cyhalothrin sprayed on mite-infested vegetation — pellet formulations where ground cover is dense; ground-based sprayers, aerial or ultra-low-volume (ULV) spraying (Service, Ch 19, pp 277–278)
Personal prophylaxis	Protective clothing; impregnation of clothes and blankets with benzyl benzoate ; permethrin -treated socks and trousers; repellents to exposed skin — diethyltoluamide (DEET) , indalone, benzyl benzoate, dimethyl carbamate (Park 28e, Ch 5, p 366; Ch 15, p 935)
Rodent (host) control	Reduce wild rodent and shrew larval hosts around habitations and worksites where practicable
Case management	Prompt tetracycline/doxycycline for febrile patients with an eschar from an endemic focus — abolishes mortality and shortens the infective period
Not available	No vaccine exists at present

Practical limits of control — the patchy, focal distribution of chiggers makes area control difficult; blanket insecticidal spraying is impracticable over cultivated land and may be environmentally unacceptable, and care is needed to avoid contaminating watercourses and food crops. Personal protection is therefore the mainstay in the field (Service, Ch 19, pp 277–278).



Stepwise control of trombiculid mites and scrub typhus

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Dorsal view of a larval trombiculid mite (chigger)** — label: *three pairs of legs with terminal paired claws; five-segmented palps forming the "false head"; rectangular/pentagonal scutum; five feathered scutal setae; one pair of sensillae; pair of eyes lateral to the scutum; feathered body hairs.*
- **Life cycle of *Leptotrombidium*** — draw as a circle bisected by a horizontal line labelled *surface of soil*. Above the line: *LARVA (active on vegetation and soil) → feeds on a vertebrate host → LARVA (engorged) → falls to ground*. Below the line: *PROTONYMPH (inactive) → DEUTONYMPH (active and predaceous) → TRITONYMPH (inactive) → ADULT (active and predaceous) → EGG → DEUTOVUM → back to larva.*

High-yield exam pointers

- **Only the larva bites** — nymph and adult are free-living soil predators; the larva feeds **once in its lifetime**, on **lymph and tissue fluid, not blood**, through the **stylostome**.
- Full sequence: (Egg) → (Deutovum) → Larva → (Protonymph) → Deutonymph → (Tritonymph) → **Adult**; total **40–75 days** (up to 12 months). Larva 0.15–0.3 mm; adult 1–2 mm.
- Larval scutum carries **5 feathered setae + 1 pair of sensillae** — the key identifying feature of medically important species.
- The **mite is the true reservoir**; maintained by **transovarial** transmission, supplemented by **co-feeding** then **transstadial** transmission. Man is a **dead-end host**.
- **Scrub typhus**: *Orientia tsutsugamushi*; incubation **10–12 days (6–21)**; **eschar** at the bite site; rash on day 5; **Weil–Felix positive with *Proteus* OX-K**; **tetracycline/doxycycline**; **no vaccine**.
- Vectors in India and Asia: *L. deliense* and *L. akamushi*; focal transmission from "**mite islands**" in fringe habitats.
- Park's arthropod table: **trombiculid mite → scrub typhus and rickettsial pox; itch mite (*Sarcoptes/Acarus scabiei*) → scabies** (Park 28e, Ch 15, p 922).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written: (1) Park's arthropod-and-disease table credits the trombiculid mite with both scrub typhus and rickettsial pox, whereas Service attributes rickettsial pox (*Rickettsia akari*) to the mouse mite *Liponyssoides sanguineus*, a mite of a different family — write scrub typhus as the trombiculid disease and mention rickettsial pox only as Park lists it. (2) Park 28e condenses the mite life history to four stages (egg, larva, nymph, adult), while the seven-stage sequence given above is Service's fuller entomological account and is the one expected in a medical-entomology answer. (3) Park still names lindane and chlordane for vegetation treatment; these organochlorines are restricted in current Indian practice, and the pyrethroid acaricides listed alongside are what is used in the field today. (4) Coverage note: the indexed corpus carries no India-specific scrub typhus incidence figure or national programme component for mite control, so no Indian burden number is quoted here.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 5 (Epidemiology of Communicable Diseases, rickettsial diseases and scrub typhus, pp 365–366) and Ch 15 (Environment and Health, arthropods of public health importance and control of ticks and mites, pp 922, 935); Service — Medical Entomology for Students, Ch 19 (Scrub typhus mites, Trombiculidae, pp 271–278), including Fig 19.2 and Fig 19.3.



Vectorial capacity

Asked in: Paper I 24-Sep-2025 — RGUHS MD Community Medicine

Definition — Vectorial capacity is the combined effect of the density of the vector population, its susceptibility to infection, its life span and its probability of feeding on man (Park 28e, Ch 5, p 322). Quantitatively it is a transmission-probability index expressing the mean number of probable infective inoculations that the whole vector population distributes from one infective human case in unit time, conventionally per day (Gilles & Warrell 4e, Ch 5, pp 99-100). It is a property of the **vector population in its environment**, and is expressly distinct from the physiological (intrinsic) capacity of the individual mosquito to transmit infection (Park 28e, Ch 5, p 322).

Mathematical expression and its components

$C = (m \times a^2 \times p^n) \div (-\log_e p)$ — where $\log_e$ is the natural logarithm (Gilles & Warrell 4e, Ch 5, p 100).

Symb ol	Parameter	What it means	How it is measured in the field
m	Vector density	Relative density of female anophelines per person	Mosquito density expressed as number of mosquitoes per man-hour catch (Park 28e, Ch 5, p 324)
a	Man-biting habit	Probability that a mosquito takes a human blood meal on a given day (host preference × biting frequency)	Human blood index — proportion of freshly-fed female anophelines whose stomach contains human blood; indicates the degree of anthropophilism (Park 28e, Ch 5, p 324)
m × a	Man-biting rate (biting density)	Average incidence of anopheline bites per day per person	Standardised vector catch on human bait (human landing catch) (Park 28e, Ch 5, p 324; Gilles & Warrell 4e, Ch 5, p 103)
a²	Biting enters twice	One bite is needed to acquire the parasite from an infective case and a second, later bite to deliver it — hence <i>a</i> is squared	—

Symb ol	Parameter	What it means	How it is measured in the field
p	Probability of daily survival	Probability that the mosquito survives through one day	Ovarian age-grading / proportion parous, an important entomological parameter of malaria epidemiology (Gilles & Warrell 4e, Ch 4, p 94)
n	Duration of sporogony (extrinsic incubation period)	Days from the infective feed until sporozoites reach the salivary glands	<i>P. falciparum</i> 9 days at 30 °C, 10 days at 25 °C, 23 days at 20 °C, not completed below 17 °C; <i>P. vivax</i> 9 days at 25 °C, 16 days at 20 °C (Service, Ch 1, p 49)
p^n	Survival through sporogony	Proportion of vectors surviving the parasite's incubation period	Derived from p and n (Gilles & Warrell 4e, Ch 5, p 100)
$1 \div (-\log_e p)$	Expectation of infective life	Mean number of days a mosquito lives, i.e. the days available for infective bites	Derived from p ; the same term forms Davidson's stability index (Gilles & Warrell 4e, Ch 5, pp 99-100)

Vectorial capacity distinguished from allied measures

Measure	Definition / formula	Key distinction
Vectorial capacity (C)	$(m a^2 p^n) \div (-\log_e p)$	Purely entomological ; the <i>potential</i> daily transmission a vector population can deliver, irrespective of how many people are actually infected
Vector competence	Intrinsic susceptibility to infection, ability to replicate the agent and ability to transmit it to another host (Park 28e, Ch 5, p 307)	A physiological property of the individual mosquito , measured in the laboratory; it is one input to vectorial capacity, not a synonym
Basic reproduction rate (R_0)	$(m a^2 b p^n) \div (-r \log_e p)$ (Gilles & Warrell 4e, Ch 5, p 100)	Adds two human-side terms (b, r) that vectorial capacity omits. $R_0 > 1$ sustains transmission; $R_0 < 1$ means diminishing risk and a tendency to unstable conditions
Stability index (Davidson)	$a \div (-\log_e p)$	An index built only on the vector's human-biting habit and expectancy of life ; separates stable from unstable malaria
Inoculation rate / entomological inoculation rate (EIR)	Man-biting rate $\times$ infective sporozoite rate (Park 28e, Ch 5, p 324; Gilles & Warrell 4e, Ch 4, pp 91-92)	The observed rate; annualised EIR reflects the incidence of new infections actually received by the human population (Gilles & Warrell 4e, Ch 5, p 103)

Determinants of the individual components

Determinant	Component chiefly affected	Effect (all Park 28e, Ch 5, p 322 unless stated)
Temperature	n, p	Optimum for parasite development in the vector 20–30 °C (68–86 °F); development ceases below 16 °C (60.8 °F); above 30 °C is lethal to the parasite
Humidity	p	Atmospheric humidity has a direct effect on the length of life of the mosquito, but no effect on the parasite
Rainfall	m	Raises humidity needed for mosquito survival; heavy rain may flush out breeding places, while drought created river-bed pools and severe epidemics in Sri Lanka
Altitude	m	Anophelines are as a rule not found above 2000–2500 metres
Host choice (anthropophily)	a	Anthropophilic species are better vectors than zoophilic ones; human blood feeds in <i>An. culicifacies</i> vary 2–80 per cent, whereas <i>An. fluviatilis</i> is highly anthropophilic
Resting habit	p	Endophily (indoor resting) makes indoor residual spray effective; exophilic vectors escape it
Time of biting	a	Most Indian anophelines feed nocturnally, between dusk and dawn; <i>Aedes</i> bite by day — this decides whether bed-nets can work
Insecticide resistance	p	Resistance status decides the choice of insecticide; reduced susceptibility of <i>An. stephensi</i> is documented in India (Service, Ch 1, pp 61-62)
Man-made breeding	m	Burrow pits, garden pools, irrigation channels, dams, roads and bridges create "man-made malaria"
Vector competence	Susceptibility term	Susceptibility to infection, replication and onward transmission — e.g. the high competence of <i>Aedes aegypti</i> and <i>Ae. albopictus</i> for dengue virus (Park 28e, Ch 5, p 307)

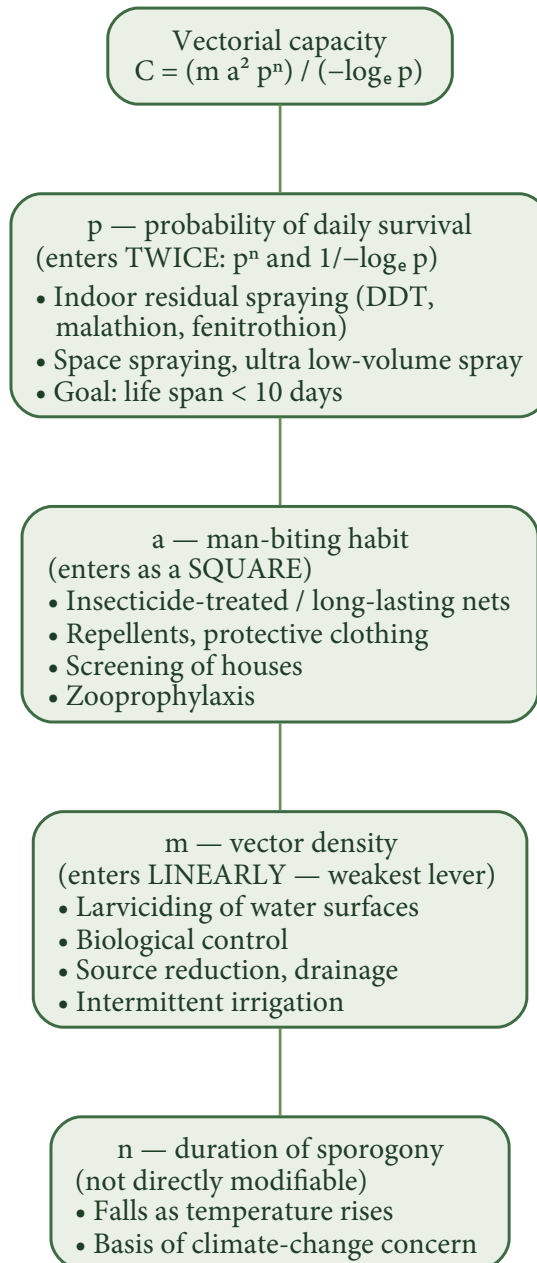
Implications

- **Longevity is the single most powerful term.** The probability of daily survival enters the formula twice — as p^n and again in the expectation of life — so vectorial capacity is exquisitely sensitive to it. The vector must live at least **10–12 days after an infective blood meal** to become infective, and the classical strategy of malaria eradication is therefore to **shorten the mosquito's life span to under 10 days** with insecticides (Park 28e, Ch 5, p 322). An average daily survival above 60 per cent is associated with stable endemicity in *An. gambiae* and *An. funestus* (Gilles & Warrell 4e, Ch 5, p 100).
- **This is why indoor residual spraying ranks first among vector-control measures.** Indoor house spraying **reduces the longevity of the vector** (Park 28e, Ch 5, p 332), and residual house spraying reduces the proportion of parous females by shortening adult longevity (Gilles & Warrell 4e, Ch 4, p 94). It remains the primary method of vector control in rural India, with anti-larval measures leading in urban areas (Park 28e, Ch 7, p 498).

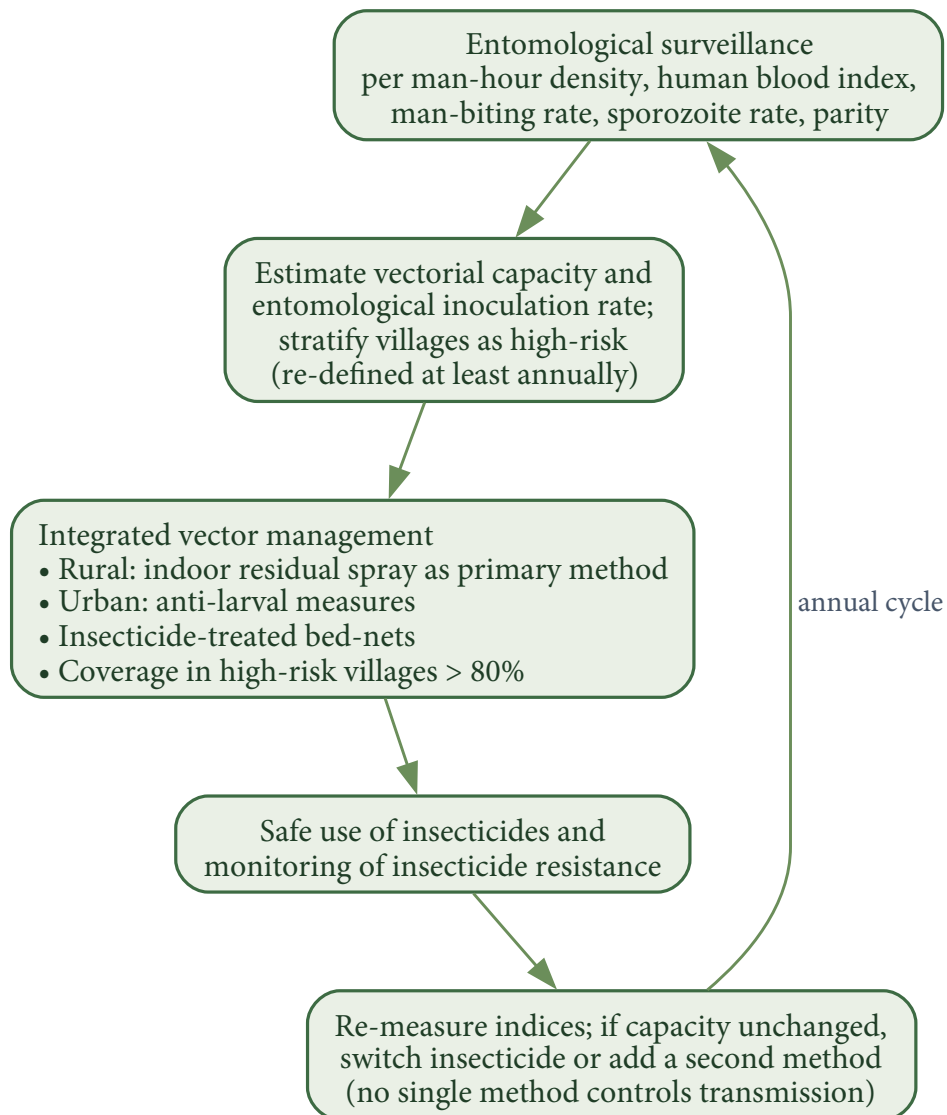
- ▶ **Man-vector contact enters as a square.** Halving the man-biting habit cuts vectorial capacity roughly fourfold. This is the quantitative justification for insecticide-treated and long-lasting insecticidal nets, repellents, protective clothing, screening of houses and zooprophylaxis (Park 28e, Ch 5, p 332).
- ▶ **Vector density is the weakest, purely linear lever.** Larviciding, biological control and source reduction (drainage, filling, deepening, flushing, water-level management, altering salt content, intermittent irrigation) act only in proportion to m , so no single method can by itself ensure control of transmission — the explicit rationale for **integrated vector control** combining bioenvironmental and personal-protection measures (Park 28e, Ch 5, p 333).
- ▶ **Transmission can persist at low vector density.** A long-lived, strongly anthropophilic species sustains transmission at densities at which a purely density-based control programme looks successful; conversely a zoophilic, short-lived species is a poor vector even in enormous numbers. Species-specific control, not blanket mosquito reduction, is what the concept demands.
- ▶ **It converts entomological surveillance into a decision tool.** Its components are precisely the routine entomological indices — per-man-hour density, human blood index, man-biting rate, sporozoite rate and parity — which together permit quantitative assessment of malaria and construction of a composite epidemiological picture (Park 28e, Ch 5, p 324).
- ▶ **Insecticide resistance is a vectorial-capacity failure.** Resistance restores daily survival and hence transmission; discontinuation of spraying has often been followed by resurgence of malaria (Park 28e, Ch 5, p 332). Integrated vector management under the national programme therefore explicitly includes safe use of insecticides and **monitoring of insecticide resistance** (Park 28e, Ch 7, p 498).
- ▶ **Climate warming raises vectorial capacity steeply.** Even a 2 °C rise shortens the extrinsic incubation period so that more infected mosquitoes are available for longer, and dehydration makes mosquitoes bite more frequently, increasing man-mosquito contact (Park 28e, Ch 5, p 307) — n falls while a rises, and both act non-linearly.
- ▶ **It underlies receptivity in the elimination phase.** Receptivity — the probability of resumption of transmission, dependent on the presence and abundance of local vectors and on favourable climatic conditions — is essentially the residual vectorial capacity of an area; with vulnerability (the probability of introduced infections) it governs the risk of re-establishment in malaria-free areas (Gilles & Warrell 4e, Ch 6, p 123).
- ▶ **The concept generalises beyond malaria.** The density, behaviour and vectorial capacity of the vector population is one of the factors that initiate and maintain a dengue epidemic (Park 28e, Ch 5, p 307), and the same logic drives integrated control of filariasis, in which no single vector-control measure applied alone achieves sustained control (Park 28e, Ch 5, pp 339-340).
- ▶ **Its limitations must be stated.** It is a *potential*, not an observed, transmission rate; it takes no account of human immunity, case treatment or heterogeneous man-vector contact; and its two most influential parameters are the hardest to measure, since human bait catches are expensive and technically difficult to replicate (Gilles & Warrell 4e, Ch 5, p 103).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **DRAWN figure 1 — "The vectorial capacity equation tree".** Draw the formula at the top and drop a labelled branch to each term. Labels: C = vectorial capacity (infective inoculations per case per day); m = vector density per person; a = man-biting habit (human blood index $\times$ biting frequency); a^2 = two bites required — one infective feed, one transmitting bite; p = probability of daily survival; n = duration of sporogony (extrinsic incubation period); p^n = proportion surviving sporogony; $1 \div (-\log_e p)$ = expectation of life in days.
- ▶ **DRAWN figure 2 — "Man-mosquito-man cycle with the measurable points".** Labels in order: Infective human case (gametocyte carrier) $\rightarrow$ Mosquito takes infective blood meal (a) $\rightarrow$ Sporogony in mosquito, n days, temperature dependent $\rightarrow$ Mosquito must survive n days (p^n) $\rightarrow$ Infective mosquito, sporozoites in salivary glands (sporozoite rate) $\rightarrow$ Second bite on a susceptible person (a) $\rightarrow$ New human infection; mark vector density (m) against the mosquito population and man-hour catch as its measurement.



Components of vectorial capacity and the control measure that attacks each, in descending order of impact



Translating vectorial capacity into integrated vector management under the national programme

High-yield exam pointers

- ▶ **Formula:** $C = (m a^2 p^n) \div (-\log_e p)$; basic reproduction rate $R_0 = (m a^2 b p^n) \div (-r \log_e p)$; Davidson's stability index = $a \div (-\log_e p)$.
- ▶ **Park's one-liner (write it verbatim):** combined effect of vector density, susceptibility to infection, life span and probability of feeding on man — distinct from the physiological capacity to transmit infection.
- ▶ **The 10-day rule:** the vector must survive 10–12 days after an infective feed; eradication strategy = shorten life span to under 10 days.
- ▶ **Temperature limits:** parasite development in the vector optimal 20–30 °C, ceases below 16 °C, lethal above 30 °C; *P. falciparum* sporogony 9 days at 30 °C but 23 days at 20 °C.
- ▶ **Four indices to name:** human blood index, sporozoite rate, mosquito density (per man-hour catch), man-biting rate; inoculation rate = man-biting rate $\times$ infective sporozoite rate.
- ▶ **Leverage order to write in the implications:** survival (squared-plus effect) > man-biting habit (squared) > density (linear) — hence indoor residual spray first, nets second, larviciding last.

- **Never confuse** vectorial capacity (population-level, potential, entomological) with vector competence (individual mosquito, physiological).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the retrieved Park 28e text names the programme the National Vector Borne Disease Control Programme (NVBDCP), which has since been re-designated the National Center for Vector Borne Diseases Control (NCVBDC); the algebraic formula, the >60 per cent daily-survival threshold for stable endemicity and the receptivity–vulnerability framing are drawn from Essential Malariology 4e, since Park 28e gives only the descriptive definition of vectorial capacity.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 3 (Principles of Epidemiology and Epidemiologic Methods), Ch 5 (Epidemiology of Communicable Diseases), Ch 7 (Health Programmes in India); Gilles & Warrell — Essential Malariology 4e — Ch 4 (The *Anopheles* vector), Ch 5 (The epidemiology of malaria), Ch 6 (Rationale and technique of malaria control); Service — Medical Entomology for Students, Ch 1.



Integrated vector control

Asked in: Paper I 15-Nov-2022 — RGUHS MD Community Medicine

Definition — Integrated vector control, now usually called **integrated vector management (IVM)**, is the utilisation of all appropriate technological and management techniques to bring about an effective degree of vector suppression in a cost-effective manner (Gilles & Warrell 4e, Ch 6, pp 179–180). It is best regarded as **any approach that takes into consideration more than one method**, whether directed at the larvae, the adults, or both (Service — Medical Entomology, Ch 1, p 40). Operationally it is an approach that avoids the excessive use of any one method (especially insecticides) and instead combines two or more methods to obtain maximum results with minimum inputs, while preventing environmental pollution with toxic chemicals and the development of insecticide resistance (Park 28e, Ch 15, p 926).

Why a single-method approach fails: the rationale for integration

Constraint	Evidence / consequence
No universal method	No single method of control is likely to provide a solution in all situations — there is no single and simple method that would ensure control of transmission (Park 28e, Ch 5, p 333); none of the vector-control measures applied alone gives sustained control of filariasis vectors (Park 28e, Ch 5, p 339)
Insecticide resistance	Vector control by insecticides alone is no longer fully effective — resistance has appeared in over 100 species of arthropods of public health importance (Park 28e, Ch 15, p 922)

Constraint	Evidence / consequence
Environmental contamination	Danger of pollution with persistent compounds has led to restricted use of many insecticides; biodegradable, less toxic compounds are replacing DDT (Park 28e, Ch 15, p 922)
Cost escalation	Switching from organochlorines to organophosphorus and carbamate compounds involves a substantial increase in cost (Park 28e, Ch 15, pp 938–939)
Biological methods are adjuncts	Biological control can be effective only when used in conjunction with other methods (Park 28e, Ch 15, p 927)
Genetic methods immature	Sterile-male technique, cytoplasmic incompatibility, chromosomal translocation, sex distortion and gene replacement are still in the "research phase" (Park 28e, Ch 15, p 927)
Permanent engineering unaffordable	Permanent environmental engineering costs several crores of rupees and is not feasible in India in the near future, so temporary/recurrent methods must also be used (Park 28e, Ch 5, p 339)

The building blocks: methods available for integration

General principles of arthropod control are environmental, chemical, biological and genetic control, plus newer methods — insect growth regulators, chemosterilants and sex attractants (pheromones) (Park 28e, Ch 15, p 922).

Component	Content and key operational specifics
Environmental control (source reduction)	Best approach, because results are likely to be permanent — elimination of breeding places, filling and drainage, planned water management, piped water supply, proper refuse and waste disposal, cleanliness in and around houses; intensive health education and political support are essential prerequisites (Park 28e, Ch 15, p 922)
Chemical — larvicides	Mineral oils (diesel, fuel oil, kerosene, mosquito larvicidal oil) 40–90 litres/hectare, weekly (mosquito life cycle $\approx$ 8 days); Paris green (copper acetoarsenite), a stomach poison used as 2% dust, particle size 20–25 microns , kills surface-feeding <i>Anopheles</i> larvae; synthetic larvicides — temephos (Abate) 56–112, fenthion 22–112, malathion 224–672, chlorpyrifos 11–16 g/hectare (Park 28e, Ch 15, pp 926–927, Table 6)
Chemical — residual sprays (anti-adult)	DDT 1–2 g of pure DDT per m², applied 1–3 times a year to walls and resting surfaces; where DDT resistance exists, malathion, propoxur (OMS-33) and to a lesser extent lindane (Park 28e, Ch 15, p 927)
Chemical — space sprays (anti-adult)	Pyrethrum extract (0.1% pyrethrin solution) 1 oz per 1,000 cubic feet , doors and windows closed for half an hour; ultra-low-volume (ULV) fogging with malathion or fenitrothion. No residual action — reduction is only temporary and reinfestation occurs quickly (Park 28e, Ch 15, p 927)
Biological control	Larvivorous fish — <i>Gambusia affinis</i> and <i>Lebistes reticulatus</i> ("Barbados millions") in burrow pits, oxidation ponds, cisterns and farm ponds; fungi of genus <i>Coelomomyces</i> ; bacteria, nematodes, protozoa and viruses under study (Park 28e, Ch 15, pp 922, 927)
Genetic control	Sterile-male technique, cytoplasmic incompatibility, chromosomal translocations, sex distortion, gene replacement. Cheaper, potentially more efficient and not subject to vector resistance , but still experimental (Park 28e, Ch 15, p 927)

Component	Content and key operational specifics
Protection against bites	Mosquito net — white material, mesh aperture not exceeding 0.0475 inch, about 150 holes per square inch; screening with copper or bronze gauze of 16 meshes to the inch; repellents — diethyltoluamide (DEET), active against <i>Culex quinquefasciatus</i> for 18–20 hours (Park 28e, Ch 15, p 927)

Principles governing integration

- **Compatibility rule** — the essential requirement is the ability to use one method that favours, or at least does not interfere with, another; e.g. larviciding of ponds combined **sequentially** with source reduction and larvivorous fish, as practised in Italy, Hawaii and Japan (Gilles & Warrell 4e, Ch 6, pp 179–180).
- **Vector-specific evidence base** — adequate knowledge of the biology, ecology and behaviour of the vector, of local agricultural practices and of aquatic animal life is necessary to avoid costly mistakes (Gilles & Warrell 4e, Ch 6, pp 179–180). Endophilic, indoor-resting vectors justify indoor residual spraying; exophilic, outdoor-resting vectors do not.
- **Human safety and safe use of insecticides** — only compounds and formulations conforming to World Health Organization (WHO) specifications; hazard is judged from the formulation, concentration, method of use, storage conditions, labelling and worker training, not from intrinsic toxicity alone (Gilles & Warrell 4e, Ch 6).
- **Community involvement** — the trend of integrated vector control is towards community involvement through well-planned, sustained measures related to socio-economic conditions (Gilles & Warrell 4e, Ch 6, pp 179–180).
- **Enduring core components** — personal protection, better sanitation, skilful agriculture and water management, source reduction, waste disposal and acceptable insecticides remain the major components of any integrated scheme (Gilles & Warrell 4e, Ch 6, pp 179–180).

Integrated vector management in India (National Vector Borne Disease Control Programme)

The National Vector Borne Disease Control Programme (NVBDCP) covers **six diseases** — malaria, filariasis, kala-azar, Japanese encephalitis (JE), dengue and chikungunya; five are mosquito-borne and kala-azar is sandfly-borne. The Directorate is the nodal agency for planning, policy, technical guidance and monitoring and evaluation under the umbrella of the National Health Mission, while States plan, implement and supervise (Park 28e, Ch 7, p 493).

Prong of the NVBDCP strategy	Content
1. Disease management	Early case detection and complete treatment, strengthening of referral services, epidemic preparedness and rapid response (Park 28e, Ch 7, p 493)
2. Integrated vector management	Indoor residual spraying (IRS) in selected high-risk areas, insecticide-treated bed-nets, larvivorous fish, anti-larval measures in urban areas, source reduction and minor environmental engineering (Park 28e, Ch 7, p 493)

Prong of the NVBDCP strategy	Content
3. Supportive interventions	Behaviour change communication, public–private partnership and inter-sectoral convergence, human resource development, operational research on drug resistance and insecticide susceptibility, monitoring and evaluation with a web-based management information system, JE vaccination and annual mass drug administration for lymphatic filariasis (Park 28e, Ch 7, p 493)

How integration is actually achieved in the programme (Park 28e, Ch 7, pp 498, 501; Ch 5, p 339):

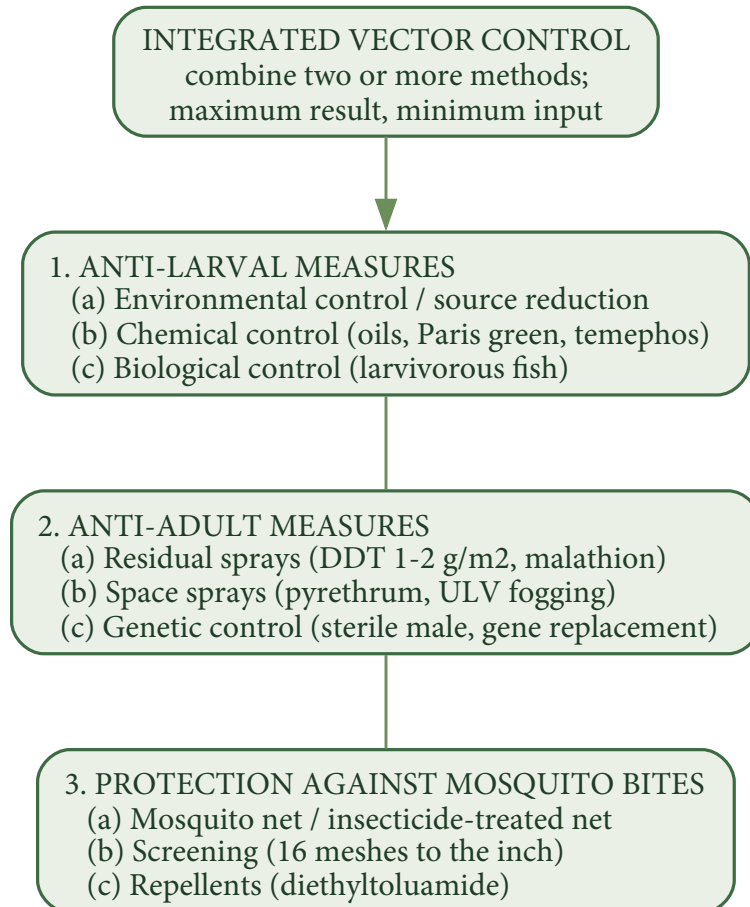
- Biological, chemical and environmental interventions of proven efficacy are used **separately or in combination as appropriate to the area**, through optimal use of resources.
- **Identical vector-control methods** are used to control malaria and leishmaniasis in rural areas, and malaria and dengue in urban areas, to achieve **cost-effectiveness and synergy**.
- IRS is the primary method in rural areas; anti-larval measures are the mainstay in urban areas. For filariasis, anti-larval work is restricted to urban areas **plus a 3-km peripheral belt**, matching the flight range of *Culex quinquefasciatus*.
- High-risk villages receive IRS and insecticide-treated nets with **coverage above 80 per cent**; high-risk areas and populations are re-defined **at least annually**.
- For JE, IVM relies on personal protection and larvivorous fish, because JE vectors are outdoor resters and IRS is not effective.
- Collaboration with public and private agencies and community participation is built in, and the package explicitly includes **safe use of insecticides and monitoring of insecticide resistance**.

Monitoring, evaluation and resistance management

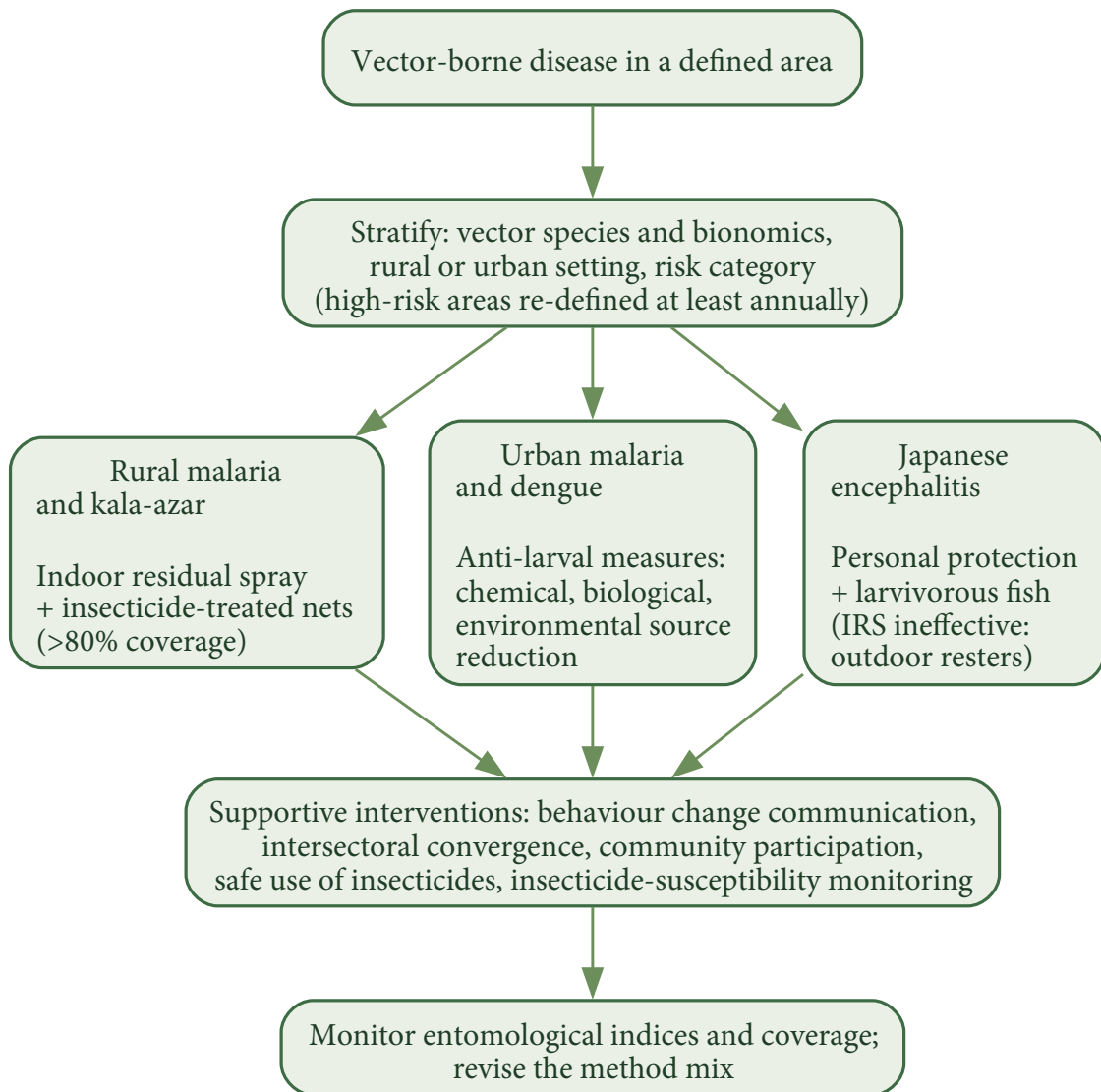
Element	Indicator / action
Entomological surveillance	<i>Aedes aegypti</i> index — percentage of houses in a defined area showing actual breeding of <i>Ae. aegypti</i> larvae; kept at zero at all ports and below 1 per cent in towns and seaports (Park 28e, Ch 15, p 925; Ch 5, pp 347–348)
International obligation	Under the International Health Regulations, airports and seaports are kept free of mosquitoes for 400 metres around their perimeter (Park 28e, Ch 15, p 925)
Susceptibility testing	Periodic tests to determine susceptibility of the vector to available insecticides, so that only effective insecticides are applied (Park 28e, Ch 15, p 927)
Resistance mechanisms	Biochemical (toxicant converted to a non-toxicant by insect enzymes) and behaviouristic; transmitted through single or multiple genes (Park 28e, Ch 15, pp 938–939)
Resistance patterns to anticipate	Organochlorine resistance splits into a DDT-and-analogues group and an HCH–dieldrin group; "double resistance" when both are present; cross-resistance between most carbamates and most organophosphorus compounds (Park 28e, Ch 15, pp 938–939)
Selection pressure	Resistance depends on the biological characteristics of the local vector population and on the type and degree of insecticide selection pressure; less persistent deposits and immigration of susceptible mosquitoes slow its appearance (Gilles & Warrell 4e, Ch 6)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- Label box for hand reproduction — "Classification of mosquito control measures" (Park 28e, Ch 15, p 926): three headed columns with labels — 1. *Anti-larval measures* (a) Environmental control (b) Chemical control (c) Biological control; 2. *Anti-adult measures* (a) Residual sprays (b) Space sprays (c) Genetic control; 3. *Protection against mosquito bites* (a) Mosquito net (b) Screening (c) Repellents.



Building blocks of integrated vector control (Park 28e, Ch 15, p 926)



Selecting the vector-control method mix under the National Vector Borne Disease Control Programme

High-yield exam pointers

- ▶ **Definition to write verbatim:** "utilisation of all appropriate technological and management techniques to bring about an effective degree of vector suppression, in a cost-effective manner."
- ▶ **Classification skeleton (Park):** anti-larval (environmental / chemical / biological) → anti-adult (residual spray / space spray / genetic) → protection against bites (net / screening / repellent).
- ▶ **Numbers examiners look for:** DDT 1–2 g/m², 1–3 times a year; mineral oil 40–90 L/hectare weekly; temephos 56–112 g/ha; screening 16 meshes/inch and net aperture ≤0.0475 inch; DEET protective 18–20 hours; IRS + ITN coverage >80 per cent in high-risk villages; 3-km peripheral belt for *Culex quinquefasciatus*; 400 m mosquito-free perimeter at ports under the International Health Regulations.
- ▶ **NVBDCP three-pronged strategy:** disease management → integrated vector management → supportive interventions; six diseases (malaria, filariasis, kala-azar, JE, dengue, chikungunya).
- ▶ **Integration example to quote:** same vector-control methods used for malaria and kala-azar in rural areas, and malaria and dengue in urban areas — cost-effectiveness and synergy.

- **Compatibility rule:** one method must favour, or at least not interfere with, another — never larvicide a pond with a compound that kills the larvivorous fish stocked in it.
- **Resistance:** over 100 arthropod species of public health importance are resistant; cross-resistance between most carbamates and most organophosphorus compounds.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The Directorate of the National Vector Borne Disease Control Programme has since been renamed the National Center for Vector Borne Diseases Control (NCVBDC); the retrieved textbook corpus uses only the older name and the older name is what the model answer above cites.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, NCVBDC programme data, ICMR, WHO insecticide-susceptibility criteria, WHO prequalified vector-control product list

Programme and policy updates (India)

- The programme directorate is now the **National Center for Vector Borne Diseases Control (NCVBDC)**, and vector-borne disease reporting runs on two parallel national tracks — NCVBDC and the Integrated Disease Surveillance Programme, which moved in 2018 from paper-based aggregate returns to real-time case-based reporting on the Integrated Health Information Platform (IHIP-IDSP). Fieldwork in two high-burden western Indian districts found the two systems operating with separate information flows, only 13 per cent of sanctioned district public health laboratories functional, and four districts with no sentinel hospital cover; harmonising outbreak criteria across the two programmes is therefore now as much a part of integrated vector management as choosing the insecticide [PMID 42245358].
- Adding vector control on top of mass drug administration for lymphatic filariasis did **not** pay for itself in a three-arm cluster-randomised trial in 36 Tamil Nadu villages: annual economic cost per resident rose from US\$0.53 for mass drug administration alone to US\$1.02 when expanded-polystyrene beads were floated on wells and cesspits and US\$1.83 when pyrethroid-impregnated curtains were added, with no statistically significant extra fall in microfilaria or antigen prevalence, while mass drug administration alone cost US\$112 per DALY averted. India's lymphatic filariasis elimination goal is now **2030**, with the disease still endemic in 345 districts [PMID 39630851; PMID 41951149].

Current national data

- NCVBDC figures record roughly **1.1 million confirmed dengue cases and 1,653 deaths in India between 2019 and 2024**, with the number of states and union territories reporting dengue rising from 8 in 2000 to 35 (28 states, 7 union territories). Analysis of 718 districts for 2010–2022 showed significant spatial clustering (global Moran's I 0.06, $p < 0.001$) and 31 high-incidence district clusters — 12 in Punjab, 5 in Tamil Nadu, 3 each in Haryana and Kerala — with the steepest upward trends in cooler and semi-arid zones, which argues for climate-sensitive, geographically targeted vector control rather than a uniform national spray calendar [PMID 42263103].

- Pan-India molecular surveillance through 45 Virus Research and Diagnostic Laboratories in 25 states and union territories tested 6,889 dengue-positive sera in **2023–2025**: every region showed more than one serotype circulating, eight states carried all four, DENV-2 predominated, and infection with more than one serotype at once was associated with more thrombocytopenia, arthralgia and haemorrhagic manifestations [PMID 42323159].

Elimination, surveillance and disease status

- Kala-azar: the Kala-azar Elimination Programme target is **fewer than 1 new case per 10,000 population**; Bangladesh reached it in 2017 and was validated by WHO in 2023, whereas **India has held the criterion for two consecutive years but has not yet been certified**. Against pre-programme levels, cases by 2023 had fallen 99.5 per cent in Bangladesh, 98.4 per cent in India and 88.5 per cent in Nepal. Transmission has not stopped: in Bihar and Jharkhand villages rK39 seroprevalence was 20–30 per cent and *Leishmania*-infected sand fly pools were still caught despite indoor residual spraying (1.4 per cent, 4 of 285 pools, and 1.2 per cent, 5 of 431 pools), mostly from vegetation and cattle enclosures rather than sprayed house walls. One Health has correspondingly entered the endgame: a seven-district survey screened 782 domestic animals and found 98 rK39-seropositive, *Leishmania* DNA on qPCR in 129 and ITS1-PCR confirmation in 81 dogs, goats, cattle and one cat — evidence that the classical anthroponotic model is incomplete and that surveillance must cover humans, sand flies and domestic animals together [PMID 41856659; PMID 41324335; PMID 42206020].
- Malaria elimination has become a **micro-level exercise**: in Sonbhadra district 18,217 cases over 2017–2024 fell by 95 per cent, annual parasite incidence dropping from 2.7 in 2017 to 0.1 in 2024, yet significant sub-district *P. falciparum* and *P. vivax* clusters persisted and annual falciparum incidence rebounded to 0.09 in 2024 after near-total suppression. Odisha alone still contributes about a quarter of India's reported malaria cases and 30 per cent of the deaths, reporting 41,971 cases in 2023 [PMID 41871706; PMID 41948398].

Guideline and legislative updates

- Susceptibility test results are graded against the **WHO criterion — susceptible $\geq$ 98 per cent mortality, possible resistance 90–97 per cent, confirmed resistance $<$ 90 per cent** — and a meta-analysis of 51 Indian studies (2000–2024) found DDT resistance in almost every state except the north-east, malathion resistance in *An. culicifacies* and *An. stephensi*, and deltamethrin resistance concentrated in Chhattisgarh and Madhya Pradesh (*An. culicifacies* mean mortality 89 per cent, 95% CI 79–94). North-eastern vectors remain largely susceptible, so the insecticide chosen for indoor residual spraying should now follow state-level resistance maps rather than one national schedule [PMID 41437021; PMID 41948398].
- **Broflanilide (VECTRON T500)**, a GABA-receptor antagonist whose mode of action is new to public health and shows no cross-resistance to existing classes, has been added to the WHO list of prequalified indoor residual spraying products. In a 16-cluster non-inferiority trial in Benin it matched clothianidin–deltamethrin for indoor human biting rate (IRR 0.54, 95% CI 0.22–1.35, $p = 0.150$) and gave 100 per cent cone-bioassay mortality on both cement and mud walls for 24 months, which supports two-yearly instead of annual spray rounds and makes genuine insecticide rotation possible [PMID 37857762].

Global and WHO developments

- ▶ A proactive integrated vector management package — community engagement, targeted outdoor residual spraying and autodissemination devices aimed at both larvae and adults — was tested against routine vector control in a cluster-randomised trial of 280 urban housing blocks and about 904,000 residents in Kuala Lumpur and Putrajaya, and **did not significantly reduce dengue incidence** (risk ratio 0.86, 95% CI 0.70–1.06; $p = 0.17$). The examination point is that integration is a rational framework, not an automatic gain — the method mix still has to match local vector bionomics [PMID 40373783].
- ▶ The strongest recent evidence for any single component remains the treated bed-net: pooled across 22 studies and 616,611 participants in nine countries, insecticide-treated nets cut *P. falciparum* incidence (RR 0.55, 95% CI 0.37–0.79) and *P. vivax* incidence (RR 0.69, 95% CI 0.51–0.94), both high-certainty, whereas evidence for indoor residual spraying was very low certainty and topical repellents moderate (RR 0.58, 95% CI 0.35–0.97) [PMID 36925174].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 15 Environment and Health — medical entomology and mosquito control, pp 922–927, 938–939; Ch 7 Health Programmes in India — National Vector Borne Disease Control Programme, pp 493, 498, 501; Ch 5 Epidemiology of Communicable Diseases — malaria and filariasis vector control, pp 333, 339, 348); Gilles & Warrell — Essential Malariology 4e (Ch 6 Rationale and technique of malaria control, pp 155–193); Service — Medical Entomology for Students (Ch 1, pp 36, 40, 61–62). Coverage note: the corpus supports the classical textbook framework of integrated vector control but does not contain the WHO Global Vector Control Response 2017–2030 or the formal five key elements of the WHO Global Strategic Framework for IVM, so those are not stated here.

Recent-advances sources: 42245358 — Informational challenges of dengue surveillance and response: a case study from India, *Front Public Health* 2026, [10.3389/fpubh.2026.1809584](https://doi.org/10.3389/fpubh.2026.1809584); 39630851 — Cost-effectiveness of vector control for supplementing mass drug administration for eliminating lymphatic filariasis in India, *PLoS Negl Trop Dis* 2024, [10.1371/journal.pntd.0011835](https://doi.org/10.1371/journal.pntd.0011835); 41951149 — Public-private partnership offers a strategic opportunity for lymphatic filariasis elimination in India by 2030, *Acta Trop* 2026, [10.1016/j.actatropica.2026.108083](https://doi.org/10.1016/j.actatropica.2026.108083) (abstract-only); 42263103 — Spatial distribution and risk assessment of dengue incidence at district level across major climatic zones in India, *PLoS One* 2026, [10.1371/journal.pone.0350325](https://doi.org/10.1371/journal.pone.0350325); 42323159 — Pan-India molecular surveillance of dengue serotypes, 2023–2025, *Int J Infect Dis* 2026, [10.1016/j.ijid.2026.108897](https://doi.org/10.1016/j.ijid.2026.108897) (abstract-only); 41856659 — Visceral leishmaniasis elimination in South Asia: lessons learnt can inform disease elimination in East Africa, *BMJ Glob Health* 2026, [10.1136/bmjgh-2026-023521](https://doi.org/10.1136/bmjgh-2026-023521); 41324335 — High seropositivity to rK39 and capture of *Leishmania*-infected sand flies in Bihar and Jharkhand emphasize the need for post-elimination surveillance of visceral leishmaniasis in India, *J Infect Dis* 2026, [10.1093/infdis/jiaf543](https://doi.org/10.1093/infdis/jiaf543); 42206020 — Insights into animal reservoir hosts for the circulation of the parasite in the sustainable phase of the National Kala-azar Elimination Programme in India, *One Health* 2026, [10.1016/j.onehlt.2026.101444](https://doi.org/10.1016/j.onehlt.2026.101444); 41871706 — Identification of micro-level malaria hotspots and its implications for achieving malaria elimination target in a remote district of India, *Acta Trop* 2026, [10.1016/j.actatropica.2026.108066](https://doi.org/10.1016/j.actatropica.2026.108066) (abstract-only); 41948398 — Mapping the insecticide resistance landscape of malaria vectors in Odisha, India (1993–2024): a scoping review, *J Trop Med* 2026, [10.1155/jotm/8879127](https://doi.org/10.1155/jotm/8879127); 41437021 — Spatio-temporal trends of insecticide resistance in Indian malaria vectors (2000–2024): a systematic review and meta-analysis, *BMC Public Health* 2025, [10.1186/s12889-025-25977-y](https://doi.org/10.1186/s12889-025-25977-y); 37857762 — Community evaluation of VECTRON T500, a broflaniide insecticide, for indoor residual spraying for malaria vector control in central Benin, *Sci Rep* 2023, [10.1038/s41598-023-45047-w](https://doi.org/10.1038/s41598-023-45047-w); 40373783 — Effectiveness of integrated vector management on the incidence of dengue in urban Malaysia: a cluster-randomised controlled trial, *Lancet Infect Dis* 2025, [10.1016/S1473-3099\(25\)00086-6](https://doi.org/10.1016/S1473-3099(25)00086-6) (abstract-only); 36925174 — Vector control for malaria prevention during humanitarian emergencies: a systematic review and meta-analysis, *Lancet Glob Health* 2023, [10.1016/S2214-109X\(23\)00044-X](https://doi.org/10.1016/S2214-109X(23)00044-X) (abstract-only). Also used: WHO list of prequalified indoor residual spraying products (broflaniide/VECTRON T500); WHO insecticide-susceptibility test criteria.

Environmental health and ecology

Safe water for all

Asked in: Paper IV Nov 2016 — RGUHS MD Community Medicine

Definition — Water for human consumption must be **safe and wholesome**: (a) free from pathogenic agents, (b) free from harmful chemicals, (c) pleasant to taste, i.e. free from colour and odour, (d) usable for domestic purposes; water failing any criterion is **polluted or contaminated**. “Safe water for all” means such water for every household in **adequate quantity, accessible on premises and continuously available**. The 34th World Health Assembly (1981) declared safe drinking water a basic element of **primary health care** (Park 28e, Ch 15, pp 860–861).

1. The public health case — why safe water for all

Indicator	Value (year)	Source
Global population with safely managed drinking water (on premises, available when needed, free from contamination)	74% (2020)	Park 28e, Ch 15, p 861
People still lacking safely managed water services	2 billion (2020) — 1.2 billion basic only, 282 million limited, 367 million unimproved sources, 122 million drinking surface water	Park 28e, Ch 15, p 861
Global population with safely managed sanitation	54% (2020); 3.6 billion insufficient	Park 28e, Ch 15, p 861
Lacking a basic handwashing facility with water and soap at home	2.3 billion (29%) (2020)	Park 28e, Ch 15, p 861
India — improved drinking water / improved sanitation	90% / 71% (2020)	Park 28e, Ch 15, p 861
Household wastewater discharged without safe treatment (128 countries)	44% (2020)	Park 28e, Ch 15, p 861

Sustainable Development Goal 6 is the current global frame for the phrase:

SDG 6 target (by 2030)	Global monitoring indicator
6.1 — universal and equitable access to safe and affordable drinking water for all	6.1 Access to a safely managed drinking-water source
6.2 — adequate and equitable sanitation and hygiene for all and an end to open defecation, with special attention to women and girls	6.2 Access to safely managed sanitation
6.3 — improve water quality, halving the proportion of untreated wastewater and increasing safe reuse	—

(Park 28e, Ch 14, pp 854–855; *safe water and sanitation were already Millennium Development Goals — Ch 15, p 861*)

Water-related diseases — the burden safe water prevents:

Group	Mechanism	Diseases
A. Biological — infective agent: viral	Ingestion of the agent in water	Hepatitis A, hepatitis E, poliomyelitis, rotavirus diarrhoea of infants (Park 28e, Ch 15, p 865)
A. Biological — infective agent: bacterial and leptospiral	Ingestion of the agent in water	Typhoid and paratyphoid fever, bacillary dysentery, <i>Escherichia coli</i> diarrhoea, cholera; leptospirosis (Weil's disease) (Park 28e, Ch 15, p 865)
A. Biological — infective agent: protozoal	Ingestion of cysts	Amoebiasis, giardiasis (Park 28e, Ch 15, p 865)
A. Biological — infective agent: helminthic	Ingestion of ova	Roundworm, threadworm, hydatid disease (Park 28e, Ch 15, p 865)
A. Biological — water-based (aquatic intermediate host)	The agent completes an obligatory stage of its life cycle in an aquatic host, so the water body is both nursery and vehicle	Snail → schistosomiasis; cyclops → guinea worm (dracunculiasis) and fish tapeworm (Park 28e, Ch 15, p 865; Sunder Lal, Ch 3, p 94)
Water-washed (water-scarce)	Too little water for personal hygiene	Trachoma, scabies, skin sepsis, lice infestation, conjunctivitis, bacillary and amoebic dysentery, worm infestations (Sunder Lal, Ch 3, p 94)
Water-related insect vector	Insects breeding in or near water	Malaria, filariasis, dengue, yellow fever, Japanese encephalitis, sleeping sickness (Sunder Lal, Ch 3, p 94)
B. Chemical	Excess or deficiency of a dissolved constituent	Dental caries (low fluoride) and dental/skeletal fluorosis (high fluoride); methaemoglobinaemia — “cyanosis in infants” (high nitrate); hardness of water appears protective against cardiovascular disease (Park 28e, Ch 15, p 865)

2. Quantity, accessibility and sources

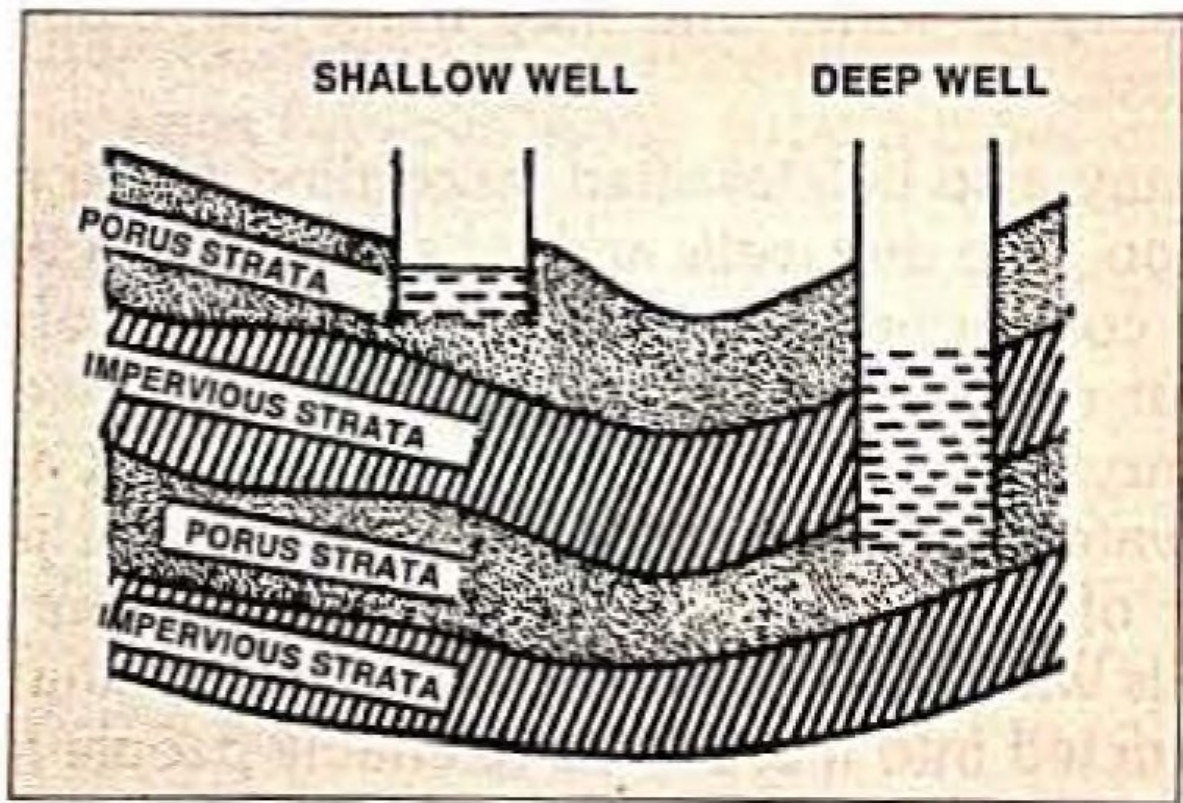


Fig 2 — Shallow and deep wells (Park's Textbook of Preventive and Social Medicine, 28e, p.863)

Element	Norm / fact
Bare physiological requirement	~2 litres per head per day — survival only (Park 28e, Ch 15, p 861)
Adequate urban domestic supply	150–200 litres per capita per day (lpcd) (Park 28e, Ch 15, p 861)
Indian rural norm (earlier)	40 lpcd, with at least one spot-source (hand pump) per 250 persons (Park 28e, Ch 7, p 573)
Indian rural norm (enhanced, for system design)	55 lpcd; at least 50% of the rural population to have water within household premises or within a 100 m radius, at least 30% with individual household connections (Park 28e, Ch 7, p 573)
Jal Jeevan Mission service level	Functional Household Tap Connection (FHTC) delivering ≥55 lpcd of prescribed quality (Park 28e, Ch 7, p 573)
Sources	1. Rain (purest in nature, soft, corrodes lead pipes). 2. Surface water — impounding reservoirs, rivers and streams, tanks/ponds/lakes; never safe without protection and treatment. 3. Ground water — shallow wells, deep wells, springs; deep wells furnish the safest water and are often the most satisfactory source (Park 28e, Ch 15, pp 862–863, 865)

3. Criteria and standards of water quality

WHO *Guidelines for Drinking-water Quality* (2011) group the variables under four heads — acceptability, microbiological, chemical, radiological — from which each country frames its own enforceable standards (Park 28e, Ch 15, p 874).

(a) Bacteriological quality — the single most important standard (Park 28e, Ch 15, pp 876–877):

Water	Organism	Guideline value
All water intended for drinking	<i>E. coli</i> or thermotolerant coliform bacteria	Must not be detectable in any 100 ml sample
Treated water entering the distribution system	<i>E. coli</i> or thermotolerant coliform bacteria	Must not be detectable in any 100 ml sample
Treated water in the distribution system	<i>E. coli</i> or thermotolerant coliform bacteria	Must not be detectable in any 100 ml sample; in large supplies, where sufficient samples are examined, must not be present in 95% of samples taken through any 12-month period

Faecal streptococci confirm *recent* pollution and, resisting drying, are used after laying new mains; *Clostridium perfringens* spores resist usual chlorination, so their presence without coliforms means remote contamination or deficient filtration; total coliforms are **not** acceptable indicators in rural tropical supplies. Virologically, **0.5 mg/L free residual chlorine for ≥ 30 min at pH 8.0** inactivates viruses (p 876). Radiological screening values: **gross alpha 0.5 becquerel (Bq)/L, gross beta 1.0 Bq/L** (p 880).

(b) Acceptability parameters (Park 28e, Ch 15, pp 874–876):

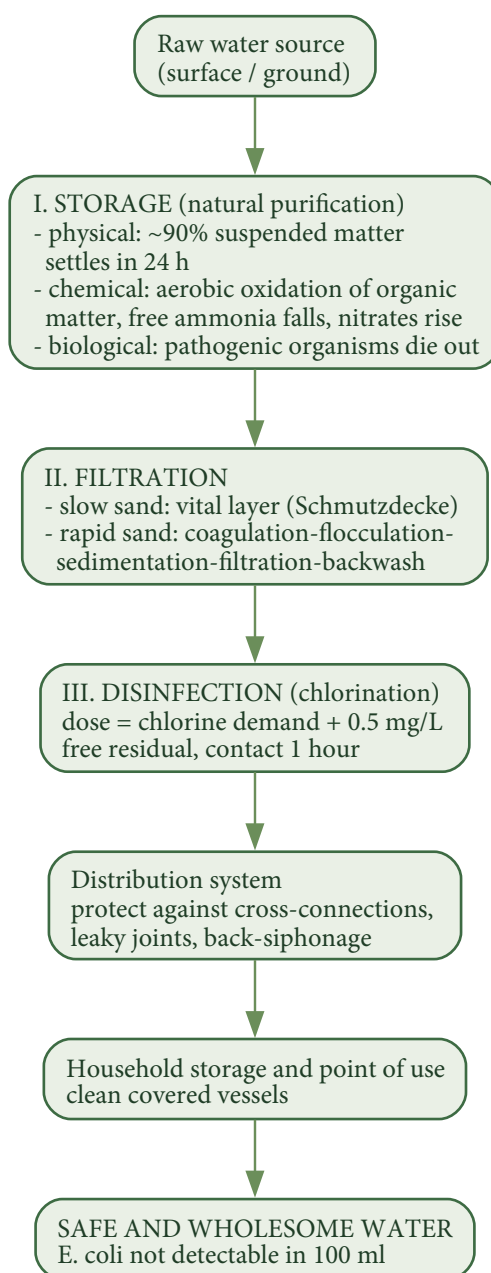
Parameter	Level likely to provoke consumer complaint / guideline
Turbidity	Median <1 NTU for effective terminal disinfection; >4 NTU visible to the naked eye
Colour	15 TCU (true colour units)
pH	Preferably <8.0 for effective chlorination
Chloride / sulphate / sodium	250 / 250 / 200 mg/L (taste, corrosion)
Iron / manganese	0.3 / 0.1 mg/L — aesthetic limits (staining of laundry and sanitary ware, undesirable taste)
Aluminium / ammonia / hydrogen sulphide / zinc	0.2 / 1.5 / 0.05 / 4 mg/L
Total dissolved solids	1000 mg/L (taste)

(c) Chemicals of health significance — guideline values, mg/L (Park 28e, Ch 15, pp 877–878; P = provisional):

Constituent	Value	Health significance
Arsenic	0.01 (P)	Chronic arsenicosis, skin lesions, cancer
Fluoride	1.5	>5 mg/L reported in India, China, Thailand → dental and skeletal fluorosis; Indian workers regard 0.5–0.8 ppm as optimum (1 ppm in temperate climates)

Constituent	Value	Health significance
Lead	0.01	Central and peripheral neurotoxicity; mostly from plumbing, solder and service connections
Nitrate (as NO ₃) / nitrite (as NO ₂)	50 / 3 (P)	Methaemoglobinaemia; bottle-fed infants <3 months most susceptible
Cadmium / mercury (total) / chromium	0.003 / 0.006 / 0.05 (P)	Cumulative toxicity
Copper / manganese / nickel / molybdenum	2 / 0.4 (P) / 0.07 / 0.07	Health-based limits for long-term exposure

4. Making water safe — purification



The multi-barrier chain that converts raw water into safe water at community scale

Ground water may need disinfection alone; surface water the full train (Park 28e, Ch 15, p 866).

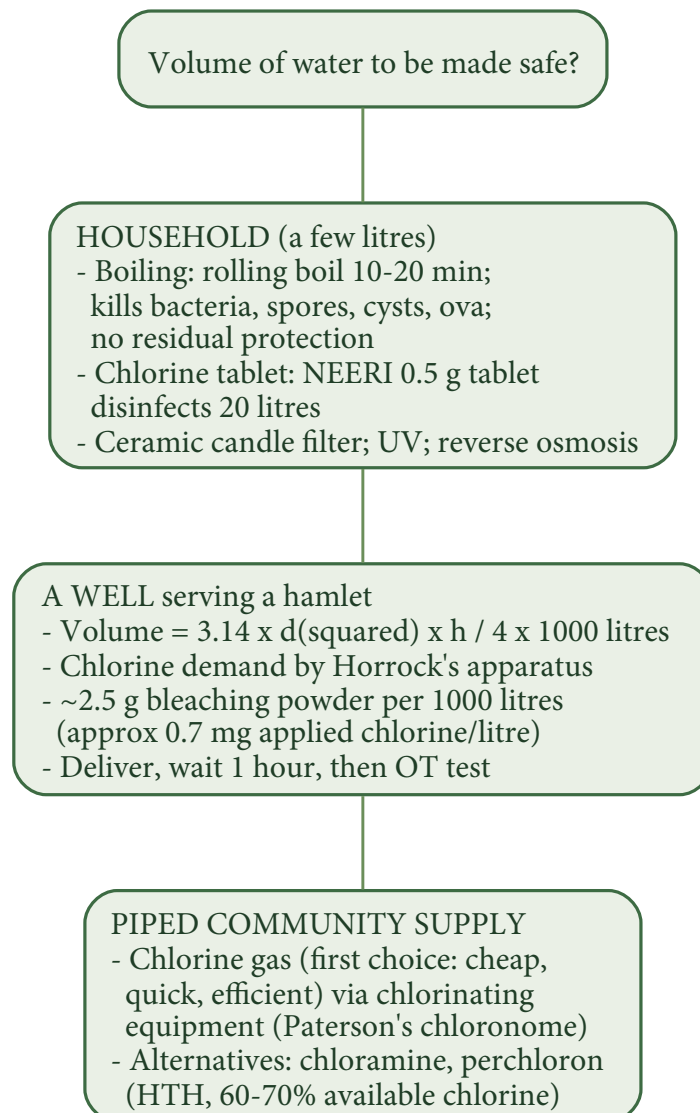
- **Slow sand filtration** — the bed acquires a slimy gelatinous **vital layer** (**Schmutzdecke**, **zoogeal or biological layer**) of algae, plankton, diatoms and bacteria extending 2–3 cm into the sand; its formation is “**ripening**”, until which the filtrate is run to waste. It is the *heart* of the filter — removing organic matter, holding back bacteria, oxidising ammoniacal nitrogen to nitrates (p 867).
- **Rapid sand filtration** — alum 5–40 mg/L or more → rapid mixing → flocculation at 2–4 rpm for ~30 min → sedimentation (detention 2–6 h) → filtration → **backwashing** ~15 min; gravity (*Paterson’s*) and pressure (*Candy’s*) types (pp 868–869).

Feature	Slow sand filter	Rapid sand (mechanical) filter
Space	Large area	Very little space
Sand bed depth	1.2 m (filter box: supernatant water 1–1.5 m, sand 1.2 m, gravel 0.30 m, filter bottom 0.16 m)	~1.0 m
Effective size of sand	0.2–0.3 mm	0.4–0.7 mm
Preliminary storage	Necessary	Not necessary — deals with raw water directly
Preliminary treatment	Plain sedimentation	Coagulation–flocculation–sedimentation, essential
Rate of filtration	2–3 million gallons per acre per day (m.g.a.d.)	200 m.g.a.d.; 40–50 times faster
Rate of filtration, metric (<i>Suryakantha</i>)	0.1–0.4 m ³ /m ² /hour	5–15 m ³ /m ² /hour
Cleaning	Scraping off the top 1–2 cm of sand	Backwashing (~15 minutes)
Loss of head allowed	4 feet (1.5 m)	6–8 feet (2–2.5 m)
Operation	Less skilled	Highly skilled
Removal of colour	Fair	Good
Removal of bacteria	99.9–99.99% of total count (<i>E. coli</i> 99–99.9%)	98–99%
Mechanism of action (<i>Suryakantha</i>)	Mainly biological	Physical and chemical
Filter run (<i>Suryakantha</i>)	3 weeks to 3 months	1 to 3 days
Post-treatment chlorination (<i>Suryakantha</i>)	Not necessary	A must

(Park rows: Park 28e, Ch 15, pp 867–869. Rows marked (*Suryakantha*): *Suryakantha*, Ch 5 (*Environment and Water*), p 49)

Chlorination acts mainly through **hypochlorous acid**, **70–80 times** more effective than the hypochlorite ion; it works best at pH ≈7 and is unreliable above pH 8.5. The five rules (Park 28e, Ch 15, pp 869–871):

Rule	Content
1	Water must be clear and free of turbidity — turbidity impedes chlorination
2	The chlorine demand (chlorine added minus residual remaining after a stated contact period, usually 60 minutes, at given temperature and pH) must be met; the point at which it is met is the break-point , beyond which free chlorine appears
3	Contact at least one hour ; chlorine has no effect on spores, protozoal cysts and helminthic ova, nor on certain viruses (poliomyelitis, viral hepatitis), except in high doses
4	Minimum free residual chlorine 0.5 mg/L
5	Correct dose = chlorine demand + 0.5 mg/L free residual
Variants	Break-point chlorination = controlled superchlorination ; superchlorination with dechlorination for heavily polluted water of fluctuating quality; the orthotolidine (OT) test measures free and combined chlorine

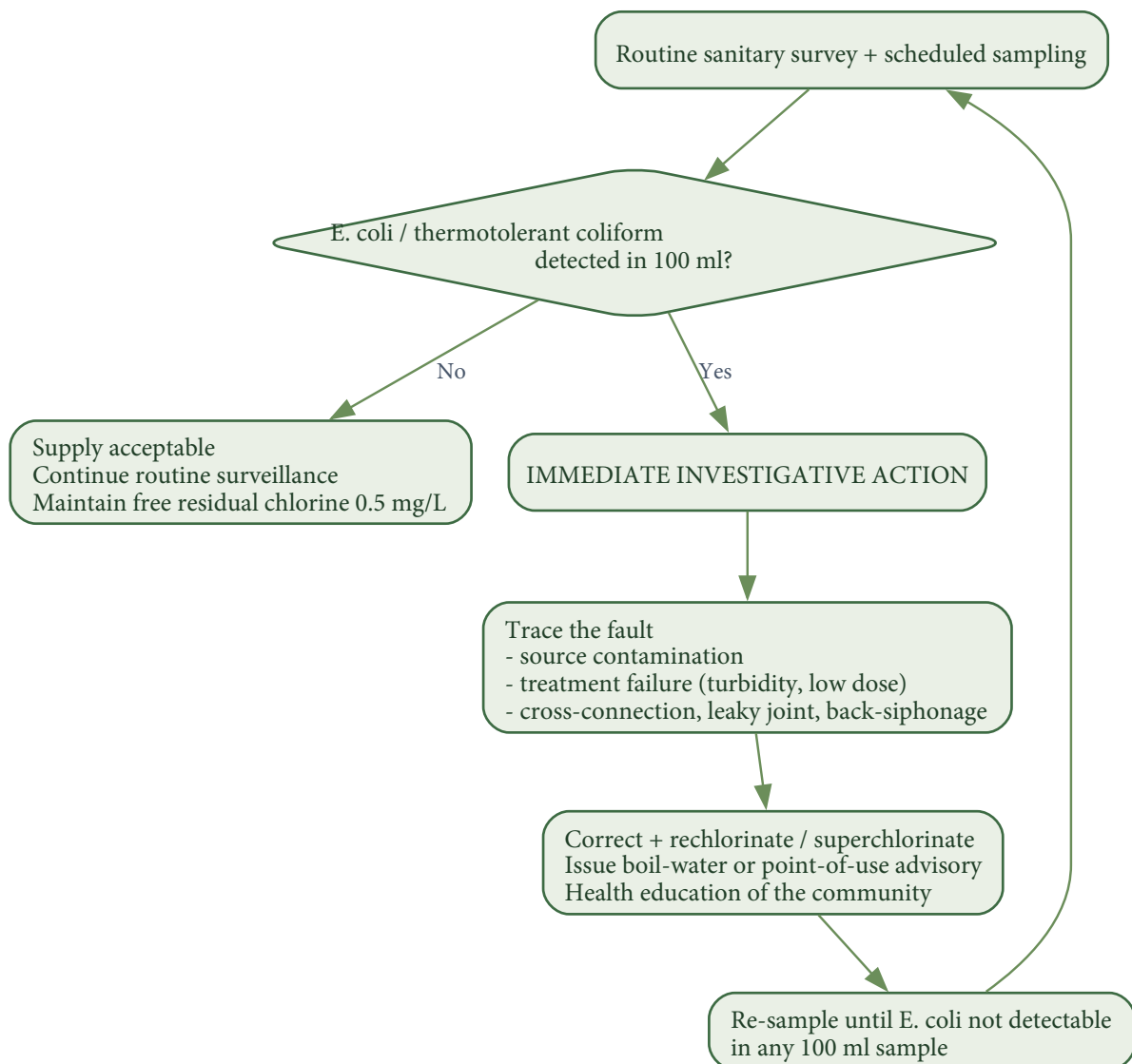


Choosing the method of disinfection by the scale of the supply

5. Keeping it safe — surveillance of drinking water quality

Surveillance is a **health** measure, not an engineering one (Park 28e, Ch 15, pp 880–881, 884):

- ▶ **Sanitary survey** — on-the-spot inspection of the *entire* supply system by a qualified person; essential for interpreting laboratory results.
- ▶ **Sampling** — aseptic, by trained staff, at WHO guideline or ICMR *Manual of Standards of Drinking Water Supplies* frequencies; iced samples analysed **within 48 hours**, labelled with date, time, source, recent rainfall and survey findings.
- ▶ **Bacteriological** — presumptive coliform (multiple-tube **most probable number**) test confirmed in brilliant green bile broth at **37°C for 48 h** (coliforms) and **44°C** (*E. coli*, indole-positive); plus faecal streptococci, *Clostridium perfringens*, colony counts.
- ▶ **Chemical** — pH, colour, turbidity, chlorides, ammonia, chlorine demand and **residual chlorine**; regular residual-chlorine measurement may in part replace bacteriological surveillance.
- ▶ **System safeguards** — approval of new sources, watershed protection, flushing of storage, certification of operators, **cross-connection control**, **back-flow prevention**, **leak detection**, well and plumbing codes, control of bottled water and ice.



Surveillance-and-response algorithm for a community drinking water supply

6. Delivering “safe water for all” in India — the programme response

Year	Milestone	Content
1948–49	Environmental Hygiene Committee	Recommended a comprehensive plan for safe water supply and sanitation
1954	National Water Supply and Sanitation Programme	First national programme
1972	Accelerated Rural Water Supply Programme	Started as a supplement to the national water supply and sanitation programme, supporting “problem villages”
1981–1990	International Drinking Water Supply and Sanitation Decade (UN)	India committed to providing safe drinking water and adequate sanitation for all by 1990 (Park 28e, Ch 23, pp 1038–1039). Targets fixed by the Government of India for the decade: 100% urban and rural water supply, 50% urban sanitation, 25% rural sanitation ; the guinea-worm eradication programme was linked to this decade (Jugal Kishore, Ch 1, p 699)
1986	National Drinking Water Mission (“Technical Mission”)	Scientific, cost-effective content to the Accelerated Rural Water Supply Programme
1987	National Water Policy	Gave drinking water the highest priority
1990	New Delhi Declaration (UN General Assembly)	Integrated management of water and waste; organisational reform and women’s participation; community management; sound financing and appropriate technology
1991	Rajiv Gandhi National Drinking Water Mission	Renaming of the 1986 mission
1999–2002	Sector Reform Project → Swajaldhara (25 Dec 2002)	Community-led, participatory; community ownership, hygiene awareness, water conservation and rainwater harvesting
2009	National Rural Drinking Water Programme (NRDWP)	Norm raised 40 → 55 lpcd; vision of “safe and adequate drinking water for all, at all times, in rural India”; Strategic Plan 2011–2022
2011	Ministry of Drinking Water and Sanitation	Departmental status raised to a Ministry
2014	Swachh Bharat Mission (2 Oct 2014; Gramin and Urban)	Sanitation and open-defecation-free villages — the essential complement to safe water; India declared open-defecation free on 2 Oct 2019
Current	Jal Jeevan Mission / Nal Jal Yojana — Department of Drinking Water and Sanitation	NRDWP restructured and subsumed into it: Functional Household Tap Connection to every rural household (“Har Ghar Jal”) by 2024, ≥55 lpcd of prescribed quality, with mandatory source-sustainability measures (recharge, grey-water reuse, rainwater harvesting) and a nationwide <i>Jan Andolan</i> for water

(Park 28e, Ch 7, pp 572–573 and Ch 23, pp 1038–1039; Jugal Kishore, Ch 1, pp 699–700, 706; AFMC Red Book of Community Medicine 2024, Ch 5, pp 127–128)

Operational objectives — safe, adequate water within reasonable distance of every rural household, communities themselves keeping surveillance on their sources, guided by **potability, reliability, sustainability, convenience, equity and consumer preference**, with village drinking-water security

plans (Jugal Kishore, Ch 1, pp 700, 706). On completion the **gram panchayat water and sanitation committee** runs the scheme (Park 28e, Ch 7, p 573).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Slow sand filter (vertical section)** — label: supernatant (raw) water 1–1.5 m, **vital layer / Schmutzdecke** 2–3 cm, graded sand bed 1.2 m (effective size 0.2–0.3 mm), gravel support 0.30 m, under-drainage of perforated pipes, filter box wall, inlet chamber, filter control (regulating) valve, venturi meter, outlet chamber.
- ▶ **Break-point chlorination curve** — label: X-axis “chlorine dose applied”, Y-axis “residual chlorine”; initial rise (demand being met), chloramine peak, fall as chloramines are destroyed, the **break-point**, then the straight line of **free residual chlorine**.

High-yield exam pointers

- ▶ **Numbers:** 2 L/day physiological; **150–200 lpcd urban**; **40 → 55 lpcd rural**; one spot-source per 250 persons; FHTC ≥ 55 lpcd by 2024.
- ▶ **Bacteriological standard:** *E. coli*/thermotolerant coliforms **not detectable in any 100 ml** at works, in distribution and at the tap; $\geq 95\%$ of samples negative over 12 months in large supplies.
- ▶ **Chlorination:** dose = **chlorine demand + 0.5 mg/L free residual**, contact **1 hour**; hypochlorous acid **70–80×** more active than hypochlorite; no effect on spores, cysts, ova, polio and hepatitis viruses.
- ▶ **Well disinfection:** volume (L) = $3.14 \times d^2 \times h \div 4 \times 1000$; **~2.5 g bleaching powder/1000 L** (≈ 0.7 mg Cl/L); Horrock’s apparatus; orthotolidine test.
- ▶ **Filters:** sand effective size **0.2–0.3 vs 0.4–0.7 mm**; **2–3 vs 200 m.g.a.d.** (Park), **0.1–0.4 vs 5–15 m³/m²/h** (Suryakantha); bacteria **99.9–99.99% vs 98–99%**.
- ▶ **Guideline values (mg/L):** arsenic 0.01, fluoride 1.5 (Indian optimum 0.5–0.8 ppm), lead 0.01, nitrate 50, nitrite 3, cadmium 0.003, mercury 0.006; gross alpha 0.5 Bq/L, beta 1.0 Bq/L.
- ▶ **SDG 6.1:** safe and affordable drinking water for all by 2030; indicator = access to a **safely managed** source.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Suryakantha gives wider effective sand sizes (0.15–0.3 and 0.6–2.0 mm) than Park’s 0.2–0.3 and 0.4–0.7 mm; quote Park, the spine. Park dates the water and sanitation decade 1981–1990, Jugal Kishore 1980–90; follow Park. Park’s “India 90% water, 71% sanitation (2020)” uses the improved-source definition, not the stricter “safely managed” one behind the 74% global figure. Park and the AFMC Red Book still place the Department of Drinking Water and Sanitation under the Ministry of Drinking Water and Sanitation, not the Ministry of Jal Shakti.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, WHO Guidelines for Drinking-water Quality (4th edition incorporating the first and second addenda, 2022), Jal Jeevan Mission (Ministry of Jal Shakti), AERB 60 µg/L and BIS 30 µg/L uranium limits

Programme and policy updates (India)

- ▶ **Jal Jeevan Mission has changed hands** — the mission now sits under the **Department of Drinking Water and Sanitation, Ministry of Jal Shakti**, not the Ministry of Drinking Water and Sanitation that the textbooks still name; write the Jal Shakti placement in the exam. [Ministry of Jal Shakti]
- ▶ **What the first evaluation of JJM could actually attribute to the mission** — districts matched between NFHS-4 (2015–16) and NFHS-5 (2019–21): districts with higher household tap-connection coverage in the early-JJM period showed on average about a **10% fall in the proportion of households where adult women fetched water from a distant source**. Reduced drudgery on adult women is the one benefit the analysis could credit to JJM. [PMID 39570840]
- ▶ **What that evaluation could *not* attribute to JJM** — nationally, households needing more than 30 minutes to fetch water fell from **6.75% to 4.87%** and the female-child fetching burden barely moved (**2.82% → 2.81%**), but these are descriptive NFHS-4 to NFHS-5 changes: the regression found **no significant association between having a household tap connection and either the female-child burden or the duration of water collection**. Coverage also remained **pro-rich** — the richest quintile's advantage over the poorest did not narrow (25.9% before JJM, 26.0% after) — although distribution did favour Scheduled Caste households. This is the standard criticism to raise. [PMID 39570840]
- ▶ **Swachh Bharat Mission — the sanitation half of “safe water for all” — now has a child-survival signal**: a quasi-experimental analysis of 35 states and 640 districts (2011–2020) found every 10-percentage-point rise in district toilet access corresponded to IMR falling by 0.9 and U5MR by 1.1 per 1000, with a threshold effect above about 30% district coverage (5.3 fewer infant and 6.7 fewer child deaths per 1000); the authors estimate roughly 60,000–70,000 infant deaths averted a year. [PMID 39223201]

Current national data

- ▶ **Rural tap-water coverage rose from 16.63% of rural households at JJM's launch in 2019 to 76.80% by June 2024** [PMID 39570840] — the single most quotable programme statistic, and the figure to lead with in place of the older 40 lpcd / one spot-source per 250 persons framing.
- ▶ **Household Consumption Expenditure Survey 2022–23** (261,746 households, all 36 states and UTs): 12.5% of households — 13.4% of people, about **162.7 million** — still have no toilet at all; rural prevalence 17.4% versus urban 1.6%, and 71% of them live in six states (Bihar, Uttar Pradesh, Odisha, Rajasthan, Madhya Pradesh, Jharkhand). NFHS shows no-toilet households fell from 39.8% (2016) to 17.8% (2021). Rajasthan declared itself open-defecation free in 2018 yet about 19% of its residents still had no toilet in 2023 — the concrete evidence that ODF declarations are administrative rather than absolute. [PMID 40536324]

Elimination, surveillance and disease status

- ▶ **Routine surveillance still fails the zero-tolerance bacteriological standard**: 592 drinking-water sources across 11 Himachal Pradesh districts showed **55.6% coliform and 9.6% faecal *E. coli*** contamination, worst in natural sources (67.1%) and public piped supplies (59.7%) and least in groundwater (31.5%) — a reminder that a tap connection is not the same as safe water. [PMID 39670946 — abstract-only, no PMC full text]

- **What the water ladder is worth, quantified:** systematic review and meta-analysis of case-control studies, classified on the WHO/UNICEF JMP service ladder, found surface water carried an OR of 3.40 (95% CI 2.52–4.58) and unimproved water 2.91 (1.21–7.02) for cholera, while basic water (OR 0.44; 0.27–0.69) and treated water (0.42; 0.27–0.65) were protective. **Piped tap water on its own showed no significant protection (OR 0.92; 0.41–2.08)**, whereas boiling (0.38; 0.17–0.84) and chlorination (0.37; 0.17–0.83) did — the strongest recent argument that point-of-use disinfection still matters after piped supply arrives. [PMID 41013364]

Guideline and legislative updates

- **The operative WHO standard is the fourth edition incorporating the first and second addenda (2022), not the 2011 guidelines quoted in the body** — cite the 2022 edition; for uranium it retains 10 Bq/L on radiological grounds alongside the 30 µg/L chemical-toxicity value. [WHO 2022; corroborated in PMID 39007979]
- **Uranium is a chemical of health significance absent from the classical Park table:** WHO sets 30 µg/L on chemical toxicity and **BIS has adopted the same 30 µg/L as a national limit** — taken up *subsequent to* IS 10500:2012, which itself prescribes only gross alpha 0.5 Bq/L and gross beta 1 Bq/L for all radionuclides together and does not name uranium separately — while India's Atomic Energy Regulatory Board applies **60 µg/L** on radiological grounds. Nationwide mapping of 55,554 samples across 403 districts in 23 states and 3 UTs found 0.2–4918 µg/L, with about **2.2% above the AERB 60 µg/L limit and about 6% above the WHO 30 µg/L value** (only ~1% above 100 µg/L); the authors argue India should retain 60 µg/L on cost–benefit and waste-management grounds. [PMID 39007979]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 15, pp 860–885; Ch 14, pp 854–856; Ch 7, pp 572–573; Ch 23, pp 1038–1039. Suryakantha — Community Medicine with Recent Advances, Ch 5, pp 46–49. Sunder Lal — Textbook of Community Medicine, Ch 3, p 94. Jugal Kishore — National Health Programmes of India, Ch 1, pp 698–706. AFMC Red Book of Community Medicine 2024, Ch 5, pp 127–128. WHO *Guidelines for Drinking-water Quality* 2011 and the ICMR *Manual of Standards of Drinking Water Supplies*, as reproduced in Park 28e.

Recent-advances sources: 39570840 — Rural drinking water supply program and societal development: evidence from the early implementation phase of India's Jal Jeevan Mission, *PLoS One* 2024, DOI [10.1371/journal.pone.0312144](https://doi.org/10.1371/journal.pone.0312144); 39223201 — Toilet construction under the Swachh Bharat Mission and infant mortality in India, *Scientific Reports* 2024, DOI [10.1038/s41598-024-71268-8](https://doi.org/10.1038/s41598-024-71268-8); 40536324 — Prevalence and burden of no-toilet households in India: an analysis of 261,746 households in 36 states/Union Territories in 2022–2023, *Global Health Action* 2025, DOI [10.1080/16549716.2025.2511351](https://doi.org/10.1080/16549716.2025.2511351); 39670946 — Microbial assessment of drinking water sources in Himachal Pradesh, India, *Indian Journal of Public Health* 2024, DOI [10.4103/ijph.ijph_372_24](https://doi.org/10.4103/ijph.ijph_372_24) (abstract-only; no PMC full text); 41013364 — Associations of water quality with cholera in case-control studies: a systematic review and meta-analysis, *BMC Infectious Diseases* 2025, DOI [10.1186/s12879-025-11533-x](https://doi.org/10.1186/s12879-025-11533-x); 39007979 — Uranium standards in drinking water: an examination from scientific and socio-economic standpoints of India, *Environmental Science and Pollution Research* 2024, DOI [10.1007/s11356-024-34352-0](https://doi.org/10.1007/s11356-024-34352-0). WHO *Guidelines for Drinking-water Quality*, fourth edition incorporating the first and second addenda, 2022. Jal Jeevan Mission, Department of Drinking Water and Sanitation, Ministry of Jal Shakti. Uranium limits as described in PMID 39007979: Atomic Energy Regulatory Board 60 µg/L (radiological) and Bureau of Indian Standards 30 µg/L (adopted subsequent to IS 10500:2012, aligned with the WHO value).



Urban solid waste management

Asked in: Paper I 06-Apr-2026 — RGUHS MD Community Medicine

Definition — Solid waste includes garbage (food waste), rubbish (paper, plastic, wood, metal, glass, throw-away containers), demolition products, sewage treatment residue, dead animals, manure and other discarded material; strictly it should not contain nightsoil (Park 28e, Ch 15, p 908). For a fully occupied 850-house metro apartment complex the feasible answer is a **decentralised, on-site, segregation-led system** — not a bigger dustbin — because the law requires the biodegradable fraction to be processed inside the premises.

1. Public health rationale and the size of the problem

Hazard of accumulated solid waste	Public health consequence (Park 28e, Ch 15, p 908)
Decomposes and favours fly breeding	Fly-borne faeco-oral disease
Attracts rodents and vermin	Rat-bite fever (direct bite), salmonellosis and leptospirosis (contaminated food or water), plague and murine typhus (via rat fleas) (Park 28e, Ch 15, p 940)
Pathogens carried back to food by flies and dust	Diarrhoeal and enteric disease
Water and soil pollution (leachate, drainage from dumps)	Contamination of surface and ground water
Unsightly heaps, bad odour	Nuisance; correlation with incidence of vector-borne disease

Estimating the load (the first practical step):

Parameter	Basis used	Estimate for 850 houses
Residents	850 houses × ~4 persons (working assumption)	≈ 3,400 persons
Total municipal solid waste (MSW)	CPCB per-capita rate 0.2–0.6 kg/person/day (2014–2015) (Solid Waste Management, Ch 23, p 251)	≈ 0.7–2.0 tonnes/day
Total MSW — real benchmark	GPRA Complex, New Moti Bagh, New Delhi: 1.5 tonnes/day from 1,100 households (Solid Waste Management, Ch 40, pp 303-304)	≈ 1.2 tonnes/day
Planning figure adopted	—	≈ 1.2–1.5 tonnes/day
Wet (biodegradable) fraction	47.4% — Manual Table 1.6 / Fig 3.3, 2005 composition (Zhu D. et al., 2008)	≈ 570–710 kg/day
Recyclable dry fraction	paper 8.1 + plastic/rubber 9.2 + rags 4.5 + glass 1.0 + metal 0.5 = 23.4% (same 2005 row)	≈ 280–350 kg/day
Inerts and other	inerts 25.2 + other 4.0 = 29.2% (same 2005 row)	≈ 350–440 kg/day

2. Legal position — what the complex must do under SWM Rules, 2016

- ▶ **Clause 4(1)(a)** — every waste generator shall **segregate and store waste in three separate streams**: bio-degradable, non-biodegradable and domestic hazardous, in suitable bins, and hand over segregated waste to authorised waste pickers or collectors (*Ch 2, p 177*).
- ▶ **Clause 4(1)(b)** — used **sanitary waste** (diapers, sanitary pads) shall be securely wrapped in the pouch supplied by the manufacturer or brand owner and **placed in the bin meant for dry / non-biodegradable waste**.
- ▶ **Clause 15(zg)(iv)** — the urban local body (ULB) shall educate generators to practise **four-stream ("BASIC+") segregation**: bio-degradable, non-biodegradable (recyclable and combustible), **sanitary waste** and domestic hazardous — the target the complex should move to once three-bin segregation is working.
- ▶ **Clause 4(2)** — **no generator shall throw, burn or bury** solid waste on streets, open public spaces outside the premises, drains or water bodies. This outlaws corner-dumping and evening leaf-burning.
- ▶ **Clause 4(3)** — all generators shall **pay the user fee** specified in the local body's bye-laws.
- ▶ **Clause 4(6) and 4(7)** — every **resident welfare association (RWA)**, and every **gated community or institution with more than 5,000 sq m area**, shall within one year and in partnership with the local body ensure source segregation, facilitate separate-stream collection, hand recyclables to authorised waste pickers or recyclers, and **process the biodegradable waste by composting or biomethanation within the premises as far as possible**; only the **residual waste** goes to the local body's collector (*Ch 2, p 175; Ch 43, p 336*).
- ▶ **Clause 15(b)** — the ULB arranges door-to-door collection; from **multi-storeyed buildings and housing complexes this may be collected at the entry gate or another designated point**.
- ▶ **Clause 15(ze)** — building-plan approval for a group housing society must incorporate **space for collection, segregation and storage** of segregated waste — the RWA can demand this space.

3. Segregation and storage at source — the foundation

Stream	Household bin (colour)	Contents	Complex-level storage
Wet / biodegradable	Covered green bin, 12–15 L for a family of five, emptied and washed daily	Cooked and uncooked food, eggshells, bones, fruit/vegetable peel, flowers, garden and plant waste	240 L (96 kg) bins; one 240 L bin serves 48 households → ≈ 18 bins for 850 houses
Dry / non-biodegradable	Covered white bin, 12–15 L	Paper, cardboard, cartons, plastic, metal, glass, rags, packaging	Dry-waste room / material recovery point within the complex
Domestic hazardous	Separate black bin (colour not fixed by the Rules; Coimbatore City Municipal Corporation uses red)	Batteries, button cells, CFL and tube lights, paints, thinners, glues, pesticide and chemical containers, mercury thermometers, expired medicines, used needles	Locked cupboard; handed over on notified days

Stream	Household bin (colour)	Contents	Complex-level storage
Sanitary waste	Wrapped in the manufacturer's pouch and placed in the dry-waste bin (Clause 4(1) (b))	Diapers, sanitary napkins, tampons, incontinence sheets	Kept in a separate bag and handed over separately and daily to the collector, to minimise manual handling

- Bin capacity must carry the waste generated in 24 hours **plus 100% spare capacity** to cover delayed clearance; plastic bag liners are undesirable (*Solid Waste Management, Ch 2, p 186*).
- 60 L (25 kg) bins serve 12 households and 120 L (50 kg) bins serve 24 households, for smaller blocks/towers.

4. On-site processing of the wet fraction — the core of the answer

Park's six methods of refuse disposal — **dumping, controlled tipping (sanitary landfill), incineration, composting, manure pits and burial** (*Park 28e, Ch 15, p 909*) — appraised for this setting:

Method	Verdict for an 850-house metro complex
Dumping	Rejected. Condemned by a WHO Expert Committee (1967) as a most insanitary method causing nuisance, health hazards and pollution; refuse is exposed to flies and rodents, blown by wind, and its drainage pollutes water (<i>Park 28e, Ch 15, p 909</i>)
Controlled tipping / sanitary landfill	Not an in-complex option; it is the ULB's function. Indian rules permit only inert rejects in a sanitary landfill, and it is the least preferred step of the hierarchy (<i>Solid Waste Management, Ch 22, p 249</i>)
Incineration	Rejected on site. Method of choice only where land is unavailable, and for hospital refuse; unpopular in India because refuse carries much fine ash needing costly pre-separation, and burning wastes badly needed manure (<i>Park 28e, Ch 15, pp 909-910</i>). On-site burning is also barred by Clause 4(2)
Composting	Method of choice. Park divides it into (1) the <i>Bangalore method</i> — anaerobic hot fermentation of refuse plus nightsoil in trenches sited ≥ 800 m from city limits, hence a town/rural method with no place inside a metro complex — and (2) mechanical composting (aerobic), complete in 4–6 weeks. Aerobic decomposition raises the mass to about 60°C or higher over several days, destroying fly eggs and larvae, weed seeds and pathogens ; the product is an odourless humus-like soil builder carrying nitrates and phosphates (<i>Park 28e, Ch 15, p 910</i>). The complex uses its modern decentralised aerobic descendants (box, bin, vermicomposting)
Manure pits	Park's rural household method: garbage, cattle dung, straw and leaves tipped into a pit and covered with earth after each day's dumping; two pits worked alternately , manure ready in 5–6 months (<i>Park 28e, Ch 15, p 910</i>). Effective and simple in rural communities; no space (and no cattle dung) in a metro block — but it is the conceptual parent of the on-site composting box
Burial	For small camps only — a 1.5 m wide, 2 m deep trench, refuse covered with 20–30 cm of earth daily

Choosing the composting technology for $\approx 0.6\text{--}0.7$ tonnes/day of wet waste. Decentralised composting handles very small quantities up to about 20 tonnes per day (TPD) and is based mainly on manual labour (*Solid Waste Management, Ch 37, p 299*):

Technology	Input capacity	Cycle time	Suitability here
Box composting (decentralised, community level)	3-tonne compost box with shed, manually loaded	40 days after the box is full	First choice — simple, manual, low capital, no power supply needed
Bin composting (decentralised, household level)	Household or small-cluster level	—	Useful add-on for individual towers
Batch aerobic digester / organic waste converter (OWC)	0.5–3.0 TPD — matches this complex exactly	20–30 min in the machine, then racked ~3 weeks (New Moti Bagh OWC: 20–30 min + 15 days curing in sprinkler-fitted trays)	Choose if space is tight and capital available; needs a three-phase power supply and periodic maintenance
Vermicomposting	Up to 50 TPD; 2 ha for 50 TPD	Table 3.6: 8–10 weeks ; under Indian conditions the cycle is about 10 weeks , or 7–8 weeks if the material is pre-composted	Feasible; uses <i>Eisenia fetida</i> , <i>Eudrilus eugeniae</i> , <i>Perionyx excavatus</i> , <i>Pheretima elongata</i> , <i>Lampito mauritii</i> ; temperature-sensitive (20–40°C) and needs shade
Biomethanation / anaerobic digestion	Expressly permitted by the Rules for on-premises processing; recovers biogas	—	Feasible where the complex can use the gas (community kitchen, security lighting)
Windrow	1–500 tonnes/day; 8 ha for 500 TPD	8 weeks	Over-sized — city-level plant
Aerated static pile	1–500 tonnes/day; 5 ha for 500 TPD	5 weeks	Over-sized — city-level plant
In-vessel	1–300 tonnes/day ; 4 ha for 500 TPD	4 weeks (3–5 days in vessel + 3 weeks to mature)	Over-sized and very costly

(Solid Waste Management, Ch 40, p 311 — Table 3.6; Ch 38, pp 300–301; Ch 40, pp 303–304)

Practical sizing: the design unit is a **3-tonne compost box** under a shed; waste is spread in **20 cm layers** around vertical perforated PVC aeration pipes, one layer a day, each layer loosely forked into the previous one, and the full box is left **40 days** (Solid Waste Management, Ch 38, pp 300–301). At ≈ 0.65 tonnes/day a 3-tonne box fills in about **4–5 days**, so with the 40-day maturation roughly $(4.5 + 40) \div 4.5 \approx 10$ **boxes** are worked in rotation, with a leachate drain and a covered curing area. Compost yield is **18–20%** from segregated organic feedstock but only **10–15%** from mixed MSW — the financial argument for strict segregation (Solid Waste Management, Ch 30, p 284). The compost goes to the complex's own gardens, which closes the loop and creates resident ownership.

5. Routing the remaining streams

Stream	Practical route from the complex
Dry recyclables (paper, plastic, metal, glass, rags)	Stored in a dedicated dry-waste room, further sub-sorted, and handed to authorised waste pickers or authorised recyclers on a fixed weekly day (Clause 4(6)/4(7)); sale proceeds part-fund the system

Stream	Practical route from the complex
Domestic hazardous waste	Handed over separately to the collector or deposited at the domestic hazardous waste deposition centre the ULB must provide at the rate of one centre per 20 sq km (Clause 15(i)); batteries may go back through retail trade
Sanitary waste	Wrapped in the manufacturer's pouch, bagged separately, handed over daily ; disposed preferably in a biomedical or MSW incinerator as directed by the State Pollution Control Board
E-waste (printers, cartridges, electronic parts)	Never mixed with wet or dry waste; routed to authorised e-waste handlers
Garden and horticultural waste	Shredded and used as the carbon-rich bulking agent in the composting boxes
Residual inert reject	Handed to the ULB's collection vehicle at the entry gate → sanitary landfill (only inerts are legally landfillable)

6. Making the system work — organisation, IEC and monitoring

- ▶ **Constitute an RWA waste-management sub-committee** and sign a written agreement with the ULB fixing the collection point (entry gate), timings, user fee and the ULB's obligation to lift and transport each segregated stream **separately** — without that, source segregation collapses (*Solid Waste Management, Ch 2, p 177*).
- ▶ **Engage and protect the workforce** — house-to-house collection is by far the best method of collecting refuse (*Park 28e, Ch 15, p 909*); employ or contract collectors (a self-help group, NGO or private firm is expressly contemplated), issue identity cards, and supply **gloves, shoes, full-body clothing and face masks**.
- ▶ **Provide the physical infrastructure** — covered segregated bins on every floor/tower, a composting shed with leachate drain, a dry-waste sorting room and a locked hazardous-waste cupboard.
- ▶ **Information, education and communication (IEC)** — refuse disposal cannot be solved without public education, as people take little interest in cleanliness outside their own homes (*Park 28e, Ch 15, p 910*): resident meetings, tower-wise "green champions", printed sorting charts on every bin, school-children campaigns, demonstration of the compost used in the complex garden.
- ▶ **Bye-laws, fees and incentives** — bye-laws can make segregation mandatory with **spot fines** (Corporation of Cochin levied Rs 250–10,000 for illegal dumping, and user charges of Rs 30–80 per household), and a "pay as you throw" variable charge rewards low generators (*Solid Waste Management, Ch 2, p 185*).
- ▶ **Monitor with simple indicators** — proportion of the 850 houses handing over segregated waste, tonnes of wet waste processed on site per day, kilograms of residual handed to the ULB per day, compost produced against the 18–20% expected yield, proportion of collection cost recovered from user fees, complaints of odour or fly nuisance, and worker injury records.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Diagram 1 — Source segregation and on-site processing layout of the apartment complex.** Labels to write: 850 houses / multiple towers; green bin — wet waste; white bin — dry waste; black (or red) bin — domestic hazardous; pouch-wrapped sanitary waste in the dry bin, bagged separately; floor-level collection point; 240 L community bins (1 per 48 households); composting shed — about 10 three-tonne boxes in rotation, 20 cm layers, vertical perforated PVC aeration pipes; leachate drain; curing area; dry-waste sorting room; locked hazardous-waste cupboard; entry-gate handover point to ULB vehicle; compost to complex gardens.

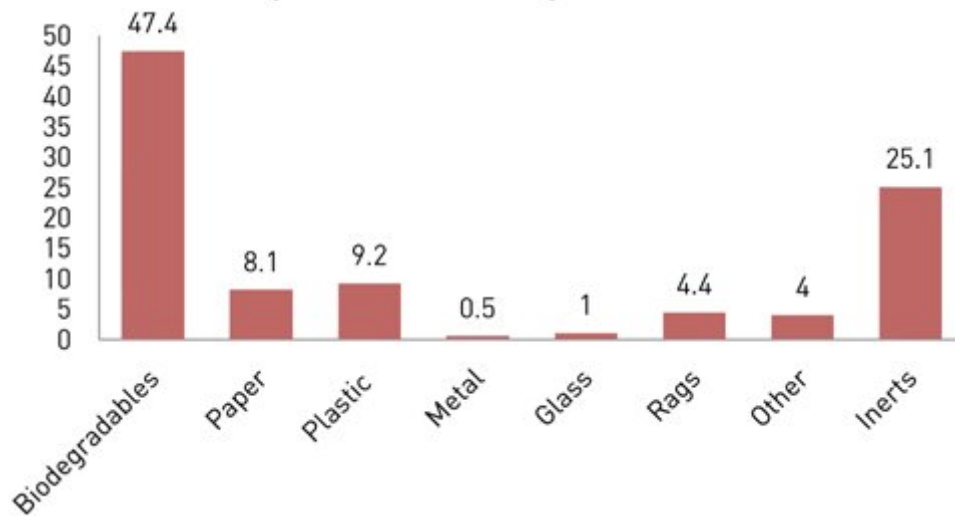
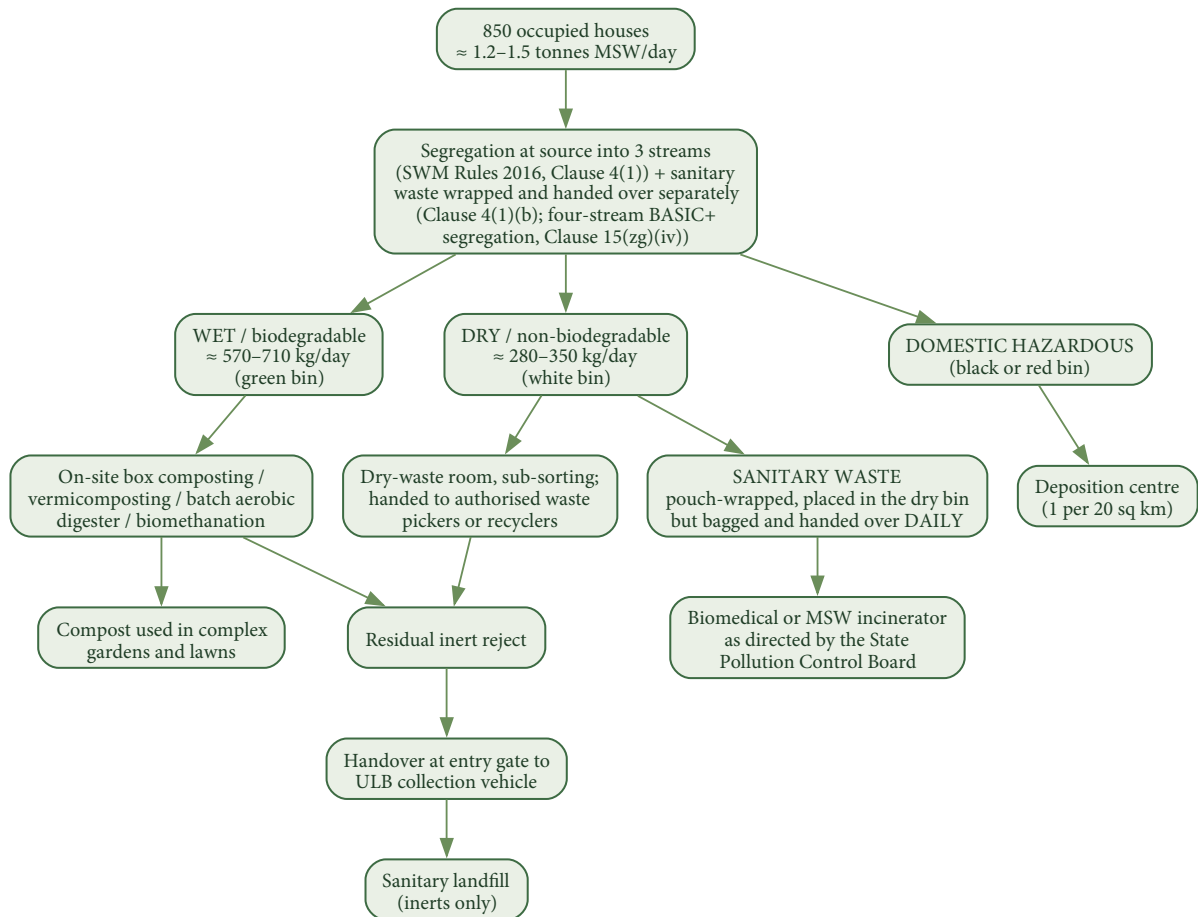
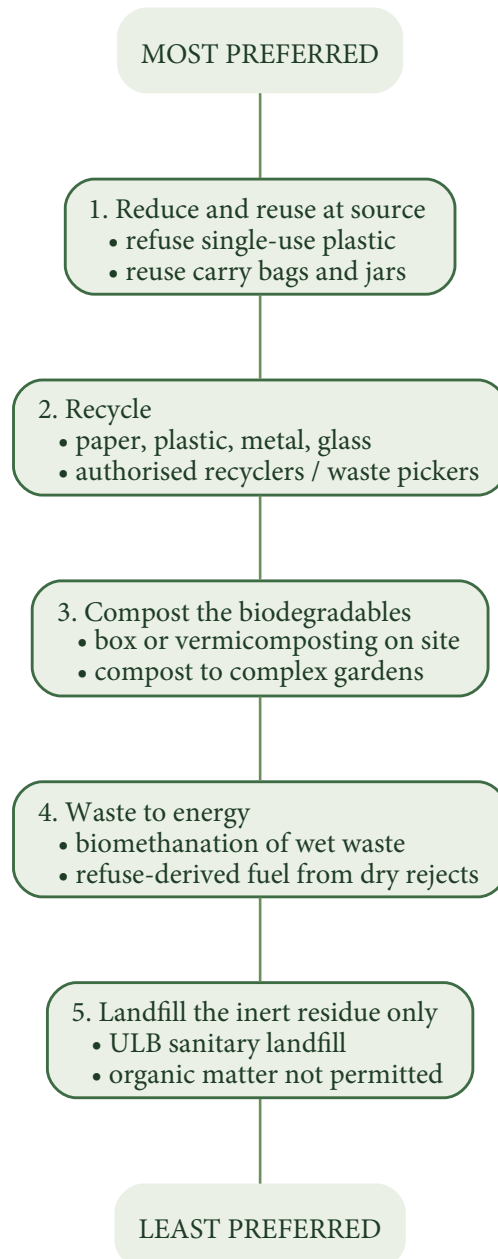


Fig 3.3 — Municipal Solid Waste Characterisation in India³ (solid waste management, p.252)



Waste pathway inside an 850-house apartment complex under SWM Rules, 2016



Integrated solid waste management hierarchy applied to the complex (most preferred at top)

High-yield exam pointers

- ▶ **Park's six methods of refuse disposal:** Dumping, Controlled tipping (sanitary landfill), Incineration, Composting, Manure pits, Burial — mnemonic "D-C-I-C-M-B". Composting itself splits into the **Bangalore method (anaerobic)** and **mechanical composting (aerobic)**; **manure pits are a separate, rural-household method** — two pits alternately, manure in 5–6 months.
- ▶ **SWM Rules 2016 numbers to quote:** three-stream segregation (Clause 4(1)(a)); sanitary waste wrapped and put in the **dry bin** (Clause 4(1)(b)); **four-stream BASIC+** segregation (Clause 15(zg)(iv)); gated communities >5,000 sq m must process biodegradables **within the premises** within one year (Clause 4(7)); domestic hazardous waste deposition centre **one per 20 sq km** (Clause 15(i)).

- ▶ **Indian MSW QUANTITY** (CPCB, 2014-15): urban India generated 1,43,449 metric tonnes per day in 2014-15; separately, CPCB puts urban India at about 51 million metric tonnes per year with a per-capita rate of 0.2–0.6 kg/day (2014-2015).
- ▶ **Indian MSW COMPOSITION** (Manual Table 1.6 / Fig 3.3, 2005 row, from Zhu D. et al., 2008 — a different source and year from the quantity figures): biodegradables 47.4%, inerts 25.2%, plastic/rubber 9.2%, paper 8.1%, rags 4.5%, other 4.0%, glass 1.0%, metal 0.5%.
- ▶ **Composting biology and yields**: temperature about 60°C or higher for several days kills fly eggs and larvae, weed seeds and pathogens; compost yield 18–20% from segregated organic feedstock and 10–15% from mixed MSW; box composting matures in 40 days; vermicomposting takes about 10 weeks in India (7–8 weeks if pre-composted).
- ▶ **Bin arithmetic**: 12–15 L household bin per family of five with 100% spare capacity; 240 L (96 kg) bin per 48 households → ≈ 18 bins per stream for 850 houses.
- ▶ **Benchmark case**: New Moti Bagh GPRA Complex, New Delhi — 1.5 tonnes/day from 1,100 households, shredder + organic waste converter (20–30 min) + 15 days curing.
- ▶ **Bangalore method detail** (quote it to show why it is a town/rural, not an apartment, solution): trench 90 cm deep (deeper slows decomposition), 1.5–2.5 m broad, 4.5–10 m long, sited ≥800 m from city limits; alternate layers of 15 cm refuse and 5 cm nightsoil until the heap rises 30 cm above ground, top layer refuse ≥25 cm, then earth-covered; heat >60°C within 7 days, persisting 2–3 weeks.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the CPCB waste-QUANTITY figures (1,43,449 tonnes per day, about 51 million tonnes per year, 0.2–0.6 kg per capita per day) are 2014-15 data, while the seven CHARACTERISATION percentages are the Manual's Table 1.6 and Figure 3.3 row for the year 2005, footnoted to Zhu D. et al. (2008) — never merge the two into one dated statement. Two numbers here are derived rather than retrieved and should be declared as assumptions in the exam: the household size of four persons, and the "about 10 composting boxes" (a 3-tonne box filling in 4–5 days at 0.65 tonnes of wet waste a day, plus the Manual's 40-day maturation). On bin colours, the Manual's text, following the SWM Rules 2016, prescribes green for wet and white for dry waste while the Manual's own Table 2.1 shows a blue dry-waste bin, and the Rules fix no colour for the domestic hazardous bin (Coimbatore City Municipal Corporation uses red). Swachh Bharat Mission-Urban, which funds such municipal solid waste work, lists "modern and scientific municipal solid waste management" as its third objective (Park 28e, Ch 7, p 574).

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, PIB India, MoHUA, Solid Waste Management Rules 2026, WHO

Programme and policy updates (India)

- ▶ **The Solid Waste Management Rules, 2026 supersede the SWM Rules, 2016** — notified 27 January 2026 under the Environment (Protection) Act, 1986 and in force from 1 April 2026. Scope widens from urban local bodies and census towns to rural local bodies, SEZs, food parks,

railways, airports and defence areas, and the framework moves from collect-and-dump to segregate-process-recover on circular-economy and polluter-pays principles. Every clause of the 2016 Rules quoted above should now be re-cited to its 2026 counterpart in an answer written after April 2026. [PIB 2026-Jan]

- ▶ **Four-stream segregation at source is now mandatory** — wet, dry, sanitary and a new **special care waste** stream (paint cans, bulbs and tube lights, batteries, mercury thermometers, expired medicines), collected by authorised agencies or deposited at designated centres. The household bin scheme becomes **green (wet), blue (dry), red (sanitary), black (special care)** — so sanitary waste leaves the dry bin and gets a bin of its own, and the 2016 "BASIC+" aspiration under Clause 15(zg)(iv) becomes binding law. [PIB 2026-Jan]
- ▶ **Bulk waste generator redefined, with extended responsibility** — any entity with **built-up area $\geq$ 20,000 sq m, water consumption $\geq$ 40,000 L/day, or waste generation $\geq$ 100 kg/day** (the 2016 trigger for gated communities was 5,000 sq m). An 850-house complex at 1.2–1.5 tonnes/day qualifies on the waste criterion several times over. Bulk generators must process wet and horticultural waste **on site** by composting or biomethanation, or procure an **Extended Bulk Waste Generator Responsibility (EBWGR) certificate**, and must register and file returns on the CPCB centralised online portal. [PIB 2026-Jan]
- ▶ **NAMASTE extended to waste pickers** — the Waste Picker Enumeration App was launched on World Environment Day 2025 to profile **2.5 lakh waste pickers** for occupational identity cards, PPE kits, Ayushman Bharat PM-JAY cover, occupational safety training, capital subsidy for collection vehicles, and support to 750 dry-waste collection centres; about 37,980 waste pickers were validated by August 2025. This is the scheme to name when asked how the complex's collectors are to be protected and formalised. [PIB 2025-Jun]

Current national data

- ▶ **Urban India now generates 1,66,096 tonnes of MSW per day, of which 1,33,164 TPD (80.17%) is processed** across 7,783 processing facilities, against roughly 16% processed in 2014 (MoHUA, Rajya Sabha reply, December 2025). Quote this in preference to the CPCB 2014-15 figure of 1,43,449 tonnes/day used in the body, and keep the two dated statements separate. [MoHUA 2025-Dec]
- ▶ **Swachh Survekshan 2024-25, the ninth edition**, was awarded on 17 July 2025: Ahmedabad, Bhopal and Lucknow led the million-plus cities, and a new **Super Swachh League** was created for cities with sustained excellence (23 cities, including Indore, Surat, Navi Mumbai and Vijayawada). Coverage has grown from 73 ULBs in 2016 to over 4,500 cities, with ODF certification and Garbage Free City star rating remaining the certification pillar. [PIB 2025-Jul]

Elimination, surveillance and disease status

- ▶ **Legacy dumpsites are now run as a mission-mode drive under DRAP, with an October 2026 target** — the **Dumpsite Remediation Accelerator Programme (DRAP)**, launched in **November 2025** under SBM-U 2.0 as a year-long accelerated push, is the mechanism that now governs legacy-waste clearance, and it sets "**Lakshya: Zero Dumpsites**" for **October 2026**. It prioritises **214 high-impact dumpsites** across 200 urban local bodies in 30 States/UTs, which together hold nearly **80% of the remaining legacy waste** (about 8.6 crore tonnes), with projects worth about ₹6,700 crore already approved for them. Position at end-2025: **1,138 of about 2,479 identified**

dumpsites fully remediated cumulatively — 459 of them during 2025 alone, across 438 cities — remediation underway at a further 1,428 sites, and over 62% of all legacy waste already processed. Alongside, the 2026 Rules require mapping and assessment of every legacy dumpsite, **time-bound biomining and bioremediation with quarterly progress reporting** through the online portal, annual landfill audits by State Pollution Control Boards, and restrict landfills to non-recyclable, non-energy-recoverable inert material; the landfill fee for unsegregated waste is deliberately set above the cost of segregating, transporting and processing segregated waste. [PIB 2026-Jan]

- **Occupational disease surveillance in the solid-waste workforce** — among 1,200 municipal waste workers enrolled in Varanasi, 71.5% were current smokeless-tobacco users against a national adult prevalence of 21.4%; of the 975 high-risk workers who underwent oral visual screening, 32.9% were screen-positive for oral potentially malignant lesions or cancer (leukoplakia commonest at 60.6% of lesions, then tobacco pouch keratosis 20.6% and oral submucous fibrosis 13.5%). Individual behavioural counselling gave the largest fall in Fagerström score (mean 2.88 to 1.63) but no participant in any arm reached the six-month abstinence that defined cessation. The practical lesson is to write oral screening and cessation into the contract with the complex's collectors, alongside gloves, boots and masks. [PMID 39462398]

Guideline and legislative updates

- **Decentralised composting outperforms centralised plants on compost maturity** — a review of household biodegradable waste systems finds centralised plants rarely produce matured compost because sorting and aeration of the substrate are inadequate, whereas decentralised units achieve both and cut the load on the municipality. Composting kitchen waste alone still yields immature compost and heavy greenhouse and odour emissions because free air space and the carbon-to-nitrogen ratio are too low and moisture too high; adding bulking agents and microbial additives lengthens the thermophilic phase and matures the compost faster. This is the evidence behind shredding garden and horticultural waste into the composting boxes rather than discarding it. [PMID 37696186]

Global and WHO developments

- **WHO's first dedicated report on solid waste and health**, *Throwing away our health* (16 December 2025), reframes waste as a public-health crisis: global MSW is projected to rise from **2.1 billion tonnes in 2020 to 3.8 billion tonnes by 2050**, open burning and inefficient incineration release **dioxins and furans**, and informal waste workers, children, pregnant women and residents near disposal sites carry the greatest risk. Reported associations include adverse birth outcomes, congenital anomalies, cardiovascular and respiratory disease and certain cancers, though the evidence is largely ecological and drawn from high-income settings. WHO's asks — apply the waste hierarchy, end open dumping and burning, expand collection, implement extended producer responsibility, and run occupational health programmes for waste workers — map directly onto SDG indicator **11.6.1** (proportion of municipal solid waste collected and managed in controlled facilities). [WHO 2025-Dec]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 15 (Environment and Health), pp 908-910 and p 940; Ch 7 (Health Programmes in India), p 574. Manual on Municipal Solid Waste Management, CPHEEO, Ministry of Urban Development (2016) — Table 1.6 (physical composition of MSW, 1996 and 2005 rows, after Zhu D. et al., 2008); Ch 2 (Clause 4 duties of waste generators, segregation, bins and colours, Kochi case study, pp 173-186); Ch 22-23 (integrated solid waste management hierarchy and CPCB quantity data, pp 249-252); Ch 30 (compost yields, p 284); Ch 37-40 (decentralised, box and vermicomposting, Table 3.6 composting technologies, and the New Moti Bagh residential-colony and batch-aerobic-digester cases, pp 299-311); Ch 43 (Clause 4(7) gated communities, p 336). Solid Waste Management Rules, 2016 (Clauses 4, 15 and 17) as reproduced in that Manual.

Recent-advances sources: 39462398 — Prevalence of Oral Potentially Malignant Lesions, Tobacco use, and Effect of Cessation Strategies among Solid Waste Management workers in Northern India: a pre-post intervention study, *BMC Oral Health* 2024, doi:[10.1186/s12903-024-05087-8](https://doi.org/10.1186/s12903-024-05087-8); 37696186 — A comprehensive review on the decentralized composting systems for household biodegradable waste management, *Journal of Environmental Management* 2023, doi:[10.1016/j.jenvman.2023.118824](https://doi.org/10.1016/j.jenvman.2023.118824) (abstract-only). Solid Waste Management Rules, 2026 (Gazette of India, notified 27 January 2026, in force 1 April 2026) via PIB India, January 2026; PIB India backgrounder, 31 January 2026, *Lakshya Zero Dumpsite: India's Drive to Eliminate City Dumpsites Under Swachh Bharat Mission* (Dumpsite Remediation Accelerator Programme launched November 2025, October 2026 target, 214 high-impact sites, and the 2025 and cumulative remediation counts), with the Ministry of Housing and Urban Affairs DRAP concept note, November 2025; PIB India, June 2025 (NAMASTE Waste Picker Enumeration App); PIB India, July 2025 (Swachh Survekshan 2024-25 awards); Ministry of Housing and Urban Affairs, Rajya Sabha reply, December 2025 (national MSW generation and processing); WHO, *Throwing away our health: the impacts of solid waste on human health*, 16 December 2025.



Biological effects of radiation

Asked in: Paper I 14-Jul-2020 — RGUHS MD Community Medicine

Definition — Ionizing radiation is radiation able to penetrate tissue and deposit its energy within it, comprising electromagnetic radiations (X-rays, gamma rays) and corpuscular radiations (alpha particles, beta particles or electrons, protons) (Park 28e, Ch 15, p 899). Its biological effects arise from ionization and consequent injury to the cell — chiefly to deoxyribonucleic acid (DNA) — and are conventionally classified as **somatic** (appearing within the lifespan of the irradiated person) and **genetic** (appearing in the offspring), or, by dose–response behaviour, as **deterministic** and **stochastic** effects.

Dose, radiation quality and the two classes of biological effect

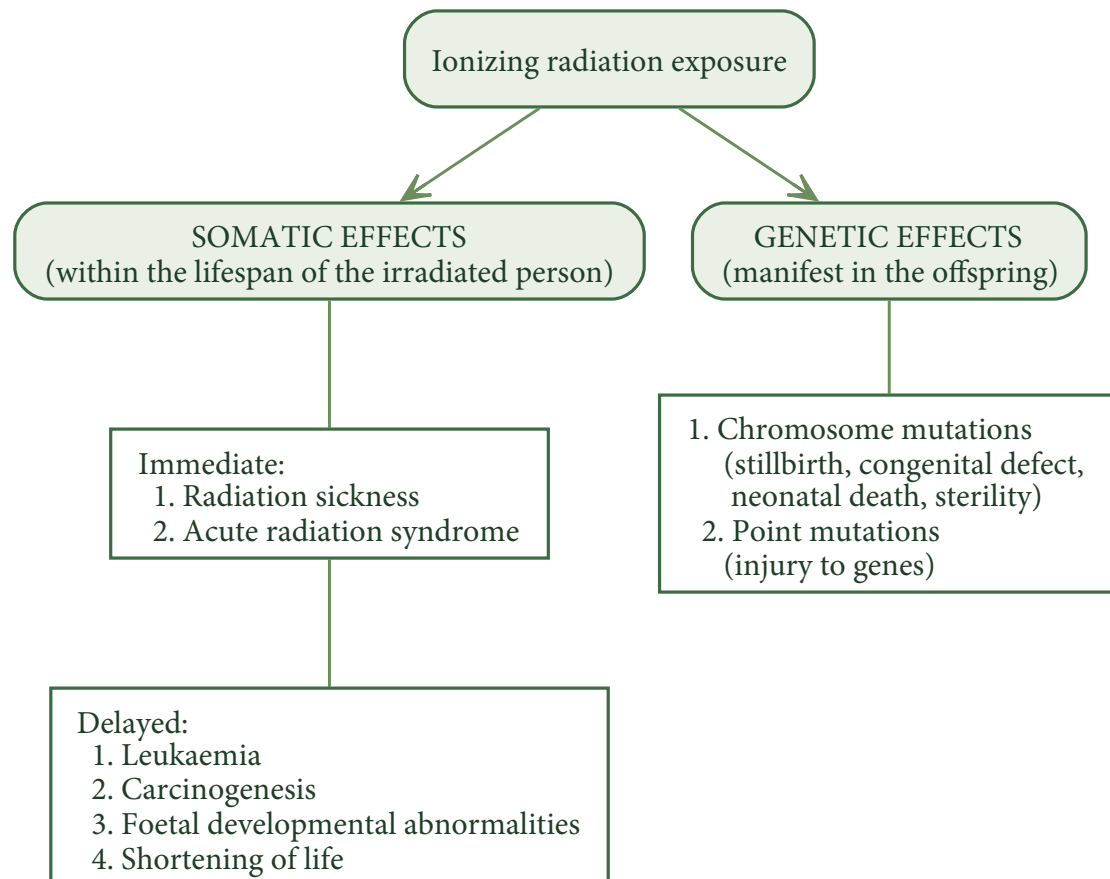
Item	Content (with the fact that must be written)
Absorbed dose (D)	Gray (Gy) — the dose that imparts 1 joule of energy to 1 kg of absorbing material; 1 rad = 0.01 Gy (Park 28e, Ch 15, p 899)
Dose equivalent / equivalent dose (H)	Sievert (Sv); $H = D \times Q$, where Q is the quality factor (in current International Commission on Radiological Protection, ICRP, usage the radiation weighting factor w_R) depending on ionization density; 1 Sv = 100 rem (Park 28e, Ch 15, p 899; Oxford Textbook of Public Health 7e, Ch 54, pp 2021–2022)
Value of Q / w _R	1 for X-rays, gamma rays and electrons — sparsely ionizing, low linear energy transfer (LET); 20 for alpha particles — densely ionizing, high LET (Park 28e, Ch 15, p 899; Oxford Textbook of Public Health 7e, Ch 54, pp 2020–2022)
Effective dose	Sum of the equivalent doses to each organ, each multiplied by that organ's tissue weighting factor w_T ; also in sievert; used when different organs receive different doses (Oxford Textbook of Public Health 7e, Ch 54, p 2021)
Activity	Becquerel (Bq) = 1 nuclear disintegration per second; formerly the curie (Ci) = 37 GBq (Park 28e, Ch 15, p 899; Oxford Textbook of Public Health 7e, Ch 54, p 2021)
Two exposure patterns	High dose over a short time → acute or short-term effects (cell killing, organ damage); low dose over an extended period → chronic or long-term effects (Park 28e, Ch 15, p 899)

Every biological effect then behaves in one of two ways with dose:

Feature	Deterministic (tissue reaction, non-stochastic)	Stochastic (probabilistic)
Mechanism	Killing of a large number of cells in the affected organ	Subtle change (mutation) within a single surviving cell
Threshold dose	Present — effect does not occur below it	Assumed absent (linear non-threshold)
Effect of increasing dose	Severity of the effect increases	Probability (incidence) increases; severity is dose-independent
Attribution	Cause and effect can be seen directly	Only inferred from an increased probability in the population

Feature	Deterministic (tissue reaction, non-stochastic)	Stochastic (probabilistic)
Examples	Acute radiation syndrome, cataract, non-malignant skin damage, suppression of haematopoiesis, damage to gametes with loss of fertility	Cancer (including leukaemia) and hereditary (genetic) effects

(Maxcy-Rosenau-Last 15e, Ch 35, p 762; Sunder Lal, Ch 11, p 643; Hunter's Diseases of Occupations, Ch 53, pp 650–651; Oxford Textbook of Public Health 7e, Ch 54, pp 2027–2028)



Classification of the biological effects of ionizing radiation

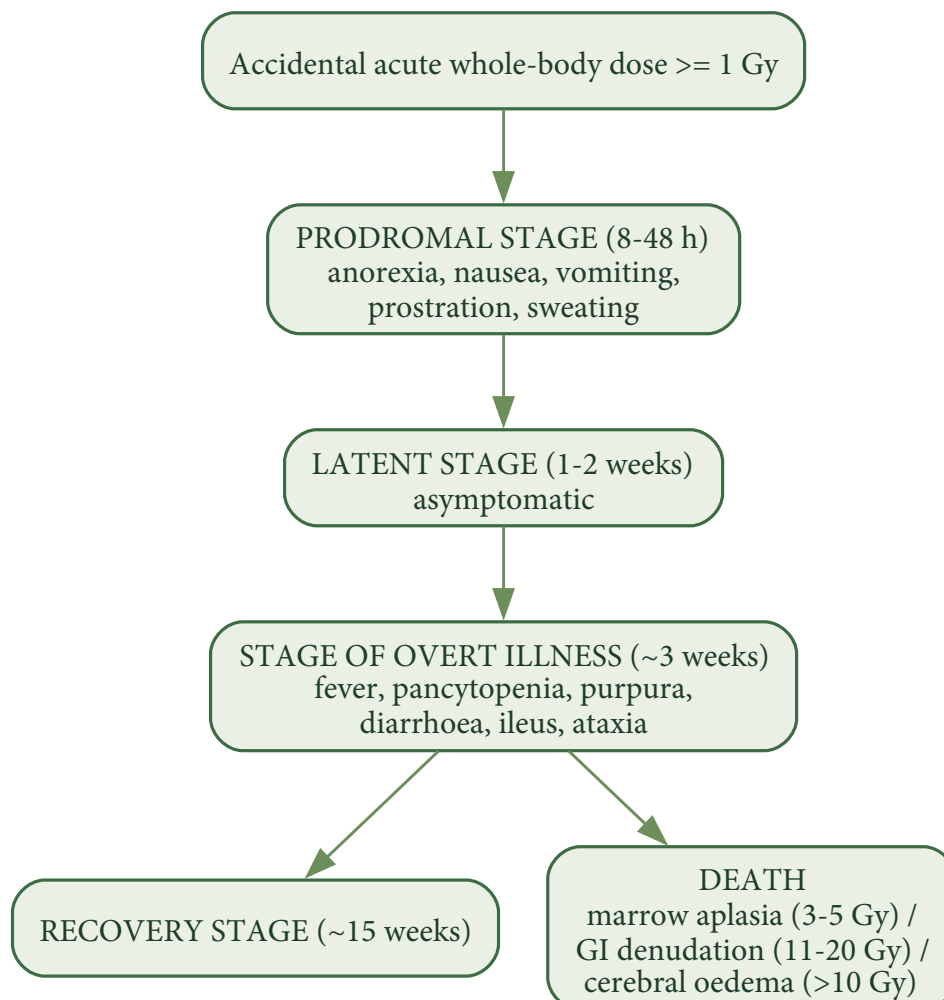
Immediate somatic effects — acute radiation syndrome

Acute effects follow heavy (about 1 Gy) or very heavy (1–9 Gy) whole-body doses received over a short period, almost always accidentally (Suryakantha, Ch 10, p 78). The clinical course runs in four stages:

- ▶ **Prodromal stage** — anorexia, nausea, vomiting, prostration, fatigue and sweating; diarrhoea and oliguria in fulminating cases. Lasts **8–48 hours**.
- ▶ **Latent stage** — asymptomatic; lasts **1–2 weeks**.
- ▶ **Stage of overt illness** — fever, anaemia, leukopenia, pancytopenia, thrombocytopenic purpura, diarrhoea, paralytic ileus, paraesthesia, motor disturbances, ataxia, disorientation and autonomic collapse. Lasts about **3 weeks**.
- ▶ **Recovery stage** — lasts about **15 weeks**. Exposure to massive doses above **10 Gy** may cause death within a day or two from cerebral oedema or cardiac failure.

Acute whole-body dose	Biological response
< 5 rad (< 0.05 Gy)	No immediate observable effect
5–50 rad (0.05–0.5 Gy)	Slight blood changes, detectable only on medical evaluation
50–150 rad (0.5–1.5 Gy)	Slight blood changes; nausea, fatigue and vomiting likely
150–1,100 rad (1.5–11 Gy)	Severe blood changes with immediate symptoms; some die about 2 weeks later. At 300–500 rad (3–5 Gy) up to half of those exposed die within 60 days without intensive care, from destruction of the blood-forming organs
1,100–2,000 rad (11–20 Gy)	Death in 100 per cent within 1–2 weeks; the gastrointestinal system is destroyed
> 2,000 rad (> 20 Gy)	Death certain; above 5,000 rad (50 Gy) the central nervous system can no longer control breathing and circulation

(Park 28e, Ch 15, pp 899–900)



Clinical course of the acute radiation syndrome after whole-body irradiation

- **Haematopoietic** — **lymphocytes are the most sensitive indicator**, 1–2 Gy halving the blood lymphocyte concentration within 48 hours; the median lethal dose within 60 days (LD_{50/60}) is

2.5–5 Gy, taken as 3 Gy with minimal treatment and 4.5 Gy with full supportive care (Hunter's, Ch 53, p 651).

- ▶ **Gastrointestinal** — LD50 about 15 Gy, derived from animal work (Hunter's, Ch 53, p 652).
- ▶ **Central nervous system** — reached only above about 50 Gy, survival usually under 48 hours (Hunter's, Ch 53, p 653).

Delayed (late) somatic effects, their thresholds and their latency

Delayed somatic effects are seen in those exposed to less than 1 Gy over a long period (Suryakantha, Ch 10, p 78).

Site / system	Delayed effect
Eye	Cataract — the earliest delayed effect; commonest with neutrons
Skin	Erythema, oedema, blisters and ulcers appearing late; still later hyperkeratosis and atrophy of the sebaceous glands; commonest with beta particles
Haematopoietic	Leukaemia, aplastic anaemia, pancytopenia
Solid organs	Cancer of the lung, skin and blood; tumour induction
Whole person	Shortening of life
Foetus	Malformation and microcephaly

(Suryakantha, Ch 10, p 78 and Ch 18, p 233; Park 28e, Ch 15, p 899)

Organ	Deterministic effect	Approximate threshold dose, ED ₀ (Gy)
Whole body	Prodromal syndrome (vomiting)	0.5
Whole body	Early death	1.5
Bone marrow	Depression of haematopoiesis	0.5
Bone marrow	Early death	1.5
Lung	Pneumonitis / early death	3–5 / 6
Skin	Prompt erythema, dry desquamation	≥ 8
Skin	Moist desquamation and necrosis	≥ 15
Thyroid	Hypothyroidism	3
Lens of the eye	Detectable opacity / cataract with visual loss	0.5–2 / 5
Testis	Temporary sterility / permanent sterility	0.15 / 3.5–6
Ovary	Permanent sterility	2.5–6
Embryo or foetus	Teratogenesis	0.10

(Hunter's Diseases of Occupations, Ch 53, Table 53.2, p 651)

Latency of the stochastic effects — radiation carcinogenesis is the principal stochastic effect, and the examiner expects it to be placed in time:

Late stochastic / late effect	Latent period after whole-body irradiation
Leukaemia	Appears first; latent period 2 years, peaking at 6–7 years
Solid tumours	Begin to appear only after about 10 years, and the incidence continues to rise for several decades
Radiation cataract	6 months to 35 years, average about 3 years

No threshold is detectable for carcinogenesis, and statistically significant increased risks exist in the Japanese atomic-bomb survivor cohort down to 150 mSv (Hunter's, Ch 53, p 652 and p 655).

Genetic effects and effects on the embryo and foetus

- ▶ Genetic effects follow **irradiation of the gonads with injury to chromosomes**; somatic effects are recognisable within the lifespan of the irradiated person, whereas genetic effects are manifest in the offspring (Park 28e, Ch 15, p 900).
- ▶ **Chromosome mutations** → stillbirths, congenital defects, neonatal deaths and sterility. **Point mutations** injure individual genes, and are held responsible for disorders such as Huntington's chorea, polycystic kidney and haemophilia (Suryakantha, Ch 10, p 78).
- ▶ Germ cells are highly radiosensitive: transient sterility begins at about **0.15 Gy in men** and about five to ten times higher in women; the threshold doses for **permanent sterility are 3.5–6 Gy for the testis and 2.5–6 Gy for the ovary** (Hunter's, Ch 53, Table 53.2, p 651), which the same chapter states narratively as "doses of 2.5–6 Gy or more are required for permanent sterility in both men and women" (Hunter's, Ch 53, pp 652–653).
- ▶ The **developmental effect depends on gestational age**: the gravest consequences of prenatal exposure are embryonic death, gross congenital malformation, growth retardation and severe mental retardation.
- ▶ **Severe mental retardation** was demonstrated in atomic-bomb survivors irradiated between the **8th and 15th weeks** after conception, and less so between the **16th and 25th weeks**; thresholds about **0.4 Gy and 0.1–0.2 Gy** respectively (Hunter's, Ch 53, p 653; Maxcy-Rosenau-Last 15e, Ch 35, p 766; Oxford Textbook of Public Health 7e, Ch 54, p 2038).

Biological effects of non-ionizing radiation

Radiation	Effects
Ultraviolet — skin, short term	Migration of melanin from the Malpighian layer into the corneum (suntan); histamine release causing erythema, oedema, blisters and ulcers; thickening of all epidermal layers (protective); synthesis of vitamin D, preventing rickets (beneficial)
Ultraviolet — skin, long term	Degeneration of skin, loss of elasticity, skin cancer (squamous cell carcinoma, rodent ulcer) ; dark-skinned persons are less susceptible. Ultraviolet radiation is an IARC Group 1 (known) human carcinogen , as is ionizing radiation

Radiation	Effects
Ultraviolet — eye	<i>Snow blindness</i> (keratitis from reflection off snow), <i>eclipse blindness</i> from direct gazing at the sun, and from artificial sources conjunctivitis, keratitis, photophobia, <i>welder's flash</i> from arc welding and corneal ulcer late
Infrared	Skin — flushing, burns, ulcers; eye — cataract . Risk groups: blast-furnace workers, blacksmiths, kiln, oven and stoker workers
Microwave	Prolonged exposure → cataract and microwave sickness (headache, giddiness, loss of memory, fatigue)
Laser	Thermal burns of skin, corneal damage, opacification of the lens and burning of the retina → cataract and blindness
Radiofrequency (radio and television transmission, mobile phones and base stations)	Absorbed superficially ; the established biological effect is tissue heating (thermal) , and prevention of excessive heating is the basis of most international exposure guidelines. IARC classifies radiofrequency electromagnetic fields as Group 2B, a possible human carcinogen — on limited human evidence from glioma and acoustic neuroma studies in mobile-phone users and limited animal (co-carcinogenicity) evidence — while epidemiological evidence for any cancer site remains inconsistent
Extremely low frequency (ELF) magnetic fields	Also IARC Group 2B, on a consistent epidemiological association with childhood leukaemia unsupported by animal and cell studies; ELF <i>electric</i> fields and static fields are Group 3 (not classifiable)

(Suryakantha, Ch 10, pp 75–76 and Ch 18, p 233; Park 28e, Ch 18, p 959; Oxford Textbook of Public Health 7e, Ch 54, pp 2005, 2008–2010 and 2040–2041)

Factors modifying the biological effect

Factor	Influence
Type of tissue	Gonads, lymph nodes, bone marrow and thyroid are the most susceptible
Type of radiation	Corpuscular radiations are more harmful than electromagnetic radiations of the same absorbed dose
Area of body exposed	The larger the area, the greater the marrow depression and the severer the injury
Dose rate and duration	Intensity of radiation and duration of exposure; protracting the exposure raises the dose needed for a deterministic effect (for skin, protraction over 1–14 days roughly doubles the threshold)
Host factors	Both mother and foetus at risk in pregnancy; children about ten times more susceptible than adults ; malnourished and debilitated persons at greater risk; individual susceptibility (grossly increased in ataxia telangiectasia)
Radiosensitisers	Drugs such as metronidazole increase susceptibility
Protective clothing	Reduces the effect (0.5 mm lead apron cuts scattered X-ray intensity by over 90 per cent)

(Suryakantha, Ch 10, p 78; Hunter's, Ch 53, pp 651–652; Park 28e, Ch 15, p 900)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Dose–response curves of the two classes of effect** — one pair of axes, dose on the x-axis and response on the y-axis. Label: *deterministic effect* — sigmoid curve starting at a **threshold dose (ED0)**, with ED50 marked on it and the y-axis labelled *severity of effect*; *stochastic effect* — straight line through the origin with **no threshold**, y-axis labelled *probability of effect*. On the low-dose end of the x-axis put **two separately named sets of ticks**: (i) actual exposure — *natural background* ≈ 2.4 mSv per year (UNSCEAR 2010; *total average individual dose including medical* ≈ 3.0 mSv per year); and (ii) regulatory limits — *public dose limit* 1 mSv per year (ICRP 2007) and *occupational limit* 20 mSv per year (ICRP 2007).
- ▶ **Time course of the acute radiation syndrome** — a horizontal time line with the four blocks labelled *prodromal* (8–48 h), *latent* (1–2 weeks), *overt illness* (~3 weeks), *recovery* (~15 weeks), and a superimposed blood-count curve labelled *lymphocytes fall first* (within 48 h), *granulocytes and platelets fall in days to weeks*, *erythrocytes fall last*.

High-yield exam pointers

- ▶ **Mnemonic for the classification** — "SG, ID": Somatic (Immediate = radiation sickness, acute radiation syndrome; Delayed = leukaemia, carcinogenesis, foetal abnormality, life shortening) and Genetic (chromosome mutations, point mutations).
- ▶ $H = D \times Q$; 1 Gy = 100 rad, 1 Sv = 100 rem, Q (radiation weighting factor wR) = 1 for X-rays/gamma/electrons and **20 for alpha particles**.
- ▶ **Deterministic = threshold + severity rises with dose; stochastic = no threshold + probability rises with dose** — cancer and hereditary effects are stochastic.
- ▶ **LD50/60 = 2.5–5 Gy** (3 Gy with minimal treatment, 4.5 Gy with full supportive care); **lymphocyte count is the earliest and most sensitive biological dosimeter**.
- ▶ **Threshold doses to quote**: **temporary sterility 0.15 Gy (male)**, **permanent sterility testis 3.5–6 Gy**, **ovary 2.5–6 Gy**, **lens opacity 0.5–2 Gy**, **skin prompt erythema ≥ 8 Gy**, **teratogenesis 0.10 Gy**, **severe mental retardation at 8–15 weeks' gestation from about 0.4 Gy**.
- ▶ **Latency**: leukaemia latent period **2 years**, **peak 6–7 years**; solid tumours after about **10 years**; radiation cataract **6 months to 35 years**, **average 3 years**.
- ▶ **Background versus limit** — **do not confuse the two**: worldwide average annual individual effective dose is **3.0 mSv**, of which **2.4 mSv is natural** (radon 1.26, external gamma 0.48, cosmic 0.39, ingestion 0.29) and about 0.6 mSv man-made, dominated by medical exposure (UNSCEAR 2010, quoted in Oxford Textbook of Public Health 7e, Ch 54, pp 2022–2023). **ICRP 2007 dose limits** — occupational effective dose **20 mSv per year averaged over 5 years (maximum 50 mSv in any one year)**; public **1 mSv in a year**; **1 mSv to the embryo/foetus** once pregnancy is declared (Oxford Textbook of Public Health 7e, Ch 54, Table 7.2.4, pp 2039–2040).
- ▶ **Detriment-adjusted nominal risk coefficient for stochastic effects = 5.7 per cent per Sv** for the general population and **4.2 per cent per Sv** for workers, of which hereditary effects contribute 0.2 and 0.12 per cent per Sv respectively (Oxford Textbook of Public Health 7e, Ch 54, pp 2038–2039).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park 28e still expresses the dose–response ladder in rad, still repeats the older phrasing that "alpha particles are ten times as harmful as X-rays, beta particles or gamma rays" (Ch 15, p 899) even though it also gives the modern quality factor of 20 used above, and quotes the older International Commission on Radiological Protection (ICRP) advice that the genetic dose to the whole population from sources additional to natural background should not exceed 5 rem over 30 years, whereas ICRP 2007 states the current limits in millisievert per year given above and ICRP 2012 lowered the cataract threshold to about 500 milligray from the 5 gray previously accepted; write the modern value in the exam and keep Park's older phrasing only as background.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 15 Environment and Health, pp 898–900; Ch 18 Occupational Health, p 959); Suryakantha — Community Medicine with Recent Advances (Ch 10 Radiation, pp 75–80; Ch 18 Occupational Health, p 233); Hunter's Diseases of Occupations (Ch 53 Ionizing Radiations, pp 650–664, including Table 53.2 threshold doses, p 651); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 35 Ionizing Radiation, pp 762–766); Oxford Textbook of Public Health 7e (Ch 54 "2 Radiation and public health", pp 1996–2041), quoting UNSCEAR 2010 dose data and ICRP 2007 and ICRP 2012 recommendations; Sunder Lal — Textbook of Community Medicine (Ch 11 Occupational Health, p 643). Coverage note: the indexed corpus yielded no captioned figure genuinely depicting the biological effects of radiation, so a "Diagram to draw" box is given in place of a textbook plate.



Biomedical Waste (Management and Handling) Rules 1998

Asked in: Paper IV 19-Nov-2021 — RGUHS MD Community Medicine

Definition — Bio-medical waste (BMW) is any waste generated during the diagnosis, treatment or immunisation of human beings or animals, or in research activities pertaining thereto, or in the production or testing of biologicals, including the categories listed in Schedule I of the Rules (Suryakantha, Ch 13, p 112; Park 28e, Ch 16, p 943). The **Bio-Medical Waste (Management and Handling) Rules, 1998** were India's first comprehensive statutory framework for such waste, and they made the head of every health-care facility — the "occupier" — legally accountable for the waste his institution generates. The burden justifies the law: **10–25 per cent of health-care waste is hazardous**, and a Bangalore survey found solid waste generation of **½–4 kg per bed per day in hospitals**, **½–2 kg per bed per day in private hospitals** and **½–1 kg per bed per day in nursing homes** (Park 28e, Ch 16, pp 943–944).

Statutory basis, scope and key definitions (1998 Rules)

Element	Provision under the 1998 Rules
---------	--------------------------------

(Suryakantha, Ch 13, p 112; notification number and date from the gazette preamble of the BMW Rules 2016, p 1.)

Schedule I — the ten categories, treatment options and colour code

Category	Contents	Treatment and disposal option	Colour code	Container
1. Human anatomical waste	Human tissues, organs, body parts, placenta, aborted foetus, amputated parts, tumours	Incineration ¹ / deep burial ²	Yellow	Plastic bag
2. Animal waste	Animal tissues, organs, body parts, carcasses, bleeding parts, fluid, blood; waste of veterinary hospitals and animal houses	Incineration ¹ / deep burial ²	Yellow	Plastic bag
3. Microbiology and biotechnology waste ⁶	Laboratory cultures, stocks, specimens of micro-organisms, live or attenuated vaccines, human and animal cell culture, toxins, dishes and devices used for transfer of cultures	Local autoclaving / microwaving / incineration ¹	Yellow or red	Plastic bag / disinfected container
4. Waste sharps	Needles, syringes, scalpels, blades, glass that may cause puncture and cuts — both used and unused	Disinfection / chemical treatment ³ / autoclaving / microwaving and mutilation and shredding ⁴	Blue or white translucent	Plastic bag / puncture-proof container
5. Discarded medicines and cytotoxic drugs	Outdated, contaminated and discarded medicines	Incineration / destruction and disposal in secured landfills	Black	Plastic bag
6. Soiled waste	Cotton, dressings, soiled plaster casts, linen, bedding and other items contaminated with blood and body fluids	Incineration / autoclaving / microwaving	Yellow or red	Disinfected container / plastic bag
7. Solid waste	Disposables other than sharps — tubings, catheters, intravenous sets and the like	Disinfection by chemical treatment ³ / autoclaving / microwaving and mutilation or shredding ⁴	Red, blue or white	Disinfected container / plastic bag
8. Liquid waste	Waste from laboratory, washing, cleaning, housekeeping and disinfecting activities	Disinfection by chemical treatment ³ and discharge into drains	Not applicable ⁵	Not applicable ⁵
9. Incineration ash	Ash from incineration of any bio-medical waste	Disposal in municipal landfill	Black	Plastic bag

Category	Contents	Treatment and disposal option	Colour code	Container
10. Chemical waste	Chemicals used in production of biologicals, in disinfection, as insecticides	Chemical treatment ³ and discharge into drains for liquids; secured landfill for solids	Black	Plastic bag

Footnotes to Schedule I — ¹ no chemical pre-treatment before incineration, and chlorinated plastics shall not be incinerated; ² deep burial is permitted **only in towns with population below 5 lakh and in rural areas**; ³ using at least **1 per cent hypochlorite solution** or an equivalent reagent that ensures disinfection; ⁴ mutilation or shredding must be such as to prevent unauthorised reuse; ⁵ liquid waste needs no container or bag; ⁶ if disinfected locally, need not be put in containers or bags. Where a category has several treatment options, the colour code follows the option actually chosen (Suryakantha, Ch 13, pp 115-116).

Schedule V — standards for the treatment and disposal technologies

Technology	Standard to be met
Incinerator (operating)	Combustion efficiency (CE) at least 99.00 per cent, computed as $CE = \%CO_2 \div (\%CO_2 + \%CO) \times 100$; primary chamber $800 \pm 50^\circ C$; secondary chamber gas residence time at least 1 second at $1050 \pm 50^\circ C$ with a minimum of 3 per cent oxygen in the stack gas (Jugal Kishore, Ch 3; BMW Rules 2016, Schedule II, para 1.A). Suryakantha adds that the waste gas leaving the secondary chamber is cooled immediately to $230^\circ C$
Incinerator (emission), at 12 per cent carbon dioxide correction	Particulate matter 150 mg/Nm ³ ; oxides of nitrogen 450 mg/Nm ³ ; hydrogen chloride 50 mg/Nm ³ ; minimum stack height 30 metres; volatile organic compounds in ash not more than 0.01 per cent
Autoclave (gravity-flow) — three permitted settings	121 °C / 15 psi / 60 minutes, or 135 °C / 31 psi / 45 minutes, or 149 °C / 52 psi / 30 minutes
Autoclave (vacuum) — after ≥3 pre-vacuum pulses	121 °C / 15 psi / 45 minutes, or 135 °C / 31 psi / 30 minutes
Autoclave — floor and validation	Under no circumstances below 121 °C, 15 psi and 30 minutes residence; must consistently kill the biological indicator <i>Bacillus (Geobacillus) stearothermophilus</i> spores, $\geq 1 \times 10^4$ spores per millilitre (raised to 1×10^6 spores by the 2016 Rules)
Microwave irradiation	Not suitable for category 1 and 2 (anatomical) waste; at maximum design capacity must show total destruction of <i>Bacillus subtilis</i> spores at a concentration of 10^4 spores
Chemical treatment	1 per cent hypochlorite , or 2 per cent bleach, or 10 g of fresh bleaching powder in one litre of water, with a contact time of 60 minutes

Technology	Standard to be met
Deep burial	Pit or trench about 2 metres deep and 1.5 metres square, on an impermeable site away from habitation, not flood- or erosion-prone, with no shallow well nearby, authorised by the prescribed authority; filled to half with waste, covered with lime to within 50 cm of the surface and the rest filled with soil; 10 cm of soil added over the waste on every occasion; supervised burial, animal-proof galvanised iron or wire-mesh cover, and a record of all pits maintained

(Suryakantha, Ch 13, pp 117-120, 123-125; autoclave settings and combustion efficiency verified against BMW Rules 2016, Schedule II, pp 16-20, which carries the 1998 Schedule V unchanged for these parameters.) Plasma pyrolysis or gasification is a separate technology with its own standard — combustion efficiency at least 99.99 per cent, chamber at 1050 ± 50 °C with gas residence ≥ 2 seconds — introduced under the 2016 Rules; it is not the incinerator standard (BMW Rules 2016, Schedule II, para 2.A, p 18).

Duties of the occupier and the regulatory machinery (1998 Rules)

- ▶ **Duty of care** — every generator or occupier must ensure that the waste generated is handled without any adverse effect on human health and the environment.
- ▶ **Treatment and disposal** — every occupier shall treat and dispose of waste as per Schedules 1 and 5, and set up the requisite facilities (incinerator, autoclave, and so on) for this purpose.
- ▶ **No mixing** — bio-medical waste shall **not be mixed with other waste**; it must be segregated at source into colour-coded containers as per Schedule 2, labelled as per Schedule 3, and transported for treatment and disposal within 48 hours of generation.
- ▶ **Transport outside the premises** must follow the labelling procedure of Schedule 4.
- ▶ **Prescribed authority** — every State and Union Territory Government shall establish an authority to authorise and implement the Rules.
- ▶ **Authorisation** — every occupier applies in **Form 1** with the prescribed fee; the authority grants or renews authorisation for **3 years** if satisfied of the occupier's capacity to handle the waste.
- ▶ **Annual report** — every occupier submits a report in **Form 2** by **31 January every year**; all handling records are maintained for verification.
- ▶ **Accident reporting** — any accident during handling is reported to the prescribed authority in **Form 3**.

(Suryakantha, Ch 13, pp 112-113.)

Current position: BMW Management Rules 2016, as amended in 2018 and 2019

The 1998 Rules stand superseded by the Bio-Medical Waste Management Rules, 2016 (G.S.R. 343(E), in force from 28 March 2016), which are themselves amended by the BMW Amendment Rules 2018 and 2019 (BMW Rules 2016, p 1; Sunder Lal, Ch 14, p 679). The 2016 Rules replaced ten categories with four colour-coded streams:

Colour and container	Waste covered	Treatment and disposal
Yellow — non-chlorinated plastic bags or containers	(a) human anatomical; (b) animal anatomical; (c) soiled waste; (d) expired and discarded medicines including cytotoxic; (e) chemical solid; (f) chemical liquid; (g) discarded linen, mattresses, bedding contaminated with blood or body fluid; (h) microbiology, biotechnology and other clinical laboratory waste	Incineration or plasma pyrolysis or deep burial*; expired cytotoxic drugs and cytotoxic-contaminated items are returned to the manufacturer or supplier, or sent to a common facility or a hazardous-waste treatment, storage and disposal facility, for incineration at >1200 °C, or encapsulation, or plasma pyrolysis at >1200 °C ; chemical liquid waste pre-treated and routed to the effluent treatment system; laboratory waste pre-sterilised on site as per National AIDS Control Organisation (NACO) or World Health Organization (WHO) guidance
Red — red non-chlorinated plastic bags or containers	Contaminated recyclable waste — tubing, bottles, intravenous tubes and sets, catheters, urine bags, syringes without needles (fixed-needle syringes go to the white sharps container), vacutainers with their needles cut , and gloves	Autoclaving or microwaving or hydroclaving, followed by shredding or mutilation; sent to registered recyclers or for energy recovery; plastic waste not to be sent to landfill
White (translucent) — puncture-proof, leak-proof, tamper-proof container	Waste sharps including metals — needles, syringes with fixed needles , needles from needle-tip cutter or burner, scalpels, blades, other contaminated sharps	Autoclaving or dry-heat sterilisation, then shredding or mutilation or encapsulation in metal or cement concrete; final disposal to authorised iron foundries, sanitary landfill or a designated concrete sharps pit
Blue — cardboard boxes with blue marking	(a) glassware — broken, discarded and contaminated glass including medicine vials and ampoules except those contaminated with cytotoxic waste; (b) metallic body implants	Disinfection (detergent wash and sodium hypochlorite) or autoclaving or microwaving or hydroclaving, then sent for recycling

* Deep burial only in rural or remote areas with no access to a common facility, with prior approval of the prescribed authority (BMW Rules 2016, Schedule I, pp 12-15).

What changed	1998 Rules	2016 Rules (with 2018–19 amendments)
Segregation	10 categories	4 colour-coded categories
Coverage	Hospitals, nursing homes, clinics, laboratories, animal houses	Additionally vaccination, blood-donation, medical and surgical camps, AYUSH (Ayurveda, Yoga and Naturopathy, Unani, Siddha and Homoeopathy) hospitals, first-aid rooms of schools, forensic and research laboratories
Waste expressly excluded	Application clause worded positively, with no exclusion list	Rule 2(2) excludes radioactive waste (Atomic Energy Act 1962), hazardous chemicals (Hazardous Chemicals Rules 1989), municipal solid waste (2000 Rules), lead-acid batteries (Batteries Rules 2001), hazardous waste (2008 Rules), e-waste (2011 Rules) and genetically engineered micro-organisms and cells (1989 Rules)

What changed	1998 Rules	2016 Rules (with 2018–19 amendments)
Chemical disinfection	At least 1% hypochlorite	At least 10% sodium hypochlorite with 30% residual chlorine for 20 minutes, or an agent giving log ₁₀ 4 reduction
Pre-treatment	Not specified	Rule 4(c) — laboratory waste, microbiological waste, blood samples and blood bags pre-treated by disinfection or sterilisation on site in the manner prescribed by WHO or NACO guidelines before hand-over
Plastics	—	Rule 4(8) — chlorinated plastic bags phased out within 2 years of 28 March 2016; revised by the BMW Amendment Rules 2018 — chlorinated plastic bags EXCEPT blood bags, and gloves, to be phased out by 27 March 2019 (Sunder Lal, Ch 14, p 679)
Tracking	—	Operator to establish bar coding and global positioning system within one year
Common facility	—	On-site treatment and disposal barred where a common bio-medical waste treatment facility is available within 75 km; an in-house incinerator is otherwise permitted only where no common facility is nearby, with prior authorisation from the State Pollution Control Board
Authorisation	Form 1, valid 3 years	Form II; one-time for non-bedded occupiers, synchronised with consents for bedded facilities
Annual report	Form 2 by 31 January	Form IV by 30 June for the preceding January–December
Accident report	Form 3	Form I within 24 hours for a major accident
Oversight	Prescribed authority	Plus BMW management committee meeting 6-monthly (facilities <30 beds designate a qualified person); records kept 5 years; appeal in Form V within 30 days

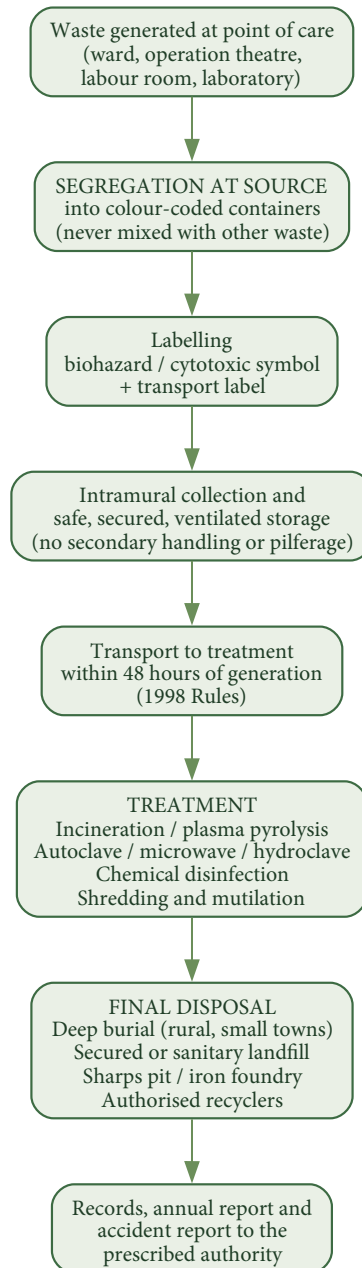
(BMW Rules 2016, pp 1-16, 25-34; Sunder Lal, Ch 14, pp 675-679.)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

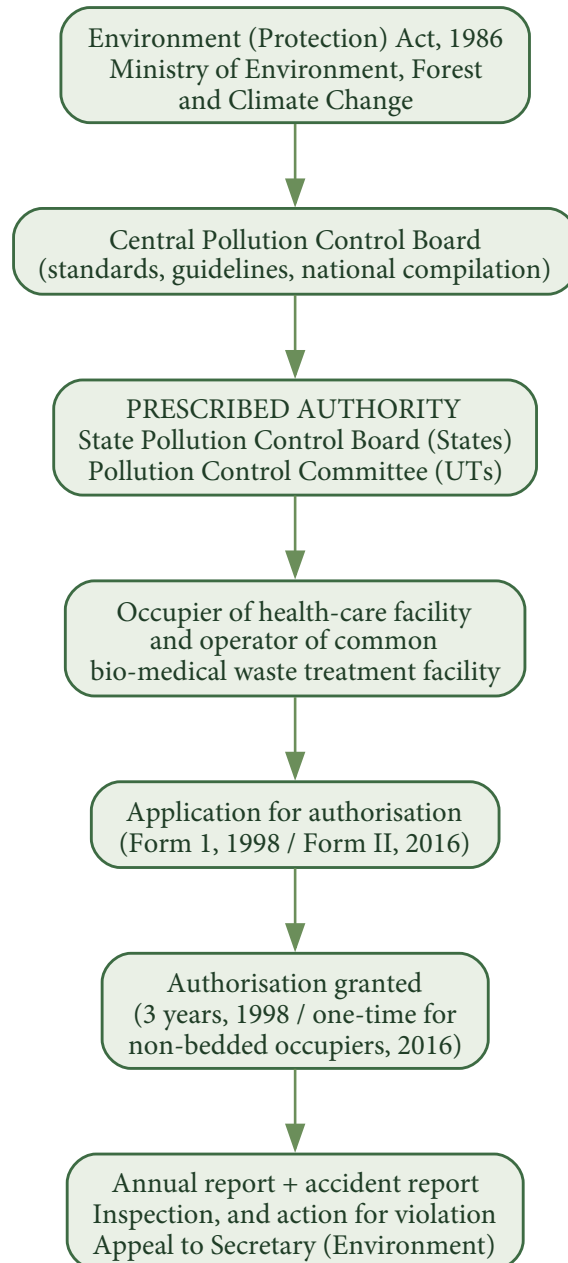
- The four colour-coded bin set (2016) — draw four foot-operated, lidded bins in a row and label them exactly: YELLOW — non-chlorinated plastic bag; RED — non-chlorinated plastic bag; WHITE (translucent) — puncture-proof, leak-proof, tamper-proof container; BLUE — cardboard box with blue marking. On each label draw the biohazard symbol, and on the yellow cytotoxic container the cytotoxic hazard symbol, marked "prominently visible and non-washable".



Fig 17.4 — Biohazard Symbol (AFMC Red Book of Community Medicine 2024, p.460)



Waste stream mandated by the Bio-Medical Waste Rules, from generation to final disposal and reporting



Regulatory and authorisation chain for bio-medical waste management in India

High-yield exam pointers

- ▶ **Statute chain:** Environment (Protection) Act 1986 → BMW (Management and Handling) Rules, S.O. 630(E) of 20 July 1998, in force 28 July 1998 (amended 2000, 2003) → BMW Management Rules, G.S.R. 343(E), 28 March 2016 (amended 2018, 2019); prescribed authority = State Pollution Control Board / Pollution Control Committee.
- ▶ **Mnemonic for the 2016 bins** — "Yellow burns, Red recycles, White is sharp, Blue is glass."
- ▶ **Numbers to quote:** combustion efficiency $\geq 99.00\%$ for incinerators versus $\geq 99.99\%$ for plasma pyrolysis; transport within 48 h of generation; authorisation 3 years (1998); annual report 31 January (1998) versus 30 June (2016); deep burial only where population < 5 lakh or rural; common facility within 75 km (2016).

- **Waste profile (WHO planning estimate):** 80% general, 15% pathological and infectious, 1% sharps, 3% chemical and pharmacological, <1% radioactive or cytotoxic (Park 28e, Ch 16, p 943).
- **Average composition of Indian hospital waste** (National Environmental Engineering Research Institute, 1997; 10 large hospitals in Mumbai, Kolkata, Delhi and Nagpur, 1993–96): general waste 53.5%, paper 15%, rags 15%, plastics 10%, glass 4%, infectious waste 1.5%, metal sharps 1% (Park 28e, Ch 16, pp 943–944).
- **Occupational protection under the 2016 Rules:** training at induction and **annually**, immunisation against **hepatitis B and tetanus**, health check-up at induction and yearly, and personal protective equipment for every waste handler.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only — the 1998 Rules are of historical and comparative interest since they were superseded by the Bio-Medical Waste Management Rules 2016 (in force 28 March 2016) and further amended in 2018 and 2019, so an examiner expects the 1998 provisions to be stated first and the current 2016 four-colour scheme with the 2018 chlorinated-plastic amendment given alongside; the list of expressly excluded wastes above is the exclusion clause of Rule 2(2) of the 2016 Rules.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, WHO health-care waste fact sheet 2024, WHO global analysis of health-care waste 2022, Central Pollution Control Board national biomedical-waste data

Programme and policy updates (India)

- **A COVID-19 layer was bolted on to the 2016 Rules and is now the standard "what came after 2016" answer** — the Central Pollution Control Board issued dedicated COVID-19 biomedical-waste guidelines on **18 March 2020** and revised them three times within four months (**19 April, 10 June and 17 July 2020**): double-layered bags from isolation and quarantine areas, dedicated bins labelled "COVID-19 waste", disinfection of bins and trolleys with **1 per cent sodium hypochlorite**, a **72-hour hold** on masks and gloves discarded by the general public before they are shredded and disposed of as dry general waste, and — once yellow-category volumes outran capacity — permission to burn biomedical waste in **hazardous-waste incinerators**. Biomedical-waste handling was declared an **essential service**, and the **COVID19BMW** mobile application made every facility and operator report waste generated and treated on a daily basis [PMID 34622401].
- **Household biomedical waste is an enforced stream, not a footnote** — Schedule I Part 2 of the 2016 Rules requires households to hand over segregated biomedical waste to municipal collectors in separate bags and obliges **urban local bodies to tie up with a common bio-medical waste treatment facility**; yet a community-based survey of 240 urban households in Nagpur (2023–24) found **no household segregating biomedical waste at all**, although 76.7 per cent segregated wet from dry domestic waste. Sanitary pads (84.2 per cent), expired medicines (81.7 per cent), dressings (57.1 per cent) and used blades (56.3 per cent) were the commonest items, and 98.3 per cent went straight into the municipal bin [PMID 42125580].

- **The Kayakalp checklist has become the working audit instrument for the Rules** — a mixed-methods clinical audit of a 2,248-bed teaching hospital in Puducherry scored **86 per cent overall compliance** across the eight Kayakalp waste domains, with **sharps management weakest at 74 per cent** and the timing of the outsourced collection service the dominant bottleneck named by staff [PMID 41836713].

Current national data

- **National quantum against installed capacity** — India generated about **619 tonnes per day** of biomedical waste (Central Pollution Control Board, 2019) against a total common-facility capacity of roughly **840 tonnes per day**: **202 common facilities** operating with 35 more under construction, treating 489 t/day, captive systems 55 t/day, and **74 t/day still going to deep burial**. Of 322,425 health-care facilities, only 153,885 held authorisation and more than **29,000 violated the Rules in 2019**; incinerator utilisation had already crossed 70 per cent in Bihar, Jammu and Kashmir, Karnataka, Kerala, Odisha, Rajasthan, Tamil Nadu and Uttar Pradesh, while Sikkim, Nagaland, Mizoram, Goa, Arunachal Pradesh, Andaman and Nicobar and Dadra and Nagar Haveli had **no common facility at all** [PMID 34622401]. The longer trend runs from **409 to 558 Mg per day between 2008 and 2017**, with Karnataka, Maharashtra, Tamil Nadu, Uttar Pradesh, Kerala, Bihar, West Bengal and Gujarat the largest generators and the quantity actually *treated* rising once the 2016 Rules came into force [PMID 38625143].
- **Nationwide compliance gap assessment (267 facilities, July–September 2023)** — the Consortium of Accredited Healthcare Organizations survey found **barcoding fully implemented in only 58 per cent**, **49 per cent still cutting needles before disposal**, **50 per cent using blue bins with blue covers** instead of the blue-marked cardboard box the Rules specify, laboratory waste autoclaved on site in 79 per cent, dedicated temporary storage in 92 per cent, and **mercury-filled devices still in use in 28 per cent** of facilities [PMID 40744379].

Elimination, surveillance and disease status

- **Occupational infection burden in waste handlers quantified** — a PRISMA meta-analysis of 14 cross-sectional studies (2,713 handlers, five from India) gave a pooled prevalence of **needlestick and sharps injury 26.4 per cent (95% CI 15.8–40.7)**, **hepatitis B 5.8 per cent (95% CI 2–15.4)** and **hepatitis C 2.4 per cent (95% CI 0.9–6.3)**; only **15.3 per cent were immunised against hepatitis B** and **57.4 per cent had received any training**, with boots used by 19 per cent and gowns by 55.4 per cent — a direct indictment of the 2016 Rules' mandatory immunisation, training and personal-protective-equipment clauses [PMID 42022636].
- **Doctors are the training and reporting blind spot** — across four tertiary teaching hospitals in Chennai (356 workers), **72.1 per cent** had correct segregation knowledge (nurses 76.3 per cent, doctors 63 per cent), accuracy collapsing to **34.5 per cent among doctors for cytotoxic and pharmaceutical waste**; **41 sharps injuries** occurred in 12 months of which 48.8 per cent were in doctors, yet only **31 per cent of doctors had attended training** against 74.2 per cent of nurses and 88.3 per cent of housekeeping staff ($\chi^2 = 71.93$, $p < 0.001$), and injury reporting was 60 per cent in doctors versus 81.3 per cent in nurses [PMID 40476130].

Guideline and legislative updates

- **Dioxins, not particulates, are now the governing emission parameter** — a systematic review of three decades of data found biomedical-waste incinerators contribute the largest single share of

the **national dioxin inventories** of most developing countries, and that a Stockholm Convention-aligned stack limit of **0.1 ng TEQ/Nm³** for dioxins and furans is now widely endorsed — a parameter entirely absent from the 1998 Schedule V, which capped only particulate matter, oxides of nitrogen and hydrogen chloride [PMID 38176262]. Correspondingly, the **secondary-chamber gas residence time was raised to at least 2 seconds** under the 2016 Rules, but most common facilities built before 2016 were designed for only 1 second, so an existing incinerator meeting the 1998 standard does **not** automatically meet the current one [PMID 34622401].

Global and WHO developments

- The WHO baseline figures have moved — the WHO health-care waste fact sheet (updated 24 October 2024) now puts the split at **85 per cent general and 15 per cent hazardous waste**, against the older teaching figure of 80 per cent general; only **61 per cent of hospitals worldwide had basic waste-management services in 2021**, falling to **25 per cent of facilities in fragile settings in 2023**. Unsafe injections in low- and middle-income countries were estimated to cause **33,800 HIV, 1.7 million hepatitis B and 315,000 hepatitis C infections** in a single year, with per-exposure transmission risk after a needlestick from an infected source of **30 per cent for hepatitis B, 1.8 per cent for hepatitis C and 0.3 per cent for HIV** [WHO 2024-Oct].
- WHO's global analysis of health-care waste (February 2022) concluded that **one in three health-care facilities worldwide does not safely manage health-care waste**, and set out a **waste-hierarchy** response — reduce, reuse and non-burn treatment ahead of incineration — for the personal protective equipment, testing and vaccination waste streams, with about a third of shipped protective equipment unable to be safely stored for want of biohazard bags [WHO 2022-Feb].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 16, Hospital Waste Management, pp 943-948); Suryakantha — Community Medicine with Recent Advances (Ch 13, Management of Hospital Waste, pp 111-125); Bio-Medical Waste Management Rules, 2016, Ministry of Environment, Forest and Climate Change, Government of India (full gazette text, pp 1-36, including the Schedule II treatment standards and the 1998 notification particulars in the preamble); Jugal Kishore — National Health Programmes of India (Ch 3, biomedical waste, pp 946-951); Sunder Lal — Textbook of Community Medicine (Ch 14, Hospital Waste Management, pp 675-679, for the BMW Amendment Rules 2018); figure of the biohazard symbol from AFMC Red Book of Community Medicine 2024 (p 460). Coverage note: the corpus holds the full 2016 gazette but only textbook summaries of the 1998 gazette, so the 1998 Schedule V values above are those the 2016 Schedule II carries forward unchanged, cross-checked between Jugal Kishore and Suryakantha.

Recent-advances sources: 34622401 — Biomedical waste generation and management during COVID-19 pandemic in India: challenges and possible management strategies, *Environmental Science and Pollution Research International* 2021, doi:[10.1007/s11356-021-16736-8](https://doi.org/10.1007/s11356-021-16736-8); 42125580 — Pattern and Disposal of Biomedical Waste Generated by Households in Urban Nagpur: A Community-Based Assessment, *Indian Journal of Community Medicine* 2026, doi:[10.4103/ijcm.ijcm.818.24](https://doi.org/10.4103/ijcm.ijcm.818.24); 41836713 — Clinical Audit of Biomedical Waste Management, Awareness, and Challenges among Health Care Workers in Segregation of Biomedical Waste in a Tertiary Care Hospital, Puducherry — A Mixed-Method Study, *Indian Journal of Community Medicine* 2025, doi:[10.4103/ijcm.ijcm.622.24](https://doi.org/10.4103/ijcm.ijcm.622.24); 38625143 — Geographical and temporal analysis of bio-medical waste management in India, *GeoJournal* 2023, doi:[10.1007/s10708-023-10854-1](https://doi.org/10.1007/s10708-023-10854-1) (abstract-only); 40744379 — Gap assessment in implementation of Biomedical Waste Management in Health Care Facilities: Nationwide survey in India, *Indian Journal of Medical Microbiology* 2025, doi:[10.1016/j.ijmmb.2025.100947](https://doi.org/10.1016/j.ijmmb.2025.100947) (abstract-only); 42022636 — Occupational Hazard Among Biomedical Waste Handlers in Low-Resource Settings: A Systematic Review and Meta-Analysis, *Health Science Reports* 2026, doi:[10.1002/hsr2.72083](https://doi.org/10.1002/hsr2.72083); 40476130 — Assessment of Biomedical Waste Management Practices and Risk Perception Among Healthcare Workers in South India, *Cureus* 2025, doi:[10.7759/cureus.83584](https://doi.org/10.7759/cureus.83584); 38176262 — Dioxins emissions from bio-medical waste incineration: a systematic review on emission factors, inventories, trends and health risk studies, *Journal of Hazardous Materials* 2023, doi:[10.1016/j.jhazmat.2023.133384](https://doi.org/10.1016/j.jhazmat.2023.133384) (abstract-only). Non-journal sources: World Health Organization, Health-care waste fact sheet, updated 24 October 2024; World Health Organization, Global analysis of health care waste in the context of COVID-19: status, impacts and recommendations, February 2022.

♦ ♦ ♦

Climate change and health

Asked in: Paper I 15-Nov-2022 — RGUHS MD Community Medicine

Definition — Carbon dioxide, methane, nitrous oxide and chlorofluorocarbons let sunlight reach the earth but block the re-radiated heat, exactly as the glass of a greenhouse does; this **greenhouse effect** holds the earth's surface temperature at about 15°C and makes life possible (Suryakantha, Ch 48, p 909). Accumulation of these **greenhouse gases** from human activity traps more heat and raises that temperature — **global warming** — and the resulting sustained shift in temperature, rainfall, sea level and extreme-weather patterns is **climate change**. Its public health importance is that it acts on the three most fundamental determinants of health: air, food and water.

Drivers and projected magnitude

Driver	Mechanism
Combustion of fossil fuels (coal, petroleum) in industry, power generation and vehicles	The major single component of carbon dioxide emission; rising continuously since the industrial revolution (Park 28e, Ch 15, p 903)
Industrialisation and urbanisation	Emission of carbon dioxide, nitrous oxide and methane, which have atmospheric lifetimes of years and so distribute globally (Park 28e, Ch 15, p 888)
Deforestation	Removes carbon sinks; greenery is the most effective carbon sink (Suryakantha, Ch 48, p 911)
Chlorofluorocarbons	Both a greenhouse gas (Suryakantha, Ch 48, p 909) and the agent of stratospheric ozone-shield depletion — depletion of the ozone layer by chlorofluorocarbon use and by the exhaust of supersonic aircraft leads to skin cancer (Suryakantha, Ch 21, p 559)
Chemical fertilisers, nuclear explosions, population growth	Add nitrous oxide and other greenhouse gases; amplify demand for energy and land (Suryakantha, Ch 48, p 909)

Projected physical change	Magnitude
Mean global surface temperature	Present value 15°C; risen by ½°C over the last 100 years. At this rate a further 2.5°C by 2050 and another 2.5°C by the end of the century; a 2°C rise is the threshold beyond which warming becomes irreversible and catastrophic (Suryakantha, Ch 48, pp 909–910). Park quotes a rise of about 3°C by 2030 (Park 28e, Ch 15, p 903)
Sea level	Rise of 0.1–0.3 metres by 2050 (Park 28e, Ch 15, p 903)
Extreme events	More frequent and more intense cyclones, heat waves, droughts, rainstorms and tidal surges (Park 28e, Ch 15, p 903; Suryakantha, Ch 48, p 911)
Rainfall distribution	Altered jet streams, prevailing winds and ocean currents make some regions wetter and others drier; temperate warming lowers soil moisture and impairs grain production (Park 28e, Ch 15, p 903)
Urban temperature	The heat island phenomenon makes cities warmer than surrounding rural areas, giving longer and more severe heat waves (Park 28e, Ch 15, p 903)

Projected physical change	Magnitude
Water security	The Intergovernmental Panel on Climate Change (IPCC) projects 150–250 million people in Africa facing water shortage, and residents of Asian megacities at great risk of river and coastal flooding (Suryakantha, Ch 48, p 911)

Health effects

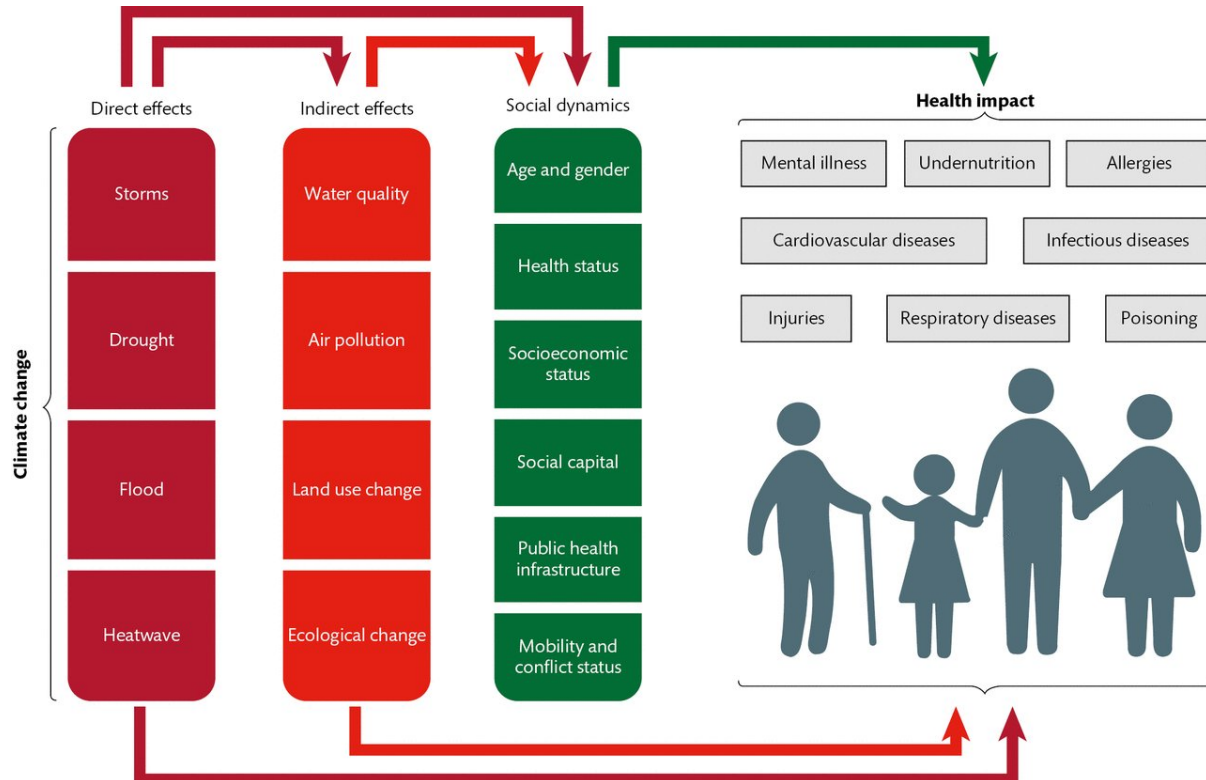


Fig 1.4 — 5 The direct and indirect effects of climate change on health and well (Oxford public health 7th ed, p.243)

Pathway	Mechanism	Health outcome
Direct — heat	A J-shaped relation between temperature and mortality, the optimal (lowest-mortality) temperature differing from one setting to another — near 0°C in far-northern Haerbin, higher nearer the equator, so populations adapt to their ecological context (Oxford Public Health 7e, Ch 14, p 551). Variability, duration, frequency and timing of heat matter, not only the maximum or average; high air pollution amplifies the damage (Ch 14, pp 552–553)	Heat stroke (failure of heat regulation, temperature to 110°F, sweating classically absent, case fatality 40% or more), heat hyperpyrexia (above 106°F), heat exhaustion, heat cramps, heat syncope (Park 28e, Ch 15, pp 902–903)
Direct — extreme weather	Storms, floods, cyclones, drought	Drowning, injury, death; mental health disorders linked to the trauma of the extreme weather events (Oxford Public Health 7e, Ch 4, p 244)

Pathway	Mechanism	Health outcome
Direct — sea level rise	Coastal inundation, storm surge, coastal destruction	Small island states and low-lying coastal countries are home to an estimated 65 million people and are already experiencing the adverse effects of climate change; one-third of that population lives on land less than 5 metres above sea level , so the risks are existential — some islands may eventually disappear, and displacement is the end result (Oxford Public Health 7e, Ch 107, pp 4054–4055)
Ecosystem — vector-borne	Warmth accelerates development of <i>Aedes aegypti</i> from the immature stage and the shift from dry to rainy season triggers dengue outbreaks (Ch 14, p 554); climate change alters the range and reproductive rate of malaria-carrying mosquitoes even at higher altitudes previously protected by cooler temperature (Ch 4, p 244), lengthening transmission seasons and altering the geographic range of vectors (Ch 107, p 4055)	Malaria, dengue, yellow fever, chikungunya and Zika (Oxford Public Health 7e, Ch 14, p 554; Ch 4, p 244; Ch 107, p 4055); Japanese encephalitis and dengue haemorrhagic fever (Suryakantha, Fig 48.1, p 910)
Ecosystem — water-borne	Warmer temperature promotes cholera bacterial growth; drought and extreme weather limit access to safe drinking water	Cholera, diarrhoea, toxic algal blooms (Oxford Public Health 7e, Ch 4, p 244; Suryakantha, Fig 48.1, p 910)
Ecosystem — food	Crop failure and falling staple production in the poorest regions; reduced world food production	Undernutrition, wasting and stunting; undernutrition already causes 3.1 million deaths a year (Oxford Public Health 7e, Ch 107, pp 4054–4055)
Ecosystem — air	Ambient air pollution and photochemical smog	Respiratory disease, asthma, allergy (Suryakantha, Ch 6, p 68)
Ecosystem — ozone-shield depletion	Thinning stratospheric ozone lets ultraviolet radiation through	Melanotic and non-melanotic skin cancer , cataract , impaired immunity predisposing to infection (Suryakantha, Ch 6, p 68)
Ecosystem — acid rain	Sulphur dioxide, carbon dioxide and nitrogen dioxide form sulphuric, carbonic and nitric acid in atmospheric water vapour	Soil and water acidification, deforestation, desertification, loss of aquatic life, destruction of food crops (Suryakantha, Ch 6, p 68)
Social	Displacement, migration, conflict; overcrowding and poor sanitation in relief settlements	Outbreaks of infectious disease, malnutrition, mental illness (Suryakantha, Fig 48.1, p 910)

Quantified burden

Estimate	Figure
Burden already incurred (World Health Organization review)	From the mid-1970s , climate change may already have contributed about 150,000 deaths and 5 million disability-adjusted life years (DALYs) per year , largely through malnutrition, diarrhoea and malaria (Oxford Public Health 7e, Ch 1, pp 164–165)

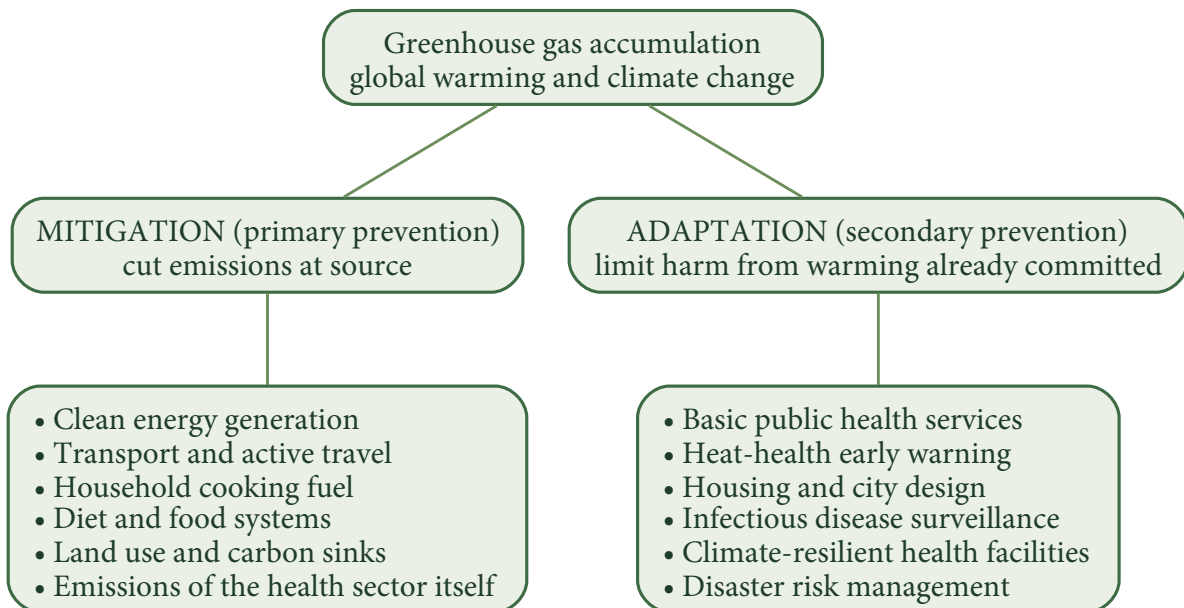
Determinants of vulnerability

Determinant	Why it raises risk	Source
Baseline health status	The World Bank's 2010 report concluded it is "the single most important determinant of both future losses, and the cost of adaptation" — so adaptation must improve the social determinants of health, including social safety nets, not merely the weather response	Oxford Public Health 7e, Ch 14, p 570
Age and sex	Older people (70 years and over) and children are generally at greatest risk. There is no consistent pattern with sex — men were at greater risk in São Paulo and Santiago, women in Brisbane	Ch 14, pp 552–553
Low socioeconomic status	In many countries, though not universally , associated with increased risk of heat- and cold-related poor health. Heat mortality varies more by socioeconomic status in the United States than in England , perhaps because air conditioning is distributed differentially in the USA whereas few English houses have artificial cooling	Ch 14, p 553
Chronic disease and obesity	Impair thermoregulation; programmes promoting physical activity and preventing unhealthy weight gain reduce heat vulnerability	Ch 14, pp 565–566
Housing and urban form	Poor housing, absence of green space and flood-prone land concentrate exposure — most heat-wave deaths occur <i>inside</i> buildings	Ch 14, p 566
Geography and occupation	Vulnerability is expressed spatially, reflecting chronic disease prevalence, housing design, poverty and district-level urban form and services; coastal and small-island populations and outdoor manual workers face the highest exposure	Ch 14, p 553; Ch 107, p 4055
Health system capacity	Preparedness of the healthcare sector decides whether a warning translates into protection	Ch 14, p 566

Climate change is thus an equity issue: those at greatest risk already carry a heavy load of disadvantage and have contributed little to the carbon dioxide now in the atmosphere (Ch 14, p 583).

Prevention and control

Two arms — **mitigation** = **primary prevention** (cutting emissions) and **adaptation** = **secondary prevention** (limiting harm from warming already committed). Both are needed: a further **0.7°C over the next 30 years** is committed by the gases already in the atmosphere (Oxford Public Health 7e, Ch 14, pp 562–564).



Two-pronged public health response to climate change — mitigation as primary prevention, adaptation as secondary prevention

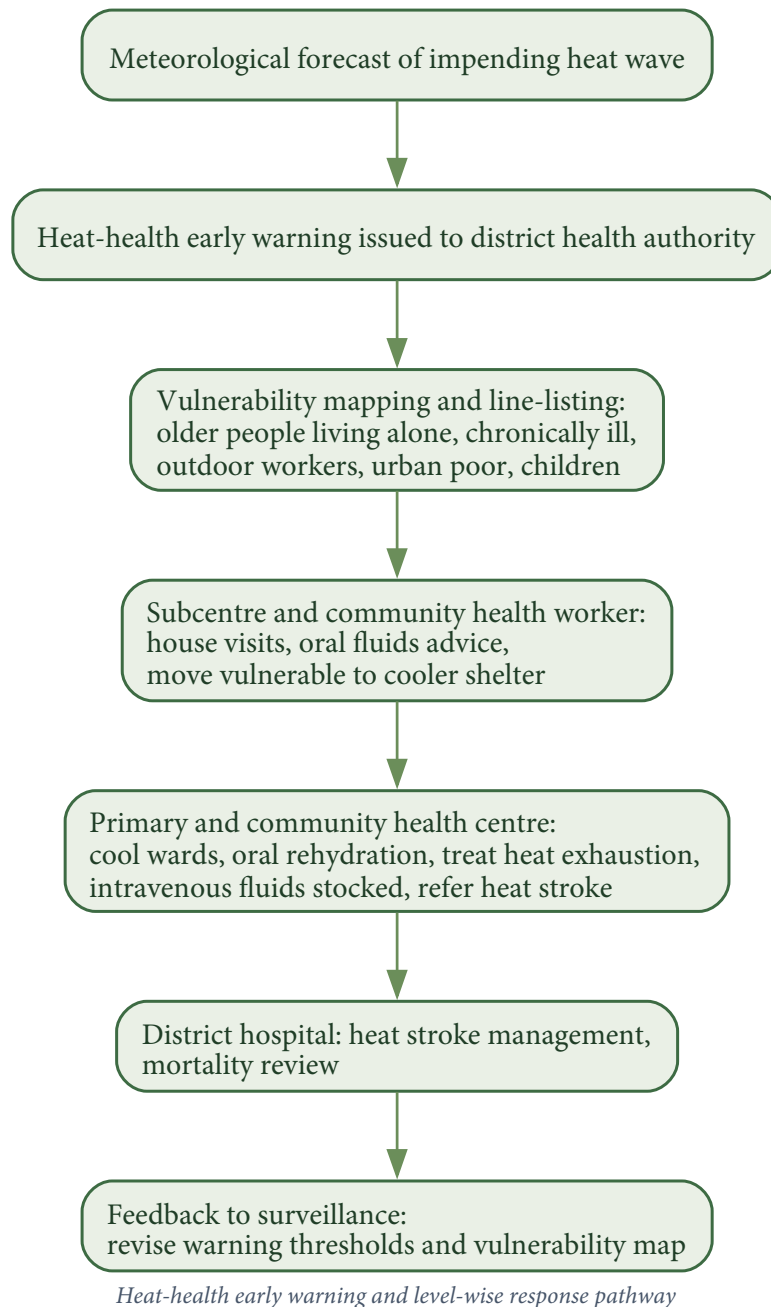
Mitigation — grounded specifics

- ▶ Emissions must fall to **about zero by 2050** for a reasonable chance of holding warming to 2°C by 2100 (Ch 14, p 579).
- ▶ The **United Nations Environment Programme (UNEP, 2011)** assessed measures proven to cut **black carbon and methane** — diesel particle filters, clean-burning biomass cook stoves, bans on open-field burning of agricultural waste. Full implementation by 2030 would save **0.5°C of warming in 2050**, prevent about **50 million tonnes a year of lost crop yields** and avoid about **2.5 million premature deaths** (Ch 14, p 576).
- ▶ Building on that, **WHO (2015)** named **four priority strategies** against short-lived climate pollutants, chosen for health co-benefits: **active travel** with rapid mass public transport; **healthy plant-based diets**; **low-emission cook stoves**; **stricter vehicle emission and efficiency standards**, including freight (Ch 14, p 576).
- ▶ Delhi modelling: low-carbon vehicles alone gave health co-benefits about **ten times smaller** than active-transport or combined strategies (Ch 14, p 579).
- ▶ Community measures: greenery, solar and wind power, community biogas, rainwater harvesting, wetland conservation, plastic-bag ban, sewage-free rivers, walking and cycling (Suryakantha, Ch 48, p 911).
- ▶ The health sector must cut **its own** emissions — hospitals off coal power, clinics held in communities (Ch 19, p 759).

Adaptation — grounded specifics

- ▶ The **most effective immediate adaptation** is **basic public health**: vaccination, essential child health services, safe water and sanitation, disaster risk management, poverty reduction; resilience must be operationalised into **each of WHO's six health system building blocks** (Ch 19, p 759).

- ▶ **Heat:** early warning systems, cooling at-risk environments such as residential care facilities for the elderly, **identifying the isolated and most vulnerable citizens**, and healthcare-sector preparedness — the package France instituted after the 2003 heat wave (Ch 14, pp 566–567).
- ▶ **Housing and city design:** most heat-wave deaths occur indoors, so energy-independent "passive survivability" — window design, insulation, ventilation, shading — outranks air conditioning, which fails in demand-driven power cuts and blunts acclimatisation; cool building and road materials, urban green space and reflective paint almost **halved** modelled heat-related emergency calls (Ch 14, p 566).
- ▶ **Infectious disease:** enhanced surveillance, epidemic early warning, mosquito surveillance and standing-water management (Ch 14, p 572).
- ▶ **Climate-resilient facilities:** assess existing health buildings, water and energy supplies for vulnerability to extreme weather and sea level rise (Ch 14, p 572).
- ▶ **Intersectoral coordination** across government, non-governmental agencies, professional groups and communities — "Think globally, act locally" (Suryakantha, Ch 48, p 911).



Indian response

- ▶ The over-arching policy is the **National Action Plan on Climate Change (NAPCC)**, 2008, of the **Ministry of Environment, Forest and Climate Change** (Solid Waste Management manual, p 597).
- ▶ Its **National Mission on Sustainable Habitat (NMSH)**, approved in 2008 under NAPCC, requires recycling strategies to avoid greenhouse gas emissions and links waste management to city sanitation planning (Solid Waste Management manual, p 55).
- ▶ The health-sector arm is the **National Programme on Climate Change and Human Health (NPCCHH)**, run by the **National Centre for Disease Control (NCDC)** under the **Ministry of Health and Family Welfare**, alongside the **National One Health Programme for Prevention and Control of Zoonoses (NOHP-PCZ)** (One Health Booklet, AIIMS, pp 5–7).

- Field data on climate-sensitive illness flow through the **Integrated Disease Surveillance Programme / Integrated Health Information Platform (IDSP/IHIP)**, supported by a National Centre on One Health under NCDC, **15 Regional Coordinators and 33 sentinel sites**, and a laboratory network under PM-ABHIM (pp 3–7).

International response

Instrument	Year	Content
UN Framework Convention on Climate Change (UNFCCC)	—	The framework whose principles the Kyoto Protocol operationalised (Suryakantha, Ch 48, p 911)
Kyoto Protocol	Adopted at Kyoto, Japan, 1997; in force 2005	Industrialised countries to cut greenhouse gas emission to at least 5.2% below 1990 levels by 2010 , with penalty for failure; supported by 141 nations, signed by 34 industrialised countries, boycotted by the USA ; permits emission trading through the Clean Development Mechanism (CDM) . Most nations failed their targets (Suryakantha, Ch 48, p 911)
World Health Day theme	2008	<i>"Protecting Health from Climate Change"</i> — WHO's recognition of climate change as a growing threat to public health (Suryakantha, Ch 48, p 911)
Copenhagen Summit	7–18 Dec 2009	Rich nations to significantly reduce greenhouse gas emissions and provide finance and technology to poor nations (Suryakantha, Ch 48, p 911)
Sendai Framework for Disaster Risk Reduction	Adopted 18 Mar 2015, for 2015–2030	Third UN World Conference, Sendai, Japan; successor to the Hyogo Framework for Action 2005–2015; non-binding, complied with voluntarily; India a signatory (Jugal Kishore, pp 670–671)
Paris Agreement (UN COP21)	Adopted Dec 2015	Holds average global warming below 2°C , aiming for no more than 1.5°C to protect low-lying island states. Not legally binding, but the first truly global climate instrument and the first to recognise both the need to protect health and the health co-benefits of well-designed climate action; delivered through nationally determined contributions , which are not yet enough on their own to hold warming to 2°C (Oxford Public Health 7e, Ch 14, pp 564, 582)
Sustainable Development Goals	Sept 2015	Disaster risk reduction is the common theme running through all three 2015 agreements; loss and damage from climate change is to be averted, minimised and addressed (Jugal Kishore, pp 670–671)

Diagrams

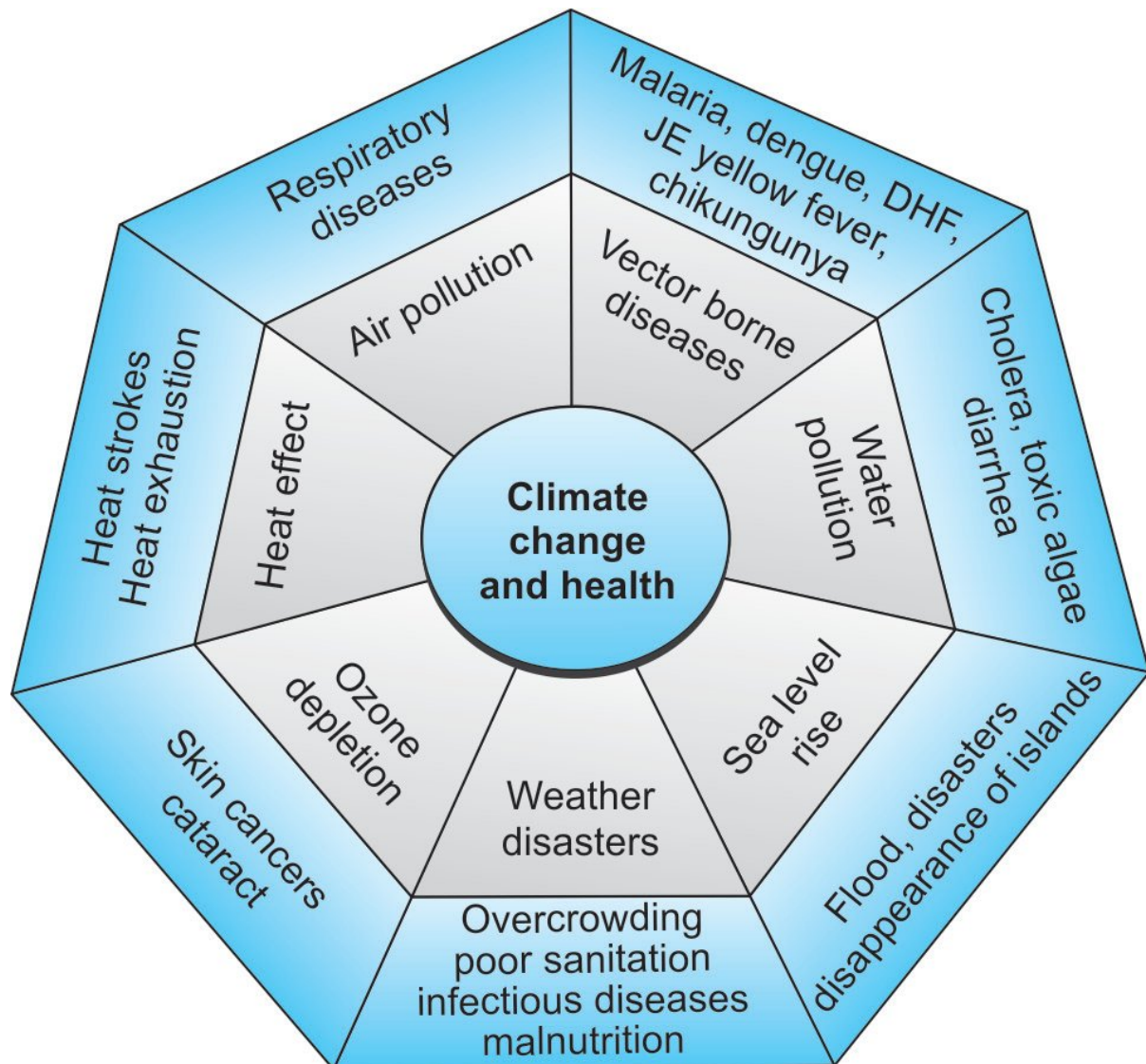


Fig 48.1 — Possible health impacts due to climate change (Suryakantha Environment Best, p.910)

High-yield exam pointers

- ▶ **Greenhouse gases:** carbon dioxide, methane, nitrous oxide, chlorofluorocarbons — surface temperature about 15°C, without which life would not exist.
- ▶ **Temperature ladder (Suryakantha):** +½°C in the last 100 years → +2.5°C by 2050 → another +2.5°C by 2100; 2°C = irreversibility threshold; net-zero emissions by 2050. Park: ~3°C by 2030, sea level 0.1–0.3 m by 2050.
- ▶ **The two burden numbers:** ~150,000 deaths and 5 million DALYs a year already (WHO, since the mid-1970s); ~250,000 extra deaths a year, 2030–2050 = 38,000 heat + 48,000 diarrhoea + 60,000 malaria + 95,000 childhood undernutrition, costing US\$2–4 billion a year by 2030.
- ▶ **Attribute correctly:** 0.5°C saved by 2050 + 2.5 million deaths averted + 50 million tonnes a year of crop loss prevented = UNEP's black-carbon-and-methane package. WHO's four strategies (active travel, plant-based diets, low-emission cook stoves, vehicle standards) are a separate, unquantified list.

- **Heat vulnerability:** greatest at 70 years and over and in children; no consistent pattern with sex; low socioeconomic status raises risk in many but not all countries.
- **Kyoto:** 1997 Japan, in force 2005, 5.2% below 1990 levels by 2010, Clean Development Mechanism, USA boycotted. **Paris 2015:** below 2°C, aiming 1.5°C, nationally determined contributions.
- **India:** NAPCC 2008 with the National Mission on Sustainable Habitat; health arm NPCCHH under NCDC, Ministry of Health and Family Welfare, with NOHP-PCZ; data via IDSP/IHIP.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: Suryakantha prints the Kyoto commitment periods as 2005–2012 and 2012–2016, but the correct dates are a first commitment period of 2008–2012 and a Doha Amendment second period of 2013–2020. The corpus carries the National Programme on Climate Change and Human Health (NPCCHH) only at naming level, so its objectives and state/district levels are not given above; add them from the current Ministry of Health and Family Welfare document if the stem asks for programme structure. In the heat-health flowchart, the subcentre–primary health centre–community health centre–district hospital ladder is the standard Indian delivery structure onto which the retrieved early-warning content is mapped.

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, MoHFW and NCDC national and state climate-health action plans, NFHS-5, NCVBDC vector-borne disease surveillance, WHO Fourteenth General Programme of Work, WHO operational framework for climate-resilient health systems, Lancet Countdown 2024, COP28 Declaration on Climate and Health, Belém Health Action Plan (COP30)

Programme and policy updates (India)

- India's climate–health architecture now extends well beyond NAPCC 2008. The **National Action Plan for Climate Change and Human Health (NAPCCHH)**, released 2018 under the Ministry of Health and Family Welfare, is now flanked by a **National Action Plan on Heat-Related Illness (2021)**, a **National Action Plan for Diseases due to Air Pollution (2021)**, a **National Health Adaptation Plan for Climate Change-Related Disaster (2019)** and a health adaptation plan for cardiopulmonary disease; states operationalise these as **State Action Plans on Climate Change and Human Health (SAPCCHH)**, tracked through a directory maintained by the National Centre for Disease Control. A framework synthesis of 10 national policies and 29 state plans found the Indian response still dominated by vector-borne disease, with cardiovascular and other non-communicable disease poorly contextualised and largely missing from climate-health surveillance; the **Ahmedabad Heat Action Plan** remains the template other states copied, but city- and region-specific temperature thresholds are still not defined [PMID 39185584]
- About **80% of the Indian population** lives in areas highly vulnerable to cyclone, flood or heatwave, and the operational recommendation is to mainstream district-level climate-vulnerability differentials into **National Health Mission District Action Plans**, the SAPCCHHs and the **National Indicator Framework** used for SDG monitoring — with heat- and flood-

resilient health facilities, vulnerability-weighted buffer stocks of drugs and supplies, and trained frontline and paramedical staff [PMID 41296661]

Current national data

- ▶ **NFHS-5 (2019–21)** linked to the ICAR–CRIDA district climate-vulnerability assessment across **575 rural districts** (154,547 children, 447,348 women) shows that highly vulnerable districts consistently underperform on the health SDG targets: odds of a child being **underweight 1.25** (95% CI 1.203–1.299), **stunted 1.15** (1.109–1.198) and **wasted 1.06** (1.017–1.112), and of a woman having a **non-institutional delivery 1.38** (1.307–1.473), each significant at $p < 0.01$ and each **independent of maternal education, prenatal care by a doctor, number of antenatal visits, birth order, the child's age and sex, and state** (household wealth was deliberately not adjusted for, as income is already embedded in the vulnerability index — the ICAR–CRIDA score is itself built from fifteen indicators across five capital endowments: natural, human, social, physical and financial) [PMID 41296661]
- ▶ Occupational heat exposure in India is stratified by **caste**. Satellite heat-stress data combined with a nationally representative labour-force survey (>100,000 individuals) for the **2019 and 2022 heat waves** showed the heat-stress dose–workhours slope for marginalised caste groups was **25–150% steeper** than for dominant caste groups at Universal Thermal Climate Index thresholds of 26–35°C, after controlling for age, sex, education and economic status — so heat adaptation planning in India must be designed around social hierarchy, not occupation alone [PMID 39918922]

Elimination, surveillance and disease status

- ▶ Vector-borne disease surveillance under the renamed **National Centre for Vector Borne Diseases Control (NCVBDC)**, feeding the IDSP network, recorded **more than 1 million dengue cases and 2,712 deaths in India between 2007 and 2022**, with **73% of all cases occurring in 2016–2022** and an overall fatality of about **2.1 per 1,000 cases** (fatality rate highest in 2007 at 12.5 per 1,000; case count highest in 2021). The steepest growth was in previously cool Himalayan states — Mizoram (growth factor 655), Jammu & Kashmir (417), Meghalaya (255), Himachal Pradesh (168) and Sikkim (74) — confirming altitudinal expansion of transmission; ensemble projections to 2030 and 2050 argue for climate- and socioeconomic-data-integrated early warning and region-specific resource allocation rather than a uniform national policy [PMID 41915702]
- ▶ Heat is now a measurable threat to **routine immunisation delivery**. Across **132.4 million doses** recorded in a provincial electronic immunisation registry in Sindh, Pakistan (2018–2024), high-temperature alert days (33–39.9°C) and heatwave days (>40°C) cut **routine outreach sessions by 13.6%** and **enhanced outreach by 21.2%**, against only 5.8% at fixed sites — a direct argument for heat-adjusted session microplanning, altered session timing and cold-chain strengthening in Universal Immunization Programme-type services [PMID 40570743]

Guideline and legislative updates

- ▶ WHO's operational framework for **climate-resilient and environmentally sustainable health systems** now runs to **ten components**, an expansion of the older six-building-block formulation, and is being used as the audit yardstick for South Asian health sectors: a mixed-method review of Nepal's climate-health policies with 12 stakeholder interviews found a sound policy base but weak implementation — poor data, weak intersectoral collaboration, insufficient capacity,

inadequate financing, and disagreement across ministries over whether climate change is an environmental or a public health problem, which blunted ownership of early-warning and monitoring systems. A separate scoping review of **67 publications from 30 countries** mapped adaptation strategies onto the six building blocks: most addressed **health workforce** (climate training, task-shifting, surge capacity) and **service delivery** (infrastructure resilience, facility preparedness, continuity of care), with governance and health-information strategies (climate-health data integration, early warning, real-time surveillance) also described, while **access to essential medicines and technologies, and financing, were the least studied** [PMID 41130166; PMID 41192337]

- A secondary analysis of the **ASPIRIN** randomised trial (11,558 nulliparous women; Indian sites at **Belagavi and Nagpur**) found each 1°C rise in **mean daily maximum wet-bulb globe temperature across gestation raised the odds of preterm birth by 5%** (adjusted OR 1.05; 95% CI 1.01–1.10). The effect was confined to the placebo arm (aOR 1.07; 1.02–1.13) and absent among aspirin recipients (aOR 1.03; 0.97–1.10); distributed lag models placed the vulnerable window at **17–19 weeks before delivery**, again only in the placebo arm — the first signal that a cheap, already-programmatic drug may buffer heat-related preterm birth. Treat it as hypothesis-generating: the aspirin × heat interaction term was **not statistically significant**, and heat was associated with perinatal mortality only in the aspirin arm (aOR 1.15; 1.05–1.26), so safety in malaria-endemic settings needs testing before any policy change [PMID 42090150]

Global and WHO developments

- The **2024 Lancet Countdown** found **10 of its 15 climate-hazard indicators** at record levels: heat-related deaths in people **over 65 years were 167% above the 1990–99 level** in 2023 (more than double the 65% rise that would have been expected had temperatures not changed), people were exposed to an extra **50 days of health-threatening heat**, and a record **512 billion potential work hours** were lost to heat, **49% above the 1990–99 average** and worth about **US\$835 billion** in lost income. Adaptation lags badly — only **28 of 84 countries (33%)** reported a heat-related-illness early warning system, and of World Meteorological Organization members **157 (81%)** provided climate services for health but only **44 (23%)** had even one climate-informed public health surveillance system. **WHO's Fourteenth General Programme of Work**, adopted May 2024, made responding to climate change its **first strategic priority**; **151 countries** have endorsed the **COP28 Declaration on Climate and Health**, and the Global Goal on Adaptation now carries a specific health target [PMID 39488222]
- **COP30 at Belém (2025)** was framed as the "implementation COP", shifting the agenda from negotiation to decentralised delivery. The **Belém Health Action Plan, launched 13 November 2025**, gives countries a practical adaptation blueprint for integrating health — explicitly including non-communicable and cardiovascular disease, historically neglected in climate-health policy — into national climate strategies, health-system preparedness and community resilience [PMID 42110323]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 15, Environment and Health, pp 888, 902–903); Suryakantha — Community Medicine with Recent Advances (Ch 48, Global Warming, pp 909–911 with Fig 48.1, p 910; Ch 21, p 559; Ch 6, p 68); Oxford Textbook of Public Health 7e (Ch 14, Climate change and human health, pp 551–583; Ch 4, p 244; Ch 19, p 759; Ch 1, pp 164–165; Ch 107, pp 4054–4055); Manual on Municipal Solid Waste Management (pp 55, 597); One Health Booklet, AIIMS Bibinagar with NCDC (pp 3–7); Jugal Kishore — National Health Programmes of India (pp 670–671); WHO World Health Day 2008; UNEP 2011 black-carbon and methane assessment and WHO 2015 short-lived climate pollutant policies, both as cited in Oxford 7e.

Recent-advances sources: 39185584 — A review of climate change and cardiovascular diseases in the Indian policy context, *Health Policy and Planning* 2024, doi:[10.1093/heapol/czae076](https://doi.org/10.1093/heapol/czae076); 41296661 — Health SDGs are at risk from climate change: evidence from India, *PLoS One* 2025, doi:[10.1371/journal.pone.0335529](https://doi.org/10.1371/journal.pone.0335529); 39918922 — Caste inequality in occupational exposure to heat waves in India, *Demography* 2025, doi:[10.1215/00703370-11803010](https://doi.org/10.1215/00703370-11803010) (abstract-only); 41915702 — Impact of projected climate and socioeconomic scenarios on state-wise annual dengue incidence in India using ensemble models, *PLoS Neglected Tropical Diseases* 2026, doi:[10.1371/journal.pntd.0014159](https://doi.org/10.1371/journal.pntd.0014159); 40570743 — Assessing the impact of heat waves on childhood immunization coverage in Sindh, Pakistan, *Vaccine* 2025, doi:[10.1016/j.vaccine.2025.127424](https://doi.org/10.1016/j.vaccine.2025.127424) (abstract-only); 41130166 — Building climate-resilient and low-carbon health systems in Nepal, *Public Health* 2025, doi:[10.1016/j.puhe.2025.106001](https://doi.org/10.1016/j.puhe.2025.106001) (abstract-only); 41192337 — A scoping review of climate-resilient health system strategies in low-resource settings, *Public Health* 2025, doi:[10.1016/j.puhe.2025.106026](https://doi.org/10.1016/j.puhe.2025.106026) (abstract-only); 42090150 — Aspirin and preterm birth among pregnant people with increased heat exposure: secondary analysis of the ASPIRIN randomized clinical trial, *JAMA Network Open* 2026, doi:[10.1001/jamanetworkopen.2026.11402](https://doi.org/10.1001/jamanetworkopen.2026.11402); 39488222 — The 2024 report of the Lancet Countdown on health and climate change, *Lancet* 2024, doi:[10.1016/S0140-6736\(24\)01822-1](https://doi.org/10.1016/S0140-6736(24)01822-1); 42110323 — Climate change and cardiovascular health: advancing resilient action beyond COP30, *Global Heart* 2026, doi:[10.5334/gh.1550](https://doi.org/10.5334/gh.1550). Also used: MoHFW National Action Plan for Climate Change and Human Health 2018 and the State Action Plans on Climate Change and Human Health directory maintained by NCDC; National Action Plan on Heat-Related Illness 2021; National Action Plan for Diseases due to Air Pollution 2021; National Health Adaptation Plan for Climate Change-Related Disaster 2019; Health Adaptation Plan for Cardiopulmonary Diseases; NFHS-5 (2019–21); ICAR–CRIDA risk and vulnerability assessment of Indian agriculture to climate change (district climate-vulnerability index); NCVBDC dengue surveillance 2007–2022; WHO Fourteenth General Programme of Work (adopted May 2024); WHO ten-component operational framework for climate-resilient and environmentally sustainable health systems; COP28 Declaration on Climate and Health (151 countries); Belém Health Action Plan, COP30, 13-Nov-2025. All ten cited PMIDs were re-checked against source on 2026-08-16: the six with PubMed Central deposits (39185584/PMC11562124, 41296661/PMC12654917, 41915702/PMC13099098, 42090150/PMC13150641, 39488222/PMC7616816, 42110323/PMC13155086) were verified

against full text; the four marked abstract-only have no PMC deposit and were verified against the PubMed abstract.



Hardness of water

Asked in: Paper I 19-Nov-2020 — RGUHS MD Community Medicine

Definition — Hardness is defined as the **soap-destroying power of water**; the consumer considers water hard when large amounts of soap are needed before lather will form (Park 28e, Ch 15, p 881). It is caused chiefly by dissolved salts of **calcium and magnesium**, which form an insoluble, sticky precipitate (scum) with soap instead of lather (Suryakantha, Ch 6, p 58). Hardness is expressed as **milligrams of calcium carbonate (CaCO₃) per litre**, and its degree depends largely on the geology of the catchment — limestone regions yield considerably hard water, granite areas yield soft water (Suryakantha, Ch 6, p 58).

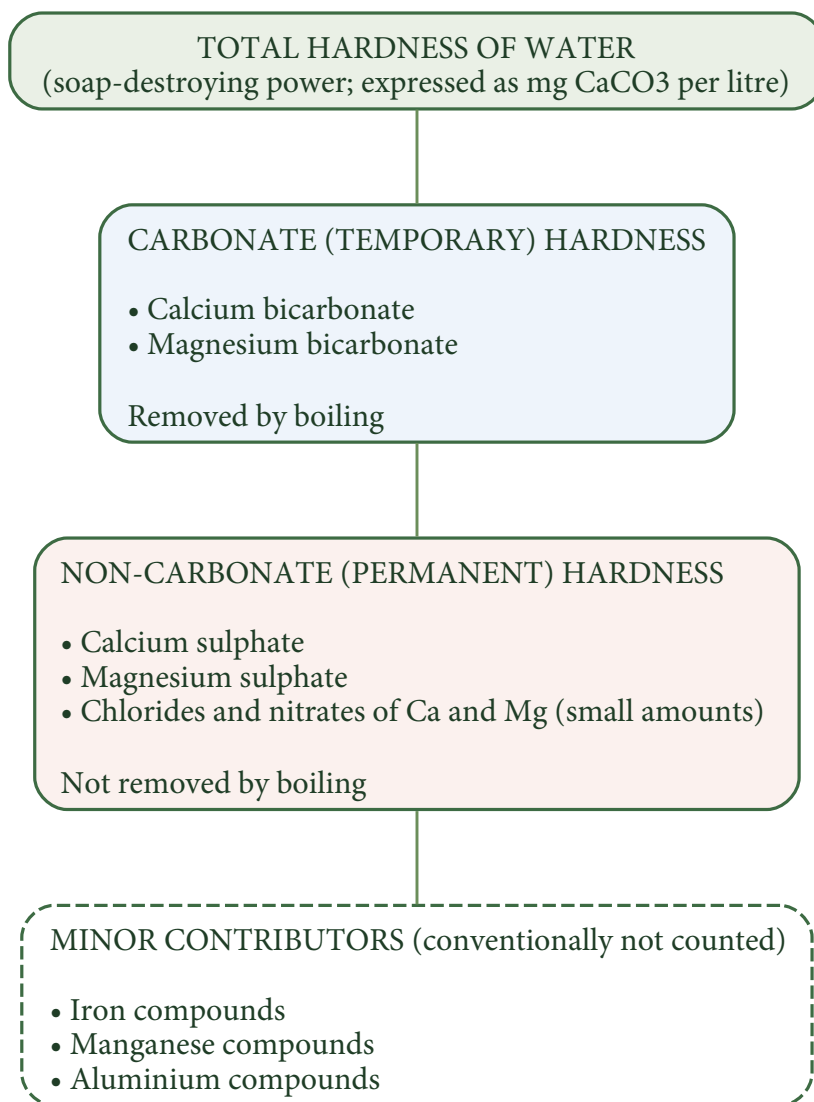
Causes of hardness

- ▶ Hardness in water is caused **mainly by four dissolved compounds** — calcium bicarbonate, magnesium bicarbonate, calcium sulphate and magnesium sulphate; the presence of **any one** of these produces hardness (Park 28e, Ch 15, p 881).
- ▶ **Chlorides and nitrates** of calcium and magnesium also cause hardness but occur generally in small amounts and are of less importance; **iron, manganese and aluminium** compounds likewise cause hardness but are present in such small amounts that it is customary not to consider them (Park 28e, Ch 15, p 881).
- ▶ Hardness depends not only on the geology of the region but also on **pollution of the source with sewage and other wastes** (Suryakantha, Ch 6, p 58).

Types of hardness

Feature	Carbonate (temporary) hardness	Non-carbonate (permanent) hardness
Former designation	Formerly designated "temporary" hardness	Formerly designated "permanent" hardness
Responsible salts	Bicarbonates of calcium and magnesium	Sulphates of calcium and magnesium ; also chlorides and nitrates in smaller amounts
Behaviour on boiling	Removed by boiling — carbon dioxide is driven off and insoluble calcium carbonate precipitates	Not removed by boiling ; persists after the water is boiled
Methods that remove it	Boiling, addition of lime, addition of sodium carbonate, permutit (base exchange) process	Addition of sodium carbonate, base exchange (permutit) process

(Park 28e, Ch 15, pp 881–882; Suryakantha, Ch 6, p 58)



Classification of hardness of water by the salts responsible

Measurement and grading

- ▶ Hardness is estimated by titration with a standardised titrant, ethylenediamine tetra-acetic acid (EDTA) (Suryakantha, Ch 6, p 58).
- ▶ Results are expressed as **milligrams of calcium carbonate per litre**, i.e. in milliequivalents per litre (mEq/L). **1 mEq/L of hardness = 50 mg of calcium carbonate per litre = 50 parts per million (ppm)**, as suggested by the World Health Organization (WHO) in its *International Standards for Drinking Water* (Suryakantha, Ch 6, p 58).
- ▶ An older expression is the **degree of hardness**; *Clark's* method of softening reckons **one degree = 14.25 ppm** (Park 28e, Ch 15, p 882). By *Clark's* method the same four grades below are expressed instead as **less than 10 per cent, 10–15 per cent, 15–30 per cent and over 30 per cent** (Suryakantha, Ch 6, p 58).

Classification	Level of hardness (mEq/L)	Equivalent (mg/L of CaCO ₃)
(a) Soft water	Less than 1	< 50
(b) Moderately hard	1–3	50–150

Classification	Level of hardness (mEq/L)	Equivalent (mg/L of CaCO ₃)
(c) Hard water	3–6	150–300
(d) Very hard water	Over 6	> 300

(Park 28e, Ch 15, Table 9, pp 881–882)

- ▶ **Drinking water should be moderately hard.** Softening is recommended only when hardness exceeds 3 mEq/L (150 mg per litre) (Park 28e, Ch 15, pp 881–882); the prescribed upper limit for total hardness in drinking water is accordingly 3 mEq/L (50–150 mg CaCO₃/L), with calcium 75 mg/L and magnesium 30 mg/L (Suryakantha, Ch 6, Table 5.2, p 61).
- ▶ **Both extremes are undesirable** — water of about 200 mg/L causes scale deposition in the distribution system with excessive soap consumption and scum formation, while soft water of less than 100 mg/L has a low buffer capacity and is more corrosive for water pipes (Park 28e, Ch 15, p 874).

Disadvantages of hardness

Sphere affected	Disadvantage
Domestic — soap	Hard water consumes more soap and detergents; the calcium and magnesium salts precipitate the soap as scum instead of lather, causing great wastage
Domestic — cooking	Adversely affects cooking; food cooked in soft water retains its natural colour and appearance
Domestic — clothing	Fabrics washed with soap in hard water do not have a long life
Industrial — boilers	On heating, carbonates precipitate and cause furring or scaling of boilers, leading to greater fuel consumption, loss of efficiency and sometimes boiler explosions
Industrial — processes	Many industrial processes are unsuited to hard water, giving rise to economic losses
Distribution system	Hardness shortens the life of pipes and fixtures, and at about 200 mg/L causes scale deposition in the distribution system
Health	Irritation of the skin and of the gastrointestinal system

(Park 28e, Ch 15, p 882; Suryakantha, Ch 6, p 58; the 200 mg/L scale-deposition figure — Park 28e, Ch 15, p 874)

Hardness and cardiovascular disease

Reports from several countries show an **inverse statistical association between the hardness of drinking water and the death rate from cardiovascular disease**: areas supplied with soft drinking water had a significantly higher prevalence of arteriosclerotic heart disease, degenerative heart disease, hypertension, sudden deaths of cardiovascular origin, or a combination of these (Park 28e, Ch 15, p 882; Suryakantha, Ch 6, p 58, where it is listed as the one *advantage of hard water*). Park is explicit that the **evidence rests solely on circumstantial evidence and statistical association**, and that further studies are in progress — so it must be written as an association, never as proof that hard water is protective (Park 28e, Ch 15, p 882).

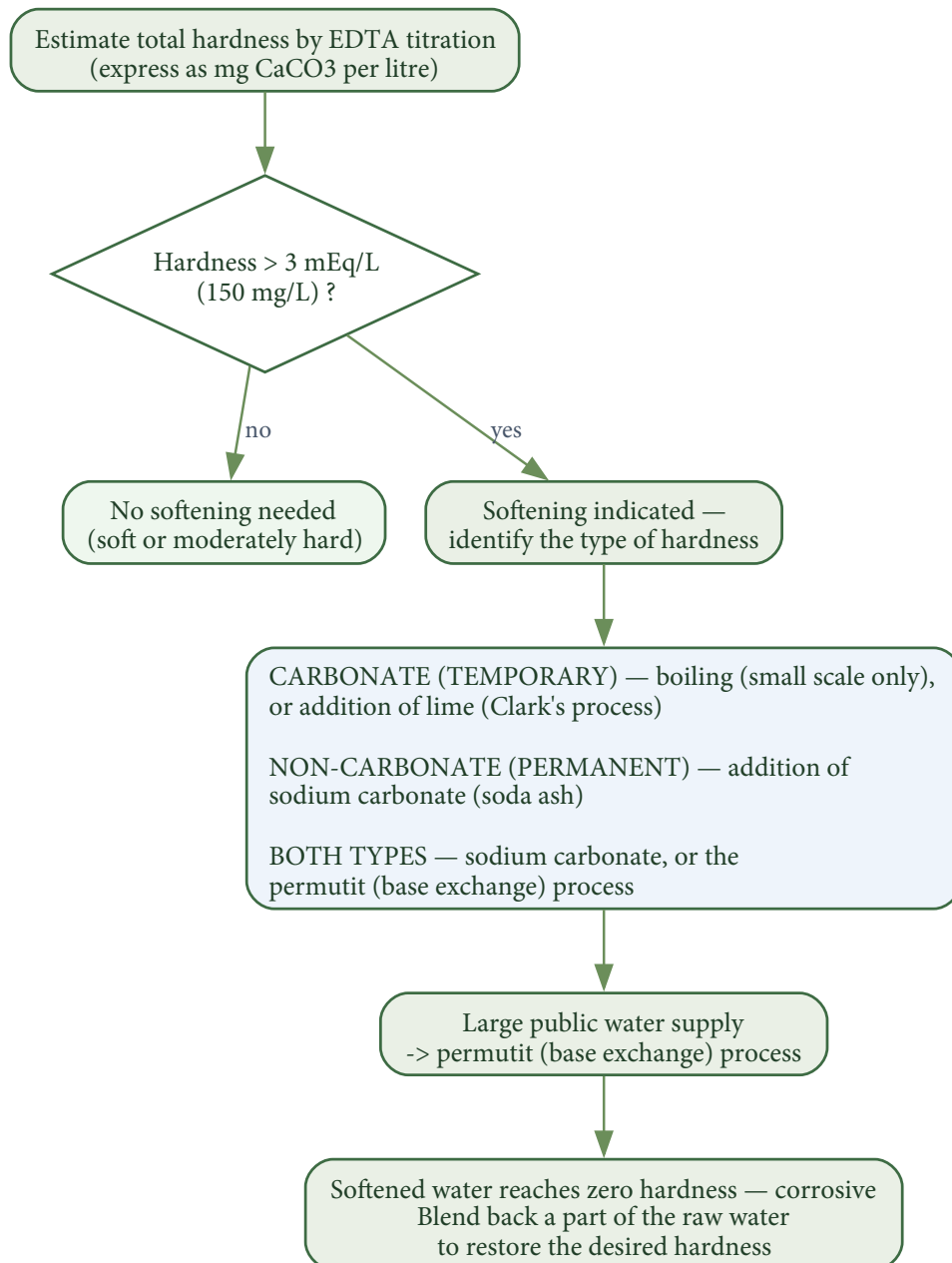
Methods for removal of hardness (softening)

Method	Principle and reaction	Hardness removed	Remarks / limitation
1. Boiling	Expels carbon dioxide — the main principle is to remove carbon dioxide gas — and precipitates insoluble calcium carbonate: $\text{Ca}(\text{HCO}_3)_2 \rightarrow \text{CaCO}_3\downarrow + \text{H}_2\text{O} + \text{CO}_2\uparrow$	Temporary only	An expensive method to soften water on a large scale; not practicable for a community supply
2. Addition of lime (<i>Clark's process</i>)	Lime absorbs carbon dioxide and precipitates insoluble calcium carbonate — $\text{Ca}(\text{OH})_2 + \text{Ca}(\text{HCO}_3)_2 \rightarrow 2\text{CaCO}_3\downarrow + 2\text{H}_2\text{O}$	Temporary; also achieves magnesium reduction and reduces total hardness	Dose in <i>Clark's</i> method: one ounce of quicklime to every 700 gallons of water for each degree (14.25 ppm) of hardness
3. Addition of sodium carbonate (soda ash)	(i) $\text{Na}_2\text{CO}_3 + \text{Ca}(\text{HCO}_3)_2 \rightarrow 2\text{NaHCO}_3 + \text{CaCO}_3\downarrow$ (ii) $\text{CaSO}_4 + \text{Na}_2\text{CO}_3 \rightarrow \text{CaCO}_3\downarrow + \text{Na}_2\text{SO}_4$	Both temporary and permanent	The classical chemical method for non-carbonate hardness
4. Base exchange (permutit) process	Sodium permutit — a complex compound of sodium, aluminium and silica — exchanges its sodium cation for the calcium and magnesium ions in the water, becoming calcium and magnesium permutit	Both temporary and permanent; can soften to zero hardness	The method used for large water supplies; regenerated with brine (see below)

(Park 28e, Ch 15, p 882; Suryakantha, Ch 6, p 58)

Base exchange (permutit) process — working detail:

- ▶ Water can be softened to **zero hardness**. Because **water of zero hardness is corrosive**, a part of the raw water is **blended back** with the softened water to secure the desired hardness (Park 28e, Ch 15, p 882).
- ▶ After the permutit has been in use for some time it **loses its effectiveness**, but it can be **regenerated by treating with a concentrated solution of sodium chloride (brine)** and washing away the soluble calcium and magnesium chlorides that are formed — which is why it is the process of choice for large public water supplies (Park 28e, Ch 15, p 882).



Decision pathway for softening a community water supply

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Draw the **permutit (base exchange) water softener** as a tall closed vertical vessel, with these exact labels:

- ▶ Hard water inlet (raw water) at the top, with inlet valve
- ▶ Free-board / water space above the bed
- ▶ Bed of **sodium permutit** (sodium–aluminium–silicate) — the base exchange medium
- ▶ Supporting layer of **graded gravel** beneath the permutit bed
- ▶ **Underdrain / collecting system** at the conical base
- ▶ **Softened water outlet** at the bottom
- ▶ **Brine (concentrated sodium chloride) inlet** for regeneration, and **waste line** carrying away soluble calcium and magnesium chlorides
- ▶ **Bypass line** carrying a part of the raw water to blend with the softened water (to avoid corrosive zero-hardness water)

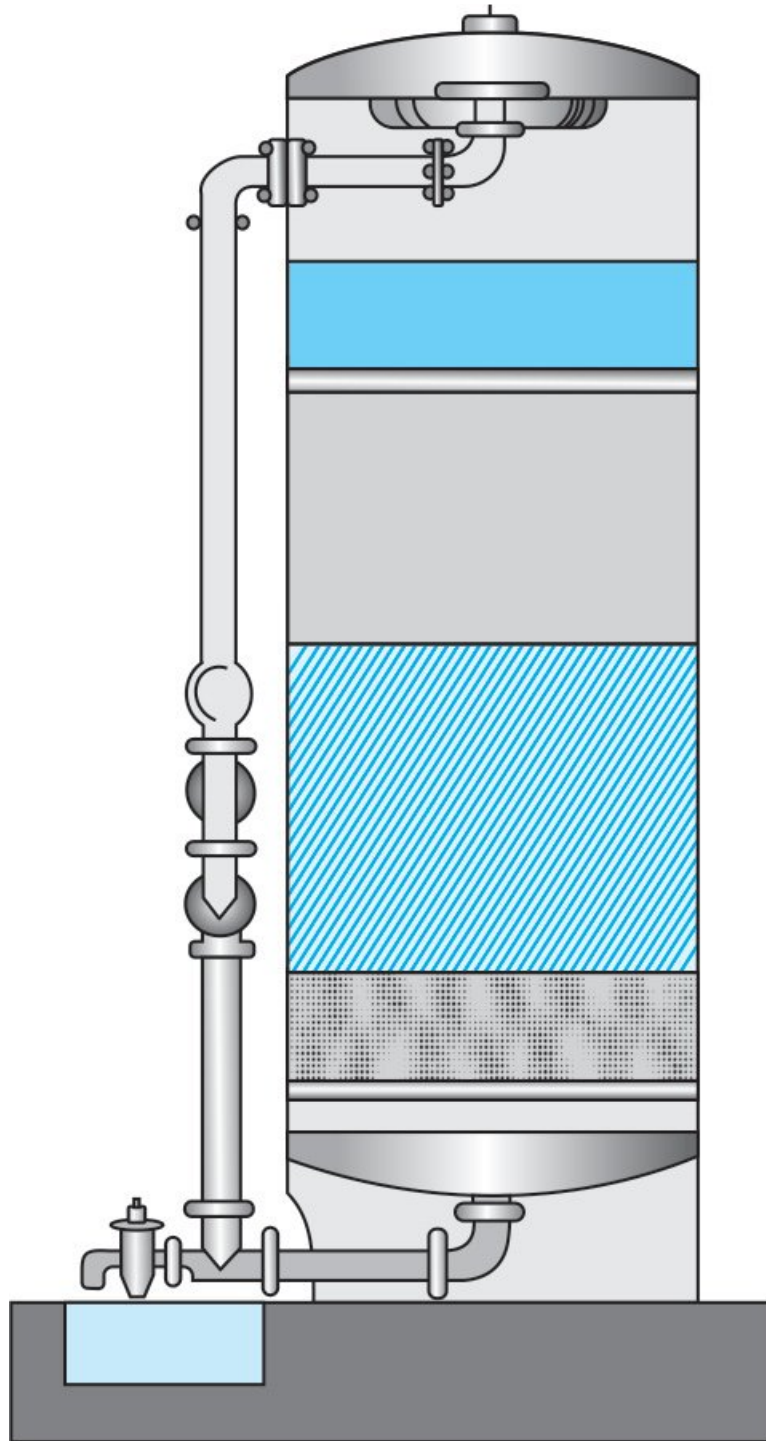


Fig 5.15 — Permutit water softener Source: Ghosh BN. *A Treatise on Hygiene and Pu* (Suryakantha Environment Best, p.59)

High-yield exam pointers

- ▶ **Definition to write first:** hardness = the soap-destroying power of water; caused mainly by four compounds — calcium and magnesium bicarbonates (carbonate/temporary) and calcium and magnesium sulphates (non-carbonate/permanent).
- ▶ **Conversion that is always asked:** 1 mEq/L = 50 mg CaCO_3 /L = 50 ppm (WHO *International Standards for Drinking Water*); estimated by EDTA titration.

- ▶ **Park's Table 9 grading:** soft <1 mEq/L (<50 mg/L) — moderately hard 1–3 (50–150) — hard 3–6 (150–300) — very hard >6 (>300 mg/L). **Drinking water should be moderately hard; soften only above 3 mEq/L (150 mg/L).**
- ▶ **Mnemonic for the four methods — "B-L-S-P":** Boiling, Lime, Soda ash (sodium carbonate), Permutit. First two for **temporary** only; last two **remove BOTH** temporary and permanent.
- ▶ **Clark's dose:** 1 ounce of quicklime per 700 gallons of water for each degree (14.25 ppm) of hardness.
- ▶ **Permutit trap:** it softens to **zero hardness**, which is **corrosive** — so blend back raw water; regenerate the exhausted bed with **brine (concentrated sodium chloride)**.
- ▶ **Disadvantages in one line:** more soap, poor cooking, short fabric life, **boiler scaling** → **fuel loss and explosions**, industrial losses, shortened life of pipes and fixtures, skin and gastrointestinal irritation.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written: Park lists exactly **four** classical methods of removing hardness — the permutit process and the base exchange process being two names for the same method rather than two separate methods — and describes membrane methods such as reverse osmosis under high-pressure treatment of drinking water rather than under removal of hardness, so the examiner's expected list remains those four.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Chapter 15 (Environment and Health), pp 874, 881–882 (definition, four causative compounds, carbonate/non-carbonate classification with the former "temporary"/"permanent" designations, Table 9 grading of hardness, disadvantages, all four methods of removal with their chemical reactions, Clark's dose, permutit regeneration, water hardness and cardiovascular diseases, and the 200 mg/L scale-deposition and <100 mg/L corrosivity figures at p 874). Suryakantha — Community Medicine with Recent Advances, Chapter 6 (Chapter-05_Environment and Water), pp 58, 59 (Fig 5.15 permutit water softener), 61 (Table 5.2 prescribed upper limits) — definition, causes, EDTA estimation, 1 mEq/L = 50 mg CaCO₃/L per the WHO *International Standards for Drinking Water*, Clark's percentage grading of the same four grades, geology of hard and soft water, the carbon-dioxide-removal principle of boiling, skin and gastrointestinal irritation, advantages of hard water, and the prescribed upper limit for total hardness.



Carbon footprint and carbon credit

Asked in: Paper I 24-Sep-2025 — RGUHS MD Community Medicine

Definition — The **carbon footprint** is the total quantity of greenhouse gases released, directly and indirectly, by an individual, household, organisation, product, event or nation, expressed as **tonnes of carbon dioxide equivalent (CO₂e)** over a stated period. A **carbon credit** is a tradable certificate representing **one tonne of CO₂ equivalent** that has been reduced, avoided or removed from the atmosphere, which the holder may sell to an emitter who has exceeded a legally set emission cap. The first is a *measure* of environmental burden; the second is an *economic instrument* that makes the reduction of that burden financially rewarding.

Basis: the greenhouse effect and the gases counted

The earth's atmospheric mantle allows short-wavelength solar radiation into the biosphere but traps outgoing long-wavelength infrared radiation, acting as an insulator — the **greenhouse effect**; without the *natural* greenhouse effect the earth's surface would be about **33°C colder** (AFMC Red Book of Community Medicine 2024, Ch 6, p 188). The **enhanced** greenhouse effect is the extra radiative forcing from human-generated gases, and it is this excess that a carbon footprint quantifies.

Greenhouse gas	Chief man-made sources	Note on accounting
Carbon dioxide (CO ₂)	Combustion of fossil fuels, deforestation	Reference gas; about 35 billion tonnes added to the atmosphere annually (AFMC 2024, Ch 6, pp 188–189)
Methane (CH ₄)	Anaerobic decay of organic waste in open dumps and landfills, paddy, livestock	Global warming potential 21 times that of CO ₂ (Solid Waste Management, Ch 40, pp 315–316)
Nitrous oxide (N ₂ O)	Nitrogen fertilisers, biomass burning	Long atmospheric lifetime
Hydrochlorofluorocarbons (HCFCs)	First-generation replacements for chlorofluorocarbons in refrigeration and foam blowing	Still contain chlorine and therefore do deplete stratospheric ozone ; being phased out under the <i>Montreal Protocol</i> , the treaty that phases out ozone-depleting compounds (Hunter's Diseases of Occupations, Ch 42, pp 397–399; Park Plus, Ch 13, p 166)
Hydrofluorocarbons (HFCs), with perfluorocarbons and sulphur hexafluoride	Refrigeration and air-conditioning, aerosol propellants, foam blowing, industry	Contain no chlorine, so they are ozone-safe , but are powerful greenhouse gases counted among the six Kyoto gases (Hunter's, Ch 39, pp 290–291); their phase-down was later brought under the Montreal Protocol by the <i>Kigali Amendment</i> , 2016

Because the gases differ in warming power, each is multiplied by its global warming potential and reported as a single CO₂-equivalent figure — this is why the footprint is always written in **tonnes CO₂e**, never in tonnes of gas.

Types of carbon footprint and how each is used

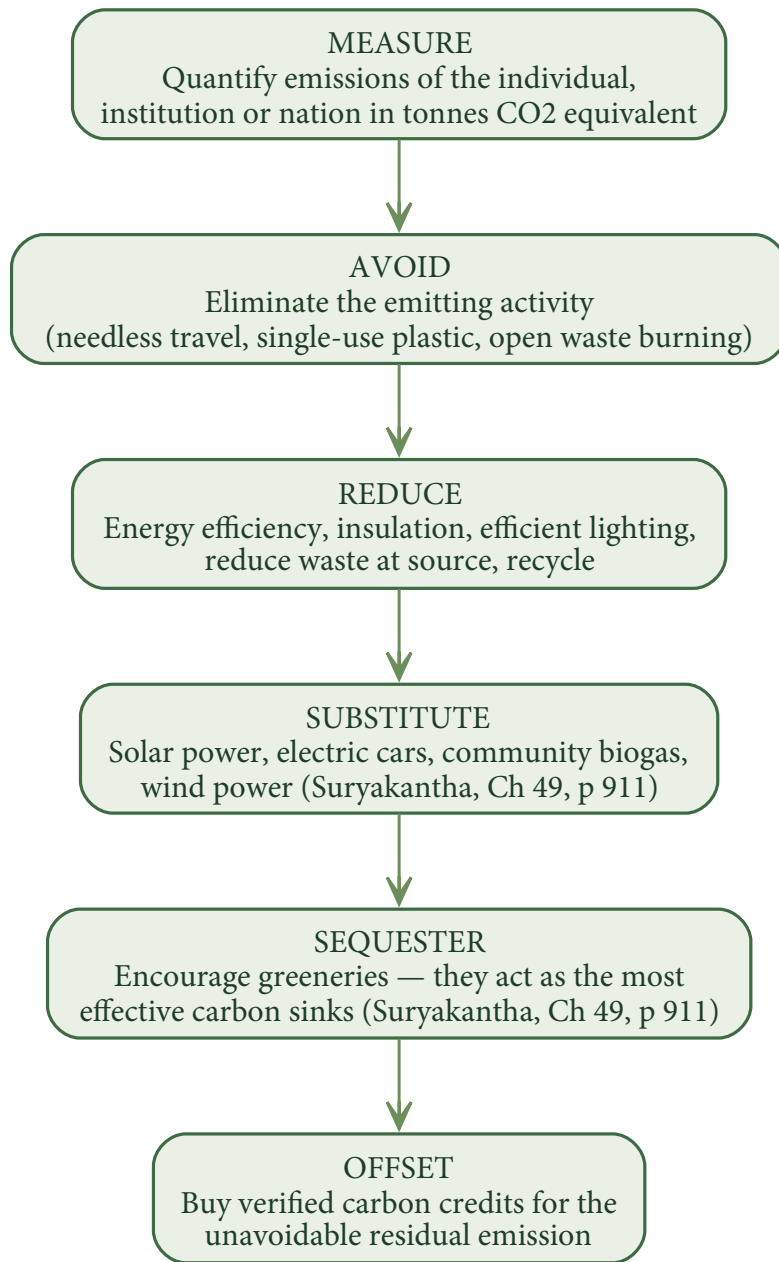
Type	What is counted	Public-health use
Individual/ household	Fuel, electricity, food, travel, goods consumed by one person	Behaviour-change and health-education messaging
Organisational (including hospitals)	Energy use of buildings, procurement, staff and patient travel, waste	Institutional greening; hospital sustainability audit
Product/service ("life-cycle")	Emissions from raw material to disposal of one item	Choice of low-emission drugs, devices, packaging
National/city	All sectors within a geographic boundary	Nationally determined contributions; city climate plans
Related concept — <i>ecological footprint</i>	Consumption expressed as hectares of biologically productive land per person (Loh and Wackernagel)	Global average 2.3 hectares/person in 2003 ; USA 9.6, Malawi 0.6 (Textbook of International Health, Ch 10, pp 517–518)

Carbon footprints of United States cities are almost **twice** those of large European cities, reflecting urban sprawl, less public transport and higher consumption — showing that the footprint is set by **urban design and policy**, not by individual behaviour alone (Textbook of International Health, Ch 10, p 518).

Public health significance

- **Climate change is the health consequence.** Continued emission is projected to raise average global surface temperature by about **3°C by 2030**, raise sea level by **0.1–0.3 metres by 2050**, and increase cyclones, heatwaves and droughts, altering vector distribution, food and water supply (Park 28e, Ch 15, p 903). Average global temperature has already risen by at least **1.1°C since 1880**, at 0.15–0.20°C per decade since 1975 (AFMC 2024, Ch 6, p 188).
- **Health care is itself a large emitter.** England, the United States and Australia estimate their health sectors at a substantial share of national emissions, arising mainly from hospital energy use, procurement of materials (particularly pharmaceuticals) and health-related transport (Oxford Textbook of Public Health 7e, Ch 14, pp 572–573).
- **Mitigation carries immediate health co-benefits.** In a Delhi modelling study, meeting emission targets through **active transport** (walking and cycling) produced health gains roughly **ten times** greater than switching to low-carbon vehicles alone (Oxford 7e, Ch 14, pp 578–579).
- **Equity.** Emissions are generated largely by high-consumption populations, but heat, flood and crop failure fall hardest on the poor — the central environmental-justice argument for footprint accounting.

Reducing the carbon footprint



Hierarchy of action on a carbon footprint — offsetting is the last step, never the first

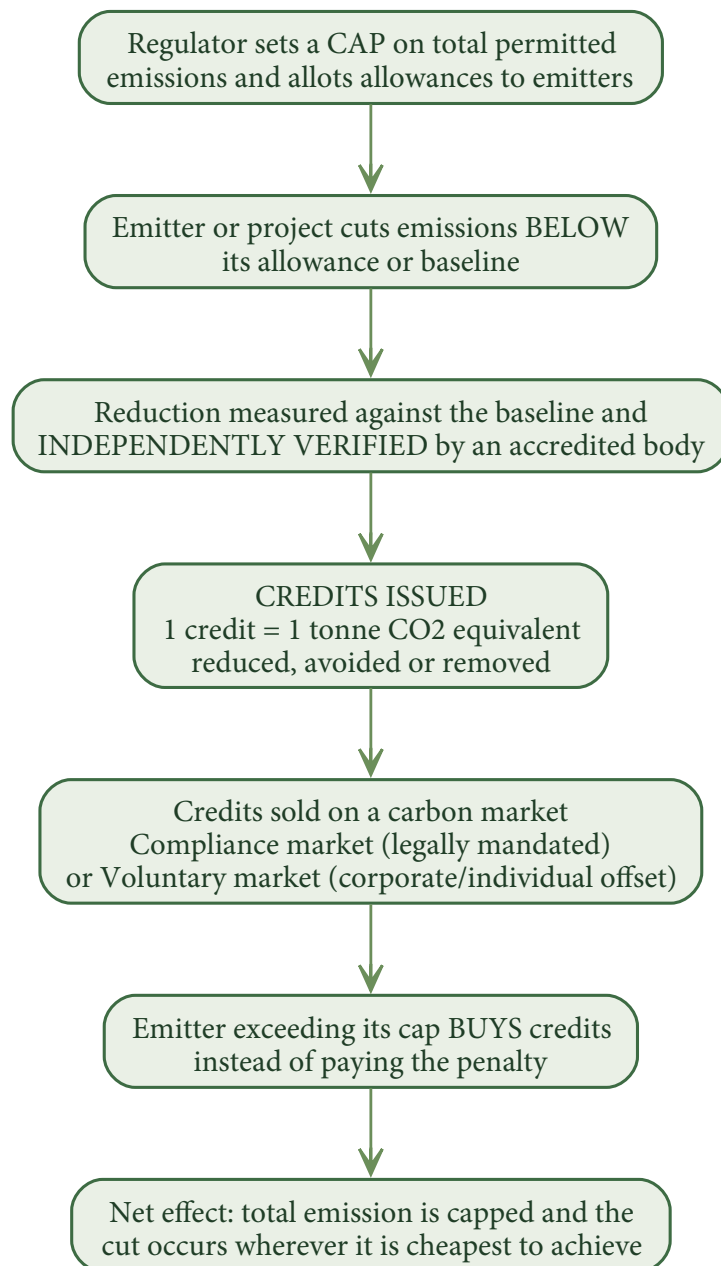
Sector	Technology	Policy instrument
Energy	Hydro, solar, wind, bioenergy	Withdraw fossil-fuel subsidies; carbon charges on fossil fuels; phase out coal
Transport	Fuel-efficient, hybrid and electric vehicles; biofuels	Shift road to rail; subsidise public transport, bike lanes, footpaths; high fuel taxes
Buildings	Efficient lighting and appliances; improved stoves; insulation; alternative refrigeration fluids	Eco-taxes, appliance labelling, building codes, eco-housing

Sector	Technology	Policy instrument
Industry	Efficient end-use equipment, heat and power recovery, recycling, carbon sequestration	Carbon taxes; tradable permits; subsidies and tax credits; performance standards
Agriculture	Soil carbon management, low-water rice, livestock and manure management, efficient nitrogen use	Incentives for organic and local production; regulated land management
Waste	Landfill methane recovery, composting, waste-to-energy, recycling	Waste taxes; composting and recycling collection systems

(Sector table adapted from *Textbook of International Health*, Ch 10, Table 10-6, pp 556–557; source IPCC 2007.)

Worked example of the size of the gain	Effect
Recycling 1 tonne of paper; recycling 1 tonne of aluminium	Avoids about 1 tonne CO ₂ e and about 9 tonnes CO ₂ e respectively (Solid Waste Management, Ch 27, pp 268–270)
NHS England, 2007–2015, under the UK Climate Change Act 2008 (80% cut required by 2050)	Greenhouse gas emissions cut by 11% despite an 18% rise in inpatient admissions, saving about GBP 30 million per year (Oxford 7e, Ch 14, pp 573–574)

Carbon credit — how it is generated and traded



Generation, verification and trading of a carbon credit (cap-and-trade principle)

Steps in earning a credit

- ▶ A project (renewable power, methane capture from a landfill, composting, afforestation, efficient cookstoves) establishes a **baseline** — the emission that would have occurred without it.
- ▶ Emission reduction is **monitored** and **independently verified**; the reduction must be *additional*, i.e. it would not have happened anyway.
- ▶ The verifying authority **issues credits**, one per tonne CO₂e.
- ▶ Credits are **traded**; the buyer sets them against its own excess emission.
- ▶ When used, the credit is **retired** so that it cannot be counted twice.

Composting of municipal organic waste is recognised by the **United Nations Framework Convention on Climate Change (UNFCCC)** as an emission-reduction methodology in waste management,

although the scope for actual financial gain by selling carbon credits is at present limited (Solid Waste Management, Ch 40, p 316).

International framework, the Kyoto mechanisms and the Indian position

Instrument	Year	Core commitment relevant to carbon credits
UN Framework Convention on Climate Change (UNFCCC)	1992	Parent treaty; sets the principles the later protocols rest on
Kyoto Protocol	Adopted at Kyoto, Japan 11 December 1997; in force 16 February 2005	Binding targets for 37 industrialised countries and the European Community, averaging a five per cent cut against 1990 levels over the five-year commitment period 2008–2012, expressed as allowed emissions or "assigned amounts"; implementation rules adopted at COP 7, 2001 as the Marrakesh Accords (Park Plus, Ch 13, p 166). Suryakantha states the target as at least 5.2% below 1990 levels by 2010 with a penalty for failure, supported by 141 nations, boycotted by the USA, signed by 34 industrialised countries (Suryakantha, Ch 49, p 911)
Paris Agreement	2015	Hold warming below 2°C, aiming for 1.5°C; ratified by 175 of 197 members; nationally determined contributions; not legally binding (Oxford 7e, Ch 14, p 564). COP 21 also established the framework for a global carbon market (Harrison 22e, Ch 474, pp 3795–3796)
COP 28, Dubai	Dec 2023	About 200 countries agreed to transition away from fossil fuels, net zero carbon pollution by 2050, triple renewable capacity and double energy efficiency by 2030; Global Cooling Pledge — 66 governments to cut cooling-related emissions by at least 68% versus 2022 levels by 2050 (AFMC 2024, Ch 6, pp 189–190)
India's targets	—	Energy independence by 2047 and Net Zero by 2070; National Green Hydrogen Mission; Pradhan Mantri Suryodaya Yojana for rooftop solar (AFMC 2024, Ch 6, p 190)

Parties must meet their Kyoto targets primarily through national measures, but the Protocol offers three additional market-based mechanisms — this is the machinery that creates the carbon credit (Park Plus, Ch 13, p 166).

Kyoto flexibility mechanism	How it works	Unit traded
Emissions trading	A party that has kept emissions below its assigned amount sells the unused allowance to a party that has exceeded its own	Assigned amount units

Kyoto flexibility mechanism	How it works	Unit traded
Clean Development Mechanism (CDM)	An industrialised country finances an emission-reduction project hosted in a developing country ; the verified reduction is credited to the financier and also transfers clean technology to the host	Certified emission reductions (CERs) — a party short of its Kyoto target buys these, i.e. buys carbon credits generated by CDM projects (Suryakantha, Ch 49, p 911)
Joint Implementation	One industrialised country with a target finances a project in another country that also has a target	Emission reduction units

The point to state precisely in the exam: what a defaulting country buys is the **certified emission reduction (a carbon credit)** produced by a CDM project — the CDM is the project-based mechanism that generates credits, not itself a commodity that can be bought. Because carbon dioxide is the principal greenhouse gas, people speak simply of trading in carbon, and **carbon is now tracked and traded like any other commodity — the "carbon market"** (Park Plus, Ch 13, p 166). Global emissions are currently about 50 Gt CO₂ equivalent per year, **roughly 14% above** the level compatible with holding warming near 2°C, and must fall to about zero by 2050 (Oxford 7e, Ch 14, p 564).

Limitations and criticisms of carbon credits

- **Additionality is hard to prove** — credit may be issued for a reduction that would have occurred anyway.
- **Leakage** — emission is displaced to an uncapped country or sector rather than removed.
- **Permanence** — forestry credits are reversed by fire, felling or disease.
- **Licence to pollute** — offsetting may substitute for genuine emission reduction at source, which is why offsetting sits at the *bottom* of the action hierarchy.
- **Price volatility and weak demand** — returns from selling credits are currently small for many genuinely useful projects, e.g. municipal composting (Solid Waste Management, Ch 40, p 316).
- **Equity concern** — credits let high-emitting nations purchase compliance from low-emitting poor nations without changing their own consumption.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **The greenhouse effect** — label: *incoming short-wavelength solar radiation, earth's surface, outgoing long-wavelength infrared radiation, layer of greenhouse gases (CO₂, CH₄, N₂O, HCFCs, HFCs), radiation trapped and re-radiated to earth, rise in surface temperature = global warming.*
- ▶ **The carbon-credit balance** — label: two boxes side by side, *Emitter above the cap (deficit — must buy)* and *Project below the baseline (surplus — may sell)*, joined by an arrow labelled *1 credit = 1 tonne CO₂ equivalent*, with *verification body* on the arrow and *cap line* drawn across both.

High-yield exam pointers

- ▶ **Unit to write:** carbon footprint = **tonnes of CO₂ equivalent (CO₂e)**; 1 carbon credit = **1 tonne CO₂e** reduced, avoided or removed.
- ▶ **Methane's global warming potential = 21 times CO₂**; about 35 billion tonnes CO₂ added to the atmosphere annually; without the natural greenhouse effect the earth would be **33°C colder**.
- ▶ **Kyoto Protocol:** adopted 11 Dec 1997, in force 16 Feb 2005, 37 industrialised countries plus the European Community, average **5% cut below 1990 levels over the commitment period 2008–2012** (Suryakantha: at least 5.2% by 2010), Marrakesh Accords 2001.
- ▶ **Three Kyoto flexibility mechanisms** — **emissions trading, Clean Development Mechanism, Joint Implementation**; the CDM yields **certified emission reductions (CERs)**, the credits a defaulting country buys. **Paris 2015:** below 2°C, aiming 1.5°C, 175 of 197 ratified.
- ▶ **India: Net Zero by 2070, energy independence by 2047** — quote both with the year.
- ▶ **Hierarchy of action mnemonic** — **MARSSO:** Measure, Avoid, Reduce, Substitute, Sequester, Offset (offset last).
- ▶ **Four criticisms to name:** additionality, leakage, permanence, "licence to pollute".

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

The retrieved corpus carries no formal textbook definition of "carbon footprint" or "carbon credit", so the two definitions, the one-tonne-carbon-dioxide-equivalent unit, the naming of Joint Implementation as the third Kyoto mechanism, the term certified emission reduction, the Kigali Amendment of 2016 and the additionality, leakage and permanence criticisms are universally accepted standard public-health content rather than direct quotations from a retrieved passage. The six-step hierarchy Measure–Avoid–Reduce–Substitute–Sequester–Offset and its MARSSO mnemonic are a teaching scaffold assembled for this answer and not a framework named in any book of the corpus, although the individual steps are grounded. Suryakantha prints the Kyoto commitment periods as 2005–2012 and 2012–2016 with a 5.2% target by 2010, whereas Park Plus gives the internationally used five-year first commitment period of 2008–2012 and an average five per cent cut — write the Park Plus dates. India's domestic Carbon Credit Trading Scheme is not present in the retrieved corpus and has therefore been omitted.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, Ministry of Power, MoEFCC, CERC, PIB, UNFCCC, WHO, MoHFW/NCDC

Programme and policy updates (India)

- ▶ **India now has its own compliance carbon market** — the gap flagged in the Note above is now closed. The *Energy Conservation (Amendment) Act 2022* gave the legal basis, and the **Carbon Credit Trading Scheme (CCTS)**, notified by the **Ministry of Power on 28 June 2023**, replaced the older *Perform, Achieve and Trade (PAT)* energy-saving-certificate scheme. The **Bureau of Energy Efficiency** is the administrator, the traded unit is a **Carbon Credit Certificate (CCC) = 1 tonne CO₂e**, and the scheme runs on two arms — a **compliance mechanism** for obligated industry and an **offset mechanism** for voluntary projects outside those sectors. The *Greenhouse Gases Emission Intensity Target Rules 2025* (notified 8 October 2025, extended 16 January 2026) then made the targets binding for about **740 entities across nine energy-intensive sectors** — aluminium, cement, chlor-alkali, pulp and paper, iron and steel, fertiliser, petroleum refining, petrochemicals and textiles — for compliance years **2025-26 and 2026-27** against a **2023-24 baseline**, covering over **700 million tonnes CO₂e**, roughly **16% of national emissions**. The trading rules are now framed too: the *Central Electricity Regulatory Commission (Terms and Conditions for Purchase and Sale of Carbon Credit Certificates) Regulations 2026*, notified **27 February 2026**, put certificate trading on the **power exchanges** in separate compliance and offset segments, with **Grid Controller of India as the registry** holding member accounts and recording every transfer. This is the same cap-and-trade machinery described above, run domestically instead of through Kyoto. [MoP 2023-Jun; MoEFCC 2025-Oct; CERC 2026-Feb]
- ▶ **India's third Nationally Determined Contribution (2031–2035)** was approved by the Union Cabinet on **25 March 2026** and communicated to the UNFCCC on **24 April 2026**: cut the **emissions intensity of GDP by 47% from the 2005 level by 2035**, reach **about 60% of installed electricity capacity from non-fossil sources by 2035**, and build an additional carbon sink of 3.5–

4.0 billion tonnes CO₂e by 2035. It supersedes the 2022 pledge (45%, about 50%, and 2.5–3 billion tonnes by 2030); **Net Zero by 2070 is unchanged**. [PIB 2026-Mar; UNFCCC 2026-Apr]

Current national data

- ▶ **India's Fourth Biennial Update Report to the UNFCCC (submitted 30 December 2024)** is the figure set to quote, always with its year: emissions intensity of GDP fell **36% between 2005 and 2020**, beating the 33–35% pledge; non-fossil sources reached **46.52% of installed electricity capacity by October 2024**; and an **additional carbon sink of 2.29 billion tonnes CO₂e was created between 2005 and 2021**. [MoEFCC 2024-Dec]
- ▶ **Non-fossil capacity has since crossed half the grid — 52.57% of installed electricity capacity from non-fossil sources by February 2026**, meeting the 50%-by-2030 pledge about five years early. Attach this figure to its own month and source, not to the Biennial Update Report: BUR-4 was submitted in December 2024 and its capacity figure stops at October 2024, so a December 2024 document cannot carry a 2026 number. [PIB 2026]

Elimination, surveillance and disease status

- ▶ **Heat-related illness is now under daily national surveillance** — the operational link between the carbon footprint and the disease it causes. Under the **National Programme on Climate Change and Human Health (NPCCHH)**, facilities from primary health centre level upward submit **daily heatstroke case and death data on the IHIP portal**, with digital line lists, for the heat season **1 March to 31 July**, running each year since **1 March 2024**, alongside the National Action Plan on Heat Related Illnesses and dedicated heat-stroke rooms in hospitals. [MoHFW/NCDC 2024]

Guideline and legislative updates

- ▶ **Carbon pricing does measurably cut emissions** — the best answer yet to the "it is only paper trading" objection. A machine-learning-assisted systematic review and meta-analysis of **80 ex-post evaluations, 483 effect sizes and 21 carbon-pricing schemes** found statistically significant reductions of **5% to 21%** in at least 17 of those schemes (**4–15% after correcting for publication bias**), even where prices were low. State the scope precisely: the review covered **explicit carbon taxes and cap-and-trade schemes only, offsetting mechanisms being excluded**, so it vindicates the *cap*, not the *offset* [PMID 38755167]. A parallel synthesis of 58 modelling studies shows the **health co-benefits** — cleaner air, active transport, affordable healthier diets — depend on design: how revenue is spent, how permits are allocated, whether prices are differentiated [PMID 39483612].
- ▶ **Offset quality is the weak link, and it is now quantified**. Among the **20 companies that retired the most credits on the voluntary carbon market between 2020 and 2023**, **87% of the offsets carried a high risk of not representing real, additional reductions**, most of them from forest-conservation and renewable-energy projects — direct empirical backing for the additionality and "licence to pollute" criticisms listed above [PMID 39127784].

Global and WHO developments

- ▶ **WHO's first Global action plan on climate change and health (2025–2028)** was adopted at the **Seventy-eighth World Health Assembly in May 2025**, committing member states to climate-resilient and low-carbon health systems, stronger surveillance and early warning, and health within climate financing. **COP30 opened at Belém on 10 November 2025**, and on its **Health Day, 13 November 2025**, the **Belém Health Action Plan was launched** — the first international

climate-*adaptation* plan written specifically for the health sector — while the cover "global mutirão" decision became the first COP decision to name the **health benefits** of climate action, alongside a call to triple adaptation finance by 2035. [WHO 2025-May; WHO 2025-Nov-13; UNFCCC 2025-Nov]

- **The Kyoto Clean Development Mechanism now has a Paris-era successor.** At COP29, Baku, November 2024, the **Article 6.4 Paris Agreement Crediting Mechanism (PACM)** was operationalised once the Supervisory Body's standards on **methodology requirements** and on **removal activities** were adopted. In the exam, write CDM and its certified emission reductions as the *Kyoto* machinery and **PACM as its Paris replacement**, with Article 6.2 covering country-to-country transfers. [UNFCCC 2024-Nov]
- **Health care's own footprint has been quantified service by service.** Health systems account for about **4.4% of global greenhouse-gas emissions**, reaching up to 10% of national emissions in some high-income countries, and **84 countries and areas have set net-zero or low-carbon health targets**; a review of **76 studies covering 151 hospital services and care pathways** found **travel, facilities and consumables** to be the dominant contributors, with **waste disposal contributing least** — so hospital greening should target energy, procurement and transport before it targets bins [PMID 39729358]. In 35,242 anaesthetics, **total intravenous anaesthesia emitted 0.4 kg CO₂e per hour against 3.1–3.8 kg for sevoflurane strategies** [PMID 40187906].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 15, Environment and Health); Park Plus — Amrutha & Vijayalakshmi (Ch 13, Environment and health, p 166, Kyoto Protocol, the three market-based mechanisms and the carbon market); Suryakantha — Community Medicine with Recent Advances (Ch 49, Global Warming, p 911); AFMC Red Book of Community Medicine 2024 (Ch 6, Climate and Health, pp 188–190, including COP 28 Dubai 2023 and India's Net Zero 2070 target); Oxford Textbook of Public Health 7e (Ch 14, Climate Change and Human Health); Textbook of International Health (Ch 10, Health and the Environment, including Table 10-6 adapted from IPCC 2007); Hunter's Diseases of Occupations (Ch 39, Gases, and Ch 42, Organic chemicals, for the ozone and greenhouse status of HCFCs and HFCs); Harrison's Principles of Internal Medicine 22e (Ch 474, Climate Change and Infectious Disease); Solid Waste Management (greenhouse gas avoidance through composting and recycling); United Nations Framework Convention on Climate Change; Kyoto Protocol 1997; Paris Agreement 2015. Coverage note: no indexed textbook in this corpus defines "carbon footprint" or "carbon credit" formally — see the Note above.

Recent-advances sources: 38755167 — Systematic review and meta-analysis of ex-post evaluations on the effectiveness of carbon pricing, *Nature Communications* 2024, doi:[10.1038/s41467-024-48512-w](https://doi.org/10.1038/s41467-024-48512-w); 39483612 — Health co-benefits and trade-offs of carbon pricing: a narrative synthesis, *Climate Policy* 2024, doi:[10.1080/14693062.2024.2356822](https://doi.org/10.1080/14693062.2024.2356822); 39127784 — Demand for low-quality offsets by major companies undermines climate integrity of the voluntary carbon market, *Nature Communications* 2024, doi:[10.1038/s41467-024-51151-w](https://doi.org/10.1038/s41467-024-51151-w); 39729358 — The Carbon Footprint of Hospital Services and Care Pathways: A State-of-the-Science Review, *Environmental Health Perspectives* 2024, doi:[10.1289/EHP14754](https://doi.org/10.1289/EHP14754); 40187906 — The carbon footprint of general anaesthesia in adult patients: a multicentre observational comparison of intravenous and inhalation anaesthetic strategies in 35,242 procedures, *British Journal of Anaesthesia* 2025, doi:[10.1016/j.bja.2025.01.043](https://doi.org/10.1016/j.bja.2025.01.043) (abstract-only). Regulatory and policy sources: Ministry of Power — Carbon Credit Trading Scheme, 28 June 2023; MoEFCC — Greenhouse Gases Emission Intensity Target Rules 2025, notified 8 October 2025 with further sectors 16 January 2026; Central Electricity Regulatory Commission (Terms and Conditions for Purchase and Sale of Carbon Credit Certificates) Regulations 2026, notified 27 February 2026; India's third Nationally Determined Contribution 2031–2035 — Cabinet approval 25 March 2026, communicated to the UNFCCC 24 April 2026; India's Fourth Biennial Update Report to the UNFCCC, 30 December 2024 (emissions intensity, 46.52% non-fossil capacity at October 2024, 2.29 bn t CO₂e sink); PIB — 52.57% non-fossil installed electricity capacity as of February 2026; MoHFW/NCDC — National Programme on Climate Change and Human Health, heat-related illness and death surveillance on IHIP from 1 March 2024; WHO — Global action plan on climate change and health 2025–2028, Seventy-eighth World Health Assembly, May 2025; WHO/PAHO — Belém Health Action Plan launched on COP30 Health Day, 13 November 2025 (COP30 opened 10 November 2025); UNFCCC COP29 Baku — Article 6.4 Paris Agreement Crediting Mechanism, November 2024.



Hospital waste management

Asked in: Paper I 21-Jun-2024 — RGUHS MD Community Medicine

Definition — **Bio-medical waste** is "any waste which is generated during the diagnosis, treatment or immunisation of human beings or animals, or research activities pertaining thereto, or in the production or testing of biologicals, or in health camps, including the categories mentioned in Schedule I" (Bio-Medical Waste Management Rules 2016, Rule 3(f)). **Hospital waste management** is the whole cradle-to-grave chain — segregation, collection, storage, transport, treatment and final disposal — carried out so that the waste harms neither health nor the environment (BMW Rules 2016, Rule 3(l)). In India it is governed by the Bio-Medical Waste Management Rules 2016, notified by the Ministry of Environment, Forest and Climate Change under the Environment (Protection) Act 1986, which replaced the Bio-Medical Waste (Management and Handling) Rules 1998 (Park 28e, Ch 16, pp 946-948).

SOURCE & COVERAGE

Every institution that generates such waste is covered — government and private hospitals, nursing homes, physicians' and dentists' clinics, dispensaries, primary health centres, medical research and training establishments, mortuaries, blood banks and collection centres, animal houses, slaughter houses, laboratories, research organisations and biotechnology production units (Park 28e, Ch 16, pp 943-944).

Item	Figure
Non-risk ("general") healthcare waste, comparable to domestic waste	75-90 per cent of all waste produced by healthcare providers (Park 28e, Ch 16, p 943)
Hazardous fraction requiring special handling	10-25 per cent (Park 28e, Ch 16, p 943)
Generation — hospitals	0.5 to 4 kg per bed per day (Bangalore survey; Park 28e, Ch 16, p 944)
Generation — private hospitals	0.5 to 2 kg per bed per day
Generation — nursing homes	0.5 to 1 kg per bed per day
Bangalore city total	~40 tonnes per day, of which 45-50 per cent infectious; segregation practised in only about 30 per cent of hospitals (Park 28e, Ch 16, p 944)

Average composition of hospital waste in India (wet-weight basis)	Per cent
General waste (food waste, sweepings from premises)	53.5
Paper	15
Rags	15
Plastics	10
Glass	4.0

Item	Figure
Infectious waste	1.5
Metal (sharps etc.)	1

Source: *National Environmental Engineering Research Institute 1997, quoted in Park 28e, Ch 16, pp 943-944.*

Health hazards and groups at risk

Exposure causes disease or injury because the waste (a) contains infectious agents, (b) contains toxic or hazardous chemicals, pathological waste or pharmaceuticals, (c) contains sharps, (d) is genotoxic, (e) is radioactive, or (f) is non-hazardous general waste (Park 28e, Ch 16, p 944).

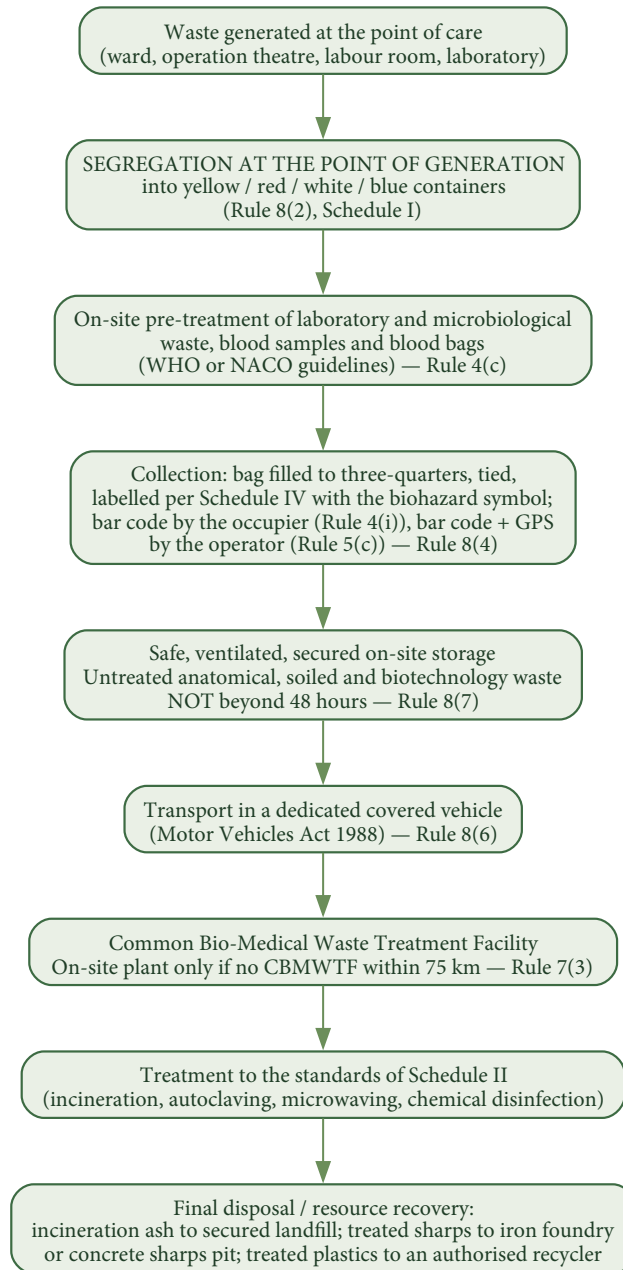
Hazard class	Mechanism and consequence
Infectious waste and sharps	Pathogens enter through a puncture, abrasion or cut in the skin, through mucous membranes, by inhalation or by ingestion. Strong evidence exists for transmission of human immunodeficiency virus (HIV) and hepatitis B and C viruses through healthcare waste; antibiotic- and disinfectant-resistant bacteria add to the hazard (Park 28e, Ch 16, p 944)
Chemical and pharmaceutical waste	Toxic, corrosive, flammable or explosive injury; expired drugs and cytotoxics are genotoxic
Genotoxic waste	Cytotoxic drugs and contaminated items — mutagenic, teratogenic, carcinogenic risk to handlers
Radioactive waste	Radiation injury from sealed sources and contaminated material
Groups at risk	Doctors, nurses, healthcare auxiliaries and maintenance staff; patients; visitors; support-service workers (laundry, waste handling, transport); workers at landfills and incinerators, including scavengers (Park 28e, Ch 16, p 944)

Steps in hospital waste management

- **Waste minimisation at source** — reduce, reuse and recycle; purchase in bulk and prefer non-chlorinated products, because the Rules require chlorinated plastic bags, gloves and blood bags to be phased out (BMW Rules 2016, Rule 4(d)).
- **Segregation at the point of generation** by the person who generates the waste, into four colour-coded bags or containers as per Schedule I — the single most important step, because mixing converts a small hazardous fraction into a wholly hazardous load (BMW Rules 2016, Rule 8(2); Suryakantha, Ch 13, p 116).
- **Collection** in thick, non-corrosive, non-chlorinated disposable plastic bags inside covered, foot-operated, hard containers placed at strategic points in every department; bags filled to a maximum of **three-quarters** of capacity, then tied and labelled (Suryakantha, Ch 13, p 116; Park 28e, Ch 16, p 948).
- **Labelling** — the biohazard symbol and, where relevant, the cytotoxic hazard symbol, prominently visible and non-washable, as specified in Schedule IV (BMW Rules 2016, Rule 8(3)). In addition, **bar code and global positioning system shall be added by the occupier and the**

common treatment facility within one year (Rule 8(4)); the corresponding duties are a **bar-code system** for bags leaving the premises, on the **occupier** (Rule 4(i)), and **bar coding together with a global positioning system**, on the **operator** of the common facility (Rule 5(c)).

- ▶ **Internal transport** by dedicated covered handcart or trolley, cleaned with disinfectant, along a route and at a time that avoids patient and visitor traffic; external transport in a roadworthy covered vehicle complying with the Motor Vehicles Act 1988 (Suryakantha, Ch 13, p 116; BMW Rules 2016, Rule 8(6)).
- ▶ **On-site storage** in a safe, ventilated, secured room away from service areas, protected from rodents and animals; **untreated human anatomical, animal anatomical, soiled and biotechnology waste must not be stored beyond forty-eight hours** (BMW Rules 2016, Rules 4(b), 8(7)).
- ▶ **Pre-treatment on site** of laboratory waste, microbiological waste, blood samples and blood bags by disinfection or sterilisation as per World Health Organization or National AIDS Control Organisation guidelines, before hand-over (BMW Rules 2016, Rule 4(c)).
- ▶ **Treatment and disposal** as per Schedule I, meeting the standards of Schedule II, at a **Common Bio-Medical Waste Treatment Facility (CBMWTF)**; no occupier may set up an on-site treatment plant if a CBMWTF is available within **75 kilometres** (BMW Rules 2016, Rule 7).



Cradle-to-grave pathway of bio-medical waste under the Bio-Medical Waste Management Rules 2016

Segregation and colour coding — Schedule I, BMW Rules 2016

Color	Type of waste	Container	Treatment and disposal
Yellow	(a) Human anatomical waste — tissues, organs, body parts, foetus below the viability period (Medical Termination of Pregnancy Act 1971); (b) animal anatomical waste — experimental carcasses, body parts, organs, tissues	Yellow non-chlorinated plastic bags	Incineration or plasma pyrolysis or deep burial* — no alternative route is listed for these two

Colour	Type of waste	Container	Treatment and disposal
Yellow	(c) Soiled waste — dressings, plaster casts, cotton swabs, bags with residual blood or blood components	Yellow non-chlorinated plastic bags	Incineration or plasma pyrolysis or deep burial*. In the absence of those facilities — autoclaving or microwaving/hydroclaving followed by shredding or mutilation, or a combination of sterilisation and shredding; treated waste sent for energy recovery
Yellow	(d) Expired or discarded medicines, including cytotoxic drugs and items contaminated with them	Yellow non-chlorinated bags or containers	Cytotoxics: returned to manufacturer, or incineration/plasma pyrolysis at >1200°C, or encapsulation. Other medicines: returned to manufacturer or incinerated. Cytotoxic vials must never be handed to an unauthorised person
Yellow	(e) Chemical solid waste; (f) chemical liquid waste (silver X-ray developing liquid, discarded formalin, aspirated body fluids, laboratory and floor washings)	Solids in yellow containers; liquids to a separate collection system leading to an effluent treatment plant	Solids — incineration, plasma pyrolysis or encapsulation at a hazardous-waste facility. Liquids — pre-treated before mixing with other wastewater; combined discharge to conform to Schedule III norms
Yellow	(g) Discarded linen, mattresses, bedding contaminated with blood or body fluid	Non-chlorinated yellow bags or suitable packing	Non-chlorinated chemical disinfection followed by incineration, plasma pyrolysis or energy recovery
Yellow	(h) Microbiology, biotechnology and other clinical laboratory waste — blood bags, cultures, stocks, live or attenuated vaccines, cell cultures, residual toxins, culture dishes	Autoclave-safe bags or containers	Pre-treat on site to sterilise with non-chlorinated chemicals as per NACO or WHO guidelines, thereafter incineration
Red	Contaminated recyclable waste — disposable tubing, bottles, intravenous sets, catheters, urine bags, syringes without needles, vacutainers with needles cut, gloves	Red non-chlorinated plastic bags or containers	Autoclaving or microwaving/hydroclaving followed by shredding or mutilation; treated waste to registered recyclers, energy recovery, plastic-to-fuel or road-making. Plastic waste must not go to landfill
White (translucent)	Waste sharps including metals — needles, fixed-needle syringes, needles from a needle-tip cutter or burner, scalpels, blades, any contaminated sharp object	Puncture-proof, leak-proof, tamper-proof container	Autoclaving or dry-heat sterilisation , then shredding or mutilation or encapsulation in metal container or cement concrete; final disposal to iron foundries (with consent of the State Pollution Control Board), sanitary landfill, or a designated concrete sharps pit

Colour	Type of waste	Container	Treatment and disposal
Blue	(a) Glassware — broken, discarded or contaminated glass, medicine vials and ampoules (except cytotoxic-contaminated); (b) metallic body implants	Cardboard boxes with blue marking	Disinfection by washing with detergent and sodium hypochlorite, or autoclaving/microwaving/hydroclaving, then recycling

\Deep burial is permitted only in rural or remote areas with no access to a CBMWTF, with prior approval of the prescribed authority (Schedule I footnote, BMW Rules 2016).*

Treatment technologies and their prescribed standards — Schedule II and Schedule I Part-2

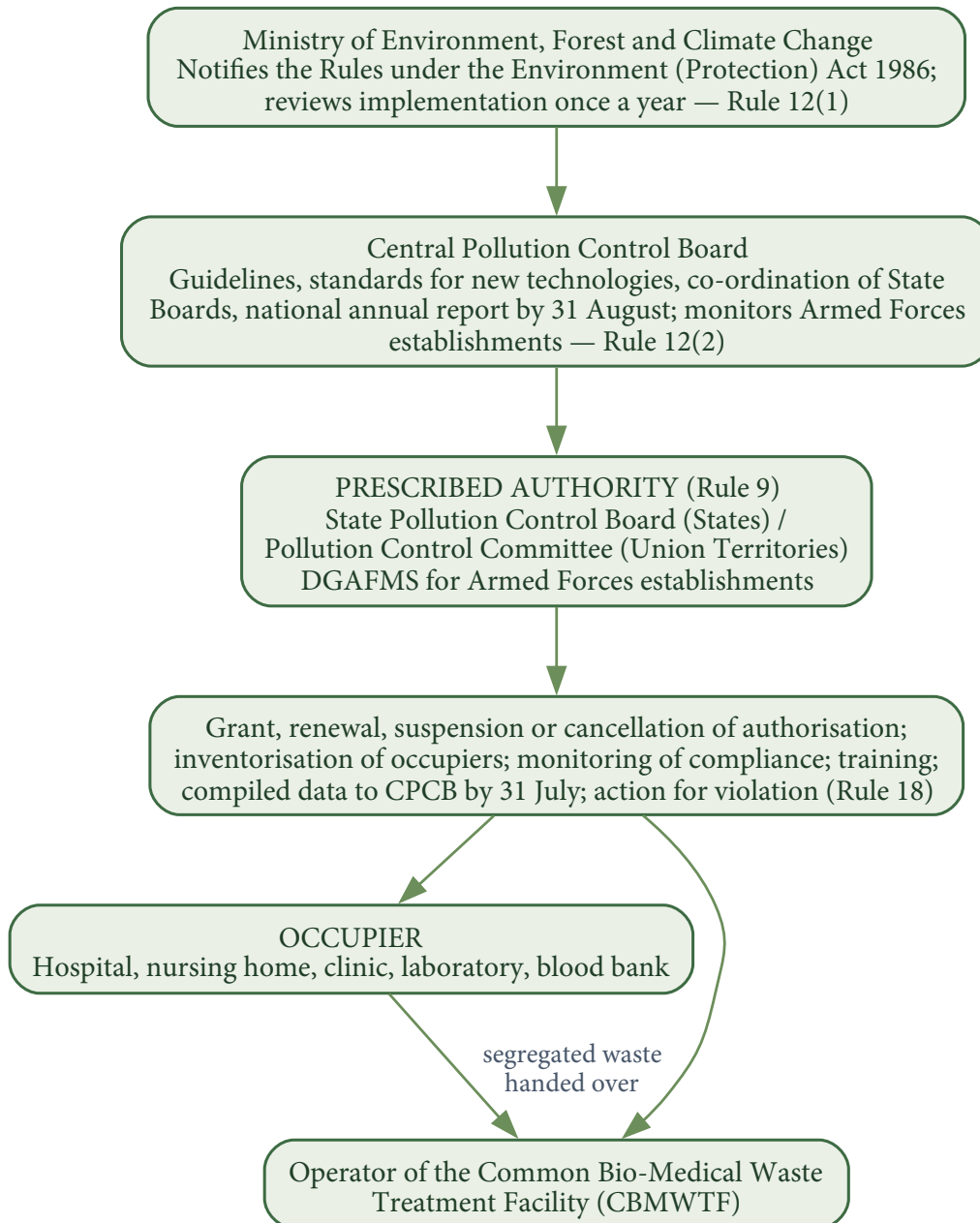
Technology	Standard laid down
Incineration (Schedule II, 1A)	Combustion efficiency at least 99.00 per cent ; primary chamber minimum 800°C ; secondary chamber minimum 1050 ± 50°C ; secondary-chamber gas residence time at least 2 seconds
Plasma pyrolysis / gasification (Schedule II, 2A)	Combustion efficiency at least 99.99 per cent — the standard that distinguishes it from incineration's 99.00 per cent; combustion chamber after gasification 1050 ± 50°C , gas residence time at least 2 seconds , minimum 3 per cent oxygen in stack gas, stack at least 30 m; chlorinated plastics and chlorine-disinfected waste must not be treated
Autoclaving — gravity flow (Schedule II, 3(1))	121°C at 15 psi for ≥60 min , or 135°C at 31 psi for ≥45 min , or 149°C at 52 psi for ≥30 min
Autoclaving — vacuum (Schedule II, 3(2))	Minimum three pre-vacuum pulses, then 121°C at 15 psi for ≥45 min or 135°C at 31 psi for ≥30 min . Under no circumstances below 121°C, 15 psi, 30 min
Autoclave validation (Schedule II, 3(5)-(7))	Biological indicator <i>Geobacillus stearothermophilus</i> spores, at least 1 × 10⁶ spores — spore test weekly, validation test 3-monthly, chemical-indicator strip with every batch
Microwaving (Schedule II, 4)	Not to be used for cytotoxic, hazardous or radioactive waste, contaminated animal carcasses, body parts or large metal items. Biological indicator <i>Bacillus atrophaeus</i> spores, at least 1 × 10⁴ spores per detachable strip
Deep burial (Schedule II, 5)	Pit or trench about 2 metres deep , half filled with waste, covered with lime to within 50 cm of the surface , then filled with soil; 10 cm of soil added each time waste is put in; site impermeable and animal-proof, authorised by the prescribed authority, records maintained; ground-water table at least 6 metres below the lower level of the pit
Chemical disinfection — reagent (Schedule I, Part-2(2))	Chemical treatment using at least 10 per cent sodium hypochlorite having 30 per cent residual chlorine for twenty minutes , or any other equivalent chemical reagent
Chemical disinfection — efficacy (Schedule II, 6)	Microbial inactivation is expressed as " log₁₀ kill "; the system must demonstrate a 4 log₁₀ reduction (99.99 per cent kill) in microbial count or greater for <i>Bacillus subtilis</i> (ATCC 19659)

Technology	Standard laid down
Dry-heat sterilisation of sharps (Schedule II, 7)	Temperature not less than 185°C for a residence period of at least 150 minutes in each cycle, with automatic recording of operating parameters; validated 3-monthly with <i>Geobacillus stearothermophilus</i> or <i>Bacillus atrophaeus</i> vials of at least 10 ⁶ spores per ml
Pre-treatment before incineration (Schedule I, Part-2(4))	There shall be no chemical pre-treatment before incineration, except for microbiological, laboratory and highly infectious waste
In-house incinerator (Schedule I, Part-2(10))	Installation of an in-house incinerator is not allowed; it may be installed only where there is no common facility nearby, after authorisation from the State Pollution Control Board

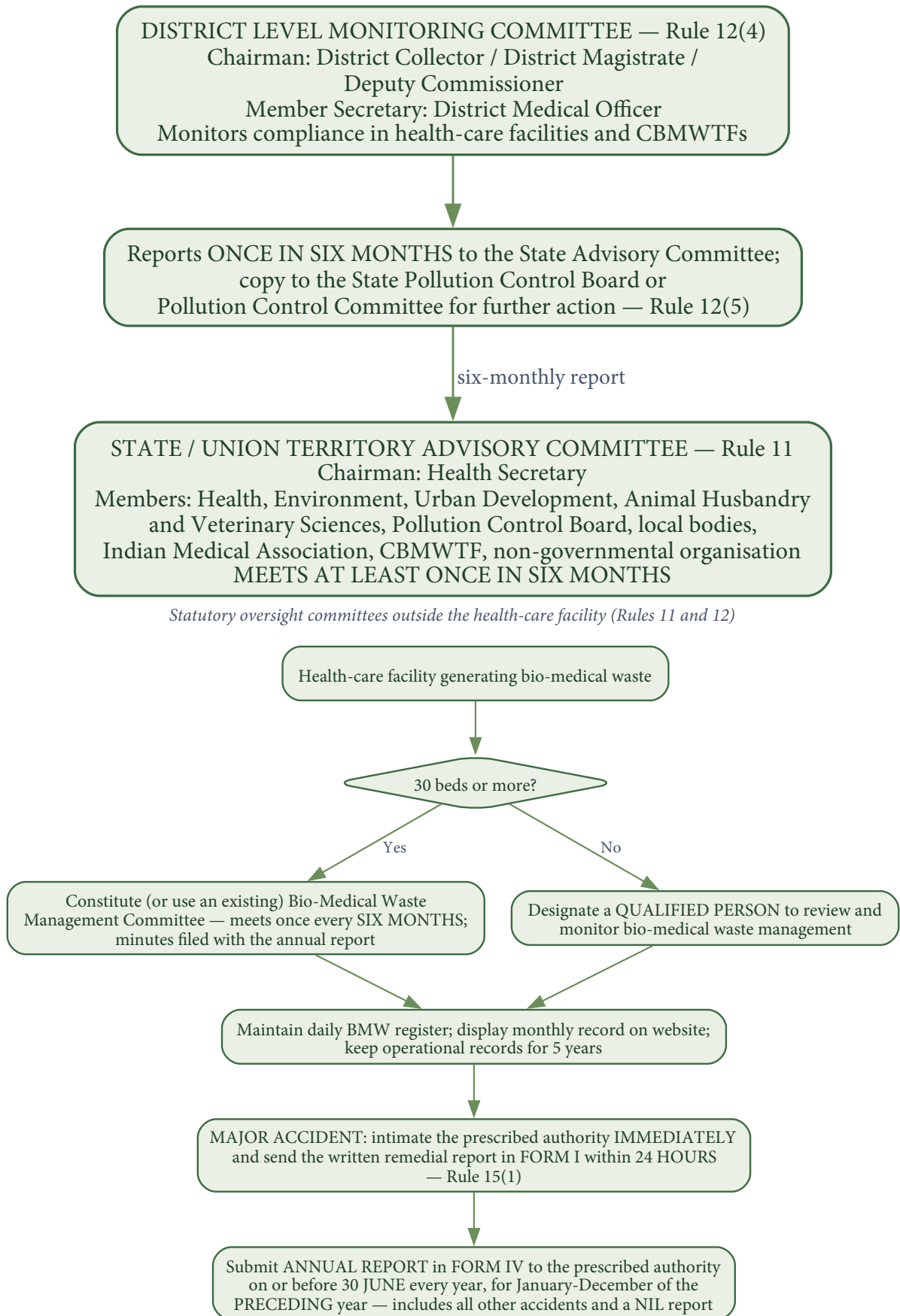
Statutory duties, regulatory framework and monitoring

Provision	Requirement under BMW Rules 2016
Occupier (Rule 3(m))	Any person with administrative control over an institution generating bio-medical waste — hospital, nursing home, clinic, dispensary, veterinary institution, animal house, laboratory, blood bank, any clinical establishment, irrespective of system of medicine
Authorisation (Rule 10)	Application in Form II to the prescribed authority; provisional authorisation in Form III; validity of a bedded facility's authorisation synchronised with its consents; one-time for non-bedded occupiers, deemed granted if not objected to within 90 days
Prescribed authority (Rule 9)	State Pollution Control Boards for States, Pollution Control Committees for Union Territories; Director General, Armed Forces Medical Services for Armed Forces establishments
Training (Rule 4(g))	All health-care workers and waste handlers trained at induction and at least once every year; numbers trained and untrained reported annually
Immunisation (Rule 4(h))	All handlers immunised against hepatitis B and tetanus
Occupational safety (Rules 4(l), 4(m))	Appropriate and adequate personal protective equipment; health check-up at induction and at least once a year, records maintained
Records and register (Rules 4(n), 4(s), 14)	Bio-medical waste register updated daily, monthly record displayed on the facility's website; operational records of incinerator/autoclave and all handling records kept for 5 years, open to inspection by the prescribed authority or the Ministry
Institutional monitoring committee (Rule 4(r))	A committee to review and monitor bio-medical waste management, meeting once every six months, minutes filed with the annual report; an establishment with fewer than 30 beds designates a qualified person instead
State / Union Territory Advisory Committee (Rule 11)	Constituted by every State or Union Territory under the chairmanship of the Health Secretary, with members from Health, Environment, Urban Development, Animal Husbandry and Veterinary Sciences, the Pollution Control Board, local bodies, the Indian Medical Association, a common facility and a non-governmental organisation; meets at least once in six months to oversee implementation. The Ministry of Defence constitutes a parallel Advisory Committee (Defence) under the Director General of Health Services of the Armed Forces

Provision	Requirement under BMW Rules 2016
District Level Monitoring Committee (Rule 12(4)-(6))	Constituted by every State or Union Territory under the chairmanship of the District Collector / District Magistrate / Deputy Commissioner , with the District Medical Officer as Member Secretary ; monitors compliance in health-care facilities and common facilities and reports once in six months to the State Advisory Committee , with a copy to the Pollution Control Board
Central monitoring (Rules 12(1)-(3))	The Ministry of Environment, Forest and Climate Change reviews implementation once a year through State Health Secretaries and the Pollution Control Boards; the Central Pollution Control Board monitors all Armed Forces health-care establishments and may inspect them after prior intimation to the Director General, Armed Forces Medical Services
Accidents (Rules 4(o), 15)	On any major accident (including one caused by fire hazard or blast) the authorised person shall intimate the prescribed authority IMMEDIATELY and forward a written report of the remedial steps taken in FORM I WITHIN 24 HOURS . All other accidents and the remedial steps taken are reported in the annual report — a nil report must still be filed
Annual report (Rule 13)	Occupier and operator submit Form IV to the prescribed authority on or before 30 June every year, covering January to December of the preceding year ; the prescribed authority sends compiled data to the Central Pollution Control Board by 31 July , and the Board to the Ministry by 31 August ; all annual reports placed online
Liability (Rule 18)	Occupier and CBMWTF operator liable for all damage to environment or public; action under sections 5 and 15 of the Environment (Protection) Act 1986



Administrative framework for bio-medical waste management in India



Institutional review and reporting requirement for a health-care facility (Rules 4(r), 13 and 15)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Label for bio-medical waste bags and containers** (Park 28e, Ch 16, Fig 2) — draw the three-crescent **biohazard symbol**, and beside it the **cytotoxic hazard symbol** with the caption "CYTOTOXIC — HANDLE WITH CARE". Add the note: *label shall be non-washable and prominently visible*.
- ▶ **Four colour-coded bins in a row** — label each bin with its colour and one example: **YELLOW** (anatomical and soiled waste, non-chlorinated bag), **RED** (contaminated recyclable plastics, non-chlorinated bag), **WHITE / translucent** (sharps, puncture-proof and leak-proof container), **BLUE** (glassware and metallic implants, cardboard box with blue marking). Mark each bin **foot-operated lid, bag filled to $\frac{3}{4}$ only**.



Biohazard symbol required on bio-medical waste bags and containers (AFMC Red Book of Community Medicine 2024, p.460)

High-yield exam pointers

- ▶ **Four colours, four streams (Y-R-W-B):** Yellow = incinerate (anatomical, soiled, expired drugs, chemical, linen, laboratory); Red = autoclave then recycle (contaminated plastics); White = sharps in puncture-proof container; Blue = glass and metallic implants for recycling.
- ▶ **The numbers that must appear:** 75-90 per cent general vs 10-25 per cent hazardous waste; 0.5-4 kg per bed per day; **48 hours** maximum storage of untreated anatomical/soiled/biotechnology waste; **75 km** CBMWTF rule; **90 days** deemed authorisation for non-bedded occupiers; records for **5 years**; committees every **6 months**; training and health check-up **yearly**; bar code and global positioning system within **1 year**.
- ▶ **The reporting calendar:** major accident — intimate **immediately**, Form I within **24 hours** (Rule 15); annual report in **Form IV by 30 June** for January-December of the preceding year; prescribed authority to Central Pollution Control Board by **31 July**; Board to Ministry by **31 August**.
- ▶ **Incinerator:** primary **800°C**, secondary **1050 ± 50°C**, residence **2 seconds**, combustion efficiency **≥99.00 per cent**; **plasma pyrolysis ≥99.99 per cent**; cytotoxics at **>1200°C**.
- ▶ **Autoclave:** **121°C / 15 psi / 60 minutes** (gravity) — never below 121°C, 15 psi, 30 minutes; indicator *Geobacillus stearothermophilus* 10⁶ spores. Microwave indicator *Bacillus atrophaeus* 10⁴ spores. **Dry heat for sharps: 185°C for 150 minutes.**
- ▶ **Chemical disinfection:** 10 per cent sodium hypochlorite with 30 per cent residual chlorine for 20 minutes, demonstrating a **4 log₁₀ reduction (99.99 per cent kill)** for *Bacillus subtilis* ATCC 19659. **Deep burial:** 2 m pit, lime to within 50 cm of surface, water table **≥6 m** below.
- ▶ **Who watches whom:** occupier's own committee (Rule 4(r)) → **District Level Monitoring Committee** under the District Collector (Rule 12) → **State Advisory Committee** under the Health Secretary (Rule 11) → Ministry review once a year (Rule 12(1)). Prescribed authority = **State Pollution Control Board / Pollution Control Committee**; parent statute = **Environment (Protection) Act 1986**; liability under **sections 5 and 15**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the older ten-category classification of the Bio-Medical Waste (Management and Handling) Rules 1998 (amended 2000 and 2003), still printed in some textbooks, has been superseded by the four colour-coded categories of Schedule I of the 2016 Rules and should not be reproduced as current; the notified 2016 text retrieved here is the operative version used above, and separate Central Pollution Control Board guidelines (Government of India, Revision 5, 26 April 2022) govern waste generated during the treatment, diagnosis or quarantine of COVID-19 patients, requiring double-layered bags labelled "COVID-19 Waste" (Park 28e, Ch 16, p 948).

Recent advances [RA]

Current as of 2026-08-16 · depth: elaborate · sources: PubMed, MoEFCC, CPCB, WHO

Programme and policy updates (India)

- ▶ **Bio-Medical Waste Management (Amendment) Rules 2026** — G.S.R. 293(E), notified 17 April 2026 by the Ministry of Environment, Forest and Climate Change, brings **Ayush** into the

statutory oversight machinery: the word "Ayush" is inserted into the list of departments represented on the **State / Union Territory Advisory Committee under Rule 11(1)**, and a **representative nominated by the State or Union Territory Department of Ayush** is added alongside the District Health Officer in the **District Level Monitoring Committee under Rule 12(6)**. The committee compositions printed in the textbooks predate this and are now incomplete. The amendment chain to quote is: principal rules G.S.R. 343(E), 28 March 2016, amended by G.S.R. 234(E), 16 March 2018; G.S.R. 129(E), 19 February 2019; G.S.R. 360(E), 10 May 2019; and G.S.R. 293(E), 17 April 2026. [MoEFCC 2026-Apr]

- **Bar coding is moving from a facility-level duty to a national tracking portal** — acting on Supreme Court directions in W.P.(C) 13029 of 1985 (I.A. 181745 of 2019 and I.A. 46339 of 2020), the Central Pollution Control Board is developing a **Centralised Barcode System for Tracking of Bio-medical Waste (CBST-BMW)**, an end-to-end platform registering common facilities and health-care facilities and tracking individual waste bags from generation to treatment. The underlying duties are unchanged — bar code by the occupier (Rule 4(i)) and bar code with global positioning system by the operator (Rule 5(c)) — but compliance is now auditable centrally. [CPCB 2024]

Current national data

- **Central Pollution Control Board Annual Report on Bio-medical Waste Management, year 2024** — 815 tonnes per day generated (743 tonnes per day in 2023), of which 810 tonnes per day treated; 4,56,538 health-care facilities (1,38,516 bedded and 3,18,022 non-bedded, totalling 31,51,294 beds), of which 3,65,604 authorised; 245 Common Bio-Medical Waste Treatment Facilities in operation with 30 under construction (234 + 30 in 2023); installed capacity 2,097 tonnes per day (incineration 1,008, autoclaving 1,089) of which only 39 per cent is utilised; Online Continuous Emission Monitoring Systems installed at 225 of 245 facilities (92 per cent); captive incinerators down from 86 to 72; 27,313 facilities in violation and 14,783 show-cause notices. **Captive treatment including deep burial persists in 25 States and Union Territories** — Himachal Pradesh alone accounting for 3,953 facilities in 2024 (4,454 in 2023) — corroborated by a state-level analysis that found deep burial still being practised in two districts despite a state prohibition. [CPCB 2024] [PMID 37820814]
- **Nationwide compliance survey of 267 health-care facilities (July to September 2023; 67 per cent accredited)** — only 58 per cent had fully implemented bar coding and 43 per cent used dedicated tags to secure bags; 49 per cent still cut needles before disposal; 50 per cent used blue bins with blue covers instead of the puncture-proof, leak-proof blue-marked containers the amended Schedule I now prescribes; mercury-filled devices were still in use in 28 per cent; laboratory waste was autoclaved in 79 per cent and a dedicated temporary storage area existed in 92 per cent. Sharps disposal, blue-bin use, pharmaceutical waste and mercury devices were the four practices varying most across the country. [PMID 40744379]

Elimination, surveillance and disease status

- **Pooled evidence now quantifies the harm caused by segregation failure** — systematic review and meta-analysis of 94 studies from low- and middle-income countries: inadequate segregation 68.4 per cent (95% CI 61.2-75.1), open dumping or uncontrolled burning the commonest disposal route at 57.1 per cent, functional incinerators in only 18.3 per cent, and a twelve-month

needle-stick injury prevalence of **42.1 per cent**. Where waste management was poor, health-care workers had **4.31 times the odds of needle-stick injury** (95% CI 3.02-6.15) and **3.45 times the odds of hepatitis B infection** (95% CI 2.10-5.68); waste handlers additionally showed raised odds of dermatological (5.12), respiratory (3.89) and gastrointestinal (2.98) illness. This converts the textbook statement that HIV and hepatitis B and C are transmissible through health-care waste into a quantified effect size. [PMID 41764427]

- **Indian occupational-exposure surveillance** — a tertiary hospital in Manipur reported a one-year needle-stick and sharps injury prevalence of **23.8 per cent**, recapping being the commonest precipitating act (28.2 per cent) and **61.2 per cent of injuries never reported** to the authorities; in southern Indian tertiary hospitals **24.1 per cent of all exposures occurred while handling or discarding bio-medical waste**, with post-exposure prophylaxis started beyond 24 hours in 15.1 per cent; and a Tamil Nadu audit of 2,593 observed disposal events found **73 per cent appropriate**, with significantly better odds for nurses (adjusted OR 1.54, 95% CI 1.06-2.23), doctors (1.60, 1.05-2.45) and inpatient areas (2.77, 1.95-3.94). Under-reporting and recapping, not ignorance of colour coding, are the operative failures. [PMID 40214315] [PMID 41802105] [PMID 36096850]

Guideline and legislative updates

- **Amendment-updated Schedule detail that supersedes the as-notified 2016 text** — chemical disinfection now reads **1 to 2 per cent sodium hypochlorite** having 30 per cent residual chlorine for twenty minutes, **not 10 per cent** (Schedule I, Part-2, item 2); "**routine mask and gown**" was added to yellow category (g); yellow (h) microbiology, biotechnology and laboratory waste now goes into autoclave, microwave or hydroclave-safe bags or containers and is pre-treated per **World Health Organization guidance and the WHO Blue Book 2014**, the National AIDS Control Organisation reference having been deleted; **blue category glassware and metallic implants now go into puncture-proof, leak-proof containers with blue marking rather than cardboard boxes**; chlorinated plastic bags **excluding blood bags** and gloves were to be phased out by **27 March 2019**, and urine bags, effluent bags, abdominal bags and chest drainage bags are expressly outside the expression "chlorinated plastic bags"; the prescribed authority reports to the Central Pollution Control Board in **Form IV A** by 31 July; bedded facilities must display the **monthly** waste record on their website; and facilities with **fewer than ten beds** were to comply with the liquid-waste output discharge standard by **31 December 2019**. [MoEFCC 2018-Mar] [MoEFCC 2019-Feb] [MoEFCC 2019-May]
- **Current technical guidance and the emission benchmark behind the non-burn preference** — the Central Pollution Control Board's **Guidelines for Common Bio-medical Waste Treatment and Disposal Facilities** were revised on **16 April 2025**, supported by a **Revision 1 gap-analysis methodology (2024)** for matching State-wise generation against available treatment capacity, and by separate guidance on waste generated during the **Universal Immunisation Programme** and on mercury waste from health-care facilities. For stack emissions the widely adopted dioxin limit is **0.1 ng TEQ/Nm³**; bio-medical waste incinerators are the dominant plant type in developing countries and contribute the largest share of their national dioxin, furan and dioxin-like polychlorinated biphenyl inventories, which is the technical argument for autoclaving and hydroclaving wherever the category permits. [CPCB 2025-Apr] [PMID 38176262]

Global and WHO developments

- ▶ **World Health Organization position, fact sheet updated 24 October 2024** — 85 per cent of health-care waste is general and 15 per cent hazardous, a tighter split than the 75-90 / 10-25 per cent range carried in the textbooks; **only 61 per cent of hospitals had basic health-care waste services (2021)** and just **25 per cent of health facilities in fragile settings (2023)**; roughly **16 billion injections** are administered worldwide each year; and after a needle-stick from an infected source the transmission risk is **30 per cent for hepatitis B, 1.8 per cent for hepatitis C and 0.3 per cent for HIV**. High-income countries generate about 0.5 kg and low-income countries about 0.2 kg of hazardous waste per bed per day. [WHO 2024-Oct]
- ▶ **Waste minimisation is being reframed as climate and circular-economy action** — a pooled analysis of 17 studies of primary hip and knee arthroplasty found **12.27 kg of waste per case (95% CI 10.88-13.66)** of which only **14.5 per cent was recyclable**, and **135.37 kg carbon dioxide equivalent per case** on full life-cycle assessment; an Indian "green dialysis" audit measured **1.29 kg of waste, 250 litres of net water and 3 kWh of electricity per session**. These figures give the first step of the management hierarchy — waste minimisation at source — a measurable environmental and financial denominator rather than an exhortation. [PMID 42143633] [PMID 38716524]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Chapter 16, Hospital Waste Management, pp 943-948; Bio-Medical Waste Management Rules 2016 (Ministry of Environment, Forest and Climate Change, Government of India) — Rules 3, 4, 5, 7, 8, 9, 10, 11, 12, 13, 14, 15, 18, Forms I and IV, and Schedules I (Parts 1 and 2), II, III and IV; Suryakantha — Community Medicine with Recent Advances, Chapter 13, Management of Hospital Waste, pp 112-117; AFMC Red Book of Community Medicine 2024 (Fig 17.4, biohazard symbol, p 460); National Environmental Engineering Research Institute 1997 (composition of hospital waste in India, quoted in Park 28e).

Recent-advances sources: 40744379 — Gap assessment in implementation of Biomedical Waste Management in Health Care Facilities: Nationwide survey in India, Indian Journal of Medical Microbiology 2025, doi:[10.1016/j.ijmmb.2025.100947](https://doi.org/10.1016/j.ijmmb.2025.100947) (abstract-only); 41764427 — Medical waste disposal practices and their health consequences in resource-limited settings: a systematic review and meta-analysis, BMC Public Health 2026, doi:[10.1186/s12889-026-26722-9](https://doi.org/10.1186/s12889-026-26722-9); 36096850 — Biomedical waste disposal practices among healthcare workers during COVID-19 pandemic in secondary and tertiary care facilities of Tamil Nadu, Indian Journal of Medical Microbiology 2022, doi:[10.1016/j.ijmmb.2022.08.011](https://doi.org/10.1016/j.ijmmb.2022.08.011); 40214315 — Needlestick and Sharp Injuries among Health Care Workers in a Tertiary Care Hospital in Manipur, India: A Cross-sectional Study, Indian Journal of Public Health 2025, doi:[10.4103/ijph.1263_23](https://doi.org/10.4103/ijph.1263_23) (abstract-only); 41802105 — Clinical profile of healthcare employees with needlestick and sharps injuries and infectious body fluid exposure in tertiary care hospitals in southern India, International Journal of Risk and Safety in Medicine 2026, doi:[10.1177/09246479261432856](https://doi.org/10.1177/09246479261432856) (abstract-only); 37820814 — Evaluation of biomedical waste generation in Himachal Pradesh before and during the Covid 19 pandemic, Science of the Total Environment 2023, doi:[10.1016/j.scitotenv.2023.167689](https://doi.org/10.1016/j.scitotenv.2023.167689) (abstract-only); 38176262 — Dioxins emissions from bio-medical waste incineration: a systematic review on emission factors, inventories, trends and health risk studies, Journal of Hazardous Materials 2023, doi:[10.1016/j.jhazmat.2023.133384](https://doi.org/10.1016/j.jhazmat.2023.133384) (abstract-only); 42143633 — Quantifying the environmental footprint of primary hip and knee arthroplasty: a systematic review and pooled-analysis of waste generation and carbon emissions, International Orthopaedics 2026, doi:[10.1007/s00264-026-06854-8](https://doi.org/10.1007/s00264-026-06854-8) (abstract-only); 38716524 — Assessment of Biomedical Waste Generation in Dialysis Units: A Prospective Observational Study — Is it Time for "Green Dialysis"?, Journal of the Association of Physicians of India 2023, doi:[10.59556/japi.71.0383](https://doi.org/10.59556/japi.71.0383) (abstract-only). Regulatory and institutional sources: Bio-Medical Waste Management (Amendment) Rules 2026, G.S.R. 293(E), 17 April 2026 (Ministry of Environment, Forest and Climate Change); Bio-Medical Waste Management (Amendment) Rules 2018, G.S.R. 234(E), 16 March 2018; Bio-Medical Waste Management (Amendment) Rules 2019, G.S.R. 129(E), 19 February 2019; Bio-Medical Waste Management (Second Amendment) Rules 2019, G.S.R. 360(E), 10 May 2019; Central Pollution Control Board, Annual Report on Bio-medical Waste Management for the year 2024; Central Pollution Control Board, Guidelines for Common Bio-medical Waste Treatment and Disposal Facilities, revised 16 April 2025, and Methodology to conduct gap analysis with respect to generation and treatment of biomedical waste, Revision 1,

2024; World Health Organization, health-care waste fact sheet, updated 24 October 2024.



Surveillance of drinking water quality

Asked in: Paper I 24-Sep-2025 — RGUHS MD Community Medicine

Definition — Surveillance of drinking water quality is the continuous and vigilant public health assessment and overview of the safety and acceptability of drinking water supplies. It is **essentially a health measure**, intended to protect the public from water-borne diseases, and is carried out by an agency independent of the water supplier (Park 28e, Ch 15, p 880; Sunder Lal, Ch 3, p 101).

Scope — activities included in the surveillance function

- ▶ **Approval of new sources**, including privately owned supplies.
- ▶ **Watershed protection** of the catchment.
- ▶ **Approval of construction and operating procedures of waterworks** — (i) disinfection of the plant and distribution system after repair or interruption of supply, (ii) periodic flushing and cleaning of storage facilities, (iii) certification of operators, (iv) regulation of chemicals used in treatment, (v) cross-connection control, back-flow prevention and leak detection.
- ▶ **Sanitary surveys**.
- ▶ **Monitoring programmes**, including provision for central and regional analytical laboratory services.
- ▶ **Development of codes of practice** for well construction, pump installation and plumbing.
- ▶ **Inspection and quality control** in bottled-water and ice manufacturing operations (Park 28e, Ch 15, p 880).

Water quality control versus water quality surveillance

Feature	Water quality control (WQC)	Water quality surveillance (WQS)
Who does it	The water supplying agency itself	An independent agency — responsibility entrusted to the health department
Purpose	Meet the quality standards set by the regulatory authority	Ensure a safe supply reaches the community; investigate and give feedback to the supplier
Typical activities	Preventive maintenance; day-to-day control; routine samples at source, conveyance, treatment and distribution, supplemented by sanitary inspection	Residual chlorine testing at consumer level, supply of chlorine tablets, <i>E. coli</i> testing, sanitary inspection of sources, outbreak monitoring, health education, training of <i>Panchayati Raj</i> Institutions
Value	Indicates source quality, treatment efficiency, seasonal variation, need for remedial action	Detects failure of control, links water data to water-borne disease data, triggers corrective action

(Sunder Lal, Ch 3, p 101)

Elements of a surveillance programme

Element	What is done	Key point
1. Sanitary survey	On-the-spot inspection and evaluation, by a qualified person, of the entire water supply system — source, catchment, treatment, storage, distribution and consumer end	Purpose is detection and correction of faults and deficiencies ; it is essential for adequate interpretation of laboratory results
2. Sampling	Collection of representative samples "with the thoroughness of a surgical operation" and with aseptic precautions, by trained personnel, in strict accordance with the methods and frequency prescribed in the WHO <i>Guidelines for Drinking-water Quality</i> or the ICMR <i>Manual of Standards of Drinking Water Supplies</i>	Results of analysis depend entirely on the quality of sampling
3. Bacteriological surveillance	Presumptive coliform test, tests for faecal streptococci and <i>Clostridium perfringens</i> , and colony (plate) count	The most sensitive element — detects faecal pollution
4. Biological examination	Microscopy for plankton — algae, fungi, yeasts, protozoa, rotifers, crustaceans, minute worms	Plankton cause objectionable taste and odour and are an index of pollution ; sediment from a Foerst centrifuge is read on a Sedgwick-Rafter cell with a Whipple micrometer
5. Chemical surveillance	Basic tests: pH, colour, turbidity, chlorides, ammonia, chlorine demand and residual chlorine	Increasingly important because of industrial and agricultural pollutants; regular measurement of chlorine residuals may in part replace bacteriological surveillance

(Park 28e, Ch 15, pp 880-881; Suryakantha, Ch 5, p 62)

Complete chemical analysis adds toxic metals, pesticides, persistent organic chemicals and radioactivity; iron and manganese are tested when raw-water levels influence treatment (Park 28e, Ch 15, p 881).

Sampling technique and transport

Purpose	Container and volume	Precautions
Physical and chemical examination	Clean neutral-glass stoppered bottle, not less than 2 litres (Winchester Quart bottle)	Rinse the bottle three times with the water, filling about one-third each time
Bacteriological examination	Sterilised neutral-glass bottle of 200-250 ml with ground-glass stopper and overlapping rim	If chlorine is present or likely, add sodium thiosulphate (0.1 ml of 3 per cent solution, or a small crystal) before sterilisation; bottle opened only at the moment of filling
Tap in regular use	—	Open the tap fully and run to waste for at least 2 minutes
Well fitted with a pump	—	Pump to waste for about 2 minutes , then collect from the delivery

Purpose	Container and volume	Precautions
River, stream or lake	—	Collect mid-stream, well below the surface, mouth facing the current
Transport and storage	—	Begin examination as soon as possible; otherwise keep in ice and analyse within 48 hours — samples not so preserved should not be accepted

(Park 28e, Ch 15, Appendix II, p 884; Suryakantha, Ch 5, p 60)

Bacteriological surveillance — the tests

- **Presumptive coliform test (multiple tube method)** — measured quantities of the sample are inoculated into tubes of MacConkey's lactose bile salt broth with **bromocresol purple** as indicator and incubated for 48 hours. From the number of tubes showing acid and gas, the **most probable number (MPN)** of coliform organisms is read from statistical tables; this is the *presumptive coliform count*.
- **Confirmatory test** — not generally required for unchlorinated water but **required for chlorinated water**. Each presumptive-positive tube is subcultured into two tubes of brilliant green bile broth: one at 37°C for **up to 48 hours** confirms coliform organisms; the other at 44°C, **inspected at 6 and 24 hours**, decides whether *E. coli* is present. *E. coli* is almost the only coliform capable of producing gas from lactose at 44°C; further confirmation is by indole production at 44°C.
- **Membrane filtration technique** — a measured volume is filtered through a cellulose-ester membrane; the retained bacteria are counted after incubating the membrane face upwards on suitable medium, giving results within about **20 hours**, far faster than the multiple tube method.
- **Faecal streptococci and *C. perfringens*** — **confirmatory** evidence of faecal pollution in doubtful cases. Faecal streptococci resist drying and are valuable after laying new mains or repairs and for detecting surface run-off pollution. *C. perfringens* spores survive longer and resist ordinary chlorination; presence **in the absence of coliforms** suggests remote faecal contamination, and in filtered supplies suggests deficient filtration.
- **Colony (plate) count** on nutrient agar at 37°C and 22°C — estimates the general bacterial content. Counts from the same source at frequent intervals are of considerable value: a **sudden rise in the colony count may give the earliest indication of contamination**.

(Park 28e, Ch 15, pp 875-876, 881)

Standards against which results are interpreted

Parameter	Standard / guideline value	Source
<i>E. coli</i> or thermotolerant coliform bacteria — all water directly intended for drinking; treated water entering the distribution system; treated water in the distribution system	Must not be detectable in any 100 ml sample ; immediate investigative action if <i>E. coli</i> is detected	WHO guideline values (Park 28e, Ch 15, p 877; Sunder Lal Table 3.1, WHO 2017)

Parameter	Standard / guideline value	Source
Total coliform bacteria, treated water in distribution system	Must not be detectable in any 100 ml sample; in large supplies where sufficient samples are examined, must not be present in 95 per cent of samples taken throughout any 12-month period	Park 28e, Ch 15, p 877
Colony count, disinfected water at the point of consumption	0 (after 2 days at 37°C) and 20 (after 3 days at 22°C)	Park 28e, Ch 15, p 881
Colony count, non-disinfected water at the point of consumption	10 (after 2 days at 37°C) and 100 (after 3 days at 22°C)	Park 28e, Ch 15, p 881
Grading of disinfected water by coliform count per 100 ml	0 = excellent; 1-3 = satisfactory; 4-10 = suspicious; >10 = unsatisfactory	Suryakantha, Ch 5, p 64
Free residual chlorine (virological safety)	0.5 mg/L after a contact of at least 30 minutes at pH below 8.0 in disinfected supplies in areas endemic for hepatitis A; 0.2 mg/L for half an hour elsewhere; raised above 0.5 mg/L throughout the system in outbreaks and emergencies	Park 28e, Ch 15, p 876; Sunder Lal, Ch 3, p 100
Turbidity	≤1 nephelometric turbidity unit (NTU) before disinfection; should always be under 5 NTU	Park 28e, Ch 15, p 876; Sunder Lal, Ch 3, p 100
Fluoride	WHO 1.5 mg/L; Bureau of Indian Standards upper limit 1 mg/L	Sunder Lal, Ch 3, p 103
Arsenic	0.01 mg/L (provisional guideline value)	Park 28e, Ch 15, p 877

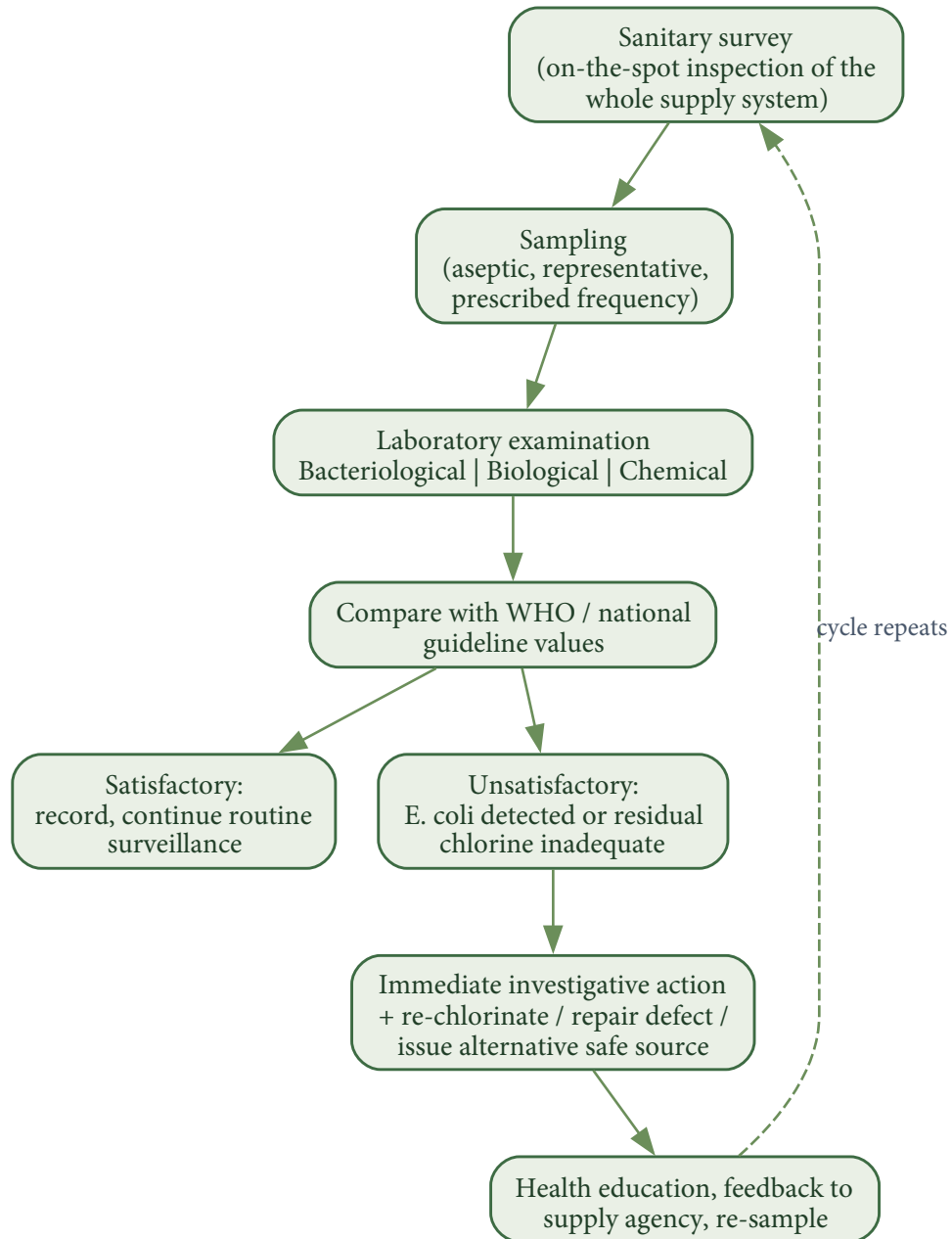
Surveillance of drinking water quality in India

- **National Rural Drinking Water Quality Monitoring and Surveillance Programme**, launched **February 2006**, seeks community participation in monitoring and surveillance of drinking water sources at grass-root level by village *Panchayats* and **Village Water and Sanitation Committees**, supported by **district laboratories**. Hydrogen sulphide (H₂S) strips are supplied to accredited social health activists (ASHAs) and auxiliary nurse midwives (ANMs) (Sunder Lal, Ch 3, p 107).
- **Minimum district-level capacity** to be developed for residual chlorine, coliform count and *E. coli* count (Sunder Lal, Ch 3, p 107).
- **Field test kits for potable water quality assurance** — portable, run on an alternative power source, and test essential physical, chemical and bacteriological parameters on the spot and in emergencies. Developed by the Defence Research Laboratory Jodhpur, the National Environmental Engineering Research Institute (NEERI) Nagpur, the Industrial Toxicology Research Centre Lucknow, the All India Institute of Hygiene and Public Health Kolkata, and the Central Scientific Instrument Organization Chandigarh (Sunder Lal, Ch 3, p 102).
- **H₂S strip test (Manja)** — a very cheap **qualitative** screen for microbiological quality: fill the bottle of dry sterile medium to the label, shake gently for 5 minutes, keep at 30-35°C (or 37°C incubator) for 18 hours; **blackening of the medium means the water is unfit** (Sunder Lal, Ch 3, p 102).

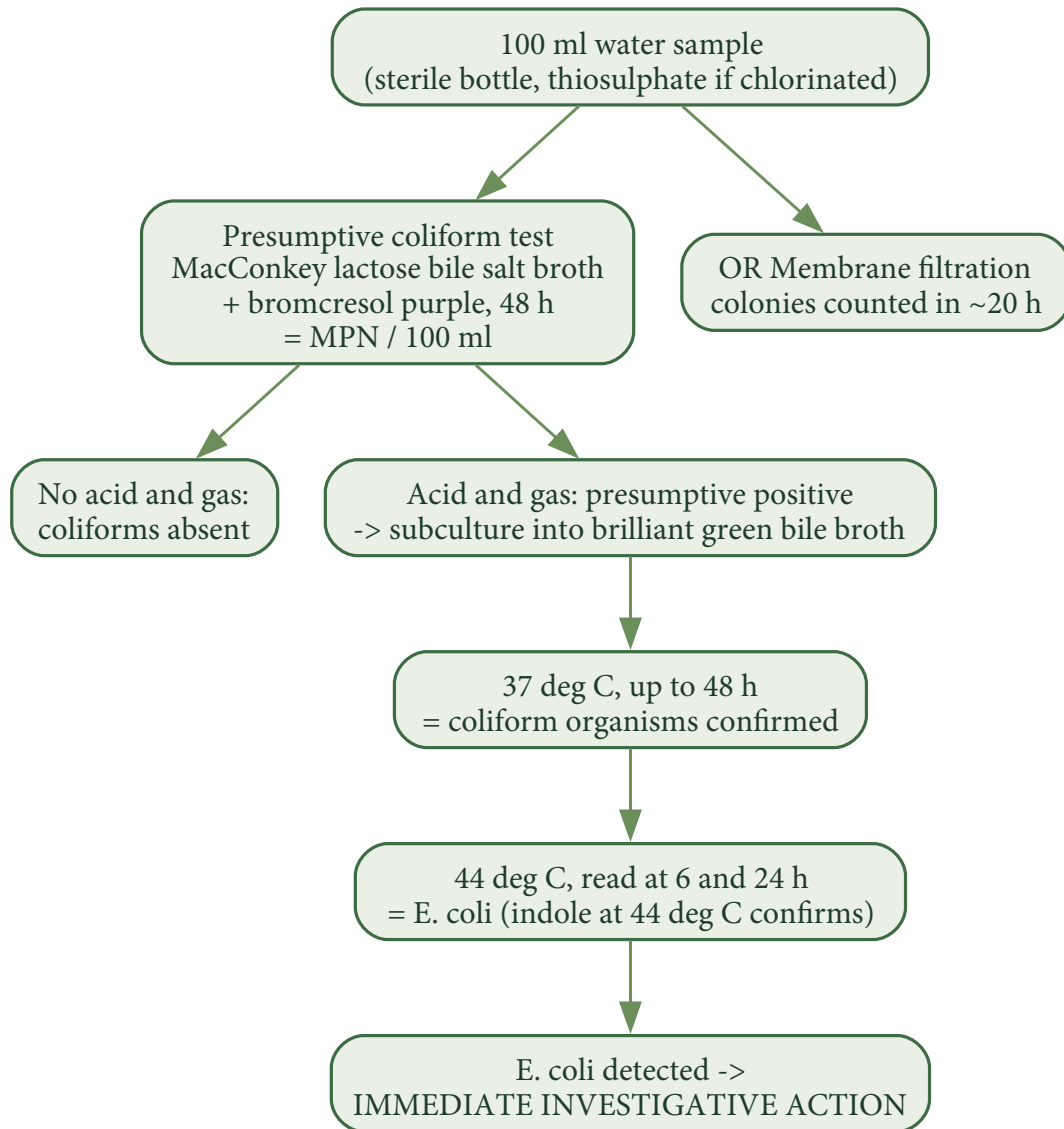
- **Programme linkage** — surveillance sits under the **Ministry of Jal Shakti**. The **Jal Jeevan Mission** (*Nal Jal Yojana, Har Ghar Jal*) aims at a Functional Household Tap Connection delivering **at least 55 litres per person per day of prescribed quality** to every rural household (Park 28e, Ch 7, p 573; Sunder Lal, Ch 3, p 107).
- **Disease-side surveillance** — the water-borne conditions under regular surveillance in the **Integrated Disease Surveillance Programme** are acute diarrhoeal disease (cholera) and typhoid, giving early warning that complements water testing (Park 28e, Ch 7, p 569).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **DRAWN figure** — the **multiple tube (MPN) presumptive coliform test**. Label: rows of test tubes containing *MacConkey's lactose bile salt broth with bromcresol purple*; inverted *Durham tube* in each; measured volumes of sample water inoculated into each row; incubation 37°C for 48 hours; tubes showing *acid (yellow) and gas*; *statistical (probability) tables* → *most probable number of coliforms per 100 ml*.



Operational cycle of drinking water quality surveillance



Bacteriological examination of a drinking water sample

High-yield exam pointers

- ▶ **Five elements of surveillance** — Sanitary survey, Sampling, Bacteriological surveillance, Biological examination, Chemical surveillance (mnemonic: S-S-B-B-C).
- ▶ *E. coli* or thermotolerant coliforms **must not be detectable in any 100 ml sample** at all three points — water intended for drinking, treated water entering the distribution system, treated water in the distribution system.
- ▶ Colony counts at the point of consumption: **disinfected 0 / 20, not disinfected 10 / 100** (37°C for 2 days / 22°C for 3 days).
- ▶ Free residual chlorine **0.5 mg/L for 30 minutes at pH <8.0**; turbidity ≤1 NTU before disinfection.
- ▶ Grading by coliforms per 100 ml: **0 excellent, 1-3 satisfactory, 4-10 suspicious, >10 unsatisfactory.**
- ▶ Sample volumes: **2 litres** (Winchester Quart) for physical/chemical; **200-250 ml** sterile for bacteriological; analyse **within 48 hours** on ice.

- India: National Rural Drinking Water Quality Monitoring and Surveillance Programme, February 2006; H₂S strip test is qualitative — black = unfit; Jal Jeevan Mission target 55 litres per capita per day through a Functional Household Tap Connection.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park's 28th edition reproduces the World Health Organization drinking water guideline values of 2011 while Sunder Lal reproduces the 2017 (fourth edition, first addendum) table; the *E. coli* requirement is identical in both, so either citation is acceptable in the examination. The corpus did not carry a population-wise minimum sampling frequency table, so no such figures are quoted here.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO Guidelines for Drinking-water Quality, WHO Guidelines for drinking-water quality: small water supplies, WHO/UNICEF Joint Monitoring Programme

Programme and policy updates (India)

- The binding constraint on Indian water surveillance is missing data, not missing standards. A district-level analysis pairing the National Water Quality Monitoring Programme with Health Management Information System diarrhoea returns for 2017–2019 found **43 per cent of India's districts reported no faecal coliform value at all** over those three years, against roughly **17 per cent household tap water connectivity as of August 2019**. The authors conclude that granular, district-level water-quality data are simply not available, so bacteriological water quality and diarrhoea cannot at present be linked to one another [PMID 38935461].
- India carries two different national uranium limits, and the examinable point is which body set which. The Bureau of Indian Standards has adopted **30 µg/L**, aligned to the WHO guideline value, while the Atomic Energy Regulatory Board retains **60 µg/L**, derived from a radiological dose of 0.1 mSv per year by the water route. In the Bhabha Atomic Research Centre nationwide survey of 55,554 samples covering **403 of India's 718 districts** across 23 states and 3 union territories, uranium ranged from **0.2 to 4918 µg/L**, with 93.9 per cent of samples below 30 µg/L and 97.8 per cent below 60 µg/L; the authors argue that 60 µg/L remains the defensible Indian limit on cost–benefit and radioactive-waste grounds. IS 10500:2012 first screens **gross alpha 0.5 Bq/L and gross beta 1 Bq/L** before individual radionuclides are pursued. Park's standards table carries no uranium row, so name both agencies and both figures [PMID 39007979].

Current national data

- Pooled dental fluorosis prevalence in India is **34.5 per cent** (95% CI 28.9–40.3) across 55 studies and 108,678 individuals, with literature searched to 29 May 2024, and **32.1 per cent** in children aged 5–18 years; the milder grades dominate (very mild 10.6 per cent, mild 9.7 per cent). Prevalence is **more than three-fold higher where water fluoride exceeds 1 ppm than where it is below** (50.3 versus 15.1 per cent), and higher in rural populations (38.1 per cent) and in western India (43.9 per cent) — field validation of the Bureau of Indian Standards 1 mg/L limit quoted in the standards table above [PMID 40741788].

- **Cholera in India, 2011–2020 is the strongest recent national dataset tying water surveillance to disease: 565 outbreaks, 45,759 cases and 263 deaths (case fatality 0.6 per cent) across 24 of the 36 states and union territories, assembled from Integrated Disease Surveillance Programme outbreak reports plus 20 published outbreak investigations. 72 per cent of cases arose in five states (Maharashtra, West Bengal, Punjab, Karnataka, Madhya Pradesh), 90 per cent of outbreaks were rural, and 61 per cent clustered in the June–September monsoon with a July peak — Tamil Nadu being the exception, peaking December–January. A transmission route was identified in 62 per cent of outbreaks, and of those 91 per cent were contaminated drinking water or unimproved sources; only outbreaks traced to damaged water pipelines showed a falling trend between 2011–2015 and 2016–2020. The WHO Global Task Force on Cholera Control targets ending cholera as a public health problem in at least 20 countries by 2030 [PMID 35565133].**

Elimination, surveillance and disease status

- **Environmental (wastewater) surveillance is now being validated against clinical disease in India and is the direction of travel beyond routine sampling.** In Vellore, Tamil Nadu, *Salmonella* Typhi was detected by multiplex quantitative PCR in 80 of 1,446 wastewater samples (5.5 per cent) from 50 sites between December 2022 and June 2023, alongside 207 blood-culture-confirmed cases; each **ten-fold rise in typhoid incidence carried 2.43 times the odds of a positive sample (95% CI 1.38–4.27)**, and spatial clustering of cases matched higher wastewater detection — supporting environmental surveillance as a tool to monitor the impact of typhoid conjugate vaccine [PMID 42419341].
- **Standardised typhoid environmental surveillance protocols now exist, field-tested over 12 months in Vellore (India) and Blantyre (Malawi) during 2021–2022.** In Vellore, **trap (Moore swab) samples outperformed grab samples — 79 of 517 (15.3 per cent) versus 39 of 520 (7.5 per cent) positive** — and *S. Typhi* was detected in months when local hospitals reported no culture-confirmed case at all, showing that environmental surveillance can map circulation where clinical surveillance is blind. Detection clustered by site and rose with catchment population, so **existing polio environmental surveillance sites, selected to represent populations above 100,000 and already implemented in many countries, may be leveraged for typhoid** — which the authors suggest as a ready, low-cost route to scale [PMID 37775091].
- **For settings without piped sewerage — the Indian rural norm — sampling doctrine extends beyond the classical grab sample into soil, sediment and passive traps.** Multi-parallel qPCR detected soil-transmitted helminth DNA in **36 per cent of environmental samples in Tamil Nadu** (25 per cent in Comé, Benin), with **soil and wastewater complementary: soil prevalence was similar to that of wastewater in India (32 of 95 versus 24 of 60) and higher than wastewater in Benin (39 of 121 versus 8 of 64)**, so both matrices should be sampled. **Drainage sediment was the most sensitive wastewater matrix** for helminths, while Moore swabs suited a broader bacterial, viral and protozoal panel [PMID 40881471].

Guideline and legislative updates

- **The WHO reference standard is now the *Guidelines for Drinking-water Quality*, fourth edition incorporating the first, second and third addenda (17 June 2026) — cite this in preference to the 2011 values reproduced in Park and the 2017 first-addendum table reproduced in Sunder Lal.**

Alongside it, WHO has issued a separate volume, *Guidelines for drinking-water quality: small water supplies* (15 February 2024), addressed to exactly the small and community systems that dominate rural India, which WHO describes as commonly facing operational, managerial, technical and resourcing constraints on delivering safe water [WHO 2026-Jun]; [WHO 2024-Feb].

- ▶ A 74-study meta-analysis found an inverse dose–response between fluoride exposure and children's IQ — pooled standardised mean difference -0.45 (95% CI -0.57 to -0.33) across 20,932 children, and 1.63 IQ points lower per 1 mg/L rise in urinary fluoride (95% CI -2.33 to -0.93), falling to 1.14 points among low-risk-of-bias studies. In the pooled drinking-water analysis the association was null below 1.5 mg/L, but it remained inverse below 1.5 mg/L when restricted to low-risk-of-bias studies, and for urinary fluoride it remained inverse at every cut-off tested. The authors state only that the findings may inform future comprehensive public-health risk–benefit assessments of fluoride exposures; for the examination the Indian 1 mg/L limit remains the safer figure to quote [PMID 39761023].

Global and WHO developments

- ▶ The 2025 WHO/UNICEF Joint Monitoring Programme report (*Progress on Household Drinking Water, Sanitation and Hygiene 2000–2024*) records 2.1 billion people — one in four — still without a safely managed drinking-water service, including 106 million who drink untreated surface water; this is the denominator against which Sustainable Development Goal target 6.1 progress and national surveillance coverage are judged [JMP 2025].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 15, pp 874-884; Ch 7, pp 569, 573); Sunder Lal — Textbook of Community Medicine (Ch 3, pp 98-107); Suryakantha — Community Medicine with Recent Advances (Ch 5, pp 60-64); Jugal Kishore — National Health Programmes of India; WHO Guidelines for Drinking-water Quality (2011 and 2017, as reproduced in the above texts); ICMR Manual of Standards of Drinking Water Supplies; National Rural Drinking Water Quality Monitoring and Surveillance Programme, 2006; Jal Jeevan Mission (target 2024).

Recent-advances sources: 38935461 — Analysing the interlinkage between diarrhoea and drinking water faecal contamination in India, *Journal of Water and Health* 2024, DOI [10.2166/wh.2024.109](https://doi.org/10.2166/wh.2024.109) (abstract-only); 39007979 — Uranium standards in drinking water: an examination from scientific and socio-economic standpoints of India, *Environmental Science and Pollution Research International* 2024, DOI [10.1007/s11356-024-34352-0](https://doi.org/10.1007/s11356-024-34352-0); 40741788 — Prevalence of Dental Fluorosis in India: A Systematic Review and Meta-Analysis, *Journal of Public Health Dentistry* 2025, DOI [10.1111/jphd.70004](https://doi.org/10.1111/jphd.70004) (abstract-only); 35565133 — Cholera Outbreaks in India, 2011-2020: A Systematic Review, *International Journal of Environmental Research and Public Health* 2022, DOI [10.3390/ijerph19095738](https://doi.org/10.3390/ijerph19095738); 42419341 — Integrating environmental and clinical surveillance for typhoid: insights from a prospective wastewater surveillance study in urban India, *The Lancet Microbe* 2026, DOI [10.1016/j.lanmic.2026.101414](https://doi.org/10.1016/j.lanmic.2026.101414) (abstract-only); 37775091 — Environmental Surveillance for *Salmonella* Typhi and its Association With Typhoid Fever Incidence in India and Malawi, *The Journal of Infectious Diseases* 2024, DOI [10.1093/infdis/jiad427](https://doi.org/10.1093/infdis/jiad427); 40881471 — Environmental surveillance of soil-transmitted helminths and other enteric pathogens in settings without networked wastewater infrastructure, *PLOS Water* 2025, DOI [10.1371/journal.pwat.0000337](https://doi.org/10.1371/journal.pwat.0000337); 39761023 — Fluoride Exposure and Children's IQ Scores: A Systematic Review and Meta-Analysis, *JAMA Pediatrics* 2025, DOI [10.1001/jamapediatrics.2024.5542](https://doi.org/10.1001/jamapediatrics.2024.5542). WHO Guidelines for Drinking-water Quality, fourth edition incorporating the first, second and third addenda, 17 June 2026; WHO Guidelines for drinking-water quality: small water supplies, 15 February 2024; WHO/UNICEF Joint Monitoring Programme, *Progress on Household Drinking Water, Sanitation and Hygiene 2000-2024*, 2025.



Global warming and its health effects

Asked in: Paper I 26-May-2025 — RGUHS MD Community Medicine

Definition — Certain atmospheric gases (carbon dioxide, methane, nitrous oxide, chlorofluorocarbons) allow solar radiation to reach the earth but prevent the re-radiated heat from escaping, acting like the glass of a greenhouse; this is the **greenhouse effect**, and without it life on earth would not exist. **Global warming** is the rise in the earth's mean surface temperature (normally about 15°C) above the desirable level caused by the accumulation of these **greenhouse gases (GHGs)** in the troposphere from human activity (Suryakantha, Ch 48, p 909). The resulting **climate change** acts on health by disturbing the three most fundamental determinants of health — air, food and water.

Causes — the greenhouse gases and where they come from

Emission of greenhouse gases has been rising ever since the industrial revolution, and the major component of carbon dioxide emission is the combustion of fossil fuels (Park 28e, Ch 15, p 903). Unlike ordinary air pollutants, these gases have atmospheric lifetimes of **years**, so they are distributed throughout a hemisphere and ultimately globally — which is why the problem is planetary and not local (Park 28e, Ch 15, p 888).

Greenhouse gas	Principal anthropogenic sources	Note
Carbon dioxide (CO ₂)	Combustion of fossil fuel — coal, petroleum products; industry; domestic combustion of coal, wood and oil; motor vehicles	The single largest contributor; long atmospheric lifetime (Park 28e, Ch 15, pp 888, 903)
Methane (CH ₄)	Paddy cultivation, livestock, biomass burning, decomposition of organic waste	Named among the gases whose accumulation traps heat in the troposphere (Suryakantha, Ch 48, p 909; Ch 6, p 68)
Nitrous oxide (N ₂ O)	Chemical (nitrogenous) fertilizers, fuel combustion	Lifetime of years — globally distributed (Park 28e, Ch 15, p 888)
Chlorofluorocarbons (CFCs)	Refrigerants, aerosol propellants, foam blowing	Both a greenhouse gas AND the agent of stratospheric ozone depletion (Suryakantha, Ch 48, p 909; Ch 6, p 68)
Tropospheric ozone (O ₃)	Photochemical reaction of vehicular hydrocarbons and oxides of nitrogen in strong sunlight	Accumulation in the troposphere adds to warming (Suryakantha, Ch 6, p 68; Park 28e, Ch 15, p 888)

Human activities driving the accumulation — the prime cause of global warming is human activity alone (Suryakantha, Ch 48, p 909):

- **Burning of fossil fuel** — coal and petroleum products; in 2007 the world consumed over 30 billion barrels of oil and 3 billion tonnes of coal.
- **Industrialization** — combustion of fuel for heat and power emits smoke, sulphur dioxide, oxides of nitrogen and fly ash; petrochemical industry adds halides (Park 28e, Ch 15, p 888).
- **Vehicular traffic** — motor vehicles are a major urban source of hydrocarbons, carbon monoxide, oxides of nitrogen and particulate matter, converted in sunlight to photochemical oxidants.

- **Urbanization** — the **heat island** phenomenon makes cities warmer than surrounding rural areas and lengthens and intensifies heat waves (Park 28e, Ch 15, p 903).
- **Deforestation** — removal of the green cover that acts as the natural carbon sink; acid rain kills trees and worsens desertification and soil erosion (Suryakantha, Ch 6, p 68).
- **Chemical fertilizers** in intensive agriculture, and **nuclear explosions**.
- **Domestic combustion** of coal, wood and biomass fuel — a major source in less developed countries.
- **Population growth** — the underlying multiplier that makes the problem a continuously increasing one (Suryakantha, Ch 48, p 909).

Historical note: the phenomenon was first described in 1824 by the French mathematician *J.B. Joseph Fourier*, who named it the "greenhouse effect"; the **Intergovernmental Panel on Climate Change (IPCC)** subsequently concluded that global warming is due to the build-up of greenhouse gases in the atmosphere (Suryakantha, Ch 48, p 909).

Magnitude of the change and its physical consequences

Parameter	Projected / observed change	Source
Mean global surface temperature	Rise of about 3°C by 2030; likely to surpass 3°C this century	Park 28e, Ch 15, p 903; Oxford Public Health 7e, Ch 3, pp 218–219
Surface temperature trend	Earth's surface temperature is about 15°C; rose ~0.5°C in the last 100 years; estimated further +2.5°C by 2050	Suryakantha, Ch 48, p 909
Threshold of irreversibility	A rise of 2°C could trigger an irreversible and catastrophic state	Suryakantha, Ch 48, p 909
Sea level	Rise of 0.1–0.3 metres by 2050	Park 28e, Ch 15, p 903
Extreme events	Increased frequency of cyclones, heat waves, droughts, rain storms and tidal surges	Park 28e, Ch 15, p 903; Suryakantha, Ch 48, p 911

Downstream physical effects: altered jet streams, prevailing winds and ocean currents shift the distribution of rainfall (some regions wetter, others drier); temperate-zone warming lowers soil moisture and impairs grain production; the distribution of vegetation and of **insect vectors of disease** changes (Park 28e, Ch 15, p 903). Melting of polar ice caps, a shift in the hydrological cycle, acid rain and increased dryness of climate follow (Suryakantha, Ch 6, p 68; Ch 48, p 909).

Effects on health

Health effects are conventionally grouped as **direct** (mediated by weather itself), **indirect / ecosystem-mediated**, and **socially mediated**; most climate-related health risks in fact operate through ecological and social processes (Oxford Public Health 7e, Ch 4, pp 243–244).

Pathway	Mechanism	Health outcome
Thermal extremes (direct)	Failure or impairment of the heat-regulating mechanism; salt and water depletion	Heat stroke (temperature may reach 110°F, delirium, coma; case-fatality 40% or more), heat hyperpyrexia (>106°F), heat exhaustion, heat cramps, heat syncope — 14 heat disorders are documented (Park 28e, Ch 15, p 902)
Extreme weather events (direct)	Storms, floods, drought, wildfire, cyclone, tidal surge	Drowning, physical injury, direct mortality; mental health disorders linked to the trauma of the event; destruction of homes, medical facilities and essential services (Oxford 7e, Ch 4, pp 243–244; Ch 107, pp 4054–4055)
Vector-borne disease	Warmer temperature and altered rainfall lengthen transmission seasons and extend the geographic and altitudinal range and reproductive rate of vectors	Malaria (new epidemics at previously protected higher altitudes), dengue, DHF, chikungunya, Japanese encephalitis, yellow fever, Zika (Suryakantha, Fig 48.1, p 910; Oxford 7e, Ch 4, p 244; Ch 107, pp 4054–4055)
Water quality and quantity	Variable rainfall reduces fresh-water supply and compromises hygiene; floods contaminate supplies and create vector breeding grounds; warmth promotes cholera vibrio growth	Cholera, diarrhoeal disease, toxic algal illness (Oxford 7e, Ch 4, p 244; Ch 107, pp 4054–4055)
Food security and nutrition	Crop failure, reduced staple production, loss of arable land, food-borne infection, civil unrest	Undernutrition, wasting and stunting in children; paradoxically obesity with mild–moderate food insecurity (Oxford 7e, Ch 4, pp 243–244; Ch 107, pp 4054–4055)
Air pollution and aeroallergens	Shared combustion sources; photochemical smog	Respiratory diseases, asthma, allergies, cardiovascular disease (Suryakantha, Fig 48.1, p 910; Oxford 7e, Ch 4, p 243)
Stratospheric ozone depletion	CFC-induced thinning of the ozone shield lets ultraviolet radiation reach the surface	Skin cancers (melanotic and non-melanotic), cataract, impairment of human immune mechanism predisposing to infection (Suryakantha, Ch 6, p 68)
Sea level rise and coastal flooding	Inundation of low-lying land and small islands	Displacement, drowning, loss of fresh water and of health infrastructure; 65 million people live in small island states and low-lying coastal countries already affected (Oxford 7e, Ch 107, pp 4054–4055)
Socially mediated	Displacement, overcrowding, poor sanitation, conflict, migration	Environmental refugees highly vulnerable to poor health; infectious disease from overcrowding; violence; widening health inequity within and between countries (Oxford 7e, Ch 6, pp 310–311; Suryakantha, Fig 48.1, p 910)

Quantified burden

Estimate	Figure	Source
Attributable burden already incurred (from the mid-1970s)	~150,000 deaths and ~5 million DALYs per year, mainly malnutrition, diarrhoea and malaria	WHO review, cited in Oxford 7e, Ch 1, pp 164–165
Additional deaths, 2030–2050	~250,000 per year — 38,000 heat exposure in the elderly, 48,000 diarrhoea, 60,000 malaria, 95,000 childhood undernutrition	WHO, cited in Oxford 7e, Ch 107, p 4054
Direct costs to health by 2030	US\$ 2–4 billion per year	Oxford 7e, Ch 107, p 4054
Water crisis in Africa	150–250 million people; Asian mega-city residents at great risk of river and coastal flooding	IPCC, cited in Suryakantha, Ch 48, p 911
Extreme-weather deaths in Europe	Projected to rise from ~3,000 to ~150,000 per year by 2100	Oxford 7e, Ch 107, pp 4054–4055

Vulnerable groups — the elderly, young children, the poor, outdoor and agricultural workers, coastal and small-island populations, and the urban poor. Mortality in the 1995 *Chicago heat wave* was a function of both individual variables (age, income) and of neighbourhood quality: the poor and elderly were at much greater risk if they lived in areas with rundown businesses, abandoned housing and degraded public spaces (Oxford 7e, Ch 14, pp 551–552). Because these impacts fall most heavily on populations already subject to social disadvantage, climate change is a driver of **health inequity** (Oxford 7e, Ch 6, pp 310–311).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- Draw the greenhouse effect diagram with these exact labels: Sun → incoming short-wave solar radiation → Earth's surface (mean 15°C) → outgoing long-wave infrared radiation → blanket of greenhouse gases (CO₂, CH₄, N₂O, CFCs) in the troposphere → heat re-radiated back to earth → rise in surface temperature = global warming; label the stratospheric ozone shield above, with CFC-induced thinning letting ultraviolet radiation through.

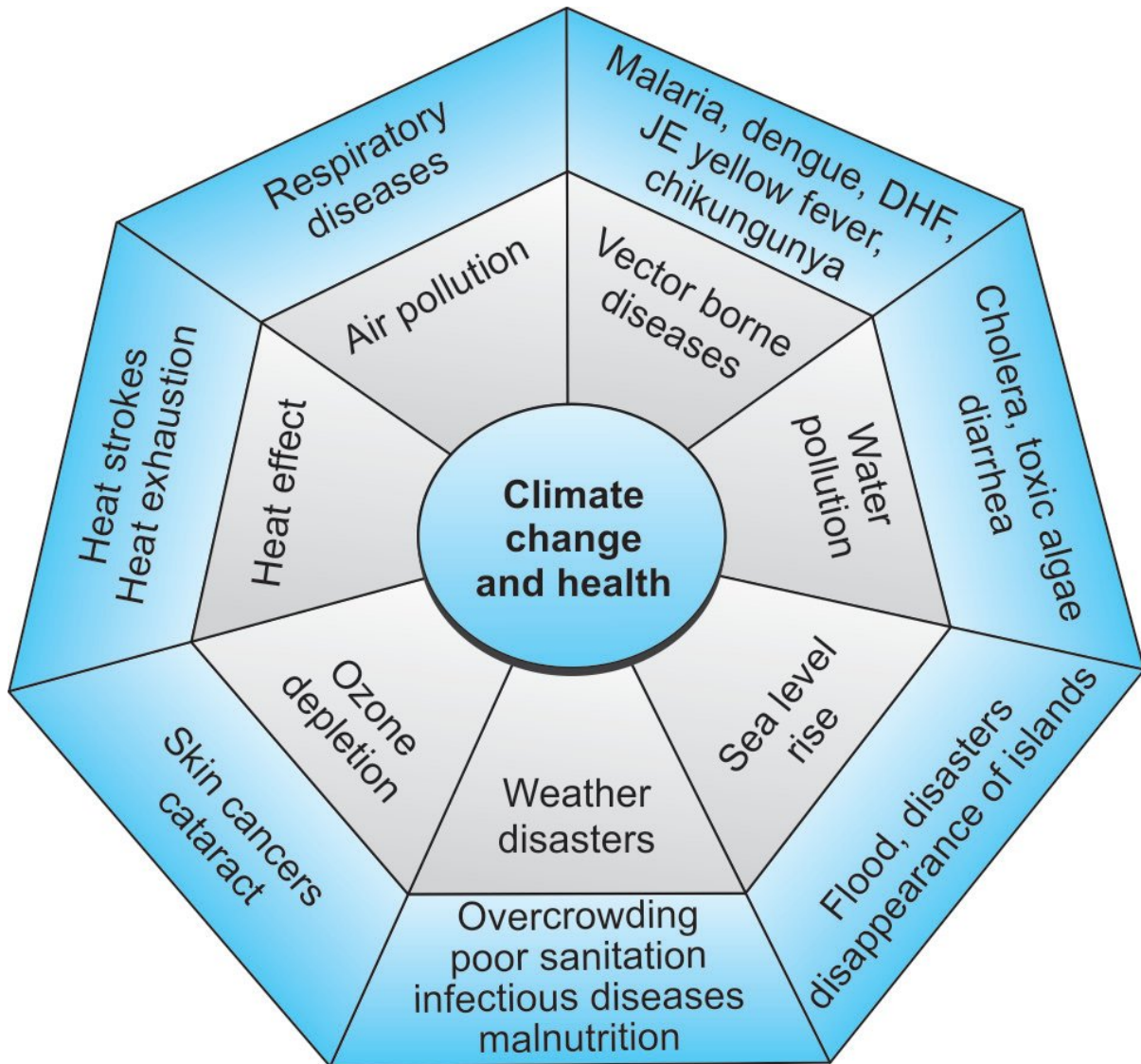


Fig 48.1 — Possible health impacts due to climate change (Suryakantha Environment Best, p.910)

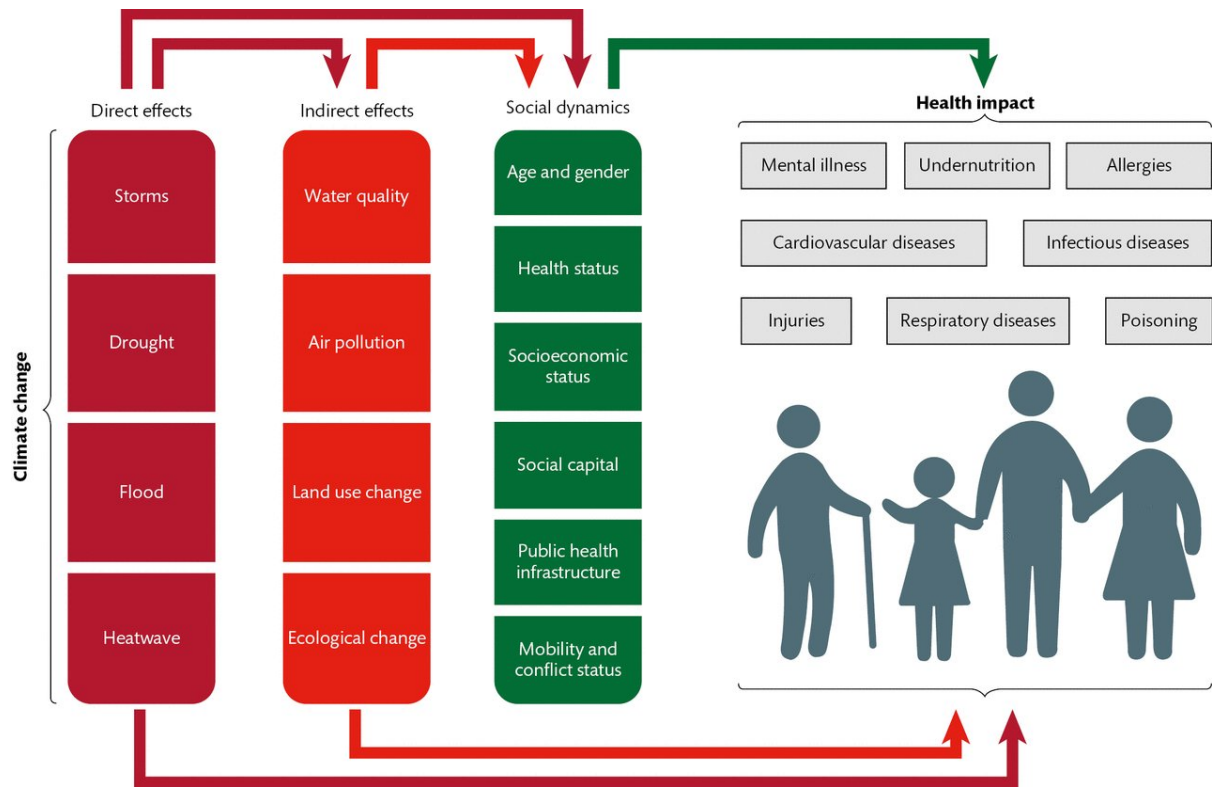
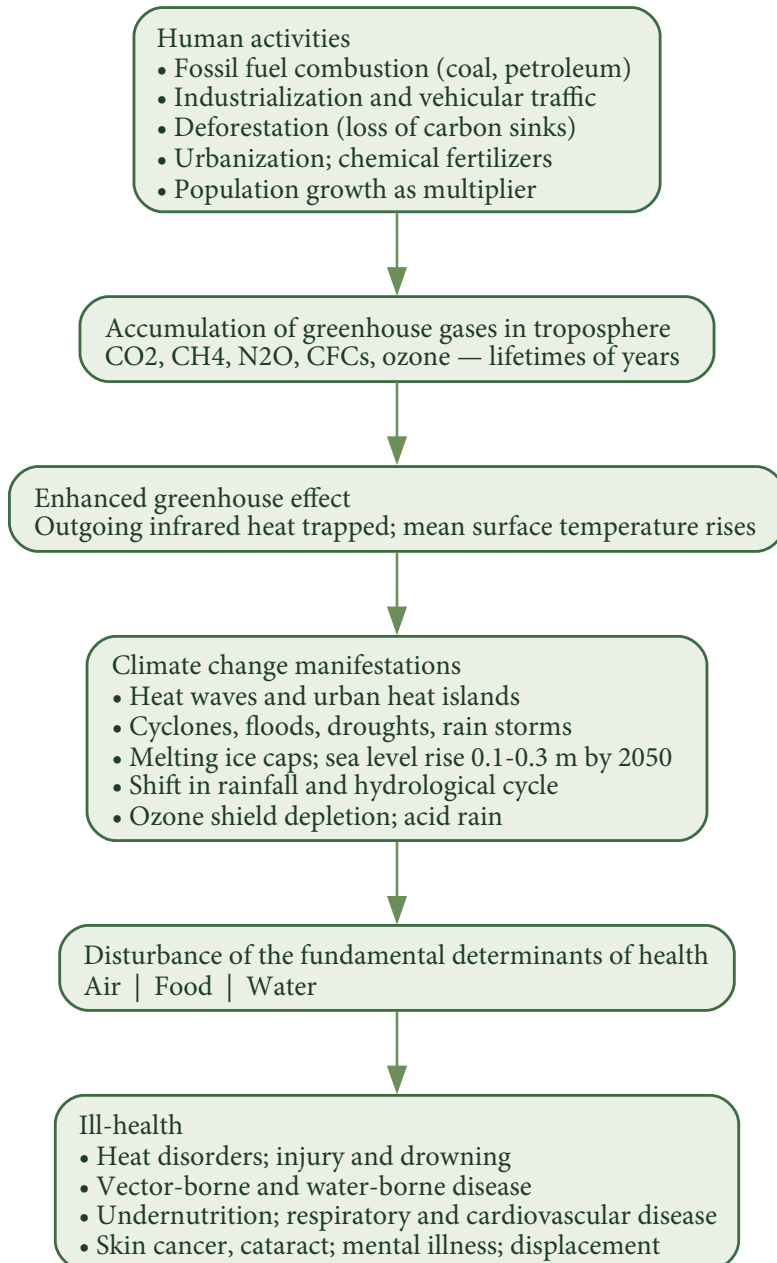
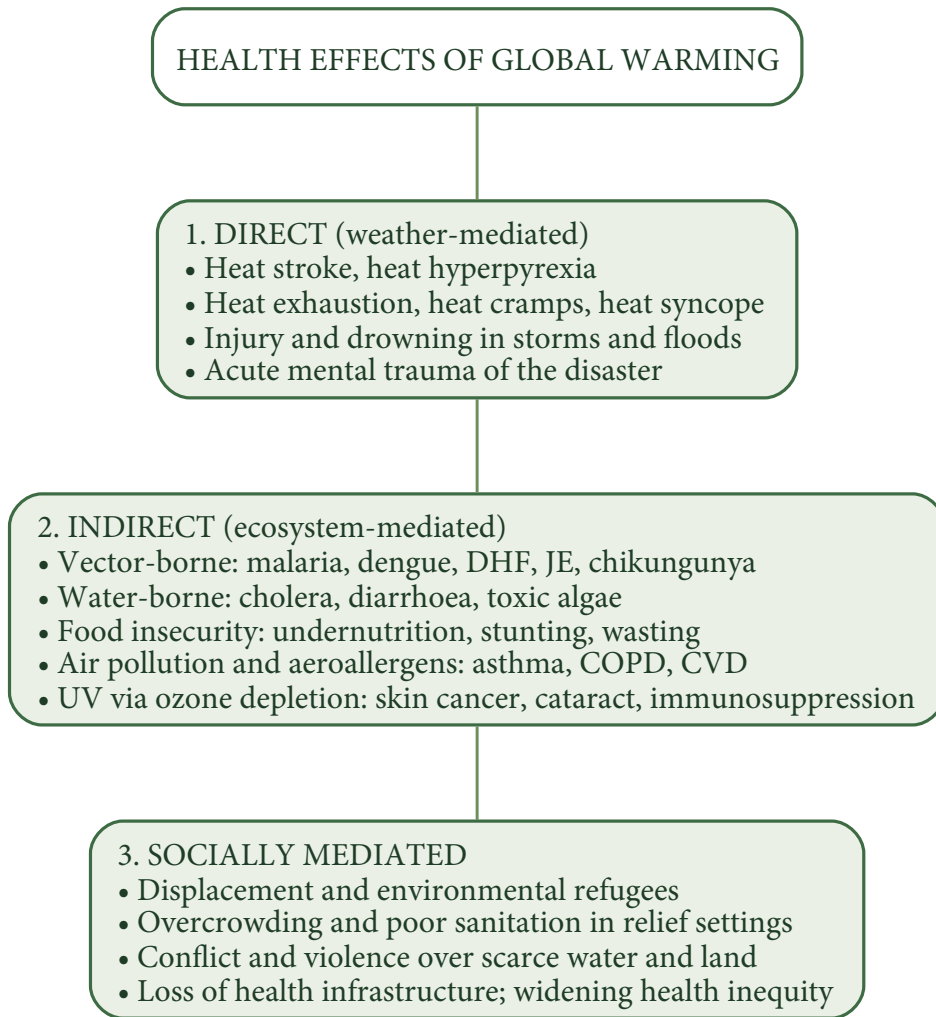


Fig 1.4 — 5 The direct and indirect effects of climate change on health and well (Oxford public health 7th ed, p.243)



Causal chain from human activity to ill-health in global warming



Classification of the health effects of global warming

High-yield exam pointers

- ▶ **Greenhouse gases mnemonic** — "CMNC": Carbon dioxide, Methane, Nitrous oxide, Chlorofluorocarbons (plus tropospheric ozone). CO₂ from fossil fuel is the largest single contributor; their atmospheric lifetime is in *years*, which is why the effect is global.
- ▶ **Numbers to write:** earth's mean surface temperature 15°C; +3°C by 2030 and sea level +0.1–0.3 m by 2050 (Park 28e); 2°C is the threshold of irreversible change; WHO estimate 250,000 extra deaths/year, 2030–2050 (38,000 heat in elderly, 48,000 diarrhoea, 60,000 malaria, 95,000 child undernutrition).
- ▶ **Five heat disorders in order of severity:** heat stroke (>110°F, sweating absent, case-fatality ≥40%) → heat hyperpyrexia (>106°F) → heat exhaustion (≤102°F, salt/water depletion) → heat cramps → heat syncope. Park lists 14 heat disorders in all.
- ▶ **Eponyms/landmarks:** J.B. Joseph Fourier (1824) named the greenhouse effect; IPCC attributed warming to GHG build-up; WHO World Health Day 2008 theme — "Protecting Health from Climate Change"; Chicago 1995 heat wave for the social patterning of heat mortality.
- ▶ **Always classify the health effects as direct / indirect (ecosystem-mediated) / socially mediated** — this framework alone earns marks and prevents a scattered list.

- **Ozone depletion is a separate but linked effect of the same CFCs** — quote its triad: skin cancer (melanotic and non-melanotic), cataract, immune impairment.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park 28e's projections (about 3°C by 2030, sea level 0.1–0.3 m by 2050) and Suryakantha's temperature figures are drawn from earlier Intergovernmental Panel on Climate Change assessments, so quote them with the textbook attached rather than as current IPCC values; the question asks only for causes and health effects, so mitigation measures (the Kyoto Protocol, carbon sinks, renewable energy, net-zero emission) and India's climate-health programme are deliberately not written here.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, MoHFW national and state action plans, NCDC, ICMR, NFHS-5

Programme and policy updates (India)

- India's climate-health response is now delivered through the **National Programme on Climate Change and Human Health (NPCCHH)**, whose plan document is the **National Action Plan for Climate Change and Human Health (NAPCCHH, 2018, MoHFW)**, layered on the older **National Action Plan on Climate Change (NAPCC, 2008, Prime Minister's Council on Climate Change and MoEF&CC)**. Beneath it sit two MoHFW vertical plans — the **National Action Plan on Heat-Related Illness (2021)** and the **National Action Plan for Diseases due to Air Pollution (2021)** — plus the **National Health Adaptation Plan for Climate Change-Related Disaster (2019)**, which comes from the **National Institute of Disaster Management**, not MoHFW. NCDC, under MoHFW, maintains the directory of **State Action Plans on Climate Change and Human Health (SAPCCHH)**: 29 of 37 states and union territories had a plan in that directory at the time of review, and only 8 of those were final versions. Name the programme and the year in the answer rather than writing vaguely about "climate cells" [PMID 39185584]
- Analysis of those 10 national policies plus 29 state plans found the Indian response still weighted towards **vector-borne and water-borne disease; cardiovascular disease** — the leading cause of death in India — remains poorly contextualised as a climate-sensitive condition, has no section of its own in the NAPCCHH, and carries explicit surveillance data in only **7 of the 29** state plans, being otherwise folded into heat-related illness or air-pollution-related cardiopulmonary disease. **Ahmedabad's Heat Action Plan** has now been evaluated against city mortality data (2002–2018): the post-plan period showed reduced immediate heat-mortality risk across maximum-temperature thresholds, but the benefit did not extend to longer lag days or to moderate heat stress [PMID 39185584] [PMID 40658206]

Current national data

- District-level NFHS-5 (2019–21) data merged with a climate-vulnerability assessment across **575 rural districts** (154,547 children, 447,348 women) showed that highly vulnerable districts consistently underperform on health SDG targets — children **1.25 times** more likely to be underweight and women **1.38 times** more likely to have a non-institutional delivery than in less vulnerable districts. About **80%** of India's population lives in areas highly vulnerable to cyclone,

flood or heatwave. Quote the malnutrition figures with the round attached — NFHS-5: stunting **36%**, underweight **32%**, wasting **19%** in under-fives [PMID 41296661]

- An Indian city time-series (Chandigarh, 2010–2015) identified a **mortality threshold of 33.8°C** maximum temperature ($p = 0.0007$) above which all-cause mortality rose significantly, though the effect size was modest (about **4%**, ~0.5 extra deaths a day); excess mortality on IMD-defined heatwave days ranged from **14% to 100%** over reference periods, risk was similar in both sexes, and those over 65 carried **1.5 times** the heat-mortality risk of the general population. Which heatwave definition is used (IMD fixed cut-off versus 90th/95th/99th percentile) changed the number of days captured from 42 to 4 — the argument for locally derived, city-level triggers in heat-health alerts [PMID 41611746]

Elimination, surveillance and disease status

- Warming cuts across the elimination programmes in a direction the textbooks do not anticipate. In a global 2000–2021 analysis, each **1°C** rise in national annual mean temperature was associated with *lower* TB burden — a 0.89% reduction in age-standardised incidence (95% CI 0.60–1.18) and a 1.61% reduction in age-standardised mortality (95% CI 1.27–1.95); but in the **South-East Asia Region the association reverses** — a 1°C rise raised age-standardised TB incidence by **1.36%** (95% CI 0.47–2.26), and by 12.25% in the Western Pacific Region. Separately and **globally**, each 1 mm rise in national annual mean precipitation raised TB mortality by **1.80%** (95% CI 0.63–2.95); that precipitation effect was significant in the African, Eastern Mediterranean and Americas regions, not in South-East Asia [PMID 40880929]
- Dengue in Bihar rose from **8,136 cases (2019) to 11,285 (2022) and 19,093 (2023)** across its 38 districts; minimum temperature and rainfall correlated most strongly with weekly incidence at a **4–6 week lag**, which is the basis for climate-informed early-warning systems and pre-emptive vector control rather than reactive outbreak response [PMID 40817799]

Guideline and legislative updates

- An ICMR-led overview of 13 systematic reviews graded heat interventions by tier: **national-level** measures (heat action plans, early-warning systems) were the most effective at reducing heat morbidity and mortality, community-level measures next, and individual air-conditioning protective but not sustainable. Adaptation produced immediate, measurable reductions; mitigation remains essential but under-evaluated in health terms [PMID 42320672]
- **Heat stress nephropathy** is now described in Indian medical literature as an emerging occupational and environmental entity, reframing much of the chronic kidney disease of unknown aetiology (CKDu) seen in agricultural workers — recurrent heat exposure with dehydration, compounded by concentration of agrochemicals and heavy metals in shrinking water sources; a useful addition to the heat-disorder list in the answer [PMID 39390883]

Global and WHO developments

- A meta-analysis of 21 studies (126,930,288 people) using **locally defined** extreme heat events found a pooled mortality risk ratio of **1.24 (95% CI 1.06–1.46)**, rising significantly over calendar time, with higher risk ratios in older adults, urban populations and low- and middle-income countries [PMID 41418864]
- A scoping review of 199 studies across 157 countries put the pooled **heat-attributable fraction at 1.18%** of all-cause mortality (95% CI 1.01–1.37) — about **800,000 deaths a year** — with higher

fractions for cardiovascular (2.15%), respiratory (3.08%) and stroke (2.71%) mortality. Evidence remains skewed to high- and upper-middle-income countries, over half the studies coming from China alone, and the authors call for climate-health surveillance systems in under-represented regions [PMID 41928226]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 15, Environment and Health, pp 886–903); Suryakantha — Community Medicine with Recent Advances (Ch 48, Global Warming, pp 909–911; Ch 6, Air and Ventilation, p 68; Fig 48.1, p 910); Oxford Textbook of Public Health 7e (Ch 14, Climate change and human health; Ch 107, Global health in the era of sustainable development, pp 4054–4056; Ch 4, pp 243–244; Ch 6, pp 310–311; Ch 1, pp 164–165); WHO estimates of the climate-attributable disease burden and WHO World Health Day 2008 theme, as cited in these texts.

Recent-advances sources: 39185584 — A review of climate change and cardiovascular diseases in the Indian policy context, Health Policy Plan 2024, doi:[10.1093/heapol/czae076](https://doi.org/10.1093/heapol/czae076); 41296661 — Health SDGs are at risk from climate change: Evidence from India, PLoS One 2025, doi:[10.1371/journal.pone.0335529](https://doi.org/10.1371/journal.pone.0335529); 41611746 — Extreme temperature events and their relationship with excess all-cause mortality in Chandigarh, India, Sci Rep 2026, doi:[10.1038/s41598-025-32614-6](https://doi.org/10.1038/s41598-025-32614-6); 40880929 — The influence and lag-effect of temperature and precipitation on the incidence and mortality of tuberculosis, 2000–2021, Front Public Health 2025, doi:[10.3389/fpubh.2025.1572422](https://doi.org/10.3389/fpubh.2025.1572422); 40817799 — Unravelling the interplay of space, time, and weather in dengue outbreaks: an insight from Bihar, India, J Vector Borne Dis 2026, doi:[10.4103/jvbd.jvbd_114_25](https://doi.org/10.4103/jvbd.jvbd_114_25) (abstract-only); 42320672 — Adaptation and mitigation approaches to reduce heat-related health risks: an overview of systematic reviews, Environ Res 2026, doi:[10.1016/j.envres.2026.125055](https://doi.org/10.1016/j.envres.2026.125055) (abstract-only); 40658206 — Comparison of biometeorological index and temperature linked mortality: a case study from Ahmedabad city, India, Int J Biometeorol 2025, doi:[10.1007/s00484-025-02959-8](https://doi.org/10.1007/s00484-025-02959-8) (abstract-only); 39390883 — Heat stress nephropathy: an emerging epidemic of global warming, J Assoc Physicians India 2024, doi:[10.59556/japi.72.0686](https://doi.org/10.59556/japi.72.0686) (abstract-only); 41418864 — Association between all-cause mortality and locally-defined extreme heat events: a global systematic review and meta-analysis, Environ Res 2025, doi:[10.1016/j.envres.2025.123566](https://doi.org/10.1016/j.envres.2025.123566) (abstract-only); 41928226 — Mapping the global health burden of climate-sensitive exposures: a systematic scoping review, Environ Health 2026, doi:[10.1186/s12940-026-01294-8](https://doi.org/10.1186/s12940-026-01294-8). Indian programme and policy documents named above, with the issuing body as given in Table 1 of PMID 39185584: NAPCC 2008 (Prime Minister's Council on Climate Change and MoEF&CC); NAPCCHH 2018, National Action Plan on Heat-Related Illness 2021 and National Action Plan for Diseases due to Air Pollution 2021 (MoHFW); National Health Adaptation Plan for Climate Change-Related Disaster 2019 (National Institute of Disaster Management); State Action Plans on Climate Change and Human Health (NCDC directory, National Programme on Climate Change and Human Health); NFHS-5 (2019–21).



Water pollution

Asked in: Paper I 02-Jan-2025 — RGUHS MD Community Medicine

Definition — Water intended for human consumption must be **safe and wholesome**, i.e. free from pathogenic agents, free from harmful chemical substances, pleasant to the taste (free from colour and odour) and usable for domestic purposes; water is said to be **polluted or contaminated when it does not fulfil these four criteria** (Park 28e, Ch 15, p 861). Water pollution therefore means any adverse change — physical, chemical or biological — in the composition or condition of water that renders it less suitable for use (Mahajan & Gupta 4e, Ch 10, p 117). The standard **indicators of pollution** are total suspended solids, biochemical oxygen demand (BOD) at 20°C, concentration of chlorides, nitrogen and phosphorus, and absence of dissolved oxygen (Park 28e, Ch 15, p 865).

Causes of water pollution

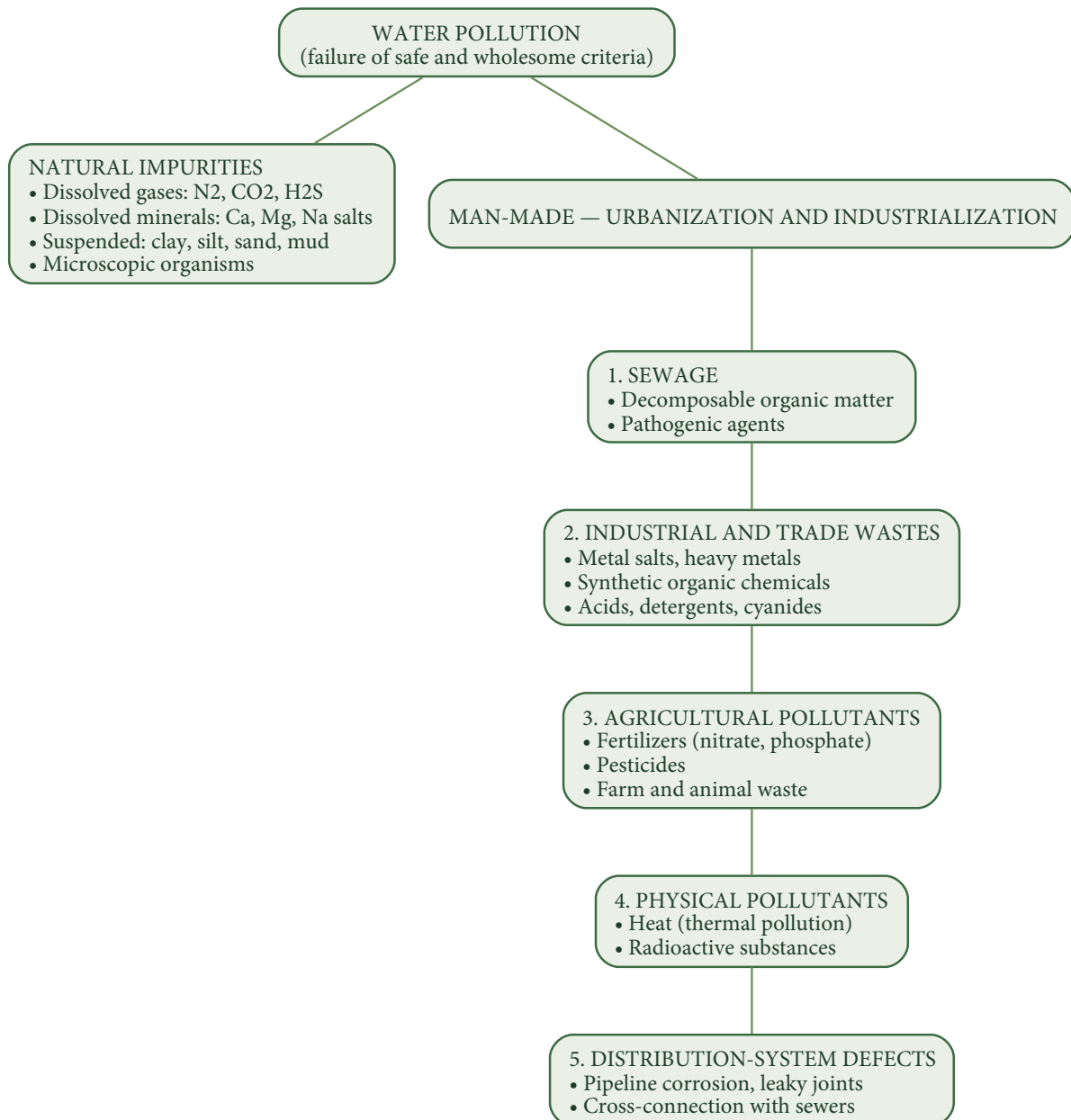
Pure uncontaminated water does not occur in nature; impurities are of two broad kinds — natural (not essentially dangerous) and man-made, the latter arising from urbanization and industrialization (Park 28e, Ch 15, p 865).

Group	Pollutants	Typical origin
Natural impurities	Dissolved gases (nitrogen, carbon dioxide, hydrogen sulphide); dissolved minerals (salts of calcium, magnesium, sodium); suspended impurities (clay, silt, sand, mud); microscopic organisms	Atmosphere, catchment area and soil (Park 28e, Ch 15, p 865)
Sewage (man-made)	Decomposable organic matter and pathogenic agents; sullage water	Human excreta, municipal sewers; the second major source overall (Park 28e, Ch 15, p 865; Mahajan & Gupta 4e, Ch 10, p 117)
Industrial and trade wastes	Toxic agents from metal salts to complex synthetic organic chemicals — mineral acids (battery, electroplating), organic acids (rayon, leather, dyeing, distilleries), soaps and synthetic detergents, gaseous pollutants (ammonia, chlorine, hydrogen sulphide, ozone), heavy metals, cyanides, dyes and pigments	Direct discharge into rivers, or into the sewer at high temperature (Park 28e, Ch 15, p 865; Mahajan & Gupta 4e, Ch 10, p 117)
Agricultural pollutants	Chemical fertilizers (nitrates, phosphates), pesticides and insecticides, farm/animal waste	Run-off and drainage from cultivated land; a cow and a pig produce 16 and 3 times as much organic matter respectively as one man (Mahajan & Gupta 4e, Ch 10, p 117)
Physical pollutants	Heat (thermal pollution) and radioactive substances	Power plants, industrial cooling water (Park 28e, Ch 15, p 865)
Social and cultural practices	Bathing, washing of animals, disposal of the dead, open defecation into water courses	Customs and habits of people living along rivers (Park 28e, Ch 15, p 862)

Group	Pollutants	Typical origin
Defects in the distribution system	Corrosion of pipelines, leaky joints, cross-connections between water-supply pipes and sewage drainage pipes	Occurs even when the source and its treatment are of a high standard (Park 28e, Ch 15, p 865)

Biological pollutants are further divided into *primary* pollutants, directly attributable to man's activity (bacteria and viruses in sewage — *Salmonella*, *Shigella*, *Vibrio cholerae*, *Leptospira*, polio and hepatitis viruses), and *corollary* pollutants, not directly attributable to human endeavour (weeds and algae) (Mahajan & Gupta 4e, Ch 10, p 117).

Rivers derive their impurities from surface washings, sewage and sullage, industrial and trade wastes and agricultural drainage. **Self-purification** by dilution, sedimentation, aeration, oxidation, sunlight and plant and animal life does occur, but these natural agencies are **not sufficient to render river water potable** (Park 28e, Ch 15, p 862).



Classification of the causes of water pollution

Effects of water pollution — I. Water-related diseases

Health may be affected by ingesting contaminated water directly or through food, and by using contaminated water for personal hygiene and recreation. Developing countries carry a heavy burden of water-related disease, the heaviest being the diarrhoeal diseases (Park 28e, Ch 15, p 865).

Category	Mechanism	Diseases
Water-borne, infective agent present — viral	Ingestion of the agent in water	Viral hepatitis A, hepatitis E, poliomyelitis, rotavirus diarrhoea in infants
— bacterial	Ingestion	Typhoid and paratyphoid fever, bacillary dysentery, <i>Escherichia coli</i> diarrhoea, cholera
— protozoal	Ingestion	Amoebiasis, giardiasis
— helminthic	Ingestion	Roundworm, threadworm, hydatid disease
— leptospiral	Contact/ingestion	<i>Weil's disease</i>
Water-borne, aquatic host present	Snail as intermediate host	Schistosomiasis
	Cyclops as intermediate host	Guineaworm, fish tapeworm
Water-washed (inadequate quantity for hygiene)	Deficient personal and domestic hygiene	Shigellosis, trachoma and conjunctivitis, ascariasis, scabies
Water-related vector-borne	Insect vectors breeding in or near water	Malaria, filaria, arboviruses, onchocerciasis, African trypanosomiasis

(Park 28e, Ch 15, pp 865–866)

Effects of water pollution — II. Chemical and long-term effects

The risk from toxic chemicals differs from the microbial risk: it arises chiefly from **cumulative effects after prolonged exposure**, particularly with heavy metals and carcinogens, rather than from acute illness (Park 28e, Ch 15, p 866).

Chemical	WHO guideline value in drinking water	Health effect of excess
Fluoride	1.5 mg/L	Dental fluorosis (mottling of enamel) above 1.5 mg/L intake; skeletal fluorosis with lifetime daily intake 3.0–6.0 mg/L or more; crippling fluorosis above 10 mg/L; genu valgum with osteoporosis of lower limbs (Park 28e, Ch 11, p 787)
Nitrate (as NO ₃)	50 mg/L	Methaemoglobinaemia — "cyanosis in infants"; bottle-fed infants under 3 months of age are most susceptible ; follows fertilizer-treated farmland run-off entering the supply
Nitrite (as NO ₂)	3 mg/L (provisional)	As above; the sum of the ratios of nitrate and nitrite to their guideline values must not exceed 1
Arsenic	0.01 mg/L (provisional)	Dissolution of minerals and ores, industrial effluent; chronic arsenicosis, carcinogenic

Chemical	WHO guideline value in drinking water	Health effect of excess
Lead	0.01 mg/L	Accumulates in skeleton; toxic to central and peripheral nervous system; young children absorb 4–5 times as much as adults; interferes with calcium and vitamin D metabolism; mostly from plumbing
Cadmium	0.003 mg/L	Accumulates in the kidney, biological half-life in man 10–35 years
Mercury (total)	0.006 mg/L	Inorganic mercury targets the kidney; methyl mercury affects mainly the central nervous system
Chromium	0.05 mg/L (provisional)	Industrial (tannery, plating) contamination
Cyanide	—	Effects on the thyroid and particularly the nervous system on long-term consumption

(Park 28e, Ch 15, pp 866–867, Table 6)

Two chemical effects run the other way and are examinable: fluoride at about **1 mg/L protects against dental caries**, and **hardness of water appears to have a beneficial effect against cardiovascular disease** (Park 28e, Ch 15, p 866). In most parts of India the fluoride content of drinking water is about 0.5 mg/L, but in fluorosis-endemic areas — Andhra Pradesh (Nellore, Nalgonda, Prakasam districts), Punjab, Haryana, Karnataka, Kerala and Tamil Nadu — it may be as high as 3–12 mg/L (Park 28e, Ch 11, pp 786–787).

Effects of water pollution — III. Ecological, aesthetic and economic

Effect	Mechanism
Eutrophication	Combined excess of nitrates and phosphates from fertilizer and farm-waste run-off causes nuisance over large stretches of water — green algal scums, dense growth of rooted plants, oxygen depletion in deeper water and bottom sediments, bad taste and odour (Mahajan & Gupta 4e, Ch 10, p 117; AFMC Red Book 2024, Ch 5, p 86)
Destruction of aquatic life	Excess free chlorine in river water destroys fish and other aquatic life, corrodes metals, and causes mucous-membrane irritation and even pulmonary oedema in man
Oil spills	The oil film over the sea surface prevents atmospheric oxygen from aerating and purifying the water; large-scale destruction of fish
Synthetic detergent foam	Alkyl benzene sulphonate is non-biodegradable and foams at very low concentration; foaming retards aeration of river water and damages aquatic flora and fauna
Failure of sewage treatment	Hot industrial effluent discharged into sewers damages the biological purification mechanism, so "treated" sewage itself pollutes the receiving river
Loss of acceptability	Colour (magenta at 0.02 ppm; iron and chromium), turbidity, foam, taste (iron, chlorine, phenols, algae) and odour — putrid from hydrogen sulphide, fishy from organic amines, earthy from humus

Effect	Mechanism
Physical pollution	Thermal pollution lowers dissolved oxygen; radioactive substances enter the aquatic food chain

(Mahajan & Gupta 4e, Ch 10, p 117)

Prevention and control of water pollution

Prevention operates at four levels — protecting the source, treating what is put in, treating what is taken out, and enforcing the law.

1. Protection of the source

- Keep the catchment area free of human habitation, animal keeping and grazing; contamination from wild animals is possible even where no habitation exists (Park 28e, Ch 15, p 862).
- Prefer **deep wells**, which tap water below the first impervious layer and furnish the safest water; convert shallow wells and springs into sanitary, well-built protected structures (Park 28e, Ch 15, pp 863–865).
- Eliminate cross-connections between water mains and sewers, repair leaky joints and control pipeline corrosion; exercise **surveillance at every point of the distribution system** (Park 28e, Ch 15, p 865).

2. Treatment of water before use — purification on a large scale (Park 28e, Ch 15, pp 866–871)

Stage	Action	Effect achieved
Storage	Impound in a reservoir; optimum storage of river water 10–14 days	About 90% of suspended impurities settle in 24 hours; total bacterial count falls by about 90% in the first 5–7 days; free ammonia falls, nitrates rise
Filtration	Slow sand (biological) or rapid sand (mechanical) filters	Removes 98–99% of bacteria besides other impurities
Disinfection	Chlorination to the break point	Minimum recommended free residual chlorine 0.5 mg/L for a contact period of 1 hour ; turbidity must be ≤ 1 NTU and pH preferably < 8.0 before disinfection
Special treatment	Defluoridation by the <i>Nalgonda technique</i> (lime + alum, then flocculation, sedimentation, filtration — developed at NEERI, Nagpur); or change to a source with 0.5–0.8 mg/L fluoride	Prevents endemic fluorosis (Park 28e, Ch 11, p 787)

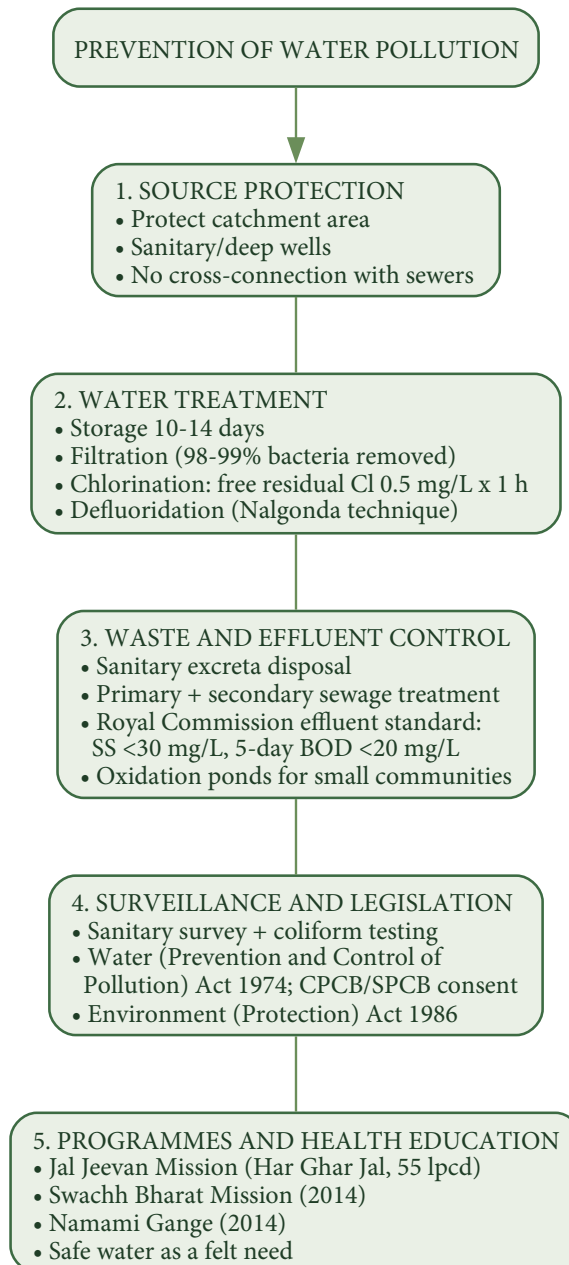
3. Treatment of what is discharged — sewage and effluent control

- **Excreta disposal**: sanitary latrines and the water-carriage system in sewered areas; end open defecation in unsewered areas.
- **Sewage treatment**: primary (screening, grit removal, plain sedimentation) followed by secondary/aerobic treatment (trickling filter or activated sludge process), then secondary sedimentation and chlorination before disposal (Park 28e, Ch 15, pp 917–918).

- **Effluent standard:** the *Royal Commission* (England, Fifth Report 1908) recommended that effluent from a sewage treatment plant should contain **not more than 30 mg/L suspended solids** and a **5-day BOD not exceeding 20 mg/L**, assuming an 8 : 1 dilution in the receiving water (Park 28e, Ch 15, p 919).
- Raw sewage should **never** be discharged into rivers; a river outfall must be preceded by purification proportionate to the dilution the river can provide.
- Low-cost options for small communities — oxidation (waste stabilization) ponds, oxidation ditches, aerated lagoons and sewage farming.

4. Monitoring, legislation and programmes

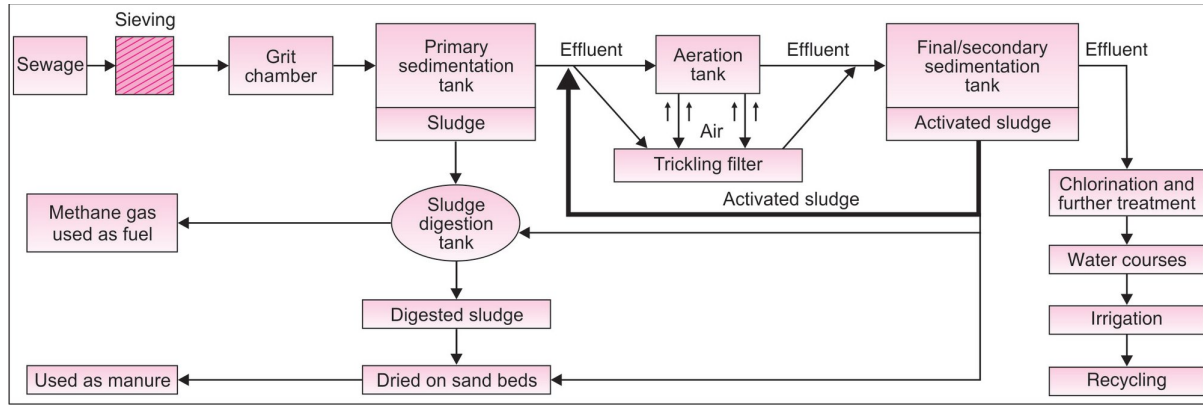
Measure	Content
Surveillance of drinking-water quality	Sanitary survey plus periodic sampling; coliform organisms are the primary bacterial indicator of faecal pollution, with faecal streptococci and <i>Clostridium perfringens</i> as supplementary indicators (Park 28e, Ch 15, p 876 and p 880)
Water (Prevention and Control of Pollution) Act, 1974	Comprehensive legislation providing a legal deterrent against water pollution; constitutes Central, State and Joint Water Boards with wide powers to control pollution (Park 28e, Ch 15, p 866)
Water (Prevention and Control of Pollution) Cess Act, 1977; Environment (Protection) Act, 1986 and EIA Rules	Industry needs "consent to establish" and "consent to operate" from the State Pollution Control Board (Manual on Municipal Solid Waste Management, Ch 5)
Jal Jeevan Mission (formerly Nal Jal Yojana / National Rural Drinking Water Programme)	Functional Household Tap Connection (<i>Har Ghar Jal</i>) to every rural household, at least 55 litres per capita per day of prescribed quality, target year 2024 (Park 28e, Ch 7, p 573)
Swachh Bharat Mission (launched 2 October 2014)	Urban and Gramin sub-missions; elimination of open defecation, solid and liquid waste management (Park 28e, Ch 7, p 574)
Namami Gange (June 2014, outlay ₹20,000 crore)	Eight pillars — sewerage treatment infrastructure, river-front development, river-surface cleaning, biodiversity, afforestation, public awareness, industrial effluent monitoring, Ganga Gram (AFMC Red Book 2024, Ch 5, p 128)
Health education	Provision of a good water supply does not by itself secure freedom from water-borne disease; people must recognise safe water as a <i>felt need</i> and give up habits that pollute water supplies (Park 28e, Ch 15, p 883)



Prevention and control of water pollution by level of action

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Flow diagram of a modern sewage treatment plant** — label exactly: Sewage → Sieving/screening → Grit chamber → Primary sedimentation tank (Sludge) → Effluent → Aeration tank / Trickling filter (Air) → Final (secondary) sedimentation tank (Activated sludge, returned) → Chlorination and further treatment → Water courses / Irrigation / Recycling; and the sludge line: Sludge → Sludge digestion tank → Methane gas used as fuel; Digested sludge → Dried on sand beds → Used as manure.



Sewage treatment processes (Sunderlal, p.115)

High-yield exam pointers

- ▶ **Safe and wholesome water = 4 criteria:** free from pathogenic agents, free from harmful chemicals, pleasant to taste (no colour/odour), usable for domestic purposes. Failure of any one = polluted water.
- ▶ **Four man-made sources (mnemonic S-I-A-P):** Sewage, Industrial and trade wastes, Agricultural pollutants, Physical pollutants (heat and radioactivity).
- ▶ **Indicators of pollution:** total suspended solids, BOD at 20°C, chlorides, nitrogen, phosphorus, absent dissolved oxygen. Untreated domestic sewage BOD ≈ 300 mg/L; ≥ 300 mg/L = "strong" sewage, ≤ 100 mg/L = "weak".
- ▶ **Royal Commission effluent standard (1908):** suspended solids ≤ 30 mg/L, 5-day BOD ≤ 20 mg/L, at 8 : 1 dilution.
- ▶ **Disinfection end-point:** free residual chlorine 0.5 mg/L after 1 hour contact; turbidity ≤ 1 NTU, pH preferably < 8.0 .
- ▶ **WHO guideline values to quote:** fluoride 1.5, nitrate 50, nitrite 3, arsenic 0.01, lead 0.01, cadmium 0.003, mercury 0.006, chromium 0.05 mg/L.
- ▶ **Law and programmes:** Water (Prevention and Control of Pollution) Act 1974; Water Cess Act 1977; Environment (Protection) Act 1986; Jal Jeevan Mission (55 lpcd, *Har Ghar Jal*, 2024); Swachh Bharat Mission (2 Oct 2014); Namami Gange (June 2014).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park 28e states the Water Act 1974 constitutes "Central and State Water Boards" — these are the bodies now functioning as the Central Pollution Control Board and the State Pollution Control Boards, which issue the "consent to establish" and "consent to operate" referred to in the waste-management manuals; the Jal Jeevan Mission target year of 2024 quoted above is the target as printed in Park 28e and in the AFMC Red Book 2024.

Recent advances [RA]

Current as of 2026-08-19 · depth: elaborate · sources: PubMed, Parliament of India, National Mission for Clean Ganga, Department of Drinking Water and Sanitation, WHO

Programme and policy updates (India)

- **Swachh Bharat Mission-Grameen has moved to Phase II ("ODF Plus", outlay ₹1.40 lakh crore)** — the operative target is no longer the household toilet but solid and **liquid (greywater) waste management** at village level, which is the village-scale water-pollution control step; a quasi-experimental national analysis of 35 states and 640 districts over 2011–2020 found districts where more than 30% of households received an SBM toilet had **5.3 lower IMR and 6.8 lower U5MR** per 1000 live births ($p<0.05$), scaling to roughly **60,000–70,000 infant deaths averted annually** [PMID 39223201; DDWS 2026]
- **Namami Gange is now far larger than the ₹20,000 crore figure carried in the textbooks** — 200 sewerage-infrastructure projects sanctioned at **₹31,810 crore**, of which **116 are completed and operational**, alongside 84 ghat and crematorium projects, river-surface cleaning at 11 locations and afforestation over 1,34,106 hectares in five states [NMCG 2026-Aug]

Current national data

- **Jal Jeevan Mission** — **119.5 million rural households had a tap connection as of October 2024**, so the 2024 universal-coverage year printed in the textbooks is best quoted as a target and not as an achievement; the binding constraint has shifted from pipe-laying to groundwater — India is the world's largest groundwater user (about **250 km³ a year** drawn for irrigation, more than China and the USA together), borewells rose from 1 million to 20 million in five decades, and in 2022 only **93% of Indians** had water within a 30-minute round trip [PMID 40160545]
- **Water Access Index built from five NFHS rounds (NFHS-1 1992–93 to NFHS-5 2019–21, 15.28 lakh households)** — on a composite of improved source, on-premises availability, 5-minute round-trip access and household water treatment, **343 districts (49%) were front runners, 142 (20%) performers and 221 (31%) aspirants, and only one district (0.1%) reached universal access**; the worst clusters are in West Bengal, Bihar, Odisha and Jharkhand — a district-level SDG 6.1 figure worth quoting in place of a bare national percentage [PMID 40499970]

Elimination, surveillance and disease status

- **Wastewater (environmental) surveillance is being extended from polio to enteric fever in India** — prospective surveillance in Vellore, Tamil Nadu (12 Dec 2022–30 Jun 2023; 1446 samples, 723 grab plus 723 Moore swabs, from 50 sites) detected *S. Typhi* in **80 (5.5%)**, more often in grab samples (8.3%) than in Moore swabs (2.8%); site positivity tracked blood-culture-confirmed typhoid incidence (Spearman $\rho=0.32$, $p=0.014$ across the 60 vaccine-trial clusters; $\rho=0.57$, $p<0.0001$ by catchment area), and each 10-fold rise in incidence raised the odds of detection **2.43-fold (95% CI 1.38–4.27)** — a pre-typhoid-conjugate-vaccine baseline [PMID 42419341]
- **Polluted water is now treated as an antimicrobial-resistance reservoir, not only a pathogen reservoir** — shotgun metagenomics of untreated wastewater across urban and rural Nagpur (422 grab samples pooled into 138 composites, February–April 2021) recovered **606 unique resistance genes**, led by drug and biocide efflux determinants (19.1%) and macrolide-lincosamide-streptogramin B genes (17.6%), with carbapenemases (*bla*NDM, *bla*KPC; 3.5%) and colistin resistance (*mcr*; 0.7%) present at lower abundance; this is the practical link between sewage discharge and NAP-AMR containment [PMID 41757357]

Guideline and legislative updates

- The Water (Prevention and Control of Pollution) Act 1974 has been amended for the first time since 1988 — the Water (Prevention and Control of Pollution) Amendment Act, 2024 (introduced in the Rajya Sabha 5 Feb 2024, passed Rajya Sabha 6 Feb 2024 and Lok Sabha 8 Feb 2024): it is the **first amendment ever to decriminalise offences** under the Act, replacing imprisonment for discharge of polluting matter and for residual violations with **monetary penalties of ₹10,000 to ₹15 lakh** imposed by an adjudicating officer of Joint Secretary (Centre) or Secretary (State) rank, with appeal to the **National Green Tribunal after depositing 10% of the penalty**; it also lets the Centre prescribe the manner of nominating State Pollution Control Board chairpersons and exempt categories of industrial plants from "consent to establish". It applies at once to **Himachal Pradesh, Rajasthan and the union territories**, other states adopting it by resolution [Parliament of India 2024-Feb]
- Systematic review of arsenic and chromium in Indian ground and surface water (63 studies, search to 8 January 2025) — arsenic ingestion carried non-carcinogenic risk (hazard quotient >1) in **49% of adult and 60% of child-focused studies** against only 12% and 14% for chromium, carcinogenic risk was higher for both metals by ingestion, and groundwater posed greater risk than surface water; the affected belt is **Punjab, West Bengal, Haryana, Delhi, Uttarakhand and Jammu & Kashmir**, so the 0.01 mg/L arsenic limit is an enforcement problem, not a standard-setting one [PMID 42555689]

Global and WHO developments

- **SDG 6.1 status (2022 data)** — 73% of the world (6 billion people) used a safely managed drinking-water service, leaving **2.2 billion without**, of whom 115 million still drink untreated surface water; **at least 1.7 billion use a faecally contaminated source** and contaminated drinking water causes about **505,000 diarrhoeal deaths a year**. Safely managed coverage must rise **six-fold** on current trends to reach universality by 2030 [WHO 2023]
- **Cheap information, not only engineering, shifts chemical exposure** — a cluster-randomised trial in **2,334 households** across 156 villages of Samastipur district, Bihar, found that low-cost information on arsenic and on household water treatment raised safe-water consumption by **7.5 percentage points at one year** in the individual-information arm (**9.3 points at two months**, against a 34% baseline rate of unsafe water); the group-information arm's 2.7-point one-year effect was **not statistically significant** and should be read with caution. Cost per DALY averted was **0.89 euros for individual and 1.84 euros for group information** — a health-education intervention with a defensible cost-effectiveness figure [PMID 41994420]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 15 (Environment and Health, water, water-related diseases, water quality criteria and standards, sewage disposal), Ch 11 (Nutrition and Health, endemic fluorosis), Ch 7 (Health Programmes in India, Jal Jeevan Mission and Swachh Bharat Mission); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e, Ch 10 (Environmental Pollution); Sunder Lal — Textbook of Community Medicine, Ch 3 (Environmental Health Problems, figure of sewage treatment processes); AFMC Red Book of Community Medicine 2024, Ch 5 (Namami Gange, eutrophication of lakes); Manual on Municipal Solid Waste Management (statutory clearances under the Water Act 1974 and Environment Protection Act 1986); WHO Guidelines for Drinking-water Quality as reproduced in Park 28e Tables 4–6.

Recent-advances sources: 39223201 — Toilet construction under the Swachh Bharat Mission and infant mortality in India, *Scientific Reports* 2024, DOI [10.1038/s41598-024-71268-8](https://doi.org/10.1038/s41598-024-71268-8); 40160545 — Improving access to safe water and groundwater sustainability, *The Lancet Regional Health – Southeast Asia* 2025, DOI [10.1016/j.lansea.2025.100558](https://doi.org/10.1016/j.lansea.2025.100558); 40499970 — Progress and determinants of household access to improved drinking water in India using a Water Access Index: insights from the National Family Health Survey towards achieving SDG 6.1, *BMJ Open* 2025, DOI [10.1136/bmjopen-2024-091269](https://doi.org/10.1136/bmjopen-2024-091269); 42419341 — Integrating environmental and clinical surveillance for typhoid: insights from a prospective wastewater surveillance study in urban India, *The Lancet Microbe* 2026, DOI [10.1016/j.lanmic.2026.101414](https://doi.org/10.1016/j.lanmic.2026.101414) (abstract-only); 41757357 — Metagenomic insights into the urban-rural variation of antimicrobial resistance and pathogen reservoirs in untreated wastewater from central India, *Frontiers in Microbiology* 2025, DOI [10.3389/fmicb.2025.1722229](https://doi.org/10.3389/fmicb.2025.1722229); 42555689 — Health risks of arsenic and chromium in groundwater and surface water in India: a systematic review, *Indian Journal of Medical Research* 2026, DOI [10.25259/IJMR_218_2026](https://doi.org/10.25259/IJMR_218_2026); 41994420 — An information intervention in India reduces consumption of arsenic-contaminated drinking water and has small effects on health, *PNAS Nexus* 2026, DOI [10.1093/pnasnexus/pgag093](https://doi.org/10.1093/pnasnexus/pgag093). Non-journal sources: The Water (Prevention and Control of Pollution) Amendment Bill/Act, 2024 — introduced in the Rajya Sabha 5 February 2024, passed Rajya Sabha 6 February 2024 and Lok Sabha 8 February 2024 (Parliament of India); National Mission for Clean Ganga, Namami Gange programme status, August 2026; Department of Drinking Water and Sanitation, Swachh Bharat Mission-Grameen Phase II (ODF Plus); WHO drinking-water fact sheet (2022 data, updated September 2023).



Air pollution — effects and control

Asked in: Paper I Nov 2016 — RGUHS MD Community Medicine

Definition — Air pollution is the presence in the air of substances (gases and particulate matter) generated by the activities of man in concentrations that interfere with human health, safety or comfort, or are injurious to vegetation and animals, and that enter other environmental media so that chemicals reach the food chain or drinking-water and constitute an additional source of human exposure (Park 28e, Ch 15, pp 887-888). Its control is ultimately an engineering problem, for which the World Health Organization (WHO) has recommended five procedures (Park 28e, Ch 15, p 893).

Effects on health

Air pollution acts on the respiratory and cardio-respiratory systems in various ways; the elderly, children, smokers and those with chronic respiratory difficulty are the most vulnerable (Park 28e, Ch 15, p 893). About 1.3 billion urban residents worldwide are exposed to levels above the recommended limits, and mortality in cities with high pollution exceeds that in relatively cleaner cities by 15-20 per cent (Park 28e, Ch 15, pp 890, 892).

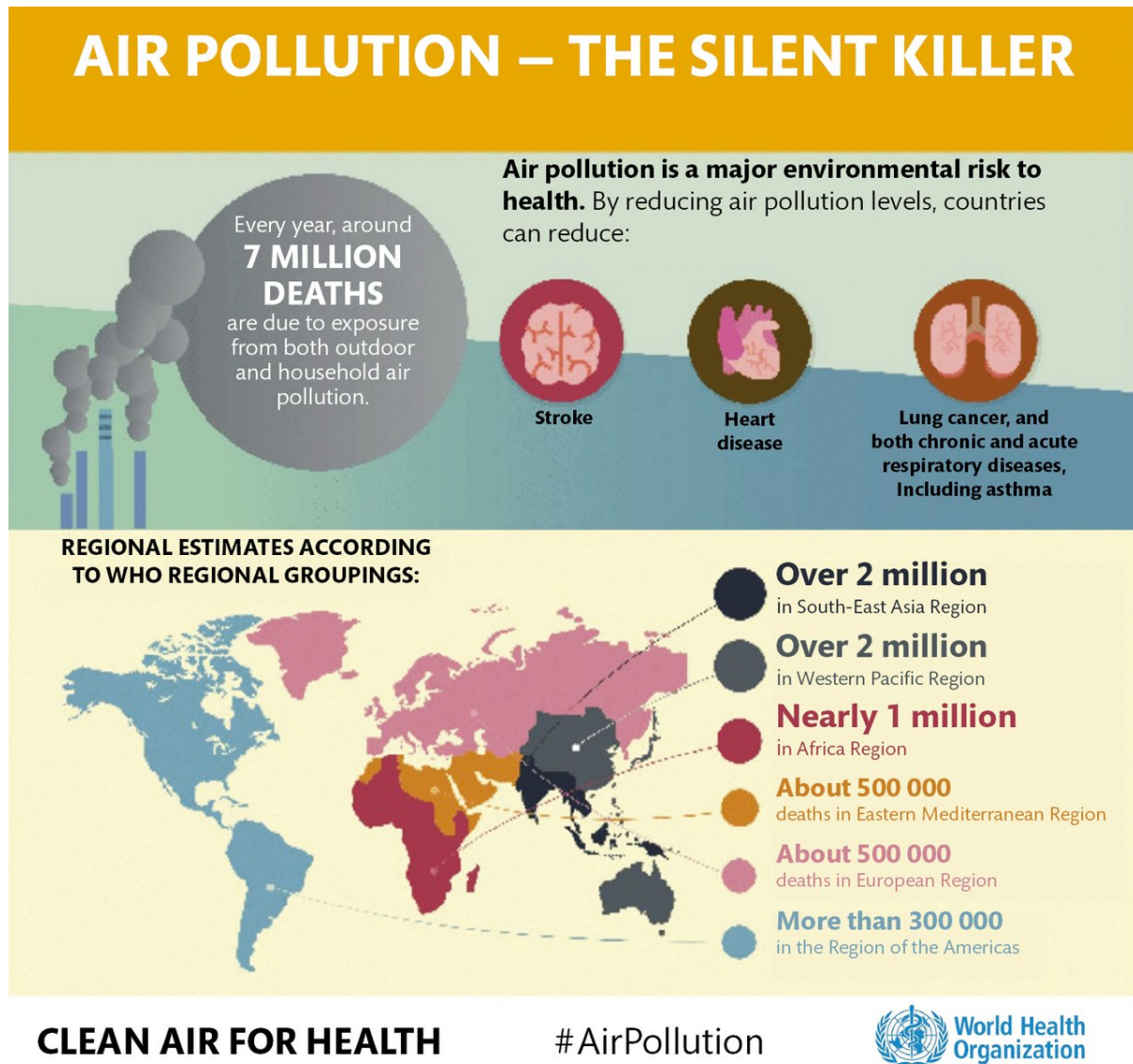


Fig 1.1 — 1 Air pollution—the silent killer. From <http://www.euro.who.int/en/hea> (Oxford public health 7th ed, p.135)

Timing	Effects	Illustrative evidence
Immediate (acute)	Borne by the respiratory system — irritation of conjunctiva, nose, throat and respiratory mucosa; acute bronchitis; acute exacerbation of bronchial asthma; when intense, death by suffocation	London disaster 1952, due to domestic coal burning; a sudden rise in air pollution is consistently followed by an immediate rise in morbidity and mortality (Park 28e, Ch 15, pp 892-893)
Delayed (chronic)	Chronic bronchitis, emphysema and chronic obstructive pulmonary disease (COPD), bronchiectasis, bronchial asthma, lung cancer, respiratory allergies	Delayed effects most commonly linked with air pollution (Park 28e, Ch 15, p 892; Suryakantha, Ch 7, p 68)
Attributable burden	Achievable reduction in urban air pollution could prevent 5 per cent of all infections and chronic respiratory diseases, averting 0.6 per cent of the global burden of disease	Park 28e, Ch 15, p 893

Pollutant-specific adverse effects (Park 28e, Ch 15, pp 889-892):

Pollutant	Principal source	Adverse effect
Oxides of nitrogen (nitric oxide ~95% of emission; nitrogen dioxide the health concern)	Automobile exhaust, coal combustion, gas stoves, kerosene space heaters	Respiratory tract irritation, bronchial hyperactivity, impaired lung defences, bronchiolitis obliterans; bronchitis in asthmatic children and reduced lung-function growth on long-term exposure
Sulphur dioxide	Power plants, smelters, oil refineries, kerosene heaters, coal combustion	Exacerbation of asthma and COPD, respiratory tract irritation, hospitalisation, death in severe exposure; people with asthma show changes in pulmonary function after exposure as short as 10 minutes
Ozone (ground level; a major constituent of photochemical smog)	Photochemical reaction of sunlight with nitrogen oxides and volatile organic compounds from vehicles and industry	Cough, substernal discomfort, bronchoconstriction, reduced exercise performance, triggering of asthma; daily mortality rises by 0.3% and mortality from heart disease by 0.4% per 10 µg/m ³ rise in ozone
Carbon monoxide	Petrol-powered vehicles (peaks coincide with morning and evening rush hours), fuel and tobacco combustion	Carboxyhaemoglobin formation — 4-12 hours needed for air and blood levels to equilibrate, hence the 8-hour averaging standard
Lead	Leaded gasoline, smelting, leaded paint, lead-glazed food vessels, some traditional medicines	Impaired neuropsychological development in children (loss of intelligence quotient, poor school performance, reduced attention span, antisocial behaviour) — believed irreversible ; anaemia, hypertension, renal impairment, immunotoxicity, reproductive toxicity; in pregnancy, miscarriage, stillbirth, preterm birth and low birth weight
Hydrocarbons	Automobile exhaust, cigarette smoke, incineration, petroleum processing	Lung cancer; take part in the reactions producing photochemical smog
Carbon dioxide	Combustion of coal, oil and gas	Not a classical pollutant, but the rise above natural level could raise global temperature enough to affect climate markedly

Household (indoor) air pollution — the attributable burden

People spend a large part of each day indoors, so indoor air pollution probably contributes more to population exposure than the outdoor environment; the major sources are indoor combustion of solid fuels, tobacco smoking, ingress of outdoor pollutants, emissions from construction materials and furnishings, and poorly maintained ventilation and air-conditioning systems (Park 28e, Ch 15, pp 890-891).

Outcome	Attributable burden from household air pollution (solid-fuel use)
All-cause	4.3 million premature deaths every year

Outcome	Attributable burden from household air pollution (solid-fuel use)
Distribution of these deaths	Stroke 34%, ischaemic heart disease 26%, COPD 22%, pneumonia 12%, lung cancer 6%
Childhood pneumonia	Exposure almost doubles the risk; over half of deaths from acute lower respiratory infection (ALRI) in children under 5 years are due to particulate matter inhaled from household solid fuels
Stroke	Nearly one quarter of all premature stroke deaths (about 1.4 million, half of them in women)
Ischaemic heart disease	About 15% of all deaths (over a million premature deaths annually)
COPD	Over one-third of premature COPD deaths in adults in low- and middle-income countries; women using solid fuels are 2.3 times as likely to have COPD as women using cleaner fuels, and indoor smoke nearly doubles the already-heightened risk in men
Lung cancer	About 17% of annual premature lung-cancer deaths in adults
Other associations	Low birth weight, tuberculosis, cataract, nasopharyngeal and laryngeal cancers

Source: Park 28e, Ch 15, pp 891-892.

Social, economic and ecological effects

Domain	Effects
Social and economic	Destruction of plant and animal life; corrosion of metals; damage to buildings; cost of cleaning, maintenance and repair; aesthetic nuisance; reduced visibility in towns; soiling and damage to clothing (Park 28e, Ch 15, p 893)
Ecosystem	Direct effects of pollutants on plants, animals and soil alter the structure and function of ecosystems, including their self-regulating ability (Park 28e, Ch 15, pp 887-888)
Acid rain	Sulphur dioxide, oxides of nitrogen and carbon dioxide dissolve in atmospheric water vapour to form sulphuric, nitric and carbonic acids — acidification of soil and water, deforestation, desertification, soil erosion, destruction of aquatic life and loss of food crops (Suryakantha, Ch 7, p 68)
Global warming	Accumulation of carbon dioxide, methane, chlorofluorocarbons and tropospheric ozone traps heat (greenhouse effect) — increased dryness, reduced world food production, melting of polar ice caps, rise in sea level and floods, smog formation, more skin cancer and cataract, and spread of tropical diseases to temperate regions (Suryakantha, Ch 7, p 68 and Ch 49, p 909)
Ozone-shield depletion	Thinning of the stratospheric ozone layer lets ultraviolet radiation through — inhibition of photosynthesis and fall in crop yield, disruption of the marine food chain, impaired human immunity, cataract, melanotic and non-melanotic skin cancers, damage to plankton, pollen grains and nitrifying soil bacteria (Suryakantha, Ch 7, p 68)
On animals	Cattle become weak and cachectic; yield of animal products falls (Suryakantha, Ch 7, p 68)

WHO recommended procedures for the prevention and control of air pollution

The control of air pollution is ultimately an engineering problem. WHO has recommended the following five procedures (Park 28e, Ch 15, p 893):

- ▶ **Containment** — prevention of the escape of toxic substances into the ambient air, achieved by engineering methods such as **enclosure, ventilation and air cleaning**; the major contribution in this field has been the development of "arresters" for the removal of contaminants. In practice this also means wet methods to suppress dust, hoods and exhaust ducts, and increasing the height of the smoke vent (Suryakantha, Ch 7, p 69).
- ▶ **Replacement** — replacing a technological process that causes air pollution with one that does not. Increased use of **electricity, solar power generation, natural gas and central heating in place of coal** has greatly helped smoke reduction; lead in petrol is being reduced and de-lead petrol is used in India. Within industry, siting plants away from human habitation on favourable topography and substituting liquefied petroleum gas (LPG) for coal are the practical forms of this measure (Park 28e, Ch 15, p 893; Suryakantha, Ch 7, p 69).
- ▶ **Dilution** — valid only so long as the load is within the **self-cleaning capacity of the environment**; some pollutants are readily removed by vegetation, and the establishment of "green belts" between industrial and residential areas is an attempt at dilution. The capacity for dilution is limited, and trouble occurs when the atmosphere is overburdened.
- ▶ **Legislation** — air pollution is controlled in many countries by suitable legislation, e.g. Clean Air Acts. Legislation covers the **height of chimneys**, powers to local authorities to carry out **investigations, research and education** concerning air pollution, creation of **smokeless zones**, and enforcement of **standards for ambient air quality**. India enacted the **Air (Prevention and Control of Pollution) Act, 1981** (Park 27e, Ch 15, p 869 — the corresponding 28e text is garbled in scan).
- ▶ **International action** — to deal with air pollution on a world-wide scale, WHO has established an **international network of laboratories** for the monitoring and study of air pollution: two international centres at **London and Washington**, three centres at **Moscow, Nagpur and Tokyo**, and 20 laboratories in various parts of the world; these centres issue warnings of air pollution where and when necessary (Park 27e, Ch 15, p 869).

Supporting general measures (Suryakantha, Ch 7, p 69; Sunder Lal, Ch 3, p 122): efficient public transport and fewer private vehicles; periodic servicing of vehicles and correct petrol-oil proportion; lead-free petrol with **catalytic converters**, which cut oxides of nitrogen, carbon monoxide and volatile organic compounds; **compressed natural gas (CNG)** for public transport (Supreme Court ruling of 5 April 2002 for Delhi); bypass roads to divert traffic; green belts; health education of the public on the hazards of air pollution and their own role in its control; and population stabilisation.

Air quality guidelines and their application in India

Pollutant / averaging time	WHO AQG 2005	WHO AQG 2021	Indian National Ambient Air Quality Standard 2009
PM _{2.5} — annual ($\mu\text{g}/\text{m}^3$)	10	5	40

Pollutant / averaging time	WHO AQG 2005	WHO AQG 2021	Indian National Ambient Air Quality Standard 2009
PM _{2.5} — 24-hour ($\mu\text{g}/\text{m}^3$)	25	15	60
PM ₁₀ — annual ($\mu\text{g}/\text{m}^3$)	20	15	60
PM ₁₀ — 24-hour ($\mu\text{g}/\text{m}^3$)	50	45	100
Ozone — 8-hour ($\mu\text{g}/\text{m}^3$)	100	100	100
Nitrogen dioxide — annual ($\mu\text{g}/\text{m}^3$)	40	10	40
Sulphur dioxide — 24-hour ($\mu\text{g}/\text{m}^3$)	20	40	80
Carbon monoxide — 8-hour (mg/m^3)	10	4 (24-hour)	2

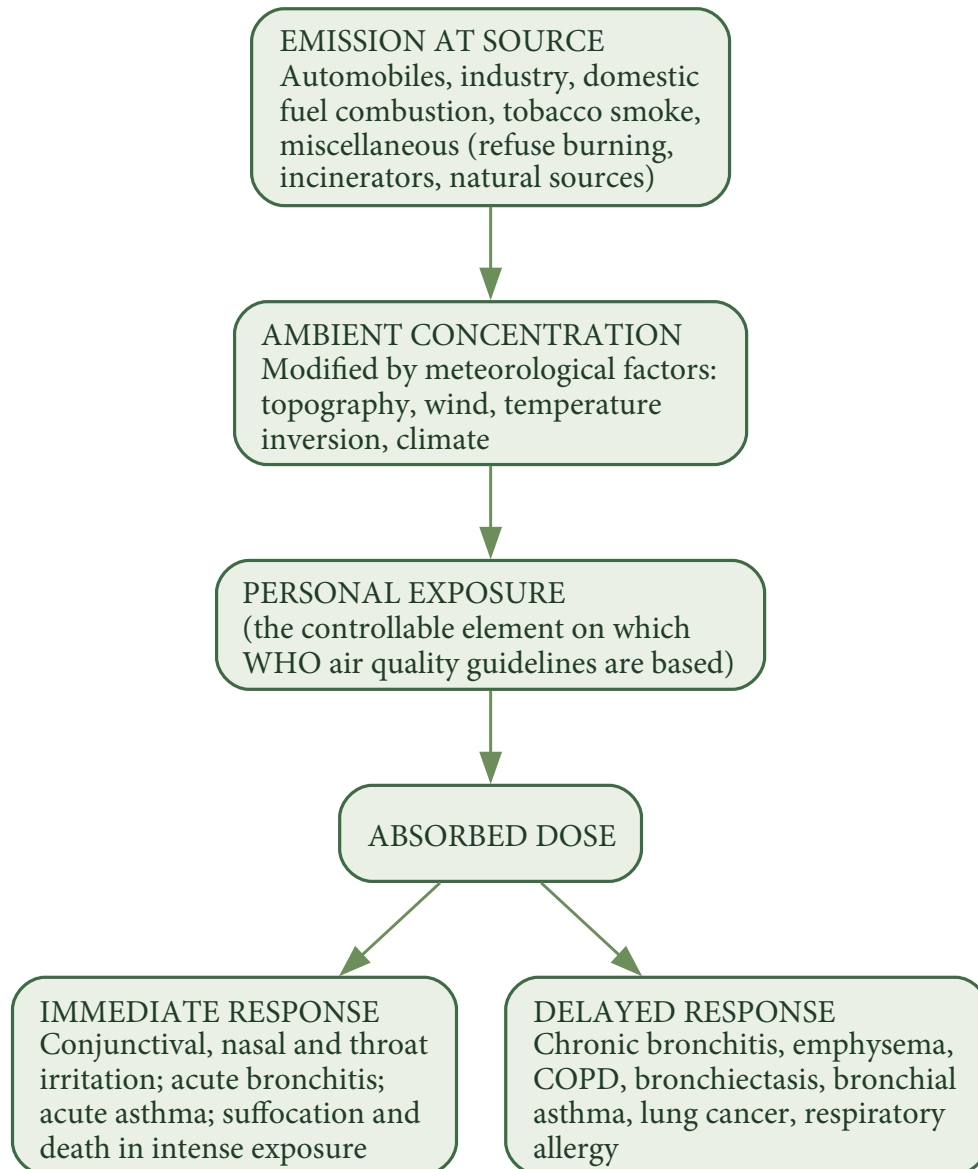
Source: Sunder Lal, Ch 3, p 123 (Table 3.3); carbon monoxide 8-hour value of $10 \text{ mg}/\text{m}^3$ also in Park 28e, Ch 15, p 892 (Table 2).

Indian instrument	Content
Air (Prevention and Control of Pollution) Act, 1981	An outcome of the 1972 Stockholm UN Conference on the Human Environment; extends to the whole of India. The Central Pollution Control Board (CPCB) improves air quality and prevents, controls or abates air pollution (Sec 16), advises government, lays down air standards (Sec 17); a Board may declare an air pollution control area (Sec 19) and restrict an industrial plant (Sec 21); no industrial plant may emit a pollutant in excess of State Board standards (Sec 22); powers of entry, seizure and testing (Sec 24); appointment of analysts and laboratories (Sec 28-30). Penalty for non-compliance: imprisonment of not less than one year six months, extendable to six years, with fine (Jugal Kishore, Ch 3, pp 958-959)
Air (Prevention and Control of Pollution) Rules 1982; Amendment Act 1987	Define Board procedures and empower central and state boards to meet grave air-pollution emergencies (Jugal Kishore, Ch 3, p 943)
National Air Quality Monitoring Programme (NAMP)	Run by CPCB with State Pollution Control Boards and the National Environmental Engineering Research Institute (NEERI), Nagpur, through 793 stations in 344 cities and towns across 28 states and 8 union territories ; four pollutants monitored — sulphur dioxide, nitrogen dioxide, respirable particulate matter (PM ₁₀) and PM _{2.5} (Sunder Lal, Ch 3, p 123)
National Clean Air Programme (NCAP), 2019	Tracker launched 2019; target of 40 per cent reduction in PM_{2.5} concentration by 2026 . NCAP report 2022 — Delhi the most polluted city (mean PM _{2.5} $99.71 \mu\text{g}/\text{m}^3$), Srinagar the cleanest ($26.33 \mu\text{g}/\text{m}^3$) (Sunder Lal, Ch 3, pp 122-123)
National Air Quality Index (AQI)	Six categories with health breakpoints — Good 0-50 (minimal impact), Satisfactory 51-100 (minor breathing discomfort in sensitive people), Moderate 101-200 (discomfort in those with lung or heart disease, children, older adults), Poor 201-300, Very poor 301-400 (respiratory illness on prolonged exposure), Severe >400 (respiratory effects even in healthy people) (Sunder Lal, Ch 3, p 124)

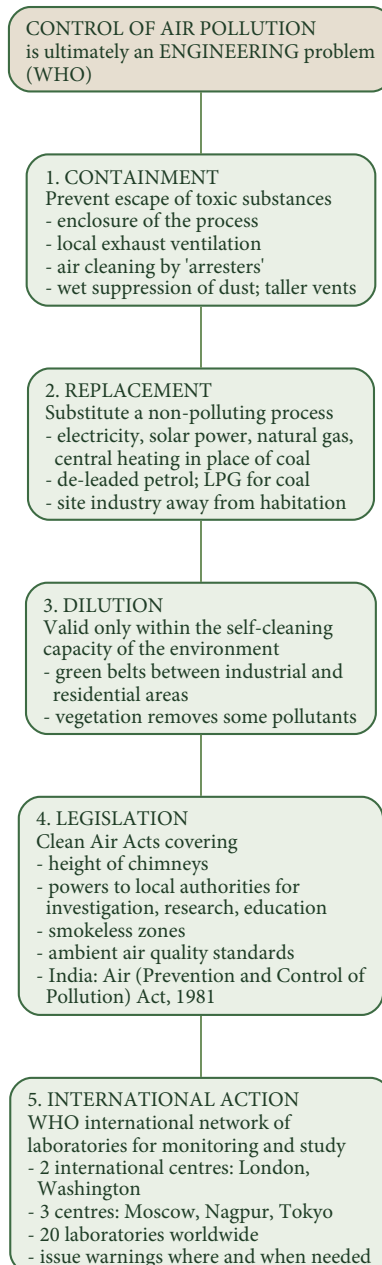
Indian instrument	Content
Global Environment Monitoring System (GEMS), WHO	Periodic monitoring of air quality by satellite and ground-level data, with acceptable standards laid down (Sunder Lal, Ch 3, p 123)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Diagram 1 — WHO health-effect wheel of air pollution (reproduce the embedded plate above by hand). Exact labels: *Air pollution — the silent killer*; central label *7 million deaths every year from outdoor and household air pollution*; four radiating outcome labels — *Stroke, Heart disease, Lung cancer, Chronic and acute respiratory disease including asthma*; regional burden labels — *South-East Asia Region over 2 million, Western Pacific Region over 2 million, Africa Region nearly 1 million, Eastern Mediterranean Region about 500 000, European Region about 500 000, Region of the Americas more than 300 000*.



Emission-to-response chain — exposure is the element WHO guidelines target, because reducing exposure lessens the dose and hence the response



The five WHO-recommended procedures for the prevention and control of air pollution (Park 28e, Ch 15, p 893)

High-yield exam pointers

- ▶ Mnemonic for the five WHO procedures — "C-R-D-L-I": **Containment, Replacement, Dilution, Legislation, International action**. Write them in this order; the stem asks for exactly these.
- ▶ Household air pollution kills **4.3 million** annually — stroke 34%, ischaemic heart disease 26%, COPD 22%, pneumonia 12%, lung cancer 6%.
- ▶ A feasible cut in urban air pollution would prevent **5% of all infections and chronic respiratory diseases** and avert **0.6% of the global burden of disease**; mortality in highly polluted cities exceeds cleaner cities by 15-20%.
- ▶ WHO air quality guideline 2021 vs 2005 for PM_{2.5}: annual 5 vs 10 µg/m³, 24-hour 15 vs 25 µg/m³; Indian National Ambient Air Quality Standard 2009 is annual 40 and 24-hour 60 µg/m³.

- ▶ **Air (Prevention and Control of Pollution) Act, 1981** — enforced through the Central and State Pollution Control Boards; NAMP runs 793 stations in 344 cities; NCAP targets a **40% cut in PM2.5 by 2026**.
- ▶ Ozone: daily mortality **+0.3%**, cardiac mortality **+0.4%** per 10 µg/m³ rise; sulphur dioxide changes pulmonary function in asthmatics after exposure as short as **10 minutes**.
- ▶ Classic episodes: Meuse Valley, Belgium 1930 (63 deaths); Donora, Pennsylvania 1948 (20 deaths); London 1952 (>4000 deaths, from domestic coal burning); Bhopal 1984 (methyl isocyanate — toxic air pollution, not photo-oxidant).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park 28e still quotes the WHO 2005 air quality guideline values (particulate matter 2.5 annual 10 µg/m³, nitrogen dioxide annual 40 µg/m³), whereas the WHO 2021 global update lowered most of them, so quote the 2021 figures with their year attached; the passage in Park 28e describing legislation and international action is badly scanned, and the wording used above was verified against the identical passage in Park 27e.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, WHO, MoEFCC, CPCB, CAQM, NCDC (NPCCHH), Pradhan Mantri Ujjwala Yojana, Jan Vishwas (Amendment of Provisions) Act 2023

Programme and policy updates (India)

- ▶ **NCAP — the target, the pollutant and the deadline have all changed.** The original goal of a 20-30 per cent cut by 2024 against a 2017-18 baseline has been superseded by a target of **up to 40 per cent reduction in PM10 — not PM2.5 — or attainment of the National Ambient Air Quality Standards, by 2025-26**, covering **130 non-attainment and million-plus cities in 24 states and union territories**; on the latest ministry statement in Parliament, 108 of these cities had lower annual PM10 than in 2017-18 (72 cities by over 20 per cent, 26 by over 40 per cent) and 14 met the standards in 2025-26 against six in 2017-18. Cities are now ranked annually by **Swachh Vayu Survekshan** (Indore first among million-plus cities in the 2025 round, announced 9 September 2025) and fund release has been made performance-linked, prioritising road-dust control, greening, vehicular emissions, crematoria and public awareness. An independent 28-city analysis found PM10 fell in 20 cities between 2018 and 2022, with roughly 62,000 excess deaths averted in 2021 relative to 2018. [MoEFCC 2026-Aug] [MoEFCC 2025-Sep] [MoEFCC 2025-May] [PMID 39970552]
- ▶ **A statutory commission and a forecast-driven graded response for Delhi-NCR.** The **Commission for Air Quality Management in the National Capital Region and Adjoining Areas Act, 2021** created a permanent statutory commission in place of the earlier court-appointed authority, and its revised **Graded Response Action Plan (GRAP)** operates in four AQI-linked stages — Stage I 'Poor' 201-300, Stage II 'Very Poor' 301-400, Stage III 'Severe' 401-450, Stage IV 'Severe+' above 450 — each invoked up to three days *in advance* on forecast rather than after the threshold is breached. This is the working example of legislation plus containment in a defined airshed, and is not in the textbook account. [CAQM 2021; revised GRAP 2022-Oct]

- ▶ **Household fuel replacement at national scale.** Pradhan Mantri Ujjwala Yojana has released over **10.3 crore** deposit-free LPG connections against a revised target of 10.6 crore, with the targeted refill subsidy continued for 2025-26 — the operational Indian form of WHO's "replacement" procedure applied to indoor solid-fuel smoke. [PMUY 2025]

Current national data

- ▶ **Air pollution killed 1.67 million Indians in 2019 — 17.8 per cent of all deaths that year** (ambient particulate matter 0.98 million, household air pollution 0.61 million, ambient ozone 0.17 million), with lost output valued at **1.36 per cent of India's GDP**. The direction of travel matters for the exam: the household air pollution death rate fell 64.2 per cent between 1990 and 2019, while the ambient particulate rate rose 115.3 per cent and the ozone rate 139.2 per cent. [PMID 33357500]
- ▶ **First Indian multi-city evidence for the acute effect.** Across ten cities and about 3.6 million deaths (2008-2019), daily mortality rose **1.42 per cent (95% CI 0.67-2.19) per 10 µg/m³ PM_{2.5}** at lag 0-1, and **3.57 per cent (2.11-5.04)** when causal instrumental-variable methods isolated locally generated pollution; **7.2 per cent of all deaths** were attributable to PM_{2.5} above the WHO 15 µg/m³ 24-hour value, about 33,600 deaths a year in those ten cities, and 99.8 per cent of city-days exceeded that value. Risk was steepest at *low* concentrations, which undercuts the argument that Indian cities are too polluted for marginal cuts to matter. [PMID 38969471]

Elimination, surveillance and disease status

- ▶ **Monitoring and illness surveillance have both been scaled up.** The manual NAMP network now runs **966 stations in 419 cities and towns across 28 states and 7 union territories** (November 2024), well beyond the older 793-station/344-city figure, and is supplemented by continuous automatic stations. On the health side, the **National Programme on Climate Change and Human Health (NPCCHH)**, approved under the National Health Mission in February 2019 with NCDC as technical nodal agency, makes air pollution one of three focus areas and runs sentinel surveillance of air-pollution-related illness (daily respiratory emergency attendances from tertiary hospitals in highly polluted cities), alongside a health adaptation plan and state action plans. [CPCB 2024-Nov] [NPCCHH 2019]
- ▶ **The list of outcomes is widening beyond lung and heart.** In older Indians, higher long-term PM_{2.5} — particularly from residential combustion — was associated with faster cognitive decline (−0.012 standard units per year per SD; 95% CI −0.021 to −0.004), while global burden work attributes **about 20 per cent of diabetes deaths and DALYs to PM_{2.5}** (13.5 per cent ambient, 6.5 per cent household). Quote these as additions to the classical respiratory-cardiovascular list. [PMID 41045631] [PMID 38310921]

Guideline and legislative updates

- ▶ **The penalty clause of the Air Act 1981 has been rewritten.** The **Jan Vishwas (Amendment of Provisions) Act, 2023** decriminalised key offences (including section 37, emission beyond prescribed standards), replacing imprisonment with monetary penalties of **not less than ₹10,000 up to ₹15 lakh** imposed by a centrally appointed **adjudicating officer**, appealable to the National Green Tribunal, with proceeds credited to the Environment Protection Fund; the Air Act amendments took effect from **1 April 2024**. Any answer still quoting "imprisonment of one year six months extendable to six years" is now out of date. [Jan Vishwas Act 2023]

Global and WHO developments

- ▶ **A first global numerical target for air pollution and health.** The Second WHO Global Conference on Air Pollution and Health (Cartagena, 25-27 March 2025) drew commitments from more than 50 countries, cities and organisations around a shared goal of halving the health impacts of air pollution by 2040; the **Seventy-eighth World Health Assembly (May 2025)** then adopted the updated road map for the health sector response, carrying the first **voluntary global target to halve the health impacts of air pollution by 2040**. This sits alongside the standing figures of roughly 7 million deaths a year and 99 per cent of the world breathing air above WHO guideline values. [WHO 2025-Mar] [WHO 2025-May]
- ▶ **Household air pollution, restated for 2021.** GBD 2021 estimates **2.67 billion people — 33.8 per cent of the world — still exposed to household air pollution**, at a mean PM_{2.5} of 84.2 µg/m³, accounting for **111 million DALYs (3.9 per cent of all DALYs)**; the age-standardised rate has fallen 63.8 per cent since 1990 but south Asia and sub-Saharan Africa remain the worst affected, and about a third of the burden is mediated through short gestation and low birth weight. [PMID 40118081]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e, Ch 15 (Environment and Health), pp 887-893; Park's Textbook of Preventive and Social Medicine 27e, Ch 15, p 869 (for the legislation and international-action passage, illegible in the 28e scan); Suryakantha — Community Medicine with Recent Advances, Ch 7 (Air and Ventilation), pp 67-69 and Ch 49 (Global Warming), p 909; Sunder Lal — Textbook of Community Medicine, Ch 3 (Environmental Health Problems), pp 122-124; Jugal Kishore — National Health Programmes of India, Ch 3, pp 943, 958-959 (Air Act 1981); Oxford Textbook of Public Health 7e, p 135 (WHO "Air pollution — the silent killer" figure). Named documents: WHO Air Quality Guidelines Global Update 2005 and WHO Global Air Quality Guidelines 2021; Indian National Ambient Air Quality Standards 2009; National Clean Air Programme 2019 (report 2022).

Recent-advances sources: 33357500 — Health and economic impact of air pollution in the states of India: the Global Burden of Disease Study 2019, *Lancet Planetary Health* 2021, doi:[10.1016/S2542-5196\(20\)30298-9](https://doi.org/10.1016/S2542-5196(20)30298-9); 38969471 — Ambient air pollution and daily mortality in ten cities of India: a causal modelling study, *Lancet Planetary Health* 2024, doi:[10.1016/S2542-5196\(24\)00114-1](https://doi.org/10.1016/S2542-5196(24)00114-1); 40118081 — Global, regional, and national burden of household air pollution, 1990-2021: a systematic analysis for the Global Burden of Disease Study 2021, *Lancet* 2025, doi:[10.1016/S0140-6736\(24\)02840-X](https://doi.org/10.1016/S0140-6736(24)02840-X); 41045631 — Particulate air pollution from different emission sources, cognitive performance, and cognitive declines in India, *Environment International* 2025, doi:[10.1016/j.envint.2025.109826](https://doi.org/10.1016/j.envint.2025.109826); 38310921 — Air pollution exposure and cardiometabolic risk, *Lancet Diabetes & Endocrinology* 2024, doi:[10.1016/S2213-8587\(23\)00361-3](https://doi.org/10.1016/S2213-8587(23)00361-3); 39970552 — Impact of the National Clean Air Programme (NCAP) on the particulate matter pollution and associated reduction in human mortalities in Indian cities, *Science of the Total Environment* 2025, doi:[10.1016/j.scitotenv.2025.178787](https://doi.org/10.1016/j.scitotenv.2025.178787) (abstract-only). Non-journal sources: WHO Second Global Conference on Air Pollution and Health, Cartagena, 25-27 March 2025; Seventy-eighth World Health Assembly, May 2025 — updated road map on the health sector response to air pollution with the voluntary 2040 target; MoEFCC statement in Rajya Sabha on NCAP progress, August 2026; Swachh Vayu Survekshan 2025 results, September 2025; MoEFCC revised NCAP fund-release guidelines, 14 May 2025; CPCB National Air Quality Monitoring Programme network status, November 2024; Commission for Air Quality Management in National Capital Region and Adjoining Areas Act 2021 and the revised Graded Response Action Plan (in force from October 2022); National Programme on Climate Change and Human Health, MoHFW/NCDC, 2019; Pradhan Mantri Ujjwala Yojana status, 2025; Jan Vishwas (Amendment of Provisions) Act 2023, Air Act amendments effective 1 April 2024.

♦ ♦ ♦

Five R's of waste management

Asked in: Paper I 02-Jan-2025 — RGUHS MD Community Medicine

Definition — The five R's — **Refuse, Reduce, Reuse, Recycle and Recover** — are a mnemonic for the **waste-management hierarchy**, the order of preference in which any waste stream should be dealt with, ending in safe disposal of the inert residue as the last resort. The integrated solid waste management (ISWM) hierarchy ranks **waste minimisation (at-source reduction and reuse)** as the most preferred strategy, then **recycling**, then **composting**, then **waste-to-energy**, with **sanitary landfill of residual inert waste as the least preferred** (Manual on Municipal Solid Waste Management, Part II §2.1, p 127). Source reduction sits at the top because it alone directly reduces the quantity of waste generated, and with it the financial and environmental cost of handling that waste.

The five R's at a glance

R	Principle	Municipal / domestic example	Health-care example
1. Refuse	Decline the product or packaging that will inevitably become waste; enforced by product bans	Full ban on plastic carry bags in Shimla , enforceable only after the Himachal Pradesh government legislated it; the same state banned non-biodegradable packaging for 25 kinds of edible foodstuff (MSWM Manual, Part II §2.1.4.1, p 130)	Refuse chlorinated plastics and mercury-containing devices; by law plastic material in any form shall not be used in any package for gutkha, pan masala and tobacco (Plastic Waste Management Rules 2016, reproduced in MSWM Manual, Part II §7.4, p 501)
2. Reduce	Cut the quantity and the toxicity of waste generated; increase product durability, reusability and reparability	Buying in bulk; refill containers; redesigning packaging to eliminate excess material while keeping strength; double-sided printing and electronic training manuals in offices (MSWM Manual, Part II §2.1.2 and §2.1.4.2, pp 127, 131)	Sound purchase policy and stock rotation to avoid expired medicines ; above all correct segregation, since mixing makes the whole bag hazardous — 75–90% of health-care waste is non-risk "general" waste and only 10–25% is hazardous (Park 28e, Ch 16, p 943)
3. Reuse	Use the same article again, in its own form, without reprocessing	Reuse of carry bags and packaging jars; reusable transport containers ; rechargeable in place of single-use batteries ; deposit-refund ("bottle bill") systems for glass bottles, aluminium cans and lead-acid batteries (MSWM Manual, Part II §2.1, pp 127–131)	Reuse of items of repeat utility after sterilisation — linen, instruments, glassware (Suryakantha, Ch 13, p 93). Disposables must never be reused — sharps and red-category plastics are shredded or mutilated precisely to prevent unauthorised reuse (BMW Rules 2016, Schedule I)
4. Recycle	Transform segregated waste into a new product or a raw material , so the original loses its identity	Paper, cardboard, metal, glass, plastic and e-waste routed through the informal kabadi / scrap-dealer sector and waste pickers to the recycling industry (MSWM Manual, Part II §3.1, pp 209–211)	Red-bag contaminated plastics — autoclave, microwave or hydroclave, then shred or mutilate, then to registered or authorised recyclers ; blue glassware disinfected or autoclaved and then sent for recycling (BMW Rules 2016, Schedule I)

R	Principle	Municipal / domestic example	Health-care example
5. Recover	Extract material or energy from what cannot be reduced, reused or recycled, before final disposal	Composting of biodegradable waste (windrow, in-vessel, vermicomposting); waste-to-energy — refuse-derived fuel (RDF), biomethanation, co-processing, incineration (MSWM Manual, Part II §2.1.2, p 127)	Treated red-category plastic sent for energy recovery , plastics-to-diesel or fuel oil , or road making ; metal sharps encapsulated and sent to iron foundries; silver recovered from X-ray film developing liquid before the effluent is pre-treated (BMW Rules 2016, Schedule I)

Refuse and Reduce — source reduction, the top of the hierarchy

Waste minimisation reduces the amount **and the toxicity** of waste produced and covers all activities that increase product durability, reusability and reparability (MSWM Manual, Part II §2.1.2, p 127). Its instruments fall into two orders.

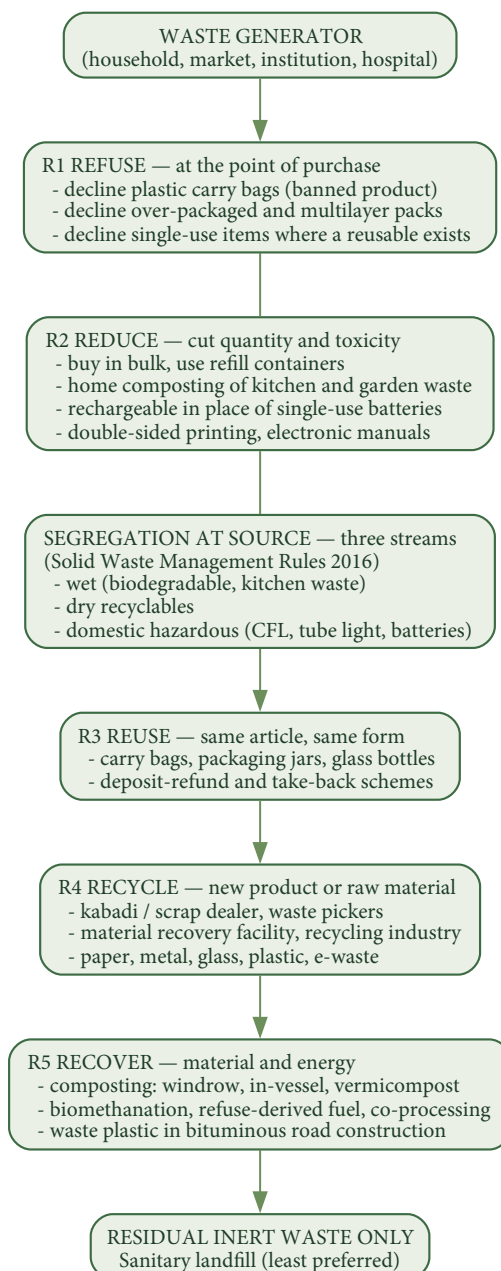
A. Instruments requiring a national or state-level directive (MSWM Manual, Part II §2.1.4.1, pp 128–130):

- ▶ **Extended producer responsibility (EPR)** — the producer is held responsible for the post-consumer stage of the product: separate collection (e-waste), reuse (deposit–refund for bottles), recycling (used cars), and storage and treatment (batteries). In India it is already applied to electronics, batteries, packaging and consumer durables.
- ▶ **Deposit–refund systems** — a disposal fee charged to the consumer and refunded when the used container is returned; manufacturers take back glass bottles, aluminium cans and lead-acid batteries by this route.
- ▶ **Quotas** — a stipulated percentage of product content or packaging to be of recycled or refillable material, with a mandatory deposit system enforced if the quota is missed.
- ▶ **Product bans** — the threat of a ban makes producers phase out undesirable materials and design for recyclability and high rates of reuse.
- ▶ **Product charges / eco-tax** — an eco-tax on polyvinyl chloride (PVC) in Belgium raised its cost and cut consumption of the polymer.
- ▶ **Eco-labelling standards** based on waste-reduction and recycling potential, and **eco-industrial parks** built around material and resource exchange between industries.

B. Instruments requiring urban local body (ULB) support (MSWM Manual, Part II §2.1.4.2, pp 130–131):

- ▶ Awareness and education programmes for residential, commercial and industrial generators, to raise participation in at-source reduction.
- ▶ **Domestic (home) composting**, which keeps food waste, leaves and garden trimmings out of the collection system altogether.
- ▶ **Material substitution** campaigns — rechargeable for single-use batteries, reusable products for disposable ones.
- ▶ **Local bans and bye-laws** on products and packaging that cannot be reused, repaired, recycled or composted.

- **Green procurement and take-back programmes** — suppliers to the municipality made responsible for taking back monitors, auto oil, batteries and paper; procurement to give preference to recyclable products.
- **On-site business assistance** — improving product design to use less material, redesigning packaging, packaging-return schemes, reusable transport containers, purchase in bulk.



The five R's applied along the municipal solid waste chain, with sanitary landfill as the terminal step (Manual on Municipal Solid Waste Management, Part II; Solid Waste Management Rules 2016)

Reuse and Recycle — keeping material in circulation

- **Recycling defined** — under the Solid Waste Management (SWM) Rules 2016, recycling is "the process of transforming segregated solid waste into a new product or a raw material for producing new products"; the recycled article loses the identity of the original (MSWM Manual, Part II §3.1, p 209; Suryakantha, Ch 13, p 93).

- ▶ **Segregation is the precondition.** The SWM Rules 2016 require the generator to separate waste into **wet, dry recyclable and domestic hazardous** streams ("basic segregation"), and require the ULB to set up a **domestic hazardous waste deposition centre, one for every twenty square kilometres or part thereof** (Solid Waste Management Rules 2016, duties of local authorities, reproduced in MSWM Manual, Part II). Errors in segregation mean infectious waste may be picked up for recycling (Suryakantha, Ch 13).
- ▶ **The informal sector is the Indian recycling system.** The kabadi / scrap-dealer network and waste pickers collect the bulk of recyclables; EPR schemes that build them in gain collection efficiency and lower supply-chain cost (MSWM Manual, Part II §2.1.4.1, p 130).
- ▶ **Benefits of recycling** — for the ULB: reduced waste volume, savings in collection, transport and disposal, longer landfill life, less environmental management effort; for the economy: fewer imported raw materials and fertilisers, and livelihoods in the recycling industry; for the environment: sustainable use of resources and less land needed for storage (MSWM Manual, Part II §3.1.1, pp 209–211).
- ▶ **Limits of recycling** — virgin plastic can be recycled only **2–3 times** before thermal degradation, so recycling delays rather than abolishes disposal; plastics are also highly resistant to biodegradation and choke urban drains, causing local flooding (MSWM Manual, Part II §7.4.2, p 505).
- ▶ **A down-cycling example** — waste polyethylene **below 60 microns**, which is not worth recycling, is cleaned, shredded and blended into bitumen heated to **160–170 °C** and stirred for 20–30 minutes for bituminous road construction; polyethylene above 60 microns goes to conventional recycling (MSWM Manual, Part II §7.4.7.1, p 509).

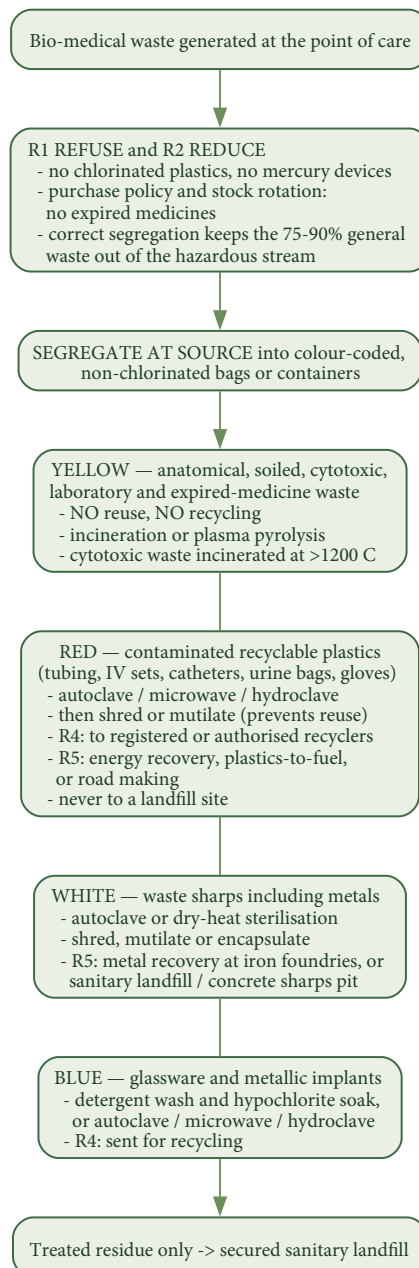
Recover — material and energy before final disposal

- ▶ **Composting** — processing biodegradable waste to recover compost, by windrow, in-vessel or vermicomposting. Park describes composting as the combined disposal of refuse and nightsoil in which heat of **about 60 °C or higher over several days** destroys fly eggs and larvae, weed seeds and pathogenic agents; the **Bangalore method** (anaerobic hot fermentation, developed at the Indian Institute of Science) is the recommended Indian variant, and the **manure pit** is its rural equivalent (Park 28e, Ch 15, p 910).
- ▶ **Energy recovery** — refuse-derived fuel (RDF), biomethanation, co-processing of the combustible non-biodegradable dry fraction, and incineration. RDF is the next desirable strategy after source reduction, reuse and recycling, and it greatly reduces the landfill space required (MSWM Manual, Part II §3.5.2, p 311).
- ▶ **Incineration** reduces waste volume by **85–95%** and is the method of choice for hospital refuse and wherever suitable land is unavailable; it remains unpopular in India because the refuse carries a high proportion of fine ash, and burning organic matter loses much-needed manure (Suryakantha, Ch 13; Park 28e, Ch 15, pp 909–910).
- ▶ **Reverse production system for e-waste** — infrastructure to recover and reuse **lead, copper, aluminium, gold, plastics, glass and wire** from discarded electronics, with manufacturers, distributors and retailers made responsible for recycling or disposal of their own products (Suryakantha, Ch 14).

- **Residual disposal, not a sixth R** — only inert residue, after recycling and reuse to the maximum extent possible, goes to a **sanitary landfill**. Open dumping was condemned by the WHO Expert Committee (1967) as "a most insanitary method of disposal of refuse because of the health hazards" (Suryakantha, Ch 12), and Park likewise holds that dumping should be outlawed and replaced by sound procedures (Park 28e, Ch 15, p 909).

Applying the five R's to bio-medical waste

For health-care waste the hierarchy is modified in one important respect: **reuse and recycling are forbidden for infectious categories** and are permitted only after treatment for the designated recyclable ones. Slaughterhouse and bio-medical waste must not be recycled but treated and disposed of appropriately (MSWM Manual, Part II §7.3, p 499).



Where each R acts in bio-medical waste — colour-coded segregation and the permitted recycling and recovery routes (Bio-Medical Waste Management Rules 2016, Schedule I)

Public health and economic case for the five R's

Argument	Supporting figure (with year)
Sheer volume	India generated 1,43,449 tonnes per day of municipal solid waste (Status report on municipal solid waste management, 2014–15), of which 40–60% is organic and 10–20% recyclable — most of it therefore amenable to the higher R's (MSWM Manual, Part II §2.1.3, p 128)
Cost avoided	At about Rs 1,000 per tonne, processing and disposal of that waste costs roughly Rs 10.7 crore per day; minimisation saves collection, treatment and disposal costs (MSWM Manual, Part II §2.1.3, p 128)
Land	A rapidly shrinking urban land bank and rising cost of land for processing and disposal; recycling gives a longer life span for landfills (MSWM Manual, Part II §2.1.3, §3.1.1, pp 128, 209–211)
Climate	Disposal releases methane and carbon dioxide on a large scale; waste minimisation cuts greenhouse-gas emissions and the associated climate impact (MSWM Manual, Part II §2.1.3, p 128)
Health-care waste composition	75–90% of health-care waste is non-risk general waste and only 10–25% is hazardous; the WHO planning estimate for developing countries is 80% general, 15% pathological and infectious, 1% sharps, 3% chemical and pharmacological, and under 1% special waste (Park 28e, Ch 16, p 943)
Segregation fails in practice	A Bangalore survey found hospitals generating 0.5–4 kg per bed per day, of which 45–50% was infectious, yet segregation of infectious from non-infectious waste was practised in only about 30% of hospitals (Park 28e, Ch 16, p 944)
Occupational risk	Pathogens in infectious waste enter through a puncture, abrasion or cut, through mucous membranes, by inhalation or by ingestion; there is strong evidence of transmission of HIV and hepatitis B and C via health-care waste, and waste handlers, transporters, landfill workers and scavengers are among the main groups at risk (Park 28e, Ch 16, p 944)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **The waste-management hierarchy pyramid** — draw as an inverted triangle, widest at the top. Labels from the top (**Most preferred**) downward: *Refuse; Reduce — at-source reduction; Reuse of products, e.g. carry bags and packaging jars; Recycle — processing non-biodegradable waste to recover commercially valuable materials (plastic, paper, metal, glass, e-waste); Recover — composting (windrow, in-vessel, vermicomposting) and waste-to-energy (refuse-derived fuel, biomethanation, co-processing, incineration); Sanitary landfill — safe disposal of inert residual waste after recycling and reuse to the maximum extent possible (Least preferred).*
- **Three-bin source segregation under the Solid Waste Management Rules 2016** — draw three bins side by side and label: *Green — wet biodegradable (kitchen waste); Blue — dry recyclables (paper, plastic, metal, glass); Third bin — domestic hazardous (compact fluorescent lamp, tube light, batteries, expired medicine), with an arrow from the third bin to Domestic hazardous waste deposition centre — one per 20 square kilometres.*

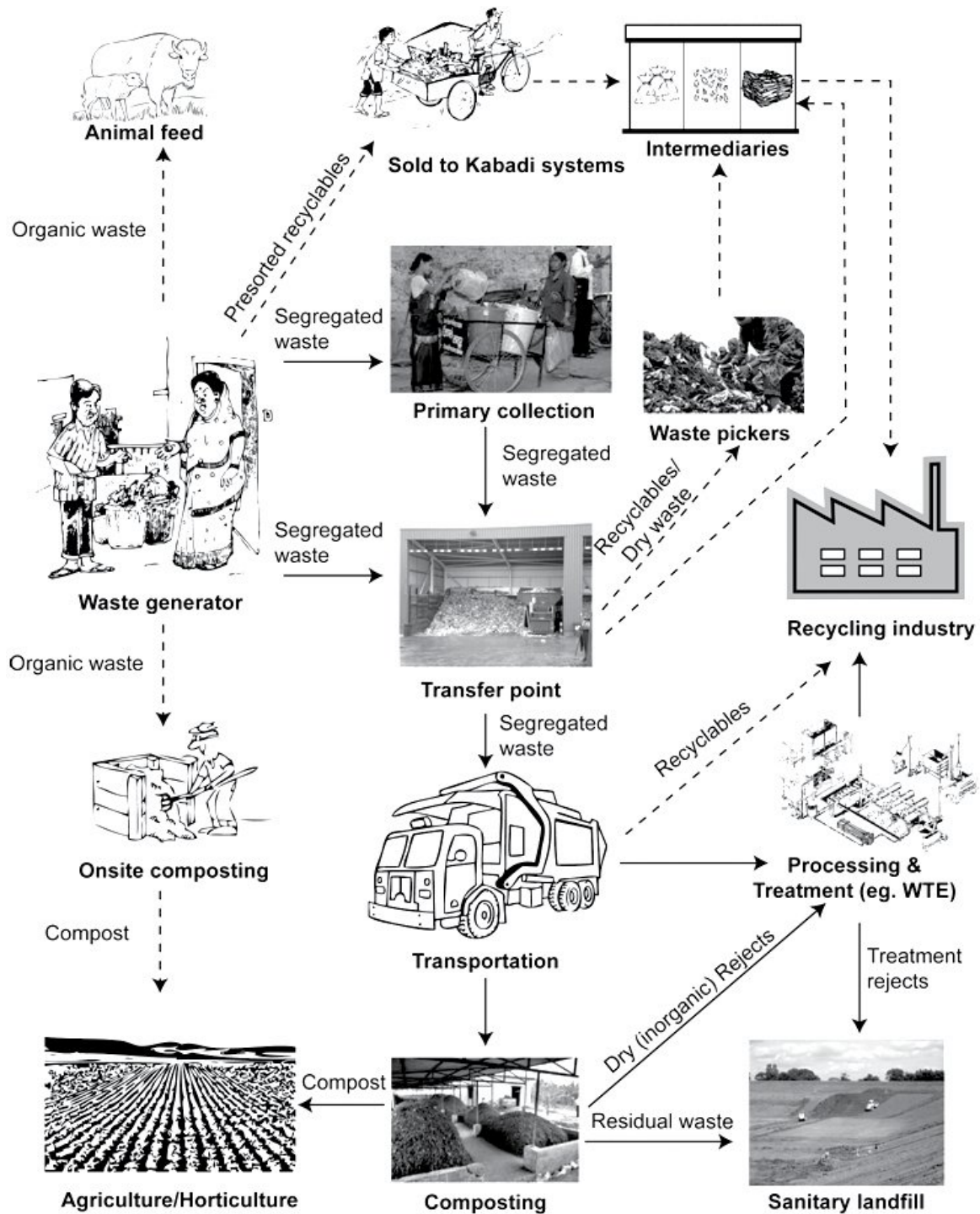


Fig 3.4 — Stages of Material Recovery in Municipal Solid Waste Management Chain: reuse as animal feed, recycling via kabadi systems and waste pickers, recovery by composting and waste-to-energy, and residual waste to sanitary landfill (solid waste management, p.255)

High-yield exam pointers

- **Order of the hierarchy, top to bottom:** Refuse → Reduce → Reuse → Recycle → Recover → safe disposal of residual inert waste in a sanitary landfill. Write the order first; the marks are for the sequence, not for the list of words.
- **Statutory anchors to name:** Solid Waste Management Rules 2016 (three-stream segregation; definition of recycling; domestic hazardous waste deposition centre **one per 20 sq km**); Plastic Waste Management Rules 2016 (carry bag **not less than 50 microns**; recycling as per IS

14534:1998; compostable bags as per IS 17088:2008); Bio-Medical Waste Management Rules 2016 (four-colour segregation, Schedule I).

- ▶ **Dated Indian figures:** 1,43,449 tonnes per day of municipal solid waste (2014–15), 40–60% organic, 10–20% recyclable, disposal cost about Rs 10.7 crore per day.
- ▶ **Health-care waste split:** 75–90% general, 10–25% hazardous (Park 28e); Bangalore survey — 0.5–4 kg per bed per day, 45–50% infectious, segregation in only 30% of hospitals.
- ▶ **Two limits worth stating:** virgin plastic can be recycled only 2–3 times; incineration cuts volume by 85–95% but destroys manurial value and suits India's high-ash, low-calorific refuse poorly.
- ▶ **One worked example per R** is what earns the "with suitable examples" marks — Shimla plastic-bag ban (Refuse), home composting (Reduce), deposit–refund bottles (Reuse), kabadi network and treated red-bag plastics to authorised recyclers (Recycle), refuse-derived fuel and metal sharps to iron foundries (Recover).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: the retrieved corpus states the hierarchy in full (at-source reduction and reuse → recycling → composting → waste-to-energy → landfill) but does not print the phrase "five R's" as a numbered list, and teachers vary the fifth R — *Rethink*, *Repair*, *Repurpose* or *Refill* is sometimes substituted for *Refuse* or *Recover*; write whichever set you were taught, keep the hierarchical order, and give one concrete example for each, because that is what the stem is marking.

Recent advances [RA]

Current as of 2026-08-19 · depth: elaborate · sources: PubMed, WHO, CPCB national waste data

Programme and policy updates (India)

- ▶ **Extended producer responsibility now runs across four streams** — India's EPR framework covers waste oils, e-waste, batteries and plastic packaging (CPCB, 2022), among the broadest coverage of the eleven developing countries scored against a waste-prevention indicator set (zero-waste index, food-loss index, EPR, education and awareness, waste prevention, plastic-bag reduction ratio); the same assessment found law running ahead of practice, with weak measured reduction and diversion. India's per-capita generation was taken as 0.3 kg/day rural and 0.55 kg/day urban, and the country's instruments listed as the Solid Waste Management Rules 2016, the Plastic Waste Management Rules 2016 and Swachh Bharat Mission (Urban) 2.0. [PMID 40173265]
- ▶ **Legacy-dumpsite remediation is now the measurable arm of the 2016 regime** — satellite time-series of the Sarona legacy dumpsite at Raipur showed land occupied rising 661.9% up to 2019, then falling 36.65% in 2019–20 and a further 78.30% in 2020–21 once the Solid Waste Management Rules 2016 were enforced and a material recovery facility began working in 2020; 4.84 acres were left to remediate. Biomining of legacy dumps is the recovery step examiners now expect alongside composting and waste-to-energy. [PMID 39877284]

Current national data

- **Quote the 2021–22 tonnage, not the 2014–15 one** — Central Pollution Control Board figures for 2021–22 give roughly **1,70,338 tonnes per day** of municipal waste generated with only about **91,512 tonnes** adequately treated, the rest going to dumping, burning or informal handling; Indian plastic consumption exceeds **20 million tonnes a year** with only about **13% recovered**, and city per-capita generation ranges 0.45–1.24 kg/person/day (most cities 0.5–0.7). Rural surveys find flexible plastics ~60% uncollected, ~20% openly burnt and ~40% dumped. [PMID 42266523]
- **The collection–processing gap, and what recovery could buy** — **stated as a projection, not an achievement** — a CPCB-based review puts national generation at about **62 million tonnes a year**, of which only **75–80% is collected and 22–28% processed**; Uttar Pradesh alone generates over **15,000 tonnes a day**. On the authors' modelling of refuse-derived fuel prepared from five UP landfill sites, RDF substituted for coal in industrial furnaces **could** divert over **30%** of solid waste from landfill and cut greenhouse-gas emissions by as much as **25%** against conventional disposal — a modelled potential, not a measured outcome. Do **not** quote the same abstract's claim of lower sulphur dioxide and oxides of nitrogen: the paper's own discussion states the opposite, that burning RDF has produced **higher** NO_x and SO_x, SO_x by about **5–10%**. The fuel actually measured was poor — gross calorific value **1250–1680 kJ/kg** across the five sites, below the 3000–4500 kJ/kg reported for comparable Indian regions and far short of the 15–25 MJ/kg a cement kiln needs — so upstream segregation, not the furnace, is the rate-limiting step. [PMID 39811298]

Elimination, surveillance and disease status

- **Nationwide audit of bio-medical waste rule compliance** — 267 health-care facilities surveyed July–September 2023 (67% accredited): only 58% had fully implemented barcoding and 43% used dedicated tags; 49% still cut needles before disposal while 51% discarded them directly into the sharps container; 79% autoclaved laboratory waste; 92% had a dedicated temporary storage area; and 28% still used mercury-filled devices. Sharps handling, blue-bin use and pharmaceutical-waste disposal were the most variable practices. [PMID 40744379]
- **Pooled occupational risk in waste handlers** — meta-analysis of 14 studies from low-resource settings: needle and sharps injury prevalence 26.4%, hepatitis B 5.8% and hepatitis C 2.4%; protective equipment use was gloves 84.9%, masks 76.9%, gowns 55.4% but boots only 19.0%; just 15.3% were immunised against hepatitis B and 57.4% had received training. This is the current quantitative version of the occupational-risk argument for segregation. [PMID 42022636]

Guideline and legislative updates

- **Household bio-medical waste** — **a statutory duty almost nobody performs** — Schedule I Part 2 of the Bio-Medical Waste Management Rules 2016 requires waste from health care at home to be kept separate and handed to the municipal collector, and obliges the urban local body to tie up with a common bio-medical waste treatment facility; in urban Nagpur (2023–24, 240 households) 76.7% separated wet from dry domestic waste but **none** segregated bio-medical waste and 98.3% put it into ordinary municipal collection. Sanitary pads (84.2%), expired medicines (81.7%), swabs and bandages (57.1%) and used blades (56.3%) were the commonest

items; 64.2% were willing to segregate but only 29.2% to return waste to a health centre. [PMID 42125580]

- **Safe drug disposal is the gap in India's AMR plan** — expired and unused antimicrobials discarded into household bins, drains and toilets reach soil, surface water and groundwater at sub-inhibitory concentrations, sustaining selection pressure and horizontal transfer of resistance genes; conventional waste and sewage treatment is not designed to remove them. India's National Action Plan on Antimicrobial Resistance 2.0 adopts a One Health frame but does not link its objectives to pharmaceutical waste, and India has no national household take-back route — Kerala's nPROUD is the first government-led state programme and a few metros have informal pharmacy collection bins — so producer responsibility and ecopharmacovigilance are proposed as the fix. [PMID 41972769]

Global and WHO developments

- **WHO has restated the health-care waste split** — the WHO health-care waste fact sheet, updated October 2024, gives about **85% general** and **15% hazardous**, which supersedes the older 75–90% / 10–25% teaching figure; only 61% of hospitals had basic health-care waste services in 2021 and just 25% of facilities in fragile settings in 2023; generation is up to 0.5 kg of hazardous waste per bed per day in high-income and about 0.2 kg in low-income countries, and WHO now prefers autoclaving or microwaving to incineration wherever feasible. [WHO 2024-Oct]
- **The global plastics treaty is a fight over this very hierarchy** — after the 2022 UN Environment Assembly resolution to negotiate a legally binding instrument and World Health Assembly resolution 76 (2023), talks resumed in August 2025 with fossil-fuel, chemical and plastics blocs pressing for a treaty confined to waste management and recycling instead of production limits; plastics contain more than 16,000 chemicals, over 4,200 of them known hazardous [PMID 40630155]. A lifecycle model attributes **83 million DALYs** to the global plastics system for 2016–40 under business as usual, and found scenarios resting on better waste management or more recycling **alone** substantially less effective than cutting primary production — the combined package (production cuts plus collection, recycling, substitution and reuse) lowered annual DALYs by 43% by 2040. This is the quantitative case for placing Refuse and Reduce above Recycle. [PMID 41610860]

SOURCE & COVERAGE

Manual on Municipal Solid Waste Management (Central Public Health and Environmental Engineering Organisation; revised by the Expert Committee, 2013–15) — Part II §2.1 Waste minimisation (pp 127–131), §3.1 Recycling and recovery (pp 209–211), §3.5.2 Refuse-derived fuel in the ISWM hierarchy (p 311), §7.3 Special waste streams (p 499), §7.4 Plastic waste management (pp 501–509), and Fig 3.4 Stages of material recovery in the municipal solid waste management chain; Solid Waste Management Rules 2016 and Plastic Waste Management Rules 2016 as reproduced therein; Park's Textbook of Preventive and Social Medicine, 28e — Chapter 15 (Environment and Health), pp 909–911, and Chapter 16 (Hospital Waste Management), pp 943–944; Bio-Medical Waste Management Rules 2016 (Ministry of Environment, Forest and Climate Change), Schedule I; Suryakantha — Community Medicine with Recent Advances, Chapter 12 (Disposal of Wastes), Chapter 13 (Management of Hospital Waste, p 93) and Chapter 14 (Electronic Waste Management). Coverage note: the corpus carries no post-2016 national municipal-solid-waste tonnage, so the most recent dated Indian figure available is the 2014–15 status report.

Recent-advances sources: 40173265 — A simplified framework for assessing waste prevention and minimisation in developing countries within the context of CE, SDGs and ESG principles, *Waste Management & Research* 2025, doi:[10.1177/0734242X251328911](https://doi.org/10.1177/0734242X251328911); 39877284 — Application of Geographic Information System and Google Earth Engine for Spatial Analysis of Land Use by the Legacy Dumpsite at Raipur, Chhattisgarh: A Time Series Analysis, *Indian Journal of Occupational and Environmental Medicine* 2024, doi:[10.4103/ijoem.ijoem.290.23](https://doi.org/10.4103/ijoem.ijoem.290.23); 42266523 — Microplastics and Human Health: A Global Public Health Crisis, *Indian Journal of Community Medicine* 2026, doi:[10.4103/ijcm.ijcm.715.25](https://doi.org/10.4103/ijcm.ijcm.715.25); 39811298 — Utilization of refuse-derived fuel in industrial applications: Insights from Uttar Pradesh, India, *Heliyon* 2025, doi:[10.1016/j.heliyon.2024.e41336](https://doi.org/10.1016/j.heliyon.2024.e41336) (full text checked; its abstract's SO₂/NO_x reduction claim is contradicted by its own discussion and is therefore not carried forward); 40744379 — Gap assessment in implementation of Biomedical Waste Management in Health Care Facilities: Nationwide survey in India, *Indian Journal of Medical Microbiology* 2025, doi:[10.1016/j.ijmm.2025.100947](https://doi.org/10.1016/j.ijmm.2025.100947) (abstract-only); 42022636 — Occupational Hazard Among Biomedical Waste Handlers in Low-Resource Settings: A Systematic Review and Meta-Analysis, *Health Science Reports* 2026, doi:[10.1002/hsr.2.72083](https://doi.org/10.1002/hsr.2.72083); 42125580 — Pattern and Disposal of Biomedical Waste Generated by Households in Urban Nagpur: A Community-Based Assessment, *Indian Journal of Community Medicine* 2026, doi:[10.4103/ijcm.ijcm.818.24](https://doi.org/10.4103/ijcm.ijcm.818.24); 41972769 — The missing thread of One Health efforts: improper drug disposal as an overlooked driver of antimicrobial resistance, *mSphere* 2026, doi:[10.1128/msphere.00586-25](https://doi.org/10.1128/msphere.00586-25); 40630155 — Negotiating the plastics treaty to protect health and the environment, *Bulletin of the World Health Organization* 2025, doi:[10.2471/BLT.25.294144](https://doi.org/10.2471/BLT.25.294144); 41610860 — Global health burdens of plastics: a lifecycle assessment model from 2016 to 2040, *The Lancet Planetary Health* 2026, doi:[10.1016/j.lanplh.2025.101406](https://doi.org/10.1016/j.lanplh.2025.101406). Plus the WHO health-care waste fact sheet, updated October 2024.



Slum dwellers — environmental health factors

Asked in: Paper I 15-Nov-2022 — RGUHS MD Community Medicine

Definition — A **slum** is a compact urban settlement of poorly built, congested households on land with inadequate infrastructure and an unhygienic environment, lacking proper sanitary and safe drinking-water facilities (Sunder Lal, Ch 2, p 68). **Environment** is "all that which is external to the individual human host", divisible into **physical, biological and psychosocial** components, any or all of which can affect health and susceptibility to illness (Park 28e, Ch 2, p 31). The slum is the one setting in which all three components are simultaneously adverse, so that health indices of slum dwellers are often **worse than those of the rural population**, despite the slum sitting inside a well-served city — "poverty in plenty" (Sunder Lal, Ch 17, p 756).

Magnitude — why the slum is a distinct epidemiological setting

Item	Figure (with year)
Urban share of India's population	31.16% (Census 2011); 34.0% (2021) (Sunder Lal, Ch 9, p 543)
Population living in slums and squatter settlements	48.8 million (1990); ~40% of them in million-plus cities (Park 28e, Ch 15, p 908)
Share of urban population living in slums	One-third or more (Sunder Lal, Ch 2, p 68); 50–60% by some accounts (Jugal Kishore, NUHM chapter, p 243)
Annual growth of slum population	~7% per year (Suryakantha, Ch 36, p 799)
Urban vital rates for comparison	Crude birth rate 16.1, crude death rate 5.1, infant mortality rate (IMR) 19 per 1,000 live births (SRS 2022) (Sunder Lal, Ch 2, p 68)
Under-5 mortality rate (U5MR): urban poor vs urban average	72.7 vs 51.9 per 1,000 live births (Jugal Kishore, p 162)
Underweight, children <3 years: urban poor vs urban average vs rural	47.1% vs 32.8% vs 45% (Jugal Kishore, p 162)
Anaemia in children: urban poor vs urban average	71.4% vs 62.9% (Jugal Kishore, p 162)
Children missing complete immunisation: urban poor vs urban average	60% vs 42% (Jugal Kishore, p 162)
Households with piped water at home: urban poor vs urban average	18.5% vs 50% (Jugal Kishore, p 162)
Institutional deliveries: urban poor vs urban average	44% vs 67.5% (Jugal Kishore, p 162)
Women with no education: urban poor vs urban average	46.8% vs 19.3% (Jugal Kishore, p 162)
Malaria in the urban poor	Twice the rate of other urban residents (Jugal Kishore, p 162)

Physical environmental factors and their health consequences

Physical factor	Defect in the slum	Health consequence
Dwelling structure and tenure	Substandard, dilapidated and crumbling houses of temporary material; homeless on pavements, railway and bus stations, discarded truck bodies and rail carriages; no security of tenure	Collapse, fire, injury; unlisted settlements excluded from civic services (Suryakantha, Ch 10, p 83)
Overcrowding	Floor-space, persons-per-room and sex-separation standards all breached; raw material of domestic industry and domestic animals kept in the living room	Respiratory infections — common cold, tuberculosis, influenza, diphtheria, bronchitis, measles, whooping cough (Park 28e, Ch 15, p 907)
Ventilation and lighting	No cross-ventilation, dark rooms, no drainage of rainwater	Rise of room temperature, excessive humidity and air stagnation, which lowers vitality and increases susceptibility; rapid droplet spread (Suryakantha, Ch 10, pp 82–83)
Indoor (household) air pollution	Combustion of solid biomass fuel indoors, tobacco smoke, emissions from construction material, poor ventilation	Women using solid fuel are 2.3 times as likely to have chronic obstructive pulmonary disease (COPD); ~17% of premature adult lung-cancer deaths, ~15% of ischaemic heart disease deaths and ~one-quarter of premature stroke deaths (~1.4 million) are attributable to household air pollution; acute lower respiratory infection deaths in under-5s (Park 28e, Ch 15, pp 890–892)
Water supply	Only 18.5% have piped water at home; intermittent public standposts, long-stored household water, contaminated shallow sources	Cholera, typhoid, viral hepatitis, diarrhoeal disease; stored water breeds <i>Aedes</i>
Excreta disposal	Open defecation, insanitary and single-pit latrines, overloaded shared community toilets	Soil-transmitted helminths, hookworm, enteric infection; women defecate at night, adding assault risk
Drainage and solid waste	Open drains, cesspools, stagnant water, uncollected refuse	Fly and mosquito breeding; malaria at twice the urban rate; dengue, filariasis (Jugal Kishore, p 162)
Vectors and rodents	Open sewers, poorly built septic tanks, stagnant water inside and outside the house	Houseflies, mosquitoes, fleas and bugs; rat infestation → plague (Park 28e, Ch 15, p 907)
Person-to-person contact and personal hygiene	Shared bathing and washing, no water for personal hygiene	Skin infections — scabies, ringworm, impetigo, leprosy (Park 28e, Ch 15, p 907)
Bad surroundings	Vicinity of cesspools, open drains, offensive trades, noisy industries, liquor shops, red-light areas	Air, water, soil and noise pollution; industries near slums often evade regulation (Suryakantha, Ch 10, p 83)

Physical factor	Defect in the slum	Health consequence
Home safety	Defects in the home and its environment; unguarded stoves, kerosene, open wiring, traffic at the doorstep	A substantial proportion of house accidents; burns, falls, road traffic injury (Park 28e, Ch 15, p 907)
Siting of the settlement	Flood-prone land, railway and canal margins, dumpsite edges	Disaster vulnerability, displacement
Net effect	—	High morbidity and mortality rates are observed wherever housing conditions are substandard (Park 28e, Ch 15, p 907)

The housing standards against which slum dwellings fail

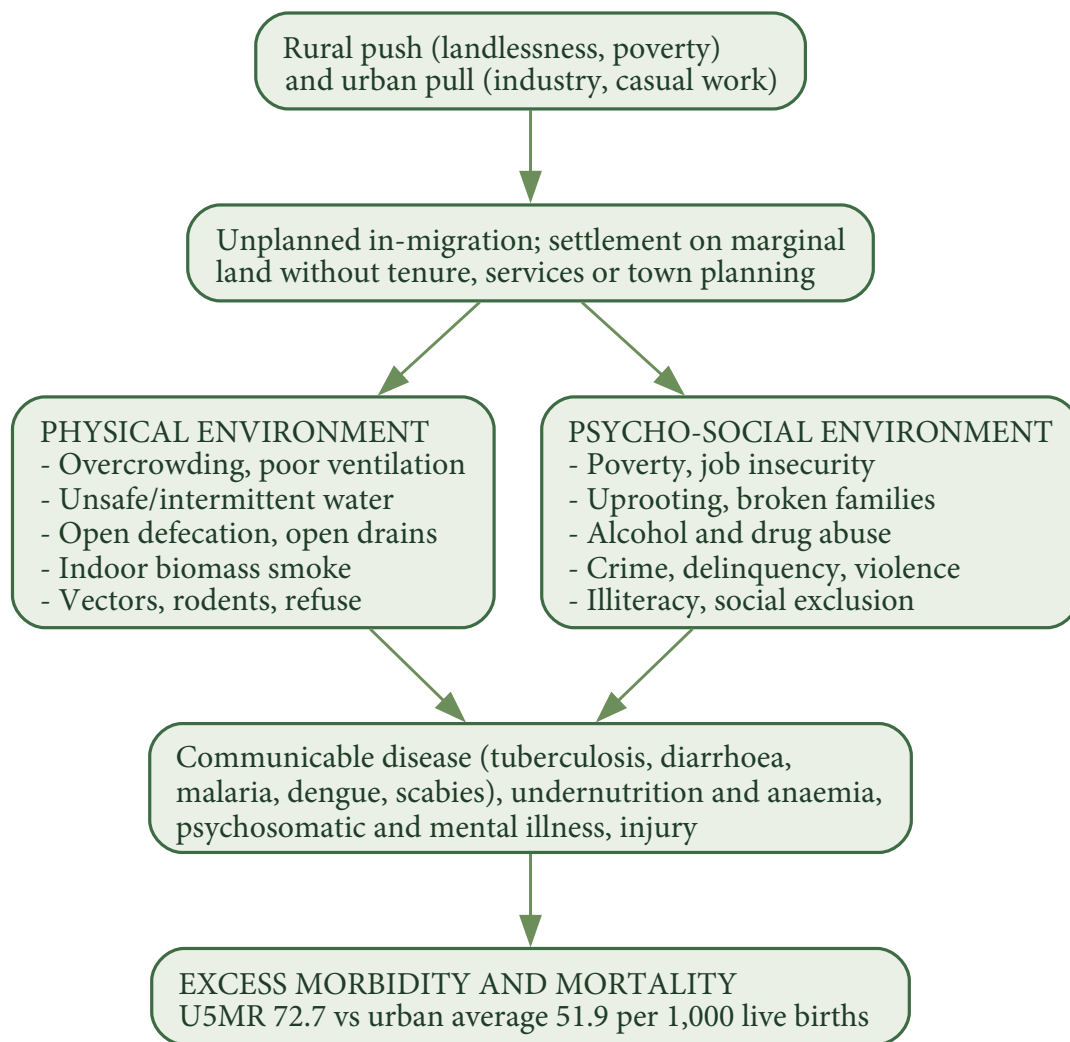
Parameter	Accepted standard (Park 28e, Ch 15, pp 906–908)
Floor area of a living room	≥120 sq ft (12 sq m) if occupied by more than one person; ≥100 sq ft (10 sq m) for a single person
Floor area per person in living rooms	Not less than 50 sq ft; optimum 100 sq ft
Cubic (air) space	≥500 cu ft per capita, preferably 1,000 cu ft (unless mechanical air replacement is provided)
Number of rooms	Not less than two living rooms, at least one of which can be closed for security
Floor space → permitted occupancy	110 sq ft (11 sq m) or more = 2 persons; 90–100 sq ft = 1½ persons; 70–90 sq ft = 1 person; 50–70 sq ft = ½ person; under 50 sq ft = nil (baby under 12 months not counted; child 1–10 years = half a unit)
Persons per room	1 room = 2 persons; 2 rooms = 3; 3 rooms = 5; 4 rooms = 7; 5 or more rooms = 10 (+2 for each further room)
Sex separation	Overcrowding exists if 2 persons over 9 years of age , of opposite sex and not husband and wife, are obliged to sleep in the same room
Rate of new house construction	10 houses per 1,000 persons per year (United Nations expert body)

Psycho-social environmental factors and their health consequences

Psychosocial environment means "those factors affecting personal health, health care and community well-being that stem from the psychosocial make-up of individuals and the structure and functions of social groups" — cultural values, customs, habits, beliefs, attitudes, morals, religion, education, lifestyles, community life, health services, and social and political organisation (Park 28e, Ch 2, p 56).

Domain	Factor operating in the slum	Health consequence
Economic	Poverty, unemployment, casual daily-wage work, economic insecurity, out-of-pocket spending on private care	Undernutrition, delayed care-seeking, impoverishment; low socioeconomic status is a risk factor common to mental disorders and other non-communicable diseases (Park 28e, Ch 20, p 987)
Migration and uprooting	Rural push and urban pull; temporary migrants; loss of the extended family; records of services held in the village of origin	Failure of adjustment producing behavioural disorders, neuroses and psychoses (Suryakantha, Ch 17, p 244); mother and infant miss antenatal, immunisation and postnatal follow-up (Jugal Kishore, p 162)
Family structure	Broken homes, unhappy marriages, desertion, changing family structure, population mobility, cruelty, rejection, neglect	Anxiety, depression, emotional stress, tension, frustration (Park 28e, Ch 20, p 987)
Chronic psychosocial stress	Prolonged anxiety, anger and frustration acting through the endocrine, autonomic and motor systems in interaction with genetic and personality factors	Psychosomatic disorders — duodenal ulcer, bronchial asthma, hypertension, coronary heart disease, mental disorders (Park 28e, Ch 2, p 56)
Substance use	Alcoholism, drug abuse, injecting drug use, gambling; urban slums and unemployed restless youth named as root causes	Human immunodeficiency virus (HIV), hepatitis B and C in injecting drug users; wage diversion, domestic violence (Sunder Lal, Ch 15, pp 687–688)
Socially deviant behaviour	Suicide, crime, violence, theft, prostitution, sex crimes, child abuse	Injury and death; sexually transmitted infections and reproductive tract infections (Suryakantha, Ch 23, pp 644–645)
Juvenile delinquency	Slum-dwelling itself listed as a cause, with broken homes, poverty, parental alcoholism and neglect, absence of recreation facilities, cheap recreation and unplanned urbanisation	Antisocial behaviour before 16 years (boys) / 18 years (girls) under the <i>Juvenile Justice Act, 1986</i> (Park 28e, Ch 9, p 690; Sunder Lal, Ch 15, pp 693–694)
Neighbourhood indicators of mental health	Park's recognised housing indicators — frequency of suicides in the neighbourhood, neglected and abandoned youth , and drug abuse including alcohol in the neighbourhood	These are the accepted markers that the psychosocial housing environment has failed (Park 28e, Ch 15, p 907)
Comfort deprivation	Thermal, acoustic, visual and spatial discomfort under crowding	Frustration, lack of sleep, anxiety and violence (Park 28e, Ch 15, p 907)
Education and health literacy	46.8% of urban-poor women have no education vs 19.3% urban average	Poor infant and young-child feeding, low demand for preventive services
Child welfare	Street children, child labour, sexual exploitation, school dropout	Injury, undernutrition, HIV and sexually transmitted infections (Jugal Kishore, p 243)

Domain	Factor operating in the slum	Health consequence
Relationship with the health system	Perceived unfriendly treatment at government hospitals, expensive private facilities, no caretaker for other children, emotionally more secure home environment	Preference for home delivery — institutional delivery only 44% vs 67.5% urban average (Jugal Kishore, p 162; Suryakantha, Ch 36, p 799)
Social exclusion	Unlisted / non-notified settlements outside civic entitlement; stigma; nuclear families with little interaction and social isolation	Loss of social support; exclusion from schemes (Sunder Lal, Ch 2, p 68)



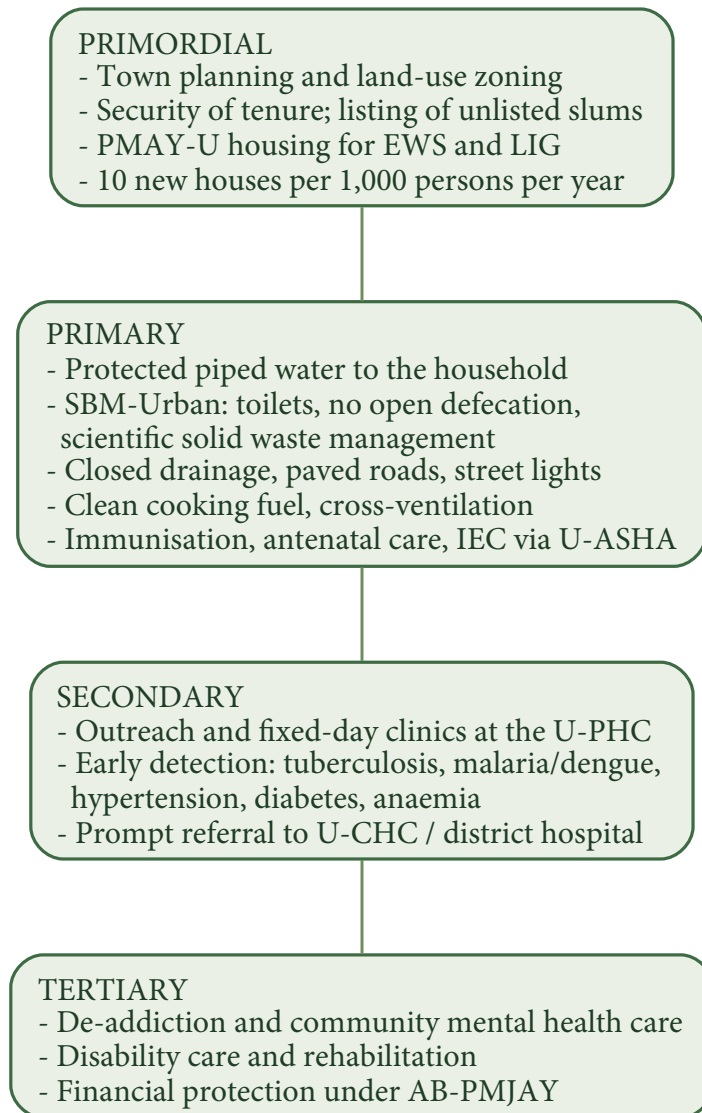
Pathway from unplanned urbanisation to excess morbidity and mortality in the slum dweller

Remedial measures

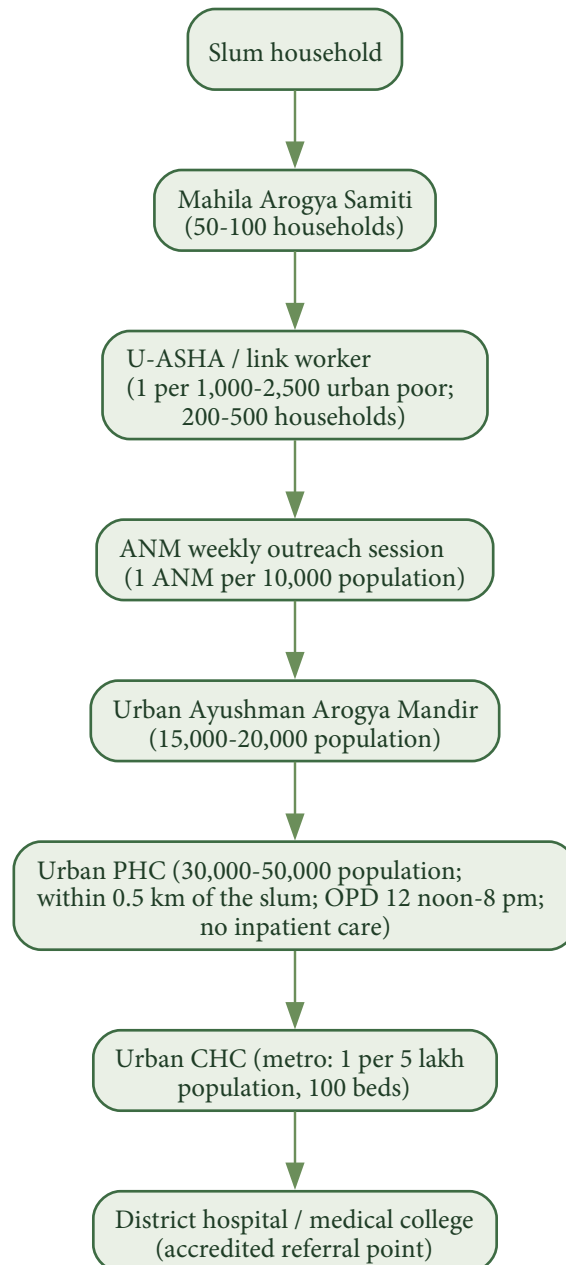
Sector	Measures
Water, sanitation and environment	Protected piped water supply with household connections; Swachh Bharat Mission–Urban (Ministry of Urban Development) — elimination of open defecation, eradication of manual scavenging, and modern scientific municipal solid waste management; individual household toilets for the 80% of urban households practising open defecation and community toilets for the remaining 20% where space is a constraint; conversion of pit latrines to sanitary latrines (Park 28e, Ch 7, p 574)
Basic amenities in existing slums	Under the Twenty Point Programme, all slums in cities of three lakh population or more are to be provided with water taps, sewers, storm-water drains, community bathing places, paved roads and street lights (Suryakantha, Ch 43, pp 889–890); in-situ upgrading with protected water supply, drainage and community latrines (Suryakantha, Ch 10, pp 83–84)
Housing and town planning	Pradhan Mantri Awas Yojana–Urban (PMAY-U) , launched 25 June 2015 by the Ministry of Housing and Urban Affairs, addressing urban housing shortage among the Economically Weaker Section and Low Income Group (Park 28e, Ch 7, p 575); good-quality tenements for the poorest of the poor; sites with loans on easy terms to the landless poor; employer-built quarters for large establishments; night shelters for street children; construction at 10 houses per 1,000 persons per year; delivery through statutory State Housing Boards, the National Buildings Organisation, the Housing and Urban Development Corporation (HUDCO) and the National Buildings Construction Corporation
Household energy and air	Clean cooking fuel and smokeless <i>chulha</i> ; kitchen separation and cross-ventilation to cut household air pollution
Health services	National Urban Health Mission (NUHM) — the definitive service-delivery remedy (norms below)
Social and behavioural	Behaviour-change communication through the urban Accredited Social Health Activist (U-ASHA) and Mahila Arogya Samiti; hygiene behaviour, HIV and sexually transmitted infection prevention, and strategies to tackle violence and alcoholism as set out in the National Slum Policy (Draft) 2002 ; de-addiction services — noting that legislation and de-addiction centres alone are insufficient (Sunder Lal, Ch 15, p 688); school enrolment and retention; rehabilitation, not merely legal action, for delinquency and child neglect under the <i>Children Act 1960</i> and <i>Juvenile Justice Act 1986</i>
Financial protection	Ayushman Bharat–Pradhan Mantri Jan Arogya Yojana , launched 23 September 2018, giving cover up to Rs 5 lakh per family per year for secondary and tertiary hospitalisation to 10.74 crore poor and vulnerable families (Park 28e, Ch 10, p 745); community-based risk pooling and insurance under NUHM
Intersectoral coordination and monitoring	Convergence with the Integrated Child Development Services, education department and Swarna Jayanti Shahari Rozgar Yojana; co-location of tuberculosis, HIV counselling and testing, AYUSH, integrated disease surveillance and vector-borne disease programmes at urban health centres; housing and slum development bodies to provide sites for new urban health centres; health management information system reporting, community-based monitoring and external evaluation (Suryakantha, Ch 36, p 803)

National Urban Health Mission — coverage and service norms (Park 28e, Ch 7, pp 535–536)

Element	Norm
Cities covered	All state capitals, district headquarters and about 779 other cities/towns with population 50,000 and above (Census 2011), phased; towns below 50,000 remain under the National Rural Health Mission
Target groups	Urban poor in listed and unlisted slums , plus homeless, street children, rickshaw pullers, construction workers, brick- and lime-kiln workers, sex workers and temporary migrants
Mahila Arogya Samiti	One per 50–100 households
U-ASHA / link worker	One per 1,000–2,500 urban poor (about 200–500 households)
Auxiliary nurse midwife (ANM)	One per 10,000 population , with weekly outreach sessions in her area
Urban Ayushman Arogya Mandir	One per 15,000–20,000 population (15th Finance Commission health grants and PM-ABHIM)
Urban Primary Health Centre (U-PHC)	For 30,000–50,000 population , sited within or within half a kilometre of a slum, serving a slum population of 25,000–30,000 ; OPD 12 noon to 8 pm ; no inpatient care; may be set up for 75,000 in very-high-density areas, or for 5,000–10,000 in isolated slum clusters
Urban Community Health Centre	In metro cities, one per 5 lakh population with 100 beds
Impact target	Reduce the urban IMR to 20 per 1,000 live births (Park 28e, Ch 7, p 538)



Remedial measures for slum health arranged by level of prevention



National Urban Health Mission service-delivery ladder for a slum household

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **The total environment of the slum household** — three overlapping circles with the slum household at the intersection. Labels: **PHYSICAL** (housing and overcrowding, water, air and indoor smoke, refuse and drainage, noise, light); **BIOLOGICAL** (mosquitoes, houseflies, fleas, bugs, rats, microbes, domestic animals); **PSYCHOSOCIAL** (poverty, migration, family structure, customs and beliefs, education, lifestyle, health services, social and political organisation). Caption the intersection "Slum household — all three components simultaneously adverse".
- ▶ **Vicious cycle of slum ill-health** — a closed loop of four boxes joined by arrows: *Poverty and insecure work* → *Overcrowded insanitary dwelling* → *Repeated infection and undernutrition* → *Loss of wages and school dropout* → back to *Poverty and insecure work*. Mark the two entry points for intervention with block arrows: "Housing and civic amenities (PMAY-U, SBM-U)" into the second box and "NUHM primary health care" into the third.

High-yield exam pointers

- ▶ **Overcrowding is judged on three criteria** — floor area, persons per room, and sex separation. Sex separation: overcrowding if **2 persons over 9 years**, opposite sex, not husband and wife, sleep in one room (Park 28e).
- ▶ **Numbers to write:** floor area per person ≥ 50 sq ft (optimum 100 sq ft); living room ≥ 120 sq ft for >1 person; cubic space 500 cu ft per capita, preferably 1,000; 110 sq ft = 2 persons; 1 room = 2 persons, 2 rooms = 3, 3 rooms = 5, 4 rooms = 7, 5+ rooms = 10.
- ▶ **Park's seven consequences of poor housing** — Respiratory infections, Skin infections, Rat infestation (plague), Arthropods, Accidents, Morbidity and mortality, Psychosocial effects.
- ▶ **NUHM norms** — Mahila Arogya Samiti 50–100 households; U-ASHA 1:1,000–2,500; ANM 1:10,000; U-AAM 15,000–20,000; U-PHC 30,000–50,000 serving 25,000–30,000 slum population with OPD 12 noon–8 pm; metro U-CHC 5 lakh / 100 beds.
- ▶ **Urban poor vs urban average** (NUHM rationale) — U5MR 72.7 vs 51.9; underweight under-3s 47.1% vs 32.8%; missing complete immunisation 60% vs 42%; piped water at home 18.5% vs 50%; institutional delivery 44% vs 67.5%.
- ▶ **Programme names with dates** — SBM launched 2 Oct 2014 (SBM-Urban: no open defecation, no manual scavenging, scientific solid waste management); PMAY-Urban 25 June 2015 for EWS and LIG; AB-PMJAY 23 Sept 2018, Rs 5 lakh per family per year.
- ▶ **Indian denominators** — urban crude birth rate 16.1, crude death rate 5.1, IMR 19 per 1,000 live births (SRS 2022); urban population 31.16% (Census 2011).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Park 28e sets the sex-separation threshold at "over 9 years of age" while some Indian texts print "over 10 years" — write Park's figure. The urban-poor-versus-urban-average comparison is the National Urban Health Mission's founding rationale table, derived from a re-analysis of the third National Family Health Survey (2005–06), and the textbooks still carry it; an equivalent slum-specific breakdown is not published in NFHS-5 (2019–21), so quote it with its survey and year attached. The Census of India's formal three-category slum definition (notified, recognised and identified slums) is not carried in the indexed corpus, so the operational description of a slum given above is used instead.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, NFHS-5, NSSO 76th round, National Urban Health Mission Framework 2023, WHO/UNICEF Joint Monitoring Programme

Programme and policy updates (India)

- ▶ **Revised NUHM Framework 2023** — the Ministry of Health and Family Welfare reissued the National Urban Health Mission framework, adding city categories that align with the Ministry of Housing and Urban Affairs and the Census 2011 classification, while keeping the urban poor, vulnerable and marginalised as the named target group. Government data for March 2023 show **5,195 urban PHCs operational**, three-quarters of them with the minimum staffing and service package and 10% functioning 24×7; NUHM reported reaching about **275 million people in 1,213 cities and towns**. Pooled analysis of roughly 203,000 urban births (DLHS 2002–08 plus NFHS 2015–21) shows the equity gain the mission was built for: urban neonatal mortality fell from 33 to 20 per 1,000 live births, and from 64 to 33 in the poorest wealth decile, halving the poorest-to-richest gap from 50 to 24 units; the richest-to-poorest gap in facility delivery narrowed from 57 to 21 percentage points. A "bottom inequality" persists — facility delivery is 77% in the poorest decile against 86% in the second poorest, and the poorest decile's caesarean rate (11%) is only just at the WHO met-need threshold. [PMID 42274885]
- ▶ **ARTEMIS cluster randomised trial (December 2022 – December 2023)** took mental health care into 60 slum clusters in New Delhi and Vijayawada, enrolling 3,739 adolescents aged 10–19. A multimedia anti-stigma campaign plus a primary-care-worker-led digital screening and treatment package improved anticipated-behaviour scores (mean difference 0.78; 95% CI 0.40 to 1.16) and lowered PHQ-9 depression scores (mean difference –0.87; 95% CI –1.66 to –0.08), though the difference in remission was not significant (OR 1.47; 95% CI 0.93 to 2.32). About 12 million Indian adolescents live in slums, so this is the first tested service model for the psychosocial morbidity described above. [PMID 42054038]

Current national data

- ▶ **NSSO 76th round** — a Tobit model of per-capita household disease incidence puts national effect estimates on the physical-environment rows above. Visible stagnant water or human faeces in or around the premises raised disease incidence most in cities (about 1.8% for each 1% rise in such nuisance, against 1.3% rural; $p < 0.001$); handwashing with soap cut it by about 2% in urban households against 0.4% in rural ones ($p < 0.001$); better garbage disposal was roughly twice as

protective in cities as in villages ($p < 0.001$); and use of a well-maintained toilet was protective in urban areas ($p < 0.01$) but ran the other way in rural ones. **Per-capita living space is the row to quote carefully** — more space per person significantly reduced disease spread in urban areas ($p < 0.001$), but at the all-India level and in rural areas more space was associated with *higher* incidence ($p < 0.001$), and the authors state that per-capita space alone does not significantly influence household disease spread. Park's overcrowding standard therefore finds support only in the urban stratum, not nationally. Coverage figures worth carrying: 96.2% of urban households have a toilet but only 77.6% have one for their exclusive use (rural 71.3% and 63.2%), and 88.3% of urban households wash hands with soap or detergent after defecation. [PMID 41310542]

- **NFHS-5 (2019–21)** — household air pollution tracks the wealth line almost exactly: 95.9% of women in the poorest quintile were exposed, wood was the cooking fuel of 70% of the poorest households, and 346 of 707 districts fell in the highest exposure band. In the same round neonatal mortality was 11.5 per 1,000 live births in urban areas against 18.3 in rural areas, yet wealth-related inequality in neonatal mortality was **greater inside urban areas** than in rural ones, with unclean cooking fuel, high parity, large family size and inadequate antenatal care the leading contributors — the statistical form of "poverty in plenty". [PMID 41102709; PMID 42354325]
- **Slum denominators to quote with their source** — about 65.5 million people and 13.9 million households live in India's urban slums, projected to reach 105.7 million; the World Health Organization attributes more than one in four under-five deaths worldwide to slum settlements. A slum versus non-slum comparison of 933 and 2,315 mother–child pairs (India DHS 2015–16) found exclusive breastfeeding lower in slums (50.0% vs 55.8%; $p = 0.004$), and 65% of the variation in early breastfeeding lay between clusters rather than between mothers — the neighbourhood, not the individual, is the unit of exposure. [PMID 41950202]

Elimination, surveillance and disease status

- **100-Day TB Elimination Campaign (7 December 2024 – 17 March 2025)** — intensified case-finding, early diagnosis, targeted screening and nutritional plus financial support, run to push the 2025 elimination goal. In Maharashtra 53 campaign districts were compared with 27 non-campaign districts using NIKSHAY data: notification fell 10% in non-campaign districts but only 1% in campaign districts, so the relative change was 10% higher where the campaign ran (difference-in-differences incidence rate ratio 1.10; 95% CI 1.03 to 1.17; $p = 0.003$). The gain was confined to rural areas (IRR 1.21) and the public sector (IRR 1.17). There was **no significant effect in the urban set-up** (IRR 0.95; 95% CI 0.89 to 1.02) or in private care (IRR 0.97) — precisely the two channels a slum dweller uses. [PMID 42457294]
- **Lymphatic filariasis** — the national elimination target has moved to **2030** and the disease remains endemic in **345 districts**, so the earlier deadlines carried by the textbooks are superseded; mass drug administration, morbidity management and disability prevention, and post-MDA surveillance are the current pillars, with public–private partnership proposed to close the operational gaps. [PMID 41951149]
- **Non-communicable disease is now a slum problem, not only an infection problem** — meta-analysis of 62 studies and 108,110 participants found slum residents 35% more likely to be

hypertensive than rural residents (OR 1.35; 95% CI 1.29 to 1.42) but 30% less likely than other urban residents (OR 0.70; 95% CI 0.51 to 0.96). Pooled prevalence in Indian slums was 26.8% for hypertension and 12.2% for type 2 diabetes, and Indian slum hypertension rose about 6.9% a year (95% CI 2.0 to 12.0) between 2001 and 2019. The care cascade is the examinable part — only about a third (33.6%) were aware, about half of those aware (51.9%) were on treatment, and only about a quarter of those treated (25.9%) had blood pressure controlled. [PMID 35210339]

Guideline and legislative updates

- **WHO/UNICEF Joint Monitoring Programme service ladder** governs how the community toilet blocks built under Swachh Bharat Mission–Urban are counted: an improved but *shared* facility ranks only as "limited", never "basic" or "safely managed", so the standard slum solution scores as a failure on paper even though over 600,000 shared facilities have been built under SBM since 2014. Field evidence pushes back — a qualitative study of 32 adults (sampled 16 male, 16 female) and 7 adolescents using two free, staffed, twice-daily-cleaned, gender-separated and lockable community blocks in Bokaro, Jharkhand found them used both by households with a home toilet and by those without: those without avoided being seen, and those with a toilet preferred the block when their own pit was shallow, unventilated, waterless or occupied. Women reported specific safety benefits (no longer walking to fields at night, with the harassment risk that carried) and menstrual-hygiene benefits (sanitary-waste bins). **The examinable caveat is the opening hours** — the two blocks ran 04:00–21:00 and 05:00–18:30 and were shut at night, and households without a home toilet said they were then forced back to open defecation for emergencies. Shared blocks are therefore complements to household toilets, not substitutes, and the authors argue the JMP ladder should add a quality-graded category for clean, staffed shared sanitation if Sustainable Development Goal target 6.2 is to be met in dense settlements. [PMID 41723228]

Global and WHO developments

- **Two findings recalibrate the physical-environment list.** The HAPIN four-country trial (India included; 3,061 infants, 11,996 infant-quarters of facility surveillance, 13,910 personal exposure measurements) found no association between measured personal PM_{2.5} and severe infant pneumonia (adjusted risk ratio 1.03; 95% CI 0.94 to 1.13 prenatal, and 0.97; 95% CI 0.87 to 1.09 postnatal), nor for carbon monoxide — clean-fuel programmes should therefore be defended on the wider household-air-pollution burden rather than on infant pneumonia alone. Separately, a scoping review of 61 studies (2012–2025) links ambient heat to low birth weight, stillbirth, neonatal mortality and congenital anomalies, with risk amplified by low socioeconomic status and neighbourhood deprivation — heat now belongs in the list of physical environmental factors acting on the slum dwelling. [PMID 41160028; PMID 42429868]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 2 Concept of Health and Disease; Ch 7 Health Programmes in India; Ch 9 Preventive Medicine in Obstetrics, Paediatrics and Geriatrics; Ch 10 Health Care of the Community; Ch 15 Environment and Health; Ch 20 Mental Health); Suryakantha — Community Medicine with Recent Advances (Ch 10 Housing; Ch 17 Occupational Health; Ch 23 Demography; Ch 26 Social Science; Ch 36 National Urban Health Mission; Ch 43 National Health Programs); Sunder Lal — Textbook of Community Medicine (Ch 2 Social and Behavioural Sciences; Ch 9 Demography and Vital Statistics; Ch 15 Mental Health; Ch 17 Healthcare of the Community); Jugul Kishore — National Health Programmes of India (National Urban Health Mission; National Slum Policy Draft 2002); Textbook of International Health (Ch 7 Societal Determinants of Health and Social Inequalities in Health). Dated national sources quoted in the text: Census of India 2011, SRS Bulletin 2022, NFHS-3 re-analysis (2005–06), Swachh Bharat Mission 2014, PMAY-Urban 2015, AB-PMJAY 2018.

Recent-advances sources: 42274885 — The Role of Public Health Services in Reducing Maternal and Newborn Health Inequalities in Urban India: A Survey Analysis of 200,000 Births Over Two Decades, Journal of Urban Health 2026, doi:[10.1007/s11524-026-01062-6](https://doi.org/10.1007/s11524-026-01062-6); 42054038 — Adolescent Mental Health Care and Stigma: The ARTEMIS Randomized Clinical Trial, JAMA Psychiatry 2026, doi:[10.1001/jamapsychiatry.2026.0603](https://doi.org/10.1001/jamapsychiatry.2026.0603) (PMC13130063 carries the structured abstract only; bullet grounded on that); 41310542 — Analyzing the health impact of substandard housing environment and access to water, sanitation and hygiene in India, BMC Public Health 2025, doi:[10.1186/s12889-025-25028-6](https://doi.org/10.1186/s12889-025-25028-6); 41102709 — Household air pollution and association with heart disease among women in India: evidence from the nationally representative survey (NFHS-5), BMC Public Health 2025, doi:[10.1186/s12889-025-24609-9](https://doi.org/10.1186/s12889-025-24609-9); 42354325 — Decomposing Wealth-Based Inequalities in Neonatal Mortality in India: Evidence from National Family Health Survey (2019–2021), International Journal of Environmental Research and Public Health 2026, doi:[10.3390/ijerph23060795](https://doi.org/10.3390/ijerph23060795); 41950202 — Prevalence and social determinants of breastfeeding practices in urban slums and urban non-slum areas in India: A comparative analysis, PLoS One 2026, doi:[10.1371/journal.pone.0323861](https://doi.org/10.1371/journal.pone.0323861); 42457294 — Effect of the '100 days TB elimination campaign' on TB notification in Maharashtra, Indian Journal of Tuberculosis 2025, doi:[10.1016/j.ijtb.2025.08.014](https://doi.org/10.1016/j.ijtb.2025.08.014) (abstract-only); 41951149 — Public-private partnership offers a strategic opportunity for lymphatic filariasis elimination in India by 2030, Acta Tropica 2026, doi:[10.1016/j.actatropica.2026.108083](https://doi.org/10.1016/j.actatropica.2026.108083) (abstract-only); 35210339 — Global prevalence and trends in hypertension and type 2 diabetes mellitus among slum residents: a systematic review and meta-analysis, BMJ Open 2022, doi:[10.1136/bmjopen-2021-052393](https://doi.org/10.1136/bmjopen-2021-052393); 41723228 — An exploratory study on shared sanitation and equity in peri-urban India, Scientific Reports 2026, doi:[10.1038/s41598-026-40069-6](https://doi.org/10.1038/s41598-026-40069-6); 41160028 — Exposure to Household Air Pollution From Biomass Cooking and Severe Pneumonia in Infants, JAMA Network Open 2025, doi:[10.1001/jamanetworkopen.2025.38721](https://doi.org/10.1001/jamanetworkopen.2025.38721); 42429868 — The Impact of Ambient Heat on Perinatal and Neonatal Health in the Context of Climate Change: A Scoping Review, Current Environmental Health Reports 2026, doi:[10.1007/s40572-026-00549-3](https://doi.org/10.1007/s40572-026-00549-3).



Town planning and community health

Asked in: Paper IV 03-Oct-2025 — RGUHS MD Community Medicine

Definition — Town (city) planning is the statutory arrangement of land use, housing, transport, utilities and open space within a defined settlement, executed through a **master plan and building regulations enforced by the urban local body**. In public health terms it is **primordial prevention applied to the physical environment**: it decides, before anyone falls ill, how densely people will live, what air they breathe, where water and sewage go, and how far they travel for care. Park treats it as a named environmental health measure — air pollution around industry "requires proper town planning and zoning to eliminate this hazard", and worker housing "must be correlated to essential community amenities and to social and sanitary facilities. Town planning and zoning are highly desirable" (Park 28e, Ch 18, pp 966-967).

The setting: what is being planned for

Item	Figure (with source year)
Definition of "urban" in India	Towns (places with a municipal corporation, municipal area committee, town committee, notified area committee or cantonment board); or all places with ≥5,000 inhabitants , a density of ≥1,000 persons per square mile (400 per sq km) , pronounced urban characteristics, and at least three-fourths of the adult male population employed in pursuits other than agriculture (Park 28e, Ch 2, p 47)
World urban population, 2020	56% overall — 81% in more developed , 68% in less developed and 43% in least developed countries (Park 28e, Ch 2, p 47)
India	About 36% of the population is urban ; Delhi is the largest city (33.8 million), then Mumbai (21.7 million), Kolkata (15.6 million), Bengaluru (14.0 million) (Park 28e, Ch 8, pp 583-584)
Housing stock, Census 2011	330.84 million houses — 220.7 million rural, 110.14 million urban (Park 28e, Ch 15, p 908)
Slums	About 48.8 million persons living in slums (1990 estimate) , roughly 40% of them in million-plus cities ; the growth of slums and squatter settlements is the chief manifestation of the urban housing crisis (Park 28e, Ch 15, p 908)
Construction norm	The United Nations expert body recommends 10 new houses per 1,000 persons per year (Park 28e, Ch 15, p 908)

Components of town planning and the health outcome each secures

Planning component	What the plan prescribes	Health gain
Land-use zoning	Division of the city into residential, industrial, commercial, institutional and transport zones , with industries, railways and marshalling yards earmarked outside residential areas (Park 28e, Ch 15, p 898); industries sited away from human habitation on favourable topography (Suryakantha, Ch 6, p 69)	Removes the source of air pollution, noise and toxic exposure from where people sleep — the single largest planning lever on non-communicable disease
Green belts and open space	Protective green belts between industry/traffic and dwellings ; house fronts not less than 15 metres from the road , the intervening space thickly planted with trees and bushes (Park 28e, Ch 15, p 898); green belts between industries and residential areas (Suryakantha, Ch 6, p 69)	Buffers noise and particulate matter; green open space is a site for physical activity and social interaction and an urban heat sink , reducing harm from climate-change-induced urban heat islands (Oxford Textbook of Public Health 7e, Ch 58, p 2160)
Road network and transport	Widening of main streets ; heavy vehicles kept off narrow and residential streets; bypass roads (Park 28e, Ch 15, p 898; Suryakantha, Ch 6, p 69)	Cuts road traffic injury, traffic noise penetration into dwellings and vehicular air pollution
Housing layout and density	Floor area, cubic space, window area, set back , plinth height and site standards written into building bye-laws (Park 28e, Ch 15, p 906)	Prevents overcrowding and its droplet-borne and contact-borne diseases
Water supply	150–200 litres per capita per day is adequate for all urban domestic purposes; a continuous 24-hour supply is desirable because intermittent low-pressure supply permits back-siphonage and has caused recorded outbreaks of typhoid (Park 28e, Ch 15, pp 861, 883)	Prevents water-borne and water-washed disease
Sewerage and drainage	Sewerage laid before industry and population are located: "there is bound to be pressure on the existing sanitation services if proper planning is not undertaken before locating industries" (Park 28e, Ch 18, p 966)	Prevents pollution of the water supply and soil contamination with parasites
Solid waste and refuse	Refuse removed from every dwelling at least daily and disposed of in a sanitary manner; sites reserved for treatment and landfill (Park 28e, Ch 15, p 906)	Prevents fly and rodent breeding, and hence plague and enteric disease
Reservation of sites for public facilities	Land reserved for schools, markets, slaughter houses, burial grounds, recreation and health facilities — for a municipal corporation these are executed by the Commissioner, Engineer and Health Officer (Park 28e, Ch 23, p 1035; Ch 12, p 841)	Determines geographical access to care; siting decides who reaches a health facility in time

Planning component	What the plan prescribes	Health gain
Industrial worker housing	Worker housing near a plant correlated with community amenities and sanitary facilities (Park 28e, Ch 18, p 967)	Prevents the industrial slum

Housing and building standards the plan enforces (Environmental Hygiene Committee, India, 1947)

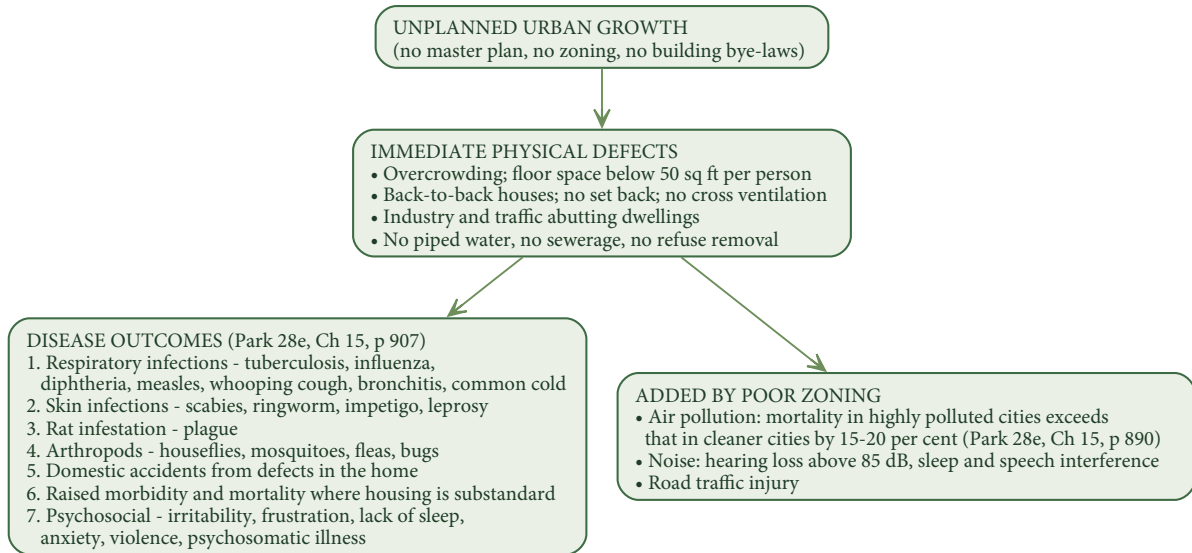
Standard	Prescribed value (Park 28e, Ch 15, p 906)
Site	Elevated above surroundings (not flooded in rains); independent access to a street of adequate width; away from mosquito and fly breeding places; away from dust, smoke, smell, excessive noise and traffic; dry, well-drained soil; subsoil water below 10 feet (3 m) . "Made-soil" (ground levelled by dumping refuse) is unsatisfactory for building for 20–25 years
Set back (open space all round the house)	Built-up area $\leq$ one-third of the total area in rural areas, up to two-thirds in urban areas , with no obstruction to lighting and ventilation
Floor area of a living room	≥ 120 sq ft (12 sq m) if occupied by more than one person; ≥ 100 sq ft (10 sq m) for a single occupant. Floor area per person ≥ 50 sq ft; optimum 100 sq ft
Cubic space	Air space ≥ 500 cubic ft per capita, preferably 1,000 cubic ft (unless mechanical replacement of air is provided)
Windows	At least 2 windows per living room, one opening directly on to open space; sill not more than 3 feet (1 m) above the ground; window area = 1/5th of floor area; doors + windows together = 2/5th of floor area
Lighting	Daylight factor should exceed 1 per cent over half the floor area
Rooms	Not fewer than two living rooms , at least one closable for security
Floor and plinth	Impermeable, smooth, crack-free, damp-proof; plinth height 2–3 feet (0.6–1 m)
Kitchen, privy, water, refuse	Separate kitchen with impervious floor, water supply, sink and drainage; a sanitary privy is a MUST in every house ; safe adequate water at all times; refuse removed daily
Noise exposure	Detached rather than continuous buildings for acoustic insulation; hearing protection for consistent exposure above 85 dB in frequency bands above 150 Hz (Park 28e, Ch 15, p 898)

Overcrowding — the measurable index of planning failure

Criterion	Standard
Floor space (Park 28e, Ch 15, p 907)	110 sq ft (11 sq m) or more — 2 persons; 90–100 sq ft (9–10 sq m) — 1½ persons; 70–90 sq ft (7–9 sq m) — 1 person; 50–70 sq ft (5–7 sq m) — ½ person; under 50 sq ft — nil. A baby under 12 months is not counted; a child aged 1–10 years counts as half a unit
Persons per room	Degree of overcrowding expressed as the number of persons in the household divided by the number of rooms per dwelling (Park 28e, Ch 15, p 907)

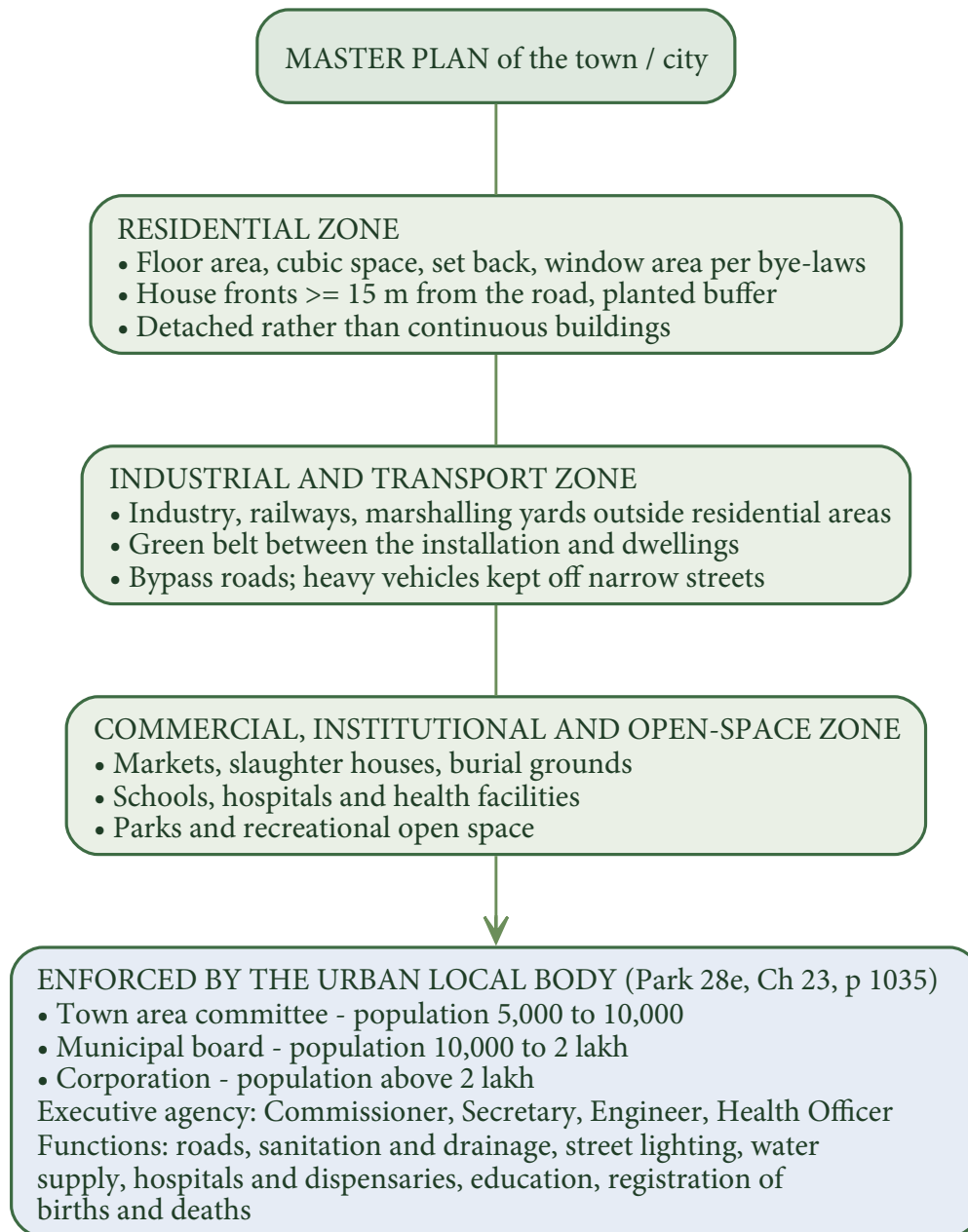
Criterion	Standard
Sex separation	Overcrowding exists if 2 persons over 9 years of age, not husband and wife, of opposite sexes, are obliged to sleep in the same room (Park 28e, Ch 15, p 907)

How defective planning produces disease



Causal chain from unplanned urban growth to the disease burden of the community

Zoning and the statutory machinery in India



Land-use zoning under a master plan and the urban local bodies that enforce it in India

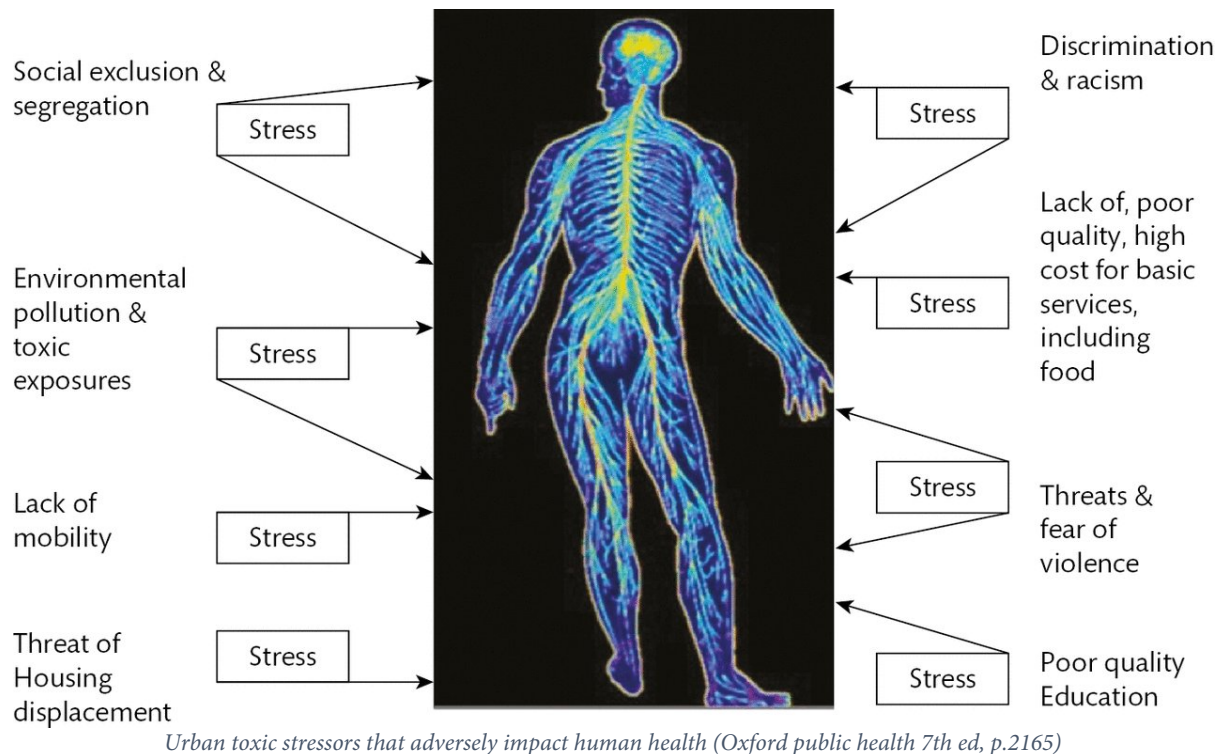
National scheme working through town planning	Content
Pradhan Mantri Awas Yojana – Urban (PMAY-U)	Launched 25 June 2015 by the Ministry of Housing and Urban Affairs, a flagship mission addressing urban housing shortage among the Economically Weaker Section and Low Income Group (Park 28e, Ch 7, p 575)

National scheme working through town planning	Content
National Urban Health Mission (NUHM)	Covers all state capitals, district headquarters and about 779 other cities/towns with population $\geq 50,000$ (Census 2011); towns below 50,000 are covered under the rural mission. Target group: the urban poor in listed and unlisted slums, homeless, street children, rickshaw pullers, construction and brick-kiln workers, sex workers and temporary migrants. Staffing norms: one ASHA/link worker per 1,000–2,500 urban poor (about 200–500 households), one Mahila Arogya Samiti per 50–100 households (250–500 population), and in metro cities a U-CHC of 100 beds per 5 lakh population (Park 28e, Ch 7, pp 535-536)
Swachh Bharat Mission (Urban)	Aims that no household practises open defecation, no new insanitary toilets are built during the mission period and pit latrines are converted to sanitary latrines; 20% of households practising open defecation are to be catered to by community toilets because of constraints of space — a planning constraint (Park 28e, Ch 7, p 574)
National housing agencies	National Buildings Organisation, National Buildings Construction Corporation, Housing and Urban Development Corporation (HUDCO), Hindustan Housing Factory (Park 28e, Ch 15, p 908)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Draw the "planned versus unplanned town" as one vertical two-column figure. Labels to write:

- ▶ Column heading left: PLANNED TOWN; right: UNPLANNED TOWN
- ▶ Left column boxes (top to bottom): Zoned — industry and railway outside residential area; Green belt, house front ≥ 15 m from road; Set back — built-up area $\leq 2/3$ of plot (urban); Floor area ≥ 50 sq ft/person, air space ≥ 500 cu ft/person; Piped water 150–200 litres per capita per day, sewerage, refuse removed daily; Reserved sites: school, park, market, U-PHC
- ▶ Right column boxes (facing each): Industry abutting dwellings; No buffer — noise and particulate matter at the doorstep; Back-to-back houses, no set back; Overcrowding — >2 persons over 9 years sharing a room; Standpost water, open drains, uncollected refuse; No reserved site — no health facility within reach
- ▶ Bottom band spanning both: Health outcome — left: low respiratory and vector-borne disease, low injury; right: tuberculosis, measles, diarrhoea, plague, scabies, domestic accidents, psychosocial morbidity



High-yield exam pointers

- ▶ **Frame town planning as primordial prevention** and quote Park's own words: air pollution around industry "requires proper town planning and zoning to eliminate this hazard" (Ch 18, p 966).
- ▶ **The Indian housing standards (EHC 1947) that must be written verbatim:** floor area ≥ 120 sq ft for >1 person and ≥ 100 sq ft for one; ≥ 50 sq ft per person, optimum 100; air space ≥ 500 cu ft, preferably 1,000; window area $1/5$ th and doors+windows $2/5$ th of floor area; daylight factor $>1\%$ over half the floor area; subsoil water below 3 m; plinth 0.6–1 m; set back — built-up area $\leq 1/3$ rural, $\leq 2/3$ urban.
- ▶ **Overcrowding, three criteria:** floor space (110 sq ft = 2 persons ... <50 sq ft = nil; child 1–10 years = half a unit), persons per room, and sex separation (2 persons >9 years, not husband and wife, of opposite sexes in one room).
- ▶ **Seven consequences of poor housing (Park 28e, Ch 15, p 907):** respiratory infections, skin infections, rat infestation (plague), arthropods, accidents, raised morbidity and mortality, psychosocial effects.
- ▶ **Numbers for zoning:** house front ≥ 15 metres from the road with a planted buffer; hearing protection above 85 dB; mortality in highly polluted cities exceeds cleaner cities by 15–20 per cent.
- ▶ **Indian denominators:** urban = $\geq 5,000$ inhabitants, ≥ 400 persons per sq km, $\geq 3/4$ adult males in non-agricultural work; urban water 150–200 lpcd; 330.84 million houses (Census 2011); new-house construction norm 10 per 1,000 persons per year.
- ▶ **Name three programmes:** PMAY-U (2015, Ministry of Housing and Urban Affairs), NUHM (cities $\geq 50,000$, Census 2011), Swachh Bharat Mission (Urban); and the three urban local bodies — town area committee, municipal board, corporation.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the *WHO Housing and Health Guidelines* address crowding, indoor cold and insulation, indoor heat, home safety and injuries, and accessibility for the disabled and elderly, and link improved housing to Sustainable Development Goal 3 (health) and Goal 11 (sustainable cities) (Sunder Lal, Ch 3, pp 108-109); the WHO *Healthy Cities* programme was created with local government agencies specifically to re-invigorate the link between city planning and public health, cities enrolling to draft "healthy city plans", and remains focused mainly in Europe (Oxford Textbook of Public Health 7e, Ch 58, p 2159).

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, National Urban Health Mission programme reports, National Sample Survey, Longitudinal Ageing Study in India, WHO Global Heat Health Information Network heat-health action plans

Programme and policy updates (India)

- **NUHM at the close of its first decade — the numbers that replace the launch-era norms in the body.** Between 2014 and 2022 the mission put 5,025 functional Urban Primary Health Centres in place against only 871 urban health posts that existed before it, mostly in the metros; by September 2022 it had deployed 78,127 urban ASHAs and formed 80,852 Mahila Arogya Samitis. Slum mapping was finished in 1,094 of 1,178 cities, facility mapping in 1,100, and vulnerable-group mapping in 826. Staffing norms as implemented: one UPHC per 50,000, one Urban CHC per 5 lakh in metros but per 2.5 lakh in smaller cities, one ANM per 10,000 and one ASHA per up to 2,500. Implementation sits with the municipal corporations in the seven megacities (Mumbai, Delhi, Kolkata, Chennai, Hyderabad, Bengaluru, Ahmedabad) and with the state health department elsewhere. National Sample Survey rounds 71 and 75 show average medical spend for non-hospitalised care in urban public facilities fell from INR 639 (2014) to INR 344 (2017-18), a 46% drop, institutional births in urban government facilities rose 42.0% to 48.3% overall and 54.0% to 62.1% in the poorest quintile, while NUHM's own budget stagnated at about INR 950 crore across 2019-20 and 2020-21 [PMID 36896811]
- **First evaluation of the Mahila Arogya Samiti, the community arm of urban planning-for-health.** In Chhattisgarh 3,700 MAS cover roughly 3 million urban poor across 20 cities; a representative survey of 401 MAS found that over three years 95.3% (95% CI 93.2-97.3) had acted on at least one of six domains and 79.8% on three or more — food security and nutrition 74.1%, sanitation 70.8%, social environment 64.1%, gender-based violence 60.8%, health services 59.1%, drinking water 56.4%. Each MAS is ten women, each nominated by about twenty contiguous households, so one MAS covers roughly 200 households — larger than the 50-100 household norm quoted in the body. The annual untied fund of INR 5,000 (about USD 60) was reported as inadequate and often late, and MAS have no formal seat at the planning or monitoring table [PMID 40349012]

Current national data

- **India's urban denominator has moved.** The urban population, put at 479 million in 2022, is projected to be 38.6% of the national total by 2036 — the figure to quote in place of the

textbook's "about 36%", and the basis on which NUHM's coverage of 779 cities is now judged [PMID 36896811]

- **Housing quality is now quantified against a hard outcome in national data.** Longitudinal Ageing Study in India, 2017-18, n = 30,632 adults aged 60 and over: a five-indicator housing quality index (building materials, sanitation, water access, cooking fuel, electricity) was associated with falls at OR 1.17 (95% CI 1.13-1.21) per unit; against good housing, moderate quality gave OR 1.32 (1.20-1.45) and poor quality OR 1.76 (1.56-1.98). Poor sanitation, inadequate water access, solid-fuel use, absent electricity, **lack of a separate bedroom or kitchen**, indoor air pollutants and household dampness were each independently associated with falls. About one-third of Indian older adults fall each year. This is the modern outcome evidence behind the Environmental Hygiene Committee 1947 standards that the answer quotes [PMID 40977273]

Elimination, surveillance and disease status

- **Urban dengue transmission has intensified sharply — a planning failure measured serologically.** Repeat serosurveys in Pune City show dengue IgG seroprevalence rising from 62.9% (95% CI 59.4-66.1) in 2009 to 88.4% (86.8-89.8) in 2019, with the annual force of infection climbing from 4.1 (3.8-4.5) to 10.9 (10.2-11.6) and DENV-3 dominant. Because over 85% of those above nine years had already been exposed by 2019, the authors argue such city serosurveys should inform readiness for dengue vaccine introduction [PMID 37742447]
- **Malaria risk in an Indian city tracks housing defects, not just vector density.** A study of 330 households in urban Chetla (Kolkata) and rural Singur, West Bengal, found malaria prevalence 6.1% urban versus 1.8% rural among individuals and 18.8% versus 6.7% at household level. Independent predictors were **mosquito breeding places on the premises** (AOR 4.93; 95% CI 1.53-15.93), **inadequate lighting in the house** (AOR 4.35; 1.78-10.64), **absence of an effective window screen** (AOR 3.19; 1.16-8.71), **no separate kitchen** (AOR 2.38; 1.06-5.34) and **five or more household members** (AOR 2.91; 1.19-7.15). The authors conclude that investment in housing programmes is itself a malaria-control measure [PMID 38287772]

Guideline and legislative updates

- **Heat-health action plans are the newest statutory instrument that town planning must serve, and India writes most of them.** A global review of 83 plans from 24 countries published 2009-2024 found 35 came from India alone — 89.7% of all lower-middle-income-country plans and 94.6% of the South-East Asia Region's — and that 72.3% of all plans appeared between 2020 and 2024. Six plans, Ahmedabad 2019 among them, prescribe **urban green space, cool roofs or urban forests** as the structural cooling measure, which is the town-planning lever sitting inside a heat plan; Indian sub-city plans named include **Thane City 2024** and **Jodhpur 2023**. Only 21.7% of plans worldwide carried any targeted mental-health intervention despite 75.9% mentioning mental health [PMID 40418305]
- **Heat is to be planned for structurally, not managed as an acute disaster.** Writing after the record Delhi summer of 2024, this position paper argues that heat-related illness and death are a public-health and economic problem requiring long-term resilience built into **urban planning and health systems**, combined with emergency preparedness — because low-income households live in dwellings that give little protection from heat and struggle to reach water. Sectors named

as necessarily involved include medical care, surveillance, risk communication, disaster preparedness, livelihoods, and adaptation and urban planning [PMID 39515351]

Global and WHO developments

- ▶ **Green space now has an effect size, which strengthens the green-belt clause of a master plan.** A scoping review of 27 studies found a consistent protective association between green space exposure and stroke-related death and disability, with **mortality hazard ratios between 0.66 and 0.95**; estimates for stroke incidence ranged **0.4 to 0.98** but were less consistent. The proposed mechanisms are precisely the ones a plan controls — moderation of ambient temperature, noise and air pollution, plus opportunity for physical activity [PMID 38219382]
- ▶ **Landscape-fire smoke, the pollution source zoning cannot fix, causes 1.53 million deaths a year.** Global assessment for 2000-19: **1.53 million all-cause deaths annually (95% empirical CI 1.24-1.82)**, including **0.45 million cardiovascular** and **0.22 million respiratory** deaths; PM2.5 accounted for 77.6% of the burden and ozone 22.4%. **Over 90% of deaths were in low- and middle-income countries**, with south Asia at 170,762 deaths per year and India among the five countries with the largest attributable death toll — the reminder that the airshed a city sits in is set beyond its own municipal boundary [PMID 39615506]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 2 — Concept of Health and Disease, p 47; Ch 7 — Health Programmes in India, pp 535-536, 574-575; Ch 8 — Demography and Family Planning, pp 583-584; Ch 12 — Medicine and Social Sciences, p 841; Ch 15 — Environment and Health, pp 861, 883, 890, 893, 896-898, 905-908; Ch 18 — Occupational Health, pp 966-967; Ch 23 — Health Planning and Management, p 1035); Suryakantha — Community Medicine with Recent Advances (Ch 6 — Air and Ventilation, p 69; Ch 7 — Noise, p 72; Ch 10 — Housing, pp 82-83); Sunder Lal — Textbook of Community Medicine (Ch 3 — Environmental Health Problems, pp 108-109); Oxford Textbook of Public Health 7e (Ch 58 — Urbanization and Health, pp 2159-2160; Ch 63 — Physical Activity and Public Health, pp 2370-2372). Named documents cited within: Environmental Hygiene Committee (India), 1947 housing standards; WHO Expert Committee criteria for healthful housing; WHO Housing and Health Guidelines; Census of India 2011; Pradhan Mantri Awas Yojana–Urban, 2015; National Urban Health Mission; Swachh Bharat Mission (Urban).

Recent-advances sources: 36896811 — Core interventions contributing to the effectiveness of the National Urban Health Mission in India, *Journal of Global Health* 2023, doi:[10.7189/jogh.13.03009](https://doi.org/10.7189/jogh.13.03009); 40349012 — Community participation through women's health collectives promoted by India's National Urban Health Mission: a realist evaluation in Chhattisgarh state, *International Journal for Equity in Health* 2025, doi:[10.1186/s12939-025-02498-z](https://doi.org/10.1186/s12939-025-02498-z); 40977273 — Housing quality and related housing environmental factors associated with falls in older adults: results from the Longitudinal Ageing Study in India, *Age and Ageing* 2025, doi:[10.1093/ageing/afaf260](https://doi.org/10.1093/ageing/afaf260) (abstract-only); 37742447 — Seroprevalence of dengue virus infection in Pune City in India, 2019: a decadal change, *Journal of Infection and Public Health* 2023, doi:[10.1016/j.jiph.2023.08.020](https://doi.org/10.1016/j.jiph.2023.08.020) (abstract-only); 38287772 — Malaria infection and associated household level risks in an urban and rural areas of West Bengal, India, *Journal of Vector Borne Diseases* 2024, doi:[10.4103/JVBD.JVBD_162_23](https://doi.org/10.4103/JVBD.JVBD_162_23) (abstract-only); 40418305 — A Critical Gap in Addressing Mental Health in Heat-Health Action Plans Worldwide, *Current Environmental Health Reports* 2025, doi:[10.1007/s40572-025-00486-7](https://doi.org/10.1007/s40572-025-00486-7); 39515351 — Responding to heat-related health risks: the urgency of an equipoise between emergency and equity, *The Lancet Planetary Health* 2024, doi:[10.1016/S2542-5196\(24\)00246-8](https://doi.org/10.1016/S2542-5196(24)00246-8) (abstract-only); 38219382 — Green space and stroke: a scoping review of the evidence, *Journal of the Neurological Sciences* 2024, doi:[10.1016/j.jns.2024.122870](https://doi.org/10.1016/j.jns.2024.122870) (abstract-only); 39615506 — Global, regional, and national mortality burden attributable to air pollution from landscape fires: a health impact assessment study, *The Lancet* 2024, doi:[10.1016/S0140-6736\(24\)02251-7](https://doi.org/10.1016/S0140-6736(24)02251-7) (abstract-only).

Coverage note: the corpus carries no chapter titled "town planning"; the answer is built from the town-planning and zoning material distributed across Park's chapters on environment (housing, noise, air pollution), occupational health and health planning, supplemented by Suryakantha and Sunder Lal, so no separate town-planning classification or Indian urban-development statute (for example the state Town and Country Planning Acts or Development Control Regulations) is quoted, as none appears in the retrieved text.



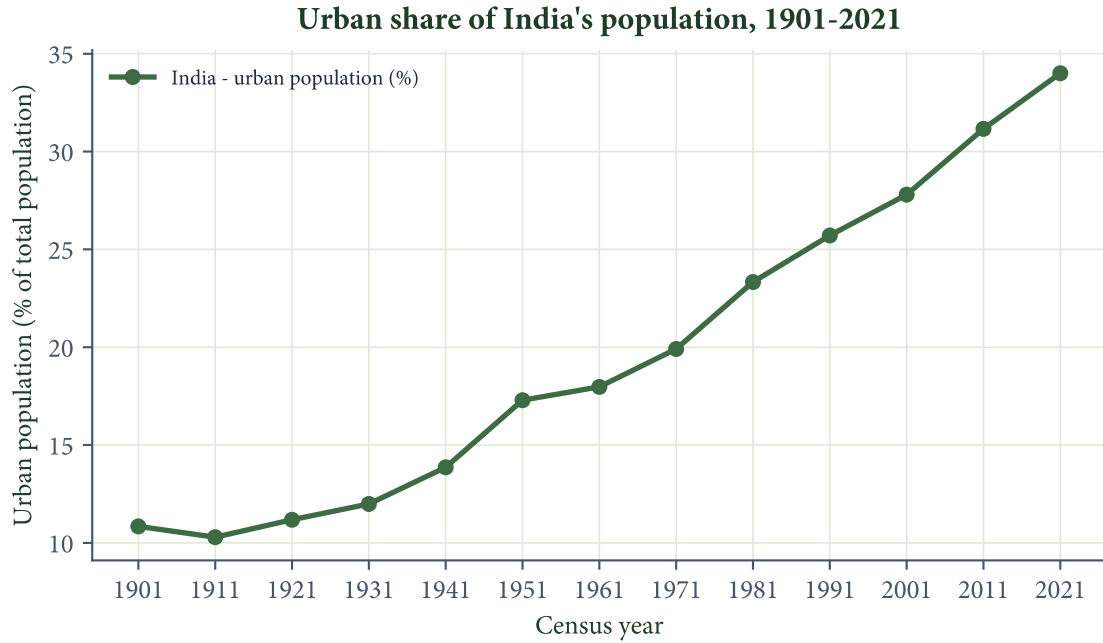
Urbanisation and its impact on health

Asked in: Paper I 29-Jul-2021 — RGUHS MD Community Medicine

Definition — Urbanization is the progressive increase in the proportion of a population residing in urban localities, together with the physical, social and economic transformation of settlements that accompanies it. In the Indian census an **urban area** means a **town** (a place with a municipal corporation, municipal area committee, town committee, notified area committee or cantonment board), **or** any place with **5,000 or more inhabitants**, a **density of not less than 1,000 persons per square mile (400 per square kilometre)**, **pronounced urban characteristics**, and **at least three-fourths of the adult male population employed in pursuits other than agriculture** (Park 28e, Ch 2, p 47; Ch 8, p 583). Urbanization is now accepted as a **determinant of health in its own right**, and of health inequity both between and within cities (Oxford Textbook of Public Health 7e, Ch 58, p 2151).

Magnitude and trend

Level	Figure (with year)	Source
World urban population	56% (2020); 55% in 2018, projected >68% by 2050	Park 28e, Ch 2, p 47; Oxford PH 7e, Ch 58, p 2151
India — urban share	31.16% (Census 2011) → 35.39% (2021) → 36% (2024)	Sunder Lal, Ch 9, p 543; Park 28e, Ch 2, p 35 and Ch 8, p 583
India — urban population size	35.7 crore (2011), projected 43.2 crore (2021)	Suryakantha, Ch 37, p 799
Population density, India	464 per km ² (2020); Delhi highest at 11,297 (2021)	Park 28e, Ch 8, p 583; Sunder Lal, Ch 9, p 546
Megacity (population ≥10 million)	Delhi 33.8 million, Mumbai 21.7, Kolkata 15.6, Bengaluru 14.0 (2024)	Suryakantha, Ch 24, p 640; Park 28e, Ch 8, p 583
Slum population — India	≥ one-third of all urban dwellers; growing ~7% per year	Sunder Lal, Ch 2, p 68; Suryakantha, Ch 37, p 799
Slum / informal-settlement dwellers — world	~1 billion, with the worst health outcomes of all urban residents	Oxford PH 7e, Ch 58, pp 2151–2152



Trend in the urban share of India's population, 1901-2021. Source: Sunder Lal, Textbook of Community Medicine, Table 9.2 (Ch 9, p 543).

Determinants of urbanization and its hazards

Driver	Content
Natural increase	Growth of the resident urban population itself (births exceeding deaths) (Suryakantha, Ch 24, pp 639–640)
Rural-to-urban migration (dominant)	Employment opportunities, attraction of better living conditions, and educational, health, transport and entertainment facilities; migrants typically settle in peri-urban slums (Suryakantha, Ch 24, pp 639–640)
Industrialization	Migration of labour to industrial sites; slums "come up like mushrooms" around them (Suryakantha, Ch 18, p 244)
Reclassification	Villages acquiring town status by crossing the census size, density and occupation criteria (Park 28e, Ch 8, p 583)

HAZARDS OF RAPID, UNPLANNED URBANIZATION

1. PHYSICAL (ENVIRONMENTAL)

- Housing: eruption of slums, overcrowding
- Air, water, soil and noise pollution
- Vector problems (domestic water storage)

2. PSYCHOLOGICAL

- Behavioural disorders
- Mental illness (neuroses, psychoses)
- Anxiety, tension, worry from stress and strain

3. SOCIAL AND ANTISOCIAL

- Alcoholism, drug abuse, gambling
- Broken homes, divorce, unemployment
- Theft, murder, sex crimes, child abuse, juvenile delinquency

4. MISCELLANEOUS

- Malnutrition, infections, epidemics
- Sexually transmitted infections including HIV
- Accidents; hypertension from stress

Classification of the hazards of rapid, unplanned urbanization

Impact on health

Domain	Mechanism in the urban setting	Health outcome (source)
Housing and overcrowding	Sub-standard, poorly ventilated, congested dwellings	Respiratory infections (common cold, tuberculosis, influenza, diphtheria, bronchitis, measles, whooping cough); skin infections (scabies, ringworm, impetigo, leprosy); rat infestation (plague); arthropod-borne disease; domestic accidents; high morbidity and mortality (Park 28e, Ch 15, p 907)
Environmental pollution	Industrial and vehicular emissions, sewage, solid waste	Air, water, soil and noise pollution affecting workers and the surrounding resident population alike (Suryakantha, Ch 27, p 729)

Domain	Mechanism in the urban setting	Health outcome (source)
Vector-borne disease	Domestic water storage and construction sites; <i>Anopheles stephensi</i> and <i>Aedes</i> breeding	Dengue risk rising with rapid urbanization (Park 28e, Ch 5); 15 large cities account for ~80% of urban malaria, of which <i>P. vivax</i> contributes 85–90% (Sunder Lal, Ch 8, p 386)
Emerging and re-emerging infections	Unplanned urbanization, overcrowding, poor sanitation, weak public health infrastructure	Diseases once regarded as "rural" have acquired serious urban dimensions (Park 28e, Ch 2, p 33; Ch 5)
Non-communicable disease	Altered diet, obesity, physical inactivity, tobacco and alcohol, occupational stress	Diabetes prevalence approximately twice that of rural populations; urbanization with globalization and ageing forms the "causes of the causes" of cardiovascular disease (Park 28e, Ch 6, pp 438, 468)
Injury	Traffic density, defective narrow roads, poor lighting, mixed traffic	Road traffic crashes commoner in urban than rural areas (Suryakantha, Ch 22, p 570); India recorded 4,49,002 crashes with 1,51,113 deaths in 2019 (Park 28e, Ch 7, p 571)
Mental and social health	Migration, loss of joint-family support, isolation, unemployment	Behavioural disorders, neuroses and psychoses, anxiety, alcoholism, drug abuse, broken homes and antisocial activity (Suryakantha, Ch 24, pp 644–645; Ch 18, p 244)
Sexual and reproductive health	Single male migrants isolated from family; slum residence	Casual sexual contact, sexually transmitted infections and HIV (Suryakantha, Ch 21, pp 478, 494)
Nutrition of the urban poor	Poverty, food insecurity, poor hygiene	Undernutrition and stunting in children of the urban poor, alongside rising childhood overweight (Oxford PH 7e, Ch 4, pp 237–238; Park 28e, Ch 2, p 36)
Health system	Infrastructure, water, sanitation and health care cannot keep pace with growth	The urban poor remain unable to find or afford services and lack political power to change their living conditions (Park 28e, Ch 2; Oxford PH 7e, Ch 58, pp 2151–2152)

Overcrowding is the single measurable core of the housing hazard and is defined by any one of three criteria (Park 28e, Ch 15, pp 906–907).

Criterion	Accepted standard
Persons per room	1 room – 2 persons; 2 rooms – 3; 3 rooms – 5; 4 rooms – 7; 5 or more rooms – 10 (add 2 for each further room)
Floor space per person	≥110 sq ft – 2 persons; 90–100 sq ft – 1½; 70–90 sq ft – 1; 50–70 sq ft – ½; under 50 sq ft – nil (infant <12 months not counted; child 1–10 years = half unit)
Sex separation	Overcrowding exists if 2 persons over 9 years of age, of opposite sex and not husband and wife, must sleep in the same room
Related housing minima	Living room floor area ≥120 sq ft for more than one occupant, ≥100 sq ft for one; air space ≥500 cubic ft per capita, preferably 1,000

The urban advantage and the intra-urban penalty

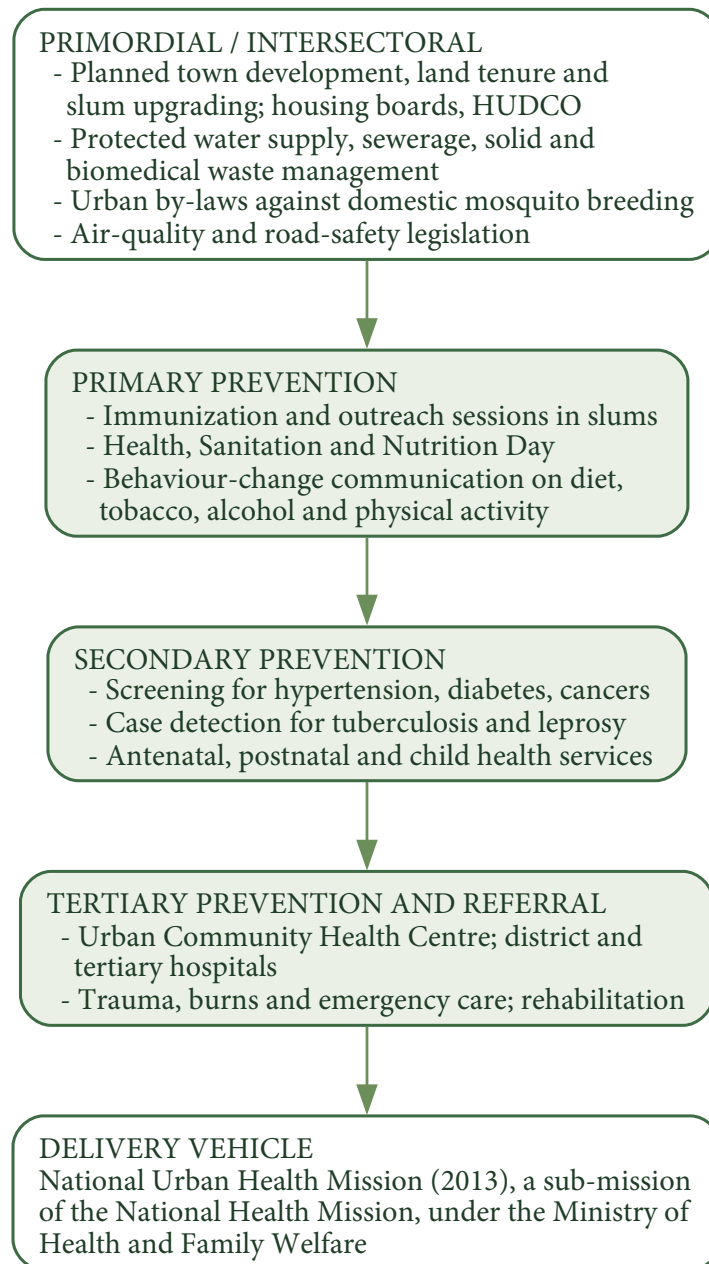
On aggregate indicators urban India fares better than rural India, and urban vital rates confirm it — crude birth rate 16.1 and crude death rate 5.1 per 1,000 population, infant mortality rate 19 per 1,000 live births (SRS 2022) (Sunder Lal, Ch 2, p 68).

Indicator (year)	Urban	Rural
Total fertility rate (2019–21)	1.6	2.1
Women aged 15–19 who have begun childbearing, % (2019–21)	3.8	7.9
Population below the national poverty line, % (2011–12)	13.7	25.7
Women who are literate, % (2019–21)	83	69.3
Antenatal care, at least 4 visits, % (2019–21)	68.1	54.2
Skilled attendance at delivery, % (2019–21)	94	81.8
Infant mortality rate per 1,000 live births (2020)	19	31
Children under 5 stunted, % (2019–21)	30.1	37.3
Children under 5 overweight, % (2019–21)	4.2	3.2
Access to improved sanitation facility, % (2019–21)	81.5	64.9

Source: Park 28e, Ch 2, pp 35–36 (National Family Health Survey-5, 2019–21).

This aggregate advantage is an **average that conceals a gradient**. No city, rich or poor, is immune to health inequity, and slum, informal-settlement and pavement dwellers frequently have health outcomes **worse than populations living in rural poverty** — the *urban penalty* (Oxford PH 7e, Ch 58, pp 2151–2152). Demographically, urban slums are deprived areas, and the plight of the slum dweller may be worse than that of the villager (Sunder Lal, Ch 9, p 543).

Public health response



Levels of action against the health impact of urbanization

- **Coverage of the National Urban Health Mission (NUHM)** — all State capitals, district headquarters and other cities/towns with a population of 50,000 and above (Census 2011), taken up in a phased manner; towns below 50,000 remain under the National Rural Health Mission (Park 28e, Ch 7, pp 535–536).
- **Target groups** — the poor living in listed and unlisted slums, and other vulnerable urban groups: the homeless, street children, rickshaw pullers, construction, brick-kiln and lime-kiln workers, sex workers and temporary migrants (Park 28e, Ch 7, p 535).
- **Community platform** — a resident **Accredited Social Health Activist (ASHA)** or link worker links the slum to the health centre and generates demand for services, while a **Mahila Arogya Samiti** of 10–12 women acts as a peer-educator group, undertakes community-based monitoring

and manages a revolving fund on an annual grant of ₹5,000 (Park 28e, Ch 7, p 535; Sunder Lal, Ch 17, p 758).

- ▶ **Service delivery** — monthly outreach sessions in the slum by auxiliary nurse midwives, an evening out-patient department at the Urban Primary Health Centre (12 noon to 8 pm, no inpatient care) and referral upward; existing Urban Primary Health Centres are being strengthened as **Ayushman Arogya Mandirs** (Park 28e, Ch 7, p 536; Sunder Lal, Ch 17, p 758).
- ▶ **Disease-specific urban components** — control of urban malaria rests primarily on enforcing **urban by-laws** that prevent mosquito breeding in domestic containers (Sunder Lal, Ch 8, p 386).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Urban health care facility pyramid under the National Urban Health Mission — draw a four-tier pyramid, widest at the base, and label from bottom upward (Park 28e, Ch 7, p 536; Sunder Lal, Ch 17, p 758):

- ▶ **Base: Community level** — ASHA / link worker (1 per 1,000–2,500 urban poor; 200–500 households) and Mahila Arogya Samiti (50–100 households)
- ▶ **Second tier: Urban Ayushman Arogya Mandir (U-AAM)** — 15,000–20,000 population; approved under 15th Finance Commission health grants and PM-ABHIM; linked to the nearest Urban Primary Health Centre
- ▶ **Third tier: Urban Primary Health Centre (U-PHC)** — ~50,000 population, sited within or within half a kilometre of a slum, covering ~25,000–30,000 slum population; may be one per 75,000 in very high-density areas, or one per 5,000–10,000 for isolated slum clusters
- ▶ **Apex: Urban Community Health Centre (U-CHC)** — in metro cities, one per 5 lakh population with 100 beds; onward referral to district and tertiary hospitals
- ▶ **Side annotation: no sub-centres exist in urban areas** (unlike the rural three-tier system)

High-yield exam pointers

- ▶ **Census definition of "urban"** (write all four limbs): statutory town or $\geq 5,000$ population + density ≥ 400 per km^2 (1,000 per square mile) + pronounced urban characteristics + $\geq 75\%$ of adult males in non-agricultural work.
- ▶ **Dated Indian figures:** urban share 31.16% (Census 2011) $\rightarrow$ 36% (2024); world 56% urban (2020) $\rightarrow$ $>68\%$ projected by 2050; slum growth $\sim 7\%$ per year; $\geq 1/3$ of urban Indians live in slums.
- ▶ **Overcrowding triad:** persons-per-room / floor-space-per-person / sex separation at **9 years** — quote the 1-room-2-persons and 110-sq-ft-2-persons ladders.
- ▶ **Urban–rural divide (NFHS-5, 2019–21):** IMR 19 vs 31; skilled attendance 94% vs 81.8%; sanitation 81.5% vs 64.9% — then state the *urban penalty*: slum dwellers can be worse off than the rural poor.
- ▶ **Diabetes prevalence is about twice the rural figure** — the classic single-line NCD marker of urbanization.
- ▶ **NUHM (2013) norms:** ASHA 1:1,000–2,500; MAS 50–100 households; U-PHC $\sim 50,000$ (25,000–30,000 slum) with 12 noon–8 pm out-patient department; U-CHC 5 lakh/100 beds in metros; U-AAM 15,000–20,000.

- Mnemonic for the impact table — "HEVEN-MISH": Housing/overcrowding, Environmental pollution, Vector-borne, Emerging infections, Non-communicable disease, Mental and social, Injury, Sexual and reproductive, Health-system failure.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: the corpus carries two slightly different values for India's urban share in 2021 (34.0% in Sunder Lal's Table 9.2, used for the graph, and 35.39% in Park 28e's urban-rural divide table), because the two draw on different projection series; quote either with its source and year attached. The dated urban-share series ends at 2021 as the decadal census due in 2021 has not been conducted.

SOURCE & COVERAGE

- Park's Textbook of Preventive and Social Medicine, 28e — Ch 2 (Concept of Health and Disease, pp 33, 35–36, 47), Ch 5, Ch 6 (p 468), Ch 7 (pp 535–536, 571), Ch 8 (p 583), Ch 15 (pp 906–908)
- Sunder Lal — Textbook of Community Medicine — Ch 2 (p 68), Ch 8 (p 386), Ch 9 (pp 543, 546), Ch 17 (p 758)
- Suryakantha — Community Medicine with Recent Advances — Ch 11 (Housing, pp 82–83), Ch 18 (Occupational Health, p 244), Ch 22 (p 570), Ch 24 (Demography, pp 639–645), Ch 27 (Social Science, p 729), Ch 37 (National Urban Health Mission, pp 799–800)
- Oxford Textbook of Public Health 7e — Ch 4 (pp 237–238), Ch 6 (pp 304–306), Ch 58 (Urbanization and Health, pp 2151–2159)
- National Family Health Survey-5 (2019–21); Sample Registration System 2022; Census of India 2011; National Urban Health Mission (2013)



Household water purification methods

Asked in: Paper I 06-Apr-2026 — RGUHS MD Community Medicine

Definition — Household (point-of-use) purification of water is the treatment of drinking water by an individual or a family at the place of consumption, as opposed to purification on a large scale at a municipal treatment works. Park considers purification of water under two headings — on a large scale and on a small scale — the small-scale group comprising **household purification of water** and **disinfection of wells** (Park 28e, Ch 15, pp 866, 871). Three methods are generally available on an individual or domestic scale — **boiling, chemical disinfection and filtration** — and they may be used singly or in combination; **ultraviolet (UV) irradiation** and **multi-stage reverse osmosis (RO)** are the appliance-based additions described for individual and institutional systems (Park 28e, Ch 15, pp 871-872).

1. Boiling

- ▶ To be effective the water must be brought to a **"rolling boil" for 10 to 20 minutes** (Park 28e, Ch 15, p 871).
- ▶ It kills **all bacteria, spores, cysts and ova** and yields **sterilized** water — no other household method sterilizes.
- ▶ It also **removes temporary hardness**, by driving off carbon dioxide and precipitating the calcium carbonate; the taste of the water is altered, but this is harmless.
- ▶ Water should be boiled **preferably in the same container in which it is to be stored**, to avoid contamination during storage.
- ▶ It is the method of choice for the individual, the household, high-risk individuals and outbreak situations, but cannot be used for a community supply (Sunder Lal, Ch 3, p 96).

2. Chemical disinfection

Agent	Available chlorine	Household preparation / dose	Stability and storage
Bleaching powder (chlorinated lime, CaOCl_2) — white amorphous powder with a pungent smell of chlorine	About 33 per cent when freshly made	Amount required for a given quantity of water is read off Park's Appendix III; the chlorine demand is measured in the field by Horrock's apparatus (Park 28e, Ch 15, p 871 and Appendix I)	Unstable — rapidly loses chlorine on exposure to air, light and moisture. Mixed with excess of lime it retains strength = " stabilized bleach ". Store in a dark, cool, dry place in a closed corrosion-resistant container and check stock strength frequently
Chlorine solution	4 kg of bleaching powder of 25 per cent available chlorine in 20 litres of water gives a 5 per cent chlorine solution	Ready-made solutions of different strengths are also marketed	Loses strength on exposure to light or on prolonged storage; keep in a dark, cool, dry place in a closed container

Agent	Available chlorine	Household preparation / dose	Stability and storage
High test hypochlorite (HTH, perchloron) — a calcium compound	60 to 70 per cent	Solutions prepared from HTH are used for water disinfection	More stable than bleaching powder; deteriorates much less on storage
Chlorine tablets (halazone, and the tablet formulated by the National Environmental Engineering Research Institute, Nagpur — 15 times better than ordinary halogen tablets)	Varies with the strength manufactured	A single 0.5 g tablet disinfects 20 litres of water (Park 28e, Ch 15, p 872; Sunder Lal, Ch 3, p 98)	Halazone tablets are costly; the Indian tablet is cheap and widely available
Iodine — for emergency disinfection only	2 per cent ethanol solution of iodine	Two drops per litre of clear water, contact time 20 to 30 minutes	Does not react with ammonia or organic compounds, so stays in its active molecular form over a wide pH range and persists longer than chlorine or bromine
Potassium permanganate	—	No longer recommended for water disinfection	A powerful oxidizing agent but an unsatisfactory disinfectant

Principles governing household chlorination

- ▶ The aim is a free residual chlorine of 0.5 mg per litre at the end of one hour's contact (Park 28e, Ch 15, p 871).
- ▶ Highly polluted and turbid waters are not suited for direct chlorination — the water must be clarified first.
- ▶ Chlorine acts best at a pH around 7; low temperature and high pH reduce its efficiency, because the disinfecting action is mainly due to undissociated hypochlorous acid (Suryakantha, Ch 5, p 50).
- ▶ Normal doses of chlorine do not destroy hepatitis virus, protozoal cysts, spores and helminth eggs; *Cryptosporidium* and some viruses are notably resistant, and superchlorination is needed for these (Sunder Lal, Ch 3, p 96).

3. Filtration through domestic filters

The essential part of a domestic filter is the "candle"; its action is purely mechanical, except in the silver-coated types (Park 28e, Ch 15, p 872; Sunder Lal, Ch 3, p 96).

Domestic filter	Filtering medium	Mode of action
Pasteur Chamberland filter	Candle of unglazed porcelain — porous tubes or bougies that can be screwed on to a tap	Purely mechanical straining; separates suspended particles including bacteria; reliable but slow

Domestic filter	Filtering medium	Mode of action
<i>Berkefeld filter</i>	Candle of kieselguhr (infusorial earth)	Purely mechanical straining
<i>Katadyn filter</i>	Filter surface coated with a silver catalyst	Bacteria coming into contact with the surface are killed by the "oligodynamic" action of silver ions liberated into the water
<i>Meta and Stellar filters</i>	Kieselguhr coated with silver, intermixed with aluminium hydrate	Pressure filters for domestic use, taking water directly from the tap (Sunder Lal, Ch 3, p 96)
Carbon-and-pad filter	Filtering pad plus activated carbon, screwed on to the tap	Traps turbidity and adsorbs organic matter, colour and odour; loses efficiency with time (Suryakantha, Ch 5, p 54)

Care of candles. Filter candles of the fine type usually remove the bacteria found in drinking water **but not the filter-passing viruses**. They become clogged with impurities and bacteria and must be **cleaned** by scrubbing with a hard brush under running water and **boiled at least once a week**; only clean water should be used with ceramic filters (Park 28e, Ch 15, p 872).

4. Ultraviolet irradiation and multi-stage reverse osmosis

- **Ultraviolet irradiation** — a film of water up to about 120 mm thick is exposed to one or several quartz mercury vapour arc lamps emitting ultraviolet radiation at a wavelength of 254 nanometres. It is effective against most micro-organisms that contaminate water supplies — bacteria, yeasts, viruses, fungi, algae and protozoa. Applications are limited to individual or institutional systems, and the water must be free from turbidity and from suspended or colloidal constituents for efficient disinfection (Park 28e, Ch 15, p 872).
- **Multi-stage reverse osmosis** makes water both chemically and microbiologically potable by reducing total dissolved solids, hardness and heavy metals, and by removing bacteria, viruses, protozoa and cysts. The **clarity cartridge** removes suspended particles such as dust, mud and sand; the **reverse osmosis cartridge** reduces total dissolved solids and hardness, removes heavy metals such as arsenic, lead and mercury, and eliminates micro-organisms (Park 28e, Ch 15, p 872). Domestic units combine a sediment filter, an activated carbon block, the reverse osmosis membrane and a silver-impregnated carbon polishing stage (Suryakantha, Ch 5, pp 54-55).
- **Solar water disinfection** — filled transparent water bottles exposed to sunlight for 6 hours disinfect the water, a method suited to resource-poor households (Sunder Lal, Ch 3, p 96).

5. Advantages and disadvantages of each method

Method	Advantages	Disadvantages
Boiling	Sterilizes — kills bacteria, spores, cysts and ova alike; removes temporary hardness; needs no apparatus, chemicals or skill; taste change is harmless; ideal in an outbreak and for high-risk individuals	No "residual protection" against subsequent microbial contamination; fuel-costly and time-consuming; impracticable beyond the household; risk of recontamination unless boiled and stored in the same container; does not remove turbidity, dissolved salts, fluoride or arsenic

Method	Advantages	Disadvantages
Chemical disinfection (chlorine compounds)	Cheap, rapid, simple; leaves a measurable free residual chlorine that both protects against and warns of recontamination; dose can be worked out in the field with Horrock's apparatus; tablets are portable and ideal for travel, camps and epidemics	Ineffective in turbid or highly polluted water; does not kill spores, cysts, helminth eggs, hepatitis virus or <i>Cryptosporidium</i> at normal doses; bleaching powder is unstable and loses strength on storage; produces chlorinous taste and odour; needs correct dosing — under-dosing fails and over-dosing is unpalatable; halazone tablets are costly
Iodine	Active over a wide pH range and in ammoniacal or organic-rich water; persists longer than chlorine or bromine; convenient for emergency use	High cost ; the element is physiologically active (thyroid activity) , so it is unsuitable for prolonged or population-wide use
Potassium permanganate	None of practical value — of historical interest only	Not a satisfactory disinfectant — may kill cholera vibrios but is of little use against other organisms; alters the colour, smell and taste of water; no longer recommended
Domestic (ceramic candle) filters	Simple, need no power or chemicals; remove turbidity and the bacteria found in drinking water; the <i>Katadyn</i> type adds bactericidal action through silver ions; activated carbon versions also remove pesticides, chlorine and odour	Do not remove filter-passing viruses; candles clog readily and a clogged candle becomes a nidus for organisms; need weekly scrubbing and boiling, and may break during cleaning; give a false sense of security ; slow output; not suitable for widespread use under Indian conditions (Park 28e, Ch 15, p 872)
Ultraviolet irradiation	Exposure is for a short period; introduces no foreign matter and produces no taste or odour ; over-exposure is harmless; the chemical composition of water is unaltered	No residual effect ; no rapid field test to assess treatment efficiency; the apparatus is expensive and needs electricity and annual lamp replacement; fails if the water is turbid
Multi-stage reverse osmosis	The only household method that also removes chemical contamination — total dissolved solids, hardness, arsenic, lead, mercury, excess fluoride — besides bacteria, viruses, protozoa and cysts	Removes desirable naturally occurring minerals; slow; wastes about four litres of water for every litre purified ; heavy initial cost and needs regular maintenance and electricity (Suryakantha, Ch 5, p 55)
Solar disinfection	Zero cost, no chemicals, suited to the poorest households	Needs 6 hours of good sunlight and clear water; small volumes only; no residual protection

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Domestic ceramic candle filter (section)** — draw a cylindrical vessel divided into an upper and a lower chamber. Label from top down: *lid*; *upper chamber — raw water*; *filter candles (porcelain in the Chamberland type, kieselguhr in the Berkefeld type)*; *partition*; *lower chamber — filtered water*; *outlet tap*. Add a side note: *Katadyn — candle surface coated with silver, oligodynamic action*.

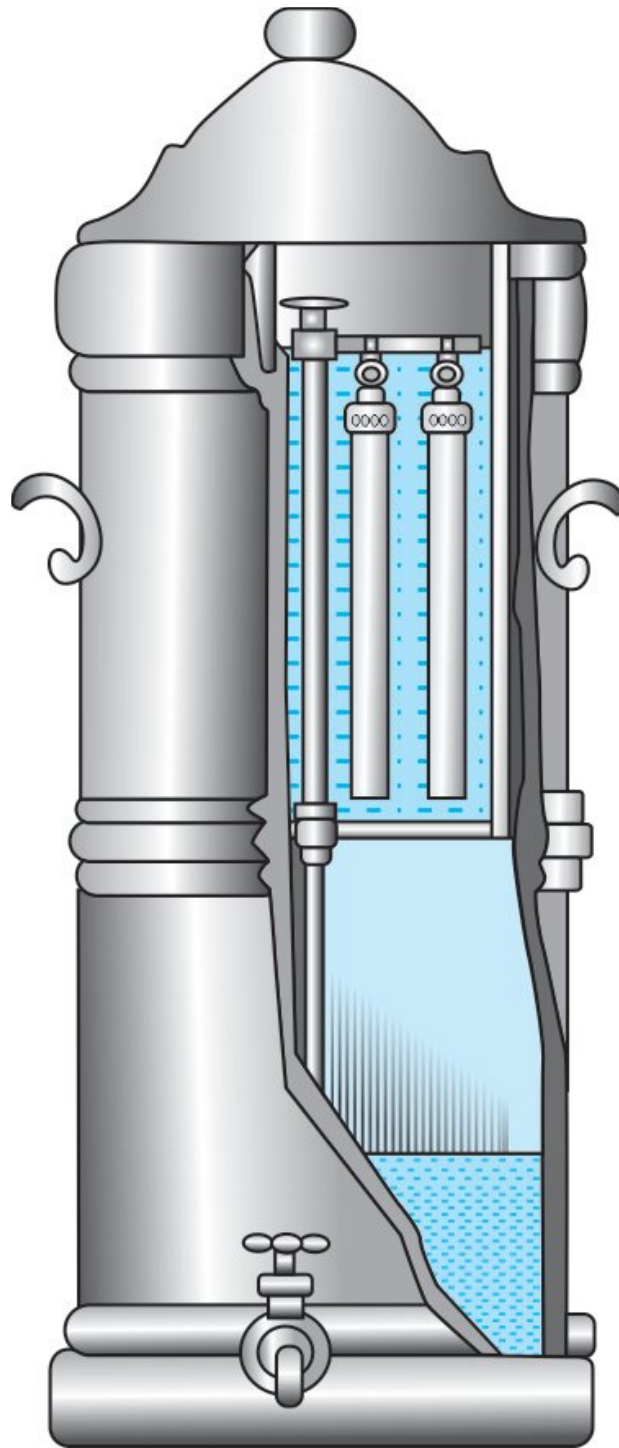


Fig 5.10 — Pasteur-Chamberland filter Fig. 5.11 Berkefeld filter Source: Ghosh BN (Suryakantha Environment Best, p.53)

HOUSEHOLD PURIFICATION OF WATER
(individual / domestic scale; methods may be used singly or in combination)

PHYSICAL - HEAT AND LIGHT

- Boiling: rolling boil for 10-20 minutes
- Solar disinfection: filled bottles in sunlight for 6 hours

CHEMICAL DISINFECTION

- Bleaching powder (33% available chlorine)
- Chlorine solution (5% from bleaching powder)
- High test hypochlorite / perchloron (60-70%)
- Chlorine tablets: 0.5 g per 20 litres
- Iodine: 2 drops of 2% solution per litre (emergency)
- Potassium permanganate: no longer recommended

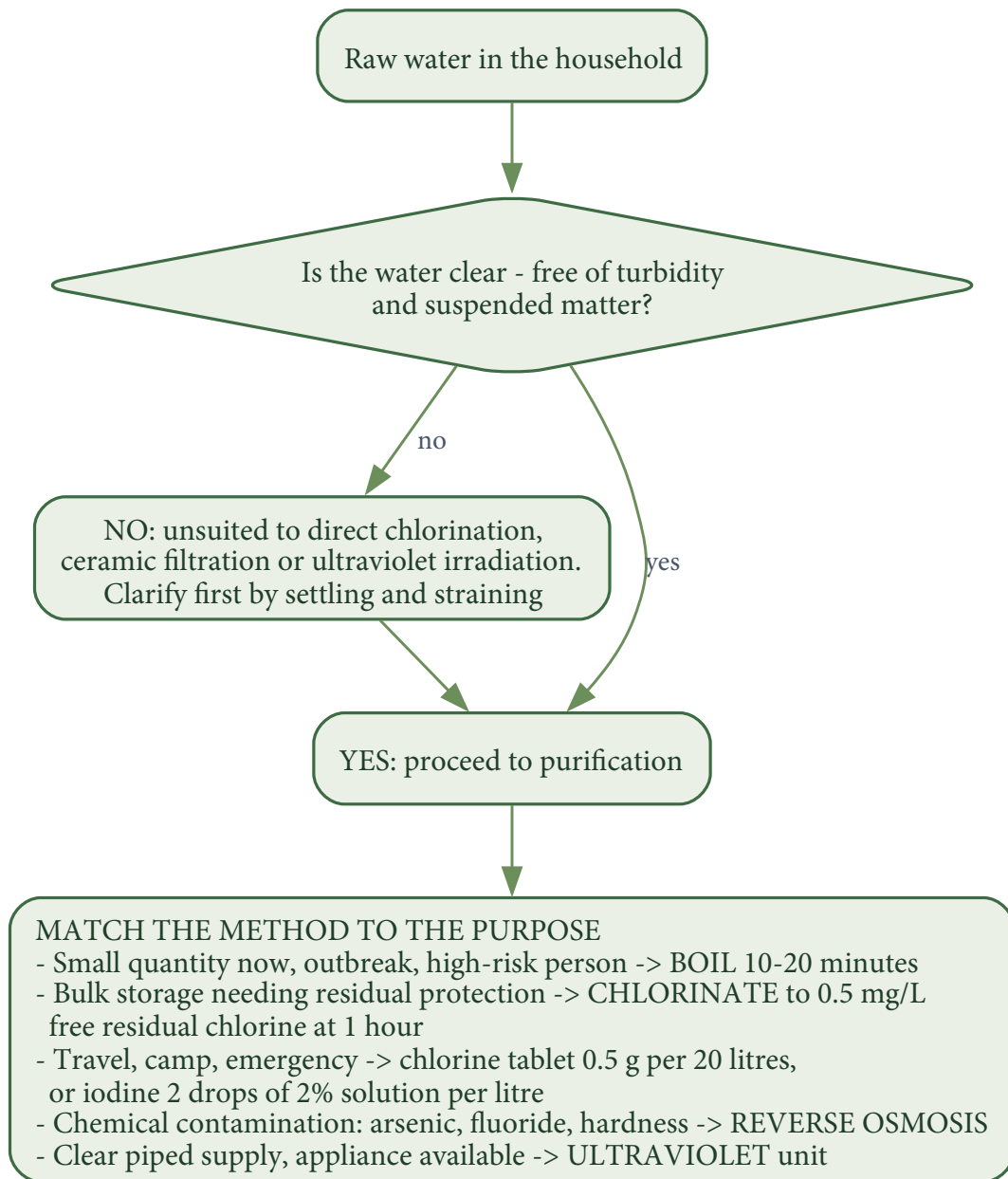
FILTRATION - DOMESTIC FILTERS

- Pasteur Chamberland: porcelain candle
- Berkefeld: kieselguhr candle
- Katadyn: silver-coated, oligodynamic action
- Meta and Stellar pressure filters
- Carbon-and-pad filter

APPLIANCE-BASED

- Ultraviolet irradiation: 254 nm, film up to 120 mm thick
- Multi-stage reverse osmosis: clarity + RO cartridge

Methods of household (point-of-use) purification of water



Choosing a household purification method by the quality of the raw water and the purpose

High-yield exam pointers

- ▶ **Three basic methods** on an individual or domestic scale — **boiling, chemical disinfection, filtration** — used singly or in combination; UV irradiation and multi-stage reverse osmosis are the appliance additions (Park 28e, Ch 15, pp 871-872).
- ▶ Boiling: **rolling boil, 10-20 minutes**; kills bacteria, spores, cysts and ova; removes temporary hardness; **no residual protection**.
- ▶ Available chlorine: bleaching powder **33 per cent**, chlorine solution **5 per cent** (4 kg of 25 per cent bleaching powder in 20 litres), high test hypochlorite / perchloron **60-70 per cent**.
- ▶ Doses to write: chlorine tablet **0.5 g per 20 litres**; iodine **2 drops of 2 per cent ethanol solution per litre**, contact **20-30 minutes**; target **free residual chlorine 0.5 mg per litre at one hour's contact**.

- ▶ Candle materials: **porcelain** = Pasteur Chamberland, **kieselguhr** (infusorial earth) = Berkefeld, **silver-coated** = Katadyn (oligodynamic action); candles remove bacteria but not filter-passing viruses; scrub and boil weekly.
- ▶ Ultraviolet: **254 nm**, film of water up to **120 mm** thick, quartz mercury vapour arc lamp; no residual effect, no field test of efficiency.
- ▶ **Horrock's apparatus** finds the dose of bleaching powder required; **stabilized bleach** = bleaching powder mixed with excess lime.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: Park states the household chlorination target as a free residual chlorine of 0.5 mg per litre after one hour's contact, while Sunder Lal quotes the World Health Organization criterion of 0.5 mg per litre after at least 30 minutes' contact at pH below 8.0, with a minimum of 0.2 mg per litre at the point of delivery — quote Park's figure in the RGUHS answer and cite the other only if asked. The available chlorine of high test hypochlorite is given as 60-70 per cent by Park and as 75 per cent by Suryakantha (Ch 5, p 50); Park's figure is the safer one to write.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 15 (Environment and Health, pp 866, 871-872, and Appendix I, Horrock's apparatus); Suryakantha — Community Medicine with Recent Advances, Ch 5 (Environment and Water, pp 49-55); Sunder Lal — Textbook of Community Medicine, Ch 3 (Environmental Health Problems, pp 96-98).



Temperature inversion

Asked in: Paper I Nov 2018 — RGUHS MD Community Medicine

Definition — Temperature inversion is a meteorological state in which the normal vertical temperature gradient of the lower atmosphere is reversed, so that a layer of cool, heavy air lies trapped near the ground beneath a layer of warmer, lighter air, and temperature therefore *rises* with increasing altitude instead of falling (Suryakantha, Ch 7, p 67). Because the normal upward convection of air is abolished, pollutants and water vapour remain locked in the breathing zone and their concentration rises steeply, the characteristic result being "smog" (Park 28e, Ch 15, p 888). It is the single most important meteorological determinant of acute air-pollution episodes, and is far more frequent in the winter months than in spring or summer.

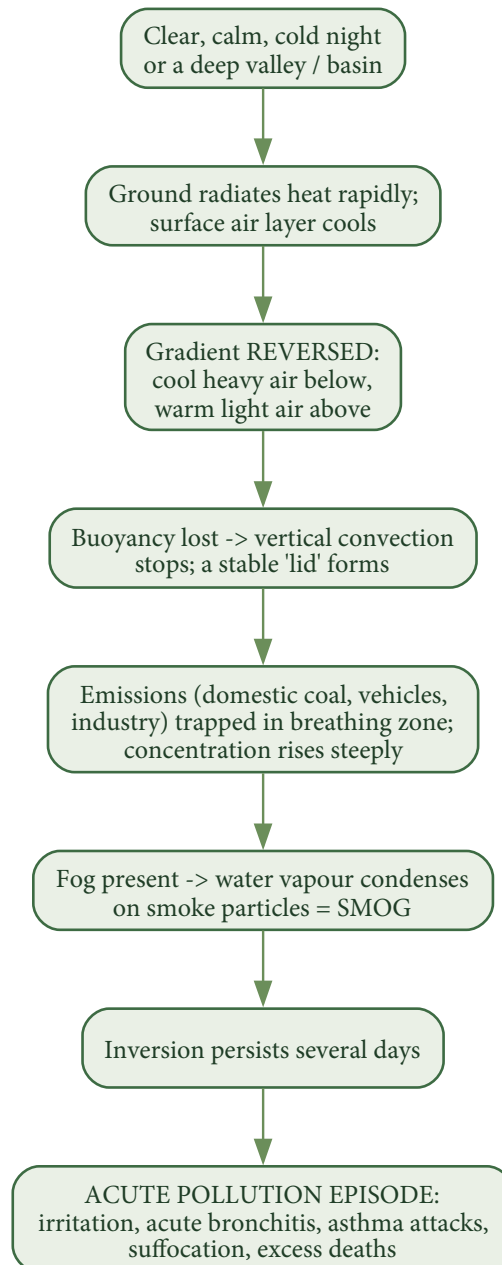
Normal vertical temperature gradient versus inversion

Feature	Normal atmosphere	Temperature inversion
Temperature with rising altitude	Falls — the environmental lapse rate, about 1.98 °C per 305 m (1,000 ft) in the troposphere (AFMC Red Book 2024, Ch 11, pp 244-246)	Rises — gradient reversed; surface layer cooler than the layer above (Suryakantha, Ch 7, p 67)

Feature	Normal atmosphere	Temperature inversion
Air near the earth's surface	Warmer and lighter than air above	Cooler and heavier than air above (AFMC Red Book 2024, Ch 15, p 400)
Vertical air movement	Warm surface air rises, expands and cools; cool upper air descends — turbulent (convective) flow	Convection abolished; little or no vertical motion — a stable "lid" over the ground layer (Park 28e, Ch 15, p 888)
Fate of pollutants	Diluted and dispersed over long distances, aided by higher horizontal wind velocity at altitude (Mahajan & Gupta 4e, Ch 5, p 64)	Locked up in the breathing zone; concentration rises steeply
Public health consequence	Self-cleansing of the atmosphere	Smog formation, acute pollution episode, excess morbidity and mortality

Mechanism and the conditions that favour it

- ▶ **Radiative cooling of the ground.** On a clear, calm night the earth's surface loses heat rapidly by infrared radiation; the thin air layer in contact with it cools below the temperature of the air above (Suryakantha, Ch 7, p 67).
- ▶ **Loss of the driving force for convection.** Warm air rises only because it is lighter than the air above it. Once the surface layer is the colder and denser one, that buoyancy is lost and the vertical exchange stops (AFMC Red Book 2024, Ch 15, p 400).
- ▶ **Topography.** Deep valleys and low-lying basins are classically affected — cold dense air drains down the slopes and pools on the valley floor, and the surrounding hills also cut off horizontal ventilation (Mahajan & Gupta 4e, Ch 5, p 64).
- ▶ **Season and time of day.** Inversions are commonest in the colder part of the year and at night or in the early morning, when solar heating of ground-level air is minimal; they are much less frequent in spring and summer (Park 28e, Ch 15, p 888).
- ▶ **Still weather.** Calm, windless anticyclonic conditions are essential — winds normally disperse and dilute pollutants, and where the topography is dominated by mountains or tall buildings the winds become weak and pollutants concentrate at the breathing zone.
- ▶ **Persistence.** An inversion may persist for several days at a time, which is precisely why a pollution load that is tolerable on ordinary days becomes lethal (Suryakantha, Ch 7, p 67).
- ▶ **Smog formation.** If fog is present under inversion conditions, water vapour condenses around suspended smoke particles: **smoke + water vapour = smog**. Intense smog is lethal.



How a temperature inversion converts routine urban emissions into a lethal smog episode

Public health significance — the classical inversion episodes

Episode (year)	Setting and cause	Outcome
Meuse Valley, Belgium (December 1930)	Deep river valley; inversion persisted 5 days	63 deaths and many cattle killed (Suryakantha, Ch 7, p 67)
Donora, Pennsylvania, USA (1948)	Valley town, industrial emissions under inversion	20 deaths; several hundred taken ill

Episode (year)	Setting and cause	Outcome
London, England (December 1952)	Domestic coal burning under a dense winter inversion fog; up to 10-fold rise in ambient sulphur dioxide, visibility down to a few metres	More than 4,000 deaths — the deadliest smog in history; deaths mainly from pneumonia, bronchitis, tuberculosis and cardiovascular failure (Handbook of Epidemiology, Ch 24, pp 954-955)

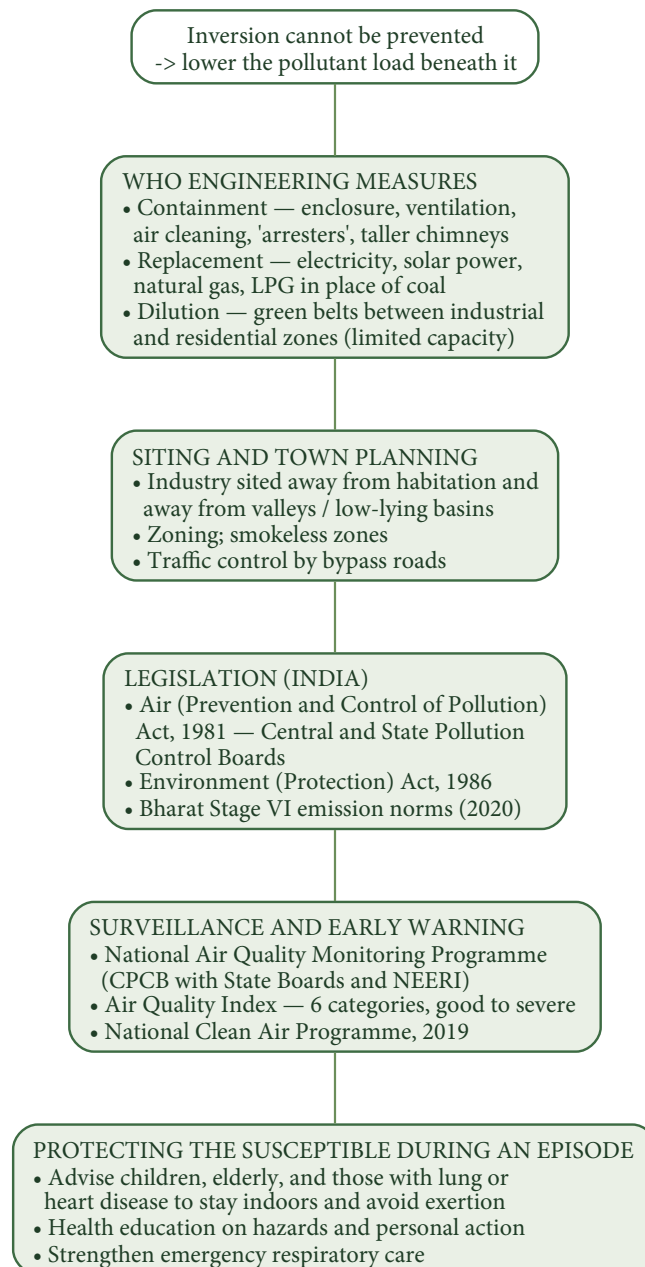
Two further points give this section its examination value. First, the London episode of 1952 is the historical starting point of quantitative risk assessment of air pollution on human health, and it forced the first British *Clean Air Act* in 1956 — legislation born directly out of an inversion. Second, air pollution under inversion takes two distinct forms: **winter (classical or "acid") smog**, a reducing smog of smoke and sulphur dioxide that accumulates because vertical dispersion is impossible, and **summer (photochemical) smog**, an oxidising smog in which ultraviolet light acts on nitrogen oxides and volatile organic compounds to generate ozone and other photo-oxidants, favoured by summer temperatures, direct sunlight and a standing air layer (Jugal Kishore, Ch 3, pp 956-957).

Health effects and susceptible groups

Timing	Effects
Immediate / acute	Irritation of conjunctiva, nose, throat and respiratory mucosa — conjunctivitis, allergic rhinitis, acute pharyngitis; acute bronchitis ; acute attacks of bronchial asthma; with intense pollution, death by suffocation (Park 28e, Ch 15, p 892; Suryakantha, Ch 7, pp 67-68)
Delayed / chronic	Chronic bronchitis, bronchiectasis, emphysema, chronic obstructive pulmonary disease (COPD), bronchial asthma, lung cancer and respiratory allergies
Highly susceptible groups	Young children; the elderly; those with pre-existing lung disease; those with heart disease (Suryakantha, Ch 7, p 67)

Prevention and control of inversion-related pollution episodes

Nothing can be done to prevent the inversion itself; public health action therefore aims at keeping the pollutant load under the "lid" as low as possible, and at protecting the susceptible when an episode occurs.



Control framework for air pollution episodes occurring under a temperature inversion

Two additional levers matter in the Indian setting. **Vehicular measures** — unleaded petrol, catalytic converters, compressed natural gas for public transport (mandated for Delhi by the Supreme Court in April 2002), periodic servicing — cut the very pollutants that accumulate under a winter inversion (Sunder Lal, Ch 3, p 122). **Programme targets** — the National Clean Air Programme, launched in 2019, now seeks a 40% reduction in PM_{2.5} concentration by 2026, against Indian national ambient air quality standards of 2009 (annual PM_{2.5} 40 µg/m³, 24-hour 60 µg/m³) (Sunder Lal, Ch 3, pp 122-123). **International action** — WHO maintains an international network of laboratories for monitoring and study of air pollution, with centres including one at Nagpur, which issue warnings of air pollution where and when necessary (Park 28e, Ch 15, p 893).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Temperature–height profile: normal atmosphere versus inversion.** Two line graphs side by side. Common axes: x-axis "Air temperature (°C)", y-axis "Height above ground (m)". Labels — Graph A "NORMAL": line sloping so temperature falls with height; annotate "lapse rate $\approx 1.98^\circ\text{C per } 305\text{ m}$ ", "warm light air at surface", "convection currents (arrows curving upward)", "pollutants dispersed". Graph B "INVERSION": line showing temperature rising with height up to the inversion layer, then falling; annotate "warm air above = inversion layer / 'lid'", "top of inversion layer", "cool heavy air at surface", "no vertical mixing (crossed-out upward arrow)", "pollutants + fog trapped = SMOG in breathing zone".
- ▶ **Valley inversion, cross-section.** Draw a V-shaped valley with a town on the floor. Labels — "hill slopes (block horizontal wind)", "cold dense air draining down slopes", "pool of cold air on valley floor", "warm air layer above (inversion lid)", "smoke plumes from chimneys flattening and spreading sideways at the inversion base", "breathing zone".

High-yield exam pointers

- ▶ One-line definition to write first: *reversal of the normal vertical temperature gradient, so that cool air is trapped near the ground under warmer air and temperature rises with altitude.*
- ▶ **Smoke + water vapour (fog) = smog.** Winter/classical smog is reducing (smoke and sulphur dioxide); summer smog is photochemical and oxidising (ozone from nitrogen oxides plus volatile organic compounds under ultraviolet light).
- ▶ Normal environmental lapse rate $\approx 1.98^\circ\text{C per } 305\text{ m (1,000 ft)}$ in the troposphere; man is directly concerned with only the lowest 8–10 km of the atmosphere.
- ▶ Episode triad to quote with numbers: **Meuse Valley 1930 (63 deaths)**, **Donora 1948 (20 deaths)**, **London 1952 (>4,000 deaths)**; London led to the British Clean Air Act, 1956.
- ▶ Predisposing conditions mnemonic — **V-W-N-C**: Valley/low-lying, Winter, Night/early morning, Calm (still, windless) weather.
- ▶ Susceptible groups: young children, the elderly, and those with chronic lung or heart disease.
- ▶ WHO control quartet: **containment, replacement, dilution, legislation** (+ international action). India: Air (Prevention and Control of Pollution) Act, 1981; National Clean Air Programme 2019 targeting 40% PM_{2.5} reduction by 2026.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the Bhopal gas tragedy of 1984 is often wrongly quoted as an inversion episode — it was acute toxic air pollution from a methyl isocyanate leak, not smog produced by trapped ambient pollutants, and should not be listed with Meuse Valley, Donora and London. Indian national ambient air quality standards still in force date from 2009 and are stated to be under revision, so quote them with that year attached.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 15, Environment and Health, pp 888-893); Suryakantha — Community Medicine with Recent Advances (Ch 6, Air and Ventilation, pp 67-69); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e (Ch 5, Physical Environment: Air, p 64); AFMC Red Book of Community Medicine 2024 (Ch 11, pp 244-247; Ch 15, p 400); Sunder Lal — Textbook of Community Medicine (Ch 3, Environmental Health Problems, pp 119-124); Handbook of Epidemiology, Ahrens & Pigeot (Ch 24, Environmental Epidemiology, pp 954-955); Jugal Kishore — National Health Programmes of India (Ch 3, pp 941, 956-958); Air (Prevention and Control of Pollution) Act, 1981; National Clean Air Programme, 2019; Indian National Ambient Air Quality Standards, 2009.



Health implications of traditional urban planning

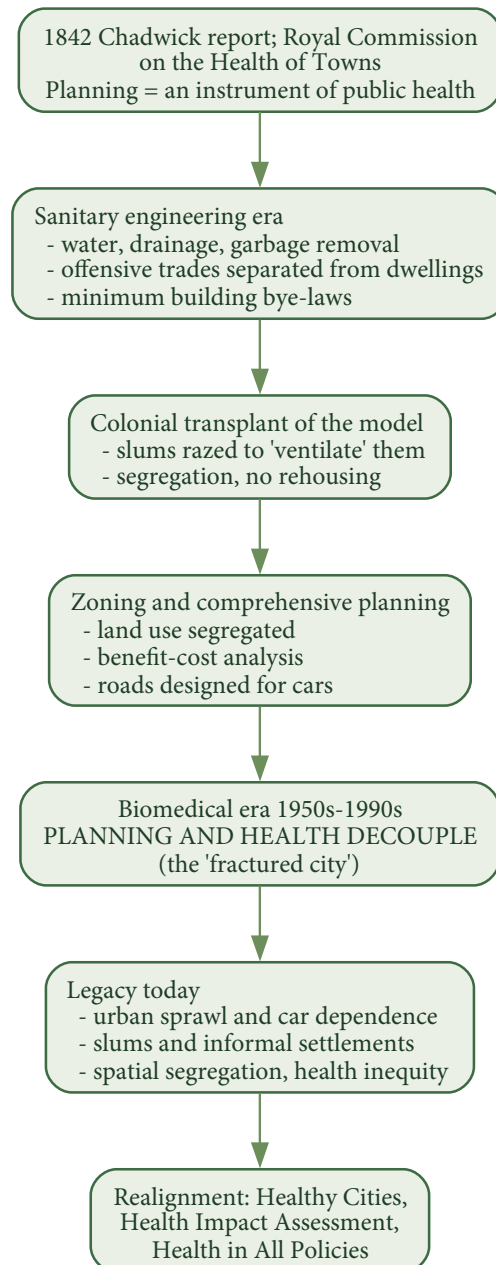
Asked in: Paper IV May 2018 — RGUHS MD Community Medicine

Definition — Traditional (conventional) urban planning is the nineteenth-century **sanitary-reform model of city building** — physically re-engineering the town to remove filth, provide water and drainage and so arrest epidemics — which subsequently narrowed into a separate technical sector concerned with setting strategic direction for cities, assessing and approving developments, land-use **zoning** and urban design, with aims no longer framed around health (Oxford Textbook of Public Health 7e, Ch 6, p 305). Its health implications are therefore double-edged: very large early gains against communicable disease, followed by a legacy of segregated land use, car dependence, slum formation and spatial segregation that today's *healthy urban planning* movement is trying to reverse.

Evolution of the planning–health relationship

Era	What traditional planning did	Health consequence
Sanitarian, 1840s–1900	<i>Villermé</i> (1820s) showed the wealthier the Parisian <i>arrondissement</i> , the lower the mortality; Chadwick's Report on the Sanitary Conditions of the Labouring Population in Great Britain, 1842 led the Royal Commission on the Health of Towns to conclude that physical changes to the city were necessary to improve the well-being of the poor	Miasma (filth/dirty air) was the ruling causal theory; sanitary commissions cleaned up urban environments and sanitary engineers piped waste away from cities into rivers and oceans — large falls in epidemic disease, but pollution merely displaced (Oxford 7e, Ch 58, pp 2154–2155)
Colonial planning	British sanitary engineers were sent to Singapore, Calcutta, Pune, Lagos and Cape Town to control outbreaks; indigenous practices were subjected to inspection, regulation and disciplinary action, including razing homes and markets to "ventilate" slums	Eviction without rehousing, segregation of indigenous populations from life-supporting resources, and no legal property rights — the historical root of today's informal settlements (Oxford 7e, Ch 58, pp 2156–2158)

Era	What traditional planning did	Health consequence
Zoning and comprehensive planning	Land-use zoning, benefit–cost analysis and long-range comprehensive planning radically altered urban governance	Residence separated from work and services; long commutes, car dependence, and exclusionary zoning that concentrates disadvantage (Oxford 7e, Ch 58, p 2156)
Biomedical era, 1950s–1990s	Public health shifted to the biomedical model and attributed disease to individuals; planning became a policy sector with its own narrow aims	The "fractured city" — planning and public health decoupled for nearly half a century (Oxford 7e, Ch 58, pp 2157–2158; Ch 6, p 305)
Present	Concepts such as "agglomeration" dominate; planning is driven by inter-city competition, local politics and perceived economic imperatives rather than health	Health and economic vitality are not in fact incompatible — well-planned, liveable, equitable cities are also more competitive economically (Oxford 7e, Ch 6, p 305)



Evolution of the planning–public health relationship, and where the traditional model broke

Health gains that traditional planning did deliver

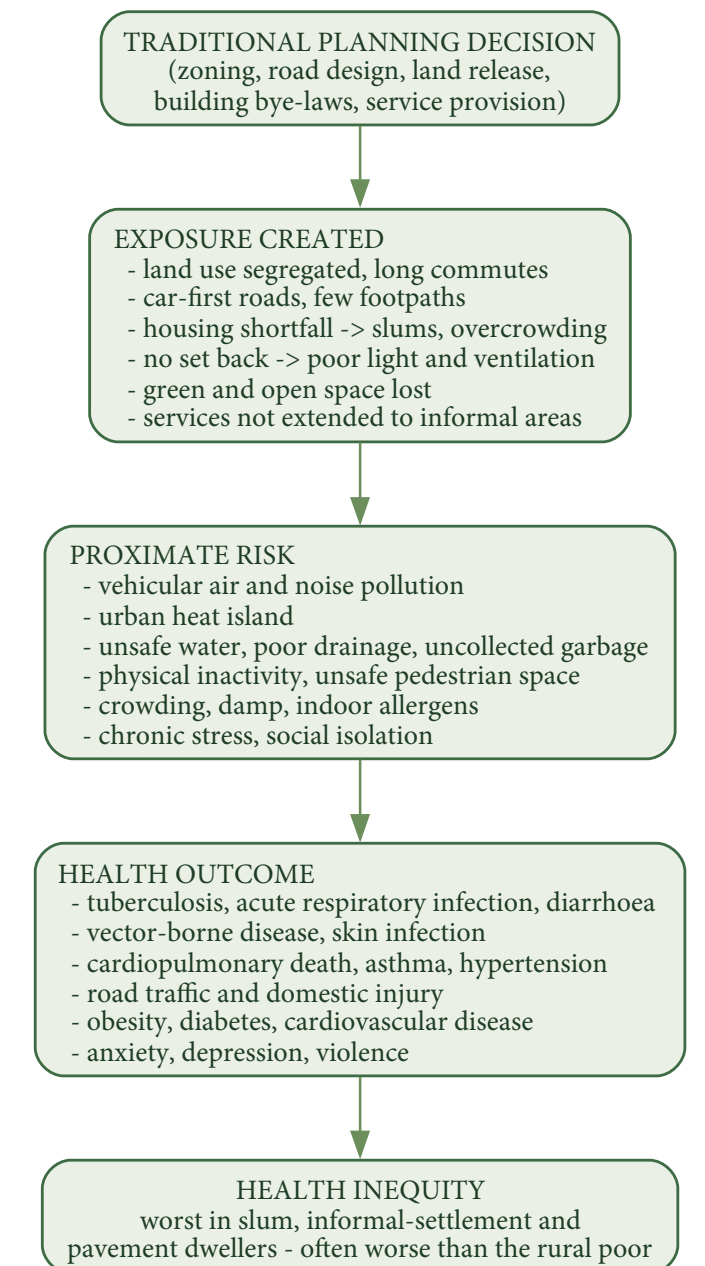
- ▶ **Control of faeco-oral and water-borne disease** through piped water supply, sewerage, drainage and municipal garbage collection — still the physical environments that matter most for urban health (Oxford 7e, Ch 58, pp 2159–2160).
- ▶ **Statutory minimum housing standards and building bye-laws**, giving enforceable floor space, ventilation, lighting and sanitation norms (Park 28e, Ch 15, pp 905–906).
- ▶ **Set-back and open-space rules**, preventing "back-to-back" houses and preserving light and ventilation (Park 28e, Ch 15, p 906).
- ▶ **Separation of offensive trades, noisy industry and cesspools from dwellings** — the legitimate public-health core of zoning (Suryakantha, Ch 10, p 83).

- **Routine census collection of housing statistics** (persons per room, cooking and sanitary facilities), enabling small-area analysis to identify neighbourhoods at high risk of crowding-related disease (Maxcy-Rosenau-Last 15e, Ch 51, p 948).

Adverse health implications, by domain

Domain	Planning failure	Health outcome
Air quality	Roads and highways designed for motor vehicles; industry and traffic not buffered from housing	Urban air pollution is linked to up to 1 million premature deaths a year ; over 90 per cent of city air pollution in developing cities is attributed to vehicle emissions from old, poorly maintained vehicles, inadequate infrastructure and low fuel quality. Urban particulate matter causes cardiovascular death and asthma ; in 2018 air pollution was the fifth leading risk factor for death globally and the third for India (mortality and disability-adjusted life years, DALYs, combined) (Oxford 7e, Ch 58, pp 2159–2160; Ch 4, p 238)
Noise	No acoustic zoning between traffic corridors, industry and dwellings	Hearing impairment, hypertension and ischaemic heart disease ; chronic noise harms sleep and temperament and can cause developmental delay in children (Oxford 7e, Ch 58, pp 2160–2162)
Heat and climate	Dense built-up area with limited green space and tree cover	Urban heat island — elevated temperature from increased heat production, reduced surface evaporation, altered thermal characteristics and rising air pollution, causing heat stroke and cardiovascular and respiratory mortality; the urban poor on precarious land are worst hit by flooding and extreme weather (Oxford 7e, Ch 4, p 238; Ch 58, pp 2161–2162)
Housing supply	Housing stock never matched migration; the UN expert body recommends 10 new houses per 1,000 persons per year	Homelessness (pavement, station and shelter dwellers), sub-standard houses and slums, dilapidated houses, and good houses in bad surroundings (Park 28e, Ch 15, p 908; Suryakantha, Ch 10, p 83)
Overcrowding	Floor-space, room and set-back norms unenforced in unplanned growth	Rapid spread of respiratory and skin infection; restricted movement, no privacy, impossible hygiene, disturbed rest and sleep, irritability, frustration and violence (Park 28e, Ch 15, p 907)
Transport and mobility	Streets built for cars, not pedestrians; poor or absent public transport	Pedestrian–motor-vehicle conflict is a leading cause of urban injury ; inadequate transport blocks the urban poor from employment and health-promoting services; highways drive vehicular pollution and suburban sprawl (Oxford 7e, Ch 58, p 2160)
Open space and activity	Highways and streets crowd out green open space; traditional transport planning asked only how to make roads safer for cars, not how "walkable" a city is	Physical inactivity and the resulting obesity, hypertension, diabetes and cardiovascular disease ; loss of a site for physical activity, social interaction and an urban heat sink (Oxford 7e, Ch 63, pp 2368, 2371; Ch 58, p 2160)
Food and basic services	Land use that does not site grocery stores, schools or clinics within reach	Poor diets; unaffordable rent forces trade-offs between housing, food and medical care (Oxford 7e, Ch 58, pp 2162–2163)

Domain	Planning failure	Health outcome
Social environment	Spatial segregation by race, ethnicity, caste or tribe, and residential segregation written into zoning	Disproportionate exposure and susceptibility to economic and social deprivation, toxic substances and hazardous conditions; concentrated disadvantage; fear of crime keeps children indoors, raising exposure to indoor air and allergens (Oxford 7e, Ch 58, pp 2161–2163)
Health equity	Services not extended to unlisted settlements	The nearly 1 billion people in slums, informal settlements or living as pavement dwellers have the worst health outcomes of all urban residents — frequently worse than populations living in rural poverty (Oxford 7e, Ch 58, pp 2151–2152)



Causal pathway from a traditional planning decision to a health outcome

The housing pathway — the standards traditional planning is judged against

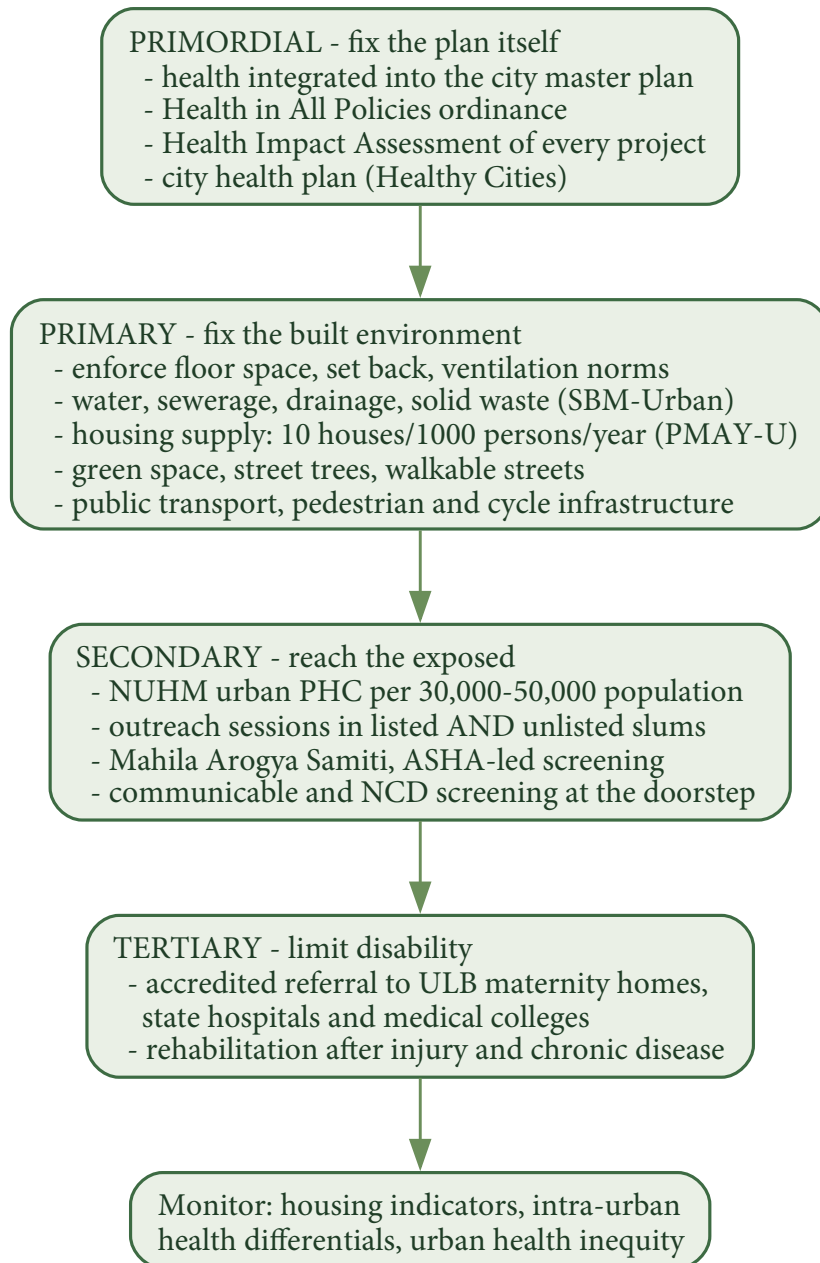
Element	Accepted standard
WHO criteria for healthful housing	Provides physical protection and shelter; provides adequately for cooking, eating, washing and excretory functions; is designed, constructed, maintained and used so as to prevent the spread of communicable diseases ; protects from hazards of exposure to noise and pollution; is free from unsafe physical arrangements and from toxic or harmful materials (Park 28e, Ch 15, pp 905–906)
Floor area of a living room	At least 120 sq ft (12 sq m) for occupancy by more than one person; at least 100 sq ft (10 sq m) for a single person (Park 28e, Ch 15, pp 905–906)
Rooms	Not less than two living rooms , at least one closable for security (Park 28e, Ch 15, pp 906–907)
Windows	At least 2 windows per living room, one opening directly onto open space; sill not more than 3 ft (1 m) above ground; window area 1/5th of floor area ; doors and windows combined 2/5th of floor area (Park 28e, Ch 15, p 906)
Set back	Open space all round the house; built-up area not more than one-third of the plot in rural areas , up to two-thirds in urban areas where land is costly (Park 28e, Ch 15, p 906)
Floor	<i>Pucca</i> , impermeable, smooth, crack-free and damp-proof; plinth 2–3 ft (0.6–1 m) high (Park 28e, Ch 15, p 906)
Overcrowding — floor space	≥110 sq ft = 2 persons; 90–100 sq ft = 1½ persons; 70–90 sq ft = 1 person; 50–70 sq ft = ½ person; under 50 sq ft = nil . A baby under 12 months is not counted; a child aged 1–10 years counts as half a unit (Park 28e, Ch 15, p 907)
Overcrowding — sex separation	Overcrowding exists if 2 persons over 9 years of age, not husband and wife, of opposite sexes are obliged to sleep in the same room (Park 28e, Ch 15, p 907)

Conditions with a demonstrated relation to sub-standard housing (Park 28e, Ch 15, p 907): (1) **respiratory infections** — common cold, tuberculosis, influenza, diphtheria, bronchitis, measles, whooping cough; (2) **skin infections** — scabies, ringworm, impetigo, leprosy; (3) **rat infestation** — plague; (4) **arthropods** — houseflies, mosquitoes, fleas and bugs; (5) **accidents**, a substantial proportion caused by a defect in the home and its environment; (6) **high morbidity and mortality**; (7) **psychosocial effects** — the sense of isolation felt in upper floors of high buildings and in densely populated urban areas, leading to neurosis and behaviour disorders.

The Indian expression, and the corrective response

Item	Current position
Urban housing stock	Census 2011 counted 330.84 million houses in all — 220.7 million rural and 110.14 million urban (Park 28e, Ch 15, p 908)
Slums	About 48.8 million persons lived in slums in 1990, ~40 per cent of them in million-plus cities ; slum population has since been growing at about 7 per cent a year (Park 28e, Ch 15, p 908; Suryakantha, Ch 36, p 799)
Urban population	35.7 crore in 2011, projected to 43.2 crore by 2021 (Suryakantha, Ch 36, p 799)

Item	Current position
National Urban Health Mission (NUHM), a sub-mission of the National Health Mission	Covers state capitals, district headquarters and other cities/towns of 50,000 population and above (Census 2011) in a phased manner; targets the poor in listed and unlisted slums and other vulnerable groups — street children, rickshaw pullers, construction, brick-kiln and lime-kiln workers, sex workers, the homeless and temporary migrants. Service norms: one Urban Primary Health Centre (U-PHC) for about 30,000–50,000 population , sited within or within half a kilometre of a slum and catering to a slum population of about 25,000–30,000 (up to 75,000 in very high-density areas); one auxiliary nurse midwife (ANM) per 10,000 population ; one Accredited Social Health Activist (ASHA) or link worker per 1,000–2,500 urban poor (about 200–500 households) ; <i>Mahila Arogya Samitis</i> for community participation (Park 28e, Ch 7, pp 535–536; Suryakantha, Ch 36, pp 799–803)
Urban Health and Wellness Centre (U-HWC)	One per 15,000–20,000 population , catering predominantly to poor and vulnerable populations in slums and similar pockets; staffed by a medical officer, staff nurse/pharmacist, male multipurpose worker and support staff (Park 28e, Ch 10, p 731)
Pradhan Mantri Awas Yojana –Urban (PMAY-U)	Launched 25 June 2015 by the Ministry of Housing and Urban Affairs to address urban housing shortage, principally for the Economically Weaker Section (Park 28e, Ch 7, p 575)
Swachh Bharat Mission–Urban	Elimination of open defecation ; eradication of manual scavenging ; modern and scientific municipal solid waste management (Park 28e, Ch 7, p 574)
Healthy Cities	Grew from the <i>Beyond Health Care</i> conference, Toronto 1984 ; the WHO Healthy Cities programme was created to re-invigorate the link between city planning and public health — cities draft a city health plan built on intersectoral partnership, community empowerment, a health information system, and tools to assess health effects (Oxford 7e, Ch 58, p 2159; Ch 99, pp 3770–3771)
Health in All Policies (HiAP)	Richmond, California became the first US city to pass a HiAP ordinance (2014), covering governance and leadership, economic development and education, safe communities, neighbourhood built environments, environmental health and justice, and accessible healthy homes and social services, with a racial-equity evaluation of every city project (Oxford 7e, Ch 58, pp 2169–2170)



Correcting the health legacy of traditional urban planning, by level of prevention

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Urban toxic stressors acting on the body** (as above, reproducible by hand). Draw a central human figure and arrow in eight stressors, each through a box marked *Stress*: on the left — *Social exclusion and segregation, Environmental pollution and toxic exposures, Lack of mobility, Threat of housing displacement*; on the right — *Discrimination, Lack of, poor quality or high cost basic services including food, Threats and fear of violence, Poor quality education*.
- **Indicators of housing**, as a three-limbed label diagram (Park 28e, Ch 15, p 907): (1) *Physical* — floor space, cubic space, room height, persons per room, rooms per dwelling, environmental quality (air, light, water, noise, sewage disposal); (2) *Economic* — cost of building, rental levels, taxes, expenditure on housing; (3) *Social* (UN inter-regional seminar on the Social Aspects of Housing, 1975) — (a) prevention of illness: frequency of illness from inadequate sewage and garbage collection, contaminated water source, insect-borne disease, overcrowding, accidents, proximity to animals, and access to a medical facility; (b) comfort: thermal, acoustic, visual, spatial; (c) mental health and social well-being: frequency of suicides, neglected and abandoned youth, drug and alcohol abuse in the neighbourhood.

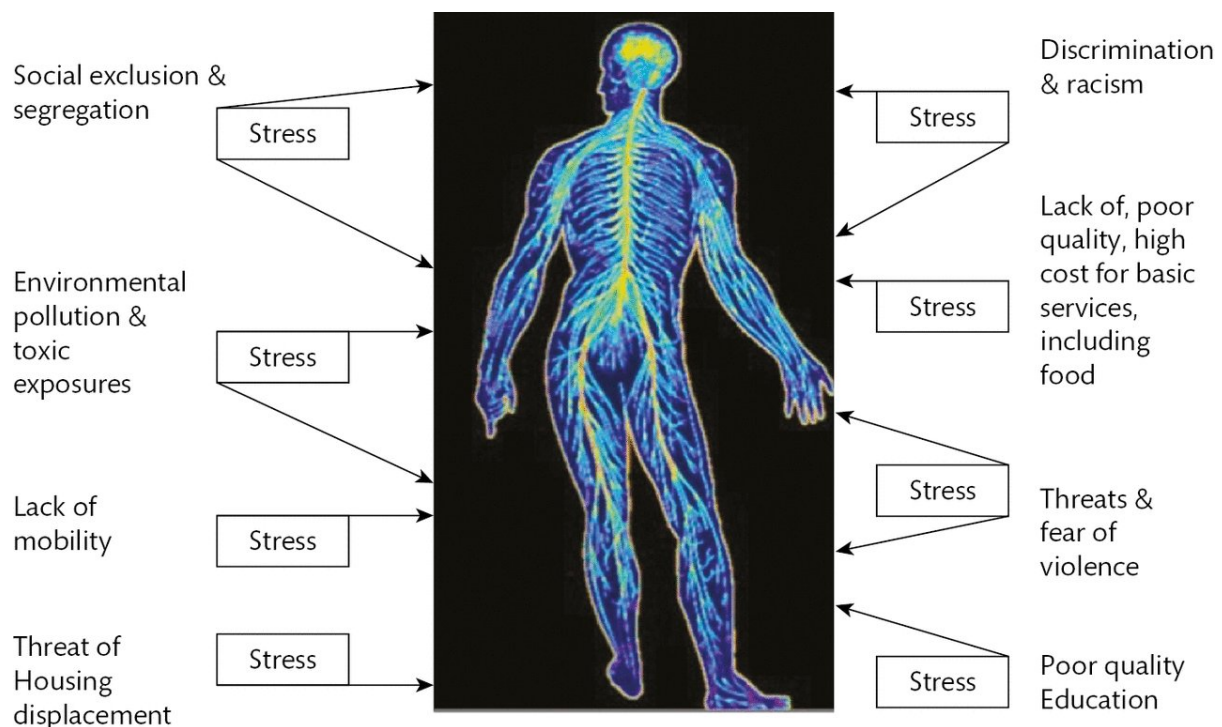


Fig 7.6 — 1 Urban toxic stressors that adversely impact human health. Corburn, J (Oxford public health 7th ed, p.2165)

High-yield exam pointers

- **The one-line thesis:** a century ago planning was an instrument of public health; it then became a separate sector concerned with zoning and urban design, and the health cost of that divorce is today's urban disease burden (Oxford 7e, Ch 6, p 305).

- **Four dated anchors:** Chadwick's Sanitary Report 1842; *Beyond Health Care* conference Toronto 1984 → WHO Healthy Cities; WHO/UN-Habitat *Hidden Cities* report 2010; Richmond, California HiAP ordinance 2014.
- **Overcrowding, both criteria:** floor space — ≥ 110 sq ft = 2 persons, 90–100 = $1\frac{1}{2}$, 70–90 = 1, 50–70 = $\frac{1}{2}$, < 50 = nil; sex separation — 2 persons **over 9 years**, not husband and wife, of opposite sexes sharing a room.
- **Housing norms to quote:** living room ≥ 120 sq ft for > 1 person, ≥ 100 sq ft for one; window area $\frac{1}{5}$ th and doors+windows $\frac{2}{5}$ th of floor area; built-up area $\leq \frac{1}{3}$ of plot rural, $\leq \frac{2}{3}$ urban; **10 new houses per 1,000 persons per year** (UN expert body).
- **Park's seven housing-linked conditions** (mnemonic R-S-R-A-A-M-P): Respiratory infections, Skin infections, Rat infestation, Arthropods, Accidents, Morbidity/mortality, Psychosocial effects.
- **Indian figures with their year:** 330.84 million houses — 110.14 million urban (Census 2011); urban population 35.7 crore (2011) → 43.2 crore (2021 projection); slum population growing ~ 7 per cent per year; NUHM covers towns of 50,000 population and above, one urban PHC per 30,000–50,000 (slum population 25,000–30,000), one urban health and wellness centre per 15,000–20,000.
- **The equity punchline:** slum and pavement dwellers have the worst health outcomes of all urban residents — **often worse than the rural poor.**

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written as textbook fact: the corpus quantifies India's slum burden from the 1990 estimate and Census 2011 and does not carry Census 2011's own slum-population figure or any later urban survey, so quote the years exactly as given; likewise, AMRUT and the Smart Cities Mission are not described in the retrieved chapters and are best mentioned only by name if the examiner invites them.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, Ministry of Housing and Urban Affairs (PMAY-U), National Center for Vector Borne Diseases Control, WHO heat-health action planning guidance, WHO Age-Friendly Cities and Communities framework

Programme and policy updates (India)

- **National Urban Health Mission — what a decade of it delivered.** NUHM began in May 2013 as a sub-mission of the NHM and runs in 779 cities; it put 5,025 functional urban PHCs in place over 2014–2022, and by September 2022 had 78,127 urban ASHAs and 80,852 Mahila Arogya Samitis, with slum mapping complete in 1,094 of 1,178 cities and vulnerable-group mapping in 826. Between 2014 and 2017–18 out-of-pocket spending for non-hospitalised care in urban government facilities fell 46 per cent (₹639 → ₹344) and institutional births in urban government facilities rose from 42.0 to 48.3 per cent (poorest quintile 54.0 → 62.1 per cent). The programme's own budget line stayed flat at about ₹950 crore in 2019–20 and 2020–21, which is the standing criticism to quote [PMID 36896811].

- **Housing policy has moved a phase on from the 2015 launch date.** Pradhan Mantri Awas Yojana –Urban is now in its second phase, PMAY-U 2.0, with a live application portal; the mission dashboard reports **172.72 lakh houses sanctioned and 99.25 lakh completed**, ₹2.12 lakh crore of central assistance committed and ₹1.81 lakh crore released, across the beneficiary-led construction, affordable housing in partnership, in-situ slum redevelopment and credit-linked subsidy verticals [MoHUA/PMAY-U 2026-Aug].

Current national data

- **Urban India, dated.** The urban share rose from **32.0 per cent (2013) to 35.8 per cent (2022)**, an average annual growth of about 2.09 per cent; the urban population is projected to approach **60 crore by 2036**, and the UN expects roughly **68 per cent of Indians to be urban by 2050**. Quote these in place of the older "43.2 crore by 2021" projection [PMID 40124872].
- **Air pollution measured against the standard, not in the abstract.** In 2019 more than **70 per cent of India's population** lived above the national annual PM2.5 standard of **40 µg/m³** and **56 per cent of households** still cooked on solid fuel; GBD 2019 attributed about **980,000 Indian deaths** to ambient PM2.5. Meeting the Indian standard alongside a halving of solid-fuel use would avert **more than 300,000 deaths a year**, rising to **500,000** at the WHO interim target of **25 µg/m³** and **675,000** at **15 µg/m³** — worth 2.2, 3.6 and 4.9 per cent of GDP respectively [PMID 41701846].
- **Heat and air pollution multiply each other — the modern form of the "fractured city" argument.** Across **ten Indian cities, 2008–2019 (about 3.6 million deaths)**, the PM2.5–mortality effect per **10 µg/m³** rose from **0.8 per cent (95% CI –0.3 to 1.9)** at the 75th temperature percentile to **4.6 per cent (2.9–6.5) at the 99th**; moving from the 75th to the 99th temperature percentile raised daily mortality **22.0 per cent (13.5–31.2)** [PMID 40245548].

Elimination, surveillance and disease status

- **The vector-borne programme has been renamed and restructured.** It now runs as the **National Center for Vector Borne Diseases Control (NCVBDC)** — not NVBDCP — covering malaria, dengue, lymphatic filariasis, kala-azar, Japanese encephalitis and chikungunya [NCVBDC 2026-Aug]. Its dengue arm is a direct verdict on urban form: across **Kolkata's 144 wards (2017–2022)** about **90 per cent of cases** fell in the post-monsoon August–November peak, lagging rainfall, temperature and humidity by **2–3 months**, with *Aedes aegypti* breeding indoors and peri-domestically and *Ae. albopictus* outdoors. Ward dengue rates tracked **built-up area (r = 0.48)** and rose with **under-construction buildings and vacant land used for dumping**, while the number of slums in a ward did not correlate — and wards above 20 cases per 10,000 grew from **six in 2017 (about 4 per cent of the city) to 52 in 2022 (about 34 per cent)** [PMID 40231455].
- **Urban malaria is a water-supply problem, not a rainfall problem.** *Anopheles stephensi* is an urban-adapted vector that breeds in artificial containers — drums, buckets, construction-site pits — and persists year-round including the dry season, so rainfall predicts transmission poorly; roughly **10–12 per cent of India's malaria** arises in urban areas. A **4,074-household survey in Ahmedabad and Surat (September–November 2023)** found only **13 per cent** of homes had continuous all-day water, about **75 per cent** of the remainder getting only two to three hours a day, while **96.2 per cent** stored water for non-drinking use. India's standing answer is the **Urban Malaria Scheme (1971)**, worked through **municipal byelaws** in towns with an annual parasitic

incidence above **2 per 1,000** and now running in **131 towns** under the National Framework for Malaria Elimination (target 2030) [PMID 41703587].

Guideline and legislative updates

- **Heat action plans, benchmarked against WHO guidance.** The WHO/Europe guidance sets **eight core elements** for a heat–health action plan: a lead coordinating body, accurate and timely alerts, a heat–health information plan, reduction of indoor heat exposure, particular care for vulnerable groups, preparedness of the health and social care system, **long-term urban planning**, and real-time surveillance and evaluation. Benchmarking **eight Indian city HAPs** against it found the weakest coverage in precisely the planning-dependent elements — care for vulnerable groups, health-system preparedness, long-term urban planning and surveillance of health outcomes — because heat is still framed as a time-limited disaster rather than a structural feature of the built city; within long-term planning, green and blue space and building design are covered but **land-use change** is least represented. India's first *State* heat response plan followed the **1998 Odisha heatwave**, and **Ahmedabad's 2013 HAP** was developed with IIPH Gandhinagar, PHFI and NRDC [PMID 41296585]. A distributed-lag mortality analysis of Ahmedabad (**2002–2018**) then tested it two ways: measured by **maximum temperature** the plan reduced immediate heat mortality across thresholds, but measured by the **Universal Thermal Climate Index (UTCI)** the benefit showed only in extreme heat stress — and on neither metric at longer lags or in moderate heat stress [PMID 40658206].

Global and WHO developments

- **Heat–health planning is now a global instrument, and India writes most of the Asian examples.** Formal plans were first developed at scale after the **2003 European heatwave (more than 35,000 deaths)**; WHO/Europe published the first comprehensive guide in 2008, updated it in 2011, and a further update was expected in 2025. A review of **83 plans from 24 countries covering about 2.2 billion people (26 per cent of the world)** found 75.9 per cent mentioned mental health but only **21.7 per cent** carried targeted mental-health interventions, and **India alone supplied 35 of the 37 South-East Asian plans** — outside India, low- and middle-income countries were markedly under-represented [PMID 40418305].
- **The WHO Age-Friendly Cities and Communities checklist gives the pedestrian critique a measurable form.** Applied to 27 locations and 214 public amenities in Bengaluru Urban, it classed **85.2 per cent of locations as not age-friendly**: 85.2 per cent lacked wheelchair-accessible pavements and **96.3 per cent had no safe pedestrian crossing**, with specialised transport for disabled persons at a single site (3.7 per cent) [PMID 42378383].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Chapter 15 (Environment and Health), pp 905–908, Chapter 7 (Health Programmes in India), pp 535–536, 574–575, Chapter 10 (Health Care of the Community), p 731; Oxford Textbook of Public Health 7e — Chapter 58 (Urbanization and health), pp 2151–2172 including Table 7.6.1 and Fig 7.6, Chapter 6 (Determinants of health: overview), pp 304–306, Chapter 4 (Public health priorities in countries undergoing economic transition), pp 237–238, Chapter 63 (Physical activity and public health), pp 2368–2371, Chapter 99 (Practice of environmental health in an era of sustainable development), pp 3770–3771; Suryakantha — Community Medicine with Recent Advances, Chapter 10 (Housing), pp 82–83 and Chapter 36 (National Urban Health Mission), pp 799–803; Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e — Chapter 51 (Housing and Health), pp 947–948. Named documents: WHO/UN-Habitat *Hidden Cities* (2010); UN inter-regional seminar on the Social Aspects of Housing (1975); Census of India 2011. Coverage note: the indexed corpus has no chapter devoted to town planning as such, so the answer is built from the urbanization-and-health, housing, and determinants chapters listed above.

Recent-advances sources: PMID 36896811 — Core interventions contributing to the effectiveness of the National Urban Health Mission in India, *J Glob Health* 2023, doi:[10.7189/jogh.13.03009](https://doi.org/10.7189/jogh.13.03009); PMID 40124872 — Urban health resilience: strategies for strengthening public health systems in response to urbanization challenges, *Indian J Community Med* 2024, doi:[10.4103/ijcm.ijcm_755_24](https://doi.org/10.4103/ijcm.ijcm_755_24); PMID 41701846 — The mortality and economic benefits of achieving air pollution standards in India, *Proc Natl Acad Sci U S A* 2026, doi:[10.1073/pnas.2522228123](https://doi.org/10.1073/pnas.2522228123); PMID 40245548 — Synergistic associations of ambient air pollution and heat on daily mortality in India, *Environ Int* 2025, doi:[10.1016/j.envint.2025.109426](https://doi.org/10.1016/j.envint.2025.109426) (abstract-only, no PMC record); PMID 40231455 — Spatio-temporal distribution of dengue cases and vectors along with the interrelationship of environmental and climatic factors in the metropolitan city, Kolkata, India, for 2017–2022, *Emerg Microbes Infect* 2025, doi:[10.1080/22221751.2025.2493924](https://doi.org/10.1080/22221751.2025.2493924); PMID 41703587 — Socioeconomic and household water management determinants of malaria and other vector-borne disease prevention in urban Gujarat, India, *Malar J* 2026, doi:[10.1186/s12936-026-05830-2](https://doi.org/10.1186/s12936-026-05830-2); PMID 41296585 — Heat action plans in eight Indian cities: knowledge gaps and opportunities for intersectoral heat governance, *Indian J Med Res* 2025, doi:[10.25259/IJMR_1019_2025](https://doi.org/10.25259/IJMR_1019_2025); PMID 40658206 — Comparison of biometeorological index and temperature linked mortality: a case study from Ahmedabad city, India, *Int J Biometeorol* 2025, doi:[10.1007/s00484-025-02959-8](https://doi.org/10.1007/s00484-025-02959-8) (abstract-only, no PMC record; the pre-/post-HAP result is quoted exactly as the abstract states it for both Tmax and UTCI); PMID 40418305 — A critical gap in addressing mental health in heat-health action plans worldwide, *Curr Environ Health Rep* 2025, doi:[10.1007/s40572-025-00486-7](https://doi.org/10.1007/s40572-025-00486-7); PMID 42378383 — A transdisciplinary assessment of age-friendly cities and communities in Bengaluru Urban, India: towards a healthy aging framework, *Health Care Manage Rev* 2026, doi:[10.1097/HMR.0000000000000485](https://doi.org/10.1097/HMR.0000000000000485) (abstract-only, no PMC record). Non-journal sources: Pradhan Mantri Awas Yojana–Urban mission dashboard, Ministry of Housing and Urban Affairs (August 2026); National Center for Vector Borne Diseases Control, Ministry of Health and

Family Welfare (August 2026).

♦ ♦ ♦

Indices of thermal comfort in comfort zone

Asked in: Paper I May 2018 — RGUHS MD Community Medicine

Definition — **Thermal comfort** exists when a person maintains a normal balance between heat production and heat loss at normal body temperature and without sweating (Park 28e, Ch 15, p 887). Since no single instrument reading captures this, an **index of thermal comfort** is an arbitrary scale that combines two or more of the four physical factors of air — temperature, humidity, air movement and radiant heat — into one number. The **comfort zone** is the range of effective temperatures over which the majority of adults feel comfortable (Park 28e, Ch 15, p 887).

Physical factors of air, and the instrument that reads each

Discomfort in a hot or crowded room is due to physical, not chemical, change in the air; these four factors together determine the **cooling power** of air on the body — "the problems of ventilation are physical not chemical; cutaneous not respiratory" (Professor Lee, quoted in Park 28e, Ch 15, pp 886-887).

Factor	Instrument	Key specification / remark
Air temperature	Dry bulb thermometer (mercury or alcohol); silvered thermometer reflects incident radiant heat and so gives a truer air temperature	Air temperature alone is never an adequate index of thermal comfort (Park 28e, Ch 15, pp 886-887, 902)
Humidity	Wet bulb thermometer; sling (whirling) psychrometer — paired wet and dry mercury thermometers on a rotating frame	Air must pass the wet bulb at about 800 ft/min (5 m/sec) for accuracy; relative humidity is then read off a psychrometric chart (Park 28e, Ch 15, p 904)
Air movement (velocity)	Kata thermometer (Hill) — alcohol thermometer, bulb 4 cm long × 1.8 cm wide, stem marked 100°F to 95°F, silvered bulb to reduce radiation error	Three types by cooling range and alcohol colour: standard 100–95°F (red), high temperature 130–125°F (dark blue), extra high 150–145°F (magenta); measures velocities too low for an anemometer (Park 28e, Ch 15, p 902; Suryakantha, Ch 12, p 93)
Mean radiant heat	Globe thermometer — hollow copper globe 15 cm (6 inches) across, matt-black outside, mercury thermometer bulb at its centre	Equilibrium in 15–20 minutes; globe temperature minus dry bulb temperature = a measure of radiant heat (Park 28e, Ch 15, pp 901-902)
All four combined	Wet globe thermometer — dial thermometer inside a blackened copper sphere fully covered with wet black cloth	Exchanges heat by conduction, convection, evaporation and radiation as a perspiring man does, so its equilibrium temperature measures the cooling capacity of the work environment (Park 28e, Ch 15, p 902)

The classical sequence of indices of thermal comfort

- ▶ **Air temperature alone** — used for a long time, but found inadequate because it ignores humidity, air movement and radiation (Park 28e, Ch 15, pp 886-887).
- ▶ **Air temperature with humidity** — the two taken together; still unsatisfactory.
- ▶ **Cooling power of air** — temperature, humidity and air movement expressed together as the "cooling power", measured by *Hill's* Kata thermometer. A **dry Kata reading of 6 and above** and a **wet Kata reading of 20 and above** were regarded as indices of thermal comfort; Kata cooling powers are now known **not** to be reliable comfort indices, though the wet Kata remains valuable in the tropics — for example in cotton mills — where sweating occurs (Park 28e, Ch 15, p 887; Suryakantha, Ch 12, p 93).
- ▶ **Effective temperature (ET) and 5. Corrected effective temperature (CET)** — the two scales at present widely used as indices of thermal comfort (Park 28e, Ch 15, p 887), compared below.

Feature	Effective temperature (ET)	Corrected effective temperature (CET)
Definition	The temperature of still, saturated air that would induce the same sensation of warmth or cold as that experienced under the conditions observed	The same definition, but the globe thermometer reading replaces the dry bulb so that radiant heat is allowed for
Factors combined	Three — air temperature, humidity, air movement	Four — air temperature, air velocity, humidity, mean radiant heat
Readings required	Dry bulb + wet bulb + air velocity	Globe + wet bulb + air velocity (velocity from the Kata thermometer)
Origin	Evolved in 1923 by <i>Houghton</i> and <i>Yaglou</i> at the Pittsburgh Laboratory of the American Society of Heating and Ventilation Engineers	An improvement over the ET index
Two clothing scales	<i>Basic</i> scale — men stripped to the waist; <i>normal</i> scale — men fully clad in indoor clothing	The same basic and normal scales are provided (AFMC Red Book 2024, Ch 6, pp 159-160)
Read from	Special charts	Prepared nomograms , entered with globe temperature, wet bulb temperature and air speed
Limitation / use	Criticised because it ignores radiation from surrounding structures	Preferred whenever a source of radiation is present (Park 28e, Ch 15, p 887)

Worked meaning: an environment whose ET is 30°C (86°F) feels subjectively the same as a saturated atmosphere at 30°C with no air movement (Park 28e, Ch 15, p 887).

Other indices of thermal comfort and heat stress

Heat stress is the burden or load of heat that must be dissipated if the body is to remain in thermal equilibrium; it is influenced by metabolic rate, air temperature, humidity, air movement and radiant temperature (Park 28e, Ch 15, p 902). Expressed as a heat balance, $S = M + C + R - E$ (metabolic heat, convection, radiation, evaporation), which should equal zero in thermal equilibrium (Suryakantha, Ch 12, p 90).

Index	What it combines / means	Interpretation
Kata cooling power	Rate of cooling in millicalories per sq cm per second = Kata factor ÷ average cooling time	Dry Kata ≥ 6 , wet Kata ≥ 20 taken as comfort; unreliable as a comfort index (Park 28e, Ch 15, pp 887, 902)
Equatorial comfort index (ECI)	The temperature of still, saturated air that is physiologically equivalent to the climate under consideration	A direct thermal-comfort equivalent scale (Park 28e, Ch 15, p 902)
Heat stress index (HSI)	Metabolic rate plus the principal channels of heat exchange between body and environment; read from nomograms; expressed as the percentage of the heat storage capacity of an average man	0 = no thermal stress; 10–30 = mild to moderate heat strain; 40–60 = severe; 70–90 = very severe; 100 = upper limit of heat tolerance (Park 28e, Ch 15, p 902)
Predicted four-hour sweat rate (P4SR) — <i>McArdle</i>	Sweat predicted over 4 hours from any combination of dry and wet bulb temperature, mean radiant temperature, air velocity and work intensity	1–3 L = comfort zone ; 3–4.5 L just tolerable ; >4.5 L intolerable ; 2.5 L in 4 h is optimal for a working man; 4.5 L in 4 h is the maximum allowable sweat rate for acclimatised healthy young men; applicable only where sweating occurs (Park 28e, Ch 15, pp 887, 902)
Oxford index (weighted dry–wet bulb, WD)	$WD = 0.15 d + 0.85 W$, where d = dry bulb and W = wet bulb temperature	Valued for simplicity and ready applicability, and useful where CET and P4SR fail because both dry and wet bulb readings are high (AFMC Red Book 2024, Ch 6, p 162)
Wet bulb globe temperature (WBGT)	Indoor: $WBGT = 0.7 WB + 0.3 GT$; outdoor: $WBGT = 0.7 WB + 0.2 GT + 0.1 DB$ (natural wet bulb, globe, dry bulb, all °C)	The most widely accepted index for assessing heat in industry (ISO 7243, BS EN 27243, ACGIH threshold limit values); reference values are set to keep deep body temperature $\leq 38^\circ\text{C}$ (Oxford Handbook of Occupational Health 2e, Ch 2, pp 60–63; Hunter's, Ch 49, p 559)
Predicted mean vote / predicted percentage dissatisfied (PMV/PPD)	Analytical calculation of thermal comfort rather than heat stress	The current international thermal-comfort standard, ISO 7730 (Hunter's, Ch 49, p 569)

Indices measure the **environment**; the corresponding **indicators of heat strain in the person** are heart rate, body temperature and the amount of sweat produced (Suryakantha, Ch 12, p 90).

The comfort zone

Comfort is a complex subjective experience depending not only on physical and physiological but also on psychological factors, so there is **no single zone of comfort valid for all people** (Park 28e, Ch 15, p 887). An individual's comfort zone shifts with the environment, degree of physical activity, racial habit, type of clothing, state of acclimatisation and season (AFMC Red Book 2024, Ch 6, p 162).

Criteria of the comfort zone (the standard industrial definition — the zone in which a worker feels comfortable while working):

Criterion	Value for the comfort zone
Corrected effective temperature (CET)	25–27°C (77–80°F)
Relative humidity (RH)	30–65 per cent
Dry Kata reading	6 and above
Wet Kata reading	20 and above
Predicted four-hour sweat rate (P4SR)	1–3 litres (average 2.5 L); applies only where sweating occurs

(Suryakantha, Ch 7, pp 65-67 and Ch 12, p 93; Ch 18, p 249)

Comfort zones evaluated in India, by corrected effective temperature (Park 28e, Ch 15, p 887):

Sensation	CET (°C)
Pleasant and cool	20
Comfortable and cool	20–25
Comfortable (the comfort zone)	25–27
Hot and uncomfortable	27–28
Extremely hot	28 +
Intolerably hot	30 +

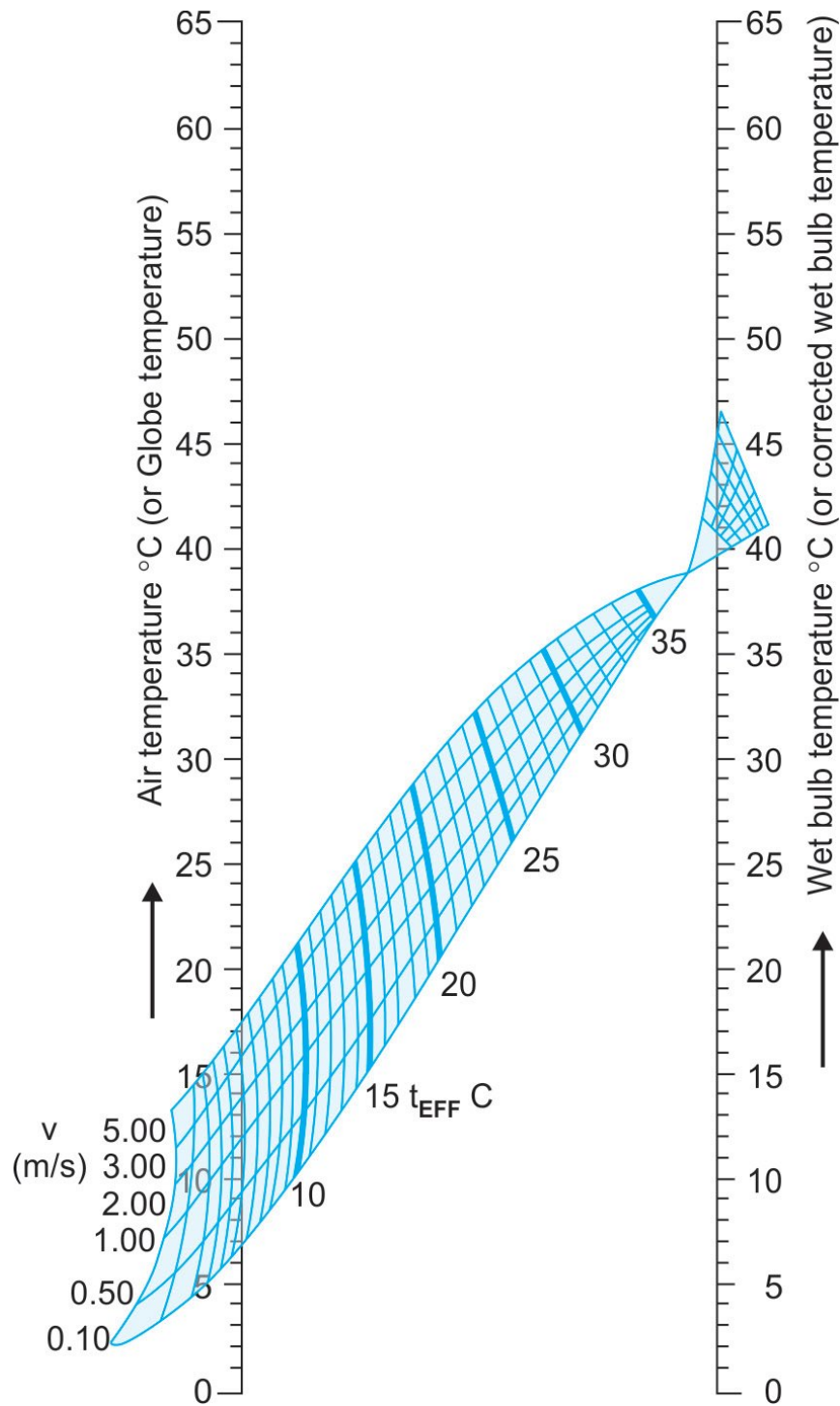
Comfort ranges quoted from other sources, for comparison (AFMC Red Book 2024, Ch 6, p 162):

Setting	Range
Temperate zones	ET 64–69°F
American Society for Heating, Refrigerating and Air-conditioning Engineers (ASHRAE)	ET 60–73°F
Indian hot weather, humidity below 75% (Defence Research Laboratories data)	Upper limit of comfort 76–80°F CET; 70–76°F CET feels comfortable and pleasant (lower limit)
Below the zone	CET below 72°F is cool and stimulating; below 69°F a definite feeling of cold in cotton clothing
Winter, woollen clothing	57–63°F ET in other countries; in India the upper limit in service battle dress is about 68°F CET, feeling warm above 70°F CET

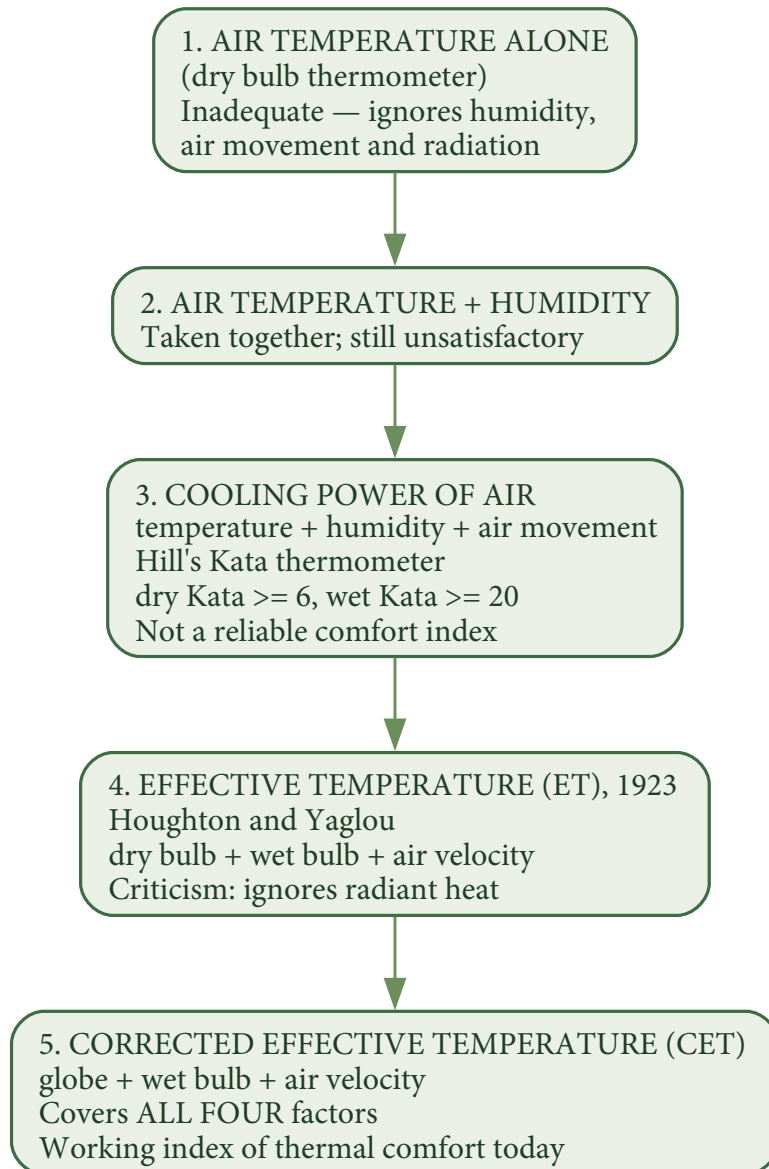
DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Labels to reproduce on the CET nomogram:

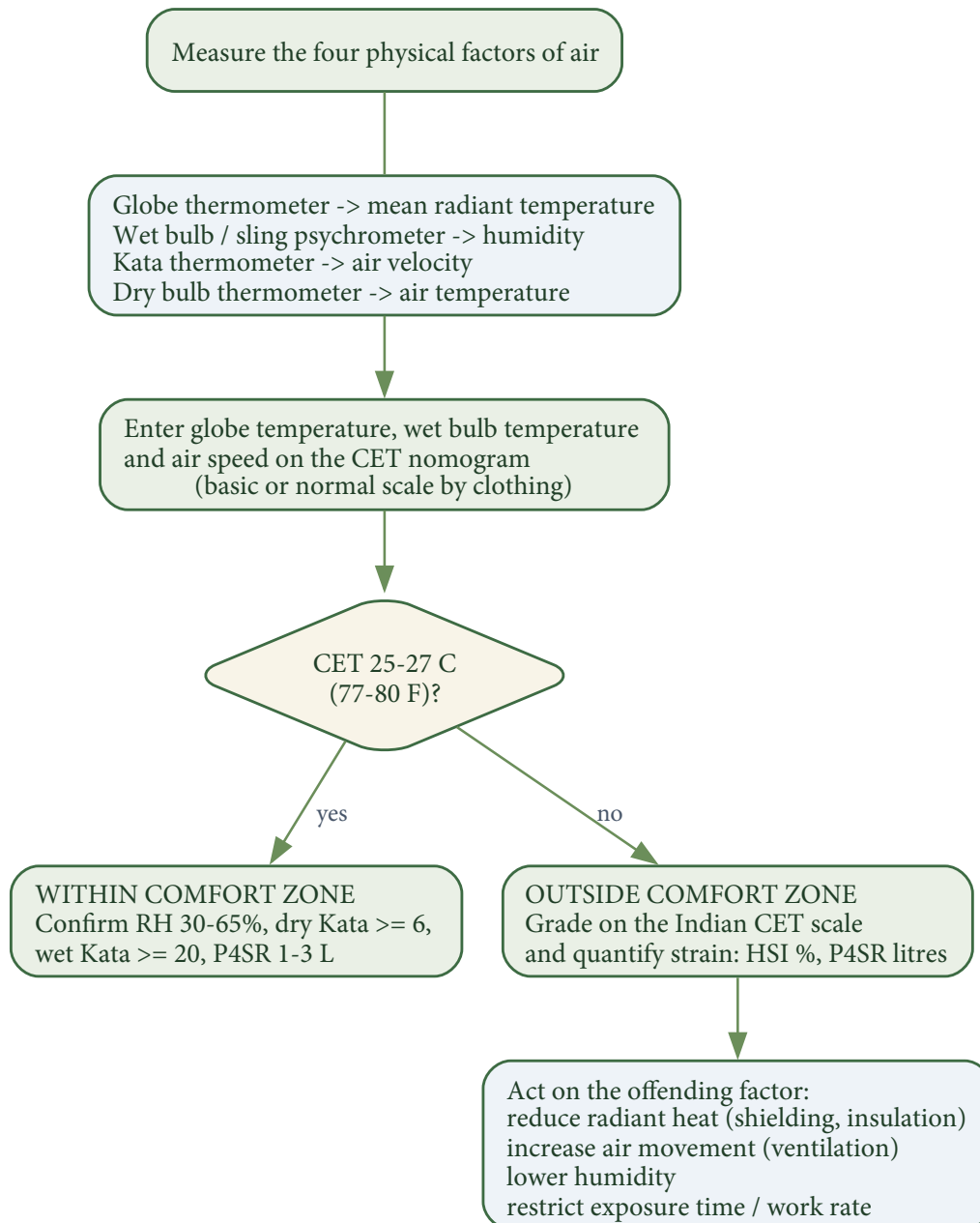
- ▶ Left vertical axis — *Air temperature °C (or globe temperature)*, scale 0 to 65
- ▶ Right vertical axis — *Wet bulb temperature °C (or corrected wet bulb temperature)*, scale 0 to 65
- ▶ Family of oblique curves inside the grid — *air velocity v (m/sec): 0.10, 0.50, 1.00, 2.00, 3.00, 5.00*
- ▶ Heavy diagonal lines crossing the grid — *effective temperature / corrected effective temperature ($t\text{-EFF}$ °C): 10, 15, 20, 25, 30, 35*
- ▶ Shade the band CET 25–27°C and mark it "*comfort zone*"



Effective temperature / corrected effective temperature nomogram for lightly clothed men (Suryakantha — Community Medicine with Recent Advances, Ch 7 (Chapter-06_Air and Ventilation), p 66)



Evolution of the indices of thermal comfort — from one variable to all four physical factors of air



Applying the indices — from instrument readings to a verdict on the comfort zone

High-yield exam pointers

- **Four physical factors** = air temperature, humidity, air movement, mean radiant heat — mnemonic *"THAR"*; ET covers the first three, CET covers all four.
- ET = temperature of still saturated air giving the same sensation of warmth/cold; *Houghton and Yaglou, 1923*, Pittsburgh Laboratory, ASHVE. CET = ET with the globe thermometer reading substituted for the dry bulb.
- **Comfort zone criteria (write all five)**: CET 25–27°C (77–80°F); RH 30–65%; dry Kata ≥6; wet Kata ≥20; P4SR 1–3 L (average 2.5 L).
- **Indian CET bands**: 20 pleasant and cool · 20–25 comfortable and cool · 25–27 comfortable · 27–28 hot and uncomfortable · 28+ extremely hot · 30+ intolerably hot.

- ▶ **P4SR:** 1–3 L comfort · 3–4.5 L just tolerable · >4.5 L intolerable; 4.5 L in 4 h = *McArdle's* maximum allowable sweat rate.
- ▶ **HSI:** 0 none · 10–30 mild-moderate · 40–60 severe · 70–90 very severe · 100 = **upper limit of heat tolerance**.
- ▶ **WBGT formulas:** indoor $0.7 \text{ WB} + 0.3 \text{ GT}$; outdoor $0.7 \text{ WB} + 0.2 \text{ GT} + 0.1 \text{ DB}$. **Oxford index** $\text{WD} = 0.15 \text{ d} + 0.85 \text{ W}$.
- ▶ **Globe thermometer minus dry bulb = radiant heat;** Kata thermometer measures air velocity too low for an anemometer.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: Park's textbook stops at effective temperature, corrected effective temperature, the equatorial comfort index, the heat stress index and the predicted four-hour sweat rate, while occupational-health practice has moved to the wet bulb globe temperature index (International Organization for Standardization standard ISO 7243) for heat stress and to the predicted mean vote / predicted percentage dissatisfied model (ISO 7730) for thermal comfort, so quote the Park indices first and add the wet bulb globe temperature index as the current industrial standard.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 15, Environment and Health, pp 886-887, 901-904); Suryakantha — Community Medicine with Recent Advances (Ch 7, Air and Ventilation, pp 65-67; Ch 12, Meteorology, pp 90-93; Ch 18, Occupational Health, p 249); AFMC Red Book of Community Medicine 2024 (Ch 6, pp 159-162); Oxford Handbook of Occupational Health 2e (Thermal environment, pp 59-63); Hunter's Diseases of Occupations (Ch 49, Heat and Cold, pp 559-569).



Indices of thermal comfort

Asked in: Paper I 21-Jun-2024 · Paper I 19-Nov-2020 (2×) — RGUHS MD Community Medicine

Definition — Considering only environmental factors, comfortable thermal conditions are those under which a person can maintain a normal balance between production and loss of heat, at normal body temperature and without sweating (Park 28e, Ch 15, p 887). Because no single physical measurement captures this, several **indices of thermal comfort** have been devised, each combining air temperature, humidity, air movement and radiant heat into a single value. The **comfort zone** is the range of effective temperatures over which the majority of adults feel comfortable; there is no unanimous single zone, since comfort is a subjective experience depending on physical, physiological and psychological factors (Park 28e, Ch 15, p 887).

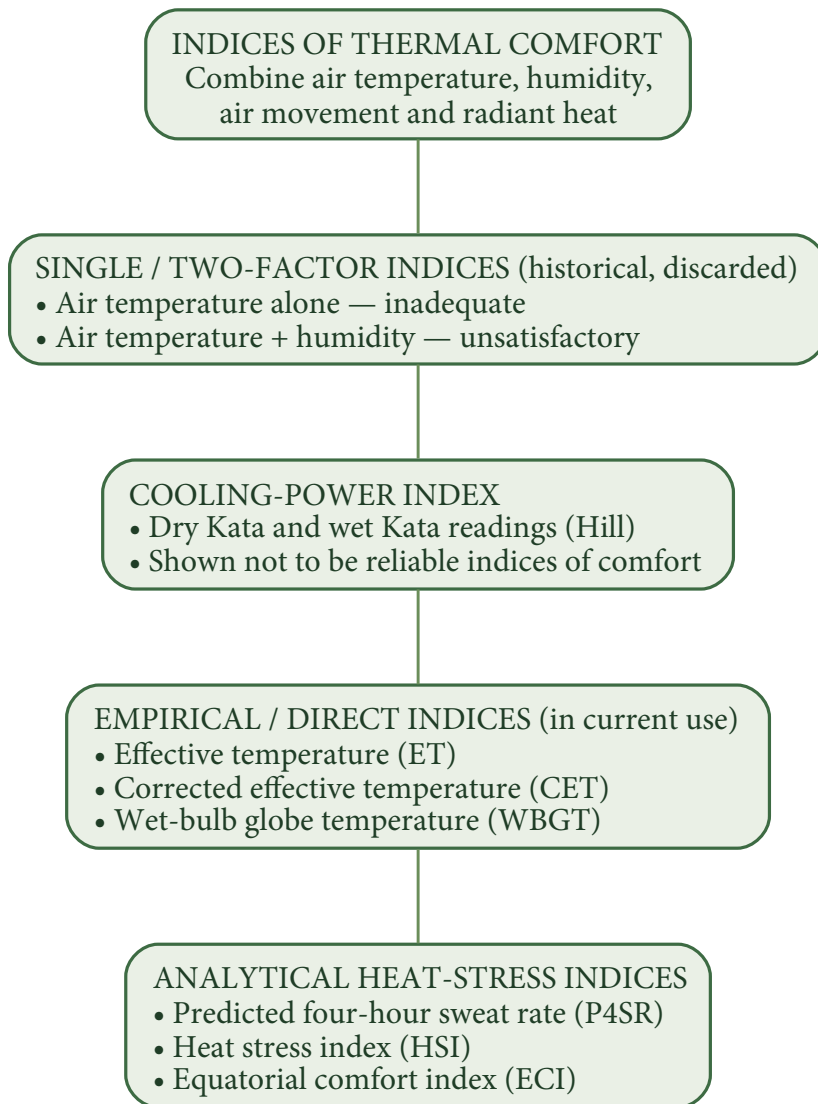
Factors determining thermal comfort

The causes of thermal discomfort are physical, not chemical — temperature, humidity, air movement and heat radiation, which together determine the **cooling power** of the air on the human body. As Professor Lee put it, *"the problems of ventilation are physical not chemical; cutaneous not respiratory"* (Park 28e, Ch 15, pp 886–887). Every index below is simply a different way of combining these same four factors.

Physical factor	Route of heat exchange it governs	Measured by
Air temperature	Convection and conduction	Dry bulb thermometer
Humidity	Evaporation of sweat	Wet bulb thermometer / psychrometer
Air movement (velocity)	Convection and evaporation	Anemometer; Kata thermometer at low velocities
Mean radiant heat	Radiation from surrounding surfaces	Globe thermometer

Heat balance basis — heat stress (S) = metabolic heat (M) + convection (C) + radiation (R) – evaporation (E); S must equal zero for the body to remain in thermal equilibrium (Suryakantha, Ch 12, p 90). *Heat stress* is the load of heat that must be dissipated to maintain that equilibrium; the **indicators of heat strain** in the exposed worker are heart rate, body temperature and the amount of sweat produced.

Evolution and classification of the indices



Classification of the indices of thermal comfort and heat stress

The indices compared

Index	Factors combined	How obtained	Comfort criterion / limit	Chief limitation
Air temperature	1 (temperature)	Dry bulb thermometer	—	Ignores humidity, movement and radiation; inadequate alone
Air temperature + humidity	2	Dry and wet bulb	—	Still unsatisfactory as an index

Index	Factors combined	How obtained	Comfort criterion / limit	Chief limitation
Cooling power (Kata)	Temperature, humidity, air movement	Kata thermometer, devised by <i>Hill</i>	Dry Kata 6 and above; wet Kata 20 and above	Kata cooling powers are not reliable indices of comfort; now used mainly as an anemometer
Effective temperature (ET)	Temperature, humidity, air movement	Nomogram of <i>Houghton and Yaglou</i> , 1923	Read off the ET scale	Ignores radiation from surrounding structures
Corrected effective temperature (CET)	All four, including mean radiant heat	Nomogram using globe temperature, wet bulb and air speed	25–27 °C (77–80 °F)	Needs a nomogram; comfort still subjectively defined
Wet-bulb globe temperature (WBGT)	Natural wet bulb, globe, dry bulb	Indoors: 0.7 WB + 0.3 GT; outdoors: 0.7 WB + 0.2 GT + 0.1 DB	Reference values set to keep deep body temperature ≤ 38 °C	A screening index; needs metabolic-rate and acclimatisation tables
Predicted four-hour sweat rate (P4SR)	Dry and wet bulb, mean radiant temperature, air velocity, work intensity	<i>McArdle's</i> chart	Comfort 1–3 L/4 h; 2.5 L optimal; 4.5 L upper limit of tolerance	Applicable only where sweating actually occurs
Heat stress index (HSI)	Metabolic rate + the principal channels of heat exchange	Nomogram; expressed as % of the heat storage capacity of an average man	0–100 scale (see below)	No index is valid across all work rates and air conditions
Equatorial comfort index (ECI)	Whole climate expressed as one temperature	Denotes the temperature of still, saturated air physiologically equivalent to the climate under consideration	—	Descriptive; little used in practice

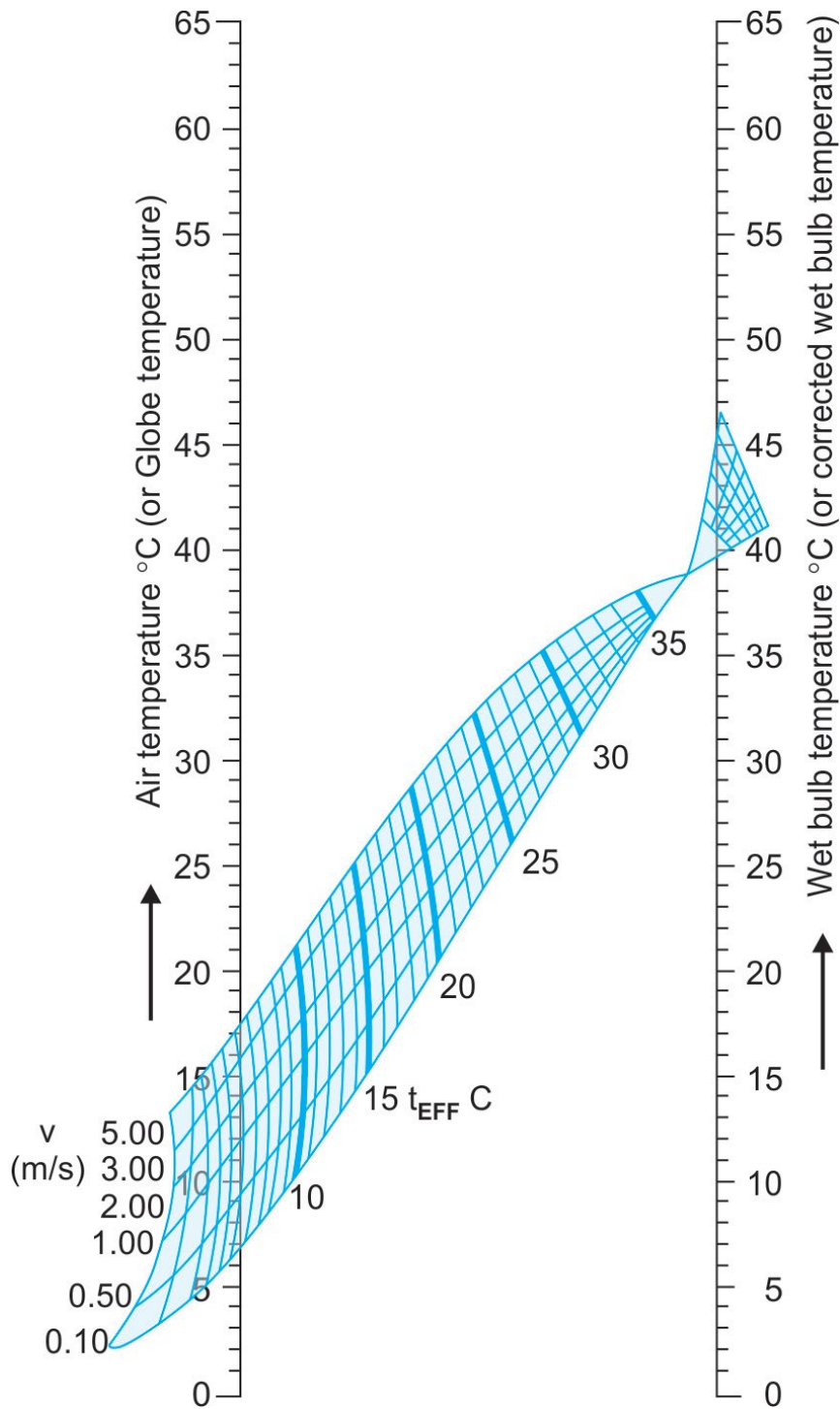
(*Park* 28e, Ch 15, pp 887, 902; *Suryakantha*, Ch 7, p 67 and Ch 12, p 90; *Hunter's Diseases of Occupations*, Ch 49, p 559; *Oxford Handbook of Occupational Health* 2e, Ch 1, p 63)

Effective temperature and corrected effective temperature — the working indices

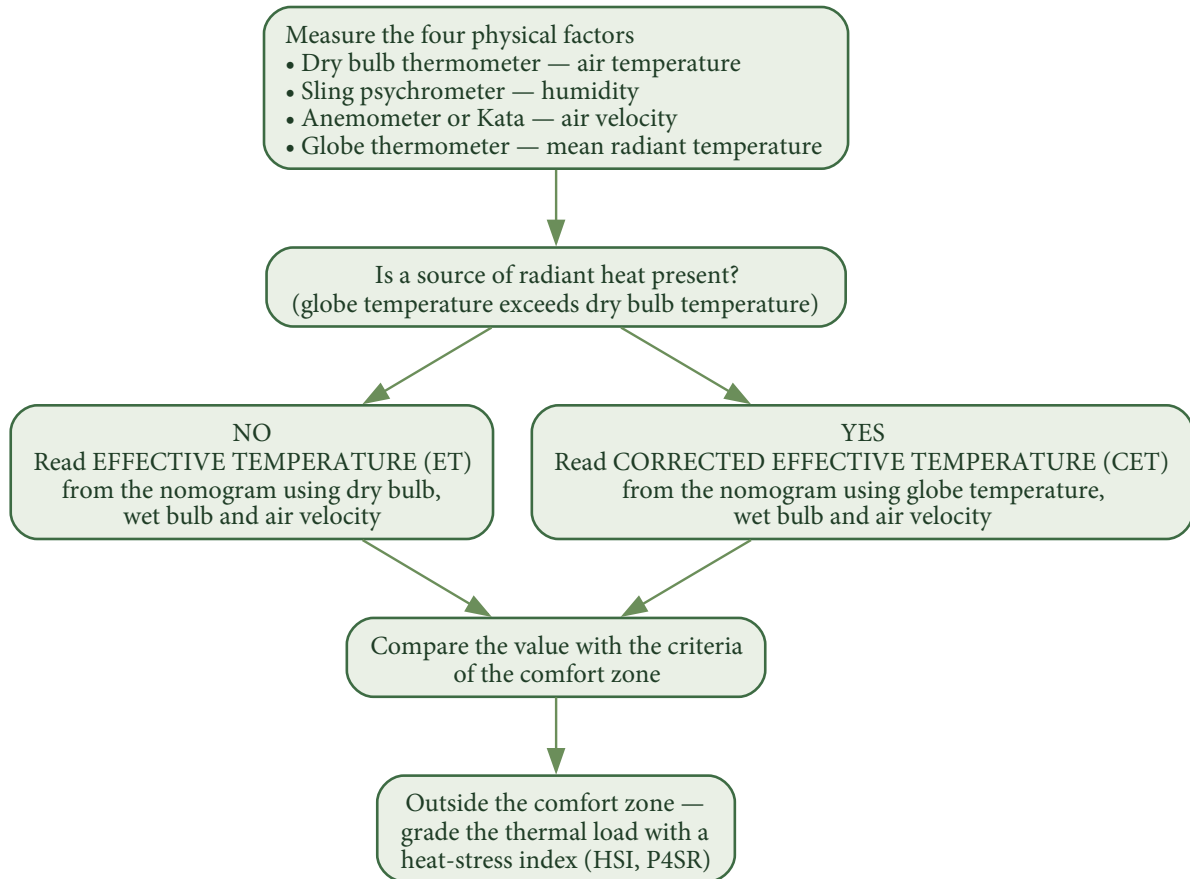
- **Effective temperature (ET)** is an *arbitrary* index which combines into a single value the effect of temperature, humidity and movement of the internal air on the sensation of warmth or cold felt by the body.
- **Numerical meaning** — the ET value is the temperature of *still, saturated* air which would induce the same sensation of warmth or cold as that experienced in the given conditions. An

environment with an ET of 30 °C (86 °F) feels exactly like a saturated, motionless atmosphere at 30 °C.

- ▶ **Origin** — evolved in 1923 by *Houghton and Yaglou* at the Pittsburgh Laboratory of the American Society of Heating and Ventilation Engineers.
- ▶ **Two scales** exist — one referring to men stripped to the waist (the *basic scale*) and one to men fully clad in indoor clothing (the *normal scale*). ET is read from a chart against the three variables (Park 28e, Ch 15, p 887; AFMC Red Book 2024, Ch 6, pp 159–160).
- ▶ **Criticism** — the ET scale ignores the effect of radiation from the surrounding structures.
- ▶ **Corrected effective temperature (CET)** removes this defect: the **globe thermometer reading is substituted for the dry bulb temperature** to allow for radiant heat, so the CET scale deals with all four factors — air temperature, air velocity, humidity and mean radiant heat.
- ▶ **Use** — whenever a source of radiation is present, CET is preferable. CET is read from prepared nomograms using globe thermometer temperature, wet bulb temperature and air speed. ET and CET are at present the indices of thermal comfort most widely used (Park 28e, Ch 15, p 887).



Effective temperature / corrected effective temperature nomogram for lightly clothed men (Suryakantha — Community Medicine with Recent Advances, Fig 6.1, p 66)



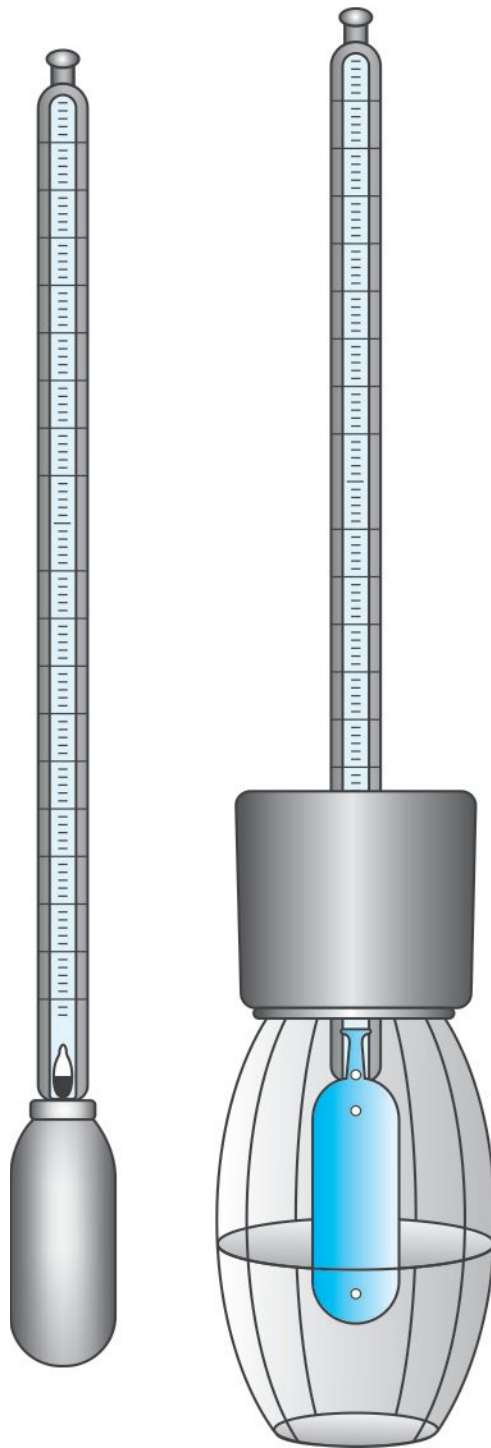
Stepwise use of the indices to assess a workplace thermal environment

Instruments used to measure the thermal environment

Instrument	Measures	Key specifications (exam-critical)
Dry bulb thermometer	Air temperature	Mounted in a <i>Stevenson screen</i> 1.20–1.80 m above ground; the screen shields it from radiant heat, direct sun and rain
Wet bulb thermometer	Humidity (with the dry bulb)	Bulb kept wet by muslin fed through a wick; reads lower than the dry bulb — the drier the air, the lower the reading; equal readings mean fully saturated air
Sling (whirling) psychrometer	Relative humidity	Two mercury thermometers on a frame, whirled at 4 revolutions/second for ~15 s to give the required air speed of about 5 m/s over the wet bulb; repeated until two successive wet bulb readings are identical
Assmann psychrometer	Wet and dry bulb temperature accurately	Portable; air drawn past the bulbs at >5 m/s by a clockwork fan; bulbs shielded from solar radiation, so usable in strong sunshine
Globe thermometer	Mean radiant temperature	Hollow copper bulb 6 inches (15 cm) in diameter, matt-black outside, with a calibrated mercury thermometer at its centre; equilibrium in 15–20 min (the <i>Helion and Crockford</i> modified globe, with a 6-volt fan stirrer, in 8–10 min). Globe temperature – dry bulb temperature = a measure of radiant heat

Instrument	Measures	Key specifications (exam-critical)
Kata thermometer	Cooling power of air; low air velocities	Greek <i>kata</i> = "down". Alcohol thermometer, bulb 4 cm long × 1.8 cm wide, stem marked 100–95 °F; wet and dry pairs; warmed above 130 °F, then time in seconds to fall 100→95 °F is taken 4 times (first discarded, last three averaged). Kata factor ÷ mean cooling time = cooling power in millicalories/cm ² /second. Standard (red) 100–95 °F, high-temperature (dark blue) 130–125 °F, extra-high (magenta) 150–145 °F; silvered bulbs reduce radiation error
Wet globe thermometer	Cooling capacity of the work environment	Dial thermometer inside a blackened copper sphere covered with wet black cloth; exchanges heat by conduction, convection, evaporation and radiation like a perspiring man
Silvered thermometer	True air temperature near radiant sources	Bright metallic surface reflects incident radiant heat
Anemometer	Air velocity	Four hemispherical cups on crossed arms with a vertical spindle; velocity read on a cyclometer in the anemometer box; recorded in m/s

(Park 28e, Ch 15, pp 901–902, 904–905)



Kata thermometer — dry (left) and wet (right) (Suryakantha — Community Medicine with Recent Advances, Fig 11.9, p 92)

Comfort zone values and the grading of heat stress

Criteria of the comfort zone (the range in which a worker feels comfortable while working):

Criterion	Value
Corrected effective temperature (CET)	25–27 °C (77–80 °F)
Relative humidity	30–65 per cent
Dry Kata reading	6 and above

Criterion	Value
Wet Kata reading	20 and above
Predicted four-hour sweat rate (P4SR)	1–3 litres (optimal 2.5 L)

(Suryakantha, Ch 7, p 67 and Ch 18, p 249)

Comfort zones evaluated in India, by CET (Park 28e, Ch 15, p 887):

Sensation	Corrected effective temperature (°C)
Pleasant and cool	20
Comfortable and cool	20–25
Comfortable	25–27
Hot and uncomfortable	27–28
Extremely hot	28 +
Intolerably hot	30 +

Grading by the analytical indices:

P4SR (litres of sweat in 4 hours)	Interpretation
1–3	Comfort zone
3–4.5	Just tolerable
4.5 +	Intolerable

Heat stress index (%)	Interpretation
0	No thermal stress
10–30	Mild to moderate heat strain
40–60	Severe heat strain
70–90	Very severe heat strain
100	Upper limit of heat tolerance

McArdle's maximum allowable sweat rate is 4.5 litres in four hours — the maximum compatible with a physiologically normal reaction in acclimatised, healthy young men on repeated heat exposure; McArdle placed a P4SR of 3 as the upper limit of the comfort zone (Park 28e, Ch 15, p 887).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Globe thermometer** — label: hollow copper sphere 15 cm (6 inches) diameter; matt-black outer coating; calibrated mercury thermometer stem; thermometer bulb at the *centre* of the globe; supporting stand. Caption the difference "globe temperature – dry bulb temperature = radiant heat".
- ▶ **Kata thermometer (dry and wet pair)** — label: alcohol column; stem graduated 100 °F and 95 °F marks; upper reservoir; silvered bulb (4 cm × 1.8 cm); wet muslin sleeve on the wet Kata; the Kata factor engraved on the stem.

High-yield exam pointers

- ▶ **Four factors** of thermal comfort: air temperature, humidity, air movement, radiant heat — "THAR"; ET covers the first three, **CET covers all four**.
- ▶ ET = *Houghton and Yaglou, 1923*; two scales — basic (stripped to waist) and normal (fully clothed); its one criticism is that it ignores radiation.
- ▶ CET substitutes the **globe thermometer reading for the dry bulb**; comfort zone CET 25–27 °C (77–80 °F), relative humidity 30–65 per cent.
- ▶ Kata: **dry ≥ 6, wet ≥ 20**; cooling power = Kata factor ÷ mean cooling time (100→95 °F), in millicalories/cm²/second.
- ▶ P4SR: comfort 1–3 L, optimal 2.5 L, upper limit of tolerance **4.5 L in 4 hours** (*McArdle*); valid only where sweating occurs.
- ▶ HSI is the **percentage of the heat storage capacity of an average man**: 0 none, 10–30 mild-moderate, 40–60 severe, 70–90 very severe, 100 upper limit of tolerance.
- ▶ WBGT formulas: indoors **0.7 WB + 0.3 GT**; outdoors **0.7 WB + 0.2 GT + 0.1 DB**; blackened globe of 150 mm (ISO 7243).
- ▶ Heat balance: $S = M + C + R - E$, and S must be zero for thermal equilibrium.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: Park's Textbook (28e) presents effective temperature and corrected effective temperature as the indices in current use, while the wet-bulb globe temperature is the index actually standardised for industry internationally (ISO 7243, British Standard BS EN 27243 and the American Conference of Governmental Industrial Hygienists threshold limit values), with reference values chosen to keep deep body temperature at or below 38 °C — so quote both, giving effective temperature and corrected effective temperature first. Suryakantha's chapter on air and ventilation gives the average predicted four-hour sweat rate as 2 litres while its meteorology and occupational health chapters and Park give 2.5 litres; the 2.5-litre figure is the safer one to write.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 15, Environment and Health — pp 886–887, 901–902, 904–905); Suryakantha — Community Medicine with Recent Advances (Ch 6 Air and Ventilation p 67, Ch 11 Meteorology pp 87–93, Ch 17 Occupational Health p 249); Hunter's Diseases of Occupations (Ch 49, Heat and Cold, p 559 — ISO 7243 wet-bulb globe temperature); Oxford Handbook of Occupational Health 2e (Assessment of the thermal environment, p 63 — WBGT equations and reference values); AFMC Red Book of Community Medicine 2024 (Ch 6, pp 159–160 — the basic and normal corrected-effective-temperature scales).



Motor vehicle air pollution control

Asked in: Paper I 12-Nov-2021 — RGUHS MD Community Medicine

Definition — Vehicular (traffic-related) air pollution is the contamination of ambient urban air by the products of incomplete combustion of petrol and diesel discharged from motor-vehicle exhausts, in concentrations that interfere with human health, safety or comfort. Motor vehicles are a major source of air pollution throughout urban areas, emitting hydrocarbons, carbon monoxide, lead, nitrogen oxides and particulate matter; in strong sunlight some of these hydrocarbons and oxides of nitrogen are converted in the atmosphere into "photochemical" pollutants of an oxidising nature (Park 28e, Ch 15, p 888). Control is ultimately an **engineering problem**, tackled through the four World Health Organization (WHO) principles of containment, replacement, dilution and legislation (Park 28e, Ch 15, p 893).

1. The premise — what motor vehicles emit and why it matters

Pollutant	Origin in the vehicle	Principal adverse effect
Carbon monoxide (CO)	Incomplete combustion of carbon-containing fuel; urban levels track traffic density, with peaks at the morning and evening rush hours (Park 28e, Ch 15, pp 888-889)	Alters human performance and cardiac function; takes 4–12 hours for air levels and blood carboxyhaemoglobin to equilibrate, so standards are set as 8-hour averages (Park 28e, Ch 15, p 889)
Oxides of nitrogen (NO _x)	Combustion; ~95% emitted as nitric oxide, oxidised to nitrogen dioxide (Park 28e, Ch 15, p 890)	Respiratory tract irritation, bronchial hyperactivity, impaired lung defences, bronchiolitis obliterans; reduced lung-function growth in children (Park 28e, Ch 15, pp 890, 892)
Hydrocarbons / volatile organic compounds	Unburnt and partially burnt fuel; evaporative losses	Lung cancer; substrate for photochemical smog (Park 28e, Ch 15, p 892)
Ground-level ozone (secondary pollutant)	Photochemical reaction of sunlight with vehicle NO _x and volatile organic compounds; highest in sunny weather (Park 28e, Ch 15, pp 889-890)	Breathing problems, triggers asthma; daily mortality rises 0.3% and cardiac mortality 0.4% per 10 µg/m ³ rise in ozone (Park 28e, Ch 15, p 890)

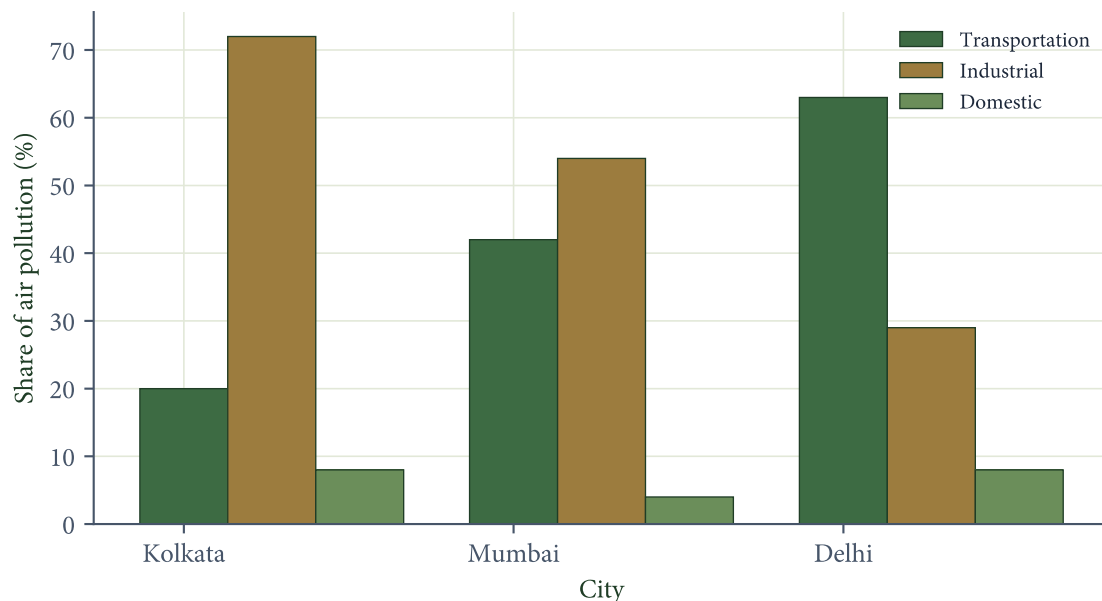
Pollutant	Origin in the vehicle	Principal adverse effect
Lead	Tetraethyl lead added to petrol as an anti-knock agent	Impaired neuropsychological development in children; children are most susceptible (Park 28e, Ch 15, p 892)
Particulate matter (PM ₁₀ , PM _{2.5}) and black smoke	Diesel engines when misused or badly adjusted emit black smoke and malodorous fumes (Park 28e, Ch 15, p 888)	Chronic obstructive pulmonary disease, ischaemic heart disease, stroke, lung cancer, lower respiratory infection

Emission profile differs by engine type (Mahajan & Gupta 4e, Ch 5, p 63):

Engine	Predominant pollutants
Two-stroke (two- and three-wheelers)	Lead compounds, carbon monoxide
Four-stroke	Hydrocarbons, oxides of nitrogen
Diesel	Smoke, odour, sulphur dioxide

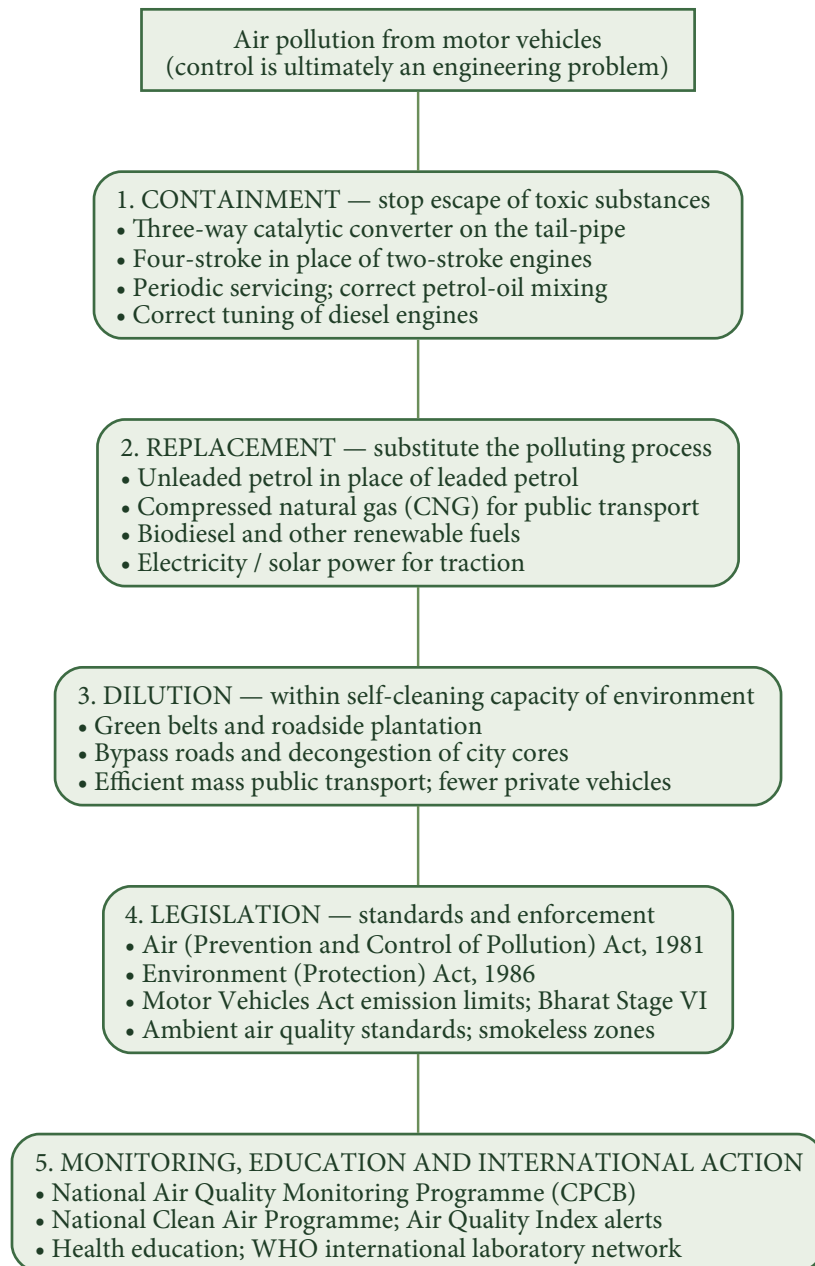
- Urbanisation and economic growth have increased vehicular traffic using both diesel and petrol, which add to oxides of nitrogen and oxides of sulphur in cities; **heavy diesel vehicles emit large quantities of respirable suspended particulate matter** (Sunder Lal, Ch 3, p 119).
- Air pollution was the **leading risk factor for premature death in India in 2019**, accounting for nearly **18% of all deaths (over 1.67 million)**; India had the highest population-weighted annual PM_{2.5} exposure at **83.2 µg/m³ in 2019**, falling to **58.1 µg/m³ in 2021** (Sunder Lal, Ch 3, p 120).
- Because combustion of petrol and diesel is the major problem, **cleaner engines and cleaner fuels** are the essential levers, and clean urban transportation would by itself greatly reduce air pollution (Maxcy-Rosenau-Last 15e, Ch 32, pp 726-727).

Relative contribution of sources to urban air pollution, Indian metropolitan cities



Transport is the single largest source in Delhi (63%). Source: Mahajan & Gupta, *Textbook of Preventive and Social Medicine* 4e, Table 5.2, data for 1990-91.

2. Framework of control — the four WHO principles applied to motor vehicles



WHO four-fold framework for the prevention and control of air pollution, applied to motor vehicles

3. Technical measures — cleaner fuel and cleaner engines

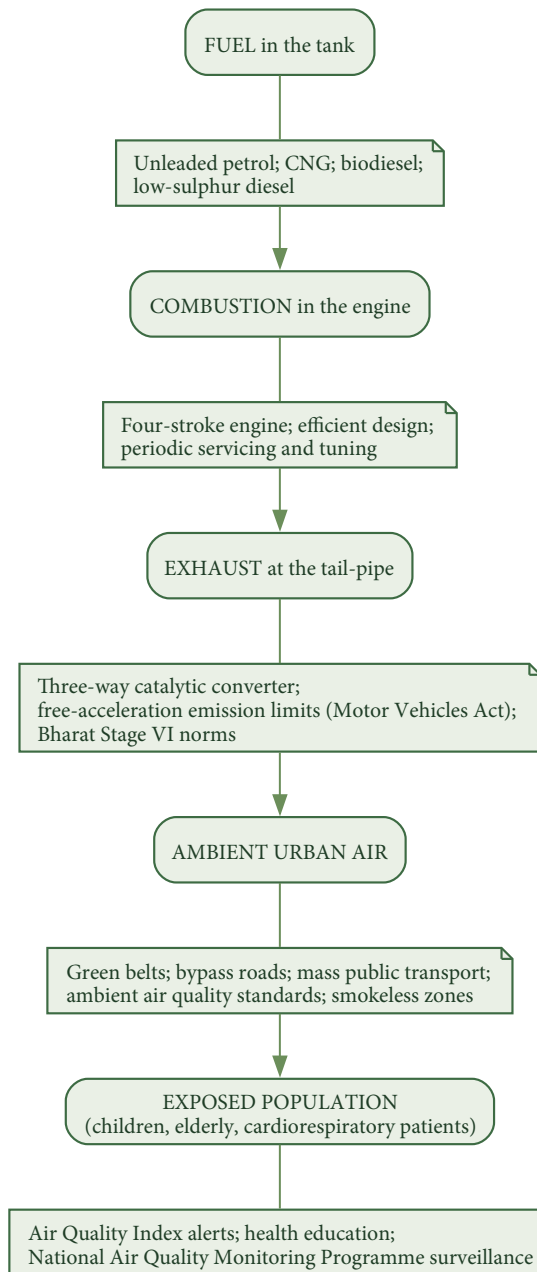
(a) Fuel substitution (replacement)

- **Unleaded petrol.** Tetraethyl lead is added to petrol at refineries to raise the octane rating and prevent knocking, but it is emitted as lead compounds in the exhaust. Indian levels were cut in stages — 0.56 g/L (Bureau of Indian Standards specification) → 0.15 g/L from June 1994 in Delhi, Mumbai, Kolkata and Chennai → **unleaded petrol (not more than 0.023 g/L) from 1 April 1995** in the four metros (Mahajan & Gupta 4e, Ch 5, p 63). Unleaded fuel is mandatory for any vehicle carrying a catalytic converter, since lead inactivates the catalyst.

- ▶ **Compressed natural gas (CNG).** Combustion of CNG is almost pollution-free. The landmark Supreme Court ruling of **5 April 2002** directed that all public transport in Delhi run on CNG, with Lucknow, Pune and Jodhpur to follow; Delhi's pollution levels fell significantly thereafter (Sunder Lal, Ch 3, p 122).
- ▶ **Biodiesel and renewable fuels,** which reduce emission of carbon dioxide, sulphur dioxide and unburnt hydrocarbons during combustion and can substitute for fossil fuels (Sunder Lal, Ch 3, p 122).
- ▶ **Electricity and solar power** in place of fossil-fuel combustion, the general replacement strategy recommended by WHO (Park 28e, Ch 15, p 893).

(b) Engine and exhaust technology (containment)

- ▶ **Three-way catalytic converter** — a device fitted near the tail-pipe, coated with the noble metals **platinum, rhodium and palladium**. It **oxidises** carbon monoxide and hydrocarbons to carbon dioxide and water vapour, and **reduces** oxides of nitrogen to atmospheric nitrogen, acting within seconds. It was made mandatory for all new cars sold in the four metros from **1 April 1995**, costs about Rs 10,000 and must be changed every **100,000 km** (Mahajan & Gupta 4e, Ch 5, pp 63-64).
- ▶ **Four-stroke in place of two-stroke engines.** Four-stroke engines are more fuel-efficient and release less unburnt or partially burnt fuel. Compared with a car, a two-wheeler with a two-stroke engine emits, **per passenger, twice the carbon monoxide and eight times the hydrocarbons** (Mahajan & Gupta 4e, Ch 5, p 63).
- ▶ **Routine maintenance** — periodic servicing of vehicles, mixing petrol and oil in the correct proportion, and correct tuning of diesel engines, which when misused or badly adjusted emit black smoke and malodorous fumes (Suryakantha, Ch 7, p 69; Park 28e, Ch 15, p 888).



The vehicular emission chain and the point at which each control measure acts

4. Legislative and regulatory measures

Instrument	What it does
Motor Vehicles Act — free-acceleration emission limits	Carbon monoxide: petrol two- and three-wheelers 4.5% by volume, four-wheelers 3%; diesel vehicles 65 Hartridge units (Mahajan & Gupta 4e, Ch 5, p 63)
Bharat Stage VI (BS-VI) vehicle emission standards	Nationwide switch initiated in April 2020, tightening permitted exhaust emissions for new vehicles (Sunder Lal, Ch 3, pp 120-121)
Air (Prevention and Control of Pollution) Act, 1981	Statutory basis for monitoring and controlling air pollution in India; also notifies noise as a pollutant (Sunder Lal, Ch 3, p 122; Park 28e, Ch 15, p 893)

Instrument	What it does
Environment (Protection) Act, 1986	Umbrella legislation strengthened after the Bhopal gas tragedy of 1984 (Sunder Lal, Ch 3, p 122)
Indian Factories Act, Smoke Nuisance Prevention Act, Industrial Zoning Order	Ancillary legal controls on emission of smoke and other pollutants (Mahajan & Gupta 4e, Ch 5)
Ambient air quality standards and smokeless zones	Clean-air legislation typically covers standards for ambient air quality, creation of smokeless zones, powers to local authorities, enforcement, research and education (Park 28e, Ch 15, p 893)

5. Traffic management, town planning and dilution

- **Efficient mass public transport with a reduction in the number of private vehicles** — the single highest-yield population measure, listed first among outdoor air-pollution controls (Sunder Lal, Ch 3, p 122).
- **Traffic control by construction of bypass roads**, diverting through-traffic away from congested city cores (Suryakantha, Ch 7, p 69).
- **Green belts and roadside plantation** to create a buffer between traffic corridors, industrial estates and residential areas; leaves absorb carbon dioxide, and dilution is a valid strategy so long as the load stays within the self-cleaning capacity of the environment (Park 28e, Ch 15, p 893; Sunder Lal, Ch 3, p 122).
- **Seamless urban breathing spaces**, as demonstrated by the Mahim Nature Park project in Mumbai (Sunder Lal, Ch 3, p 122).
- **Land-use planning** that separates high-traffic corridors from schools, hospitals and housing.

6. Monitoring, surveillance and health education

Element	Content
National Air Quality Monitoring Programme (NAMP)	Run by the Central Pollution Control Board with State Pollution Control Boards and the National Environmental Engineering Research Institute (NEERI), Nagpur; 793 stations across 344 cities and towns in 28 states and 8 union territories ; monitors sulphur dioxide, nitrogen dioxide, PM10 and PM2.5 (Sunder Lal, Ch 3, p 123)
National Clean Air Programme (NCAP)	Released 2019 ; revised target of 40% reduction in PM2.5 concentration by 2026 ; NCAP tracker launched 2019 (Sunder Lal, Ch 3, p 123)
Indian Air Quality Index (AQI)	Converts multi-pollutant data into one number, name and colour, in six categories — Good (0–50), Satisfactory (51–100), Moderate (101–200), Poor (201–300), Very poor (301–400), Severe (>400) (Sunder Lal, Ch 3, p 123)
International action	WHO has established an international network of laboratories for the monitoring and study of air pollution (Park 28e, Ch 15, p 893)
Health education	Public education on the hazards of vehicular pollution and on the citizen's own role — car-pooling, walking and cycling for short trips, switching off engines at signals, compliance with servicing (Suryakantha, Ch 7, p 69)

Air quality standards against which vehicular control is judged (Sunder Lal, Ch 3, p 123):

Pollutant	Averaging time	WHO guideline 2005	WHO guideline 2021	Indian national standard 2009
PM _{2.5} (µg/m ³)	Annual	10	5	40
PM _{2.5} (µg/m ³)	24-hour	25	15	60
PM ₁₀ (µg/m ³)	Annual	20	15	60
Nitrogen dioxide (µg/m ³)	Annual	40	10	40
Nitrogen dioxide (µg/m ³)	24-hour	200 (1 hour)	25	80
Ozone (µg/m ³)	8-hour	100	100	100

WHO guideline values for carbon monoxide are 100 mg/m³ for 15 minutes, 60 mg/m³ for 30 minutes, 30 mg/m³ for 1 hour and 10 mg/m³ for 8 hours (Park 28e, Ch 15, p 892).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Diagram 1 — Three-way catalytic converter (the key containment device). Labels: exhaust manifold; inlet gases — carbon monoxide, hydrocarbons, oxides of nitrogen; ceramic honeycomb monolith; washcoat bearing platinum, rhodium and palladium; reduction stage (oxides of nitrogen → atmospheric nitrogen); oxidation stage (carbon monoxide and hydrocarbons → carbon dioxide + water vapour); outlet gases — carbon dioxide, water vapour, nitrogen; tail-pipe. Add the caption "requires unleaded petrol — lead inactivates the catalyst".

Diagram 2 — Formation of photochemical smog. Labels: vehicle exhaust; oxides of nitrogen + volatile organic compounds; strong sunlight (ultraviolet); secondary photochemical oxidants — ground-level ozone; effects — eye and throat irritation, asthma exacerbation, reduced visibility.

High-yield exam pointers

- ▶ WHO four principles (mnemonic C-R-D-L): Containment, Replacement, Dilution, Legislation — write these as the spine of any air-pollution control answer.
- ▶ Indian free-acceleration carbon monoxide limits: petrol two/three-wheelers 4.5% by volume, four-wheelers 3%; diesel 65 Hartridge units.
- ▶ Lead in petrol, staged reduction: 0.56 g/L → 0.15 g/L (June 1994) → ≤0.023 g/L unleaded (1 April 1995) in the four metros; catalytic converter mandatory for new cars in the metros from the same date.
- ▶ Two-stroke two-wheeler vs car, per passenger: 2× carbon monoxide, 8× hydrocarbons.
- ▶ Dates to quote: Air Act 1981; Environment (Protection) Act 1986; Supreme Court CNG order 5 April 2002; Bharat Stage VI April 2020; NCAP 2019 with a 40% PM_{2.5} reduction target by 2026.
- ▶ Carbon monoxide is expressed as an 8-hour average because air–blood carboxyhaemoglobin equilibrium takes 4–12 hours.

- **Burden:** air pollution was India's leading risk factor for premature death in 2019 — ~18% of all deaths, >1.67 million.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: the free-acceleration carbon monoxide limits and the lead-in-petrol and catalytic-converter dates quoted above come from Mahajan & Gupta 4e and describe the pre-Bharat Stage regulatory position of the mid-1990s, which the later Bharat Stage I–VI norms have since progressively superseded; the source-contribution percentages in the bar chart are the 1990–91 figures with year-2000 projections given in that text. The retrieved corpus does not cover the newer vehicular measures (electric-vehicle policy, scrappage of overage vehicles, the Graded Response Action Plan), so these have deliberately not been stated.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, Global Burden of Disease Study 2023, WHO Global Air Quality Guidelines 2021, IARC — European Code Against Cancer 5th edition, Directive (EU) 2024/2881, National Clean Air Programme

Programme and policy updates (India)

- **First national evaluation of the National Clean Air Programme.** NCAP, launched 2019, set a 20–30% cut in PM₁₀ by 2024–25 against a 2017–18 baseline (the target was later revised to 40% by 2025–26 — see the NCAP row in section 6). Central Pollution Control Board measurements across 28 cities for 2018–2022 showed PM₁₀ falling by roughly 15–60% in 20 of them, a decline credited with **62,219 excess deaths avoided** across those 20 cities in 2021 compared with 2018. PM₁₀ nonetheless *rose* in 8 cities, including Kolkata and Durgapur. Indo-Gangetic Plain and central Indian cities still ran at 80–120 µg/m³ against about 40 µg/m³ in peninsular cities such as Hubballi. [PMID 39970552]
- **Vehicular control must be planned at airshed, not city, scale.** GAINS modelling for Uttar Pradesh (base year 2020, projected to 2035) found about **44% of population-weighted PM_{2.5} exposure originates outside the state**, and that **about 40% of total ambient PM_{2.5} comes from secondary particle formation** rather than from anything emitted directly. Under current policies, 2035 exposure stays far above the national standards; adding measures across the residential, industrial, transport and agricultural sectors together would cut exposure substantially and at relatively low cost [PMID 42034361]. That secondary fraction is what makes the finding a vehicle argument: road transport is one of the key sectors emitting nitrogen oxides, and nitrogen oxides react in the atmosphere to form secondary fine particles [PMID 41542811] — the same precursor chemistry as the photochemical smog set out in section 1 of this answer.

Current national data

- **Global Burden of Disease 2023 — India carries the heaviest ambient particulate burden among the BRICS group.** Ambient PM_{2.5} caused **2.955 million deaths across BRICS in 2023 (59% of the global total)**; India had the highest age-standardised mortality rate at **89.68 per 100,000** (estimated annual percentage change +2.24) and the highest age-standardised DALY rate at **2,262.68 per 100,000**, both still rising while G7 rates fell steadily. Ambient PM_{2.5} moved from the 12th to the **6th leading risk factor for DALYs globally between 1990 and 2023**; the leading

attributable causes in BRICS were ischaemic heart disease, chronic obstructive pulmonary disease and intracerebral haemorrhage. [PMID 42331512]

- **National exceedance and attributable mortality, 2019–2023.** All 271 Indian cities studied exceeded the WHO 2021 annual PM_{2.5} guideline of 5 µg/m³ and 175 (63.6%) exceeded the Indian national standard; 17 of 31 states and union territories breached the national standard, and all 31 breached the WHO value. Attributable premature deaths were led by cardiovascular disease (1,945,584; 43.5% of the total), then ischaemic heart disease (1,252,188), chronic obstructive pulmonary disease (725,367) and stroke (552,974), concentrated in the Indo-Gangetic Plain. Delhi is singled out as the city needing the single largest PM_{2.5} reduction of any in India, and the country as a whole is put at an estimated 10–20 years away from the WHO guideline of 5 µg/m³, assuming sustained annual reduction rates of 20% down to 10%. [PMID 41388116]
- **Indian cohort data temper the modelled mortality estimates.** The Million Death Study linked satellite-derived PM_{2.5} to 212,573 deaths at ages 15–69 and, after spatial adjustment, found a significant excess only for stroke — RR 1.09 (95% credible interval 1.04–1.14) per 10 µg/m³ — with no significant excess for chronic respiratory disease, ischaemic heart disease or total mortality. The authors caution against transferring risk coefficients derived mainly from high-income countries directly to India. A useful balancing point when the burden figures in the body of this answer are quoted. [PMID 36102642]

Elimination, surveillance and disease status

- **Hotspot surveillance identifies 170 polluted cities outside the NCAP list.** Getis-Ord Gi* geospatial analysis of modelled 1 km × 1 km annual PM_{2.5} for 2008–2019 found consistent hotspots covering 9.9% of India's area but housing 16% of its population, emerging hotspots over a further 2.6% of area (4.9% of population) and declining hotspots over 2.6% (3.4%). Rajasthan had the largest consistent-hotspot area, Uttar Pradesh the most densely populated, Maharashtra the most emerging and declining, and Kolkata the highest proportion among the three largest cities. Critically, 170 cities with consistent or emerging hotspots lie outside the NCAP non-attainment city list — a direct argument for widening programme coverage. [PMID 39510231]
- **Traffic pollutants are now linked to disease well beyond the respiratory tract.** A systematic review and meta-analysis of 32 studies found incident dementia significantly associated with PM_{2.5} (HR 1.08, 95% CI 1.02–1.14 per 5 µg/m³; 21 studies, n = 24,030,527), nitrogen dioxide (HR 1.03, 1.01–1.05 per 10 µg/m³; 16 studies) and black carbon — HR 1.13 (1.01–1.27) per 1 µg/m³, from six studies — with no significant excess for PM₁₀, nitrogen oxides or annual ozone [PMID 40716448]. In a prospective cohort of 512,724 Chinese adults, each 10 µg/m³ of annual nitrogen dioxide raised cardiovascular (HR 1.04, 1.02–1.05), hypertensive (1.08, 1.05–1.11), ischaemic stroke (1.08, 1.06–1.11) and asthma (1.15, 1.04–1.27) incidence; that paper treats nitrogen dioxide as a hallmark of transport-related air pollution and argues the results justify low-emission zones, tighter vehicle emission standards and a shift to clean energy [PMID 39643329].

Guideline and legislative updates

- ▶ **European Code Against Cancer, 5th edition (IARC/WHO) adds a dedicated air-pollution recommendation.** For the first time the Code carries 14 parallel recommendations aimed at **policymakers** alongside those for individuals. Recommendation 11 advises using public transport, walking or cycling in place of a car, choosing low-traffic routes, and keeping the home free of coal and wood smoke; the policymaker set asks governments to align legal limit values with the WHO 2021 guidelines, cut motorised traffic through spatial planning, and **not site new schools or nursing homes beside busy roads**. Supporting facts: IARC classified outdoor air pollution and particulate matter as Group 1 carcinogens in 2013, diesel engine exhaust is an established carcinogen and **petrol (gasoline) engine emissions have now also been so designated**; WHO-commissioned reviews give lung-cancer mortality RR 1.07 (1.04–1.10) per 10 $\mu\text{g}/\text{m}^3$ NO₂ and 1.09 (1.05–1.13) per 10 $\mu\text{g}/\text{m}^3$ PM_{2.5}. Directive (EU) 2024/2881 of 23 October 2024 sets 2030 limits of PM_{2.5} 10 $\mu\text{g}/\text{m}^3$ and NO₂ 20 $\mu\text{g}/\text{m}^3$ — still double the WHO values. Ultrafine particles, diesel exhaust, benzene and NO₂ run up to **twice as high on major roads as on nearby minor roads, falling to urban background about 50 m away**, which is the quantitative basis for green buffers, set-backs and route choice. [PMID 41542811]
- ▶ **Which transport measures actually deliver health gains.** A PRISMA-guided systematic review of 10 studies published 2011–2024 assessed five families of national or regional instrument: fuel taxes, incentives to buy electric vehicles, compulsory inspection-and-maintenance testing, retirement of diesel vehicles, and measures that shift people onto buses, trains, bicycles and their own feet. Taken together and across multiple settings, these policies were associated with lower PM_{2.5} and with premature deaths, respiratory and cardiovascular hospital admissions and healthcare spending avoided. Benefits were largest and most consistent for **integrated packages** — regulatory and technological measures acting at source (fuel-quality rules, vehicle electrification, diesel retirement) bundled with fiscal or behavioural instruments — rather than for any single measure acting alone. Most of the evidence comes from high-income countries, with little from low- and middle-income regions, which limits how far the findings carry to countries with different transport systems and resources. India sits in that under-evidenced group, though the review itself names no country. [PMID 42137951]

Global and WHO developments

- ▶ **South Asia's shared airshed lacks any binding regional instrument.** Nearly two billion people live in a region where air quality routinely exceeds WHO guidelines by **10 to 15 times**, with about **two million premature deaths a year** and World Bank welfare losses estimated above **5% of regional GDP**. The **Malé Declaration on Control and Prevention of Air Pollution (1998)** remains voluntary and under-resourced, and national programmes — India's NCAP, Bangladesh's National Air Quality Management Plan, Nepal's Air Quality Management Action Plan — are not regionally integrated. A dual-track architecture has been proposed: a UN-anchored South Asian Transboundary Air Pollution Protocol under UNEP, WHO and UNESCAP carrying legally enforceable emission standards and harmonised monitoring, plus a SAARC Prime Ministers' Council on Air Quality for political leadership and rapid response to cross-border haze. [PMID 41302574]

Grounding note: PMIDs 42331512, 36102642, 39510231, 39643329, 41542811 and 41302574 have been checked against the full text in PubMed Central. PMIDs 39970552, 42034361, 41388116, 40716448 and

42137951 are paywalled with no PubMed Central full text; those bullets are limited to what the published abstract states, and nothing has been inferred beyond it.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 15, Environment and Health); Sunder Lal — Textbook of Community Medicine (Ch 3, Environmental Health Problems); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e (Ch 5, Physical Environment: Air — "Reduction of Vehicular Pollution"); Suryakantha — Community Medicine with Recent Advances (Ch 6, Air and Ventilation); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 32, Pulmonary Responses to Gases and Particles). Named documents cited within these texts: Air (Prevention and Control of Pollution) Act 1981; Environment (Protection) Act 1986; Bharat Stage VI emission norms 2020; National Clean Air Programme 2019; National Air Quality Monitoring Programme (Central Pollution Control Board); WHO Air Quality Guidelines 2021; Indian National Ambient Air Quality Standards 2009.

Recent-advances sources: 39970552 — Impact of the National Clean Air Programme (NCAP) on the particulate matter pollution and associated reduction in human mortalities in Indian cities, *Science of the Total Environment* 2025;968:178787, doi:[10.1016/j.scitotenv.2025.178787](https://doi.org/10.1016/j.scitotenv.2025.178787) (abstract-only); 42034361 — Cost-effective mitigation strategies for air quality management in Uttar Pradesh, India, *Journal of Environmental Management* 2026;405:129784, doi:[10.1016/j.jenvman.2026.129784](https://doi.org/10.1016/j.jenvman.2026.129784) (abstract-only); 42331512 — Impact of ambient fine particulate matter (PM_{2.5}) pollution on disease burden in BRICS from 1990 to 2023: evidence from the Global Burden of Disease Study 2023, *BMJ Global Health* 2026, doi:[10.1136/bmjgh-2026-023674](https://doi.org/10.1136/bmjgh-2026-023674); 41388116 — Evaluating the national burden of PM_{2.5} in India: a comprehensive study of spatiotemporal distribution at state and city levels, non-carcinogenic health hazards, and premature mortality, *Environmental Monitoring and Assessment* 2025;198(1):43, doi:[10.1007/s10661-025-14860-3](https://doi.org/10.1007/s10661-025-14860-3) (abstract-only); 36102642 — Mortality Associated with Ambient PM_{2.5} Exposure in India: Results from the Million Death Study, *Environmental Health Perspectives* 2022, doi:[10.1289/EHP9538](https://doi.org/10.1289/EHP9538); 39510231 — Nationwide analysis of air pollution hotspots across India: A spatiotemporal PM_{2.5} trend analysis (2008–2019), *Environmental Research* 2024, doi:[10.1016/j.envres.2024.120276](https://doi.org/10.1016/j.envres.2024.120276); 40716448 — Long-term air pollution exposure and incident dementia: a systematic review and meta-analysis, *The Lancet Planetary Health* 2025;9(7):101266, doi:[10.1016/S2542-5196\(25\)00118-4](https://doi.org/10.1016/S2542-5196(25)00118-4) (abstract-only); 39643329 — Associations of long-term nitrogen dioxide exposure with a wide spectrum of diseases: a prospective cohort study of 0.5 million Chinese adults, *The Lancet Public Health* 2024, doi:[10.1016/S2468-2667\(24\)00264-0](https://doi.org/10.1016/S2468-2667(24)00264-0); 41542811 — European Code Against Cancer, 5th edition — outdoor and indoor air pollution and cancer, *Molecular Oncology* 2026, doi:[10.1002/1878-0261.70184](https://doi.org/10.1002/1878-0261.70184); 42137951 — Air quality and public health co-benefits of national transport policies, *Reviews on Environmental Health* 2026;41(2):193-209, doi:[10.1515/reveh-2025-0170](https://doi.org/10.1515/reveh-2025-0170) (abstract-only); 41302574 — Binding Multilateral Framework for South Asian Air Pollution Control: An Urgent Call for SAARC-UN Cooperation, *International Journal of Environmental Research and Public Health* 2025, doi:[10.3390/ijerph22111628](https://doi.org/10.3390/ijerph22111628). Named documents cited within these sources: WHO Global Air Quality Guidelines 2021; IARC Monographs — outdoor air pollution and particulate matter classified Group 1 (2013); European Code Against Cancer, 5th edition

(IARC/WHO 2026); Directive (EU) 2024/2881 on ambient air quality, 23 October 2024; National Clean Air Programme 2019; Global Burden of Disease Study 2023; Malé Declaration on Control and Prevention of Air Pollution 1998.



Standards of ventilation

Asked in: Paper I 16-Dec-2023 — RGUHS MD Community Medicine

Definition — The modern concept of **ventilation** implies not only the replacement of vitiated air by a supply of fresh outdoor air, but also **control of the quality of the incoming air** with regard to its temperature, humidity and purity, so as to provide a thermal environment that is comfortable and free from the risk of infection (Park 28e, Ch 15, p 893); vitiated air is the stagnant, warm and moist air produced by human occupancy of a closed room (Suryakantha, Ch 6 Air and Ventilation, p 69). **Standards of ventilation** are the minimum quantitative requirements — of fresh air supply, air change, cubic space, floor space, openings and thermal comfort — laid down to secure this.

Why the standards are needed, and the basis on which they were fixed

- ▶ Human occupancy vitiates the air of a room, and unless that air is replaced it adversely affects the comfort, health and efficiency of the occupants. Those in insufficiently ventilated rooms experience a feeling of **suffocation or discomfort**, and complain of **headache, drowsiness and inability to concentrate**; on prolonged exposure there is in addition the risk of **droplet infection and lowered resistance to disease** (Park 28e, Ch 15, p 886). Every standard set out below exists to prevent this, and each is a working field rule rather than a physiological constant.
- ▶ Fixing standards of ventilation is admittedly a **matter of much difficulty**, and **most of the standards were based on the efficiency of ventilation in removing body odour** — not on the oxygen requirement of the occupants (Park 28e, Ch 15, p 893).
- ▶ The older **carbon dioxide theory** — that the discomfort of a crowded room is due to a rise in carbon dioxide and a fall in oxygen — has been **refuted**. Oxygen may fall to **18 per cent** and carbon dioxide rise above **5 per cent** without deleterious effect, provided the "**cooling power**" of the air remains satisfactory (Park 28e, Ch 15, pp 886, 893-894).
- ▶ The causes of discomfort are therefore **physical, not chemical** — air temperature, humidity, air movement and heat radiation, which together determine the cooling power of the air with respect to the human body. As Professor Lee put it, "*The problems of ventilation are physical, not chemical; cutaneous, not respiratory*" (Park 28e, Ch 15, pp 886-887).
- ▶ The classic illustration is the *Black Hole of Kolkata* — 146 prisoners confined in a room 18 × 14 × 10 feet with two small windows that were adequate to supply all their oxygen needs, of whom only 23 survived; the deaths were concluded to be due to changes in the **physical** condition of the air leading to **heat retention**, not to lack of oxygen (Park 28e, Ch 15, p 886).

The standards of ventilation

These are the six quantitative standards an examiner expects to see stated with their figures and their derivation.

Standard	Prescribed value	Basis / caveat
Fresh air supply (cubic space)	Different workers advocated a minimal fresh air supply ranging from 300 to 3,000 cubic feet (c.ft) per person per hour; the widely quoted standard is De Chaumont's 3,000 c.ft per person per hour	Derived with carbon dioxide as the index: air seemed fresh so long as respiratory carbon dioxide was not more than 2 parts in 10,000; an average person expires 0.6 c.ft of carbon dioxide per hour and the permissible impurity is 0.0002 c.ft in 1 c.ft of air, so $0.6 \div 0.0002 = 3,000$ c.ft. This standard is no longer followed (Park 28e, Ch 15, pp 893-894)
Formula for the fresh air to be delivered	$d = e \div p$ — d = fresh air to be delivered to the room per hour, e = carbon dioxide exhaled per person per hour (0.6 c.ft), p = limit of respiratory carbon dioxide per c.ft of air (0.0002)	A child requires about 200 c.ft of air per hour (Suryakantha, Ch 6, p 69)
Air change (the operative modern standard)	Living rooms 2–3 air changes per hour; work rooms and assemblies 4–6 per hour; changing the air more than 6 times per hour produces an unpleasant draught and is to be avoided	The standard is set by the type of room, not by the system used. Air change is more important than cubic space. Air changes per hour = total hourly air supply to the room $\div$ cubic capacity of the room (Park 28e, Ch 15, pp 893-894)
Cubic space per person	1,000–1,200 c.ft per person is quite sufficient; in housing, an air space of at least 500 c.ft per capita, preferably 1,000 c.ft, unless mechanical replacement of air is provided	Follows from the air-change concept and replaces De Chaumont's figure (Park 28e, Ch 15, pp 894, 906)
Room height	Roof height not less than 10 feet (3 m) in the absence of air conditioning; heights above 10–12 feet are not counted when computing cubic space	Heights in excess of 10–12 feet are ineffective for ventilation because the products of respiration accumulate in the lower levels (Park 28e, Ch 15, pp 894, 906)
Floor space per person	Optimum 50–100 sq.ft per person, never less than 50 sq.ft; a living room should be ≥ 120 sq.ft (12 sq.m) if occupied by more than one person and ≥ 100 sq.ft (10 sq.m) for a single occupant	Floor space is even more important than cubic space; below 50 sq.ft per person there is overcrowding, which favours the spread of droplet infections (Park 28e, Ch 15, pp 894, 906; Suryakantha, Ch 6, p 69)
Windows and openings	At least 2 windows per living room, at least one opening directly onto an open space; placed not more than 3 feet (1 m) above floor level (sill at about 2½ feet); window area = 1/5 of floor area; doors + windows combined = 2/5 of floor area; ventilating grills near the ceiling = 2 per cent of floor area	Inlets placed low and outlets high, with doors and windows facing each other, secure cross-ventilation; back-to-back houses, which permit none, are not allowed (Park 28e, Ch 15, pp 894, 906; Suryakantha, Ch 10 Housing, p 82)

Standard	Prescribed value	Basis / caveat
Thermal comfort	Corrected effective temperature (CET) 25–27°C (77–80°F) = "comfortable" and predicted four-hour sweat rate (P4SR) 1–3 litres = comfort zone, on the comfort zones evaluated in India (Park 28e, Ch 15, p 887); relative humidity 30–65 per cent (Suryakantha, Ch 6, p 67 and Ch 11 Meteorology, p 93); dry Kata ≥6, wet Kata ≥20	These express the cooling power of the air, which the chemical standards ignore. Caveat: the dry and wet Kata readings were regarded as indices of thermal comfort, but further research has shown the Kata cooling powers are not reliable indices of comfort conditions — effective temperature and CET are the scales now widely used (Park 28e, Ch 15, p 887)

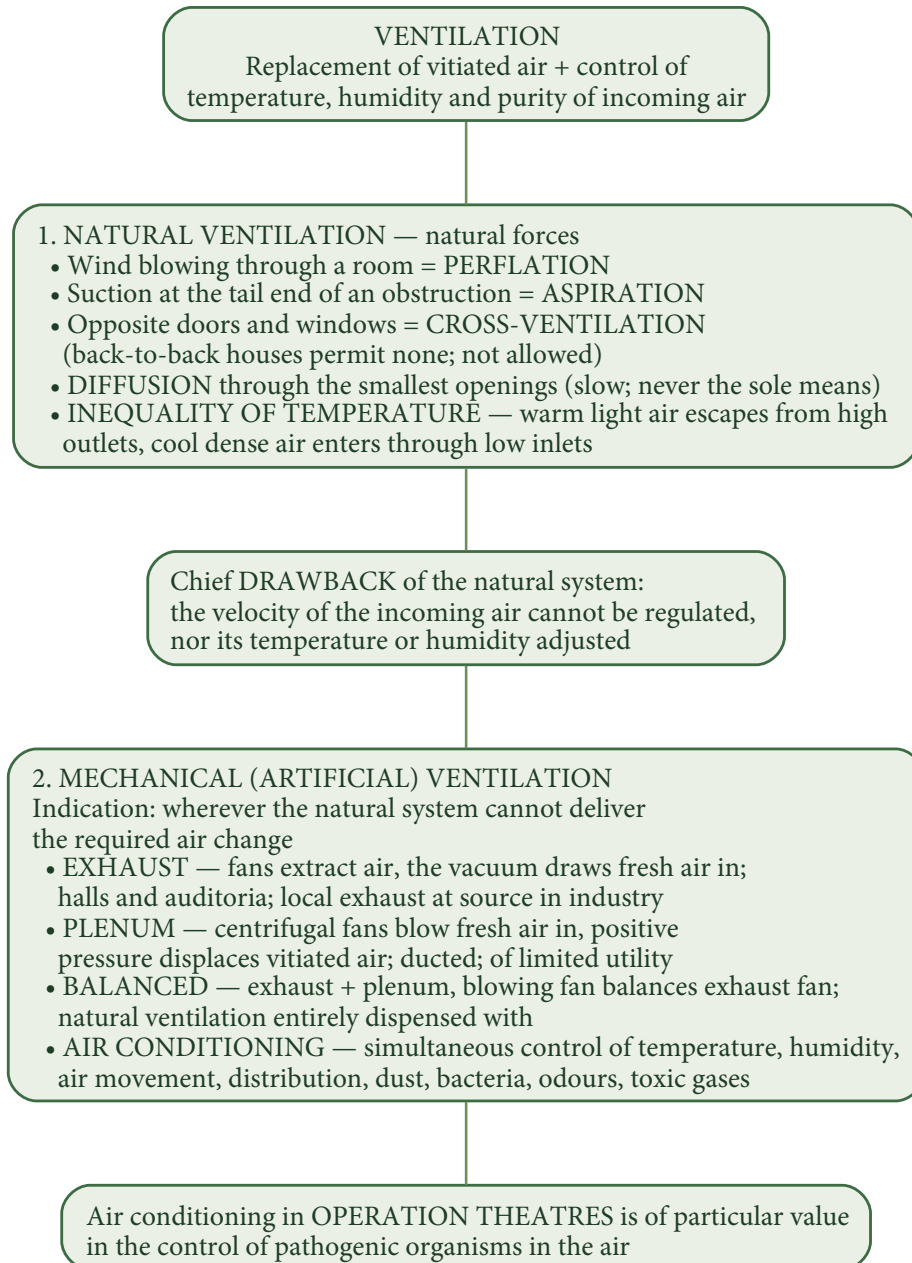
Applied standards: occupancy, institutions and the factory

The same figures are applied in the field as occupancy limits, and separately as institutional and statutory requirements. Note carefully which of the factory figures is law and which is only a recommendation — this is where most answers go wrong.

Setting	Standard
Dwelling — floor area per person (Park scale)	≥110 sq.ft (11 sq.m) → 2 persons; 90–100 sq.ft (9–10 sq.m) → 1½ persons; 70–90 sq.ft (7–9 sq.m) → 1 person; 50–70 sq.ft (5–7 sq.m) → ½ person; under 50 sq.ft (5 sq.m) → nil. A baby under 12 months is not counted; a child aged 1–10 years counts as half a unit (Park 28e, Ch 15, pp 906–907)
Variant of the same scale	Suryakantha prints the second band as 90–110 sq.ft → 1½ persons (Suryakantha, Ch 10, p 82); write Park's 90–100 sq.ft in the examination
Dwelling — persons per room	1 room → 2 persons; 2 rooms → 3; 3 rooms → 5; 4 rooms → 7; 5 or more rooms → 10 persons, with 2 more for each additional room (Suryakantha, Ch 10, pp 82–83)
Sex separation	Overcrowding is considered to exist if 2 persons over 9 years of age, not husband and wife, of opposite sexes are obliged to sleep in the same room (Park 28e, Ch 15, p 907) — Suryakantha gives the age as over 10 years (Ch 10, pp 82–83)
General hospital	150 sq.ft of floor area per person (Suryakantha, Ch 6, p 69)
Infectious diseases hospital	200 sq.ft of floor area per person (Suryakantha, Ch 6, p 69)
Factory — STATUTORY (Factories Act, 1948)	A minimum of 500 c.ft of air space for each worker has been prescribed, space more than 14 feet above ground level not being taken into account; a lower minimum applies to factories installed before the 1948 Act (Park 28e, Ch 18, pp 969–970)
Factory — RECOMMENDED general ventilation (not a provision of the Act)	It has been recommended that in every room of a factory, ventilating openings be provided in the proportion of 5 sq.ft for each worker employed in that room, the openings being such as to admit a continued supply of fresh air; and that rooms where dust is generated have an efficient exhaust ventilation system (Park 28e, Ch 18, p 969; Park 27e, Ch 18, p 945; Suryakantha, Ch 17 Occupational Health, p 249)

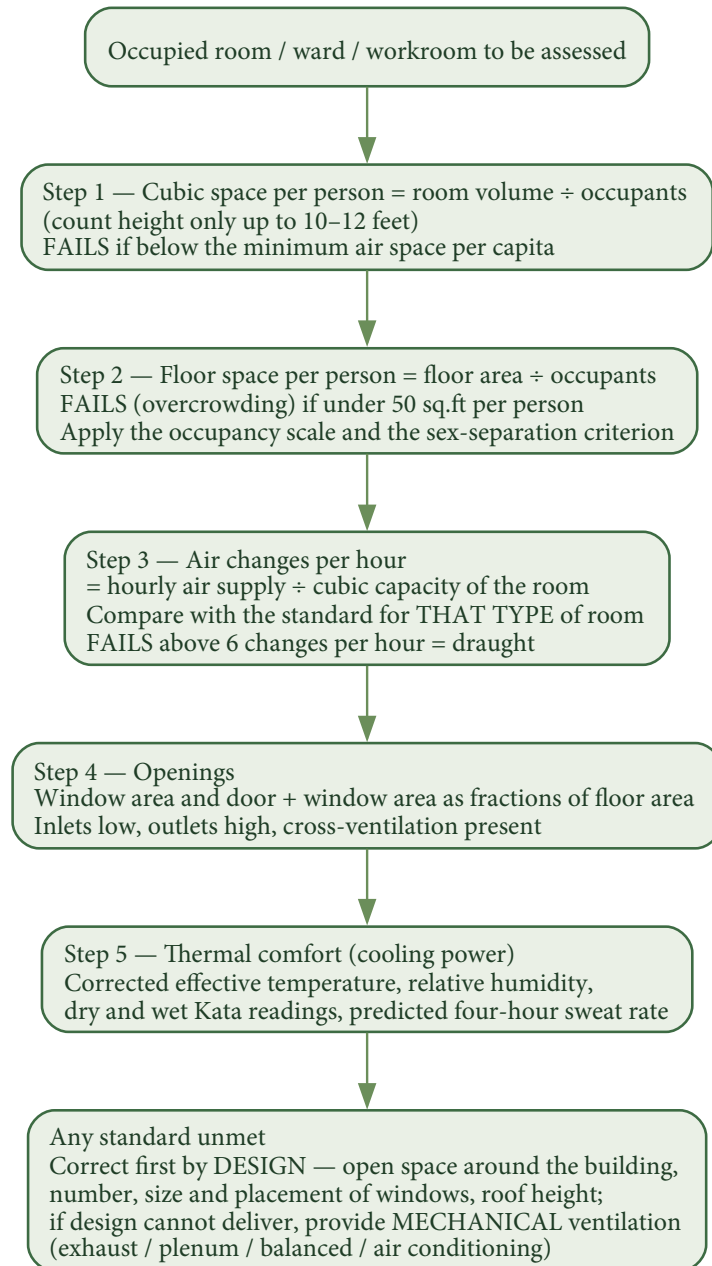
Systems by which the standards are met

Natural ventilation is the simplest system for small dwellings, schools and offices; mechanical ventilation is resorted to wherever the natural system cannot deliver the required air change.



Systems of ventilation — the forces used, the drawback of the natural system and the indications for the mechanical system

Assessing a room against the standards



Stepwise field assessment of whether an occupied room meets the standards of ventilation

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Natural ventilation of a living room (section view)** — label: *low-level inlet* (cool, dense outdoor air entering); *high-level outlet / ventilating grill near the ceiling* (warm, light vitiated air escaping); *perflation* (wind blowing straight through); *aspiration* (suction at the tail end of an obstruction); *cross-ventilation* between two opposite windows; *window sill 2½–3 feet above floor*; *window area = 1/5 of floor area*; *roof height 10–12 feet (space above this not counted)*; *floor space 50–100 sq.ft per person*.

High-yield exam pointers

- ▶ Mnemonic for the standards — "C-A-F-O-T": Cubic space, Air change, Floor space, Openings (windows), Thermal comfort.
- ▶ **De Chaumont: 3,000 c.ft per person per hour** = $0.6 \div 0.0002$ (carbon dioxide 0.6 c.ft exhaled per hour $\div$ permissible 0.0002 c.ft per c.ft of air, i.e. 2 parts in 10,000) — quote the derivation, then say it is **no longer followed**.
- ▶ **Air changes per hour** = **hourly air supply $\div$ cubic capacity of the room**, and the standard goes by the **room type**: living room 2–3, work room and assembly 4–6, above 6 = draught.
- ▶ Working figures: cubic space **1,000–1,200 c.ft per person** (housing minimum 500 c.ft, preferably 1,000); floor space **50–100 sq.ft**; height counted only up to **10–12 feet**.
- ▶ Overcrowding by Park's floor-area scale — ≥ 110 sq.ft $\rightarrow$ 2 persons; 90–100 $\rightarrow$ 1½; 70–90 $\rightarrow$ 1; 50–70 $\rightarrow$ ½; $< 50 \rightarrow$ nil; sex separation = **2 persons over 9 years of age**, not husband and wife, of opposite sexes in one sleeping room.
- ▶ Factory — separate the law from the advice: the **Factories Act, 1948 prescribes only 500 c.ft of air space per worker** (space above 14 feet not counted; a lower minimum for pre-1948 factories). The **5 sq.ft of ventilating opening per worker** and the **exhaust ventilation where dust is generated** are **recommendations**, not provisions of the Act.
- ▶ Comfort zone: **corrected effective temperature 25–27°C**, relative humidity **30–65 per cent**, **dry Kata ≥ 6** , **wet Kata ≥ 20** , **P4SR 1–3 litres** — adding that Kata cooling powers are not reliable indices of comfort.
- ▶ One-line clincher: "*The problems of ventilation are physical, not chemical; cutaneous, not respiratory*" (Lee) — with the **Black Hole of Kolkata** as the illustration.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the textbook standards are printed in imperial units (cubic feet, square feet) and are reproduced here as such. Where the two prescribed books differ, Park's figures are given first because they are the ones RGUHS marks against — Park's floor-area band is 90–100 square feet for 1½ persons and its sex-separation age is over 9 years, while Suryakantha prints 90–110 square feet and over 10 years. The scanned text of Park 28e is partly corrupt at two places, so the 5 square feet of ventilating opening per worker was read through Park 27e (Ch 18, p 945) and Suryakantha, and the exact lower air-space minimum for factories installed before the 1948 Act is illegible in the scan and is therefore stated only as "a lower minimum".

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Ch 15 (Environment and Health), pp 886-887, 893-894, 906-907; Ch 18 (Occupational Health), pp 969-970. Park's Textbook of Preventive and Social Medicine, 27e — Ch 15, pp 869-870 and Ch 18, p 945 (consulted only to read through a corrupt scan of the same passages in the 28e). Suryakantha — Community Medicine with Recent Advances: Ch 6 (Air and Ventilation) pp 67, 69; Ch 10 (Housing) pp 82-83; Ch 11 (Meteorology) p 93; Ch 17 (Occupational Health) p 249. Factories Act, 1948 (air-space provision, as quoted in Park 28e, Ch 18, pp 969-970).



Types of disinfection

Asked in: Paper I 16-Dec-2023 — RGUHS MD Community Medicine

Definition — Disinfection is the thermal or chemical destruction of pathogens and other types of microorganisms. It is less lethal than **sterilization**, because it destroys most recognised pathogenic microorganisms but not necessarily all microbial forms, notably not bacterial spores (Park 28e, Ch 3, pp 152–153). In communicable disease control it is a supplementary measure to environmental sanitation: the agent is destroyed at the point where it leaves the host, and the chain of transmission is thereby broken.

Disinfection is classified along three axes — by the **time** at which it is applied, by the **nature of the agent** used, and by the **level of kill** achieved. All three are described below.

Terms that must be distinguished first

Term	Meaning	Applied to
Disinfection	Thermal or chemical destruction of pathogens; spores may survive	Inanimate objects, excreta, surfaces
Sterilization	Validated process rendering a product free of all viable microorganisms, including bacterial spores	Instruments, dressings, culture media

Term	Meaning	Applied to
Disinfectant	Chemical (sometimes physical) agent that destroys pathogens but might not kill spores	Inanimate objects only
Antiseptic	Substance that prevents or arrests the growth or action of microorganisms; in effect a disinfectant at a lower, tissue-safe concentration	Living tissue — skin, mucous membrane
Asepsis	Prevention of contact with microorganisms	A technique, not an agent
Sanitizer	Agent reducing bacterial contaminants to safe levels as judged by public health requirements	Inanimate objects
Germicide	Agent that destroys microorganisms, especially pathogens	Both
Detergent (surfactant)	Surface-cleaning agent with no antimicrobial claim; acts by lowering surface tension	Skin, surfaces
Cleaning	Removal of visible soil, blood, protein and debris with detergent or enzyme cleaner and water, preparing the item for further decontamination	Instruments, devices, equipment
Deodorant	Substance that suppresses or neutralises bad odours, e.g. lime, bleaching powder	Latrines, drains, cattle sheds
Disinfestation	Destruction of ectoparasites — lice, ticks, mites (the agent is a <i>disinfestant</i> , e.g. BHC)	Persons, clothing, premises

(Park 28e, Ch 3, pp 152–153; Suryakantha, Ch 19 — Epidemiology of Infectious Diseases, pp 314–316)

Type A — classification by the time of application

This is the classification the examiner expects first. Three types are recognised, defined by their timing in relation to the infectious patient.

Type	What it is	When applied	Examples / materials covered
1. Precurrent (prophylactic) disinfection	Disinfection carried out as a routine preventive measure, before any case has occurred, to prevent the onset of disease	Continuously, as part of normal public health practice	Chlorination of drinking water; pasteurization of milk; handwashing; surgical scrub before operating and after examining a patient; sterilization of vaccines and of instruments before use
2. Concurrent (concomitant) disinfection	Application of disinfective measures as soon as possible after infectious material is discharged from the body, or after articles are soiled by such discharges — all personal contact with the discharge being prevented before disinfection	Throughout the course of the illness, as long as the patient remains a source of infection	Urine, faeces, vomit, sputum, saliva and other body discharges; linen, clothes, towels, dressings, utensils, bedding; hands, aprons, gloves, bowls and instruments used on the patient

Type	What it is	When applied	Examples / materials covered
3. Terminal disinfection	Application of disinfective measures after the patient has died, been removed to hospital, been discharged, or has ceased to be a source of infection, or after isolation practices are discontinued	Once, at the end of the illness or isolation	All articles left behind, plus floor, walls, furniture and curtains of the ward; also the ambulance, wheelchair and stretcher used by the patient

Terminal disinfection is further subdivided (AFMC Red Book of Community Medicine 2024, Ch 27, p 771):

- **Local** — disinfection of the bedding and bedsheet used by the patient, the walls, floor, furniture including kit box, shelf and lockers with their contents, and every surface or article within **2 metres all round the bed**.
- **Complete** — disinfection of the **whole room and all its contents**; reserved for cholera, pulmonary tuberculosis, puerperal sepsis and pneumonic plague.

Two linked principles decide how much terminal work is actually needed. Concurrent disinfection destroys the maximum number of pathogens at the moment their virulence and infectivity are highest, so properly performed concurrent disinfection markedly reduces the requirement for terminal disinfection; terminal disinfection is complementary to it, not a substitute. Park 28e adds that terminal disinfection is now scarcely practised — **terminal cleaning**, together with airing and sunning of rooms, furniture and bedding, is considered adequate (Park 28e, Ch 3, p 153).

Type B — classification by the nature of the disinfecting agent

Disinfecting agents fall into three groups: **natural, physical and chemical** (Suryakantha, Ch 19, p 316).

Natural and physical agents (Park 28e, Ch 3, pp 153–154; Suryakantha, Ch 19, pp 316–317):

Group	Agent	Conditions required	Chief use	Limitation
Natural	Sunlight	Direct, continuous exposure for several hours; ultraviolet rays act by coagulating bacterial protoplasm	Linen, bedding, mattresses, furniture	Ultraviolet rays do not penetrate glass; unreliable
Natural	Air (airing)	Acts by drying / evaporation of moisture	Rooms, bedding	Vagaries of weather — cannot be depended on
Physical	Burning / incineration	Best done in an incinerator, never in the open; sawdust, paper or kerosene aid combustion	Contaminated dressings, rags, swabs, disposables, faeces	Destroys the article
Physical	Hot air (hot-air oven)	160–180 °C for at least one hour to kill spores	Glassware, syringes, swabs, dressings, oils, vaseline, sharp instruments	No penetrating power — unsuitable for bulky articles such as mattresses; destroys plastic and rubber

Group	Agent	Conditions required	Chief use	Limitation
Physical	Boiling	Rolling boil 5–10 minutes; about 30 minutes for linen, utensils and bedpans; 1% soap plus 0.3% washing soda enhance the effect	Small instruments not used for subcutaneous insertion, linen, rubber gloves	Kills bacteria but not spores or viruses ; slow; woollen and thick fabrics shrink; fixes albuminous stains
Physical	Autoclaving (saturated steam under pressure)	122 °C at 15 lb/sq inch (1 kg/sq cm); acts by giving off latent heat	Linen, dressings, gloves, syringes, instruments, culture media — the most effective method	Not suitable for plastics and sharp instruments
Physical	Ionizing radiation	Gamma rays from Cobalt-60; articles pre-packed and irradiated, remaining sterile until opened	Bandages, dressings, catgut, plastic syringes, drip sets, catheters, copper-T	Costly; needs special packing and equipment
Physical	Ultraviolet radiation	Maximum bactericidal effect at 240–280 nm; mercury vapour lamps emit over 90% of radiation at 253.7 nm	Drinking water, air, contact lenses, titanium implants	Poor penetration; spores relatively resistant
Physical	Filtration	Membrane filters of cellulose and other polymers	Heat-labile liquids and solutions; household water filters (Berkefeld, Pasteur–Chamberland, Katadyn)	Earthenware, asbestos and sintered-glass filters retain bacteria but not viruses

Chemical agents, grouped as solid, liquid, gaseous and miscellaneous (Park 28e, Ch 3, pp 154–155; Suryakantha, Ch 19, pp 318–320):

Chemical group	Representative agents	Working strength	Chief public health use
Phenol and coal-tar derivatives	Phenol (carbolic acid) — the standard against which germicidal activity is compared	10% crude phenol; 5% for floors	Faeces, floors, drains; pure phenol is caustic and is not used as an antiseptic
	Cresol — 3–10 times as powerful as phenol, yet not toxic to tissue	5–10% (50 ml/litre)	Faeces and urine; soaking soiled linen before boiling
	Saponified cresol emulsions — <i>Lysol, IZAL, Cyllin</i>	Lysol 5–10%, IZAL 8%, Cyllin 17% phenol	Hands, clothes, excreta, hospital equipment, drains and latrines
	Chloroxylenol (<i>Dettol</i>) — 70% phenol but weakened by organic matter	As marketed	Surgical antiseptic, hand lotion, mouthwash

Chemical group	Representative agents	Working strength	Chief public health use
	Chlorhexidine (<i>Hibitane</i>)	0.5% aqueous or alcoholic; 1% cream	Skin antiseptic; creams for burns
	Hexachlorophene	Incorporated in soaps	Skin antiseptics, obstetric practice
Halogens — chlorine (solid)	Bleaching powder (chlorinated lime, CaOCl_2); fresh powder has 33.3% available chlorine ; unstable, stored in a brown air-tight bottle in a cool dark place	5% solution (50 g/litre) for excreta, 1 hour contact	Water, faeces and urine; also a deodorant
	Hypochlorites — sodium (household bleach, 5.25–6.15%), calcium	1% sodium hypochlorite for floors and high-contact surfaces	Broad spectrum, fast, no toxic residue, unaffected by water hardness
	Halazone and other chlorine tablets — 25% available chlorine	One 4 mg tablet per litre, 30–60 minutes	Small-scale disinfection of drinking water
	Sodium dichloroisocyanurate, chloramine-T, chlorine dioxide	As directed	Retain chlorine longer, giving a more prolonged bactericidal effect
Halogens — iodine	Tincture of iodine (2% in 70% alcohol); iodophors, e.g. povidone-iodine (<i>Betadine</i>)	As marketed	Skin and wound antiseptics; iodophors are non-irritant and do not stain
Alcohols (liquid)	Ethyl alcohol (methylated spirit), isopropyl alcohol	60–90% v/v ; activity falls sharply below 50%	Hand and skin disinfection, thermometers, table tops; not sporicidal ; inflammable and evaporates fast
Quaternary ammonium compounds	Cetrimide (<i>Cetavlon</i>); <i>Savlon</i> = cetrimide + chlorhexidine; <i>Zephiran</i>	Cetrimide 1%; <i>savlon</i> 1 in 6 in spirit	Cleaning-cum-disinfection of roadside wounds, instruments, gloves, clinical thermometers
Aldehydes	Formaldehyde; commercial formalin = 40% aqueous formaldehyde	2–3% solution (20–30 ml of 40% formalin per litre) for spraying; 10% for excreta	Rooms, walls, furniture; fumigation of blankets, beds and books that cannot be boiled
Gaseous	Ethylene oxide at 55–60 °C, mixed with 12% carbon dioxide (it is explosive), relative humidity 33%	Chamber process	Heat-sensitive articles — fabrics, plastic and cardiac equipment, books; process hard to control, so discouraged when alternatives exist

Chemical group	Representative agents	Working strength	Chief public health use
	Sulphur dioxide (burning sulphur)	Requires moisture; inert when dry	More useful as an insecticide and rodenticide
Oxidizing agents	Potassium permanganate; hydrogen peroxide; peracetic acid	H ₂ O ₂ 3% for wounds; peracetic acid <100 ppm kills bacteria, fungi and yeast in under 5 minutes (200–250 ppm if organic matter present)	Fruits and vegetables, swimming pools; hospital surfaces; high-level disinfection
Miscellaneous — metals and dyes	Silver (oligodynamic action in the Katadyn filter; silver nitrate for ophthalmia neonatorum), mercury; acridine dyes (acriflavine, proflavine), roseline dyes (gentian violet, brilliant green), fluorescein dyes	As formulated	Water purification, burns dressings, wound antiseptics
Lime	Quicklime; milk of lime — 10–20% aqueous suspension (1 part lime to 4 parts water)	2 hours contact for excreta	The cheapest of all disinfectants; lime wash for walls; deodorant in cattle sheds and latrines

Type C — classification by the level of kill, and how the method is chosen

High-level disinfection is the process of destroying all microorganisms, although some spores may survive, particularly if present in large numbers to begin with (Suryakantha, Ch 19, p 322). It is achieved by continuous rolling boiling for **20–30 minutes** — the simplest and most reliable way of freeing an article of most pathogens including HIV and hepatitis B virus when autoclaving is not available — or by exposure to ethylene oxide or formaldehyde vapour, or by prolonged soaking in activated glutaraldehyde or carbolic acid.

The intended use of the article decides the level required. Items that come into contact with intact mucous membrane without penetrating it — vaginal specula, proctoscope, laryngoscope, flexible fibre-optic endoscope — should ideally be sterilized, and must receive high-level disinfection if sterilization is impossible. Syringes, needles and other skin-piercing instruments may be reused after high-level disinfection by boiling only when autoclaving is impossible; **chemical disinfectants must never be used to make syringes and needles safe for reuse.**

Factors that decide whether a given method actually works (Suryakantha, Ch 19, p 318; Park 28e, Ch 3, pp 155–156):

- ▶ Strength (concentration) of the disinfectant and the nature of the solvent.
- ▶ Nature and number of the organisms — spore-forming organisms are the most resistant.
- ▶ Presence of organic matter such as blood, sputum or faeces, which inactivates many agents.
- ▶ Contact time allowed and the temperature of the mixture.

- Adequacy of prior **cleaning** — failure to clean leaves a high bioburden, protein load and salt concentration, all of which reduce efficacy.
- Biofilm accumulation, lumen length and diameter, restricted flow and device design, all of which impede contact between the agent and the microbe.

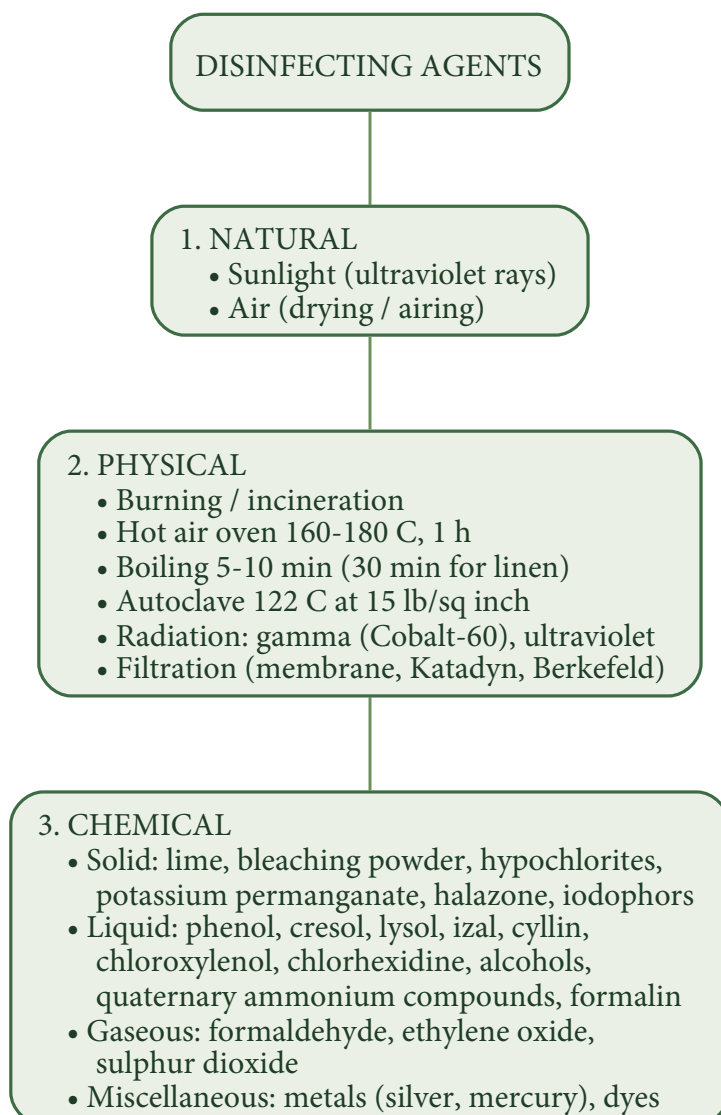
Applied disinfection procedures by material

Material	Agent and strength	Contact time and disposal
Faeces and urine	Equal volume of any one of: bleaching powder 50 g/litre (5%), crude phenol 100 ml/litre (10%), cresol 50 ml/litre (5%), formalin 100 ml/litre (10%); faeces broken up with a stick	1–2 hours, then emptied into a water closet or buried (Park 28e, Ch 3, Table 46)
Faeces — if none of these is available	Equal amount of quicklime or freshly prepared milk of lime (1 lime : 4 water); failing that, a bucket of boiling water, covered until cool	2 hours
Bedpans and urinals	Ideally steam-disinfected; otherwise 2.5% cresol after cleaning	1 hour
Sputum (especially tuberculosis)	Received in gauze or paper handkerchiefs and burnt; large volumes boiled or autoclaved 20 minutes at 20 lb; or spat into a cup half-filled with 5% cresol	Cup left standing 1 hour before emptying
Linen and clothes soiled with blood or dejecta	Soak in 10% formalin or 2.5% cresol or 5% phenol, then wash with soap and water, sun-dry and iron; alternatively boil 30 minutes or steam	Full soak period before washing
Room / isolation ward / operation theatre	Thorough cleaning, airing and direct sunlight; chemical — chlorinated lime leaving ≥ 25 ppm free chlorine, or formaldehyde solution $\geq 1\%$, or 2.5% cresol, sprayed or mopped (spray walls from below upwards)	Solution to remain in contact at least 4 hours before final washing
Room — fumigation	Formaldehyde gas: boil 500 ml formalin in 1 litre water per 30 m ³ , or pour that formalin–water mixture over 170–200 g potassium permanganate crystals (never add the permanganate into formalin — violent explosion)	Room kept closed 6–12 hours ; most effective at high temperature and relative humidity 80–90%
Clinical thermometer	Surgical spirit, or savlon 1 in 6 in spirit; a hot-water swab must never be used	Immersion
Leather, fur, woollen and silk fabrics; books	Exposure to formaldehyde gas or hot air	Fumigation cycle
Utensils and crockery	Boiling 5–10 minutes; cutlery that would be blunted by boiling — 10% formalin or 2% lysol	10 minutes, then wash
Used needles and syringes	Needle not detached; disinfectant aspirated into the syringe; immersed horizontally in a puncture-proof tray	At least 30 minutes , then mutilated mechanically and incinerated or deep-buried

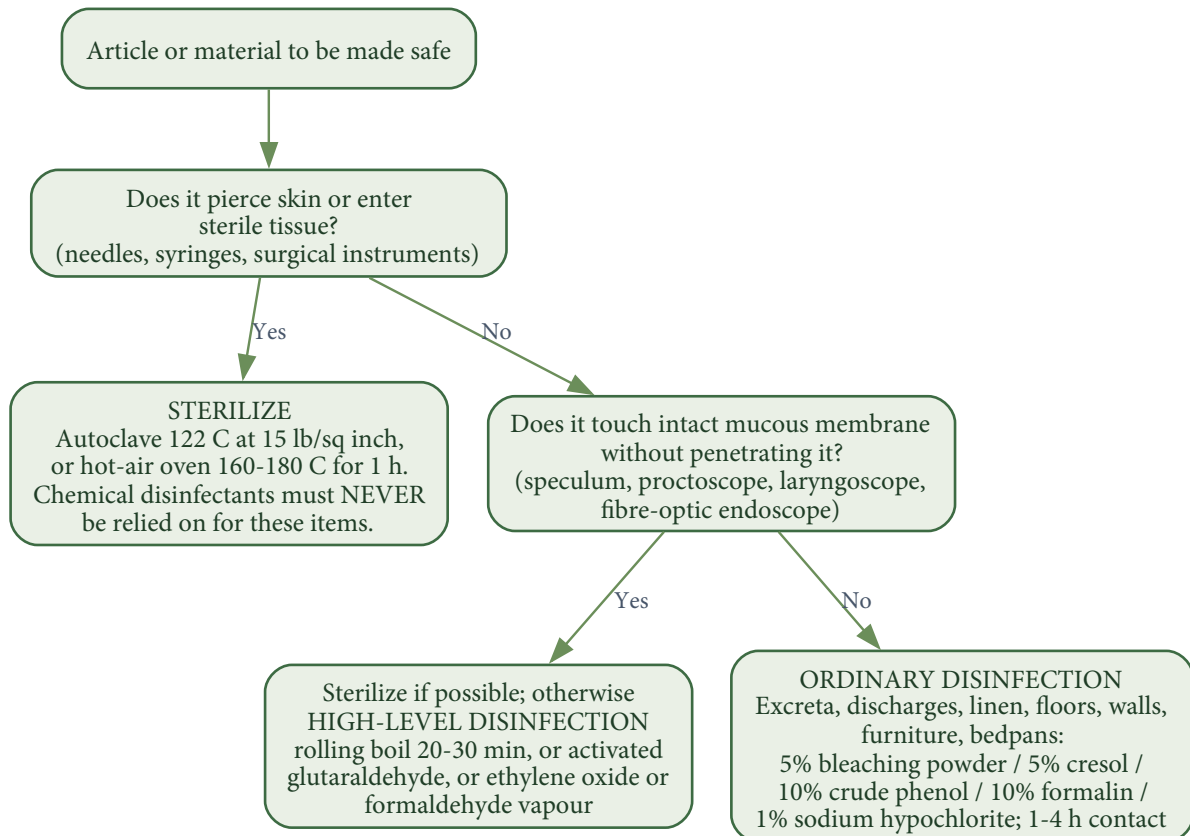
DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

Diagram — the three types of disinfection on the timeline of an infectious illness. Draw a single horizontal time line of one patient's illness and mark three blocks above it, with these exact labels:

- ▶ **Before onset — PRECURRENT (PROPHYLACTIC) DISINFECTION:** chlorination of water, pasteurization of milk, handwashing, sterilization of instruments and vaccines.
- ▶ **Throughout the illness — CONCURRENT (CONCOMITANT) DISINFECTION:** faeces, urine, vomit, sputum; linen, clothes, dressings, utensils; hands, gloves, aprons.
- ▶ **After death, discharge or recovery — TERMINAL DISINFECTION:** articles left behind, floor, walls, furniture, curtains, ambulance and stretcher; marked *local* (within 2 m of the bed) and *complete* (whole room and contents).
- ▶ Add an arrow from the concurrent block to the terminal block labelled "good concurrent disinfection reduces the need for terminal disinfection".



Classification of disinfecting agents — natural, physical and chemical



Choosing the method - the intended use of the article decides sterilization, high-level disinfection or ordinary disinfection

High-yield exam pointers

- ▶ **Three types by timing — mnemonic "P-C-T":** Precurrent (prophylactic, before the case), Concurrent (during illness), Terminal (after death or discharge; *local* within 2 m of the bed, *complete* for the whole room).
- ▶ **Three groups of agents:** natural (sunlight, air) — physical (burning, hot air, boiling, autoclave, radiation, filtration) — chemical (solid, liquid, gaseous, miscellaneous).
- ▶ **Numbers that must be written:** hot-air oven 160–180 °C for 1 hour; boiling 5–10 minutes (30 minutes for linen); autoclave 122 °C at 15 lb/sq inch; high-level disinfection by boiling 20–30 minutes; ultraviolet maximum kill 240–280 nm (mercury lamp 253.7 nm).
- ▶ **Excreta disinfection (Park Table 46), per litre:** bleaching powder 50 g (5%), crude phenol 100 ml (10%), cresol 50 ml (5%), formalin 100 ml (10%) — equal volume, contact 1–2 hours. Fresh bleaching powder = 33.3% available chlorine.
- ▶ **Fumigation:** 500 ml formalin + 1 litre water per 30 m³, room closed 6–12 hours, best at relative humidity 80–90%.
- ▶ **Standardization of a disinfectant:** Rideal–Walker (carbolic) coefficient — potency compared with phenol using *Salmonella typhi*; the Chick–Martin test improves on it by testing in the presence of organic matter such as yeast and excreta (Suryakantha, Ch 19, p 320).
- ▶ **Two negatives worth a mark each:** terminal disinfection is now scarcely practised — terminal cleaning with airing and sunning suffices; and chemical disinfectants must never be used to make syringes and needles fit for reuse.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only: the Indian standard texts retrieved describe "high-level disinfection" explicitly but do not set out the intermediate-level and low-level categories of the Spaulding scheme, so those tiers have been omitted rather than quoted from memory; where Park 28e and Suryakantha differ on a contact time (room fumigation 6–12 hours in Park, 10–12 hours and preferably 24 hours in Suryakantha), Park's figure has been given as the spine value.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, WHO Fungal Priority Pathogens List, WHO Infection Prevention and Control Assessment Framework (IPCAF), Biomedical Waste Management Rules 2016

Programme and policy updates (India)

- ▶ Disinfection is now assessed as one component of a hospital **infection prevention and control (IPC) programme**, not as a stand-alone ritual. A multi-centric observational study of 31 tertiary institutions across 15 states and Union Territories found HAI training in 74.5% and antibiotic policies in 61.4%, but only 27.5% benchmarked their HAI data, 53.3% had isolation rooms and 48.9% had outbreak surge plans — the weak links are surveillance and benchmarking rather than the disinfectants themselves [PMID 41448550].
- ▶ A WHO-aligned IPC package delivered across 11 Indian private tertiary hospitals (2022–23) lifted the IPCAF score from a median of 78% to 97% ($p = 0.004$) and hand-hygiene compliance from 65% to 88% ($p = 0.004$), with parallel gains in CAUTI, CLABSI, SSI and VAP bundle compliance. A few hospitals still slipped at particular hand-hygiene moments — aseptic procedures, body-fluid exposure and contact with patient surroundings — and were pulled back by focused retraining, real-time audit feedback and point-of-care reminders; **regular training with ongoing monitoring, not a one-off exercise, is what sustains precurrent disinfection** [PMID 41814356].

Current national data

- ▶ Applying the WHO IPCAF tool in 2019 to 32 public and private tertiary facilities in the national HAI surveillance network gave a median score of 620, with 13% of hospitals at basic, 28% at intermediate and 59% at advanced IPC level; multimodal strategy had the widest spread of scores and IPC guidelines the narrowest [PMID 34600081].
- ▶ A nationwide survey of 267 health-care facilities (2023) on **Biomedical Waste Management Rules 2016** compliance found 49% still cutting needles before disposal, only 58% fully barcoding waste, 79% autoclaving laboratory waste, and 28% still using mercury-filled devices — a direct audit of the disposal end of concurrent disinfection [PMID 40744379].

Elimination, surveillance and disease status

- ▶ **Candida auris** has re-opened the choice of chemical agent. Benzalkonium chloride and other quaternary ammonium compounds give MICs of 16–32 mg/L against an epidemiological cutoff of 32 mg/L and are not recommended for decontamination; 2% chlorhexidine fails on contaminated surfaces while 4% works; **sodium hypochlorite at about 1% for 5–10 minutes kills**

the organism completely, and hydrogen peroxide — vapour 8 g/m³ or 0.5–1.4% solution — gives 96–100% kill; 5% phenol and 2% glutaraldehyde work but need prolonged contact. The organism survives 7–14 days on plastic, stainless steel, textile and walls, and biofilm plus efflux-pump activation drives tolerance [PMID 41853952].

- The systematic review underpinning the WHO Fungal Priority Pathogens List placed *C. auris* in the critical-priority group: candidaemia mortality 29–62%, fluconazole resistance as high as 87–100%, median hospital stays of 46–68 days in the longer series, and isolation reported from over 50 countries across six continents between its discovery in 2009 and 2023. Control rests on reinforced hand hygiene, patient isolation with contact precautions and screening, and **daily plus terminal cleaning and disinfection of the environment and shared equipment using an appropriate disinfectant** [PMID 38935900].

Guideline and legislative updates

- **Ultraviolet-C is moving from air disinfection to device disinfection.** A scoping review of eight studies on semicritical devices (flexible endoscopes without a working channel, rigid otorhinolaryngological endoscopes, and transthoracic, abdominopelvic, duplex, rectal and vaginal ultrasound probes) used disinfectors mostly at 254 nm with exposures of 25 seconds to 15 minutes; bacterial reduction was not inferior to existing methods. Every study still performed mechanical pre-cleaning first, so UV-C supplements rather than replaces cleaning [PMID 41318532].
- Vaporised hydrogen peroxide generated from **sodium percarbonate granules** achieved high-level disinfection of gastrointestinal endoscope lumens within 30 minutes with a >5 log reduction of *Bacillus* spores, matching a peracetic-acid automated reprocessor (91.6% vs 100%, *p* = 0.068) at a fraction of the disinfectant cost and with no water use. Raising 3% bovine albumin organic load significantly reduced efficacy — confirming the textbook rule that **cleaning must precede disinfection** [PMID 40248380].

Global and WHO developments

- A systematic review of cholera case-control studies (67 appraised, 41 pooled) quantified what water disinfection is worth: **boiling OR 0.38 (95% CI 0.17–0.84)**, **chlorination OR 0.37 (0.17–0.83)**, any household treatment OR 0.42 (0.27–0.65), against untreated water OR 2.51 (2.03–3.10) and surface water OR 3.40 (2.52–4.58). Sachet water carried *higher* odds (OR 1.69, 1.13–2.52) — packaged water sitting at the top of the JMP ladder is not automatically safe, because contamination enters during collection, packaging and distribution [PMID 41013364].
- A systematic review of 36 studies of household point-of-use chlorination in low- and middle-income countries found chlorine-confirmed use ranging from 1.5% to 100%, declining on average over time and tracking the frequency of staff contact — sample-size-weighted median use was 84% with weekly household visits, 47% with monthly and 11% when visits were less frequent. Relying on the individual household **cannot deliver the population-level benefit of centralised piped chlorination** [PMID 36715546].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e — Chapter 3, Principles of Epidemiology and Epidemiologic Methods, section on Disinfection (definitions, types of disinfection, natural, physical and chemical agents, Table 46, recommended disinfection procedures); Suryakantha — Community Medicine with Recent Advances, Chapter 19, Epidemiology of Infectious Diseases (classification of disinfectants, solid/liquid/gaseous/miscellaneous chemical disinfectants, disinfection procedures, high-level disinfection, Rideal–Walker and Chick–Martin standardization); AFMC Red Book of Community Medicine 2024, Chapter 27 (local versus complete terminal disinfection).

Recent-advances sources: 41448550 — A multi-centric study on the organization of infection prevention and control programmes in tertiary-care public-sector hospitals of India, *J Hosp Infect* 2025, doi:[10.1016/j.jhin.2025.11.047](https://doi.org/10.1016/j.jhin.2025.11.047) (abstract-only); 41814356 — Customizing infection prevention and control modules for combating healthcare-acquired infections in low-resource hospitals, *Antimicrob Resist Infect Control* 2026, doi:[10.1186/s13756-026-01727-6](https://doi.org/10.1186/s13756-026-01727-6); 34600081 — Self-reported survey on infection prevention and control structures in healthcare facilities part of a national level healthcare associated infection surveillance network in India, *Am J Infect Control* 2021, doi:[10.1016/j.ajic.2021.09.019](https://doi.org/10.1016/j.ajic.2021.09.019) (abstract-only); 40744379 — Gap assessment in implementation of Biomedical Waste Management in Health Care Facilities: Nationwide survey in India, *Indian J Med Microbiol* 2025, doi:[10.1016/j.ijmmb.2025.100947](https://doi.org/10.1016/j.ijmmb.2025.100947) (abstract-only); 41853952 — Biocide Response of *Candida auris*, *Mycoses* 2026, doi:[10.1111/myc.70169](https://doi.org/10.1111/myc.70169); 38935900 — *Candida auris* — a systematic review to inform the World Health Organization fungal priority pathogens list, *Med Mycol* 2024, doi:[10.1093/mmy/myae042](https://doi.org/10.1093/mmy/myae042); 41318532 — Is UV-C light disinfection suitable as a disinfection method for medical devices: a scoping review, *Antimicrob Resist Infect Control* 2025, doi:[10.1186/s13756-025-01669-5](https://doi.org/10.1186/s13756-025-01669-5); 40248380 — Using vaporized hydrogen peroxide for anhydrous disinfection of gastrointestinal endoscopes, *World J Gastroenterol* 2025, doi:[10.3748/wjg.v31.i14.103921](https://doi.org/10.3748/wjg.v31.i14.103921); 41013364 — Associations of water quality with cholera in case-control studies: a systematic review and meta-analysis, *BMC Infect Dis* 2025, doi:[10.1186/s12879-025-11533-x](https://doi.org/10.1186/s12879-025-11533-x); 36715546 — Adoption of Point-of-Use Chlorination for Household Drinking Water Treatment: A Systematic Review, *Environ Health Perspect* 2023, doi:[10.1289/EHP10839](https://doi.org/10.1289/EHP10839). Also used: WHO Fungal Priority Pathogens List (2022), WHO Infection Prevention and Control Assessment Framework (IPCAF), Biomedical Waste Management Rules 2016 (India).

♦ ♦ ♦

Water conservation

Asked in: Paper I 12-Nov-2021 — RGUHS MD Community Medicine

Definition — **Water conservation** means the **protection of water resources** so that precious water reserves are preserved, and it is needed because the rate of water extraction is now exceeding the replenishment that takes place by natural processes, mainly recharge due to rain (Park 28e, Ch 15, Appendix IV, p 885). Park treats it under exactly **two heads** — (a) **prevention of wastage** and (b) **water harvesting**. It is a public-health and not merely an engineering concern, because **water quantity is as important as water quality**: without abundant water in or near the home, hygiene becomes difficult or impossible (Park 28e, Ch 15, p 860).

Why conservation is needed — the drivers of water stress

Driver	What it does	Grounding figure
Finite freshwater stock	Only 3 per cent of all water on earth is fresh, the rest 97 per cent is in oceans; most fresh water is locked in ice caps, glaciers and deep aquifers, so only about 1 per cent is easily accessible through lakes, rivers and shallow aquifers	Sunder Lal, Ch 3, p 92
Declining rainfall with rapid urbanisation and industrialisation	Growing water shortage in several areas; ponds, lakes and available resources are shrinking, throwing more pressure on sub-soil water resources	Park 28e, Ch 15, p 885
Extraction exceeding recharge	Alarming fall in sub-soil water levels in and around several cities; development of agriculture dependent on tube wells has further worsened the situation	Park 28e, Ch 15, p 885
Agriculture as the dominant user	India uses 70–90 per cent of all water for agriculture, 7 per cent for industry, 3 per cent for domestic use (worldwide 69, 23 and 8 per cent)	Sunder Lal, Ch 3, p 92
Dependence on groundwater	80 per cent of rural India's water needs are met by groundwater and 20 per cent by surface water — the reverse of urban India; about 50 per cent of the country's irrigation uses groundwater	Mahajan & Gupta 4e, Ch 6, p 67
Rainwater run to waste	A vast quantity of rainwater is normally discharged into drains; only a meagre 8 per cent of rainwater is currently harvested in India	Park 28e, Ch 15, p 885; Sunder Lal, Ch 3, p 105
Pollution shrinking the usable stock	Agriculture is the biggest polluter — indiscriminate insecticide and pesticide use contaminates aquifers and surface water; over 4 million hectares of high-quality land have been abandoned to salinisation and water-logging from excess irrigation	Sunder Lal, Ch 3, p 93

Health stake — diarrhoea and intestinal worm infestations account for **10 per cent of the total burden of disease** in developing countries, and an inadequate supply also raises the risk of schistosomiasis, skin and eye infections and guineaworm disease; in **2020, 90 per cent** of India's population had access to improved drinking water and **71 per cent** to adequate sanitation (Park 28e, Ch 15, p 860 and Table 1, p 861).

How much water must be conserved for — requirement and Indian norms

Norm	Value	Source
Basic physiological requirement (survival only)	about 2 litres per head per day	Park 28e, Ch 15, p 861
Adequate supply for all urban domestic purposes	150–200 litres per capita per day (lpcd)	Park 28e, Ch 15, p 861
Earlier rural target for India	40 lpcd, with at least one spot-source for every 250 persons	Park 28e, Ch 15, p 861; Ch 7, p 573
Enhanced rural design service level (12th Plan)	55 lpcd; at least 50 per cent of the rural population to have water within household premises or within a 100 metre radius, at least 30 per cent by individual household connection	Park 28e, Ch 7, p 573
Jal Jeevan Mission (also known as the Nal Jal Yojana)	Functional Household Tap Connection (FHTC) delivering at least 55 litres per person per day of water meeting prescribed quality standards, to every rural household by 2024	Park 28e, Ch 7, p 573

Methods of water conservation

A. Prevention of wastage — Park lists exactly three measures (Park 28e, Ch 15, Appendix IV, p 885):

- **Widespread awareness** about the economical use of water needs to be developed among people; it has to be propagated that people should make an effort not to waste water and thereby help reduce consumption of the invaluable water resource.
- **Efficient water management** can substantially reduce the total water requirement of communities.
- **Domestic consumption** can be reduced by individuals cultivating **better habits in kitchen and bathroom use**, avoiding **free running of water**.

Caution — **continuous supply is a safety measure, not a wastage-prevention measure**. Intermittent, low-pressure supply empties the pipes, creates negative pressure that back-siphons bacteria and foul gases through leaky joints, and forces unsafe household storage; outbreaks of **typhoid and relapsing fever** have been traced to it, and the **WHO Expert Committee (1965)** recommended that it be avoided. Park therefore calls 24-hour flowing water desirable — but adds that it "may entail some wastage of water through misuse" (Park 28e, Ch 15, p 883).

B. **Water harvesting** — "extremely important to preserve and build up underground water reserves in urban and semi-urban areas", where considerable water is drawn out by tube wells for domestic consumption (Park 28e, Ch 15, p 885). Two routes:

Route	What is done	Product
Recharge (indirect use)	Rainwater from rooftops and courtyards is diverted into soaking pits or trenches instead of drains; it is also viable to clean and filter this water and divert it into existing tube wells or wells	Adds to the underground reserves; raises the sub-soil water table (Park 28e, Ch 15, p 885)

Route	What is done	Product
Storage (direct use)	Rainwater collected from rooftop and ground catchment is stored in tanks, containers and below-ground tanks for drinking and domestic use — practised at <i>Dwarka</i> (Gujarat) and <i>Churu</i> (Rajasthan)	Soft, clean water, but must be protected against contamination (Mahajan & Gupta 4e, Ch 6, p 67)

Construction of a recharge pit (Park 28e, Ch 15, Fig 11, p 885) — 1. dig a suitably large pit; 2. fill it in layers with big stones, followed by gravel and then sand; 3. bring collected rooftop rainwater into the pit by PVC pipes; 4. **the pit itself is the filter** — the rainwater is filtered as it passes through these layers; 5. it then travels by a PVC pipe connecting the **bottom of the pit** into the nearby well or tube well. Economical designs are available from the **Central Ground Water Board (CGWB)** and UNICEF.

C. Reviving traditional and community structures (Sunder Lal, Ch 3, pp 104-105) — village ponds and *dobas*, **step wells**, earthen **check dams**, **watersheds**, field trenches, **desilting** of water bodies and retaining walls around ponds. The process must begin at village level, with gram panchayats conserving water locally so that it becomes a people's movement — "catch the rain water when and where it falls".

D. Recycling and reuse — recycling of **treated effluents for horticulture, irrigation and other non-domestic purposes**, alongside financially viable sewerage systems in big cities and important pilgrimage and tourist centres (Park 28e, Ch 7, p 573). A **dual water supply** — filtered water for personal use through one set of pipes, unfiltered water for flushing toilets and washing roads through another — once existed in Kolkata, but it is a **historical practice now disapproved**: people may mistake one for the other through ignorance, the **WHO Expert Committee (1965) strongly disapproved of supplying two kinds of water**, and **cross-connection constitutes a serious health hazard** (Park 28e, Ch 15, p 883).

E. Demand-side and inter-sectoral measures (Sunder Lal, Ch 3, p 105) — **changing the cropping pattern** in drought-hit areas; **rationalising free electricity and free water to agriculture** to check groundwater depletion; **intensified afforestation**; and integrating the water conservation programme with agriculture policy and other relevant policies.

Programme and policy support in India

Programme	Year and agency	Relevance to conservation
National Water Supply and Sanitation Programme	Launched 1954 by the Government of India as part of the Health Plan	To assist the States to provide adequate water supply and sanitation facilities in the entire country; provision made in successive Five Year Plans (Park 28e, Ch 15, p 883)
Accelerated Rural Water Supply Programme (ARWSP)	Started 1972, 100 per cent central grant-in-aid	Supplement to the above, targeted at <i>problem villages</i> (Park 28e, Ch 7, p 572; Sunder Lal, Ch 3, p 106)
Swajaldhara	Launched 25 December 2002; two components (Swajaldhara I at gram panchayat / block level, Swajaldhara II at district level)	Community-led participatory programme that explicitly encourages water conservation practices along with rainwater harvesting , and scales up conservation through rainwater harvesting and ground water recharge systems for sustained supply (Park 28e, Ch 7, p 573)

Programme	Year and agency	Relevance to conservation
Jal Shakti Abhiyan	First phase from 15 July 2019 in 256 water-stressed districts of 766, Jal Shakti Mantralaya	Reviving water bodies, recycling water, reducing excessive groundwater use for agriculture, saving every drop for drinking water (Sunder Lal, Ch 3, p 105)
Jal Jeevan Mission (Nal Jal Yojana)	Ministry of Jal Shakti (merged water resources and drinking water ministries); target 2024	Safe and reliable tap water to every rural household at 55 lpcd through a Functional Household Tap Connection (Park 28e, Ch 7, p 573)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Rooftop rainwater recharge pit (draw and label)** — rain cloud; rooftop and courtyard catchment; gutter; PVC down-pipe; a pit filled in three layers labelled from top down sand → gravel → big stones; a PVC pipe from the bottom of the pit running to the adjacent tube well / dug well; below it the water table and the shaded aquifer, with an upward arrow marked "recharge".

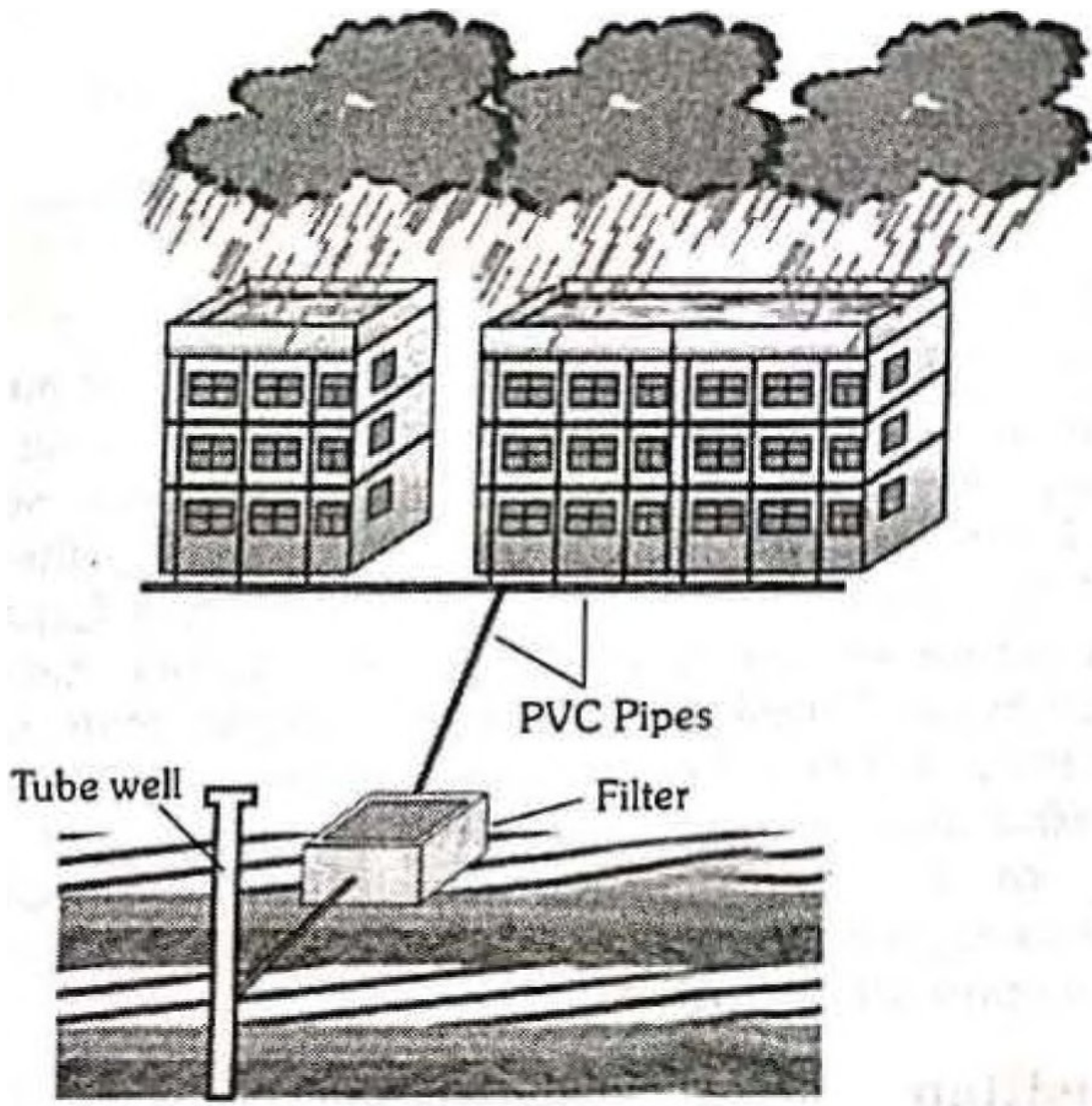
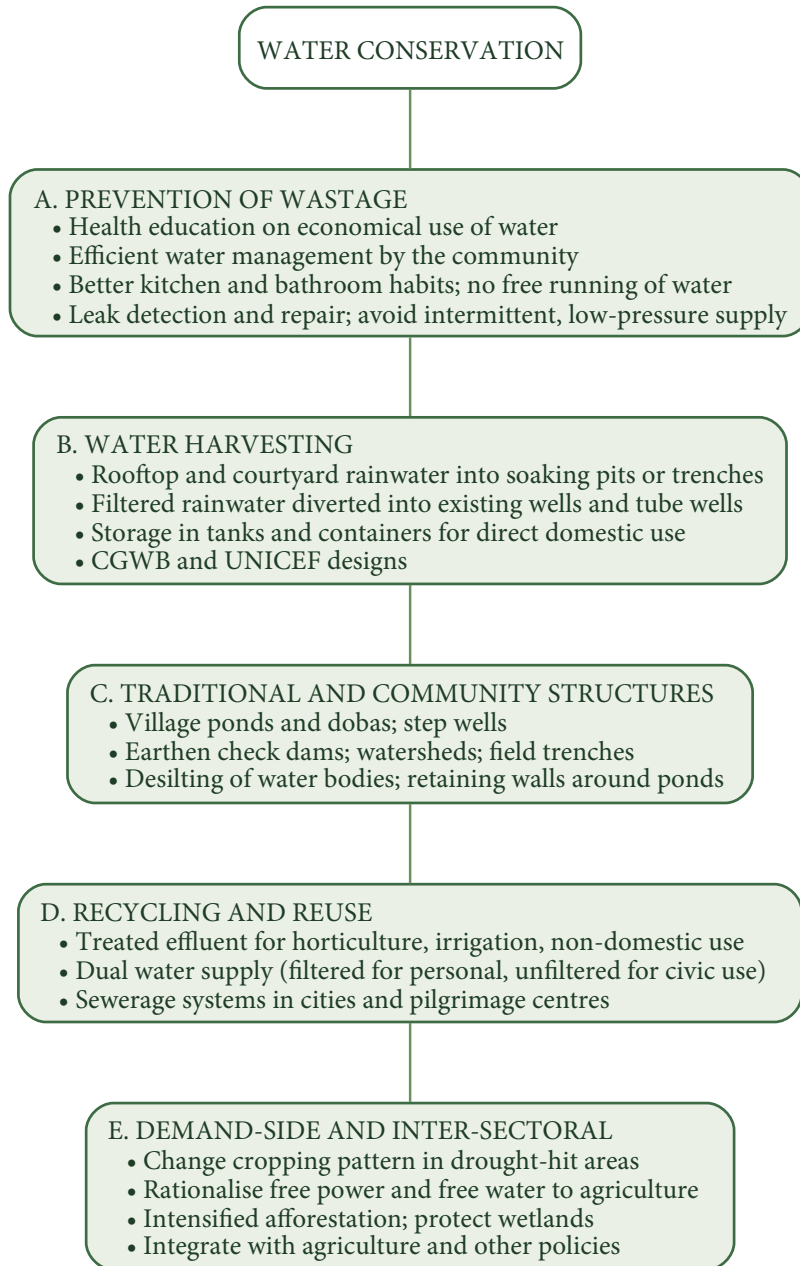
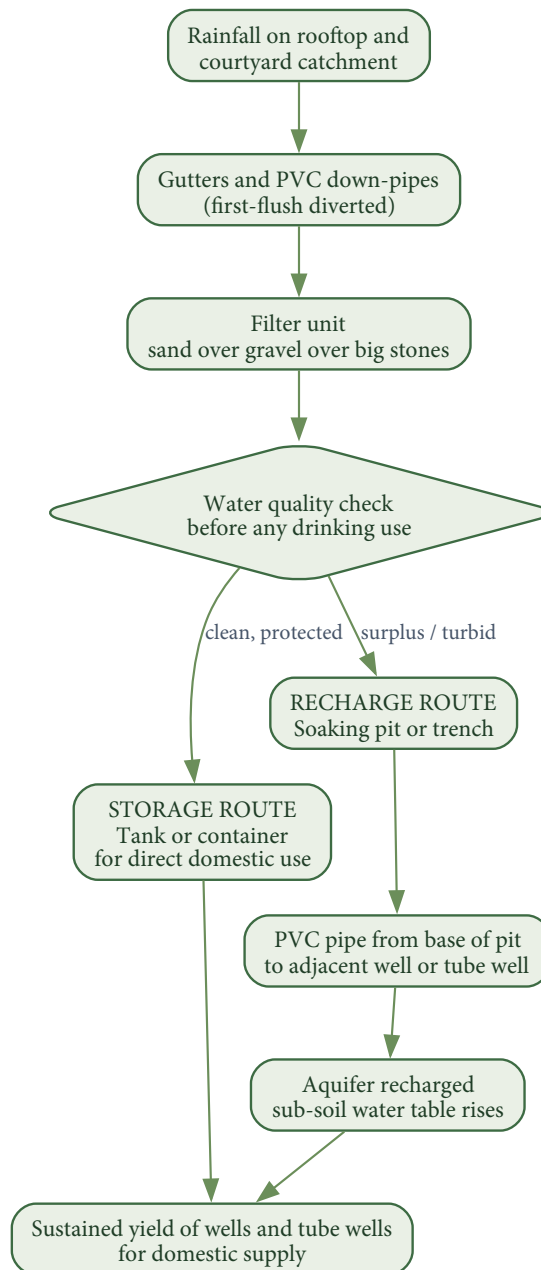


Fig 11 — Water harvesting a tube well (Park's Textbook of Preventive and Social Medicine, 28e, p.885)



The five heads of water conservation



Pathway of rooftop rainwater harvesting — the storage route and the recharge route

High-yield exam pointers

- Park's two heads — water conservation = (a) **prevention of wastage** + (b) **water harvesting** (Park 28e, Ch 15, Appendix IV, p 885). Open the answer with this.
- **Prevention of wastage has exactly three items** — awareness of economical use, efficient water management to cut the community's total requirement, better kitchen and bathroom habits with no free running of water. Nothing else belongs here.
- **Water requirement ladder** — 2 lpcd physiological survival → 40 lpcd old rural norm (one source per 250 persons) → 55 lpcd current design norm and Jal Jeevan Mission FHTC → 150–200 lpcd adequate urban domestic supply.

- ▶ **Availability figures** — freshwater 3 per cent of earth's water, only 1 per cent easily accessible; India's use 70–90 per cent agriculture : 7 per cent industry : 3 per cent domestic; rural India 80 per cent groundwater, urban 80 per cent surface water; only 8 per cent of rainwater harvested.
- ▶ **Recharge pit layers, top to bottom** — sand → gravel → big stones; rooftop water in by PVC pipe, out from the pit base to the well or tube well; designs from **Central Ground Water Board (CGWB)** and UNICEF.
- ▶ **Dates to quote** — National Water Supply and Sanitation Programme 1954; ARWSP 1972; Swajaldhara 25 December 2002; Jal Shakti Abhiyan 15 July 2019 in 256 of 766 districts; Jal Jeevan Mission target 2024.
- ▶ **Two traps to avoid** — never write continuous 24-hour supply or dual water supply as conservation measures; both attach to the **WHO Expert Committee (1965)**, the first recommended, the second disapproved.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only — Park's Appendix IV covers only prevention of wastage and water harvesting, so the traditional structures, recycling and demand-side measures above are grounded in Sunder Lal and Park's health-programmes chapter instead; the indexed corpus does not carry the National Water Policy 2012 or the Atal Bhujal Yojana, and the Jal Jeevan Mission target year of 2024 is quoted as printed in Park 28e.

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 15, Environment and Health — Water, pp 860-861; distribution of water and the National Water Supply and Sanitation Programme, p 883; Appendix IV "Water Conservation" with Fig 11, p 885; Ch 7, Health Programmes in India, pp 572-573); Sunder Lal — Textbook of Community Medicine (Ch 3, Environmental Health Problems, pp 92-93, 104-106); Mahajan & Gupta — Textbook of Preventive and Social Medicine 4e (Ch 6, Physical Environment: Water, p 67).

♦ ♦ ♦

Water quality standards

Asked in: Paper I 12-Nov-2019 — RGUHS MD Community Medicine

Definition — Water for human consumption must be **safe and wholesome**: (a) free from pathogenic agents, (b) free from harmful chemical substances, (c) pleasant to the taste, that is free from colour and odour, and (d) usable for domestic purposes; water failing any of these is **polluted or contaminated** (Park 28e, Ch 15, p 861). **Water quality standards** are the exposure limits for bacteriological, viral, chemical and physical agents adopted by governments or appropriate authorities, and therefore **carry legal force**; their purpose is to minimise all known health hazards, since it is impossible to prevent all pollution (Park 28e, Ch 15, p 874).

1. Criteria, guidelines and standards — the framework

The World Health Organization (WHO) published the *Guidelines for Drinking-water Quality* — volume 1 in 1993 and volume 2 in 1996 (second edition), and volumes I and II in 2011 (fourth edition), superseding those of 1984. They are meant to be used by countries as a **basis for developing their own standards**, applied in the context of prevailing **environmental, social, economic and cultural conditions** (Park 28e, Ch 15, p 874).

Term	What it means	Force
Criterion	A qualitative or quantitative characteristic used to judge whether water is fit for a stated use	Scientific/technical basis only
Guideline value	The concentration of a constituent recommended by WHO as safe for drinking water, derived by expert consensus by applying uncertainty factors (interspecies, intraspecies, study adequacy, nature of toxicity) to the lowest no-observed-adverse-effect level (NOAEL) or lowest-observed-adverse-effect level (LOAEL) — to be adapted nationally	Advisory; not enforceable
Standard	A guideline value formally adopted by a government or appropriate authority as the permissible limit	Legally enforceable
Provisional (P) value	Given where evidence is limited or the value is at the limit of what treatment or analysis can achieve — e.g. arsenic, chromium, manganese, nitrite	Advisory, to be revised

The 2011 guidelines cover **four groups of variables** — I Acceptability, II Microbiological, III Chemical and IV Radiological aspects. Where conditions do not yet permit full compliance, national authorities should set **medium-term targets for progressive improvement** rather than abandon the standard (Park 28e, Ch 15, pp 874, 877).

2. Acceptability standards (physical and organoleptic)

The consumer judges water by its physical characteristics, so a supply must be not only safe but pleasing in appearance, taste and odour — an unsatisfactory one undermines confidence and drives consumers to **less safe alternative sources** (Park 28e, Ch 15, p 874).

Constituent	Level likely to provoke consumer complaint / standard	Public health significance
Turbidity	>4 nephelometric turbidity units (NTU) is noticeable to the naked eye; median <1 NTU required for effective terminal disinfection	Interferes with disinfection and with microbiological determination
Colour	Guideline value up to 15 true colour units (TCU)	Coloured organic matter (humic substances), iron, manganese, industrial waste
Taste and odour	Should be acceptable; no health-based guideline value	May signal pollution, treatment malfunction, or harmful substances
Temperature	No guideline value (control impracticable)	Cool water more palatable; high temperature promotes microbial growth and corrosion
pH	Acceptable range 6.5–8.5; preferably <8.0 for effective chlorine disinfection	pH <7 corrodes distribution metals (leaches lead); above pH 8 chlorine disinfection efficiency falls progressively
Total dissolved solids	1000 mg/L	Taste
Hardness (as CaCO ₃)	Ca taste threshold 100–300 mg/L; >200 mg/L causes scale, scum and excess soap use; <100 mg/L low buffering and corrosive	Drinking water should be moderately hard
Chloride	Standard 200 mg/L; maximum permissible 600 mg/L; complaint level 250 mg/L	Taste, corrosion; indicator of sewage or sea-water contamination
Sulphate	250 mg/L	Taste, corrosion
Sodium	200 mg/L	Taste
Iron	0.3 mg/L	Stains laundry and sanitary ware
Manganese	0.1 mg/L (health-based provisional value 0.4 mg/L)	Staining
Copper	1 mg/L (health-based provisional value 2 mg/L)	Staining
Aluminium	0.2 mg/L	Deposition of aluminium hydroxide floc, discolouration
Ammonia	1.5 mg/L (natural levels usually <0.2 mg/L)	Odour and taste; indicator of bacterial, sewage and animal-waste pollution
Hydrogen sulphide	0.05 mg/L (taste/odour threshold 0.05–0.1 mg/L)	Rotten-egg odour; sulphate reduction in stagnant mains
Zinc	4 mg/L	Appearance, taste; opalescent, greasy film on boiling

(Park 28e, Ch 15, Table 4, pp 874-876.)

Classification of hardness (Park 28e, Ch 15, Table 9, pp 881-882) — softening is recommended when hardness exceeds 3 mEq/L (150 mg/L):

Grade	Hardness (mEq/L)	Hardness (mg/L)
Soft water	<1	<50
Moderately hard	1–3	50–150
Hard water	3–6	150–300
Very hard water	>6	>300

3. Microbiological standards

The **coliform group** is the bacterial indicator of faecal contamination — faecal (typically *Escherichia coli*) and non-faecal (*Klebsiella aerogenes*); **all coliforms are assumed faecal unless a non-faecal origin is proved**. They are chosen because they resist the forces of natural purification better than the water-borne pathogens, so their presence implies probable presence of intestinal pathogens (Park 28e, Ch 15, p 876).

Water sample	Organism	Guideline value
All water intended for drinking	<i>E. coli</i> or thermotolerant coliform bacteria	Must not be detectable in any 100 mL sample
Treated water entering the distribution system	<i>E. coli</i> or thermotolerant coliform bacteria	Must not be detectable in any 100 mL sample
	Total coliform bacteria	Must not be detectable in any 100 mL sample
Treated water in the distribution system	<i>E. coli</i> or thermotolerant coliform bacteria	Must not be detectable in any 100 mL sample
	Total coliform bacteria	Must not be detectable in any 100 mL sample; in large supplies , where sufficient samples are examined, must not be present in 95% of samples taken throughout any 12-month period

(Park 28e, Ch 15, Table 5, pp 876-877.) **Immediate investigative action must be taken if *E. coli* is detected.** Where *E. coli* estimation is not feasible, a thermotolerant coliform count is an acceptable alternative. **Total coliforms are not acceptable indicators of the sanitary quality of rural supplies, particularly in the tropics**, where bacteria of no sanitary significance occur in almost all untreated supplies.

Indicator organism	Interpretation
<i>E. coli</i>	The most precise indicator of recent faecal pollution
Faecal streptococci	Occur regularly in faeces but in much smaller numbers than <i>E. coli</i> ; in doubtful cases their presence is important confirmatory evidence of recent faecal pollution ; highly resistant to drying, so valuable after laying new mains or repairs, and for detecting surface run-off pollution

Indicator organism	Interpretation
<i>Clostridium perfringens</i>	Spores survive longer than coliforms and resist chlorination at usual waterworks doses; presence in the absence of coliforms suggests remote faecal contamination ; presence in filtered supplies indicates deficient filtration

Other microbiological standards (Park 28e, Ch 15, pp 876-877, 881):

- ▶ **Virological** — water must be free of viruses infectious for man; **0.5 mg/L free chlorine residual for at least 30 minutes at pH 8.0** inactivates virus, and is insisted upon in all disinfected supplies in areas suspected of **hepatitis A endemicity**.
- ▶ **Protozoal and helminthic** — no pathogenic intestinal protozoa (*Entamoeba histolytica*, *Giardia* spp., rarely *Balantidium coli*). Chlorine has **no effect on spores, protozoal cysts and helminthic ova** except in higher doses.
- ▶ **Colony (plate) count at the point of consumption** — disinfected supply **0** after 2 days at 37 °C and **20** after 3 days at 22 °C; non-disinfected supply **10** and **100**. A sudden rise in the count from the same source gives the **earliest indication of contamination**.
- ▶ **How compliance is judged** — the **presumptive coliform count** (multiple tube method) giving the most probable number (MPN) per 100 mL, plus tests for faecal streptococci and *C. perfringens*; confirmation in brilliant green bile broth at 37 °C for coliforms and at 44 °C for *E. coli*, almost the only coliform producing gas from lactose at 44 °C. **Membrane filtration** is the standard alternative.

4. Chemical and radiological standards

Few chemicals cause acute illness except after massive accidental contamination, when the water usually becomes undrinkable first on taste, odour and appearance. Chemicals given guideline values are those **hazardous to health, detected relatively frequently and in relatively high concentrations**, and those harmful after **prolonged exposure** — especially cumulative and carcinogenic substances (Park 28e, Ch 15, p 877).

Constituent	WHO guideline value (mg/L)	Note
Arsenic	0.01 (P)	Arsenicosis — pigmentation, keratosis, skin cancer
Barium	0.7	
Boron	2.4	
Cadmium	0.003	
Chromium	0.05 (P)	
Copper	2	
Fluoride	1.5	Optimum for drinking 0.5–0.8 mg/L; deficiency → dental caries, excess → dental and skeletal fluorosis

Constituent	WHO guideline value (mg/L)	Note
Lead	0.01	Most lead in drinking water arises from plumbing in buildings ; central and peripheral neurotoxicity; Group B possible human carcinogen
Manganese	0.4 (P)	
Mercury (total)	0.006	Inorganic → kidney; methyl mercury → central nervous system
Molybdenum	0.07	
Nickel	0.07	
Nitrate (as NO ₃)	50	Guideline set solely to prevent methaemoglobinaemia ; bottle-fed infants under 3 months most susceptible
Nitrite (as NO ₂)	3 (P)	
Selenium	0.04	
Polynuclear aromatic hydrocarbons	0.0002 (0.2 µg/L)	Carcinogenic; removed along with turbidity

(Park 28e, Ch 15, Table 6, pp 877-878; polynuclear aromatic hydrocarbons — Suryakantha, Ch 5, p 61.)
(P) = provisional guideline value.

Guideline values are also laid down for **organic constituents and pesticides**, expressed in micrograms per litre (Park 28e, Ch 15, p 879):

Organic constituent	µg/L	Pesticide	µg/L
Carbon tetrachloride	2	Aldrin / dieldrin	0.03
Dichloromethane	20	Heptachlor and heptachlor epoxide	0.03
1,1-dichloroethene	30	Chlordane	0.2
1,2-dichloroethene	50	DDT	2
Benzene	10	Lindane	2
Toluene	700	Hexachlorobenzene	1
Xylenes	500	Pentachlorophenol	9 (P)
Ethylbenzene	300	Methoxychlor	20
Styrene	20	2,4-D	30
Benzo[a]pyrene	0.7		

Radiological standards — the proposed guideline values are **gross alpha activity 0.5 becquerel per litre (Bq/L)** and **gross beta activity 1.0 Bq/L**; one becquerel is one nuclear disintegration per second. These represent values below which water may be considered potable **without any further radiological examination** (Park 28e, Ch 15, p 880).

5. Chemical indicators of sanitary pollution

Beyond the health-based limits, a set of chemical determinations is used to date and characterise faecal pollution — the classical basis of a "chemical analysis of water" report.

Determination	Prescribed limit	Sanitary interpretation
Chlorides	200 mg/L (maximum permissible 600 mg/L)	Once present, rarely changes — a permanent indicator of pollution with sewage or sea water; local normal range must first be established
Free and saline ammonia	0.05 mg/L	Decomposition of organic matter — recent sewage contamination
Albuminoid ammonia (organic ammonia)	0.1 mg/L	Decomposable organic matter yet to be oxidised
Nitrites	Zero in potable water	Recent pollution by putrefaction; except in deep-well waters, where nitrate is reduced by ferrous salts — nitrite-containing water without iron should be condemned
Nitrates	1.0 mg/L	End product of oxidation of animal matter; high nitrate with very low ammonia and absent nitrite indicates remote (old) contamination
Oxygen absorbed	1 mg/L	Oxygen used to oxidise organic matter; not reliable alone — must be read with free and albuminoid ammonia
Dissolved oxygen	Not less than 5 mg/L	Depletion allows reduction of nitrate to nitrite and sulphate to sulphide, producing odour

(Suryakantha, Ch 5, pp 61-62.)

6. Applying the standards — surveillance of drinking-water quality

Surveillance of drinking water is **essentially a health measure**, intended to protect the public from water-borne disease (Park 28e, Ch 15, p 880).

- ▶ **Sanitary survey** — an on-the-spot inspection and evaluation, by a qualified person, of the **entire water supply system**, to detect and correct faults and deficiencies; it is **essential for interpreting laboratory results** and can never be replaced by them.
- ▶ **Sampling** — done "with the thoroughness of a surgical operation" and the same aseptic precautions, by trained personnel, in strict accordance with the methods and **frequency** laid down in the WHO guidelines or the Indian Council of Medical Research (ICMR) *Manual of Standards of Quality for Drinking Water Supplies*.
- ▶ **Laboratory examination** — the bacteriological battery above, plus residual chlorine estimation and chemical analysis, extended where indicated to toxic metals, pesticides, persistent organic chemicals and radioactivity.
- ▶ **Residual chlorine** — the **operational standard** — minimum free residual chlorine **0.5 mg/L after one hour** of contact; the correct dose = **chlorine demand of that water + 0.5 mg/L**, the margin of safety against contamination during storage and distribution. In emergencies the

residual is raised to **0.2–0.5 mg/L**, since low pressure lets pollutants infiltrate the mains (Park 28e, Ch 15, pp 869-871; Ch 17, p 953).

- **Statutory and administrative functions** — approval of new sources including private supplies; watershed protection; approval of waterworks construction and operation (disinfection after repair, cleaning of storage, certification of operators, cross-connection and back-flow control); central and regional monitoring laboratories; codes of practice for wells, pumps and plumbing; quality control of **bottled water and ice**.

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **The four domains of drinking-water quality standards** — draw a central box "SAFE AND WHOLESOME WATER" with four boxes radiating from it, labelled **I Acceptability** (turbidity <1 NTU, colour 15 TCU, taste and odour acceptable, pH 6.5–8.5), **II Microbiological** (*E. coli* not detectable in any 100 mL), **III Chemical** (arsenic 0.01, fluoride 1.5, lead 0.01, nitrate 50 mg/L), **IV Radiological** (gross alpha 0.5 Bq/L, gross beta 1.0 Bq/L). Beneath, a base bar labelled "Surveillance — sanitary survey + sampling + laboratory + residual chlorine 0.5 mg/L for 1 hour".

WHO Guidelines for Drinking-water Quality, 4th ed (2011)
adapted nationally, in the context of prevailing environmental,
social, economic and cultural conditions, into ENFORCEABLE STANDARDS

I. ACCEPTABILITY ASPECTS

Physical: turbidity (<1 NTU for disinfection), colour (15 TCU),
taste and odour (no value), temperature (no value)
Inorganic: pH 6.5-8.5, hardness, chloride 200 mg/L, sulphate 250,
sodium 200, iron 0.3, manganese 0.1, ammonia, zinc 4,
total dissolved solids 1000 mg/L

II. MICROBIOLOGICAL ASPECTS

Bacteriological: E. coli / thermotolerant coliform NOT DETECTABLE
in any 100 mL; total coliform absent (95% of samples/12 months
in large supplies)
Supplementary indicators: faecal streptococci, C. perfringens
Virological: free chlorine 0.5 mg/L, 30 min, pH 8.0
Protozoal: E. histolytica, Giardia spp, Balantidium coli absent

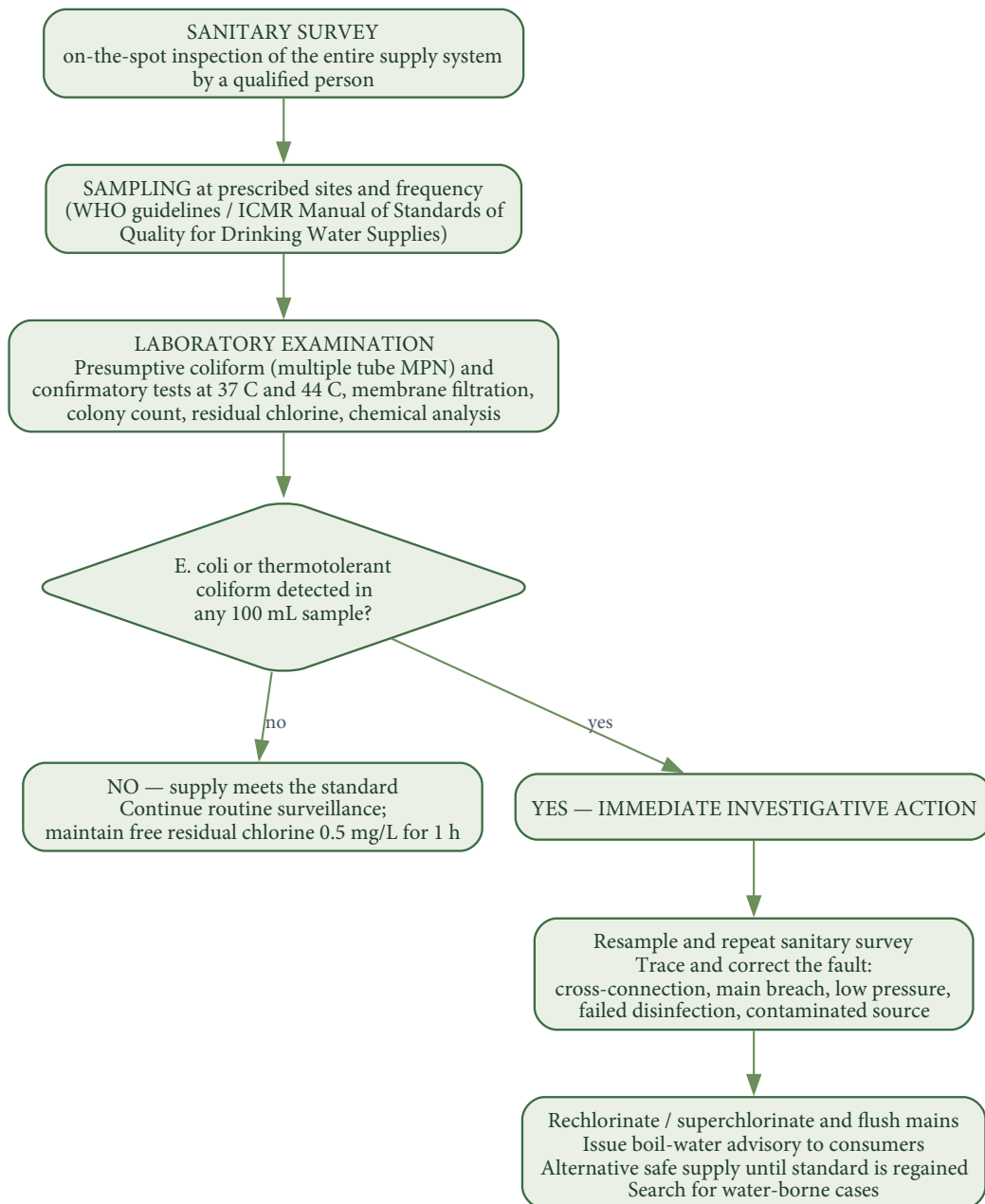
III. CHEMICAL ASPECTS

Arsenic 0.01 (P) Cadmium 0.003 Fluoride 1.5
Lead 0.01 Mercury 0.006 Nitrate (as NO₃) 50
Nitrite 3 (P) Selenium 0.04 Chromium 0.05 (P)
(values in mg/L; P = provisional)

IV. RADIOLOGICAL ASPECTS

Gross alpha activity 0.5 Bq/L
Gross beta activity 1.0 Bq/L
Below these, potable without further radiological examination

The WHO Guidelines for Drinking-water Quality (2011) — four groups of variables and the parameters standardised under each



Surveillance of drinking-water quality — from sanitary survey to corrective action when a standard is breached

High-yield exam pointers

- **Safe and wholesome water (4 criteria):** free of pathogens; free of harmful chemicals; pleasant to taste; usable for domestic purposes. **Standard = guideline + legal force.** WHO 2011, four groups: Acceptability, Microbiological, Chemical, Radiological.
- **The most quoted standard:** *E. coli* or thermotolerant coliforms **not detectable in any 100 mL sample** — in all drinking water, in treated water entering the distribution system and within it; **immediate investigative action** if detected.
- **Numbers to write:** turbidity <1 NTU (median, for terminal disinfection), 4 NTU visible; colour 15 TCU; pH 6.5–8.5; chloride 200 (max 600) mg/L; total dissolved solids 1000 mg/L; hardness soft <50, moderately hard 50–150, hard 150–300, very hard >300 mg/L, soften above 150 mg/L.

- **Chemical values (mg/L):** arsenic 0.01, cadmium 0.003, fluoride 1.5 (optimum 0.5–0.8), lead 0.01, mercury 0.006, nitrate as NO_3 50, nitrite 3, selenium 0.04, chromium 0.05. **Radiological:** gross alpha 0.5 Bq/L, gross beta 1.0 Bq/L.
- **Chlorination standard:** free residual chlorine 0.5 mg/L after 1 hour contact (0.5 mg/L for 30 minutes at pH 8.0 for viruses); correct dose = chlorine demand + 0.5 mg/L; ineffective against spores, cysts and ova.
- **Pollution-dating triad:** free ammonia and nitrites = **recent** pollution; nitrates high with ammonia low and nitrite absent = **remote** pollution; chlorides = **permanent** indicator; *C. perfringens* without coliforms = remote faecal contamination.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only — the Bureau of Indian Standards (BIS) has fixed the upper limit of **fluoride in drinking water at 1 mg/L** for India, lower than the WHO value of 1.5 mg/L, because fluorosis is endemic in 19 states (Sunder Lal, Ch 3, pp 102-104); the two nitrate figures in this answer are not in conflict, as 50 mg/L (as NO_3) is the WHO **health-based** guideline against methaemoglobinaemia while the 1.0 mg/L quoted in the sanitary-chemistry table is the classical **pollution-indicator** limit; the corpus indexed here does not carry the full BIS IS 10500 acceptable-versus-permissible table, so only those Indian values that were retrievable are quoted.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, WHO Guidelines for Drinking-water Quality 2026, WHO small water supplies guideline 2024, WHO/UNICEF Joint Monitoring Programme, ICMR, BIS IS 10500:2012, NFHS-5

Programme and policy updates (India)

- **Jal Jeevan Mission (Har Ghar Jal)**, launched August 2019, is now the operational vehicle through which drinking-water standards are applied in rural India: it promises a **functional household tap connection** delivering **55 litres per capita per day** of water conforming to BIS IS 10500, and runs through a three-tier structure — State Water and Sanitation Mission, District Water and Sanitation Mission and a village-level Water and Sanitation Committee (*Pani Samiti*). Rural coverage rose from **16.63% of households in 2019 to 76.80% by June 2024**, the target being SDG 6.1 [PMID 39570840].

Current national data

- Comparing NFHS-4 (2015–16, 421,665 rural households) with NFHS-5 (2019–21, 390,214 rural households), tap water on the premises rose from 18.33% to 21.62%, households where adult women fetched water from a distant source fell from 82.25% to 75.89%, and those spending >30 minutes collecting water fell from 6.75% to 4.87%; coverage remained **pro-rich** — in the early-mission period the richest wealth quintile had a **26.0 percentage-point higher probability** of a household tap connection than the poorest [PMID 39570840].
- A systematic review of 63 Indian studies in the *Indian Journal of Medical Research* (2026) found the arsenic ingestion hazard quotient >1 in **49% of adult-focused** and **60% of child-focused** studies, and **unacceptable carcinogenic risk ($\geq 1 \times 10^{-4}$) from ingestion** in **72% of adult** and **69%**

of child studies; groundwater carried greater non-carcinogenic risk than surface water, the affected states being Punjab, West Bengal, Haryana, Delhi, Uttarakhand and Jammu & Kashmir, and children were the more vulnerable group because they drink more water per unit body weight [PMID 42555689].

- Field data from North 24 Parganas, West Bengal (2026) showed 63.5% of the 521 people surveyed drinking water graded poor by a water quality index computed against BIS limits, with fluoride above 1 mg/L for 76% and arsenic above 0.01 mg/L for 65% of them; crucially, perception diverged from measurement — 184 individuals wrongly believed unsafe water was safe versus 15 the reverse (McNemar χ^2 12.25, $p = 0.0005$), which is the practical argument for displaying test results to the community rather than merely filing them [PMID 42086805].

Elimination, surveillance and disease status

- The Rourkela (Odisha) cholera outbreak of December 2023, investigated across 34 cases of acute diarrhoeal disease, was attributed to piped-water contamination following unseasonal rainfall — *Vibrio cholerae* in 35% and diarrhoeagenic *E. coli* in 23.5% of samples, the El Tor 7PET strain identified in that area for the first time, virulence genes *ctxAB*, *toxR*, *tcpA*, *ace* and *hlyA* in 100% of isolates, and complete ciprofloxacin resistance carried with an SXT integrative conjugative element — a worked example of climate-linked distribution-system failure compounded by antimicrobial resistance [PMID 40744378] (abstract-only).
- Geogenic fluoride remains a live problem: in the Pambar river basin of south India groundwater fluoride ranged 0.30–4.65 mg/L across north-east monsoon and post-monsoon sampling, against the WHO guideline of 1.5 mg/L, with a combined hazard index >1 in 49% of samples for children, 47% teenagers, 45% women and 39% men and oral intake the dominant exposure route; managed aquifer recharge is proposed as the in-situ remedy and reverse osmosis or nanofiltration as the ex-situ one [PMID 41935562] (abstract-only).

Guideline and legislative updates

- WHO has issued the fourth edition of the *Guidelines for Drinking-water Quality* incorporating the first, second and third addenda (17 June 2026) — this, not the 2011 fourth edition quoted in most textbooks, is the current reference. The addenda changed actual numbers: new guideline values for manganese, microcystins, cylindrospermopsins and saxitoxins, revised values for tetrachloroethene and trichloroethene, and revised assessments for chromium, nickel, silver, asbestos, bentazone and iodine [WHO 2022-Mar]. The third addendum adds prioritisation matrices for microbial water quality testing, revised pathogen-specific fact sheets, updated water-safety-plan and sanitary-inspection guidance, explicit policy attention to small supplies, and revised guidance on pesticides used for vector control in drinking water [WHO 2026-Jun].
- For India the enforceable standard is BIS IS 10500:2012, and its limits come in two tiers: an acceptable limit that a supply should ordinarily meet, and a higher permissible limit that may be used only in the absence of an alternate source. For arsenic these are 0.01 and 0.05 mg/L, for fluoride 1.0 and 1.5 mg/L, while lead (0.01) and cadmium (0.003) carry no relaxation at all — so where no alternative source exists, water containing five times the WHO arsenic guideline value can still be lawfully supplied [BIS IS 10500:2012]. That relaxation is not what is driving the West Bengal figures, however: Central Ground Water Board monitoring for 2019–21 found several blocks of North 24 Parganas above both the WHO guideline of 0.01 mg/L and the BIS

permissible limit of 0.05 mg/L, so those exceedances breach the Indian standard on its own terms [PMID 42086805].

Global and WHO developments

- ▶ **WHO *Guidelines for drinking-water quality: small water supplies* (15 February 2024)** replaces the 1997 Volume 3 and covers household, community and professionally managed small supplies — the categories that dominate rural India. Its three pillars are **health-based standards adapted locally, preventive risk management through water safety plans and sanitary inspection, and independent surveillance**, formally shifting the emphasis from end-product testing to catchment-to-consumer risk management [WHO 2024-Feb].
- ▶ On SDG 6.1, the WHO/UNICEF Joint Monitoring Programme reported **6.0 billion people using safely managed drinking-water services in 2022** (improved source, on premises, available when needed, free from contamination), **1.5 billion with only basic services** and **2.2 billion without safely managed services**; achieving universal safely managed coverage by 2030 requires a **six-fold acceleration** of current progress [WHO/UNICEF JMP 2022].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 15, Environment and Health — "Water Quality: Criteria and Standards", pp 861, 869-871, 874-882; Ch 17, Disaster Management, p 953); Suryakantha — Community Medicine with Recent Advances (Ch 5, Environment and Water, pp 60-63); Sunder Lal — Textbook of Community Medicine (Ch 3, Environmental Health Problems, pp 102-104); WHO Guidelines for Drinking-water Quality, 4th edition, 2011, as reproduced in Park 28e; ICMR Manual of Standards of Quality for Drinking Water Supplies (cited in Park 28e for sampling method and frequency). Coverage note: no captioned figure in the indexed corpus genuinely depicts water quality standards, so the answer uses a label box and two flowcharts instead of an embedded plate.

Recent-advances sources: 39570840 — Rural drinking water supply program and societal development: evidence from the early implementation phase of India's Jal Jeevan Mission, PLoS One 2024, doi:[10.1371/journal.pone.0312144](https://doi.org/10.1371/journal.pone.0312144); 42555689 — Health risks of arsenic and chromium in groundwater and surface water in India: a systematic review, Indian J Med Res 2026, doi:[10.25259/IJMR.218.2026](https://doi.org/10.25259/IJMR.218.2026); 42086805 — Perceived versus measured drinking water quality in rural West Bengal, India, Sci Rep 2026, doi:[10.1038/s41598-026-50626-8](https://doi.org/10.1038/s41598-026-50626-8); 40744378 — Unraveling the 2023 cholera outbreak in Rourkela municipal corporation of Odisha, India, Indian J Med Microbiol 2025, doi:[10.1016/j.ijmmb.2025.100944](https://doi.org/10.1016/j.ijmmb.2025.100944) (abstract-only); 41935562 — Sources of geogenic groundwater fluoride and associated health hazards in Pambar River basin of south India, Environ Res 2026, doi:[10.1016/j.envres.2026.124407](https://doi.org/10.1016/j.envres.2026.124407) (abstract-only). WHO, *Guidelines for drinking-water quality: fourth edition incorporating the first, second and third addenda*, 17 June 2026 (ISBN 978-92-4-012122-5), and the fourth edition incorporating the first and second addenda, 21 March 2022; WHO, *Guidelines for drinking-water quality: small water supplies*, 15 February 2024; WHO/UNICEF Joint Monitoring Programme drinking-water estimates for 2022; Bureau of Indian Standards, IS 10500:2012 *Drinking Water — Specification* (second revision), acceptable limits and permissible limits in the absence of an alternate source, cited directly; NFHS-4/NFHS-5 estimates as reported in PMID 39570840.

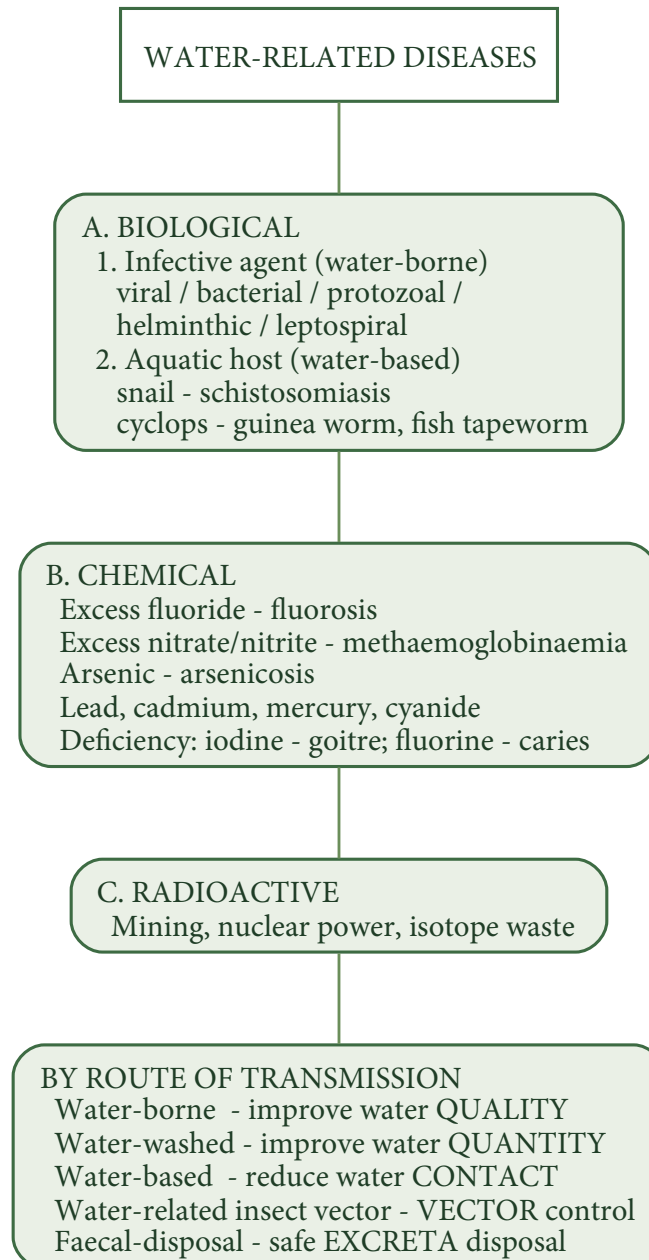
♦ ♦ ♦

Water-related diseases

Asked in: Paper I 29-Jul-2021 — RGUHS MD Community Medicine

Definition — **Water-related diseases** are all illnesses in which water participates in the causal chain: ingestion of contaminated water directly or through food, use of contaminated water for personal hygiene and recreation, an inadequate *quantity* of water, an aquatic intermediate host, or insect vectors breeding in or near water. The term is deliberately wider than the classical **water-borne diseases**, which form only one of its subsets (Park 28e, Ch 15, p 865).

Classification I — by nature of the agent (Park)



Classification of water-related diseases by the nature of the agent (Park 28e, Ch 15, p 865)

Classification II — by route of transmission (White et al. 1972) with the intervention that works

Category	Mechanism	Examples	Intervention with the greatest effect
Water-borne	Water is a passive vehicle ; pathogen ingested in drinking water. Depends also on poor sanitation	Cholera, typhoid, bacillary dysentery, viral hepatitis A and E, leptospirosis, giardiasis, gastroenteritis, poliomyelitis	Improve water quality (protection of source, treatment, chlorination)
Water-washed (water-scarce)	Due to lack of water for washing; poor personal hygiene favours spread	Scabies, skin sepsis and ulcers, yaws, lice and typhus, trachoma, conjunctivitis, bacillary and amoebic dysentery, salmonellosis, worm infestations	Improve water quantity and accessibility ; hand washing with soap
Water-based	An essential part of the agent's life cycle occurs in an aquatic intermediate host ; spread by contact with or ingestion of water	Schistosomiasis (snail), dracunculiasis (cyclops)	Reduce contact with and contamination of surface water; snail and cyclops control
Water-related insect vector	Transmitted by insects that breed in or live close to water	Malaria, lymphatic filariasis, dengue, yellow fever, encephalitides, onchocerciasis, African trypanosomiasis (sleeping sickness)	Source reduction and vector control; siting dwellings away from breeding sites
Faecal-disposal diseases	Agent acquired by eating uncooked fish and other food from faecally polluted water	Clonorchiasis, diphylobothriasis, fasciolopsiasis, paragonimiasis	Safe excreta disposal; cooking of fish and aquatic plants

(Oxford Textbook of Public Health 7e, Ch 2.7, p 479; AFMC Red Book of Community Medicine 2024, Ch 5, pp 92–93)

Chemical water-related disease — the Indian priority

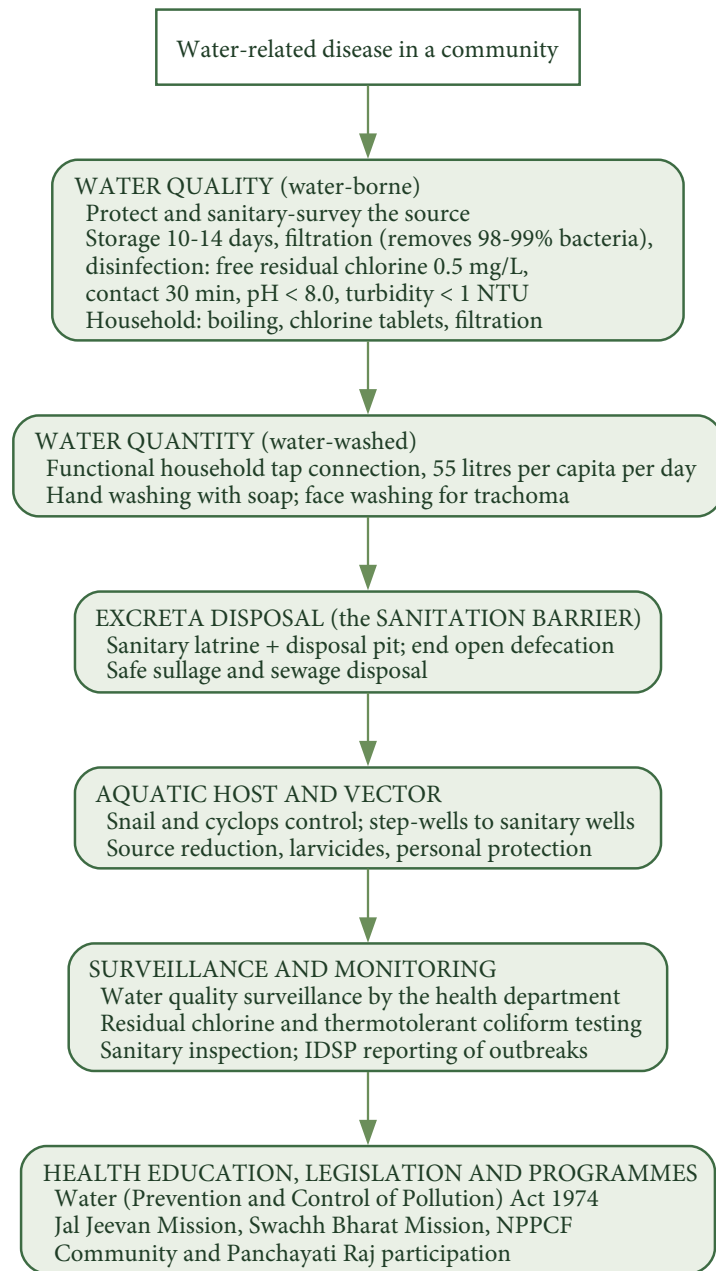
Constituent	Disease produced	WHO guideline value (Park 28e, Ch 15, Table 6)	Indian relevance
Fluoride	Deficiency → dental caries; excess → dental (mottled enamel) and skeletal fluorosis	1.5 mg/L	Bureau of Indian Standards upper limit 1 mg/L ; fluorosis endemic in 19 states (Sunder Lal, Ch 3, pp 102–104)
Nitrate (as NO ₃)	Methaemoglobinaemia — "cyanosis in the infant"; from fertiliser run-off into surface water	50 mg/L (nitrite as NO ₂ 3 mg/L, provisional)	Agricultural belts with intensive fertiliser use
Arsenic	Arsenicosis — hyperkeratosis, pigmentation, skin cancer	0.01 mg/L (provisional)	Ganga–Brahmaputra plain (West Bengal, Bihar)

Constituent	Disease produced	WHO guideline value (Park 28e, Ch 15, Table 6)	Indian relevance
Lead / cadmium / mercury	Cumulative toxicity, carcinogenic and teratogenic effects	0.01 / 0.003 / 0.006 mg/L	Industrial effluent, corrosion of pipes
Hardness	Appears to have a beneficial effect against cardiovascular disease	No health-based value	Park 28e, Ch 15, p 866

Problem statement — burden

- ▶ Developing countries carry the heaviest burden of water-related disease, the largest single component being the **diarrhoeal diseases** (Park 28e, Ch 15, p 865).
- ▶ Diarrhoeal disease kills about **4,43,832 children under 5 years** and **50,851 children aged 5–9 years**, and accounts for **8 per cent of all under-five deaths** worldwide; under-fives suffer about **1.7 billion episodes a year**, 123 million clinic visits, 9 million hospitalisations and a loss of **62 million disability-adjusted life years (DALYs)** (Park 28e, Ch 5).
- ▶ In **India**, acute diarrhoeal disease accounts for about **10 per cent of deaths in the under-5 age group**; in **2022** about **5.5 million cases with 246 deaths** were reported (Park 28e, Ch 5).
- ▶ Nearly **90 per cent** of diarrhoea deaths worldwide are attributable to unsafe water, inadequate sanitation and poor hygiene; in **2015**, 844 million people lacked a basic drinking-water service and 2.3 billion lacked a basic sanitation service (Oxford Textbook of Public Health 7e).
- ▶ Improvements in **water, sanitation and hygiene (WASH)** could reduce this burden by an estimated **58 per cent**; improved water supply alone reduces diarrhoea morbidity by 6–25 per cent, and water plus sanitation reduces cholera morbidity by 80–100 per cent (Oxford 7e, Ch 2.7; Sunder Lal, Ch 3 and Ch 8).
- ▶ **NFHS-5 (2019–21), India: 95.9 per cent** of households use an improved source of drinking water, but only **32.9 per cent** have water piped into the dwelling or yard; **58 per cent do not treat water before drinking** (66 per cent rural); **69.3 per cent** use an improved, non-shared sanitation facility and **19 per cent** have no facility at all.

Prevention and control



Control of water-related disease - breaking the chain at each transmission route

Standards to be met and monitored

Indicator	Standard	Source
<i>E. coli</i> or thermotolerant coliform bacteria	Must not be detectable in any 100 mL sample — in all water intended for drinking, in treated water entering the distribution system, and in treated water within the distribution system	Park 28e, Ch 15, Table 5
Free residual chlorine	0.5 mg/L after ≥30 minutes contact at pH <8.0 (insisted upon where hepatitis A is endemic); 0.2 mg/L for half an hour elsewhere	Park 28e, Ch 15, p 876

Indicator	Standard	Source
Free residual chlorine in a cholera-affected area	0.5 mg/L at all sampling points in a piped system; 1.0 mg/L at standposts ; 2.0 mg/L in tanker trucks at filling	Sunder Lal, Ch 8, pp 410–411
Turbidity	<5 nephelometric turbidity units (NTU); ideally <1 NTU, and ≤1 NTU before disinfection	Sunder Lal, Ch 3, p 101; Park 28e, Ch 15
pH	<8.0 for effective chlorination	Sunder Lal, Ch 3, p 101

Investigation of a water-borne outbreak

- ▶ **Verify the diagnosis and confirm the existence of an outbreak** against the expected (endemic) level for that place and season.
- ▶ **Define a case** and prepare a **line list**; conduct active house-to-house search with the auxiliary nurse midwife (ANM) and accredited social health activist (ASHA).
- ▶ **Describe by time, place and person** — plot an **epidemic curve** (an explosive single-peaked curve within one incubation period suggests a common point source) and **spot-map** cases against the water distribution network.
- ▶ **Formulate a hypothesis** on the source and the point of contamination; look actively for *other* water-borne diseases, since the same faecal contamination causes more than one (Sathe & Sathe, Ch 5, p 58).
- ▶ **Test the hypothesis** — sanitary survey of the source, storage and distribution system for leaky joints and cross-connections with sewer lines, and for sewer lines running on top of or parallel to water mains (Sathe & Sathe, Ch 24, pp 265–266); water samples for thermotolerant coliform count and residual chlorine; clinical specimens (rectal swab in Cary-Blair medium for cholera).
- ▶ **Institute control measures without waiting for laboratory confirmation** — condemn the suspected source, super-chlorinate wells (twice daily during an epidemic), distribute chlorine tablets and oral rehydration salts, treat cases, and health-educate.
- ▶ **Report under the Integrated Disease Surveillance Programme (IDSP)** and submit the final report after the outbreak is over; follow with long-term provision of safe water and sanitary latrines.

(Sathe & Sathe, Ch 5 and Ch 24; Sunder Lal, Ch 8; Suryakantha, Ch 20)

National programmes and legislation

Programme / law	Year / status	Key content
Jal Jeevan Mission (Har Ghar Jal / Nal Jal Yojana)	Target year 2024	Functional Household Tap Connection (FHTC) delivering at least 55 litres per person per day of prescribed quality to every rural household by 2024; rural service norm raised from 40 to 55 litres per capita per day; source sustainability, greywater management, rainwater harvesting. Restructured from, and subsumed, the National Rural Drinking Water Programme (Park 28e, Ch 7, p 573)

Programme / law	Year / status	Key content
Swachh Bharat Mission	Launched 2 October 2014	Two sub-missions — Gramin (Ministry of Jal Shakti) and Urban; open-defecation-free gram panchayats, conversion of insanitary and single-pit latrines, solid and liquid waste management (Park 28e, Ch 7, pp 573–574)
National Programme for Prevention and Control of Fluorosis (NPPCF)	2008 (XIth Five-Year Plan)	Surveillance of fluorosis, training of health personnel, health education, district-level diagnostic and fluoride-testing laboratories, management and rehabilitation of cases; implemented in 163 districts of 19 states and 1 union territory , with 85 district laboratories by 2020–21 (Sunder Lal, Ch 3, p 104)
Water (Prevention and Control of Pollution) Act	1974, amended 1988	Constitutes Central, State and Joint Water Pollution Control Boards with powers of entry, sampling and prosecution (Park 28e, Ch 15; AFMC Red Book 2024, Ch 5)
National Viral Hepatitis Control Programme (NVHCP)	Rolled out 28 July 2018, under the National Health Mission	Covers the water-borne hepatitis A and E alongside B and C: aims to eliminate hepatitis C by 2030, reduce morbidity and mortality from hepatitis B and C, and reduce risk and morbidity from hepatitis A and E; National Centre for Disease Control, Delhi is the nodal centre (Sunder Lal, Ch 8, pp 417–418)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **The sanitation barrier** — draw a circle labelled **EXCRETA** on the left, five arrows fanning out to circles labelled **WATER, FINGERS, FLIES, SOIL, FOOD**, and a circle on the right labelled **PROTECTED HOST**. Draw one thick vertical line between excreta and the five routes, labelled **SANITATION BARRIER** (a sanitary latrine plus a disposal pit). Caption: *Sanitation barrier to transmission of faecal-borne diseases* (Park 28e, Ch 15, Fig 2, p 911).
- **The five-F transmission diagram** — the same circle of **FAECES** leading through **FLUIDS, FINGERS, FLIES, FIELDS/SOIL → FOOD → NEW HOST**, with the interruption points labelled *sanitation, water quality, water quantity, hand washing*.

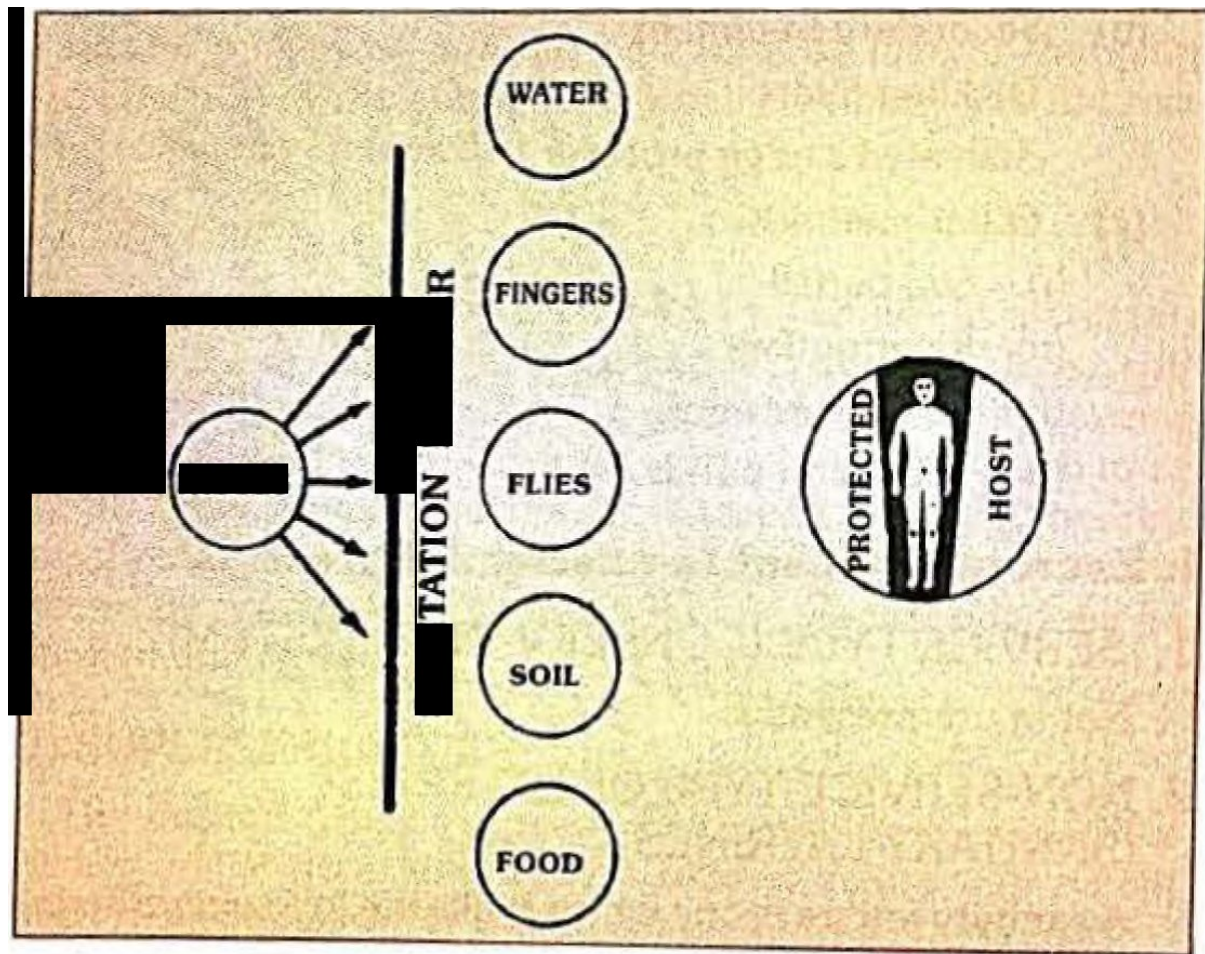


Fig 2 — Sanitation barrier to transmission of faecal-borne diseases (1) (Park's Textbook of Preventive and Social Medicine, 28e, p.911)

High-yield exam pointers

- ▶ Mnemonic for the transmission-route classification: **BORNE – WASHED – BASED – VECTOR – FISH** (water-borne, water-washed, water-based, water-related insect vector, faecal-disposal); attribute it to *White, Bradley and White (1972)*.
- ▶ The single discriminator to write: **water-borne = poor water QUALITY**; **water-washed = insufficient water QUANTITY**. Marks are lost when these two are confused.
- ▶ Bacteriological standard verbatim: *E. coli* or thermotolerant coliform **must not be detectable in any 100 mL sample**; free residual chlorine **0.5 mg/L for 30 minutes at pH <8.0**, turbidity <1 NTU before disinfection.
- ▶ Chemical cut-offs: fluoride WHO 1.5 mg/L, BIS 1 mg/L; nitrate (as NO₃) 50 mg/L; arsenic and lead 0.01 mg/L.
- ▶ Dated Indian figures: NFHS-5 (2019–21) — improved drinking-water source 95.9%, improved non-shared sanitation 69.3%, no facility 19%, water untreated before drinking 58%; acute diarrhoeal disease, India 2022 — 5.5 million cases, 246 deaths, about 10% of under-5 deaths.
- ▶ Jal Jeevan Mission entitlement: 55 litres per capita per day through a Functional Household Tap Connection, to every rural household by 2024.
- ▶ Guinea-worm disease, the completed Indian water-based disease eradication: last case **July 1996** (village Aau, Jodhpur district, Rajasthan); **zero cases since August 1996**; the International

Commission for the Certification of Dracunculiasis Eradication recommended certification in **February 2000**, and India was **certified free of dracunculiasis in 2000** (Park 28e, Ch 7, p 572; Sunder Lal, Ch 8, p 419).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not to be written in the exam: the Jal Jeevan Mission subsumed the earlier National Rural Drinking Water Programme, which itself descends from the National Water Supply and Sanitation Programme (1954), the Accelerated Rural Water Supply Programme (1972), the Rajiv Gandhi National Drinking Water Mission (1991) and Swajaldhara (2002) — give the current name first and the former name in brackets once if you mention the lineage. Control of endemic fluorosis and of arsenicosis are separate national efforts (the National Programme for Prevention and Control of Fluorosis and the drinking-water quality sub-missions), not components of the National Viral Hepatitis Control Programme. Suryakantha additionally records that the World Health Organization formally declared India a guinea-worm-free country on 15 February 2001, after the February 2000 certification recommendation.

SOURCE & COVERAGE

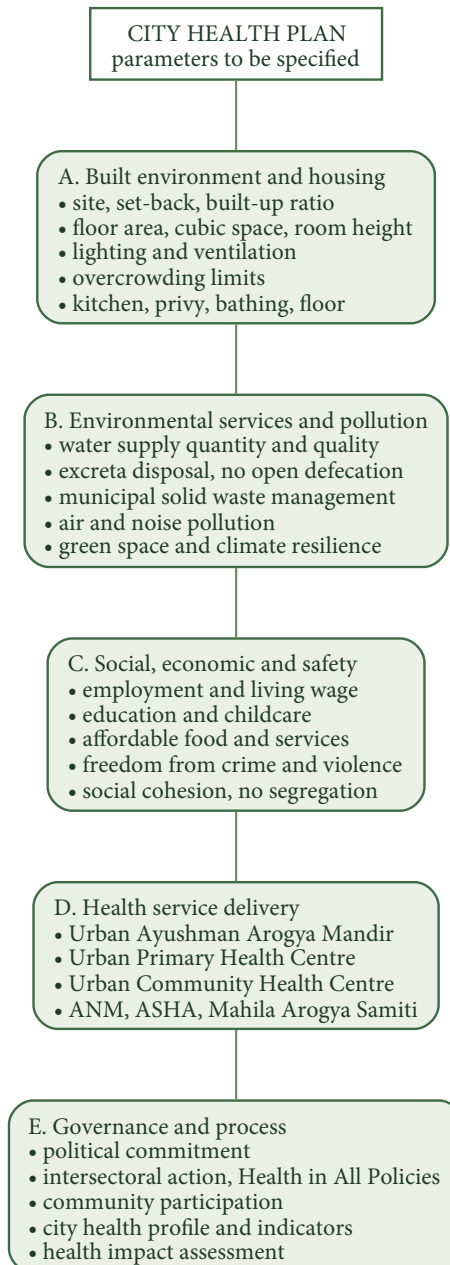
Park's Textbook of Preventive and Social Medicine 28e (Ch 5 Epidemiology of Communicable Diseases; Ch 7 Health Programmes in India, pp 572–574; Ch 15 Environment and Health, pp 861–880, including Tables 5 and 6 and Fig 2 p 911); Oxford Textbook of Public Health 7e (Ch 2.7 Water and Sanitation, pp 479–486); Sunder Lal — Textbook of Community Medicine (Ch 3 Environmental Health Problems, pp 94–104; Ch 8 Epidemiology of Communicable and Non-Communicable Diseases, pp 410–419); AFMC Red Book of Community Medicine 2024 (Ch 5 Water Supply, pp 92–93); Suryakantha — Community Medicine with Recent Advances (Ch 20 Epidemiology of Communicable Diseases; Ch 43 National Health Programmes, p 851); Sathe & Sathe — Epidemiology and Management for Health Care (Ch 5 Investigation of an Outbreak, pp 51–58; Ch 24 Control of Vehicle-Borne Diseases, pp 265–266); National Family Health Survey-5 India Report (2019–21).



Planning healthy cities

Asked in: Paper IV 18-Nov-2019 — RGUHS MD Community Medicine

Definition — A healthy city is "one that is continually creating and strengthening those community resources which enable people to mutually support each other in performing all the functions of life and achieving their maximum potential" (Hancock and Duhl, 1988) (Textbook of International Health, Ch 10, p 560). The World Health Organization (WHO) Healthy Cities project began as a small WHO initiative arising from the *Beyond Healthcare* conference at Toronto, Canada in 1984 and grew into a worldwide movement in which enrolled cities draft a **city health plan** that re-links town planning to public health (Oxford Textbook of Public Health 7e, Ch 99, pp 3770–3771; Ch 58, p 2159). Planning parameters are therefore not health-department parameters alone: they are measurable specifications spread across the built environment, environmental services, the social and economic environment, health service delivery, and governance. India is now 36 per cent urban (2024) (Park 28e, Ch 8, pp 583–584), which makes these parameters an operational, not an academic, agenda.



The five parameter domains to be specified in a city health plan

A. Built-environment and housing parameters

The WHO Expert Committee criteria for healthful housing are the physiological, protective and psychosocial specification a city's housing stock must satisfy: physical protection and shelter; adequate provision for cooking, eating, washing and excretory functions; design and use that prevent the spread of communicable diseases; protection from noise and pollution; freedom from unsafe construction and toxic materials; and encouragement of personal and community development and social relationships (Park 28e, Ch 15, pp 905–906). These translate into the Environmental Hygiene Committee (1947) numeric standards used in India.

Parameter	Standard to be planned for
Site	Elevated, not liable to flooding; independent access to a street of adequate width; away from mosquito and fly breeding places; away from dust, smoke, smell, excessive noise and traffic; dry, well-drained soil; subsoil water below 10 ft (3 m); made-soil unsuitable for 20–25 years (Park 28e, Ch 15, p 906)
Set-back / built-up ratio	Open space all round the dwelling; built-up area $\leq$ one-third of plot in rural areas, up to two-thirds in urban areas where land is costly (Park 28e, Ch 15, p 906)
Floor area	Living room $\geq$ 120 sq ft (12 sq m) if occupied by more than one person; $\geq$ 100 sq ft (10 sq m) for a single occupant; floor area per person $\geq$ 50 sq ft, optimum 100 sq ft (Park 28e, Ch 15, p 906)
Cubic space	$\geq$ 500 c ft per capita, preferably 1,000 c ft, unless mechanical air replacement is provided (Park 28e, Ch 15, p 906)
Rooms and roof	$\geq$ 2 living rooms, at least one closable for security; roof height $\geq$ 10 ft (3 m) in the absence of air-conditioning (Park 28e, Ch 15, pp 906–907)
Windows / lighting	$\geq$ 2 windows per living room, at least one opening on to open space; window area = 1/5 of floor area, doors + windows = 2/5; sill $\leq$ 3 ft (1 m) above floor; daylight factor $>$ 1 per cent over half the floor area (Park 28e, Ch 15, pp 906–907)
Overcrowding — persons per room	1 room = 2 persons; 2 rooms = 3; 3 rooms = 5; 4 rooms = 7; 5 or more rooms = 10 (+2 for each further room) (Park 28e, Ch 15, pp 906–907)
Overcrowding — floor space	$\geq$ 110 sq ft = 2 persons; 90–100 sq ft = 1½; 70–90 sq ft = 1; 50–70 sq ft = ½; under 50 sq ft = nil. Infant $<$ 12 months not counted; child 1–10 years = half a unit (Park 28e, Ch 15, pp 906–907)
Sex separation	Overcrowding exists if 2 persons over 9 years of age, of opposite sexes and not husband and wife, must sleep in the same room (Park 28e, Ch 15, pp 906–907)
Kitchen, privy, bathing	Separate kitchen with impervious floor, water supply, sink and drainage; a sanitary privy exclusive to and readily accessible from each house; exclusive bathing and washing facility with privacy (Park 28e, Ch 15, pp 906–907)
Floor and walls	Impermeable, smooth, crack-free, damp-proof floor; plinth 2–3 ft (0.6–1 m); walls weather-resistant, low heat capacity, not harbouring rats and vermin (Park 28e, Ch 15, pp 906–907)
Housing supply	New house construction at 10 houses per 1,000 persons per year (United Nations expert body) (Park 28e, Ch 15, p 908)

B. Environmental service and pollution parameters

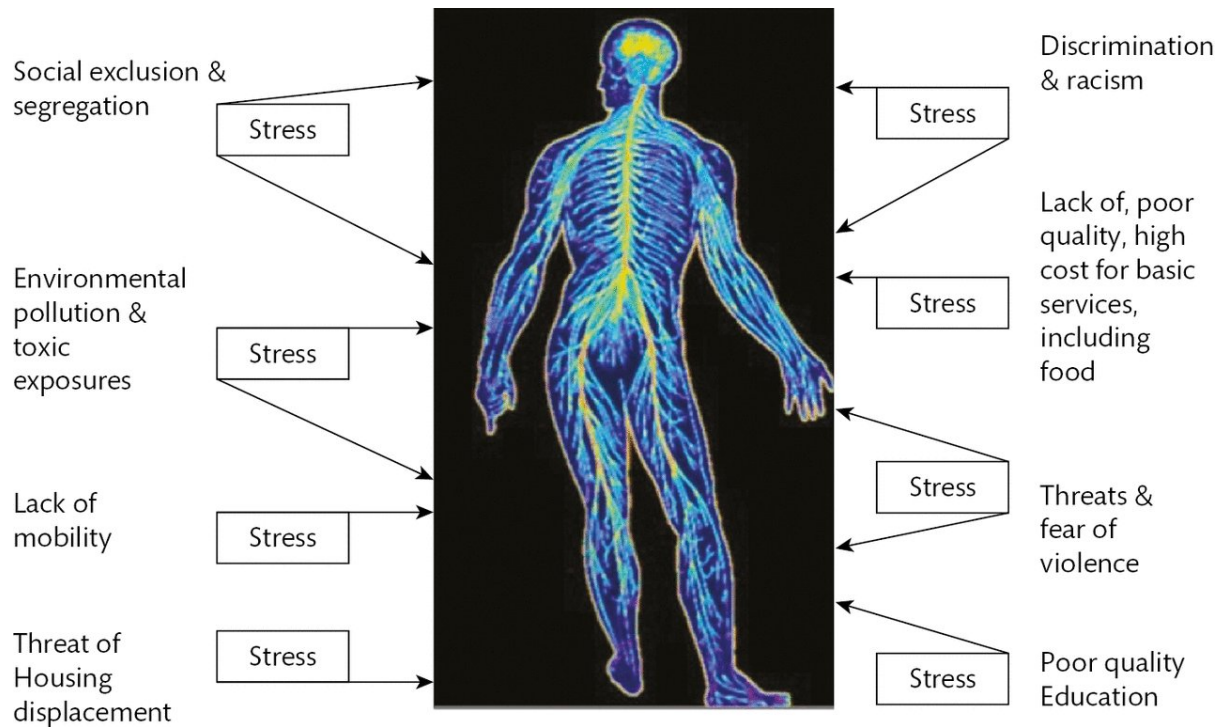
Parameter	Standard / action to be planned
Water supply	150–200 litres per capita per day is adequate for all urban domestic purposes; source safe yield must serve the population at the end of a 10–50 year design period (Park 28e, Ch 15, p 861)
Excreta disposal	Elimination of open defecation and eradication of manual scavenging; household toilets with conversion of insanitary latrines to pour-flush, community toilets for congested and informal settlements, and public toilets for the floating population — Swachh Bharat Mission-Urban (Park 28e, Ch 7, p 574)

Parameter	Standard / action to be planned
Solid waste	Modern and scientific municipal solid waste management, with capacity augmentation of urban local bodies and an enabling environment for private participation in capital and operating expenditure (Park 28e, Ch 7, p 574)
Air quality	Vehicle emissions exacerbate respiratory disease and raise cardiopulmonary mortality; urban particulate matter is associated with cardiovascular death and asthma (Oxford Textbook of Public Health 7e, Ch 58, pp 2161–2162)
Noise	Chronic noise exposure harms sleep, temperament, hearing and blood pressure, and can cause developmental delay in children (Oxford Textbook of Public Health 7e, Ch 58, pp 2161–2162)
Green and open space	Trees and green space remove air pollution and mitigate the urban heat island; clean, safe parks increase physical activity (Oxford Textbook of Public Health 7e, Ch 58, pp 2161–2162)
Transport	Safe roadways, sidewalks and bicycle lanes reduce pedestrian injury and facilitate physical activity; public transit provides access to employment, education, parks and health care (Oxford Textbook of Public Health 7e, Ch 58, pp 2161–2162)
Climate resilience	Heat islands are commoner in dense areas with limited green space; extreme weather worsens the vulnerability of the urban poor on precarious land; climate change is increasing urban dengue, malaria and Zika (Oxford Textbook of Public Health 7e, Ch 58, p 2162)

C. Social, economic and safety parameters

Parameter	Health consequence if unaddressed
Affordable, safe, stable, socially integrated housing	Crowded, substandard housing raises risk of infection, respiratory disease, fires and stress; unaffordable rent forces trade-offs between housing, food and medical care; residential segregation concentrates disadvantage (Oxford Textbook of Public Health 7e, Ch 58, pp 2162–2163)
Employment — meaningful, safe, living wage	Unemployment is a source of chronic stress; higher income is associated with better overall health, reduced mortality and higher emotional stability (Oxford Textbook of Public Health 7e, Ch 58, pp 2162–2163)
Childcare, education, health-care access	Quality childcare raises later educational attainment and earnings; education improves health literacy; timely primary care prevents serious illness (Oxford Textbook of Public Health 7e, Ch 58, pp 2162–2163)
Affordable, quality goods and services	Neighbourhood grocery stores support nutritious diets; local financial institutions help families build wealth (Oxford Textbook of Public Health 7e, Ch 58, pp 2162–2163)
Protection from crime and violence	Fear, stress, anxiety and unhealthy coping — overeating, smoking, alcohol and drug abuse; fear of crime keeps children indoors, raising indoor allergen exposure (Oxford Textbook of Public Health 7e, Ch 58, pp 2162–2163)
Social cohesion and political power	Support buffers stress, prevents isolation and reduces risk of early death; severed social ties contribute to low birth weight (Oxford Textbook of Public Health 7e, Ch 58, pp 2162–2163)

Parameter	Health consequence if unaddressed
Freedom from spatial segregation	Segregated communities carry disproportionate exposure and susceptibility to deprivation, toxic substances and hazardous conditions (Oxford Textbook of Public Health 7e, Ch 58, pp 2161–2162)



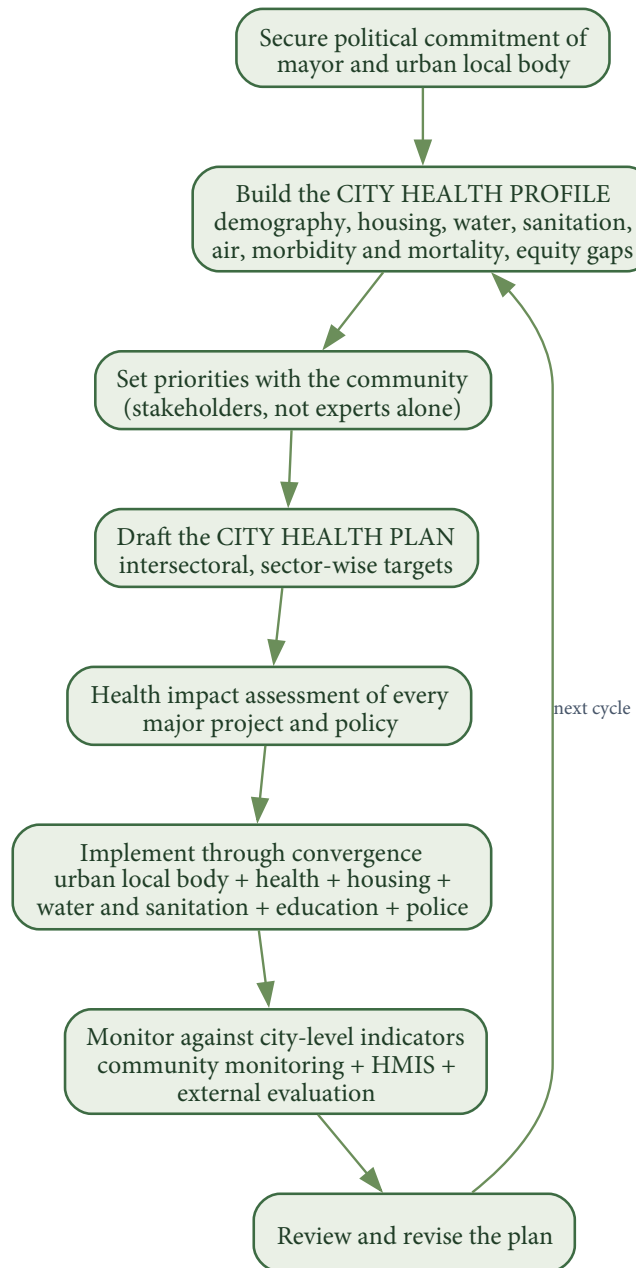
Urban stressors a healthy-city plan must address (Oxford public health 7th ed, p.2165)

D. Health service delivery parameters (Indian norms, National Urban Health Mission)

Level	Population norm and function
Community volunteer — ASHA / Link Worker	1 per 1,000–2,500 urban poor population, covering 200–500 households (Park 28e, Ch 7, pp 535–536)
Mahila Arogya Samiti (MAS)	1 per 50–100 households, with an annual grant of Rs 5,000 (Park 28e, Ch 7, pp 535–536)
Auxiliary nurse midwife (ANM)	1 per 10,000 population, with weekly outreach sessions in her area (Park 28e, Ch 7, p 536)
Urban Ayushman Arogya Mandir (U-AAM)	1 per 15,000–20,000 population, linked to the nearest Urban Primary Health Centre (Park 28e, Ch 7, p 536)
Urban Primary Health Centre (U-PHC)	1 per 30,000–50,000, sited within or within half a kilometre of a slum, serving 25,000–30,000 slum population; up to 75,000 in very high density areas, or 5,000–10,000 for isolated slum clusters (Park 28e, Ch 7, p 536)
Urban Community Health Centre (U-CHC)	30–50 bedded referral unit per 2,50,000 population in cities above 5 lakh, acting as a satellite hospital for 4–5 U-PHCs (Park 28e, Ch 7, p 536)

E. Governance and process parameters

- ▶ **Political commitment and a common vision** — the movement's defining emphasis is on process: building political commitment locally to push health higher up the urban policy agenda (Oxford Textbook of Public Health 7e, Ch 99, p 3771).
- ▶ **Stakeholder involvement and community empowerment** — a wide range of stakeholders including local communities, so that the agenda is empowered and participatory rather than expert-led (Oxford Textbook of Public Health 7e, Ch 99, p 3771).
- ▶ **A written city health plan** — strategy development within local government, based on intersectoral partnerships and stakeholder engagement (Oxford Textbook of Public Health 7e, Ch 99, p 3771).
- ▶ **Health information systems and indicators** — indicators turn data into usable information, give a synthesis view of environment and health trends, track progress, prioritise problems and support evaluation (Oxford Textbook of Public Health 7e, Ch 99, p 3771 and p 3763).
- ▶ **Integration of health into other urban policies, with health impact assessment** — health must fit into the planning processes of other sectors rather than the reverse; Health in All Policies makes this explicit, as in Richmond, California's 2014 ordinance (Oxford Textbook of Public Health 7e, Ch 99, p 3771; Ch 58, pp 2169–2170).
- ▶ **Four goals every healthy-city project must attain** — assure organisational capacity, enhance information, establish initiatives, and create networks, through public, private and voluntary partnerships (Maxcy-Rosenau-Last 15e, Ch 75, p 1273).
- ▶ **Sustainability principles for the city as a system** — local management (local planning and accountability), policy integration (intersectoral collaboration on shared responsibility), ecosystems thinking (energy, natural resources and waste as chains of activity requiring restoration and closure), and cooperation and partnership across levels and interests (Textbook of International Health, Ch 10, pp 560–561).
- ▶ **Monitoring and evaluation** — city and state urban health missions monitor through community-based monitoring, a health management information system for reporting and feedback, and external evaluation (Suryakantha, Ch 37, p 803).



The healthy-city planning cycle, from city health profile to review

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Domains of a city health plan (wheel or box diagram).** Labels: *Built environment and housing; Water, sanitation and solid waste; Air, noise, green space and climate; Transport and road safety; Employment, education and food security; Safety from crime and violence; Social cohesion and equity; Urban health services (U-AAM → U-PHC → U-CHC); Governance — political commitment, intersectoral action, community participation, indicators.*
- ▶ **Housing standards diagram (section through a dwelling and plot).** Labels: *set-back (open space all round); built-up area $\leq 2/3$ of plot in urban areas; plinth 0.6–1 m; roof height ≥ 3 m; window area = $1/5$ floor area; window sill ≤ 1 m; floor area ≥ 12 sq m per living room; cubic space ≥ 500 c ft per person; separate kitchen; sanitary privy; subsoil water > 3 m below site.*

High-yield exam pointers

- ▶ **Definition to write verbatim:** Hancock and Duhl, 1988 — "continually creating and strengthening those community resources..."; project born of the *Beyond Healthcare* conference, Toronto, 1984, WHO European Region.
- ▶ **Mnemonic for the five domains — "BEST-G":** Built environment, Environmental services, Social and economic, Treatment (health services), Governance.
- ▶ **Numbers examiners look for:** water 150–200 litres per capita per day (urban); floor area ≥ 12 sq m per living room, ≥ 10 sq m single occupant; cubic space ≥ 500 c ft, preferably 1,000 c ft; window area $1/5$ of floor area; daylight factor > 1 per cent; subsoil water > 3 m below site; built-up area $\leq 2/3$ of plot in urban areas; new houses 10 per 1,000 persons per year (United Nations).
- ▶ **Overcrowding, all three criteria:** persons per room (1 room = 2, 2 = 3, 3 = 5, 4 = 7, 5+ = 10); floor space (≥ 110 sq ft = 2 persons, down to < 50 sq ft = nil); sex separation (2 persons over 9 years, opposite sex, not husband and wife, in one room).
- ▶ **National Urban Health Mission ladder:** U-AAM 15,000–20,000 → U-PHC 30,000–50,000 (within half a km of the slum) → U-CHC 30–50 beds per 2.5 lakh; ASHA 1 per 1,000–2,500 urban poor; MAS per 50–100 households.
- ▶ **Four project goals (Maxcy-Rosenau-Last):** organisational capacity, information, initiatives, networks.
- ▶ **Dated Indian figure:** India was 36 per cent urban in 2024; Swachh Bharat Mission-Urban objectives — end open defecation, eradicate manual scavenging, scientific municipal solid waste management.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Coverage note: the retrieved corpus grounds the healthy-city concept, its process parameters and every housing, water, sanitation and service norm quoted above, but it does not carry the often-quoted WHO list of "eleven qualities of a healthy city", which has therefore been deliberately omitted rather than reproduced from memory.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, WHO heat-health action plan guidance (eight core elements), WHO AWaRe classification, NFHS-4/NFHS-5, CAMS 2022–23

Programme and policy updates (India)

- ▶ The Comprehensive Annual Modular Survey 2022–23 (NSS 79th round, 302,086 households) found publicly funded health insurance raised outpatient use (adjusted OR 1.404, 95% CI 1.328–1.484) and inpatient use (adjusted OR 1.168, 95% CI 1.128–1.210) and lowered inpatient catastrophic expenditure at the 20 per cent threshold (adjusted OR 0.790, 95% CI 0.737–0.846), but outpatient out-of-pocket spending stayed high even among the insured — the authors' remedy is to strengthen primary care through the Ayushman Arogya Mandirs, so the service-delivery parameter, not the insurance card, is the binding constraint in a city health plan [PMID 42167946].
- ▶ First evaluation of the Mahila Arogya Samiti, the National Urban Health Mission's community-participation mechanism: 401 MAS surveyed across all 20 mission cities of Chhattisgarh, of which 95.3 per cent (95% CI 93.2–97.3) had acted on at least one of six domains and 79.8 per cent on three or more — food security and nutrition 74.1 per cent, sanitation 70.8 per cent, social environment 64.1 per cent, gender-based violence 60.8 per cent, health services 59.1 per cent, drinking water 56.4 per cent; the Rs 5,000 untied grant and the absence of any formal seat at the planning and monitoring table were the stated limits [PMID 40349012].
- ▶ A *Lancet Global Health* Series paper on south Asian cities re-frames the urban health-service parameter: geographical access to primary care is easy in cities, but financial hardship and poor quality persist for low-income residents, and the prescribed response is effective and resourced urban local bodies, increased public financing, and delivery models built with low-income communities and the providers who already serve them [PMID 39178875].

Current national data

- ▶ NFHS-4 (2015–16) to NFHS-5 (2019–21): non-contributory public health insurance in urban India rose from 19.6 per cent to 27.6 per cent of households, an increase of 8.0 percentage points (95% CI 7.3–8.7) against 11.7 percentage points (95% CI 11.0–12.4) rurally — urban coverage stays the lower of the two, and the rise was not larger in the states that had actually implemented PM-JAY between rounds [PMID 37640493].
- ▶ Air-quality target-setting for Indian cities now carries a costed mortality estimate: from a 2019 baseline in which over 70 per cent of Indians breathed annual PM_{2.5} above 40 µg/m³ and 56 per cent of households burned solid fuel, meeting India's own 40 µg/m³ standard alongside a halving of solid-fuel use would avert more than 300,000 deaths a year, with the WHO interim targets of

25 and 15 $\mu\text{g}/\text{m}^3$ averting 500,000 and 675,000 — economic benefits of 2.2, 3.6 and 4.9 per cent of GDP respectively [PMID 41701846].

Elimination, surveillance and disease status

- Community antimicrobial-resistance surveillance in the urban health training centre area of a medical college in Ahmedabad (2024) gives a city-level indicator for the national AMR agenda: 33.5 per cent of eligible residents had consumed an antibiotic in the preceding three months, 59.9 per cent of the 152 antibiotics used fell in the WHO AWaRe *Watch* group, and 37.5 per cent were obtained without any prescription [PMID 42024912].
- Heatwave hotspots mapped across India for 1981–2020 give the surveillance layer a city health profile should now carry: mean summer maximum temperature rose 1.0 ± 0.12 °C, the spatial extent of hotspots expanded roughly 1.5-fold between 1981–2000 and 2001–2020, and combined hazard–vulnerability–exposure risk is now highest over the central plains, the southeastern coast, Gangetic West Bengal and parts of the central Indo-Gangetic plain [PMID 41673178].

Guideline and legislative updates

- Eight Indian city heat action plans (Surat, Ahmedabad, Rajkot, Thane, Jodhpur, Delhi, Vadodara, Bhubaneswar) benchmarked against the WHO eight core elements of heat–health action planning: best covered were the health information plan, the lead coordinating body and alert systems, while the three elements that map directly onto healthy-city planning — particular care for vulnerable groups, real-time surveillance, and long-term urban planning — were the least represented; Thane covered the most sub-elements (38), Vadodara the fewest (17) [PMID 41296585].
- Policy analysis of 19 state heat action plans plus 16 State Action Plans on Climate Change covering all states and union territories: 12 of 35 plans (33 per cent) contained nothing at all for pregnant or lactating women, and of 102 unique strategies only 40 targeted them directly, most being behaviour-change advice rather than structural or built-environment measures (Karnataka 16 unique strategies, Telangana 11, Delhi 10) [PMID 41403386].

Global and WHO developments

- Global review of 83 heat–health action plans from 24 countries, covering about 2.2 billion people or 26 per cent of the world's population: 75.9 per cent mentioned mental health but only 31.3 per cent described its specific impacts and just 21.7 per cent carried any targeted intervention; no low-income country had a plan, and India supplied 35 of the 39 plans from lower-middle-income countries [PMID 40418305].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 7 Health Programmes in India — National Urban Health Mission and Swachh Bharat Mission-Urban; Ch 8 Demography and Family Planning; Ch 15 Environment and Health — water requirement, criteria for healthful housing, housing standards, overcrowding); Oxford Textbook of Public Health 7e (Ch 58 Urbanization and Health; Ch 99 The Practice of Environmental Health in an Era of Sustainable Development); Textbook of International Health (Ch 10 Health and the Environment — Hancock and Duhl definition, principles of sustainable cities); Maxcy-Rosenau-Last Public Health and Preventive Medicine 15e (Ch 75 Structure and Function of the Public Health System); Suryakantha — Community Medicine with Recent Advances (Ch 37 National Urban Health Mission). Environmental Hygiene Committee (1947) housing standards as reproduced in Park 28e.

Recent-advances sources: 42167946 — Is publicly funded health insurance associated with healthcare utilisation and financial protection in India? Evidence from a cross-sectional household survey (CAMS 2022–2023), *BMJ Open* 2026, doi:[10.1136/bmjopen-2025-106382](https://doi.org/10.1136/bmjopen-2025-106382); 40349012 — Community participation through women's health collectives promoted by India's National Urban Health Mission: a realist evaluation in Chhattisgarh state, *International Journal for Equity in Health* 2025, doi:[10.1186/s12939-025-02498-z](https://doi.org/10.1186/s12939-025-02498-z); 39178875 — Improving urban health through primary health care in south Asia, *The Lancet Global Health* 2024, doi:[10.1016/S2214-109X\(24\)00121-9](https://doi.org/10.1016/S2214-109X(24)00121-9) (abstract-only); 37640493 — Public health insurance coverage in India before and after PM-JAY: repeated cross-sectional analysis of nationally representative survey data, *BMJ Global Health* 2023, doi:[10.1136/bmjgh-2023-012725](https://doi.org/10.1136/bmjgh-2023-012725); 41701846 — The mortality and economic benefits of achieving air pollution standards in India, *Proceedings of the National Academy of Sciences* 2026, doi:[10.1073/pnas.2522228123](https://doi.org/10.1073/pnas.2522228123) (abstract-only); 42024912 — A community-based study of antibiotic consumption in an urban health training centre area of Ahmedabad city, *Indian Journal of Medical Research* 2026, doi:[10.25259/IJMR_452_2025](https://doi.org/10.25259/IJMR_452_2025); 41673178 — Intensification and spatial shifts of heatwave hotspot across India under climate and ENSO influences with health risk assessment, *Scientific Reports* 2026, doi:[10.1038/s41598-026-38289-x](https://doi.org/10.1038/s41598-026-38289-x); 41296585 — Heat action plans in eight Indian cities: knowledge gaps and opportunities for intersectoral heat governance, *Indian Journal of Medical Research* 2025, doi:[10.25259/IJMR_1019_2025](https://doi.org/10.25259/IJMR_1019_2025); 41403386 — Safeguarding maternal health against extreme heat in India: a policy analysis of state-level heat action plans, *International Journal of Gynaecology and Obstetrics* 2025, doi:[10.1002/ijgo.70747](https://doi.org/10.1002/ijgo.70747); 40418305 — A critical gap in addressing mental health in heat-health action plans worldwide, *Current Environmental Health Reports* 2025, doi:[10.1007/s40572-025-00486-7](https://doi.org/10.1007/s40572-025-00486-7). Also used: WHO Regional Office for Europe guidance on heat-health action planning (eight core elements); WHO AWaRe antibiotic classification; NFHS-4 (2015–16) and NFHS-5 (2019–21); Comprehensive Annual Modular Survey 2022–23 (NSS 79th round).

♦ ♦ ♦

Air pollution control in city planning

Asked in: Paper IV 19-Nov-2021 — RGUHS MD Community Medicine

Definition — Air pollution is the presence in ambient air of substances (gases, particulates, vapours, biological agents) in concentrations harmful to human health, animals, vegetation or property. Its control is ultimately an engineering problem (Park 28e, Ch 15, p 893), so it is settled far more at the town-planning and zoning stage than by remedial action later: measures must be instituted in the planning stage itself, and proper town planning and zoning is what eliminates the hazard (Park 28e, Ch 18, pp 966-967).

1. Determinants that fix a city's air quality — the parameters planned against

Parameter	What it contributes	Planning implication
Automobiles	Hydrocarbons, carbon monoxide, lead, nitrogen oxides, particulate matter; in strong sunlight hydrocarbons + nitrogen oxides form "photochemical" oxidant pollutants (Park 28e, Ch 15, p 888)	Road network, traffic routing, fuel and emission norms
Industry	Fuel combustion gives smoke, sulphur dioxide, nitrogen oxides and fly ash; petrochemical units give hydrogen fluoride, hydrochloric acid, organic halides (Park 28e, Ch 15, p 888)	Siting, buffer zone, stack and arrester specification
Domestic combustion	Coal, wood or oil burning gives smoke, dust, sulphur dioxide, nitrogen oxides — the cause of the London disaster of 1952 (Park 28e, Ch 15, p 888)	Piped clean fuel/electricity, kitchen ventilation
Tobacco smoke	The most direct and important source affecting the health of many people, including passive smokers (Park 28e, Ch 15, p 888)	Smoke-free public places in the city bye-laws
Miscellaneous	Burning refuse, incinerators, pesticide spraying, wind-borne dust, fungi, moulds, bacteria, nuclear energy programmes (Park 28e, Ch 15, p 888)	Solid-waste system, no open burning
Topography and air movement	Winds disperse and dilute pollutants; where mountains (or tall buildings) dominate, winds become weak and calm and pollutants concentrate in the breathing zone (Park 28e, Ch 15, p 888)	Site selection, building height and street orientation
Temperature inversion	Rapid cooling of the lower air layers traps pollutants and water vapour at low levels, producing smog; commoner in winter than in spring or summer (Park 28e, Ch 15, p 888; Suryakantha, Ch 7, p 67)	Avoid deep valleys/basins; plan winter episode response

2. Site, land-use and zoning parameters

Parameter	Specification to be written into the plan
Site selection	Reject sites where topography impedes air movement (deep valleys, basins ringed by hills) — the reversed temperature gradient locks pollutants in, and such inversions may persist for several days, causing acute episodes of respiratory illness and death in children, the elderly and those with lung or heart disease (Suryakantha, Ch 7, p 67)

Parameter	Specification to be written into the plan
Zoning	Divide the city into residential, industrial and transport zones , with the areas concerned with industry and transport separated from where people live (Park 28e, Ch 15, p 898; Suryakantha, Ch 8, p 72)
Industrial siting	Industries located far away from human habitations, where the topography of the soil is favourable, and downwind of the residential zone, with regulation ensured (Suryakantha, Ch 7, p 69; Sunder Lal, Ch 3, p 122)
Green belts	Trees and plants grown between the industries and the residential areas , the leaves absorbing carbon dioxide and giving out oxygen (Suryakantha, Ch 7, p 69) — the planned dilution device, since some pollutants are readily removed by vegetation (Park 28e, Ch 15, p 893); breathing spaces such as the <i>Mahim Nature Park</i> project, Mumbai, have demonstrably lowered pollution levels (Sunder Lal, Ch 3, p 122)
Housing set back from main streets	House fronts not less than 15 metres from the road, the intervening space thickly planted with trees and bushes (Park 28e, Ch 15, p 898)
Set-back and built-up ceiling	Open space all round the house so that lighting and ventilation are not obstructed; built-up area not to exceed one-third of total area in rural areas, up to two-thirds in urban areas (Park 28e, Ch 15, p 906)
Air space and openings in the building bye-laws	Air space at least 500 cubic feet per capita, preferably 1,000 ; window area one-fifth of floor area, doors plus windows two-fifths ; at least two windows per living room, one opening directly on to an open space (Park 28e, Ch 15, p 906)
Ventilation standard	<i>de Chaumont's</i> 3,000 cubic feet of fresh air per hour per person is the historical standard and is no longer followed ; air change matters more than cubic space — plan for 2–3 air changes per hour in living rooms and 4–6 in work rooms and assemblies (never more than 6, which produces a draught), with 1,000–1,200 cubic feet per person now held sufficient (Park 28e, Ch 15, pp 893–894)
Worker housing and sanitation	Housing near a plant correlated with community amenities; sewage disposal planned before industries are located, else existing services are overwhelmed (Park 28e, Ch 18, pp 966–967)

3. Traffic and transport parameters

Parameter	Specification to be written into the plan
Bypass roads	Heavy vehicles routed away from narrow streets and residential areas (Suryakantha, Ch 7, p 69; Ch 8, p 72)
Street geometry	Widen main streets; plan roads and land use so traffic flows on major routes while bypassing urban residential areas (Park 28e, Ch 15, p 898; Sunder Lal, Ch 8, p 511)
Modal priority	Walking, cycling and public transport remain the predominant modes of travel in India; the needs of pedestrians and cyclists must prevail because of their vulnerability, and mixed chaotic traffic must be segregated (Sunder Lal, Ch 8, p 511). An efficient public transport system with fewer private vehicles is the primary control (Sunder Lal, Ch 3, p 122)

Parameter	Specification to be written into the plan
Fuel and emission norms	Unleaded (de-leaded) petrol and catalytic converters , which cut oxides of nitrogen, carbon monoxide and volatile organic compounds (Park 28e, Ch 15, p 893; Sunder Lal, Ch 3, p 122); compressed natural gas for public transport, which brought Delhi's pollution down significantly after the Supreme Court ruling of 5 April 2002 (Sunder Lal, Ch 3, p 122); the current norm is the switch to Bharat Stage VI (BS-VI) emission standards initiated in April 2020 (Sunder Lal, Ch 3, pp 120-121)
Vehicle upkeep	Periodic servicing and correct petrol-oil proportions written into the transport bye-laws (Suryakantha, Ch 7, p 69)

4. Energy, industrial-emission and waste parameters

The WHO sequence for prevention and control of air pollution is **containment, replacement, dilution, legislation and international action** (Park 28e, Ch 15, p 893).

Parameter	Specification to be written into the plan
Containment	Prevent escape of toxic substances into ambient air by enclosure, ventilation and air cleaning; arresters for removal of contaminants (Park 28e, Ch 15, p 893). Wet methods to suppress dust, hoods and exhaust pipes, and increased height of the smoke vent (Suryakantha, Ch 7, p 69)
Emission-control hardware	Electrostatic precipitators and bag filters for particulates (dust, ash, soot); scrubbers reduce sulphur dioxide emission by up to 95 per cent ; catalytic converters cut oxides of nitrogen (Sunder Lal, Ch 3, p 122)
Replacement	Replace the polluting process with one that does not pollute — electricity, solar power, natural gas and central heating in place of coal; liquefied petroleum gas (smokeless fuel) domestically (Park 28e, Ch 15, p 893; Suryakantha, Ch 7, p 69); non-conventional sources — solar cookers, wind energy, biogas plants (Sunder Lal, Ch 3, p 123)
Dilution	Valid only within the self-cleaning capacity of the environment — green belts are the planned dilution device, but capacity is limited and trouble occurs when the atmosphere is overburdened (Park 28e, Ch 15, p 893)
Household air pollution	Clean cooking fuel in the kitchen, with subsidised liquefied petroleum gas for poor households under <i>Pradhan Mantri Ujjwala Yojana</i> , since 70–80 per cent of households use unprocessed biomass fuel; plus town and country planning for parks and open spaces (Sunder Lal, Ch 8, pp 339-340, 500-501)
Solid waste	No open burning; waste-to-energy or incineration only to the stack standards of Schedule II(C), Solid Waste Management Rules 2016 — particulates 50 mg/Nm ³ , sulphur dioxide 200 mg/Nm ³ , carbon monoxide 100 mg/Nm ³ (half-hourly), total organic carbon 20 mg/Nm ³ , hydrogen fluoride 4 mg/Nm ³ (Solid Waste Management, Ch 43, pp 330-331)
Combustion temperature	Minimum gas-phase combustion temperature 850 °C with ≥2 seconds residence time after the last air supply (Solid Waste Management, Ch 41, pp 319-320); twin-chamber incinerators must be designed for a minimum 950 °C in the secondary combustion chamber , again with ≥2 seconds gas residence time (Solid Waste Management, Ch 43, p 331)

Parameter	Specification to be written into the plan
Landfill	Buffer zone of no development around any solid waste processing and disposal facility exceeding five tonnes per day of installed capacity, and the sites incorporated in the Town Planning Department's land-use plans (Solid Waste Management, Ch 48, pp 392-393); vegetative cover of locally adapted non-edible perennials, roots not penetrating beyond 30 cm, with green belts all round the boundary (Ch 48, p 396); landfill gas utilised or flared, passive venting only where neither is possible; methane at site not to exceed 25 per cent of the lower explosive limit (Ch 48, p 396)

5. Air-quality standards, indices and monitoring parameters

The best indicators of air pollution are **sulphur dioxide, smoke and suspended particles**, monitored daily at several sites with results collated by a central agency (Park 28e, Ch 15, p 891).

Index / parameter	How measured	Value / remark
Sulphur dioxide	Lead-peroxide device; estimated in all air pollution surveys	Major contaminant of urban and industrial areas (Park 28e, Ch 15, p 891)
Smoke (soiling) index	Known volume of air filtered through white filter paper; stain read on a photoelectric meter	Micrograms per cubic metre averaged over time, or coefficient-of-haze units per 1,000 linear feet of air (Park 28e, Ch 15, p 891)
Grit and dust	Deposit gauges analysed monthly; midjet impinger for suspended dust	Expressed as mg per cubic metre (Park 28e, Ch 15, p 891; Suryakantha, Ch 7, pp 68-69)
Coefficient of haze	Amount of smoke or other aerosol per cubic metre of air	Used particularly in the USA (Park 28e, Ch 15, p 891)
Air pollution index	(10 × sulphur dioxide) + (2 × carbon monoxide), both ppm by volume, + (2 × coefficient of haze)	Alarm when the value rises from its usual ~12 to 50 or more (Park 28e, Ch 15, p 891)
Indian Air Quality Index (IND AQI)	Combines complex air quality data of several pollutants into a single number, nomenclature and colour	The index a city actually reports against; six bands below (Sunder Lal, Ch 3, p 123)

National Ambient Air Quality Standards, 2009 (Central Pollution Control Board) with the World Health Organization guideline levels for comparison — all values $\mu\text{g}/\text{m}^3$ except carbon monoxide (mg/m^3):

Pollutant	Averaging time	WHO AQG 2005	WHO AQG 2021	Indian national standard 2009
PM _{2.5}	Annual	10	5	40
PM _{2.5}	24-hour	25	15	60
PM ₁₀	Annual	20	15	60
PM ₁₀	24-hour	50	45	100
Ozone	8-hour	100	100	100
Nitrogen dioxide	Annual	40	10	40

Pollutant	Averaging time	WHO AQG 2005	WHO AQG 2021	Indian national standard 2009
Nitrogen dioxide	24-hour	200 (1-hour)	25	80
Sulphur dioxide	24-hour	20	40	80
Carbon monoxide (mg/m ³)	24-hour	10 (8-hour)	4	2 (8-hour)

(Sunder Lal, Ch 3, p 123, Table 3.3. 24-hour values are the 99th percentile, i.e. 3–4 exceedance days a year. Only the national standards have legal value and they are under revision; the WHO guidelines are not legally binding.)

Indian Air Quality Index band	Possible health impact
Good (0–50)	Minimal impact
Satisfactory (51–100)	Minor breathing discomfort to sensitive people
Moderate (101–200)	Breathing discomfort in people with lung or heart disease, children, older adults
Poor (201–300)	Breathing discomfort on prolonged exposure
Very poor (301–400)	Respiratory illness on prolonged exposure
Severe (>400)	Respiratory effects even in healthy people

(Sunder Lal, Ch 3, pp 123–124, Table 3.4, after the Central Pollution Control Board, Ministry of Environment, Forest and Climate Change.)

- **Monitoring machinery — current network (2022–23 position):** under the **National Air Quality Monitoring Programme**, the Central Pollution Control Board with the State Pollution Control Boards and Committees and the **National Environmental Engineering Research Institute, Nagpur** monitors ambient air through **793 stations covering 344 cities and towns in 28 states and 8 union territories**, regularly measuring sulphur dioxide, oxides of nitrogen as nitrogen dioxide, PM10 (respirable suspended particulate matter) and PM2.5 (Sunder Lal, Ch 3, p 123). Park's older account of the same programme, sponsored by the Board **since 1990**, describes a 14-year database from only **10 cities** — Ahmedabad, Mumbai, Kolkata, Delhi, Hyderabad, Jaipur, Kanpur, Kochi, Chennai, Nagpur — in all of which suspended particulate matter exceeded the Board's standards most of the time, the respirable fraction being 30–60 per cent and invariably higher at industrial sites (Park 28e, Ch 15, p 892).
- **Site-specific monitoring in the plan** — ambient air at waste processing, recycling and landfill sites monitored **six times a year for cities above 5 million population**, four times for 1–5 million, twice for 0.1–1 million (Solid Waste Management, Ch 48, pp 385–386, 397–398).

6. Legislative, programme and administrative parameters

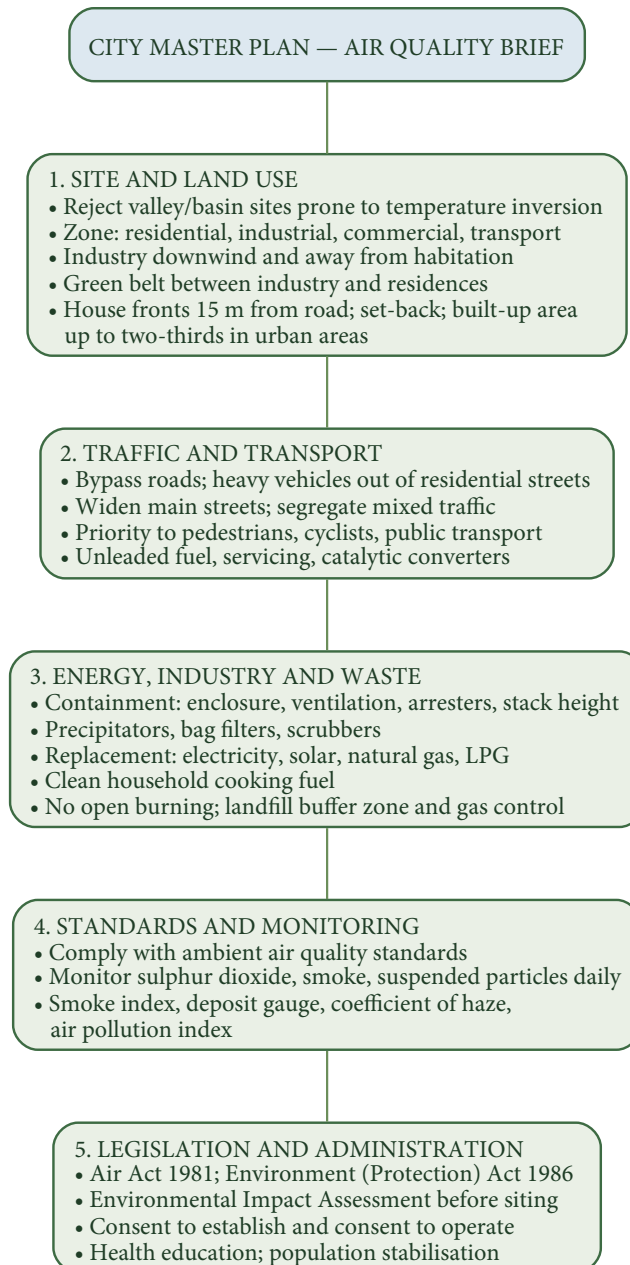
- **National Clean Air Programme (NCAP), 2019** — India's flagship urban air-quality programme, released in **2019** by the Government of India (Ministry of Environment, Forest and Climate Change) with a view to **reducing PM2.5 by 2024**; the **NCAP tracker** was launched the same year, and the target has since been revised to a **40 per cent reduction in PM2.5 concentration by 2026**,

delivered through city-specific clean-air action plans in cities not meeting the national standards ("non-attainment" cities). By the NCAP report of 2022, Delhi was the most polluted city with a mean PM_{2.5} of 99.71 µg/m³, followed by Faridabad 95.64 and Ghaziabad 91.25; Srinagar was cleanest at 26.33 (Sunder Lal, Ch 3, pp 122-124). In the national plan the heading "Improving air quality — National Clean Air Programme" carries the clean household fuel component alongside legislation and education (Sunder Lal, Ch 8, pp 339-340).

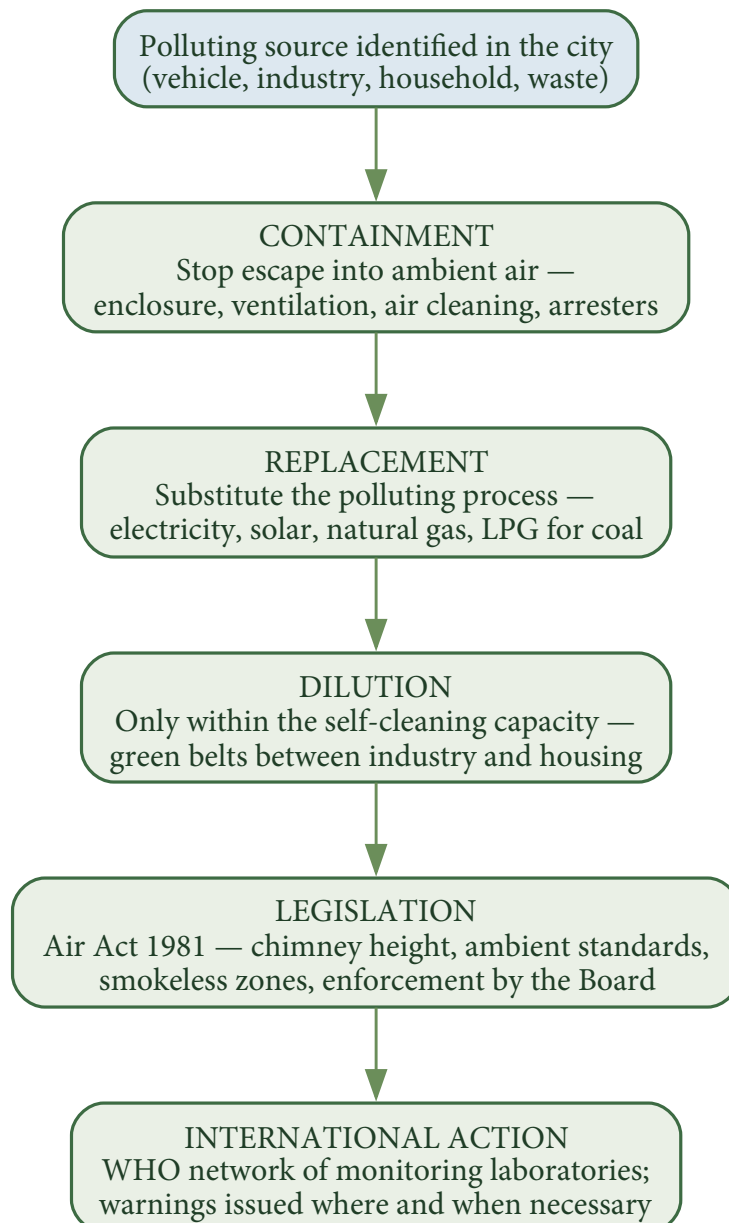
- ▶ **The Air (Prevention and Control of Pollution) Act, 1981** — the central statute; it covers height of chimneys, powers of local authorities to investigate and research, standards for ambient air quality, creation of **smokeless zones**, and enforcement (Park 28e, Ch 15, p 893; Sunder Lal, Ch 3, p 122).
- ▶ **The Environment (Protection) Act, 1986 and Rules, with the Environmental Impact Assessment Rules, 2006** — statutory clearance before a facility is sited (Solid Waste Management, Ch 74, p 469).
- ▶ **State Pollution Control Board clearances** — "consent to establish" and "consent to operate" for every polluting installation, with land-use clearance from the revenue authority (same source).
- ▶ **Supporting statutes** — Factories Act 1948, Motor Vehicles Act (1938, amended 1988; Rules 1989), Energy Conservation Act 2001, Public Liability Insurance Act 1991, and smoke-nuisance legislation (Suryakantha, Ch 7, p 69; Solid Waste Management, Ch 74, p 469).
- ▶ **Health education and population stabilisation** — educating the public on the hazards of air pollution and their own role in preventing it, with population stabilisation as the long-term general measure (Suryakantha, Ch 7, p 69).
- ▶ **International action** — the WHO international network of laboratories for monitoring and study of air pollution (a centre at Nagpur among them) issues warnings where and when necessary (Park 28e, Ch 15, p 893); the Global Environment Monitoring System of WHO monitors air quality by satellite and ground data (Sunder Lal, Ch 3, p 123).

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Zoning plan of a planned city (label exactly):** prevailing wind direction arrow; industrial zone on the **leeward/downwind edge**; **green belt (buffer plantation)** between industrial and residential zones; residential zone; civic core; **bypass/ring road** carrying heavy vehicles outside the residential zone; arterial road with **15-metre set-back and tree-planted strip** in front of houses; parks and open spaces; solid-waste landfill with **no-development buffer zone** sited downwind and away from habitation; air quality monitoring stations marked in residential and industrial zones.
- ▶ **Temperature inversion over a city (label exactly):** normal condition — warm light air near ground rises, cool heavy upper air descends, pollutants dispersed; inversion condition — cool air trapped near ground, warm layer above acting as a lid, "inversion layer", pollutants locked in the breathing zone, fog + smoke = **smog**, valley/tall-building topography restricting wind.



WHO sequence for prevention and control of air pollution, applied to a city plan



Statutory pathway before a polluting facility is sited in the city plan

High-yield exam pointers

- **Five WHO control procedures (mnemonic C-R-D-L-I):** Containment, Replacement, Dilution, Legislation, International action (Park 28e, Ch 15, p 893).
- **Air pollution index (USA)** = $10 \times \text{SO}_2 + 2 \times \text{CO}$ (ppm by volume) + $2 \times \text{coefficient of haze}$; alarm at **50 or more** (usual ~12). Best indicators of air pollution = sulphur dioxide, smoke, suspended particles.
- **Planning numbers:** house fronts ≥ 15 m from the road with tree-planted strip; built-up area $\leq$ **one-third** (rural), up to **two-thirds** (urban); air space **500–1,000 cubic feet per capita**; window area **one-fifth** of floor area; **2–3 air changes/hour** in living rooms and **4–6** in work rooms (de Chaumont's 3,000 c.ft/hour is the superseded historical standard).

- **Standards:** Indian national standards 2009 — PM10 60 annual / 100 24-hourly $\mu\text{g}/\text{m}^3$; PM2.5 40 / 60; SO₂ 80 24-hourly; NO₂ 40 / 80. WHO AQG 2021 — PM2.5 5 annual / 15 24-hour, PM10 15 annual (2005 values were PM2.5 10/25, PM10 20/50).
- **Programme + statute:** National Clean Air Programme 2019 — original target PM2.5 reduction by 2024, revised to 40 per cent reduction in PM2.5 by 2026; Air (Prevention and Control of Pollution) Act 1981; Environment (Protection) Act 1986; Bharat Stage VI emission norms from April 2020.
- **Monitoring:** National Air Quality Monitoring Programme, Central Pollution Control Board since 1990, now 793 stations in 344 cities/towns, 28 states and 8 union territories, measuring SO₂, NO₂, PM10, PM2.5; Indian Air Quality Index in six bands, good (0–50) to severe (>400).
- **Why it matters:** mortality in cities with high pollution exceeds that in relatively cleaner cities by 15–20 per cent (Park 28e, p 890); air pollution caused 6.67 million deaths worldwide in 2019 (State of Global Air 2020), and 39 of the world's 50 most polluted cities in 2022 were in India (Sunder Lal, Ch 3, pp 120–121).

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

Coverage note — the corpus supports every figure above; only the phrase "city-specific clean-air action plans for non-attainment cities" is standard programme knowledge not spelled out in the retrieved passages, and Park's 15-metre set-back and wide green-belt separation of residential areas from main streets appear under "careful planning of cities" in the noise-control section, where the same zoning device serves both noise and air pollution.

Recent advances [RA]

Current as of 2026-08-17 · depth: elaborate · sources: PubMed, National Clean Air Programme (MoEFCC), Central Pollution Control Board, WHO Air Quality Guidelines 2021, WHA78 road map on air pollution and health 2025, EU Ambient Air Quality Directive 2024/2881, European Environment Agency

Programme and policy updates (India)

- **The NCAP target and its city list have both moved since the textbook account.** Launched by the Ministry of Environment, Forest and Climate Change in January 2019 with a 20–30 per cent particulate reduction by 2024 against a 2017–18 baseline, the programme was re-set to a 40 per cent reduction in particulate concentration, or attainment of the National Ambient Air Quality Standards, by 2025–26; the Central Pollution Control Board list has also grown from the original 122 to 131 cities across 24 states and union territories that breached the standards for five consecutive years. [PMID 39434902] [PMID 40157486]
- **The programme is working, but not uniformly.** Across 28 Board-monitored cities, 20 recorded PM10 falls of 15–60 per cent between 2018 and 2022, with 62,219 excess deaths averted in 20 cities in 2021 relative to 2018; eight cities including Kolkata and Durgapur were worse than in 2018, and Indo-Gangetic and central Indian cities still sit at 80–120 $\mu\text{g}/\text{m}^3$ PM10 against about 40 $\mu\text{g}/\text{m}^3$ in peninsular cities such as Hubballi. [PMID 39970552]
- **City-level forecasting shows which planning measures actually deliver.** Among Uttar Pradesh's 15 non-attainment cities, Bareilly (>70 per cent), Raebareli (58), Moradabad (55), Ghaziabad

(48), Agra (41) and Varanasi (40) are projected to meet the target by 2025–26, while Gorakhpur (+50 per cent) and Prayagraj (+32 per cent) are projected to worsen; the separator was capacity — added continuous ambient air quality monitoring stations, real-time public reporting, road-dust management, vehicular emission checks and industrial enforcement — versus continuing open biomass burning, unmanaged construction and mass-gathering traffic in the failing cities. [PMID 39434902]

Current national data

- ▶ **First multi-city Indian mortality evidence (ten cities, 2008–2019, about 3.6 million deaths):** daily mortality rose 1.4 per cent (95% CI 0.7–2.2) per 10 $\mu\text{g}/\text{m}^3$ PM2.5, and 3.6 per cent (2.1–5.0) on causal instrumental-variable analysis isolating locally generated pollution; 7.2 per cent (4.2–10.1) of all deaths — roughly 33,600 a year in those ten cities — were attributable to PM2.5 above the WHO 24-hour guideline of 15 $\mu\text{g}/\text{m}^3$, which was exceeded on 99.8 per cent of city-days. Restricting to days below the Indian 24-hour standard of 60 $\mu\text{g}/\text{m}^3$ **doubled** the risk to 2.7 per cent (1.7–3.6) per 10 $\mu\text{g}/\text{m}^3$, so the 2009 standard is not a health-protective threshold. [PMID 38969471]
- ▶ **What compliance would buy.** At the 2019 baseline over 70 per cent of Indians breathed annual PM2.5 above 40 $\mu\text{g}/\text{m}^3$ and 56 per cent of households burned solid fuel; meeting the Indian annual standard of 40 $\mu\text{g}/\text{m}^3$ would avert 118,000 deaths a year, the WHO interim target of 25 $\mu\text{g}/\text{m}^3$ 264,000, and 15 $\mu\text{g}/\text{m}^3$ 397,000. Pairing each with a halving of solid-fuel households raises these to 310,000, 501,000 and 679,000 deaths averted — worth 2.2, 3.6 and 4.9 per cent of GDP — because the exposure–response curve is concave, making ambient and household measures synergistic rather than additive. [PMID 41701846]
- ▶ **Heat and particulates act together, which merges two city plans into one.** In the same ten cities the PM2.5 effect rose from 0.8 per cent (95% CI –0.3 to 1.9) per 10 $\mu\text{g}/\text{m}^3$ at the 75th temperature percentile to 4.6 per cent (2.9–6.5) at the 99th, and the mortality risk of heat rose from 8.3 per cent (2.2–14.9) when PM2.5 was 20 $\mu\text{g}/\text{m}^3$ to 63.9 per cent (38.7–93.7) at 100 $\mu\text{g}/\text{m}^3$. [PMID 40245548]

Elimination, surveillance and disease status

- ▶ **Satellite surveillance now supplements the ground station network.** Tropospheric nitrogen dioxide vertical column density over the 131 non-attainment cities rose steadily from 2005 to 2023 ($1\text{--}4 \times 10^{13}$ molecules cm^{-2} per year), but has fallen about **1.9 per cent per year since 2017**, the steepest decline being over the Indo-Gangetic Plain, attributed to Bharat Stage VI norms, electric-vehicle uptake and NCAP enforcement; 2023 column values were back down to lockdown-year levels. [PMID 40157486]
- ▶ **Crop-residue burning remains the seasonal driver, and community awareness is the gap.** Seasonal rice-residue burning contributes up to about 26 per cent of Delhi's winter air pollution; in a rural National Capital Region survey in Ballabgarh, Haryana ($n = 1,813$, response 90.7 per cent), only 33 per cent named stubble burning as a cause of air pollution and **90 per cent knew of no government scheme** promoting alternatives such as the Happy Seeder or zero-till drill. [PMID 38933804]

Guideline and legislative updates

- **India's decadal review of the National Ambient Air Quality Standards is under way**, and the multi-city Indian mortality data are explicitly offered to it: the 2009 standards remain far looser than the WHO 2021 guidelines (24-hour PM_{2.5} 60 versus 15 µg/m³), and Graded Response Action Plans that trigger only around 150 µg/m³ capture little of the avoidable mortality, which accrues well below that. Internationally, the EU's revised Ambient Air Quality Directive (2024/2881) of 23 October 2024 moved limit values toward the WHO 2021 guidelines, with a target of cutting premature deaths by more than 55 per cent by 2030 against 2005. [PMID 38969471] [PMID 41941397]

Global and WHO developments

- **The 78th World Health Assembly (May 2025) approved a global road map on air pollution and health, aiming to halve premature deaths from air pollution by 2040.** About 99 per cent of the world's population now lives where the WHO 2021 guideline values are exceeded; ambient air pollution causes over 4.2 million deaths a year and household air pollution about 3.2 million, with 2.4 billion people still cooking and heating on polluting fuels. The WHO Ambient Air Quality Database version 6.1 (January 2024) holds ground measurements from over 7,000 human settlements in more than 120 member states, and on 1 December 2025 the European Environment Agency reported 182,000 premature deaths in the EU-27 in 2023 from PM_{2.5} above WHO levels, with EU PM_{2.5} and PM₁₀ emissions down 38 and 36 per cent between 2005 and 2023. [PMID 41941397]

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine, 28e (Ch 15, Environment and Health, pp 888-894, 898, 906-907; Ch 18, Occupational Health, pp 966-967); Sunder Lal — Textbook of Community Medicine (Ch 3, Environmental Health Problems, pp 120-125, including Table 3.3 National Ambient Air Quality Standards 2009 with WHO Air Quality Guideline levels 2005 and 2021, and Table 3.4 Indian Air Quality Index; Ch 8, pp 339-340, 500-501, 511); Suryakantha — Community Medicine with Recent Advances (Ch 7, Air and Ventilation, pp 67-69; Ch 8, Noise, p 72); Manual on Municipal Solid Waste Management (Ch 41, pp 319-320; Ch 43, pp 330-332; Ch 48, pp 385-386, 392-393, 396-398; Ch 74, p 469), incorporating the Solid Waste Management Rules 2016; National Clean Air Programme 2019 and NCAP report 2022; World Health Organization Air Quality Guidelines 2005 and revised guidelines September 2021; State of Global Air 2020; The Air (Prevention and Control of Pollution) Act 1981; The Environment (Protection) Act 1986 and Environmental Impact Assessment Rules 2006.

Recent-advances sources: 39434902 — Assessing the impact of the National Clean Air Programme in Uttar Pradesh's non-attainment cities: a prophet model time series analysis, *Lancet Reg Health Southeast Asia* 2024, doi:[10.1016/j.lansea.2024.100486](https://doi.org/10.1016/j.lansea.2024.100486); 39970552 — Impact of the National Clean Air Programme (NCAP) on the particulate matter pollution and associated reduction in human mortalities in Indian cities, *Science of the Total Environment* 2025, doi:[10.1016/j.scitotenv.2025.178787](https://doi.org/10.1016/j.scitotenv.2025.178787) (abstract-only); 38969471 — Ambient air pollution and daily mortality in ten cities of India: a causal modelling study, *Lancet Planetary Health* 2024, doi:[10.1016/S2542-5196\(24\)00114-1](https://doi.org/10.1016/S2542-5196(24)00114-1); 41701846 — The mortality and economic benefits of achieving air pollution standards in India, *Proceedings of the National Academy of Sciences USA* 2026, doi:[10.1073/pnas.2522228123](https://doi.org/10.1073/pnas.2522228123); 40245548 — Synergistic associations of ambient air pollution and heat on daily mortality in India, *Environment International* 2025, doi:[10.1016/j.envint.2025.109426](https://doi.org/10.1016/j.envint.2025.109426) (abstract-only); 40157486 — Changing tropospheric NO₂ dynamics across Indian air pollution hotspots, *Environmental Pollution* 2025, doi:[10.1016/j.envpol.2025.126160](https://doi.org/10.1016/j.envpol.2025.126160) (abstract-only); 38933804 — Perception and Practices Regarding Stubble Burning in Rural Community of Haryana, *Indian Journal of Community Medicine* 2024, doi:[10.4103/ijcm.ijcm_820_22](https://doi.org/10.4103/ijcm.ijcm_820_22); 41941397 — A Review of the Increasing Impact and Effects of Air Pollution Throughout Life and Before Birth, *Medical Science Monitor* 2026, doi:[10.12659/MSM.953109](https://doi.org/10.12659/MSM.953109). Non-journal sources: WHA78 global road map on air pollution and health, May 2025; WHO Ambient Air Quality Database version 6.1, January 2024; EU Ambient Air Quality Directive 2024/2881, 23 October 2024; European Environment Agency statement, 1 December 2025.



Ideal disinfectant and types of disinfection

Asked in: Paper I 07-May-2022 — RGUHS MD Community Medicine

Definition — Disinfection is the thermal or chemical destruction of pathogens and other types of micro-organisms; it is **less lethal than sterilization**, because it destroys most recognised pathogenic organisms but not necessarily all microbial forms, notably bacterial spores. A **disinfectant** is usually a chemical (sometimes a physical) agent that destroys disease-causing pathogens on **inanimate objects**, but might not kill bacterial spores (Park 28e, Ch 3, pp 152-153).

Allied terms distinguished

Term	Meaning (Park 28e, Ch 3, pp 152-153)
Sterilization	Validated process rendering a product free of all forms of viable micro-organisms including bacterial spores
Disinfection	Thermal or chemical destruction of pathogens; spores may survive
Antiseptic	Substance that prevents or arrests the growth or action of micro-organisms; used especially for preparations applied topically to living tissue
Asepsis	Prevention of contact with micro-organisms
Sanitizer	Agent reducing bacterial contaminants to safe levels as judged by public health requirements ; used on inanimate objects
Germicide	Agent that destroys micro-organisms, especially pathogenic organisms
Cleaning	Removal of adherent visible soil, blood, protein and debris from surfaces, crevices, joints and lumens by a manual or mechanical process, preparing items for safe handling or further decontamination
Deodorant	Substance that suppresses or neutralises bad odour, e.g., lime and bleaching powder

Properties of an ideal disinfectant

No single agent satisfies every criterion — "An ideal disinfectant may be a distant dream but one that can fulfill a certain purpose and suit a certain situation can always be identified" (Suryakantha, Ch 19, p 317). Park lists **thirteen criteria** (Park 28e, Ch 3, p 153); rows 14-16 are the additional criteria Suryakantha states (Ch 19, p 317).

#	Property	What it means in practice
1	Broad spectrum	Wide antimicrobial spectrum; ideally kills all pathogens
2	Fast acting	Produces a rapid kill within a practical contact time
3	Not affected by environmental factors	Active in the presence of organic matter (blood, sputum, faeces) and compatible with soaps, detergents and other chemicals met in use
4	Non-toxic	Not harmful to the user or the patient
5	Surface compatibility	Does not corrode instruments and metallic surfaces; does not deteriorate cloth, rubber or plastics; does not bleach or stain articles

#	Property	What it means in practice
6	Residual effect	Leaves an antimicrobial film on the treated surface
7	Easy to use	Simple application with clear label directions
8	Odourless	Pleasant odour or no odour, so routine use is not resisted
9	Economical	Not prohibitively high in cost
10	Solubility	Soluble in water
11	Stability	Stable both in concentrate and in use-dilution
12	Cleaner	Possesses good cleaning properties
13	Environmentally friendly	Does not damage the environment on disposal
14	Penetrating and consistently reliable	Miscible with water, highly penetrable, reliable batch to batch
15	Active in both acid and alkaline media	Efficacy independent of the pH of the material treated
16	Safe, cheap and effective	Stable and free of unpleasant smell; kills all pathogens yet is not harmful to man

Factors that modify the action of a disinfectant — the action of a disinfectant depends on six things (Suryakantha, Ch 19, p 318):

- ▶ **Strength (concentration)** of the solution.
- ▶ **Nature of the solvent** in which it is made up.
- ▶ **Nature and number of organisms** — spores and mycobacteria are far more resistant than vegetative bacteria.
- ▶ **Presence of organic matter** — blood, pus and faeces inactivate many agents, which is why *cleaning must always precede disinfection*.
- ▶ **Time given to act** — the contact period.
- ▶ **Temperature of the mixture**.

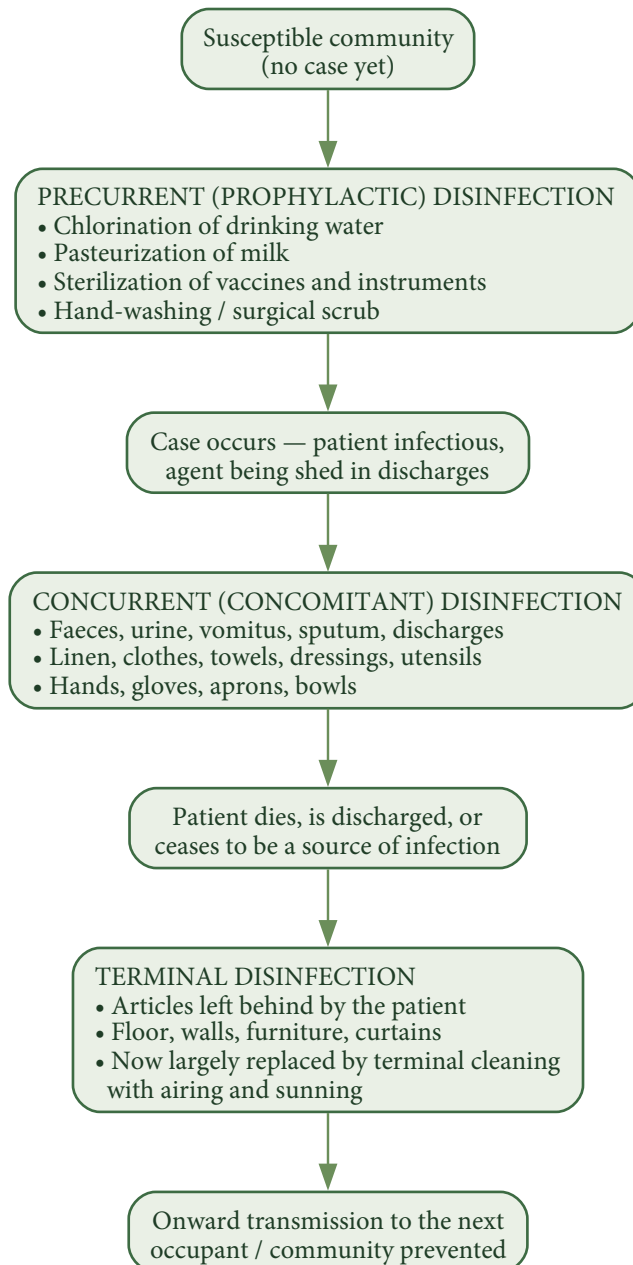
Standardization — bactericidal activity is compared with phenol using *Salmonella typhi* as the test organism and expressed as the **Rideal-Walker coefficient** (carbolic coefficient); a coefficient of 10 means the agent is 10 times as potent as phenol against *S. typhi* under laboratory conditions in the absence of organic matter. The **Chick-Martin test** is an improvement on the Rideal-Walker test: it estimates germicidal power **in the presence of organic material** such as yeast and excreta, and so predicts field performance better (Suryakantha, Ch 19, p 320). In practice, a coal-tar disinfectant with a Rideal-Walker coefficient of **10 or more** (such as cresol) is chosen, and one with a coefficient **below 5 should not be used** (Park 28e, Ch 5, pp 291-292).

Types of disinfection

Three types, classified by **when** the disinfective measure is applied in relation to the illness (Park 28e, Ch 3, p 153).

Type	Definition	Timing	Examples
Precurrent (prophylactic) disinfection	Disinfection carried out as a routine preventive measure, before any case occurs	Continuous, community-wide	Chlorination of drinking water; pasteurization of milk; hand-washing
Concurrent (concomitant) disinfection	Application of disinfective measures as soon as possible after the discharge of infectious material from the body of an infected person, or after articles are soiled with such discharges — the agent is destroyed as soon as it is released from the body, stopping further spread	Throughout the course of the illness	Urine, faeces, vomit; contaminated linen, clothes, dressings; hands, aprons, gloves
Terminal disinfection	Application of disinfective measures after the patient has been removed by death or to hospital, or has ceased to be a source of infection , or after isolation practices are discontinued	Once, at the end of the episode	Articles left behind, plus the room, furniture and bedding

Current practice caveat — terminal disinfection is now **scarcely practised**; *terminal cleaning* with airing and sunning of rooms, furniture and bedding is considered adequate (Park 28e, Ch 3, p 153).

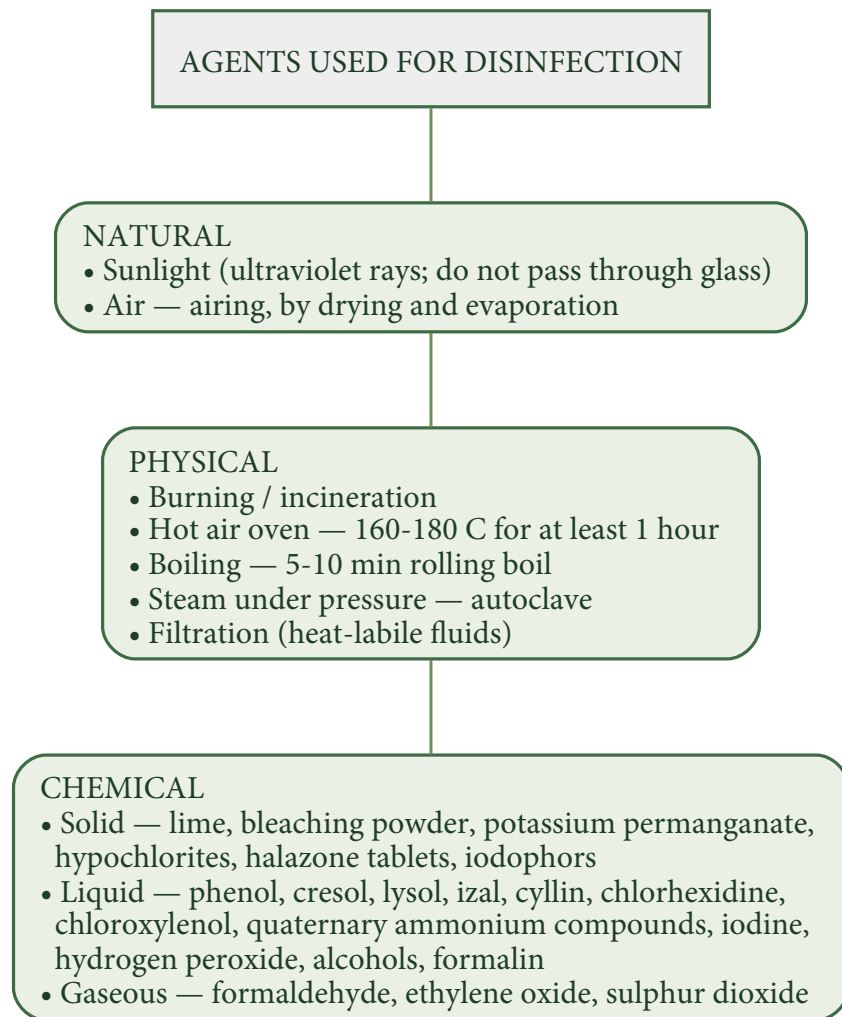


The three types of disinfection placed along the course of an infectious illness

Agents used for disinfection and their application in practice

Group	Agents	Key working detail
Natural	Sunlight; air (airing)	Ultraviolet rays of sunlight (which do not penetrate glass) are particularly lethal to bacteria and some viruses — linen, bedding and furniture are disinfected by several hours of direct sun. Airing acts by drying and evaporation. Because of their vagaries, natural agents cannot be totally depended upon (Park 28e, Ch 3, p 153)

Group	Agents	Key working detail
Physical	Burning (incineration); hot air; boiling; steam under pressure (autoclave); filtration	Hot air oven at 160-180°C for at least 1 hour kills spores but has no penetrating power (unsuitable for mattresses). Boiling for 5-10 minutes (rolling boil) kills bacteria but not spores or viruses ; boilers reach only just above 90°C, so the >100°C needed for spores is unattainable in them. The autoclave attains 122°C under 15 lb/sq inch (1 kg/sq cm) by giving off latent heat; absolute sterility needs over 135°C , and sputum is autoclaved for 20 minutes at 20 lb (Park 28e, Ch 3, pp 153-154, 156). Filtration is used for heat-labile fluids
Chemical — solid	Lime, bleaching powder, potassium permanganate, hypochlorites, halazone tablets, iodophors	Fresh bleaching powder contains about 33% available chlorine but is unstable, losing chlorine on exposure to air, light and moisture — hence its action is rapid but brief and stocks must be checked. Roughly 2.5 g disinfects 1,000 litres of water (~0.7 mg applied chlorine per litre). Milk of lime (1 lime : 4 water) disinfects excreta (Park 28e, Ch 15, pp 871-873; Suryakantha, Ch 19, p 318)
Chemical — liquid	Coal-tar group (phenol, cresol, lysol, izal, cyllin, hexachlorophene, chlorhexidine, chloroxylenol); quaternary ammonium compounds; iodine; hydrogen peroxide; alcohols; formalin	Phenol is the reference standard for germicidal activity but is caustic and not used as an antiseptic . Crude phenol (phenyl, a phenol-cresol mixture) — 10% for faeces, 5% for mopping floors and cleaning drains . Cresol is 3-10 times phenol; lysol 5-10 times; izal 8 times (special action on the coli-typhoid group); cyllin 17 times (drains and latrines) (Park 28e, Ch 3, p 154; Suryakantha, Ch 19, p 319)
Chemical — gaseous	Formaldehyde, ethylene oxide, sulphur dioxide	Commercial formalin is a 40% aqueous solution of formaldehyde . For fumigation, formaldehyde is generated by boiling 500 ml formalin with 1 litre water per 30 cubic metres of space, or by adding 170-200 g potassium permanganate to that mixture; the room is kept closed 6-12 hours, and the process works best at high temperature and 80-90% relative humidity (Park 28e, Ch 3, p 157)



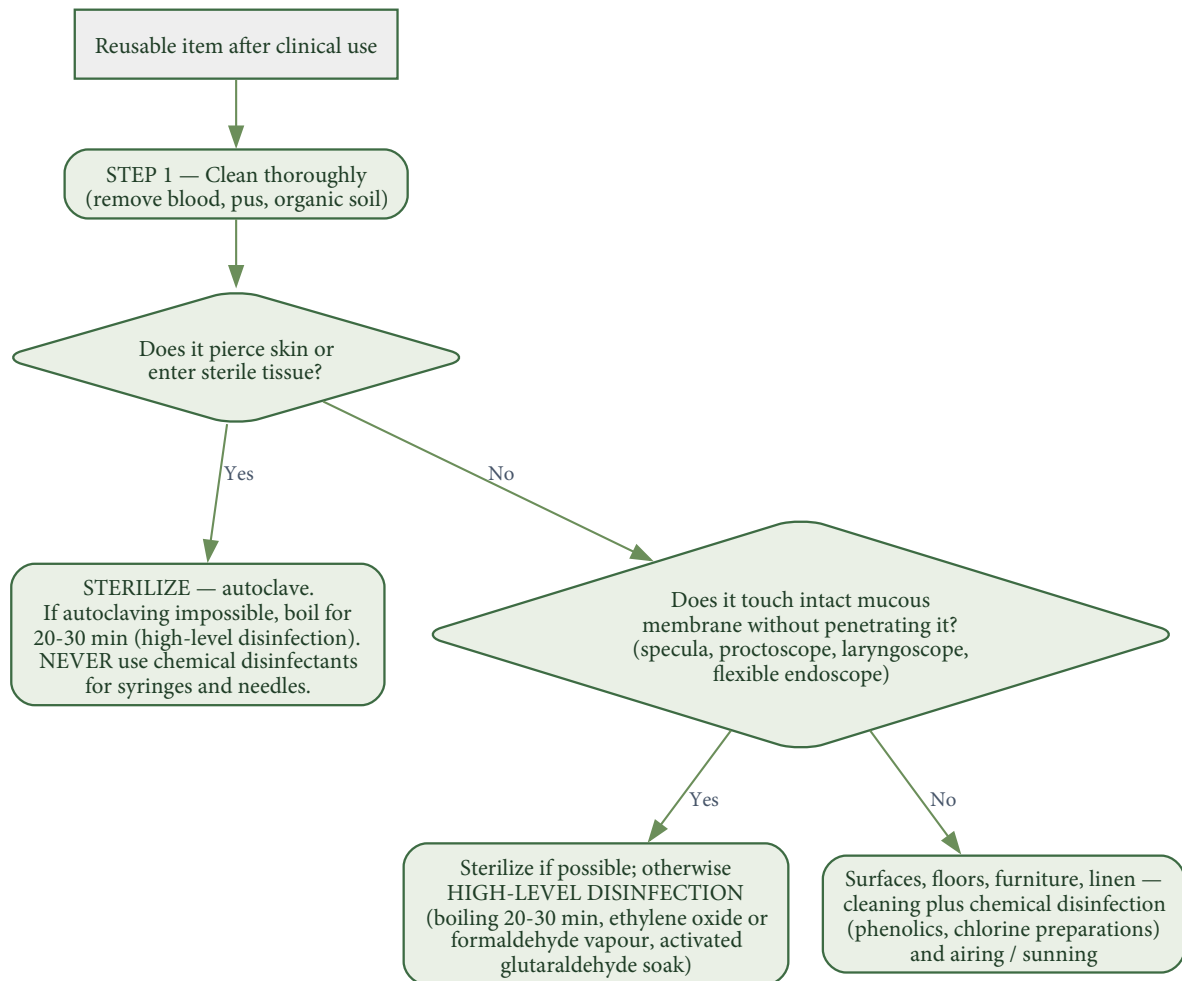
Classification of the agents used for disinfection

Material to be disinfected	Recommended procedure
Faeces and urine	Collect in impervious vessels; add an equal volume of one of the agents of Park's Table 46 — bleaching powder 50 g per litre = 5% , crude phenol 100 ml = 10%, cresol 50 ml = 5%, formalin 100 ml = 10%. Break the faeces up with a stick, stand 1-2 hours , then empty into a water closet or bury. If none is available, add an equal amount of quicklime or milk of lime (1 part lime to 4 parts water) and leave 2 hours (Park 28e, Ch 3, p 156, Table 46)
Bedpans and urinals	Ideally steam disinfected ; alternatively 2.5% cresol for one hour after cleaning (Park 28e, Ch 3, p 156)
Sputum	Best received in gauze or paper handkerchiefs and destroyed by burning . Where the amount is large (tuberculosis hospitals), boil or autoclave for 20 minutes at 20 lb pressure ; or have the patient spit into a cup half filled with 5% cresol , stand one hour when full. 5% phenol or 4.8% chloroxylenol is also used (Park 28e, Ch 3, pp 156-157)
Room and hard surfaces	Thorough cleaning, airing and several hours of direct sunlight usually suffice. For chemical disinfection, spray or mop with chlorinated lime leaving ≥25 ppm free chlorine , or ≥1% formaldehyde solution , or 2.5% cresol — contact for at least 4 hours before final washing (Park 28e, Ch 3, p 157)

Material to be disinfected	Recommended procedure
Linen and clothes soiled with dejecta	First wash in cold water containing a disinfectant such as 2.5% cresol , then boil about 30 minutes with frequent stirring (1% soap and 0.3% washing soda enhance the effect); or soak in 10% formalin, 2.5% cresol or 5% phenol , then wash, sun-dry and iron (Park 28e, Ch 3, p 154; Suryakantha, Ch 19, p 322)
Drinking water	Ensure a free residual chlorine of 0.5 mg/litre at the end of one hour contact; chlorine has no effect on spores, protozoal cysts or helminth ova except in higher doses (Park 28e, Ch 15, pp 870-871)

High-level disinfection and choice of process

High-level disinfection is a process of destroying all micro-organisms, though **some spores may survive**, particularly if present in large numbers to begin with. It is achieved by **continuous boiling for 20-30 minutes** — the simplest and most reliable way of freeing an article from most pathogens, including **human immunodeficiency virus (HIV) and hepatitis B virus (HBV)**, where autoclaving is not possible — or by exposure to vapours of ethylene oxide or formaldehyde, or prolonged soaking in activated glutaraldehyde or carbolic acid. Instruments that touch intact mucous membrane without penetrating it (vaginal specula, proctoscope, laryngoscope, fibre-optic endoscope) should ideally be sterilized, and must at minimum receive high-level disinfection. Syringes, needles and other skin-piercing instruments may likewise be boiled when autoclaving is impossible, but **chemical disinfectants must never be used for syringes and needles** (Suryakantha, Ch 19, p 322).



Choosing between sterilization and high-level disinfection for a reusable item

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- **Autoclave** — label the plate above: chamber, steam inlet, pressure gauge, safety valve, air/steam discharge. Annotate 122°C at 15 lb/sq inch, and 20 minutes at 20 lb for sputum.
- **Ladder of microbial kill** — draw three ascending steps and label them exactly: (1) **Cleaning** — physical removal of soil, blood, protein and debris; (2) **High-level disinfection** — destroys all micro-organisms, *some spores may survive*; (3) **Sterilization** — free of all viable micro-organisms *including bacterial spores*. Mark the arrow alongside as "increasing lethality".

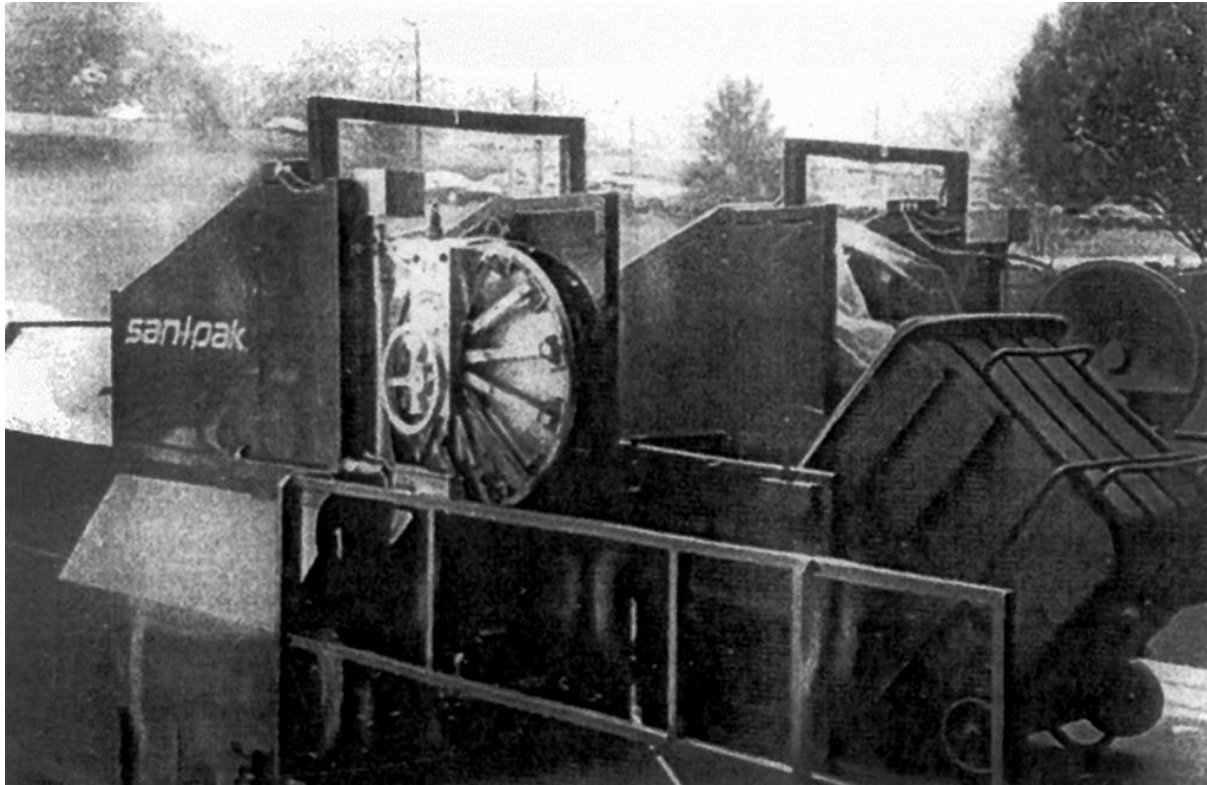


Fig 13.4 — Autoclave (Suryakantha Environment Best, p.119)

High-yield exam pointers

- ▶ **Park's 13 criteria** — **group them, don't list them**: what it **kills** (broad spectrum, fast acting) → what **resists** it (unaffected by organic matter, soaps, detergents) → what it must **not harm** (non-toxic, surface-compatible, environment-friendly) → how it **behaves in use** (residual effect, easy, odourless, economical, soluble, stable, cleans). Suryakantha adds: penetrating, reliable, active in acid and alkaline media.
- ▶ **Three types of disinfection** — **P-C-T**: Precurrent (prophylactic, before any case), Concurrent (during the illness), Terminal (after death, removal or loss of infectivity). Terminal disinfection is now scarcely practised — terminal *cleaning* suffices.
- ▶ **Two tests to name**: **Rideal-Walker** (carbolic) **coefficient** — potency against *Salmonella typhi* relative to phenol, **without** organic matter; **Chick-Martin test** — the improvement that measures germicidal power **in the presence of** organic material. Use a disinfectant with a Rideal-Walker coefficient ≥ 10 ; never one below 5.
- ▶ **Relative potencies vs phenol**: cresol 3-10 $\times$, lysol 5-10 $\times$, izal 8 $\times$ (coli-typhoid group), cyllin 17 $\times$ (drains and latrines).
- ▶ **Numbers to write**: hot air oven 160-180°C for ≥ 1 hour; boiling 5-10 min (bacteria only), 20-30 min (high-level disinfection); autoclave 122°C at 15 lb/sq inch, sputum 20 min at 20 lb; bleaching powder 33% available chlorine; **excreta** — **equal volume of 5% bleaching powder (50 g/L) / 10% crude phenol / 5% cresol / 10% formalin for 1-2 hours**; room surfaces ≥ 25 ppm free chlorine or 2.5% cresol for 4 hours; drinking water 0.5 mg/L free residual chlorine for 1 hour.
- ▶ **Two rules that earn marks**: cleaning must precede disinfection because organic matter inactivates disinfectants; chemical disinfectants must **never** be used for syringes and needles.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only — Park 28e places disinfection inside the chapter on principles of epidemiology (control of the reservoir), so the same question may be framed as part of "control of communicable disease"; Suryakantha prints the excreta dilution as "8 per cent" for the identical recipe of 50 g bleaching powder in 1 litre, which is arithmetically 5 per cent and is given as 5 per cent both in Park's Table 46 and in Suryakantha's own bleaching-powder section, so the 5 per cent figure has been used here; the retrieved corpus supports the term "high-level disinfection" but does not carry the *Spaulding* critical / semi-critical / non-critical wording or any definition of low- or intermediate-level disinfection, so neither has been used.

SOURCE & COVERAGE

- ▶ Park's Textbook of Preventive and Social Medicine, 28e — Ch 3 (Principles of Epidemiology and Epidemiologic Methods), pp 152-157 incl. Table 46; Ch 5 (Epidemiology of Communicable Diseases), pp 291-292; Ch 15 (Environment and Health), pp 869-873
- ▶ Suryakantha — Community Medicine with Recent Advances — Ch 19 (Epidemiology of Infectious Diseases), pp 316-322; Ch 13 (Management of Hospital Waste), p 119; Ch 5 (Environment and Water), pp 49-50



Solid waste (refuse) — sources and disposal

Asked in: Paper I 07-May-2022 — RGUHS MD Community Medicine

Definition — **Solid wastes** ("refuse") are the useless, unwanted and discarded solid materials of community living: garbage (food wastes), rubbish (paper, plastics, wood, metal, throw-away containers, glass), demolition products, sewage treatment residue, dead animals, manure and other discarded material. Strictly they should not contain nightsoil, though in India nightsoil is not uncommonly found in refuse. Daily output depends on dietary habits, life style, living standards and the degree of urbanisation and industrialisation; **per capita daily solid waste ranges between 0.25 and 2.5 kg in different countries** (Park 28e, Ch 15, p 908).

SOURCE & COVERAGE

Source / type	Constituents	Sanitary importance
Street refuse — collected by the street-cleansing service or by scavenging	Leaves, straw, paper, animal droppings and litter of all kinds	Bulky, largely dry; adds to sweeping load where there is no house-to-house collection
Market refuse — collected from markets	Large proportion of putrescible vegetable and animal matter	Ferments rapidly; prime fly-breeding and odour source; needs daily removal
Stable litter — collected from stables	Mainly animal droppings and left-over animal feed	High manurial value; attracts flies; best composted
Industrial refuse	Ranges from completely inert material such as calcium carbonate to highly toxic and explosive compounds	Cannot go into the municipal stream unsorted; needs separate hazardous-waste handling
Domestic refuse — <i>ash</i>	Residue from fire used for cooking and heating	Inert; the high fine-ash fraction of Indian refuse is why incineration is difficult here
Domestic refuse — <i>rubbish</i>	Paper, clothing, bits of wood, metal, glass, dust and dirt	Largely recyclable dry fraction
Domestic refuse — <i>garbage</i>	Waste matter arising from the preparation, cooking and consumption of food — waste food, vegetable peelings, other organic matter	Ferments on storage, so needs quick removal and disposal ; the fraction that determines fly breeding

(All rows: Park 28e, Ch 15, p 908; Suryakantha, Ch 12, p 94)

Accumulated refuse is a health hazard because it decomposes and favours fly breeding, attracts rodents and vermin, carries pathogens back to food through flies and dust, pollutes water and soil, and is an odour nuisance (Park 28e, Ch 15, p 908). Urban India generated **1,43,449 metric tonnes of municipal solid waste per day in 2014–15** — about **51 million metric tonnes a year at 0.2–0.6 kg per capita per day**, of which **40–60 per cent is organic and 10–20 per cent recyclable** (Central Pollution Control Board, 2014–15; Manual on Municipal Solid Waste Management, Ch 3, p 251).

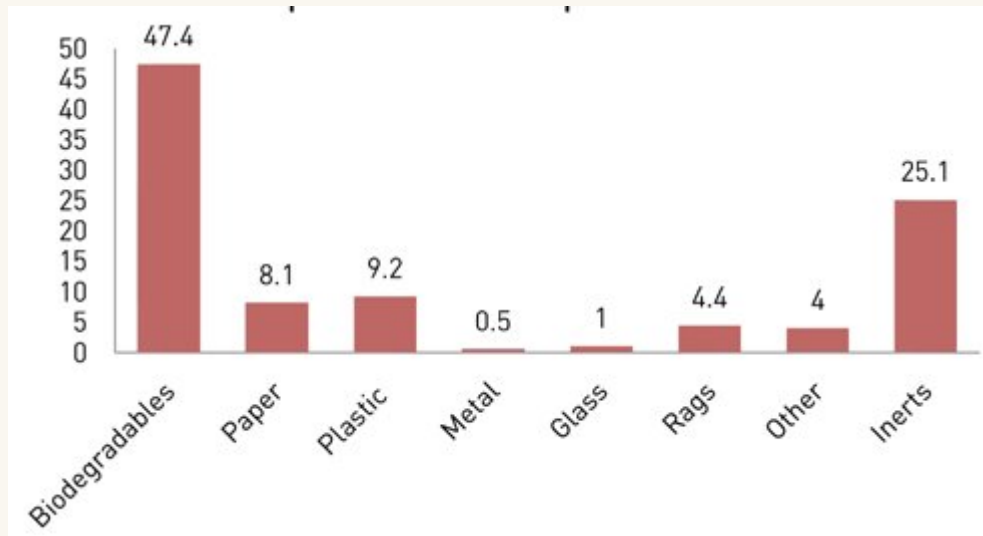


Fig 3.3 — Municipal Solid Waste Characterisation in India³ (solid waste management, p.252)

Methods of disposal — overview

There is no single method of refuse disposal equally suitable in all circumstances; the choice is governed by local factors such as cost and the availability of land and labour (Park 28e, Ch 15, p 909).

Method	Principle	Best suited to	Main limitation
(a) Dumping	Refuse tipped into low-lying land; bacterial action reduces volume and converts it to humus; reclaimed land leased out for cultivation	Dry refuse; land reclamation (Kolkata disposes of its refuse by dumping)	Condemned by the WHO Expert Committee (1967) as "a most insanitary method of disposal of refuse" — to be outlawed and replaced by sound procedures
(b) Controlled tipping / sanitary landfill	Refuse placed in a prepared trench or area, compacted and covered with earth at the end of each working day	Method of choice where suitable land is available	Needs land near the city and vehicles; leachate risk
(c) Incineration	Hygienic disposal by burning	Where suitable land is not available; hospital refuse	Fine ash in Indian refuse hampers burning; costly; loses manurial value
(d) Composting	Combined disposal of refuse with nightsoil or sludge into humus-like compost	Municipalities with population over 100,000 (Environmental Hygiene Committee, 1949)	Needs land, labour and a market for the compost
(e) Manure pits	Household pit filled daily and earth-covered	Rural areas with no collection system	Individual householder effort; two pits needed
(f) Burial	Daily trench filling with earth cover	Small camps	Only for small numbers; not a municipal option

(Park 28e, Ch 15, pp 909–910; Suryakantha, Ch 12, p 95)

Controlled tipping (sanitary landfill)

It differs from ordinary dumping in that the material is placed in a trench or prepared area, **adequately compacted, and covered with earth at the end of the working day**. “Modified sanitary landfill” is the term for operations where compaction and covering are done only **once or twice a week**. Three methods are used (Park 28e, Ch 15, p 909).

Method	Terrain / indication	Specifications
Trench method	Where level ground is available	Trench 2–3 m (6–10 ft) deep and 4–12 m (12–36 ft) wide; refuse compacted and covered with excavated earth. With compacted refuse filled to a depth of 2 m (6 ft), one acre of land per year is required for 10,000 population
Ramp method	Where the terrain is moderately sloping	Some excavation is done to secure the covering material
Area method	For land depressions, disused quarries and clay pits	Refuse deposited, packed and consolidated in uniform layers up to 2–2.5 m (6–8 ft) deep; each layer sealed on its exposed surface with a mud cover at least 30 cm (12 inches) thick ; often needs supplemental earth from outside

Changes in buried refuse (Park 28e, Ch 15, p 909):

- ▶ Chemical, bacteriological and physical changes occur in the buried refuse.
- ▶ Temperature rises to **over 60°C within 7 days**, killing all pathogens and hastening decomposition.
- ▶ It then takes **2 to 3 weeks to cool down**.
- ▶ Complete decomposition into an innocuous mass takes **4 to 6 months**; the built-up land is called “**made soil**”, fit for gardening (Suryakantha, Ch 12, p 95).
- ▶ Earth sealing prevents fly and rodent infestation and suppresses smell and dust; the bulldozer, which spreads, trims and lays top soil, has mechanised the method.
- ▶ **Tipping of refuse into water must never be done** — decomposing organic matter creates an odour nuisance.

Composting

Composting is the **combined disposal of refuse with nightsoil or sludge** — organic matter breaks down under bacterial action into a stable humus-like **compost** of considerable manurial value; by-products are carbon dioxide, water and heat. The heat, **about 60°C or higher over several days**, **destroys fly eggs and larvae, weed seeds and pathogenic agents**, so the compost carries few or no disease-producing organisms and is a good soil builder containing small amounts of nitrates and phosphates (Park 28e, Ch 15, p 910).

Feature	Bangalore method (anaerobic; hot fermentation process)	Mechanical composting (aerobic)
Origin	Developed under the Indian Council of Agricultural Research at the Indian Institute of Science, Bangalore ; recommended for disposal of town wastes and nightsoil	In vogue in Holland, Germany, Switzerland, Israel; pilot plants offered by Delhi, Nagpur, Mumbai, Chennai, Pune, Allahabad, Hyderabad, Lucknow, Kanpur

Feature	Bangalore method (anaerobic; hot fermentation process)	Mechanical composting (aerobic)
Site	Trenches not less than 800 m (½ mile) from city limits	Mechanical plant
Dimensions	90 cm (3 ft) deep, 1.5–2.5 m (5–8 ft) broad, 4.5–10 m (15–30 ft) long; depths greater than 90 cm are not recommended because decomposition is slow	—
Procedure	First a layer of refuse 15 cm (6 in) at the bottom, then nightsoil 5 cm (2 in); alternate layers of 15 cm and 5 cm until the heap rises 30 cm (1 ft) above ground level; top layer refuse, at least 25 cm (9 in) thick; heap covered with excavated earth	Refuse first cleared of salvageable material (rags, bones, metal, glass), pulverised to less than 2 inches, mixed with sewage, sludge or nightsoil in a rotating machine and incubated
Controls	Correct layering; if properly laid, a man's legs will not sink when walking over the compost mass	Carbon–nitrogen ratio, temperature, moisture, pH and aeration
Time	Heat over 60°C within 7 days, persisting 2–3 weeks; compost ready in 4–6 months	Entire process complete in 4–6 weeks

(Park 28e, Ch 15, p 910; Suryakantha, Ch 12, p 95)

Incineration, manure pits and burial

- **Incineration** — hygienic disposal by burning; the method of choice where suitable land is not available, and hospital refuse, being particularly dangerous, is best disposed of by incineration. Practised in industrialised countries, especially large cities, for want of land.
- **Why it is not popular in India** — refuse here carries a fair proportion of fine ash that makes burning difficult, so dust or ash must first be separated; this means heavy outlay and manipulative difficulty in the incinerator, and burning is a loss to the community of much-needed manure (Park 28e, Ch 15, pp 909–910).
- **Manure pits** — the answer for rural India, where there is no collection system and refuse thrown around houses grossly pollutes the soil. The householder digs a pit into which garbage, cattle dung, straw and leaves are dumped and covered with earth after each day's dumping; two pits are needed, one in use while the other is closed, and in 5 to 6 months the refuse becomes manure for the field (Park 28e, Ch 15, p 910).
- **Burial** — for small camps. A trench 1.5 m wide and 2 m deep is dug and the day's refuse covered with 20–30 cm of earth; when the level reaches 40 cm from ground level the trench is filled, compacted, and a new one dug. Contents are removed after 4–6 months for the fields, and 1 m of trench per 20 persons fills in about a week (Park 28e, Ch 15, p 910). Unlike composting, only refuse — not human excreta — is buried (Suryakantha, Ch 12, p 95).

Statutory framework and current practice in India

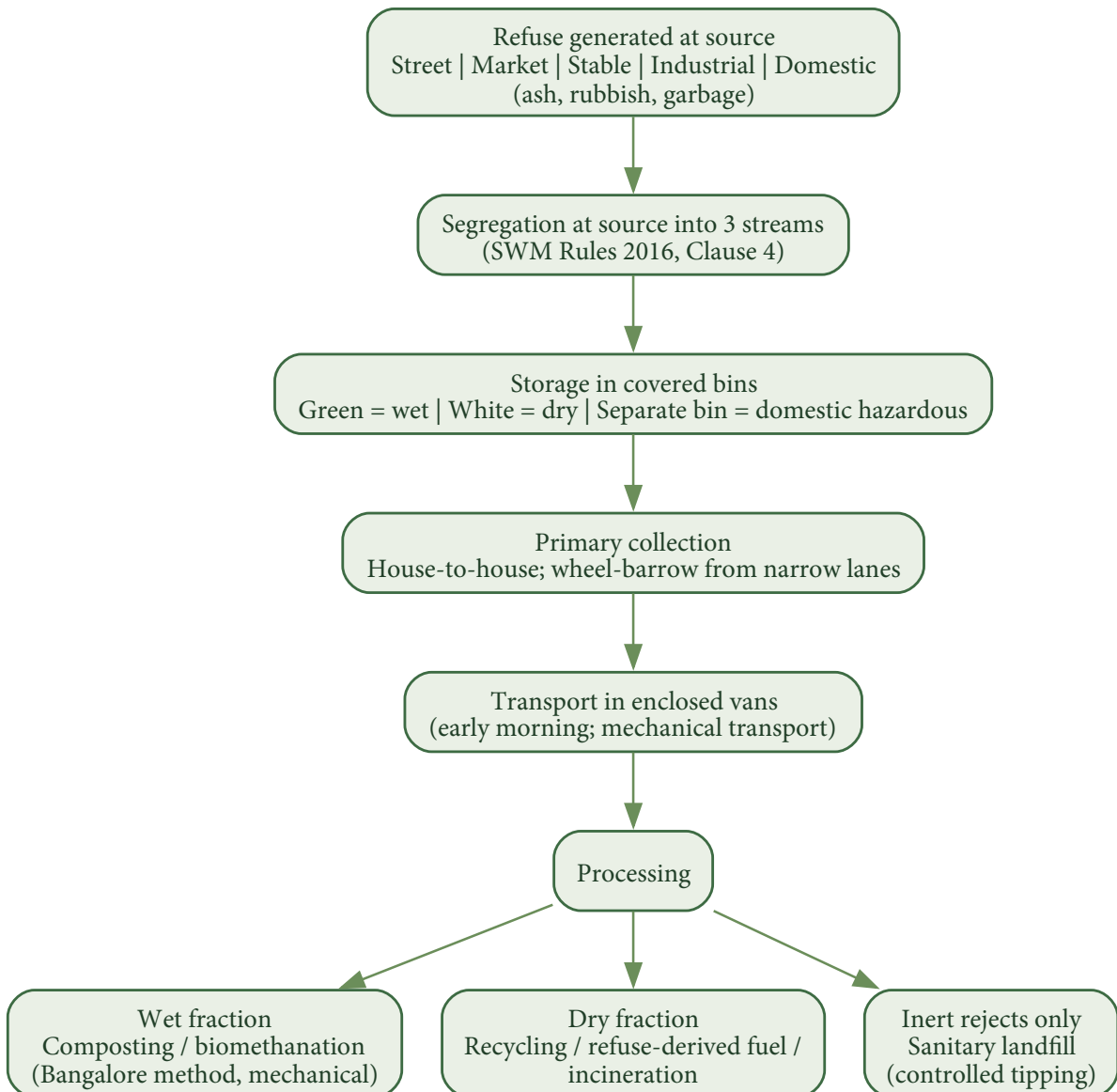
- **Solid Waste Management Rules, 2016, Clause 4** — every waste generator shall segregate and store waste in three streams: bio-degradable, non-bio-degradable and domestic hazardous, and hand it to authorised waste pickers or collectors; used sanitary waste is securely wrapped and placed in the dry bin.

- ▶ **Bin colours** — wet waste in a **covered green bin**, dry waste in a **covered white bin**; no colour is fixed for domestic hazardous waste, so the urban local body decides.
- ▶ **Clause 15** — local authorities must **stop landfilling of mixed waste**, allow **only non-usable, non-recyclable, non-biodegradable, non-combustible, non-reactive inert waste and processing rejects** into a sanitary landfill, and **bio-mine or bio-remediate** old dumpsites.
- ▶ **Collection** — house-to-house collection is by far the best method; the **Environmental Hygiene Committee (1949)** advised collecting from individual houses and not only from public bins, replacing **open refuse carts with enclosed vans**, and mechanical transport wherever possible. Refuse disposal cannot be solved without **public education**.

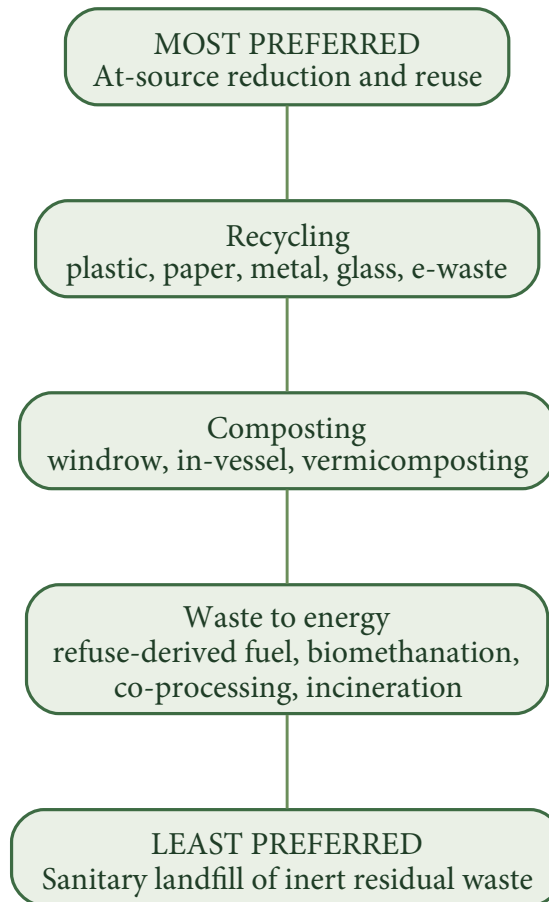
(Manual on Municipal Solid Waste Management, Ch 2, pp 175–186 and Ch 4, p 391; Park 28e, Ch 15, p 909; Suryakantha, Ch 12, p 94)

DIAGRAMS TO DRAW — IMAGE SUGGESTIONS

- ▶ **Diagram 1 — Trench method of controlled tipping (cross-section)**. Labels: *ground level; trench 2–3 m deep; trench 4–12 m wide; compacted refuse layer; daily earth cover; made soil above; temperature over 60°C within 7 days.*
- ▶ **Diagram 2 — Bangalore method trench (cross-section)**. Labels: *trench 90 cm deep, 1.5–2.5 m broad, 4.5–10 m long; bottom layer refuse 15 cm; nightsoil 5 cm; alternate layers; top layer refuse 25 cm; heap 30 cm above ground level; earth cover; sited 800 m from city limits.*



Municipal refuse management chain — from generation to final disposal under the Solid Waste Management Rules, 2016



Integrated Solid Waste Management hierarchy — most preferred to least preferred (Manual on Municipal Solid Waste Management, Ch 3, p 249)

High-yield exam pointers

- ▶ **Mnemonic, Park's six methods — “D-C-I-C-M-B”:** Dumping, Controlled tipping, Incineration, Composting, Manure pits, Burial. Five sources: street, market, stable litter, industrial, domestic (= ash + rubbish + garbage).
- ▶ **Numbers:** per capita 0.25–2.5 kg/day; trench method 2–3 m deep × 4–12 m wide, **1 acre/year per 10,000 population**; area method 2–2.5 m layers with 30 cm mud seal; buried refuse **>60°C in 7 days**, cools in 2–3 weeks, decomposed in 4–6 months.
- ▶ **Bangalore method** = anaerobic hot fermentation, ICAR at the Indian Institute of Science; trench **90 cm deep**; layers refuse **15 cm** : nightsoil **5 cm**; top layer refuse **25 cm**; ready in **4–6 months**. **Mechanical composting** = aerobic, pulverise to <2 inches, ready in **4–6 weeks**.
- ▶ **Environmental Hygiene Committee (1949):** house-to-house collection, enclosed vans for open carts, composting for municipalities over **100,000 population**.
- ▶ **Incineration** = choice when land is unavailable and for **hospital refuse**; unpopular in India — fine ash, cost, loss of manure.
- ▶ **Solid Waste Management Rules, 2016:** three streams at source — bio-degradable, non-bio-degradable, domestic hazardous; only inert rejects to landfill; old dumpsites **bio-mined / bio-remediated**.
- ▶ **Indian denominator (CPCB 2014–15):** 1,43,449 metric tonnes/day urban municipal solid waste, **40–60 per cent organic**.

NOTE — FOR REFERENCE, NOT TO BE WRITTEN IN THE EXAM

For reference only, not written in the exam: the quantum figures quoted are Central Pollution Control Board estimates for 2014–15, the latest in the retrieved corpus, and Suryakantha gives 4–6 months for manure pits where Park gives 5–6 months.

Recent advances [RA]

Current as of 2026-08-18 · depth: elaborate · sources: PubMed, WHO health-care waste fact sheet 2024

Programme and policy updates (India)

- ▶ **Bio-mining is now the working route out of legacy dumpsites.** At the 47.7-acre Ariyamangalam dumpyard, Tiruchirappalli, landfill mining began in 2019 in compliance with the Solid Waste Management Rules, 2016 — excavation, stabilisation with a bioculture that shrinks the heap by 35–40 per cent, trommel screening into soil-like material, refuse-derived fuel, recyclables and inerts, then engineered disposal. Soil-like material is **40–80 per cent** of the recovered fraction, and **77 per cent of India's waste is still open-dumped**, which is the gap this route is meant to close [PMID 40995675].
- ▶ **The informal waste sector is being brought into formal governance.** A weighted vulnerability index across Nagpur, Mumbai, Ghaziabad and Muzaffarnagar — built on wages, working hours, gender mix and use of protective equipment — ranked **Muzaffarnagar most vulnerable (0.71) and Mumbai least (0.25)**, and argues for rights-based integration of waste pickers into municipal systems rather than leaving collection to an unprotected workforce [PMID 41852047].
- ▶ **Swachh Bharat Mission now has hard outcome data.** Across **640 districts over 2011–2020**, districts where more than 30 per cent of households received a toilet under the mission recorded **5.3 fewer infant deaths per 1,000 live births ($p < 0.05$)**, scaling to an estimated **60,000–70,000 infant deaths averted annually** [PMID 39223201].

Current national data

- ▶ **Open defecation, NFHS-4 to NFHS-5:** rural households practising open defecation fell from **55 per cent in 2015–16 to 27 per cent in 2019–21**, and about **one-fifth** of the simultaneous gain in rural child height is statistically attributable to that fall [PMID 40138892].
- ▶ **Occupational burden in the municipal waste workforce:** among **1,200 solid-waste workers** screened in Varanasi between July 2022 and August 2023, **71.5 per cent** used smokeless tobacco and **32.9 per cent** of those screened were positive for oral potentially malignant lesions or oral cancer, leukoplakia commonest (**60.6 per cent**) — the figure to quote for sanitation-worker health [PMID 39462398].
- ▶ **Groundwater around dumpsites (2025 three-city study):** near Kolkata's Dhapa dumpsite the nitrate hazard quotient exceeded 1 for **100 per cent of children in every season**, and the heavy-metal pollution index exceeded threshold in all Kolkata samples; Bengaluru showed nitrate risk rising from 20 per cent of adults pre-monsoon to 66.7 per cent post-monsoon — leachate harm measured, which is precisely what controlled tipping exists to prevent [PMID 41390581].

Elimination, surveillance and disease status

- ▶ **Refuse handling is now a quantified dengue risk factor.** In a Delhi hotspot survey of 347 households, living within 200 m of a garbage collection site raised household odds of dengue 7.26-fold (95% CI 2.55–20.68), and relying on common-site or self-dumping instead of door-to-door collection raised it 2.72-fold (95% CI 1.22–6.03) [PMID 39043792].
- ▶ **Discarded solid waste ranks third as an Aedes breeding habitat.** Across 3,400 Delhi houses and 27,776 water-holding containers (2018–19), the breeding preference ratio was highest for clay pots (4.33) and metal containers (3.94), with solid waste next at 3.30 — the entomological case for treating uncollected refuse as vector-control work, not merely a nuisance [PMID 37641822].

Guideline and legislative updates

- ▶ **Health-care waste classification was corrected after COVID-19.** A WHO analysis (February 2022) restated that only 10–15 per cent of routine health-care waste is hazardous; classifying all COVID-19 waste as infectious almost quadrupled waste volumes in New Delhi at the May 2021 peak. About 30 per cent of health facilities worldwide, and 50 per cent in least-developed countries, do not segregate hazardous from general waste, and health care contributes roughly 4–5 per cent of global greenhouse-gas emissions [PMID 35123679].

Global and WHO developments

- ▶ **WHO's current position on health-care waste (October 2024):** 15 per cent of health-care waste is hazardous; only 61 per cent of hospitals had basic waste management services in 2021, falling to 25 per cent of facilities in fragile settings (2023); non-burn treatment — autoclaving, steam, microwaving — is preferred to incineration wherever feasible, a shift from the textbook position that incineration is the automatic answer for hospital refuse [WHO 2024-Oct].

SOURCE & COVERAGE

Park's Textbook of Preventive and Social Medicine 28e (Ch 15, pp 908–910); Suryakantha — Community Medicine with Recent Advances (Ch 12, Disposal of Wastes, pp 94–95); Manual on Municipal Solid Waste Management with the Solid Waste Management Rules, 2016 (Ch 2, pp 175–186; Ch 3, pp 249–252; Ch 4, p 391); Central Pollution Control Board municipal solid waste estimates, 2014–15; Report of the Environmental Hygiene Committee, Government of India, 1949.

Recent-advances sources: 40995675 — Assessment of phytotoxicity in soil-like material recovered from landfill mining of legacy waste in open dumpsites, Waste Manag Res 2025, doi:[10.1177/0734242X251364645](https://doi.org/10.1177/0734242X251364645); 41852047 — Vulnerability assessment of informal waste workers in India and its implications towards sustainable waste management policy, Waste Manag Res 2026, doi:[10.1177/0734242X261425150](https://doi.org/10.1177/0734242X261425150) (abstract-only); 39223201 — Toilet construction under the Swachh Bharat Mission and infant mortality in India, Sci Rep 2024, doi:[10.1038/s41598-024-71268-8](https://doi.org/10.1038/s41598-024-71268-8); 40138892 — Changes in child height and open defecation in rural India: understanding improvements between the two most recent demographic surveys, Econ Hum Biol 2025, doi:[10.1016/j.ehb.2025.101484](https://doi.org/10.1016/j.ehb.2025.101484) (abstract-only); 39462398 — Prevalence of oral potentially malignant lesions, tobacco use, and effect of cessation strategies among solid waste management workers in Northern India, BMC Oral Health 2024, doi:[10.1186/s12903-024-05087-8](https://doi.org/10.1186/s12903-024-05087-8); 41390581 — Seasonal dynamics of groundwater pollution and health risks in municipal solid waste-affected urban settlements of Bengaluru, Kolkata and Durgapur, India, Environ Geochem Health 2025, doi:[10.1007/s10653-025-02928-5](https://doi.org/10.1007/s10653-025-02928-5) (abstract-only); 39043792 — Socio-ecological predictors of dengue in high incidence area of Delhi, India, Sci Rep 2024, doi:[10.1038/s41598-024-67909-7](https://doi.org/10.1038/s41598-024-67909-7); 37641822 — Aedes aegypti container preference for oviposition and its possible implications for dengue vector surveillance in Delhi, India, Epidemiol Health 2023, doi:[10.4178/epih.e2023073](https://doi.org/10.4178/epih.e2023073); 35123679 — WHO concerned over COVID-19 health-care waste, Lancet 2022, doi:[10.1016/S0140-6736\(22\)00225-2](https://doi.org/10.1016/S0140-6736(22)00225-2); WHO health-care waste fact sheet, October 2024.

♦ ♦ ♦